

計算問題 —小テスト集—

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0.1 はじめに

本書は小テスト用のプリント集です。

学校では知識を与えることが主になりがちですが、計算能力等の能力が年々落ちてきているように感じています。数学では計算の遅さや不正確さが原因で話の理解が遅れていくこともあり、計算能力の不足は学習面に悪影響があります。そこで、脳を鍛え計算能力を身に付けさせるために行っていた小テストをまとめたものが本書です。

本書の目的は次のようになります。

目的 脳を鍛えること

元々は学校での授業で行う目的で作成していますが、塾での勉強や自習学習での利用、企業での研修、高齢者施設などでのボケ防止等でも利用出来るかもしれません。

PDF は GitHub でも公開中です。 https://github.com/tyoji/math_workbook

0.2 使い方

次のような使い方を想定していますが、学習者の状況に応じて自由に利用してもらえればと思います。

- (1) 問題のページを印刷する
- (2) 黒板に 1 問目の解き方を解説し答えを書いておく
- (3) 解き方や解答等について質問を受け付ける
- (4) 質問がなくなれば、プリントを配布し解かせる
- (5) わからなければ、黒板を見て 1 問目を丸写しし、2 問目以降は数字を変えて同じように解く
- (6) 時間 (5 分から 10 分程度) 経過したら終了
- (7) 隣の人とプリントを交換し採点をさせる
- (8) 採点が終われば回収し、点数の集計をする

注意事項

- 生徒の能力を教師側が決めつけない。
- 小テストとして行うため、解いている時間は一人で静かに解かせる。一人で考えることこそが脳を鍛える。
- 問題を解くことを強要しない。
- 解けていなくてもあれこれ言わない。

プリントを配布前に第 1 問目の解説をし解答を書いておきます。これは、解き方がわからない人や勘違いしている人が小テストの時間を無駄に過ごさないためです。全員が一定数解けるようになったのであれば、省略してもいいです。

質問を受け付けるのも小テスト中に静かに取り組ませるためです。わからないなりに自力で考えることは脳の発達を促します。

小テストはわからない人が黒板を見ながら考えることに意味が有るので、邪魔になるような行為をするのであれば注意をします。

小テストの時間は解き終わらないぐらいの時間設定をします。具体的には教師が必死に解いてギリギリ解き終わるぐらいからその2倍ぐらいの時間を設定します。脳を鍛えることが目的の為、限界まで脳を使わせるように簡単に解き終わらないようにします。

採点は別の人のプリントを行います。これは、他人がどのように解いているかを知り自分の解き方を見直すきっかけにする為と、読めない文字を書いている事を指摘し他人に読めるように書くことを促す為です。学力の低い生徒が定期テストで読めない文字を書くことが多く、早めに指摘しておく意味を持ちます。

また、解答を提示する方法はいくつかあります。

- 口頭で読み上げる
- 黒板に書く
- 印刷して配布する

上にある方法がより教師生徒に負担が大きく、下にある方法が負担は軽いです。特に口頭で解答を読み上げると生徒は「聞く → 確認 → 丸付け」という事を繰り返します。聞く能力も鍛えられますし、採点時間が全員同じぐらいですから時間配分もしやすくなります。解答を読み上げる側も大変ではありますが。

答案を集めて集計をし、どの程度計算ができるかを確認します。

これを何度も繰り返すと多くの生徒が解き終わるようになります。

0.2.1 仕組み

人は猿が進化した動物なので、群れを作る性質を持ちます。具体的には、みんなと同じであろうとすることと、人よりも優れた存在になろうとすることです。

そこで、全員に同じ問題を何度も解かせるとサボっている生徒が出来なくてそれ以外の生徒が点数が伸びてきます。出来ていない生徒が少数になれば、その事を伝えてあげると周りにおいていかれることを恐れて取り組むようになります。

採点を隣の人と交換して行うもの同様の意味を持ちます。自分が出来ていないことを自分以外の人が知ることになるので、何とかしようと取り組むようになります。

こういった生徒のフォローの為に、テスト配布前に問題解説と質問受付を行います。

知識や技術の習得には本人の努力が不可欠です。他人から勉強をするように言われても本人の意思がなければ身につけません。そこで、勉強を促すのは自分自身やクラスメイト等からの緩やかなプレッシャーに任せます。教師側からの強要することは無いようにすることで、過度のプレッシャーが掛からないようになります。

目的は脳を鍛えることなので、静かに考える事が出来れば目的は達成です。考えた答えが正答でも誤答でも構いません。問題解説と小テストを繰り返す内に少しずつ正答に近づいていきます。この効果はどんなに勉強が出来ない生徒でもあらわれます。根気強く繰り返す必要があるかもしれませんが、全員同じように脳に負荷をかけて下さい。

第 1 章

数

1.1 自然数

1.1.1 加法

自然数の和を求める問題。

(1) $46 + 24$	(31) $31 + 91$	(61) $67 + 85$	(91) $44 + 95$
(2) $26 + 33$	(32) $43 + 35$	(62) $98 + 64$	(92) $48 + 99$
(3) $87 + 8$	(33) $94 + 89$	(63) $75 + 49$	(93) $43 + 83$
(4) $71 + 91$	(34) $22 + 23$	(64) $86 + 59$	(94) $42 + 23$
(5) $7 + 50$	(35) $58 + 20$	(65) $47 + 71$	(95) $19 + 29$
(6) $76 + 36$	(36) $7 + 60$	(66) $45 + 61$	(96) $11 + 72$
(7) $26 + 46$	(37) $67 + 72$	(67) $40 + 63$	(97) $59 + 21$
(8) $94 + 96$	(38) $15 + 3$	(68) $81 + 59$	(98) $59 + 66$
(9) $96 + 40$	(39) $2 + 82$	(69) $9 + 56$	(99) $24 + 43$
(10) $13 + 23$	(40) $74 + 66$	(70) $37 + 36$	(100) $85 + 82$
(11) $51 + 75$	(41) $46 + 0$	(71) $61 + 89$	(101) $26 + 89$
(12) $84 + 77$	(42) $18 + 92$	(72) $36 + 54$	(102) $83 + 67$
(13) $97 + 26$	(43) $98 + 82$	(73) $4 + 38$	(103) $41 + 91$
(14) $39 + 17$	(44) $37 + 85$	(74) $83 + 1$	(104) $54 + 69$
(15) $24 + 80$	(45) $0 + 59$	(75) $17 + 48$	(105) $82 + 99$
(16) $64 + 97$	(46) $9 + 87$	(76) $52 + 80$	(106) $5 + 10$
(17) $3 + 47$	(47) $16 + 22$	(77) $34 + 43$	(107) $75 + 24$
(18) $52 + 87$	(48) $52 + 18$	(78) $69 + 73$	(108) $41 + 12$
(19) $100 + 92$	(49) $49 + 74$	(79) $49 + 26$	(109) $8 + 56$
(20) $82 + 68$	(50) $47 + 4$	(80) $6 + 66$	(110) $34 + 52$
(21) $18 + 12$	(51) $70 + 22$	(81) $93 + 39$	(111) $91 + 78$
(22) $70 + 94$	(52) $36 + 93$	(82) $20 + 35$	(112) $30 + 57$
(23) $84 + 89$	(53) $94 + 21$	(83) $41 + 6$	(113) $55 + 10$
(24) $78 + 98$	(54) $2 + 73$	(84) $30 + 13$	(114) $81 + 95$
(25) $97 + 27$	(55) $50 + 55$	(85) $71 + 47$	(115) $93 + 69$
(26) $29 + 42$	(56) $67 + 38$	(86) $26 + 17$	(116) $47 + 49$
(27) $67 + 8$	(57) $66 + 88$	(87) $17 + 14$	(117) $66 + 6$
(28) $26 + 55$	(58) $58 + 91$	(88) $32 + 65$	(118) $44 + 52$
(29) $30 + 5$	(59) $48 + 99$	(89) $96 + 26$	(119) $75 + 79$
(30) $80 + 4$	(60) $74 + 91$	(90) $24 + 47$	(120) $41 + 58$

(1) $1 + 74$	(31) $67 + 7$	(61) $55 + 34$	(91) $10 + 10$
(2) $12 + 46$	(32) $36 + 69$	(62) $80 + 69$	(92) $38 + 58$
(3) $18 + 38$	(33) $48 + 29$	(63) $88 + 51$	(93) $26 + 16$
(4) $42 + 7$	(34) $87 + 18$	(64) $85 + 100$	(94) $16 + 3$
(5) $44 + 62$	(35) $89 + 92$	(65) $5 + 23$	(95) $32 + 24$
(6) $66 + 94$	(36) $23 + 51$	(66) $66 + 83$	(96) $68 + 96$
(7) $96 + 69$	(37) $62 + 95$	(67) $63 + 19$	(97) $10 + 42$
(8) $20 + 100$	(38) $22 + 3$	(68) $15 + 23$	(98) $83 + 54$
(9) $70 + 15$	(39) $31 + 22$	(69) $48 + 44$	(99) $100 + 48$
(10) $14 + 35$	(40) $95 + 98$	(70) $87 + 41$	(100) $59 + 38$
(11) $90 + 42$	(41) $48 + 96$	(71) $70 + 76$	(101) $17 + 58$
(12) $57 + 76$	(42) $42 + 66$	(72) $21 + 25$	(102) $6 + 65$
(13) $82 + 88$	(43) $58 + 15$	(73) $95 + 19$	(103) $63 + 15$
(14) $48 + 74$	(44) $4 + 96$	(74) $57 + 34$	(104) $9 + 63$
(15) $23 + 84$	(45) $60 + 98$	(75) $13 + 1$	(105) $89 + 42$
(16) $77 + 52$	(46) $27 + 82$	(76) $22 + 10$	(106) $95 + 22$
(17) $72 + 54$	(47) $82 + 12$	(77) $30 + 8$	(107) $73 + 85$
(18) $17 + 65$	(48) $85 + 60$	(78) $78 + 49$	(108) $0 + 99$
(19) $62 + 71$	(49) $39 + 57$	(79) $88 + 6$	(109) $53 + 60$
(20) $74 + 79$	(50) $49 + 98$	(80) $48 + 93$	(110) $16 + 6$
(21) $89 + 89$	(51) $1 + 4$	(81) $42 + 59$	(111) $49 + 22$
(22) $26 + 36$	(52) $85 + 90$	(82) $58 + 91$	(112) $30 + 28$
(23) $62 + 67$	(53) $100 + 23$	(83) $1 + 31$	(113) $20 + 7$
(24) $55 + 17$	(54) $53 + 97$	(84) $62 + 92$	(114) $16 + 40$
(25) $77 + 68$	(55) $21 + 11$	(85) $57 + 100$	(115) $38 + 54$
(26) $2 + 100$	(56) $18 + 53$	(86) $85 + 20$	(116) $32 + 23$
(27) $47 + 92$	(57) $13 + 16$	(87) $30 + 31$	(117) $41 + 93$
(28) $6 + 49$	(58) $65 + 34$	(88) $57 + 60$	(118) $95 + 1$
(29) $62 + 89$	(59) $70 + 8$	(89) $3 + 30$	(119) $29 + 7$
(30) $43 + 93$	(60) $63 + 17$	(90) $83 + 38$	(120) $5 + 28$

(1) $86 + 70$	(31) $61 + 81$	(61) $93 + 57$	(91) $6 + 3$
(2) $3 + 98$	(32) $61 + 56$	(62) $7 + 41$	(92) $100 + 57$
(3) $66 + 54$	(33) $53 + 13$	(63) $93 + 47$	(93) $95 + 5$
(4) $97 + 60$	(34) $52 + 81$	(64) $18 + 59$	(94) $95 + 40$
(5) $45 + 91$	(35) $25 + 24$	(65) $78 + 35$	(95) $52 + 39$
(6) $0 + 100$	(36) $47 + 17$	(66) $80 + 28$	(96) $2 + 97$
(7) $73 + 76$	(37) $86 + 28$	(67) $11 + 32$	(97) $75 + 19$
(8) $40 + 17$	(38) $41 + 59$	(68) $8 + 85$	(98) $61 + 6$
(9) $100 + 49$	(39) $97 + 41$	(69) $13 + 76$	(99) $87 + 60$
(10) $61 + 25$	(40) $1 + 24$	(70) $37 + 31$	(100) $56 + 75$
(11) $45 + 62$	(41) $15 + 72$	(71) $82 + 98$	(101) $84 + 40$
(12) $2 + 29$	(42) $19 + 36$	(72) $57 + 38$	(102) $51 + 59$
(13) $69 + 2$	(43) $48 + 69$	(73) $72 + 3$	(103) $56 + 82$
(14) $96 + 15$	(44) $55 + 100$	(74) $45 + 2$	(104) $7 + 95$
(15) $20 + 38$	(45) $50 + 92$	(75) $88 + 69$	(105) $2 + 63$
(16) $54 + 26$	(46) $66 + 11$	(76) $3 + 33$	(106) $42 + 41$
(17) $25 + 17$	(47) $22 + 84$	(77) $38 + 19$	(107) $3 + 28$
(18) $38 + 72$	(48) $28 + 41$	(78) $19 + 73$	(108) $27 + 38$
(19) $40 + 56$	(49) $54 + 40$	(79) $32 + 57$	(109) $21 + 12$
(20) $9 + 58$	(50) $65 + 20$	(80) $88 + 86$	(110) $23 + 82$
(21) $56 + 84$	(51) $48 + 52$	(81) $77 + 42$	(111) $35 + 42$
(22) $27 + 50$	(52) $11 + 8$	(82) $80 + 77$	(112) $23 + 79$
(23) $94 + 58$	(53) $46 + 46$	(83) $100 + 2$	(113) $77 + 2$
(24) $99 + 12$	(54) $45 + 79$	(84) $100 + 6$	(114) $92 + 82$
(25) $4 + 78$	(55) $16 + 57$	(85) $57 + 47$	(115) $100 + 82$
(26) $40 + 16$	(56) $85 + 74$	(86) $10 + 34$	(116) $39 + 57$
(27) $95 + 67$	(57) $56 + 38$	(87) $74 + 64$	(117) $32 + 90$
(28) $11 + 77$	(58) $11 + 54$	(88) $2 + 44$	(118) $16 + 42$
(29) $48 + 54$	(59) $46 + 42$	(89) $33 + 51$	(119) $49 + 24$
(30) $2 + 1$	(60) $26 + 58$	(90) $61 + 76$	(120) $23 + 100$

(1) $77 + 46$	(31) $95 + 76$	(61) $11 + 100$	(91) $18 + 15$
(2) $27 + 7$	(32) $29 + 35$	(62) $75 + 33$	(92) $68 + 5$
(3) $51 + 30$	(33) $51 + 86$	(63) $56 + 43$	(93) $18 + 85$
(4) $31 + 12$	(34) $55 + 45$	(64) $37 + 23$	(94) $36 + 51$
(5) $84 + 30$	(35) $79 + 51$	(65) $31 + 50$	(95) $49 + 33$
(6) $41 + 59$	(36) $93 + 56$	(66) $21 + 88$	(96) $81 + 15$
(7) $87 + 67$	(37) $25 + 54$	(67) $79 + 92$	(97) $91 + 6$
(8) $28 + 71$	(38) $35 + 15$	(68) $85 + 71$	(98) $76 + 45$
(9) $52 + 68$	(39) $78 + 83$	(69) $72 + 3$	(99) $46 + 50$
(10) $24 + 18$	(40) $87 + 87$	(70) $85 + 82$	(100) $8 + 51$
(11) $75 + 82$	(41) $49 + 60$	(71) $48 + 48$	(101) $27 + 17$
(12) $97 + 1$	(42) $97 + 58$	(72) $84 + 89$	(102) $45 + 73$
(13) $35 + 20$	(43) $46 + 51$	(73) $99 + 26$	(103) $89 + 15$
(14) $94 + 43$	(44) $42 + 37$	(74) $20 + 82$	(104) $36 + 59$
(15) $28 + 98$	(45) $70 + 79$	(75) $80 + 85$	(105) $96 + 86$
(16) $100 + 29$	(46) $28 + 65$	(76) $88 + 72$	(106) $73 + 2$
(17) $94 + 1$	(47) $12 + 3$	(77) $68 + 39$	(107) $44 + 16$
(18) $18 + 54$	(48) $32 + 5$	(78) $79 + 64$	(108) $29 + 59$
(19) $65 + 87$	(49) $72 + 91$	(79) $68 + 29$	(109) $35 + 46$
(20) $89 + 69$	(50) $66 + 54$	(80) $54 + 78$	(110) $10 + 69$
(21) $60 + 97$	(51) $93 + 75$	(81) $38 + 49$	(111) $18 + 83$
(22) $83 + 15$	(52) $48 + 12$	(82) $38 + 90$	(112) $14 + 72$
(23) $76 + 17$	(53) $79 + 38$	(83) $40 + 8$	(113) $51 + 7$
(24) $74 + 99$	(54) $81 + 22$	(84) $52 + 82$	(114) $19 + 6$
(25) $35 + 88$	(55) $17 + 16$	(85) $45 + 83$	(115) $31 + 39$
(26) $30 + 93$	(56) $1 + 64$	(86) $62 + 91$	(116) $21 + 27$
(27) $88 + 62$	(57) $25 + 54$	(87) $39 + 48$	(117) $14 + 92$
(28) $75 + 21$	(58) $5 + 35$	(88) $95 + 28$	(118) $33 + 70$
(29) $95 + 74$	(59) $47 + 83$	(89) $54 + 62$	(119) $34 + 60$
(30) $84 + 45$	(60) $88 + 4$	(90) $3 + 83$	(120) $38 + 43$

(1) $76 + 55$	(31) $91 + 56$	(61) $7 + 33$	(91) $100 + 31$
(2) $1 + 79$	(32) $68 + 15$	(62) $97 + 78$	(92) $23 + 4$
(3) $24 + 32$	(33) $58 + 92$	(63) $37 + 64$	(93) $30 + 34$
(4) $61 + 48$	(34) $77 + 66$	(64) $69 + 51$	(94) $86 + 75$
(5) $10 + 89$	(35) $99 + 24$	(65) $21 + 43$	(95) $35 + 57$
(6) $35 + 75$	(36) $23 + 84$	(66) $4 + 78$	(96) $20 + 51$
(7) $39 + 98$	(37) $70 + 74$	(67) $69 + 97$	(97) $96 + 65$
(8) $91 + 69$	(38) $9 + 38$	(68) $14 + 97$	(98) $28 + 53$
(9) $26 + 55$	(39) $95 + 61$	(69) $68 + 68$	(99) $94 + 62$
(10) $47 + 78$	(40) $4 + 4$	(70) $36 + 14$	(100) $9 + 97$
(11) $40 + 5$	(41) $89 + 5$	(71) $88 + 2$	(101) $20 + 58$
(12) $8 + 63$	(42) $8 + 88$	(72) $41 + 20$	(102) $13 + 22$
(13) $98 + 51$	(43) $48 + 62$	(73) $80 + 37$	(103) $71 + 30$
(14) $30 + 46$	(44) $0 + 62$	(74) $40 + 56$	(104) $25 + 100$
(15) $17 + 80$	(45) $73 + 43$	(75) $73 + 90$	(105) $98 + 54$
(16) $25 + 21$	(46) $71 + 72$	(76) $91 + 33$	(106) $37 + 32$
(17) $25 + 85$	(47) $87 + 4$	(77) $28 + 95$	(107) $56 + 36$
(18) $26 + 4$	(48) $61 + 86$	(78) $97 + 0$	(108) $57 + 71$
(19) $67 + 24$	(49) $49 + 91$	(79) $64 + 42$	(109) $58 + 86$
(20) $93 + 51$	(50) $46 + 92$	(80) $91 + 87$	(110) $32 + 11$
(21) $79 + 7$	(51) $99 + 85$	(81) $61 + 65$	(111) $42 + 86$
(22) $69 + 30$	(52) $13 + 6$	(82) $61 + 15$	(112) $37 + 66$
(23) $82 + 10$	(53) $28 + 49$	(83) $95 + 40$	(113) $90 + 33$
(24) $61 + 51$	(54) $29 + 46$	(84) $9 + 3$	(114) $7 + 47$
(25) $9 + 91$	(55) $56 + 80$	(85) $15 + 60$	(115) $46 + 53$
(26) $77 + 33$	(56) $69 + 50$	(86) $49 + 49$	(116) $12 + 2$
(27) $7 + 69$	(57) $81 + 66$	(87) $10 + 45$	(117) $90 + 100$
(28) $44 + 32$	(58) $24 + 38$	(88) $76 + 88$	(118) $35 + 66$
(29) $88 + 11$	(59) $47 + 41$	(89) $44 + 88$	(119) $72 + 45$
(30) $79 + 33$	(60) $33 + 20$	(90) $15 + 85$	(120) $55 + 92$

(1) $30 + 63$	(31) $87 + 80$	(61) $29 + 75$	(91) $53 + 99$
(2) $21 + 74$	(32) $55 + 30$	(62) $69 + 78$	(92) $76 + 24$
(3) $97 + 27$	(33) $36 + 4$	(63) $57 + 43$	(93) $96 + 71$
(4) $28 + 6$	(34) $3 + 52$	(64) $40 + 1$	(94) $81 + 55$
(5) $2 + 65$	(35) $16 + 99$	(65) $61 + 47$	(95) $58 + 81$
(6) $88 + 55$	(36) $9 + 84$	(66) $27 + 27$	(96) $97 + 9$
(7) $23 + 83$	(37) $83 + 6$	(67) $43 + 43$	(97) $0 + 46$
(8) $7 + 32$	(38) $96 + 46$	(68) $100 + 37$	(98) $44 + 87$
(9) $87 + 33$	(39) $26 + 37$	(69) $69 + 44$	(99) $50 + 83$
(10) $29 + 30$	(40) $34 + 92$	(70) $52 + 55$	(100) $24 + 97$
(11) $76 + 8$	(41) $63 + 30$	(71) $12 + 85$	(101) $85 + 40$
(12) $48 + 65$	(42) $60 + 20$	(72) $15 + 100$	(102) $71 + 96$
(13) $27 + 12$	(43) $73 + 63$	(73) $88 + 45$	(103) $88 + 19$
(14) $35 + 43$	(44) $94 + 11$	(74) $23 + 58$	(104) $62 + 45$
(15) $56 + 78$	(45) $85 + 1$	(75) $17 + 83$	(105) $11 + 100$
(16) $19 + 57$	(46) $72 + 20$	(76) $56 + 95$	(106) $73 + 67$
(17) $45 + 46$	(47) $98 + 79$	(77) $96 + 15$	(107) $32 + 17$
(18) $56 + 72$	(48) $8 + 24$	(78) $96 + 24$	(108) $32 + 95$
(19) $51 + 54$	(49) $65 + 26$	(79) $84 + 95$	(109) $18 + 14$
(20) $46 + 13$	(50) $100 + 33$	(80) $48 + 7$	(110) $59 + 85$
(21) $12 + 7$	(51) $11 + 9$	(81) $52 + 45$	(111) $23 + 32$
(22) $31 + 100$	(52) $16 + 72$	(82) $48 + 83$	(112) $38 + 100$
(23) $24 + 26$	(53) $84 + 74$	(83) $95 + 61$	(113) $10 + 88$
(24) $56 + 96$	(54) $49 + 90$	(84) $97 + 29$	(114) $87 + 61$
(25) $97 + 20$	(55) $80 + 89$	(85) $34 + 82$	(115) $41 + 31$
(26) $55 + 57$	(56) $51 + 16$	(86) $91 + 99$	(116) $37 + 51$
(27) $40 + 93$	(57) $47 + 57$	(87) $64 + 83$	(117) $79 + 17$
(28) $63 + 26$	(58) $95 + 35$	(88) $13 + 88$	(118) $62 + 5$
(29) $54 + 71$	(59) $70 + 96$	(89) $12 + 48$	(119) $87 + 25$
(30) $78 + 15$	(60) $97 + 1$	(90) $98 + 63$	(120) $20 + 25$

(1) $91 + 29$	(31) $88 + 78$	(61) $7 + 45$	(91) $90 + 57$
(2) $13 + 15$	(32) $78 + 77$	(62) $70 + 73$	(92) $27 + 83$
(3) $92 + 59$	(33) $53 + 78$	(63) $8 + 88$	(93) $51 + 33$
(4) $1 + 46$	(34) $73 + 40$	(64) $20 + 70$	(94) $68 + 26$
(5) $14 + 90$	(35) $51 + 96$	(65) $73 + 51$	(95) $79 + 81$
(6) $79 + 97$	(36) $71 + 25$	(66) $98 + 38$	(96) $6 + 16$
(7) $82 + 5$	(37) $72 + 14$	(67) $12 + 97$	(97) $33 + 71$
(8) $46 + 80$	(38) $62 + 7$	(68) $32 + 63$	(98) $64 + 32$
(9) $56 + 34$	(39) $92 + 48$	(69) $73 + 28$	(99) $34 + 12$
(10) $18 + 50$	(40) $2 + 7$	(70) $90 + 23$	(100) $33 + 77$
(11) $97 + 69$	(41) $17 + 35$	(71) $34 + 73$	(101) $65 + 18$
(12) $17 + 6$	(42) $13 + 56$	(72) $6 + 87$	(102) $19 + 68$
(13) $38 + 16$	(43) $94 + 3$	(73) $53 + 24$	(103) $28 + 90$
(14) $13 + 45$	(44) $60 + 57$	(74) $21 + 61$	(104) $68 + 40$
(15) $40 + 20$	(45) $15 + 28$	(75) $57 + 46$	(105) $70 + 48$
(16) $87 + 68$	(46) $83 + 70$	(76) $33 + 15$	(106) $66 + 67$
(17) $4 + 34$	(47) $14 + 84$	(77) $31 + 0$	(107) $43 + 90$
(18) $25 + 4$	(48) $1 + 74$	(78) $72 + 78$	(108) $84 + 2$
(19) $42 + 22$	(49) $57 + 72$	(79) $88 + 28$	(109) $26 + 48$
(20) $24 + 38$	(50) $84 + 87$	(80) $53 + 50$	(110) $78 + 15$
(21) $61 + 38$	(51) $39 + 71$	(81) $39 + 9$	(111) $95 + 32$
(22) $26 + 97$	(52) $77 + 67$	(82) $63 + 95$	(112) $76 + 88$
(23) $96 + 75$	(53) $79 + 85$	(83) $73 + 73$	(113) $68 + 32$
(24) $71 + 47$	(54) $76 + 87$	(84) $64 + 3$	(114) $24 + 39$
(25) $54 + 45$	(55) $83 + 46$	(85) $71 + 100$	(115) $99 + 43$
(26) $78 + 72$	(56) $20 + 39$	(86) $46 + 43$	(116) $22 + 2$
(27) $14 + 13$	(57) $35 + 68$	(87) $11 + 86$	(117) $94 + 63$
(28) $54 + 29$	(58) $72 + 1$	(88) $39 + 83$	(118) $40 + 49$
(29) $50 + 34$	(59) $98 + 87$	(89) $25 + 15$	(119) $29 + 88$
(30) $60 + 76$	(60) $3 + 39$	(90) $85 + 69$	(120) $96 + 79$

(1) $61 + 16$	(31) $77 + 30$	(61) $64 + 90$	(91) $72 + 78$
(2) $11 + 13$	(32) $20 + 54$	(62) $53 + 80$	(92) $36 + 33$
(3) $26 + 73$	(33) $83 + 81$	(63) $78 + 99$	(93) $81 + 85$
(4) $97 + 96$	(34) $86 + 51$	(64) $43 + 80$	(94) $75 + 54$
(5) $75 + 6$	(35) $25 + 44$	(65) $40 + 19$	(95) $70 + 32$
(6) $93 + 40$	(36) $33 + 76$	(66) $76 + 40$	(96) $89 + 38$
(7) $70 + 74$	(37) $55 + 83$	(67) $37 + 83$	(97) $100 + 35$
(8) $31 + 38$	(38) $10 + 32$	(68) $36 + 91$	(98) $71 + 6$
(9) $71 + 29$	(39) $36 + 98$	(69) $38 + 87$	(99) $90 + 4$
(10) $57 + 58$	(40) $87 + 63$	(70) $94 + 12$	(100) $18 + 70$
(11) $18 + 16$	(41) $76 + 75$	(71) $49 + 99$	(101) $82 + 64$
(12) $48 + 92$	(42) $48 + 32$	(72) $100 + 59$	(102) $56 + 9$
(13) $74 + 23$	(43) $43 + 68$	(73) $79 + 2$	(103) $53 + 63$
(14) $33 + 86$	(44) $8 + 51$	(74) $85 + 100$	(104) $45 + 6$
(15) $65 + 46$	(45) $91 + 97$	(75) $66 + 33$	(105) $89 + 33$
(16) $89 + 45$	(46) $47 + 97$	(76) $84 + 39$	(106) $10 + 7$
(17) $85 + 88$	(47) $96 + 98$	(77) $83 + 60$	(107) $87 + 64$
(18) $43 + 18$	(48) $55 + 10$	(78) $32 + 22$	(108) $41 + 78$
(19) $56 + 2$	(49) $2 + 93$	(79) $80 + 20$	(109) $15 + 48$
(20) $81 + 95$	(50) $57 + 70$	(80) $86 + 9$	(110) $57 + 43$
(21) $72 + 55$	(51) $56 + 33$	(81) $59 + 81$	(111) $81 + 78$
(22) $12 + 30$	(52) $47 + 33$	(82) $24 + 38$	(112) $0 + 53$
(23) $11 + 50$	(53) $46 + 60$	(83) $34 + 39$	(113) $98 + 55$
(24) $52 + 11$	(54) $44 + 99$	(84) $20 + 74$	(114) $5 + 32$
(25) $14 + 86$	(55) $7 + 18$	(85) $55 + 77$	(115) $77 + 37$
(26) $59 + 14$	(56) $94 + 46$	(86) $51 + 95$	(116) $0 + 39$
(27) $33 + 70$	(57) $87 + 78$	(87) $53 + 76$	(117) $89 + 2$
(28) $37 + 95$	(58) $56 + 24$	(88) $4 + 39$	(118) $9 + 10$
(29) $95 + 28$	(59) $90 + 66$	(89) $17 + 42$	(119) $89 + 49$
(30) $79 + 11$	(60) $48 + 66$	(90) $51 + 17$	(120) $57 + 60$

(1) $1 + 13$	(31) $74 + 83$	(61) $37 + 45$	(91) $11 + 82$
(2) $54 + 60$	(32) $17 + 94$	(62) $6 + 48$	(92) $14 + 84$
(3) $84 + 99$	(33) $91 + 4$	(63) $16 + 59$	(93) $8 + 16$
(4) $96 + 96$	(34) $76 + 32$	(64) $47 + 96$	(94) $29 + 17$
(5) $69 + 60$	(35) $39 + 44$	(65) $48 + 35$	(95) $46 + 19$
(6) $94 + 61$	(36) $73 + 59$	(66) $23 + 100$	(96) $61 + 61$
(7) $7 + 65$	(37) $84 + 21$	(67) $20 + 52$	(97) $40 + 42$
(8) $32 + 36$	(38) $68 + 14$	(68) $62 + 62$	(98) $10 + 12$
(9) $51 + 49$	(39) $79 + 68$	(69) $90 + 22$	(99) $41 + 98$
(10) $2 + 94$	(40) $16 + 46$	(70) $38 + 39$	(100) $54 + 5$
(11) $76 + 18$	(41) $91 + 11$	(71) $61 + 41$	(101) $81 + 78$
(12) $59 + 79$	(42) $23 + 100$	(72) $3 + 96$	(102) $9 + 72$
(13) $13 + 36$	(43) $27 + 79$	(73) $30 + 92$	(103) $14 + 46$
(14) $93 + 69$	(44) $70 + 55$	(74) $33 + 75$	(104) $31 + 15$
(15) $60 + 66$	(45) $41 + 54$	(75) $82 + 42$	(105) $61 + 66$
(16) $28 + 100$	(46) $57 + 9$	(76) $85 + 53$	(106) $55 + 45$
(17) $70 + 91$	(47) $50 + 96$	(77) $85 + 11$	(107) $45 + 60$
(18) $72 + 27$	(48) $8 + 56$	(78) $80 + 15$	(108) $55 + 68$
(19) $81 + 29$	(49) $7 + 73$	(79) $87 + 37$	(109) $9 + 88$
(20) $39 + 45$	(50) $28 + 61$	(80) $58 + 94$	(110) $93 + 40$
(21) $74 + 3$	(51) $2 + 71$	(81) $4 + 15$	(111) $56 + 27$
(22) $53 + 92$	(52) $94 + 99$	(82) $70 + 33$	(112) $28 + 46$
(23) $35 + 81$	(53) $15 + 51$	(83) $59 + 62$	(113) $44 + 52$
(24) $24 + 93$	(54) $33 + 57$	(84) $16 + 33$	(114) $61 + 46$
(25) $92 + 17$	(55) $64 + 27$	(85) $73 + 41$	(115) $36 + 73$
(26) $56 + 10$	(56) $82 + 65$	(86) $47 + 86$	(116) $49 + 45$
(27) $53 + 43$	(57) $20 + 60$	(87) $44 + 89$	(117) $15 + 34$
(28) $94 + 28$	(58) $96 + 88$	(88) $9 + 23$	(118) $35 + 90$
(29) $77 + 44$	(59) $99 + 65$	(89) $72 + 32$	(119) $60 + 9$
(30) $23 + 62$	(60) $88 + 82$	(90) $98 + 5$	(120) $67 + 37$

(1) $31 + 76$	(31) $29 + 29$	(61) $54 + 35$	(91) $86 + 68$
(2) $37 + 59$	(32) $8 + 68$	(62) $17 + 56$	(92) $88 + 9$
(3) $43 + 72$	(33) $93 + 59$	(63) $48 + 100$	(93) $1 + 35$
(4) $80 + 42$	(34) $32 + 28$	(64) $9 + 99$	(94) $22 + 3$
(5) $20 + 73$	(35) $77 + 84$	(65) $78 + 60$	(95) $84 + 92$
(6) $100 + 46$	(36) $59 + 5$	(66) $14 + 25$	(96) $42 + 74$
(7) $5 + 48$	(37) $18 + 62$	(67) $24 + 71$	(97) $15 + 74$
(8) $78 + 13$	(38) $16 + 23$	(68) $87 + 67$	(98) $36 + 41$
(9) $43 + 37$	(39) $27 + 91$	(69) $70 + 51$	(99) $0 + 13$
(10) $17 + 67$	(40) $2 + 95$	(70) $83 + 84$	(100) $25 + 32$
(11) $67 + 1$	(41) $7 + 69$	(71) $23 + 78$	(101) $17 + 28$
(12) $33 + 88$	(42) $6 + 91$	(72) $82 + 76$	(102) $71 + 83$
(13) $5 + 87$	(43) $86 + 87$	(73) $83 + 85$	(103) $7 + 69$
(14) $71 + 63$	(44) $89 + 92$	(74) $99 + 98$	(104) $12 + 64$
(15) $45 + 92$	(45) $35 + 68$	(75) $25 + 42$	(105) $43 + 89$
(16) $67 + 68$	(46) $49 + 75$	(76) $1 + 20$	(106) $68 + 14$
(17) $96 + 6$	(47) $82 + 19$	(77) $54 + 1$	(107) $76 + 98$
(18) $30 + 56$	(48) $84 + 44$	(78) $86 + 38$	(108) $20 + 87$
(19) $61 + 16$	(49) $15 + 37$	(79) $58 + 9$	(109) $95 + 71$
(20) $65 + 50$	(50) $80 + 12$	(80) $81 + 97$	(110) $65 + 97$
(21) $27 + 63$	(51) $35 + 85$	(81) $15 + 86$	(111) $18 + 71$
(22) $23 + 79$	(52) $4 + 21$	(82) $21 + 28$	(112) $8 + 18$
(23) $33 + 9$	(53) $55 + 73$	(83) $93 + 50$	(113) $83 + 59$
(24) $59 + 83$	(54) $16 + 67$	(84) $36 + 13$	(114) $34 + 44$
(25) $49 + 68$	(55) $63 + 15$	(85) $55 + 79$	(115) $89 + 92$
(26) $50 + 60$	(56) $58 + 97$	(86) $52 + 40$	(116) $7 + 31$
(27) $92 + 64$	(57) $36 + 52$	(87) $62 + 55$	(117) $27 + 38$
(28) $55 + 56$	(58) $42 + 58$	(88) $37 + 56$	(118) $53 + 90$
(29) $58 + 74$	(59) $45 + 99$	(89) $12 + 76$	(119) $82 + 36$
(30) $22 + 23$	(60) $1 + 70$	(90) $73 + 61$	(120) $88 + 8$

(1) $94 + 95$	(31) $71 + 26$	(61) $71 + 91$	(91) $76 + 97$
(2) $56 + 48$	(32) $46 + 47$	(62) $60 + 64$	(92) $50 + 41$
(3) $21 + 18$	(33) $30 + 83$	(63) $69 + 3$	(93) $27 + 32$
(4) $98 + 48$	(34) $99 + 65$	(64) $38 + 63$	(94) $17 + 76$
(5) $29 + 94$	(35) $98 + 77$	(65) $2 + 77$	(95) $7 + 26$
(6) $7 + 71$	(36) $64 + 79$	(66) $12 + 4$	(96) $60 + 94$
(7) $55 + 40$	(37) $12 + 54$	(67) $35 + 72$	(97) $41 + 50$
(8) $0 + 99$	(38) $35 + 45$	(68) $71 + 15$	(98) $77 + 12$
(9) $64 + 63$	(39) $14 + 44$	(69) $68 + 38$	(99) $19 + 32$
(10) $96 + 96$	(40) $88 + 92$	(70) $87 + 49$	(100) $79 + 94$
(11) $51 + 79$	(41) $76 + 93$	(71) $93 + 100$	(101) $32 + 84$
(12) $34 + 34$	(42) $3 + 36$	(72) $94 + 2$	(102) $36 + 18$
(13) $14 + 90$	(43) $83 + 26$	(73) $66 + 80$	(103) $37 + 25$
(14) $80 + 69$	(44) $59 + 90$	(74) $39 + 30$	(104) $75 + 36$
(15) $32 + 4$	(45) $38 + 60$	(75) $63 + 29$	(105) $92 + 45$
(16) $19 + 20$	(46) $61 + 3$	(76) $96 + 45$	(106) $19 + 50$
(17) $74 + 97$	(47) $55 + 46$	(77) $10 + 6$	(107) $52 + 95$
(18) $96 + 78$	(48) $72 + 86$	(78) $47 + 91$	(108) $30 + 27$
(19) $14 + 60$	(49) $45 + 71$	(79) $97 + 43$	(109) $23 + 88$
(20) $14 + 13$	(50) $78 + 98$	(80) $31 + 40$	(110) $54 + 16$
(21) $2 + 78$	(51) $70 + 13$	(81) $52 + 48$	(111) $60 + 84$
(22) $20 + 53$	(52) $80 + 11$	(82) $61 + 55$	(112) $74 + 55$
(23) $14 + 37$	(53) $2 + 22$	(83) $16 + 11$	(113) $52 + 13$
(24) $22 + 14$	(54) $15 + 99$	(84) $20 + 84$	(114) $22 + 52$
(25) $28 + 18$	(55) $36 + 85$	(85) $50 + 69$	(115) $79 + 76$
(26) $19 + 74$	(56) $14 + 2$	(86) $1 + 31$	(116) $28 + 90$
(27) $35 + 77$	(57) $91 + 41$	(87) $32 + 78$	(117) $26 + 7$
(28) $58 + 5$	(58) $74 + 84$	(88) $27 + 5$	(118) $47 + 67$
(29) $62 + 30$	(59) $19 + 51$	(89) $28 + 1$	(119) $77 + 37$
(30) $52 + 29$	(60) $26 + 44$	(90) $8 + 3$	(120) $35 + 14$

(1) $11 + 68$	(31) $51 + 42$	(61) $64 + 67$	(91) $22 + 33$
(2) $45 + 1$	(32) $12 + 93$	(62) $86 + 30$	(92) $100 + 38$
(3) $66 + 69$	(33) $62 + 90$	(63) $42 + 52$	(93) $100 + 28$
(4) $12 + 18$	(34) $45 + 6$	(64) $26 + 88$	(94) $43 + 97$
(5) $80 + 81$	(35) $68 + 89$	(65) $17 + 54$	(95) $92 + 50$
(6) $66 + 51$	(36) $54 + 86$	(66) $74 + 13$	(96) $44 + 15$
(7) $60 + 19$	(37) $6 + 93$	(67) $0 + 79$	(97) $66 + 37$
(8) $62 + 29$	(38) $22 + 5$	(68) $50 + 60$	(98) $92 + 100$
(9) $45 + 1$	(39) $99 + 77$	(69) $72 + 78$	(99) $61 + 81$
(10) $95 + 43$	(40) $20 + 55$	(70) $59 + 27$	(100) $69 + 13$
(11) $99 + 61$	(41) $26 + 67$	(71) $94 + 98$	(101) $4 + 59$
(12) $43 + 87$	(42) $64 + 78$	(72) $95 + 1$	(102) $22 + 93$
(13) $8 + 11$	(43) $88 + 47$	(73) $12 + 70$	(103) $81 + 77$
(14) $56 + 40$	(44) $22 + 17$	(74) $32 + 47$	(104) $34 + 78$
(15) $86 + 50$	(45) $44 + 83$	(75) $73 + 94$	(105) $77 + 67$
(16) $96 + 59$	(46) $48 + 28$	(76) $47 + 47$	(106) $14 + 32$
(17) $94 + 49$	(47) $25 + 65$	(77) $26 + 3$	(107) $53 + 48$
(18) $23 + 27$	(48) $74 + 56$	(78) $17 + 76$	(108) $20 + 53$
(19) $82 + 3$	(49) $96 + 88$	(79) $56 + 86$	(109) $86 + 60$
(20) $18 + 89$	(50) $62 + 100$	(80) $12 + 97$	(110) $24 + 47$
(21) $86 + 25$	(51) $0 + 74$	(81) $56 + 5$	(111) $27 + 27$
(22) $24 + 35$	(52) $35 + 49$	(82) $87 + 14$	(112) $15 + 19$
(23) $6 + 85$	(53) $69 + 42$	(83) $12 + 3$	(113) $3 + 24$
(24) $89 + 42$	(54) $4 + 79$	(84) $82 + 38$	(114) $100 + 4$
(25) $9 + 42$	(55) $25 + 10$	(85) $30 + 32$	(115) $61 + 18$
(26) $48 + 58$	(56) $54 + 10$	(86) $67 + 88$	(116) $73 + 94$
(27) $69 + 65$	(57) $86 + 29$	(87) $39 + 86$	(117) $9 + 36$
(28) $66 + 8$	(58) $80 + 27$	(88) $47 + 46$	(118) $89 + 81$
(29) $1 + 49$	(59) $18 + 6$	(89) $20 + 28$	(119) $85 + 34$
(30) $23 + 55$	(60) $70 + 55$	(90) $20 + 46$	(120) $33 + 56$

(1) $56 + 60$	(31) $97 + 18$	(61) $91 + 25$	(91) $3 + 33$
(2) $38 + 13$	(32) $46 + 38$	(62) $93 + 36$	(92) $22 + 8$
(3) $43 + 58$	(33) $85 + 81$	(63) $100 + 63$	(93) $65 + 56$
(4) $67 + 37$	(34) $4 + 79$	(64) $76 + 37$	(94) $19 + 81$
(5) $69 + 50$	(35) $44 + 39$	(65) $59 + 57$	(95) $70 + 2$
(6) $71 + 28$	(36) $66 + 72$	(66) $85 + 56$	(96) $5 + 64$
(7) $10 + 76$	(37) $70 + 88$	(67) $66 + 75$	(97) $15 + 54$
(8) $100 + 42$	(38) $55 + 24$	(68) $66 + 83$	(98) $4 + 61$
(9) $8 + 93$	(39) $98 + 65$	(69) $72 + 41$	(99) $82 + 97$
(10) $55 + 41$	(40) $95 + 82$	(70) $59 + 85$	(100) $10 + 13$
(11) $88 + 36$	(41) $39 + 66$	(71) $4 + 88$	(101) $65 + 2$
(12) $66 + 3$	(42) $32 + 16$	(72) $99 + 36$	(102) $4 + 83$
(13) $83 + 68$	(43) $58 + 24$	(73) $57 + 71$	(103) $51 + 88$
(14) $66 + 17$	(44) $30 + 6$	(74) $56 + 100$	(104) $43 + 32$
(15) $58 + 64$	(45) $55 + 77$	(75) $73 + 94$	(105) $33 + 14$
(16) $9 + 81$	(46) $98 + 26$	(76) $79 + 70$	(106) $90 + 90$
(17) $34 + 62$	(47) $92 + 77$	(77) $20 + 90$	(107) $19 + 26$
(18) $15 + 88$	(48) $13 + 73$	(78) $83 + 14$	(108) $52 + 54$
(19) $39 + 60$	(49) $67 + 21$	(79) $79 + 7$	(109) $27 + 72$
(20) $52 + 68$	(50) $98 + 95$	(80) $26 + 86$	(110) $54 + 2$
(21) $54 + 75$	(51) $21 + 15$	(81) $13 + 19$	(111) $56 + 89$
(22) $14 + 57$	(52) $14 + 36$	(82) $73 + 61$	(112) $80 + 4$
(23) $81 + 95$	(53) $88 + 43$	(83) $6 + 39$	(113) $4 + 28$
(24) $52 + 40$	(54) $46 + 3$	(84) $36 + 45$	(114) $74 + 71$
(25) $56 + 22$	(55) $32 + 64$	(85) $39 + 95$	(115) $72 + 15$
(26) $33 + 66$	(56) $77 + 76$	(86) $27 + 57$	(116) $3 + 19$
(27) $68 + 29$	(57) $23 + 26$	(87) $99 + 81$	(117) $95 + 76$
(28) $53 + 19$	(58) $24 + 7$	(88) $12 + 95$	(118) $43 + 83$
(29) $11 + 32$	(59) $54 + 91$	(89) $100 + 46$	(119) $19 + 74$
(30) $19 + 14$	(60) $32 + 37$	(90) $70 + 51$	(120) $30 + 83$

(1) $82 + 65$	(31) $69 + 65$	(61) $31 + 57$	(91) $61 + 15$
(2) $22 + 65$	(32) $13 + 68$	(62) $48 + 27$	(92) $58 + 39$
(3) $61 + 46$	(33) $74 + 65$	(63) $76 + 98$	(93) $23 + 73$
(4) $94 + 45$	(34) $58 + 81$	(64) $77 + 59$	(94) $47 + 65$
(5) $54 + 68$	(35) $12 + 17$	(65) $1 + 4$	(95) $92 + 45$
(6) $9 + 100$	(36) $43 + 99$	(66) $54 + 83$	(96) $20 + 21$
(7) $82 + 82$	(37) $75 + 9$	(67) $35 + 65$	(97) $39 + 27$
(8) $23 + 67$	(38) $17 + 13$	(68) $53 + 54$	(98) $79 + 74$
(9) $3 + 5$	(39) $28 + 90$	(69) $17 + 91$	(99) $39 + 62$
(10) $96 + 61$	(40) $76 + 88$	(70) $51 + 25$	(100) $5 + 31$
(11) $28 + 41$	(41) $11 + 21$	(71) $100 + 4$	(101) $27 + 62$
(12) $80 + 38$	(42) $63 + 19$	(72) $39 + 30$	(102) $89 + 45$
(13) $83 + 49$	(43) $32 + 34$	(73) $94 + 9$	(103) $59 + 76$
(14) $49 + 61$	(44) $30 + 92$	(74) $33 + 65$	(104) $87 + 7$
(15) $68 + 21$	(45) $39 + 8$	(75) $53 + 28$	(105) $19 + 54$
(16) $34 + 80$	(46) $72 + 60$	(76) $38 + 59$	(106) $23 + 37$
(17) $75 + 69$	(47) $88 + 1$	(77) $5 + 61$	(107) $91 + 35$
(18) $87 + 30$	(48) $52 + 90$	(78) $24 + 61$	(108) $28 + 43$
(19) $2 + 25$	(49) $58 + 25$	(79) $3 + 75$	(109) $12 + 89$
(20) $91 + 53$	(50) $89 + 88$	(80) $81 + 50$	(110) $12 + 47$
(21) $67 + 11$	(51) $15 + 46$	(81) $9 + 89$	(111) $78 + 53$
(22) $46 + 45$	(52) $35 + 92$	(82) $66 + 93$	(112) $20 + 64$
(23) $92 + 12$	(53) $57 + 55$	(83) $9 + 26$	(113) $58 + 17$
(24) $63 + 64$	(54) $9 + 60$	(84) $33 + 70$	(114) $30 + 57$
(25) $98 + 85$	(55) $10 + 14$	(85) $13 + 64$	(115) $12 + 88$
(26) $52 + 30$	(56) $0 + 48$	(86) $37 + 79$	(116) $25 + 72$
(27) $57 + 28$	(57) $13 + 73$	(87) $87 + 29$	(117) $65 + 95$
(28) $76 + 87$	(58) $57 + 38$	(88) $71 + 72$	(118) $79 + 13$
(29) $66 + 13$	(59) $26 + 37$	(89) $5 + 60$	(119) $16 + 17$
(30) $37 + 50$	(60) $81 + 66$	(90) $34 + 42$	(120) $29 + 63$

(1) $69 + 55$	(31) $71 + 48$	(61) $14 + 30$	(91) $9 + 74$
(2) $75 + 63$	(32) $13 + 88$	(62) $61 + 95$	(92) $83 + 55$
(3) $39 + 86$	(33) $48 + 57$	(63) $32 + 72$	(93) $52 + 41$
(4) $54 + 40$	(34) $36 + 56$	(64) $23 + 26$	(94) $98 + 27$
(5) $57 + 12$	(35) $100 + 94$	(65) $20 + 40$	(95) $68 + 93$
(6) $9 + 34$	(36) $67 + 97$	(66) $10 + 57$	(96) $29 + 24$
(7) $63 + 29$	(37) $23 + 68$	(67) $78 + 96$	(97) $36 + 89$
(8) $23 + 48$	(38) $87 + 69$	(68) $55 + 83$	(98) $11 + 43$
(9) $49 + 98$	(39) $80 + 81$	(69) $5 + 80$	(99) $56 + 51$
(10) $90 + 73$	(40) $10 + 86$	(70) $92 + 13$	(100) $13 + 89$
(11) $63 + 31$	(41) $12 + 16$	(71) $37 + 35$	(101) $66 + 19$
(12) $50 + 46$	(42) $85 + 49$	(72) $26 + 64$	(102) $94 + 34$
(13) $32 + 81$	(43) $6 + 94$	(73) $70 + 63$	(103) $84 + 96$
(14) $70 + 78$	(44) $48 + 25$	(74) $29 + 28$	(104) $71 + 70$
(15) $68 + 85$	(45) $25 + 33$	(75) $62 + 33$	(105) $81 + 54$
(16) $31 + 86$	(46) $83 + 89$	(76) $10 + 16$	(106) $92 + 77$
(17) $75 + 23$	(47) $70 + 66$	(77) $87 + 54$	(107) $46 + 26$
(18) $100 + 39$	(48) $16 + 16$	(78) $9 + 41$	(108) $86 + 20$
(19) $3 + 6$	(49) $29 + 68$	(79) $66 + 45$	(109) $52 + 42$
(20) $70 + 8$	(50) $33 + 42$	(80) $70 + 9$	(110) $73 + 2$
(21) $37 + 14$	(51) $87 + 95$	(81) $84 + 91$	(111) $90 + 11$
(22) $36 + 43$	(52) $1 + 96$	(82) $77 + 8$	(112) $48 + 78$
(23) $92 + 35$	(53) $41 + 1$	(83) $26 + 75$	(113) $87 + 58$
(24) $29 + 50$	(54) $78 + 41$	(84) $62 + 25$	(114) $44 + 52$
(25) $45 + 3$	(55) $50 + 8$	(85) $67 + 50$	(115) $59 + 15$
(26) $64 + 2$	(56) $2 + 84$	(86) $40 + 22$	(116) $20 + 67$
(27) $94 + 62$	(57) $60 + 100$	(87) $23 + 17$	(117) $54 + 40$
(28) $85 + 29$	(58) $16 + 5$	(88) $42 + 1$	(118) $3 + 95$
(29) $91 + 79$	(59) $47 + 0$	(89) $30 + 97$	(119) $59 + 97$
(30) $69 + 88$	(60) $59 + 91$	(90) $52 + 12$	(120) $23 + 36$

(1) $83 + 62$	(31) $4 + 21$	(61) $36 + 52$	(91) $64 + 90$
(2) $86 + 11$	(32) $51 + 23$	(62) $31 + 7$	(92) $36 + 14$
(3) $90 + 80$	(33) $61 + 42$	(63) $7 + 36$	(93) $74 + 22$
(4) $16 + 55$	(34) $45 + 94$	(64) $91 + 57$	(94) $38 + 62$
(5) $76 + 99$	(35) $48 + 46$	(65) $61 + 93$	(95) $50 + 81$
(6) $36 + 93$	(36) $96 + 80$	(66) $14 + 42$	(96) $9 + 96$
(7) $9 + 88$	(37) $63 + 72$	(67) $28 + 3$	(97) $83 + 21$
(8) $68 + 16$	(38) $4 + 64$	(68) $17 + 45$	(98) $69 + 17$
(9) $93 + 35$	(39) $98 + 37$	(69) $44 + 61$	(99) $90 + 33$
(10) $93 + 37$	(40) $70 + 97$	(70) $39 + 20$	(100) $96 + 65$
(11) $87 + 65$	(41) $52 + 85$	(71) $59 + 59$	(101) $48 + 30$
(12) $38 + 23$	(42) $1 + 98$	(72) $3 + 88$	(102) $40 + 66$
(13) $72 + 56$	(43) $65 + 45$	(73) $32 + 72$	(103) $81 + 77$
(14) $40 + 30$	(44) $79 + 8$	(74) $60 + 54$	(104) $60 + 97$
(15) $94 + 11$	(45) $49 + 82$	(75) $90 + 8$	(105) $19 + 96$
(16) $55 + 80$	(46) $65 + 38$	(76) $15 + 45$	(106) $25 + 51$
(17) $2 + 24$	(47) $59 + 61$	(77) $21 + 28$	(107) $25 + 38$
(18) $47 + 18$	(48) $51 + 66$	(78) $8 + 60$	(108) $78 + 63$
(19) $69 + 35$	(49) $85 + 11$	(79) $44 + 70$	(109) $76 + 72$
(20) $35 + 99$	(50) $22 + 6$	(80) $53 + 6$	(110) $75 + 64$
(21) $26 + 82$	(51) $40 + 77$	(81) $68 + 4$	(111) $85 + 69$
(22) $57 + 30$	(52) $11 + 25$	(82) $29 + 22$	(112) $95 + 4$
(23) $10 + 55$	(53) $39 + 49$	(83) $24 + 35$	(113) $78 + 55$
(24) $40 + 89$	(54) $11 + 76$	(84) $47 + 31$	(114) $40 + 63$
(25) $79 + 72$	(55) $35 + 66$	(85) $56 + 88$	(115) $91 + 82$
(26) $76 + 48$	(56) $86 + 92$	(86) $45 + 44$	(116) $83 + 37$
(27) $88 + 47$	(57) $10 + 70$	(87) $56 + 84$	(117) $41 + 33$
(28) $72 + 75$	(58) $78 + 92$	(88) $87 + 80$	(118) $1 + 66$
(29) $38 + 21$	(59) $51 + 23$	(89) $98 + 73$	(119) $29 + 19$
(30) $48 + 52$	(60) $62 + 28$	(90) $12 + 56$	(120) $83 + 41$

(1) $26 + 27$	(31) $18 + 99$	(61) $17 + 58$	(91) $41 + 57$
(2) $83 + 48$	(32) $30 + 67$	(62) $97 + 55$	(92) $32 + 63$
(3) $1 + 55$	(33) $15 + 51$	(63) $67 + 43$	(93) $16 + 51$
(4) $69 + 71$	(34) $25 + 42$	(64) $50 + 99$	(94) $31 + 33$
(5) $1 + 72$	(35) $75 + 98$	(65) $58 + 69$	(95) $2 + 100$
(6) $94 + 70$	(36) $81 + 4$	(66) $90 + 41$	(96) $47 + 51$
(7) $43 + 64$	(37) $96 + 73$	(67) $90 + 20$	(97) $49 + 61$
(8) $82 + 85$	(38) $27 + 35$	(68) $7 + 46$	(98) $48 + 7$
(9) $47 + 73$	(39) $49 + 94$	(69) $80 + 30$	(99) $75 + 59$
(10) $55 + 43$	(40) $36 + 75$	(70) $33 + 95$	(100) $27 + 95$
(11) $4 + 53$	(41) $15 + 22$	(71) $95 + 73$	(101) $39 + 69$
(12) $27 + 28$	(42) $76 + 44$	(72) $11 + 29$	(102) $52 + 70$
(13) $92 + 6$	(43) $73 + 3$	(73) $54 + 22$	(103) $23 + 43$
(14) $25 + 3$	(44) $22 + 35$	(74) $67 + 79$	(104) $31 + 41$
(15) $29 + 43$	(45) $21 + 66$	(75) $64 + 83$	(105) $99 + 11$
(16) $99 + 75$	(46) $57 + 91$	(76) $49 + 86$	(106) $44 + 53$
(17) $9 + 17$	(47) $88 + 10$	(77) $22 + 26$	(107) $30 + 4$
(18) $18 + 94$	(48) $42 + 43$	(78) $14 + 56$	(108) $48 + 44$
(19) $70 + 51$	(49) $31 + 78$	(79) $71 + 53$	(109) $3 + 24$
(20) $72 + 50$	(50) $27 + 2$	(80) $62 + 3$	(110) $48 + 73$
(21) $8 + 27$	(51) $51 + 73$	(81) $26 + 32$	(111) $75 + 37$
(22) $9 + 35$	(52) $90 + 86$	(82) $86 + 57$	(112) $28 + 71$
(23) $49 + 56$	(53) $35 + 98$	(83) $78 + 58$	(113) $14 + 16$
(24) $69 + 75$	(54) $9 + 79$	(84) $25 + 52$	(114) $28 + 13$
(25) $72 + 5$	(55) $87 + 78$	(85) $22 + 60$	(115) $75 + 89$
(26) $43 + 86$	(56) $55 + 83$	(86) $6 + 20$	(116) $2 + 74$
(27) $52 + 55$	(57) $24 + 54$	(87) $15 + 37$	(117) $98 + 99$
(28) $19 + 91$	(58) $93 + 93$	(88) $2 + 92$	(118) $96 + 62$
(29) $47 + 55$	(59) $19 + 9$	(89) $16 + 37$	(119) $43 + 73$
(30) $54 + 38$	(60) $5 + 96$	(90) $8 + 69$	(120) $38 + 4$

(1) $58 + 6$	(31) $36 + 45$	(61) $43 + 17$	(91) $86 + 54$
(2) $80 + 84$	(32) $25 + 14$	(62) $94 + 30$	(92) $11 + 50$
(3) $46 + 16$	(33) $96 + 94$	(63) $3 + 96$	(93) $55 + 22$
(4) $15 + 96$	(34) $52 + 56$	(64) $62 + 12$	(94) $2 + 14$
(5) $2 + 12$	(35) $62 + 97$	(65) $84 + 50$	(95) $74 + 29$
(6) $39 + 82$	(36) $38 + 94$	(66) $85 + 63$	(96) $47 + 32$
(7) $44 + 84$	(37) $44 + 59$	(67) $69 + 58$	(97) $31 + 92$
(8) $70 + 24$	(38) $82 + 76$	(68) $18 + 11$	(98) $44 + 14$
(9) $81 + 54$	(39) $6 + 57$	(69) $34 + 98$	(99) $16 + 60$
(10) $69 + 3$	(40) $87 + 70$	(70) $32 + 70$	(100) $67 + 42$
(11) $21 + 95$	(41) $66 + 56$	(71) $39 + 69$	(101) $39 + 5$
(12) $7 + 61$	(42) $69 + 97$	(72) $16 + 38$	(102) $73 + 38$
(13) $89 + 75$	(43) $61 + 80$	(73) $27 + 32$	(103) $30 + 17$
(14) $49 + 99$	(44) $89 + 30$	(74) $32 + 5$	(104) $47 + 74$
(15) $54 + 19$	(45) $20 + 13$	(75) $84 + 35$	(105) $57 + 100$
(16) $55 + 75$	(46) $99 + 88$	(76) $72 + 95$	(106) $86 + 35$
(17) $23 + 34$	(47) $37 + 51$	(77) $91 + 83$	(107) $86 + 78$
(18) $48 + 97$	(48) $35 + 50$	(78) $49 + 27$	(108) $96 + 53$
(19) $87 + 2$	(49) $58 + 80$	(79) $55 + 98$	(109) $56 + 47$
(20) $85 + 29$	(50) $78 + 66$	(80) $8 + 3$	(110) $74 + 18$
(21) $53 + 42$	(51) $75 + 41$	(81) $23 + 10$	(111) $52 + 5$
(22) $20 + 76$	(52) $3 + 3$	(82) $67 + 78$	(112) $49 + 74$
(23) $48 + 57$	(53) $32 + 1$	(83) $71 + 83$	(113) $68 + 58$
(24) $3 + 20$	(54) $58 + 64$	(84) $27 + 62$	(114) $58 + 55$
(25) $29 + 16$	(55) $37 + 71$	(85) $83 + 57$	(115) $85 + 55$
(26) $84 + 91$	(56) $18 + 50$	(86) $47 + 1$	(116) $65 + 4$
(27) $65 + 59$	(57) $52 + 54$	(87) $83 + 38$	(117) $21 + 87$
(28) $55 + 90$	(58) $41 + 72$	(88) $53 + 38$	(118) $73 + 23$
(29) $85 + 12$	(59) $76 + 66$	(89) $79 + 40$	(119) $77 + 96$
(30) $69 + 58$	(60) $14 + 66$	(90) $6 + 93$	(120) $27 + 8$

(1) $26 + 97$	(31) $16 + 46$	(61) $98 + 92$	(91) $66 + 11$
(2) $76 + 7$	(32) $16 + 83$	(62) $99 + 16$	(92) $95 + 66$
(3) $13 + 49$	(33) $79 + 14$	(63) $37 + 53$	(93) $81 + 21$
(4) $76 + 69$	(34) $57 + 19$	(64) $93 + 16$	(94) $11 + 69$
(5) $98 + 32$	(35) $74 + 55$	(65) $71 + 45$	(95) $40 + 75$
(6) $85 + 100$	(36) $52 + 4$	(66) $6 + 83$	(96) $65 + 18$
(7) $17 + 45$	(37) $67 + 20$	(67) $92 + 23$	(97) $83 + 74$
(8) $43 + 46$	(38) $43 + 84$	(68) $37 + 98$	(98) $87 + 50$
(9) $63 + 43$	(39) $100 + 16$	(69) $79 + 45$	(99) $49 + 56$
(10) $72 + 28$	(40) $71 + 6$	(70) $7 + 18$	(100) $9 + 64$
(11) $85 + 82$	(41) $25 + 69$	(71) $37 + 84$	(101) $31 + 6$
(12) $35 + 10$	(42) $80 + 81$	(72) $65 + 27$	(102) $32 + 13$
(13) $41 + 94$	(43) $18 + 89$	(73) $25 + 52$	(103) $9 + 69$
(14) $19 + 74$	(44) $33 + 89$	(74) $2 + 82$	(104) $46 + 14$
(15) $44 + 9$	(45) $80 + 69$	(75) $7 + 7$	(105) $76 + 20$
(16) $93 + 53$	(46) $33 + 98$	(76) $17 + 27$	(106) $76 + 78$
(17) $12 + 92$	(47) $7 + 62$	(77) $92 + 90$	(107) $50 + 31$
(18) $16 + 45$	(48) $48 + 42$	(78) $92 + 59$	(108) $18 + 23$
(19) $100 + 21$	(49) $28 + 54$	(79) $55 + 9$	(109) $28 + 19$
(20) $24 + 88$	(50) $44 + 32$	(80) $19 + 20$	(110) $68 + 59$
(21) $56 + 87$	(51) $28 + 60$	(81) $9 + 30$	(111) $54 + 89$
(22) $5 + 67$	(52) $18 + 62$	(82) $67 + 63$	(112) $43 + 31$
(23) $85 + 33$	(53) $73 + 48$	(83) $32 + 33$	(113) $80 + 14$
(24) $6 + 81$	(54) $2 + 44$	(84) $65 + 70$	(114) $52 + 57$
(25) $17 + 55$	(55) $98 + 18$	(85) $16 + 39$	(115) $65 + 70$
(26) $64 + 24$	(56) $70 + 96$	(86) $74 + 47$	(116) $69 + 28$
(27) $85 + 40$	(57) $3 + 84$	(87) $63 + 43$	(117) $74 + 30$
(28) $6 + 28$	(58) $28 + 59$	(88) $31 + 42$	(118) $71 + 97$
(29) $31 + 16$	(59) $24 + 20$	(89) $37 + 2$	(119) $15 + 69$
(30) $74 + 17$	(60) $67 + 65$	(90) $8 + 20$	(120) $73 + 84$

(1) $17 + 33$	(31) $90 + 10$	(61) $1 + 81$	(91) $34 + 16$
(2) $35 + 61$	(32) $5 + 34$	(62) $94 + 99$	(92) $32 + 72$
(3) $60 + 27$	(33) $39 + 49$	(63) $59 + 7$	(93) $14 + 53$
(4) $27 + 68$	(34) $10 + 47$	(64) $0 + 23$	(94) $84 + 3$
(5) $8 + 20$	(35) $7 + 41$	(65) $60 + 60$	(95) $63 + 56$
(6) $89 + 77$	(36) $65 + 9$	(66) $96 + 18$	(96) $96 + 51$
(7) $21 + 94$	(37) $0 + 69$	(67) $0 + 86$	(97) $55 + 79$
(8) $83 + 52$	(38) $39 + 46$	(68) $84 + 86$	(98) $96 + 99$
(9) $37 + 99$	(39) $49 + 47$	(69) $58 + 62$	(99) $19 + 77$
(10) $19 + 59$	(40) $42 + 80$	(70) $23 + 17$	(100) $24 + 6$
(11) $41 + 32$	(41) $83 + 52$	(71) $86 + 17$	(101) $64 + 66$
(12) $89 + 42$	(42) $42 + 14$	(72) $92 + 57$	(102) $99 + 55$
(13) $8 + 94$	(43) $24 + 70$	(73) $13 + 77$	(103) $18 + 37$
(14) $76 + 29$	(44) $98 + 43$	(74) $58 + 25$	(104) $67 + 55$
(15) $41 + 71$	(45) $82 + 47$	(75) $62 + 38$	(105) $20 + 90$
(16) $40 + 10$	(46) $2 + 39$	(76) $70 + 68$	(106) $61 + 55$
(17) $1 + 89$	(47) $36 + 87$	(77) $43 + 89$	(107) $44 + 93$
(18) $62 + 93$	(48) $41 + 43$	(78) $98 + 61$	(108) $39 + 21$
(19) $22 + 4$	(49) $13 + 15$	(79) $1 + 19$	(109) $39 + 6$
(20) $98 + 90$	(50) $66 + 0$	(80) $90 + 24$	(110) $77 + 88$
(21) $56 + 39$	(51) $54 + 3$	(81) $97 + 48$	(111) $11 + 58$
(22) $55 + 88$	(52) $57 + 51$	(82) $56 + 18$	(112) $28 + 60$
(23) $43 + 11$	(53) $65 + 15$	(83) $0 + 23$	(113) $79 + 32$
(24) $7 + 6$	(54) $72 + 18$	(84) $81 + 95$	(114) $86 + 8$
(25) $14 + 54$	(55) $31 + 15$	(85) $65 + 83$	(115) $92 + 37$
(26) $65 + 90$	(56) $33 + 3$	(86) $36 + 46$	(116) $44 + 68$
(27) $25 + 85$	(57) $66 + 91$	(87) $98 + 10$	(117) $56 + 47$
(28) $99 + 52$	(58) $45 + 38$	(88) $34 + 58$	(118) $10 + 95$
(29) $88 + 44$	(59) $29 + 24$	(89) $39 + 1$	(119) $7 + 77$
(30) $64 + 51$	(60) $7 + 76$	(90) $8 + 7$	(120) $48 + 10$

(1) $46 + 24 = 70$	(31) $31 + 91 = 122$	(61) $67 + 85 = 152$	(91) $44 + 95 = 139$
(2) $26 + 33 = 59$	(32) $43 + 35 = 78$	(62) $98 + 64 = 162$	(92) $48 + 99 = 147$
(3) $87 + 8 = 95$	(33) $94 + 89 = 183$	(63) $75 + 49 = 124$	(93) $43 + 83 = 126$
(4) $71 + 91 = 162$	(34) $22 + 23 = 45$	(64) $86 + 59 = 145$	(94) $42 + 23 = 65$
(5) $7 + 50 = 57$	(35) $58 + 20 = 78$	(65) $47 + 71 = 118$	(95) $19 + 29 = 48$
(6) $76 + 36 = 112$	(36) $7 + 60 = 67$	(66) $45 + 61 = 106$	(96) $11 + 72 = 83$
(7) $26 + 46 = 72$	(37) $67 + 72 = 139$	(67) $40 + 63 = 103$	(97) $59 + 21 = 80$
(8) $94 + 96 = 190$	(38) $15 + 3 = 18$	(68) $81 + 59 = 140$	(98) $59 + 66 = 125$
(9) $96 + 40 = 136$	(39) $2 + 82 = 84$	(69) $9 + 56 = 65$	(99) $24 + 43 = 67$
(10) $13 + 23 = 36$	(40) $74 + 66 = 140$	(70) $37 + 36 = 73$	(100) $85 + 82 = 167$
(11) $51 + 75 = 126$	(41) $46 + 0 = 46$	(71) $61 + 89 = 150$	(101) $26 + 89 = 115$
(12) $84 + 77 = 161$	(42) $18 + 92 = 110$	(72) $36 + 54 = 90$	(102) $83 + 67 = 150$
(13) $97 + 26 = 123$	(43) $98 + 82 = 180$	(73) $4 + 38 = 42$	(103) $41 + 91 = 132$
(14) $39 + 17 = 56$	(44) $37 + 85 = 122$	(74) $83 + 1 = 84$	(104) $54 + 69 = 123$
(15) $24 + 80 = 104$	(45) $0 + 59 = 59$	(75) $17 + 48 = 65$	(105) $82 + 99 = 181$
(16) $64 + 97 = 161$	(46) $9 + 87 = 96$	(76) $52 + 80 = 132$	(106) $5 + 10 = 15$
(17) $3 + 47 = 50$	(47) $16 + 22 = 38$	(77) $34 + 43 = 77$	(107) $75 + 24 = 99$
(18) $52 + 87 = 139$	(48) $52 + 18 = 70$	(78) $69 + 73 = 142$	(108) $41 + 12 = 53$
(19) $100 + 92 = 192$	(49) $49 + 74 = 123$	(79) $49 + 26 = 75$	(109) $8 + 56 = 64$
(20) $82 + 68 = 150$	(50) $47 + 4 = 51$	(80) $6 + 66 = 72$	(110) $34 + 52 = 86$
(21) $18 + 12 = 30$	(51) $70 + 22 = 92$	(81) $93 + 39 = 132$	(111) $91 + 78 = 169$
(22) $70 + 94 = 164$	(52) $36 + 93 = 129$	(82) $20 + 35 = 55$	(112) $30 + 57 = 87$
(23) $84 + 89 = 173$	(53) $94 + 21 = 115$	(83) $41 + 6 = 47$	(113) $55 + 10 = 65$
(24) $78 + 98 = 176$	(54) $2 + 73 = 75$	(84) $30 + 13 = 43$	(114) $81 + 95 = 176$
(25) $97 + 27 = 124$	(55) $50 + 55 = 105$	(85) $71 + 47 = 118$	(115) $93 + 69 = 162$
(26) $29 + 42 = 71$	(56) $67 + 38 = 105$	(86) $26 + 17 = 43$	(116) $47 + 49 = 96$
(27) $67 + 8 = 75$	(57) $66 + 88 = 154$	(87) $17 + 14 = 31$	(117) $66 + 6 = 72$
(28) $26 + 55 = 81$	(58) $58 + 91 = 149$	(88) $32 + 65 = 97$	(118) $44 + 52 = 96$
(29) $30 + 5 = 35$	(59) $48 + 99 = 147$	(89) $96 + 26 = 122$	(119) $75 + 79 = 154$
(30) $80 + 4 = 84$	(60) $74 + 91 = 165$	(90) $24 + 47 = 71$	(120) $41 + 58 = 99$

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|----------------------|-----------------------|-----------------------|-----------------------|
| (1) $1 + 74 = 75$ | (31) $67 + 7 = 74$ | (61) $55 + 34 = 89$ | (91) $10 + 10 = 20$ |
| (2) $12 + 46 = 58$ | (32) $36 + 69 = 105$ | (62) $80 + 69 = 149$ | (92) $38 + 58 = 96$ |
| (3) $18 + 38 = 56$ | (33) $48 + 29 = 77$ | (63) $88 + 51 = 139$ | (93) $26 + 16 = 42$ |
| (4) $42 + 7 = 49$ | (34) $87 + 18 = 105$ | (64) $85 + 100 = 185$ | (94) $16 + 3 = 19$ |
| (5) $44 + 62 = 106$ | (35) $89 + 92 = 181$ | (65) $5 + 23 = 28$ | (95) $32 + 24 = 56$ |
| (6) $66 + 94 = 160$ | (36) $23 + 51 = 74$ | (66) $66 + 83 = 149$ | (96) $68 + 96 = 164$ |
| (7) $96 + 69 = 165$ | (37) $62 + 95 = 157$ | (67) $63 + 19 = 82$ | (97) $10 + 42 = 52$ |
| (8) $20 + 100 = 120$ | (38) $22 + 3 = 25$ | (68) $15 + 23 = 38$ | (98) $83 + 54 = 137$ |
| (9) $70 + 15 = 85$ | (39) $31 + 22 = 53$ | (69) $48 + 44 = 92$ | (99) $100 + 48 = 148$ |
| (10) $14 + 35 = 49$ | (40) $95 + 98 = 193$ | (70) $87 + 41 = 128$ | (100) $59 + 38 = 97$ |
| (11) $90 + 42 = 132$ | (41) $48 + 96 = 144$ | (71) $70 + 76 = 146$ | (101) $17 + 58 = 75$ |
| (12) $57 + 76 = 133$ | (42) $42 + 66 = 108$ | (72) $21 + 25 = 46$ | (102) $6 + 65 = 71$ |
| (13) $82 + 88 = 170$ | (43) $58 + 15 = 73$ | (73) $95 + 19 = 114$ | (103) $63 + 15 = 78$ |
| (14) $48 + 74 = 122$ | (44) $4 + 96 = 100$ | (74) $57 + 34 = 91$ | (104) $9 + 63 = 72$ |
| (15) $23 + 84 = 107$ | (45) $60 + 98 = 158$ | (75) $13 + 1 = 14$ | (105) $89 + 42 = 131$ |
| (16) $77 + 52 = 129$ | (46) $27 + 82 = 109$ | (76) $22 + 10 = 32$ | (106) $95 + 22 = 117$ |
| (17) $72 + 54 = 126$ | (47) $82 + 12 = 94$ | (77) $30 + 8 = 38$ | (107) $73 + 85 = 158$ |
| (18) $17 + 65 = 82$ | (48) $85 + 60 = 145$ | (78) $78 + 49 = 127$ | (108) $0 + 99 = 99$ |
| (19) $62 + 71 = 133$ | (49) $39 + 57 = 96$ | (79) $88 + 6 = 94$ | (109) $53 + 60 = 113$ |
| (20) $74 + 79 = 153$ | (50) $49 + 98 = 147$ | (80) $48 + 93 = 141$ | (110) $16 + 6 = 22$ |
| (21) $89 + 89 = 178$ | (51) $1 + 4 = 5$ | (81) $42 + 59 = 101$ | (111) $49 + 22 = 71$ |
| (22) $26 + 36 = 62$ | (52) $85 + 90 = 175$ | (82) $58 + 91 = 149$ | (112) $30 + 28 = 58$ |
| (23) $62 + 67 = 129$ | (53) $100 + 23 = 123$ | (83) $1 + 31 = 32$ | (113) $20 + 7 = 27$ |
| (24) $55 + 17 = 72$ | (54) $53 + 97 = 150$ | (84) $62 + 92 = 154$ | (114) $16 + 40 = 56$ |
| (25) $77 + 68 = 145$ | (55) $21 + 11 = 32$ | (85) $57 + 100 = 157$ | (115) $38 + 54 = 92$ |
| (26) $2 + 100 = 102$ | (56) $18 + 53 = 71$ | (86) $85 + 20 = 105$ | (116) $32 + 23 = 55$ |
| (27) $47 + 92 = 139$ | (57) $13 + 16 = 29$ | (87) $30 + 31 = 61$ | (117) $41 + 93 = 134$ |
| (28) $6 + 49 = 55$ | (58) $65 + 34 = 99$ | (88) $57 + 60 = 117$ | (118) $95 + 1 = 96$ |
| (29) $62 + 89 = 151$ | (59) $70 + 8 = 78$ | (89) $3 + 30 = 33$ | (119) $29 + 7 = 36$ |
| (30) $43 + 93 = 136$ | (60) $63 + 17 = 80$ | (90) $83 + 38 = 121$ | (120) $5 + 28 = 33$ |

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|----------------------|-----------------------|----------------------|------------------------|
| (1) $86 + 70 = 156$ | (31) $61 + 81 = 142$ | (61) $93 + 57 = 150$ | (91) $6 + 3 = 9$ |
| (2) $3 + 98 = 101$ | (32) $61 + 56 = 117$ | (62) $7 + 41 = 48$ | (92) $100 + 57 = 157$ |
| (3) $66 + 54 = 120$ | (33) $53 + 13 = 66$ | (63) $93 + 47 = 140$ | (93) $95 + 5 = 100$ |
| (4) $97 + 60 = 157$ | (34) $52 + 81 = 133$ | (64) $18 + 59 = 77$ | (94) $95 + 40 = 135$ |
| (5) $45 + 91 = 136$ | (35) $25 + 24 = 49$ | (65) $78 + 35 = 113$ | (95) $52 + 39 = 91$ |
| (6) $0 + 100 = 100$ | (36) $47 + 17 = 64$ | (66) $80 + 28 = 108$ | (96) $2 + 97 = 99$ |
| (7) $73 + 76 = 149$ | (37) $86 + 28 = 114$ | (67) $11 + 32 = 43$ | (97) $75 + 19 = 94$ |
| (8) $40 + 17 = 57$ | (38) $41 + 59 = 100$ | (68) $8 + 85 = 93$ | (98) $61 + 6 = 67$ |
| (9) $100 + 49 = 149$ | (39) $97 + 41 = 138$ | (69) $13 + 76 = 89$ | (99) $87 + 60 = 147$ |
| (10) $61 + 25 = 86$ | (40) $1 + 24 = 25$ | (70) $37 + 31 = 68$ | (100) $56 + 75 = 131$ |
| (11) $45 + 62 = 107$ | (41) $15 + 72 = 87$ | (71) $82 + 98 = 180$ | (101) $84 + 40 = 124$ |
| (12) $2 + 29 = 31$ | (42) $19 + 36 = 55$ | (72) $57 + 38 = 95$ | (102) $51 + 59 = 110$ |
| (13) $69 + 2 = 71$ | (43) $48 + 69 = 117$ | (73) $72 + 3 = 75$ | (103) $56 + 82 = 138$ |
| (14) $96 + 15 = 111$ | (44) $55 + 100 = 155$ | (74) $45 + 2 = 47$ | (104) $7 + 95 = 102$ |
| (15) $20 + 38 = 58$ | (45) $50 + 92 = 142$ | (75) $88 + 69 = 157$ | (105) $2 + 63 = 65$ |
| (16) $54 + 26 = 80$ | (46) $66 + 11 = 77$ | (76) $3 + 33 = 36$ | (106) $42 + 41 = 83$ |
| (17) $25 + 17 = 42$ | (47) $22 + 84 = 106$ | (77) $38 + 19 = 57$ | (107) $3 + 28 = 31$ |
| (18) $38 + 72 = 110$ | (48) $28 + 41 = 69$ | (78) $19 + 73 = 92$ | (108) $27 + 38 = 65$ |
| (19) $40 + 56 = 96$ | (49) $54 + 40 = 94$ | (79) $32 + 57 = 89$ | (109) $21 + 12 = 33$ |
| (20) $9 + 58 = 67$ | (50) $65 + 20 = 85$ | (80) $88 + 86 = 174$ | (110) $23 + 82 = 105$ |
| (21) $56 + 84 = 140$ | (51) $48 + 52 = 100$ | (81) $77 + 42 = 119$ | (111) $35 + 42 = 77$ |
| (22) $27 + 50 = 77$ | (52) $11 + 8 = 19$ | (82) $80 + 77 = 157$ | (112) $23 + 79 = 102$ |
| (23) $94 + 58 = 152$ | (53) $46 + 46 = 92$ | (83) $100 + 2 = 102$ | (113) $77 + 2 = 79$ |
| (24) $99 + 12 = 111$ | (54) $45 + 79 = 124$ | (84) $100 + 6 = 106$ | (114) $92 + 82 = 174$ |
| (25) $4 + 78 = 82$ | (55) $16 + 57 = 73$ | (85) $57 + 47 = 104$ | (115) $100 + 82 = 182$ |
| (26) $40 + 16 = 56$ | (56) $85 + 74 = 159$ | (86) $10 + 34 = 44$ | (116) $39 + 57 = 96$ |
| (27) $95 + 67 = 162$ | (57) $56 + 38 = 94$ | (87) $74 + 64 = 138$ | (117) $32 + 90 = 122$ |
| (28) $11 + 77 = 88$ | (58) $11 + 54 = 65$ | (88) $2 + 44 = 46$ | (118) $16 + 42 = 58$ |
| (29) $48 + 54 = 102$ | (59) $46 + 42 = 88$ | (89) $33 + 51 = 84$ | (119) $49 + 24 = 73$ |
| (30) $2 + 1 = 3$ | (60) $26 + 58 = 84$ | (90) $61 + 76 = 137$ | (120) $23 + 100 = 123$ |

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|-----------------------|----------------------|-----------------------|-----------------------|
| (1) $77 + 46 = 123$ | (31) $95 + 76 = 171$ | (61) $11 + 100 = 111$ | (91) $18 + 15 = 33$ |
| (2) $27 + 7 = 34$ | (32) $29 + 35 = 64$ | (62) $75 + 33 = 108$ | (92) $68 + 5 = 73$ |
| (3) $51 + 30 = 81$ | (33) $51 + 86 = 137$ | (63) $56 + 43 = 99$ | (93) $18 + 85 = 103$ |
| (4) $31 + 12 = 43$ | (34) $55 + 45 = 100$ | (64) $37 + 23 = 60$ | (94) $36 + 51 = 87$ |
| (5) $84 + 30 = 114$ | (35) $79 + 51 = 130$ | (65) $31 + 50 = 81$ | (95) $49 + 33 = 82$ |
| (6) $41 + 59 = 100$ | (36) $93 + 56 = 149$ | (66) $21 + 88 = 109$ | (96) $81 + 15 = 96$ |
| (7) $87 + 67 = 154$ | (37) $25 + 54 = 79$ | (67) $79 + 92 = 171$ | (97) $91 + 6 = 97$ |
| (8) $28 + 71 = 99$ | (38) $35 + 15 = 50$ | (68) $85 + 71 = 156$ | (98) $76 + 45 = 121$ |
| (9) $52 + 68 = 120$ | (39) $78 + 83 = 161$ | (69) $72 + 3 = 75$ | (99) $46 + 50 = 96$ |
| (10) $24 + 18 = 42$ | (40) $87 + 87 = 174$ | (70) $85 + 82 = 167$ | (100) $8 + 51 = 59$ |
| (11) $75 + 82 = 157$ | (41) $49 + 60 = 109$ | (71) $48 + 48 = 96$ | (101) $27 + 17 = 44$ |
| (12) $97 + 1 = 98$ | (42) $97 + 58 = 155$ | (72) $84 + 89 = 173$ | (102) $45 + 73 = 118$ |
| (13) $35 + 20 = 55$ | (43) $46 + 51 = 97$ | (73) $99 + 26 = 125$ | (103) $89 + 15 = 104$ |
| (14) $94 + 43 = 137$ | (44) $42 + 37 = 79$ | (74) $20 + 82 = 102$ | (104) $36 + 59 = 95$ |
| (15) $28 + 98 = 126$ | (45) $70 + 79 = 149$ | (75) $80 + 85 = 165$ | (105) $96 + 86 = 182$ |
| (16) $100 + 29 = 129$ | (46) $28 + 65 = 93$ | (76) $88 + 72 = 160$ | (106) $73 + 2 = 75$ |
| (17) $94 + 1 = 95$ | (47) $12 + 3 = 15$ | (77) $68 + 39 = 107$ | (107) $44 + 16 = 60$ |
| (18) $18 + 54 = 72$ | (48) $32 + 5 = 37$ | (78) $79 + 64 = 143$ | (108) $29 + 59 = 88$ |
| (19) $65 + 87 = 152$ | (49) $72 + 91 = 163$ | (79) $68 + 29 = 97$ | (109) $35 + 46 = 81$ |
| (20) $89 + 69 = 158$ | (50) $66 + 54 = 120$ | (80) $54 + 78 = 132$ | (110) $10 + 69 = 79$ |
| (21) $60 + 97 = 157$ | (51) $93 + 75 = 168$ | (81) $38 + 49 = 87$ | (111) $18 + 83 = 101$ |
| (22) $83 + 15 = 98$ | (52) $48 + 12 = 60$ | (82) $38 + 90 = 128$ | (112) $14 + 72 = 86$ |
| (23) $76 + 17 = 93$ | (53) $79 + 38 = 117$ | (83) $40 + 8 = 48$ | (113) $51 + 7 = 58$ |
| (24) $74 + 99 = 173$ | (54) $81 + 22 = 103$ | (84) $52 + 82 = 134$ | (114) $19 + 6 = 25$ |
| (25) $35 + 88 = 123$ | (55) $17 + 16 = 33$ | (85) $45 + 83 = 128$ | (115) $31 + 39 = 70$ |
| (26) $30 + 93 = 123$ | (56) $1 + 64 = 65$ | (86) $62 + 91 = 153$ | (116) $21 + 27 = 48$ |
| (27) $88 + 62 = 150$ | (57) $25 + 54 = 79$ | (87) $39 + 48 = 87$ | (117) $14 + 92 = 106$ |
| (28) $75 + 21 = 96$ | (58) $5 + 35 = 40$ | (88) $95 + 28 = 123$ | (118) $33 + 70 = 103$ |
| (29) $95 + 74 = 169$ | (59) $47 + 83 = 130$ | (89) $54 + 62 = 116$ | (119) $34 + 60 = 94$ |
| (30) $84 + 45 = 129$ | (60) $88 + 4 = 92$ | (90) $3 + 83 = 86$ | (120) $38 + 43 = 81$ |

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|----------------------|----------------------|----------------------|------------------------|
| (1) $76 + 55 = 131$ | (31) $91 + 56 = 147$ | (61) $7 + 33 = 40$ | (91) $100 + 31 = 131$ |
| (2) $1 + 79 = 80$ | (32) $68 + 15 = 83$ | (62) $97 + 78 = 175$ | (92) $23 + 4 = 27$ |
| (3) $24 + 32 = 56$ | (33) $58 + 92 = 150$ | (63) $37 + 64 = 101$ | (93) $30 + 34 = 64$ |
| (4) $61 + 48 = 109$ | (34) $77 + 66 = 143$ | (64) $69 + 51 = 120$ | (94) $86 + 75 = 161$ |
| (5) $10 + 89 = 99$ | (35) $99 + 24 = 123$ | (65) $21 + 43 = 64$ | (95) $35 + 57 = 92$ |
| (6) $35 + 75 = 110$ | (36) $23 + 84 = 107$ | (66) $4 + 78 = 82$ | (96) $20 + 51 = 71$ |
| (7) $39 + 98 = 137$ | (37) $70 + 74 = 144$ | (67) $69 + 97 = 166$ | (97) $96 + 65 = 161$ |
| (8) $91 + 69 = 160$ | (38) $9 + 38 = 47$ | (68) $14 + 97 = 111$ | (98) $28 + 53 = 81$ |
| (9) $26 + 55 = 81$ | (39) $95 + 61 = 156$ | (69) $68 + 68 = 136$ | (99) $94 + 62 = 156$ |
| (10) $47 + 78 = 125$ | (40) $4 + 4 = 8$ | (70) $36 + 14 = 50$ | (100) $9 + 97 = 106$ |
| (11) $40 + 5 = 45$ | (41) $89 + 5 = 94$ | (71) $88 + 2 = 90$ | (101) $20 + 58 = 78$ |
| (12) $8 + 63 = 71$ | (42) $8 + 88 = 96$ | (72) $41 + 20 = 61$ | (102) $13 + 22 = 35$ |
| (13) $98 + 51 = 149$ | (43) $48 + 62 = 110$ | (73) $80 + 37 = 117$ | (103) $71 + 30 = 101$ |
| (14) $30 + 46 = 76$ | (44) $0 + 62 = 62$ | (74) $40 + 56 = 96$ | (104) $25 + 100 = 125$ |
| (15) $17 + 80 = 97$ | (45) $73 + 43 = 116$ | (75) $73 + 90 = 163$ | (105) $98 + 54 = 152$ |
| (16) $25 + 21 = 46$ | (46) $71 + 72 = 143$ | (76) $91 + 33 = 124$ | (106) $37 + 32 = 69$ |
| (17) $25 + 85 = 110$ | (47) $87 + 4 = 91$ | (77) $28 + 95 = 123$ | (107) $56 + 36 = 92$ |
| (18) $26 + 4 = 30$ | (48) $61 + 86 = 147$ | (78) $97 + 0 = 97$ | (108) $57 + 71 = 128$ |
| (19) $67 + 24 = 91$ | (49) $49 + 91 = 140$ | (79) $64 + 42 = 106$ | (109) $58 + 86 = 144$ |
| (20) $93 + 51 = 144$ | (50) $46 + 92 = 138$ | (80) $91 + 87 = 178$ | (110) $32 + 11 = 43$ |
| (21) $79 + 7 = 86$ | (51) $99 + 85 = 184$ | (81) $61 + 65 = 126$ | (111) $42 + 86 = 128$ |
| (22) $69 + 30 = 99$ | (52) $13 + 6 = 19$ | (82) $61 + 15 = 76$ | (112) $37 + 66 = 103$ |
| (23) $82 + 10 = 92$ | (53) $28 + 49 = 77$ | (83) $95 + 40 = 135$ | (113) $90 + 33 = 123$ |
| (24) $61 + 51 = 112$ | (54) $29 + 46 = 75$ | (84) $9 + 3 = 12$ | (114) $7 + 47 = 54$ |
| (25) $9 + 91 = 100$ | (55) $56 + 80 = 136$ | (85) $15 + 60 = 75$ | (115) $46 + 53 = 99$ |
| (26) $77 + 33 = 110$ | (56) $69 + 50 = 119$ | (86) $49 + 49 = 98$ | (116) $12 + 2 = 14$ |
| (27) $7 + 69 = 76$ | (57) $81 + 66 = 147$ | (87) $10 + 45 = 55$ | (117) $90 + 100 = 190$ |
| (28) $44 + 32 = 76$ | (58) $24 + 38 = 62$ | (88) $76 + 88 = 164$ | (118) $35 + 66 = 101$ |
| (29) $88 + 11 = 99$ | (59) $47 + 41 = 88$ | (89) $44 + 88 = 132$ | (119) $72 + 45 = 117$ |
| (30) $79 + 33 = 112$ | (60) $33 + 20 = 53$ | (90) $15 + 85 = 100$ | (120) $55 + 92 = 147$ |

(1) $30 + 63 = 93$	(31) $87 + 80 = 167$	(61) $29 + 75 = 104$	(91) $53 + 99 = 152$
(2) $21 + 74 = 95$	(32) $55 + 30 = 85$	(62) $69 + 78 = 147$	(92) $76 + 24 = 100$
(3) $97 + 27 = 124$	(33) $36 + 4 = 40$	(63) $57 + 43 = 100$	(93) $96 + 71 = 167$
(4) $28 + 6 = 34$	(34) $3 + 52 = 55$	(64) $40 + 1 = 41$	(94) $81 + 55 = 136$
(5) $2 + 65 = 67$	(35) $16 + 99 = 115$	(65) $61 + 47 = 108$	(95) $58 + 81 = 139$
(6) $88 + 55 = 143$	(36) $9 + 84 = 93$	(66) $27 + 27 = 54$	(96) $97 + 9 = 106$
(7) $23 + 83 = 106$	(37) $83 + 6 = 89$	(67) $43 + 43 = 86$	(97) $0 + 46 = 46$
(8) $7 + 32 = 39$	(38) $96 + 46 = 142$	(68) $100 + 37 = 137$	(98) $44 + 87 = 131$
(9) $87 + 33 = 120$	(39) $26 + 37 = 63$	(69) $69 + 44 = 113$	(99) $50 + 83 = 133$
(10) $29 + 30 = 59$	(40) $34 + 92 = 126$	(70) $52 + 55 = 107$	(100) $24 + 97 = 121$
(11) $76 + 8 = 84$	(41) $63 + 30 = 93$	(71) $12 + 85 = 97$	(101) $85 + 40 = 125$
(12) $48 + 65 = 113$	(42) $60 + 20 = 80$	(72) $15 + 100 = 115$	(102) $71 + 96 = 167$
(13) $27 + 12 = 39$	(43) $73 + 63 = 136$	(73) $88 + 45 = 133$	(103) $88 + 19 = 107$
(14) $35 + 43 = 78$	(44) $94 + 11 = 105$	(74) $23 + 58 = 81$	(104) $62 + 45 = 107$
(15) $56 + 78 = 134$	(45) $85 + 1 = 86$	(75) $17 + 83 = 100$	(105) $11 + 100 = 111$
(16) $19 + 57 = 76$	(46) $72 + 20 = 92$	(76) $56 + 95 = 151$	(106) $73 + 67 = 140$
(17) $45 + 46 = 91$	(47) $98 + 79 = 177$	(77) $96 + 15 = 111$	(107) $32 + 17 = 49$
(18) $56 + 72 = 128$	(48) $8 + 24 = 32$	(78) $96 + 24 = 120$	(108) $32 + 95 = 127$
(19) $51 + 54 = 105$	(49) $65 + 26 = 91$	(79) $84 + 95 = 179$	(109) $18 + 14 = 32$
(20) $46 + 13 = 59$	(50) $100 + 33 = 133$	(80) $48 + 7 = 55$	(110) $59 + 85 = 144$
(21) $12 + 7 = 19$	(51) $11 + 9 = 20$	(81) $52 + 45 = 97$	(111) $23 + 32 = 55$
(22) $31 + 100 = 131$	(52) $16 + 72 = 88$	(82) $48 + 83 = 131$	(112) $38 + 100 = 138$
(23) $24 + 26 = 50$	(53) $84 + 74 = 158$	(83) $95 + 61 = 156$	(113) $10 + 88 = 98$
(24) $56 + 96 = 152$	(54) $49 + 90 = 139$	(84) $97 + 29 = 126$	(114) $87 + 61 = 148$
(25) $97 + 20 = 117$	(55) $80 + 89 = 169$	(85) $34 + 82 = 116$	(115) $41 + 31 = 72$
(26) $55 + 57 = 112$	(56) $51 + 16 = 67$	(86) $91 + 99 = 190$	(116) $37 + 51 = 88$
(27) $40 + 93 = 133$	(57) $47 + 57 = 104$	(87) $64 + 83 = 147$	(117) $79 + 17 = 96$
(28) $63 + 26 = 89$	(58) $95 + 35 = 130$	(88) $13 + 88 = 101$	(118) $62 + 5 = 67$
(29) $54 + 71 = 125$	(59) $70 + 96 = 166$	(89) $12 + 48 = 60$	(119) $87 + 25 = 112$
(30) $78 + 15 = 93$	(60) $97 + 1 = 98$	(90) $98 + 63 = 161$	(120) $20 + 25 = 45$

(1) $91 + 29 = 120$	(31) $88 + 78 = 166$	(61) $7 + 45 = 52$	(91) $90 + 57 = 147$
(2) $13 + 15 = 28$	(32) $78 + 77 = 155$	(62) $70 + 73 = 143$	(92) $27 + 83 = 110$
(3) $92 + 59 = 151$	(33) $53 + 78 = 131$	(63) $8 + 88 = 96$	(93) $51 + 33 = 84$
(4) $1 + 46 = 47$	(34) $73 + 40 = 113$	(64) $20 + 70 = 90$	(94) $68 + 26 = 94$
(5) $14 + 90 = 104$	(35) $51 + 96 = 147$	(65) $73 + 51 = 124$	(95) $79 + 81 = 160$
(6) $79 + 97 = 176$	(36) $71 + 25 = 96$	(66) $98 + 38 = 136$	(96) $6 + 16 = 22$
(7) $82 + 5 = 87$	(37) $72 + 14 = 86$	(67) $12 + 97 = 109$	(97) $33 + 71 = 104$
(8) $46 + 80 = 126$	(38) $62 + 7 = 69$	(68) $32 + 63 = 95$	(98) $64 + 32 = 96$
(9) $56 + 34 = 90$	(39) $92 + 48 = 140$	(69) $73 + 28 = 101$	(99) $34 + 12 = 46$
(10) $18 + 50 = 68$	(40) $2 + 7 = 9$	(70) $90 + 23 = 113$	(100) $33 + 77 = 110$
(11) $97 + 69 = 166$	(41) $17 + 35 = 52$	(71) $34 + 73 = 107$	(101) $65 + 18 = 83$
(12) $17 + 6 = 23$	(42) $13 + 56 = 69$	(72) $6 + 87 = 93$	(102) $19 + 68 = 87$
(13) $38 + 16 = 54$	(43) $94 + 3 = 97$	(73) $53 + 24 = 77$	(103) $28 + 90 = 118$
(14) $13 + 45 = 58$	(44) $60 + 57 = 117$	(74) $21 + 61 = 82$	(104) $68 + 40 = 108$
(15) $40 + 20 = 60$	(45) $15 + 28 = 43$	(75) $57 + 46 = 103$	(105) $70 + 48 = 118$
(16) $87 + 68 = 155$	(46) $83 + 70 = 153$	(76) $33 + 15 = 48$	(106) $66 + 67 = 133$
(17) $4 + 34 = 38$	(47) $14 + 84 = 98$	(77) $31 + 0 = 31$	(107) $43 + 90 = 133$
(18) $25 + 4 = 29$	(48) $1 + 74 = 75$	(78) $72 + 78 = 150$	(108) $84 + 2 = 86$
(19) $42 + 22 = 64$	(49) $57 + 72 = 129$	(79) $88 + 28 = 116$	(109) $26 + 48 = 74$
(20) $24 + 38 = 62$	(50) $84 + 87 = 171$	(80) $53 + 50 = 103$	(110) $78 + 15 = 93$
(21) $61 + 38 = 99$	(51) $39 + 71 = 110$	(81) $39 + 9 = 48$	(111) $95 + 32 = 127$
(22) $26 + 97 = 123$	(52) $77 + 67 = 144$	(82) $63 + 95 = 158$	(112) $76 + 88 = 164$
(23) $96 + 75 = 171$	(53) $79 + 85 = 164$	(83) $73 + 73 = 146$	(113) $68 + 32 = 100$
(24) $71 + 47 = 118$	(54) $76 + 87 = 163$	(84) $64 + 3 = 67$	(114) $24 + 39 = 63$
(25) $54 + 45 = 99$	(55) $83 + 46 = 129$	(85) $71 + 100 = 171$	(115) $99 + 43 = 142$
(26) $78 + 72 = 150$	(56) $20 + 39 = 59$	(86) $46 + 43 = 89$	(116) $22 + 2 = 24$
(27) $14 + 13 = 27$	(57) $35 + 68 = 103$	(87) $11 + 86 = 97$	(117) $94 + 63 = 157$
(28) $54 + 29 = 83$	(58) $72 + 1 = 73$	(88) $39 + 83 = 122$	(118) $40 + 49 = 89$
(29) $50 + 34 = 84$	(59) $98 + 87 = 185$	(89) $25 + 15 = 40$	(119) $29 + 88 = 117$
(30) $60 + 76 = 136$	(60) $3 + 39 = 42$	(90) $85 + 69 = 154$	(120) $96 + 79 = 175$

(1) $61 + 16 = 77$	(31) $77 + 30 = 107$	(61) $64 + 90 = 154$	(91) $72 + 78 = 150$
(2) $11 + 13 = 24$	(32) $20 + 54 = 74$	(62) $53 + 80 = 133$	(92) $36 + 33 = 69$
(3) $26 + 73 = 99$	(33) $83 + 81 = 164$	(63) $78 + 99 = 177$	(93) $81 + 85 = 166$
(4) $97 + 96 = 193$	(34) $86 + 51 = 137$	(64) $43 + 80 = 123$	(94) $75 + 54 = 129$
(5) $75 + 6 = 81$	(35) $25 + 44 = 69$	(65) $40 + 19 = 59$	(95) $70 + 32 = 102$
(6) $93 + 40 = 133$	(36) $33 + 76 = 109$	(66) $76 + 40 = 116$	(96) $89 + 38 = 127$
(7) $70 + 74 = 144$	(37) $55 + 83 = 138$	(67) $37 + 83 = 120$	(97) $100 + 35 = 135$
(8) $31 + 38 = 69$	(38) $10 + 32 = 42$	(68) $36 + 91 = 127$	(98) $71 + 6 = 77$
(9) $71 + 29 = 100$	(39) $36 + 98 = 134$	(69) $38 + 87 = 125$	(99) $90 + 4 = 94$
(10) $57 + 58 = 115$	(40) $87 + 63 = 150$	(70) $94 + 12 = 106$	(100) $18 + 70 = 88$
(11) $18 + 16 = 34$	(41) $76 + 75 = 151$	(71) $49 + 99 = 148$	(101) $82 + 64 = 146$
(12) $48 + 92 = 140$	(42) $48 + 32 = 80$	(72) $100 + 59 = 159$	(102) $56 + 9 = 65$
(13) $74 + 23 = 97$	(43) $43 + 68 = 111$	(73) $79 + 2 = 81$	(103) $53 + 63 = 116$
(14) $33 + 86 = 119$	(44) $8 + 51 = 59$	(74) $85 + 100 = 185$	(104) $45 + 6 = 51$
(15) $65 + 46 = 111$	(45) $91 + 97 = 188$	(75) $66 + 33 = 99$	(105) $89 + 33 = 122$
(16) $89 + 45 = 134$	(46) $47 + 97 = 144$	(76) $84 + 39 = 123$	(106) $10 + 7 = 17$
(17) $85 + 88 = 173$	(47) $96 + 98 = 194$	(77) $83 + 60 = 143$	(107) $87 + 64 = 151$
(18) $43 + 18 = 61$	(48) $55 + 10 = 65$	(78) $32 + 22 = 54$	(108) $41 + 78 = 119$
(19) $56 + 2 = 58$	(49) $2 + 93 = 95$	(79) $80 + 20 = 100$	(109) $15 + 48 = 63$
(20) $81 + 95 = 176$	(50) $57 + 70 = 127$	(80) $86 + 9 = 95$	(110) $57 + 43 = 100$
(21) $72 + 55 = 127$	(51) $56 + 33 = 89$	(81) $59 + 81 = 140$	(111) $81 + 78 = 159$
(22) $12 + 30 = 42$	(52) $47 + 33 = 80$	(82) $24 + 38 = 62$	(112) $0 + 53 = 53$
(23) $11 + 50 = 61$	(53) $46 + 60 = 106$	(83) $34 + 39 = 73$	(113) $98 + 55 = 153$
(24) $52 + 11 = 63$	(54) $44 + 99 = 143$	(84) $20 + 74 = 94$	(114) $5 + 32 = 37$
(25) $14 + 86 = 100$	(55) $7 + 18 = 25$	(85) $55 + 77 = 132$	(115) $77 + 37 = 114$
(26) $59 + 14 = 73$	(56) $94 + 46 = 140$	(86) $51 + 95 = 146$	(116) $0 + 39 = 39$
(27) $33 + 70 = 103$	(57) $87 + 78 = 165$	(87) $53 + 76 = 129$	(117) $89 + 2 = 91$
(28) $37 + 95 = 132$	(58) $56 + 24 = 80$	(88) $4 + 39 = 43$	(118) $9 + 10 = 19$
(29) $95 + 28 = 123$	(59) $90 + 66 = 156$	(89) $17 + 42 = 59$	(119) $89 + 49 = 138$
(30) $79 + 11 = 90$	(60) $48 + 66 = 114$	(90) $51 + 17 = 68$	(120) $57 + 60 = 117$

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| (1) $1 + 13 = 14$ | (31) $74 + 83 = 157$ | (61) $37 + 45 = 82$ | (91) $11 + 82 = 93$ |
| (2) $54 + 60 = 114$ | (32) $17 + 94 = 111$ | (62) $6 + 48 = 54$ | (92) $14 + 84 = 98$ |
| (3) $84 + 99 = 183$ | (33) $91 + 4 = 95$ | (63) $16 + 59 = 75$ | (93) $8 + 16 = 24$ |
| (4) $96 + 96 = 192$ | (34) $76 + 32 = 108$ | (64) $47 + 96 = 143$ | (94) $29 + 17 = 46$ |
| (5) $69 + 60 = 129$ | (35) $39 + 44 = 83$ | (65) $48 + 35 = 83$ | (95) $46 + 19 = 65$ |
| (6) $94 + 61 = 155$ | (36) $73 + 59 = 132$ | (66) $23 + 100 = 123$ | (96) $61 + 61 = 122$ |
| (7) $7 + 65 = 72$ | (37) $84 + 21 = 105$ | (67) $20 + 52 = 72$ | (97) $40 + 42 = 82$ |
| (8) $32 + 36 = 68$ | (38) $68 + 14 = 82$ | (68) $62 + 62 = 124$ | (98) $10 + 12 = 22$ |
| (9) $51 + 49 = 100$ | (39) $79 + 68 = 147$ | (69) $90 + 22 = 112$ | (99) $41 + 98 = 139$ |
| (10) $2 + 94 = 96$ | (40) $16 + 46 = 62$ | (70) $38 + 39 = 77$ | (100) $54 + 5 = 59$ |
| (11) $76 + 18 = 94$ | (41) $91 + 11 = 102$ | (71) $61 + 41 = 102$ | (101) $81 + 78 = 159$ |
| (12) $59 + 79 = 138$ | (42) $23 + 100 = 123$ | (72) $3 + 96 = 99$ | (102) $9 + 72 = 81$ |
| (13) $13 + 36 = 49$ | (43) $27 + 79 = 106$ | (73) $30 + 92 = 122$ | (103) $14 + 46 = 60$ |
| (14) $93 + 69 = 162$ | (44) $70 + 55 = 125$ | (74) $33 + 75 = 108$ | (104) $31 + 15 = 46$ |
| (15) $60 + 66 = 126$ | (45) $41 + 54 = 95$ | (75) $82 + 42 = 124$ | (105) $61 + 66 = 127$ |
| (16) $28 + 100 = 128$ | (46) $57 + 9 = 66$ | (76) $85 + 53 = 138$ | (106) $55 + 45 = 100$ |
| (17) $70 + 91 = 161$ | (47) $50 + 96 = 146$ | (77) $85 + 11 = 96$ | (107) $45 + 60 = 105$ |
| (18) $72 + 27 = 99$ | (48) $8 + 56 = 64$ | (78) $80 + 15 = 95$ | (108) $55 + 68 = 123$ |
| (19) $81 + 29 = 110$ | (49) $7 + 73 = 80$ | (79) $87 + 37 = 124$ | (109) $9 + 88 = 97$ |
| (20) $39 + 45 = 84$ | (50) $28 + 61 = 89$ | (80) $58 + 94 = 152$ | (110) $93 + 40 = 133$ |
| (21) $74 + 3 = 77$ | (51) $2 + 71 = 73$ | (81) $4 + 15 = 19$ | (111) $56 + 27 = 83$ |
| (22) $53 + 92 = 145$ | (52) $94 + 99 = 193$ | (82) $70 + 33 = 103$ | (112) $28 + 46 = 74$ |
| (23) $35 + 81 = 116$ | (53) $15 + 51 = 66$ | (83) $59 + 62 = 121$ | (113) $44 + 52 = 96$ |
| (24) $24 + 93 = 117$ | (54) $33 + 57 = 90$ | (84) $16 + 33 = 49$ | (114) $61 + 46 = 107$ |
| (25) $92 + 17 = 109$ | (55) $64 + 27 = 91$ | (85) $73 + 41 = 114$ | (115) $36 + 73 = 109$ |
| (26) $56 + 10 = 66$ | (56) $82 + 65 = 147$ | (86) $47 + 86 = 133$ | (116) $49 + 45 = 94$ |
| (27) $53 + 43 = 96$ | (57) $20 + 60 = 80$ | (87) $44 + 89 = 133$ | (117) $15 + 34 = 49$ |
| (28) $94 + 28 = 122$ | (58) $96 + 88 = 184$ | (88) $9 + 23 = 32$ | (118) $35 + 90 = 125$ |
| (29) $77 + 44 = 121$ | (59) $99 + 65 = 164$ | (89) $72 + 32 = 104$ | (119) $60 + 9 = 69$ |
| (30) $23 + 62 = 85$ | (60) $88 + 82 = 170$ | (90) $98 + 5 = 103$ | (120) $67 + 37 = 104$ |

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| (1) $31 + 76 = 107$ | (31) $29 + 29 = 58$ | (61) $54 + 35 = 89$ | (91) $86 + 68 = 154$ |
| (2) $37 + 59 = 96$ | (32) $8 + 68 = 76$ | (62) $17 + 56 = 73$ | (92) $88 + 9 = 97$ |
| (3) $43 + 72 = 115$ | (33) $93 + 59 = 152$ | (63) $48 + 100 = 148$ | (93) $1 + 35 = 36$ |
| (4) $80 + 42 = 122$ | (34) $32 + 28 = 60$ | (64) $9 + 99 = 108$ | (94) $22 + 3 = 25$ |
| (5) $20 + 73 = 93$ | (35) $77 + 84 = 161$ | (65) $78 + 60 = 138$ | (95) $84 + 92 = 176$ |
| (6) $100 + 46 = 146$ | (36) $59 + 5 = 64$ | (66) $14 + 25 = 39$ | (96) $42 + 74 = 116$ |
| (7) $5 + 48 = 53$ | (37) $18 + 62 = 80$ | (67) $24 + 71 = 95$ | (97) $15 + 74 = 89$ |
| (8) $78 + 13 = 91$ | (38) $16 + 23 = 39$ | (68) $87 + 67 = 154$ | (98) $36 + 41 = 77$ |
| (9) $43 + 37 = 80$ | (39) $27 + 91 = 118$ | (69) $70 + 51 = 121$ | (99) $0 + 13 = 13$ |
| (10) $17 + 67 = 84$ | (40) $2 + 95 = 97$ | (70) $83 + 84 = 167$ | (100) $25 + 32 = 57$ |
| (11) $67 + 1 = 68$ | (41) $7 + 69 = 76$ | (71) $23 + 78 = 101$ | (101) $17 + 28 = 45$ |
| (12) $33 + 88 = 121$ | (42) $6 + 91 = 97$ | (72) $82 + 76 = 158$ | (102) $71 + 83 = 154$ |
| (13) $5 + 87 = 92$ | (43) $86 + 87 = 173$ | (73) $83 + 85 = 168$ | (103) $7 + 69 = 76$ |
| (14) $71 + 63 = 134$ | (44) $89 + 92 = 181$ | (74) $99 + 98 = 197$ | (104) $12 + 64 = 76$ |
| (15) $45 + 92 = 137$ | (45) $35 + 68 = 103$ | (75) $25 + 42 = 67$ | (105) $43 + 89 = 132$ |
| (16) $67 + 68 = 135$ | (46) $49 + 75 = 124$ | (76) $1 + 20 = 21$ | (106) $68 + 14 = 82$ |
| (17) $96 + 6 = 102$ | (47) $82 + 19 = 101$ | (77) $54 + 1 = 55$ | (107) $76 + 98 = 174$ |
| (18) $30 + 56 = 86$ | (48) $84 + 44 = 128$ | (78) $86 + 38 = 124$ | (108) $20 + 87 = 107$ |
| (19) $61 + 16 = 77$ | (49) $15 + 37 = 52$ | (79) $58 + 9 = 67$ | (109) $95 + 71 = 166$ |
| (20) $65 + 50 = 115$ | (50) $80 + 12 = 92$ | (80) $81 + 97 = 178$ | (110) $65 + 97 = 162$ |
| (21) $27 + 63 = 90$ | (51) $35 + 85 = 120$ | (81) $15 + 86 = 101$ | (111) $18 + 71 = 89$ |
| (22) $23 + 79 = 102$ | (52) $4 + 21 = 25$ | (82) $21 + 28 = 49$ | (112) $8 + 18 = 26$ |
| (23) $33 + 9 = 42$ | (53) $55 + 73 = 128$ | (83) $93 + 50 = 143$ | (113) $83 + 59 = 142$ |
| (24) $59 + 83 = 142$ | (54) $16 + 67 = 83$ | (84) $36 + 13 = 49$ | (114) $34 + 44 = 78$ |
| (25) $49 + 68 = 117$ | (55) $63 + 15 = 78$ | (85) $55 + 79 = 134$ | (115) $89 + 92 = 181$ |
| (26) $50 + 60 = 110$ | (56) $58 + 97 = 155$ | (86) $52 + 40 = 92$ | (116) $7 + 31 = 38$ |
| (27) $92 + 64 = 156$ | (57) $36 + 52 = 88$ | (87) $62 + 55 = 117$ | (117) $27 + 38 = 65$ |
| (28) $55 + 56 = 111$ | (58) $42 + 58 = 100$ | (88) $37 + 56 = 93$ | (118) $53 + 90 = 143$ |
| (29) $58 + 74 = 132$ | (59) $45 + 99 = 144$ | (89) $12 + 76 = 88$ | (119) $82 + 36 = 118$ |
| (30) $22 + 23 = 45$ | (60) $1 + 70 = 71$ | (90) $73 + 61 = 134$ | (120) $88 + 8 = 96$ |

(1) $94 + 95 = 189$	(31) $71 + 26 = 97$	(61) $71 + 91 = 162$	(91) $76 + 97 = 173$
(2) $56 + 48 = 104$	(32) $46 + 47 = 93$	(62) $60 + 64 = 124$	(92) $50 + 41 = 91$
(3) $21 + 18 = 39$	(33) $30 + 83 = 113$	(63) $69 + 3 = 72$	(93) $27 + 32 = 59$
(4) $98 + 48 = 146$	(34) $99 + 65 = 164$	(64) $38 + 63 = 101$	(94) $17 + 76 = 93$
(5) $29 + 94 = 123$	(35) $98 + 77 = 175$	(65) $2 + 77 = 79$	(95) $7 + 26 = 33$
(6) $7 + 71 = 78$	(36) $64 + 79 = 143$	(66) $12 + 4 = 16$	(96) $60 + 94 = 154$
(7) $55 + 40 = 95$	(37) $12 + 54 = 66$	(67) $35 + 72 = 107$	(97) $41 + 50 = 91$
(8) $0 + 99 = 99$	(38) $35 + 45 = 80$	(68) $71 + 15 = 86$	(98) $77 + 12 = 89$
(9) $64 + 63 = 127$	(39) $14 + 44 = 58$	(69) $68 + 38 = 106$	(99) $19 + 32 = 51$
(10) $96 + 96 = 192$	(40) $88 + 92 = 180$	(70) $87 + 49 = 136$	(100) $79 + 94 = 173$
(11) $51 + 79 = 130$	(41) $76 + 93 = 169$	(71) $93 + 100 = 193$	(101) $32 + 84 = 116$
(12) $34 + 34 = 68$	(42) $3 + 36 = 39$	(72) $94 + 2 = 96$	(102) $36 + 18 = 54$
(13) $14 + 90 = 104$	(43) $83 + 26 = 109$	(73) $66 + 80 = 146$	(103) $37 + 25 = 62$
(14) $80 + 69 = 149$	(44) $59 + 90 = 149$	(74) $39 + 30 = 69$	(104) $75 + 36 = 111$
(15) $32 + 4 = 36$	(45) $38 + 60 = 98$	(75) $63 + 29 = 92$	(105) $92 + 45 = 137$
(16) $19 + 20 = 39$	(46) $61 + 3 = 64$	(76) $96 + 45 = 141$	(106) $19 + 50 = 69$
(17) $74 + 97 = 171$	(47) $55 + 46 = 101$	(77) $10 + 6 = 16$	(107) $52 + 95 = 147$
(18) $96 + 78 = 174$	(48) $72 + 86 = 158$	(78) $47 + 91 = 138$	(108) $30 + 27 = 57$
(19) $14 + 60 = 74$	(49) $45 + 71 = 116$	(79) $97 + 43 = 140$	(109) $23 + 88 = 111$
(20) $14 + 13 = 27$	(50) $78 + 98 = 176$	(80) $31 + 40 = 71$	(110) $54 + 16 = 70$
(21) $2 + 78 = 80$	(51) $70 + 13 = 83$	(81) $52 + 48 = 100$	(111) $60 + 84 = 144$
(22) $20 + 53 = 73$	(52) $80 + 11 = 91$	(82) $61 + 55 = 116$	(112) $74 + 55 = 129$
(23) $14 + 37 = 51$	(53) $2 + 22 = 24$	(83) $16 + 11 = 27$	(113) $52 + 13 = 65$
(24) $22 + 14 = 36$	(54) $15 + 99 = 114$	(84) $20 + 84 = 104$	(114) $22 + 52 = 74$
(25) $28 + 18 = 46$	(55) $36 + 85 = 121$	(85) $50 + 69 = 119$	(115) $79 + 76 = 155$
(26) $19 + 74 = 93$	(56) $14 + 2 = 16$	(86) $1 + 31 = 32$	(116) $28 + 90 = 118$
(27) $35 + 77 = 112$	(57) $91 + 41 = 132$	(87) $32 + 78 = 110$	(117) $26 + 7 = 33$
(28) $58 + 5 = 63$	(58) $74 + 84 = 158$	(88) $27 + 5 = 32$	(118) $47 + 67 = 114$
(29) $62 + 30 = 92$	(59) $19 + 51 = 70$	(89) $28 + 1 = 29$	(119) $77 + 37 = 114$
(30) $52 + 29 = 81$	(60) $26 + 44 = 70$	(90) $8 + 3 = 11$	(120) $35 + 14 = 49$

(1) $11 + 68 = 79$	(31) $51 + 42 = 93$	(61) $64 + 67 = 131$	(91) $22 + 33 = 55$
(2) $45 + 1 = 46$	(32) $12 + 93 = 105$	(62) $86 + 30 = 116$	(92) $100 + 38 = 138$
(3) $66 + 69 = 135$	(33) $62 + 90 = 152$	(63) $42 + 52 = 94$	(93) $100 + 28 = 128$
(4) $12 + 18 = 30$	(34) $45 + 6 = 51$	(64) $26 + 88 = 114$	(94) $43 + 97 = 140$
(5) $80 + 81 = 161$	(35) $68 + 89 = 157$	(65) $17 + 54 = 71$	(95) $92 + 50 = 142$
(6) $66 + 51 = 117$	(36) $54 + 86 = 140$	(66) $74 + 13 = 87$	(96) $44 + 15 = 59$
(7) $60 + 19 = 79$	(37) $6 + 93 = 99$	(67) $0 + 79 = 79$	(97) $66 + 37 = 103$
(8) $62 + 29 = 91$	(38) $22 + 5 = 27$	(68) $50 + 60 = 110$	(98) $92 + 100 = 192$
(9) $45 + 1 = 46$	(39) $99 + 77 = 176$	(69) $72 + 78 = 150$	(99) $61 + 81 = 142$
(10) $95 + 43 = 138$	(40) $20 + 55 = 75$	(70) $59 + 27 = 86$	(100) $69 + 13 = 82$
(11) $99 + 61 = 160$	(41) $26 + 67 = 93$	(71) $94 + 98 = 192$	(101) $4 + 59 = 63$
(12) $43 + 87 = 130$	(42) $64 + 78 = 142$	(72) $95 + 1 = 96$	(102) $22 + 93 = 115$
(13) $8 + 11 = 19$	(43) $88 + 47 = 135$	(73) $12 + 70 = 82$	(103) $81 + 77 = 158$
(14) $56 + 40 = 96$	(44) $22 + 17 = 39$	(74) $32 + 47 = 79$	(104) $34 + 78 = 112$
(15) $86 + 50 = 136$	(45) $44 + 83 = 127$	(75) $73 + 94 = 167$	(105) $77 + 67 = 144$
(16) $96 + 59 = 155$	(46) $48 + 28 = 76$	(76) $47 + 47 = 94$	(106) $14 + 32 = 46$
(17) $94 + 49 = 143$	(47) $25 + 65 = 90$	(77) $26 + 3 = 29$	(107) $53 + 48 = 101$
(18) $23 + 27 = 50$	(48) $74 + 56 = 130$	(78) $17 + 76 = 93$	(108) $20 + 53 = 73$
(19) $82 + 3 = 85$	(49) $96 + 88 = 184$	(79) $56 + 86 = 142$	(109) $86 + 60 = 146$
(20) $18 + 89 = 107$	(50) $62 + 100 = 162$	(80) $12 + 97 = 109$	(110) $24 + 47 = 71$
(21) $86 + 25 = 111$	(51) $0 + 74 = 74$	(81) $56 + 5 = 61$	(111) $27 + 27 = 54$
(22) $24 + 35 = 59$	(52) $35 + 49 = 84$	(82) $87 + 14 = 101$	(112) $15 + 19 = 34$
(23) $6 + 85 = 91$	(53) $69 + 42 = 111$	(83) $12 + 3 = 15$	(113) $3 + 24 = 27$
(24) $89 + 42 = 131$	(54) $4 + 79 = 83$	(84) $82 + 38 = 120$	(114) $100 + 4 = 104$
(25) $9 + 42 = 51$	(55) $25 + 10 = 35$	(85) $30 + 32 = 62$	(115) $61 + 18 = 79$
(26) $48 + 58 = 106$	(56) $54 + 10 = 64$	(86) $67 + 88 = 155$	(116) $73 + 94 = 167$
(27) $69 + 65 = 134$	(57) $86 + 29 = 115$	(87) $39 + 86 = 125$	(117) $9 + 36 = 45$
(28) $66 + 8 = 74$	(58) $80 + 27 = 107$	(88) $47 + 46 = 93$	(118) $89 + 81 = 170$
(29) $1 + 49 = 50$	(59) $18 + 6 = 24$	(89) $20 + 28 = 48$	(119) $85 + 34 = 119$
(30) $23 + 55 = 78$	(60) $70 + 55 = 125$	(90) $20 + 46 = 66$	(120) $33 + 56 = 89$

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|----------------------|----------------------|-----------------------|-----------------------|
| (1) $56 + 60 = 116$ | (31) $97 + 18 = 115$ | (61) $91 + 25 = 116$ | (91) $3 + 33 = 36$ |
| (2) $38 + 13 = 51$ | (32) $46 + 38 = 84$ | (62) $93 + 36 = 129$ | (92) $22 + 8 = 30$ |
| (3) $43 + 58 = 101$ | (33) $85 + 81 = 166$ | (63) $100 + 63 = 163$ | (93) $65 + 56 = 121$ |
| (4) $67 + 37 = 104$ | (34) $4 + 79 = 83$ | (64) $76 + 37 = 113$ | (94) $19 + 81 = 100$ |
| (5) $69 + 50 = 119$ | (35) $44 + 39 = 83$ | (65) $59 + 57 = 116$ | (95) $70 + 2 = 72$ |
| (6) $71 + 28 = 99$ | (36) $66 + 72 = 138$ | (66) $85 + 56 = 141$ | (96) $5 + 64 = 69$ |
| (7) $10 + 76 = 86$ | (37) $70 + 88 = 158$ | (67) $66 + 75 = 141$ | (97) $15 + 54 = 69$ |
| (8) $100 + 42 = 142$ | (38) $55 + 24 = 79$ | (68) $66 + 83 = 149$ | (98) $4 + 61 = 65$ |
| (9) $8 + 93 = 101$ | (39) $98 + 65 = 163$ | (69) $72 + 41 = 113$ | (99) $82 + 97 = 179$ |
| (10) $55 + 41 = 96$ | (40) $95 + 82 = 177$ | (70) $59 + 85 = 144$ | (100) $10 + 13 = 23$ |
| (11) $88 + 36 = 124$ | (41) $39 + 66 = 105$ | (71) $4 + 88 = 92$ | (101) $65 + 2 = 67$ |
| (12) $66 + 3 = 69$ | (42) $32 + 16 = 48$ | (72) $99 + 36 = 135$ | (102) $4 + 83 = 87$ |
| (13) $83 + 68 = 151$ | (43) $58 + 24 = 82$ | (73) $57 + 71 = 128$ | (103) $51 + 88 = 139$ |
| (14) $66 + 17 = 83$ | (44) $30 + 6 = 36$ | (74) $56 + 100 = 156$ | (104) $43 + 32 = 75$ |
| (15) $58 + 64 = 122$ | (45) $55 + 77 = 132$ | (75) $73 + 94 = 167$ | (105) $33 + 14 = 47$ |
| (16) $9 + 81 = 90$ | (46) $98 + 26 = 124$ | (76) $79 + 70 = 149$ | (106) $90 + 90 = 180$ |
| (17) $34 + 62 = 96$ | (47) $92 + 77 = 169$ | (77) $20 + 90 = 110$ | (107) $19 + 26 = 45$ |
| (18) $15 + 88 = 103$ | (48) $13 + 73 = 86$ | (78) $83 + 14 = 97$ | (108) $52 + 54 = 106$ |
| (19) $39 + 60 = 99$ | (49) $67 + 21 = 88$ | (79) $79 + 7 = 86$ | (109) $27 + 72 = 99$ |
| (20) $52 + 68 = 120$ | (50) $98 + 95 = 193$ | (80) $26 + 86 = 112$ | (110) $54 + 2 = 56$ |
| (21) $54 + 75 = 129$ | (51) $21 + 15 = 36$ | (81) $13 + 19 = 32$ | (111) $56 + 89 = 145$ |
| (22) $14 + 57 = 71$ | (52) $14 + 36 = 50$ | (82) $73 + 61 = 134$ | (112) $80 + 4 = 84$ |
| (23) $81 + 95 = 176$ | (53) $88 + 43 = 131$ | (83) $6 + 39 = 45$ | (113) $4 + 28 = 32$ |
| (24) $52 + 40 = 92$ | (54) $46 + 3 = 49$ | (84) $36 + 45 = 81$ | (114) $74 + 71 = 145$ |
| (25) $56 + 22 = 78$ | (55) $32 + 64 = 96$ | (85) $39 + 95 = 134$ | (115) $72 + 15 = 87$ |
| (26) $33 + 66 = 99$ | (56) $77 + 76 = 153$ | (86) $27 + 57 = 84$ | (116) $3 + 19 = 22$ |
| (27) $68 + 29 = 97$ | (57) $23 + 26 = 49$ | (87) $99 + 81 = 180$ | (117) $95 + 76 = 171$ |
| (28) $53 + 19 = 72$ | (58) $24 + 7 = 31$ | (88) $12 + 95 = 107$ | (118) $43 + 83 = 126$ |
| (29) $11 + 32 = 43$ | (59) $54 + 91 = 145$ | (89) $100 + 46 = 146$ | (119) $19 + 74 = 93$ |
| (30) $19 + 14 = 33$ | (60) $32 + 37 = 69$ | (90) $70 + 51 = 121$ | (120) $30 + 83 = 113$ |

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|----------------------|----------------------|----------------------|-----------------------|
| (1) $82 + 65 = 147$ | (31) $69 + 65 = 134$ | (61) $31 + 57 = 88$ | (91) $61 + 15 = 76$ |
| (2) $22 + 65 = 87$ | (32) $13 + 68 = 81$ | (62) $48 + 27 = 75$ | (92) $58 + 39 = 97$ |
| (3) $61 + 46 = 107$ | (33) $74 + 65 = 139$ | (63) $76 + 98 = 174$ | (93) $23 + 73 = 96$ |
| (4) $94 + 45 = 139$ | (34) $58 + 81 = 139$ | (64) $77 + 59 = 136$ | (94) $47 + 65 = 112$ |
| (5) $54 + 68 = 122$ | (35) $12 + 17 = 29$ | (65) $1 + 4 = 5$ | (95) $92 + 45 = 137$ |
| (6) $9 + 100 = 109$ | (36) $43 + 99 = 142$ | (66) $54 + 83 = 137$ | (96) $20 + 21 = 41$ |
| (7) $82 + 82 = 164$ | (37) $75 + 9 = 84$ | (67) $35 + 65 = 100$ | (97) $39 + 27 = 66$ |
| (8) $23 + 67 = 90$ | (38) $17 + 13 = 30$ | (68) $53 + 54 = 107$ | (98) $79 + 74 = 153$ |
| (9) $3 + 5 = 8$ | (39) $28 + 90 = 118$ | (69) $17 + 91 = 108$ | (99) $39 + 62 = 101$ |
| (10) $96 + 61 = 157$ | (40) $76 + 88 = 164$ | (70) $51 + 25 = 76$ | (100) $5 + 31 = 36$ |
| (11) $28 + 41 = 69$ | (41) $11 + 21 = 32$ | (71) $100 + 4 = 104$ | (101) $27 + 62 = 89$ |
| (12) $80 + 38 = 118$ | (42) $63 + 19 = 82$ | (72) $39 + 30 = 69$ | (102) $89 + 45 = 134$ |
| (13) $83 + 49 = 132$ | (43) $32 + 34 = 66$ | (73) $94 + 9 = 103$ | (103) $59 + 76 = 135$ |
| (14) $49 + 61 = 110$ | (44) $30 + 92 = 122$ | (74) $33 + 65 = 98$ | (104) $87 + 7 = 94$ |
| (15) $68 + 21 = 89$ | (45) $39 + 8 = 47$ | (75) $53 + 28 = 81$ | (105) $19 + 54 = 73$ |
| (16) $34 + 80 = 114$ | (46) $72 + 60 = 132$ | (76) $38 + 59 = 97$ | (106) $23 + 37 = 60$ |
| (17) $75 + 69 = 144$ | (47) $88 + 1 = 89$ | (77) $5 + 61 = 66$ | (107) $91 + 35 = 126$ |
| (18) $87 + 30 = 117$ | (48) $52 + 90 = 142$ | (78) $24 + 61 = 85$ | (108) $28 + 43 = 71$ |
| (19) $2 + 25 = 27$ | (49) $58 + 25 = 83$ | (79) $3 + 75 = 78$ | (109) $12 + 89 = 101$ |
| (20) $91 + 53 = 144$ | (50) $89 + 88 = 177$ | (80) $81 + 50 = 131$ | (110) $12 + 47 = 59$ |
| (21) $67 + 11 = 78$ | (51) $15 + 46 = 61$ | (81) $9 + 89 = 98$ | (111) $78 + 53 = 131$ |
| (22) $46 + 45 = 91$ | (52) $35 + 92 = 127$ | (82) $66 + 93 = 159$ | (112) $20 + 64 = 84$ |
| (23) $92 + 12 = 104$ | (53) $57 + 55 = 112$ | (83) $9 + 26 = 35$ | (113) $58 + 17 = 75$ |
| (24) $63 + 64 = 127$ | (54) $9 + 60 = 69$ | (84) $33 + 70 = 103$ | (114) $30 + 57 = 87$ |
| (25) $98 + 85 = 183$ | (55) $10 + 14 = 24$ | (85) $13 + 64 = 77$ | (115) $12 + 88 = 100$ |
| (26) $52 + 30 = 82$ | (56) $0 + 48 = 48$ | (86) $37 + 79 = 116$ | (116) $25 + 72 = 97$ |
| (27) $57 + 28 = 85$ | (57) $13 + 73 = 86$ | (87) $87 + 29 = 116$ | (117) $65 + 95 = 160$ |
| (28) $76 + 87 = 163$ | (58) $57 + 38 = 95$ | (88) $71 + 72 = 143$ | (118) $79 + 13 = 92$ |
| (29) $66 + 13 = 79$ | (59) $26 + 37 = 63$ | (89) $5 + 60 = 65$ | (119) $16 + 17 = 33$ |
| (30) $37 + 50 = 87$ | (60) $81 + 66 = 147$ | (90) $34 + 42 = 76$ | (120) $29 + 63 = 92$ |

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|-----------------------|-----------------------|----------------------|-----------------------|
| (1) $69 + 55 = 124$ | (31) $71 + 48 = 119$ | (61) $14 + 30 = 44$ | (91) $9 + 74 = 83$ |
| (2) $75 + 63 = 138$ | (32) $13 + 88 = 101$ | (62) $61 + 95 = 156$ | (92) $83 + 55 = 138$ |
| (3) $39 + 86 = 125$ | (33) $48 + 57 = 105$ | (63) $32 + 72 = 104$ | (93) $52 + 41 = 93$ |
| (4) $54 + 40 = 94$ | (34) $36 + 56 = 92$ | (64) $23 + 26 = 49$ | (94) $98 + 27 = 125$ |
| (5) $57 + 12 = 69$ | (35) $100 + 94 = 194$ | (65) $20 + 40 = 60$ | (95) $68 + 93 = 161$ |
| (6) $9 + 34 = 43$ | (36) $67 + 97 = 164$ | (66) $10 + 57 = 67$ | (96) $29 + 24 = 53$ |
| (7) $63 + 29 = 92$ | (37) $23 + 68 = 91$ | (67) $78 + 96 = 174$ | (97) $36 + 89 = 125$ |
| (8) $23 + 48 = 71$ | (38) $87 + 69 = 156$ | (68) $55 + 83 = 138$ | (98) $11 + 43 = 54$ |
| (9) $49 + 98 = 147$ | (39) $80 + 81 = 161$ | (69) $5 + 80 = 85$ | (99) $56 + 51 = 107$ |
| (10) $90 + 73 = 163$ | (40) $10 + 86 = 96$ | (70) $92 + 13 = 105$ | (100) $13 + 89 = 102$ |
| (11) $63 + 31 = 94$ | (41) $12 + 16 = 28$ | (71) $37 + 35 = 72$ | (101) $66 + 19 = 85$ |
| (12) $50 + 46 = 96$ | (42) $85 + 49 = 134$ | (72) $26 + 64 = 90$ | (102) $94 + 34 = 128$ |
| (13) $32 + 81 = 113$ | (43) $6 + 94 = 100$ | (73) $70 + 63 = 133$ | (103) $84 + 96 = 180$ |
| (14) $70 + 78 = 148$ | (44) $48 + 25 = 73$ | (74) $29 + 28 = 57$ | (104) $71 + 70 = 141$ |
| (15) $68 + 85 = 153$ | (45) $25 + 33 = 58$ | (75) $62 + 33 = 95$ | (105) $81 + 54 = 135$ |
| (16) $31 + 86 = 117$ | (46) $83 + 89 = 172$ | (76) $10 + 16 = 26$ | (106) $92 + 77 = 169$ |
| (17) $75 + 23 = 98$ | (47) $70 + 66 = 136$ | (77) $87 + 54 = 141$ | (107) $46 + 26 = 72$ |
| (18) $100 + 39 = 139$ | (48) $16 + 16 = 32$ | (78) $9 + 41 = 50$ | (108) $86 + 20 = 106$ |
| (19) $3 + 6 = 9$ | (49) $29 + 68 = 97$ | (79) $66 + 45 = 111$ | (109) $52 + 42 = 94$ |
| (20) $70 + 8 = 78$ | (50) $33 + 42 = 75$ | (80) $70 + 9 = 79$ | (110) $73 + 2 = 75$ |
| (21) $37 + 14 = 51$ | (51) $87 + 95 = 182$ | (81) $84 + 91 = 175$ | (111) $90 + 11 = 101$ |
| (22) $36 + 43 = 79$ | (52) $1 + 96 = 97$ | (82) $77 + 8 = 85$ | (112) $48 + 78 = 126$ |
| (23) $92 + 35 = 127$ | (53) $41 + 1 = 42$ | (83) $26 + 75 = 101$ | (113) $87 + 58 = 145$ |
| (24) $29 + 50 = 79$ | (54) $78 + 41 = 119$ | (84) $62 + 25 = 87$ | (114) $44 + 52 = 96$ |
| (25) $45 + 3 = 48$ | (55) $50 + 8 = 58$ | (85) $67 + 50 = 117$ | (115) $59 + 15 = 74$ |
| (26) $64 + 2 = 66$ | (56) $2 + 84 = 86$ | (86) $40 + 22 = 62$ | (116) $20 + 67 = 87$ |
| (27) $94 + 62 = 156$ | (57) $60 + 100 = 160$ | (87) $23 + 17 = 40$ | (117) $54 + 40 = 94$ |
| (28) $85 + 29 = 114$ | (58) $16 + 5 = 21$ | (88) $42 + 1 = 43$ | (118) $3 + 95 = 98$ |
| (29) $91 + 79 = 170$ | (59) $47 + 0 = 47$ | (89) $30 + 97 = 127$ | (119) $59 + 97 = 156$ |
| (30) $69 + 88 = 157$ | (60) $59 + 91 = 150$ | (90) $52 + 12 = 64$ | (120) $23 + 36 = 59$ |

(1) $83 + 62 = 145$	(31) $4 + 21 = 25$	(61) $36 + 52 = 88$	(91) $64 + 90 = 154$
(2) $86 + 11 = 97$	(32) $51 + 23 = 74$	(62) $31 + 7 = 38$	(92) $36 + 14 = 50$
(3) $90 + 80 = 170$	(33) $61 + 42 = 103$	(63) $7 + 36 = 43$	(93) $74 + 22 = 96$
(4) $16 + 55 = 71$	(34) $45 + 94 = 139$	(64) $91 + 57 = 148$	(94) $38 + 62 = 100$
(5) $76 + 99 = 175$	(35) $48 + 46 = 94$	(65) $61 + 93 = 154$	(95) $50 + 81 = 131$
(6) $36 + 93 = 129$	(36) $96 + 80 = 176$	(66) $14 + 42 = 56$	(96) $9 + 96 = 105$
(7) $9 + 88 = 97$	(37) $63 + 72 = 135$	(67) $28 + 3 = 31$	(97) $83 + 21 = 104$
(8) $68 + 16 = 84$	(38) $4 + 64 = 68$	(68) $17 + 45 = 62$	(98) $69 + 17 = 86$
(9) $93 + 35 = 128$	(39) $98 + 37 = 135$	(69) $44 + 61 = 105$	(99) $90 + 33 = 123$
(10) $93 + 37 = 130$	(40) $70 + 97 = 167$	(70) $39 + 20 = 59$	(100) $96 + 65 = 161$
(11) $87 + 65 = 152$	(41) $52 + 85 = 137$	(71) $59 + 59 = 118$	(101) $48 + 30 = 78$
(12) $38 + 23 = 61$	(42) $1 + 98 = 99$	(72) $3 + 88 = 91$	(102) $40 + 66 = 106$
(13) $72 + 56 = 128$	(43) $65 + 45 = 110$	(73) $32 + 72 = 104$	(103) $81 + 77 = 158$
(14) $40 + 30 = 70$	(44) $79 + 8 = 87$	(74) $60 + 54 = 114$	(104) $60 + 97 = 157$
(15) $94 + 11 = 105$	(45) $49 + 82 = 131$	(75) $90 + 8 = 98$	(105) $19 + 96 = 115$
(16) $55 + 80 = 135$	(46) $65 + 38 = 103$	(76) $15 + 45 = 60$	(106) $25 + 51 = 76$
(17) $2 + 24 = 26$	(47) $59 + 61 = 120$	(77) $21 + 28 = 49$	(107) $25 + 38 = 63$
(18) $47 + 18 = 65$	(48) $51 + 66 = 117$	(78) $8 + 60 = 68$	(108) $78 + 63 = 141$
(19) $69 + 35 = 104$	(49) $85 + 11 = 96$	(79) $44 + 70 = 114$	(109) $76 + 72 = 148$
(20) $35 + 99 = 134$	(50) $22 + 6 = 28$	(80) $53 + 6 = 59$	(110) $75 + 64 = 139$
(21) $26 + 82 = 108$	(51) $40 + 77 = 117$	(81) $68 + 4 = 72$	(111) $85 + 69 = 154$
(22) $57 + 30 = 87$	(52) $11 + 25 = 36$	(82) $29 + 22 = 51$	(112) $95 + 4 = 99$
(23) $10 + 55 = 65$	(53) $39 + 49 = 88$	(83) $24 + 35 = 59$	(113) $78 + 55 = 133$
(24) $40 + 89 = 129$	(54) $11 + 76 = 87$	(84) $47 + 31 = 78$	(114) $40 + 63 = 103$
(25) $79 + 72 = 151$	(55) $35 + 66 = 101$	(85) $56 + 88 = 144$	(115) $91 + 82 = 173$
(26) $76 + 48 = 124$	(56) $86 + 92 = 178$	(86) $45 + 44 = 89$	(116) $83 + 37 = 120$
(27) $88 + 47 = 135$	(57) $10 + 70 = 80$	(87) $56 + 84 = 140$	(117) $41 + 33 = 74$
(28) $72 + 75 = 147$	(58) $78 + 92 = 170$	(88) $87 + 80 = 167$	(118) $1 + 66 = 67$
(29) $38 + 21 = 59$	(59) $51 + 23 = 74$	(89) $98 + 73 = 171$	(119) $29 + 19 = 48$
(30) $48 + 52 = 100$	(60) $62 + 28 = 90$	(90) $12 + 56 = 68$	(120) $83 + 41 = 124$

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|----------------------|----------------------|----------------------|-----------------------|
| (1) $26 + 27 = 53$ | (31) $18 + 99 = 117$ | (61) $17 + 58 = 75$ | (91) $41 + 57 = 98$ |
| (2) $83 + 48 = 131$ | (32) $30 + 67 = 97$ | (62) $97 + 55 = 152$ | (92) $32 + 63 = 95$ |
| (3) $1 + 55 = 56$ | (33) $15 + 51 = 66$ | (63) $67 + 43 = 110$ | (93) $16 + 51 = 67$ |
| (4) $69 + 71 = 140$ | (34) $25 + 42 = 67$ | (64) $50 + 99 = 149$ | (94) $31 + 33 = 64$ |
| (5) $1 + 72 = 73$ | (35) $75 + 98 = 173$ | (65) $58 + 69 = 127$ | (95) $2 + 100 = 102$ |
| (6) $94 + 70 = 164$ | (36) $81 + 4 = 85$ | (66) $90 + 41 = 131$ | (96) $47 + 51 = 98$ |
| (7) $43 + 64 = 107$ | (37) $96 + 73 = 169$ | (67) $90 + 20 = 110$ | (97) $49 + 61 = 110$ |
| (8) $82 + 85 = 167$ | (38) $27 + 35 = 62$ | (68) $7 + 46 = 53$ | (98) $48 + 7 = 55$ |
| (9) $47 + 73 = 120$ | (39) $49 + 94 = 143$ | (69) $80 + 30 = 110$ | (99) $75 + 59 = 134$ |
| (10) $55 + 43 = 98$ | (40) $36 + 75 = 111$ | (70) $33 + 95 = 128$ | (100) $27 + 95 = 122$ |
| (11) $4 + 53 = 57$ | (41) $15 + 22 = 37$ | (71) $95 + 73 = 168$ | (101) $39 + 69 = 108$ |
| (12) $27 + 28 = 55$ | (42) $76 + 44 = 120$ | (72) $11 + 29 = 40$ | (102) $52 + 70 = 122$ |
| (13) $92 + 6 = 98$ | (43) $73 + 3 = 76$ | (73) $54 + 22 = 76$ | (103) $23 + 43 = 66$ |
| (14) $25 + 3 = 28$ | (44) $22 + 35 = 57$ | (74) $67 + 79 = 146$ | (104) $31 + 41 = 72$ |
| (15) $29 + 43 = 72$ | (45) $21 + 66 = 87$ | (75) $64 + 83 = 147$ | (105) $99 + 11 = 110$ |
| (16) $99 + 75 = 174$ | (46) $57 + 91 = 148$ | (76) $49 + 86 = 135$ | (106) $44 + 53 = 97$ |
| (17) $9 + 17 = 26$ | (47) $88 + 10 = 98$ | (77) $22 + 26 = 48$ | (107) $30 + 4 = 34$ |
| (18) $18 + 94 = 112$ | (48) $42 + 43 = 85$ | (78) $14 + 56 = 70$ | (108) $48 + 44 = 92$ |
| (19) $70 + 51 = 121$ | (49) $31 + 78 = 109$ | (79) $71 + 53 = 124$ | (109) $3 + 24 = 27$ |
| (20) $72 + 50 = 122$ | (50) $27 + 2 = 29$ | (80) $62 + 3 = 65$ | (110) $48 + 73 = 121$ |
| (21) $8 + 27 = 35$ | (51) $51 + 73 = 124$ | (81) $26 + 32 = 58$ | (111) $75 + 37 = 112$ |
| (22) $9 + 35 = 44$ | (52) $90 + 86 = 176$ | (82) $86 + 57 = 143$ | (112) $28 + 71 = 99$ |
| (23) $49 + 56 = 105$ | (53) $35 + 98 = 133$ | (83) $78 + 58 = 136$ | (113) $14 + 16 = 30$ |
| (24) $69 + 75 = 144$ | (54) $9 + 79 = 88$ | (84) $25 + 52 = 77$ | (114) $28 + 13 = 41$ |
| (25) $72 + 5 = 77$ | (55) $87 + 78 = 165$ | (85) $22 + 60 = 82$ | (115) $75 + 89 = 164$ |
| (26) $43 + 86 = 129$ | (56) $55 + 83 = 138$ | (86) $6 + 20 = 26$ | (116) $2 + 74 = 76$ |
| (27) $52 + 55 = 107$ | (57) $24 + 54 = 78$ | (87) $15 + 37 = 52$ | (117) $98 + 99 = 197$ |
| (28) $19 + 91 = 110$ | (58) $93 + 93 = 186$ | (88) $2 + 92 = 94$ | (118) $96 + 62 = 158$ |
| (29) $47 + 55 = 102$ | (59) $19 + 9 = 28$ | (89) $16 + 37 = 53$ | (119) $43 + 73 = 116$ |
| (30) $54 + 38 = 92$ | (60) $5 + 96 = 101$ | (90) $8 + 69 = 77$ | (120) $38 + 4 = 42$ |

(1) $58 + 6 = 64$	(31) $36 + 45 = 81$	(61) $43 + 17 = 60$	(91) $86 + 54 = 140$
(2) $80 + 84 = 164$	(32) $25 + 14 = 39$	(62) $94 + 30 = 124$	(92) $11 + 50 = 61$
(3) $46 + 16 = 62$	(33) $96 + 94 = 190$	(63) $3 + 96 = 99$	(93) $55 + 22 = 77$
(4) $15 + 96 = 111$	(34) $52 + 56 = 108$	(64) $62 + 12 = 74$	(94) $2 + 14 = 16$
(5) $2 + 12 = 14$	(35) $62 + 97 = 159$	(65) $84 + 50 = 134$	(95) $74 + 29 = 103$
(6) $39 + 82 = 121$	(36) $38 + 94 = 132$	(66) $85 + 63 = 148$	(96) $47 + 32 = 79$
(7) $44 + 84 = 128$	(37) $44 + 59 = 103$	(67) $69 + 58 = 127$	(97) $31 + 92 = 123$
(8) $70 + 24 = 94$	(38) $82 + 76 = 158$	(68) $18 + 11 = 29$	(98) $44 + 14 = 58$
(9) $81 + 54 = 135$	(39) $6 + 57 = 63$	(69) $34 + 98 = 132$	(99) $16 + 60 = 76$
(10) $69 + 3 = 72$	(40) $87 + 70 = 157$	(70) $32 + 70 = 102$	(100) $67 + 42 = 109$
(11) $21 + 95 = 116$	(41) $66 + 56 = 122$	(71) $39 + 69 = 108$	(101) $39 + 5 = 44$
(12) $7 + 61 = 68$	(42) $69 + 97 = 166$	(72) $16 + 38 = 54$	(102) $73 + 38 = 111$
(13) $89 + 75 = 164$	(43) $61 + 80 = 141$	(73) $27 + 32 = 59$	(103) $30 + 17 = 47$
(14) $49 + 99 = 148$	(44) $89 + 30 = 119$	(74) $32 + 5 = 37$	(104) $47 + 74 = 121$
(15) $54 + 19 = 73$	(45) $20 + 13 = 33$	(75) $84 + 35 = 119$	(105) $57 + 100 = 157$
(16) $55 + 75 = 130$	(46) $99 + 88 = 187$	(76) $72 + 95 = 167$	(106) $86 + 35 = 121$
(17) $23 + 34 = 57$	(47) $37 + 51 = 88$	(77) $91 + 83 = 174$	(107) $86 + 78 = 164$
(18) $48 + 97 = 145$	(48) $35 + 50 = 85$	(78) $49 + 27 = 76$	(108) $96 + 53 = 149$
(19) $87 + 2 = 89$	(49) $58 + 80 = 138$	(79) $55 + 98 = 153$	(109) $56 + 47 = 103$
(20) $85 + 29 = 114$	(50) $78 + 66 = 144$	(80) $8 + 3 = 11$	(110) $74 + 18 = 92$
(21) $53 + 42 = 95$	(51) $75 + 41 = 116$	(81) $23 + 10 = 33$	(111) $52 + 5 = 57$
(22) $20 + 76 = 96$	(52) $3 + 3 = 6$	(82) $67 + 78 = 145$	(112) $49 + 74 = 123$
(23) $48 + 57 = 105$	(53) $32 + 1 = 33$	(83) $71 + 83 = 154$	(113) $68 + 58 = 126$
(24) $3 + 20 = 23$	(54) $58 + 64 = 122$	(84) $27 + 62 = 89$	(114) $58 + 55 = 113$
(25) $29 + 16 = 45$	(55) $37 + 71 = 108$	(85) $83 + 57 = 140$	(115) $85 + 55 = 140$
(26) $84 + 91 = 175$	(56) $18 + 50 = 68$	(86) $47 + 1 = 48$	(116) $65 + 4 = 69$
(27) $65 + 59 = 124$	(57) $52 + 54 = 106$	(87) $83 + 38 = 121$	(117) $21 + 87 = 108$
(28) $55 + 90 = 145$	(58) $41 + 72 = 113$	(88) $53 + 38 = 91$	(118) $73 + 23 = 96$
(29) $85 + 12 = 97$	(59) $76 + 66 = 142$	(89) $79 + 40 = 119$	(119) $77 + 96 = 173$
(30) $69 + 58 = 127$	(60) $14 + 66 = 80$	(90) $6 + 93 = 99$	(120) $27 + 8 = 35$

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|-----------------------|-----------------------|----------------------|-----------------------|
| (1) $26 + 97 = 123$ | (31) $16 + 46 = 62$ | (61) $98 + 92 = 190$ | (91) $66 + 11 = 77$ |
| (2) $76 + 7 = 83$ | (32) $16 + 83 = 99$ | (62) $99 + 16 = 115$ | (92) $95 + 66 = 161$ |
| (3) $13 + 49 = 62$ | (33) $79 + 14 = 93$ | (63) $37 + 53 = 90$ | (93) $81 + 21 = 102$ |
| (4) $76 + 69 = 145$ | (34) $57 + 19 = 76$ | (64) $93 + 16 = 109$ | (94) $11 + 69 = 80$ |
| (5) $98 + 32 = 130$ | (35) $74 + 55 = 129$ | (65) $71 + 45 = 116$ | (95) $40 + 75 = 115$ |
| (6) $85 + 100 = 185$ | (36) $52 + 4 = 56$ | (66) $6 + 83 = 89$ | (96) $65 + 18 = 83$ |
| (7) $17 + 45 = 62$ | (37) $67 + 20 = 87$ | (67) $92 + 23 = 115$ | (97) $83 + 74 = 157$ |
| (8) $43 + 46 = 89$ | (38) $43 + 84 = 127$ | (68) $37 + 98 = 135$ | (98) $87 + 50 = 137$ |
| (9) $63 + 43 = 106$ | (39) $100 + 16 = 116$ | (69) $79 + 45 = 124$ | (99) $49 + 56 = 105$ |
| (10) $72 + 28 = 100$ | (40) $71 + 6 = 77$ | (70) $7 + 18 = 25$ | (100) $9 + 64 = 73$ |
| (11) $85 + 82 = 167$ | (41) $25 + 69 = 94$ | (71) $37 + 84 = 121$ | (101) $31 + 6 = 37$ |
| (12) $35 + 10 = 45$ | (42) $80 + 81 = 161$ | (72) $65 + 27 = 92$ | (102) $32 + 13 = 45$ |
| (13) $41 + 94 = 135$ | (43) $18 + 89 = 107$ | (73) $25 + 52 = 77$ | (103) $9 + 69 = 78$ |
| (14) $19 + 74 = 93$ | (44) $33 + 89 = 122$ | (74) $2 + 82 = 84$ | (104) $46 + 14 = 60$ |
| (15) $44 + 9 = 53$ | (45) $80 + 69 = 149$ | (75) $7 + 7 = 14$ | (105) $76 + 20 = 96$ |
| (16) $93 + 53 = 146$ | (46) $33 + 98 = 131$ | (76) $17 + 27 = 44$ | (106) $76 + 78 = 154$ |
| (17) $12 + 92 = 104$ | (47) $7 + 62 = 69$ | (77) $92 + 90 = 182$ | (107) $50 + 31 = 81$ |
| (18) $16 + 45 = 61$ | (48) $48 + 42 = 90$ | (78) $92 + 59 = 151$ | (108) $18 + 23 = 41$ |
| (19) $100 + 21 = 121$ | (49) $28 + 54 = 82$ | (79) $55 + 9 = 64$ | (109) $28 + 19 = 47$ |
| (20) $24 + 88 = 112$ | (50) $44 + 32 = 76$ | (80) $19 + 20 = 39$ | (110) $68 + 59 = 127$ |
| (21) $56 + 87 = 143$ | (51) $28 + 60 = 88$ | (81) $9 + 30 = 39$ | (111) $54 + 89 = 143$ |
| (22) $5 + 67 = 72$ | (52) $18 + 62 = 80$ | (82) $67 + 63 = 130$ | (112) $43 + 31 = 74$ |
| (23) $85 + 33 = 118$ | (53) $73 + 48 = 121$ | (83) $32 + 33 = 65$ | (113) $80 + 14 = 94$ |
| (24) $6 + 81 = 87$ | (54) $2 + 44 = 46$ | (84) $65 + 70 = 135$ | (114) $52 + 57 = 109$ |
| (25) $17 + 55 = 72$ | (55) $98 + 18 = 116$ | (85) $16 + 39 = 55$ | (115) $65 + 70 = 135$ |
| (26) $64 + 24 = 88$ | (56) $70 + 96 = 166$ | (86) $74 + 47 = 121$ | (116) $69 + 28 = 97$ |
| (27) $85 + 40 = 125$ | (57) $3 + 84 = 87$ | (87) $63 + 43 = 106$ | (117) $74 + 30 = 104$ |
| (28) $6 + 28 = 34$ | (58) $28 + 59 = 87$ | (88) $31 + 42 = 73$ | (118) $71 + 97 = 168$ |
| (29) $31 + 16 = 47$ | (59) $24 + 20 = 44$ | (89) $37 + 2 = 39$ | (119) $15 + 69 = 84$ |
| (30) $74 + 17 = 91$ | (60) $67 + 65 = 132$ | (90) $8 + 20 = 28$ | (120) $73 + 84 = 157$ |

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|----------------------|----------------------|----------------------|-----------------------|
| (1) $17 + 33 = 50$ | (31) $90 + 10 = 100$ | (61) $1 + 81 = 82$ | (91) $34 + 16 = 50$ |
| (2) $35 + 61 = 96$ | (32) $5 + 34 = 39$ | (62) $94 + 99 = 193$ | (92) $32 + 72 = 104$ |
| (3) $60 + 27 = 87$ | (33) $39 + 49 = 88$ | (63) $59 + 7 = 66$ | (93) $14 + 53 = 67$ |
| (4) $27 + 68 = 95$ | (34) $10 + 47 = 57$ | (64) $0 + 23 = 23$ | (94) $84 + 3 = 87$ |
| (5) $8 + 20 = 28$ | (35) $7 + 41 = 48$ | (65) $60 + 60 = 120$ | (95) $63 + 56 = 119$ |
| (6) $89 + 77 = 166$ | (36) $65 + 9 = 74$ | (66) $96 + 18 = 114$ | (96) $96 + 51 = 147$ |
| (7) $21 + 94 = 115$ | (37) $0 + 69 = 69$ | (67) $0 + 86 = 86$ | (97) $55 + 79 = 134$ |
| (8) $83 + 52 = 135$ | (38) $39 + 46 = 85$ | (68) $84 + 86 = 170$ | (98) $96 + 99 = 195$ |
| (9) $37 + 99 = 136$ | (39) $49 + 47 = 96$ | (69) $58 + 62 = 120$ | (99) $19 + 77 = 96$ |
| (10) $19 + 59 = 78$ | (40) $42 + 80 = 122$ | (70) $23 + 17 = 40$ | (100) $24 + 6 = 30$ |
| (11) $41 + 32 = 73$ | (41) $83 + 52 = 135$ | (71) $86 + 17 = 103$ | (101) $64 + 66 = 130$ |
| (12) $89 + 42 = 131$ | (42) $42 + 14 = 56$ | (72) $92 + 57 = 149$ | (102) $99 + 55 = 154$ |
| (13) $8 + 94 = 102$ | (43) $24 + 70 = 94$ | (73) $13 + 77 = 90$ | (103) $18 + 37 = 55$ |
| (14) $76 + 29 = 105$ | (44) $98 + 43 = 141$ | (74) $58 + 25 = 83$ | (104) $67 + 55 = 122$ |
| (15) $41 + 71 = 112$ | (45) $82 + 47 = 129$ | (75) $62 + 38 = 100$ | (105) $20 + 90 = 110$ |
| (16) $40 + 10 = 50$ | (46) $2 + 39 = 41$ | (76) $70 + 68 = 138$ | (106) $61 + 55 = 116$ |
| (17) $1 + 89 = 90$ | (47) $36 + 87 = 123$ | (77) $43 + 89 = 132$ | (107) $44 + 93 = 137$ |
| (18) $62 + 93 = 155$ | (48) $41 + 43 = 84$ | (78) $98 + 61 = 159$ | (108) $39 + 21 = 60$ |
| (19) $22 + 4 = 26$ | (49) $13 + 15 = 28$ | (79) $1 + 19 = 20$ | (109) $39 + 6 = 45$ |
| (20) $98 + 90 = 188$ | (50) $66 + 0 = 66$ | (80) $90 + 24 = 114$ | (110) $77 + 88 = 165$ |
| (21) $56 + 39 = 95$ | (51) $54 + 3 = 57$ | (81) $97 + 48 = 145$ | (111) $11 + 58 = 69$ |
| (22) $55 + 88 = 143$ | (52) $57 + 51 = 108$ | (82) $56 + 18 = 74$ | (112) $28 + 60 = 88$ |
| (23) $43 + 11 = 54$ | (53) $65 + 15 = 80$ | (83) $0 + 23 = 23$ | (113) $79 + 32 = 111$ |
| (24) $7 + 6 = 13$ | (54) $72 + 18 = 90$ | (84) $81 + 95 = 176$ | (114) $86 + 8 = 94$ |
| (25) $14 + 54 = 68$ | (55) $31 + 15 = 46$ | (85) $65 + 83 = 148$ | (115) $92 + 37 = 129$ |
| (26) $65 + 90 = 155$ | (56) $33 + 3 = 36$ | (86) $36 + 46 = 82$ | (116) $44 + 68 = 112$ |
| (27) $25 + 85 = 110$ | (57) $66 + 91 = 157$ | (87) $98 + 10 = 108$ | (117) $56 + 47 = 103$ |
| (28) $99 + 52 = 151$ | (58) $45 + 38 = 83$ | (88) $34 + 58 = 92$ | (118) $10 + 95 = 105$ |
| (29) $88 + 44 = 132$ | (59) $29 + 24 = 53$ | (89) $39 + 1 = 40$ | (119) $7 + 77 = 84$ |
| (30) $64 + 51 = 115$ | (60) $7 + 76 = 83$ | (90) $8 + 7 = 15$ | (120) $48 + 10 = 58$ |

1.1.2 減法

自然数の差を求める問題。

(1) 48 - 16	(31) 40 - 24	(61) 79 - 54	(91) 38 - 32
(2) 28 - 3	(32) 79 - 66	(62) 60 - 46	(92) 72 - 58
(3) 49 - 3	(33) 74 - 26	(63) 17 - 15	(93) 94 - 47
(4) 57 - 54	(34) 25 - 4	(64) 98 - 18	(94) 13 - 0
(5) 93 - 6	(35) 86 - 51	(65) 58 - 25	(95) 75 - 66
(6) 60 - 36	(36) 76 - 65	(66) 67 - 15	(96) 84 - 27
(7) 82 - 43	(37) 97 - 23	(67) 65 - 22	(97) 85 - 31
(8) 38 - 8	(38) 56 - 52	(68) 16 - 3	(98) 96 - 18
(9) 37 - 35	(39) 9 - 3	(69) 85 - 74	(99) 66 - 48
(10) 83 - 52	(40) 80 - 31	(70) 78 - 29	(100) 80 - 67
(11) 37 - 33	(41) 54 - 53	(71) 98 - 57	(101) 48 - 36
(12) 77 - 5	(42) 96 - 33	(72) 76 - 33	(102) 16 - 13
(13) 95 - 33	(43) 91 - 47	(73) 82 - 10	(103) 79 - 66
(14) 97 - 83	(44) 64 - 19	(74) 83 - 7	(104) 67 - 50
(15) 89 - 42	(45) 39 - 12	(75) 70 - 21	(105) 90 - 70
(16) 49 - 6	(46) 47 - 29	(76) 74 - 2	(106) 62 - 46
(17) 66 - 2	(47) 70 - 4	(77) 29 - 22	(107) 80 - 71
(18) 98 - 66	(48) 4 - 0	(78) 55 - 37	(108) 76 - 15
(19) 70 - 31	(49) 61 - 43	(79) 74 - 10	(109) 34 - 25
(20) 72 - 1	(50) 94 - 24	(80) 25 - 13	(110) 69 - 46
(21) 78 - 55	(51) 66 - 61	(81) 58 - 50	(111) 72 - 22
(22) 79 - 48	(52) 41 - 36	(82) 57 - 23	(112) 46 - 1
(23) 42 - 37	(53) 78 - 13	(83) 67 - 39	(113) 56 - 42
(24) 74 - 2	(54) 99 - 98	(84) 87 - 43	(114) 90 - 81
(25) 92 - 39	(55) 44 - 7	(85) 77 - 34	(115) 60 - 55
(26) 88 - 67	(56) 100 - 73	(86) 68 - 26	(116) 76 - 4
(27) 78 - 4	(57) 43 - 20	(87) 29 - 11	(117) 40 - 2
(28) 28 - 20	(58) 76 - 28	(88) 98 - 72	(118) 65 - 30
(29) 95 - 91	(59) 98 - 24	(89) 60 - 8	(119) 80 - 56
(30) 32 - 25	(60) 66 - 40	(90) 84 - 58	(120) 94 - 71

(1) 96 - 85	(31) 76 - 47	(61) 31 - 12	(91) 91 - 56
(2) 66 - 26	(32) 30 - 7	(62) 80 - 40	(92) 100 - 55
(3) 91 - 60	(33) 62 - 14	(63) 52 - 47	(93) 24 - 8
(4) 93 - 13	(34) 68 - 3	(64) 72 - 3	(94) 95 - 71
(5) 26 - 1	(35) 78 - 11	(65) 28 - 3	(95) 24 - 13
(6) 76 - 36	(36) 90 - 66	(66) 54 - 12	(96) 33 - 17
(7) 85 - 53	(37) 98 - 90	(67) 75 - 22	(97) 52 - 35
(8) 63 - 57	(38) 60 - 33	(68) 89 - 27	(98) 98 - 46
(9) 87 - 54	(39) 96 - 45	(69) 90 - 9	(99) 76 - 47
(10) 26 - 23	(40) 43 - 31	(70) 89 - 73	(100) 94 - 62
(11) 33 - 4	(41) 36 - 5	(71) 75 - 66	(101) 91 - 45
(12) 85 - 58	(42) 69 - 56	(72) 94 - 58	(102) 55 - 18
(13) 38 - 5	(43) 63 - 3	(73) 90 - 5	(103) 23 - 3
(14) 48 - 23	(44) 68 - 50	(74) 36 - 32	(104) 43 - 18
(15) 96 - 60	(45) 46 - 25	(75) 79 - 22	(105) 82 - 33
(16) 46 - 9	(46) 26 - 5	(76) 26 - 23	(106) 97 - 66
(17) 99 - 50	(47) 24 - 5	(77) 95 - 86	(107) 50 - 26
(18) 35 - 1	(48) 86 - 63	(78) 78 - 40	(108) 77 - 22
(19) 41 - 25	(49) 59 - 31	(79) 23 - 2	(109) 98 - 32
(20) 55 - 21	(50) 89 - 22	(80) 91 - 32	(110) 96 - 71
(21) 46 - 27	(51) 47 - 11	(81) 63 - 54	(111) 79 - 76
(22) 71 - 28	(52) 58 - 9	(82) 82 - 7	(112) 87 - 40
(23) 73 - 66	(53) 53 - 22	(83) 41 - 13	(113) 57 - 48
(24) 83 - 77	(54) 19 - 5	(84) 95 - 27	(114) 64 - 46
(25) 67 - 32	(55) 64 - 30	(85) 36 - 10	(115) 50 - 3
(26) 37 - 21	(56) 32 - 29	(86) 65 - 44	(116) 100 - 98
(27) 92 - 56	(57) 91 - 57	(87) 84 - 79	(117) 48 - 6
(28) 62 - 39	(58) 73 - 4	(88) 52 - 15	(118) 63 - 23
(29) 91 - 59	(59) 69 - 51	(89) 26 - 21	(119) 98 - 1
(30) 36 - 34	(60) 74 - 5	(90) 31 - 10	(120) 87 - 37

(1) 64 - 50	(31) 95 - 52	(61) 75 - 25	(91) 79 - 18
(2) 42 - 15	(32) 77 - 57	(62) 77 - 48	(92) 33 - 19
(3) 61 - 27	(33) 86 - 20	(63) 83 - 57	(93) 64 - 31
(4) 46 - 17	(34) 66 - 38	(64) 89 - 56	(94) 49 - 39
(5) 35 - 21	(35) 95 - 89	(65) 23 - 13	(95) 68 - 55
(6) 77 - 44	(36) 85 - 13	(66) 75 - 36	(96) 93 - 70
(7) 53 - 12	(37) 52 - 49	(67) 79 - 37	(97) 56 - 40
(8) 63 - 7	(38) 29 - 21	(68) 75 - 70	(98) 81 - 81
(9) 57 - 17	(39) 54 - 51	(69) 65 - 49	(99) 93 - 52
(10) 72 - 63	(40) 92 - 57	(70) 28 - 20	(100) 43 - 36
(11) 99 - 2	(41) 40 - 14	(71) 92 - 81	(101) 90 - 29
(12) 73 - 32	(42) 50 - 2	(72) 31 - 15	(102) 89 - 9
(13) 42 - 24	(43) 100 - 73	(73) 87 - 83	(103) 36 - 0
(14) 50 - 28	(44) 37 - 26	(74) 86 - 14	(104) 34 - 29
(15) 55 - 39	(45) 65 - 25	(75) 30 - 3	(105) 50 - 33
(16) 88 - 31	(46) 82 - 38	(76) 91 - 33	(106) 82 - 39
(17) 72 - 30	(47) 42 - 8	(77) 59 - 14	(107) 55 - 30
(18) 54 - 7	(48) 58 - 32	(78) 94 - 52	(108) 82 - 9
(19) 73 - 26	(49) 97 - 91	(79) 87 - 56	(109) 90 - 9
(20) 13 - 3	(50) 97 - 44	(80) 93 - 88	(110) 64 - 40
(21) 74 - 73	(51) 40 - 22	(81) 89 - 50	(111) 38 - 33
(22) 60 - 13	(52) 87 - 0	(82) 55 - 36	(112) 57 - 33
(23) 61 - 54	(53) 41 - 38	(83) 65 - 22	(113) 82 - 69
(24) 30 - 22	(54) 17 - 5	(84) 87 - 55	(114) 51 - 19
(25) 65 - 52	(55) 88 - 19	(85) 63 - 37	(115) 95 - 89
(26) 43 - 25	(56) 96 - 47	(86) 62 - 9	(116) 62 - 51
(27) 46 - 43	(57) 68 - 51	(87) 91 - 38	(117) 85 - 77
(28) 64 - 26	(58) 84 - 47	(88) 41 - 20	(118) 12 - 10
(29) 56 - 36	(59) 81 - 67	(89) 79 - 66	(119) 88 - 52
(30) 90 - 85	(60) 22 - 4	(90) 30 - 26	(120) 76 - 14

(1) 60 - 17	(31) 82 - 45	(61) 73 - 36	(91) 64 - 59
(2) 86 - 59	(32) 79 - 10	(62) 62 - 26	(92) 66 - 7
(3) 70 - 20	(33) 77 - 66	(63) 17 - 9	(93) 55 - 53
(4) 72 - 51	(34) 72 - 4	(64) 91 - 73	(94) 81 - 36
(5) 44 - 2	(35) 89 - 1	(65) 84 - 8	(95) 93 - 69
(6) 77 - 47	(36) 19 - 11	(66) 59 - 33	(96) 31 - 29
(7) 35 - 30	(37) 64 - 29	(67) 67 - 5	(97) 86 - 27
(8) 27 - 4	(38) 21 - 19	(68) 65 - 48	(98) 50 - 28
(9) 86 - 11	(39) 98 - 12	(69) 88 - 67	(99) 92 - 63
(10) 89 - 59	(40) 97 - 89	(70) 2 - 1	(100) 100 - 24
(11) 91 - 62	(41) 96 - 11	(71) 38 - 28	(101) 70 - 67
(12) 25 - 7	(42) 90 - 50	(72) 61 - 54	(102) 77 - 63
(13) 92 - 80	(43) 68 - 42	(73) 21 - 8	(103) 71 - 56
(14) 72 - 62	(44) 98 - 93	(74) 52 - 21	(104) 44 - 18
(15) 92 - 48	(45) 66 - 12	(75) 63 - 11	(105) 16 - 13
(16) 58 - 34	(46) 82 - 73	(76) 85 - 3	(106) 40 - 12
(17) 83 - 81	(47) 59 - 16	(77) 40 - 4	(107) 30 - 26
(18) 34 - 19	(48) 42 - 20	(78) 59 - 13	(108) 78 - 66
(19) 90 - 68	(49) 72 - 59	(79) 81 - 41	(109) 28 - 17
(20) 68 - 8	(50) 100 - 30	(80) 97 - 71	(110) 77 - 32
(21) 100 - 75	(51) 45 - 24	(81) 91 - 44	(111) 74 - 5
(22) 45 - 1	(52) 93 - 10	(82) 67 - 62	(112) 76 - 32
(23) 91 - 5	(53) 72 - 57	(83) 75 - 12	(113) 74 - 58
(24) 77 - 45	(54) 54 - 7	(84) 67 - 30	(114) 88 - 40
(25) 25 - 9	(55) 91 - 68	(85) 60 - 17	(115) 61 - 4
(26) 55 - 35	(56) 78 - 27	(86) 83 - 74	(116) 14 - 4
(27) 87 - 58	(57) 38 - 38	(87) 82 - 29	(117) 52 - 24
(28) 68 - 26	(58) 77 - 58	(88) 46 - 40	(118) 99 - 37
(29) 75 - 15	(59) 80 - 56	(89) 78 - 72	(119) 30 - 28
(30) 42 - 36	(60) 45 - 45	(90) 90 - 11	(120) 68 - 64

(1) 93 - 20	(31) 54 - 5	(61) 41 - 25	(91) 94 - 62
(2) 92 - 39	(32) 58 - 27	(62) 81 - 4	(92) 43 - 18
(3) 43 - 20	(33) 78 - 42	(63) 11 - 11	(93) 91 - 52
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(6) 52 - 52	(36) 70 - 53	(66) 94 - 13	(96) 46 - 10
(7) 97 - 70	(37) 63 - 35	(67) 74 - 41	(97) 48 - 13
(8) 39 - 23	(38) 94 - 61	(68) 58 - 45	(98) 66 - 31
(9) 57 - 40	(39) 82 - 9	(69) 98 - 4	(99) 78 - 2
(10) 76 - 74	(40) 49 - 44	(70) 77 - 44	(100) 63 - 43
(11) 88 - 39	(41) 60 - 18	(71) 81 - 34	(101) 45 - 15
(12) 41 - 31	(42) 83 - 5	(72) 75 - 72	(102) 58 - 29
(13) 82 - 8	(43) 53 - 19	(73) 41 - 30	(103) 43 - 22
(14) 94 - 10	(44) 16 - 6	(74) 40 - 38	(104) 57 - 8
(15) 69 - 16	(45) 73 - 14	(75) 94 - 11	(105) 82 - 33
(16) 6 - 0	(46) 53 - 28	(76) 26 - 26	(106) 30 - 28
(17) 84 - 39	(47) 75 - 43	(77) 83 - 17	(107) 97 - 17
(18) 96 - 44	(48) 96 - 1	(78) 66 - 32	(108) 67 - 22
(19) 20 - 4	(49) 68 - 43	(79) 48 - 24	(109) 94 - 11
(20) 100 - 91	(50) 99 - 68	(80) 69 - 44	(110) 80 - 61
(21) 67 - 59	(51) 68 - 34	(81) 93 - 41	(111) 75 - 48
(22) 30 - 30	(52) 76 - 26	(82) 80 - 57	(112) 75 - 22
(23) 98 - 69	(53) 87 - 3	(83) 45 - 44	(113) 85 - 19
(24) 93 - 55	(54) 87 - 58	(84) 47 - 29	(114) 32 - 24
(25) 32 - 12	(55) 53 - 14	(85) 72 - 52	(115) 36 - 26
(26) 62 - 47	(56) 98 - 6	(86) 93 - 51	(116) 55 - 49
(27) 79 - 60	(57) 65 - 16	(87) 99 - 70	(117) 66 - 62
(28) 93 - 52	(58) 85 - 53	(88) 99 - 6	(118) 73 - 41
(29) 51 - 35	(59) 64 - 26	(89) 18 - 17	(119) 85 - 79
(30) 34 - 11	(60) 87 - 76	(90) 52 - 0	(120) 39 - 15

(1) 90 - 84	(31) 96 - 52	(61) 74 - 20	(91) 98 - 1
(2) 45 - 20	(32) 70 - 4	(62) 88 - 65	(92) 26 - 5
(3) 76 - 6	(33) 99 - 30	(63) 55 - 10	(93) 98 - 84
(4) 56 - 5	(34) 65 - 21	(64) 88 - 80	(94) 59 - 45
(5) 90 - 84	(35) 78 - 76	(65) 70 - 58	(95) 76 - 66
(6) 80 - 49	(36) 79 - 3	(66) 81 - 72	(96) 80 - 71
(7) 5 - 3	(37) 93 - 8	(67) 83 - 19	(97) 75 - 56
(8) 85 - 30	(38) 35 - 25	(68) 97 - 86	(98) 90 - 33
(9) 87 - 76	(39) 81 - 60	(69) 84 - 64	(99) 77 - 1
(10) 75 - 22	(40) 42 - 17	(70) 72 - 16	(100) 45 - 25
(11) 100 - 29	(41) 78 - 42	(71) 86 - 77	(101) 27 - 27
(12) 92 - 5	(42) 22 - 15	(72) 58 - 11	(102) 41 - 14
(13) 61 - 45	(43) 31 - 17	(73) 80 - 64	(103) 96 - 37
(14) 94 - 40	(44) 95 - 25	(74) 81 - 44	(104) 42 - 16
(15) 5 - 5	(45) 55 - 29	(75) 89 - 20	(105) 99 - 98
(16) 86 - 70	(46) 86 - 63	(76) 80 - 15	(106) 69 - 11
(17) 13 - 8	(47) 100 - 88	(77) 99 - 2	(107) 70 - 12
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(19) 82 - 37	(49) 98 - 94	(79) 79 - 52	(109) 41 - 29
(20) 91 - 60	(50) 86 - 59	(80) 78 - 41	(110) 73 - 27
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(23) 94 - 63	(53) 88 - 42	(83) 50 - 23	(113) 77 - 17
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(25) 90 - 77	(55) 68 - 47	(85) 65 - 22	(115) 45 - 15
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(27) 83 - 45	(57) 75 - 14	(87) 96 - 2	(117) 44 - 40
(28) 56 - 51	(58) 72 - 10	(88) 97 - 23	(118) 14 - 1
(29) 55 - 6	(59) 55 - 18	(89) 94 - 4	(119) 73 - 37
(30) 29 - 2	(60) 41 - 15	(90) 92 - 53	(120) 81 - 44

(1) 85 - 77	(31) 68 - 26	(61) 79 - 24	(91) 47 - 25
(2) 54 - 50	(32) 21 - 6	(62) 100 - 58	(92) 53 - 37
(3) 30 - 2	(33) 56 - 42	(63) 68 - 24	(93) 73 - 20
(4) 99 - 81	(34) 75 - 35	(64) 88 - 61	(94) 95 - 59
(5) 59 - 31	(35) 65 - 1	(65) 45 - 9	(95) 82 - 38
(6) 67 - 15	(36) 68 - 26	(66) 20 - 13	(96) 84 - 69
(7) 72 - 18	(37) 57 - 43	(67) 100 - 4	(97) 70 - 44
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(9) 81 - 6	(39) 78 - 71	(69) 62 - 58	(99) 53 - 6
(10) 68 - 32	(40) 99 - 61	(70) 54 - 4	(100) 88 - 56
(11) 78 - 60	(41) 38 - 5	(71) 57 - 13	(101) 85 - 48
(12) 96 - 54	(42) 93 - 23	(72) 97 - 50	(102) 83 - 27
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(15) 17 - 15	(45) 90 - 36	(75) 91 - 72	(105) 52 - 42
(16) 19 - 10	(46) 52 - 45	(76) 82 - 43	(106) 78 - 35
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(28) 22 - 5	(58) 15 - 9	(88) 73 - 38	(118) 48 - 42
(29) 86 - 33	(59) 96 - 68	(89) 100 - 17	(119) 81 - 1
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(10) 88 - 4	(40) 82 - 61	(70) 69 - 24	(100) 96 - 9
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(12) 75 - 68	(42) 74 - 56	(72) 51 - 18	(102) 56 - 11
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(17) 76 - 76	(47) 91 - 43	(77) 99 - 74	(107) 26 - 26
(18) 25 - 20	(48) 71 - 1	(78) 67 - 22	(108) 71 - 33
(19) 98 - 26	(49) 69 - 63	(79) 99 - 24	(109) 100 - 37
(20) 93 - 73	(50) 37 - 21	(80) 100 - 37	(110) 62 - 0
(21) 89 - 38	(51) 28 - 20	(81) 60 - 18	(111) 60 - 39
(22) 87 - 39	(52) 61 - 10	(82) 17 - 8	(112) 58 - 40
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(24) 95 - 6	(54) 67 - 37	(84) 53 - 9	(114) 77 - 66
(25) 80 - 47	(55) 62 - 46	(85) 62 - 47	(115) 60 - 60
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(27) 75 - 31	(57) 33 - 20	(87) 85 - 44	(117) 77 - 29
(28) 92 - 70	(58) 68 - 42	(88) 66 - 45	(118) 87 - 78
(29) 71 - 48	(59) 86 - 42	(89) 42 - 14	(119) 49 - 21
(30) 100 - 24	(60) 44 - 3	(90) 54 - 19	(120) 84 - 48

(1) 76 - 52	(31) 70 - 18	(61) 53 - 40	(91) 58 - 29
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(3) 43 - 5	(33) 75 - 59	(63) 41 - 11	(93) 65 - 20
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(5) 56 - 33	(35) 84 - 30	(65) 56 - 36	(95) 88 - 51
(6) 72 - 11	(36) 47 - 18	(66) 69 - 54	(96) 62 - 51
(7) 100 - 35	(37) 70 - 41	(67) 85 - 31	(97) 94 - 67
(8) 99 - 84	(38) 86 - 20	(68) 87 - 66	(98) 30 - 24
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(13) 63 - 11	(43) 89 - 1	(73) 98 - 17	(103) 74 - 66
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(15) 67 - 41	(45) 86 - 10	(75) 44 - 23	(105) 67 - 47
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(19) 96 - 3	(49) 73 - 62	(79) 83 - 22	(109) 55 - 6
(20) 63 - 35	(50) 91 - 10	(80) 78 - 36	(110) 85 - 59
(21) 47 - 31	(51) 96 - 15	(81) 75 - 18	(111) 34 - 3
(22) 91 - 68	(52) 68 - 10	(82) 77 - 63	(112) 83 - 19
(23) 98 - 47	(53) 19 - 7	(83) 65 - 21	(113) 89 - 17
(24) 72 - 2	(54) 97 - 86	(84) 74 - 11	(114) 70 - 39
(25) 98 - 19	(55) 71 - 8	(85) 99 - 10	(115) 100 - 35
(26) 86 - 57	(56) 96 - 54	(86) 67 - 20	(116) 48 - 1
(27) 55 - 13	(57) 27 - 17	(87) 61 - 40	(117) 78 - 40
(28) 92 - 87	(58) 95 - 10	(88) 12 - 5	(118) 83 - 78
(29) 83 - 23	(59) 95 - 67	(89) 80 - 19	(119) 97 - 84
(30) 81 - 2	(60) 81 - 60	(90) 99 - 26	(120) 18 - 2

(1) 51 - 29	(31) 80 - 8	(61) 75 - 47	(91) 79 - 68
(2) 63 - 5	(32) 67 - 6	(62) 80 - 25	(92) 87 - 32
(3) 63 - 49	(33) 76 - 45	(63) 97 - 23	(93) 78 - 63
(4) 68 - 20	(34) 81 - 54	(64) 88 - 15	(94) 46 - 3
(5) 94 - 57	(35) 52 - 46	(65) 60 - 48	(95) 97 - 48
(6) 81 - 57	(36) 77 - 39	(66) 34 - 28	(96) 61 - 46
(7) 97 - 85	(37) 49 - 48	(67) 84 - 39	(97) 95 - 84
(8) 35 - 15	(38) 95 - 38	(68) 68 - 5	(98) 93 - 28
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(10) 68 - 51	(40) 37 - 18	(70) 58 - 20	(100) 83 - 17
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(12) 26 - 10	(42) 98 - 87	(72) 60 - 56	(102) 5 - 4
(13) 89 - 35	(43) 50 - 47	(73) 26 - 10	(103) 30 - 7
(14) 39 - 25	(44) 78 - 71	(74) 90 - 8	(104) 89 - 30
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(16) 30 - 22	(46) 94 - 42	(76) 97 - 34	(106) 86 - 83
(17) 89 - 15	(47) 93 - 10	(77) 45 - 45	(107) 18 - 12
(18) 61 - 6	(48) 47 - 42	(78) 93 - 67	(108) 70 - 39
(19) 63 - 1	(49) 52 - 42	(79) 26 - 20	(109) 39 - 9
(20) 40 - 39	(50) 55 - 35	(80) 68 - 19	(110) 60 - 37
(21) 29 - 6	(51) 54 - 8	(81) 99 - 3	(111) 48 - 46
(22) 36 - 13	(52) 76 - 8	(82) 79 - 74	(112) 80 - 6
(23) 95 - 41	(53) 91 - 42	(83) 87 - 7	(113) 39 - 34
(24) 71 - 44	(54) 47 - 12	(84) 67 - 41	(114) 48 - 41
(25) 96 - 42	(55) 12 - 4	(85) 58 - 22	(115) 98 - 87
(26) 97 - 55	(56) 59 - 9	(86) 73 - 70	(116) 90 - 31
(27) 24 - 23	(57) 54 - 41	(87) 74 - 50	(117) 87 - 73
(28) 84 - 48	(58) 76 - 26	(88) 95 - 70	(118) 88 - 48
(29) 98 - 67	(59) 39 - 4	(89) 65 - 35	(119) 18 - 12
(30) 91 - 13	(60) 66 - 41	(90) 97 - 6	(120) 75 - 46

(1) 63 - 22	(31) 72 - 24	(61) 97 - 88	(91) 90 - 23
(2) 40 - 39	(32) 62 - 51	(62) 41 - 19	(92) 94 - 64
(3) 45 - 39	(33) 18 - 11	(63) 75 - 57	(93) 18 - 16
(4) 71 - 12	(34) 63 - 44	(64) 33 - 11	(94) 59 - 30
(5) 83 - 56	(35) 63 - 57	(65) 40 - 15	(95) 82 - 13
(6) 51 - 5	(36) 84 - 63	(66) 96 - 25	(96) 83 - 26
(7) 78 - 73	(37) 99 - 25	(67) 65 - 64	(97) 22 - 18
(8) 86 - 9	(38) 98 - 53	(68) 74 - 52	(98) 96 - 73
(9) 59 - 39	(39) 85 - 40	(69) 99 - 8	(99) 58 - 16
(10) 90 - 61	(40) 69 - 33	(70) 60 - 43	(100) 48 - 7
(11) 85 - 81	(41) 59 - 54	(71) 54 - 12	(101) 50 - 27
(12) 84 - 73	(42) 62 - 50	(72) 53 - 36	(102) 16 - 14
(13) 92 - 87	(43) 29 - 25	(73) 99 - 4	(103) 59 - 29
(14) 93 - 8	(44) 100 - 62	(74) 29 - 24	(104) 47 - 17
(15) 50 - 37	(45) 98 - 84	(75) 70 - 64	(105) 93 - 86
(16) 67 - 57	(46) 47 - 4	(76) 39 - 11	(106) 74 - 9
(17) 98 - 30	(47) 89 - 51	(77) 12 - 5	(107) 44 - 32
(18) 52 - 35	(48) 97 - 33	(78) 45 - 14	(108) 47 - 31
(19) 65 - 19	(49) 83 - 70	(79) 54 - 18	(109) 83 - 11
(20) 64 - 46	(50) 67 - 19	(80) 9 - 1	(110) 73 - 4
(21) 76 - 48	(51) 94 - 11	(81) 90 - 83	(111) 82 - 52
(22) 91 - 59	(52) 64 - 49	(82) 71 - 7	(112) 97 - 72
(23) 68 - 42	(53) 92 - 80	(83) 93 - 38	(113) 85 - 82
(24) 30 - 6	(54) 65 - 49	(84) 60 - 22	(114) 70 - 65
(25) 78 - 46	(55) 93 - 42	(85) 88 - 75	(115) 64 - 49
(26) 60 - 13	(56) 92 - 23	(86) 68 - 7	(116) 85 - 16
(27) 89 - 38	(57) 78 - 33	(87) 83 - 34	(117) 70 - 27
(28) 97 - 47	(58) 86 - 27	(88) 70 - 16	(118) 25 - 9
(29) 58 - 1	(59) 71 - 66	(89) 98 - 24	(119) 44 - 18
(30) 76 - 11	(60) 77 - 71	(90) 26 - 3	(120) 80 - 48

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|---------------------|----------------------|---------------------|----------------------|
| (1) $48 - 16 = 32$ | (31) $40 - 24 = 16$ | (61) $79 - 54 = 25$ | (91) $38 - 32 = 6$ |
| (2) $28 - 3 = 25$ | (32) $79 - 66 = 13$ | (62) $60 - 46 = 14$ | (92) $72 - 58 = 14$ |
| (3) $49 - 3 = 46$ | (33) $74 - 26 = 48$ | (63) $17 - 15 = 2$ | (93) $94 - 47 = 47$ |
| (4) $57 - 54 = 3$ | (34) $25 - 4 = 21$ | (64) $98 - 18 = 80$ | (94) $13 - 0 = 13$ |
| (5) $93 - 6 = 87$ | (35) $86 - 51 = 35$ | (65) $58 - 25 = 33$ | (95) $75 - 66 = 9$ |
| (6) $60 - 36 = 24$ | (36) $76 - 65 = 11$ | (66) $67 - 15 = 52$ | (96) $84 - 27 = 57$ |
| (7) $82 - 43 = 39$ | (37) $97 - 23 = 74$ | (67) $65 - 22 = 43$ | (97) $85 - 31 = 54$ |
| (8) $38 - 8 = 30$ | (38) $56 - 52 = 4$ | (68) $16 - 3 = 13$ | (98) $96 - 18 = 78$ |
| (9) $37 - 35 = 2$ | (39) $9 - 3 = 6$ | (69) $85 - 74 = 11$ | (99) $66 - 48 = 18$ |
| (10) $83 - 52 = 31$ | (40) $80 - 31 = 49$ | (70) $78 - 29 = 49$ | (100) $80 - 67 = 13$ |
| (11) $37 - 33 = 4$ | (41) $54 - 53 = 1$ | (71) $98 - 57 = 41$ | (101) $48 - 36 = 12$ |
| (12) $77 - 5 = 72$ | (42) $96 - 33 = 63$ | (72) $76 - 33 = 43$ | (102) $16 - 13 = 3$ |
| (13) $95 - 33 = 62$ | (43) $91 - 47 = 44$ | (73) $82 - 10 = 72$ | (103) $79 - 66 = 13$ |
| (14) $97 - 83 = 14$ | (44) $64 - 19 = 45$ | (74) $83 - 7 = 76$ | (104) $67 - 50 = 17$ |
| (15) $89 - 42 = 47$ | (45) $39 - 12 = 27$ | (75) $70 - 21 = 49$ | (105) $90 - 70 = 20$ |
| (16) $49 - 6 = 43$ | (46) $47 - 29 = 18$ | (76) $74 - 2 = 72$ | (106) $62 - 46 = 16$ |
| (17) $66 - 2 = 64$ | (47) $70 - 4 = 66$ | (77) $29 - 22 = 7$ | (107) $80 - 71 = 9$ |
| (18) $98 - 66 = 32$ | (48) $4 - 0 = 4$ | (78) $55 - 37 = 18$ | (108) $76 - 15 = 61$ |
| (19) $70 - 31 = 39$ | (49) $61 - 43 = 18$ | (79) $74 - 10 = 64$ | (109) $34 - 25 = 9$ |
| (20) $72 - 1 = 71$ | (50) $94 - 24 = 70$ | (80) $25 - 13 = 12$ | (110) $69 - 46 = 23$ |
| (21) $78 - 55 = 23$ | (51) $66 - 61 = 5$ | (81) $58 - 50 = 8$ | (111) $72 - 22 = 50$ |
| (22) $79 - 48 = 31$ | (52) $41 - 36 = 5$ | (82) $57 - 23 = 34$ | (112) $46 - 1 = 45$ |
| (23) $42 - 37 = 5$ | (53) $78 - 13 = 65$ | (83) $67 - 39 = 28$ | (113) $56 - 42 = 14$ |
| (24) $74 - 2 = 72$ | (54) $99 - 98 = 1$ | (84) $87 - 43 = 44$ | (114) $90 - 81 = 9$ |
| (25) $92 - 39 = 53$ | (55) $44 - 7 = 37$ | (85) $77 - 34 = 43$ | (115) $60 - 55 = 5$ |
| (26) $88 - 67 = 21$ | (56) $100 - 73 = 27$ | (86) $68 - 26 = 42$ | (116) $76 - 4 = 72$ |
| (27) $78 - 4 = 74$ | (57) $43 - 20 = 23$ | (87) $29 - 11 = 18$ | (117) $40 - 2 = 38$ |
| (28) $28 - 20 = 8$ | (58) $76 - 28 = 48$ | (88) $98 - 72 = 26$ | (118) $65 - 30 = 35$ |
| (29) $95 - 91 = 4$ | (59) $98 - 24 = 74$ | (89) $60 - 8 = 52$ | (119) $80 - 56 = 24$ |
| (30) $32 - 25 = 7$ | (60) $66 - 40 = 26$ | (90) $84 - 58 = 26$ | (120) $94 - 71 = 23$ |

- | | | | |
|---------------------|---------------------|---------------------|----------------------|
| (1) $96 - 85 = 11$ | (31) $76 - 47 = 29$ | (61) $31 - 12 = 19$ | (91) $91 - 56 = 35$ |
| (2) $66 - 26 = 40$ | (32) $30 - 7 = 23$ | (62) $80 - 40 = 40$ | (92) $100 - 55 = 45$ |
| (3) $91 - 60 = 31$ | (33) $62 - 14 = 48$ | (63) $52 - 47 = 5$ | (93) $24 - 8 = 16$ |
| (4) $93 - 13 = 80$ | (34) $68 - 3 = 65$ | (64) $72 - 3 = 69$ | (94) $95 - 71 = 24$ |
| (5) $26 - 1 = 25$ | (35) $78 - 11 = 67$ | (65) $28 - 3 = 25$ | (95) $24 - 13 = 11$ |
| (6) $76 - 36 = 40$ | (36) $90 - 66 = 24$ | (66) $54 - 12 = 42$ | (96) $33 - 17 = 16$ |
| (7) $85 - 53 = 32$ | (37) $98 - 90 = 8$ | (67) $75 - 22 = 53$ | (97) $52 - 35 = 17$ |
| (8) $63 - 57 = 6$ | (38) $60 - 33 = 27$ | (68) $89 - 27 = 62$ | (98) $98 - 46 = 52$ |
| (9) $87 - 54 = 33$ | (39) $96 - 45 = 51$ | (69) $90 - 9 = 81$ | (99) $76 - 47 = 29$ |
| (10) $26 - 23 = 3$ | (40) $43 - 31 = 12$ | (70) $89 - 73 = 16$ | (100) $94 - 62 = 32$ |
| (11) $33 - 4 = 29$ | (41) $36 - 5 = 31$ | (71) $75 - 66 = 9$ | (101) $91 - 45 = 46$ |
| (12) $85 - 58 = 27$ | (42) $69 - 56 = 13$ | (72) $94 - 58 = 36$ | (102) $55 - 18 = 37$ |
| (13) $38 - 5 = 33$ | (43) $63 - 3 = 60$ | (73) $90 - 5 = 85$ | (103) $23 - 3 = 20$ |
| (14) $48 - 23 = 25$ | (44) $68 - 50 = 18$ | (74) $36 - 32 = 4$ | (104) $43 - 18 = 25$ |
| (15) $96 - 60 = 36$ | (45) $46 - 25 = 21$ | (75) $79 - 22 = 57$ | (105) $82 - 33 = 49$ |
| (16) $46 - 9 = 37$ | (46) $26 - 5 = 21$ | (76) $26 - 23 = 3$ | (106) $97 - 66 = 31$ |
| (17) $99 - 50 = 49$ | (47) $24 - 5 = 19$ | (77) $95 - 86 = 9$ | (107) $50 - 26 = 24$ |
| (18) $35 - 1 = 34$ | (48) $86 - 63 = 23$ | (78) $78 - 40 = 38$ | (108) $77 - 22 = 55$ |
| (19) $41 - 25 = 16$ | (49) $59 - 31 = 28$ | (79) $23 - 2 = 21$ | (109) $98 - 32 = 66$ |
| (20) $55 - 21 = 34$ | (50) $89 - 22 = 67$ | (80) $91 - 32 = 59$ | (110) $96 - 71 = 25$ |
| (21) $46 - 27 = 19$ | (51) $47 - 11 = 36$ | (81) $63 - 54 = 9$ | (111) $79 - 76 = 3$ |
| (22) $71 - 28 = 43$ | (52) $58 - 9 = 49$ | (82) $82 - 7 = 75$ | (112) $87 - 40 = 47$ |
| (23) $73 - 66 = 7$ | (53) $53 - 22 = 31$ | (83) $41 - 13 = 28$ | (113) $57 - 48 = 9$ |
| (24) $83 - 77 = 6$ | (54) $19 - 5 = 14$ | (84) $95 - 27 = 68$ | (114) $64 - 46 = 18$ |
| (25) $67 - 32 = 35$ | (55) $64 - 30 = 34$ | (85) $36 - 10 = 26$ | (115) $50 - 3 = 47$ |
| (26) $37 - 21 = 16$ | (56) $32 - 29 = 3$ | (86) $65 - 44 = 21$ | (116) $100 - 98 = 2$ |
| (27) $92 - 56 = 36$ | (57) $91 - 57 = 34$ | (87) $84 - 79 = 5$ | (117) $48 - 6 = 42$ |
| (28) $62 - 39 = 23$ | (58) $73 - 4 = 69$ | (88) $52 - 15 = 37$ | (118) $63 - 23 = 40$ |
| (29) $91 - 59 = 32$ | (59) $69 - 51 = 18$ | (89) $26 - 21 = 5$ | (119) $98 - 1 = 97$ |
| (30) $36 - 34 = 2$ | (60) $74 - 5 = 69$ | (90) $31 - 10 = 21$ | (120) $87 - 37 = 50$ |

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|---------------------|----------------------|---------------------|----------------------|
| (1) $64 - 50 = 14$ | (31) $95 - 52 = 43$ | (61) $75 - 25 = 50$ | (91) $79 - 18 = 61$ |
| (2) $42 - 15 = 27$ | (32) $77 - 57 = 20$ | (62) $77 - 48 = 29$ | (92) $33 - 19 = 14$ |
| (3) $61 - 27 = 34$ | (33) $86 - 20 = 66$ | (63) $83 - 57 = 26$ | (93) $64 - 31 = 33$ |
| (4) $46 - 17 = 29$ | (34) $66 - 38 = 28$ | (64) $89 - 56 = 33$ | (94) $49 - 39 = 10$ |
| (5) $35 - 21 = 14$ | (35) $95 - 89 = 6$ | (65) $23 - 13 = 10$ | (95) $68 - 55 = 13$ |
| (6) $77 - 44 = 33$ | (36) $85 - 13 = 72$ | (66) $75 - 36 = 39$ | (96) $93 - 70 = 23$ |
| (7) $53 - 12 = 41$ | (37) $52 - 49 = 3$ | (67) $79 - 37 = 42$ | (97) $56 - 40 = 16$ |
| (8) $63 - 7 = 56$ | (38) $29 - 21 = 8$ | (68) $75 - 70 = 5$ | (98) $81 - 81 = 0$ |
| (9) $57 - 17 = 40$ | (39) $54 - 51 = 3$ | (69) $65 - 49 = 16$ | (99) $93 - 52 = 41$ |
| (10) $72 - 63 = 9$ | (40) $92 - 57 = 35$ | (70) $28 - 20 = 8$ | (100) $43 - 36 = 7$ |
| (11) $99 - 2 = 97$ | (41) $40 - 14 = 26$ | (71) $92 - 81 = 11$ | (101) $90 - 29 = 61$ |
| (12) $73 - 32 = 41$ | (42) $50 - 2 = 48$ | (72) $31 - 15 = 16$ | (102) $89 - 9 = 80$ |
| (13) $42 - 24 = 18$ | (43) $100 - 73 = 27$ | (73) $87 - 83 = 4$ | (103) $36 - 0 = 36$ |
| (14) $50 - 28 = 22$ | (44) $37 - 26 = 11$ | (74) $86 - 14 = 72$ | (104) $34 - 29 = 5$ |
| (15) $55 - 39 = 16$ | (45) $65 - 25 = 40$ | (75) $30 - 3 = 27$ | (105) $50 - 33 = 17$ |
| (16) $88 - 31 = 57$ | (46) $82 - 38 = 44$ | (76) $91 - 33 = 58$ | (106) $82 - 39 = 43$ |
| (17) $72 - 30 = 42$ | (47) $42 - 8 = 34$ | (77) $59 - 14 = 45$ | (107) $55 - 30 = 25$ |
| (18) $54 - 7 = 47$ | (48) $58 - 32 = 26$ | (78) $94 - 52 = 42$ | (108) $82 - 9 = 73$ |
| (19) $73 - 26 = 47$ | (49) $97 - 91 = 6$ | (79) $87 - 56 = 31$ | (109) $90 - 9 = 81$ |
| (20) $13 - 3 = 10$ | (50) $97 - 44 = 53$ | (80) $93 - 88 = 5$ | (110) $64 - 40 = 24$ |
| (21) $74 - 73 = 1$ | (51) $40 - 22 = 18$ | (81) $89 - 50 = 39$ | (111) $38 - 33 = 5$ |
| (22) $60 - 13 = 47$ | (52) $87 - 0 = 87$ | (82) $55 - 36 = 19$ | (112) $57 - 33 = 24$ |
| (23) $61 - 54 = 7$ | (53) $41 - 38 = 3$ | (83) $65 - 22 = 43$ | (113) $82 - 69 = 13$ |
| (24) $30 - 22 = 8$ | (54) $17 - 5 = 12$ | (84) $87 - 55 = 32$ | (114) $51 - 19 = 32$ |
| (25) $65 - 52 = 13$ | (55) $88 - 19 = 69$ | (85) $63 - 37 = 26$ | (115) $95 - 89 = 6$ |
| (26) $43 - 25 = 18$ | (56) $96 - 47 = 49$ | (86) $62 - 9 = 53$ | (116) $62 - 51 = 11$ |
| (27) $46 - 43 = 3$ | (57) $68 - 51 = 17$ | (87) $91 - 38 = 53$ | (117) $85 - 77 = 8$ |
| (28) $64 - 26 = 38$ | (58) $84 - 47 = 37$ | (88) $41 - 20 = 21$ | (118) $12 - 10 = 2$ |
| (29) $56 - 36 = 20$ | (59) $81 - 67 = 14$ | (89) $79 - 66 = 13$ | (119) $88 - 52 = 36$ |
| (30) $90 - 85 = 5$ | (60) $22 - 4 = 18$ | (90) $30 - 26 = 4$ | (120) $76 - 14 = 62$ |

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|----------------------|----------------------|---------------------|-----------------------|
| (1) $60 - 17 = 43$ | (31) $82 - 45 = 37$ | (61) $73 - 36 = 37$ | (91) $64 - 59 = 5$ |
| (2) $86 - 59 = 27$ | (32) $79 - 10 = 69$ | (62) $62 - 26 = 36$ | (92) $66 - 7 = 59$ |
| (3) $70 - 20 = 50$ | (33) $77 - 66 = 11$ | (63) $17 - 9 = 8$ | (93) $55 - 53 = 2$ |
| (4) $72 - 51 = 21$ | (34) $72 - 4 = 68$ | (64) $91 - 73 = 18$ | (94) $81 - 36 = 45$ |
| (5) $44 - 2 = 42$ | (35) $89 - 1 = 88$ | (65) $84 - 8 = 76$ | (95) $93 - 69 = 24$ |
| (6) $77 - 47 = 30$ | (36) $19 - 11 = 8$ | (66) $59 - 33 = 26$ | (96) $31 - 29 = 2$ |
| (7) $35 - 30 = 5$ | (37) $64 - 29 = 35$ | (67) $67 - 5 = 62$ | (97) $86 - 27 = 59$ |
| (8) $27 - 4 = 23$ | (38) $21 - 19 = 2$ | (68) $65 - 48 = 17$ | (98) $50 - 28 = 22$ |
| (9) $86 - 11 = 75$ | (39) $98 - 12 = 86$ | (69) $88 - 67 = 21$ | (99) $92 - 63 = 29$ |
| (10) $89 - 59 = 30$ | (40) $97 - 89 = 8$ | (70) $2 - 1 = 1$ | (100) $100 - 24 = 76$ |
| (11) $91 - 62 = 29$ | (41) $96 - 11 = 85$ | (71) $38 - 28 = 10$ | (101) $70 - 67 = 3$ |
| (12) $25 - 7 = 18$ | (42) $90 - 50 = 40$ | (72) $61 - 54 = 7$ | (102) $77 - 63 = 14$ |
| (13) $92 - 80 = 12$ | (43) $68 - 42 = 26$ | (73) $21 - 8 = 13$ | (103) $71 - 56 = 15$ |
| (14) $72 - 62 = 10$ | (44) $98 - 93 = 5$ | (74) $52 - 21 = 31$ | (104) $44 - 18 = 26$ |
| (15) $92 - 48 = 44$ | (45) $66 - 12 = 54$ | (75) $63 - 11 = 52$ | (105) $16 - 13 = 3$ |
| (16) $58 - 34 = 24$ | (46) $82 - 73 = 9$ | (76) $85 - 3 = 82$ | (106) $40 - 12 = 28$ |
| (17) $83 - 81 = 2$ | (47) $59 - 16 = 43$ | (77) $40 - 4 = 36$ | (107) $30 - 26 = 4$ |
| (18) $34 - 19 = 15$ | (48) $42 - 20 = 22$ | (78) $59 - 13 = 46$ | (108) $78 - 66 = 12$ |
| (19) $90 - 68 = 22$ | (49) $72 - 59 = 13$ | (79) $81 - 41 = 40$ | (109) $28 - 17 = 11$ |
| (20) $68 - 8 = 60$ | (50) $100 - 30 = 70$ | (80) $97 - 71 = 26$ | (110) $77 - 32 = 45$ |
| (21) $100 - 75 = 25$ | (51) $45 - 24 = 21$ | (81) $91 - 44 = 47$ | (111) $74 - 5 = 69$ |
| (22) $45 - 1 = 44$ | (52) $93 - 10 = 83$ | (82) $67 - 62 = 5$ | (112) $76 - 32 = 44$ |
| (23) $91 - 5 = 86$ | (53) $72 - 57 = 15$ | (83) $75 - 12 = 63$ | (113) $74 - 58 = 16$ |
| (24) $77 - 45 = 32$ | (54) $54 - 7 = 47$ | (84) $67 - 30 = 37$ | (114) $88 - 40 = 48$ |
| (25) $25 - 9 = 16$ | (55) $91 - 68 = 23$ | (85) $60 - 17 = 43$ | (115) $61 - 4 = 57$ |
| (26) $55 - 35 = 20$ | (56) $78 - 27 = 51$ | (86) $83 - 74 = 9$ | (116) $14 - 4 = 10$ |
| (27) $87 - 58 = 29$ | (57) $38 - 38 = 0$ | (87) $82 - 29 = 53$ | (117) $52 - 24 = 28$ |
| (28) $68 - 26 = 42$ | (58) $77 - 58 = 19$ | (88) $46 - 40 = 6$ | (118) $99 - 37 = 62$ |
| (29) $75 - 15 = 60$ | (59) $80 - 56 = 24$ | (89) $78 - 72 = 6$ | (119) $30 - 28 = 2$ |
| (30) $42 - 36 = 6$ | (60) $45 - 45 = 0$ | (90) $90 - 11 = 79$ | (120) $68 - 64 = 4$ |

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|----------------------|---------------------|---------------------|----------------------|
| (1) $93 - 20 = 73$ | (31) $54 - 5 = 49$ | (61) $41 - 25 = 16$ | (91) $94 - 62 = 32$ |
| (2) $92 - 39 = 53$ | (32) $58 - 27 = 31$ | (62) $81 - 4 = 77$ | (92) $43 - 18 = 25$ |
| (3) $43 - 20 = 23$ | (33) $78 - 42 = 36$ | (63) $11 - 11 = 0$ | (93) $91 - 52 = 39$ |
| (4) $46 - 33 = 13$ | (34) $72 - 29 = 43$ | (64) $89 - 24 = 65$ | (94) $100 - 35 = 65$ |
| (5) $61 - 44 = 17$ | (35) $80 - 73 = 7$ | (65) $85 - 1 = 84$ | (95) $74 - 71 = 3$ |
| (6) $86 - 55 = 31$ | (36) $94 - 9 = 85$ | (66) $73 - 30 = 43$ | (96) $80 - 6 = 74$ |
| (7) $66 - 36 = 30$ | (37) $92 - 33 = 59$ | (67) $48 - 27 = 21$ | (97) $36 - 32 = 4$ |
| (8) $80 - 63 = 17$ | (38) $94 - 41 = 53$ | (68) $54 - 52 = 2$ | (98) $71 - 19 = 52$ |
| (9) $83 - 48 = 35$ | (39) $66 - 48 = 18$ | (69) $62 - 1 = 61$ | (99) $95 - 86 = 9$ |
| (10) $19 - 11 = 8$ | (40) $51 - 5 = 46$ | (70) $45 - 3 = 42$ | (100) $73 - 56 = 17$ |
| (11) $67 - 31 = 36$ | (41) $68 - 12 = 56$ | (71) $91 - 44 = 47$ | (101) $48 - 19 = 29$ |
| (12) $43 - 28 = 15$ | (42) $63 - 14 = 49$ | (72) $77 - 5 = 72$ | (102) $80 - 78 = 2$ |
| (13) $82 - 57 = 25$ | (43) $89 - 11 = 78$ | (73) $82 - 9 = 73$ | (103) $22 - 3 = 19$ |
| (14) $77 - 12 = 65$ | (44) $66 - 17 = 49$ | (74) $81 - 73 = 8$ | (104) $34 - 13 = 21$ |
| (15) $80 - 49 = 31$ | (45) $85 - 47 = 38$ | (75) $20 - 18 = 2$ | (105) $75 - 32 = 43$ |
| (16) $3 - 1 = 2$ | (46) $69 - 20 = 49$ | (76) $27 - 14 = 13$ | (106) $71 - 49 = 22$ |
| (17) $83 - 20 = 63$ | (47) $76 - 75 = 1$ | (77) $20 - 7 = 13$ | (107) $75 - 11 = 64$ |
| (18) $67 - 4 = 63$ | (48) $13 - 3 = 10$ | (78) $88 - 53 = 35$ | (108) $91 - 78 = 13$ |
| (19) $60 - 19 = 41$ | (49) $23 - 22 = 1$ | (79) $29 - 13 = 16$ | (109) $29 - 9 = 20$ |
| (20) $2 - 0 = 2$ | (50) $57 - 24 = 33$ | (80) $95 - 32 = 63$ | (110) $92 - 19 = 73$ |
| (21) $52 - 4 = 48$ | (51) $49 - 11 = 38$ | (81) $17 - 8 = 9$ | (111) $34 - 22 = 12$ |
| (22) $34 - 7 = 27$ | (52) $91 - 90 = 1$ | (82) $7 - 4 = 3$ | (112) $91 - 77 = 14$ |
| (23) $98 - 31 = 67$ | (53) $39 - 4 = 35$ | (83) $96 - 88 = 8$ | (113) $90 - 68 = 22$ |
| (24) $66 - 56 = 10$ | (54) $99 - 32 = 67$ | (84) $95 - 71 = 24$ | (114) $73 - 0 = 73$ |
| (25) $42 - 25 = 17$ | (55) $94 - 45 = 49$ | (85) $40 - 11 = 29$ | (115) $88 - 57 = 31$ |
| (26) $91 - 56 = 35$ | (56) $74 - 67 = 7$ | (86) $63 - 44 = 19$ | (116) $31 - 27 = 4$ |
| (27) $100 - 79 = 21$ | (57) $90 - 60 = 30$ | (87) $69 - 16 = 53$ | (117) $87 - 36 = 51$ |
| (28) $59 - 13 = 46$ | (58) $5 - 4 = 1$ | (88) $84 - 21 = 63$ | (118) $45 - 29 = 16$ |
| (29) $97 - 56 = 41$ | (59) $76 - 12 = 64$ | (89) $58 - 31 = 27$ | (119) $77 - 29 = 48$ |
| (30) $80 - 59 = 21$ | (60) $69 - 31 = 38$ | (90) $39 - 28 = 11$ | (120) $50 - 18 = 32$ |

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|---------------------|---------------------|---------------------|----------------------|
| (1) $28 - 14 = 14$ | (31) $79 - 18 = 61$ | (61) $18 - 12 = 6$ | (91) $70 - 14 = 56$ |
| (2) $6 - 5 = 1$ | (32) $64 - 10 = 54$ | (62) $89 - 44 = 45$ | (92) $35 - 31 = 4$ |
| (3) $34 - 14 = 20$ | (33) $90 - 2 = 88$ | (63) $58 - 54 = 4$ | (93) $89 - 24 = 65$ |
| (4) $38 - 7 = 31$ | (34) $62 - 44 = 18$ | (64) $98 - 73 = 25$ | (94) $88 - 23 = 65$ |
| (5) $45 - 3 = 42$ | (35) $10 - 10 = 0$ | (65) $88 - 19 = 69$ | (95) $91 - 64 = 27$ |
| (6) $83 - 4 = 79$ | (36) $3 - 3 = 0$ | (66) $44 - 33 = 11$ | (96) $88 - 58 = 30$ |
| (7) $88 - 49 = 39$ | (37) $84 - 71 = 13$ | (67) $68 - 28 = 40$ | (97) $97 - 52 = 45$ |
| (8) $45 - 24 = 21$ | (38) $71 - 36 = 35$ | (68) $97 - 81 = 16$ | (98) $66 - 26 = 40$ |
| (9) $72 - 46 = 26$ | (39) $63 - 11 = 52$ | (69) $94 - 41 = 53$ | (99) $96 - 20 = 76$ |
| (10) $69 - 53 = 16$ | (40) $67 - 48 = 19$ | (70) $46 - 14 = 32$ | (100) $73 - 7 = 66$ |
| (11) $54 - 8 = 46$ | (41) $55 - 3 = 52$ | (71) $82 - 47 = 35$ | (101) $66 - 42 = 24$ |
| (12) $49 - 11 = 38$ | (42) $98 - 32 = 66$ | (72) $71 - 16 = 55$ | (102) $98 - 32 = 66$ |
| (13) $43 - 29 = 14$ | (43) $99 - 84 = 15$ | (73) $83 - 4 = 79$ | (103) $56 - 27 = 29$ |
| (14) $44 - 21 = 23$ | (44) $94 - 76 = 18$ | (74) $98 - 43 = 55$ | (104) $62 - 25 = 37$ |
| (15) $40 - 20 = 20$ | (45) $92 - 18 = 74$ | (75) $74 - 62 = 12$ | (105) $89 - 38 = 51$ |
| (16) $47 - 26 = 21$ | (46) $61 - 3 = 58$ | (76) $31 - 29 = 2$ | (106) $83 - 18 = 65$ |
| (17) $57 - 43 = 14$ | (47) $100 - 3 = 97$ | (77) $91 - 9 = 82$ | (107) $64 - 45 = 19$ |
| (18) $18 - 12 = 6$ | (48) $77 - 36 = 41$ | (78) $79 - 35 = 44$ | (108) $76 - 65 = 11$ |
| (19) $92 - 37 = 55$ | (49) $92 - 31 = 61$ | (79) $32 - 25 = 7$ | (109) $78 - 70 = 8$ |
| (20) $45 - 28 = 17$ | (50) $43 - 39 = 4$ | (80) $57 - 1 = 56$ | (110) $96 - 21 = 75$ |
| (21) $45 - 38 = 7$ | (51) $67 - 56 = 11$ | (81) $97 - 6 = 91$ | (111) $75 - 35 = 40$ |
| (22) $98 - 27 = 71$ | (52) $77 - 41 = 36$ | (82) $27 - 12 = 15$ | (112) $78 - 17 = 61$ |
| (23) $94 - 78 = 16$ | (53) $49 - 1 = 48$ | (83) $72 - 71 = 1$ | (113) $84 - 48 = 36$ |
| (24) $17 - 12 = 5$ | (54) $30 - 4 = 26$ | (84) $93 - 5 = 88$ | (114) $83 - 41 = 42$ |
| (25) $34 - 10 = 24$ | (55) $94 - 40 = 54$ | (85) $32 - 23 = 9$ | (115) $73 - 46 = 27$ |
| (26) $91 - 23 = 68$ | (56) $65 - 36 = 29$ | (86) $81 - 3 = 78$ | (116) $16 - 4 = 12$ |
| (27) $82 - 30 = 52$ | (57) $20 - 18 = 2$ | (87) $49 - 40 = 9$ | (117) $59 - 7 = 52$ |
| (28) $70 - 4 = 66$ | (58) $44 - 2 = 42$ | (88) $94 - 2 = 92$ | (118) $83 - 46 = 37$ |
| (29) $63 - 4 = 59$ | (59) $60 - 5 = 55$ | (89) $96 - 65 = 31$ | (119) $36 - 34 = 2$ |
| (30) $90 - 31 = 59$ | (60) $93 - 79 = 14$ | (90) $64 - 33 = 31$ | (120) $94 - 70 = 24$ |

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|---------------------|----------------------|---------------------|----------------------|
| (1) $29 - 18 = 11$ | (31) $79 - 42 = 37$ | (61) $45 - 7 = 38$ | (91) $17 - 0 = 17$ |
| (2) $91 - 76 = 15$ | (32) $89 - 44 = 45$ | (62) $60 - 41 = 19$ | (92) $68 - 44 = 24$ |
| (3) $74 - 28 = 46$ | (33) $89 - 45 = 44$ | (63) $67 - 7 = 60$ | (93) $59 - 39 = 20$ |
| (4) $86 - 64 = 22$ | (34) $77 - 35 = 42$ | (64) $67 - 18 = 49$ | (94) $97 - 58 = 39$ |
| (5) $72 - 37 = 35$ | (35) $86 - 44 = 42$ | (65) $29 - 3 = 26$ | (95) $32 - 7 = 25$ |
| (6) $84 - 16 = 68$ | (36) $58 - 9 = 49$ | (66) $40 - 29 = 11$ | (96) $25 - 3 = 22$ |
| (7) $74 - 11 = 63$ | (37) $73 - 55 = 18$ | (67) $98 - 1 = 97$ | (97) $49 - 12 = 37$ |
| (8) $81 - 77 = 4$ | (38) $77 - 5 = 72$ | (68) $8 - 2 = 6$ | (98) $100 - 22 = 78$ |
| (9) $46 - 17 = 29$ | (39) $63 - 38 = 25$ | (69) $78 - 58 = 20$ | (99) $69 - 55 = 14$ |
| (10) $42 - 24 = 18$ | (40) $24 - 22 = 2$ | (70) $84 - 65 = 19$ | (100) $100 - 94 = 6$ |
| (11) $93 - 43 = 50$ | (41) $78 - 21 = 57$ | (71) $60 - 44 = 16$ | (101) $77 - 65 = 12$ |
| (12) $94 - 68 = 26$ | (42) $67 - 15 = 52$ | (72) $24 - 20 = 4$ | (102) $12 - 7 = 5$ |
| (13) $89 - 72 = 17$ | (43) $98 - 69 = 29$ | (73) $70 - 69 = 1$ | (103) $80 - 59 = 21$ |
| (14) $84 - 26 = 58$ | (44) $81 - 28 = 53$ | (74) $75 - 30 = 45$ | (104) $95 - 92 = 3$ |
| (15) $39 - 25 = 14$ | (45) $87 - 59 = 28$ | (75) $95 - 93 = 2$ | (105) $84 - 67 = 17$ |
| (16) $88 - 72 = 16$ | (46) $48 - 6 = 42$ | (76) $65 - 60 = 5$ | (106) $95 - 36 = 59$ |
| (17) $82 - 7 = 75$ | (47) $100 - 46 = 54$ | (77) $78 - 65 = 13$ | (107) $79 - 52 = 27$ |
| (18) $22 - 13 = 9$ | (48) $65 - 19 = 46$ | (78) $62 - 16 = 46$ | (108) $80 - 50 = 30$ |
| (19) $6 - 4 = 2$ | (49) $98 - 53 = 45$ | (79) $66 - 8 = 58$ | (109) $79 - 21 = 58$ |
| (20) $94 - 52 = 42$ | (50) $20 - 12 = 8$ | (80) $3 - 0 = 3$ | (110) $58 - 56 = 2$ |
| (21) $60 - 2 = 58$ | (51) $69 - 58 = 11$ | (81) $26 - 21 = 5$ | (111) $71 - 17 = 54$ |
| (22) $68 - 66 = 2$ | (52) $53 - 33 = 20$ | (82) $64 - 20 = 44$ | (112) $16 - 3 = 13$ |
| (23) $72 - 48 = 24$ | (53) $60 - 48 = 12$ | (83) $76 - 40 = 36$ | (113) $88 - 12 = 76$ |
| (24) $68 - 27 = 41$ | (54) $63 - 51 = 12$ | (84) $26 - 14 = 12$ | (114) $66 - 57 = 9$ |
| (25) $71 - 69 = 2$ | (55) $70 - 47 = 23$ | (85) $99 - 84 = 15$ | (115) $50 - 8 = 42$ |
| (26) $7 - 5 = 2$ | (56) $80 - 37 = 43$ | (86) $53 - 36 = 17$ | (116) $37 - 16 = 21$ |
| (27) $50 - 30 = 20$ | (57) $8 - 5 = 3$ | (87) $94 - 81 = 13$ | (117) $76 - 24 = 52$ |
| (28) $90 - 88 = 2$ | (58) $49 - 18 = 31$ | (88) $58 - 23 = 35$ | (118) $99 - 19 = 80$ |
| (29) $50 - 18 = 32$ | (59) $84 - 37 = 47$ | (89) $45 - 31 = 14$ | (119) $81 - 0 = 81$ |
| (30) $65 - 6 = 59$ | (60) $81 - 65 = 16$ | (90) $19 - 16 = 3$ | (120) $98 - 18 = 80$ |

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|----------------------|---------------------|---------------------|----------------------|
| (1) $37 - 12 = 25$ | (31) $85 - 3 = 82$ | (61) $74 - 1 = 73$ | (91) $64 - 58 = 6$ |
| (2) $51 - 15 = 36$ | (32) $99 - 46 = 53$ | (62) $13 - 11 = 2$ | (92) $72 - 67 = 5$ |
| (3) $96 - 94 = 2$ | (33) $27 - 10 = 17$ | (63) $25 - 18 = 7$ | (93) $79 - 64 = 15$ |
| (4) $98 - 67 = 31$ | (34) $91 - 59 = 32$ | (64) $29 - 17 = 12$ | (94) $97 - 82 = 15$ |
| (5) $78 - 53 = 25$ | (35) $74 - 65 = 9$ | (65) $24 - 16 = 8$ | (95) $38 - 16 = 22$ |
| (6) $99 - 93 = 6$ | (36) $42 - 29 = 13$ | (66) $75 - 1 = 74$ | (96) $79 - 64 = 15$ |
| (7) $79 - 37 = 42$ | (37) $59 - 45 = 14$ | (67) $69 - 20 = 49$ | (97) $58 - 32 = 26$ |
| (8) $52 - 38 = 14$ | (38) $92 - 52 = 40$ | (68) $64 - 58 = 6$ | (98) $23 - 2 = 21$ |
| (9) $70 - 36 = 34$ | (39) $55 - 11 = 44$ | (69) $58 - 22 = 36$ | (99) $84 - 31 = 53$ |
| (10) $76 - 58 = 18$ | (40) $85 - 71 = 14$ | (70) $85 - 35 = 50$ | (100) $88 - 33 = 55$ |
| (11) $48 - 18 = 30$ | (41) $99 - 50 = 49$ | (71) $78 - 8 = 70$ | (101) $20 - 8 = 12$ |
| (12) $68 - 39 = 29$ | (42) $94 - 12 = 82$ | (72) $76 - 9 = 67$ | (102) $70 - 58 = 12$ |
| (13) $30 - 22 = 8$ | (43) $71 - 1 = 70$ | (73) $86 - 15 = 71$ | (103) $39 - 27 = 12$ |
| (14) $85 - 39 = 46$ | (44) $56 - 8 = 48$ | (74) $85 - 51 = 34$ | (104) $72 - 4 = 68$ |
| (15) $79 - 1 = 78$ | (45) $51 - 7 = 44$ | (75) $54 - 21 = 33$ | (105) $85 - 62 = 23$ |
| (16) $67 - 14 = 53$ | (46) $39 - 13 = 26$ | (76) $77 - 9 = 68$ | (106) $73 - 21 = 52$ |
| (17) $94 - 32 = 62$ | (47) $91 - 14 = 77$ | (77) $47 - 2 = 45$ | (107) $67 - 15 = 52$ |
| (18) $89 - 88 = 1$ | (48) $53 - 31 = 22$ | (78) $37 - 24 = 13$ | (108) $53 - 10 = 43$ |
| (19) $47 - 43 = 4$ | (49) $91 - 84 = 7$ | (79) $88 - 26 = 62$ | (109) $40 - 15 = 25$ |
| (20) $100 - 65 = 35$ | (50) $66 - 43 = 23$ | (80) $75 - 75 = 0$ | (110) $69 - 60 = 9$ |
| (21) $91 - 42 = 49$ | (51) $81 - 9 = 72$ | (81) $98 - 26 = 72$ | (111) $98 - 39 = 59$ |
| (22) $55 - 8 = 47$ | (52) $30 - 11 = 19$ | (82) $76 - 4 = 72$ | (112) $77 - 54 = 23$ |
| (23) $50 - 38 = 12$ | (53) $99 - 89 = 10$ | (83) $88 - 80 = 8$ | (113) $89 - 10 = 79$ |
| (24) $56 - 11 = 45$ | (54) $62 - 26 = 36$ | (84) $25 - 20 = 5$ | (114) $74 - 32 = 42$ |
| (25) $91 - 54 = 37$ | (55) $78 - 66 = 12$ | (85) $64 - 47 = 17$ | (115) $65 - 39 = 26$ |
| (26) $45 - 30 = 15$ | (56) $86 - 8 = 78$ | (86) $38 - 26 = 12$ | (116) $91 - 82 = 9$ |
| (27) $48 - 41 = 7$ | (57) $40 - 13 = 27$ | (87) $43 - 40 = 3$ | (117) $81 - 17 = 64$ |
| (28) $15 - 11 = 4$ | (58) $57 - 36 = 21$ | (88) $51 - 41 = 10$ | (118) $73 - 43 = 30$ |
| (29) $31 - 11 = 20$ | (59) $90 - 86 = 4$ | (89) $50 - 0 = 50$ | (119) $1 - 0 = 1$ |
| (30) $57 - 31 = 26$ | (60) $75 - 71 = 4$ | (90) $41 - 18 = 23$ | (120) $88 - 40 = 48$ |

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|---------------------|----------------------|---------------------|----------------------|
| (1) $32 - 29 = 3$ | (31) $34 - 15 = 19$ | (61) $94 - 0 = 94$ | (91) $94 - 20 = 74$ |
| (2) $50 - 48 = 2$ | (32) $36 - 19 = 17$ | (62) $70 - 3 = 67$ | (92) $95 - 83 = 12$ |
| (3) $82 - 59 = 23$ | (33) $85 - 25 = 60$ | (63) $81 - 20 = 61$ | (93) $95 - 32 = 63$ |
| (4) $38 - 29 = 9$ | (34) $36 - 26 = 10$ | (64) $57 - 32 = 25$ | (94) $81 - 37 = 44$ |
| (5) $19 - 18 = 1$ | (35) $86 - 6 = 80$ | (65) $94 - 45 = 49$ | (95) $64 - 21 = 43$ |
| (6) $47 - 3 = 44$ | (36) $93 - 62 = 31$ | (66) $78 - 61 = 17$ | (96) $99 - 1 = 98$ |
| (7) $95 - 46 = 49$ | (37) $54 - 40 = 14$ | (67) $84 - 52 = 32$ | (97) $98 - 24 = 74$ |
| (8) $49 - 22 = 27$ | (38) $62 - 4 = 58$ | (68) $64 - 31 = 33$ | (98) $91 - 77 = 14$ |
| (9) $53 - 11 = 42$ | (39) $86 - 77 = 9$ | (69) $98 - 23 = 75$ | (99) $86 - 46 = 40$ |
| (10) $72 - 62 = 10$ | (40) $92 - 58 = 34$ | (70) $40 - 40 = 0$ | (100) $81 - 46 = 35$ |
| (11) $77 - 31 = 46$ | (41) $78 - 49 = 29$ | (71) $47 - 7 = 40$ | (101) $67 - 44 = 23$ |
| (12) $98 - 91 = 7$ | (42) $92 - 31 = 61$ | (72) $43 - 33 = 10$ | (102) $46 - 39 = 7$ |
| (13) $79 - 20 = 59$ | (43) $99 - 15 = 84$ | (73) $99 - 73 = 26$ | (103) $70 - 16 = 54$ |
| (14) $63 - 5 = 58$ | (44) $64 - 9 = 55$ | (74) $40 - 39 = 1$ | (104) $86 - 79 = 7$ |
| (15) $53 - 27 = 26$ | (45) $50 - 12 = 38$ | (75) $61 - 14 = 47$ | (105) $54 - 26 = 28$ |
| (16) $59 - 33 = 26$ | (46) $33 - 3 = 30$ | (76) $69 - 58 = 11$ | (106) $83 - 25 = 58$ |
| (17) $50 - 36 = 14$ | (47) $71 - 41 = 30$ | (77) $93 - 21 = 72$ | (107) $87 - 29 = 58$ |
| (18) $69 - 31 = 38$ | (48) $92 - 23 = 69$ | (78) $22 - 11 = 11$ | (108) $78 - 37 = 41$ |
| (19) $73 - 15 = 58$ | (49) $76 - 13 = 63$ | (79) $85 - 68 = 17$ | (109) $76 - 29 = 47$ |
| (20) $68 - 39 = 29$ | (50) $82 - 58 = 24$ | (80) $74 - 58 = 16$ | (110) $75 - 35 = 40$ |
| (21) $43 - 35 = 8$ | (51) $90 - 35 = 55$ | (81) $51 - 34 = 17$ | (111) $92 - 5 = 87$ |
| (22) $73 - 15 = 58$ | (52) $7 - 5 = 2$ | (82) $42 - 21 = 21$ | (112) $83 - 83 = 0$ |
| (23) $66 - 58 = 8$ | (53) $83 - 8 = 75$ | (83) $51 - 18 = 33$ | (113) $95 - 4 = 91$ |
| (24) $52 - 21 = 31$ | (54) $64 - 35 = 29$ | (84) $20 - 8 = 12$ | (114) $8 - 6 = 2$ |
| (25) $28 - 13 = 15$ | (55) $91 - 3 = 88$ | (85) $46 - 13 = 33$ | (115) $58 - 53 = 5$ |
| (26) $63 - 37 = 26$ | (56) $49 - 44 = 5$ | (86) $31 - 15 = 16$ | (116) $69 - 62 = 7$ |
| (27) $37 - 16 = 21$ | (57) $85 - 71 = 14$ | (87) $92 - 47 = 45$ | (117) $10 - 9 = 1$ |
| (28) $39 - 9 = 30$ | (58) $49 - 26 = 23$ | (88) $45 - 34 = 11$ | (118) $35 - 33 = 2$ |
| (29) $48 - 10 = 38$ | (59) $100 - 90 = 10$ | (89) $73 - 39 = 34$ | (119) $27 - 8 = 19$ |
| (30) $57 - 31 = 26$ | (60) $98 - 81 = 17$ | (90) $86 - 11 = 75$ | (120) $50 - 11 = 39$ |

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|----------------------|----------------------|----------------------|-----------------------|
| (1) $63 - 38 = 25$ | (31) $63 - 60 = 3$ | (61) $92 - 0 = 92$ | (91) $58 - 33 = 25$ |
| (2) $12 - 5 = 7$ | (32) $67 - 60 = 7$ | (62) $88 - 52 = 36$ | (92) $49 - 45 = 4$ |
| (3) $54 - 5 = 49$ | (33) $94 - 72 = 22$ | (63) $91 - 27 = 64$ | (93) $70 - 58 = 12$ |
| (4) $90 - 72 = 18$ | (34) $67 - 36 = 31$ | (64) $52 - 7 = 45$ | (94) $43 - 19 = 24$ |
| (5) $61 - 56 = 5$ | (35) $83 - 50 = 33$ | (65) $91 - 83 = 8$ | (95) $67 - 17 = 50$ |
| (6) $69 - 41 = 28$ | (36) $100 - 83 = 17$ | (66) $96 - 1 = 95$ | (96) $56 - 42 = 14$ |
| (7) $98 - 48 = 50$ | (37) $42 - 22 = 20$ | (67) $60 - 25 = 35$ | (97) $79 - 26 = 53$ |
| (8) $36 - 23 = 13$ | (38) $65 - 33 = 32$ | (68) $96 - 50 = 46$ | (98) $85 - 76 = 9$ |
| (9) $30 - 16 = 14$ | (39) $38 - 0 = 38$ | (69) $75 - 72 = 3$ | (99) $72 - 32 = 40$ |
| (10) $76 - 58 = 18$ | (40) $96 - 37 = 59$ | (70) $39 - 2 = 37$ | (100) $43 - 21 = 22$ |
| (11) $32 - 25 = 7$ | (41) $77 - 47 = 30$ | (71) $15 - 8 = 7$ | (101) $94 - 31 = 63$ |
| (12) $88 - 26 = 62$ | (42) $33 - 8 = 25$ | (72) $46 - 25 = 21$ | (102) $100 - 68 = 32$ |
| (13) $65 - 10 = 55$ | (43) $39 - 33 = 6$ | (73) $66 - 63 = 3$ | (103) $84 - 47 = 37$ |
| (14) $91 - 89 = 2$ | (44) $50 - 0 = 50$ | (74) $63 - 32 = 31$ | (104) $76 - 31 = 45$ |
| (15) $78 - 51 = 27$ | (45) $72 - 28 = 44$ | (75) $53 - 2 = 51$ | (105) $41 - 2 = 39$ |
| (16) $94 - 85 = 9$ | (46) $87 - 38 = 49$ | (76) $74 - 32 = 42$ | (106) $100 - 19 = 81$ |
| (17) $48 - 25 = 23$ | (47) $27 - 9 = 18$ | (77) $76 - 67 = 9$ | (107) $89 - 9 = 80$ |
| (18) $93 - 21 = 72$ | (48) $95 - 67 = 28$ | (78) $63 - 43 = 20$ | (108) $51 - 27 = 24$ |
| (19) $72 - 47 = 25$ | (49) $27 - 25 = 2$ | (79) $29 - 23 = 6$ | (109) $74 - 53 = 21$ |
| (20) $86 - 15 = 71$ | (50) $18 - 1 = 17$ | (80) $40 - 19 = 21$ | (110) $51 - 50 = 1$ |
| (21) $54 - 45 = 9$ | (51) $48 - 18 = 30$ | (81) $48 - 15 = 33$ | (111) $82 - 60 = 22$ |
| (22) $81 - 11 = 70$ | (52) $54 - 46 = 8$ | (82) $72 - 26 = 46$ | (112) $73 - 0 = 73$ |
| (23) $70 - 64 = 6$ | (53) $84 - 29 = 55$ | (83) $100 - 57 = 43$ | (113) $73 - 25 = 48$ |
| (24) $18 - 13 = 5$ | (54) $77 - 57 = 20$ | (84) $97 - 63 = 34$ | (114) $98 - 86 = 12$ |
| (25) $69 - 43 = 26$ | (55) $56 - 26 = 30$ | (85) $92 - 58 = 34$ | (115) $9 - 7 = 2$ |
| (26) $64 - 47 = 17$ | (56) $46 - 6 = 40$ | (86) $79 - 32 = 47$ | (116) $96 - 82 = 14$ |
| (27) $21 - 1 = 20$ | (57) $98 - 91 = 7$ | (87) $6 - 1 = 5$ | (117) $52 - 21 = 31$ |
| (28) $25 - 22 = 3$ | (58) $93 - 58 = 35$ | (88) $91 - 15 = 76$ | (118) $73 - 57 = 16$ |
| (29) $100 - 40 = 60$ | (59) $69 - 10 = 59$ | (89) $97 - 96 = 1$ | (119) $98 - 66 = 32$ |
| (30) $50 - 19 = 31$ | (60) $96 - 40 = 56$ | (90) $76 - 57 = 19$ | (120) $80 - 53 = 27$ |

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|---------------------|----------------------|---------------------|----------------------|
| (1) $76 - 8 = 68$ | (31) $45 - 23 = 22$ | (61) $34 - 11 = 23$ | (91) $56 - 45 = 11$ |
| (2) $93 - 92 = 1$ | (32) $38 - 24 = 14$ | (62) $20 - 11 = 9$ | (92) $71 - 6 = 65$ |
| (3) $96 - 60 = 36$ | (33) $31 - 24 = 7$ | (63) $81 - 70 = 11$ | (93) $83 - 72 = 11$ |
| (4) $82 - 10 = 72$ | (34) $34 - 5 = 29$ | (64) $84 - 25 = 59$ | (94) $42 - 31 = 11$ |
| (5) $49 - 22 = 27$ | (35) $67 - 67 = 0$ | (65) $98 - 3 = 95$ | (95) $71 - 60 = 11$ |
| (6) $69 - 2 = 67$ | (36) $98 - 24 = 74$ | (66) $35 - 29 = 6$ | (96) $98 - 22 = 76$ |
| (7) $78 - 22 = 56$ | (37) $71 - 34 = 37$ | (67) $70 - 57 = 13$ | (97) $42 - 18 = 24$ |
| (8) $61 - 6 = 55$ | (38) $97 - 58 = 39$ | (68) $85 - 83 = 2$ | (98) $84 - 41 = 43$ |
| (9) $54 - 40 = 14$ | (39) $38 - 17 = 21$ | (69) $40 - 8 = 32$ | (99) $98 - 19 = 79$ |
| (10) $86 - 13 = 73$ | (40) $73 - 33 = 40$ | (70) $22 - 15 = 7$ | (100) $96 - 49 = 47$ |
| (11) $88 - 49 = 39$ | (41) $85 - 14 = 71$ | (71) $39 - 34 = 5$ | (101) $13 - 11 = 2$ |
| (12) $68 - 15 = 53$ | (42) $23 - 2 = 21$ | (72) $53 - 51 = 2$ | (102) $40 - 6 = 34$ |
| (13) $79 - 10 = 69$ | (43) $99 - 51 = 48$ | (73) $43 - 4 = 39$ | (103) $25 - 2 = 23$ |
| (14) $87 - 32 = 55$ | (44) $77 - 17 = 60$ | (74) $70 - 27 = 43$ | (104) $35 - 20 = 15$ |
| (15) $92 - 76 = 16$ | (45) $94 - 72 = 22$ | (75) $45 - 2 = 43$ | (105) $77 - 48 = 29$ |
| (16) $94 - 29 = 65$ | (46) $96 - 9 = 87$ | (76) $33 - 22 = 11$ | (106) $25 - 11 = 14$ |
| (17) $88 - 23 = 65$ | (47) $38 - 35 = 3$ | (77) $90 - 59 = 31$ | (107) $77 - 46 = 31$ |
| (18) $65 - 60 = 5$ | (48) $65 - 22 = 43$ | (78) $86 - 50 = 36$ | (108) $96 - 69 = 27$ |
| (19) $45 - 28 = 17$ | (49) $49 - 46 = 3$ | (79) $50 - 48 = 2$ | (109) $30 - 21 = 9$ |
| (20) $37 - 8 = 29$ | (50) $100 - 77 = 23$ | (80) $97 - 87 = 10$ | (110) $61 - 33 = 28$ |
| (21) $64 - 45 = 19$ | (51) $74 - 51 = 23$ | (81) $9 - 3 = 6$ | (111) $30 - 18 = 12$ |
| (22) $88 - 67 = 21$ | (52) $86 - 68 = 18$ | (82) $81 - 56 = 25$ | (112) $36 - 21 = 15$ |
| (23) $15 - 13 = 2$ | (53) $33 - 20 = 13$ | (83) $81 - 53 = 28$ | (113) $20 - 10 = 10$ |
| (24) $69 - 69 = 0$ | (54) $42 - 4 = 38$ | (84) $77 - 20 = 57$ | (114) $30 - 18 = 12$ |
| (25) $48 - 28 = 20$ | (55) $35 - 17 = 18$ | (85) $88 - 87 = 1$ | (115) $96 - 88 = 8$ |
| (26) $84 - 79 = 5$ | (56) $88 - 34 = 54$ | (86) $75 - 74 = 1$ | (116) $46 - 18 = 28$ |
| (27) $85 - 37 = 48$ | (57) $63 - 40 = 23$ | (87) $58 - 16 = 42$ | (117) $44 - 17 = 27$ |
| (28) $87 - 1 = 86$ | (58) $28 - 25 = 3$ | (88) $71 - 19 = 52$ | (118) $26 - 7 = 19$ |
| (29) $59 - 27 = 32$ | (59) $51 - 5 = 46$ | (89) $57 - 32 = 25$ | (119) $13 - 6 = 7$ |
| (30) $75 - 11 = 64$ | (60) $63 - 25 = 38$ | (90) $14 - 3 = 11$ | (120) $74 - 65 = 9$ |

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|----------------------|---------------------|---------------------|-----------------------|
| (1) $15 - 14 = 1$ | (31) $96 - 13 = 83$ | (61) $25 - 15 = 10$ | (91) $16 - 6 = 10$ |
| (2) $75 - 14 = 61$ | (32) $96 - 75 = 21$ | (62) $30 - 8 = 22$ | (92) $58 - 11 = 47$ |
| (3) $31 - 20 = 11$ | (33) $39 - 14 = 25$ | (63) $91 - 7 = 84$ | (93) $88 - 15 = 73$ |
| (4) $93 - 46 = 47$ | (34) $58 - 17 = 41$ | (64) $97 - 8 = 89$ | (94) $51 - 50 = 1$ |
| (5) $75 - 38 = 37$ | (35) $85 - 51 = 34$ | (65) $72 - 17 = 55$ | (95) $3 - 0 = 3$ |
| (6) $65 - 36 = 29$ | (36) $42 - 42 = 0$ | (66) $71 - 68 = 3$ | (96) $95 - 38 = 57$ |
| (7) $30 - 19 = 11$ | (37) $85 - 6 = 79$ | (67) $75 - 38 = 37$ | (97) $95 - 67 = 28$ |
| (8) $90 - 87 = 3$ | (38) $46 - 9 = 37$ | (68) $48 - 22 = 26$ | (98) $71 - 19 = 52$ |
| (9) $45 - 4 = 41$ | (39) $54 - 21 = 33$ | (69) $73 - 21 = 52$ | (99) $63 - 41 = 22$ |
| (10) $76 - 40 = 36$ | (40) $35 - 9 = 26$ | (70) $81 - 17 = 64$ | (100) $66 - 33 = 33$ |
| (11) $94 - 30 = 64$ | (41) $76 - 38 = 38$ | (71) $98 - 10 = 88$ | (101) $63 - 47 = 16$ |
| (12) $99 - 60 = 39$ | (42) $63 - 62 = 1$ | (72) $66 - 6 = 60$ | (102) $52 - 26 = 26$ |
| (13) $67 - 6 = 61$ | (43) $83 - 31 = 52$ | (73) $66 - 46 = 20$ | (103) $98 - 78 = 20$ |
| (14) $99 - 57 = 42$ | (44) $79 - 47 = 32$ | (74) $30 - 2 = 28$ | (104) $90 - 61 = 29$ |
| (15) $89 - 22 = 67$ | (45) $57 - 22 = 35$ | (75) $99 - 31 = 68$ | (105) $22 - 21 = 1$ |
| (16) $91 - 45 = 46$ | (46) $35 - 31 = 4$ | (76) $90 - 33 = 57$ | (106) $23 - 9 = 14$ |
| (17) $97 - 29 = 68$ | (47) $77 - 55 = 22$ | (77) $96 - 68 = 28$ | (107) $41 - 17 = 24$ |
| (18) $19 - 7 = 12$ | (48) $97 - 91 = 6$ | (78) $93 - 67 = 26$ | (108) $92 - 6 = 86$ |
| (19) $72 - 18 = 54$ | (49) $39 - 32 = 7$ | (79) $91 - 75 = 16$ | (109) $85 - 25 = 60$ |
| (20) $93 - 18 = 75$ | (50) $34 - 10 = 24$ | (80) $72 - 15 = 57$ | (110) $93 - 4 = 89$ |
| (21) $28 - 28 = 0$ | (51) $60 - 33 = 27$ | (81) $60 - 10 = 50$ | (111) $12 - 7 = 5$ |
| (22) $89 - 83 = 6$ | (52) $92 - 48 = 44$ | (82) $82 - 14 = 68$ | (112) $62 - 18 = 44$ |
| (23) $92 - 10 = 82$ | (53) $26 - 21 = 5$ | (83) $97 - 84 = 13$ | (113) $73 - 57 = 16$ |
| (24) $78 - 47 = 31$ | (54) $45 - 37 = 8$ | (84) $98 - 36 = 62$ | (114) $53 - 46 = 7$ |
| (25) $84 - 14 = 70$ | (55) $91 - 28 = 63$ | (85) $95 - 50 = 45$ | (115) $56 - 47 = 9$ |
| (26) $21 - 7 = 14$ | (56) $70 - 41 = 29$ | (86) $84 - 38 = 46$ | (116) $100 - 62 = 38$ |
| (27) $100 - 57 = 43$ | (57) $21 - 7 = 14$ | (87) $97 - 7 = 90$ | (117) $79 - 74 = 5$ |
| (28) $57 - 55 = 2$ | (58) $40 - 8 = 32$ | (88) $51 - 2 = 49$ | (118) $13 - 10 = 3$ |
| (29) $99 - 63 = 36$ | (59) $76 - 27 = 49$ | (89) $77 - 13 = 64$ | (119) $91 - 41 = 50$ |
| (30) $13 - 11 = 2$ | (60) $14 - 14 = 0$ | (90) $90 - 83 = 7$ | (120) $83 - 50 = 33$ |

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|---------------------|---------------------|----------------------|----------------------|
| (1) $52 - 8 = 44$ | (31) $31 - 12 = 19$ | (61) $74 - 56 = 18$ | (91) $39 - 16 = 23$ |
| (2) $88 - 15 = 73$ | (32) $81 - 46 = 35$ | (62) $91 - 66 = 25$ | (92) $82 - 68 = 14$ |
| (3) $49 - 41 = 8$ | (33) $90 - 14 = 76$ | (63) $97 - 41 = 56$ | (93) $55 - 27 = 28$ |
| (4) $94 - 22 = 72$ | (34) $92 - 91 = 1$ | (64) $40 - 36 = 4$ | (94) $100 - 6 = 94$ |
| (5) $22 - 18 = 4$ | (35) $91 - 12 = 79$ | (65) $58 - 3 = 55$ | (95) $83 - 7 = 76$ |
| (6) $58 - 6 = 52$ | (36) $41 - 28 = 13$ | (66) $44 - 27 = 17$ | (96) $27 - 4 = 23$ |
| (7) $8 - 1 = 7$ | (37) $20 - 12 = 8$ | (67) $100 - 83 = 17$ | (97) $39 - 22 = 17$ |
| (8) $99 - 1 = 98$ | (38) $66 - 8 = 58$ | (68) $76 - 22 = 54$ | (98) $98 - 97 = 1$ |
| (9) $91 - 27 = 64$ | (39) $74 - 65 = 9$ | (69) $97 - 74 = 23$ | (99) $73 - 28 = 45$ |
| (10) $79 - 59 = 20$ | (40) $75 - 61 = 14$ | (70) $86 - 17 = 69$ | (100) $21 - 11 = 10$ |
| (11) $92 - 48 = 44$ | (41) $54 - 36 = 18$ | (71) $84 - 35 = 49$ | (101) $93 - 68 = 25$ |
| (12) $64 - 47 = 17$ | (42) $25 - 17 = 8$ | (72) $56 - 11 = 45$ | (102) $88 - 77 = 11$ |
| (13) $38 - 10 = 28$ | (43) $69 - 66 = 3$ | (73) $99 - 66 = 33$ | (103) $51 - 18 = 33$ |
| (14) $75 - 0 = 75$ | (44) $100 - 8 = 92$ | (74) $81 - 53 = 28$ | (104) $85 - 18 = 67$ |
| (15) $90 - 25 = 65$ | (45) $84 - 48 = 36$ | (75) $93 - 38 = 55$ | (105) $91 - 70 = 21$ |
| (16) $75 - 21 = 54$ | (46) $99 - 13 = 86$ | (76) $56 - 19 = 37$ | (106) $53 - 40 = 13$ |
| (17) $84 - 41 = 43$ | (47) $30 - 16 = 14$ | (77) $61 - 34 = 27$ | (107) $96 - 62 = 34$ |
| (18) $62 - 21 = 41$ | (48) $15 - 0 = 15$ | (78) $73 - 23 = 50$ | (108) $82 - 6 = 76$ |
| (19) $95 - 11 = 84$ | (49) $71 - 12 = 59$ | (79) $93 - 15 = 78$ | (109) $73 - 31 = 42$ |
| (20) $89 - 46 = 43$ | (50) $38 - 1 = 37$ | (80) $60 - 9 = 51$ | (110) $92 - 55 = 37$ |
| (21) $80 - 38 = 42$ | (51) $70 - 24 = 46$ | (81) $38 - 7 = 31$ | (111) $85 - 21 = 64$ |
| (22) $9 - 5 = 4$ | (52) $39 - 35 = 4$ | (82) $73 - 68 = 5$ | (112) $23 - 5 = 18$ |
| (23) $53 - 17 = 36$ | (53) $72 - 62 = 10$ | (83) $98 - 64 = 34$ | (113) $73 - 15 = 58$ |
| (24) $89 - 67 = 22$ | (54) $7 - 7 = 0$ | (84) $99 - 9 = 90$ | (114) $89 - 27 = 62$ |
| (25) $58 - 45 = 13$ | (55) $50 - 24 = 26$ | (85) $23 - 18 = 5$ | (115) $38 - 23 = 15$ |
| (26) $84 - 32 = 52$ | (56) $71 - 3 = 68$ | (86) $18 - 5 = 13$ | (116) $92 - 67 = 25$ |
| (27) $96 - 17 = 79$ | (57) $41 - 34 = 7$ | (87) $96 - 25 = 71$ | (117) $88 - 18 = 70$ |
| (28) $78 - 53 = 25$ | (58) $23 - 14 = 9$ | (88) $39 - 27 = 12$ | (118) $9 - 7 = 2$ |
| (29) $84 - 38 = 46$ | (59) $74 - 3 = 71$ | (89) $67 - 31 = 36$ | (119) $83 - 41 = 42$ |
| (30) $95 - 17 = 78$ | (60) $32 - 28 = 4$ | (90) $68 - 7 = 61$ | (120) $8 - 8 = 0$ |

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|---------------------|---------------------|---------------------|----------------------|
| (1) $99 - 36 = 63$ | (31) $41 - 34 = 7$ | (61) $67 - 64 = 3$ | (91) $82 - 6 = 76$ |
| (2) $81 - 47 = 34$ | (32) $94 - 39 = 55$ | (62) $13 - 7 = 6$ | (92) $90 - 45 = 45$ |
| (3) $68 - 11 = 57$ | (33) $46 - 15 = 31$ | (63) $46 - 3 = 43$ | (93) $33 - 31 = 2$ |
| (4) $33 - 31 = 2$ | (34) $80 - 45 = 35$ | (64) $79 - 78 = 1$ | (94) $88 - 72 = 16$ |
| (5) $47 - 1 = 46$ | (35) $39 - 26 = 13$ | (65) $96 - 74 = 22$ | (95) $70 - 22 = 48$ |
| (6) $52 - 52 = 0$ | (36) $70 - 53 = 17$ | (66) $94 - 13 = 81$ | (96) $46 - 10 = 36$ |
| (7) $97 - 70 = 27$ | (37) $63 - 35 = 28$ | (67) $74 - 41 = 33$ | (97) $48 - 13 = 35$ |
| (8) $39 - 23 = 16$ | (38) $94 - 61 = 33$ | (68) $58 - 45 = 13$ | (98) $66 - 31 = 35$ |
| (9) $57 - 40 = 17$ | (39) $82 - 9 = 73$ | (69) $98 - 4 = 94$ | (99) $78 - 2 = 76$ |
| (10) $76 - 74 = 2$ | (40) $49 - 44 = 5$ | (70) $77 - 44 = 33$ | (100) $63 - 43 = 20$ |
| (11) $88 - 39 = 49$ | (41) $60 - 18 = 42$ | (71) $81 - 34 = 47$ | (101) $45 - 15 = 30$ |
| (12) $41 - 31 = 10$ | (42) $83 - 5 = 78$ | (72) $75 - 72 = 3$ | (102) $58 - 29 = 29$ |
| (13) $82 - 8 = 74$ | (43) $53 - 19 = 34$ | (73) $41 - 30 = 11$ | (103) $43 - 22 = 21$ |
| (14) $94 - 10 = 84$ | (44) $16 - 6 = 10$ | (74) $40 - 38 = 2$ | (104) $57 - 8 = 49$ |
| (15) $69 - 16 = 53$ | (45) $73 - 14 = 59$ | (75) $94 - 11 = 83$ | (105) $82 - 33 = 49$ |
| (16) $6 - 0 = 6$ | (46) $53 - 28 = 25$ | (76) $26 - 26 = 0$ | (106) $30 - 28 = 2$ |
| (17) $84 - 39 = 45$ | (47) $75 - 43 = 32$ | (77) $83 - 17 = 66$ | (107) $97 - 17 = 80$ |
| (18) $96 - 44 = 52$ | (48) $96 - 1 = 95$ | (78) $66 - 32 = 34$ | (108) $67 - 22 = 45$ |
| (19) $20 - 4 = 16$ | (49) $68 - 43 = 25$ | (79) $48 - 24 = 24$ | (109) $94 - 11 = 83$ |
| (20) $100 - 91 = 9$ | (50) $99 - 68 = 31$ | (80) $69 - 44 = 25$ | (110) $80 - 61 = 19$ |
| (21) $67 - 59 = 8$ | (51) $68 - 34 = 34$ | (81) $93 - 41 = 52$ | (111) $75 - 48 = 27$ |
| (22) $30 - 30 = 0$ | (52) $76 - 26 = 50$ | (82) $80 - 57 = 23$ | (112) $75 - 22 = 53$ |
| (23) $98 - 69 = 29$ | (53) $87 - 3 = 84$ | (83) $45 - 44 = 1$ | (113) $85 - 19 = 66$ |
| (24) $93 - 55 = 38$ | (54) $87 - 58 = 29$ | (84) $47 - 29 = 18$ | (114) $32 - 24 = 8$ |
| (25) $32 - 12 = 20$ | (55) $53 - 14 = 39$ | (85) $72 - 52 = 20$ | (115) $36 - 26 = 10$ |
| (26) $62 - 47 = 15$ | (56) $98 - 6 = 92$ | (86) $93 - 51 = 42$ | (116) $55 - 49 = 6$ |
| (27) $79 - 60 = 19$ | (57) $65 - 16 = 49$ | (87) $99 - 70 = 29$ | (117) $66 - 62 = 4$ |
| (28) $93 - 52 = 41$ | (58) $85 - 53 = 32$ | (88) $99 - 6 = 93$ | (118) $73 - 41 = 32$ |
| (29) $51 - 35 = 16$ | (59) $64 - 26 = 38$ | (89) $18 - 17 = 1$ | (119) $85 - 79 = 6$ |
| (30) $34 - 11 = 23$ | (60) $87 - 76 = 11$ | (90) $52 - 0 = 52$ | (120) $39 - 15 = 24$ |

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|----------------------|----------------------|---------------------|----------------------|
| (1) $90 - 84 = 6$ | (31) $96 - 52 = 44$ | (61) $74 - 20 = 54$ | (91) $98 - 1 = 97$ |
| (2) $45 - 20 = 25$ | (32) $70 - 4 = 66$ | (62) $88 - 65 = 23$ | (92) $26 - 5 = 21$ |
| (3) $76 - 6 = 70$ | (33) $99 - 30 = 69$ | (63) $55 - 10 = 45$ | (93) $98 - 84 = 14$ |
| (4) $56 - 5 = 51$ | (34) $65 - 21 = 44$ | (64) $88 - 80 = 8$ | (94) $59 - 45 = 14$ |
| (5) $90 - 84 = 6$ | (35) $78 - 76 = 2$ | (65) $70 - 58 = 12$ | (95) $76 - 66 = 10$ |
| (6) $80 - 49 = 31$ | (36) $79 - 3 = 76$ | (66) $81 - 72 = 9$ | (96) $80 - 71 = 9$ |
| (7) $5 - 3 = 2$ | (37) $93 - 8 = 85$ | (67) $83 - 19 = 64$ | (97) $75 - 56 = 19$ |
| (8) $85 - 30 = 55$ | (38) $35 - 25 = 10$ | (68) $97 - 86 = 11$ | (98) $90 - 33 = 57$ |
| (9) $87 - 76 = 11$ | (39) $81 - 60 = 21$ | (69) $84 - 64 = 20$ | (99) $77 - 1 = 76$ |
| (10) $75 - 22 = 53$ | (40) $42 - 17 = 25$ | (70) $72 - 16 = 56$ | (100) $45 - 25 = 20$ |
| (11) $100 - 29 = 71$ | (41) $78 - 42 = 36$ | (71) $86 - 77 = 9$ | (101) $27 - 27 = 0$ |
| (12) $92 - 5 = 87$ | (42) $22 - 15 = 7$ | (72) $58 - 11 = 47$ | (102) $41 - 14 = 27$ |
| (13) $61 - 45 = 16$ | (43) $31 - 17 = 14$ | (73) $80 - 64 = 16$ | (103) $96 - 37 = 59$ |
| (14) $94 - 40 = 54$ | (44) $95 - 25 = 70$ | (74) $81 - 44 = 37$ | (104) $42 - 16 = 26$ |
| (15) $5 - 5 = 0$ | (45) $55 - 29 = 26$ | (75) $89 - 20 = 69$ | (105) $99 - 98 = 1$ |
| (16) $86 - 70 = 16$ | (46) $86 - 63 = 23$ | (76) $80 - 15 = 65$ | (106) $69 - 11 = 58$ |
| (17) $13 - 8 = 5$ | (47) $100 - 88 = 12$ | (77) $99 - 2 = 97$ | (107) $70 - 12 = 58$ |
| (18) $65 - 20 = 45$ | (48) $56 - 48 = 8$ | (78) $72 - 43 = 29$ | (108) $95 - 0 = 95$ |
| (19) $82 - 37 = 45$ | (49) $98 - 94 = 4$ | (79) $79 - 52 = 27$ | (109) $41 - 29 = 12$ |
| (20) $91 - 60 = 31$ | (50) $86 - 59 = 27$ | (80) $78 - 41 = 37$ | (110) $73 - 27 = 46$ |
| (21) $82 - 5 = 77$ | (51) $79 - 9 = 70$ | (81) $92 - 78 = 14$ | (111) $68 - 11 = 57$ |
| (22) $8 - 4 = 4$ | (52) $60 - 4 = 56$ | (82) $50 - 40 = 10$ | (112) $77 - 45 = 32$ |
| (23) $94 - 63 = 31$ | (53) $88 - 42 = 46$ | (83) $50 - 23 = 27$ | (113) $77 - 17 = 60$ |
| (24) $63 - 51 = 12$ | (54) $64 - 22 = 42$ | (84) $18 - 11 = 7$ | (114) $77 - 57 = 20$ |
| (25) $90 - 77 = 13$ | (55) $68 - 47 = 21$ | (85) $65 - 22 = 43$ | (115) $45 - 15 = 30$ |
| (26) $64 - 21 = 43$ | (56) $92 - 40 = 52$ | (86) $28 - 5 = 23$ | (116) $74 - 66 = 8$ |
| (27) $83 - 45 = 38$ | (57) $75 - 14 = 61$ | (87) $96 - 2 = 94$ | (117) $44 - 40 = 4$ |
| (28) $56 - 51 = 5$ | (58) $72 - 10 = 62$ | (88) $97 - 23 = 74$ | (118) $14 - 1 = 13$ |
| (29) $55 - 6 = 49$ | (59) $55 - 18 = 37$ | (89) $94 - 4 = 90$ | (119) $73 - 37 = 36$ |
| (30) $29 - 2 = 27$ | (60) $41 - 15 = 26$ | (90) $92 - 53 = 39$ | (120) $81 - 44 = 37$ |

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|---------------------|---------------------|----------------------|-----------------------|
| (1) $85 - 77 = 8$ | (31) $68 - 26 = 42$ | (61) $79 - 24 = 55$ | (91) $47 - 25 = 22$ |
| (2) $54 - 50 = 4$ | (32) $21 - 6 = 15$ | (62) $100 - 58 = 42$ | (92) $53 - 37 = 16$ |
| (3) $30 - 2 = 28$ | (33) $56 - 42 = 14$ | (63) $68 - 24 = 44$ | (93) $73 - 20 = 53$ |
| (4) $99 - 81 = 18$ | (34) $75 - 35 = 40$ | (64) $88 - 61 = 27$ | (94) $95 - 59 = 36$ |
| (5) $59 - 31 = 28$ | (35) $65 - 1 = 64$ | (65) $45 - 9 = 36$ | (95) $82 - 38 = 44$ |
| (6) $67 - 15 = 52$ | (36) $68 - 26 = 42$ | (66) $20 - 13 = 7$ | (96) $84 - 69 = 15$ |
| (7) $72 - 18 = 54$ | (37) $57 - 43 = 14$ | (67) $100 - 4 = 96$ | (97) $70 - 44 = 26$ |
| (8) $36 - 0 = 36$ | (38) $14 - 12 = 2$ | (68) $96 - 24 = 72$ | (98) $33 - 27 = 6$ |
| (9) $81 - 6 = 75$ | (39) $78 - 71 = 7$ | (69) $62 - 58 = 4$ | (99) $53 - 6 = 47$ |
| (10) $68 - 32 = 36$ | (40) $99 - 61 = 38$ | (70) $54 - 4 = 50$ | (100) $88 - 56 = 32$ |
| (11) $78 - 60 = 18$ | (41) $38 - 5 = 33$ | (71) $57 - 13 = 44$ | (101) $85 - 48 = 37$ |
| (12) $96 - 54 = 42$ | (42) $93 - 23 = 70$ | (72) $97 - 50 = 47$ | (102) $83 - 27 = 56$ |
| (13) $6 - 6 = 0$ | (43) $69 - 32 = 37$ | (73) $81 - 78 = 3$ | (103) $87 - 16 = 71$ |
| (14) $94 - 69 = 25$ | (44) $75 - 54 = 21$ | (74) $75 - 67 = 8$ | (104) $76 - 62 = 14$ |
| (15) $17 - 15 = 2$ | (45) $90 - 36 = 54$ | (75) $91 - 72 = 19$ | (105) $52 - 42 = 10$ |
| (16) $19 - 10 = 9$ | (46) $52 - 45 = 7$ | (76) $82 - 43 = 39$ | (106) $78 - 35 = 43$ |
| (17) $89 - 78 = 11$ | (47) $56 - 34 = 22$ | (77) $90 - 67 = 23$ | (107) $81 - 54 = 27$ |
| (18) $65 - 49 = 16$ | (48) $95 - 83 = 12$ | (78) $78 - 28 = 50$ | (108) $69 - 60 = 9$ |
| (19) $98 - 24 = 74$ | (49) $49 - 33 = 16$ | (79) $78 - 47 = 31$ | (109) $58 - 26 = 32$ |
| (20) $79 - 71 = 8$ | (50) $68 - 31 = 37$ | (80) $67 - 37 = 30$ | (110) $35 - 8 = 27$ |
| (21) $88 - 21 = 67$ | (51) $79 - 61 = 18$ | (81) $90 - 62 = 28$ | (111) $85 - 54 = 31$ |
| (22) $64 - 26 = 38$ | (52) $61 - 31 = 30$ | (82) $55 - 29 = 26$ | (112) $3 - 1 = 2$ |
| (23) $48 - 0 = 48$ | (53) $62 - 59 = 3$ | (83) $94 - 68 = 26$ | (113) $44 - 14 = 30$ |
| (24) $65 - 27 = 38$ | (54) $66 - 26 = 40$ | (84) $43 - 40 = 3$ | (114) $82 - 4 = 78$ |
| (25) $63 - 33 = 30$ | (55) $97 - 35 = 62$ | (85) $31 - 24 = 7$ | (115) $52 - 40 = 12$ |
| (26) $95 - 43 = 52$ | (56) $45 - 12 = 33$ | (86) $50 - 38 = 12$ | (116) $34 - 5 = 29$ |
| (27) $73 - 3 = 70$ | (57) $97 - 38 = 59$ | (87) $49 - 45 = 4$ | (117) $90 - 8 = 82$ |
| (28) $22 - 5 = 17$ | (58) $15 - 9 = 6$ | (88) $73 - 38 = 35$ | (118) $48 - 42 = 6$ |
| (29) $86 - 33 = 53$ | (59) $96 - 68 = 28$ | (89) $100 - 17 = 83$ | (119) $81 - 1 = 80$ |
| (30) $85 - 48 = 37$ | (60) $54 - 31 = 23$ | (90) $77 - 56 = 21$ | (120) $100 - 13 = 87$ |

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|----------------------|---------------------|----------------------|-----------------------|
| (1) $21 - 6 = 15$ | (31) $89 - 65 = 24$ | (61) $87 - 71 = 16$ | (91) $94 - 50 = 44$ |
| (2) $67 - 38 = 29$ | (32) $56 - 37 = 19$ | (62) $100 - 32 = 68$ | (92) $46 - 40 = 6$ |
| (3) $77 - 20 = 57$ | (33) $95 - 41 = 54$ | (63) $98 - 30 = 68$ | (93) $13 - 13 = 0$ |
| (4) $80 - 32 = 48$ | (34) $51 - 6 = 45$ | (64) $99 - 15 = 84$ | (94) $54 - 17 = 37$ |
| (5) $79 - 1 = 78$ | (35) $66 - 45 = 21$ | (65) $96 - 56 = 40$ | (95) $47 - 46 = 1$ |
| (6) $75 - 27 = 48$ | (36) $96 - 82 = 14$ | (66) $51 - 49 = 2$ | (96) $85 - 69 = 16$ |
| (7) $61 - 27 = 34$ | (37) $68 - 13 = 55$ | (67) $18 - 3 = 15$ | (97) $74 - 6 = 68$ |
| (8) $98 - 72 = 26$ | (38) $82 - 71 = 11$ | (68) $69 - 14 = 55$ | (98) $70 - 12 = 58$ |
| (9) $61 - 18 = 43$ | (39) $60 - 40 = 20$ | (69) $94 - 64 = 30$ | (99) $82 - 33 = 49$ |
| (10) $88 - 4 = 84$ | (40) $82 - 61 = 21$ | (70) $69 - 24 = 45$ | (100) $96 - 9 = 87$ |
| (11) $73 - 32 = 41$ | (41) $39 - 39 = 0$ | (71) $59 - 0 = 59$ | (101) $73 - 2 = 71$ |
| (12) $75 - 68 = 7$ | (42) $74 - 56 = 18$ | (72) $51 - 18 = 33$ | (102) $56 - 11 = 45$ |
| (13) $100 - 71 = 29$ | (43) $92 - 59 = 33$ | (73) $55 - 13 = 42$ | (103) $88 - 2 = 86$ |
| (14) $32 - 20 = 12$ | (44) $65 - 45 = 20$ | (74) $62 - 47 = 15$ | (104) $57 - 18 = 39$ |
| (15) $70 - 38 = 32$ | (45) $17 - 12 = 5$ | (75) $71 - 71 = 0$ | (105) $74 - 8 = 66$ |
| (16) $71 - 63 = 8$ | (46) $68 - 58 = 10$ | (76) $10 - 3 = 7$ | (106) $60 - 19 = 41$ |
| (17) $76 - 76 = 0$ | (47) $91 - 43 = 48$ | (77) $99 - 74 = 25$ | (107) $26 - 26 = 0$ |
| (18) $25 - 20 = 5$ | (48) $71 - 1 = 70$ | (78) $67 - 22 = 45$ | (108) $71 - 33 = 38$ |
| (19) $98 - 26 = 72$ | (49) $69 - 63 = 6$ | (79) $99 - 24 = 75$ | (109) $100 - 37 = 63$ |
| (20) $93 - 73 = 20$ | (50) $37 - 21 = 16$ | (80) $100 - 37 = 63$ | (110) $62 - 0 = 62$ |
| (21) $89 - 38 = 51$ | (51) $28 - 20 = 8$ | (81) $60 - 18 = 42$ | (111) $60 - 39 = 21$ |
| (22) $87 - 39 = 48$ | (52) $61 - 10 = 51$ | (82) $17 - 8 = 9$ | (112) $58 - 40 = 18$ |
| (23) $96 - 86 = 10$ | (53) $70 - 41 = 29$ | (83) $86 - 69 = 17$ | (113) $52 - 52 = 0$ |
| (24) $95 - 6 = 89$ | (54) $67 - 37 = 30$ | (84) $53 - 9 = 44$ | (114) $77 - 66 = 11$ |
| (25) $80 - 47 = 33$ | (55) $62 - 46 = 16$ | (85) $62 - 47 = 15$ | (115) $60 - 60 = 0$ |
| (26) $68 - 10 = 58$ | (56) $35 - 20 = 15$ | (86) $98 - 76 = 22$ | (116) $82 - 26 = 56$ |
| (27) $75 - 31 = 44$ | (57) $33 - 20 = 13$ | (87) $85 - 44 = 41$ | (117) $77 - 29 = 48$ |
| (28) $92 - 70 = 22$ | (58) $68 - 42 = 26$ | (88) $66 - 45 = 21$ | (118) $87 - 78 = 9$ |
| (29) $71 - 48 = 23$ | (59) $86 - 42 = 44$ | (89) $42 - 14 = 28$ | (119) $49 - 21 = 28$ |
| (30) $100 - 24 = 76$ | (60) $44 - 3 = 41$ | (90) $54 - 19 = 35$ | (120) $84 - 48 = 36$ |

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|---------------------|----------------------|----------------------|-----------------------|
| (1) $76 - 52 = 24$ | (31) $70 - 18 = 52$ | (61) $53 - 40 = 13$ | (91) $58 - 29 = 29$ |
| (2) $56 - 9 = 47$ | (32) $68 - 58 = 10$ | (62) $65 - 3 = 62$ | (92) $100 - 94 = 6$ |
| (3) $43 - 5 = 38$ | (33) $75 - 59 = 16$ | (63) $41 - 11 = 30$ | (93) $65 - 20 = 45$ |
| (4) $80 - 38 = 42$ | (34) $97 - 24 = 73$ | (64) $91 - 79 = 12$ | (94) $83 - 37 = 46$ |
| (5) $56 - 33 = 23$ | (35) $84 - 30 = 54$ | (65) $56 - 36 = 20$ | (95) $88 - 51 = 37$ |
| (6) $72 - 11 = 61$ | (36) $47 - 18 = 29$ | (66) $69 - 54 = 15$ | (96) $62 - 51 = 11$ |
| (7) $100 - 35 = 65$ | (37) $70 - 41 = 29$ | (67) $85 - 31 = 54$ | (97) $94 - 67 = 27$ |
| (8) $99 - 84 = 15$ | (38) $86 - 20 = 66$ | (68) $87 - 66 = 21$ | (98) $30 - 24 = 6$ |
| (9) $39 - 35 = 4$ | (39) $95 - 33 = 62$ | (69) $90 - 2 = 88$ | (99) $77 - 6 = 71$ |
| (10) $90 - 11 = 79$ | (40) $49 - 6 = 43$ | (70) $70 - 18 = 52$ | (100) $67 - 36 = 31$ |
| (11) $67 - 44 = 23$ | (41) $76 - 15 = 61$ | (71) $40 - 15 = 25$ | (101) $98 - 73 = 25$ |
| (12) $53 - 52 = 1$ | (42) $88 - 58 = 30$ | (72) $47 - 42 = 5$ | (102) $80 - 54 = 26$ |
| (13) $63 - 11 = 52$ | (43) $89 - 1 = 88$ | (73) $98 - 17 = 81$ | (103) $74 - 66 = 8$ |
| (14) $95 - 89 = 6$ | (44) $52 - 32 = 20$ | (74) $46 - 44 = 2$ | (104) $88 - 42 = 46$ |
| (15) $67 - 41 = 26$ | (45) $86 - 10 = 76$ | (75) $44 - 23 = 21$ | (105) $67 - 47 = 20$ |
| (16) $40 - 38 = 2$ | (46) $95 - 73 = 22$ | (76) $83 - 10 = 73$ | (106) $23 - 8 = 15$ |
| (17) $26 - 21 = 5$ | (47) $100 - 88 = 12$ | (77) $100 - 56 = 44$ | (107) $55 - 11 = 44$ |
| (18) $62 - 26 = 36$ | (48) $68 - 11 = 57$ | (78) $23 - 7 = 16$ | (108) $43 - 29 = 14$ |
| (19) $96 - 3 = 93$ | (49) $73 - 62 = 11$ | (79) $83 - 22 = 61$ | (109) $55 - 6 = 49$ |
| (20) $63 - 35 = 28$ | (50) $91 - 10 = 81$ | (80) $78 - 36 = 42$ | (110) $85 - 59 = 26$ |
| (21) $47 - 31 = 16$ | (51) $96 - 15 = 81$ | (81) $75 - 18 = 57$ | (111) $34 - 3 = 31$ |
| (22) $91 - 68 = 23$ | (52) $68 - 10 = 58$ | (82) $77 - 63 = 14$ | (112) $83 - 19 = 64$ |
| (23) $98 - 47 = 51$ | (53) $19 - 7 = 12$ | (83) $65 - 21 = 44$ | (113) $89 - 17 = 72$ |
| (24) $72 - 2 = 70$ | (54) $97 - 86 = 11$ | (84) $74 - 11 = 63$ | (114) $70 - 39 = 31$ |
| (25) $98 - 19 = 79$ | (55) $71 - 8 = 63$ | (85) $99 - 10 = 89$ | (115) $100 - 35 = 65$ |
| (26) $86 - 57 = 29$ | (56) $96 - 54 = 42$ | (86) $67 - 20 = 47$ | (116) $48 - 1 = 47$ |
| (27) $55 - 13 = 42$ | (57) $27 - 17 = 10$ | (87) $61 - 40 = 21$ | (117) $78 - 40 = 38$ |
| (28) $92 - 87 = 5$ | (58) $95 - 10 = 85$ | (88) $12 - 5 = 7$ | (118) $83 - 78 = 5$ |
| (29) $83 - 23 = 60$ | (59) $95 - 67 = 28$ | (89) $80 - 19 = 61$ | (119) $97 - 84 = 13$ |
| (30) $81 - 2 = 79$ | (60) $81 - 60 = 21$ | (90) $99 - 26 = 73$ | (120) $18 - 2 = 16$ |

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|---------------------|---------------------|---------------------|----------------------|
| (1) $51 - 29 = 22$ | (31) $80 - 8 = 72$ | (61) $75 - 47 = 28$ | (91) $79 - 68 = 11$ |
| (2) $63 - 5 = 58$ | (32) $67 - 6 = 61$ | (62) $80 - 25 = 55$ | (92) $87 - 32 = 55$ |
| (3) $63 - 49 = 14$ | (33) $76 - 45 = 31$ | (63) $97 - 23 = 74$ | (93) $78 - 63 = 15$ |
| (4) $68 - 20 = 48$ | (34) $81 - 54 = 27$ | (64) $88 - 15 = 73$ | (94) $46 - 3 = 43$ |
| (5) $94 - 57 = 37$ | (35) $52 - 46 = 6$ | (65) $60 - 48 = 12$ | (95) $97 - 48 = 49$ |
| (6) $81 - 57 = 24$ | (36) $77 - 39 = 38$ | (66) $34 - 28 = 6$ | (96) $61 - 46 = 15$ |
| (7) $97 - 85 = 12$ | (37) $49 - 48 = 1$ | (67) $84 - 39 = 45$ | (97) $95 - 84 = 11$ |
| (8) $35 - 15 = 20$ | (38) $95 - 38 = 57$ | (68) $68 - 5 = 63$ | (98) $93 - 28 = 65$ |
| (9) $36 - 12 = 24$ | (39) $74 - 55 = 19$ | (69) $76 - 29 = 47$ | (99) $86 - 10 = 76$ |
| (10) $68 - 51 = 17$ | (40) $37 - 18 = 19$ | (70) $58 - 20 = 38$ | (100) $83 - 17 = 66$ |
| (11) $82 - 23 = 59$ | (41) $80 - 29 = 51$ | (71) $85 - 21 = 64$ | (101) $10 - 1 = 9$ |
| (12) $26 - 10 = 16$ | (42) $98 - 87 = 11$ | (72) $60 - 56 = 4$ | (102) $5 - 4 = 1$ |
| (13) $89 - 35 = 54$ | (43) $50 - 47 = 3$ | (73) $26 - 10 = 16$ | (103) $30 - 7 = 23$ |
| (14) $39 - 25 = 14$ | (44) $78 - 71 = 7$ | (74) $90 - 8 = 82$ | (104) $89 - 30 = 59$ |
| (15) $80 - 66 = 14$ | (45) $48 - 47 = 1$ | (75) $51 - 50 = 1$ | (105) $98 - 20 = 78$ |
| (16) $30 - 22 = 8$ | (46) $94 - 42 = 52$ | (76) $97 - 34 = 63$ | (106) $86 - 83 = 3$ |
| (17) $89 - 15 = 74$ | (47) $93 - 10 = 83$ | (77) $45 - 45 = 0$ | (107) $18 - 12 = 6$ |
| (18) $61 - 6 = 55$ | (48) $47 - 42 = 5$ | (78) $93 - 67 = 26$ | (108) $70 - 39 = 31$ |
| (19) $63 - 1 = 62$ | (49) $52 - 42 = 10$ | (79) $26 - 20 = 6$ | (109) $39 - 9 = 30$ |
| (20) $40 - 39 = 1$ | (50) $55 - 35 = 20$ | (80) $68 - 19 = 49$ | (110) $60 - 37 = 23$ |
| (21) $29 - 6 = 23$ | (51) $54 - 8 = 46$ | (81) $99 - 3 = 96$ | (111) $48 - 46 = 2$ |
| (22) $36 - 13 = 23$ | (52) $76 - 8 = 68$ | (82) $79 - 74 = 5$ | (112) $80 - 6 = 74$ |
| (23) $95 - 41 = 54$ | (53) $91 - 42 = 49$ | (83) $87 - 7 = 80$ | (113) $39 - 34 = 5$ |
| (24) $71 - 44 = 27$ | (54) $47 - 12 = 35$ | (84) $67 - 41 = 26$ | (114) $48 - 41 = 7$ |
| (25) $96 - 42 = 54$ | (55) $12 - 4 = 8$ | (85) $58 - 22 = 36$ | (115) $98 - 87 = 11$ |
| (26) $97 - 55 = 42$ | (56) $59 - 9 = 50$ | (86) $73 - 70 = 3$ | (116) $90 - 31 = 59$ |
| (27) $24 - 23 = 1$ | (57) $54 - 41 = 13$ | (87) $74 - 50 = 24$ | (117) $87 - 73 = 14$ |
| (28) $84 - 48 = 36$ | (58) $76 - 26 = 50$ | (88) $95 - 70 = 25$ | (118) $88 - 48 = 40$ |
| (29) $98 - 67 = 31$ | (59) $39 - 4 = 35$ | (89) $65 - 35 = 30$ | (119) $18 - 12 = 6$ |
| (30) $91 - 13 = 78$ | (60) $66 - 41 = 25$ | (90) $97 - 6 = 91$ | (120) $75 - 46 = 29$ |

- | | | | |
|---------------------|----------------------|---------------------|----------------------|
| (1) $63 - 22 = 41$ | (31) $72 - 24 = 48$ | (61) $97 - 88 = 9$ | (91) $90 - 23 = 67$ |
| (2) $40 - 39 = 1$ | (32) $62 - 51 = 11$ | (62) $41 - 19 = 22$ | (92) $94 - 64 = 30$ |
| (3) $45 - 39 = 6$ | (33) $18 - 11 = 7$ | (63) $75 - 57 = 18$ | (93) $18 - 16 = 2$ |
| (4) $71 - 12 = 59$ | (34) $63 - 44 = 19$ | (64) $33 - 11 = 22$ | (94) $59 - 30 = 29$ |
| (5) $83 - 56 = 27$ | (35) $63 - 57 = 6$ | (65) $40 - 15 = 25$ | (95) $82 - 13 = 69$ |
| (6) $51 - 5 = 46$ | (36) $84 - 63 = 21$ | (66) $96 - 25 = 71$ | (96) $83 - 26 = 57$ |
| (7) $78 - 73 = 5$ | (37) $99 - 25 = 74$ | (67) $65 - 64 = 1$ | (97) $22 - 18 = 4$ |
| (8) $86 - 9 = 77$ | (38) $98 - 53 = 45$ | (68) $74 - 52 = 22$ | (98) $96 - 73 = 23$ |
| (9) $59 - 39 = 20$ | (39) $85 - 40 = 45$ | (69) $99 - 8 = 91$ | (99) $58 - 16 = 42$ |
| (10) $90 - 61 = 29$ | (40) $69 - 33 = 36$ | (70) $60 - 43 = 17$ | (100) $48 - 7 = 41$ |
| (11) $85 - 81 = 4$ | (41) $59 - 54 = 5$ | (71) $54 - 12 = 42$ | (101) $50 - 27 = 23$ |
| (12) $84 - 73 = 11$ | (42) $62 - 50 = 12$ | (72) $53 - 36 = 17$ | (102) $16 - 14 = 2$ |
| (13) $92 - 87 = 5$ | (43) $29 - 25 = 4$ | (73) $99 - 4 = 95$ | (103) $59 - 29 = 30$ |
| (14) $93 - 8 = 85$ | (44) $100 - 62 = 38$ | (74) $29 - 24 = 5$ | (104) $47 - 17 = 30$ |
| (15) $50 - 37 = 13$ | (45) $98 - 84 = 14$ | (75) $70 - 64 = 6$ | (105) $93 - 86 = 7$ |
| (16) $67 - 57 = 10$ | (46) $47 - 4 = 43$ | (76) $39 - 11 = 28$ | (106) $74 - 9 = 65$ |
| (17) $98 - 30 = 68$ | (47) $89 - 51 = 38$ | (77) $12 - 5 = 7$ | (107) $44 - 32 = 12$ |
| (18) $52 - 35 = 17$ | (48) $97 - 33 = 64$ | (78) $45 - 14 = 31$ | (108) $47 - 31 = 16$ |
| (19) $65 - 19 = 46$ | (49) $83 - 70 = 13$ | (79) $54 - 18 = 36$ | (109) $83 - 11 = 72$ |
| (20) $64 - 46 = 18$ | (50) $67 - 19 = 48$ | (80) $9 - 1 = 8$ | (110) $73 - 4 = 69$ |
| (21) $76 - 48 = 28$ | (51) $94 - 11 = 83$ | (81) $90 - 83 = 7$ | (111) $82 - 52 = 30$ |
| (22) $91 - 59 = 32$ | (52) $64 - 49 = 15$ | (82) $71 - 7 = 64$ | (112) $97 - 72 = 25$ |
| (23) $68 - 42 = 26$ | (53) $92 - 80 = 12$ | (83) $93 - 38 = 55$ | (113) $85 - 82 = 3$ |
| (24) $30 - 6 = 24$ | (54) $65 - 49 = 16$ | (84) $60 - 22 = 38$ | (114) $70 - 65 = 5$ |
| (25) $78 - 46 = 32$ | (55) $93 - 42 = 51$ | (85) $88 - 75 = 13$ | (115) $64 - 49 = 15$ |
| (26) $60 - 13 = 47$ | (56) $92 - 23 = 69$ | (86) $68 - 7 = 61$ | (116) $85 - 16 = 69$ |
| (27) $89 - 38 = 51$ | (57) $78 - 33 = 45$ | (87) $83 - 34 = 49$ | (117) $70 - 27 = 43$ |
| (28) $97 - 47 = 50$ | (58) $86 - 27 = 59$ | (88) $70 - 16 = 54$ | (118) $25 - 9 = 16$ |
| (29) $58 - 1 = 57$ | (59) $71 - 66 = 5$ | (89) $98 - 24 = 74$ | (119) $44 - 18 = 26$ |
| (30) $76 - 11 = 65$ | (60) $77 - 71 = 6$ | (90) $26 - 3 = 23$ | (120) $80 - 48 = 32$ |

1.1.3 乗法

自然数の積を求める問題。

(1) 9×6	(31) 1×5	(61) 6×4	(91) 4×9
(2) 4×5	(32) 1×3	(62) 6×8	(92) 2×3
(3) 5×1	(33) 5×3	(63) 4×8	(93) 4×7
(4) 2×8	(34) 10×6	(64) 8×5	(94) 8×1
(5) 6×0	(35) 1×8	(65) 6×2	(95) 7×3
(6) 10×1	(36) 5×5	(66) 1×1	(96) 7×4
(7) 7×4	(37) 8×4	(67) 10×10	(97) 4×4
(8) 5×6	(38) 4×8	(68) 3×5	(98) 3×2
(9) 10×10	(39) 0×3	(69) 6×9	(99) 4×1
(10) 1×5	(40) 5×3	(70) 7×6	(100) 4×10
(11) 5×6	(41) 7×10	(71) 4×2	(101) 9×5
(12) 4×10	(42) 6×10	(72) 6×1	(102) 5×10
(13) 5×5	(43) 7×6	(73) 6×10	(103) 10×5
(14) 6×4	(44) 9×5	(74) 6×5	(104) 2×8
(15) 2×2	(45) 6×5	(75) 1×10	(105) 1×4
(16) 9×2	(46) 2×6	(76) 4×0	(106) 7×4
(17) 5×1	(47) 10×7	(77) 2×7	(107) 6×6
(18) 10×10	(48) 9×3	(78) 2×3	(108) 6×5
(19) 8×9	(49) 8×7	(79) 2×5	(109) 3×10
(20) 5×5	(50) 2×5	(80) 3×8	(110) 5×10
(21) 2×1	(51) 2×7	(81) 6×7	(111) 1×3
(22) 10×2	(52) 1×10	(82) 2×6	(112) 5×6
(23) 6×5	(53) 4×7	(83) 8×8	(113) 6×8
(24) 9×4	(54) 3×9	(84) 7×5	(114) 2×3
(25) 3×10	(55) 1×6	(85) 2×7	(115) 1×0
(26) 8×6	(56) 5×5	(86) 3×0	(116) 1×9
(27) 10×9	(57) 8×9	(87) 4×9	(117) 8×0
(28) 8×3	(58) 5×8	(88) 5×10	(118) 5×2
(29) 8×7	(59) 4×6	(89) 10×1	(119) 2×0
(30) 5×3	(60) 2×5	(90) 1×2	(120) 6×4

(1) 2×7	(31) 2×0	(61) 8×4	(91) 8×9
(2) 10×4	(32) 2×4	(62) 6×3	(92) 0×10
(3) 3×7	(33) 1×10	(63) 4×6	(93) 8×2
(4) 7×1	(34) 1×4	(64) 1×8	(94) 8×5
(5) 5×4	(35) 2×8	(65) 7×2	(95) 1×5
(6) 5×7	(36) 3×4	(66) 7×7	(96) 1×8
(7) 7×3	(37) 0×4	(67) 5×10	(97) 9×2
(8) 8×1	(38) 4×9	(68) 3×4	(98) 3×0
(9) 6×6	(39) 1×3	(69) 6×1	(99) 8×9
(10) 2×10	(40) 9×8	(70) 7×2	(100) 5×4
(11) 10×7	(41) 7×0	(71) 7×3	(101) 1×8
(12) 0×10	(42) 2×3	(72) 6×8	(102) 1×10
(13) 1×6	(43) 10×1	(73) 6×2	(103) 1×2
(14) 10×7	(44) 4×3	(74) 4×6	(104) 4×2
(15) 8×5	(45) 10×7	(75) 10×5	(105) 5×0
(16) 10×8	(46) 9×6	(76) 5×3	(106) 7×3
(17) 7×10	(47) 3×0	(77) 3×8	(107) 9×5
(18) 7×9	(48) 2×0	(78) 10×2	(108) 0×2
(19) 9×0	(49) 7×6	(79) 1×8	(109) 7×4
(20) 8×2	(50) 4×5	(80) 9×7	(110) 8×5
(21) 6×4	(51) 5×10	(81) 3×9	(111) 1×8
(22) 2×0	(52) 1×3	(82) 8×7	(112) 6×8
(23) 8×10	(53) 7×8	(83) 4×5	(113) 7×4
(24) 9×5	(54) 4×6	(84) 9×3	(114) 5×9
(25) 3×9	(55) 1×9	(85) 4×5	(115) 2×10
(26) 4×10	(56) 2×8	(86) 4×0	(116) 2×7
(27) 6×9	(57) 1×7	(87) 7×2	(117) 9×2
(28) 9×10	(58) 3×2	(88) 6×7	(118) 9×7
(29) 9×9	(59) 10×5	(89) 10×9	(119) 6×4
(30) 9×2	(60) 0×6	(90) 7×2	(120) 1×9

(1) 4×9	(31) 1×4	(61) 10×6	(91) 2×2
(2) 1×3	(32) 1×6	(62) 1×8	(92) 2×3
(3) 7×10	(33) 2×10	(63) 7×10	(93) 6×8
(4) 7×7	(34) 9×10	(64) 4×3	(94) 5×4
(5) 2×2	(35) 5×4	(65) 4×9	(95) 5×8
(6) 6×10	(36) 6×2	(66) 8×1	(96) 6×3
(7) 2×8	(37) 4×7	(67) 6×3	(97) 2×1
(8) 1×6	(38) 6×1	(68) 10×4	(98) 5×2
(9) 2×8	(39) 7×2	(69) 6×7	(99) 2×4
(10) 4×8	(40) 5×9	(70) 1×1	(100) 4×4
(11) 8×5	(41) 0×7	(71) 2×8	(101) 7×0
(12) 8×4	(42) 9×2	(72) 1×2	(102) 5×3
(13) 3×4	(43) 2×8	(73) 9×0	(103) 5×2
(14) 6×10	(44) 10×7	(74) 10×9	(104) 7×6
(15) 6×5	(45) 5×1	(75) 7×1	(105) 9×1
(16) 7×6	(46) 1×4	(76) 9×10	(106) 4×7
(17) 8×8	(47) 8×4	(77) 2×9	(107) 6×2
(18) 0×9	(48) 2×6	(78) 8×8	(108) 3×3
(19) 0×5	(49) 7×9	(79) 2×5	(109) 2×7
(20) 2×9	(50) 6×7	(80) 9×1	(110) 8×3
(21) 0×6	(51) 2×8	(81) 10×7	(111) 4×1
(22) 10×8	(52) 6×5	(82) 1×2	(112) 6×7
(23) 5×5	(53) 10×6	(83) 4×6	(113) 9×1
(24) 3×6	(54) 8×7	(84) 2×9	(114) 2×3
(25) 0×9	(55) 7×2	(85) 3×3	(115) 4×0
(26) 0×10	(56) 2×9	(86) 10×5	(116) 1×5
(27) 10×8	(57) 3×3	(87) 8×9	(117) 1×1
(28) 2×2	(58) 9×2	(88) 8×9	(118) 7×7
(29) 2×2	(59) 6×4	(89) 3×8	(119) 3×3
(30) 0×1	(60) 1×9	(90) 7×6	(120) 1×6

(1) 3×8	(31) 5×0	(61) 2×7	(91) 7×2
(2) 9×4	(32) 1×3	(62) 4×2	(92) 10×8
(3) 6×8	(33) 3×0	(63) 2×9	(93) 8×2
(4) 5×1	(34) 2×8	(64) 4×3	(94) 3×5
(5) 10×5	(35) 1×8	(65) 3×7	(95) 1×8
(6) 8×8	(36) 2×2	(66) 1×3	(96) 3×8
(7) 3×1	(37) 10×3	(67) 6×3	(97) 8×1
(8) 5×1	(38) 5×8	(68) 7×6	(98) 3×3
(9) 0×3	(39) 8×5	(69) 1×10	(99) 7×1
(10) 9×8	(40) 7×7	(70) 6×9	(100) 10×1
(11) 9×2	(41) 10×1	(71) 2×8	(101) 1×1
(12) 0×10	(42) 4×6	(72) 10×3	(102) 10×10
(13) 9×8	(43) 8×1	(73) 3×4	(103) 4×5
(14) 6×1	(44) 8×7	(74) 8×7	(104) 10×7
(15) 8×3	(45) 3×5	(75) 6×10	(105) 5×3
(16) 10×7	(46) 1×2	(76) 6×4	(106) 9×8
(17) 5×7	(47) 3×10	(77) 7×1	(107) 9×1
(18) 10×5	(48) 2×2	(78) 3×1	(108) 5×4
(19) 3×6	(49) 4×0	(79) 0×7	(109) 0×1
(20) 6×5	(50) 7×7	(80) 5×1	(110) 3×2
(21) 6×9	(51) 8×5	(81) 3×9	(111) 9×5
(22) 5×9	(52) 5×4	(82) 6×8	(112) 5×7
(23) 2×3	(53) 9×9	(83) 8×5	(113) 0×0
(24) 1×2	(54) 1×7	(84) 6×8	(114) 6×10
(25) 5×8	(55) 8×8	(85) 5×7	(115) 7×1
(26) 10×2	(56) 9×0	(86) 3×1	(116) 1×3
(27) 7×0	(57) 1×3	(87) 3×1	(117) 1×5
(28) 0×4	(58) 6×5	(88) 1×8	(118) 8×5
(29) 1×1	(59) 9×0	(89) 7×3	(119) 0×4
(30) 0×2	(60) 8×3	(90) 8×2	(120) 3×7

(1) 2×1	(31) 7×0	(61) 8×4	(91) 1×1
(2) 9×2	(32) 9×4	(62) 3×1	(92) 9×2
(3) 0×4	(33) 4×7	(63) 5×4	(93) 8×3
(4) 3×1	(34) 3×7	(64) 10×9	(94) 6×2
(5) 8×9	(35) 7×9	(65) 3×6	(95) 9×6
(6) 6×10	(36) 2×1	(66) 10×10	(96) 10×0
(7) 2×7	(37) 3×10	(67) 0×9	(97) 7×3
(8) 7×10	(38) 2×8	(68) 9×4	(98) 9×5
(9) 7×0	(39) 5×2	(69) 2×3	(99) 7×9
(10) 4×8	(40) 2×2	(70) 5×8	(100) 6×0
(11) 6×0	(41) 9×8	(71) 3×6	(101) 0×7
(12) 9×5	(42) 2×1	(72) 10×6	(102) 8×1
(13) 6×4	(43) 6×7	(73) 0×9	(103) 0×6
(14) 6×1	(44) 8×4	(74) 5×9	(104) 0×9
(15) 3×7	(45) 3×5	(75) 5×0	(105) 6×2
(16) 3×5	(46) 5×1	(76) 5×2	(106) 0×9
(17) 10×8	(47) 2×7	(77) 2×2	(107) 7×3
(18) 9×9	(48) 5×4	(78) 1×6	(108) 8×9
(19) 5×7	(49) 3×9	(79) 6×10	(109) 8×1
(20) 0×3	(50) 3×7	(80) 10×6	(110) 1×9
(21) 10×3	(51) 1×4	(81) 9×8	(111) 1×4
(22) 10×0	(52) 9×9	(82) 4×5	(112) 7×0
(23) 5×4	(53) 2×2	(83) 5×10	(113) 0×4
(24) 7×4	(54) 6×2	(84) 8×7	(114) 9×10
(25) 2×9	(55) 5×2	(85) 6×9	(115) 1×4
(26) 7×5	(56) 10×8	(86) 10×3	(116) 1×10
(27) 4×0	(57) 0×4	(87) 4×0	(117) 10×1
(28) 9×8	(58) 0×9	(88) 1×5	(118) 4×4
(29) 7×9	(59) 9×3	(89) 10×1	(119) 0×8
(30) 9×1	(60) 10×5	(90) 8×3	(120) 4×1

(1) 3×5	(31) 1×5	(61) 8×7	(91) 9×0
(2) 0×7	(32) 2×8	(62) 10×9	(92) 9×8
(3) 9×7	(33) 3×0	(63) 5×5	(93) 2×10
(4) 3×0	(34) 3×6	(64) 2×10	(94) 1×7
(5) 9×1	(35) 4×5	(65) 4×1	(95) 1×4
(6) 3×9	(36) 8×2	(66) 6×10	(96) 7×10
(7) 7×9	(37) 8×8	(67) 3×7	(97) 2×1
(8) 5×3	(38) 7×10	(68) 0×1	(98) 7×7
(9) 9×10	(39) 6×5	(69) 5×2	(99) 10×6
(10) 10×9	(40) 10×3	(70) 1×5	(100) 0×9
(11) 8×2	(41) 0×5	(71) 4×4	(101) 3×3
(12) 6×7	(42) 5×6	(72) 4×5	(102) 8×3
(13) 6×9	(43) 10×2	(73) 4×9	(103) 8×8
(14) 1×6	(44) 8×5	(74) 2×3	(104) 10×1
(15) 10×2	(45) 6×5	(75) 3×4	(105) 3×7
(16) 1×8	(46) 5×7	(76) 7×8	(106) 3×5
(17) 4×5	(47) 2×2	(77) 10×3	(107) 10×8
(18) 8×9	(48) 5×7	(78) 2×4	(108) 1×3
(19) 2×5	(49) 9×2	(79) 3×6	(109) 1×8
(20) 1×2	(50) 4×7	(80) 6×3	(110) 1×5
(21) 0×9	(51) 10×0	(81) 10×1	(111) 6×5
(22) 10×1	(52) 7×6	(82) 1×6	(112) 1×1
(23) 1×3	(53) 6×3	(83) 9×6	(113) 6×7
(24) 4×6	(54) 10×8	(84) 0×5	(114) 1×10
(25) 8×9	(55) 1×3	(85) 9×4	(115) 2×10
(26) 6×0	(56) 9×9	(86) 3×8	(116) 8×6
(27) 9×8	(57) 4×1	(87) 6×10	(117) 9×8
(28) 6×6	(58) 6×2	(88) 7×3	(118) 1×3
(29) 9×7	(59) 6×0	(89) 1×3	(119) 7×2
(30) 8×5	(60) 9×8	(90) 5×3	(120) 8×0

(1) 2×10	(31) 4×8	(61) 9×5	(91) 6×8
(2) 7×2	(32) 7×3	(62) 7×10	(92) 4×9
(3) 2×4	(33) 5×1	(63) 10×7	(93) 8×1
(4) 6×7	(34) 2×9	(64) 2×5	(94) 4×1
(5) 1×6	(35) 4×5	(65) 3×4	(95) 8×5
(6) 2×5	(36) 6×6	(66) 6×1	(96) 7×0
(7) 10×9	(37) 7×5	(67) 3×9	(97) 3×1
(8) 6×10	(38) 8×9	(68) 10×8	(98) 9×8
(9) 1×5	(39) 2×0	(69) 6×5	(99) 5×4
(10) 2×6	(40) 7×2	(70) 0×0	(100) 2×1
(11) 8×2	(41) 7×3	(71) 4×4	(101) 3×10
(12) 9×2	(42) 0×4	(72) 2×3	(102) 5×4
(13) 10×9	(43) 2×8	(73) 2×3	(103) 0×9
(14) 7×6	(44) 3×5	(74) 4×1	(104) 6×1
(15) 0×9	(45) 7×2	(75) 2×9	(105) 8×10
(16) 10×3	(46) 1×8	(76) 2×6	(106) 9×8
(17) 2×6	(47) 6×2	(77) 4×3	(107) 3×8
(18) 5×6	(48) 4×7	(78) 9×9	(108) 6×9
(19) 9×6	(49) 3×9	(79) 0×6	(109) 10×6
(20) 9×8	(50) 2×5	(80) 9×4	(110) 3×6
(21) 9×9	(51) 9×4	(81) 4×1	(111) 1×4
(22) 1×5	(52) 10×7	(82) 9×3	(112) 6×6
(23) 6×0	(53) 7×2	(83) 6×8	(113) 10×0
(24) 6×1	(54) 9×1	(84) 8×0	(114) 8×8
(25) 10×10	(55) 7×2	(85) 6×1	(115) 0×9
(26) 9×7	(56) 4×9	(86) 6×6	(116) 1×1
(27) 10×2	(57) 4×0	(87) 9×1	(117) 1×5
(28) 9×10	(58) 8×4	(88) 1×1	(118) 1×9
(29) 4×5	(59) 4×8	(89) 2×4	(119) 0×0
(30) 6×6	(60) 2×4	(90) 7×7	(120) 10×3

(1) 4×6	(31) 9×5	(61) 10×7	(91) 0×1
(2) 9×7	(32) 9×6	(62) 2×0	(92) 7×2
(3) 5×9	(33) 1×4	(63) 5×3	(93) 3×8
(4) 10×10	(34) 8×0	(64) 10×6	(94) 7×3
(5) 2×2	(35) 10×8	(65) 8×8	(95) 0×7
(6) 2×6	(36) 2×7	(66) 10×8	(96) 6×9
(7) 2×9	(37) 4×2	(67) 5×6	(97) 1×1
(8) 2×0	(38) 5×3	(68) 2×2	(98) 2×5
(9) 4×4	(39) 3×2	(69) 7×7	(99) 7×8
(10) 10×9	(40) 10×8	(70) 6×7	(100) 7×1
(11) 6×4	(41) 10×2	(71) 1×0	(101) 10×6
(12) 6×1	(42) 4×2	(72) 10×2	(102) 8×6
(13) 10×8	(43) 10×10	(73) 7×8	(103) 2×6
(14) 3×4	(44) 5×4	(74) 4×7	(104) 6×9
(15) 8×0	(45) 3×7	(75) 4×7	(105) 7×6
(16) 10×4	(46) 2×2	(76) 2×1	(106) 2×10
(17) 0×8	(47) 9×0	(77) 7×2	(107) 9×8
(18) 8×3	(48) 2×3	(78) 1×1	(108) 6×6
(19) 9×3	(49) 5×7	(79) 7×6	(109) 6×5
(20) 6×7	(50) 4×5	(80) 8×7	(110) 7×8
(21) 9×2	(51) 4×1	(81) 8×8	(111) 6×7
(22) 10×3	(52) 0×9	(82) 4×8	(112) 3×5
(23) 9×6	(53) 1×8	(83) 9×9	(113) 4×4
(24) 10×1	(54) 1×0	(84) 1×2	(114) 9×9
(25) 6×0	(55) 7×4	(85) 10×10	(115) 7×10
(26) 3×3	(56) 4×6	(86) 1×9	(116) 6×7
(27) 3×10	(57) 3×6	(87) 9×6	(117) 6×9
(28) 10×1	(58) 3×6	(88) 1×10	(118) 1×7
(29) 10×1	(59) 9×4	(89) 1×2	(119) 1×9
(30) 5×5	(60) 9×2	(90) 7×4	(120) 10×3

(1) 6×4	(31) 6×2	(61) 2×9	(91) 0×1
(2) 5×2	(32) 1×8	(62) 7×7	(92) 5×10
(3) 8×8	(33) 5×5	(63) 9×4	(93) 6×7
(4) 2×9	(34) 7×10	(64) 0×9	(94) 3×6
(5) 6×8	(35) 3×6	(65) 8×5	(95) 10×3
(6) 1×1	(36) 3×5	(66) 9×1	(96) 3×2
(7) 4×1	(37) 7×2	(67) 8×9	(97) 2×4
(8) 2×6	(38) 4×3	(68) 5×9	(98) 1×2
(9) 4×10	(39) 5×7	(69) 5×10	(99) 3×8
(10) 7×6	(40) 1×6	(70) 3×7	(100) 3×6
(11) 2×5	(41) 6×9	(71) 2×1	(101) 10×0
(12) 8×4	(42) 2×5	(72) 6×6	(102) 8×4
(13) 2×5	(43) 2×4	(73) 5×3	(103) 4×1
(14) 2×9	(44) 9×0	(74) 1×5	(104) 9×6
(15) 8×9	(45) 9×10	(75) 0×5	(105) 5×10
(16) 10×9	(46) 2×9	(76) 8×8	(106) 4×2
(17) 1×9	(47) 4×9	(77) 4×8	(107) 0×1
(18) 6×6	(48) 1×5	(78) 7×8	(108) 9×10
(19) 7×8	(49) 4×9	(79) 1×3	(109) 2×4
(20) 4×2	(50) 1×1	(80) 2×7	(110) 5×6
(21) 5×7	(51) 9×8	(81) 9×1	(111) 6×10
(22) 9×2	(52) 10×6	(82) 8×6	(112) 3×6
(23) 2×6	(53) 2×8	(83) 3×8	(113) 10×9
(24) 1×9	(54) 3×4	(84) 6×9	(114) 5×0
(25) 3×4	(55) 1×6	(85) 1×7	(115) 6×1
(26) 1×4	(56) 8×9	(86) 0×2	(116) 3×2
(27) 7×2	(57) 0×3	(87) 2×5	(117) 9×1
(28) 3×1	(58) 8×9	(88) 10×8	(118) 7×7
(29) 8×2	(59) 4×6	(89) 6×0	(119) 2×3
(30) 3×5	(60) 0×4	(90) 9×7	(120) 2×2

(1) 6×3	(31) 4×4	(61) 1×1	(91) 2×7
(2) 7×9	(32) 7×8	(62) 5×4	(92) 3×3
(3) 3×5	(33) 4×9	(63) 7×4	(93) 3×1
(4) 5×1	(34) 7×10	(64) 10×7	(94) 5×9
(5) 9×5	(35) 1×5	(65) 4×4	(95) 7×8
(6) 1×3	(36) 4×6	(66) 2×2	(96) 5×1
(7) 4×5	(37) 9×2	(67) 2×3	(97) 10×7
(8) 9×6	(38) 6×0	(68) 8×9	(98) 2×2
(9) 0×0	(39) 2×5	(69) 4×7	(99) 7×6
(10) 10×1	(40) 5×4	(70) 0×6	(100) 10×0
(11) 7×5	(41) 3×4	(71) 5×4	(101) 10×5
(12) 4×10	(42) 0×10	(72) 1×2	(102) 9×8
(13) 0×3	(43) 9×1	(73) 5×4	(103) 1×8
(14) 5×5	(44) 6×6	(74) 5×9	(104) 3×1
(15) 3×7	(45) 5×3	(75) 4×2	(105) 1×6
(16) 4×4	(46) 0×10	(76) 8×6	(106) 5×2
(17) 10×7	(47) 9×3	(77) 5×6	(107) 2×9
(18) 1×9	(48) 8×1	(78) 0×8	(108) 9×3
(19) 1×8	(49) 3×6	(79) 10×3	(109) 4×10
(20) 8×4	(50) 10×8	(80) 8×7	(110) 1×2
(21) 3×3	(51) 2×5	(81) 0×6	(111) 3×5
(22) 7×1	(52) 8×2	(82) 8×7	(112) 2×7
(23) 8×9	(53) 8×9	(83) 8×7	(113) 6×1
(24) 10×3	(54) 1×2	(84) 1×6	(114) 9×2
(25) 10×1	(55) 6×10	(85) 0×8	(115) 4×5
(26) 5×5	(56) 2×9	(86) 7×4	(116) 4×8
(27) 8×6	(57) 9×3	(87) 9×9	(117) 7×6
(28) 6×1	(58) 4×2	(88) 9×9	(118) 7×1
(29) 4×2	(59) 5×10	(89) 10×9	(119) 2×10
(30) 10×2	(60) 9×10	(90) 8×3	(120) 6×8

(1) 4×4	(31) 7×7	(61) 8×7	(91) 9×5
(2) 4×9	(32) 8×1	(62) 8×1	(92) 4×1
(3) 0×4	(33) 0×4	(63) 5×8	(93) 7×10
(4) 8×9	(34) 6×4	(64) 2×5	(94) 1×2
(5) 0×10	(35) 0×2	(65) 0×0	(95) 1×0
(6) 6×6	(36) 6×0	(66) 3×3	(96) 2×5
(7) 8×9	(37) 10×7	(67) 9×9	(97) 0×7
(8) 5×4	(38) 5×3	(68) 9×4	(98) 6×2
(9) 3×10	(39) 5×7	(69) 4×10	(99) 1×9
(10) 3×10	(40) 5×2	(70) 6×7	(100) 7×9
(11) 0×8	(41) 10×1	(71) 2×4	(101) 10×3
(12) 7×10	(42) 6×9	(72) 10×6	(102) 0×10
(13) 7×9	(43) 8×1	(73) 5×7	(103) 7×7
(14) 3×2	(44) 7×6	(74) 10×6	(104) 1×6
(15) 9×0	(45) 0×9	(75) 10×1	(105) 4×5
(16) 7×4	(46) 7×7	(76) 2×7	(106) 7×1
(17) 4×2	(47) 10×7	(77) 3×2	(107) 6×6
(18) 10×10	(48) 1×3	(78) 0×9	(108) 5×9
(19) 6×10	(49) 10×7	(79) 8×10	(109) 1×4
(20) 1×9	(50) 2×6	(80) 5×4	(110) 5×8
(21) 10×2	(51) 7×6	(81) 8×9	(111) 1×8
(22) 3×1	(52) 3×8	(82) 9×4	(112) 7×3
(23) 6×7	(53) 5×1	(83) 6×8	(113) 9×9
(24) 3×2	(54) 7×4	(84) 9×7	(114) 9×2
(25) 9×5	(55) 1×1	(85) 1×1	(115) 4×10
(26) 4×3	(56) 10×10	(86) 9×8	(116) 10×6
(27) 2×7	(57) 8×1	(87) 7×9	(117) 10×3
(28) 9×1	(58) 7×3	(88) 4×9	(118) 8×2
(29) 7×8	(59) 3×7	(89) 1×7	(119) 3×0
(30) 10×8	(60) 8×8	(90) 7×3	(120) 2×9

(1) 10×4	(31) 10×1	(61) 7×6	(91) 3×3
(2) 4×1	(32) 7×9	(62) 8×3	(92) 3×8
(3) 9×9	(33) 4×8	(63) 8×7	(93) 10×7
(4) 10×4	(34) 4×2	(64) 10×9	(94) 3×7
(5) 3×2	(35) 3×5	(65) 10×6	(95) 3×5
(6) 5×6	(36) 1×3	(66) 8×10	(96) 7×10
(7) 1×5	(37) 7×7	(67) 9×10	(97) 8×7
(8) 7×0	(38) 8×2	(68) 8×1	(98) 10×6
(9) 6×6	(39) 10×3	(69) 5×10	(99) 3×5
(10) 9×3	(40) 6×6	(70) 5×9	(100) 9×1
(11) 2×8	(41) 6×1	(71) 8×5	(101) 3×9
(12) 6×1	(42) 1×8	(72) 1×6	(102) 7×6
(13) 9×0	(43) 3×1	(73) 2×3	(103) 10×7
(14) 9×6	(44) 2×0	(74) 4×7	(104) 0×4
(15) 1×10	(45) 2×3	(75) 4×3	(105) 8×2
(16) 2×9	(46) 10×8	(76) 8×3	(106) 2×10
(17) 3×9	(47) 0×3	(77) 2×5	(107) 2×1
(18) 6×4	(48) 10×9	(78) 4×7	(108) 10×0
(19) 6×9	(49) 1×5	(79) 9×9	(109) 10×1
(20) 5×4	(50) 8×7	(80) 0×7	(110) 5×2
(21) 5×2	(51) 5×6	(81) 0×7	(111) 1×7
(22) 7×2	(52) 8×4	(82) 4×9	(112) 9×1
(23) 2×10	(53) 4×5	(83) 10×10	(113) 0×6
(24) 10×3	(54) 10×3	(84) 7×8	(114) 7×10
(25) 5×3	(55) 3×1	(85) 4×1	(115) 4×10
(26) 2×2	(56) 1×2	(86) 2×1	(116) 5×10
(27) 2×4	(57) 4×0	(87) 5×9	(117) 9×9
(28) 8×4	(58) 3×3	(88) 8×9	(118) 9×2
(29) 9×4	(59) 1×5	(89) 4×3	(119) 10×0
(30) 4×9	(60) 6×3	(90) 10×8	(120) 9×6

(1) 2×8	(31) 6×2	(61) 0×1	(91) 9×5
(2) 2×10	(32) 3×1	(62) 2×1	(92) 3×5
(3) 6×8	(33) 7×7	(63) 4×8	(93) 1×0
(4) 10×10	(34) 9×3	(64) 4×1	(94) 2×3
(5) 7×1	(35) 9×0	(65) 3×7	(95) 3×10
(6) 6×5	(36) 1×6	(66) 2×3	(96) 9×10
(7) 8×8	(37) 1×6	(67) 0×7	(97) 8×1
(8) 9×5	(38) 10×0	(68) 5×2	(98) 1×4
(9) 8×7	(39) 3×2	(69) 6×2	(99) 2×4
(10) 9×2	(40) 2×6	(70) 1×7	(100) 6×4
(11) 3×0	(41) 5×4	(71) 2×2	(101) 4×4
(12) 4×4	(42) 3×6	(72) 2×3	(102) 1×6
(13) 7×1	(43) 2×8	(73) 6×5	(103) 3×4
(14) 5×0	(44) 8×10	(74) 6×8	(104) 6×3
(15) 5×9	(45) 8×4	(75) 9×2	(105) 5×3
(16) 1×0	(46) 8×5	(76) 6×1	(106) 10×4
(17) 3×7	(47) 4×7	(77) 1×6	(107) 8×4
(18) 6×7	(48) 6×6	(78) 1×8	(108) 7×3
(19) 6×1	(49) 7×9	(79) 8×0	(109) 8×3
(20) 2×3	(50) 10×9	(80) 7×1	(110) 4×4
(21) 0×5	(51) 1×3	(81) 8×2	(111) 3×8
(22) 7×9	(52) 0×6	(82) 7×9	(112) 2×8
(23) 10×9	(53) 1×10	(83) 0×9	(113) 1×1
(24) 4×8	(54) 10×4	(84) 4×1	(114) 5×1
(25) 10×2	(55) 8×8	(85) 7×1	(115) 9×1
(26) 6×3	(56) 8×4	(86) 2×6	(116) 2×1
(27) 8×4	(57) 1×8	(87) 4×1	(117) 2×1
(28) 1×1	(58) 6×5	(88) 3×4	(118) 2×8
(29) 0×6	(59) 10×6	(89) 9×10	(119) 0×2
(30) 4×10	(60) 0×7	(90) 9×10	(120) 4×4

(1) 1×4	(31) 10×8	(61) 7×8	(91) 9×5
(2) 2×9	(32) 5×0	(62) 8×8	(92) 4×4
(3) 2×2	(33) 9×4	(63) 1×3	(93) 10×9
(4) 10×8	(34) 4×3	(64) 0×1	(94) 2×2
(5) 5×6	(35) 7×5	(65) 2×3	(95) 1×1
(6) 6×6	(36) 1×3	(66) 2×2	(96) 8×5
(7) 7×2	(37) 7×9	(67) 1×3	(97) 7×5
(8) 0×7	(38) 4×2	(68) 9×4	(98) 0×2
(9) 2×5	(39) 7×9	(69) 3×6	(99) 5×10
(10) 8×7	(40) 7×4	(70) 9×1	(100) 1×0
(11) 4×6	(41) 3×6	(71) 7×8	(101) 0×6
(12) 7×4	(42) 4×3	(72) 2×5	(102) 8×1
(13) 6×10	(43) 0×4	(73) 10×8	(103) 4×1
(14) 8×8	(44) 7×1	(74) 8×6	(104) 4×10
(15) 10×10	(45) 3×8	(75) 7×6	(105) 2×10
(16) 5×0	(46) 4×7	(76) 5×2	(106) 7×5
(17) 4×0	(47) 6×10	(77) 8×4	(107) 0×0
(18) 4×7	(48) 10×3	(78) 5×8	(108) 10×3
(19) 6×2	(49) 8×6	(79) 1×7	(109) 2×2
(20) 4×1	(50) 7×5	(80) 9×4	(110) 5×6
(21) 4×8	(51) 6×1	(81) 10×5	(111) 5×8
(22) 6×10	(52) 4×0	(82) 10×1	(112) 2×4
(23) 9×9	(53) 5×2	(83) 6×3	(113) 8×0
(24) 4×6	(54) 3×9	(84) 10×2	(114) 10×3
(25) 3×1	(55) 7×4	(85) 8×8	(115) 2×4
(26) 3×4	(56) 1×9	(86) 5×10	(116) 2×9
(27) 9×10	(57) 7×7	(87) 4×10	(117) 3×7
(28) 10×2	(58) 6×7	(88) 8×4	(118) 9×10
(29) 4×9	(59) 9×10	(89) 9×8	(119) 3×2
(30) 7×9	(60) 4×6	(90) 0×7	(120) 6×6

(1) 8×9	(31) 10×2	(61) 0×9	(91) 5×2
(2) 5×9	(32) 3×2	(62) 9×2	(92) 6×4
(3) 7×10	(33) 5×8	(63) 10×1	(93) 1×4
(4) 4×8	(34) 2×3	(64) 8×1	(94) 4×6
(5) 4×1	(35) 4×5	(65) 8×3	(95) 9×9
(6) 7×8	(36) 8×10	(66) 3×5	(96) 4×4
(7) 5×9	(37) 8×6	(67) 10×8	(97) 6×5
(8) 9×7	(38) 9×9	(68) 1×10	(98) 4×9
(9) 3×4	(39) 6×9	(69) 1×1	(99) 6×10
(10) 4×10	(40) 1×3	(70) 10×8	(100) 0×7
(11) 4×9	(41) 4×5	(71) 8×4	(101) 4×9
(12) 6×9	(42) 3×5	(72) 9×3	(102) 10×3
(13) 1×8	(43) 4×5	(73) 3×10	(103) 7×1
(14) 10×5	(44) 2×3	(74) 2×9	(104) 5×10
(15) 1×10	(45) 4×10	(75) 9×8	(105) 9×9
(16) 0×2	(46) 3×8	(76) 7×7	(106) 4×1
(17) 8×6	(47) 1×7	(77) 6×9	(107) 2×8
(18) 10×5	(48) 0×1	(78) 8×5	(108) 2×0
(19) 10×10	(49) 8×2	(79) 6×8	(109) 6×10
(20) 2×6	(50) 7×7	(80) 10×6	(110) 10×5
(21) 9×6	(51) 0×3	(81) 2×10	(111) 6×4
(22) 1×10	(52) 9×10	(82) 1×8	(112) 6×10
(23) 10×2	(53) 5×7	(83) 2×1	(113) 0×6
(24) 2×9	(54) 8×6	(84) 5×0	(114) 3×4
(25) 10×2	(55) 7×2	(85) 3×0	(115) 7×4
(26) 6×9	(56) 10×6	(86) 8×8	(116) 0×10
(27) 1×4	(57) 4×4	(87) 7×5	(117) 3×9
(28) 4×1	(58) 5×1	(88) 5×9	(118) 4×6
(29) 6×9	(59) 10×9	(89) 5×9	(119) 5×2
(30) 0×7	(60) 3×4	(90) 6×8	(120) 3×1

(1) 1×9	(31) 9×3	(61) 4×1	(91) 1×3
(2) 8×6	(32) 5×3	(62) 3×2	(92) 7×4
(3) 5×3	(33) 5×2	(63) 8×7	(93) 9×7
(4) 5×9	(34) 7×2	(64) 7×1	(94) 10×0
(5) 8×10	(35) 10×4	(65) 7×6	(95) 6×0
(6) 1×3	(36) 7×8	(66) 0×8	(96) 3×2
(7) 4×5	(37) 10×7	(67) 3×5	(97) 8×2
(8) 9×4	(38) 8×7	(68) 7×9	(98) 3×8
(9) 5×3	(39) 4×0	(69) 1×2	(99) 10×7
(10) 7×9	(40) 2×8	(70) 1×9	(100) 10×6
(11) 10×5	(41) 9×10	(71) 6×8	(101) 3×2
(12) 2×6	(42) 10×6	(72) 9×10	(102) 0×9
(13) 6×0	(43) 8×2	(73) 4×9	(103) 4×10
(14) 3×4	(44) 2×10	(74) 10×2	(104) 8×7
(15) 1×9	(45) 5×0	(75) 4×7	(105) 8×1
(16) 2×3	(46) 10×6	(76) 9×2	(106) 9×7
(17) 10×2	(47) 4×7	(77) 5×8	(107) 3×8
(18) 10×3	(48) 6×1	(78) 10×2	(108) 4×3
(19) 6×1	(49) 5×2	(79) 6×5	(109) 5×3
(20) 8×1	(50) 3×9	(80) 10×2	(110) 7×5
(21) 9×9	(51) 6×8	(81) 0×5	(111) 3×4
(22) 3×2	(52) 5×2	(82) 2×8	(112) 6×6
(23) 7×6	(53) 1×9	(83) 6×4	(113) 2×4
(24) 7×3	(54) 0×8	(84) 6×6	(114) 1×9
(25) 2×3	(55) 1×7	(85) 3×6	(115) 9×5
(26) 10×3	(56) 3×8	(86) 5×1	(116) 2×7
(27) 10×10	(57) 9×5	(87) 0×2	(117) 7×6
(28) 6×7	(58) 2×5	(88) 5×1	(118) 6×5
(29) 8×4	(59) 0×4	(89) 2×10	(119) 5×7
(30) 6×5	(60) 4×7	(90) 7×7	(120) 4×4

(1) 7×10	(31) 7×6	(61) 2×3	(91) 2×5
(2) 7×6	(32) 4×4	(62) 5×2	(92) 10×8
(3) 0×0	(33) 1×6	(63) 6×2	(93) 4×5
(4) 2×9	(34) 5×8	(64) 3×2	(94) 2×6
(5) 10×3	(35) 3×7	(65) 3×2	(95) 6×9
(6) 3×8	(36) 4×5	(66) 9×10	(96) 9×4
(7) 7×9	(37) 4×10	(67) 1×4	(97) 10×6
(8) 9×0	(38) 5×6	(68) 9×3	(98) 6×0
(9) 2×1	(39) 6×8	(69) 4×8	(99) 2×9
(10) 2×10	(40) 2×9	(70) 5×8	(100) 1×2
(11) 2×3	(41) 0×1	(71) 2×6	(101) 3×8
(12) 10×3	(42) 8×10	(72) 9×3	(102) 6×5
(13) 10×7	(43) 2×5	(73) 5×4	(103) 9×10
(14) 3×8	(44) 7×4	(74) 0×4	(104) 4×8
(15) 10×7	(45) 6×3	(75) 1×1	(105) 3×3
(16) 10×10	(46) 6×3	(76) 4×2	(106) 10×8
(17) 7×3	(47) 9×7	(77) 10×10	(107) 5×1
(18) 9×2	(48) 8×5	(78) 4×6	(108) 7×8
(19) 1×3	(49) 8×6	(79) 7×1	(109) 4×10
(20) 8×4	(50) 2×9	(80) 9×6	(110) 1×3
(21) 9×5	(51) 10×1	(81) 2×9	(111) 10×7
(22) 6×2	(52) 10×9	(82) 10×10	(112) 2×6
(23) 1×1	(53) 4×5	(83) 1×5	(113) 10×4
(24) 6×6	(54) 9×3	(84) 7×6	(114) 4×8
(25) 3×10	(55) 4×7	(85) 4×6	(115) 7×9
(26) 7×4	(56) 8×3	(86) 3×0	(116) 5×0
(27) 9×4	(57) 2×2	(87) 6×2	(117) 4×2
(28) 2×1	(58) 2×6	(88) 7×2	(118) 1×8
(29) 0×3	(59) 1×10	(89) 4×5	(119) 8×2
(30) 10×3	(60) 1×2	(90) 9×6	(120) 5×7

(1) 3×5	(31) 2×5	(61) 2×3	(91) 3×4
(2) 5×1	(32) 7×10	(62) 9×6	(92) 3×4
(3) 2×3	(33) 0×1	(63) 5×3	(93) 1×10
(4) 7×6	(34) 10×1	(64) 5×5	(94) 10×1
(5) 5×7	(35) 5×3	(65) 9×1	(95) 0×5
(6) 5×9	(36) 2×7	(66) 2×3	(96) 6×8
(7) 7×5	(37) 7×8	(67) 10×4	(97) 3×5
(8) 3×2	(38) 4×6	(68) 4×2	(98) 6×6
(9) 5×2	(39) 5×8	(69) 0×1	(99) 7×4
(10) 7×4	(40) 1×3	(70) 3×2	(100) 3×8
(11) 3×4	(41) 6×1	(71) 4×5	(101) 7×2
(12) 5×1	(42) 10×7	(72) 0×1	(102) 9×3
(13) 7×4	(43) 8×7	(73) 7×10	(103) 4×5
(14) 6×3	(44) 2×6	(74) 10×5	(104) 7×0
(15) 10×3	(45) 10×7	(75) 8×10	(105) 0×6
(16) 6×2	(46) 8×4	(76) 3×4	(106) 9×9
(17) 6×6	(47) 1×7	(77) 5×8	(107) 2×2
(18) 0×1	(48) 5×2	(78) 6×7	(108) 7×2
(19) 7×8	(49) 8×9	(79) 0×0	(109) 6×4
(20) 9×4	(50) 3×8	(80) 1×3	(110) 9×2
(21) 2×4	(51) 7×2	(81) 7×1	(111) 2×9
(22) 1×9	(52) 7×7	(82) 3×9	(112) 3×1
(23) 2×4	(53) 3×0	(83) 8×10	(113) 1×9
(24) 2×4	(54) 10×1	(84) 6×9	(114) 9×7
(25) 9×8	(55) 3×2	(85) 2×4	(115) 4×7
(26) 10×1	(56) 10×10	(86) 0×4	(116) 1×6
(27) 5×1	(57) 7×7	(87) 3×8	(117) 10×7
(28) 2×8	(58) 1×5	(88) 8×4	(118) 5×2
(29) 1×8	(59) 8×6	(89) 7×8	(119) 1×8
(30) 7×6	(60) 5×6	(90) 4×1	(120) 6×7

(1) 2×7	(31) 3×2	(61) 5×5	(91) 6×6
(2) 3×1	(32) 3×6	(62) 3×4	(92) 8×7
(3) 9×10	(33) 1×7	(63) 6×5	(93) 10×7
(4) 1×3	(34) 9×6	(64) 8×10	(94) 3×6
(5) 1×6	(35) 7×6	(65) 6×1	(95) 1×5
(6) 1×10	(36) 6×0	(66) 0×10	(96) 6×6
(7) 5×2	(37) 3×2	(67) 1×8	(97) 1×2
(8) 1×10	(38) 3×1	(68) 8×4	(98) 3×5
(9) 8×4	(39) 4×1	(69) 1×8	(99) 8×3
(10) 9×6	(40) 7×7	(70) 6×3	(100) 10×3
(11) 10×3	(41) 5×10	(71) 2×6	(101) 1×7
(12) 9×7	(42) 4×5	(72) 1×10	(102) 8×7
(13) 6×8	(43) 3×7	(73) 8×9	(103) 4×9
(14) 7×2	(44) 9×10	(74) 8×9	(104) 10×2
(15) 0×8	(45) 3×8	(75) 9×2	(105) 4×1
(16) 5×9	(46) 1×5	(76) 7×0	(106) 7×3
(17) 10×2	(47) 8×4	(77) 0×9	(107) 10×5
(18) 5×8	(48) 10×7	(78) 2×1	(108) 3×9
(19) 1×4	(49) 10×8	(79) 1×1	(109) 6×7
(20) 5×6	(50) 1×10	(80) 3×8	(110) 9×5
(21) 1×7	(51) 4×2	(81) 9×1	(111) 4×0
(22) 3×8	(52) 4×9	(82) 2×3	(112) 0×10
(23) 6×10	(53) 0×7	(83) 8×9	(113) 7×8
(24) 9×2	(54) 2×6	(84) 3×3	(114) 9×3
(25) 6×0	(55) 1×0	(85) 9×9	(115) 3×2
(26) 10×6	(56) 10×2	(86) 3×7	(116) 2×8
(27) 5×2	(57) 10×5	(87) 6×2	(117) 4×1
(28) 9×4	(58) 7×3	(88) 3×4	(118) 0×2
(29) 2×10	(59) 5×2	(89) 7×4	(119) 1×9
(30) 5×1	(60) 4×3	(90) 9×6	(120) 2×3

(1) 3×3	(31) 6×2	(61) 8×5	(91) 7×8
(2) 9×4	(32) 10×9	(62) 8×10	(92) 4×3
(3) 7×1	(33) 4×8	(63) 5×10	(93) 4×2
(4) 2×1	(34) 6×6	(64) 0×7	(94) 9×6
(5) 3×3	(35) 6×6	(65) 10×8	(95) 3×2
(6) 2×2	(36) 4×0	(66) 5×7	(96) 4×3
(7) 1×9	(37) 4×3	(67) 5×7	(97) 3×3
(8) 6×8	(38) 7×2	(68) 5×7	(98) 6×9
(9) 9×6	(39) 9×1	(69) 7×7	(99) 0×3
(10) 7×2	(40) 6×6	(70) 3×4	(100) 9×1
(11) 4×9	(41) 8×9	(71) 3×3	(101) 3×0
(12) 9×9	(42) 6×10	(72) 9×5	(102) 8×9
(13) 6×6	(43) 7×2	(73) 5×4	(103) 1×8
(14) 1×0	(44) 8×1	(74) 8×0	(104) 9×3
(15) 6×6	(45) 2×7	(75) 1×9	(105) 0×7
(16) 7×3	(46) 2×10	(76) 4×3	(106) 3×2
(17) 4×3	(47) 2×9	(77) 4×1	(107) 10×8
(18) 0×8	(48) 8×8	(78) 3×9	(108) 7×3
(19) 6×3	(49) 7×3	(79) 7×7	(109) 0×5
(20) 5×2	(50) 2×5	(80) 10×8	(110) 7×7
(21) 5×6	(51) 8×0	(81) 9×8	(111) 5×5
(22) 7×7	(52) 7×9	(82) 7×1	(112) 9×1
(23) 9×10	(53) 6×4	(83) 3×4	(113) 8×1
(24) 1×3	(54) 9×10	(84) 2×5	(114) 8×7
(25) 7×8	(55) 3×8	(85) 1×4	(115) 7×10
(26) 8×5	(56) 5×2	(86) 2×4	(116) 6×7
(27) 7×3	(57) 6×3	(87) 4×3	(117) 3×8
(28) 5×10	(58) 3×4	(88) 1×8	(118) 2×3
(29) 7×5	(59) 5×5	(89) 10×6	(119) 4×2
(30) 1×10	(60) 10×7	(90) 3×1	(120) 3×5

(1) $9 \times 6 = 54$	(31) $1 \times 5 = 5$	(61) $6 \times 4 = 24$	(91) $4 \times 9 = 36$
(2) $4 \times 5 = 20$	(32) $1 \times 3 = 3$	(62) $6 \times 8 = 48$	(92) $2 \times 3 = 6$
(3) $5 \times 1 = 5$	(33) $5 \times 3 = 15$	(63) $4 \times 8 = 32$	(93) $4 \times 7 = 28$
(4) $2 \times 8 = 16$	(34) $10 \times 6 = 60$	(64) $8 \times 5 = 40$	(94) $8 \times 1 = 8$
(5) $6 \times 0 = 0$	(35) $1 \times 8 = 8$	(65) $6 \times 2 = 12$	(95) $7 \times 3 = 21$
(6) $10 \times 1 = 10$	(36) $5 \times 5 = 25$	(66) $1 \times 1 = 1$	(96) $7 \times 4 = 28$
(7) $7 \times 4 = 28$	(37) $8 \times 4 = 32$	(67) $10 \times 10 = 100$	(97) $4 \times 4 = 16$
(8) $5 \times 6 = 30$	(38) $4 \times 8 = 32$	(68) $3 \times 5 = 15$	(98) $3 \times 2 = 6$
(9) $10 \times 10 = 100$	(39) $0 \times 3 = 0$	(69) $6 \times 9 = 54$	(99) $4 \times 1 = 4$
(10) $1 \times 5 = 5$	(40) $5 \times 3 = 15$	(70) $7 \times 6 = 42$	(100) $4 \times 10 = 40$
(11) $5 \times 6 = 30$	(41) $7 \times 10 = 70$	(71) $4 \times 2 = 8$	(101) $9 \times 5 = 45$
(12) $4 \times 10 = 40$	(42) $6 \times 10 = 60$	(72) $6 \times 1 = 6$	(102) $5 \times 10 = 50$
(13) $5 \times 5 = 25$	(43) $7 \times 6 = 42$	(73) $6 \times 10 = 60$	(103) $10 \times 5 = 50$
(14) $6 \times 4 = 24$	(44) $9 \times 5 = 45$	(74) $6 \times 5 = 30$	(104) $2 \times 8 = 16$
(15) $2 \times 2 = 4$	(45) $6 \times 5 = 30$	(75) $1 \times 10 = 10$	(105) $1 \times 4 = 4$
(16) $9 \times 2 = 18$	(46) $2 \times 6 = 12$	(76) $4 \times 0 = 0$	(106) $7 \times 4 = 28$
(17) $5 \times 1 = 5$	(47) $10 \times 7 = 70$	(77) $2 \times 7 = 14$	(107) $6 \times 6 = 36$
(18) $10 \times 10 = 100$	(48) $9 \times 3 = 27$	(78) $2 \times 3 = 6$	(108) $6 \times 5 = 30$
(19) $8 \times 9 = 72$	(49) $8 \times 7 = 56$	(79) $2 \times 5 = 10$	(109) $3 \times 10 = 30$
(20) $5 \times 5 = 25$	(50) $2 \times 5 = 10$	(80) $3 \times 8 = 24$	(110) $5 \times 10 = 50$
(21) $2 \times 1 = 2$	(51) $2 \times 7 = 14$	(81) $6 \times 7 = 42$	(111) $1 \times 3 = 3$
(22) $10 \times 2 = 20$	(52) $1 \times 10 = 10$	(82) $2 \times 6 = 12$	(112) $5 \times 6 = 30$
(23) $6 \times 5 = 30$	(53) $4 \times 7 = 28$	(83) $8 \times 8 = 64$	(113) $6 \times 8 = 48$
(24) $9 \times 4 = 36$	(54) $3 \times 9 = 27$	(84) $7 \times 5 = 35$	(114) $2 \times 3 = 6$
(25) $3 \times 10 = 30$	(55) $1 \times 6 = 6$	(85) $2 \times 7 = 14$	(115) $1 \times 0 = 0$
(26) $8 \times 6 = 48$	(56) $5 \times 5 = 25$	(86) $3 \times 0 = 0$	(116) $1 \times 9 = 9$
(27) $10 \times 9 = 90$	(57) $8 \times 9 = 72$	(87) $4 \times 9 = 36$	(117) $8 \times 0 = 0$
(28) $8 \times 3 = 24$	(58) $5 \times 8 = 40$	(88) $5 \times 10 = 50$	(118) $5 \times 2 = 10$
(29) $8 \times 7 = 56$	(59) $4 \times 6 = 24$	(89) $10 \times 1 = 10$	(119) $2 \times 0 = 0$
(30) $5 \times 3 = 15$	(60) $2 \times 5 = 10$	(90) $1 \times 2 = 2$	(120) $6 \times 4 = 24$

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|-------------------------|-------------------------|-------------------------|--------------------------|
| (1) $2 \times 7 = 14$ | (31) $2 \times 0 = 0$ | (61) $8 \times 4 = 32$ | (91) $8 \times 9 = 72$ |
| (2) $10 \times 4 = 40$ | (32) $2 \times 4 = 8$ | (62) $6 \times 3 = 18$ | (92) $0 \times 10 = 0$ |
| (3) $3 \times 7 = 21$ | (33) $1 \times 10 = 10$ | (63) $4 \times 6 = 24$ | (93) $8 \times 2 = 16$ |
| (4) $7 \times 1 = 7$ | (34) $1 \times 4 = 4$ | (64) $1 \times 8 = 8$ | (94) $8 \times 5 = 40$ |
| (5) $5 \times 4 = 20$ | (35) $2 \times 8 = 16$ | (65) $7 \times 2 = 14$ | (95) $1 \times 5 = 5$ |
| (6) $5 \times 7 = 35$ | (36) $3 \times 4 = 12$ | (66) $7 \times 7 = 49$ | (96) $1 \times 8 = 8$ |
| (7) $7 \times 3 = 21$ | (37) $0 \times 4 = 0$ | (67) $5 \times 10 = 50$ | (97) $9 \times 2 = 18$ |
| (8) $8 \times 1 = 8$ | (38) $4 \times 9 = 36$ | (68) $3 \times 4 = 12$ | (98) $3 \times 0 = 0$ |
| (9) $6 \times 6 = 36$ | (39) $1 \times 3 = 3$ | (69) $6 \times 1 = 6$ | (99) $8 \times 9 = 72$ |
| (10) $2 \times 10 = 20$ | (40) $9 \times 8 = 72$ | (70) $7 \times 2 = 14$ | (100) $5 \times 4 = 20$ |
| (11) $10 \times 7 = 70$ | (41) $7 \times 0 = 0$ | (71) $7 \times 3 = 21$ | (101) $1 \times 8 = 8$ |
| (12) $0 \times 10 = 0$ | (42) $2 \times 3 = 6$ | (72) $6 \times 8 = 48$ | (102) $1 \times 10 = 10$ |
| (13) $1 \times 6 = 6$ | (43) $10 \times 1 = 10$ | (73) $6 \times 2 = 12$ | (103) $1 \times 2 = 2$ |
| (14) $10 \times 7 = 70$ | (44) $4 \times 3 = 12$ | (74) $4 \times 6 = 24$ | (104) $4 \times 2 = 8$ |
| (15) $8 \times 5 = 40$ | (45) $10 \times 7 = 70$ | (75) $10 \times 5 = 50$ | (105) $5 \times 0 = 0$ |
| (16) $10 \times 8 = 80$ | (46) $9 \times 6 = 54$ | (76) $5 \times 3 = 15$ | (106) $7 \times 3 = 21$ |
| (17) $7 \times 10 = 70$ | (47) $3 \times 0 = 0$ | (77) $3 \times 8 = 24$ | (107) $9 \times 5 = 45$ |
| (18) $7 \times 9 = 63$ | (48) $2 \times 0 = 0$ | (78) $10 \times 2 = 20$ | (108) $0 \times 2 = 0$ |
| (19) $9 \times 0 = 0$ | (49) $7 \times 6 = 42$ | (79) $1 \times 8 = 8$ | (109) $7 \times 4 = 28$ |
| (20) $8 \times 2 = 16$ | (50) $4 \times 5 = 20$ | (80) $9 \times 7 = 63$ | (110) $8 \times 5 = 40$ |
| (21) $6 \times 4 = 24$ | (51) $5 \times 10 = 50$ | (81) $3 \times 9 = 27$ | (111) $1 \times 8 = 8$ |
| (22) $2 \times 0 = 0$ | (52) $1 \times 3 = 3$ | (82) $8 \times 7 = 56$ | (112) $6 \times 8 = 48$ |
| (23) $8 \times 10 = 80$ | (53) $7 \times 8 = 56$ | (83) $4 \times 5 = 20$ | (113) $7 \times 4 = 28$ |
| (24) $9 \times 5 = 45$ | (54) $4 \times 6 = 24$ | (84) $9 \times 3 = 27$ | (114) $5 \times 9 = 45$ |
| (25) $3 \times 9 = 27$ | (55) $1 \times 9 = 9$ | (85) $4 \times 5 = 20$ | (115) $2 \times 10 = 20$ |
| (26) $4 \times 10 = 40$ | (56) $2 \times 8 = 16$ | (86) $4 \times 0 = 0$ | (116) $2 \times 7 = 14$ |
| (27) $6 \times 9 = 54$ | (57) $1 \times 7 = 7$ | (87) $7 \times 2 = 14$ | (117) $9 \times 2 = 18$ |
| (28) $9 \times 10 = 90$ | (58) $3 \times 2 = 6$ | (88) $6 \times 7 = 42$ | (118) $9 \times 7 = 63$ |
| (29) $9 \times 9 = 81$ | (59) $10 \times 5 = 50$ | (89) $10 \times 9 = 90$ | (119) $6 \times 4 = 24$ |
| (30) $9 \times 2 = 18$ | (60) $0 \times 6 = 0$ | (90) $7 \times 2 = 14$ | (120) $1 \times 9 = 9$ |

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|-------------------------|-------------------------|-------------------------|-------------------------|
| (1) $4 \times 9 = 36$ | (31) $1 \times 4 = 4$ | (61) $10 \times 6 = 60$ | (91) $2 \times 2 = 4$ |
| (2) $1 \times 3 = 3$ | (32) $1 \times 6 = 6$ | (62) $1 \times 8 = 8$ | (92) $2 \times 3 = 6$ |
| (3) $7 \times 10 = 70$ | (33) $2 \times 10 = 20$ | (63) $7 \times 10 = 70$ | (93) $6 \times 8 = 48$ |
| (4) $7 \times 7 = 49$ | (34) $9 \times 10 = 90$ | (64) $4 \times 3 = 12$ | (94) $5 \times 4 = 20$ |
| (5) $2 \times 2 = 4$ | (35) $5 \times 4 = 20$ | (65) $4 \times 9 = 36$ | (95) $5 \times 8 = 40$ |
| (6) $6 \times 10 = 60$ | (36) $6 \times 2 = 12$ | (66) $8 \times 1 = 8$ | (96) $6 \times 3 = 18$ |
| (7) $2 \times 8 = 16$ | (37) $4 \times 7 = 28$ | (67) $6 \times 3 = 18$ | (97) $2 \times 1 = 2$ |
| (8) $1 \times 6 = 6$ | (38) $6 \times 1 = 6$ | (68) $10 \times 4 = 40$ | (98) $5 \times 2 = 10$ |
| (9) $2 \times 8 = 16$ | (39) $7 \times 2 = 14$ | (69) $6 \times 7 = 42$ | (99) $2 \times 4 = 8$ |
| (10) $4 \times 8 = 32$ | (40) $5 \times 9 = 45$ | (70) $1 \times 1 = 1$ | (100) $4 \times 4 = 16$ |
| (11) $8 \times 5 = 40$ | (41) $0 \times 7 = 0$ | (71) $2 \times 8 = 16$ | (101) $7 \times 0 = 0$ |
| (12) $8 \times 4 = 32$ | (42) $9 \times 2 = 18$ | (72) $1 \times 2 = 2$ | (102) $5 \times 3 = 15$ |
| (13) $3 \times 4 = 12$ | (43) $2 \times 8 = 16$ | (73) $9 \times 0 = 0$ | (103) $5 \times 2 = 10$ |
| (14) $6 \times 10 = 60$ | (44) $10 \times 7 = 70$ | (74) $10 \times 9 = 90$ | (104) $7 \times 6 = 42$ |
| (15) $6 \times 5 = 30$ | (45) $5 \times 1 = 5$ | (75) $7 \times 1 = 7$ | (105) $9 \times 1 = 9$ |
| (16) $7 \times 6 = 42$ | (46) $1 \times 4 = 4$ | (76) $9 \times 10 = 90$ | (106) $4 \times 7 = 28$ |
| (17) $8 \times 8 = 64$ | (47) $8 \times 4 = 32$ | (77) $2 \times 9 = 18$ | (107) $6 \times 2 = 12$ |
| (18) $0 \times 9 = 0$ | (48) $2 \times 6 = 12$ | (78) $8 \times 8 = 64$ | (108) $3 \times 3 = 9$ |
| (19) $0 \times 5 = 0$ | (49) $7 \times 9 = 63$ | (79) $2 \times 5 = 10$ | (109) $2 \times 7 = 14$ |
| (20) $2 \times 9 = 18$ | (50) $6 \times 7 = 42$ | (80) $9 \times 1 = 9$ | (110) $8 \times 3 = 24$ |
| (21) $0 \times 6 = 0$ | (51) $2 \times 8 = 16$ | (81) $10 \times 7 = 70$ | (111) $4 \times 1 = 4$ |
| (22) $10 \times 8 = 80$ | (52) $6 \times 5 = 30$ | (82) $1 \times 2 = 2$ | (112) $6 \times 7 = 42$ |
| (23) $5 \times 5 = 25$ | (53) $10 \times 6 = 60$ | (83) $4 \times 6 = 24$ | (113) $9 \times 1 = 9$ |
| (24) $3 \times 6 = 18$ | (54) $8 \times 7 = 56$ | (84) $2 \times 9 = 18$ | (114) $2 \times 3 = 6$ |
| (25) $0 \times 9 = 0$ | (55) $7 \times 2 = 14$ | (85) $3 \times 3 = 9$ | (115) $4 \times 0 = 0$ |
| (26) $0 \times 10 = 0$ | (56) $2 \times 9 = 18$ | (86) $10 \times 5 = 50$ | (116) $1 \times 5 = 5$ |
| (27) $10 \times 8 = 80$ | (57) $3 \times 3 = 9$ | (87) $8 \times 9 = 72$ | (117) $1 \times 1 = 1$ |
| (28) $2 \times 2 = 4$ | (58) $9 \times 2 = 18$ | (88) $8 \times 9 = 72$ | (118) $7 \times 7 = 49$ |
| (29) $2 \times 2 = 4$ | (59) $6 \times 4 = 24$ | (89) $3 \times 8 = 24$ | (119) $3 \times 3 = 9$ |
| (30) $0 \times 1 = 0$ | (60) $1 \times 9 = 9$ | (90) $7 \times 6 = 42$ | (120) $1 \times 6 = 6$ |

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|-------------------------|-------------------------|-------------------------|----------------------------|
| (1) $3 \times 8 = 24$ | (31) $5 \times 0 = 0$ | (61) $2 \times 7 = 14$ | (91) $7 \times 2 = 14$ |
| (2) $9 \times 4 = 36$ | (32) $1 \times 3 = 3$ | (62) $4 \times 2 = 8$ | (92) $10 \times 8 = 80$ |
| (3) $6 \times 8 = 48$ | (33) $3 \times 0 = 0$ | (63) $2 \times 9 = 18$ | (93) $8 \times 2 = 16$ |
| (4) $5 \times 1 = 5$ | (34) $2 \times 8 = 16$ | (64) $4 \times 3 = 12$ | (94) $3 \times 5 = 15$ |
| (5) $10 \times 5 = 50$ | (35) $1 \times 8 = 8$ | (65) $3 \times 7 = 21$ | (95) $1 \times 8 = 8$ |
| (6) $8 \times 8 = 64$ | (36) $2 \times 2 = 4$ | (66) $1 \times 3 = 3$ | (96) $3 \times 8 = 24$ |
| (7) $3 \times 1 = 3$ | (37) $10 \times 3 = 30$ | (67) $6 \times 3 = 18$ | (97) $8 \times 1 = 8$ |
| (8) $5 \times 1 = 5$ | (38) $5 \times 8 = 40$ | (68) $7 \times 6 = 42$ | (98) $3 \times 3 = 9$ |
| (9) $0 \times 3 = 0$ | (39) $8 \times 5 = 40$ | (69) $1 \times 10 = 10$ | (99) $7 \times 1 = 7$ |
| (10) $9 \times 8 = 72$ | (40) $7 \times 7 = 49$ | (70) $6 \times 9 = 54$ | (100) $10 \times 1 = 10$ |
| (11) $9 \times 2 = 18$ | (41) $10 \times 1 = 10$ | (71) $2 \times 8 = 16$ | (101) $1 \times 1 = 1$ |
| (12) $0 \times 10 = 0$ | (42) $4 \times 6 = 24$ | (72) $10 \times 3 = 30$ | (102) $10 \times 10 = 100$ |
| (13) $9 \times 8 = 72$ | (43) $8 \times 1 = 8$ | (73) $3 \times 4 = 12$ | (103) $4 \times 5 = 20$ |
| (14) $6 \times 1 = 6$ | (44) $8 \times 7 = 56$ | (74) $8 \times 7 = 56$ | (104) $10 \times 7 = 70$ |
| (15) $8 \times 3 = 24$ | (45) $3 \times 5 = 15$ | (75) $6 \times 10 = 60$ | (105) $5 \times 3 = 15$ |
| (16) $10 \times 7 = 70$ | (46) $1 \times 2 = 2$ | (76) $6 \times 4 = 24$ | (106) $9 \times 8 = 72$ |
| (17) $5 \times 7 = 35$ | (47) $3 \times 10 = 30$ | (77) $7 \times 1 = 7$ | (107) $9 \times 1 = 9$ |
| (18) $10 \times 5 = 50$ | (48) $2 \times 2 = 4$ | (78) $3 \times 1 = 3$ | (108) $5 \times 4 = 20$ |
| (19) $3 \times 6 = 18$ | (49) $4 \times 0 = 0$ | (79) $0 \times 7 = 0$ | (109) $0 \times 1 = 0$ |
| (20) $6 \times 5 = 30$ | (50) $7 \times 7 = 49$ | (80) $5 \times 1 = 5$ | (110) $3 \times 2 = 6$ |
| (21) $6 \times 9 = 54$ | (51) $8 \times 5 = 40$ | (81) $3 \times 9 = 27$ | (111) $9 \times 5 = 45$ |
| (22) $5 \times 9 = 45$ | (52) $5 \times 4 = 20$ | (82) $6 \times 8 = 48$ | (112) $5 \times 7 = 35$ |
| (23) $2 \times 3 = 6$ | (53) $9 \times 9 = 81$ | (83) $8 \times 5 = 40$ | (113) $0 \times 0 = 0$ |
| (24) $1 \times 2 = 2$ | (54) $1 \times 7 = 7$ | (84) $6 \times 8 = 48$ | (114) $6 \times 10 = 60$ |
| (25) $5 \times 8 = 40$ | (55) $8 \times 8 = 64$ | (85) $5 \times 7 = 35$ | (115) $7 \times 1 = 7$ |
| (26) $10 \times 2 = 20$ | (56) $9 \times 0 = 0$ | (86) $3 \times 1 = 3$ | (116) $1 \times 3 = 3$ |
| (27) $7 \times 0 = 0$ | (57) $1 \times 3 = 3$ | (87) $3 \times 1 = 3$ | (117) $1 \times 5 = 5$ |
| (28) $0 \times 4 = 0$ | (58) $6 \times 5 = 30$ | (88) $1 \times 8 = 8$ | (118) $8 \times 5 = 40$ |
| (29) $1 \times 1 = 1$ | (59) $9 \times 0 = 0$ | (89) $7 \times 3 = 21$ | (119) $0 \times 4 = 0$ |
| (30) $0 \times 2 = 0$ | (60) $8 \times 3 = 24$ | (90) $8 \times 2 = 16$ | (120) $3 \times 7 = 21$ |

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|-------------------------|-------------------------|---------------------------|--------------------------|
| (1) $2 \times 1 = 2$ | (31) $7 \times 0 = 0$ | (61) $8 \times 4 = 32$ | (91) $1 \times 1 = 1$ |
| (2) $9 \times 2 = 18$ | (32) $9 \times 4 = 36$ | (62) $3 \times 1 = 3$ | (92) $9 \times 2 = 18$ |
| (3) $0 \times 4 = 0$ | (33) $4 \times 7 = 28$ | (63) $5 \times 4 = 20$ | (93) $8 \times 3 = 24$ |
| (4) $3 \times 1 = 3$ | (34) $3 \times 7 = 21$ | (64) $10 \times 9 = 90$ | (94) $6 \times 2 = 12$ |
| (5) $8 \times 9 = 72$ | (35) $7 \times 9 = 63$ | (65) $3 \times 6 = 18$ | (95) $9 \times 6 = 54$ |
| (6) $6 \times 10 = 60$ | (36) $2 \times 1 = 2$ | (66) $10 \times 10 = 100$ | (96) $10 \times 0 = 0$ |
| (7) $2 \times 7 = 14$ | (37) $3 \times 10 = 30$ | (67) $0 \times 9 = 0$ | (97) $7 \times 3 = 21$ |
| (8) $7 \times 10 = 70$ | (38) $2 \times 8 = 16$ | (68) $9 \times 4 = 36$ | (98) $9 \times 5 = 45$ |
| (9) $7 \times 0 = 0$ | (39) $5 \times 2 = 10$ | (69) $2 \times 3 = 6$ | (99) $7 \times 9 = 63$ |
| (10) $4 \times 8 = 32$ | (40) $2 \times 2 = 4$ | (70) $5 \times 8 = 40$ | (100) $6 \times 0 = 0$ |
| (11) $6 \times 0 = 0$ | (41) $9 \times 8 = 72$ | (71) $3 \times 6 = 18$ | (101) $0 \times 7 = 0$ |
| (12) $9 \times 5 = 45$ | (42) $2 \times 1 = 2$ | (72) $10 \times 6 = 60$ | (102) $8 \times 1 = 8$ |
| (13) $6 \times 4 = 24$ | (43) $6 \times 7 = 42$ | (73) $0 \times 9 = 0$ | (103) $0 \times 6 = 0$ |
| (14) $6 \times 1 = 6$ | (44) $8 \times 4 = 32$ | (74) $5 \times 9 = 45$ | (104) $0 \times 9 = 0$ |
| (15) $3 \times 7 = 21$ | (45) $3 \times 5 = 15$ | (75) $5 \times 0 = 0$ | (105) $6 \times 2 = 12$ |
| (16) $3 \times 5 = 15$ | (46) $5 \times 1 = 5$ | (76) $5 \times 2 = 10$ | (106) $0 \times 9 = 0$ |
| (17) $10 \times 8 = 80$ | (47) $2 \times 7 = 14$ | (77) $2 \times 2 = 4$ | (107) $7 \times 3 = 21$ |
| (18) $9 \times 9 = 81$ | (48) $5 \times 4 = 20$ | (78) $1 \times 6 = 6$ | (108) $8 \times 9 = 72$ |
| (19) $5 \times 7 = 35$ | (49) $3 \times 9 = 27$ | (79) $6 \times 10 = 60$ | (109) $8 \times 1 = 8$ |
| (20) $0 \times 3 = 0$ | (50) $3 \times 7 = 21$ | (80) $10 \times 6 = 60$ | (110) $1 \times 9 = 9$ |
| (21) $10 \times 3 = 30$ | (51) $1 \times 4 = 4$ | (81) $9 \times 8 = 72$ | (111) $1 \times 4 = 4$ |
| (22) $10 \times 0 = 0$ | (52) $9 \times 9 = 81$ | (82) $4 \times 5 = 20$ | (112) $7 \times 0 = 0$ |
| (23) $5 \times 4 = 20$ | (53) $2 \times 2 = 4$ | (83) $5 \times 10 = 50$ | (113) $0 \times 4 = 0$ |
| (24) $7 \times 4 = 28$ | (54) $6 \times 2 = 12$ | (84) $8 \times 7 = 56$ | (114) $9 \times 10 = 90$ |
| (25) $2 \times 9 = 18$ | (55) $5 \times 2 = 10$ | (85) $6 \times 9 = 54$ | (115) $1 \times 4 = 4$ |
| (26) $7 \times 5 = 35$ | (56) $10 \times 8 = 80$ | (86) $10 \times 3 = 30$ | (116) $1 \times 10 = 10$ |
| (27) $4 \times 0 = 0$ | (57) $0 \times 4 = 0$ | (87) $4 \times 0 = 0$ | (117) $10 \times 1 = 10$ |
| (28) $9 \times 8 = 72$ | (58) $0 \times 9 = 0$ | (88) $1 \times 5 = 5$ | (118) $4 \times 4 = 16$ |
| (29) $7 \times 9 = 63$ | (59) $9 \times 3 = 27$ | (89) $10 \times 1 = 10$ | (119) $0 \times 8 = 0$ |
| (30) $9 \times 1 = 9$ | (60) $10 \times 5 = 50$ | (90) $8 \times 3 = 24$ | (120) $4 \times 1 = 4$ |

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|-------------------------|-------------------------|-------------------------|--------------------------|
| (1) $3 \times 5 = 15$ | (31) $1 \times 5 = 5$ | (61) $8 \times 7 = 56$ | (91) $9 \times 0 = 0$ |
| (2) $0 \times 7 = 0$ | (32) $2 \times 8 = 16$ | (62) $10 \times 9 = 90$ | (92) $9 \times 8 = 72$ |
| (3) $9 \times 7 = 63$ | (33) $3 \times 0 = 0$ | (63) $5 \times 5 = 25$ | (93) $2 \times 10 = 20$ |
| (4) $3 \times 0 = 0$ | (34) $3 \times 6 = 18$ | (64) $2 \times 10 = 20$ | (94) $1 \times 7 = 7$ |
| (5) $9 \times 1 = 9$ | (35) $4 \times 5 = 20$ | (65) $4 \times 1 = 4$ | (95) $1 \times 4 = 4$ |
| (6) $3 \times 9 = 27$ | (36) $8 \times 2 = 16$ | (66) $6 \times 10 = 60$ | (96) $7 \times 10 = 70$ |
| (7) $7 \times 9 = 63$ | (37) $8 \times 8 = 64$ | (67) $3 \times 7 = 21$ | (97) $2 \times 1 = 2$ |
| (8) $5 \times 3 = 15$ | (38) $7 \times 10 = 70$ | (68) $0 \times 1 = 0$ | (98) $7 \times 7 = 49$ |
| (9) $9 \times 10 = 90$ | (39) $6 \times 5 = 30$ | (69) $5 \times 2 = 10$ | (99) $10 \times 6 = 60$ |
| (10) $10 \times 9 = 90$ | (40) $10 \times 3 = 30$ | (70) $1 \times 5 = 5$ | (100) $0 \times 9 = 0$ |
| (11) $8 \times 2 = 16$ | (41) $0 \times 5 = 0$ | (71) $4 \times 4 = 16$ | (101) $3 \times 3 = 9$ |
| (12) $6 \times 7 = 42$ | (42) $5 \times 6 = 30$ | (72) $4 \times 5 = 20$ | (102) $8 \times 3 = 24$ |
| (13) $6 \times 9 = 54$ | (43) $10 \times 2 = 20$ | (73) $4 \times 9 = 36$ | (103) $8 \times 8 = 64$ |
| (14) $1 \times 6 = 6$ | (44) $8 \times 5 = 40$ | (74) $2 \times 3 = 6$ | (104) $10 \times 1 = 10$ |
| (15) $10 \times 2 = 20$ | (45) $6 \times 5 = 30$ | (75) $3 \times 4 = 12$ | (105) $3 \times 7 = 21$ |
| (16) $1 \times 8 = 8$ | (46) $5 \times 7 = 35$ | (76) $7 \times 8 = 56$ | (106) $3 \times 5 = 15$ |
| (17) $4 \times 5 = 20$ | (47) $2 \times 2 = 4$ | (77) $10 \times 3 = 30$ | (107) $10 \times 8 = 80$ |
| (18) $8 \times 9 = 72$ | (48) $5 \times 7 = 35$ | (78) $2 \times 4 = 8$ | (108) $1 \times 3 = 3$ |
| (19) $2 \times 5 = 10$ | (49) $9 \times 2 = 18$ | (79) $3 \times 6 = 18$ | (109) $1 \times 8 = 8$ |
| (20) $1 \times 2 = 2$ | (50) $4 \times 7 = 28$ | (80) $6 \times 3 = 18$ | (110) $1 \times 5 = 5$ |
| (21) $0 \times 9 = 0$ | (51) $10 \times 0 = 0$ | (81) $10 \times 1 = 10$ | (111) $6 \times 5 = 30$ |
| (22) $10 \times 1 = 10$ | (52) $7 \times 6 = 42$ | (82) $1 \times 6 = 6$ | (112) $1 \times 1 = 1$ |
| (23) $1 \times 3 = 3$ | (53) $6 \times 3 = 18$ | (83) $9 \times 6 = 54$ | (113) $6 \times 7 = 42$ |
| (24) $4 \times 6 = 24$ | (54) $10 \times 8 = 80$ | (84) $0 \times 5 = 0$ | (114) $1 \times 10 = 10$ |
| (25) $8 \times 9 = 72$ | (55) $1 \times 3 = 3$ | (85) $9 \times 4 = 36$ | (115) $2 \times 10 = 20$ |
| (26) $6 \times 0 = 0$ | (56) $9 \times 9 = 81$ | (86) $3 \times 8 = 24$ | (116) $8 \times 6 = 48$ |
| (27) $9 \times 8 = 72$ | (57) $4 \times 1 = 4$ | (87) $6 \times 10 = 60$ | (117) $9 \times 8 = 72$ |
| (28) $6 \times 6 = 36$ | (58) $6 \times 2 = 12$ | (88) $7 \times 3 = 21$ | (118) $1 \times 3 = 3$ |
| (29) $9 \times 7 = 63$ | (59) $6 \times 0 = 0$ | (89) $1 \times 3 = 3$ | (119) $7 \times 2 = 14$ |
| (30) $8 \times 5 = 40$ | (60) $9 \times 8 = 72$ | (90) $5 \times 3 = 15$ | (120) $8 \times 0 = 0$ |

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|---------------------------|-------------------------|-------------------------|--------------------------|
| (1) $2 \times 10 = 20$ | (31) $4 \times 8 = 32$ | (61) $9 \times 5 = 45$ | (91) $6 \times 8 = 48$ |
| (2) $7 \times 2 = 14$ | (32) $7 \times 3 = 21$ | (62) $7 \times 10 = 70$ | (92) $4 \times 9 = 36$ |
| (3) $2 \times 4 = 8$ | (33) $5 \times 1 = 5$ | (63) $10 \times 7 = 70$ | (93) $8 \times 1 = 8$ |
| (4) $6 \times 7 = 42$ | (34) $2 \times 9 = 18$ | (64) $2 \times 5 = 10$ | (94) $4 \times 1 = 4$ |
| (5) $1 \times 6 = 6$ | (35) $4 \times 5 = 20$ | (65) $3 \times 4 = 12$ | (95) $8 \times 5 = 40$ |
| (6) $2 \times 5 = 10$ | (36) $6 \times 6 = 36$ | (66) $6 \times 1 = 6$ | (96) $7 \times 0 = 0$ |
| (7) $10 \times 9 = 90$ | (37) $7 \times 5 = 35$ | (67) $3 \times 9 = 27$ | (97) $3 \times 1 = 3$ |
| (8) $6 \times 10 = 60$ | (38) $8 \times 9 = 72$ | (68) $10 \times 8 = 80$ | (98) $9 \times 8 = 72$ |
| (9) $1 \times 5 = 5$ | (39) $2 \times 0 = 0$ | (69) $6 \times 5 = 30$ | (99) $5 \times 4 = 20$ |
| (10) $2 \times 6 = 12$ | (40) $7 \times 2 = 14$ | (70) $0 \times 0 = 0$ | (100) $2 \times 1 = 2$ |
| (11) $8 \times 2 = 16$ | (41) $7 \times 3 = 21$ | (71) $4 \times 4 = 16$ | (101) $3 \times 10 = 30$ |
| (12) $9 \times 2 = 18$ | (42) $0 \times 4 = 0$ | (72) $2 \times 3 = 6$ | (102) $5 \times 4 = 20$ |
| (13) $10 \times 9 = 90$ | (43) $2 \times 8 = 16$ | (73) $2 \times 3 = 6$ | (103) $0 \times 9 = 0$ |
| (14) $7 \times 6 = 42$ | (44) $3 \times 5 = 15$ | (74) $4 \times 1 = 4$ | (104) $6 \times 1 = 6$ |
| (15) $0 \times 9 = 0$ | (45) $7 \times 2 = 14$ | (75) $2 \times 9 = 18$ | (105) $8 \times 10 = 80$ |
| (16) $10 \times 3 = 30$ | (46) $1 \times 8 = 8$ | (76) $2 \times 6 = 12$ | (106) $9 \times 8 = 72$ |
| (17) $2 \times 6 = 12$ | (47) $6 \times 2 = 12$ | (77) $4 \times 3 = 12$ | (107) $3 \times 8 = 24$ |
| (18) $5 \times 6 = 30$ | (48) $4 \times 7 = 28$ | (78) $9 \times 9 = 81$ | (108) $6 \times 9 = 54$ |
| (19) $9 \times 6 = 54$ | (49) $3 \times 9 = 27$ | (79) $0 \times 6 = 0$ | (109) $10 \times 6 = 60$ |
| (20) $9 \times 8 = 72$ | (50) $2 \times 5 = 10$ | (80) $9 \times 4 = 36$ | (110) $3 \times 6 = 18$ |
| (21) $9 \times 9 = 81$ | (51) $9 \times 4 = 36$ | (81) $4 \times 1 = 4$ | (111) $1 \times 4 = 4$ |
| (22) $1 \times 5 = 5$ | (52) $10 \times 7 = 70$ | (82) $9 \times 3 = 27$ | (112) $6 \times 6 = 36$ |
| (23) $6 \times 0 = 0$ | (53) $7 \times 2 = 14$ | (83) $6 \times 8 = 48$ | (113) $10 \times 0 = 0$ |
| (24) $6 \times 1 = 6$ | (54) $9 \times 1 = 9$ | (84) $8 \times 0 = 0$ | (114) $8 \times 8 = 64$ |
| (25) $10 \times 10 = 100$ | (55) $7 \times 2 = 14$ | (85) $6 \times 1 = 6$ | (115) $0 \times 9 = 0$ |
| (26) $9 \times 7 = 63$ | (56) $4 \times 9 = 36$ | (86) $6 \times 6 = 36$ | (116) $1 \times 1 = 1$ |
| (27) $10 \times 2 = 20$ | (57) $4 \times 0 = 0$ | (87) $9 \times 1 = 9$ | (117) $1 \times 5 = 5$ |
| (28) $9 \times 10 = 90$ | (58) $8 \times 4 = 32$ | (88) $1 \times 1 = 1$ | (118) $1 \times 9 = 9$ |
| (29) $4 \times 5 = 20$ | (59) $4 \times 8 = 32$ | (89) $2 \times 4 = 8$ | (119) $0 \times 0 = 0$ |
| (30) $6 \times 6 = 36$ | (60) $2 \times 4 = 8$ | (90) $7 \times 7 = 49$ | (120) $10 \times 3 = 30$ |

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|--------------------------|---------------------------|---------------------------|--------------------------|
| (1) $4 \times 6 = 24$ | (31) $9 \times 5 = 45$ | (61) $10 \times 7 = 70$ | (91) $0 \times 1 = 0$ |
| (2) $9 \times 7 = 63$ | (32) $9 \times 6 = 54$ | (62) $2 \times 0 = 0$ | (92) $7 \times 2 = 14$ |
| (3) $5 \times 9 = 45$ | (33) $1 \times 4 = 4$ | (63) $5 \times 3 = 15$ | (93) $3 \times 8 = 24$ |
| (4) $10 \times 10 = 100$ | (34) $8 \times 0 = 0$ | (64) $10 \times 6 = 60$ | (94) $7 \times 3 = 21$ |
| (5) $2 \times 2 = 4$ | (35) $10 \times 8 = 80$ | (65) $8 \times 8 = 64$ | (95) $0 \times 7 = 0$ |
| (6) $2 \times 6 = 12$ | (36) $2 \times 7 = 14$ | (66) $10 \times 8 = 80$ | (96) $6 \times 9 = 54$ |
| (7) $2 \times 9 = 18$ | (37) $4 \times 2 = 8$ | (67) $5 \times 6 = 30$ | (97) $1 \times 1 = 1$ |
| (8) $2 \times 0 = 0$ | (38) $5 \times 3 = 15$ | (68) $2 \times 2 = 4$ | (98) $2 \times 5 = 10$ |
| (9) $4 \times 4 = 16$ | (39) $3 \times 2 = 6$ | (69) $7 \times 7 = 49$ | (99) $7 \times 8 = 56$ |
| (10) $10 \times 9 = 90$ | (40) $10 \times 8 = 80$ | (70) $6 \times 7 = 42$ | (100) $7 \times 1 = 7$ |
| (11) $6 \times 4 = 24$ | (41) $10 \times 2 = 20$ | (71) $1 \times 0 = 0$ | (101) $10 \times 6 = 60$ |
| (12) $6 \times 1 = 6$ | (42) $4 \times 2 = 8$ | (72) $10 \times 2 = 20$ | (102) $8 \times 6 = 48$ |
| (13) $10 \times 8 = 80$ | (43) $10 \times 10 = 100$ | (73) $7 \times 8 = 56$ | (103) $2 \times 6 = 12$ |
| (14) $3 \times 4 = 12$ | (44) $5 \times 4 = 20$ | (74) $4 \times 7 = 28$ | (104) $6 \times 9 = 54$ |
| (15) $8 \times 0 = 0$ | (45) $3 \times 7 = 21$ | (75) $4 \times 7 = 28$ | (105) $7 \times 6 = 42$ |
| (16) $10 \times 4 = 40$ | (46) $2 \times 2 = 4$ | (76) $2 \times 1 = 2$ | (106) $2 \times 10 = 20$ |
| (17) $0 \times 8 = 0$ | (47) $9 \times 0 = 0$ | (77) $7 \times 2 = 14$ | (107) $9 \times 8 = 72$ |
| (18) $8 \times 3 = 24$ | (48) $2 \times 3 = 6$ | (78) $1 \times 1 = 1$ | (108) $6 \times 6 = 36$ |
| (19) $9 \times 3 = 27$ | (49) $5 \times 7 = 35$ | (79) $7 \times 6 = 42$ | (109) $6 \times 5 = 30$ |
| (20) $6 \times 7 = 42$ | (50) $4 \times 5 = 20$ | (80) $8 \times 7 = 56$ | (110) $7 \times 8 = 56$ |
| (21) $9 \times 2 = 18$ | (51) $4 \times 1 = 4$ | (81) $8 \times 8 = 64$ | (111) $6 \times 7 = 42$ |
| (22) $10 \times 3 = 30$ | (52) $0 \times 9 = 0$ | (82) $4 \times 8 = 32$ | (112) $3 \times 5 = 15$ |
| (23) $9 \times 6 = 54$ | (53) $1 \times 8 = 8$ | (83) $9 \times 9 = 81$ | (113) $4 \times 4 = 16$ |
| (24) $10 \times 1 = 10$ | (54) $1 \times 0 = 0$ | (84) $1 \times 2 = 2$ | (114) $9 \times 9 = 81$ |
| (25) $6 \times 0 = 0$ | (55) $7 \times 4 = 28$ | (85) $10 \times 10 = 100$ | (115) $7 \times 10 = 70$ |
| (26) $3 \times 3 = 9$ | (56) $4 \times 6 = 24$ | (86) $1 \times 9 = 9$ | (116) $6 \times 7 = 42$ |
| (27) $3 \times 10 = 30$ | (57) $3 \times 6 = 18$ | (87) $9 \times 6 = 54$ | (117) $6 \times 9 = 54$ |
| (28) $10 \times 1 = 10$ | (58) $3 \times 6 = 18$ | (88) $1 \times 10 = 10$ | (118) $1 \times 7 = 7$ |
| (29) $10 \times 1 = 10$ | (59) $9 \times 4 = 36$ | (89) $1 \times 2 = 2$ | (119) $1 \times 9 = 9$ |
| (30) $5 \times 5 = 25$ | (60) $9 \times 2 = 18$ | (90) $7 \times 4 = 28$ | (120) $10 \times 3 = 30$ |

(1) $6 \times 4 = 24$	(31) $6 \times 2 = 12$	(61) $2 \times 9 = 18$	(91) $0 \times 1 = 0$
(2) $5 \times 2 = 10$	(32) $1 \times 8 = 8$	(62) $7 \times 7 = 49$	(92) $5 \times 10 = 50$
(3) $8 \times 8 = 64$	(33) $5 \times 5 = 25$	(63) $9 \times 4 = 36$	(93) $6 \times 7 = 42$
(4) $2 \times 9 = 18$	(34) $7 \times 10 = 70$	(64) $0 \times 9 = 0$	(94) $3 \times 6 = 18$
(5) $6 \times 8 = 48$	(35) $3 \times 6 = 18$	(65) $8 \times 5 = 40$	(95) $10 \times 3 = 30$
(6) $1 \times 1 = 1$	(36) $3 \times 5 = 15$	(66) $9 \times 1 = 9$	(96) $3 \times 2 = 6$
(7) $4 \times 1 = 4$	(37) $7 \times 2 = 14$	(67) $8 \times 9 = 72$	(97) $2 \times 4 = 8$
(8) $2 \times 6 = 12$	(38) $4 \times 3 = 12$	(68) $5 \times 9 = 45$	(98) $1 \times 2 = 2$
(9) $4 \times 10 = 40$	(39) $5 \times 7 = 35$	(69) $5 \times 10 = 50$	(99) $3 \times 8 = 24$
(10) $7 \times 6 = 42$	(40) $1 \times 6 = 6$	(70) $3 \times 7 = 21$	(100) $3 \times 6 = 18$
(11) $2 \times 5 = 10$	(41) $6 \times 9 = 54$	(71) $2 \times 1 = 2$	(101) $10 \times 0 = 0$
(12) $8 \times 4 = 32$	(42) $2 \times 5 = 10$	(72) $6 \times 6 = 36$	(102) $8 \times 4 = 32$
(13) $2 \times 5 = 10$	(43) $2 \times 4 = 8$	(73) $5 \times 3 = 15$	(103) $4 \times 1 = 4$
(14) $2 \times 9 = 18$	(44) $9 \times 0 = 0$	(74) $1 \times 5 = 5$	(104) $9 \times 6 = 54$
(15) $8 \times 9 = 72$	(45) $9 \times 10 = 90$	(75) $0 \times 5 = 0$	(105) $5 \times 10 = 50$
(16) $10 \times 9 = 90$	(46) $2 \times 9 = 18$	(76) $8 \times 8 = 64$	(106) $4 \times 2 = 8$
(17) $1 \times 9 = 9$	(47) $4 \times 9 = 36$	(77) $4 \times 8 = 32$	(107) $0 \times 1 = 0$
(18) $6 \times 6 = 36$	(48) $1 \times 5 = 5$	(78) $7 \times 8 = 56$	(108) $9 \times 10 = 90$
(19) $7 \times 8 = 56$	(49) $4 \times 9 = 36$	(79) $1 \times 3 = 3$	(109) $2 \times 4 = 8$
(20) $4 \times 2 = 8$	(50) $1 \times 1 = 1$	(80) $2 \times 7 = 14$	(110) $5 \times 6 = 30$
(21) $5 \times 7 = 35$	(51) $9 \times 8 = 72$	(81) $9 \times 1 = 9$	(111) $6 \times 10 = 60$
(22) $9 \times 2 = 18$	(52) $10 \times 6 = 60$	(82) $8 \times 6 = 48$	(112) $3 \times 6 = 18$
(23) $2 \times 6 = 12$	(53) $2 \times 8 = 16$	(83) $3 \times 8 = 24$	(113) $10 \times 9 = 90$
(24) $1 \times 9 = 9$	(54) $3 \times 4 = 12$	(84) $6 \times 9 = 54$	(114) $5 \times 0 = 0$
(25) $3 \times 4 = 12$	(55) $1 \times 6 = 6$	(85) $1 \times 7 = 7$	(115) $6 \times 1 = 6$
(26) $1 \times 4 = 4$	(56) $8 \times 9 = 72$	(86) $0 \times 2 = 0$	(116) $3 \times 2 = 6$
(27) $7 \times 2 = 14$	(57) $0 \times 3 = 0$	(87) $2 \times 5 = 10$	(117) $9 \times 1 = 9$
(28) $3 \times 1 = 3$	(58) $8 \times 9 = 72$	(88) $10 \times 8 = 80$	(118) $7 \times 7 = 49$
(29) $8 \times 2 = 16$	(59) $4 \times 6 = 24$	(89) $6 \times 0 = 0$	(119) $2 \times 3 = 6$
(30) $3 \times 5 = 15$	(60) $0 \times 4 = 0$	(90) $9 \times 7 = 63$	(120) $2 \times 2 = 4$

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|-------------------------|-------------------------|-------------------------|--------------------------|
| (1) $6 \times 3 = 18$ | (31) $4 \times 4 = 16$ | (61) $1 \times 1 = 1$ | (91) $2 \times 7 = 14$ |
| (2) $7 \times 9 = 63$ | (32) $7 \times 8 = 56$ | (62) $5 \times 4 = 20$ | (92) $3 \times 3 = 9$ |
| (3) $3 \times 5 = 15$ | (33) $4 \times 9 = 36$ | (63) $7 \times 4 = 28$ | (93) $3 \times 1 = 3$ |
| (4) $5 \times 1 = 5$ | (34) $7 \times 10 = 70$ | (64) $10 \times 7 = 70$ | (94) $5 \times 9 = 45$ |
| (5) $9 \times 5 = 45$ | (35) $1 \times 5 = 5$ | (65) $4 \times 4 = 16$ | (95) $7 \times 8 = 56$ |
| (6) $1 \times 3 = 3$ | (36) $4 \times 6 = 24$ | (66) $2 \times 2 = 4$ | (96) $5 \times 1 = 5$ |
| (7) $4 \times 5 = 20$ | (37) $9 \times 2 = 18$ | (67) $2 \times 3 = 6$ | (97) $10 \times 7 = 70$ |
| (8) $9 \times 6 = 54$ | (38) $6 \times 0 = 0$ | (68) $8 \times 9 = 72$ | (98) $2 \times 2 = 4$ |
| (9) $0 \times 0 = 0$ | (39) $2 \times 5 = 10$ | (69) $4 \times 7 = 28$ | (99) $7 \times 6 = 42$ |
| (10) $10 \times 1 = 10$ | (40) $5 \times 4 = 20$ | (70) $0 \times 6 = 0$ | (100) $10 \times 0 = 0$ |
| (11) $7 \times 5 = 35$ | (41) $3 \times 4 = 12$ | (71) $5 \times 4 = 20$ | (101) $10 \times 5 = 50$ |
| (12) $4 \times 10 = 40$ | (42) $0 \times 10 = 0$ | (72) $1 \times 2 = 2$ | (102) $9 \times 8 = 72$ |
| (13) $0 \times 3 = 0$ | (43) $9 \times 1 = 9$ | (73) $5 \times 4 = 20$ | (103) $1 \times 8 = 8$ |
| (14) $5 \times 5 = 25$ | (44) $6 \times 6 = 36$ | (74) $5 \times 9 = 45$ | (104) $3 \times 1 = 3$ |
| (15) $3 \times 7 = 21$ | (45) $5 \times 3 = 15$ | (75) $4 \times 2 = 8$ | (105) $1 \times 6 = 6$ |
| (16) $4 \times 4 = 16$ | (46) $0 \times 10 = 0$ | (76) $8 \times 6 = 48$ | (106) $5 \times 2 = 10$ |
| (17) $10 \times 7 = 70$ | (47) $9 \times 3 = 27$ | (77) $5 \times 6 = 30$ | (107) $2 \times 9 = 18$ |
| (18) $1 \times 9 = 9$ | (48) $8 \times 1 = 8$ | (78) $0 \times 8 = 0$ | (108) $9 \times 3 = 27$ |
| (19) $1 \times 8 = 8$ | (49) $3 \times 6 = 18$ | (79) $10 \times 3 = 30$ | (109) $4 \times 10 = 40$ |
| (20) $8 \times 4 = 32$ | (50) $10 \times 8 = 80$ | (80) $8 \times 7 = 56$ | (110) $1 \times 2 = 2$ |
| (21) $3 \times 3 = 9$ | (51) $2 \times 5 = 10$ | (81) $0 \times 6 = 0$ | (111) $3 \times 5 = 15$ |
| (22) $7 \times 1 = 7$ | (52) $8 \times 2 = 16$ | (82) $8 \times 7 = 56$ | (112) $2 \times 7 = 14$ |
| (23) $8 \times 9 = 72$ | (53) $8 \times 9 = 72$ | (83) $8 \times 7 = 56$ | (113) $6 \times 1 = 6$ |
| (24) $10 \times 3 = 30$ | (54) $1 \times 2 = 2$ | (84) $1 \times 6 = 6$ | (114) $9 \times 2 = 18$ |
| (25) $10 \times 1 = 10$ | (55) $6 \times 10 = 60$ | (85) $0 \times 8 = 0$ | (115) $4 \times 5 = 20$ |
| (26) $5 \times 5 = 25$ | (56) $2 \times 9 = 18$ | (86) $7 \times 4 = 28$ | (116) $4 \times 8 = 32$ |
| (27) $8 \times 6 = 48$ | (57) $9 \times 3 = 27$ | (87) $9 \times 9 = 81$ | (117) $7 \times 6 = 42$ |
| (28) $6 \times 1 = 6$ | (58) $4 \times 2 = 8$ | (88) $9 \times 9 = 81$ | (118) $7 \times 1 = 7$ |
| (29) $4 \times 2 = 8$ | (59) $5 \times 10 = 50$ | (89) $10 \times 9 = 90$ | (119) $2 \times 10 = 20$ |
| (30) $10 \times 2 = 20$ | (60) $9 \times 10 = 90$ | (90) $8 \times 3 = 24$ | (120) $6 \times 8 = 48$ |

(1) $4 \times 4 = 16$	(31) $7 \times 7 = 49$	(61) $8 \times 7 = 56$	(91) $9 \times 5 = 45$
(2) $4 \times 9 = 36$	(32) $8 \times 1 = 8$	(62) $8 \times 1 = 8$	(92) $4 \times 1 = 4$
(3) $0 \times 4 = 0$	(33) $0 \times 4 = 0$	(63) $5 \times 8 = 40$	(93) $7 \times 10 = 70$
(4) $8 \times 9 = 72$	(34) $6 \times 4 = 24$	(64) $2 \times 5 = 10$	(94) $1 \times 2 = 2$
(5) $0 \times 10 = 0$	(35) $0 \times 2 = 0$	(65) $0 \times 0 = 0$	(95) $1 \times 0 = 0$
(6) $6 \times 6 = 36$	(36) $6 \times 0 = 0$	(66) $3 \times 3 = 9$	(96) $2 \times 5 = 10$
(7) $8 \times 9 = 72$	(37) $10 \times 7 = 70$	(67) $9 \times 9 = 81$	(97) $0 \times 7 = 0$
(8) $5 \times 4 = 20$	(38) $5 \times 3 = 15$	(68) $9 \times 4 = 36$	(98) $6 \times 2 = 12$
(9) $3 \times 10 = 30$	(39) $5 \times 7 = 35$	(69) $4 \times 10 = 40$	(99) $1 \times 9 = 9$
(10) $3 \times 10 = 30$	(40) $5 \times 2 = 10$	(70) $6 \times 7 = 42$	(100) $7 \times 9 = 63$
(11) $0 \times 8 = 0$	(41) $10 \times 1 = 10$	(71) $2 \times 4 = 8$	(101) $10 \times 3 = 30$
(12) $7 \times 10 = 70$	(42) $6 \times 9 = 54$	(72) $10 \times 6 = 60$	(102) $0 \times 10 = 0$
(13) $7 \times 9 = 63$	(43) $8 \times 1 = 8$	(73) $5 \times 7 = 35$	(103) $7 \times 7 = 49$
(14) $3 \times 2 = 6$	(44) $7 \times 6 = 42$	(74) $10 \times 6 = 60$	(104) $1 \times 6 = 6$
(15) $9 \times 0 = 0$	(45) $0 \times 9 = 0$	(75) $10 \times 1 = 10$	(105) $4 \times 5 = 20$
(16) $7 \times 4 = 28$	(46) $7 \times 7 = 49$	(76) $2 \times 7 = 14$	(106) $7 \times 1 = 7$
(17) $4 \times 2 = 8$	(47) $10 \times 7 = 70$	(77) $3 \times 2 = 6$	(107) $6 \times 6 = 36$
(18) $10 \times 10 = 100$	(48) $1 \times 3 = 3$	(78) $0 \times 9 = 0$	(108) $5 \times 9 = 45$
(19) $6 \times 10 = 60$	(49) $10 \times 7 = 70$	(79) $8 \times 10 = 80$	(109) $1 \times 4 = 4$
(20) $1 \times 9 = 9$	(50) $2 \times 6 = 12$	(80) $5 \times 4 = 20$	(110) $5 \times 8 = 40$
(21) $10 \times 2 = 20$	(51) $7 \times 6 = 42$	(81) $8 \times 9 = 72$	(111) $1 \times 8 = 8$
(22) $3 \times 1 = 3$	(52) $3 \times 8 = 24$	(82) $9 \times 4 = 36$	(112) $7 \times 3 = 21$
(23) $6 \times 7 = 42$	(53) $5 \times 1 = 5$	(83) $6 \times 8 = 48$	(113) $9 \times 9 = 81$
(24) $3 \times 2 = 6$	(54) $7 \times 4 = 28$	(84) $9 \times 7 = 63$	(114) $9 \times 2 = 18$
(25) $9 \times 5 = 45$	(55) $1 \times 1 = 1$	(85) $1 \times 1 = 1$	(115) $4 \times 10 = 40$
(26) $4 \times 3 = 12$	(56) $10 \times 10 = 100$	(86) $9 \times 8 = 72$	(116) $10 \times 6 = 60$
(27) $2 \times 7 = 14$	(57) $8 \times 1 = 8$	(87) $7 \times 9 = 63$	(117) $10 \times 3 = 30$
(28) $9 \times 1 = 9$	(58) $7 \times 3 = 21$	(88) $4 \times 9 = 36$	(118) $8 \times 2 = 16$
(29) $7 \times 8 = 56$	(59) $3 \times 7 = 21$	(89) $1 \times 7 = 7$	(119) $3 \times 0 = 0$
(30) $10 \times 8 = 80$	(60) $8 \times 8 = 64$	(90) $7 \times 3 = 21$	(120) $2 \times 9 = 18$

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|-------------------------|-------------------------|---------------------------|--------------------------|
| (1) $10 \times 4 = 40$ | (31) $10 \times 1 = 10$ | (61) $7 \times 6 = 42$ | (91) $3 \times 3 = 9$ |
| (2) $4 \times 1 = 4$ | (32) $7 \times 9 = 63$ | (62) $8 \times 3 = 24$ | (92) $3 \times 8 = 24$ |
| (3) $9 \times 9 = 81$ | (33) $4 \times 8 = 32$ | (63) $8 \times 7 = 56$ | (93) $10 \times 7 = 70$ |
| (4) $10 \times 4 = 40$ | (34) $4 \times 2 = 8$ | (64) $10 \times 9 = 90$ | (94) $3 \times 7 = 21$ |
| (5) $3 \times 2 = 6$ | (35) $3 \times 5 = 15$ | (65) $10 \times 6 = 60$ | (95) $3 \times 5 = 15$ |
| (6) $5 \times 6 = 30$ | (36) $1 \times 3 = 3$ | (66) $8 \times 10 = 80$ | (96) $7 \times 10 = 70$ |
| (7) $1 \times 5 = 5$ | (37) $7 \times 7 = 49$ | (67) $9 \times 10 = 90$ | (97) $8 \times 7 = 56$ |
| (8) $7 \times 0 = 0$ | (38) $8 \times 2 = 16$ | (68) $8 \times 1 = 8$ | (98) $10 \times 6 = 60$ |
| (9) $6 \times 6 = 36$ | (39) $10 \times 3 = 30$ | (69) $5 \times 10 = 50$ | (99) $3 \times 5 = 15$ |
| (10) $9 \times 3 = 27$ | (40) $6 \times 6 = 36$ | (70) $5 \times 9 = 45$ | (100) $9 \times 1 = 9$ |
| (11) $2 \times 8 = 16$ | (41) $6 \times 1 = 6$ | (71) $8 \times 5 = 40$ | (101) $3 \times 9 = 27$ |
| (12) $6 \times 1 = 6$ | (42) $1 \times 8 = 8$ | (72) $1 \times 6 = 6$ | (102) $7 \times 6 = 42$ |
| (13) $9 \times 0 = 0$ | (43) $3 \times 1 = 3$ | (73) $2 \times 3 = 6$ | (103) $10 \times 7 = 70$ |
| (14) $9 \times 6 = 54$ | (44) $2 \times 0 = 0$ | (74) $4 \times 7 = 28$ | (104) $0 \times 4 = 0$ |
| (15) $1 \times 10 = 10$ | (45) $2 \times 3 = 6$ | (75) $4 \times 3 = 12$ | (105) $8 \times 2 = 16$ |
| (16) $2 \times 9 = 18$ | (46) $10 \times 8 = 80$ | (76) $8 \times 3 = 24$ | (106) $2 \times 10 = 20$ |
| (17) $3 \times 9 = 27$ | (47) $0 \times 3 = 0$ | (77) $2 \times 5 = 10$ | (107) $2 \times 1 = 2$ |
| (18) $6 \times 4 = 24$ | (48) $10 \times 9 = 90$ | (78) $4 \times 7 = 28$ | (108) $10 \times 0 = 0$ |
| (19) $6 \times 9 = 54$ | (49) $1 \times 5 = 5$ | (79) $9 \times 9 = 81$ | (109) $10 \times 1 = 10$ |
| (20) $5 \times 4 = 20$ | (50) $8 \times 7 = 56$ | (80) $0 \times 7 = 0$ | (110) $5 \times 2 = 10$ |
| (21) $5 \times 2 = 10$ | (51) $5 \times 6 = 30$ | (81) $0 \times 7 = 0$ | (111) $1 \times 7 = 7$ |
| (22) $7 \times 2 = 14$ | (52) $8 \times 4 = 32$ | (82) $4 \times 9 = 36$ | (112) $9 \times 1 = 9$ |
| (23) $2 \times 10 = 20$ | (53) $4 \times 5 = 20$ | (83) $10 \times 10 = 100$ | (113) $0 \times 6 = 0$ |
| (24) $10 \times 3 = 30$ | (54) $10 \times 3 = 30$ | (84) $7 \times 8 = 56$ | (114) $7 \times 10 = 70$ |
| (25) $5 \times 3 = 15$ | (55) $3 \times 1 = 3$ | (85) $4 \times 1 = 4$ | (115) $4 \times 10 = 40$ |
| (26) $2 \times 2 = 4$ | (56) $1 \times 2 = 2$ | (86) $2 \times 1 = 2$ | (116) $5 \times 10 = 50$ |
| (27) $2 \times 4 = 8$ | (57) $4 \times 0 = 0$ | (87) $5 \times 9 = 45$ | (117) $9 \times 9 = 81$ |
| (28) $8 \times 4 = 32$ | (58) $3 \times 3 = 9$ | (88) $8 \times 9 = 72$ | (118) $9 \times 2 = 18$ |
| (29) $9 \times 4 = 36$ | (59) $1 \times 5 = 5$ | (89) $4 \times 3 = 12$ | (119) $10 \times 0 = 0$ |
| (30) $4 \times 9 = 36$ | (60) $6 \times 3 = 18$ | (90) $10 \times 8 = 80$ | (120) $9 \times 6 = 54$ |

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|--------------------------|-------------------------|-------------------------|--------------------------|
| (1) $2 \times 8 = 16$ | (31) $6 \times 2 = 12$ | (61) $0 \times 1 = 0$ | (91) $9 \times 5 = 45$ |
| (2) $2 \times 10 = 20$ | (32) $3 \times 1 = 3$ | (62) $2 \times 1 = 2$ | (92) $3 \times 5 = 15$ |
| (3) $6 \times 8 = 48$ | (33) $7 \times 7 = 49$ | (63) $4 \times 8 = 32$ | (93) $1 \times 0 = 0$ |
| (4) $10 \times 10 = 100$ | (34) $9 \times 3 = 27$ | (64) $4 \times 1 = 4$ | (94) $2 \times 3 = 6$ |
| (5) $7 \times 1 = 7$ | (35) $9 \times 0 = 0$ | (65) $3 \times 7 = 21$ | (95) $3 \times 10 = 30$ |
| (6) $6 \times 5 = 30$ | (36) $1 \times 6 = 6$ | (66) $2 \times 3 = 6$ | (96) $9 \times 10 = 90$ |
| (7) $8 \times 8 = 64$ | (37) $1 \times 6 = 6$ | (67) $0 \times 7 = 0$ | (97) $8 \times 1 = 8$ |
| (8) $9 \times 5 = 45$ | (38) $10 \times 0 = 0$ | (68) $5 \times 2 = 10$ | (98) $1 \times 4 = 4$ |
| (9) $8 \times 7 = 56$ | (39) $3 \times 2 = 6$ | (69) $6 \times 2 = 12$ | (99) $2 \times 4 = 8$ |
| (10) $9 \times 2 = 18$ | (40) $2 \times 6 = 12$ | (70) $1 \times 7 = 7$ | (100) $6 \times 4 = 24$ |
| (11) $3 \times 0 = 0$ | (41) $5 \times 4 = 20$ | (71) $2 \times 2 = 4$ | (101) $4 \times 4 = 16$ |
| (12) $4 \times 4 = 16$ | (42) $3 \times 6 = 18$ | (72) $2 \times 3 = 6$ | (102) $1 \times 6 = 6$ |
| (13) $7 \times 1 = 7$ | (43) $2 \times 8 = 16$ | (73) $6 \times 5 = 30$ | (103) $3 \times 4 = 12$ |
| (14) $5 \times 0 = 0$ | (44) $8 \times 10 = 80$ | (74) $6 \times 8 = 48$ | (104) $6 \times 3 = 18$ |
| (15) $5 \times 9 = 45$ | (45) $8 \times 4 = 32$ | (75) $9 \times 2 = 18$ | (105) $5 \times 3 = 15$ |
| (16) $1 \times 0 = 0$ | (46) $8 \times 5 = 40$ | (76) $6 \times 1 = 6$ | (106) $10 \times 4 = 40$ |
| (17) $3 \times 7 = 21$ | (47) $4 \times 7 = 28$ | (77) $1 \times 6 = 6$ | (107) $8 \times 4 = 32$ |
| (18) $6 \times 7 = 42$ | (48) $6 \times 6 = 36$ | (78) $1 \times 8 = 8$ | (108) $7 \times 3 = 21$ |
| (19) $6 \times 1 = 6$ | (49) $7 \times 9 = 63$ | (79) $8 \times 0 = 0$ | (109) $8 \times 3 = 24$ |
| (20) $2 \times 3 = 6$ | (50) $10 \times 9 = 90$ | (80) $7 \times 1 = 7$ | (110) $4 \times 4 = 16$ |
| (21) $0 \times 5 = 0$ | (51) $1 \times 3 = 3$ | (81) $8 \times 2 = 16$ | (111) $3 \times 8 = 24$ |
| (22) $7 \times 9 = 63$ | (52) $0 \times 6 = 0$ | (82) $7 \times 9 = 63$ | (112) $2 \times 8 = 16$ |
| (23) $10 \times 9 = 90$ | (53) $1 \times 10 = 10$ | (83) $0 \times 9 = 0$ | (113) $1 \times 1 = 1$ |
| (24) $4 \times 8 = 32$ | (54) $10 \times 4 = 40$ | (84) $4 \times 1 = 4$ | (114) $5 \times 1 = 5$ |
| (25) $10 \times 2 = 20$ | (55) $8 \times 8 = 64$ | (85) $7 \times 1 = 7$ | (115) $9 \times 1 = 9$ |
| (26) $6 \times 3 = 18$ | (56) $8 \times 4 = 32$ | (86) $2 \times 6 = 12$ | (116) $2 \times 1 = 2$ |
| (27) $8 \times 4 = 32$ | (57) $1 \times 8 = 8$ | (87) $4 \times 1 = 4$ | (117) $2 \times 1 = 2$ |
| (28) $1 \times 1 = 1$ | (58) $6 \times 5 = 30$ | (88) $3 \times 4 = 12$ | (118) $2 \times 8 = 16$ |
| (29) $0 \times 6 = 0$ | (59) $10 \times 6 = 60$ | (89) $9 \times 10 = 90$ | (119) $0 \times 2 = 0$ |
| (30) $4 \times 10 = 40$ | (60) $0 \times 7 = 0$ | (90) $9 \times 10 = 90$ | (120) $4 \times 4 = 16$ |

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|---------------------------|-------------------------|-------------------------|--------------------------|
| (1) $1 \times 4 = 4$ | (31) $10 \times 8 = 80$ | (61) $7 \times 8 = 56$ | (91) $9 \times 5 = 45$ |
| (2) $2 \times 9 = 18$ | (32) $5 \times 0 = 0$ | (62) $8 \times 8 = 64$ | (92) $4 \times 4 = 16$ |
| (3) $2 \times 2 = 4$ | (33) $9 \times 4 = 36$ | (63) $1 \times 3 = 3$ | (93) $10 \times 9 = 90$ |
| (4) $10 \times 8 = 80$ | (34) $4 \times 3 = 12$ | (64) $0 \times 1 = 0$ | (94) $2 \times 2 = 4$ |
| (5) $5 \times 6 = 30$ | (35) $7 \times 5 = 35$ | (65) $2 \times 3 = 6$ | (95) $1 \times 1 = 1$ |
| (6) $6 \times 6 = 36$ | (36) $1 \times 3 = 3$ | (66) $2 \times 2 = 4$ | (96) $8 \times 5 = 40$ |
| (7) $7 \times 2 = 14$ | (37) $7 \times 9 = 63$ | (67) $1 \times 3 = 3$ | (97) $7 \times 5 = 35$ |
| (8) $0 \times 7 = 0$ | (38) $4 \times 2 = 8$ | (68) $9 \times 4 = 36$ | (98) $0 \times 2 = 0$ |
| (9) $2 \times 5 = 10$ | (39) $7 \times 9 = 63$ | (69) $3 \times 6 = 18$ | (99) $5 \times 10 = 50$ |
| (10) $8 \times 7 = 56$ | (40) $7 \times 4 = 28$ | (70) $9 \times 1 = 9$ | (100) $1 \times 0 = 0$ |
| (11) $4 \times 6 = 24$ | (41) $3 \times 6 = 18$ | (71) $7 \times 8 = 56$ | (101) $0 \times 6 = 0$ |
| (12) $7 \times 4 = 28$ | (42) $4 \times 3 = 12$ | (72) $2 \times 5 = 10$ | (102) $8 \times 1 = 8$ |
| (13) $6 \times 10 = 60$ | (43) $0 \times 4 = 0$ | (73) $10 \times 8 = 80$ | (103) $4 \times 1 = 4$ |
| (14) $8 \times 8 = 64$ | (44) $7 \times 1 = 7$ | (74) $8 \times 6 = 48$ | (104) $4 \times 10 = 40$ |
| (15) $10 \times 10 = 100$ | (45) $3 \times 8 = 24$ | (75) $7 \times 6 = 42$ | (105) $2 \times 10 = 20$ |
| (16) $5 \times 0 = 0$ | (46) $4 \times 7 = 28$ | (76) $5 \times 2 = 10$ | (106) $7 \times 5 = 35$ |
| (17) $4 \times 0 = 0$ | (47) $6 \times 10 = 60$ | (77) $8 \times 4 = 32$ | (107) $0 \times 0 = 0$ |
| (18) $4 \times 7 = 28$ | (48) $10 \times 3 = 30$ | (78) $5 \times 8 = 40$ | (108) $10 \times 3 = 30$ |
| (19) $6 \times 2 = 12$ | (49) $8 \times 6 = 48$ | (79) $1 \times 7 = 7$ | (109) $2 \times 2 = 4$ |
| (20) $4 \times 1 = 4$ | (50) $7 \times 5 = 35$ | (80) $9 \times 4 = 36$ | (110) $5 \times 6 = 30$ |
| (21) $4 \times 8 = 32$ | (51) $6 \times 1 = 6$ | (81) $10 \times 5 = 50$ | (111) $5 \times 8 = 40$ |
| (22) $6 \times 10 = 60$ | (52) $4 \times 0 = 0$ | (82) $10 \times 1 = 10$ | (112) $2 \times 4 = 8$ |
| (23) $9 \times 9 = 81$ | (53) $5 \times 2 = 10$ | (83) $6 \times 3 = 18$ | (113) $8 \times 0 = 0$ |
| (24) $4 \times 6 = 24$ | (54) $3 \times 9 = 27$ | (84) $10 \times 2 = 20$ | (114) $10 \times 3 = 30$ |
| (25) $3 \times 1 = 3$ | (55) $7 \times 4 = 28$ | (85) $8 \times 8 = 64$ | (115) $2 \times 4 = 8$ |
| (26) $3 \times 4 = 12$ | (56) $1 \times 9 = 9$ | (86) $5 \times 10 = 50$ | (116) $2 \times 9 = 18$ |
| (27) $9 \times 10 = 90$ | (57) $7 \times 7 = 49$ | (87) $4 \times 10 = 40$ | (117) $3 \times 7 = 21$ |
| (28) $10 \times 2 = 20$ | (58) $6 \times 7 = 42$ | (88) $8 \times 4 = 32$ | (118) $9 \times 10 = 90$ |
| (29) $4 \times 9 = 36$ | (59) $9 \times 10 = 90$ | (89) $9 \times 8 = 72$ | (119) $3 \times 2 = 6$ |
| (30) $7 \times 9 = 63$ | (60) $4 \times 6 = 24$ | (90) $0 \times 7 = 0$ | (120) $6 \times 6 = 36$ |

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|---------------------------|-------------------------|-------------------------|--------------------------|
| (1) $8 \times 9 = 72$ | (31) $10 \times 2 = 20$ | (61) $0 \times 9 = 0$ | (91) $5 \times 2 = 10$ |
| (2) $5 \times 9 = 45$ | (32) $3 \times 2 = 6$ | (62) $9 \times 2 = 18$ | (92) $6 \times 4 = 24$ |
| (3) $7 \times 10 = 70$ | (33) $5 \times 8 = 40$ | (63) $10 \times 1 = 10$ | (93) $1 \times 4 = 4$ |
| (4) $4 \times 8 = 32$ | (34) $2 \times 3 = 6$ | (64) $8 \times 1 = 8$ | (94) $4 \times 6 = 24$ |
| (5) $4 \times 1 = 4$ | (35) $4 \times 5 = 20$ | (65) $8 \times 3 = 24$ | (95) $9 \times 9 = 81$ |
| (6) $7 \times 8 = 56$ | (36) $8 \times 10 = 80$ | (66) $3 \times 5 = 15$ | (96) $4 \times 4 = 16$ |
| (7) $5 \times 9 = 45$ | (37) $8 \times 6 = 48$ | (67) $10 \times 8 = 80$ | (97) $6 \times 5 = 30$ |
| (8) $9 \times 7 = 63$ | (38) $9 \times 9 = 81$ | (68) $1 \times 10 = 10$ | (98) $4 \times 9 = 36$ |
| (9) $3 \times 4 = 12$ | (39) $6 \times 9 = 54$ | (69) $1 \times 1 = 1$ | (99) $6 \times 10 = 60$ |
| (10) $4 \times 10 = 40$ | (40) $1 \times 3 = 3$ | (70) $10 \times 8 = 80$ | (100) $0 \times 7 = 0$ |
| (11) $4 \times 9 = 36$ | (41) $4 \times 5 = 20$ | (71) $8 \times 4 = 32$ | (101) $4 \times 9 = 36$ |
| (12) $6 \times 9 = 54$ | (42) $3 \times 5 = 15$ | (72) $9 \times 3 = 27$ | (102) $10 \times 3 = 30$ |
| (13) $1 \times 8 = 8$ | (43) $4 \times 5 = 20$ | (73) $3 \times 10 = 30$ | (103) $7 \times 1 = 7$ |
| (14) $10 \times 5 = 50$ | (44) $2 \times 3 = 6$ | (74) $2 \times 9 = 18$ | (104) $5 \times 10 = 50$ |
| (15) $1 \times 10 = 10$ | (45) $4 \times 10 = 40$ | (75) $9 \times 8 = 72$ | (105) $9 \times 9 = 81$ |
| (16) $0 \times 2 = 0$ | (46) $3 \times 8 = 24$ | (76) $7 \times 7 = 49$ | (106) $4 \times 1 = 4$ |
| (17) $8 \times 6 = 48$ | (47) $1 \times 7 = 7$ | (77) $6 \times 9 = 54$ | (107) $2 \times 8 = 16$ |
| (18) $10 \times 5 = 50$ | (48) $0 \times 1 = 0$ | (78) $8 \times 5 = 40$ | (108) $2 \times 0 = 0$ |
| (19) $10 \times 10 = 100$ | (49) $8 \times 2 = 16$ | (79) $6 \times 8 = 48$ | (109) $6 \times 10 = 60$ |
| (20) $2 \times 6 = 12$ | (50) $7 \times 7 = 49$ | (80) $10 \times 6 = 60$ | (110) $10 \times 5 = 50$ |
| (21) $9 \times 6 = 54$ | (51) $0 \times 3 = 0$ | (81) $2 \times 10 = 20$ | (111) $6 \times 4 = 24$ |
| (22) $1 \times 10 = 10$ | (52) $9 \times 10 = 90$ | (82) $1 \times 8 = 8$ | (112) $6 \times 10 = 60$ |
| (23) $10 \times 2 = 20$ | (53) $5 \times 7 = 35$ | (83) $2 \times 1 = 2$ | (113) $0 \times 6 = 0$ |
| (24) $2 \times 9 = 18$ | (54) $8 \times 6 = 48$ | (84) $5 \times 0 = 0$ | (114) $3 \times 4 = 12$ |
| (25) $10 \times 2 = 20$ | (55) $7 \times 2 = 14$ | (85) $3 \times 0 = 0$ | (115) $7 \times 4 = 28$ |
| (26) $6 \times 9 = 54$ | (56) $10 \times 6 = 60$ | (86) $8 \times 8 = 64$ | (116) $0 \times 10 = 0$ |
| (27) $1 \times 4 = 4$ | (57) $4 \times 4 = 16$ | (87) $7 \times 5 = 35$ | (117) $3 \times 9 = 27$ |
| (28) $4 \times 1 = 4$ | (58) $5 \times 1 = 5$ | (88) $5 \times 9 = 45$ | (118) $4 \times 6 = 24$ |
| (29) $6 \times 9 = 54$ | (59) $10 \times 9 = 90$ | (89) $5 \times 9 = 45$ | (119) $5 \times 2 = 10$ |
| (30) $0 \times 7 = 0$ | (60) $3 \times 4 = 12$ | (90) $6 \times 8 = 48$ | (120) $3 \times 1 = 3$ |

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|---------------------------|-------------------------|-------------------------|--------------------------|
| (1) $1 \times 9 = 9$ | (31) $9 \times 3 = 27$ | (61) $4 \times 1 = 4$ | (91) $1 \times 3 = 3$ |
| (2) $8 \times 6 = 48$ | (32) $5 \times 3 = 15$ | (62) $3 \times 2 = 6$ | (92) $7 \times 4 = 28$ |
| (3) $5 \times 3 = 15$ | (33) $5 \times 2 = 10$ | (63) $8 \times 7 = 56$ | (93) $9 \times 7 = 63$ |
| (4) $5 \times 9 = 45$ | (34) $7 \times 2 = 14$ | (64) $7 \times 1 = 7$ | (94) $10 \times 0 = 0$ |
| (5) $8 \times 10 = 80$ | (35) $10 \times 4 = 40$ | (65) $7 \times 6 = 42$ | (95) $6 \times 0 = 0$ |
| (6) $1 \times 3 = 3$ | (36) $7 \times 8 = 56$ | (66) $0 \times 8 = 0$ | (96) $3 \times 2 = 6$ |
| (7) $4 \times 5 = 20$ | (37) $10 \times 7 = 70$ | (67) $3 \times 5 = 15$ | (97) $8 \times 2 = 16$ |
| (8) $9 \times 4 = 36$ | (38) $8 \times 7 = 56$ | (68) $7 \times 9 = 63$ | (98) $3 \times 8 = 24$ |
| (9) $5 \times 3 = 15$ | (39) $4 \times 0 = 0$ | (69) $1 \times 2 = 2$ | (99) $10 \times 7 = 70$ |
| (10) $7 \times 9 = 63$ | (40) $2 \times 8 = 16$ | (70) $1 \times 9 = 9$ | (100) $10 \times 6 = 60$ |
| (11) $10 \times 5 = 50$ | (41) $9 \times 10 = 90$ | (71) $6 \times 8 = 48$ | (101) $3 \times 2 = 6$ |
| (12) $2 \times 6 = 12$ | (42) $10 \times 6 = 60$ | (72) $9 \times 10 = 90$ | (102) $0 \times 9 = 0$ |
| (13) $6 \times 0 = 0$ | (43) $8 \times 2 = 16$ | (73) $4 \times 9 = 36$ | (103) $4 \times 10 = 40$ |
| (14) $3 \times 4 = 12$ | (44) $2 \times 10 = 20$ | (74) $10 \times 2 = 20$ | (104) $8 \times 7 = 56$ |
| (15) $1 \times 9 = 9$ | (45) $5 \times 0 = 0$ | (75) $4 \times 7 = 28$ | (105) $8 \times 1 = 8$ |
| (16) $2 \times 3 = 6$ | (46) $10 \times 6 = 60$ | (76) $9 \times 2 = 18$ | (106) $9 \times 7 = 63$ |
| (17) $10 \times 2 = 20$ | (47) $4 \times 7 = 28$ | (77) $5 \times 8 = 40$ | (107) $3 \times 8 = 24$ |
| (18) $10 \times 3 = 30$ | (48) $6 \times 1 = 6$ | (78) $10 \times 2 = 20$ | (108) $4 \times 3 = 12$ |
| (19) $6 \times 1 = 6$ | (49) $5 \times 2 = 10$ | (79) $6 \times 5 = 30$ | (109) $5 \times 3 = 15$ |
| (20) $8 \times 1 = 8$ | (50) $3 \times 9 = 27$ | (80) $10 \times 2 = 20$ | (110) $7 \times 5 = 35$ |
| (21) $9 \times 9 = 81$ | (51) $6 \times 8 = 48$ | (81) $0 \times 5 = 0$ | (111) $3 \times 4 = 12$ |
| (22) $3 \times 2 = 6$ | (52) $5 \times 2 = 10$ | (82) $2 \times 8 = 16$ | (112) $6 \times 6 = 36$ |
| (23) $7 \times 6 = 42$ | (53) $1 \times 9 = 9$ | (83) $6 \times 4 = 24$ | (113) $2 \times 4 = 8$ |
| (24) $7 \times 3 = 21$ | (54) $0 \times 8 = 0$ | (84) $6 \times 6 = 36$ | (114) $1 \times 9 = 9$ |
| (25) $2 \times 3 = 6$ | (55) $1 \times 7 = 7$ | (85) $3 \times 6 = 18$ | (115) $9 \times 5 = 45$ |
| (26) $10 \times 3 = 30$ | (56) $3 \times 8 = 24$ | (86) $5 \times 1 = 5$ | (116) $2 \times 7 = 14$ |
| (27) $10 \times 10 = 100$ | (57) $9 \times 5 = 45$ | (87) $0 \times 2 = 0$ | (117) $7 \times 6 = 42$ |
| (28) $6 \times 7 = 42$ | (58) $2 \times 5 = 10$ | (88) $5 \times 1 = 5$ | (118) $6 \times 5 = 30$ |
| (29) $8 \times 4 = 32$ | (59) $0 \times 4 = 0$ | (89) $2 \times 10 = 20$ | (119) $5 \times 7 = 35$ |
| (30) $6 \times 5 = 30$ | (60) $4 \times 7 = 28$ | (90) $7 \times 7 = 49$ | (120) $4 \times 4 = 16$ |

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|---------------------------|-------------------------|---------------------------|--------------------------|
| (1) $7 \times 10 = 70$ | (31) $7 \times 6 = 42$ | (61) $2 \times 3 = 6$ | (91) $2 \times 5 = 10$ |
| (2) $7 \times 6 = 42$ | (32) $4 \times 4 = 16$ | (62) $5 \times 2 = 10$ | (92) $10 \times 8 = 80$ |
| (3) $0 \times 0 = 0$ | (33) $1 \times 6 = 6$ | (63) $6 \times 2 = 12$ | (93) $4 \times 5 = 20$ |
| (4) $2 \times 9 = 18$ | (34) $5 \times 8 = 40$ | (64) $3 \times 2 = 6$ | (94) $2 \times 6 = 12$ |
| (5) $10 \times 3 = 30$ | (35) $3 \times 7 = 21$ | (65) $3 \times 2 = 6$ | (95) $6 \times 9 = 54$ |
| (6) $3 \times 8 = 24$ | (36) $4 \times 5 = 20$ | (66) $9 \times 10 = 90$ | (96) $9 \times 4 = 36$ |
| (7) $7 \times 9 = 63$ | (37) $4 \times 10 = 40$ | (67) $1 \times 4 = 4$ | (97) $10 \times 6 = 60$ |
| (8) $9 \times 0 = 0$ | (38) $5 \times 6 = 30$ | (68) $9 \times 3 = 27$ | (98) $6 \times 0 = 0$ |
| (9) $2 \times 1 = 2$ | (39) $6 \times 8 = 48$ | (69) $4 \times 8 = 32$ | (99) $2 \times 9 = 18$ |
| (10) $2 \times 10 = 20$ | (40) $2 \times 9 = 18$ | (70) $5 \times 8 = 40$ | (100) $1 \times 2 = 2$ |
| (11) $2 \times 3 = 6$ | (41) $0 \times 1 = 0$ | (71) $2 \times 6 = 12$ | (101) $3 \times 8 = 24$ |
| (12) $10 \times 3 = 30$ | (42) $8 \times 10 = 80$ | (72) $9 \times 3 = 27$ | (102) $6 \times 5 = 30$ |
| (13) $10 \times 7 = 70$ | (43) $2 \times 5 = 10$ | (73) $5 \times 4 = 20$ | (103) $9 \times 10 = 90$ |
| (14) $3 \times 8 = 24$ | (44) $7 \times 4 = 28$ | (74) $0 \times 4 = 0$ | (104) $4 \times 8 = 32$ |
| (15) $10 \times 7 = 70$ | (45) $6 \times 3 = 18$ | (75) $1 \times 1 = 1$ | (105) $3 \times 3 = 9$ |
| (16) $10 \times 10 = 100$ | (46) $6 \times 3 = 18$ | (76) $4 \times 2 = 8$ | (106) $10 \times 8 = 80$ |
| (17) $7 \times 3 = 21$ | (47) $9 \times 7 = 63$ | (77) $10 \times 10 = 100$ | (107) $5 \times 1 = 5$ |
| (18) $9 \times 2 = 18$ | (48) $8 \times 5 = 40$ | (78) $4 \times 6 = 24$ | (108) $7 \times 8 = 56$ |
| (19) $1 \times 3 = 3$ | (49) $8 \times 6 = 48$ | (79) $7 \times 1 = 7$ | (109) $4 \times 10 = 40$ |
| (20) $8 \times 4 = 32$ | (50) $2 \times 9 = 18$ | (80) $9 \times 6 = 54$ | (110) $1 \times 3 = 3$ |
| (21) $9 \times 5 = 45$ | (51) $10 \times 1 = 10$ | (81) $2 \times 9 = 18$ | (111) $10 \times 7 = 70$ |
| (22) $6 \times 2 = 12$ | (52) $10 \times 9 = 90$ | (82) $10 \times 10 = 100$ | (112) $2 \times 6 = 12$ |
| (23) $1 \times 1 = 1$ | (53) $4 \times 5 = 20$ | (83) $1 \times 5 = 5$ | (113) $10 \times 4 = 40$ |
| (24) $6 \times 6 = 36$ | (54) $9 \times 3 = 27$ | (84) $7 \times 6 = 42$ | (114) $4 \times 8 = 32$ |
| (25) $3 \times 10 = 30$ | (55) $4 \times 7 = 28$ | (85) $4 \times 6 = 24$ | (115) $7 \times 9 = 63$ |
| (26) $7 \times 4 = 28$ | (56) $8 \times 3 = 24$ | (86) $3 \times 0 = 0$ | (116) $5 \times 0 = 0$ |
| (27) $9 \times 4 = 36$ | (57) $2 \times 2 = 4$ | (87) $6 \times 2 = 12$ | (117) $4 \times 2 = 8$ |
| (28) $2 \times 1 = 2$ | (58) $2 \times 6 = 12$ | (88) $7 \times 2 = 14$ | (118) $1 \times 8 = 8$ |
| (29) $0 \times 3 = 0$ | (59) $1 \times 10 = 10$ | (89) $4 \times 5 = 20$ | (119) $8 \times 2 = 16$ |
| (30) $10 \times 3 = 30$ | (60) $1 \times 2 = 2$ | (90) $9 \times 6 = 54$ | (120) $5 \times 7 = 35$ |

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|-------------------------|---------------------------|-------------------------|--------------------------|
| (1) $3 \times 5 = 15$ | (31) $2 \times 5 = 10$ | (61) $2 \times 3 = 6$ | (91) $3 \times 4 = 12$ |
| (2) $5 \times 1 = 5$ | (32) $7 \times 10 = 70$ | (62) $9 \times 6 = 54$ | (92) $3 \times 4 = 12$ |
| (3) $2 \times 3 = 6$ | (33) $0 \times 1 = 0$ | (63) $5 \times 3 = 15$ | (93) $1 \times 10 = 10$ |
| (4) $7 \times 6 = 42$ | (34) $10 \times 1 = 10$ | (64) $5 \times 5 = 25$ | (94) $10 \times 1 = 10$ |
| (5) $5 \times 7 = 35$ | (35) $5 \times 3 = 15$ | (65) $9 \times 1 = 9$ | (95) $0 \times 5 = 0$ |
| (6) $5 \times 9 = 45$ | (36) $2 \times 7 = 14$ | (66) $2 \times 3 = 6$ | (96) $6 \times 8 = 48$ |
| (7) $7 \times 5 = 35$ | (37) $7 \times 8 = 56$ | (67) $10 \times 4 = 40$ | (97) $3 \times 5 = 15$ |
| (8) $3 \times 2 = 6$ | (38) $4 \times 6 = 24$ | (68) $4 \times 2 = 8$ | (98) $6 \times 6 = 36$ |
| (9) $5 \times 2 = 10$ | (39) $5 \times 8 = 40$ | (69) $0 \times 1 = 0$ | (99) $7 \times 4 = 28$ |
| (10) $7 \times 4 = 28$ | (40) $1 \times 3 = 3$ | (70) $3 \times 2 = 6$ | (100) $3 \times 8 = 24$ |
| (11) $3 \times 4 = 12$ | (41) $6 \times 1 = 6$ | (71) $4 \times 5 = 20$ | (101) $7 \times 2 = 14$ |
| (12) $5 \times 1 = 5$ | (42) $10 \times 7 = 70$ | (72) $0 \times 1 = 0$ | (102) $9 \times 3 = 27$ |
| (13) $7 \times 4 = 28$ | (43) $8 \times 7 = 56$ | (73) $7 \times 10 = 70$ | (103) $4 \times 5 = 20$ |
| (14) $6 \times 3 = 18$ | (44) $2 \times 6 = 12$ | (74) $10 \times 5 = 50$ | (104) $7 \times 0 = 0$ |
| (15) $10 \times 3 = 30$ | (45) $10 \times 7 = 70$ | (75) $8 \times 10 = 80$ | (105) $0 \times 6 = 0$ |
| (16) $6 \times 2 = 12$ | (46) $8 \times 4 = 32$ | (76) $3 \times 4 = 12$ | (106) $9 \times 9 = 81$ |
| (17) $6 \times 6 = 36$ | (47) $1 \times 7 = 7$ | (77) $5 \times 8 = 40$ | (107) $2 \times 2 = 4$ |
| (18) $0 \times 1 = 0$ | (48) $5 \times 2 = 10$ | (78) $6 \times 7 = 42$ | (108) $7 \times 2 = 14$ |
| (19) $7 \times 8 = 56$ | (49) $8 \times 9 = 72$ | (79) $0 \times 0 = 0$ | (109) $6 \times 4 = 24$ |
| (20) $9 \times 4 = 36$ | (50) $3 \times 8 = 24$ | (80) $1 \times 3 = 3$ | (110) $9 \times 2 = 18$ |
| (21) $2 \times 4 = 8$ | (51) $7 \times 2 = 14$ | (81) $7 \times 1 = 7$ | (111) $2 \times 9 = 18$ |
| (22) $1 \times 9 = 9$ | (52) $7 \times 7 = 49$ | (82) $3 \times 9 = 27$ | (112) $3 \times 1 = 3$ |
| (23) $2 \times 4 = 8$ | (53) $3 \times 0 = 0$ | (83) $8 \times 10 = 80$ | (113) $1 \times 9 = 9$ |
| (24) $2 \times 4 = 8$ | (54) $10 \times 1 = 10$ | (84) $6 \times 9 = 54$ | (114) $9 \times 7 = 63$ |
| (25) $9 \times 8 = 72$ | (55) $3 \times 2 = 6$ | (85) $2 \times 4 = 8$ | (115) $4 \times 7 = 28$ |
| (26) $10 \times 1 = 10$ | (56) $10 \times 10 = 100$ | (86) $0 \times 4 = 0$ | (116) $1 \times 6 = 6$ |
| (27) $5 \times 1 = 5$ | (57) $7 \times 7 = 49$ | (87) $3 \times 8 = 24$ | (117) $10 \times 7 = 70$ |
| (28) $2 \times 8 = 16$ | (58) $1 \times 5 = 5$ | (88) $8 \times 4 = 32$ | (118) $5 \times 2 = 10$ |
| (29) $1 \times 8 = 8$ | (59) $8 \times 6 = 48$ | (89) $7 \times 8 = 56$ | (119) $1 \times 8 = 8$ |
| (30) $7 \times 6 = 42$ | (60) $5 \times 6 = 30$ | (90) $4 \times 1 = 4$ | (120) $6 \times 7 = 42$ |

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|-------------------------|-------------------------|-------------------------|--------------------------|
| (1) $2 \times 7 = 14$ | (31) $3 \times 2 = 6$ | (61) $5 \times 5 = 25$ | (91) $6 \times 6 = 36$ |
| (2) $3 \times 1 = 3$ | (32) $3 \times 6 = 18$ | (62) $3 \times 4 = 12$ | (92) $8 \times 7 = 56$ |
| (3) $9 \times 10 = 90$ | (33) $1 \times 7 = 7$ | (63) $6 \times 5 = 30$ | (93) $10 \times 7 = 70$ |
| (4) $1 \times 3 = 3$ | (34) $9 \times 6 = 54$ | (64) $8 \times 10 = 80$ | (94) $3 \times 6 = 18$ |
| (5) $1 \times 6 = 6$ | (35) $7 \times 6 = 42$ | (65) $6 \times 1 = 6$ | (95) $1 \times 5 = 5$ |
| (6) $1 \times 10 = 10$ | (36) $6 \times 0 = 0$ | (66) $0 \times 10 = 0$ | (96) $6 \times 6 = 36$ |
| (7) $5 \times 2 = 10$ | (37) $3 \times 2 = 6$ | (67) $1 \times 8 = 8$ | (97) $1 \times 2 = 2$ |
| (8) $1 \times 10 = 10$ | (38) $3 \times 1 = 3$ | (68) $8 \times 4 = 32$ | (98) $3 \times 5 = 15$ |
| (9) $8 \times 4 = 32$ | (39) $4 \times 1 = 4$ | (69) $1 \times 8 = 8$ | (99) $8 \times 3 = 24$ |
| (10) $9 \times 6 = 54$ | (40) $7 \times 7 = 49$ | (70) $6 \times 3 = 18$ | (100) $10 \times 3 = 30$ |
| (11) $10 \times 3 = 30$ | (41) $5 \times 10 = 50$ | (71) $2 \times 6 = 12$ | (101) $1 \times 7 = 7$ |
| (12) $9 \times 7 = 63$ | (42) $4 \times 5 = 20$ | (72) $1 \times 10 = 10$ | (102) $8 \times 7 = 56$ |
| (13) $6 \times 8 = 48$ | (43) $3 \times 7 = 21$ | (73) $8 \times 9 = 72$ | (103) $4 \times 9 = 36$ |
| (14) $7 \times 2 = 14$ | (44) $9 \times 10 = 90$ | (74) $8 \times 9 = 72$ | (104) $10 \times 2 = 20$ |
| (15) $0 \times 8 = 0$ | (45) $3 \times 8 = 24$ | (75) $9 \times 2 = 18$ | (105) $4 \times 1 = 4$ |
| (16) $5 \times 9 = 45$ | (46) $1 \times 5 = 5$ | (76) $7 \times 0 = 0$ | (106) $7 \times 3 = 21$ |
| (17) $10 \times 2 = 20$ | (47) $8 \times 4 = 32$ | (77) $0 \times 9 = 0$ | (107) $10 \times 5 = 50$ |
| (18) $5 \times 8 = 40$ | (48) $10 \times 7 = 70$ | (78) $2 \times 1 = 2$ | (108) $3 \times 9 = 27$ |
| (19) $1 \times 4 = 4$ | (49) $10 \times 8 = 80$ | (79) $1 \times 1 = 1$ | (109) $6 \times 7 = 42$ |
| (20) $5 \times 6 = 30$ | (50) $1 \times 10 = 10$ | (80) $3 \times 8 = 24$ | (110) $9 \times 5 = 45$ |
| (21) $1 \times 7 = 7$ | (51) $4 \times 2 = 8$ | (81) $9 \times 1 = 9$ | (111) $4 \times 0 = 0$ |
| (22) $3 \times 8 = 24$ | (52) $4 \times 9 = 36$ | (82) $2 \times 3 = 6$ | (112) $0 \times 10 = 0$ |
| (23) $6 \times 10 = 60$ | (53) $0 \times 7 = 0$ | (83) $8 \times 9 = 72$ | (113) $7 \times 8 = 56$ |
| (24) $9 \times 2 = 18$ | (54) $2 \times 6 = 12$ | (84) $3 \times 3 = 9$ | (114) $9 \times 3 = 27$ |
| (25) $6 \times 0 = 0$ | (55) $1 \times 0 = 0$ | (85) $9 \times 9 = 81$ | (115) $3 \times 2 = 6$ |
| (26) $10 \times 6 = 60$ | (56) $10 \times 2 = 20$ | (86) $3 \times 7 = 21$ | (116) $2 \times 8 = 16$ |
| (27) $5 \times 2 = 10$ | (57) $10 \times 5 = 50$ | (87) $6 \times 2 = 12$ | (117) $4 \times 1 = 4$ |
| (28) $9 \times 4 = 36$ | (58) $7 \times 3 = 21$ | (88) $3 \times 4 = 12$ | (118) $0 \times 2 = 0$ |
| (29) $2 \times 10 = 20$ | (59) $5 \times 2 = 10$ | (89) $7 \times 4 = 28$ | (119) $1 \times 9 = 9$ |
| (30) $5 \times 1 = 5$ | (60) $4 \times 3 = 12$ | (90) $9 \times 6 = 54$ | (120) $2 \times 3 = 6$ |

(1) $3 \times 3 = 9$	(31) $6 \times 2 = 12$	(61) $8 \times 5 = 40$	(91) $7 \times 8 = 56$
(2) $9 \times 4 = 36$	(32) $10 \times 9 = 90$	(62) $8 \times 10 = 80$	(92) $4 \times 3 = 12$
(3) $7 \times 1 = 7$	(33) $4 \times 8 = 32$	(63) $5 \times 10 = 50$	(93) $4 \times 2 = 8$
(4) $2 \times 1 = 2$	(34) $6 \times 6 = 36$	(64) $0 \times 7 = 0$	(94) $9 \times 6 = 54$
(5) $3 \times 3 = 9$	(35) $6 \times 6 = 36$	(65) $10 \times 8 = 80$	(95) $3 \times 2 = 6$
(6) $2 \times 2 = 4$	(36) $4 \times 0 = 0$	(66) $5 \times 7 = 35$	(96) $4 \times 3 = 12$
(7) $1 \times 9 = 9$	(37) $4 \times 3 = 12$	(67) $5 \times 7 = 35$	(97) $3 \times 3 = 9$
(8) $6 \times 8 = 48$	(38) $7 \times 2 = 14$	(68) $5 \times 7 = 35$	(98) $6 \times 9 = 54$
(9) $9 \times 6 = 54$	(39) $9 \times 1 = 9$	(69) $7 \times 7 = 49$	(99) $0 \times 3 = 0$
(10) $7 \times 2 = 14$	(40) $6 \times 6 = 36$	(70) $3 \times 4 = 12$	(100) $9 \times 1 = 9$
(11) $4 \times 9 = 36$	(41) $8 \times 9 = 72$	(71) $3 \times 3 = 9$	(101) $3 \times 0 = 0$
(12) $9 \times 9 = 81$	(42) $6 \times 10 = 60$	(72) $9 \times 5 = 45$	(102) $8 \times 9 = 72$
(13) $6 \times 6 = 36$	(43) $7 \times 2 = 14$	(73) $5 \times 4 = 20$	(103) $1 \times 8 = 8$
(14) $1 \times 0 = 0$	(44) $8 \times 1 = 8$	(74) $8 \times 0 = 0$	(104) $9 \times 3 = 27$
(15) $6 \times 6 = 36$	(45) $2 \times 7 = 14$	(75) $1 \times 9 = 9$	(105) $0 \times 7 = 0$
(16) $7 \times 3 = 21$	(46) $2 \times 10 = 20$	(76) $4 \times 3 = 12$	(106) $3 \times 2 = 6$
(17) $4 \times 3 = 12$	(47) $2 \times 9 = 18$	(77) $4 \times 1 = 4$	(107) $10 \times 8 = 80$
(18) $0 \times 8 = 0$	(48) $8 \times 8 = 64$	(78) $3 \times 9 = 27$	(108) $7 \times 3 = 21$
(19) $6 \times 3 = 18$	(49) $7 \times 3 = 21$	(79) $7 \times 7 = 49$	(109) $0 \times 5 = 0$
(20) $5 \times 2 = 10$	(50) $2 \times 5 = 10$	(80) $10 \times 8 = 80$	(110) $7 \times 7 = 49$
(21) $5 \times 6 = 30$	(51) $8 \times 0 = 0$	(81) $9 \times 8 = 72$	(111) $5 \times 5 = 25$
(22) $7 \times 7 = 49$	(52) $7 \times 9 = 63$	(82) $7 \times 1 = 7$	(112) $9 \times 1 = 9$
(23) $9 \times 10 = 90$	(53) $6 \times 4 = 24$	(83) $3 \times 4 = 12$	(113) $8 \times 1 = 8$
(24) $1 \times 3 = 3$	(54) $9 \times 10 = 90$	(84) $2 \times 5 = 10$	(114) $8 \times 7 = 56$
(25) $7 \times 8 = 56$	(55) $3 \times 8 = 24$	(85) $1 \times 4 = 4$	(115) $7 \times 10 = 70$
(26) $8 \times 5 = 40$	(56) $5 \times 2 = 10$	(86) $2 \times 4 = 8$	(116) $6 \times 7 = 42$
(27) $7 \times 3 = 21$	(57) $6 \times 3 = 18$	(87) $4 \times 3 = 12$	(117) $3 \times 8 = 24$
(28) $5 \times 10 = 50$	(58) $3 \times 4 = 12$	(88) $1 \times 8 = 8$	(118) $2 \times 3 = 6$
(29) $7 \times 5 = 35$	(59) $5 \times 5 = 25$	(89) $10 \times 6 = 60$	(119) $4 \times 2 = 8$
(30) $1 \times 10 = 10$	(60) $10 \times 7 = 70$	(90) $3 \times 1 = 3$	(120) $3 \times 5 = 15$

1.1.4 除法

自然数の商と剰余を求める問題。

(1) $59 \div 3$

(19) $64 \div 7$

(37) $68 \div 6$

(55) $33 \div 1$

(2) $74 \div 4$

(20) $70 \div 1$

(38) $81 \div 2$

(56) $70 \div 10$

(3) $62 \div 4$

(21) $86 \div 9$

(39) $27 \div 7$

(57) $32 \div 8$

(4) $84 \div 5$

(22) $62 \div 9$

(40) $37 \div 10$

(58) $33 \div 2$

(5) $76 \div 6$

(23) $1 \div 10$

(41) $62 \div 8$

(59) $12 \div 5$

(6) $1 \div 2$

(24) $86 \div 4$

(42) $36 \div 5$

(60) $19 \div 8$

(7) $48 \div 9$

(25) $73 \div 7$

(43) $22 \div 7$

(61) $37 \div 1$

(8) $53 \div 6$

(26) $59 \div 6$

(44) $22 \div 5$

(62) $75 \div 10$

(9) $85 \div 10$

(27) $45 \div 1$

(45) $47 \div 3$

(63) $34 \div 4$

(10) $21 \div 3$

(28) $24 \div 8$

(46) $89 \div 2$

(64) $18 \div 5$

(11) $71 \div 9$

(29) $7 \div 3$

(47) $87 \div 2$

(65) $60 \div 3$

(12) $24 \div 1$

(30) $48 \div 9$

(48) $14 \div 4$

(66) $22 \div 2$

(13) $22 \div 1$

(31) $39 \div 9$

(49) $2 \div 8$

(67) $40 \div 4$

(14) $22 \div 2$

(32) $79 \div 10$

(50) $1 \div 2$

(68) $91 \div 1$

(15) $18 \div 2$

(33) $4 \div 6$

(51) $49 \div 5$

(69) $89 \div 10$

(16) $70 \div 7$

(34) $56 \div 2$

(52) $34 \div 9$

(70) $1 \div 3$

(17) $9 \div 5$

(35) $50 \div 2$

(53) $70 \div 2$

(71) $86 \div 6$

(18) $38 \div 1$

(36) $58 \div 1$

(54) $76 \div 2$

(72) $46 \div 9$

(1) $50 \div 5$

(19) $25 \div 1$

(37) $100 \div 6$

(55) $89 \div 8$

(2) $100 \div 8$

(20) $87 \div 10$

(38) $2 \div 6$

(56) $66 \div 4$

(3) $65 \div 8$

(21) $46 \div 8$

(39) $12 \div 7$

(57) $55 \div 4$

(4) $25 \div 9$

(22) $72 \div 6$

(40) $32 \div 10$

(58) $64 \div 1$

(5) $88 \div 4$

(23) $42 \div 6$

(41) $76 \div 9$

(59) $38 \div 3$

(6) $23 \div 4$

(24) $47 \div 8$

(42) $44 \div 6$

(60) $25 \div 5$

(7) $18 \div 9$

(25) $38 \div 2$

(43) $52 \div 10$

(61) $77 \div 1$

(8) $12 \div 6$

(26) $54 \div 6$

(44) $24 \div 6$

(62) $21 \div 3$

(9) $60 \div 4$

(27) $55 \div 9$

(45) $47 \div 8$

(63) $67 \div 4$

(10) $58 \div 9$

(28) $57 \div 2$

(46) $96 \div 10$

(64) $89 \div 10$

(11) $31 \div 6$

(29) $64 \div 1$

(47) $76 \div 8$

(65) $81 \div 7$

(12) $53 \div 1$

(30) $24 \div 4$

(48) $10 \div 10$

(66) $30 \div 3$

(13) $79 \div 6$

(31) $98 \div 1$

(49) $38 \div 3$

(67) $26 \div 5$

(14) $57 \div 1$

(32) $9 \div 4$

(50) $75 \div 4$

(68) $40 \div 3$

(15) $87 \div 4$

(33) $58 \div 3$

(51) $67 \div 4$

(69) $83 \div 7$

(16) $35 \div 7$

(34) $35 \div 4$

(52) $4 \div 3$

(70) $20 \div 10$

(17) $31 \div 1$

(35) $38 \div 6$

(53) $22 \div 8$

(71) $10 \div 5$

(18) $86 \div 3$

(36) $33 \div 8$

(54) $8 \div 2$

(72) $47 \div 1$

- | | | | |
|-------------------|-------------------|-------------------|-------------------|
| (1) $86 \div 5$ | (19) $71 \div 4$ | (37) $8 \div 2$ | (55) $78 \div 9$ |
| (2) $30 \div 5$ | (20) $84 \div 7$ | (38) $71 \div 4$ | (56) $29 \div 6$ |
| (3) $99 \div 1$ | (21) $79 \div 10$ | (39) $78 \div 8$ | (57) $46 \div 4$ |
| (4) $1 \div 4$ | (22) $46 \div 8$ | (40) $74 \div 9$ | (58) $34 \div 9$ |
| (5) $52 \div 8$ | (23) $52 \div 4$ | (41) $66 \div 10$ | (59) $4 \div 7$ |
| (6) $4 \div 9$ | (24) $43 \div 5$ | (42) $66 \div 6$ | (60) $9 \div 5$ |
| (7) $8 \div 6$ | (25) $54 \div 1$ | (43) $75 \div 8$ | (61) $38 \div 3$ |
| (8) $100 \div 1$ | (26) $50 \div 5$ | (44) $84 \div 1$ | (62) $50 \div 7$ |
| (9) $42 \div 1$ | (27) $7 \div 7$ | (45) $54 \div 1$ | (63) $16 \div 1$ |
| (10) $56 \div 8$ | (28) $86 \div 9$ | (46) $79 \div 4$ | (64) $12 \div 9$ |
| (11) $9 \div 3$ | (29) $75 \div 3$ | (47) $55 \div 4$ | (65) $66 \div 6$ |
| (12) $61 \div 4$ | (30) $26 \div 1$ | (48) $99 \div 6$ | (66) $88 \div 6$ |
| (13) $62 \div 4$ | (31) $50 \div 10$ | (49) $99 \div 10$ | (67) $9 \div 1$ |
| (14) $71 \div 3$ | (32) $58 \div 9$ | (50) $83 \div 3$ | (68) $40 \div 8$ |
| (15) $44 \div 4$ | (33) $18 \div 7$ | (51) $63 \div 5$ | (69) $93 \div 1$ |
| (16) $97 \div 4$ | (34) $19 \div 1$ | (52) $62 \div 8$ | (70) $41 \div 5$ |
| (17) $41 \div 10$ | (35) $92 \div 6$ | (53) $42 \div 2$ | (71) $20 \div 10$ |
| (18) $76 \div 6$ | (36) $46 \div 5$ | (54) $26 \div 4$ | (72) $1 \div 1$ |

(1) $33 \div 7$	(19) $98 \div 8$	(37) $41 \div 6$	(55) $12 \div 4$
(2) $34 \div 6$	(20) $88 \div 2$	(38) $83 \div 10$	(56) $85 \div 10$
(3) $63 \div 8$	(21) $90 \div 9$	(39) $58 \div 4$	(57) $29 \div 3$
(4) $94 \div 3$	(22) $51 \div 9$	(40) $51 \div 10$	(58) $87 \div 9$
(5) $21 \div 5$	(23) $65 \div 1$	(41) $28 \div 1$	(59) $2 \div 5$
(6) $30 \div 8$	(24) $61 \div 2$	(42) $11 \div 5$	(60) $50 \div 1$
(7) $21 \div 8$	(25) $27 \div 8$	(43) $30 \div 6$	(61) $47 \div 10$
(8) $0 \div 8$	(26) $35 \div 4$	(44) $17 \div 1$	(62) $21 \div 2$
(9) $88 \div 6$	(27) $90 \div 6$	(45) $93 \div 1$	(63) $83 \div 10$
(10) $5 \div 1$	(28) $52 \div 3$	(46) $38 \div 5$	(64) $34 \div 9$
(11) $38 \div 6$	(29) $71 \div 10$	(47) $27 \div 9$	(65) $12 \div 7$
(12) $50 \div 4$	(30) $34 \div 6$	(48) $57 \div 8$	(66) $16 \div 5$
(13) $2 \div 10$	(31) $20 \div 7$	(49) $83 \div 3$	(67) $99 \div 5$
(14) $85 \div 1$	(32) $1 \div 5$	(50) $69 \div 7$	(68) $55 \div 1$
(15) $83 \div 5$	(33) $84 \div 4$	(51) $59 \div 2$	(69) $18 \div 6$
(16) $31 \div 8$	(34) $30 \div 6$	(52) $44 \div 2$	(70) $26 \div 5$
(17) $42 \div 10$	(35) $4 \div 7$	(53) $6 \div 8$	(71) $50 \div 9$
(18) $78 \div 4$	(36) $15 \div 4$	(54) $14 \div 7$	(72) $0 \div 9$

(1) $6 \div 1$	(19) $48 \div 10$	(37) $5 \div 1$	(55) $23 \div 9$
(2) $19 \div 4$	(20) $84 \div 7$	(38) $74 \div 10$	(56) $87 \div 2$
(3) $91 \div 6$	(21) $1 \div 4$	(39) $46 \div 3$	(57) $83 \div 2$
(4) $90 \div 3$	(22) $52 \div 2$	(40) $96 \div 4$	(58) $58 \div 2$
(5) $79 \div 8$	(23) $46 \div 5$	(41) $54 \div 8$	(59) $75 \div 8$
(6) $52 \div 9$	(24) $63 \div 10$	(42) $33 \div 9$	(60) $28 \div 1$
(7) $51 \div 7$	(25) $67 \div 6$	(43) $77 \div 4$	(61) $36 \div 5$
(8) $100 \div 2$	(26) $27 \div 10$	(44) $37 \div 6$	(62) $48 \div 2$
(9) $79 \div 3$	(27) $62 \div 9$	(45) $37 \div 9$	(63) $36 \div 3$
(10) $7 \div 9$	(28) $39 \div 1$	(46) $64 \div 7$	(64) $66 \div 3$
(11) $30 \div 5$	(29) $87 \div 9$	(47) $77 \div 4$	(65) $21 \div 6$
(12) $97 \div 8$	(30) $34 \div 1$	(48) $33 \div 2$	(66) $89 \div 3$
(13) $55 \div 9$	(31) $17 \div 2$	(49) $70 \div 9$	(67) $47 \div 7$
(14) $7 \div 10$	(32) $42 \div 2$	(50) $84 \div 5$	(68) $38 \div 8$
(15) $63 \div 8$	(33) $60 \div 9$	(51) $32 \div 2$	(69) $7 \div 10$
(16) $3 \div 10$	(34) $9 \div 1$	(52) $12 \div 1$	(70) $34 \div 2$
(17) $6 \div 8$	(35) $41 \div 10$	(53) $36 \div 10$	(71) $75 \div 3$
(18) $66 \div 5$	(36) $89 \div 2$	(54) $79 \div 2$	(72) $78 \div 4$

- | | | | |
|-------------------|-------------------|-------------------|-------------------|
| (1) $51 \div 10$ | (19) $78 \div 1$ | (37) $52 \div 7$ | (55) $27 \div 8$ |
| (2) $21 \div 5$ | (20) $30 \div 9$ | (38) $34 \div 6$ | (56) $92 \div 3$ |
| (3) $97 \div 5$ | (21) $36 \div 9$ | (39) $52 \div 2$ | (57) $38 \div 9$ |
| (4) $54 \div 10$ | (22) $93 \div 7$ | (40) $12 \div 10$ | (58) $12 \div 9$ |
| (5) $32 \div 4$ | (23) $16 \div 9$ | (41) $91 \div 10$ | (59) $80 \div 6$ |
| (6) $92 \div 1$ | (24) $99 \div 9$ | (42) $60 \div 8$ | (60) $7 \div 9$ |
| (7) $59 \div 9$ | (25) $75 \div 7$ | (43) $71 \div 6$ | (61) $96 \div 10$ |
| (8) $12 \div 8$ | (26) $3 \div 7$ | (44) $5 \div 5$ | (62) $0 \div 9$ |
| (9) $84 \div 6$ | (27) $88 \div 10$ | (45) $17 \div 8$ | (63) $6 \div 4$ |
| (10) $72 \div 5$ | (28) $7 \div 3$ | (46) $9 \div 5$ | (64) $88 \div 10$ |
| (11) $52 \div 5$ | (29) $27 \div 5$ | (47) $41 \div 4$ | (65) $85 \div 8$ |
| (12) $99 \div 5$ | (30) $35 \div 3$ | (48) $49 \div 7$ | (66) $31 \div 3$ |
| (13) $63 \div 5$ | (31) $33 \div 9$ | (49) $73 \div 3$ | (67) $48 \div 8$ |
| (14) $18 \div 10$ | (32) $53 \div 9$ | (50) $26 \div 9$ | (68) $62 \div 7$ |
| (15) $50 \div 1$ | (33) $64 \div 9$ | (51) $74 \div 3$ | (69) $100 \div 3$ |
| (16) $57 \div 1$ | (34) $69 \div 6$ | (52) $4 \div 10$ | (70) $56 \div 7$ |
| (17) $96 \div 3$ | (35) $39 \div 5$ | (53) $38 \div 5$ | (71) $69 \div 5$ |
| (18) $36 \div 8$ | (36) $16 \div 3$ | (54) $58 \div 4$ | (72) $10 \div 3$ |

(1) $80 \div 10$	(19) $20 \div 8$	(37) $55 \div 5$	(55) $75 \div 6$
(2) $99 \div 7$	(20) $58 \div 7$	(38) $81 \div 3$	(56) $67 \div 1$
(3) $12 \div 1$	(21) $13 \div 10$	(39) $5 \div 8$	(57) $15 \div 4$
(4) $17 \div 3$	(22) $14 \div 6$	(40) $89 \div 5$	(58) $73 \div 4$
(5) $38 \div 9$	(23) $26 \div 10$	(41) $72 \div 3$	(59) $59 \div 7$
(6) $22 \div 4$	(24) $55 \div 8$	(42) $18 \div 8$	(60) $78 \div 5$
(7) $4 \div 6$	(25) $13 \div 8$	(43) $82 \div 8$	(61) $31 \div 7$
(8) $75 \div 1$	(26) $78 \div 10$	(44) $71 \div 3$	(62) $65 \div 8$
(9) $38 \div 10$	(27) $97 \div 1$	(45) $82 \div 4$	(63) $42 \div 5$
(10) $93 \div 9$	(28) $27 \div 10$	(46) $48 \div 4$	(64) $50 \div 3$
(11) $73 \div 3$	(29) $0 \div 2$	(47) $74 \div 1$	(65) $5 \div 3$
(12) $74 \div 6$	(30) $50 \div 7$	(48) $15 \div 8$	(66) $77 \div 6$
(13) $21 \div 10$	(31) $83 \div 8$	(49) $29 \div 7$	(67) $28 \div 1$
(14) $27 \div 10$	(32) $98 \div 8$	(50) $96 \div 5$	(68) $24 \div 7$
(15) $55 \div 2$	(33) $72 \div 7$	(51) $64 \div 1$	(69) $22 \div 1$
(16) $21 \div 8$	(34) $40 \div 7$	(52) $74 \div 9$	(70) $12 \div 3$
(17) $8 \div 10$	(35) $83 \div 1$	(53) $54 \div 8$	(71) $89 \div 3$
(18) $29 \div 1$	(36) $57 \div 3$	(54) $9 \div 3$	(72) $13 \div 9$

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|-------------------|-------------------|------------------|-------------------|
| (1) $38 \div 1$ | (19) $91 \div 6$ | (37) $65 \div 7$ | (55) $67 \div 9$ |
| (2) $59 \div 5$ | (20) $51 \div 4$ | (38) $25 \div 7$ | (56) $95 \div 7$ |
| (3) $33 \div 5$ | (21) $0 \div 5$ | (39) $68 \div 3$ | (57) $66 \div 6$ |
| (4) $89 \div 8$ | (22) $49 \div 10$ | (40) $97 \div 2$ | (58) $62 \div 10$ |
| (5) $70 \div 7$ | (23) $92 \div 6$ | (41) $41 \div 9$ | (59) $84 \div 8$ |
| (6) $39 \div 3$ | (24) $1 \div 6$ | (42) $41 \div 9$ | (60) $50 \div 2$ |
| (7) $22 \div 1$ | (25) $12 \div 2$ | (43) $97 \div 8$ | (61) $81 \div 8$ |
| (8) $76 \div 7$ | (26) $37 \div 2$ | (44) $39 \div 3$ | (62) $91 \div 4$ |
| (9) $59 \div 6$ | (27) $77 \div 8$ | (45) $21 \div 8$ | (63) $61 \div 7$ |
| (10) $79 \div 1$ | (28) $29 \div 5$ | (46) $68 \div 3$ | (64) $14 \div 8$ |
| (11) $42 \div 1$ | (29) $75 \div 10$ | (47) $77 \div 1$ | (65) $67 \div 3$ |
| (12) $33 \div 10$ | (30) $11 \div 10$ | (48) $21 \div 4$ | (66) $98 \div 4$ |
| (13) $58 \div 6$ | (31) $68 \div 1$ | (49) $80 \div 1$ | (67) $52 \div 9$ |
| (14) $20 \div 1$ | (32) $19 \div 3$ | (50) $35 \div 4$ | (68) $70 \div 6$ |
| (15) $81 \div 8$ | (33) $8 \div 9$ | (51) $77 \div 2$ | (69) $86 \div 6$ |
| (16) $61 \div 2$ | (34) $84 \div 7$ | (52) $97 \div 4$ | (70) $23 \div 8$ |
| (17) $9 \div 10$ | (35) $37 \div 4$ | (53) $78 \div 2$ | (71) $48 \div 8$ |
| (18) $43 \div 5$ | (36) $78 \div 10$ | (54) $25 \div 4$ | (72) $63 \div 6$ |

(1) $42 \div 8$

(19) $32 \div 2$

(37) $68 \div 4$

(55) $92 \div 1$

(2) $1 \div 5$

(20) $97 \div 4$

(38) $57 \div 2$

(56) $50 \div 9$

(3) $44 \div 3$

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(62) $13 \div 9$

(9) $17 \div 1$

(27) $24 \div 2$

(45) $21 \div 1$

(63) $40 \div 7$

(10) $92 \div 5$

(28) $81 \div 10$

(46) $66 \div 2$

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(11) $2 \div 1$

(29) $36 \div 1$

(47) $39 \div 8$

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(12) $21 \div 6$

(30) $27 \div 1$

(48) $67 \div 10$

(66) $80 \div 2$

(13) $70 \div 9$

(31) $8 \div 10$

(49) $30 \div 7$

(67) $58 \div 4$

(14) $81 \div 6$

(32) $98 \div 4$

(50) $76 \div 2$

(68) $85 \div 4$

(15) $45 \div 6$

(33) $79 \div 3$

(51) $100 \div 3$

(69) $81 \div 10$

(16) $23 \div 1$

(34) $27 \div 4$

(52) $62 \div 9$

(70) $64 \div 6$

(17) $60 \div 8$

(35) $56 \div 4$

(53) $86 \div 8$

(71) $78 \div 8$

(18) $7 \div 3$

(36) $68 \div 3$

(54) $60 \div 1$

(72) $97 \div 6$

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(19) $24 \div 5$

(37) $65 \div 2$

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(2) $76 \div 2$

(20) $95 \div 2$

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(3) $57 \div 6$

(21) $94 \div 10$

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(57) $34 \div 10$

(4) $1 \div 9$

(22) $65 \div 8$

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(5) $41 \div 2$

(23) $64 \div 9$

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(6) $75 \div 1$

(24) $10 \div 1$

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(7) $30 \div 8$

(25) $9 \div 2$

(43) $23 \div 1$

(61) $43 \div 10$

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(9) $86 \div 7$

(27) $23 \div 3$

(45) $21 \div 2$

(63) $48 \div 1$

(10) $59 \div 9$

(28) $47 \div 10$

(46) $28 \div 4$

(64) $100 \div 4$

(11) $75 \div 3$

(29) $52 \div 4$

(47) $33 \div 4$

(65) $50 \div 10$

(12) $45 \div 2$

(30) $57 \div 5$

(48) $36 \div 9$

(66) $66 \div 10$

(13) $91 \div 9$

(31) $98 \div 5$

(49) $15 \div 3$

(67) $46 \div 6$

(14) $21 \div 9$

(32) $84 \div 3$

(50) $0 \div 2$

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(15) $84 \div 3$

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(51) $93 \div 3$

(69) $82 \div 8$

(16) $51 \div 7$

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(17) $68 \div 6$

(35) $78 \div 3$

(53) $65 \div 5$

(71) $16 \div 1$

(18) $37 \div 6$

(36) $58 \div 8$

(54) $96 \div 5$

(72) $50 \div 7$

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
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商 9, 余り 1 | (37) $68 \div 6$
商 11, 余り 2 | (55) $33 \div 1$
商 33, 余り 0 |
| (2) $74 \div 4$
商 18, 余り 2 | (20) $70 \div 1$
商 70, 余り 0 | (38) $81 \div 2$
商 40, 余り 1 | (56) $70 \div 10$
商 7, 余り 0 |
| (3) $62 \div 4$
商 15, 余り 2 | (21) $86 \div 9$
商 9, 余り 5 | (39) $27 \div 7$
商 3, 余り 6 | (57) $32 \div 8$
商 4, 余り 0 |
| (4) $84 \div 5$
商 16, 余り 4 | (22) $62 \div 9$
商 6, 余り 8 | (40) $37 \div 10$
商 3, 余り 7 | (58) $33 \div 2$
商 16, 余り 1 |
| (5) $76 \div 6$
商 12, 余り 4 | (23) $1 \div 10$
商 0, 余り 1 | (41) $62 \div 8$
商 7, 余り 6 | (59) $12 \div 5$
商 2, 余り 2 |
| (6) $1 \div 2$
商 0, 余り 1 | (24) $86 \div 4$
商 21, 余り 2 | (42) $36 \div 5$
商 7, 余り 1 | (60) $19 \div 8$
商 2, 余り 3 |
| (7) $48 \div 9$
商 5, 余り 3 | (25) $73 \div 7$
商 10, 余り 3 | (43) $22 \div 7$
商 3, 余り 1 | (61) $37 \div 1$
商 37, 余り 0 |
| (8) $53 \div 6$
商 8, 余り 5 | (26) $59 \div 6$
商 9, 余り 5 | (44) $22 \div 5$
商 4, 余り 2 | (62) $75 \div 10$
商 7, 余り 5 |
| (9) $85 \div 10$
商 8, 余り 5 | (27) $45 \div 1$
商 45, 余り 0 | (45) $47 \div 3$
商 15, 余り 2 | (63) $34 \div 4$
商 8, 余り 2 |
| (10) $21 \div 3$
商 7, 余り 0 | (28) $24 \div 8$
商 3, 余り 0 | (46) $89 \div 2$
商 44, 余り 1 | (64) $18 \div 5$
商 3, 余り 3 |
| (11) $71 \div 9$
商 7, 余り 8 | (29) $7 \div 3$
商 2, 余り 1 | (47) $87 \div 2$
商 43, 余り 1 | (65) $60 \div 3$
商 20, 余り 0 |
| (12) $24 \div 1$
商 24, 余り 0 | (30) $48 \div 9$
商 5, 余り 3 | (48) $14 \div 4$
商 3, 余り 2 | (66) $22 \div 2$
商 11, 余り 0 |
| (13) $22 \div 1$
商 22, 余り 0 | (31) $39 \div 9$
商 4, 余り 3 | (49) $2 \div 8$
商 0, 余り 2 | (67) $40 \div 4$
商 10, 余り 0 |
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商 11, 余り 0 | (32) $79 \div 10$
商 7, 余り 9 | (50) $1 \div 2$
商 0, 余り 1 | (68) $91 \div 1$
商 91, 余り 0 |
| (15) $18 \div 2$
商 9, 余り 0 | (33) $4 \div 6$
商 0, 余り 4 | (51) $49 \div 5$
商 9, 余り 4 | (69) $89 \div 10$
商 8, 余り 9 |
| (16) $70 \div 7$
商 10, 余り 0 | (34) $56 \div 2$
商 28, 余り 0 | (52) $34 \div 9$
商 3, 余り 7 | (70) $1 \div 3$
商 0, 余り 1 |
| (17) $9 \div 5$
商 1, 余り 4 | (35) $50 \div 2$
商 25, 余り 0 | (53) $70 \div 2$
商 35, 余り 0 | (71) $86 \div 6$
商 14, 余り 2 |
| (18) $38 \div 1$
商 38, 余り 0 | (36) $58 \div 1$
商 58, 余り 0 | (54) $76 \div 2$
商 38, 余り 0 | (72) $46 \div 9$
商 5, 余り 1 |

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|--------------------------------|--------------------------------|---------------------------------|--------------------------------|
| (1) $50 \div 5$
商 10, 余り 0 | (19) $25 \div 1$
商 25, 余り 0 | (37) $100 \div 6$
商 16, 余り 4 | (55) $89 \div 8$
商 11, 余り 1 |
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商 13, 余り 3 |
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商 64, 余り 0 |
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商 12, 余り 2 |
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商 5, 余り 3 | (24) $47 \div 8$
商 5, 余り 7 | (42) $44 \div 6$
商 7, 余り 2 | (60) $25 \div 5$
商 5, 余り 0 |
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商 5, 余り 2 | (61) $77 \div 1$
商 77, 余り 0 |
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商 9, 余り 0 | (44) $24 \div 6$
商 4, 余り 0 | (62) $21 \div 3$
商 7, 余り 0 |
| (9) $60 \div 4$
商 15, 余り 0 | (27) $55 \div 9$
商 6, 余り 1 | (45) $47 \div 8$
商 5, 余り 7 | (63) $67 \div 4$
商 16, 余り 3 |
| (10) $58 \div 9$
商 6, 余り 4 | (28) $57 \div 2$
商 28, 余り 1 | (46) $96 \div 10$
商 9, 余り 6 | (64) $89 \div 10$
商 8, 余り 9 |
| (11) $31 \div 6$
商 5, 余り 1 | (29) $64 \div 1$
商 64, 余り 0 | (47) $76 \div 8$
商 9, 余り 4 | (65) $81 \div 7$
商 11, 余り 4 |
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商 53, 余り 0 | (30) $24 \div 4$
商 6, 余り 0 | (48) $10 \div 10$
商 1, 余り 0 | (66) $30 \div 3$
商 10, 余り 0 |
| (13) $79 \div 6$
商 13, 余り 1 | (31) $98 \div 1$
商 98, 余り 0 | (49) $38 \div 3$
商 12, 余り 2 | (67) $26 \div 5$
商 5, 余り 1 |
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商 57, 余り 0 | (32) $9 \div 4$
商 2, 余り 1 | (50) $75 \div 4$
商 18, 余り 3 | (68) $40 \div 3$
商 13, 余り 1 |
| (15) $87 \div 4$
商 21, 余り 3 | (33) $58 \div 3$
商 19, 余り 1 | (51) $67 \div 4$
商 16, 余り 3 | (69) $83 \div 7$
商 11, 余り 6 |
| (16) $35 \div 7$
商 5, 余り 0 | (34) $35 \div 4$
商 8, 余り 3 | (52) $4 \div 3$
商 1, 余り 1 | (70) $20 \div 10$
商 2, 余り 0 |
| (17) $31 \div 1$
商 31, 余り 0 | (35) $38 \div 6$
商 6, 余り 2 | (53) $22 \div 8$
商 2, 余り 6 | (71) $10 \div 5$
商 2, 余り 0 |
| (18) $86 \div 3$
商 28, 余り 2 | (36) $33 \div 8$
商 4, 余り 1 | (54) $8 \div 2$
商 4, 余り 0 | (72) $47 \div 1$
商 47, 余り 0 |

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|---------------------------------|--------------------------------|--------------------------------|--------------------------------|
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商 17, 余り 1 | (19) $71 \div 4$
商 17, 余り 3 | (37) $8 \div 2$
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商 8, 余り 6 |
| (2) $30 \div 5$
商 6, 余り 0 | (20) $84 \div 7$
商 12, 余り 0 | (38) $71 \div 4$
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商 4, 余り 5 |
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商 7, 余り 9 | (39) $78 \div 8$
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商 11, 余り 2 |
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商 5, 余り 6 | (40) $74 \div 9$
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商 3, 余り 7 |
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商 6, 余り 4 | (23) $52 \div 4$
商 13, 余り 0 | (41) $66 \div 10$
商 6, 余り 6 | (59) $4 \div 7$
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| (6) $4 \div 9$
商 0, 余り 4 | (24) $43 \div 5$
商 8, 余り 3 | (42) $66 \div 6$
商 11, 余り 0 | (60) $9 \div 5$
商 1, 余り 4 |
| (7) $8 \div 6$
商 1, 余り 2 | (25) $54 \div 1$
商 54, 余り 0 | (43) $75 \div 8$
商 9, 余り 3 | (61) $38 \div 3$
商 12, 余り 2 |
| (8) $100 \div 1$
商 100, 余り 0 | (26) $50 \div 5$
商 10, 余り 0 | (44) $84 \div 1$
商 84, 余り 0 | (62) $50 \div 7$
商 7, 余り 1 |
| (9) $42 \div 1$
商 42, 余り 0 | (27) $7 \div 7$
商 1, 余り 0 | (45) $54 \div 1$
商 54, 余り 0 | (63) $16 \div 1$
商 16, 余り 0 |
| (10) $56 \div 8$
商 7, 余り 0 | (28) $86 \div 9$
商 9, 余り 5 | (46) $79 \div 4$
商 19, 余り 3 | (64) $12 \div 9$
商 1, 余り 3 |
| (11) $9 \div 3$
商 3, 余り 0 | (29) $75 \div 3$
商 25, 余り 0 | (47) $55 \div 4$
商 13, 余り 3 | (65) $66 \div 6$
商 11, 余り 0 |
| (12) $61 \div 4$
商 15, 余り 1 | (30) $26 \div 1$
商 26, 余り 0 | (48) $99 \div 6$
商 16, 余り 3 | (66) $88 \div 6$
商 14, 余り 4 |
| (13) $62 \div 4$
商 15, 余り 2 | (31) $50 \div 10$
商 5, 余り 0 | (49) $99 \div 10$
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商 9, 余り 0 |
| (14) $71 \div 3$
商 23, 余り 2 | (32) $58 \div 9$
商 6, 余り 4 | (50) $83 \div 3$
商 27, 余り 2 | (68) $40 \div 8$
商 5, 余り 0 |
| (15) $44 \div 4$
商 11, 余り 0 | (33) $18 \div 7$
商 2, 余り 4 | (51) $63 \div 5$
商 12, 余り 3 | (69) $93 \div 1$
商 93, 余り 0 |
| (16) $97 \div 4$
商 24, 余り 1 | (34) $19 \div 1$
商 19, 余り 0 | (52) $62 \div 8$
商 7, 余り 6 | (70) $41 \div 5$
商 8, 余り 1 |
| (17) $41 \div 10$
商 4, 余り 1 | (35) $92 \div 6$
商 15, 余り 2 | (53) $42 \div 2$
商 21, 余り 0 | (71) $20 \div 10$
商 2, 余り 0 |
| (18) $76 \div 6$
商 12, 余り 4 | (36) $46 \div 5$
商 9, 余り 1 | (54) $26 \div 4$
商 6, 余り 2 | (72) $1 \div 1$
商 1, 余り 0 |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
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商 4, 余り 5 | (19) $98 \div 8$
商 12, 余り 2 | (37) $41 \div 6$
商 6, 余り 5 | (55) $12 \div 4$
商 3, 余り 0 |
| (2) $34 \div 6$
商 5, 余り 4 | (20) $88 \div 2$
商 44, 余り 0 | (38) $83 \div 10$
商 8, 余り 3 | (56) $85 \div 10$
商 8, 余り 5 |
| (3) $63 \div 8$
商 7, 余り 7 | (21) $90 \div 9$
商 10, 余り 0 | (39) $58 \div 4$
商 14, 余り 2 | (57) $29 \div 3$
商 9, 余り 2 |
| (4) $94 \div 3$
商 31, 余り 1 | (22) $51 \div 9$
商 5, 余り 6 | (40) $51 \div 10$
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商 9, 余り 6 |
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商 65, 余り 0 | (41) $28 \div 1$
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商 3, 余り 6 | (24) $61 \div 2$
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商 2, 余り 1 | (60) $50 \div 1$
商 50, 余り 0 |
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商 3, 余り 3 | (43) $30 \div 6$
商 5, 余り 0 | (61) $47 \div 10$
商 4, 余り 7 |
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商 10, 余り 1 |
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商 14, 余り 4 | (27) $90 \div 6$
商 15, 余り 0 | (45) $93 \div 1$
商 93, 余り 0 | (63) $83 \div 10$
商 8, 余り 3 |
| (10) $5 \div 1$
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商 17, 余り 1 | (46) $38 \div 5$
商 7, 余り 3 | (64) $34 \div 9$
商 3, 余り 7 |
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商 7, 余り 1 | (47) $27 \div 9$
商 3, 余り 0 | (65) $12 \div 7$
商 1, 余り 5 |
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商 12, 余り 2 | (30) $34 \div 6$
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商 7, 余り 1 | (66) $16 \div 5$
商 3, 余り 1 |
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商 2, 余り 6 | (49) $83 \div 3$
商 27, 余り 2 | (67) $99 \div 5$
商 19, 余り 4 |
| (14) $85 \div 1$
商 85, 余り 0 | (32) $1 \div 5$
商 0, 余り 1 | (50) $69 \div 7$
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商 55, 余り 0 |
| (15) $83 \div 5$
商 16, 余り 3 | (33) $84 \div 4$
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商 3, 余り 0 |
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商 0, 余り 4 | (53) $6 \div 8$
商 0, 余り 6 | (71) $50 \div 9$
商 5, 余り 5 |
| (18) $78 \div 4$
商 19, 余り 2 | (36) $15 \div 4$
商 3, 余り 3 | (54) $14 \div 7$
商 2, 余り 0 | (72) $0 \div 9$
商 0, 余り 0 |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
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商 4, 余り 8 | (37) $5 \div 1$
商 5, 余り 0 | (55) $23 \div 9$
商 2, 余り 5 |
| (2) $19 \div 4$
商 4, 余り 3 | (20) $84 \div 7$
商 12, 余り 0 | (38) $74 \div 10$
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| (7) $51 \div 7$
商 7, 余り 2 | (25) $67 \div 6$
商 11, 余り 1 | (43) $77 \div 4$
商 19, 余り 1 | (61) $36 \div 5$
商 7, 余り 1 |
| (8) $100 \div 2$
商 50, 余り 0 | (26) $27 \div 10$
商 2, 余り 7 | (44) $37 \div 6$
商 6, 余り 1 | (62) $48 \div 2$
商 24, 余り 0 |
| (9) $79 \div 3$
商 26, 余り 1 | (27) $62 \div 9$
商 6, 余り 8 | (45) $37 \div 9$
商 4, 余り 1 | (63) $36 \div 3$
商 12, 余り 0 |
| (10) $7 \div 9$
商 0, 余り 7 | (28) $39 \div 1$
商 39, 余り 0 | (46) $64 \div 7$
商 9, 余り 1 | (64) $66 \div 3$
商 22, 余り 0 |
| (11) $30 \div 5$
商 6, 余り 0 | (29) $87 \div 9$
商 9, 余り 6 | (47) $77 \div 4$
商 19, 余り 1 | (65) $21 \div 6$
商 3, 余り 3 |
| (12) $97 \div 8$
商 12, 余り 1 | (30) $34 \div 1$
商 34, 余り 0 | (48) $33 \div 2$
商 16, 余り 1 | (66) $89 \div 3$
商 29, 余り 2 |
| (13) $55 \div 9$
商 6, 余り 1 | (31) $17 \div 2$
商 8, 余り 1 | (49) $70 \div 9$
商 7, 余り 7 | (67) $47 \div 7$
商 6, 余り 5 |
| (14) $7 \div 10$
商 0, 余り 7 | (32) $42 \div 2$
商 21, 余り 0 | (50) $84 \div 5$
商 16, 余り 4 | (68) $38 \div 8$
商 4, 余り 6 |
| (15) $63 \div 8$
商 7, 余り 7 | (33) $60 \div 9$
商 6, 余り 6 | (51) $32 \div 2$
商 16, 余り 0 | (69) $7 \div 10$
商 0, 余り 7 |
| (16) $3 \div 10$
商 0, 余り 3 | (34) $9 \div 1$
商 9, 余り 0 | (52) $12 \div 1$
商 12, 余り 0 | (70) $34 \div 2$
商 17, 余り 0 |
| (17) $6 \div 8$
商 0, 余り 6 | (35) $41 \div 10$
商 4, 余り 1 | (53) $36 \div 10$
商 3, 余り 6 | (71) $75 \div 3$
商 25, 余り 0 |
| (18) $66 \div 5$
商 13, 余り 1 | (36) $89 \div 2$
商 44, 余り 1 | (54) $79 \div 2$
商 39, 余り 1 | (72) $78 \div 4$
商 19, 余り 2 |

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|--------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $51 \div 10$
商 5, 余り 1 | (19) $78 \div 1$
商 78, 余り 0 | (37) $52 \div 7$
商 7, 余り 3 | (55) $27 \div 8$
商 3, 余り 3 |
| (2) $21 \div 5$
商 4, 余り 1 | (20) $30 \div 9$
商 3, 余り 3 | (38) $34 \div 6$
商 5, 余り 4 | (56) $92 \div 3$
商 30, 余り 2 |
| (3) $97 \div 5$
商 19, 余り 2 | (21) $36 \div 9$
商 4, 余り 0 | (39) $52 \div 2$
商 26, 余り 0 | (57) $38 \div 9$
商 4, 余り 2 |
| (4) $54 \div 10$
商 5, 余り 4 | (22) $93 \div 7$
商 13, 余り 2 | (40) $12 \div 10$
商 1, 余り 2 | (58) $12 \div 9$
商 1, 余り 3 |
| (5) $32 \div 4$
商 8, 余り 0 | (23) $16 \div 9$
商 1, 余り 7 | (41) $91 \div 10$
商 9, 余り 1 | (59) $80 \div 6$
商 13, 余り 2 |
| (6) $92 \div 1$
商 92, 余り 0 | (24) $99 \div 9$
商 11, 余り 0 | (42) $60 \div 8$
商 7, 余り 4 | (60) $7 \div 9$
商 0, 余り 7 |
| (7) $59 \div 9$
商 6, 余り 5 | (25) $75 \div 7$
商 10, 余り 5 | (43) $71 \div 6$
商 11, 余り 5 | (61) $96 \div 10$
商 9, 余り 6 |
| (8) $12 \div 8$
商 1, 余り 4 | (26) $3 \div 7$
商 0, 余り 3 | (44) $5 \div 5$
商 1, 余り 0 | (62) $0 \div 9$
商 0, 余り 0 |
| (9) $84 \div 6$
商 14, 余り 0 | (27) $88 \div 10$
商 8, 余り 8 | (45) $17 \div 8$
商 2, 余り 1 | (63) $6 \div 4$
商 1, 余り 2 |
| (10) $72 \div 5$
商 14, 余り 2 | (28) $7 \div 3$
商 2, 余り 1 | (46) $9 \div 5$
商 1, 余り 4 | (64) $88 \div 10$
商 8, 余り 8 |
| (11) $52 \div 5$
商 10, 余り 2 | (29) $27 \div 5$
商 5, 余り 2 | (47) $41 \div 4$
商 10, 余り 1 | (65) $85 \div 8$
商 10, 余り 5 |
| (12) $99 \div 5$
商 19, 余り 4 | (30) $35 \div 3$
商 11, 余り 2 | (48) $49 \div 7$
商 7, 余り 0 | (66) $31 \div 3$
商 10, 余り 1 |
| (13) $63 \div 5$
商 12, 余り 3 | (31) $33 \div 9$
商 3, 余り 6 | (49) $73 \div 3$
商 24, 余り 1 | (67) $48 \div 8$
商 6, 余り 0 |
| (14) $18 \div 10$
商 1, 余り 8 | (32) $53 \div 9$
商 5, 余り 8 | (50) $26 \div 9$
商 2, 余り 8 | (68) $62 \div 7$
商 8, 余り 6 |
| (15) $50 \div 1$
商 50, 余り 0 | (33) $64 \div 9$
商 7, 余り 1 | (51) $74 \div 3$
商 24, 余り 2 | (69) $100 \div 3$
商 33, 余り 1 |
| (16) $57 \div 1$
商 57, 余り 0 | (34) $69 \div 6$
商 11, 余り 3 | (52) $4 \div 10$
商 0, 余り 4 | (70) $56 \div 7$
商 8, 余り 0 |
| (17) $96 \div 3$
商 32, 余り 0 | (35) $39 \div 5$
商 7, 余り 4 | (53) $38 \div 5$
商 7, 余り 3 | (71) $69 \div 5$
商 13, 余り 4 |
| (18) $36 \div 8$
商 4, 余り 4 | (36) $16 \div 3$
商 5, 余り 1 | (54) $58 \div 4$
商 14, 余り 2 | (72) $10 \div 3$
商 3, 余り 1 |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $80 \div 10$
商 8, 余り 0 | (19) $20 \div 8$
商 2, 余り 4 | (37) $55 \div 5$
商 11, 余り 0 | (55) $75 \div 6$
商 12, 余り 3 |
| (2) $99 \div 7$
商 14, 余り 1 | (20) $58 \div 7$
商 8, 余り 2 | (38) $81 \div 3$
商 27, 余り 0 | (56) $67 \div 1$
商 67, 余り 0 |
| (3) $12 \div 1$
商 12, 余り 0 | (21) $13 \div 10$
商 1, 余り 3 | (39) $5 \div 8$
商 0, 余り 5 | (57) $15 \div 4$
商 3, 余り 3 |
| (4) $17 \div 3$
商 5, 余り 2 | (22) $14 \div 6$
商 2, 余り 2 | (40) $89 \div 5$
商 17, 余り 4 | (58) $73 \div 4$
商 18, 余り 1 |
| (5) $38 \div 9$
商 4, 余り 2 | (23) $26 \div 10$
商 2, 余り 6 | (41) $72 \div 3$
商 24, 余り 0 | (59) $59 \div 7$
商 8, 余り 3 |
| (6) $22 \div 4$
商 5, 余り 2 | (24) $55 \div 8$
商 6, 余り 7 | (42) $18 \div 8$
商 2, 余り 2 | (60) $78 \div 5$
商 15, 余り 3 |
| (7) $4 \div 6$
商 0, 余り 4 | (25) $13 \div 8$
商 1, 余り 5 | (43) $82 \div 8$
商 10, 余り 2 | (61) $31 \div 7$
商 4, 余り 3 |
| (8) $75 \div 1$
商 75, 余り 0 | (26) $78 \div 10$
商 7, 余り 8 | (44) $71 \div 3$
商 23, 余り 2 | (62) $65 \div 8$
商 8, 余り 1 |
| (9) $38 \div 10$
商 3, 余り 8 | (27) $97 \div 1$
商 97, 余り 0 | (45) $82 \div 4$
商 20, 余り 2 | (63) $42 \div 5$
商 8, 余り 2 |
| (10) $93 \div 9$
商 10, 余り 3 | (28) $27 \div 10$
商 2, 余り 7 | (46) $48 \div 4$
商 12, 余り 0 | (64) $50 \div 3$
商 16, 余り 2 |
| (11) $73 \div 3$
商 24, 余り 1 | (29) $0 \div 2$
商 0, 余り 0 | (47) $74 \div 1$
商 74, 余り 0 | (65) $5 \div 3$
商 1, 余り 2 |
| (12) $74 \div 6$
商 12, 余り 2 | (30) $50 \div 7$
商 7, 余り 1 | (48) $15 \div 8$
商 1, 余り 7 | (66) $77 \div 6$
商 12, 余り 5 |
| (13) $21 \div 10$
商 2, 余り 1 | (31) $83 \div 8$
商 10, 余り 3 | (49) $29 \div 7$
商 4, 余り 1 | (67) $28 \div 1$
商 28, 余り 0 |
| (14) $27 \div 10$
商 2, 余り 7 | (32) $98 \div 8$
商 12, 余り 2 | (50) $96 \div 5$
商 19, 余り 1 | (68) $24 \div 7$
商 3, 余り 3 |
| (15) $55 \div 2$
商 27, 余り 1 | (33) $72 \div 7$
商 10, 余り 2 | (51) $64 \div 1$
商 64, 余り 0 | (69) $22 \div 1$
商 22, 余り 0 |
| (16) $21 \div 8$
商 2, 余り 5 | (34) $40 \div 7$
商 5, 余り 5 | (52) $74 \div 9$
商 8, 余り 2 | (70) $12 \div 3$
商 4, 余り 0 |
| (17) $8 \div 10$
商 0, 余り 8 | (35) $83 \div 1$
商 83, 余り 0 | (53) $54 \div 8$
商 6, 余り 6 | (71) $89 \div 3$
商 29, 余り 2 |
| (18) $29 \div 1$
商 29, 余り 0 | (36) $57 \div 3$
商 19, 余り 0 | (54) $9 \div 3$
商 3, 余り 0 | (72) $13 \div 9$
商 1, 余り 4 |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $38 \div 1$
商 38, 余り 0 | (19) $91 \div 6$
商 15, 余り 1 | (37) $65 \div 7$
商 9, 余り 2 | (55) $67 \div 9$
商 7, 余り 4 |
| (2) $59 \div 5$
商 11, 余り 4 | (20) $51 \div 4$
商 12, 余り 3 | (38) $25 \div 7$
商 3, 余り 4 | (56) $95 \div 7$
商 13, 余り 4 |
| (3) $33 \div 5$
商 6, 余り 3 | (21) $0 \div 5$
商 0, 余り 0 | (39) $68 \div 3$
商 22, 余り 2 | (57) $66 \div 6$
商 11, 余り 0 |
| (4) $89 \div 8$
商 11, 余り 1 | (22) $49 \div 10$
商 4, 余り 9 | (40) $97 \div 2$
商 48, 余り 1 | (58) $62 \div 10$
商 6, 余り 2 |
| (5) $70 \div 7$
商 10, 余り 0 | (23) $92 \div 6$
商 15, 余り 2 | (41) $41 \div 9$
商 4, 余り 5 | (59) $84 \div 8$
商 10, 余り 4 |
| (6) $39 \div 3$
商 13, 余り 0 | (24) $1 \div 6$
商 0, 余り 1 | (42) $41 \div 9$
商 4, 余り 5 | (60) $50 \div 2$
商 25, 余り 0 |
| (7) $22 \div 1$
商 22, 余り 0 | (25) $12 \div 2$
商 6, 余り 0 | (43) $97 \div 8$
商 12, 余り 1 | (61) $81 \div 8$
商 10, 余り 1 |
| (8) $76 \div 7$
商 10, 余り 6 | (26) $37 \div 2$
商 18, 余り 1 | (44) $39 \div 3$
商 13, 余り 0 | (62) $91 \div 4$
商 22, 余り 3 |
| (9) $59 \div 6$
商 9, 余り 5 | (27) $77 \div 8$
商 9, 余り 5 | (45) $21 \div 8$
商 2, 余り 5 | (63) $61 \div 7$
商 8, 余り 5 |
| (10) $79 \div 1$
商 79, 余り 0 | (28) $29 \div 5$
商 5, 余り 4 | (46) $68 \div 3$
商 22, 余り 2 | (64) $14 \div 8$
商 1, 余り 6 |
| (11) $42 \div 1$
商 42, 余り 0 | (29) $75 \div 10$
商 7, 余り 5 | (47) $77 \div 1$
商 77, 余り 0 | (65) $67 \div 3$
商 22, 余り 1 |
| (12) $33 \div 10$
商 3, 余り 3 | (30) $11 \div 10$
商 1, 余り 1 | (48) $21 \div 4$
商 5, 余り 1 | (66) $98 \div 4$
商 24, 余り 2 |
| (13) $58 \div 6$
商 9, 余り 4 | (31) $68 \div 1$
商 68, 余り 0 | (49) $80 \div 1$
商 80, 余り 0 | (67) $52 \div 9$
商 5, 余り 7 |
| (14) $20 \div 1$
商 20, 余り 0 | (32) $19 \div 3$
商 6, 余り 1 | (50) $35 \div 4$
商 8, 余り 3 | (68) $70 \div 6$
商 11, 余り 4 |
| (15) $81 \div 8$
商 10, 余り 1 | (33) $8 \div 9$
商 0, 余り 8 | (51) $77 \div 2$
商 38, 余り 1 | (69) $86 \div 6$
商 14, 余り 2 |
| (16) $61 \div 2$
商 30, 余り 1 | (34) $84 \div 7$
商 12, 余り 0 | (52) $97 \div 4$
商 24, 余り 1 | (70) $23 \div 8$
商 2, 余り 7 |
| (17) $9 \div 10$
商 0, 余り 9 | (35) $37 \div 4$
商 9, 余り 1 | (53) $78 \div 2$
商 39, 余り 0 | (71) $48 \div 8$
商 6, 余り 0 |
| (18) $43 \div 5$
商 8, 余り 3 | (36) $78 \div 10$
商 7, 余り 8 | (54) $25 \div 4$
商 6, 余り 1 | (72) $63 \div 6$
商 10, 余り 3 |

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|--------------------------------|--------------------------------|---------------------------------|---------------------------------|
| (1) $42 \div 8$
商 5, 余り 2 | (19) $32 \div 2$
商 16, 余り 0 | (37) $68 \div 4$
商 17, 余り 0 | (55) $92 \div 1$
商 92, 余り 0 |
| (2) $1 \div 5$
商 0, 余り 1 | (20) $97 \div 4$
商 24, 余り 1 | (38) $57 \div 2$
商 28, 余り 1 | (56) $50 \div 9$
商 5, 余り 5 |
| (3) $44 \div 3$
商 14, 余り 2 | (21) $2 \div 10$
商 0, 余り 2 | (39) $52 \div 8$
商 6, 余り 4 | (57) $6 \div 1$
商 6, 余り 0 |
| (4) $12 \div 2$
商 6, 余り 0 | (22) $4 \div 8$
商 0, 余り 4 | (40) $48 \div 2$
商 24, 余り 0 | (58) $86 \div 1$
商 86, 余り 0 |
| (5) $58 \div 5$
商 11, 余り 3 | (23) $96 \div 4$
商 24, 余り 0 | (41) $85 \div 3$
商 28, 余り 1 | (59) $46 \div 1$
商 46, 余り 0 |
| (6) $7 \div 10$
商 0, 余り 7 | (24) $9 \div 3$
商 3, 余り 0 | (42) $69 \div 7$
商 9, 余り 6 | (60) $87 \div 4$
商 21, 余り 3 |
| (7) $75 \div 2$
商 37, 余り 1 | (25) $76 \div 3$
商 25, 余り 1 | (43) $70 \div 5$
商 14, 余り 0 | (61) $79 \div 10$
商 7, 余り 9 |
| (8) $5 \div 1$
商 5, 余り 0 | (26) $91 \div 3$
商 30, 余り 1 | (44) $45 \div 3$
商 15, 余り 0 | (62) $100 \div 5$
商 20, 余り 0 |
| (9) $87 \div 2$
商 43, 余り 1 | (27) $72 \div 8$
商 9, 余り 0 | (45) $100 \div 2$
商 50, 余り 0 | (63) $69 \div 7$
商 9, 余り 6 |
| (10) $30 \div 6$
商 5, 余り 0 | (28) $35 \div 2$
商 17, 余り 1 | (46) $85 \div 4$
商 21, 余り 1 | (64) $19 \div 4$
商 4, 余り 3 |
| (11) $22 \div 10$
商 2, 余り 2 | (29) $57 \div 1$
商 57, 余り 0 | (47) $7 \div 10$
商 0, 余り 7 | (65) $11 \div 5$
商 2, 余り 1 |
| (12) $48 \div 1$
商 48, 余り 0 | (30) $58 \div 9$
商 6, 余り 4 | (48) $4 \div 1$
商 4, 余り 0 | (66) $83 \div 6$
商 13, 余り 5 |
| (13) $77 \div 9$
商 8, 余り 5 | (31) $40 \div 4$
商 10, 余り 0 | (49) $11 \div 6$
商 1, 余り 5 | (67) $71 \div 5$
商 14, 余り 1 |
| (14) $55 \div 10$
商 5, 余り 5 | (32) $21 \div 6$
商 3, 余り 3 | (50) $21 \div 7$
商 3, 余り 0 | (68) $33 \div 1$
商 33, 余り 0 |
| (15) $7 \div 1$
商 7, 余り 0 | (33) $16 \div 10$
商 1, 余り 6 | (51) $79 \div 6$
商 13, 余り 1 | (69) $41 \div 3$
商 13, 余り 2 |
| (16) $74 \div 3$
商 24, 余り 2 | (34) $10 \div 4$
商 2, 余り 2 | (52) $82 \div 3$
商 27, 余り 1 | (70) $25 \div 5$
商 5, 余り 0 |
| (17) $65 \div 5$
商 13, 余り 0 | (35) $47 \div 5$
商 9, 余り 2 | (53) $53 \div 4$
商 13, 余り 1 | (71) $95 \div 8$
商 11, 余り 7 |
| (18) $82 \div 3$
商 27, 余り 1 | (36) $5 \div 5$
商 1, 余り 0 | (54) $39 \div 6$
商 6, 余り 3 | (72) $36 \div 1$
商 36, 余り 0 |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $41 \div 4$
商 10, 余り 1 | (19) $67 \div 4$
商 16, 余り 3 | (37) $18 \div 4$
商 4, 余り 2 | (55) $4 \div 10$
商 0, 余り 4 |
| (2) $25 \div 8$
商 3, 余り 1 | (20) $78 \div 6$
商 13, 余り 0 | (38) $4 \div 2$
商 2, 余り 0 | (56) $90 \div 9$
商 10, 余り 0 |
| (3) $99 \div 7$
商 14, 余り 1 | (21) $46 \div 6$
商 7, 余り 4 | (39) $64 \div 5$
商 12, 余り 4 | (57) $66 \div 6$
商 11, 余り 0 |
| (4) $86 \div 8$
商 10, 余り 6 | (22) $27 \div 2$
商 13, 余り 1 | (40) $10 \div 5$
商 2, 余り 0 | (58) $24 \div 5$
商 4, 余り 4 |
| (5) $36 \div 9$
商 4, 余り 0 | (23) $92 \div 3$
商 30, 余り 2 | (41) $47 \div 7$
商 6, 余り 5 | (59) $6 \div 3$
商 2, 余り 0 |
| (6) $12 \div 6$
商 2, 余り 0 | (24) $37 \div 5$
商 7, 余り 2 | (42) $52 \div 3$
商 17, 余り 1 | (60) $3 \div 8$
商 0, 余り 3 |
| (7) $95 \div 4$
商 23, 余り 3 | (25) $35 \div 8$
商 4, 余り 3 | (43) $11 \div 4$
商 2, 余り 3 | (61) $7 \div 5$
商 1, 余り 2 |
| (8) $20 \div 9$
商 2, 余り 2 | (26) $95 \div 6$
商 15, 余り 5 | (44) $65 \div 5$
商 13, 余り 0 | (62) $13 \div 10$
商 1, 余り 3 |
| (9) $10 \div 5$
商 2, 余り 0 | (27) $43 \div 2$
商 21, 余り 1 | (45) $95 \div 4$
商 23, 余り 3 | (63) $27 \div 3$
商 9, 余り 0 |
| (10) $67 \div 4$
商 16, 余り 3 | (28) $4 \div 7$
商 0, 余り 4 | (46) $42 \div 5$
商 8, 余り 2 | (64) $17 \div 4$
商 4, 余り 1 |
| (11) $8 \div 5$
商 1, 余り 3 | (29) $89 \div 7$
商 12, 余り 5 | (47) $67 \div 7$
商 9, 余り 4 | (65) $52 \div 1$
商 52, 余り 0 |
| (12) $66 \div 1$
商 66, 余り 0 | (30) $61 \div 4$
商 15, 余り 1 | (48) $12 \div 2$
商 6, 余り 0 | (66) $83 \div 2$
商 41, 余り 1 |
| (13) $65 \div 5$
商 13, 余り 0 | (31) $17 \div 1$
商 17, 余り 0 | (49) $29 \div 4$
商 7, 余り 1 | (67) $88 \div 1$
商 88, 余り 0 |
| (14) $9 \div 4$
商 2, 余り 1 | (32) $70 \div 7$
商 10, 余り 0 | (50) $25 \div 4$
商 6, 余り 1 | (68) $98 \div 4$
商 24, 余り 2 |
| (15) $58 \div 2$
商 29, 余り 0 | (33) $8 \div 3$
商 2, 余り 2 | (51) $39 \div 6$
商 6, 余り 3 | (69) $96 \div 8$
商 12, 余り 0 |
| (16) $2 \div 8$
商 0, 余り 2 | (34) $63 \div 10$
商 6, 余り 3 | (52) $82 \div 1$
商 82, 余り 0 | (70) $17 \div 8$
商 2, 余り 1 |
| (17) $48 \div 9$
商 5, 余り 3 | (35) $28 \div 2$
商 14, 余り 0 | (53) $79 \div 5$
商 15, 余り 4 | (71) $36 \div 3$
商 12, 余り 0 |
| (18) $77 \div 8$
商 9, 余り 5 | (36) $23 \div 1$
商 23, 余り 0 | (54) $46 \div 4$
商 11, 余り 2 | (72) $44 \div 8$
商 5, 余り 4 |

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|--------------------------------|---------------------------------|--------------------------------|---------------------------------|
| (1) $51 \div 7$
商 7, 余り 2 | (19) $97 \div 10$
商 9, 余り 7 | (37) $37 \div 10$
商 3, 余り 7 | (55) $80 \div 2$
商 40, 余り 0 |
| (2) $98 \div 8$
商 12, 余り 2 | (20) $77 \div 3$
商 25, 余り 2 | (38) $26 \div 1$
商 26, 余り 0 | (56) $59 \div 8$
商 7, 余り 3 |
| (3) $20 \div 1$
商 20, 余り 0 | (21) $18 \div 10$
商 1, 余り 8 | (39) $44 \div 10$
商 4, 余り 4 | (57) $38 \div 5$
商 7, 余り 3 |
| (4) $12 \div 8$
商 1, 余り 4 | (22) $41 \div 3$
商 13, 余り 2 | (40) $84 \div 6$
商 14, 余り 0 | (58) $88 \div 6$
商 14, 余り 4 |
| (5) $86 \div 4$
商 21, 余り 2 | (23) $31 \div 2$
商 15, 余り 1 | (41) $51 \div 9$
商 5, 余り 6 | (59) $95 \div 7$
商 13, 余り 4 |
| (6) $99 \div 9$
商 11, 余り 0 | (24) $66 \div 6$
商 11, 余り 0 | (42) $56 \div 3$
商 18, 余り 2 | (60) $16 \div 2$
商 8, 余り 0 |
| (7) $3 \div 1$
商 3, 余り 0 | (25) $73 \div 4$
商 18, 余り 1 | (43) $45 \div 3$
商 15, 余り 0 | (61) $35 \div 8$
商 4, 余り 3 |
| (8) $28 \div 7$
商 4, 余り 0 | (26) $63 \div 5$
商 12, 余り 3 | (44) $82 \div 10$
商 8, 余り 2 | (62) $100 \div 8$
商 12, 余り 4 |
| (9) $97 \div 8$
商 12, 余り 1 | (27) $11 \div 5$
商 2, 余り 1 | (45) $58 \div 4$
商 14, 余り 2 | (63) $85 \div 4$
商 21, 余り 1 |
| (10) $28 \div 6$
商 4, 余り 4 | (28) $65 \div 4$
商 16, 余り 1 | (46) $40 \div 10$
商 4, 余り 0 | (64) $78 \div 5$
商 15, 余り 3 |
| (11) $40 \div 8$
商 5, 余り 0 | (29) $70 \div 6$
商 11, 余り 4 | (47) $41 \div 5$
商 8, 余り 1 | (65) $13 \div 3$
商 4, 余り 1 |
| (12) $26 \div 5$
商 5, 余り 1 | (30) $53 \div 6$
商 8, 余り 5 | (48) $98 \div 2$
商 49, 余り 0 | (66) $15 \div 2$
商 7, 余り 1 |
| (13) $52 \div 1$
商 52, 余り 0 | (31) $29 \div 4$
商 7, 余り 1 | (49) $65 \div 9$
商 7, 余り 2 | (67) $50 \div 5$
商 10, 余り 0 |
| (14) $7 \div 9$
商 0, 余り 7 | (32) $52 \div 5$
商 10, 余り 2 | (50) $11 \div 9$
商 1, 余り 2 | (68) $79 \div 10$
商 7, 余り 9 |
| (15) $2 \div 7$
商 0, 余り 2 | (33) $40 \div 9$
商 4, 余り 4 | (51) $64 \div 7$
商 9, 余り 1 | (69) $52 \div 5$
商 10, 余り 2 |
| (16) $65 \div 1$
商 65, 余り 0 | (34) $15 \div 5$
商 3, 余り 0 | (52) $82 \div 6$
商 13, 余り 4 | (70) $54 \div 9$
商 6, 余り 0 |
| (17) $40 \div 9$
商 4, 余り 4 | (35) $100 \div 8$
商 12, 余り 4 | (53) $87 \div 5$
商 17, 余り 2 | (71) $5 \div 8$
商 0, 余り 5 |
| (18) $2 \div 6$
商 0, 余り 2 | (36) $8 \div 10$
商 0, 余り 8 | (54) $51 \div 2$
商 25, 余り 1 | (72) $11 \div 3$
商 3, 余り 2 |

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|--------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $9 \div 6$
商 1, 余り 3 | (19) $25 \div 5$
商 5, 余り 0 | (37) $39 \div 3$
商 13, 余り 0 | (55) $21 \div 6$
商 3, 余り 3 |
| (2) $64 \div 6$
商 10, 余り 4 | (20) $81 \div 3$
商 27, 余り 0 | (38) $7 \div 3$
商 2, 余り 1 | (56) $6 \div 7$
商 0, 余り 6 |
| (3) $22 \div 5$
商 4, 余り 2 | (21) $23 \div 8$
商 2, 余り 7 | (39) $79 \div 10$
商 7, 余り 9 | (57) $3 \div 3$
商 1, 余り 0 |
| (4) $35 \div 10$
商 3, 余り 5 | (22) $96 \div 10$
商 9, 余り 6 | (40) $14 \div 10$
商 1, 余り 4 | (58) $45 \div 8$
商 5, 余り 5 |
| (5) $85 \div 3$
商 28, 余り 1 | (23) $24 \div 10$
商 2, 余り 4 | (41) $82 \div 7$
商 11, 余り 5 | (59) $82 \div 2$
商 41, 余り 0 |
| (6) $99 \div 9$
商 11, 余り 0 | (24) $17 \div 3$
商 5, 余り 2 | (42) $52 \div 7$
商 7, 余り 3 | (60) $9 \div 6$
商 1, 余り 3 |
| (7) $17 \div 10$
商 1, 余り 7 | (25) $34 \div 10$
商 3, 余り 4 | (43) $19 \div 10$
商 1, 余り 9 | (61) $82 \div 10$
商 8, 余り 2 |
| (8) $76 \div 7$
商 10, 余り 6 | (26) $6 \div 8$
商 0, 余り 6 | (44) $68 \div 3$
商 22, 余り 2 | (62) $12 \div 8$
商 1, 余り 4 |
| (9) $97 \div 6$
商 16, 余り 1 | (27) $59 \div 5$
商 11, 余り 4 | (45) $90 \div 1$
商 90, 余り 0 | (63) $99 \div 1$
商 99, 余り 0 |
| (10) $1 \div 7$
商 0, 余り 1 | (28) $14 \div 4$
商 3, 余り 2 | (46) $84 \div 8$
商 10, 余り 4 | (64) $4 \div 8$
商 0, 余り 4 |
| (11) $6 \div 8$
商 0, 余り 6 | (29) $54 \div 1$
商 54, 余り 0 | (47) $81 \div 4$
商 20, 余り 1 | (65) $7 \div 2$
商 3, 余り 1 |
| (12) $77 \div 5$
商 15, 余り 2 | (30) $97 \div 6$
商 16, 余り 1 | (48) $5 \div 7$
商 0, 余り 5 | (66) $40 \div 8$
商 5, 余り 0 |
| (13) $27 \div 8$
商 3, 余り 3 | (31) $30 \div 7$
商 4, 余り 2 | (49) $37 \div 6$
商 6, 余り 1 | (67) $35 \div 3$
商 11, 余り 2 |
| (14) $65 \div 10$
商 6, 余り 5 | (32) $49 \div 7$
商 7, 余り 0 | (50) $83 \div 6$
商 13, 余り 5 | (68) $24 \div 10$
商 2, 余り 4 |
| (15) $92 \div 6$
商 15, 余り 2 | (33) $16 \div 8$
商 2, 余り 0 | (51) $6 \div 6$
商 1, 余り 0 | (69) $4 \div 4$
商 1, 余り 0 |
| (16) $85 \div 6$
商 14, 余り 1 | (34) $70 \div 4$
商 17, 余り 2 | (52) $40 \div 8$
商 5, 余り 0 | (70) $38 \div 8$
商 4, 余り 6 |
| (17) $64 \div 2$
商 32, 余り 0 | (35) $67 \div 9$
商 7, 余り 4 | (53) $40 \div 9$
商 4, 余り 4 | (71) $79 \div 1$
商 79, 余り 0 |
| (18) $86 \div 3$
商 28, 余り 2 | (36) $16 \div 8$
商 2, 余り 0 | (54) $92 \div 4$
商 23, 余り 0 | (72) $100 \div 8$
商 12, 余り 4 |

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|--------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $2 \div 6$
商 0, 余り 2 | (19) $77 \div 2$
商 38, 余り 1 | (37) $15 \div 8$
商 1, 余り 7 | (55) $86 \div 4$
商 21, 余り 2 |
| (2) $38 \div 10$
商 3, 余り 8 | (20) $83 \div 2$
商 41, 余り 1 | (38) $48 \div 3$
商 16, 余り 0 | (56) $28 \div 9$
商 3, 余り 1 |
| (3) $92 \div 8$
商 11, 余り 4 | (21) $71 \div 9$
商 7, 余り 8 | (39) $67 \div 5$
商 13, 余り 2 | (57) $75 \div 5$
商 15, 余り 0 |
| (4) $46 \div 6$
商 7, 余り 4 | (22) $32 \div 2$
商 16, 余り 0 | (40) $75 \div 7$
商 10, 余り 5 | (58) $15 \div 5$
商 3, 余り 0 |
| (5) $90 \div 1$
商 90, 余り 0 | (23) $95 \div 6$
商 15, 余り 5 | (41) $27 \div 5$
商 5, 余り 2 | (59) $99 \div 8$
商 12, 余り 3 |
| (6) $5 \div 1$
商 5, 余り 0 | (24) $46 \div 4$
商 11, 余り 2 | (42) $37 \div 1$
商 37, 余り 0 | (60) $56 \div 4$
商 14, 余り 0 |
| (7) $56 \div 10$
商 5, 余り 6 | (25) $61 \div 9$
商 6, 余り 7 | (43) $2 \div 7$
商 0, 余り 2 | (61) $89 \div 6$
商 14, 余り 5 |
| (8) $44 \div 9$
商 4, 余り 8 | (26) $93 \div 10$
商 9, 余り 3 | (44) $70 \div 3$
商 23, 余り 1 | (62) $18 \div 5$
商 3, 余り 3 |
| (9) $2 \div 8$
商 0, 余り 2 | (27) $56 \div 10$
商 5, 余り 6 | (45) $4 \div 7$
商 0, 余り 4 | (63) $68 \div 10$
商 6, 余り 8 |
| (10) $78 \div 4$
商 19, 余り 2 | (28) $69 \div 2$
商 34, 余り 1 | (46) $19 \div 5$
商 3, 余り 4 | (64) $53 \div 10$
商 5, 余り 3 |
| (11) $66 \div 1$
商 66, 余り 0 | (29) $94 \div 5$
商 18, 余り 4 | (47) $26 \div 10$
商 2, 余り 6 | (65) $33 \div 5$
商 6, 余り 3 |
| (12) $29 \div 10$
商 2, 余り 9 | (30) $93 \div 7$
商 13, 余り 2 | (48) $22 \div 2$
商 11, 余り 0 | (66) $30 \div 1$
商 30, 余り 0 |
| (13) $48 \div 8$
商 6, 余り 0 | (31) $91 \div 9$
商 10, 余り 1 | (49) $40 \div 3$
商 13, 余り 1 | (67) $92 \div 8$
商 11, 余り 4 |
| (14) $16 \div 4$
商 4, 余り 0 | (32) $59 \div 9$
商 6, 余り 5 | (50) $69 \div 10$
商 6, 余り 9 | (68) $100 \div 7$
商 14, 余り 2 |
| (15) $75 \div 7$
商 10, 余り 5 | (33) $65 \div 1$
商 65, 余り 0 | (51) $6 \div 2$
商 3, 余り 0 | (69) $58 \div 8$
商 7, 余り 2 |
| (16) $46 \div 9$
商 5, 余り 1 | (34) $12 \div 7$
商 1, 余り 5 | (52) $74 \div 4$
商 18, 余り 2 | (70) $16 \div 4$
商 4, 余り 0 |
| (17) $58 \div 5$
商 11, 余り 3 | (35) $45 \div 3$
商 15, 余り 0 | (53) $59 \div 1$
商 59, 余り 0 | (71) $37 \div 4$
商 9, 余り 1 |
| (18) $33 \div 7$
商 4, 余り 5 | (36) $54 \div 4$
商 13, 余り 2 | (54) $73 \div 9$
商 8, 余り 1 | (72) $89 \div 8$
商 11, 余り 1 |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $100 \div 4$
商 25, 余り 0 | (19) $79 \div 8$
商 9, 余り 7 | (37) $82 \div 9$
商 9, 余り 1 | (55) $21 \div 1$
商 21, 余り 0 |
| (2) $75 \div 10$
商 7, 余り 5 | (20) $53 \div 7$
商 7, 余り 4 | (38) $80 \div 3$
商 26, 余り 2 | (56) $71 \div 7$
商 10, 余り 1 |
| (3) $29 \div 5$
商 5, 余り 4 | (21) $1 \div 10$
商 0, 余り 1 | (39) $3 \div 3$
商 1, 余り 0 | (57) $66 \div 3$
商 22, 余り 0 |
| (4) $82 \div 6$
商 13, 余り 4 | (22) $27 \div 5$
商 5, 余り 2 | (40) $66 \div 7$
商 9, 余り 3 | (58) $94 \div 8$
商 11, 余り 6 |
| (5) $30 \div 10$
商 3, 余り 0 | (23) $16 \div 2$
商 8, 余り 0 | (41) $67 \div 9$
商 7, 余り 4 | (59) $68 \div 9$
商 7, 余り 5 |
| (6) $81 \div 3$
商 27, 余り 0 | (24) $55 \div 8$
商 6, 余り 7 | (42) $81 \div 8$
商 10, 余り 1 | (60) $51 \div 1$
商 51, 余り 0 |
| (7) $58 \div 4$
商 14, 余り 2 | (25) $25 \div 2$
商 12, 余り 1 | (43) $42 \div 3$
商 14, 余り 0 | (61) $74 \div 6$
商 12, 余り 2 |
| (8) $8 \div 8$
商 1, 余り 0 | (26) $52 \div 7$
商 7, 余り 3 | (44) $93 \div 5$
商 18, 余り 3 | (62) $23 \div 5$
商 4, 余り 3 |
| (9) $58 \div 1$
商 58, 余り 0 | (27) $27 \div 3$
商 9, 余り 0 | (45) $51 \div 6$
商 8, 余り 3 | (63) $79 \div 1$
商 79, 余り 0 |
| (10) $38 \div 2$
商 19, 余り 0 | (28) $29 \div 10$
商 2, 余り 9 | (46) $77 \div 5$
商 15, 余り 2 | (64) $66 \div 9$
商 7, 余り 3 |
| (11) $58 \div 6$
商 9, 余り 4 | (29) $1 \div 8$
商 0, 余り 1 | (47) $71 \div 5$
商 14, 余り 1 | (65) $28 \div 5$
商 5, 余り 3 |
| (12) $70 \div 1$
商 70, 余り 0 | (30) $23 \div 9$
商 2, 余り 5 | (48) $56 \div 8$
商 7, 余り 0 | (66) $7 \div 10$
商 0, 余り 7 |
| (13) $74 \div 9$
商 8, 余り 2 | (31) $41 \div 3$
商 13, 余り 2 | (49) $4 \div 10$
商 0, 余り 4 | (67) $82 \div 5$
商 16, 余り 2 |
| (14) $58 \div 9$
商 6, 余り 4 | (32) $69 \div 4$
商 17, 余り 1 | (50) $81 \div 8$
商 10, 余り 1 | (68) $11 \div 9$
商 1, 余り 2 |
| (15) $73 \div 1$
商 73, 余り 0 | (33) $8 \div 1$
商 8, 余り 0 | (51) $69 \div 10$
商 6, 余り 9 | (69) $14 \div 8$
商 1, 余り 6 |
| (16) $5 \div 10$
商 0, 余り 5 | (34) $42 \div 4$
商 10, 余り 2 | (52) $29 \div 9$
商 3, 余り 2 | (70) $95 \div 2$
商 47, 余り 1 |
| (17) $17 \div 5$
商 3, 余り 2 | (35) $42 \div 1$
商 42, 余り 0 | (53) $80 \div 8$
商 10, 余り 0 | (71) $49 \div 1$
商 49, 余り 0 |
| (18) $37 \div 7$
商 5, 余り 2 | (36) $93 \div 7$
商 13, 余り 2 | (54) $42 \div 9$
商 4, 余り 6 | (72) $79 \div 3$
商 26, 余り 1 |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $74 \div 6$
商 12, 余り 2 | (19) $97 \div 3$
商 32, 余り 1 | (37) $74 \div 8$
商 9, 余り 2 | (55) $81 \div 9$
商 9, 余り 0 |
| (2) $3 \div 7$
商 0, 余り 3 | (20) $71 \div 10$
商 7, 余り 1 | (38) $41 \div 8$
商 5, 余り 1 | (56) $40 \div 5$
商 8, 余り 0 |
| (3) $63 \div 5$
商 12, 余り 3 | (21) $62 \div 3$
商 20, 余り 2 | (39) $66 \div 8$
商 8, 余り 2 | (57) $10 \div 5$
商 2, 余り 0 |
| (4) $21 \div 6$
商 3, 余り 3 | (22) $37 \div 3$
商 12, 余り 1 | (40) $47 \div 3$
商 15, 余り 2 | (58) $16 \div 3$
商 5, 余り 1 |
| (5) $99 \div 10$
商 9, 余り 9 | (23) $85 \div 4$
商 21, 余り 1 | (41) $13 \div 2$
商 6, 余り 1 | (59) $37 \div 1$
商 37, 余り 0 |
| (6) $26 \div 1$
商 26, 余り 0 | (24) $11 \div 4$
商 2, 余り 3 | (42) $2 \div 5$
商 0, 余り 2 | (60) $78 \div 6$
商 13, 余り 0 |
| (7) $95 \div 4$
商 23, 余り 3 | (25) $32 \div 5$
商 6, 余り 2 | (43) $68 \div 7$
商 9, 余り 5 | (61) $46 \div 1$
商 46, 余り 0 |
| (8) $15 \div 7$
商 2, 余り 1 | (26) $38 \div 1$
商 38, 余り 0 | (44) $15 \div 1$
商 15, 余り 0 | (62) $74 \div 7$
商 10, 余り 4 |
| (9) $20 \div 3$
商 6, 余り 2 | (27) $85 \div 2$
商 42, 余り 1 | (45) $98 \div 8$
商 12, 余り 2 | (63) $81 \div 7$
商 11, 余り 4 |
| (10) $45 \div 6$
商 7, 余り 3 | (28) $17 \div 8$
商 2, 余り 1 | (46) $82 \div 8$
商 10, 余り 2 | (64) $66 \div 4$
商 16, 余り 2 |
| (11) $20 \div 1$
商 20, 余り 0 | (29) $58 \div 6$
商 9, 余り 4 | (47) $62 \div 3$
商 20, 余り 2 | (65) $18 \div 2$
商 9, 余り 0 |
| (12) $77 \div 8$
商 9, 余り 5 | (30) $20 \div 4$
商 5, 余り 0 | (48) $39 \div 10$
商 3, 余り 9 | (66) $59 \div 10$
商 5, 余り 9 |
| (13) $89 \div 2$
商 44, 余り 1 | (31) $94 \div 4$
商 23, 余り 2 | (49) $13 \div 2$
商 6, 余り 1 | (67) $48 \div 10$
商 4, 余り 8 |
| (14) $77 \div 7$
商 11, 余り 0 | (32) $12 \div 10$
商 1, 余り 2 | (50) $9 \div 2$
商 4, 余り 1 | (68) $35 \div 4$
商 8, 余り 3 |
| (15) $22 \div 2$
商 11, 余り 0 | (33) $78 \div 10$
商 7, 余り 8 | (51) $10 \div 4$
商 2, 余り 2 | (69) $20 \div 3$
商 6, 余り 2 |
| (16) $76 \div 10$
商 7, 余り 6 | (34) $84 \div 6$
商 14, 余り 0 | (52) $75 \div 1$
商 75, 余り 0 | (70) $89 \div 9$
商 9, 余り 8 |
| (17) $3 \div 2$
商 1, 余り 1 | (35) $71 \div 2$
商 35, 余り 1 | (53) $35 \div 5$
商 7, 余り 0 | (71) $87 \div 7$
商 12, 余り 3 |
| (18) $30 \div 3$
商 10, 余り 0 | (36) $61 \div 2$
商 30, 余り 1 | (54) $15 \div 5$
商 3, 余り 0 | (72) $40 \div 4$
商 10, 余り 0 |

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|----------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $47 \div 4$
商 11, 余り 3 | (19) $96 \div 4$
商 24, 余り 0 | (37) $84 \div 2$
商 42, 余り 0 | (55) $1 \div 7$
商 0, 余り 1 |
| (2) $30 \div 9$
商 3, 余り 3 | (20) $81 \div 5$
商 16, 余り 1 | (38) $77 \div 8$
商 9, 余り 5 | (56) $23 \div 5$
商 4, 余り 3 |
| (3) $80 \div 8$
商 10, 余り 0 | (21) $17 \div 10$
商 1, 余り 7 | (39) $87 \div 5$
商 17, 余り 2 | (57) $55 \div 7$
商 7, 余り 6 |
| (4) $38 \div 6$
商 6, 余り 2 | (22) $57 \div 10$
商 5, 余り 7 | (40) $74 \div 7$
商 10, 余り 4 | (58) $72 \div 8$
商 9, 余り 0 |
| (5) $11 \div 8$
商 1, 余り 3 | (23) $85 \div 5$
商 17, 余り 0 | (41) $52 \div 9$
商 5, 余り 7 | (59) $61 \div 6$
商 10, 余り 1 |
| (6) $51 \div 6$
商 8, 余り 3 | (24) $43 \div 10$
商 4, 余り 3 | (42) $69 \div 4$
商 17, 余り 1 | (60) $75 \div 10$
商 7, 余り 5 |
| (7) $88 \div 7$
商 12, 余り 4 | (25) $90 \div 10$
商 9, 余り 0 | (43) $1 \div 9$
商 0, 余り 1 | (61) $23 \div 1$
商 23, 余り 0 |
| (8) $48 \div 3$
商 16, 余り 0 | (26) $84 \div 5$
商 16, 余り 4 | (44) $38 \div 4$
商 9, 余り 2 | (62) $50 \div 1$
商 50, 余り 0 |
| (9) $24 \div 6$
商 4, 余り 0 | (27) $76 \div 4$
商 19, 余り 0 | (45) $28 \div 9$
商 3, 余り 1 | (63) $20 \div 7$
商 2, 余り 6 |
| (10) $85 \div 5$
商 17, 余り 0 | (28) $8 \div 7$
商 1, 余り 1 | (46) $96 \div 6$
商 16, 余り 0 | (64) $75 \div 7$
商 10, 余り 5 |
| (11) $19 \div 5$
商 3, 余り 4 | (29) $90 \div 4$
商 22, 余り 2 | (47) $97 \div 1$
商 97, 余り 0 | (65) $72 \div 5$
商 14, 余り 2 |
| (12) $40 \div 6$
商 6, 余り 4 | (30) $77 \div 9$
商 8, 余り 5 | (48) $1 \div 7$
商 0, 余り 1 | (66) $64 \div 9$
商 7, 余り 1 |
| (13) $100 \div 10$
商 10, 余り 0 | (31) $26 \div 5$
商 5, 余り 1 | (49) $31 \div 7$
商 4, 余り 3 | (67) $29 \div 5$
商 5, 余り 4 |
| (14) $7 \div 7$
商 1, 余り 0 | (32) $81 \div 10$
商 8, 余り 1 | (50) $99 \div 5$
商 19, 余り 4 | (68) $91 \div 3$
商 30, 余り 1 |
| (15) $10 \div 6$
商 1, 余り 4 | (33) $89 \div 9$
商 9, 余り 8 | (51) $13 \div 4$
商 3, 余り 1 | (69) $79 \div 9$
商 8, 余り 7 |
| (16) $68 \div 6$
商 11, 余り 2 | (34) $29 \div 4$
商 7, 余り 1 | (52) $26 \div 8$
商 3, 余り 2 | (70) $67 \div 9$
商 7, 余り 4 |
| (17) $68 \div 10$
商 6, 余り 8 | (35) $49 \div 10$
商 4, 余り 9 | (53) $10 \div 1$
商 10, 余り 0 | (71) $9 \div 4$
商 2, 余り 1 |
| (18) $76 \div 2$
商 38, 余り 0 | (36) $65 \div 4$
商 16, 余り 1 | (54) $95 \div 9$
商 10, 余り 5 | (72) $45 \div 9$
商 5, 余り 0 |

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|--------------------------------|--------------------------------|---------------------------------|--------------------------------|
| (1) $92 \div 1$
商 92, 余り 0 | (19) $46 \div 2$
商 23, 余り 0 | (37) $9 \div 4$
商 2, 余り 1 | (55) $72 \div 9$
商 8, 余り 0 |
| (2) $19 \div 8$
商 2, 余り 3 | (20) $70 \div 10$
商 7, 余り 0 | (38) $84 \div 5$
商 16, 余り 4 | (56) $17 \div 3$
商 5, 余り 2 |
| (3) $25 \div 3$
商 8, 余り 1 | (21) $3 \div 10$
商 0, 余り 3 | (39) $14 \div 1$
商 14, 余り 0 | (57) $65 \div 10$
商 6, 余り 5 |
| (4) $41 \div 7$
商 5, 余り 6 | (22) $65 \div 4$
商 16, 余り 1 | (40) $5 \div 2$
商 2, 余り 1 | (58) $14 \div 6$
商 2, 余り 2 |
| (5) $96 \div 3$
商 32, 余り 0 | (23) $90 \div 1$
商 90, 余り 0 | (41) $23 \div 5$
商 4, 余り 3 | (59) $17 \div 3$
商 5, 余り 2 |
| (6) $79 \div 5$
商 15, 余り 4 | (24) $97 \div 8$
商 12, 余り 1 | (42) $18 \div 8$
商 2, 余り 2 | (60) $43 \div 4$
商 10, 余り 3 |
| (7) $64 \div 6$
商 10, 余り 4 | (25) $68 \div 3$
商 22, 余り 2 | (43) $99 \div 8$
商 12, 余り 3 | (61) $12 \div 7$
商 1, 余り 5 |
| (8) $75 \div 9$
商 8, 余り 3 | (26) $18 \div 2$
商 9, 余り 0 | (44) $77 \div 9$
商 8, 余り 5 | (62) $72 \div 2$
商 36, 余り 0 |
| (9) $54 \div 8$
商 6, 余り 6 | (27) $41 \div 7$
商 5, 余り 6 | (45) $90 \div 1$
商 90, 余り 0 | (63) $68 \div 5$
商 13, 余り 3 |
| (10) $45 \div 10$
商 4, 余り 5 | (28) $70 \div 9$
商 7, 余り 7 | (46) $81 \div 8$
商 10, 余り 1 | (64) $41 \div 8$
商 5, 余り 1 |
| (11) $68 \div 1$
商 68, 余り 0 | (29) $95 \div 8$
商 11, 余り 7 | (47) $100 \div 2$
商 50, 余り 0 | (65) $16 \div 7$
商 2, 余り 2 |
| (12) $37 \div 4$
商 9, 余り 1 | (30) $50 \div 10$
商 5, 余り 0 | (48) $38 \div 5$
商 7, 余り 3 | (66) $73 \div 7$
商 10, 余り 3 |
| (13) $34 \div 10$
商 3, 余り 4 | (31) $49 \div 6$
商 8, 余り 1 | (49) $30 \div 3$
商 10, 余り 0 | (67) $65 \div 9$
商 7, 余り 2 |
| (14) $98 \div 7$
商 14, 余り 0 | (32) $0 \div 9$
商 0, 余り 0 | (50) $93 \div 4$
商 23, 余り 1 | (68) $51 \div 4$
商 12, 余り 3 |
| (15) $59 \div 6$
商 9, 余り 5 | (33) $67 \div 3$
商 22, 余り 1 | (51) $42 \div 9$
商 4, 余り 6 | (69) $41 \div 8$
商 5, 余り 1 |
| (16) $56 \div 6$
商 9, 余り 2 | (34) $76 \div 3$
商 25, 余り 1 | (52) $44 \div 4$
商 11, 余り 0 | (70) $85 \div 5$
商 17, 余り 0 |
| (17) $47 \div 10$
商 4, 余り 7 | (35) $53 \div 10$
商 5, 余り 3 | (53) $13 \div 5$
商 2, 余り 3 | (71) $53 \div 5$
商 10, 余り 3 |
| (18) $99 \div 6$
商 16, 余り 3 | (36) $52 \div 2$
商 26, 余り 0 | (54) $73 \div 8$
商 9, 余り 1 | (72) $88 \div 2$
商 44, 余り 0 |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $26 \div 10$
商 2, 余り 6 | (19) $39 \div 5$
商 7, 余り 4 | (37) $19 \div 7$
商 2, 余り 5 | (55) $43 \div 8$
商 5, 余り 3 |
| (2) $43 \div 2$
商 21, 余り 1 | (20) $73 \div 7$
商 10, 余り 3 | (38) $17 \div 10$
商 1, 余り 7 | (56) $10 \div 7$
商 1, 余り 3 |
| (3) $61 \div 7$
商 8, 余り 5 | (21) $66 \div 1$
商 66, 余り 0 | (39) $80 \div 8$
商 10, 余り 0 | (57) $92 \div 6$
商 15, 余り 2 |
| (4) $81 \div 2$
商 40, 余り 1 | (22) $9 \div 6$
商 1, 余り 3 | (40) $67 \div 3$
商 22, 余り 1 | (58) $26 \div 6$
商 4, 余り 2 |
| (5) $21 \div 8$
商 2, 余り 5 | (23) $45 \div 4$
商 11, 余り 1 | (41) $39 \div 5$
商 7, 余り 4 | (59) $95 \div 6$
商 15, 余り 5 |
| (6) $69 \div 1$
商 69, 余り 0 | (24) $72 \div 1$
商 72, 余り 0 | (42) $18 \div 5$
商 3, 余り 3 | (60) $77 \div 7$
商 11, 余り 0 |
| (7) $11 \div 1$
商 11, 余り 0 | (25) $2 \div 6$
商 0, 余り 2 | (43) $46 \div 9$
商 5, 余り 1 | (61) $31 \div 7$
商 4, 余り 3 |
| (8) $60 \div 4$
商 15, 余り 0 | (26) $4 \div 1$
商 4, 余り 0 | (44) $59 \div 8$
商 7, 余り 3 | (62) $86 \div 3$
商 28, 余り 2 |
| (9) $1 \div 7$
商 0, 余り 1 | (27) $91 \div 2$
商 45, 余り 1 | (45) $68 \div 10$
商 6, 余り 8 | (63) $6 \div 5$
商 1, 余り 1 |
| (10) $69 \div 9$
商 7, 余り 6 | (28) $67 \div 4$
商 16, 余り 3 | (46) $78 \div 9$
商 8, 余り 6 | (64) $61 \div 4$
商 15, 余り 1 |
| (11) $66 \div 3$
商 22, 余り 0 | (29) $90 \div 6$
商 15, 余り 0 | (47) $82 \div 3$
商 27, 余り 1 | (65) $57 \div 9$
商 6, 余り 3 |
| (12) $1 \div 10$
商 0, 余り 1 | (30) $84 \div 5$
商 16, 余り 4 | (48) $20 \div 6$
商 3, 余り 2 | (66) $32 \div 5$
商 6, 余り 2 |
| (13) $87 \div 10$
商 8, 余り 7 | (31) $31 \div 5$
商 6, 余り 1 | (49) $73 \div 7$
商 10, 余り 3 | (67) $67 \div 4$
商 16, 余り 3 |
| (14) $67 \div 2$
商 33, 余り 1 | (32) $39 \div 3$
商 13, 余り 0 | (50) $81 \div 5$
商 16, 余り 1 | (68) $5 \div 6$
商 0, 余り 5 |
| (15) $13 \div 3$
商 4, 余り 1 | (33) $42 \div 6$
商 7, 余り 0 | (51) $3 \div 7$
商 0, 余り 3 | (69) $44 \div 10$
商 4, 余り 4 |
| (16) $84 \div 8$
商 10, 余り 4 | (34) $28 \div 10$
商 2, 余り 8 | (52) $18 \div 10$
商 1, 余り 8 | (70) $82 \div 9$
商 9, 余り 1 |
| (17) $43 \div 9$
商 4, 余り 7 | (35) $23 \div 3$
商 7, 余り 2 | (53) $92 \div 3$
商 30, 余り 2 | (71) $81 \div 8$
商 10, 余り 1 |
| (18) $70 \div 6$
商 11, 余り 4 | (36) $77 \div 4$
商 19, 余り 1 | (54) $92 \div 3$
商 30, 余り 2 | (72) $55 \div 9$
商 6, 余り 1 |

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|--------------------------------|--------------------------------|---------------------------------|--------------------------------|
| (1) $2 \div 3$
商 0, 余り 2 | (19) $32 \div 9$
商 3, 余り 5 | (37) $24 \div 3$
商 8, 余り 0 | (55) $29 \div 8$
商 3, 余り 5 |
| (2) $7 \div 7$
商 1, 余り 0 | (20) $42 \div 3$
商 14, 余り 0 | (38) $82 \div 7$
商 11, 余り 5 | (56) $76 \div 9$
商 8, 余り 4 |
| (3) $29 \div 6$
商 4, 余り 5 | (21) $9 \div 2$
商 4, 余り 1 | (39) $10 \div 8$
商 1, 余り 2 | (57) $81 \div 4$
商 20, 余り 1 |
| (4) $78 \div 3$
商 26, 余り 0 | (22) $50 \div 10$
商 5, 余り 0 | (40) $76 \div 6$
商 12, 余り 4 | (58) $10 \div 8$
商 1, 余り 2 |
| (5) $14 \div 6$
商 2, 余り 2 | (23) $7 \div 6$
商 1, 余り 1 | (41) $51 \div 3$
商 17, 余り 0 | (59) $49 \div 7$
商 7, 余り 0 |
| (6) $16 \div 4$
商 4, 余り 0 | (24) $65 \div 8$
商 8, 余り 1 | (42) $78 \div 3$
商 26, 余り 0 | (60) $15 \div 5$
商 3, 余り 0 |
| (7) $13 \div 2$
商 6, 余り 1 | (25) $36 \div 8$
商 4, 余り 4 | (43) $88 \div 6$
商 14, 余り 4 | (61) $6 \div 8$
商 0, 余り 6 |
| (8) $44 \div 1$
商 44, 余り 0 | (26) $87 \div 8$
商 10, 余り 7 | (44) $66 \div 8$
商 8, 余り 2 | (62) $13 \div 9$
商 1, 余り 4 |
| (9) $17 \div 1$
商 17, 余り 0 | (27) $24 \div 2$
商 12, 余り 0 | (45) $21 \div 1$
商 21, 余り 0 | (63) $40 \div 7$
商 5, 余り 5 |
| (10) $92 \div 5$
商 18, 余り 2 | (28) $81 \div 10$
商 8, 余り 1 | (46) $66 \div 2$
商 33, 余り 0 | (64) $38 \div 4$
商 9, 余り 2 |
| (11) $2 \div 1$
商 2, 余り 0 | (29) $36 \div 1$
商 36, 余り 0 | (47) $39 \div 8$
商 4, 余り 7 | (65) $69 \div 5$
商 13, 余り 4 |
| (12) $21 \div 6$
商 3, 余り 3 | (30) $27 \div 1$
商 27, 余り 0 | (48) $67 \div 10$
商 6, 余り 7 | (66) $80 \div 2$
商 40, 余り 0 |
| (13) $70 \div 9$
商 7, 余り 7 | (31) $8 \div 10$
商 0, 余り 8 | (49) $30 \div 7$
商 4, 余り 2 | (67) $58 \div 4$
商 14, 余り 2 |
| (14) $81 \div 6$
商 13, 余り 3 | (32) $98 \div 4$
商 24, 余り 2 | (50) $76 \div 2$
商 38, 余り 0 | (68) $85 \div 4$
商 21, 余り 1 |
| (15) $45 \div 6$
商 7, 余り 3 | (33) $79 \div 3$
商 26, 余り 1 | (51) $100 \div 3$
商 33, 余り 1 | (69) $81 \div 10$
商 8, 余り 1 |
| (16) $23 \div 1$
商 23, 余り 0 | (34) $27 \div 4$
商 6, 余り 3 | (52) $62 \div 9$
商 6, 余り 8 | (70) $64 \div 6$
商 10, 余り 4 |
| (17) $60 \div 8$
商 7, 余り 4 | (35) $56 \div 4$
商 14, 余り 0 | (53) $86 \div 8$
商 10, 余り 6 | (71) $78 \div 8$
商 9, 余り 6 |
| (18) $7 \div 3$
商 2, 余り 1 | (36) $68 \div 3$
商 22, 余り 2 | (54) $60 \div 1$
商 60, 余り 0 | (72) $97 \div 6$
商 16, 余り 1 |

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|--------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $5 \div 6$
商 0, 余り 5 | (19) $24 \div 5$
商 4, 余り 4 | (37) $65 \div 2$
商 32, 余り 1 | (55) $67 \div 2$
商 33, 余り 1 |
| (2) $76 \div 2$
商 38, 余り 0 | (20) $95 \div 2$
商 47, 余り 1 | (38) $54 \div 10$
商 5, 余り 4 | (56) $33 \div 7$
商 4, 余り 5 |
| (3) $57 \div 6$
商 9, 余り 3 | (21) $94 \div 10$
商 9, 余り 4 | (39) $33 \div 3$
商 11, 余り 0 | (57) $34 \div 10$
商 3, 余り 4 |
| (4) $1 \div 9$
商 0, 余り 1 | (22) $65 \div 8$
商 8, 余り 1 | (40) $73 \div 2$
商 36, 余り 1 | (58) $96 \div 1$
商 96, 余り 0 |
| (5) $41 \div 2$
商 20, 余り 1 | (23) $64 \div 9$
商 7, 余り 1 | (41) $53 \div 4$
商 13, 余り 1 | (59) $95 \div 10$
商 9, 余り 5 |
| (6) $75 \div 1$
商 75, 余り 0 | (24) $10 \div 1$
商 10, 余り 0 | (42) $87 \div 9$
商 9, 余り 6 | (60) $81 \div 6$
商 13, 余り 3 |
| (7) $30 \div 8$
商 3, 余り 6 | (25) $9 \div 2$
商 4, 余り 1 | (43) $23 \div 1$
商 23, 余り 0 | (61) $43 \div 10$
商 4, 余り 3 |
| (8) $52 \div 3$
商 17, 余り 1 | (26) $85 \div 5$
商 17, 余り 0 | (44) $32 \div 6$
商 5, 余り 2 | (62) $13 \div 4$
商 3, 余り 1 |
| (9) $86 \div 7$
商 12, 余り 2 | (27) $23 \div 3$
商 7, 余り 2 | (45) $21 \div 2$
商 10, 余り 1 | (63) $48 \div 1$
商 48, 余り 0 |
| (10) $59 \div 9$
商 6, 余り 5 | (28) $47 \div 10$
商 4, 余り 7 | (46) $28 \div 4$
商 7, 余り 0 | (64) $100 \div 4$
商 25, 余り 0 |
| (11) $75 \div 3$
商 25, 余り 0 | (29) $52 \div 4$
商 13, 余り 0 | (47) $33 \div 4$
商 8, 余り 1 | (65) $50 \div 10$
商 5, 余り 0 |
| (12) $45 \div 2$
商 22, 余り 1 | (30) $57 \div 5$
商 11, 余り 2 | (48) $36 \div 9$
商 4, 余り 0 | (66) $66 \div 10$
商 6, 余り 6 |
| (13) $91 \div 9$
商 10, 余り 1 | (31) $98 \div 5$
商 19, 余り 3 | (49) $15 \div 3$
商 5, 余り 0 | (67) $46 \div 6$
商 7, 余り 4 |
| (14) $21 \div 9$
商 2, 余り 3 | (32) $84 \div 3$
商 28, 余り 0 | (50) $0 \div 2$
商 0, 余り 0 | (68) $70 \div 4$
商 17, 余り 2 |
| (15) $84 \div 3$
商 28, 余り 0 | (33) $86 \div 2$
商 43, 余り 0 | (51) $93 \div 3$
商 31, 余り 0 | (69) $82 \div 8$
商 10, 余り 2 |
| (16) $51 \div 7$
商 7, 余り 2 | (34) $27 \div 5$
商 5, 余り 2 | (52) $69 \div 3$
商 23, 余り 0 | (70) $55 \div 2$
商 27, 余り 1 |
| (17) $68 \div 6$
商 11, 余り 2 | (35) $78 \div 3$
商 26, 余り 0 | (53) $65 \div 5$
商 13, 余り 0 | (71) $16 \div 1$
商 16, 余り 0 |
| (18) $37 \div 6$
商 6, 余り 1 | (36) $58 \div 8$
商 7, 余り 2 | (54) $96 \div 5$
商 19, 余り 1 | (72) $50 \div 7$
商 7, 余り 1 |

1.2 整数

1.2.1 加減法

整数の和や差を求める問題。

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|--------------------|--------------------|--------------------|---------------------|
| (1) $-27 - 61$ | (20) $67 + (-8)$ | (39) $-60 - 8$ | (58) $28 + 93$ |
| (2) $-19 - (-83)$ | (21) $-8 - (-63)$ | (40) $-82 + 26$ | (59) $-42 - (-84)$ |
| (3) $-84 + (-79)$ | (22) $-78 + (-39)$ | (41) $17 - 38$ | (60) $66 - 36$ |
| (4) $-65 - 46$ | (23) $-36 - 51$ | (42) $-70 + (-43)$ | (61) $-96 - (-95)$ |
| (5) $83 - 66$ | (24) $88 + 26$ | (43) $40 - (-73)$ | (62) $-100 + (-23)$ |
| (6) $-77 + (-95)$ | (25) $-84 - 18$ | (44) $26 + 45$ | (63) $-85 + (-69)$ |
| (7) $-8 - (-18)$ | (26) $-8 + 42$ | (45) $11 + (-86)$ | (64) $-34 + 89$ |
| (8) $5 + (-73)$ | (27) $-51 + 41$ | (46) $38 + 88$ | (65) $-78 + 45$ |
| (9) $-17 + 37$ | (28) $50 + 86$ | (47) $67 - 35$ | (66) $6 - (-56)$ |
| (10) $-95 - (-11)$ | (29) $16 - 60$ | (48) $-79 - (-58)$ | (67) $6 - (-40)$ |
| (11) $64 + 37$ | (30) $-1 + (-16)$ | (49) $-15 - 68$ | (68) $67 + (-39)$ |
| (12) $-56 + 76$ | (31) $69 + (-30)$ | (50) $-3 - 59$ | (69) $89 - (-90)$ |
| (13) $-79 - 46$ | (32) $-51 + 84$ | (51) $-86 - 26$ | (70) $-45 + (-34)$ |
| (14) $-10 - (-1)$ | (33) $-99 - (-79)$ | (52) $72 + (-66)$ | (71) $-38 + 47$ |
| (15) $-30 - (-27)$ | (34) $86 + (-24)$ | (53) $72 - 79$ | (72) $36 - 34$ |
| (16) $7 - 16$ | (35) $-52 + 41$ | (54) $-36 - 70$ | (73) $64 + 88$ |
| (17) $-35 + 3$ | (36) $-5 + 47$ | (55) $61 + (-25)$ | (74) $-38 + 32$ |
| (18) $-13 + 80$ | (37) $29 - 15$ | (56) $-53 - (-66)$ | (75) $9 + (-87)$ |
| (19) $-79 + (-38)$ | (38) $29 + (-40)$ | (57) $-83 + (-98)$ | (76) $59 + 45$ |

(1) $-3 + (-46)$	(20) $10 + (-59)$	(39) $-40 - (-40)$	(58) $-18 + 28$
(2) $2 - 21$	(21) $-53 + 37$	(40) $74 - (-33)$	(59) $-69 + (-10)$
(3) $64 - (-42)$	(22) $-95 - (-44)$	(41) $68 - (-57)$	(60) $70 + (-22)$
(4) $31 - 77$	(23) $28 + (-13)$	(42) $-68 - 4$	(61) $82 - 39$
(5) $-7 - 20$	(24) $-13 + 98$	(43) $-36 + (-11)$	(62) $33 + 88$
(6) $-41 + (-44)$	(25) $-85 - 75$	(44) $-75 + 42$	(63) $-54 - 41$
(7) $-27 - (-96)$	(26) $-42 - (-72)$	(45) $49 + (-4)$	(64) $7 + (-70)$
(8) $-77 - (-6)$	(27) $51 + (-56)$	(46) $61 - 47$	(65) $-38 + 81$
(9) $46 + (-32)$	(28) $36 - (-34)$	(47) $-26 - 21$	(66) $50 + 37$
(10) $-33 + 99$	(29) $-81 + (-79)$	(48) $-74 + 29$	(67) $88 + 94$
(11) $-65 + (-37)$	(30) $-31 + 4$	(49) $-71 - (-90)$	(68) $82 - (-41)$
(12) $95 + 41$	(31) $-83 - (-63)$	(50) $95 - (-37)$	(69) $-62 + 90$
(13) $-100 + (-11)$	(32) $34 + 48$	(51) $66 - 61$	(70) $85 - (-18)$
(14) $33 - (-81)$	(33) $-78 - 73$	(52) $96 + (-67)$	(71) $96 - 25$
(15) $90 - 17$	(34) $-34 + (-65)$	(53) $58 + 18$	(72) $-10 - 35$
(16) $28 + 13$	(35) $-90 + 24$	(54) $64 - (-16)$	(73) $88 - 100$
(17) $77 - 7$	(36) $29 - 27$	(55) $-7 + (-17)$	(74) $-12 + 96$
(18) $-67 + 89$	(37) $81 - (-65)$	(56) $22 + (-57)$	(75) $64 - 54$
(19) $-84 + 13$	(38) $-88 + (-33)$	(57) $14 + 14$	(76) $45 + 41$

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|--------------------|---------------------|--------------------|---------------------|
| (1) $-92 + (-6)$ | (20) $-8 - 82$ | (39) $-73 - (-91)$ | (58) $-11 - (-71)$ |
| (2) $-63 + (-87)$ | (21) $78 + (-24)$ | (40) $-61 + (-77)$ | (59) $89 + (-4)$ |
| (3) $-33 + (-36)$ | (22) $50 - 56$ | (41) $-68 - (-14)$ | (60) $-71 - 23$ |
| (4) $-100 + 14$ | (23) $61 - (-93)$ | (42) $48 + (-77)$ | (61) $-100 + (-26)$ |
| (5) $-70 - 82$ | (24) $-54 - 73$ | (43) $-7 + (-50)$ | (62) $92 + 29$ |
| (6) $-31 - 58$ | (25) $39 - 74$ | (44) $70 + 19$ | (63) $-70 - (-36)$ |
| (7) $-65 + 33$ | (26) $-53 - 46$ | (45) $24 + 4$ | (64) $-65 - 70$ |
| (8) $41 - 41$ | (27) $-68 - (-100)$ | (46) $-63 - 46$ | (65) $41 - (-99)$ |
| (9) $77 + 95$ | (28) $-21 - (-20)$ | (47) $-74 - 3$ | (66) $-41 - 45$ |
| (10) $81 + 4$ | (29) $-23 - (-4)$ | (48) $37 - (-91)$ | (67) $-52 + 82$ |
| (11) $-29 + (-92)$ | (30) $89 + 100$ | (49) $-39 - 73$ | (68) $51 + (-61)$ |
| (12) $42 + 20$ | (31) $10 - 13$ | (50) $-92 - 29$ | (69) $-21 - (-49)$ |
| (13) $-87 - 13$ | (32) $-86 - (-91)$ | (51) $89 + (-53)$ | (70) $57 + 37$ |
| (14) $94 + (-56)$ | (33) $14 + 59$ | (52) $99 - 75$ | (71) $59 - 50$ |
| (15) $75 - 65$ | (34) $86 - 66$ | (53) $-74 + (-13)$ | (72) $-35 + (-81)$ |
| (16) $82 + 78$ | (35) $-4 - (-82)$ | (54) $-37 + (-80)$ | (73) $-72 - 91$ |
| (17) $-25 + 87$ | (36) $20 - 73$ | (55) $-95 - 81$ | (74) $10 - (-26)$ |
| (18) $-14 + 44$ | (37) $11 - (-67)$ | (56) $76 + (-98)$ | (75) $35 + (-59)$ |
| (19) $-50 + (-45)$ | (38) $-22 - 20$ | (57) $-37 - (-3)$ | (76) $100 - (-79)$ |

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|--------------------|--------------------|---------------------|--------------------|
| (1) $88 + 26$ | (20) $-94 + (-14)$ | (39) $73 - 60$ | (58) $-81 - (-32)$ |
| (2) $-99 - (-95)$ | (21) $-22 - (-35)$ | (40) $-69 + (-51)$ | (59) $61 - (-20)$ |
| (3) $18 - 92$ | (22) $-25 + 52$ | (41) $83 - (-44)$ | (60) $53 - 89$ |
| (4) $-41 - (-93)$ | (23) $-94 - (-8)$ | (42) $-100 + (-97)$ | (61) $84 + 35$ |
| (5) $0 + 82$ | (24) $-72 - (-67)$ | (43) $9 + 37$ | (62) $74 + (-49)$ |
| (6) $-31 - 87$ | (25) $34 + 43$ | (44) $73 + 96$ | (63) $-69 - 65$ |
| (7) $-48 - 35$ | (26) $80 - 52$ | (45) $-48 - 16$ | (64) $-12 + (-64)$ |
| (8) $15 + 82$ | (27) $-32 + (-95)$ | (46) $2 + 71$ | (65) $43 + 61$ |
| (9) $-54 + 68$ | (28) $11 - (-72)$ | (47) $-6 - (-88)$ | (66) $19 - 49$ |
| (10) $-98 + (-45)$ | (29) $2 + 96$ | (48) $-89 + (-86)$ | (67) $-9 - (-30)$ |
| (11) $19 + (-97)$ | (30) $-16 - 56$ | (49) $-16 + 75$ | (68) $-66 + (-58)$ |
| (12) $-6 - (-26)$ | (31) $20 + (-32)$ | (50) $-93 - (-94)$ | (69) $64 - (-89)$ |
| (13) $-24 + 95$ | (32) $84 - (-12)$ | (51) $-79 - 57$ | (70) $-52 - (-8)$ |
| (14) $-81 + (-88)$ | (33) $-98 - (-38)$ | (52) $-72 - 79$ | (71) $-68 - (-68)$ |
| (15) $62 - 4$ | (34) $91 + 85$ | (53) $20 + 93$ | (72) $14 + 31$ |
| (16) $83 - (-8)$ | (35) $-95 - (-12)$ | (54) $42 - (-95)$ | (73) $92 + 86$ |
| (17) $11 + 25$ | (36) $69 - (-90)$ | (55) $-80 + (-58)$ | (74) $-97 - 68$ |
| (18) $91 + (-58)$ | (37) $-93 - 28$ | (56) $12 - (-74)$ | (75) $57 - 15$ |
| (19) $37 - (-80)$ | (38) $-79 - (-32)$ | (57) $98 - (-23)$ | (76) $-31 - (-13)$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $-55 - (-76)$ | (20) $20 - (-8)$ | (39) $-92 - 64$ | (58) $33 - (-23)$ |
| (2) $4 + (-15)$ | (21) $57 - (-64)$ | (40) $93 - 45$ | (59) $20 - (-47)$ |
| (3) $-90 + 55$ | (22) $-82 + (-59)$ | (41) $64 - 91$ | (60) $48 - (-11)$ |
| (4) $27 - (-51)$ | (23) $-34 - 4$ | (42) $43 - 10$ | (61) $79 + (-46)$ |
| (5) $92 + 14$ | (24) $-73 + 64$ | (43) $-37 - (-76)$ | (62) $-81 - (-18)$ |
| (6) $60 + 57$ | (25) $-86 + (-92)$ | (44) $-90 + (-45)$ | (63) $82 + (-3)$ |
| (7) $100 + 26$ | (26) $-41 - 39$ | (45) $-26 + (-10)$ | (64) $-46 - (-52)$ |
| (8) $71 - (-46)$ | (27) $-41 - (-74)$ | (46) $-43 + 59$ | (65) $66 - (-3)$ |
| (9) $88 + 62$ | (28) $-83 + 59$ | (47) $-35 - (-1)$ | (66) $71 + (-24)$ |
| (10) $-2 + (-25)$ | (29) $91 - 55$ | (48) $15 + (-14)$ | (67) $-77 + (-90)$ |
| (11) $17 - (-87)$ | (30) $45 + 79$ | (49) $13 + 29$ | (68) $11 - 80$ |
| (12) $47 - 7$ | (31) $65 + 42$ | (50) $-40 - (-19)$ | (69) $-63 - (-12)$ |
| (13) $49 - 38$ | (32) $61 + (-35)$ | (51) $-82 + (-58)$ | (70) $-98 + 19$ |
| (14) $45 + (-92)$ | (33) $-90 - 75$ | (52) $-77 - (-81)$ | (71) $66 + (-42)$ |
| (15) $72 - (-55)$ | (34) $78 + (-56)$ | (53) $50 - (-47)$ | (72) $-78 - (-25)$ |
| (16) $-42 - (-61)$ | (35) $-72 - (-93)$ | (54) $-49 - 97$ | (73) $64 - 15$ |
| (17) $57 - 89$ | (36) $2 - (-65)$ | (55) $-55 - (-57)$ | (74) $71 - (-57)$ |
| (18) $-62 - 90$ | (37) $-95 + (-53)$ | (56) $85 - (-12)$ | (75) $57 + 91$ |
| (19) $-46 - 100$ | (38) $80 + 50$ | (57) $-73 - (-29)$ | (76) $57 - 13$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $-72 - 43$ | (20) $-16 + (-77)$ | (39) $-52 + 8$ | (58) $73 - 6$ |
| (2) $-36 + (-20)$ | (21) $11 + (-27)$ | (40) $57 + (-11)$ | (59) $68 - (-6)$ |
| (3) $64 + (-17)$ | (22) $-45 + (-28)$ | (41) $16 - 12$ | (60) $-62 - (-9)$ |
| (4) $94 - (-56)$ | (23) $-31 - 39$ | (42) $37 + 52$ | (61) $60 + (-90)$ |
| (5) $-96 - 10$ | (24) $-1 - 67$ | (43) $59 - 61$ | (62) $49 - (-77)$ |
| (6) $-34 - 2$ | (25) $-84 + 72$ | (44) $-1 - 4$ | (63) $11 - (-15)$ |
| (7) $-32 - 29$ | (26) $15 + (-76)$ | (45) $-68 - 31$ | (64) $41 + (-76)$ |
| (8) $42 - (-67)$ | (27) $-30 - (-4)$ | (46) $96 - (-20)$ | (65) $96 + (-37)$ |
| (9) $-38 - 97$ | (28) $-6 - (-61)$ | (47) $46 - 45$ | (66) $-51 + 32$ |
| (10) $46 - 12$ | (29) $13 + 31$ | (48) $-67 - (-35)$ | (67) $96 + 64$ |
| (11) $10 + (-47)$ | (30) $8 + (-10)$ | (49) $27 - (-76)$ | (68) $-44 - 57$ |
| (12) $80 - (-42)$ | (31) $66 - 98$ | (50) $62 - (-81)$ | (69) $-95 + (-38)$ |
| (13) $-68 + 48$ | (32) $-41 - 66$ | (51) $-42 + (-13)$ | (70) $-82 - (-59)$ |
| (14) $-78 + 32$ | (33) $-99 + (-77)$ | (52) $-19 + (-62)$ | (71) $-3 + (-25)$ |
| (15) $93 - 84$ | (34) $-5 - 92$ | (53) $85 + 100$ | (72) $90 + (-55)$ |
| (16) $66 - 86$ | (35) $98 - (-53)$ | (54) $85 - 35$ | (73) $63 - (-55)$ |
| (17) $-92 + 50$ | (36) $-48 + (-17)$ | (55) $42 - 5$ | (74) $25 + 15$ |
| (18) $-65 - (-49)$ | (37) $95 - 0$ | (56) $54 + (-100)$ | (75) $54 - (-74)$ |
| (19) $-97 - (-65)$ | (38) $44 - 48$ | (57) $-1 - (-90)$ | (76) $27 + 47$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $81 - (-85)$ | (20) $4 + 13$ | (39) $77 + (-32)$ | (58) $97 + (-15)$ |
| (2) $38 - 85$ | (21) $19 - 83$ | (40) $-69 + 0$ | (59) $56 + 55$ |
| (3) $-4 - (-26)$ | (22) $87 - (-12)$ | (41) $1 - 60$ | (60) $-92 - (-93)$ |
| (4) $11 + 95$ | (23) $-7 - (-6)$ | (42) $-65 - (-64)$ | (61) $-83 + (-25)$ |
| (5) $68 - (-27)$ | (24) $-51 + 31$ | (43) $-37 + 100$ | (62) $-36 - (-39)$ |
| (6) $59 - 38$ | (25) $65 - 76$ | (44) $-97 + 75$ | (63) $-10 - (-11)$ |
| (7) $-35 + (-35)$ | (26) $17 - (-96)$ | (45) $46 + 28$ | (64) $80 + 58$ |
| (8) $35 + (-50)$ | (27) $67 + 5$ | (46) $-2 + 10$ | (65) $-91 + (-16)$ |
| (9) $82 - 56$ | (28) $76 + (-80)$ | (47) $2 - 37$ | (66) $15 - 13$ |
| (10) $-22 - (-73)$ | (29) $-94 + 48$ | (48) $15 + 63$ | (67) $-71 - (-81)$ |
| (11) $89 - 5$ | (30) $48 - 40$ | (49) $-42 - 3$ | (68) $96 + (-19)$ |
| (12) $-82 - (-24)$ | (31) $-24 - (-52)$ | (50) $78 - (-79)$ | (69) $6 - (-67)$ |
| (13) $14 - (-20)$ | (32) $-66 - (-11)$ | (51) $16 - 60$ | (70) $85 + (-78)$ |
| (14) $74 - 24$ | (33) $-60 + 79$ | (52) $52 - (-41)$ | (71) $-6 + 85$ |
| (15) $-34 - 93$ | (34) $-96 + 68$ | (53) $82 + (-12)$ | (72) $-22 + 42$ |
| (16) $-67 - (-44)$ | (35) $-33 - (-61)$ | (54) $-45 - 98$ | (73) $-41 + (-77)$ |
| (17) $-95 - (-14)$ | (36) $95 - 43$ | (55) $77 - 8$ | (74) $-38 - 49$ |
| (18) $92 + 44$ | (37) $13 + (-39)$ | (56) $-62 + 94$ | (75) $-66 - (-71)$ |
| (19) $86 + (-50)$ | (38) $-100 + 57$ | (57) $-51 - (-93)$ | (76) $-69 - (-3)$ |

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|--------------------|--------------------|---------------------|--------------------|
| (1) $48 + 30$ | (20) $14 + (-84)$ | (39) $97 + (-7)$ | (58) $-41 - (-49)$ |
| (2) $15 + 3$ | (21) $-80 + (-1)$ | (40) $-72 + 20$ | (59) $-20 + 99$ |
| (3) $-35 + 27$ | (22) $-89 - (-99)$ | (41) $-52 + (-56)$ | (60) $-59 + 92$ |
| (4) $37 + (-21)$ | (23) $-55 + (-49)$ | (42) $60 + 12$ | (61) $80 + 4$ |
| (5) $7 - (-58)$ | (24) $74 - (-49)$ | (43) $-50 + (-53)$ | (62) $-95 - 12$ |
| (6) $-70 - 37$ | (25) $-70 + (-15)$ | (44) $-46 - 4$ | (63) $69 + (-21)$ |
| (7) $-31 - (-25)$ | (26) $-71 + (-94)$ | (45) $94 - (-72)$ | (64) $-50 - (-40)$ |
| (8) $-92 + (-12)$ | (27) $-30 + 99$ | (46) $-49 - (-59)$ | (65) $85 + 26$ |
| (9) $7 - 78$ | (28) $-98 + (-99)$ | (47) $20 + 94$ | (66) $94 - 58$ |
| (10) $-41 - 25$ | (29) $-64 - 95$ | (48) $-80 + 16$ | (67) $-83 + (-43)$ |
| (11) $-47 - (-53)$ | (30) $-62 - (-73)$ | (49) $1 + (-23)$ | (68) $-7 + (-97)$ |
| (12) $-77 + (-32)$ | (31) $48 - 82$ | (50) $15 - (-29)$ | (69) $83 + 18$ |
| (13) $-74 + 19$ | (32) $-1 - (-86)$ | (51) $3 + 31$ | (70) $66 - 47$ |
| (14) $-38 + (-48)$ | (33) $-1 + 64$ | (52) $-12 - 52$ | (71) $17 + 98$ |
| (15) $57 - (-25)$ | (34) $-55 - 93$ | (53) $-84 + 38$ | (72) $4 - (-26)$ |
| (16) $51 + (-10)$ | (35) $77 + 44$ | (54) $57 - 97$ | (73) $24 + (-1)$ |
| (17) $25 + 61$ | (36) $-9 - (-24)$ | (55) $91 - (-63)$ | (74) $-80 - 97$ |
| (18) $52 + 10$ | (37) $-76 + (-87)$ | (56) $-100 + (-30)$ | (75) $49 + 9$ |
| (19) $96 - 27$ | (38) $-8 - (-2)$ | (57) $46 - (-15)$ | (76) $-84 + 73$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $-88 - (-94)$ | (20) $84 + (-61)$ | (39) $31 + 46$ | (58) $14 - (-37)$ |
| (2) $91 + (-30)$ | (21) $21 - 70$ | (40) $16 + (-67)$ | (59) $-3 - (-40)$ |
| (3) $-64 - 96$ | (22) $-59 - 96$ | (41) $-89 - (-65)$ | (60) $-21 + (-39)$ |
| (4) $-63 - 69$ | (23) $15 - (-98)$ | (42) $-78 - (-80)$ | (61) $-96 + 44$ |
| (5) $-72 - (-28)$ | (24) $8 - 82$ | (43) $68 - 13$ | (62) $6 + 29$ |
| (6) $-87 + 75$ | (25) $6 + 79$ | (44) $21 - (-35)$ | (63) $-19 + (-9)$ |
| (7) $-99 + (-29)$ | (26) $-44 + (-53)$ | (45) $24 + 40$ | (64) $42 + 64$ |
| (8) $81 - (-87)$ | (27) $-46 - (-71)$ | (46) $36 + (-9)$ | (65) $84 + 58$ |
| (9) $-8 - (-84)$ | (28) $-34 - (-25)$ | (47) $-46 + 63$ | (66) $-90 - 96$ |
| (10) $89 - 44$ | (29) $-48 - 92$ | (48) $46 + 41$ | (67) $-95 - (-34)$ |
| (11) $78 + 20$ | (30) $-13 - 7$ | (49) $-76 - (-29)$ | (68) $-80 - (-33)$ |
| (12) $83 + 99$ | (31) $32 + (-89)$ | (50) $-39 + (-58)$ | (69) $6 + 8$ |
| (13) $-12 - (-39)$ | (32) $-41 - (-56)$ | (51) $50 + 45$ | (70) $57 + (-95)$ |
| (14) $79 + (-60)$ | (33) $44 - (-72)$ | (52) $65 + 76$ | (71) $-40 - 35$ |
| (15) $-16 - 82$ | (34) $42 + 5$ | (53) $14 - 88$ | (72) $2 - (-99)$ |
| (16) $9 + (-48)$ | (35) $37 - 91$ | (54) $1 - (-39)$ | (73) $-95 + (-25)$ |
| (17) $52 - (-2)$ | (36) $-52 + 3$ | (55) $-20 + 35$ | (74) $7 - (-82)$ |
| (18) $-28 - 15$ | (37) $-51 + (-3)$ | (56) $49 - 11$ | (75) $-29 + 5$ |
| (19) $79 + 19$ | (38) $97 + (-53)$ | (57) $-73 - (-57)$ | (76) $12 + 37$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $70 - 41$ | (20) $-75 + 11$ | (39) $-34 - 63$ | (58) $70 + (-17)$ |
| (2) $75 - 69$ | (21) $87 - (-11)$ | (40) $30 - 65$ | (59) $-51 + 9$ |
| (3) $-5 - (-13)$ | (22) $38 - 62$ | (41) $17 - (-27)$ | (60) $18 + 75$ |
| (4) $49 + 68$ | (23) $7 - (-68)$ | (42) $61 + (-55)$ | (61) $-10 + (-17)$ |
| (5) $-78 + (-12)$ | (24) $-6 - 3$ | (43) $93 - (-45)$ | (62) $-22 + 24$ |
| (6) $69 + (-90)$ | (25) $59 - (-51)$ | (44) $42 + (-14)$ | (63) $-28 - 27$ |
| (7) $73 + 57$ | (26) $89 - 96$ | (45) $-61 - (-18)$ | (64) $80 + (-6)$ |
| (8) $68 + 83$ | (27) $73 + 83$ | (46) $-7 + (-64)$ | (65) $-67 + (-54)$ |
| (9) $49 + (-85)$ | (28) $70 - 11$ | (47) $-91 + 53$ | (66) $58 + (-37)$ |
| (10) $-60 - (-6)$ | (29) $27 + 19$ | (48) $23 + 7$ | (67) $-22 + (-75)$ |
| (11) $-98 - (-37)$ | (30) $32 - 54$ | (49) $-5 + (-7)$ | (68) $3 - 87$ |
| (12) $-77 + (-5)$ | (31) $-56 - 15$ | (50) $-21 - (-56)$ | (69) $-47 - (-3)$ |
| (13) $-13 + 53$ | (32) $-99 + (-50)$ | (51) $-17 + (-27)$ | (70) $54 - 21$ |
| (14) $-28 + 17$ | (33) $-31 + 33$ | (52) $84 - (-44)$ | (71) $-63 + (-34)$ |
| (15) $14 + (-9)$ | (34) $70 + 27$ | (53) $-4 - 19$ | (72) $-67 + (-54)$ |
| (16) $32 - 0$ | (35) $30 - (-69)$ | (54) $64 - 51$ | (73) $-21 - 39$ |
| (17) $63 + (-17)$ | (36) $89 + 61$ | (55) $-59 - (-50)$ | (74) $47 - (-93)$ |
| (18) $83 - (-55)$ | (37) $-85 + 9$ | (56) $19 - 25$ | (75) $88 - 18$ |
| (19) $-17 + 59$ | (38) $53 - 85$ | (57) $99 + (-69)$ | (76) $-21 - (-59)$ |

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|--------------------|--------------------|--------------------|---------------------|
| (1) $88 - 51$ | (20) $-84 - 42$ | (39) $-21 - 9$ | (58) $43 + 91$ |
| (2) $15 + 85$ | (21) $36 - (-63)$ | (40) $-63 - (-36)$ | (59) $-19 + (-47)$ |
| (3) $-91 - 32$ | (22) $52 + 11$ | (41) $1 - (-12)$ | (60) $-96 - (-94)$ |
| (4) $-65 - (-36)$ | (23) $100 - 53$ | (42) $0 - 25$ | (61) $-92 - 41$ |
| (5) $-70 - (-39)$ | (24) $78 - (-2)$ | (43) $48 - (-41)$ | (62) $-19 - (-44)$ |
| (6) $-58 + 10$ | (25) $-42 - (-68)$ | (44) $-69 + (-43)$ | (63) $-69 - (-74)$ |
| (7) $49 - (-36)$ | (26) $-66 + (-81)$ | (45) $-23 + (-72)$ | (64) $-86 + (-48)$ |
| (8) $59 - 12$ | (27) $22 + (-7)$ | (46) $-42 + 86$ | (65) $-85 - (-46)$ |
| (9) $70 - (-24)$ | (28) $-1 + (-65)$ | (47) $-52 + 43$ | (66) $42 + 76$ |
| (10) $-29 - 42$ | (29) $67 + 50$ | (48) $-56 + 39$ | (67) $25 + 45$ |
| (11) $100 + (-46)$ | (30) $74 + 33$ | (49) $-62 + 93$ | (68) $42 - 70$ |
| (12) $54 - 74$ | (31) $0 + (-39)$ | (50) $34 - (-7)$ | (69) $-100 + (-87)$ |
| (13) $36 - 70$ | (32) $96 + (-6)$ | (51) $44 - 67$ | (70) $84 + 45$ |
| (14) $51 + 92$ | (33) $-54 - (-97)$ | (52) $-55 + 46$ | (71) $-68 + (-45)$ |
| (15) $-43 - 33$ | (34) $92 + 75$ | (53) $72 - 15$ | (72) $70 + (-97)$ |
| (16) $73 - 1$ | (35) $6 - 96$ | (54) $44 - 36$ | (73) $39 - (-6)$ |
| (17) $-18 + (-19)$ | (36) $16 - (-49)$ | (55) $27 + (-30)$ | (74) $-59 - (-73)$ |
| (18) $31 - (-12)$ | (37) $53 - (-60)$ | (56) $1 + (-65)$ | (75) $-77 + (-46)$ |
| (19) $49 - 68$ | (38) $-16 - 73$ | (57) $53 + 14$ | (76) $1 - (-38)$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $-8 + 27$ | (20) $-44 + (-2)$ | (39) $96 + 8$ | (58) $74 - (-56)$ |
| (2) $-87 + (-97)$ | (21) $86 - 11$ | (40) $57 - (-60)$ | (59) $-69 + 92$ |
| (3) $-49 + 72$ | (22) $93 - (-7)$ | (41) $11 - 88$ | (60) $98 - 58$ |
| (4) $-3 - 92$ | (23) $52 - 70$ | (42) $34 - 65$ | (61) $-82 - 90$ |
| (5) $1 + (-32)$ | (24) $-21 + (-22)$ | (43) $21 - (-71)$ | (62) $71 - 1$ |
| (6) $0 - (-74)$ | (25) $-20 + 67$ | (44) $-74 + 44$ | (63) $-44 + 57$ |
| (7) $-38 + (-32)$ | (26) $44 - 19$ | (45) $-50 - 78$ | (64) $-95 - (-75)$ |
| (8) $95 - (-20)$ | (27) $-13 + 94$ | (46) $-21 + 95$ | (65) $96 + 78$ |
| (9) $78 - 26$ | (28) $-90 - 49$ | (47) $71 + (-99)$ | (66) $6 - 62$ |
| (10) $99 - (-29)$ | (29) $7 - (-99)$ | (48) $53 + 72$ | (67) $24 + (-54)$ |
| (11) $-26 - (-84)$ | (30) $-93 + 5$ | (49) $11 - (-70)$ | (68) $-62 - 52$ |
| (12) $14 + 63$ | (31) $-52 + 92$ | (50) $3 - (-30)$ | (69) $-28 - (-49)$ |
| (13) $44 - (-46)$ | (32) $-66 - (-52)$ | (51) $8 - 94$ | (70) $-89 + 12$ |
| (14) $-87 + (-72)$ | (33) $58 - (-60)$ | (52) $-70 + (-84)$ | (71) $-35 - 59$ |
| (15) $-31 + 47$ | (34) $59 - 38$ | (53) $65 + (-44)$ | (72) $-95 + 90$ |
| (16) $-84 + 21$ | (35) $-22 - (-90)$ | (54) $30 - 43$ | (73) $85 - (-9)$ |
| (17) $100 - (-88)$ | (36) $27 - (-7)$ | (55) $86 - 22$ | (74) $-11 - (-45)$ |
| (18) $11 - 18$ | (37) $45 - (-62)$ | (56) $-56 + (-68)$ | (75) $-19 + (-45)$ |
| (19) $-71 + 99$ | (38) $-34 - (-94)$ | (57) $85 + 72$ | (76) $-72 + (-31)$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $84 - (-98)$ | (20) $35 + (-3)$ | (39) $-48 + 6$ | (58) $-58 + 23$ |
| (2) $17 + 13$ | (21) $-15 + 45$ | (40) $-97 + 70$ | (59) $-53 - (-57)$ |
| (3) $-5 - (-45)$ | (22) $-11 - (-95)$ | (41) $-25 - 61$ | (60) $88 + 71$ |
| (4) $-52 + 58$ | (23) $-17 - (-78)$ | (42) $81 + 86$ | (61) $56 - (-95)$ |
| (5) $18 - 59$ | (24) $41 - (-36)$ | (43) $-2 + 55$ | (62) $-46 - 35$ |
| (6) $-40 - (-98)$ | (25) $95 - 26$ | (44) $-18 + (-72)$ | (63) $5 - (-15)$ |
| (7) $-87 + (-82)$ | (26) $8 + 45$ | (45) $73 - 69$ | (64) $-41 - (-73)$ |
| (8) $-18 - (-68)$ | (27) $-42 - (-70)$ | (46) $-98 + 89$ | (65) $14 + 88$ |
| (9) $-75 + 12$ | (28) $61 + (-27)$ | (47) $-82 - 9$ | (66) $2 + (-37)$ |
| (10) $-56 + 10$ | (29) $71 - 48$ | (48) $78 + (-1)$ | (67) $52 + (-97)$ |
| (11) $43 + (-100)$ | (30) $-18 - 68$ | (49) $44 + 12$ | (68) $-28 + (-87)$ |
| (12) $-43 - 25$ | (31) $-55 - (-32)$ | (50) $-97 + (-42)$ | (69) $-45 + 4$ |
| (13) $24 - 71$ | (32) $76 - 10$ | (51) $6 + 23$ | (70) $-82 - (-22)$ |
| (14) $13 - 75$ | (33) $-32 + (-77)$ | (52) $9 + (-2)$ | (71) $-34 - 31$ |
| (15) $-21 + (-35)$ | (34) $23 - (-22)$ | (53) $69 - 82$ | (72) $-13 + (-78)$ |
| (16) $65 - (-71)$ | (35) $45 - 75$ | (54) $39 + (-21)$ | (73) $47 + (-56)$ |
| (17) $88 + (-42)$ | (36) $-40 + 88$ | (55) $-43 + (-75)$ | (74) $87 - (-1)$ |
| (18) $91 + (-50)$ | (37) $83 - 22$ | (56) $41 - (-12)$ | (75) $-31 + (-11)$ |
| (19) $38 + (-75)$ | (38) $76 - 44$ | (57) $65 + (-72)$ | (76) $-29 - 48$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $12 + 41$ | (20) $63 + (-86)$ | (39) $-60 - 18$ | (58) $35 + (-44)$ |
| (2) $-22 - (-18)$ | (21) $39 + 68$ | (40) $-51 + (-88)$ | (59) $-92 - (-90)$ |
| (3) $-16 + (-28)$ | (22) $-4 - 52$ | (41) $-65 - (-48)$ | (60) $-78 - (-39)$ |
| (4) $15 + (-72)$ | (23) $-31 + 68$ | (42) $-63 + (-79)$ | (61) $64 - (-51)$ |
| (5) $-64 + 3$ | (24) $34 + (-8)$ | (43) $-91 - (-20)$ | (62) $19 - 84$ |
| (6) $0 - 27$ | (25) $-78 + 78$ | (44) $79 - (-46)$ | (63) $34 - (-7)$ |
| (7) $-28 - 51$ | (26) $89 - (-54)$ | (45) $-11 - (-6)$ | (64) $3 - (-65)$ |
| (8) $-34 + (-97)$ | (27) $-36 + (-1)$ | (46) $34 - 98$ | (65) $48 - (-10)$ |
| (9) $45 + 93$ | (28) $53 + (-54)$ | (47) $-46 + 57$ | (66) $-96 - (-51)$ |
| (10) $61 + 42$ | (29) $23 - 31$ | (48) $-72 - 53$ | (67) $45 - (-22)$ |
| (11) $75 + 98$ | (30) $-37 - (-76)$ | (49) $-71 - 38$ | (68) $-28 - 80$ |
| (12) $47 - 90$ | (31) $-59 - 98$ | (50) $-84 - (-40)$ | (69) $-26 + (-41)$ |
| (13) $-96 - (-35)$ | (32) $1 - (-27)$ | (51) $-18 + (-89)$ | (70) $-91 - 84$ |
| (14) $-30 - (-89)$ | (33) $-98 - 35$ | (52) $91 + 83$ | (71) $-79 + 58$ |
| (15) $15 + 55$ | (34) $14 - 35$ | (53) $-66 + 23$ | (72) $37 - (-79)$ |
| (16) $57 - (-87)$ | (35) $-33 + (-6)$ | (54) $27 - (-31)$ | (73) $48 + (-9)$ |
| (17) $-38 - 11$ | (36) $-35 - (-23)$ | (55) $44 + 100$ | (74) $-68 + (-22)$ |
| (18) $6 - 36$ | (37) $-57 + 10$ | (56) $-89 - (-26)$ | (75) $12 - (-25)$ |
| (19) $57 - 83$ | (38) $66 - 55$ | (57) $-16 + 60$ | (76) $-13 + (-43)$ |

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|--------------------|---------------------|--------------------|--------------------|
| (1) $-89 - (-40)$ | (20) $-18 - (-35)$ | (39) $43 - 69$ | (58) $47 - (-16)$ |
| (2) $25 - 84$ | (21) $80 + (-45)$ | (40) $-66 + 6$ | (59) $65 + (-20)$ |
| (3) $77 - (-64)$ | (22) $-25 + 58$ | (41) $-92 - (-11)$ | (60) $34 + 44$ |
| (4) $-98 - (-22)$ | (23) $42 + (-72)$ | (42) $-35 + (-6)$ | (61) $44 - 32$ |
| (5) $50 - (-34)$ | (24) $2 - (-3)$ | (43) $63 - 3$ | (62) $-86 - (-25)$ |
| (6) $53 + 100$ | (25) $-27 + 97$ | (44) $-78 + (-32)$ | (63) $-56 - 82$ |
| (7) $-94 + 20$ | (26) $-38 + (-85)$ | (45) $80 - (-53)$ | (64) $-56 + (-20)$ |
| (8) $-13 + 90$ | (27) $57 + 23$ | (46) $-91 - (-48)$ | (65) $10 + (-58)$ |
| (9) $-19 + (-40)$ | (28) $34 + 64$ | (47) $39 + (-54)$ | (66) $-56 - (-73)$ |
| (10) $73 + 31$ | (29) $19 - (-30)$ | (48) $26 + 26$ | (67) $-88 + (-89)$ |
| (11) $45 + 11$ | (30) $-82 + 24$ | (49) $-95 - 96$ | (68) $-32 + 55$ |
| (12) $61 + (-77)$ | (31) $29 + 35$ | (50) $-76 - (-13)$ | (69) $49 - 62$ |
| (13) $-32 + (-92)$ | (32) $79 - 13$ | (51) $-85 + 59$ | (70) $6 + (-34)$ |
| (14) $-72 + (-22)$ | (33) $-83 - 45$ | (52) $44 + 65$ | (71) $5 + 53$ |
| (15) $-47 - 98$ | (34) $-100 - (-70)$ | (53) $52 - 24$ | (72) $32 - (-70)$ |
| (16) $53 + (-24)$ | (35) $-5 - 89$ | (54) $-58 - (-89)$ | (73) $12 - (-30)$ |
| (17) $80 + (-8)$ | (36) $-6 + 96$ | (55) $24 - (-75)$ | (74) $-82 - (-72)$ |
| (18) $-86 - (-77)$ | (37) $-31 + (-82)$ | (56) $44 - 79$ | (75) $-27 - 14$ |
| (19) $-92 + (-31)$ | (38) $-28 + 55$ | (57) $-88 - (-99)$ | (76) $28 - 74$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $-17 - 18$ | (20) $1 + (-58)$ | (39) $84 - (-98)$ | (58) $55 + (-13)$ |
| (2) $-8 + 80$ | (21) $38 + (-54)$ | (40) $-16 - (-79)$ | (59) $-58 + 1$ |
| (3) $-60 + 84$ | (22) $90 + 2$ | (41) $-48 - (-23)$ | (60) $-74 + (-35)$ |
| (4) $-1 - (-74)$ | (23) $92 + 46$ | (42) $-25 - 50$ | (61) $-76 + 29$ |
| (5) $-88 - 77$ | (24) $-91 - (-66)$ | (43) $61 + 40$ | (62) $85 - (-3)$ |
| (6) $85 + (-36)$ | (25) $-16 - 53$ | (44) $-27 - 89$ | (63) $-42 - 70$ |
| (7) $92 - 14$ | (26) $-65 + (-27)$ | (45) $-48 - (-4)$ | (64) $-24 + 100$ |
| (8) $-100 + 69$ | (27) $-92 + 22$ | (46) $-29 - (-16)$ | (65) $-7 + 96$ |
| (9) $39 + 68$ | (28) $-40 - 36$ | (47) $-67 - (-21)$ | (66) $-36 - (-14)$ |
| (10) $5 - (-66)$ | (29) $5 + 55$ | (48) $-34 - (-47)$ | (67) $-91 - 34$ |
| (11) $-82 + 66$ | (30) $-26 + (-13)$ | (49) $-99 + 84$ | (68) $22 + 5$ |
| (12) $44 + 98$ | (31) $69 + 1$ | (50) $-67 + 13$ | (69) $33 - (-76)$ |
| (13) $-3 + (-54)$ | (32) $35 - 54$ | (51) $-3 - 50$ | (70) $34 - (-32)$ |
| (14) $-65 - (-12)$ | (33) $-73 - 74$ | (52) $38 - 41$ | (71) $-72 + 7$ |
| (15) $-100 - 93$ | (34) $-67 + (-80)$ | (53) $41 + (-39)$ | (72) $47 - 53$ |
| (16) $-39 - (-69)$ | (35) $65 + (-51)$ | (54) $-19 + (-42)$ | (73) $63 + (-84)$ |
| (17) $-9 - (-62)$ | (36) $6 - 50$ | (55) $24 - (-9)$ | (74) $-51 + (-15)$ |
| (18) $30 + (-32)$ | (37) $-63 + 54$ | (56) $-58 - (-22)$ | (75) $-26 + 73$ |
| (19) $-88 + 49$ | (38) $-55 + 30$ | (57) $-15 - (-49)$ | (76) $-76 + 76$ |

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|--------------------|---------------------|--------------------|---------------------|
| (1) $-79 - 58$ | (20) $31 + (-53)$ | (39) $-93 + (-19)$ | (58) $84 + 74$ |
| (2) $70 + 45$ | (21) $-12 + 50$ | (40) $-20 - (-32)$ | (59) $-9 + (-92)$ |
| (3) $-60 + 78$ | (22) $86 - (-69)$ | (41) $-70 - 15$ | (60) $-5 - 78$ |
| (4) $2 - (-53)$ | (23) $72 - (-96)$ | (42) $5 - 51$ | (61) $52 - 57$ |
| (5) $-29 + (-4)$ | (24) $80 + (-53)$ | (43) $78 - 2$ | (62) $-92 - (-35)$ |
| (6) $-75 - 97$ | (25) $-58 + 20$ | (44) $90 + 57$ | (63) $14 + (-89)$ |
| (7) $-80 + (-32)$ | (26) $-100 + (-20)$ | (45) $-3 - (-24)$ | (64) $83 + 69$ |
| (8) $66 + (-5)$ | (27) $-35 + (-54)$ | (46) $-95 - 88$ | (65) $77 + (-50)$ |
| (9) $-22 - (-63)$ | (28) $-43 + (-44)$ | (47) $-45 - (-95)$ | (66) $38 + (-59)$ |
| (10) $95 + 64$ | (29) $-70 + 81$ | (48) $57 + (-36)$ | (67) $90 + (-21)$ |
| (11) $-16 + (-24)$ | (30) $18 - (-53)$ | (49) $-98 + (-72)$ | (68) $15 - 12$ |
| (12) $-8 - (-14)$ | (31) $76 + 100$ | (50) $46 + (-79)$ | (69) $71 + 68$ |
| (13) $16 - 86$ | (32) $-58 + (-12)$ | (51) $-29 + 7$ | (70) $-100 + (-17)$ |
| (14) $-77 - 97$ | (33) $50 - 72$ | (52) $-85 - (-30)$ | (71) $-61 + (-94)$ |
| (15) $-71 + 14$ | (34) $0 - (-35)$ | (53) $69 + (-33)$ | (72) $-74 - (-89)$ |
| (16) $-14 - 66$ | (35) $14 - 82$ | (54) $29 - 39$ | (73) $64 + 100$ |
| (17) $-9 - (-60)$ | (36) $67 + 10$ | (55) $-94 + 95$ | (74) $-91 - 78$ |
| (18) $-92 + (-48)$ | (37) $-75 + 17$ | (56) $-50 + 0$ | (75) $84 + (-62)$ |
| (19) $27 + (-69)$ | (38) $-69 - 38$ | (57) $-1 + 31$ | (76) $-74 + (-9)$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $65 + 16$ | (20) $11 - 79$ | (39) $56 + 87$ | (58) $-28 - 58$ |
| (2) $-3 - (-10)$ | (21) $19 - 56$ | (40) $-13 - 84$ | (59) $-6 - (-8)$ |
| (3) $-17 - 64$ | (22) $93 - (-95)$ | (41) $-36 + (-15)$ | (60) $-97 + (-28)$ |
| (4) $-95 + (-14)$ | (23) $-12 + (-31)$ | (42) $-56 + 40$ | (61) $33 + 63$ |
| (5) $86 - (-58)$ | (24) $42 - 100$ | (43) $-15 - (-92)$ | (62) $43 - 0$ |
| (6) $-13 + 65$ | (25) $-64 - (-8)$ | (44) $30 - 84$ | (63) $73 + 18$ |
| (7) $91 + 62$ | (26) $-74 - 58$ | (45) $-30 + (-52)$ | (64) $100 + (-78)$ |
| (8) $-11 + (-98)$ | (27) $-88 - 66$ | (46) $-89 + 53$ | (65) $68 + 65$ |
| (9) $3 + 63$ | (28) $77 + 73$ | (47) $-76 + 76$ | (66) $44 - (-100)$ |
| (10) $-25 - (-92)$ | (29) $82 - (-57)$ | (48) $19 - (-65)$ | (67) $19 - (-92)$ |
| (11) $-51 + (-24)$ | (30) $91 - (-100)$ | (49) $-61 + 4$ | (68) $72 - (-22)$ |
| (12) $69 - 86$ | (31) $-20 - (-12)$ | (50) $-93 + (-3)$ | (69) $-54 + 93$ |
| (13) $-56 + 92$ | (32) $-51 + 43$ | (51) $-48 + (-26)$ | (70) $-90 - 39$ |
| (14) $61 - (-52)$ | (33) $-9 + 10$ | (52) $26 + (-19)$ | (71) $35 - (-90)$ |
| (15) $-39 - 95$ | (34) $44 - (-54)$ | (53) $52 - 28$ | (72) $65 + (-55)$ |
| (16) $-68 - 100$ | (35) $-53 + (-54)$ | (54) $14 - 18$ | (73) $-40 - 96$ |
| (17) $-91 - 100$ | (36) $21 + (-88)$ | (55) $76 - (-31)$ | (74) $-61 + 11$ |
| (18) $72 - 12$ | (37) $-7 + 59$ | (56) $-73 - 69$ | (75) $70 + (-42)$ |
| (19) $-71 - (-66)$ | (38) $96 + 60$ | (57) $-1 - (-18)$ | (76) $-68 + 1$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $85 - 87$ | (20) $99 + (-38)$ | (39) $-55 + 69$ | (58) $26 - (-57)$ |
| (2) $-29 + (-62)$ | (21) $-56 - (-53)$ | (40) $61 - (-39)$ | (59) $-59 - (-73)$ |
| (3) $-91 - (-32)$ | (22) $46 + 16$ | (41) $-28 + 80$ | (60) $-68 + (-49)$ |
| (4) $-45 + (-76)$ | (23) $-42 + 22$ | (42) $22 + (-44)$ | (61) $-100 - 0$ |
| (5) $99 + 100$ | (24) $39 + 18$ | (43) $64 + 47$ | (62) $-18 + 64$ |
| (6) $22 - (-68)$ | (25) $27 - 2$ | (44) $54 - (-87)$ | (63) $8 + (-20)$ |
| (7) $24 - (-5)$ | (26) $82 - (-65)$ | (45) $-100 + 22$ | (64) $-38 - (-24)$ |
| (8) $19 + 35$ | (27) $90 - 56$ | (46) $14 - 70$ | (65) $-50 + 24$ |
| (9) $92 + 53$ | (28) $72 + 14$ | (47) $-94 - 61$ | (66) $-78 - 45$ |
| (10) $-26 - 21$ | (29) $2 - 82$ | (48) $57 - (-88)$ | (67) $1 + (-15)$ |
| (11) $36 - 16$ | (30) $-39 + (-39)$ | (49) $0 + (-29)$ | (68) $46 + 57$ |
| (12) $-33 + (-77)$ | (31) $13 - 98$ | (50) $-21 + 18$ | (69) $-18 - 85$ |
| (13) $-9 - (-9)$ | (32) $85 + 54$ | (51) $-37 + (-1)$ | (70) $-12 + (-3)$ |
| (14) $-82 + 91$ | (33) $-98 + (-68)$ | (52) $-22 - (-73)$ | (71) $-37 + (-44)$ |
| (15) $89 + 45$ | (34) $-4 + (-11)$ | (53) $-69 - (-15)$ | (72) $-72 - (-89)$ |
| (16) $79 - (-64)$ | (35) $39 - (-86)$ | (54) $-78 - (-93)$ | (73) $-78 + 76$ |
| (17) $9 + (-41)$ | (36) $-38 + (-68)$ | (55) $44 + 8$ | (74) $-84 - (-65)$ |
| (18) $-55 + (-68)$ | (37) $-44 + 36$ | (56) $-65 + 47$ | (75) $-25 + (-24)$ |
| (19) $5 + 47$ | (38) $29 + (-34)$ | (57) $-83 + (-16)$ | (76) $42 - (-55)$ |

(1) $-49 + 2$

(20) $-69 - 33$

(39) $-7 + 84$

(58) $-73 - 57$

(2) $47 - (-49)$

(21) $5 + 45$

(40) $-59 - (-70)$

(59) $58 + (-18)$

(3) $-65 - (-71)$

(22) $-78 + 1$

(41) $9 - 29$

(60) $19 + (-9)$

(4) $86 - 39$

(23) $-52 - (-11)$

(42) $45 - (-75)$

(61) $52 + 22$

(5) $32 + 38$

(24) $-46 + 31$

(43) $-25 + (-10)$

(62) $-33 - (-88)$

(6) $4 - 66$

(25) $-48 - 46$

(44) $-78 - 18$

(63) $67 + (-75)$

(7) $-26 - 27$

(26) $-18 - (-46)$

(45) $-75 - 35$

(64) $-16 - (-20)$

(8) $-46 - (-15)$

(27) $-28 - 55$

(46) $-58 + 70$

(65) $-84 - 52$

(9) $61 - 41$

(28) $55 + (-72)$

(47) $-67 + 16$

(66) $-3 + (-96)$

(10) $-54 - 3$

(29) $38 - 34$

(48) $66 - (-19)$

(67) $16 - (-35)$

(11) $79 - (-36)$

(30) $-63 - 80$

(49) $-18 + 15$

(68) $98 - 46$

(12) $9 + (-17)$

(31) $-72 - (-44)$

(50) $-25 + 42$

(69) $10 - (-47)$

(13) $-14 - (-24)$

(32) $75 - 74$

(51) $-82 - 61$

(70) $-46 - (-79)$

(14) $5 - (-20)$

(33) $66 - (-63)$

(52) $18 + 61$

(71) $83 - 76$

(15) $-6 - 47$

(34) $-69 - 15$

(53) $-54 - 43$

(72) $25 + (-45)$

(16) $57 - (-79)$

(35) $15 - (-9)$

(54) $74 + (-99)$

(73) $-23 + 97$

(17) $8 + 58$

(36) $-88 + 82$

(55) $-54 + 97$

(74) $48 - (-6)$

(18) $-12 + (-98)$

(37) $72 + 91$

(56) $-66 + 39$

(75) $23 - 18$

(19) $-82 - (-12)$

(38) $83 + 34$

(57) $-36 - 64$

(76) $20 - (-83)$

- | | | | |
|--------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $-27 - 61$
$= -88$ | (20) $67 + (-8)$
$= 59$ | (39) $-60 - 8$
$= -68$ | (58) $28 + 93$
$= 121$ |
| (2) $-19 - (-83)$
$= 64$ | (21) $-8 - (-63)$
$= 55$ | (40) $-82 + 26$
$= -56$ | (59) $-42 - (-84)$
$= 42$ |
| (3) $-84 + (-79)$
$= -163$ | (22) $-78 + (-39)$
$= -117$ | (41) $17 - 38$
$= -21$ | (60) $66 - 36$
$= 30$ |
| (4) $-65 - 46$
$= -111$ | (23) $-36 - 51$
$= -87$ | (42) $-70 + (-43)$
$= -113$ | (61) $-96 - (-95)$
$= -1$ |
| (5) $83 - 66$
$= 17$ | (24) $88 + 26$
$= 114$ | (43) $40 - (-73)$
$= 113$ | (62) $-100 + (-23)$
$= -123$ |
| (6) $-77 + (-95)$
$= -172$ | (25) $-84 - 18$
$= -102$ | (44) $26 + 45$
$= 71$ | (63) $-85 + (-69)$
$= -154$ |
| (7) $-8 - (-18)$
$= 10$ | (26) $-8 + 42$
$= 34$ | (45) $11 + (-86)$
$= -75$ | (64) $-34 + 89$
$= 55$ |
| (8) $5 + (-73)$
$= -68$ | (27) $-51 + 41$
$= -10$ | (46) $38 + 88$
$= 126$ | (65) $-78 + 45$
$= -33$ |
| (9) $-17 + 37$
$= 20$ | (28) $50 + 86$
$= 136$ | (47) $67 - 35$
$= 32$ | (66) $6 - (-56)$
$= 62$ |
| (10) $-95 - (-11)$
$= -84$ | (29) $16 - 60$
$= -44$ | (48) $-79 - (-58)$
$= -21$ | (67) $6 - (-40)$
$= 46$ |
| (11) $64 + 37$
$= 101$ | (30) $-1 + (-16)$
$= -17$ | (49) $-15 - 68$
$= -83$ | (68) $67 + (-39)$
$= 28$ |
| (12) $-56 + 76$
$= 20$ | (31) $69 + (-30)$
$= 39$ | (50) $-3 - 59$
$= -62$ | (69) $89 - (-90)$
$= 179$ |
| (13) $-79 - 46$
$= -125$ | (32) $-51 + 84$
$= 33$ | (51) $-86 - 26$
$= -112$ | (70) $-45 + (-34)$
$= -79$ |
| (14) $-10 - (-1)$
$= -9$ | (33) $-99 - (-79)$
$= -20$ | (52) $72 + (-66)$
$= 6$ | (71) $-38 + 47$
$= 9$ |
| (15) $-30 - (-27)$
$= -3$ | (34) $86 + (-24)$
$= 62$ | (53) $72 - 79$
$= -7$ | (72) $36 - 34$
$= 2$ |
| (16) $7 - 16$
$= -9$ | (35) $-52 + 41$
$= -11$ | (54) $-36 - 70$
$= -106$ | (73) $64 + 88$
$= 152$ |
| (17) $-35 + 3$
$= -32$ | (36) $-5 + 47$
$= 42$ | (55) $61 + (-25)$
$= 36$ | (74) $-38 + 32$
$= -6$ |
| (18) $-13 + 80$
$= 67$ | (37) $29 - 15$
$= 14$ | (56) $-53 - (-66)$
$= 13$ | (75) $9 + (-87)$
$= -78$ |
| (19) $-79 + (-38)$
$= -117$ | (38) $29 + (-40)$
$= -11$ | (57) $-83 + (-98)$
$= -181$ | (76) $59 + 45$
$= 104$ |

(1) $-3 + (-46)$ $= -49$	(20) $10 + (-59)$ $= -49$	(39) $-40 - (-40)$ $= 0$	(58) $-18 + 28$ $= 10$
(2) $2 - 21$ $= -19$	(21) $-53 + 37$ $= -16$	(40) $74 - (-33)$ $= 107$	(59) $-69 + (-10)$ $= -79$
(3) $64 - (-42)$ $= 106$	(22) $-95 - (-44)$ $= -51$	(41) $68 - (-57)$ $= 125$	(60) $70 + (-22)$ $= 48$
(4) $31 - 77$ $= -46$	(23) $28 + (-13)$ $= 15$	(42) $-68 - 4$ $= -72$	(61) $82 - 39$ $= 43$
(5) $-7 - 20$ $= -27$	(24) $-13 + 98$ $= 85$	(43) $-36 + (-11)$ $= -47$	(62) $33 + 88$ $= 121$
(6) $-41 + (-44)$ $= -85$	(25) $-85 - 75$ $= -160$	(44) $-75 + 42$ $= -33$	(63) $-54 - 41$ $= -95$
(7) $-27 - (-96)$ $= 69$	(26) $-42 - (-72)$ $= 30$	(45) $49 + (-4)$ $= 45$	(64) $7 + (-70)$ $= -63$
(8) $-77 - (-6)$ $= -71$	(27) $51 + (-56)$ $= -5$	(46) $61 - 47$ $= 14$	(65) $-38 + 81$ $= 43$
(9) $46 + (-32)$ $= 14$	(28) $36 - (-34)$ $= 70$	(47) $-26 - 21$ $= -47$	(66) $50 + 37$ $= 87$
(10) $-33 + 99$ $= 66$	(29) $-81 + (-79)$ $= -160$	(48) $-74 + 29$ $= -45$	(67) $88 + 94$ $= 182$
(11) $-65 + (-37)$ $= -102$	(30) $-31 + 4$ $= -27$	(49) $-71 - (-90)$ $= 19$	(68) $82 - (-41)$ $= 123$
(12) $95 + 41$ $= 136$	(31) $-83 - (-63)$ $= -20$	(50) $95 - (-37)$ $= 132$	(69) $-62 + 90$ $= 28$
(13) $-100 + (-11)$ $= -111$	(32) $34 + 48$ $= 82$	(51) $66 - 61$ $= 5$	(70) $85 - (-18)$ $= 103$
(14) $33 - (-81)$ $= 114$	(33) $-78 - 73$ $= -151$	(52) $96 + (-67)$ $= 29$	(71) $96 - 25$ $= 71$
(15) $90 - 17$ $= 73$	(34) $-34 + (-65)$ $= -99$	(53) $58 + 18$ $= 76$	(72) $-10 - 35$ $= -45$
(16) $28 + 13$ $= 41$	(35) $-90 + 24$ $= -66$	(54) $64 - (-16)$ $= 80$	(73) $88 - 100$ $= -12$
(17) $77 - 7$ $= 70$	(36) $29 - 27$ $= 2$	(55) $-7 + (-17)$ $= -24$	(74) $-12 + 96$ $= 84$
(18) $-67 + 89$ $= 22$	(37) $81 - (-65)$ $= 146$	(56) $22 + (-57)$ $= -35$	(75) $64 - 54$ $= 10$
(19) $-84 + 13$ $= -71$	(38) $-88 + (-33)$ $= -121$	(57) $14 + 14$ $= 28$	(76) $45 + 41$ $= 86$

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|--------------------------------|-------------------------------|--------------------------------|---------------------------------|
| (1) $-92 + (-6)$
$= -98$ | (20) $-8 - 82$
$= -90$ | (39) $-73 - (-91)$
$= 18$ | (58) $-11 - (-71)$
$= 60$ |
| (2) $-63 + (-87)$
$= -150$ | (21) $78 + (-24)$
$= 54$ | (40) $-61 + (-77)$
$= -138$ | (59) $89 + (-4)$
$= 85$ |
| (3) $-33 + (-36)$
$= -69$ | (22) $50 - 56$
$= -6$ | (41) $-68 - (-14)$
$= -54$ | (60) $-71 - 23$
$= -94$ |
| (4) $-100 + 14$
$= -86$ | (23) $61 - (-93)$
$= 154$ | (42) $48 + (-77)$
$= -29$ | (61) $-100 + (-26)$
$= -126$ |
| (5) $-70 - 82$
$= -152$ | (24) $-54 - 73$
$= -127$ | (43) $-7 + (-50)$
$= -57$ | (62) $92 + 29$
$= 121$ |
| (6) $-31 - 58$
$= -89$ | (25) $39 - 74$
$= -35$ | (44) $70 + 19$
$= 89$ | (63) $-70 - (-36)$
$= -34$ |
| (7) $-65 + 33$
$= -32$ | (26) $-53 - 46$
$= -99$ | (45) $24 + 4$
$= 28$ | (64) $-65 - 70$
$= -135$ |
| (8) $41 - 41$
$= 0$ | (27) $-68 - (-100)$
$= 32$ | (46) $-63 - 46$
$= -109$ | (65) $41 - (-99)$
$= 140$ |
| (9) $77 + 95$
$= 172$ | (28) $-21 - (-20)$
$= -1$ | (47) $-74 - 3$
$= -77$ | (66) $-41 - 45$
$= -86$ |
| (10) $81 + 4$
$= 85$ | (29) $-23 - (-4)$
$= -19$ | (48) $37 - (-91)$
$= 128$ | (67) $-52 + 82$
$= 30$ |
| (11) $-29 + (-92)$
$= -121$ | (30) $89 + 100$
$= 189$ | (49) $-39 - 73$
$= -112$ | (68) $51 + (-61)$
$= -10$ |
| (12) $42 + 20$
$= 62$ | (31) $10 - 13$
$= -3$ | (50) $-92 - 29$
$= -121$ | (69) $-21 - (-49)$
$= 28$ |
| (13) $-87 - 13$
$= -100$ | (32) $-86 - (-91)$
$= 5$ | (51) $89 + (-53)$
$= 36$ | (70) $57 + 37$
$= 94$ |
| (14) $94 + (-56)$
$= 38$ | (33) $14 + 59$
$= 73$ | (52) $99 - 75$
$= 24$ | (71) $59 - 50$
$= 9$ |
| (15) $75 - 65$
$= 10$ | (34) $86 - 66$
$= 20$ | (53) $-74 + (-13)$
$= -87$ | (72) $-35 + (-81)$
$= -116$ |
| (16) $82 + 78$
$= 160$ | (35) $-4 - (-82)$
$= 78$ | (54) $-37 + (-80)$
$= -117$ | (73) $-72 - 91$
$= -163$ |
| (17) $-25 + 87$
$= 62$ | (36) $20 - 73$
$= -53$ | (55) $-95 - 81$
$= -176$ | (74) $10 - (-26)$
$= 36$ |
| (18) $-14 + 44$
$= 30$ | (37) $11 - (-67)$
$= 78$ | (56) $76 + (-98)$
$= -22$ | (75) $35 + (-59)$
$= -24$ |
| (19) $-50 + (-45)$
$= -95$ | (38) $-22 - 20$
$= -42$ | (57) $-37 - (-3)$
$= -34$ | (76) $100 - (-79)$
$= 179$ |

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|--------------------------------|--------------------------------|---------------------------------|--------------------------------|
| (1) $88 + 26$
$= 114$ | (20) $-94 + (-14)$
$= -108$ | (39) $73 - 60$
$= 13$ | (58) $-81 - (-32)$
$= -49$ |
| (2) $-99 - (-95)$
$= -4$ | (21) $-22 - (-35)$
$= 13$ | (40) $-69 + (-51)$
$= -120$ | (59) $61 - (-20)$
$= 81$ |
| (3) $18 - 92$
$= -74$ | (22) $-25 + 52$
$= 27$ | (41) $83 - (-44)$
$= 127$ | (60) $53 - 89$
$= -36$ |
| (4) $-41 - (-93)$
$= 52$ | (23) $-94 - (-8)$
$= -86$ | (42) $-100 + (-97)$
$= -197$ | (61) $84 + 35$
$= 119$ |
| (5) $0 + 82$
$= 82$ | (24) $-72 - (-67)$
$= -5$ | (43) $9 + 37$
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$= 25$ |
| (6) $-31 - 87$
$= -118$ | (25) $34 + 43$
$= 77$ | (44) $73 + 96$
$= 169$ | (63) $-69 - 65$
$= -134$ |
| (7) $-48 - 35$
$= -83$ | (26) $80 - 52$
$= 28$ | (45) $-48 - 16$
$= -64$ | (64) $-12 + (-64)$
$= -76$ |
| (8) $15 + 82$
$= 97$ | (27) $-32 + (-95)$
$= -127$ | (46) $2 + 71$
$= 73$ | (65) $43 + 61$
$= 104$ |
| (9) $-54 + 68$
$= 14$ | (28) $11 - (-72)$
$= 83$ | (47) $-6 - (-88)$
$= 82$ | (66) $19 - 49$
$= -30$ |
| (10) $-98 + (-45)$
$= -143$ | (29) $2 + 96$
$= 98$ | (48) $-89 + (-86)$
$= -175$ | (67) $-9 - (-30)$
$= 21$ |
| (11) $19 + (-97)$
$= -78$ | (30) $-16 - 56$
$= -72$ | (49) $-16 + 75$
$= 59$ | (68) $-66 + (-58)$
$= -124$ |
| (12) $-6 - (-26)$
$= 20$ | (31) $20 + (-32)$
$= -12$ | (50) $-93 - (-94)$
$= 1$ | (69) $64 - (-89)$
$= 153$ |
| (13) $-24 + 95$
$= 71$ | (32) $84 - (-12)$
$= 96$ | (51) $-79 - 57$
$= -136$ | (70) $-52 - (-8)$
$= -44$ |
| (14) $-81 + (-88)$
$= -169$ | (33) $-98 - (-38)$
$= -60$ | (52) $-72 - 79$
$= -151$ | (71) $-68 - (-68)$
$= 0$ |
| (15) $62 - 4$
$= 58$ | (34) $91 + 85$
$= 176$ | (53) $20 + 93$
$= 113$ | (72) $14 + 31$
$= 45$ |
| (16) $83 - (-8)$
$= 91$ | (35) $-95 - (-12)$
$= -83$ | (54) $42 - (-95)$
$= 137$ | (73) $92 + 86$
$= 178$ |
| (17) $11 + 25$
$= 36$ | (36) $69 - (-90)$
$= 159$ | (55) $-80 + (-58)$
$= -138$ | (74) $-97 - 68$
$= -165$ |
| (18) $91 + (-58)$
$= 33$ | (37) $-93 - 28$
$= -121$ | (56) $12 - (-74)$
$= 86$ | (75) $57 - 15$
$= 42$ |
| (19) $37 - (-80)$
$= 117$ | (38) $-79 - (-32)$
$= -47$ | (57) $98 - (-23)$
$= 121$ | (76) $-31 - (-13)$
$= -18$ |

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|------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $-55 - (-76)$
$= 21$ | (20) $20 - (-8)$
$= 28$ | (39) $-92 - 64$
$= -156$ | (58) $33 - (-23)$
$= 56$ |
| (2) $4 + (-15)$
$= -11$ | (21) $57 - (-64)$
$= 121$ | (40) $93 - 45$
$= 48$ | (59) $20 - (-47)$
$= 67$ |
| (3) $-90 + 55$
$= -35$ | (22) $-82 + (-59)$
$= -141$ | (41) $64 - 91$
$= -27$ | (60) $48 - (-11)$
$= 59$ |
| (4) $27 - (-51)$
$= 78$ | (23) $-34 - 4$
$= -38$ | (42) $43 - 10$
$= 33$ | (61) $79 + (-46)$
$= 33$ |
| (5) $92 + 14$
$= 106$ | (24) $-73 + 64$
$= -9$ | (43) $-37 - (-76)$
$= 39$ | (62) $-81 - (-18)$
$= -63$ |
| (6) $60 + 57$
$= 117$ | (25) $-86 + (-92)$
$= -178$ | (44) $-90 + (-45)$
$= -135$ | (63) $82 + (-3)$
$= 79$ |
| (7) $100 + 26$
$= 126$ | (26) $-41 - 39$
$= -80$ | (45) $-26 + (-10)$
$= -36$ | (64) $-46 - (-52)$
$= 6$ |
| (8) $71 - (-46)$
$= 117$ | (27) $-41 - (-74)$
$= 33$ | (46) $-43 + 59$
$= 16$ | (65) $66 - (-3)$
$= 69$ |
| (9) $88 + 62$
$= 150$ | (28) $-83 + 59$
$= -24$ | (47) $-35 - (-1)$
$= -34$ | (66) $71 + (-24)$
$= 47$ |
| (10) $-2 + (-25)$
$= -27$ | (29) $91 - 55$
$= 36$ | (48) $15 + (-14)$
$= 1$ | (67) $-77 + (-90)$
$= -167$ |
| (11) $17 - (-87)$
$= 104$ | (30) $45 + 79$
$= 124$ | (49) $13 + 29$
$= 42$ | (68) $11 - 80$
$= -69$ |
| (12) $47 - 7$
$= 40$ | (31) $65 + 42$
$= 107$ | (50) $-40 - (-19)$
$= -21$ | (69) $-63 - (-12)$
$= -51$ |
| (13) $49 - 38$
$= 11$ | (32) $61 + (-35)$
$= 26$ | (51) $-82 + (-58)$
$= -140$ | (70) $-98 + 19$
$= -79$ |
| (14) $45 + (-92)$
$= -47$ | (33) $-90 - 75$
$= -165$ | (52) $-77 - (-81)$
$= 4$ | (71) $66 + (-42)$
$= 24$ |
| (15) $72 - (-55)$
$= 127$ | (34) $78 + (-56)$
$= 22$ | (53) $50 - (-47)$
$= 97$ | (72) $-78 - (-25)$
$= -53$ |
| (16) $-42 - (-61)$
$= 19$ | (35) $-72 - (-93)$
$= 21$ | (54) $-49 - 97$
$= -146$ | (73) $64 - 15$
$= 49$ |
| (17) $57 - 89$
$= -32$ | (36) $2 - (-65)$
$= 67$ | (55) $-55 - (-57)$
$= 2$ | (74) $71 - (-57)$
$= 128$ |
| (18) $-62 - 90$
$= -152$ | (37) $-95 + (-53)$
$= -148$ | (56) $85 - (-12)$
$= 97$ | (75) $57 + 91$
$= 148$ |
| (19) $-46 - 100$
$= -146$ | (38) $80 + 50$
$= 130$ | (57) $-73 - (-29)$
$= -44$ | (76) $57 - 13$
$= 44$ |

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|-------------------------------|--------------------------------|-------------------------------|--------------------------------|
| (1) $-72 - 43$
$= -115$ | (20) $-16 + (-77)$
$= -93$ | (39) $-52 + 8$
$= -44$ | (58) $73 - 6$
$= 67$ |
| (2) $-36 + (-20)$
$= -56$ | (21) $11 + (-27)$
$= -16$ | (40) $57 + (-11)$
$= 46$ | (59) $68 - (-6)$
$= 74$ |
| (3) $64 + (-17)$
$= 47$ | (22) $-45 + (-28)$
$= -73$ | (41) $16 - 12$
$= 4$ | (60) $-62 - (-9)$
$= -53$ |
| (4) $94 - (-56)$
$= 150$ | (23) $-31 - 39$
$= -70$ | (42) $37 + 52$
$= 89$ | (61) $60 + (-90)$
$= -30$ |
| (5) $-96 - 10$
$= -106$ | (24) $-1 - 67$
$= -68$ | (43) $59 - 61$
$= -2$ | (62) $49 - (-77)$
$= 126$ |
| (6) $-34 - 2$
$= -36$ | (25) $-84 + 72$
$= -12$ | (44) $-1 - 4$
$= -5$ | (63) $11 - (-15)$
$= 26$ |
| (7) $-32 - 29$
$= -61$ | (26) $15 + (-76)$
$= -61$ | (45) $-68 - 31$
$= -99$ | (64) $41 + (-76)$
$= -35$ |
| (8) $42 - (-67)$
$= 109$ | (27) $-30 - (-4)$
$= -26$ | (46) $96 - (-20)$
$= 116$ | (65) $96 + (-37)$
$= 59$ |
| (9) $-38 - 97$
$= -135$ | (28) $-6 - (-61)$
$= 55$ | (47) $46 - 45$
$= 1$ | (66) $-51 + 32$
$= -19$ |
| (10) $46 - 12$
$= 34$ | (29) $13 + 31$
$= 44$ | (48) $-67 - (-35)$
$= -32$ | (67) $96 + 64$
$= 160$ |
| (11) $10 + (-47)$
$= -37$ | (30) $8 + (-10)$
$= -2$ | (49) $27 - (-76)$
$= 103$ | (68) $-44 - 57$
$= -101$ |
| (12) $80 - (-42)$
$= 122$ | (31) $66 - 98$
$= -32$ | (50) $62 - (-81)$
$= 143$ | (69) $-95 + (-38)$
$= -133$ |
| (13) $-68 + 48$
$= -20$ | (32) $-41 - 66$
$= -107$ | (51) $-42 + (-13)$
$= -55$ | (70) $-82 - (-59)$
$= -23$ |
| (14) $-78 + 32$
$= -46$ | (33) $-99 + (-77)$
$= -176$ | (52) $-19 + (-62)$
$= -81$ | (71) $-3 + (-25)$
$= -28$ |
| (15) $93 - 84$
$= 9$ | (34) $-5 - 92$
$= -97$ | (53) $85 + 100$
$= 185$ | (72) $90 + (-55)$
$= 35$ |
| (16) $66 - 86$
$= -20$ | (35) $98 - (-53)$
$= 151$ | (54) $85 - 35$
$= 50$ | (73) $63 - (-55)$
$= 118$ |
| (17) $-92 + 50$
$= -42$ | (36) $-48 + (-17)$
$= -65$ | (55) $42 - 5$
$= 37$ | (74) $25 + 15$
$= 40$ |
| (18) $-65 - (-49)$
$= -16$ | (37) $95 - 0$
$= 95$ | (56) $54 + (-100)$
$= -46$ | (75) $54 - (-74)$
$= 128$ |
| (19) $-97 - (-65)$
$= -32$ | (38) $44 - 48$
$= -4$ | (57) $-1 - (-90)$
$= 89$ | (76) $27 + 47$
$= 74$ |

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|-----------------------------|-----------------------------|----------------------------|------------------------------|
| (1) $81 - (-85)$
= 166 | (20) $4 + 13$
= 17 | (39) $77 + (-32)$
= 45 | (58) $97 + (-15)$
= 82 |
| (2) $38 - 85$
= -47 | (21) $19 - 83$
= -64 | (40) $-69 + 0$
= -69 | (59) $56 + 55$
= 111 |
| (3) $-4 - (-26)$
= 22 | (22) $87 - (-12)$
= 99 | (41) $1 - 60$
= -59 | (60) $-92 - (-93)$
= 1 |
| (4) $11 + 95$
= 106 | (23) $-7 - (-6)$
= -1 | (42) $-65 - (-64)$
= -1 | (61) $-83 + (-25)$
= -108 |
| (5) $68 - (-27)$
= 95 | (24) $-51 + 31$
= -20 | (43) $-37 + 100$
= 63 | (62) $-36 - (-39)$
= 3 |
| (6) $59 - 38$
= 21 | (25) $65 - 76$
= -11 | (44) $-97 + 75$
= -22 | (63) $-10 - (-11)$
= 1 |
| (7) $-35 + (-35)$
= -70 | (26) $17 - (-96)$
= 113 | (45) $46 + 28$
= 74 | (64) $80 + 58$
= 138 |
| (8) $35 + (-50)$
= -15 | (27) $67 + 5$
= 72 | (46) $-2 + 10$
= 8 | (65) $-91 + (-16)$
= -107 |
| (9) $82 - 56$
= 26 | (28) $76 + (-80)$
= -4 | (47) $2 - 37$
= -35 | (66) $15 - 13$
= 2 |
| (10) $-22 - (-73)$
= 51 | (29) $-94 + 48$
= -46 | (48) $15 + 63$
= 78 | (67) $-71 - (-81)$
= 10 |
| (11) $89 - 5$
= 84 | (30) $48 - 40$
= 8 | (49) $-42 - 3$
= -45 | (68) $96 + (-19)$
= 77 |
| (12) $-82 - (-24)$
= -58 | (31) $-24 - (-52)$
= 28 | (50) $78 - (-79)$
= 157 | (69) $6 - (-67)$
= 73 |
| (13) $14 - (-20)$
= 34 | (32) $-66 - (-11)$
= -55 | (51) $16 - 60$
= -44 | (70) $85 + (-78)$
= 7 |
| (14) $74 - 24$
= 50 | (33) $-60 + 79$
= 19 | (52) $52 - (-41)$
= 93 | (71) $-6 + 85$
= 79 |
| (15) $-34 - 93$
= -127 | (34) $-96 + 68$
= -28 | (53) $82 + (-12)$
= 70 | (72) $-22 + 42$
= 20 |
| (16) $-67 - (-44)$
= -23 | (35) $-33 - (-61)$
= 28 | (54) $-45 - 98$
= -143 | (73) $-41 + (-77)$
= -118 |
| (17) $-95 - (-14)$
= -81 | (36) $95 - 43$
= 52 | (55) $77 - 8$
= 69 | (74) $-38 - 49$
= -87 |
| (18) $92 + 44$
= 136 | (37) $13 + (-39)$
= -26 | (56) $-62 + 94$
= 32 | (75) $-66 - (-71)$
= 5 |
| (19) $86 + (-50)$
= 36 | (38) $-100 + 57$
= -43 | (57) $-51 - (-93)$
= 42 | (76) $-69 - (-3)$
= -66 |

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|--------------------------------|--------------------------------|---------------------------------|--------------------------------|
| (1) $48 + 30$
$= 78$ | (20) $14 + (-84)$
$= -70$ | (39) $97 + (-7)$
$= 90$ | (58) $-41 - (-49)$
$= 8$ |
| (2) $15 + 3$
$= 18$ | (21) $-80 + (-1)$
$= -81$ | (40) $-72 + 20$
$= -52$ | (59) $-20 + 99$
$= 79$ |
| (3) $-35 + 27$
$= -8$ | (22) $-89 - (-99)$
$= 10$ | (41) $-52 + (-56)$
$= -108$ | (60) $-59 + 92$
$= 33$ |
| (4) $37 + (-21)$
$= 16$ | (23) $-55 + (-49)$
$= -104$ | (42) $60 + 12$
$= 72$ | (61) $80 + 4$
$= 84$ |
| (5) $7 - (-58)$
$= 65$ | (24) $74 - (-49)$
$= 123$ | (43) $-50 + (-53)$
$= -103$ | (62) $-95 - 12$
$= -107$ |
| (6) $-70 - 37$
$= -107$ | (25) $-70 + (-15)$
$= -85$ | (44) $-46 - 4$
$= -50$ | (63) $69 + (-21)$
$= 48$ |
| (7) $-31 - (-25)$
$= -6$ | (26) $-71 + (-94)$
$= -165$ | (45) $94 - (-72)$
$= 166$ | (64) $-50 - (-40)$
$= -10$ |
| (8) $-92 + (-12)$
$= -104$ | (27) $-30 + 99$
$= 69$ | (46) $-49 - (-59)$
$= 10$ | (65) $85 + 26$
$= 111$ |
| (9) $7 - 78$
$= -71$ | (28) $-98 + (-99)$
$= -197$ | (47) $20 + 94$
$= 114$ | (66) $94 - 58$
$= 36$ |
| (10) $-41 - 25$
$= -66$ | (29) $-64 - 95$
$= -159$ | (48) $-80 + 16$
$= -64$ | (67) $-83 + (-43)$
$= -126$ |
| (11) $-47 - (-53)$
$= 6$ | (30) $-62 - (-73)$
$= 11$ | (49) $1 + (-23)$
$= -22$ | (68) $-7 + (-97)$
$= -104$ |
| (12) $-77 + (-32)$
$= -109$ | (31) $48 - 82$
$= -34$ | (50) $15 - (-29)$
$= 44$ | (69) $83 + 18$
$= 101$ |
| (13) $-74 + 19$
$= -55$ | (32) $-1 - (-86)$
$= 85$ | (51) $3 + 31$
$= 34$ | (70) $66 - 47$
$= 19$ |
| (14) $-38 + (-48)$
$= -86$ | (33) $-1 + 64$
$= 63$ | (52) $-12 - 52$
$= -64$ | (71) $17 + 98$
$= 115$ |
| (15) $57 - (-25)$
$= 82$ | (34) $-55 - 93$
$= -148$ | (53) $-84 + 38$
$= -46$ | (72) $4 - (-26)$
$= 30$ |
| (16) $51 + (-10)$
$= 41$ | (35) $77 + 44$
$= 121$ | (54) $57 - 97$
$= -40$ | (73) $24 + (-1)$
$= 23$ |
| (17) $25 + 61$
$= 86$ | (36) $-9 - (-24)$
$= 15$ | (55) $91 - (-63)$
$= 154$ | (74) $-80 - 97$
$= -177$ |
| (18) $52 + 10$
$= 62$ | (37) $-76 + (-87)$
$= -163$ | (56) $-100 + (-30)$
$= -130$ | (75) $49 + 9$
$= 58$ |
| (19) $96 - 27$
$= 69$ | (38) $-8 - (-2)$
$= -6$ | (57) $46 - (-15)$
$= 61$ | (76) $-84 + 73$
$= -11$ |

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|-----------------------------|-----------------------------|-----------------------------|------------------------------|
| (1) $-88 - (-94)$
= 6 | (20) $84 + (-61)$
= 23 | (39) $31 + 46$
= 77 | (58) $14 - (-37)$
= 51 |
| (2) $91 + (-30)$
= 61 | (21) $21 - 70$
= -49 | (40) $16 + (-67)$
= -51 | (59) $-3 - (-40)$
= 37 |
| (3) $-64 - 96$
= -160 | (22) $-59 - 96$
= -155 | (41) $-89 - (-65)$
= -24 | (60) $-21 + (-39)$
= -60 |
| (4) $-63 - 69$
= -132 | (23) $15 - (-98)$
= 113 | (42) $-78 - (-80)$
= 2 | (61) $-96 + 44$
= -52 |
| (5) $-72 - (-28)$
= -44 | (24) $8 - 82$
= -74 | (43) $68 - 13$
= 55 | (62) $6 + 29$
= 35 |
| (6) $-87 + 75$
= -12 | (25) $6 + 79$
= 85 | (44) $21 - (-35)$
= 56 | (63) $-19 + (-9)$
= -28 |
| (7) $-99 + (-29)$
= -128 | (26) $-44 + (-53)$
= -97 | (45) $24 + 40$
= 64 | (64) $42 + 64$
= 106 |
| (8) $81 - (-87)$
= 168 | (27) $-46 - (-71)$
= 25 | (46) $36 + (-9)$
= 27 | (65) $84 + 58$
= 142 |
| (9) $-8 - (-84)$
= 76 | (28) $-34 - (-25)$
= -9 | (47) $-46 + 63$
= 17 | (66) $-90 - 96$
= -186 |
| (10) $89 - 44$
= 45 | (29) $-48 - 92$
= -140 | (48) $46 + 41$
= 87 | (67) $-95 - (-34)$
= -61 |
| (11) $78 + 20$
= 98 | (30) $-13 - 7$
= -20 | (49) $-76 - (-29)$
= -47 | (68) $-80 - (-33)$
= -47 |
| (12) $83 + 99$
= 182 | (31) $32 + (-89)$
= -57 | (50) $-39 + (-58)$
= -97 | (69) $6 + 8$
= 14 |
| (13) $-12 - (-39)$
= 27 | (32) $-41 - (-56)$
= 15 | (51) $50 + 45$
= 95 | (70) $57 + (-95)$
= -38 |
| (14) $79 + (-60)$
= 19 | (33) $44 - (-72)$
= 116 | (52) $65 + 76$
= 141 | (71) $-40 - 35$
= -75 |
| (15) $-16 - 82$
= -98 | (34) $42 + 5$
= 47 | (53) $14 - 88$
= -74 | (72) $2 - (-99)$
= 101 |
| (16) $9 + (-48)$
= -39 | (35) $37 - 91$
= -54 | (54) $1 - (-39)$
= 40 | (73) $-95 + (-25)$
= -120 |
| (17) $52 - (-2)$
= 54 | (36) $-52 + 3$
= -49 | (55) $-20 + 35$
= 15 | (74) $7 - (-82)$
= 89 |
| (18) $-28 - 15$
= -43 | (37) $-51 + (-3)$
= -54 | (56) $49 - 11$
= 38 | (75) $-29 + 5$
= -24 |
| (19) $79 + 19$
= 98 | (38) $97 + (-53)$
= 44 | (57) $-73 - (-57)$
= -16 | (76) $12 + 37$
= 49 |

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|-------------------------------|--------------------------------|-------------------------------|--------------------------------|
| (1) $70 - 41$
$= 29$ | (20) $-75 + 11$
$= -64$ | (39) $-34 - 63$
$= -97$ | (58) $70 + (-17)$
$= 53$ |
| (2) $75 - 69$
$= 6$ | (21) $87 - (-11)$
$= 98$ | (40) $30 - 65$
$= -35$ | (59) $-51 + 9$
$= -42$ |
| (3) $-5 - (-13)$
$= 8$ | (22) $38 - 62$
$= -24$ | (41) $17 - (-27)$
$= 44$ | (60) $18 + 75$
$= 93$ |
| (4) $49 + 68$
$= 117$ | (23) $7 - (-68)$
$= 75$ | (42) $61 + (-55)$
$= 6$ | (61) $-10 + (-17)$
$= -27$ |
| (5) $-78 + (-12)$
$= -90$ | (24) $-6 - 3$
$= -9$ | (43) $93 - (-45)$
$= 138$ | (62) $-22 + 24$
$= 2$ |
| (6) $69 + (-90)$
$= -21$ | (25) $59 - (-51)$
$= 110$ | (44) $42 + (-14)$
$= 28$ | (63) $-28 - 27$
$= -55$ |
| (7) $73 + 57$
$= 130$ | (26) $89 - 96$
$= -7$ | (45) $-61 - (-18)$
$= -43$ | (64) $80 + (-6)$
$= 74$ |
| (8) $68 + 83$
$= 151$ | (27) $73 + 83$
$= 156$ | (46) $-7 + (-64)$
$= -71$ | (65) $-67 + (-54)$
$= -121$ |
| (9) $49 + (-85)$
$= -36$ | (28) $70 - 11$
$= 59$ | (47) $-91 + 53$
$= -38$ | (66) $58 + (-37)$
$= 21$ |
| (10) $-60 - (-6)$
$= -54$ | (29) $27 + 19$
$= 46$ | (48) $23 + 7$
$= 30$ | (67) $-22 + (-75)$
$= -97$ |
| (11) $-98 - (-37)$
$= -61$ | (30) $32 - 54$
$= -22$ | (49) $-5 + (-7)$
$= -12$ | (68) $3 - 87$
$= -84$ |
| (12) $-77 + (-5)$
$= -82$ | (31) $-56 - 15$
$= -71$ | (50) $-21 - (-56)$
$= 35$ | (69) $-47 - (-3)$
$= -44$ |
| (13) $-13 + 53$
$= 40$ | (32) $-99 + (-50)$
$= -149$ | (51) $-17 + (-27)$
$= -44$ | (70) $54 - 21$
$= 33$ |
| (14) $-28 + 17$
$= -11$ | (33) $-31 + 33$
$= 2$ | (52) $84 - (-44)$
$= 128$ | (71) $-63 + (-34)$
$= -97$ |
| (15) $14 + (-9)$
$= 5$ | (34) $70 + 27$
$= 97$ | (53) $-4 - 19$
$= -23$ | (72) $-67 + (-54)$
$= -121$ |
| (16) $32 - 0$
$= 32$ | (35) $30 - (-69)$
$= 99$ | (54) $64 - 51$
$= 13$ | (73) $-21 - 39$
$= -60$ |
| (17) $63 + (-17)$
$= 46$ | (36) $89 + 61$
$= 150$ | (55) $-59 - (-50)$
$= -9$ | (74) $47 - (-93)$
$= 140$ |
| (18) $83 - (-55)$
$= 138$ | (37) $-85 + 9$
$= -76$ | (56) $19 - 25$
$= -6$ | (75) $88 - 18$
$= 70$ |
| (19) $-17 + 59$
$= 42$ | (38) $53 - 85$
$= -32$ | (57) $99 + (-69)$
$= 30$ | (76) $-21 - (-59)$
$= 38$ |

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|-------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $88 - 51$
$= 37$ | (20) $-84 - 42$
$= -126$ | (39) $-21 - 9$
$= -30$ | (58) $43 + 91$
$= 134$ |
| (2) $15 + 85$
$= 100$ | (21) $36 - (-63)$
$= 99$ | (40) $-63 - (-36)$
$= -27$ | (59) $-19 + (-47)$
$= -66$ |
| (3) $-91 - 32$
$= -123$ | (22) $52 + 11$
$= 63$ | (41) $1 - (-12)$
$= 13$ | (60) $-96 - (-94)$
$= -2$ |
| (4) $-65 - (-36)$
$= -29$ | (23) $100 - 53$
$= 47$ | (42) $0 - 25$
$= -25$ | (61) $-92 - 41$
$= -133$ |
| (5) $-70 - (-39)$
$= -31$ | (24) $78 - (-2)$
$= 80$ | (43) $48 - (-41)$
$= 89$ | (62) $-19 - (-44)$
$= 25$ |
| (6) $-58 + 10$
$= -48$ | (25) $-42 - (-68)$
$= 26$ | (44) $-69 + (-43)$
$= -112$ | (63) $-69 - (-74)$
$= 5$ |
| (7) $49 - (-36)$
$= 85$ | (26) $-66 + (-81)$
$= -147$ | (45) $-23 + (-72)$
$= -95$ | (64) $-86 + (-48)$
$= -134$ |
| (8) $59 - 12$
$= 47$ | (27) $22 + (-7)$
$= 15$ | (46) $-42 + 86$
$= 44$ | (65) $-85 - (-46)$
$= -39$ |
| (9) $70 - (-24)$
$= 94$ | (28) $-1 + (-65)$
$= -66$ | (47) $-52 + 43$
$= -9$ | (66) $42 + 76$
$= 118$ |
| (10) $-29 - 42$
$= -71$ | (29) $67 + 50$
$= 117$ | (48) $-56 + 39$
$= -17$ | (67) $25 + 45$
$= 70$ |
| (11) $100 + (-46)$
$= 54$ | (30) $74 + 33$
$= 107$ | (49) $-62 + 93$
$= 31$ | (68) $42 - 70$
$= -28$ |
| (12) $54 - 74$
$= -20$ | (31) $0 + (-39)$
$= -39$ | (50) $34 - (-7)$
$= 41$ | (69) $-100 + (-87)$
$= -187$ |
| (13) $36 - 70$
$= -34$ | (32) $96 + (-6)$
$= 90$ | (51) $44 - 67$
$= -23$ | (70) $84 + 45$
$= 129$ |
| (14) $51 + 92$
$= 143$ | (33) $-54 - (-97)$
$= 43$ | (52) $-55 + 46$
$= -9$ | (71) $-68 + (-45)$
$= -113$ |
| (15) $-43 - 33$
$= -76$ | (34) $92 + 75$
$= 167$ | (53) $72 - 15$
$= 57$ | (72) $70 + (-97)$
$= -27$ |
| (16) $73 - 1$
$= 72$ | (35) $6 - 96$
$= -90$ | (54) $44 - 36$
$= 8$ | (73) $39 - (-6)$
$= 45$ |
| (17) $-18 + (-19)$
$= -37$ | (36) $16 - (-49)$
$= 65$ | (55) $27 + (-30)$
$= -3$ | (74) $-59 - (-73)$
$= 14$ |
| (18) $31 - (-12)$
$= 43$ | (37) $53 - (-60)$
$= 113$ | (56) $1 + (-65)$
$= -64$ | (75) $-77 + (-46)$
$= -123$ |
| (19) $49 - 68$
$= -19$ | (38) $-16 - 73$
$= -89$ | (57) $53 + 14$
$= 67$ | (76) $1 - (-38)$
$= 39$ |

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|--------------------------------|-------------------------------|--------------------------------|--------------------------------|
| (1) $-8 + 27$
$= 19$ | (20) $-44 + (-2)$
$= -46$ | (39) $96 + 8$
$= 104$ | (58) $74 - (-56)$
$= 130$ |
| (2) $-87 + (-97)$
$= -184$ | (21) $86 - 11$
$= 75$ | (40) $57 - (-60)$
$= 117$ | (59) $-69 + 92$
$= 23$ |
| (3) $-49 + 72$
$= 23$ | (22) $93 - (-7)$
$= 100$ | (41) $11 - 88$
$= -77$ | (60) $98 - 58$
$= 40$ |
| (4) $-3 - 92$
$= -95$ | (23) $52 - 70$
$= -18$ | (42) $34 - 65$
$= -31$ | (61) $-82 - 90$
$= -172$ |
| (5) $1 + (-32)$
$= -31$ | (24) $-21 + (-22)$
$= -43$ | (43) $21 - (-71)$
$= 92$ | (62) $71 - 1$
$= 70$ |
| (6) $0 - (-74)$
$= 74$ | (25) $-20 + 67$
$= 47$ | (44) $-74 + 44$
$= -30$ | (63) $-44 + 57$
$= 13$ |
| (7) $-38 + (-32)$
$= -70$ | (26) $44 - 19$
$= 25$ | (45) $-50 - 78$
$= -128$ | (64) $-95 - (-75)$
$= -20$ |
| (8) $95 - (-20)$
$= 115$ | (27) $-13 + 94$
$= 81$ | (46) $-21 + 95$
$= 74$ | (65) $96 + 78$
$= 174$ |
| (9) $78 - 26$
$= 52$ | (28) $-90 - 49$
$= -139$ | (47) $71 + (-99)$
$= -28$ | (66) $6 - 62$
$= -56$ |
| (10) $99 - (-29)$
$= 128$ | (29) $7 - (-99)$
$= 106$ | (48) $53 + 72$
$= 125$ | (67) $24 + (-54)$
$= -30$ |
| (11) $-26 - (-84)$
$= 58$ | (30) $-93 + 5$
$= -88$ | (49) $11 - (-70)$
$= 81$ | (68) $-62 - 52$
$= -114$ |
| (12) $14 + 63$
$= 77$ | (31) $-52 + 92$
$= 40$ | (50) $3 - (-30)$
$= 33$ | (69) $-28 - (-49)$
$= 21$ |
| (13) $44 - (-46)$
$= 90$ | (32) $-66 - (-52)$
$= -14$ | (51) $8 - 94$
$= -86$ | (70) $-89 + 12$
$= -77$ |
| (14) $-87 + (-72)$
$= -159$ | (33) $58 - (-60)$
$= 118$ | (52) $-70 + (-84)$
$= -154$ | (71) $-35 - 59$
$= -94$ |
| (15) $-31 + 47$
$= 16$ | (34) $59 - 38$
$= 21$ | (53) $65 + (-44)$
$= 21$ | (72) $-95 + 90$
$= -5$ |
| (16) $-84 + 21$
$= -63$ | (35) $-22 - (-90)$
$= 68$ | (54) $30 - 43$
$= -13$ | (73) $85 - (-9)$
$= 94$ |
| (17) $100 - (-88)$
$= 188$ | (36) $27 - (-7)$
$= 34$ | (55) $86 - 22$
$= 64$ | (74) $-11 - (-45)$
$= 34$ |
| (18) $11 - 18$
$= -7$ | (37) $45 - (-62)$
$= 107$ | (56) $-56 + (-68)$
$= -124$ | (75) $-19 + (-45)$
$= -64$ |
| (19) $-71 + 99$
$= 28$ | (38) $-34 - (-94)$
$= 60$ | (57) $85 + 72$
$= 157$ | (76) $-72 + (-31)$
$= -103$ |

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|-------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $84 - (-98)$
$= 182$ | (20) $35 + (-3)$
$= 32$ | (39) $-48 + 6$
$= -42$ | (58) $-58 + 23$
$= -35$ |
| (2) $17 + 13$
$= 30$ | (21) $-15 + 45$
$= 30$ | (40) $-97 + 70$
$= -27$ | (59) $-53 - (-57)$
$= 4$ |
| (3) $-5 - (-45)$
$= 40$ | (22) $-11 - (-95)$
$= 84$ | (41) $-25 - 61$
$= -86$ | (60) $88 + 71$
$= 159$ |
| (4) $-52 + 58$
$= 6$ | (23) $-17 - (-78)$
$= 61$ | (42) $81 + 86$
$= 167$ | (61) $56 - (-95)$
$= 151$ |
| (5) $18 - 59$
$= -41$ | (24) $41 - (-36)$
$= 77$ | (43) $-2 + 55$
$= 53$ | (62) $-46 - 35$
$= -81$ |
| (6) $-40 - (-98)$
$= 58$ | (25) $95 - 26$
$= 69$ | (44) $-18 + (-72)$
$= -90$ | (63) $5 - (-15)$
$= 20$ |
| (7) $-87 + (-82)$
$= -169$ | (26) $8 + 45$
$= 53$ | (45) $73 - 69$
$= 4$ | (64) $-41 - (-73)$
$= 32$ |
| (8) $-18 - (-68)$
$= 50$ | (27) $-42 - (-70)$
$= 28$ | (46) $-98 + 89$
$= -9$ | (65) $14 + 88$
$= 102$ |
| (9) $-75 + 12$
$= -63$ | (28) $61 + (-27)$
$= 34$ | (47) $-82 - 9$
$= -91$ | (66) $2 + (-37)$
$= -35$ |
| (10) $-56 + 10$
$= -46$ | (29) $71 - 48$
$= 23$ | (48) $78 + (-1)$
$= 77$ | (67) $52 + (-97)$
$= -45$ |
| (11) $43 + (-100)$
$= -57$ | (30) $-18 - 68$
$= -86$ | (49) $44 + 12$
$= 56$ | (68) $-28 + (-87)$
$= -115$ |
| (12) $-43 - 25$
$= -68$ | (31) $-55 - (-32)$
$= -23$ | (50) $-97 + (-42)$
$= -139$ | (69) $-45 + 4$
$= -41$ |
| (13) $24 - 71$
$= -47$ | (32) $76 - 10$
$= 66$ | (51) $6 + 23$
$= 29$ | (70) $-82 - (-22)$
$= -60$ |
| (14) $13 - 75$
$= -62$ | (33) $-32 + (-77)$
$= -109$ | (52) $9 + (-2)$
$= 7$ | (71) $-34 - 31$
$= -65$ |
| (15) $-21 + (-35)$
$= -56$ | (34) $23 - (-22)$
$= 45$ | (53) $69 - 82$
$= -13$ | (72) $-13 + (-78)$
$= -91$ |
| (16) $65 - (-71)$
$= 136$ | (35) $45 - 75$
$= -30$ | (54) $39 + (-21)$
$= 18$ | (73) $47 + (-56)$
$= -9$ |
| (17) $88 + (-42)$
$= 46$ | (36) $-40 + 88$
$= 48$ | (55) $-43 + (-75)$
$= -118$ | (74) $87 - (-1)$
$= 88$ |
| (18) $91 + (-50)$
$= 41$ | (37) $83 - 22$
$= 61$ | (56) $41 - (-12)$
$= 53$ | (75) $-31 + (-11)$
$= -42$ |
| (19) $38 + (-75)$
$= -37$ | (38) $76 - 44$
$= 32$ | (57) $65 + (-72)$
$= -7$ | (76) $-29 - 48$
$= -77$ |

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|-------------------------------|-------------------------------|--------------------------------|-------------------------------|
| (1) $12 + 41$
$= 53$ | (20) $63 + (-86)$
$= -23$ | (39) $-60 - 18$
$= -78$ | (58) $35 + (-44)$
$= -9$ |
| (2) $-22 - (-18)$
$= -4$ | (21) $39 + 68$
$= 107$ | (40) $-51 + (-88)$
$= -139$ | (59) $-92 - (-90)$
$= -2$ |
| (3) $-16 + (-28)$
$= -44$ | (22) $-4 - 52$
$= -56$ | (41) $-65 - (-48)$
$= -17$ | (60) $-78 - (-39)$
$= -39$ |
| (4) $15 + (-72)$
$= -57$ | (23) $-31 + 68$
$= 37$ | (42) $-63 + (-79)$
$= -142$ | (61) $64 - (-51)$
$= 115$ |
| (5) $-64 + 3$
$= -61$ | (24) $34 + (-8)$
$= 26$ | (43) $-91 - (-20)$
$= -71$ | (62) $19 - 84$
$= -65$ |
| (6) $0 - 27$
$= -27$ | (25) $-78 + 78$
$= 0$ | (44) $79 - (-46)$
$= 125$ | (63) $34 - (-7)$
$= 41$ |
| (7) $-28 - 51$
$= -79$ | (26) $89 - (-54)$
$= 143$ | (45) $-11 - (-6)$
$= -5$ | (64) $3 - (-65)$
$= 68$ |
| (8) $-34 + (-97)$
$= -131$ | (27) $-36 + (-1)$
$= -37$ | (46) $34 - 98$
$= -64$ | (65) $48 - (-10)$
$= 58$ |
| (9) $45 + 93$
$= 138$ | (28) $53 + (-54)$
$= -1$ | (47) $-46 + 57$
$= 11$ | (66) $-96 - (-51)$
$= -45$ |
| (10) $61 + 42$
$= 103$ | (29) $23 - 31$
$= -8$ | (48) $-72 - 53$
$= -125$ | (67) $45 - (-22)$
$= 67$ |
| (11) $75 + 98$
$= 173$ | (30) $-37 - (-76)$
$= 39$ | (49) $-71 - 38$
$= -109$ | (68) $-28 - 80$
$= -108$ |
| (12) $47 - 90$
$= -43$ | (31) $-59 - 98$
$= -157$ | (50) $-84 - (-40)$
$= -44$ | (69) $-26 + (-41)$
$= -67$ |
| (13) $-96 - (-35)$
$= -61$ | (32) $1 - (-27)$
$= 28$ | (51) $-18 + (-89)$
$= -107$ | (70) $-91 - 84$
$= -175$ |
| (14) $-30 - (-89)$
$= 59$ | (33) $-98 - 35$
$= -133$ | (52) $91 + 83$
$= 174$ | (71) $-79 + 58$
$= -21$ |
| (15) $15 + 55$
$= 70$ | (34) $14 - 35$
$= -21$ | (53) $-66 + 23$
$= -43$ | (72) $37 - (-79)$
$= 116$ |
| (16) $57 - (-87)$
$= 144$ | (35) $-33 + (-6)$
$= -39$ | (54) $27 - (-31)$
$= 58$ | (73) $48 + (-9)$
$= 39$ |
| (17) $-38 - 11$
$= -49$ | (36) $-35 - (-23)$
$= -12$ | (55) $44 + 100$
$= 144$ | (74) $-68 + (-22)$
$= -90$ |
| (18) $6 - 36$
$= -30$ | (37) $-57 + 10$
$= -47$ | (56) $-89 - (-26)$
$= -63$ | (75) $12 - (-25)$
$= 37$ |
| (19) $57 - 83$
$= -26$ | (38) $66 - 55$
$= 11$ | (57) $-16 + 60$
$= 44$ | (76) $-13 + (-43)$
$= -56$ |

(1) $-89 - (-40)$ $= -49$	(20) $-18 - (-35)$ $= 17$	(39) $43 - 69$ $= -26$	(58) $47 - (-16)$ $= 63$
(2) $25 - 84$ $= -59$	(21) $80 + (-45)$ $= 35$	(40) $-66 + 6$ $= -60$	(59) $65 + (-20)$ $= 45$
(3) $77 - (-64)$ $= 141$	(22) $-25 + 58$ $= 33$	(41) $-92 - (-11)$ $= -81$	(60) $34 + 44$ $= 78$
(4) $-98 - (-22)$ $= -76$	(23) $42 + (-72)$ $= -30$	(42) $-35 + (-6)$ $= -41$	(61) $44 - 32$ $= 12$
(5) $50 - (-34)$ $= 84$	(24) $2 - (-3)$ $= 5$	(43) $63 - 3$ $= 60$	(62) $-86 - (-25)$ $= -61$
(6) $53 + 100$ $= 153$	(25) $-27 + 97$ $= 70$	(44) $-78 + (-32)$ $= -110$	(63) $-56 - 82$ $= -138$
(7) $-94 + 20$ $= -74$	(26) $-38 + (-85)$ $= -123$	(45) $80 - (-53)$ $= 133$	(64) $-56 + (-20)$ $= -76$
(8) $-13 + 90$ $= 77$	(27) $57 + 23$ $= 80$	(46) $-91 - (-48)$ $= -43$	(65) $10 + (-58)$ $= -48$
(9) $-19 + (-40)$ $= -59$	(28) $34 + 64$ $= 98$	(47) $39 + (-54)$ $= -15$	(66) $-56 - (-73)$ $= 17$
(10) $73 + 31$ $= 104$	(29) $19 - (-30)$ $= 49$	(48) $26 + 26$ $= 52$	(67) $-88 + (-89)$ $= -177$
(11) $45 + 11$ $= 56$	(30) $-82 + 24$ $= -58$	(49) $-95 - 96$ $= -191$	(68) $-32 + 55$ $= 23$
(12) $61 + (-77)$ $= -16$	(31) $29 + 35$ $= 64$	(50) $-76 - (-13)$ $= -63$	(69) $49 - 62$ $= -13$
(13) $-32 + (-92)$ $= -124$	(32) $79 - 13$ $= 66$	(51) $-85 + 59$ $= -26$	(70) $6 + (-34)$ $= -28$
(14) $-72 + (-22)$ $= -94$	(33) $-83 - 45$ $= -128$	(52) $44 + 65$ $= 109$	(71) $5 + 53$ $= 58$
(15) $-47 - 98$ $= -145$	(34) $-100 - (-70)$ $= -30$	(53) $52 - 24$ $= 28$	(72) $32 - (-70)$ $= 102$
(16) $53 + (-24)$ $= 29$	(35) $-5 - 89$ $= -94$	(54) $-58 - (-89)$ $= 31$	(73) $12 - (-30)$ $= 42$
(17) $80 + (-8)$ $= 72$	(36) $-6 + 96$ $= 90$	(55) $24 - (-75)$ $= 99$	(74) $-82 - (-72)$ $= -10$
(18) $-86 - (-77)$ $= -9$	(37) $-31 + (-82)$ $= -113$	(56) $44 - 79$ $= -35$	(75) $-27 - 14$ $= -41$
(19) $-92 + (-31)$ $= -123$	(38) $-28 + 55$ $= 27$	(57) $-88 - (-99)$ $= 11$	(76) $28 - 74$ $= -46$

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|-------------------------------|--------------------------------|-------------------------------|--------------------------------|
| (1) $-17 - 18$
$= -35$ | (20) $1 + (-58)$
$= -57$ | (39) $84 - (-98)$
$= 182$ | (58) $55 + (-13)$
$= 42$ |
| (2) $-8 + 80$
$= 72$ | (21) $38 + (-54)$
$= -16$ | (40) $-16 - (-79)$
$= 63$ | (59) $-58 + 1$
$= -57$ |
| (3) $-60 + 84$
$= 24$ | (22) $90 + 2$
$= 92$ | (41) $-48 - (-23)$
$= -25$ | (60) $-74 + (-35)$
$= -109$ |
| (4) $-1 - (-74)$
$= 73$ | (23) $92 + 46$
$= 138$ | (42) $-25 - 50$
$= -75$ | (61) $-76 + 29$
$= -47$ |
| (5) $-88 - 77$
$= -165$ | (24) $-91 - (-66)$
$= -25$ | (43) $61 + 40$
$= 101$ | (62) $85 - (-3)$
$= 88$ |
| (6) $85 + (-36)$
$= 49$ | (25) $-16 - 53$
$= -69$ | (44) $-27 - 89$
$= -116$ | (63) $-42 - 70$
$= -112$ |
| (7) $92 - 14$
$= 78$ | (26) $-65 + (-27)$
$= -92$ | (45) $-48 - (-4)$
$= -44$ | (64) $-24 + 100$
$= 76$ |
| (8) $-100 + 69$
$= -31$ | (27) $-92 + 22$
$= -70$ | (46) $-29 - (-16)$
$= -13$ | (65) $-7 + 96$
$= 89$ |
| (9) $39 + 68$
$= 107$ | (28) $-40 - 36$
$= -76$ | (47) $-67 - (-21)$
$= -46$ | (66) $-36 - (-14)$
$= -22$ |
| (10) $5 - (-66)$
$= 71$ | (29) $5 + 55$
$= 60$ | (48) $-34 - (-47)$
$= 13$ | (67) $-91 - 34$
$= -125$ |
| (11) $-82 + 66$
$= -16$ | (30) $-26 + (-13)$
$= -39$ | (49) $-99 + 84$
$= -15$ | (68) $22 + 5$
$= 27$ |
| (12) $44 + 98$
$= 142$ | (31) $69 + 1$
$= 70$ | (50) $-67 + 13$
$= -54$ | (69) $33 - (-76)$
$= 109$ |
| (13) $-3 + (-54)$
$= -57$ | (32) $35 - 54$
$= -19$ | (51) $-3 - 50$
$= -53$ | (70) $34 - (-32)$
$= 66$ |
| (14) $-65 - (-12)$
$= -53$ | (33) $-73 - 74$
$= -147$ | (52) $38 - 41$
$= -3$ | (71) $-72 + 7$
$= -65$ |
| (15) $-100 - 93$
$= -193$ | (34) $-67 + (-80)$
$= -147$ | (53) $41 + (-39)$
$= 2$ | (72) $47 - 53$
$= -6$ |
| (16) $-39 - (-69)$
$= 30$ | (35) $65 + (-51)$
$= 14$ | (54) $-19 + (-42)$
$= -61$ | (73) $63 + (-84)$
$= -21$ |
| (17) $-9 - (-62)$
$= 53$ | (36) $6 - 50$
$= -44$ | (55) $24 - (-9)$
$= 33$ | (74) $-51 + (-15)$
$= -66$ |
| (18) $30 + (-32)$
$= -2$ | (37) $-63 + 54$
$= -9$ | (56) $-58 - (-22)$
$= -36$ | (75) $-26 + 73$
$= 47$ |
| (19) $-88 + 49$
$= -39$ | (38) $-55 + 30$
$= -25$ | (57) $-15 - (-49)$
$= 34$ | (76) $-76 + 76$
$= 0$ |

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|--------------------------------|---------------------------------|--------------------------------|---------------------------------|
| (1) $-79 - 58$
$= -137$ | (20) $31 + (-53)$
$= -22$ | (39) $-93 + (-19)$
$= -112$ | (58) $84 + 74$
$= 158$ |
| (2) $70 + 45$
$= 115$ | (21) $-12 + 50$
$= 38$ | (40) $-20 - (-32)$
$= 12$ | (59) $-9 + (-92)$
$= -101$ |
| (3) $-60 + 78$
$= 18$ | (22) $86 - (-69)$
$= 155$ | (41) $-70 - 15$
$= -85$ | (60) $-5 - 78$
$= -83$ |
| (4) $2 - (-53)$
$= 55$ | (23) $72 - (-96)$
$= 168$ | (42) $5 - 51$
$= -46$ | (61) $52 - 57$
$= -5$ |
| (5) $-29 + (-4)$
$= -33$ | (24) $80 + (-53)$
$= 27$ | (43) $78 - 2$
$= 76$ | (62) $-92 - (-35)$
$= -57$ |
| (6) $-75 - 97$
$= -172$ | (25) $-58 + 20$
$= -38$ | (44) $90 + 57$
$= 147$ | (63) $14 + (-89)$
$= -75$ |
| (7) $-80 + (-32)$
$= -112$ | (26) $-100 + (-20)$
$= -120$ | (45) $-3 - (-24)$
$= 21$ | (64) $83 + 69$
$= 152$ |
| (8) $66 + (-5)$
$= 61$ | (27) $-35 + (-54)$
$= -89$ | (46) $-95 - 88$
$= -183$ | (65) $77 + (-50)$
$= 27$ |
| (9) $-22 - (-63)$
$= 41$ | (28) $-43 + (-44)$
$= -87$ | (47) $-45 - (-95)$
$= 50$ | (66) $38 + (-59)$
$= -21$ |
| (10) $95 + 64$
$= 159$ | (29) $-70 + 81$
$= 11$ | (48) $57 + (-36)$
$= 21$ | (67) $90 + (-21)$
$= 69$ |
| (11) $-16 + (-24)$
$= -40$ | (30) $18 - (-53)$
$= 71$ | (49) $-98 + (-72)$
$= -170$ | (68) $15 - 12$
$= 3$ |
| (12) $-8 - (-14)$
$= 6$ | (31) $76 + 100$
$= 176$ | (50) $46 + (-79)$
$= -33$ | (69) $71 + 68$
$= 139$ |
| (13) $16 - 86$
$= -70$ | (32) $-58 + (-12)$
$= -70$ | (51) $-29 + 7$
$= -22$ | (70) $-100 + (-17)$
$= -117$ |
| (14) $-77 - 97$
$= -174$ | (33) $50 - 72$
$= -22$ | (52) $-85 - (-30)$
$= -55$ | (71) $-61 + (-94)$
$= -155$ |
| (15) $-71 + 14$
$= -57$ | (34) $0 - (-35)$
$= 35$ | (53) $69 + (-33)$
$= 36$ | (72) $-74 - (-89)$
$= 15$ |
| (16) $-14 - 66$
$= -80$ | (35) $14 - 82$
$= -68$ | (54) $29 - 39$
$= -10$ | (73) $64 + 100$
$= 164$ |
| (17) $-9 - (-60)$
$= 51$ | (36) $67 + 10$
$= 77$ | (55) $-94 + 95$
$= 1$ | (74) $-91 - 78$
$= -169$ |
| (18) $-92 + (-48)$
$= -140$ | (37) $-75 + 17$
$= -58$ | (56) $-50 + 0$
$= -50$ | (75) $84 + (-62)$
$= 22$ |
| (19) $27 + (-69)$
$= -42$ | (38) $-69 - 38$
$= -107$ | (57) $-1 + 31$
$= 30$ | (76) $-74 + (-9)$
$= -83$ |

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|-------------------------------|--------------------------------|-------------------------------|--------------------------------|
| (1) $65 + 16$
$= 81$ | (20) $11 - 79$
$= -68$ | (39) $56 + 87$
$= 143$ | (58) $-28 - 58$
$= -86$ |
| (2) $-3 - (-10)$
$= 7$ | (21) $19 - 56$
$= -37$ | (40) $-13 - 84$
$= -97$ | (59) $-6 - (-8)$
$= 2$ |
| (3) $-17 - 64$
$= -81$ | (22) $93 - (-95)$
$= 188$ | (41) $-36 + (-15)$
$= -51$ | (60) $-97 + (-28)$
$= -125$ |
| (4) $-95 + (-14)$
$= -109$ | (23) $-12 + (-31)$
$= -43$ | (42) $-56 + 40$
$= -16$ | (61) $33 + 63$
$= 96$ |
| (5) $86 - (-58)$
$= 144$ | (24) $42 - 100$
$= -58$ | (43) $-15 - (-92)$
$= 77$ | (62) $43 - 0$
$= 43$ |
| (6) $-13 + 65$
$= 52$ | (25) $-64 - (-8)$
$= -56$ | (44) $30 - 84$
$= -54$ | (63) $73 + 18$
$= 91$ |
| (7) $91 + 62$
$= 153$ | (26) $-74 - 58$
$= -132$ | (45) $-30 + (-52)$
$= -82$ | (64) $100 + (-78)$
$= 22$ |
| (8) $-11 + (-98)$
$= -109$ | (27) $-88 - 66$
$= -154$ | (46) $-89 + 53$
$= -36$ | (65) $68 + 65$
$= 133$ |
| (9) $3 + 63$
$= 66$ | (28) $77 + 73$
$= 150$ | (47) $-76 + 76$
$= 0$ | (66) $44 - (-100)$
$= 144$ |
| (10) $-25 - (-92)$
$= 67$ | (29) $82 - (-57)$
$= 139$ | (48) $19 - (-65)$
$= 84$ | (67) $19 - (-92)$
$= 111$ |
| (11) $-51 + (-24)$
$= -75$ | (30) $91 - (-100)$
$= 191$ | (49) $-61 + 4$
$= -57$ | (68) $72 - (-22)$
$= 94$ |
| (12) $69 - 86$
$= -17$ | (31) $-20 - (-12)$
$= -8$ | (50) $-93 + (-3)$
$= -96$ | (69) $-54 + 93$
$= 39$ |
| (13) $-56 + 92$
$= 36$ | (32) $-51 + 43$
$= -8$ | (51) $-48 + (-26)$
$= -74$ | (70) $-90 - 39$
$= -129$ |
| (14) $61 - (-52)$
$= 113$ | (33) $-9 + 10$
$= 1$ | (52) $26 + (-19)$
$= 7$ | (71) $35 - (-90)$
$= 125$ |
| (15) $-39 - 95$
$= -134$ | (34) $44 - (-54)$
$= 98$ | (53) $52 - 28$
$= 24$ | (72) $65 + (-55)$
$= 10$ |
| (16) $-68 - 100$
$= -168$ | (35) $-53 + (-54)$
$= -107$ | (54) $14 - 18$
$= -4$ | (73) $-40 - 96$
$= -136$ |
| (17) $-91 - 100$
$= -191$ | (36) $21 + (-88)$
$= -67$ | (55) $76 - (-31)$
$= 107$ | (74) $-61 + 11$
$= -50$ |
| (18) $72 - 12$
$= 60$ | (37) $-7 + 59$
$= 52$ | (56) $-73 - 69$
$= -142$ | (75) $70 + (-42)$
$= 28$ |
| (19) $-71 - (-66)$
$= -5$ | (38) $96 + 60$
$= 156$ | (57) $-1 - (-18)$
$= 17$ | (76) $-68 + 1$
$= -67$ |

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|--------------------------------|--------------------------------|-------------------------------|--------------------------------|
| (1) $85 - 87$
$= -2$ | (20) $99 + (-38)$
$= 61$ | (39) $-55 + 69$
$= 14$ | (58) $26 - (-57)$
$= 83$ |
| (2) $-29 + (-62)$
$= -91$ | (21) $-56 - (-53)$
$= -3$ | (40) $61 - (-39)$
$= 100$ | (59) $-59 - (-73)$
$= 14$ |
| (3) $-91 - (-32)$
$= -59$ | (22) $46 + 16$
$= 62$ | (41) $-28 + 80$
$= 52$ | (60) $-68 + (-49)$
$= -117$ |
| (4) $-45 + (-76)$
$= -121$ | (23) $-42 + 22$
$= -20$ | (42) $22 + (-44)$
$= -22$ | (61) $-100 - 0$
$= -100$ |
| (5) $99 + 100$
$= 199$ | (24) $39 + 18$
$= 57$ | (43) $64 + 47$
$= 111$ | (62) $-18 + 64$
$= 46$ |
| (6) $22 - (-68)$
$= 90$ | (25) $27 - 2$
$= 25$ | (44) $54 - (-87)$
$= 141$ | (63) $8 + (-20)$
$= -12$ |
| (7) $24 - (-5)$
$= 29$ | (26) $82 - (-65)$
$= 147$ | (45) $-100 + 22$
$= -78$ | (64) $-38 - (-24)$
$= -14$ |
| (8) $19 + 35$
$= 54$ | (27) $90 - 56$
$= 34$ | (46) $14 - 70$
$= -56$ | (65) $-50 + 24$
$= -26$ |
| (9) $92 + 53$
$= 145$ | (28) $72 + 14$
$= 86$ | (47) $-94 - 61$
$= -155$ | (66) $-78 - 45$
$= -123$ |
| (10) $-26 - 21$
$= -47$ | (29) $2 - 82$
$= -80$ | (48) $57 - (-88)$
$= 145$ | (67) $1 + (-15)$
$= -14$ |
| (11) $36 - 16$
$= 20$ | (30) $-39 + (-39)$
$= -78$ | (49) $0 + (-29)$
$= -29$ | (68) $46 + 57$
$= 103$ |
| (12) $-33 + (-77)$
$= -110$ | (31) $13 - 98$
$= -85$ | (50) $-21 + 18$
$= -3$ | (69) $-18 - 85$
$= -103$ |
| (13) $-9 - (-9)$
$= 0$ | (32) $85 + 54$
$= 139$ | (51) $-37 + (-1)$
$= -38$ | (70) $-12 + (-3)$
$= -15$ |
| (14) $-82 + 91$
$= 9$ | (33) $-98 + (-68)$
$= -166$ | (52) $-22 - (-73)$
$= 51$ | (71) $-37 + (-44)$
$= -81$ |
| (15) $89 + 45$
$= 134$ | (34) $-4 + (-11)$
$= -15$ | (53) $-69 - (-15)$
$= -54$ | (72) $-72 - (-89)$
$= 17$ |
| (16) $79 - (-64)$
$= 143$ | (35) $39 - (-86)$
$= 125$ | (54) $-78 - (-93)$
$= 15$ | (73) $-78 + 76$
$= -2$ |
| (17) $9 + (-41)$
$= -32$ | (36) $-38 + (-68)$
$= -106$ | (55) $44 + 8$
$= 52$ | (74) $-84 - (-65)$
$= -19$ |
| (18) $-55 + (-68)$
$= -123$ | (37) $-44 + 36$
$= -8$ | (56) $-65 + 47$
$= -18$ | (75) $-25 + (-24)$
$= -49$ |
| (19) $5 + 47$
$= 52$ | (38) $29 + (-34)$
$= -5$ | (57) $-83 + (-16)$
$= -99$ | (76) $42 - (-55)$
$= 97$ |

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|--------------------------------|-------------------------------|-------------------------------|------------------------------|
| (1) $-49 + 2$
$= -47$ | (20) $-69 - 33$
$= -102$ | (39) $-7 + 84$
$= 77$ | (58) $-73 - 57$
$= -130$ |
| (2) $47 - (-49)$
$= 96$ | (21) $5 + 45$
$= 50$ | (40) $-59 - (-70)$
$= 11$ | (59) $58 + (-18)$
$= 40$ |
| (3) $-65 - (-71)$
$= 6$ | (22) $-78 + 1$
$= -77$ | (41) $9 - 29$
$= -20$ | (60) $19 + (-9)$
$= 10$ |
| (4) $86 - 39$
$= 47$ | (23) $-52 - (-11)$
$= -41$ | (42) $45 - (-75)$
$= 120$ | (61) $52 + 22$
$= 74$ |
| (5) $32 + 38$
$= 70$ | (24) $-46 + 31$
$= -15$ | (43) $-25 + (-10)$
$= -35$ | (62) $-33 - (-88)$
$= 55$ |
| (6) $4 - 66$
$= -62$ | (25) $-48 - 46$
$= -94$ | (44) $-78 - 18$
$= -96$ | (63) $67 + (-75)$
$= -8$ |
| (7) $-26 - 27$
$= -53$ | (26) $-18 - (-46)$
$= 28$ | (45) $-75 - 35$
$= -110$ | (64) $-16 - (-20)$
$= 4$ |
| (8) $-46 - (-15)$
$= -31$ | (27) $-28 - 55$
$= -83$ | (46) $-58 + 70$
$= 12$ | (65) $-84 - 52$
$= -136$ |
| (9) $61 - 41$
$= 20$ | (28) $55 + (-72)$
$= -17$ | (47) $-67 + 16$
$= -51$ | (66) $-3 + (-96)$
$= -99$ |
| (10) $-54 - 3$
$= -57$ | (29) $38 - 34$
$= 4$ | (48) $66 - (-19)$
$= 85$ | (67) $16 - (-35)$
$= 51$ |
| (11) $79 - (-36)$
$= 115$ | (30) $-63 - 80$
$= -143$ | (49) $-18 + 15$
$= -3$ | (68) $98 - 46$
$= 52$ |
| (12) $9 + (-17)$
$= -8$ | (31) $-72 - (-44)$
$= -28$ | (50) $-25 + 42$
$= 17$ | (69) $10 - (-47)$
$= 57$ |
| (13) $-14 - (-24)$
$= 10$ | (32) $75 - 74$
$= 1$ | (51) $-82 - 61$
$= -143$ | (70) $-46 - (-79)$
$= 33$ |
| (14) $5 - (-20)$
$= 25$ | (33) $66 - (-63)$
$= 129$ | (52) $18 + 61$
$= 79$ | (71) $83 - 76$
$= 7$ |
| (15) $-6 - 47$
$= -53$ | (34) $-69 - 15$
$= -84$ | (53) $-54 - 43$
$= -97$ | (72) $25 + (-45)$
$= -20$ |
| (16) $57 - (-79)$
$= 136$ | (35) $15 - (-9)$
$= 24$ | (54) $74 + (-99)$
$= -25$ | (73) $-23 + 97$
$= 74$ |
| (17) $8 + 58$
$= 66$ | (36) $-88 + 82$
$= -6$ | (55) $-54 + 97$
$= 43$ | (74) $48 - (-6)$
$= 54$ |
| (18) $-12 + (-98)$
$= -110$ | (37) $72 + 91$
$= 163$ | (56) $-66 + 39$
$= -27$ | (75) $23 - 18$
$= 5$ |
| (19) $-82 - (-12)$
$= -70$ | (38) $83 + 34$
$= 117$ | (57) $-36 - 64$
$= -100$ | (76) $20 - (-83)$
$= 103$ |

1.2.2 乗法

整数の積を求める問題。

(1) $-7 \times (-7)$	(31) $-8 \times (-2)$	(61) $-8 \times (-3)$	(91) $5 \times (-7)$
(2) $7 \times (-1)$	(32) 7×3	(62) 7×2	(92) $-7 \times (-9)$
(3) $8 \times (-6)$	(33) $9 \times (-9)$	(63) $8 \times (-3)$	(93) $-2 \times (-3)$
(4) -7×5	(34) 5×3	(64) $-1 \times (-9)$	(94) -7×0
(5) -5×6	(35) $7 \times (-9)$	(65) 5×2	(95) $3 \times (-2)$
(6) $8 \times (-8)$	(36) -7×4	(66) -8×6	(96) $-2 \times (-1)$
(7) $7 \times (-2)$	(37) -2×9	(67) -3×8	(97) $-9 \times (-2)$
(8) 0×6	(38) $4 \times (-7)$	(68) 0×7	(98) $4 \times (-9)$
(9) -9×8	(39) -9×1	(69) $-1 \times (-8)$	(99) -1×5
(10) -2×5	(40) $-6 \times (-6)$	(70) 4×9	(100) $-1 \times (-1)$
(11) 6×7	(41) 7×6	(71) -6×1	(101) 3×4
(12) $2 \times (-9)$	(42) -3×1	(72) $-7 \times (-1)$	(102) $-8 \times (-3)$
(13) 3×2	(43) $7 \times (-2)$	(73) -4×0	(103) $7 \times (-4)$
(14) -3×1	(44) $-5 \times (-8)$	(74) $4 \times (-6)$	(104) -5×2
(15) 1×4	(45) -9×7	(75) $5 \times (-2)$	(105) 1×9
(16) -6×6	(46) -3×6	(76) 9×3	(106) -3×2
(17) -7×3	(47) 7×9	(77) -7×4	(107) $9 \times (-8)$
(18) 4×1	(48) $-9 \times (-1)$	(78) $4 \times (-1)$	(108) -1×4
(19) 3×2	(49) $0 \times (-1)$	(79) -9×3	(109) -6×1
(20) $4 \times (-1)$	(50) $8 \times (-1)$	(80) -4×2	(110) $-1 \times (-6)$
(21) -6×0	(51) -9×4	(81) 5×8	(111) $6 \times (-4)$
(22) $8 \times (-9)$	(52) -7×3	(82) $2 \times (-3)$	(112) $-8 \times (-7)$
(23) $-8 \times (-9)$	(53) $-8 \times (-2)$	(83) -5×9	(113) $-9 \times (-7)$
(24) 0×4	(54) -1×7	(84) -9×5	(114) -8×7
(25) $7 \times (-3)$	(55) $-5 \times (-9)$	(85) $0 \times (-9)$	(115) $1 \times (-3)$
(26) 0×1	(56) $-9 \times (-8)$	(86) $4 \times (-6)$	(116) 2×5
(27) $-8 \times (-5)$	(57) $-6 \times (-3)$	(87) $-6 \times (-5)$	(117) $4 \times (-9)$
(28) -9×8	(58) $1 \times (-2)$	(88) $5 \times (-5)$	(118) 1×9
(29) $-9 \times (-6)$	(59) $-1 \times (-3)$	(89) -8×5	(119) $6 \times (-3)$
(30) $4 \times (-8)$	(60) -5×8	(90) 3×2	(120) 9×6

(1) 8×9	(31) 0×8	(61) -2×0	(91) 5×4
(2) $-4 \times (-8)$	(32) $3 \times (-9)$	(62) 6×6	(92) 6×2
(3) -6×6	(33) 8×4	(63) $4 \times (-9)$	(93) 7×6
(4) $1 \times (-4)$	(34) 5×2	(64) $-7 \times (-2)$	(94) $4 \times (-6)$
(5) $-5 \times (-3)$	(35) $-4 \times (-8)$	(65) $-5 \times (-8)$	(95) -8×4
(6) 0×7	(36) $-3 \times (-8)$	(66) 6×7	(96) $3 \times (-3)$
(7) -7×3	(37) $2 \times (-1)$	(67) 5×1	(97) $2 \times (-2)$
(8) $-9 \times (-9)$	(38) -6×8	(68) $-1 \times (-6)$	(98) -4×8
(9) -1×3	(39) $0 \times (-7)$	(69) $-2 \times (-2)$	(99) $8 \times (-7)$
(10) -1×9	(40) 7×2	(70) 1×2	(100) $6 \times (-6)$
(11) -7×0	(41) -3×3	(71) 7×2	(101) 1×5
(12) 2×8	(42) $5 \times (-4)$	(72) -8×1	(102) -2×9
(13) 8×8	(43) -6×7	(73) -1×2	(103) -3×5
(14) 7×3	(44) 1×2	(74) $0 \times (-7)$	(104) 5×0
(15) $-8 \times (-5)$	(45) 9×6	(75) $-5 \times (-3)$	(105) $-2 \times (-2)$
(16) $0 \times (-7)$	(46) -7×3	(76) -2×1	(106) $7 \times (-9)$
(17) 3×5	(47) -2×8	(77) $-2 \times (-1)$	(107) $7 \times (-3)$
(18) 0×8	(48) -1×2	(78) $7 \times (-6)$	(108) 4×3
(19) $-6 \times (-7)$	(49) $-2 \times (-1)$	(79) $-1 \times (-6)$	(109) -1×9
(20) -3×8	(50) $-7 \times (-2)$	(80) -5×9	(110) -8×2
(21) $-4 \times (-2)$	(51) $3 \times (-5)$	(81) -5×5	(111) $3 \times (-8)$
(22) -8×7	(52) $-4 \times (-8)$	(82) 0×1	(112) -6×8
(23) $-9 \times (-4)$	(53) 5×1	(83) $8 \times (-3)$	(113) $-2 \times (-4)$
(24) -9×5	(54) -9×8	(84) $3 \times (-6)$	(114) -7×2
(25) -1×0	(55) 4×9	(85) 3×1	(115) 3×3
(26) -2×4	(56) 0×4	(86) -9×1	(116) -7×1
(27) $-7 \times (-8)$	(57) 6×5	(87) $-2 \times (-1)$	(117) $1 \times (-6)$
(28) 2×4	(58) 5×7	(88) $8 \times (-6)$	(118) 7×4
(29) -3×2	(59) 8×8	(89) -9×0	(119) 6×6
(30) $8 \times (-4)$	(60) $5 \times (-9)$	(90) 9×1	(120) $0 \times (-1)$

- | | | | |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) 0×9 | (31) -4×8 | (61) 7×5 | (91) -7×0 |
| (2) $6 \times (-7)$ | (32) $7 \times (-8)$ | (62) $8 \times (-2)$ | (92) -2×4 |
| (3) $-6 \times (-8)$ | (33) -9×6 | (63) $-5 \times (-7)$ | (93) 2×5 |
| (4) $-6 \times (-7)$ | (34) $0 \times (-3)$ | (64) $2 \times (-3)$ | (94) $-9 \times (-4)$ |
| (5) $-9 \times (-1)$ | (35) -5×4 | (65) $2 \times (-4)$ | (95) $-8 \times (-7)$ |
| (6) -3×4 | (36) 2×0 | (66) -2×5 | (96) -9×7 |
| (7) -8×7 | (37) $7 \times (-4)$ | (67) $-9 \times (-1)$ | (97) 6×6 |
| (8) $-4 \times (-1)$ | (38) $-6 \times (-1)$ | (68) -5×2 | (98) $5 \times (-3)$ |
| (9) -8×0 | (39) $-5 \times (-4)$ | (69) $3 \times (-9)$ | (99) $4 \times (-7)$ |
| (10) $-4 \times (-4)$ | (40) $-5 \times (-2)$ | (70) 7×0 | (100) $8 \times (-8)$ |
| (11) $-7 \times (-3)$ | (41) $-7 \times (-7)$ | (71) $-3 \times (-6)$ | (101) $8 \times (-4)$ |
| (12) 5×3 | (42) $-9 \times (-6)$ | (72) -7×7 | (102) $-3 \times (-2)$ |
| (13) -1×7 | (43) $-1 \times (-8)$ | (73) -7×9 | (103) -5×0 |
| (14) $-7 \times (-3)$ | (44) 1×6 | (74) $-9 \times (-4)$ | (104) 4×8 |
| (15) 7×4 | (45) 4×7 | (75) $7 \times (-6)$ | (105) $0 \times (-5)$ |
| (16) $4 \times (-2)$ | (46) 5×7 | (76) $-7 \times (-9)$ | (106) $-8 \times (-6)$ |
| (17) $-3 \times (-1)$ | (47) $9 \times (-1)$ | (77) -7×7 | (107) 0×9 |
| (18) -9×9 | (48) $0 \times (-9)$ | (78) $-1 \times (-6)$ | (108) $1 \times (-1)$ |
| (19) 9×5 | (49) 8×4 | (79) $-5 \times (-9)$ | (109) 7×2 |
| (20) 3×9 | (50) -4×5 | (80) $2 \times (-4)$ | (110) 8×6 |
| (21) 2×4 | (51) $-9 \times (-1)$ | (81) -8×9 | (111) 0×1 |
| (22) 5×4 | (52) $-4 \times (-2)$ | (82) -5×3 | (112) $-9 \times (-3)$ |
| (23) -5×5 | (53) $3 \times (-8)$ | (83) $1 \times (-3)$ | (113) $7 \times (-1)$ |
| (24) $-8 \times (-1)$ | (54) 1×9 | (84) $-1 \times (-1)$ | (114) $5 \times (-8)$ |
| (25) $-3 \times (-4)$ | (55) -2×4 | (85) $-7 \times (-7)$ | (115) -4×2 |
| (26) $-9 \times (-8)$ | (56) -9×6 | (86) -4×3 | (116) -2×0 |
| (27) $7 \times (-2)$ | (57) -2×9 | (87) $7 \times (-6)$ | (117) 1×8 |
| (28) $0 \times (-2)$ | (58) 5×6 | (88) 2×5 | (118) -5×3 |
| (29) 6×9 | (59) -9×0 | (89) 7×3 | (119) $2 \times (-7)$ |
| (30) 4×1 | (60) $-6 \times (-7)$ | (90) -3×8 | (120) -9×5 |

(1) -9×0	(31) 4×7	(61) $-6 \times (-9)$	(91) -3×8
(2) $-7 \times (-1)$	(32) 7×4	(62) 1×7	(92) -9×0
(3) -5×8	(33) $3 \times (-2)$	(63) 5×3	(93) 1×7
(4) -9×0	(34) 4×8	(64) 2×6	(94) 9×7
(5) 3×1	(35) 4×2	(65) 4×9	(95) -4×4
(6) $-3 \times (-3)$	(36) 2×8	(66) $2 \times (-7)$	(96) -9×7
(7) -5×0	(37) 8×0	(67) 4×7	(97) $-5 \times (-7)$
(8) -1×5	(38) $8 \times (-4)$	(68) -3×8	(98) 2×0
(9) -5×1	(39) $-3 \times (-9)$	(69) -1×0	(99) 4×9
(10) -4×7	(40) -3×1	(70) $9 \times (-8)$	(100) $7 \times (-6)$
(11) $2 \times (-8)$	(41) $-4 \times (-2)$	(71) 6×4	(101) 1×5
(12) $2 \times (-9)$	(42) -7×0	(72) -5×7	(102) -4×4
(13) -1×0	(43) 4×2	(73) -7×9	(103) 5×2
(14) 2×7	(44) -3×9	(74) -9×7	(104) $-5 \times (-8)$
(15) $-2 \times (-7)$	(45) 9×3	(75) -1×6	(105) $9 \times (-1)$
(16) $8 \times (-2)$	(46) -1×8	(76) -9×9	(106) 6×0
(17) $9 \times (-6)$	(47) -1×3	(77) 0×8	(107) 1×4
(18) 8×5	(48) 6×3	(78) $3 \times (-8)$	(108) $7 \times (-3)$
(19) $8 \times (-3)$	(49) -3×1	(79) $-9 \times (-9)$	(109) 0×1
(20) 6×0	(50) $-8 \times (-9)$	(80) -5×7	(110) -6×5
(21) $-9 \times (-7)$	(51) $-6 \times (-2)$	(81) $7 \times (-2)$	(111) 9×6
(22) $5 \times (-4)$	(52) $7 \times (-5)$	(82) $0 \times (-6)$	(112) -5×8
(23) -1×2	(53) -7×6	(83) 3×2	(113) 8×3
(24) $8 \times (-1)$	(54) -7×2	(84) -5×0	(114) -3×9
(25) $4 \times (-1)$	(55) 6×9	(85) -2×5	(115) -3×2
(26) -7×5	(56) $1 \times (-3)$	(86) $2 \times (-4)$	(116) -7×0
(27) $1 \times (-7)$	(57) -2×2	(87) -4×0	(117) $4 \times (-9)$
(28) 9×8	(58) $7 \times (-4)$	(88) 6×9	(118) $0 \times (-4)$
(29) -2×1	(59) $6 \times (-9)$	(89) $-4 \times (-7)$	(119) $-9 \times (-3)$
(30) $-7 \times (-1)$	(60) $-4 \times (-9)$	(90) $-4 \times (-1)$	(120) $-5 \times (-7)$

(1) $2 \times (-3)$	(31) $-1 \times (-4)$	(61) 9×0	(91) 4×7
(2) $7 \times (-9)$	(32) -6×2	(62) $-5 \times (-5)$	(92) $-1 \times (-5)$
(3) 2×6	(33) 8×0	(63) $6 \times (-1)$	(93) $8 \times (-9)$
(4) 1×9	(34) $-3 \times (-4)$	(64) 3×2	(94) $4 \times (-1)$
(5) $9 \times (-2)$	(35) -4×3	(65) -1×5	(95) $7 \times (-5)$
(6) -2×8	(36) $6 \times (-9)$	(66) -5×6	(96) $-2 \times (-8)$
(7) -2×0	(37) -2×8	(67) $-7 \times (-6)$	(97) 2×3
(8) $-6 \times (-4)$	(38) 7×9	(68) -7×8	(98) 8×0
(9) $0 \times (-1)$	(39) $8 \times (-6)$	(69) $-3 \times (-3)$	(99) -2×1
(10) $5 \times (-3)$	(40) $-7 \times (-5)$	(70) $2 \times (-1)$	(100) 0×0
(11) -3×5	(41) $1 \times (-4)$	(71) $-2 \times (-3)$	(101) $-7 \times (-7)$
(12) 1×5	(42) 4×1	(72) $4 \times (-7)$	(102) -8×9
(13) -8×9	(43) $3 \times (-6)$	(73) 7×5	(103) 8×8
(14) -5×1	(44) $8 \times (-4)$	(74) 2×3	(104) 1×8
(15) $8 \times (-5)$	(45) -7×8	(75) 1×3	(105) $-6 \times (-9)$
(16) $-1 \times (-5)$	(46) -2×6	(76) -1×6	(106) -5×3
(17) 0×5	(47) $8 \times (-7)$	(77) 4×4	(107) -5×2
(18) -9×2	(48) $-6 \times (-5)$	(78) $-3 \times (-7)$	(108) $-5 \times (-2)$
(19) $-4 \times (-1)$	(49) 4×8	(79) $9 \times (-1)$	(109) $-8 \times (-8)$
(20) $7 \times (-6)$	(50) 8×5	(80) $1 \times (-3)$	(110) 5×0
(21) $6 \times (-6)$	(51) $7 \times (-9)$	(81) 8×9	(111) $0 \times (-3)$
(22) $9 \times (-6)$	(52) 0×1	(82) $3 \times (-9)$	(112) 0×6
(23) 0×1	(53) $-4 \times (-6)$	(83) -5×9	(113) $8 \times (-1)$
(24) $-2 \times (-6)$	(54) $3 \times (-4)$	(84) $7 \times (-3)$	(114) 5×4
(25) $2 \times (-6)$	(55) $8 \times (-5)$	(85) -5×0	(115) 7×1
(26) -9×9	(56) $-1 \times (-8)$	(86) $3 \times (-8)$	(116) $8 \times (-2)$
(27) -3×6	(57) $-7 \times (-4)$	(87) $1 \times (-6)$	(117) -1×1
(28) 6×0	(58) -6×3	(88) $-3 \times (-2)$	(118) 3×7
(29) -5×9	(59) 6×8	(89) $3 \times (-2)$	(119) $6 \times (-4)$
(30) $-9 \times (-5)$	(60) $-7 \times (-8)$	(90) $3 \times (-6)$	(120) $-4 \times (-7)$

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|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-4 \times (-4)$ | (31) $-2 \times (-6)$ | (61) $-6 \times (-2)$ | (91) -3×8 |
| (2) 6×8 | (32) -6×0 | (62) $6 \times (-1)$ | (92) -7×1 |
| (3) -6×8 | (33) 3×6 | (63) $-8 \times (-9)$ | (93) -8×0 |
| (4) $1 \times (-9)$ | (34) 6×7 | (64) $0 \times (-5)$ | (94) 6×1 |
| (5) -6×2 | (35) $-6 \times (-6)$ | (65) $-7 \times (-5)$ | (95) $0 \times (-5)$ |
| (6) $5 \times (-3)$ | (36) $-5 \times (-1)$ | (66) $2 \times (-6)$ | (96) 7×9 |
| (7) $7 \times (-1)$ | (37) -2×8 | (67) 9×8 | (97) 6×9 |
| (8) 3×7 | (38) 9×5 | (68) $2 \times (-5)$ | (98) 9×6 |
| (9) 7×0 | (39) -3×4 | (69) 2×9 | (99) -2×6 |
| (10) $-1 \times (-6)$ | (40) 4×2 | (70) 4×7 | (100) $-6 \times (-2)$ |
| (11) -1×7 | (41) $9 \times (-3)$ | (71) $-7 \times (-8)$ | (101) 9×8 |
| (12) $2 \times (-4)$ | (42) -9×8 | (72) $8 \times (-8)$ | (102) -6×1 |
| (13) $2 \times (-6)$ | (43) $1 \times (-3)$ | (73) -3×5 | (103) $1 \times (-6)$ |
| (14) $5 \times (-1)$ | (44) 9×4 | (74) -2×9 | (104) -8×3 |
| (15) 0×0 | (45) $1 \times (-2)$ | (75) $7 \times (-3)$ | (105) 1×0 |
| (16) -1×1 | (46) 0×0 | (76) $-4 \times (-9)$ | (106) 7×8 |
| (17) $7 \times (-6)$ | (47) -6×5 | (77) $-9 \times (-8)$ | (107) 9×7 |
| (18) 2×5 | (48) $7 \times (-1)$ | (78) $5 \times (-6)$ | (108) -4×6 |
| (19) $-6 \times (-2)$ | (49) -8×6 | (79) $0 \times (-5)$ | (109) 2×7 |
| (20) 5×7 | (50) -7×7 | (80) 4×2 | (110) 3×1 |
| (21) $5 \times (-4)$ | (51) -3×5 | (81) $5 \times (-4)$ | (111) $0 \times (-5)$ |
| (22) 1×2 | (52) 3×0 | (82) $-9 \times (-1)$ | (112) $-1 \times (-1)$ |
| (23) -5×8 | (53) 1×4 | (83) -1×8 | (113) $-1 \times (-6)$ |
| (24) $4 \times (-8)$ | (54) $1 \times (-9)$ | (84) $-8 \times (-2)$ | (114) 9×8 |
| (25) $-2 \times (-6)$ | (55) 3×9 | (85) $5 \times (-4)$ | (115) $4 \times (-3)$ |
| (26) 3×0 | (56) -9×0 | (86) $-6 \times (-6)$ | (116) -9×7 |
| (27) $-6 \times (-9)$ | (57) $-9 \times (-7)$ | (87) -5×3 | (117) -7×0 |
| (28) 6×0 | (58) 7×0 | (88) $-6 \times (-3)$ | (118) $3 \times (-8)$ |
| (29) $-4 \times (-2)$ | (59) $6 \times (-8)$ | (89) -9×0 | (119) 1×3 |
| (30) $-9 \times (-3)$ | (60) $-8 \times (-9)$ | (90) -9×6 | (120) $1 \times (-9)$ |

(1) $4 \times (-3)$	(31) $-7 \times (-3)$	(61) $-9 \times (-7)$	(91) $0 \times (-2)$
(2) $-4 \times (-6)$	(32) -4×0	(62) 5×9	(92) $-7 \times (-9)$
(3) $-2 \times (-6)$	(33) $9 \times (-8)$	(63) 6×0	(93) 5×9
(4) $9 \times (-8)$	(34) $-4 \times (-9)$	(64) $3 \times (-9)$	(94) $6 \times (-5)$
(5) 3×5	(35) $2 \times (-4)$	(65) 7×1	(95) -6×7
(6) $5 \times (-9)$	(36) 0×8	(66) 3×4	(96) 6×9
(7) -4×0	(37) $-5 \times (-6)$	(67) -5×7	(97) $2 \times (-7)$
(8) $4 \times (-6)$	(38) $-6 \times (-9)$	(68) -5×0	(98) 2×6
(9) 2×0	(39) $-1 \times (-1)$	(69) $-4 \times (-7)$	(99) 6×9
(10) $-1 \times (-8)$	(40) $9 \times (-8)$	(70) $-7 \times (-2)$	(100) 1×8
(11) -9×8	(41) -1×7	(71) 7×6	(101) $8 \times (-3)$
(12) 4×4	(42) -7×6	(72) $3 \times (-9)$	(102) $-5 \times (-9)$
(13) $-5 \times (-8)$	(43) $2 \times (-9)$	(73) 7×5	(103) -5×5
(14) -3×4	(44) $6 \times (-3)$	(74) -1×4	(104) -9×5
(15) $3 \times (-8)$	(45) -9×8	(75) $8 \times (-5)$	(105) 9×5
(16) 3×0	(46) 3×1	(76) $-7 \times (-6)$	(106) 8×1
(17) -6×7	(47) 8×8	(77) $5 \times (-9)$	(107) $8 \times (-7)$
(18) $-3 \times (-9)$	(48) $6 \times (-5)$	(78) $0 \times (-7)$	(108) 4×9
(19) $-5 \times (-7)$	(49) $-5 \times (-1)$	(79) $-2 \times (-2)$	(109) -2×2
(20) $-3 \times (-4)$	(50) -2×3	(80) -5×1	(110) $-7 \times (-7)$
(21) 5×6	(51) 5×8	(81) $-7 \times (-5)$	(111) 2×6
(22) $4 \times (-5)$	(52) 4×3	(82) $-5 \times (-2)$	(112) -3×1
(23) $8 \times (-9)$	(53) 3×3	(83) $-3 \times (-7)$	(113) $2 \times (-8)$
(24) -4×2	(54) $-9 \times (-1)$	(84) -8×4	(114) -6×7
(25) -4×5	(55) 8×0	(85) -1×1	(115) 9×8
(26) 9×4	(56) $8 \times (-8)$	(86) -8×5	(116) 9×1
(27) 3×7	(57) 4×8	(87) -4×5	(117) -7×4
(28) $-5 \times (-8)$	(58) $-1 \times (-5)$	(88) -6×4	(118) 9×8
(29) -4×4	(59) -4×2	(89) $-8 \times (-6)$	(119) $-6 \times (-8)$
(30) 8×2	(60) 8×6	(90) 2×2	(120) -6×3

(1) $-1 \times (-5)$	(31) $1 \times (-6)$	(61) 1×1	(91) $-8 \times (-3)$
(2) -5×7	(32) $-1 \times (-1)$	(62) $-8 \times (-7)$	(92) 1×4
(3) 2×6	(33) -9×1	(63) -3×5	(93) 5×1
(4) -6×8	(34) $-5 \times (-1)$	(64) $5 \times (-8)$	(94) $-6 \times (-4)$
(5) $6 \times (-9)$	(35) -3×4	(65) $-5 \times (-7)$	(95) $0 \times (-8)$
(6) $1 \times (-1)$	(36) -3×5	(66) -5×4	(96) $2 \times (-4)$
(7) $2 \times (-3)$	(37) 0×5	(67) $-9 \times (-4)$	(97) 7×2
(8) $-1 \times (-3)$	(38) 2×5	(68) 4×6	(98) 1×7
(9) $-7 \times (-7)$	(39) $-9 \times (-5)$	(69) $-5 \times (-8)$	(99) $7 \times (-3)$
(10) $-5 \times (-9)$	(40) 9×6	(70) -9×7	(100) $3 \times (-1)$
(11) -7×2	(41) -7×0	(71) -1×7	(101) 2×1
(12) $4 \times (-9)$	(42) $-8 \times (-7)$	(72) 8×5	(102) 3×3
(13) -9×3	(43) -3×0	(73) -7×7	(103) $-3 \times (-6)$
(14) $-8 \times (-4)$	(44) $2 \times (-1)$	(74) $-4 \times (-9)$	(104) 3×9
(15) -7×2	(45) $-2 \times (-3)$	(75) 2×1	(105) 0×2
(16) -5×1	(46) 6×6	(76) -3×0	(106) 4×0
(17) -6×7	(47) $-1 \times (-4)$	(77) $7 \times (-1)$	(107) 0×0
(18) 1×2	(48) 0×6	(78) $9 \times (-2)$	(108) $-4 \times (-2)$
(19) 2×1	(49) $9 \times (-7)$	(79) $8 \times (-7)$	(109) -2×4
(20) 2×7	(50) 0×5	(80) 9×5	(110) $4 \times (-1)$
(21) 2×0	(51) $8 \times (-5)$	(81) 6×5	(111) $-9 \times (-2)$
(22) $8 \times (-9)$	(52) $-1 \times (-5)$	(82) 0×6	(112) -2×2
(23) $-7 \times (-4)$	(53) 3×1	(83) $-2 \times (-4)$	(113) $6 \times (-5)$
(24) 7×7	(54) $2 \times (-5)$	(84) -9×3	(114) $-8 \times (-8)$
(25) -1×9	(55) $-6 \times (-9)$	(85) $2 \times (-5)$	(115) $-8 \times (-5)$
(26) $7 \times (-3)$	(56) 1×0	(86) -7×6	(116) $-9 \times (-1)$
(27) -3×5	(57) $2 \times (-6)$	(87) $1 \times (-8)$	(117) $2 \times (-6)$
(28) $6 \times (-9)$	(58) $2 \times (-9)$	(88) $-6 \times (-6)$	(118) $1 \times (-5)$
(29) 0×4	(59) -1×9	(89) 3×7	(119) $7 \times (-7)$
(30) $8 \times (-7)$	(60) -8×4	(90) $8 \times (-3)$	(120) $5 \times (-6)$

(1) -3×8	(31) 7×7	(61) $7 \times (-9)$	(91) $2 \times (-9)$
(2) -5×0	(32) 9×4	(62) $-5 \times (-2)$	(92) -7×6
(3) $6 \times (-6)$	(33) 2×9	(63) $-7 \times (-4)$	(93) $0 \times (-8)$
(4) -8×2	(34) $-6 \times (-5)$	(64) -7×0	(94) -1×8
(5) -3×9	(35) $4 \times (-3)$	(65) -2×4	(95) $-8 \times (-5)$
(6) -5×7	(36) $-1 \times (-3)$	(66) $4 \times (-5)$	(96) 8×3
(7) $4 \times (-8)$	(37) $1 \times (-6)$	(67) -6×8	(97) -3×0
(8) $1 \times (-1)$	(38) 0×4	(68) 1×6	(98) $4 \times (-8)$
(9) 1×4	(39) -9×0	(69) 6×9	(99) 9×4
(10) $9 \times (-4)$	(40) $-2 \times (-7)$	(70) $4 \times (-6)$	(100) $4 \times (-4)$
(11) 1×6	(41) -4×9	(71) 5×1	(101) $-1 \times (-7)$
(12) -9×2	(42) 3×4	(72) 2×2	(102) $3 \times (-4)$
(13) -3×3	(43) $-3 \times (-2)$	(73) $4 \times (-4)$	(103) $6 \times (-4)$
(14) $-2 \times (-5)$	(44) $9 \times (-1)$	(74) -4×9	(104) -2×5
(15) $-1 \times (-8)$	(45) 6×2	(75) 7×6	(105) $5 \times (-1)$
(16) $8 \times (-4)$	(46) -4×9	(76) $8 \times (-4)$	(106) $-1 \times (-8)$
(17) $-3 \times (-5)$	(47) $8 \times (-4)$	(77) $-3 \times (-8)$	(107) 8×5
(18) -6×4	(48) $-2 \times (-4)$	(78) $-2 \times (-7)$	(108) 4×8
(19) $-5 \times (-6)$	(49) $0 \times (-7)$	(79) $1 \times (-2)$	(109) -1×3
(20) 0×3	(50) $2 \times (-8)$	(80) 3×8	(110) $-8 \times (-3)$
(21) $-1 \times (-6)$	(51) 4×4	(81) 7×2	(111) $-3 \times (-6)$
(22) $9 \times (-2)$	(52) $-9 \times (-6)$	(82) -2×8	(112) 9×1
(23) -9×4	(53) 2×6	(83) $-9 \times (-6)$	(113) 2×6
(24) 2×1	(54) $-8 \times (-5)$	(84) 8×3	(114) $-5 \times (-1)$
(25) 8×9	(55) $-1 \times (-5)$	(85) 6×5	(115) $-5 \times (-3)$
(26) 7×7	(56) 4×1	(86) 7×8	(116) $-9 \times (-1)$
(27) 6×7	(57) 7×9	(87) $-1 \times (-7)$	(117) 9×8
(28) -7×8	(58) $6 \times (-7)$	(88) $8 \times (-9)$	(118) $-4 \times (-4)$
(29) $-9 \times (-5)$	(59) $7 \times (-8)$	(89) -2×4	(119) -8×4
(30) 9×1	(60) $0 \times (-4)$	(90) -2×9	(120) -4×6

(1) -5×2	(31) $8 \times (-1)$	(61) -6×1	(91) -2×4
(2) -7×6	(32) $-5 \times (-3)$	(62) -5×7	(92) $6 \times (-7)$
(3) $1 \times (-4)$	(33) $8 \times (-3)$	(63) $6 \times (-6)$	(93) -2×0
(4) 6×9	(34) -6×4	(64) $0 \times (-1)$	(94) -2×2
(5) 4×9	(35) 0×9	(65) 2×4	(95) -5×7
(6) $7 \times (-2)$	(36) $-4 \times (-8)$	(66) 4×7	(96) -5×7
(7) 8×3	(37) $7 \times (-9)$	(67) $-4 \times (-9)$	(97) 7×0
(8) 3×2	(38) $-8 \times (-3)$	(68) -2×8	(98) $2 \times (-2)$
(9) -4×9	(39) $-3 \times (-2)$	(69) -4×7	(99) -3×7
(10) $8 \times (-8)$	(40) 9×5	(70) 0×5	(100) $5 \times (-6)$
(11) $-1 \times (-4)$	(41) 9×1	(71) -8×6	(101) -6×4
(12) $-3 \times (-2)$	(42) $8 \times (-6)$	(72) -2×1	(102) 8×7
(13) $-7 \times (-2)$	(43) $-1 \times (-7)$	(73) $1 \times (-9)$	(103) $4 \times (-2)$
(14) 2×7	(44) $9 \times (-7)$	(74) -1×6	(104) -3×7
(15) -6×9	(45) $-4 \times (-3)$	(75) $3 \times (-7)$	(105) 1×8
(16) 1×0	(46) 2×0	(76) $-5 \times (-3)$	(106) $8 \times (-2)$
(17) $5 \times (-9)$	(47) 1×9	(77) 9×8	(107) $-6 \times (-5)$
(18) $9 \times (-9)$	(48) 6×2	(78) $1 \times (-7)$	(108) $-4 \times (-6)$
(19) 7×6	(49) -1×0	(79) -1×2	(109) 9×6
(20) -6×3	(50) $-1 \times (-1)$	(80) -2×9	(110) $4 \times (-1)$
(21) -3×4	(51) -2×2	(81) -1×4	(111) $-3 \times (-2)$
(22) -1×6	(52) $4 \times (-2)$	(82) $2 \times (-2)$	(112) $9 \times (-9)$
(23) $-4 \times (-8)$	(53) -7×3	(83) -9×1	(113) $-8 \times (-5)$
(24) $-4 \times (-8)$	(54) $0 \times (-7)$	(84) $-4 \times (-3)$	(114) -4×5
(25) -3×0	(55) $-3 \times (-8)$	(85) $-1 \times (-2)$	(115) $-6 \times (-3)$
(26) 8×5	(56) $-9 \times (-2)$	(86) $4 \times (-6)$	(116) $2 \times (-6)$
(27) -9×8	(57) $4 \times (-3)$	(87) 6×1	(117) -8×5
(28) $-4 \times (-2)$	(58) 0×5	(88) $4 \times (-9)$	(118) $3 \times (-1)$
(29) $6 \times (-6)$	(59) -3×1	(89) $-8 \times (-6)$	(119) 8×6
(30) -2×4	(60) $0 \times (-4)$	(90) $-8 \times (-3)$	(120) 1×0

(1) -2×9	(31) $1 \times (-2)$	(61) 3×7	(91) $1 \times (-5)$
(2) 8×7	(32) -5×4	(62) $-8 \times (-8)$	(92) $-2 \times (-6)$
(3) $6 \times (-2)$	(33) $2 \times (-3)$	(63) $-4 \times (-5)$	(93) -6×8
(4) $6 \times (-8)$	(34) -8×9	(64) $9 \times (-2)$	(94) 2×6
(5) 7×4	(35) $-7 \times (-7)$	(65) $-4 \times (-4)$	(95) $2 \times (-9)$
(6) $-1 \times (-2)$	(36) $8 \times (-5)$	(66) $6 \times (-4)$	(96) -8×9
(7) $-4 \times (-3)$	(37) $-8 \times (-5)$	(67) $-3 \times (-2)$	(97) $9 \times (-8)$
(8) $3 \times (-2)$	(38) 8×0	(68) 4×8	(98) $7 \times (-8)$
(9) -4×6	(39) 4×0	(69) $1 \times (-2)$	(99) $-6 \times (-6)$
(10) $5 \times (-3)$	(40) $-1 \times (-9)$	(70) $-4 \times (-5)$	(100) $8 \times (-4)$
(11) $4 \times (-3)$	(41) 8×1	(71) -3×4	(101) -1×0
(12) $3 \times (-3)$	(42) 6×9	(72) 8×2	(102) $-2 \times (-1)$
(13) $-4 \times (-4)$	(43) 1×7	(73) $-7 \times (-7)$	(103) $3 \times (-8)$
(14) 0×4	(44) $-9 \times (-2)$	(74) $-6 \times (-4)$	(104) -8×2
(15) 8×3	(45) -1×7	(75) -3×5	(105) -1×2
(16) -2×3	(46) $5 \times (-3)$	(76) 7×2	(106) -1×9
(17) 6×2	(47) $-9 \times (-2)$	(77) $6 \times (-2)$	(107) 5×5
(18) -2×9	(48) 7×1	(78) $-2 \times (-7)$	(108) 5×4
(19) $0 \times (-1)$	(49) -8×8	(79) $-1 \times (-9)$	(109) $4 \times (-7)$
(20) $0 \times (-6)$	(50) -1×7	(80) $-2 \times (-7)$	(110) $-2 \times (-4)$
(21) 7×0	(51) -7×8	(81) -2×9	(111) $4 \times (-7)$
(22) -3×8	(52) $-1 \times (-7)$	(82) $3 \times (-8)$	(112) $9 \times (-3)$
(23) -5×1	(53) $2 \times (-4)$	(83) 2×3	(113) $-4 \times (-7)$
(24) $-4 \times (-7)$	(54) 6×6	(84) 5×2	(114) $-8 \times (-6)$
(25) -9×2	(55) -6×6	(85) 1×1	(115) -4×6
(26) $5 \times (-9)$	(56) -9×8	(86) 5×4	(116) $7 \times (-1)$
(27) $8 \times (-4)$	(57) 7×9	(87) $3 \times (-7)$	(117) $-8 \times (-3)$
(28) 6×3	(58) $3 \times (-6)$	(88) 8×7	(118) -1×8
(29) 0×2	(59) $9 \times (-5)$	(89) $-6 \times (-9)$	(119) $-3 \times (-4)$
(30) $-3 \times (-8)$	(60) $5 \times (-8)$	(90) -5×5	(120) $4 \times (-4)$

(1) $0 \times (-7)$	(31) 8×8	(61) -5×6	(91) 4×1
(2) $2 \times (-2)$	(32) -4×0	(62) 9×5	(92) $-2 \times (-5)$
(3) $-7 \times (-2)$	(33) 9×1	(63) 3×5	(93) $5 \times (-8)$
(4) 4×2	(34) $-4 \times (-3)$	(64) 7×8	(94) 8×3
(5) $0 \times (-1)$	(35) 5×3	(65) $8 \times (-7)$	(95) $1 \times (-7)$
(6) -8×1	(36) $-4 \times (-9)$	(66) -1×2	(96) $0 \times (-7)$
(7) $4 \times (-1)$	(37) $4 \times (-9)$	(67) $-8 \times (-4)$	(97) $6 \times (-4)$
(8) -3×9	(38) 9×3	(68) 9×7	(98) 2×3
(9) $2 \times (-4)$	(39) -9×3	(69) $-5 \times (-1)$	(99) $0 \times (-6)$
(10) $-3 \times (-6)$	(40) $-7 \times (-7)$	(70) $4 \times (-2)$	(100) 8×0
(11) 8×5	(41) -3×0	(71) $-2 \times (-7)$	(101) $9 \times (-4)$
(12) $-5 \times (-9)$	(42) $-2 \times (-4)$	(72) -4×7	(102) 0×4
(13) $-6 \times (-2)$	(43) -6×5	(73) 0×5	(103) -4×5
(14) $-8 \times (-8)$	(44) $5 \times (-7)$	(74) $5 \times (-2)$	(104) 3×5
(15) 0×6	(45) -5×1	(75) -4×0	(105) 9×3
(16) 8×8	(46) 6×2	(76) $-4 \times (-1)$	(106) 6×0
(17) -5×2	(47) 1×8	(77) $-1 \times (-6)$	(107) 2×2
(18) $3 \times (-7)$	(48) $-1 \times (-1)$	(78) -5×4	(108) -1×7
(19) $5 \times (-6)$	(49) $-9 \times (-7)$	(79) -5×4	(109) -4×5
(20) 5×9	(50) -4×6	(80) $-4 \times (-4)$	(110) $7 \times (-1)$
(21) 0×0	(51) 7×5	(81) $-6 \times (-4)$	(111) $3 \times (-3)$
(22) -3×4	(52) -2×7	(82) $8 \times (-2)$	(112) 8×7
(23) 5×5	(53) 7×9	(83) $-2 \times (-2)$	(113) -2×9
(24) -5×4	(54) $0 \times (-7)$	(84) -9×8	(114) $9 \times (-1)$
(25) $-1 \times (-1)$	(55) $-7 \times (-8)$	(85) $-5 \times (-5)$	(115) $3 \times (-4)$
(26) -6×0	(56) 9×2	(86) $-7 \times (-7)$	(116) 2×3
(27) -9×9	(57) $3 \times (-8)$	(87) $4 \times (-6)$	(117) $1 \times (-7)$
(28) -2×6	(58) $2 \times (-2)$	(88) 5×8	(118) $-6 \times (-9)$
(29) $-4 \times (-3)$	(59) $-2 \times (-5)$	(89) $5 \times (-5)$	(119) 5×2
(30) 5×3	(60) $-6 \times (-8)$	(90) 8×4	(120) -9×6

- | | | | |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-4 \times (-7)$ | (31) $-9 \times (-5)$ | (61) -8×8 | (91) $-6 \times (-6)$ |
| (2) 6×9 | (32) -3×8 | (62) $4 \times (-7)$ | (92) -7×3 |
| (3) -7×3 | (33) $4 \times (-3)$ | (63) 8×0 | (93) 0×7 |
| (4) 7×4 | (34) $-6 \times (-8)$ | (64) $3 \times (-3)$ | (94) -6×1 |
| (5) 2×8 | (35) 6×5 | (65) $4 \times (-3)$ | (95) 4×6 |
| (6) $4 \times (-7)$ | (36) 8×8 | (66) $9 \times (-9)$ | (96) 0×4 |
| (7) -5×3 | (37) 2×1 | (67) 2×9 | (97) $-9 \times (-5)$ |
| (8) -8×0 | (38) $-6 \times (-5)$ | (68) 9×3 | (98) $4 \times (-8)$ |
| (9) -3×8 | (39) $6 \times (-8)$ | (69) -2×4 | (99) 9×2 |
| (10) $-4 \times (-2)$ | (40) $-6 \times (-3)$ | (70) $-7 \times (-1)$ | (100) $9 \times (-7)$ |
| (11) 1×7 | (41) $-7 \times (-9)$ | (71) -1×6 | (101) 7×8 |
| (12) 2×8 | (42) $0 \times (-4)$ | (72) 1×1 | (102) -3×6 |
| (13) $2 \times (-8)$ | (43) $1 \times (-5)$ | (73) $-8 \times (-9)$ | (103) $-4 \times (-7)$ |
| (14) -8×4 | (44) 1×4 | (74) 5×7 | (104) -3×5 |
| (15) 0×7 | (45) -6×4 | (75) -3×8 | (105) 2×7 |
| (16) 2×9 | (46) 4×6 | (76) $7 \times (-4)$ | (106) $-6 \times (-6)$ |
| (17) $-3 \times (-2)$ | (47) $4 \times (-5)$ | (77) $-8 \times (-4)$ | (107) $3 \times (-9)$ |
| (18) 3×2 | (48) 7×8 | (78) -1×4 | (108) $-6 \times (-9)$ |
| (19) -4×7 | (49) $-5 \times (-5)$ | (79) $6 \times (-4)$ | (109) 9×3 |
| (20) $-3 \times (-1)$ | (50) 2×6 | (80) $6 \times (-6)$ | (110) 7×7 |
| (21) 1×3 | (51) $-5 \times (-8)$ | (81) $0 \times (-3)$ | (111) 7×6 |
| (22) $-1 \times (-5)$ | (52) 9×7 | (82) $7 \times (-7)$ | (112) 2×7 |
| (23) -2×6 | (53) $-4 \times (-6)$ | (83) -5×2 | (113) $-2 \times (-3)$ |
| (24) -4×2 | (54) -9×1 | (84) $7 \times (-6)$ | (114) 1×4 |
| (25) 1×0 | (55) 0×8 | (85) 9×6 | (115) 2×8 |
| (26) 5×5 | (56) $-4 \times (-3)$ | (86) 7×6 | (116) 5×8 |
| (27) $2 \times (-4)$ | (57) 2×6 | (87) 6×3 | (117) 6×1 |
| (28) -6×8 | (58) 0×4 | (88) 5×8 | (118) 8×8 |
| (29) $-7 \times (-9)$ | (59) -5×6 | (89) $-6 \times (-2)$ | (119) 8×6 |
| (30) -5×2 | (60) $-4 \times (-1)$ | (90) $-6 \times (-1)$ | (120) 3×5 |

(1) $4 \times (-7)$	(31) 0×7	(61) -6×7	(91) $3 \times (-3)$
(2) $-8 \times (-7)$	(32) $-1 \times (-6)$	(62) -7×5	(92) 9×1
(3) $-4 \times (-1)$	(33) 6×5	(63) $7 \times (-9)$	(93) $-3 \times (-8)$
(4) -8×5	(34) 6×5	(64) $7 \times (-3)$	(94) 9×6
(5) $-2 \times (-3)$	(35) $6 \times (-8)$	(65) $4 \times (-8)$	(95) -9×1
(6) $-9 \times (-4)$	(36) 5×7	(66) $2 \times (-5)$	(96) 6×7
(7) $8 \times (-7)$	(37) $2 \times (-5)$	(67) $2 \times (-7)$	(97) -6×9
(8) -1×2	(38) $1 \times (-9)$	(68) 7×6	(98) $5 \times (-2)$
(9) $9 \times (-9)$	(39) 6×9	(69) 8×1	(99) 1×6
(10) $7 \times (-7)$	(40) $4 \times (-9)$	(70) 5×0	(100) 4×3
(11) 6×7	(41) 9×3	(71) 2×2	(101) 3×9
(12) $2 \times (-9)$	(42) $5 \times (-5)$	(72) $2 \times (-8)$	(102) $-5 \times (-6)$
(13) $-1 \times (-8)$	(43) -9×7	(73) -1×5	(103) $-3 \times (-3)$
(14) -8×8	(44) -9×6	(74) $5 \times (-1)$	(104) 9×0
(15) $5 \times (-8)$	(45) $-2 \times (-1)$	(75) 8×6	(105) $9 \times (-3)$
(16) -2×6	(46) $8 \times (-1)$	(76) 5×0	(106) $-2 \times (-8)$
(17) $1 \times (-8)$	(47) $-9 \times (-4)$	(77) -9×0	(107) -6×0
(18) 3×7	(48) $8 \times (-3)$	(78) 4×2	(108) $6 \times (-4)$
(19) $-3 \times (-2)$	(49) $6 \times (-9)$	(79) $4 \times (-2)$	(109) $-9 \times (-1)$
(20) $8 \times (-1)$	(50) $6 \times (-6)$	(80) 5×4	(110) $3 \times (-7)$
(21) -7×9	(51) $-9 \times (-7)$	(81) 8×5	(111) $-8 \times (-4)$
(22) $-1 \times (-3)$	(52) 6×2	(82) 2×1	(112) $-3 \times (-7)$
(23) $9 \times (-1)$	(53) -6×2	(83) -7×0	(113) -6×2
(24) 8×5	(54) -7×0	(84) 0×3	(114) $3 \times (-6)$
(25) $-6 \times (-9)$	(55) $8 \times (-7)$	(85) $6 \times (-1)$	(115) 6×3
(26) $6 \times (-5)$	(56) 8×0	(86) -8×0	(116) 0×8
(27) $0 \times (-4)$	(57) $3 \times (-4)$	(87) $-7 \times (-8)$	(117) $8 \times (-9)$
(28) $-1 \times (-8)$	(58) -4×7	(88) -8×8	(118) $5 \times (-1)$
(29) 1×4	(59) -3×8	(89) $-2 \times (-7)$	(119) $-6 \times (-2)$
(30) $-5 \times (-8)$	(60) $-2 \times (-9)$	(90) 5×9	(120) 7×8

(1) $7 \times (-2)$	(31) -9×7	(61) -3×8	(91) -3×0
(2) -7×0	(32) $-7 \times (-3)$	(62) $-5 \times (-5)$	(92) 3×0
(3) 3×9	(33) $8 \times (-8)$	(63) $-5 \times (-7)$	(93) 3×1
(4) -3×5	(34) $2 \times (-4)$	(64) $-7 \times (-9)$	(94) $-9 \times (-6)$
(5) $-1 \times (-3)$	(35) -8×5	(65) 2×5	(95) 3×5
(6) 6×0	(36) $1 \times (-3)$	(66) -5×9	(96) -1×3
(7) 1×1	(37) $0 \times (-4)$	(67) 6×1	(97) $-7 \times (-7)$
(8) -1×4	(38) $1 \times (-3)$	(68) 1×9	(98) -1×8
(9) $-1 \times (-8)$	(39) -7×2	(69) $-9 \times (-8)$	(99) $-3 \times (-9)$
(10) $0 \times (-4)$	(40) $3 \times (-7)$	(70) $-3 \times (-8)$	(100) $-8 \times (-6)$
(11) -2×7	(41) $5 \times (-1)$	(71) 9×9	(101) -6×6
(12) $3 \times (-9)$	(42) 7×1	(72) 8×7	(102) $-4 \times (-6)$
(13) $-2 \times (-6)$	(43) 8×2	(73) 3×4	(103) 9×8
(14) -4×1	(44) $9 \times (-7)$	(74) $-8 \times (-1)$	(104) 8×4
(15) $3 \times (-3)$	(45) 5×4	(75) $7 \times (-9)$	(105) 7×6
(16) $4 \times (-2)$	(46) $-1 \times (-8)$	(76) -8×2	(106) 8×2
(17) 3×9	(47) $-9 \times (-3)$	(77) $8 \times (-7)$	(107) $3 \times (-7)$
(18) -9×6	(48) $-1 \times (-1)$	(78) $-8 \times (-4)$	(108) 2×9
(19) 6×5	(49) 4×9	(79) $-7 \times (-9)$	(109) $6 \times (-9)$
(20) -9×5	(50) $1 \times (-1)$	(80) $-1 \times (-7)$	(110) -7×7
(21) 6×5	(51) 1×5	(81) -3×9	(111) $1 \times (-3)$
(22) $9 \times (-3)$	(52) 6×0	(82) 4×3	(112) $-7 \times (-3)$
(23) $6 \times (-7)$	(53) -1×8	(83) $-2 \times (-9)$	(113) 8×4
(24) $-8 \times (-9)$	(54) -2×3	(84) $-9 \times (-7)$	(114) $-5 \times (-7)$
(25) -1×2	(55) 9×9	(85) 5×4	(115) -1×4
(26) 1×8	(56) 0×5	(86) -2×8	(116) 9×7
(27) $-2 \times (-5)$	(57) $-8 \times (-1)$	(87) $-2 \times (-4)$	(117) $-6 \times (-3)$
(28) $0 \times (-9)$	(58) $-2 \times (-3)$	(88) $6 \times (-4)$	(118) -5×3
(29) $8 \times (-2)$	(59) -5×0	(89) -9×2	(119) $9 \times (-1)$
(30) $-7 \times (-6)$	(60) $4 \times (-3)$	(90) $5 \times (-1)$	(120) -7×5

(1) -2×6	(31) $1 \times (-8)$	(61) -5×0	(91) $5 \times (-4)$
(2) 9×9	(32) $-3 \times (-2)$	(62) 8×5	(92) $-4 \times (-7)$
(3) 0×2	(33) $-7 \times (-5)$	(63) 3×0	(93) 4×3
(4) -7×1	(34) $-7 \times (-1)$	(64) -1×0	(94) -9×8
(5) -1×9	(35) $8 \times (-3)$	(65) $6 \times (-7)$	(95) -7×4
(6) -4×3	(36) -4×9	(66) -8×0	(96) $-2 \times (-7)$
(7) 6×9	(37) 4×1	(67) $3 \times (-3)$	(97) $8 \times (-9)$
(8) 2×9	(38) $-7 \times (-1)$	(68) $-5 \times (-5)$	(98) -7×7
(9) $9 \times (-8)$	(39) $0 \times (-5)$	(69) -3×9	(99) 0×0
(10) $-8 \times (-4)$	(40) $-1 \times (-7)$	(70) $-3 \times (-5)$	(100) $5 \times (-5)$
(11) -6×4	(41) 6×6	(71) 2×2	(101) -7×2
(12) $-8 \times (-8)$	(42) 0×4	(72) -7×2	(102) -8×8
(13) -9×7	(43) $1 \times (-1)$	(73) $3 \times (-4)$	(103) -9×2
(14) 9×6	(44) $8 \times (-7)$	(74) -7×5	(104) -6×9
(15) $-3 \times (-3)$	(45) -1×0	(75) -7×3	(105) -2×9
(16) $7 \times (-2)$	(46) $7 \times (-4)$	(76) 4×8	(106) $-6 \times (-4)$
(17) $2 \times (-3)$	(47) 1×3	(77) $7 \times (-9)$	(107) 4×2
(18) 3×5	(48) $9 \times (-3)$	(78) $1 \times (-3)$	(108) 3×0
(19) -4×3	(49) $-2 \times (-2)$	(79) -2×5	(109) -4×7
(20) -6×9	(50) 6×3	(80) $-7 \times (-5)$	(110) $-4 \times (-8)$
(21) 7×4	(51) -9×7	(81) $-6 \times (-5)$	(111) -6×3
(22) 3×0	(52) -5×4	(82) $4 \times (-3)$	(112) $3 \times (-6)$
(23) $5 \times (-6)$	(53) -5×9	(83) 6×4	(113) $5 \times (-8)$
(24) $-8 \times (-5)$	(54) 6×1	(84) $9 \times (-3)$	(114) 2×9
(25) $-4 \times (-4)$	(55) $-7 \times (-4)$	(85) $2 \times (-8)$	(115) $-9 \times (-3)$
(26) $-4 \times (-9)$	(56) $-1 \times (-4)$	(86) $9 \times (-7)$	(116) 7×4
(27) 9×5	(57) $-2 \times (-5)$	(87) $0 \times (-4)$	(117) $-7 \times (-9)$
(28) -7×8	(58) $2 \times (-3)$	(88) -6×3	(118) $-9 \times (-2)$
(29) $9 \times (-6)$	(59) $6 \times (-4)$	(89) $-6 \times (-8)$	(119) 3×0
(30) 0×8	(60) -2×2	(90) -4×6	(120) -8×2

(1) $4 \times (-4)$	(31) $2 \times (-8)$	(61) -9×3	(91) 0×0
(2) $-5 \times (-5)$	(32) -1×3	(62) -4×3	(92) 3×0
(3) 4×4	(33) $1 \times (-4)$	(63) $-2 \times (-3)$	(93) $-4 \times (-7)$
(4) $9 \times (-4)$	(34) 5×1	(64) $0 \times (-4)$	(94) 4×2
(5) $-4 \times (-5)$	(35) $-3 \times (-4)$	(65) -2×4	(95) 1×5
(6) 5×0	(36) 1×7	(66) -4×0	(96) 9×3
(7) -7×8	(37) $-4 \times (-9)$	(67) 4×2	(97) 6×8
(8) $9 \times (-2)$	(38) $4 \times (-2)$	(68) 9×0	(98) -1×2
(9) -2×9	(39) $-2 \times (-6)$	(69) $-1 \times (-4)$	(99) $5 \times (-7)$
(10) -7×1	(40) 2×3	(70) $9 \times (-2)$	(100) 3×9
(11) -4×4	(41) $-9 \times (-1)$	(71) 2×2	(101) $6 \times (-1)$
(12) $-8 \times (-1)$	(42) $-8 \times (-2)$	(72) $-5 \times (-1)$	(102) 7×8
(13) 5×5	(43) $-8 \times (-3)$	(73) 4×4	(103) $-4 \times (-5)$
(14) -4×7	(44) $-3 \times (-8)$	(74) $-1 \times (-6)$	(104) $2 \times (-9)$
(15) $-6 \times (-6)$	(45) $-6 \times (-1)$	(75) 9×1	(105) -2×1
(16) $9 \times (-6)$	(46) 2×5	(76) $6 \times (-2)$	(106) $-3 \times (-2)$
(17) 2×2	(47) $-6 \times (-6)$	(77) $-6 \times (-6)$	(107) $6 \times (-2)$
(18) $-1 \times (-5)$	(48) $-5 \times (-8)$	(78) $-9 \times (-2)$	(108) $6 \times (-4)$
(19) -7×8	(49) $-1 \times (-1)$	(79) -7×3	(109) $-4 \times (-9)$
(20) 7×8	(50) $2 \times (-1)$	(80) 7×8	(110) $7 \times (-9)$
(21) $-8 \times (-8)$	(51) -4×0	(81) -8×8	(111) 2×5
(22) 9×7	(52) -1×9	(82) $2 \times (-8)$	(112) -6×0
(23) 5×3	(53) $-4 \times (-7)$	(83) -8×1	(113) 7×9
(24) 6×8	(54) -8×0	(84) 6×9	(114) -2×7
(25) 8×9	(55) 6×1	(85) 3×3	(115) 0×7
(26) $8 \times (-5)$	(56) 6×0	(86) -5×9	(116) $-8 \times (-4)$
(27) $3 \times (-1)$	(57) $7 \times (-7)$	(87) 1×5	(117) 6×9
(28) $-7 \times (-2)$	(58) 5×1	(88) 7×1	(118) 0×8
(29) $3 \times (-8)$	(59) $-9 \times (-1)$	(89) $2 \times (-2)$	(119) $1 \times (-4)$
(30) $-3 \times (-7)$	(60) $1 \times (-6)$	(90) $1 \times (-1)$	(120) $7 \times (-9)$

- | | | | |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $8 \times (-9)$ | (31) 3×4 | (61) $3 \times (-5)$ | (91) 0×6 |
| (2) -4×6 | (32) 2×6 | (62) $1 \times (-6)$ | (92) 0×8 |
| (3) -6×6 | (33) -3×0 | (63) $7 \times (-9)$ | (93) 3×4 |
| (4) $7 \times (-7)$ | (34) $-1 \times (-9)$ | (64) -1×4 | (94) $2 \times (-8)$ |
| (5) $7 \times (-3)$ | (35) $7 \times (-2)$ | (65) $-1 \times (-9)$ | (95) $-6 \times (-3)$ |
| (6) -4×3 | (36) -1×1 | (66) 7×5 | (96) 1×7 |
| (7) $-9 \times (-2)$ | (37) 4×9 | (67) 3×7 | (97) 0×8 |
| (8) 2×4 | (38) 8×5 | (68) 6×5 | (98) $-7 \times (-5)$ |
| (9) 1×8 | (39) 8×1 | (69) $8 \times (-4)$ | (99) 4×4 |
| (10) 0×9 | (40) -3×1 | (70) $-8 \times (-9)$ | (100) $-8 \times (-4)$ |
| (11) -6×2 | (41) $8 \times (-2)$ | (71) 1×3 | (101) $9 \times (-2)$ |
| (12) $7 \times (-5)$ | (42) 4×9 | (72) -2×9 | (102) 3×9 |
| (13) -6×9 | (43) -3×9 | (73) -7×9 | (103) -7×1 |
| (14) 2×8 | (44) $7 \times (-2)$ | (74) -1×4 | (104) $9 \times (-5)$ |
| (15) 2×7 | (45) $-3 \times (-6)$ | (75) $6 \times (-5)$ | (105) -1×5 |
| (16) -8×9 | (46) -4×9 | (76) $6 \times (-3)$ | (106) 8×5 |
| (17) $2 \times (-6)$ | (47) 7×2 | (77) $-5 \times (-9)$ | (107) $6 \times (-1)$ |
| (18) -1×2 | (48) -7×8 | (78) 9×3 | (108) 0×9 |
| (19) $0 \times (-7)$ | (49) $-5 \times (-8)$ | (79) 3×8 | (109) $1 \times (-6)$ |
| (20) -2×1 | (50) $-1 \times (-4)$ | (80) 4×9 | (110) $-1 \times (-7)$ |
| (21) $5 \times (-2)$ | (51) $-4 \times (-1)$ | (81) -4×4 | (111) 7×3 |
| (22) -1×5 | (52) 9×5 | (82) $5 \times (-4)$ | (112) -6×7 |
| (23) 0×7 | (53) $9 \times (-5)$ | (83) $-1 \times (-9)$ | (113) -8×7 |
| (24) $-6 \times (-5)$ | (54) $3 \times (-7)$ | (84) -4×4 | (114) $-6 \times (-8)$ |
| (25) $1 \times (-8)$ | (55) -7×5 | (85) $-8 \times (-6)$ | (115) $-7 \times (-8)$ |
| (26) $7 \times (-6)$ | (56) -4×6 | (86) $-2 \times (-7)$ | (116) $0 \times (-5)$ |
| (27) $-8 \times (-1)$ | (57) 5×3 | (87) $-5 \times (-8)$ | (117) 4×4 |
| (28) 7×2 | (58) $-9 \times (-1)$ | (88) -9×2 | (118) -1×0 |
| (29) $1 \times (-6)$ | (59) $5 \times (-2)$ | (89) $2 \times (-5)$ | (119) $-7 \times (-1)$ |
| (30) 1×6 | (60) -3×6 | (90) -2×9 | (120) -5×7 |

(1) $-7 \times (-6)$	(31) $4 \times (-9)$	(61) 5×5	(91) $3 \times (-1)$
(2) 6×9	(32) 0×8	(62) 2×8	(92) 5×6
(3) $9 \times (-5)$	(33) 7×5	(63) $-7 \times (-2)$	(93) -2×8
(4) $4 \times (-7)$	(34) -3×1	(64) $3 \times (-3)$	(94) 5×3
(5) -8×8	(35) $1 \times (-7)$	(65) $-9 \times (-6)$	(95) $5 \times (-2)$
(6) -4×6	(36) -1×5	(66) $-3 \times (-1)$	(96) $4 \times (-9)$
(7) -5×5	(37) 7×2	(67) -6×7	(97) $5 \times (-7)$
(8) $-5 \times (-7)$	(38) $2 \times (-9)$	(68) $-5 \times (-2)$	(98) -7×2
(9) $-7 \times (-5)$	(39) 4×2	(69) 6×1	(99) 0×3
(10) 9×9	(40) 8×6	(70) 9×7	(100) $-8 \times (-1)$
(11) $-6 \times (-3)$	(41) 0×9	(71) -2×1	(101) 7×0
(12) -5×5	(42) $-7 \times (-3)$	(72) 0×7	(102) 1×0
(13) $-1 \times (-7)$	(43) $2 \times (-5)$	(73) $3 \times (-6)$	(103) $9 \times (-3)$
(14) -3×1	(44) 0×9	(74) -5×4	(104) $-3 \times (-8)$
(15) -6×6	(45) $1 \times (-9)$	(75) $-2 \times (-2)$	(105) $8 \times (-2)$
(16) $4 \times (-2)$	(46) 2×7	(76) $2 \times (-4)$	(106) -8×5
(17) $-3 \times (-6)$	(47) 4×0	(77) -8×4	(107) -3×3
(18) $5 \times (-3)$	(48) $9 \times (-5)$	(78) $-4 \times (-8)$	(108) 0×8
(19) $-9 \times (-1)$	(49) $1 \times (-8)$	(79) -7×6	(109) $4 \times (-6)$
(20) $-1 \times (-2)$	(50) -1×4	(80) $6 \times (-4)$	(110) $8 \times (-4)$
(21) $0 \times (-2)$	(51) $5 \times (-1)$	(81) 9×9	(111) 7×2
(22) $2 \times (-7)$	(52) 0×5	(82) -6×4	(112) $3 \times (-5)$
(23) -1×8	(53) $7 \times (-9)$	(83) -5×8	(113) 2×1
(24) -7×5	(54) $4 \times (-8)$	(84) -4×7	(114) 8×2
(25) 8×5	(55) $7 \times (-4)$	(85) -8×9	(115) -7×7
(26) $-3 \times (-5)$	(56) -5×9	(86) $-1 \times (-6)$	(116) -3×1
(27) -1×3	(57) -1×7	(87) $-3 \times (-3)$	(117) -8×4
(28) $8 \times (-9)$	(58) 6×5	(88) $9 \times (-3)$	(118) 0×1
(29) $-6 \times (-1)$	(59) -8×2	(89) 7×3	(119) $5 \times (-1)$
(30) -4×8	(60) -1×6	(90) 6×5	(120) $-5 \times (-2)$

(1) 7×7	(31) $3 \times (-3)$	(61) $3 \times (-4)$	(91) 3×9
(2) $-6 \times (-6)$	(32) $6 \times (-7)$	(62) -4×0	(92) 7×1
(3) 6×3	(33) -3×0	(63) $-3 \times (-7)$	(93) $8 \times (-6)$
(4) $-7 \times (-1)$	(34) $-1 \times (-1)$	(64) -8×7	(94) $7 \times (-3)$
(5) -4×8	(35) $-5 \times (-5)$	(65) $6 \times (-7)$	(95) -2×8
(6) $1 \times (-3)$	(36) $-1 \times (-7)$	(66) $-9 \times (-6)$	(96) $-5 \times (-2)$
(7) 9×2	(37) $-3 \times (-6)$	(67) $-9 \times (-3)$	(97) 0×8
(8) 8×8	(38) $-5 \times (-4)$	(68) -2×2	(98) -4×0
(9) $-1 \times (-3)$	(39) 2×7	(69) -8×8	(99) $1 \times (-2)$
(10) $-2 \times (-7)$	(40) $-7 \times (-3)$	(70) 7×9	(100) $6 \times (-7)$
(11) 5×7	(41) 2×2	(71) -3×9	(101) $4 \times (-4)$
(12) 7×9	(42) -1×1	(72) $3 \times (-4)$	(102) 5×6
(13) $6 \times (-1)$	(43) -1×3	(73) 9×8	(103) 4×4
(14) 9×9	(44) $-6 \times (-3)$	(74) $-7 \times (-4)$	(104) $2 \times (-3)$
(15) 6×2	(45) -9×9	(75) 8×6	(105) $1 \times (-9)$
(16) $5 \times (-2)$	(46) $-6 \times (-8)$	(76) -3×8	(106) $0 \times (-1)$
(17) 5×2	(47) $-8 \times (-8)$	(77) $-8 \times (-7)$	(107) -3×3
(18) -5×7	(48) $2 \times (-8)$	(78) -2×1	(108) $3 \times (-7)$
(19) -6×9	(49) $-8 \times (-9)$	(79) -9×5	(109) $-1 \times (-4)$
(20) $5 \times (-8)$	(50) $5 \times (-9)$	(80) $-1 \times (-3)$	(110) -1×7
(21) -4×8	(51) $-6 \times (-5)$	(81) $-7 \times (-1)$	(111) 7×5
(22) -9×8	(52) -2×5	(82) -8×8	(112) -4×5
(23) 6×2	(53) 8×1	(83) $-4 \times (-9)$	(113) -3×3
(24) $9 \times (-3)$	(54) $9 \times (-3)$	(84) $-3 \times (-5)$	(114) $-1 \times (-9)$
(25) -5×1	(55) $2 \times (-6)$	(85) 0×1	(115) -3×4
(26) $5 \times (-7)$	(56) -4×0	(86) 2×1	(116) $7 \times (-2)$
(27) $0 \times (-6)$	(57) -3×9	(87) $-3 \times (-9)$	(117) $-6 \times (-6)$
(28) 7×7	(58) -2×9	(88) $-3 \times (-8)$	(118) $5 \times (-2)$
(29) 8×0	(59) 2×0	(89) -2×8	(119) -2×4
(30) -8×2	(60) 1×3	(90) $-9 \times (-4)$	(120) 3×0

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-7 \times (-7) = 49$ | (31) $-8 \times (-2) = 16$ | (61) $-8 \times (-3) = 24$ | (91) $5 \times (-7) = -35$ |
| (2) $7 \times (-1) = -7$ | (32) $7 \times 3 = 21$ | (62) $7 \times 2 = 14$ | (92) $-7 \times (-9) = 63$ |
| (3) $8 \times (-6) = -48$ | (33) $9 \times (-9) = -81$ | (63) $8 \times (-3) = -24$ | (93) $-2 \times (-3) = 6$ |
| (4) $-7 \times 5 = -35$ | (34) $5 \times 3 = 15$ | (64) $-1 \times (-9) = 9$ | (94) $-7 \times 0 = 0$ |
| (5) $-5 \times 6 = -30$ | (35) $7 \times (-9) = -63$ | (65) $5 \times 2 = 10$ | (95) $3 \times (-2) = -6$ |
| (6) $8 \times (-8) = -64$ | (36) $-7 \times 4 = -28$ | (66) $-8 \times 6 = -48$ | (96) $-2 \times (-1) = 2$ |
| (7) $7 \times (-2) = -14$ | (37) $-2 \times 9 = -18$ | (67) $-3 \times 8 = -24$ | (97) $-9 \times (-2) = 18$ |
| (8) $0 \times 6 = 0$ | (38) $4 \times (-7) = -28$ | (68) $0 \times 7 = 0$ | (98) $4 \times (-9) = -36$ |
| (9) $-9 \times 8 = -72$ | (39) $-9 \times 1 = -9$ | (69) $-1 \times (-8) = 8$ | (99) $-1 \times 5 = -5$ |
| (10) $-2 \times 5 = -10$ | (40) $-6 \times (-6) = 36$ | (70) $4 \times 9 = 36$ | (100) $-1 \times (-1) = 1$ |
| (11) $6 \times 7 = 42$ | (41) $7 \times 6 = 42$ | (71) $-6 \times 1 = -6$ | (101) $3 \times 4 = 12$ |
| (12) $2 \times (-9) = -18$ | (42) $-3 \times 1 = -3$ | (72) $-7 \times (-1) = 7$ | (102) $-8 \times (-3) = 24$ |
| (13) $3 \times 2 = 6$ | (43) $7 \times (-2) = -14$ | (73) $-4 \times 0 = 0$ | (103) $7 \times (-4) = -28$ |
| (14) $-3 \times 1 = -3$ | (44) $-5 \times (-8) = 40$ | (74) $4 \times (-6) = -24$ | (104) $-5 \times 2 = -10$ |
| (15) $1 \times 4 = 4$ | (45) $-9 \times 7 = -63$ | (75) $5 \times (-2) = -10$ | (105) $1 \times 9 = 9$ |
| (16) $-6 \times 6 = -36$ | (46) $-3 \times 6 = -18$ | (76) $9 \times 3 = 27$ | (106) $-3 \times 2 = -6$ |
| (17) $-7 \times 3 = -21$ | (47) $7 \times 9 = 63$ | (77) $-7 \times 4 = -28$ | (107) $9 \times (-8) = -72$ |
| (18) $4 \times 1 = 4$ | (48) $-9 \times (-1) = 9$ | (78) $4 \times (-1) = -4$ | (108) $-1 \times 4 = -4$ |
| (19) $3 \times 2 = 6$ | (49) $0 \times (-1) = 0$ | (79) $-9 \times 3 = -27$ | (109) $-6 \times 1 = -6$ |
| (20) $4 \times (-1) = -4$ | (50) $8 \times (-1) = -8$ | (80) $-4 \times 2 = -8$ | (110) $-1 \times (-6) = 6$ |
| (21) $-6 \times 0 = 0$ | (51) $-9 \times 4 = -36$ | (81) $5 \times 8 = 40$ | (111) $6 \times (-4) = -24$ |
| (22) $8 \times (-9) = -72$ | (52) $-7 \times 3 = -21$ | (82) $2 \times (-3) = -6$ | (112) $-8 \times (-7) = 56$ |
| (23) $-8 \times (-9) = 72$ | (53) $-8 \times (-2) = 16$ | (83) $-5 \times 9 = -45$ | (113) $-9 \times (-7) = 63$ |
| (24) $0 \times 4 = 0$ | (54) $-1 \times 7 = -7$ | (84) $-9 \times 5 = -45$ | (114) $-8 \times 7 = -56$ |
| (25) $7 \times (-3) = -21$ | (55) $-5 \times (-9) = 45$ | (85) $0 \times (-9) = 0$ | (115) $1 \times (-3) = -3$ |
| (26) $0 \times 1 = 0$ | (56) $-9 \times (-8) = 72$ | (86) $4 \times (-6) = -24$ | (116) $2 \times 5 = 10$ |
| (27) $-8 \times (-5) = 40$ | (57) $-6 \times (-3) = 18$ | (87) $-6 \times (-5) = 30$ | (117) $4 \times (-9) = -36$ |
| (28) $-9 \times 8 = -72$ | (58) $1 \times (-2) = -2$ | (88) $5 \times (-5) = -25$ | (118) $1 \times 9 = 9$ |
| (29) $-9 \times (-6) = 54$ | (59) $-1 \times (-3) = 3$ | (89) $-8 \times 5 = -40$ | (119) $6 \times (-3) = -18$ |
| (30) $4 \times (-8) = -32$ | (60) $-5 \times 8 = -40$ | (90) $3 \times 2 = 6$ | (120) $9 \times 6 = 54$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $8 \times 9 = 72$ | (31) $0 \times 8 = 0$ | (61) $-2 \times 0 = 0$ | (91) $5 \times 4 = 20$ |
| (2) $-4 \times (-8) = 32$ | (32) $3 \times (-9) = -27$ | (62) $6 \times 6 = 36$ | (92) $6 \times 2 = 12$ |
| (3) $-6 \times 6 = -36$ | (33) $8 \times 4 = 32$ | (63) $4 \times (-9) = -36$ | (93) $7 \times 6 = 42$ |
| (4) $1 \times (-4) = -4$ | (34) $5 \times 2 = 10$ | (64) $-7 \times (-2) = 14$ | (94) $4 \times (-6) = -24$ |
| (5) $-5 \times (-3) = 15$ | (35) $-4 \times (-8) = 32$ | (65) $-5 \times (-8) = 40$ | (95) $-8 \times 4 = -32$ |
| (6) $0 \times 7 = 0$ | (36) $-3 \times (-8) = 24$ | (66) $6 \times 7 = 42$ | (96) $3 \times (-3) = -9$ |
| (7) $-7 \times 3 = -21$ | (37) $2 \times (-1) = -2$ | (67) $5 \times 1 = 5$ | (97) $2 \times (-2) = -4$ |
| (8) $-9 \times (-9) = 81$ | (38) $-6 \times 8 = -48$ | (68) $-1 \times (-6) = 6$ | (98) $-4 \times 8 = -32$ |
| (9) $-1 \times 3 = -3$ | (39) $0 \times (-7) = 0$ | (69) $-2 \times (-2) = 4$ | (99) $8 \times (-7) = -56$ |
| (10) $-1 \times 9 = -9$ | (40) $7 \times 2 = 14$ | (70) $1 \times 2 = 2$ | (100) $6 \times (-6) = -36$ |
| (11) $-7 \times 0 = 0$ | (41) $-3 \times 3 = -9$ | (71) $7 \times 2 = 14$ | (101) $1 \times 5 = 5$ |
| (12) $2 \times 8 = 16$ | (42) $5 \times (-4) = -20$ | (72) $-8 \times 1 = -8$ | (102) $-2 \times 9 = -18$ |
| (13) $8 \times 8 = 64$ | (43) $-6 \times 7 = -42$ | (73) $-1 \times 2 = -2$ | (103) $-3 \times 5 = -15$ |
| (14) $7 \times 3 = 21$ | (44) $1 \times 2 = 2$ | (74) $0 \times (-7) = 0$ | (104) $5 \times 0 = 0$ |
| (15) $-8 \times (-5) = 40$ | (45) $9 \times 6 = 54$ | (75) $-5 \times (-3) = 15$ | (105) $-2 \times (-2) = 4$ |
| (16) $0 \times (-7) = 0$ | (46) $-7 \times 3 = -21$ | (76) $-2 \times 1 = -2$ | (106) $7 \times (-9) = -63$ |
| (17) $3 \times 5 = 15$ | (47) $-2 \times 8 = -16$ | (77) $-2 \times (-1) = 2$ | (107) $7 \times (-3) = -21$ |
| (18) $0 \times 8 = 0$ | (48) $-1 \times 2 = -2$ | (78) $7 \times (-6) = -42$ | (108) $4 \times 3 = 12$ |
| (19) $-6 \times (-7) = 42$ | (49) $-2 \times (-1) = 2$ | (79) $-1 \times (-6) = 6$ | (109) $-1 \times 9 = -9$ |
| (20) $-3 \times 8 = -24$ | (50) $-7 \times (-2) = 14$ | (80) $-5 \times 9 = -45$ | (110) $-8 \times 2 = -16$ |
| (21) $-4 \times (-2) = 8$ | (51) $3 \times (-5) = -15$ | (81) $-5 \times 5 = -25$ | (111) $3 \times (-8) = -24$ |
| (22) $-8 \times 7 = -56$ | (52) $-4 \times (-8) = 32$ | (82) $0 \times 1 = 0$ | (112) $-6 \times 8 = -48$ |
| (23) $-9 \times (-4) = 36$ | (53) $5 \times 1 = 5$ | (83) $8 \times (-3) = -24$ | (113) $-2 \times (-4) = 8$ |
| (24) $-9 \times 5 = -45$ | (54) $-9 \times 8 = -72$ | (84) $3 \times (-6) = -18$ | (114) $-7 \times 2 = -14$ |
| (25) $-1 \times 0 = 0$ | (55) $4 \times 9 = 36$ | (85) $3 \times 1 = 3$ | (115) $3 \times 3 = 9$ |
| (26) $-2 \times 4 = -8$ | (56) $0 \times 4 = 0$ | (86) $-9 \times 1 = -9$ | (116) $-7 \times 1 = -7$ |
| (27) $-7 \times (-8) = 56$ | (57) $6 \times 5 = 30$ | (87) $-2 \times (-1) = 2$ | (117) $1 \times (-6) = -6$ |
| (28) $2 \times 4 = 8$ | (58) $5 \times 7 = 35$ | (88) $8 \times (-6) = -48$ | (118) $7 \times 4 = 28$ |
| (29) $-3 \times 2 = -6$ | (59) $8 \times 8 = 64$ | (89) $-9 \times 0 = 0$ | (119) $6 \times 6 = 36$ |
| (30) $8 \times (-4) = -32$ | (60) $5 \times (-9) = -45$ | (90) $9 \times 1 = 9$ | (120) $0 \times (-1) = 0$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $0 \times 9 = 0$ | (31) $-4 \times 8 = -32$ | (61) $7 \times 5 = 35$ | (91) $-7 \times 0 = 0$ |
| (2) $6 \times (-7) = -42$ | (32) $7 \times (-8) = -56$ | (62) $8 \times (-2) = -16$ | (92) $-2 \times 4 = -8$ |
| (3) $-6 \times (-8) = 48$ | (33) $-9 \times 6 = -54$ | (63) $-5 \times (-7) = 35$ | (93) $2 \times 5 = 10$ |
| (4) $-6 \times (-7) = 42$ | (34) $0 \times (-3) = 0$ | (64) $2 \times (-3) = -6$ | (94) $-9 \times (-4) = 36$ |
| (5) $-9 \times (-1) = 9$ | (35) $-5 \times 4 = -20$ | (65) $2 \times (-4) = -8$ | (95) $-8 \times (-7) = 56$ |
| (6) $-3 \times 4 = -12$ | (36) $2 \times 0 = 0$ | (66) $-2 \times 5 = -10$ | (96) $-9 \times 7 = -63$ |
| (7) $-8 \times 7 = -56$ | (37) $7 \times (-4) = -28$ | (67) $-9 \times (-1) = 9$ | (97) $6 \times 6 = 36$ |
| (8) $-4 \times (-1) = 4$ | (38) $-6 \times (-1) = 6$ | (68) $-5 \times 2 = -10$ | (98) $5 \times (-3) = -15$ |
| (9) $-8 \times 0 = 0$ | (39) $-5 \times (-4) = 20$ | (69) $3 \times (-9) = -27$ | (99) $4 \times (-7) = -28$ |
| (10) $-4 \times (-4) = 16$ | (40) $-5 \times (-2) = 10$ | (70) $7 \times 0 = 0$ | (100) $8 \times (-8) = -64$ |
| (11) $-7 \times (-3) = 21$ | (41) $-7 \times (-7) = 49$ | (71) $-3 \times (-6) = 18$ | (101) $8 \times (-4) = -32$ |
| (12) $5 \times 3 = 15$ | (42) $-9 \times (-6) = 54$ | (72) $-7 \times 7 = -49$ | (102) $-3 \times (-2) = 6$ |
| (13) $-1 \times 7 = -7$ | (43) $-1 \times (-8) = 8$ | (73) $-7 \times 9 = -63$ | (103) $-5 \times 0 = 0$ |
| (14) $-7 \times (-3) = 21$ | (44) $1 \times 6 = 6$ | (74) $-9 \times (-4) = 36$ | (104) $4 \times 8 = 32$ |
| (15) $7 \times 4 = 28$ | (45) $4 \times 7 = 28$ | (75) $7 \times (-6) = -42$ | (105) $0 \times (-5) = 0$ |
| (16) $4 \times (-2) = -8$ | (46) $5 \times 7 = 35$ | (76) $-7 \times (-9) = 63$ | (106) $-8 \times (-6) = 48$ |
| (17) $-3 \times (-1) = 3$ | (47) $9 \times (-1) = -9$ | (77) $-7 \times 7 = -49$ | (107) $0 \times 9 = 0$ |
| (18) $-9 \times 9 = -81$ | (48) $0 \times (-9) = 0$ | (78) $-1 \times (-6) = 6$ | (108) $1 \times (-1) = -1$ |
| (19) $9 \times 5 = 45$ | (49) $8 \times 4 = 32$ | (79) $-5 \times (-9) = 45$ | (109) $7 \times 2 = 14$ |
| (20) $3 \times 9 = 27$ | (50) $-4 \times 5 = -20$ | (80) $2 \times (-4) = -8$ | (110) $8 \times 6 = 48$ |
| (21) $2 \times 4 = 8$ | (51) $-9 \times (-1) = 9$ | (81) $-8 \times 9 = -72$ | (111) $0 \times 1 = 0$ |
| (22) $5 \times 4 = 20$ | (52) $-4 \times (-2) = 8$ | (82) $-5 \times 3 = -15$ | (112) $-9 \times (-3) = 27$ |
| (23) $-5 \times 5 = -25$ | (53) $3 \times (-8) = -24$ | (83) $1 \times (-3) = -3$ | (113) $7 \times (-1) = -7$ |
| (24) $-8 \times (-1) = 8$ | (54) $1 \times 9 = 9$ | (84) $-1 \times (-1) = 1$ | (114) $5 \times (-8) = -40$ |
| (25) $-3 \times (-4) = 12$ | (55) $-2 \times 4 = -8$ | (85) $-7 \times (-7) = 49$ | (115) $-4 \times 2 = -8$ |
| (26) $-9 \times (-8) = 72$ | (56) $-9 \times 6 = -54$ | (86) $-4 \times 3 = -12$ | (116) $-2 \times 0 = 0$ |
| (27) $7 \times (-2) = -14$ | (57) $-2 \times 9 = -18$ | (87) $7 \times (-6) = -42$ | (117) $1 \times 8 = 8$ |
| (28) $0 \times (-2) = 0$ | (58) $5 \times 6 = 30$ | (88) $2 \times 5 = 10$ | (118) $-5 \times 3 = -15$ |
| (29) $6 \times 9 = 54$ | (59) $-9 \times 0 = 0$ | (89) $7 \times 3 = 21$ | (119) $2 \times (-7) = -14$ |
| (30) $4 \times 1 = 4$ | (60) $-6 \times (-7) = 42$ | (90) $-3 \times 8 = -24$ | (120) $-9 \times 5 = -45$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-9 \times 0 = 0$ | (31) $4 \times 7 = 28$ | (61) $-6 \times (-9) = 54$ | (91) $-3 \times 8 = -24$ |
| (2) $-7 \times (-1) = 7$ | (32) $7 \times 4 = 28$ | (62) $1 \times 7 = 7$ | (92) $-9 \times 0 = 0$ |
| (3) $-5 \times 8 = -40$ | (33) $3 \times (-2) = -6$ | (63) $5 \times 3 = 15$ | (93) $1 \times 7 = 7$ |
| (4) $-9 \times 0 = 0$ | (34) $4 \times 8 = 32$ | (64) $2 \times 6 = 12$ | (94) $9 \times 7 = 63$ |
| (5) $3 \times 1 = 3$ | (35) $4 \times 2 = 8$ | (65) $4 \times 9 = 36$ | (95) $-4 \times 4 = -16$ |
| (6) $-3 \times (-3) = 9$ | (36) $2 \times 8 = 16$ | (66) $2 \times (-7) = -14$ | (96) $-9 \times 7 = -63$ |
| (7) $-5 \times 0 = 0$ | (37) $8 \times 0 = 0$ | (67) $4 \times 7 = 28$ | (97) $-5 \times (-7) = 35$ |
| (8) $-1 \times 5 = -5$ | (38) $8 \times (-4) = -32$ | (68) $-3 \times 8 = -24$ | (98) $2 \times 0 = 0$ |
| (9) $-5 \times 1 = -5$ | (39) $-3 \times (-9) = 27$ | (69) $-1 \times 0 = 0$ | (99) $4 \times 9 = 36$ |
| (10) $-4 \times 7 = -28$ | (40) $-3 \times 1 = -3$ | (70) $9 \times (-8) = -72$ | (100) $7 \times (-6) = -42$ |
| (11) $2 \times (-8) = -16$ | (41) $-4 \times (-2) = 8$ | (71) $6 \times 4 = 24$ | (101) $1 \times 5 = 5$ |
| (12) $2 \times (-9) = -18$ | (42) $-7 \times 0 = 0$ | (72) $-5 \times 7 = -35$ | (102) $-4 \times 4 = -16$ |
| (13) $-1 \times 0 = 0$ | (43) $4 \times 2 = 8$ | (73) $-7 \times 9 = -63$ | (103) $5 \times 2 = 10$ |
| (14) $2 \times 7 = 14$ | (44) $-3 \times 9 = -27$ | (74) $-9 \times 7 = -63$ | (104) $-5 \times (-8) = 40$ |
| (15) $-2 \times (-7) = 14$ | (45) $9 \times 3 = 27$ | (75) $-1 \times 6 = -6$ | (105) $9 \times (-1) = -9$ |
| (16) $8 \times (-2) = -16$ | (46) $-1 \times 8 = -8$ | (76) $-9 \times 9 = -81$ | (106) $6 \times 0 = 0$ |
| (17) $9 \times (-6) = -54$ | (47) $-1 \times 3 = -3$ | (77) $0 \times 8 = 0$ | (107) $1 \times 4 = 4$ |
| (18) $8 \times 5 = 40$ | (48) $6 \times 3 = 18$ | (78) $3 \times (-8) = -24$ | (108) $7 \times (-3) = -21$ |
| (19) $8 \times (-3) = -24$ | (49) $-3 \times 1 = -3$ | (79) $-9 \times (-9) = 81$ | (109) $0 \times 1 = 0$ |
| (20) $6 \times 0 = 0$ | (50) $-8 \times (-9) = 72$ | (80) $-5 \times 7 = -35$ | (110) $-6 \times 5 = -30$ |
| (21) $-9 \times (-7) = 63$ | (51) $-6 \times (-2) = 12$ | (81) $7 \times (-2) = -14$ | (111) $9 \times 6 = 54$ |
| (22) $5 \times (-4) = -20$ | (52) $7 \times (-5) = -35$ | (82) $0 \times (-6) = 0$ | (112) $-5 \times 8 = -40$ |
| (23) $-1 \times 2 = -2$ | (53) $-7 \times 6 = -42$ | (83) $3 \times 2 = 6$ | (113) $8 \times 3 = 24$ |
| (24) $8 \times (-1) = -8$ | (54) $-7 \times 2 = -14$ | (84) $-5 \times 0 = 0$ | (114) $-3 \times 9 = -27$ |
| (25) $4 \times (-1) = -4$ | (55) $6 \times 9 = 54$ | (85) $-2 \times 5 = -10$ | (115) $-3 \times 2 = -6$ |
| (26) $-7 \times 5 = -35$ | (56) $1 \times (-3) = -3$ | (86) $2 \times (-4) = -8$ | (116) $-7 \times 0 = 0$ |
| (27) $1 \times (-7) = -7$ | (57) $-2 \times 2 = -4$ | (87) $-4 \times 0 = 0$ | (117) $4 \times (-9) = -36$ |
| (28) $9 \times 8 = 72$ | (58) $7 \times (-4) = -28$ | (88) $6 \times 9 = 54$ | (118) $0 \times (-4) = 0$ |
| (29) $-2 \times 1 = -2$ | (59) $6 \times (-9) = -54$ | (89) $-4 \times (-7) = 28$ | (119) $-9 \times (-3) = 27$ |
| (30) $-7 \times (-1) = 7$ | (60) $-4 \times (-9) = 36$ | (90) $-4 \times (-1) = 4$ | (120) $-5 \times (-7) = 35$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $2 \times (-3) = -6$ | (31) $-1 \times (-4) = 4$ | (61) $9 \times 0 = 0$ | (91) $4 \times 7 = 28$ |
| (2) $7 \times (-9) = -63$ | (32) $-6 \times 2 = -12$ | (62) $-5 \times (-5) = 25$ | (92) $-1 \times (-5) = 5$ |
| (3) $2 \times 6 = 12$ | (33) $8 \times 0 = 0$ | (63) $6 \times (-1) = -6$ | (93) $8 \times (-9) = -72$ |
| (4) $1 \times 9 = 9$ | (34) $-3 \times (-4) = 12$ | (64) $3 \times 2 = 6$ | (94) $4 \times (-1) = -4$ |
| (5) $9 \times (-2) = -18$ | (35) $-4 \times 3 = -12$ | (65) $-1 \times 5 = -5$ | (95) $7 \times (-5) = -35$ |
| (6) $-2 \times 8 = -16$ | (36) $6 \times (-9) = -54$ | (66) $-5 \times 6 = -30$ | (96) $-2 \times (-8) = 16$ |
| (7) $-2 \times 0 = 0$ | (37) $-2 \times 8 = -16$ | (67) $-7 \times (-6) = 42$ | (97) $2 \times 3 = 6$ |
| (8) $-6 \times (-4) = 24$ | (38) $7 \times 9 = 63$ | (68) $-7 \times 8 = -56$ | (98) $8 \times 0 = 0$ |
| (9) $0 \times (-1) = 0$ | (39) $8 \times (-6) = -48$ | (69) $-3 \times (-3) = 9$ | (99) $-2 \times 1 = -2$ |
| (10) $5 \times (-3) = -15$ | (40) $-7 \times (-5) = 35$ | (70) $2 \times (-1) = -2$ | (100) $0 \times 0 = 0$ |
| (11) $-3 \times 5 = -15$ | (41) $1 \times (-4) = -4$ | (71) $-2 \times (-3) = 6$ | (101) $-7 \times (-7) = 49$ |
| (12) $1 \times 5 = 5$ | (42) $4 \times 1 = 4$ | (72) $4 \times (-7) = -28$ | (102) $-8 \times 9 = -72$ |
| (13) $-8 \times 9 = -72$ | (43) $3 \times (-6) = -18$ | (73) $7 \times 5 = 35$ | (103) $8 \times 8 = 64$ |
| (14) $-5 \times 1 = -5$ | (44) $8 \times (-4) = -32$ | (74) $2 \times 3 = 6$ | (104) $1 \times 8 = 8$ |
| (15) $8 \times (-5) = -40$ | (45) $-7 \times 8 = -56$ | (75) $1 \times 3 = 3$ | (105) $-6 \times (-9) = 54$ |
| (16) $-1 \times (-5) = 5$ | (46) $-2 \times 6 = -12$ | (76) $-1 \times 6 = -6$ | (106) $-5 \times 3 = -15$ |
| (17) $0 \times 5 = 0$ | (47) $8 \times (-7) = -56$ | (77) $4 \times 4 = 16$ | (107) $-5 \times 2 = -10$ |
| (18) $-9 \times 2 = -18$ | (48) $-6 \times (-5) = 30$ | (78) $-3 \times (-7) = 21$ | (108) $-5 \times (-2) = 10$ |
| (19) $-4 \times (-1) = 4$ | (49) $4 \times 8 = 32$ | (79) $9 \times (-1) = -9$ | (109) $-8 \times (-8) = 64$ |
| (20) $7 \times (-6) = -42$ | (50) $8 \times 5 = 40$ | (80) $1 \times (-3) = -3$ | (110) $5 \times 0 = 0$ |
| (21) $6 \times (-6) = -36$ | (51) $7 \times (-9) = -63$ | (81) $8 \times 9 = 72$ | (111) $0 \times (-3) = 0$ |
| (22) $9 \times (-6) = -54$ | (52) $0 \times 1 = 0$ | (82) $3 \times (-9) = -27$ | (112) $0 \times 6 = 0$ |
| (23) $0 \times 1 = 0$ | (53) $-4 \times (-6) = 24$ | (83) $-5 \times 9 = -45$ | (113) $8 \times (-1) = -8$ |
| (24) $-2 \times (-6) = 12$ | (54) $3 \times (-4) = -12$ | (84) $7 \times (-3) = -21$ | (114) $5 \times 4 = 20$ |
| (25) $2 \times (-6) = -12$ | (55) $8 \times (-5) = -40$ | (85) $-5 \times 0 = 0$ | (115) $7 \times 1 = 7$ |
| (26) $-9 \times 9 = -81$ | (56) $-1 \times (-8) = 8$ | (86) $3 \times (-8) = -24$ | (116) $8 \times (-2) = -16$ |
| (27) $-3 \times 6 = -18$ | (57) $-7 \times (-4) = 28$ | (87) $1 \times (-6) = -6$ | (117) $-1 \times 1 = -1$ |
| (28) $6 \times 0 = 0$ | (58) $-6 \times 3 = -18$ | (88) $-3 \times (-2) = 6$ | (118) $3 \times 7 = 21$ |
| (29) $-5 \times 9 = -45$ | (59) $6 \times 8 = 48$ | (89) $3 \times (-2) = -6$ | (119) $6 \times (-4) = -24$ |
| (30) $-9 \times (-5) = 45$ | (60) $-7 \times (-8) = 56$ | (90) $3 \times (-6) = -18$ | (120) $-4 \times (-7) = 28$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-4 \times (-4) = 16$ | (31) $-2 \times (-6) = 12$ | (61) $-6 \times (-2) = 12$ | (91) $-3 \times 8 = -24$ |
| (2) $6 \times 8 = 48$ | (32) $-6 \times 0 = 0$ | (62) $6 \times (-1) = -6$ | (92) $-7 \times 1 = -7$ |
| (3) $-6 \times 8 = -48$ | (33) $3 \times 6 = 18$ | (63) $-8 \times (-9) = 72$ | (93) $-8 \times 0 = 0$ |
| (4) $1 \times (-9) = -9$ | (34) $6 \times 7 = 42$ | (64) $0 \times (-5) = 0$ | (94) $6 \times 1 = 6$ |
| (5) $-6 \times 2 = -12$ | (35) $-6 \times (-6) = 36$ | (65) $-7 \times (-5) = 35$ | (95) $0 \times (-5) = 0$ |
| (6) $5 \times (-3) = -15$ | (36) $-5 \times (-1) = 5$ | (66) $2 \times (-6) = -12$ | (96) $7 \times 9 = 63$ |
| (7) $7 \times (-1) = -7$ | (37) $-2 \times 8 = -16$ | (67) $9 \times 8 = 72$ | (97) $6 \times 9 = 54$ |
| (8) $3 \times 7 = 21$ | (38) $9 \times 5 = 45$ | (68) $2 \times (-5) = -10$ | (98) $9 \times 6 = 54$ |
| (9) $7 \times 0 = 0$ | (39) $-3 \times 4 = -12$ | (69) $2 \times 9 = 18$ | (99) $-2 \times 6 = -12$ |
| (10) $-1 \times (-6) = 6$ | (40) $4 \times 2 = 8$ | (70) $4 \times 7 = 28$ | (100) $-6 \times (-2) = 12$ |
| (11) $-1 \times 7 = -7$ | (41) $9 \times (-3) = -27$ | (71) $-7 \times (-8) = 56$ | (101) $9 \times 8 = 72$ |
| (12) $2 \times (-4) = -8$ | (42) $-9 \times 8 = -72$ | (72) $8 \times (-8) = -64$ | (102) $-6 \times 1 = -6$ |
| (13) $2 \times (-6) = -12$ | (43) $1 \times (-3) = -3$ | (73) $-3 \times 5 = -15$ | (103) $1 \times (-6) = -6$ |
| (14) $5 \times (-1) = -5$ | (44) $9 \times 4 = 36$ | (74) $-2 \times 9 = -18$ | (104) $-8 \times 3 = -24$ |
| (15) $0 \times 0 = 0$ | (45) $1 \times (-2) = -2$ | (75) $7 \times (-3) = -21$ | (105) $1 \times 0 = 0$ |
| (16) $-1 \times 1 = -1$ | (46) $0 \times 0 = 0$ | (76) $-4 \times (-9) = 36$ | (106) $7 \times 8 = 56$ |
| (17) $7 \times (-6) = -42$ | (47) $-6 \times 5 = -30$ | (77) $-9 \times (-8) = 72$ | (107) $9 \times 7 = 63$ |
| (18) $2 \times 5 = 10$ | (48) $7 \times (-1) = -7$ | (78) $5 \times (-6) = -30$ | (108) $-4 \times 6 = -24$ |
| (19) $-6 \times (-2) = 12$ | (49) $-8 \times 6 = -48$ | (79) $0 \times (-5) = 0$ | (109) $2 \times 7 = 14$ |
| (20) $5 \times 7 = 35$ | (50) $-7 \times 7 = -49$ | (80) $4 \times 2 = 8$ | (110) $3 \times 1 = 3$ |
| (21) $5 \times (-4) = -20$ | (51) $-3 \times 5 = -15$ | (81) $5 \times (-4) = -20$ | (111) $0 \times (-5) = 0$ |
| (22) $1 \times 2 = 2$ | (52) $3 \times 0 = 0$ | (82) $-9 \times (-1) = 9$ | (112) $-1 \times (-1) = 1$ |
| (23) $-5 \times 8 = -40$ | (53) $1 \times 4 = 4$ | (83) $-1 \times 8 = -8$ | (113) $-1 \times (-6) = 6$ |
| (24) $4 \times (-8) = -32$ | (54) $1 \times (-9) = -9$ | (84) $-8 \times (-2) = 16$ | (114) $9 \times 8 = 72$ |
| (25) $-2 \times (-6) = 12$ | (55) $3 \times 9 = 27$ | (85) $5 \times (-4) = -20$ | (115) $4 \times (-3) = -12$ |
| (26) $3 \times 0 = 0$ | (56) $-9 \times 0 = 0$ | (86) $-6 \times (-6) = 36$ | (116) $-9 \times 7 = -63$ |
| (27) $-6 \times (-9) = 54$ | (57) $-9 \times (-7) = 63$ | (87) $-5 \times 3 = -15$ | (117) $-7 \times 0 = 0$ |
| (28) $6 \times 0 = 0$ | (58) $7 \times 0 = 0$ | (88) $-6 \times (-3) = 18$ | (118) $3 \times (-8) = -24$ |
| (29) $-4 \times (-2) = 8$ | (59) $6 \times (-8) = -48$ | (89) $-9 \times 0 = 0$ | (119) $1 \times 3 = 3$ |
| (30) $-9 \times (-3) = 27$ | (60) $-8 \times (-9) = 72$ | (90) $-9 \times 6 = -54$ | (120) $1 \times (-9) = -9$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $4 \times (-3) = -12$ | (31) $-7 \times (-3) = 21$ | (61) $-9 \times (-7) = 63$ | (91) $0 \times (-2) = 0$ |
| (2) $-4 \times (-6) = 24$ | (32) $-4 \times 0 = 0$ | (62) $5 \times 9 = 45$ | (92) $-7 \times (-9) = 63$ |
| (3) $-2 \times (-6) = 12$ | (33) $9 \times (-8) = -72$ | (63) $6 \times 0 = 0$ | (93) $5 \times 9 = 45$ |
| (4) $9 \times (-8) = -72$ | (34) $-4 \times (-9) = 36$ | (64) $3 \times (-9) = -27$ | (94) $6 \times (-5) = -30$ |
| (5) $3 \times 5 = 15$ | (35) $2 \times (-4) = -8$ | (65) $7 \times 1 = 7$ | (95) $-6 \times 7 = -42$ |
| (6) $5 \times (-9) = -45$ | (36) $0 \times 8 = 0$ | (66) $3 \times 4 = 12$ | (96) $6 \times 9 = 54$ |
| (7) $-4 \times 0 = 0$ | (37) $-5 \times (-6) = 30$ | (67) $-5 \times 7 = -35$ | (97) $2 \times (-7) = -14$ |
| (8) $4 \times (-6) = -24$ | (38) $-6 \times (-9) = 54$ | (68) $-5 \times 0 = 0$ | (98) $2 \times 6 = 12$ |
| (9) $2 \times 0 = 0$ | (39) $-1 \times (-1) = 1$ | (69) $-4 \times (-7) = 28$ | (99) $6 \times 9 = 54$ |
| (10) $-1 \times (-8) = 8$ | (40) $9 \times (-8) = -72$ | (70) $-7 \times (-2) = 14$ | (100) $1 \times 8 = 8$ |
| (11) $-9 \times 8 = -72$ | (41) $-1 \times 7 = -7$ | (71) $7 \times 6 = 42$ | (101) $8 \times (-3) = -24$ |
| (12) $4 \times 4 = 16$ | (42) $-7 \times 6 = -42$ | (72) $3 \times (-9) = -27$ | (102) $-5 \times (-9) = 45$ |
| (13) $-5 \times (-8) = 40$ | (43) $2 \times (-9) = -18$ | (73) $7 \times 5 = 35$ | (103) $-5 \times 5 = -25$ |
| (14) $-3 \times 4 = -12$ | (44) $6 \times (-3) = -18$ | (74) $-1 \times 4 = -4$ | (104) $-9 \times 5 = -45$ |
| (15) $3 \times (-8) = -24$ | (45) $-9 \times 8 = -72$ | (75) $8 \times (-5) = -40$ | (105) $9 \times 5 = 45$ |
| (16) $3 \times 0 = 0$ | (46) $3 \times 1 = 3$ | (76) $-7 \times (-6) = 42$ | (106) $8 \times 1 = 8$ |
| (17) $-6 \times 7 = -42$ | (47) $8 \times 8 = 64$ | (77) $5 \times (-9) = -45$ | (107) $8 \times (-7) = -56$ |
| (18) $-3 \times (-9) = 27$ | (48) $6 \times (-5) = -30$ | (78) $0 \times (-7) = 0$ | (108) $4 \times 9 = 36$ |
| (19) $-5 \times (-7) = 35$ | (49) $-5 \times (-1) = 5$ | (79) $-2 \times (-2) = 4$ | (109) $-2 \times 2 = -4$ |
| (20) $-3 \times (-4) = 12$ | (50) $-2 \times 3 = -6$ | (80) $-5 \times 1 = -5$ | (110) $-7 \times (-7) = 49$ |
| (21) $5 \times 6 = 30$ | (51) $5 \times 8 = 40$ | (81) $-7 \times (-5) = 35$ | (111) $2 \times 6 = 12$ |
| (22) $4 \times (-5) = -20$ | (52) $4 \times 3 = 12$ | (82) $-5 \times (-2) = 10$ | (112) $-3 \times 1 = -3$ |
| (23) $8 \times (-9) = -72$ | (53) $3 \times 3 = 9$ | (83) $-3 \times (-7) = 21$ | (113) $2 \times (-8) = -16$ |
| (24) $-4 \times 2 = -8$ | (54) $-9 \times (-1) = 9$ | (84) $-8 \times 4 = -32$ | (114) $-6 \times 7 = -42$ |
| (25) $-4 \times 5 = -20$ | (55) $8 \times 0 = 0$ | (85) $-1 \times 1 = -1$ | (115) $9 \times 8 = 72$ |
| (26) $9 \times 4 = 36$ | (56) $8 \times (-8) = -64$ | (86) $-8 \times 5 = -40$ | (116) $9 \times 1 = 9$ |
| (27) $3 \times 7 = 21$ | (57) $4 \times 8 = 32$ | (87) $-4 \times 5 = -20$ | (117) $-7 \times 4 = -28$ |
| (28) $-5 \times (-8) = 40$ | (58) $-1 \times (-5) = 5$ | (88) $-6 \times 4 = -24$ | (118) $9 \times 8 = 72$ |
| (29) $-4 \times 4 = -16$ | (59) $-4 \times 2 = -8$ | (89) $-8 \times (-6) = 48$ | (119) $-6 \times (-8) = 48$ |
| (30) $8 \times 2 = 16$ | (60) $8 \times 6 = 48$ | (90) $2 \times 2 = 4$ | (120) $-6 \times 3 = -18$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-1 \times (-5) = 5$ | (31) $1 \times (-6) = -6$ | (61) $1 \times 1 = 1$ | (91) $-8 \times (-3) = 24$ |
| (2) $-5 \times 7 = -35$ | (32) $-1 \times (-1) = 1$ | (62) $-8 \times (-7) = 56$ | (92) $1 \times 4 = 4$ |
| (3) $2 \times 6 = 12$ | (33) $-9 \times 1 = -9$ | (63) $-3 \times 5 = -15$ | (93) $5 \times 1 = 5$ |
| (4) $-6 \times 8 = -48$ | (34) $-5 \times (-1) = 5$ | (64) $5 \times (-8) = -40$ | (94) $-6 \times (-4) = 24$ |
| (5) $6 \times (-9) = -54$ | (35) $-3 \times 4 = -12$ | (65) $-5 \times (-7) = 35$ | (95) $0 \times (-8) = 0$ |
| (6) $1 \times (-1) = -1$ | (36) $-3 \times 5 = -15$ | (66) $-5 \times 4 = -20$ | (96) $2 \times (-4) = -8$ |
| (7) $2 \times (-3) = -6$ | (37) $0 \times 5 = 0$ | (67) $-9 \times (-4) = 36$ | (97) $7 \times 2 = 14$ |
| (8) $-1 \times (-3) = 3$ | (38) $2 \times 5 = 10$ | (68) $4 \times 6 = 24$ | (98) $1 \times 7 = 7$ |
| (9) $-7 \times (-7) = 49$ | (39) $-9 \times (-5) = 45$ | (69) $-5 \times (-8) = 40$ | (99) $7 \times (-3) = -21$ |
| (10) $-5 \times (-9) = 45$ | (40) $9 \times 6 = 54$ | (70) $-9 \times 7 = -63$ | (100) $3 \times (-1) = -3$ |
| (11) $-7 \times 2 = -14$ | (41) $-7 \times 0 = 0$ | (71) $-1 \times 7 = -7$ | (101) $2 \times 1 = 2$ |
| (12) $4 \times (-9) = -36$ | (42) $-8 \times (-7) = 56$ | (72) $8 \times 5 = 40$ | (102) $3 \times 3 = 9$ |
| (13) $-9 \times 3 = -27$ | (43) $-3 \times 0 = 0$ | (73) $-7 \times 7 = -49$ | (103) $-3 \times (-6) = 18$ |
| (14) $-8 \times (-4) = 32$ | (44) $2 \times (-1) = -2$ | (74) $-4 \times (-9) = 36$ | (104) $3 \times 9 = 27$ |
| (15) $-7 \times 2 = -14$ | (45) $-2 \times (-3) = 6$ | (75) $2 \times 1 = 2$ | (105) $0 \times 2 = 0$ |
| (16) $-5 \times 1 = -5$ | (46) $6 \times 6 = 36$ | (76) $-3 \times 0 = 0$ | (106) $4 \times 0 = 0$ |
| (17) $-6 \times 7 = -42$ | (47) $-1 \times (-4) = 4$ | (77) $7 \times (-1) = -7$ | (107) $0 \times 0 = 0$ |
| (18) $1 \times 2 = 2$ | (48) $0 \times 6 = 0$ | (78) $9 \times (-2) = -18$ | (108) $-4 \times (-2) = 8$ |
| (19) $2 \times 1 = 2$ | (49) $9 \times (-7) = -63$ | (79) $8 \times (-7) = -56$ | (109) $-2 \times 4 = -8$ |
| (20) $2 \times 7 = 14$ | (50) $0 \times 5 = 0$ | (80) $9 \times 5 = 45$ | (110) $4 \times (-1) = -4$ |
| (21) $2 \times 0 = 0$ | (51) $8 \times (-5) = -40$ | (81) $6 \times 5 = 30$ | (111) $-9 \times (-2) = 18$ |
| (22) $8 \times (-9) = -72$ | (52) $-1 \times (-5) = 5$ | (82) $0 \times 6 = 0$ | (112) $-2 \times 2 = -4$ |
| (23) $-7 \times (-4) = 28$ | (53) $3 \times 1 = 3$ | (83) $-2 \times (-4) = 8$ | (113) $6 \times (-5) = -30$ |
| (24) $7 \times 7 = 49$ | (54) $2 \times (-5) = -10$ | (84) $-9 \times 3 = -27$ | (114) $-8 \times (-8) = 64$ |
| (25) $-1 \times 9 = -9$ | (55) $-6 \times (-9) = 54$ | (85) $2 \times (-5) = -10$ | (115) $-8 \times (-5) = 40$ |
| (26) $7 \times (-3) = -21$ | (56) $1 \times 0 = 0$ | (86) $-7 \times 6 = -42$ | (116) $-9 \times (-1) = 9$ |
| (27) $-3 \times 5 = -15$ | (57) $2 \times (-6) = -12$ | (87) $1 \times (-8) = -8$ | (117) $2 \times (-6) = -12$ |
| (28) $6 \times (-9) = -54$ | (58) $2 \times (-9) = -18$ | (88) $-6 \times (-6) = 36$ | (118) $1 \times (-5) = -5$ |
| (29) $0 \times 4 = 0$ | (59) $-1 \times 9 = -9$ | (89) $3 \times 7 = 21$ | (119) $7 \times (-7) = -49$ |
| (30) $8 \times (-7) = -56$ | (60) $-8 \times 4 = -32$ | (90) $8 \times (-3) = -24$ | (120) $5 \times (-6) = -30$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-3 \times 8 = -24$ | (31) $7 \times 7 = 49$ | (61) $7 \times (-9) = -63$ | (91) $2 \times (-9) = -18$ |
| (2) $-5 \times 0 = 0$ | (32) $9 \times 4 = 36$ | (62) $-5 \times (-2) = 10$ | (92) $-7 \times 6 = -42$ |
| (3) $6 \times (-6) = -36$ | (33) $2 \times 9 = 18$ | (63) $-7 \times (-4) = 28$ | (93) $0 \times (-8) = 0$ |
| (4) $-8 \times 2 = -16$ | (34) $-6 \times (-5) = 30$ | (64) $-7 \times 0 = 0$ | (94) $-1 \times 8 = -8$ |
| (5) $-3 \times 9 = -27$ | (35) $4 \times (-3) = -12$ | (65) $-2 \times 4 = -8$ | (95) $-8 \times (-5) = 40$ |
| (6) $-5 \times 7 = -35$ | (36) $-1 \times (-3) = 3$ | (66) $4 \times (-5) = -20$ | (96) $8 \times 3 = 24$ |
| (7) $4 \times (-8) = -32$ | (37) $1 \times (-6) = -6$ | (67) $-6 \times 8 = -48$ | (97) $-3 \times 0 = 0$ |
| (8) $1 \times (-1) = -1$ | (38) $0 \times 4 = 0$ | (68) $1 \times 6 = 6$ | (98) $4 \times (-8) = -32$ |
| (9) $1 \times 4 = 4$ | (39) $-9 \times 0 = 0$ | (69) $6 \times 9 = 54$ | (99) $9 \times 4 = 36$ |
| (10) $9 \times (-4) = -36$ | (40) $-2 \times (-7) = 14$ | (70) $4 \times (-6) = -24$ | (100) $4 \times (-4) = -16$ |
| (11) $1 \times 6 = 6$ | (41) $-4 \times 9 = -36$ | (71) $5 \times 1 = 5$ | (101) $-1 \times (-7) = 7$ |
| (12) $-9 \times 2 = -18$ | (42) $3 \times 4 = 12$ | (72) $2 \times 2 = 4$ | (102) $3 \times (-4) = -12$ |
| (13) $-3 \times 3 = -9$ | (43) $-3 \times (-2) = 6$ | (73) $4 \times (-4) = -16$ | (103) $6 \times (-4) = -24$ |
| (14) $-2 \times (-5) = 10$ | (44) $9 \times (-1) = -9$ | (74) $-4 \times 9 = -36$ | (104) $-2 \times 5 = -10$ |
| (15) $-1 \times (-8) = 8$ | (45) $6 \times 2 = 12$ | (75) $7 \times 6 = 42$ | (105) $5 \times (-1) = -5$ |
| (16) $8 \times (-4) = -32$ | (46) $-4 \times 9 = -36$ | (76) $8 \times (-4) = -32$ | (106) $-1 \times (-8) = 8$ |
| (17) $-3 \times (-5) = 15$ | (47) $8 \times (-4) = -32$ | (77) $-3 \times (-8) = 24$ | (107) $8 \times 5 = 40$ |
| (18) $-6 \times 4 = -24$ | (48) $-2 \times (-4) = 8$ | (78) $-2 \times (-7) = 14$ | (108) $4 \times 8 = 32$ |
| (19) $-5 \times (-6) = 30$ | (49) $0 \times (-7) = 0$ | (79) $1 \times (-2) = -2$ | (109) $-1 \times 3 = -3$ |
| (20) $0 \times 3 = 0$ | (50) $2 \times (-8) = -16$ | (80) $3 \times 8 = 24$ | (110) $-8 \times (-3) = 24$ |
| (21) $-1 \times (-6) = 6$ | (51) $4 \times 4 = 16$ | (81) $7 \times 2 = 14$ | (111) $-3 \times (-6) = 18$ |
| (22) $9 \times (-2) = -18$ | (52) $-9 \times (-6) = 54$ | (82) $-2 \times 8 = -16$ | (112) $9 \times 1 = 9$ |
| (23) $-9 \times 4 = -36$ | (53) $2 \times 6 = 12$ | (83) $-9 \times (-6) = 54$ | (113) $2 \times 6 = 12$ |
| (24) $2 \times 1 = 2$ | (54) $-8 \times (-5) = 40$ | (84) $8 \times 3 = 24$ | (114) $-5 \times (-1) = 5$ |
| (25) $8 \times 9 = 72$ | (55) $-1 \times (-5) = 5$ | (85) $6 \times 5 = 30$ | (115) $-5 \times (-3) = 15$ |
| (26) $7 \times 7 = 49$ | (56) $4 \times 1 = 4$ | (86) $7 \times 8 = 56$ | (116) $-9 \times (-1) = 9$ |
| (27) $6 \times 7 = 42$ | (57) $7 \times 9 = 63$ | (87) $-1 \times (-7) = 7$ | (117) $9 \times 8 = 72$ |
| (28) $-7 \times 8 = -56$ | (58) $6 \times (-7) = -42$ | (88) $8 \times (-9) = -72$ | (118) $-4 \times (-4) = 16$ |
| (29) $-9 \times (-5) = 45$ | (59) $7 \times (-8) = -56$ | (89) $-2 \times 4 = -8$ | (119) $-8 \times 4 = -32$ |
| (30) $9 \times 1 = 9$ | (60) $0 \times (-4) = 0$ | (90) $-2 \times 9 = -18$ | (120) $-4 \times 6 = -24$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-5 \times 2 = -10$ | (31) $8 \times (-1) = -8$ | (61) $-6 \times 1 = -6$ | (91) $-2 \times 4 = -8$ |
| (2) $-7 \times 6 = -42$ | (32) $-5 \times (-3) = 15$ | (62) $-5 \times 7 = -35$ | (92) $6 \times (-7) = -42$ |
| (3) $1 \times (-4) = -4$ | (33) $8 \times (-3) = -24$ | (63) $6 \times (-6) = -36$ | (93) $-2 \times 0 = 0$ |
| (4) $6 \times 9 = 54$ | (34) $-6 \times 4 = -24$ | (64) $0 \times (-1) = 0$ | (94) $-2 \times 2 = -4$ |
| (5) $4 \times 9 = 36$ | (35) $0 \times 9 = 0$ | (65) $2 \times 4 = 8$ | (95) $-5 \times 7 = -35$ |
| (6) $7 \times (-2) = -14$ | (36) $-4 \times (-8) = 32$ | (66) $4 \times 7 = 28$ | (96) $-5 \times 7 = -35$ |
| (7) $8 \times 3 = 24$ | (37) $7 \times (-9) = -63$ | (67) $-4 \times (-9) = 36$ | (97) $7 \times 0 = 0$ |
| (8) $3 \times 2 = 6$ | (38) $-8 \times (-3) = 24$ | (68) $-2 \times 8 = -16$ | (98) $2 \times (-2) = -4$ |
| (9) $-4 \times 9 = -36$ | (39) $-3 \times (-2) = 6$ | (69) $-4 \times 7 = -28$ | (99) $-3 \times 7 = -21$ |
| (10) $8 \times (-8) = -64$ | (40) $9 \times 5 = 45$ | (70) $0 \times 5 = 0$ | (100) $5 \times (-6) = -30$ |
| (11) $-1 \times (-4) = 4$ | (41) $9 \times 1 = 9$ | (71) $-8 \times 6 = -48$ | (101) $-6 \times 4 = -24$ |
| (12) $-3 \times (-2) = 6$ | (42) $8 \times (-6) = -48$ | (72) $-2 \times 1 = -2$ | (102) $8 \times 7 = 56$ |
| (13) $-7 \times (-2) = 14$ | (43) $-1 \times (-7) = 7$ | (73) $1 \times (-9) = -9$ | (103) $4 \times (-2) = -8$ |
| (14) $2 \times 7 = 14$ | (44) $9 \times (-7) = -63$ | (74) $-1 \times 6 = -6$ | (104) $-3 \times 7 = -21$ |
| (15) $-6 \times 9 = -54$ | (45) $-4 \times (-3) = 12$ | (75) $3 \times (-7) = -21$ | (105) $1 \times 8 = 8$ |
| (16) $1 \times 0 = 0$ | (46) $2 \times 0 = 0$ | (76) $-5 \times (-3) = 15$ | (106) $8 \times (-2) = -16$ |
| (17) $5 \times (-9) = -45$ | (47) $1 \times 9 = 9$ | (77) $9 \times 8 = 72$ | (107) $-6 \times (-5) = 30$ |
| (18) $9 \times (-9) = -81$ | (48) $6 \times 2 = 12$ | (78) $1 \times (-7) = -7$ | (108) $-4 \times (-6) = 24$ |
| (19) $7 \times 6 = 42$ | (49) $-1 \times 0 = 0$ | (79) $-1 \times 2 = -2$ | (109) $9 \times 6 = 54$ |
| (20) $-6 \times 3 = -18$ | (50) $-1 \times (-1) = 1$ | (80) $-2 \times 9 = -18$ | (110) $4 \times (-1) = -4$ |
| (21) $-3 \times 4 = -12$ | (51) $-2 \times 2 = -4$ | (81) $-1 \times 4 = -4$ | (111) $-3 \times (-2) = 6$ |
| (22) $-1 \times 6 = -6$ | (52) $4 \times (-2) = -8$ | (82) $2 \times (-2) = -4$ | (112) $9 \times (-9) = -81$ |
| (23) $-4 \times (-8) = 32$ | (53) $-7 \times 3 = -21$ | (83) $-9 \times 1 = -9$ | (113) $-8 \times (-5) = 40$ |
| (24) $-4 \times (-8) = 32$ | (54) $0 \times (-7) = 0$ | (84) $-4 \times (-3) = 12$ | (114) $-4 \times 5 = -20$ |
| (25) $-3 \times 0 = 0$ | (55) $-3 \times (-8) = 24$ | (85) $-1 \times (-2) = 2$ | (115) $-6 \times (-3) = 18$ |
| (26) $8 \times 5 = 40$ | (56) $-9 \times (-2) = 18$ | (86) $4 \times (-6) = -24$ | (116) $2 \times (-6) = -12$ |
| (27) $-9 \times 8 = -72$ | (57) $4 \times (-3) = -12$ | (87) $6 \times 1 = 6$ | (117) $-8 \times 5 = -40$ |
| (28) $-4 \times (-2) = 8$ | (58) $0 \times 5 = 0$ | (88) $4 \times (-9) = -36$ | (118) $3 \times (-1) = -3$ |
| (29) $6 \times (-6) = -36$ | (59) $-3 \times 1 = -3$ | (89) $-8 \times (-6) = 48$ | (119) $8 \times 6 = 48$ |
| (30) $-2 \times 4 = -8$ | (60) $0 \times (-4) = 0$ | (90) $-8 \times (-3) = 24$ | (120) $1 \times 0 = 0$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-2 \times 9 = -18$ | (31) $1 \times (-2) = -2$ | (61) $3 \times 7 = 21$ | (91) $1 \times (-5) = -5$ |
| (2) $8 \times 7 = 56$ | (32) $-5 \times 4 = -20$ | (62) $-8 \times (-8) = 64$ | (92) $-2 \times (-6) = 12$ |
| (3) $6 \times (-2) = -12$ | (33) $2 \times (-3) = -6$ | (63) $-4 \times (-5) = 20$ | (93) $-6 \times 8 = -48$ |
| (4) $6 \times (-8) = -48$ | (34) $-8 \times 9 = -72$ | (64) $9 \times (-2) = -18$ | (94) $2 \times 6 = 12$ |
| (5) $7 \times 4 = 28$ | (35) $-7 \times (-7) = 49$ | (65) $-4 \times (-4) = 16$ | (95) $2 \times (-9) = -18$ |
| (6) $-1 \times (-2) = 2$ | (36) $8 \times (-5) = -40$ | (66) $6 \times (-4) = -24$ | (96) $-8 \times 9 = -72$ |
| (7) $-4 \times (-3) = 12$ | (37) $-8 \times (-5) = 40$ | (67) $-3 \times (-2) = 6$ | (97) $9 \times (-8) = -72$ |
| (8) $3 \times (-2) = -6$ | (38) $8 \times 0 = 0$ | (68) $4 \times 8 = 32$ | (98) $7 \times (-8) = -56$ |
| (9) $-4 \times 6 = -24$ | (39) $4 \times 0 = 0$ | (69) $1 \times (-2) = -2$ | (99) $-6 \times (-6) = 36$ |
| (10) $5 \times (-3) = -15$ | (40) $-1 \times (-9) = 9$ | (70) $-4 \times (-5) = 20$ | (100) $8 \times (-4) = -32$ |
| (11) $4 \times (-3) = -12$ | (41) $8 \times 1 = 8$ | (71) $-3 \times 4 = -12$ | (101) $-1 \times 0 = 0$ |
| (12) $3 \times (-3) = -9$ | (42) $6 \times 9 = 54$ | (72) $8 \times 2 = 16$ | (102) $-2 \times (-1) = 2$ |
| (13) $-4 \times (-4) = 16$ | (43) $1 \times 7 = 7$ | (73) $-7 \times (-7) = 49$ | (103) $3 \times (-8) = -24$ |
| (14) $0 \times 4 = 0$ | (44) $-9 \times (-2) = 18$ | (74) $-6 \times (-4) = 24$ | (104) $-8 \times 2 = -16$ |
| (15) $8 \times 3 = 24$ | (45) $-1 \times 7 = -7$ | (75) $-3 \times 5 = -15$ | (105) $-1 \times 2 = -2$ |
| (16) $-2 \times 3 = -6$ | (46) $5 \times (-3) = -15$ | (76) $7 \times 2 = 14$ | (106) $-1 \times 9 = -9$ |
| (17) $6 \times 2 = 12$ | (47) $-9 \times (-2) = 18$ | (77) $6 \times (-2) = -12$ | (107) $5 \times 5 = 25$ |
| (18) $-2 \times 9 = -18$ | (48) $7 \times 1 = 7$ | (78) $-2 \times (-7) = 14$ | (108) $5 \times 4 = 20$ |
| (19) $0 \times (-1) = 0$ | (49) $-8 \times 8 = -64$ | (79) $-1 \times (-9) = 9$ | (109) $4 \times (-7) = -28$ |
| (20) $0 \times (-6) = 0$ | (50) $-1 \times 7 = -7$ | (80) $-2 \times (-7) = 14$ | (110) $-2 \times (-4) = 8$ |
| (21) $7 \times 0 = 0$ | (51) $-7 \times 8 = -56$ | (81) $-2 \times 9 = -18$ | (111) $4 \times (-7) = -28$ |
| (22) $-3 \times 8 = -24$ | (52) $-1 \times (-7) = 7$ | (82) $3 \times (-8) = -24$ | (112) $9 \times (-3) = -27$ |
| (23) $-5 \times 1 = -5$ | (53) $2 \times (-4) = -8$ | (83) $2 \times 3 = 6$ | (113) $-4 \times (-7) = 28$ |
| (24) $-4 \times (-7) = 28$ | (54) $6 \times 6 = 36$ | (84) $5 \times 2 = 10$ | (114) $-8 \times (-6) = 48$ |
| (25) $-9 \times 2 = -18$ | (55) $-6 \times 6 = -36$ | (85) $1 \times 1 = 1$ | (115) $-4 \times 6 = -24$ |
| (26) $5 \times (-9) = -45$ | (56) $-9 \times 8 = -72$ | (86) $5 \times 4 = 20$ | (116) $7 \times (-1) = -7$ |
| (27) $8 \times (-4) = -32$ | (57) $7 \times 9 = 63$ | (87) $3 \times (-7) = -21$ | (117) $-8 \times (-3) = 24$ |
| (28) $6 \times 3 = 18$ | (58) $3 \times (-6) = -18$ | (88) $8 \times 7 = 56$ | (118) $-1 \times 8 = -8$ |
| (29) $0 \times 2 = 0$ | (59) $9 \times (-5) = -45$ | (89) $-6 \times (-9) = 54$ | (119) $-3 \times (-4) = 12$ |
| (30) $-3 \times (-8) = 24$ | (60) $5 \times (-8) = -40$ | (90) $-5 \times 5 = -25$ | (120) $4 \times (-4) = -16$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $0 \times (-7) = 0$ | (31) $8 \times 8 = 64$ | (61) $-5 \times 6 = -30$ | (91) $4 \times 1 = 4$ |
| (2) $2 \times (-2) = -4$ | (32) $-4 \times 0 = 0$ | (62) $9 \times 5 = 45$ | (92) $-2 \times (-5) = 10$ |
| (3) $-7 \times (-2) = 14$ | (33) $9 \times 1 = 9$ | (63) $3 \times 5 = 15$ | (93) $5 \times (-8) = -40$ |
| (4) $4 \times 2 = 8$ | (34) $-4 \times (-3) = 12$ | (64) $7 \times 8 = 56$ | (94) $8 \times 3 = 24$ |
| (5) $0 \times (-1) = 0$ | (35) $5 \times 3 = 15$ | (65) $8 \times (-7) = -56$ | (95) $1 \times (-7) = -7$ |
| (6) $-8 \times 1 = -8$ | (36) $-4 \times (-9) = 36$ | (66) $-1 \times 2 = -2$ | (96) $0 \times (-7) = 0$ |
| (7) $4 \times (-1) = -4$ | (37) $4 \times (-9) = -36$ | (67) $-8 \times (-4) = 32$ | (97) $6 \times (-4) = -24$ |
| (8) $-3 \times 9 = -27$ | (38) $9 \times 3 = 27$ | (68) $9 \times 7 = 63$ | (98) $2 \times 3 = 6$ |
| (9) $2 \times (-4) = -8$ | (39) $-9 \times 3 = -27$ | (69) $-5 \times (-1) = 5$ | (99) $0 \times (-6) = 0$ |
| (10) $-3 \times (-6) = 18$ | (40) $-7 \times (-7) = 49$ | (70) $4 \times (-2) = -8$ | (100) $8 \times 0 = 0$ |
| (11) $8 \times 5 = 40$ | (41) $-3 \times 0 = 0$ | (71) $-2 \times (-7) = 14$ | (101) $9 \times (-4) = -36$ |
| (12) $-5 \times (-9) = 45$ | (42) $-2 \times (-4) = 8$ | (72) $-4 \times 7 = -28$ | (102) $0 \times 4 = 0$ |
| (13) $-6 \times (-2) = 12$ | (43) $-6 \times 5 = -30$ | (73) $0 \times 5 = 0$ | (103) $-4 \times 5 = -20$ |
| (14) $-8 \times (-8) = 64$ | (44) $5 \times (-7) = -35$ | (74) $5 \times (-2) = -10$ | (104) $3 \times 5 = 15$ |
| (15) $0 \times 6 = 0$ | (45) $-5 \times 1 = -5$ | (75) $-4 \times 0 = 0$ | (105) $9 \times 3 = 27$ |
| (16) $8 \times 8 = 64$ | (46) $6 \times 2 = 12$ | (76) $-4 \times (-1) = 4$ | (106) $6 \times 0 = 0$ |
| (17) $-5 \times 2 = -10$ | (47) $1 \times 8 = 8$ | (77) $-1 \times (-6) = 6$ | (107) $2 \times 2 = 4$ |
| (18) $3 \times (-7) = -21$ | (48) $-1 \times (-1) = 1$ | (78) $-5 \times 4 = -20$ | (108) $-1 \times 7 = -7$ |
| (19) $5 \times (-6) = -30$ | (49) $-9 \times (-7) = 63$ | (79) $-5 \times 4 = -20$ | (109) $-4 \times 5 = -20$ |
| (20) $5 \times 9 = 45$ | (50) $-4 \times 6 = -24$ | (80) $-4 \times (-4) = 16$ | (110) $7 \times (-1) = -7$ |
| (21) $0 \times 0 = 0$ | (51) $7 \times 5 = 35$ | (81) $-6 \times (-4) = 24$ | (111) $3 \times (-3) = -9$ |
| (22) $-3 \times 4 = -12$ | (52) $-2 \times 7 = -14$ | (82) $8 \times (-2) = -16$ | (112) $8 \times 7 = 56$ |
| (23) $5 \times 5 = 25$ | (53) $7 \times 9 = 63$ | (83) $-2 \times (-2) = 4$ | (113) $-2 \times 9 = -18$ |
| (24) $-5 \times 4 = -20$ | (54) $0 \times (-7) = 0$ | (84) $-9 \times 8 = -72$ | (114) $9 \times (-1) = -9$ |
| (25) $-1 \times (-1) = 1$ | (55) $-7 \times (-8) = 56$ | (85) $-5 \times (-5) = 25$ | (115) $3 \times (-4) = -12$ |
| (26) $-6 \times 0 = 0$ | (56) $9 \times 2 = 18$ | (86) $-7 \times (-7) = 49$ | (116) $2 \times 3 = 6$ |
| (27) $-9 \times 9 = -81$ | (57) $3 \times (-8) = -24$ | (87) $4 \times (-6) = -24$ | (117) $1 \times (-7) = -7$ |
| (28) $-2 \times 6 = -12$ | (58) $2 \times (-2) = -4$ | (88) $5 \times 8 = 40$ | (118) $-6 \times (-9) = 54$ |
| (29) $-4 \times (-3) = 12$ | (59) $-2 \times (-5) = 10$ | (89) $5 \times (-5) = -25$ | (119) $5 \times 2 = 10$ |
| (30) $5 \times 3 = 15$ | (60) $-6 \times (-8) = 48$ | (90) $8 \times 4 = 32$ | (120) $-9 \times 6 = -54$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-4 \times (-7) = 28$ | (31) $-9 \times (-5) = 45$ | (61) $-8 \times 8 = -64$ | (91) $-6 \times (-6) = 36$ |
| (2) $6 \times 9 = 54$ | (32) $-3 \times 8 = -24$ | (62) $4 \times (-7) = -28$ | (92) $-7 \times 3 = -21$ |
| (3) $-7 \times 3 = -21$ | (33) $4 \times (-3) = -12$ | (63) $8 \times 0 = 0$ | (93) $0 \times 7 = 0$ |
| (4) $7 \times 4 = 28$ | (34) $-6 \times (-8) = 48$ | (64) $3 \times (-3) = -9$ | (94) $-6 \times 1 = -6$ |
| (5) $2 \times 8 = 16$ | (35) $6 \times 5 = 30$ | (65) $4 \times (-3) = -12$ | (95) $4 \times 6 = 24$ |
| (6) $4 \times (-7) = -28$ | (36) $8 \times 8 = 64$ | (66) $9 \times (-9) = -81$ | (96) $0 \times 4 = 0$ |
| (7) $-5 \times 3 = -15$ | (37) $2 \times 1 = 2$ | (67) $2 \times 9 = 18$ | (97) $-9 \times (-5) = 45$ |
| (8) $-8 \times 0 = 0$ | (38) $-6 \times (-5) = 30$ | (68) $9 \times 3 = 27$ | (98) $4 \times (-8) = -32$ |
| (9) $-3 \times 8 = -24$ | (39) $6 \times (-8) = -48$ | (69) $-2 \times 4 = -8$ | (99) $9 \times 2 = 18$ |
| (10) $-4 \times (-2) = 8$ | (40) $-6 \times (-3) = 18$ | (70) $-7 \times (-1) = 7$ | (100) $9 \times (-7) = -63$ |
| (11) $1 \times 7 = 7$ | (41) $-7 \times (-9) = 63$ | (71) $-1 \times 6 = -6$ | (101) $7 \times 8 = 56$ |
| (12) $2 \times 8 = 16$ | (42) $0 \times (-4) = 0$ | (72) $1 \times 1 = 1$ | (102) $-3 \times 6 = -18$ |
| (13) $2 \times (-8) = -16$ | (43) $1 \times (-5) = -5$ | (73) $-8 \times (-9) = 72$ | (103) $-4 \times (-7) = 28$ |
| (14) $-8 \times 4 = -32$ | (44) $1 \times 4 = 4$ | (74) $5 \times 7 = 35$ | (104) $-3 \times 5 = -15$ |
| (15) $0 \times 7 = 0$ | (45) $-6 \times 4 = -24$ | (75) $-3 \times 8 = -24$ | (105) $2 \times 7 = 14$ |
| (16) $2 \times 9 = 18$ | (46) $4 \times 6 = 24$ | (76) $7 \times (-4) = -28$ | (106) $-6 \times (-6) = 36$ |
| (17) $-3 \times (-2) = 6$ | (47) $4 \times (-5) = -20$ | (77) $-8 \times (-4) = 32$ | (107) $3 \times (-9) = -27$ |
| (18) $3 \times 2 = 6$ | (48) $7 \times 8 = 56$ | (78) $-1 \times 4 = -4$ | (108) $-6 \times (-9) = 54$ |
| (19) $-4 \times 7 = -28$ | (49) $-5 \times (-5) = 25$ | (79) $6 \times (-4) = -24$ | (109) $9 \times 3 = 27$ |
| (20) $-3 \times (-1) = 3$ | (50) $2 \times 6 = 12$ | (80) $6 \times (-6) = -36$ | (110) $7 \times 7 = 49$ |
| (21) $1 \times 3 = 3$ | (51) $-5 \times (-8) = 40$ | (81) $0 \times (-3) = 0$ | (111) $7 \times 6 = 42$ |
| (22) $-1 \times (-5) = 5$ | (52) $9 \times 7 = 63$ | (82) $7 \times (-7) = -49$ | (112) $2 \times 7 = 14$ |
| (23) $-2 \times 6 = -12$ | (53) $-4 \times (-6) = 24$ | (83) $-5 \times 2 = -10$ | (113) $-2 \times (-3) = 6$ |
| (24) $-4 \times 2 = -8$ | (54) $-9 \times 1 = -9$ | (84) $7 \times (-6) = -42$ | (114) $1 \times 4 = 4$ |
| (25) $1 \times 0 = 0$ | (55) $0 \times 8 = 0$ | (85) $9 \times 6 = 54$ | (115) $2 \times 8 = 16$ |
| (26) $5 \times 5 = 25$ | (56) $-4 \times (-3) = 12$ | (86) $7 \times 6 = 42$ | (116) $5 \times 8 = 40$ |
| (27) $2 \times (-4) = -8$ | (57) $2 \times 6 = 12$ | (87) $6 \times 3 = 18$ | (117) $6 \times 1 = 6$ |
| (28) $-6 \times 8 = -48$ | (58) $0 \times 4 = 0$ | (88) $5 \times 8 = 40$ | (118) $8 \times 8 = 64$ |
| (29) $-7 \times (-9) = 63$ | (59) $-5 \times 6 = -30$ | (89) $-6 \times (-2) = 12$ | (119) $8 \times 6 = 48$ |
| (30) $-5 \times 2 = -10$ | (60) $-4 \times (-1) = 4$ | (90) $-6 \times (-1) = 6$ | (120) $3 \times 5 = 15$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $4 \times (-7) = -28$ | (31) $0 \times 7 = 0$ | (61) $-6 \times 7 = -42$ | (91) $3 \times (-3) = -9$ |
| (2) $-8 \times (-7) = 56$ | (32) $-1 \times (-6) = 6$ | (62) $-7 \times 5 = -35$ | (92) $9 \times 1 = 9$ |
| (3) $-4 \times (-1) = 4$ | (33) $6 \times 5 = 30$ | (63) $7 \times (-9) = -63$ | (93) $-3 \times (-8) = 24$ |
| (4) $-8 \times 5 = -40$ | (34) $6 \times 5 = 30$ | (64) $7 \times (-3) = -21$ | (94) $9 \times 6 = 54$ |
| (5) $-2 \times (-3) = 6$ | (35) $6 \times (-8) = -48$ | (65) $4 \times (-8) = -32$ | (95) $-9 \times 1 = -9$ |
| (6) $-9 \times (-4) = 36$ | (36) $5 \times 7 = 35$ | (66) $2 \times (-5) = -10$ | (96) $6 \times 7 = 42$ |
| (7) $8 \times (-7) = -56$ | (37) $2 \times (-5) = -10$ | (67) $2 \times (-7) = -14$ | (97) $-6 \times 9 = -54$ |
| (8) $-1 \times 2 = -2$ | (38) $1 \times (-9) = -9$ | (68) $7 \times 6 = 42$ | (98) $5 \times (-2) = -10$ |
| (9) $9 \times (-9) = -81$ | (39) $6 \times 9 = 54$ | (69) $8 \times 1 = 8$ | (99) $1 \times 6 = 6$ |
| (10) $7 \times (-7) = -49$ | (40) $4 \times (-9) = -36$ | (70) $5 \times 0 = 0$ | (100) $4 \times 3 = 12$ |
| (11) $6 \times 7 = 42$ | (41) $9 \times 3 = 27$ | (71) $2 \times 2 = 4$ | (101) $3 \times 9 = 27$ |
| (12) $2 \times (-9) = -18$ | (42) $5 \times (-5) = -25$ | (72) $2 \times (-8) = -16$ | (102) $-5 \times (-6) = 30$ |
| (13) $-1 \times (-8) = 8$ | (43) $-9 \times 7 = -63$ | (73) $-1 \times 5 = -5$ | (103) $-3 \times (-3) = 9$ |
| (14) $-8 \times 8 = -64$ | (44) $-9 \times 6 = -54$ | (74) $5 \times (-1) = -5$ | (104) $9 \times 0 = 0$ |
| (15) $5 \times (-8) = -40$ | (45) $-2 \times (-1) = 2$ | (75) $8 \times 6 = 48$ | (105) $9 \times (-3) = -27$ |
| (16) $-2 \times 6 = -12$ | (46) $8 \times (-1) = -8$ | (76) $5 \times 0 = 0$ | (106) $-2 \times (-8) = 16$ |
| (17) $1 \times (-8) = -8$ | (47) $-9 \times (-4) = 36$ | (77) $-9 \times 0 = 0$ | (107) $-6 \times 0 = 0$ |
| (18) $3 \times 7 = 21$ | (48) $8 \times (-3) = -24$ | (78) $4 \times 2 = 8$ | (108) $6 \times (-4) = -24$ |
| (19) $-3 \times (-2) = 6$ | (49) $6 \times (-9) = -54$ | (79) $4 \times (-2) = -8$ | (109) $-9 \times (-1) = 9$ |
| (20) $8 \times (-1) = -8$ | (50) $6 \times (-6) = -36$ | (80) $5 \times 4 = 20$ | (110) $3 \times (-7) = -21$ |
| (21) $-7 \times 9 = -63$ | (51) $-9 \times (-7) = 63$ | (81) $8 \times 5 = 40$ | (111) $-8 \times (-4) = 32$ |
| (22) $-1 \times (-3) = 3$ | (52) $6 \times 2 = 12$ | (82) $2 \times 1 = 2$ | (112) $-3 \times (-7) = 21$ |
| (23) $9 \times (-1) = -9$ | (53) $-6 \times 2 = -12$ | (83) $-7 \times 0 = 0$ | (113) $-6 \times 2 = -12$ |
| (24) $8 \times 5 = 40$ | (54) $-7 \times 0 = 0$ | (84) $0 \times 3 = 0$ | (114) $3 \times (-6) = -18$ |
| (25) $-6 \times (-9) = 54$ | (55) $8 \times (-7) = -56$ | (85) $6 \times (-1) = -6$ | (115) $6 \times 3 = 18$ |
| (26) $6 \times (-5) = -30$ | (56) $8 \times 0 = 0$ | (86) $-8 \times 0 = 0$ | (116) $0 \times 8 = 0$ |
| (27) $0 \times (-4) = 0$ | (57) $3 \times (-4) = -12$ | (87) $-7 \times (-8) = 56$ | (117) $8 \times (-9) = -72$ |
| (28) $-1 \times (-8) = 8$ | (58) $-4 \times 7 = -28$ | (88) $-8 \times 8 = -64$ | (118) $5 \times (-1) = -5$ |
| (29) $1 \times 4 = 4$ | (59) $-3 \times 8 = -24$ | (89) $-2 \times (-7) = 14$ | (119) $-6 \times (-2) = 12$ |
| (30) $-5 \times (-8) = 40$ | (60) $-2 \times (-9) = 18$ | (90) $5 \times 9 = 45$ | (120) $7 \times 8 = 56$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $7 \times (-2) = -14$ | (31) $-9 \times 7 = -63$ | (61) $-3 \times 8 = -24$ | (91) $-3 \times 0 = 0$ |
| (2) $-7 \times 0 = 0$ | (32) $-7 \times (-3) = 21$ | (62) $-5 \times (-5) = 25$ | (92) $3 \times 0 = 0$ |
| (3) $3 \times 9 = 27$ | (33) $8 \times (-8) = -64$ | (63) $-5 \times (-7) = 35$ | (93) $3 \times 1 = 3$ |
| (4) $-3 \times 5 = -15$ | (34) $2 \times (-4) = -8$ | (64) $-7 \times (-9) = 63$ | (94) $-9 \times (-6) = 54$ |
| (5) $-1 \times (-3) = 3$ | (35) $-8 \times 5 = -40$ | (65) $2 \times 5 = 10$ | (95) $3 \times 5 = 15$ |
| (6) $6 \times 0 = 0$ | (36) $1 \times (-3) = -3$ | (66) $-5 \times 9 = -45$ | (96) $-1 \times 3 = -3$ |
| (7) $1 \times 1 = 1$ | (37) $0 \times (-4) = 0$ | (67) $6 \times 1 = 6$ | (97) $-7 \times (-7) = 49$ |
| (8) $-1 \times 4 = -4$ | (38) $1 \times (-3) = -3$ | (68) $1 \times 9 = 9$ | (98) $-1 \times 8 = -8$ |
| (9) $-1 \times (-8) = 8$ | (39) $-7 \times 2 = -14$ | (69) $-9 \times (-8) = 72$ | (99) $-3 \times (-9) = 27$ |
| (10) $0 \times (-4) = 0$ | (40) $3 \times (-7) = -21$ | (70) $-3 \times (-8) = 24$ | (100) $-8 \times (-6) = 48$ |
| (11) $-2 \times 7 = -14$ | (41) $5 \times (-1) = -5$ | (71) $9 \times 9 = 81$ | (101) $-6 \times 6 = -36$ |
| (12) $3 \times (-9) = -27$ | (42) $7 \times 1 = 7$ | (72) $8 \times 7 = 56$ | (102) $-4 \times (-6) = 24$ |
| (13) $-2 \times (-6) = 12$ | (43) $8 \times 2 = 16$ | (73) $3 \times 4 = 12$ | (103) $9 \times 8 = 72$ |
| (14) $-4 \times 1 = -4$ | (44) $9 \times (-7) = -63$ | (74) $-8 \times (-1) = 8$ | (104) $8 \times 4 = 32$ |
| (15) $3 \times (-3) = -9$ | (45) $5 \times 4 = 20$ | (75) $7 \times (-9) = -63$ | (105) $7 \times 6 = 42$ |
| (16) $4 \times (-2) = -8$ | (46) $-1 \times (-8) = 8$ | (76) $-8 \times 2 = -16$ | (106) $8 \times 2 = 16$ |
| (17) $3 \times 9 = 27$ | (47) $-9 \times (-3) = 27$ | (77) $8 \times (-7) = -56$ | (107) $3 \times (-7) = -21$ |
| (18) $-9 \times 6 = -54$ | (48) $-1 \times (-1) = 1$ | (78) $-8 \times (-4) = 32$ | (108) $2 \times 9 = 18$ |
| (19) $6 \times 5 = 30$ | (49) $4 \times 9 = 36$ | (79) $-7 \times (-9) = 63$ | (109) $6 \times (-9) = -54$ |
| (20) $-9 \times 5 = -45$ | (50) $1 \times (-1) = -1$ | (80) $-1 \times (-7) = 7$ | (110) $-7 \times 7 = -49$ |
| (21) $6 \times 5 = 30$ | (51) $1 \times 5 = 5$ | (81) $-3 \times 9 = -27$ | (111) $1 \times (-3) = -3$ |
| (22) $9 \times (-3) = -27$ | (52) $6 \times 0 = 0$ | (82) $4 \times 3 = 12$ | (112) $-7 \times (-3) = 21$ |
| (23) $6 \times (-7) = -42$ | (53) $-1 \times 8 = -8$ | (83) $-2 \times (-9) = 18$ | (113) $8 \times 4 = 32$ |
| (24) $-8 \times (-9) = 72$ | (54) $-2 \times 3 = -6$ | (84) $-9 \times (-7) = 63$ | (114) $-5 \times (-7) = 35$ |
| (25) $-1 \times 2 = -2$ | (55) $9 \times 9 = 81$ | (85) $5 \times 4 = 20$ | (115) $-1 \times 4 = -4$ |
| (26) $1 \times 8 = 8$ | (56) $0 \times 5 = 0$ | (86) $-2 \times 8 = -16$ | (116) $9 \times 7 = 63$ |
| (27) $-2 \times (-5) = 10$ | (57) $-8 \times (-1) = 8$ | (87) $-2 \times (-4) = 8$ | (117) $-6 \times (-3) = 18$ |
| (28) $0 \times (-9) = 0$ | (58) $-2 \times (-3) = 6$ | (88) $6 \times (-4) = -24$ | (118) $-5 \times 3 = -15$ |
| (29) $8 \times (-2) = -16$ | (59) $-5 \times 0 = 0$ | (89) $-9 \times 2 = -18$ | (119) $9 \times (-1) = -9$ |
| (30) $-7 \times (-6) = 42$ | (60) $4 \times (-3) = -12$ | (90) $5 \times (-1) = -5$ | (120) $-7 \times 5 = -35$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-2 \times 6 = -12$ | (31) $1 \times (-8) = -8$ | (61) $-5 \times 0 = 0$ | (91) $5 \times (-4) = -20$ |
| (2) $9 \times 9 = 81$ | (32) $-3 \times (-2) = 6$ | (62) $8 \times 5 = 40$ | (92) $-4 \times (-7) = 28$ |
| (3) $0 \times 2 = 0$ | (33) $-7 \times (-5) = 35$ | (63) $3 \times 0 = 0$ | (93) $4 \times 3 = 12$ |
| (4) $-7 \times 1 = -7$ | (34) $-7 \times (-1) = 7$ | (64) $-1 \times 0 = 0$ | (94) $-9 \times 8 = -72$ |
| (5) $-1 \times 9 = -9$ | (35) $8 \times (-3) = -24$ | (65) $6 \times (-7) = -42$ | (95) $-7 \times 4 = -28$ |
| (6) $-4 \times 3 = -12$ | (36) $-4 \times 9 = -36$ | (66) $-8 \times 0 = 0$ | (96) $-2 \times (-7) = 14$ |
| (7) $6 \times 9 = 54$ | (37) $4 \times 1 = 4$ | (67) $3 \times (-3) = -9$ | (97) $8 \times (-9) = -72$ |
| (8) $2 \times 9 = 18$ | (38) $-7 \times (-1) = 7$ | (68) $-5 \times (-5) = 25$ | (98) $-7 \times 7 = -49$ |
| (9) $9 \times (-8) = -72$ | (39) $0 \times (-5) = 0$ | (69) $-3 \times 9 = -27$ | (99) $0 \times 0 = 0$ |
| (10) $-8 \times (-4) = 32$ | (40) $-1 \times (-7) = 7$ | (70) $-3 \times (-5) = 15$ | (100) $5 \times (-5) = -25$ |
| (11) $-6 \times 4 = -24$ | (41) $6 \times 6 = 36$ | (71) $2 \times 2 = 4$ | (101) $-7 \times 2 = -14$ |
| (12) $-8 \times (-8) = 64$ | (42) $0 \times 4 = 0$ | (72) $-7 \times 2 = -14$ | (102) $-8 \times 8 = -64$ |
| (13) $-9 \times 7 = -63$ | (43) $1 \times (-1) = -1$ | (73) $3 \times (-4) = -12$ | (103) $-9 \times 2 = -18$ |
| (14) $9 \times 6 = 54$ | (44) $8 \times (-7) = -56$ | (74) $-7 \times 5 = -35$ | (104) $-6 \times 9 = -54$ |
| (15) $-3 \times (-3) = 9$ | (45) $-1 \times 0 = 0$ | (75) $-7 \times 3 = -21$ | (105) $-2 \times 9 = -18$ |
| (16) $7 \times (-2) = -14$ | (46) $7 \times (-4) = -28$ | (76) $4 \times 8 = 32$ | (106) $-6 \times (-4) = 24$ |
| (17) $2 \times (-3) = -6$ | (47) $1 \times 3 = 3$ | (77) $7 \times (-9) = -63$ | (107) $4 \times 2 = 8$ |
| (18) $3 \times 5 = 15$ | (48) $9 \times (-3) = -27$ | (78) $1 \times (-3) = -3$ | (108) $3 \times 0 = 0$ |
| (19) $-4 \times 3 = -12$ | (49) $-2 \times (-2) = 4$ | (79) $-2 \times 5 = -10$ | (109) $-4 \times 7 = -28$ |
| (20) $-6 \times 9 = -54$ | (50) $6 \times 3 = 18$ | (80) $-7 \times (-5) = 35$ | (110) $-4 \times (-8) = 32$ |
| (21) $7 \times 4 = 28$ | (51) $-9 \times 7 = -63$ | (81) $-6 \times (-5) = 30$ | (111) $-6 \times 3 = -18$ |
| (22) $3 \times 0 = 0$ | (52) $-5 \times 4 = -20$ | (82) $4 \times (-3) = -12$ | (112) $3 \times (-6) = -18$ |
| (23) $5 \times (-6) = -30$ | (53) $-5 \times 9 = -45$ | (83) $6 \times 4 = 24$ | (113) $5 \times (-8) = -40$ |
| (24) $-8 \times (-5) = 40$ | (54) $6 \times 1 = 6$ | (84) $9 \times (-3) = -27$ | (114) $2 \times 9 = 18$ |
| (25) $-4 \times (-4) = 16$ | (55) $-7 \times (-4) = 28$ | (85) $2 \times (-8) = -16$ | (115) $-9 \times (-3) = 27$ |
| (26) $-4 \times (-9) = 36$ | (56) $-1 \times (-4) = 4$ | (86) $9 \times (-7) = -63$ | (116) $7 \times 4 = 28$ |
| (27) $9 \times 5 = 45$ | (57) $-2 \times (-5) = 10$ | (87) $0 \times (-4) = 0$ | (117) $-7 \times (-9) = 63$ |
| (28) $-7 \times 8 = -56$ | (58) $2 \times (-3) = -6$ | (88) $-6 \times 3 = -18$ | (118) $-9 \times (-2) = 18$ |
| (29) $9 \times (-6) = -54$ | (59) $6 \times (-4) = -24$ | (89) $-6 \times (-8) = 48$ | (119) $3 \times 0 = 0$ |
| (30) $0 \times 8 = 0$ | (60) $-2 \times 2 = -4$ | (90) $-4 \times 6 = -24$ | (120) $-8 \times 2 = -16$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $4 \times (-4) = -16$ | (31) $2 \times (-8) = -16$ | (61) $-9 \times 3 = -27$ | (91) $0 \times 0 = 0$ |
| (2) $-5 \times (-5) = 25$ | (32) $-1 \times 3 = -3$ | (62) $-4 \times 3 = -12$ | (92) $3 \times 0 = 0$ |
| (3) $4 \times 4 = 16$ | (33) $1 \times (-4) = -4$ | (63) $-2 \times (-3) = 6$ | (93) $-4 \times (-7) = 28$ |
| (4) $9 \times (-4) = -36$ | (34) $5 \times 1 = 5$ | (64) $0 \times (-4) = 0$ | (94) $4 \times 2 = 8$ |
| (5) $-4 \times (-5) = 20$ | (35) $-3 \times (-4) = 12$ | (65) $-2 \times 4 = -8$ | (95) $1 \times 5 = 5$ |
| (6) $5 \times 0 = 0$ | (36) $1 \times 7 = 7$ | (66) $-4 \times 0 = 0$ | (96) $9 \times 3 = 27$ |
| (7) $-7 \times 8 = -56$ | (37) $-4 \times (-9) = 36$ | (67) $4 \times 2 = 8$ | (97) $6 \times 8 = 48$ |
| (8) $9 \times (-2) = -18$ | (38) $4 \times (-2) = -8$ | (68) $9 \times 0 = 0$ | (98) $-1 \times 2 = -2$ |
| (9) $-2 \times 9 = -18$ | (39) $-2 \times (-6) = 12$ | (69) $-1 \times (-4) = 4$ | (99) $5 \times (-7) = -35$ |
| (10) $-7 \times 1 = -7$ | (40) $2 \times 3 = 6$ | (70) $9 \times (-2) = -18$ | (100) $3 \times 9 = 27$ |
| (11) $-4 \times 4 = -16$ | (41) $-9 \times (-1) = 9$ | (71) $2 \times 2 = 4$ | (101) $6 \times (-1) = -6$ |
| (12) $-8 \times (-1) = 8$ | (42) $-8 \times (-2) = 16$ | (72) $-5 \times (-1) = 5$ | (102) $7 \times 8 = 56$ |
| (13) $5 \times 5 = 25$ | (43) $-8 \times (-3) = 24$ | (73) $4 \times 4 = 16$ | (103) $-4 \times (-5) = 20$ |
| (14) $-4 \times 7 = -28$ | (44) $-3 \times (-8) = 24$ | (74) $-1 \times (-6) = 6$ | (104) $2 \times (-9) = -18$ |
| (15) $-6 \times (-6) = 36$ | (45) $-6 \times (-1) = 6$ | (75) $9 \times 1 = 9$ | (105) $-2 \times 1 = -2$ |
| (16) $9 \times (-6) = -54$ | (46) $2 \times 5 = 10$ | (76) $6 \times (-2) = -12$ | (106) $-3 \times (-2) = 6$ |
| (17) $2 \times 2 = 4$ | (47) $-6 \times (-6) = 36$ | (77) $-6 \times (-6) = 36$ | (107) $6 \times (-2) = -12$ |
| (18) $-1 \times (-5) = 5$ | (48) $-5 \times (-8) = 40$ | (78) $-9 \times (-2) = 18$ | (108) $6 \times (-4) = -24$ |
| (19) $-7 \times 8 = -56$ | (49) $-1 \times (-1) = 1$ | (79) $-7 \times 3 = -21$ | (109) $-4 \times (-9) = 36$ |
| (20) $7 \times 8 = 56$ | (50) $2 \times (-1) = -2$ | (80) $7 \times 8 = 56$ | (110) $7 \times (-9) = -63$ |
| (21) $-8 \times (-8) = 64$ | (51) $-4 \times 0 = 0$ | (81) $-8 \times 8 = -64$ | (111) $2 \times 5 = 10$ |
| (22) $9 \times 7 = 63$ | (52) $-1 \times 9 = -9$ | (82) $2 \times (-8) = -16$ | (112) $-6 \times 0 = 0$ |
| (23) $5 \times 3 = 15$ | (53) $-4 \times (-7) = 28$ | (83) $-8 \times 1 = -8$ | (113) $7 \times 9 = 63$ |
| (24) $6 \times 8 = 48$ | (54) $-8 \times 0 = 0$ | (84) $6 \times 9 = 54$ | (114) $-2 \times 7 = -14$ |
| (25) $8 \times 9 = 72$ | (55) $6 \times 1 = 6$ | (85) $3 \times 3 = 9$ | (115) $0 \times 7 = 0$ |
| (26) $8 \times (-5) = -40$ | (56) $6 \times 0 = 0$ | (86) $-5 \times 9 = -45$ | (116) $-8 \times (-4) = 32$ |
| (27) $3 \times (-1) = -3$ | (57) $7 \times (-7) = -49$ | (87) $1 \times 5 = 5$ | (117) $6 \times 9 = 54$ |
| (28) $-7 \times (-2) = 14$ | (58) $5 \times 1 = 5$ | (88) $7 \times 1 = 7$ | (118) $0 \times 8 = 0$ |
| (29) $3 \times (-8) = -24$ | (59) $-9 \times (-1) = 9$ | (89) $2 \times (-2) = -4$ | (119) $1 \times (-4) = -4$ |
| (30) $-3 \times (-7) = 21$ | (60) $1 \times (-6) = -6$ | (90) $1 \times (-1) = -1$ | (120) $7 \times (-9) = -63$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $8 \times (-9) = -72$ | (31) $3 \times 4 = 12$ | (61) $3 \times (-5) = -15$ | (91) $0 \times 6 = 0$ |
| (2) $-4 \times 6 = -24$ | (32) $2 \times 6 = 12$ | (62) $1 \times (-6) = -6$ | (92) $0 \times 8 = 0$ |
| (3) $-6 \times 6 = -36$ | (33) $-3 \times 0 = 0$ | (63) $7 \times (-9) = -63$ | (93) $3 \times 4 = 12$ |
| (4) $7 \times (-7) = -49$ | (34) $-1 \times (-9) = 9$ | (64) $-1 \times 4 = -4$ | (94) $2 \times (-8) = -16$ |
| (5) $7 \times (-3) = -21$ | (35) $7 \times (-2) = -14$ | (65) $-1 \times (-9) = 9$ | (95) $-6 \times (-3) = 18$ |
| (6) $-4 \times 3 = -12$ | (36) $-1 \times 1 = -1$ | (66) $7 \times 5 = 35$ | (96) $1 \times 7 = 7$ |
| (7) $-9 \times (-2) = 18$ | (37) $4 \times 9 = 36$ | (67) $3 \times 7 = 21$ | (97) $0 \times 8 = 0$ |
| (8) $2 \times 4 = 8$ | (38) $8 \times 5 = 40$ | (68) $6 \times 5 = 30$ | (98) $-7 \times (-5) = 35$ |
| (9) $1 \times 8 = 8$ | (39) $8 \times 1 = 8$ | (69) $8 \times (-4) = -32$ | (99) $4 \times 4 = 16$ |
| (10) $0 \times 9 = 0$ | (40) $-3 \times 1 = -3$ | (70) $-8 \times (-9) = 72$ | (100) $-8 \times (-4) = 32$ |
| (11) $-6 \times 2 = -12$ | (41) $8 \times (-2) = -16$ | (71) $1 \times 3 = 3$ | (101) $9 \times (-2) = -18$ |
| (12) $7 \times (-5) = -35$ | (42) $4 \times 9 = 36$ | (72) $-2 \times 9 = -18$ | (102) $3 \times 9 = 27$ |
| (13) $-6 \times 9 = -54$ | (43) $-3 \times 9 = -27$ | (73) $-7 \times 9 = -63$ | (103) $-7 \times 1 = -7$ |
| (14) $2 \times 8 = 16$ | (44) $7 \times (-2) = -14$ | (74) $-1 \times 4 = -4$ | (104) $9 \times (-5) = -45$ |
| (15) $2 \times 7 = 14$ | (45) $-3 \times (-6) = 18$ | (75) $6 \times (-5) = -30$ | (105) $-1 \times 5 = -5$ |
| (16) $-8 \times 9 = -72$ | (46) $-4 \times 9 = -36$ | (76) $6 \times (-3) = -18$ | (106) $8 \times 5 = 40$ |
| (17) $2 \times (-6) = -12$ | (47) $7 \times 2 = 14$ | (77) $-5 \times (-9) = 45$ | (107) $6 \times (-1) = -6$ |
| (18) $-1 \times 2 = -2$ | (48) $-7 \times 8 = -56$ | (78) $9 \times 3 = 27$ | (108) $0 \times 9 = 0$ |
| (19) $0 \times (-7) = 0$ | (49) $-5 \times (-8) = 40$ | (79) $3 \times 8 = 24$ | (109) $1 \times (-6) = -6$ |
| (20) $-2 \times 1 = -2$ | (50) $-1 \times (-4) = 4$ | (80) $4 \times 9 = 36$ | (110) $-1 \times (-7) = 7$ |
| (21) $5 \times (-2) = -10$ | (51) $-4 \times (-1) = 4$ | (81) $-4 \times 4 = -16$ | (111) $7 \times 3 = 21$ |
| (22) $-1 \times 5 = -5$ | (52) $9 \times 5 = 45$ | (82) $5 \times (-4) = -20$ | (112) $-6 \times 7 = -42$ |
| (23) $0 \times 7 = 0$ | (53) $9 \times (-5) = -45$ | (83) $-1 \times (-9) = 9$ | (113) $-8 \times 7 = -56$ |
| (24) $-6 \times (-5) = 30$ | (54) $3 \times (-7) = -21$ | (84) $-4 \times 4 = -16$ | (114) $-6 \times (-8) = 48$ |
| (25) $1 \times (-8) = -8$ | (55) $-7 \times 5 = -35$ | (85) $-8 \times (-6) = 48$ | (115) $-7 \times (-8) = 56$ |
| (26) $7 \times (-6) = -42$ | (56) $-4 \times 6 = -24$ | (86) $-2 \times (-7) = 14$ | (116) $0 \times (-5) = 0$ |
| (27) $-8 \times (-1) = 8$ | (57) $5 \times 3 = 15$ | (87) $-5 \times (-8) = 40$ | (117) $4 \times 4 = 16$ |
| (28) $7 \times 2 = 14$ | (58) $-9 \times (-1) = 9$ | (88) $-9 \times 2 = -18$ | (118) $-1 \times 0 = 0$ |
| (29) $1 \times (-6) = -6$ | (59) $5 \times (-2) = -10$ | (89) $2 \times (-5) = -10$ | (119) $-7 \times (-1) = 7$ |
| (30) $1 \times 6 = 6$ | (60) $-3 \times 6 = -18$ | (90) $-2 \times 9 = -18$ | (120) $-5 \times 7 = -35$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-7 \times (-6) = 42$ | (31) $4 \times (-9) = -36$ | (61) $5 \times 5 = 25$ | (91) $3 \times (-1) = -3$ |
| (2) $6 \times 9 = 54$ | (32) $0 \times 8 = 0$ | (62) $2 \times 8 = 16$ | (92) $5 \times 6 = 30$ |
| (3) $9 \times (-5) = -45$ | (33) $7 \times 5 = 35$ | (63) $-7 \times (-2) = 14$ | (93) $-2 \times 8 = -16$ |
| (4) $4 \times (-7) = -28$ | (34) $-3 \times 1 = -3$ | (64) $3 \times (-3) = -9$ | (94) $5 \times 3 = 15$ |
| (5) $-8 \times 8 = -64$ | (35) $1 \times (-7) = -7$ | (65) $-9 \times (-6) = 54$ | (95) $5 \times (-2) = -10$ |
| (6) $-4 \times 6 = -24$ | (36) $-1 \times 5 = -5$ | (66) $-3 \times (-1) = 3$ | (96) $4 \times (-9) = -36$ |
| (7) $-5 \times 5 = -25$ | (37) $7 \times 2 = 14$ | (67) $-6 \times 7 = -42$ | (97) $5 \times (-7) = -35$ |
| (8) $-5 \times (-7) = 35$ | (38) $2 \times (-9) = -18$ | (68) $-5 \times (-2) = 10$ | (98) $-7 \times 2 = -14$ |
| (9) $-7 \times (-5) = 35$ | (39) $4 \times 2 = 8$ | (69) $6 \times 1 = 6$ | (99) $0 \times 3 = 0$ |
| (10) $9 \times 9 = 81$ | (40) $8 \times 6 = 48$ | (70) $9 \times 7 = 63$ | (100) $-8 \times (-1) = 8$ |
| (11) $-6 \times (-3) = 18$ | (41) $0 \times 9 = 0$ | (71) $-2 \times 1 = -2$ | (101) $7 \times 0 = 0$ |
| (12) $-5 \times 5 = -25$ | (42) $-7 \times (-3) = 21$ | (72) $0 \times 7 = 0$ | (102) $1 \times 0 = 0$ |
| (13) $-1 \times (-7) = 7$ | (43) $2 \times (-5) = -10$ | (73) $3 \times (-6) = -18$ | (103) $9 \times (-3) = -27$ |
| (14) $-3 \times 1 = -3$ | (44) $0 \times 9 = 0$ | (74) $-5 \times 4 = -20$ | (104) $-3 \times (-8) = 24$ |
| (15) $-6 \times 6 = -36$ | (45) $1 \times (-9) = -9$ | (75) $-2 \times (-2) = 4$ | (105) $8 \times (-2) = -16$ |
| (16) $4 \times (-2) = -8$ | (46) $2 \times 7 = 14$ | (76) $2 \times (-4) = -8$ | (106) $-8 \times 5 = -40$ |
| (17) $-3 \times (-6) = 18$ | (47) $4 \times 0 = 0$ | (77) $-8 \times 4 = -32$ | (107) $-3 \times 3 = -9$ |
| (18) $5 \times (-3) = -15$ | (48) $9 \times (-5) = -45$ | (78) $-4 \times (-8) = 32$ | (108) $0 \times 8 = 0$ |
| (19) $-9 \times (-1) = 9$ | (49) $1 \times (-8) = -8$ | (79) $-7 \times 6 = -42$ | (109) $4 \times (-6) = -24$ |
| (20) $-1 \times (-2) = 2$ | (50) $-1 \times 4 = -4$ | (80) $6 \times (-4) = -24$ | (110) $8 \times (-4) = -32$ |
| (21) $0 \times (-2) = 0$ | (51) $5 \times (-1) = -5$ | (81) $9 \times 9 = 81$ | (111) $7 \times 2 = 14$ |
| (22) $2 \times (-7) = -14$ | (52) $0 \times 5 = 0$ | (82) $-6 \times 4 = -24$ | (112) $3 \times (-5) = -15$ |
| (23) $-1 \times 8 = -8$ | (53) $7 \times (-9) = -63$ | (83) $-5 \times 8 = -40$ | (113) $2 \times 1 = 2$ |
| (24) $-7 \times 5 = -35$ | (54) $4 \times (-8) = -32$ | (84) $-4 \times 7 = -28$ | (114) $8 \times 2 = 16$ |
| (25) $8 \times 5 = 40$ | (55) $7 \times (-4) = -28$ | (85) $-8 \times 9 = -72$ | (115) $-7 \times 7 = -49$ |
| (26) $-3 \times (-5) = 15$ | (56) $-5 \times 9 = -45$ | (86) $-1 \times (-6) = 6$ | (116) $-3 \times 1 = -3$ |
| (27) $-1 \times 3 = -3$ | (57) $-1 \times 7 = -7$ | (87) $-3 \times (-3) = 9$ | (117) $-8 \times 4 = -32$ |
| (28) $8 \times (-9) = -72$ | (58) $6 \times 5 = 30$ | (88) $9 \times (-3) = -27$ | (118) $0 \times 1 = 0$ |
| (29) $-6 \times (-1) = 6$ | (59) $-8 \times 2 = -16$ | (89) $7 \times 3 = 21$ | (119) $5 \times (-1) = -5$ |
| (30) $-4 \times 8 = -32$ | (60) $-1 \times 6 = -6$ | (90) $6 \times 5 = 30$ | (120) $-5 \times (-2) = 10$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $7 \times 7 = 49$ | (31) $3 \times (-3) = -9$ | (61) $3 \times (-4) = -12$ | (91) $3 \times 9 = 27$ |
| (2) $-6 \times (-6) = 36$ | (32) $6 \times (-7) = -42$ | (62) $-4 \times 0 = 0$ | (92) $7 \times 1 = 7$ |
| (3) $6 \times 3 = 18$ | (33) $-3 \times 0 = 0$ | (63) $-3 \times (-7) = 21$ | (93) $8 \times (-6) = -48$ |
| (4) $-7 \times (-1) = 7$ | (34) $-1 \times (-1) = 1$ | (64) $-8 \times 7 = -56$ | (94) $7 \times (-3) = -21$ |
| (5) $-4 \times 8 = -32$ | (35) $-5 \times (-5) = 25$ | (65) $6 \times (-7) = -42$ | (95) $-2 \times 8 = -16$ |
| (6) $1 \times (-3) = -3$ | (36) $-1 \times (-7) = 7$ | (66) $-9 \times (-6) = 54$ | (96) $-5 \times (-2) = 10$ |
| (7) $9 \times 2 = 18$ | (37) $-3 \times (-6) = 18$ | (67) $-9 \times (-3) = 27$ | (97) $0 \times 8 = 0$ |
| (8) $8 \times 8 = 64$ | (38) $-5 \times (-4) = 20$ | (68) $-2 \times 2 = -4$ | (98) $-4 \times 0 = 0$ |
| (9) $-1 \times (-3) = 3$ | (39) $2 \times 7 = 14$ | (69) $-8 \times 8 = -64$ | (99) $1 \times (-2) = -2$ |
| (10) $-2 \times (-7) = 14$ | (40) $-7 \times (-3) = 21$ | (70) $7 \times 9 = 63$ | (100) $6 \times (-7) = -42$ |
| (11) $5 \times 7 = 35$ | (41) $2 \times 2 = 4$ | (71) $-3 \times 9 = -27$ | (101) $4 \times (-4) = -16$ |
| (12) $7 \times 9 = 63$ | (42) $-1 \times 1 = -1$ | (72) $3 \times (-4) = -12$ | (102) $5 \times 6 = 30$ |
| (13) $6 \times (-1) = -6$ | (43) $-1 \times 3 = -3$ | (73) $9 \times 8 = 72$ | (103) $4 \times 4 = 16$ |
| (14) $9 \times 9 = 81$ | (44) $-6 \times (-3) = 18$ | (74) $-7 \times (-4) = 28$ | (104) $2 \times (-3) = -6$ |
| (15) $6 \times 2 = 12$ | (45) $-9 \times 9 = -81$ | (75) $8 \times 6 = 48$ | (105) $1 \times (-9) = -9$ |
| (16) $5 \times (-2) = -10$ | (46) $-6 \times (-8) = 48$ | (76) $-3 \times 8 = -24$ | (106) $0 \times (-1) = 0$ |
| (17) $5 \times 2 = 10$ | (47) $-8 \times (-8) = 64$ | (77) $-8 \times (-7) = 56$ | (107) $-3 \times 3 = -9$ |
| (18) $-5 \times 7 = -35$ | (48) $2 \times (-8) = -16$ | (78) $-2 \times 1 = -2$ | (108) $3 \times (-7) = -21$ |
| (19) $-6 \times 9 = -54$ | (49) $-8 \times (-9) = 72$ | (79) $-9 \times 5 = -45$ | (109) $-1 \times (-4) = 4$ |
| (20) $5 \times (-8) = -40$ | (50) $5 \times (-9) = -45$ | (80) $-1 \times (-3) = 3$ | (110) $-1 \times 7 = -7$ |
| (21) $-4 \times 8 = -32$ | (51) $-6 \times (-5) = 30$ | (81) $-7 \times (-1) = 7$ | (111) $7 \times 5 = 35$ |
| (22) $-9 \times 8 = -72$ | (52) $-2 \times 5 = -10$ | (82) $-8 \times 8 = -64$ | (112) $-4 \times 5 = -20$ |
| (23) $6 \times 2 = 12$ | (53) $8 \times 1 = 8$ | (83) $-4 \times (-9) = 36$ | (113) $-3 \times 3 = -9$ |
| (24) $9 \times (-3) = -27$ | (54) $9 \times (-3) = -27$ | (84) $-3 \times (-5) = 15$ | (114) $-1 \times (-9) = 9$ |
| (25) $-5 \times 1 = -5$ | (55) $2 \times (-6) = -12$ | (85) $0 \times 1 = 0$ | (115) $-3 \times 4 = -12$ |
| (26) $5 \times (-7) = -35$ | (56) $-4 \times 0 = 0$ | (86) $2 \times 1 = 2$ | (116) $7 \times (-2) = -14$ |
| (27) $0 \times (-6) = 0$ | (57) $-3 \times 9 = -27$ | (87) $-3 \times (-9) = 27$ | (117) $-6 \times (-6) = 36$ |
| (28) $7 \times 7 = 49$ | (58) $-2 \times 9 = -18$ | (88) $-3 \times (-8) = 24$ | (118) $5 \times (-2) = -10$ |
| (29) $8 \times 0 = 0$ | (59) $2 \times 0 = 0$ | (89) $-2 \times 8 = -16$ | (119) $-2 \times 4 = -8$ |
| (30) $-8 \times 2 = -16$ | (60) $1 \times 3 = 3$ | (90) $-9 \times (-4) = 36$ | (120) $3 \times 0 = 0$ |

1.2.3 除法

整数の商と剰余を求める問題。

(1) $-8 \div 5$

(19) $-74 \div 7$

(37) $90 \div 8$

(55) $96 \div 6$

(2) $-54 \div 8$

(20) $-47 \div 6$

(38) $-50 \div 7$

(56) $78 \div 3$

(3) $-28 \div 7$

(21) $-15 \div 1$

(39) $76 \div 1$

(57) $-10 \div 3$

(4) $80 \div 4$

(22) $-55 \div 2$

(40) $-92 \div 9$

(58) $58 \div 6$

(5) $93 \div 3$

(23) $-30 \div 2$

(41) $-66 \div 6$

(59) $65 \div 2$

(6) $80 \div 10$

(24) $-95 \div 3$

(42) $-41 \div 9$

(60) $60 \div 3$

(7) $-99 \div 4$

(25) $7 \div 2$

(43) $-1 \div 1$

(61) $5 \div 9$

(8) $73 \div 6$

(26) $74 \div 4$

(44) $33 \div 10$

(62) $-64 \div 7$

(9) $65 \div 9$

(27) $-3 \div 6$

(45) $60 \div 8$

(63) $-67 \div 8$

(10) $48 \div 3$

(28) $20 \div 2$

(46) $42 \div 8$

(64) $45 \div 8$

(11) $-6 \div 7$

(29) $-47 \div 10$

(47) $-90 \div 7$

(65) $-33 \div 6$

(12) $32 \div 9$

(30) $74 \div 9$

(48) $-69 \div 3$

(66) $-91 \div 8$

(13) $72 \div 6$

(31) $96 \div 5$

(49) $37 \div 1$

(67) $-73 \div 2$

(14) $-40 \div 6$

(32) $21 \div 6$

(50) $98 \div 10$

(68) $92 \div 5$

(15) $-31 \div 1$

(33) $24 \div 8$

(51) $78 \div 2$

(69) $27 \div 5$

(16) $-47 \div 10$

(34) $-68 \div 6$

(52) $76 \div 2$

(70) $-67 \div 5$

(17) $60 \div 3$

(35) $10 \div 8$

(53) $-76 \div 7$

(71) $-93 \div 1$

(18) $61 \div 4$

(36) $-89 \div 8$

(54) $15 \div 6$

(72) $-97 \div 3$

(1) $19 \div 5$

(19) $-52 \div 5$

(37) $91 \div 3$

(55) $-94 \div 6$

(2) $63 \div 4$

(20) $81 \div 8$

(38) $98 \div 3$

(56) $-99 \div 6$

(3) $-71 \div 7$

(21) $-91 \div 3$

(39) $-6 \div 3$

(57) $-27 \div 5$

(4) $61 \div 10$

(22) $-64 \div 7$

(40) $-86 \div 5$

(58) $82 \div 5$

(5) $-24 \div 5$

(23) $-34 \div 7$

(41) $-70 \div 6$

(59) $64 \div 1$

(6) $28 \div 3$

(24) $31 \div 6$

(42) $71 \div 10$

(60) $65 \div 5$

(7) $0 \div 9$

(25) $84 \div 6$

(43) $64 \div 2$

(61) $-99 \div 3$

(8) $63 \div 9$

(26) $88 \div 8$

(44) $-23 \div 7$

(62) $100 \div 8$

(9) $55 \div 9$

(27) $54 \div 2$

(45) $-6 \div 4$

(63) $-78 \div 3$

(10) $-20 \div 3$

(28) $58 \div 5$

(46) $35 \div 3$

(64) $-96 \div 7$

(11) $85 \div 10$

(29) $28 \div 3$

(47) $88 \div 7$

(65) $-93 \div 4$

(12) $-4 \div 3$

(30) $39 \div 5$

(48) $-58 \div 7$

(66) $-25 \div 3$

(13) $26 \div 5$

(31) $-87 \div 6$

(49) $-43 \div 3$

(67) $-56 \div 8$

(14) $-24 \div 6$

(32) $-100 \div 2$

(50) $-82 \div 9$

(68) $91 \div 6$

(15) $-89 \div 2$

(33) $-87 \div 3$

(51) $18 \div 8$

(69) $-94 \div 3$

(16) $22 \div 8$

(34) $99 \div 1$

(52) $83 \div 7$

(70) $1 \div 8$

(17) $-41 \div 10$

(35) $62 \div 4$

(53) $-65 \div 6$

(71) $-43 \div 3$

(18) $-51 \div 8$

(36) $-71 \div 7$

(54) $72 \div 8$

(72) $84 \div 8$

(1) $-96 \div 5$

(19) $4 \div 3$

(37) $52 \div 8$

(55) $26 \div 4$

(2) $-96 \div 9$

(20) $56 \div 2$

(38) $-16 \div 8$

(56) $65 \div 9$

(3) $15 \div 7$

(21) $14 \div 9$

(39) $72 \div 7$

(57) $-33 \div 2$

(4) $-11 \div 4$

(22) $-16 \div 1$

(40) $22 \div 6$

(58) $-62 \div 7$

(5) $-40 \div 1$

(23) $68 \div 5$

(41) $-3 \div 10$

(59) $-12 \div 3$

(6) $-61 \div 4$

(24) $-58 \div 2$

(42) $-86 \div 10$

(60) $54 \div 1$

(7) $41 \div 6$

(25) $3 \div 3$

(43) $45 \div 10$

(61) $-74 \div 9$

(8) $-93 \div 1$

(26) $64 \div 7$

(44) $66 \div 6$

(62) $-8 \div 3$

(9) $90 \div 3$

(27) $-93 \div 9$

(45) $49 \div 3$

(63) $-68 \div 2$

(10) $-21 \div 3$

(28) $32 \div 5$

(46) $47 \div 6$

(64) $-67 \div 2$

(11) $71 \div 5$

(29) $89 \div 7$

(47) $53 \div 5$

(65) $72 \div 1$

(12) $79 \div 8$

(30) $-11 \div 4$

(48) $-63 \div 8$

(66) $71 \div 10$

(13) $68 \div 7$

(31) $34 \div 3$

(49) $66 \div 8$

(67) $92 \div 8$

(14) $27 \div 4$

(32) $-26 \div 9$

(50) $-36 \div 6$

(68) $-25 \div 2$

(15) $24 \div 7$

(33) $-58 \div 10$

(51) $89 \div 4$

(69) $-93 \div 3$

(16) $-5 \div 8$

(34) $-99 \div 7$

(52) $26 \div 8$

(70) $31 \div 10$

(17) $95 \div 7$

(35) $-19 \div 9$

(53) $-23 \div 9$

(71) $90 \div 2$

(18) $-16 \div 5$

(36) $-26 \div 4$

(54) $-40 \div 10$

(72) $49 \div 5$

- | | | | |
|-------------------|--------------------|-------------------|-------------------|
| (1) $34 \div 3$ | (19) $21 \div 2$ | (37) $-20 \div 3$ | (55) $95 \div 2$ |
| (2) $37 \div 3$ | (20) $39 \div 1$ | (38) $-24 \div 5$ | (56) $73 \div 9$ |
| (3) $-8 \div 1$ | (21) $19 \div 8$ | (39) $7 \div 5$ | (57) $67 \div 7$ |
| (4) $-6 \div 7$ | (22) $-47 \div 10$ | (40) $-45 \div 2$ | (58) $56 \div 5$ |
| (5) $57 \div 5$ | (23) $85 \div 1$ | (41) $4 \div 10$ | (59) $-73 \div 5$ |
| (6) $31 \div 1$ | (24) $-69 \div 8$ | (42) $14 \div 6$ | (60) $50 \div 3$ |
| (7) $28 \div 6$ | (25) $55 \div 2$ | (43) $24 \div 9$ | (61) $-52 \div 8$ |
| (8) $64 \div 10$ | (26) $-40 \div 2$ | (44) $89 \div 2$ | (62) $35 \div 3$ |
| (9) $84 \div 9$ | (27) $-56 \div 9$ | (45) $31 \div 3$ | (63) $-80 \div 9$ |
| (10) $0 \div 6$ | (28) $35 \div 1$ | (46) $-38 \div 3$ | (64) $64 \div 7$ |
| (11) $-18 \div 4$ | (29) $72 \div 9$ | (47) $-37 \div 2$ | (65) $12 \div 1$ |
| (12) $0 \div 6$ | (30) $-60 \div 10$ | (48) $32 \div 5$ | (66) $-31 \div 4$ |
| (13) $-39 \div 5$ | (31) $-25 \div 5$ | (49) $74 \div 1$ | (67) $83 \div 8$ |
| (14) $-67 \div 1$ | (32) $89 \div 2$ | (50) $-82 \div 5$ | (68) $-98 \div 2$ |
| (15) $31 \div 2$ | (33) $88 \div 2$ | (51) $-32 \div 8$ | (69) $36 \div 1$ |
| (16) $-68 \div 8$ | (34) $-55 \div 3$ | (52) $-73 \div 4$ | (70) $-26 \div 4$ |
| (17) $54 \div 2$ | (35) $-79 \div 7$ | (53) $33 \div 4$ | (71) $46 \div 6$ |
| (18) $77 \div 10$ | (36) $4 \div 8$ | (54) $-85 \div 5$ | (72) $-12 \div 8$ |

- | | | | |
|-------------------|--------------------|-------------------|-------------------|
| (1) $44 \div 10$ | (19) $2 \div 7$ | (37) $38 \div 10$ | (55) $77 \div 10$ |
| (2) $60 \div 5$ | (20) $71 \div 4$ | (38) $-69 \div 8$ | (56) $45 \div 7$ |
| (3) $23 \div 8$ | (21) $-46 \div 5$ | (39) $1 \div 7$ | (57) $7 \div 7$ |
| (4) $10 \div 10$ | (22) $91 \div 4$ | (40) $-87 \div 3$ | (58) $80 \div 8$ |
| (5) $55 \div 8$ | (23) $-43 \div 1$ | (41) $-64 \div 1$ | (59) $76 \div 8$ |
| (6) $56 \div 5$ | (24) $-34 \div 8$ | (42) $-21 \div 5$ | (60) $59 \div 5$ |
| (7) $-9 \div 6$ | (25) $-99 \div 7$ | (43) $-77 \div 2$ | (61) $74 \div 6$ |
| (8) $-49 \div 1$ | (26) $-23 \div 3$ | (44) $43 \div 1$ | (62) $-48 \div 6$ |
| (9) $32 \div 5$ | (27) $78 \div 6$ | (45) $13 \div 5$ | (63) $80 \div 6$ |
| (10) $-42 \div 2$ | (28) $-74 \div 7$ | (46) $-1 \div 3$ | (64) $-80 \div 2$ |
| (11) $-75 \div 9$ | (29) $-100 \div 4$ | (47) $-73 \div 1$ | (65) $-89 \div 8$ |
| (12) $64 \div 3$ | (30) $-25 \div 3$ | (48) $62 \div 6$ | (66) $12 \div 1$ |
| (13) $75 \div 5$ | (31) $-1 \div 1$ | (49) $-29 \div 5$ | (67) $-47 \div 3$ |
| (14) $-82 \div 6$ | (32) $66 \div 8$ | (50) $42 \div 6$ | (68) $-82 \div 2$ |
| (15) $26 \div 3$ | (33) $28 \div 7$ | (51) $44 \div 4$ | (69) $23 \div 4$ |
| (16) $64 \div 3$ | (34) $49 \div 4$ | (52) $32 \div 2$ | (70) $-43 \div 4$ |
| (17) $10 \div 1$ | (35) $-68 \div 3$ | (53) $-42 \div 5$ | (71) $84 \div 10$ |
| (18) $19 \div 5$ | (36) $85 \div 5$ | (54) $22 \div 10$ | (72) $94 \div 5$ |

- | | | | |
|-------------------|-------------------|--------------------|-------------------|
| (1) $54 \div 10$ | (19) $-64 \div 3$ | (37) $-43 \div 1$ | (55) $-22 \div 4$ |
| (2) $-94 \div 7$ | (20) $90 \div 2$ | (38) $10 \div 10$ | (56) $-76 \div 7$ |
| (3) $14 \div 2$ | (21) $-67 \div 4$ | (39) $-66 \div 2$ | (57) $59 \div 4$ |
| (4) $82 \div 3$ | (22) $-77 \div 4$ | (40) $26 \div 10$ | (58) $-32 \div 3$ |
| (5) $18 \div 9$ | (23) $43 \div 1$ | (41) $-26 \div 4$ | (59) $-14 \div 7$ |
| (6) $-16 \div 6$ | (24) $96 \div 7$ | (42) $48 \div 10$ | (60) $-53 \div 2$ |
| (7) $-43 \div 6$ | (25) $-88 \div 6$ | (43) $-32 \div 3$ | (61) $-45 \div 8$ |
| (8) $61 \div 10$ | (26) $-49 \div 1$ | (44) $-6 \div 5$ | (62) $36 \div 5$ |
| (9) $100 \div 10$ | (27) $48 \div 2$ | (45) $-30 \div 6$ | (63) $37 \div 9$ |
| (10) $79 \div 1$ | (28) $71 \div 10$ | (46) $-75 \div 2$ | (64) $-57 \div 1$ |
| (11) $-80 \div 6$ | (29) $-19 \div 4$ | (47) $-76 \div 10$ | (65) $58 \div 5$ |
| (12) $28 \div 1$ | (30) $40 \div 9$ | (48) $-82 \div 4$ | (66) $-48 \div 1$ |
| (13) $-14 \div 4$ | (31) $67 \div 9$ | (49) $-89 \div 8$ | (67) $27 \div 9$ |
| (14) $-75 \div 2$ | (32) $23 \div 3$ | (50) $27 \div 8$ | (68) $28 \div 6$ |
| (15) $-72 \div 7$ | (33) $56 \div 9$ | (51) $-88 \div 9$ | (69) $-77 \div 6$ |
| (16) $-68 \div 1$ | (34) $40 \div 8$ | (52) $-20 \div 3$ | (70) $40 \div 10$ |
| (17) $-26 \div 3$ | (35) $-22 \div 9$ | (53) $12 \div 9$ | (71) $-65 \div 4$ |
| (18) $25 \div 6$ | (36) $4 \div 3$ | (54) $15 \div 6$ | (72) $41 \div 5$ |

(1) $29 \div 7$

(19) $3 \div 10$

(37) $-12 \div 2$

(55) $83 \div 8$

(2) $1 \div 9$

(20) $66 \div 7$

(38) $50 \div 7$

(56) $30 \div 4$

(3) $-81 \div 7$

(21) $54 \div 9$

(39) $22 \div 2$

(57) $-39 \div 1$

(4) $38 \div 3$

(22) $-11 \div 1$

(40) $59 \div 5$

(58) $10 \div 7$

(5) $-92 \div 9$

(23) $83 \div 3$

(41) $-9 \div 3$

(59) $-91 \div 10$

(6) $68 \div 5$

(24) $-70 \div 9$

(42) $-93 \div 6$

(60) $-76 \div 4$

(7) $36 \div 10$

(25) $63 \div 7$

(43) $-87 \div 2$

(61) $35 \div 7$

(8) $-63 \div 1$

(26) $57 \div 5$

(44) $-85 \div 9$

(62) $59 \div 7$

(9) $9 \div 6$

(27) $-13 \div 8$

(45) $93 \div 5$

(63) $11 \div 2$

(10) $-56 \div 8$

(28) $28 \div 7$

(46) $-73 \div 4$

(64) $64 \div 1$

(11) $56 \div 10$

(29) $28 \div 7$

(47) $-21 \div 3$

(65) $31 \div 4$

(12) $54 \div 10$

(30) $84 \div 10$

(48) $45 \div 8$

(66) $73 \div 9$

(13) $26 \div 10$

(31) $24 \div 3$

(49) $-1 \div 8$

(67) $-29 \div 10$

(14) $16 \div 1$

(32) $-20 \div 8$

(50) $-14 \div 8$

(68) $-84 \div 6$

(15) $59 \div 2$

(33) $18 \div 7$

(51) $68 \div 2$

(69) $-6 \div 7$

(16) $2 \div 2$

(34) $-78 \div 3$

(52) $61 \div 2$

(70) $87 \div 3$

(17) $13 \div 8$

(35) $21 \div 5$

(53) $86 \div 2$

(71) $92 \div 10$

(18) $61 \div 9$

(36) $58 \div 8$

(54) $-10 \div 3$

(72) $56 \div 1$

(1) $17 \div 10$

(19) $57 \div 3$

(37) $24 \div 10$

(55) $22 \div 4$

(2) $-77 \div 2$

(20) $45 \div 9$

(38) $-73 \div 5$

(56) $100 \div 9$

(3) $31 \div 9$

(21) $4 \div 8$

(39) $84 \div 9$

(57) $-52 \div 1$

(4) $29 \div 1$

(22) $79 \div 1$

(40) $-96 \div 8$

(58) $-14 \div 2$

(5) $23 \div 1$

(23) $59 \div 3$

(41) $-33 \div 2$

(59) $-2 \div 6$

(6) $-18 \div 10$

(24) $-80 \div 4$

(42) $-38 \div 2$

(60) $43 \div 1$

(7) $-92 \div 10$

(25) $-77 \div 9$

(43) $100 \div 10$

(61) $-82 \div 10$

(8) $65 \div 10$

(26) $-88 \div 7$

(44) $-69 \div 2$

(62) $-4 \div 10$

(9) $-82 \div 2$

(27) $-78 \div 6$

(45) $43 \div 6$

(63) $4 \div 9$

(10) $14 \div 2$

(28) $47 \div 10$

(46) $59 \div 9$

(64) $99 \div 10$

(11) $-95 \div 7$

(29) $27 \div 10$

(47) $92 \div 7$

(65) $-22 \div 7$

(12) $-13 \div 9$

(30) $-10 \div 5$

(48) $19 \div 6$

(66) $-38 \div 5$

(13) $34 \div 7$

(31) $-96 \div 10$

(49) $-89 \div 5$

(67) $-26 \div 5$

(14) $69 \div 3$

(32) $79 \div 10$

(50) $-69 \div 5$

(68) $56 \div 10$

(15) $92 \div 1$

(33) $52 \div 2$

(51) $-27 \div 5$

(69) $-6 \div 4$

(16) $-30 \div 9$

(34) $94 \div 7$

(52) $81 \div 4$

(70) $48 \div 2$

(17) $30 \div 2$

(35) $-24 \div 3$

(53) $-35 \div 6$

(71) $-36 \div 10$

(18) $6 \div 9$

(36) $78 \div 9$

(54) $5 \div 7$

(72) $-27 \div 10$

(1) $-37 \div 9$

(19) $-58 \div 10$

(37) $74 \div 10$

(55) $-35 \div 3$

(2) $-75 \div 8$

(20) $78 \div 9$

(38) $62 \div 1$

(56) $87 \div 6$

(3) $94 \div 7$

(21) $-11 \div 8$

(39) $78 \div 8$

(57) $-60 \div 6$

(4) $45 \div 9$

(22) $33 \div 3$

(40) $15 \div 6$

(58) $-5 \div 1$

(5) $5 \div 2$

(23) $16 \div 10$

(41) $-50 \div 1$

(59) $31 \div 8$

(6) $-19 \div 5$

(24) $-9 \div 6$

(42) $-60 \div 4$

(60) $-72 \div 7$

(7) $91 \div 8$

(25) $66 \div 7$

(43) $2 \div 3$

(61) $58 \div 6$

(8) $65 \div 4$

(26) $-36 \div 4$

(44) $80 \div 2$

(62) $-38 \div 2$

(9) $39 \div 2$

(27) $-14 \div 10$

(45) $8 \div 1$

(63) $89 \div 3$

(10) $-45 \div 6$

(28) $-78 \div 10$

(46) $79 \div 1$

(64) $-31 \div 1$

(11) $27 \div 8$

(29) $-100 \div 7$

(47) $68 \div 1$

(65) $-65 \div 5$

(12) $-87 \div 6$

(30) $-18 \div 7$

(48) $-41 \div 6$

(66) $16 \div 10$

(13) $-12 \div 3$

(31) $85 \div 6$

(49) $-17 \div 7$

(67) $-20 \div 4$

(14) $5 \div 3$

(32) $85 \div 2$

(50) $-85 \div 6$

(68) $39 \div 3$

(15) $-61 \div 6$

(33) $-1 \div 7$

(51) $-25 \div 2$

(69) $-33 \div 3$

(16) $60 \div 7$

(34) $-86 \div 9$

(52) $-2 \div 6$

(70) $41 \div 9$

(17) $20 \div 1$

(35) $82 \div 4$

(53) $34 \div 5$

(71) $-26 \div 8$

(18) $-61 \div 2$

(36) $-8 \div 6$

(54) $-23 \div 10$

(72) $76 \div 2$

(1) $-72 \div 5$

(19) $-17 \div 4$

(37) $-30 \div 3$

(55) $39 \div 6$

(2) $96 \div 1$

(20) $-3 \div 9$

(38) $89 \div 8$

(56) $36 \div 10$

(3) $68 \div 4$

(21) $-66 \div 5$

(39) $-24 \div 8$

(57) $48 \div 2$

(4) $44 \div 2$

(22) $33 \div 6$

(40) $-90 \div 9$

(58) $55 \div 7$

(5) $7 \div 1$

(23) $99 \div 5$

(41) $-32 \div 2$

(59) $71 \div 1$

(6) $22 \div 4$

(24) $-70 \div 7$

(42) $-67 \div 7$

(60) $-90 \div 8$

(7) $-30 \div 4$

(25) $58 \div 9$

(43) $20 \div 2$

(61) $75 \div 2$

(8) $-8 \div 6$

(26) $12 \div 8$

(44) $43 \div 1$

(62) $-20 \div 9$

(9) $86 \div 7$

(27) $-66 \div 9$

(45) $-78 \div 1$

(63) $-83 \div 7$

(10) $-86 \div 10$

(28) $-75 \div 9$

(46) $12 \div 2$

(64) $66 \div 10$

(11) $-68 \div 6$

(29) $63 \div 8$

(47) $74 \div 10$

(65) $-12 \div 4$

(12) $-44 \div 9$

(30) $-15 \div 5$

(48) $10 \div 5$

(66) $-84 \div 2$

(13) $63 \div 7$

(31) $-4 \div 4$

(49) $44 \div 10$

(67) $98 \div 10$

(14) $67 \div 8$

(32) $-54 \div 3$

(50) $-97 \div 4$

(68) $-20 \div 1$

(15) $-43 \div 8$

(33) $77 \div 10$

(51) $-18 \div 2$

(69) $39 \div 1$

(16) $25 \div 2$

(34) $20 \div 6$

(52) $50 \div 8$

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| (2) $-54 \div 8$
商 -7 , 余り 2 | (20) $-47 \div 6$
商 -8 , 余り 1 | (38) $-50 \div 7$
商 -8 , 余り 6 | (56) $78 \div 3$
商 26 , 余り 0 |
| (3) $-28 \div 7$
商 -4 , 余り 0 | (21) $-15 \div 1$
商 -15 , 余り 0 | (39) $76 \div 1$
商 76 , 余り 0 | (57) $-10 \div 3$
商 -4 , 余り 2 |
| (4) $80 \div 4$
商 20 , 余り 0 | (22) $-55 \div 2$
商 -28 , 余り 1 | (40) $-92 \div 9$
商 -11 , 余り 7 | (58) $58 \div 6$
商 9 , 余り 4 |
| (5) $93 \div 3$
商 31 , 余り 0 | (23) $-30 \div 2$
商 -15 , 余り 0 | (41) $-66 \div 6$
商 -11 , 余り 0 | (59) $65 \div 2$
商 32 , 余り 1 |
| (6) $80 \div 10$
商 8 , 余り 0 | (24) $-95 \div 3$
商 -32 , 余り 1 | (42) $-41 \div 9$
商 -5 , 余り 4 | (60) $60 \div 3$
商 20 , 余り 0 |
| (7) $-99 \div 4$
商 -25 , 余り 1 | (25) $7 \div 2$
商 3 , 余り 1 | (43) $-1 \div 1$
商 -1 , 余り 0 | (61) $5 \div 9$
商 0 , 余り 5 |
| (8) $73 \div 6$
商 12 , 余り 1 | (26) $74 \div 4$
商 18 , 余り 2 | (44) $33 \div 10$
商 3 , 余り 3 | (62) $-64 \div 7$
商 -10 , 余り 6 |
| (9) $65 \div 9$
商 7 , 余り 2 | (27) $-3 \div 6$
商 -1 , 余り 3 | (45) $60 \div 8$
商 7 , 余り 4 | (63) $-67 \div 8$
商 -9 , 余り 5 |
| (10) $48 \div 3$
商 16 , 余り 0 | (28) $20 \div 2$
商 10 , 余り 0 | (46) $42 \div 8$
商 5 , 余り 2 | (64) $45 \div 8$
商 5 , 余り 5 |
| (11) $-6 \div 7$
商 -1 , 余り 1 | (29) $-47 \div 10$
商 -5 , 余り 3 | (47) $-90 \div 7$
商 -13 , 余り 1 | (65) $-33 \div 6$
商 -6 , 余り 3 |
| (12) $32 \div 9$
商 3 , 余り 5 | (30) $74 \div 9$
商 8 , 余り 2 | (48) $-69 \div 3$
商 -23 , 余り 0 | (66) $-91 \div 8$
商 -12 , 余り 5 |
| (13) $72 \div 6$
商 12 , 余り 0 | (31) $96 \div 5$
商 19 , 余り 1 | (49) $37 \div 1$
商 37 , 余り 0 | (67) $-73 \div 2$
商 -37 , 余り 1 |
| (14) $-40 \div 6$
商 -7 , 余り 2 | (32) $21 \div 6$
商 3 , 余り 3 | (50) $98 \div 10$
商 9 , 余り 8 | (68) $92 \div 5$
商 18 , 余り 2 |
| (15) $-31 \div 1$
商 -31 , 余り 0 | (33) $24 \div 8$
商 3 , 余り 0 | (51) $78 \div 2$
商 39 , 余り 0 | (69) $27 \div 5$
商 5 , 余り 2 |
| (16) $-47 \div 10$
商 -5 , 余り 3 | (34) $-68 \div 6$
商 -12 , 余り 4 | (52) $76 \div 2$
商 38 , 余り 0 | (70) $-67 \div 5$
商 -14 , 余り 3 |
| (17) $60 \div 3$
商 20 , 余り 0 | (35) $10 \div 8$
商 1 , 余り 2 | (53) $-76 \div 7$
商 -11 , 余り 1 | (71) $-93 \div 1$
商 -93 , 余り 0 |
| (18) $61 \div 4$
商 15 , 余り 1 | (36) $-89 \div 8$
商 -12 , 余り 7 | (54) $15 \div 6$
商 2 , 余り 3 | (72) $-97 \div 3$
商 -33 , 余り 2 |

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|----------------------------------|-----------------------------------|----------------------------------|----------------------------------|
| (1) $19 \div 5$
商 3, 余り 4 | (19) $-52 \div 5$
商 -11, 余り 3 | (37) $91 \div 3$
商 30, 余り 1 | (55) $-94 \div 6$
商 -16, 余り 2 |
| (2) $63 \div 4$
商 15, 余り 3 | (20) $81 \div 8$
商 10, 余り 1 | (38) $98 \div 3$
商 32, 余り 2 | (56) $-99 \div 6$
商 -17, 余り 3 |
| (3) $-71 \div 7$
商 -11, 余り 6 | (21) $-91 \div 3$
商 -31, 余り 2 | (39) $-6 \div 3$
商 -2, 余り 0 | (57) $-27 \div 5$
商 -6, 余り 3 |
| (4) $61 \div 10$
商 6, 余り 1 | (22) $-64 \div 7$
商 -10, 余り 6 | (40) $-86 \div 5$
商 -18, 余り 4 | (58) $82 \div 5$
商 16, 余り 2 |
| (5) $-24 \div 5$
商 -5, 余り 1 | (23) $-34 \div 7$
商 -5, 余り 1 | (41) $-70 \div 6$
商 -12, 余り 2 | (59) $64 \div 1$
商 64, 余り 0 |
| (6) $28 \div 3$
商 9, 余り 1 | (24) $31 \div 6$
商 5, 余り 1 | (42) $71 \div 10$
商 7, 余り 1 | (60) $65 \div 5$
商 13, 余り 0 |
| (7) $0 \div 9$
商 0, 余り 0 | (25) $84 \div 6$
商 14, 余り 0 | (43) $64 \div 2$
商 32, 余り 0 | (61) $-99 \div 3$
商 -33, 余り 0 |
| (8) $63 \div 9$
商 7, 余り 0 | (26) $88 \div 8$
商 11, 余り 0 | (44) $-23 \div 7$
商 -4, 余り 5 | (62) $100 \div 8$
商 12, 余り 4 |
| (9) $55 \div 9$
商 6, 余り 1 | (27) $54 \div 2$
商 27, 余り 0 | (45) $-6 \div 4$
商 -2, 余り 2 | (63) $-78 \div 3$
商 -26, 余り 0 |
| (10) $-20 \div 3$
商 -7, 余り 1 | (28) $58 \div 5$
商 11, 余り 3 | (46) $35 \div 3$
商 11, 余り 2 | (64) $-96 \div 7$
商 -14, 余り 2 |
| (11) $85 \div 10$
商 8, 余り 5 | (29) $28 \div 3$
商 9, 余り 1 | (47) $88 \div 7$
商 12, 余り 4 | (65) $-93 \div 4$
商 -24, 余り 3 |
| (12) $-4 \div 3$
商 -2, 余り 2 | (30) $39 \div 5$
商 7, 余り 4 | (48) $-58 \div 7$
商 -9, 余り 5 | (66) $-25 \div 3$
商 -9, 余り 2 |
| (13) $26 \div 5$
商 5, 余り 1 | (31) $-87 \div 6$
商 -15, 余り 3 | (49) $-43 \div 3$
商 -15, 余り 2 | (67) $-56 \div 8$
商 -7, 余り 0 |
| (14) $-24 \div 6$
商 -4, 余り 0 | (32) $-100 \div 2$
商 -50, 余り 0 | (50) $-82 \div 9$
商 -10, 余り 8 | (68) $91 \div 6$
商 15, 余り 1 |
| (15) $-89 \div 2$
商 -45, 余り 1 | (33) $-87 \div 3$
商 -29, 余り 0 | (51) $18 \div 8$
商 2, 余り 2 | (69) $-94 \div 3$
商 -32, 余り 2 |
| (16) $22 \div 8$
商 2, 余り 6 | (34) $99 \div 1$
商 99, 余り 0 | (52) $83 \div 7$
商 11, 余り 6 | (70) $1 \div 8$
商 0, 余り 1 |
| (17) $-41 \div 10$
商 -5, 余り 9 | (35) $62 \div 4$
商 15, 余り 2 | (53) $-65 \div 6$
商 -11, 余り 1 | (71) $-43 \div 3$
商 -15, 余り 2 |
| (18) $-51 \div 8$
商 -7, 余り 5 | (36) $-71 \div 7$
商 -11, 余り 6 | (54) $72 \div 8$
商 9, 余り 0 | (72) $84 \div 8$
商 10, 余り 4 |

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|--------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| (1) $-96 \div 5$
商 -20 , 余り 4 | (19) $4 \div 3$
商 1 , 余り 1 | (37) $52 \div 8$
商 6 , 余り 4 | (55) $26 \div 4$
商 6 , 余り 2 |
| (2) $-96 \div 9$
商 -11 , 余り 3 | (20) $56 \div 2$
商 28 , 余り 0 | (38) $-16 \div 8$
商 -2 , 余り 0 | (56) $65 \div 9$
商 7 , 余り 2 |
| (3) $15 \div 7$
商 2 , 余り 1 | (21) $14 \div 9$
商 1 , 余り 5 | (39) $72 \div 7$
商 10 , 余り 2 | (57) $-33 \div 2$
商 -17 , 余り 1 |
| (4) $-11 \div 4$
商 -3 , 余り 1 | (22) $-16 \div 1$
商 -16 , 余り 0 | (40) $22 \div 6$
商 3 , 余り 4 | (58) $-62 \div 7$
商 -9 , 余り 1 |
| (5) $-40 \div 1$
商 -40 , 余り 0 | (23) $68 \div 5$
商 13 , 余り 3 | (41) $-3 \div 10$
商 -1 , 余り 7 | (59) $-12 \div 3$
商 -4 , 余り 0 |
| (6) $-61 \div 4$
商 -16 , 余り 3 | (24) $-58 \div 2$
商 -29 , 余り 0 | (42) $-86 \div 10$
商 -9 , 余り 4 | (60) $54 \div 1$
商 54 , 余り 0 |
| (7) $41 \div 6$
商 6 , 余り 5 | (25) $3 \div 3$
商 1 , 余り 0 | (43) $45 \div 10$
商 4 , 余り 5 | (61) $-74 \div 9$
商 -9 , 余り 7 |
| (8) $-93 \div 1$
商 -93 , 余り 0 | (26) $64 \div 7$
商 9 , 余り 1 | (44) $66 \div 6$
商 11 , 余り 0 | (62) $-8 \div 3$
商 -3 , 余り 1 |
| (9) $90 \div 3$
商 30 , 余り 0 | (27) $-93 \div 9$
商 -11 , 余り 6 | (45) $49 \div 3$
商 16 , 余り 1 | (63) $-68 \div 2$
商 -34 , 余り 0 |
| (10) $-21 \div 3$
商 -7 , 余り 0 | (28) $32 \div 5$
商 6 , 余り 2 | (46) $47 \div 6$
商 7 , 余り 5 | (64) $-67 \div 2$
商 -34 , 余り 1 |
| (11) $71 \div 5$
商 14 , 余り 1 | (29) $89 \div 7$
商 12 , 余り 5 | (47) $53 \div 5$
商 10 , 余り 3 | (65) $72 \div 1$
商 72 , 余り 0 |
| (12) $79 \div 8$
商 9 , 余り 7 | (30) $-11 \div 4$
商 -3 , 余り 1 | (48) $-63 \div 8$
商 -8 , 余り 1 | (66) $71 \div 10$
商 7 , 余り 1 |
| (13) $68 \div 7$
商 9 , 余り 5 | (31) $34 \div 3$
商 11 , 余り 1 | (49) $66 \div 8$
商 8 , 余り 2 | (67) $92 \div 8$
商 11 , 余り 4 |
| (14) $27 \div 4$
商 6 , 余り 3 | (32) $-26 \div 9$
商 -3 , 余り 1 | (50) $-36 \div 6$
商 -6 , 余り 0 | (68) $-25 \div 2$
商 -13 , 余り 1 |
| (15) $24 \div 7$
商 3 , 余り 3 | (33) $-58 \div 10$
商 -6 , 余り 2 | (51) $89 \div 4$
商 22 , 余り 1 | (69) $-93 \div 3$
商 -31 , 余り 0 |
| (16) $-5 \div 8$
商 -1 , 余り 3 | (34) $-99 \div 7$
商 -15 , 余り 6 | (52) $26 \div 8$
商 3 , 余り 2 | (70) $31 \div 10$
商 3 , 余り 1 |
| (17) $95 \div 7$
商 13 , 余り 4 | (35) $-19 \div 9$
商 -3 , 余り 8 | (53) $-23 \div 9$
商 -3 , 余り 4 | (71) $90 \div 2$
商 45 , 余り 0 |
| (18) $-16 \div 5$
商 -4 , 余り 4 | (36) $-26 \div 4$
商 -7 , 余り 2 | (54) $-40 \div 10$
商 -4 , 余り 0 | (72) $49 \div 5$
商 9 , 余り 4 |

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|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| (1) $34 \div 3$
商 11, 余り 1 | (19) $21 \div 2$
商 10, 余り 1 | (37) $-20 \div 3$
商 -7, 余り 1 | (55) $95 \div 2$
商 47, 余り 1 |
| (2) $37 \div 3$
商 12, 余り 1 | (20) $39 \div 1$
商 39, 余り 0 | (38) $-24 \div 5$
商 -5, 余り 1 | (56) $73 \div 9$
商 8, 余り 1 |
| (3) $-8 \div 1$
商 -8, 余り 0 | (21) $19 \div 8$
商 2, 余り 3 | (39) $7 \div 5$
商 1, 余り 2 | (57) $67 \div 7$
商 9, 余り 4 |
| (4) $-6 \div 7$
商 -1, 余り 1 | (22) $-47 \div 10$
商 -5, 余り 3 | (40) $-45 \div 2$
商 -23, 余り 1 | (58) $56 \div 5$
商 11, 余り 1 |
| (5) $57 \div 5$
商 11, 余り 2 | (23) $85 \div 1$
商 85, 余り 0 | (41) $4 \div 10$
商 0, 余り 4 | (59) $-73 \div 5$
商 -15, 余り 2 |
| (6) $31 \div 1$
商 31, 余り 0 | (24) $-69 \div 8$
商 -9, 余り 3 | (42) $14 \div 6$
商 2, 余り 2 | (60) $50 \div 3$
商 16, 余り 2 |
| (7) $28 \div 6$
商 4, 余り 4 | (25) $55 \div 2$
商 27, 余り 1 | (43) $24 \div 9$
商 2, 余り 6 | (61) $-52 \div 8$
商 -7, 余り 4 |
| (8) $64 \div 10$
商 6, 余り 4 | (26) $-40 \div 2$
商 -20, 余り 0 | (44) $89 \div 2$
商 44, 余り 1 | (62) $35 \div 3$
商 11, 余り 2 |
| (9) $84 \div 9$
商 9, 余り 3 | (27) $-56 \div 9$
商 -7, 余り 7 | (45) $31 \div 3$
商 10, 余り 1 | (63) $-80 \div 9$
商 -9, 余り 1 |
| (10) $0 \div 6$
商 0, 余り 0 | (28) $35 \div 1$
商 35, 余り 0 | (46) $-38 \div 3$
商 -13, 余り 1 | (64) $64 \div 7$
商 9, 余り 1 |
| (11) $-18 \div 4$
商 -5, 余り 2 | (29) $72 \div 9$
商 8, 余り 0 | (47) $-37 \div 2$
商 -19, 余り 1 | (65) $12 \div 1$
商 12, 余り 0 |
| (12) $0 \div 6$
商 0, 余り 0 | (30) $-60 \div 10$
商 -6, 余り 0 | (48) $32 \div 5$
商 6, 余り 2 | (66) $-31 \div 4$
商 -8, 余り 1 |
| (13) $-39 \div 5$
商 -8, 余り 1 | (31) $-25 \div 5$
商 -5, 余り 0 | (49) $74 \div 1$
商 74, 余り 0 | (67) $83 \div 8$
商 10, 余り 3 |
| (14) $-67 \div 1$
商 -67, 余り 0 | (32) $89 \div 2$
商 44, 余り 1 | (50) $-82 \div 5$
商 -17, 余り 3 | (68) $-98 \div 2$
商 -49, 余り 0 |
| (15) $31 \div 2$
商 15, 余り 1 | (33) $88 \div 2$
商 44, 余り 0 | (51) $-32 \div 8$
商 -4, 余り 0 | (69) $36 \div 1$
商 36, 余り 0 |
| (16) $-68 \div 8$
商 -9, 余り 4 | (34) $-55 \div 3$
商 -19, 余り 2 | (52) $-73 \div 4$
商 -19, 余り 3 | (70) $-26 \div 4$
商 -7, 余り 2 |
| (17) $54 \div 2$
商 27, 余り 0 | (35) $-79 \div 7$
商 -12, 余り 5 | (53) $33 \div 4$
商 8, 余り 1 | (71) $46 \div 6$
商 7, 余り 4 |
| (18) $77 \div 10$
商 7, 余り 7 | (36) $4 \div 8$
商 0, 余り 4 | (54) $-85 \div 5$
商 -17, 余り 0 | (72) $-12 \div 8$
商 -2, 余り 4 |

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|----------------------------------|-----------------------------------|----------------------------------|----------------------------------|
| (1) $44 \div 10$
商 4, 余り 4 | (19) $2 \div 7$
商 0, 余り 2 | (37) $38 \div 10$
商 3, 余り 8 | (55) $77 \div 10$
商 7, 余り 7 |
| (2) $60 \div 5$
商 12, 余り 0 | (20) $71 \div 4$
商 17, 余り 3 | (38) $-69 \div 8$
商 -9, 余り 3 | (56) $45 \div 7$
商 6, 余り 3 |
| (3) $23 \div 8$
商 2, 余り 7 | (21) $-46 \div 5$
商 -10, 余り 4 | (39) $1 \div 7$
商 0, 余り 1 | (57) $7 \div 7$
商 1, 余り 0 |
| (4) $10 \div 10$
商 1, 余り 0 | (22) $91 \div 4$
商 22, 余り 3 | (40) $-87 \div 3$
商 -29, 余り 0 | (58) $80 \div 8$
商 10, 余り 0 |
| (5) $55 \div 8$
商 6, 余り 7 | (23) $-43 \div 1$
商 -43, 余り 0 | (41) $-64 \div 1$
商 -64, 余り 0 | (59) $76 \div 8$
商 9, 余り 4 |
| (6) $56 \div 5$
商 11, 余り 1 | (24) $-34 \div 8$
商 -5, 余り 6 | (42) $-21 \div 5$
商 -5, 余り 4 | (60) $59 \div 5$
商 11, 余り 4 |
| (7) $-9 \div 6$
商 -2, 余り 3 | (25) $-99 \div 7$
商 -15, 余り 6 | (43) $-77 \div 2$
商 -39, 余り 1 | (61) $74 \div 6$
商 12, 余り 2 |
| (8) $-49 \div 1$
商 -49, 余り 0 | (26) $-23 \div 3$
商 -8, 余り 1 | (44) $43 \div 1$
商 43, 余り 0 | (62) $-48 \div 6$
商 -8, 余り 0 |
| (9) $32 \div 5$
商 6, 余り 2 | (27) $78 \div 6$
商 13, 余り 0 | (45) $13 \div 5$
商 2, 余り 3 | (63) $80 \div 6$
商 13, 余り 2 |
| (10) $-42 \div 2$
商 -21, 余り 0 | (28) $-74 \div 7$
商 -11, 余り 3 | (46) $-1 \div 3$
商 -1, 余り 2 | (64) $-80 \div 2$
商 -40, 余り 0 |
| (11) $-75 \div 9$
商 -9, 余り 6 | (29) $-100 \div 4$
商 -25, 余り 0 | (47) $-73 \div 1$
商 -73, 余り 0 | (65) $-89 \div 8$
商 -12, 余り 7 |
| (12) $64 \div 3$
商 21, 余り 1 | (30) $-25 \div 3$
商 -9, 余り 2 | (48) $62 \div 6$
商 10, 余り 2 | (66) $12 \div 1$
商 12, 余り 0 |
| (13) $75 \div 5$
商 15, 余り 0 | (31) $-1 \div 1$
商 -1, 余り 0 | (49) $-29 \div 5$
商 -6, 余り 1 | (67) $-47 \div 3$
商 -16, 余り 1 |
| (14) $-82 \div 6$
商 -14, 余り 2 | (32) $66 \div 8$
商 8, 余り 2 | (50) $42 \div 6$
商 7, 余り 0 | (68) $-82 \div 2$
商 -41, 余り 0 |
| (15) $26 \div 3$
商 8, 余り 2 | (33) $28 \div 7$
商 4, 余り 0 | (51) $44 \div 4$
商 11, 余り 0 | (69) $23 \div 4$
商 5, 余り 3 |
| (16) $64 \div 3$
商 21, 余り 1 | (34) $49 \div 4$
商 12, 余り 1 | (52) $32 \div 2$
商 16, 余り 0 | (70) $-43 \div 4$
商 -11, 余り 1 |
| (17) $10 \div 1$
商 10, 余り 0 | (35) $-68 \div 3$
商 -23, 余り 1 | (53) $-42 \div 5$
商 -9, 余り 3 | (71) $84 \div 10$
商 8, 余り 4 |
| (18) $19 \div 5$
商 3, 余り 4 | (36) $85 \div 5$
商 17, 余り 0 | (54) $22 \div 10$
商 2, 余り 2 | (72) $94 \div 5$
商 18, 余り 4 |

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|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| (1) $54 \div 10$
商 5, 余り 4 | (19) $-64 \div 3$
商 - 22, 余り 2 | (37) $-43 \div 1$
商 - 43, 余り 0 | (55) $-22 \div 4$
商 - 6, 余り 2 |
| (2) $-94 \div 7$
商 - 14, 余り 4 | (20) $90 \div 2$
商 45, 余り 0 | (38) $10 \div 10$
商 1, 余り 0 | (56) $-76 \div 7$
商 - 11, 余り 1 |
| (3) $14 \div 2$
商 7, 余り 0 | (21) $-67 \div 4$
商 - 17, 余り 1 | (39) $-66 \div 2$
商 - 33, 余り 0 | (57) $59 \div 4$
商 14, 余り 3 |
| (4) $82 \div 3$
商 27, 余り 1 | (22) $-77 \div 4$
商 - 20, 余り 3 | (40) $26 \div 10$
商 2, 余り 6 | (58) $-32 \div 3$
商 - 11, 余り 1 |
| (5) $18 \div 9$
商 2, 余り 0 | (23) $43 \div 1$
商 43, 余り 0 | (41) $-26 \div 4$
商 - 7, 余り 2 | (59) $-14 \div 7$
商 - 2, 余り 0 |
| (6) $-16 \div 6$
商 - 3, 余り 2 | (24) $96 \div 7$
商 13, 余り 5 | (42) $48 \div 10$
商 4, 余り 8 | (60) $-53 \div 2$
商 - 27, 余り 1 |
| (7) $-43 \div 6$
商 - 8, 余り 5 | (25) $-88 \div 6$
商 - 15, 余り 2 | (43) $-32 \div 3$
商 - 11, 余り 1 | (61) $-45 \div 8$
商 - 6, 余り 3 |
| (8) $61 \div 10$
商 6, 余り 1 | (26) $-49 \div 1$
商 - 49, 余り 0 | (44) $-6 \div 5$
商 - 2, 余り 4 | (62) $36 \div 5$
商 7, 余り 1 |
| (9) $100 \div 10$
商 10, 余り 0 | (27) $48 \div 2$
商 24, 余り 0 | (45) $-30 \div 6$
商 - 5, 余り 0 | (63) $37 \div 9$
商 4, 余り 1 |
| (10) $79 \div 1$
商 79, 余り 0 | (28) $71 \div 10$
商 7, 余り 1 | (46) $-75 \div 2$
商 - 38, 余り 1 | (64) $-57 \div 1$
商 - 57, 余り 0 |
| (11) $-80 \div 6$
商 - 14, 余り 4 | (29) $-19 \div 4$
商 - 5, 余り 1 | (47) $-76 \div 10$
商 - 8, 余り 4 | (65) $58 \div 5$
商 11, 余り 3 |
| (12) $28 \div 1$
商 28, 余り 0 | (30) $40 \div 9$
商 4, 余り 4 | (48) $-82 \div 4$
商 - 21, 余り 2 | (66) $-48 \div 1$
商 - 48, 余り 0 |
| (13) $-14 \div 4$
商 - 4, 余り 2 | (31) $67 \div 9$
商 7, 余り 4 | (49) $-89 \div 8$
商 - 12, 余り 7 | (67) $27 \div 9$
商 3, 余り 0 |
| (14) $-75 \div 2$
商 - 38, 余り 1 | (32) $23 \div 3$
商 7, 余り 2 | (50) $27 \div 8$
商 3, 余り 3 | (68) $28 \div 6$
商 4, 余り 4 |
| (15) $-72 \div 7$
商 - 11, 余り 5 | (33) $56 \div 9$
商 6, 余り 2 | (51) $-88 \div 9$
商 - 10, 余り 2 | (69) $-77 \div 6$
商 - 13, 余り 1 |
| (16) $-68 \div 1$
商 - 68, 余り 0 | (34) $40 \div 8$
商 5, 余り 0 | (52) $-20 \div 3$
商 - 7, 余り 1 | (70) $40 \div 10$
商 4, 余り 0 |
| (17) $-26 \div 3$
商 - 9, 余り 1 | (35) $-22 \div 9$
商 - 3, 余り 5 | (53) $12 \div 9$
商 1, 余り 3 | (71) $-65 \div 4$
商 - 17, 余り 3 |
| (18) $25 \div 6$
商 4, 余り 1 | (36) $4 \div 3$
商 1, 余り 1 | (54) $15 \div 6$
商 2, 余り 3 | (72) $41 \div 5$
商 8, 余り 1 |

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|---------------------------------|----------------------------------|----------------------------------|-----------------------------------|
| (1) $29 \div 7$
商 4, 余り 1 | (19) $3 \div 10$
商 0, 余り 3 | (37) $-12 \div 2$
商 -6, 余り 0 | (55) $83 \div 8$
商 10, 余り 3 |
| (2) $1 \div 9$
商 0, 余り 1 | (20) $66 \div 7$
商 9, 余り 3 | (38) $50 \div 7$
商 7, 余り 1 | (56) $30 \div 4$
商 7, 余り 2 |
| (3) $-81 \div 7$
商 -12, 余り 3 | (21) $54 \div 9$
商 6, 余り 0 | (39) $22 \div 2$
商 11, 余り 0 | (57) $-39 \div 1$
商 -39, 余り 0 |
| (4) $38 \div 3$
商 12, 余り 2 | (22) $-11 \div 1$
商 -11, 余り 0 | (40) $59 \div 5$
商 11, 余り 4 | (58) $10 \div 7$
商 1, 余り 3 |
| (5) $-92 \div 9$
商 -11, 余り 7 | (23) $83 \div 3$
商 27, 余り 2 | (41) $-9 \div 3$
商 -3, 余り 0 | (59) $-91 \div 10$
商 -10, 余り 9 |
| (6) $68 \div 5$
商 13, 余り 3 | (24) $-70 \div 9$
商 -8, 余り 2 | (42) $-93 \div 6$
商 -16, 余り 3 | (60) $-76 \div 4$
商 -19, 余り 0 |
| (7) $36 \div 10$
商 3, 余り 6 | (25) $63 \div 7$
商 9, 余り 0 | (43) $-87 \div 2$
商 -44, 余り 1 | (61) $35 \div 7$
商 5, 余り 0 |
| (8) $-63 \div 1$
商 -63, 余り 0 | (26) $57 \div 5$
商 11, 余り 2 | (44) $-85 \div 9$
商 -10, 余り 5 | (62) $59 \div 7$
商 8, 余り 3 |
| (9) $9 \div 6$
商 1, 余り 3 | (27) $-13 \div 8$
商 -2, 余り 3 | (45) $93 \div 5$
商 18, 余り 3 | (63) $11 \div 2$
商 5, 余り 1 |
| (10) $-56 \div 8$
商 -7, 余り 0 | (28) $28 \div 7$
商 4, 余り 0 | (46) $-73 \div 4$
商 -19, 余り 3 | (64) $64 \div 1$
商 64, 余り 0 |
| (11) $56 \div 10$
商 5, 余り 6 | (29) $28 \div 7$
商 4, 余り 0 | (47) $-21 \div 3$
商 -7, 余り 0 | (65) $31 \div 4$
商 7, 余り 3 |
| (12) $54 \div 10$
商 5, 余り 4 | (30) $84 \div 10$
商 8, 余り 4 | (48) $45 \div 8$
商 5, 余り 5 | (66) $73 \div 9$
商 8, 余り 1 |
| (13) $26 \div 10$
商 2, 余り 6 | (31) $24 \div 3$
商 8, 余り 0 | (49) $-1 \div 8$
商 -1, 余り 7 | (67) $-29 \div 10$
商 -3, 余り 1 |
| (14) $16 \div 1$
商 16, 余り 0 | (32) $-20 \div 8$
商 -3, 余り 4 | (50) $-14 \div 8$
商 -2, 余り 2 | (68) $-84 \div 6$
商 -14, 余り 0 |
| (15) $59 \div 2$
商 29, 余り 1 | (33) $18 \div 7$
商 2, 余り 4 | (51) $68 \div 2$
商 34, 余り 0 | (69) $-6 \div 7$
商 -1, 余り 1 |
| (16) $2 \div 2$
商 1, 余り 0 | (34) $-78 \div 3$
商 -26, 余り 0 | (52) $61 \div 2$
商 30, 余り 1 | (70) $87 \div 3$
商 29, 余り 0 |
| (17) $13 \div 8$
商 1, 余り 5 | (35) $21 \div 5$
商 4, 余り 1 | (53) $86 \div 2$
商 43, 余り 0 | (71) $92 \div 10$
商 9, 余り 2 |
| (18) $61 \div 9$
商 6, 余り 7 | (36) $58 \div 8$
商 7, 余り 2 | (54) $-10 \div 3$
商 -4, 余り 2 | (72) $56 \div 1$
商 56, 余り 0 |

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|----------------------------------|-----------------------------------|----------------------------------|----------------------------------|
| (1) $17 \div 10$
商 1, 余り 7 | (19) $57 \div 3$
商 19, 余り 0 | (37) $24 \div 10$
商 2, 余り 4 | (55) $22 \div 4$
商 5, 余り 2 |
| (2) $-77 \div 2$
商 -39, 余り 1 | (20) $45 \div 9$
商 5, 余り 0 | (38) $-73 \div 5$
商 -15, 余り 2 | (56) $100 \div 9$
商 11, 余り 1 |
| (3) $31 \div 9$
商 3, 余り 4 | (21) $4 \div 8$
商 0, 余り 4 | (39) $84 \div 9$
商 9, 余り 3 | (57) $-52 \div 1$
商 -52, 余り 0 |
| (4) $29 \div 1$
商 29, 余り 0 | (22) $79 \div 1$
商 79, 余り 0 | (40) $-96 \div 8$
商 -12, 余り 0 | (58) $-14 \div 2$
商 -7, 余り 0 |
| (5) $23 \div 1$
商 23, 余り 0 | (23) $59 \div 3$
商 19, 余り 2 | (41) $-33 \div 2$
商 -17, 余り 1 | (59) $-2 \div 6$
商 -1, 余り 4 |
| (6) $-18 \div 10$
商 -2, 余り 2 | (24) $-80 \div 4$
商 -20, 余り 0 | (42) $-38 \div 2$
商 -19, 余り 0 | (60) $43 \div 1$
商 43, 余り 0 |
| (7) $-92 \div 10$
商 -10, 余り 8 | (25) $-77 \div 9$
商 -9, 余り 4 | (43) $100 \div 10$
商 10, 余り 0 | (61) $-82 \div 10$
商 -9, 余り 8 |
| (8) $65 \div 10$
商 6, 余り 5 | (26) $-88 \div 7$
商 -13, 余り 3 | (44) $-69 \div 2$
商 -35, 余り 1 | (62) $-4 \div 10$
商 -1, 余り 6 |
| (9) $-82 \div 2$
商 -41, 余り 0 | (27) $-78 \div 6$
商 -13, 余り 0 | (45) $43 \div 6$
商 7, 余り 1 | (63) $4 \div 9$
商 0, 余り 4 |
| (10) $14 \div 2$
商 7, 余り 0 | (28) $47 \div 10$
商 4, 余り 7 | (46) $59 \div 9$
商 6, 余り 5 | (64) $99 \div 10$
商 9, 余り 9 |
| (11) $-95 \div 7$
商 -14, 余り 3 | (29) $27 \div 10$
商 2, 余り 7 | (47) $92 \div 7$
商 13, 余り 1 | (65) $-22 \div 7$
商 -4, 余り 6 |
| (12) $-13 \div 9$
商 -2, 余り 5 | (30) $-10 \div 5$
商 -2, 余り 0 | (48) $19 \div 6$
商 3, 余り 1 | (66) $-38 \div 5$
商 -8, 余り 2 |
| (13) $34 \div 7$
商 4, 余り 6 | (31) $-96 \div 10$
商 -10, 余り 4 | (49) $-89 \div 5$
商 -18, 余り 1 | (67) $-26 \div 5$
商 -6, 余り 4 |
| (14) $69 \div 3$
商 23, 余り 0 | (32) $79 \div 10$
商 7, 余り 9 | (50) $-69 \div 5$
商 -14, 余り 1 | (68) $56 \div 10$
商 5, 余り 6 |
| (15) $92 \div 1$
商 92, 余り 0 | (33) $52 \div 2$
商 26, 余り 0 | (51) $-27 \div 5$
商 -6, 余り 3 | (69) $-6 \div 4$
商 -2, 余り 2 |
| (16) $-30 \div 9$
商 -4, 余り 6 | (34) $94 \div 7$
商 13, 余り 3 | (52) $81 \div 4$
商 20, 余り 1 | (70) $48 \div 2$
商 24, 余り 0 |
| (17) $30 \div 2$
商 15, 余り 0 | (35) $-24 \div 3$
商 -8, 余り 0 | (53) $-35 \div 6$
商 -6, 余り 1 | (71) $-36 \div 10$
商 -4, 余り 4 |
| (18) $6 \div 9$
商 0, 余り 6 | (36) $78 \div 9$
商 8, 余り 6 | (54) $5 \div 7$
商 0, 余り 5 | (72) $-27 \div 10$
商 -3, 余り 3 |

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|---------------------------------------|--|---------------------------------------|---------------------------------------|
| (1) $-37 \div 9$
商 -5 , 余り 8 | (19) $-58 \div 10$
商 -6 , 余り 2 | (37) $74 \div 10$
商 7 , 余り 4 | (55) $-35 \div 3$
商 -12 , 余り 1 |
| (2) $-75 \div 8$
商 -10 , 余り 5 | (20) $78 \div 9$
商 8 , 余り 6 | (38) $62 \div 1$
商 62 , 余り 0 | (56) $87 \div 6$
商 14 , 余り 3 |
| (3) $94 \div 7$
商 13 , 余り 3 | (21) $-11 \div 8$
商 -2 , 余り 5 | (39) $78 \div 8$
商 9 , 余り 6 | (57) $-60 \div 6$
商 -10 , 余り 0 |
| (4) $45 \div 9$
商 5 , 余り 0 | (22) $33 \div 3$
商 11 , 余り 0 | (40) $15 \div 6$
商 2 , 余り 3 | (58) $-5 \div 1$
商 -5 , 余り 0 |
| (5) $5 \div 2$
商 2 , 余り 1 | (23) $16 \div 10$
商 1 , 余り 6 | (41) $-50 \div 1$
商 -50 , 余り 0 | (59) $31 \div 8$
商 3 , 余り 7 |
| (6) $-19 \div 5$
商 -4 , 余り 1 | (24) $-9 \div 6$
商 -2 , 余り 3 | (42) $-60 \div 4$
商 -15 , 余り 0 | (60) $-72 \div 7$
商 -11 , 余り 5 |
| (7) $91 \div 8$
商 11 , 余り 3 | (25) $66 \div 7$
商 9 , 余り 3 | (43) $2 \div 3$
商 0 , 余り 2 | (61) $58 \div 6$
商 9 , 余り 4 |
| (8) $65 \div 4$
商 16 , 余り 1 | (26) $-36 \div 4$
商 -9 , 余り 0 | (44) $80 \div 2$
商 40 , 余り 0 | (62) $-38 \div 2$
商 -19 , 余り 0 |
| (9) $39 \div 2$
商 19 , 余り 1 | (27) $-14 \div 10$
商 -2 , 余り 6 | (45) $8 \div 1$
商 8 , 余り 0 | (63) $89 \div 3$
商 29 , 余り 2 |
| (10) $-45 \div 6$
商 -8 , 余り 3 | (28) $-78 \div 10$
商 -8 , 余り 2 | (46) $79 \div 1$
商 79 , 余り 0 | (64) $-31 \div 1$
商 -31 , 余り 0 |
| (11) $27 \div 8$
商 3 , 余り 3 | (29) $-100 \div 7$
商 -15 , 余り 5 | (47) $68 \div 1$
商 68 , 余り 0 | (65) $-65 \div 5$
商 -13 , 余り 0 |
| (12) $-87 \div 6$
商 -15 , 余り 3 | (30) $-18 \div 7$
商 -3 , 余り 3 | (48) $-41 \div 6$
商 -7 , 余り 1 | (66) $16 \div 10$
商 1 , 余り 6 |
| (13) $-12 \div 3$
商 -4 , 余り 0 | (31) $85 \div 6$
商 14 , 余り 1 | (49) $-17 \div 7$
商 -3 , 余り 4 | (67) $-20 \div 4$
商 -5 , 余り 0 |
| (14) $5 \div 3$
商 1 , 余り 2 | (32) $85 \div 2$
商 42 , 余り 1 | (50) $-85 \div 6$
商 -15 , 余り 5 | (68) $39 \div 3$
商 13 , 余り 0 |
| (15) $-61 \div 6$
商 -11 , 余り 5 | (33) $-1 \div 7$
商 -1 , 余り 6 | (51) $-25 \div 2$
商 -13 , 余り 1 | (69) $-33 \div 3$
商 -11 , 余り 0 |
| (16) $60 \div 7$
商 8 , 余り 4 | (34) $-86 \div 9$
商 -10 , 余り 4 | (52) $-2 \div 6$
商 -1 , 余り 4 | (70) $41 \div 9$
商 4 , 余り 5 |
| (17) $20 \div 1$
商 20 , 余り 0 | (35) $82 \div 4$
商 20 , 余り 2 | (53) $34 \div 5$
商 6 , 余り 4 | (71) $-26 \div 8$
商 -4 , 余り 6 |
| (18) $-61 \div 2$
商 -31 , 余り 1 | (36) $-8 \div 6$
商 -2 , 余り 4 | (54) $-23 \div 10$
商 -3 , 余り 7 | (72) $76 \div 2$
商 38 , 余り 0 |

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|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| (1) $-72 \div 5$
商 -15 , 余り 3 | (19) $-17 \div 4$
商 -5 , 余り 3 | (37) $-30 \div 3$
商 -10 , 余り 0 | (55) $39 \div 6$
商 6 , 余り 3 |
| (2) $96 \div 1$
商 96 , 余り 0 | (20) $-3 \div 9$
商 -1 , 余り 6 | (38) $89 \div 8$
商 11 , 余り 1 | (56) $36 \div 10$
商 3 , 余り 6 |
| (3) $68 \div 4$
商 17 , 余り 0 | (21) $-66 \div 5$
商 -14 , 余り 4 | (39) $-24 \div 8$
商 -3 , 余り 0 | (57) $48 \div 2$
商 24 , 余り 0 |
| (4) $44 \div 2$
商 22 , 余り 0 | (22) $33 \div 6$
商 5 , 余り 3 | (40) $-90 \div 9$
商 -10 , 余り 0 | (58) $55 \div 7$
商 7 , 余り 6 |
| (5) $7 \div 1$
商 7 , 余り 0 | (23) $99 \div 5$
商 19 , 余り 4 | (41) $-32 \div 2$
商 -16 , 余り 0 | (59) $71 \div 1$
商 71 , 余り 0 |
| (6) $22 \div 4$
商 5 , 余り 2 | (24) $-70 \div 7$
商 -10 , 余り 0 | (42) $-67 \div 7$
商 -10 , 余り 3 | (60) $-90 \div 8$
商 -12 , 余り 6 |
| (7) $-30 \div 4$
商 -8 , 余り 2 | (25) $58 \div 9$
商 6 , 余り 4 | (43) $20 \div 2$
商 10 , 余り 0 | (61) $75 \div 2$
商 37 , 余り 1 |
| (8) $-8 \div 6$
商 -2 , 余り 4 | (26) $12 \div 8$
商 1 , 余り 4 | (44) $43 \div 1$
商 43 , 余り 0 | (62) $-20 \div 9$
商 -3 , 余り 7 |
| (9) $86 \div 7$
商 12 , 余り 2 | (27) $-66 \div 9$
商 -8 , 余り 6 | (45) $-78 \div 1$
商 -78 , 余り 0 | (63) $-83 \div 7$
商 -12 , 余り 1 |
| (10) $-86 \div 10$
商 -9 , 余り 4 | (28) $-75 \div 9$
商 -9 , 余り 6 | (46) $12 \div 2$
商 6 , 余り 0 | (64) $66 \div 10$
商 6 , 余り 6 |
| (11) $-68 \div 6$
商 -12 , 余り 4 | (29) $63 \div 8$
商 7 , 余り 7 | (47) $74 \div 10$
商 7 , 余り 4 | (65) $-12 \div 4$
商 -3 , 余り 0 |
| (12) $-44 \div 9$
商 -5 , 余り 1 | (30) $-15 \div 5$
商 -3 , 余り 0 | (48) $10 \div 5$
商 2 , 余り 0 | (66) $-84 \div 2$
商 -42 , 余り 0 |
| (13) $63 \div 7$
商 9 , 余り 0 | (31) $-4 \div 4$
商 -1 , 余り 0 | (49) $44 \div 10$
商 4 , 余り 4 | (67) $98 \div 10$
商 9 , 余り 8 |
| (14) $67 \div 8$
商 8 , 余り 3 | (32) $-54 \div 3$
商 -18 , 余り 0 | (50) $-97 \div 4$
商 -25 , 余り 3 | (68) $-20 \div 1$
商 -20 , 余り 0 |
| (15) $-43 \div 8$
商 -6 , 余り 5 | (33) $77 \div 10$
商 7 , 余り 7 | (51) $-18 \div 2$
商 -9 , 余り 0 | (69) $39 \div 1$
商 39 , 余り 0 |
| (16) $25 \div 2$
商 12 , 余り 1 | (34) $20 \div 6$
商 3 , 余り 2 | (52) $50 \div 8$
商 6 , 余り 2 | (70) $-78 \div 7$
商 -12 , 余り 6 |
| (17) $4 \div 8$
商 0 , 余り 4 | (35) $-4 \div 3$
商 -2 , 余り 2 | (53) $39 \div 3$
商 13 , 余り 0 | (71) $72 \div 1$
商 72 , 余り 0 |
| (18) $-4 \div 8$
商 -1 , 余り 4 | (36) $-33 \div 3$
商 -11 , 余り 0 | (54) $87 \div 6$
商 14 , 余り 3 | (72) $-45 \div 3$
商 -15 , 余り 0 |

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|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| (1) $-56 \div 6$
商 -10 , 余り 4 | (19) $-69 \div 10$
商 -7 , 余り 1 | (37) $-70 \div 4$
商 -18 , 余り 2 | (55) $-36 \div 9$
商 -4 , 余り 0 |
| (2) $92 \div 1$
商 92 , 余り 0 | (20) $-34 \div 3$
商 -12 , 余り 2 | (38) $45 \div 8$
商 5 , 余り 5 | (56) $73 \div 2$
商 36 , 余り 1 |
| (3) $-16 \div 8$
商 -2 , 余り 0 | (21) $34 \div 4$
商 8 , 余り 2 | (39) $3 \div 2$
商 1 , 余り 1 | (57) $-56 \div 5$
商 -12 , 余り 4 |
| (4) $-34 \div 4$
商 -9 , 余り 2 | (22) $59 \div 1$
商 59 , 余り 0 | (40) $-40 \div 2$
商 -20 , 余り 0 | (58) $-81 \div 3$
商 -27 , 余り 0 |
| (5) $-76 \div 4$
商 -19 , 余り 0 | (23) $-91 \div 5$
商 -19 , 余り 4 | (41) $41 \div 5$
商 8 , 余り 1 | (59) $93 \div 6$
商 15 , 余り 3 |
| (6) $35 \div 8$
商 4 , 余り 3 | (24) $-78 \div 10$
商 -8 , 余り 2 | (42) $9 \div 1$
商 9 , 余り 0 | (60) $-67 \div 3$
商 -23 , 余り 2 |
| (7) $-82 \div 8$
商 -11 , 余り 6 | (25) $71 \div 2$
商 35 , 余り 1 | (43) $-6 \div 5$
商 -2 , 余り 4 | (61) $-48 \div 2$
商 -24 , 余り 0 |
| (8) $-33 \div 5$
商 -7 , 余り 2 | (26) $-51 \div 4$
商 -13 , 余り 1 | (44) $15 \div 10$
商 1 , 余り 5 | (62) $0 \div 6$
商 0 , 余り 0 |
| (9) $-96 \div 9$
商 -11 , 余り 3 | (27) $-67 \div 1$
商 -67 , 余り 0 | (45) $-71 \div 2$
商 -36 , 余り 1 | (63) $47 \div 1$
商 47 , 余り 0 |
| (10) $-92 \div 7$
商 -14 , 余り 6 | (28) $71 \div 4$
商 17 , 余り 3 | (46) $31 \div 5$
商 6 , 余り 1 | (64) $-70 \div 8$
商 -9 , 余り 2 |
| (11) $26 \div 5$
商 5 , 余り 1 | (29) $-88 \div 4$
商 -22 , 余り 0 | (47) $-73 \div 5$
商 -15 , 余り 2 | (65) $12 \div 8$
商 1 , 余り 4 |
| (12) $-45 \div 10$
商 -5 , 余り 5 | (30) $-20 \div 3$
商 -7 , 余り 1 | (48) $15 \div 7$
商 2 , 余り 1 | (66) $-10 \div 7$
商 -2 , 余り 4 |
| (13) $-60 \div 2$
商 -30 , 余り 0 | (31) $-77 \div 3$
商 -26 , 余り 1 | (49) $-92 \div 6$
商 -16 , 余り 4 | (67) $-43 \div 4$
商 -11 , 余り 1 |
| (14) $-70 \div 2$
商 -35 , 余り 0 | (32) $-46 \div 3$
商 -16 , 余り 2 | (50) $-94 \div 2$
商 -47 , 余り 0 | (68) $-82 \div 6$
商 -14 , 余り 2 |
| (15) $4 \div 1$
商 4 , 余り 0 | (33) $57 \div 10$
商 5 , 余り 7 | (51) $79 \div 1$
商 79 , 余り 0 | (69) $51 \div 9$
商 5 , 余り 6 |
| (16) $89 \div 9$
商 9 , 余り 8 | (34) $23 \div 6$
商 3 , 余り 5 | (52) $89 \div 4$
商 22 , 余り 1 | (70) $100 \div 9$
商 11 , 余り 1 |
| (17) $63 \div 6$
商 10 , 余り 3 | (35) $-53 \div 9$
商 -6 , 余り 1 | (53) $33 \div 5$
商 6 , 余り 3 | (71) $29 \div 2$
商 14 , 余り 1 |
| (18) $1 \div 5$
商 0 , 余り 1 | (36) $30 \div 5$
商 6 , 余り 0 | (54) $31 \div 2$
商 15 , 余り 1 | (72) $75 \div 3$
商 25 , 余り 0 |

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|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| (1) $48 \div 9$
商 5, 余り 3 | (19) $-66 \div 3$
商 - 22, 余り 0 | (37) $-16 \div 6$
商 - 3, 余り 2 | (55) $96 \div 5$
商 19, 余り 1 |
| (2) $-32 \div 10$
商 - 4, 余り 8 | (20) $71 \div 6$
商 11, 余り 5 | (38) $-63 \div 9$
商 - 7, 余り 0 | (56) $45 \div 2$
商 22, 余り 1 |
| (3) $-1 \div 7$
商 - 1, 余り 6 | (21) $88 \div 6$
商 14, 余り 4 | (39) $-55 \div 1$
商 - 55, 余り 0 | (57) $67 \div 1$
商 67, 余り 0 |
| (4) $77 \div 6$
商 12, 余り 5 | (22) $-31 \div 10$
商 - 4, 余り 9 | (40) $87 \div 5$
商 17, 余り 2 | (58) $4 \div 4$
商 1, 余り 0 |
| (5) $53 \div 7$
商 7, 余り 4 | (23) $76 \div 9$
商 8, 余り 4 | (41) $-12 \div 7$
商 - 2, 余り 2 | (59) $69 \div 8$
商 8, 余り 5 |
| (6) $42 \div 10$
商 4, 余り 2 | (24) $74 \div 1$
商 74, 余り 0 | (42) $-12 \div 9$
商 - 2, 余り 6 | (60) $70 \div 5$
商 14, 余り 0 |
| (7) $8 \div 5$
商 1, 余り 3 | (25) $13 \div 10$
商 1, 余り 3 | (43) $67 \div 8$
商 8, 余り 3 | (61) $31 \div 7$
商 4, 余り 3 |
| (8) $42 \div 6$
商 7, 余り 0 | (26) $-73 \div 5$
商 - 15, 余り 2 | (44) $-88 \div 3$
商 - 30, 余り 2 | (62) $-14 \div 2$
商 - 7, 余り 0 |
| (9) $-33 \div 3$
商 - 11, 余り 0 | (27) $73 \div 5$
商 14, 余り 3 | (45) $-61 \div 5$
商 - 13, 余り 4 | (63) $-63 \div 4$
商 - 16, 余り 1 |
| (10) $34 \div 7$
商 4, 余り 6 | (28) $-93 \div 6$
商 - 16, 余り 3 | (46) $82 \div 6$
商 13, 余り 4 | (64) $-66 \div 4$
商 - 17, 余り 2 |
| (11) $-59 \div 4$
商 - 15, 余り 1 | (29) $5 \div 9$
商 0, 余り 5 | (47) $-50 \div 7$
商 - 8, 余り 6 | (65) $30 \div 5$
商 6, 余り 0 |
| (12) $69 \div 10$
商 6, 余り 9 | (30) $68 \div 2$
商 34, 余り 0 | (48) $91 \div 1$
商 91, 余り 0 | (66) $8 \div 6$
商 1, 余り 2 |
| (13) $-75 \div 10$
商 - 8, 余り 5 | (31) $84 \div 7$
商 12, 余り 0 | (49) $-39 \div 5$
商 - 8, 余り 1 | (67) $-45 \div 2$
商 - 23, 余り 1 |
| (14) $4 \div 6$
商 0, 余り 4 | (32) $-86 \div 4$
商 - 22, 余り 2 | (50) $-67 \div 9$
商 - 8, 余り 5 | (68) $-20 \div 10$
商 - 2, 余り 0 |
| (15) $-22 \div 6$
商 - 4, 余り 2 | (33) $88 \div 10$
商 8, 余り 8 | (51) $0 \div 8$
商 0, 余り 0 | (69) $-99 \div 4$
商 - 25, 余り 1 |
| (16) $-55 \div 5$
商 - 11, 余り 0 | (34) $-15 \div 5$
商 - 3, 余り 0 | (52) $-32 \div 5$
商 - 7, 余り 3 | (70) $-83 \div 1$
商 - 83, 余り 0 |
| (17) $-33 \div 2$
商 - 17, 余り 1 | (35) $-22 \div 9$
商 - 3, 余り 5 | (53) $-22 \div 2$
商 - 11, 余り 0 | (71) $95 \div 2$
商 47, 余り 1 |
| (18) $-31 \div 9$
商 - 4, 余り 5 | (36) $-46 \div 4$
商 - 12, 余り 2 | (54) $-27 \div 6$
商 - 5, 余り 3 | (72) $95 \div 9$
商 10, 余り 5 |

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|---------------------------------------|---|---------------------------------------|--|
| (1) $-62 \div 3$
商 -21 , 余り 1 | (19) $-92 \div 1$
商 -92 , 余り 0 | (37) $22 \div 10$
商 2 , 余り 2 | (55) $93 \div 4$
商 23 , 余り 1 |
| (2) $51 \div 4$
商 12 , 余り 3 | (20) $36 \div 6$
商 6 , 余り 0 | (38) $2 \div 6$
商 0 , 余り 2 | (56) $88 \div 6$
商 14 , 余り 4 |
| (3) $38 \div 5$
商 7 , 余り 3 | (21) $62 \div 4$
商 15 , 余り 2 | (39) $61 \div 2$
商 30 , 余り 1 | (57) $62 \div 4$
商 15 , 余り 2 |
| (4) $76 \div 8$
商 9 , 余り 4 | (22) $-91 \div 4$
商 -23 , 余り 1 | (40) $97 \div 1$
商 97 , 余り 0 | (58) $-22 \div 1$
商 -22 , 余り 0 |
| (5) $-53 \div 7$
商 -8 , 余り 3 | (23) $3 \div 6$
商 0 , 余り 3 | (41) $-59 \div 3$
商 -20 , 余り 1 | (59) $38 \div 7$
商 5 , 余り 3 |
| (6) $-49 \div 6$
商 -9 , 余り 5 | (24) $67 \div 2$
商 33 , 余り 1 | (42) $-15 \div 2$
商 -8 , 余り 1 | (60) $-98 \div 5$
商 -20 , 余り 2 |
| (7) $-62 \div 2$
商 -31 , 余り 0 | (25) $68 \div 2$
商 34 , 余り 0 | (43) $29 \div 1$
商 29 , 余り 0 | (61) $-11 \div 5$
商 -3 , 余り 4 |
| (8) $63 \div 3$
商 21 , 余り 0 | (26) $87 \div 5$
商 17 , 余り 2 | (44) $57 \div 5$
商 11 , 余り 2 | (62) $-100 \div 4$
商 -25 , 余り 0 |
| (9) $90 \div 5$
商 18 , 余り 0 | (27) $89 \div 3$
商 29 , 余り 2 | (45) $34 \div 7$
商 4 , 余り 6 | (63) $3 \div 4$
商 0 , 余り 3 |
| (10) $-56 \div 7$
商 -8 , 余り 0 | (28) $3 \div 1$
商 3 , 余り 0 | (46) $-15 \div 8$
商 -2 , 余り 1 | (64) $93 \div 8$
商 11 , 余り 5 |
| (11) $90 \div 7$
商 12 , 余り 6 | (29) $-2 \div 8$
商 -1 , 余り 6 | (47) $73 \div 4$
商 18 , 余り 1 | (65) $-51 \div 7$
商 -8 , 余り 5 |
| (12) $52 \div 5$
商 10 , 余り 2 | (30) $-36 \div 9$
商 -4 , 余り 0 | (48) $-99 \div 7$
商 -15 , 余り 6 | (66) $-94 \div 8$
商 -12 , 余り 2 |
| (13) $72 \div 7$
商 10 , 余り 2 | (31) $-100 \div 1$
商 -100 , 余り 0 | (49) $-38 \div 9$
商 -5 , 余り 7 | (67) $43 \div 6$
商 7 , 余り 1 |
| (14) $-77 \div 7$
商 -11 , 余り 0 | (32) $6 \div 1$
商 6 , 余り 0 | (50) $7 \div 1$
商 7 , 余り 0 | (68) $-70 \div 8$
商 -9 , 余り 2 |
| (15) $10 \div 1$
商 10 , 余り 0 | (33) $-1 \div 2$
商 -1 , 余り 1 | (51) $-49 \div 7$
商 -7 , 余り 0 | (69) $-93 \div 7$
商 -14 , 余り 5 |
| (16) $10 \div 4$
商 2 , 余り 2 | (34) $59 \div 9$
商 6 , 余り 5 | (52) $71 \div 5$
商 14 , 余り 1 | (70) $15 \div 5$
商 3 , 余り 0 |
| (17) $-51 \div 2$
商 -26 , 余り 1 | (35) $-27 \div 7$
商 -4 , 余り 1 | (53) $35 \div 8$
商 4 , 余り 3 | (71) $-85 \div 7$
商 -13 , 余り 6 |
| (18) $-24 \div 2$
商 -12 , 余り 0 | (36) $-50 \div 8$
商 -7 , 余り 6 | (54) $-92 \div 4$
商 -23 , 余り 0 | (72) $-20 \div 8$
商 -3 , 余り 4 |

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|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| (1) $-87 \div 8$
商 -11 , 余り 1 | (19) $-15 \div 3$
商 -5 , 余り 0 | (37) $-16 \div 8$
商 -2 , 余り 0 | (55) $-61 \div 2$
商 -31 , 余り 1 |
| (2) $50 \div 3$
商 16 , 余り 2 | (20) $87 \div 4$
商 21 , 余り 3 | (38) $-30 \div 7$
商 -5 , 余り 5 | (56) $-61 \div 8$
商 -8 , 余り 3 |
| (3) $-7 \div 4$
商 -2 , 余り 1 | (21) $59 \div 6$
商 9 , 余り 5 | (39) $64 \div 6$
商 10 , 余り 4 | (57) $-93 \div 8$
商 -12 , 余り 3 |
| (4) $71 \div 8$
商 8 , 余り 7 | (22) $5 \div 10$
商 0 , 余り 5 | (40) $-44 \div 9$
商 -5 , 余り 1 | (58) $73 \div 3$
商 24 , 余り 1 |
| (5) $-29 \div 6$
商 -5 , 余り 1 | (23) $-16 \div 10$
商 -2 , 余り 4 | (41) $15 \div 9$
商 1 , 余り 6 | (59) $-17 \div 3$
商 -6 , 余り 1 |
| (6) $29 \div 5$
商 5 , 余り 4 | (24) $-82 \div 10$
商 -9 , 余り 8 | (42) $58 \div 7$
商 8 , 余り 2 | (60) $37 \div 7$
商 5 , 余り 2 |
| (7) $56 \div 6$
商 9 , 余り 2 | (25) $-16 \div 4$
商 -4 , 余り 0 | (43) $-81 \div 9$
商 -9 , 余り 0 | (61) $-74 \div 6$
商 -13 , 余り 4 |
| (8) $-57 \div 6$
商 -10 , 余り 3 | (26) $-46 \div 5$
商 -10 , 余り 4 | (44) $-1 \div 6$
商 -1 , 余り 5 | (62) $-82 \div 6$
商 -14 , 余り 2 |
| (9) $96 \div 4$
商 24 , 余り 0 | (27) $14 \div 2$
商 7 , 余り 0 | (45) $78 \div 6$
商 13 , 余り 0 | (63) $6 \div 2$
商 3 , 余り 0 |
| (10) $-98 \div 4$
商 -25 , 余り 2 | (28) $96 \div 9$
商 10 , 余り 6 | (46) $-91 \div 7$
商 -13 , 余り 0 | (64) $8 \div 7$
商 1 , 余り 1 |
| (11) $2 \div 6$
商 0 , 余り 2 | (29) $80 \div 5$
商 16 , 余り 0 | (47) $70 \div 6$
商 11 , 余り 4 | (65) $-91 \div 3$
商 -31 , 余り 2 |
| (12) $92 \div 10$
商 9 , 余り 2 | (30) $-56 \div 4$
商 -14 , 余り 0 | (48) $-71 \div 8$
商 -9 , 余り 1 | (66) $40 \div 1$
商 40 , 余り 0 |
| (13) $57 \div 6$
商 9 , 余り 3 | (31) $93 \div 4$
商 23 , 余り 1 | (49) $-98 \div 8$
商 -13 , 余り 6 | (67) $81 \div 2$
商 40 , 余り 1 |
| (14) $-42 \div 3$
商 -14 , 余り 0 | (32) $-96 \div 8$
商 -12 , 余り 0 | (50) $42 \div 9$
商 4 , 余り 6 | (68) $-31 \div 6$
商 -6 , 余り 5 |
| (15) $83 \div 10$
商 8 , 余り 3 | (33) $79 \div 3$
商 26 , 余り 1 | (51) $-43 \div 7$
商 -7 , 余り 6 | (69) $65 \div 6$
商 10 , 余り 5 |
| (16) $-44 \div 6$
商 -8 , 余り 4 | (34) $-55 \div 2$
商 -28 , 余り 1 | (52) $-83 \div 6$
商 -14 , 余り 1 | (70) $7 \div 4$
商 1 , 余り 3 |
| (17) $-42 \div 9$
商 -5 , 余り 3 | (35) $-90 \div 5$
商 -18 , 余り 0 | (53) $-80 \div 5$
商 -16 , 余り 0 | (71) $89 \div 2$
商 44 , 余り 1 |
| (18) $9 \div 1$
商 9 , 余り 0 | (36) $89 \div 8$
商 11 , 余り 1 | (54) $-71 \div 5$
商 -15 , 余り 4 | (72) $-37 \div 10$
商 -4 , 余り 3 |

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|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| (1) $29 \div 1$
商 29, 余り 0 | (19) $-51 \div 6$
商 -9, 余り 3 | (37) $63 \div 10$
商 6, 余り 3 | (55) $82 \div 1$
商 82, 余り 0 |
| (2) $53 \div 10$
商 5, 余り 3 | (20) $79 \div 6$
商 13, 余り 1 | (38) $74 \div 2$
商 37, 余り 0 | (56) $-12 \div 4$
商 -3, 余り 0 |
| (3) $-98 \div 4$
商 -25, 余り 2 | (21) $97 \div 7$
商 13, 余り 6 | (39) $46 \div 5$
商 9, 余り 1 | (57) $93 \div 3$
商 31, 余り 0 |
| (4) $-99 \div 10$
商 -10, 余り 1 | (22) $99 \div 7$
商 14, 余り 1 | (40) $73 \div 5$
商 14, 余り 3 | (58) $-28 \div 2$
商 -14, 余り 0 |
| (5) $3 \div 2$
商 1, 余り 1 | (23) $63 \div 6$
商 10, 余り 3 | (41) $55 \div 9$
商 6, 余り 1 | (59) $-26 \div 3$
商 -9, 余り 1 |
| (6) $85 \div 7$
商 12, 余り 1 | (24) $50 \div 6$
商 8, 余り 2 | (42) $94 \div 4$
商 23, 余り 2 | (60) $-11 \div 1$
商 -11, 余り 0 |
| (7) $69 \div 2$
商 34, 余り 1 | (25) $-2 \div 9$
商 -1, 余り 7 | (43) $98 \div 2$
商 49, 余り 0 | (61) $31 \div 3$
商 10, 余り 1 |
| (8) $92 \div 9$
商 10, 余り 2 | (26) $59 \div 4$
商 14, 余り 3 | (44) $30 \div 1$
商 30, 余り 0 | (62) $8 \div 4$
商 2, 余り 0 |
| (9) $1 \div 10$
商 0, 余り 1 | (27) $-92 \div 6$
商 -16, 余り 4 | (45) $-14 \div 4$
商 -4, 余り 2 | (63) $-58 \div 1$
商 -58, 余り 0 |
| (10) $67 \div 8$
商 8, 余り 3 | (28) $5 \div 1$
商 5, 余り 0 | (46) $-96 \div 8$
商 -12, 余り 0 | (64) $-86 \div 6$
商 -15, 余り 4 |
| (11) $-36 \div 8$
商 -5, 余り 4 | (29) $-58 \div 6$
商 -10, 余り 2 | (47) $-74 \div 1$
商 -74, 余り 0 | (65) $-73 \div 8$
商 -10, 余り 7 |
| (12) $-71 \div 6$
商 -12, 余り 1 | (30) $-96 \div 7$
商 -14, 余り 2 | (48) $0 \div 6$
商 0, 余り 0 | (66) $-45 \div 6$
商 -8, 余り 3 |
| (13) $-75 \div 7$
商 -11, 余り 2 | (31) $9 \div 8$
商 1, 余り 1 | (49) $52 \div 10$
商 5, 余り 2 | (67) $-63 \div 3$
商 -21, 余り 0 |
| (14) $73 \div 4$
商 18, 余り 1 | (32) $64 \div 2$
商 32, 余り 0 | (50) $-86 \div 5$
商 -18, 余り 4 | (68) $-71 \div 4$
商 -18, 余り 1 |
| (15) $-6 \div 10$
商 -1, 余り 4 | (33) $82 \div 7$
商 11, 余り 5 | (51) $17 \div 4$
商 4, 余り 1 | (69) $49 \div 10$
商 4, 余り 9 |
| (16) $40 \div 4$
商 10, 余り 0 | (34) $51 \div 7$
商 7, 余り 2 | (52) $-69 \div 5$
商 -14, 余り 1 | (70) $-19 \div 10$
商 -2, 余り 1 |
| (17) $-1 \div 1$
商 -1, 余り 0 | (35) $83 \div 3$
商 27, 余り 2 | (53) $40 \div 3$
商 13, 余り 1 | (71) $69 \div 4$
商 17, 余り 1 |
| (18) $34 \div 6$
商 5, 余り 4 | (36) $95 \div 7$
商 13, 余り 4 | (54) $14 \div 6$
商 2, 余り 2 | (72) $-43 \div 10$
商 -5, 余り 7 |

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|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| (1) $86 \div 3$
商 28, 余り 2 | (19) $-99 \div 4$
商 -25, 余り 1 | (37) $40 \div 3$
商 13, 余り 1 | (55) $-29 \div 5$
商 -6, 余り 1 |
| (2) $71 \div 4$
商 17, 余り 3 | (20) $13 \div 4$
商 3, 余り 1 | (38) $59 \div 6$
商 9, 余り 5 | (56) $-6 \div 8$
商 -1, 余り 2 |
| (3) $35 \div 8$
商 4, 余り 3 | (21) $-37 \div 3$
商 -13, 余り 2 | (39) $-70 \div 2$
商 -35, 余り 0 | (57) $88 \div 7$
商 12, 余り 4 |
| (4) $78 \div 6$
商 13, 余り 0 | (22) $-92 \div 2$
商 -46, 余り 0 | (40) $18 \div 1$
商 18, 余り 0 | (58) $42 \div 10$
商 4, 余り 2 |
| (5) $-71 \div 9$
商 -8, 余り 1 | (23) $54 \div 2$
商 27, 余り 0 | (41) $-38 \div 9$
商 -5, 余り 7 | (59) $-29 \div 7$
商 -5, 余り 6 |
| (6) $-58 \div 2$
商 -29, 余り 0 | (24) $100 \div 4$
商 25, 余り 0 | (42) $-66 \div 2$
商 -33, 余り 0 | (60) $1 \div 6$
商 0, 余り 1 |
| (7) $58 \div 7$
商 8, 余り 2 | (25) $-79 \div 4$
商 -20, 余り 1 | (43) $-92 \div 7$
商 -14, 余り 6 | (61) $-45 \div 8$
商 -6, 余り 3 |
| (8) $-77 \div 7$
商 -11, 余り 0 | (26) $-18 \div 1$
商 -18, 余り 0 | (44) $47 \div 2$
商 23, 余り 1 | (62) $81 \div 6$
商 13, 余り 3 |
| (9) $3 \div 2$
商 1, 余り 1 | (27) $53 \div 7$
商 7, 余り 4 | (45) $-75 \div 9$
商 -9, 余り 6 | (63) $70 \div 9$
商 7, 余り 7 |
| (10) $97 \div 8$
商 12, 余り 1 | (28) $-98 \div 4$
商 -25, 余り 2 | (46) $-92 \div 6$
商 -16, 余り 4 | (64) $42 \div 6$
商 7, 余り 0 |
| (11) $-62 \div 7$
商 -9, 余り 1 | (29) $43 \div 1$
商 43, 余り 0 | (47) $92 \div 10$
商 9, 余り 2 | (65) $-99 \div 1$
商 -99, 余り 0 |
| (12) $-79 \div 7$
商 -12, 余り 5 | (30) $-90 \div 8$
商 -12, 余り 6 | (48) $-32 \div 3$
商 -11, 余り 1 | (66) $15 \div 3$
商 5, 余り 0 |
| (13) $-48 \div 2$
商 -24, 余り 0 | (31) $44 \div 1$
商 44, 余り 0 | (49) $9 \div 1$
商 9, 余り 0 | (67) $27 \div 5$
商 5, 余り 2 |
| (14) $90 \div 10$
商 9, 余り 0 | (32) $42 \div 2$
商 21, 余り 0 | (50) $32 \div 7$
商 4, 余り 4 | (68) $56 \div 1$
商 56, 余り 0 |
| (15) $76 \div 4$
商 19, 余り 0 | (33) $-68 \div 4$
商 -17, 余り 0 | (51) $-10 \div 3$
商 -4, 余り 2 | (69) $-45 \div 2$
商 -23, 余り 1 |
| (16) $-85 \div 1$
商 -85, 余り 0 | (34) $25 \div 4$
商 6, 余り 1 | (52) $88 \div 5$
商 17, 余り 3 | (70) $-17 \div 2$
商 -9, 余り 1 |
| (17) $-13 \div 10$
商 -2, 余り 7 | (35) $3 \div 6$
商 0, 余り 3 | (53) $-85 \div 4$
商 -22, 余り 3 | (71) $29 \div 4$
商 7, 余り 1 |
| (18) $12 \div 1$
商 12, 余り 0 | (36) $-60 \div 10$
商 -6, 余り 0 | (54) $92 \div 2$
商 46, 余り 0 | (72) $-24 \div 8$
商 -3, 余り 0 |

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|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| (1) $35 \div 10$
商 3, 余り 5 | (19) $-68 \div 5$
商 -14, 余り 2 | (37) $90 \div 3$
商 30, 余り 0 | (55) $-44 \div 5$
商 -9, 余り 1 |
| (2) $38 \div 10$
商 3, 余り 8 | (20) $-87 \div 8$
商 -11, 余り 1 | (38) $28 \div 4$
商 7, 余り 0 | (56) $41 \div 4$
商 10, 余り 1 |
| (3) $-62 \div 3$
商 -21, 余り 1 | (21) $82 \div 7$
商 11, 余り 5 | (39) $1 \div 4$
商 0, 余り 1 | (57) $-53 \div 3$
商 -18, 余り 1 |
| (4) $-51 \div 2$
商 -26, 余り 1 | (22) $-2 \div 4$
商 -1, 余り 2 | (40) $-35 \div 8$
商 -5, 余り 5 | (58) $-52 \div 1$
商 -52, 余り 0 |
| (5) $65 \div 5$
商 13, 余り 0 | (23) $-97 \div 1$
商 -97, 余り 0 | (41) $-97 \div 3$
商 -33, 余り 2 | (59) $18 \div 4$
商 4, 余り 2 |
| (6) $-87 \div 4$
商 -22, 余り 1 | (24) $-27 \div 10$
商 -3, 余り 3 | (42) $-32 \div 9$
商 -4, 余り 4 | (60) $-28 \div 9$
商 -4, 余り 8 |
| (7) $84 \div 5$
商 16, 余り 4 | (25) $92 \div 7$
商 13, 余り 1 | (43) $34 \div 10$
商 3, 余り 4 | (61) $72 \div 2$
商 36, 余り 0 |
| (8) $25 \div 5$
商 5, 余り 0 | (26) $-89 \div 9$
商 -10, 余り 1 | (44) $-10 \div 6$
商 -2, 余り 2 | (62) $-72 \div 10$
商 -8, 余り 8 |
| (9) $-87 \div 5$
商 -18, 余り 3 | (27) $2 \div 8$
商 0, 余り 2 | (45) $-52 \div 7$
商 -8, 余り 4 | (63) $-8 \div 7$
商 -2, 余り 6 |
| (10) $91 \div 5$
商 18, 余り 1 | (28) $-77 \div 5$
商 -16, 余り 3 | (46) $-55 \div 8$
商 -7, 余り 1 | (64) $9 \div 7$
商 1, 余り 2 |
| (11) $-82 \div 2$
商 -41, 余り 0 | (29) $-33 \div 8$
商 -5, 余り 7 | (47) $-74 \div 7$
商 -11, 余り 3 | (65) $-9 \div 4$
商 -3, 余り 3 |
| (12) $-82 \div 2$
商 -41, 余り 0 | (30) $50 \div 9$
商 5, 余り 5 | (48) $-92 \div 1$
商 -92, 余り 0 | (66) $-74 \div 1$
商 -74, 余り 0 |
| (13) $-66 \div 6$
商 -11, 余り 0 | (31) $-37 \div 9$
商 -5, 余り 8 | (49) $68 \div 6$
商 11, 余り 2 | (67) $-37 \div 10$
商 -4, 余り 3 |
| (14) $18 \div 6$
商 3, 余り 0 | (32) $-2 \div 8$
商 -1, 余り 6 | (50) $40 \div 10$
商 4, 余り 0 | (68) $-6 \div 5$
商 -2, 余り 4 |
| (15) $-86 \div 9$
商 -10, 余り 4 | (33) $83 \div 1$
商 83, 余り 0 | (51) $82 \div 3$
商 27, 余り 1 | (69) $44 \div 1$
商 44, 余り 0 |
| (16) $99 \div 5$
商 19, 余り 4 | (34) $92 \div 9$
商 10, 余り 2 | (52) $3 \div 1$
商 3, 余り 0 | (70) $41 \div 1$
商 41, 余り 0 |
| (17) $93 \div 10$
商 9, 余り 3 | (35) $43 \div 7$
商 6, 余り 1 | (53) $-77 \div 3$
商 -26, 余り 1 | (71) $35 \div 10$
商 3, 余り 5 |
| (18) $-34 \div 4$
商 -9, 余り 2 | (36) $-22 \div 4$
商 -6, 余り 2 | (54) $-81 \div 2$
商 -41, 余り 1 | (72) $47 \div 9$
商 5, 余り 2 |

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|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| (1) $84 \div 1$
商 84, 余り 0 | (19) $69 \div 6$
商 11, 余り 3 | (37) $42 \div 2$
商 21, 余り 0 | (55) $-17 \div 3$
商 -6, 余り 1 |
| (2) $-14 \div 3$
商 -5, 余り 1 | (20) $-20 \div 2$
商 -10, 余り 0 | (38) $37 \div 9$
商 4, 余り 1 | (56) $-75 \div 5$
商 -15, 余り 0 |
| (3) $86 \div 8$
商 10, 余り 6 | (21) $46 \div 8$
商 5, 余り 6 | (39) $-38 \div 4$
商 -10, 余り 2 | (57) $-2 \div 1$
商 -2, 余り 0 |
| (4) $-89 \div 6$
商 -15, 余り 1 | (22) $-86 \div 2$
商 -43, 余り 0 | (40) $-36 \div 4$
商 -9, 余り 0 | (58) $5 \div 5$
商 1, 余り 0 |
| (5) $5 \div 8$
商 0, 余り 5 | (23) $86 \div 9$
商 9, 余り 5 | (41) $15 \div 8$
商 1, 余り 7 | (59) $21 \div 7$
商 3, 余り 0 |
| (6) $-67 \div 4$
商 -17, 余り 1 | (24) $89 \div 9$
商 9, 余り 8 | (42) $56 \div 9$
商 6, 余り 2 | (60) $20 \div 7$
商 2, 余り 6 |
| (7) $89 \div 7$
商 12, 余り 5 | (25) $18 \div 7$
商 2, 余り 4 | (43) $-57 \div 10$
商 -6, 余り 3 | (61) $-1 \div 7$
商 -1, 余り 6 |
| (8) $82 \div 5$
商 16, 余り 2 | (26) $-56 \div 7$
商 -8, 余り 0 | (44) $-72 \div 6$
商 -12, 余り 0 | (62) $87 \div 10$
商 8, 余り 7 |
| (9) $-5 \div 7$
商 -1, 余り 2 | (27) $-81 \div 9$
商 -9, 余り 0 | (45) $-79 \div 3$
商 -27, 余り 2 | (63) $-9 \div 8$
商 -2, 余り 7 |
| (10) $22 \div 1$
商 22, 余り 0 | (28) $-88 \div 4$
商 -22, 余り 0 | (46) $-36 \div 4$
商 -9, 余り 0 | (64) $-9 \div 2$
商 -5, 余り 1 |
| (11) $96 \div 5$
商 19, 余り 1 | (29) $-91 \div 9$
商 -11, 余り 8 | (47) $12 \div 1$
商 12, 余り 0 | (65) $-57 \div 10$
商 -6, 余り 3 |
| (12) $-87 \div 2$
商 -44, 余り 1 | (30) $100 \div 7$
商 14, 余り 2 | (48) $43 \div 9$
商 4, 余り 7 | (66) $-51 \div 10$
商 -6, 余り 9 |
| (13) $-31 \div 10$
商 -4, 余り 9 | (31) $-92 \div 3$
商 -31, 余り 1 | (49) $-62 \div 2$
商 -31, 余り 0 | (67) $-40 \div 8$
商 -5, 余り 0 |
| (14) $46 \div 6$
商 7, 余り 4 | (32) $36 \div 4$
商 9, 余り 0 | (50) $56 \div 5$
商 11, 余り 1 | (68) $63 \div 8$
商 7, 余り 7 |
| (15) $62 \div 6$
商 10, 余り 2 | (33) $-73 \div 3$
商 -25, 余り 2 | (51) $2 \div 4$
商 0, 余り 2 | (69) $-16 \div 5$
商 -4, 余り 4 |
| (16) $47 \div 6$
商 7, 余り 5 | (34) $63 \div 1$
商 63, 余り 0 | (52) $-26 \div 10$
商 -3, 余り 4 | (70) $73 \div 4$
商 18, 余り 1 |
| (17) $57 \div 2$
商 28, 余り 1 | (35) $15 \div 2$
商 7, 余り 1 | (53) $-65 \div 9$
商 -8, 余り 7 | (71) $39 \div 1$
商 39, 余り 0 |
| (18) $21 \div 8$
商 2, 余り 5 | (36) $-39 \div 5$
商 -8, 余り 1 | (54) $75 \div 9$
商 8, 余り 3 | (72) $98 \div 9$
商 10, 余り 8 |

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|-----------------------------------|----------------------------------|----------------------------------|------------------------------------|
| (1) $37 \div 7$
商 5, 余り 2 | (19) $81 \div 10$
商 8, 余り 1 | (37) $30 \div 1$
商 30, 余り 0 | (55) $-17 \div 4$
商 -5, 余り 3 |
| (2) $71 \div 5$
商 14, 余り 1 | (20) $11 \div 10$
商 1, 余り 1 | (38) $-13 \div 2$
商 -7, 余り 1 | (56) $7 \div 7$
商 1, 余り 0 |
| (3) $5 \div 2$
商 2, 余り 1 | (21) $-10 \div 2$
商 -5, 余り 0 | (39) $-37 \div 7$
商 -6, 余り 5 | (57) $9 \div 8$
商 1, 余り 1 |
| (4) $30 \div 4$
商 7, 余り 2 | (22) $74 \div 7$
商 10, 余り 4 | (40) $-8 \div 10$
商 -1, 余り 2 | (58) $-100 \div 10$
商 -10, 余り 0 |
| (5) $50 \div 9$
商 5, 余り 5 | (23) $-92 \div 4$
商 -23, 余り 0 | (41) $97 \div 4$
商 24, 余り 1 | (59) $-28 \div 1$
商 -28, 余り 0 |
| (6) $-78 \div 8$
商 -10, 余り 2 | (24) $-50 \div 7$
商 -8, 余り 6 | (42) $-62 \div 6$
商 -11, 余り 4 | (60) $76 \div 1$
商 76, 余り 0 |
| (7) $97 \div 6$
商 16, 余り 1 | (25) $-68 \div 9$
商 -8, 余り 4 | (43) $13 \div 4$
商 3, 余り 1 | (61) $-61 \div 3$
商 -21, 余り 2 |
| (8) $10 \div 5$
商 2, 余り 0 | (26) $-17 \div 8$
商 -3, 余り 7 | (44) $84 \div 7$
商 12, 余り 0 | (62) $99 \div 1$
商 99, 余り 0 |
| (9) $11 \div 1$
商 11, 余り 0 | (27) $70 \div 7$
商 10, 余り 0 | (45) $55 \div 5$
商 11, 余り 0 | (63) $-52 \div 4$
商 -13, 余り 0 |
| (10) $33 \div 10$
商 3, 余り 3 | (28) $88 \div 1$
商 88, 余り 0 | (46) $-59 \div 10$
商 -6, 余り 1 | (64) $30 \div 4$
商 7, 余り 2 |
| (11) $10 \div 4$
商 2, 余り 2 | (29) $97 \div 7$
商 13, 余り 6 | (47) $-79 \div 2$
商 -40, 余り 1 | (65) $-86 \div 10$
商 -9, 余り 4 |
| (12) $-70 \div 5$
商 -14, 余り 0 | (30) $63 \div 4$
商 15, 余り 3 | (48) $78 \div 6$
商 13, 余り 0 | (66) $81 \div 4$
商 20, 余り 1 |
| (13) $72 \div 1$
商 72, 余り 0 | (31) $-71 \div 5$
商 -15, 余り 4 | (49) $94 \div 7$
商 13, 余り 3 | (67) $-33 \div 4$
商 -9, 余り 3 |
| (14) $-97 \div 10$
商 -10, 余り 3 | (32) $32 \div 9$
商 3, 余り 5 | (50) $49 \div 5$
商 9, 余り 4 | (68) $-5 \div 1$
商 -5, 余り 0 |
| (15) $57 \div 6$
商 9, 余り 3 | (33) $-15 \div 10$
商 -2, 余り 5 | (51) $-19 \div 9$
商 -3, 余り 8 | (69) $21 \div 6$
商 3, 余り 3 |
| (16) $-63 \div 5$
商 -13, 余り 2 | (34) $23 \div 4$
商 5, 余り 3 | (52) $74 \div 1$
商 74, 余り 0 | (70) $41 \div 4$
商 10, 余り 1 |
| (17) $-28 \div 7$
商 -4, 余り 0 | (35) $41 \div 3$
商 13, 余り 2 | (53) $-25 \div 9$
商 -3, 余り 2 | (71) $47 \div 4$
商 11, 余り 3 |
| (18) $50 \div 8$
商 6, 余り 2 | (36) $58 \div 5$
商 11, 余り 3 | (54) $-82 \div 9$
商 -10, 余り 8 | (72) $-19 \div 1$
商 -19, 余り 0 |

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|----------------------------------|----------------------------------|----------------------------------|-----------------------------------|
| (1) $42 \div 3$
商 14, 余り 0 | (19) $-38 \div 3$
商 -13, 余り 1 | (37) $-36 \div 7$
商 -6, 余り 6 | (55) $-91 \div 10$
商 -10, 余り 9 |
| (2) $-57 \div 5$
商 -12, 余り 3 | (20) $61 \div 10$
商 6, 余り 1 | (38) $-42 \div 1$
商 -42, 余り 0 | (56) $35 \div 7$
商 5, 余り 0 |
| (3) $-3 \div 9$
商 -1, 余り 6 | (21) $88 \div 1$
商 88, 余り 0 | (39) $-5 \div 1$
商 -5, 余り 0 | (57) $84 \div 2$
商 42, 余り 0 |
| (4) $-38 \div 1$
商 -38, 余り 0 | (22) $55 \div 5$
商 11, 余り 0 | (40) $-36 \div 2$
商 -18, 余り 0 | (58) $-97 \div 6$
商 -17, 余り 5 |
| (5) $-32 \div 2$
商 -16, 余り 0 | (23) $-9 \div 4$
商 -3, 余り 3 | (41) $-84 \div 10$
商 -9, 余り 6 | (59) $89 \div 6$
商 14, 余り 5 |
| (6) $-73 \div 8$
商 -10, 余り 7 | (24) $56 \div 1$
商 56, 余り 0 | (42) $-7 \div 10$
商 -1, 余り 3 | (60) $96 \div 3$
商 32, 余り 0 |
| (7) $87 \div 1$
商 87, 余り 0 | (25) $-11 \div 9$
商 -2, 余り 7 | (43) $-41 \div 10$
商 -5, 余り 9 | (61) $55 \div 1$
商 55, 余り 0 |
| (8) $-81 \div 9$
商 -9, 余り 0 | (26) $-75 \div 10$
商 -8, 余り 5 | (44) $95 \div 4$
商 23, 余り 3 | (62) $-92 \div 1$
商 -92, 余り 0 |
| (9) $54 \div 7$
商 7, 余り 5 | (27) $-17 \div 9$
商 -2, 余り 1 | (45) $-28 \div 10$
商 -3, 余り 2 | (63) $-78 \div 10$
商 -8, 余り 2 |
| (10) $-32 \div 7$
商 -5, 余り 3 | (28) $92 \div 1$
商 92, 余り 0 | (46) $-78 \div 9$
商 -9, 余り 3 | (64) $61 \div 7$
商 8, 余り 5 |
| (11) $99 \div 3$
商 33, 余り 0 | (29) $93 \div 8$
商 11, 余り 5 | (47) $-93 \div 8$
商 -12, 余り 3 | (65) $40 \div 4$
商 10, 余り 0 |
| (12) $-55 \div 3$
商 -19, 余り 2 | (30) $-5 \div 10$
商 -1, 余り 5 | (48) $22 \div 5$
商 4, 余り 2 | (66) $59 \div 9$
商 6, 余り 5 |
| (13) $-17 \div 2$
商 -9, 余り 1 | (31) $-25 \div 7$
商 -4, 余り 3 | (49) $97 \div 4$
商 24, 余り 1 | (67) $-25 \div 8$
商 -4, 余り 7 |
| (14) $-32 \div 2$
商 -16, 余り 0 | (32) $-71 \div 1$
商 -71, 余り 0 | (50) $-6 \div 2$
商 -3, 余り 0 | (68) $85 \div 1$
商 85, 余り 0 |
| (15) $78 \div 8$
商 9, 余り 6 | (33) $56 \div 4$
商 14, 余り 0 | (51) $-79 \div 9$
商 -9, 余り 2 | (69) $44 \div 5$
商 8, 余り 4 |
| (16) $53 \div 2$
商 26, 余り 1 | (34) $-65 \div 4$
商 -17, 余り 3 | (52) $27 \div 3$
商 9, 余り 0 | (70) $64 \div 9$
商 7, 余り 1 |
| (17) $-78 \div 4$
商 -20, 余り 2 | (35) $5 \div 2$
商 2, 余り 1 | (53) $-97 \div 5$
商 -20, 余り 3 | (71) $96 \div 1$
商 96, 余り 0 |
| (18) $-19 \div 1$
商 -19, 余り 0 | (36) $100 \div 2$
商 50, 余り 0 | (54) $-53 \div 4$
商 -14, 余り 3 | (72) $-82 \div 10$
商 -9, 余り 8 |

1.3 有理数

1.3.1 加減法

有理数の和や差を求める問題。

(1) $-3 + \left(-\frac{3}{4}\right)$

(11) $\frac{3}{4} - \frac{8}{3}$

(21) $2 + 3$

(31) $\frac{1}{8} + \left(-\frac{1}{10}\right)$

(2) $-\frac{10}{9} + \left(-\frac{2}{3}\right)$

(12) $-5 - \frac{7}{4}$

(22) $1 + 1$

(32) $-\frac{1}{7} - \frac{3}{2}$

(3) $-\frac{4}{7} - \left(-\frac{1}{2}\right)$

(13) $-6 + \left(-\frac{1}{3}\right)$

(23) $-\frac{8}{3} + 3$

(33) $1 + \frac{8}{3}$

(4) $9 - \left(-\frac{8}{7}\right)$

(14) $-\frac{7}{4} + \frac{8}{3}$

(24) $\frac{2}{5} + \left(-\frac{5}{3}\right)$

(34) $-3 + \frac{2}{3}$

(5) $\frac{3}{5} + (-3)$

(15) $-\frac{3}{7} - \left(-\frac{5}{4}\right)$

(25) $\frac{3}{5} - \frac{4}{3}$

(35) $-2 - \frac{3}{7}$

(6) $-1 - \frac{3}{4}$

(16) $-\frac{4}{5} + (-1)$

(26) $2 + \frac{8}{3}$

(36) $\frac{7}{3} + (-1)$

(7) $\frac{3}{10} + (-5)$

(17) $\frac{1}{5} + \left(-\frac{4}{5}\right)$

(27) $-\frac{4}{9} + \left(-\frac{4}{3}\right)$

(37) $-\frac{5}{3} - \frac{1}{9}$

(8) $-\frac{8}{7} + \frac{4}{3}$

(18) $-\frac{5}{2} - \left(-\frac{7}{8}\right)$

(28) $-\frac{2}{5} + \left(-\frac{1}{3}\right)$

(38) $\frac{4}{5} + 7$

(9) $-5 - \frac{2}{3}$

(19) $-\frac{2}{3} + \frac{9}{4}$

(29) $\frac{7}{8} - \frac{8}{9}$

(39) $-\frac{2}{3} - \left(-\frac{1}{3}\right)$

(10) $\frac{1}{7} - \frac{4}{3}$

(20) $-\frac{1}{3} + \frac{3}{2}$

(30) $-\frac{5}{6} + (-3)$

(40) $-\frac{9}{5} + 1$

(1) $\frac{5}{3} - (-3)$

(11) $-\frac{1}{8} - \left(-\frac{1}{3}\right)$

(21) $\frac{1}{6} + \left(-\frac{1}{7}\right)$

(31) $-\frac{1}{3} + \left(-\frac{1}{2}\right)$

(2) $-\frac{4}{9} + 10$

(12) $-\frac{8}{5} - \frac{3}{2}$

(22) $\frac{4}{7} - \frac{2}{3}$

(32) $\frac{9}{8} - \left(-\frac{5}{7}\right)$

(3) $-\frac{9}{5} - \left(-\frac{1}{4}\right)$

(13) $-\frac{1}{2} + \frac{3}{5}$

(23) $3 - \frac{3}{5}$

(33) $-3 + (-2)$

(4) $\frac{7}{5} - \frac{1}{5}$

(14) $-5 - \left(-\frac{1}{3}\right)$

(24) $\frac{5}{2} + \left(-\frac{1}{2}\right)$

(34) $-\frac{1}{4} - \frac{3}{2}$

(5) $\frac{7}{3} + \left(-\frac{5}{8}\right)$

(15) $-\frac{3}{7} + \frac{3}{5}$

(25) $-\frac{3}{7} - \frac{8}{9}$

(35) $9 - \frac{4}{9}$

(6) $\frac{4}{9} + 5$

(16) $-\frac{1}{2} + \left(-\frac{7}{5}\right)$

(26) $\frac{2}{7} + (-8)$

(36) $-\frac{5}{9} + \left(-\frac{5}{2}\right)$

(7) $\frac{3}{2} + \frac{10}{7}$

(17) $-2 - \left(-\frac{5}{3}\right)$

(27) $-\frac{1}{2} + (-7)$

(37) $-3 - \left(-\frac{2}{3}\right)$

(8) $-\frac{1}{9} - (-2)$

(18) $\frac{1}{8} + 6$

(28) $-\frac{4}{5} + 6$

(38) $\frac{1}{2} + \frac{6}{7}$

(9) $\frac{1}{7} + \left(-\frac{1}{3}\right)$

(19) $-\frac{3}{7} + \frac{4}{7}$

(29) $\frac{10}{3} - \left(-\frac{6}{7}\right)$

(39) $\frac{2}{3} + 2$

(10) $-\frac{1}{8} - 5$

(20) $\frac{1}{4} - \left(-\frac{3}{7}\right)$

(30) $\frac{5}{3} + \frac{4}{3}$

(40) $-6 - \frac{3}{2}$

(1) $\frac{8}{7} - \frac{2}{7}$

(11) $-2 + \frac{1}{4}$

(21) $-6 - \frac{8}{9}$

(31) $4 - 1$

(2) $-\frac{3}{2} - \left(-\frac{6}{7}\right)$

(12) $-\frac{8}{5} + \frac{1}{9}$

(22) $-\frac{2}{5} + \left(-\frac{1}{2}\right)$

(32) $\frac{2}{7} + \frac{1}{8}$

(3) $-\frac{1}{6} + \frac{7}{8}$

(13) $-\frac{9}{5} - \frac{8}{9}$

(23) $-\frac{3}{4} + \frac{1}{4}$

(33) $-\frac{5}{4} - \frac{1}{7}$

(4) $\frac{2}{3} + \left(-\frac{1}{10}\right)$

(14) $-1 + (-3)$

(24) $1 - \left(-\frac{3}{2}\right)$

(34) $\frac{2}{3} + \left(-\frac{4}{3}\right)$

(5) $-1 + \left(-\frac{4}{9}\right)$

(15) $\frac{1}{3} - (-8)$

(25) $-1 + 3$

(35) $-\frac{7}{10} - (-7)$

(6) $\frac{1}{6} - \left(-\frac{5}{2}\right)$

(16) $-10 + \frac{6}{5}$

(26) $-\frac{8}{5} + (-2)$

(36) $\frac{1}{2} - \frac{5}{6}$

(7) $-2 - \frac{5}{7}$

(17) $-\frac{7}{2} + 1$

(27) $-\frac{10}{3} - \frac{9}{2}$

(37) $\frac{3}{8} - \left(-\frac{7}{9}\right)$

(8) $\frac{1}{2} - \left(-\frac{1}{3}\right)$

(18) $-\frac{3}{7} - \frac{5}{7}$

(28) $1 + \left(-\frac{4}{7}\right)$

(38) $\frac{1}{3} - (-1)$

(9) $1 - \left(-\frac{1}{10}\right)$

(19) $\frac{7}{3} - 2$

(29) $2 + \left(-\frac{9}{2}\right)$

(39) $1 - \frac{2}{3}$

(10) $-\frac{5}{4} + \left(-\frac{1}{3}\right)$

(20) $\frac{1}{2} + (-8)$

(30) $-\frac{2}{7} + \frac{7}{10}$

(40) $\frac{7}{4} + 1$

(1) $\frac{1}{3} - 2$

(11) $10 - \left(-\frac{3}{2}\right)$

(21) $-8 + (-2)$

(31) $\frac{9}{4} + 1$

(2) $\frac{1}{4} + 3$

(12) $-\frac{5}{4} + \frac{7}{9}$

(22) $\frac{7}{8} - 5$

(32) $-5 - 1$

(3) $\frac{3}{10} + \left(-\frac{5}{2}\right)$

(13) $-1 + \left(-\frac{2}{3}\right)$

(23) $\frac{3}{2} + \frac{2}{9}$

(33) $-\frac{5}{2} - (-1)$

(4) $\frac{3}{5} - \frac{4}{9}$

(14) $\frac{3}{8} + 10$

(24) $\frac{4}{7} - \frac{7}{5}$

(34) $-\frac{7}{4} + \left(-\frac{1}{2}\right)$

(5) $-\frac{4}{3} + \frac{3}{8}$

(15) $-\frac{5}{4} + \left(-\frac{6}{5}\right)$

(25) $-8 + 2$

(35) $\frac{7}{9} - 1$

(6) $-\frac{5}{8} + \left(-\frac{4}{7}\right)$

(16) $-\frac{4}{7} - \left(-\frac{8}{3}\right)$

(26) $\frac{4}{5} - 1$

(36) $-5 - \left(-\frac{1}{2}\right)$

(7) $-1 + \left(-\frac{1}{5}\right)$

(17) $\frac{1}{2} + \frac{5}{8}$

(27) $3 - \frac{4}{7}$

(37) $-\frac{1}{4} + \left(-\frac{1}{7}\right)$

(8) $\frac{7}{5} - \left(-\frac{5}{2}\right)$

(18) $-\frac{1}{3} - \left(-\frac{1}{10}\right)$

(28) $5 + \left(-\frac{1}{8}\right)$

(38) $\frac{5}{2} - (-3)$

(9) $-1 - (-1)$

(19) $1 - \left(-\frac{1}{4}\right)$

(29) $4 - 4$

(39) $\frac{2}{3} + \frac{9}{8}$

(10) $-\frac{5}{8} - \frac{1}{3}$

(20) $-1 - (-1)$

(30) $\frac{7}{4} - (-5)$

(40) $\frac{7}{10} - \left(-\frac{9}{5}\right)$

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(11) $\frac{8}{7} + \left(-\frac{8}{5}\right)$

(21) $-\frac{3}{8} + \left(-\frac{4}{7}\right)$

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(2) $-1 - \left(-\frac{10}{7}\right)$

(12) $\frac{6}{5} - \left(-\frac{1}{4}\right)$

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(34) $-\frac{1}{8} - 3$

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(15) $-5 - \frac{4}{7}$

(25) $1 + \left(-\frac{3}{8}\right)$

(35) $\frac{7}{9} + \frac{7}{3}$

(6) $-\frac{10}{7} + \left(-\frac{7}{6}\right)$

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(36) $-\frac{3}{4} + \frac{1}{2}$

(7) $-5 + \left(-\frac{2}{7}\right)$

(17) $\frac{4}{3} + \frac{1}{7}$

(27) $-\frac{10}{3} - 2$

(37) $-\frac{2}{3} - \frac{2}{3}$

(8) $7 + \left(-\frac{8}{3}\right)$

(18) $-\frac{5}{7} + 1$

(28) $-\frac{1}{6} - \frac{1}{8}$

(38) $-\frac{5}{9} - \frac{6}{5}$

(9) $-3 + 1$

(19) $\frac{5}{2} + \frac{3}{8}$

(29) $\frac{10}{9} - \frac{5}{2}$

(39) $\frac{3}{8} - \frac{1}{2}$

(10) $2 + (-3)$

(20) $\frac{2}{3} - \left(-\frac{3}{4}\right)$

(30) $-1 - \frac{3}{5}$

(40) $-\frac{1}{7} + \frac{5}{8}$

(1) $-\frac{1}{2} - \frac{4}{3}$

(11) $-\frac{8}{9} + \frac{1}{5}$

(21) $-10 + (-4)$

(31) $-\frac{1}{5} - \left(-\frac{1}{2}\right)$

(2) $3 - (-1)$

(12) $\frac{3}{5} - \frac{2}{5}$

(22) $\frac{7}{3} - \left(-\frac{4}{7}\right)$

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(3) $\frac{1}{2} - 2$

(13) $-\frac{5}{6} + \frac{9}{8}$

(23) $-\frac{3}{8} + \frac{1}{7}$

(33) $-\frac{3}{5} + 8$

(4) $-\frac{1}{2} + \left(-\frac{4}{5}\right)$

(14) $-\frac{5}{4} + \frac{4}{5}$

(24) $-\frac{1}{3} - (-6)$

(34) $\frac{7}{10} + \frac{4}{9}$

(5) $-\frac{1}{2} + 1$

(15) $-\frac{3}{10} + \frac{2}{7}$

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(6) $-1 + \frac{5}{4}$

(16) $\frac{3}{2} - \frac{5}{2}$

(26) $\frac{3}{2} + \left(-\frac{3}{5}\right)$

(36) $\frac{1}{5} + \left(-\frac{2}{3}\right)$

(7) $-\frac{7}{9} + 4$

(17) $\frac{5}{3} - 10$

(27) $-3 + \frac{4}{3}$

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(8) $-8 - \left(-\frac{3}{4}\right)$

(18) $\frac{7}{3} - \frac{10}{3}$

(28) $3 - \frac{5}{6}$

(38) $-7 + \left(-\frac{7}{2}\right)$

(9) $-\frac{6}{7} + 4$

(19) $\frac{1}{10} - \frac{1}{3}$

(29) $-\frac{3}{2} - \left(-\frac{1}{8}\right)$

(39) $-\frac{1}{4} - \left(-\frac{9}{10}\right)$

(10) $1 + \frac{1}{2}$

(20) $\frac{5}{6} + (-10)$

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(40) $\frac{6}{5} + (-1)$

(1) $-\frac{5}{7} + \left(-\frac{7}{10}\right)$

(11) $1 + 1$

(21) $-1 + (-1)$

(31) $-\frac{3}{4} - 1$

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(12) $-\frac{7}{8} + (-3)$

(22) $\frac{3}{5} - \frac{7}{2}$

(32) $\frac{10}{3} + (-4)$

(3) $-\frac{7}{2} + \frac{7}{6}$

(13) $\frac{2}{7} + \left(-\frac{1}{3}\right)$

(23) $-\frac{9}{2} - \frac{3}{4}$

(33) $\frac{3}{2} - \frac{7}{3}$

(4) $-\frac{5}{9} + \frac{3}{4}$

(14) $-\frac{7}{5} + \frac{4}{3}$

(24) $9 + 1$

(34) $\frac{6}{7} + \left(-\frac{10}{7}\right)$

(5) $-\frac{8}{5} - \left(-\frac{1}{10}\right)$

(15) $-\frac{2}{7} - (-5)$

(25) $\frac{1}{8} - \frac{5}{3}$

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(6) $-2 - \frac{8}{7}$

(16) $-\frac{3}{10} - \left(-\frac{3}{5}\right)$

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(36) $-\frac{1}{2} - \left(-\frac{7}{9}\right)$

(7) $-\frac{7}{9} + \left(-\frac{4}{9}\right)$

(17) $-\frac{1}{3} + \frac{2}{5}$

(27) $\frac{5}{6} - (-2)$

(37) $-1 + \frac{2}{7}$

(28) $-\frac{4}{3} + \frac{1}{2}$

(8) $3 + \left(-\frac{5}{4}\right)$

(18) $-\frac{1}{2} - 2$

(38) $-\frac{7}{2} + (-1)$

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(19) $\frac{7}{6} + (-1)$

(29) $-\frac{8}{3} + \left(-\frac{1}{2}\right)$

(39) $-\frac{8}{3} + \left(-\frac{4}{7}\right)$

(30) $-\frac{9}{2} + \frac{9}{2}$

(20) $-1 + (-2)$

(10) $\frac{2}{3} - \frac{8}{3}$

(40) $-\frac{1}{3} + 1$

(1) $-\frac{3}{4} - \frac{9}{4}$

(11) $-2 - \left(-\frac{5}{9}\right)$

(21) $\frac{2}{5} - 5$

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(2) $\frac{4}{3} + \left(-\frac{7}{2}\right)$

(12) $-10 + \frac{10}{9}$

(22) $-\frac{10}{3} + \left(-\frac{2}{7}\right)$

(32) $-\frac{1}{3} + \left(-\frac{9}{8}\right)$

(3) $\frac{10}{7} + \frac{5}{3}$

(13) $-1 + (-1)$

(23) $9 + \frac{2}{3}$

(33) $\frac{9}{8} - \frac{7}{2}$

(4) $-1 - \left(-\frac{6}{5}\right)$

(14) $3 + \frac{1}{5}$

(24) $2 + \left(-\frac{7}{9}\right)$

(34) $-\frac{1}{3} - \frac{7}{10}$

(5) $-\frac{4}{3} - (-2)$

(15) $3 + \frac{5}{9}$

(25) $-\frac{4}{3} - \frac{1}{3}$

(35) $7 + \frac{5}{8}$

(6) $\frac{2}{5} - \frac{5}{3}$

(16) $\frac{5}{3} - \frac{9}{4}$

(26) $-\frac{3}{8} - \left(-\frac{1}{2}\right)$

(36) $-3 - \left(-\frac{1}{2}\right)$

(7) $-\frac{3}{5} + \frac{2}{3}$

(17) $10 + \left(-\frac{9}{4}\right)$

(27) $2 + \left(-\frac{5}{3}\right)$

(37) $-\frac{8}{5} - \left(-\frac{3}{10}\right)$

(8) $\frac{3}{2} - \left(-\frac{5}{7}\right)$

(18) $\frac{9}{5} + \left(-\frac{6}{5}\right)$

(28) $4 + \left(-\frac{1}{4}\right)$

(38) $\frac{4}{5} - 2$

(9) $\frac{5}{2} + \frac{2}{5}$

(19) $-\frac{1}{2} + 1$

(29) $\frac{3}{2} + \frac{2}{3}$

(39) $\frac{2}{7} + \frac{8}{7}$

(10) $\frac{1}{10} + \left(-\frac{4}{5}\right)$

(20) $\frac{3}{4} + \frac{1}{7}$

(30) $\frac{7}{9} - (-1)$

(40) $-\frac{9}{2} - 1$

(1) $2 - \frac{3}{2}$

(11) $\frac{3}{4} + (-1)$

(21) $\frac{8}{3} + 1$

(31) $-\frac{5}{3} + \frac{2}{7}$

(2) $-\frac{6}{5} + \left(-\frac{9}{2}\right)$

(12) $\frac{4}{9} - \frac{1}{9}$

(22) $\frac{9}{2} - \left(-\frac{1}{9}\right)$

(32) $-\frac{3}{2} + \frac{7}{8}$

(3) $1 + \left(-\frac{7}{4}\right)$

(13) $\frac{2}{5} + \left(-\frac{10}{7}\right)$

(23) $-1 + \frac{1}{3}$

(33) $-\frac{3}{7} - \frac{4}{5}$

(4) $-5 + \frac{5}{7}$

(14) $-4 + \left(-\frac{5}{4}\right)$

(24) $-\frac{3}{2} - (-3)$

(34) $2 - 2$

(5) $-\frac{6}{5} + \frac{5}{2}$

(15) $-\frac{5}{2} - \frac{4}{7}$

(25) $\frac{4}{5} + (-1)$

(35) $\frac{2}{3} - \left(-\frac{5}{3}\right)$

(6) $-4 - 3$

(16) $-1 - (-8)$

(26) $\frac{5}{9} + \left(-\frac{9}{7}\right)$

(36) $\frac{5}{4} + \left(-\frac{2}{3}\right)$

(7) $-\frac{1}{2} - \left(-\frac{7}{4}\right)$

(17) $\frac{3}{2} - \frac{7}{4}$

(27) $-\frac{7}{10} + \frac{1}{6}$

(37) $\frac{8}{7} + \frac{3}{2}$

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(18) $-\frac{9}{10} + \frac{7}{6}$

(28) $-2 + \frac{5}{3}$

(38) $-1 + \left(-\frac{1}{3}\right)$

(9) $\frac{10}{9} + \frac{5}{6}$

(19) $-\frac{1}{3} + \left(-\frac{5}{9}\right)$

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(10) $\frac{3}{10} + \frac{9}{5}$

(20) $2 - \frac{5}{3}$

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(11) $-\frac{1}{6} + 1$

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(23) $-9 - \left(-\frac{9}{8}\right)$

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(4) $\frac{5}{2} - \frac{1}{3}$

(14) $-7 - \frac{7}{6}$

(24) $-\frac{4}{7} + \frac{7}{9}$

(34) $\frac{3}{5} + \frac{8}{9}$

(5) $\frac{9}{2} + 4$

(15) $1 - \frac{1}{3}$

(25) $9 + 1$

(35) $\frac{3}{4} + \left(-\frac{9}{10}\right)$

(6) $-\frac{1}{5} - \frac{2}{3}$

(16) $\frac{1}{8} - \frac{2}{3}$

(26) $-\frac{6}{5} + \frac{6}{7}$

(36) $-\frac{8}{7} + \frac{3}{2}$

(7) $-\frac{10}{9} + \frac{9}{2}$

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(8) $1 + \frac{1}{2}$

(18) $\frac{5}{6} + 3$

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(19) $\frac{1}{2} - \left(-\frac{9}{2}\right)$

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(33) $\frac{8}{3} + (-6)$

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(14) $\frac{2}{5} - \left(-\frac{7}{4}\right)$

(24) $\frac{1}{2} + \frac{4}{7}$

(34) $-\frac{6}{5} - \left(-\frac{1}{2}\right)$

(5) $-\frac{5}{2} + \left(-\frac{9}{4}\right)$

(15) $8 - 2$

(25) $\frac{1}{7} - 5$

(35) $\frac{3}{2} - \frac{1}{7}$

(6) $-1 + \frac{4}{3}$

(16) $2 + 1$

(26) $-\frac{9}{8} - \left(-\frac{7}{6}\right)$

(36) $\frac{3}{10} + \frac{2}{7}$

(7) $\frac{10}{7} - \left(-\frac{7}{9}\right)$

(17) $-\frac{8}{7} - \frac{5}{4}$

(27) $\frac{2}{3} - \frac{8}{9}$

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(32) $2 + \left(-\frac{1}{3}\right)$

(3) $\frac{2}{5} + \left(-\frac{9}{4}\right)$

(13) $\frac{1}{2} + \frac{3}{2}$

(23) $\frac{4}{3} - \left(-\frac{8}{5}\right)$

(33) $-\frac{5}{2} + \left(-\frac{1}{2}\right)$

(4) $\frac{1}{3} - \frac{9}{4}$

(14) $-\frac{3}{10} - 1$

(24) $-\frac{1}{3} - \left(-\frac{5}{7}\right)$

(34) $-\frac{5}{2} - \left(-\frac{5}{6}\right)$

(5) $-\frac{1}{8} - \frac{5}{8}$

(15) $1 - \frac{9}{7}$

(25) $-\frac{1}{9} + \left(-\frac{7}{8}\right)$

(35) $-5 + \left(-\frac{4}{5}\right)$

(6) $-\frac{1}{9} - (-5)$

(16) $-\frac{10}{7} + (-1)$

(26) $\frac{2}{5} - (-10)$

(36) $-\frac{3}{5} + \left(-\frac{9}{2}\right)$

(7) $\frac{3}{10} + (-1)$

(17) $\frac{1}{2} + \frac{5}{2}$

(27) $-3 + 2$

(37) $\frac{8}{5} - \frac{3}{5}$

(8) $-\frac{8}{7} - 8$

(18) $\frac{9}{4} - (-2)$

(28) $\frac{7}{8} - \frac{4}{5}$

(38) $\frac{5}{7} - \left(-\frac{9}{7}\right)$

(9) $\frac{4}{5} - \frac{3}{5}$

(19) $\frac{1}{6} - (-2)$

(29) $-\frac{5}{4} + \frac{3}{10}$

(39) $1 + \frac{1}{3}$

(10) $-\frac{5}{9} - \frac{8}{9}$

(20) $\frac{5}{3} + \frac{7}{3}$

(30) $-1 - \frac{1}{2}$

(40) $-\frac{6}{7} + 1$

(1) $-\frac{5}{3} - (-3)$

(11) $\frac{5}{9} - \left(-\frac{3}{5}\right)$

(21) $1 + \frac{5}{3}$

(31) $\frac{8}{5} + \frac{9}{7}$

(2) $-\frac{3}{7} + \frac{5}{4}$

(12) $\frac{5}{2} + 1$

(22) $-1 - (-7)$

(32) $\frac{5}{9} + \left(-\frac{5}{2}\right)$

(3) $-1 - \left(-\frac{3}{7}\right)$

(13) $-\frac{1}{2} - (-1)$

(23) $\frac{10}{3} - \left(-\frac{7}{5}\right)$

(33) $-\frac{9}{7} - \left(-\frac{9}{4}\right)$

(4) $-\frac{9}{8} + \frac{4}{9}$

(14) $-\frac{9}{5} + \left(-\frac{5}{3}\right)$

(24) $-\frac{1}{4} + \left(-\frac{4}{3}\right)$

(34) $-\frac{7}{10} + \frac{10}{3}$

(5) $-2 - 1$

(15) $-\frac{5}{3} - \left(-\frac{5}{4}\right)$

(25) $-\frac{4}{9} + \left(-\frac{4}{9}\right)$

(35) $-\frac{1}{10} - \left(-\frac{9}{10}\right)$

(6) $-\frac{3}{4} - \frac{3}{10}$

(16) $\frac{7}{6} + 10$

(26) $\frac{10}{3} + \frac{3}{5}$

(36) $-\frac{2}{9} - \left(-\frac{3}{10}\right)$

(7) $\frac{10}{3} - \left(-\frac{3}{4}\right)$

(17) $-\frac{9}{10} - 7$

(27) $\frac{8}{5} + (-1)$

(37) $-1 + \frac{1}{4}$

(8) $-\frac{4}{9} + \left(-\frac{4}{5}\right)$

(18) $\frac{10}{3} - \left(-\frac{9}{8}\right)$

(28) $-\frac{1}{2} + \left(-\frac{7}{8}\right)$

(38) $\frac{10}{7} - \frac{2}{9}$

(9) $-6 - \left(-\frac{9}{4}\right)$

(19) $-\frac{6}{7} - \left(-\frac{8}{9}\right)$

(29) $-\frac{1}{7} - \frac{8}{3}$

(39) $2 + \frac{4}{3}$

(10) $-\frac{4}{7} + 10$

(20) $\frac{1}{5} + (-5)$

(30) $6 - \frac{1}{2}$

(40) $-\frac{3}{8} + \left(-\frac{2}{7}\right)$

(1) $\frac{5}{4} - \left(-\frac{5}{4}\right)$

(11) $-5 + \left(-\frac{6}{7}\right)$

(21) $\frac{7}{6} - (-1)$

(31) $\frac{1}{2} - 5$

(2) $1 - (-1)$

(12) $-\frac{2}{5} + 1$

(22) $-\frac{2}{5} - \left(-\frac{1}{6}\right)$

(32) $-3 - \left(-\frac{2}{9}\right)$

(3) $3 + \frac{10}{3}$

(13) $-\frac{1}{3} - \frac{8}{3}$

(23) $-\frac{6}{7} + \frac{1}{2}$

(33) $\frac{1}{7} + 1$

(4) $\frac{2}{5} + (-6)$

(14) $\frac{5}{2} + 2$

(24) $\frac{3}{4} + \left(-\frac{1}{7}\right)$

(34) $\frac{2}{3} + \frac{4}{9}$

(5) $\frac{1}{2} + \frac{1}{4}$

(15) $-1 + \frac{3}{2}$

(25) $\frac{1}{2} - \frac{1}{3}$

(35) $-\frac{1}{4} - \frac{1}{7}$

(6) $\frac{1}{3} - \frac{4}{7}$

(16) $-\frac{5}{7} - \left(-\frac{4}{5}\right)$

(26) $-5 - (-4)$

(36) $\frac{5}{8} - \frac{6}{5}$

(7) $-5 - \left(-\frac{4}{5}\right)$

(17) $\frac{1}{2} + \frac{1}{2}$

(27) $5 - 3$

(37) $\frac{7}{6} + (-2)$

(8) $\frac{7}{6} + \left(-\frac{8}{7}\right)$

(18) $1 + \frac{9}{8}$

(28) $\frac{8}{7} + (-1)$

(38) $\frac{1}{7} - \left(-\frac{2}{5}\right)$

(9) $-2 + \frac{3}{4}$

(19) $\frac{8}{7} + \frac{7}{10}$

(29) $1 - \frac{1}{3}$

(39) $-\frac{10}{3} + \frac{3}{2}$

(10) $4 - \left(-\frac{5}{6}\right)$

(20) $-\frac{7}{8} - 1$

(30) $\frac{2}{3} + \frac{9}{4}$

(40) $\frac{4}{5} - \frac{9}{8}$

(1) $\frac{3}{4} - (-2)$

(11) $-\frac{2}{3} - \left(-\frac{3}{5}\right)$

(21) $-\frac{1}{7} - (-9)$

(31) $\frac{3}{8} - \left(-\frac{1}{5}\right)$

(2) $\frac{2}{7} - \left(-\frac{8}{5}\right)$

(12) $-\frac{5}{8} + \frac{3}{8}$

(22) $-3 - \left(-\frac{4}{9}\right)$

(32) $-3 - 1$

(3) $-\frac{5}{7} - \left(-\frac{8}{7}\right)$

(13) $1 - \frac{9}{10}$

(23) $\frac{1}{2} + \frac{5}{4}$

(33) $-4 - \left(-\frac{1}{10}\right)$

(4) $-\frac{3}{2} + \frac{1}{3}$

(14) $-\frac{1}{2} - \frac{8}{3}$

(24) $-\frac{10}{9} - (-1)$

(34) $\frac{5}{2} - \left(-\frac{10}{3}\right)$

(5) $-\frac{2}{3} - \left(-\frac{7}{6}\right)$

(15) $1 - \left(-\frac{3}{2}\right)$

(25) $-\frac{3}{8} - 1$

(35) $-8 - \left(-\frac{4}{7}\right)$

(6) $-8 - \left(-\frac{5}{9}\right)$

(16) $-1 + (-5)$

(26) $7 + 1$

(36) $-\frac{5}{4} - \left(-\frac{5}{4}\right)$

(7) $2 - \left(-\frac{3}{4}\right)$

(17) $\frac{1}{4} + \left(-\frac{5}{4}\right)$

(27) $\frac{3}{7} - \frac{9}{5}$

(37) $\frac{7}{3} - \frac{5}{4}$

(18) $\frac{1}{10} - \left(-\frac{9}{8}\right)$

(28) $-4 - \frac{3}{5}$

(38) $4 - \frac{9}{4}$

(8) $-\frac{3}{5} + \frac{9}{7}$

(19) $\frac{7}{8} - (-7)$

(29) $1 - 1$

(39) $-\frac{7}{8} + (-2)$

(9) $-\frac{2}{3} - \left(-\frac{5}{3}\right)$

(20) $\frac{5}{2} - 8$

(30) $\frac{2}{3} + 1$

(40) $\frac{8}{9} - \frac{8}{9}$

(10) $\frac{7}{10} + \left(-\frac{9}{4}\right)$

(1) $-\frac{7}{5} + \left(-\frac{10}{9}\right)$

(11) $-\frac{1}{2} + \frac{9}{4}$

(21) $1 + \frac{1}{3}$

(31) $-\frac{7}{5} + \left(-\frac{2}{3}\right)$

(2) $-\frac{6}{5} - \frac{9}{7}$

(12) $-9 - \left(-\frac{5}{4}\right)$

(22) $\frac{5}{2} - (-1)$

(32) $3 + \frac{3}{5}$

(3) $-\frac{5}{2} - \frac{3}{4}$

(13) $-\frac{9}{7} + \left(-\frac{8}{9}\right)$

(23) $-5 - \left(-\frac{6}{7}\right)$

(33) $\frac{4}{3} + 10$

(4) $2 + \left(-\frac{4}{3}\right)$

(14) $\frac{10}{7} + \frac{1}{2}$

(24) $-\frac{1}{2} + \left(-\frac{1}{8}\right)$

(34) $-\frac{5}{9} - 1$

(5) $-1 - \left(-\frac{1}{3}\right)$

(15) $-\frac{8}{3} + (-3)$

(25) $\frac{3}{10} - \left(-\frac{4}{5}\right)$

(35) $2 - \frac{1}{5}$

(6) $\frac{5}{7} + \frac{9}{10}$

(16) $\frac{5}{4} - 4$

(26) $\frac{2}{3} - \left(-\frac{5}{8}\right)$

(36) $\frac{5}{3} - (-4)$

(7) $\frac{1}{4} + \left(-\frac{3}{8}\right)$

(17) $1 + \frac{3}{7}$

(27) $-\frac{1}{2} + \frac{5}{4}$

(37) $-3 + 1$

(8) $-\frac{3}{5} + \left(-\frac{2}{9}\right)$

(18) $-\frac{3}{4} + (-2)$

(28) $-\frac{10}{9} - \left(-\frac{4}{9}\right)$

(38) $-2 + \left(-\frac{1}{3}\right)$

(9) $\frac{6}{7} + 5$

(19) $5 - \left(-\frac{3}{5}\right)$

(29) $-\frac{2}{3} + \frac{9}{2}$

(39) $2 - 7$

(40) $\frac{1}{3} - \frac{3}{4}$

(10) $\frac{5}{9} - (-6)$

(20) $\frac{4}{9} + \left(-\frac{3}{8}\right)$

(30) $1 - \left(-\frac{4}{5}\right)$

(1) $-\frac{1}{3} + (-1)$

(11) $1 - \left(-\frac{1}{5}\right)$

(21) $-8 + \left(-\frac{5}{9}\right)$

(31) $-\frac{4}{5} - \left(-\frac{3}{8}\right)$

(2) $\frac{8}{3} - \left(-\frac{1}{2}\right)$

(12) $-\frac{3}{2} - \frac{5}{3}$

(22) $-4 - \frac{7}{8}$

(32) $\frac{7}{3} - \left(-\frac{9}{5}\right)$

(3) $\frac{3}{10} + \left(-\frac{1}{5}\right)$

(13) $-\frac{5}{2} + \frac{9}{7}$

(23) $\frac{3}{4} - \left(-\frac{1}{2}\right)$

(33) $-\frac{9}{8} + \left(-\frac{1}{7}\right)$

(4) $\frac{2}{5} - \left(-\frac{3}{2}\right)$

(14) $-\frac{3}{10} - 10$

(24) $\frac{1}{8} + \left(-\frac{4}{9}\right)$

(34) $1 + \left(-\frac{4}{3}\right)$

(5) $-1 + (-3)$

(15) $\frac{4}{5} + \frac{5}{3}$

(25) $-\frac{2}{3} - \frac{5}{4}$

(35) $-2 - \frac{3}{2}$

(6) $\frac{4}{3} + \frac{1}{7}$

(16) $-\frac{7}{4} + \left(-\frac{8}{7}\right)$

(26) $-2 - \left(-\frac{5}{4}\right)$

(36) $-\frac{4}{3} + (-1)$

(7) $-\frac{1}{2} - \frac{1}{4}$

(17) $-\frac{5}{6} - \left(-\frac{7}{6}\right)$

(27) $2 - \left(-\frac{2}{5}\right)$

(37) $\frac{9}{4} - (-1)$

(8) $1 + \left(-\frac{4}{9}\right)$

(18) $2 - \left(-\frac{1}{9}\right)$

(28) $-\frac{1}{3} - \left(-\frac{9}{4}\right)$

(38) $\frac{1}{4} + \frac{1}{10}$

(9) $-1 + \left(-\frac{5}{4}\right)$

(19) $\frac{5}{4} - \left(-\frac{7}{3}\right)$

(29) $1 + \frac{1}{2}$

(39) $\frac{2}{7} - (-1)$

(10) $-\frac{6}{5} - \left(-\frac{1}{10}\right)$

(20) $\frac{5}{2} - 4$

(30) $-2 - 4$

(40) $-8 - \frac{10}{9}$

(1) $3 - \frac{1}{4}$

(11) $-\frac{9}{7} + \left(-\frac{9}{5}\right)$

(21) $-\frac{1}{9} - \frac{5}{7}$

(31) $\frac{1}{6} + \frac{7}{4}$

(2) $-4 - 1$

(12) $-\frac{7}{10} + (-1)$

(22) $\frac{5}{7} - \frac{5}{3}$

(32) $-1 + \left(-\frac{4}{9}\right)$

(3) $-\frac{4}{5} - \left(-\frac{9}{7}\right)$

(13) $\frac{6}{7} - \left(-\frac{4}{5}\right)$

(23) $3 + \left(-\frac{1}{4}\right)$

(33) $-\frac{2}{9} + \left(-\frac{2}{3}\right)$

(4) $-\frac{2}{3} + \frac{1}{9}$

(14) $\frac{1}{3} + (-2)$

(24) $\frac{9}{5} + \left(-\frac{5}{7}\right)$

(34) $\frac{4}{3} - 2$

(5) $-5 + \frac{3}{7}$

(15) $-5 - \frac{8}{9}$

(25) $-\frac{2}{5} + \frac{10}{9}$

(35) $-2 - (-1)$

(6) $\frac{3}{2} - 1$

(16) $-\frac{7}{4} + \left(-\frac{10}{7}\right)$

(26) $\frac{10}{9} + \left(-\frac{1}{9}\right)$

(36) $\frac{2}{7} + \frac{3}{4}$

(7) $\frac{7}{2} + \frac{1}{2}$

(17) $2 + \frac{1}{2}$

(27) $-\frac{5}{6} + \left(-\frac{1}{2}\right)$

(37) $-\frac{3}{2} - \frac{9}{2}$

(8) $\frac{5}{6} - 4$

(18) $\frac{1}{2} + \frac{2}{3}$

(28) $-1 + \frac{7}{4}$

(38) $\frac{7}{3} + \frac{6}{7}$

(9) $\frac{9}{8} - (-1)$

(19) $-\frac{1}{6} - \frac{3}{2}$

(29) $-\frac{10}{7} - \left(-\frac{1}{2}\right)$

(39) $-1 + \frac{5}{7}$

(10) $\frac{6}{7} + \frac{3}{4}$

(20) $-\frac{4}{7} + \frac{7}{5}$

(30) $\frac{9}{10} + \frac{1}{2}$

(40) $1 - \frac{3}{7}$

(1) $9 - \frac{7}{3}$

(11) $-\frac{5}{8} - 3$

(21) $\frac{3}{5} - \frac{10}{9}$

(31) $\frac{5}{7} - \left(-\frac{7}{4}\right)$

(2) $-\frac{4}{5} - \left(-\frac{3}{10}\right)$

(12) $-\frac{7}{3} + \left(-\frac{1}{9}\right)$

(22) $\frac{9}{10} - \frac{1}{4}$

(32) $1 + \frac{5}{4}$

(3) $-\frac{3}{4} + \left(-\frac{7}{3}\right)$

(13) $6 + \left(-\frac{1}{4}\right)$

(23) $-\frac{5}{2} - \left(-\frac{5}{2}\right)$

(33) $\frac{7}{2} - \frac{1}{3}$

(4) $-\frac{3}{7} - \left(-\frac{3}{4}\right)$

(14) $\frac{8}{3} + (-5)$

(24) $-\frac{1}{2} + \left(-\frac{6}{5}\right)$

(34) $-\frac{4}{3} - \frac{2}{9}$

(5) $2 + (-10)$

(15) $\frac{1}{6} + \left(-\frac{1}{7}\right)$

(25) $-\frac{4}{3} + 2$

(35) $-\frac{1}{5} + \frac{4}{5}$

(6) $-\frac{5}{6} - \frac{1}{2}$

(16) $2 - \frac{10}{7}$

(26) $-3 + \left(-\frac{2}{5}\right)$

(36) $\frac{5}{3} + \left(-\frac{5}{7}\right)$

(7) $-\frac{3}{8} + \left(-\frac{7}{2}\right)$

(17) $-\frac{6}{5} + \frac{2}{3}$

(27) $-\frac{9}{5} + \left(-\frac{1}{2}\right)$

(37) $\frac{3}{2} - \left(-\frac{8}{9}\right)$

(8) $\frac{10}{7} + \frac{9}{7}$

(18) $-\frac{6}{5} - \frac{3}{4}$

(28) $-\frac{1}{5} + \frac{9}{2}$

(38) $\frac{1}{5} - \left(-\frac{5}{3}\right)$

(9) $-6 - 1$

(19) $\frac{3}{5} - 1$

(29) $-\frac{8}{9} - (-2)$

(39) $-\frac{4}{5} - (-5)$

(10) $\frac{1}{4} - (-2)$

(20) $-\frac{1}{2} + \frac{1}{4}$

(30) $-\frac{1}{4} - \left(-\frac{3}{5}\right)$

(40) $\frac{6}{5} + \left(-\frac{1}{4}\right)$

(1) $\frac{1}{3} - (-5)$

(11) $\frac{7}{9} + (-1)$

(21) $-1 + \frac{10}{3}$

(31) $-\frac{7}{4} - \left(-\frac{4}{3}\right)$

(2) $\frac{9}{4} + \left(-\frac{7}{6}\right)$

(12) $-\frac{2}{5} - \frac{9}{4}$

(22) $-4 + \frac{9}{8}$

(32) $1 + \left(-\frac{1}{2}\right)$

(3) $-3 - \frac{5}{9}$

(13) $\frac{2}{5} + \left(-\frac{5}{3}\right)$

(23) $-\frac{5}{3} - \frac{1}{3}$

(33) $-3 + \frac{4}{7}$

(4) $\frac{3}{2} - \left(-\frac{7}{5}\right)$

(14) $-\frac{3}{5} - \frac{2}{7}$

(24) $-\frac{3}{5} - \left(-\frac{6}{7}\right)$

(34) $\frac{2}{3} + \frac{8}{3}$

(5) $9 + \frac{3}{2}$

(15) $\frac{4}{5} - \left(-\frac{2}{9}\right)$

(25) $-\frac{1}{6} + \left(-\frac{3}{2}\right)$

(35) $5 + \frac{1}{5}$

(6) $\frac{1}{2} - \left(-\frac{4}{7}\right)$

(16) $-\frac{7}{2} - \frac{2}{7}$

(26) $-1 - \left(-\frac{7}{10}\right)$

(36) $1 - (-2)$

(7) $-\frac{3}{4} - \frac{7}{2}$

(17) $\frac{9}{8} - \frac{1}{2}$

(27) $-10 + (-4)$

(37) $\frac{4}{5} + \left(-\frac{8}{5}\right)$

(8) $\frac{5}{2} - \frac{1}{9}$

(18) $-\frac{5}{3} - \left(-\frac{9}{5}\right)$

(28) $-\frac{5}{9} + \frac{1}{9}$

(38) $7 + \left(-\frac{5}{7}\right)$

(9) $\frac{5}{8} + \frac{10}{3}$

(19) $\frac{10}{7} + 3$

(29) $-\frac{2}{3} + \left(-\frac{4}{5}\right)$

(39) $\frac{5}{7} + \frac{3}{4}$

(10) $\frac{5}{3} - \left(-\frac{7}{4}\right)$

(20) $\frac{5}{7} - \frac{1}{4}$

(30) $-2 - \left(-\frac{1}{9}\right)$

(40) $1 + (-1)$

$$(1) -3 + \left(-\frac{3}{4}\right) \\ = -\frac{15}{4}$$

$$(2) -\frac{10}{9} + \left(-\frac{2}{3}\right) \\ = -\frac{16}{9}$$

$$(3) -\frac{4}{7} - \left(-\frac{1}{2}\right) \\ = -\frac{1}{14}$$

$$(4) 9 - \left(-\frac{8}{7}\right) \\ = \frac{71}{7}$$

$$(5) \frac{3}{5} + (-3) \\ = -\frac{12}{5}$$

$$(6) -1 - \frac{3}{4} \\ = -\frac{7}{4}$$

$$(7) \frac{3}{10} + (-5) \\ = -\frac{47}{10}$$

$$(8) -\frac{8}{7} + \frac{4}{3} \\ = \frac{4}{21}$$

$$(9) -5 - \frac{2}{3} \\ = -\frac{17}{3}$$

$$(10) \frac{1}{7} - \frac{4}{3} \\ = -\frac{25}{21}$$

$$(11) \frac{3}{4} - \frac{8}{3} \\ = -\frac{23}{12}$$

$$(12) -5 - \frac{7}{4} \\ = -\frac{27}{4}$$

$$(13) -6 + \left(-\frac{1}{3}\right) \\ = -\frac{19}{3}$$

$$(14) -\frac{7}{4} + \frac{8}{3} \\ = \frac{11}{12}$$

$$(15) -\frac{3}{7} - \left(-\frac{5}{4}\right) \\ = \frac{23}{28}$$

$$(16) -\frac{4}{5} + (-1) \\ = -\frac{9}{5}$$

$$(17) \frac{1}{5} + \left(-\frac{4}{5}\right) \\ = -\frac{3}{5}$$

$$(18) -\frac{5}{2} - \left(-\frac{7}{8}\right) \\ = -\frac{13}{8}$$

$$(19) -\frac{2}{3} + \frac{9}{4} \\ = \frac{19}{12}$$

$$(20) -\frac{1}{3} + \frac{3}{2} \\ = \frac{5}{6}$$

$$(21) 2 + 3 \\ = 5$$

$$(22) 1 + 1 \\ = 2$$

$$(23) -\frac{8}{3} + 3 \\ = \frac{1}{3}$$

$$(24) \frac{2}{5} + \left(-\frac{5}{3}\right) \\ = -\frac{19}{15}$$

$$(25) \frac{3}{5} - \frac{4}{3} \\ = -\frac{11}{15}$$

$$(26) 2 + \frac{8}{3} \\ = \frac{14}{3}$$

$$(27) -\frac{4}{9} + \left(-\frac{4}{3}\right) \\ = -\frac{16}{9}$$

$$(28) -\frac{2}{5} + \left(-\frac{1}{3}\right) \\ = -\frac{11}{15}$$

$$(29) \frac{7}{8} - \frac{8}{9} \\ = -\frac{1}{72}$$

$$(30) -\frac{5}{6} + (-3) \\ = -\frac{23}{6}$$

$$(31) \frac{1}{8} + \left(-\frac{1}{10}\right) \\ = \frac{1}{40}$$

$$(32) -\frac{1}{7} - \frac{3}{2} \\ = -\frac{23}{14}$$

$$(33) 1 + \frac{8}{3} \\ = \frac{11}{3}$$

$$(34) -3 + \frac{2}{3} \\ = -\frac{7}{3}$$

$$(35) -2 - \frac{3}{7} \\ = -\frac{17}{7}$$

$$(36) \frac{7}{3} + (-1) \\ = \frac{4}{3}$$

$$(37) -\frac{5}{3} - \frac{1}{9} \\ = -\frac{16}{9}$$

$$(38) \frac{4}{5} + 7 \\ = \frac{39}{5}$$

$$(39) -\frac{2}{3} - \left(-\frac{1}{3}\right) \\ = -\frac{1}{3}$$

$$(40) -\frac{9}{5} + 1 \\ = -\frac{4}{5}$$

$$(1) \frac{5}{3} - (-3) \\ = \frac{14}{3}$$

$$(11) -\frac{1}{8} - \left(-\frac{1}{3}\right) \\ = \frac{5}{24}$$

$$(21) \frac{1}{6} + \left(-\frac{1}{7}\right) \\ = \frac{1}{42}$$

$$(31) -\frac{1}{3} + \left(-\frac{1}{2}\right) \\ = -\frac{5}{6}$$

$$(2) -\frac{4}{9} + 10 \\ = \frac{86}{9}$$

$$(12) -\frac{8}{5} - \frac{3}{2} \\ = -\frac{31}{10}$$

$$(22) \frac{4}{7} - \frac{2}{3} \\ = -\frac{2}{21}$$

$$(32) \frac{9}{8} - \left(-\frac{5}{7}\right) \\ = \frac{103}{56}$$

$$(3) -\frac{9}{5} - \left(-\frac{1}{4}\right) \\ = -\frac{31}{20}$$

$$(13) -\frac{1}{2} + \frac{3}{5} \\ = \frac{1}{10}$$

$$(23) 3 - \frac{3}{5} \\ = \frac{12}{5}$$

$$(33) -3 + (-2) \\ = -5$$

$$(4) \frac{7}{5} - \frac{1}{5} \\ = \frac{6}{5}$$

$$(14) -5 - \left(-\frac{1}{3}\right) \\ = -\frac{14}{3}$$

$$(24) \frac{5}{2} + \left(-\frac{1}{2}\right) \\ = 2$$

$$(34) -\frac{1}{4} - \frac{3}{2} \\ = -\frac{7}{4}$$

$$(5) \frac{7}{3} + \left(-\frac{5}{8}\right) \\ = \frac{41}{24}$$

$$(15) -\frac{3}{7} + \frac{3}{5} \\ = \frac{3}{35}$$

$$(25) -\frac{3}{7} - \frac{8}{9} \\ = -\frac{83}{63}$$

$$(35) 9 - \frac{4}{9} \\ = \frac{77}{9}$$

$$(6) \frac{4}{9} + 5 \\ = \frac{49}{9}$$

$$(16) -\frac{1}{2} + \left(-\frac{7}{5}\right) \\ = -\frac{19}{10}$$

$$(26) \frac{2}{7} + (-8) \\ = -\frac{54}{7}$$

$$(36) -\frac{5}{9} + \left(-\frac{5}{2}\right) \\ = -\frac{55}{18}$$

$$(7) \frac{3}{2} + \frac{10}{7} \\ = \frac{41}{14}$$

$$(17) -2 - \left(-\frac{5}{3}\right) \\ = -\frac{1}{3}$$

$$(27) -\frac{1}{2} + (-7) \\ = -\frac{15}{2}$$

$$(37) -3 - \left(-\frac{2}{3}\right) \\ = -\frac{7}{3}$$

$$(8) -\frac{1}{9} - (-2) \\ = \frac{17}{9}$$

$$(18) \frac{1}{8} + 6 \\ = \frac{49}{8}$$

$$(28) -\frac{4}{5} + 6 \\ = \frac{26}{5}$$

$$(38) \frac{1}{2} + \frac{6}{7} \\ = \frac{19}{14}$$

$$(9) \frac{1}{7} + \left(-\frac{1}{3}\right) \\ = -\frac{4}{21}$$

$$(19) -\frac{3}{7} + \frac{4}{7} \\ = \frac{1}{7}$$

$$(29) \frac{10}{3} - \left(-\frac{6}{7}\right) \\ = \frac{88}{21}$$

$$(39) \frac{2}{3} + 2 \\ = \frac{8}{3}$$

$$(10) -\frac{1}{8} - 5 \\ = -\frac{41}{8}$$

$$(20) \frac{1}{4} - \left(-\frac{3}{7}\right) \\ = \frac{19}{28}$$

$$(30) \frac{5}{3} + \frac{4}{3} \\ = 3$$

$$(40) -6 - \frac{3}{2} \\ = -\frac{15}{2}$$

$$(1) \frac{8}{7} - \frac{2}{7} \\ = \frac{6}{7}$$

$$(11) -2 + \frac{1}{4} \\ = -\frac{7}{4}$$

$$(21) -6 - \frac{8}{9} \\ = -\frac{62}{9}$$

$$(31) 4 - 1 \\ = 3$$

$$(2) -\frac{3}{2} - \left(-\frac{6}{7}\right) \\ = -\frac{9}{14}$$

$$(12) -\frac{8}{5} + \frac{1}{9} \\ = -\frac{67}{45}$$

$$(22) -\frac{2}{5} + \left(-\frac{1}{2}\right) \\ = -\frac{9}{10}$$

$$(32) \frac{2}{7} + \frac{1}{8} \\ = \frac{23}{56}$$

$$(3) -\frac{1}{6} + \frac{7}{8} \\ = \frac{17}{24}$$

$$(13) -\frac{9}{5} - \frac{8}{9} \\ = -\frac{121}{45}$$

$$(23) -\frac{3}{4} + \frac{1}{4} \\ = -\frac{1}{2}$$

$$(33) -\frac{5}{4} - \frac{1}{7} \\ = -\frac{39}{28}$$

$$(4) \frac{2}{3} + \left(-\frac{1}{10}\right) \\ = \frac{17}{30}$$

$$(14) -1 + (-3) \\ = -4$$

$$(24) 1 - \left(-\frac{3}{2}\right) \\ = \frac{5}{2}$$

$$(34) \frac{2}{3} + \left(-\frac{4}{3}\right) \\ = -\frac{2}{3}$$

$$(5) -1 + \left(-\frac{4}{9}\right) \\ = -\frac{13}{9}$$

$$(15) \frac{1}{3} - (-8) \\ = \frac{25}{3}$$

$$(25) -1 + 3 \\ = 2$$

$$(35) -\frac{7}{10} - (-7) \\ = \frac{10}{10}$$

$$(6) \frac{1}{6} - \left(-\frac{5}{2}\right) \\ = \frac{8}{3}$$

$$(16) -10 + \frac{6}{5} \\ = -\frac{44}{5}$$

$$(26) -\frac{8}{5} + (-2) \\ = -\frac{18}{5}$$

$$(36) \frac{1}{2} - \frac{5}{6} \\ = -\frac{1}{3}$$

$$(7) -2 - \frac{5}{7} \\ = -\frac{19}{7}$$

$$(17) -\frac{7}{2} + 1 \\ = -\frac{5}{2}$$

$$(27) -\frac{10}{3} - \frac{9}{2} \\ = -\frac{47}{6}$$

$$(37) \frac{3}{8} - \left(-\frac{7}{9}\right) \\ = \frac{83}{72}$$

$$(8) \frac{1}{2} - \left(-\frac{1}{3}\right) \\ = \frac{5}{6}$$

$$(18) -\frac{3}{7} - \frac{5}{7} \\ = -\frac{8}{7}$$

$$(28) 1 + \left(-\frac{4}{7}\right) \\ = \frac{3}{7}$$

$$(38) \frac{1}{3} - (-1) \\ = \frac{4}{3}$$

$$(9) 1 - \left(-\frac{1}{10}\right) \\ = \frac{11}{10}$$

$$(19) \frac{7}{3} - 2 \\ = \frac{1}{3}$$

$$(29) 2 + \left(-\frac{9}{2}\right) \\ = -\frac{5}{2}$$

$$(39) 1 - \frac{2}{3} \\ = \frac{1}{3}$$

$$(10) -\frac{5}{4} + \left(-\frac{1}{3}\right) \\ = -\frac{19}{12}$$

$$(20) \frac{1}{2} + (-8) \\ = -\frac{15}{2}$$

$$(30) -\frac{2}{7} + \frac{7}{10} \\ = \frac{29}{70}$$

$$(40) \frac{7}{4} + 1 \\ = \frac{11}{4}$$

$$(1) \frac{1}{3} - 2 \\ = -\frac{5}{3}$$

$$(11) 10 - \left(-\frac{3}{2}\right) \\ = \frac{23}{2}$$

$$(21) -8 + (-2) \\ = -10$$

$$(31) \frac{9}{4} + 1 \\ = \frac{13}{4}$$

$$(2) \frac{1}{4} + 3 \\ = \frac{13}{4}$$

$$(12) -\frac{5}{4} + \frac{7}{9} \\ = -\frac{17}{36}$$

$$(22) \frac{7}{8} - 5 \\ = -\frac{33}{8}$$

$$(32) -5 - 1 \\ = -6$$

$$(3) \frac{3}{10} + \left(-\frac{5}{2}\right) \\ = -\frac{11}{5}$$

$$(13) -1 + \left(-\frac{2}{3}\right) \\ = -\frac{5}{3}$$

$$(23) \frac{3}{2} + \frac{2}{9} \\ = \frac{31}{18}$$

$$(33) -\frac{5}{2} - (-1) \\ = -\frac{3}{2}$$

$$(4) \frac{3}{5} - \frac{4}{9} \\ = \frac{7}{45}$$

$$(14) \frac{3}{8} + 10 \\ = \frac{83}{8}$$

$$(24) \frac{4}{7} - \frac{7}{5} \\ = -\frac{29}{35}$$

$$(34) -\frac{7}{4} + \left(-\frac{1}{2}\right) \\ = -\frac{9}{4}$$

$$(5) -\frac{4}{3} + \frac{3}{8} \\ = -\frac{23}{24}$$

$$(15) -\frac{5}{4} + \left(-\frac{6}{5}\right) \\ = -\frac{49}{20}$$

$$(25) -8 + 2 \\ = -6$$

$$(35) \frac{7}{9} - 1 \\ = -\frac{2}{9}$$

$$(6) -\frac{5}{8} + \left(-\frac{4}{7}\right) \\ = -\frac{67}{56}$$

$$(16) -\frac{4}{7} - \left(-\frac{8}{3}\right) \\ = \frac{44}{21}$$

$$(26) \frac{4}{5} - 1 \\ = -\frac{1}{5}$$

$$(36) -5 - \left(-\frac{1}{2}\right) \\ = -\frac{9}{2}$$

$$(7) -1 + \left(-\frac{1}{5}\right) \\ = -\frac{6}{5}$$

$$(17) \frac{1}{2} + \frac{5}{8} \\ = \frac{9}{8}$$

$$(27) 3 - \frac{4}{7} \\ = \frac{17}{7}$$

$$(37) -\frac{1}{4} + \left(-\frac{1}{7}\right) \\ = -\frac{11}{28}$$

$$(8) \frac{7}{5} - \left(-\frac{5}{2}\right) \\ = \frac{39}{10}$$

$$(18) -\frac{1}{3} - \left(-\frac{1}{10}\right) \\ = -\frac{7}{30}$$

$$(28) 5 + \left(-\frac{1}{8}\right) \\ = \frac{39}{8}$$

$$(38) \frac{5}{2} - (-3) \\ = \frac{11}{2}$$

$$(9) -1 - (-1) \\ = 0$$

$$(19) 1 - \left(-\frac{1}{4}\right) \\ = \frac{5}{4}$$

$$(29) 4 - 4 \\ = 0$$

$$(39) \frac{2}{3} + \frac{9}{8} \\ = \frac{43}{24}$$

$$(10) -\frac{5}{8} - \frac{1}{3} \\ = -\frac{23}{24}$$

$$(20) -1 - (-1) \\ = 0$$

$$(30) \frac{7}{4} - (-5) \\ = \frac{27}{4}$$

$$(40) \frac{7}{10} - \left(-\frac{9}{5}\right) \\ = \frac{5}{2}$$

$$(1) -5 - \left(-\frac{6}{5}\right) \\ = -\frac{19}{5}$$

$$(11) \frac{8}{7} + \left(-\frac{8}{5}\right) \\ = -\frac{16}{35}$$

$$(21) -\frac{3}{8} + \left(-\frac{4}{7}\right) \\ = -\frac{53}{56}$$

$$(31) -1 - \frac{3}{5} \\ = -\frac{8}{5}$$

$$(2) -1 - \left(-\frac{10}{7}\right) \\ = \frac{3}{7}$$

$$(12) \frac{6}{5} - \left(-\frac{1}{4}\right) \\ = \frac{29}{20}$$

$$(22) -\frac{9}{10} - (-1) \\ = \frac{1}{10}$$

$$(32) 1 - \left(-\frac{5}{3}\right) \\ = \frac{8}{3}$$

$$(3) -\frac{3}{4} + \left(-\frac{1}{4}\right) \\ = -1$$

$$(13) -9 - \left(-\frac{1}{2}\right) \\ = -\frac{17}{2}$$

$$(23) -\frac{2}{5} + (-2) \\ = -\frac{12}{5}$$

$$(33) \frac{9}{8} - (-3) \\ = \frac{33}{8}$$

$$(4) -5 - \frac{9}{10} \\ = -\frac{59}{10}$$

$$(14) \frac{2}{5} - \left(-\frac{9}{2}\right) \\ = \frac{49}{10}$$

$$(24) -\frac{5}{8} - \frac{4}{9} \\ = -\frac{77}{72}$$

$$(34) -\frac{1}{8} - 3 \\ = -\frac{25}{8}$$

$$(5) \frac{1}{9} + \frac{10}{7} \\ = \frac{97}{63}$$

$$(15) -5 - \frac{4}{7} \\ = -\frac{39}{7}$$

$$(25) 1 + \left(-\frac{3}{8}\right) \\ = \frac{5}{8}$$

$$(35) \frac{7}{9} + \frac{7}{3} \\ = \frac{28}{9}$$

$$(6) -\frac{10}{7} + \left(-\frac{7}{6}\right) \\ = -\frac{109}{42}$$

$$(16) -\frac{1}{4} - \frac{2}{3} \\ = -\frac{11}{12}$$

$$(26) 1 - (-3) \\ = 4$$

$$(36) -\frac{3}{4} + \frac{1}{2} \\ = -\frac{1}{4}$$

$$(7) -5 + \left(-\frac{2}{7}\right) \\ = -\frac{37}{7}$$

$$(17) \frac{4}{3} + \frac{1}{7} \\ = \frac{31}{21}$$

$$(27) -\frac{10}{3} - 2 \\ = -\frac{16}{3}$$

$$(37) -\frac{2}{3} - \frac{2}{3} \\ = -\frac{4}{3}$$

$$(8) 7 + \left(-\frac{8}{3}\right) \\ = \frac{13}{3}$$

$$(18) -\frac{5}{7} + 1 \\ = \frac{2}{7}$$

$$(28) -\frac{1}{6} - \frac{1}{8} \\ = -\frac{7}{24}$$

$$(38) -\frac{5}{9} - \frac{6}{5} \\ = -\frac{79}{45}$$

$$(9) -3 + 1 \\ = -2$$

$$(19) \frac{5}{2} + \frac{3}{8} \\ = \frac{23}{8}$$

$$(29) \frac{10}{9} - \frac{5}{2} \\ = -\frac{1}{18}$$

$$(39) \frac{3}{8} - \frac{1}{2} \\ = -\frac{1}{8}$$

$$(10) 2 + (-3) \\ = -1$$

$$(20) \frac{2}{3} - \left(-\frac{3}{4}\right) \\ = \frac{17}{12}$$

$$(30) -1 - \frac{3}{5} \\ = -\frac{8}{5}$$

$$(40) -\frac{1}{7} + \frac{5}{8} \\ = \frac{27}{56}$$

$$(1) -\frac{1}{2} - \frac{4}{3} \\ = -\frac{11}{6}$$

$$(2) 3 - (-1) \\ = 4$$

$$(3) \frac{1}{2} - 2 \\ = -\frac{3}{2}$$

$$(4) -\frac{1}{2} + \left(-\frac{4}{5}\right) \\ = -\frac{13}{10}$$

$$(5) -\frac{1}{2} + 1 \\ = \frac{1}{2}$$

$$(6) -1 + \frac{5}{4} \\ = \frac{1}{4}$$

$$(7) -\frac{7}{9} + 4 \\ = \frac{29}{9}$$

$$(8) -8 - \left(-\frac{3}{4}\right) \\ = -\frac{29}{4}$$

$$(9) -\frac{6}{7} + 4 \\ = \frac{22}{7}$$

$$(10) 1 + \frac{1}{2} \\ = \frac{3}{2}$$

$$(11) -\frac{8}{9} + \frac{1}{5} \\ = -\frac{31}{45}$$

$$(12) \frac{3}{5} - \frac{2}{5} \\ = \frac{1}{5}$$

$$(13) -\frac{5}{6} + \frac{9}{8} \\ = \frac{7}{24}$$

$$(14) -\frac{5}{4} + \frac{4}{5} \\ = -\frac{9}{20}$$

$$(15) -\frac{3}{10} + \frac{2}{7} \\ = -\frac{1}{70}$$

$$(16) \frac{3}{2} - \frac{5}{2} \\ = -1$$

$$(17) \frac{5}{3} - 10 \\ = -\frac{25}{3}$$

$$(18) \frac{7}{3} - \frac{10}{3} \\ = -1$$

$$(19) \frac{1}{10} - \frac{1}{3} \\ = -\frac{7}{30}$$

$$(20) \frac{5}{6} + (-10) \\ = -\frac{55}{6}$$

$$(21) -10 + (-4) \\ = -14$$

$$(22) \frac{7}{3} - \left(-\frac{4}{7}\right) \\ = \frac{61}{21}$$

$$(23) -\frac{3}{8} + \frac{1}{7} \\ = -\frac{13}{56}$$

$$(24) -\frac{1}{3} - (-6) \\ = \frac{17}{3}$$

$$(25) -\frac{2}{3} - \left(-\frac{4}{3}\right) \\ = \frac{2}{3}$$

$$(26) \frac{3}{2} + \left(-\frac{3}{5}\right) \\ = \frac{9}{10}$$

$$(27) -3 + \frac{4}{3} \\ = -\frac{5}{3}$$

$$(28) 3 - \frac{5}{6} \\ = \frac{13}{6}$$

$$(29) -\frac{3}{2} - \left(-\frac{1}{8}\right) \\ = -\frac{11}{8}$$

$$(30) -\frac{2}{9} - \left(-\frac{1}{9}\right) \\ = -\frac{1}{9}$$

$$(31) -\frac{1}{5} - \left(-\frac{1}{2}\right) \\ = \frac{3}{10}$$

$$(32) -\frac{1}{4} + \left(-\frac{7}{2}\right) \\ = -\frac{15}{4}$$

$$(33) -\frac{3}{5} + 8 \\ = \frac{37}{5}$$

$$(34) \frac{7}{10} + \frac{4}{9} \\ = \frac{103}{90}$$

$$(35) -\frac{1}{5} + \frac{5}{2} \\ = \frac{23}{10}$$

$$(36) \frac{1}{5} + \left(-\frac{2}{3}\right) \\ = -\frac{7}{15}$$

$$(37) -\frac{7}{5} - \left(-\frac{4}{5}\right) \\ = -\frac{3}{5}$$

$$(38) -7 + \left(-\frac{7}{2}\right) \\ = -\frac{21}{2}$$

$$(39) -\frac{1}{4} - \left(-\frac{9}{10}\right) \\ = \frac{13}{20}$$

$$(40) \frac{6}{5} + (-1) \\ = \frac{1}{5}$$

$$(1) -\frac{5}{7} + \left(-\frac{7}{10}\right) \\ = -\frac{99}{70}$$

$$(2) -7 - \left(-\frac{2}{5}\right) \\ = -\frac{33}{5}$$

$$(3) -\frac{7}{2} + \frac{7}{6} \\ = -\frac{7}{3}$$

$$(4) -\frac{5}{9} + \frac{3}{4} \\ = \frac{7}{36}$$

$$(5) -\frac{8}{5} - \left(-\frac{1}{10}\right) \\ = -\frac{3}{2}$$

$$(6) -2 - \frac{8}{7} \\ = -\frac{22}{7}$$

$$(7) -\frac{7}{9} + \left(-\frac{4}{9}\right) \\ = -\frac{11}{9}$$

$$(8) 3 + \left(-\frac{5}{4}\right) \\ = \frac{7}{4}$$

$$(9) -\frac{3}{10} - \frac{1}{2} \\ = -\frac{4}{5}$$

$$(10) \frac{2}{3} - \frac{8}{3} \\ = -2$$

$$(11) 1 + 1 \\ = 2$$

$$(12) -\frac{7}{8} + (-3) \\ = -\frac{31}{8}$$

$$(13) \frac{2}{7} + \left(-\frac{1}{3}\right) \\ = -\frac{1}{21}$$

$$(14) -\frac{7}{5} + \frac{4}{3} \\ = -\frac{1}{15}$$

$$(15) -\frac{2}{7} - (-5) \\ = \frac{33}{7}$$

$$(16) -\frac{3}{10} - \left(-\frac{3}{5}\right) \\ = \frac{3}{10}$$

$$(17) -\frac{1}{3} + \frac{2}{5} \\ = \frac{1}{15}$$

$$(18) -\frac{1}{2} - 2 \\ = -\frac{5}{2}$$

$$(19) \frac{7}{6} + (-1) \\ = \frac{1}{6}$$

$$(20) -1 + (-2) \\ = -3$$

$$(21) -1 + (-1) \\ = -2$$

$$(22) \frac{3}{5} - \frac{7}{2} \\ = -\frac{29}{10}$$

$$(23) -\frac{9}{2} - \frac{3}{4} \\ = -\frac{21}{4}$$

$$(24) 9 + 1 \\ = 10$$

$$(25) \frac{1}{8} - \frac{5}{3} \\ = -\frac{37}{24}$$

$$(26) \frac{5}{3} + \frac{3}{4} \\ = \frac{29}{12}$$

$$(27) \frac{5}{6} - (-2) \\ = \frac{17}{6}$$

$$(28) -\frac{4}{3} + \frac{1}{2} \\ = -\frac{5}{6}$$

$$(29) -\frac{8}{3} + \left(-\frac{1}{2}\right) \\ = -\frac{19}{6}$$

$$(30) -\frac{9}{2} + \frac{9}{2} \\ = 0$$

$$(31) -\frac{3}{4} - 1 \\ = -\frac{7}{4}$$

$$(32) \frac{10}{3} + (-4) \\ = -\frac{2}{3}$$

$$(33) \frac{3}{2} - \frac{7}{3} \\ = -\frac{5}{6}$$

$$(34) \frac{6}{7} + \left(-\frac{10}{7}\right) \\ = -\frac{4}{7}$$

$$(35) \frac{5}{3} + \frac{4}{5} \\ = \frac{37}{15}$$

$$(36) -\frac{1}{2} - \left(-\frac{7}{9}\right) \\ = \frac{5}{18}$$

$$(37) -1 + \frac{2}{7} \\ = -\frac{5}{7}$$

$$(38) -\frac{7}{2} + (-1) \\ = -\frac{9}{2}$$

$$(39) -\frac{8}{3} + \left(-\frac{4}{7}\right) \\ = -\frac{68}{21}$$

$$(40) -\frac{1}{3} + 1 \\ = \frac{2}{3}$$

$$(1) \quad -\frac{3}{4} - \frac{9}{4} \\ = -3$$

$$(11) \quad -2 - \left(-\frac{5}{9}\right) \\ = -\frac{13}{9}$$

$$(21) \quad \frac{2}{5} - 5 \\ = -\frac{23}{5}$$

$$(31) \quad \frac{1}{2} - \left(-\frac{7}{8}\right) \\ = \frac{11}{8}$$

$$(2) \quad \frac{4}{3} + \left(-\frac{7}{2}\right) \\ = -\frac{13}{6}$$

$$(12) \quad -10 + \frac{10}{9} \\ = -\frac{80}{9}$$

$$(22) \quad -\frac{10}{3} + \left(-\frac{2}{7}\right) \\ = -\frac{76}{21}$$

$$(32) \quad -\frac{1}{3} + \left(-\frac{9}{8}\right) \\ = -\frac{35}{24}$$

$$(3) \quad \frac{10}{7} + \frac{5}{3} \\ = \frac{65}{21}$$

$$(13) \quad -1 + (-1) \\ = -2$$

$$(23) \quad 9 + \frac{2}{3} \\ = \frac{29}{3}$$

$$(33) \quad \frac{9}{8} - \frac{7}{2} \\ = -\frac{19}{8}$$

$$(4) \quad -1 - \left(-\frac{6}{5}\right) \\ = \frac{1}{5}$$

$$(14) \quad 3 + \frac{1}{5} \\ = \frac{16}{5}$$

$$(24) \quad 2 + \left(-\frac{7}{9}\right) \\ = \frac{11}{9}$$

$$(34) \quad -\frac{1}{3} - \frac{7}{10} \\ = -\frac{31}{30}$$

$$(5) \quad -\frac{4}{3} - (-2) \\ = \frac{2}{3}$$

$$(15) \quad 3 + \frac{5}{9} \\ = \frac{32}{9}$$

$$(25) \quad -\frac{4}{3} - \frac{1}{3} \\ = -\frac{5}{3}$$

$$(35) \quad 7 + \frac{5}{8} \\ = \frac{61}{8}$$

$$(6) \quad \frac{2}{5} - \frac{5}{3} \\ = -\frac{19}{15}$$

$$(16) \quad \frac{5}{3} - \frac{9}{4} \\ = -\frac{7}{12}$$

$$(26) \quad -\frac{3}{8} - \left(-\frac{1}{2}\right) \\ = \frac{1}{8}$$

$$(36) \quad -3 - \left(-\frac{1}{2}\right) \\ = -\frac{5}{2}$$

$$(7) \quad -\frac{3}{5} + \frac{2}{3} \\ = \frac{1}{15}$$

$$(17) \quad 10 + \left(-\frac{9}{4}\right) \\ = \frac{31}{4}$$

$$(27) \quad 2 + \left(-\frac{5}{3}\right) \\ = \frac{1}{3}$$

$$(37) \quad -\frac{8}{5} - \left(-\frac{3}{10}\right) \\ = -\frac{13}{10}$$

$$(8) \quad \frac{3}{2} - \left(-\frac{5}{7}\right) \\ = \frac{31}{14}$$

$$(18) \quad \frac{9}{5} + \left(-\frac{6}{5}\right) \\ = \frac{3}{5}$$

$$(28) \quad 4 + \left(-\frac{1}{4}\right) \\ = \frac{15}{4}$$

$$(38) \quad \frac{4}{5} - 2 \\ = -\frac{6}{5}$$

$$(9) \quad \frac{5}{2} + \frac{2}{5} \\ = \frac{29}{10}$$

$$(19) \quad -\frac{1}{2} + 1 \\ = \frac{1}{2}$$

$$(29) \quad \frac{3}{2} + \frac{2}{3} \\ = \frac{13}{6}$$

$$(39) \quad \frac{2}{7} + \frac{8}{7} \\ = \frac{10}{7}$$

$$(10) \quad \frac{1}{10} + \left(-\frac{4}{5}\right) \\ = -\frac{7}{10}$$

$$(20) \quad \frac{3}{4} + \frac{1}{7} \\ = \frac{25}{28}$$

$$(30) \quad \frac{7}{9} - (-1) \\ = \frac{16}{9}$$

$$(40) \quad -\frac{9}{2} - 1 \\ = -\frac{11}{2}$$

$$(1) 2 - \frac{3}{2}$$

$$= \frac{1}{2}$$

$$(11) \frac{3}{4} + (-1)$$

$$= -\frac{1}{4}$$

$$(21) \frac{8}{3} + 1$$

$$= \frac{11}{3}$$

$$(31) -\frac{5}{3} + \frac{2}{7}$$

$$= -\frac{29}{21}$$

$$(2) -\frac{6}{5} + \left(-\frac{9}{2}\right)$$

$$= -\frac{57}{10}$$

$$(12) \frac{4}{9} - \frac{1}{9}$$

$$= \frac{1}{3}$$

$$(22) \frac{9}{2} - \left(-\frac{1}{9}\right)$$

$$= \frac{83}{18}$$

$$(32) -\frac{3}{2} + \frac{7}{8}$$

$$= -\frac{5}{8}$$

$$(3) 1 + \left(-\frac{7}{4}\right)$$

$$= -\frac{3}{4}$$

$$(13) \frac{2}{5} + \left(-\frac{10}{7}\right)$$

$$= -\frac{36}{35}$$

$$(23) -1 + \frac{1}{3}$$

$$= -\frac{2}{3}$$

$$(33) -\frac{3}{7} - \frac{4}{5}$$

$$= -\frac{43}{35}$$

$$(4) -5 + \frac{5}{7}$$

$$= -\frac{30}{7}$$

$$(14) -4 + \left(-\frac{5}{4}\right)$$

$$= -\frac{21}{4}$$

$$(24) -\frac{3}{2} - (-3)$$

$$= \frac{3}{2}$$

$$(34) 2 - 2$$

$$= 0$$

$$(5) -\frac{6}{5} + \frac{5}{2}$$

$$= \frac{13}{10}$$

$$(15) -\frac{5}{2} - \frac{4}{7}$$

$$= -\frac{43}{14}$$

$$(25) \frac{4}{5} + (-1)$$

$$= -\frac{1}{5}$$

$$(35) \frac{2}{3} - \left(-\frac{5}{3}\right)$$

$$= \frac{7}{3}$$

$$(6) -4 - 3$$

$$= -7$$

$$(16) -1 - (-8)$$

$$= 7$$

$$(26) \frac{5}{9} + \left(-\frac{9}{7}\right)$$

$$= -\frac{46}{63}$$

$$(36) \frac{5}{4} + \left(-\frac{2}{3}\right)$$

$$= \frac{7}{12}$$

$$(7) -\frac{1}{2} - \left(-\frac{7}{4}\right)$$

$$= \frac{5}{4}$$

$$(17) \frac{3}{2} - \frac{7}{4}$$

$$= -\frac{1}{4}$$

$$(27) -\frac{7}{10} + \frac{1}{6}$$

$$= -\frac{8}{15}$$

$$(37) \frac{8}{7} + \frac{3}{2}$$

$$= \frac{37}{14}$$

$$(8) 5 - \left(-\frac{3}{10}\right)$$

$$= \frac{53}{10}$$

$$(18) -\frac{9}{10} + \frac{7}{6}$$

$$= \frac{1}{15}$$

$$(28) -2 + \frac{5}{3}$$

$$= -\frac{1}{3}$$

$$(38) -1 + \left(-\frac{1}{3}\right)$$

$$= -\frac{4}{3}$$

$$(9) \frac{10}{9} + \frac{5}{6}$$

$$= \frac{35}{18}$$

$$(19) -\frac{1}{3} + \left(-\frac{5}{9}\right)$$

$$= -\frac{8}{9}$$

$$(29) \frac{2}{5} + \left(-\frac{3}{4}\right)$$

$$= -\frac{7}{20}$$

$$(39) 2 + \frac{10}{3}$$

$$= \frac{16}{3}$$

$$(10) \frac{3}{10} + \frac{9}{5}$$

$$= \frac{21}{10}$$

$$(20) 2 - \frac{5}{3}$$

$$= \frac{1}{3}$$

$$(30) -\frac{1}{2} + \left(-\frac{3}{2}\right)$$

$$= -2$$

$$(40) -1 - (-6)$$

$$= 5$$

$$(1) -\frac{1}{2} + \frac{5}{7}$$

$$= -\frac{3}{14}$$

$$(11) -\frac{1}{6} + 1$$

$$= \frac{5}{6}$$

$$(21) \frac{3}{5} - (-1)$$

$$= \frac{8}{5}$$

$$(31) \frac{2}{5} + \left(-\frac{4}{3}\right)$$

$$= -\frac{14}{15}$$

$$(2) \frac{1}{4} - \frac{9}{2}$$

$$= -\frac{17}{4}$$

$$(12) \frac{4}{5} + \frac{2}{5}$$

$$= \frac{6}{5}$$

$$(22) -\frac{9}{8} + 1$$

$$= -\frac{1}{8}$$

$$(32) \frac{4}{3} - \left(-\frac{3}{2}\right)$$

$$= \frac{17}{6}$$

$$(3) -3 - \left(-\frac{9}{2}\right)$$

$$= \frac{3}{2}$$

$$(13) -2 + \frac{1}{3}$$

$$= -\frac{5}{3}$$

$$(23) -9 - \left(-\frac{9}{8}\right)$$

$$= -\frac{63}{8}$$

$$(33) \frac{7}{5} + \left(-\frac{5}{3}\right)$$

$$= -\frac{4}{15}$$

$$(4) \frac{5}{2} - \frac{1}{3}$$

$$= \frac{13}{6}$$

$$(14) -7 - \frac{7}{6}$$

$$= -\frac{49}{6}$$

$$(24) -\frac{4}{7} + \frac{7}{9}$$

$$= \frac{13}{63}$$

$$(34) \frac{3}{5} + \frac{8}{9}$$

$$= \frac{67}{45}$$

$$(5) \frac{9}{2} + 4$$

$$= \frac{17}{2}$$

$$(15) 1 - \frac{1}{3}$$

$$= \frac{2}{3}$$

$$(25) 9 + 1$$

$$= 10$$

$$(35) \frac{3}{4} + \left(-\frac{9}{10}\right)$$

$$= -\frac{3}{20}$$

$$(6) -\frac{1}{5} - \frac{2}{3}$$

$$= -\frac{13}{15}$$

$$(16) \frac{1}{8} - \frac{2}{3}$$

$$= -\frac{13}{24}$$

$$(26) -\frac{6}{5} + \frac{6}{7}$$

$$= -\frac{12}{35}$$

$$(36) -\frac{8}{7} + \frac{3}{2}$$

$$= \frac{5}{14}$$

$$(7) -\frac{10}{9} + \frac{9}{2}$$

$$= \frac{61}{18}$$

$$(17) \frac{5}{4} - \frac{5}{9}$$

$$= \frac{25}{36}$$

$$(27) -\frac{1}{2} + \frac{1}{10}$$

$$= -\frac{2}{5}$$

$$(37) -\frac{6}{5} - (-10)$$

$$= \frac{44}{5}$$

$$(8) 1 + \frac{1}{2}$$

$$= \frac{3}{2}$$

$$(18) \frac{5}{6} + 3$$

$$= \frac{23}{6}$$

$$(28) 1 - \left(-\frac{9}{10}\right)$$

$$= \frac{19}{10}$$

$$(38) -\frac{9}{5} - \left(-\frac{1}{2}\right)$$

$$= -\frac{13}{10}$$

$$(9) -\frac{9}{7} + \frac{1}{10}$$

$$= -\frac{83}{70}$$

$$(19) \frac{1}{2} - \left(-\frac{9}{2}\right)$$

$$= 5$$

$$(29) \frac{7}{8} - \left(-\frac{1}{10}\right)$$

$$= \frac{39}{40}$$

$$(39) \frac{5}{2} - \frac{9}{4}$$

$$= \frac{1}{4}$$

$$(10) \frac{4}{7} + \frac{1}{5}$$

$$= \frac{27}{35}$$

$$(20) -2 + \frac{2}{5}$$

$$= -\frac{8}{5}$$

$$(30) -\frac{2}{5} - (-8)$$

$$= \frac{38}{5}$$

$$(40) -\frac{5}{7} + \left(-\frac{9}{2}\right)$$

$$= -\frac{73}{14}$$

$$(1) -9 + \left(-\frac{9}{4}\right) \\ = -\frac{45}{4}$$

$$(11) -\frac{1}{7} - \left(-\frac{1}{2}\right) \\ = \frac{5}{14}$$

$$(21) \frac{10}{3} - \frac{8}{5} \\ = \frac{26}{15}$$

$$(31) -\frac{1}{5} + (-10) \\ = -\frac{51}{5}$$

$$(2) 5 - 8 \\ = -3$$

$$(12) \frac{3}{7} - \frac{1}{4} \\ = \frac{5}{28}$$

$$(22) \frac{3}{7} - 1 \\ = -\frac{4}{7}$$

$$(32) -\frac{9}{10} + \left(-\frac{7}{3}\right) \\ = -\frac{97}{30}$$

$$(3) \frac{2}{5} + \frac{3}{2} \\ = \frac{19}{10}$$

$$(13) -\frac{1}{2} + \left(-\frac{3}{5}\right) \\ = -\frac{11}{10}$$

$$(23) -\frac{7}{6} - \left(-\frac{4}{5}\right) \\ = -\frac{11}{30}$$

$$(33) \frac{8}{3} + (-6) \\ = -\frac{10}{3}$$

$$(4) 7 + \frac{2}{5} \\ = \frac{37}{5}$$

$$(14) \frac{2}{5} - \left(-\frac{7}{4}\right) \\ = \frac{43}{20}$$

$$(24) \frac{1}{2} + \frac{4}{7} \\ = \frac{15}{14}$$

$$(34) -\frac{6}{5} - \left(-\frac{1}{2}\right) \\ = -\frac{7}{10}$$

$$(5) -\frac{5}{2} + \left(-\frac{9}{4}\right) \\ = -\frac{19}{4}$$

$$(15) 8 - 2 \\ = 6$$

$$(25) \frac{1}{7} - 5 \\ = -\frac{34}{7}$$

$$(35) \frac{3}{2} - \frac{1}{7} \\ = \frac{19}{14}$$

$$(6) -1 + \frac{4}{3} \\ = \frac{1}{3}$$

$$(16) 2 + 1 \\ = 3$$

$$(26) -\frac{9}{8} - \left(-\frac{7}{6}\right) \\ = \frac{1}{24}$$

$$(36) \frac{3}{10} + \frac{2}{7} \\ = \frac{41}{70}$$

$$(7) \frac{10}{7} - \left(-\frac{7}{9}\right) \\ = \frac{139}{63}$$

$$(17) -\frac{8}{7} - \frac{5}{4} \\ = -\frac{67}{28}$$

$$(27) \frac{2}{3} - \frac{8}{9} \\ = -\frac{2}{9}$$

$$(37) \frac{6}{5} - (-1) \\ = \frac{11}{5}$$

$$(8) \frac{3}{2} - 1 \\ = \frac{1}{2}$$

$$(18) \frac{4}{3} - \frac{1}{3} \\ = 1$$

$$(28) \frac{1}{7} + \left(-\frac{9}{10}\right) \\ = -\frac{53}{70}$$

$$(38) -\frac{4}{5} - \left(-\frac{9}{2}\right) \\ = \frac{37}{10}$$

$$(9) -1 - \frac{7}{3} \\ = -\frac{10}{3}$$

$$(19) \frac{10}{3} - \left(-\frac{1}{2}\right) \\ = \frac{23}{6}$$

$$(29) \frac{5}{8} + \left(-\frac{1}{8}\right) \\ = \frac{1}{2}$$

$$(39) -\frac{2}{3} - (-2) \\ = \frac{4}{3}$$

$$(10) 9 + \left(-\frac{1}{9}\right) \\ = \frac{80}{9}$$

$$(20) -2 + \left(-\frac{1}{8}\right) \\ = -\frac{17}{8}$$

$$(30) \frac{5}{2} + \frac{3}{4} \\ = \frac{13}{4}$$

$$(40) -2 + \left(-\frac{7}{9}\right) \\ = -\frac{25}{9}$$

$$(1) \ 5 + (-3) \\ = 2$$

$$(11) \ -\frac{4}{7} - \left(-\frac{4}{7}\right) \\ = 0$$

$$(21) \ -\frac{3}{5} + \frac{5}{4} \\ = \frac{13}{20}$$

$$(31) \ \frac{7}{6} - \frac{2}{3} \\ = \frac{1}{2}$$

$$(2) \ -\frac{1}{5} - \frac{1}{4} \\ = -\frac{9}{20}$$

$$(12) \ -1 - \frac{1}{5} \\ = -\frac{6}{5}$$

$$(22) \ -\frac{2}{3} + \left(-\frac{5}{8}\right) \\ = -\frac{31}{24}$$

$$(32) \ 2 + \left(-\frac{1}{3}\right) \\ = \frac{5}{3}$$

$$(3) \ \frac{2}{5} + \left(-\frac{9}{4}\right) \\ = -\frac{37}{20}$$

$$(13) \ \frac{1}{2} + \frac{3}{2} \\ = 2$$

$$(23) \ \frac{4}{3} - \left(-\frac{8}{5}\right) \\ = \frac{44}{15}$$

$$(33) \ -\frac{5}{2} + \left(-\frac{1}{2}\right) \\ = -3$$

$$(4) \ \frac{1}{3} - \frac{9}{4} \\ = -\frac{23}{12}$$

$$(14) \ -\frac{3}{10} - 1 \\ = -\frac{13}{10}$$

$$(24) \ -\frac{1}{3} - \left(-\frac{5}{7}\right) \\ = \frac{8}{21}$$

$$(34) \ -\frac{5}{2} - \left(-\frac{5}{6}\right) \\ = -\frac{5}{3}$$

$$(5) \ -\frac{1}{8} - \frac{5}{8} \\ = -\frac{3}{4}$$

$$(15) \ 1 - \frac{9}{7} \\ = -\frac{2}{7}$$

$$(25) \ -\frac{1}{9} + \left(-\frac{7}{8}\right) \\ = -\frac{71}{72}$$

$$(35) \ -5 + \left(-\frac{4}{5}\right) \\ = -\frac{29}{5}$$

$$(6) \ -\frac{1}{9} - (-5) \\ = \frac{44}{9}$$

$$(16) \ -\frac{10}{7} + (-1) \\ = -\frac{17}{7}$$

$$(26) \ \frac{2}{5} - (-10) \\ = \frac{52}{5}$$

$$(36) \ -\frac{3}{5} + \left(-\frac{9}{2}\right) \\ = -\frac{51}{10}$$

$$(7) \ \frac{3}{10} + (-1) \\ = -\frac{7}{10}$$

$$(17) \ \frac{1}{2} + \frac{5}{2} \\ = 3$$

$$(27) \ -3 + 2 \\ = -1$$

$$(37) \ \frac{8}{5} - \frac{3}{5} \\ = 1$$

$$(8) \ -\frac{8}{7} - 8 \\ = -\frac{64}{7}$$

$$(18) \ \frac{9}{4} - (-2) \\ = \frac{17}{4}$$

$$(28) \ \frac{7}{8} - \frac{4}{5} \\ = \frac{3}{40}$$

$$(38) \ \frac{5}{7} - \left(-\frac{9}{7}\right) \\ = 2$$

$$(9) \ \frac{4}{5} - \frac{3}{5} \\ = \frac{1}{5}$$

$$(19) \ \frac{1}{6} - (-2) \\ = \frac{13}{6}$$

$$(29) \ -\frac{5}{4} + \frac{3}{10} \\ = -\frac{19}{20}$$

$$(39) \ 1 + \frac{1}{3} \\ = \frac{4}{3}$$

$$(10) \ -\frac{5}{9} - \frac{8}{9} \\ = -\frac{13}{9}$$

$$(20) \ \frac{5}{3} + \frac{7}{3} \\ = 4$$

$$(30) \ -1 - \frac{1}{2} \\ = -\frac{3}{2}$$

$$(40) \ -\frac{6}{7} + 1 \\ = \frac{1}{7}$$

$$(1) -\frac{5}{3} - (-3) \\ = \frac{4}{3}$$

$$(11) \frac{5}{9} - \left(-\frac{3}{5}\right) \\ = \frac{52}{45}$$

$$(21) 1 + \frac{5}{3} \\ = \frac{8}{3}$$

$$(31) \frac{8}{5} + \frac{9}{7} \\ = \frac{101}{35}$$

$$(2) -\frac{3}{7} + \frac{5}{4} \\ = \frac{23}{28}$$

$$(12) \frac{5}{2} + 1 \\ = \frac{7}{2}$$

$$(22) -1 - (-7) \\ = 6$$

$$(32) \frac{5}{9} + \left(-\frac{5}{2}\right) \\ = -\frac{35}{18}$$

$$(3) -1 - \left(-\frac{3}{7}\right) \\ = -\frac{4}{7}$$

$$(13) -\frac{1}{2} - (-1) \\ = \frac{1}{2}$$

$$(23) \frac{10}{3} - \left(-\frac{7}{5}\right) \\ = \frac{71}{15}$$

$$(33) -\frac{9}{7} - \left(-\frac{9}{4}\right) \\ = \frac{27}{28}$$

$$(4) -\frac{9}{8} + \frac{4}{9} \\ = -\frac{49}{72}$$

$$(14) -\frac{9}{5} + \left(-\frac{5}{3}\right) \\ = -\frac{52}{15}$$

$$(24) -\frac{1}{4} + \left(-\frac{4}{3}\right) \\ = -\frac{19}{12}$$

$$(34) -\frac{7}{10} + \frac{10}{3} \\ = \frac{37}{30}$$

$$(5) -2 - 1 \\ = -3$$

$$(15) -\frac{5}{3} - \left(-\frac{5}{4}\right) \\ = -\frac{5}{12}$$

$$(25) -\frac{4}{9} + \left(-\frac{4}{9}\right) \\ = -\frac{8}{9}$$

$$(35) -\frac{1}{10} - \left(-\frac{9}{10}\right) \\ = \frac{4}{5}$$

$$(6) -\frac{3}{4} - \frac{3}{10} \\ = -\frac{21}{20}$$

$$(16) \frac{7}{6} + 10 \\ = \frac{67}{6}$$

$$(26) \frac{10}{3} + \frac{3}{5} \\ = \frac{59}{15}$$

$$(36) -\frac{2}{9} - \left(-\frac{3}{10}\right) \\ = \frac{7}{90}$$

$$(7) \frac{10}{3} - \left(-\frac{3}{4}\right) \\ = \frac{49}{12}$$

$$(17) -\frac{9}{10} - 7 \\ = -\frac{79}{10}$$

$$(27) \frac{8}{5} + (-1) \\ = \frac{3}{5}$$

$$(37) -1 + \frac{1}{4} \\ = -\frac{3}{4}$$

$$(8) -\frac{4}{9} + \left(-\frac{4}{5}\right) \\ = -\frac{56}{45}$$

$$(18) \frac{10}{3} - \left(-\frac{9}{8}\right) \\ = \frac{107}{24}$$

$$(28) -\frac{1}{2} + \left(-\frac{7}{8}\right) \\ = -\frac{11}{8}$$

$$(38) \frac{10}{7} - \frac{2}{9} \\ = \frac{76}{63}$$

$$(9) -6 - \left(-\frac{9}{4}\right) \\ = -\frac{15}{4}$$

$$(19) -\frac{6}{7} - \left(-\frac{8}{9}\right) \\ = \frac{2}{63}$$

$$(29) -\frac{1}{7} - \frac{8}{3} \\ = -\frac{59}{21}$$

$$(39) 2 + \frac{4}{3} \\ = \frac{10}{3}$$

$$(10) -\frac{4}{7} + 10 \\ = \frac{66}{7}$$

$$(20) \frac{1}{5} + (-5) \\ = -\frac{24}{5}$$

$$(30) 6 - \frac{1}{2} \\ = \frac{11}{2}$$

$$(40) -\frac{3}{8} + \left(-\frac{2}{7}\right) \\ = -\frac{37}{56}$$

$$(1) \frac{5}{4} - \left(-\frac{5}{4}\right) \\ = \frac{5}{2}$$

$$(11) -5 + \left(-\frac{6}{7}\right) \\ = -\frac{41}{7}$$

$$(21) \frac{7}{6} - (-1) \\ = \frac{13}{6}$$

$$(31) \frac{1}{2} - 5 \\ = -\frac{9}{2}$$

$$(2) 1 - (-1) \\ = 2$$

$$(12) -\frac{2}{5} + 1 \\ = \frac{3}{5}$$

$$(22) -\frac{2}{5} - \left(-\frac{1}{6}\right) \\ = -\frac{7}{30}$$

$$(32) -3 - \left(-\frac{2}{9}\right) \\ = -\frac{25}{9}$$

$$(3) 3 + \frac{10}{3} \\ = \frac{19}{3}$$

$$(13) -\frac{1}{3} - \frac{8}{3} \\ = -3$$

$$(23) -\frac{6}{7} + \frac{1}{2} \\ = -\frac{5}{14}$$

$$(33) \frac{1}{7} + 1 \\ = \frac{8}{7}$$

$$(4) \frac{2}{5} + (-6) \\ = -\frac{28}{5}$$

$$(14) \frac{5}{2} + 2 \\ = \frac{9}{2}$$

$$(24) \frac{3}{4} + \left(-\frac{1}{7}\right) \\ = \frac{17}{28}$$

$$(34) \frac{2}{3} + \frac{4}{9} \\ = \frac{10}{9}$$

$$(5) \frac{1}{2} + \frac{1}{4} \\ = \frac{3}{4}$$

$$(15) -1 + \frac{3}{2} \\ = \frac{1}{2}$$

$$(25) \frac{1}{2} - \frac{1}{3} \\ = \frac{1}{6}$$

$$(35) -\frac{1}{4} - \frac{1}{7} \\ = -\frac{11}{28}$$

$$(6) \frac{1}{3} - \frac{4}{7} \\ = -\frac{11}{21}$$

$$(16) -\frac{5}{7} - \left(-\frac{4}{5}\right) \\ = \frac{3}{35}$$

$$(26) -5 - (-4) \\ = -1$$

$$(36) \frac{5}{8} - \frac{6}{5} \\ = -\frac{23}{40}$$

$$(7) -5 - \left(-\frac{4}{5}\right) \\ = -\frac{21}{5}$$

$$(17) \frac{1}{2} + \frac{1}{2} \\ = 1$$

$$(27) 5 - 3 \\ = 2$$

$$(37) \frac{7}{6} + (-2) \\ = -\frac{5}{6}$$

$$(8) \frac{7}{6} + \left(-\frac{8}{7}\right) \\ = \frac{1}{42}$$

$$(18) 1 + \frac{9}{8} \\ = \frac{17}{8}$$

$$(28) \frac{8}{7} + (-1) \\ = \frac{1}{7}$$

$$(38) \frac{1}{7} - \left(-\frac{2}{5}\right) \\ = \frac{19}{35}$$

$$(9) -2 + \frac{3}{4} \\ = -\frac{5}{4}$$

$$(19) \frac{8}{7} + \frac{7}{10} \\ = \frac{129}{70}$$

$$(29) 1 - \frac{1}{3} \\ = \frac{2}{3}$$

$$(39) -\frac{10}{3} + \frac{3}{2} \\ = -\frac{11}{6}$$

$$(10) 4 - \left(-\frac{5}{6}\right) \\ = \frac{29}{6}$$

$$(20) -\frac{7}{8} - 1 \\ = -\frac{15}{8}$$

$$(30) \frac{2}{3} + \frac{9}{4} \\ = \frac{35}{12}$$

$$(40) \frac{4}{5} - \frac{9}{8} \\ = -\frac{13}{40}$$

$$(1) \frac{3}{4} - (-2) \\ = \frac{11}{4}$$

$$(11) -\frac{2}{3} - \left(-\frac{3}{5}\right) \\ = -\frac{1}{15}$$

$$(21) -\frac{1}{7} - (-9) \\ = \frac{62}{7}$$

$$(31) \frac{3}{8} - \left(-\frac{1}{5}\right) \\ = \frac{23}{40}$$

$$(2) \frac{2}{7} - \left(-\frac{8}{5}\right) \\ = \frac{66}{35}$$

$$(12) -\frac{5}{8} + \frac{3}{8} \\ = -\frac{1}{4}$$

$$(22) -3 - \left(-\frac{4}{9}\right) \\ = -\frac{23}{9}$$

$$(32) -3 - 1 \\ = -4$$

$$(3) -\frac{5}{7} - \left(-\frac{8}{7}\right) \\ = \frac{3}{7}$$

$$(13) 1 - \frac{9}{10} \\ = \frac{1}{10}$$

$$(23) \frac{1}{2} + \frac{5}{4} \\ = \frac{7}{4}$$

$$(33) -4 - \left(-\frac{1}{10}\right) \\ = -\frac{39}{10}$$

$$(4) -\frac{3}{2} + \frac{1}{3} \\ = -\frac{7}{6}$$

$$(14) -\frac{1}{2} - \frac{8}{3} \\ = -\frac{19}{6}$$

$$(24) -\frac{10}{9} - (-1) \\ = -\frac{1}{9}$$

$$(34) \frac{5}{2} - \left(-\frac{10}{3}\right) \\ = \frac{35}{6}$$

$$(5) -\frac{2}{3} - \left(-\frac{7}{6}\right) \\ = \frac{1}{2}$$

$$(15) 1 - \left(-\frac{3}{2}\right) \\ = \frac{5}{2}$$

$$(25) -\frac{3}{8} - 1 \\ = -\frac{11}{8}$$

$$(35) -8 - \left(-\frac{4}{7}\right) \\ = -\frac{52}{7}$$

$$(6) -8 - \left(-\frac{5}{9}\right) \\ = -\frac{67}{9}$$

$$(16) -1 + (-5) \\ = -6$$

$$(26) 7 + 1 \\ = 8$$

$$(36) -\frac{5}{4} - \left(-\frac{5}{4}\right) \\ = 0$$

$$(7) 2 - \left(-\frac{3}{4}\right) \\ = \frac{11}{4}$$

$$(17) \frac{1}{4} + \left(-\frac{5}{4}\right) \\ = -1$$

$$(27) \frac{3}{7} - \frac{9}{5} \\ = -\frac{48}{35}$$

$$(37) \frac{7}{3} - \frac{5}{4} \\ = \frac{13}{12}$$

$$(18) \frac{1}{10} - \left(-\frac{9}{8}\right) \\ = \frac{49}{40}$$

$$(28) -4 - \frac{3}{5} \\ = -\frac{23}{5}$$

$$(38) 4 - \frac{9}{4} \\ = \frac{7}{4}$$

$$(8) -\frac{3}{5} + \frac{9}{7} \\ = \frac{24}{35}$$

$$(19) \frac{7}{8} - (-7) \\ = \frac{63}{8}$$

$$(29) 1 - 1 \\ = 0$$

$$(39) -\frac{7}{8} + (-2) \\ = -\frac{23}{8}$$

$$(9) -\frac{2}{3} - \left(-\frac{5}{3}\right) \\ = 1$$

$$(20) \frac{5}{2} - 8 \\ = -\frac{11}{2}$$

$$(30) \frac{2}{3} + 1 \\ = \frac{5}{3}$$

$$(40) \frac{8}{9} - \frac{8}{9} \\ = 0$$

$$(10) \frac{7}{10} + \left(-\frac{9}{4}\right) \\ = -\frac{31}{20}$$

$$(1) -\frac{7}{5} + \left(-\frac{10}{9}\right) \\ = -\frac{113}{45}$$

$$(11) -\frac{1}{2} + \frac{9}{4} \\ = \frac{7}{4}$$

$$(21) 1 + \frac{1}{3} \\ = \frac{4}{3}$$

$$(31) -\frac{7}{5} + \left(-\frac{2}{3}\right) \\ = -\frac{31}{15}$$

$$(2) -\frac{6}{5} - \frac{9}{7} \\ = -\frac{87}{35}$$

$$(12) -9 - \left(-\frac{5}{4}\right) \\ = -\frac{31}{4}$$

$$(22) \frac{5}{2} - (-1) \\ = \frac{7}{2}$$

$$(32) 3 + \frac{3}{5} \\ = \frac{18}{5}$$

$$(3) -\frac{5}{2} - \frac{3}{4} \\ = -\frac{13}{4}$$

$$(13) -\frac{9}{7} + \left(-\frac{8}{9}\right) \\ = -\frac{137}{63}$$

$$(23) -5 - \left(-\frac{6}{7}\right) \\ = -\frac{29}{7}$$

$$(33) \frac{4}{3} + 10 \\ = \frac{34}{3}$$

$$(4) 2 + \left(-\frac{4}{3}\right) \\ = \frac{2}{3}$$

$$(14) \frac{10}{7} + \frac{1}{2} \\ = \frac{27}{14}$$

$$(24) -\frac{1}{2} + \left(-\frac{1}{8}\right) \\ = -\frac{5}{8}$$

$$(34) -\frac{5}{9} - 1 \\ = -\frac{14}{9}$$

$$(5) -1 - \left(-\frac{1}{3}\right) \\ = -\frac{2}{3}$$

$$(15) -\frac{8}{3} + (-3) \\ = -\frac{17}{3}$$

$$(25) \frac{3}{10} - \left(-\frac{4}{5}\right) \\ = \frac{11}{10}$$

$$(35) 2 - \frac{1}{5} \\ = \frac{9}{5}$$

$$(6) \frac{5}{7} + \frac{9}{10} \\ = \frac{113}{70}$$

$$(16) \frac{5}{4} - 4 \\ = -\frac{11}{4}$$

$$(26) \frac{2}{3} - \left(-\frac{5}{8}\right) \\ = \frac{31}{24}$$

$$(36) \frac{5}{3} - (-4) \\ = \frac{17}{3}$$

$$(7) \frac{1}{4} + \left(-\frac{3}{8}\right) \\ = -\frac{1}{8}$$

$$(17) 1 + \frac{3}{7} \\ = \frac{10}{7}$$

$$(27) -\frac{1}{2} + \frac{5}{4} \\ = \frac{3}{4}$$

$$(37) -3 + 1 \\ = -2$$

$$(8) -\frac{3}{5} + \left(-\frac{2}{9}\right) \\ = -\frac{37}{45}$$

$$(18) -\frac{3}{4} + (-2) \\ = -\frac{11}{4}$$

$$(28) -\frac{10}{9} - \left(-\frac{4}{9}\right) \\ = -\frac{2}{3}$$

$$(38) -2 + \left(-\frac{1}{3}\right) \\ = -\frac{7}{3}$$

$$(9) \frac{6}{7} + 5 \\ = \frac{41}{7}$$

$$(19) 5 - \left(-\frac{3}{5}\right) \\ = \frac{28}{5}$$

$$(29) -\frac{2}{3} + \frac{9}{2} \\ = \frac{23}{6}$$

$$(39) 2 - 7 \\ = -5$$

$$(10) \frac{5}{9} - (-6) \\ = \frac{59}{9}$$

$$(20) \frac{4}{9} + \left(-\frac{3}{8}\right) \\ = \frac{5}{72}$$

$$(30) 1 - \left(-\frac{4}{5}\right) \\ = \frac{9}{5}$$

$$(40) \frac{1}{3} - \frac{3}{4} \\ = -\frac{5}{12}$$

$$(1) -\frac{1}{3} + (-1) \\ = -\frac{4}{3}$$

$$(11) 1 - \left(-\frac{1}{5}\right) \\ = \frac{6}{5}$$

$$(21) -8 + \left(-\frac{5}{9}\right) \\ = -\frac{77}{9}$$

$$(31) -\frac{4}{5} - \left(-\frac{3}{8}\right) \\ = -\frac{17}{40}$$

$$(2) \frac{8}{3} - \left(-\frac{1}{2}\right) \\ = \frac{19}{6}$$

$$(12) -\frac{3}{2} - \frac{5}{3} \\ = -\frac{19}{6}$$

$$(22) -4 - \frac{7}{8} \\ = -\frac{39}{8}$$

$$(32) \frac{7}{3} - \left(-\frac{9}{5}\right) \\ = \frac{62}{15}$$

$$(3) \frac{3}{10} + \left(-\frac{1}{5}\right) \\ = \frac{1}{10}$$

$$(13) -\frac{5}{2} + \frac{9}{7} \\ = -\frac{17}{14}$$

$$(23) \frac{3}{4} - \left(-\frac{1}{2}\right) \\ = \frac{5}{4}$$

$$(33) -\frac{9}{8} + \left(-\frac{1}{7}\right) \\ = -\frac{71}{56}$$

$$(4) \frac{2}{5} - \left(-\frac{3}{2}\right) \\ = \frac{19}{10}$$

$$(14) -\frac{3}{10} - 10 \\ = -\frac{103}{10}$$

$$(24) \frac{1}{8} + \left(-\frac{4}{9}\right) \\ = -\frac{23}{72}$$

$$(34) 1 + \left(-\frac{4}{3}\right) \\ = -\frac{1}{3}$$

$$(5) -1 + (-3) \\ = -4$$

$$(15) \frac{4}{5} + \frac{5}{3} \\ = \frac{37}{15}$$

$$(25) -\frac{2}{3} - \frac{5}{4} \\ = -\frac{23}{12}$$

$$(35) -2 - \frac{3}{2} \\ = -\frac{7}{2}$$

$$(6) \frac{4}{3} + \frac{1}{7} \\ = \frac{31}{21}$$

$$(16) -\frac{7}{4} + \left(-\frac{8}{7}\right) \\ = -\frac{81}{28}$$

$$(26) -2 - \left(-\frac{5}{4}\right) \\ = -\frac{3}{4}$$

$$(36) -\frac{4}{3} + (-1) \\ = -\frac{7}{3}$$

$$(7) -\frac{1}{2} - \frac{1}{4} \\ = -\frac{3}{4}$$

$$(17) -\frac{5}{6} - \left(-\frac{7}{6}\right) \\ = \frac{1}{3}$$

$$(27) 2 - \left(-\frac{2}{5}\right) \\ = \frac{12}{5}$$

$$(37) \frac{9}{4} - (-1) \\ = \frac{13}{4}$$

$$(8) 1 + \left(-\frac{4}{9}\right) \\ = \frac{5}{9}$$

$$(18) 2 - \left(-\frac{1}{9}\right) \\ = \frac{19}{9}$$

$$(28) -\frac{1}{3} - \left(-\frac{9}{4}\right) \\ = \frac{23}{12}$$

$$(38) \frac{1}{4} + \frac{1}{10} \\ = \frac{7}{20}$$

$$(9) -1 + \left(-\frac{5}{4}\right) \\ = -\frac{9}{4}$$

$$(19) \frac{5}{4} - \left(-\frac{7}{3}\right) \\ = \frac{43}{12}$$

$$(29) 1 + \frac{1}{2} \\ = \frac{3}{2}$$

$$(39) \frac{2}{7} - (-1) \\ = \frac{9}{7}$$

$$(10) -\frac{6}{5} - \left(-\frac{1}{10}\right) \\ = -\frac{11}{10}$$

$$(20) \frac{5}{2} - 4 \\ = -\frac{3}{2}$$

$$(30) -2 - 4 \\ = -6$$

$$(40) -8 - \frac{10}{9} \\ = -\frac{82}{9}$$

$$(1) 3 - \frac{1}{4} \\ = \frac{11}{4}$$

$$(11) -\frac{9}{7} + \left(-\frac{9}{5}\right) \\ = -\frac{108}{35}$$

$$(21) -\frac{1}{9} - \frac{5}{7} \\ = -\frac{52}{63}$$

$$(31) \frac{1}{6} + \frac{7}{4} \\ = \frac{23}{12}$$

$$(2) -4 - 1 \\ = -5$$

$$(12) -\frac{7}{10} + (-1) \\ = -\frac{17}{10}$$

$$(22) \frac{5}{7} - \frac{5}{3} \\ = -\frac{20}{21}$$

$$(32) -1 + \left(-\frac{4}{9}\right) \\ = -\frac{13}{9}$$

$$(3) -\frac{4}{5} - \left(-\frac{9}{7}\right) \\ = \frac{17}{35}$$

$$(13) \frac{6}{7} - \left(-\frac{4}{5}\right) \\ = \frac{58}{35}$$

$$(23) 3 + \left(-\frac{1}{4}\right) \\ = \frac{11}{4}$$

$$(33) -\frac{2}{9} + \left(-\frac{2}{3}\right) \\ = -\frac{8}{9}$$

$$(4) -\frac{2}{3} + \frac{1}{9} \\ = -\frac{5}{9}$$

$$(14) \frac{1}{3} + (-2) \\ = -\frac{5}{3}$$

$$(24) \frac{9}{5} + \left(-\frac{5}{7}\right) \\ = \frac{38}{35}$$

$$(34) \frac{4}{3} - 2 \\ = -\frac{2}{3}$$

$$(5) -5 + \frac{3}{7} \\ = -\frac{32}{7}$$

$$(15) -5 - \frac{8}{9} \\ = -\frac{53}{9}$$

$$(25) -\frac{2}{5} + \frac{10}{9} \\ = \frac{32}{45}$$

$$(35) -2 - (-1) \\ = -1$$

$$(6) \frac{3}{2} - 1 \\ = \frac{1}{2}$$

$$(16) -\frac{7}{4} + \left(-\frac{10}{7}\right) \\ = -\frac{89}{28}$$

$$(26) \frac{10}{9} + \left(-\frac{1}{9}\right) \\ = 1$$

$$(36) \frac{2}{7} + \frac{3}{4} \\ = \frac{29}{28}$$

$$(7) \frac{7}{2} + \frac{1}{2} \\ = 4$$

$$(17) 2 + \frac{1}{2} \\ = \frac{5}{2}$$

$$(27) -\frac{5}{6} + \left(-\frac{1}{2}\right) \\ = -\frac{4}{3}$$

$$(37) -\frac{3}{2} - \frac{9}{2} \\ = -6$$

$$(8) \frac{5}{6} - 4 \\ = -\frac{19}{6}$$

$$(18) \frac{1}{2} + \frac{2}{3} \\ = \frac{7}{6}$$

$$(28) -1 + \frac{7}{4} \\ = \frac{3}{4}$$

$$(38) \frac{7}{3} + \frac{6}{7} \\ = \frac{67}{21}$$

$$(9) \frac{9}{8} - (-1) \\ = \frac{17}{8}$$

$$(19) -\frac{1}{6} - \frac{3}{2} \\ = -\frac{5}{3}$$

$$(29) -\frac{10}{7} - \left(-\frac{1}{2}\right) \\ = -\frac{13}{14}$$

$$(39) -1 + \frac{5}{7} \\ = -\frac{2}{7}$$

$$(10) \frac{6}{7} + \frac{3}{4} \\ = \frac{45}{28}$$

$$(20) -\frac{4}{7} + \frac{7}{5} \\ = \frac{29}{35}$$

$$(30) \frac{9}{10} + \frac{1}{2} \\ = \frac{7}{5}$$

$$(40) 1 - \frac{3}{7} \\ = \frac{4}{7}$$

$$(1) 9 - \frac{7}{3}$$

$$= \frac{20}{3}$$

$$(11) -\frac{5}{8} - 3$$

$$= -\frac{29}{8}$$

$$(21) \frac{3}{5} - \frac{10}{9}$$

$$= -\frac{23}{45}$$

$$(31) \frac{5}{7} - \left(-\frac{7}{4}\right)$$

$$= \frac{69}{28}$$

$$(2) -\frac{4}{5} - \left(-\frac{3}{10}\right)$$

$$= -\frac{1}{2}$$

$$(12) -\frac{7}{3} + \left(-\frac{1}{9}\right)$$

$$= -\frac{22}{9}$$

$$(22) \frac{9}{10} - \frac{1}{4}$$

$$= \frac{13}{20}$$

$$(32) 1 + \frac{5}{4}$$

$$= \frac{9}{4}$$

$$(3) -\frac{3}{4} + \left(-\frac{7}{3}\right)$$

$$= -\frac{37}{12}$$

$$(13) 6 + \left(-\frac{1}{4}\right)$$

$$= \frac{23}{4}$$

$$(23) -\frac{5}{2} - \left(-\frac{5}{2}\right)$$

$$= 0$$

$$(33) \frac{7}{2} - \frac{1}{3}$$

$$= \frac{19}{6}$$

$$(4) -\frac{3}{7} - \left(-\frac{3}{4}\right)$$

$$= \frac{9}{28}$$

$$(14) \frac{8}{3} + (-5)$$

$$= -\frac{7}{3}$$

$$(24) -\frac{1}{2} + \left(-\frac{6}{5}\right)$$

$$= -\frac{17}{10}$$

$$(34) -\frac{4}{3} - \frac{2}{9}$$

$$= -\frac{14}{9}$$

$$(5) 2 + (-10)$$

$$= -8$$

$$(15) \frac{1}{6} + \left(-\frac{1}{7}\right)$$

$$= \frac{1}{42}$$

$$(25) -\frac{4}{3} + 2$$

$$= \frac{2}{3}$$

$$(35) -\frac{1}{5} + \frac{4}{5}$$

$$= \frac{3}{5}$$

$$(6) -\frac{5}{6} - \frac{1}{2}$$

$$= -\frac{4}{3}$$

$$(16) 2 - \frac{10}{7}$$

$$= \frac{4}{7}$$

$$(26) -3 + \left(-\frac{2}{5}\right)$$

$$= -\frac{17}{5}$$

$$(36) \frac{5}{3} + \left(-\frac{5}{7}\right)$$

$$= \frac{20}{21}$$

$$(7) -\frac{3}{8} + \left(-\frac{7}{2}\right)$$

$$= -\frac{31}{8}$$

$$(17) -\frac{6}{5} + \frac{2}{3}$$

$$= -\frac{8}{15}$$

$$(27) -\frac{9}{5} + \left(-\frac{1}{2}\right)$$

$$= -\frac{23}{10}$$

$$(37) \frac{3}{2} - \left(-\frac{8}{9}\right)$$

$$= \frac{43}{18}$$

$$(8) \frac{10}{7} + \frac{9}{7}$$

$$= \frac{19}{7}$$

$$(18) -\frac{6}{5} - \frac{3}{4}$$

$$= -\frac{39}{20}$$

$$(28) -\frac{1}{5} + \frac{9}{2}$$

$$= \frac{43}{10}$$

$$(38) \frac{1}{5} - \left(-\frac{5}{3}\right)$$

$$= \frac{28}{15}$$

$$(9) -6 - 1$$

$$= -7$$

$$(19) \frac{3}{5} - 1$$

$$= -\frac{2}{5}$$

$$(29) -\frac{8}{9} - (-2)$$

$$= \frac{10}{9}$$

$$(39) -\frac{4}{5} - (-5)$$

$$= \frac{21}{5}$$

$$(10) \frac{1}{4} - (-2)$$

$$= \frac{9}{4}$$

$$(20) -\frac{1}{2} + \frac{1}{4}$$

$$= -\frac{1}{4}$$

$$(30) -\frac{1}{4} - \left(-\frac{3}{5}\right)$$

$$= \frac{7}{20}$$

$$(40) \frac{6}{5} + \left(-\frac{1}{4}\right)$$

$$= \frac{19}{20}$$

$$(1) \frac{1}{3} - (-5) \\ = \frac{16}{3}$$

$$(11) \frac{7}{9} + (-1) \\ = -\frac{2}{9}$$

$$(21) -1 + \frac{10}{3} \\ = \frac{7}{3}$$

$$(31) -\frac{7}{4} - \left(-\frac{4}{3}\right) \\ = -\frac{5}{12}$$

$$(2) \frac{9}{4} + \left(-\frac{7}{6}\right) \\ = \frac{13}{12}$$

$$(12) -\frac{2}{5} - \frac{9}{4} \\ = -\frac{53}{20}$$

$$(22) -4 + \frac{9}{8} \\ = -\frac{23}{8}$$

$$(32) 1 + \left(-\frac{1}{2}\right) \\ = \frac{1}{2}$$

$$(3) -3 - \frac{5}{9} \\ = -\frac{32}{9}$$

$$(13) \frac{2}{5} + \left(-\frac{5}{3}\right) \\ = -\frac{19}{15}$$

$$(23) -\frac{5}{3} - \frac{1}{3} \\ = -2$$

$$(33) -3 + \frac{4}{7} \\ = -\frac{17}{7}$$

$$(4) \frac{3}{2} - \left(-\frac{7}{5}\right) \\ = \frac{29}{10}$$

$$(14) -\frac{3}{5} - \frac{2}{7} \\ = -\frac{31}{35}$$

$$(24) -\frac{3}{5} - \left(-\frac{6}{7}\right) \\ = \frac{9}{35}$$

$$(34) \frac{2}{3} + \frac{8}{3} \\ = \frac{10}{3}$$

$$(5) 9 + \frac{3}{2} \\ = \frac{21}{2}$$

$$(15) \frac{4}{5} - \left(-\frac{2}{9}\right) \\ = \frac{46}{45}$$

$$(25) -\frac{1}{6} + \left(-\frac{3}{2}\right) \\ = -\frac{5}{3}$$

$$(35) 5 + \frac{1}{5} \\ = \frac{26}{5}$$

$$(6) \frac{1}{2} - \left(-\frac{4}{7}\right) \\ = \frac{15}{14}$$

$$(16) -\frac{7}{2} - \frac{2}{7} \\ = -\frac{53}{14}$$

$$(26) -1 - \left(-\frac{7}{10}\right) \\ = -\frac{3}{10}$$

$$(36) 1 - (-2) \\ = 3$$

$$(7) -\frac{3}{4} - \frac{7}{2} \\ = -\frac{17}{4}$$

$$(17) \frac{9}{8} - \frac{1}{2} \\ = \frac{5}{8}$$

$$(27) -10 + (-4) \\ = -14$$

$$(37) \frac{4}{5} + \left(-\frac{8}{5}\right) \\ = -\frac{4}{5}$$

$$(8) \frac{5}{2} - \frac{1}{9} \\ = \frac{43}{18}$$

$$(18) -\frac{5}{3} - \left(-\frac{9}{5}\right) \\ = \frac{2}{15}$$

$$(28) -\frac{5}{9} + \frac{1}{9} \\ = -\frac{4}{9}$$

$$(38) 7 + \left(-\frac{5}{7}\right) \\ = \frac{44}{7}$$

$$(9) \frac{5}{8} + \frac{10}{3} \\ = \frac{95}{24}$$

$$(19) \frac{10}{7} + 3 \\ = \frac{31}{7}$$

$$(29) -\frac{2}{3} + \left(-\frac{4}{5}\right) \\ = -\frac{22}{15}$$

$$(39) \frac{5}{7} + \frac{3}{4} \\ = \frac{41}{28}$$

$$(10) \frac{5}{3} - \left(-\frac{7}{4}\right) \\ = \frac{41}{12}$$

$$(20) \frac{5}{7} - \frac{1}{4} \\ = \frac{13}{28}$$

$$(30) -2 - \left(-\frac{1}{9}\right) \\ = -\frac{17}{9}$$

$$(40) 1 + (-1) \\ = 0$$

1.3.2 乗除法

有理数の積や商を求める問題。

(1) $-\frac{9}{10} \times \frac{3}{10}$

(11) $\frac{1}{2} \times \left(-\frac{5}{3}\right)$

(21) $\frac{4}{5} \times 1$

(31) -1×1

(2) $-8 \div \frac{8}{7}$

(12) $-\frac{7}{8} \times \frac{9}{4}$

(22) $-\frac{1}{5} \div \left(-\frac{5}{2}\right)$

(32) $\frac{5}{2} \div \frac{1}{2}$

(3) $-2 \times \left(-\frac{9}{7}\right)$

(13) $-4 \div \frac{1}{4}$

(23) $-\frac{6}{7} \times \frac{1}{2}$

(33) $-9 \times (-3)$

(4) $\frac{4}{5} \times (-2)$

(14) $-\frac{2}{3} \times \frac{1}{5}$

(24) $\frac{1}{6} \div 10$

(34) $-8 \div \frac{7}{9}$

(5) $\frac{9}{10} \div \left(-\frac{3}{2}\right)$

(15) $\frac{10}{9} \div 1$

(25) $\frac{2}{5} \div 4$

(35) $-\frac{5}{8} \times \left(-\frac{3}{4}\right)$

(6) $\frac{4}{5} \div \left(-\frac{2}{3}\right)$

(16) $\frac{2}{9} \times \frac{5}{2}$

(26) $\frac{1}{3} \times 1$

(36) $\frac{1}{2} \div 2$

(7) $-\frac{1}{6} \times \frac{7}{9}$

(17) $\frac{4}{3} \times \frac{1}{8}$

(27) $\frac{1}{4} \div \frac{3}{8}$

(37) $-\frac{9}{4} \div (-2)$

(8) $\frac{7}{3} \div 9$

(18) $-\frac{3}{2} \div \frac{2}{3}$

(28) $\frac{3}{2} \times \left(-\frac{2}{9}\right)$

(38) $\frac{5}{2} \div \frac{1}{7}$

(9) $-2 \times \frac{2}{3}$

(19) $\frac{9}{8} \div \frac{4}{5}$

(29) $-\frac{5}{4} \div \left(-\frac{1}{10}\right)$

(39) $-\frac{8}{3} \div \frac{7}{10}$

(10) $-\frac{4}{5} \times \frac{1}{2}$

(20) $\frac{3}{2} \div \frac{3}{4}$

(30) $-2 \times \frac{8}{9}$

(40) $-\frac{9}{10} \div \left(-\frac{5}{3}\right)$

(1) $-\frac{5}{3} \div \left(-\frac{5}{4}\right)$

(11) $\frac{1}{5} \div \frac{2}{5}$

(21) $1 \div 6$

(31) $\frac{5}{3} \times \left(-\frac{6}{5}\right)$

(2) $-\frac{7}{5} \div \left(-\frac{4}{3}\right)$

(12) $\frac{3}{10} \times \frac{3}{4}$

(22) $-\frac{5}{7} \div \left(-\frac{8}{3}\right)$

(32) $1 \div \frac{2}{3}$

(3) $-\frac{2}{3} \times \left(-\frac{5}{3}\right)$

(13) $\frac{1}{5} \times \left(-\frac{10}{9}\right)$

(23) $-\frac{5}{2} \div 3$

(33) $-\frac{5}{2} \div \left(-\frac{2}{7}\right)$

(4) $2 \div \left(-\frac{1}{3}\right)$

(14) $-1 \times \left(-\frac{1}{2}\right)$

(24) $-\frac{2}{5} \div (-8)$

(34) $-\frac{1}{3} \div (-1)$

(5) $5 \div \left(-\frac{1}{4}\right)$

(15) $\frac{9}{7} \div (-1)$

(25) $5 \times \frac{5}{8}$

(35) 1×1

(6) $\frac{3}{4} \div \left(-\frac{5}{2}\right)$

(16) $2 \div 1$

(26) $\frac{1}{9} \div \left(-\frac{3}{2}\right)$

(36) $-\frac{7}{10} \div \left(-\frac{7}{8}\right)$

(7) $-\frac{1}{4} \times \left(-\frac{7}{10}\right)$

(17) $-1 \times \left(-\frac{2}{3}\right)$

(27) $\frac{7}{5} \div \left(-\frac{1}{3}\right)$

(37) 2×3

(8) $\frac{1}{2} \div \left(-\frac{1}{9}\right)$

(18) $-\frac{5}{9} \div \frac{5}{2}$

(28) $-5 \div (-1)$

(38) $\frac{1}{4} \times \frac{10}{9}$

(9) $-\frac{7}{10} \times \left(-\frac{1}{2}\right)$

(19) $-5 \times \left(-\frac{1}{3}\right)$

(29) $2 \div 2$

(39) $3 \times \frac{3}{7}$

(10) $3 \div \frac{4}{9}$

(20) $-\frac{8}{9} \div \left(-\frac{10}{9}\right)$

(30) $1 \div \left(-\frac{1}{5}\right)$

(40) $\frac{3}{4} \div \left(-\frac{1}{2}\right)$

(1) $\frac{4}{7} \div \left(-\frac{1}{5}\right)$

(11) $-1 \times (-2)$

(21) $\frac{10}{9} \times 2$

(31) $-\frac{3}{5} \times 3$

(2) $-\frac{1}{5} \div \frac{1}{5}$

(12) $-1 \div \left(-\frac{3}{8}\right)$

(22) $-\frac{9}{8} \times \left(-\frac{4}{3}\right)$

(32) $-1 \times \frac{2}{3}$

(3) $-\frac{1}{3} \times \frac{5}{4}$

(13) $\frac{3}{2} \div 4$

(23) $-1 \times \left(-\frac{9}{4}\right)$

(33) $1 \times \frac{1}{5}$

(4) $-\frac{4}{3} \div (-2)$

(14) $-\frac{1}{4} \times \left(-\frac{7}{3}\right)$

(24) $\frac{6}{7} \times \left(-\frac{2}{7}\right)$

(34) $1 \div \frac{1}{5}$

(5) $2 \div \frac{9}{8}$

(15) $\frac{5}{3} \div \left(-\frac{10}{7}\right)$

(25) $-\frac{7}{3} \div (-3)$

(35) $-\frac{2}{5} \times \left(-\frac{6}{5}\right)$

(6) $-\frac{8}{7} \div \left(-\frac{6}{5}\right)$

(16) $-\frac{1}{3} \times 2$

(26) $-1 \div \left(-\frac{9}{5}\right)$

(36) $-5 \times \frac{10}{9}$

(7) $-\frac{8}{5} \times (-3)$

(17) $\frac{9}{2} \div \left(-\frac{4}{7}\right)$

(27) $-1 \div \frac{1}{2}$

(37) $-\frac{1}{2} \div \left(-\frac{2}{3}\right)$

(8) $4 \div \frac{1}{10}$

(18) $\frac{10}{9} \times \left(-\frac{5}{7}\right)$

(28) $8 \times \left(-\frac{5}{3}\right)$

(38) $\frac{3}{4} \div \left(-\frac{3}{5}\right)$

(9) $\frac{5}{4} \div \left(-\frac{8}{7}\right)$

(19) $\frac{3}{10} \div \left(-\frac{1}{2}\right)$

(29) $8 \div \left(-\frac{1}{2}\right)$

(39) $-\frac{1}{5} \times (-2)$

(10) $\frac{3}{2} \div \left(-\frac{3}{5}\right)$

(20) $-\frac{4}{7} \times \left(-\frac{5}{3}\right)$

(30) $-\frac{4}{3} \div \frac{2}{3}$

(40) $\frac{2}{3} \times \left(-\frac{7}{3}\right)$

(1) $\frac{4}{7} \times \left(-\frac{1}{2}\right)$

(11) $-\frac{5}{8} \div \left(-\frac{2}{3}\right)$

(21) $-\frac{3}{7} \times \left(-\frac{5}{8}\right)$

(31) $1 \times \left(-\frac{8}{3}\right)$

(2) $\frac{1}{3} \div \frac{10}{7}$

(12) $\frac{2}{5} \times 4$

(22) $-\frac{3}{2} \times \frac{9}{5}$

(32) $\frac{1}{5} \div \left(-\frac{3}{2}\right)$

(3) $\frac{9}{5} \div 3$

(13) $-\frac{9}{5} \div \frac{9}{5}$

(23) $\frac{8}{9} \times \left(-\frac{1}{5}\right)$

(33) 3×9

(4) $\frac{6}{5} \div \left(-\frac{3}{5}\right)$

(14) $\frac{1}{7} \div \frac{6}{7}$

(24) $-\frac{4}{5} \times 2$

(34) $\frac{1}{5} \div (-5)$

(5) $1 \div (-2)$

(15) $-1 \div \frac{1}{4}$

(25) $\frac{3}{2} \div \frac{1}{2}$

(35) $\frac{7}{8} \div \left(-\frac{7}{4}\right)$

(6) $\frac{9}{5} \times 4$

(16) $-1 \times \left(-\frac{7}{9}\right)$

(26) $-9 \times \left(-\frac{5}{2}\right)$

(36) $-\frac{7}{2} \div (-4)$

(7) $\frac{4}{9} \div 1$

(17) $\frac{3}{5} \div \left(-\frac{2}{7}\right)$

(27) $-\frac{1}{3} \div \frac{4}{9}$

(37) $\frac{5}{8} \div \frac{3}{4}$

(8) $-\frac{1}{6} \times \left(-\frac{1}{5}\right)$

(18) $4 \div \frac{1}{9}$

(28) $\frac{6}{7} \div \frac{2}{3}$

(38) $-1 \div 3$

(9) $-\frac{10}{3} \div (-3)$

(19) $\frac{5}{8} \times \left(-\frac{1}{2}\right)$

(29) $\frac{3}{8} \div \frac{3}{8}$

(39) $-\frac{3}{2} \div \frac{1}{8}$

(10) $-\frac{5}{6} \div (-3)$

(20) $3 \div \frac{8}{5}$

(30) $-\frac{5}{4} \div \frac{1}{3}$

(40) 7×1

(1) $\frac{4}{3} \times (-1)$

(11) $1 \div \left(-\frac{1}{8}\right)$

(21) $\frac{5}{3} \times \frac{7}{4}$

(31) $-\frac{9}{2} \times \frac{9}{7}$

(2) $\frac{3}{4} \div 1$

(12) $1 \div \frac{2}{3}$

(22) $-\frac{5}{2} \div \left(-\frac{3}{4}\right)$

(32) $-\frac{8}{5} \div \left(-\frac{1}{4}\right)$

(3) $\frac{5}{7} \times \left(-\frac{5}{3}\right)$

(13) $\frac{1}{4} \div 1$

(23) $-5 \div 1$

(33) $\frac{9}{2} \times 1$

(4) $-\frac{1}{2} \div \frac{3}{4}$

(14) $\frac{5}{4} \times \left(-\frac{7}{2}\right)$

(24) $-\frac{5}{4} \div \frac{3}{2}$

(34) $1 \times \left(-\frac{6}{5}\right)$

(5) $-1 \div \frac{4}{3}$

(15) $\frac{5}{4} \div \left(-\frac{3}{5}\right)$

(25) $-\frac{7}{10} \times \left(-\frac{3}{8}\right)$

(35) $-\frac{4}{5} \times \left(-\frac{3}{5}\right)$

(6) $\frac{5}{8} \times \frac{5}{2}$

(16) $-\frac{2}{3} \div \frac{1}{8}$

(26) $-2 \div \left(-\frac{1}{4}\right)$

(36) $\frac{1}{2} \div \frac{7}{4}$

(7) $4 \times \left(-\frac{6}{7}\right)$

(17) $-\frac{1}{2} \times \frac{7}{2}$

(27) $-\frac{1}{4} \times (-1)$

(37) $-\frac{7}{6} \div \left(-\frac{5}{8}\right)$

(8) $-\frac{9}{2} \div \frac{2}{5}$

(18) $1 \times \left(-\frac{9}{7}\right)$

(28) $\frac{5}{3} \times \left(-\frac{8}{5}\right)$

(38) $\frac{4}{3} \times (-5)$

(9) 9×1

(19) $\frac{4}{5} \times \frac{6}{7}$

(29) $-\frac{10}{3} \div (-3)$

(39) $9 \div \frac{9}{8}$

(10) $\frac{4}{3} \div 1$

(20) $-1 \div (-2)$

(30) $-\frac{3}{2} \div \frac{3}{5}$

(40) $-\frac{1}{4} \times \frac{4}{9}$

(1) $-\frac{9}{8} \times \left(-\frac{8}{3}\right)$

(11) $-7 \div \left(-\frac{8}{7}\right)$

(21) $-\frac{4}{5} \times 3$

(31) $\frac{4}{3} \div \frac{10}{7}$

(2) $7 \div 1$

(12) $-\frac{5}{4} \times \frac{3}{5}$

(22) $\frac{8}{7} \div \frac{3}{5}$

(32) $\frac{10}{7} \div \frac{7}{10}$

(3) $-1 \div 5$

(13) $-\frac{5}{2} \times \frac{1}{2}$

(23) $\frac{5}{8} \times \frac{1}{2}$

(33) $1 \div \frac{9}{2}$

(4) -2×1

(14) $\frac{7}{9} \div 3$

(24) $-2 \div \frac{2}{5}$

(34) $-9 \times \frac{3}{10}$

(5) $-\frac{2}{3} \div \frac{7}{3}$

(15) $\frac{5}{4} \times \frac{1}{3}$

(25) $-\frac{9}{5} \div \frac{5}{2}$

(35) $-\frac{5}{7} \div \frac{9}{10}$

(6) $\frac{1}{6} \times \left(-\frac{3}{10}\right)$

(16) $\frac{9}{10} \div \left(-\frac{1}{9}\right)$

(26) $-8 \div \left(-\frac{1}{6}\right)$

(36) $\frac{9}{4} \times \left(-\frac{9}{8}\right)$

(7) $-\frac{7}{10} \times \frac{1}{5}$

(27) $-\frac{4}{5} \div \left(-\frac{3}{5}\right)$

(37) $\frac{9}{10} \div \left(-\frac{3}{8}\right)$

(8) $\frac{5}{9} \times 1$

(17) $-\frac{9}{8} \div 5$

(28) $-\frac{1}{10} \times 2$

(38) $-3 \div \frac{9}{5}$

(9) $\frac{4}{5} \div (-3)$

(18) $\frac{5}{7} \div 3$

(29) $-\frac{7}{6} \div \left(-\frac{7}{4}\right)$

(39) $\frac{5}{7} \div \left(-\frac{5}{3}\right)$

(10) $2 \div \frac{4}{3}$

(19) $-\frac{1}{10} \times \frac{3}{5}$

(30) $-\frac{5}{2} \times \frac{5}{4}$

(40) $-\frac{6}{5} \times \left(-\frac{10}{9}\right)$

(20) $-2 \div \frac{1}{7}$

(1) $-2 \div \frac{4}{3}$

(11) $-2 \times \left(-\frac{8}{9}\right)$

(21) $-7 \div \frac{5}{4}$

(31) $2 \times \left(-\frac{7}{2}\right)$

(2) $\frac{2}{9} \div 2$

(12) 4×4

(22) $-\frac{1}{6} \times (-1)$

(32) $\frac{1}{3} \div \left(-\frac{1}{2}\right)$

(3) $\frac{7}{2} \div \frac{5}{8}$

(13) $\frac{4}{3} \div \left(-\frac{5}{6}\right)$

(23) $\frac{1}{2} \times (-1)$

(33) $-\frac{2}{5} \times \left(-\frac{7}{10}\right)$

(4) $\frac{5}{6} \times \frac{5}{4}$

(14) $\frac{7}{5} \div 2$

(24) $8 \div \left(-\frac{1}{6}\right)$

(34) $-\frac{5}{9} \times \left(-\frac{7}{4}\right)$

(5) $3 \div (-6)$

(15) $5 \times \frac{7}{3}$

(25) $\frac{10}{9} \times 1$

(35) $-\frac{8}{3} \times \frac{3}{5}$

(6) $1 \times (-1)$

(16) $\frac{1}{5} \times \frac{6}{7}$

(26) $\frac{1}{5} \div \left(-\frac{9}{4}\right)$

(36) $1 \div \left(-\frac{7}{3}\right)$

(7) $\frac{3}{4} \times \frac{3}{8}$

(17) $-\frac{7}{5} \times (-2)$

(27) $-2 \div 4$

(37) $-\frac{4}{9} \times \frac{5}{9}$

(8) $-\frac{10}{9} \div \left(-\frac{4}{5}\right)$

(18) $\frac{7}{9} \times \left(-\frac{9}{2}\right)$

(28) $\frac{9}{4} \times 1$

(38) $-\frac{2}{5} \div 2$

(9) $-\frac{7}{2} \times \left(-\frac{4}{3}\right)$

(19) $-\frac{4}{5} \times (-1)$

(29) -1×3

(39) $\frac{8}{5} \div \left(-\frac{3}{8}\right)$

(10) $1 \times \frac{1}{2}$

(20) $-\frac{2}{9} \div \frac{4}{5}$

(30) $-2 \div \left(-\frac{6}{7}\right)$

(40) $\frac{4}{3} \div (-10)$

(1) $-\frac{9}{10} \times 6$

(11) $-\frac{7}{10} \div \frac{1}{5}$

(21) $\frac{1}{9} \div (-3)$

(31) $2 \div \frac{7}{6}$

(2) $-\frac{7}{3} \times \frac{5}{4}$

(12) $-\frac{6}{5} \div \frac{10}{9}$

(22) $-\frac{1}{2} \times \left(-\frac{8}{3}\right)$

(32) $\frac{1}{3} \div \frac{10}{9}$

(3) $-\frac{9}{10} \times \frac{7}{5}$

(13) $-\frac{4}{5} \times \left(-\frac{2}{3}\right)$

(23) $\frac{8}{7} \times \left(-\frac{3}{10}\right)$

(33) $1 \times \frac{7}{5}$

(4) $\frac{2}{5} \div \frac{3}{2}$

(14) $\frac{9}{4} \times \left(-\frac{1}{8}\right)$

(24) $-\frac{1}{3} \times \frac{1}{5}$

(34) $1 \times \frac{4}{5}$

(5) $-\frac{10}{7} \div 2$

(15) $\frac{10}{9} \times (-1)$

(25) $1 \div \frac{3}{2}$

(35) $-\frac{3}{4} \times \left(-\frac{1}{4}\right)$

(6) $\frac{4}{3} \times \left(-\frac{1}{4}\right)$

(16) $-2 \div \left(-\frac{9}{4}\right)$

(26) $\frac{1}{5} \times \frac{1}{4}$

(36) $\frac{2}{3} \times \left(-\frac{2}{9}\right)$

(7) $-6 \div (-8)$

(17) $\frac{7}{2} \div \frac{5}{6}$

(27) $-\frac{1}{7} \div \frac{6}{7}$

(37) $-1 \div \left(-\frac{10}{9}\right)$

(8) $-\frac{5}{2} \div \frac{9}{2}$

(18) $-\frac{9}{10} \times \frac{7}{2}$

(28) $\frac{1}{3} \times \frac{4}{5}$

(38) $-1 \times (-1)$

(9) $\frac{7}{8} \div \frac{9}{7}$

(19) $-2 \div \left(-\frac{8}{9}\right)$

(29) $\frac{8}{9} \times (-4)$

(39) $\frac{5}{8} \div 3$

(10) $-\frac{2}{9} \times \left(-\frac{5}{4}\right)$

(20) $-3 \div 5$

(30) $\frac{7}{9} \div \frac{4}{5}$

(40) $-\frac{9}{7} \div \frac{3}{7}$

(1) $\frac{1}{4} \div 5$

(11) $\frac{1}{2} \times \frac{1}{9}$

(21) $\frac{4}{3} \div \left(-\frac{1}{2}\right)$

(31) $-2 \times \left(-\frac{1}{2}\right)$

(2) $-\frac{8}{3} \times 1$

(12) $1 \div (-4)$

(22) $2 \times \left(-\frac{8}{5}\right)$

(32) $\frac{4}{7} \times \left(-\frac{2}{3}\right)$

(3) $\frac{1}{2} \div 2$

(13) $\frac{3}{8} \times \left(-\frac{8}{5}\right)$

(23) $\frac{1}{9} \times \frac{8}{7}$

(33) $-1 \div \left(-\frac{10}{9}\right)$

(4) $-9 \times \left(-\frac{6}{7}\right)$

(14) $2 \times \frac{9}{4}$

(24) $-\frac{8}{9} \div \left(-\frac{7}{2}\right)$

(34) $-\frac{2}{5} \div \left(-\frac{3}{4}\right)$

(5) $3 \times \frac{1}{8}$

(15) $\frac{3}{5} \div \frac{9}{4}$

(25) $\frac{7}{8} \div \left(-\frac{4}{7}\right)$

(35) $1 \times \left(-\frac{4}{3}\right)$

(6) $\frac{7}{3} \div \frac{5}{7}$

(16) $-\frac{3}{5} \div 4$

(26) $-\frac{3}{5} \div 1$

(36) $-\frac{8}{7} \times 2$

(7) $-\frac{2}{3} \div \frac{7}{5}$

(17) $-\frac{10}{7} \times \frac{2}{3}$

(27) $\frac{4}{5} \div \frac{7}{6}$

(37) $\frac{1}{3} \div \left(-\frac{7}{4}\right)$

(8) $-1 \times \frac{1}{2}$

(18) $\frac{1}{2} \div (-1)$

(28) $-\frac{1}{6} \div \frac{9}{10}$

(38) $\frac{4}{9} \times \left(-\frac{7}{10}\right)$

(9) $\frac{2}{5} \div \frac{5}{8}$

(19) $10 \times \left(-\frac{9}{8}\right)$

(29) $\frac{3}{5} \div \left(-\frac{8}{7}\right)$

(39) $-\frac{4}{7} \times 5$

(10) $-\frac{3}{4} \times \frac{1}{9}$

(20) $-\frac{1}{3} \times \left(-\frac{7}{10}\right)$

(30) $-\frac{2}{3} \div 4$

(40) $\frac{10}{7} \div \frac{7}{3}$

(1) $-\frac{3}{2} \times 2$

(11) $\frac{8}{5} \times 3$

(21) $\frac{4}{5} \times \left(-\frac{7}{9}\right)$

(31) $10 \div 10$

(2) $-\frac{4}{3} \div \left(-\frac{1}{2}\right)$

(12) $\frac{1}{3} \times \left(-\frac{9}{4}\right)$

(22) $-\frac{3}{10} \times (-5)$

(32) $-\frac{3}{10} \div \frac{5}{2}$

(3) $\frac{8}{9} \times \left(-\frac{8}{5}\right)$

(13) $\frac{4}{7} \div \frac{4}{3}$

(23) $-1 \div \frac{3}{8}$

(33) $\frac{5}{9} \times \left(-\frac{3}{2}\right)$

(4) $-\frac{5}{7} \div (-6)$

(14) $\frac{1}{5} \times (-4)$

(24) $-\frac{1}{3} \times \left(-\frac{8}{3}\right)$

(34) $-\frac{5}{7} \times \frac{9}{7}$

(5) $-2 \div \frac{3}{2}$

(15) $-4 \div \left(-\frac{3}{4}\right)$

(25) $-6 \times \frac{1}{5}$

(35) $-\frac{1}{5} \div 5$

(6) $\frac{3}{2} \div \frac{2}{3}$

(16) $-1 \div \frac{7}{10}$

(26) $\frac{4}{3} \times \frac{5}{3}$

(36) $-2 \div \left(-\frac{1}{4}\right)$

(7) $-\frac{7}{10} \div (-8)$

(17) $-\frac{6}{5} \div \left(-\frac{9}{10}\right)$

(27) $10 \times \frac{4}{3}$

(37) $-2 \div \left(-\frac{8}{3}\right)$

(8) $-\frac{1}{7} \times \frac{7}{9}$

(18) $\frac{2}{5} \div 1$

(28) $\frac{4}{9} \times \frac{8}{3}$

(38) $-\frac{1}{6} \div \left(-\frac{1}{5}\right)$

(9) $-\frac{10}{7} \times \left(-\frac{1}{4}\right)$

(19) $-\frac{7}{3} \times \left(-\frac{5}{7}\right)$

(29) $-1 \times \left(-\frac{1}{10}\right)$

(39) $-\frac{4}{5} \times \frac{5}{3}$

(10) $\frac{5}{3} \times \frac{7}{8}$

(20) $-\frac{1}{2} \times \frac{9}{4}$

(30) $-\frac{6}{7} \div 8$

(40) $-\frac{3}{7} \times \frac{7}{9}$

(1) $-2 \div \frac{3}{2}$

(11) $1 \div \left(-\frac{5}{3}\right)$

(21) $1 \times \frac{9}{8}$

(31) $\frac{1}{7} \div 1$

(2) $\frac{3}{5} \div \frac{1}{7}$

(12) $-\frac{3}{10} \div \left(-\frac{5}{2}\right)$

(22) $\frac{4}{5} \times \frac{5}{4}$

(32) $1 \div \frac{4}{3}$

(3) $-\frac{3}{5} \times (-1)$

(13) $7 \div \left(-\frac{3}{2}\right)$

(23) $\frac{3}{2} \div \left(-\frac{5}{4}\right)$

(33) $-\frac{9}{2} \times \frac{3}{8}$

(4) $1 \div \left(-\frac{8}{3}\right)$

(14) $-1 \times \left(-\frac{7}{5}\right)$

(24) $4 \times \frac{5}{3}$

(34) $-\frac{9}{5} \times (-2)$

(5) $\frac{1}{4} \div \frac{1}{4}$

(15) $-8 \div 1$

(25) 2×4

(35) $\frac{4}{9} \times \frac{1}{4}$

(6) $-4 \times \left(-\frac{3}{4}\right)$

(16) $3 \div 7$

(26) $-\frac{2}{5} \times \left(-\frac{2}{5}\right)$

(36) $\frac{3}{10} \div \frac{3}{5}$

(7) $\frac{2}{7} \div \left(-\frac{3}{7}\right)$

(17) $\frac{3}{5} \times \left(-\frac{7}{10}\right)$

(27) $-\frac{4}{7} \div \left(-\frac{5}{3}\right)$

(37) $-\frac{9}{10} \times 1$

(8) $-\frac{6}{5} \times \frac{3}{4}$

(18) $-\frac{7}{9} \times \frac{9}{5}$

(28) $\frac{5}{3} \div \left(-\frac{2}{3}\right)$

(38) $\frac{10}{7} \div \frac{1}{6}$

(9) $\frac{1}{9} \times \frac{9}{5}$

(19) $1 \times \frac{3}{2}$

(29) $1 \div \frac{9}{10}$

(39) $\frac{1}{3} \div \frac{1}{4}$

(10) $-\frac{1}{2} \div \left(-\frac{3}{2}\right)$

(20) $-\frac{1}{3} \times (-4)$

(30) $2 \div \frac{4}{9}$

(40) $\frac{3}{2} \times \left(-\frac{1}{5}\right)$

(1) $-\frac{2}{3} \times (-2)$

(11) $-\frac{7}{9} \times \left(-\frac{4}{7}\right)$

(21) $-\frac{2}{9} \times \left(-\frac{3}{5}\right)$

(31) $\frac{9}{10} \times \left(-\frac{1}{2}\right)$

(2) $-\frac{5}{9} \div \left(-\frac{9}{5}\right)$

(12) $-\frac{1}{2} \times \frac{1}{10}$

(22) $-\frac{7}{6} \div \left(-\frac{9}{4}\right)$

(32) $\frac{10}{3} \times \frac{9}{7}$

(3) $\frac{7}{4} \times \left(-\frac{5}{4}\right)$

(13) $\frac{1}{7} \div \frac{10}{3}$

(23) $-\frac{8}{9} \times \left(-\frac{1}{5}\right)$

(33) $-\frac{9}{4} \div \frac{8}{5}$

(4) $2 \div \frac{5}{8}$

(14) $5 \times \left(-\frac{5}{4}\right)$

(24) $\frac{9}{8} \times \left(-\frac{4}{5}\right)$

(34) $\frac{5}{3} \div \frac{1}{9}$

(5) $-\frac{1}{2} \div (-4)$

(15) $\frac{8}{3} \times \frac{3}{2}$

(25) $4 \div (-1)$

(35) $1 \times \left(-\frac{4}{7}\right)$

(6) $\frac{1}{2} \div 4$

(16) $1 \div \frac{4}{3}$

(26) $-\frac{2}{3} \times 3$

(36) $\frac{3}{8} \div \left(-\frac{7}{3}\right)$

(7) $\frac{7}{5} \times \frac{3}{8}$

(17) $-1 \times \left(-\frac{4}{5}\right)$

(27) $-1 \div \left(-\frac{1}{9}\right)$

(37) $-\frac{3}{5} \div (-7)$

(8) $\frac{4}{3} \times \left(-\frac{4}{5}\right)$

(18) $-\frac{3}{7} \times \left(-\frac{5}{4}\right)$

(28) $\frac{3}{10} \div \left(-\frac{5}{9}\right)$

(38) $-\frac{7}{2} \times \left(-\frac{9}{4}\right)$

(9) $-\frac{3}{2} \div \left(-\frac{8}{7}\right)$

(19) $-\frac{1}{6} \div \frac{2}{3}$

(29) $3 \div \frac{10}{3}$

(39) $3 \div (-1)$

(10) $\frac{1}{2} \times \left(-\frac{3}{5}\right)$

(20) $-\frac{1}{3} \div \left(-\frac{1}{2}\right)$

(30) $-\frac{9}{5} \div \left(-\frac{2}{3}\right)$

(40) $-9 \div \frac{1}{10}$

(1) $\frac{4}{5} \div (-2)$

(11) $-\frac{9}{7} \div \left(-\frac{7}{8}\right)$

(21) $-\frac{5}{8} \times \left(-\frac{7}{4}\right)$

(31) $\frac{1}{2} \times \frac{9}{2}$

(2) $\frac{7}{9} \div \frac{5}{4}$

(12) $-\frac{5}{8} \times \frac{3}{2}$

(22) $-9 \times \left(-\frac{5}{9}\right)$

(32) $\frac{1}{4} \times \left(-\frac{5}{3}\right)$

(3) $-\frac{10}{3} \div \frac{2}{9}$

(13) $5 \div \frac{3}{2}$

(23) $-5 \times \left(-\frac{9}{8}\right)$

(33) $-\frac{9}{8} \div 1$

(4) $-\frac{4}{7} \div \left(-\frac{2}{7}\right)$

(14) $\frac{5}{3} \div \left(-\frac{3}{4}\right)$

(24) $-\frac{6}{5} \times \left(-\frac{9}{2}\right)$

(34) $\frac{2}{5} \times \frac{8}{9}$

(5) $\frac{7}{10} \div 1$

(15) $-\frac{7}{6} \div \frac{2}{5}$

(25) $\frac{3}{7} \div \frac{6}{7}$

(35) $\frac{3}{7} \div \left(-\frac{1}{3}\right)$

(6) $-\frac{1}{10} \div \left(-\frac{10}{9}\right)$

(16) $\frac{5}{7} \div \frac{3}{5}$

(26) $-\frac{3}{5} \div \left(-\frac{6}{7}\right)$

(36) $\frac{4}{5} \times \frac{1}{2}$

(7) $\frac{5}{2} \div \frac{2}{9}$

(17) $\frac{3}{10} \div \left(-\frac{6}{5}\right)$

(27) $-1 \div \frac{7}{8}$

(37) $-2 \div \frac{3}{2}$

(8) $\frac{10}{3} \times \left(-\frac{1}{3}\right)$

(18) $\frac{8}{3} \times \left(-\frac{10}{9}\right)$

(28) $1 \div \left(-\frac{6}{5}\right)$

(38) $\frac{2}{5} \div \frac{1}{6}$

(9) $-\frac{9}{8} \div \left(-\frac{2}{3}\right)$

(19) $\frac{1}{2} \times \left(-\frac{3}{2}\right)$

(29) $\frac{2}{3} \times \frac{1}{8}$

(39) $-1 \times \frac{4}{3}$

(10) $-2 \div 2$

(20) $-\frac{1}{7} \div \left(-\frac{2}{9}\right)$

(30) $\frac{5}{2} \times \frac{2}{3}$

(40) $-\frac{1}{2} \div 1$

(1) $\frac{3}{5} \times \left(-\frac{3}{10}\right)$

(11) $\frac{3}{5} \times \left(-\frac{4}{7}\right)$

(21) $\frac{5}{8} \times \left(-\frac{1}{3}\right)$

(31) $10 \times \left(-\frac{2}{5}\right)$

(2) $\frac{3}{4} \div \left(-\frac{5}{8}\right)$

(12) $\frac{10}{7} \times 10$

(22) $\frac{3}{8} \div \left(-\frac{3}{4}\right)$

(32) $-\frac{4}{9} \times \frac{7}{6}$

(3) $-\frac{9}{2} \times (-4)$

(13) $\frac{5}{3} \div \left(-\frac{9}{8}\right)$

(23) $-\frac{10}{3} \div \left(-\frac{9}{10}\right)$

(33) $-\frac{9}{8} \div \frac{3}{2}$

(4) $1 \times \left(-\frac{8}{5}\right)$

(14) $\frac{4}{9} \div \left(-\frac{4}{3}\right)$

(24) $\frac{3}{2} \times \frac{4}{5}$

(34) $\frac{3}{5} \div \left(-\frac{1}{2}\right)$

(5) $-3 \times (-2)$

(15) $\frac{5}{9} \times \frac{5}{3}$

(25) $\frac{7}{6} \div \frac{5}{2}$

(35) $-1 \times \frac{7}{6}$

(6) $-\frac{10}{7} \times \frac{9}{10}$

(16) $-\frac{5}{9} \times \frac{1}{7}$

(26) $\frac{1}{3} \div \frac{1}{5}$

(36) $\frac{2}{7} \div \left(-\frac{5}{6}\right)$

(7) $-\frac{8}{3} \div 1$

(17) $\frac{4}{3} \times \frac{1}{2}$

(27) $\frac{3}{5} \times \frac{1}{3}$

(37) $\frac{1}{2} \div \frac{6}{7}$

(8) -4×4

(18) $1 \div \left(-\frac{7}{6}\right)$

(28) $-\frac{5}{2} \times (-1)$

(38) $-\frac{9}{7} \div 1$

(9) $-\frac{1}{3} \div 1$

(19) $\frac{2}{5} \div 1$

(29) $-6 \times \left(-\frac{5}{2}\right)$

(39) $-\frac{6}{7} \div \frac{3}{4}$

(10) $-\frac{1}{3} \times (-6)$

(20) $-\frac{8}{9} \div \left(-\frac{5}{4}\right)$

(30) $-\frac{7}{3} \times (-4)$

(40) $\frac{3}{5} \div \frac{5}{8}$

(1) $1 \times \left(-\frac{4}{9}\right)$

(11) $-\frac{3}{5} \div (-3)$

(21) $1 \times \frac{7}{4}$

(31) $2 \times \frac{2}{3}$

(2) $-\frac{4}{7} \div \frac{3}{7}$

(12) $\frac{9}{8} \div 3$

(22) $-\frac{4}{5} \times \frac{2}{5}$

(32) $-9 \times \frac{5}{6}$

(3) $-2 \times \frac{10}{7}$

(13) $\frac{7}{3} \div \left(-\frac{9}{4}\right)$

(23) $-1 \times \left(-\frac{1}{2}\right)$

(33) $-\frac{2}{7} \times \frac{5}{9}$

(4) $1 \times (-5)$

(14) $-1 \times \left(-\frac{1}{2}\right)$

(24) $-\frac{7}{8} \times (-3)$

(34) $-\frac{3}{2} \times \left(-\frac{5}{9}\right)$

(5) $1 \div \left(-\frac{9}{8}\right)$

(15) $5 \div (-4)$

(25) $\frac{10}{7} \times \frac{5}{2}$

(35) $-\frac{5}{8} \times \frac{7}{8}$

(6) $\frac{1}{5} \div \frac{4}{5}$

(16) $\frac{7}{10} \div (-9)$

(26) $-\frac{4}{9} \times \left(-\frac{1}{4}\right)$

(36) $5 \times \frac{4}{5}$

(7) $-\frac{9}{8} \times \frac{4}{7}$

(17) $\frac{2}{7} \div \frac{4}{9}$

(27) $-\frac{10}{9} \times (-1)$

(37) $\frac{5}{2} \div \frac{3}{2}$

(8) $-\frac{1}{4} \div \left(-\frac{3}{2}\right)$

(18) $\frac{7}{9} \times 5$

(28) $-\frac{8}{5} \times \frac{1}{3}$

(38) $\frac{5}{2} \times \left(-\frac{3}{4}\right)$

(9) $-\frac{3}{7} \div \left(-\frac{2}{9}\right)$

(19) $\frac{1}{7} \times \frac{8}{3}$

(29) $\frac{8}{7} \times \frac{1}{4}$

(39) $\frac{3}{5} \div \left(-\frac{1}{6}\right)$

(10) $-\frac{5}{4} \div \frac{9}{7}$

(20) $\frac{1}{3} \div \left(-\frac{5}{8}\right)$

(30) $-\frac{5}{3} \times \left(-\frac{5}{3}\right)$

(40) $1 \div \left(-\frac{4}{5}\right)$

(1) $\frac{4}{9} \times \left(-\frac{1}{8}\right)$

(11) $-1 \div \left(-\frac{5}{4}\right)$

(21) $\frac{3}{2} \times \left(-\frac{1}{2}\right)$

(31) $7 \times \left(-\frac{1}{2}\right)$

(2) $1 \times \left(-\frac{7}{3}\right)$

(12) $1 \times \frac{3}{5}$

(22) $-1 \div \frac{2}{5}$

(32) $\frac{7}{8} \div \frac{8}{5}$

(3) $9 \div \frac{8}{3}$

(13) $\frac{7}{2} \div \left(-\frac{2}{5}\right)$

(23) $-\frac{1}{3} \times 4$

(33) $-2 \div \left(-\frac{9}{8}\right)$

(4) $-\frac{7}{5} \div 3$

(14) $-\frac{4}{5} \times 1$

(24) $-\frac{7}{3} \div (-8)$

(34) $-\frac{3}{5} \times \left(-\frac{8}{9}\right)$

(5) $-\frac{5}{8} \div \frac{10}{3}$

(15) $\frac{9}{8} \times \left(-\frac{9}{4}\right)$

(25) $-2 \times \left(-\frac{1}{2}\right)$

(35) $\frac{5}{4} \times \frac{1}{10}$

(6) $-\frac{5}{4} \times \frac{8}{9}$

(16) $-6 \div \left(-\frac{5}{2}\right)$

(26) $1 \div \left(-\frac{7}{4}\right)$

(36) $-\frac{1}{2} \times \left(-\frac{1}{9}\right)$

(7) $1 \div 2$

(17) $\frac{3}{10} \times \left(-\frac{7}{5}\right)$

(27) $1 \times \left(-\frac{1}{2}\right)$

(37) $\frac{3}{2} \div \frac{3}{8}$

(8) $\frac{10}{3} \div (-6)$

(18) $-\frac{9}{5} \div (-3)$

(28) $-\frac{5}{9} \div \frac{4}{7}$

(38) $\frac{2}{3} \times \left(-\frac{5}{4}\right)$

(9) $-5 \div \left(-\frac{1}{9}\right)$

(19) $-\frac{8}{5} \div (-2)$

(29) $\frac{1}{6} \times \left(-\frac{7}{10}\right)$

(39) $-\frac{4}{3} \times \left(-\frac{1}{5}\right)$

(10) $-3 \times \frac{3}{7}$

(20) $\frac{3}{4} \times \left(-\frac{7}{5}\right)$

(30) $\frac{1}{5} \div \frac{3}{2}$

(40) $-1 \div \frac{1}{5}$

(1) $-\frac{10}{7} \times \frac{3}{5}$

(11) 8×2

(21) $-1 \times \left(-\frac{5}{3}\right)$

(31) $-\frac{1}{4} \times \left(-\frac{9}{2}\right)$

(2) $\frac{8}{3} \times \left(-\frac{3}{2}\right)$

(12) $-\frac{2}{3} \times \left(-\frac{1}{7}\right)$

(22) $-\frac{10}{7} \times \left(-\frac{10}{7}\right)$

(32) $\frac{4}{5} \times \frac{7}{6}$

(3) $-\frac{5}{3} \div (-1)$

(13) $-\frac{2}{3} \div \left(-\frac{7}{10}\right)$

(23) $-\frac{5}{9} \times \frac{2}{3}$

(33) $1 \div \left(-\frac{3}{4}\right)$

(4) $3 \div \left(-\frac{5}{2}\right)$

(14) $-\frac{7}{2} \times \frac{7}{2}$

(24) $-\frac{9}{4} \div 1$

(34) $3 \div \left(-\frac{3}{5}\right)$

(5) $2 \times \left(-\frac{9}{5}\right)$

(15) $\frac{5}{2} \times (-3)$

(25) $\frac{2}{9} \times \frac{1}{9}$

(35) $1 \times \left(-\frac{8}{9}\right)$

(6) $\frac{3}{8} \times \frac{10}{9}$

(16) $\frac{1}{7} \div (-1)$

(26) $\frac{1}{4} \times (-2)$

(36) $-\frac{10}{3} \div (-1)$

(7) $1 \div \frac{2}{3}$

(17) $-\frac{2}{9} \times (-2)$

(27) $\frac{8}{7} \div \frac{7}{6}$

(37) $\frac{9}{4} \div \left(-\frac{9}{10}\right)$

(8) $-1 \div \frac{3}{5}$

(18) $\frac{1}{9} \times (-1)$

(28) $\frac{5}{4} \div (-1)$

(38) $-\frac{5}{2} \times \frac{5}{2}$

(9) $-3 \div \frac{3}{2}$

(19) $\frac{1}{7} \times \left(-\frac{3}{4}\right)$

(29) $8 \div (-3)$

(39) $4 \times \frac{1}{2}$

(10) $-\frac{4}{5} \times \frac{3}{4}$

(20) $\frac{5}{2} \div \frac{6}{7}$

(30) $1 \times \left(-\frac{2}{7}\right)$

(40) $\frac{7}{10} \times \frac{6}{5}$

(1) $-\frac{9}{7} \div \frac{1}{9}$

(11) $-3 \div \frac{2}{9}$

(21) $\frac{5}{3} \div \frac{1}{2}$

(31) $-\frac{2}{9} \times \frac{7}{9}$

(2) $\frac{8}{9} \times 2$

(12) $-\frac{3}{2} \div \frac{3}{10}$

(22) $\frac{1}{8} \div \left(-\frac{5}{2}\right)$

(32) $\frac{3}{5} \div \frac{10}{9}$

(3) $\frac{3}{2} \div \left(-\frac{7}{2}\right)$

(13) $\frac{1}{3} \times \left(-\frac{2}{3}\right)$

(23) $-\frac{5}{3} \div \frac{10}{9}$

(33) $1 \div \left(-\frac{1}{2}\right)$

(4) $\frac{1}{2} \times \left(-\frac{2}{9}\right)$

(14) $\frac{5}{9} \div \left(-\frac{4}{3}\right)$

(24) $-\frac{5}{3} \times \frac{7}{4}$

(34) $-\frac{1}{6} \div 3$

(5) $-9 \times (-1)$

(15) $-\frac{2}{5} \times (-3)$

(25) $-1 \times \frac{1}{4}$

(35) $-\frac{3}{5} \div \left(-\frac{1}{10}\right)$

(6) $\frac{9}{10} \div \frac{2}{3}$

(16) $-\frac{4}{5} \times (-3)$

(26) $\frac{3}{4} \div \frac{9}{5}$

(36) $-\frac{1}{5} \times \left(-\frac{10}{9}\right)$

(7) $\frac{1}{7} \div 2$

(17) $-\frac{8}{3} \div \frac{1}{9}$

(27) $-\frac{1}{9} \times 9$

(37) $\frac{4}{5} \times \left(-\frac{2}{7}\right)$

(8) $-\frac{4}{5} \times 1$

(18) $-2 \div (-3)$

(28) $\frac{1}{8} \times \frac{2}{9}$

(38) $\frac{1}{5} \times \left(-\frac{8}{9}\right)$

(9) $\frac{3}{2} \times \frac{7}{9}$

(19) $-\frac{5}{4} \div \left(-\frac{2}{3}\right)$

(29) $\frac{7}{2} \div \frac{1}{5}$

(39) $-\frac{7}{9} \times \left(-\frac{1}{2}\right)$

(10) $-\frac{1}{2} \div \left(-\frac{1}{9}\right)$

(20) $\frac{1}{2} \div (-1)$

(30) $\frac{2}{5} \div (-4)$

(40) $-\frac{9}{4} \div \left(-\frac{2}{5}\right)$

(1) $1 \times \frac{5}{2}$

(11) $-2 \div \frac{2}{3}$

(21) $\frac{10}{7} \div (-1)$

(31) $-\frac{1}{2} \div \left(-\frac{7}{10}\right)$

(2) $10 \div \left(-\frac{10}{3}\right)$

(12) $-6 \times \left(-\frac{1}{2}\right)$

(22) $\frac{5}{7} \times \frac{5}{8}$

(32) $-\frac{2}{3} \times \frac{1}{3}$

(3) $\frac{3}{2} \div (-3)$

(13) $-\frac{7}{5} \times \left(-\frac{9}{5}\right)$

(23) $\frac{1}{3} \times \left(-\frac{6}{5}\right)$

(33) $-\frac{10}{9} \div \frac{3}{10}$

(4) $\frac{3}{4} \times \left(-\frac{1}{3}\right)$

(14) 1×5

(24) $4 \div \frac{2}{5}$

(34) $\frac{4}{9} \times \left(-\frac{1}{4}\right)$

(5) $-\frac{3}{5} \div \frac{7}{5}$

(15) $-2 \div \left(-\frac{1}{5}\right)$

(25) $-\frac{4}{7} \div (-6)$

(35) $-\frac{5}{2} \div \left(-\frac{3}{5}\right)$

(6) $-6 \div \frac{2}{3}$

(16) $-\frac{1}{2} \times (-3)$

(26) $-\frac{5}{3} \times 2$

(36) $\frac{1}{4} \div \left(-\frac{1}{2}\right)$

(7) $-\frac{1}{5} \times \frac{8}{7}$

(17) $-2 \div 5$

(27) $-\frac{2}{7} \times \frac{3}{8}$

(37) $1 \times \frac{10}{7}$

(8) $-\frac{2}{7} \times \left(-\frac{1}{3}\right)$

(18) $-\frac{9}{10} \div 2$

(28) $\frac{8}{3} \times (-1)$

(38) $\frac{1}{4} \times \left(-\frac{3}{5}\right)$

(9) $-\frac{2}{3} \times \left(-\frac{7}{10}\right)$

(19) $-\frac{5}{4} \div \frac{2}{3}$

(29) $-\frac{3}{4} \times (-1)$

(39) $-\frac{5}{3} \div \frac{4}{5}$

(10) $-\frac{6}{5} \times 1$

(20) $-1 \times \left(-\frac{6}{7}\right)$

(30) $4 \div \frac{4}{7}$

(40) $1 \times \left(-\frac{7}{2}\right)$

(1) $-\frac{7}{5} \div (-1)$

(11) $-\frac{3}{2} \times 1$

(21) $-\frac{1}{4} \div \left(-\frac{2}{5}\right)$

(31) $\frac{1}{8} \div \frac{3}{8}$

(2) $-\frac{1}{7} \times \frac{9}{5}$

(12) $-\frac{2}{7} \times \left(-\frac{4}{9}\right)$

(22) $\frac{1}{5} \div 1$

(32) $-\frac{8}{3} \div \frac{4}{7}$

(3) $\frac{5}{7} \div (-5)$

(13) $-\frac{1}{10} \times \left(-\frac{5}{4}\right)$

(23) $\frac{3}{2} \times \frac{5}{4}$

(33) $\frac{10}{7} \div \frac{3}{2}$

(4) $-\frac{2}{7} \times \frac{2}{3}$

(14) $\frac{8}{7} \times \left(-\frac{3}{5}\right)$

(24) $\frac{8}{3} \times 1$

(34) $1 \times (-1)$

(5) $-\frac{5}{7} \times 5$

(15) $1 \times \left(-\frac{7}{4}\right)$

(25) $-\frac{4}{9} \div \frac{5}{7}$

(35) $-\frac{8}{5} \times (-3)$

(6) $-\frac{4}{3} \div \frac{4}{5}$

(16) $\frac{5}{4} \times \left(-\frac{9}{7}\right)$

(26) $\frac{3}{2} \times \frac{5}{7}$

(36) $-1 \div (-7)$

(7) $2 \div \frac{7}{3}$

(17) $-\frac{7}{6} \times \left(-\frac{3}{2}\right)$

(27) $\frac{2}{3} \div (-1)$

(37) $\frac{5}{3} \times (-1)$

(8) $-\frac{6}{5} \times \frac{10}{9}$

(18) $\frac{1}{5} \times \left(-\frac{5}{7}\right)$

(28) $-\frac{2}{5} \times \left(-\frac{2}{5}\right)$

(38) $-\frac{1}{10} \div \left(-\frac{8}{5}\right)$

(9) $\frac{5}{2} \times \frac{4}{5}$

(19) $\frac{3}{2} \div \frac{7}{4}$

(29) $\frac{3}{5} \times 3$

(39) $\frac{1}{4} \times (-2)$

(10) $\frac{3}{2} \times \frac{1}{3}$

(20) $\frac{4}{3} \div (-2)$

(30) $\frac{7}{2} \div \frac{3}{5}$

(40) $1 \times \left(-\frac{5}{8}\right)$

$$(1) -\frac{9}{10} \times \frac{3}{10} \\ = -\frac{27}{100}$$

$$(11) \frac{1}{2} \times \left(-\frac{5}{3}\right) \\ = -\frac{5}{6}$$

$$(21) \frac{4}{5} \times 1 \\ = \frac{4}{5}$$

$$(31) -1 \times 1 \\ = -1$$

$$(2) -8 \div \frac{8}{7} \\ = -7$$

$$(12) -\frac{7}{8} \times \frac{9}{4} \\ = -\frac{63}{32}$$

$$(22) -\frac{1}{5} \div \left(-\frac{5}{2}\right) \\ = \frac{2}{25}$$

$$(32) \frac{5}{2} \div \frac{1}{2} \\ = 5$$

$$(3) -2 \times \left(-\frac{9}{7}\right) \\ = \frac{18}{7}$$

$$(13) -4 \div \frac{1}{4} \\ = -16$$

$$(23) -\frac{6}{7} \times \frac{1}{2} \\ = -\frac{3}{7}$$

$$(33) -9 \times (-3) \\ = 27$$

$$(4) \frac{4}{5} \times (-2) \\ = -\frac{8}{5}$$

$$(14) -\frac{2}{3} \times \frac{1}{5} \\ = -\frac{2}{15}$$

$$(24) \frac{1}{6} \div 10 \\ = \frac{1}{60}$$

$$(34) -8 \div \frac{7}{9} \\ = -\frac{72}{7}$$

$$(5) \frac{9}{10} \div \left(-\frac{3}{2}\right) \\ = -\frac{3}{5}$$

$$(15) \frac{10}{9} \div 1 \\ = \frac{10}{9}$$

$$(25) \frac{2}{5} \div 4 \\ = \frac{1}{10}$$

$$(35) -\frac{5}{8} \times \left(-\frac{3}{4}\right) \\ = \frac{15}{32}$$

$$(6) \frac{4}{5} \div \left(-\frac{2}{3}\right) \\ = -\frac{6}{5}$$

$$(16) \frac{2}{9} \times \frac{5}{2} \\ = \frac{5}{9}$$

$$(26) \frac{1}{3} \times 1 \\ = \frac{1}{3}$$

$$(36) \frac{1}{2} \div 2 \\ = \frac{1}{4}$$

$$(7) -\frac{1}{6} \times \frac{7}{9} \\ = -\frac{7}{54}$$

$$(17) \frac{4}{3} \times \frac{1}{8} \\ = \frac{1}{6}$$

$$(27) \frac{1}{4} \div \frac{3}{8} \\ = \frac{2}{3}$$

$$(37) -\frac{9}{4} \div (-2) \\ = \frac{9}{8}$$

$$(8) \frac{7}{3} \div 9 \\ = \frac{7}{27}$$

$$(18) -\frac{3}{2} \div \frac{2}{3} \\ = -\frac{9}{4}$$

$$(28) \frac{3}{2} \times \left(-\frac{2}{9}\right) \\ = -\frac{1}{3}$$

$$(38) \frac{5}{2} \div \frac{1}{7} \\ = \frac{35}{2}$$

$$(9) -2 \times \frac{2}{3} \\ = -\frac{4}{3}$$

$$(19) \frac{9}{8} \div \frac{4}{5} \\ = \frac{45}{32}$$

$$(29) -\frac{5}{4} \div \left(-\frac{1}{10}\right) \\ = \frac{25}{2}$$

$$(39) -\frac{8}{3} \div \frac{7}{10} \\ = -\frac{80}{21}$$

$$(10) -\frac{4}{5} \times \frac{1}{2} \\ = -\frac{2}{5}$$

$$(20) \frac{3}{2} \div \frac{3}{4} \\ = 2$$

$$(30) -2 \times \frac{8}{9} \\ = -\frac{16}{9}$$

$$(40) -\frac{9}{10} \div \left(-\frac{5}{3}\right) \\ = \frac{27}{50}$$

$$(1) -\frac{5}{3} \div \left(-\frac{5}{4}\right) \\ = \frac{4}{3}$$

$$(11) \frac{1}{5} \div \frac{2}{5} \\ = \frac{1}{2}$$

$$(21) 1 \div 6 \\ = \frac{1}{6}$$

$$(31) \frac{5}{3} \times \left(-\frac{6}{5}\right) \\ = -2$$

$$(2) -\frac{7}{5} \div \left(-\frac{4}{3}\right) \\ = \frac{21}{20}$$

$$(12) \frac{3}{10} \times \frac{3}{4} \\ = \frac{9}{40}$$

$$(22) -\frac{5}{7} \div \left(-\frac{8}{3}\right) \\ = \frac{15}{56}$$

$$(32) 1 \div \frac{2}{3} \\ = \frac{3}{2}$$

$$(3) -\frac{2}{3} \times \left(-\frac{5}{3}\right) \\ = \frac{10}{9}$$

$$(13) \frac{1}{5} \times \left(-\frac{10}{9}\right) \\ = -\frac{2}{9}$$

$$(23) -\frac{5}{2} \div 3 \\ = -\frac{5}{6}$$

$$(33) -\frac{5}{2} \div \left(-\frac{2}{7}\right) \\ = \frac{35}{4}$$

$$(4) 2 \div \left(-\frac{1}{3}\right) \\ = -6$$

$$(14) -1 \times \left(-\frac{1}{2}\right) \\ = \frac{1}{2}$$

$$(24) -\frac{2}{5} \div (-8) \\ = \frac{1}{20}$$

$$(34) -\frac{1}{3} \div (-1) \\ = \frac{1}{3}$$

$$(5) 5 \div \left(-\frac{1}{4}\right) \\ = -20$$

$$(15) \frac{9}{7} \div (-1) \\ = -\frac{9}{7}$$

$$(25) 5 \times \frac{5}{8} \\ = \frac{25}{8}$$

$$(35) 1 \times 1 \\ = 1$$

$$(6) \frac{3}{4} \div \left(-\frac{5}{2}\right) \\ = -\frac{3}{10}$$

$$(16) 2 \div 1 \\ = 2$$

$$(26) \frac{1}{9} \div \left(-\frac{3}{2}\right) \\ = -\frac{2}{27}$$

$$(36) -\frac{7}{10} \div \left(-\frac{7}{8}\right) \\ = \frac{4}{5}$$

$$(7) -\frac{1}{4} \times \left(-\frac{7}{10}\right) \\ = \frac{7}{40}$$

$$(17) -1 \times \left(-\frac{2}{3}\right) \\ = \frac{2}{3}$$

$$(27) \frac{7}{5} \div \left(-\frac{1}{3}\right) \\ = -\frac{21}{5}$$

$$(37) 2 \times 3 \\ = 6$$

$$(8) \frac{1}{2} \div \left(-\frac{1}{9}\right) \\ = -\frac{9}{2}$$

$$(18) -\frac{5}{9} \div \frac{5}{2} \\ = -\frac{2}{9}$$

$$(28) -5 \div (-1) \\ = 5$$

$$(38) \frac{1}{4} \times \frac{10}{9} \\ = \frac{5}{18}$$

$$(9) -\frac{7}{10} \times \left(-\frac{1}{2}\right) \\ = \frac{7}{20}$$

$$(19) -5 \times \left(-\frac{1}{3}\right) \\ = \frac{5}{3}$$

$$(29) 2 \div 2 \\ = 1$$

$$(39) 3 \times \frac{3}{7} \\ = \frac{9}{7}$$

$$(10) 3 \div \frac{4}{9} \\ = \frac{27}{4}$$

$$(20) -\frac{8}{9} \div \left(-\frac{10}{9}\right) \\ = \frac{4}{5}$$

$$(30) 1 \div \left(-\frac{1}{5}\right) \\ = -5$$

$$(40) \frac{3}{4} \div \left(-\frac{1}{2}\right) \\ = -\frac{3}{2}$$

$$(1) \frac{4}{7} \div \left(-\frac{1}{5}\right) \\ = -\frac{20}{7}$$

$$(11) -1 \times (-2) \\ = 2$$

$$(21) \frac{10}{9} \times 2 \\ = \frac{20}{9}$$

$$(31) -\frac{3}{5} \times 3 \\ = -\frac{9}{5}$$

$$(2) -\frac{1}{5} \div \frac{1}{5} \\ = -1$$

$$(12) -1 \div \left(-\frac{3}{8}\right) \\ = \frac{8}{3}$$

$$(22) -\frac{9}{8} \times \left(-\frac{4}{3}\right) \\ = \frac{3}{2}$$

$$(32) -1 \times \frac{2}{3} \\ = -\frac{2}{3}$$

$$(3) -\frac{1}{3} \times \frac{5}{4} \\ = -\frac{5}{12}$$

$$(13) \frac{3}{2} \div 4 \\ = \frac{3}{8}$$

$$(23) -1 \times \left(-\frac{9}{4}\right) \\ = \frac{9}{4}$$

$$(33) 1 \times \frac{1}{5} \\ = \frac{1}{5}$$

$$(4) -\frac{4}{3} \div (-2) \\ = \frac{2}{3}$$

$$(14) -\frac{1}{4} \times \left(-\frac{7}{3}\right) \\ = \frac{7}{12}$$

$$(24) \frac{6}{7} \times \left(-\frac{2}{7}\right) \\ = -\frac{12}{49}$$

$$(34) 1 \div \frac{1}{5} \\ = 5$$

$$(5) 2 \div \frac{9}{8} \\ = \frac{16}{9}$$

$$(15) \frac{5}{3} \div \left(-\frac{10}{7}\right) \\ = -\frac{7}{6}$$

$$(25) -\frac{7}{3} \div (-3) \\ = \frac{7}{9}$$

$$(35) -\frac{2}{5} \times \left(-\frac{6}{5}\right) \\ = \frac{12}{25}$$

$$(6) -\frac{8}{7} \div \left(-\frac{6}{5}\right) \\ = \frac{20}{21}$$

$$(16) -\frac{1}{3} \times 2 \\ = -\frac{2}{3}$$

$$(26) -1 \div \left(-\frac{9}{5}\right) \\ = \frac{5}{9}$$

$$(36) -5 \times \frac{10}{9} \\ = -\frac{50}{9}$$

$$(7) -\frac{8}{5} \times (-3) \\ = \frac{24}{5}$$

$$(17) \frac{9}{2} \div \left(-\frac{4}{7}\right) \\ = -\frac{63}{8}$$

$$(27) -1 \div \frac{1}{2} \\ = -2$$

$$(37) -\frac{1}{2} \div \left(-\frac{2}{3}\right) \\ = \frac{3}{4}$$

$$(8) 4 \div \frac{1}{10} \\ = 40$$

$$(18) \frac{10}{9} \times \left(-\frac{5}{7}\right) \\ = -\frac{50}{63}$$

$$(28) 8 \times \left(-\frac{5}{3}\right) \\ = -\frac{40}{3}$$

$$(38) \frac{3}{4} \div \left(-\frac{3}{5}\right) \\ = -\frac{5}{4}$$

$$(9) \frac{5}{4} \div \left(-\frac{8}{7}\right) \\ = -\frac{35}{32}$$

$$(19) \frac{3}{10} \div \left(-\frac{1}{2}\right) \\ = -\frac{3}{5}$$

$$(29) 8 \div \left(-\frac{1}{2}\right) \\ = -16$$

$$(39) -\frac{1}{5} \times (-2) \\ = \frac{2}{5}$$

$$(10) \frac{3}{2} \div \left(-\frac{3}{5}\right) \\ = -\frac{5}{2}$$

$$(20) -\frac{4}{7} \times \left(-\frac{5}{3}\right) \\ = \frac{20}{21}$$

$$(30) -\frac{4}{3} \div \frac{2}{3} \\ = -2$$

$$(40) \frac{2}{3} \times \left(-\frac{7}{3}\right) \\ = -\frac{14}{9}$$

$$(1) \frac{4}{7} \times \left(-\frac{1}{2}\right) \\ = -\frac{2}{7}$$

$$(11) -\frac{5}{8} \div \left(-\frac{2}{3}\right) \\ = \frac{15}{16}$$

$$(21) -\frac{3}{7} \times \left(-\frac{5}{8}\right) \\ = \frac{15}{56}$$

$$(31) 1 \times \left(-\frac{8}{3}\right) \\ = -\frac{8}{3}$$

$$(2) \frac{1}{3} \div \frac{10}{7} \\ = \frac{7}{30}$$

$$(12) \frac{2}{5} \times 4 \\ = \frac{8}{5}$$

$$(22) -\frac{3}{2} \times \frac{9}{5} \\ = -\frac{27}{10}$$

$$(32) \frac{1}{5} \div \left(-\frac{3}{2}\right) \\ = -\frac{2}{15}$$

$$(3) \frac{9}{5} \div 3 \\ = \frac{3}{5}$$

$$(13) -\frac{9}{5} \div \frac{9}{5} \\ = -1$$

$$(23) \frac{8}{9} \times \left(-\frac{1}{5}\right) \\ = -\frac{8}{45}$$

$$(33) 3 \times 9 \\ = 27$$

$$(4) \frac{6}{5} \div \left(-\frac{3}{5}\right) \\ = -2$$

$$(14) \frac{1}{7} \div \frac{6}{7} \\ = \frac{1}{6}$$

$$(24) -\frac{4}{5} \times 2 \\ = -\frac{8}{5}$$

$$(34) \frac{1}{5} \div (-5) \\ = -\frac{1}{25}$$

$$(5) 1 \div (-2) \\ = -\frac{1}{2}$$

$$(15) -1 \div \frac{1}{4} \\ = -4$$

$$(25) \frac{3}{2} \div \frac{1}{2} \\ = 3$$

$$(35) \frac{7}{8} \div \left(-\frac{7}{4}\right) \\ = -\frac{1}{2}$$

$$(6) \frac{9}{5} \times 4 \\ = \frac{36}{5}$$

$$(16) -1 \times \left(-\frac{7}{9}\right) \\ = \frac{7}{9}$$

$$(26) -9 \times \left(-\frac{5}{2}\right) \\ = \frac{45}{2}$$

$$(36) -\frac{7}{2} \div (-4) \\ = \frac{7}{8}$$

$$(7) \frac{4}{9} \div 1 \\ = \frac{4}{9}$$

$$(17) \frac{3}{5} \div \left(-\frac{2}{7}\right) \\ = -\frac{21}{10}$$

$$(27) -\frac{1}{3} \div \frac{4}{9} \\ = -\frac{3}{4}$$

$$(37) \frac{5}{8} \div \frac{3}{4} \\ = \frac{5}{6}$$

$$(8) -\frac{1}{6} \times \left(-\frac{1}{5}\right) \\ = \frac{1}{30}$$

$$(18) 4 \div \frac{1}{9} \\ = 36$$

$$(28) \frac{6}{7} \div \frac{2}{3} \\ = \frac{9}{7}$$

$$(38) -1 \div 3 \\ = -\frac{1}{3}$$

$$(9) -\frac{10}{3} \div (-3) \\ = \frac{10}{9}$$

$$(19) \frac{5}{8} \times \left(-\frac{1}{2}\right) \\ = -\frac{5}{16}$$

$$(29) \frac{3}{8} \div \frac{3}{8} \\ = 1$$

$$(39) -\frac{3}{2} \div \frac{1}{8} \\ = -12$$

$$(10) -\frac{5}{6} \div (-3) \\ = \frac{5}{18}$$

$$(20) 3 \div \frac{8}{5} \\ = \frac{15}{8}$$

$$(30) -\frac{5}{4} \div \frac{1}{3} \\ = -\frac{15}{4}$$

$$(40) 7 \times 1 \\ = 7$$

$$(1) \frac{4}{3} \times (-1) \\ = -\frac{4}{3}$$

$$(11) 1 \div \left(-\frac{1}{8}\right) \\ = -8$$

$$(21) \frac{5}{3} \times \frac{7}{4} \\ = \frac{35}{12}$$

$$(31) -\frac{9}{2} \times \frac{9}{7} \\ = -\frac{81}{14}$$

$$(2) \frac{3}{4} \div 1 \\ = \frac{3}{4}$$

$$(12) 1 \div \frac{2}{3} \\ = \frac{3}{2}$$

$$(22) -\frac{5}{2} \div \left(-\frac{3}{4}\right) \\ = \frac{10}{3}$$

$$(32) -\frac{8}{5} \div \left(-\frac{1}{4}\right) \\ = \frac{32}{5}$$

$$(3) \frac{5}{7} \times \left(-\frac{5}{3}\right) \\ = -\frac{25}{21}$$

$$(13) \frac{1}{4} \div 1 \\ = \frac{1}{4}$$

$$(23) -5 \div 1 \\ = -5$$

$$(33) \frac{9}{2} \times 1 \\ = \frac{9}{2}$$

$$(4) -\frac{1}{2} \div \frac{3}{4} \\ = -\frac{2}{3}$$

$$(14) \frac{5}{4} \times \left(-\frac{7}{2}\right) \\ = -\frac{35}{8}$$

$$(24) -\frac{5}{4} \div \frac{3}{2} \\ = -\frac{5}{6}$$

$$(34) 1 \times \left(-\frac{6}{5}\right) \\ = -\frac{6}{5}$$

$$(5) -1 \div \frac{4}{3} \\ = -\frac{3}{4}$$

$$(15) \frac{5}{4} \div \left(-\frac{3}{5}\right) \\ = -\frac{25}{12}$$

$$(25) -\frac{7}{10} \times \left(-\frac{3}{8}\right) \\ = \frac{21}{80}$$

$$(35) -\frac{4}{5} \times \left(-\frac{3}{5}\right) \\ = \frac{12}{25}$$

$$(6) \frac{5}{8} \times \frac{5}{2} \\ = \frac{25}{16}$$

$$(16) -\frac{2}{3} \div \frac{1}{8} \\ = -\frac{16}{3}$$

$$(26) -2 \div \left(-\frac{1}{4}\right) \\ = 8$$

$$(36) \frac{1}{2} \div \frac{7}{4} \\ = \frac{2}{7}$$

$$(7) 4 \times \left(-\frac{6}{7}\right) \\ = -\frac{24}{7}$$

$$(17) -\frac{1}{2} \times \frac{7}{2} \\ = -\frac{7}{4}$$

$$(27) -\frac{1}{4} \times (-1) \\ = \frac{1}{4}$$

$$(37) -\frac{7}{6} \div \left(-\frac{5}{8}\right) \\ = \frac{28}{15}$$

$$(8) -\frac{9}{2} \div \frac{2}{5} \\ = -\frac{45}{4}$$

$$(18) 1 \times \left(-\frac{9}{7}\right) \\ = -\frac{9}{7}$$

$$(28) \frac{5}{3} \times \left(-\frac{8}{5}\right) \\ = -\frac{8}{3}$$

$$(38) \frac{4}{3} \times (-5) \\ = -\frac{20}{3}$$

$$(9) 9 \times 1 \\ = 9$$

$$(19) \frac{4}{5} \times \frac{6}{7} \\ = \frac{24}{35}$$

$$(29) -\frac{10}{3} \div (-3) \\ = \frac{10}{9}$$

$$(39) 9 \div \frac{9}{8} \\ = 8$$

$$(10) \frac{4}{3} \div 1 \\ = \frac{4}{3}$$

$$(20) -1 \div (-2) \\ = \frac{1}{2}$$

$$(30) -\frac{3}{2} \div \frac{3}{5} \\ = -\frac{5}{2}$$

$$(40) -\frac{1}{4} \times \frac{4}{9} \\ = -\frac{1}{9}$$

$$(1) -\frac{9}{8} \times \left(-\frac{8}{3}\right) \\ = 3$$

$$(11) -7 \div \left(-\frac{8}{7}\right) \\ = \frac{49}{8}$$

$$(21) -\frac{4}{5} \times 3 \\ = -\frac{12}{5}$$

$$(31) \frac{4}{3} \div \frac{10}{7} \\ = \frac{14}{15}$$

$$(2) 7 \div 1 \\ = 7$$

$$(12) -\frac{5}{4} \times \frac{3}{5} \\ = -\frac{3}{4}$$

$$(22) \frac{8}{7} \div \frac{3}{5} \\ = \frac{40}{21}$$

$$(32) \frac{10}{7} \div \frac{7}{10} \\ = \frac{100}{49}$$

$$(3) -1 \div 5 \\ = -\frac{1}{5}$$

$$(13) -\frac{5}{2} \times \frac{1}{2} \\ = -\frac{5}{4}$$

$$(23) \frac{5}{8} \times \frac{1}{2} \\ = \frac{5}{16}$$

$$(33) 1 \div \frac{9}{2} \\ = \frac{2}{9}$$

$$(4) -2 \times 1 \\ = -2$$

$$(14) \frac{7}{9} \div 3 \\ = \frac{7}{27}$$

$$(24) -2 \div \frac{2}{5} \\ = -5$$

$$(34) -9 \times \frac{3}{10} \\ = -\frac{27}{10}$$

$$(5) -\frac{2}{3} \div \frac{7}{3} \\ = -\frac{2}{7}$$

$$(15) \frac{5}{4} \times \frac{1}{3} \\ = \frac{5}{12}$$

$$(25) -\frac{9}{5} \div \frac{5}{2} \\ = -\frac{18}{25}$$

$$(35) -\frac{5}{7} \div \frac{9}{10} \\ = -\frac{50}{63}$$

$$(6) \frac{1}{6} \times \left(-\frac{3}{10}\right) \\ = -\frac{1}{20}$$

$$(16) \frac{9}{10} \div \left(-\frac{1}{9}\right) \\ = -\frac{81}{10}$$

$$(26) -8 \div \left(-\frac{1}{6}\right) \\ = 48$$

$$(36) \frac{9}{4} \times \left(-\frac{9}{8}\right) \\ = -\frac{81}{32}$$

$$(7) -\frac{7}{10} \times \frac{1}{5} \\ = -\frac{7}{50}$$

$$(17) -\frac{9}{8} \div 5 \\ = -\frac{9}{40}$$

$$(27) -\frac{4}{5} \div \left(-\frac{3}{5}\right) \\ = \frac{4}{3}$$

$$(37) \frac{9}{10} \div \left(-\frac{3}{8}\right) \\ = -\frac{12}{5}$$

$$(8) \frac{5}{9} \times 1 \\ = \frac{5}{9}$$

$$(18) \frac{5}{7} \div 3 \\ = \frac{5}{21}$$

$$(28) -\frac{1}{10} \times 2 \\ = -\frac{1}{5}$$

$$(38) -3 \div \frac{9}{5} \\ = -\frac{5}{3}$$

$$(9) \frac{4}{5} \div (-3) \\ = -\frac{4}{15}$$

$$(19) -\frac{1}{10} \times \frac{3}{5} \\ = -\frac{3}{50}$$

$$(29) -\frac{7}{6} \div \left(-\frac{7}{4}\right) \\ = \frac{2}{3}$$

$$(39) \frac{5}{7} \div \left(-\frac{5}{3}\right) \\ = -\frac{3}{7}$$

$$(10) 2 \div \frac{4}{3} \\ = \frac{3}{2}$$

$$(20) -2 \div \frac{1}{7} \\ = -14$$

$$(30) -\frac{5}{2} \times \frac{5}{4} \\ = -\frac{25}{8}$$

$$(40) -\frac{6}{5} \times \left(-\frac{10}{9}\right) \\ = \frac{4}{3}$$

$$(1) -2 \div \frac{4}{3} \\ = -\frac{3}{2}$$

$$(11) -2 \times \left(-\frac{8}{9}\right) \\ = \frac{16}{9}$$

$$(21) -7 \div \frac{5}{4} \\ = -\frac{28}{5}$$

$$(31) 2 \times \left(-\frac{7}{2}\right) \\ = -7$$

$$(2) \frac{2}{9} \div 2 \\ = \frac{1}{9}$$

$$(12) 4 \times 4 \\ = 16$$

$$(22) -\frac{1}{6} \times (-1) \\ = \frac{1}{6}$$

$$(32) \frac{1}{3} \div \left(-\frac{1}{2}\right) \\ = -\frac{2}{3}$$

$$(3) \frac{7}{2} \div \frac{5}{8} \\ = \frac{28}{5}$$

$$(13) \frac{4}{3} \div \left(-\frac{5}{6}\right) \\ = -\frac{8}{5}$$

$$(23) \frac{1}{2} \times (-1) \\ = -\frac{1}{2}$$

$$(33) -\frac{2}{5} \times \left(-\frac{7}{10}\right) \\ = \frac{7}{25}$$

$$(4) \frac{5}{6} \times \frac{5}{4} \\ = \frac{25}{24}$$

$$(14) \frac{7}{5} \div 2 \\ = \frac{7}{10}$$

$$(24) 8 \div \left(-\frac{1}{6}\right) \\ = -48$$

$$(34) -\frac{5}{9} \times \left(-\frac{7}{4}\right) \\ = \frac{35}{36}$$

$$(5) 3 \div (-6) \\ = -\frac{1}{2}$$

$$(15) 5 \times \frac{7}{3} \\ = \frac{35}{3}$$

$$(25) \frac{10}{9} \times 1 \\ = \frac{10}{9}$$

$$(35) -\frac{8}{3} \times \frac{3}{5} \\ = -\frac{8}{5}$$

$$(6) 1 \times (-1) \\ = -1$$

$$(16) \frac{1}{5} \times \frac{6}{7} \\ = \frac{6}{35}$$

$$(26) \frac{1}{5} \div \left(-\frac{9}{4}\right) \\ = -\frac{4}{45}$$

$$(36) 1 \div \left(-\frac{7}{3}\right) \\ = -\frac{3}{7}$$

$$(7) \frac{3}{4} \times \frac{3}{8} \\ = \frac{9}{32}$$

$$(17) -\frac{7}{5} \times (-2) \\ = \frac{14}{5}$$

$$(27) -2 \div 4 \\ = -\frac{1}{2}$$

$$(37) -\frac{4}{9} \times \frac{5}{9} \\ = -\frac{20}{81}$$

$$(8) -\frac{10}{9} \div \left(-\frac{4}{5}\right) \\ = \frac{25}{18}$$

$$(18) \frac{7}{9} \times \left(-\frac{9}{2}\right) \\ = -\frac{7}{2}$$

$$(28) \frac{9}{4} \times 1 \\ = \frac{9}{4}$$

$$(38) -\frac{2}{5} \div 2 \\ = -\frac{1}{5}$$

$$(9) -\frac{7}{2} \times \left(-\frac{4}{3}\right) \\ = \frac{14}{3}$$

$$(19) -\frac{4}{5} \times (-1) \\ = \frac{4}{5}$$

$$(29) -1 \times 3 \\ = -3$$

$$(39) \frac{8}{5} \div \left(-\frac{3}{8}\right) \\ = -\frac{64}{15}$$

$$(10) 1 \times \frac{1}{2} \\ = \frac{1}{2}$$

$$(20) -\frac{2}{9} \div \frac{4}{5} \\ = -\frac{5}{18}$$

$$(30) -2 \div \left(-\frac{6}{7}\right) \\ = \frac{7}{3}$$

$$(40) \frac{4}{3} \div (-10) \\ = -\frac{2}{15}$$

$$(1) -\frac{9}{10} \times 6$$

$$= -\frac{27}{5}$$

$$(11) -\frac{7}{10} \div \frac{1}{5}$$

$$= -\frac{7}{2}$$

$$(21) \frac{1}{9} \div (-3)$$

$$= -\frac{1}{27}$$

$$(31) 2 \div \frac{7}{6}$$

$$= \frac{12}{7}$$

$$(2) -\frac{7}{3} \times \frac{5}{4}$$

$$= -\frac{35}{12}$$

$$(12) -\frac{6}{5} \div \frac{10}{9}$$

$$= -\frac{27}{25}$$

$$(22) -\frac{1}{2} \times \left(-\frac{8}{3}\right)$$

$$= \frac{4}{3}$$

$$(32) \frac{1}{3} \div \frac{10}{9}$$

$$= \frac{3}{10}$$

$$(3) -\frac{9}{10} \times \frac{7}{5}$$

$$= -\frac{63}{50}$$

$$(13) -\frac{4}{5} \times \left(-\frac{2}{3}\right)$$

$$= \frac{8}{15}$$

$$(23) \frac{8}{7} \times \left(-\frac{3}{10}\right)$$

$$= -\frac{12}{35}$$

$$(33) 1 \times \frac{7}{5}$$

$$= \frac{7}{5}$$

$$(4) \frac{2}{5} \div \frac{3}{2}$$

$$= \frac{4}{15}$$

$$(14) \frac{9}{4} \times \left(-\frac{1}{8}\right)$$

$$= -\frac{9}{32}$$

$$(24) -\frac{1}{3} \times \frac{1}{5}$$

$$= -\frac{1}{15}$$

$$(34) 1 \times \frac{4}{5}$$

$$= \frac{4}{5}$$

$$(5) -\frac{10}{7} \div 2$$

$$= -\frac{5}{7}$$

$$(15) \frac{10}{9} \times (-1)$$

$$= -\frac{10}{9}$$

$$(25) 1 \div \frac{3}{2}$$

$$= \frac{2}{3}$$

$$(35) -\frac{3}{4} \times \left(-\frac{1}{4}\right)$$

$$= \frac{3}{16}$$

$$(6) \frac{4}{3} \times \left(-\frac{1}{4}\right)$$

$$= -\frac{1}{3}$$

$$(16) -2 \div \left(-\frac{9}{4}\right)$$

$$= \frac{8}{9}$$

$$(26) \frac{1}{5} \times \frac{1}{4}$$

$$= \frac{1}{20}$$

$$(36) \frac{2}{3} \times \left(-\frac{2}{9}\right)$$

$$= -\frac{4}{27}$$

$$(7) -6 \div (-8)$$

$$= \frac{3}{4}$$

$$(17) \frac{7}{2} \div \frac{5}{6}$$

$$= \frac{21}{5}$$

$$(27) -\frac{1}{7} \div \frac{6}{7}$$

$$= -\frac{1}{6}$$

$$(37) -1 \div \left(-\frac{10}{9}\right)$$

$$= \frac{9}{10}$$

$$(8) -\frac{5}{2} \div \frac{9}{2}$$

$$= -\frac{5}{9}$$

$$(18) -\frac{9}{10} \times \frac{7}{2}$$

$$= -\frac{63}{20}$$

$$(28) \frac{1}{3} \times \frac{4}{5}$$

$$= \frac{4}{15}$$

$$(38) -1 \times (-1)$$

$$= 1$$

$$(9) \frac{7}{8} \div \frac{9}{7}$$

$$= \frac{49}{72}$$

$$(19) -2 \div \left(-\frac{8}{9}\right)$$

$$= \frac{9}{4}$$

$$(29) \frac{8}{9} \times (-4)$$

$$= -\frac{32}{9}$$

$$(39) \frac{5}{8} \div 3$$

$$= \frac{5}{24}$$

$$(10) -\frac{2}{9} \times \left(-\frac{5}{4}\right)$$

$$= \frac{5}{18}$$

$$(20) -3 \div 5$$

$$= -\frac{3}{5}$$

$$(30) \frac{7}{9} \div \frac{4}{5}$$

$$= \frac{35}{36}$$

$$(40) -\frac{9}{7} \div \frac{3}{7}$$

$$= -3$$

$$(1) \frac{1}{4} \div 5 \\ = \frac{1}{20}$$

$$(11) \frac{1}{2} \times \frac{1}{9} \\ = \frac{1}{18}$$

$$(21) \frac{4}{3} \div \left(-\frac{1}{2}\right) \\ = -\frac{8}{3}$$

$$(31) -2 \times \left(-\frac{1}{2}\right) \\ = 1$$

$$(2) -\frac{8}{3} \times 1 \\ = -\frac{8}{3}$$

$$(12) 1 \div (-4) \\ = -\frac{1}{4}$$

$$(22) 2 \times \left(-\frac{8}{5}\right) \\ = -\frac{16}{5}$$

$$(32) \frac{4}{7} \times \left(-\frac{2}{3}\right) \\ = -\frac{8}{21}$$

$$(3) \frac{1}{2} \div 2 \\ = \frac{1}{4}$$

$$(13) \frac{3}{8} \times \left(-\frac{8}{5}\right) \\ = -\frac{3}{5}$$

$$(23) \frac{1}{9} \times \frac{8}{7} \\ = \frac{8}{63}$$

$$(33) -1 \div \left(-\frac{10}{9}\right) \\ = \frac{9}{10}$$

$$(4) -9 \times \left(-\frac{6}{7}\right) \\ = \frac{54}{7}$$

$$(14) 2 \times \frac{9}{4} \\ = \frac{9}{2}$$

$$(24) -\frac{8}{9} \div \left(-\frac{7}{2}\right) \\ = \frac{16}{63}$$

$$(34) -\frac{2}{5} \div \left(-\frac{3}{4}\right) \\ = \frac{8}{15}$$

$$(5) 3 \times \frac{1}{8} \\ = \frac{3}{8}$$

$$(15) \frac{3}{5} \div \frac{4}{4} \\ = \frac{4}{15}$$

$$(25) \frac{7}{8} \div \left(-\frac{4}{7}\right) \\ = -\frac{49}{32}$$

$$(35) 1 \times \left(-\frac{4}{3}\right) \\ = -\frac{4}{3}$$

$$(6) \frac{7}{3} \div \frac{5}{7} \\ = \frac{49}{15}$$

$$(16) -\frac{3}{5} \div 4 \\ = -\frac{3}{20}$$

$$(26) -\frac{3}{5} \div 1 \\ = -\frac{3}{5}$$

$$(36) -\frac{8}{7} \times 2 \\ = -\frac{16}{7}$$

$$(7) -\frac{2}{3} \div \frac{7}{5} \\ = -\frac{10}{21}$$

$$(17) -\frac{10}{7} \times \frac{2}{3} \\ = -\frac{20}{21}$$

$$(27) \frac{4}{5} \div \frac{7}{6} \\ = \frac{24}{35}$$

$$(37) \frac{1}{3} \div \left(-\frac{7}{4}\right) \\ = -\frac{4}{21}$$

$$(8) -1 \times \frac{1}{2} \\ = -\frac{1}{2}$$

$$(18) \frac{1}{2} \div (-1) \\ = -\frac{1}{2}$$

$$(28) -\frac{1}{6} \div \frac{9}{10} \\ = -\frac{5}{27}$$

$$(38) \frac{4}{9} \times \left(-\frac{7}{10}\right) \\ = -\frac{14}{45}$$

$$(9) \frac{2}{5} \div \frac{5}{8} \\ = \frac{16}{25}$$

$$(19) 10 \times \left(-\frac{9}{8}\right) \\ = -\frac{45}{4}$$

$$(29) \frac{3}{5} \div \left(-\frac{8}{7}\right) \\ = -\frac{21}{40}$$

$$(39) -\frac{4}{7} \times 5 \\ = -\frac{20}{7}$$

$$(10) -\frac{3}{4} \times \frac{1}{9} \\ = -\frac{1}{12}$$

$$(20) -\frac{1}{3} \times \left(-\frac{7}{10}\right) \\ = \frac{7}{30}$$

$$(30) -\frac{2}{3} \div 4 \\ = -\frac{1}{6}$$

$$(40) \frac{10}{7} \div \frac{7}{3} \\ = \frac{30}{49}$$

$$(1) -\frac{3}{2} \times 2 \\ = -3$$

$$(11) \frac{8}{5} \times 3 \\ = \frac{24}{5}$$

$$(21) \frac{4}{5} \times \left(-\frac{7}{9}\right) \\ = -\frac{28}{45}$$

$$(31) 10 \div 10 \\ = 1$$

$$(2) -\frac{4}{3} \div \left(-\frac{1}{2}\right) \\ = \frac{8}{3}$$

$$(12) \frac{1}{3} \times \left(-\frac{9}{4}\right) \\ = -\frac{3}{4}$$

$$(22) -\frac{3}{10} \times (-5) \\ = \frac{3}{2}$$

$$(32) -\frac{3}{10} \div \frac{5}{2} \\ = -\frac{3}{25}$$

$$(3) \frac{8}{9} \times \left(-\frac{8}{5}\right) \\ = -\frac{64}{45}$$

$$(13) \frac{4}{7} \div \frac{4}{3} \\ = \frac{3}{7}$$

$$(23) -1 \div \frac{3}{8} \\ = -\frac{8}{3}$$

$$(33) \frac{5}{9} \times \left(-\frac{3}{2}\right) \\ = -\frac{5}{6}$$

$$(4) -\frac{5}{7} \div (-6) \\ = \frac{5}{42}$$

$$(14) \frac{1}{5} \times (-4) \\ = -\frac{4}{5}$$

$$(24) -\frac{1}{3} \times \left(-\frac{8}{3}\right) \\ = \frac{8}{9}$$

$$(34) -\frac{5}{7} \times \frac{9}{7} \\ = -\frac{45}{49}$$

$$(5) -2 \div \frac{3}{2} \\ = -\frac{4}{3}$$

$$(15) -4 \div \left(-\frac{3}{4}\right) \\ = \frac{16}{3}$$

$$(25) -6 \times \frac{1}{5} \\ = -\frac{6}{5}$$

$$(35) -\frac{1}{5} \div 5 \\ = -\frac{1}{25}$$

$$(6) \frac{3}{2} \div \frac{2}{3} \\ = \frac{9}{4}$$

$$(16) -1 \div \frac{7}{10} \\ = -\frac{10}{7}$$

$$(26) \frac{4}{3} \times \frac{5}{3} \\ = \frac{20}{9}$$

$$(36) -2 \div \left(-\frac{1}{4}\right) \\ = 8$$

$$(7) -\frac{7}{10} \div (-8) \\ = \frac{7}{80}$$

$$(17) -\frac{6}{5} \div \left(-\frac{9}{10}\right) \\ = \frac{4}{3}$$

$$(27) 10 \times \frac{4}{3} \\ = \frac{40}{3}$$

$$(37) -2 \div \left(-\frac{8}{3}\right) \\ = \frac{3}{4}$$

$$(8) -\frac{1}{7} \times \frac{7}{9} \\ = -\frac{1}{9}$$

$$(18) \frac{2}{5} \div 1 \\ = \frac{2}{5}$$

$$(28) \frac{4}{9} \times \frac{8}{3} \\ = \frac{32}{27}$$

$$(38) -\frac{1}{6} \div \left(-\frac{1}{5}\right) \\ = \frac{5}{6}$$

$$(9) -\frac{10}{7} \times \left(-\frac{1}{4}\right) \\ = \frac{5}{14}$$

$$(19) -\frac{7}{3} \times \left(-\frac{5}{7}\right) \\ = \frac{5}{3}$$

$$(29) -1 \times \left(-\frac{1}{10}\right) \\ = \frac{1}{10}$$

$$(39) -\frac{4}{5} \times \frac{5}{3} \\ = -\frac{4}{3}$$

$$(10) \frac{5}{3} \times \frac{7}{8} \\ = \frac{35}{24}$$

$$(20) -\frac{1}{2} \times \frac{9}{4} \\ = -\frac{9}{8}$$

$$(30) -\frac{6}{7} \div 8 \\ = -\frac{3}{28}$$

$$(40) -\frac{3}{7} \times \frac{7}{9} \\ = -\frac{1}{3}$$

$$(1) -2 \div \frac{3}{2}$$

$$= -\frac{4}{3}$$

$$(11) 1 \div \left(-\frac{5}{3}\right)$$

$$= -\frac{3}{5}$$

$$(21) 1 \times \frac{9}{8}$$

$$= \frac{9}{8}$$

$$(31) \frac{1}{7} \div 1$$

$$= \frac{1}{7}$$

$$(2) \frac{3}{5} \div \frac{1}{7}$$

$$= \frac{21}{5}$$

$$(12) -\frac{3}{10} \div \left(-\frac{5}{2}\right)$$

$$= \frac{3}{25}$$

$$(22) \frac{4}{5} \times \frac{5}{4}$$

$$= 1$$

$$(32) 1 \div \frac{4}{3}$$

$$= \frac{3}{4}$$

$$(3) -\frac{3}{5} \times (-1)$$

$$= \frac{3}{5}$$

$$(13) 7 \div \left(-\frac{3}{2}\right)$$

$$= -\frac{14}{3}$$

$$(23) \frac{3}{2} \div \left(-\frac{5}{4}\right)$$

$$= -\frac{6}{5}$$

$$(33) -\frac{9}{2} \times \frac{3}{8}$$

$$= -\frac{27}{16}$$

$$(4) 1 \div \left(-\frac{8}{3}\right)$$

$$= -\frac{3}{8}$$

$$(14) -1 \times \left(-\frac{7}{5}\right)$$

$$= \frac{7}{5}$$

$$(24) 4 \times \frac{5}{3}$$

$$= \frac{20}{3}$$

$$(34) -\frac{9}{5} \times (-2)$$

$$= \frac{18}{5}$$

$$(5) \frac{1}{4} \div \frac{1}{4}$$

$$= 1$$

$$(15) -8 \div 1$$

$$= -8$$

$$(25) 2 \times 4$$

$$= 8$$

$$(35) \frac{4}{9} \times \frac{1}{4}$$

$$= \frac{1}{9}$$

$$(6) -4 \times \left(-\frac{3}{4}\right)$$

$$= 3$$

$$(16) 3 \div \frac{7}{3}$$

$$= \frac{9}{7}$$

$$(26) -\frac{2}{5} \times \left(-\frac{2}{5}\right)$$

$$= \frac{4}{25}$$

$$(36) \frac{3}{10} \div \frac{3}{5}$$

$$= \frac{1}{2}$$

$$(7) \frac{2}{7} \div \left(-\frac{3}{7}\right)$$

$$= -\frac{2}{3}$$

$$(17) \frac{3}{5} \times \left(-\frac{7}{10}\right)$$

$$= -\frac{21}{50}$$

$$(27) -\frac{4}{7} \div \left(-\frac{5}{3}\right)$$

$$= \frac{12}{35}$$

$$(37) -\frac{9}{10} \times 1$$

$$= -\frac{9}{10}$$

$$(8) -\frac{6}{5} \times \frac{3}{4}$$

$$= -\frac{9}{10}$$

$$(18) -\frac{7}{9} \times \frac{9}{5}$$

$$= -\frac{7}{5}$$

$$(28) \frac{5}{3} \div \left(-\frac{2}{3}\right)$$

$$= -\frac{5}{2}$$

$$(38) \frac{10}{7} \div \frac{1}{6}$$

$$= \frac{60}{7}$$

$$(9) \frac{1}{9} \times \frac{9}{5}$$

$$= \frac{1}{5}$$

$$(19) 1 \times \frac{3}{2}$$

$$= \frac{3}{2}$$

$$(29) 1 \div \frac{9}{10}$$

$$= \frac{10}{9}$$

$$(39) \frac{1}{3} \div \frac{1}{4}$$

$$= \frac{4}{3}$$

$$(10) -\frac{1}{2} \div \left(-\frac{3}{2}\right)$$

$$= \frac{1}{3}$$

$$(20) -\frac{1}{3} \times (-4)$$

$$= \frac{4}{3}$$

$$(30) 2 \div \frac{4}{9}$$

$$= \frac{9}{2}$$

$$(40) \frac{3}{2} \times \left(-\frac{1}{5}\right)$$

$$= -\frac{3}{10}$$

$$(1) -\frac{2}{3} \times (-2) \\ = \frac{4}{3}$$

$$(11) -\frac{7}{9} \times \left(-\frac{4}{7}\right) \\ = \frac{4}{9}$$

$$(21) -\frac{2}{9} \times \left(-\frac{3}{5}\right) \\ = \frac{2}{15}$$

$$(31) \frac{9}{10} \times \left(-\frac{1}{2}\right) \\ = -\frac{9}{20}$$

$$(2) -\frac{5}{9} \div \left(-\frac{9}{5}\right) \\ = \frac{25}{81}$$

$$(12) -\frac{1}{2} \times \frac{1}{10} \\ = -\frac{1}{20}$$

$$(22) -\frac{7}{6} \div \left(-\frac{9}{4}\right) \\ = \frac{14}{27}$$

$$(32) \frac{10}{3} \times \frac{9}{7} \\ = \frac{30}{7}$$

$$(3) \frac{7}{4} \times \left(-\frac{5}{4}\right) \\ = -\frac{35}{16}$$

$$(13) \frac{1}{7} \div \frac{10}{3} \\ = \frac{3}{70}$$

$$(23) -\frac{8}{9} \times \left(-\frac{1}{5}\right) \\ = \frac{8}{45}$$

$$(33) -\frac{9}{4} \div \frac{8}{5} \\ = -\frac{45}{32}$$

$$(4) 2 \div \frac{5}{8} \\ = \frac{16}{5}$$

$$(14) 5 \times \left(-\frac{5}{4}\right) \\ = -\frac{25}{4}$$

$$(24) \frac{9}{8} \times \left(-\frac{4}{5}\right) \\ = -\frac{9}{10}$$

$$(34) \frac{5}{3} \div \frac{1}{9} \\ = 15$$

$$(5) -\frac{1}{2} \div (-4) \\ = \frac{1}{8}$$

$$(15) \frac{8}{3} \times \frac{3}{2} \\ = 4$$

$$(25) 4 \div (-1) \\ = -4$$

$$(35) 1 \times \left(-\frac{4}{7}\right) \\ = -\frac{4}{7}$$

$$(6) \frac{1}{2} \div 4 \\ = \frac{1}{8}$$

$$(16) 1 \div \frac{4}{3} \\ = \frac{3}{4}$$

$$(26) -\frac{2}{3} \times 3 \\ = -2$$

$$(36) \frac{3}{8} \div \left(-\frac{7}{3}\right) \\ = -\frac{9}{56}$$

$$(7) \frac{7}{5} \times \frac{3}{8} \\ = \frac{21}{40}$$

$$(17) -1 \times \left(-\frac{4}{5}\right) \\ = \frac{4}{5}$$

$$(27) -1 \div \left(-\frac{1}{9}\right) \\ = 9$$

$$(37) -\frac{3}{5} \div (-7) \\ = \frac{3}{35}$$

$$(8) \frac{4}{3} \times \left(-\frac{4}{5}\right) \\ = -\frac{16}{15}$$

$$(18) -\frac{3}{7} \times \left(-\frac{5}{4}\right) \\ = \frac{15}{28}$$

$$(28) \frac{3}{10} \div \left(-\frac{5}{9}\right) \\ = -\frac{27}{50}$$

$$(38) -\frac{7}{2} \times \left(-\frac{9}{4}\right) \\ = \frac{63}{8}$$

$$(9) -\frac{3}{2} \div \left(-\frac{8}{7}\right) \\ = \frac{21}{16}$$

$$(19) -\frac{1}{6} \div \frac{2}{3} \\ = -\frac{1}{4}$$

$$(29) 3 \div \frac{10}{3} \\ = \frac{9}{10}$$

$$(39) 3 \div (-1) \\ = -3$$

$$(10) \frac{1}{2} \times \left(-\frac{3}{5}\right) \\ = -\frac{3}{10}$$

$$(20) -\frac{1}{3} \div \left(-\frac{1}{2}\right) \\ = \frac{2}{3}$$

$$(30) -\frac{9}{5} \div \left(-\frac{2}{3}\right) \\ = \frac{27}{10}$$

$$(40) -9 \div \frac{1}{10} \\ = -90$$

$$(1) \frac{4}{5} \div (-2) \\ = -\frac{2}{5}$$

$$(11) -\frac{9}{7} \div \left(-\frac{7}{8}\right) \\ = \frac{72}{49}$$

$$(21) -\frac{5}{8} \times \left(-\frac{7}{4}\right) \\ = \frac{35}{32}$$

$$(31) \frac{1}{2} \times \frac{9}{2} \\ = \frac{9}{4}$$

$$(2) \frac{7}{9} \div \frac{5}{4} \\ = \frac{28}{45}$$

$$(12) -\frac{5}{8} \times \frac{3}{2} \\ = -\frac{15}{16}$$

$$(22) -9 \times \left(-\frac{5}{9}\right) \\ = 5$$

$$(32) \frac{1}{4} \times \left(-\frac{5}{3}\right) \\ = -\frac{5}{12}$$

$$(3) -\frac{10}{3} \div \frac{2}{9} \\ = -15$$

$$(13) 5 \div \frac{3}{2} \\ = \frac{10}{3}$$

$$(23) -5 \times \left(-\frac{9}{8}\right) \\ = \frac{45}{8}$$

$$(33) -\frac{9}{8} \div 1 \\ = -\frac{9}{8}$$

$$(4) -\frac{4}{7} \div \left(-\frac{2}{7}\right) \\ = 2$$

$$(14) \frac{5}{3} \div \left(-\frac{3}{4}\right) \\ = -\frac{20}{9}$$

$$(24) -\frac{6}{5} \times \left(-\frac{9}{2}\right) \\ = \frac{27}{5}$$

$$(34) \frac{2}{5} \times \frac{8}{9} \\ = \frac{16}{45}$$

$$(5) \frac{7}{10} \div 1 \\ = \frac{7}{10}$$

$$(15) -\frac{7}{6} \div \frac{2}{5} \\ = -\frac{35}{12}$$

$$(25) \frac{3}{7} \div \frac{6}{7} \\ = \frac{1}{2}$$

$$(35) \frac{3}{7} \div \left(-\frac{1}{3}\right) \\ = -\frac{9}{7}$$

$$(6) -\frac{1}{10} \div \left(-\frac{10}{9}\right) \\ = \frac{9}{100}$$

$$(16) \frac{5}{7} \div \frac{3}{5} \\ = \frac{25}{21}$$

$$(26) -\frac{3}{5} \div \left(-\frac{6}{7}\right) \\ = \frac{7}{10}$$

$$(36) \frac{4}{5} \times \frac{1}{2} \\ = \frac{2}{5}$$

$$(7) \frac{5}{2} \div \frac{2}{9} \\ = \frac{45}{4}$$

$$(17) \frac{3}{10} \div \left(-\frac{6}{5}\right) \\ = -\frac{1}{4}$$

$$(27) -1 \div \frac{7}{8} \\ = -\frac{8}{7}$$

$$(37) -2 \div \frac{3}{2} \\ = -\frac{4}{3}$$

$$(8) \frac{10}{3} \times \left(-\frac{1}{3}\right) \\ = -\frac{10}{9}$$

$$(18) \frac{8}{3} \times \left(-\frac{10}{9}\right) \\ = -\frac{80}{27}$$

$$(28) 1 \div \left(-\frac{6}{5}\right) \\ = -\frac{5}{6}$$

$$(38) \frac{2}{5} \div \frac{1}{6} \\ = \frac{12}{5}$$

$$(9) -\frac{9}{8} \div \left(-\frac{2}{3}\right) \\ = \frac{27}{16}$$

$$(19) \frac{1}{2} \times \left(-\frac{3}{2}\right) \\ = -\frac{3}{4}$$

$$(29) \frac{2}{3} \times \frac{1}{8} \\ = \frac{1}{12}$$

$$(39) -1 \times \frac{4}{3} \\ = -\frac{4}{3}$$

$$(10) -2 \div 2 \\ = -1$$

$$(20) -\frac{1}{7} \div \left(-\frac{2}{9}\right) \\ = \frac{9}{14}$$

$$(30) \frac{5}{2} \times \frac{2}{3} \\ = \frac{5}{3}$$

$$(40) -\frac{1}{2} \div 1 \\ = -\frac{1}{2}$$

$$(1) \frac{3}{5} \times \left(-\frac{3}{10}\right) \\ = -\frac{9}{50}$$

$$(11) \frac{3}{5} \times \left(-\frac{4}{7}\right) \\ = -\frac{12}{35}$$

$$(21) \frac{5}{8} \times \left(-\frac{1}{3}\right) \\ = -\frac{5}{24}$$

$$(31) 10 \times \left(-\frac{2}{5}\right) \\ = -4$$

$$(2) \frac{3}{4} \div \left(-\frac{5}{8}\right) \\ = -\frac{6}{5}$$

$$(12) \frac{10}{7} \times 10 \\ = \frac{100}{7}$$

$$(22) \frac{3}{8} \div \left(-\frac{3}{4}\right) \\ = -\frac{1}{2}$$

$$(32) -\frac{4}{9} \times \frac{7}{6} \\ = -\frac{14}{27}$$

$$(3) -\frac{9}{2} \times (-4) \\ = 18$$

$$(13) \frac{5}{3} \div \left(-\frac{9}{8}\right) \\ = -\frac{40}{27}$$

$$(23) -\frac{10}{3} \div \left(-\frac{9}{10}\right) \\ = \frac{100}{27}$$

$$(33) -\frac{9}{8} \div \frac{3}{2} \\ = -\frac{3}{4}$$

$$(4) 1 \times \left(-\frac{8}{5}\right) \\ = -\frac{8}{5}$$

$$(14) \frac{4}{9} \div \left(-\frac{4}{3}\right) \\ = -\frac{1}{3}$$

$$(24) \frac{3}{2} \times \frac{4}{5} \\ = \frac{6}{5}$$

$$(34) \frac{3}{5} \div \left(-\frac{1}{2}\right) \\ = -\frac{6}{5}$$

$$(5) -3 \times (-2) \\ = 6$$

$$(15) \frac{5}{9} \times \frac{5}{3} \\ = \frac{25}{27}$$

$$(25) \frac{7}{6} \div \frac{5}{2} \\ = \frac{7}{15}$$

$$(35) -1 \times \frac{7}{6} \\ = -\frac{7}{6}$$

$$(6) -\frac{10}{7} \times \frac{9}{10} \\ = -\frac{9}{7}$$

$$(16) -\frac{5}{9} \times \frac{1}{7} \\ = -\frac{5}{63}$$

$$(26) \frac{1}{3} \div \frac{1}{5} \\ = \frac{5}{3}$$

$$(36) \frac{2}{7} \div \left(-\frac{5}{6}\right) \\ = -\frac{12}{35}$$

$$(7) -\frac{8}{3} \div 1 \\ = -\frac{8}{3}$$

$$(17) \frac{4}{3} \times \frac{1}{2} \\ = \frac{2}{3}$$

$$(27) \frac{3}{5} \times \frac{1}{3} \\ = \frac{1}{5}$$

$$(37) \frac{1}{2} \div \frac{6}{7} \\ = \frac{7}{12}$$

$$(8) -4 \times 4 \\ = -16$$

$$(18) 1 \div \left(-\frac{7}{6}\right) \\ = -\frac{6}{7}$$

$$(28) -\frac{5}{2} \times (-1) \\ = \frac{5}{2}$$

$$(38) -\frac{9}{7} \div 1 \\ = -\frac{9}{7}$$

$$(9) -\frac{1}{3} \div 1 \\ = -\frac{1}{3}$$

$$(19) \frac{2}{5} \div 1 \\ = \frac{2}{5}$$

$$(29) -6 \times \left(-\frac{5}{2}\right) \\ = 15$$

$$(39) -\frac{6}{7} \div \frac{3}{4} \\ = -\frac{8}{7}$$

$$(10) -\frac{1}{3} \times (-6) \\ = 2$$

$$(20) -\frac{8}{9} \div \left(-\frac{5}{4}\right) \\ = \frac{32}{45}$$

$$(30) -\frac{7}{3} \times (-4) \\ = \frac{28}{3}$$

$$(40) \frac{3}{5} \div \frac{5}{8} \\ = \frac{24}{25}$$

$$(1) 1 \times \left(-\frac{4}{9}\right) \\ = -\frac{4}{9}$$

$$(11) -\frac{3}{5} \div (-3) \\ = \frac{1}{5}$$

$$(21) 1 \times \frac{7}{4} \\ = \frac{7}{4}$$

$$(31) 2 \times \frac{2}{3} \\ = \frac{4}{3}$$

$$(2) -\frac{4}{7} \div \frac{3}{7} \\ = -\frac{4}{3}$$

$$(12) \frac{9}{8} \div 3 \\ = \frac{3}{8}$$

$$(22) -\frac{4}{5} \times \frac{2}{5} \\ = -\frac{8}{25}$$

$$(32) -9 \times \frac{5}{6} \\ = -\frac{15}{2}$$

$$(3) -2 \times \frac{10}{7} \\ = -\frac{20}{7}$$

$$(13) \frac{7}{3} \div \left(-\frac{9}{4}\right) \\ = -\frac{28}{27}$$

$$(23) -1 \times \left(-\frac{1}{2}\right) \\ = \frac{1}{2}$$

$$(33) -\frac{2}{7} \times \frac{5}{9} \\ = -\frac{10}{63}$$

$$(4) 1 \times (-5) \\ = -5$$

$$(14) -1 \times \left(-\frac{1}{2}\right) \\ = \frac{1}{2}$$

$$(24) -\frac{7}{8} \times (-3) \\ = \frac{21}{8}$$

$$(34) -\frac{3}{2} \times \left(-\frac{5}{9}\right) \\ = \frac{5}{6}$$

$$(5) 1 \div \left(-\frac{9}{8}\right) \\ = -\frac{8}{9}$$

$$(15) 5 \div (-4) \\ = -\frac{5}{4}$$

$$(25) \frac{10}{7} \times \frac{5}{2} \\ = \frac{25}{7}$$

$$(35) -\frac{5}{8} \times \frac{7}{8} \\ = -\frac{35}{64}$$

$$(6) \frac{1}{5} \div \frac{4}{5} \\ = \frac{1}{4}$$

$$(16) \frac{7}{10} \div (-9) \\ = -\frac{7}{90}$$

$$(26) -\frac{4}{9} \times \left(-\frac{1}{4}\right) \\ = \frac{1}{9}$$

$$(36) 5 \times \frac{4}{5} \\ = 4$$

$$(7) -\frac{9}{8} \times \frac{4}{7} \\ = -\frac{9}{14}$$

$$(17) \frac{2}{7} \div \frac{4}{9} \\ = \frac{9}{14}$$

$$(27) -\frac{10}{9} \times (-1) \\ = \frac{10}{9}$$

$$(37) \frac{5}{2} \div \frac{3}{2} \\ = \frac{5}{3}$$

$$(8) -\frac{1}{4} \div \left(-\frac{3}{2}\right) \\ = \frac{1}{6}$$

$$(18) \frac{7}{9} \times 5 \\ = \frac{35}{9}$$

$$(28) -\frac{8}{5} \times \frac{1}{3} \\ = -\frac{8}{15}$$

$$(38) \frac{5}{2} \times \left(-\frac{3}{4}\right) \\ = -\frac{15}{8}$$

$$(9) -\frac{3}{7} \div \left(-\frac{2}{9}\right) \\ = \frac{27}{14}$$

$$(19) \frac{1}{7} \times \frac{8}{3} \\ = \frac{8}{21}$$

$$(29) \frac{8}{7} \times \frac{1}{4} \\ = \frac{2}{7}$$

$$(39) \frac{3}{5} \div \left(-\frac{1}{6}\right) \\ = -\frac{18}{5}$$

$$(10) -\frac{5}{4} \div \frac{9}{7} \\ = -\frac{35}{36}$$

$$(20) \frac{1}{3} \div \left(-\frac{5}{8}\right) \\ = -\frac{8}{15}$$

$$(30) -\frac{5}{3} \times \left(-\frac{5}{3}\right) \\ = \frac{25}{9}$$

$$(40) 1 \div \left(-\frac{4}{5}\right) \\ = -\frac{5}{4}$$

$$(1) \frac{4}{9} \times \left(-\frac{1}{8}\right) \\ = -\frac{1}{18}$$

$$(11) -1 \div \left(-\frac{5}{4}\right) \\ = \frac{4}{5}$$

$$(21) \frac{3}{2} \times \left(-\frac{1}{2}\right) \\ = -\frac{3}{4}$$

$$(31) 7 \times \left(-\frac{1}{2}\right) \\ = -\frac{7}{2}$$

$$(2) 1 \times \left(-\frac{7}{3}\right) \\ = -\frac{7}{3}$$

$$(12) 1 \times \frac{3}{5} \\ = \frac{3}{5}$$

$$(22) -1 \div \frac{2}{5} \\ = -\frac{5}{2}$$

$$(32) \frac{7}{8} \div \frac{8}{5} \\ = \frac{35}{64}$$

$$(3) 9 \div \frac{8}{3} \\ = \frac{27}{8}$$

$$(13) \frac{7}{2} \div \left(-\frac{2}{5}\right) \\ = -\frac{35}{4}$$

$$(23) -\frac{1}{3} \times 4 \\ = -\frac{4}{3}$$

$$(33) -2 \div \left(-\frac{9}{8}\right) \\ = \frac{16}{9}$$

$$(4) -\frac{7}{5} \div 3 \\ = -\frac{7}{15}$$

$$(14) -\frac{4}{5} \times 1 \\ = -\frac{4}{5}$$

$$(24) -\frac{7}{3} \div (-8) \\ = \frac{7}{24}$$

$$(34) -\frac{3}{5} \times \left(-\frac{8}{9}\right) \\ = \frac{8}{15}$$

$$(5) -\frac{5}{8} \div \frac{10}{3} \\ = -\frac{15}{16}$$

$$(15) \frac{9}{8} \times \left(-\frac{9}{4}\right) \\ = -\frac{81}{32}$$

$$(25) -2 \times \left(-\frac{1}{2}\right) \\ = 1$$

$$(35) \frac{5}{4} \times \frac{1}{10} \\ = \frac{1}{8}$$

$$(6) -\frac{5}{4} \times \frac{8}{9} \\ = -\frac{10}{9}$$

$$(16) -6 \div \left(-\frac{5}{2}\right) \\ = \frac{12}{5}$$

$$(26) 1 \div \left(-\frac{7}{4}\right) \\ = -\frac{4}{7}$$

$$(36) -\frac{1}{2} \times \left(-\frac{1}{9}\right) \\ = \frac{1}{18}$$

$$(7) 1 \div 2 \\ = \frac{1}{2}$$

$$(17) \frac{3}{10} \times \left(-\frac{7}{5}\right) \\ = -\frac{21}{50}$$

$$(27) 1 \times \left(-\frac{1}{2}\right) \\ = -\frac{1}{2}$$

$$(37) \frac{3}{2} \div \frac{3}{8} \\ = 4$$

$$(8) \frac{10}{3} \div (-6) \\ = -\frac{5}{9}$$

$$(18) -\frac{9}{5} \div (-3) \\ = \frac{3}{5}$$

$$(28) -\frac{5}{9} \div \frac{4}{7} \\ = -\frac{35}{36}$$

$$(38) \frac{2}{3} \times \left(-\frac{5}{4}\right) \\ = -\frac{5}{6}$$

$$(9) -5 \div \left(-\frac{1}{9}\right) \\ = 45$$

$$(19) -\frac{8}{5} \div (-2) \\ = \frac{4}{5}$$

$$(29) \frac{1}{6} \times \left(-\frac{7}{10}\right) \\ = -\frac{7}{60}$$

$$(39) -\frac{4}{3} \times \left(-\frac{1}{5}\right) \\ = \frac{4}{15}$$

$$(10) -3 \times \frac{3}{7} \\ = -\frac{9}{7}$$

$$(20) \frac{3}{4} \times \left(-\frac{7}{5}\right) \\ = -\frac{21}{20}$$

$$(30) \frac{1}{5} \div \frac{3}{2} \\ = \frac{2}{15}$$

$$(40) -1 \div \frac{1}{5} \\ = -5$$

$$(1) -\frac{10}{7} \times \frac{3}{5} \\ = -\frac{6}{7}$$

$$(11) 8 \times 2 \\ = 16$$

$$(21) -1 \times \left(-\frac{5}{3}\right) \\ = \frac{5}{3}$$

$$(31) -\frac{1}{4} \times \left(-\frac{9}{2}\right) \\ = \frac{9}{8}$$

$$(2) \frac{8}{3} \times \left(-\frac{3}{2}\right) \\ = -4$$

$$(12) -\frac{2}{3} \times \left(-\frac{1}{7}\right) \\ = \frac{2}{21}$$

$$(22) -\frac{10}{7} \times \left(-\frac{10}{7}\right) \\ = \frac{100}{49}$$

$$(32) \frac{4}{5} \times \frac{7}{6} \\ = \frac{14}{15}$$

$$(3) -\frac{5}{3} \div (-1) \\ = \frac{5}{3}$$

$$(13) -\frac{2}{3} \div \left(-\frac{7}{10}\right) \\ = \frac{20}{21}$$

$$(23) -\frac{5}{9} \times \frac{2}{3} \\ = -\frac{10}{27}$$

$$(33) 1 \div \left(-\frac{3}{4}\right) \\ = -\frac{4}{3}$$

$$(4) 3 \div \left(-\frac{5}{2}\right) \\ = -\frac{6}{5}$$

$$(14) -\frac{7}{2} \times \frac{7}{2} \\ = -\frac{49}{4}$$

$$(24) -\frac{9}{4} \div 1 \\ = -\frac{9}{4}$$

$$(34) 3 \div \left(-\frac{3}{5}\right) \\ = -5$$

$$(5) 2 \times \left(-\frac{9}{5}\right) \\ = -\frac{18}{5}$$

$$(15) \frac{5}{2} \times (-3) \\ = -\frac{15}{2}$$

$$(25) \frac{2}{9} \times \frac{1}{9} \\ = \frac{2}{81}$$

$$(35) 1 \times \left(-\frac{8}{9}\right) \\ = -\frac{8}{9}$$

$$(6) \frac{3}{8} \times \frac{10}{9} \\ = \frac{5}{12}$$

$$(16) \frac{1}{7} \div (-1) \\ = -\frac{1}{7}$$

$$(26) \frac{1}{4} \times (-2) \\ = -\frac{1}{2}$$

$$(36) -\frac{10}{3} \div (-1) \\ = \frac{10}{3}$$

$$(7) 1 \div \frac{2}{3} \\ = \frac{3}{2}$$

$$(17) -\frac{2}{9} \times (-2) \\ = \frac{4}{9}$$

$$(27) \frac{8}{7} \div \frac{7}{6} \\ = \frac{48}{49}$$

$$(37) \frac{9}{4} \div \left(-\frac{9}{10}\right) \\ = -\frac{5}{2}$$

$$(8) -1 \div \frac{3}{5} \\ = -\frac{5}{3}$$

$$(18) \frac{1}{9} \times (-1) \\ = -\frac{1}{9}$$

$$(28) \frac{5}{4} \div (-1) \\ = -\frac{5}{4}$$

$$(38) -\frac{5}{2} \times \frac{5}{2} \\ = -\frac{25}{4}$$

$$(9) -3 \div \frac{3}{2} \\ = -2$$

$$(19) \frac{1}{7} \times \left(-\frac{3}{4}\right) \\ = -\frac{3}{28}$$

$$(29) 8 \div (-3) \\ = -\frac{8}{3}$$

$$(39) 4 \times \frac{1}{2} \\ = 2$$

$$(10) -\frac{4}{5} \times \frac{3}{4} \\ = -\frac{3}{5}$$

$$(20) \frac{5}{2} \div \frac{6}{7} \\ = \frac{35}{12}$$

$$(30) 1 \times \left(-\frac{2}{7}\right) \\ = -\frac{2}{7}$$

$$(40) \frac{7}{10} \times \frac{6}{5} \\ = \frac{21}{25}$$

$$(1) -\frac{9}{7} \div \frac{1}{9}$$

$$= -\frac{81}{7}$$

$$(11) -3 \div \frac{2}{9}$$

$$= -\frac{27}{2}$$

$$(21) \frac{5}{3} \div \frac{1}{2}$$

$$= \frac{10}{3}$$

$$(31) -\frac{2}{9} \times \frac{7}{9}$$

$$= -\frac{14}{81}$$

$$(2) \frac{8}{9} \times 2$$

$$= \frac{16}{9}$$

$$(12) -\frac{3}{2} \div \frac{3}{10}$$

$$= -5$$

$$(22) \frac{1}{8} \div \left(-\frac{5}{2}\right)$$

$$= -\frac{1}{20}$$

$$(32) \frac{3}{5} \div \frac{10}{9}$$

$$= \frac{27}{50}$$

$$(3) \frac{3}{2} \div \left(-\frac{7}{2}\right)$$

$$= -\frac{3}{7}$$

$$(13) \frac{1}{3} \times \left(-\frac{2}{3}\right)$$

$$= -\frac{2}{9}$$

$$(23) -\frac{5}{3} \div \frac{10}{9}$$

$$= -\frac{3}{2}$$

$$(33) 1 \div \left(-\frac{1}{2}\right)$$

$$= -2$$

$$(4) \frac{1}{2} \times \left(-\frac{2}{9}\right)$$

$$= -\frac{1}{9}$$

$$(14) \frac{5}{9} \div \left(-\frac{4}{3}\right)$$

$$= -\frac{5}{12}$$

$$(24) -\frac{5}{3} \times \frac{7}{4}$$

$$= -\frac{35}{12}$$

$$(34) -\frac{1}{6} \div 3$$

$$= -\frac{1}{18}$$

$$(5) -9 \times (-1)$$

$$= 9$$

$$(15) -\frac{2}{5} \times (-3)$$

$$= \frac{6}{5}$$

$$(25) -1 \times \frac{1}{4}$$

$$= -\frac{1}{4}$$

$$(35) -\frac{3}{5} \div \left(-\frac{1}{10}\right)$$

$$= 6$$

$$(6) \frac{9}{10} \div \frac{2}{3}$$

$$= \frac{27}{20}$$

$$(16) -\frac{4}{5} \times (-3)$$

$$= \frac{12}{5}$$

$$(26) \frac{3}{4} \div \frac{9}{5}$$

$$= \frac{5}{12}$$

$$(36) -\frac{1}{5} \times \left(-\frac{10}{9}\right)$$

$$= \frac{2}{9}$$

$$(7) \frac{1}{7} \div 2$$

$$= \frac{1}{14}$$

$$(17) -\frac{8}{3} \div \frac{1}{9}$$

$$= -24$$

$$(27) -\frac{1}{9} \times 9$$

$$= -1$$

$$(37) \frac{4}{5} \times \left(-\frac{2}{7}\right)$$

$$= -\frac{8}{35}$$

$$(8) -\frac{4}{5} \times 1$$

$$= -\frac{4}{5}$$

$$(18) -\frac{2}{2} \div (-3)$$

$$= \frac{3}{3}$$

$$(28) \frac{1}{8} \times \frac{2}{9}$$

$$= \frac{1}{36}$$

$$(38) \frac{1}{5} \times \left(-\frac{8}{9}\right)$$

$$= -\frac{8}{45}$$

$$(9) \frac{3}{2} \times \frac{7}{9}$$

$$= \frac{7}{6}$$

$$(19) -\frac{5}{4} \div \left(-\frac{2}{3}\right)$$

$$= \frac{15}{8}$$

$$(29) \frac{7}{2} \div \frac{1}{5}$$

$$= \frac{35}{2}$$

$$(39) -\frac{7}{9} \times \left(-\frac{1}{2}\right)$$

$$= \frac{7}{18}$$

$$(10) -\frac{1}{2} \div \left(-\frac{1}{9}\right)$$

$$= \frac{9}{2}$$

$$(20) \frac{1}{2} \div (-1)$$

$$= -\frac{1}{2}$$

$$(30) \frac{2}{5} \div (-4)$$

$$= -\frac{1}{10}$$

$$(40) -\frac{9}{4} \div \left(-\frac{2}{5}\right)$$

$$= \frac{45}{8}$$

$$(1) 1 \times \frac{5}{2} \\ = \frac{5}{2}$$

$$(2) 10 \div \left(-\frac{10}{3}\right) \\ = -3$$

$$(3) \frac{3}{2} \div (-3) \\ = -\frac{1}{2}$$

$$(4) \frac{3}{4} \times \left(-\frac{1}{3}\right) \\ = -\frac{1}{4}$$

$$(5) -\frac{3}{5} \div \frac{7}{5} \\ = -\frac{3}{7}$$

$$(6) -6 \div \frac{2}{3} \\ = -9$$

$$(7) -\frac{1}{5} \times \frac{8}{7} \\ = -\frac{8}{35}$$

$$(8) -\frac{2}{7} \times \left(-\frac{1}{3}\right) \\ = \frac{2}{21}$$

$$(9) -\frac{2}{3} \times \left(-\frac{7}{10}\right) \\ = \frac{7}{15}$$

$$(10) -\frac{6}{5} \times 1 \\ = -\frac{6}{5}$$

$$(11) -2 \div \frac{2}{3} \\ = -3$$

$$(12) -6 \times \left(-\frac{1}{2}\right) \\ = 3$$

$$(13) -\frac{7}{5} \times \left(-\frac{9}{5}\right) \\ = \frac{63}{25}$$

$$(14) 1 \times 5 \\ = 5$$

$$(15) -2 \div \left(-\frac{1}{5}\right) \\ = 10$$

$$(16) -\frac{1}{2} \times (-3) \\ = \frac{3}{2}$$

$$(17) -2 \div 5 \\ = -\frac{2}{5}$$

$$(18) -\frac{9}{10} \div 2 \\ = -\frac{9}{20}$$

$$(19) -\frac{5}{4} \div \frac{2}{3} \\ = -\frac{15}{8}$$

$$(20) -1 \times \left(-\frac{6}{7}\right) \\ = \frac{6}{7}$$

$$(21) \frac{10}{7} \div (-1) \\ = -\frac{10}{7}$$

$$(22) \frac{5}{7} \times \frac{5}{8} \\ = \frac{25}{56}$$

$$(23) \frac{1}{3} \times \left(-\frac{6}{5}\right) \\ = -\frac{2}{5}$$

$$(24) 4 \div \frac{2}{5} \\ = 10$$

$$(25) -\frac{4}{7} \div (-6) \\ = \frac{2}{21}$$

$$(26) -\frac{5}{3} \times 2 \\ = -\frac{10}{3}$$

$$(27) -\frac{2}{7} \times \frac{3}{8} \\ = -\frac{3}{28}$$

$$(28) \frac{8}{3} \times (-1) \\ = -\frac{8}{3}$$

$$(29) -\frac{3}{4} \times (-1) \\ = \frac{3}{4}$$

$$(30) 4 \div \frac{4}{7} \\ = 7$$

$$(31) -\frac{1}{2} \div \left(-\frac{7}{10}\right) \\ = \frac{5}{7}$$

$$(32) -\frac{2}{3} \times \frac{1}{3} \\ = -\frac{2}{9}$$

$$(33) -\frac{10}{9} \div \frac{3}{10} \\ = -\frac{100}{27}$$

$$(34) \frac{4}{9} \times \left(-\frac{1}{4}\right) \\ = -\frac{1}{9}$$

$$(35) -\frac{5}{2} \div \left(-\frac{3}{5}\right) \\ = \frac{25}{6}$$

$$(36) \frac{1}{4} \div \left(-\frac{1}{2}\right) \\ = -\frac{1}{2}$$

$$(37) 1 \times \frac{10}{7} \\ = \frac{10}{7}$$

$$(38) \frac{1}{4} \times \left(-\frac{3}{5}\right) \\ = -\frac{3}{20}$$

$$(39) -\frac{5}{3} \div \frac{4}{5} \\ = -\frac{25}{12}$$

$$(40) 1 \times \left(-\frac{7}{2}\right) \\ = -\frac{7}{2}$$

$$(1) -\frac{7}{5} \div (-1) \\ = \frac{7}{5}$$

$$(11) -\frac{3}{2} \times 1 \\ = -\frac{3}{2}$$

$$(21) -\frac{1}{4} \div \left(-\frac{2}{5}\right) \\ = \frac{5}{8}$$

$$(31) \frac{1}{8} \div \frac{3}{8} \\ = \frac{1}{3}$$

$$(2) -\frac{1}{7} \times \frac{9}{5} \\ = -\frac{9}{35}$$

$$(12) -\frac{2}{7} \times \left(-\frac{4}{9}\right) \\ = \frac{8}{63}$$

$$(22) \frac{1}{5} \div 1 \\ = \frac{1}{5}$$

$$(32) -\frac{8}{3} \div \frac{4}{7} \\ = -\frac{14}{3}$$

$$(3) \frac{5}{7} \div (-5) \\ = -\frac{1}{7}$$

$$(13) -\frac{1}{10} \times \left(-\frac{5}{4}\right) \\ = \frac{1}{8}$$

$$(23) \frac{3}{2} \times \frac{5}{4} \\ = \frac{15}{8}$$

$$(33) \frac{10}{7} \div \frac{3}{2} \\ = \frac{20}{21}$$

$$(4) -\frac{2}{7} \times \frac{2}{3} \\ = -\frac{4}{21}$$

$$(14) \frac{8}{7} \times \left(-\frac{3}{5}\right) \\ = -\frac{24}{35}$$

$$(24) \frac{8}{3} \times 1 \\ = \frac{8}{3}$$

$$(34) 1 \times (-1) \\ = -1$$

$$(5) -\frac{5}{7} \times 5 \\ = -\frac{25}{7}$$

$$(15) 1 \times \left(-\frac{7}{4}\right) \\ = -\frac{7}{4}$$

$$(25) -\frac{4}{9} \div \frac{5}{7} \\ = -\frac{28}{45}$$

$$(35) -\frac{8}{5} \times (-3) \\ = \frac{24}{5}$$

$$(6) -\frac{4}{3} \div \frac{4}{5} \\ = -\frac{5}{3}$$

$$(16) \frac{5}{4} \times \left(-\frac{9}{7}\right) \\ = -\frac{45}{28}$$

$$(26) \frac{3}{2} \times \frac{5}{7} \\ = \frac{15}{14}$$

$$(36) -1 \div (-7) \\ = \frac{1}{7}$$

$$(7) 2 \div \frac{7}{3} \\ = \frac{6}{7}$$

$$(17) -\frac{7}{6} \times \left(-\frac{3}{2}\right) \\ = \frac{7}{4}$$

$$(27) \frac{2}{3} \div (-1) \\ = -\frac{2}{3}$$

$$(37) \frac{5}{3} \times (-1) \\ = -\frac{5}{3}$$

$$(8) -\frac{6}{5} \times \frac{10}{9} \\ = -\frac{4}{3}$$

$$(18) \frac{1}{5} \times \left(-\frac{5}{7}\right) \\ = -\frac{1}{7}$$

$$(28) -\frac{2}{5} \times \left(-\frac{2}{5}\right) \\ = \frac{4}{25}$$

$$(38) -\frac{1}{10} \div \left(-\frac{8}{5}\right) \\ = \frac{1}{16}$$

$$(9) \frac{5}{2} \times \frac{4}{5} \\ = 2$$

$$(19) \frac{3}{2} \div \frac{7}{4} \\ = \frac{6}{7}$$

$$(29) \frac{3}{5} \times 3 \\ = \frac{9}{5}$$

$$(39) \frac{1}{4} \times (-2) \\ = -\frac{1}{2}$$

$$(10) \frac{3}{2} \times \frac{1}{3} \\ = \frac{1}{2}$$

$$(20) \frac{4}{3} \div (-2) \\ = -\frac{2}{3}$$

$$(30) \frac{7}{2} \div \frac{3}{5} \\ = \frac{35}{6}$$

$$(40) 1 \times \left(-\frac{5}{8}\right) \\ = -\frac{5}{8}$$

1.4 実数

1.4.1 分母の有理化

分母の有理化を行う問題。

$$(1) \frac{3\sqrt{2}+3}{-4\sqrt{2}-2}$$

$$(6) \frac{-4\sqrt{5}+2}{-2\sqrt{5}-3}$$

$$(11) \frac{-2-3\sqrt{3}}{-1+4\sqrt{3}}$$

$$(2) \frac{-5\sqrt{2}+\sqrt{5}}{\sqrt{2}+4\sqrt{5}}$$

$$(7) \frac{-\sqrt{3}+1}{3\sqrt{3}}$$

$$(12) \frac{-\sqrt{2}-2}{-2\sqrt{2}+1}$$

$$(3) \frac{-1+3\sqrt{5}}{-4-\sqrt{5}}$$

$$(8) \frac{2+2\sqrt{2}}{-3+2\sqrt{2}}$$

$$(13) \frac{-3\sqrt{3}-6}{-5\sqrt{3}+4}$$

$$(4) \frac{2\sqrt{5}+1}{-2\sqrt{5}+5}$$

$$(9) \frac{-2\sqrt{3}-\sqrt{5}}{-5\sqrt{3}+3\sqrt{5}}$$

$$(14) \frac{3\sqrt{2}-\sqrt{5}}{5\sqrt{2}-3\sqrt{5}}$$

$$(5) \frac{5\sqrt{5}-3}{3\sqrt{5}+1}$$

$$(10) \frac{2\sqrt{2}-1}{-2\sqrt{2}+5}$$

$$(15) \frac{-5+4\sqrt{3}}{5+5\sqrt{3}}$$

$$(1) \frac{2\sqrt{3}+5}{-2\sqrt{3}+1}$$

$$(6) \frac{-\sqrt{2}+3}{-4\sqrt{2}}$$

$$(11) \frac{-\sqrt{3}+\sqrt{5}}{2\sqrt{3}-\sqrt{5}}$$

$$(2) \frac{5}{4+\sqrt{5}}$$

$$(7) \frac{-4-\sqrt{3}}{3+2\sqrt{3}}$$

$$(12) \frac{-2+4\sqrt{3}}{-10+3\sqrt{3}}$$

$$(3) \frac{5\sqrt{5}+4\sqrt{2}}{-2\sqrt{5}-\sqrt{2}}$$

$$(8) \frac{-4\sqrt{5}-5\sqrt{2}}{-2\sqrt{5}-2\sqrt{2}}$$

$$(13) \frac{-3\sqrt{3}-2\sqrt{5}}{3\sqrt{3}+5\sqrt{5}}$$

$$(4) \frac{-5\sqrt{5}}{3\sqrt{5}-\sqrt{2}}$$

$$(9) \frac{3\sqrt{2}+10}{2\sqrt{2}+8}$$

$$(14) \frac{5\sqrt{5}+2}{4\sqrt{5}+10}$$

$$(5) \frac{3\sqrt{3}+10}{5\sqrt{3}+2}$$

$$(10) \frac{-3}{-1+4\sqrt{5}}$$

$$(15) \frac{-5\sqrt{5}}{2\sqrt{2}-4\sqrt{5}}$$

$$(1) \frac{8 - \sqrt{3}}{8 - 4\sqrt{3}}$$

$$(6) \frac{\sqrt{5} - 10}{2\sqrt{5}}$$

$$(11) \frac{\sqrt{3} - 6}{-5\sqrt{3}}$$

$$(2) \frac{-4 - 3\sqrt{3}}{4 - 5\sqrt{3}}$$

$$(7) \frac{\sqrt{3} - 2}{\sqrt{3} + 8}$$

$$(12) \frac{-4\sqrt{2} - 3}{-4\sqrt{2} - 4}$$

$$(3) \frac{2 + 3\sqrt{5}}{-10 + 2\sqrt{5}}$$

$$(8) \frac{-3\sqrt{3} + 3}{-5\sqrt{3}}$$

$$(13) \frac{5 + 3\sqrt{5}}{1 + 5\sqrt{5}}$$

$$(4) \frac{-5\sqrt{2} + 4\sqrt{5}}{-4\sqrt{5}}$$

$$(9) \frac{3\sqrt{3}}{5\sqrt{5} - 2\sqrt{3}}$$

$$(14) \frac{8 + 2\sqrt{5}}{-10 + \sqrt{5}}$$

$$(5) \frac{-5\sqrt{5} - 5}{5\sqrt{5} - 4}$$

$$(10) \frac{5\sqrt{2} - 4\sqrt{5}}{-5\sqrt{2} + 5\sqrt{5}}$$

$$(15) \frac{-2\sqrt{3} - 5}{-3\sqrt{3} + 5}$$

$$(1) \frac{-\sqrt{3}}{5\sqrt{3}-4}$$

$$(6) \frac{4-\sqrt{5}}{6+4\sqrt{5}}$$

$$(11) \frac{4+5\sqrt{2}}{-4-\sqrt{2}}$$

$$(2) \frac{-5\sqrt{5}-2\sqrt{2}}{5\sqrt{5}+4\sqrt{2}}$$

$$(7) \frac{3+\sqrt{2}}{-1+\sqrt{2}}$$

$$(12) \frac{-4\sqrt{3}+5\sqrt{2}}{-2\sqrt{3}+5\sqrt{2}}$$

$$(3) \frac{3\sqrt{5}+10}{-2\sqrt{5}-10}$$

$$(8) \frac{8}{-3\sqrt{3}+4}$$

$$(13) \frac{-4\sqrt{3}+4}{\sqrt{3}+4}$$

$$(4) \frac{2\sqrt{3}+4}{\sqrt{3}+2}$$

$$(9) \frac{-\sqrt{2}-2}{\sqrt{2}+4}$$

$$(14) \frac{4-3\sqrt{3}}{-10-3\sqrt{3}}$$

$$(5) \frac{5\sqrt{2}+3\sqrt{3}}{\sqrt{3}}$$

$$(10) \frac{-6+2\sqrt{5}}{2-\sqrt{5}}$$

$$(15) \frac{1}{-4\sqrt{3}+4}$$

$$(1) \frac{5 - 3\sqrt{2}}{\sqrt{2}}$$

$$(6) \frac{-2\sqrt{3} + 2}{-3\sqrt{3} + 2}$$

$$(11) \frac{-4\sqrt{2} + 2}{-5\sqrt{2} + 10}$$

$$(2) \frac{-\sqrt{5} - \sqrt{2}}{\sqrt{2}}$$

$$(7) \frac{-\sqrt{3} + 2\sqrt{2}}{2\sqrt{3} - \sqrt{2}}$$

$$(12) \frac{-2\sqrt{3} - 1}{-2\sqrt{3} - 2}$$

$$(3) \frac{-10 - 5\sqrt{2}}{2 + \sqrt{2}}$$

$$(8) \frac{-2}{2 + 5\sqrt{2}}$$

$$(13) \frac{-5\sqrt{2} - 3}{-\sqrt{2}}$$

$$(4) \frac{4\sqrt{3} - 5\sqrt{5}}{-\sqrt{3} - 3\sqrt{5}}$$

$$(9) \frac{-3\sqrt{3} + \sqrt{2}}{-5\sqrt{3} - 4\sqrt{2}}$$

$$(14) \frac{-\sqrt{3} - 10}{-2\sqrt{3} - 6}$$

$$(5) \frac{2\sqrt{3} - 5}{3\sqrt{3}}$$

$$(10) \frac{10 - 5\sqrt{3}}{2 - \sqrt{3}}$$

$$(15) \frac{-2 + \sqrt{5}}{8 - \sqrt{5}}$$

$$(1) \frac{-4\sqrt{3} + 3\sqrt{5}}{2\sqrt{3} + 4\sqrt{5}}$$

$$(6) \frac{\sqrt{2} - 3}{2\sqrt{2} - 4}$$

$$(11) \frac{-3\sqrt{2} + \sqrt{5}}{-4\sqrt{2} - 2\sqrt{5}}$$

$$(2) \frac{-5\sqrt{3} - 6}{-3\sqrt{3}}$$

$$(7) \frac{3\sqrt{3} - 6}{\sqrt{3} - 8}$$

$$(12) \frac{2}{10 + 5\sqrt{3}}$$

$$(3) \frac{-5 + 2\sqrt{3}}{3 + 5\sqrt{3}}$$

$$(8) \frac{4}{-8 + 3\sqrt{3}}$$

$$(13) \frac{\sqrt{2} - 1}{3\sqrt{2} + 4}$$

$$(4) \frac{3\sqrt{3} + 6}{3\sqrt{3} + 2}$$

$$(9) \frac{1 - \sqrt{2}}{3 - 4\sqrt{2}}$$

$$(14) \frac{5 + 2\sqrt{3}}{-1 + 2\sqrt{3}}$$

$$(5) \frac{2\sqrt{3} - 6}{-3\sqrt{3} + 2}$$

$$(10) \frac{-4\sqrt{2} - 2\sqrt{5}}{-3\sqrt{2}}$$

$$(15) \frac{\sqrt{2} - 1}{4\sqrt{2} - 4}$$

$$(1) \frac{4\sqrt{2} + \sqrt{5}}{\sqrt{2} - \sqrt{5}}$$

$$(6) \frac{2\sqrt{5}}{2\sqrt{5} - 5}$$

$$(11) \frac{-3\sqrt{5} - 1}{\sqrt{5} - 1}$$

$$(2) \frac{\sqrt{3}}{2\sqrt{2} - \sqrt{3}}$$

$$(7) \frac{2 + 5\sqrt{2}}{10 - 4\sqrt{2}}$$

$$(12) \frac{-2\sqrt{3} - 5\sqrt{5}}{3\sqrt{5}}$$

$$(3) \frac{4\sqrt{2} + 5\sqrt{5}}{3\sqrt{2} + 4\sqrt{5}}$$

$$(8) \frac{-\sqrt{5} - 2}{3\sqrt{5} - 4}$$

$$(13) \frac{2\sqrt{2} + 5\sqrt{3}}{-3\sqrt{2} + 4\sqrt{3}}$$

$$(4) \frac{5\sqrt{2} + 2\sqrt{5}}{-5\sqrt{2} + 2\sqrt{5}}$$

$$(9) \frac{5\sqrt{2} + 2\sqrt{3}}{-\sqrt{2} - 4\sqrt{3}}$$

$$(14) \frac{-10 + 4\sqrt{2}}{-3\sqrt{2}}$$

$$(5) \frac{2\sqrt{3}}{4\sqrt{3} + 5\sqrt{5}}$$

$$(10) \frac{-2 - \sqrt{2}}{-8 - 5\sqrt{2}}$$

$$(15) \frac{-\sqrt{2} - 4\sqrt{5}}{-3\sqrt{2} - 5\sqrt{5}}$$

$$(1) \frac{-\sqrt{5}-5\sqrt{2}}{4\sqrt{5}}$$

$$(6) \frac{-\sqrt{5}+2}{-2\sqrt{5}-4}$$

$$(11) \frac{-4\sqrt{3}-2}{-3\sqrt{3}-4}$$

$$(2) \frac{-1-\sqrt{2}}{-2-\sqrt{2}}$$

$$(7) \frac{-4\sqrt{5}-5}{-3\sqrt{5}}$$

$$(12) \frac{-1-3\sqrt{5}}{2-2\sqrt{5}}$$

$$(3) \frac{4-\sqrt{2}}{3+\sqrt{2}}$$

$$(8) \frac{6+4\sqrt{3}}{6+\sqrt{3}}$$

$$(13) \frac{2\sqrt{3}}{-\sqrt{3}+5\sqrt{2}}$$

$$(4) \frac{-2\sqrt{5}}{5+2\sqrt{5}}$$

$$(9) \frac{-\sqrt{2}+5\sqrt{3}}{-4\sqrt{2}-2\sqrt{3}}$$

$$(14) \frac{2\sqrt{2}+3\sqrt{3}}{-5\sqrt{2}+3\sqrt{3}}$$

$$(5) \frac{8-\sqrt{2}}{2+5\sqrt{2}}$$

$$(10) \frac{-4\sqrt{5}+3\sqrt{3}}{2\sqrt{3}}$$

$$(15) \frac{-2\sqrt{5}-4\sqrt{2}}{-3\sqrt{2}}$$

$$(1) \frac{-3\sqrt{2} + 5\sqrt{5}}{4\sqrt{5}}$$

$$(6) \frac{-1 - 5\sqrt{5}}{-2 + 3\sqrt{5}}$$

$$(11) \frac{-5\sqrt{3} - 3\sqrt{5}}{-3\sqrt{3} - 2\sqrt{5}}$$

$$(2) \frac{-3\sqrt{3}}{\sqrt{3} - 5\sqrt{5}}$$

$$(7) \frac{1}{2\sqrt{5} + 1}$$

$$(12) \frac{-6 - \sqrt{5}}{-6 - \sqrt{5}}$$

$$(3) \frac{-4\sqrt{5}}{-5\sqrt{5} - \sqrt{2}}$$

$$(8) \frac{2\sqrt{2} + 6}{5\sqrt{2} - 4}$$

$$(13) \frac{-4 + 4\sqrt{2}}{-1 - \sqrt{2}}$$

$$(4) \frac{3\sqrt{2} - 10}{-4\sqrt{2} - 6}$$

$$(9) \frac{2\sqrt{3} - \sqrt{5}}{-3\sqrt{3} - 5\sqrt{5}}$$

$$(14) \frac{3\sqrt{2} + 3\sqrt{5}}{3\sqrt{2} + 2\sqrt{5}}$$

$$(5) \frac{5\sqrt{2}}{4\sqrt{2} + 6}$$

$$(10) \frac{-4\sqrt{5}}{\sqrt{5} + \sqrt{3}}$$

$$(15) \frac{\sqrt{3} + 5}{2\sqrt{3} + 4}$$

$$(1) \frac{8 - 2\sqrt{5}}{8 + 5\sqrt{5}}$$

$$(6) \frac{4\sqrt{5} - 2\sqrt{2}}{-2\sqrt{5} - 5\sqrt{2}}$$

$$(11) \frac{2\sqrt{2} - 1}{-2\sqrt{2} - 2}$$

$$(2) \frac{2 + 2\sqrt{2}}{2 - 5\sqrt{2}}$$

$$(7) \frac{-4\sqrt{5} - 2\sqrt{3}}{\sqrt{5} - 5\sqrt{3}}$$

$$(12) \frac{-1 + \sqrt{5}}{-2 - 4\sqrt{5}}$$

$$(3) \frac{5\sqrt{5}}{5\sqrt{5} - 3}$$

$$(8) \frac{-4}{-\sqrt{3} - 2}$$

$$(13) \frac{\sqrt{5} + 8}{-4\sqrt{5}}$$

$$(4) \frac{-\sqrt{2} + 3}{2\sqrt{2} - 2}$$

$$(9) \frac{-4\sqrt{3} + 10}{-\sqrt{3} + 4}$$

$$(14) \frac{-4 - 5\sqrt{5}}{1 - 3\sqrt{5}}$$

$$(5) \frac{-1}{2\sqrt{3} - 4}$$

$$(10) \frac{3\sqrt{2}}{-5\sqrt{3} + 4\sqrt{2}}$$

$$(15) \frac{\sqrt{2} - 1}{\sqrt{2} - 3}$$

$$(1) \frac{5 - 2\sqrt{3}}{3 - \sqrt{3}}$$

$$(6) \frac{5\sqrt{5} - 5\sqrt{2}}{-\sqrt{5} - \sqrt{2}}$$

$$(11) \frac{4\sqrt{2} - 4\sqrt{5}}{-3\sqrt{2} + 3\sqrt{5}}$$

$$(2) \frac{-5}{-2\sqrt{5} + 4}$$

$$(7) \frac{3\sqrt{2} + 2\sqrt{3}}{-2\sqrt{2}}$$

$$(12) \frac{-3\sqrt{3} + 6}{2\sqrt{3} - 10}$$

$$(3) \frac{-4}{2\sqrt{5} - 1}$$

$$(8) \frac{\sqrt{5}}{-\sqrt{5} - \sqrt{3}}$$

$$(13) \frac{2\sqrt{2} + 3\sqrt{3}}{-4\sqrt{2} + \sqrt{3}}$$

$$(4) \frac{\sqrt{3} + 4}{-2\sqrt{3} + 2}$$

$$(9) \frac{-6 - \sqrt{3}}{-8 - 3\sqrt{3}}$$

$$(14) \frac{-3\sqrt{2} - 4}{-4\sqrt{2} - 2}$$

$$(5) \frac{\sqrt{5} + 2}{4\sqrt{5} - 4}$$

$$(10) \frac{1 - \sqrt{5}}{3 - 2\sqrt{5}}$$

$$(15) \frac{4\sqrt{2}}{\sqrt{3} + 3\sqrt{2}}$$

$$(1) \frac{\sqrt{5} - \sqrt{2}}{-\sqrt{2}}$$

$$(6) \frac{-\sqrt{3} + 2\sqrt{5}}{2\sqrt{3} + 2\sqrt{5}}$$

$$(11) \frac{\sqrt{3} + 4\sqrt{5}}{-4\sqrt{3} - \sqrt{5}}$$

$$(2) \frac{-\sqrt{3} + 5\sqrt{2}}{\sqrt{3} + 2\sqrt{2}}$$

$$(7) \frac{-2\sqrt{5} - 3\sqrt{2}}{\sqrt{5} - \sqrt{2}}$$

$$(12) \frac{\sqrt{5}}{2\sqrt{5} + 4}$$

$$(3) \frac{6 + 5\sqrt{5}}{10 + 3\sqrt{5}}$$

$$(8) \frac{-5\sqrt{2} + 8}{-\sqrt{2} + 2}$$

$$(13) \frac{2\sqrt{5} - 3\sqrt{2}}{-2\sqrt{5} + 2\sqrt{2}}$$

$$(4) \frac{-2 + 4\sqrt{3}}{8 - \sqrt{3}}$$

$$(9) \frac{-4\sqrt{5}}{-5\sqrt{5} - 3\sqrt{2}}$$

$$(14) \frac{-\sqrt{2} - 2}{-\sqrt{2} - 5}$$

$$(5) \frac{-3\sqrt{3}}{\sqrt{3} + 3\sqrt{5}}$$

$$(10) \frac{-2\sqrt{5}}{3\sqrt{3} - 3\sqrt{5}}$$

$$(15) \frac{5\sqrt{2} + 2\sqrt{3}}{5\sqrt{2} + \sqrt{3}}$$

$$(1) \frac{3 - \sqrt{2}}{4 + 3\sqrt{2}}$$

$$(6) \frac{\sqrt{5}}{\sqrt{5}}$$

$$(11) \frac{-4\sqrt{5} + 2\sqrt{3}}{3\sqrt{5} + 3\sqrt{3}}$$

$$(2) \frac{-\sqrt{3}}{-\sqrt{3} - \sqrt{5}}$$

$$(7) \frac{-2\sqrt{3}}{5\sqrt{3} + 5}$$

$$(12) \frac{4\sqrt{3} + 3\sqrt{5}}{-4\sqrt{5}}$$

$$(3) \frac{-2 - \sqrt{5}}{4 - 4\sqrt{5}}$$

$$(8) \frac{3}{3 + \sqrt{2}}$$

$$(13) \frac{\sqrt{5}}{2\sqrt{3} - \sqrt{5}}$$

$$(4) \frac{-3\sqrt{2} + 6}{4\sqrt{2} - 8}$$

$$(9) \frac{4\sqrt{2} + 2\sqrt{5}}{\sqrt{2} - 5\sqrt{5}}$$

$$(14) \frac{-3\sqrt{2} + \sqrt{3}}{4\sqrt{2} + 3\sqrt{3}}$$

$$(5) \frac{-4 + 5\sqrt{3}}{5 + 3\sqrt{3}}$$

$$(10) \frac{-6 - 5\sqrt{5}}{4 + 5\sqrt{5}}$$

$$(15) \frac{3}{\sqrt{3} + 1}$$

$$(1) \frac{-3\sqrt{3} - 4\sqrt{2}}{3\sqrt{3} - 2\sqrt{2}}$$

$$(6) \frac{6 + 3\sqrt{5}}{-2 + 4\sqrt{5}}$$

$$(11) \frac{3\sqrt{2} - 3\sqrt{3}}{-3\sqrt{2} + \sqrt{3}}$$

$$(2) \frac{5 - \sqrt{5}}{-4 + 4\sqrt{5}}$$

$$(7) \frac{-5\sqrt{5} - 5}{-3\sqrt{5} - 3}$$

$$(12) \frac{-3\sqrt{3} - 5\sqrt{2}}{-3\sqrt{3} - \sqrt{2}}$$

$$(3) \frac{6 + 2\sqrt{2}}{-4 - 3\sqrt{2}}$$

$$(8) \frac{-5\sqrt{2}}{-5\sqrt{3} + 4\sqrt{2}}$$

$$(13) \frac{-4}{-10 - 3\sqrt{2}}$$

$$(4) \frac{-2 - 4\sqrt{5}}{-1 + 3\sqrt{5}}$$

$$(9) \frac{-2\sqrt{3} - 3\sqrt{2}}{2\sqrt{3} + \sqrt{2}}$$

$$(14) \frac{\sqrt{2} - 4}{-\sqrt{2} + 2}$$

$$(5) \frac{\sqrt{2} - 5\sqrt{3}}{\sqrt{2} - 2\sqrt{3}}$$

$$(10) \frac{2\sqrt{3} - 3\sqrt{2}}{3\sqrt{3} + 2\sqrt{2}}$$

$$(15) \frac{-1 - 4\sqrt{5}}{4 + 2\sqrt{5}}$$

$$(1) \frac{-2}{\sqrt{2}-2}$$

$$(6) \frac{-\sqrt{2}+4\sqrt{5}}{\sqrt{5}}$$

$$(11) \frac{4\sqrt{3}+1}{\sqrt{3}-3}$$

$$(2) \frac{-2-4\sqrt{2}}{-2+5\sqrt{2}}$$

$$(7) \frac{-5}{-5-4\sqrt{2}}$$

$$(12) \frac{3\sqrt{3}+\sqrt{5}}{4\sqrt{3}-\sqrt{5}}$$

$$(3) \frac{3+4\sqrt{5}}{2\sqrt{5}}$$

$$(8) \frac{2+\sqrt{3}}{5\sqrt{3}}$$

$$(13) \frac{-5\sqrt{2}-2\sqrt{3}}{2\sqrt{2}+3\sqrt{3}}$$

$$(4) \frac{5\sqrt{3}-4}{-3\sqrt{3}-1}$$

$$(9) \frac{-5\sqrt{2}}{10-2\sqrt{2}}$$

$$(14) \frac{\sqrt{2}+2\sqrt{3}}{-\sqrt{2}}$$

$$(5) \frac{-1+3\sqrt{5}}{-4\sqrt{5}}$$

$$(10) \frac{-4\sqrt{3}+5\sqrt{2}}{5\sqrt{3}-4\sqrt{2}}$$

$$(15) \frac{-2-3\sqrt{5}}{8-4\sqrt{5}}$$

$$(1) \frac{2}{1+4\sqrt{3}}$$

$$(6) \frac{-4+4\sqrt{2}}{4-\sqrt{2}}$$

$$(11) \frac{-\sqrt{3}}{4\sqrt{3}}$$

$$(2) \frac{3\sqrt{5}-4\sqrt{2}}{-2\sqrt{5}+5\sqrt{2}}$$

$$(7) \frac{2\sqrt{2}}{-2\sqrt{5}-\sqrt{2}}$$

$$(12) \frac{\sqrt{5}+2\sqrt{2}}{2\sqrt{2}}$$

$$(3) \frac{-4\sqrt{5}}{3\sqrt{2}-3\sqrt{5}}$$

$$(8) \frac{\sqrt{2}+8}{-3\sqrt{2}-4}$$

$$(13) \frac{-4\sqrt{2}-3}{-\sqrt{2}+2}$$

$$(4) \frac{\sqrt{2}+2}{-\sqrt{2}+3}$$

$$(9) \frac{-2-3\sqrt{2}}{3-4\sqrt{2}}$$

$$(14) \frac{5-5\sqrt{5}}{4+5\sqrt{5}}$$

$$(5) \frac{\sqrt{3}+8}{-3\sqrt{3}-4}$$

$$(10) \frac{\sqrt{5}+4}{3\sqrt{5}+5}$$

$$(15) \frac{3\sqrt{5}+6}{5\sqrt{5}}$$

$$(1) \frac{3\sqrt{3} + 5\sqrt{2}}{-3\sqrt{3} + \sqrt{2}}$$

$$(6) \frac{-5 + \sqrt{5}}{-3 - 2\sqrt{5}}$$

$$(11) \frac{-1 - \sqrt{2}}{-4 + \sqrt{2}}$$

$$(2) \frac{-2}{-\sqrt{2} + 2}$$

$$(7) \frac{3\sqrt{3} + \sqrt{5}}{5\sqrt{3} + 2\sqrt{5}}$$

$$(12) \frac{\sqrt{3}}{-6 - 2\sqrt{3}}$$

$$(3) \frac{1}{-4 - \sqrt{5}}$$

$$(8) \frac{-4\sqrt{3} - 3\sqrt{2}}{-3\sqrt{2}}$$

$$(13) \frac{3\sqrt{5} - \sqrt{3}}{-2\sqrt{5} + 5\sqrt{3}}$$

$$(4) \frac{-6 + \sqrt{5}}{-6 + 2\sqrt{5}}$$

$$(9) \frac{5 + \sqrt{5}}{2 + 5\sqrt{5}}$$

$$(14) \frac{-5\sqrt{2} - 8}{-2\sqrt{2} - 4}$$

$$(5) \frac{2\sqrt{5} + 4}{-5\sqrt{5} + 3}$$

$$(10) \frac{-3\sqrt{5} + 5\sqrt{2}}{2\sqrt{5} + 3\sqrt{2}}$$

$$(15) \frac{5\sqrt{5} - 4\sqrt{3}}{-3\sqrt{5} - 3\sqrt{3}}$$

$$(1) \frac{\sqrt{2}}{-5\sqrt{2} - 3\sqrt{5}}$$

$$(6) \frac{-2 + \sqrt{3}}{-4\sqrt{3}}$$

$$(11) \frac{\sqrt{5} + 10}{-\sqrt{5} - 6}$$

$$(2) \frac{-2 + 2\sqrt{5}}{-1 + \sqrt{5}}$$

$$(7) \frac{4\sqrt{3} + 3\sqrt{5}}{-\sqrt{3} - 5\sqrt{5}}$$

$$(12) \frac{3\sqrt{5} - 10}{-4\sqrt{5} - 6}$$

$$(3) \frac{3\sqrt{5} + 3\sqrt{2}}{\sqrt{5} + \sqrt{2}}$$

$$(8) \frac{5 - 5\sqrt{2}}{2 + 5\sqrt{2}}$$

$$(13) \frac{4\sqrt{5} + 5\sqrt{3}}{\sqrt{5} - \sqrt{3}}$$

$$(4) \frac{-3\sqrt{5} - 2}{5\sqrt{5} - 3}$$

$$(9) \frac{-4\sqrt{5} + 5\sqrt{2}}{-2\sqrt{5} - 5\sqrt{2}}$$

$$(14) \frac{5 + 3\sqrt{2}}{-3 - 2\sqrt{2}}$$

$$(5) \frac{\sqrt{3} + \sqrt{5}}{-4\sqrt{3} + 4\sqrt{5}}$$

$$(10) \frac{4\sqrt{5} + 2\sqrt{3}}{3\sqrt{5} - 3\sqrt{3}}$$

$$(15) \frac{\sqrt{3} + 2}{-2\sqrt{3}}$$

$$(1) \frac{2\sqrt{5} - 3\sqrt{3}}{-4\sqrt{5} - 3\sqrt{3}}$$

$$(6) \frac{4\sqrt{3} + \sqrt{5}}{4\sqrt{5}}$$

$$(11) \frac{2 - 5\sqrt{5}}{3 - \sqrt{5}}$$

$$(2) \frac{2\sqrt{2} + 2\sqrt{5}}{-\sqrt{2} + \sqrt{5}}$$

$$(7) \frac{5 + 3\sqrt{2}}{-3 - 3\sqrt{2}}$$

$$(12) \frac{\sqrt{3} + 2\sqrt{5}}{-2\sqrt{3} + 2\sqrt{5}}$$

$$(3) \frac{4\sqrt{3} - 3\sqrt{2}}{4\sqrt{3} - 3\sqrt{2}}$$

$$(8) \frac{-2\sqrt{5} + 2\sqrt{2}}{5\sqrt{5} - 3\sqrt{2}}$$

$$(13) \frac{-4 + 5\sqrt{3}}{-1 - 3\sqrt{3}}$$

$$(4) \frac{-3\sqrt{2} - 8}{4\sqrt{2} - 6}$$

$$(9) \frac{3\sqrt{2} + 8}{-4\sqrt{2}}$$

$$(14) \frac{5 - 4\sqrt{3}}{-4 + 4\sqrt{3}}$$

$$(5) \frac{2\sqrt{5} + 2}{3\sqrt{5} - 4}$$

$$(10) \frac{-2 + 4\sqrt{2}}{3 + 3\sqrt{2}}$$

$$(15) \frac{3 - 2\sqrt{5}}{3 - 3\sqrt{5}}$$

$$(1) \frac{3\sqrt{3}-2}{\sqrt{3}}$$

$$(6) \frac{\sqrt{2}}{1+\sqrt{2}}$$

$$(11) \frac{-5\sqrt{3}-1}{-3\sqrt{3}-4}$$

$$(2) \frac{3\sqrt{2}}{3\sqrt{2}+4\sqrt{3}}$$

$$(7) \frac{4+\sqrt{5}}{1+2\sqrt{5}}$$

$$(12) \frac{-10+\sqrt{3}}{-6+3\sqrt{3}}$$

$$(3) \frac{5\sqrt{5}-\sqrt{3}}{\sqrt{5}+2\sqrt{3}}$$

$$(8) \frac{-3}{5\sqrt{5}+3}$$

$$(13) \frac{-2-4\sqrt{3}}{-1+5\sqrt{3}}$$

$$(4) \frac{-2\sqrt{2}+2}{2\sqrt{2}-1}$$

$$(9) \frac{2\sqrt{3}-1}{4\sqrt{3}+2}$$

$$(14) \frac{3\sqrt{2}}{4-3\sqrt{2}}$$

$$(5) \frac{2}{-\sqrt{2}+2}$$

$$(10) \frac{-\sqrt{5}-2}{-3\sqrt{5}+10}$$

$$(15) \frac{-4\sqrt{5}}{5\sqrt{5}}$$

$$\begin{aligned}(1) \quad & \frac{3\sqrt{2}+3}{-4\sqrt{2}-2} \\ &= \frac{-9-3\sqrt{2}}{14}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{-4\sqrt{5}+2}{-2\sqrt{5}-3} \\ &= \frac{46-16\sqrt{5}}{11}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-2-3\sqrt{3}}{-1+4\sqrt{3}} \\ &= \frac{-11\sqrt{3}-38}{47}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{-5\sqrt{2}+\sqrt{5}}{\sqrt{2}+4\sqrt{5}} \\ &= \frac{-7\sqrt{10}+10}{26}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{-\sqrt{3}+1}{3\sqrt{3}} \\ &= \frac{\sqrt{3}-3}{9}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-\sqrt{2}-2}{-2\sqrt{2}+1} \\ &= \frac{6+5\sqrt{2}}{7}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{-1+3\sqrt{5}}{-4-\sqrt{5}} \\ &= \frac{-13\sqrt{5}+19}{11}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{2+2\sqrt{2}}{-3+2\sqrt{2}} \\ &= -10\sqrt{2}-14\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-3\sqrt{3}-6}{-5\sqrt{3}+4} \\ &= \frac{69+42\sqrt{3}}{59}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{2\sqrt{5}+1}{-2\sqrt{5}+5} \\ &= \frac{25+12\sqrt{5}}{5}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{-2\sqrt{3}-\sqrt{5}}{-5\sqrt{3}+3\sqrt{5}} \\ &= \frac{45+11\sqrt{15}}{30}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{3\sqrt{2}-\sqrt{5}}{5\sqrt{2}-3\sqrt{5}} \\ &= \frac{4\sqrt{10}+15}{5}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{5\sqrt{5}-3}{3\sqrt{5}+1} \\ &= \frac{-7\sqrt{5}+39}{22}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{2\sqrt{2}-1}{-2\sqrt{2}+5} \\ &= \frac{8\sqrt{2}+3}{17}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{-5+4\sqrt{3}}{5+5\sqrt{3}} \\ &= \frac{-9\sqrt{3}+17}{10}\end{aligned}$$

$$\begin{aligned}(1) \quad & \frac{2\sqrt{3}+5}{-2\sqrt{3}+1} \\ &= \frac{-12\sqrt{3}-17}{11}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{-\sqrt{2}+3}{-4\sqrt{2}} \\ &= \frac{-3\sqrt{2}+2}{8}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-\sqrt{3}+\sqrt{5}}{2\sqrt{3}-\sqrt{5}} \\ &= \frac{\sqrt{15}-1}{7}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{5}{4+\sqrt{5}} \\ &= \frac{-5\sqrt{5}+20}{11}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{-4-\sqrt{3}}{3+2\sqrt{3}} \\ &= \frac{6-5\sqrt{3}}{3}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-2+4\sqrt{3}}{-10+3\sqrt{3}} \\ &= \frac{-16-34\sqrt{3}}{73}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{5\sqrt{5}+4\sqrt{2}}{-2\sqrt{5}-\sqrt{2}} \\ &= \frac{-14-\sqrt{10}}{6}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{-4\sqrt{5}-5\sqrt{2}}{-2\sqrt{5}-2\sqrt{2}} \\ &= \frac{\sqrt{10}+10}{6}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-3\sqrt{3}-2\sqrt{5}}{3\sqrt{3}+5\sqrt{5}} \\ &= \frac{-23-9\sqrt{15}}{98}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{-5\sqrt{5}}{3\sqrt{5}-\sqrt{2}} \\ &= \frac{-5\sqrt{10}-75}{43}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{3\sqrt{2}+10}{2\sqrt{2}+8} \\ &= \frac{17+\sqrt{2}}{14}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{5\sqrt{5}+2}{4\sqrt{5}+10} \\ &= \frac{21\sqrt{5}-40}{10}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{3\sqrt{3}+10}{5\sqrt{3}+2} \\ &= \frac{44\sqrt{3}+25}{71}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{-3}{-1+4\sqrt{5}} \\ &= \frac{-3-12\sqrt{5}}{79}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{-5\sqrt{5}}{2\sqrt{2}-4\sqrt{5}} \\ &= \frac{5\sqrt{10}+50}{36}\end{aligned}$$

$$\begin{aligned}(1) \quad & \frac{8 - \sqrt{3}}{8 - 4\sqrt{3}} \\ &= \frac{13 + 6\sqrt{3}}{4}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{\sqrt{5} - 10}{2\sqrt{5}} \\ &= \frac{1 - 2\sqrt{5}}{2}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{\sqrt{3} - 6}{-5\sqrt{3}} \\ &= \frac{2\sqrt{3} - 1}{5}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{-4 - 3\sqrt{3}}{4 - 5\sqrt{3}} \\ &= \frac{61 + 32\sqrt{3}}{59}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{\sqrt{3} - 2}{\sqrt{3} + 8} \\ &= \frac{10\sqrt{3} - 19}{61}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-4\sqrt{2} - 3}{-4\sqrt{2} - 4} \\ &= \frac{-\sqrt{2} + 5}{4}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{2 + 3\sqrt{5}}{-10 + 2\sqrt{5}} \\ &= \frac{-25 - 17\sqrt{5}}{40}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{-3\sqrt{3} + 3}{-5\sqrt{3}} \\ &= \frac{3 - \sqrt{3}}{5}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{5 + 3\sqrt{5}}{1 + 5\sqrt{5}} \\ &= \frac{11\sqrt{5} + 35}{62}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{-5\sqrt{2} + 4\sqrt{5}}{-4\sqrt{5}} \\ &= \frac{-4 + \sqrt{10}}{4}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{3\sqrt{3}}{5\sqrt{5} - 2\sqrt{3}} \\ &= \frac{18 + 15\sqrt{15}}{113}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{8 + 2\sqrt{5}}{-10 + \sqrt{5}} \\ &= \frac{-90 - 28\sqrt{5}}{95}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{-5\sqrt{5} - 5}{5\sqrt{5} - 4} \\ &= \frac{-145 - 45\sqrt{5}}{109}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{5\sqrt{2} - 4\sqrt{5}}{-5\sqrt{2} + 5\sqrt{5}} \\ &= \frac{\sqrt{10} - 10}{15}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{-2\sqrt{3} - 5}{-3\sqrt{3} + 5} \\ &= \frac{25\sqrt{3} + 43}{2}\end{aligned}$$

$$\begin{aligned}(1) \quad & \frac{-\sqrt{3}}{5\sqrt{3}-4} \\ &= \frac{-15-4\sqrt{3}}{59}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{4-\sqrt{5}}{6+4\sqrt{5}} \\ &= \frac{-2+\sqrt{5}}{2}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{4+5\sqrt{2}}{-4-\sqrt{2}} \\ &= \frac{-3-8\sqrt{2}}{7}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{-5\sqrt{5}-2\sqrt{2}}{5\sqrt{5}+4\sqrt{2}} \\ &= \frac{-109+10\sqrt{10}}{93}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{3+\sqrt{2}}{-1+\sqrt{2}} \\ &= 4\sqrt{2}+5\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-4\sqrt{3}+5\sqrt{2}}{-2\sqrt{3}+5\sqrt{2}} \\ &= \frac{13-5\sqrt{6}}{19}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{3\sqrt{5}+10}{-2\sqrt{5}-10} \\ &= \frac{-\sqrt{5}-7}{8}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{8}{-3\sqrt{3}+4} \\ &= \frac{-24\sqrt{3}-32}{11}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-4\sqrt{3}+4}{\sqrt{3}+4} \\ &= \frac{-20\sqrt{3}+28}{13}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{2\sqrt{3}+4}{\sqrt{3}+2} \\ &= 2\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{-\sqrt{2}-2}{\sqrt{2}+4} \\ &= \frac{-\sqrt{2}-3}{7}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{4-3\sqrt{3}}{-10-3\sqrt{3}} \\ &= \frac{-67+42\sqrt{3}}{73}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{5\sqrt{2}+3\sqrt{3}}{\sqrt{3}} \\ &= \frac{9+5\sqrt{6}}{3}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{-6+2\sqrt{5}}{2-\sqrt{5}} \\ &= 2\sqrt{5}+2\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{1}{-4\sqrt{3}+4} \\ &= \frac{-1-\sqrt{3}}{8}\end{aligned}$$

$$\begin{aligned}(1) \quad & \frac{5-3\sqrt{2}}{\sqrt{2}} \\ &= \frac{-6+5\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{-2\sqrt{3}+2}{-3\sqrt{3}+2} \\ &= \frac{-2\sqrt{3}+14}{23}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-4\sqrt{2}+2}{-5\sqrt{2}+10} \\ &= \frac{-2-3\sqrt{2}}{5}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{-\sqrt{5}-\sqrt{2}}{\sqrt{2}} \\ &= \frac{-2-\sqrt{10}}{2}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{-\sqrt{3}+2\sqrt{2}}{2\sqrt{3}-\sqrt{2}} \\ &= \frac{3\sqrt{6}-2}{10}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-2\sqrt{3}-1}{-2\sqrt{3}-2} \\ &= \frac{5-\sqrt{3}}{4}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{-10-5\sqrt{2}}{2+\sqrt{2}} \\ &= -5\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{-2}{2+5\sqrt{2}} \\ &= \frac{2-5\sqrt{2}}{23}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-5\sqrt{2}-3}{-\sqrt{2}} \\ &= \frac{10+3\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{4\sqrt{3}-5\sqrt{5}}{-\sqrt{3}-3\sqrt{5}} \\ &= \frac{87-17\sqrt{15}}{42}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{-3\sqrt{3}+\sqrt{2}}{-5\sqrt{3}-4\sqrt{2}} \\ &= \frac{53-17\sqrt{6}}{43}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{-\sqrt{3}-10}{-2\sqrt{3}-6} \\ &= \frac{27-7\sqrt{3}}{12}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{2\sqrt{3}-5}{3\sqrt{3}} \\ &= \frac{6-5\sqrt{3}}{9}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{10-5\sqrt{3}}{2-\sqrt{3}} \\ &= 5\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{-2+\sqrt{5}}{8-\sqrt{5}} \\ &= \frac{-11+6\sqrt{5}}{59}\end{aligned}$$

$$\begin{aligned}(1) \quad & \frac{-4\sqrt{3} + 3\sqrt{5}}{2\sqrt{3} + 4\sqrt{5}} \\ &= \frac{42 - 11\sqrt{15}}{34}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{\sqrt{2} - 3}{2\sqrt{2} - 4} \\ &= \frac{4 + \sqrt{2}}{4}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-3\sqrt{2} + \sqrt{5}}{-4\sqrt{2} - 2\sqrt{5}} \\ &= \frac{17 - 5\sqrt{10}}{6}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{-5\sqrt{3} - 6}{-3\sqrt{3}} \\ &= \frac{5 + 2\sqrt{3}}{3}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{3\sqrt{3} - 6}{\sqrt{3} - 8} \\ &= \frac{39 - 18\sqrt{3}}{61}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{2}{10 + 5\sqrt{3}} \\ &= \frac{-2\sqrt{3} + 4}{5}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{-5 + 2\sqrt{3}}{3 + 5\sqrt{3}} \\ &= \frac{-31\sqrt{3} + 45}{66}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{4}{-8 + 3\sqrt{3}} \\ &= \frac{-32 - 12\sqrt{3}}{37}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{\sqrt{2} - 1}{3\sqrt{2} + 4} \\ &= \frac{10 - 7\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{3\sqrt{3} + 6}{3\sqrt{3} + 2} \\ &= \frac{15 + 12\sqrt{3}}{23}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{1 - \sqrt{2}}{3 - 4\sqrt{2}} \\ &= \frac{5 - \sqrt{2}}{23}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{5 + 2\sqrt{3}}{-1 + 2\sqrt{3}} \\ &= \frac{17 + 12\sqrt{3}}{11}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{2\sqrt{3} - 6}{-3\sqrt{3} + 2} \\ &= \frac{14\sqrt{3} - 6}{23}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{-4\sqrt{2} - 2\sqrt{5}}{-3\sqrt{2}} \\ &= \frac{4 + \sqrt{10}}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{\sqrt{2} - 1}{4\sqrt{2} - 4} \\ &= \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(1) \quad & \frac{4\sqrt{2} + \sqrt{5}}{\sqrt{2} - \sqrt{5}} \\ &= \frac{-13 - 5\sqrt{10}}{3}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{2\sqrt{5}}{2\sqrt{5} - 5} \\ &= -2\sqrt{5} - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-3\sqrt{5} - 1}{\sqrt{5} - 1} \\ &= -\sqrt{5} - 4\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{\sqrt{3}}{2\sqrt{2} - \sqrt{3}} \\ &= \frac{3 + 2\sqrt{6}}{5}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{2 + 5\sqrt{2}}{10 - 4\sqrt{2}} \\ &= \frac{30 + 29\sqrt{2}}{34}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-2\sqrt{3} - 5\sqrt{5}}{3\sqrt{5}} \\ &= \frac{-25 - 2\sqrt{15}}{15}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{4\sqrt{2} + 5\sqrt{5}}{3\sqrt{2} + 4\sqrt{5}} \\ &= \frac{76 + \sqrt{10}}{62}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{-\sqrt{5} - 2}{3\sqrt{5} - 4} \\ &= \frac{-23 - 10\sqrt{5}}{29}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{2\sqrt{2} + 5\sqrt{3}}{-3\sqrt{2} + 4\sqrt{3}} \\ &= \frac{72 + 23\sqrt{6}}{30}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{5\sqrt{2} + 2\sqrt{5}}{-5\sqrt{2} + 2\sqrt{5}} \\ &= \frac{-7 - 2\sqrt{10}}{3}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{5\sqrt{2} + 2\sqrt{3}}{-\sqrt{2} - 4\sqrt{3}} \\ &= \frac{-7 - 9\sqrt{6}}{23}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{-10 + 4\sqrt{2}}{-3\sqrt{2}} \\ &= \frac{-4 + 5\sqrt{2}}{3}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{2\sqrt{3}}{4\sqrt{3} + 5\sqrt{5}} \\ &= \frac{10\sqrt{15} - 24}{77}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{-2 - \sqrt{2}}{-8 - 5\sqrt{2}} \\ &= \frac{-\sqrt{2} + 3}{7}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{-\sqrt{2} - 4\sqrt{5}}{-3\sqrt{2} - 5\sqrt{5}} \\ &= \frac{94 - 7\sqrt{10}}{107}\end{aligned}$$

$$(1) \frac{-\sqrt{5}-5\sqrt{2}}{4\sqrt{5}} \\ = \frac{-1-\sqrt{10}}{4}$$

$$(6) \frac{-\sqrt{5}+2}{-2\sqrt{5}-4} \\ = \frac{-4\sqrt{5}+9}{2}$$

$$(11) \frac{-4\sqrt{3}-2}{-3\sqrt{3}-4} \\ = \frac{-10\sqrt{3}+28}{11}$$

$$(2) \frac{-1-\sqrt{2}}{-2-\sqrt{2}} \\ = \frac{\sqrt{2}}{2}$$

$$(7) \frac{-4\sqrt{5}-5}{-3\sqrt{5}} \\ = \frac{4+\sqrt{5}}{3}$$

$$(12) \frac{-1-3\sqrt{5}}{2-2\sqrt{5}} \\ = \frac{\sqrt{5}+4}{2}$$

$$(3) \frac{4-\sqrt{2}}{3+\sqrt{2}} \\ = -\sqrt{2}+2$$

$$(8) \frac{6+4\sqrt{3}}{6+\sqrt{3}} \\ = \frac{6\sqrt{3}+8}{11}$$

$$(13) \frac{2\sqrt{3}}{-\sqrt{3}+5\sqrt{2}} \\ = \frac{6+10\sqrt{6}}{47}$$

$$(4) \frac{-2\sqrt{5}}{5+2\sqrt{5}} \\ = -2\sqrt{5}+4$$

$$(9) \frac{-\sqrt{2}+5\sqrt{3}}{-4\sqrt{2}-2\sqrt{3}} \\ = \frac{-11\sqrt{6}+19}{10}$$

$$(14) \frac{2\sqrt{2}+3\sqrt{3}}{-5\sqrt{2}+3\sqrt{3}} \\ = \frac{-21\sqrt{6}-47}{23}$$

$$(5) \frac{8-\sqrt{2}}{2+5\sqrt{2}} \\ = \frac{-13+21\sqrt{2}}{23}$$

$$(10) \frac{-4\sqrt{5}+3\sqrt{3}}{2\sqrt{3}} \\ = \frac{-4\sqrt{15}+9}{6}$$

$$(15) \frac{-2\sqrt{5}-4\sqrt{2}}{-3\sqrt{2}} \\ = \frac{4+\sqrt{10}}{3}$$

$$(1) \frac{-3\sqrt{2} + 5\sqrt{5}}{4\sqrt{5}} \\ = \frac{25 - 3\sqrt{10}}{20}$$

$$(6) \frac{-1 - 5\sqrt{5}}{-2 + 3\sqrt{5}} \\ = \frac{-13\sqrt{5} - 77}{41}$$

$$(11) \frac{-5\sqrt{3} - 3\sqrt{5}}{-3\sqrt{3} - 2\sqrt{5}} \\ = \frac{15 - \sqrt{15}}{7}$$

$$(2) \frac{-3\sqrt{3}}{\sqrt{3} - 5\sqrt{5}} \\ = \frac{9 + 15\sqrt{15}}{122}$$

$$(7) \frac{1}{2\sqrt{5} + 1} \\ = \frac{2\sqrt{5} - 1}{19}$$

$$(12) \frac{-6 - \sqrt{5}}{-6 - \sqrt{5}} \\ = 1$$

$$(3) \frac{-4\sqrt{5}}{-5\sqrt{5} - \sqrt{2}} \\ = \frac{100 - 4\sqrt{10}}{123}$$

$$(8) \frac{2\sqrt{2} + 6}{5\sqrt{2} - 4} \\ = \frac{19\sqrt{2} + 22}{17}$$

$$(13) \frac{-4 + 4\sqrt{2}}{-1 - \sqrt{2}} \\ = 8\sqrt{2} - 12$$

$$(4) \frac{3\sqrt{2} - 10}{-4\sqrt{2} - 6} \\ = \frac{42 - 29\sqrt{2}}{2}$$

$$(9) \frac{2\sqrt{3} - \sqrt{5}}{-3\sqrt{3} - 5\sqrt{5}} \\ = \frac{-13\sqrt{15} + 43}{98}$$

$$(14) \frac{3\sqrt{2} + 3\sqrt{5}}{3\sqrt{2} + 2\sqrt{5}} \\ = \frac{12 - 3\sqrt{10}}{2}$$

$$(5) \frac{5\sqrt{2}}{4\sqrt{2} + 6} \\ = \frac{15\sqrt{2} - 20}{2}$$

$$(10) \frac{-4\sqrt{5}}{\sqrt{5} + \sqrt{3}} \\ = -10 + 2\sqrt{15}$$

$$(15) \frac{\sqrt{3} + 5}{2\sqrt{3} + 4} \\ = \frac{7 - 3\sqrt{3}}{2}$$

$$\begin{aligned} (1) \quad & \frac{8 - 2\sqrt{5}}{8 + 5\sqrt{5}} \\ &= \frac{56\sqrt{5} - 114}{61} \end{aligned}$$

$$\begin{aligned} (6) \quad & \frac{4\sqrt{5} - 2\sqrt{2}}{-2\sqrt{5} - 5\sqrt{2}} \\ &= \frac{10 - 4\sqrt{10}}{5} \end{aligned}$$

$$\begin{aligned} (11) \quad & \frac{2\sqrt{2} - 1}{-2\sqrt{2} - 2} \\ &= \frac{-5 + 3\sqrt{2}}{2} \end{aligned}$$

$$\begin{aligned} (2) \quad & \frac{2 + 2\sqrt{2}}{2 - 5\sqrt{2}} \\ &= \frac{-12 - 7\sqrt{2}}{23} \end{aligned}$$

$$\begin{aligned} (7) \quad & \frac{-4\sqrt{5} - 2\sqrt{3}}{\sqrt{5} - 5\sqrt{3}} \\ &= \frac{11\sqrt{15} + 25}{35} \end{aligned}$$

$$\begin{aligned} (12) \quad & \frac{-1 + \sqrt{5}}{-2 - 4\sqrt{5}} \\ &= \frac{3\sqrt{5} - 11}{38} \end{aligned}$$

$$\begin{aligned} (3) \quad & \frac{5\sqrt{5}}{5\sqrt{5} - 3} \\ &= \frac{15\sqrt{5} + 125}{116} \end{aligned}$$

$$\begin{aligned} (8) \quad & \frac{-4}{-\sqrt{3} - 2} \\ &= -4\sqrt{3} + 8 \end{aligned}$$

$$\begin{aligned} (13) \quad & \frac{\sqrt{5} + 8}{-4\sqrt{5}} \\ &= \frac{-5 - 8\sqrt{5}}{20} \end{aligned}$$

$$\begin{aligned} (4) \quad & \frac{-\sqrt{2} + 3}{2\sqrt{2} - 2} \\ &= \frac{1 + 2\sqrt{2}}{2} \end{aligned}$$

$$\begin{aligned} (9) \quad & \frac{-4\sqrt{3} + 10}{-\sqrt{3} + 4} \\ &= \frac{28 - 6\sqrt{3}}{13} \end{aligned}$$

$$\begin{aligned} (14) \quad & \frac{-4 - 5\sqrt{5}}{1 - 3\sqrt{5}} \\ &= \frac{17\sqrt{5} + 79}{44} \end{aligned}$$

$$\begin{aligned} (5) \quad & \frac{-1}{2\sqrt{3} - 4} \\ &= \frac{\sqrt{3} + 2}{2} \end{aligned}$$

$$\begin{aligned} (10) \quad & \frac{3\sqrt{2}}{-5\sqrt{3} + 4\sqrt{2}} \\ &= \frac{-15\sqrt{6} - 24}{43} \end{aligned}$$

$$\begin{aligned} (15) \quad & \frac{\sqrt{2} - 1}{\sqrt{2} - 3} \\ &= \frac{1 - 2\sqrt{2}}{7} \end{aligned}$$

$$(1) \frac{5-2\sqrt{3}}{3-\sqrt{3}} \\ = \frac{-\sqrt{3}+9}{6}$$

$$(6) \frac{5\sqrt{5}-5\sqrt{2}}{-\sqrt{5}-\sqrt{2}} \\ = \frac{-35+10\sqrt{10}}{3}$$

$$(11) \frac{4\sqrt{2}-4\sqrt{5}}{-3\sqrt{2}+3\sqrt{5}} \\ = \frac{-4}{3}$$

$$(2) \frac{-5}{-2\sqrt{5}+4} \\ = \frac{10+5\sqrt{5}}{2}$$

$$(7) \frac{3\sqrt{2}+2\sqrt{3}}{-2\sqrt{2}} \\ = \frac{-\sqrt{6}-3}{2}$$

$$(12) \frac{-3\sqrt{3}+6}{2\sqrt{3}-10} \\ = \frac{9\sqrt{3}-21}{44}$$

$$(3) \frac{-4}{2\sqrt{5}-1} \\ = \frac{-4-8\sqrt{5}}{19}$$

$$(8) \frac{\sqrt{5}}{-\sqrt{5}-\sqrt{3}} \\ = \frac{\sqrt{15}-5}{2}$$

$$(13) \frac{2\sqrt{2}+3\sqrt{3}}{-4\sqrt{2}+\sqrt{3}} \\ = \frac{-14\sqrt{6}-25}{29}$$

$$(4) \frac{\sqrt{3}+4}{-2\sqrt{3}+2} \\ = \frac{-7-5\sqrt{3}}{4}$$

$$(9) \frac{-6-\sqrt{3}}{-8-3\sqrt{3}} \\ = \frac{39-10\sqrt{3}}{37}$$

$$(14) \frac{-3\sqrt{2}-4}{-4\sqrt{2}-2} \\ = \frac{8+5\sqrt{2}}{14}$$

$$(5) \frac{\sqrt{5}+2}{4\sqrt{5}-4} \\ = \frac{3\sqrt{5}+7}{16}$$

$$(10) \frac{1-\sqrt{5}}{3-2\sqrt{5}} \\ = \frac{7+\sqrt{5}}{11}$$

$$(15) \frac{4\sqrt{2}}{\sqrt{3}+3\sqrt{2}} \\ = \frac{-4\sqrt{6}+24}{15}$$

$$(1) \frac{\sqrt{5} - \sqrt{2}}{-\sqrt{2}} \\ = \frac{-\sqrt{10} + 2}{2}$$

$$(6) \frac{-\sqrt{3} + 2\sqrt{5}}{2\sqrt{3} + 2\sqrt{5}} \\ = \frac{13 - 3\sqrt{15}}{4}$$

$$(11) \frac{\sqrt{3} + 4\sqrt{5}}{-4\sqrt{3} - \sqrt{5}} \\ = \frac{-15\sqrt{15} + 8}{43}$$

$$(2) \frac{-\sqrt{3} + 5\sqrt{2}}{\sqrt{3} + 2\sqrt{2}} \\ = \frac{-7\sqrt{6} + 23}{5}$$

$$(7) \frac{-2\sqrt{5} - 3\sqrt{2}}{\sqrt{5} - \sqrt{2}} \\ = \frac{-5\sqrt{10} - 16}{3}$$

$$(12) \frac{\sqrt{5}}{2\sqrt{5} + 4} \\ = \frac{-2\sqrt{5} + 5}{2}$$

$$(3) \frac{6 + 5\sqrt{5}}{10 + 3\sqrt{5}} \\ = \frac{-15 + 32\sqrt{5}}{55}$$

$$(8) \frac{-5\sqrt{2} + 8}{-\sqrt{2} + 2} \\ = -\sqrt{2} + 3$$

$$(13) \frac{2\sqrt{5} - 3\sqrt{2}}{-2\sqrt{5} + 2\sqrt{2}} \\ = \frac{-4 + \sqrt{10}}{6}$$

$$(4) \frac{-2 + 4\sqrt{3}}{8 - \sqrt{3}} \\ = \frac{30\sqrt{3} - 4}{61}$$

$$(9) \frac{-4\sqrt{5}}{-5\sqrt{5} - 3\sqrt{2}} \\ = \frac{-12\sqrt{10} + 100}{107}$$

$$(14) \frac{-\sqrt{2} - 2}{-\sqrt{2} - 5} \\ = \frac{3\sqrt{2} + 8}{23}$$

$$(5) \frac{-3\sqrt{3}}{\sqrt{3} + 3\sqrt{5}} \\ = \frac{-3\sqrt{15} + 3}{14}$$

$$(10) \frac{-2\sqrt{5}}{3\sqrt{3} - 3\sqrt{5}} \\ = \frac{5 + \sqrt{15}}{3}$$

$$(15) \frac{5\sqrt{2} + 2\sqrt{3}}{5\sqrt{2} + \sqrt{3}} \\ = \frac{44 + 5\sqrt{6}}{47}$$

$$(1) \frac{3 - \sqrt{2}}{4 + 3\sqrt{2}} \\ = \frac{-18 + 13\sqrt{2}}{2}$$

$$(6) \frac{\sqrt{5}}{\sqrt{5}} \\ = 1$$

$$(11) \frac{-4\sqrt{5} + 2\sqrt{3}}{3\sqrt{5} + 3\sqrt{3}} \\ = \frac{-13 + 3\sqrt{15}}{3}$$

$$(2) \frac{-\sqrt{3}}{-\sqrt{3} - \sqrt{5}} \\ = \frac{\sqrt{15} - 3}{2}$$

$$(7) \frac{-2\sqrt{3}}{5\sqrt{3} + 5} \\ = \frac{\sqrt{3} - 3}{5}$$

$$(12) \frac{4\sqrt{3} + 3\sqrt{5}}{-4\sqrt{5}} \\ = \frac{-15 - 4\sqrt{15}}{20}$$

$$(3) \frac{-2 - \sqrt{5}}{4 - 4\sqrt{5}} \\ = \frac{3\sqrt{5} + 7}{16}$$

$$(8) \frac{3}{3 + \sqrt{2}} \\ = \frac{-3\sqrt{2} + 9}{7}$$

$$(13) \frac{\sqrt{5}}{2\sqrt{3} - \sqrt{5}} \\ = \frac{5 + 2\sqrt{15}}{7}$$

$$(4) \frac{-3\sqrt{2} + 6}{4\sqrt{2} - 8} \\ = \frac{-3}{4}$$

$$(9) \frac{4\sqrt{2} + 2\sqrt{5}}{\sqrt{2} - 5\sqrt{5}} \\ = \frac{-22\sqrt{10} - 58}{123}$$

$$(14) \frac{-3\sqrt{2} + \sqrt{3}}{4\sqrt{2} + 3\sqrt{3}} \\ = \frac{-33 + 13\sqrt{6}}{5}$$

$$(5) \frac{-4 + 5\sqrt{3}}{5 + 3\sqrt{3}} \\ = \frac{65 - 37\sqrt{3}}{2}$$

$$(10) \frac{-6 - 5\sqrt{5}}{4 + 5\sqrt{5}} \\ = \frac{-101 - 10\sqrt{5}}{109}$$

$$(15) \frac{3}{\sqrt{3} + 1} \\ = \frac{3\sqrt{3} - 3}{2}$$

$$(1) \frac{-3\sqrt{3} - 4\sqrt{2}}{3\sqrt{3} - 2\sqrt{2}} \\ = \frac{-18\sqrt{6} - 43}{19}$$

$$(6) \frac{6 + 3\sqrt{5}}{-2 + 4\sqrt{5}} \\ = \frac{15\sqrt{5} + 36}{38}$$

$$(11) \frac{3\sqrt{2} - 3\sqrt{3}}{-3\sqrt{2} + \sqrt{3}} \\ = \frac{2\sqrt{6} - 3}{5}$$

$$(2) \frac{5 - \sqrt{5}}{-4 + 4\sqrt{5}} \\ = \frac{\sqrt{5}}{4}$$

$$(7) \frac{-5\sqrt{5} - 5}{-3\sqrt{5} - 3} \\ = \frac{5}{3}$$

$$(12) \frac{-3\sqrt{3} - 5\sqrt{2}}{-3\sqrt{3} - \sqrt{2}} \\ = \frac{12\sqrt{6} + 17}{25}$$

$$(3) \frac{6 + 2\sqrt{2}}{-4 - 3\sqrt{2}} \\ = 6 - 5\sqrt{2}$$

$$(8) \frac{-5\sqrt{2}}{-5\sqrt{3} + 4\sqrt{2}} \\ = \frac{40 + 25\sqrt{6}}{43}$$

$$(13) \frac{-4}{-10 - 3\sqrt{2}} \\ = \frac{20 - 6\sqrt{2}}{41}$$

$$(4) \frac{-2 - 4\sqrt{5}}{-1 + 3\sqrt{5}} \\ = \frac{-5\sqrt{5} - 31}{22}$$

$$(9) \frac{-2\sqrt{3} - 3\sqrt{2}}{2\sqrt{3} + \sqrt{2}} \\ = \frac{-3 - 2\sqrt{6}}{5}$$

$$(14) \frac{\sqrt{2} - 4}{-\sqrt{2} + 2} \\ = -3 - \sqrt{2}$$

$$(5) \frac{\sqrt{2} - 5\sqrt{3}}{\sqrt{2} - 2\sqrt{3}} \\ = \frac{28 + 3\sqrt{6}}{10}$$

$$(10) \frac{2\sqrt{3} - 3\sqrt{2}}{3\sqrt{3} + 2\sqrt{2}} \\ = \frac{-13\sqrt{6} + 30}{19}$$

$$(15) \frac{-1 - 4\sqrt{5}}{4 + 2\sqrt{5}} \\ = \frac{-18 + 7\sqrt{5}}{2}$$

$$(1) \frac{-2}{\sqrt{2}-2} \\ = 2 + \sqrt{2}$$

$$(6) \frac{-\sqrt{2}+4\sqrt{5}}{\sqrt{5}} \\ = \frac{20-\sqrt{10}}{5}$$

$$(11) \frac{4\sqrt{3}+1}{\sqrt{3}-3} \\ = \frac{-15-13\sqrt{3}}{6}$$

$$(2) \frac{-2-4\sqrt{2}}{-2+5\sqrt{2}} \\ = \frac{-9\sqrt{2}-22}{23}$$

$$(7) \frac{-5}{-5-4\sqrt{2}} \\ = \frac{-25+20\sqrt{2}}{7}$$

$$(12) \frac{3\sqrt{3}+\sqrt{5}}{4\sqrt{3}-\sqrt{5}} \\ = \frac{7\sqrt{15}+41}{43}$$

$$(3) \frac{3+4\sqrt{5}}{2\sqrt{5}} \\ = \frac{3\sqrt{5}+20}{10}$$

$$(8) \frac{2+\sqrt{3}}{5\sqrt{3}} \\ = \frac{3+2\sqrt{3}}{15}$$

$$(13) \frac{-5\sqrt{2}-2\sqrt{3}}{2\sqrt{2}+3\sqrt{3}} \\ = \frac{-11\sqrt{6}+2}{19}$$

$$(4) \frac{5\sqrt{3}-4}{-3\sqrt{3}-1} \\ = \frac{-49+17\sqrt{3}}{26}$$

$$(9) \frac{-5\sqrt{2}}{10-2\sqrt{2}} \\ = \frac{-10-25\sqrt{2}}{46}$$

$$(14) \frac{\sqrt{2}+2\sqrt{3}}{-\sqrt{2}} \\ = -1-\sqrt{6}$$

$$(5) \frac{-1+3\sqrt{5}}{-4\sqrt{5}} \\ = \frac{-15+\sqrt{5}}{20}$$

$$(10) \frac{-4\sqrt{3}+5\sqrt{2}}{5\sqrt{3}-4\sqrt{2}} \\ = \frac{9\sqrt{6}-20}{43}$$

$$(15) \frac{-2-3\sqrt{5}}{8-4\sqrt{5}} \\ = \frac{19+8\sqrt{5}}{4}$$

$$\begin{aligned}(1) \quad & \frac{2}{1+4\sqrt{3}} \\ &= \frac{8\sqrt{3}-2}{47}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{-4+4\sqrt{2}}{4-\sqrt{2}} \\ &= \frac{-4+6\sqrt{2}}{7}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-\sqrt{3}}{4\sqrt{3}} \\ &= \frac{-1}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{3\sqrt{5}-4\sqrt{2}}{-2\sqrt{5}+5\sqrt{2}} \\ &= \frac{-10+7\sqrt{10}}{30}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{2\sqrt{2}}{-2\sqrt{5}-\sqrt{2}} \\ &= \frac{2-2\sqrt{10}}{9}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{\sqrt{5}+2\sqrt{2}}{2\sqrt{2}} \\ &= \frac{4+\sqrt{10}}{4}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{-4\sqrt{5}}{3\sqrt{2}-3\sqrt{5}} \\ &= \frac{20+4\sqrt{10}}{9}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{\sqrt{2}+8}{-3\sqrt{2}-4} \\ &= -10\sqrt{2}+13\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-4\sqrt{2}-3}{-\sqrt{2}+2} \\ &= \frac{-14-11\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{\sqrt{2}+2}{-\sqrt{2}+3} \\ &= \frac{5\sqrt{2}+8}{7}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{-2-3\sqrt{2}}{3-4\sqrt{2}} \\ &= \frac{30+17\sqrt{2}}{23}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{5-5\sqrt{5}}{4+5\sqrt{5}} \\ &= \frac{45\sqrt{5}-145}{109}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{\sqrt{3}+8}{-3\sqrt{3}-4} \\ &= \frac{23-20\sqrt{3}}{11}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{\sqrt{5}+4}{3\sqrt{5}+5} \\ &= \frac{-5+7\sqrt{5}}{20}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{3\sqrt{5}+6}{5\sqrt{5}} \\ &= \frac{15+6\sqrt{5}}{25}\end{aligned}$$

$$(1) \frac{3\sqrt{3} + 5\sqrt{2}}{-3\sqrt{3} + \sqrt{2}} \\ = \frac{-18\sqrt{6} - 37}{25}$$

$$(6) \frac{-5 + \sqrt{5}}{-3 - 2\sqrt{5}} \\ = \frac{-25 + 13\sqrt{5}}{11}$$

$$(11) \frac{-1 - \sqrt{2}}{-4 + \sqrt{2}} \\ = \frac{6 + 5\sqrt{2}}{14}$$

$$(2) \frac{-2}{-\sqrt{2} + 2} \\ = -2 - \sqrt{2}$$

$$(7) \frac{3\sqrt{3} + \sqrt{5}}{5\sqrt{3} + 2\sqrt{5}} \\ = \frac{35 - \sqrt{15}}{55}$$

$$(12) \frac{\sqrt{3}}{-6 - 2\sqrt{3}} \\ = \frac{-\sqrt{3} + 1}{4}$$

$$(3) \frac{1}{-4 - \sqrt{5}} \\ = \frac{\sqrt{5} - 4}{11}$$

$$(8) \frac{-4\sqrt{3} - 3\sqrt{2}}{-3\sqrt{2}} \\ = \frac{2\sqrt{6} + 3}{3}$$

$$(13) \frac{3\sqrt{5} - \sqrt{3}}{-2\sqrt{5} + 5\sqrt{3}} \\ = \frac{13\sqrt{15} + 15}{55}$$

$$(4) \frac{-6 + \sqrt{5}}{-6 + 2\sqrt{5}} \\ = \frac{3\sqrt{5} + 13}{8}$$

$$(9) \frac{5 + \sqrt{5}}{2 + 5\sqrt{5}} \\ = \frac{15 + 23\sqrt{5}}{121}$$

$$(14) \frac{-5\sqrt{2} - 8}{-2\sqrt{2} - 4} \\ = \frac{\sqrt{2} + 3}{2}$$

$$(5) \frac{2\sqrt{5} + 4}{-5\sqrt{5} + 3} \\ = \frac{-31 - 13\sqrt{5}}{58}$$

$$(10) \frac{-3\sqrt{5} + 5\sqrt{2}}{2\sqrt{5} + 3\sqrt{2}} \\ = \frac{-60 + 19\sqrt{10}}{2}$$

$$(15) \frac{5\sqrt{5} - 4\sqrt{3}}{-3\sqrt{5} - 3\sqrt{3}} \\ = \frac{-37 + 9\sqrt{15}}{6}$$

$$(1) \frac{\sqrt{2}}{-5\sqrt{2} - 3\sqrt{5}} \\ = \frac{3\sqrt{10} - 10}{5}$$

$$(6) \frac{-2 + \sqrt{3}}{-4\sqrt{3}} \\ = \frac{-3 + 2\sqrt{3}}{12}$$

$$(11) \frac{\sqrt{5} + 10}{-\sqrt{5} - 6} \\ = \frac{4\sqrt{5} - 55}{31}$$

$$(2) \frac{-2 + 2\sqrt{5}}{-1 + \sqrt{5}} \\ = 2$$

$$(7) \frac{4\sqrt{3} + 3\sqrt{5}}{-\sqrt{3} - 5\sqrt{5}} \\ = \frac{-17\sqrt{15} - 63}{122}$$

$$(12) \frac{3\sqrt{5} - 10}{-4\sqrt{5} - 6} \\ = \frac{-60 + 29\sqrt{5}}{22}$$

$$(3) \frac{3\sqrt{5} + 3\sqrt{2}}{\sqrt{5} + \sqrt{2}} \\ = 3$$

$$(8) \frac{5 - 5\sqrt{2}}{2 + 5\sqrt{2}} \\ = \frac{-60 + 35\sqrt{2}}{46}$$

$$(13) \frac{4\sqrt{5} + 5\sqrt{3}}{\sqrt{5} - \sqrt{3}} \\ = \frac{35 + 9\sqrt{15}}{2}$$

$$(4) \frac{-3\sqrt{5} - 2}{5\sqrt{5} - 3} \\ = \frac{-19\sqrt{5} - 81}{116}$$

$$(9) \frac{-4\sqrt{5} + 5\sqrt{2}}{-2\sqrt{5} - 5\sqrt{2}} \\ = -3 + \sqrt{10}$$

$$(14) \frac{5 + 3\sqrt{2}}{-3 - 2\sqrt{2}} \\ = -3 + \sqrt{2}$$

$$(5) \frac{\sqrt{3} + \sqrt{5}}{-4\sqrt{3} + 4\sqrt{5}} \\ = \frac{\sqrt{15} + 4}{4}$$

$$(10) \frac{4\sqrt{5} + 2\sqrt{3}}{3\sqrt{5} - 3\sqrt{3}} \\ = \frac{13 + 3\sqrt{15}}{3}$$

$$(15) \frac{\sqrt{3} + 2}{-2\sqrt{3}} \\ = \frac{-3 - 2\sqrt{3}}{6}$$

$$\begin{aligned}(1) \quad & \frac{2\sqrt{5} - 3\sqrt{3}}{-4\sqrt{5} - 3\sqrt{3}} \\ &= \frac{-67 + 18\sqrt{15}}{53}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{4\sqrt{3} + \sqrt{5}}{4\sqrt{5}} \\ &= \frac{5 + 4\sqrt{15}}{20}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{2 - 5\sqrt{5}}{3 - \sqrt{5}} \\ &= \frac{-13\sqrt{5} - 19}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{2\sqrt{2} + 2\sqrt{5}}{-\sqrt{2} + \sqrt{5}} \\ &= \frac{4\sqrt{10} + 14}{3}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{5 + 3\sqrt{2}}{-3 - 3\sqrt{2}} \\ &= \frac{-1 - 2\sqrt{2}}{3}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{\sqrt{3} + 2\sqrt{5}}{-2\sqrt{3} + 2\sqrt{5}} \\ &= \frac{3\sqrt{15} + 13}{4}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{4\sqrt{3} - 3\sqrt{2}}{4\sqrt{3} - 3\sqrt{2}} \\ &= 1\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{-2\sqrt{5} + 2\sqrt{2}}{5\sqrt{5} - 3\sqrt{2}} \\ &= \frac{4\sqrt{10} - 38}{107}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-4 + 5\sqrt{3}}{-1 - 3\sqrt{3}} \\ &= \frac{-49 + 17\sqrt{3}}{26}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{-3\sqrt{2} - 8}{4\sqrt{2} - 6} \\ &= \frac{36 + 25\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{3\sqrt{2} + 8}{-4\sqrt{2}} \\ &= \frac{-3 - 4\sqrt{2}}{4}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{5 - 4\sqrt{3}}{-4 + 4\sqrt{3}} \\ &= \frac{\sqrt{3} - 7}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{2\sqrt{5} + 2}{3\sqrt{5} - 4} \\ &= \frac{38 + 14\sqrt{5}}{29}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{-2 + 4\sqrt{2}}{3 + 3\sqrt{2}} \\ &= \frac{10 - 6\sqrt{2}}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{3 - 2\sqrt{5}}{3 - 3\sqrt{5}} \\ &= \frac{7 - \sqrt{5}}{12}\end{aligned}$$

$$\begin{aligned}(1) \quad & \frac{3\sqrt{3}-2}{\sqrt{3}} \\ &= \frac{9-2\sqrt{3}}{3}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{\sqrt{2}}{1+\sqrt{2}} \\ &= -\sqrt{2}+2\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-5\sqrt{3}-1}{-3\sqrt{3}-4} \\ &= \frac{-17\sqrt{3}+41}{11}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{3\sqrt{2}}{3\sqrt{2}+4\sqrt{3}} \\ &= \frac{2\sqrt{6}-3}{5}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{4+\sqrt{5}}{1+2\sqrt{5}} \\ &= \frac{6+7\sqrt{5}}{19}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-10+\sqrt{3}}{-6+3\sqrt{3}} \\ &= \frac{17+8\sqrt{3}}{3}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{5\sqrt{5}-\sqrt{3}}{\sqrt{5}+2\sqrt{3}} \\ &= \frac{-31+11\sqrt{15}}{7}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{-3}{5\sqrt{5}+3} \\ &= \frac{-15\sqrt{5}+9}{116}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-2-4\sqrt{3}}{-1+5\sqrt{3}} \\ &= \frac{-31-7\sqrt{3}}{37}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{-2\sqrt{2}+2}{2\sqrt{2}-1} \\ &= \frac{-6+2\sqrt{2}}{7}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{2\sqrt{3}-1}{4\sqrt{3}+2} \\ &= \frac{13-4\sqrt{3}}{22}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{3\sqrt{2}}{4-3\sqrt{2}} \\ &= -9-6\sqrt{2}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{2}{-\sqrt{2}+2} \\ &= 2+\sqrt{2}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{-\sqrt{5}-2}{-3\sqrt{5}+10} \\ &= \frac{-16\sqrt{5}-35}{55}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{-4\sqrt{5}}{5\sqrt{5}} \\ &= \frac{-4}{5}\end{aligned}$$

1.5 複素数

1.5.1 加減法

複素数の和や差を求める問題。

(1) $(-10 - 2i) + (-5 - 4i)$

(20) $(-5 - i) - (-9 - 3i)$

(39) $(-3 + 4i) + (8 + 5i)$

(2) $(2 + 7i) - (3 - 8i)$

(21) $(2 + 5i) + (2 - 10i)$

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(6) $(5 - 5i) + (7 + 5i)$

(25) $(-8 + 9i) - (-9 + 7i)$

(44) $(6 + 10i) - (8 - 4i)$

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(46) $(-2 + 3i) - (-8 - 10i)$

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(24) $(4 + 8i) + (-1 + 5i)$

(43) $(8 + 4i) + (-5 + 3i)$

(6) $(10 + 3i) + (-7 - 8i)$

(25) $(-9 - 10i) + (-3 + 8i)$

(44) $(8 - 6i) - (-7 - 4i)$

(7) $(-8 + 7i) - (-6 - 7i)$

(26) $(3 + 10i) + (-9 - 4i)$

(45) $(4 - 10i) + (-9 - 10i)$

(8) $(-2 + 8i) - (6 - i)$

(27) $(9 - 4i) - (-8 - 9i)$

(46) $(3 + 6i) - (-10 + 7i)$

(9) $(-5 + 5i) - (-4 + i)$

(28) $(1 - 3i) + (8 - 3i)$

(47) $(-9 + 10i) + (-8 - 7i)$

(10) $(-5 + 6i) - (2 - 3i)$

(29) $(-4 - 7i) - (9 + 2i)$

(48) $(1 + 6i) + (-8 - 8i)$

(11) $(6 + 2i) + (-4 + 8i)$

(30) $(8 + 7i) - (-1 + 3i)$

(49) $(7 + 6i) - (9 + 8i)$

(12) $(-10 + 6i) - (9 - 10i)$

(31) $(-3 + 2i) - (6 + 8i)$

(50) $(-8 + 10i) - (-7 - 10i)$

(13) $(6 + i) - (7 + 4i)$

(32) $(7 - 9i) - (-10 - 6i)$

(51) $(9 - 6i) - (3 + 10i)$

(14) $(-7 - 2i) + (10 + 7i)$

(33) $(-5 + 8i) + (4 - 6i)$

(52) $(-1 + 5i) + (8 + i)$

(15) $(8 - 6i) + (-7 - 6i)$

(34) $(7 - 5i) + (8 + 10i)$

(53) $(6 + i) + (-5 + 10i)$

(16) $(-1 + 5i) - (-2 - 10i)$

(35) $(10 + 2i) + (7 + 10i)$

(54) $(-6 + i) + (8 - 6i)$

(17) $(-6 + 10i) + (9 - 7i)$

(36) $(9 + 6i) + (7 - 2i)$

(55) $(6 - 5i) - (7 + 6i)$

(18) $(-6 + 3i) - (-7 + 3i)$

(37) $(-7 - 8i) + (-4 + 7i)$

(56) $(2 - 6i) + (8 + 5i)$

(19) $(-5 - i) - (7 + 6i)$

(38) $(-2 + 4i) + (9 + 9i)$

(57) $(3 - 7i) - (7 - 6i)$

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|-------------------------------|-------------------------------|--------------------------------|
| (1) $(9 - 3i) + (9 + 5i)$ | (20) $(-3 + 7i) + (10 - 2i)$ | (39) $(-6 + 6i) - (-8 + 4i)$ |
| (2) $(-5 + 6i) - (-8 - i)$ | (21) $(-10 - 7i) + (-1 + 7i)$ | (40) $(8 - 2i) - (2 - 7i)$ |
| (3) $(-7 + 4i) - (-4 - 8i)$ | (22) $(-3 - 10i) + (-8 + 2i)$ | (41) $(-7 + 7i) + (8 + 9i)$ |
| (4) $(-9 - 9i) + (1 - 4i)$ | (23) $(3 + 2i) + (-3 + 7i)$ | (42) $(10 - 2i) + (10 - 10i)$ |
| (5) $(7 - i) + (6 - 9i)$ | (24) $(-10 + 9i) - (-8 + 5i)$ | (43) $(9 - 7i) - (10 - 5i)$ |
| (6) $(5 + i) - (-2 + 4i)$ | (25) $(8 + 10i) + (-10 - 4i)$ | (44) $(-9 - 8i) - (-4 + 5i)$ |
| (7) $(-9 - 10i) - (1 + 3i)$ | (26) $(-8 + 3i) - (-3 - 7i)$ | (45) $(5 + 7i) - (1 - 8i)$ |
| (8) $(-2 + 10i) - (-5 - 5i)$ | (27) $(6 + 7i) - (-1 - i)$ | (46) $(6 + 6i) - (-10 - 4i)$ |
| (9) $(-4 + 9i) - (-4 + 8i)$ | (28) $(-8 + 10i) + (-3 - 9i)$ | (47) $(1 - 8i) - (-10 + 2i)$ |
| (10) $(-1 + 7i) - (-4 + 7i)$ | (29) $(4 - 9i) + (-10 + 6i)$ | (48) $(-10 - 8i) + (6 - 6i)$ |
| (11) $(-8 + 4i) - (3 - 8i)$ | (30) $(-7 + 3i) + (1 + i)$ | (49) $(-8 - 7i) + (4 + 3i)$ |
| (12) $(9 - i) - (-8 + 4i)$ | (31) $(-8 - 3i) - (7 - 7i)$ | (50) $(5 - 10i) + (7 - 9i)$ |
| (13) $(-9 + 2i) + (-9 + i)$ | (32) $(3 + 7i) + (9 + 6i)$ | (51) $(8 - 9i) + (1 - 3i)$ |
| (14) $(-9 + 5i) - (6 - i)$ | (33) $(-2 - i) + (-10 + 9i)$ | (52) $(3 + 6i) - (6 + 6i)$ |
| (15) $(-9 - 2i) - (-5 - 10i)$ | (34) $(-2 + 8i) - (9 - 5i)$ | (53) $(-3 + 10i) - (10 + i)$ |
| (16) $(8 - 4i) + (8 - 4i)$ | (35) $(-4 + 3i) - (6 + 8i)$ | (54) $(-3 + 8i) - (-5 - 10i)$ |
| (17) $(-9 + 10i) - (-8 + 2i)$ | (36) $(-10 + 8i) + (-3 + 6i)$ | (55) $(6 - 7i) + (-3 + i)$ |
| (18) $(8 + 10i) + (2 + 10i)$ | (37) $(-4 + 7i) - (-2 + 9i)$ | (56) $(9 - 7i) + (-7 + 10i)$ |
| (19) $(-5 + 3i) + (-8 - 4i)$ | (38) $(-1 - 6i) - (-4 + 2i)$ | (57) $(-10 - 2i) + (-5 + 10i)$ |

(1) $(6 + 8i) + (-9 - 10i)$

(20) $(-6 - 6i) - (-2 - 8i)$

(39) $(3 + 8i) - (-4 - i)$

(2) $(-4 - 9i) + (4 - 2i)$

(21) $(10 - 6i) - (-8 + 9i)$

(40) $(-6 - i) + (-9 + 6i)$

(3) $(6 - 2i) + (-4 - 5i)$

(22) $(-9 + i) + (6 + 7i)$

(41) $(-2 - 7i) + (-3 + i)$

(4) $(5 + 8i) - (8 + i)$

(23) $(9 - 8i) - (7 + 5i)$

(42) $(-3 + i) + (1 + 2i)$

(5) $(-7 + 2i) - (10 - 8i)$

(24) $(7 - 6i) - (4 + 4i)$

(43) $(-6 + 8i) + (6 + 4i)$

(6) $(-7 - i) - (10 - 6i)$

(25) $(10 - 2i) - (9 + 7i)$

(44) $(3 + 5i) + (-7 - 6i)$

(7) $(5 + 8i) - (-10 + i)$

(26) $(-9 - 5i) + (3 - 8i)$

(45) $(-9 - 2i) + (-8 - 3i)$

(8) $(-9 + 3i) - (10 - 4i)$

(27) $(-3 + 6i) - (-7 - 7i)$

(46) $(7 + 7i) - (-5 + 8i)$

(9) $(4 + 8i) + (-3 + 5i)$

(28) $(5 - 9i) - (-8 + 8i)$

(47) $(-9 - 2i) + (8 - 5i)$

(10) $(-10 - 10i) + (5 + i)$

(29) $(4 + 2i) + (4 - 9i)$

(48) $(6 + 2i) + (-3 + i)$

(11) $(-2 + 10i) + (-9 - i)$

(30) $(1 - 6i) + (-4 + 3i)$

(49) $(-1 - 10i) - (-3 + 2i)$

(12) $(5 + 4i) + (10 - 8i)$

(31) $(9 + 7i) - (5 + 2i)$

(50) $(5 + 6i) + (-4 - 9i)$

(13) $(-1 + 5i) + (4 - i)$

(32) $(3 + 8i) - (-1 + 4i)$

(51) $(-10 + 6i) - (4 + 4i)$

(14) $(2 + 9i) - (-4 - i)$

(33) $(6 - 6i) + (4 + 5i)$

(52) $(8 + 6i) - (-1 + 10i)$

(15) $(-9 + 6i) - (-3 - 4i)$

(34) $(-3 + 8i) + (-3 - 7i)$

(53) $(-5 + 6i) - (-2 - 3i)$

(16) $(3 + 6i) - (2 - 4i)$

(35) $(1 + 6i) - (-9 + 4i)$

(54) $(-10 + 4i) + (-4 - 10i)$

(17) $(-5 - 3i) - (6 - 10i)$

(36) $(-9 + 5i) - (1 + 6i)$

(55) $(7 + i) - (-3 + 3i)$

(18) $(2 + 6i) - (4 + 2i)$

(37) $(3 + 8i) - (1 + 7i)$

(56) $(2 - 6i) - (8 + 4i)$

(19) $(-2 + 4i) - (-5 + 9i)$

(38) $(10 - 3i) + (-1 - 5i)$

(57) $(-6 + 3i) + (3 + 8i)$

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|------------------------------|--------------------------------|-------------------------------|
| (1) $(-7 + 4i) - (-8 - 10i)$ | (20) $(-7 + 7i) + (-2 - 8i)$ | (39) $(6 + 9i) + (7 + 5i)$ |
| (2) $(-1 - i) - (-8 + 3i)$ | (21) $(-6 - i) + (9 + 7i)$ | (40) $(6 + 10i) - (7 - 5i)$ |
| (3) $(-6 + 7i) + (-3 + i)$ | (22) $(-5 - 4i) - (-3 - 7i)$ | (41) $(9 - 4i) + (1 + 10i)$ |
| (4) $(-3 - 2i) - (5 - 7i)$ | (23) $(-5 + 2i) - (-5 - 2i)$ | (42) $(-3 + 2i) + (-9 - 10i)$ |
| (5) $(1 + 7i) - (-10 - 4i)$ | (24) $(-7 - 2i) + (2 + 7i)$ | (43) $(-10 - 8i) - (6 + 6i)$ |
| (6) $(5 + i) + (-1 + 2i)$ | (25) $(-6 - 3i) + (-9 - i)$ | (44) $(-2 + 5i) + (-6 + 2i)$ |
| (7) $(-3 - 10i) - (9 + 9i)$ | (26) $(-3 + 3i) + (3 - 7i)$ | (45) $(7 + 4i) + (-4 - 8i)$ |
| (8) $(8 + 10i) - (8 + 3i)$ | (27) $(-2 + 5i) + (10 - 2i)$ | (46) $(8 - 6i) + (-7 + 10i)$ |
| (9) $(9 - 10i) - (10 - 6i)$ | (28) $(-2 + 3i) - (4 + 8i)$ | (47) $(4 + 10i) + (-8 - 2i)$ |
| (10) $(-2 - 8i) - (-8 - 7i)$ | (29) $(-5 - i) - (8 + 8i)$ | (48) $(7 + i) - (10 + 8i)$ |
| (11) $(-8 + 3i) - (4 - i)$ | (30) $(1 + i) + (-6 + 2i)$ | (49) $(-8 + 3i) - (2 - 2i)$ |
| (12) $(1 - 8i) + (-7 + 7i)$ | (31) $(10 - 7i) - (6 - 6i)$ | (50) $(-8 - 2i) + (-5 + 8i)$ |
| (13) $(5 + 4i) + (-8 + 2i)$ | (32) $(9 + i) + (10 + 10i)$ | (51) $(1 + 10i) - (-8 - 4i)$ |
| (14) $(3 + 4i) + (-1 + 2i)$ | (33) $(2 - 2i) - (8 + 7i)$ | (52) $(-6 + i) - (4 + 5i)$ |
| (15) $(-9 + 9i) + (4 - 8i)$ | (34) $(2 - 4i) + (5 - 2i)$ | (53) $(-3 - 10i) + (-8 - 6i)$ |
| (16) $(-3 - i) + (6 + 7i)$ | (35) $(-7 + 7i) + (-10 - 6i)$ | (54) $(-1 - 3i) + (6 - 8i)$ |
| (17) $(4 - 5i) - (-5 + 7i)$ | (36) $(6 - 2i) + (-3 - 2i)$ | (55) $(5 - 5i) - (10 + 3i)$ |
| (18) $(2 - 8i) + (-1 - 3i)$ | (37) $(-7 - 4i) - (-9 - 5i)$ | (56) $(-1 - 10i) - (-2 + 9i)$ |
| (19) $(-9 - 5i) + (-3 - 2i)$ | (38) $(-5 + 8i) + (-10 - 10i)$ | (57) $(1 + i) - (-5 - 8i)$ |

(1) $(-6 - 10i) + (-2 + 3i)$

(20) $(-4 + i) + (3 + 9i)$

(39) $(5 - 8i) + (9 - i)$

(2) $(-7 + 4i) - (-5 - 5i)$

(21) $(-8 - 3i) - (-3 + 6i)$

(40) $(-7 - 10i) + (-10 + 5i)$

(3) $(2 - 3i) + (-10 - i)$

(22) $(2 - 8i) + (4 + 6i)$

(41) $(-4 - i) - (4 + 7i)$

(4) $(7 - 4i) + (-3 + 8i)$

(23) $(9 + 7i) + (2 + 2i)$

(42) $(8 + 4i) + (5 - 10i)$

(5) $(-4 - 6i) - (-2 + 9i)$

(24) $(6 + 9i) - (-10 - 6i)$

(43) $(-9 - 10i) + (10 - 4i)$

(6) $(-2 - 10i) - (-1 - i)$

(25) $(2 + 7i) + (-2 - 5i)$

(44) $(4 + 6i) + (4 - 5i)$

(7) $(-1 - i) + (-5 - 10i)$

(26) $(-8 - 9i) + (3 - 10i)$

(45) $(-7 + 9i) + (-10 + 6i)$

(8) $(7 + 3i) + (6 - 4i)$

(27) $(6 + 4i) - (5 + 2i)$

(46) $(7 + 3i) + (-8 + 3i)$

(9) $(1 - 4i) - (-7 + 2i)$

(28) $(-1 - 10i) - (-6 + 10i)$

(47) $(-2 + 2i) + (6 + 6i)$

(10) $(-6 - 9i) + (1 - 9i)$

(29) $(6 - 6i) + (-8 - 8i)$

(48) $(8 + 7i) + (-9 - 2i)$

(11) $(9 + 10i) + (9 - 3i)$

(30) $(9 - 10i) + (1 + 6i)$

(49) $(-3 + 6i) + (7 + 5i)$

(12) $(-3 + 10i) + (5 - 6i)$

(31) $(-8 - 7i) - (-6 + 2i)$

(50) $(9 + 8i) - (-5 + 6i)$

(13) $(-1 + i) - (-2 - 7i)$

(32) $(-2 - 4i) + (1 - 2i)$

(51) $(8 + 7i) - (7 - 9i)$

(14) $(6 - i) - (-10 + 3i)$

(33) $(3 - 2i) + (6 + 7i)$

(52) $(3 - 2i) - (-6 - 8i)$

(15) $(10 + 10i) - (-3 - 3i)$

(34) $(5 + 2i) + (-3 + 5i)$

(53) $(3 + 6i) + (-7 - 7i)$

(16) $(2 - 2i) - (4 + 6i)$

(35) $(-5 + 2i) + (-7 - 3i)$

(54) $(-4 + 8i) - (-10 - i)$

(17) $(10 - 7i) - (-4 + 4i)$

(36) $(-8 - 8i) + (5 + 5i)$

(55) $(2 - 10i) + (-4 + 9i)$

(18) $(-3 + 3i) + (7 + 10i)$

(37) $(-8 - 3i) - (2 + 6i)$

(56) $(-4 + 10i) - (8 + 7i)$

(19) $(-1 - 5i) - (1 - i)$

(38) $(6 + i) - (-4 + 7i)$

(57) $(4 - 8i) + (8 - 4i)$

(1) $(-1 - 4i) - (-3 - 9i)$

(20) $(3 - 8i) + (9 - i)$

(39) $(-6 + 10i) + (-6 + 2i)$

(2) $(-5 + 3i) + (3 - 8i)$

(21) $(-3 + 6i) - (-1 - 2i)$

(40) $(-10 + 10i) + (-10 + 10i)$

(3) $(1 + 4i) + (9 + 5i)$

(22) $(-3 + 10i) - (-6 - 9i)$

(41) $(-1 + 4i) + (6 - 8i)$

(4) $(-2 + 7i) + (3 - 9i)$

(23) $(-6 + 2i) + (-2 + 10i)$

(42) $(1 - 3i) + (10 + 3i)$

(5) $(3 - 7i) - (-2 + i)$

(24) $(-8 - 7i) + (4 - 5i)$

(43) $(3 - 6i) - (-4 + 10i)$

(6) $(8 + 5i) + (-3 + 5i)$

(25) $(-8 + 9i) + (6 - 9i)$

(44) $(9 - 4i) + (-10 + 10i)$

(7) $(10 + 10i) - (-10 - 7i)$

(26) $(6 + 7i) - (-6 - 8i)$

(45) $(5 + 3i) + (3 - 6i)$

(8) $(-2 + 9i) + (10 + 3i)$

(27) $(-3 - 3i) - (7 - 8i)$

(46) $(5 - 8i) - (-4 + 10i)$

(9) $(7 - 5i) + (-8 - 9i)$

(28) $(-7 + 6i) - (3 - 5i)$

(47) $(2 - 10i) + (-4 - 10i)$

(10) $(10 - 7i) + (-7 + 8i)$

(29) $(-1 + i) + (-9 - 4i)$

(48) $(-5 - 8i) + (9 - 9i)$

(11) $(5 + 3i) + (-8 + 2i)$

(30) $(-5 + 7i) - (3 + 4i)$

(49) $(4 + i) + (7 + 9i)$

(12) $(7 + 6i) + (5 - 6i)$

(31) $(-7 + 4i) + (-8 - 7i)$

(50) $(8 - 9i) - (1 - 8i)$

(13) $(-9 - i) - (-10 + 2i)$

(32) $(10 + 7i) + (9 - i)$

(51) $(-5 - 2i) - (6 - 6i)$

(14) $(-7 - i) - (-1 - 8i)$

(33) $(6 + 5i) + (-9 - 2i)$

(52) $(9 - 6i) - (-1 + 2i)$

(15) $(-10 - 7i) - (1 + 8i)$

(34) $(-3 - 9i) - (2 - 9i)$

(53) $(-5 - 9i) - (2 + 2i)$

(16) $(-10 - i) + (8 - 9i)$

(35) $(9 - 3i) + (2 - 4i)$

(54) $(-9 - 8i) + (-1 + 5i)$

(17) $(10 + 3i) + (-2 + i)$

(36) $(9 + 3i) + (-5 - 2i)$

(55) $(-8 - 6i) - (-9 + 3i)$

(18) $(-2 - i) + (5 + 8i)$

(37) $(8 + 6i) + (-10 + 5i)$

(56) $(-8 - 7i) - (-4 + 7i)$

(19) $(-8 - 2i) - (-5 - 6i)$

(38) $(-7 - i) - (2 + 10i)$

(57) $(5 - 10i) - (-8 + 9i)$

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|--|--|--|
| (1) $(-10 - 2i) + (-5 - 4i)$
$= -15 - 6i$ | (20) $(-5 - i) - (-9 - 3i)$
$= 4 + 2i$ | (39) $(-3 + 4i) + (8 + 5i)$
$= 5 + 9i$ |
| (2) $(2 + 7i) - (3 - 8i)$
$= -1 + 15i$ | (21) $(2 + 5i) + (2 - 10i)$
$= 4 - 5i$ | (40) $(3 + 5i) + (-6 - 3i)$
$= -3 + 2i$ |
| (3) $(-10 + 2i) - (-9 + 8i)$
$= -1 - 6i$ | (22) $(-10 - 10i) + (-8 + 3i)$
$= -18 - 7i$ | (41) $(7 - 8i) - (9 + 10i)$
$= -2 - 18i$ |
| (4) $(-3 - 4i) + (-10 + 5i)$
$= -13 + i$ | (23) $(1 - i) + (-10 - 5i)$
$= -9 - 6i$ | (42) $(-9 - 9i) + (-5 - 7i)$
$= -14 - 16i$ |
| (5) $(4 - 6i) + (-6 - 9i)$
$= -2 - 15i$ | (24) $(-1 - 4i) - (-10 + 6i)$
$= 9 - 10i$ | (43) $(-9 + 6i) + (8 + 6i)$
$= -1 + 12i$ |
| (6) $(-4 + 4i) - (4 + 2i)$
$= -8 + 2i$ | (25) $(-10 - 6i) + (10 + 3i)$
$= -3i$ | (44) $(-4 - i) - (10 + i)$
$= -14 - 2i$ |
| (7) $(-1 + 3i) - (-4 - i)$
$= 3 + 4i$ | (26) $(2 + 4i) + (3 + 6i)$
$= 5 + 10i$ | (45) $(6 + 2i) - (-6 - i)$
$= 12 + 3i$ |
| (8) $(4 + 5i) + (2 - i)$
$= 6 + 4i$ | (27) $(4 + 8i) + (9 + 5i)$
$= 13 + 13i$ | (46) $(4 + 8i) + (-8 + i)$
$= -4 + 9i$ |
| (9) $(4 + 9i) + (5 + 4i)$
$= 9 + 13i$ | (28) $(-2 - 5i) - (-2 + 7i)$
$= -12i$ | (47) $(3 - i) + (2 - i)$
$= 5 - 2i$ |
| (10) $(-10 - 9i) + (-6 - 8i)$
$= -16 - 17i$ | (29) $(1 + 10i) + (-8 + 5i)$
$= -7 + 15i$ | (48) $(1 - 4i) - (-1 + 2i)$
$= 2 - 6i$ |
| (11) $(-1 + 4i) + (-4 - 4i)$
$= -5$ | (30) $(8 - 10i) - (-4 - 2i)$
$= 12 - 8i$ | (49) $(4 - 9i) - (-4 - 2i)$
$= 8 - 7i$ |
| (12) $(-9 + 6i) - (-7 - 9i)$
$= -2 + 15i$ | (31) $(5 - 2i) - (5 - 10i)$
$= 8i$ | (50) $(-10 - 9i) + (-7 - 5i)$
$= -17 - 14i$ |
| (13) $(-1 - 2i) - (1 + 3i)$
$= -2 - 5i$ | (32) $(-10 + 7i) - (3 + 2i)$
$= -13 + 5i$ | (51) $(-9 - 8i) - (-10 - 5i)$
$= 1 - 3i$ |
| (14) $(-1 - 2i) + (-8 + 9i)$
$= -9 + 7i$ | (33) $(-9 - 9i) - (3 + 8i)$
$= -12 - 17i$ | (52) $(9 - 10i) + (-10 - 2i)$
$= -1 - 12i$ |
| (15) $(5 - 6i) + (-4 + 3i)$
$= 1 - 3i$ | (34) $(10 + 10i) - (10 - 8i)$
$= 18i$ | (53) $(1 - 2i) + (7 + 3i)$
$= 8 + i$ |
| (16) $(-7 - i) + (-5 - i)$
$= -12 - 2i$ | (35) $(5 + i) + (6 - 3i)$
$= 11 - 2i$ | (54) $(6 + 5i) + (-4 - 10i)$
$= 2 - 5i$ |
| (17) $(-2 - 7i) + (2 - 2i)$
$= -9i$ | (36) $(-9 + 2i) - (1 - 4i)$
$= -10 + 6i$ | (55) $(8 - 6i) - (-10 - 2i)$
$= 18 - 4i$ |
| (18) $(-2 + 3i) - (-10 - 8i)$
$= 8 + 11i$ | (37) $(7 + 5i) + (-4 + 9i)$
$= 3 + 14i$ | (56) $(-7 - i) + (4 - 5i)$
$= -3 - 6i$ |
| (19) $(-9 - 4i) - (9 + 9i)$
$= -18 - 13i$ | (38) $(3 - i) + (-4 + 5i)$
$= -1 + 4i$ | (57) $(-1 + 8i) + (3 - 4i)$
$= 2 + 4i$ |

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|--|---|--|
| (1) $(4 - 6i) + (-2 - 7i)$
$= 2 - 13i$ | (20) $(3 - 9i) - (8 - 2i)$
$= -5 - 7i$ | (39) $(1 + i) - (-7 + 6i)$
$= 8 - 5i$ |
| (2) $(4 + 6i) + (-9 + 8i)$
$= -5 + 14i$ | (21) $(7 + 6i) + (1 - 5i)$
$= 8 + i$ | (40) $(1 + 2i) + (1 + 9i)$
$= 2 + 11i$ |
| (3) $(7 - 5i) - (-4 - 4i)$
$= 11 - i$ | (22) $(-6 - 10i) - (2 - i)$
$= -8 - 9i$ | (41) $(8 - 8i) - (4 - 5i)$
$= 4 - 3i$ |
| (4) $(-7 - i) + (-6 + 2i)$
$= -13 + i$ | (23) $(8 + 4i) + (5 + 4i)$
$= 13 + 8i$ | (42) $(-10 - 2i) + (6 + 8i)$
$= -4 + 6i$ |
| (5) $(3 + i) - (-7 + 4i)$
$= 10 - 3i$ | (24) $(3 - 5i) + (10 + 3i)$
$= 13 - 2i$ | (43) $(-8 - 8i) - (-6 + i)$
$= -2 - 9i$ |
| (6) $(-6 + 3i) - (6 + 2i)$
$= -12 + i$ | (25) $(-7 + 9i) + (1 - 9i)$
$= -6$ | (44) $(6 - i) + (-10 + 3i)$
$= -4 + 2i$ |
| (7) $(-1 - 9i) - (-6 - i)$
$= 5 - 8i$ | (26) $(-7 - 5i) + (-9 + i)$
$= -16 - 4i$ | (45) $(-2 + i) + (-4 + 4i)$
$= -6 + 5i$ |
| (8) $(-5 + 3i) + (3 - 8i)$
$= -2 - 5i$ | (27) $(-3 + 5i) + (-8 - 10i)$
$= -11 - 5i$ | (46) $(-5 + 5i) - (7 - 2i)$
$= -12 + 7i$ |
| (9) $(-4 + 2i) + (-1 + 10i)$
$= -5 + 12i$ | (28) $(1 + 7i) + (-6 + 10i)$
$= -5 + 17i$ | (47) $(-2 - 3i) - (7 + 3i)$
$= -9 - 6i$ |
| (10) $(6 + 6i) - (6 - 8i)$
$= 14i$ | (29) $(-2 - 3i) + (4 - 2i)$
$= 2 - 5i$ | (48) $(9 - 10i) - (9 + 2i)$
$= -12i$ |
| (11) $(8 + 4i) - (2 + 10i)$
$= 6 - 6i$ | (30) $(-9 + 9i) + (4 + 6i)$
$= -5 + 15i$ | (49) $(-7 + 9i) - (5 + 6i)$
$= -12 + 3i$ |
| (12) $(-9 - i) + (9 - 5i)$
$= -6i$ | (31) $(3 - 7i) - (2 - 5i)$
$= 1 - 2i$ | (50) $(-2 + 10i) + (3 + 5i)$
$= 1 + 15i$ |
| (13) $(-10 + 5i) + (-9 - 6i)$
$= -19 - i$ | (32) $(-7 + 5i) - (-8 + 4i)$
$= 1 + i$ | (51) $(9 + 4i) + (-8 + i)$
$= 1 + 5i$ |
| (14) $(4 + 3i) - (-1 + 2i)$
$= 5 + i$ | (33) $(5 + 5i) + (-7 + 2i)$
$= -2 + 7i$ | (52) $(10 - 6i) - (6 + 5i)$
$= 4 - 11i$ |
| (15) $(3 + i) + (-4 + i)$
$= -1 + 2i$ | (34) $(4 + 2i) + (-6 + 6i)$
$= -2 + 8i$ | (53) $(10 + i) + (1 - 9i)$
$= 11 - 8i$ |
| (16) $(-5 - 2i) - (-1 + 3i)$
$= -4 - 5i$ | (35) $(10 + 5i) + (5 + 4i)$
$= 15 + 9i$ | (54) $(2 - 8i) + (6 - 7i)$
$= 8 - 15i$ |
| (17) $(8 + 9i) - (-3 + 10i)$
$= 11 - i$ | (36) $(4 + 4i) - (-8 - 7i)$
$= 12 + 11i$ | (55) $(-5 - 10i) - (7 + i)$
$= -12 - 11i$ |
| (18) $(-4 - i) - (10 - 3i)$
$= -14 + 2i$ | (37) $(4 + 6i) + (-4 + 5i)$
$= 11i$ | (56) $(-7 - 2i) + (2 - 6i)$
$= -5 - 8i$ |
| (19) $(10 + 10i) - (10 + 7i)$
$= 3i$ | (38) $(3 - 7i) + (-10 - 10i)$
$= -7 - 17i$ | (57) $(7 + i) - (-7 - 3i)$
$= 14 + 4i$ |

- | | | |
|--|--|--|
| (1) $(10 + 7i) + (6 - 8i)$
$= 16 - i$ | (20) $(7 - i) + (5 + 3i)$
$= 12 + 2i$ | (39) $(10 - 8i) - (-3 + 7i)$
$= 13 - 15i$ |
| (2) $(2 + 9i) + (-8 + 4i)$
$= -6 + 13i$ | (21) $(-8 - 9i) + (-1 - 7i)$
$= -9 - 16i$ | (40) $(7 + 10i) - (-6 + 9i)$
$= 13 + i$ |
| (3) $(-4 - 2i) - (6 + 7i)$
$= -10 - 9i$ | (22) $(-8 + 4i) - (-2 - i)$
$= -6 + 5i$ | (41) $(-3 + 9i) - (2 - 8i)$
$= -5 + 17i$ |
| (4) $(-10 - 6i) - (3 - 5i)$
$= -13 - i$ | (23) $(-6 + 3i) + (-7 - i)$
$= -13 + 2i$ | (42) $(6 + 6i) - (10 - 5i)$
$= -4 + 11i$ |
| (5) $(-7 + i) + (-10 - 4i)$
$= -17 - 3i$ | (24) $(10 + 4i) - (9 - 3i)$
$= 1 + 7i$ | (43) $(2 + 8i) - (5 - i)$
$= -3 + 9i$ |
| (6) $(-7 + 7i) + (-5 + 7i)$
$= -12 + 14i$ | (25) $(-2 - 6i) + (6 - 6i)$
$= 4 - 12i$ | (44) $(4 + 5i) + (9 - 3i)$
$= 13 + 2i$ |
| (7) $(-6 - 6i) - (-3 + 4i)$
$= -3 - 10i$ | (26) $(1 - 2i) + (-9 - 7i)$
$= -8 - 9i$ | (45) $(5 - i) - (-1 + 6i)$
$= 6 - 7i$ |
| (8) $(7 + 4i) + (2 - 7i)$
$= 9 - 3i$ | (27) $(-5 + 3i) - (-6 - 5i)$
$= 1 + 8i$ | (46) $(5 - 3i) + (4 + 8i)$
$= 9 + 5i$ |
| (9) $(-4 - 4i) + (10 - 5i)$
$= 6 - 9i$ | (28) $(6 - 9i) - (4 - i)$
$= 2 - 8i$ | (47) $(-1 - 9i) + (7 - 4i)$
$= 6 - 13i$ |
| (10) $(5 + 2i) + (-1 + 3i)$
$= 4 + 5i$ | (29) $(-1 - 9i) - (1 + 7i)$
$= -2 - 16i$ | (48) $(-2 - 5i) - (-5 + 9i)$
$= 3 - 14i$ |
| (11) $(2 - 2i) + (-9 + i)$
$= -7 - i$ | (30) $(4 + i) - (1 + 10i)$
$= 3 - 9i$ | (49) $(7 + 4i) - (6 - 10i)$
$= 1 + 14i$ |
| (12) $(6 + i) + (2 - 4i)$
$= 8 - 3i$ | (31) $(4 + 3i) - (-4 + 2i)$
$= 8 + i$ | (50) $(10 + 10i) + (3 - 10i)$
$= 13$ |
| (13) $(10 - 7i) + (7 - 7i)$
$= 17 - 14i$ | (32) $(9 + 9i) - (-3 + 3i)$
$= 12 + 6i$ | (51) $(3 + 2i) - (-6 + 10i)$
$= 9 - 8i$ |
| (14) $(8 - 10i) + (-2 - 6i)$
$= 6 - 16i$ | (33) $(5 - 7i) + (3 - 4i)$
$= 8 - 11i$ | (52) $(-8 + i) + (-4 + 8i)$
$= -12 + 9i$ |
| (15) $(9 + 2i) - (8 + 5i)$
$= 1 - 3i$ | (34) $(4 - 4i) + (-6 - 9i)$
$= -2 - 13i$ | (53) $(-8 + 3i) + (-2 - 3i)$
$= -10$ |
| (16) $(9 - 4i) + (9 - i)$
$= 18 - 5i$ | (35) $(-2 + i) + (1 - 8i)$
$= -1 - 7i$ | (54) $(3 - 3i) - (4 - 10i)$
$= -1 + 7i$ |
| (17) $(-3 + 2i) - (4 + 7i)$
$= -7 - 5i$ | (36) $(5 + 7i) - (1 - 6i)$
$= 4 + 13i$ | (55) $(7 + 9i) + (10 + 4i)$
$= 17 + 13i$ |
| (18) $(-10 + 5i) + (7 - 6i)$
$= -3 - i$ | (37) $(4 + 6i) + (7 + 6i)$
$= 11 + 12i$ | (56) $(6 + 5i) + (-2 + 4i)$
$= 4 + 9i$ |
| (19) $(1 + 10i) - (-5 - 3i)$
$= 6 + 13i$ | (38) $(-9 + 9i) - (6 + 9i)$
$= -15$ | (57) $(9 - 10i) + (3 + 7i)$
$= 12 - 3i$ |

- | | | |
|--|--|---|
| (1) $(10 - 6i) - (-9 - 8i)$
$= 19 + 2i$ | (20) $(7 + 7i) + (2 - i)$
$= 9 + 6i$ | (39) $(6 - 4i) - (9 - 2i)$
$= -3 - 2i$ |
| (2) $(-7 + 7i) + (-8 - 6i)$
$= -15 + i$ | (21) $(1 - 6i) + (8 - 8i)$
$= 9 - 14i$ | (40) $(7 - i) + (9 + 10i)$
$= 16 + 9i$ |
| (3) $(9 + i) - (-9 + 6i)$
$= 18 - 5i$ | (22) $(-8 - 5i) + (-9 + 6i)$
$= -17 + i$ | (41) $(-3 - 3i) - (7 + 5i)$
$= -10 - 8i$ |
| (4) $(5 + 5i) + (4 + 3i)$
$= 9 + 8i$ | (23) $(4 - 9i) + (5 - 6i)$
$= 9 - 15i$ | (42) $(-5 + 5i) - (5 + 7i)$
$= -10 - 2i$ |
| (5) $(-9 + 2i) + (-3 - 10i)$
$= -12 - 8i$ | (24) $(10 + 4i) - (-6 + 6i)$
$= 16 - 2i$ | (43) $(-8 + i) + (-3 + 7i)$
$= -11 + 8i$ |
| (6) $(-3 - 4i) - (3 - 8i)$
$= -6 + 4i$ | (25) $(8 + 2i) - (-1 - 7i)$
$= 9 + 9i$ | (44) $(-9 - 9i) - (6 + 10i)$
$= -15 - 19i$ |
| (7) $(6 + 3i) - (-8 + 10i)$
$= 14 - 7i$ | (26) $(10 - 6i) + (7 + 7i)$
$= 17 + i$ | (45) $(-10 + 6i) - (2 - 2i)$
$= -12 + 8i$ |
| (8) $(-9 - 2i) + (10 + 9i)$
$= 1 + 7i$ | (27) $(-1 - 3i) - (3 + 2i)$
$= -4 - 5i$ | (46) $(7 + 2i) + (-9 + 4i)$
$= -2 + 6i$ |
| (9) $(-4 - 4i) + (-10 + 8i)$
$= -14 + 4i$ | (28) $(-2 - 10i) - (-4 + 2i)$
$= 2 - 12i$ | (47) $(10 - i) + (6 + 8i)$
$= 16 + 7i$ |
| (10) $(1 - 10i) + (-1 - 2i)$
$= -12i$ | (29) $(-5 - 5i) - (3 + 7i)$
$= -8 - 12i$ | (48) $(5 + 8i) - (-9 + 5i)$
$= 14 + 3i$ |
| (11) $(-2 + 8i) + (-10 - 10i)$
$= -12 - 2i$ | (30) $(-3 + 4i) + (-7 + i)$
$= -10 + 5i$ | (49) $(-6 - 4i) - (6 - 8i)$
$= -12 + 4i$ |
| (12) $(9 - 3i) - (4 + 9i)$
$= 5 - 12i$ | (31) $(10 + 2i) - (-9 + 6i)$
$= 19 - 4i$ | (50) $(7 - i) + (-5 - 7i)$
$= 2 - 8i$ |
| (13) $(9 - i) - (-3 - i)$
$= 12$ | (32) $(5 + 5i) - (8 + i)$
$= -3 + 4i$ | (51) $(6 + 2i) - (-2 - 2i)$
$= 8 + 4i$ |
| (14) $(-6 - 9i) - (3 - 9i)$
$= -9$ | (33) $(-5 - 4i) - (-4 - 7i)$
$= -1 + 3i$ | (52) $(-1 + 6i) + (8 - i)$
$= 7 + 5i$ |
| (15) $(-4 + 4i) + (-9 + 2i)$
$= -13 + 6i$ | (34) $(-8 + 2i) - (4 - 2i)$
$= -12 + 4i$ | (53) $(-3 + 8i) + (-2 - 10i)$
$= -5 - 2i$ |
| (16) $(6 + 8i) + (-10 + 5i)$
$= -4 + 13i$ | (35) $(-1 - 3i) + (8 - i)$
$= 7 - 4i$ | (54) $(4 + 6i) - (10 + i)$
$= -6 + 5i$ |
| (17) $(7 - 7i) - (-8 + i)$
$= 15 - 8i$ | (36) $(-3 + 9i) + (-9 + 5i)$
$= -12 + 14i$ | (55) $(10 - 9i) + (1 + 4i)$
$= 11 - 5i$ |
| (18) $(-8 + 9i) - (-2 + 4i)$
$= -6 + 5i$ | (37) $(-9 - 6i) - (7 - 8i)$
$= -16 + 2i$ | (56) $(-5 + 10i) - (-4 - 6i)$
$= -1 + 16i$ |
| (19) $(-8 - 2i) - (-4 - 8i)$
$= -4 + 6i$ | (38) $(-1 + 7i) + (-10 - 10i)$
$= -11 - 3i$ | (57) $(-1 - 2i) - (-3 + 6i)$
$= 2 - 8i$ |

- | | | |
|---|---|---|
| (1) $(10 - 3i) - (6 + 7i)$
$= 4 - 10i$ | (20) $(-8 - 10i) + (5 - 10i)$
$= -3 - 20i$ | (39) $(7 + i) - (-10 + 10i)$
$= 17 - 9i$ |
| (2) $(7 - 9i) - (-6 - 7i)$
$= 13 - 2i$ | (21) $(-9 + 8i) + (4 - 4i)$
$= -5 + 4i$ | (40) $(-9 - 10i) + (-3 + 4i)$
$= -12 - 6i$ |
| (3) $(-7 + 7i) - (-9 - i)$
$= 2 + 8i$ | (22) $(-1 + 5i) - (-8 + 7i)$
$= 7 - 2i$ | (41) $(1 - 2i) + (5 - 6i)$
$= 6 - 8i$ |
| (4) $(9 + 10i) - (-1 + 3i)$
$= 10 + 7i$ | (23) $(-5 + 6i) + (8 - 8i)$
$= 3 - 2i$ | (42) $(4 - 3i) - (1 + 9i)$
$= 3 - 12i$ |
| (5) $(-2 + 7i) - (-7 + i)$
$= 5 + 6i$ | (24) $(-2 + 4i) + (7 + 4i)$
$= 5 + 8i$ | (43) $(-5 + 9i) + (-8 + 4i)$
$= -13 + 13i$ |
| (6) $(-7 + 8i) - (10 - 4i)$
$= -17 + 12i$ | (25) $(-8 - 3i) - (-6 + 9i)$
$= -2 - 12i$ | (44) $(10 - i) + (-5 - 10i)$
$= 5 - 11i$ |
| (7) $(2 - 4i) - (-6 + 5i)$
$= 8 - 9i$ | (26) $(-3 - 10i) + (-5 - 4i)$
$= -8 - 14i$ | (45) $(9 + 6i) + (1 - 8i)$
$= 10 - 2i$ |
| (8) $(-4 - 7i) - (-10 + 6i)$
$= 6 - 13i$ | (27) $(-1 - 7i) - (2 - 5i)$
$= -3 - 2i$ | (46) $(8 - 5i) - (8 - 5i)$
$= 0$ |
| (9) $(3 + 7i) - (10 - 2i)$
$= -7 + 9i$ | (28) $(9 + 7i) - (-8 - 7i)$
$= 17 + 14i$ | (47) $(-2 - 6i) + (-4 + i)$
$= -6 - 5i$ |
| (10) $(1 - 7i) + (-10 - i)$
$= -9 - 8i$ | (29) $(-5 - 7i) + (-9 - 9i)$
$= -14 - 16i$ | (48) $(-6 + i) - (1 - 9i)$
$= -7 + 10i$ |
| (11) $(-5 - 9i) + (-8 - 6i)$
$= -13 - 15i$ | (30) $(9 + 10i) - (7 + i)$
$= 2 + 9i$ | (49) $(4 + i) + (-7 + 3i)$
$= -3 + 4i$ |
| (12) $(5 - 9i) + (7 + 10i)$
$= 12 + i$ | (31) $(4 + 7i) - (1 + 9i)$
$= 3 - 2i$ | (50) $(3 - 9i) - (10 + 9i)$
$= -7 - 18i$ |
| (13) $(4 - 9i) + (10 - 2i)$
$= 14 - 11i$ | (32) $(5 + 4i) - (8 + 3i)$
$= -3 + i$ | (51) $(-10 + 5i) - (4 - i)$
$= -14 + 6i$ |
| (14) $(5 - 4i) - (7 + 8i)$
$= -2 - 12i$ | (33) $(-7 + 10i) + (-8 - 2i)$
$= -15 + 8i$ | (52) $(-7 + 3i) - (-1 - 6i)$
$= -6 + 9i$ |
| (15) $(1 + 6i) - (1 - 7i)$
$= 13i$ | (34) $(-2 + 7i) - (3 + 6i)$
$= -5 + i$ | (53) $(-7 + 8i) + (8 + 7i)$
$= 1 + 15i$ |
| (16) $(-3 - 9i) - (1 + 7i)$
$= -4 - 16i$ | (35) $(1 + 10i) - (8 - 7i)$
$= -7 + 17i$ | (54) $(9 - 2i) + (2 - 3i)$
$= 11 - 5i$ |
| (17) $(5 - 3i) - (3 - 9i)$
$= 2 + 6i$ | (36) $(-5 + 2i) + (-10 - i)$
$= -15 + i$ | (55) $(1 + 7i) + (9 + 5i)$
$= 10 + 12i$ |
| (18) $(-7 + 6i) + (2 + 4i)$
$= -5 + 10i$ | (37) $(-3 + 2i) + (4 - 9i)$
$= 1 - 7i$ | (56) $(7 - 4i) - (5 + 2i)$
$= 2 - 6i$ |
| (19) $(3 + 8i) + (2 + 10i)$
$= 5 + 18i$ | (38) $(6 - 2i) - (6 - 3i)$
$= i$ | (57) $(4 + 8i) + (2 + 8i)$
$= 6 + 16i$ |

- | | | |
|--|--|--|
| (1) $(-3 - 3i) - (3 + i)$
$= -6 - 4i$ | (20) $(4 + 6i) - (1 + 2i)$
$= 3 + 4i$ | (39) $(2 - 3i) - (2 - 7i)$
$= 4i$ |
| (2) $(4 - 5i) + (8 - i)$
$= 12 - 6i$ | (21) $(-8 + 3i) + (-3 - i)$
$= -11 + 2i$ | (40) $(3 + 2i) + (10 - 5i)$
$= 13 - 3i$ |
| (3) $(3 + 6i) + (-7 + 10i)$
$= -4 + 16i$ | (22) $(1 - i) + (-10 - 3i)$
$= -9 - 4i$ | (41) $(2 + 4i) - (7 + 4i)$
$= -5$ |
| (4) $(1 + 10i) - (10 + 4i)$
$= -9 + 6i$ | (23) $(-4 + i) - (4 + 3i)$
$= -8 - 2i$ | (42) $(-1 - 5i) + (-2 - 3i)$
$= -3 - 8i$ |
| (5) $(-5 - 5i) + (-4 - 7i)$
$= -9 - 12i$ | (24) $(8 + 8i) - (-4 - 4i)$
$= 12 + 12i$ | (43) $(-6 + 9i) - (-10 - i)$
$= 4 + 10i$ |
| (6) $(-9 - 4i) + (-4 + 4i)$
$= -13$ | (25) $(5 + 10i) + (7 - i)$
$= 12 + 9i$ | (44) $(10 - 9i) - (-5 + 9i)$
$= 15 - 18i$ |
| (7) $(8 - 5i) - (-4 - 2i)$
$= 12 - 3i$ | (26) $(7 + 10i) + (9 - 3i)$
$= 16 + 7i$ | (45) $(-4 - i) + (10 - 8i)$
$= 6 - 9i$ |
| (8) $(1 + 3i) + (-8 - 9i)$
$= -7 - 6i$ | (27) $(3 - 10i) - (-8 - i)$
$= 11 - 9i$ | (46) $(2 + 8i) - (5 + 10i)$
$= -3 - 2i$ |
| (9) $(3 - 10i) + (-2 - 10i)$
$= 1 - 20i$ | (28) $(-6 + 3i) + (-1 - 10i)$
$= -7 - 7i$ | (47) $(9 + 8i) + (8 - 8i)$
$= 17$ |
| (10) $(-4 + 3i) + (7 + i)$
$= 3 + 4i$ | (29) $(4 + 6i) - (-9 + 8i)$
$= 13 - 2i$ | (48) $(-6 + 6i) - (-9 + 10i)$
$= 3 - 4i$ |
| (11) $(-8 - 6i) - (1 + i)$
$= -9 - 7i$ | (30) $(-3 - 7i) + (3 + 3i)$
$= -4i$ | (49) $(2 - 9i) - (-2 - 2i)$
$= 4 - 7i$ |
| (12) $(8 - 3i) + (-7 + 9i)$
$= 1 + 6i$ | (31) $(7 - i) + (-9 - 10i)$
$= -2 - 11i$ | (50) $(-7 + 9i) + (-6 - 7i)$
$= -13 + 2i$ |
| (13) $(-3 - 7i) + (-9 + 6i)$
$= -12 - i$ | (32) $(2 - i) - (10 - 7i)$
$= -8 + 6i$ | (51) $(-10 - 5i) - (-2 + 4i)$
$= -8 - 9i$ |
| (14) $(-7 + 10i) - (-1 + 5i)$
$= -6 + 5i$ | (33) $(2 - i) + (10 + 10i)$
$= 12 + 9i$ | (52) $(10 + 8i) - (1 - 4i)$
$= 9 + 12i$ |
| (15) $(-4 - 7i) - (8 - 10i)$
$= -12 + 3i$ | (34) $(10 + 7i) - (-2 - 6i)$
$= 12 + 13i$ | (53) $(-7 + 7i) - (2 + 8i)$
$= -9 - i$ |
| (16) $(-2 + 9i) + (-1 - 2i)$
$= -3 + 7i$ | (35) $(10 + 10i) + (-7 + i)$
$= 3 + 11i$ | (54) $(2 - i) + (9 + i)$
$= 11$ |
| (17) $(-7 + i) + (-3 - 5i)$
$= -10 - 4i$ | (36) $(1 + i) + (6 + 7i)$
$= 7 + 8i$ | (55) $(9 - 7i) + (7 - 8i)$
$= 16 - 15i$ |
| (18) $(-1 + 9i) - (-7 - 10i)$
$= 6 + 19i$ | (37) $(4 + 7i) - (1 - 10i)$
$= 3 + 17i$ | (56) $(-4 - 4i) + (4 - 7i)$
$= -11i$ |
| (19) $(4 - i) + (8 - 7i)$
$= 12 - 8i$ | (38) $(5 - 6i) + (-7 - 4i)$
$= -2 - 10i$ | (57) $(4 + 9i) - (-8 + 9i)$
$= 12$ |

- | | | |
|--|--|--|
| (1) $(-4 + 8i) + (1 + i)$
$= -3 + 9i$ | (20) $(-1 - 2i) - (5 + 3i)$
$= -6 - 5i$ | (39) $(1 - 8i) - (4 + 7i)$
$= -3 - 15i$ |
| (2) $(-1 - i) + (1 + 9i)$
$= 8i$ | (21) $(-4 - i) - (7 - 10i)$
$= -11 + 9i$ | (40) $(4 - 3i) + (-2 + 4i)$
$= 2 + i$ |
| (3) $(-2 - 7i) - (-8 - 3i)$
$= 6 - 4i$ | (22) $(2 + 7i) - (-7 - 2i)$
$= 9 + 9i$ | (41) $(-6 + 3i) + (-5 + 10i)$
$= -11 + 13i$ |
| (4) $(-9 - 6i) - (-9 - i)$
$= -5i$ | (23) $(7 - 4i) - (-10 + 7i)$
$= 17 - 11i$ | (42) $(-6 - 8i) - (-7 + 3i)$
$= 1 - 11i$ |
| (5) $(-1 - i) + (5 + 5i)$
$= 4 + 4i$ | (24) $(2 + 8i) + (3 + 2i)$
$= 5 + 10i$ | (43) $(-4 - 10i) - (4 + 2i)$
$= -8 - 12i$ |
| (6) $(5 - 5i) + (7 + 5i)$
$= 12$ | (25) $(-8 + 9i) - (-9 + 7i)$
$= 1 + 2i$ | (44) $(6 + 10i) - (8 - 4i)$
$= -2 + 14i$ |
| (7) $(-8 - 5i) + (8 - 7i)$
$= -12i$ | (26) $(-6 + 4i) + (-3 - 8i)$
$= -9 - 4i$ | (45) $(-9 + 2i) - (-8 + 8i)$
$= -1 - 6i$ |
| (8) $(2 + i) - (-5 + 3i)$
$= 7 - 2i$ | (27) $(-7 + 8i) + (1 - 10i)$
$= -6 - 2i$ | (46) $(-2 + 3i) - (-8 - 10i)$
$= 6 + 13i$ |
| (9) $(10 - i) + (-6 - 8i)$
$= 4 - 9i$ | (28) $(-7 - 8i) - (-10 - 10i)$
$= 3 + 2i$ | (47) $(-10 + 4i) + (7 - 8i)$
$= -3 - 4i$ |
| (10) $(-10 + 9i) + (1 - 2i)$
$= -9 + 7i$ | (29) $(-3 + 9i) - (3 + 10i)$
$= -6 - i$ | (48) $(-3 + 8i) - (7 + 10i)$
$= -10 - 2i$ |
| (11) $(-4 - 10i) - (8 - 3i)$
$= -12 - 7i$ | (30) $(4 - 7i) + (8 + 7i)$
$= 12$ | (49) $(-1 + 8i) - (-2 + 7i)$
$= 1 + i$ |
| (12) $(6 - 5i) - (10 + 5i)$
$= -4 - 10i$ | (31) $(-10 + 3i) + (-2 - 4i)$
$= -12 - i$ | (50) $(5 - 4i) - (5 - i)$
$= -3i$ |
| (13) $(-6 - 5i) - (-2 - 5i)$
$= -4$ | (32) $(10 - 2i) - (-1 + 6i)$
$= 11 - 8i$ | (51) $(1 - i) - (9 + 5i)$
$= -8 - 6i$ |
| (14) $(-5 - 8i) + (9 + 3i)$
$= 4 - 5i$ | (33) $(10 - 9i) - (-3 + 6i)$
$= 13 - 15i$ | (52) $(-6 + i) + (7 + 8i)$
$= 1 + 9i$ |
| (15) $(8 - 9i) - (8 - 2i)$
$= -7i$ | (34) $(-8 - 3i) - (-4 - 4i)$
$= -4 + i$ | (53) $(-3 + 3i) - (5 + 7i)$
$= -8 - 4i$ |
| (16) $(2 - 10i) - (-5 + 5i)$
$= 7 - 15i$ | (35) $(-1 - 4i) - (-5 - i)$
$= 4 - 3i$ | (54) $(8 - i) - (-9 - 9i)$
$= 17 + 8i$ |
| (17) $(1 + 2i) + (9 + 2i)$
$= 10 + 4i$ | (36) $(-3 + 7i) + (7 + 4i)$
$= 4 + 11i$ | (55) $(-6 + 6i) + (8 + 8i)$
$= 2 + 14i$ |
| (18) $(2 - 10i) - (-3 + 3i)$
$= 5 - 13i$ | (37) $(10 - 7i) - (-1 - 3i)$
$= 11 - 4i$ | (56) $(10 - 8i) + (7 - 9i)$
$= 17 - 17i$ |
| (19) $(-4 + 9i) + (1 - 10i)$
$= -3 - i$ | (38) $(-9 + 9i) + (9 + 4i)$
$= 13i$ | (57) $(1 - 7i) - (1 - 7i)$
$= 0$ |

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|--|---|--|
| (1) $(-7 - 10i) - (-8 + 7i)$
$= 1 - 17i$ | (20) $(-5 - 10i) - (9 - 7i)$
$= -14 - 3i$ | (39) $(-9 + i) - (-4 - 6i)$
$= -5 + 7i$ |
| (2) $(-2 + 7i) + (10 - 6i)$
$= 8 + i$ | (21) $(2 + 9i) - (9 + 3i)$
$= -7 + 6i$ | (40) $(4 - i) - (-9 - 6i)$
$= 13 + 5i$ |
| (3) $(4 + 9i) + (6 - 9i)$
$= 10$ | (22) $(-8 + i) + (2 - 7i)$
$= -6 - 6i$ | (41) $(-2 - 10i) + (3 - 3i)$
$= 1 - 13i$ |
| (4) $(-3 + 4i) - (8 + 6i)$
$= -11 - 2i$ | (23) $(1 - 7i) - (7 + 4i)$
$= -6 - 11i$ | (42) $(-10 - 8i) - (-7 + 6i)$
$= -3 - 14i$ |
| (5) $(3 + 8i) - (2 - 10i)$
$= 1 + 18i$ | (24) $(-10 + 3i) + (-10 - 2i)$
$= -20 + i$ | (43) $(-3 + 8i) + (7 - 8i)$
$= 4$ |
| (6) $(-9 - 9i) - (-7 + 9i)$
$= -2 - 18i$ | (25) $(9 + 4i) - (-6 + 2i)$
$= 15 + 2i$ | (44) $(9 - 7i) + (-1 + 5i)$
$= 8 - 2i$ |
| (7) $(3 + i) - (-5 + 3i)$
$= 8 - 2i$ | (26) $(3 - 10i) + (9 - 4i)$
$= 12 - 14i$ | (45) $(-5 - 9i) + (6 - i)$
$= 1 - 10i$ |
| (8) $(2 - 6i) - (2 + 3i)$
$= -9i$ | (27) $(-9 - i) - (6 - 8i)$
$= -15 + 7i$ | (46) $(-3 - 10i) + (-4 - 2i)$
$= -7 - 12i$ |
| (9) $(4 + 9i) + (-6 + 3i)$
$= -2 + 12i$ | (28) $(-6 - 7i) + (5 - 7i)$
$= -1 - 14i$ | (47) $(6 - 7i) + (6 - 5i)$
$= 12 - 12i$ |
| (10) $(-9 + 6i) - (1 + 8i)$
$= -10 - 2i$ | (29) $(5 - 3i) - (2 + 10i)$
$= 3 - 13i$ | (48) $(1 - 8i) + (6 - 2i)$
$= 7 - 10i$ |
| (11) $(3 + 6i) + (-8 - 5i)$
$= -5 + i$ | (30) $(-9 - 4i) - (-1 - 6i)$
$= -8 + 2i$ | (49) $(-9 + 10i) + (-8 + 3i)$
$= -17 + 13i$ |
| (12) $(-3 - 4i) - (-1 + 9i)$
$= -2 - 13i$ | (31) $(-3 + 7i) + (2 + 6i)$
$= -1 + 13i$ | (50) $(-4 + 3i) - (9 + 6i)$
$= -13 - 3i$ |
| (13) $(3 + 2i) + (-2 + 3i)$
$= 1 + 5i$ | (32) $(-1 - 10i) - (10 - 10i)$
$= -11$ | (51) $(-7 - 8i) - (-9 + 7i)$
$= 2 - 15i$ |
| (14) $(-9 - 10i) + (-9 - 4i)$
$= -18 - 14i$ | (33) $(-1 + i) + (-4 + 3i)$
$= -5 + 4i$ | (52) $(5 + 5i) - (-3 - 9i)$
$= 8 + 14i$ |
| (15) $(-3 - 10i) - (-4 + 7i)$
$= 1 - 17i$ | (34) $(-8 + 6i) + (2 - 4i)$
$= -6 + 2i$ | (53) $(-1 - i) - (4 + 4i)$
$= -5 - 5i$ |
| (16) $(10 - 7i) + (-4 + i)$
$= 6 - 6i$ | (35) $(-10 - 6i) - (-2 + 8i)$
$= -8 - 14i$ | (54) $(10 - 7i) + (-7 + i)$
$= 3 - 6i$ |
| (17) $(-10 - 5i) + (-4 - 9i)$
$= -14 - 14i$ | (36) $(-8 - 7i) - (10 + 8i)$
$= -18 - 15i$ | (55) $(3 + 4i) + (-6 + 8i)$
$= -3 + 12i$ |
| (18) $(7 - i) + (-7 - 3i)$
$= -4i$ | (37) $(-2 + 8i) - (5 - 2i)$
$= -7 + 10i$ | (56) $(10 + 2i) + (7 + i)$
$= 17 + 3i$ |
| (19) $(-6 - 9i) + (-1 - 5i)$
$= -7 - 14i$ | (38) $(-4 - 4i) - (-7 - 5i)$
$= 3 + i$ | (57) $(-7 + i) - (8 + 2i)$
$= -15 - i$ |

- | | | |
|--|--|---|
| (1) $(-2 + i) - (-1 - 5i)$
$= -1 + 6i$ | (20) $(-8 - 4i) + (-8 - 9i)$
$= -16 - 13i$ | (39) $(5 - 8i) - (-2 - 3i)$
$= 7 - 5i$ |
| (2) $(-6 - 5i) - (8 + 8i)$
$= -14 - 13i$ | (21) $(-4 - 10i) + (-3 - 4i)$
$= -7 - 14i$ | (40) $(-7 + 5i) - (5 + 4i)$
$= -12 + i$ |
| (3) $(-6 + 10i) + (-1 - 8i)$
$= -7 + 2i$ | (22) $(3 + 10i) - (10 + 8i)$
$= -7 + 2i$ | (41) $(-8 - 6i) - (-8 - 9i)$
$= 3i$ |
| (4) $(7 - 10i) + (9 + i)$
$= 16 - 9i$ | (23) $(10 + 2i) - (1 - 7i)$
$= 9 + 9i$ | (42) $(3 + 8i) + (9 - 10i)$
$= 12 - 2i$ |
| (5) $(-6 - 6i) - (-9 - 5i)$
$= 3 - i$ | (24) $(-7 + 8i) + (-8 + 4i)$
$= -15 + 12i$ | (43) $(-3 - i) - (-9 - 8i)$
$= 6 + 7i$ |
| (6) $(-6 - 7i) - (-10 - 8i)$
$= 4 + i$ | (25) $(5 + 9i) + (3 - 2i)$
$= 8 + 7i$ | (44) $(5 + 9i) + (4 - 10i)$
$= 9 - i$ |
| (7) $(5 + i) + (4 - 6i)$
$= 9 - 5i$ | (26) $(3 - 4i) + (-10 + 9i)$
$= -7 + 5i$ | (45) $(-4 + 7i) - (-8 + 9i)$
$= 4 - 2i$ |
| (8) $(1 - 5i) + (3 - 10i)$
$= 4 - 15i$ | (27) $(6 + 3i) - (-6 + 9i)$
$= 12 - 6i$ | (46) $(5 - 4i) - (-2 + 4i)$
$= 7 - 8i$ |
| (9) $(3 + 5i) + (-2 + 3i)$
$= 1 + 8i$ | (28) $(2 + 2i) + (-9 - 9i)$
$= -7 - 7i$ | (47) $(-10 - 2i) + (2 - 10i)$
$= -8 - 12i$ |
| (10) $(2 - 7i) - (-7 + 4i)$
$= 9 - 11i$ | (29) $(-6 - i) - (4 + i)$
$= -10 - 2i$ | (48) $(3 + 4i) + (3 + 2i)$
$= 6 + 6i$ |
| (11) $(6 + 3i) - (10 - 6i)$
$= -4 + 9i$ | (30) $(-3 - 8i) + (-1 - 9i)$
$= -4 - 17i$ | (49) $(-9 - 6i) + (10 - 7i)$
$= 1 - 13i$ |
| (12) $(10 - 3i) + (-9 - 5i)$
$= 1 - 8i$ | (31) $(-9 + 7i) - (-2 + 2i)$
$= -7 + 5i$ | (50) $(-9 - 4i) - (4 - 6i)$
$= -13 + 2i$ |
| (13) $(-5 + 4i) + (-10 - i)$
$= -15 + 3i$ | (32) $(2 - 4i) + (6 + 9i)$
$= 8 + 5i$ | (51) $(-1 - 4i) + (-9 + 6i)$
$= -10 + 2i$ |
| (14) $(10 - 6i) - (-1 + 9i)$
$= 11 - 15i$ | (33) $(2 - 7i) + (3 - 10i)$
$= 5 - 17i$ | (52) $(-8 - 3i) - (-2 + 6i)$
$= -6 - 9i$ |
| (15) $(5 - 9i) - (6 - 4i)$
$= -1 - 5i$ | (34) $(-9 - 4i) + (5 + i)$
$= -4 - 3i$ | (53) $(-4 - 7i) + (9 + 8i)$
$= 5 + i$ |
| (16) $(-5 - i) - (-3 + 7i)$
$= -2 - 8i$ | (35) $(-3 - 10i) + (1 + 8i)$
$= -2 - 2i$ | (54) $(-1 + 4i) + (10 - 10i)$
$= 9 - 6i$ |
| (17) $(-8 + 4i) + (2 + 6i)$
$= -6 + 10i$ | (36) $(-6 + 9i) + (-10 + 8i)$
$= -16 + 17i$ | (55) $(8 + 5i) + (-2 - 10i)$
$= 6 - 5i$ |
| (18) $(2 + 10i) - (3 + 4i)$
$= -1 + 6i$ | (37) $(-6 + 2i) - (2 + 4i)$
$= -8 - 2i$ | (56) $(6 - 4i) - (2 - 8i)$
$= 4 + 4i$ |
| (19) $(3 - 4i) + (2 - 3i)$
$= 5 - 7i$ | (38) $(-7 - 3i) + (5 + 4i)$
$= -2 + i$ | (57) $(10 - 9i) + (-3 + 9i)$
$= 7$ |

- | | | |
|--|--|---|
| (1) $(10 + 6i) - (9 - 9i)$
$= 1 + 15i$ | (20) $(2 - 3i) + (-4 - 8i)$
$= -2 - 11i$ | (39) $(6 + 3i) - (-3 + 9i)$
$= 9 - 6i$ |
| (2) $(-6 - 4i) - (9 + 2i)$
$= -15 - 6i$ | (21) $(6 + 6i) + (2 + 3i)$
$= 8 + 9i$ | (40) $(-10 - 3i) - (10 - 7i)$
$= -20 + 4i$ |
| (3) $(-4 + 5i) - (6 - 5i)$
$= -10 + 10i$ | (22) $(6 + 5i) - (-1 + 3i)$
$= 7 + 2i$ | (41) $(2 - 8i) - (3 - 6i)$
$= -1 - 2i$ |
| (4) $(-9 - 5i) - (-6 - 9i)$
$= -3 + 4i$ | (23) $(-10 - 3i) - (-3 + 10i)$
$= -7 - 13i$ | (42) $(-7 - i) + (4 + 10i)$
$= -3 + 9i$ |
| (5) $(10 + 5i) + (7 - 10i)$
$= 17 - 5i$ | (24) $(-2 + 7i) - (-1 + 2i)$
$= -1 + 5i$ | (43) $(-5 - 8i) + (-1 - 8i)$
$= -6 - 16i$ |
| (6) $(9 + 9i) + (8 + 7i)$
$= 17 + 16i$ | (25) $(-3 - 10i) - (3 - 7i)$
$= -6 - 3i$ | (44) $(10 + i) - (-3 - 2i)$
$= 13 + 3i$ |
| (7) $(-8 - 10i) - (-10 + 7i)$
$= 2 - 17i$ | (26) $(-10 + 2i) + (10 - 9i)$
$= -7i$ | (45) $(-9 - 7i) - (4 + 5i)$
$= -13 - 12i$ |
| (8) $(-1 + 3i) - (-6 - i)$
$= 5 + 4i$ | (27) $(9 + 7i) + (-6 - 6i)$
$= 3 + i$ | (46) $(-4 - 5i) + (-5 + 4i)$
$= -9 - i$ |
| (9) $(5 + i) + (-5 - 10i)$
$= -9i$ | (28) $(6 - 2i) - (3 + 10i)$
$= 3 - 12i$ | (47) $(7 - 9i) - (-4 + 9i)$
$= 11 - 18i$ |
| (10) $(-5 - 3i) - (1 + 3i)$
$= -6 - 6i$ | (29) $(1 + 6i) + (9 - 6i)$
$= 10$ | (48) $(5 - 5i) + (-2 + 2i)$
$= 3 - 3i$ |
| (11) $(4 - 6i) - (6 + 10i)$
$= -2 - 16i$ | (30) $(7 - 8i) + (3 + 10i)$
$= 10 + 2i$ | (49) $(-7 - 7i) + (4 - 3i)$
$= -3 - 10i$ |
| (12) $(3 - i) - (-9 + 6i)$
$= 12 - 7i$ | (31) $(-3 + 3i) + (-5 + 10i)$
$= -8 + 13i$ | (50) $(-4 - 7i) + (-2 + 3i)$
$= -6 - 4i$ |
| (13) $(4 + 9i) - (7 - 9i)$
$= -3 + 18i$ | (32) $(-8 - 5i) + (-1 + 8i)$
$= -9 + 3i$ | (51) $(3 - 6i) - (10 + i)$
$= -7 - 7i$ |
| (14) $(-4 + 8i) - (-7 + 5i)$
$= 3 + 3i$ | (33) $(-5 + 8i) - (-5 - 6i)$
$= 14i$ | (52) $(-7 - 4i) + (1 - 4i)$
$= -6 - 8i$ |
| (15) $(9 + 8i) - (8 - 10i)$
$= 1 + 18i$ | (34) $(-10 + 9i) - (7 - 6i)$
$= -17 + 15i$ | (53) $(9 + i) + (2 - 9i)$
$= 11 - 8i$ |
| (16) $(7 - i) - (9 - 2i)$
$= -2 + i$ | (35) $(1 - 7i) - (-8 - 6i)$
$= 9 - i$ | (54) $(3 + 7i) + (-6 + 2i)$
$= -3 + 9i$ |
| (17) $(-1 + 4i) + (-2 - 2i)$
$= -3 + 2i$ | (36) $(-7 + 8i) - (-8 + 6i)$
$= 1 + 2i$ | (55) $(1 - 10i) + (8 + 5i)$
$= 9 - 5i$ |
| (18) $(3 - 7i) + (-5 + 2i)$
$= -2 - 5i$ | (37) $(-6 + 7i) - (5 + 3i)$
$= -11 + 4i$ | (56) $(8 - 10i) + (8 + 3i)$
$= 16 - 7i$ |
| (19) $(-3 + 4i) - (-4 + 10i)$
$= 1 - 6i$ | (38) $(-7 + 2i) - (10 - 8i)$
$= -17 + 10i$ | (57) $(9 + 2i) - (4 + 7i)$
$= 5 - 5i$ |

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|--|---|---|
| (1) $(1 - 10i) - (10 + 5i)$
$= -9 - 15i$ | (20) $(-3 + 2i) + (2 + 7i)$
$= -1 + 9i$ | (39) $(-2 + 8i) - (-9 + 10i)$
$= 7 - 2i$ |
| (2) $(3 + i) + (10 - 6i)$
$= 13 - 5i$ | (21) $(9 - i) - (-8 + 2i)$
$= 17 - 3i$ | (40) $(-6 + 9i) - (-9 + 10i)$
$= 3 - i$ |
| (3) $(-4 + 2i) - (8 + i)$
$= -12 + i$ | (22) $(-7 - 3i) + (5 - 6i)$
$= -2 - 9i$ | (41) $(-4 + 5i) - (-3 - 10i)$
$= -1 + 15i$ |
| (4) $(-4 + 10i) - (9 + 5i)$
$= -13 + 5i$ | (23) $(-10 - 5i) - (-1 + 6i)$
$= -9 - 11i$ | (42) $(6 + 8i) - (-1 + 8i)$
$= 7$ |
| (5) $(1 + 7i) - (7 - 9i)$
$= -6 + 16i$ | (24) $(-7 + 4i) + (-8 - 4i)$
$= -15$ | (43) $(-8 - 7i) + (-8 - 2i)$
$= -16 - 9i$ |
| (6) $(10 + 4i) + (-1 + 4i)$
$= 9 + 8i$ | (25) $(5 - 2i) - (8 + 3i)$
$= -3 - 5i$ | (44) $(4 - 6i) + (2 + 10i)$
$= 6 + 4i$ |
| (7) $(1 + 8i) - (2 - 2i)$
$= -1 + 10i$ | (26) $(-10 + 3i) - (-3 - 4i)$
$= -7 + 7i$ | (45) $(-3 - 2i) + (4 - 4i)$
$= 1 - 6i$ |
| (8) $(9 + 7i) - (8 - 7i)$
$= 1 + 14i$ | (27) $(2 + 10i) + (1 - 7i)$
$= 3 + 3i$ | (46) $(6 - 10i) + (-10 + 5i)$
$= -4 - 5i$ |
| (9) $(6 - i) + (-10 - 6i)$
$= -4 - 7i$ | (28) $(3 - 5i) - (7 - 9i)$
$= -4 + 4i$ | (47) $(-7 - 10i) - (-1 + 7i)$
$= -6 - 17i$ |
| (10) $(-2 + 5i) - (7 + 4i)$
$= -9 + i$ | (29) $(-7 - 6i) - (10 - i)$
$= -17 - 5i$ | (48) $(-1 - 4i) + (7 + 8i)$
$= 6 + 4i$ |
| (11) $(1 - 3i) - (10 - 4i)$
$= -9 + i$ | (30) $(10 - 7i) - (-9 - i)$
$= 19 - 6i$ | (49) $(6 - 10i) - (3 - 6i)$
$= 3 - 4i$ |
| (12) $(-7 - 8i) + (1 - 9i)$
$= -6 - 17i$ | (31) $(6 - i) + (-9 - 8i)$
$= -3 - 9i$ | (50) $(-8 + 9i) - (-4 - 8i)$
$= -4 + 17i$ |
| (13) $(2 - 5i) - (7 + 5i)$
$= -5 - 10i$ | (32) $(-10 - 6i) + (5 + 2i)$
$= -5 - 4i$ | (51) $(-1 - 3i) + (-4 + 7i)$
$= -5 + 4i$ |
| (14) $(7 + 4i) + (5 + i)$
$= 12 + 5i$ | (33) $(-2 - 3i) - (-4 + 10i)$
$= 2 - 13i$ | (52) $(8 + 10i) - (1 + 6i)$
$= 7 + 4i$ |
| (15) $(2 + i) - (-6 + 8i)$
$= 8 - 7i$ | (34) $(7 - 6i) + (10 + 9i)$
$= 17 + 3i$ | (53) $(4 + 6i) - (-1 + i)$
$= 5 + 5i$ |
| (16) $(3 - i) + (-8 + 5i)$
$= -5 + 4i$ | (35) $(4 - 10i) + (-10 + 2i)$
$= -6 - 8i$ | (54) $(9 + i) + (9 - 6i)$
$= 18 - 5i$ |
| (17) $(10 - 9i) - (-1 + 3i)$
$= 11 - 12i$ | (36) $(8 + 5i) - (7 - 10i)$
$= 1 + 15i$ | (55) $(10 - 4i) + (-8 - i)$
$= 2 - 5i$ |
| (18) $(-3 - 4i) + (-8 + 4i)$
$= -11$ | (37) $(-8 + 4i) + (5 - 4i)$
$= -3$ | (56) $(-7 - 2i) - (-6 - 5i)$
$= -1 + 3i$ |
| (19) $(6 - 7i) - (5 - 4i)$
$= 1 - 3i$ | (38) $(9 + 5i) - (8 + 5i)$
$= 1$ | (57) $(1 - 9i) + (9 + 7i)$
$= 10 - 2i$ |

- | | | |
|---|--|---|
| (1) $(-2 + 7i) - (8 - i)$
$= -10 + 8i$ | (20) $(4 + 9i) - (10 + 10i)$
$= -6 - i$ | (39) $(-3 + 7i) + (-10 + i)$
$= -13 + 8i$ |
| (2) $(1 - 10i) - (-1 + 3i)$
$= 2 - 13i$ | (21) $(-6 - 5i) - (-6 - 9i)$
$= 4i$ | (40) $(10 - 10i) - (-9 + 9i)$
$= 19 - 19i$ |
| (3) $(10 - 4i) + (-4 - 2i)$
$= 6 - 6i$ | (22) $(-3 - 8i) + (10 + 3i)$
$= 7 - 5i$ | (41) $(-7 + 10i) + (-9 - 10i)$
$= -16$ |
| (4) $(5 + 2i) + (9 - 8i)$
$= 14 - 6i$ | (23) $(-2 - 6i) + (-8 - i)$
$= -10 - 7i$ | (42) $(-7 + 10i) - (-2 + i)$
$= -5 + 9i$ |
| (5) $(3 + 8i) - (-3 + 6i)$
$= 6 + 2i$ | (24) $(-7 - 9i) - (1 + 8i)$
$= -8 - 17i$ | (43) $(-6 + 2i) + (4 - 6i)$
$= -2 - 4i$ |
| (6) $(4 - 6i) + (-7 + 8i)$
$= -3 + 2i$ | (25) $(5 - 5i) + (-5 + 5i)$
$= 0$ | (44) $(-3 - 10i) + (6 + 5i)$
$= 3 - 5i$ |
| (7) $(8 + 2i) - (-6 - 7i)$
$= 14 + 9i$ | (26) $(-1 + 5i) + (9 + 8i)$
$= 8 + 13i$ | (45) $(-7 + 6i) - (-6 + 4i)$
$= -1 + 2i$ |
| (8) $(-5 - 8i) - (-5 + 8i)$
$= -16i$ | (27) $(-2 + i) + (-9 + 2i)$
$= -11 + 3i$ | (46) $(1 - 4i) + (-7 + i)$
$= -6 - 3i$ |
| (9) $(7 + 5i) - (5 - 5i)$
$= 2 + 10i$ | (28) $(-2 + 2i) - (-10 - 5i)$
$= 8 + 7i$ | (47) $(-2 - 3i) - (-8 + 8i)$
$= 6 - 11i$ |
| (10) $(-1 + 8i) + (-6 + 10i)$
$= -7 + 18i$ | (29) $(-6 + 5i) - (-4 + 7i)$
$= -2 - 2i$ | (48) $(4 + i) - (-2 + 10i)$
$= 6 - 9i$ |
| (11) $(-4 + i) - (-10 + 4i)$
$= 6 - 3i$ | (30) $(-1 - 2i) - (2 - 10i)$
$= -3 + 8i$ | (49) $(-3 - 4i) - (-5 + i)$
$= 2 - 5i$ |
| (12) $(-5 - 3i) - (-10 - 10i)$
$= 5 + 7i$ | (31) $(-6 - 8i) - (-9 - 9i)$
$= 3 + i$ | (50) $(4 + 3i) - (-2 - 3i)$
$= 6 + 6i$ |
| (13) $(-1 - 6i) + (9 - 8i)$
$= 8 - 14i$ | (32) $(-3 + 9i) - (-3 + 2i)$
$= 7i$ | (51) $(8 - i) - (-2 + 4i)$
$= 10 - 5i$ |
| (14) $(8 - 4i) - (8 - 6i)$
$= 2i$ | (33) $(-10 + 5i) - (2 + 8i)$
$= -12 - 3i$ | (52) $(3 + 8i) - (-1 + 2i)$
$= 4 + 6i$ |
| (15) $(1 - 7i) - (10 - 10i)$
$= -9 + 3i$ | (34) $(-6 + 4i) - (-1 + 8i)$
$= -5 - 4i$ | (53) $(-6 - 5i) - (6 + 2i)$
$= -12 - 7i$ |
| (16) $(-2 + 4i) - (4 + 7i)$
$= -6 - 3i$ | (35) $(-2 - 10i) - (-7 + 3i)$
$= 5 - 13i$ | (54) $(5 - 10i) + (-6 + i)$
$= -1 - 9i$ |
| (17) $(8 - 5i) - (-4 + 3i)$
$= 12 - 8i$ | (36) $(7 + 3i) + (-6 - 4i)$
$= 1 - i$ | (55) $(-2 + 7i) - (-9 + 9i)$
$= 7 - 2i$ |
| (18) $(-4 + 2i) - (-1 - 8i)$
$= -3 + 10i$ | (37) $(-1 - 6i) + (4 - 6i)$
$= 3 - 12i$ | (56) $(-5 + 6i) + (5 + 9i)$
$= 15i$ |
| (19) $(4 - 10i) - (-4 - 9i)$
$= 8 - i$ | (38) $(-3 + 6i) - (1 + 8i)$
$= -4 - 2i$ | (57) $(-6 - 3i) - (-5 - 6i)$
$= -1 + 3i$ |

- | | | |
|---|---|--|
| (1) $(1 + 6i) + (2 + i)$
$= 3 + 7i$ | (20) $(-5 + 5i) + (9 - 7i)$
$= 4 - 2i$ | (39) $(3 + 8i) - (-7 + 2i)$
$= 10 + 6i$ |
| (2) $(-3 - 8i) + (4 + 2i)$
$= 1 - 6i$ | (21) $(10 + 6i) + (-4 + 5i)$
$= 6 + 11i$ | (40) $(-6 - 4i) - (10 + i)$
$= -16 - 5i$ |
| (3) $(-9 + 8i) - (8 - i)$
$= -17 + 9i$ | (22) $(2 - 5i) - (2 - 9i)$
$= 4i$ | (41) $(-1 - 3i) - (6 - 6i)$
$= -7 + 3i$ |
| (4) $(9 - 9i) - (-10 + 3i)$
$= 19 - 12i$ | (23) $(4 - i) - (10 + 9i)$
$= -6 - 10i$ | (42) $(-5 + 9i) - (3 + 9i)$
$= -8$ |
| (5) $(7 + 4i) + (-10 + i)$
$= -3 + 5i$ | (24) $(-10 + 7i) - (-1 + 9i)$
$= -9 - 2i$ | (43) $(-6 - 3i) - (3 - 10i)$
$= -9 + 7i$ |
| (6) $(-8 - 9i) - (5 + 5i)$
$= -13 - 14i$ | (25) $(7 - 4i) + (-8 - 9i)$
$= -1 - 13i$ | (44) $(2 - 10i) + (7 - i)$
$= 9 - 11i$ |
| (7) $(-2 - 3i) + (-3 + 2i)$
$= -5 - i$ | (26) $(3 - 9i) - (1 + 5i)$
$= 2 - 14i$ | (45) $(-3 + 5i) - (6 + 10i)$
$= -9 - 5i$ |
| (8) $(-5 - 4i) + (-7 - 10i)$
$= -12 - 14i$ | (27) $(-7 - 8i) - (-1 - 9i)$
$= -6 + i$ | (46) $(2 + 9i) - (1 - 5i)$
$= 1 + 14i$ |
| (9) $(3 + i) - (-5 + 10i)$
$= 8 - 9i$ | (28) $(8 + 7i) - (-5 + 8i)$
$= 13 - i$ | (47) $(-4 - 2i) + (-4 + 5i)$
$= -8 + 3i$ |
| (10) $(1 - 3i) + (6 - 3i)$
$= 7 - 6i$ | (29) $(-8 + 2i) + (-8 + 2i)$
$= -16 + 4i$ | (48) $(-1 + 5i) + (8 + 3i)$
$= 7 + 8i$ |
| (11) $(-8 - 3i) - (10 - 6i)$
$= -18 + 3i$ | (30) $(5 + i) + (5 + i)$
$= 10 + 2i$ | (49) $(9 - 9i) + (6 + 4i)$
$= 15 - 5i$ |
| (12) $(8 - 4i) - (9 + 9i)$
$= -1 - 13i$ | (31) $(-4 + 5i) + (-7 - 7i)$
$= -11 - 2i$ | (50) $(-10 - 3i) - (-4 + 10i)$
$= -6 - 13i$ |
| (13) $(8 + 2i) - (-6 - 3i)$
$= 14 + 5i$ | (32) $(-6 - 4i) - (3 - 4i)$
$= -9$ | (51) $(-8 + 6i) + (4 - 6i)$
$= -4$ |
| (14) $(2 + 5i) + (-1 - 6i)$
$= 1 - i$ | (33) $(4 - 9i) + (-3 + 9i)$
$= 1$ | (52) $(-2 + 8i) - (-6 + 9i)$
$= 4 - i$ |
| (15) $(-4 + 7i) + (6 - 4i)$
$= 2 + 3i$ | (34) $(6 - 10i) - (-1 + 10i)$
$= 7 - 20i$ | (53) $(4 + 5i) - (7 + 7i)$
$= -3 - 2i$ |
| (16) $(-9 + 5i) - (-6 + i)$
$= -3 + 4i$ | (35) $(-5 - 5i) - (10 + 7i)$
$= -15 - 12i$ | (54) $(-3 - 7i) - (-6 + 2i)$
$= 3 - 9i$ |
| (17) $(-4 + 10i) - (4 + 8i)$
$= -8 + 2i$ | (36) $(2 - 9i) + (-3 - 10i)$
$= -1 - 19i$ | (55) $(-4 + i) - (7 - 5i)$
$= -11 + 6i$ |
| (18) $(8 - 10i) + (-3 - 2i)$
$= 5 - 12i$ | (37) $(5 + 7i) - (-8 + 8i)$
$= 13 - i$ | (56) $(-2 - i) - (-9 + 2i)$
$= 7 - 3i$ |
| (19) $(-2 - 8i) - (-8 - 9i)$
$= 6 + i$ | (38) $(1 + 9i) - (-3 + 5i)$
$= 4 + 4i$ | (57) $(-9 - 8i) + (-8 - 2i)$
$= -17 - 10i$ |

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|--|--|--|
| (1) $(5 + 6i) - (-7 + 10i)$
$= 12 - 4i$ | (20) $(-9 + 10i) - (-9 - 3i)$
$= 13i$ | (39) $(4 + 4i) - (-6 + 10i)$
$= 10 - 6i$ |
| (2) $(-8 - 10i) - (3 - 5i)$
$= -11 - 5i$ | (21) $(1 + i) - (6 + 6i)$
$= -5 - 5i$ | (40) $(3 - 8i) + (4 - 6i)$
$= 7 - 14i$ |
| (3) $(6 - 10i) - (-8 - 7i)$
$= 14 - 3i$ | (22) $(-3 - 2i) + (-8 - 6i)$
$= -11 - 8i$ | (41) $(-9 + 2i) - (-9 - 6i)$
$= 8i$ |
| (4) $(4 + 7i) + (-6 - 3i)$
$= -2 + 4i$ | (23) $(-10 + 8i) + (7 - 7i)$
$= -3 + i$ | (42) $(6 - i) + (-3 - 8i)$
$= 3 - 9i$ |
| (5) $(-3 - 3i) + (4 + 9i)$
$= 1 + 6i$ | (24) $(9 - 4i) - (-4 + 10i)$
$= 13 - 14i$ | (43) $(3 + 5i) + (4 + 6i)$
$= 7 + 11i$ |
| (6) $(9 + 3i) + (10 - 5i)$
$= 19 - 2i$ | (25) $(9 + 7i) + (10 + 2i)$
$= 19 + 9i$ | (44) $(5 - 6i) + (8 - 2i)$
$= 13 - 8i$ |
| (7) $(-4 + 6i) + (-5 - 9i)$
$= -9 - 3i$ | (26) $(3 - 10i) + (-5 + 5i)$
$= -2 - 5i$ | (45) $(-2 + 6i) + (-7 + 4i)$
$= -9 + 10i$ |
| (8) $(7 - 6i) + (5 + 10i)$
$= 12 + 4i$ | (27) $(-10 + i) - (2 + 5i)$
$= -12 - 4i$ | (46) $(5 - 5i) - (9 + 2i)$
$= -4 - 7i$ |
| (9) $(-8 - 4i) + (8 + 6i)$
$= 2i$ | (28) $(-8 - 5i) - (8 + 8i)$
$= -16 - 13i$ | (47) $(-1 + i) - (-1 + 9i)$
$= -8i$ |
| (10) $(2 + i) + (7 - 4i)$
$= 9 - 3i$ | (29) $(-4 + 10i) + (6 - 2i)$
$= 2 + 8i$ | (48) $(-3 - 6i) - (10 - 5i)$
$= -13 - i$ |
| (11) $(-5 + i) + (-3 + 8i)$
$= -8 + 9i$ | (30) $(6 + 8i) - (4 + 8i)$
$= 2$ | (49) $(9 + 2i) + (8 - 5i)$
$= 17 - 3i$ |
| (12) $(-8 + 10i) - (-9 - 9i)$
$= 1 + 19i$ | (31) $(2 + 6i) + (10 + 5i)$
$= 12 + 11i$ | (50) $(10 - 3i) + (-2 + 10i)$
$= 8 + 7i$ |
| (13) $(-7 + 6i) + (-5 - 9i)$
$= -12 - 3i$ | (32) $(3 + 7i) - (5 + 7i)$
$= -2$ | (51) $(6 + 8i) - (10 - 3i)$
$= -4 + 11i$ |
| (14) $(-4 - 10i) - (-6 - 8i)$
$= 2 - 2i$ | (33) $(-9 + 2i) + (-5 + 3i)$
$= -14 + 5i$ | (52) $(-10 + i) + (-5 + 3i)$
$= -15 + 4i$ |
| (15) $(-7 - 3i) + (-10 - 8i)$
$= -17 - 11i$ | (34) $(-4 + 2i) - (-3 + 6i)$
$= -1 - 4i$ | (53) $(-1 + 3i) + (-8 - 3i)$
$= -9$ |
| (16) $(1 - i) + (10 - 9i)$
$= 11 - 10i$ | (35) $(7 - 2i) + (-2 + 2i)$
$= 5$ | (54) $(2 - i) + (-3 + 2i)$
$= -1 + i$ |
| (17) $(9 + 9i) + (-10 - 2i)$
$= -1 + 7i$ | (36) $(9 + 8i) + (-4 - i)$
$= 5 + 7i$ | (55) $(7 - 8i) + (-8 - 5i)$
$= -1 - 13i$ |
| (18) $(-1 + 9i) + (-10 + 2i)$
$= -11 + 11i$ | (37) $(1 + 4i) - (1 + 3i)$
$= i$ | (56) $(5 - 2i) + (1 - 4i)$
$= 6 - 6i$ |
| (19) $(4 - 2i) - (3 - 9i)$
$= 1 + 7i$ | (38) $(-4 - 4i) + (4 - i)$
$= -5i$ | (57) $(1 + 6i) + (-8 - 7i)$
$= -7 - i$ |

- | | | |
|--|---|--|
| (1) $(2 - 10i) - (-1 - 4i)$
$= 3 - 6i$ | (20) $(6 + 8i) + (4 - 8i)$
$= 10$ | (39) $(7 - 10i) + (-6 - 10i)$
$= 1 - 20i$ |
| (2) $(-1 - 8i) + (-3 + 7i)$
$= -4 - i$ | (21) $(-9 - 10i) + (-8 - i)$
$= -17 - 11i$ | (40) $(-6 - 9i) - (-9 - 5i)$
$= 3 - 4i$ |
| (3) $(6 - 6i) + (3 + 7i)$
$= 9 + i$ | (22) $(1 - 3i) - (-6 + 3i)$
$= 7 - 6i$ | (41) $(-10 - 8i) - (-5 + 8i)$
$= -5 - 16i$ |
| (4) $(-9 - i) + (-3 + 4i)$
$= -12 + 3i$ | (23) $(-5 - 7i) - (6 + 3i)$
$= -11 - 10i$ | (42) $(-3 + 6i) - (9 + 2i)$
$= -12 + 4i$ |
| (5) $(-10 + 7i) - (5 - 5i)$
$= -15 + 12i$ | (24) $(4 + 8i) + (-1 + 5i)$
$= 3 + 13i$ | (43) $(8 + 4i) + (-5 + 3i)$
$= 3 + 7i$ |
| (6) $(10 + 3i) + (-7 - 8i)$
$= 3 - 5i$ | (25) $(-9 - 10i) + (-3 + 8i)$
$= -12 - 2i$ | (44) $(8 - 6i) - (-7 - 4i)$
$= 15 - 2i$ |
| (7) $(-8 + 7i) - (-6 - 7i)$
$= -2 + 14i$ | (26) $(3 + 10i) + (-9 - 4i)$
$= -6 + 6i$ | (45) $(4 - 10i) + (-9 - 10i)$
$= -5 - 20i$ |
| (8) $(-2 + 8i) - (6 - i)$
$= -8 + 9i$ | (27) $(9 - 4i) - (-8 - 9i)$
$= 17 + 5i$ | (46) $(3 + 6i) - (-10 + 7i)$
$= 13 - i$ |
| (9) $(-5 + 5i) - (-4 + i)$
$= -1 + 4i$ | (28) $(1 - 3i) + (8 - 3i)$
$= 9 - 6i$ | (47) $(-9 + 10i) + (-8 - 7i)$
$= -17 + 3i$ |
| (10) $(-5 + 6i) - (2 - 3i)$
$= -7 + 9i$ | (29) $(-4 - 7i) - (9 + 2i)$
$= -13 - 9i$ | (48) $(1 + 6i) + (-8 - 8i)$
$= -7 - 2i$ |
| (11) $(6 + 2i) + (-4 + 8i)$
$= 2 + 10i$ | (30) $(8 + 7i) - (-1 + 3i)$
$= 9 + 4i$ | (49) $(7 + 6i) - (9 + 8i)$
$= -2 - 2i$ |
| (12) $(-10 + 6i) - (9 - 10i)$
$= -19 + 16i$ | (31) $(-3 + 2i) - (6 + 8i)$
$= -9 - 6i$ | (50) $(-8 + 10i) - (-7 - 10i)$
$= -1 + 20i$ |
| (13) $(6 + i) - (7 + 4i)$
$= -1 - 3i$ | (32) $(7 - 9i) - (-10 - 6i)$
$= 17 - 3i$ | (51) $(9 - 6i) - (3 + 10i)$
$= 6 - 16i$ |
| (14) $(-7 - 2i) + (10 + 7i)$
$= 3 + 5i$ | (33) $(-5 + 8i) + (4 - 6i)$
$= -1 + 2i$ | (52) $(-1 + 5i) + (8 + i)$
$= 7 + 6i$ |
| (15) $(8 - 6i) + (-7 - 6i)$
$= 1 - 12i$ | (34) $(7 - 5i) + (8 + 10i)$
$= 15 + 5i$ | (53) $(6 + i) + (-5 + 10i)$
$= 1 + 11i$ |
| (16) $(-1 + 5i) - (-2 - 10i)$
$= 1 + 15i$ | (35) $(10 + 2i) + (7 + 10i)$
$= 17 + 12i$ | (54) $(-6 + i) + (8 - 6i)$
$= 2 - 5i$ |
| (17) $(-6 + 10i) + (9 - 7i)$
$= 3 + 3i$ | (36) $(9 + 6i) + (7 - 2i)$
$= 16 + 4i$ | (55) $(6 - 5i) - (7 + 6i)$
$= -1 - 11i$ |
| (18) $(-6 + 3i) - (-7 + 3i)$
$= 1$ | (37) $(-7 - 8i) + (-4 + 7i)$
$= -11 - i$ | (56) $(2 - 6i) + (8 + 5i)$
$= 10 - i$ |
| (19) $(-5 - i) - (7 + 6i)$
$= -12 - 7i$ | (38) $(-2 + 4i) + (9 + 9i)$
$= 7 + 13i$ | (57) $(3 - 7i) - (7 - 6i)$
$= -4 - i$ |

- | | | |
|--|--|--|
| (1) $(9 - 3i) + (9 + 5i)$
$= 18 + 2i$ | (20) $(-3 + 7i) + (10 - 2i)$
$= 7 + 5i$ | (39) $(-6 + 6i) - (-8 + 4i)$
$= 2 + 2i$ |
| (2) $(-5 + 6i) - (-8 - i)$
$= 3 + 7i$ | (21) $(-10 - 7i) + (-1 + 7i)$
$= -11$ | (40) $(8 - 2i) - (2 - 7i)$
$= 6 + 5i$ |
| (3) $(-7 + 4i) - (-4 - 8i)$
$= -3 + 12i$ | (22) $(-3 - 10i) + (-8 + 2i)$
$= -11 - 8i$ | (41) $(-7 + 7i) + (8 + 9i)$
$= 1 + 16i$ |
| (4) $(-9 - 9i) + (1 - 4i)$
$= -8 - 13i$ | (23) $(3 + 2i) + (-3 + 7i)$
$= 9i$ | (42) $(10 - 2i) + (10 - 10i)$
$= 20 - 12i$ |
| (5) $(7 - i) + (6 - 9i)$
$= 13 - 10i$ | (24) $(-10 + 9i) - (-8 + 5i)$
$= -2 + 4i$ | (43) $(9 - 7i) - (10 - 5i)$
$= -1 - 2i$ |
| (6) $(5 + i) - (-2 + 4i)$
$= 7 - 3i$ | (25) $(8 + 10i) + (-10 - 4i)$
$= -2 + 6i$ | (44) $(-9 - 8i) - (-4 + 5i)$
$= -5 - 13i$ |
| (7) $(-9 - 10i) - (1 + 3i)$
$= -10 - 13i$ | (26) $(-8 + 3i) - (-3 - 7i)$
$= -5 + 10i$ | (45) $(5 + 7i) - (1 - 8i)$
$= 4 + 15i$ |
| (8) $(-2 + 10i) - (-5 - 5i)$
$= 3 + 15i$ | (27) $(6 + 7i) - (-1 - i)$
$= 7 + 8i$ | (46) $(6 + 6i) - (-10 - 4i)$
$= 16 + 10i$ |
| (9) $(-4 + 9i) - (-4 + 8i)$
$= i$ | (28) $(-8 + 10i) + (-3 - 9i)$
$= -11 + i$ | (47) $(1 - 8i) - (-10 + 2i)$
$= 11 - 10i$ |
| (10) $(-1 + 7i) - (-4 + 7i)$
$= 3$ | (29) $(4 - 9i) + (-10 + 6i)$
$= -6 - 3i$ | (48) $(-10 - 8i) + (6 - 6i)$
$= -4 - 14i$ |
| (11) $(-8 + 4i) - (3 - 8i)$
$= -11 + 12i$ | (30) $(-7 + 3i) + (1 + i)$
$= -6 + 4i$ | (49) $(-8 - 7i) + (4 + 3i)$
$= -4 - 4i$ |
| (12) $(9 - i) - (-8 + 4i)$
$= 17 - 5i$ | (31) $(-8 - 3i) - (7 - 7i)$
$= -15 + 4i$ | (50) $(5 - 10i) + (7 - 9i)$
$= 12 - 19i$ |
| (13) $(-9 + 2i) + (-9 + i)$
$= -18 + 3i$ | (32) $(3 + 7i) + (9 + 6i)$
$= 12 + 13i$ | (51) $(8 - 9i) + (1 - 3i)$
$= 9 - 12i$ |
| (14) $(-9 + 5i) - (6 - i)$
$= -15 + 6i$ | (33) $(-2 - i) + (-10 + 9i)$
$= -12 + 8i$ | (52) $(3 + 6i) - (6 + 6i)$
$= -3$ |
| (15) $(-9 - 2i) - (-5 - 10i)$
$= -4 + 8i$ | (34) $(-2 + 8i) - (9 - 5i)$
$= -11 + 13i$ | (53) $(-3 + 10i) - (10 + i)$
$= -13 + 9i$ |
| (16) $(8 - 4i) + (8 - 4i)$
$= 16 - 8i$ | (35) $(-4 + 3i) - (6 + 8i)$
$= -10 - 5i$ | (54) $(-3 + 8i) - (-5 - 10i)$
$= 2 + 18i$ |
| (17) $(-9 + 10i) - (-8 + 2i)$
$= -1 + 8i$ | (36) $(-10 + 8i) + (-3 + 6i)$
$= -13 + 14i$ | (55) $(6 - 7i) + (-3 + i)$
$= 3 - 6i$ |
| (18) $(8 + 10i) + (2 + 10i)$
$= 10 + 20i$ | (37) $(-4 + 7i) - (-2 + 9i)$
$= -2 - 2i$ | (56) $(9 - 7i) + (-7 + 10i)$
$= 2 + 3i$ |
| (19) $(-5 + 3i) + (-8 - 4i)$
$= -13 - i$ | (38) $(-1 - 6i) - (-4 + 2i)$
$= 3 - 8i$ | (57) $(-10 - 2i) + (-5 + 10i)$
$= -15 + 8i$ |

- | | | |
|--|--|--|
| (1) $(6 + 8i) + (-9 - 10i)$
$= -3 - 2i$ | (20) $(-6 - 6i) - (-2 - 8i)$
$= -4 + 2i$ | (39) $(3 + 8i) - (-4 - i)$
$= 7 + 9i$ |
| (2) $(-4 - 9i) + (4 - 2i)$
$= -11i$ | (21) $(10 - 6i) - (-8 + 9i)$
$= 18 - 15i$ | (40) $(-6 - i) + (-9 + 6i)$
$= -15 + 5i$ |
| (3) $(6 - 2i) + (-4 - 5i)$
$= 2 - 7i$ | (22) $(-9 + i) + (6 + 7i)$
$= -3 + 8i$ | (41) $(-2 - 7i) + (-3 + i)$
$= -5 - 6i$ |
| (4) $(5 + 8i) - (8 + i)$
$= -3 + 7i$ | (23) $(9 - 8i) - (7 + 5i)$
$= 2 - 13i$ | (42) $(-3 + i) + (1 + 2i)$
$= -2 + 3i$ |
| (5) $(-7 + 2i) - (10 - 8i)$
$= -17 + 10i$ | (24) $(7 - 6i) - (4 + 4i)$
$= 3 - 10i$ | (43) $(-6 + 8i) + (6 + 4i)$
$= 12i$ |
| (6) $(-7 - i) - (10 - 6i)$
$= -17 + 5i$ | (25) $(10 - 2i) - (9 + 7i)$
$= 1 - 9i$ | (44) $(3 + 5i) + (-7 - 6i)$
$= -4 - i$ |
| (7) $(5 + 8i) - (-10 + i)$
$= 15 + 7i$ | (26) $(-9 - 5i) + (3 - 8i)$
$= -6 - 13i$ | (45) $(-9 - 2i) + (-8 - 3i)$
$= -17 - 5i$ |
| (8) $(-9 + 3i) - (10 - 4i)$
$= -19 + 7i$ | (27) $(-3 + 6i) - (-7 - 7i)$
$= 4 + 13i$ | (46) $(7 + 7i) - (-5 + 8i)$
$= 12 - i$ |
| (9) $(4 + 8i) + (-3 + 5i)$
$= 1 + 13i$ | (28) $(5 - 9i) - (-8 + 8i)$
$= 13 - 17i$ | (47) $(-9 - 2i) + (8 - 5i)$
$= -1 - 7i$ |
| (10) $(-10 - 10i) + (5 + i)$
$= -5 - 9i$ | (29) $(4 + 2i) + (4 - 9i)$
$= 8 - 7i$ | (48) $(6 + 2i) + (-3 + i)$
$= 3 + 3i$ |
| (11) $(-2 + 10i) + (-9 - i)$
$= -11 + 9i$ | (30) $(1 - 6i) + (-4 + 3i)$
$= -3 - 3i$ | (49) $(-1 - 10i) - (-3 + 2i)$
$= 2 - 12i$ |
| (12) $(5 + 4i) + (10 - 8i)$
$= 15 - 4i$ | (31) $(9 + 7i) - (5 + 2i)$
$= 4 + 5i$ | (50) $(5 + 6i) + (-4 - 9i)$
$= 1 - 3i$ |
| (13) $(-1 + 5i) + (4 - i)$
$= 3 + 4i$ | (32) $(3 + 8i) - (-1 + 4i)$
$= 4 + 4i$ | (51) $(-10 + 6i) - (4 + 4i)$
$= -14 + 2i$ |
| (14) $(2 + 9i) - (-4 - i)$
$= 6 + 10i$ | (33) $(6 - 6i) + (4 + 5i)$
$= 10 - i$ | (52) $(8 + 6i) - (-1 + 10i)$
$= 9 - 4i$ |
| (15) $(-9 + 6i) - (-3 - 4i)$
$= -6 + 10i$ | (34) $(-3 + 8i) + (-3 - 7i)$
$= -6 + i$ | (53) $(-5 + 6i) - (-2 - 3i)$
$= -3 + 9i$ |
| (16) $(3 + 6i) - (2 - 4i)$
$= 1 + 10i$ | (35) $(1 + 6i) - (-9 + 4i)$
$= 10 + 2i$ | (54) $(-10 + 4i) + (-4 - 10i)$
$= -14 - 6i$ |
| (17) $(-5 - 3i) - (6 - 10i)$
$= -11 + 7i$ | (36) $(-9 + 5i) - (1 + 6i)$
$= -10 - i$ | (55) $(7 + i) - (-3 + 3i)$
$= 10 - 2i$ |
| (18) $(2 + 6i) - (4 + 2i)$
$= -2 + 4i$ | (37) $(3 + 8i) - (1 + 7i)$
$= 2 + i$ | (56) $(2 - 6i) - (8 + 4i)$
$= -6 - 10i$ |
| (19) $(-2 + 4i) - (-5 + 9i)$
$= 3 - 5i$ | (38) $(10 - 3i) + (-1 - 5i)$
$= 9 - 8i$ | (57) $(-6 + 3i) + (3 + 8i)$
$= -3 + 11i$ |

$$(1) (-7 + 4i) - (-8 - 10i) \\ = 1 + 14i$$

$$(2) (-1 - i) - (-8 + 3i) \\ = 7 - 4i$$

$$(3) (-6 + 7i) + (-3 + i) \\ = -9 + 8i$$

$$(4) (-3 - 2i) - (5 - 7i) \\ = -8 + 5i$$

$$(5) (1 + 7i) - (-10 - 4i) \\ = 11 + 11i$$

$$(6) (5 + i) + (-1 + 2i) \\ = 4 + 3i$$

$$(7) (-3 - 10i) - (9 + 9i) \\ = -12 - 19i$$

$$(8) (8 + 10i) - (8 + 3i) \\ = 7i$$

$$(9) (9 - 10i) - (10 - 6i) \\ = -1 - 4i$$

$$(10) (-2 - 8i) - (-8 - 7i) \\ = 6 - i$$

$$(11) (-8 + 3i) - (4 - i) \\ = -12 + 4i$$

$$(12) (1 - 8i) + (-7 + 7i) \\ = -6 - i$$

$$(13) (5 + 4i) + (-8 + 2i) \\ = -3 + 6i$$

$$(14) (3 + 4i) + (-1 + 2i) \\ = 2 + 6i$$

$$(15) (-9 + 9i) + (4 - 8i) \\ = -5 + i$$

$$(16) (-3 - i) + (6 + 7i) \\ = 3 + 6i$$

$$(17) (4 - 5i) - (-5 + 7i) \\ = 9 - 12i$$

$$(18) (2 - 8i) + (-1 - 3i) \\ = 1 - 11i$$

$$(19) (-9 - 5i) + (-3 - 2i) \\ = -12 - 7i$$

$$(20) (-7 + 7i) + (-2 - 8i) \\ = -9 - i$$

$$(21) (-6 - i) + (9 + 7i) \\ = 3 + 6i$$

$$(22) (-5 - 4i) - (-3 - 7i) \\ = -2 + 3i$$

$$(23) (-5 + 2i) - (-5 - 2i) \\ = 4i$$

$$(24) (-7 - 2i) + (2 + 7i) \\ = -5 + 5i$$

$$(25) (-6 - 3i) + (-9 - i) \\ = -15 - 4i$$

$$(26) (-3 + 3i) + (3 - 7i) \\ = -4i$$

$$(27) (-2 + 5i) + (10 - 2i) \\ = 8 + 3i$$

$$(28) (-2 + 3i) - (4 + 8i) \\ = -6 - 5i$$

$$(29) (-5 - i) - (8 + 8i) \\ = -13 - 9i$$

$$(30) (1 + i) + (-6 + 2i) \\ = -5 + 3i$$

$$(31) (10 - 7i) - (6 - 6i) \\ = 4 - i$$

$$(32) (9 + i) + (10 + 10i) \\ = 19 + 11i$$

$$(33) (2 - 2i) - (8 + 7i) \\ = -6 - 9i$$

$$(34) (2 - 4i) + (5 - 2i) \\ = 7 - 6i$$

$$(35) (-7 + 7i) + (-10 - 6i) \\ = -17 + i$$

$$(36) (6 - 2i) + (-3 - 2i) \\ = 3 - 4i$$

$$(37) (-7 - 4i) - (-9 - 5i) \\ = 2 + i$$

$$(38) (-5 + 8i) + (-10 - 10i) \\ = -15 - 2i$$

$$(39) (6 + 9i) + (7 + 5i) \\ = 13 + 14i$$

$$(40) (6 + 10i) - (7 - 5i) \\ = -1 + 15i$$

$$(41) (9 - 4i) + (1 + 10i) \\ = 10 + 6i$$

$$(42) (-3 + 2i) + (-9 - 10i) \\ = -12 - 8i$$

$$(43) (-10 - 8i) - (6 + 6i) \\ = -16 - 14i$$

$$(44) (-2 + 5i) + (-6 + 2i) \\ = -8 + 7i$$

$$(45) (7 + 4i) + (-4 - 8i) \\ = 3 - 4i$$

$$(46) (8 - 6i) + (-7 + 10i) \\ = 1 + 4i$$

$$(47) (4 + 10i) + (-8 - 2i) \\ = -4 + 8i$$

$$(48) (7 + i) - (10 + 8i) \\ = -3 - 7i$$

$$(49) (-8 + 3i) - (2 - 2i) \\ = -10 + 5i$$

$$(50) (-8 - 2i) + (-5 + 8i) \\ = -13 + 6i$$

$$(51) (1 + 10i) - (-8 - 4i) \\ = 9 + 14i$$

$$(52) (-6 + i) - (4 + 5i) \\ = -10 - 4i$$

$$(53) (-3 - 10i) + (-8 - 6i) \\ = -11 - 16i$$

$$(54) (-1 - 3i) + (6 - 8i) \\ = 5 - 11i$$

$$(55) (5 - 5i) - (10 + 3i) \\ = -5 - 8i$$

$$(56) (-1 - 10i) - (-2 + 9i) \\ = 1 - 19i$$

$$(57) (1 + i) - (-5 - 8i) \\ = 6 + 9i$$

- | | | |
|---|---|--|
| (1) $(-6 - 10i) + (-2 + 3i)$
$= -8 - 7i$ | (20) $(-4 + i) + (3 + 9i)$
$= -1 + 10i$ | (39) $(5 - 8i) + (9 - i)$
$= 14 - 9i$ |
| (2) $(-7 + 4i) - (-5 - 5i)$
$= -2 + 9i$ | (21) $(-8 - 3i) - (-3 + 6i)$
$= -5 - 9i$ | (40) $(-7 - 10i) + (-10 + 5i)$
$= -17 - 5i$ |
| (3) $(2 - 3i) + (-10 - i)$
$= -8 - 4i$ | (22) $(2 - 8i) + (4 + 6i)$
$= 6 - 2i$ | (41) $(-4 - i) - (4 + 7i)$
$= -8 - 8i$ |
| (4) $(7 - 4i) + (-3 + 8i)$
$= 4 + 4i$ | (23) $(9 + 7i) + (2 + 2i)$
$= 11 + 9i$ | (42) $(8 + 4i) + (5 - 10i)$
$= 13 - 6i$ |
| (5) $(-4 - 6i) - (-2 + 9i)$
$= -2 - 15i$ | (24) $(6 + 9i) - (-10 - 6i)$
$= 16 + 15i$ | (43) $(-9 - 10i) + (10 - 4i)$
$= 1 - 14i$ |
| (6) $(-2 - 10i) - (-1 - i)$
$= -1 - 9i$ | (25) $(2 + 7i) + (-2 - 5i)$
$= 2i$ | (44) $(4 + 6i) + (4 - 5i)$
$= 8 + i$ |
| (7) $(-1 - i) + (-5 - 10i)$
$= -6 - 11i$ | (26) $(-8 - 9i) + (3 - 10i)$
$= -5 - 19i$ | (45) $(-7 + 9i) + (-10 + 6i)$
$= -17 + 15i$ |
| (8) $(7 + 3i) + (6 - 4i)$
$= 13 - i$ | (27) $(6 + 4i) - (5 + 2i)$
$= 1 + 2i$ | (46) $(7 + 3i) + (-8 + 3i)$
$= -1 + 6i$ |
| (9) $(1 - 4i) - (-7 + 2i)$
$= 8 - 6i$ | (28) $(-1 - 10i) - (-6 + 10i)$
$= 5 - 20i$ | (47) $(-2 + 2i) + (6 + 6i)$
$= 4 + 8i$ |
| (10) $(-6 - 9i) + (1 - 9i)$
$= -5 - 18i$ | (29) $(6 - 6i) + (-8 - 8i)$
$= -2 - 14i$ | (48) $(8 + 7i) + (-9 - 2i)$
$= -1 + 5i$ |
| (11) $(9 + 10i) + (9 - 3i)$
$= 18 + 7i$ | (30) $(9 - 10i) + (1 + 6i)$
$= 10 - 4i$ | (49) $(-3 + 6i) + (7 + 5i)$
$= 4 + 11i$ |
| (12) $(-3 + 10i) + (5 - 6i)$
$= 2 + 4i$ | (31) $(-8 - 7i) - (-6 + 2i)$
$= -2 - 9i$ | (50) $(9 + 8i) - (-5 + 6i)$
$= 14 + 2i$ |
| (13) $(-1 + i) - (-2 - 7i)$
$= 1 + 8i$ | (32) $(-2 - 4i) + (1 - 2i)$
$= -1 - 6i$ | (51) $(8 + 7i) - (7 - 9i)$
$= 1 + 16i$ |
| (14) $(6 - i) - (-10 + 3i)$
$= 16 - 4i$ | (33) $(3 - 2i) + (6 + 7i)$
$= 9 + 5i$ | (52) $(3 - 2i) - (-6 - 8i)$
$= 9 + 6i$ |
| (15) $(10 + 10i) - (-3 - 3i)$
$= 13 + 13i$ | (34) $(5 + 2i) + (-3 + 5i)$
$= 2 + 7i$ | (53) $(3 + 6i) + (-7 - 7i)$
$= -4 - i$ |
| (16) $(2 - 2i) - (4 + 6i)$
$= -2 - 8i$ | (35) $(-5 + 2i) + (-7 - 3i)$
$= -12 - i$ | (54) $(-4 + 8i) - (-10 - i)$
$= 6 + 9i$ |
| (17) $(10 - 7i) - (-4 + 4i)$
$= 14 - 11i$ | (36) $(-8 - 8i) + (5 + 5i)$
$= -3 - 3i$ | (55) $(2 - 10i) + (-4 + 9i)$
$= -2 - i$ |
| (18) $(-3 + 3i) + (7 + 10i)$
$= 4 + 13i$ | (37) $(-8 - 3i) - (2 + 6i)$
$= -10 - 9i$ | (56) $(-4 + 10i) - (8 + 7i)$
$= -12 + 3i$ |
| (19) $(-1 - 5i) - (1 - i)$
$= -2 - 4i$ | (38) $(6 + i) - (-4 + 7i)$
$= 10 - 6i$ | (57) $(4 - 8i) + (8 - 4i)$
$= 12 - 12i$ |

- | | | |
|---|---|---|
| (1) $(-1 - 4i) - (-3 - 9i)$
$= 2 + 5i$ | (20) $(3 - 8i) + (9 - i)$
$= 12 - 9i$ | (39) $(-6 + 10i) + (-6 + 2i)$
$= -12 + 12i$ |
| (2) $(-5 + 3i) + (3 - 8i)$
$= -2 - 5i$ | (21) $(-3 + 6i) - (-1 - 2i)$
$= -2 + 8i$ | (40) $(-10 + 10i) + (-10 + 10i)$
$= -20 + 20i$ |
| (3) $(1 + 4i) + (9 + 5i)$
$= 10 + 9i$ | (22) $(-3 + 10i) - (-6 - 9i)$
$= 3 + 19i$ | (41) $(-1 + 4i) + (6 - 8i)$
$= 5 - 4i$ |
| (4) $(-2 + 7i) + (3 - 9i)$
$= 1 - 2i$ | (23) $(-6 + 2i) + (-2 + 10i)$
$= -8 + 12i$ | (42) $(1 - 3i) + (10 + 3i)$
$= 11$ |
| (5) $(3 - 7i) - (-2 + i)$
$= 5 - 8i$ | (24) $(-8 - 7i) + (4 - 5i)$
$= -4 - 12i$ | (43) $(3 - 6i) - (-4 + 10i)$
$= 7 - 16i$ |
| (6) $(8 + 5i) + (-3 + 5i)$
$= 5 + 10i$ | (25) $(-8 + 9i) + (6 - 9i)$
$= -2$ | (44) $(9 - 4i) + (-10 + 10i)$
$= -1 + 6i$ |
| (7) $(10 + 10i) - (-10 - 7i)$
$= 20 + 17i$ | (26) $(6 + 7i) - (-6 - 8i)$
$= 12 + 15i$ | (45) $(5 + 3i) + (3 - 6i)$
$= 8 - 3i$ |
| (8) $(-2 + 9i) + (10 + 3i)$
$= 8 + 12i$ | (27) $(-3 - 3i) - (7 - 8i)$
$= -10 + 5i$ | (46) $(5 - 8i) - (-4 + 10i)$
$= 9 - 18i$ |
| (9) $(7 - 5i) + (-8 - 9i)$
$= -1 - 14i$ | (28) $(-7 + 6i) - (3 - 5i)$
$= -10 + 11i$ | (47) $(2 - 10i) + (-4 - 10i)$
$= -2 - 20i$ |
| (10) $(10 - 7i) + (-7 + 8i)$
$= 3 + i$ | (29) $(-1 + i) + (-9 - 4i)$
$= -10 - 3i$ | (48) $(-5 - 8i) + (9 - 9i)$
$= 4 - 17i$ |
| (11) $(5 + 3i) + (-8 + 2i)$
$= -3 + 5i$ | (30) $(-5 + 7i) - (3 + 4i)$
$= -8 + 3i$ | (49) $(4 + i) + (7 + 9i)$
$= 11 + 10i$ |
| (12) $(7 + 6i) + (5 - 6i)$
$= 12$ | (31) $(-7 + 4i) + (-8 - 7i)$
$= -15 - 3i$ | (50) $(8 - 9i) - (1 - 8i)$
$= 7 - i$ |
| (13) $(-9 - i) - (-10 + 2i)$
$= 1 - 3i$ | (32) $(10 + 7i) + (9 - i)$
$= 19 + 6i$ | (51) $(-5 - 2i) - (6 - 6i)$
$= -11 + 4i$ |
| (14) $(-7 - i) - (-1 - 8i)$
$= -6 + 7i$ | (33) $(6 + 5i) + (-9 - 2i)$
$= -3 + 3i$ | (52) $(9 - 6i) - (-1 + 2i)$
$= 10 - 8i$ |
| (15) $(-10 - 7i) - (1 + 8i)$
$= -11 - 15i$ | (34) $(-3 - 9i) - (2 - 9i)$
$= -5$ | (53) $(-5 - 9i) - (2 + 2i)$
$= -7 - 11i$ |
| (16) $(-10 - i) + (8 - 9i)$
$= -2 - 10i$ | (35) $(9 - 3i) + (2 - 4i)$
$= 11 - 7i$ | (54) $(-9 - 8i) + (-1 + 5i)$
$= -10 - 3i$ |
| (17) $(10 + 3i) + (-2 + i)$
$= 8 + 4i$ | (36) $(9 + 3i) + (-5 - 2i)$
$= 4 + i$ | (55) $(-8 - 6i) - (-9 + 3i)$
$= 1 - 9i$ |
| (18) $(-2 - i) + (5 + 8i)$
$= 3 + 7i$ | (37) $(8 + 6i) + (-10 + 5i)$
$= -2 + 11i$ | (56) $(-8 - 7i) - (-4 + 7i)$
$= -4 - 14i$ |
| (19) $(-8 - 2i) - (-5 - 6i)$
$= -3 + 4i$ | (38) $(-7 - i) - (2 + 10i)$
$= -9 - 11i$ | (57) $(5 - 10i) - (-8 + 9i)$
$= 13 - 19i$ |

1.5.2 乗法

複素数の積を求める問題。

(1) $(-3 - 3i)(-10 + 6i)$

(16) $(-8 - 7i)(-7 + 5i)$

(31) $(-10 - 9i)(-6 + 5i)$

(2) $(-3 - 6i)(-1 - 5i)$

(17) $(3 + 9i)(4 + 3i)$

(32) $(-6 + 4i)(4 - 2i)$

(3) $(10 + 9i)(-4 - 5i)$

(18) $(-2 - i)(-7 + 4i)$

(33) $(10 + i)(-4 - i)$

(4) $(-7 - 8i)(-5 + 2i)$

(19) $(10 - 7i)(5 + 2i)$

(34) $(-10 - 8i)(6 + 6i)$

(5) $(-3 + 8i)(-2 + 7i)$

(20) $(5 + 4i)(-2 + 7i)$

(35) $(-3 - 4i)(-6 + 10i)$

(6) $(4 + 9i)(-6 + 3i)$

(21) $(1 - i)(1 - 5i)$

(36) $(10 + 2i)(5 - 8i)$

(7) $(6 + 7i)(-2 + 8i)$

(22) $(10 + 10i)(-9 - 9i)$

(37) $(4 - 6i)(3 - 10i)$

(8) $(7 + 3i)(-10 - 4i)$

(23) $(-7 - 4i)(-3 - 8i)$

(38) $(-7 + 2i)(-10 + 6i)$

(9) $(5 + 10i)(-2 - 4i)$

(24) $(10 + i)(7 - 6i)$

(39) $(2 - 3i)(6 + i)$

(10) $(-4 - i)(-6 + 5i)$

(25) $(-4 + 3i)(-5 - 6i)$

(40) $(-5 - 3i)(-4 - 3i)$

(11) $(-1 - 4i)(4 - 8i)$

(26) $(1 - 9i)(-2 + 5i)$

(41) $(-7 - 6i)(-10 + 6i)$

(12) $(-10 - 4i)(7 + 7i)$

(27) $(-9 - 10i)(-7 - 3i)$

(42) $(-5 - 5i)(-6 - 7i)$

(13) $(5 - 10i)(5 + 6i)$

(28) $(-2 + i)(8 + i)$

(43) $(-10 + 3i)(-3 + 4i)$

(14) $(5 + 10i)(4 - 7i)$

(29) $(-6 + 10i)(-7 + 8i)$

(44) $(4 + 4i)(-4 + 6i)$

(15) $(9 + 4i)(-4 + 2i)$

(30) $(-2 - 4i)(2 - 6i)$

(45) $(8 - i)(-5 - 7i)$

(1) $(-2 - 6i)(-9 - i)$

(16) $(-8 - 3i)(-10 - 7i)$

(31) $(2 - 6i)(-9 - 6i)$

(2) $(9 - 5i)(4 + 6i)$

(17) $(6 + 10i)(-4 - 7i)$

(32) $(-8 + 2i)(-6 - 8i)$

(3) $(-2 - 3i)(4 - 5i)$

(18) $(-2 - 8i)(-7 + 8i)$

(33) $(2 + 6i)(7 + 2i)$

(4) $(10 - i)(-6 - 3i)$

(19) $(-10 + 2i)(-8 + 4i)$

(34) $(-5 + 3i)(-5 + 8i)$

(5) $(7 - 3i)(10 + 2i)$

(20) $(1 + 9i)(-4 + 7i)$

(35) $(-5 - 5i)(10 - 3i)$

(6) $(-9 + 10i)(-9 - 8i)$

(21) $(3 + 8i)(9 - 7i)$

(36) $(1 - 2i)(10 - 2i)$

(7) $(8 - 2i)(-7 - 8i)$

(22) $(2 - 3i)(-4 + 7i)$

(37) $(-4 + 4i)(3 + 3i)$

(8) $(-10 - 10i)(5 + 3i)$

(23) $(-4 + 3i)(-2 + 5i)$

(38) $(-10 + 9i)(8 + 4i)$

(9) $(5 + 2i)(6 + 3i)$

(24) $(2 + 8i)(-3 + 7i)$

(39) $(2 - 6i)(-7 + i)$

(10) $(-8 - 10i)(-2 - i)$

(25) $(-4 + 2i)(-1 - 2i)$

(40) $(1 - 6i)(-7 + 6i)$

(11) $(-4 + 2i)(-9 - 8i)$

(26) $(9 + 6i)(-2 - 6i)$

(41) $(-6 + 2i)(-4 + 6i)$

(12) $(1 + 9i)(-7 - 6i)$

(27) $(-6 + 7i)(5 - 9i)$

(42) $(-6 + 8i)(-7 + 6i)$

(13) $(-8 + 3i)(1 - 9i)$

(28) $(-7 + 4i)(-4 - 10i)$

(43) $(-7 + i)(5 - 4i)$

(14) $(6 - i)(6 + i)$

(29) $(-10 - 9i)(-9 + 9i)$

(44) $(8 + 2i)(10 + 10i)$

(15) $(-2 + 10i)(1 + 8i)$

(30) $(5 - 6i)(-10 + 8i)$

(45) $(2 + 6i)(4 + 3i)$

(1) $(-10 - 10i)(-9 - 5i)$

(16) $(6 + 9i)(6 - i)$

(31) $(3 - 3i)(-3 + i)$

(2) $(3 + 7i)(8 + 9i)$

(17) $(7 + 5i)(9 + 5i)$

(32) $(-3 + 10i)(-4 - i)$

(3) $(-5 + i)(2 + 7i)$

(18) $(-3 + 10i)(2 - 8i)$

(33) $(1 + 10i)(-4 - 8i)$

(4) $(9 - i)(1 - 4i)$

(19) $(-5 - i)(10 - i)$

(34) $(-2 + 9i)(4 + 8i)$

(5) $(-1 + i)(-10 + 3i)$

(20) $(-10 + 8i)(4 + 5i)$

(35) $(1 + 8i)(-10 - 6i)$

(6) $(7 - 9i)(-3 - i)$

(21) $(-8 + 7i)(7 + 3i)$

(36) $(-5 + i)(-9 - 2i)$

(7) $(10 + i)(9 + 3i)$

(22) $(-3 - 4i)(-1 + 8i)$

(37) $(7 - 2i)(-10 + 2i)$

(8) $(6 - 9i)(10 + 2i)$

(23) $(1 + 3i)(-4 - 5i)$

(38) $(7 - 2i)(3 + 3i)$

(9) $(-10 + 5i)(-1 - i)$

(24) $(-2 - 8i)(7 - 5i)$

(39) $(10 - 3i)(-10 - 8i)$

(10) $(-4 - 9i)(-3 - 4i)$

(25) $(7 + 3i)(6 + 2i)$

(40) $(3 - 7i)(2 + 10i)$

(11) $(-1 - 5i)(5 + 4i)$

(26) $(2 + 10i)(-4 - 2i)$

(41) $(3 + 6i)(5 - 9i)$

(12) $(-7 - 9i)(10 - 6i)$

(27) $(4 - 2i)(-7 - 7i)$

(42) $(-1 + 6i)(-9 + 9i)$

(13) $(10 + 2i)(-6 + 10i)$

(28) $(-10 + 9i)(-9 + 9i)$

(43) $(4 - i)(3 - i)$

(14) $(9 + 3i)(-7 - 4i)$

(29) $(2 - 2i)(-6 + 9i)$

(44) $(-9 + 10i)(3 - 3i)$

(15) $(3 - 7i)(-2 - 10i)$

(30) $(5 + 2i)(-9 - 3i)$

(45) $(10 - 2i)(9 - 5i)$

(1) $(-8 - 7i)(2 - 4i)$

(16) $(1 + 5i)(1 - 6i)$

(31) $(10 + 4i)(-6 + 7i)$

(2) $(-8 + 8i)(-1 + 2i)$

(17) $(-10 + 7i)(-9 - 4i)$

(32) $(-6 + 10i)(3 - 8i)$

(3) $(-7 - 7i)(8 + 3i)$

(18) $(6 + 6i)(8 - 7i)$

(33) $(1 - 9i)(-10 + 6i)$

(4) $(-1 - 8i)(-3 + 7i)$

(19) $(9 + 4i)(3 - 5i)$

(34) $(6 - 10i)(2 + i)$

(5) $(9 - 2i)(8 + i)$

(20) $(4 + 5i)(1 + 8i)$

(35) $(-4 - 2i)(-2 - 3i)$

(6) $(5 + i)(2 - 6i)$

(21) $(-3 + 3i)(-4 + 3i)$

(36) $(8 + 8i)(3 + 7i)$

(7) $(1 + 5i)(1 + 7i)$

(22) $(-8 - 3i)(3 + 7i)$

(37) $(6 - 3i)(-9 - 9i)$

(8) $(2 + 4i)(-9 - i)$

(23) $(2 + 5i)(3 + i)$

(38) $(-6 - 10i)(-1 - 7i)$

(9) $(6 + 8i)(-2 + 6i)$

(24) $(-8 - 4i)(-10 + 2i)$

(39) $(-7 + i)(-6 + 4i)$

(10) $(4 + 4i)(8 - 5i)$

(25) $(4 + 8i)(6 - 7i)$

(40) $(-9 + 4i)(3 + 10i)$

(11) $(-5 + 9i)(-9 - 7i)$

(26) $(10 - 9i)(3 + 6i)$

(41) $(3 + 5i)(1 + 6i)$

(12) $(10 - i)(-9 - 7i)$

(27) $(-7 + 10i)(6 - 5i)$

(42) $(9 + i)(-4 + 6i)$

(13) $(-7 - 6i)(2 + 3i)$

(28) $(10 - 4i)(-4 - 6i)$

(43) $(-10 - 3i)(4 + 9i)$

(14) $(3 + 9i)(-2 - 6i)$

(29) $(-9 + 4i)(7 + 2i)$

(44) $(4 - 2i)(-6 + 9i)$

(15) $(2 + 3i)(-1 + 3i)$

(30) $(-5 + 3i)(-1 + 8i)$

(45) $(-7 + 3i)(-3 + 2i)$

(1) $(-1 + i)(8 - 3i)$

(16) $(-3 - 3i)(4 - 6i)$

(31) $(7 - 7i)(7 - 3i)$

(2) $(8 + i)(6 - 10i)$

(17) $(9 + 2i)(-6 + i)$

(32) $(3 + 3i)(-5 + 5i)$

(3) $(-9 + 5i)(8 + 9i)$

(18) $(-5 - 10i)(-7 - 3i)$

(33) $(4 + 2i)(-4 - 7i)$

(4) $(-8 - 7i)(-1 - 10i)$

(19) $(-6 - 9i)(1 + 9i)$

(34) $(-3 - 10i)(9 - 10i)$

(5) $(-3 + 9i)(6 + 10i)$

(20) $(10 - 3i)(-9 - 10i)$

(35) $(-8 + 2i)(4 + 6i)$

(6) $(-1 + 2i)(-2 + 2i)$

(21) $(-5 - i)(-6 + 7i)$

(36) $(-7 - i)(-9 - 5i)$

(7) $(6 - 4i)(1 + 8i)$

(22) $(4 - 8i)(-4 - i)$

(37) $(7 + 10i)(4 - 10i)$

(8) $(5 - 9i)(1 + 3i)$

(23) $(1 - i)(2 - 6i)$

(38) $(-4 - 2i)(-9 - i)$

(9) $(4 + 2i)(-6 + i)$

(24) $(5 + 6i)(7 - 4i)$

(39) $(10 + 7i)(2 + 4i)$

(10) $(-10 + i)(9 + 10i)$

(25) $(4 - 8i)(-10 + 8i)$

(40) $(-7 - 7i)(-6 + i)$

(11) $(-10 + i)(8 + 8i)$

(26) $(-5 - 5i)(9 + 7i)$

(41) $(-7 + 7i)(-7 + i)$

(12) $(10 - 3i)(5 + 7i)$

(27) $(-10 + 9i)(-7 + i)$

(42) $(8 + 7i)(2 - 6i)$

(13) $(8 + 9i)(-6 - 9i)$

(28) $(-6 - 10i)(3 - 4i)$

(43) $(-2 - 10i)(7 + 7i)$

(14) $(-9 - 5i)(4 + 5i)$

(29) $(9 + 4i)(2 + 6i)$

(44) $(7 - i)(4 - 5i)$

(15) $(-4 + 6i)(4 - 3i)$

(30) $(-6 - 9i)(9 - 3i)$

(45) $(3 - 5i)(7 + 3i)$

(1) $(3 + 6i)(6 + 9i)$

(16) $(-7 - 3i)(-10 - 4i)$

(31) $(2 + 2i)(-9 - 5i)$

(2) $(5 + 3i)(-1 - 3i)$

(17) $(3 + 7i)(-9 + 9i)$

(32) $(-5 - 5i)(-10 - 7i)$

(3) $(7 + 7i)(4 - 8i)$

(18) $(-9 - i)(2 + 5i)$

(33) $(1 + 5i)(3 - 2i)$

(4) $(-9 + 6i)(-9 + 9i)$

(19) $(9 - 2i)(-1 + 4i)$

(34) $(9 - 3i)(-7 - 7i)$

(5) $(-6 - 2i)(9 + 9i)$

(20) $(4 - 4i)(-2 - 7i)$

(35) $(-3 + 2i)(4 + 5i)$

(6) $(-3 - 3i)(-8 + 8i)$

(21) $(7 - i)(4 - 6i)$

(36) $(-5 + 10i)(-3 - 9i)$

(7) $(8 - 5i)(3 - 6i)$

(22) $(6 + 4i)(-4 - 5i)$

(37) $(1 - 5i)(-2 - 4i)$

(8) $(8 - 3i)(9 + 10i)$

(23) $(2 - 7i)(4 + 7i)$

(38) $(-9 - 5i)(9 - 3i)$

(9) $(10 - 8i)(-5 + 3i)$

(24) $(-2 + 3i)(-8 + 2i)$

(39) $(3 - i)(10 + 9i)$

(10) $(-9 - 8i)(-6 - 9i)$

(25) $(-1 + 2i)(-3 - 8i)$

(40) $(-7 - 9i)(7 - 10i)$

(11) $(7 - 10i)(5 + 7i)$

(26) $(7 - 6i)(7 - 7i)$

(41) $(3 - 3i)(-8 - 5i)$

(12) $(4 - 2i)(-2 + 5i)$

(27) $(4 - 7i)(-4 + 6i)$

(42) $(6 + 4i)(-5 - 7i)$

(13) $(-5 - 4i)(1 + 2i)$

(28) $(8 - 6i)(-9 + 10i)$

(43) $(8 + 8i)(4 - 5i)$

(14) $(-9 - 4i)(-4 + i)$

(29) $(9 - 6i)(-9 + 7i)$

(44) $(5 - i)(-7 - 6i)$

(15) $(-7 - 4i)(-5 - i)$

(30) $(-7 + 3i)(3 - 2i)$

(45) $(-1 + 2i)(1 + 8i)$

(1) $(5 - 4i)(9 + 2i)$

(16) $(-8 + 8i)(-7 + 10i)$

(31) $(5 - 10i)(-5 + 4i)$

(2) $(10 + 9i)(-7 - 7i)$

(17) $(-4 - 3i)(-7 - 8i)$

(32) $(-3 + 5i)(6 + 9i)$

(3) $(9 + 6i)(2 + 7i)$

(18) $(6 - 5i)(5 - 10i)$

(33) $(5 + 7i)(6 + 3i)$

(4) $(-1 + 2i)(1 + 3i)$

(19) $(8 + 5i)(10 + 7i)$

(34) $(3 + 3i)(-1 - i)$

(5) $(-6 + 7i)(1 + 5i)$

(20) $(-1 - 3i)(-2 - i)$

(35) $(4 + 4i)(2 + 10i)$

(6) $(6 + 5i)(-1 + 6i)$

(21) $(5 - 4i)(-8 + 8i)$

(36) $(6 - 2i)(1 - 7i)$

(7) $(-5 + i)(-4 - 2i)$

(22) $(6 + 8i)(5 + 5i)$

(37) $(7 - 6i)(-5 + i)$

(8) $(6 - 10i)(-10 + 9i)$

(23) $(1 - 7i)(-1 - 3i)$

(38) $(-6 - 4i)(5 - 5i)$

(9) $(-6 - 4i)(-7 + 3i)$

(24) $(-6 - 8i)(-5 - 4i)$

(39) $(3 - 7i)(-1 + 3i)$

(10) $(-9 + 8i)(-7 - 8i)$

(25) $(4 - i)(-10 + 6i)$

(40) $(9 - 6i)(4 - 2i)$

(11) $(-4 - 5i)(2 - 9i)$

(26) $(2 - 9i)(1 + 10i)$

(41) $(3 + 4i)(-6 + 6i)$

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(28) $(-9 - 4i)(7 - 7i)$

(43) $(-4 - 7i)(-7 - i)$

(14) $(5 - 6i)(2 + 7i)$

(29) $(-5 + i)(2 + 5i)$

(44) $(-1 + 9i)(-10 + 2i)$

(15) $(8 - 6i)(-7 - 3i)$

(30) $(6 - 8i)(-5 + 3i)$

(45) $(7 - 3i)(6 - 6i)$

(1) $(-10 - 4i)(8 - 2i)$

(16) $(-5 + 8i)(4 - 6i)$

(31) $(-3 - 3i)(4 - 5i)$

(2) $(9 + 6i)(-5 + 6i)$

(17) $(-8 + 6i)(-3 + 8i)$

(32) $(4 + 7i)(7 + 7i)$

(3) $(2 + 5i)(-2 - 8i)$

(18) $(-4 - 5i)(-7 - 7i)$

(33) $(5 + 5i)(6 + 10i)$

(4) $(9 + 4i)(10 - 6i)$

(19) $(3 + 10i)(3 + i)$

(34) $(5 - 3i)(-7 + i)$

(5) $(3 - 7i)(-4 + 6i)$

(20) $(10 - 8i)(3 - 2i)$

(35) $(1 + 5i)(8 - 7i)$

(6) $(3 - 10i)(10 - 6i)$

(21) $(-2 - 7i)(9 + 3i)$

(36) $(-7 + 6i)(9 + 7i)$

(7) $(-2 + 4i)(1 + 7i)$

(22) $(4 - 5i)(4 - 9i)$

(37) $(-4 + 4i)(-4 - 7i)$

(8) $(2 - 9i)(10 - 4i)$

(23) $(3 + 3i)(-7 - 2i)$

(38) $(1 + 10i)(-2 - 8i)$

(9) $(-10 - 7i)(9 - 10i)$

(24) $(4 + 10i)(8 + 2i)$

(39) $(5 - 4i)(-6 - i)$

(10) $(5 - 9i)(-6 + 5i)$

(25) $(-5 - i)(-6 + 6i)$

(40) $(5 + 4i)(4 - 10i)$

(11) $(-7 + i)(5 - 5i)$

(26) $(-8 - 9i)(5 - 7i)$

(41) $(10 + 8i)(-2 + 10i)$

(12) $(-3 + 8i)(2 + 5i)$

(27) $(2 - 4i)(-8 + 8i)$

(42) $(3 - 8i)(-7 + 4i)$

(13) $(9 - 6i)(-3 - 8i)$

(28) $(-6 + 7i)(10 - 2i)$

(43) $(-2 - i)(6 + 6i)$

(14) $(-7 + 10i)(-10 + 5i)$

(29) $(9 - 7i)(5 - i)$

(44) $(10 + 4i)(5 - 9i)$

(15) $(9 - 6i)(6 - 10i)$

(30) $(5 + 2i)(8 + 3i)$

(45) $(3 + 8i)(10 + 4i)$

(1) $(7 - 3i)(5 + 3i)$

(16) $(2 + 8i)(9 + 7i)$

(31) $(-2 + 7i)(-1 + i)$

(2) $(-4 + 7i)(1 + 4i)$

(17) $(-5 - 6i)(-8 + 10i)$

(32) $(-5 + 2i)(3 - 4i)$

(3) $(-7 + 5i)(5 - i)$

(18) $(-10 + 3i)(1 + 8i)$

(33) $(9 - 5i)(2 + 4i)$

(4) $(10 + 3i)(4 - 10i)$

(19) $(-7 - 9i)(-7 - 3i)$

(34) $(1 + i)(-1 - i)$

(5) $(9 + 6i)(8 + i)$

(20) $(-9 - 9i)(1 - 10i)$

(35) $(9 - 10i)(-7 + 10i)$

(6) $(9 + 4i)(-6 - 10i)$

(21) $(1 + 6i)(6 - 8i)$

(36) $(4 - 10i)(1 + 2i)$

(7) $(-5 - 2i)(-2 - 2i)$

(22) $(-7 - 2i)(-1 + 2i)$

(37) $(6 + 4i)(-3 + 2i)$

(8) $(-6 + 6i)(6 + 7i)$

(23) $(-2 + 7i)(-7 + 6i)$

(38) $(-10 - 8i)(9 - 9i)$

(9) $(10 - 7i)(9 - 4i)$

(24) $(1 + 9i)(6 - 8i)$

(39) $(3 - 9i)(-4 + 3i)$

(10) $(3 - 2i)(1 - 10i)$

(25) $(-2 + 10i)(-8 + 10i)$

(40) $(5 + 7i)(-8 + 10i)$

(11) $(-8 + 5i)(-2 + 3i)$

(26) $(-7 - 4i)(-6 + i)$

(41) $(-5 - 4i)(-5 - 5i)$

(12) $(-9 + 6i)(4 - 7i)$

(27) $(-7 - 9i)(3 + 7i)$

(42) $(-10 - 7i)(-1 - 8i)$

(13) $(8 + 9i)(-9 - 3i)$

(28) $(-5 - 5i)(-2 - 9i)$

(43) $(10 + 8i)(-9 - 9i)$

(14) $(4 + 9i)(-1 + 4i)$

(29) $(-5 + 10i)(1 - 5i)$

(44) $(-9 + 3i)(-4 + 2i)$

(15) $(-5 - 8i)(8 - 6i)$

(30) $(-8 + 9i)(3 + 8i)$

(45) $(8 - 8i)(4 - 7i)$

$$(1) \quad (-3-3i)(-10+6i) \\ = 48+12i$$

$$(2) \quad (-3-6i)(-1-5i) \\ = -27+21i$$

$$(3) \quad (10+9i)(-4-5i) \\ = 5-86i$$

$$(4) \quad (-7-8i)(-5+2i) \\ = 51+26i$$

$$(5) \quad (-3+8i)(-2+7i) \\ = -50-37i$$

$$(6) \quad (4+9i)(-6+3i) \\ = -51-42i$$

$$(7) \quad (6+7i)(-2+8i) \\ = -68+34i$$

$$(8) \quad (7+3i)(-10-4i) \\ = -58-58i$$

$$(9) \quad (5+10i)(-2-4i) \\ = 30-40i$$

$$(10) \quad (-4-i)(-6+5i) \\ = 29-14i$$

$$(11) \quad (-1-4i)(4-8i) \\ = -36-8i$$

$$(12) \quad (-10-4i)(7+7i) \\ = -42-98i$$

$$(13) \quad (5-10i)(5+6i) \\ = 85-20i$$

$$(14) \quad (5+10i)(4-7i) \\ = 90+5i$$

$$(15) \quad (9+4i)(-4+2i) \\ = -44+2i$$

$$(16) \quad (-8-7i)(-7+5i) \\ = 91+9i$$

$$(17) \quad (3+9i)(4+3i) \\ = -15+45i$$

$$(18) \quad (-2-i)(-7+4i) \\ = 18-i$$

$$(19) \quad (10-7i)(5+2i) \\ = 64-15i$$

$$(20) \quad (5+4i)(-2+7i) \\ = -38+27i$$

$$(21) \quad (1-i)(1-5i) \\ = -4-6i$$

$$(22) \quad (10+10i)(-9-9i) \\ = -180i$$

$$(23) \quad (-7-4i)(-3-8i) \\ = -11+68i$$

$$(24) \quad (10+i)(7-6i) \\ = 76-53i$$

$$(25) \quad (-4+3i)(-5-6i) \\ = 38+9i$$

$$(26) \quad (1-9i)(-2+5i) \\ = 43+23i$$

$$(27) \quad (-9-10i)(-7-3i) \\ = 33+97i$$

$$(28) \quad (-2+i)(8+i) \\ = -17+6i$$

$$(29) \quad (-6+10i)(-7+8i) \\ = -38-118i$$

$$(30) \quad (-2-4i)(2-6i) \\ = -28+4i$$

$$(31) \quad (-10-9i)(-6+5i) \\ = 105+4i$$

$$(32) \quad (-6+4i)(4-2i) \\ = -16+28i$$

$$(33) \quad (10+i)(-4-i) \\ = -39-14i$$

$$(34) \quad (-10-8i)(6+6i) \\ = -12-108i$$

$$(35) \quad (-3-4i)(-6+10i) \\ = 58-6i$$

$$(36) \quad (10+2i)(5-8i) \\ = 66-70i$$

$$(37) \quad (4-6i)(3-10i) \\ = -48-58i$$

$$(38) \quad (-7+2i)(-10+6i) \\ = 58-62i$$

$$(39) \quad (2-3i)(6+i) \\ = 15-16i$$

$$(40) \quad (-5-3i)(-4-3i) \\ = 11+27i$$

$$(41) \quad (-7-6i)(-10+6i) \\ = 106+18i$$

$$(42) \quad (-5-5i)(-6-7i) \\ = -5+65i$$

$$(43) \quad (-10+3i)(-3+4i) \\ = 18-49i$$

$$(44) \quad (4+4i)(-4+6i) \\ = -40+8i$$

$$(45) \quad (8-i)(-5-7i) \\ = -47-51i$$

$$(1) \quad (-2 - 6i)(-9 - i) \\ = 12 + 56i$$

$$(2) \quad (9 - 5i)(4 + 6i) \\ = 66 + 34i$$

$$(3) \quad (-2 - 3i)(4 - 5i) \\ = -23 - 2i$$

$$(4) \quad (10 - i)(-6 - 3i) \\ = -63 - 24i$$

$$(5) \quad (7 - 3i)(10 + 2i) \\ = 76 - 16i$$

$$(6) \quad (-9 + 10i)(-9 - 8i) \\ = 161 - 18i$$

$$(7) \quad (8 - 2i)(-7 - 8i) \\ = -72 - 50i$$

$$(8) \quad (-10 - 10i)(5 + 3i) \\ = -20 - 80i$$

$$(9) \quad (5 + 2i)(6 + 3i) \\ = 24 + 27i$$

$$(10) \quad (-8 - 10i)(-2 - i) \\ = 6 + 28i$$

$$(11) \quad (-4 + 2i)(-9 - 8i) \\ = 52 + 14i$$

$$(12) \quad (1 + 9i)(-7 - 6i) \\ = 47 - 69i$$

$$(13) \quad (-8 + 3i)(1 - 9i) \\ = 19 + 75i$$

$$(14) \quad (6 - i)(6 + i) \\ = 37$$

$$(15) \quad (-2 + 10i)(1 + 8i) \\ = -82 - 6i$$

$$(16) \quad (-8 - 3i)(-10 - 7i) \\ = 59 + 86i$$

$$(17) \quad (6 + 10i)(-4 - 7i) \\ = 46 - 82i$$

$$(18) \quad (-2 - 8i)(-7 + 8i) \\ = 78 + 40i$$

$$(19) \quad (-10 + 2i)(-8 + 4i) \\ = 72 - 56i$$

$$(20) \quad (1 + 9i)(-4 + 7i) \\ = -67 - 29i$$

$$(21) \quad (3 + 8i)(9 - 7i) \\ = 83 + 51i$$

$$(22) \quad (2 - 3i)(-4 + 7i) \\ = 13 + 26i$$

$$(23) \quad (-4 + 3i)(-2 + 5i) \\ = -7 - 26i$$

$$(24) \quad (2 + 8i)(-3 + 7i) \\ = -62 - 10i$$

$$(25) \quad (-4 + 2i)(-1 - 2i) \\ = 8 + 6i$$

$$(26) \quad (9 + 6i)(-2 - 6i) \\ = 18 - 66i$$

$$(27) \quad (-6 + 7i)(5 - 9i) \\ = 33 + 89i$$

$$(28) \quad (-7 + 4i)(-4 - 10i) \\ = 68 + 54i$$

$$(29) \quad (-10 - 9i)(-9 + 9i) \\ = 171 - 9i$$

$$(30) \quad (5 - 6i)(-10 + 8i) \\ = -2 + 100i$$

$$(31) \quad (2 - 6i)(-9 - 6i) \\ = -54 + 42i$$

$$(32) \quad (-8 + 2i)(-6 - 8i) \\ = 64 + 52i$$

$$(33) \quad (2 + 6i)(7 + 2i) \\ = 2 + 46i$$

$$(34) \quad (-5 + 3i)(-5 + 8i) \\ = 1 - 55i$$

$$(35) \quad (-5 - 5i)(10 - 3i) \\ = -65 - 35i$$

$$(36) \quad (1 - 2i)(10 - 2i) \\ = 6 - 22i$$

$$(37) \quad (-4 + 4i)(3 + 3i) \\ = -24$$

$$(38) \quad (-10 + 9i)(8 + 4i) \\ = -116 + 32i$$

$$(39) \quad (2 - 6i)(-7 + i) \\ = -8 + 44i$$

$$(40) \quad (1 - 6i)(-7 + 6i) \\ = 29 + 48i$$

$$(41) \quad (-6 + 2i)(-4 + 6i) \\ = 12 - 44i$$

$$(42) \quad (-6 + 8i)(-7 + 6i) \\ = -6 - 92i$$

$$(43) \quad (-7 + i)(5 - 4i) \\ = -31 + 33i$$

$$(44) \quad (8 + 2i)(10 + 10i) \\ = 60 + 100i$$

$$(45) \quad (2 + 6i)(4 + 3i) \\ = -10 + 30i$$

$$(1) (-10 - 10i)(-9 - 5i) \\ = 40 + 140i$$

$$(2) (3 + 7i)(8 + 9i) \\ = -39 + 83i$$

$$(3) (-5 + i)(2 + 7i) \\ = -17 - 33i$$

$$(4) (9 - i)(1 - 4i) \\ = 5 - 37i$$

$$(5) (-1 + i)(-10 + 3i) \\ = 7 - 13i$$

$$(6) (7 - 9i)(-3 - i) \\ = -30 + 20i$$

$$(7) (10 + i)(9 + 3i) \\ = 87 + 39i$$

$$(8) (6 - 9i)(10 + 2i) \\ = 78 - 78i$$

$$(9) (-10 + 5i)(-1 - i) \\ = 15 + 5i$$

$$(10) (-4 - 9i)(-3 - 4i) \\ = -24 + 43i$$

$$(11) (-1 - 5i)(5 + 4i) \\ = 15 - 29i$$

$$(12) (-7 - 9i)(10 - 6i) \\ = -124 - 48i$$

$$(13) (10 + 2i)(-6 + 10i) \\ = -80 + 88i$$

$$(14) (9 + 3i)(-7 - 4i) \\ = -51 - 57i$$

$$(15) (3 - 7i)(-2 - 10i) \\ = -76 - 16i$$

$$(16) (6 + 9i)(6 - i) \\ = 45 + 48i$$

$$(17) (7 + 5i)(9 + 5i) \\ = 38 + 80i$$

$$(18) (-3 + 10i)(2 - 8i) \\ = 74 + 44i$$

$$(19) (-5 - i)(10 - i) \\ = -51 - 5i$$

$$(20) (-10 + 8i)(4 + 5i) \\ = -80 - 18i$$

$$(21) (-8 + 7i)(7 + 3i) \\ = -77 + 25i$$

$$(22) (-3 - 4i)(-1 + 8i) \\ = 35 - 20i$$

$$(23) (1 + 3i)(-4 - 5i) \\ = 11 - 17i$$

$$(24) (-2 - 8i)(7 - 5i) \\ = -54 - 46i$$

$$(25) (7 + 3i)(6 + 2i) \\ = 36 + 32i$$

$$(26) (2 + 10i)(-4 - 2i) \\ = 12 - 44i$$

$$(27) (4 - 2i)(-7 - 7i) \\ = -42 - 14i$$

$$(28) (-10 + 9i)(-9 + 9i) \\ = 9 - 171i$$

$$(29) (2 - 2i)(-6 + 9i) \\ = 6 + 30i$$

$$(30) (5 + 2i)(-9 - 3i) \\ = -39 - 33i$$

$$(31) (3 - 3i)(-3 + i) \\ = -6 + 12i$$

$$(32) (-3 + 10i)(-4 - i) \\ = 22 - 37i$$

$$(33) (1 + 10i)(-4 - 8i) \\ = 76 - 48i$$

$$(34) (-2 + 9i)(4 + 8i) \\ = -80 + 20i$$

$$(35) (1 + 8i)(-10 - 6i) \\ = 38 - 86i$$

$$(36) (-5 + i)(-9 - 2i) \\ = 47 + i$$

$$(37) (7 - 2i)(-10 + 2i) \\ = -66 + 34i$$

$$(38) (7 - 2i)(3 + 3i) \\ = 27 + 15i$$

$$(39) (10 - 3i)(-10 - 8i) \\ = -124 - 50i$$

$$(40) (3 - 7i)(2 + 10i) \\ = 76 + 16i$$

$$(41) (3 + 6i)(5 - 9i) \\ = 69 + 3i$$

$$(42) (-1 + 6i)(-9 + 9i) \\ = -45 - 63i$$

$$(43) (4 - i)(3 - i) \\ = 11 - 7i$$

$$(44) (-9 + 10i)(3 - 3i) \\ = 3 + 57i$$

$$(45) (10 - 2i)(9 - 5i) \\ = 80 - 68i$$

$$(1) \quad (-8 - 7i)(2 - 4i) \\ = -44 + 18i$$

$$(2) \quad (-8 + 8i)(-1 + 2i) \\ = -8 - 24i$$

$$(3) \quad (-7 - 7i)(8 + 3i) \\ = -35 - 77i$$

$$(4) \quad (-1 - 8i)(-3 + 7i) \\ = 59 + 17i$$

$$(5) \quad (9 - 2i)(8 + i) \\ = 74 - 7i$$

$$(6) \quad (5 + i)(2 - 6i) \\ = 16 - 28i$$

$$(7) \quad (1 + 5i)(1 + 7i) \\ = -34 + 12i$$

$$(8) \quad (2 + 4i)(-9 - i) \\ = -14 - 38i$$

$$(9) \quad (6 + 8i)(-2 + 6i) \\ = -60 + 20i$$

$$(10) \quad (4 + 4i)(8 - 5i) \\ = 52 + 12i$$

$$(11) \quad (-5 + 9i)(-9 - 7i) \\ = 108 - 46i$$

$$(12) \quad (10 - i)(-9 - 7i) \\ = -97 - 61i$$

$$(13) \quad (-7 - 6i)(2 + 3i) \\ = 4 - 33i$$

$$(14) \quad (3 + 9i)(-2 - 6i) \\ = 48 - 36i$$

$$(15) \quad (2 + 3i)(-1 + 3i) \\ = -11 + 3i$$

$$(16) \quad (1 + 5i)(1 - 6i) \\ = 31 - i$$

$$(17) \quad (-10 + 7i)(-9 - 4i) \\ = 118 - 23i$$

$$(18) \quad (6 + 6i)(8 - 7i) \\ = 90 + 6i$$

$$(19) \quad (9 + 4i)(3 - 5i) \\ = 47 - 33i$$

$$(20) \quad (4 + 5i)(1 + 8i) \\ = -36 + 37i$$

$$(21) \quad (-3 + 3i)(-4 + 3i) \\ = 3 - 21i$$

$$(22) \quad (-8 - 3i)(3 + 7i) \\ = -3 - 65i$$

$$(23) \quad (2 + 5i)(3 + i) \\ = 1 + 17i$$

$$(24) \quad (-8 - 4i)(-10 + 2i) \\ = 88 + 24i$$

$$(25) \quad (4 + 8i)(6 - 7i) \\ = 80 + 20i$$

$$(26) \quad (10 - 9i)(3 + 6i) \\ = 84 + 33i$$

$$(27) \quad (-7 + 10i)(6 - 5i) \\ = 8 + 95i$$

$$(28) \quad (10 - 4i)(-4 - 6i) \\ = -64 - 44i$$

$$(29) \quad (-9 + 4i)(7 + 2i) \\ = -71 + 10i$$

$$(30) \quad (-5 + 3i)(-1 + 8i) \\ = -19 - 43i$$

$$(31) \quad (10 + 4i)(-6 + 7i) \\ = -88 + 46i$$

$$(32) \quad (-6 + 10i)(3 - 8i) \\ = 62 + 78i$$

$$(33) \quad (1 - 9i)(-10 + 6i) \\ = 44 + 96i$$

$$(34) \quad (6 - 10i)(2 + i) \\ = 22 - 14i$$

$$(35) \quad (-4 - 2i)(-2 - 3i) \\ = 2 + 16i$$

$$(36) \quad (8 + 8i)(3 + 7i) \\ = -32 + 80i$$

$$(37) \quad (6 - 3i)(-9 - 9i) \\ = -81 - 27i$$

$$(38) \quad (-6 - 10i)(-1 - 7i) \\ = -64 + 52i$$

$$(39) \quad (-7 + i)(-6 + 4i) \\ = 38 - 34i$$

$$(40) \quad (-9 + 4i)(3 + 10i) \\ = -67 - 78i$$

$$(41) \quad (3 + 5i)(1 + 6i) \\ = -27 + 23i$$

$$(42) \quad (9 + i)(-4 + 6i) \\ = -42 + 50i$$

$$(43) \quad (-10 - 3i)(4 + 9i) \\ = -13 - 102i$$

$$(44) \quad (4 - 2i)(-6 + 9i) \\ = -6 + 48i$$

$$(45) \quad (-7 + 3i)(-3 + 2i) \\ = 15 - 23i$$

$$(1) \quad (-1 + i)(8 - 3i) \\ = -5 + 11i$$

$$(2) \quad (8 + i)(6 - 10i) \\ = 58 - 74i$$

$$(3) \quad (-9 + 5i)(8 + 9i) \\ = -117 - 41i$$

$$(4) \quad (-8 - 7i)(-1 - 10i) \\ = -62 + 87i$$

$$(5) \quad (-3 + 9i)(6 + 10i) \\ = -108 + 24i$$

$$(6) \quad (-1 + 2i)(-2 + 2i) \\ = -2 - 6i$$

$$(7) \quad (6 - 4i)(1 + 8i) \\ = 38 + 44i$$

$$(8) \quad (5 - 9i)(1 + 3i) \\ = 32 + 6i$$

$$(9) \quad (4 + 2i)(-6 + i) \\ = -26 - 8i$$

$$(10) \quad (-10 + i)(9 + 10i) \\ = -100 - 91i$$

$$(11) \quad (-10 + i)(8 + 8i) \\ = -88 - 72i$$

$$(12) \quad (10 - 3i)(5 + 7i) \\ = 71 + 55i$$

$$(13) \quad (8 + 9i)(-6 - 9i) \\ = 33 - 126i$$

$$(14) \quad (-9 - 5i)(4 + 5i) \\ = -11 - 65i$$

$$(15) \quad (-4 + 6i)(4 - 3i) \\ = 2 + 36i$$

$$(16) \quad (-3 - 3i)(4 - 6i) \\ = -30 + 6i$$

$$(17) \quad (9 + 2i)(-6 + i) \\ = -56 - 3i$$

$$(18) \quad (-5 - 10i)(-7 - 3i) \\ = 5 + 85i$$

$$(19) \quad (-6 - 9i)(1 + 9i) \\ = 75 - 63i$$

$$(20) \quad (10 - 3i)(-9 - 10i) \\ = -120 - 73i$$

$$(21) \quad (-5 - i)(-6 + 7i) \\ = 37 - 29i$$

$$(22) \quad (4 - 8i)(-4 - i) \\ = -24 + 28i$$

$$(23) \quad (1 - i)(2 - 6i) \\ = -4 - 8i$$

$$(24) \quad (5 + 6i)(7 - 4i) \\ = 59 + 22i$$

$$(25) \quad (4 - 8i)(-10 + 8i) \\ = 24 + 112i$$

$$(26) \quad (-5 - 5i)(9 + 7i) \\ = -10 - 80i$$

$$(27) \quad (-10 + 9i)(-7 + i) \\ = 61 - 73i$$

$$(28) \quad (-6 - 10i)(3 - 4i) \\ = -58 - 6i$$

$$(29) \quad (9 + 4i)(2 + 6i) \\ = -6 + 62i$$

$$(30) \quad (-6 - 9i)(9 - 3i) \\ = -81 - 63i$$

$$(31) \quad (7 - 7i)(7 - 3i) \\ = 28 - 70i$$

$$(32) \quad (3 + 3i)(-5 + 5i) \\ = -30$$

$$(33) \quad (4 + 2i)(-4 - 7i) \\ = -2 - 36i$$

$$(34) \quad (-3 - 10i)(9 - 10i) \\ = -127 - 60i$$

$$(35) \quad (-8 + 2i)(4 + 6i) \\ = -44 - 40i$$

$$(36) \quad (-7 - i)(-9 - 5i) \\ = 58 + 44i$$

$$(37) \quad (7 + 10i)(4 - 10i) \\ = 128 - 30i$$

$$(38) \quad (-4 - 2i)(-9 - i) \\ = 34 + 22i$$

$$(39) \quad (10 + 7i)(2 + 4i) \\ = -8 + 54i$$

$$(40) \quad (-7 - 7i)(-6 + i) \\ = 49 + 35i$$

$$(41) \quad (-7 + 7i)(-7 + i) \\ = 42 - 56i$$

$$(42) \quad (8 + 7i)(2 - 6i) \\ = 58 - 34i$$

$$(43) \quad (-2 - 10i)(7 + 7i) \\ = 56 - 84i$$

$$(44) \quad (7 - i)(4 - 5i) \\ = 23 - 39i$$

$$(45) \quad (3 - 5i)(7 + 3i) \\ = 36 - 26i$$

$$(1) (3 + 6i)(6 + 9i) \\ = -36 + 63i$$

$$(2) (5 + 3i)(-1 - 3i) \\ = 4 - 18i$$

$$(3) (7 + 7i)(4 - 8i) \\ = 84 - 28i$$

$$(4) (-9 + 6i)(-9 + 9i) \\ = 27 - 135i$$

$$(5) (-6 - 2i)(9 + 9i) \\ = -36 - 72i$$

$$(6) (-3 - 3i)(-8 + 8i) \\ = 48$$

$$(7) (8 - 5i)(3 - 6i) \\ = -6 - 63i$$

$$(8) (8 - 3i)(9 + 10i) \\ = 102 + 53i$$

$$(9) (10 - 8i)(-5 + 3i) \\ = -26 + 70i$$

$$(10) (-9 - 8i)(-6 - 9i) \\ = -18 + 129i$$

$$(11) (7 - 10i)(5 + 7i) \\ = 105 - i$$

$$(12) (4 - 2i)(-2 + 5i) \\ = 2 + 24i$$

$$(13) (-5 - 4i)(1 + 2i) \\ = 3 - 14i$$

$$(14) (-9 - 4i)(-4 + i) \\ = 40 + 7i$$

$$(15) (-7 - 4i)(-5 - i) \\ = 31 + 27i$$

$$(16) (-7 - 3i)(-10 - 4i) \\ = 58 + 58i$$

$$(17) (3 + 7i)(-9 + 9i) \\ = -90 - 36i$$

$$(18) (-9 - i)(2 + 5i) \\ = -13 - 47i$$

$$(19) (9 - 2i)(-1 + 4i) \\ = -1 + 38i$$

$$(20) (4 - 4i)(-2 - 7i) \\ = -36 - 20i$$

$$(21) (7 - i)(4 - 6i) \\ = 22 - 46i$$

$$(22) (6 + 4i)(-4 - 5i) \\ = -4 - 46i$$

$$(23) (2 - 7i)(4 + 7i) \\ = 57 - 14i$$

$$(24) (-2 + 3i)(-8 + 2i) \\ = 10 - 28i$$

$$(25) (-1 + 2i)(-3 - 8i) \\ = 19 + 2i$$

$$(26) (7 - 6i)(7 - 7i) \\ = 7 - 91i$$

$$(27) (4 - 7i)(-4 + 6i) \\ = 26 + 52i$$

$$(28) (8 - 6i)(-9 + 10i) \\ = -12 + 134i$$

$$(29) (9 - 6i)(-9 + 7i) \\ = -39 + 117i$$

$$(30) (-7 + 3i)(3 - 2i) \\ = -15 + 23i$$

$$(31) (2 + 2i)(-9 - 5i) \\ = -8 - 28i$$

$$(32) (-5 - 5i)(-10 - 7i) \\ = 15 + 85i$$

$$(33) (1 + 5i)(3 - 2i) \\ = 13 + 13i$$

$$(34) (9 - 3i)(-7 - 7i) \\ = -84 - 42i$$

$$(35) (-3 + 2i)(4 + 5i) \\ = -22 - 7i$$

$$(36) (-5 + 10i)(-3 - 9i) \\ = 105 + 15i$$

$$(37) (1 - 5i)(-2 - 4i) \\ = -22 + 6i$$

$$(38) (-9 - 5i)(9 - 3i) \\ = -96 - 18i$$

$$(39) (3 - i)(10 + 9i) \\ = 39 + 17i$$

$$(40) (-7 - 9i)(7 - 10i) \\ = -139 + 7i$$

$$(41) (3 - 3i)(-8 - 5i) \\ = -39 + 9i$$

$$(42) (6 + 4i)(-5 - 7i) \\ = -2 - 62i$$

$$(43) (8 + 8i)(4 - 5i) \\ = 72 - 8i$$

$$(44) (5 - i)(-7 - 6i) \\ = -41 - 23i$$

$$(45) (-1 + 2i)(1 + 8i) \\ = -17 - 6i$$

$$(1) (5 - 4i)(9 + 2i) \\ = 53 - 26i$$

$$(2) (10 + 9i)(-7 - 7i) \\ = -7 - 133i$$

$$(3) (9 + 6i)(2 + 7i) \\ = -24 + 75i$$

$$(4) (-1 + 2i)(1 + 3i) \\ = -7 - i$$

$$(5) (-6 + 7i)(1 + 5i) \\ = -41 - 23i$$

$$(6) (6 + 5i)(-1 + 6i) \\ = -36 + 31i$$

$$(7) (-5 + i)(-4 - 2i) \\ = 22 + 6i$$

$$(8) (6 - 10i)(-10 + 9i) \\ = 30 + 154i$$

$$(9) (-6 - 4i)(-7 + 3i) \\ = 54 + 10i$$

$$(10) (-9 + 8i)(-7 - 8i) \\ = 127 + 16i$$

$$(11) (-4 - 5i)(2 - 9i) \\ = -53 + 26i$$

$$(12) (10 - 9i)(-8 + 3i) \\ = -53 + 102i$$

$$(13) (6 + 10i)(4 + 8i) \\ = -56 + 88i$$

$$(14) (-3 + i)(-8 - 10i) \\ = 34 + 22i$$

$$(15) (5 - 9i)(8 - 4i) \\ = 4 - 92i$$

$$(16) (-8 + 8i)(-7 + 10i) \\ = -24 - 136i$$

$$(17) (-4 - 3i)(-7 - 8i) \\ = 4 + 53i$$

$$(18) (6 - 5i)(5 - 10i) \\ = -20 - 85i$$

$$(19) (8 + 5i)(10 + 7i) \\ = 45 + 106i$$

$$(20) (-1 - 3i)(-2 - i) \\ = -1 + 7i$$

$$(21) (5 - 4i)(-8 + 8i) \\ = -8 + 72i$$

$$(22) (6 + 8i)(5 + 5i) \\ = -10 + 70i$$

$$(23) (1 - 7i)(-1 - 3i) \\ = -22 + 4i$$

$$(24) (-6 - 8i)(-5 - 4i) \\ = -2 + 64i$$

$$(25) (4 - i)(-10 + 6i) \\ = -34 + 34i$$

$$(26) (2 - 9i)(1 + 10i) \\ = 92 + 11i$$

$$(27) (5 + 5i)(6 - 5i) \\ = 55 + 5i$$

$$(28) (-7 - i)(-5 + 6i) \\ = 41 - 37i$$

$$(29) (5 - 3i)(-1 + 5i) \\ = 10 + 28i$$

$$(30) (6 + 5i)(-6 - 4i) \\ = -16 - 54i$$

$$(31) (5 - 10i)(-5 + 4i) \\ = 15 + 70i$$

$$(32) (-3 + 5i)(6 + 9i) \\ = -63 + 3i$$

$$(33) (5 + 7i)(6 + 3i) \\ = 9 + 57i$$

$$(34) (3 + 3i)(-1 - i) \\ = -6i$$

$$(35) (4 + 4i)(2 + 10i) \\ = -32 + 48i$$

$$(36) (6 - 2i)(1 - 7i) \\ = -8 - 44i$$

$$(37) (7 - 6i)(-5 + i) \\ = -29 + 37i$$

$$(38) (-6 - 4i)(5 - 5i) \\ = -50 + 10i$$

$$(39) (3 - 7i)(-1 + 3i) \\ = 18 + 16i$$

$$(40) (9 - 6i)(4 - 2i) \\ = 24 - 42i$$

$$(41) (3 + 4i)(-6 + 6i) \\ = -42 - 6i$$

$$(42) (10 + 5i)(-4 - 4i) \\ = -20 - 60i$$

$$(43) (-3 - 10i)(-6 - 5i) \\ = -32 + 75i$$

$$(44) (2 - 10i)(-6 + 8i) \\ = 68 + 76i$$

$$(45) (-3 + 3i)(-10 - 4i) \\ = 42 - 18i$$

$$(1) \quad (-8 + 3i)(10 + 4i) \\ = -92 - 2i$$

$$(2) \quad (5 + 8i)(-7 - i) \\ = -27 - 61i$$

$$(3) \quad (-10 - 6i)(-10 + 9i) \\ = 154 - 30i$$

$$(4) \quad (10 + 2i)(7 - 8i) \\ = 86 - 66i$$

$$(5) \quad (-3 + 3i)(1 + i) \\ = -6$$

$$(6) \quad (10 - 7i)(7 - 4i) \\ = 42 - 89i$$

$$(7) \quad (-2 - 2i)(2 - 5i) \\ = -14 + 6i$$

$$(8) \quad (6 + i)(-2 + 4i) \\ = -16 + 22i$$

$$(9) \quad (-1 + 4i)(1 - 3i) \\ = 11 + 7i$$

$$(10) \quad (6 + 5i)(4 + i) \\ = 19 + 26i$$

$$(11) \quad (-7 + i)(6 + 9i) \\ = -51 - 57i$$

$$(12) \quad (-5 - 6i)(-4 - 10i) \\ = -40 + 74i$$

$$(13) \quad (7 - 3i)(-9 - i) \\ = -66 + 20i$$

$$(14) \quad (-4 + 6i)(3 - 6i) \\ = 24 + 42i$$

$$(15) \quad (-10 - 7i)(9 - 6i) \\ = -132 - 3i$$

$$(16) \quad (-3 - 4i)(10 + 10i) \\ = 10 - 70i$$

$$(17) \quad (1 - i)(10 + 8i) \\ = 18 - 2i$$

$$(18) \quad (4 + 4i)(1 + 4i) \\ = -12 + 20i$$

$$(19) \quad (6 - 6i)(-3 - 5i) \\ = -48 - 12i$$

$$(20) \quad (-3 + 3i)(1 - 3i) \\ = 6 + 12i$$

$$(21) \quad (10 + 2i)(-9 - 9i) \\ = -72 - 108i$$

$$(22) \quad (-2 + 4i)(8 + 8i) \\ = -48 + 16i$$

$$(23) \quad (-6 - 6i)(5 + 4i) \\ = -6 - 54i$$

$$(24) \quad (-7 - 7i)(1 - 9i) \\ = -70 + 56i$$

$$(25) \quad (-1 - 8i)(-4 + 4i) \\ = 36 + 28i$$

$$(26) \quad (-7 - i)(-2 - 2i) \\ = 12 + 16i$$

$$(27) \quad (3 + 3i)(2 + 9i) \\ = -21 + 33i$$

$$(28) \quad (-9 - 6i)(-2 + 2i) \\ = 30 - 6i$$

$$(29) \quad (8 - 6i)(-2 - 3i) \\ = -34 - 12i$$

$$(30) \quad (-3 + 6i)(7 + 4i) \\ = -45 + 30i$$

$$(31) \quad (2 + 8i)(-2 - 3i) \\ = 20 - 22i$$

$$(32) \quad (-2 + 4i)(5 + 4i) \\ = -26 + 12i$$

$$(33) \quad (-6 + 9i)(-10 - 10i) \\ = 150 - 30i$$

$$(34) \quad (1 - 5i)(4 + 10i) \\ = 54 - 10i$$

$$(35) \quad (8 - 6i)(6 - 7i) \\ = 6 - 92i$$

$$(36) \quad (-8 + 7i)(-8 + 10i) \\ = -6 - 136i$$

$$(37) \quad (-3 + 6i)(-1 - 6i) \\ = 39 + 12i$$

$$(38) \quad (7 - i)(1 + 5i) \\ = 12 + 34i$$

$$(39) \quad (-10 + 10i)(-9 - 7i) \\ = 160 - 20i$$

$$(40) \quad (8 + 2i)(-3 + 9i) \\ = -42 + 66i$$

$$(41) \quad (5 - 6i)(9 - 9i) \\ = -9 - 99i$$

$$(42) \quad (10 + 6i)(-4 - 8i) \\ = 8 - 104i$$

$$(43) \quad (-8 + 8i)(10 + i) \\ = -88 + 72i$$

$$(44) \quad (-4 - 10i)(-7 + 2i) \\ = 48 + 62i$$

$$(45) \quad (-7 + 9i)(4 - 10i) \\ = 62 + 106i$$

$$(1) \quad (-6 - 4i)(-4 - 6i) \\ = 52i$$

$$(2) \quad (-2 + 9i)(7 + 8i) \\ = -86 + 47i$$

$$(3) \quad (1 + 4i)(-2 + 7i) \\ = -30 - i$$

$$(4) \quad (4 + i)(5 + 5i) \\ = 15 + 25i$$

$$(5) \quad (-10 + 10i)(-8 + 9i) \\ = -10 - 170i$$

$$(6) \quad (-4 + 2i)(-8 - 5i) \\ = 42 + 4i$$

$$(7) \quad (-5 - 8i)(10 - 10i) \\ = -130 - 30i$$

$$(8) \quad (2 - 8i)(6 + 4i) \\ = 44 - 40i$$

$$(9) \quad (-3 - 2i)(9 - 6i) \\ = -39$$

$$(10) \quad (10 + 7i)(-1 + 9i) \\ = -73 + 83i$$

$$(11) \quad (3 - 4i)(8 - 6i) \\ = -50i$$

$$(12) \quad (2 + 6i)(9 + 7i) \\ = -24 + 68i$$

$$(13) \quad (5 - 2i)(-9 + 4i) \\ = -37 + 38i$$

$$(14) \quad (5 - 8i)(-5 - 5i) \\ = -65 + 15i$$

$$(15) \quad (-1 - 3i)(10 + 10i) \\ = 20 - 40i$$

$$(16) \quad (-6 - 7i)(8 + 5i) \\ = -13 - 86i$$

$$(17) \quad (-8 + 10i)(6 + 2i) \\ = -68 + 44i$$

$$(18) \quad (8 + 5i)(9 - 10i) \\ = 122 - 35i$$

$$(19) \quad (-9 + 4i)(-1 - 7i) \\ = 37 + 59i$$

$$(20) \quad (-3 + 8i)(4 - 10i) \\ = 68 + 62i$$

$$(21) \quad (9 + 3i)(10 - 6i) \\ = 108 - 24i$$

$$(22) \quad (7 - 10i)(-4 - 4i) \\ = -68 + 12i$$

$$(23) \quad (-9 - 8i)(-5 + 10i) \\ = 125 - 50i$$

$$(24) \quad (-10 + i)(-6 + 4i) \\ = 56 - 46i$$

$$(25) \quad (9 - 7i)(-3 - 9i) \\ = -90 - 60i$$

$$(26) \quad (2 + 3i)(-4 + 3i) \\ = -17 - 6i$$

$$(27) \quad (-9 + 8i)(-10 + 9i) \\ = 18 - 161i$$

$$(28) \quad (-7 + 9i)(2 - 7i) \\ = 49 + 67i$$

$$(29) \quad (2 - i)(-9 + 3i) \\ = -15 + 15i$$

$$(30) \quad (-8 + 6i)(-8 - 7i) \\ = 106 + 8i$$

$$(31) \quad (7 + 3i)(-1 - 10i) \\ = 23 - 73i$$

$$(32) \quad (2 - 6i)(-6 + 5i) \\ = 18 + 46i$$

$$(33) \quad (-5 - 2i)(5 - 6i) \\ = -37 + 20i$$

$$(34) \quad (-4 - 10i)(-5 + 3i) \\ = 50 + 38i$$

$$(35) \quad (-10 + 8i)(-3 + 3i) \\ = 6 - 54i$$

$$(36) \quad (-5 - 5i)(-5 - 5i) \\ = 50i$$

$$(37) \quad (-1 - 2i)(10 + 3i) \\ = -4 - 23i$$

$$(38) \quad (-2 - 3i)(9 + 4i) \\ = -6 - 35i$$

$$(39) \quad (-6 - i)(-2 - 5i) \\ = 7 + 32i$$

$$(40) \quad (3 + 9i)(-2 + 2i) \\ = -24 - 12i$$

$$(41) \quad (-3 - 2i)(-2 - 7i) \\ = -8 + 25i$$

$$(42) \quad (7 - 5i)(4 - 8i) \\ = -12 - 76i$$

$$(43) \quad (-1 - 8i)(7 - 5i) \\ = -47 - 51i$$

$$(44) \quad (-5 - 6i)(1 + 7i) \\ = 37 - 41i$$

$$(45) \quad (-4 + 7i)(4 + 6i) \\ = -58 + 4i$$

$$(1) (8 - 6i)(5 - 7i) \\ = -2 - 86i$$

$$(2) (-2 + 7i)(-4 - 9i) \\ = 71 - 10i$$

$$(3) (-7 + 4i)(1 + i) \\ = -11 - 3i$$

$$(4) (-4 - 10i)(-5 - 5i) \\ = -30 + 70i$$

$$(5) (-8 + 5i)(-2 + 7i) \\ = -19 - 66i$$

$$(6) (-6 + 4i)(9 - 8i) \\ = -22 + 84i$$

$$(7) (7 + 2i)(-1 - 9i) \\ = 11 - 65i$$

$$(8) (-10 - 2i)(1 - 6i) \\ = -22 + 58i$$

$$(9) (9 + 5i)(3 + 10i) \\ = -23 + 105i$$

$$(10) (-2 + 5i)(-3 - 3i) \\ = 21 - 9i$$

$$(11) (-10 - 5i)(-4 - 5i) \\ = 15 + 70i$$

$$(12) (-5 + 2i)(-3 + 5i) \\ = 5 - 31i$$

$$(13) (-9 - 8i)(-6 - 9i) \\ = -18 + 129i$$

$$(14) (-6 + 7i)(-1 + 6i) \\ = -36 - 43i$$

$$(15) (-4 - 8i)(-2 - 3i) \\ = -16 + 28i$$

$$(16) (7 + 7i)(-10 + 3i) \\ = -91 - 49i$$

$$(17) (7 + 8i)(9 + 4i) \\ = 31 + 100i$$

$$(18) (-8 + 8i)(3 - 9i) \\ = 48 + 96i$$

$$(19) (-5 - 3i)(9 + 9i) \\ = -18 - 72i$$

$$(20) (3 - 3i)(6 - i) \\ = 15 - 21i$$

$$(21) (-9 + 9i)(-8 - 9i) \\ = 153 + 9i$$

$$(22) (-5 + 9i)(5 - 5i) \\ = 20 + 70i$$

$$(23) (-9 + 4i)(-5 + 6i) \\ = 21 - 74i$$

$$(24) (-5 + 6i)(-3 + 7i) \\ = -27 - 53i$$

$$(25) (1 + 7i)(8 - 2i) \\ = 22 + 54i$$

$$(26) (10 + 2i)(6 + 8i) \\ = 44 + 92i$$

$$(27) (-1 + 9i)(8 - 5i) \\ = 37 + 77i$$

$$(28) (-1 - 7i)(7 - 10i) \\ = -77 - 39i$$

$$(29) (-9 + 9i)(10 - 10i) \\ = 180i$$

$$(30) (10 - 5i)(-5 - 5i) \\ = -75 - 25i$$

$$(31) (9 + 4i)(4 + 10i) \\ = -4 + 106i$$

$$(32) (-4 - 6i)(8 - 10i) \\ = -92 - 8i$$

$$(33) (1 + 6i)(8 + 10i) \\ = -52 + 58i$$

$$(34) (10 + 2i)(-3 + 6i) \\ = -42 + 54i$$

$$(35) (-2 - 10i)(3 - 10i) \\ = -106 - 10i$$

$$(36) (-7 - 4i)(7 - 8i) \\ = -81 + 28i$$

$$(37) (-6 - 6i)(9 + 8i) \\ = -6 - 102i$$

$$(38) (10 - 5i)(7 + 10i) \\ = 120 + 65i$$

$$(39) (9 - 6i)(3 + 4i) \\ = 51 + 18i$$

$$(40) (-2 + 10i)(-6 + 3i) \\ = -18 - 66i$$

$$(41) (-7 - 10i)(5 + 5i) \\ = 15 - 85i$$

$$(42) (5 + 8i)(-8 + 10i) \\ = -120 - 14i$$

$$(43) (2 + 4i)(10 - 4i) \\ = 36 + 32i$$

$$(44) (4 + 5i)(10 - 4i) \\ = 60 + 34i$$

$$(45) (9 + 2i)(4 + i) \\ = 34 + 17i$$

$$(1) (10 + 2i)(10 - 9i) \\ = 118 - 70i$$

$$(2) (7 + 5i)(-5 + 2i) \\ = -45 - 11i$$

$$(3) (7 + 3i)(8 + 7i) \\ = 35 + 73i$$

$$(4) (2 + 3i)(-7 + 7i) \\ = -35 - 7i$$

$$(5) (6 + 5i)(7 + 8i) \\ = 2 + 83i$$

$$(6) (4 + 2i)(-2 - i) \\ = -6 - 8i$$

$$(7) (2 - 2i)(-1 - 6i) \\ = -14 - 10i$$

$$(8) (4 - 3i)(2 + 10i) \\ = 38 + 34i$$

$$(9) (8 + 2i)(4 - 6i) \\ = 44 - 40i$$

$$(10) (3 + 6i)(-3 + 5i) \\ = -39 - 3i$$

$$(11) (7 + i)(10 - 5i) \\ = 75 - 25i$$

$$(12) (7 + 5i)(4 - 3i) \\ = 43 - i$$

$$(13) (4 + 10i)(2 - 6i) \\ = 68 - 4i$$

$$(14) (-1 + 6i)(5 + 5i) \\ = -35 + 25i$$

$$(15) (10 + 4i)(-6 + 10i) \\ = -100 + 76i$$

$$(16) (1 - 2i)(2 + 2i) \\ = 6 - 2i$$

$$(17) (3 + 7i)(-10 + 6i) \\ = -72 - 52i$$

$$(18) (1 + 10i)(7 + 3i) \\ = -23 + 73i$$

$$(19) (-7 + 2i)(5 + 4i) \\ = -43 - 18i$$

$$(20) (-3 + 9i)(-8 + 5i) \\ = -21 - 87i$$

$$(21) (-5 + 6i)(-5 + 8i) \\ = -23 - 70i$$

$$(22) (-7 + 10i)(-10 - 5i) \\ = 120 - 65i$$

$$(23) (10 + 6i)(-6 + 10i) \\ = -120 + 64i$$

$$(24) (4 + 10i)(-5 - 10i) \\ = 80 - 90i$$

$$(25) (-6 + 9i)(9 - 6i) \\ = 117i$$

$$(26) (9 - 10i)(-2 + 4i) \\ = 22 + 56i$$

$$(27) (-3 + 8i)(-2 - 8i) \\ = 70 + 8i$$

$$(28) (8 - i)(-7 - 9i) \\ = -65 - 65i$$

$$(29) (9 - 4i)(7 + 6i) \\ = 87 + 26i$$

$$(30) (-9 - 8i)(-9 - 6i) \\ = 33 + 126i$$

$$(31) (5 + 8i)(2 - 10i) \\ = 90 - 34i$$

$$(32) (9 - 10i)(2 - 3i) \\ = -12 - 47i$$

$$(33) (10 - 8i)(8 + i) \\ = 88 - 54i$$

$$(34) (-10 + 4i)(5 - 10i) \\ = -10 + 120i$$

$$(35) (3 - 7i)(8 - 8i) \\ = -32 - 80i$$

$$(36) (8 - 2i)(-4 - 4i) \\ = -40 - 24i$$

$$(37) (5 - 5i)(-10 + 10i) \\ = 100i$$

$$(38) (5 - 8i)(-6 - i) \\ = -38 + 43i$$

$$(39) (-9 + 10i)(-8 + 10i) \\ = -28 - 170i$$

$$(40) (4 - 5i)(-4 + 6i) \\ = 14 + 44i$$

$$(41) (8 - 9i)(-2 + 5i) \\ = 29 + 58i$$

$$(42) (9 - 7i)(3 + 5i) \\ = 62 + 24i$$

$$(43) (8 + 7i)(-8 - 2i) \\ = -50 - 72i$$

$$(44) (-6 + 6i)(8 + 6i) \\ = -84 + 12i$$

$$(45) (-7 - 2i)(-4 - 2i) \\ = 24 + 22i$$

$$(1) \quad (-3 - 7i)(8 + 5i) \\ = 11 - 71i$$

$$(2) \quad (7 + 7i)(2 + 8i) \\ = -42 + 70i$$

$$(3) \quad (9 - 7i)(-8 - 5i) \\ = -107 + 11i$$

$$(4) \quad (9 + 5i)(6 + i) \\ = 49 + 39i$$

$$(5) \quad (7 + 5i)(3 - 7i) \\ = 56 - 34i$$

$$(6) \quad (10 + 3i)(-1 + 4i) \\ = -22 + 37i$$

$$(7) \quad (-8 + i)(3 + 9i) \\ = -33 - 69i$$

$$(8) \quad (4 - 9i)(10 - 7i) \\ = -23 - 118i$$

$$(9) \quad (4 + i)(-9 + 5i) \\ = -41 + 11i$$

$$(10) \quad (6 - 8i)(5 - i) \\ = 22 - 46i$$

$$(11) \quad (9 - 5i)(2 + 3i) \\ = 33 + 17i$$

$$(12) \quad (5 - 3i)(10 + 2i) \\ = 56 - 20i$$

$$(13) \quad (-3 - 10i)(-7 + 8i) \\ = 101 + 46i$$

$$(14) \quad (10 + i)(-6 + 7i) \\ = -67 + 64i$$

$$(15) \quad (3 - i)(10 + i) \\ = 31 - 7i$$

$$(16) \quad (-1 - 7i)(-2 - 9i) \\ = -61 + 23i$$

$$(17) \quad (4 - 10i)(2 + 2i) \\ = 28 - 12i$$

$$(18) \quad (-8 + i)(-3 - 2i) \\ = 26 + 13i$$

$$(19) \quad (-6 - 7i)(-9 + i) \\ = 61 + 57i$$

$$(20) \quad (3 - 10i)(1 + 2i) \\ = 23 - 4i$$

$$(21) \quad (1 + 6i)(7 + 7i) \\ = -35 + 49i$$

$$(22) \quad (2 - i)(-8 - 3i) \\ = -19 + 2i$$

$$(23) \quad (10 - 2i)(-10 + 3i) \\ = -94 + 50i$$

$$(24) \quad (-6 + 6i)(3 - 5i) \\ = 12 + 48i$$

$$(25) \quad (5 + i)(6 + 3i) \\ = 27 + 21i$$

$$(26) \quad (-8 + i)(-7 + 5i) \\ = 51 - 47i$$

$$(27) \quad (2 + 4i)(2 + 9i) \\ = -32 + 26i$$

$$(28) \quad (-6 + 8i)(10 + 2i) \\ = -76 + 68i$$

$$(29) \quad (-9 + 6i)(4 - 10i) \\ = 24 + 114i$$

$$(30) \quad (9 + 3i)(-9 - 9i) \\ = -54 - 108i$$

$$(31) \quad (-6 - 6i)(-1 + 8i) \\ = 54 - 42i$$

$$(32) \quad (-2 + 10i)(5 + 5i) \\ = -60 + 40i$$

$$(33) \quad (6 - 6i)(-1 + 2i) \\ = 6 + 18i$$

$$(34) \quad (10 + 5i)(9 - 6i) \\ = 120 - 15i$$

$$(35) \quad (3 + 7i)(1 + 2i) \\ = -11 + 13i$$

$$(36) \quad (-3 + 9i)(3 + 10i) \\ = -99 - 3i$$

$$(37) \quad (-4 - 10i)(1 - 8i) \\ = -84 + 22i$$

$$(38) \quad (3 - 7i)(-3 - i) \\ = -16 + 18i$$

$$(39) \quad (-5 + 10i)(6 + 9i) \\ = -120 + 15i$$

$$(40) \quad (8 + 4i)(7 + 4i) \\ = 40 + 60i$$

$$(41) \quad (-2 - 8i)(6 + i) \\ = -4 - 50i$$

$$(42) \quad (7 - 7i)(8 - i) \\ = 49 - 63i$$

$$(43) \quad (-2 - 8i)(-10 + 10i) \\ = 100 + 60i$$

$$(44) \quad (7 - 2i)(8 - 9i) \\ = 38 - 79i$$

$$(45) \quad (-8 + i)(-3 - 10i) \\ = 34 + 77i$$

$$(1) (3 + 8i)(1 - 9i) \\ = 75 - 19i$$

$$(2) (-8 + 8i)(1 - 5i) \\ = 32 + 48i$$

$$(3) (5 - 10i)(-3 - i) \\ = -25 + 25i$$

$$(4) (-1 + 7i)(9 - 10i) \\ = 61 + 73i$$

$$(5) (6 + 8i)(10 - 4i) \\ = 92 + 56i$$

$$(6) (9 + 5i)(-10 - 4i) \\ = -70 - 86i$$

$$(7) (-4 + 6i)(-9 - 2i) \\ = 48 - 46i$$

$$(8) (-5 - 10i)(-3 + 4i) \\ = 55 + 10i$$

$$(9) (-1 - 9i)(1 - 10i) \\ = -91 + i$$

$$(10) (-4 + 9i)(-3 + 3i) \\ = -15 - 39i$$

$$(11) (7 + 5i)(-7 - 3i) \\ = -34 - 56i$$

$$(12) (3 + 3i)(-3 - 2i) \\ = -3 - 15i$$

$$(13) (-9 - 4i)(3 + 10i) \\ = 13 - 102i$$

$$(14) (9 - 10i)(-4 + 8i) \\ = 44 + 112i$$

$$(15) (7 + 6i)(-9 + 9i) \\ = -117 + 9i$$

$$(16) (-10 + 5i)(6 - 4i) \\ = -40 + 70i$$

$$(17) (10 - 3i)(5 - i) \\ = 47 - 25i$$

$$(18) (-1 + 8i)(-1 - 8i) \\ = 65$$

$$(19) (-8 - 8i)(-8 - 6i) \\ = 16 + 112i$$

$$(20) (2 + i)(8 - 10i) \\ = 26 - 12i$$

$$(21) (-7 - 8i)(-3 - 5i) \\ = -19 + 59i$$

$$(22) (8 - i)(3 - 4i) \\ = 20 - 35i$$

$$(23) (5 + 7i)(-5 + i) \\ = -32 - 30i$$

$$(24) (-6 + 7i)(-1 - 5i) \\ = 41 + 23i$$

$$(25) (-3 + 2i)(-6 + 8i) \\ = 2 - 36i$$

$$(26) (-3 + 5i)(1 - 8i) \\ = 37 + 29i$$

$$(27) (-1 + 10i)(-7 - 6i) \\ = 67 - 64i$$

$$(28) (3 - 6i)(10 + 7i) \\ = 72 - 39i$$

$$(29) (4 - 5i)(-7 + 5i) \\ = -3 + 55i$$

$$(30) (2 + 4i)(-8 - 5i) \\ = 4 - 42i$$

$$(31) (-2 + 8i)(1 + 6i) \\ = -50 - 4i$$

$$(32) (-6 + 5i)(1 + 6i) \\ = -36 - 31i$$

$$(33) (-5 + 9i)(9 - 2i) \\ = -27 + 91i$$

$$(34) (-7 + 10i)(3 + 9i) \\ = -111 - 33i$$

$$(35) (-8 - 5i)(4 - 7i) \\ = -67 + 36i$$

$$(36) (6 + 6i)(-8 + 8i) \\ = -96$$

$$(37) (-5 - 8i)(-5 + 3i) \\ = 49 + 25i$$

$$(38) (-6 + 9i)(-7 - 5i) \\ = 87 - 33i$$

$$(39) (-8 - 6i)(-3 + i) \\ = 30 + 10i$$

$$(40) (7 + 5i)(-9 - 10i) \\ = -13 - 115i$$

$$(41) (-2 - 2i)(9 - i) \\ = -20 - 16i$$

$$(42) (10 - 3i)(1 - 8i) \\ = -14 - 83i$$

$$(43) (9 + 5i)(-7 - 7i) \\ = -28 - 98i$$

$$(44) (-3 + 2i)(6 + 2i) \\ = -22 + 6i$$

$$(45) (10 + 9i)(-10 - 8i) \\ = -28 - 170i$$

$$(1) \quad (-9 + 4i)(8 - 8i) \\ = -40 + 104i$$

$$(2) \quad (4 - 7i)(-6 + 4i) \\ = 4 + 58i$$

$$(3) \quad (-7 - 4i)(-2 - 2i) \\ = 6 + 22i$$

$$(4) \quad (9 - 4i)(2 - 5i) \\ = -2 - 53i$$

$$(5) \quad (-5 + 2i)(1 + 5i) \\ = -15 - 23i$$

$$(6) \quad (-6 + 7i)(6 - 3i) \\ = -15 + 60i$$

$$(7) \quad (4 + 9i)(-8 - 6i) \\ = 22 - 96i$$

$$(8) \quad (-1 + 4i)(10 - 10i) \\ = 30 + 50i$$

$$(9) \quad (2 + 2i)(-4 - 8i) \\ = 8 - 24i$$

$$(10) \quad (-2 - 2i)(2 + 8i) \\ = 12 - 20i$$

$$(11) \quad (-7 - 3i)(-5 - 9i) \\ = 8 + 78i$$

$$(12) \quad (-9 - 2i)(-8 + 9i) \\ = 90 - 65i$$

$$(13) \quad (9 - 3i)(-4 - 2i) \\ = -42 - 6i$$

$$(14) \quad (1 - i)(-3 - 6i) \\ = -9 - 3i$$

$$(15) \quad (-3 - 6i)(3 - 6i) \\ = -45$$

$$(16) \quad (-8 + i)(-9 - 8i) \\ = 80 + 55i$$

$$(17) \quad (4 + 3i)(-5 + 9i) \\ = -47 + 21i$$

$$(18) \quad (1 + 4i)(-9 - 3i) \\ = 3 - 39i$$

$$(19) \quad (-2 + 8i)(10 + 2i) \\ = -36 + 76i$$

$$(20) \quad (6 + 9i)(-7 + 3i) \\ = -69 - 45i$$

$$(21) \quad (-4 + 4i)(-4 + 10i) \\ = -24 - 56i$$

$$(22) \quad (-5 + 4i)(5 + 3i) \\ = -37 + 5i$$

$$(23) \quad (8 + 9i)(-9 + 3i) \\ = -99 - 57i$$

$$(24) \quad (-6 + 9i)(2 - 9i) \\ = 69 + 72i$$

$$(25) \quad (-7 + 9i)(6 - 2i) \\ = -24 + 68i$$

$$(26) \quad (-7 + 10i)(8 - 3i) \\ = -26 + 101i$$

$$(27) \quad (4 - i)(-10 + 9i) \\ = -31 + 46i$$

$$(28) \quad (-5 - 7i)(10 - 4i) \\ = -78 - 50i$$

$$(29) \quad (5 + 6i)(-2 - 8i) \\ = 38 - 52i$$

$$(30) \quad (-7 - 9i)(4 + i) \\ = -19 - 43i$$

$$(31) \quad (8 + 5i)(-6 + 4i) \\ = -68 + 2i$$

$$(32) \quad (-8 - 6i)(-5 - 7i) \\ = -2 + 86i$$

$$(33) \quad (3 - 6i)(-5 - 4i) \\ = -39 + 18i$$

$$(34) \quad (4 + i)(-3 + 8i) \\ = -20 + 29i$$

$$(35) \quad (-9 - 7i)(-1 + 9i) \\ = 72 - 74i$$

$$(36) \quad (-3 - 4i)(-2 - 2i) \\ = -2 + 14i$$

$$(37) \quad (4 - 3i)(5 + 9i) \\ = 47 + 21i$$

$$(38) \quad (9 - 7i)(-4 + 6i) \\ = 6 + 82i$$

$$(39) \quad (9 + 8i)(5 - 7i) \\ = 101 - 23i$$

$$(40) \quad (9 - 2i)(3 - 10i) \\ = 7 - 96i$$

$$(41) \quad (2 - i)(1 - 10i) \\ = -8 - 21i$$

$$(42) \quad (7 - 5i)(9 - 7i) \\ = 28 - 94i$$

$$(43) \quad (2 - 9i)(10 - 3i) \\ = -7 - 96i$$

$$(44) \quad (8 + 4i)(-6 + 2i) \\ = -56 - 8i$$

$$(45) \quad (3 - 4i)(8 - 2i) \\ = 16 - 38i$$

$$(1) (2 - 9i)(9 + 9i) \\ = 99 - 63i$$

$$(2) (10 + 4i)(-3 + 6i) \\ = -54 + 48i$$

$$(3) (-10 + 5i)(4 - 6i) \\ = -10 + 80i$$

$$(4) (10 + 8i)(9 + 2i) \\ = 74 + 92i$$

$$(5) (-8 + 6i)(6 - 9i) \\ = 6 + 108i$$

$$(6) (3 - 3i)(-3 + 7i) \\ = 12 + 30i$$

$$(7) (9 + i)(6 + 8i) \\ = 46 + 78i$$

$$(8) (-7 + 2i)(2 - 2i) \\ = -10 + 18i$$

$$(9) (-10 + 8i)(-9 + 3i) \\ = 66 - 102i$$

$$(10) (4 + i)(7 + 2i) \\ = 26 + 15i$$

$$(11) (-10 + 9i)(-4 + 4i) \\ = 4 - 76i$$

$$(12) (-6 - 9i)(-3 - 4i) \\ = -18 + 51i$$

$$(13) (4 + 4i)(10 + 9i) \\ = 4 + 76i$$

$$(14) (5 - 2i)(-10 - 5i) \\ = -60 - 5i$$

$$(15) (-1 - 4i)(-9 + 8i) \\ = 41 + 28i$$

$$(16) (-6 + 8i)(2 - i) \\ = -4 + 22i$$

$$(17) (2 + 9i)(5 + 3i) \\ = -17 + 51i$$

$$(18) (7 - 3i)(-1 - 6i) \\ = -25 - 39i$$

$$(19) (4 + 6i)(-1 + 3i) \\ = -22 + 6i$$

$$(20) (6 - 5i)(7 + 7i) \\ = 77 + 7i$$

$$(21) (-6 - 5i)(7 - 10i) \\ = -92 + 25i$$

$$(22) (1 - 2i)(-9 - 6i) \\ = -21 + 12i$$

$$(23) (5 - 8i)(3 + 6i) \\ = 63 + 6i$$

$$(24) (4 - 9i)(1 + 3i) \\ = 31 + 3i$$

$$(25) (-10 - 3i)(-3 + 2i) \\ = 36 - 11i$$

$$(26) (10 + 7i)(-8 - 10i) \\ = -10 - 156i$$

$$(27) (1 - i)(1 + 10i) \\ = 11 + 9i$$

$$(28) (6 + i)(4 - i) \\ = 25 - 2i$$

$$(29) (3 + 2i)(-4 + 5i) \\ = -22 + 7i$$

$$(30) (3 + 2i)(-2 + 2i) \\ = -10 + 2i$$

$$(31) (-3 + 8i)(-10 - 3i) \\ = 54 - 71i$$

$$(32) (-3 + 3i)(-5 - 7i) \\ = 36 + 6i$$

$$(33) (-7 + 2i)(-10 + 3i) \\ = 64 - 41i$$

$$(34) (-10 - 8i)(-9 + 6i) \\ = 138 + 12i$$

$$(35) (3 + 3i)(8 + i) \\ = 21 + 27i$$

$$(36) (-10 + 10i)(5 - 6i) \\ = 10 + 110i$$

$$(37) (6 - 7i)(1 - 6i) \\ = -36 - 43i$$

$$(38) (8 - 3i)(-8 + 8i) \\ = -40 + 88i$$

$$(39) (-5 - 7i)(10 - 5i) \\ = -85 - 45i$$

$$(40) (3 - 4i)(-1 + 4i) \\ = 13 + 16i$$

$$(41) (8 - 3i)(2 + 7i) \\ = 37 + 50i$$

$$(42) (-9 - 3i)(6 + 10i) \\ = -24 - 108i$$

$$(43) (2 - 2i)(5 + 10i) \\ = 30 + 10i$$

$$(44) (-2 - 7i)(-6 + 9i) \\ = 75 + 24i$$

$$(45) (5 + 7i)(-5 + i) \\ = -32 - 30i$$

$$(1) \quad (-8 + i)(5 + 5i) \\ = -45 - 35i$$

$$(2) \quad (-5 + 5i)(-8 + 7i) \\ = 5 - 75i$$

$$(3) \quad (-8 + 8i)(-2 + 9i) \\ = -56 - 88i$$

$$(4) \quad (-9 - 4i)(-5 - 6i) \\ = 21 + 74i$$

$$(5) \quad (2 - 7i)(8 + 3i) \\ = 37 - 50i$$

$$(6) \quad (-5 + 3i)(-10 - 5i) \\ = 65 - 5i$$

$$(7) \quad (1 - 3i)(-7 + 8i) \\ = 17 + 29i$$

$$(8) \quad (2 - 5i)(-8 + 3i) \\ = -1 + 46i$$

$$(9) \quad (4 - 4i)(9 + 2i) \\ = 44 - 28i$$

$$(10) \quad (-4 - 4i)(-6 - 6i) \\ = 48i$$

$$(11) \quad (-2 + 7i)(-4 + i) \\ = 1 - 30i$$

$$(12) \quad (10 + 3i)(-2 + 7i) \\ = -41 + 64i$$

$$(13) \quad (-6 - 2i)(-5 - 6i) \\ = 18 + 46i$$

$$(14) \quad (-6 + 7i)(10 - 8i) \\ = -4 + 118i$$

$$(15) \quad (1 - 3i)(10 - 4i) \\ = -2 - 34i$$

$$(16) \quad (6 + 4i)(7 - i) \\ = 46 + 22i$$

$$(17) \quad (-9 - 3i)(5 - i) \\ = -48 - 6i$$

$$(18) \quad (3 + 10i)(-6 - 10i) \\ = 82 - 90i$$

$$(19) \quad (2 - 3i)(-10 + 7i) \\ = 1 + 44i$$

$$(20) \quad (2 + 3i)(9 + 6i) \\ = 39i$$

$$(21) \quad (1 + 6i)(1 + 6i) \\ = -35 + 12i$$

$$(22) \quad (9 + 7i)(-10 - i) \\ = -83 - 79i$$

$$(23) \quad (1 + i)(10 + 4i) \\ = 6 + 14i$$

$$(24) \quad (-2 - 10i)(4 + i) \\ = 2 - 42i$$

$$(25) \quad (5 + i)(5 + 2i) \\ = 23 + 15i$$

$$(26) \quad (3 - 5i)(10 + 9i) \\ = 75 - 23i$$

$$(27) \quad (1 + 5i)(-9 - 9i) \\ = 36 - 54i$$

$$(28) \quad (6 - 8i)(-9 - 5i) \\ = -94 + 42i$$

$$(29) \quad (-9 + 2i)(1 + 6i) \\ = -21 - 52i$$

$$(30) \quad (1 - 10i)(8 - 7i) \\ = -62 - 87i$$

$$(31) \quad (6 + 4i)(-5 + 2i) \\ = -38 - 8i$$

$$(32) \quad (-10 - 5i)(-4 - 10i) \\ = -10 + 120i$$

$$(33) \quad (10 - 7i)(9 - 5i) \\ = 55 - 113i$$

$$(34) \quad (-6 + 9i)(-6 + 4i) \\ = -78i$$

$$(35) \quad (6 + 10i)(2 - 2i) \\ = 32 + 8i$$

$$(36) \quad (-1 + 3i)(4 - 7i) \\ = 17 + 19i$$

$$(37) \quad (-1 + 5i)(1 - 2i) \\ = 9 + 7i$$

$$(38) \quad (-1 - 9i)(4 - 10i) \\ = -94 - 26i$$

$$(39) \quad (2 - 5i)(-6 - 2i) \\ = -22 + 26i$$

$$(40) \quad (-5 - 7i)(-7 - 2i) \\ = 21 + 59i$$

$$(41) \quad (9 - 8i)(6 + 2i) \\ = 70 - 30i$$

$$(42) \quad (-5 + 2i)(10 + 6i) \\ = -62 - 10i$$

$$(43) \quad (6 + 6i)(-7 - 4i) \\ = -18 - 66i$$

$$(44) \quad (-9 + 10i)(1 - 6i) \\ = 51 + 64i$$

$$(45) \quad (-2 - 9i)(-1 - i) \\ = -7 + 11i$$

$$(1) \quad (-2-i)(-8-4i) \\ = 12 + 16i$$

$$(2) \quad (5+3i)(-1+i) \\ = -8 + 2i$$

$$(3) \quad (6+5i)(-8-7i) \\ = -13 - 82i$$

$$(4) \quad (-6+10i)(1-4i) \\ = 34 + 34i$$

$$(5) \quad (-3-9i)(2-8i) \\ = -78 + 6i$$

$$(6) \quad (10+4i)(-1+5i) \\ = -30 + 46i$$

$$(7) \quad (-7-3i)(-2-7i) \\ = -7 + 55i$$

$$(8) \quad (2-7i)(-8+10i) \\ = 54 + 76i$$

$$(9) \quad (-3-8i)(5-7i) \\ = -71 - 19i$$

$$(10) \quad (-4+6i)(-2+3i) \\ = -10 - 24i$$

$$(11) \quad (-10+6i)(10+9i) \\ = -154 - 30i$$

$$(12) \quad (-10-3i)(1+3i) \\ = -1 - 33i$$

$$(13) \quad (6-4i)(-7-5i) \\ = -62 - 2i$$

$$(14) \quad (7+4i)(9+8i) \\ = 31 + 92i$$

$$(15) \quad (-9+4i)(-4+6i) \\ = 12 - 70i$$

$$(16) \quad (3+6i)(10+8i) \\ = -18 + 84i$$

$$(17) \quad (2-3i)(4+2i) \\ = 14 - 8i$$

$$(18) \quad (-8-3i)(4-6i) \\ = -50 + 36i$$

$$(19) \quad (-5+9i)(-4+10i) \\ = -70 - 86i$$

$$(20) \quad (-7-i)(-2-10i) \\ = 4 + 72i$$

$$(21) \quad (2-3i)(6-3i) \\ = 3 - 24i$$

$$(22) \quad (-2+3i)(2+4i) \\ = -16 - 2i$$

$$(23) \quad (-7+4i)(-2+4i) \\ = -2 - 36i$$

$$(24) \quad (3+5i)(5-6i) \\ = 45 + 7i$$

$$(25) \quad (3+8i)(-7+10i) \\ = -101 - 26i$$

$$(26) \quad (1+i)(-2-2i) \\ = -4i$$

$$(27) \quad (-7-10i)(9-8i) \\ = -143 - 34i$$

$$(28) \quad (-2-3i)(-4+2i) \\ = 14 + 8i$$

$$(29) \quad (7-3i)(1+3i) \\ = 16 + 18i$$

$$(30) \quad (5-9i)(-1+5i) \\ = 40 + 34i$$

$$(31) \quad (-7+10i)(-3-i) \\ = 31 - 23i$$

$$(32) \quad (-8+6i)(10+2i) \\ = -92 + 44i$$

$$(33) \quad (-6-7i)(-7+10i) \\ = 112 - 11i$$

$$(34) \quad (10+10i)(-8-i) \\ = -70 - 90i$$

$$(35) \quad (-1+i)(-3-3i) \\ = 6$$

$$(36) \quad (1-7i)(-5+5i) \\ = 30 + 40i$$

$$(37) \quad (-2+3i)(-3-7i) \\ = 27 + 5i$$

$$(38) \quad (-4-5i)(6+8i) \\ = 16 - 62i$$

$$(39) \quad (10-6i)(7+4i) \\ = 94 - 2i$$

$$(40) \quad (10-3i)(-6+3i) \\ = -51 + 48i$$

$$(41) \quad (8-10i)(-1+2i) \\ = 12 + 26i$$

$$(42) \quad (2-2i)(7-6i) \\ = 2 - 26i$$

$$(43) \quad (7+i)(-3-6i) \\ = -15 - 45i$$

$$(44) \quad (5+7i)(-4-8i) \\ = 36 - 68i$$

$$(45) \quad (-4+6i)(5-10i) \\ = 40 + 70i$$

$$(1) \quad (-1 - 9i)(-3 - 9i) \\ = -78 + 36i$$

$$(2) \quad (10 - 4i)(-3 + 3i) \\ = -18 + 42i$$

$$(3) \quad (-10 - 9i)(-5 - 4i) \\ = 14 + 85i$$

$$(4) \quad (9 - 3i)(1 + 2i) \\ = 15 + 15i$$

$$(5) \quad (9 - 4i)(6 + 5i) \\ = 74 + 21i$$

$$(6) \quad (7 + 6i)(8 - 4i) \\ = 80 + 20i$$

$$(7) \quad (-1 + 10i)(2 - i) \\ = 8 + 21i$$

$$(8) \quad (8 + 7i)(-1 + 2i) \\ = -22 + 9i$$

$$(9) \quad (-3 - i)(2 + 10i) \\ = 4 - 32i$$

$$(10) \quad (5 + 2i)(-8 + 4i) \\ = -48 + 4i$$

$$(11) \quad (2 - 6i)(-8 - 2i) \\ = -28 + 44i$$

$$(12) \quad (-3 - 4i)(1 - 10i) \\ = -43 + 26i$$

$$(13) \quad (5 + 8i)(5 + 8i) \\ = -39 + 80i$$

$$(14) \quad (5 - 6i)(2 + 7i) \\ = 52 + 23i$$

$$(15) \quad (8 - 6i)(-7 - 3i) \\ = -74 + 18i$$

$$(16) \quad (-2 - 5i)(-2 + 3i) \\ = 19 + 4i$$

$$(17) \quad (8 + 9i)(-10 - i) \\ = -71 - 98i$$

$$(18) \quad (6 - 4i)(-9 + 3i) \\ = -42 + 54i$$

$$(19) \quad (7 - 8i)(-1 + 9i) \\ = 65 + 71i$$

$$(20) \quad (-4 + 4i)(7 + 6i) \\ = -52 + 4i$$

$$(21) \quad (9 - 5i)(10 - 9i) \\ = 45 - 131i$$

$$(22) \quad (9 + 5i)(-8 + 10i) \\ = -122 + 50i$$

$$(23) \quad (-8 - 6i)(8 - 10i) \\ = -124 + 32i$$

$$(24) \quad (-10 - 4i)(-2 - 2i) \\ = 12 + 28i$$

$$(25) \quad (-2 - 7i)(2 - 9i) \\ = -67 + 4i$$

$$(26) \quad (7 + i)(4 + 6i) \\ = 22 + 46i$$

$$(27) \quad (10 - 8i)(-10 - 6i) \\ = -148 + 20i$$

$$(28) \quad (-9 - 4i)(7 - 7i) \\ = -91 + 35i$$

$$(29) \quad (-5 + i)(2 + 5i) \\ = -15 - 23i$$

$$(30) \quad (6 - 8i)(-5 + 3i) \\ = -6 + 58i$$

$$(31) \quad (1 + 5i)(8 - 4i) \\ = 28 + 36i$$

$$(32) \quad (2 + 9i)(-2 + 10i) \\ = -94 + 2i$$

$$(33) \quad (9 - 9i)(8 - 6i) \\ = 18 - 126i$$

$$(34) \quad (1 + 2i)(1 - 4i) \\ = 9 - 2i$$

$$(35) \quad (-6 + 6i)(-2 + 2i) \\ = -24i$$

$$(36) \quad (7 + 2i)(1 + 8i) \\ = -9 + 58i$$

$$(37) \quad (-4 - 2i)(6 - 3i) \\ = -30$$

$$(38) \quad (4 + 10i)(-7 - 8i) \\ = 52 - 102i$$

$$(39) \quad (1 - 6i)(1 + 7i) \\ = 43 + i$$

$$(40) \quad (1 - 7i)(-5 - 10i) \\ = -75 + 25i$$

$$(41) \quad (9 + i)(5 - 7i) \\ = 52 - 58i$$

$$(42) \quad (10 - 6i)(4 + 7i) \\ = 82 + 46i$$

$$(43) \quad (-4 - 7i)(-7 - i) \\ = 21 + 53i$$

$$(44) \quad (-1 + 9i)(-10 + 2i) \\ = -8 - 92i$$

$$(45) \quad (7 - 3i)(6 - 6i) \\ = 24 - 60i$$

$$(1) \quad (-10 - 4i)(8 - 2i) \\ = -88 - 12i$$

$$(2) \quad (9 + 6i)(-5 + 6i) \\ = -81 + 24i$$

$$(3) \quad (2 + 5i)(-2 - 8i) \\ = 36 - 26i$$

$$(4) \quad (9 + 4i)(10 - 6i) \\ = 114 - 14i$$

$$(5) \quad (3 - 7i)(-4 + 6i) \\ = 30 + 46i$$

$$(6) \quad (3 - 10i)(10 - 6i) \\ = -30 - 118i$$

$$(7) \quad (-2 + 4i)(1 + 7i) \\ = -30 - 10i$$

$$(8) \quad (2 - 9i)(10 - 4i) \\ = -16 - 98i$$

$$(9) \quad (-10 - 7i)(9 - 10i) \\ = -160 + 37i$$

$$(10) \quad (5 - 9i)(-6 + 5i) \\ = 15 + 79i$$

$$(11) \quad (-7 + i)(5 - 5i) \\ = -30 + 40i$$

$$(12) \quad (-3 + 8i)(2 + 5i) \\ = -46 + i$$

$$(13) \quad (9 - 6i)(-3 - 8i) \\ = -75 - 54i$$

$$(14) \quad (-7 + 10i)(-10 + 5i) \\ = 20 - 135i$$

$$(15) \quad (9 - 6i)(6 - 10i) \\ = -6 - 126i$$

$$(16) \quad (-5 + 8i)(4 - 6i) \\ = 28 + 62i$$

$$(17) \quad (-8 + 6i)(-3 + 8i) \\ = -24 - 82i$$

$$(18) \quad (-4 - 5i)(-7 - 7i) \\ = -7 + 63i$$

$$(19) \quad (3 + 10i)(3 + i) \\ = -1 + 33i$$

$$(20) \quad (10 - 8i)(3 - 2i) \\ = 14 - 44i$$

$$(21) \quad (-2 - 7i)(9 + 3i) \\ = 3 - 69i$$

$$(22) \quad (4 - 5i)(4 - 9i) \\ = -29 - 56i$$

$$(23) \quad (3 + 3i)(-7 - 2i) \\ = -15 - 27i$$

$$(24) \quad (4 + 10i)(8 + 2i) \\ = 12 + 88i$$

$$(25) \quad (-5 - i)(-6 + 6i) \\ = 36 - 24i$$

$$(26) \quad (-8 - 9i)(5 - 7i) \\ = -103 + 11i$$

$$(27) \quad (2 - 4i)(-8 + 8i) \\ = 16 + 48i$$

$$(28) \quad (-6 + 7i)(10 - 2i) \\ = -46 + 82i$$

$$(29) \quad (9 - 7i)(5 - i) \\ = 38 - 44i$$

$$(30) \quad (5 + 2i)(8 + 3i) \\ = 34 + 31i$$

$$(31) \quad (-3 - 3i)(4 - 5i) \\ = -27 + 3i$$

$$(32) \quad (4 + 7i)(7 + 7i) \\ = -21 + 77i$$

$$(33) \quad (5 + 5i)(6 + 10i) \\ = -20 + 80i$$

$$(34) \quad (5 - 3i)(-7 + i) \\ = -32 + 26i$$

$$(35) \quad (1 + 5i)(8 - 7i) \\ = 43 + 33i$$

$$(36) \quad (-7 + 6i)(9 + 7i) \\ = -105 + 5i$$

$$(37) \quad (-4 + 4i)(-4 - 7i) \\ = 44 + 12i$$

$$(38) \quad (1 + 10i)(-2 - 8i) \\ = 78 - 28i$$

$$(39) \quad (5 - 4i)(-6 - i) \\ = -34 + 19i$$

$$(40) \quad (5 + 4i)(4 - 10i) \\ = 60 - 34i$$

$$(41) \quad (10 + 8i)(-2 + 10i) \\ = -100 + 84i$$

$$(42) \quad (3 - 8i)(-7 + 4i) \\ = 11 + 68i$$

$$(43) \quad (-2 - i)(6 + 6i) \\ = -6 - 18i$$

$$(44) \quad (10 + 4i)(5 - 9i) \\ = 86 - 70i$$

$$(45) \quad (3 + 8i)(10 + 4i) \\ = -2 + 92i$$

$$(1) (7 - 3i)(5 + 3i) \\ = 44 + 6i$$

$$(2) (-4 + 7i)(1 + 4i) \\ = -32 - 9i$$

$$(3) (-7 + 5i)(5 - i) \\ = -30 + 32i$$

$$(4) (10 + 3i)(4 - 10i) \\ = 70 - 88i$$

$$(5) (9 + 6i)(8 + i) \\ = 66 + 57i$$

$$(6) (9 + 4i)(-6 - 10i) \\ = -14 - 114i$$

$$(7) (-5 - 2i)(-2 - 2i) \\ = 6 + 14i$$

$$(8) (-6 + 6i)(6 + 7i) \\ = -78 - 6i$$

$$(9) (10 - 7i)(9 - 4i) \\ = 62 - 103i$$

$$(10) (3 - 2i)(1 - 10i) \\ = -17 - 32i$$

$$(11) (-8 + 5i)(-2 + 3i) \\ = 1 - 34i$$

$$(12) (-9 + 6i)(4 - 7i) \\ = 6 + 87i$$

$$(13) (8 + 9i)(-9 - 3i) \\ = -45 - 105i$$

$$(14) (4 + 9i)(-1 + 4i) \\ = -40 + 7i$$

$$(15) (-5 - 8i)(8 - 6i) \\ = -88 - 34i$$

$$(16) (2 + 8i)(9 + 7i) \\ = -38 + 86i$$

$$(17) (-5 - 6i)(-8 + 10i) \\ = 100 - 2i$$

$$(18) (-10 + 3i)(1 + 8i) \\ = -34 - 77i$$

$$(19) (-7 - 9i)(-7 - 3i) \\ = 22 + 84i$$

$$(20) (-9 - 9i)(1 - 10i) \\ = -99 + 81i$$

$$(21) (1 + 6i)(6 - 8i) \\ = 54 + 28i$$

$$(22) (-7 - 2i)(-1 + 2i) \\ = 11 - 12i$$

$$(23) (-2 + 7i)(-7 + 6i) \\ = -28 - 61i$$

$$(24) (1 + 9i)(6 - 8i) \\ = 78 + 46i$$

$$(25) (-2 + 10i)(-8 + 10i) \\ = -84 - 100i$$

$$(26) (-7 - 4i)(-6 + i) \\ = 46 + 17i$$

$$(27) (-7 - 9i)(3 + 7i) \\ = 42 - 76i$$

$$(28) (-5 - 5i)(-2 - 9i) \\ = -35 + 55i$$

$$(29) (-5 + 10i)(1 - 5i) \\ = 45 + 35i$$

$$(30) (-8 + 9i)(3 + 8i) \\ = -96 - 37i$$

$$(31) (-2 + 7i)(-1 + i) \\ = -5 - 9i$$

$$(32) (-5 + 2i)(3 - 4i) \\ = -7 + 26i$$

$$(33) (9 - 5i)(2 + 4i) \\ = 38 + 26i$$

$$(34) (1 + i)(-1 - i) \\ = -2i$$

$$(35) (9 - 10i)(-7 + 10i) \\ = 37 + 160i$$

$$(36) (4 - 10i)(1 + 2i) \\ = 24 - 2i$$

$$(37) (6 + 4i)(-3 + 2i) \\ = -26$$

$$(38) (-10 - 8i)(9 - 9i) \\ = -162 + 18i$$

$$(39) (3 - 9i)(-4 + 3i) \\ = 15 + 45i$$

$$(40) (5 + 7i)(-8 + 10i) \\ = -110 - 6i$$

$$(41) (-5 - 4i)(-5 - 5i) \\ = 5 + 45i$$

$$(42) (-10 - 7i)(-1 - 8i) \\ = -46 + 87i$$

$$(43) (10 + 8i)(-9 - 9i) \\ = -18 - 162i$$

$$(44) (-9 + 3i)(-4 + 2i) \\ = 30 - 30i$$

$$(45) (8 - 8i)(4 - 7i) \\ = -24 - 88i$$

1.5.3 除法

複素数の商を求める問題。

(1) $(-3 + 2i) \div (4 + i)$

(11) $(4 + 3i) \div (5 + i)$

(21) $(4 + 2i) \div (-3 + 3i)$

(2) $(-4 - 5i) \div (5 - 5i)$

(12) $(-1 + 5i) \div (3 - 3i)$

(22) $(-4 - 4i) \div (-3 - 5i)$

(3) $(-1 - i) \div (-2 - 5i)$

(13) $(5 + 3i) \div (-2 + i)$

(23) $(1 + 5i) \div (-1 + 4i)$

(4) $(2 + 3i) \div (3 + 2i)$

(14) $(1 - 5i) \div (5 - 4i)$

(24) $(-1 - 2i) \div (-1 - 3i)$

(5) $(-2 - 2i) \div (-5 + 4i)$

(15) $(5 - i) \div (-2 - 2i)$

(25) $(-5 + 4i) \div (5 + i)$

(6) $(-4 - i) \div (4 - i)$

(16) $(-2 + 4i) \div (1 - 3i)$

(26) $(4 + i) \div (-5 - i)$

(7) $(-3 + 2i) \div (4 - 2i)$

(17) $(-1 - 3i) \div (-4 - 5i)$

(27) $(2 + i) \div (2 + 5i)$

(8) $(3 - 4i) \div (-2 - 5i)$

(18) $(3 + i) \div (-1 + 2i)$

(28) $(1 - i) \div (-3 + 2i)$

(9) $(-3 - 3i) \div (-4 - 5i)$

(19) $(-4 - 5i) \div (1 + 4i)$

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(30) $(2 - 5i) \div (1 - i)$

(1) $(-2 - 3i) \div (3 + 5i)$

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$$(22) (5 - 2i) \div (-3 - 4i)$$

$$(3) (2 + i) \div (4 + 2i)$$

$$(13) (3 + i) \div (2 + 4i)$$

$$(23) (2 + 2i) \div (4 + 2i)$$

$$(4) (2 + 2i) \div (-5 + 5i)$$

$$(14) (-2 - 2i) \div (-5 - i)$$

$$(24) (-3 - 5i) \div (4 - 3i)$$

$$(5) (3 + 4i) \div (5 - i)$$

$$(15) (-1 - i) \div (2 + 3i)$$

$$(25) (-5 - i) \div (2 + i)$$

$$(6) (-1 - 4i) \div (2 + 3i)$$

$$(16) (-2 + 4i) \div (-4 + 3i)$$

$$(26) (-4 - 3i) \div (5 - 3i)$$

$$(7) (3 - i) \div (3 + 4i)$$

$$(17) (1 + 4i) \div (-1 + i)$$

$$(27) (4 - 3i) \div (5 + i)$$

$$(8) (2 + 5i) \div (-3 - 4i)$$

$$(18) (5 - 4i) \div (1 - 2i)$$

$$(28) (1 - 3i) \div (5 - 2i)$$

$$(9) (3 + 2i) \div (-3 - 2i)$$

$$(19) (-4 + i) \div (-1 - i)$$

$$(29) (-3 - 2i) \div (3 - 5i)$$

$$(10) (2 - 4i) \div (-3 + 4i)$$

$$(20) (-1 - 4i) \div (-2 + 5i)$$

$$(30) (2 + 5i) \div (4 + 4i)$$

$$(1) (1 + 3i) \div (-5 - 5i)$$

$$(11) (1 - 3i) \div (1 - 4i)$$

$$(21) (-5 + 3i) \div (-1 + i)$$

$$(2) (1 - 3i) \div (-5 + 5i)$$

$$(12) (-5 - 3i) \div (1 + i)$$

$$(22) (-1 + i) \div (-3 - 5i)$$

$$(3) (3 + 4i) \div (4 - 2i)$$

$$(13) (4 - i) \div (4 - 2i)$$

$$(23) (-1 - i) \div (-5 + 2i)$$

$$(4) (5 + 5i) \div (-2 - 2i)$$

$$(14) (1 + i) \div (5 - 5i)$$

$$(24) (-5 + 4i) \div (-4 - 3i)$$

$$(5) (-5 + 5i) \div (1 + 3i)$$

$$(15) (4 - i) \div (-1 + 5i)$$

$$(25) (-2 - 2i) \div (-3 + 3i)$$

$$(6) (-2 + 5i) \div (-1 + i)$$

$$(16) (-3 - 4i) \div (2 + i)$$

$$(26) (1 - i) \div (-3 - 4i)$$

$$(7) (2 - 2i) \div (3 + i)$$

$$(17) (-2 + 2i) \div (-4 - 4i)$$

$$(27) (3 - 4i) \div (-5 - 3i)$$

$$(8) (3 + 3i) \div (4 + 2i)$$

$$(18) (1 + 5i) \div (1 - i)$$

$$(28) (3 - 5i) \div (-1 - 3i)$$

$$(9) (2 + 2i) \div (-3 - 2i)$$

$$(19) (-4 - i) \div (1 - 5i)$$

$$(29) (-1 + 2i) \div (-1 + 3i)$$

$$(10) (-4 + 4i) \div (-3 - 5i)$$

$$(20) (-2 + 4i) \div (4 + 4i)$$

$$(30) (-5 - 2i) \div (2 + 3i)$$

$$(1) (4 - i) \div (1 - 5i)$$

$$(11) (4 + 4i) \div (-5 - 2i)$$

$$(21) (4 + 5i) \div (3 + 2i)$$

$$(2) (1 - 4i) \div (4 + 5i)$$

$$(12) (-5 + i) \div (3 - 4i)$$

$$(22) (2 + 4i) \div (1 - i)$$

$$(3) (-4 + i) \div (3 + 2i)$$

$$(13) (-5 - 3i) \div (4 + 2i)$$

$$(23) (-3 + 2i) \div (2 - 2i)$$

$$(4) (3 + 4i) \div (-3 + i)$$

$$(14) (-1 + 5i) \div (-2 + 2i)$$

$$(24) (1 - 2i) \div (-4 - i)$$

$$(5) (3 - 5i) \div (4 + 2i)$$

$$(15) (-3 - 2i) \div (-5 + 3i)$$

$$(25) (-4 - i) \div (-1 - 4i)$$

$$(6) (2 - 4i) \div (-5 + 2i)$$

$$(16) (-4 - 4i) \div (-1 + 2i)$$

$$(26) (4 + i) \div (1 + 3i)$$

$$(7) (3 + 4i) \div (1 - 2i)$$

$$(17) (5 + 5i) \div (5 - 3i)$$

$$(27) (-3 + 3i) \div (-2 + i)$$

$$(8) (-3 + 2i) \div (3 + 4i)$$

$$(18) (5 + 3i) \div (2 - 3i)$$

$$(28) (5 + 3i) \div (-5 + 5i)$$

$$(9) (1 - 4i) \div (4 + 4i)$$

$$(19) (2 + 5i) \div (5 - 4i)$$

$$(29) (5 - i) \div (2 - 3i)$$

$$(10) (3 - 4i) \div (4 + 3i)$$

$$(20) (5 - 3i) \div (3 + 4i)$$

$$(30) (4 + 5i) \div (3 + 2i)$$

$$(1) (2 + 4i) \div (-5 - 3i)$$

$$(11) (3 - 3i) \div (-1 + 3i)$$

$$(21) (-5 + i) \div (-3 + 2i)$$

$$(2) (1 - i) \div (2 + i)$$

$$(12) (-3 + 2i) \div (-2 - 5i)$$

$$(22) (-3 + 5i) \div (2 + 4i)$$

$$(3) (-5 + 2i) \div (-3 + 2i)$$

$$(13) (3 + 5i) \div (4 - 4i)$$

$$(23) (-3 - i) \div (-4 + i)$$

$$(4) (-1 + 3i) \div (4 + 3i)$$

$$(14) (-3 + 3i) \div (1 - i)$$

$$(24) (-3 + i) \div (-5 - 3i)$$

$$(5) (-3 + 2i) \div (4 + i)$$

$$(15) (-2 - 5i) \div (-1 - 2i)$$

$$(25) (-3 - 2i) \div (1 + 4i)$$

$$(6) (2 - i) \div (-2 - 4i)$$

$$(16) (1 - 3i) \div (-2 + 2i)$$

$$(26) (4 + 5i) \div (4 + 5i)$$

$$(7) (-4 - 4i) \div (1 + 5i)$$

$$(17) (-2 - 4i) \div (4 - 5i)$$

$$(27) (-3 - 3i) \div (-2 + 3i)$$

$$(8) (-1 - i) \div (3 - 4i)$$

$$(18) (-4 + 5i) \div (-1 - 5i)$$

$$(28) (-2 + 3i) \div (-1 - 3i)$$

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$$(19) (-4 + 5i) \div (2 + i)$$

$$(29) (5 + i) \div (-5 + 3i)$$

$$(10) (-5 - 5i) \div (1 + 5i)$$

$$(20) (5 + 3i) \div (4 - 3i)$$

$$(30) (-2 - 5i) \div (-5 - 4i)$$

(1) $(-4 + i) \div (-2 + 2i)$

(11) $(-1 - i) \div (3 + i)$

(21) $(1 + 2i) \div (-3 - i)$

(2) $(-1 - i) \div (-4 - 5i)$

(12) $(-1 + 3i) \div (-3 + 3i)$

(22) $(-1 + 2i) \div (-5 - i)$

(3) $(-4 + i) \div (-4 + i)$

(13) $(-2 + 2i) \div (2 - 4i)$

(23) $(-1 - i) \div (-1 + i)$

(4) $(3 - i) \div (1 - 5i)$

(14) $(2 - 2i) \div (5 - 2i)$

(24) $(-1 - 5i) \div (5 + 3i)$

(5) $(-1 - 2i) \div (1 + 4i)$

(15) $(4 + i) \div (-4 - 3i)$

(25) $(-1 - i) \div (-4 - i)$

(6) $(4 - 3i) \div (-1 - 3i)$

(16) $(3 + 3i) \div (-2 - i)$

(26) $(3 + 5i) \div (3 - 2i)$

(7) $(-1 + 3i) \div (-4 - i)$

(17) $(4 - 3i) \div (-2 - 2i)$

(27) $(-5 + 5i) \div (4 - 3i)$

(8) $(-3 + 5i) \div (4 + 3i)$

(18) $(1 - 4i) \div (-1 + 4i)$

(28) $(-2 - 3i) \div (-1 - 2i)$

(9) $(-2 + 3i) \div (-4 + 4i)$

(19) $(-4 + 4i) \div (4 + 4i)$

(29) $(-5 - 4i) \div (-1 - i)$

(10) $(-1 + i) \div (1 + 2i)$

(20) $(-4 - 4i) \div (3 + i)$

(30) $(1 - 2i) \div (-4 - 4i)$

(1) $(-1 - 3i) \div (-3 + 2i)$

(11) $(2 - 5i) \div (-4 - 5i)$

(21) $(-2 - 3i) \div (3 - 2i)$

(2) $(3 - 2i) \div (5 + i)$

(12) $(5 + 5i) \div (-5 + 3i)$

(22) $(3 + 2i) \div (2 - 4i)$

(3) $(2 + 5i) \div (1 - 5i)$

(13) $(3 - 5i) \div (5 + 2i)$

(23) $(5 + i) \div (2 - i)$

(4) $(-4 + 5i) \div (-5 + 2i)$

(14) $(5 - i) \div (2 - 5i)$

(24) $(3 - 5i) \div (-2 - 2i)$

(5) $(5 - i) \div (-4 - 2i)$

(15) $(-3 - i) \div (1 - 3i)$

(25) $(3 - 5i) \div (1 + i)$

(6) $(-3 + 2i) \div (-5 - i)$

(16) $(3 - i) \div (-5 + 2i)$

(26) $(-2 - 3i) \div (1 + 2i)$

(7) $(-5 - 2i) \div (-2 - 3i)$

(17) $(1 - 3i) \div (4 + 2i)$

(27) $(-1 - 3i) \div (5 - i)$

(8) $(3 + i) \div (1 - i)$

(18) $(-2 + 5i) \div (-3 - 3i)$

(28) $(5 + i) \div (4 + 5i)$

(9) $(-1 - 4i) \div (-4 - 5i)$

(19) $(3 - 5i) \div (-2 - 5i)$

(29) $(2 - i) \div (3 + 3i)$

(10) $(5 - 4i) \div (-4 - 5i)$

(20) $(4 + 5i) \div (3 + 4i)$

(30) $(-5 - 5i) \div (1 + 4i)$

(1) $(2 + i) \div (-3 + 4i)$

(11) $(5 - 3i) \div (-1 + 5i)$

(21) $(1 - 3i) \div (-1 - 5i)$

(2) $(2 + i) \div (2 + 3i)$

(12) $(-3 - i) \div (2 + 4i)$

(22) $(-1 - 4i) \div (-2 - 3i)$

(3) $(-2 + i) \div (-5 + 4i)$

(13) $(-4 - i) \div (-2 + 3i)$

(23) $(-1 - 3i) \div (-1 - 3i)$

(4) $(5 + 3i) \div (-3 + 2i)$

(14) $(1 + 4i) \div (4 - 5i)$

(24) $(1 - 2i) \div (3 + 5i)$

(5) $(-4 + 2i) \div (-2 - 2i)$

(15) $(2 + i) \div (-2 - 4i)$

(25) $(4 + 5i) \div (-4 - 4i)$

(6) $(-1 - 5i) \div (2 - i)$

(16) $(-1 + 4i) \div (-2 + 4i)$

(26) $(-1 + 5i) \div (-2 + 5i)$

(7) $(-5 + 2i) \div (2 + 4i)$

(17) $(3 - 3i) \div (-1 + i)$

(27) $(-2 - 4i) \div (1 - 5i)$

(8) $(-5 - 2i) \div (-3 - 2i)$

(18) $(-2 + 5i) \div (-1 + 2i)$

(28) $(2 + 3i) \div (-2 - 5i)$

(9) $(4 - 2i) \div (2 + 3i)$

(19) $(-1 - 3i) \div (-2 - 5i)$

(29) $(2 - 3i) \div (-3 + i)$

(10) $(-1 + 5i) \div (2 - 5i)$

(20) $(4 - 5i) \div (1 + 5i)$

(30) $(1 + 5i) \div (-5 + 4i)$

$$(1) (-3 + 2i) \div (4 + i) \\ = -\frac{10}{17} + \frac{11}{17}i$$

$$(11) (4 + 3i) \div (5 + i) \\ = \frac{23}{26} + \frac{11}{26}i$$

$$(21) (4 + 2i) \div (-3 + 3i) \\ = -\frac{1}{3} - i$$

$$(2) (-4 - 5i) \div (5 - 5i) \\ = \frac{1}{10} - \frac{9}{10}i$$

$$(12) (-1 + 5i) \div (3 - 3i) \\ = -1 + \frac{2}{3}i$$

$$(22) (-4 - 4i) \div (-3 - 5i) \\ = \frac{16}{17} - \frac{4}{17}i$$

$$(3) (-1 - i) \div (-2 - 5i) \\ = \frac{7}{29} - \frac{3}{29}i$$

$$(13) (5 + 3i) \div (-2 + i) \\ = -\frac{7}{5} - \frac{11}{5}i$$

$$(23) (1 + 5i) \div (-1 + 4i) \\ = \frac{19}{17} - \frac{9}{17}i$$

$$(4) (2 + 3i) \div (3 + 2i) \\ = \frac{12}{13} + \frac{5}{13}i$$

$$(14) (1 - 5i) \div (5 - 4i) \\ = \frac{25}{41} - \frac{21}{41}i$$

$$(24) (-1 - 2i) \div (-1 - 3i) \\ = \frac{7}{10} - \frac{1}{10}i$$

$$(5) (-2 - 2i) \div (-5 + 4i) \\ = \frac{2}{41} + \frac{18}{41}i$$

$$(15) (5 - i) \div (-2 - 2i) \\ = -1 + \frac{3}{2}i$$

$$(25) (-5 + 4i) \div (5 + i) \\ = -\frac{21}{26} + \frac{25}{26}i$$

$$(6) (-4 - i) \div (4 - i) \\ = -\frac{15}{17} - \frac{8}{17}i$$

$$(16) (-2 + 4i) \div (1 - 3i) \\ = -\frac{7}{5} - \frac{1}{5}i$$

$$(26) (4 + i) \div (-5 - i) \\ = -\frac{21}{26} - \frac{1}{26}i$$

$$(7) (-3 + 2i) \div (4 - 2i) \\ = -\frac{4}{5} + \frac{1}{10}i$$

$$(17) (-1 - 3i) \div (-4 - 5i) \\ = \frac{19}{41} + \frac{7}{41}i$$

$$(27) (2 + i) \div (2 + 5i) \\ = \frac{9}{29} - \frac{8}{29}i$$

$$(8) (3 - 4i) \div (-2 - 5i) \\ = \frac{14}{29} + \frac{23}{29}i$$

$$(18) (3 + i) \div (-1 + 2i) \\ = -\frac{1}{5} - \frac{7}{5}i$$

$$(28) (1 - i) \div (-3 + 2i) \\ = -\frac{5}{13} + \frac{1}{13}i$$

$$(9) (-3 - 3i) \div (-4 - 5i) \\ = \frac{27}{41} - \frac{3}{41}i$$

$$(19) (-4 - 5i) \div (1 + 4i) \\ = -\frac{24}{17} + \frac{11}{17}i$$

$$(29) (4 + 4i) \div (-5 + 2i) \\ = -\frac{12}{29} - \frac{28}{29}i$$

$$(10) (-2 - 3i) \div (3 - 4i) \\ = \frac{6}{25} - \frac{17}{25}i$$

$$(20) (5 - i) \div (-1 + 4i) \\ = -\frac{9}{17} - \frac{19}{17}i$$

$$(30) (2 - 5i) \div (1 - i) \\ = \frac{7}{2} - \frac{3}{2}i$$

$$(1) \quad (-2-3i) \div (3+5i) \\ = -\frac{21}{34} + \frac{1}{34}i$$

$$(11) \quad (-2+i) \div (-5+3i) \\ = \frac{13}{34} + \frac{1}{34}i$$

$$(21) \quad (3+5i) \div (-5+2i) \\ = -\frac{5}{29} - \frac{31}{29}i$$

$$(2) \quad (3+i) \div (-3+i) \\ = -\frac{4}{5} - \frac{3}{5}i$$

$$(12) \quad (2+4i) \div (-2+2i) \\ = \frac{1}{2} - \frac{3}{2}i$$

$$(22) \quad (-1+4i) \div (-3-4i) \\ = -\frac{13}{25} - \frac{16}{25}i$$

$$(3) \quad (5-2i) \div (5+i) \\ = \frac{23}{26} - \frac{15}{26}i$$

$$(13) \quad (-1+2i) \div (2+3i) \\ = \frac{4}{13} + \frac{7}{13}i$$

$$(23) \quad (-5-5i) \div (2+i) \\ = -3-i$$

$$(4) \quad (-1-4i) \div (-5-5i) \\ = \frac{1}{2} + \frac{3}{10}i$$

$$(14) \quad (5+5i) \div (1+3i) \\ = 2-i$$

$$(24) \quad (1+i) \div (-3-3i) \\ = -\frac{1}{3}$$

$$(5) \quad (-2+2i) \div (-4+i) \\ = \frac{10}{17} - \frac{6}{17}i$$

$$(15) \quad (2+i) \div (1-2i) \\ = i$$

$$(25) \quad (-4+2i) \div (4-4i) \\ = -\frac{3}{4} - \frac{1}{4}i$$

$$(6) \quad (2+4i) \div (-1+2i) \\ = \frac{6}{5} - \frac{8}{5}i$$

$$(16) \quad (-1+5i) \div (-1+2i) \\ = \frac{11}{5} - \frac{3}{5}i$$

$$(26) \quad (5-2i) \div (2+2i) \\ = \frac{3}{4} - \frac{7}{4}i$$

$$(7) \quad (4+4i) \div (-1-i) \\ = -4$$

$$(17) \quad (1-4i) \div (-1-4i) \\ = \frac{15}{17} + \frac{8}{17}i$$

$$(27) \quad (1+4i) \div (-1+2i) \\ = \frac{7}{5} - \frac{6}{5}i$$

$$(8) \quad (1-4i) \div (3-4i) \\ = \frac{19}{25} - \frac{8}{25}i$$

$$(18) \quad (3+5i) \div (4-5i) \\ = -\frac{13}{41} + \frac{35}{41}i$$

$$(28) \quad (-3+i) \div (-1+4i) \\ = \frac{7}{17} + \frac{11}{17}i$$

$$(9) \quad (-5-5i) \div (5-i) \\ = -\frac{10}{13} - \frac{15}{13}i$$

$$(19) \quad (4-3i) \div (-2-4i) \\ = \frac{1}{5} + \frac{11}{10}i$$

$$(29) \quad (-4-3i) \div (-3-i) \\ = \frac{3}{2} + \frac{1}{2}i$$

$$(10) \quad (-5-i) \div (3+5i) \\ = -\frac{10}{17} + \frac{11}{17}i$$

$$(20) \quad (-5+5i) \div (-4+5i) \\ = \frac{45}{41} + \frac{5}{41}i$$

$$(30) \quad (3+2i) \div (3-i) \\ = \frac{7}{10} + \frac{9}{10}i$$

$$(1) (1-i) \div (1-5i) \\ = \frac{3}{13} + \frac{2}{13}i$$

$$(11) (-2-i) \div (5-2i) \\ = -\frac{8}{29} - \frac{9}{29}i$$

$$(21) (3-5i) \div (-4-i) \\ = -\frac{7}{17} + \frac{23}{17}i$$

$$(2) (-4+3i) \div (-4+4i) \\ = \frac{7}{8} + \frac{1}{8}i$$

$$(12) (1+5i) \div (3+2i) \\ = 1+i$$

$$(22) (-3+3i) \div (-3-4i) \\ = -\frac{3}{25} - \frac{21}{25}i$$

$$(3) (-4-4i) \div (2+3i) \\ = -\frac{20}{13} + \frac{4}{13}i$$

$$(13) (2-5i) \div (5+2i) \\ = -i$$

$$(23) (-4-5i) \div (-1-4i) \\ = \frac{24}{17} - \frac{11}{17}i$$

$$(4) (2-2i) \div (1+5i) \\ = -\frac{4}{13} - \frac{6}{13}i$$

$$(14) (-4+3i) \div (-2+4i) \\ = 1 + \frac{1}{2}i$$

$$(24) (-4+3i) \div (-2-5i) \\ = -\frac{7}{29} - \frac{26}{29}i$$

$$(5) (-4-i) \div (-5-2i) \\ = \frac{22}{29} - \frac{3}{29}i$$

$$(15) (2-3i) \div (1+2i) \\ = -\frac{4}{5} - \frac{7}{5}i$$

$$(25) (3-5i) \div (3-5i) \\ = 1$$

$$(6) (-4+i) \div (1-i) \\ = -\frac{5}{2} - \frac{3}{2}i$$

$$(16) (-1-4i) \div (-1-i) \\ = \frac{5}{2} + \frac{3}{2}i$$

$$(26) (-3-2i) \div (-4+5i) \\ = \frac{2}{41} + \frac{23}{41}i$$

$$(7) (-2-3i) \div (1-3i) \\ = \frac{7}{10} - \frac{9}{10}i$$

$$(17) (2+4i) \div (1+2i) \\ = 2$$

$$(27) (-3+i) \div (3-3i) \\ = -\frac{2}{3} - \frac{1}{3}i$$

$$(8) (1-i) \div (-1-3i) \\ = \frac{1}{5} + \frac{2}{5}i$$

$$(18) (-4-i) \div (-3-4i) \\ = \frac{16}{25} - \frac{13}{25}i$$

$$(28) (-1-3i) \div (-2-5i) \\ = \frac{17}{29} + \frac{1}{29}i$$

$$(9) (-5-4i) \div (2-2i) \\ = -\frac{1}{4} - \frac{9}{4}i$$

$$(19) (5+5i) \div (-4-3i) \\ = -\frac{7}{5} - \frac{1}{5}i$$

$$(29) (-2-5i) \div (-2-4i) \\ = \frac{6}{5} + \frac{1}{10}i$$

$$(10) (3-3i) \div (5-5i) \\ = \frac{3}{5}$$

$$(20) (2-5i) \div (-1+5i) \\ = -\frac{27}{26} - \frac{5}{26}i$$

$$(30) (-3-5i) \div (-3+4i) \\ = -\frac{11}{25} + \frac{27}{25}i$$

$$(1) (2+i) \div (5+2i) \\ = \frac{12}{29} + \frac{1}{29}i$$

$$(11) (-1+2i) \div (5+3i) \\ = \frac{1}{34} + \frac{13}{34}i$$

$$(21) (1+2i) \div (4+2i) \\ = \frac{2}{5} + \frac{3}{10}i$$

$$(2) (-5+3i) \div (-5+4i) \\ = \frac{37}{41} + \frac{5}{41}i$$

$$(12) (1-2i) \div (-4+5i) \\ = -\frac{14}{41} + \frac{3}{41}i$$

$$(22) (-1-5i) \div (1+4i) \\ = -\frac{21}{17} - \frac{1}{17}i$$

$$(3) (-2-2i) \div (-5-2i) \\ = \frac{14}{29} + \frac{6}{29}i$$

$$(13) (-2+2i) \div (3-3i) \\ = -\frac{2}{3}$$

$$(23) (-5+3i) \div (-1+2i) \\ = \frac{11}{5} + \frac{7}{5}i$$

$$(4) (3-i) \div (4-i) \\ = \frac{13}{17} - \frac{1}{17}i$$

$$(14) (-1+2i) \div (-1+2i) \\ = 1$$

$$(24) (-3+i) \div (2+2i) \\ = -\frac{1}{2} + i$$

$$(5) (3-i) \div (-2+3i) \\ = -\frac{9}{13} - \frac{7}{13}i$$

$$(15) (-3-5i) \div (2-2i) \\ = \frac{1}{2} - 2i$$

$$(25) (2+4i) \div (-3-5i) \\ = -\frac{13}{17} - \frac{1}{17}i$$

$$(6) (5-i) \div (-5-5i) \\ = -\frac{2}{5} + \frac{3}{5}i$$

$$(16) (2+i) \div (-5-2i) \\ = -\frac{12}{29} - \frac{1}{29}i$$

$$(26) (-4+5i) \div (4+4i) \\ = \frac{1}{8} + \frac{9}{8}i$$

$$(7) (2-4i) \div (-2-4i) \\ = \frac{3}{5} + \frac{4}{5}i$$

$$(17) (1-3i) \div (-4+5i) \\ = -\frac{19}{41} + \frac{7}{41}i$$

$$(27) (2-2i) \div (2-4i) \\ = \frac{3}{5} + \frac{1}{5}i$$

$$(8) (-1-5i) \div (-4+2i) \\ = -\frac{3}{10} + \frac{11}{10}i$$

$$(18) (5+5i) \div (-5-5i) \\ = -1$$

$$(28) (1-5i) \div (-4-5i) \\ = \frac{21}{41} + \frac{25}{41}i$$

$$(9) (4+4i) \div (2+5i) \\ = \frac{28}{29} - \frac{12}{29}i$$

$$(19) (-2-4i) \div (5-5i) \\ = \frac{1}{5} - \frac{3}{5}i$$

$$(29) (3+2i) \div (-4-4i) \\ = -\frac{5}{8} + \frac{1}{8}i$$

$$(10) (-3-i) \div (1-i) \\ = -1 - 2i$$

$$(20) (-5+5i) \div (-2+2i) \\ = \frac{5}{2}$$

$$(30) (1-i) \div (5+3i) \\ = \frac{1}{17} - \frac{4}{17}i$$

$$(1) \quad (-1 + 2i) \div (4 + 5i) \\ = \frac{6}{41} + \frac{13}{41}i$$

$$(11) \quad (1 - 5i) \div (-5 - 4i) \\ = \frac{15}{41} + \frac{29}{41}i$$

$$(21) \quad (1 - i) \div (-5 + 4i) \\ = -\frac{9}{41} + \frac{1}{41}i$$

$$(2) \quad (4 - 3i) \div (1 - 5i) \\ = \frac{19}{26} + \frac{17}{26}i$$

$$(12) \quad (-2 - 3i) \div (-2 + i) \\ = \frac{1}{5} + \frac{8}{5}i$$

$$(22) \quad (3 + 3i) \div (5 + 2i) \\ = \frac{21}{29} + \frac{9}{29}i$$

$$(3) \quad (-4 - 2i) \div (2 - 2i) \\ = -\frac{1}{2} - \frac{3}{2}i$$

$$(13) \quad (3 + 5i) \div (4 - 5i) \\ = -\frac{13}{41} + \frac{35}{41}i$$

$$(23) \quad (5 - 3i) \div (5 + 4i) \\ = \frac{13}{41} - \frac{35}{41}i$$

$$(4) \quad (5 - 2i) \div (3 + 4i) \\ = \frac{7}{25} - \frac{26}{25}i$$

$$(14) \quad (1 + 2i) \div (2 + 5i) \\ = \frac{12}{29} - \frac{1}{29}i$$

$$(24) \quad (-2 - 2i) \div (3 - 3i) \\ = -\frac{2}{3}i$$

$$(5) \quad (2 - 3i) \div (1 - 3i) \\ = \frac{11}{10} + \frac{3}{10}i$$

$$(15) \quad (5 + 4i) \div (4 - 3i) \\ = \frac{8}{25} + \frac{31}{25}i$$

$$(25) \quad (5 - i) \div (2 + 3i) \\ = \frac{7}{13} - \frac{17}{13}i$$

$$(6) \quad (4 + 4i) \div (5 - 2i) \\ = \frac{12}{29} + \frac{28}{29}i$$

$$(16) \quad (2 - 2i) \div (-3 - 2i) \\ = -\frac{2}{13} + \frac{10}{13}i$$

$$(26) \quad (1 + i) \div (-5 - i) \\ = -\frac{3}{13} - \frac{2}{13}i$$

$$(7) \quad (2 - 4i) \div (5 - i) \\ = \frac{7}{13} - \frac{9}{13}i$$

$$(17) \quad (-3 + 5i) \div (4 - 3i) \\ = -\frac{27}{25} + \frac{11}{25}i$$

$$(27) \quad (-2 + 5i) \div (5 - 4i) \\ = -\frac{30}{41} + \frac{17}{41}i$$

$$(8) \quad (2 + 4i) \div (5 + 3i) \\ = \frac{11}{17} + \frac{7}{17}i$$

$$(18) \quad (3 - 3i) \div (3 + 2i) \\ = \frac{3}{13} - \frac{15}{13}i$$

$$(28) \quad (-3 - i) \div (-3 + i) \\ = \frac{4}{5} + \frac{3}{5}i$$

$$(9) \quad (-3 + 3i) \div (4 + i) \\ = -\frac{9}{17} + \frac{15}{17}i$$

$$(19) \quad (5 - 2i) \div (2 - 5i) \\ = \frac{20}{29} + \frac{21}{29}i$$

$$(29) \quad (4 - i) \div (-3 + i) \\ = -\frac{13}{10} - \frac{1}{10}i$$

$$(10) \quad (2 + 5i) \div (-2 - i) \\ = -\frac{9}{5} - \frac{8}{5}i$$

$$(20) \quad (5 - 3i) \div (4 - 3i) \\ = \frac{29}{25} + \frac{3}{25}i$$

$$(30) \quad (5 - 4i) \div (-3 + 4i) \\ = -\frac{31}{25} - \frac{8}{25}i$$

$$(1) (3 + 4i) \div (2 - 5i) \\ = -\frac{14}{29} + \frac{23}{29}i$$

$$(11) (4 + 4i) \div (-4 - 2i) \\ = -\frac{6}{5} - \frac{2}{5}i$$

$$(21) (-5 + 4i) \div (2 - 5i) \\ = -\frac{30}{29} - \frac{17}{29}i$$

$$(2) (2 - 2i) \div (-4 + 5i) \\ = -\frac{18}{41} - \frac{2}{41}i$$

$$(12) (-5 + 3i) \div (1 - i) \\ = -4 - i$$

$$(22) (4 + 4i) \div (-2 - 2i) \\ = -2$$

$$(3) (-4 + 5i) \div (-1 - i) \\ = -\frac{1}{2} - \frac{9}{2}i$$

$$(13) (3 + 3i) \div (1 - i) \\ = 3i$$

$$(23) (4 - 2i) \div (2 - 2i) \\ = \frac{3}{2} + \frac{1}{2}i$$

$$(4) (3 - 2i) \div (-2 + 4i) \\ = -\frac{7}{10} - \frac{2}{5}i$$

$$(14) (3 + 4i) \div (4 - 3i) \\ = i$$

$$(24) (5 + 2i) \div (-5 + 3i) \\ = -\frac{19}{34} - \frac{25}{34}i$$

$$(5) (-4 + 4i) \div (1 - 2i) \\ = -\frac{12}{5} - \frac{4}{5}i$$

$$(15) (-1 + i) \div (-2 - i) \\ = \frac{1}{5} - \frac{3}{5}i$$

$$(25) (3 + i) \div (-2 + 3i) \\ = -\frac{3}{13} - \frac{11}{13}i$$

$$(6) (-2 + 2i) \div (5 + i) \\ = -\frac{4}{13} + \frac{6}{13}i$$

$$(16) (1 + i) \div (-2 + 3i) \\ = \frac{1}{13} - \frac{5}{13}i$$

$$(26) (3 + 5i) \div (1 + 3i) \\ = \frac{9}{5} - \frac{2}{5}i$$

$$(7) (-3 - 5i) \div (5 + 5i) \\ = -\frac{4}{5} - \frac{1}{5}i$$

$$(17) (-4 - 4i) \div (-4 - 4i) \\ = 1$$

$$(27) (-4 - 5i) \div (3 - i) \\ = -\frac{7}{10} - \frac{19}{10}i$$

$$(8) (-5 - i) \div (4 + 5i) \\ = -\frac{25}{41} + \frac{21}{41}i$$

$$(18) (-3 - 2i) \div (-2 - 4i) \\ = \frac{7}{10} - \frac{2}{5}i$$

$$(28) (-5 - 3i) \div (-5 + 2i) \\ = \frac{19}{29} + \frac{25}{29}i$$

$$(9) (4 + 5i) \div (-5 + 4i) \\ = -i$$

$$(19) (-1 + 4i) \div (-3 + 3i) \\ = \frac{5}{6} - \frac{1}{2}i$$

$$(29) (-4 - 3i) \div (-1 - i) \\ = \frac{7}{2} - \frac{1}{2}i$$

$$(10) (4 - 4i) \div (-1 - i) \\ = 4i$$

$$(20) (-3 - 4i) \div (-5 + 3i) \\ = \frac{3}{34} + \frac{29}{34}i$$

$$(30) (2 - 4i) \div (5 + 5i) \\ = -\frac{1}{5} - \frac{3}{5}i$$

$$(1) (4 - 5i) \div (5 + 5i) \\ = -\frac{1}{10} - \frac{9}{10}i$$

$$(11) (2 - 2i) \div (-5 - 3i) \\ = -\frac{2}{17} + \frac{8}{17}i$$

$$(21) (-4 - 4i) \div (-5 + 3i) \\ = \frac{4}{17} + \frac{16}{17}i$$

$$(2) (2 - i) \div (5 - i) \\ = \frac{11}{26} - \frac{3}{26}i$$

$$(12) (-3 + 4i) \div (-1 + i) \\ = \frac{7}{2} - \frac{1}{2}i$$

$$(22) (-4 - i) \div (-3 + 5i) \\ = \frac{7}{34} + \frac{23}{34}i$$

$$(3) (5 - 2i) \div (1 - i) \\ = \frac{7}{2} + \frac{3}{2}i$$

$$(13) (3 - 4i) \div (-2 + i) \\ = -2 + i$$

$$(23) (-5 - 4i) \div (-1 + i) \\ = \frac{1}{2} + \frac{9}{2}i$$

$$(4) (2 - 5i) \div (3 - 3i) \\ = \frac{7}{6} - \frac{1}{2}i$$

$$(14) (-3 + 2i) \div (-2 + 4i) \\ = \frac{7}{10} + \frac{2}{5}i$$

$$(24) (-4 + 5i) \div (-2 + 5i) \\ = \frac{33}{29} + \frac{10}{29}i$$

$$(5) (1 + 2i) \div (1 - i) \\ = -\frac{1}{2} + \frac{3}{2}i$$

$$(15) (-4 - 4i) \div (3 + 4i) \\ = -\frac{28}{25} + \frac{4}{25}i$$

$$(25) (-2 + 3i) \div (5 - i) \\ = -\frac{1}{2} + \frac{1}{2}i$$

$$(6) (3 - 3i) \div (2 + i) \\ = \frac{3}{5} - \frac{9}{5}i$$

$$(16) (3 - 3i) \div (-5 - 2i) \\ = -\frac{9}{29} + \frac{21}{29}i$$

$$(26) (3 + 3i) \div (3 - 5i) \\ = -\frac{3}{17} + \frac{12}{17}i$$

$$(7) (5 + 2i) \div (1 + 4i) \\ = \frac{13}{17} - \frac{18}{17}i$$

$$(17) (-4 + 2i) \div (-4 + 4i) \\ = \frac{3}{4} + \frac{1}{4}i$$

$$(27) (-3 - 3i) \div (1 - 4i) \\ = \frac{9}{17} - \frac{15}{17}i$$

$$(8) (2 - 4i) \div (2 + 3i) \\ = -\frac{8}{13} - \frac{14}{13}i$$

$$(18) (-3 + i) \div (-2 - i) \\ = 1 - i$$

$$(28) (-5 + 3i) \div (4 - 4i) \\ = -1 - \frac{1}{4}i$$

$$(9) (-1 + 2i) \div (2 + 5i) \\ = \frac{8}{29} + \frac{9}{29}i$$

$$(19) (3 + 3i) \div (5 - 2i) \\ = \frac{9}{29} + \frac{21}{29}i$$

$$(29) (4 - 5i) \div (-2 - 4i) \\ = \frac{3}{5} + \frac{13}{10}i$$

$$(10) (5 + 3i) \div (-4 - i) \\ = -\frac{23}{17} - \frac{7}{17}i$$

$$(20) (1 - i) \div (-4 + 3i) \\ = -\frac{7}{25} + \frac{1}{25}i$$

$$(30) (-2 + 4i) \div (5 - 5i) \\ = -\frac{3}{5} + \frac{1}{5}i$$

$$(1) (2 + 4i) \div (-5 + 2i) \\ = -\frac{2}{29} - \frac{24}{29}i$$

$$(11) (-3 - 5i) \div (-5 - 4i) \\ = \frac{35}{41} + \frac{13}{41}i$$

$$(21) (5 - 4i) \div (1 - 3i) \\ = \frac{17}{10} + \frac{11}{10}i$$

$$(2) (-1 + 3i) \div (2 - 4i) \\ = -\frac{7}{10} + \frac{1}{10}i$$

$$(12) (4 + 2i) \div (5 + 3i) \\ = \frac{13}{17} - \frac{1}{17}i$$

$$(22) (-5 + 2i) \div (-3 + 3i) \\ = \frac{7}{6} + \frac{1}{2}i$$

$$(3) (-1 + i) \div (1 - 2i) \\ = -\frac{3}{5} - \frac{1}{5}i$$

$$(13) (3 - 2i) \div (-5 - 2i) \\ = -\frac{11}{29} + \frac{16}{29}i$$

$$(23) (-5 - 5i) \div (5 - 5i) \\ = -i$$

$$(4) (2 + 3i) \div (-1 + 2i) \\ = \frac{4}{5} - \frac{7}{5}i$$

$$(14) (4 - i) \div (-4 - 4i) \\ = -\frac{3}{8} + \frac{5}{8}i$$

$$(24) (1 + 3i) \div (2 + 4i) \\ = \frac{7}{10} + \frac{1}{10}i$$

$$(5) (2 + i) \div (-1 - i) \\ = -\frac{3}{2} + \frac{1}{2}i$$

$$(15) (-3 - 4i) \div (5 - 2i) \\ = -\frac{7}{29} - \frac{26}{29}i$$

$$(25) (-1 - 5i) \div (3 + 4i) \\ = -\frac{23}{25} - \frac{11}{25}i$$

$$(6) (-5 - i) \div (-4 + 4i) \\ = \frac{1}{2} + \frac{3}{4}i$$

$$(16) (-1 + 4i) \div (1 + 2i) \\ = \frac{7}{5} + \frac{6}{5}i$$

$$(26) (-5 - i) \div (2 + 2i) \\ = -\frac{3}{2} + i$$

$$(7) (-4 - i) \div (-1 - 5i) \\ = \frac{9}{26} - \frac{19}{26}i$$

$$(17) (-1 + 4i) \div (4 - 3i) \\ = -\frac{16}{25} + \frac{13}{25}i$$

$$(27) (1 + 4i) \div (-4 - 5i) \\ = -\frac{24}{41} - \frac{11}{41}i$$

$$(8) (3 - 4i) \div (-1 - 4i) \\ = \frac{13}{17} + \frac{16}{17}i$$

$$(18) (-4 - 4i) \div (-5 + i) \\ = \frac{8}{13} + \frac{12}{13}i$$

$$(28) (-2 - 4i) \div (-4 - i) \\ = \frac{12}{17} + \frac{14}{17}i$$

$$(9) (5 - 4i) \div (2 - 4i) \\ = \frac{13}{10} + \frac{3}{5}i$$

$$(19) (-3 - 4i) \div (3 + 3i) \\ = -\frac{7}{6} - \frac{1}{6}i$$

$$(29) (3 - i) \div (5 - 3i) \\ = \frac{9}{17} + \frac{2}{17}i$$

$$(10) (3 + 4i) \div (-4 - 2i) \\ = -1 - \frac{1}{2}i$$

$$(20) (3 - 3i) \div (5 + i) \\ = \frac{6}{13} - \frac{9}{13}i$$

$$(30) (-5 + 4i) \div (4 - 2i) \\ = -\frac{7}{5} + \frac{3}{10}i$$

$$(1) (5+2i) \div (2-3i) \\ = \frac{4}{13} + \frac{19}{13}i$$

$$(11) (-1-i) \div (3-i) \\ = -\frac{1}{5} - \frac{2}{5}i$$

$$(21) (5+2i) \div (5+2i) \\ = 1$$

$$(2) (-1-5i) \div (-2-i) \\ = \frac{7}{5} + \frac{9}{5}i$$

$$(12) (2-3i) \div (2-5i) \\ = \frac{19}{29} + \frac{4}{29}i$$

$$(22) (-2-i) \div (4-5i) \\ = -\frac{3}{41} - \frac{14}{41}i$$

$$(3) (-5+3i) \div (4-4i) \\ = -1 - \frac{1}{4}i$$

$$(13) (-2+5i) \div (1-3i) \\ = -\frac{17}{10} - \frac{1}{10}i$$

$$(23) (-1-3i) \div (5+2i) \\ = -\frac{11}{29} - \frac{13}{29}i$$

$$(4) (-5+2i) \div (-3+2i) \\ = \frac{19}{13} + \frac{4}{13}i$$

$$(14) (2-3i) \div (5+3i) \\ = \frac{1}{34} - \frac{21}{34}i$$

$$(24) (3-2i) \div (5-2i) \\ = \frac{19}{29} - \frac{4}{29}i$$

$$(5) (-3+i) \div (-5+4i) \\ = \frac{19}{41} + \frac{7}{41}i$$

$$(15) (5-i) \div (1-3i) \\ = \frac{4}{5} + \frac{7}{5}i$$

$$(25) (-3+5i) \div (1+4i) \\ = 1 + i$$

$$(6) (3-4i) \div (-3-2i) \\ = -\frac{1}{13} + \frac{18}{13}i$$

$$(16) (-5+3i) \div (-2-5i) \\ = -\frac{5}{29} - \frac{31}{29}i$$

$$(26) (5+5i) \div (-1-3i) \\ = -2 + i$$

$$(7) (-5-i) \div (-3-5i) \\ = \frac{10}{17} - \frac{11}{17}i$$

$$(17) (5-2i) \div (-2+2i) \\ = -\frac{7}{4} - \frac{3}{4}i$$

$$(27) (-2+3i) \div (1+2i) \\ = \frac{4}{5} + \frac{7}{5}i$$

$$(8) (-1+i) \div (-2-5i) \\ = -\frac{3}{29} - \frac{7}{29}i$$

$$(18) (-4-5i) \div (4-i) \\ = -\frac{11}{17} - \frac{24}{17}i$$

$$(28) (3+4i) \div (-4+4i) \\ = \frac{1}{8} - \frac{7}{8}i$$

$$(9) (-4+i) \div (-4+2i) \\ = \frac{9}{10} + \frac{1}{5}i$$

$$(19) (3+5i) \div (-2+5i) \\ = \frac{19}{29} - \frac{25}{29}i$$

$$(29) (5+2i) \div (-5-5i) \\ = -\frac{7}{10} + \frac{3}{10}i$$

$$(10) (1-5i) \div (1+4i) \\ = -\frac{19}{17} - \frac{9}{17}i$$

$$(20) (-4+5i) \div (-1-4i) \\ = -\frac{16}{17} - \frac{21}{17}i$$

$$(30) (4+i) \div (-4+i) \\ = -\frac{15}{17} - \frac{8}{17}i$$

$$(1) (3 + 5i) \div (-5 - 4i) \\ = -\frac{35}{41} - \frac{13}{41}i$$

$$(11) (2 + 5i) \div (-5 + i) \\ = -\frac{5}{26} - \frac{27}{26}i$$

$$(21) (-5 + i) \div (-2 + 3i) \\ = 1 + i$$

$$(2) (4 - 2i) \div (1 + 4i) \\ = -\frac{4}{17} - \frac{18}{17}i$$

$$(12) (1 + 3i) \div (1 - i) \\ = -1 + 2i$$

$$(22) (-2 + 4i) \div (-2 + 4i) \\ = 1$$

$$(3) (-5 + 4i) \div (-5 + 5i) \\ = \frac{9}{10} + \frac{1}{10}i$$

$$(13) (5 + 4i) \div (-5 + 5i) \\ = -\frac{1}{10} - \frac{9}{10}i$$

$$(23) (2 + 4i) \div (4 + 3i) \\ = \frac{4}{5} + \frac{2}{5}i$$

$$(4) (1 + 4i) \div (-5 - 4i) \\ = -\frac{21}{41} - \frac{16}{41}i$$

$$(14) (-4 + 2i) \div (5 - 5i) \\ = -\frac{3}{5} - \frac{1}{5}i$$

$$(24) (3 - 3i) \div (-1 - i) \\ = 3i$$

$$(5) (2 + 4i) \div (5 - 2i) \\ = \frac{2}{29} + \frac{24}{29}i$$

$$(15) (1 - 2i) \div (4 - 4i) \\ = \frac{3}{8} - \frac{1}{8}i$$

$$(25) (4 - 5i) \div (4 - 4i) \\ = \frac{9}{8} - \frac{1}{8}i$$

$$(6) (-2 + 4i) \div (-3 + 5i) \\ = \frac{13}{17} - \frac{1}{17}i$$

$$(16) (4 - 4i) \div (2 - i) \\ = \frac{12}{5} - \frac{4}{5}i$$

$$(26) (5 + 5i) \div (-4 - i) \\ = -\frac{25}{17} - \frac{15}{17}i$$

$$(7) (-3 + 3i) \div (-2 - 4i) \\ = -\frac{3}{10} - \frac{9}{10}i$$

$$(17) (-3 - 3i) \div (3 - 2i) \\ = -\frac{3}{13} - \frac{15}{13}i$$

$$(27) (2 + 5i) \div (-2 + 4i) \\ = \frac{4}{5} - \frac{9}{10}i$$

$$(8) (1 + 4i) \div (3 + i) \\ = \frac{7}{10} + \frac{11}{10}i$$

$$(18) (-5 - i) \div (-5 + i) \\ = \frac{12}{13} + \frac{5}{13}i$$

$$(28) (4 + i) \div (-5 + 3i) \\ = -\frac{1}{2} - \frac{1}{2}i$$

$$(9) (-1 - 3i) \div (4 - i) \\ = -\frac{1}{17} - \frac{13}{17}i$$

$$(19) (1 + 5i) \div (-3 + 2i) \\ = \frac{7}{13} - \frac{17}{13}i$$

$$(29) (4 - i) \div (-2 + 2i) \\ = -\frac{5}{4} - \frac{3}{4}i$$

$$(10) (-1 + 5i) \div (1 + i) \\ = 2 + 3i$$

$$(20) (-1 - 3i) \div (-5 + i) \\ = \frac{1}{13} + \frac{8}{13}i$$

$$(30) (2 - 5i) \div (-3 + 3i) \\ = -\frac{7}{6} + \frac{1}{2}i$$

$$(1) (-4 + 3i) \div (5 - i) \\ = -\frac{23}{26} + \frac{11}{26}i$$

$$(11) (1 - i) \div (-2 + 4i) \\ = -\frac{3}{10} - \frac{1}{10}i$$

$$(21) (5 - 4i) \div (2 - 5i) \\ = \frac{30}{29} + \frac{17}{29}i$$

$$(2) (-4 - i) \div (3 + 4i) \\ = -\frac{16}{25} + \frac{13}{25}i$$

$$(12) (2 - 5i) \div (-4 + i) \\ = -\frac{13}{17} + \frac{18}{17}i$$

$$(22) (-5 + 2i) \div (1 + 4i) \\ = \frac{3}{17} + \frac{22}{17}i$$

$$(3) (1 + 5i) \div (-4 + i) \\ = \frac{1}{17} - \frac{21}{17}i$$

$$(13) (-5 + i) \div (5 + 5i) \\ = -\frac{2}{5} + \frac{3}{5}i$$

$$(23) (1 - 4i) \div (1 - 3i) \\ = \frac{13}{10} - \frac{1}{10}i$$

$$(4) (3 - 3i) \div (-1 + 2i) \\ = -\frac{9}{5} - \frac{3}{5}i$$

$$(14) (3 + 4i) \div (-1 + i) \\ = \frac{1}{2} - \frac{7}{2}i$$

$$(24) (-5 - 3i) \div (2 + 4i) \\ = -\frac{11}{10} + \frac{7}{10}i$$

$$(5) (2 + 4i) \div (-5 + 4i) \\ = \frac{6}{41} - \frac{28}{41}i$$

$$(15) (1 + 3i) \div (4 - 4i) \\ = -\frac{1}{4} + \frac{1}{2}i$$

$$(25) (-5 - 2i) \div (3 + 4i) \\ = -\frac{23}{25} + \frac{14}{25}i$$

$$(6) (1 + 5i) \div (5 - 3i) \\ = -\frac{5}{17} + \frac{14}{17}i$$

$$(16) (2 - i) \div (-4 - i) \\ = -\frac{7}{17} + \frac{6}{17}i$$

$$(26) (3 - 5i) \div (2 - 5i) \\ = \frac{31}{29} + \frac{5}{29}i$$

$$(7) (1 - 2i) \div (-3 - 2i) \\ = \frac{1}{13} + \frac{8}{13}i$$

$$(17) (-4 - 4i) \div (-5 + 4i) \\ = \frac{4}{41} + \frac{36}{41}i$$

$$(27) (5 + 3i) \div (-3 - 5i) \\ = -\frac{15}{17} + \frac{8}{17}i$$

$$(8) (1 + i) \div (-3 - 2i) \\ = -\frac{5}{13} - \frac{1}{13}i$$

$$(18) (4 + 3i) \div (-4 - 5i) \\ = -\frac{31}{41} + \frac{8}{41}i$$

$$(28) (3 - i) \div (-3 + 3i) \\ = -\frac{2}{3} - \frac{1}{3}i$$

$$(9) (3 - 5i) \div (-3 - 2i) \\ = \frac{1}{13} + \frac{21}{13}i$$

$$(19) (1 + 5i) \div (5 - i) \\ = i$$

$$(29) (1 + 5i) \div (-3 - 5i) \\ = -\frac{14}{17} - \frac{5}{17}i$$

$$(10) (-3 + 5i) \div (-3 + 4i) \\ = \frac{29}{25} - \frac{3}{25}i$$

$$(20) (4 - 3i) \div (1 + 5i) \\ = -\frac{11}{26} - \frac{23}{26}i$$

$$(30) (2 - 4i) \div (-3 - i) \\ = -\frac{1}{5} + \frac{7}{5}i$$

$$(1) (4 + 3i) \div (3 - i) \\ = \frac{9}{10} + \frac{13}{10}i$$

$$(11) (-5 - 4i) \div (-5 - 5i) \\ = \frac{9}{10} - \frac{1}{10}i$$

$$(21) (1 - 2i) \div (-5 - i) \\ = -\frac{3}{26} + \frac{11}{26}i$$

$$(2) (2 - 4i) \div (-3 - 4i) \\ = \frac{2}{5} + \frac{4}{5}i$$

$$(12) (-5 - 4i) \div (2 + 5i) \\ = -\frac{30}{29} + \frac{17}{29}i$$

$$(22) (-5 + i) \div (3 + 4i) \\ = -\frac{11}{25} + \frac{23}{25}i$$

$$(3) (2 + 3i) \div (-3 - i) \\ = -\frac{9}{10} - \frac{7}{10}i$$

$$(13) (-1 - 5i) \div (-1 + 3i) \\ = -\frac{7}{5} + \frac{4}{5}i$$

$$(23) (-4 + 4i) \div (-4 - 5i) \\ = -\frac{4}{41} - \frac{36}{41}i$$

$$(4) (2 - 4i) \div (-1 - 2i) \\ = \frac{6}{5} + \frac{8}{5}i$$

$$(14) (1 + 5i) \div (-3 - 2i) \\ = -1 - i$$

$$(24) (-5 - i) \div (-3 - 3i) \\ = 1 - \frac{2}{3}i$$

$$(5) (4 + i) \div (1 + 2i) \\ = \frac{6}{5} - \frac{7}{5}i$$

$$(15) (-2 - 2i) \div (4 + i) \\ = -\frac{10}{17} - \frac{6}{17}i$$

$$(25) (3 - i) \div (1 + i) \\ = 1 - 2i$$

$$(6) (-4 - 5i) \div (3 + 2i) \\ = -\frac{22}{13} - \frac{7}{13}i$$

$$(16) (-3 + 4i) \div (-5 + 5i) \\ = \frac{7}{10} - \frac{1}{10}i$$

$$(26) (-1 - 2i) \div (-3 + 3i) \\ = -\frac{1}{6} + \frac{1}{2}i$$

$$(7) (-1 + i) \div (-1 + 3i) \\ = \frac{2}{5} + \frac{1}{5}i$$

$$(17) (-4 + 3i) \div (-1 + 5i) \\ = \frac{19}{26} + \frac{17}{26}i$$

$$(27) (-5 - 3i) \div (-5 - 3i) \\ = 1$$

$$(8) (4 + 3i) \div (-4 - 3i) \\ = -1$$

$$(18) (-5 - 4i) \div (2 + 3i) \\ = -\frac{22}{13} + \frac{7}{13}i$$

$$(28) (4 - 5i) \div (-3 - 4i) \\ = \frac{8}{25} + \frac{31}{25}i$$

$$(9) (1 + 3i) \div (2 + 2i) \\ = 1 + \frac{1}{2}i$$

$$(19) (3 + 4i) \div (5 - 5i) \\ = -\frac{1}{10} + \frac{7}{10}i$$

$$(29) (-2 - 5i) \div (1 - 2i) \\ = \frac{8}{5} - \frac{9}{5}i$$

$$(10) (-5 + 2i) \div (-4 + 3i) \\ = \frac{26}{25} + \frac{7}{25}i$$

$$(20) (-4 - 4i) \div (-4 - 3i) \\ = \frac{28}{25} + \frac{4}{25}i$$

$$(30) (4 + 5i) \div (-4 + i) \\ = -\frac{11}{17} - \frac{24}{17}i$$

$$(1) \quad (-4 - 3i) \div (-5 + 2i) \\ = \frac{14}{29} + \frac{23}{29}i$$

$$(11) \quad (4 + 5i) \div (5 - 4i) \\ = i$$

$$(21) \quad (-4 - i) \div (-4 + 2i) \\ = \frac{7}{10} + \frac{3}{5}i$$

$$(2) \quad (-2 - i) \div (5 - 2i) \\ = -\frac{8}{29} - \frac{9}{29}i$$

$$(12) \quad (-3 - 3i) \div (-5 - 5i) \\ = \frac{3}{5}$$

$$(22) \quad (-4 - 3i) \div (-1 - 5i) \\ = \frac{19}{26} - \frac{17}{26}i$$

$$(3) \quad (-1 + 4i) \div (1 + i) \\ = \frac{3}{2} + \frac{5}{2}i$$

$$(13) \quad (-1 + 2i) \div (-1 - 3i) \\ = -\frac{1}{2} - \frac{1}{2}i$$

$$(23) \quad (-2 - 3i) \div (4 - 2i) \\ = -\frac{1}{10} - \frac{4}{5}i$$

$$(4) \quad (5 - 2i) \div (5 + 4i) \\ = \frac{17}{41} - \frac{30}{41}i$$

$$(14) \quad (-5 + 3i) \div (2 - 3i) \\ = -\frac{19}{13} - \frac{9}{13}i$$

$$(24) \quad (-5 - 4i) \div (-5 + 2i) \\ = \frac{17}{29} + \frac{30}{29}i$$

$$(5) \quad (-2 - 2i) \div (-5 + i) \\ = \frac{4}{13} + \frac{6}{13}i$$

$$(15) \quad (4 + 3i) \div (5 + 3i) \\ = \frac{29}{34} + \frac{3}{34}i$$

$$(25) \quad (2 + 2i) \div (4 - i) \\ = \frac{6}{17} + \frac{10}{17}i$$

$$(6) \quad (-1 - 5i) \div (2 - 4i) \\ = \frac{9}{10} - \frac{7}{10}i$$

$$(16) \quad (2 - i) \div (3 + 5i) \\ = \frac{1}{34} - \frac{13}{34}i$$

$$(26) \quad (3 + 2i) \div (2 + 3i) \\ = \frac{12}{13} - \frac{5}{13}i$$

$$(7) \quad (5 + 2i) \div (-2 + i) \\ = -\frac{8}{5} - \frac{9}{5}i$$

$$(17) \quad (-4 + 4i) \div (5 - 4i) \\ = -\frac{36}{41} + \frac{4}{41}i$$

$$(27) \quad (-5 - 4i) \div (3 + 5i) \\ = -\frac{35}{34} + \frac{13}{34}i$$

$$(8) \quad (4 + i) \div (-5 + 5i) \\ = -\frac{3}{10} - \frac{1}{2}i$$

$$(18) \quad (5 - 4i) \div (4 - 4i) \\ = \frac{9}{8} + \frac{1}{8}i$$

$$(28) \quad (3 + i) \div (-1 + i) \\ = -1 - 2i$$

$$(9) \quad (-5 - 4i) \div (-3 - 5i) \\ = \frac{35}{34} - \frac{13}{34}i$$

$$(19) \quad (1 + 2i) \div (4 + 2i) \\ = \frac{2}{5} + \frac{3}{10}i$$

$$(29) \quad (-5 + 4i) \div (-2 + i) \\ = \frac{14}{5} - \frac{3}{5}i$$

$$(10) \quad (1 + i) \div (4 + i) \\ = \frac{5}{17} + \frac{3}{17}i$$

$$(20) \quad (-5 + 2i) \div (-1 - i) \\ = \frac{3}{2} - \frac{7}{2}i$$

$$(30) \quad (-2 - 5i) \div (-4 + i) \\ = \frac{3}{17} + \frac{22}{17}i$$

$$(1) (-5 + 4i) \div (1 - 5i) \\ = -\frac{25}{26} - \frac{21}{26}i$$

$$(11) (-1 - 5i) \div (1 + i) \\ = -3 - 2i$$

$$(21) (-3 - 2i) \div (5 + 2i) \\ = -\frac{19}{29} - \frac{4}{29}i$$

$$(2) (-3 + 5i) \div (-2 - 2i) \\ = -\frac{1}{2} - 2i$$

$$(12) (4 - i) \div (3 + 3i) \\ = \frac{1}{2} - \frac{5}{6}i$$

$$(22) (5 - 2i) \div (-3 - 4i) \\ = -\frac{7}{25} + \frac{26}{25}i$$

$$(3) (2 + i) \div (4 + 2i) \\ = \frac{1}{2}$$

$$(13) (3 + i) \div (2 + 4i) \\ = \frac{1}{2} - \frac{1}{2}i$$

$$(23) (2 + 2i) \div (4 + 2i) \\ = \frac{3}{5} + \frac{1}{5}i$$

$$(4) (2 + 2i) \div (-5 + 5i) \\ = -\frac{2}{5}i$$

$$(14) (-2 - 2i) \div (-5 - i) \\ = \frac{6}{13} + \frac{4}{13}i$$

$$(24) (-3 - 5i) \div (4 - 3i) \\ = \frac{3}{25} - \frac{29}{25}i$$

$$(5) (3 + 4i) \div (5 - i) \\ = \frac{11}{26} + \frac{23}{26}i$$

$$(15) (-1 - i) \div (2 + 3i) \\ = -\frac{5}{13} + \frac{1}{13}i$$

$$(25) (-5 - i) \div (2 + i) \\ = -\frac{11}{5} + \frac{3}{5}i$$

$$(6) (-1 - 4i) \div (2 + 3i) \\ = -\frac{14}{13} - \frac{5}{13}i$$

$$(16) (-2 + 4i) \div (-4 + 3i) \\ = \frac{4}{5} - \frac{2}{5}i$$

$$(26) (-4 - 3i) \div (5 - 3i) \\ = -\frac{11}{34} - \frac{27}{34}i$$

$$(7) (3 - i) \div (3 + 4i) \\ = \frac{1}{5} - \frac{3}{5}i$$

$$(17) (1 + 4i) \div (-1 + i) \\ = \frac{3}{2} - \frac{5}{2}i$$

$$(27) (4 - 3i) \div (5 + i) \\ = \frac{17}{26} - \frac{19}{26}i$$

$$(8) (2 + 5i) \div (-3 - 4i) \\ = -\frac{26}{25} - \frac{7}{25}i$$

$$(18) (5 - 4i) \div (1 - 2i) \\ = \frac{13}{5} + \frac{6}{5}i$$

$$(28) (1 - 3i) \div (5 - 2i) \\ = \frac{11}{29} - \frac{13}{29}i$$

$$(9) (3 + 2i) \div (-3 - 2i) \\ = -1$$

$$(19) (-4 + i) \div (-1 - i) \\ = \frac{3}{2} - \frac{5}{2}i$$

$$(29) (-3 - 2i) \div (3 - 5i) \\ = \frac{1}{34} - \frac{21}{34}i$$

$$(10) (2 - 4i) \div (-3 + 4i) \\ = -\frac{22}{25} + \frac{4}{25}i$$

$$(20) (-1 - 4i) \div (-2 + 5i) \\ = -\frac{18}{29} + \frac{13}{29}i$$

$$(30) (2 + 5i) \div (4 + 4i) \\ = \frac{7}{8} + \frac{3}{8}i$$

$$(1) (1+3i) \div (-5-5i) \\ = -\frac{2}{5} - \frac{1}{5}i$$

$$(11) (1-3i) \div (1-4i) \\ = \frac{13}{17} + \frac{1}{17}i$$

$$(21) (-5+3i) \div (-1+i) \\ = 4+i$$

$$(2) (1-3i) \div (-5+5i) \\ = -\frac{2}{5} + \frac{1}{5}i$$

$$(12) (-5-3i) \div (1+i) \\ = -4+i$$

$$(22) (-1+i) \div (-3-5i) \\ = -\frac{1}{17} - \frac{4}{17}i$$

$$(3) (3+4i) \div (4-2i) \\ = \frac{1}{5} + \frac{11}{10}i$$

$$(13) (4-i) \div (4-2i) \\ = \frac{9}{10} + \frac{1}{5}i$$

$$(23) (-1-i) \div (-5+2i) \\ = \frac{3}{29} + \frac{7}{29}i$$

$$(4) (5+5i) \div (-2-2i) \\ = -\frac{5}{2}$$

$$(14) (1+i) \div (5-5i) \\ = \frac{1}{5}i$$

$$(24) (-5+4i) \div (-4-3i) \\ = \frac{8}{25} - \frac{31}{25}i$$

$$(5) (-5+5i) \div (1+3i) \\ = 1+2i$$

$$(15) (4-i) \div (-1+5i) \\ = -\frac{9}{26} - \frac{19}{26}i$$

$$(25) (-2-2i) \div (-3+3i) \\ = \frac{2}{3}i$$

$$(6) (-2+5i) \div (-1+i) \\ = \frac{7}{2} - \frac{3}{2}i$$

$$(16) (-3-4i) \div (2+i) \\ = -2-i$$

$$(26) (1-i) \div (-3-4i) \\ = \frac{1}{25} + \frac{7}{25}i$$

$$(7) (2-2i) \div (3+i) \\ = \frac{2}{5} - \frac{4}{5}i$$

$$(17) (-2+2i) \div (-4-4i) \\ = -\frac{1}{2}i$$

$$(27) (3-4i) \div (-5-3i) \\ = -\frac{3}{34} + \frac{29}{34}i$$

$$(8) (3+3i) \div (4+2i) \\ = \frac{9}{10} + \frac{3}{10}i$$

$$(18) (1+5i) \div (1-i) \\ = -2+3i$$

$$(28) (3-5i) \div (-1-3i) \\ = \frac{6}{5} + \frac{7}{5}i$$

$$(9) (2+2i) \div (-3-2i) \\ = -\frac{10}{13} - \frac{2}{13}i$$

$$(19) (-4-i) \div (1-5i) \\ = \frac{1}{26} - \frac{21}{26}i$$

$$(29) (-1+2i) \div (-1+3i) \\ = \frac{7}{10} + \frac{1}{10}i$$

$$(10) (-4+4i) \div (-3-5i) \\ = -\frac{4}{17} - \frac{16}{17}i$$

$$(20) (-2+4i) \div (4+4i) \\ = \frac{1}{4} + \frac{3}{4}i$$

$$(30) (-5-2i) \div (2+3i) \\ = -\frac{16}{13} + \frac{11}{13}i$$

$$(1) (4-i) \div (1-5i) \\ = \frac{9}{26} + \frac{19}{26}i$$

$$(11) (4+4i) \div (-5-2i) \\ = -\frac{28}{29} - \frac{12}{29}i$$

$$(21) (4+5i) \div (3+2i) \\ = \frac{22}{13} + \frac{7}{13}i$$

$$(2) (1-4i) \div (4+5i) \\ = -\frac{16}{41} - \frac{21}{41}i$$

$$(12) (-5+i) \div (3-4i) \\ = -\frac{19}{25} - \frac{17}{25}i$$

$$(22) (2+4i) \div (1-i) \\ = -1 + 3i$$

$$(3) (-4+i) \div (3+2i) \\ = -\frac{10}{13} + \frac{11}{13}i$$

$$(13) (-5-3i) \div (4+2i) \\ = -\frac{13}{10} - \frac{1}{10}i$$

$$(23) (-3+2i) \div (2-2i) \\ = -\frac{5}{4} - \frac{1}{4}i$$

$$(4) (3+4i) \div (-3+i) \\ = -\frac{1}{2} - \frac{3}{2}i$$

$$(14) (-1+5i) \div (-2+2i) \\ = \frac{3}{2} - i$$

$$(24) (1-2i) \div (-4-i) \\ = -\frac{2}{17} + \frac{9}{17}i$$

$$(5) (3-5i) \div (4+2i) \\ = \frac{1}{10} - \frac{13}{10}i$$

$$(15) (-3-2i) \div (-5+3i) \\ = \frac{9}{34} + \frac{19}{34}i$$

$$(25) (-4-i) \div (-1-4i) \\ = \frac{8}{17} - \frac{15}{17}i$$

$$(6) (2-4i) \div (-5+2i) \\ = -\frac{18}{29} + \frac{16}{29}i$$

$$(16) (-4-4i) \div (-1+2i) \\ = -\frac{4}{5} + \frac{12}{5}i$$

$$(26) (4+i) \div (1+3i) \\ = \frac{7}{10} - \frac{11}{10}i$$

$$(7) (3+4i) \div (1-2i) \\ = -1 + 2i$$

$$(17) (5+5i) \div (5-3i) \\ = \frac{5}{17} + \frac{20}{17}i$$

$$(27) (-3+3i) \div (-2+i) \\ = \frac{9}{5} - \frac{3}{5}i$$

$$(8) (-3+2i) \div (3+4i) \\ = -\frac{1}{25} + \frac{18}{25}i$$

$$(18) (5+3i) \div (2-3i) \\ = \frac{1}{13} + \frac{21}{13}i$$

$$(28) (5+3i) \div (-5+5i) \\ = -\frac{1}{5} - \frac{4}{5}i$$

$$(9) (1-4i) \div (4+4i) \\ = -\frac{3}{8} - \frac{5}{8}i$$

$$(19) (2+5i) \div (5-4i) \\ = -\frac{10}{41} + \frac{33}{41}i$$

$$(29) (5-i) \div (2-3i) \\ = 1 + i$$

$$(10) (3-4i) \div (4+3i) \\ = -i$$

$$(20) (5-3i) \div (3+4i) \\ = \frac{3}{25} - \frac{29}{25}i$$

$$(30) (4+5i) \div (3+2i) \\ = \frac{22}{13} + \frac{7}{13}i$$

$$(1) (2+4i) \div (-5-3i) \\ = -\frac{11}{17} - \frac{7}{17}i$$

$$(11) (3-3i) \div (-1+3i) \\ = -\frac{6}{5} - \frac{3}{5}i$$

$$(21) (-5+i) \div (-3+2i) \\ = \frac{17}{13} + \frac{7}{13}i$$

$$(2) (1-i) \div (2+i) \\ = \frac{1}{5} - \frac{3}{5}i$$

$$(12) (-3+2i) \div (-2-5i) \\ = -\frac{4}{29} - \frac{19}{29}i$$

$$(22) (-3+5i) \div (2+4i) \\ = \frac{7}{10} + \frac{11}{10}i$$

$$(3) (-5+2i) \div (-3+2i) \\ = \frac{19}{13} + \frac{4}{13}i$$

$$(13) (3+5i) \div (4-4i) \\ = -\frac{1}{4} + i$$

$$(23) (-3-i) \div (-4+i) \\ = \frac{11}{17} + \frac{7}{17}i$$

$$(4) (-1+3i) \div (4+3i) \\ = \frac{1}{5} + \frac{3}{5}i$$

$$(14) (-3+3i) \div (1-i) \\ = -3$$

$$(24) (-3+i) \div (-5-3i) \\ = \frac{6}{17} - \frac{7}{17}i$$

$$(5) (-3+2i) \div (4+i) \\ = -\frac{10}{17} + \frac{11}{17}i$$

$$(15) (-2-5i) \div (-1-2i) \\ = \frac{12}{5} + \frac{1}{5}i$$

$$(25) (-3-2i) \div (1+4i) \\ = -\frac{11}{17} + \frac{10}{17}i$$

$$(6) (2-i) \div (-2-4i) \\ = \frac{1}{2}i$$

$$(16) (1-3i) \div (-2+2i) \\ = -1 + \frac{1}{2}i$$

$$(26) (4+5i) \div (4+5i) \\ = 1$$

$$(7) (-4-4i) \div (1+5i) \\ = -\frac{12}{13} + \frac{8}{13}i$$

$$(17) (-2-4i) \div (4-5i) \\ = \frac{12}{41} - \frac{26}{41}i$$

$$(27) (-3-3i) \div (-2+3i) \\ = -\frac{3}{13} + \frac{15}{13}i$$

$$(8) (-1-i) \div (3-4i) \\ = \frac{1}{25} - \frac{7}{25}i$$

$$(18) (-4+5i) \div (-1-5i) \\ = -\frac{21}{26} - \frac{25}{26}i$$

$$(28) (-2+3i) \div (-1-3i) \\ = -\frac{7}{10} - \frac{9}{10}i$$

$$(9) (3+2i) \div (-2-2i) \\ = -\frac{5}{4} + \frac{1}{4}i$$

$$(19) (-4+5i) \div (2+i) \\ = -\frac{3}{5} + \frac{14}{5}i$$

$$(29) (5+i) \div (-5+3i) \\ = -\frac{11}{17} - \frac{10}{17}i$$

$$(10) (-5-5i) \div (1+5i) \\ = -\frac{15}{13} + \frac{10}{13}i$$

$$(20) (5+3i) \div (4-3i) \\ = \frac{11}{25} + \frac{27}{25}i$$

$$(30) (-2-5i) \div (-5-4i) \\ = \frac{30}{41} + \frac{17}{41}i$$

$$(1) \quad (-4 + i) \div (-2 + 2i) \\ = \frac{5}{4} + \frac{3}{4}i$$

$$(11) \quad (-1 - i) \div (3 + i) \\ = -\frac{2}{5} - \frac{1}{5}i$$

$$(21) \quad (1 + 2i) \div (-3 - i) \\ = -\frac{1}{2} - \frac{1}{2}i$$

$$(2) \quad (-1 - i) \div (-4 - 5i) \\ = \frac{9}{41} - \frac{1}{41}i$$

$$(12) \quad (-1 + 3i) \div (-3 + 3i) \\ = \frac{2}{3} - \frac{1}{3}i$$

$$(22) \quad (-1 + 2i) \div (-5 - i) \\ = \frac{3}{26} - \frac{11}{26}i$$

$$(3) \quad (-4 + i) \div (-4 + i) \\ = 1$$

$$(13) \quad (-2 + 2i) \div (2 - 4i) \\ = -\frac{3}{5} - \frac{1}{5}i$$

$$(23) \quad (-1 - i) \div (-1 + i) \\ = i$$

$$(4) \quad (3 - i) \div (1 - 5i) \\ = \frac{4}{13} + \frac{7}{13}i$$

$$(14) \quad (2 - 2i) \div (5 - 2i) \\ = \frac{14}{29} - \frac{6}{29}i$$

$$(24) \quad (-1 - 5i) \div (5 + 3i) \\ = -\frac{10}{17} - \frac{11}{17}i$$

$$(5) \quad (-1 - 2i) \div (1 + 4i) \\ = -\frac{9}{17} + \frac{2}{17}i$$

$$(15) \quad (4 + i) \div (-4 - 3i) \\ = -\frac{19}{25} + \frac{8}{25}i$$

$$(25) \quad (-1 - i) \div (-4 - i) \\ = \frac{5}{17} + \frac{3}{17}i$$

$$(6) \quad (4 - 3i) \div (-1 - 3i) \\ = \frac{1}{2} + \frac{3}{2}i$$

$$(16) \quad (3 + 3i) \div (-2 - i) \\ = -\frac{9}{5} - \frac{3}{5}i$$

$$(26) \quad (3 + 5i) \div (3 - 2i) \\ = -\frac{1}{13} + \frac{21}{13}i$$

$$(7) \quad (-1 + 3i) \div (-4 - i) \\ = \frac{1}{17} - \frac{13}{17}i$$

$$(17) \quad (4 - 3i) \div (-2 - 2i) \\ = -\frac{1}{4} + \frac{7}{4}i$$

$$(27) \quad (-5 + 5i) \div (4 - 3i) \\ = -\frac{7}{5} + \frac{1}{5}i$$

$$(8) \quad (-3 + 5i) \div (4 + 3i) \\ = \frac{3}{25} + \frac{29}{25}i$$

$$(18) \quad (1 - 4i) \div (-1 + 4i) \\ = -1$$

$$(28) \quad (-2 - 3i) \div (-1 - 2i) \\ = \frac{8}{5} - \frac{1}{5}i$$

$$(9) \quad (-2 + 3i) \div (-4 + 4i) \\ = \frac{5}{8} - \frac{1}{8}i$$

$$(19) \quad (-4 + 4i) \div (4 + 4i) \\ = i$$

$$(29) \quad (-5 - 4i) \div (-1 - i) \\ = \frac{9}{2} - \frac{1}{2}i$$

$$(10) \quad (-1 + i) \div (1 + 2i) \\ = \frac{1}{5} + \frac{3}{5}i$$

$$(20) \quad (-4 - 4i) \div (3 + i) \\ = -\frac{8}{5} - \frac{4}{5}i$$

$$(30) \quad (1 - 2i) \div (-4 - 4i) \\ = \frac{1}{8} + \frac{3}{8}i$$

$$(1) \quad (-1 - 3i) \div (-3 + 2i) \\ = -\frac{3}{13} + \frac{11}{13}i$$

$$(11) \quad (2 - 5i) \div (-4 - 5i) \\ = \frac{17}{41} + \frac{30}{41}i$$

$$(21) \quad (-2 - 3i) \div (3 - 2i) \\ = -i$$

$$(2) \quad (3 - 2i) \div (5 + i) \\ = \frac{1}{2} - \frac{1}{2}i$$

$$(12) \quad (5 + 5i) \div (-5 + 3i) \\ = -\frac{5}{17} - \frac{20}{17}i$$

$$(22) \quad (3 + 2i) \div (2 - 4i) \\ = -\frac{1}{10} + \frac{4}{5}i$$

$$(3) \quad (2 + 5i) \div (1 - 5i) \\ = -\frac{23}{26} + \frac{15}{26}i$$

$$(13) \quad (3 - 5i) \div (5 + 2i) \\ = \frac{5}{29} - \frac{31}{29}i$$

$$(23) \quad (5 + i) \div (2 - i) \\ = \frac{9}{5} + \frac{7}{5}i$$

$$(4) \quad (-4 + 5i) \div (-5 + 2i) \\ = \frac{30}{29} - \frac{17}{29}i$$

$$(14) \quad (5 - i) \div (2 - 5i) \\ = \frac{15}{29} + \frac{23}{29}i$$

$$(24) \quad (3 - 5i) \div (-2 - 2i) \\ = \frac{1}{2} + 2i$$

$$(5) \quad (5 - i) \div (-4 - 2i) \\ = -\frac{9}{10} + \frac{7}{10}i$$

$$(15) \quad (-3 - i) \div (1 - 3i) \\ = -i$$

$$(25) \quad (3 - 5i) \div (1 + i) \\ = -1 - 4i$$

$$(6) \quad (-3 + 2i) \div (-5 - i) \\ = \frac{1}{2} - \frac{1}{2}i$$

$$(16) \quad (3 - i) \div (-5 + 2i) \\ = -\frac{17}{29} - \frac{1}{29}i$$

$$(26) \quad (-2 - 3i) \div (1 + 2i) \\ = -\frac{8}{5} + \frac{1}{5}i$$

$$(7) \quad (-5 - 2i) \div (-2 - 3i) \\ = \frac{16}{13} - \frac{11}{13}i$$

$$(17) \quad (1 - 3i) \div (4 + 2i) \\ = -\frac{1}{10} - \frac{7}{10}i$$

$$(27) \quad (-1 - 3i) \div (5 - i) \\ = -\frac{1}{13} - \frac{8}{13}i$$

$$(8) \quad (3 + i) \div (1 - i) \\ = 1 + 2i$$

$$(18) \quad (-2 + 5i) \div (-3 - 3i) \\ = -\frac{1}{2} - \frac{7}{6}i$$

$$(28) \quad (5 + i) \div (4 + 5i) \\ = \frac{25}{41} - \frac{21}{41}i$$

$$(9) \quad (-1 - 4i) \div (-4 - 5i) \\ = \frac{24}{41} + \frac{11}{41}i$$

$$(19) \quad (3 - 5i) \div (-2 - 5i) \\ = \frac{19}{29} + \frac{25}{29}i$$

$$(29) \quad (2 - i) \div (3 + 3i) \\ = \frac{1}{6} - \frac{1}{2}i$$

$$(10) \quad (5 - 4i) \div (-4 - 5i) \\ = i$$

$$(20) \quad (4 + 5i) \div (3 + 4i) \\ = \frac{32}{25} - \frac{1}{25}i$$

$$(30) \quad (-5 - 5i) \div (1 + 4i) \\ = -\frac{25}{17} + \frac{15}{17}i$$

$$(1) (2+i) \div (-3+4i) \\ = -\frac{2}{25} - \frac{11}{25}i$$

$$(11) (5-3i) \div (-1+5i) \\ = -\frac{10}{13} - \frac{11}{13}i$$

$$(21) (1-3i) \div (-1-5i) \\ = \frac{7}{13} + \frac{4}{13}i$$

$$(2) (2+i) \div (2+3i) \\ = \frac{7}{13} - \frac{4}{13}i$$

$$(12) (-3-i) \div (2+4i) \\ = -\frac{1}{2} + \frac{1}{2}i$$

$$(22) (-1-4i) \div (-2-3i) \\ = \frac{14}{13} + \frac{5}{13}i$$

$$(3) (-2+i) \div (-5+4i) \\ = \frac{14}{41} + \frac{3}{41}i$$

$$(13) (-4-i) \div (-2+3i) \\ = \frac{5}{13} + \frac{14}{13}i$$

$$(23) (-1-3i) \div (-1-3i) \\ = 1$$

$$(4) (5+3i) \div (-3+2i) \\ = -\frac{9}{13} - \frac{19}{13}i$$

$$(14) (1+4i) \div (4-5i) \\ = -\frac{16}{41} + \frac{21}{41}i$$

$$(24) (1-2i) \div (3+5i) \\ = -\frac{7}{34} - \frac{11}{34}i$$

$$(5) (-4+2i) \div (-2-2i) \\ = \frac{1}{2} - \frac{3}{2}i$$

$$(15) (2+i) \div (-2-4i) \\ = -\frac{2}{5} + \frac{3}{10}i$$

$$(25) (4+5i) \div (-4-4i) \\ = -\frac{9}{8} - \frac{1}{8}i$$

$$(6) (-1-5i) \div (2-i) \\ = \frac{3}{5} - \frac{11}{5}i$$

$$(16) (-1+4i) \div (-2+4i) \\ = \frac{9}{10} - \frac{1}{5}i$$

$$(26) (-1+5i) \div (-2+5i) \\ = \frac{27}{29} - \frac{5}{29}i$$

$$(7) (-5+2i) \div (2+4i) \\ = -\frac{1}{10} + \frac{6}{5}i$$

$$(17) (3-3i) \div (-1+i) \\ = -3$$

$$(27) (-2-4i) \div (1-5i) \\ = \frac{9}{13} - \frac{7}{13}i$$

$$(8) (-5-2i) \div (-3-2i) \\ = \frac{19}{13} - \frac{4}{13}i$$

$$(18) (-2+5i) \div (-1+2i) \\ = \frac{12}{5} - \frac{1}{5}i$$

$$(28) (2+3i) \div (-2-5i) \\ = -\frac{19}{29} + \frac{4}{29}i$$

$$(9) (4-2i) \div (2+3i) \\ = \frac{2}{13} - \frac{16}{13}i$$

$$(19) (-1-3i) \div (-2-5i) \\ = \frac{17}{29} + \frac{1}{29}i$$

$$(29) (2-3i) \div (-3+i) \\ = -\frac{9}{10} + \frac{7}{10}i$$

$$(10) (-1+5i) \div (2-5i) \\ = -\frac{27}{29} + \frac{5}{29}i$$

$$(20) (4-5i) \div (1+5i) \\ = -\frac{21}{26} - \frac{25}{26}i$$

$$(30) (1+5i) \div (-5+4i) \\ = \frac{15}{41} - \frac{29}{41}i$$

1.6 最大公約数

ユークリッドの互除法を利用した最大公約数を求める問題。

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第 2 章

多項式

2.1 加減法

2 次多項式の和や差を求める問題。

(1) $(9x^2 + 8x - 7) - (7x^2 - 5x + 2)$

(20) $(-7x^2 + 8x - 6) + (-7x^2 + 3x - 9)$

(2) $(-8x^2 - 2x - 9) + (6x^2 - 3x + 2)$

(21) $(4x^2 + x + 6) + (-7x^2 - 3x + 5)$

(3) $(x^2 - 8x + 7) + (-9x^2 - 2x - 2)$

(22) $(7x^2 - 4x - 3) + (-3x^2 - 4x - 9)$

(4) $(3x^2 + 8x + 5) + (-6x^2 - 9x + 7)$

(23) $(6x^2 + 3x + 4) - (-2x^2 - x - 4)$

(5) $(8x^2 + 5x - 7) + (x^2 + 2)$

(24) $(9x^2 + 2x - 1) + (-6x^2 - x - 6)$

(6) $(7x^2 - 8x + 5) + (-6x^2 + 6x + 9)$

(25) $(7x^2 - 4x + 9) - (x^2 + 4x + 2)$

(7) $(-6x^2 - 9x + 4) + (9x^2 + 3x - 4)$

(26) $(3x^2 - 8x + 8) - (4x^2 + 4x + 9)$

(8) $(-6x^2 - 5x + 1) - (3x^2 + 5x - 6)$

(27) $(4x^2 + 3x + 6) + (2x^2 - 7x - 2)$

(9) $(2x^2 - 8x + 5) + (6x^2 - 8x + 1)$

(28) $(-3x^2 + x + 6) - (7x^2 + 2x - 9)$

(10) $(-7x^2 - 5x + 7) + (5x^2 + 5x + 8)$

(29) $(3x^2 - x + 1) - (-4x^2 - 3x + 1)$

(11) $(-9x^2 + 3x + 1) - (-9x^2 + x + 4)$

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(14) $(7x^2 - 5x + 7) - (-5x^2 - 9x + 8)$

(33) $(x^2 - 2) + (2x^2 + 5x + 8)$

(15) $(-9x^2 - 7x) + (6x^2 + 8x - 8)$

(34) $(-4x^2 - 8x - 2) - (-3x^2 + x - 4)$

(16) $(-5x^2 + 9x - 8) + (4x^2 - 5x - 3)$

(35) $(-8x^2 + 9x + 8) + (8x^2 + 4x + 8)$

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(4) $(6x^2 - 4x - 9) - (6x^2 + 8x + 8)$

(23) $(-4x^2 + 8x) - (2x^2 - 6x + 9)$

(5) $(-8x^2 + 8x - 8) - (-6x^2 + 9x + 9)$

(24) $(9x^2 + 6x + 7) + (-2x^2 + 2x + 4)$

(6) $(6x^2 + 7x + 1) + (-9x^2 + 4x + 4)$

(25) $(-9x^2 - 6x + 5) + (5x^2 - 4x - 3)$

(7) $(-5x^2 - 6x + 4) - (-6x^2 - 3x - 2)$

(26) $(-3x^2 - 9x + 6) - (-9x^2 + 9x + 8)$

(8) $(-9x^2 + 4x + 7) - (-4x^2 - 3x + 5)$

(27) $(9x^2 + 4) + (-3x^2 + 4x + 7)$

(9) $(-8x^2 + 2x + 3) + (-9x^2 - 6)$

(28) $(6x^2 - 9x) + (-4x^2 - 1)$

(10) $(-7x^2 + 7x - 3) + (7x^2 + 3x - 7)$

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(33) $(-4x^2 - x - 1) - (-x^2 + 7x + 1)$

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(2) $(7x^2 - 9x + 1) - (-9x^2 + 3x + 1)$

(21) $(9x^2 + 6x - 5) + (-8x^2 - 7x + 1)$

(3) $(-6x^2 - 7x - 2) - (-9x^2 - 6x + 3)$

(22) $(3x^2 + 3x - 9) - (-2x^2 + 8x - 8)$

(4) $(6x^2 + 9x) + (7x^2 - 7x + 3)$

(23) $(x^2 + 7x - 7) - (8x^2 + 6x - 1)$

(5) $(-5x^2 + 7x + 5) - (-9x^2 - x + 7)$

(24) $(7x^2 - x + 7) + (4x^2 - 9x - 2)$

(6) $(9x^2 + 8x) - (2x^2 - 5x + 7)$

(25) $(-8x^2 + 6x - 9) + (4x^2 - x - 6)$

(7) $(2x^2 + 9) - (-8x^2 + 4x + 5)$

(26) $(6x^2 + 6x - 4) + (4x^2 - 4x + 7)$

(8) $(-9x^2 - 5x - 2) + (9x^2 + 2)$

(27) $(3x^2 - 8x + 2) - (5x^2 + 9x - 3)$

(9) $(-3x^2 - 9x + 9) + (3x^2 - 2x - 3)$

(28) $(7x^2 + 3x + 9) - (-x^2 + 8x - 2)$

(10) $(5x^2 - 8x - 7) - (2x^2 - 6x + 4)$

(29) $(-7x^2 + 6x - 4) + (x^2 - 4)$

(11) $(7x^2 - 2x - 8) + (-8x^2 + 7x + 7)$

(30) $(3x^2 - 5x - 5) + (-7x^2 + 3x + 1)$

(12) $(9x^2 + 5x - 3) - (3x^2 - 3x - 3)$

(31) $(6x^2 + 5x - 5) - (x^2 - 7x + 2)$

(13) $(-3x^2 - 5x + 4) + (-6x^2 + 4x - 8)$

(32) $(-7x^2 - 7x - 6) + (-6x^2 + 4x - 9)$

(14) $(2x^2 - 6x + 8) + (-2x^2 - 2x)$

(33) $(-7x^2 + 7x + 7) - (-2x^2 + 7x + 1)$

(15) $(-6x^2 - 2x - 5) - (5x^2 - 9x)$

(34) $(-7x^2 - 5x + 2) - (7x^2 - 3x - 7)$

(16) $(-8x^2 - 6x - 6) - (-x^2 + 7x - 8)$

(35) $(4x^2 + 8x + 3) - (-9x^2 - x + 2)$

(17) $(6x^2 - 7x + 8) + (x^2 + 4x + 7)$

(36) $(-3x^2 + x - 2) - (7x^2 + x - 2)$

(18) $(5x^2 - 3x - 4) - (x^2 + 2x + 6)$

(37) $(-8x^2 + 3x - 6) - (-8x^2 + 9x - 4)$

(19) $(3x^2 + 3x - 3) + (-7x^2 + 9x - 1)$

(38) $(x^2 + 6x - 9) + (-4x^2 - 5x - 5)$

(1) $(-x^2 + 6x + 3) - (5x^2 - 5x + 5)$

(20) $(-3x^2 + 9) - (-4x^2 + 4x + 6)$

(2) $(7x^2 - 8x + 2) + (-8x^2 + 5x + 2)$

(21) $(-5x^2 - 7x - 7) - (x^2 + 2x - 7)$

(3) $(-2x^2 + 8x + 9) + (3x^2 + 3x + 6)$

(22) $(-8x^2 - 9x + 3) + (3x^2 - x + 2)$

(4) $(-3x^2 + 4x + 1) + (-8x^2 - 8x + 1)$

(23) $(-8x^2 + 4x) - (-3x^2 - x - 1)$

(5) $(-8x^2 + 8x + 9) - (7x^2 + 8x + 2)$

(24) $(-5x^2 - 8x + 8) + (4x^2 - 2x - 1)$

(6) $(6x^2 - 5x - 7) - (-2x^2 - 3x - 2)$

(25) $(-4x^2 - 6x) + (7x^2 + x - 2)$

(7) $(5x^2 + 4x + 2) - (x^2 + 5x + 6)$

(26) $(5x^2 - 5x - 9) + (-8x^2 + 5)$

(8) $(-3x^2 - 3x - 8) + (-4x^2 + 8x - 6)$

(27) $(6x^2 - 4x - 8) - (-x^2 - 9x + 8)$

(9) $(-9x^2 + 9x + 9) + (8x^2 + 9x - 8)$

(28) $(-9x^2 - 5x - 1) - (-9x^2 + 6x - 1)$

(10) $(-8x^2 - 2x - 2) + (-3x^2 + 6x + 5)$

(29) $(8x^2 + 9x - 8) - (7x^2 + 5)$

(11) $(-6x^2 + 9x) - (-2x^2 + 3x - 1)$

(30) $(-5x^2 + 9x - 8) + (-x^2 - 7x + 3)$

(12) $(3x^2 + 8x - 8) + (-7x^2 + 3x + 6)$

(31) $(-9x^2 - 6x + 7) + (-8x^2 + 2x - 3)$

(13) $(3x^2 - 2x - 1) - (7x^2 - x - 9)$

(32) $(-6x^2 - 6x + 5) + (-6x^2 + 8x + 8)$

(14) $(-2x^2 - 6x - 3) + (-9x^2 - 9x + 2)$

(33) $(2x^2 + 3) - (9x^2 - 2x + 8)$

(15) $(-4x^2 + 3x - 5) + (-4x^2 - 3x - 2)$

(34) $(-3x^2 - 3x + 7) - (-9x^2 + 4x - 4)$

(16) $(-8x^2 + 5x - 9) - (2x^2 - 3x + 8)$

(35) $(-4x^2 + 7x) + (2x^2 - 3x + 5)$

(17) $(3x^2 + 7) - (3x^2 - 4x + 8)$

(36) $(5x^2 + 4x - 9) - (5x^2 + 5x + 9)$

(18) $(8x^2 + 3x + 7) - (8x^2 - 7x - 5)$

(37) $(2x^2 - 6x + 8) + (9x^2 + 5x)$

(19) $(4x^2 + 4x - 4) - (-6x^2 - x - 8)$

(38) $(8x^2 + 4x - 8) - (8x^2 + 9x + 1)$

(1) $(7x^2 + 8x + 7) - (-x^2 - 3x - 4)$

(20) $(-4x^2 - 3x - 9) - (-6x^2 + 5x + 4)$

(2) $(-9x^2 + 4x - 2) - (7x^2 + 3x - 1)$

(21) $(7x^2 + x) - (-x^2 - 4x - 6)$

(3) $(3x^2 - 2x) - (-2x^2 - 6x - 8)$

(22) $(-4x^2 - 4x + 1) - (-4x^2 + 9x - 4)$

(4) $(-9x^2 + 2x + 9) + (-7x^2 + 2x - 8)$

(23) $(-2x^2 + 9) - (-2x^2 - 6x + 9)$

(5) $(x^2 - 7x + 2) - (-7x^2 + 6x - 6)$

(24) $(-3x^2 - 4x + 5) - (-8x^2 - 2x - 6)$

(6) $(-x^2 - 5x - 8) + (-8x^2 - 2x - 3)$

(25) $(-7x^2 + 8x - 3) - (-4x^2 + 6x + 1)$

(7) $(7x^2 + 7x - 3) + (7x^2 + 5)$

(26) $(-7x^2 - 2x - 9) + (3x^2)$

(8) $(x^2 - 9x - 8) - (2x^2 - x - 3)$

(27) $(-7x^2 + 8x + 5) + (-9x^2 - x + 8)$

(9) $(6x^2 + 3) + (-9x^2 + 4x - 2)$

(28) $(-5x^2 - 4x - 8) + (8x^2 + 7x)$

(10) $(7x^2 - 7x + 3) - (5x^2 - 8x + 2)$

(29) $(-2x^2 + 6x + 8) - (9x^2 - 7x + 3)$

(11) $(9x^2 - 4x) + (3x^2 + 6x + 3)$

(30) $(-7x^2 + x) + (9x^2 + 3x - 5)$

(12) $(2x^2 - 8x - 2) - (6x^2 - 8x - 1)$

(31) $(3x^2 - 9x - 2) - (-9x^2 - 2x + 6)$

(13) $(x^2 - 4x + 8) - (7x^2 - 6)$

(32) $(-4x^2 - 9x - 8) - (-5x^2 + 6x + 1)$

(14) $(x^2 - 3x - 5) + (-2x^2 - 2x)$

(33) $(-x^2 + 3x + 9) - (-7x^2 + 6x)$

(15) $(-6x^2 - 8x - 1) - (7x^2 - x - 6)$

(34) $(-8x^2 + 6x - 3) + (-6x^2 - 3x + 2)$

(16) $(4x^2 - x + 8) + (-3x^2 + 4x - 8)$

(35) $(9x^2 + 9x - 3) + (5x^2 + 9x)$

(17) $(x^2 - 8x + 6) - (-5x^2 - 5x + 2)$

(36) $(-6x^2 - 5x - 2) + (-9x^2 - 8x + 5)$

(18) $(9x^2 + 8x + 1) + (9x^2 + x)$

(37) $(-4x^2 + 7x - 3) - (-2x^2 - 5x - 2)$

(19) $(2x^2 + 9x - 6) + (8x^2 + 4x + 7)$

(38) $(-7x^2 - 4x - 5) + (-4x^2 + x + 5)$

(1) $(2x^2 - x + 7) + (5x^2 + 2x + 8)$

(20) $(-9x^2 - 7x + 1) + (2x^2 + x - 7)$

(2) $(3x^2 + 9x + 1) - (-9x^2 - 3x + 1)$

(21) $(7x^2 - 8x) - (2x^2 - 3x - 4)$

(3) $(-4x^2 + 3x + 3) - (8x^2 - 5x - 4)$

(22) $(9x^2 + 8x) - (-2x^2 - x + 3)$

(4) $(6x^2 + 7x - 3) + (2x^2 + 8x + 1)$

(23) $(-2x^2 - 2x - 1) - (9x^2 - 5x - 5)$

(5) $(6x^2 - 5x - 6) - (9x^2 + x - 9)$

(24) $(6x^2 + 4x + 5) - (-3x^2 + 5x + 5)$

(6) $(-6x^2 - 7x - 5) - (2x^2 - 8x - 5)$

(25) $(-6x^2 - 2x - 9) - (-5x^2 - 9x + 3)$

(7) $(-8x^2 + 5x - 5) - (-9x^2 + 8x - 4)$

(26) $(-9x^2 - 9x + 3) + (-9x^2 + 5x + 8)$

(8) $(7x^2 - 4) - (-9x^2 - 7x + 9)$

(27) $(-7x^2 + 9x - 2) + (9x^2 + 6x - 9)$

(9) $(-6x^2 + 8x + 7) + (-4x^2 - 2x)$

(28) $(-7x^2 + 5x - 3) + (7x^2 - 5x)$

(10) $(9x^2 - 7) - (-4x^2 + 8x)$

(29) $(-6x^2 - 6x - 3) + (3x^2 - 6x - 1)$

(11) $(9x^2 - 3x - 9) + (-3x^2 - 3x - 6)$

(30) $(-2x^2 - 5x + 9) - (4x^2 - 6x - 3)$

(12) $(x^2 - 4x + 4) - (-8x^2 + 7x - 7)$

(31) $(-9x^2 - 7x + 5) + (8x^2 + 2x - 4)$

(13) $(-6x^2 + 8x + 1) + (-7x^2 - 4x + 2)$

(32) $(-3x^2 - 2x + 4) + (4x^2 - 2)$

(14) $(8x^2 - 4x - 5) - (-3x^2 + 3x - 9)$

(33) $(-8x^2 + 2) - (3x^2 - 5x - 5)$

(15) $(-5x^2 + 3x) + (3x^2 - 5x + 3)$

(34) $(4x^2 - 7x + 2) + (x^2 - 3x - 5)$

(16) $(-6x^2 + 2x + 3) - (-4x^2 - 4x + 8)$

(35) $(-5x^2 + 3x - 1) - (5x^2 - 2x - 8)$

(17) $(-4x^2 - 3x - 5) - (-3x^2 + 9x - 5)$

(36) $(-x^2 + 8x) + (-8x^2 - 2x - 7)$

(18) $(7x^2 - 6x - 6) + (-4x^2 - 2x + 7)$

(37) $(5x^2 + 7x - 5) - (-7x^2 - 4x + 8)$

(19) $(8x^2 + x + 9) - (x^2 - 8x + 9)$

(38) $(-x^2 - 7x - 8) - (-9x^2 + 3x - 8)$

- (1) $(9x^2 + 8x - 7) - (7x^2 - 5x + 2)$
 $= 2x^2 + 13x - 9$
- (2) $(-8x^2 - 2x - 9) + (6x^2 - 3x + 2)$
 $= -2x^2 - 5x - 7$
- (3) $(x^2 - 8x + 7) + (-9x^2 - 2x - 2)$
 $= -8x^2 - 10x + 5$
- (4) $(3x^2 + 8x + 5) + (-6x^2 - 9x + 7)$
 $= -3x^2 - x + 12$
- (5) $(8x^2 + 5x - 7) + (x^2 + 2)$
 $= 9x^2 + 5x - 5$
- (6) $(7x^2 - 8x + 5) + (-6x^2 + 6x + 9)$
 $= x^2 - 2x + 14$
- (7) $(-6x^2 - 9x + 4) + (9x^2 + 3x - 4)$
 $= 3x^2 - 6x$
- (8) $(-6x^2 - 5x + 1) - (3x^2 + 5x - 6)$
 $= -9x^2 - 10x + 7$
- (9) $(2x^2 - 8x + 5) + (6x^2 - 8x + 1)$
 $= 8x^2 - 16x + 6$
- (10) $(-7x^2 - 5x + 7) + (5x^2 + 5x + 8)$
 $= -2x^2 + 15$
- (11) $(-9x^2 + 3x + 1) - (-9x^2 + x + 4)$
 $= 2x - 3$
- (12) $(-7x^2 + 5) - (-8x^2 + 9x - 2)$
 $= x^2 - 9x + 7$
- (13) $(-x^2 - 8x - 3) - (4x^2 + 2x + 4)$
 $= -5x^2 - 10x - 7$
- (14) $(7x^2 - 5x + 7) - (-5x^2 - 9x + 8)$
 $= 12x^2 + 4x - 1$
- (15) $(-9x^2 - 7x) + (6x^2 + 8x - 8)$
 $= -3x^2 + x - 8$
- (16) $(-5x^2 + 9x - 8) + (4x^2 - 5x - 3)$
 $= -x^2 + 4x - 11$
- (17) $(-5x^2 - x - 9) - (-8x^2 - x - 4)$
 $= 3x^2 - 5$
- (18) $(2x^2 + 5x - 6) - (-6x^2 + 8x + 6)$
 $= 8x^2 - 3x - 12$
- (19) $(7x^2 - 8x - 7) + (-x^2 + 2x - 8)$
 $= 6x^2 - 6x - 15$
- (20) $(-7x^2 + 8x - 6) + (-7x^2 + 3x - 9)$
 $= -14x^2 + 11x - 15$
- (21) $(4x^2 + x + 6) + (-7x^2 - 3x + 5)$
 $= -3x^2 - 2x + 11$
- (22) $(7x^2 - 4x - 3) + (-3x^2 - 4x - 9)$
 $= 4x^2 - 8x - 12$
- (23) $(6x^2 + 3x + 4) - (-2x^2 - x - 4)$
 $= 8x^2 + 4x + 8$
- (24) $(9x^2 + 2x - 1) + (-6x^2 - x - 6)$
 $= 3x^2 + x - 7$
- (25) $(7x^2 - 4x + 9) - (x^2 + 4x + 2)$
 $= 6x^2 - 8x + 7$
- (26) $(3x^2 - 8x + 8) - (4x^2 + 4x + 9)$
 $= -x^2 - 12x - 1$
- (27) $(4x^2 + 3x + 6) + (2x^2 - 7x - 2)$
 $= 6x^2 - 4x + 4$
- (28) $(-3x^2 + x + 6) - (7x^2 + 2x - 9)$
 $= -10x^2 - x + 15$
- (29) $(3x^2 - x + 1) - (-4x^2 - 3x + 1)$
 $= 7x^2 + 2x$
- (30) $(8x^2 - 2x + 2) - (4x^2 + 2x + 2)$
 $= 4x^2 - 4x$
- (31) $(9x^2 + 4) - (4x^2 + 5x + 3)$
 $= 5x^2 - 5x + 1$
- (32) $(-6x^2 - 8x + 4) + (8x^2 + 2x + 9)$
 $= 2x^2 - 6x + 13$
- (33) $(x^2 - 2) + (2x^2 + 5x + 8)$
 $= 3x^2 + 5x + 6$
- (34) $(-4x^2 - 8x - 2) - (-3x^2 + x - 4)$
 $= -x^2 - 9x + 2$
- (35) $(-8x^2 + 9x + 8) + (8x^2 + 4x + 8)$
 $= 13x + 16$
- (36) $(-4x^2 + 9x + 7) + (-3x^2 + x - 5)$
 $= -7x^2 + 10x + 2$
- (37) $(-9x^2 + 4x - 7) + (-2x^2 + 8x + 1)$
 $= -11x^2 + 12x - 6$
- (38) $(2x^2 + x + 6) + (4x^2 - 7x + 9)$
 $= 6x^2 - 6x + 15$

$$(1) \quad (-5x^2 - 7x + 2) + (-7x^2 + 2x - 5) \\ = -12x^2 - 5x - 3$$

$$(2) \quad (-3x^2 + 7x + 5) + (8x^2 - 2x - 3) \\ = 5x^2 + 5x + 2$$

$$(3) \quad (-6x^2 + 8x + 3) - (4x^2 - 3x + 8) \\ = -10x^2 + 11x - 5$$

$$(4) \quad (-6x^2 - 5x - 2) + (8x^2 - 3x + 6) \\ = 2x^2 - 8x + 4$$

$$(5) \quad (5x^2 + 6x + 1) - (-x^2 - 9x + 1) \\ = 6x^2 + 15x$$

$$(6) \quad (7x^2 + x + 1) - (x^2 - 2x - 6) \\ = 6x^2 + 3x + 7$$

$$(7) \quad (-x^2 + 7x + 9) + (-9x^2 + 8x + 8) \\ = -10x^2 + 15x + 17$$

$$(8) \quad (x^2 - x - 6) - (-x^2 + 4x - 9) \\ = 2x^2 - 5x + 3$$

$$(9) \quad (3x^2 + 7x - 3) + (-x^2 - 8x + 1) \\ = 2x^2 - x - 2$$

$$(10) \quad (-5x^2 - 6x - 8) + (6x^2 - 4x - 6) \\ = x^2 - 10x - 14$$

$$(11) \quad (4x^2 + 5x + 7) - (-9x^2 - 7x + 9) \\ = 13x^2 + 12x - 2$$

$$(12) \quad (7x^2 + 2x - 9) + (-6x^2 + 4x - 3) \\ = x^2 + 6x - 12$$

$$(13) \quad (x^2 - 4x - 4) + (-4x^2 - 9x + 1) \\ = -3x^2 - 13x - 3$$

$$(14) \quad (-2x^2 + 2x - 1) + (2x^2 + 4x + 6) \\ = 6x + 5$$

$$(15) \quad (-9x^2 - x + 8) - (-7x^2 - 9x - 5) \\ = -2x^2 + 8x + 13$$

$$(16) \quad (-6x^2 - 3x - 5) - (-3x^2 + 5x + 5) \\ = -3x^2 - 8x - 10$$

$$(17) \quad (-6x^2 + 6x + 8) - (8x^2 - x + 2) \\ = -14x^2 + 7x + 6$$

$$(18) \quad (-4x^2 + 8) + (-8x^2 + 6x - 6) \\ = -12x^2 + 6x + 2$$

$$(19) \quad (3x^2 + 3x - 4) + (-8x^2 + 7x + 6) \\ = -5x^2 + 10x + 2$$

$$(20) \quad (5x^2 - x + 6) - (-7x^2 + 2x - 1) \\ = 12x^2 - 3x + 7$$

$$(21) \quad (6x^2 + 7x + 7) - (-5x^2 - 2x - 8) \\ = 11x^2 + 9x + 15$$

$$(22) \quad (-x^2 + x + 7) + (4x^2 - 5x + 1) \\ = 3x^2 - 4x + 8$$

$$(23) \quad (7x^2 + 3x + 5) + (-2x^2 - 7x - 4) \\ = 5x^2 - 4x + 1$$

$$(24) \quad (-6x^2 - 4x + 9) + (-7x^2 + 7x - 7) \\ = -13x^2 + 3x + 2$$

$$(25) \quad (x^2 + 3x + 1) + (4x^2 - 4x - 4) \\ = 5x^2 - x - 3$$

$$(26) \quad (7x^2 - 8x + 4) + (-8x^2 - 8x - 9) \\ = -x^2 - 16x - 5$$

$$(27) \quad (-4x^2 - 4x - 6) - (3x^2 + 9x + 2) \\ = -7x^2 - 13x - 8$$

$$(28) \quad (-5x^2 - 3x - 9) - (-9x^2 + 9x + 2) \\ = 4x^2 - 12x - 11$$

$$(29) \quad (6x^2 + 9x + 9) + (-3x^2 + 7x + 5) \\ = 3x^2 + 16x + 14$$

$$(30) \quad (8x^2 + 2x + 2) - (-8x^2 + 6x - 6) \\ = 16x^2 - 4x + 8$$

$$(31) \quad (-x^2 + 3x + 9) - (x^2 - 6x - 4) \\ = -2x^2 + 9x + 13$$

$$(32) \quad (-9x^2 - 7x + 5) + (4x^2 + 6x - 1) \\ = -5x^2 - x + 4$$

$$(33) \quad (7x^2 + 9x - 6) + (-6x^2 + 4x + 5) \\ = x^2 + 13x - 1$$

$$(34) \quad (-6x^2 + x + 5) - (-9x^2 - 4x - 6) \\ = 3x^2 + 5x + 11$$

$$(35) \quad (9x^2 - 6x) + (-2x^2 - 9x - 7) \\ = 7x^2 - 15x - 7$$

$$(36) \quad (7x^2 + 3x - 3) + (-3x^2 - 7x + 5) \\ = 4x^2 - 4x + 2$$

$$(37) \quad (-8x^2 - 9) + (x^2 - 8x - 6) \\ = -7x^2 - 8x - 15$$

$$(38) \quad (2x^2 - 6x + 7) + (x^2 - 7x - 3) \\ = 3x^2 - 13x + 4$$

$$(1) (4x^2 + 7x - 1) - (-7x^2 + 4x + 4) \\ = 11x^2 + 3x - 5$$

$$(2) (9x^2 + 6x - 1) - (-x^2 - 2x + 7) \\ = 10x^2 + 8x - 8$$

$$(3) (5x^2 - 7x - 6) - (3x^2 + x + 3) \\ = 2x^2 - 8x - 9$$

$$(4) (-7x^2 + 7x - 5) - (-9x^2 - 8x - 3) \\ = 2x^2 + 15x - 2$$

$$(5) (4x^2 + 2x) + (3x^2 - 9x + 2) \\ = 7x^2 - 7x + 2$$

$$(6) (9x^2 + 8x + 4) - (-2x^2 + 8x - 2) \\ = 11x^2 + 6$$

$$(7) (2x^2 + x + 7) + (9x^2 + 2x + 7) \\ = 11x^2 + 3x + 14$$

$$(8) (-3x^2 + 9x + 7) + (2x^2 - 6x - 4) \\ = -x^2 + 3x + 3$$

$$(9) (-6x^2 - x - 1) + (-8x^2 - 9x - 3) \\ = -14x^2 - 10x - 4$$

$$(10) (x^2 + 5x - 3) + (-2x^2 + 2x) \\ = -x^2 + 7x - 3$$

$$(11) (3x^2 - 6x - 5) - (4x^2 - 8x + 2) \\ = -x^2 + 2x - 7$$

$$(12) (-8x^2 - 8) - (-2x^2 - 4x - 2) \\ = -6x^2 + 4x - 6$$

$$(13) (3x^2 - 4x - 2) + (-x^2 - 9x - 8) \\ = 2x^2 - 13x - 10$$

$$(14) (-6x^2 + 5x - 2) - (-7x^2 - 2x - 6) \\ = x^2 + 7x + 4$$

$$(15) (-9x^2 - 6x + 5) + (-4x^2 - 6x - 9) \\ = -13x^2 - 12x - 4$$

$$(16) (-4x^2 - x - 7) + (7x^2 + 5x + 7) \\ = 3x^2 + 4x$$

$$(17) (-8x^2 + 3x + 8) + (-2x^2 - 5x + 2) \\ = -10x^2 - 2x + 10$$

$$(18) (4x^2 - 9) - (-x^2 - 9x - 6) \\ = 5x^2 + 9x - 3$$

$$(19) (-x^2 + 8x + 5) - (8x^2 + 7x - 9) \\ = -9x^2 + x + 14$$

$$(20) (-2x^2 - 3x - 4) - (-4x^2 + 7x - 3) \\ = 2x^2 - 10x - 1$$

$$(21) (3x^2 + 9x + 3) - (6x^2 - 3x - 1) \\ = -3x^2 + 12x + 4$$

$$(22) (-7x^2 - 9x + 5) + (-3x^2 + 7x + 7) \\ = -10x^2 - 2x + 12$$

$$(23) (8x^2 - 5x + 3) + (x^2 - 4x - 3) \\ = 9x^2 - 9x$$

$$(24) (-3x^2 - 3x - 1) - (5x^2 - 5x + 9) \\ = -8x^2 + 2x - 10$$

$$(25) (-8x^2 + 7x - 3) - (-8x^2 + 7x - 7) \\ = 4$$

$$(26) (-3x^2 + 3x + 8) + (3x^2 + 5x - 7) \\ = 8x + 1$$

$$(27) (3x^2 - 3x + 2) + (-6x^2 + x - 8) \\ = -3x^2 - 2x - 6$$

$$(28) (x^2 - 3x - 9) - (9x^2 - 4x + 5) \\ = -8x^2 + x - 14$$

$$(29) (-9x^2 - 9x - 3) - (3x^2 + 5x + 5) \\ = -12x^2 - 14x - 8$$

$$(30) (-4x^2 + x + 8) - (9x^2 + 9x + 9) \\ = -13x^2 - 8x - 1$$

$$(31) (-5x^2 + 6x + 9) + (-6x^2 - 7x - 3) \\ = -11x^2 - x + 6$$

$$(32) (8x^2 + 6x - 9) - (3x^2 - 4x - 9) \\ = 5x^2 + 10x$$

$$(33) (-7x^2 + 2x) - (-8x^2 - 2x + 1) \\ = x^2 + 4x - 1$$

$$(34) (-3x^2 - 2x - 7) - (8x^2 - 6x + 8) \\ = -11x^2 + 4x - 15$$

$$(35) (-x^2 - 9x + 3) + (-2x^2 + 3x - 6) \\ = -3x^2 - 6x - 3$$

$$(36) (-4x^2 - 4x - 7) + (-2x^2 + 7x + 5) \\ = -6x^2 + 3x - 2$$

$$(37) (9x^2 - 5x + 2) + (5x^2 - 5x - 7) \\ = 14x^2 - 10x - 5$$

$$(38) (-4x^2 + 4x + 2) + (2x^2 + 2x + 4) \\ = -2x^2 + 6x + 6$$

- (1) $(3x^2 - 9x) - (x^2 + 4x - 9)$
 $= 2x^2 - 13x + 9$
- (2) $(9x^2 + 9x + 1) + (5x^2 + 8x + 6)$
 $= 14x^2 + 17x + 7$
- (3) $(6x^2 + 6x - 6) - (-7x^2 + 8x - 2)$
 $= 13x^2 - 2x - 4$
- (4) $(-5x^2 - 2x + 3) + (8x^2 + x - 2)$
 $= 3x^2 - x + 1$
- (5) $(4x^2 - 4x + 8) - (-4x^2 + 7x - 6)$
 $= 8x^2 - 11x + 14$
- (6) $(-2x^2 - 5x + 8) + (8x^2 - 7x)$
 $= 6x^2 - 12x + 8$
- (7) $(4x^2 + 7x - 9) + (x^2 - 5x - 2)$
 $= 5x^2 + 2x - 11$
- (8) $(7x^2 - 2x - 1) + (-4x^2 + 4x + 5)$
 $= 3x^2 + 2x + 4$
- (9) $(9x^2 + 8x - 3) + (-3x^2 + 8x - 5)$
 $= 6x^2 + 16x - 8$
- (10) $(5x^2 + 9x + 5) - (8x^2 + 9x + 4)$
 $= -3x^2 + 1$
- (11) $(5x^2 + 5x + 7) - (-4x^2 + 4x - 3)$
 $= 9x^2 + x + 10$
- (12) $(9x^2 + 4x - 9) + (-5x^2 - x - 3)$
 $= 4x^2 + 3x - 12$
- (13) $(2x^2 - x - 4) - (-5x^2 + x - 1)$
 $= 7x^2 - 2x - 3$
- (14) $(7x^2 - x + 4) - (-4x^2 - 8x + 3)$
 $= 11x^2 + 7x + 1$
- (15) $(-x^2 - 5x - 6) + (9x^2 + 8x)$
 $= 8x^2 + 3x - 6$
- (16) $(2x^2 - x - 7) + (2x^2 - 4x + 4)$
 $= 4x^2 - 5x - 3$
- (17) $(4x^2 - 7x + 5) - (-5x^2 - x + 1)$
 $= 9x^2 - 6x + 4$
- (18) $(5x^2 - 7x) + (-8x^2 - 6x - 9)$
 $= -3x^2 - 13x - 9$
- (19) $(5x^2 + 3x - 4) - (x^2 + 2x + 4)$
 $= 4x^2 + x - 8$
- (20) $(-4x^2 - 5x - 1) + (5x^2 + 9x + 4)$
 $= x^2 + 4x + 3$
- (21) $(-x^2 - 3x + 1) + (-4x^2 + 7x + 5)$
 $= -5x^2 + 4x + 6$
- (22) $(x^2 + 5x + 4) - (-x^2 - 3x + 8)$
 $= 2x^2 + 8x - 4$
- (23) $(4x^2 - 8x - 9) - (6x^2 - 6x + 9)$
 $= -2x^2 - 2x - 18$
- (24) $(-3x^2 + 8x - 8) - (-3x^2 - 3x + 3)$
 $= 11x - 11$
- (25) $(7x^2 + 2x + 6) - (9x^2 - 6x - 1)$
 $= -2x^2 + 8x + 7$
- (26) $(8x^2 + 9) - (-5x^2 + 5x - 7)$
 $= 13x^2 - 5x + 16$
- (27) $(-6x^2 - 2x + 3) - (-3x^2 + x)$
 $= -3x^2 - 3x + 3$
- (28) $(-9x^2 + x - 6) + (3x^2 - 3x + 4)$
 $= -6x^2 - 2x - 2$
- (29) $(8x^2 - 9x - 3) + (-6x^2 + 7x + 1)$
 $= 2x^2 - 2x - 2$
- (30) $(-3x^2 - 7x + 8) + (-8x^2 + 6x - 8)$
 $= -11x^2 - x$
- (31) $(6x^2 + x + 1) + (-7x^2 - x - 5)$
 $= -x^2 - 4$
- (32) $(-6x^2 + 3x + 4) - (5x^2 + 7x + 2)$
 $= -11x^2 - 4x + 2$
- (33) $(-3x^2 - 6x) + (-8x^2 - 7x - 1)$
 $= -11x^2 - 13x - 1$
- (34) $(4x^2 + 3x) - (6x^2 + 5x - 8)$
 $= -2x^2 - 2x + 8$
- (35) $(-2x^2 - 4x - 9) - (-8x^2 + 8x - 1)$
 $= 6x^2 - 12x - 8$
- (36) $(-7x^2 + x - 1) - (5x^2 - 7x + 9)$
 $= -12x^2 + 8x - 10$
- (37) $(6x^2 + 2x - 1) + (-5x^2 + 6x)$
 $= x^2 + 8x - 1$
- (38) $(-3x^2 + 4x - 6) + (7x^2 - 6x + 7)$
 $= 4x^2 - 2x + 1$

$$(1) \quad (-6x^2 + 3x + 7) - (9x^2 + 8x - 9) \\ = -15x^2 - 5x + 16$$

$$(2) \quad (-5x^2 + 5x - 1) - (5x^2 + 6x) \\ = -10x^2 - x - 1$$

$$(3) \quad (-9x^2 - 6x + 4) - (2x^2 - 7) \\ = -11x^2 - 6x + 11$$

$$(4) \quad (4x^2 + 9x - 6) + (-8x^2 - 4x + 2) \\ = -4x^2 + 5x - 4$$

$$(5) \quad (6x^2 + 6x - 2) - (4x^2 - 9x + 4) \\ = 2x^2 + 15x - 6$$

$$(6) \quad (x^2 + 5x + 8) + (6x^2 - 9x - 6) \\ = 7x^2 - 4x + 2$$

$$(7) \quad (x^2 + 8x - 6) - (-6x^2 - 6x + 1) \\ = 7x^2 + 14x - 7$$

$$(8) \quad (2x^2 - 9x - 9) - (-5x^2 + 6x - 4) \\ = 7x^2 - 15x - 5$$

$$(9) \quad (-2x^2 + 5x + 3) - (8x^2 + 7x - 2) \\ = -10x^2 - 2x + 5$$

$$(10) \quad (-7x^2 + x - 1) - (3x^2 + 4x - 1) \\ = -10x^2 - 3x$$

$$(11) \quad (-9x^2 - 3x - 7) + (-8x^2 - 2x - 8) \\ = -17x^2 - 5x - 15$$

$$(12) \quad (-3x^2 + 6x + 6) + (-x^2 - 5x - 6) \\ = -4x^2 + x$$

$$(13) \quad (-3x^2 - 4x + 2) - (-3x^2 + 5x - 7) \\ = -9x + 9$$

$$(14) \quad (2x^2 - 8x + 7) - (3x^2 + 7x) \\ = -x^2 - 15x + 7$$

$$(15) \quad (8x^2 - 5x - 6) + (6x^2 + 7x + 7) \\ = 14x^2 + 2x + 1$$

$$(16) \quad (6x^2 + 7x + 3) + (9x^2 + 3x - 4) \\ = 15x^2 + 10x - 1$$

$$(17) \quad (8x^2 + 5x + 5) + (9x^2 + 7x + 7) \\ = 17x^2 + 12x + 12$$

$$(18) \quad (-7x^2 - 9x + 3) + (-3x^2 + 9x + 8) \\ = -10x^2 + 11$$

$$(19) \quad (-4x^2 - 5x + 1) + (5x^2 - 2x + 2) \\ = x^2 - 7x + 3$$

$$(20) \quad (-2x^2 - 4x - 8) - (6x^2 - 6x - 4) \\ = -8x^2 + 2x - 4$$

$$(21) \quad (7x^2 - 7x + 3) - (-3x^2 - x + 9) \\ = 10x^2 - 6x - 6$$

$$(22) \quad (x^2 + 6x - 9) - (-x^2 - 2x - 3) \\ = 2x^2 + 8x - 6$$

$$(23) \quad (9x^2 + 3x - 2) + (-2x^2 - 4x - 4) \\ = 7x^2 - x - 6$$

$$(24) \quad (-8x^2 - 6x + 9) + (5x^2 + x - 5) \\ = -3x^2 - 5x + 4$$

$$(25) \quad (-x^2 - 6x + 6) - (6x^2 - 9x + 6) \\ = -7x^2 + 3x$$

$$(26) \quad (5x^2 - 2) - (4x^2 + 2x - 7) \\ = x^2 - 2x + 5$$

$$(27) \quad (-8x^2 - 3) - (-3x^2 + 3x - 6) \\ = -5x^2 - 3x + 3$$

$$(28) \quad (-x^2 + 8x) + (-5x^2 + x - 5) \\ = -6x^2 + 9x - 5$$

$$(29) \quad (6x^2 - 9x + 8) + (-3x^2 - 9x - 6) \\ = 3x^2 - 18x + 2$$

$$(30) \quad (6x^2 + 3x + 6) - (-4x^2 - 4x + 1) \\ = 10x^2 + 7x + 5$$

$$(31) \quad (7x^2 + 7) + (5x^2 + 4x + 2) \\ = 12x^2 + 4x + 9$$

$$(32) \quad (-3x^2 + 8x + 3) - (7x^2 - 9x - 4) \\ = -10x^2 + 17x + 7$$

$$(33) \quad (-3x^2 - 8) - (-9x^2 - 6x - 8) \\ = 6x^2 + 6x$$

$$(34) \quad (7x^2 + 4x) + (3x^2 + 4x - 8) \\ = 10x^2 + 8x - 8$$

$$(35) \quad (4x^2 - 4x + 9) - (-2x^2 - 2x + 4) \\ = 6x^2 - 2x + 5$$

$$(36) \quad (-x^2 + 5x + 5) - (-7x^2 + x + 6) \\ = 6x^2 + 4x - 1$$

$$(37) \quad (-9x^2 - 9x) - (-7x^2 + 9x - 3) \\ = -2x^2 - 18x + 3$$

$$(38) \quad (3x^2 - 9x - 5) - (-3x^2 - 9x - 5) \\ = 6x^2$$

$$(1) \quad (-4x^2 - 9x - 9) - (-4x^2 - x - 8) \\ = -8x - 1$$

$$(2) \quad (-4x^2 - 6x + 3) - (9x^2 - 8x + 3) \\ = -13x^2 + 2x$$

$$(3) \quad (-3x^2 + 4x + 3) + (3x^2 - 9x - 4) \\ = -5x - 1$$

$$(4) \quad (5x^2 + 9x + 6) - (8x^2 + 4x - 2) \\ = -3x^2 + 5x + 8$$

$$(5) \quad (4x^2 - x - 6) + (8x^2 - 5x + 4) \\ = 12x^2 - 6x - 2$$

$$(6) \quad (-6x^2 - 2x + 8) - (9x^2 + 6x + 3) \\ = -15x^2 - 8x + 5$$

$$(7) \quad (-6x^2 - 4x - 1) - (9x^2 + 4x + 6) \\ = -15x^2 - 8x - 7$$

$$(8) \quad (9x^2 - 5x + 9) - (2x^2 - 4x + 2) \\ = 7x^2 - x + 7$$

$$(9) \quad (-x^2 + 6x - 4) - (-4x^2 - 4x + 1) \\ = 3x^2 + 10x - 5$$

$$(10) \quad (-8x^2 - 5x + 2) + (-6x^2 + 6x - 7) \\ = -14x^2 + x - 5$$

$$(11) \quad (-5x^2 + 8x + 4) + (-9x^2 + 8x - 7) \\ = -14x^2 + 16x - 3$$

$$(12) \quad (9x^2 + x - 3) + (4x^2 - 2x - 4) \\ = 13x^2 - x - 7$$

$$(13) \quad (-9x^2 + 8x) + (-6x^2 - 7x - 8) \\ = -15x^2 + x - 8$$

$$(14) \quad (7x^2) + (9x^2 - 6) \\ = 16x^2 - 6$$

$$(15) \quad (x^2 + 5) + (-7x^2 + 7x + 2) \\ = -6x^2 + 7x + 7$$

$$(16) \quad (9x^2 + 8x + 2) - (8x^2 - 3x) \\ = x^2 + 11x + 2$$

$$(17) \quad (-5x^2 - 5x - 6) + (-7x^2 - 4x + 7) \\ = -12x^2 - 9x + 1$$

$$(18) \quad (-x^2 + 5x - 3) - (-6x^2 + 8x + 2) \\ = 5x^2 - 3x - 5$$

$$(19) \quad (-x^2 - 4x - 3) + (2x^2 - 4x - 8) \\ = x^2 - 8x - 11$$

$$(20) \quad (-7x^2 - 2x + 3) + (-2x^2 + 6x - 8) \\ = -9x^2 + 4x - 5$$

$$(21) \quad (-2x^2 - 9x + 7) - (-9x^2 - 2x + 9) \\ = 7x^2 - 7x - 2$$

$$(22) \quad (3x^2 + 9x + 4) - (-5x^2 + 7x - 5) \\ = 8x^2 + 2x + 9$$

$$(23) \quad (-4x^2 + 7x + 9) + (-5x^2 + 8x - 3) \\ = -9x^2 + 15x + 6$$

$$(24) \quad (9x^2 + x) - (2x^2 + 4x + 9) \\ = 7x^2 - 3x - 9$$

$$(25) \quad (-3x^2 - x + 4) - (3x^2 + 3x - 2) \\ = -6x^2 - 4x + 6$$

$$(26) \quad (-5x^2 - 3x - 5) + (7x^2 + 4x - 1) \\ = 2x^2 + x - 6$$

$$(27) \quad (4x^2 - 5x + 1) + (6x^2 + 4x + 7) \\ = 10x^2 - x + 8$$

$$(28) \quad (-4x^2 + 6x + 5) - (-5x^2 - x + 2) \\ = x^2 + 7x + 3$$

$$(29) \quad (3x^2 - 4x + 9) - (x^2 + 2x + 5) \\ = 2x^2 - 6x + 4$$

$$(30) \quad (x^2 + 7x - 2) + (-x^2 - x - 5) \\ = 6x - 7$$

$$(31) \quad (x^2 + 7x - 4) + (2x^2 - 7x - 7) \\ = 3x^2 - 11$$

$$(32) \quad (8x^2 - 7x + 6) - (9x^2 + 4x - 4) \\ = -x^2 - 11x + 10$$

$$(33) \quad (2x^2 + 7x + 7) - (-x^2 + 9x - 1) \\ = 3x^2 - 2x + 8$$

$$(34) \quad (-9x^2 - 3x + 8) + (7x^2 + 8x + 3) \\ = -2x^2 + 5x + 11$$

$$(35) \quad (-8x^2 + 2x - 4) + (6x^2 + 8x - 7) \\ = -2x^2 + 10x - 11$$

$$(36) \quad (5x^2 + 9x + 4) - (-3x^2 + 8x) \\ = 8x^2 + x + 4$$

$$(37) \quad (3x^2 - 9x + 2) + (-6x^2 - 3x + 2) \\ = -3x^2 - 12x + 4$$

$$(38) \quad (2x^2 - 3) + (x^2 + x - 1) \\ = 3x^2 + x - 4$$

- | | |
|---|--|
| (1) $(3x^2 - 4x - 2) + (2x^2 + 6x + 6)$
$= 5x^2 + 2x + 4$ | (20) $(6x^2 + 6x - 9) + (-2x^2 + 4x - 5)$
$= 4x^2 + 10x - 14$ |
| (2) $(6x^2 + 2x + 9) - (2x^2 - 2x + 9)$
$= 4x^2 + 4x$ | (21) $(7x^2 + 7x - 5) + (9x^2)$
$= 16x^2 + 7x - 5$ |
| (3) $(x^2 - 3x + 6) - (x^2 - 7x + 8)$
$= 4x - 2$ | (22) $(-3x^2) - (-3x^2 + 5x + 1)$
$= -5x - 1$ |
| (4) $(-9x^2 - 7x - 2) - (3x^2 + 4)$
$= -12x^2 - 7x - 6$ | (23) $(-4x^2 + 3x - 8) + (9x^2 - 9x + 4)$
$= 5x^2 - 6x - 4$ |
| (5) $(-5x^2 - 4x - 4) - (6x^2 + 4x + 7)$
$= -11x^2 - 8x - 11$ | (24) $(5x^2 + 5x - 1) + (5x^2 - 3x + 5)$
$= 10x^2 + 2x + 4$ |
| (6) $(6x^2 + 4x) + (-8x^2 - 2x - 4)$
$= -2x^2 + 2x - 4$ | (25) $(-9x^2 + 2x - 2) - (6x^2 + 7x - 5)$
$= -15x^2 - 5x + 3$ |
| (7) $(9x^2 + 4x + 2) - (3x^2 - 5x - 1)$
$= 6x^2 + 9x + 3$ | (26) $(-3x^2 - 7x - 3) - (x^2 - 3x - 8)$
$= -4x^2 - 4x + 5$ |
| (8) $(-3x^2 + 5x + 7) + (-9x^2 - 3x - 3)$
$= -12x^2 + 2x + 4$ | (27) $(6x^2 + 5x - 9) - (-3x^2 - x + 3)$
$= 9x^2 + 6x - 12$ |
| (9) $(-7x^2 + 2x - 2) + (x^2 - 6x + 2)$
$= -6x^2 - 4x$ | (28) $(2x^2 - 7x + 2) - (4x^2 - 4x - 8)$
$= -2x^2 - 3x + 10$ |
| (10) $(5x^2 + 6x - 7) - (-6x^2 + 4x + 1)$
$= 11x^2 + 2x - 8$ | (29) $(4x^2 + x - 3) + (-7x^2 + 9x + 4)$
$= -3x^2 + 10x + 1$ |
| (11) $(-4x^2 - 9x - 9) - (-5x^2 - 7x - 4)$
$= x^2 - 2x - 5$ | (30) $(-4x^2 + 4x) - (-6x^2 - 9x + 2)$
$= 2x^2 + 13x - 2$ |
| (12) $(-6x^2 - 9x + 9) - (-x^2 + 6x - 5)$
$= -5x^2 - 15x + 14$ | (31) $(-4x^2 + 3x + 4) + (3x^2 - 7x - 2)$
$= -x^2 - 4x + 2$ |
| (13) $(-3x^2 + 9x - 4) + (-7x^2 + x + 8)$
$= -10x^2 + 10x + 4$ | (32) $(8x^2 - 5x + 5) - (8x^2 + 5)$
$= -5x$ |
| (14) $(-x^2 - 7x + 2) + (5x^2 - 5x - 8)$
$= 4x^2 - 12x - 6$ | (33) $(3x^2 - 7x - 5) + (-6x^2 - 4x + 7)$
$= -3x^2 - 11x + 2$ |
| (15) $(-6x^2 - 5x - 5) + (3x^2 - 2x)$
$= -3x^2 - 7x - 5$ | (34) $(5x^2 - 2x + 3) - (2x^2 - x - 7)$
$= 3x^2 - x + 10$ |
| (16) $(-x^2 - 7x + 2) - (-8x^2 - 3x + 9)$
$= 7x^2 - 4x - 7$ | (35) $(-6x^2 + 3x + 1) + (-8x^2 - 3x - 9)$
$= -14x^2 - 8$ |
| (17) $(-5x^2 + 8x + 8) + (6x^2 + x - 4)$
$= x^2 + 9x + 4$ | (36) $(6x^2 + x - 6) - (-4x^2 - 7x - 6)$
$= 10x^2 + 8x$ |
| (18) $(6x^2 - 8x + 9) - (6x^2 - 8x + 2)$
$= 7$ | (37) $(-2x^2 + 4x - 2) - (-8x^2 + 8x + 6)$
$= 6x^2 - 4x - 8$ |
| (19) $(3x^2 - 4x - 5) + (-3x^2 + 9x - 4)$
$= 5x - 9$ | (38) $(x^2 - 5x + 3) - (-5x^2 + 4x - 7)$
$= 6x^2 - 9x + 10$ |

- (1) $(8x^2 - 9x + 8) + (6x^2 - 6x - 6)$
 $= 14x^2 - 15x + 2$
- (2) $(9x^2 - 3x) + (-8x^2 - 4x - 8)$
 $= x^2 - 7x - 8$
- (3) $(-2x^2 - 4x - 3) + (-x^2 + 6x + 3)$
 $= -3x^2 + 2x$
- (4) $(6x^2 - 7x - 1) + (4x^2 - 3x + 3)$
 $= 10x^2 - 10x + 2$
- (5) $(-9x^2 + 5x) - (2x^2 + 6x - 3)$
 $= -11x^2 - x + 3$
- (6) $(8x^2 + 9x - 5) - (8x^2 + 9x - 5)$
 $=$
- (7) $(-8x^2 - 4x + 3) + (4x^2 + 5x - 7)$
 $= -4x^2 + x - 4$
- (8) $(9x^2 - 6x - 4) + (-x^2 - 7x + 4)$
 $= 8x^2 - 13x$
- (9) $(-8x^2 + x - 6) + (5x^2 + 4x - 3)$
 $= -3x^2 + 5x - 9$
- (10) $(3x^2 - 2x - 6) - (-8x^2 - 6x - 8)$
 $= 11x^2 + 4x + 2$
- (11) $(8x^2 - x + 7) + (-9x^2 + 2x - 8)$
 $= -x^2 + x - 1$
- (12) $(-9x^2 - 3x + 6) + (-8x^2 + 8x - 7)$
 $= -17x^2 + 5x - 1$
- (13) $(-8x^2 + x - 9) - (6x^2 - 7x + 8)$
 $= -14x^2 + 8x - 17$
- (14) $(-9x^2 - 2x - 3) + (-2x^2 + 7x + 3)$
 $= -11x^2 + 5x$
- (15) $(-x^2 - 7x - 6) - (-9x^2 - 5x - 9)$
 $= 8x^2 - 2x + 3$
- (16) $(9x^2 - 3x + 9) + (-x^2 + 4x)$
 $= 8x^2 + x + 9$
- (17) $(x^2 + 3x + 4) + (-2x^2 + x - 1)$
 $= -x^2 + 4x + 3$
- (18) $(5x^2 - 2x - 3) + (x^2 - 3x - 9)$
 $= 6x^2 - 5x - 12$
- (19) $(4x^2 + 3x) + (9x^2 + 7x - 6)$
 $= 13x^2 + 10x - 6$
- (20) $(-5x^2 - 9x + 2) + (7x^2 - 2x - 9)$
 $= 2x^2 - 11x - 7$
- (21) $(9x^2 + 3x + 7) - (-3x^2 + 3x - 9)$
 $= 12x^2 + 16$
- (22) $(-x^2 - 4x - 8) + (4x^2 + 5x - 4)$
 $= 3x^2 + x - 12$
- (23) $(4x^2 - 7x + 7) + (2x^2 - 3x + 5)$
 $= 6x^2 - 10x + 12$
- (24) $(-9x^2 - 5x + 3) + (-6x^2 + 6x - 1)$
 $= -15x^2 + x + 2$
- (25) $(-6x^2 - 2x + 9) + (-3x^2 + 9)$
 $= -9x^2 - 2x + 18$
- (26) $(6x^2 - 3x + 6) - (-8x^2 - 8x + 8)$
 $= 14x^2 + 5x - 2$
- (27) $(-6x^2 - 4x + 8) - (-5x^2 + 4x + 9)$
 $= -x^2 - 8x - 1$
- (28) $(8x^2 - 7x - 4) + (-6x^2 + 7x + 4)$
 $= 2x^2$
- (29) $(-2x^2 - 9x + 2) - (6x^2 + x + 2)$
 $= -8x^2 - 10x$
- (30) $(-2x^2 + 7x - 8) - (-2x^2 + 7x + 5)$
 $= -13$
- (31) $(6x^2 + x - 6) - (7x^2 - 8x + 4)$
 $= -x^2 + 9x - 10$
- (32) $(6x^2 - 5x + 6) - (-5x^2 - 6x + 6)$
 $= 11x^2 + x$
- (33) $(-9x^2 - x + 8) + (3x^2 + 6x - 8)$
 $= -6x^2 + 5x$
- (34) $(-7x^2 + 6x - 4) - (-8x^2 - 5x - 9)$
 $= x^2 + 11x + 5$
- (35) $(-7x^2 + 5x + 7) - (-5x^2 + 6x - 4)$
 $= -2x^2 - x + 11$
- (36) $(-7x^2 - 6x - 1) - (x^2 - 2x + 9)$
 $= -8x^2 - 4x - 10$
- (37) $(8x^2 - 3x - 7) + (-x^2 - 6x)$
 $= 7x^2 - 9x - 7$
- (38) $(7x^2 + x - 3) - (-x^2 - x - 5)$
 $= 8x^2 + 2x + 2$

- (1) $(7x^2 - 6x + 3) - (-2x^2 + x - 7)$
 $= 9x^2 - 7x + 10$
- (2) $(7x^2 + 7x - 3) + (-7x^2 + 8x + 8)$
 $= 15x + 5$
- (3) $(-2x^2 + 8x + 8) - (-6x^2 + 9x + 7)$
 $= 4x^2 - x + 1$
- (4) $(6x^2 - 4x - 9) - (6x^2 + 8x + 8)$
 $= -12x - 17$
- (5) $(-8x^2 + 8x - 8) - (-6x^2 + 9x + 9)$
 $= -2x^2 - x - 17$
- (6) $(6x^2 + 7x + 1) + (-9x^2 + 4x + 4)$
 $= -3x^2 + 11x + 5$
- (7) $(-5x^2 - 6x + 4) - (-6x^2 - 3x - 2)$
 $= x^2 - 3x + 6$
- (8) $(-9x^2 + 4x + 7) - (-4x^2 - 3x + 5)$
 $= -5x^2 + 7x + 2$
- (9) $(-8x^2 + 2x + 3) + (-9x^2 - 6)$
 $= -17x^2 + 2x - 3$
- (10) $(-7x^2 + 7x - 3) + (7x^2 + 3x - 7)$
 $= 10x - 10$
- (11) $(6x^2 - 3x - 2) - (-6x^2 + x)$
 $= 12x^2 - 4x - 2$
- (12) $(-4x^2 + 3x + 4) + (6x^2 - 4x)$
 $= 2x^2 - x + 4$
- (13) $(-2x^2 - x + 1) + (4x^2 + 7x - 8)$
 $= 2x^2 + 6x - 7$
- (14) $(7x^2 + 5x - 9) - (-7x^2 - 7x - 6)$
 $= 14x^2 + 12x - 3$
- (15) $(7x^2 - 6x + 6) - (9x^2 - 3x)$
 $= -2x^2 - 3x + 6$
- (16) $(3x^2 + 5x - 4) - (5x^2 - 8x + 1)$
 $= -2x^2 + 13x - 5$
- (17) $(4x^2 + 3x - 5) + (-7x^2 - 8x + 2)$
 $= -3x^2 - 5x - 3$
- (18) $(5x^2 + 9x - 5) + (3x^2 - x + 5)$
 $= 8x^2 + 8x$
- (19) $(-7x^2 + 3x - 3) - (-5x^2 - 2x - 8)$
 $= -2x^2 + 5x + 5$
- (20) $(8x^2 + 8x - 5) + (5x^2 - 6x + 8)$
 $= 13x^2 + 2x + 3$
- (21) $(-x^2 - 7x + 5) - (x^2 - 5x - 7)$
 $= -2x^2 - 2x + 12$
- (22) $(7x^2 - 2) - (3x^2 - x - 2)$
 $= 4x^2 + x$
- (23) $(-4x^2 + 8x) - (2x^2 - 6x + 9)$
 $= -6x^2 + 14x - 9$
- (24) $(9x^2 + 6x + 7) + (-2x^2 + 2x + 4)$
 $= 7x^2 + 8x + 11$
- (25) $(-9x^2 - 6x + 5) + (5x^2 - 4x - 3)$
 $= -4x^2 - 10x + 2$
- (26) $(-3x^2 - 9x + 6) - (-9x^2 + 9x + 8)$
 $= 6x^2 - 18x - 2$
- (27) $(9x^2 + 4) + (-3x^2 + 4x + 7)$
 $= 6x^2 + 4x + 11$
- (28) $(6x^2 - 9x) + (-4x^2 - 1)$
 $= 2x^2 - 9x - 1$
- (29) $(6x^2 - 5x) + (5x^2 + x + 6)$
 $= 11x^2 - 4x + 6$
- (30) $(-2x^2 + 5x - 7) - (-8x^2 - 7x + 8)$
 $= 6x^2 + 12x - 15$
- (31) $(-7x^2 - 7x + 2) + (-4x^2 - 3x - 2)$
 $= -11x^2 - 10x$
- (32) $(8x^2 + 5x + 4) - (6x^2 - 7x + 5)$
 $= 2x^2 + 12x - 1$
- (33) $(-4x^2 - x - 1) - (-x^2 + 7x + 1)$
 $= -3x^2 - 8x - 2$
- (34) $(-5x^2 + 3) - (7x^2 + x - 6)$
 $= -12x^2 - x + 9$
- (35) $(x^2 + 8) - (3x^2 - 4x + 1)$
 $= -2x^2 + 4x + 7$
- (36) $(-7x^2 - 2x - 6) + (2x^2 - 9x + 5)$
 $= -5x^2 - 11x - 1$
- (37) $(-6x^2 - 9x + 8) + (3x^2 + 5x + 6)$
 $= -3x^2 - 4x + 14$
- (38) $(-3x^2 + 8x - 3) - (-7x^2 + 9x + 7)$
 $= 4x^2 - x - 10$

- (1) $(-8x^2 - 6x - 1) - (3x^2 - 2)$
 $= -11x^2 - 6x + 1$
- (2) $(-3x^2 + 6) + (2x^2 - 6x)$
 $= -x^2 - 6x + 6$
- (3) $(7x^2 + x) - (-6x^2 - x + 3)$
 $= 13x^2 + 2x - 3$
- (4) $(4x^2 + 8x + 8) + (8x^2 - x + 7)$
 $= 12x^2 + 7x + 15$
- (5) $(2x^2 + 3x - 8) + (4x^2 - 2x - 8)$
 $= 6x^2 + x - 16$
- (6) $(4x^2 + 7x + 1) - (-3x^2 - 4x - 7)$
 $= 7x^2 + 11x + 8$
- (7) $(-3x^2 - 5x - 4) - (-2x^2 + 5x + 4)$
 $= -x^2 - 10x - 8$
- (8) $(-2x^2 + 3x - 3) + (-3x^2 + 9x - 2)$
 $= -5x^2 + 12x - 5$
- (9) $(-2x^2 + 5x + 9) + (4x^2 - 7x - 8)$
 $= 2x^2 - 2x + 1$
- (10) $(7x^2 - 4x + 8) + (-3x^2 + x - 9)$
 $= 4x^2 - 3x - 1$
- (11) $(3x^2 - 5x) + (-5x^2 - 7x - 8)$
 $= -2x^2 - 12x - 8$
- (12) $(3x^2 - 9x - 6) + (7x^2 + 5)$
 $= 10x^2 - 9x - 1$
- (13) $(-7x^2 + 7x) - (4x^2 + 7x - 3)$
 $= -11x^2 + 3$
- (14) $(-2x^2 + 3x + 9) - (8x^2 + 2x + 4)$
 $= -10x^2 + x + 5$
- (15) $(-9x^2 - 4x + 5) + (-6x^2 - x - 8)$
 $= -15x^2 - 5x - 3$
- (16) $(-8x^2 - 3x - 9) + (-6x^2 - 5x + 1)$
 $= -14x^2 - 8x - 8$
- (17) $(-9x^2 + 5) - (4x^2 + 4x + 9)$
 $= -13x^2 - 4x - 4$
- (18) $(6x^2 + x + 1) - (x^2 + 7x - 5)$
 $= 5x^2 - 6x + 6$
- (19) $(7x^2 + x + 7) - (8x^2 + 7x)$
 $= -x^2 - 6x + 7$
- (20) $(-9x^2 - 8) + (9x^2 - 5x - 6)$
 $= -5x - 14$
- (21) $(2x^2 - 8) - (8x^2 + 7x + 3)$
 $= -6x^2 - 7x - 11$
- (22) $(9x^2 - 9x + 5) - (-3x^2 - 8x - 4)$
 $= 12x^2 - x + 9$
- (23) $(9x^2 + 8x) - (-5x^2 + 6x + 5)$
 $= 14x^2 + 2x - 5$
- (24) $(9x^2 - x + 8) - (-7x^2 - 5x - 4)$
 $= 16x^2 + 4x + 12$
- (25) $(3x^2 + 8x - 2) + (-x^2 + 8x + 7)$
 $= 2x^2 + 16x + 5$
- (26) $(2x^2 - 8x + 5) + (6x^2 - 8x - 9)$
 $= 8x^2 - 16x - 4$
- (27) $(-8x^2 - 2) - (-9x^2 - 6x - 2)$
 $= x^2 + 6x$
- (28) $(-x^2 + 8x - 5) - (-4x^2 - 4x + 7)$
 $= 3x^2 + 12x - 12$
- (29) $(-x^2 + 5x) - (3x^2 + x + 1)$
 $= -4x^2 + 4x - 1$
- (30) $(-7x^2 + 7x + 6) + (6x^2 + x - 7)$
 $= -x^2 + 8x - 1$
- (31) $(-6x^2 - 2) + (-x^2 + x + 5)$
 $= -7x^2 + x + 3$
- (32) $(-5x^2 + 4x - 5) + (3x^2 - 3x)$
 $= -2x^2 + x - 5$
- (33) $(8x^2 + 8x - 8) - (-7x^2 - 3x)$
 $= 15x^2 + 11x - 8$
- (34) $(2x^2 + 4x - 7) - (-6x^2 - 9x + 2)$
 $= 8x^2 + 13x - 9$
- (35) $(6x^2 - 8x + 6) + (-5x^2 + 9x - 3)$
 $= x^2 + x + 3$
- (36) $(5x^2 - 7x + 6) - (8x^2 - 3x - 5)$
 $= -3x^2 - 4x + 11$
- (37) $(-9x^2 - 3x) - (-4x^2 + 9x + 7)$
 $= -5x^2 - 12x - 7$
- (38) $(5x^2 - 6x + 4) + (4x^2 - 4)$
 $= 9x^2 - 6x$

- (1) $(9x^2 - 5x - 5) - (6x^2 + 6x - 8)$
 $= 3x^2 - 11x + 3$
- (2) $(-5x^2 - 8x + 6) + (6x^2 + 5x)$
 $= x^2 - 3x + 6$
- (3) $(x^2 - 8x - 6) - (-3x^2 + x - 8)$
 $= 4x^2 - 9x + 2$
- (4) $(-7x^2 - 4x + 5) + (-4x^2 - 5x + 7)$
 $= -11x^2 - 9x + 12$
- (5) $(8x^2 - 9x - 3) - (9x^2 - 8x - 9)$
 $= -x^2 - x + 6$
- (6) $(8x^2 + 4x + 6) - (-9x^2 - 3x)$
 $= 17x^2 + 7x + 6$
- (7) $(-4x^2 + 9x - 7) - (-6x^2 + 6x - 5)$
 $= 2x^2 + 3x - 2$
- (8) $(4x^2 - 5x - 6) + (-9x^2 - 4x + 9)$
 $= -5x^2 - 9x + 3$
- (9) $(x^2 + 2x + 2) - (-6x^2 - 2x + 6)$
 $= 7x^2 + 4x - 4$
- (10) $(-7x^2 - 5x - 8) - (-7x^2 - 9x + 1)$
 $= 4x - 9$
- (11) $(5x^2 - 2x + 3) + (-8x^2 + 6x)$
 $= -3x^2 + 4x + 3$
- (12) $(-4x^2 - 8x + 1) + (-9x^2 + 8x - 9)$
 $= -13x^2 - 8$
- (13) $(7x^2 - 3x - 3) + (6x^2 + 4x + 3)$
 $= 13x^2 + x$
- (14) $(-2x^2 + 3x - 9) - (6x^2 + 2x + 5)$
 $= -8x^2 + x - 14$
- (15) $(-9x^2 + 8x + 4) + (-7x^2 + 3x - 6)$
 $= -16x^2 + 11x - 2$
- (16) $(-2x^2 + 6x - 1) + (-x^2 + 2x - 6)$
 $= -3x^2 + 8x - 7$
- (17) $(-x^2 - 5x + 4) + (-5x^2 - 4x + 7)$
 $= -6x^2 - 9x + 11$
- (18) $(-3x^2 + 6x + 9) - (x^2 + 7x - 5)$
 $= -4x^2 - x + 14$
- (19) $(-8x^2 + 4x - 1) + (9x^2 + 8x - 1)$
 $= x^2 + 12x - 2$
- (20) $(-8x^2 - 4x) + (7x^2 + 6x)$
 $= -x^2 + 2x$
- (21) $(-7x^2 + 8x) + (2x^2 - 9x - 8)$
 $= -5x^2 - x - 8$
- (22) $(7x^2 + 7x - 6) + (4x^2 + 4x + 1)$
 $= 11x^2 + 11x - 5$
- (23) $(x^2 - 5x - 6) + (-3x^2 - 4x - 8)$
 $= -2x^2 - 9x - 14$
- (24) $(5x^2 + 2x - 7) + (-3x^2 + 4x + 3)$
 $= 2x^2 + 6x - 4$
- (25) $(7x^2 + x - 2) - (3x^2 - 3x + 8)$
 $= 4x^2 + 4x - 10$
- (26) $(9x^2 + 2x + 6) - (-8x^2 + 6x - 6)$
 $= 17x^2 - 4x + 12$
- (27) $(-4x^2 - 4x + 3) + (7x^2 - 4x - 5)$
 $= 3x^2 - 8x - 2$
- (28) $(4x^2 + 8x) - (-4x^2 + 7x + 4)$
 $= 8x^2 + x - 4$
- (29) $(-8x^2 - 4x - 6) - (-7x^2 + 5x - 3)$
 $= -x^2 - 9x - 3$
- (30) $(2x^2 - 3x + 9) + (-9x^2 + 3x - 8)$
 $= -7x^2 + 1$
- (31) $(-9x^2 - 2x - 5) + (x^2 - 3x + 7)$
 $= -8x^2 - 5x + 2$
- (32) $(-3x^2 - 3x + 6) - (-7x^2 + 4x + 5)$
 $= 4x^2 - 7x + 1$
- (33) $(2x^2 - 3x + 2) - (3x^2 - 2x + 4)$
 $= -x^2 - x - 2$
- (34) $(7x^2 + x - 6) + (-4x^2 - 4x + 9)$
 $= 3x^2 - 3x + 3$
- (35) $(-8x^2 + 4x - 6) + (9x^2 - 8x - 4)$
 $= x^2 - 4x - 10$
- (36) $(6x^2 + 9x - 3) + (-8x^2 - 6x - 6)$
 $= -2x^2 + 3x - 9$
- (37) $(-3x^2 - 2x + 6) - (7x^2 - 2x)$
 $= -10x^2 + 6$
- (38) $(-x^2 + 7x - 5) - (2x^2 + 5x + 5)$
 $= -3x^2 + 2x - 10$

- | | |
|--|---|
| (1) $(4x^2 + 8x - 6) + (5x^2 + 7x + 3)$
$= 9x^2 + 15x - 3$ | (20) $(-5x^2 - 2x + 1) + (-8x^2 + 4x - 5)$
$= -13x^2 + 2x - 4$ |
| (2) $(6x^2 - 8x - 5) + (8x^2 + x - 5)$
$= 14x^2 - 7x - 10$ | (21) $(4x^2 - 2x - 8) + (6x^2 + 3x - 9)$
$= 10x^2 + x - 17$ |
| (3) $(7x^2 + 6x + 2) + (-x^2 - 6x + 4)$
$= 6x^2 + 6$ | (22) $(9x^2 - 4x - 6) - (-3x^2 + 4x - 4)$
$= 12x^2 - 8x - 2$ |
| (4) $(7x^2 + 8x - 7) + (-5x^2 + 8x + 2)$
$= 2x^2 + 16x - 5$ | (23) $(x^2 - 2x - 5) - (-8x^2 - 7x - 8)$
$= 9x^2 + 5x + 3$ |
| (5) $(8x^2) + (-4x^2 - 9x - 2)$
$= 4x^2 - 9x - 2$ | (24) $(9x^2 + 7x - 9) - (-3x^2 + 9x + 4)$
$= 12x^2 - 2x - 13$ |
| (6) $(-7x^2 - 7x + 4) + (-4x^2 + 9)$
$= -11x^2 - 7x + 13$ | (25) $(7x^2 + 7x - 3) - (-x^2 - 6x - 5)$
$= 8x^2 + 13x + 2$ |
| (7) $(5x^2 + 4x - 3) + (9x^2 + 2x + 5)$
$= 14x^2 + 6x + 2$ | (26) $(2x^2 - 4x) + (-6x^2 + 2x + 8)$
$= -4x^2 - 2x + 8$ |
| (8) $(-4x^2 - x + 9) - (-x^2 - 8x + 5)$
$= -3x^2 + 7x + 4$ | (27) $(2x^2 + 5x - 1) - (4x^2 + 3x - 9)$
$= -2x^2 + 2x + 8$ |
| (9) $(5x^2 - 8x + 1) + (4x^2 - x + 5)$
$= 9x^2 - 9x + 6$ | (28) $(3x^2 - 8x + 5) - (3x^2 + 2x - 5)$
$= -10x + 10$ |
| (10) $(-5x^2 - 8x + 2) + (-7x^2 - 6x - 7)$
$= -12x^2 - 14x - 5$ | (29) $(2x^2 - 4x + 8) + (7x^2 + 5x + 4)$
$= 9x^2 + x + 12$ |
| (11) $(5x^2 - 6x + 3) - (3x^2 - 8x + 3)$
$= 2x^2 + 2x$ | (30) $(-4x^2 + 9x - 2) - (-x^2 - 3x - 7)$
$= -3x^2 + 12x + 5$ |
| (12) $(4x^2 + 2x + 8) + (2x^2 - 8x)$
$= 6x^2 - 6x + 8$ | (31) $(9x^2 + 9x - 7) - (7x^2 + 8x + 9)$
$= 2x^2 + x - 16$ |
| (13) $(3x^2 - x - 3) + (-7x^2 + 4x - 6)$
$= -4x^2 + 3x - 9$ | (32) $(-3x^2 - 7x + 1) - (-7x^2 - x + 3)$
$= 4x^2 - 6x - 2$ |
| (14) $(9x^2 + 9x - 1) + (x^2 - 6x + 1)$
$= 10x^2 + 3x$ | (33) $(7x^2 - 2x + 2) - (-x^2 - 3x + 5)$
$= 8x^2 + x - 3$ |
| (15) $(6x^2 - x + 6) + (2x^2 - 3x + 4)$
$= 8x^2 - 4x + 10$ | (34) $(4x^2) - (-7x^2 - 4x - 4)$
$= 11x^2 + 4x + 4$ |
| (16) $(4x^2 - 7x - 3) - (-8x^2 - 6)$
$= 12x^2 - 7x + 3$ | (35) $(-5x^2 - 4x + 1) - (-2x^2 + x - 8)$
$= -3x^2 - 5x + 9$ |
| (17) $(-x^2 + 6x + 9) - (x^2 + 4x + 9)$
$= -2x^2 + 2x$ | (36) $(8x^2 + 2x - 3) - (9x^2 - 2x - 6)$
$= -x^2 + 4x + 3$ |
| (18) $(2x^2 + 8x + 6) + (4x^2 - x - 7)$
$= 6x^2 + 7x - 1$ | (37) $(-9x^2 + 2x - 4) - (-3x^2 + 2x - 3)$
$= -6x^2 - 1$ |
| (19) $(2x^2 - 4x - 9) + (-2x^2 - 2x - 1)$
$= -6x - 10$ | (38) $(-2x^2 + 5x + 6) + (7x^2 + 9x)$
$= 5x^2 + 14x + 6$ |

- (1) $(-6x^2 - x - 2) - (7x^2 - 5x - 6)$
 $= -13x^2 + 4x + 4$
- (2) $(9x^2 + 9x - 8) - (-8x^2 + 2x + 4)$
 $= 17x^2 + 7x - 12$
- (3) $(5x^2 - 4x + 1) - (8x^2 - 9x + 2)$
 $= -3x^2 + 5x - 1$
- (4) $(-4x^2 + 4x - 5) - (9x^2 + 6x - 5)$
 $= -13x^2 - 2x$
- (5) $(-3x^2 + 5x + 7) - (-7x^2 - 6x + 6)$
 $= 4x^2 + 11x + 1$
- (6) $(5x^2 + 7x + 7) + (4x^2 - 8x + 9)$
 $= 9x^2 - x + 16$
- (7) $(-7x^2 + 6x - 6) - (-3x^2 - 5x - 5)$
 $= -4x^2 + 11x - 1$
- (8) $(-3x^2 + 4x - 6) + (-6x^2 - 4x - 7)$
 $= -9x^2 - 13$
- (9) $(-2x^2 + 3x - 2) - (-5x^2 - 4x + 6)$
 $= 3x^2 + 7x - 8$
- (10) $(2x^2 + 8x - 1) + (5x^2 - x + 8)$
 $= 7x^2 + 7x + 7$
- (11) $(8x^2 - 9x + 9) - (-x^2 - 4x + 2)$
 $= 9x^2 - 5x + 7$
- (12) $(2x^2 + 3x - 3) - (x^2 - 4x - 4)$
 $= x^2 + 7x + 1$
- (13) $(6x^2 - x + 1) + (-9x^2 + 9x - 8)$
 $= -3x^2 + 8x - 7$
- (14) $(-3x^2 + 3x + 9) + (-2x^2 - 2x + 2)$
 $= -5x^2 + x + 11$
- (15) $(-3x^2 + 5x + 4) - (2x^2 + 9x + 9)$
 $= -5x^2 - 4x - 5$
- (16) $(-9x^2 - 5x + 2) + (-x^2 - 6x - 7)$
 $= -10x^2 - 11x - 5$
- (17) $(-4x^2 + 8x - 8) - (3x^2 - x + 1)$
 $= -7x^2 + 9x - 9$
- (18) $(-4x^2 - 3x - 2) - (-8x^2 + 3x - 2)$
 $= 4x^2 - 6x$
- (19) $(-4x^2 + 2x - 5) + (-6x^2 + x + 2)$
 $= -10x^2 + 3x - 3$
- (20) $(-6x^2 - 4x - 1) + (6x^2 + x + 5)$
 $= -3x + 4$
- (21) $(-5x^2 - 4x - 8) - (x^2 + 8x + 1)$
 $= -6x^2 - 12x - 9$
- (22) $(-4x^2 - 8x - 2) - (4x^2 + 5x - 8)$
 $= -8x^2 - 13x + 6$
- (23) $(-x^2 + 4x - 8) - (8x^2 - 2x + 3)$
 $= -9x^2 + 6x - 11$
- (24) $(-7x^2 + 4x - 5) - (-4x^2 - 4x + 4)$
 $= -3x^2 + 8x - 9$
- (25) $(-6x^2 - 5x - 6) + (-7x^2 + 8x - 1)$
 $= -13x^2 + 3x - 7$
- (26) $(x^2 + 4x - 8) + (7x^2 + 6x)$
 $= 8x^2 + 10x - 8$
- (27) $(-4x^2 + 9x - 1) - (-8x^2 + x - 1)$
 $= 4x^2 + 8x$
- (28) $(x^2 + 5x - 3) + (-x^2 - 6x + 4)$
 $= -x + 1$
- (29) $(-2x^2 - 8x - 6) + (-4x^2 + 3x + 3)$
 $= -6x^2 - 5x - 3$
- (30) $(-5x^2 - 2x + 6) - (2x^2 - 2)$
 $= -7x^2 - 2x + 8$
- (31) $(-2x^2 - 3x) - (-2x^2 + 6x + 9)$
 $= -9x - 9$
- (32) $(6x^2 - 9) - (x^2 - 5x - 9)$
 $= 5x^2 + 5x$
- (33) $(-9x^2 - 2x + 6) - (-8x^2 + 7x - 2)$
 $= -x^2 - 9x + 8$
- (34) $(2x^2 - 6x + 9) - (-4x^2 - 2x - 1)$
 $= 6x^2 - 4x + 10$
- (35) $(-6x^2 + 5x + 1) + (8x^2 - 5x - 9)$
 $= 2x^2 - 8$
- (36) $(5x^2 + 4x + 1) + (8x^2 + 6x + 3)$
 $= 13x^2 + 10x + 4$
- (37) $(3x^2 - 6x + 6) - (-2x^2 - 5x - 5)$
 $= 5x^2 - x + 11$
- (38) $(-4x^2 + 9x + 3) - (7x^2 + 2x + 5)$
 $= -11x^2 + 7x - 2$

- (1) $(3x^2 - 8x) - (-4x^2 + 6x)$
 $= 7x^2 - 14x$
- (2) $(-9x^2 - 8x + 7) - (8x^2 - 3x + 8)$
 $= -17x^2 - 5x - 1$
- (3) $(-9x^2) - (-6x^2 - 9x + 5)$
 $= -3x^2 + 9x - 5$
- (4) $(-9x^2 - 9x - 5) - (-8x^2 + 9x - 5)$
 $= -x^2 - 18x$
- (5) $(7x^2 + x - 2) + (-9x^2 - 5x - 1)$
 $= -2x^2 - 4x - 3$
- (6) $(-2x^2 + x - 7) + (2x^2 + 4)$
 $= x - 3$
- (7) $(-6x^2 + 6x - 9) + (-8x^2 - 6x - 6)$
 $= -14x^2 - 15$
- (8) $(-x^2 - 7x - 6) + (-8x^2 - 6x + 6)$
 $= -9x^2 - 13x$
- (9) $(6x^2 + 6x - 6) + (-5x^2 - 8x - 8)$
 $= x^2 - 2x - 14$
- (10) $(-x^2 + 9x + 3) + (2x^2 - 3x - 2)$
 $= x^2 + 6x + 1$
- (11) $(-5x^2 - x - 1) + (-6x^2 + 2)$
 $= -11x^2 - x + 1$
- (12) $(-7x^2 - x + 2) + (7x^2 - 4x - 1)$
 $= -5x + 1$
- (13) $(-9x^2 - 9x) + (-4x^2 + 5x - 2)$
 $= -13x^2 - 4x - 2$
- (14) $(-5x^2 - 5x + 4) + (2x^2 + x - 3)$
 $= -3x^2 - 4x + 1$
- (15) $(-2x^2 + 4x + 2) - (-3x^2 - x - 1)$
 $= x^2 + 5x + 3$
- (16) $(-7x^2 + 3x - 2) + (7x^2 - 3x - 6)$
 $= -8$
- (17) $(9x^2 - 5x + 8) - (4x^2 + 2x + 5)$
 $= 5x^2 - 7x + 3$
- (18) $(-6x^2 - 5x - 4) + (-5x^2 + 2x - 5)$
 $= -11x^2 - 3x - 9$
- (19) $(9x^2 - 8x + 5) + (8x^2 + 9x + 6)$
 $= 17x^2 + x + 11$
- (20) $(6x^2 + x - 3) + (-3x^2 + 4x + 6)$
 $= 3x^2 + 5x + 3$
- (21) $(3x^2 + 4x - 4) + (2x^2 - 6)$
 $= 5x^2 + 4x - 10$
- (22) $(-2x^2 + 7x + 8) + (2x^2 + 3x - 2)$
 $= 10x + 6$
- (23) $(3x^2 - 8x + 2) - (-6x^2 + 6x - 6)$
 $= 9x^2 - 14x + 8$
- (24) $(-8x^2 + 2x - 1) - (5x^2 - 9x + 6)$
 $= -13x^2 + 11x - 7$
- (25) $(-9x^2 - 8x - 9) - (2x^2 + 5x - 4)$
 $= -11x^2 - 13x - 5$
- (26) $(-3x^2 - 7x - 3) + (-5x^2 + 4x + 5)$
 $= -8x^2 - 3x + 2$
- (27) $(2x^2 - 8x - 1) + (-6x^2 + 9x - 7)$
 $= -4x^2 + x - 8$
- (28) $(-x^2 + 5x + 5) + (-3x^2 - 7x + 3)$
 $= -4x^2 - 2x + 8$
- (29) $(2x^2 - 3x - 1) - (-2x^2 - 2x + 6)$
 $= 4x^2 - x - 7$
- (30) $(-4x^2 + 9x - 6) - (-9x^2 - 3x + 8)$
 $= 5x^2 + 12x - 14$
- (31) $(3x^2 + 2x) + (3x^2 + 4x - 6)$
 $= 6x^2 + 6x - 6$
- (32) $(8x^2 + 4x - 2) - (-6x^2 + 8x + 5)$
 $= 14x^2 - 4x - 7$
- (33) $(2x^2 + 7x + 6) - (-4x^2 - 8x + 9)$
 $= 6x^2 + 15x - 3$
- (34) $(x^2 + 4x - 4) + (4x^2 - 5)$
 $= 5x^2 + 4x - 9$
- (35) $(7x^2 - 5x + 7) + (3x^2 - 7x - 2)$
 $= 10x^2 - 12x + 5$
- (36) $(-8x^2 - x + 4) + (8x^2 + 6x - 6)$
 $= 5x - 2$
- (37) $(-6x^2 - x + 2) + (3x^2 - 4x - 6)$
 $= -3x^2 - 5x - 4$
- (38) $(-9x^2 + 5x + 8) + (3x^2 + x + 3)$
 $= -6x^2 + 6x + 11$

- (1) $(2x^2 - 7x + 5) + (-7x^2 - 4x)$
 $= -5x^2 - 11x + 5$
- (2) $(9x^2 - 9x + 5) - (-5x^2 + 4x - 2)$
 $= 14x^2 - 13x + 7$
- (3) $(6x^2 + 7x - 4) - (3x^2 + 9x - 4)$
 $= 3x^2 - 2x$
- (4) $(8x^2 - 6x - 3) - (3x^2 + 3x - 8)$
 $= 5x^2 - 9x + 5$
- (5) $(-9x^2 - 5x + 4) - (-7x^2 + 5x - 4)$
 $= -2x^2 - 10x + 8$
- (6) $(-x^2 + 5x - 2) - (5x^2 - 4x + 1)$
 $= -6x^2 + 9x - 3$
- (7) $(9x^2 - 4x - 4) - (-4x^2 - 6x + 8)$
 $= 13x^2 + 2x - 12$
- (8) $(-6x^2 - 5x - 1) - (-x^2 - 9x + 3)$
 $= -5x^2 + 4x - 4$
- (9) $(x^2 + 9x - 4) + (6x^2 + 8x + 2)$
 $= 7x^2 + 17x - 2$
- (10) $(2x^2 + 8x - 5) - (3x^2 - 3x - 7)$
 $= -x^2 + 11x + 2$
- (11) $(-3x^2 + 4x + 7) - (3x^2 - 5x + 3)$
 $= -6x^2 + 9x + 4$
- (12) $(3x^2 + 5x + 8) + (5x^2 - 3x + 1)$
 $= 8x^2 + 2x + 9$
- (13) $(9x^2 - 7x + 8) - (-2x^2 - 1)$
 $= 11x^2 - 7x + 9$
- (14) $(-3x^2 - 5x + 1) - (-x^2 - 6)$
 $= -2x^2 - 5x + 7$
- (15) $(-7x^2 + x - 6) + (6x^2 + 9x + 3)$
 $= -x^2 + 10x - 3$
- (16) $(4x^2 + 6x - 1) - (-x^2 + 5x)$
 $= 5x^2 + x - 1$
- (17) $(8x^2 + 7x + 5) - (4x^2 + 4x - 4)$
 $= 4x^2 + 3x + 9$
- (18) $(4x^2 + 4x + 6) - (-4x^2 - x + 4)$
 $= 8x^2 + 5x + 2$
- (19) $(-4x^2 - 9x + 5) + (6x^2 - 6x + 3)$
 $= 2x^2 - 15x + 8$
- (20) $(x^2 + 5x - 9) - (-3x^2 - 7x - 3)$
 $= 4x^2 + 12x - 6$
- (21) $(-9x^2 - 8x + 3) + (7x^2 - 3x + 8)$
 $= -2x^2 - 11x + 11$
- (22) $(-6x^2 - 7x - 8) - (6x^2 + 9x - 9)$
 $= -12x^2 - 16x + 1$
- (23) $(-6x^2 - 8x - 6) + (x^2 + x + 3)$
 $= -5x^2 - 7x - 3$
- (24) $(-4x^2 + 6x + 1) - (5x^2 + 7x)$
 $= -9x^2 - x + 1$
- (25) $(6x^2 - 4x - 2) - (2x^2 - 8x - 8)$
 $= 4x^2 + 4x + 6$
- (26) $(7x^2 - 9x + 6) + (5x^2 + 4x + 6)$
 $= 12x^2 - 5x + 12$
- (27) $(-x^2 - 3x - 8) - (-3x^2 + 7x - 4)$
 $= 2x^2 - 10x - 4$
- (28) $(-x^2 + 2x - 3) - (-2x^2 + 2x + 4)$
 $= x^2 - 7$
- (29) $(-3x^2 - 8x + 6) + (-2x^2 + 5x + 2)$
 $= -5x^2 - 3x + 8$
- (30) $(-6x^2 + 8x + 6) - (4x^2 - 6x + 3)$
 $= -10x^2 + 14x + 3$
- (31) $(-4x^2 - 5x + 8) - (-8x^2 + 6x + 7)$
 $= 4x^2 - 11x + 1$
- (32) $(5x^2) + (9x^2 + 3x)$
 $= 14x^2 + 3x$
- (33) $(2x^2 + 3) - (5x^2 + 5x + 5)$
 $= -3x^2 - 5x - 2$
- (34) $(7x^2 + 3x - 6) - (-8x^2 + 4x + 4)$
 $= 15x^2 - x - 10$
- (35) $(8x^2 - 6x + 1) + (x^2 + 2x - 9)$
 $= 9x^2 - 4x - 8$
- (36) $(-x^2 - 5x - 6) - (x^2 + x + 3)$
 $= -2x^2 - 6x - 9$
- (37) $(-8x^2 + x + 1) + (3x^2 - x + 8)$
 $= -5x^2 + 9$
- (38) $(-4x^2 - 4x + 6) - (2x^2 + 9x - 6)$
 $= -6x^2 - 13x + 12$

- (1) $(-x^2 - x - 8) - (3x^2 - 2x + 4)$
 $= -4x^2 + x - 12$
- (2) $(8x^2 + 5x - 2) - (2x^2 + 4x + 1)$
 $= 6x^2 + x - 3$
- (3) $(-5x^2 - 7x) - (-8x^2 + 9x - 3)$
 $= 3x^2 - 16x + 3$
- (4) $(-5x^2 + 6x + 9) + (-8x^2 + 9x - 4)$
 $= -13x^2 + 15x + 5$
- (5) $(6x^2 - 8x + 7) - (3x^2 + 3x - 6)$
 $= 3x^2 - 11x + 13$
- (6) $(-3x^2 - x - 7) + (4x^2 - 3x - 4)$
 $= x^2 - 4x - 11$
- (7) $(-7x^2 + 3x) + (4x^2 + x - 8)$
 $= -3x^2 + 4x - 8$
- (8) $(9x^2 - 5x - 4) - (-7x^2 - 5x + 2)$
 $= 16x^2 - 6$
- (9) $(-2x^2 - 5x) - (-3x^2 + 4x + 5)$
 $= x^2 - 9x - 5$
- (10) $(-4x^2 - 2x - 3) + (-7x^2 - x - 1)$
 $= -11x^2 - 3x - 4$
- (11) $(5x^2 - x - 6) + (x^2 + 8x - 9)$
 $= 6x^2 + 7x - 15$
- (12) $(-5x^2 + 7x + 7) + (3x^2 - x - 7)$
 $= -2x^2 + 6x$
- (13) $(x^2 + x + 2) - (4x^2 + 8x - 8)$
 $= -3x^2 - 7x + 10$
- (14) $(3x^2 - 4x + 2) - (-8x^2 + 3x + 9)$
 $= 11x^2 - 7x - 7$
- (15) $(-4x^2 - 5x + 3) + (3x^2 + 2x - 1)$
 $= -x^2 - 3x + 2$
- (16) $(-3x^2 + 7x + 8) - (-2x^2 + 8x - 2)$
 $= -x^2 - x + 10$
- (17) $(4x^2 - 2x - 5) - (-7x^2 - 9x - 8)$
 $= 11x^2 + 7x + 3$
- (18) $(2x^2 - x + 1) - (7x^2 + 8x + 9)$
 $= -5x^2 - 9x - 8$
- (19) $(4x^2 - 8x - 1) + (-4x^2 - 8x - 8)$
 $= -16x - 9$
- (20) $(x^2 + 3x + 6) + (-9x^2 + 9x + 9)$
 $= -8x^2 + 12x + 15$
- (21) $(8x^2 - 5x + 8) - (3x^2 - 6x)$
 $= 5x^2 + x + 8$
- (22) $(x^2 + 7x - 9) + (-8x^2 + 8x - 5)$
 $= -7x^2 + 15x - 14$
- (23) $(8x^2 + 6) + (2x^2 + 9x - 3)$
 $= 10x^2 + 9x + 3$
- (24) $(-3x^2 - 3x + 8) - (8x^2 + 5x + 5)$
 $= -11x^2 - 8x + 3$
- (25) $(-x^2 - 8) + (9x^2 - 7x)$
 $= 8x^2 - 7x - 8$
- (26) $(-8x^2 - 2x - 1) + (-7x^2 - 3x + 7)$
 $= -15x^2 - 5x + 6$
- (27) $(6x^2 + 8x + 1) + (-8x^2 - 4x - 1)$
 $= -2x^2 + 4x$
- (28) $(7x^2 - 7x + 5) + (9x^2 + 7x - 8)$
 $= 16x^2 - 3$
- (29) $(-8x^2 - 9x + 6) + (-4x^2 - 7x + 4)$
 $= -12x^2 - 16x + 10$
- (30) $(9x^2 + 2x - 8) - (-9x^2 + 2x + 2)$
 $= 18x^2 - 10$
- (31) $(3x^2 + 3x + 4) - (x^2 - 7x)$
 $= 2x^2 + 10x + 4$
- (32) $(7x^2 - x + 7) - (4x^2 + 7x + 1)$
 $= 3x^2 - 8x + 6$
- (33) $(-4x^2 - 8x + 8) + (4x^2 + 2x - 4)$
 $= -6x + 4$
- (34) $(7x^2 + 7x - 3) + (-2x^2 - 2x + 3)$
 $= 5x^2 + 5x$
- (35) $(-x^2 - x + 5) + (-6x^2 - x - 4)$
 $= -7x^2 - 2x + 1$
- (36) $(2x^2 + 9x + 7) - (-6x^2 + 6x - 9)$
 $= 8x^2 + 3x + 16$
- (37) $(5x^2 + 9x - 2) + (8x^2 + 9x + 8)$
 $= 13x^2 + 18x + 6$
- (38) $(3x^2 - 8x - 4) - (-7x^2 + 4x - 7)$
 $= 10x^2 - 12x + 3$

- | | |
|--|--|
| (1) $(2x^2 - 4x - 9) - (2x^2 - 2x - 5)$
$= -2x - 4$ | (20) $(-3x^2 + 6x + 6) - (-8x^2 + 3x + 9)$
$= 5x^2 + 3x - 3$ |
| (2) $(7x^2 - 9x + 1) - (-9x^2 + 3x + 1)$
$= 16x^2 - 12x$ | (21) $(9x^2 + 6x - 5) + (-8x^2 - 7x + 1)$
$= x^2 - x - 4$ |
| (3) $(-6x^2 - 7x - 2) - (-9x^2 - 6x + 3)$
$= 3x^2 - x - 5$ | (22) $(3x^2 + 3x - 9) - (-2x^2 + 8x - 8)$
$= 5x^2 - 5x - 1$ |
| (4) $(6x^2 + 9x) + (7x^2 - 7x + 3)$
$= 13x^2 + 2x + 3$ | (23) $(x^2 + 7x - 7) - (8x^2 + 6x - 1)$
$= -7x^2 + x - 6$ |
| (5) $(-5x^2 + 7x + 5) - (-9x^2 - x + 7)$
$= 4x^2 + 8x - 2$ | (24) $(7x^2 - x + 7) + (4x^2 - 9x - 2)$
$= 11x^2 - 10x + 5$ |
| (6) $(9x^2 + 8x) - (2x^2 - 5x + 7)$
$= 7x^2 + 13x - 7$ | (25) $(-8x^2 + 6x - 9) + (4x^2 - x - 6)$
$= -4x^2 + 5x - 15$ |
| (7) $(2x^2 + 9) - (-8x^2 + 4x + 5)$
$= 10x^2 - 4x + 4$ | (26) $(6x^2 + 6x - 4) + (4x^2 - 4x + 7)$
$= 10x^2 + 2x + 3$ |
| (8) $(-9x^2 - 5x - 2) + (9x^2 + 2)$
$= -5x$ | (27) $(3x^2 - 8x + 2) - (5x^2 + 9x - 3)$
$= -2x^2 - 17x + 5$ |
| (9) $(-3x^2 - 9x + 9) + (3x^2 - 2x - 3)$
$= -11x + 6$ | (28) $(7x^2 + 3x + 9) - (-x^2 + 8x - 2)$
$= 8x^2 - 5x + 11$ |
| (10) $(5x^2 - 8x - 7) - (2x^2 - 6x + 4)$
$= 3x^2 - 2x - 11$ | (29) $(-7x^2 + 6x - 4) + (x^2 - 4)$
$= -6x^2 + 6x - 8$ |
| (11) $(7x^2 - 2x - 8) + (-8x^2 + 7x + 7)$
$= -x^2 + 5x - 1$ | (30) $(3x^2 - 5x - 5) + (-7x^2 + 3x + 1)$
$= -4x^2 - 2x - 4$ |
| (12) $(9x^2 + 5x - 3) - (3x^2 - 3x - 3)$
$= 6x^2 + 8x$ | (31) $(6x^2 + 5x - 5) - (x^2 - 7x + 2)$
$= 5x^2 + 12x - 7$ |
| (13) $(-3x^2 - 5x + 4) + (-6x^2 + 4x - 8)$
$= -9x^2 - x - 4$ | (32) $(-7x^2 - 7x - 6) + (-6x^2 + 4x - 9)$
$= -13x^2 - 3x - 15$ |
| (14) $(2x^2 - 6x + 8) + (-2x^2 - 2x)$
$= -8x + 8$ | (33) $(-7x^2 + 7x + 7) - (-2x^2 + 7x + 1)$
$= -5x^2 + 6$ |
| (15) $(-6x^2 - 2x - 5) - (5x^2 - 9x)$
$= -11x^2 + 7x - 5$ | (34) $(-7x^2 - 5x + 2) - (7x^2 - 3x - 7)$
$= -14x^2 - 2x + 9$ |
| (16) $(-8x^2 - 6x - 6) - (-x^2 + 7x - 8)$
$= -7x^2 - 13x + 2$ | (35) $(4x^2 + 8x + 3) - (-9x^2 - x + 2)$
$= 13x^2 + 9x + 1$ |
| (17) $(6x^2 - 7x + 8) + (x^2 + 4x + 7)$
$= 7x^2 - 3x + 15$ | (36) $(-3x^2 + x - 2) - (7x^2 + x - 2)$
$= -10x^2$ |
| (18) $(5x^2 - 3x - 4) - (x^2 + 2x + 6)$
$= 4x^2 - 5x - 10$ | (37) $(-8x^2 + 3x - 6) - (-8x^2 + 9x - 4)$
$= -6x - 2$ |
| (19) $(3x^2 + 3x - 3) + (-7x^2 + 9x - 1)$
$= -4x^2 + 12x - 4$ | (38) $(x^2 + 6x - 9) + (-4x^2 - 5x - 5)$
$= -3x^2 + x - 14$ |

- (1) $(-x^2 + 6x + 3) - (5x^2 - 5x + 5)$
 $= -6x^2 + 11x - 2$
- (2) $(7x^2 - 8x + 2) + (-8x^2 + 5x + 2)$
 $= -x^2 - 3x + 4$
- (3) $(-2x^2 + 8x + 9) + (3x^2 + 3x + 6)$
 $= x^2 + 11x + 15$
- (4) $(-3x^2 + 4x + 1) + (-8x^2 - 8x + 1)$
 $= -11x^2 - 4x + 2$
- (5) $(-8x^2 + 8x + 9) - (7x^2 + 8x + 2)$
 $= -15x^2 + 7$
- (6) $(6x^2 - 5x - 7) - (-2x^2 - 3x - 2)$
 $= 8x^2 - 2x - 5$
- (7) $(5x^2 + 4x + 2) - (x^2 + 5x + 6)$
 $= 4x^2 - x - 4$
- (8) $(-3x^2 - 3x - 8) + (-4x^2 + 8x - 6)$
 $= -7x^2 + 5x - 14$
- (9) $(-9x^2 + 9x + 9) + (8x^2 + 9x - 8)$
 $= -x^2 + 18x + 1$
- (10) $(-8x^2 - 2x - 2) + (-3x^2 + 6x + 5)$
 $= -11x^2 + 4x + 3$
- (11) $(-6x^2 + 9x) - (-2x^2 + 3x - 1)$
 $= -4x^2 + 6x + 1$
- (12) $(3x^2 + 8x - 8) + (-7x^2 + 3x + 6)$
 $= -4x^2 + 11x - 2$
- (13) $(3x^2 - 2x - 1) - (7x^2 - x - 9)$
 $= -4x^2 - x + 8$
- (14) $(-2x^2 - 6x - 3) + (-9x^2 - 9x + 2)$
 $= -11x^2 - 15x - 1$
- (15) $(-4x^2 + 3x - 5) + (-4x^2 - 3x - 2)$
 $= -8x^2 - 7$
- (16) $(-8x^2 + 5x - 9) - (2x^2 - 3x + 8)$
 $= -10x^2 + 8x - 17$
- (17) $(3x^2 + 7) - (3x^2 - 4x + 8)$
 $= 4x - 1$
- (18) $(8x^2 + 3x + 7) - (8x^2 - 7x - 5)$
 $= 10x + 12$
- (19) $(4x^2 + 4x - 4) - (-6x^2 - x - 8)$
 $= 10x^2 + 5x + 4$
- (20) $(-3x^2 + 9) - (-4x^2 + 4x + 6)$
 $= x^2 - 4x + 3$
- (21) $(-5x^2 - 7x - 7) - (x^2 + 2x - 7)$
 $= -6x^2 - 9x$
- (22) $(-8x^2 - 9x + 3) + (3x^2 - x + 2)$
 $= -5x^2 - 10x + 5$
- (23) $(-8x^2 + 4x) - (-3x^2 - x - 1)$
 $= -5x^2 + 5x + 1$
- (24) $(-5x^2 - 8x + 8) + (4x^2 - 2x - 1)$
 $= -x^2 - 10x + 7$
- (25) $(-4x^2 - 6x) + (7x^2 + x - 2)$
 $= 3x^2 - 5x - 2$
- (26) $(5x^2 - 5x - 9) + (-8x^2 + 5)$
 $= -3x^2 - 5x - 4$
- (27) $(6x^2 - 4x - 8) - (-x^2 - 9x + 8)$
 $= 7x^2 + 5x - 16$
- (28) $(-9x^2 - 5x - 1) - (-9x^2 + 6x - 1)$
 $= -11x$
- (29) $(8x^2 + 9x - 8) - (7x^2 + 5)$
 $= x^2 + 9x - 13$
- (30) $(-5x^2 + 9x - 8) + (-x^2 - 7x + 3)$
 $= -6x^2 + 2x - 5$
- (31) $(-9x^2 - 6x + 7) + (-8x^2 + 2x - 3)$
 $= -17x^2 - 4x + 4$
- (32) $(-6x^2 - 6x + 5) + (-6x^2 + 8x + 8)$
 $= -12x^2 + 2x + 13$
- (33) $(2x^2 + 3) - (9x^2 - 2x + 8)$
 $= -7x^2 + 2x - 5$
- (34) $(-3x^2 - 3x + 7) - (-9x^2 + 4x - 4)$
 $= 6x^2 - 7x + 11$
- (35) $(-4x^2 + 7x) + (2x^2 - 3x + 5)$
 $= -2x^2 + 4x + 5$
- (36) $(5x^2 + 4x - 9) - (5x^2 + 5x + 9)$
 $= -x - 18$
- (37) $(2x^2 - 6x + 8) + (9x^2 + 5x)$
 $= 11x^2 - x + 8$
- (38) $(8x^2 + 4x - 8) - (8x^2 + 9x + 1)$
 $= -5x - 9$

- (1) $(7x^2 + 8x + 7) - (-x^2 - 3x - 4)$
 $= 8x^2 + 11x + 11$
- (2) $(-9x^2 + 4x - 2) - (7x^2 + 3x - 1)$
 $= -16x^2 + x - 1$
- (3) $(3x^2 - 2x) - (-2x^2 - 6x - 8)$
 $= 5x^2 + 4x + 8$
- (4) $(-9x^2 + 2x + 9) + (-7x^2 + 2x - 8)$
 $= -16x^2 + 4x + 1$
- (5) $(x^2 - 7x + 2) - (-7x^2 + 6x - 6)$
 $= 8x^2 - 13x + 8$
- (6) $(-x^2 - 5x - 8) + (-8x^2 - 2x - 3)$
 $= -9x^2 - 7x - 11$
- (7) $(7x^2 + 7x - 3) + (7x^2 + 5)$
 $= 14x^2 + 7x + 2$
- (8) $(x^2 - 9x - 8) - (2x^2 - x - 3)$
 $= -x^2 - 8x - 5$
- (9) $(6x^2 + 3) + (-9x^2 + 4x - 2)$
 $= -3x^2 + 4x + 1$
- (10) $(7x^2 - 7x + 3) - (5x^2 - 8x + 2)$
 $= 2x^2 + x + 1$
- (11) $(9x^2 - 4x) + (3x^2 + 6x + 3)$
 $= 12x^2 + 2x + 3$
- (12) $(2x^2 - 8x - 2) - (6x^2 - 8x - 1)$
 $= -4x^2 - 1$
- (13) $(x^2 - 4x + 8) - (7x^2 - 6)$
 $= -6x^2 - 4x + 14$
- (14) $(x^2 - 3x - 5) + (-2x^2 - 2x)$
 $= -x^2 - 5x - 5$
- (15) $(-6x^2 - 8x - 1) - (7x^2 - x - 6)$
 $= -13x^2 - 7x + 5$
- (16) $(4x^2 - x + 8) + (-3x^2 + 4x - 8)$
 $= x^2 + 3x$
- (17) $(x^2 - 8x + 6) - (-5x^2 - 5x + 2)$
 $= 6x^2 - 3x + 4$
- (18) $(9x^2 + 8x + 1) + (9x^2 + x)$
 $= 18x^2 + 9x + 1$
- (19) $(2x^2 + 9x - 6) + (8x^2 + 4x + 7)$
 $= 10x^2 + 13x + 1$
- (20) $(-4x^2 - 3x - 9) - (-6x^2 + 5x + 4)$
 $= 2x^2 - 8x - 13$
- (21) $(7x^2 + x) - (-x^2 - 4x - 6)$
 $= 8x^2 + 5x + 6$
- (22) $(-4x^2 - 4x + 1) - (-4x^2 + 9x - 4)$
 $= -13x + 5$
- (23) $(-2x^2 + 9) - (-2x^2 - 6x + 9)$
 $= 6x$
- (24) $(-3x^2 - 4x + 5) - (-8x^2 - 2x - 6)$
 $= 5x^2 - 2x + 11$
- (25) $(-7x^2 + 8x - 3) - (-4x^2 + 6x + 1)$
 $= -3x^2 + 2x - 4$
- (26) $(-7x^2 - 2x - 9) + (3x^2)$
 $= -4x^2 - 2x - 9$
- (27) $(-7x^2 + 8x + 5) + (-9x^2 - x + 8)$
 $= -16x^2 + 7x + 13$
- (28) $(-5x^2 - 4x - 8) + (8x^2 + 7x)$
 $= 3x^2 + 3x - 8$
- (29) $(-2x^2 + 6x + 8) - (9x^2 - 7x + 3)$
 $= -11x^2 + 13x + 5$
- (30) $(-7x^2 + x) + (9x^2 + 3x - 5)$
 $= 2x^2 + 4x - 5$
- (31) $(3x^2 - 9x - 2) - (-9x^2 - 2x + 6)$
 $= 12x^2 - 7x - 8$
- (32) $(-4x^2 - 9x - 8) - (-5x^2 + 6x + 1)$
 $= x^2 - 15x - 9$
- (33) $(-x^2 + 3x + 9) - (-7x^2 + 6x)$
 $= 6x^2 - 3x + 9$
- (34) $(-8x^2 + 6x - 3) + (-6x^2 - 3x + 2)$
 $= -14x^2 + 3x - 1$
- (35) $(9x^2 + 9x - 3) + (5x^2 + 9x)$
 $= 14x^2 + 18x - 3$
- (36) $(-6x^2 - 5x - 2) + (-9x^2 - 8x + 5)$
 $= -15x^2 - 13x + 3$
- (37) $(-4x^2 + 7x - 3) - (-2x^2 - 5x - 2)$
 $= -2x^2 + 12x - 1$
- (38) $(-7x^2 - 4x - 5) + (-4x^2 + x + 5)$
 $= -11x^2 - 3x$

- (1) $(2x^2 - x + 7) + (5x^2 + 2x + 8)$
 $= 7x^2 + x + 15$
- (2) $(3x^2 + 9x + 1) - (-9x^2 - 3x + 1)$
 $= 12x^2 + 12x$
- (3) $(-4x^2 + 3x + 3) - (8x^2 - 5x - 4)$
 $= -12x^2 + 8x + 7$
- (4) $(6x^2 + 7x - 3) + (2x^2 + 8x + 1)$
 $= 8x^2 + 15x - 2$
- (5) $(6x^2 - 5x - 6) - (9x^2 + x - 9)$
 $= -3x^2 - 6x + 3$
- (6) $(-6x^2 - 7x - 5) - (2x^2 - 8x - 5)$
 $= -8x^2 + x$
- (7) $(-8x^2 + 5x - 5) - (-9x^2 + 8x - 4)$
 $= x^2 - 3x - 1$
- (8) $(7x^2 - 4) - (-9x^2 - 7x + 9)$
 $= 16x^2 + 7x - 13$
- (9) $(-6x^2 + 8x + 7) + (-4x^2 - 2x)$
 $= -10x^2 + 6x + 7$
- (10) $(9x^2 - 7) - (-4x^2 + 8x)$
 $= 13x^2 - 8x - 7$
- (11) $(9x^2 - 3x - 9) + (-3x^2 - 3x - 6)$
 $= 6x^2 - 6x - 15$
- (12) $(x^2 - 4x + 4) - (-8x^2 + 7x - 7)$
 $= 9x^2 - 11x + 11$
- (13) $(-6x^2 + 8x + 1) + (-7x^2 - 4x + 2)$
 $= -13x^2 + 4x + 3$
- (14) $(8x^2 - 4x - 5) - (-3x^2 + 3x - 9)$
 $= 11x^2 - 7x + 4$
- (15) $(-5x^2 + 3x) + (3x^2 - 5x + 3)$
 $= -2x^2 - 2x + 3$
- (16) $(-6x^2 + 2x + 3) - (-4x^2 - 4x + 8)$
 $= -2x^2 + 6x - 5$
- (17) $(-4x^2 - 3x - 5) - (-3x^2 + 9x - 5)$
 $= -x^2 - 12x$
- (18) $(7x^2 - 6x - 6) + (-4x^2 - 2x + 7)$
 $= 3x^2 - 8x + 1$
- (19) $(8x^2 + x + 9) - (x^2 - 8x + 9)$
 $= 7x^2 + 9x$
- (20) $(-9x^2 - 7x + 1) + (2x^2 + x - 7)$
 $= -7x^2 - 6x - 6$
- (21) $(7x^2 - 8x) - (2x^2 - 3x - 4)$
 $= 5x^2 - 5x + 4$
- (22) $(9x^2 + 8x) - (-2x^2 - x + 3)$
 $= 11x^2 + 9x - 3$
- (23) $(-2x^2 - 2x - 1) - (9x^2 - 5x - 5)$
 $= -11x^2 + 3x + 4$
- (24) $(6x^2 + 4x + 5) - (-3x^2 + 5x + 5)$
 $= 9x^2 - x$
- (25) $(-6x^2 - 2x - 9) - (-5x^2 - 9x + 3)$
 $= -x^2 + 7x - 12$
- (26) $(-9x^2 - 9x + 3) + (-9x^2 + 5x + 8)$
 $= -18x^2 - 4x + 11$
- (27) $(-7x^2 + 9x - 2) + (9x^2 + 6x - 9)$
 $= 2x^2 + 15x - 11$
- (28) $(-7x^2 + 5x - 3) + (7x^2 - 5x)$
 $= -3$
- (29) $(-6x^2 - 6x - 3) + (3x^2 - 6x - 1)$
 $= -3x^2 - 12x - 4$
- (30) $(-2x^2 - 5x + 9) - (4x^2 - 6x - 3)$
 $= -6x^2 + x + 12$
- (31) $(-9x^2 - 7x + 5) + (8x^2 + 2x - 4)$
 $= -x^2 - 5x + 1$
- (32) $(-3x^2 - 2x + 4) + (4x^2 - 2)$
 $= x^2 - 2x + 2$
- (33) $(-8x^2 + 2) - (3x^2 - 5x - 5)$
 $= -11x^2 + 5x + 7$
- (34) $(4x^2 - 7x + 2) + (x^2 - 3x - 5)$
 $= 5x^2 - 10x - 3$
- (35) $(-5x^2 + 3x - 1) - (5x^2 - 2x - 8)$
 $= -10x^2 + 5x + 7$
- (36) $(-x^2 + 8x) + (-8x^2 - 2x - 7)$
 $= -9x^2 + 6x - 7$
- (37) $(5x^2 + 7x - 5) - (-7x^2 - 4x + 8)$
 $= 12x^2 + 11x - 13$
- (38) $(-x^2 - 7x - 8) - (-9x^2 + 3x - 8)$
 $= 8x^2 - 10x$

2.2 乗法

1 次多項式同士の積を求める問題。

(1) $(7x + 4)(6x + 3)$

(16) $(7x + 4)(3x - 8)$

(31) $(x - 9)(8x - 7)$

(2) $(9x - 4)(7x + 5)$

(17) $(x + 9)(3x + 1)$

(32) $(2x - 8)(3x - 4)$

(3) $(5x - 2)(7x + 7)$

(18) $(5x - 4)(x - 9)$

(33) $(3x + 8)(5x - 2)$

(4) $(9x - 6)(4x + 5)$

(19) $(3x - 3)(3x - 6)$

(34) $(6x + 9)(6x - 8)$

(5) $(2x - 7)(4x + 3)$

(20) $(9x + 1)(6x + 7)$

(35) $(3x + 7)(3x - 1)$

(6) $(4x - 5)(2x - 7)$

(21) $(2x + 6)(4x - 4)$

(36) $(x + 9)(7x - 2)$

(7) $(9x - 4)(5x + 6)$

(22) $(3x - 7)(x + 1)$

(37) $(x + 3)(8x - 5)$

(8) $(7x - 9)(3x + 5)$

(23) $(3x - 3)(9x + 4)$

(38) $(7x + 2)(4x - 1)$

(9) $(8x + 2)(4x - 5)$

(24) $(7x - 2)(9x + 9)$

(39) $(6x - 5)(x + 4)$

(10) $(5x + 9)(6x + 3)$

(25) $(2x + 5)(6x - 9)$

(40) $(9x - 1)(3x - 7)$

(11) $(2x + 4)(5x + 4)$

(26) $(9x + 1)(7x + 5)$

(41) $(3x + 2)(5x + 7)$

(12) $(6x - 6)(x - 7)$

(27) $(x + 1)(5x - 4)$

(42) $(9x - 5)(7x + 1)$

(13) $(2x + 3)(4x - 3)$

(28) $(4x - 4)(9x + 3)$

(43) $(4x - 6)(3x + 5)$

(14) $(2x + 7)(5x + 3)$

(29) $(8x + 6)(2x - 9)$

(44) $(6x + 7)(9x + 8)$

(15) $(x + 9)(2x + 1)$

(30) $(9x - 6)(2x + 3)$

(45) $(2x + 5)(8x + 5)$

(1) $(9x + 7)(8x + 5)$

(16) $(7x + 7)(6x - 4)$

(31) $(8x - 3)(x + 9)$

(2) $(4x - 3)(x - 9)$

(17) $(9x + 1)(8x + 6)$

(32) $(6x - 2)(5x + 9)$

(3) $(9x - 1)(8x - 3)$

(18) $(9x - 1)(9x + 9)$

(33) $(9x + 1)(5x + 7)$

(4) $(3x - 2)(6x - 4)$

(19) $(5x + 7)(8x + 7)$

(34) $(5x - 4)(2x + 2)$

(5) $(4x + 2)(4x - 4)$

(20) $(4x + 7)(5x + 4)$

(35) $(9x - 8)(2x + 7)$

(6) $(8x + 9)(6x - 4)$

(21) $(6x + 5)(3x - 9)$

(36) $(8x + 9)(3x - 5)$

(7) $(x + 9)(6x + 1)$

(22) $(7x + 4)(8x + 2)$

(37) $(x - 4)(7x + 5)$

(8) $(9x - 5)(8x - 8)$

(23) $(2x - 9)(x + 4)$

(38) $(3x - 2)(6x - 7)$

(9) $(8x - 4)(9x - 2)$

(24) $(6x + 9)(6x - 8)$

(39) $(4x + 7)(9x + 2)$

(10) $(7x - 7)(4x + 9)$

(25) $(3x - 9)(2x + 5)$

(40) $(7x + 6)(x + 6)$

(11) $(6x + 8)(7x - 3)$

(26) $(5x - 1)(5x - 7)$

(41) $(8x - 2)(3x + 3)$

(12) $(8x + 5)(7x - 4)$

(27) $(x - 1)(2x - 9)$

(42) $(9x + 7)(4x - 5)$

(13) $(6x + 2)(6x + 1)$

(28) $(3x - 4)(2x - 7)$

(43) $(2x + 3)(9x - 4)$

(14) $(9x - 9)(5x - 5)$

(29) $(6x - 7)(x - 5)$

(44) $(6x - 4)(3x + 4)$

(15) $(4x + 2)(7x - 5)$

(30) $(8x + 8)(5x - 8)$

(45) $(9x - 6)(x - 3)$

(1) $(8x + 6)(9x + 5)$

(16) $(6x - 4)(3x - 9)$

(31) $(8x - 9)(5x - 4)$

(2) $(4x + 9)(5x + 2)$

(17) $(x + 3)(2x + 1)$

(32) $(7x - 5)(8x + 2)$

(3) $(3x + 4)(2x - 2)$

(18) $(x - 9)(2x + 5)$

(33) $(3x - 9)(5x - 5)$

(4) $(3x + 1)(5x - 2)$

(19) $(9x - 1)(2x - 5)$

(34) $(8x - 9)^2$

(5) $(4x + 6)(5x + 9)$

(20) $(8x - 9)(6x + 4)$

(35) $(9x + 8)(7x - 5)$

(6) $(6x + 2)(2x + 3)$

(21) $(7x - 4)(4x - 4)$

(36) $(4x + 5)(9x - 6)$

(7) $(9x - 7)(7x + 9)$

(22) $(2x + 9)(6x - 5)$

(37) $(4x - 4)(x - 7)$

(8) $(4x - 1)(9x - 6)$

(23) $(8x + 6)(2x - 1)$

(38) $(5x + 1)(6x - 5)$

(9) $(8x + 3)(2x - 1)$

(24) $(7x + 9)(3x + 8)$

(39) $(x - 2)(9x - 8)$

(10) $(7x - 8)(x - 3)$

(25) $(7x + 2)(2x - 7)$

(40) $(x - 5)(2x - 5)$

(11) $(5x + 7)(9x + 8)$

(26) $(5x - 7)(7x - 1)$

(41) $(9x + 2)(2x + 1)$

(12) $(3x - 9)(8x - 5)$

(27) $(3x - 7)(8x + 2)$

(42) $(x - 7)(2x - 5)$

(13) $(5x + 7)(x - 4)$

(28) $(x + 9)(9x - 3)$

(43) $(4x + 3)(x + 9)$

(14) $(7x - 1)(9x - 1)$

(29) $(8x + 4)(2x + 2)$

(44) $(x - 5)(3x - 7)$

(15) $(8x - 8)(3x - 4)$

(30) $(4x + 5)(6x - 7)$

(45) $(9x + 8)(3x - 3)$

(1) $(8x + 5)(9x + 8)$

(16) $(2x - 5)(2x - 6)$

(31) $(2x + 7)(7x - 5)$

(2) $(2x + 9)(7x - 4)$

(17) $(9x + 4)(7x + 1)$

(32) $(8x + 4)(8x + 7)$

(3) $(7x - 6)(9x + 7)$

(18) $(8x - 8)(5x + 2)$

(33) $(2x - 8)(x + 6)$

(4) $(2x + 8)(4x + 1)$

(19) $(x + 2)(7x - 8)$

(34) $(3x - 9)(7x - 4)$

(5) $(8x - 8)(8x + 5)$

(20) $(x + 8)(9x - 8)$

(35) $(2x + 1)(5x - 5)$

(6) $(2x + 4)(6x + 2)$

(21) $(8x + 4)(9x + 7)$

(36) $(8x + 5)(3x + 3)$

(7) $(4x - 9)(2x - 8)$

(22) $(8x + 9)(8x + 5)$

(37) $(2x + 1)(5x - 4)$

(8) $(x - 5)(6x - 8)$

(23) $(x - 6)(8x - 8)$

(38) $(9x + 3)(6x + 8)$

(9) $(6x + 3)(x - 8)$

(24) $(2x - 7)(8x + 2)$

(39) $(2x + 6)(7x - 3)$

(10) $(9x - 5)(2x - 2)$

(25) $(9x - 6)(2x - 4)$

(40) $(5x + 6)(7x + 6)$

(11) $(6x + 8)(2x + 7)$

(26) $(8x + 2)(3x - 7)$

(41) $(7x + 6)(5x + 9)$

(12) $(9x - 7)(6x - 4)$

(27) $(2x + 3)(2x + 9)$

(42) $(3x + 4)(2x + 7)$

(13) $(8x - 9)(7x + 8)$

(28) $(x + 7)(9x + 9)$

(43) $(7x - 2)(7x + 1)$

(14) $(8x - 1)(8x - 4)$

(29) $(7x - 8)(4x + 5)$

(44) $(2x - 4)(6x - 8)$

(15) $(8x - 6)(6x - 1)$

(30) $(8x + 2)(4x + 1)$

(45) $(3x - 1)(2x + 8)$

(1) $(6x + 3)(6x - 9)$

(16) $(2x - 2)(4x - 2)$

(31) $(4x + 6)(5x + 1)$

(2) $(2x - 6)(6x - 9)$

(17) $(3x - 9)(9x + 6)$

(32) $(9x + 1)(3x - 7)$

(3) $(x + 1)(x - 3)$

(18) $(2x + 4)(2x + 3)$

(33) $(5x + 8)(2x + 9)$

(4) $(2x - 8)(6x - 1)$

(19) $(x + 2)(4x - 5)$

(34) $(2x + 2)(3x - 7)$

(5) $(2x - 6)(6x + 2)$

(20) $(x - 4)(4x + 2)$

(35) $(7x + 2)(7x + 4)$

(6) $(2x - 8)(5x - 1)$

(21) $(7x + 6)(8x - 9)$

(36) $(9x - 9)(2x - 1)$

(7) $(2x + 9)(4x + 5)$

(22) $(5x - 9)(5x + 6)$

(37) $(9x - 3)(6x - 8)$

(8) $(4x - 2)(x + 9)$

(23) $(8x + 7)(6x + 8)$

(38) $(4x + 7)(5x + 4)$

(9) $(5x + 3)(7x - 6)$

(24) $(7x - 7)(9x - 9)$

(39) $(9x + 9)(5x - 8)$

(10) $(4x + 8)(5x + 1)$

(25) $(6x - 5)(9x - 4)$

(40) $(x - 1)(3x + 1)$

(11) $(7x + 5)(7x - 1)$

(26) $(4x + 1)(5x - 2)$

(41) $(x - 6)(4x - 6)$

(12) $(6x + 8)(9x - 5)$

(27) $(8x - 9)(8x - 3)$

(42) $(9x + 5)(6x - 3)$

(13) $(2x + 1)(x + 4)$

(28) $(3x + 7)(5x - 6)$

(43) $(8x + 2)(7x - 3)$

(14) $(3x - 8)(4x + 5)$

(29) $(3x - 2)(6x - 3)$

(44) $(3x - 7)(4x + 1)$

(15) $(2x + 6)(7x + 6)$

(30) $(x - 5)(4x - 3)$

(45) $(8x + 4)(7x - 1)$

(1) $(4x + 6)(x + 1)$

(16) $(2x + 9)(7x + 2)$

(31) $(4x + 7)(3x + 4)$

(2) $(8x + 6)(8x + 2)$

(17) $(5x - 9)(9x + 2)$

(32) $(6x + 8)(8x + 6)$

(3) $(6x - 2)(4x - 8)$

(18) $(8x + 7)(2x + 1)$

(33) $(9x - 7)(7x - 4)$

(4) $(3x - 7)(5x + 4)$

(19) $(6x - 7)(5x + 8)$

(34) $(x - 5)(5x + 7)$

(5) $(5x - 2)(7x + 2)$

(20) $(x + 9)(x - 4)$

(35) $(3x - 1)(6x + 5)$

(6) $(5x + 7)(2x + 6)$

(21) $(3x - 3)(9x - 6)$

(36) $(x - 6)(7x + 4)$

(7) $(4x - 3)(7x - 8)$

(22) $(2x + 2)(4x + 9)$

(37) $(8x - 2)(7x - 8)$

(8) $(6x + 6)(7x - 4)$

(23) $(3x + 4)(x - 8)$

(38) $(7x - 7)(3x + 6)$

(9) $(9x - 9)(8x - 7)$

(24) $(8x + 6)(5x - 8)$

(39) $(6x + 2)(3x - 1)$

(10) $(x - 8)(7x + 3)$

(25) $(6x - 7)(5x + 4)$

(40) $(3x - 7)(7x + 5)$

(11) $(x - 8)(5x + 8)$

(26) $(4x + 1)(4x - 5)$

(41) $(9x - 6)(6x - 6)$

(12) $(3x - 5)(3x - 1)$

(27) $(5x + 1)(3x - 9)$

(42) $(5x - 2)(2x - 6)$

(13) $(8x - 2)(2x + 7)$

(28) $(5x - 7)(8x - 8)$

(43) $(8x - 4)(x - 7)$

(14) $(4x + 4)(7x - 1)$

(29) $(x - 5)(2x + 6)$

(44) $(4x + 1)(x + 5)$

(15) $(2x - 6)(3x + 4)$

(30) $(3x + 6)(2x + 4)$

(45) $(x - 1)(6x + 3)$

(1) $(6x - 6)(5x - 9)$

(16) $(5x + 6)(8x + 8)$

(31) $(2x + 1)(5x - 8)$

(2) $(7x + 7)(2x + 4)$

(17) $(4x + 7)(8x + 5)$

(32) $(7x + 5)(9x - 9)$

(3) $(2x + 4)(3x + 3)$

(18) $(2x - 7)(2x + 8)$

(33) $(7x + 4)(5x + 4)$

(4) $(7x + 6)(5x - 6)$

(19) $(6x - 3)(4x + 4)$

(34) $(x + 1)(8x - 9)$

(5) $(x - 3)(4x + 5)$

(20) $(x - 2)(2x - 1)$

(35) $(9x + 2)(6x + 1)$

(6) $(6x - 3)(3x + 1)$

(21) $(2x - 4)(8x - 3)$

(36) $(3x - 4)(6x + 9)$

(7) $(9x + 4)(3x - 4)$

(22) $(x - 3)(6x + 2)$

(37) $(2x + 1)(9x - 2)$

(8) $(7x - 7)(5x + 4)$

(23) $(6x - 1)(8x + 1)$

(38) $(8x + 9)(5x - 9)$

(9) $(3x + 9)(6x - 4)$

(24) $(7x + 1)(x - 3)$

(39) $(7x + 8)(4x - 6)$

(10) $(2x - 4)(5x - 3)$

(25) $(6x + 6)(6x + 3)$

(40) $(9x + 3)(5x - 3)$

(11) $(9x + 8)(x - 5)$

(26) $(7x + 9)(3x - 2)$

(41) $(7x - 5)(9x + 2)$

(12) $(3x - 2)(2x + 1)$

(27) $(9x - 7)(5x + 7)$

(42) $(6x - 4)(6x + 7)$

(13) $(2x - 9)(9x - 1)$

(28) $(9x + 6)(9x + 8)$

(43) $(8x - 4)(9x - 5)$

(14) $(5x - 8)(x + 5)$

(29) $(4x - 7)(2x + 2)$

(44) $(x + 7)(x - 5)$

(15) $(7x - 4)(7x - 6)$

(30) $(9x - 4)(8x - 4)$

(45) $(5x - 6)(5x + 8)$

(1) $(3x - 7)(4x + 4)$

(16) $(7x + 7)(5x - 4)$

(31) $(3x - 1)(x + 5)$

(2) $(8x - 2)(9x + 2)$

(17) $(9x - 5)(2x - 5)$

(32) $(5x - 4)(4x + 8)$

(3) $(x + 4)(2x + 2)$

(18) $(3x - 8)(2x - 9)$

(33) $(2x - 5)(9x + 2)$

(4) $(x + 5)(3x + 3)$

(19) $(x + 7)(6x - 3)$

(34) $(9x + 3)(8x + 5)$

(5) $(4x + 9)(4x - 3)$

(20) $(2x - 1)(5x + 7)$

(35) $(x - 2)(6x + 7)$

(6) $(2x - 2)(5x - 6)$

(21) $(8x + 4)(5x + 2)$

(36) $(2x + 4)(7x + 9)$

(7) $(4x - 9)(3x - 3)$

(22) $(7x + 2)(4x - 6)$

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(8) $(6x - 8)(4x + 3)$

(23) $(3x - 1)(4x + 1)$

(38) $(8x + 8)(2x - 5)$

(9) $(4x - 7)(6x + 3)$

(24) $(x + 8)(x + 5)$

(39) $(6x + 3)(8x + 2)$

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(25) $(x + 6)(4x + 9)$

(40) $(9x + 9)(9x - 3)$

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(40) $(x - 9)(7x + 8)$

(11) $(8x + 3)(4x - 7)$

(26) $(9x - 6)(2x + 3)$

(41) $(3x + 2)(8x - 1)$

(12) $(6x + 1)(4x + 7)$

(27) $(2x + 3)(6x - 2)$

(42) $(3x - 1)(7x - 9)$

(13) $(6x - 7)(x - 1)$

(28) $(4x - 2)(x - 5)$

(43) $(3x + 5)(2x + 5)$

(14) $(2x + 9)(9x + 1)$

(29) $(9x - 5)(3x + 8)$

(44) $(x + 8)(9x + 8)$

(15) $(x - 8)(x + 5)$

(30) $(x - 7)(7x - 1)$

(45) $(x - 9)(8x + 3)$

(1) $(4x - 7)(7x - 2)$

(16) $(3x + 1)(x + 4)$

(31) $(9x - 9)(2x + 3)$

(2) $(x - 2)(4x + 3)$

(17) $(4x + 1)(x - 8)$

(32) $(5x - 3)(9x + 8)$

(3) $(9x + 6)(3x + 1)$

(18) $(7x - 4)(x + 7)$

(33) $(8x + 8)(4x - 5)$

(4) $(2x + 7)(5x - 4)$

(19) $(6x + 2)(2x - 6)$

(34) $(8x + 4)(4x - 2)$

(5) $(5x - 9)(6x - 4)$

(20) $(3x + 7)(3x + 6)$

(35) $(8x + 9)(x + 6)$

(6) $(5x - 1)(9x + 6)$

(21) $(7x - 8)(2x + 5)$

(36) $(2x - 3)(x + 8)$

(7) $(4x + 3)(8x + 3)$

(22) $(3x + 3)(8x - 5)$

(37) $(2x + 5)(9x - 8)$

(8) $(2x - 7)(6x + 7)$

(23) $(8x + 1)(x + 6)$

(38) $(2x + 5)(4x - 5)$

(9) $(2x + 8)(7x + 4)$

(24) $(9x - 9)(9x + 4)$

(39) $(6x - 2)(9x + 8)$

(10) $(4x + 9)(5x - 7)$

(25) $(7x - 7)(3x - 4)$

(40) $(7x + 2)(6x - 2)$

(11) $(7x - 2)(3x - 4)$

(26) $(5x + 1)(3x - 2)$

(41) $(2x - 7)(8x + 6)$

(12) $(2x - 3)(3x + 1)$

(27) $(2x + 2)(7x + 5)$

(42) $(x + 2)(6x - 7)$

(13) $(2x + 4)(3x - 9)$

(28) $(3x + 1)(x + 1)$

(43) $(9x - 2)(5x - 4)$

(14) $(6x - 3)(5x - 7)$

(29) $(7x + 3)(5x - 8)$

(44) $(5x - 5)(8x - 3)$

(15) $(9x + 3)(x - 3)$

(30) $(2x + 7)(3x - 2)$

(45) $(3x - 7)(8x - 7)$

$$(1) (7x+4)(6x+3) \\ = 42x^2 + 45x + 12$$

$$(2) (9x-4)(7x+5) \\ = 63x^2 + 17x - 20$$

$$(3) (5x-2)(7x+7) \\ = 35x^2 + 21x - 14$$

$$(4) (9x-6)(4x+5) \\ = 36x^2 + 21x - 30$$

$$(5) (2x-7)(4x+3) \\ = 8x^2 - 22x - 21$$

$$(6) (4x-5)(2x-7) \\ = 8x^2 - 38x + 35$$

$$(7) (9x-4)(5x+6) \\ = 45x^2 + 34x - 24$$

$$(8) (7x-9)(3x+5) \\ = 21x^2 + 8x - 45$$

$$(9) (8x+2)(4x-5) \\ = 32x^2 - 32x - 10$$

$$(10) (5x+9)(6x+3) \\ = 30x^2 + 69x + 27$$

$$(11) (2x+4)(5x+4) \\ = 10x^2 + 28x + 16$$

$$(12) (6x-6)(x-7) \\ = 6x^2 - 48x + 42$$

$$(13) (2x+3)(4x-3) \\ = 8x^2 + 6x - 9$$

$$(14) (2x+7)(5x+3) \\ = 10x^2 + 41x + 21$$

$$(15) (x+9)(2x+1) \\ = 2x^2 + 19x + 9$$

$$(16) (7x+4)(3x-8) \\ = 21x^2 - 44x - 32$$

$$(17) (x+9)(3x+1) \\ = 3x^2 + 28x + 9$$

$$(18) (5x-4)(x-9) \\ = 5x^2 - 49x + 36$$

$$(19) (3x-3)(3x-6) \\ = 9x^2 - 27x + 18$$

$$(20) (9x+1)(6x+7) \\ = 54x^2 + 69x + 7$$

$$(21) (2x+6)(4x-4) \\ = 8x^2 + 16x - 24$$

$$(22) (3x-7)(x+1) \\ = 3x^2 - 4x - 7$$

$$(23) (3x-3)(9x+4) \\ = 27x^2 - 15x - 12$$

$$(24) (7x-2)(9x+9) \\ = 63x^2 + 45x - 18$$

$$(25) (2x+5)(6x-9) \\ = 12x^2 + 12x - 45$$

$$(26) (9x+1)(7x+5) \\ = 63x^2 + 52x + 5$$

$$(27) (x+1)(5x-4) \\ = 5x^2 + x - 4$$

$$(28) (4x-4)(9x+3) \\ = 36x^2 - 24x - 12$$

$$(29) (8x+6)(2x-9) \\ = 16x^2 - 60x - 54$$

$$(30) (9x-6)(2x+3) \\ = 18x^2 + 15x - 18$$

$$(31) (x-9)(8x-7) \\ = 8x^2 - 79x + 63$$

$$(32) (2x-8)(3x-4) \\ = 6x^2 - 32x + 32$$

$$(33) (3x+8)(5x-2) \\ = 15x^2 + 34x - 16$$

$$(34) (6x+9)(6x-8) \\ = 36x^2 + 6x - 72$$

$$(35) (3x+7)(3x-1) \\ = 9x^2 + 18x - 7$$

$$(36) (x+9)(7x-2) \\ = 7x^2 + 61x - 18$$

$$(37) (x+3)(8x-5) \\ = 8x^2 + 19x - 15$$

$$(38) (7x+2)(4x-1) \\ = 28x^2 + x - 2$$

$$(39) (6x-5)(x+4) \\ = 6x^2 + 19x - 20$$

$$(40) (9x-1)(3x-7) \\ = 27x^2 - 66x + 7$$

$$(41) (3x+2)(5x+7) \\ = 15x^2 + 31x + 14$$

$$(42) (9x-5)(7x+1) \\ = 63x^2 - 26x - 5$$

$$(43) (4x-6)(3x+5) \\ = 12x^2 + 2x - 30$$

$$(44) (6x+7)(9x+8) \\ = 54x^2 + 111x + 56$$

$$(45) (2x+5)(8x+5) \\ = 16x^2 + 50x + 25$$

$$(1) (9x+7)(8x+5) \\ = 72x^2 + 101x + 35$$

$$(16) (7x+7)(6x-4) \\ = 42x^2 + 14x - 28$$

$$(31) (8x-3)(x+9) \\ = 8x^2 + 69x - 27$$

$$(2) (4x-3)(x-9) \\ = 4x^2 - 39x + 27$$

$$(17) (9x+1)(8x+6) \\ = 72x^2 + 62x + 6$$

$$(32) (6x-2)(5x+9) \\ = 30x^2 + 44x - 18$$

$$(3) (9x-1)(8x-3) \\ = 72x^2 - 35x + 3$$

$$(18) (9x-1)(9x+9) \\ = 81x^2 + 72x - 9$$

$$(33) (9x+1)(5x+7) \\ = 45x^2 + 68x + 7$$

$$(4) (3x-2)(6x-4) \\ = 18x^2 - 24x + 8$$

$$(19) (5x+7)(8x+7) \\ = 40x^2 + 91x + 49$$

$$(34) (5x-4)(2x+2) \\ = 10x^2 + 2x - 8$$

$$(5) (4x+2)(4x-4) \\ = 16x^2 - 8x - 8$$

$$(20) (4x+7)(5x+4) \\ = 20x^2 + 51x + 28$$

$$(35) (9x-8)(2x+7) \\ = 18x^2 + 47x - 56$$

$$(6) (8x+9)(6x-4) \\ = 48x^2 + 22x - 36$$

$$(21) (6x+5)(3x-9) \\ = 18x^2 - 39x - 45$$

$$(36) (8x+9)(3x-5) \\ = 24x^2 - 13x - 45$$

$$(7) (x+9)(6x+1) \\ = 6x^2 + 55x + 9$$

$$(22) (7x+4)(8x+2) \\ = 56x^2 + 46x + 8$$

$$(37) (x-4)(7x+5) \\ = 7x^2 - 23x - 20$$

$$(8) (9x-5)(8x-8) \\ = 72x^2 - 112x + 40$$

$$(23) (2x-9)(x+4) \\ = 2x^2 - x - 36$$

$$(38) (3x-2)(6x-7) \\ = 18x^2 - 33x + 14$$

$$(9) (8x-4)(9x-2) \\ = 72x^2 - 52x + 8$$

$$(24) (6x+9)(6x-8) \\ = 36x^2 + 6x - 72$$

$$(39) (4x+7)(9x+2) \\ = 36x^2 + 71x + 14$$

$$(10) (7x-7)(4x+9) \\ = 28x^2 + 35x - 63$$

$$(25) (3x-9)(2x+5) \\ = 6x^2 - 3x - 45$$

$$(40) (7x+6)(x+6) \\ = 7x^2 + 48x + 36$$

$$(11) (6x+8)(7x-3) \\ = 42x^2 + 38x - 24$$

$$(26) (5x-1)(5x-7) \\ = 25x^2 - 40x + 7$$

$$(41) (8x-2)(3x+3) \\ = 24x^2 + 18x - 6$$

$$(12) (8x+5)(7x-4) \\ = 56x^2 + 3x - 20$$

$$(27) (x-1)(2x-9) \\ = 2x^2 - 11x + 9$$

$$(42) (9x+7)(4x-5) \\ = 36x^2 - 17x - 35$$

$$(13) (6x+2)(6x+1) \\ = 36x^2 + 18x + 2$$

$$(28) (3x-4)(2x-7) \\ = 6x^2 - 29x + 28$$

$$(43) (2x+3)(9x-4) \\ = 18x^2 + 19x - 12$$

$$(14) (9x-9)(5x-5) \\ = 45x^2 - 90x + 45$$

$$(29) (6x-7)(x-5) \\ = 6x^2 - 37x + 35$$

$$(44) (6x-4)(3x+4) \\ = 18x^2 + 12x - 16$$

$$(15) (4x+2)(7x-5) \\ = 28x^2 - 6x - 10$$

$$(30) (8x+8)(5x-8) \\ = 40x^2 - 24x - 64$$

$$(45) (9x-6)(x-3) \\ = 9x^2 - 33x + 18$$

$$(1) (8x+6)(9x+5) \\ = 72x^2 + 94x + 30$$

$$(16) (6x-4)(3x-9) \\ = 18x^2 - 66x + 36$$

$$(31) (8x-9)(5x-4) \\ = 40x^2 - 77x + 36$$

$$(2) (4x+9)(5x+2) \\ = 20x^2 + 53x + 18$$

$$(17) (x+3)(2x+1) \\ = 2x^2 + 7x + 3$$

$$(32) (7x-5)(8x+2) \\ = 56x^2 - 26x - 10$$

$$(3) (3x+4)(2x-2) \\ = 6x^2 + 2x - 8$$

$$(18) (x-9)(2x+5) \\ = 2x^2 - 13x - 45$$

$$(33) (3x-9)(5x-5) \\ = 15x^2 - 60x + 45$$

$$(4) (3x+1)(5x-2) \\ = 15x^2 - x - 2$$

$$(19) (9x-1)(2x-5) \\ = 18x^2 - 47x + 5$$

$$(34) (8x-9)^2 \\ = 64x^2 - 144x + 81$$

$$(5) (4x+6)(5x+9) \\ = 20x^2 + 66x + 54$$

$$(20) (8x-9)(6x+4) \\ = 48x^2 - 22x - 36$$

$$(35) (9x+8)(7x-5) \\ = 63x^2 + 11x - 40$$

$$(6) (6x+2)(2x+3) \\ = 12x^2 + 22x + 6$$

$$(21) (7x-4)(4x-4) \\ = 28x^2 - 44x + 16$$

$$(36) (4x+5)(9x-6) \\ = 36x^2 + 21x - 30$$

$$(7) (9x-7)(7x+9) \\ = 63x^2 + 32x - 63$$

$$(22) (2x+9)(6x-5) \\ = 12x^2 + 44x - 45$$

$$(37) (4x-4)(x-7) \\ = 4x^2 - 32x + 28$$

$$(8) (4x-1)(9x-6) \\ = 36x^2 - 33x + 6$$

$$(23) (8x+6)(2x-1) \\ = 16x^2 + 4x - 6$$

$$(38) (5x+1)(6x-5) \\ = 30x^2 - 19x - 5$$

$$(9) (8x+3)(2x-1) \\ = 16x^2 - 2x - 3$$

$$(24) (7x+9)(3x+8) \\ = 21x^2 + 83x + 72$$

$$(39) (x-2)(9x-8) \\ = 9x^2 - 26x + 16$$

$$(10) (7x-8)(x-3) \\ = 7x^2 - 29x + 24$$

$$(25) (7x+2)(2x-7) \\ = 14x^2 - 45x - 14$$

$$(40) (x-5)(2x-5) \\ = 2x^2 - 15x + 25$$

$$(11) (5x+7)(9x+8) \\ = 45x^2 + 103x + 56$$

$$(26) (5x-7)(7x-1) \\ = 35x^2 - 54x + 7$$

$$(41) (9x+2)(2x+1) \\ = 18x^2 + 13x + 2$$

$$(12) (3x-9)(8x-5) \\ = 24x^2 - 87x + 45$$

$$(27) (3x-7)(8x+2) \\ = 24x^2 - 50x - 14$$

$$(42) (x-7)(2x-5) \\ = 2x^2 - 19x + 35$$

$$(13) (5x+7)(x-4) \\ = 5x^2 - 13x - 28$$

$$(28) (x+9)(9x-3) \\ = 9x^2 + 78x - 27$$

$$(43) (4x+3)(x+9) \\ = 4x^2 + 39x + 27$$

$$(14) (7x-1)(9x-1) \\ = 63x^2 - 16x + 1$$

$$(29) (8x+4)(2x+2) \\ = 16x^2 + 24x + 8$$

$$(44) (x-5)(3x-7) \\ = 3x^2 - 22x + 35$$

$$(15) (8x-8)(3x-4) \\ = 24x^2 - 56x + 32$$

$$(30) (4x+5)(6x-7) \\ = 24x^2 + 2x - 35$$

$$(45) (9x+8)(3x-3) \\ = 27x^2 - 3x - 24$$

$$(1) (8x+5)(9x+8) \\ = 72x^2 + 109x + 40$$

$$(16) (2x-5)(2x-6) \\ = 4x^2 - 22x + 30$$

$$(31) (2x+7)(7x-5) \\ = 14x^2 + 39x - 35$$

$$(2) (2x+9)(7x-4) \\ = 14x^2 + 55x - 36$$

$$(17) (9x+4)(7x+1) \\ = 63x^2 + 37x + 4$$

$$(32) (8x+4)(8x+7) \\ = 64x^2 + 88x + 28$$

$$(3) (7x-6)(9x+7) \\ = 63x^2 - 5x - 42$$

$$(18) (8x-8)(5x+2) \\ = 40x^2 - 24x - 16$$

$$(33) (2x-8)(x+6) \\ = 2x^2 + 4x - 48$$

$$(4) (2x+8)(4x+1) \\ = 8x^2 + 34x + 8$$

$$(19) (x+2)(7x-8) \\ = 7x^2 + 6x - 16$$

$$(34) (3x-9)(7x-4) \\ = 21x^2 - 75x + 36$$

$$(5) (8x-8)(8x+5) \\ = 64x^2 - 24x - 40$$

$$(20) (x+8)(9x-8) \\ = 9x^2 + 64x - 64$$

$$(35) (2x+1)(5x-5) \\ = 10x^2 - 5x - 5$$

$$(6) (2x+4)(6x+2) \\ = 12x^2 + 28x + 8$$

$$(21) (8x+4)(9x+7) \\ = 72x^2 + 92x + 28$$

$$(36) (8x+5)(3x+3) \\ = 24x^2 + 39x + 15$$

$$(7) (4x-9)(2x-8) \\ = 8x^2 - 50x + 72$$

$$(22) (8x+9)(8x+5) \\ = 64x^2 + 112x + 45$$

$$(37) (2x+1)(5x-4) \\ = 10x^2 - 3x - 4$$

$$(8) (x-5)(6x-8) \\ = 6x^2 - 38x + 40$$

$$(23) (x-6)(8x-8) \\ = 8x^2 - 56x + 48$$

$$(38) (9x+3)(6x+8) \\ = 54x^2 + 90x + 24$$

$$(9) (6x+3)(x-8) \\ = 6x^2 - 45x - 24$$

$$(24) (2x-7)(8x+2) \\ = 16x^2 - 52x - 14$$

$$(39) (2x+6)(7x-3) \\ = 14x^2 + 36x - 18$$

$$(10) (9x-5)(2x-2) \\ = 18x^2 - 28x + 10$$

$$(25) (9x-6)(2x-4) \\ = 18x^2 - 48x + 24$$

$$(40) (5x+6)(7x+6) \\ = 35x^2 + 72x + 36$$

$$(11) (6x+8)(2x+7) \\ = 12x^2 + 58x + 56$$

$$(26) (8x+2)(3x-7) \\ = 24x^2 - 50x - 14$$

$$(41) (7x+6)(5x+9) \\ = 35x^2 + 93x + 54$$

$$(12) (9x-7)(6x-4) \\ = 54x^2 - 78x + 28$$

$$(27) (2x+3)(2x+9) \\ = 4x^2 + 24x + 27$$

$$(42) (3x+4)(2x+7) \\ = 6x^2 + 29x + 28$$

$$(13) (8x-9)(7x+8) \\ = 56x^2 + x - 72$$

$$(28) (x+7)(9x+9) \\ = 9x^2 + 72x + 63$$

$$(43) (7x-2)(7x+1) \\ = 49x^2 - 7x - 2$$

$$(14) (8x-1)(8x-4) \\ = 64x^2 - 40x + 4$$

$$(29) (7x-8)(4x+5) \\ = 28x^2 + 3x - 40$$

$$(44) (2x-4)(6x-8) \\ = 12x^2 - 40x + 32$$

$$(15) (8x-6)(6x-1) \\ = 48x^2 - 44x + 6$$

$$(30) (8x+2)(4x+1) \\ = 32x^2 + 16x + 2$$

$$(45) (3x-1)(2x+8) \\ = 6x^2 + 22x - 8$$

$$(1) (6x+3)(6x-9) \\ = 36x^2 - 36x - 27$$

$$(16) (2x-2)(4x-2) \\ = 8x^2 - 12x + 4$$

$$(31) (4x+6)(5x+1) \\ = 20x^2 + 34x + 6$$

$$(2) (2x-6)(6x-9) \\ = 12x^2 - 54x + 54$$

$$(17) (3x-9)(9x+6) \\ = 27x^2 - 63x - 54$$

$$(32) (9x+1)(3x-7) \\ = 27x^2 - 60x - 7$$

$$(3) (x+1)(x-3) \\ = x^2 - 2x - 3$$

$$(18) (2x+4)(2x+3) \\ = 4x^2 + 14x + 12$$

$$(33) (5x+8)(2x+9) \\ = 10x^2 + 61x + 72$$

$$(4) (2x-8)(6x-1) \\ = 12x^2 - 50x + 8$$

$$(19) (x+2)(4x-5) \\ = 4x^2 + 3x - 10$$

$$(34) (2x+2)(3x-7) \\ = 6x^2 - 8x - 14$$

$$(5) (2x-6)(6x+2) \\ = 12x^2 - 32x - 12$$

$$(20) (x-4)(4x+2) \\ = 4x^2 - 14x - 8$$

$$(35) (7x+2)(7x+4) \\ = 49x^2 + 42x + 8$$

$$(6) (2x-8)(5x-1) \\ = 10x^2 - 42x + 8$$

$$(21) (7x+6)(8x-9) \\ = 56x^2 - 15x - 54$$

$$(36) (9x-9)(2x-1) \\ = 18x^2 - 27x + 9$$

$$(7) (2x+9)(4x+5) \\ = 8x^2 + 46x + 45$$

$$(22) (5x-9)(5x+6) \\ = 25x^2 - 15x - 54$$

$$(37) (9x-3)(6x-8) \\ = 54x^2 - 90x + 24$$

$$(8) (4x-2)(x+9) \\ = 4x^2 + 34x - 18$$

$$(23) (8x+7)(6x+8) \\ = 48x^2 + 106x + 56$$

$$(38) (4x+7)(5x+4) \\ = 20x^2 + 51x + 28$$

$$(9) (5x+3)(7x-6) \\ = 35x^2 - 9x - 18$$

$$(24) (7x-7)(9x-9) \\ = 63x^2 - 126x + 63$$

$$(39) (9x+9)(5x-8) \\ = 45x^2 - 27x - 72$$

$$(10) (4x+8)(5x+1) \\ = 20x^2 + 44x + 8$$

$$(25) (6x-5)(9x-4) \\ = 54x^2 - 69x + 20$$

$$(40) (x-1)(3x+1) \\ = 3x^2 - 2x - 1$$

$$(11) (7x+5)(7x-1) \\ = 49x^2 + 28x - 5$$

$$(26) (4x+1)(5x-2) \\ = 20x^2 - 3x - 2$$

$$(41) (x-6)(4x-6) \\ = 4x^2 - 30x + 36$$

$$(12) (6x+8)(9x-5) \\ = 54x^2 + 42x - 40$$

$$(27) (8x-9)(8x-3) \\ = 64x^2 - 96x + 27$$

$$(42) (9x+5)(6x-3) \\ = 54x^2 + 3x - 15$$

$$(13) (2x+1)(x+4) \\ = 2x^2 + 9x + 4$$

$$(28) (3x+7)(5x-6) \\ = 15x^2 + 17x - 42$$

$$(43) (8x+2)(7x-3) \\ = 56x^2 - 10x - 6$$

$$(14) (3x-8)(4x+5) \\ = 12x^2 - 17x - 40$$

$$(29) (3x-2)(6x-3) \\ = 18x^2 - 21x + 6$$

$$(44) (3x-7)(4x+1) \\ = 12x^2 - 25x - 7$$

$$(15) (2x+6)(7x+6) \\ = 14x^2 + 54x + 36$$

$$(30) (x-5)(4x-3) \\ = 4x^2 - 23x + 15$$

$$(45) (8x+4)(7x-1) \\ = 56x^2 + 20x - 4$$

$$(1) (4x+6)(x+1) \\ = 4x^2 + 10x + 6$$

$$(16) (2x+9)(7x+2) \\ = 14x^2 + 67x + 18$$

$$(31) (4x+7)(3x+4) \\ = 12x^2 + 37x + 28$$

$$(2) (8x+6)(8x+2) \\ = 64x^2 + 64x + 12$$

$$(17) (5x-9)(9x+2) \\ = 45x^2 - 71x - 18$$

$$(32) (6x+8)(8x+6) \\ = 48x^2 + 100x + 48$$

$$(3) (6x-2)(4x-8) \\ = 24x^2 - 56x + 16$$

$$(18) (8x+7)(2x+1) \\ = 16x^2 + 22x + 7$$

$$(33) (9x-7)(7x-4) \\ = 63x^2 - 85x + 28$$

$$(4) (3x-7)(5x+4) \\ = 15x^2 - 23x - 28$$

$$(19) (6x-7)(5x+8) \\ = 30x^2 + 13x - 56$$

$$(34) (x-5)(5x+7) \\ = 5x^2 - 18x - 35$$

$$(5) (5x-2)(7x+2) \\ = 35x^2 - 4x - 4$$

$$(20) (x+9)(x-4) \\ = x^2 + 5x - 36$$

$$(35) (3x-1)(6x+5) \\ = 18x^2 + 9x - 5$$

$$(6) (5x+7)(2x+6) \\ = 10x^2 + 44x + 42$$

$$(21) (3x-3)(9x-6) \\ = 27x^2 - 45x + 18$$

$$(36) (x-6)(7x+4) \\ = 7x^2 - 38x - 24$$

$$(7) (4x-3)(7x-8) \\ = 28x^2 - 53x + 24$$

$$(22) (2x+2)(4x+9) \\ = 8x^2 + 26x + 18$$

$$(37) (8x-2)(7x-8) \\ = 56x^2 - 78x + 16$$

$$(8) (6x+6)(7x-4) \\ = 42x^2 + 18x - 24$$

$$(23) (3x+4)(x-8) \\ = 3x^2 - 20x - 32$$

$$(38) (7x-7)(3x+6) \\ = 21x^2 + 21x - 42$$

$$(9) (9x-9)(8x-7) \\ = 72x^2 - 135x + 63$$

$$(24) (8x+6)(5x-8) \\ = 40x^2 - 34x - 48$$

$$(39) (6x+2)(3x-1) \\ = 18x^2 - 2$$

$$(10) (x-8)(7x+3) \\ = 7x^2 - 53x - 24$$

$$(25) (6x-7)(5x+4) \\ = 30x^2 - 11x - 28$$

$$(40) (3x-7)(7x+5) \\ = 21x^2 - 34x - 35$$

$$(11) (x-8)(5x+8) \\ = 5x^2 - 32x - 64$$

$$(26) (4x+1)(4x-5) \\ = 16x^2 - 16x - 5$$

$$(41) (9x-6)(6x-6) \\ = 54x^2 - 90x + 36$$

$$(12) (3x-5)(3x-1) \\ = 9x^2 - 18x + 5$$

$$(27) (5x+1)(3x-9) \\ = 15x^2 - 42x - 9$$

$$(42) (5x-2)(2x-6) \\ = 10x^2 - 34x + 12$$

$$(13) (8x-2)(2x+7) \\ = 16x^2 + 52x - 14$$

$$(28) (5x-7)(8x-8) \\ = 40x^2 - 96x + 56$$

$$(43) (8x-4)(x-7) \\ = 8x^2 - 60x + 28$$

$$(14) (4x+4)(7x-1) \\ = 28x^2 + 24x - 4$$

$$(29) (x-5)(2x+6) \\ = 2x^2 - 4x - 30$$

$$(44) (4x+1)(x+5) \\ = 4x^2 + 21x + 5$$

$$(15) (2x-6)(3x+4) \\ = 6x^2 - 10x - 24$$

$$(30) (3x+6)(2x+4) \\ = 6x^2 + 24x + 24$$

$$(45) (x-1)(6x+3) \\ = 6x^2 - 3x - 3$$

$$(1) (6x-6)(5x-9) \\ = 30x^2 - 84x + 54$$

$$(16) (5x+6)(8x+8) \\ = 40x^2 + 88x + 48$$

$$(31) (2x+1)(5x-8) \\ = 10x^2 - 11x - 8$$

$$(2) (7x+7)(2x+4) \\ = 14x^2 + 42x + 28$$

$$(17) (4x+7)(8x+5) \\ = 32x^2 + 76x + 35$$

$$(32) (7x+5)(9x-9) \\ = 63x^2 - 18x - 45$$

$$(3) (2x+4)(3x+3) \\ = 6x^2 + 18x + 12$$

$$(18) (2x-7)(2x+8) \\ = 4x^2 + 2x - 56$$

$$(33) (7x+4)(5x+4) \\ = 35x^2 + 48x + 16$$

$$(4) (7x+6)(5x-6) \\ = 35x^2 - 12x - 36$$

$$(19) (6x-3)(4x+4) \\ = 24x^2 + 12x - 12$$

$$(34) (x+1)(8x-9) \\ = 8x^2 - x - 9$$

$$(5) (x-3)(4x+5) \\ = 4x^2 - 7x - 15$$

$$(20) (x-2)(2x-1) \\ = 2x^2 - 5x + 2$$

$$(35) (9x+2)(6x+1) \\ = 54x^2 + 21x + 2$$

$$(6) (6x-3)(3x+1) \\ = 18x^2 - 3x - 3$$

$$(21) (2x-4)(8x-3) \\ = 16x^2 - 38x + 12$$

$$(36) (3x-4)(6x+9) \\ = 18x^2 + 3x - 36$$

$$(7) (9x+4)(3x-4) \\ = 27x^2 - 24x - 16$$

$$(22) (x-3)(6x+2) \\ = 6x^2 - 16x - 6$$

$$(37) (2x+1)(9x-2) \\ = 18x^2 + 5x - 2$$

$$(8) (7x-7)(5x+4) \\ = 35x^2 - 7x - 28$$

$$(23) (6x-1)(8x+1) \\ = 48x^2 - 2x - 1$$

$$(38) (8x+9)(5x-9) \\ = 40x^2 - 27x - 81$$

$$(9) (3x+9)(6x-4) \\ = 18x^2 + 42x - 36$$

$$(24) (7x+1)(x-3) \\ = 7x^2 - 20x - 3$$

$$(39) (7x+8)(4x-6) \\ = 28x^2 - 10x - 48$$

$$(10) (2x-4)(5x-3) \\ = 10x^2 - 26x + 12$$

$$(25) (6x+6)(6x+3) \\ = 36x^2 + 54x + 18$$

$$(40) (9x+3)(5x-3) \\ = 45x^2 - 12x - 9$$

$$(11) (9x+8)(x-5) \\ = 9x^2 - 37x - 40$$

$$(26) (7x+9)(3x-2) \\ = 21x^2 + 13x - 18$$

$$(41) (7x-5)(9x+2) \\ = 63x^2 - 31x - 10$$

$$(12) (3x-2)(2x+1) \\ = 6x^2 - x - 2$$

$$(27) (9x-7)(5x+7) \\ = 45x^2 + 28x - 49$$

$$(42) (6x-4)(6x+7) \\ = 36x^2 + 18x - 28$$

$$(13) (2x-9)(9x-1) \\ = 18x^2 - 83x + 9$$

$$(28) (9x+6)(9x+8) \\ = 81x^2 + 126x + 48$$

$$(43) (8x-4)(9x-5) \\ = 72x^2 - 76x + 20$$

$$(14) (5x-8)(x+5) \\ = 5x^2 + 17x - 40$$

$$(29) (4x-7)(2x+2) \\ = 8x^2 - 6x - 14$$

$$(44) (x+7)(x-5) \\ = x^2 + 2x - 35$$

$$(15) (7x-4)(7x-6) \\ = 49x^2 - 70x + 24$$

$$(30) (9x-4)(8x-4) \\ = 72x^2 - 68x + 16$$

$$(45) (5x-6)(5x+8) \\ = 25x^2 + 10x - 48$$

$$(1) (3x-7)(4x+4) \\ = 12x^2 - 16x - 28$$

$$(2) (8x-2)(9x+2) \\ = 72x^2 - 2x - 4$$

$$(3) (x+4)(2x+2) \\ = 2x^2 + 10x + 8$$

$$(4) (x+5)(3x+3) \\ = 3x^2 + 18x + 15$$

$$(5) (4x+9)(4x-3) \\ = 16x^2 + 24x - 27$$

$$(6) (2x-2)(5x-6) \\ = 10x^2 - 22x + 12$$

$$(7) (4x-9)(3x-3) \\ = 12x^2 - 39x + 27$$

$$(8) (6x-8)(4x+3) \\ = 24x^2 - 14x - 24$$

$$(9) (4x-7)(6x+3) \\ = 24x^2 - 30x - 21$$

$$(10) (8x-7)(5x+9) \\ = 40x^2 + 37x - 63$$

$$(11) (3x-7)(8x+6) \\ = 24x^2 - 38x - 42$$

$$(12) (5x+8)(9x-6) \\ = 45x^2 + 42x - 48$$

$$(13) (8x-2)(5x+6) \\ = 40x^2 + 38x - 12$$

$$(14) (5x-8)(7x-3) \\ = 35x^2 - 71x + 24$$

$$(15) (5x+2)(9x-7) \\ = 45x^2 - 17x - 14$$

$$(16) (7x+7)(5x-4) \\ = 35x^2 + 7x - 28$$

$$(17) (9x-5)(2x-5) \\ = 18x^2 - 55x + 25$$

$$(18) (3x-8)(2x-9) \\ = 6x^2 - 43x + 72$$

$$(19) (x+7)(6x-3) \\ = 6x^2 + 39x - 21$$

$$(20) (2x-1)(5x+7) \\ = 10x^2 + 9x - 7$$

$$(21) (8x+4)(5x+2) \\ = 40x^2 + 36x + 8$$

$$(22) (7x+2)(4x-6) \\ = 28x^2 - 34x - 12$$

$$(23) (3x-1)(4x+1) \\ = 12x^2 - x - 1$$

$$(24) (x+8)(x+5) \\ = x^2 + 13x + 40$$

$$(25) (x+6)(4x+9) \\ = 4x^2 + 33x + 54$$

$$(26) (4x-3)(9x+3) \\ = 36x^2 - 15x - 9$$

$$(27) (6x+7)(7x-5) \\ = 42x^2 + 19x - 35$$

$$(28) (8x+7)(4x+7) \\ = 32x^2 + 84x + 49$$

$$(29) (8x+7)(5x+1) \\ = 40x^2 + 43x + 7$$

$$(30) (5x+1)(2x+7) \\ = 10x^2 + 37x + 7$$

$$(31) (3x-1)(x+5) \\ = 3x^2 + 14x - 5$$

$$(32) (5x-4)(4x+8) \\ = 20x^2 + 24x - 32$$

$$(33) (2x-5)(9x+2) \\ = 18x^2 - 41x - 10$$

$$(34) (9x+3)(8x+5) \\ = 72x^2 + 69x + 15$$

$$(35) (x-2)(6x+7) \\ = 6x^2 - 5x - 14$$

$$(36) (2x+4)(7x+9) \\ = 14x^2 + 46x + 36$$

$$(37) (5x+1)(8x+3) \\ = 40x^2 + 23x + 3$$

$$(38) (8x+8)(2x-5) \\ = 16x^2 - 24x - 40$$

$$(39) (6x+3)(8x+2) \\ = 48x^2 + 36x + 6$$

$$(40) (9x+9)(9x-3) \\ = 81x^2 + 54x - 27$$

$$(41) (7x-5)(7x-2) \\ = 49x^2 - 49x + 10$$

$$(42) (2x+2)(x+6) \\ = 2x^2 + 14x + 12$$

$$(43) (2x+1)(8x+3) \\ = 16x^2 + 14x + 3$$

$$(44) (2x-4)(3x+1) \\ = 6x^2 - 10x - 4$$

$$(45) (4x+4)(7x+4) \\ = 28x^2 + 44x + 16$$

$$(1) (2x+7)(9x-2) \\ = 18x^2 + 59x - 14$$

$$(2) (3x+8)(7x-4) \\ = 21x^2 + 44x - 32$$

$$(3) (5x-2)(3x+4) \\ = 15x^2 + 14x - 8$$

$$(4) (7x+4)(4x+9) \\ = 28x^2 + 79x + 36$$

$$(5) (7x+1)(4x+6) \\ = 28x^2 + 46x + 6$$

$$(6) (3x-2)(5x-8) \\ = 15x^2 - 34x + 16$$

$$(7) (2x+9)(2x-6) \\ = 4x^2 + 6x - 54$$

$$(8) (4x+6)(3x-5) \\ = 12x^2 - 2x - 30$$

$$(9) (6x+2)(4x+1) \\ = 24x^2 + 14x + 2$$

$$(10) (x+6)(x+4) \\ = x^2 + 10x + 24$$

$$(11) (9x+3)(6x-7) \\ = 54x^2 - 45x - 21$$

$$(12) (5x+6)(6x-9) \\ = 30x^2 - 9x - 54$$

$$(13) (3x+1)(7x-2) \\ = 21x^2 + x - 2$$

$$(14) (8x-1)(x+7) \\ = 8x^2 + 55x - 7$$

$$(15) (x+5)(7x-7) \\ = 7x^2 + 28x - 35$$

$$(16) (9x+1)(8x-4) \\ = 72x^2 - 28x - 4$$

$$(17) (4x-9)(x-7) \\ = 4x^2 - 37x + 63$$

$$(18) (5x-8)(8x+5) \\ = 40x^2 - 39x - 40$$

$$(19) (x+5)(9x-5) \\ = 9x^2 + 40x - 25$$

$$(20) (6x-8)(4x-2) \\ = 24x^2 - 44x + 16$$

$$(21) (4x+3)(6x-2) \\ = 24x^2 + 10x - 6$$

$$(22) (9x+8)(9x+2) \\ = 81x^2 + 90x + 16$$

$$(23) (2x-4)(8x+8) \\ = 16x^2 - 16x - 32$$

$$(24) (5x+3)(5x-4) \\ = 25x^2 - 5x - 12$$

$$(25) (2x+3)(7x-7) \\ = 14x^2 + 7x - 21$$

$$(26) (3x-2)(3x+2) \\ = 9x^2 - 4$$

$$(27) (5x-7)(8x-8) \\ = 40x^2 - 96x + 56$$

$$(28) (2x-7)(3x-9) \\ = 6x^2 - 39x + 63$$

$$(29) (4x+6)(4x+2) \\ = 16x^2 + 32x + 12$$

$$(30) (6x+5)(2x-7) \\ = 12x^2 - 32x - 35$$

$$(31) (4x-9)(7x-7) \\ = 28x^2 - 91x + 63$$

$$(32) (3x-2)(6x+4) \\ = 18x^2 - 8$$

$$(33) (2x-5)(7x-7) \\ = 14x^2 - 49x + 35$$

$$(34) (2x+7)(8x-1) \\ = 16x^2 + 54x - 7$$

$$(35) (8x+8)(7x+8) \\ = 56x^2 + 120x + 64$$

$$(36) (8x-3)(2x+5) \\ = 16x^2 + 34x - 15$$

$$(37) (3x+5)(9x-1) \\ = 27x^2 + 42x - 5$$

$$(38) (9x-2)(x+1) \\ = 9x^2 + 7x - 2$$

$$(39) (8x-4)(9x+5) \\ = 72x^2 + 4x - 20$$

$$(40) (7x-4)(5x-5) \\ = 35x^2 - 55x + 20$$

$$(41) (3x-3)(4x-9) \\ = 12x^2 - 39x + 27$$

$$(42) (8x-5)(9x-2) \\ = 72x^2 - 61x + 10$$

$$(43) (8x-9)(3x+6) \\ = 24x^2 + 21x - 54$$

$$(44) (6x-4)(7x-7) \\ = 42x^2 - 70x + 28$$

$$(45) (4x-4)(5x-4) \\ = 20x^2 - 36x + 16$$

$$(1) (3x+1)(7x+7) \\ = 21x^2 + 28x + 7$$

$$(2) (8x+4)(x+9) \\ = 8x^2 + 76x + 36$$

$$(3) (5x-9)(9x+7) \\ = 45x^2 - 46x - 63$$

$$(4) (5x+5)(8x-7) \\ = 40x^2 + 5x - 35$$

$$(5) (2x-9)(7x+9) \\ = 14x^2 - 45x - 81$$

$$(6) (8x-7)(5x-5) \\ = 40x^2 - 75x + 35$$

$$(7) (3x+1)(5x+6) \\ = 15x^2 + 23x + 6$$

$$(8) (7x-6)(6x+7) \\ = 42x^2 + 13x - 42$$

$$(9) (4x+2)(3x+8) \\ = 12x^2 + 38x + 16$$

$$(10) (7x+8)(6x+5) \\ = 42x^2 + 83x + 40$$

$$(11) (8x-5)(2x+6) \\ = 16x^2 + 38x - 30$$

$$(12) (6x-2)(6x+3) \\ = 36x^2 + 6x - 6$$

$$(13) (9x-4)(3x-1) \\ = 27x^2 - 21x + 4$$

$$(14) (4x+6)(4x+2) \\ = 16x^2 + 32x + 12$$

$$(15) (8x-1)(8x+1) \\ = 64x^2 - 1$$

$$(16) (6x-9)(5x+4) \\ = 30x^2 - 21x - 36$$

$$(17) (6x-6)(x+6) \\ = 6x^2 + 30x - 36$$

$$(18) (8x-6)(2x-1) \\ = 16x^2 - 20x + 6$$

$$(19) (5x+1)(7x+1) \\ = 35x^2 + 12x + 1$$

$$(20) (6x-3)(7x-4) \\ = 42x^2 - 45x + 12$$

$$(21) (4x-6)(5x-6) \\ = 20x^2 - 54x + 36$$

$$(22) (6x+6)(3x+3) \\ = 18x^2 + 36x + 18$$

$$(23) (6x-2)(7x+1) \\ = 42x^2 - 8x - 2$$

$$(24) (6x-9)(5x+3) \\ = 30x^2 - 27x - 27$$

$$(25) (5x+1)(3x+1) \\ = 15x^2 + 8x + 1$$

$$(26) (3x-1)(9x-8) \\ = 27x^2 - 33x + 8$$

$$(27) (8x-3)(7x+6) \\ = 56x^2 + 27x - 18$$

$$(28) (6x-4)(3x+4) \\ = 18x^2 + 12x - 16$$

$$(29) (5x+3)(6x-5) \\ = 30x^2 - 7x - 15$$

$$(30) (6x-2)(3x-9) \\ = 18x^2 - 60x + 18$$

$$(31) (x+8)(7x-8) \\ = 7x^2 + 48x - 64$$

$$(32) (8x-9)(x-7) \\ = 8x^2 - 65x + 63$$

$$(33) (2x+3)(x-1) \\ = 2x^2 + x - 3$$

$$(34) (x-3)(8x+4) \\ = 8x^2 - 20x - 12$$

$$(35) (6x-7)(6x-6) \\ = 36x^2 - 78x + 42$$

$$(36) (6x+1)(5x-7) \\ = 30x^2 - 37x - 7$$

$$(37) (7x-6)(4x-3) \\ = 28x^2 - 45x + 18$$

$$(38) (6x-2)(5x+6) \\ = 30x^2 + 26x - 12$$

$$(39) (7x-9)(5x+3) \\ = 35x^2 - 24x - 27$$

$$(40) (4x+3)(x+7) \\ = 4x^2 + 31x + 21$$

$$(41) (x+3)(7x-9) \\ = 7x^2 + 12x - 27$$

$$(42) (4x+8)(8x+8) \\ = 32x^2 + 96x + 64$$

$$(43) (4x-1)(4x+8) \\ = 16x^2 + 28x - 8$$

$$(44) (8x-3)(3x-1) \\ = 24x^2 - 17x + 3$$

$$(45) (7x+3)(x-9) \\ = 7x^2 - 60x - 27$$

$$(1) (4x - 2)(7x + 9) \\ = 28x^2 + 22x - 18$$

$$(2) (5x - 2)(3x - 8) \\ = 15x^2 - 46x + 16$$

$$(3) (8x + 7)(5x + 1) \\ = 40x^2 + 43x + 7$$

$$(4) (8x - 4)(8x - 8) \\ = 64x^2 - 96x + 32$$

$$(5) (2x + 4)(4x + 1) \\ = 8x^2 + 18x + 4$$

$$(6) (8x - 1)(6x - 7) \\ = 48x^2 - 62x + 7$$

$$(7) (4x - 4)(5x + 6) \\ = 20x^2 + 4x - 24$$

$$(8) (7x - 6)(8x + 6) \\ = 56x^2 - 6x - 36$$

$$(9) (7x + 5)(4x + 4) \\ = 28x^2 + 48x + 20$$

$$(10) (6x + 3)(5x + 6) \\ = 30x^2 + 51x + 18$$

$$(11) (4x - 8)(x + 6) \\ = 4x^2 + 16x - 48$$

$$(12) (2x - 4)(6x - 6) \\ = 12x^2 - 36x + 24$$

$$(13) (8x + 7)(8x - 3) \\ = 64x^2 + 32x - 21$$

$$(14) (2x + 7)(4x - 1) \\ = 8x^2 + 26x - 7$$

$$(15) (2x - 8)(8x + 2) \\ = 16x^2 - 60x - 16$$

$$(16) (8x + 7)(2x - 2) \\ = 16x^2 - 2x - 14$$

$$(17) (2x - 9)(x - 6) \\ = 2x^2 - 21x + 54$$

$$(18) (4x + 5)(8x - 8) \\ = 32x^2 + 8x - 40$$

$$(19) (2x + 4)(2x + 5) \\ = 4x^2 + 18x + 20$$

$$(20) (6x - 6)(x - 3) \\ = 6x^2 - 24x + 18$$

$$(21) (5x + 5)(7x + 8) \\ = 35x^2 + 75x + 40$$

$$(22) (5x + 4)(x + 3) \\ = 5x^2 + 19x + 12$$

$$(23) (8x + 3)(x + 5) \\ = 8x^2 + 43x + 15$$

$$(24) (7x + 2)(5x - 3) \\ = 35x^2 - 11x - 6$$

$$(25) (9x - 8)(9x + 9) \\ = 81x^2 + 9x - 72$$

$$(26) (x + 8)(4x - 1) \\ = 4x^2 + 31x - 8$$

$$(27) (3x + 6)(x - 5) \\ = 3x^2 - 9x - 30$$

$$(28) (5x - 5)(4x + 8) \\ = 20x^2 + 20x - 40$$

$$(29) (2x - 9)(3x + 6) \\ = 6x^2 - 15x - 54$$

$$(30) (9x - 1)(x - 9) \\ = 9x^2 - 82x + 9$$

$$(31) (9x - 5)(4x - 1) \\ = 36x^2 - 29x + 5$$

$$(32) (6x - 9)(2x + 3) \\ = 12x^2 - 27$$

$$(33) (7x - 3)(5x + 7) \\ = 35x^2 + 34x - 21$$

$$(34) (6x + 6)(3x + 8) \\ = 18x^2 + 66x + 48$$

$$(35) (3x - 3)(4x - 4) \\ = 12x^2 - 24x + 12$$

$$(36) (2x + 7)(2x + 4) \\ = 4x^2 + 22x + 28$$

$$(37) (4x + 3)(2x + 9) \\ = 8x^2 + 42x + 27$$

$$(38) (9x + 8)(8x - 2) \\ = 72x^2 + 46x - 16$$

$$(39) (4x + 6)(7x + 3) \\ = 28x^2 + 54x + 18$$

$$(40) (7x - 1)(7x + 6) \\ = 49x^2 + 35x - 6$$

$$(41) (9x + 2)(x - 9) \\ = 9x^2 - 79x - 18$$

$$(42) (6x - 2)(3x - 6) \\ = 18x^2 - 42x + 12$$

$$(43) (3x + 7)(2x + 4) \\ = 6x^2 + 26x + 28$$

$$(44) (9x - 6)(8x - 2) \\ = 72x^2 - 66x + 12$$

$$(45) (2x + 7)(5x + 4) \\ = 10x^2 + 43x + 28$$

$$(1) (2x+4)(x-1) \\ = 2x^2 + 2x - 4$$

$$(2) (7x-5)(7x+9) \\ = 49x^2 + 28x - 45$$

$$(3) (x+8)^2 \\ = x^2 + 16x + 64$$

$$(4) (9x+5)(6x-4) \\ = 54x^2 - 6x - 20$$

$$(5) (9x-6)(5x+6) \\ = 45x^2 + 24x - 36$$

$$(6) (4x-6)(7x+3) \\ = 28x^2 - 30x - 18$$

$$(7) (8x+4)(3x+3) \\ = 24x^2 + 36x + 12$$

$$(8) (4x+7)(8x+9) \\ = 32x^2 + 92x + 63$$

$$(9) (5x-4)(6x+6) \\ = 30x^2 + 6x - 24$$

$$(10) (2x-9)(3x-5) \\ = 6x^2 - 37x + 45$$

$$(11) (9x+4)(2x-4) \\ = 18x^2 - 28x - 16$$

$$(12) (5x+6)(x-8) \\ = 5x^2 - 34x - 48$$

$$(13) (7x+1)(x-4) \\ = 7x^2 - 27x - 4$$

$$(14) (4x-8)(5x+8) \\ = 20x^2 - 8x - 64$$

$$(15) (5x-1)(3x-4) \\ = 15x^2 - 23x + 4$$

$$(16) (x-2)(7x+3) \\ = 7x^2 - 11x - 6$$

$$(17) (7x-3)(3x-2) \\ = 21x^2 - 23x + 6$$

$$(18) (2x+8)(7x-9) \\ = 14x^2 + 38x - 72$$

$$(19) (4x+8)(4x+7) \\ = 16x^2 + 60x + 56$$

$$(20) (6x-7)(7x+4) \\ = 42x^2 - 25x - 28$$

$$(21) (6x+5)(7x+9) \\ = 42x^2 + 89x + 45$$

$$(22) (4x+8)(9x-8) \\ = 36x^2 + 40x - 64$$

$$(23) (5x+7)(4x+6) \\ = 20x^2 + 58x + 42$$

$$(24) (4x+1)(2x-2) \\ = 8x^2 - 6x - 2$$

$$(25) (5x+7)(7x-6) \\ = 35x^2 + 19x - 42$$

$$(26) (8x-7)(2x+3) \\ = 16x^2 + 10x - 21$$

$$(27) (9x+6)(8x-7) \\ = 72x^2 - 15x - 42$$

$$(28) (3x+7)(4x+1) \\ = 12x^2 + 31x + 7$$

$$(29) (7x-7)(x+3) \\ = 7x^2 + 14x - 21$$

$$(30) (8x-8)(5x-1) \\ = 40x^2 - 48x + 8$$

$$(31) (6x+6)(4x+3) \\ = 24x^2 + 42x + 18$$

$$(32) (4x+2)(4x+9) \\ = 16x^2 + 44x + 18$$

$$(33) (8x+2)(5x-5) \\ = 40x^2 - 30x - 10$$

$$(34) (8x-1)(2x+1) \\ = 16x^2 + 6x - 1$$

$$(35) (4x-2)(8x+8) \\ = 32x^2 + 16x - 16$$

$$(36) (x-8)(4x-9) \\ = 4x^2 - 41x + 72$$

$$(37) (4x-5)(x+9) \\ = 4x^2 + 31x - 45$$

$$(38) (7x+5)(8x-3) \\ = 56x^2 + 19x - 15$$

$$(39) (2x+9)(3x+9) \\ = 6x^2 + 45x + 81$$

$$(40) (2x+4)(2x-5) \\ = 4x^2 - 2x - 20$$

$$(41) (2x-8)(2x+6) \\ = 4x^2 - 4x - 48$$

$$(42) (x+9)(6x+8) \\ = 6x^2 + 62x + 72$$

$$(43) (6x-9)(3x+7) \\ = 18x^2 + 15x - 63$$

$$(44) (9x-1)(3x-8) \\ = 27x^2 - 75x + 8$$

$$(45) (4x+9)(6x-6) \\ = 24x^2 + 30x - 54$$

$$(1) (6x+3)(4x+2) \\ = 24x^2 + 24x + 6$$

$$(2) (6x+6)(2x-5) \\ = 12x^2 - 18x - 30$$

$$(3) (x+4)(7x-3) \\ = 7x^2 + 25x - 12$$

$$(4) (3x-2)(5x+9) \\ = 15x^2 + 17x - 18$$

$$(5) (5x+5)(9x+8) \\ = 45x^2 + 85x + 40$$

$$(6) (5x+2)(9x+4) \\ = 45x^2 + 38x + 8$$

$$(7) (x+5)(9x-7) \\ = 9x^2 + 38x - 35$$

$$(8) (6x+2)(x+2) \\ = 6x^2 + 14x + 4$$

$$(9) (3x-6)(5x+3) \\ = 15x^2 - 21x - 18$$

$$(10) (x+4)(x-5) \\ = x^2 - x - 20$$

$$(11) (3x+8)(2x-5) \\ = 6x^2 + x - 40$$

$$(12) (x+3)(7x-9) \\ = 7x^2 + 12x - 27$$

$$(13) (6x+6)(7x+7) \\ = 42x^2 + 84x + 42$$

$$(14) (7x+5)(5x-2) \\ = 35x^2 + 11x - 10$$

$$(15) (5x-5)(3x+5) \\ = 15x^2 + 10x - 25$$

$$(16) (3x-4)(7x+4) \\ = 21x^2 - 16x - 16$$

$$(17) (x-3)(9x-6) \\ = 9x^2 - 33x + 18$$

$$(18) (7x+6)(6x-8) \\ = 42x^2 - 20x - 48$$

$$(19) (x-5)(2x-3) \\ = 2x^2 - 13x + 15$$

$$(20) (4x+5)(6x-5) \\ = 24x^2 + 10x - 25$$

$$(21) (4x+4)(6x-6) \\ = 24x^2 - 24$$

$$(22) (9x+9)(9x-6) \\ = 81x^2 + 27x - 54$$

$$(23) (5x-2)(x-6) \\ = 5x^2 - 32x + 12$$

$$(24) (9x-7)(3x+1) \\ = 27x^2 - 12x - 7$$

$$(25) (8x-7)(5x+1) \\ = 40x^2 - 27x - 7$$

$$(26) (9x-7)(x-9) \\ = 9x^2 - 88x + 63$$

$$(27) (8x+8)(8x-9) \\ = 64x^2 - 8x - 72$$

$$(28) (2x-8)(6x+7) \\ = 12x^2 - 34x - 56$$

$$(29) (5x+6)(x-1) \\ = 5x^2 + x - 6$$

$$(30) (9x-3)(3x+7) \\ = 27x^2 + 54x - 21$$

$$(31) (3x-3)(x+2) \\ = 3x^2 + 3x - 6$$

$$(32) (6x-8)(2x+9) \\ = 12x^2 + 38x - 72$$

$$(33) (5x-9)(6x-6) \\ = 30x^2 - 84x + 54$$

$$(34) (2x-7)(6x+3) \\ = 12x^2 - 36x - 21$$

$$(35) (7x-7)(4x-6) \\ = 28x^2 - 70x + 42$$

$$(36) (8x-5)(9x-6) \\ = 72x^2 - 93x + 30$$

$$(37) (x-5)(4x+6) \\ = 4x^2 - 14x - 30$$

$$(38) (2x+8)(x-8) \\ = 2x^2 - 8x - 64$$

$$(39) (7x-9)(8x+2) \\ = 56x^2 - 58x - 18$$

$$(40) (2x+4)(5x-1) \\ = 10x^2 + 18x - 4$$

$$(41) (5x+3)(3x+7) \\ = 15x^2 + 44x + 21$$

$$(42) (5x+1)(3x+8) \\ = 15x^2 + 43x + 8$$

$$(43) (9x-6)(2x+1) \\ = 18x^2 - 3x - 6$$

$$(44) (6x-7)(x+5) \\ = 6x^2 + 23x - 35$$

$$(45) (2x-1)(7x-1) \\ = 14x^2 - 9x + 1$$

$$(1) (2x+9)(5x-8) \\ = 10x^2 + 29x - 72$$

$$(2) (3x-5)(5x+3) \\ = 15x^2 - 16x - 15$$

$$(3) (9x-6)(7x+3) \\ = 63x^2 - 15x - 18$$

$$(4) (9x-7)(2x-3) \\ = 18x^2 - 41x + 21$$

$$(5) (2x+8)(6x-5) \\ = 12x^2 + 38x - 40$$

$$(6) (4x-6)(5x+4) \\ = 20x^2 - 14x - 24$$

$$(7) (7x+7)(8x+7) \\ = 56x^2 + 105x + 49$$

$$(8) (3x-9)(4x+5) \\ = 12x^2 - 21x - 45$$

$$(9) (5x+7)(x+1) \\ = 5x^2 + 12x + 7$$

$$(10) (7x-2)(8x-6) \\ = 56x^2 - 58x + 12$$

$$(11) (3x-1)(7x-3) \\ = 21x^2 - 16x + 3$$

$$(12) (9x-2)(x-9) \\ = 9x^2 - 83x + 18$$

$$(13) (x-4)(2x-4) \\ = 2x^2 - 12x + 16$$

$$(14) (4x-9)(7x+6) \\ = 28x^2 - 39x - 54$$

$$(15) (4x+9)(7x-3) \\ = 28x^2 + 51x - 27$$

$$(16) (4x+3)(7x+2) \\ = 28x^2 + 29x + 6$$

$$(17) (9x-6)(4x-2) \\ = 36x^2 - 42x + 12$$

$$(18) (9x-3)(3x-7) \\ = 27x^2 - 72x + 21$$

$$(19) (2x-2)(9x+9) \\ = 18x^2 - 18$$

$$(20) (7x-7)(3x+9) \\ = 21x^2 + 42x - 63$$

$$(21) (x-5)(8x+9) \\ = 8x^2 - 31x - 45$$

$$(22) (x-1)(6x+4) \\ = 6x^2 - 2x - 4$$

$$(23) (3x-6)(4x+9) \\ = 12x^2 + 3x - 54$$

$$(24) (3x-6)(8x-2) \\ = 24x^2 - 54x + 12$$

$$(25) (9x-2)(x+9) \\ = 9x^2 + 79x - 18$$

$$(26) (9x+2)(2x+5) \\ = 18x^2 + 49x + 10$$

$$(27) (3x-5)(x-3) \\ = 3x^2 - 14x + 15$$

$$(28) (7x-2)(3x+8) \\ = 21x^2 + 50x - 16$$

$$(29) (3x-1)(8x+3) \\ = 24x^2 + x - 3$$

$$(30) (8x-9)(6x-8) \\ = 48x^2 - 118x + 72$$

$$(31) (3x+3)(9x+3) \\ = 27x^2 + 36x + 9$$

$$(32) (9x-6)(7x-9) \\ = 63x^2 - 123x + 54$$

$$(33) (2x-4)(8x-8) \\ = 16x^2 - 48x + 32$$

$$(34) (3x+3)(4x-5) \\ = 12x^2 - 3x - 15$$

$$(35) (3x-9)(x+1) \\ = 3x^2 - 6x - 9$$

$$(36) (5x+8)(9x-9) \\ = 45x^2 + 27x - 72$$

$$(37) (x+4)(3x-7) \\ = 3x^2 + 5x - 28$$

$$(38) (x+2)(6x+3) \\ = 6x^2 + 15x + 6$$

$$(39) (9x+2)(9x-7) \\ = 81x^2 - 45x - 14$$

$$(40) (5x+5)(4x+9) \\ = 20x^2 + 65x + 45$$

$$(41) (5x-4)(7x+7) \\ = 35x^2 + 7x - 28$$

$$(42) (9x+5)(6x+7) \\ = 54x^2 + 93x + 35$$

$$(43) (8x+5)(2x-2) \\ = 16x^2 - 6x - 10$$

$$(44) (x-7)(6x-4) \\ = 6x^2 - 46x + 28$$

$$(45) (8x+6)(2x-9) \\ = 16x^2 - 60x - 54$$

$$(1) (6x - 6)(7x + 7) \\ = 42x^2 - 42$$

$$(2) (7x - 2)(4x + 5) \\ = 28x^2 + 27x - 10$$

$$(3) (4x - 7)(6x + 6) \\ = 24x^2 - 18x - 42$$

$$(4) (7x - 7)(x - 4) \\ = 7x^2 - 35x + 28$$

$$(5) (8x + 7)(8x - 6) \\ = 64x^2 + 8x - 42$$

$$(6) (3x - 9)(x - 6) \\ = 3x^2 - 27x + 54$$

$$(7) (9x + 3)(x + 5) \\ = 9x^2 + 48x + 15$$

$$(8) (7x + 5)(7x + 1) \\ = 49x^2 + 42x + 5$$

$$(9) (7x - 6)(x - 2) \\ = 7x^2 - 20x + 12$$

$$(10) (5x - 6)(9x - 2) \\ = 45x^2 - 64x + 12$$

$$(11) (4x - 5)(2x - 9) \\ = 8x^2 - 46x + 45$$

$$(12) (3x - 2)(4x + 4) \\ = 12x^2 + 4x - 8$$

$$(13) (2x - 3)(5x - 2) \\ = 10x^2 - 19x + 6$$

$$(14) (2x + 1)(9x - 9) \\ = 18x^2 - 9x - 9$$

$$(15) (3x + 1)(7x + 5) \\ = 21x^2 + 22x + 5$$

$$(16) (9x - 6)(5x - 9) \\ = 45x^2 - 111x + 54$$

$$(17) (3x + 2)^2 \\ = 9x^2 + 12x + 4$$

$$(18) (8x - 3)(5x - 7) \\ = 40x^2 - 71x + 21$$

$$(19) (2x + 1)(3x + 2) \\ = 6x^2 + 7x + 2$$

$$(20) (6x + 5)(7x + 2) \\ = 42x^2 + 47x + 10$$

$$(21) (9x - 7)(7x + 5) \\ = 63x^2 - 4x - 35$$

$$(22) (6x - 3)(6x + 3) \\ = 36x^2 - 9$$

$$(23) (6x - 9)(9x - 1) \\ = 54x^2 - 87x + 9$$

$$(24) (5x - 5)(6x - 2) \\ = 30x^2 - 40x + 10$$

$$(25) (2x + 2)(9x + 8) \\ = 18x^2 + 34x + 16$$

$$(26) (9x - 2)(3x - 3) \\ = 27x^2 - 33x + 6$$

$$(27) (6x - 8)(3x + 5) \\ = 18x^2 + 6x - 40$$

$$(28) (7x - 7)(3x - 7) \\ = 21x^2 - 70x + 49$$

$$(29) (8x + 1)(5x + 4) \\ = 40x^2 + 37x + 4$$

$$(30) (9x - 5)(x + 2) \\ = 9x^2 + 13x - 10$$

$$(31) (2x + 6)(8x + 2) \\ = 16x^2 + 52x + 12$$

$$(32) (3x + 8)(x + 7) \\ = 3x^2 + 29x + 56$$

$$(33) (x + 2)(3x - 9) \\ = 3x^2 - 3x - 18$$

$$(34) (9x + 8)(3x - 4) \\ = 27x^2 - 12x - 32$$

$$(35) (5x - 6)(8x - 3) \\ = 40x^2 - 63x + 18$$

$$(36) (3x + 3)(8x + 9) \\ = 24x^2 + 51x + 27$$

$$(37) (5x + 7)(8x - 3) \\ = 40x^2 + 41x - 21$$

$$(38) (8x - 5)(5x + 8) \\ = 40x^2 + 39x - 40$$

$$(39) (4x + 4)(7x - 6) \\ = 28x^2 + 4x - 24$$

$$(40) (9x + 9)(8x + 7) \\ = 72x^2 + 135x + 63$$

$$(41) (3x - 1)(7x - 1) \\ = 21x^2 - 10x + 1$$

$$(42) (x - 8)(9x - 8) \\ = 9x^2 - 80x + 64$$

$$(43) (3x + 5)(8x + 9) \\ = 24x^2 + 67x + 45$$

$$(44) (5x - 3)(5x - 9) \\ = 25x^2 - 60x + 27$$

$$(45) (4x - 6)(3x - 5) \\ = 12x^2 - 38x + 30$$

$$(1) (9x+9)(9x+6) \\ = 81x^2 + 135x + 54$$

$$(2) (2x+9)(5x-6) \\ = 10x^2 + 33x - 54$$

$$(3) (x-3)(3x-6) \\ = 3x^2 - 15x + 18$$

$$(4) (x-1)(5x-3) \\ = 5x^2 - 8x + 3$$

$$(5) (8x+9)(3x-1) \\ = 24x^2 + 19x - 9$$

$$(6) (x+7)(6x-1) \\ = 6x^2 + 41x - 7$$

$$(7) (7x-7)(6x-5) \\ = 42x^2 - 77x + 35$$

$$(8) (6x-7)(3x+7) \\ = 18x^2 + 21x - 49$$

$$(9) (4x+4)(9x+5) \\ = 36x^2 + 56x + 20$$

$$(10) (4x+9)(4x-2) \\ = 16x^2 + 28x - 18$$

$$(11) (2x-9)(7x-5) \\ = 14x^2 - 73x + 45$$

$$(12) (4x-4)(8x+4) \\ = 32x^2 - 16x - 16$$

$$(13) (9x-1)(x+3) \\ = 9x^2 + 26x - 3$$

$$(14) (8x+9)(9x+4) \\ = 72x^2 + 113x + 36$$

$$(15) (7x-1)(7x+7) \\ = 49x^2 + 42x - 7$$

$$(16) (5x+6)(6x-1) \\ = 30x^2 + 31x - 6$$

$$(17) (4x-8)(5x+4) \\ = 20x^2 - 24x - 32$$

$$(18) (5x-4)(2x+7) \\ = 10x^2 + 27x - 28$$

$$(19) (4x+5)(9x+7) \\ = 36x^2 + 73x + 35$$

$$(20) (8x-3)(x+3) \\ = 8x^2 + 21x - 9$$

$$(21) (5x-5)(3x+5) \\ = 15x^2 + 10x - 25$$

$$(22) (x-3)(3x-8) \\ = 3x^2 - 17x + 24$$

$$(23) (3x-8)(x+6) \\ = 3x^2 + 10x - 48$$

$$(24) (7x-8)(3x-1) \\ = 21x^2 - 31x + 8$$

$$(25) (7x-1)(6x-7) \\ = 42x^2 - 55x + 7$$

$$(26) (5x-6)(4x+4) \\ = 20x^2 - 4x - 24$$

$$(27) (7x-5)(5x+2) \\ = 35x^2 - 11x - 10$$

$$(28) (6x-8)(6x+1) \\ = 36x^2 - 42x - 8$$

$$(29) (9x+8)(8x+6) \\ = 72x^2 + 118x + 48$$

$$(30) (3x-8)(9x-2) \\ = 27x^2 - 78x + 16$$

$$(31) (4x+7)(3x+6) \\ = 12x^2 + 45x + 42$$

$$(32) (x-8)(x+4) \\ = x^2 - 4x - 32$$

$$(33) (3x-7)(6x+9) \\ = 18x^2 - 15x - 63$$

$$(34) (x+8)(3x-8) \\ = 3x^2 + 16x - 64$$

$$(35) (2x+3)(4x-7) \\ = 8x^2 - 2x - 21$$

$$(36) (5x-3)(5x-5) \\ = 25x^2 - 40x + 15$$

$$(37) (3x-2)(4x+3) \\ = 12x^2 + x - 6$$

$$(38) (5x+3)(4x-3) \\ = 20x^2 - 3x - 9$$

$$(39) (4x+2)(2x+7) \\ = 8x^2 + 32x + 14$$

$$(40) (6x+9)(5x+5) \\ = 30x^2 + 75x + 45$$

$$(41) (4x+4)(4x-6) \\ = 16x^2 - 8x - 24$$

$$(42) (3x+7)(5x-8) \\ = 15x^2 + 11x - 56$$

$$(43) (8x+9)(6x-4) \\ = 48x^2 + 22x - 36$$

$$(44) (9x+9)(6x-7) \\ = 54x^2 - 9x - 63$$

$$(45) (9x-8)^2 \\ = 81x^2 - 144x + 64$$

$$(1) (4x-1)(8x-9) \\ = 32x^2 - 44x + 9$$

$$(16) (3x-5)(5x-1) \\ = 15x^2 - 28x + 5$$

$$(31) (4x-3)(5x-8) \\ = 20x^2 - 47x + 24$$

$$(2) (4x+8)(6x-5) \\ = 24x^2 + 28x - 40$$

$$(17) (5x+4)(7x-9) \\ = 35x^2 - 17x - 36$$

$$(32) (5x+4)(9x-8) \\ = 45x^2 - 4x - 32$$

$$(3) (6x-4)(8x+3) \\ = 48x^2 - 14x - 12$$

$$(18) (6x-7)(4x+4) \\ = 24x^2 - 4x - 28$$

$$(33) (6x+1)(3x+9) \\ = 18x^2 + 57x + 9$$

$$(4) (2x-5)(9x-5) \\ = 18x^2 - 55x + 25$$

$$(19) (8x-9)(6x-2) \\ = 48x^2 - 70x + 18$$

$$(34) (9x+1)(9x+9) \\ = 81x^2 + 90x + 9$$

$$(5) (7x-4)(8x+9) \\ = 56x^2 + 31x - 36$$

$$(20) (4x-9)(2x+9) \\ = 8x^2 + 18x - 81$$

$$(35) (4x-3)(6x+5) \\ = 24x^2 + 2x - 15$$

$$(6) (6x-3)(4x+7) \\ = 24x^2 + 30x - 21$$

$$(21) (9x-3)(6x+6) \\ = 54x^2 + 36x - 18$$

$$(36) (x+8)(5x-1) \\ = 5x^2 + 39x - 8$$

$$(7) (6x-5)(7x+7) \\ = 42x^2 + 7x - 35$$

$$(22) (2x-4)(2x+3) \\ = 4x^2 - 2x - 12$$

$$(37) (6x+3)(9x-3) \\ = 54x^2 + 9x - 9$$

$$(8) (5x+5)(9x-5) \\ = 45x^2 + 20x - 25$$

$$(23) (7x-6)(3x+2) \\ = 21x^2 - 4x - 12$$

$$(38) (8x-8)(7x+7) \\ = 56x^2 - 56$$

$$(9) (6x-9)(2x-8) \\ = 12x^2 - 66x + 72$$

$$(24) (9x+6)(5x-9) \\ = 45x^2 - 51x - 54$$

$$(39) (3x-4)(2x+2) \\ = 6x^2 - 2x - 8$$

$$(10) (8x+1)(7x-8) \\ = 56x^2 - 57x - 8$$

$$(25) (6x+6)(x-1) \\ = 6x^2 - 6$$

$$(40) (3x-2)(6x-3) \\ = 18x^2 - 21x + 6$$

$$(11) (6x+4)(8x-9) \\ = 48x^2 - 22x - 36$$

$$(26) (8x+3)(6x+6) \\ = 48x^2 + 66x + 18$$

$$(41) (7x+1)(3x+5) \\ = 21x^2 + 38x + 5$$

$$(12) (9x-9)(3x-5) \\ = 27x^2 - 72x + 45$$

$$(27) (3x-9)(8x-2) \\ = 24x^2 - 78x + 18$$

$$(42) (4x+5)(3x+8) \\ = 12x^2 + 47x + 40$$

$$(13) (6x-9)(8x-9) \\ = 48x^2 - 126x + 81$$

$$(28) (5x+7)(6x+8) \\ = 30x^2 + 82x + 56$$

$$(43) (2x+2)(8x+8) \\ = 16x^2 + 32x + 16$$

$$(14) (4x+6)(3x-6) \\ = 12x^2 - 6x - 36$$

$$(29) (5x-8)(x+3) \\ = 5x^2 + 7x - 24$$

$$(44) (x-1)(3x-5) \\ = 3x^2 - 8x + 5$$

$$(15) (4x+1)(9x-1) \\ = 36x^2 + 5x - 1$$

$$(30) (2x+9)(2x+8) \\ = 4x^2 + 34x + 72$$

$$(45) (4x+6)(5x+8) \\ = 20x^2 + 62x + 48$$

$$(1) (5x+3)(6x-9) \\ = 30x^2 - 27x - 27$$

$$(16) (x+1)(2x+6) \\ = 2x^2 + 8x + 6$$

$$(31) (3x-3)(4x+5) \\ = 12x^2 + 3x - 15$$

$$(2) (x+4)(8x-5) \\ = 8x^2 + 27x - 20$$

$$(17) (3x-2)(4x+9) \\ = 12x^2 + 19x - 18$$

$$(32) (3x-5)(7x+8) \\ = 21x^2 - 11x - 40$$

$$(3) (3x+6)(7x+4) \\ = 21x^2 + 54x + 24$$

$$(18) (3x-9)(4x-9) \\ = 12x^2 - 63x + 81$$

$$(33) (4x+7)(9x-1) \\ = 36x^2 + 59x - 7$$

$$(4) (9x+6)(8x+7) \\ = 72x^2 + 111x + 42$$

$$(19) (4x+5)(6x-5) \\ = 24x^2 + 10x - 25$$

$$(34) (5x+7)(6x-7) \\ = 30x^2 + 7x - 49$$

$$(5) (2x-6)(2x-1) \\ = 4x^2 - 14x + 6$$

$$(20) (9x-9)(8x-4) \\ = 72x^2 - 108x + 36$$

$$(35) (2x+8)(x+4) \\ = 2x^2 + 16x + 32$$

$$(6) (x-8)(x-1) \\ = x^2 - 9x + 8$$

$$(21) (4x+1)(3x-1) \\ = 12x^2 - x - 1$$

$$(36) (4x-5)(4x-8) \\ = 16x^2 - 52x + 40$$

$$(7) (3x-6)(4x-6) \\ = 12x^2 - 42x + 36$$

$$(22) (x-8)(7x-6) \\ = 7x^2 - 62x + 48$$

$$(37) (8x+3)(3x+5) \\ = 24x^2 + 49x + 15$$

$$(8) (7x+1)(x-1) \\ = 7x^2 - 6x - 1$$

$$(23) (9x-1)(5x-9) \\ = 45x^2 - 86x + 9$$

$$(38) (2x-1)(6x-3) \\ = 12x^2 - 12x + 3$$

$$(9) (x+1)(3x-4) \\ = 3x^2 - x - 4$$

$$(24) (6x-9)(9x+4) \\ = 54x^2 - 57x - 36$$

$$(39) (7x+2)(3x-9) \\ = 21x^2 - 57x - 18$$

$$(10) (7x-1)(6x+3) \\ = 42x^2 + 15x - 3$$

$$(25) (5x-7)(8x+8) \\ = 40x^2 - 16x - 56$$

$$(40) (8x+1)(7x+5) \\ = 56x^2 + 47x + 5$$

$$(11) (4x+8)(6x+7) \\ = 24x^2 + 76x + 56$$

$$(26) (7x+3)(3x+4) \\ = 21x^2 + 37x + 12$$

$$(41) (x-8)(6x-1) \\ = 6x^2 - 49x + 8$$

$$(12) (5x+4)(x-7) \\ = 5x^2 - 31x - 28$$

$$(27) (7x+7)(9x+8) \\ = 63x^2 + 119x + 56$$

$$(42) (7x+3)(5x+9) \\ = 35x^2 + 78x + 27$$

$$(13) (3x-4)(5x-9) \\ = 15x^2 - 47x + 36$$

$$(28) (6x+9)(2x-5) \\ = 12x^2 - 12x - 45$$

$$(43) (2x-7)(2x-2) \\ = 4x^2 - 18x + 14$$

$$(14) (9x-4)(8x+6) \\ = 72x^2 + 22x - 24$$

$$(29) (5x+1)(7x+6) \\ = 35x^2 + 37x + 6$$

$$(44) (7x+9)(5x-3) \\ = 35x^2 + 24x - 27$$

$$(15) (6x-9)(2x+2) \\ = 12x^2 - 6x - 18$$

$$(30) (9x-6)(7x+3) \\ = 63x^2 - 15x - 18$$

$$(45) (6x+1)(6x+8) \\ = 36x^2 + 54x + 8$$

$$(1) (4x+7)(3x-6) \\ = 12x^2 - 3x - 42$$

$$(16) (9x-4)(4x+4) \\ = 36x^2 + 20x - 16$$

$$(31) (8x-2)(9x-5) \\ = 72x^2 - 58x + 10$$

$$(2) (3x-8)(9x-1) \\ = 27x^2 - 75x + 8$$

$$(17) (8x+1)(x+4) \\ = 8x^2 + 33x + 4$$

$$(32) (8x-5)(x+8) \\ = 8x^2 + 59x - 40$$

$$(3) (7x-5)(6x-5) \\ = 42x^2 - 65x + 25$$

$$(18) (9x+4)(x+5) \\ = 9x^2 + 49x + 20$$

$$(33) (x-6)(7x+3) \\ = 7x^2 - 39x - 18$$

$$(4) (6x+1)(x+6) \\ = 6x^2 + 37x + 6$$

$$(19) (x+5)(9x+1) \\ = 9x^2 + 46x + 5$$

$$(34) (2x+4)(7x-3) \\ = 14x^2 + 22x - 12$$

$$(5) (6x+9)(5x-4) \\ = 30x^2 + 21x - 36$$

$$(20) (9x-2)(4x-8) \\ = 36x^2 - 80x + 16$$

$$(35) (4x-5)(6x-4) \\ = 24x^2 - 46x + 20$$

$$(6) (3x-1)(x+9) \\ = 3x^2 + 26x - 9$$

$$(21) (4x+4)(x-6) \\ = 4x^2 - 20x - 24$$

$$(36) (3x+2)(x-6) \\ = 3x^2 - 16x - 12$$

$$(7) (7x-3)(5x+9) \\ = 35x^2 + 48x - 27$$

$$(22) (5x-7)(4x+1) \\ = 20x^2 - 23x - 7$$

$$(37) (7x+8)(x+9) \\ = 7x^2 + 71x + 72$$

$$(8) (7x-3)(8x+1) \\ = 56x^2 - 17x - 3$$

$$(23) (8x-9)(7x-1) \\ = 56x^2 - 71x + 9$$

$$(38) (3x-6)(3x+2) \\ = 9x^2 - 12x - 12$$

$$(9) (5x+7)(7x-5) \\ = 35x^2 + 24x - 35$$

$$(24) (6x+9)(8x+7) \\ = 48x^2 + 114x + 63$$

$$(39) (3x-2)(3x-8) \\ = 9x^2 - 30x + 16$$

$$(10) (9x+2)(3x-3) \\ = 27x^2 - 21x - 6$$

$$(25) (3x+5)(x+1) \\ = 3x^2 + 8x + 5$$

$$(40) (x-9)(7x+8) \\ = 7x^2 - 55x - 72$$

$$(11) (8x+3)(4x-7) \\ = 32x^2 - 44x - 21$$

$$(26) (9x-6)(2x+3) \\ = 18x^2 + 15x - 18$$

$$(41) (3x+2)(8x-1) \\ = 24x^2 + 13x - 2$$

$$(12) (6x+1)(4x+7) \\ = 24x^2 + 46x + 7$$

$$(27) (2x+3)(6x-2) \\ = 12x^2 + 14x - 6$$

$$(42) (3x-1)(7x-9) \\ = 21x^2 - 34x + 9$$

$$(13) (6x-7)(x-1) \\ = 6x^2 - 13x + 7$$

$$(28) (4x-2)(x-5) \\ = 4x^2 - 22x + 10$$

$$(43) (3x+5)(2x+5) \\ = 6x^2 + 25x + 25$$

$$(14) (2x+9)(9x+1) \\ = 18x^2 + 83x + 9$$

$$(29) (9x-5)(3x+8) \\ = 27x^2 + 57x - 40$$

$$(44) (x+8)(9x+8) \\ = 9x^2 + 80x + 64$$

$$(15) (x-8)(x+5) \\ = x^2 - 3x - 40$$

$$(30) (x-7)(7x-1) \\ = 7x^2 - 50x + 7$$

$$(45) (x-9)(8x+3) \\ = 8x^2 - 69x - 27$$

$$(1) (4x-7)(7x-2) \\ = 28x^2 - 57x + 14$$

$$(16) (3x+1)(x+4) \\ = 3x^2 + 13x + 4$$

$$(31) (9x-9)(2x+3) \\ = 18x^2 + 9x - 27$$

$$(2) (x-2)(4x+3) \\ = 4x^2 - 5x - 6$$

$$(17) (4x+1)(x-8) \\ = 4x^2 - 31x - 8$$

$$(32) (5x-3)(9x+8) \\ = 45x^2 + 13x - 24$$

$$(3) (9x+6)(3x+1) \\ = 27x^2 + 27x + 6$$

$$(18) (7x-4)(x+7) \\ = 7x^2 + 45x - 28$$

$$(33) (8x+8)(4x-5) \\ = 32x^2 - 8x - 40$$

$$(4) (2x+7)(5x-4) \\ = 10x^2 + 27x - 28$$

$$(19) (6x+2)(2x-6) \\ = 12x^2 - 32x - 12$$

$$(34) (8x+4)(4x-2) \\ = 32x^2 - 8$$

$$(5) (5x-9)(6x-4) \\ = 30x^2 - 74x + 36$$

$$(20) (3x+7)(3x+6) \\ = 9x^2 + 39x + 42$$

$$(35) (8x+9)(x+6) \\ = 8x^2 + 57x + 54$$

$$(6) (5x-1)(9x+6) \\ = 45x^2 + 21x - 6$$

$$(21) (7x-8)(2x+5) \\ = 14x^2 + 19x - 40$$

$$(36) (2x-3)(x+8) \\ = 2x^2 + 13x - 24$$

$$(7) (4x+3)(8x+3) \\ = 32x^2 + 36x + 9$$

$$(22) (3x+3)(8x-5) \\ = 24x^2 + 9x - 15$$

$$(37) (2x+5)(9x-8) \\ = 18x^2 + 29x - 40$$

$$(8) (2x-7)(6x+7) \\ = 12x^2 - 28x - 49$$

$$(23) (8x+1)(x+6) \\ = 8x^2 + 49x + 6$$

$$(38) (2x+5)(4x-5) \\ = 8x^2 + 10x - 25$$

$$(9) (2x+8)(7x+4) \\ = 14x^2 + 64x + 32$$

$$(24) (9x-9)(9x+4) \\ = 81x^2 - 45x - 36$$

$$(39) (6x-2)(9x+8) \\ = 54x^2 + 30x - 16$$

$$(10) (4x+9)(5x-7) \\ = 20x^2 + 17x - 63$$

$$(25) (7x-7)(3x-4) \\ = 21x^2 - 49x + 28$$

$$(40) (7x+2)(6x-2) \\ = 42x^2 - 2x - 4$$

$$(11) (7x-2)(3x-4) \\ = 21x^2 - 34x + 8$$

$$(26) (5x+1)(3x-2) \\ = 15x^2 - 7x - 2$$

$$(41) (2x-7)(8x+6) \\ = 16x^2 - 44x - 42$$

$$(12) (2x-3)(3x+1) \\ = 6x^2 - 7x - 3$$

$$(27) (2x+2)(7x+5) \\ = 14x^2 + 24x + 10$$

$$(42) (x+2)(6x-7) \\ = 6x^2 + 5x - 14$$

$$(13) (2x+4)(3x-9) \\ = 6x^2 - 6x - 36$$

$$(28) (3x+1)(x+1) \\ = 3x^2 + 4x + 1$$

$$(43) (9x-2)(5x-4) \\ = 45x^2 - 46x + 8$$

$$(14) (6x-3)(5x-7) \\ = 30x^2 - 57x + 21$$

$$(29) (7x+3)(5x-8) \\ = 35x^2 - 41x - 24$$

$$(44) (5x-5)(8x-3) \\ = 40x^2 - 55x + 15$$

$$(15) (9x+3)(x-3) \\ = 9x^2 - 24x - 9$$

$$(30) (2x+7)(3x-2) \\ = 6x^2 + 17x - 14$$

$$(45) (3x-7)(8x-7) \\ = 24x^2 - 77x + 49$$

2.3 除法

3 次多項式を 2 次多項式で割った商と剰余を求める問題。

$$(1) \quad (-12x^3 + 26x^2 + 16x + 2) \div (2x^2 - 5x - 1)$$

$$(6) \quad (-4x^3 - 27x^2 + 8x + 40) \div (x^2 + 8x + 8)$$

$$(2) \quad (45x^3 - 126x^2 + 153x - 72) \div (5x^2 - 9x + 8)$$

$$(7) \quad (-72x^3 + 10x^2 + 34x - 8) \div (9x^2 + x - 4)$$

$$(3) \quad (-28x^3 - 15x^2 - 3x - 35) \div (7x^2 - 5x + 7)$$

$$(8) \quad (18x^3 - 66x^2 + 92x - 48) \div (3x^2 - 7x + 6)$$

$$(4) \quad (-24x^3 - 90x^2 - 97x - 36) \div (6x^2 + 9x + 4)$$

$$(9) \quad (-35x^3 + 69x^2 - 53x + 35) \div (7x^2 - 4x + 5)$$

$$(5) \quad (-14x^3 - 54x^2 - 43x - 6) \div (2x^2 + 6x + 1)$$

$$(10) \quad (-3x^3 + 5x + 14) \div (3x^2 + 6x + 7)$$

$$(1) \quad (-56x^3 - 34x^2 - 34x - 30) \div (7x^2 - x + 5)$$

$$(6) \quad (-81x^3 + 45x^2 + 5x - 2) \div (9x^2 - 7x + 1)$$

$$(2) \quad (9x^3 + 85x^2 + 34x - 18) \div (9x^2 + 4x - 2)$$

$$(7) \quad (-24x^3 - 22x^2 - 16x - 3) \div (6x^2 + 4x + 3)$$

$$(3) \quad (64x^3 - 104x^2 + 8x + 20) \div (8x^2 - 8x - 4)$$

$$(8) \quad (-8x^3 + 16x^2 + 40x - 48) \div (4x^2 + 4x - 8)$$

$$(4) \quad (-21x^3 - 10x^2 + 17x + 6) \div (7x^2 + x - 6)$$

$$(9) \quad (35x^3 + 52x^2 + 58x + 63) \div (5x^2 + x + 7)$$

$$(5) \quad (-28x^3 + 52x^2 - 20x - 4) \div (7x^2 - 6x - 1)$$

$$(10) \quad (-24x^3 + 76x^2 + 8x - 20) \div (3x^2 - 8x - 5)$$

(1) $(3x^3 - 17x^2 - 8x + 12) \div (x^2 - 5x - 6)$

(6) $(-30x^3 - 48x^2 - 58x - 24) \div (6x^2 + 6x + 8)$

(2) $(-6x^3 - 39x^2 - 14x + 24) \div (6x^2 + 3x - 4)$

(7) $(25x^3 - 85x^2 + 37x + 63) \div (5x^2 - 8x - 7)$

(3) $(-35x^3 - 17x^2 + 12x + 4) \div (5x^2 + x - 2)$

(8) $(24x^3 + 48x^2) \div (6x^2)$

(4) $(-32x^3 + 72x^2 - 96x + 56) \div (4x^2 - 5x + 7)$

(9) $(-30x^3 + 32x^2 + 34x - 28) \div (5x^2 - 2x - 7)$

(5) $(-28x^3 - 20x^2 + 8x) \div (4x^2 + 4x)$

(10) $(6x^3 + 59x^2 + 52x + 63) \div (6x^2 + 5x + 7)$

(1) $(12x^3 + 24x^2 + 15x + 3) \div (4x^2 + 4x + 1)$

(6) $(8x^3 - 36x^2 + 13x + 12) \div (8x^2 - 4x - 3)$

(2) $(27x^3 + 63x^2 - 75x - 63) \div (9x^2 - 6x - 7)$

(7) $(14x^3 + 36x^2 + 20x + 8) \div (7x^2 + 4x + 2)$

(3) $(32x^3 - 24x^2 - 44x - 12) \div (4x^2 - 5x - 3)$

(8) $(56x^3 - 4x^2 + 48x + 32) \div (7x^2 - 4x + 8)$

(4) $(54x^3 + 75x^2 + 48x + 12) \div (9x^2 + 8x + 4)$

(9) $(9x^3 + 38x^2 - 107x + 56) \div (x^2 + 5x - 8)$

(5) $(-30x^3 + 12x^2 + 54x) \div (5x^2 - 2x - 9)$

(10) $(-54x^3 + 57x^2 + 79x + 8) \div (6x^2 - 7x - 8)$

$$(1) (27x^3 - 84x^2 + 90x - 9) \div (3x^2 - 9x + 9)$$

$$(6) (9x^3 + 7x^2 + x + 3) \div (9x^2 - 2x + 3)$$

$$(2) (-6x^3 + 7x^2 + 15x + 4) \div (2x^2 - 3x - 4)$$

$$(7) (45x^3 + 59x^2 + 15x + 1) \div (5x^2 + 6x + 1)$$

$$(3) (-27x^3 + 72x^2 - 15x - 14) \div (9x^2 - 3x - 2)$$

$$(8) (6x^3 + 53x^2 + 15x - 4) \div (x^2 + 9x + 4)$$

$$(4) (12x^3 + 40x^2 + 66x + 18) \div (2x^2 + 6x + 9)$$

$$(9) (8x^3 + 27x^2 - 21x - 4) \div (8x^2 - 5x - 1)$$

$$(5) (8x^3 - 16x^2 - 14x + 10) \div (4x^2 + 2x - 2)$$

$$(10) (-40x^3 - 16x^2 + 64x) \div (5x^2 + 2x - 8)$$

(1) $(35x^3 + 52x^2 - 37x - 14) \div (5x^2 + 6x - 7)$

(6) $(-7x^3 - 40x^2 + 67x + 28) \div (7x^2 - 9x - 4)$

(2) $(-6x^3 + 18x^2 + 8x - 24) \div (6x^2 - 8)$

(7) $(-6x^3 + 34x^2 - 62x + 28) \div (x^2 - 5x + 7)$

(3) $(-2x^3 - 12x^2 - 13x + 15) \div (2x^2 + 6x - 5)$

(8) $(-45x^3 + 7x^2 + 37x - 15) \div (9x^2 + 4x - 5)$

(4) $(8x^3 - 4x^2 + 20x - 24) \div (2x^2 + x + 6)$

(9) $(-56x^3 + 121x^2 - 105x + 54) \div (8x^2 - 7x + 6)$

(5) $(56x^3 - 99x^2 + 112x - 45) \div (7x^2 - 8x + 9)$

(10) $(-9x^3 - 45x^2) \div (x^2 + 5x)$

$$(1) (18x^3 + 20x^2 - 5x - 24) \div (2x^2 + 4x + 3)$$

$$(6) (12x^3 - 3x^2 - 10x + 56) \div (3x^2 - 6x + 8)$$

$$(2) (21x^3 + 33x^2 - 12) \div (7x^2 + 4x - 4)$$

$$(7) (35x^3 - 94x^2 + 77x - 18) \div (5x^2 - 7x + 2)$$

$$(3) (-6x^3 - 58x^2 - 60x - 16) \div (x^2 + 9x + 4)$$

$$(8) (36x^3 + 39x^2 - 10x + 56) \div (9x^2 - 6x + 8)$$

$$(4) (-36x^3 + 6x^2 + 12x) \div (6x^2 - 4x)$$

$$(9) (-21x^3 - 42x^2 - 36x - 15) \div (7x^2 + 7x + 5)$$

$$(5) (-7x^3 + 43x^2 - 38x - 10) \div (7x^2 - 8x - 2)$$

$$(10) (20x^3 - 57x^2 + 85x - 72) \div (4x^2 - 5x + 9)$$

(1) $(-63x^3 + 17x^2 + 13x + 5) \div (9x^2 + 4x + 1)$

(6) $(-30x^3 - 8x^2 + 82x - 24) \div (5x^2 + 8x - 3)$

(2) $(12x^3 - 11x^2 + 11x - 20) \div (3x^2 + x + 4)$

(7) $(72x^3 + 46x^2 - 78x + 8) \div (8x^2 + 6x - 8)$

(3) $(-8x^3 + 26x^2 + 10x + 56) \div (4x^2 + 3x + 7)$

(8) $(6x^3 - 21x^2 + 18x + 15) \div (x^2 - 4x + 5)$

(4) $(-5x^3 - 22x^2 + 10x - 25) \div (5x^2 - 3x + 5)$

(9) $(-36x^3 - 27x^2 + 45x + 18) \div (4x^2 + 7x + 2)$

(5) $(-49x^3 - 56x^2 - 63x) \div (7x^2 + 8x + 9)$

(10) $(24x^3 - 38x^2 + 17x - 42) \div (6x^2 + x + 6)$

$$(1) (9x^3 - 4x + 35) \div (3x^2 - 5x + 7)$$

$$(6) (27x^3 + 33x^2 - 78x + 28) \div (3x^2 + 6x - 4)$$

$$(2) (35x^3 + 80x^2 - 4x - 63) \div (5x^2 + 5x - 7)$$

$$(7) (-6x^3 - 13x^2 + 90x - 63) \div (2x^2 + 9x - 9)$$

$$(3) (-20x^3 - 44x^2 + 16) \div (5x^2 + x - 2)$$

$$(8) (-45x^3 + 58x^2 + 7x) \div (9x^2 + x)$$

$$(4) (-18x^3 + 69x^2 - 63x + 15) \div (3x^2 - 9x + 3)$$

$$(9) (42x^3 - 54x^2 + 12x) \div (6x^2 - 6x)$$

$$(5) (-21x^3 - 54x^2 - 39x - 30) \div (7x^2 + 4x + 5)$$

$$(10) (-45x^3 - 21x^2 + 11x - 1) \div (9x^2 + 6x - 1)$$

(1) $(20x^3 + 13x^2 - 36x - 21) \div (4x^2 - 3x - 3)$

(6) $(72x^3 + 37x^2 + 19x + 15) \div (9x^2 - x + 3)$

(2) $(32x^3 + 36x^2 - 26x - 12) \div (8x^2 - 3x - 2)$

(7) $(54x^3 + 6x^2) \div (6x^2)$

(3) $(56x^3 + 45x^2 + 37x + 12) \div (8x^2 + 3x + 4)$

(8) $(64x^3 - 64x^2 - 56x) \div (8x^2 - 8x - 7)$

(4) $(42x^3 + 90x^2 + 34x - 16) \div (6x^2 + 6x - 2)$

(9) $(-6x^3 + 17x^2 + 49) \div (3x^2 + 2x + 7)$

(5) $(-16x^3 + 46x^2 + 82x - 40) \div (8x^2 + 9x - 5)$

(10) $(14x^3 + 70x^2 + 72x + 64) \div (7x^2 + 7x + 8)$

$$(1) (45x^3 + 47x^2 + 30x + 8) \div (5x^2 + 3x + 2)$$

$$(6) (48x^3 + 70x^2 - 30x - 16) \div (8x^2 + 9x - 8)$$

$$(2) (-28x^3 - 27x^2 - 11x + 6) \div (4x^2 + 5x + 3)$$

$$(7) (-64x^3 + 88x^2 + 42x - 45) \div (8x^2 - 6x - 9)$$

$$(3) (63x^3 - 21x^2 - 69x - 18) \div (7x^2 - 7x - 3)$$

$$(8) (8x^3 + 52x^2 + 88x + 32) \div (x^2 + 6x + 8)$$

$$(4) (56x^3 - 103x^2 - 18x + 45) \div (8x^2 - 9x - 9)$$

$$(9) (8x^3 + 36x^2 + 44x + 8) \div (2x^2 + 5x + 1)$$

$$(5) (30x^3 + 46x^2 + 50x + 56) \div (5x^2 + x + 7)$$

$$(10) (-32x^3 + 4x^2 + 39x + 15) \div (4x^2 - 3x - 3)$$

$$(1) (-5x^3 - 4x^2 - 30x - 24) \div (x^2 + 6)$$

$$(6) (25x^3 + 15x^2 - 74x - 36) \div (5x^2 - 6x - 4)$$

$$(2) (64x^3 - 112x^2 - 3x + 30) \div (8x^2 - 9x - 6)$$

$$(7) (8x^3 + 32x^2 - 8x) \div (x^2 + 4x - 1)$$

$$(3) (6x^3 + 53x^2 - 8x - 40) \div (x^2 + 8x - 8)$$

$$(8) (-6x^3 + 51x + 9) \div (2x^2 + 6x + 1)$$

$$(4) (10x^3 - 15x^2 + 5x) \div (2x^2 - 3x + 1)$$

$$(9) (-15x^3 + 19x^2 + 59x - 8) \div (5x^2 + 7x - 1)$$

$$(5) (-25x^3 - 45x^2 - 10x + 8) \div (5x^2 + 5x - 2)$$

$$(10) (-18x^3 + 34x^2 - 6x + 20) \div (9x^2 + x + 5)$$

(1) $(-2x^3 + x^2 + 32x + 14) \div (x^2 - 4x - 2)$

(6) $(63x^3 - 95x^2 + 54x - 8) \div (7x^2 - 9x + 4)$

(2) $(-5x^3 - 12x^2 + 4x + 16) \div (5x^2 + 2x - 8)$

(7) $(24x^3 + 84x^2 + 60x + 72) \div (8x^2 + 4x + 8)$

(3) $(-10x^3 + 13x^2 + 30x) \div (5x^2 + 6x)$

(8) $(-30x^3 + 29x^2 + 15x - 25) \div (5x^2 - 9x + 5)$

(4) $(-30x^3 - 15x^2 + 35x) \div (6x^2 + 3x - 7)$

(9) $(54x^3 + 6x^2 - 13x + 2) \div (6x^2 + 2x - 1)$

(5) $(-63x^3 - 130x^2 - 63x) \div (7x^2 + 9x)$

(10) $(-42x^3 + 57x^2 - 21x + 45) \div (7x^2 + x + 5)$

$$(1) (4x^3 - 36x^2 + 20x) \div (x^2 - 9x + 5)$$

$$(6) (-21x^3 - 22x^2 + 58x + 45) \div (3x^2 + x - 9)$$

$$(2) (49x^3 + 14x^2 - 78x + 27) \div (7x^2 + 5x - 9)$$

$$(7) (16x^3 + 50x^2 + 54x + 6) \div (2x^2 + 6x + 6)$$

$$(3) (5x^3 + 40x^2 + 10x) \div (x^2 + 8x + 2)$$

$$(8) (-18x^3 + 60x^2 - 12x - 48) \div (3x^2 - 6x - 6)$$

$$(4) (18x^3 - 30x^2 - 24x + 24) \div (3x^2 - 8x + 4)$$

$$(9) (-72x^3 - 4x^2 + 101x + 45) \div (8x^2 - 4x - 9)$$

$$(5) (36x^3 + 45x^2 - 20x - 25) \div (9x^2 - 5)$$

$$(10) (32x^3 + 20x^2 + 66x + 8) \div (4x^2 + 2x + 8)$$

(1) $(-40x^3 - 58x^2 - 20x - 2) \div (5x^2 + 6x + 1)$

(6) $(-63x^3 - 85x^2 - 91x - 36) \div (9x^2 + 7x + 9)$

(2) $(-48x^3 + 8x^2 + 48x - 20) \div (6x^2 + 2x - 5)$

(7) $(10x^3 + 20x^2 - 22x + 24) \div (5x^2 - 5x + 4)$

(3) $(-7x^3 - 2x^2 + 11x + 6) \div (7x^2 - 5x - 6)$

(8) $(8x^3 - 14x^2 - 52x + 45) \div (2x^2 - 8x + 5)$

(4) $(-48x^3 + 72x^2 + 16x - 20) \div (6x^2 - 6x - 5)$

(9) $(12x^3 - 20x^2 - 34x - 42) \div (6x^2 + 8x + 7)$

(5) $(-2x^3 + x^2 - 6x) \div (2x^2 - x + 6)$

(10) $(30x^3 - 31x^2 - 60x - 9) \div (5x^2 - 6x - 9)$

(1) $(-12x^3 - 38x^2 + 64x - 36) \div (6x^2 - 8x + 4)$

(6) $(10x^3 + 13x^2 + 39x + 28) \div (2x^2 + x + 7)$

(2) $(-5x^3 - 25x^2 + 5x) \div (x^2 + 5x - 1)$

(7) $(40x^3 - 31x^2 + 62x - 21) \div (5x^2 - 2x + 7)$

(3) $(42x^3 - 21x^2 - 49x) \div (6x^2 - 3x - 7)$

(8) $(12x^3 - 50x^2 - 26x + 36) \div (6x^2 + 2x - 4)$

(4) $(24x^3 - 52x^2 + 47x - 45) \div (8x^2 - 4x + 9)$

(9) $(54x^3 - 63x^2 - 54x + 63) \div (9x^2 - 9)$

(5) $(28x^3 + 22x^2 + 18x + 4) \div (4x^2 + 2x + 2)$

(10) $(14x^3 + 79x^2 + 100x + 32) \div (2x^2 + 9x + 4)$

(1) $(36x^3 - 79x^2 - 35x + 28) \div (4x^2 - 7x - 7)$

(6) $(9x^3 + 12x^2 + 9x + 42) \div (3x^2 - 2x + 7)$

(2) $(-27x^3 + 15x^2 - 46x + 48) \div (3x^2 + x + 6)$

(7) $(4x^3 - 16x^2 + 4x - 16) \div (4x^2 + 4)$

(3) $(-6x^3 + 22x^2 - 34x - 14) \div (x^2 - 4x + 7)$

(8) $(8x^3 - 4x^2 - 6x + 2) \div (4x^2 + 2x - 1)$

(4) $(8x^3 - 33x^2 - 31x - 20) \div (8x^2 + 7x + 4)$

(9) $(48x^3 - 108x^2 + 70x - 12) \div (6x^2 - 9x + 2)$

(5) $(-24x^3 + 8x^2 - 2x + 2) \div (6x^2 + x + 1)$

(10) $(-12x^3 - 56x^2 - 53x - 14) \div (6x^2 + 7x + 2)$

(1) $(-12x^3 - 18x^2 + 48x + 9) \div (2x^2 + 6x + 1)$

(6) $(-6x^3 - 61x^2 - 56x + 63) \div (6x^2 + 7x - 7)$

(2) $(24x^3 + 22x^2 - 66x - 8) \div (4x^2 + 9x + 1)$

(7) $(12x^3 + 6x^2 - 30x - 18) \div (3x^2 - 3x - 3)$

(3) $(20x^3 - 16x^2 - 32x) \div (5x^2 - 4x - 8)$

(8) $(15x^3 - 38x^2 + 48x - 32) \div (5x^2 - 6x + 8)$

(4) $(-4x^3 - 16x^2 - 12x) \div (4x^2 + 4x)$

(9) $(42x^3 - 30x^2 + 36x - 48) \div (7x^2 + 2x + 8)$

(5) $(36x^3 + 54x^2 + 74x + 45) \div (6x^2 + 4x + 9)$

(10) $(-36x^3 + 76x^2 + x - 1) \div (4x^2 - 8x - 1)$

$$(1) (8x^3 - 2x^2 + x + 2) \div (4x^2 - 3x + 2)$$

$$(6) (48x^3 + 56x^2 + 8x) \div (6x^2 + 7x + 1)$$

$$(2) (48x^3 - 28x^2 + 28x - 8) \div (8x^2 - 2x + 4)$$

$$(7) (9x^3 - 45x^2 + 36) \div (x^2 - 4x - 4)$$

$$(3) (-5x^3 + 22x^2 + 16x - 5) \div (5x^2 + 3x - 1)$$

$$(8) (36x^3 - 54x^2 + 56x - 49) \div (6x^2 - 2x + 7)$$

$$(4) (-63x^3 - 36x^2 - 18x - 45) \div (7x^2 - 3x + 5)$$

$$(9) (18x^3 - 6x^2 + 12x - 4) \div (3x^2 + 2)$$

$$(5) (36x^3 + 48x^2 - 5x - 14) \div (6x^2 + x - 2)$$

$$(10) (-20x^3 + 25x^2 + 27x - 8) \div (5x^2 - 5x - 8)$$

$$(1) (-24x^3 + 7x^2 + 3x + 2) \div (8x^2 + 3x + 1)$$

$$(6) (5x^3 + x^2 - 6x) \div (x^2 - x)$$

$$(2) (14x^3 - 64x^2 + 38x - 24) \div (7x^2 - 4x + 3)$$

$$(7) (-15x^3 - 12x^2 - 18x) \div (5x^2 + 4x + 6)$$

$$(3) (-8x^3 - 12x^2 + 88x - 64) \div (2x^2 + 7x - 8)$$

$$(8) (4x^3 + 8x^2 - 2x) \div (2x^2 + 4x - 1)$$

$$(4) (-48x^3 - 24x^2 - 32x - 16) \div (6x^2 + 4)$$

$$(9) (24x^3 - 46x^2 - 6x + 4) \div (4x^2 - 9x + 2)$$

$$(5) (42x^3 + 30x^2 - 63x - 45) \div (6x^2 - 9)$$

$$(10) (18x^3 + 12x^2 - 13x - 2) \div (6x^2 + 8x + 1)$$

(1) $(-12x^3 + 26x^2 + 16x + 2) \div (2x^2 - 5x - 1)$
 商: $-6x - 2$, 剰余: $-x + 2$

(6) $(-4x^3 - 27x^2 + 8x + 40) \div (x^2 + 8x + 8)$
 商: $-4x + 5$, 剰余: $-6x + 1$

(2) $(45x^3 - 126x^2 + 153x - 72) \div (5x^2 - 9x + 8)$
 商: $9x - 9$, 剰余: $x - 4$

(7) $(-72x^3 + 10x^2 + 34x - 8) \div (9x^2 + x - 4)$
 商: $-8x + 2$, 剰余: $-3x$

(3) $(-28x^3 - 15x^2 - 3x - 35) \div (7x^2 - 5x + 7)$
 商: $-4x - 5$, 剰余: $-4x - 4$

(8) $(18x^3 - 66x^2 + 92x - 48) \div (3x^2 - 7x + 6)$
 商: $6x - 8$, 剰余: 1

(4) $(-24x^3 - 90x^2 - 97x - 36) \div (6x^2 + 9x + 4)$
 商: $-4x - 9$, 剰余: $4x + 5$

(9) $(-35x^3 + 69x^2 - 53x + 35) \div (7x^2 - 4x + 5)$
 商: $-5x + 7$, 剰余: $-x + 7$

(5) $(-14x^3 - 54x^2 - 43x - 6) \div (2x^2 + 6x + 1)$
 商: $-7x - 6$, 剰余: $6x - 4$

(10) $(-3x^3 + 5x + 14) \div (3x^2 + 6x + 7)$
 商: $-x + 2$, 剰余: $5x - 8$

$$(1) (-56x^3 - 34x^2 - 34x - 30) \div (7x^2 - x + 5)$$

商: $-8x - 6$, 剰余: $5x - 2$

$$(6) (-81x^3 + 45x^2 + 5x - 2) \div (9x^2 - 7x + 1)$$

商: $-9x - 2$, 剰余: -9

$$(2) (9x^3 + 85x^2 + 34x - 18) \div (9x^2 + 4x - 2)$$

商: $x + 9$, 剰余: $3x - 2$

$$(7) (-24x^3 - 22x^2 - 16x - 3) \div (6x^2 + 4x + 3)$$

商: $-4x - 1$, 剰余: $2x - 2$

$$(3) (64x^3 - 104x^2 + 8x + 20) \div (8x^2 - 8x - 4)$$

商: $8x - 5$, 剰余: $3x - 2$

$$(8) (-8x^3 + 16x^2 + 40x - 48) \div (4x^2 + 4x - 8)$$

商: $-2x + 6$, 剰余: $-7x$

$$(4) (-21x^3 - 10x^2 + 17x + 6) \div (7x^2 + x - 6)$$

商: $-3x - 1$, 剰余: $-4x + 6$

$$(9) (35x^3 + 52x^2 + 58x + 63) \div (5x^2 + x + 7)$$

商: $7x + 9$, 剰余: $4x + 1$

$$(5) (-28x^3 + 52x^2 - 20x - 4) \div (7x^2 - 6x - 1)$$

商: $-4x + 4$, 剰余: -3

$$(10) (-24x^3 + 76x^2 + 8x - 20) \div (3x^2 - 8x - 5)$$

商: $-8x + 4$, 剰余: $3x - 8$

$$(1) (3x^3 - 17x^2 - 8x + 12) \div (x^2 - 5x - 6)$$

商 : $3x - 2$, 剰余 : -3

$$(6) (-30x^3 - 48x^2 - 58x - 24) \div (6x^2 + 6x + 8)$$

商 : $-5x - 3$, 剰余 : $-9x - 9$

$$(2) (-6x^3 - 39x^2 - 14x + 24) \div (6x^2 + 3x - 4)$$

商 : $-x - 6$, 剰余 : $-5x - 6$

$$(7) (25x^3 - 85x^2 + 37x + 63) \div (5x^2 - 8x - 7)$$

商 : $5x - 9$, 剰余 : $6x - 1$

$$(3) (-35x^3 - 17x^2 + 12x + 4) \div (5x^2 + x - 2)$$

商 : $-7x - 2$, 剰余 : $6x - 7$

$$(8) (24x^3 + 48x^2) \div (6x^2)$$

商 : $4x + 8$, 剰余 : 2

$$(4) (-32x^3 + 72x^2 - 96x + 56) \div (4x^2 - 5x + 7)$$

商 : $-8x + 8$, 剰余 : $-2x + 3$

$$(9) (-30x^3 + 32x^2 + 34x - 28) \div (5x^2 - 2x - 7)$$

商 : $-6x + 4$, 剰余 : $-5x + 2$

$$(5) (-28x^3 - 20x^2 + 8x) \div (4x^2 + 4x)$$

商 : $-7x + 2$, 剰余 : $9x + 6$

$$(10) (6x^3 + 59x^2 + 52x + 63) \div (6x^2 + 5x + 7)$$

商 : $x + 9$, 剰余 : $-x - 2$

$$(1) (12x^3 + 24x^2 + 15x + 3) \div (4x^2 + 4x + 1)$$

商 : $3x + 3$, 剰余 : $-4x + 4$

$$(6) (8x^3 - 36x^2 + 13x + 12) \div (8x^2 - 4x - 3)$$

商 : $x - 4$, 剰余 : $-5x + 7$

$$(2) (27x^3 + 63x^2 - 75x - 63) \div (9x^2 - 6x - 7)$$

商 : $3x + 9$, 剰余 : $2x + 4$

$$(7) (14x^3 + 36x^2 + 20x + 8) \div (7x^2 + 4x + 2)$$

商 : $2x + 4$, 剰余 : $4x + 4$

$$(3) (32x^3 - 24x^2 - 44x - 12) \div (4x^2 - 5x - 3)$$

商 : $8x + 4$, 剰余 : $-3x + 8$

$$(8) (56x^3 - 4x^2 + 48x + 32) \div (7x^2 - 4x + 8)$$

商 : $8x + 4$, 剰余 : $-x - 8$

$$(4) (54x^3 + 75x^2 + 48x + 12) \div (9x^2 + 8x + 4)$$

商 : $6x + 3$, 剰余 : $-8x + 5$

$$(9) (9x^3 + 38x^2 - 107x + 56) \div (x^2 + 5x - 8)$$

商 : $9x - 7$, 剰余 : $-2x + 7$

$$(5) (-30x^3 + 12x^2 + 54x) \div (5x^2 - 2x - 9)$$

商 : $-6x$, 剰余 : $-2x + 1$

$$(10) (-54x^3 + 57x^2 + 79x + 8) \div (6x^2 - 7x - 8)$$

商 : $-9x - 1$, 剰余 : $-8x + 5$

$$(1) (27x^3 - 84x^2 + 90x - 9) \div (3x^2 - 9x + 9)$$

商 : $9x - 1$, 剰余 : $-9x - 5$

$$(6) (9x^3 + 7x^2 + x + 3) \div (9x^2 - 2x + 3)$$

商 : $x + 1$, 剰余 : $6x - 7$

$$(2) (-6x^3 + 7x^2 + 15x + 4) \div (2x^2 - 3x - 4)$$

商 : $-3x - 1$, 剰余 : $-4x - 2$

$$(7) (45x^3 + 59x^2 + 15x + 1) \div (5x^2 + 6x + 1)$$

商 : $9x + 1$, 剰余 : $9x - 6$

$$(3) (-27x^3 + 72x^2 - 15x - 14) \div (9x^2 - 3x - 2)$$

商 : $-3x + 7$, 剰余 : $-7x + 5$

$$(8) (6x^3 + 53x^2 + 15x - 4) \div (x^2 + 9x + 4)$$

商 : $6x - 1$, 剰余 : $-3x - 7$

$$(4) (12x^3 + 40x^2 + 66x + 18) \div (2x^2 + 6x + 9)$$

商 : $6x + 2$, 剰余 : $-x + 4$

$$(9) (8x^3 + 27x^2 - 21x - 4) \div (8x^2 - 5x - 1)$$

商 : $x + 4$, 剰余 : $x + 3$

$$(5) (8x^3 - 16x^2 - 14x + 10) \div (4x^2 + 2x - 2)$$

商 : $2x - 5$, 剰余 : $9x + 6$

$$(10) (-40x^3 - 16x^2 + 64x) \div (5x^2 + 2x - 8)$$

商 : $-8x$, 剰余 : $-8x - 3$

(1) $(35x^3 + 52x^2 - 37x - 14) \div (5x^2 + 6x - 7)$
 商 : $7x + 2$, 剰余 : -2

(6) $(-7x^3 - 40x^2 + 67x + 28) \div (7x^2 - 9x - 4)$
 商 : $-x - 7$, 剰余 : 4

(2) $(-6x^3 + 18x^2 + 8x - 24) \div (6x^2 - 8)$
 商 : $-x + 3$, 剰余 : $-2x + 3$

(7) $(-6x^3 + 34x^2 - 62x + 28) \div (x^2 - 5x + 7)$
 商 : $-6x + 4$, 剰余 : $-4x + 9$

(3) $(-2x^3 - 12x^2 - 13x + 15) \div (2x^2 + 6x - 5)$
 商 : $-x - 3$, 剰余 : $-6x + 9$

(8) $(-45x^3 + 7x^2 + 37x - 15) \div (9x^2 + 4x - 5)$
 商 : $-5x + 3$, 剰余 : $-2x + 7$

(4) $(8x^3 - 4x^2 + 20x - 24) \div (2x^2 + x + 6)$
 商 : $4x - 4$, 剰余 : $x + 4$

(9) $(-56x^3 + 121x^2 - 105x + 54) \div (8x^2 - 7x + 6)$
 商 : $-7x + 9$, 剰余 : $9x + 6$

(5) $(56x^3 - 99x^2 + 112x - 45) \div (7x^2 - 8x + 9)$
 商 : $8x - 5$, 剰余 : $-x - 2$

(10) $(-9x^3 - 45x^2) \div (x^2 + 5x)$
 商 : $-9x$, 剰余 : $6x + 3$

$$(1) (18x^3 + 20x^2 - 5x - 24) \div (2x^2 + 4x + 3)$$

商 : $9x - 8$, 剰余 : $4x - 8$

$$(6) (12x^3 - 3x^2 - 10x + 56) \div (3x^2 - 6x + 8)$$

商 : $4x + 7$, 剰余 : $8x - 9$

$$(2) (21x^3 + 33x^2 - 12) \div (7x^2 + 4x - 4)$$

商 : $3x + 3$, 剰余 : $x + 1$

$$(7) (35x^3 - 94x^2 + 77x - 18) \div (5x^2 - 7x + 2)$$

商 : $7x - 9$, 剰余 : $-2x + 8$

$$(3) (-6x^3 - 58x^2 - 60x - 16) \div (x^2 + 9x + 4)$$

商 : $-6x - 4$, 剰余 : $5x + 7$

$$(8) (36x^3 + 39x^2 - 10x + 56) \div (9x^2 - 6x + 8)$$

商 : $4x + 7$, 剰余 : $8x + 1$

$$(4) (-36x^3 + 6x^2 + 12x) \div (6x^2 - 4x)$$

商 : $-6x - 3$, 剰余 : 3

$$(9) (-21x^3 - 42x^2 - 36x - 15) \div (7x^2 + 7x + 5)$$

商 : $-3x - 3$, 剰余 : $-8x + 5$

$$(5) (-7x^3 + 43x^2 - 38x - 10) \div (7x^2 - 8x - 2)$$

商 : $-x + 5$, 剰余 : $3x + 4$

$$(10) (20x^3 - 57x^2 + 85x - 72) \div (4x^2 - 5x + 9)$$

商 : $5x - 8$, 剰余 : $-x + 5$

(1) $(-63x^3 + 17x^2 + 13x + 5) \div (9x^2 + 4x + 1)$
 商 : $-7x + 5$, 剰余 : $-x - 2$

(6) $(-30x^3 - 8x^2 + 82x - 24) \div (5x^2 + 8x - 3)$
 商 : $-6x + 8$, 剰余 : 2

(2) $(12x^3 - 11x^2 + 11x - 20) \div (3x^2 + x + 4)$
 商 : $4x - 5$, 剰余 : $-3x + 2$

(7) $(72x^3 + 46x^2 - 78x + 8) \div (8x^2 + 6x - 8)$
 商 : $9x - 1$, 剰余 : $-4x - 1$

(3) $(-8x^3 + 26x^2 + 10x + 56) \div (4x^2 + 3x + 7)$
 商 : $-2x + 8$, 剰余 : $x + 9$

(8) $(6x^3 - 21x^2 + 18x + 15) \div (x^2 - 4x + 5)$
 商 : $6x + 3$, 剰余 : $x + 2$

(4) $(-5x^3 - 22x^2 + 10x - 25) \div (5x^2 - 3x + 5)$
 商 : $-x - 5$, 剰余 : $-x + 8$

(9) $(-36x^3 - 27x^2 + 45x + 18) \div (4x^2 + 7x + 2)$
 商 : $-9x + 9$, 剰余 : $-6x - 7$

(5) $(-49x^3 - 56x^2 - 63x) \div (7x^2 + 8x + 9)$
 商 : $-7x$, 剰余 : $-6x + 7$

(10) $(24x^3 - 38x^2 + 17x - 42) \div (6x^2 + x + 6)$
 商 : $4x - 7$, 剰余 : $2x - 4$

$$(1) (9x^3 - 4x + 35) \div (3x^2 - 5x + 7)$$

商 : $3x + 5$, 剰余 : $-6x - 5$

$$(6) (27x^3 + 33x^2 - 78x + 28) \div (3x^2 + 6x - 4)$$

商 : $9x - 7$, 剰余 : $9x + 4$

$$(2) (35x^3 + 80x^2 - 4x - 63) \div (5x^2 + 5x - 7)$$

商 : $7x + 9$, 剰余 : $-6x - 7$

$$(7) (-6x^3 - 13x^2 + 90x - 63) \div (2x^2 + 9x - 9)$$

商 : $-3x + 7$, 剰余 : $-9x - 7$

$$(3) (-20x^3 - 44x^2 + 16) \div (5x^2 + x - 2)$$

商 : $-4x - 8$, 剰余 : $4x + 6$

$$(8) (-45x^3 + 58x^2 + 7x) \div (9x^2 + x)$$

商 : $-5x + 7$, 剰余 : $-3x + 5$

$$(4) (-18x^3 + 69x^2 - 63x + 15) \div (3x^2 - 9x + 3)$$

商 : $-6x + 5$, 剰余 : $4x + 3$

$$(9) (42x^3 - 54x^2 + 12x) \div (6x^2 - 6x)$$

商 : $7x - 2$, 剰余 : $8x + 8$

$$(5) (-21x^3 - 54x^2 - 39x - 30) \div (7x^2 + 4x + 5)$$

商 : $-3x - 6$, 剰余 : $-6x + 4$

$$(10) (-45x^3 - 21x^2 + 11x - 1) \div (9x^2 + 6x - 1)$$

商 : $-5x + 1$, 剰余 : $-x + 1$

$$(1) (20x^3 + 13x^2 - 36x - 21) \div (4x^2 - 3x - 3)$$

商 : $5x + 7$, 剰余 : $8x + 4$

$$(6) (72x^3 + 37x^2 + 19x + 15) \div (9x^2 - x + 3)$$

商 : $8x + 5$, 剰余 : $2x + 7$

$$(2) (32x^3 + 36x^2 - 26x - 12) \div (8x^2 - 3x - 2)$$

商 : $4x + 6$, 剰余 : $9x + 4$

$$(7) (54x^3 + 6x^2) \div (6x^2)$$

商 : $9x + 1$, 剰余 : $x - 7$

$$(3) (56x^3 + 45x^2 + 37x + 12) \div (8x^2 + 3x + 4)$$

商 : $7x + 3$, 剰余 : $-6x - 6$

$$(8) (64x^3 - 64x^2 - 56x) \div (8x^2 - 8x - 7)$$

商 : $8x$, 剰余 : $-3x + 6$

$$(4) (42x^3 + 90x^2 + 34x - 16) \div (6x^2 + 6x - 2)$$

商 : $7x + 8$, 剰余 : $4x + 9$

$$(9) (-6x^3 + 17x^2 + 49) \div (3x^2 + 2x + 7)$$

商 : $-2x + 7$, 剰余 : $3x - 4$

$$(5) (-16x^3 + 46x^2 + 82x - 40) \div (8x^2 + 9x - 5)$$

商 : $-2x + 8$, 剰余 : $-6x + 1$

$$(10) (14x^3 + 70x^2 + 72x + 64) \div (7x^2 + 7x + 8)$$

商 : $2x + 8$, 剰余 : $5x - 9$

$$(1) (45x^3 + 47x^2 + 30x + 8) \div (5x^2 + 3x + 2)$$

商 : $9x + 4$, 剰余 : $5x + 3$

$$(6) (48x^3 + 70x^2 - 30x - 16) \div (8x^2 + 9x - 8)$$

商 : $6x + 2$, 剰余 : $4x - 8$

$$(2) (-28x^3 - 27x^2 - 11x + 6) \div (4x^2 + 5x + 3)$$

商 : $-7x + 2$, 剰余 : $6x - 3$

$$(7) (-64x^3 + 88x^2 + 42x - 45) \div (8x^2 - 6x - 9)$$

商 : $-8x + 5$, 剰余 : $5x + 6$

$$(3) (63x^3 - 21x^2 - 69x - 18) \div (7x^2 - 7x - 3)$$

商 : $9x + 6$, 剰余 : $-7x - 1$

$$(8) (8x^3 + 52x^2 + 88x + 32) \div (x^2 + 6x + 8)$$

商 : $8x + 4$, 剰余 : $8x - 6$

$$(4) (56x^3 - 103x^2 - 18x + 45) \div (8x^2 - 9x - 9)$$

商 : $7x - 5$, 剰余 : $3x + 4$

$$(9) (8x^3 + 36x^2 + 44x + 8) \div (2x^2 + 5x + 1)$$

商 : $4x + 8$, 剰余 : $4x - 5$

$$(5) (30x^3 + 46x^2 + 50x + 56) \div (5x^2 + x + 7)$$

商 : $6x + 8$, 剰余 : $7x - 8$

$$(10) (-32x^3 + 4x^2 + 39x + 15) \div (4x^2 - 3x - 3)$$

商 : $-8x - 5$, 剰余 : $8x - 9$

$$(1) (-5x^3 - 4x^2 - 30x - 24) \div (x^2 + 6)$$

商: $-5x - 4$, 剰余: $6x + 2$

$$(6) (25x^3 + 15x^2 - 74x - 36) \div (5x^2 - 6x - 4)$$

商: $5x + 9$, 剰余: $3x - 2$

$$(2) (64x^3 - 112x^2 - 3x + 30) \div (8x^2 - 9x - 6)$$

商: $8x - 5$, 剰余: $5x + 2$

$$(7) (8x^3 + 32x^2 - 8x) \div (x^2 + 4x - 1)$$

商: $8x$, 剰余: $-7x - 5$

$$(3) (6x^3 + 53x^2 - 8x - 40) \div (x^2 + 8x - 8)$$

商: $6x + 5$, 剰余: $7x - 5$

$$(8) (-6x^3 + 51x + 9) \div (2x^2 + 6x + 1)$$

商: $-3x + 9$, 剰余: $-3x - 4$

$$(4) (10x^3 - 15x^2 + 5x) \div (2x^2 - 3x + 1)$$

商: $5x$, 剰余: $4x - 6$

$$(9) (-15x^3 + 19x^2 + 59x - 8) \div (5x^2 + 7x - 1)$$

商: $-3x + 8$, 剰余: $4x + 1$

$$(5) (-25x^3 - 45x^2 - 10x + 8) \div (5x^2 + 5x - 2)$$

商: $-5x - 4$, 剰余: $2x + 1$

$$(10) (-18x^3 + 34x^2 - 6x + 20) \div (9x^2 + x + 5)$$

商: $-2x + 4$, 剰余: $-6x + 9$

$$(1) (-2x^3 + x^2 + 32x + 14) \div (x^2 - 4x - 2)$$

商 : $-2x - 7$, 剰余 : $2x + 4$

$$(6) (63x^3 - 95x^2 + 54x - 8) \div (7x^2 - 9x + 4)$$

商 : $9x - 2$, 剰余 : $-3x - 8$

$$(2) (-5x^3 - 12x^2 + 4x + 16) \div (5x^2 + 2x - 8)$$

商 : $-x - 2$, 剰余 : 4

$$(7) (24x^3 + 84x^2 + 60x + 72) \div (8x^2 + 4x + 8)$$

商 : $3x + 9$, 剰余 : $2x + 7$

$$(3) (-10x^3 + 13x^2 + 30x) \div (5x^2 + 6x)$$

商 : $-2x + 5$, 剰余 : $-2x + 4$

$$(8) (-30x^3 + 29x^2 + 15x - 25) \div (5x^2 - 9x + 5)$$

商 : $-6x - 5$, 剰余 : -5

$$(4) (-30x^3 - 15x^2 + 35x) \div (6x^2 + 3x - 7)$$

商 : $-5x$, 剰余 : $-3x - 7$

$$(9) (54x^3 + 6x^2 - 13x + 2) \div (6x^2 + 2x - 1)$$

商 : $9x - 2$, 剰余 : $7x + 7$

$$(5) (-63x^3 - 130x^2 - 63x) \div (7x^2 + 9x)$$

商 : $-9x - 7$, 剰余 : $-3x + 6$

$$(10) (-42x^3 + 57x^2 - 21x + 45) \div (7x^2 + x + 5)$$

商 : $-6x + 9$, 剰余 : $-4x + 6$

$$(1) (4x^3 - 36x^2 + 20x) \div (x^2 - 9x + 5)$$

商 : $4x$, 剰余 : $x + 1$

$$(6) (-21x^3 - 22x^2 + 58x + 45) \div (3x^2 + x - 9)$$

商 : $-7x - 5$, 剰余 : $-2x - 3$

$$(2) (49x^3 + 14x^2 - 78x + 27) \div (7x^2 + 5x - 9)$$

商 : $7x - 3$, 剰余 : $3x + 7$

$$(7) (16x^3 + 50x^2 + 54x + 6) \div (2x^2 + 6x + 6)$$

商 : $8x + 1$, 剰余 : $-6x - 3$

$$(3) (5x^3 + 40x^2 + 10x) \div (x^2 + 8x + 2)$$

商 : $5x$, 剰余 : $-2x + 8$

$$(8) (-18x^3 + 60x^2 - 12x - 48) \div (3x^2 - 6x - 6)$$

商 : $-6x + 8$, 剰余 : -6

$$(4) (18x^3 - 30x^2 - 24x + 24) \div (3x^2 - 8x + 4)$$

商 : $6x + 6$, 剰余 : $x + 6$

$$(9) (-72x^3 - 4x^2 + 101x + 45) \div (8x^2 - 4x - 9)$$

商 : $-9x - 5$, 剰余 : $4x - 5$

$$(5) (36x^3 + 45x^2 - 20x - 25) \div (9x^2 - 5)$$

商 : $4x + 5$, 剰余 : $-7x + 5$

$$(10) (32x^3 + 20x^2 + 66x + 8) \div (4x^2 + 2x + 8)$$

商 : $8x + 1$, 剰余 : $-7x + 1$

$$(1) (-40x^3 - 58x^2 - 20x - 2) \div (5x^2 + 6x + 1)$$

商: $-8x - 2$, 剰余: $7x + 6$

$$(6) (-63x^3 - 85x^2 - 91x - 36) \div (9x^2 + 7x + 9)$$

商: $-7x - 4$, 剰余: $4x - 1$

$$(2) (-48x^3 + 8x^2 + 48x - 20) \div (6x^2 + 2x - 5)$$

商: $-8x + 4$, 剰余: $-9x - 3$

$$(7) (10x^3 + 20x^2 - 22x + 24) \div (5x^2 - 5x + 4)$$

商: $2x + 6$, 剰余: $-6x - 1$

$$(3) (-7x^3 - 2x^2 + 11x + 6) \div (7x^2 - 5x - 6)$$

商: $-x - 1$, 剰余: $8x + 6$

$$(8) (8x^3 - 14x^2 - 52x + 45) \div (2x^2 - 8x + 5)$$

商: $4x + 9$, 剰余: $2x + 9$

$$(4) (-48x^3 + 72x^2 + 16x - 20) \div (6x^2 - 6x - 5)$$

商: $-8x + 4$, 剰余: $-6x - 4$

$$(9) (12x^3 - 20x^2 - 34x - 42) \div (6x^2 + 8x + 7)$$

商: $2x - 6$, 剰余: $8x + 9$

$$(5) (-2x^3 + x^2 - 6x) \div (2x^2 - x + 6)$$

商: $-x$, 剰余: $-2x + 9$

$$(10) (30x^3 - 31x^2 - 60x - 9) \div (5x^2 - 6x - 9)$$

商: $6x + 1$, 剰余: $-3x - 1$

$$(1) (-12x^3 - 38x^2 + 64x - 36) \div (6x^2 - 8x + 4)$$

商: $-2x - 9$, 剰余: $7x + 5$

$$(6) (10x^3 + 13x^2 + 39x + 28) \div (2x^2 + x + 7)$$

商: $5x + 4$, 剰余: $-7x + 9$

$$(2) (-5x^3 - 25x^2 + 5x) \div (x^2 + 5x - 1)$$

商: $-5x$, 剰余: $-7x + 3$

$$(7) (40x^3 - 31x^2 + 62x - 21) \div (5x^2 - 2x + 7)$$

商: $8x - 3$, 剰余: $-3x - 2$

$$(3) (42x^3 - 21x^2 - 49x) \div (6x^2 - 3x - 7)$$

商: $7x$, 剰余: $2x + 9$

$$(8) (12x^3 - 50x^2 - 26x + 36) \div (6x^2 + 2x - 4)$$

商: $2x - 9$, 剰余: $-7x - 7$

$$(4) (24x^3 - 52x^2 + 47x - 45) \div (8x^2 - 4x + 9)$$

商: $3x - 5$, 剰余: $3x + 2$

$$(9) (54x^3 - 63x^2 - 54x + 63) \div (9x^2 - 9)$$

商: $6x - 7$, 剰余: $x - 5$

$$(5) (28x^3 + 22x^2 + 18x + 4) \div (4x^2 + 2x + 2)$$

商: $7x + 2$, 剰余: $8x + 4$

$$(10) (14x^3 + 79x^2 + 100x + 32) \div (2x^2 + 9x + 4)$$

商: $7x + 8$, 剰余: $-3x + 2$

$$(1) (36x^3 - 79x^2 - 35x + 28) \div (4x^2 - 7x - 7)$$

商 : $9x - 4$, 剰余 : $-8x + 8$

$$(6) (9x^3 + 12x^2 + 9x + 42) \div (3x^2 - 2x + 7)$$

商 : $3x + 6$, 剰余 : $7x - 9$

$$(2) (-27x^3 + 15x^2 - 46x + 48) \div (3x^2 + x + 6)$$

商 : $-9x + 8$, 剰余 : $-2x + 1$

$$(7) (4x^3 - 16x^2 + 4x - 16) \div (4x^2 + 4)$$

商 : $x - 4$, 剰余 : $-6x + 4$

$$(3) (-6x^3 + 22x^2 - 34x - 14) \div (x^2 - 4x + 7)$$

商 : $-6x - 2$, 剰余 : $-8x + 3$

$$(8) (8x^3 - 4x^2 - 6x + 2) \div (4x^2 + 2x - 1)$$

商 : $2x - 2$, 剰余 : $-2x - 9$

$$(4) (8x^3 - 33x^2 - 31x - 20) \div (8x^2 + 7x + 4)$$

商 : $x - 5$, 剰余 : $2x - 3$

$$(9) (48x^3 - 108x^2 + 70x - 12) \div (6x^2 - 9x + 2)$$

商 : $8x - 6$, 剰余 : $9x + 2$

$$(5) (-24x^3 + 8x^2 - 2x + 2) \div (6x^2 + x + 1)$$

商 : $-4x + 2$, 剰余 : $5x + 5$

$$(10) (-12x^3 - 56x^2 - 53x - 14) \div (6x^2 + 7x + 2)$$

商 : $-2x - 7$, 剰余 : $-8x + 6$

$$(1) (-12x^3 - 18x^2 + 48x + 9) \div (2x^2 + 6x + 1)$$

商: $-6x + 9$, 剰余: $-9x - 4$

$$(6) (-6x^3 - 61x^2 - 56x + 63) \div (6x^2 + 7x - 7)$$

商: $-x - 9$, 剰余: -5

$$(2) (24x^3 + 22x^2 - 66x - 8) \div (4x^2 + 9x + 1)$$

商: $6x - 8$, 剰余: $-3x - 6$

$$(7) (12x^3 + 6x^2 - 30x - 18) \div (3x^2 - 3x - 3)$$

商: $4x + 6$, 剰余:

$$(3) (20x^3 - 16x^2 - 32x) \div (5x^2 - 4x - 8)$$

商: $4x$, 剰余: $-4x + 4$

$$(8) (15x^3 - 38x^2 + 48x - 32) \div (5x^2 - 6x + 8)$$

商: $3x - 4$, 剰余: $-5x + 2$

$$(4) (-4x^3 - 16x^2 - 12x) \div (4x^2 + 4x)$$

商: $-x - 3$, 剰余: $-9x + 8$

$$(9) (42x^3 - 30x^2 + 36x - 48) \div (7x^2 + 2x + 8)$$

商: $6x - 6$, 剰余: $6x + 2$

$$(5) (36x^3 + 54x^2 + 74x + 45) \div (6x^2 + 4x + 9)$$

商: $6x + 5$, 剰余: $-4x + 2$

$$(10) (-36x^3 + 76x^2 + x - 1) \div (4x^2 - 8x - 1)$$

商: $-9x + 1$, 剰余: $-5x + 8$

$$(1) (8x^3 - 2x^2 + x + 2) \div (4x^2 - 3x + 2)$$

商 : $2x + 1$, 剰余 : $8x - 1$

$$(6) (48x^3 + 56x^2 + 8x) \div (6x^2 + 7x + 1)$$

商 : $8x$, 剰余 : $-6x + 7$

$$(2) (48x^3 - 28x^2 + 28x - 8) \div (8x^2 - 2x + 4)$$

商 : $6x - 2$, 剰余 : $-5x - 5$

$$(7) (9x^3 - 45x^2 + 36) \div (x^2 - 4x - 4)$$

商 : $9x - 9$, 剰余 : $-7x + 5$

$$(3) (-5x^3 + 22x^2 + 16x - 5) \div (5x^2 + 3x - 1)$$

商 : $-x + 5$, 剰余 : $-7x - 6$

$$(8) (36x^3 - 54x^2 + 56x - 49) \div (6x^2 - 2x + 7)$$

商 : $6x - 7$, 剰余 : $9x - 6$

$$(4) (-63x^3 - 36x^2 - 18x - 45) \div (7x^2 - 3x + 5)$$

商 : $-9x - 9$, 剰余 : $-7x - 3$

$$(9) (18x^3 - 6x^2 + 12x - 4) \div (3x^2 + 2)$$

商 : $6x - 2$, 剰余 : $-8x - 6$

$$(5) (36x^3 + 48x^2 - 5x - 14) \div (6x^2 + x - 2)$$

商 : $6x + 7$, 剰余 : $x + 4$

$$(10) (-20x^3 + 25x^2 + 27x - 8) \div (5x^2 - 5x - 8)$$

商 : $-4x + 1$, 剰余 : $8x + 8$

$$(1) (-24x^3 + 7x^2 + 3x + 2) \div (8x^2 + 3x + 1)$$

商: $-3x + 2$, 剰余: $8x - 7$

$$(6) (5x^3 + x^2 - 6x) \div (x^2 - x)$$

商: $5x + 6$, 剰余: $-6x$

$$(2) (14x^3 - 64x^2 + 38x - 24) \div (7x^2 - 4x + 3)$$

商: $2x - 8$, 剰余: $-9x + 4$

$$(7) (-15x^3 - 12x^2 - 18x) \div (5x^2 + 4x + 6)$$

商: $-3x$, 剰余: $-9x + 9$

$$(3) (-8x^3 - 12x^2 + 88x - 64) \div (2x^2 + 7x - 8)$$

商: $-4x + 8$, 剰余: $x - 7$

$$(8) (4x^3 + 8x^2 - 2x) \div (2x^2 + 4x - 1)$$

商: $2x$, 剰余: $4x - 2$

$$(4) (-48x^3 - 24x^2 - 32x - 16) \div (6x^2 + 4)$$

商: $-8x - 4$, 剰余: $x + 6$

$$(9) (24x^3 - 46x^2 - 6x + 4) \div (4x^2 - 9x + 2)$$

商: $6x + 2$, 剰余: 8

$$(5) (42x^3 + 30x^2 - 63x - 45) \div (6x^2 - 9)$$

商: $7x + 5$, 剰余: $-4x - 3$

$$(10) (18x^3 + 12x^2 - 13x - 2) \div (6x^2 + 8x + 1)$$

商: $3x - 2$, 剰余: $2x - 1$

2.4 因数分解 (2 次式)

2 次多項式の因数分解をする問題。

(1) $5x^2 - 4x - 9$

(11) $5x^2 - 29x + 20$

(21) $14x^2 - 73x + 45$

(2) $x^2 - 6x - 7$

(12) $2x^2 - x - 1$

(22) $27x^2 - 24x + 5$

(3) $5x^2 + 6x - 8$

(13) $2x^2 - 15x + 7$

(23) $5x^2 + 14x + 9$

(4) $12x^2 - 13x - 4$

(14) $7x^2 + 3x - 4$

(24) $12x^2 - 11x - 5$

(5) $3x^2 - 11x - 4$

(15) $x^2 + 4x - 5$

(25) $2x^2 + 3x - 5$

(6) $36x^2 - 89x + 18$

(16) $21x^2 - 2x - 8$

(26) $2x^2 - 15x + 7$

(7) $2x^2 + 21x + 54$

(17) $14x^2 - 19x - 3$

(27) $10x^2 + 13x - 3$

(8) $12x^2 - 17x - 5$

(18) $3x^2 - x - 2$

(28) $4x^2 + 9x - 28$

(9) $3x^2 - 4x - 7$

(19) $2x^2 - 11x + 5$

(29) $3x^2 - 2x - 16$

(10) $16x^2 + 32x + 7$

(20) $5x^2 + 33x + 40$

(30) $6x^2 + 25x + 21$

(1) $3x^2 - 7x + 4$

(11) $18x^2 + 79x - 9$

(21) $6x^2 + 5x - 6$

(2) $4x^2 - 33x + 8$

(12) $14x^2 + 23x - 30$

(22) $10x^2 + 27x + 18$

(3) $18x^2 - 21x + 5$

(13) $2x^2 + 13x - 7$

(23) $3x^2 - 26x + 16$

(4) $2x^2 - 23x + 45$

(14) $21x^2 + 23x + 6$

(24) $30x^2 + 67x + 35$

(5) $3x^2 - 7x + 2$

(15) $7x^2 - 16x + 9$

(25) $16x^2 - 1$

(6) $35x^2 + 53x - 18$

(16) $25x^2 - 55x + 18$

(26) $2x^2 + 9x + 4$

(7) $42x^2 + 19x + 2$

(17) $7x^2 - 29x + 4$

(27) $4x^2 + 8x - 21$

(8) $15x^2 + 47x + 28$

(18) $2x^2 + 3x + 1$

(28) $2x^2 + 13x - 24$

(9) $3x^2 + 13x + 14$

(19) $5x^2 - 12x - 32$

(29) $x^2 + 9x + 18$

(10) $2x^2 + 5x + 2$

(20) $8x^2 + 41x + 36$

(30) $32x^2 + 84x + 27$

(1) $24x^2 - 22x - 7$

(11) $5x^2 - x - 6$

(21) $3x^2 - x - 2$

(2) $7x^2 + x - 6$

(12) $8x^2 + 9x - 14$

(22) $14x^2 + 45x - 81$

(3) $7x^2 + 15x + 2$

(13) $3x^2 + 13x + 4$

(23) $5x^2 + 3x - 8$

(4) $2x^2 + 5x - 3$

(14) $63x^2 + 4x - 35$

(24) $x^2 - x - 2$

(5) $25x^2 + 25x + 4$

(15) $9x^2 - 10x + 1$

(25) $45x^2 - 14x + 1$

(6) $21x^2 + 11x - 6$

(16) $2x^2 + 3x - 9$

(26) $49x^2 + 49x - 18$

(7) $6x^2 - 17x + 10$

(17) $8x^2 - 7x - 18$

(27) $56x^2 - 45x + 9$

(8) $2x^2 - 9x + 4$

(18) $8x^2 + 13x + 5$

(28) $36x^2 - 13x - 40$

(9) $12x^2 + 7x + 1$

(19) $42x^2 + 13x + 1$

(29) $18x^2 + 37x + 15$

(10) $5x^2 - 28x - 49$

(20) $48x^2 + 94x + 45$

(30) $2x^2 - 3x - 9$

(1) $5x^2 + 12x - 9$

(11) $x^2 + 2x + 1$

(21) $14x^2 + x - 3$

(2) $5x^2 - 12x + 7$

(12) $9x^2 + 12x - 32$

(22) $3x^2 - x - 10$

(3) $20x^2 + 3x - 9$

(13) $5x^2 - x - 4$

(23) $28x^2 - 75x + 27$

(4) $9x^2 + 6x + 1$

(14) $10x^2 - 27x + 5$

(24) $3x^2 + 4x + 1$

(5) $16x^2 - 16x - 21$

(15) $10x^2 - 21x + 8$

(25) $20x^2 + 43x + 21$

(6) $9x^2 - 3x - 20$

(16) $40x^2 + 99x + 56$

(26) $6x^2 + 19x + 3$

(7) $3x^2 + 2x - 8$

(17) $5x^2 - 16x + 12$

(27) $2x^2 - 3x + 1$

(8) $3x^2 - 8x + 4$

(18) $5x^2 - 2x - 3$

(28) $x^2 + 10x + 24$

(9) $2x^2 + 5x - 3$

(19) $2x^2 + 3x - 9$

(29) $3x^2 + 25x - 18$

(10) $x^2 - 3x - 4$

(20) $24x^2 + 25x + 6$

(30) $3x^2 - 10x + 7$

(1) $21x^2 + 25x - 4$

(11) $4x^2 - 11x + 7$

(21) $4x^2 + 5x - 9$

(2) $x^2 - 6x - 7$

(12) $3x^2 - 20x + 12$

(22) $5x^2 - 3x - 2$

(3) $36x^2 + 23x - 3$

(13) $18x^2 + 83x + 9$

(23) $7x^2 - 17x + 6$

(4) $35x^2 - 31x - 18$

(14) $25x^2 + 55x + 24$

(24) $28x^2 - 57x + 27$

(5) $4x^2 + 3x - 7$

(15) $14x^2 - 9x - 8$

(25) $24x^2 + 26x - 63$

(6) $10x^2 + 9x - 7$

(16) $24x^2 + 14x - 3$

(26) $10x^2 + 39x + 35$

(7) $2x^2 - x - 3$

(17) $9x^2 + 11x + 2$

(27) $12x^2 - 7x + 1$

(8) $27x^2 + 39x + 4$

(18) $x^2 - 3x + 2$

(28) $24x^2 - 17x - 20$

(9) $24x^2 - 29x + 7$

(19) $49x^2 + 56x + 15$

(29) $7x^2 + x - 8$

(10) $15x^2 - 44x + 21$

(20) $35x^2 + 32x - 12$

(30) $3x^2 - 11x + 8$

(1) $49x^2 + 35x - 36$

(11) $14x^2 - 3x - 2$

(21) $27x^2 - 30x + 7$

(2) $45x^2 - 76x + 32$

(12) $2x^2 + x - 1$

(22) $14x^2 - 25x + 6$

(3) $10x^2 + 3x - 1$

(13) $10x^2 - 23x + 12$

(23) $4x^2 + 12x - 27$

(4) $15x^2 - 22x - 9$

(14) $24x^2 - 17x + 3$

(24) $9x^2 - 37x + 4$

(5) $14x^2 + 15x + 4$

(15) $64x^2 + 80x + 21$

(25) $4x^2 + 11x - 3$

(6) $30x^2 + 73x + 40$

(16) $2x^2 + 11x + 15$

(26) $15x^2 + 28x - 32$

(7) $2x^2 + 17x - 9$

(17) $x^2 - 36$

(27) $3x^2 - 4x - 7$

(8) $7x^2 + 26x - 45$

(18) $x^2 + 3x + 2$

(28) $7x^2 + 43x - 42$

(9) $8x^2 - 2x - 1$

(19) $5x^2 - 31x - 28$

(29) $56x^2 - 19x - 15$

(10) $x^2 + 3x + 2$

(20) $x^2 + 3x + 2$

(30) $12x^2 - 35x + 25$

(1) $7x^2 + 2x - 5$

(11) $3x^2 + 2x - 1$

(21) $8x^2 - 3x - 5$

(2) $5x^2 + 6x + 1$

(12) $2x^2 + 3x + 1$

(22) $7x^2 - 5x - 2$

(3) $x^2 - 1$

(13) $20x^2 - 11x - 4$

(23) $20x^2 + 9x + 1$

(4) $56x^2 + 17x - 28$

(14) $3x^2 - x - 2$

(24) $3x^2 - 11x + 6$

(5) $3x^2 + 13x - 10$

(15) $x^2 - 14x + 48$

(25) $30x^2 + 31x - 6$

(6) $42x^2 - 41x + 5$

(16) $72x^2 + 113x + 36$

(26) $16x^2 - 9$

(7) $5x^2 - 27x + 10$

(17) $56x^2 + 15x - 54$

(27) $2x^2 + 7x + 5$

(8) $x^2 + 4x + 3$

(18) $8x^2 + 6x + 1$

(28) $3x^2 - 13x + 4$

(9) $7x^2 + 23x - 20$

(19) $2x^2 - 3x - 5$

(29) $25x^2 + 5x - 6$

(10) $9x^2 + 83x + 18$

(20) $x^2 - 4x + 3$

(30) $18x^2 + 95x + 63$

(1) $9x^2 - 5x - 4$

(11) $9x^2 + 30x + 16$

(21) $20x^2 + 17x - 24$

(2) $6x^2 + 7x - 3$

(12) $45x^2 + 34x + 5$

(22) $18x^2 + 51x + 35$

(3) $21x^2 - 10x - 24$

(13) $56x^2 - 73x + 21$

(23) $12x^2 - 29x + 14$

(4) $3x^2 - 4x + 1$

(14) $9x^2 + 18x - 16$

(24) $x^2 - 10x + 16$

(5) $42x^2 - 25x - 28$

(15) $24x^2 - 61x - 8$

(25) $28x^2 - 23x + 4$

(6) $6x^2 + 13x - 5$

(16) $4x^2 - 13x + 3$

(26) $20x^2 - x - 1$

(7) $x^2 + 4x + 3$

(17) $14x^2 - 13x + 3$

(27) $2x^2 + 3x + 1$

(8) $6x^2 + 17x + 7$

(18) $x^2 - 5x + 4$

(28) $x^2 - x - 2$

(9) $14x^2 - 59x + 35$

(19) $x^2 + 6x - 27$

(29) $6x^2 + 23x + 15$

(10) $4x^2 + 5x - 6$

(20) $4x^2 + 12x + 5$

(30) $8x^2 + 15x - 2$

(1) $2x^2 - x - 1$

(11) $32x^2 - 28x - 15$

(21) $21x^2 + 34x + 9$

(2) $56x^2 + 41x - 7$

(12) $15x^2 - 22x + 8$

(22) $8x^2 + 10x - 7$

(3) $24x^2 + 11x + 1$

(13) $x^2 + 3x + 2$

(23) $5x^2 - 11x + 6$

(4) $15x^2 + 22x + 8$

(14) $56x^2 + 15x + 1$

(24) $5x^2 + 12x + 7$

(5) $x^2 - 3x - 18$

(15) $10x^2 + 3x - 4$

(25) $3x^2 - 8x - 3$

(6) $15x^2 + 49x + 40$

(16) $81x^2 - 45x - 14$

(26) $56x^2 - 65x + 14$

(7) $3x^2 + 2x - 5$

(17) $6x^2 - 13x + 6$

(27) $35x^2 - 66x + 27$

(8) $x^2 - 6x - 16$

(18) $16x^2 + 2x - 5$

(28) $18x^2 + 9x + 1$

(9) $10x^2 + x - 2$

(19) $x^2 - 3x + 2$

(29) $24x^2 + 29x - 4$

(10) $9x^2 - 11x + 2$

(20) $x^2 + 11x + 24$

(30) $5x^2 - 4x - 1$

(1) $18x^2 + 27x + 7$

(11) $4x^2 + x - 3$

(21) $45x^2 + 112x + 64$

(2) $12x^2 + x - 6$

(12) $3x^2 - 10x + 7$

(22) $7x^2 + 10x + 3$

(3) $6x^2 - 47x - 8$

(13) $40x^2 + 21x - 49$

(23) $x^2 + 6x + 9$

(4) $15x^2 + x - 6$

(14) $16x^2 + 10x - 9$

(24) $7x^2 + 24x - 16$

(5) $6x^2 + x - 15$

(15) $4x^2 + 27x - 40$

(25) $9x^2 + 11x + 2$

(6) $6x^2 - 11x - 10$

(16) $20x^2 - 31x - 7$

(26) $35x^2 - 24x - 35$

(7) $x^2 + x - 2$

(17) $4x^2 + 19x + 12$

(27) $56x^2 - 59x + 15$

(8) $2x^2 + 5x + 3$

(18) $9x^2 - 9x + 2$

(28) $x^2 - 12x + 27$

(9) $x^2 - 5x + 6$

(19) $2x^2 + 7x - 9$

(29) $x^2 + 3x - 4$

(10) $72x^2 + 41x - 45$

(20) $21x^2 + 11x - 2$

(30) $7x^2 - 27x + 18$

(1) $10x^2 - 31x + 24$

(11) $5x^2 + 9x - 18$

(21) $7x^2 + 27x + 18$

(2) $9x^2 - 27x + 8$

(12) $24x^2 + x - 10$

(22) $4x^2 - 23x - 72$

(3) $8x^2 - 26x + 21$

(13) $18x^2 + 23x + 7$

(23) $21x^2 - 20x + 4$

(4) $32x^2 + 92x + 63$

(14) $14x^2 - 27x + 9$

(24) $6x^2 - 17x - 28$

(5) $4x^2 - 3x - 1$

(15) $4x^2 + 3x - 27$

(25) $x^2 - x - 2$

(6) $2x^2 - x - 10$

(16) $4x^2 - 4x - 3$

(26) $7x^2 + 22x + 16$

(7) $3x^2 + x - 2$

(17) $12x^2 - 4x - 1$

(27) $72x^2 - 137x + 63$

(8) $27x^2 - 93x + 56$

(18) $7x^2 + 39x - 18$

(28) $15x^2 + 4x - 32$

(9) $7x^2 - 52x + 21$

(19) $7x^2 + 2x - 5$

(29) $9x^2 + 9x - 10$

(10) $9x^2 - 4x - 5$

(20) $9x^2 - 8x - 1$

(30) $24x^2 + 2x - 15$

(1) $18x^2 + 3x - 28$

(11) $x^2 - 5x + 4$

(21) $5x^2 - 8x + 3$

(2) $7x^2 - 58x + 63$

(12) $49x^2 + 42x - 16$

(22) $49x^2 - 35x + 6$

(3) $36x^2 - 25x + 4$

(13) $x^2 - 4$

(23) $25x^2 - 25x + 4$

(4) $12x^2 + 17x + 6$

(14) $5x^2 + 4x - 1$

(24) $35x^2 - 44x - 7$

(5) $7x^2 + 2x - 9$

(15) $18x^2 + 45x + 28$

(25) $12x^2 + 7x + 1$

(6) $x^2 + x - 56$

(16) $6x^2 - 19x + 8$

(26) $x^2 - 5x + 6$

(7) $x^2 + 4x - 21$

(17) $27x^2 + 48x + 5$

(27) $40x^2 - x - 15$

(8) $3x^2 - 17x - 28$

(18) $4x^2 - 39x + 27$

(28) $2x^2 - 7x + 5$

(9) $2x^2 + 17x + 8$

(19) $x^2 + 10x + 9$

(29) $8x^2 + 33x + 27$

(10) $63x^2 - 44x + 5$

(20) $8x^2 - x - 9$

(30) $4x^2 - 5x - 6$

(1) $3x^2 + 5x + 2$

(11) $3x^2 - 10x + 8$

(21) $3x^2 - x - 4$

(2) $14x^2 - 25x + 9$

(12) $9x^2 + 9x - 4$

(22) $21x^2 - 29x + 10$

(3) $56x^2 + 47x - 18$

(13) $4x^2 + 9x + 5$

(23) $x^2 - x - 12$

(4) $2x^2 + 3x - 2$

(14) $9x^2 + 23x - 12$

(24) $5x^2 - 8x + 3$

(5) $3x^2 - 2x - 21$

(15) $4x^2 - 25x + 6$

(25) $5x^2 + 7x + 2$

(6) $3x^2 + 22x + 24$

(16) $72x^2 + 7x - 49$

(26) $9x^2 - 85x + 36$

(7) $25x^2 - 35x - 18$

(17) $2x^2 - 5x - 7$

(27) $36x^2 + 79x + 28$

(8) $5x^2 + 8x - 4$

(18) $63x^2 - 2x - 48$

(28) $24x^2 - 55x + 25$

(9) $12x^2 - 17x + 6$

(19) $6x^2 + 17x + 12$

(29) $2x^2 + x - 1$

(10) $10x^2 - 11x - 6$

(20) $2x^2 + 15x - 8$

(30) $4x^2 + 19x + 12$

(1) $12x^2 + 17x + 6$

(11) $8x^2 + 39x + 28$

(21) $21x^2 - 41x + 18$

(2) $14x^2 - 53x + 45$

(12) $15x^2 - x - 28$

(22) $5x^2 - 13x + 8$

(3) $3x^2 + 2x - 1$

(13) $14x^2 - 11x - 9$

(23) $4x^2 - 12x + 5$

(4) $2x^2 + x - 3$

(14) $3x^2 + 20x + 12$

(24) $14x^2 - 3x - 5$

(5) $16x^2 + 18x + 5$

(15) $3x^2 + 7x - 6$

(25) $4x^2 + 19x + 12$

(6) $4x^2 + 12x + 9$

(16) $8x^2 - 25x + 18$

(26) $3x^2 - 11x + 6$

(7) $9x^2 + 7x - 2$

(17) $6x^2 + 5x - 1$

(27) $36x^2 - 47x + 15$

(8) $3x^2 - 4x - 7$

(18) $15x^2 - x - 6$

(28) $35x^2 - 86x + 48$

(9) $8x^2 - 10x - 25$

(19) $x^2 - 8x + 7$

(29) $40x^2 - 77x + 36$

(10) $7x^2 + 11x + 4$

(20) $16x^2 + 38x + 21$

(30) $49x^2 - 21x - 40$

(1) $4x^2 - 5x + 1$

(11) $x^2 + 6x - 7$

(21) $10x^2 - 33x - 54$

(2) $x^2 + 6x + 5$

(12) $6x^2 - 17x + 7$

(22) $14x^2 + 19x - 40$

(3) $6x^2 - 13x + 7$

(13) $5x^2 - 3x - 2$

(23) $x^2 - 5x + 4$

(4) $12x^2 + 25x + 12$

(14) $x^2 - 1$

(24) $18x^2 + 3x - 28$

(5) $2x^2 + 21x + 49$

(15) $3x^2 + 7x + 2$

(25) $x^2 + 2x - 3$

(6) $7x^2 + 23x + 18$

(16) $5x^2 + 12x + 4$

(26) $8x^2 - 22x - 63$

(7) $8x^2 - 15x + 7$

(17) $20x^2 + 69x + 54$

(27) $7x^2 - 16x - 15$

(8) $4x^2 + 19x + 21$

(18) $4x^2 - 4x - 3$

(28) $14x^2 - x - 3$

(9) $3x^2 - 13x + 14$

(19) $63x^2 - 62x + 15$

(29) $3x^2 + 13x + 14$

(10) $x^2 + 7x - 8$

(20) $x^2 - x - 12$

(30) $x^2 + 3x - 4$

(1) $27x^2 + 12x - 32$

(11) $27x^2 + 33x + 8$

(21) $8x^2 + 9x - 14$

(2) $6x^2 + 31x + 18$

(12) $10x^2 + 9x - 7$

(22) $12x^2 + 8x - 7$

(3) $15x^2 - 29x + 8$

(13) $5x^2 - 3x - 14$

(23) $4x^2 + 4x - 15$

(4) $14x^2 + 23x + 8$

(14) $18x^2 + 89x + 36$

(24) $63x^2 - 8x - 15$

(5) $12x^2 + 25x - 7$

(15) $3x^2 - 16x + 5$

(25) $20x^2 + 27x - 14$

(6) $6x^2 + 5x - 6$

(16) $81x^2 - 54x - 7$

(26) $10x^2 - 21x + 8$

(7) $3x^2 - 16x - 35$

(17) $7x^2 + 10x - 8$

(27) $4x^2 - 41x + 72$

(8) $x^2 - 1$

(18) $7x^2 + 2x - 5$

(28) $16x^2 + 6x - 27$

(9) $15x^2 - 22x - 9$

(19) $27x^2 - 21x - 20$

(29) $28x^2 + 19x + 3$

(10) $2x^2 - 3x + 1$

(20) $3x^2 + 17x - 28$

(30) $3x^2 - 20x - 7$

(1) $21x^2 + 23x + 6$

(11) $21x^2 - 13x - 20$

(21) $x^2 - 2x + 1$

(2) $9x^2 + 6x + 1$

(12) $4x^2 - x - 5$

(22) $9x^2 + 38x - 35$

(3) $3x^2 + 4x - 7$

(13) $28x^2 + 81x + 56$

(23) $4x^2 + 5x - 6$

(4) $14x^2 - 3x - 2$

(14) $24x^2 - 22x + 3$

(24) $x^2 - 6x - 7$

(5) $16x^2 - 22x - 3$

(15) $2x^2 + x - 3$

(25) $4x^2 - 27x - 7$

(6) $4x^2 + 11x - 3$

(16) $6x^2 - 11x - 2$

(26) $30x^2 + x - 1$

(7) $8x^2 - 14x - 9$

(17) $18x^2 - 11x + 1$

(27) $48x^2 + 50x + 7$

(8) $20x^2 - 29x - 36$

(18) $5x^2 - 31x - 28$

(28) $20x^2 - 61x + 45$

(9) $2x^2 - 3x + 1$

(19) $x^2 - 2x + 1$

(29) $7x^2 + 12x + 5$

(10) $4x^2 + 13x + 3$

(20) $2x^2 + 5x + 3$

(30) $8x^2 + 33x + 27$

(1) $18x^2 + 19x - 12$

(11) $15x^2 + 7x - 4$

(21) $9x^2 - 17x - 2$

(2) $9x^2 - 9x + 2$

(12) $5x^2 + 53x + 72$

(22) $2x^2 + 5x + 3$

(3) $2x^2 + x - 3$

(13) $40x^2 + 49x + 15$

(23) $9x^2 - 18x + 8$

(4) $24x^2 + 7x - 6$

(14) $x^2 - 5x + 4$

(24) $14x^2 + 45x + 25$

(5) $x^2 + 3x - 4$

(15) $10x^2 - 37x + 7$

(25) $7x^2 - 5x - 2$

(6) $x^2 + x - 56$

(16) $24x^2 - 13x - 2$

(26) $x^2 + 7x - 8$

(7) $9x^2 + x - 8$

(17) $32x^2 - 60x - 27$

(27) $64x^2 - 81$

(8) $24x^2 + 22x - 7$

(18) $4x^2 - 19x - 5$

(28) $2x^2 + 5x - 7$

(9) $20x^2 + 39x + 7$

(19) $36x^2 + 5x - 1$

(29) $4x^2 - 3x - 27$

(10) $14x^2 + 3x - 2$

(20) $6x^2 - 29x + 20$

(30) $7x^2 - 54x - 16$

(1) $45x^2 - 92x + 32$

(11) $18x^2 - 17x - 15$

(21) $5x^2 - 37x - 72$

(2) $72x^2 - 83x + 21$

(12) $8x^2 - 3x - 5$

(22) $32x^2 - 4x - 15$

(3) $42x^2 - x - 30$

(13) $8x^2 + 15x + 7$

(23) $4x^2 + 13x - 12$

(4) $2x^2 + 11x + 9$

(14) $x^2 - 4x + 3$

(24) $28x^2 + 69x + 35$

(5) $25x^2 + 5x - 6$

(15) $7x^2 + 16x + 9$

(25) $28x^2 + 15x + 2$

(6) $9x^2 - 4x - 5$

(16) $2x^2 + 9x + 10$

(26) $32x^2 - 60x + 7$

(7) $3x^2 - 5x - 8$

(17) $35x^2 + 9x - 18$

(27) $16x^2 + 26x + 3$

(8) $36x^2 + 97x + 36$

(18) $5x^2 + 13x - 6$

(28) $3x^2 - 8x + 5$

(9) $32x^2 - 12x - 9$

(19) $5x^2 + 19x + 18$

(29) $36x^2 - 49x + 5$

(10) $10x^2 - 23x + 12$

(20) $2x^2 - x - 10$

(30) $12x^2 - 31x + 7$

(1) $45x^2 - 53x + 14$

(11) $6x^2 + 35x - 6$

(21) $56x^2 - 9x - 35$

(2) $24x^2 - 11x - 18$

(12) $4x^2 + 5x - 6$

(22) $7x^2 - 29x + 24$

(3) $8x^2 + 14x + 5$

(13) $2x^2 + 3x - 2$

(23) $40x^2 - 39x + 9$

(4) $15x^2 + 4x - 32$

(14) $3x^2 + 5x - 2$

(24) $36x^2 + 95x + 56$

(5) $4x^2 + 21x + 5$

(15) $6x^2 + 29x + 28$

(25) $x^2 + 2x - 15$

(6) $20x^2 + 17x - 10$

(16) $16x^2 - 74x + 63$

(26) $3x^2 - 5x - 2$

(7) $81x^2 + 36x - 32$

(17) $3x^2 + 32x + 64$

(27) $2x^2 + x - 21$

(8) $5x^2 + 14x - 3$

(18) $16x^2 + 2x - 5$

(28) $36x^2 + 37x - 10$

(9) $5x^2 + 4x - 1$

(19) $4x^2 + 17x + 15$

(29) $10x^2 + 43x + 45$

(10) $6x^2 + 13x - 8$

(20) $24x^2 - 43x + 18$

(30) $x^2 + 2x + 1$

$$\begin{aligned} (1) \quad & 5x^2 - 4x - 9 \\ &= (x + 1)(5x - 9) \end{aligned}$$

$$\begin{aligned} (11) \quad & 5x^2 - 29x + 20 \\ &= (x - 5)(5x - 4) \end{aligned}$$

$$\begin{aligned} (21) \quad & 14x^2 - 73x + 45 \\ &= (2x - 9)(7x - 5) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^2 - 6x - 7 \\ &= (x + 1)(x - 7) \end{aligned}$$

$$\begin{aligned} (12) \quad & 2x^2 - x - 1 \\ &= (2x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 27x^2 - 24x + 5 \\ &= (3x - 1)(9x - 5) \end{aligned}$$

$$\begin{aligned} (3) \quad & 5x^2 + 6x - 8 \\ &= (5x - 4)(x + 2) \end{aligned}$$

$$\begin{aligned} (13) \quad & 2x^2 - 15x + 7 \\ &= (2x - 1)(x - 7) \end{aligned}$$

$$\begin{aligned} (23) \quad & 5x^2 + 14x + 9 \\ &= (5x + 9)(x + 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 12x^2 - 13x - 4 \\ &= (4x + 1)(3x - 4) \end{aligned}$$

$$\begin{aligned} (14) \quad & 7x^2 + 3x - 4 \\ &= (7x - 4)(x + 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 12x^2 - 11x - 5 \\ &= (4x - 5)(3x + 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^2 - 11x - 4 \\ &= (x - 4)(3x + 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & x^2 + 4x - 5 \\ &= (x - 1)(x + 5) \end{aligned}$$

$$\begin{aligned} (25) \quad & 2x^2 + 3x - 5 \\ &= (x - 1)(2x + 5) \end{aligned}$$

$$\begin{aligned} (6) \quad & 36x^2 - 89x + 18 \\ &= (4x - 9)(9x - 2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 21x^2 - 2x - 8 \\ &= (3x - 2)(7x + 4) \end{aligned}$$

$$\begin{aligned} (26) \quad & 2x^2 - 15x + 7 \\ &= (x - 7)(2x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^2 + 21x + 54 \\ &= (x + 6)(2x + 9) \end{aligned}$$

$$\begin{aligned} (17) \quad & 14x^2 - 19x - 3 \\ &= (2x - 3)(7x + 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 10x^2 + 13x - 3 \\ &= (5x - 1)(2x + 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 12x^2 - 17x - 5 \\ &= (4x + 1)(3x - 5) \end{aligned}$$

$$\begin{aligned} (18) \quad & 3x^2 - x - 2 \\ &= (3x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 4x^2 + 9x - 28 \\ &= (4x - 7)(x + 4) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^2 - 4x - 7 \\ &= (3x - 7)(x + 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 2x^2 - 11x + 5 \\ &= (2x - 1)(x - 5) \end{aligned}$$

$$\begin{aligned} (29) \quad & 3x^2 - 2x - 16 \\ &= (x + 2)(3x - 8) \end{aligned}$$

$$\begin{aligned} (10) \quad & 16x^2 + 32x + 7 \\ &= (4x + 7)(4x + 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 5x^2 + 33x + 40 \\ &= (5x + 8)(x + 5) \end{aligned}$$

$$\begin{aligned} (30) \quad & 6x^2 + 25x + 21 \\ &= (6x + 7)(x + 3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^2 - 7x + 4 \\ &= (x - 1)(3x - 4) \end{aligned}$$

$$\begin{aligned} (11) \quad & 18x^2 + 79x - 9 \\ &= (2x + 9)(9x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 6x^2 + 5x - 6 \\ &= (3x - 2)(2x + 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 4x^2 - 33x + 8 \\ &= (x - 8)(4x - 1) \end{aligned}$$

$$\begin{aligned} (12) \quad & 14x^2 + 23x - 30 \\ &= (7x - 6)(2x + 5) \end{aligned}$$

$$\begin{aligned} (22) \quad & 10x^2 + 27x + 18 \\ &= (2x + 3)(5x + 6) \end{aligned}$$

$$\begin{aligned} (3) \quad & 18x^2 - 21x + 5 \\ &= (6x - 5)(3x - 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 2x^2 + 13x - 7 \\ &= (2x - 1)(x + 7) \end{aligned}$$

$$\begin{aligned} (23) \quad & 3x^2 - 26x + 16 \\ &= (x - 8)(3x - 2) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^2 - 23x + 45 \\ &= (x - 9)(2x - 5) \end{aligned}$$

$$\begin{aligned} (14) \quad & 21x^2 + 23x + 6 \\ &= (3x + 2)(7x + 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & 30x^2 + 67x + 35 \\ &= (6x + 5)(5x + 7) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^2 - 7x + 2 \\ &= (x - 2)(3x - 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 7x^2 - 16x + 9 \\ &= (7x - 9)(x - 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 16x^2 - 1 \\ &= (4x - 1)(4x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 35x^2 + 53x - 18 \\ &= (5x + 9)(7x - 2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 25x^2 - 55x + 18 \\ &= (5x - 2)(5x - 9) \end{aligned}$$

$$\begin{aligned} (26) \quad & 2x^2 + 9x + 4 \\ &= (x + 4)(2x + 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 42x^2 + 19x + 2 \\ &= (6x + 1)(7x + 2) \end{aligned}$$

$$\begin{aligned} (17) \quad & 7x^2 - 29x + 4 \\ &= (7x - 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (27) \quad & 4x^2 + 8x - 21 \\ &= (2x + 7)(2x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 15x^2 + 47x + 28 \\ &= (5x + 4)(3x + 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & 2x^2 + 3x + 1 \\ &= (x + 1)(2x + 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 2x^2 + 13x - 24 \\ &= (2x - 3)(x + 8) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^2 + 13x + 14 \\ &= (x + 2)(3x + 7) \end{aligned}$$

$$\begin{aligned} (19) \quad & 5x^2 - 12x - 32 \\ &= (5x + 8)(x - 4) \end{aligned}$$

$$\begin{aligned} (29) \quad & x^2 + 9x + 18 \\ &= (x + 6)(x + 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^2 + 5x + 2 \\ &= (2x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (20) \quad & 8x^2 + 41x + 36 \\ &= (x + 4)(8x + 9) \end{aligned}$$

$$\begin{aligned} (30) \quad & 32x^2 + 84x + 27 \\ &= (4x + 9)(8x + 3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 24x^2 - 22x - 7 \\ &= (4x + 1)(6x - 7) \end{aligned}$$

$$\begin{aligned} (11) \quad & 5x^2 - x - 6 \\ &= (5x - 6)(x + 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 3x^2 - x - 2 \\ &= (x - 1)(3x + 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 7x^2 + x - 6 \\ &= (x + 1)(7x - 6) \end{aligned}$$

$$\begin{aligned} (12) \quad & 8x^2 + 9x - 14 \\ &= (x + 2)(8x - 7) \end{aligned}$$

$$\begin{aligned} (22) \quad & 14x^2 + 45x - 81 \\ &= (7x - 9)(2x + 9) \end{aligned}$$

$$\begin{aligned} (3) \quad & 7x^2 + 15x + 2 \\ &= (7x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (13) \quad & 3x^2 + 13x + 4 \\ &= (3x + 1)(x + 4) \end{aligned}$$

$$\begin{aligned} (23) \quad & 5x^2 + 3x - 8 \\ &= (5x + 8)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^2 + 5x - 3 \\ &= (x + 3)(2x - 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 63x^2 + 4x - 35 \\ &= (7x - 5)(9x + 7) \end{aligned}$$

$$\begin{aligned} (24) \quad & x^2 - x - 2 \\ &= (x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 25x^2 + 25x + 4 \\ &= (5x + 4)(5x + 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 9x^2 - 10x + 1 \\ &= (x - 1)(9x - 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 45x^2 - 14x + 1 \\ &= (5x - 1)(9x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 21x^2 + 11x - 6 \\ &= (3x - 1)(7x + 6) \end{aligned}$$

$$\begin{aligned} (16) \quad & 2x^2 + 3x - 9 \\ &= (2x - 3)(x + 3) \end{aligned}$$

$$\begin{aligned} (26) \quad & 49x^2 + 49x - 18 \\ &= (7x - 2)(7x + 9) \end{aligned}$$

$$\begin{aligned} (7) \quad & 6x^2 - 17x + 10 \\ &= (x - 2)(6x - 5) \end{aligned}$$

$$\begin{aligned} (17) \quad & 8x^2 - 7x - 18 \\ &= (8x + 9)(x - 2) \end{aligned}$$

$$\begin{aligned} (27) \quad & 56x^2 - 45x + 9 \\ &= (7x - 3)(8x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 2x^2 - 9x + 4 \\ &= (2x - 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (18) \quad & 8x^2 + 13x + 5 \\ &= (x + 1)(8x + 5) \end{aligned}$$

$$\begin{aligned} (28) \quad & 36x^2 - 13x - 40 \\ &= (9x + 8)(4x - 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 12x^2 + 7x + 1 \\ &= (4x + 1)(3x + 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 42x^2 + 13x + 1 \\ &= (6x + 1)(7x + 1) \end{aligned}$$

$$\begin{aligned} (29) \quad & 18x^2 + 37x + 15 \\ &= (2x + 3)(9x + 5) \end{aligned}$$

$$\begin{aligned} (10) \quad & 5x^2 - 28x - 49 \\ &= (5x + 7)(x - 7) \end{aligned}$$

$$\begin{aligned} (20) \quad & 48x^2 + 94x + 45 \\ &= (8x + 9)(6x + 5) \end{aligned}$$

$$\begin{aligned} (30) \quad & 2x^2 - 3x - 9 \\ &= (x - 3)(2x + 3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 5x^2 + 12x - 9 \\ & = (5x - 3)(x + 3) \end{aligned}$$

$$\begin{aligned} (11) \quad & x^2 + 2x + 1 \\ & = (x + 1)^2 \end{aligned}$$

$$\begin{aligned} (21) \quad & 14x^2 + x - 3 \\ & = (2x + 1)(7x - 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 5x^2 - 12x + 7 \\ & = (5x - 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (12) \quad & 9x^2 + 12x - 32 \\ & = (3x - 4)(3x + 8) \end{aligned}$$

$$\begin{aligned} (22) \quad & 3x^2 - x - 10 \\ & = (x - 2)(3x + 5) \end{aligned}$$

$$\begin{aligned} (3) \quad & 20x^2 + 3x - 9 \\ & = (4x + 3)(5x - 3) \end{aligned}$$

$$\begin{aligned} (13) \quad & 5x^2 - x - 4 \\ & = (5x + 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (23) \quad & 28x^2 - 75x + 27 \\ & = (4x - 9)(7x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 9x^2 + 6x + 1 \\ & = (3x + 1)^2 \end{aligned}$$

$$\begin{aligned} (14) \quad & 10x^2 - 27x + 5 \\ & = (2x - 5)(5x - 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 3x^2 + 4x + 1 \\ & = (x + 1)(3x + 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 16x^2 - 16x - 21 \\ & = (4x - 7)(4x + 3) \end{aligned}$$

$$\begin{aligned} (15) \quad & 10x^2 - 21x + 8 \\ & = (2x - 1)(5x - 8) \end{aligned}$$

$$\begin{aligned} (25) \quad & 20x^2 + 43x + 21 \\ & = (5x + 7)(4x + 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 9x^2 - 3x - 20 \\ & = (3x + 4)(3x - 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 40x^2 + 99x + 56 \\ & = (5x + 8)(8x + 7) \end{aligned}$$

$$\begin{aligned} (26) \quad & 6x^2 + 19x + 3 \\ & = (6x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^2 + 2x - 8 \\ & = (x + 2)(3x - 4) \end{aligned}$$

$$\begin{aligned} (17) \quad & 5x^2 - 16x + 12 \\ & = (x - 2)(5x - 6) \end{aligned}$$

$$\begin{aligned} (27) \quad & 2x^2 - 3x + 1 \\ & = (2x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^2 - 8x + 4 \\ & = (3x - 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (18) \quad & 5x^2 - 2x - 3 \\ & = (5x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & x^2 + 10x + 24 \\ & = (x + 6)(x + 4) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^2 + 5x - 3 \\ & = (x + 3)(2x - 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 2x^2 + 3x - 9 \\ & = (2x - 3)(x + 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 3x^2 + 25x - 18 \\ & = (x + 9)(3x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^2 - 3x - 4 \\ & = (x + 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (20) \quad & 24x^2 + 25x + 6 \\ & = (8x + 3)(3x + 2) \end{aligned}$$

$$\begin{aligned} (30) \quad & 3x^2 - 10x + 7 \\ & = (x - 1)(3x - 7) \end{aligned}$$

$$\begin{aligned} (1) \quad & 21x^2 + 25x - 4 \\ & = (3x + 4)(7x - 1) \end{aligned}$$

$$\begin{aligned} (11) \quad & 4x^2 - 11x + 7 \\ & = (4x - 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 4x^2 + 5x - 9 \\ & = (4x + 9)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^2 - 6x - 7 \\ & = (x + 1)(x - 7) \end{aligned}$$

$$\begin{aligned} (12) \quad & 3x^2 - 20x + 12 \\ & = (x - 6)(3x - 2) \end{aligned}$$

$$\begin{aligned} (22) \quad & 5x^2 - 3x - 2 \\ & = (5x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 36x^2 + 23x - 3 \\ & = (4x + 3)(9x - 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 18x^2 + 83x + 9 \\ & = (9x + 1)(2x + 9) \end{aligned}$$

$$\begin{aligned} (23) \quad & 7x^2 - 17x + 6 \\ & = (x - 2)(7x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 35x^2 - 31x - 18 \\ & = (7x - 9)(5x + 2) \end{aligned}$$

$$\begin{aligned} (14) \quad & 25x^2 + 55x + 24 \\ & = (5x + 8)(5x + 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & 28x^2 - 57x + 27 \\ & = (4x - 3)(7x - 9) \end{aligned}$$

$$\begin{aligned} (5) \quad & 4x^2 + 3x - 7 \\ & = (x - 1)(4x + 7) \end{aligned}$$

$$\begin{aligned} (15) \quad & 14x^2 - 9x - 8 \\ & = (2x + 1)(7x - 8) \end{aligned}$$

$$\begin{aligned} (25) \quad & 24x^2 + 26x - 63 \\ & = (6x - 7)(4x + 9) \end{aligned}$$

$$\begin{aligned} (6) \quad & 10x^2 + 9x - 7 \\ & = (5x + 7)(2x - 1) \end{aligned}$$

$$\begin{aligned} (16) \quad & 24x^2 + 14x - 3 \\ & = (4x + 3)(6x - 1) \end{aligned}$$

$$\begin{aligned} (26) \quad & 10x^2 + 39x + 35 \\ & = (2x + 5)(5x + 7) \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^2 - x - 3 \\ & = (x + 1)(2x - 3) \end{aligned}$$

$$\begin{aligned} (17) \quad & 9x^2 + 11x + 2 \\ & = (9x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 12x^2 - 7x + 1 \\ & = (4x - 1)(3x - 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 27x^2 + 39x + 4 \\ & = (9x + 1)(3x + 4) \end{aligned}$$

$$\begin{aligned} (18) \quad & x^2 - 3x + 2 \\ & = (x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (28) \quad & 24x^2 - 17x - 20 \\ & = (3x - 4)(8x + 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 24x^2 - 29x + 7 \\ & = (8x - 7)(3x - 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 49x^2 + 56x + 15 \\ & = (7x + 5)(7x + 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 7x^2 + x - 8 \\ & = (7x + 8)(x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 15x^2 - 44x + 21 \\ & = (5x - 3)(3x - 7) \end{aligned}$$

$$\begin{aligned} (20) \quad & 35x^2 + 32x - 12 \\ & = (5x + 6)(7x - 2) \end{aligned}$$

$$\begin{aligned} (30) \quad & 3x^2 - 11x + 8 \\ & = (x - 1)(3x - 8) \end{aligned}$$

$$\begin{aligned} (1) \quad & 49x^2 + 35x - 36 \\ & = (7x - 4)(7x + 9) \end{aligned}$$

$$\begin{aligned} (11) \quad & 14x^2 - 3x - 2 \\ & = (2x - 1)(7x + 2) \end{aligned}$$

$$\begin{aligned} (21) \quad & 27x^2 - 30x + 7 \\ & = (9x - 7)(3x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 45x^2 - 76x + 32 \\ & = (5x - 4)(9x - 8) \end{aligned}$$

$$\begin{aligned} (12) \quad & 2x^2 + x - 1 \\ & = (x + 1)(2x - 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 14x^2 - 25x + 6 \\ & = (7x - 2)(2x - 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 10x^2 + 3x - 1 \\ & = (2x + 1)(5x - 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 10x^2 - 23x + 12 \\ & = (2x - 3)(5x - 4) \end{aligned}$$

$$\begin{aligned} (23) \quad & 4x^2 + 12x - 27 \\ & = (2x + 9)(2x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 15x^2 - 22x - 9 \\ & = (5x - 9)(3x + 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 24x^2 - 17x + 3 \\ & = (8x - 3)(3x - 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 9x^2 - 37x + 4 \\ & = (9x - 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (5) \quad & 14x^2 + 15x + 4 \\ & = (2x + 1)(7x + 4) \end{aligned}$$

$$\begin{aligned} (15) \quad & 64x^2 + 80x + 21 \\ & = (8x + 3)(8x + 7) \end{aligned}$$

$$\begin{aligned} (25) \quad & 4x^2 + 11x - 3 \\ & = (4x - 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 30x^2 + 73x + 40 \\ & = (5x + 8)(6x + 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 2x^2 + 11x + 15 \\ & = (x + 3)(2x + 5) \end{aligned}$$

$$\begin{aligned} (26) \quad & 15x^2 + 28x - 32 \\ & = (5x - 4)(3x + 8) \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^2 + 17x - 9 \\ & = (x + 9)(2x - 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & x^2 - 36 \\ & = (x + 6)(x - 6) \end{aligned}$$

$$\begin{aligned} (27) \quad & 3x^2 - 4x - 7 \\ & = (3x - 7)(x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 7x^2 + 26x - 45 \\ & = (7x - 9)(x + 5) \end{aligned}$$

$$\begin{aligned} (18) \quad & x^2 + 3x + 2 \\ & = (x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 7x^2 + 43x - 42 \\ & = (x + 7)(7x - 6) \end{aligned}$$

$$\begin{aligned} (9) \quad & 8x^2 - 2x - 1 \\ & = (2x - 1)(4x + 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 5x^2 - 31x - 28 \\ & = (x - 7)(5x + 4) \end{aligned}$$

$$\begin{aligned} (29) \quad & 56x^2 - 19x - 15 \\ & = (7x - 5)(8x + 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^2 + 3x + 2 \\ & = (x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (20) \quad & x^2 + 3x + 2 \\ & = (x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (30) \quad & 12x^2 - 35x + 25 \\ & = (4x - 5)(3x - 5) \end{aligned}$$

$$\begin{aligned} (1) \quad & 7x^2 + 2x - 5 \\ &= (7x - 5)(x + 1) \end{aligned}$$

$$\begin{aligned} (11) \quad & 3x^2 + 2x - 1 \\ &= (x + 1)(3x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 8x^2 - 3x - 5 \\ &= (8x + 5)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 5x^2 + 6x + 1 \\ &= (x + 1)(5x + 1) \end{aligned}$$

$$\begin{aligned} (12) \quad & 2x^2 + 3x + 1 \\ &= (x + 1)(2x + 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 7x^2 - 5x - 2 \\ &= (x - 1)(7x + 2) \end{aligned}$$

$$\begin{aligned} (3) \quad & x^2 - 1 \\ &= (x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 20x^2 - 11x - 4 \\ &= (5x - 4)(4x + 1) \end{aligned}$$

$$\begin{aligned} (23) \quad & 20x^2 + 9x + 1 \\ &= (4x + 1)(5x + 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 56x^2 + 17x - 28 \\ &= (7x - 4)(8x + 7) \end{aligned}$$

$$\begin{aligned} (14) \quad & 3x^2 - x - 2 \\ &= (3x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 3x^2 - 11x + 6 \\ &= (3x - 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^2 + 13x - 10 \\ &= (3x - 2)(x + 5) \end{aligned}$$

$$\begin{aligned} (15) \quad & x^2 - 14x + 48 \\ &= (x - 6)(x - 8) \end{aligned}$$

$$\begin{aligned} (25) \quad & 30x^2 + 31x - 6 \\ &= (5x + 6)(6x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 42x^2 - 41x + 5 \\ &= (7x - 1)(6x - 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 72x^2 + 113x + 36 \\ &= (8x + 9)(9x + 4) \end{aligned}$$

$$\begin{aligned} (26) \quad & 16x^2 - 9 \\ &= (4x + 3)(4x - 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & 5x^2 - 27x + 10 \\ &= (x - 5)(5x - 2) \end{aligned}$$

$$\begin{aligned} (17) \quad & 56x^2 + 15x - 54 \\ &= (7x - 6)(8x + 9) \end{aligned}$$

$$\begin{aligned} (27) \quad & 2x^2 + 7x + 5 \\ &= (x + 1)(2x + 5) \end{aligned}$$

$$\begin{aligned} (8) \quad & x^2 + 4x + 3 \\ &= (x + 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (18) \quad & 8x^2 + 6x + 1 \\ &= (2x + 1)(4x + 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 3x^2 - 13x + 4 \\ &= (3x - 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (9) \quad & 7x^2 + 23x - 20 \\ &= (x + 4)(7x - 5) \end{aligned}$$

$$\begin{aligned} (19) \quad & 2x^2 - 3x - 5 \\ &= (x + 1)(2x - 5) \end{aligned}$$

$$\begin{aligned} (29) \quad & 25x^2 + 5x - 6 \\ &= (5x + 3)(5x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 9x^2 + 83x + 18 \\ &= (9x + 2)(x + 9) \end{aligned}$$

$$\begin{aligned} (20) \quad & x^2 - 4x + 3 \\ &= (x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (30) \quad & 18x^2 + 95x + 63 \\ &= (9x + 7)(2x + 9) \end{aligned}$$

$$\begin{aligned} (1) \quad & 9x^2 - 5x - 4 \\ &= (x - 1)(9x + 4) \end{aligned}$$

$$\begin{aligned} (11) \quad & 9x^2 + 30x + 16 \\ &= (3x + 8)(3x + 2) \end{aligned}$$

$$\begin{aligned} (21) \quad & 20x^2 + 17x - 24 \\ &= (4x - 3)(5x + 8) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^2 + 7x - 3 \\ &= (3x - 1)(2x + 3) \end{aligned}$$

$$\begin{aligned} (12) \quad & 45x^2 + 34x + 5 \\ &= (9x + 5)(5x + 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 18x^2 + 51x + 35 \\ &= (6x + 7)(3x + 5) \end{aligned}$$

$$\begin{aligned} (3) \quad & 21x^2 - 10x - 24 \\ &= (3x - 4)(7x + 6) \end{aligned}$$

$$\begin{aligned} (13) \quad & 56x^2 - 73x + 21 \\ &= (7x - 3)(8x - 7) \end{aligned}$$

$$\begin{aligned} (23) \quad & 12x^2 - 29x + 14 \\ &= (3x - 2)(4x - 7) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^2 - 4x + 1 \\ &= (3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 9x^2 + 18x - 16 \\ &= (3x - 2)(3x + 8) \end{aligned}$$

$$\begin{aligned} (24) \quad & x^2 - 10x + 16 \\ &= (x - 2)(x - 8) \end{aligned}$$

$$\begin{aligned} (5) \quad & 42x^2 - 25x - 28 \\ &= (7x + 4)(6x - 7) \end{aligned}$$

$$\begin{aligned} (15) \quad & 24x^2 - 61x - 8 \\ &= (3x - 8)(8x + 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 28x^2 - 23x + 4 \\ &= (7x - 4)(4x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 6x^2 + 13x - 5 \\ &= (3x - 1)(2x + 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 4x^2 - 13x + 3 \\ &= (x - 3)(4x - 1) \end{aligned}$$

$$\begin{aligned} (26) \quad & 20x^2 - x - 1 \\ &= (4x - 1)(5x + 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^2 + 4x + 3 \\ &= (x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (17) \quad & 14x^2 - 13x + 3 \\ &= (7x - 3)(2x - 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 2x^2 + 3x + 1 \\ &= (2x + 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^2 + 17x + 7 \\ &= (2x + 1)(3x + 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & x^2 - 5x + 4 \\ &= (x - 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & x^2 - x - 2 \\ &= (x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 14x^2 - 59x + 35 \\ &= (7x - 5)(2x - 7) \end{aligned}$$

$$\begin{aligned} (19) \quad & x^2 + 6x - 27 \\ &= (x + 9)(x - 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 6x^2 + 23x + 15 \\ &= (x + 3)(6x + 5) \end{aligned}$$

$$\begin{aligned} (10) \quad & 4x^2 + 5x - 6 \\ &= (x + 2)(4x - 3) \end{aligned}$$

$$\begin{aligned} (20) \quad & 4x^2 + 12x + 5 \\ &= (2x + 5)(2x + 1) \end{aligned}$$

$$\begin{aligned} (30) \quad & 8x^2 + 15x - 2 \\ &= (8x - 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 2x^2 - x - 1 \\ &= (x - 1)(2x + 1) \end{aligned}$$

$$\begin{aligned} (11) \quad & 32x^2 - 28x - 15 \\ &= (4x - 5)(8x + 3) \end{aligned}$$

$$\begin{aligned} (21) \quad & 21x^2 + 34x + 9 \\ &= (3x + 1)(7x + 9) \end{aligned}$$

$$\begin{aligned} (2) \quad & 56x^2 + 41x - 7 \\ &= (7x - 1)(8x + 7) \end{aligned}$$

$$\begin{aligned} (12) \quad & 15x^2 - 22x + 8 \\ &= (3x - 2)(5x - 4) \end{aligned}$$

$$\begin{aligned} (22) \quad & 8x^2 + 10x - 7 \\ &= (4x + 7)(2x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 24x^2 + 11x + 1 \\ &= (3x + 1)(8x + 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & x^2 + 3x + 2 \\ &= (x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (23) \quad & 5x^2 - 11x + 6 \\ &= (x - 1)(5x - 6) \end{aligned}$$

$$\begin{aligned} (4) \quad & 15x^2 + 22x + 8 \\ &= (3x + 2)(5x + 4) \end{aligned}$$

$$\begin{aligned} (14) \quad & 56x^2 + 15x + 1 \\ &= (8x + 1)(7x + 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 5x^2 + 12x + 7 \\ &= (5x + 7)(x + 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^2 - 3x - 18 \\ &= (x + 3)(x - 6) \end{aligned}$$

$$\begin{aligned} (15) \quad & 10x^2 + 3x - 4 \\ &= (5x + 4)(2x - 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 3x^2 - 8x - 3 \\ &= (3x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 15x^2 + 49x + 40 \\ &= (5x + 8)(3x + 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 81x^2 - 45x - 14 \\ &= (9x + 2)(9x - 7) \end{aligned}$$

$$\begin{aligned} (26) \quad & 56x^2 - 65x + 14 \\ &= (8x - 7)(7x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^2 + 2x - 5 \\ &= (x - 1)(3x + 5) \end{aligned}$$

$$\begin{aligned} (17) \quad & 6x^2 - 13x + 6 \\ &= (2x - 3)(3x - 2) \end{aligned}$$

$$\begin{aligned} (27) \quad & 35x^2 - 66x + 27 \\ &= (7x - 9)(5x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & x^2 - 6x - 16 \\ &= (x - 8)(x + 2) \end{aligned}$$

$$\begin{aligned} (18) \quad & 16x^2 + 2x - 5 \\ &= (8x + 5)(2x - 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 18x^2 + 9x + 1 \\ &= (3x + 1)(6x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 10x^2 + x - 2 \\ &= (5x - 2)(2x + 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & x^2 - 3x + 2 \\ &= (x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (29) \quad & 24x^2 + 29x - 4 \\ &= (8x - 1)(3x + 4) \end{aligned}$$

$$\begin{aligned} (10) \quad & 9x^2 - 11x + 2 \\ &= (9x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & x^2 + 11x + 24 \\ &= (x + 3)(x + 8) \end{aligned}$$

$$\begin{aligned} (30) \quad & 5x^2 - 4x - 1 \\ &= (x - 1)(5x + 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 18x^2 + 27x + 7 \\ & = (3x + 1)(6x + 7) \end{aligned}$$

$$\begin{aligned} (11) \quad & 4x^2 + x - 3 \\ & = (4x - 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 45x^2 + 112x + 64 \\ & = (5x + 8)(9x + 8) \end{aligned}$$

$$\begin{aligned} (2) \quad & 12x^2 + x - 6 \\ & = (3x - 2)(4x + 3) \end{aligned}$$

$$\begin{aligned} (12) \quad & 3x^2 - 10x + 7 \\ & = (3x - 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 7x^2 + 10x + 3 \\ & = (x + 1)(7x + 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 6x^2 - 47x - 8 \\ & = (x - 8)(6x + 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 40x^2 + 21x - 49 \\ & = (8x - 7)(5x + 7) \end{aligned}$$

$$\begin{aligned} (23) \quad & x^2 + 6x + 9 \\ & = (x + 3)^2 \end{aligned}$$

$$\begin{aligned} (4) \quad & 15x^2 + x - 6 \\ & = (5x - 3)(3x + 2) \end{aligned}$$

$$\begin{aligned} (14) \quad & 16x^2 + 10x - 9 \\ & = (2x - 1)(8x + 9) \end{aligned}$$

$$\begin{aligned} (24) \quad & 7x^2 + 24x - 16 \\ & = (x + 4)(7x - 4) \end{aligned}$$

$$\begin{aligned} (5) \quad & 6x^2 + x - 15 \\ & = (3x + 5)(2x - 3) \end{aligned}$$

$$\begin{aligned} (15) \quad & 4x^2 + 27x - 40 \\ & = (4x - 5)(x + 8) \end{aligned}$$

$$\begin{aligned} (25) \quad & 9x^2 + 11x + 2 \\ & = (9x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 6x^2 - 11x - 10 \\ & = (2x - 5)(3x + 2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 20x^2 - 31x - 7 \\ & = (5x + 1)(4x - 7) \end{aligned}$$

$$\begin{aligned} (26) \quad & 35x^2 - 24x - 35 \\ & = (5x - 7)(7x + 5) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^2 + x - 2 \\ & = (x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & 4x^2 + 19x + 12 \\ & = (x + 4)(4x + 3) \end{aligned}$$

$$\begin{aligned} (27) \quad & 56x^2 - 59x + 15 \\ & = (7x - 3)(8x - 5) \end{aligned}$$

$$\begin{aligned} (8) \quad & 2x^2 + 5x + 3 \\ & = (2x + 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (18) \quad & 9x^2 - 9x + 2 \\ & = (3x - 2)(3x - 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & x^2 - 12x + 27 \\ & = (x - 3)(x - 9) \end{aligned}$$

$$\begin{aligned} (9) \quad & x^2 - 5x + 6 \\ & = (x - 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (19) \quad & 2x^2 + 7x - 9 \\ & = (x - 1)(2x + 9) \end{aligned}$$

$$\begin{aligned} (29) \quad & x^2 + 3x - 4 \\ & = (x - 1)(x + 4) \end{aligned}$$

$$\begin{aligned} (10) \quad & 72x^2 + 41x - 45 \\ & = (8x + 9)(9x - 5) \end{aligned}$$

$$\begin{aligned} (20) \quad & 21x^2 + 11x - 2 \\ & = (7x - 1)(3x + 2) \end{aligned}$$

$$\begin{aligned} (30) \quad & 7x^2 - 27x + 18 \\ & = (7x - 6)(x - 3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 10x^2 - 31x + 24 \\ & = (2x - 3)(5x - 8) \end{aligned}$$

$$\begin{aligned} (11) \quad & 5x^2 + 9x - 18 \\ & = (5x - 6)(x + 3) \end{aligned}$$

$$\begin{aligned} (21) \quad & 7x^2 + 27x + 18 \\ & = (7x + 6)(x + 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 9x^2 - 27x + 8 \\ & = (3x - 8)(3x - 1) \end{aligned}$$

$$\begin{aligned} (12) \quad & 24x^2 + x - 10 \\ & = (3x + 2)(8x - 5) \end{aligned}$$

$$\begin{aligned} (22) \quad & 4x^2 - 23x - 72 \\ & = (x - 8)(4x + 9) \end{aligned}$$

$$\begin{aligned} (3) \quad & 8x^2 - 26x + 21 \\ & = (2x - 3)(4x - 7) \end{aligned}$$

$$\begin{aligned} (13) \quad & 18x^2 + 23x + 7 \\ & = (9x + 7)(2x + 1) \end{aligned}$$

$$\begin{aligned} (23) \quad & 21x^2 - 20x + 4 \\ & = (3x - 2)(7x - 2) \end{aligned}$$

$$\begin{aligned} (4) \quad & 32x^2 + 92x + 63 \\ & = (4x + 7)(8x + 9) \end{aligned}$$

$$\begin{aligned} (14) \quad & 14x^2 - 27x + 9 \\ & = (7x - 3)(2x - 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & 6x^2 - 17x - 28 \\ & = (6x + 7)(x - 4) \end{aligned}$$

$$\begin{aligned} (5) \quad & 4x^2 - 3x - 1 \\ & = (4x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 4x^2 + 3x - 27 \\ & = (4x - 9)(x + 3) \end{aligned}$$

$$\begin{aligned} (25) \quad & x^2 - x - 2 \\ & = (x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^2 - x - 10 \\ & = (2x - 5)(x + 2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 4x^2 - 4x - 3 \\ & = (2x - 3)(2x + 1) \end{aligned}$$

$$\begin{aligned} (26) \quad & 7x^2 + 22x + 16 \\ & = (7x + 8)(x + 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^2 + x - 2 \\ & = (x + 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (17) \quad & 12x^2 - 4x - 1 \\ & = (6x + 1)(2x - 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 72x^2 - 137x + 63 \\ & = (8x - 9)(9x - 7) \end{aligned}$$

$$\begin{aligned} (8) \quad & 27x^2 - 93x + 56 \\ & = (9x - 7)(3x - 8) \end{aligned}$$

$$\begin{aligned} (18) \quad & 7x^2 + 39x - 18 \\ & = (7x - 3)(x + 6) \end{aligned}$$

$$\begin{aligned} (28) \quad & 15x^2 + 4x - 32 \\ & = (3x - 4)(5x + 8) \end{aligned}$$

$$\begin{aligned} (9) \quad & 7x^2 - 52x + 21 \\ & = (x - 7)(7x - 3) \end{aligned}$$

$$\begin{aligned} (19) \quad & 7x^2 + 2x - 5 \\ & = (7x - 5)(x + 1) \end{aligned}$$

$$\begin{aligned} (29) \quad & 9x^2 + 9x - 10 \\ & = (3x + 5)(3x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 9x^2 - 4x - 5 \\ & = (9x + 5)(x - 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 9x^2 - 8x - 1 \\ & = (x - 1)(9x + 1) \end{aligned}$$

$$\begin{aligned} (30) \quad & 24x^2 + 2x - 15 \\ & = (4x - 3)(6x + 5) \end{aligned}$$

$$\begin{aligned} (1) \quad & 18x^2 + 3x - 28 \\ & = (3x + 4)(6x - 7) \end{aligned}$$

$$\begin{aligned} (11) \quad & x^2 - 5x + 4 \\ & = (x - 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 5x^2 - 8x + 3 \\ & = (5x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 7x^2 - 58x + 63 \\ & = (7x - 9)(x - 7) \end{aligned}$$

$$\begin{aligned} (12) \quad & 49x^2 + 42x - 16 \\ & = (7x + 8)(7x - 2) \end{aligned}$$

$$\begin{aligned} (22) \quad & 49x^2 - 35x + 6 \\ & = (7x - 3)(7x - 2) \end{aligned}$$

$$\begin{aligned} (3) \quad & 36x^2 - 25x + 4 \\ & = (4x - 1)(9x - 4) \end{aligned}$$

$$\begin{aligned} (13) \quad & x^2 - 4 \\ & = (x + 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (23) \quad & 25x^2 - 25x + 4 \\ & = (5x - 1)(5x - 4) \end{aligned}$$

$$\begin{aligned} (4) \quad & 12x^2 + 17x + 6 \\ & = (3x + 2)(4x + 3) \end{aligned}$$

$$\begin{aligned} (14) \quad & 5x^2 + 4x - 1 \\ & = (x + 1)(5x - 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 35x^2 - 44x - 7 \\ & = (7x + 1)(5x - 7) \end{aligned}$$

$$\begin{aligned} (5) \quad & 7x^2 + 2x - 9 \\ & = (7x + 9)(x - 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 18x^2 + 45x + 28 \\ & = (3x + 4)(6x + 7) \end{aligned}$$

$$\begin{aligned} (25) \quad & 12x^2 + 7x + 1 \\ & = (4x + 1)(3x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & x^2 + x - 56 \\ & = (x + 8)(x - 7) \end{aligned}$$

$$\begin{aligned} (16) \quad & 6x^2 - 19x + 8 \\ & = (2x - 1)(3x - 8) \end{aligned}$$

$$\begin{aligned} (26) \quad & x^2 - 5x + 6 \\ & = (x - 3)(x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^2 + 4x - 21 \\ & = (x + 7)(x - 3) \end{aligned}$$

$$\begin{aligned} (17) \quad & 27x^2 + 48x + 5 \\ & = (3x + 5)(9x + 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 40x^2 - x - 15 \\ & = (8x - 5)(5x + 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^2 - 17x - 28 \\ & = (3x + 4)(x - 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & 4x^2 - 39x + 27 \\ & = (4x - 3)(x - 9) \end{aligned}$$

$$\begin{aligned} (28) \quad & 2x^2 - 7x + 5 \\ & = (x - 1)(2x - 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^2 + 17x + 8 \\ & = (2x + 1)(x + 8) \end{aligned}$$

$$\begin{aligned} (19) \quad & x^2 + 10x + 9 \\ & = (x + 9)(x + 1) \end{aligned}$$

$$\begin{aligned} (29) \quad & 8x^2 + 33x + 27 \\ & = (8x + 9)(x + 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 63x^2 - 44x + 5 \\ & = (7x - 1)(9x - 5) \end{aligned}$$

$$\begin{aligned} (20) \quad & 8x^2 - x - 9 \\ & = (x + 1)(8x - 9) \end{aligned}$$

$$\begin{aligned} (30) \quad & 4x^2 - 5x - 6 \\ & = (4x + 3)(x - 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^2 + 5x + 2 \\ &= (x + 1)(3x + 2) \end{aligned}$$

$$\begin{aligned} (11) \quad & 3x^2 - 10x + 8 \\ &= (x - 2)(3x - 4) \end{aligned}$$

$$\begin{aligned} (21) \quad & 3x^2 - x - 4 \\ &= (x + 1)(3x - 4) \end{aligned}$$

$$\begin{aligned} (2) \quad & 14x^2 - 25x + 9 \\ &= (2x - 1)(7x - 9) \end{aligned}$$

$$\begin{aligned} (12) \quad & 9x^2 + 9x - 4 \\ &= (3x + 4)(3x - 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 21x^2 - 29x + 10 \\ &= (3x - 2)(7x - 5) \end{aligned}$$

$$\begin{aligned} (3) \quad & 56x^2 + 47x - 18 \\ &= (7x - 2)(8x + 9) \end{aligned}$$

$$\begin{aligned} (13) \quad & 4x^2 + 9x + 5 \\ &= (4x + 5)(x + 1) \end{aligned}$$

$$\begin{aligned} (23) \quad & x^2 - x - 12 \\ &= (x + 3)(x - 4) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^2 + 3x - 2 \\ &= (x + 2)(2x - 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 9x^2 + 23x - 12 \\ &= (9x - 4)(x + 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & 5x^2 - 8x + 3 \\ &= (5x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^2 - 2x - 21 \\ &= (x - 3)(3x + 7) \end{aligned}$$

$$\begin{aligned} (15) \quad & 4x^2 - 25x + 6 \\ &= (4x - 1)(x - 6) \end{aligned}$$

$$\begin{aligned} (25) \quad & 5x^2 + 7x + 2 \\ &= (x + 1)(5x + 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 3x^2 + 22x + 24 \\ &= (3x + 4)(x + 6) \end{aligned}$$

$$\begin{aligned} (16) \quad & 72x^2 + 7x - 49 \\ &= (8x + 7)(9x - 7) \end{aligned}$$

$$\begin{aligned} (26) \quad & 9x^2 - 85x + 36 \\ &= (x - 9)(9x - 4) \end{aligned}$$

$$\begin{aligned} (7) \quad & 25x^2 - 35x - 18 \\ &= (5x + 2)(5x - 9) \end{aligned}$$

$$\begin{aligned} (17) \quad & 2x^2 - 5x - 7 \\ &= (2x - 7)(x + 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 36x^2 + 79x + 28 \\ &= (9x + 4)(4x + 7) \end{aligned}$$

$$\begin{aligned} (8) \quad & 5x^2 + 8x - 4 \\ &= (5x - 2)(x + 2) \end{aligned}$$

$$\begin{aligned} (18) \quad & 63x^2 - 2x - 48 \\ &= (7x + 6)(9x - 8) \end{aligned}$$

$$\begin{aligned} (28) \quad & 24x^2 - 55x + 25 \\ &= (8x - 5)(3x - 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 12x^2 - 17x + 6 \\ &= (3x - 2)(4x - 3) \end{aligned}$$

$$\begin{aligned} (19) \quad & 6x^2 + 17x + 12 \\ &= (3x + 4)(2x + 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 2x^2 + x - 1 \\ &= (x + 1)(2x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 10x^2 - 11x - 6 \\ &= (5x + 2)(2x - 3) \end{aligned}$$

$$\begin{aligned} (20) \quad & 2x^2 + 15x - 8 \\ &= (2x - 1)(x + 8) \end{aligned}$$

$$\begin{aligned} (30) \quad & 4x^2 + 19x + 12 \\ &= (4x + 3)(x + 4) \end{aligned}$$

$$\begin{aligned} (1) \quad & 12x^2 + 17x + 6 \\ &= (3x + 2)(4x + 3) \end{aligned}$$

$$\begin{aligned} (11) \quad & 8x^2 + 39x + 28 \\ &= (8x + 7)(x + 4) \end{aligned}$$

$$\begin{aligned} (21) \quad & 21x^2 - 41x + 18 \\ &= (3x - 2)(7x - 9) \end{aligned}$$

$$\begin{aligned} (2) \quad & 14x^2 - 53x + 45 \\ &= (7x - 9)(2x - 5) \end{aligned}$$

$$\begin{aligned} (12) \quad & 15x^2 - x - 28 \\ &= (5x - 7)(3x + 4) \end{aligned}$$

$$\begin{aligned} (22) \quad & 5x^2 - 13x + 8 \\ &= (5x - 8)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 3x^2 + 2x - 1 \\ &= (3x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 14x^2 - 11x - 9 \\ &= (2x + 1)(7x - 9) \end{aligned}$$

$$\begin{aligned} (23) \quad & 4x^2 - 12x + 5 \\ &= (2x - 5)(2x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^2 + x - 3 \\ &= (2x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 3x^2 + 20x + 12 \\ &= (x + 6)(3x + 2) \end{aligned}$$

$$\begin{aligned} (24) \quad & 14x^2 - 3x - 5 \\ &= (7x - 5)(2x + 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 16x^2 + 18x + 5 \\ &= (2x + 1)(8x + 5) \end{aligned}$$

$$\begin{aligned} (15) \quad & 3x^2 + 7x - 6 \\ &= (3x - 2)(x + 3) \end{aligned}$$

$$\begin{aligned} (25) \quad & 4x^2 + 19x + 12 \\ &= (4x + 3)(x + 4) \end{aligned}$$

$$\begin{aligned} (6) \quad & 4x^2 + 12x + 9 \\ &= (2x + 3)^2 \end{aligned}$$

$$\begin{aligned} (16) \quad & 8x^2 - 25x + 18 \\ &= (8x - 9)(x - 2) \end{aligned}$$

$$\begin{aligned} (26) \quad & 3x^2 - 11x + 6 \\ &= (x - 3)(3x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 9x^2 + 7x - 2 \\ &= (9x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & 6x^2 + 5x - 1 \\ &= (x + 1)(6x - 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 36x^2 - 47x + 15 \\ &= (9x - 5)(4x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^2 - 4x - 7 \\ &= (x + 1)(3x - 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & 15x^2 - x - 6 \\ &= (3x - 2)(5x + 3) \end{aligned}$$

$$\begin{aligned} (28) \quad & 35x^2 - 86x + 48 \\ &= (7x - 6)(5x - 8) \end{aligned}$$

$$\begin{aligned} (9) \quad & 8x^2 - 10x - 25 \\ &= (2x - 5)(4x + 5) \end{aligned}$$

$$\begin{aligned} (19) \quad & x^2 - 8x + 7 \\ &= (x - 1)(x - 7) \end{aligned}$$

$$\begin{aligned} (29) \quad & 40x^2 - 77x + 36 \\ &= (8x - 9)(5x - 4) \end{aligned}$$

$$\begin{aligned} (10) \quad & 7x^2 + 11x + 4 \\ &= (7x + 4)(x + 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 16x^2 + 38x + 21 \\ &= (8x + 7)(2x + 3) \end{aligned}$$

$$\begin{aligned} (30) \quad & 49x^2 - 21x - 40 \\ &= (7x + 5)(7x - 8) \end{aligned}$$

$$\begin{aligned} (1) \quad & 4x^2 - 5x + 1 \\ &= (4x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (11) \quad & x^2 + 6x - 7 \\ &= (x + 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 10x^2 - 33x - 54 \\ &= (5x + 6)(2x - 9) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^2 + 6x + 5 \\ &= (x + 1)(x + 5) \end{aligned}$$

$$\begin{aligned} (12) \quad & 6x^2 - 17x + 7 \\ &= (2x - 1)(3x - 7) \end{aligned}$$

$$\begin{aligned} (22) \quad & 14x^2 + 19x - 40 \\ &= (2x + 5)(7x - 8) \end{aligned}$$

$$\begin{aligned} (3) \quad & 6x^2 - 13x + 7 \\ &= (6x - 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 5x^2 - 3x - 2 \\ &= (x - 1)(5x + 2) \end{aligned}$$

$$\begin{aligned} (23) \quad & x^2 - 5x + 4 \\ &= (x - 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 12x^2 + 25x + 12 \\ &= (3x + 4)(4x + 3) \end{aligned}$$

$$\begin{aligned} (14) \quad & x^2 - 1 \\ &= (x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 18x^2 + 3x - 28 \\ &= (6x - 7)(3x + 4) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^2 + 21x + 49 \\ &= (x + 7)(2x + 7) \end{aligned}$$

$$\begin{aligned} (15) \quad & 3x^2 + 7x + 2 \\ &= (x + 2)(3x + 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & x^2 + 2x - 3 \\ &= (x - 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 7x^2 + 23x + 18 \\ &= (7x + 9)(x + 2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 5x^2 + 12x + 4 \\ &= (x + 2)(5x + 2) \end{aligned}$$

$$\begin{aligned} (26) \quad & 8x^2 - 22x - 63 \\ &= (4x + 7)(2x - 9) \end{aligned}$$

$$\begin{aligned} (7) \quad & 8x^2 - 15x + 7 \\ &= (8x - 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & 20x^2 + 69x + 54 \\ &= (5x + 6)(4x + 9) \end{aligned}$$

$$\begin{aligned} (27) \quad & 7x^2 - 16x - 15 \\ &= (7x + 5)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 4x^2 + 19x + 21 \\ &= (x + 3)(4x + 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & 4x^2 - 4x - 3 \\ &= (2x - 3)(2x + 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 14x^2 - x - 3 \\ &= (2x - 1)(7x + 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^2 - 13x + 14 \\ &= (x - 2)(3x - 7) \end{aligned}$$

$$\begin{aligned} (19) \quad & 63x^2 - 62x + 15 \\ &= (9x - 5)(7x - 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 3x^2 + 13x + 14 \\ &= (x + 2)(3x + 7) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^2 + 7x - 8 \\ &= (x + 8)(x - 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & x^2 - x - 12 \\ &= (x + 3)(x - 4) \end{aligned}$$

$$\begin{aligned} (30) \quad & x^2 + 3x - 4 \\ &= (x + 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 27x^2 + 12x - 32 \\ & = (9x - 8)(3x + 4) \end{aligned}$$

$$\begin{aligned} (11) \quad & 27x^2 + 33x + 8 \\ & = (9x + 8)(3x + 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 8x^2 + 9x - 14 \\ & = (8x - 7)(x + 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^2 + 31x + 18 \\ & = (3x + 2)(2x + 9) \end{aligned}$$

$$\begin{aligned} (12) \quad & 10x^2 + 9x - 7 \\ & = (2x - 1)(5x + 7) \end{aligned}$$

$$\begin{aligned} (22) \quad & 12x^2 + 8x - 7 \\ & = (6x + 7)(2x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 15x^2 - 29x + 8 \\ & = (5x - 8)(3x - 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 5x^2 - 3x - 14 \\ & = (x - 2)(5x + 7) \end{aligned}$$

$$\begin{aligned} (23) \quad & 4x^2 + 4x - 15 \\ & = (2x + 5)(2x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 14x^2 + 23x + 8 \\ & = (7x + 8)(2x + 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 18x^2 + 89x + 36 \\ & = (2x + 9)(9x + 4) \end{aligned}$$

$$\begin{aligned} (24) \quad & 63x^2 - 8x - 15 \\ & = (7x + 3)(9x - 5) \end{aligned}$$

$$\begin{aligned} (5) \quad & 12x^2 + 25x - 7 \\ & = (3x + 7)(4x - 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 3x^2 - 16x + 5 \\ & = (x - 5)(3x - 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 20x^2 + 27x - 14 \\ & = (5x - 2)(4x + 7) \end{aligned}$$

$$\begin{aligned} (6) \quad & 6x^2 + 5x - 6 \\ & = (3x - 2)(2x + 3) \end{aligned}$$

$$\begin{aligned} (16) \quad & 81x^2 - 54x - 7 \\ & = (9x - 7)(9x + 1) \end{aligned}$$

$$\begin{aligned} (26) \quad & 10x^2 - 21x + 8 \\ & = (2x - 1)(5x - 8) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^2 - 16x - 35 \\ & = (3x + 5)(x - 7) \end{aligned}$$

$$\begin{aligned} (17) \quad & 7x^2 + 10x - 8 \\ & = (7x - 4)(x + 2) \end{aligned}$$

$$\begin{aligned} (27) \quad & 4x^2 - 41x + 72 \\ & = (x - 8)(4x - 9) \end{aligned}$$

$$\begin{aligned} (8) \quad & x^2 - 1 \\ & = (x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (18) \quad & 7x^2 + 2x - 5 \\ & = (x + 1)(7x - 5) \end{aligned}$$

$$\begin{aligned} (28) \quad & 16x^2 + 6x - 27 \\ & = (8x - 9)(2x + 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 15x^2 - 22x - 9 \\ & = (5x - 9)(3x + 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 27x^2 - 21x - 20 \\ & = (3x - 4)(9x + 5) \end{aligned}$$

$$\begin{aligned} (29) \quad & 28x^2 + 19x + 3 \\ & = (7x + 3)(4x + 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^2 - 3x + 1 \\ & = (2x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 3x^2 + 17x - 28 \\ & = (x + 7)(3x - 4) \end{aligned}$$

$$\begin{aligned} (30) \quad & 3x^2 - 20x - 7 \\ & = (x - 7)(3x + 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 21x^2 + 23x + 6 \\ &= (7x + 3)(3x + 2) \end{aligned}$$

$$\begin{aligned} (11) \quad & 21x^2 - 13x - 20 \\ &= (3x - 4)(7x + 5) \end{aligned}$$

$$\begin{aligned} (21) \quad & x^2 - 2x + 1 \\ &= (x - 1)^2 \end{aligned}$$

$$\begin{aligned} (2) \quad & 9x^2 + 6x + 1 \\ &= (3x + 1)^2 \end{aligned}$$

$$\begin{aligned} (12) \quad & 4x^2 - x - 5 \\ &= (4x - 5)(x + 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 9x^2 + 38x - 35 \\ &= (x + 5)(9x - 7) \end{aligned}$$

$$\begin{aligned} (3) \quad & 3x^2 + 4x - 7 \\ &= (x - 1)(3x + 7) \end{aligned}$$

$$\begin{aligned} (13) \quad & 28x^2 + 81x + 56 \\ &= (4x + 7)(7x + 8) \end{aligned}$$

$$\begin{aligned} (23) \quad & 4x^2 + 5x - 6 \\ &= (4x - 3)(x + 2) \end{aligned}$$

$$\begin{aligned} (4) \quad & 14x^2 - 3x - 2 \\ &= (7x + 2)(2x - 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 24x^2 - 22x + 3 \\ &= (6x - 1)(4x - 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & x^2 - 6x - 7 \\ &= (x - 7)(x + 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 16x^2 - 22x - 3 \\ &= (8x + 1)(2x - 3) \end{aligned}$$

$$\begin{aligned} (15) \quad & 2x^2 + x - 3 \\ &= (2x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 4x^2 - 27x - 7 \\ &= (x - 7)(4x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 4x^2 + 11x - 3 \\ &= (4x - 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (16) \quad & 6x^2 - 11x - 2 \\ &= (x - 2)(6x + 1) \end{aligned}$$

$$\begin{aligned} (26) \quad & 30x^2 + x - 1 \\ &= (5x + 1)(6x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 8x^2 - 14x - 9 \\ &= (4x - 9)(2x + 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & 18x^2 - 11x + 1 \\ &= (2x - 1)(9x - 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 48x^2 + 50x + 7 \\ &= (8x + 7)(6x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 20x^2 - 29x - 36 \\ &= (5x + 4)(4x - 9) \end{aligned}$$

$$\begin{aligned} (18) \quad & 5x^2 - 31x - 28 \\ &= (5x + 4)(x - 7) \end{aligned}$$

$$\begin{aligned} (28) \quad & 20x^2 - 61x + 45 \\ &= (5x - 9)(4x - 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^2 - 3x + 1 \\ &= (x - 1)(2x - 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & x^2 - 2x + 1 \\ &= (x - 1)^2 \end{aligned}$$

$$\begin{aligned} (29) \quad & 7x^2 + 12x + 5 \\ &= (7x + 5)(x + 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 4x^2 + 13x + 3 \\ &= (x + 3)(4x + 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 2x^2 + 5x + 3 \\ &= (2x + 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (30) \quad & 8x^2 + 33x + 27 \\ &= (8x + 9)(x + 3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 18x^2 + 19x - 12 \\ & = (9x - 4)(2x + 3) \end{aligned}$$

$$\begin{aligned} (11) \quad & 15x^2 + 7x - 4 \\ & = (5x + 4)(3x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 9x^2 - 17x - 2 \\ & = (x - 2)(9x + 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 9x^2 - 9x + 2 \\ & = (3x - 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (12) \quad & 5x^2 + 53x + 72 \\ & = (5x + 8)(x + 9) \end{aligned}$$

$$\begin{aligned} (22) \quad & 2x^2 + 5x + 3 \\ & = (x + 1)(2x + 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 2x^2 + x - 3 \\ & = (x - 1)(2x + 3) \end{aligned}$$

$$\begin{aligned} (13) \quad & 40x^2 + 49x + 15 \\ & = (8x + 5)(5x + 3) \end{aligned}$$

$$\begin{aligned} (23) \quad & 9x^2 - 18x + 8 \\ & = (3x - 2)(3x - 4) \end{aligned}$$

$$\begin{aligned} (4) \quad & 24x^2 + 7x - 6 \\ & = (8x - 3)(3x + 2) \end{aligned}$$

$$\begin{aligned} (14) \quad & x^2 - 5x + 4 \\ & = (x - 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (24) \quad & 14x^2 + 45x + 25 \\ & = (7x + 5)(2x + 5) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^2 + 3x - 4 \\ & = (x + 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 10x^2 - 37x + 7 \\ & = (5x - 1)(2x - 7) \end{aligned}$$

$$\begin{aligned} (25) \quad & 7x^2 - 5x - 2 \\ & = (x - 1)(7x + 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & x^2 + x - 56 \\ & = (x - 7)(x + 8) \end{aligned}$$

$$\begin{aligned} (16) \quad & 24x^2 - 13x - 2 \\ & = (8x + 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (26) \quad & x^2 + 7x - 8 \\ & = (x - 1)(x + 8) \end{aligned}$$

$$\begin{aligned} (7) \quad & 9x^2 + x - 8 \\ & = (x + 1)(9x - 8) \end{aligned}$$

$$\begin{aligned} (17) \quad & 32x^2 - 60x - 27 \\ & = (4x - 9)(8x + 3) \end{aligned}$$

$$\begin{aligned} (27) \quad & 64x^2 - 81 \\ & = (8x + 9)(8x - 9) \end{aligned}$$

$$\begin{aligned} (8) \quad & 24x^2 + 22x - 7 \\ & = (4x - 1)(6x + 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & 4x^2 - 19x - 5 \\ & = (x - 5)(4x + 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 2x^2 + 5x - 7 \\ & = (2x + 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 20x^2 + 39x + 7 \\ & = (4x + 7)(5x + 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 36x^2 + 5x - 1 \\ & = (9x - 1)(4x + 1) \end{aligned}$$

$$\begin{aligned} (29) \quad & 4x^2 - 3x - 27 \\ & = (4x + 9)(x - 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 14x^2 + 3x - 2 \\ & = (7x - 2)(2x + 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 6x^2 - 29x + 20 \\ & = (x - 4)(6x - 5) \end{aligned}$$

$$\begin{aligned} (30) \quad & 7x^2 - 54x - 16 \\ & = (x - 8)(7x + 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 45x^2 - 92x + 32 \\ & = (5x - 8)(9x - 4) \end{aligned}$$

$$\begin{aligned} (11) \quad & 18x^2 - 17x - 15 \\ & = (9x + 5)(2x - 3) \end{aligned}$$

$$\begin{aligned} (21) \quad & 5x^2 - 37x - 72 \\ & = (5x + 8)(x - 9) \end{aligned}$$

$$\begin{aligned} (2) \quad & 72x^2 - 83x + 21 \\ & = (8x - 3)(9x - 7) \end{aligned}$$

$$\begin{aligned} (12) \quad & 8x^2 - 3x - 5 \\ & = (x - 1)(8x + 5) \end{aligned}$$

$$\begin{aligned} (22) \quad & 32x^2 - 4x - 15 \\ & = (4x - 3)(8x + 5) \end{aligned}$$

$$\begin{aligned} (3) \quad & 42x^2 - x - 30 \\ & = (6x + 5)(7x - 6) \end{aligned}$$

$$\begin{aligned} (13) \quad & 8x^2 + 15x + 7 \\ & = (x + 1)(8x + 7) \end{aligned}$$

$$\begin{aligned} (23) \quad & 4x^2 + 13x - 12 \\ & = (x + 4)(4x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^2 + 11x + 9 \\ & = (x + 1)(2x + 9) \end{aligned}$$

$$\begin{aligned} (14) \quad & x^2 - 4x + 3 \\ & = (x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & 28x^2 + 69x + 35 \\ & = (4x + 7)(7x + 5) \end{aligned}$$

$$\begin{aligned} (5) \quad & 25x^2 + 5x - 6 \\ & = (5x + 3)(5x - 2) \end{aligned}$$

$$\begin{aligned} (15) \quad & 7x^2 + 16x + 9 \\ & = (x + 1)(7x + 9) \end{aligned}$$

$$\begin{aligned} (25) \quad & 28x^2 + 15x + 2 \\ & = (4x + 1)(7x + 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 9x^2 - 4x - 5 \\ & = (x - 1)(9x + 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 2x^2 + 9x + 10 \\ & = (x + 2)(2x + 5) \end{aligned}$$

$$\begin{aligned} (26) \quad & 32x^2 - 60x + 7 \\ & = (4x - 7)(8x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^2 - 5x - 8 \\ & = (3x - 8)(x + 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & 35x^2 + 9x - 18 \\ & = (5x - 3)(7x + 6) \end{aligned}$$

$$\begin{aligned} (27) \quad & 16x^2 + 26x + 3 \\ & = (2x + 3)(8x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 36x^2 + 97x + 36 \\ & = (9x + 4)(4x + 9) \end{aligned}$$

$$\begin{aligned} (18) \quad & 5x^2 + 13x - 6 \\ & = (5x - 2)(x + 3) \end{aligned}$$

$$\begin{aligned} (28) \quad & 3x^2 - 8x + 5 \\ & = (x - 1)(3x - 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 32x^2 - 12x - 9 \\ & = (4x - 3)(8x + 3) \end{aligned}$$

$$\begin{aligned} (19) \quad & 5x^2 + 19x + 18 \\ & = (x + 2)(5x + 9) \end{aligned}$$

$$\begin{aligned} (29) \quad & 36x^2 - 49x + 5 \\ & = (4x - 5)(9x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 10x^2 - 23x + 12 \\ & = (2x - 3)(5x - 4) \end{aligned}$$

$$\begin{aligned} (20) \quad & 2x^2 - x - 10 \\ & = (x + 2)(2x - 5) \end{aligned}$$

$$\begin{aligned} (30) \quad & 12x^2 - 31x + 7 \\ & = (4x - 1)(3x - 7) \end{aligned}$$

$$\begin{aligned} (1) \quad & 45x^2 - 53x + 14 \\ & = (5x - 2)(9x - 7) \end{aligned}$$

$$\begin{aligned} (11) \quad & 6x^2 + 35x - 6 \\ & = (6x - 1)(x + 6) \end{aligned}$$

$$\begin{aligned} (21) \quad & 56x^2 - 9x - 35 \\ & = (7x + 5)(8x - 7) \end{aligned}$$

$$\begin{aligned} (2) \quad & 24x^2 - 11x - 18 \\ & = (8x - 9)(3x + 2) \end{aligned}$$

$$\begin{aligned} (12) \quad & 4x^2 + 5x - 6 \\ & = (4x - 3)(x + 2) \end{aligned}$$

$$\begin{aligned} (22) \quad & 7x^2 - 29x + 24 \\ & = (7x - 8)(x - 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 8x^2 + 14x + 5 \\ & = (4x + 5)(2x + 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 2x^2 + 3x - 2 \\ & = (2x - 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (23) \quad & 40x^2 - 39x + 9 \\ & = (8x - 3)(5x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 15x^2 + 4x - 32 \\ & = (5x + 8)(3x - 4) \end{aligned}$$

$$\begin{aligned} (14) \quad & 3x^2 + 5x - 2 \\ & = (x + 2)(3x - 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 36x^2 + 95x + 56 \\ & = (9x + 8)(4x + 7) \end{aligned}$$

$$\begin{aligned} (5) \quad & 4x^2 + 21x + 5 \\ & = (4x + 1)(x + 5) \end{aligned}$$

$$\begin{aligned} (15) \quad & 6x^2 + 29x + 28 \\ & = (3x + 4)(2x + 7) \end{aligned}$$

$$\begin{aligned} (25) \quad & x^2 + 2x - 15 \\ & = (x - 3)(x + 5) \end{aligned}$$

$$\begin{aligned} (6) \quad & 20x^2 + 17x - 10 \\ & = (4x + 5)(5x - 2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 16x^2 - 74x + 63 \\ & = (2x - 7)(8x - 9) \end{aligned}$$

$$\begin{aligned} (26) \quad & 3x^2 - 5x - 2 \\ & = (x - 2)(3x + 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 81x^2 + 36x - 32 \\ & = (9x - 4)(9x + 8) \end{aligned}$$

$$\begin{aligned} (17) \quad & 3x^2 + 32x + 64 \\ & = (x + 8)(3x + 8) \end{aligned}$$

$$\begin{aligned} (27) \quad & 2x^2 + x - 21 \\ & = (2x + 7)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 5x^2 + 14x - 3 \\ & = (x + 3)(5x - 1) \end{aligned}$$

$$\begin{aligned} (18) \quad & 16x^2 + 2x - 5 \\ & = (8x + 5)(2x - 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 36x^2 + 37x - 10 \\ & = (4x + 5)(9x - 2) \end{aligned}$$

$$\begin{aligned} (9) \quad & 5x^2 + 4x - 1 \\ & = (x + 1)(5x - 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 4x^2 + 17x + 15 \\ & = (4x + 5)(x + 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 10x^2 + 43x + 45 \\ & = (2x + 5)(5x + 9) \end{aligned}$$

$$\begin{aligned} (10) \quad & 6x^2 + 13x - 8 \\ & = (3x + 8)(2x - 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 24x^2 - 43x + 18 \\ & = (3x - 2)(8x - 9) \end{aligned}$$

$$\begin{aligned} (30) \quad & x^2 + 2x + 1 \\ & = (x + 1)^2 \end{aligned}$$

2.5 因数分解 (3 次式)

3 次多項式の因数分解をする問題。

(1) $3x^3 - 5x^2 + x + 1$

(6) $2x^3 - x^2 - 2x + 1$

(2) $x^3 + x^2 - x - 1$

(7) $x^3 - 2x^2 - 5x + 6$

(3) $9x^3 - 3x^2 - 5x - 1$

(8) $6x^3 - 5x^2 - 3x + 2$

(4) $x^3 + 3x^2 - x - 3$

(9) $12x^3 + 20x^2 - x - 6$

(5) $2x^3 - 3x^2 - 11x + 6$

(10) $2x^3 - 5x^2 - x + 6$

(1) $4x^3 - 4x^2 - x + 1$

(6) $18x^3 + 9x^2 - 23x + 6$

(2) $6x^3 + x^2 - 11x - 6$

(7) $2x^3 - 7x^2 + 8x - 3$

(3) $x^3 - 3x^2 - x + 3$

(8) $6x^3 + 17x^2 - 5x - 6$

(4) $3x^3 + 7x^2 + 5x + 1$

(9) $9x^3 + 21x^2 - 17x + 3$

(5) $2x^3 + 7x^2 + 7x + 2$

(10) $3x^3 - 4x^2 - x + 2$

(1) $9x^3 - 21x^2 + 16x - 4$

(6) $2x^3 + 7x^2 - 9$

(2) $3x^3 + x^2 - 12x - 4$

(7) $9x^3 + 21x^2 - 17x + 3$

(3) $x^3 + x^2 - 9x - 9$

(8) $3x^3 - 7x^2 + 5x - 1$

(4) $9x^3 - 3x^2 - 8x + 4$

(9) $2x^3 - 3x^2 - 18x + 27$

(5) $x^3 + 3x^2 - x - 3$

(10) $4x^3 - 4x^2 - x + 1$

(1) $4x^3 + 4x^2 - 5x - 3$

(6) $6x^3 + 11x^2 + 6x + 1$

(2) $x^3 - 5x^2 + 7x - 3$

(7) $x^3 + 2x^2 - x - 2$

(3) $27x^3 + 45x^2 + 24x + 4$

(8) $2x^3 + 5x^2 - 6x - 9$

(4) $3x^3 + x^2 - 27x - 9$

(9) $4x^3 - 8x^2 + x + 3$

(5) $x^3 - x^2 - 4x + 4$

(10) $9x^3 + 18x^2 + 11x + 2$

(1) $4x^3 - 4x^2 - 5x + 3$

(6) $9x^3 + 24x^2 + 13x + 2$

(2) $2x^3 - 11x^2 + 20x - 12$

(7) $4x^3 - 8x^2 + x + 3$

(3) $12x^3 - 4x^2 - 3x + 1$

(8) $8x^3 - 4x^2 - 18x + 9$

(4) $3x^3 - 5x^2 - 4x + 4$

(9) $2x^3 + 3x^2 - 11x - 6$

(5) $2x^3 - 7x^2 + 4x + 4$

(10) $2x^3 + x^2 - 2x - 1$

(1) $x^3 - 7x + 6$

(6) $6x^3 - 5x^2 - 2x + 1$

(2) $2x^3 - 5x^2 - x + 6$

(7) $4x^3 + 12x^2 - 9x - 27$

(3) $6x^3 - 5x^2 - 33x - 18$

(8) $2x^3 + 7x^2 - 9$

(4) $4x^3 - 4x^2 - 5x + 3$

(9) $4x^3 - 8x^2 + x + 3$

(5) $9x^3 - 9x^2 - x + 1$

(10) $2x^3 + 3x^2 - 5x - 6$

(1) $3x^3 - 8x^2 - 5x + 6$

(6) $3x^3 + 17x^2 + 21x - 9$

(2) $2x^3 - 3x^2 - 2x + 3$

(7) $3x^3 + 10x^2 + x - 6$

(3) $2x^3 - x^2 - 4x + 3$

(8) $x^3 - 3x + 2$

(4) $6x^3 - x^2 - 5x + 2$

(9) $x^3 - 3x^2 - 9x + 27$

(5) $9x^3 - 30x^2 + 28x - 8$

(10) $3x^3 + x^2 - 8x + 4$

(1) $6x^3 - 5x^2 - 8x + 3$

(6) $x^3 - 4x^2 + x + 6$

(2) $2x^3 + 5x^2 + 4x + 1$

(7) $x^3 - 3x - 2$

(3) $2x^3 + 3x^2 - 5x - 6$

(8) $2x^3 + 9x^2 + 13x + 6$

(4) $3x^3 + 5x^2 - 4x - 4$

(9) $x^3 - x^2 - 5x - 3$

(5) $2x^3 + 7x^2 + 8x + 3$

(10) $3x^3 - 10x^2 + 9x - 2$

(1) $2x^3 + 7x^2 + 7x + 2$

(6) $2x^3 + 3x^2 - 8x + 3$

(2) $18x^3 + 15x^2 - 4x - 4$

(7) $2x^3 - 5x^2 - x + 6$

(3) $3x^3 - 19x^2 + 33x - 9$

(8) $x^3 + x^2 - x - 1$

(4) $12x^3 - 4x^2 - 3x + 1$

(9) $6x^3 - 19x^2 + 19x - 6$

(5) $4x^3 - 4x^2 - 21x - 9$

(10) $2x^3 - 9x^2 + 13x - 6$

(1) $3x^3 + 10x^2 + x - 6$

(6) $12x^3 - 20x^2 - x + 6$

(2) $6x^3 - 7x^2 - 9x - 2$

(7) $6x^3 - 19x^2 + 19x - 6$

(3) $2x^3 - 7x^2 + 2x + 3$

(8) $6x^3 + x^2 - 4x + 1$

(4) $6x^3 - 17x^2 - 4x + 3$

(9) $6x^3 - 11x^2 + 6x - 1$

(5) $6x^3 + x^2 - 11x - 6$

(10) $6x^3 - 19x^2 + 19x - 6$

(1) $3x^3 - 7x^2 - 7x + 3$

(6) $x^3 - 2x^2 - 5x + 6$

(2) $8x^3 + 4x^2 - 18x - 9$

(7) $6x^3 - x^2 - 5x + 2$

(3) $12x^3 + 28x^2 + 17x + 3$

(8) $6x^3 - x^2 - 10x - 3$

(4) $2x^3 + 7x^2 + 2x - 3$

(9) $3x^3 + x^2 - 12x - 4$

(5) $3x^3 + 10x^2 + 9x + 2$

(10) $6x^3 + 13x^2 + x - 2$

(1) $18x^3 + 21x^2 - 13x - 6$

(6) $2x^3 - 9x^2 + 13x - 6$

(2) $3x^3 + 10x^2 + 9x + 2$

(7) $4x^3 - 4x^2 - x + 1$

(3) $6x^3 + 17x^2 + 4x - 12$

(8) $3x^3 - 7x^2 + 5x - 1$

(4) $4x^3 - 4x^2 - 5x + 3$

(9) $2x^3 - 3x^2 - 8x - 3$

(5) $x^3 - 4x^2 + 5x - 2$

(10) $x^3 - 4x^2 + 5x - 2$

(1) $9x^3 - 18x^2 - 4x + 8$

(6) $3x^3 - 8x^2 - 5x + 6$

(2) $6x^3 - 5x^2 - 3x + 2$

(7) $2x^3 + 3x^2 - 11x - 6$

(3) $4x^3 - 13x + 6$

(8) $9x^3 + 21x^2 + 16x + 4$

(4) $3x^3 + 4x^2 - 13x + 6$

(9) $3x^3 + 2x^2 - 3x - 2$

(5) $3x^3 + 13x^2 + 13x + 3$

(10) $4x^3 - 3x - 1$

(1) $x^3 - x^2 - 9x + 9$

(6) $18x^3 - 27x^2 + 13x - 2$

(2) $4x^3 + 16x^2 + 9x - 9$

(7) $3x^3 - x^2 - 3x + 1$

(3) $4x^3 + 4x^2 - 7x + 2$

(8) $x^3 + 3x^2 - x - 3$

(4) $2x^3 + 5x^2 - x - 6$

(9) $6x^3 - 11x^2 - x + 6$

(5) $18x^3 + 15x^2 - 16x + 3$

(10) $2x^3 + 5x^2 - 4x - 3$

(1) $2x^3 + x^2 - 5x + 2$

(6) $4x^3 - 8x^2 + 5x - 1$

(2) $6x^3 - x^2 - 20x + 12$

(7) $3x^3 + 19x^2 + 33x + 9$

(3) $6x^3 + 11x^2 - x - 6$

(8) $6x^3 + 25x^2 + 23x + 6$

(4) $9x^3 - 18x^2 - x + 2$

(9) $6x^3 + 17x^2 - 5x - 6$

(5) $2x^3 + 7x^2 + 7x + 2$

(10) $x^3 - 3x^2 + 3x - 1$

(1) $18x^3 + 15x^2 - x - 2$

(6) $4x^3 + 8x^2 + x - 3$

(2) $6x^3 + 13x^2 + 9x + 2$

(7) $3x^3 - 4x^2 - 17x + 6$

(3) $9x^3 + 18x^2 - x - 2$

(8) $3x^3 - 2x^2 - 3x + 2$

(4) $x^3 - 3x^2 + 4$

(9) $2x^3 + x^2 - 2x - 1$

(5) $9x^3 + 9x^2 - 4x - 4$

(10) $9x^3 - 6x^2 - 5x + 2$

(1) $3x^3 - x^2 - 3x + 1$

(6) $4x^3 - 8x^2 - 3x + 9$

(2) $x^3 + 3x^2 + 3x + 1$

(7) $6x^3 + 7x^2 - 1$

(3) $9x^3 + 18x^2 - x - 2$

(8) $6x^3 + 17x^2 + 14x + 3$

(4) $3x^3 - 7x^2 - 7x + 3$

(9) $3x^3 + 14x^2 + 17x + 6$

(5) $3x^3 + x^2 - 3x - 1$

(10) $x^3 + x^2 - x - 1$

(1) $x^3 - 2x^2 - x + 2$

(6) $2x^3 - 3x^2 - 2x + 3$

(2) $12x^3 - 8x^2 - 13x - 3$

(7) $x^3 - 2x^2 - 5x + 6$

(3) $4x^3 + 16x^2 + 19x + 6$

(8) $4x^3 - 8x^2 + x + 3$

(4) $x^3 - x^2 - 9x + 9$

(9) $2x^3 + x^2 - 18x - 9$

(5) $3x^3 + x^2 - 3x - 1$

(10) $x^3 + x^2 - x - 1$

(1) $2x^3 + x^2 - 7x - 6$

(6) $2x^3 - x^2 - 7x + 6$

(2) $6x^3 - x^2 - 11x + 6$

(7) $12x^3 - 8x^2 - 13x - 3$

(3) $9x^3 - 18x^2 - 4x + 8$

(8) $9x^3 - 27x^2 - 4x + 12$

(4) $x^3 + x^2 - 9x - 9$

(9) $3x^3 - 13x^2 + 8x + 12$

(5) $x^3 + x^2 - 4x - 4$

(10) $3x^3 - 11x^2 + 5x + 3$

(1) $2x^3 - x^2 - 4x + 3$

(6) $x^3 + 3x^2 - x - 3$

(2) $3x^3 - 5x^2 - 11x - 3$

(7) $9x^3 + 3x^2 - 5x + 1$

(3) $12x^3 - 16x^2 - 7x + 6$

(8) $3x^3 + 7x^2 + 5x + 1$

(4) $9x^3 - 9x^2 - x + 1$

(9) $3x^3 + x^2 - 3x - 1$

(5) $9x^3 - 9x^2 - 4x + 4$

(10) $4x^3 - 7x - 3$

$$\begin{aligned} (1) \quad & 3x^3 - 5x^2 + x + 1 \\ &= (3x + 1)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 - x^2 - 2x + 1 \\ &= (2x - 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^3 + x^2 - x - 1 \\ &= (x + 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^3 - 2x^2 - 5x + 6 \\ &= (x + 2)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 9x^3 - 3x^2 - 5x - 1 \\ &= (3x + 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 - 5x^2 - 3x + 2 \\ &= (2x - 1)(3x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & x^3 + 3x^2 - x - 3 \\ &= (x + 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 12x^3 + 20x^2 - x - 6 \\ &= (2x + 3)(2x - 1)(3x + 2) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^3 - 3x^2 - 11x + 6 \\ &= (2x - 1)(x + 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^3 - 5x^2 - x + 6 \\ &= (2x - 3)(x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 4x^3 - 4x^2 - x + 1 \\ &= (2x + 1)(2x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 18x^3 + 9x^2 - 23x + 6 \\ &= (2x + 3)(3x - 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 + x^2 - 11x - 6 \\ &= (2x - 3)(3x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^3 - 7x^2 + 8x - 3 \\ &= (2x - 3)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (3) \quad & x^3 - 3x^2 - x + 3 \\ &= (x + 1)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 + 17x^2 - 5x - 6 \\ &= (2x + 1)(3x - 2)(x + 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 + 7x^2 + 5x + 1 \\ &= (3x + 1)(x + 1)^2 \end{aligned}$$

$$\begin{aligned} (9) \quad & 9x^3 + 21x^2 - 17x + 3 \\ &= (3x - 1)^2(x + 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^3 + 7x^2 + 7x + 2 \\ &= (2x + 1)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 3x^3 - 4x^2 - x + 2 \\ &= (3x + 2)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (1) \quad & 9x^3 - 21x^2 + 16x - 4 \\ &= (3x - 2)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 + 7x^2 - 9 \\ &= (2x + 3)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 3x^3 + x^2 - 12x - 4 \\ &= (3x + 1)(x + 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 9x^3 + 21x^2 - 17x + 3 \\ &= (3x - 1)^2(x + 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & x^3 + x^2 - 9x - 9 \\ &= (x + 1)(x + 3)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^3 - 7x^2 + 5x - 1 \\ &= (3x - 1)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (4) \quad & 9x^3 - 3x^2 - 8x + 4 \\ &= (3x - 2)^2(x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^3 - 3x^2 - 18x + 27 \\ &= (2x - 3)(x + 3)(x - 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^3 + 3x^2 - x - 3 \\ &= (x + 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 4x^3 - 4x^2 - x + 1 \\ &= (2x + 1)(2x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 4x^3 + 4x^2 - 5x - 3 \\ &= (2x + 1)(2x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 6x^3 + 11x^2 + 6x + 1 \\ &= (2x + 1)(3x + 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^3 - 5x^2 + 7x - 3 \\ &= (x - 1)^2(x - 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^3 + 2x^2 - x - 2 \\ &= (x + 1)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 27x^3 + 45x^2 + 24x + 4 \\ &= (3x + 1)(3x + 2)^2 \end{aligned}$$

$$\begin{aligned} (8) \quad & 2x^3 + 5x^2 - 6x - 9 \\ &= (2x - 3)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 + x^2 - 27x - 9 \\ &= (3x + 1)(x + 3)(x - 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 4x^3 - 8x^2 + x + 3 \\ &= (2x + 1)(2x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^3 - x^2 - 4x + 4 \\ &= (x + 2)(x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 9x^3 + 18x^2 + 11x + 2 \\ &= (3x + 1)(3x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 4x^3 - 4x^2 - 5x + 3 \\ &= (2x - 1)(2x - 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 9x^3 + 24x^2 + 13x + 2 \\ &= (3x + 1)^2(x + 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 2x^3 - 11x^2 + 20x - 12 \\ &= (2x - 3)(x - 2)^2 \end{aligned}$$

$$\begin{aligned} (7) \quad & 4x^3 - 8x^2 + x + 3 \\ &= (2x + 1)(2x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 12x^3 - 4x^2 - 3x + 1 \\ &= (2x + 1)(2x - 1)(3x - 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 8x^3 - 4x^2 - 18x + 9 \\ &= (2x + 3)(2x - 1)(2x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 - 5x^2 - 4x + 4 \\ &= (3x - 2)(x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^3 + 3x^2 - 11x - 6 \\ &= (2x + 1)(x + 3)(x - 2) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^3 - 7x^2 + 4x + 4 \\ &= (2x + 1)(x - 2)^2 \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^3 + x^2 - 2x - 1 \\ &= (2x + 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & x^3 - 7x + 6 \\ &= (x + 3)(x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 6x^3 - 5x^2 - 2x + 1 \\ &= (2x + 1)(3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 2x^3 - 5x^2 - x + 6 \\ &= (2x - 3)(x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 4x^3 + 12x^2 - 9x - 27 \\ &= (2x + 3)(2x - 3)(x + 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 6x^3 - 5x^2 - 33x - 18 \\ &= (2x + 3)(3x + 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 2x^3 + 7x^2 - 9 \\ &= (2x + 3)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 4x^3 - 4x^2 - 5x + 3 \\ &= (2x - 1)(2x - 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 4x^3 - 8x^2 + x + 3 \\ &= (2x + 1)(2x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 9x^3 - 9x^2 - x + 1 \\ &= (3x + 1)(3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^3 + 3x^2 - 5x - 6 \\ &= (2x - 3)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^3 - 8x^2 - 5x + 6 \\ &= (3x - 2)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 3x^3 + 17x^2 + 21x - 9 \\ &= (3x - 1)(x + 3)^2 \end{aligned}$$

$$\begin{aligned} (2) \quad & 2x^3 - 3x^2 - 2x + 3 \\ &= (2x - 3)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^3 + 10x^2 + x - 6 \\ &= (3x - 2)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 2x^3 - x^2 - 4x + 3 \\ &= (2x + 3)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (8) \quad & x^3 - 3x + 2 \\ &= (x + 2)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (4) \quad & 6x^3 - x^2 - 5x + 2 \\ &= (2x - 1)(3x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & x^3 - 3x^2 - 9x + 27 \\ &= (x + 3)(x - 3)^2 \end{aligned}$$

$$\begin{aligned} (5) \quad & 9x^3 - 30x^2 + 28x - 8 \\ &= (3x - 2)^2(x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 3x^3 + x^2 - 8x + 4 \\ &= (3x - 2)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 6x^3 - 5x^2 - 8x + 3 \\ &= (2x - 3)(3x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & x^3 - 4x^2 + x + 6 \\ &= (x + 1)(x - 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 2x^3 + 5x^2 + 4x + 1 \\ &= (2x + 1)(x + 1)^2 \end{aligned}$$

$$\begin{aligned} (7) \quad & x^3 - 3x - 2 \\ &= (x + 1)^2(x - 2) \end{aligned}$$

$$\begin{aligned} (3) \quad & 2x^3 + 3x^2 - 5x - 6 \\ &= (2x - 3)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 2x^3 + 9x^2 + 13x + 6 \\ &= (2x + 3)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 + 5x^2 - 4x - 4 \\ &= (3x + 2)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & x^3 - x^2 - 5x - 3 \\ &= (x + 1)^2(x - 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^3 + 7x^2 + 8x + 3 \\ &= (2x + 3)(x + 1)^2 \end{aligned}$$

$$\begin{aligned} (10) \quad & 3x^3 - 10x^2 + 9x - 2 \\ &= (3x - 1)(x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 2x^3 + 7x^2 + 7x + 2 \\ &= (2x + 1)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 + 3x^2 - 8x + 3 \\ &= (2x - 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 18x^3 + 15x^2 - 4x - 4 \\ &= (2x - 1)(3x + 2)^2 \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^3 - 5x^2 - x + 6 \\ &= (2x - 3)(x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (3) \quad & 3x^3 - 19x^2 + 33x - 9 \\ &= (3x - 1)(x - 3)^2 \end{aligned}$$

$$\begin{aligned} (8) \quad & x^3 + x^2 - x - 1 \\ &= (x + 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 12x^3 - 4x^2 - 3x + 1 \\ &= (2x + 1)(2x - 1)(3x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 6x^3 - 19x^2 + 19x - 6 \\ &= (2x - 3)(3x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 4x^3 - 4x^2 - 21x - 9 \\ &= (2x + 1)(2x + 3)(x - 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^3 - 9x^2 + 13x - 6 \\ &= (2x - 3)(x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^3 + 10x^2 + x - 6 \\ &= (3x - 2)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 12x^3 - 20x^2 - x + 6 \\ &= (2x + 1)(2x - 3)(3x - 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 - 7x^2 - 9x - 2 \\ &= (2x + 1)(3x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 6x^3 - 19x^2 + 19x - 6 \\ &= (2x - 3)(3x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 2x^3 - 7x^2 + 2x + 3 \\ &= (2x + 1)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 + x^2 - 4x + 1 \\ &= (2x - 1)(3x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 6x^3 - 17x^2 - 4x + 3 \\ &= (2x + 1)(3x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 6x^3 - 11x^2 + 6x - 1 \\ &= (2x - 1)(3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 6x^3 + x^2 - 11x - 6 \\ &= (2x - 3)(3x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 6x^3 - 19x^2 + 19x - 6 \\ &= (2x - 3)(3x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^3 - 7x^2 - 7x + 3 \\ &= (3x - 1)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & x^3 - 2x^2 - 5x + 6 \\ &= (x + 2)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 8x^3 + 4x^2 - 18x - 9 \\ &= (2x + 1)(2x + 3)(2x - 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & 6x^3 - x^2 - 5x + 2 \\ &= (2x - 1)(3x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 12x^3 + 28x^2 + 17x + 3 \\ &= (2x + 1)(2x + 3)(3x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 - x^2 - 10x - 3 \\ &= (2x - 3)(3x + 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^3 + 7x^2 + 2x - 3 \\ &= (2x - 1)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^3 + x^2 - 12x - 4 \\ &= (3x + 1)(x + 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^3 + 10x^2 + 9x + 2 \\ &= (3x + 1)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 6x^3 + 13x^2 + x - 2 \\ &= (2x + 1)(3x - 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 18x^3 + 21x^2 - 13x - 6 \\ & = (2x + 3)(3x + 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 - 9x^2 + 13x - 6 \\ & = (2x - 3)(x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 3x^3 + 10x^2 + 9x + 2 \\ & = (3x + 1)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 4x^3 - 4x^2 - x + 1 \\ & = (2x + 1)(2x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 6x^3 + 17x^2 + 4x - 12 \\ & = (2x + 3)(3x - 2)(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^3 - 7x^2 + 5x - 1 \\ & = (3x - 1)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (4) \quad & 4x^3 - 4x^2 - 5x + 3 \\ & = (2x - 1)(2x - 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^3 - 3x^2 - 8x - 3 \\ & = (2x + 1)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^3 - 4x^2 + 5x - 2 \\ & = (x - 1)^2(x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^3 - 4x^2 + 5x - 2 \\ & = (x - 1)^2(x - 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 9x^3 - 18x^2 - 4x + 8 \\ &= (3x + 2)(3x - 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 3x^3 - 8x^2 - 5x + 6 \\ &= (3x - 2)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 - 5x^2 - 3x + 2 \\ &= (2x - 1)(3x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^3 + 3x^2 - 11x - 6 \\ &= (2x + 1)(x + 3)(x - 2) \end{aligned}$$

$$\begin{aligned} (3) \quad & 4x^3 - 13x + 6 \\ &= (2x - 1)(2x - 3)(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 9x^3 + 21x^2 + 16x + 4 \\ &= (3x + 2)^2(x + 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 + 4x^2 - 13x + 6 \\ &= (3x - 2)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^3 + 2x^2 - 3x - 2 \\ &= (3x + 2)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^3 + 13x^2 + 13x + 3 \\ &= (3x + 1)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 4x^3 - 3x - 1 \\ &= (2x + 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & x^3 - x^2 - 9x + 9 \\ &= (x + 3)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 18x^3 - 27x^2 + 13x - 2 \\ &= (2x - 1)(3x - 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 4x^3 + 16x^2 + 9x - 9 \\ &= (2x + 3)(2x - 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^3 - x^2 - 3x + 1 \\ &= (3x - 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 4x^3 + 4x^2 - 7x + 2 \\ &= (2x - 1)^2(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & x^3 + 3x^2 - x - 3 \\ &= (x + 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^3 + 5x^2 - x - 6 \\ &= (2x + 3)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 6x^3 - 11x^2 - x + 6 \\ &= (2x - 3)(3x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 18x^3 + 15x^2 - 16x + 3 \\ &= (2x + 3)(3x - 1)^2 \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^3 + 5x^2 - 4x - 3 \\ &= (2x + 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 2x^3 + x^2 - 5x + 2 \\ &= (2x - 1)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 4x^3 - 8x^2 + 5x - 1 \\ &= (2x - 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 - x^2 - 20x + 12 \\ &= (2x - 3)(3x - 2)(x + 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^3 + 19x^2 + 33x + 9 \\ &= (3x + 1)(x + 3)^2 \end{aligned}$$

$$\begin{aligned} (3) \quad & 6x^3 + 11x^2 - x - 6 \\ &= (2x + 3)(3x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 + 25x^2 + 23x + 6 \\ &= (2x + 1)(3x + 2)(x + 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 9x^3 - 18x^2 - x + 2 \\ &= (3x + 1)(3x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (9) \quad & 6x^3 + 17x^2 - 5x - 6 \\ &= (2x + 1)(3x - 2)(x + 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^3 + 7x^2 + 7x + 2 \\ &= (2x + 1)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^3 - 3x^2 + 3x - 1 \\ &= (x - 1)^3 \end{aligned}$$

$$\begin{aligned} (1) \quad & 18x^3 + 15x^2 - x - 2 \\ &= (2x + 1)(3x + 2)(3x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 4x^3 + 8x^2 + x - 3 \\ &= (2x + 3)(2x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 + 13x^2 + 9x + 2 \\ &= (2x + 1)(3x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^3 - 4x^2 - 17x + 6 \\ &= (3x - 1)(x + 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 9x^3 + 18x^2 - x - 2 \\ &= (3x + 1)(3x - 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^3 - 2x^2 - 3x + 2 \\ &= (3x - 2)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & x^3 - 3x^2 + 4 \\ &= (x + 1)(x - 2)^2 \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^3 + x^2 - 2x - 1 \\ &= (2x + 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 9x^3 + 9x^2 - 4x - 4 \\ &= (3x + 2)(3x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 9x^3 - 6x^2 - 5x + 2 \\ &= (3x + 2)(3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^3 - x^2 - 3x + 1 \\ &= (3x - 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 4x^3 - 8x^2 - 3x + 9 \\ &= (2x - 3)^2(x + 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^3 + 3x^2 + 3x + 1 \\ &= (x + 1)^3 \end{aligned}$$

$$\begin{aligned} (7) \quad & 6x^3 + 7x^2 - 1 \\ &= (2x + 1)(3x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 9x^3 + 18x^2 - x - 2 \\ &= (3x + 1)(3x - 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 + 17x^2 + 14x + 3 \\ &= (2x + 3)(3x + 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 - 7x^2 - 7x + 3 \\ &= (3x - 1)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^3 + 14x^2 + 17x + 6 \\ &= (3x + 2)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^3 + x^2 - 3x - 1 \\ &= (3x + 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^3 + x^2 - x - 1 \\ &= (x + 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & x^3 - 2x^2 - x + 2 \\ &= (x+1)(x-1)(x-2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 - 3x^2 - 2x + 3 \\ &= (2x-3)(x+1)(x-1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 12x^3 - 8x^2 - 13x - 3 \\ &= (2x+1)(2x-3)(3x+1) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^3 - 2x^2 - 5x + 6 \\ &= (x+2)(x-1)(x-3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 4x^3 + 16x^2 + 19x + 6 \\ &= (2x+1)(2x+3)(x+2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 4x^3 - 8x^2 + x + 3 \\ &= (2x+1)(2x-3)(x-1) \end{aligned}$$

$$\begin{aligned} (4) \quad & x^3 - x^2 - 9x + 9 \\ &= (x+3)(x-1)(x-3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^3 + x^2 - 18x - 9 \\ &= (2x+1)(x+3)(x-3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^3 + x^2 - 3x - 1 \\ &= (3x+1)(x+1)(x-1) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^3 + x^2 - x - 1 \\ &= (x+1)^2(x-1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 2x^3 + x^2 - 7x - 6 \\ &= (2x + 3)(x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 - x^2 - 7x + 6 \\ &= (2x - 3)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 - x^2 - 11x + 6 \\ &= (2x + 3)(3x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 12x^3 - 8x^2 - 13x - 3 \\ &= (2x + 1)(2x - 3)(3x + 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 9x^3 - 18x^2 - 4x + 8 \\ &= (3x + 2)(3x - 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 9x^3 - 27x^2 - 4x + 12 \\ &= (3x + 2)(3x - 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & x^3 + x^2 - 9x - 9 \\ &= (x + 1)(x + 3)(x - 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^3 - 13x^2 + 8x + 12 \\ &= (3x + 2)(x - 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^3 + x^2 - 4x - 4 \\ &= (x + 1)(x + 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 3x^3 - 11x^2 + 5x + 3 \\ &= (3x + 1)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 2x^3 - x^2 - 4x + 3 \\ &= (2x + 3)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (6) \quad & x^3 + 3x^2 - x - 3 \\ &= (x + 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 3x^3 - 5x^2 - 11x - 3 \\ &= (3x + 1)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & 9x^3 + 3x^2 - 5x + 1 \\ &= (3x - 1)^2(x + 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 12x^3 - 16x^2 - 7x + 6 \\ &= (2x - 1)(2x - 3)(3x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^3 + 7x^2 + 5x + 1 \\ &= (3x + 1)(x + 1)^2 \end{aligned}$$

$$\begin{aligned} (4) \quad & 9x^3 - 9x^2 - x + 1 \\ &= (3x + 1)(3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^3 + x^2 - 3x - 1 \\ &= (3x + 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 9x^3 - 9x^2 - 4x + 4 \\ &= (3x + 2)(3x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 4x^3 - 7x - 3 \\ &= (2x + 1)(2x - 3)(x + 1) \end{aligned}$$

2.6 代入 (易しい)

2 次多項式に整数を代入する問題。

$$(1) \begin{aligned} f(x) &= -5x^2 - 2x + 5 \\ f(5) \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 4x^2 - 3x + 4 \\ f(1) \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(5) \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 3x^2 + 3x - 2 \\ f(1) \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 3x^2 - 2x + 3 \\ f(-2) \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 5x^2 - 4x + 4 \\ f(-2) \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4x^2 + 4x - 4 \\ f(3) \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 3x^2 - 2x + 1 \\ f(2) \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -3x^2 - 5x - 1 \\ f(3) \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 - 2x + 2 \\ f(-2) \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 3x^2 + 4x - 2 \\ f(4) \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -x^2 - x + 1 \\ f(2) \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -x^2 - 4x - 5 \\ f(1) \end{aligned}$$

$$(20) \begin{aligned} f(x) &= x^2 - 4x - 1 \\ f(-3) \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -5x^2 + 2x - 2 \\ f(-4) \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3x^2 + 3x - 1 \\ f(-3) \end{aligned}$$

$$(21) \begin{aligned} f(x) &= x^2 + 2x + 4 \\ f(-1) \end{aligned}$$

$$(36) \begin{aligned} f(x) &= 5x^2 + 3x - 4 \\ f(-1) \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -3x^2 + x - 4 \\ f(-4) \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -5x^2 - 3x + 4 \\ f(-5) \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -5x^2 + 3x + 4 \\ f(2) \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 2x^2 - 5x + 2 \\ f(-2) \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 5x^2 - 2x + 2 \\ f(-1) \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 2x^2 - x + 4 \\ f(3) \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -x^2 - 3x + 3 \\ f(3) \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -2x^2 - 2x + 5 \\ f(3) \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -3x^2 + 4x + 4 \\ f(-1) \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -5x^2 - 4x - 1 \\ f(-2) \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 4x^2 + 4x - 2 \\ f(4) \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 5x^2 + 2x + 5 \\ f(1) \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 - 3x + 4 \\ f(3) \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 5x^2 - 4x + 2 \\ f(2) \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -x^2 + 3x - 5 \\ f(4) \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -3x^2 + x + 1 \\ f(-2) \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 2x^2 + 5x - 2 \\ f(-2) \end{aligned}$$

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$$(43) \begin{aligned} f(x) &= x^2 - 4x - 2 \\ f(-3) \end{aligned}$$

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$$(29) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(-1) \end{aligned}$$

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$$(18) \begin{aligned} f(x) &= x^2 + 4x + 5 \\ f(4) \end{aligned}$$

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$$(44) \begin{aligned} f(x) &= x^2 - 5x + 5 \\ f(-3) \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -5x^2 - 2x + 5 \\ f(5) \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -2x^2 - 4x + 5 \\ f(-2) \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 2x^2 - 2x - 2 \\ f(-3) \end{aligned}$$

$$(1) \quad f(x) = -3x^2 - 5x - 2 \\ f(5)$$

$$(2) \quad f(x) = -2x^2 - 2x - 5 \\ f(-2)$$

$$(3) \quad f(x) = 4x^2 + 4x + 2 \\ f(2)$$

$$(4) \quad f(x) = 3x^2 - 2x - 1 \\ f(-5)$$

$$(5) \quad f(x) = -5x^2 - x - 1 \\ f(3)$$

$$(6) \quad f(x) = -2x^2 + 2x + 1 \\ f(-5)$$

$$(7) \quad f(x) = -4x^2 + 4x + 3 \\ f(-5)$$

$$(8) \quad f(x) = 3x^2 + 2x - 3 \\ f(1)$$

$$(9) \quad f(x) = -5x^2 - x - 5 \\ f(-3)$$

$$(10) \quad f(x) = 5x^2 + 4x + 5 \\ f(2)$$

$$(11) \quad f(x) = 5x^2 + 2x + 2 \\ f(1)$$

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$$(13) \begin{aligned} f(x) &= -2x^2 - 4x + 5 \\ f(-3) \end{aligned}$$

$$(28) \begin{aligned} f(x) &= x^2 - 3x - 3 \\ f(2) \end{aligned}$$

$$(43) \begin{aligned} f(x) &= x^2 - 5x - 1 \\ f(-4) \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -x^2 - 4x - 1 \\ f(2) \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 3x^2 - 3x - 4 \\ f(-5) \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -x^2 - 2x - 1 \\ f(5) \end{aligned}$$

$$(15) \begin{aligned} f(x) &= 5x^2 - x - 5 \\ f(-4) \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 2x^2 - x - 5 \\ f(2) \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 5x^2 + x + 3 \\ f(-2) \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -5x^2 - 2x + 5 \\ f(5) &= -130 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 4x^2 - 3x + 4 \\ f(1) &= 5 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(5) &= -131 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 3x^2 + 3x - 2 \\ f(1) &= 4 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 3x^2 - 2x + 3 \\ f(-2) &= 19 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 5x^2 - 4x + 4 \\ f(-2) &= 32 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4x^2 + 4x - 4 \\ f(3) &= -28 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 3x^2 - 2x + 1 \\ f(2) &= 9 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -3x^2 - 5x - 1 \\ f(3) &= -43 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 - 2x + 2 \\ f(-2) &= 18 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 3x^2 + 4x - 2 \\ f(4) &= 62 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -x^2 - x + 1 \\ f(2) &= -5 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -x^2 - 4x - 5 \\ f(1) &= -10 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= x^2 - 4x - 1 \\ f(-3) &= 20 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -5x^2 + 2x - 2 \\ f(-4) &= -90 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3x^2 + 3x - 1 \\ f(-3) &= 17 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= x^2 + 2x + 4 \\ f(-1) &= 3 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= 5x^2 + 3x - 4 \\ f(-1) &= -2 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -3x^2 + x - 4 \\ f(-4) &= -56 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -5x^2 - 3x + 4 \\ f(-5) &= -106 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -5x^2 + 3x + 4 \\ f(2) &= -10 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 2x^2 - 5x + 2 \\ f(-2) &= 20 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 5x^2 - 2x + 2 \\ f(-1) &= 9 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 2x^2 - x + 4 \\ f(3) &= 19 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -x^2 - 3x + 3 \\ f(3) &= -15 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -2x^2 - 2x + 5 \\ f(3) &= -19 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -3x^2 + 4x + 4 \\ f(-1) &= -3 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -5x^2 - 4x - 1 \\ f(-2) &= -13 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 4x^2 + 4x - 2 \\ f(4) &= 78 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 5x^2 + 2x + 5 \\ f(1) &= 12 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 - 3x + 4 \\ f(3) &= 40 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 5x^2 - 4x + 2 \\ f(2) &= 14 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -x^2 + 3x - 5 \\ f(4) &= -9 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -3x^2 + x + 1 \\ f(-2) &= -13 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 2x^2 + 5x - 2 \\ f(-2) &= -4 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 3x^2 + 3x - 4 \\ f(1) &= 2 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -4x^2 + 3x - 5 \\ f(-3) &= -50 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= x^2 - 3x - 2 \\ f(3) &= -2 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= x^2 - 4x - 2 \\ f(-3) &= 19 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -4x^2 + 3x - 3 \\ f(-4) &= -79 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(-1) &= -5 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= 5x^2 + 2x + 5 \\ f(3) &= 56 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -4x^2 + x - 3 \\ f(3) &= -36 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 2x^2 - 5x - 5 \\ f(4) &= 7 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 3x^2 - 2x + 5 \\ f(1) &= 6 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 5x^2 - 3x + 1 \\ f(2) &= 15 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 3x^2 - x + 4 \\ f(-5) &= 84 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= x^2 - 4x - 1 \\ f(-4) &= 31 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -4x^2 - 2x + 4 \\ f(4) &= -68 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -3x^2 + 5x - 2 \\ f(3) &= -14 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 2x^2 - 4x + 2 \\ f(5) &= 32 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 3x^2 - 3x + 1 \\ f(5) &= 61 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= x^2 + 4x + 5 \\ f(4) &= 37 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -2x^2 + 4x + 1 \\ f(-4) &= -47 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -x^2 + 4x - 1 \\ f(2) &= 3 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -2x^2 - 4x + 1 \\ f(3) &= -29 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -3x^2 + 3x - 3 \\ f(-3) &= -39 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 5x^2 - 3x - 3 \\ f(1) &= -1 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -4x^2 + 2x + 3 \\ f(-4) &= -69 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 2x^2 + 2x - 2 \\ f(2) &= 10 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= x^2 + 4x + 5 \\ f(4) &= 37 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -2x^2 + 5x - 2 \\ f(5) &= -27 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -2x^2 - 3x - 3 \\ f(-2) &= -5 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 4x^2 + 5x - 1 \\ f(4) &= 83 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -2x^2 + 2x + 4 \\ f(-1) &= 0 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -5x^2 - 2x + 2 \\ f(-1) &= -1 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2x^2 - 3x - 3 \\ f(4) &= -47 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -4x^2 - 4x + 5 \\ f(-3) &= -19 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= -2x^2 - 4x + 5 \\ f(-2) &= 5 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -3x^2 - 2x + 4 \\ f(1) &= -1 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -x^2 + 4x + 2 \\ f(4) &= 2 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -2x^2 - 2x + 2 \\ f(4) &= -38 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= x^2 + 2x + 5 \\ f(-1) &= 4 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 2x^2 + x - 1 \\ f(2) &= 9 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 2x^2 - 2x + 2 \\ f(3) &= 14 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -4x^2 + 3x + 2 \\ f(3) &= -25 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 5x^2 + 3x + 4 \\ f(5) &= 144 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= x^2 + x + 1 \\ f(-2) &= 3 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -5x^2 - 4x + 3 \\ f(5) &= -142 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -x^2 + 2x + 5 \\ f(2) &= 5 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 4x^2 - 4x + 5 \\ f(-1) &= 13 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= x^2 - 4x + 5 \\ f(3) &= 2 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 2x^2 + 2x + 5 \\ f(4) &= 45 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -4x^2 + 4x - 2 \\ f(-2) &= -26 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 5x^2 - 4x - 4 \\ f(1) &= -3 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -2x^2 + 3x - 2 \\ f(4) &= -22 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -2x^2 + 5x - 3 \\ f(-2) &= -21 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -x^2 - 3x - 2 \\ f(3) &= -20 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -x^2 - 5x - 5 \\ f(-4) &= -1 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -2x^2 - 3x + 1 \\ f(-4) &= -19 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 5x^2 - 4x - 5 \\ f(3) &= 28 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= -3x^2 - 4x - 2 \\ f(-3) &= -17 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 5x^2 - 5x + 5 \\ f(3) &= 35 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= x^2 + 3x + 2 \\ f(-5) &= 12 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -4x^2 - 3x - 3 \\ f(-4) &= -55 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= x^2 - 5x + 2 \\ f(-2) &= 16 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -5x^2 - 5x + 5 \\ f(1) &= -5 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -x^2 - 3x - 2 \\ f(2) &= -12 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= x^2 + 5x + 3 \\ f(-2) &= -3 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 2x^2 + x + 2 \\ f(3) &= 23 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 5x^2 - 3x + 2 \\ f(1) &= 4 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= x^2 + 4x - 4 \\ f(-2) &= -8 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -5x^2 + 3x - 5 \\ f(3) &= -41 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= 3x^2 - x + 3 \\ f(2) &= 13 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 2x^2 - 2x + 2 \\ f(-1) &= 6 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -3x^2 + 5x + 3 \\ f(-3) &= -39 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= 2x^2 - x + 2 \\ f(3) &= 17 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -3x^2 - 5x - 1 \\ f(1) &= -9 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 5x^2 - 2x - 5 \\ f(1) &= -2 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -x^2 + 5x - 1 \\ f(2) &= 5 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 3x^2 - 4x - 2 \\ f(-4) &= 62 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -5x^2 + 4x + 4 \\ f(5) &= -101 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 2x^2 + 5x + 4 \\ f(-5) &= 29 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= x^2 - x + 2 \\ f(-1) &= 4 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -5x^2 + x - 2 \\ f(-1) &= -8 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 3x^2 + 5x - 5 \\ f(-1) &= -7 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= 5x^2 - 2x - 2 \\ f(1) &= 1 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= x^2 + 3x - 5 \\ f(4) &= 23 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -4x^2 - 5x - 4 \\ f(2) &= -30 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 4x^2 - 5x - 1 \\ f(4) &= 43 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 + 5x + 4 \\ f(-2) &= 14 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 4x^2 - 2x + 5 \\ f(-3) &= 47 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -4x^2 + 2x - 3 \\ f(4) &= -59 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -4x^2 + 3x - 2 \\ f(-5) &= -117 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 5x^2 + 4x + 5 \\ f(4) &= 101 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 5x^2 - 4x + 1 \\ f(5) &= 106 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -x^2 + 2x - 4 \\ f(-4) &= -28 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 2x^2 - 4x - 2 \\ f(-3) &= 28 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -3x^2 + 4x - 2 \\ f(-3) &= -41 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -5x^2 - x + 4 \\ f(-3) &= -38 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 4x^2 + 5x - 2 \\ f(2) &= 24 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -3x^2 + 3x - 5 \\ f(-5) &= -95 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -4x^2 - x - 2 \\ f(3) &= -41 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -4x^2 - 3x + 3 \\ f(-2) &= -7 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -x^2 - 3x + 3 \\ f(2) &= -7 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 3x^2 - 2x - 4 \\ f(-5) &= 81 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 4x^2 - 2x + 2 \\ f(4) &= 58 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 5x^2 + 2x - 2 \\ f(-5) &= 113 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -x^2 - 3x + 1 \\ f(5) &= -39 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 2x^2 + 3x + 5 \\ f(-4) &= 25 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -2x^2 - 2x - 5 \\ f(-5) &= -45 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -5x^2 - 4x + 3 \\ f(2) &= -25 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 3x^2 + x + 2 \\ f(-5) &= 72 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -2x^2 + 5x + 2 \\ f(-3) &= -31 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 2x^2 + 3x - 5 \\ f(4) &= 39 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 3x^2 + 4x + 4 \\ f(2) &= 24 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -x^2 - 4x - 4 \\ f(3) &= -25 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 4x^2 + 5x - 3 \\ f(-4) &= 41 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= x^2 - x + 5 \\ f(1) &= 5 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 3x^2 - 5x + 1 \\ f(1) &= -1 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3x^2 - 3x + 4 \\ f(-2) &= 22 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -3x^2 - x - 3 \\ f(1) &= -7 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= 3x^2 - 5x + 2 \\ f(5) &= 52 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -3x^2 - 5x - 5 \\ f(3) &= -47 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -3x^2 - 4x - 5 \\ f(-4) &= -37 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -2x^2 + 2x - 2 \\ f(-4) &= -42 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -4x^2 + 3x - 4 \\ f(1) &= -5 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 4x^2 + x - 2 \\ f(3) &= 37 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= -4x^2 + 5x + 3 \\ f(3) &= -18 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 2x^2 - 3x - 5 \\ f(-5) &= 60 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -5x^2 - 5x + 4 \\ f(4) &= -96 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= 2x^2 - 2x - 1 \\ f(-2) &= 11 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 2x^2 + 5x - 3 \\ f(-4) &= 9 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -4x^2 + 5x + 1 \\ f(-4) &= -83 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 2x^2 + 3x + 3 \\ f(-4) &= 23 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -4x^2 + 2x + 1 \\ f(3) &= -29 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 4x^2 - x + 5 \\ f(2) &= 19 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 4x^2 - 5x + 4 \\ f(-3) &= 55 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -3x^2 + x - 4 \\ f(2) &= -14 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -3x^2 - 2x + 4 \\ f(-2) &= -4 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 4x^2 + 3x + 3 \\ f(-2) &= 13 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -x^2 + x + 2 \\ f(3) &= -4 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= -5x^2 - 2x + 4 \\ f(-2) &= -12 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -4x^2 - 5x + 1 \\ f(-1) &= 2 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 5x^2 - x - 3 \\ f(1) &= 1 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -2x^2 + 4x - 3 \\ f(2) &= -3 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= x^2 + 4x + 4 \\ f(1) &= 9 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -4x^2 - 2x - 2 \\ f(2) &= -22 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 2x^2 + 3x + 1 \\ f(3) &= 28 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -3x^2 + 4x + 1 \\ f(-4) &= -63 \end{aligned}$$

$$(1) \quad f(x) = -2x^2 - 4x - 1 \\ f(1) = -7$$

$$(2) \quad f(x) = 5x^2 + 5x + 3 \\ f(-1) = 3$$

$$(3) \quad f(x) = x^2 + 2x + 3 \\ f(3) = 18$$

$$(4) \quad f(x) = x^2 - 5x + 1 \\ f(4) = -3$$

$$(5) \quad f(x) = x^2 + x - 1 \\ f(-2) = 1$$

$$(6) \quad f(x) = 3x^2 - x + 2 \\ f(-4) = 54$$

$$(7) \quad f(x) = -3x^2 - 4x + 4 \\ f(2) = -16$$

$$(8) \quad f(x) = -x^2 + 5x - 2 \\ f(3) = 4$$

$$(9) \quad f(x) = -5x^2 - 3x - 2 \\ f(5) = -142$$

$$(10) \quad f(x) = 4x^2 - x + 1 \\ f(3) = 34$$

$$(11) \quad f(x) = -3x^2 + 3x + 4 \\ f(1) = 4$$

$$(12) \quad f(x) = 5x^2 - 2x + 3 \\ f(-4) = 91$$

$$(13) \quad f(x) = 2x^2 - 5x - 3 \\ f(4) = 9$$

$$(14) \quad f(x) = 2x^2 + x + 4 \\ f(-4) = 32$$

$$(15) \quad f(x) = -4x^2 - 2x - 1 \\ f(5) = -111$$

$$(16) \quad f(x) = 4x^2 + 5x - 1 \\ f(2) = 25$$

$$(17) \quad f(x) = -2x^2 + x + 2 \\ f(3) = -13$$

$$(18) \quad f(x) = 3x^2 - 4x - 1 \\ f(-2) = 19$$

$$(19) \quad f(x) = -5x^2 + 4x - 5 \\ f(-5) = -150$$

$$(20) \quad f(x) = 4x^2 + x - 4 \\ f(-5) = 91$$

$$(21) \quad f(x) = 3x^2 - 4x + 1 \\ f(-2) = 21$$

$$(22) \quad f(x) = 4x^2 + 4x - 3 \\ f(-1) = -3$$

$$(23) \quad f(x) = -2x^2 - 4x + 4 \\ f(-4) = -12$$

$$(24) \quad f(x) = -3x^2 - 5x + 4 \\ f(4) = -64$$

$$(25) \quad f(x) = 4x^2 + 5x - 2 \\ f(3) = 49$$

$$(26) \quad f(x) = 5x^2 - 3x - 5 \\ f(-3) = 49$$

$$(27) \quad f(x) = 4x^2 - 4x + 2 \\ f(5) = 82$$

$$(28) \quad f(x) = -3x^2 + 5x + 3 \\ f(2) = 1$$

$$(29) \quad f(x) = -5x^2 - x + 2 \\ f(4) = -82$$

$$(30) \quad f(x) = -x^2 + 2x + 2 \\ f(-2) = -6$$

$$(31) \quad f(x) = 2x^2 - 4x - 3 \\ f(-1) = 3$$

$$(32) \quad f(x) = 2x^2 + 3x - 1 \\ f(1) = 4$$

$$(33) \quad f(x) = -5x^2 - 2x + 5 \\ f(3) = -46$$

$$(34) \quad f(x) = -2x^2 + 4x - 5 \\ f(-4) = -53$$

$$(35) \quad f(x) = 2x^2 - x + 2 \\ f(3) = 17$$

$$(36) \quad f(x) = 5x^2 - 5x - 2 \\ f(-4) = 98$$

$$(37) \quad f(x) = 2x^2 + 5x + 3 \\ f(-5) = 28$$

$$(38) \quad f(x) = 3x^2 - 4x - 5 \\ f(-2) = 15$$

$$(39) \quad f(x) = 2x^2 + 3x + 2 \\ f(5) = 67$$

$$(40) \quad f(x) = -4x^2 - 5x + 3 \\ f(-5) = -72$$

$$(41) \quad f(x) = 3x^2 + 5x + 1 \\ f(-1) = -1$$

$$(42) \quad f(x) = -2x^2 - 5x + 1 \\ f(-2) = 3$$

$$(43) \quad f(x) = x^2 + 4x + 1 \\ f(-3) = -2$$

$$(44) \quad f(x) = 5x^2 + x - 2 \\ f(2) = 20$$

$$(45) \quad f(x) = 2x^2 + 3x + 4 \\ f(3) = 31$$

$$(1) \begin{aligned} f(x) &= 3x^2 - x + 4 \\ f(3) &= 28 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= x^2 + x + 3 \\ f(-4) &= 15 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= x^2 - 2x + 4 \\ f(-2) &= 12 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -4x^2 - 2x - 1 \\ f(1) &= -7 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 4x^2 - 5x - 1 \\ f(-3) &= 50 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 4x^2 - 5x - 5 \\ f(2) &= 1 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 3x^2 + 2x - 3 \\ f(1) &= 2 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -x^2 + 4x + 1 \\ f(4) &= 1 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -x^2 + 5x - 1 \\ f(-5) &= -51 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5x^2 - 2x + 2 \\ f(5) &= -133 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 4x^2 + 5x + 4 \\ f(4) &= 88 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= 2x^2 + x - 3 \\ f(-5) &= 42 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 2x^2 + 4x - 5 \\ f(-1) &= -7 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= 5x^2 + 4x - 5 \\ f(-5) &= 100 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -2x^2 + 3x + 4 \\ f(2) &= 2 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3x^2 + 5x - 5 \\ f(-4) &= 23 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= x^2 + 4x + 5 \\ f(-5) &= 10 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -2x^2 - 4x - 3 \\ f(3) &= -33 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -2x^2 - 2x + 2 \\ f(3) &= -22 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -3x^2 - 4x - 2 \\ f(4) &= -66 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -x^2 + 3x - 2 \\ f(4) &= -6 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 2x^2 + 3x + 3 \\ f(-3) &= 12 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -x^2 - x - 2 \\ f(-1) &= -2 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= -3x^2 + 4x + 5 \\ f(3) &= -10 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 3x^2 - x - 1 \\ f(-3) &= 29 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 3x^2 + x - 5 \\ f(-1) &= -3 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= 2x^2 - 4x + 3 \\ f(1) &= 1 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -x^2 + x - 1 \\ f(2) &= -3 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 5x^2 + 4x - 5 \\ f(-1) &= -4 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 3x^2 + 3x + 4 \\ f(4) &= 64 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -2x^2 + 3x + 4 \\ f(-2) &= -10 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -5x^2 + 5x - 5 \\ f(-3) &= -65 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 5x^2 + x + 5 \\ f(2) &= 27 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= x^2 - 3x + 5 \\ f(-1) &= 9 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 3x^2 + 3x + 4 \\ f(4) &= 64 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 4x^2 + 4x - 2 \\ f(-1) &= -2 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -3x^2 + 4x + 2 \\ f(-2) &= -18 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 2x^2 + x - 5 \\ f(-2) &= 1 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -x^2 - x - 1 \\ f(3) &= -13 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= x^2 - x + 2 \\ f(-1) &= 4 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 5x^2 - 5x - 3 \\ f(3) &= 27 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -x^2 - 3x + 5 \\ f(4) &= -23 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= 2x^2 - x + 3 \\ f(-1) &= 6 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -2x^2 + 4x - 4 \\ f(5) &= -34 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -3x^2 - x - 5 \\ f(-2) &= -15 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -2x^2 + 3x + 5 \\ f(2) &= 3 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 4x^2 - 2x + 3 \\ f(5) &= 93 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 3x^2 + 3x + 3 \\ f(-2) &= 9 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5x^2 + 3x - 2 \\ f(4) &= -70 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -x^2 + 4x - 2 \\ f(-5) &= -47 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3x^2 + 5x + 3 \\ f(3) &= 45 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -3x^2 + 4x - 1 \\ f(-5) &= -96 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= x^2 + 5x - 2 \\ f(-5) &= -2 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 3x^2 + 3x + 5 \\ f(1) &= 11 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -3x^2 - 5x + 5 \\ f(-4) &= -23 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 - 2x + 5 \\ f(-2) &= 29 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= 3x^2 + 4x + 4 \\ f(-1) &= 3 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -2x^2 + 3x - 3 \\ f(2) &= -5 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 2x^2 + 2x - 5 \\ f(5) &= 55 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= x^2 + 2x + 3 \\ f(-5) &= 18 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 4x^2 + 4x + 3 \\ f(-4) &= 51 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -4x^2 + 3x - 4 \\ f(2) &= -14 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -4x^2 + 5x - 1 \\ f(1) &= 0 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 5x^2 - 5x - 4 \\ f(-3) &= 56 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= x^2 - 5x - 5 \\ f(-4) &= 31 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= 5x^2 + 3x + 2 \\ f(5) &= 142 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= 3x^2 + 3x - 4 \\ f(3) &= 32 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -4x^2 - x - 5 \\ f(-2) &= -19 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 3x^2 - 2x - 4 \\ f(-3) &= 29 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 3x^2 - 2x - 5 \\ f(1) &= -4 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -3x^2 + 5x - 2 \\ f(1) &= 0 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 4x^2 - x + 4 \\ f(-2) &= 22 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= -3x^2 - 3x + 4 \\ f(-3) &= -14 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -2x^2 - 3x - 5 \\ f(5) &= -70 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -x^2 - 2x - 3 \\ f(3) &= -18 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -4x^2 + 3x + 5 \\ f(5) &= -80 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 5x^2 + 4x + 4 \\ f(-3) &= 37 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -5x^2 + 4x - 4 \\ f(-4) &= -100 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -4x^2 - 2x - 2 \\ f(-4) &= -58 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -4x^2 - x + 3 \\ f(-2) &= -11 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -5x^2 - 2x - 3 \\ f(1) &= -10 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 3x^2 - x + 4 \\ f(-1) &= 8 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 4x^2 + 2x + 1 \\ f(5) &= 111 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -x^2 - 5x + 3 \\ f(3) &= -21 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 5x^2 + 5x - 5 \\ f(4) &= 95 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -3x^2 + 3x + 1 \\ f(-3) &= -35 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 3x^2 - 5x - 5 \\ f(-2) &= 17 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 3x^2 + 5x + 5 \\ f(-1) &= 3 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -5x^2 + 3x - 5 \\ f(2) &= -19 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 2x^2 - 3x + 3 \\ f(-5) &= 68 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 2x^2 - 4x - 3 \\ f(4) &= 13 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 3x^2 + 5x + 3 \\ f(-2) &= 5 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -3x^2 + 3x + 1 \\ f(5) &= -59 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 3x^2 - 4x + 2 \\ f(-5) &= 97 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 5x^2 - 2x - 5 \\ f(2) &= 11 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 2x^2 - 5x - 5 \\ f(-4) &= 47 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -3x^2 + 2x + 4 \\ f(2) &= -4 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -4x^2 + x - 5 \\ f(2) &= -19 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= 5x^2 + 3x + 1 \\ f(-5) &= 111 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5x^2 - 5x - 4 \\ f(2) &= -34 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 3x^2 - 2x + 4 \\ f(3) &= 25 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= 4x^2 + 5x - 4 \\ f(5) &= 121 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -5x^2 + 5x - 2 \\ f(-1) &= -12 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -4x^2 + 5x + 1 \\ f(4) &= -43 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 4x^2 - 3x + 3 \\ f(2) &= 13 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -x^2 + 4x - 3 \\ f(1) &= 0 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -x^2 - 3x + 1 \\ f(3) &= -17 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= 5x^2 + 4x - 1 \\ f(1) &= 8 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 3x^2 + 3x + 1 \\ f(-5) &= 61 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -2x^2 - 3x - 3 \\ f(5) &= -68 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 2x^2 + 2x - 1 \\ f(-3) &= 11 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2x^2 - 5x - 2 \\ f(1) &= -9 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -5x^2 - x + 1 \\ f(-5) &= -119 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= x^2 - x - 3 \\ f(3) &= 3 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4x^2 - 3x - 4 \\ f(4) &= -80 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -2x^2 + 3x - 1 \\ f(2) &= -3 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= 2x^2 + 3x - 1 \\ f(4) &= 43 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 3x^2 - 2x + 2 \\ f(-1) &= 7 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -3x^2 + 4x + 3 \\ f(-3) &= -36 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 3x^2 + 3x - 1 \\ f(3) &= 35 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -x^2 + x + 2 \\ f(3) &= -4 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -2x^2 - 5x + 3 \\ f(-4) &= -9 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 3x^2 + 4x + 4 \\ f(-5) &= 59 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= 4x^2 + x - 5 \\ f(3) &= 34 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -2x^2 - x + 4 \\ f(-3) &= -11 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -3x^2 + 4x + 5 \\ f(3) &= -10 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= 2x^2 - 2x - 3 \\ f(-2) &= 9 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 3x^2 - 2x + 1 \\ f(-3) &= 34 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 2x^2 - 5x - 4 \\ f(1) &= -7 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -5x^2 - 5x - 2 \\ f(-4) &= -62 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -3x^2 - 2x + 2 \\ f(5) &= -83 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -x^2 - 4x + 4 \\ f(-1) &= 7 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= 2x^2 - 4x - 3 \\ f(-5) &= 67 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 5x^2 - 5x + 4 \\ f(5) &= 104 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 5x^2 - 2x - 5 \\ f(-4) &= 83 \end{aligned}$$

$$(1) \quad f(x) = -2x^2 + 3x - 5 \\ f(-4) = -49$$

$$(2) \quad f(x) = x^2 - 3x + 1 \\ f(-3) = 19$$

$$(3) \quad f(x) = -2x^2 - x - 2 \\ f(3) = -23$$

$$(4) \quad f(x) = 4x^2 + 4x - 1 \\ f(-5) = 79$$

$$(5) \quad f(x) = -2x^2 - 2x + 5 \\ f(-3) = -7$$

$$(6) \quad f(x) = 2x^2 + x - 5 \\ f(4) = 31$$

$$(7) \quad f(x) = 4x^2 - x - 5 \\ f(-1) = 0$$

$$(8) \quad f(x) = 3x^2 - 2x - 4 \\ f(1) = -3$$

$$(9) \quad f(x) = 5x^2 + 5x - 4 \\ f(-4) = 56$$

$$(10) \quad f(x) = -x^2 - 5x + 3 \\ f(-4) = 7$$

$$(11) \quad f(x) = 2x^2 + 4x - 3 \\ f(-1) = -5$$

$$(12) \quad f(x) = -2x^2 - 3x + 2 \\ f(-3) = -7$$

$$(13) \quad f(x) = x^2 - 5x + 3 \\ f(2) = -3$$

$$(14) \quad f(x) = -3x^2 + 2x + 5 \\ f(-4) = -51$$

$$(15) \quad f(x) = -5x^2 - 2x + 5 \\ f(5) = -130$$

$$(16) \quad f(x) = -3x^2 + x - 2 \\ f(1) = -4$$

$$(17) \quad f(x) = -2x^2 - 5x + 2 \\ f(2) = -16$$

$$(18) \quad f(x) = 2x^2 + x - 4 \\ f(-3) = 11$$

$$(19) \quad f(x) = -3x^2 - 3x + 2 \\ f(2) = -16$$

$$(20) \quad f(x) = 3x^2 + 2x - 5 \\ f(-1) = -4$$

$$(21) \quad f(x) = 5x^2 + 5x - 1 \\ f(5) = 149$$

$$(22) \quad f(x) = 2x^2 + x - 2 \\ f(-1) = -1$$

$$(23) \quad f(x) = 4x^2 + 3x - 1 \\ f(1) = 6$$

$$(24) \quad f(x) = x^2 - 5x + 3 \\ f(4) = -1$$

$$(25) \quad f(x) = -5x^2 + 4x + 2 \\ f(3) = -31$$

$$(26) \quad f(x) = -x^2 - x + 5 \\ f(5) = -25$$

$$(27) \quad f(x) = 3x^2 - 5x - 1 \\ f(-2) = 21$$

$$(28) \quad f(x) = x^2 - 2x + 3 \\ f(-3) = 18$$

$$(29) \quad f(x) = -2x^2 + 5x + 1 \\ f(3) = -2$$

$$(30) \quad f(x) = -2x^2 - 4x + 5 \\ f(-2) = 5$$

$$(31) \quad f(x) = -x^2 + 3x + 2 \\ f(3) = 2$$

$$(32) \quad f(x) = -3x^2 + x + 3 \\ f(-3) = -27$$

$$(33) \quad f(x) = x^2 - 5x - 1 \\ f(-2) = 13$$

$$(34) \quad f(x) = 2x^2 - 3x + 5 \\ f(2) = 7$$

$$(35) \quad f(x) = -2x^2 + 2x + 2 \\ f(-1) = -2$$

$$(36) \quad f(x) = 3x^2 + 3x + 1 \\ f(5) = 91$$

$$(37) \quad f(x) = -2x^2 - 2x + 5 \\ f(2) = -7$$

$$(38) \quad f(x) = -x^2 + 5x - 5 \\ f(-5) = -55$$

$$(39) \quad f(x) = -2x^2 + 2x + 3 \\ f(-5) = -57$$

$$(40) \quad f(x) = x^2 - 3x + 3 \\ f(-5) = 43$$

$$(41) \quad f(x) = -3x^2 + 2x + 4 \\ f(-2) = -12$$

$$(42) \quad f(x) = -x^2 + 5x + 3 \\ f(2) = 9$$

$$(43) \quad f(x) = -4x^2 - 4x - 1 \\ f(3) = -49$$

$$(44) \quad f(x) = x^2 - 5x + 5 \\ f(-3) = 29$$

$$(45) \quad f(x) = 2x^2 - 2x - 2 \\ f(-3) = 22$$

$$(1) \begin{aligned} f(x) &= -3x^2 - 5x - 2 \\ f(5) &= -102 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 5x^2 + 3x - 1 \\ f(3) &= 53 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -3x^2 + 5x + 3 \\ f(-2) &= -19 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -2x^2 - 2x - 5 \\ f(-2) &= -9 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 4x^2 + 4x + 2 \\ f(-4) &= 50 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 5x^2 + 2x - 2 \\ f(2) &= 22 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 4x^2 + 4x + 2 \\ f(2) &= 26 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 2x^2 - 2x + 3 \\ f(3) &= 15 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -4x^2 - x + 1 \\ f(-5) &= -94 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 - 2x - 1 \\ f(-5) &= 84 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -3x^2 - x + 5 \\ f(-5) &= -65 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -5x^2 + 4x - 2 \\ f(-2) &= -30 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(3) &= -49 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= 4x^2 - 3x - 1 \\ f(-5) &= 114 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 2x^2 - 3x - 2 \\ f(1) &= -3 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -2x^2 + 2x + 1 \\ f(-5) &= -59 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -3x^2 - 4x - 3 \\ f(-4) &= -35 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -x^2 + 3x + 1 \\ f(2) &= 3 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -4x^2 + 4x + 3 \\ f(-5) &= -117 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= x^2 - 4x - 2 \\ f(-1) &= 3 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 2x^2 - 2x + 3 \\ f(2) &= 7 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 3x^2 + 2x - 3 \\ f(1) &= 2 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 2x^2 + x - 2 \\ f(-5) &= 43 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= -2x^2 - 5x - 5 \\ f(4) &= -57 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -5x^2 - x - 5 \\ f(-3) &= -47 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 2x^2 + x - 1 \\ f(-1) &= 0 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= 2x^2 - 4x + 3 \\ f(1) &= 1 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 5x^2 + 4x + 5 \\ f(2) &= 33 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 3x^2 + 3x + 2 \\ f(1) &= 8 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= x^2 + 3x + 4 \\ f(3) &= 22 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 + 2x + 2 \\ f(1) &= 9 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -2x^2 + 5x - 3 \\ f(-5) &= -78 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 5x^2 + 5x + 2 \\ f(-4) &= 62 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= x^2 - 3x + 3 \\ f(-1) &= 7 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -4x^2 + 2x + 5 \\ f(3) &= -25 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= x^2 + 2x - 4 \\ f(-4) &= 4 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -5x^2 + 2x - 1 \\ f(-3) &= -52 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 2x^2 - 5x + 4 \\ f(4) &= 16 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -4x^2 + 5x - 2 \\ f(4) &= -46 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -2x^2 - x - 3 \\ f(3) &= -24 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -x^2 - 4x - 2 \\ f(3) &= -23 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -x^2 + 4x - 5 \\ f(-4) &= -37 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -2x^2 + 2x - 2 \\ f(4) &= -26 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -4x^2 - x - 3 \\ f(-5) &= -98 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 3x^2 - x + 2 \\ f(2) &= 12 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -x^2 + 4x + 4 \\ f(-3) &= -17 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 3x^2 - 3x - 2 \\ f(5) &= 58 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -4x^2 + x + 3 \\ f(2) &= -11 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -x^2 + 3x + 1 \\ f(-4) &= -27 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 4x^2 - 5x + 3 \\ f(5) &= 78 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 3x^2 - 3x - 4 \\ f(4) &= 32 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -2x^2 - x - 1 \\ f(3) &= -22 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 4x^2 + 2x + 3 \\ f(-3) &= 33 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= x^2 + 3x + 1 \\ f(-1) &= -1 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 + 5x - 3 \\ f(-3) &= 9 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= x^2 + 2x + 3 \\ f(-5) &= 18 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= 3x^2 - 4x + 5 \\ f(4) &= 37 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -5x^2 - 5x - 3 \\ f(4) &= -103 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= 3x^2 + x + 4 \\ f(-2) &= 14 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= x^2 + 3x + 1 \\ f(1) &= 5 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 2x^2 + 2x - 4 \\ f(-3) &= 8 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -x^2 + 2x - 3 \\ f(4) &= -11 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -4x^2 + 5x - 2 \\ f(3) &= -23 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -5x^2 - 4x + 2 \\ f(-1) &= 1 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= 4x^2 + 4x - 4 \\ f(-4) &= 44 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -3x^2 - 5x + 5 \\ f(4) &= -63 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -4x^2 + 4x + 1 \\ f(-5) &= -119 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 4x^2 + 3x - 5 \\ f(3) &= 40 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= x^2 + 5x - 1 \\ f(1) &= 5 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4x^2 - 2x - 5 \\ f(1) &= -11 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -5x^2 - 5x - 2 \\ f(5) &= -152 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -x^2 + 3x - 2 \\ f(-1) &= -6 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 2x^2 - 4x - 2 \\ f(-3) &= 28 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -2x^2 + 2x + 3 \\ f(2) &= -1 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= -3x^2 - x - 1 \\ f(1) &= -5 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 + 3x - 4 \\ f(-3) &= 32 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -4x^2 + 5x + 4 \\ f(-4) &= -80 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -2x^2 - 4x - 1 \\ f(1) &= -7 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -5x^2 - 5x - 4 \\ f(4) &= -104 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= x^2 - x - 5 \\ f(5) &= 15 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 3x^2 - 3x - 1 \\ f(-1) &= 5 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= 3x^2 + 3x + 4 \\ f(3) &= 40 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= x^2 - 3x - 2 \\ f(4) &= 2 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 2x^2 + 2x + 2 \\ f(2) &= 14 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -4x^2 - 5x + 2 \\ f(-2) &= -4 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -2x^2 - 3x - 1 \\ f(-5) &= -36 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= 3x^2 - 2x - 3 \\ f(-4) &= 53 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -x^2 + 2x - 4 \\ f(1) &= -3 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 4x^2 + 2x + 2 \\ f(3) &= 44 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 3x^2 + 4x + 5 \\ f(3) &= 44 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -x^2 - 2x + 4 \\ f(3) &= -11 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= -3x^2 + 4x - 3 \\ f(4) &= -35 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 3x^2 + x - 2 \\ f(4) &= 50 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 2x^2 + x + 1 \\ f(1) &= 4 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -2x^2 + x + 1 \\ f(3) &= -14 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -4x^2 + 5x + 5 \\ f(-3) &= -46 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 4x^2 - x - 4 \\ f(-1) &= 1 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -4x^2 + 3x - 5 \\ f(5) &= -90 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -5x^2 - 5x + 3 \\ f(4) &= -97 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5x^2 - x + 2 \\ f(2) &= -20 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -4x^2 + x - 4 \\ f(1) &= -7 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -4x^2 + 3x - 2 \\ f(1) &= -3 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -4x^2 - 4x + 2 \\ f(-5) &= -78 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -x^2 - 5x - 5 \\ f(-5) &= -5 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -4x^2 + 2x + 4 \\ f(-4) &= -68 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 4x^2 - x + 2 \\ f(4) &= 62 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -4x^2 - x + 3 \\ f(-2) &= -11 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -x^2 + 3x - 4 \\ f(-1) &= -8 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= x^2 + 3x - 5 \\ f(3) &= 13 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -5x^2 - x - 3 \\ f(1) &= -9 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 3x^2 + x + 2 \\ f(-3) &= 26 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -5x^2 + 4x - 2 \\ f(2) &= -14 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -3x^2 - 4x - 3 \\ f(-2) &= -7 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 3x^2 + x + 1 \\ f(-1) &= 3 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 4x^2 + 4x - 5 \\ f(-5) &= 75 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -5x^2 - 5x + 1 \\ f(5) &= -149 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -x^2 + 2x - 2 \\ f(3) &= -5 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -3x^2 + 3x - 2 \\ f(3) &= -20 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 2x^2 - 5x + 2 \\ f(1) &= -1 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= -4x^2 + x + 4 \\ f(-5) &= -101 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 3x^2 + x - 2 \\ f(5) &= 78 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 4x^2 - 4x + 2 \\ f(-5) &= 122 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 4x^2 - 2x + 3 \\ f(4) &= 59 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= 5x^2 + 4x - 4 \\ f(3) &= 53 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -x^2 - 5x + 4 \\ f(4) &= -32 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 3x^2 + 5x + 4 \\ f(-3) &= 16 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -x^2 - 4x + 1 \\ f(3) &= -20 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 5x^2 - 5x + 3 \\ f(-1) &= 13 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 2x^2 - 2x + 1 \\ f(-3) &= 25 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 5x^2 + 2x + 4 \\ f(2) &= 28 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 3x^2 + 3x - 1 \\ f(3) &= 35 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -4x^2 - 3x - 4 \\ f(-3) &= -31 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -4x^2 - 5x + 5 \\ f(-5) &= -70 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -x^2 + 5x + 2 \\ f(-3) &= -22 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= x^2 + 3x + 4 \\ f(-5) &= 14 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 2x^2 + 3x - 2 \\ f(-5) &= 33 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= x^2 - x + 1 \\ f(-3) &= 13 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 2x^2 + 2x + 3 \\ f(4) &= 43 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 5x^2 + 4x - 4 \\ f(5) &= 141 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 5x^2 - 2x + 5 \\ f(4) &= 77 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -x^2 - x - 1 \\ f(4) &= -21 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -5x^2 + 4x + 1 \\ f(5) &= -104 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -5x^2 + 3x + 5 \\ f(5) &= -105 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= 3x^2 + 4x - 1 \\ f(4) &= 63 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 + 4x - 5 \\ f(-2) &= -1 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -2x^2 + x + 1 \\ f(1) &= 0 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -2x^2 + 4x + 3 \\ f(-3) &= -27 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 3x^2 - x + 5 \\ f(2) &= 15 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= 5x^2 - 3x - 3 \\ f(-2) &= 23 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -4x^2 + 3x + 5 \\ f(-5) &= -110 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= x^2 - 3x - 3 \\ f(-5) &= 37 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -x^2 + 4x + 4 \\ f(2) &= 8 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -5x^2 + 5x + 3 \\ f(5) &= -97 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= x^2 + x - 1 \\ f(-2) &= 1 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -x^2 + 5x - 5 \\ f(5) &= -5 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -x^2 - 3x - 4 \\ f(-5) &= -14 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -4x^2 - x - 1 \\ f(3) &= -40 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 3x^2 + 2x - 2 \\ f(-5) &= 63 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 4x^2 - 2x + 4 \\ f(-4) &= 76 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -2x^2 + 2x - 1 \\ f(-4) &= -41 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -2x^2 + 5x + 4 \\ f(4) &= -8 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -5x^2 + 5x - 4 \\ f(-2) &= -34 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -x^2 - 3x - 1 \\ f(-3) &= -1 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -2x^2 - 2x + 1 \\ f(1) &= -3 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 3x^2 - 2x + 4 \\ f(5) &= 69 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -3x^2 - 4x - 3 \\ f(-4) &= -35 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -5x^2 + 3x - 3 \\ f(1) &= -5 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 2x^2 + 4x + 2 \\ f(1) &= 8 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(-4) &= -77 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -3x^2 + 4x + 2 \\ f(5) &= -53 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -4x^2 - x + 5 \\ f(-3) &= -28 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -5x^2 + 2x + 4 \\ f(1) &= 1 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= -2x^2 - 4x + 3 \\ f(-1) &= 5 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -4x^2 - 3x - 1 \\ f(-2) &= -11 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 2x^2 + 2x - 1 \\ f(4) &= 39 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 2x^2 + 2x - 3 \\ f(3) &= 21 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -3x^2 + 3x - 1 \\ f(-3) &= -37 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= 5x^2 + 2x - 1 \\ f(-2) &= 15 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -2x^2 + x + 2 \\ f(-3) &= -19 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -4x^2 + x - 4 \\ f(-4) &= -72 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3x^2 - 2x + 5 \\ f(2) &= -11 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 4x^2 + 4x - 3 \\ f(2) &= 21 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 3x^2 + 4x + 4 \\ f(-1) &= 3 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -5x^2 + x + 4 \\ f(-2) &= -18 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -5x^2 - 3x - 1 \\ f(-4) &= -69 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -4x^2 - 3x - 2 \\ f(3) &= -47 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4x^2 + 4x - 5 \\ f(-1) &= -13 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -3x^2 + 4x - 4 \\ f(2) &= -8 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -4x^2 + 2x + 4 \\ f(-4) &= -68 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -4x^2 + x - 4 \\ f(-1) &= -9 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -x^2 + 4x + 4 \\ f(-2) &= -8 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -2x^2 - 4x + 5 \\ f(-5) &= -25 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -4x^2 - 5x + 2 \\ f(4) &= -82 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -3x^2 + 2x + 1 \\ f(5) &= -64 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -4x^2 + 5x + 1 \\ f(3) &= -20 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -x^2 + 4x + 3 \\ f(-2) &= -9 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -x^2 + 5x + 5 \\ f(-5) &= -45 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -2x^2 - x - 5 \\ f(2) &= -15 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -2x^2 - 2x + 5 \\ f(-2) &= 1 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -3x^2 - 4x - 5 \\ f(-5) &= -60 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= x^2 - x - 5 \\ f(-4) &= 15 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 3x^2 - 3x - 1 \\ f(4) &= 35 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 5x^2 - 5x - 5 \\ f(1) &= -5 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 3x^2 - 5x - 3 \\ f(4) &= 25 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -x^2 - 4x + 5 \\ f(1) &= 0 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= x^2 + x + 4 \\ f(-1) &= 4 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -5x^2 + 3x + 4 \\ f(4) &= -64 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -3x^2 + 3x - 1 \\ f(-1) &= -7 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= x^2 - 3x - 3 \\ f(2) &= -5 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 5x^2 - 3x - 3 \\ f(-5) &= 137 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -3x^2 - 4x + 1 \\ f(-2) &= -3 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= x^2 - 2x + 2 \\ f(-1) &= 5 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -2x^2 - x - 2 \\ f(1) &= -5 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= 5x^2 - x - 4 \\ f(-2) &= 18 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -x^2 + 3x - 4 \\ f(2) &= -2 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -x^2 - x - 5 \\ f(5) &= -35 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= 3x^2 - 4x + 5 \\ f(4) &= 37 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 5x^2 - 2x - 3 \\ f(-5) &= 132 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -x^2 - 4x - 2 \\ f(-2) &= 2 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 5x^2 + x + 5 \\ f(5) &= 135 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -3x^2 + 3x + 5 \\ f(4) &= -31 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= x^2 - 3x + 3 \\ f(-1) &= 7 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -5x^2 + x + 4 \\ f(2) &= -14 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -4x^2 - 5x - 2 \\ f(-3) &= -23 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -5x^2 - 4x - 4 \\ f(-5) &= -109 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -4x^2 - 4x - 5 \\ f(1) &= -13 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= -2x^2 - x - 5 \\ f(5) &= -60 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -4x^2 - 4x + 1 \\ f(-4) &= -47 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 4x^2 - 3x - 2 \\ f(-2) &= 20 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -5x^2 - 3x + 3 \\ f(-1) &= 1 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -3x^2 + 3x + 4 \\ f(4) &= -32 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 5x^2 - 5x + 4 \\ f(-5) &= 154 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= x^2 - 5x - 2 \\ f(-4) &= 34 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= x^2 - x - 3 \\ f(4) &= 9 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5x^2 + 4x + 4 \\ f(4) &= -60 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -2x^2 - 3x - 5 \\ f(5) &= -70 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -2x^2 + 2x - 5 \\ f(-2) &= -17 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -x^2 + 5x - 5 \\ f(1) &= -1 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -x^2 - 5x + 4 \\ f(-5) &= 4 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= x^2 - 3x + 1 \\ f(4) &= 5 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 4x^2 + 3x - 5 \\ f(4) &= 71 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -x^2 - x + 3 \\ f(2) &= -3 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -2x^2 + x - 5 \\ f(3) &= -20 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -x^2 - 3x + 2 \\ f(3) &= -16 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= 5x^2 + x + 4 \\ f(1) &= 10 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 4x^2 + 5x + 4 \\ f(4) &= 88 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2x^2 - 2x - 2 \\ f(-2) &= -6 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -3x^2 - 5x + 5 \\ f(1) &= -3 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 3x^2 + 2x - 1 \\ f(1) &= 4 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -3x^2 + 4x - 4 \\ f(5) &= -59 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -5x^2 - 5x - 5 \\ f(2) &= -35 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -x^2 + 2x + 2 \\ f(-4) &= -22 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -5x^2 - 2x - 5 \\ f(-2) &= -21 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -4x^2 + 2x + 1 \\ f(1) &= -1 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 3x^2 - 2x - 4 \\ f(-4) &= 52 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -5x^2 + 4x - 3 \\ f(3) &= -36 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -x^2 + 5x + 3 \\ f(2) &= 9 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 2x^2 - 2x + 3 \\ f(5) &= 43 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= 2x^2 + 4x + 5 \\ f(4) &= 53 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -4x^2 + 5x + 1 \\ f(-4) &= -83 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -x^2 - x - 4 \\ f(1) &= -6 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= 2x^2 + 2x + 3 \\ f(-5) &= 43 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 4x^2 + 4x + 1 \\ f(-3) &= 25 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 2x^2 + 5x - 1 \\ f(-2) &= -3 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= x^2 - 4x + 2 \\ f(1) &= -1 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 5x^2 - 2x + 4 \\ f(5) &= 119 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= 5x^2 - x + 4 \\ f(-4) &= 88 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= x^2 - 4x + 1 \\ f(2) &= -3 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -5x^2 - 3x - 4 \\ f(5) &= -144 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -5x^2 + 2x - 2 \\ f(-1) &= -9 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -x^2 + 2x + 3 \\ f(-3) &= -12 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= -3x^2 + x + 5 \\ f(-2) &= -9 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 4x^2 - 2x - 4 \\ f(5) &= 86 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -4x^2 + 3x + 4 \\ f(-3) &= -41 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -2x^2 - 3x + 4 \\ f(5) &= -61 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -4x^2 + 5x - 3 \\ f(-4) &= -87 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4x^2 - 4x + 2 \\ f(3) &= -46 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -4x^2 - 2x - 5 \\ f(-3) &= -35 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -3x^2 + 5x + 1 \\ f(-4) &= -67 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 2x^2 + x + 1 \\ f(-3) &= 16 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -3x^2 + x + 5 \\ f(1) &= 3 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -3x^2 + 2x + 1 \\ f(-3) &= -32 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 4x^2 + x - 2 \\ f(1) &= 3 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -5x^2 - 5x - 1 \\ f(-2) &= -11 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 5x^2 - 3x + 2 \\ f(-4) &= 94 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 4x^2 + 2x - 1 \\ f(-4) &= 55 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -3x^2 - x + 3 \\ f(-1) &= 1 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -2x^2 - 4x + 2 \\ f(2) &= -14 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -2x^2 - 3x - 3 \\ f(-4) &= -23 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -5x^2 + 2x + 4 \\ f(-5) &= -131 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -4x^2 - 4x + 4 \\ f(-2) &= -4 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2x^2 + x - 5 \\ f(1) &= -6 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 2x^2 - x - 2 \\ f(3) &= 13 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 3x^2 - x + 5 \\ f(1) &= 7 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4x^2 - 3x - 3 \\ f(-5) &= -88 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= x^2 + 5x + 4 \\ f(-2) &= -2 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= x^2 - x + 4 \\ f(-4) &= 24 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 4x^2 + 4x + 1 \\ f(-2) &= 9 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 5x^2 - 4x + 3 \\ f(2) &= 15 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 5x^2 - 2x - 2 \\ f(-3) &= 49 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -5x^2 + 3x - 1 \\ f(-2) &= -27 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -4x^2 - 3x - 2 \\ f(-2) &= -12 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -2x^2 - 3x - 2 \\ f(4) &= -46 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -5x^2 - x - 4 \\ f(4) &= -88 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -5x^2 - 4x - 1 \\ f(2) &= -29 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -4x^2 - 5x - 5 \\ f(-4) &= -49 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -x^2 + x - 1 \\ f(3) &= -7 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= -x^2 + 3x + 4 \\ f(-4) &= -24 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 3x^2 + 5x - 4 \\ f(5) &= 96 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 4x^2 + 4x - 5 \\ f(-3) &= 19 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 3x^2 + x + 4 \\ f(5) &= 84 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= 4x^2 - 5x + 2 \\ f(-1) &= 11 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -3x^2 - 4x + 5 \\ f(-3) &= -10 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -2x^2 - x + 4 \\ f(5) &= -51 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -4x^2 + 2x - 2 \\ f(-4) &= -74 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3x^2 - 5x + 2 \\ f(3) &= -40 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= -4x^2 + 2x + 2 \\ f(2) &= -10 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 4x^2 + 3x - 4 \\ f(-4) &= 48 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -x^2 - 4x + 2 \\ f(-4) &= 2 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -5x^2 + x - 1 \\ f(1) &= -5 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 5x^2 + 5x + 2 \\ f(-4) &= 62 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4x^2 + 2x - 4 \\ f(4) &= -60 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -4x^2 - 5x - 4 \\ f(-1) &= -3 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= 4x^2 + x + 4 \\ f(-3) &= 37 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 - x - 3 \\ f(-1) &= 1 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 3x^2 - 4x + 5 \\ f(2) &= 9 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -2x^2 + 4x - 3 \\ f(-5) &= -73 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 4x^2 - 3x - 2 \\ f(-4) &= 74 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -4x^2 + 3x + 2 \\ f(2) &= -8 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -x^2 + 4x - 2 \\ f(2) &= 2 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -3x^2 + 3x - 2 \\ f(1) &= -2 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -5x^2 + 5x + 2 \\ f(-1) &= -8 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= x^2 - 5x + 4 \\ f(-2) &= 18 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 4x^2 - 5x + 5 \\ f(3) &= 26 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -5x^2 + 3x + 4 \\ f(-1) &= -4 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 5x^2 + 3x + 5 \\ f(-2) &= 19 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 4x^2 - 5x - 5 \\ f(2) &= 1 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 5x^2 - 3x + 1 \\ f(-2) &= 27 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= -3x^2 + 5x - 3 \\ f(2) &= -5 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -2x^2 + x - 3 \\ f(-4) &= -39 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 3x^2 - 4x - 3 \\ f(-5) &= 92 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -4x^2 - 4x - 5 \\ f(1) &= -13 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 5x^2 + x - 4 \\ f(2) &= 18 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= x^2 - 2x + 4 \\ f(5) &= 19 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= -3x^2 + 2x - 5 \\ f(4) &= -45 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -4x^2 - x + 3 \\ f(3) &= -36 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -4x^2 - x + 2 \\ f(2) &= -16 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 3x^2 - 2x - 2 \\ f(-5) &= 83 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -x^2 + 3x + 2 \\ f(1) &= 4 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -3x^2 + 2x + 3 \\ f(1) &= 2 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 3x^2 - 2x + 5 \\ f(-1) &= 10 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= 4x^2 + 4x - 4 \\ f(-2) &= 4 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 3x^2 - x + 2 \\ f(-5) &= 82 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 4x^2 + 4x + 5 \\ f(-4) &= 53 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -2x^2 + 4x - 4 \\ f(-4) &= -52 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 4x^2 - 5x + 3 \\ f(2) &= 9 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -4x^2 + 4x - 2 \\ f(4) &= -50 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -2x^2 - 3x - 4 \\ f(-2) &= -6 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 2x^2 + x + 2 \\ f(-3) &= 17 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 4x^2 + 5x + 2 \\ f(-4) &= 46 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 5x^2 - 2x - 3 \\ f(-3) &= 48 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 3x^2 + 2x + 1 \\ f(-4) &= 41 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= x^2 - 2x - 3 \\ f(-1) &= 0 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -2x^2 + 3x - 1 \\ f(1) &= 0 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -4x^2 - 4x + 3 \\ f(-3) &= -21 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -5x^2 + 4x + 2 \\ f(-5) &= -143 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -3x^2 + 5x - 3 \\ f(-3) &= -45 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 5x^2 + 5x + 3 \\ f(-1) &= 3 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= 2x^2 + 2x - 3 \\ f(4) &= 37 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 4x^2 - x + 2 \\ f(-5) &= 107 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 2x^2 - 5x + 3 \\ f(-5) &= 78 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -x^2 - 5x - 2 \\ f(3) &= -26 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -5x^2 + 2x + 4 \\ f(-3) &= -47 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -4x^2 + 3x + 3 \\ f(-5) &= -112 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -3x^2 - 5x - 4 \\ f(1) &= -12 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3x^2 + 3x + 4 \\ f(4) &= 64 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -2x^2 - x - 3 \\ f(-4) &= -31 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -5x^2 - 3x + 1 \\ f(-1) &= -1 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 3x^2 - 2x + 2 \\ f(5) &= 67 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -x^2 - 2x + 4 \\ f(-4) &= -4 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -4x^2 - 3x + 5 \\ f(-4) &= -47 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2x^2 - x + 5 \\ f(5) &= -50 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -x^2 - 4x - 2 \\ f(-1) &= 1 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 4x^2 - 2x + 2 \\ f(2) &= 14 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 2x^2 - 3x + 5 \\ f(-1) &= 10 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -x^2 + x - 4 \\ f(3) &= -10 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -2x^2 + 2x + 5 \\ f(-4) &= -35 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 4x^2 - 5x - 5 \\ f(-5) &= 120 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -4x^2 - 2x - 3 \\ f(2) &= -23 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= -x^2 + x + 1 \\ f(2) &= -1 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 3x^2 + 4x + 5 \\ f(-1) &= 4 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= x^2 - 4x - 3 \\ f(-5) &= 42 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 4x^2 + 5x + 1 \\ f(-5) &= 76 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -3x^2 + 4x + 1 \\ f(-4) &= -63 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 2x^2 + 3x + 1 \\ f(-1) &= 0 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -2x^2 - 4x - 4 \\ f(1) &= -10 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -4x^2 - x - 2 \\ f(1) &= -7 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 4x^2 - 2x - 4 \\ f(-1) &= 2 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 3x^2 - 2x + 4 \\ f(-4) &= 60 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -3x^2 + 2x + 3 \\ f(-2) &= -13 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 3x^2 - 5x + 1 \\ f(3) &= 13 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= 2x^2 + 2x + 5 \\ f(5) &= 65 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -x^2 + x - 4 \\ f(-2) &= -10 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 3x^2 + 2x + 2 \\ f(-4) &= 42 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 4x^2 + x - 2 \\ f(-4) &= 58 \end{aligned}$$

$$(1) \quad f(x) = x^2 + 3x + 3$$

$$f(-3) = 3$$

$$(16) \quad f(x) = 4x^2 + 5x - 2$$

$$f(1) = 7$$

$$(31) \quad f(x) = 2x^2 - 2x - 5$$

$$f(4) = 19$$

$$(2) \quad f(x) = 5x^2 - 2x - 4$$

$$f(-3) = 47$$

$$(17) \quad f(x) = 2x^2 + 4x + 2$$

$$f(-4) = 18$$

$$(32) \quad f(x) = -4x^2 - x - 2$$

$$f(5) = -107$$

$$(3) \quad f(x) = -2x^2 - 3x - 4$$

$$f(1) = -9$$

$$(18) \quad f(x) = -2x^2 + x - 5$$

$$f(4) = -33$$

$$(33) \quad f(x) = -2x^2 + 4x + 3$$

$$f(-4) = -45$$

$$(4) \quad f(x) = 5x^2 + 2x - 4$$

$$f(-4) = 68$$

$$(19) \quad f(x) = 3x^2 + 3x - 1$$

$$f(5) = 89$$

$$(34) \quad f(x) = -x^2 + 2x + 3$$

$$f(-5) = -32$$

$$(5) \quad f(x) = -3x^2 + 3x - 3$$

$$f(-1) = -9$$

$$(20) \quad f(x) = -2x^2 + 4x + 1$$

$$f(1) = 3$$

$$(35) \quad f(x) = -4x^2 + 4x + 1$$

$$f(-3) = -47$$

$$(6) \quad f(x) = -5x^2 + 3x + 2$$

$$f(1) = 0$$

$$(21) \quad f(x) = -3x^2 + 3x - 3$$

$$f(2) = -9$$

$$(36) \quad f(x) = -2x^2 + 4x - 5$$

$$f(-4) = -53$$

$$(7) \quad f(x) = -2x^2 + 3x - 5$$

$$f(4) = -25$$

$$(22) \quad f(x) = 5x^2 - x - 4$$

$$f(5) = 116$$

$$(37) \quad f(x) = 5x^2 + x + 5$$

$$f(-2) = 23$$

$$(8) \quad f(x) = 5x^2 + 5x - 1$$

$$f(4) = 99$$

$$(23) \quad f(x) = -3x^2 + 5x + 4$$

$$f(-5) = -96$$

$$(38) \quad f(x) = -2x^2 + 4x + 4$$

$$f(-5) = -66$$

$$(9) \quad f(x) = -x^2 - x - 2$$

$$f(-5) = -22$$

$$(24) \quad f(x) = 5x^2 - 2x - 4$$

$$f(5) = 111$$

$$(39) \quad f(x) = -3x^2 + 5x - 4$$

$$f(-5) = -104$$

$$(10) \quad f(x) = -x^2 - x + 5$$

$$f(-5) = -15$$

$$(25) \quad f(x) = 5x^2 + 3x + 5$$

$$f(3) = 59$$

$$(40) \quad f(x) = x^2 - 2x + 3$$

$$f(1) = 2$$

$$(11) \quad f(x) = 5x^2 - 3x + 4$$

$$f(-5) = 144$$

$$(26) \quad f(x) = -5x^2 - 3x - 5$$

$$f(1) = -13$$

$$(41) \quad f(x) = -5x^2 - 5x + 2$$

$$f(-4) = -58$$

$$(12) \quad f(x) = 3x^2 - 3x + 1$$

$$f(-5) = 91$$

$$(27) \quad f(x) = -x^2 + 3x - 5$$

$$f(5) = -15$$

$$(42) \quad f(x) = 2x^2 - 4x + 2$$

$$f(-5) = 72$$

$$(13) \quad f(x) = 3x^2 - 5x + 4$$

$$f(-1) = 12$$

$$(28) \quad f(x) = -x^2 - x + 2$$

$$f(4) = -18$$

$$(43) \quad f(x) = -3x^2 + 5x + 2$$

$$f(4) = -26$$

$$(14) \quad f(x) = -3x^2 - 3x + 5$$

$$f(3) = -31$$

$$(29) \quad f(x) = -5x^2 + 2x - 5$$

$$f(-4) = -93$$

$$(44) \quad f(x) = 5x^2 + 5x + 5$$

$$f(4) = 105$$

$$(15) \quad f(x) = 3x^2 - 2x - 4$$

$$f(-5) = 81$$

$$(30) \quad f(x) = x^2 + 4x - 5$$

$$f(3) = 16$$

$$(45) \quad f(x) = -x^2 + 2x + 1$$

$$f(2) = 1$$

$$(1) \quad f(x) = -x^2 + 3x + 1 \\ f(-3) = -17$$

$$(16) \quad f(x) = -3x^2 + x - 2 \\ f(4) = -46$$

$$(31) \quad f(x) = -2x^2 - 3x + 5 \\ f(-5) = -30$$

$$(2) \quad f(x) = 2x^2 + x - 4 \\ f(-4) = 24$$

$$(17) \quad f(x) = 2x^2 + 5x - 5 \\ f(2) = 13$$

$$(32) \quad f(x) = -x^2 - 4x - 2 \\ f(-4) = -2$$

$$(3) \quad f(x) = -4x^2 + 5x + 5 \\ f(2) = -1$$

$$(18) \quad f(x) = 5x^2 + 2x - 5 \\ f(-5) = 110$$

$$(33) \quad f(x) = x^2 - 4x - 2 \\ f(1) = -5$$

$$(4) \quad f(x) = -5x^2 - x + 3 \\ f(5) = -127$$

$$(19) \quad f(x) = -2x^2 - 3x - 2 \\ f(4) = -46$$

$$(34) \quad f(x) = -3x^2 - 3x + 3 \\ f(4) = -57$$

$$(5) \quad f(x) = 3x^2 - 3x - 4 \\ f(-3) = 32$$

$$(20) \quad f(x) = -2x^2 + x - 1 \\ f(3) = -16$$

$$(35) \quad f(x) = -x^2 + 4x + 4 \\ f(-3) = -17$$

$$(6) \quad f(x) = 5x^2 - 4x + 3 \\ f(-5) = 148$$

$$(21) \quad f(x) = -3x^2 - 2x + 5 \\ f(-1) = 4$$

$$(36) \quad f(x) = -5x^2 - x - 5 \\ f(-4) = -81$$

$$(7) \quad f(x) = 3x^2 - x + 1 \\ f(2) = 11$$

$$(22) \quad f(x) = 5x^2 - 2x - 1 \\ f(-2) = 23$$

$$(37) \quad f(x) = -2x^2 + 3x - 1 \\ f(-5) = -66$$

$$(8) \quad f(x) = -2x^2 + 4x - 2 \\ f(1) = 0$$

$$(23) \quad f(x) = -4x^2 - 4x + 2 \\ f(4) = -78$$

$$(38) \quad f(x) = 5x^2 + x - 3 \\ f(5) = 127$$

$$(9) \quad f(x) = 4x^2 - x + 4 \\ f(-2) = 22$$

$$(24) \quad f(x) = 2x^2 - 2x - 4 \\ f(-1) = 0$$

$$(39) \quad f(x) = -4x^2 - 2x - 1 \\ f(1) = -7$$

$$(10) \quad f(x) = -3x^2 + 4x + 2 \\ f(1) = 3$$

$$(25) \quad f(x) = -4x^2 - 4x + 3 \\ f(-2) = -5$$

$$(40) \quad f(x) = -5x^2 + 5x + 2 \\ f(-5) = -148$$

$$(11) \quad f(x) = -2x^2 - 4x + 1 \\ f(-3) = -5$$

$$(26) \quad f(x) = -2x^2 + 2x + 2 \\ f(1) = 2$$

$$(41) \quad f(x) = 2x^2 - 2x - 3 \\ f(-1) = 1$$

$$(12) \quad f(x) = 2x^2 + 4x - 1 \\ f(3) = 29$$

$$(27) \quad f(x) = 4x^2 + 5x + 5 \\ f(5) = 130$$

$$(42) \quad f(x) = -3x^2 + x + 3 \\ f(-3) = -27$$

$$(13) \quad f(x) = -2x^2 - 4x + 5 \\ f(-3) = -1$$

$$(28) \quad f(x) = x^2 - 3x - 3 \\ f(2) = -5$$

$$(43) \quad f(x) = x^2 - 5x - 1 \\ f(-4) = 35$$

$$(14) \quad f(x) = -x^2 - 4x - 1 \\ f(2) = -13$$

$$(29) \quad f(x) = 3x^2 - 3x - 4 \\ f(-5) = 86$$

$$(44) \quad f(x) = -x^2 - 2x - 1 \\ f(5) = -36$$

$$(15) \quad f(x) = 5x^2 - x - 5 \\ f(-4) = 79$$

$$(30) \quad f(x) = 2x^2 - x - 5 \\ f(2) = 1$$

$$(45) \quad f(x) = 5x^2 + x + 3 \\ f(-2) = 21$$

2.7 代入 (難しい)

平方完成の式に整数を代入する問題。

$$(1) \begin{array}{l} f(x) = (x+2)^2 - 9 \\ f(4) \end{array}$$

$$(6) \begin{array}{l} f(x) = -5\left(x + \frac{1}{2}\right)^2 + \frac{9}{4} \\ f(-3) \end{array}$$

$$(2) \begin{array}{l} f(x) = -2\left(x - \frac{5}{4}\right)^2 - \frac{15}{8} \\ f(-5) \end{array}$$

$$(7) \begin{array}{l} f(x) = -3\left(x - \frac{2}{3}\right)^2 + \frac{1}{3} \\ f(3) \end{array}$$

$$(3) \begin{array}{l} f(x) = -5\left(x - \frac{1}{5}\right)^2 - \frac{9}{5} \\ f(-2) \end{array}$$

$$(8) \begin{array}{l} f(x) = (x+2)^2 - 1 \\ f(1) \end{array}$$

$$(4) \begin{array}{l} f(x) = 4\left(x - \frac{3}{8}\right)^2 - \frac{9}{16} \\ f(-3) \end{array}$$

$$(9) \begin{array}{l} f(x) = 3\left(x + \frac{1}{2}\right)^2 + \frac{13}{4} \\ f(4) \end{array}$$

$$(5) \begin{array}{l} f(x) = -5\left(x + \frac{1}{2}\right)^2 - \frac{15}{4} \\ f(5) \end{array}$$

$$(10) \begin{array}{l} f(x) = -2\left(x - \frac{3}{4}\right)^2 + \frac{9}{8} \\ f(3) \end{array}$$

$$\begin{array}{l} (1) \ f(x) = \left(x - \frac{3}{2}\right)^2 + \frac{3}{4} \\ \quad f(2) \end{array}$$

$$\begin{array}{l} (6) \ f(x) = \left(x + \frac{1}{2}\right)^2 - \frac{21}{4} \\ \quad f(-2) \end{array}$$

$$\begin{array}{l} (2) \ f(x) = 4\left(x + \frac{1}{2}\right)^2 + 4 \\ \quad f(4) \end{array}$$

$$\begin{array}{l} (7) \ f(x) = \left(x - \frac{1}{2}\right)^2 - \frac{17}{4} \\ \quad f(-1) \end{array}$$

$$\begin{array}{l} (3) \ f(x) = -3\left(x + \frac{2}{3}\right)^2 + \frac{4}{3} \\ \quad f(-2) \end{array}$$

$$\begin{array}{l} (8) \ f(x) = -(x + 2)^2 + 7 \\ \quad f(4) \end{array}$$

$$\begin{array}{l} (4) \ f(x) = -4\left(x - \frac{1}{2}\right)^2 + 6 \\ \quad f(-5) \end{array}$$

$$\begin{array}{l} (9) \ f(x) = 3\left(x - \frac{1}{3}\right)^2 + \frac{5}{3} \\ \quad f(0) \end{array}$$

$$\begin{array}{l} (5) \ f(x) = -3\left(x - \frac{1}{2}\right)^2 + \frac{11}{4} \\ \quad f(0) \end{array}$$

$$\begin{array}{l} (10) \ f(x) = 2(x - 1)^2 - 1 \\ \quad f(-5) \end{array}$$

$$(1) \begin{array}{l} f(x) = 3\left(x + \frac{1}{3}\right)^2 - \frac{10}{3} \\ f(-3) \end{array}$$

$$(6) \begin{array}{l} f(x) = -5\left(x + \frac{1}{10}\right)^2 - \frac{99}{20} \\ f(-2) \end{array}$$

$$(2) \begin{array}{l} f(x) = 5\left(x - \frac{1}{2}\right)^2 + \frac{7}{4} \\ f(-5) \end{array}$$

$$(7) \begin{array}{l} f(x) = (x - 1)^2 + 3 \\ f(-3) \end{array}$$

$$(3) \begin{array}{l} f(x) = -\left(x - \frac{5}{2}\right)^2 + \frac{37}{4} \\ f(1) \end{array}$$

$$(8) \begin{array}{l} f(x) = -2\left(x + \frac{3}{4}\right)^2 + \frac{9}{8} \\ f(1) \end{array}$$

$$(4) \begin{array}{l} f(x) = 2\left(x + \frac{1}{4}\right)^2 + \frac{23}{8} \\ f(3) \end{array}$$

$$(9) \begin{array}{l} f(x) = -4\left(x + \frac{1}{2}\right)^2 - 3 \\ f(2) \end{array}$$

$$(5) \begin{array}{l} f(x) = -2\left(x - \frac{5}{4}\right)^2 - \frac{7}{8} \\ f(2) \end{array}$$

$$(10) \begin{array}{l} f(x) = 3\left(x - \frac{2}{3}\right)^2 - \frac{10}{3} \\ f(4) \end{array}$$

$$(1) \begin{array}{l} f(x) = -4 \left(x + \frac{3}{8} \right)^2 - \frac{71}{16} \\ f(-3) \end{array}$$

$$(6) \begin{array}{l} f(x) = \left(x - \frac{3}{2} \right)^2 + \frac{11}{4} \\ f(-4) \end{array}$$

$$(2) \begin{array}{l} f(x) = 5 \left(x - \frac{2}{5} \right)^2 - \frac{19}{5} \\ f(-2) \end{array}$$

$$(7) \begin{array}{l} f(x) = 3 \left(x - \frac{5}{6} \right)^2 + \frac{23}{12} \\ f(5) \end{array}$$

$$(3) \begin{array}{l} f(x) = -3 \left(x - \frac{1}{3} \right)^2 + \frac{7}{3} \\ f(-1) \end{array}$$

$$(8) \begin{array}{l} f(x) = (x + 2)^2 + 1 \\ f(3) \end{array}$$

$$(4) \begin{array}{l} f(x) = -(x - 1)^2 \\ f(0) \end{array}$$

$$(9) \begin{array}{l} f(x) = -4 \left(x + \frac{1}{4} \right)^2 - \frac{19}{4} \\ f(5) \end{array}$$

$$(5) \begin{array}{l} f(x) = -4 \left(x - \frac{1}{2} \right)^2 + 6 \\ f(-4) \end{array}$$

$$(10) \begin{array}{l} f(x) = 2 \left(x - \frac{5}{4} \right)^2 - \frac{49}{8} \\ f(5) \end{array}$$

$$\begin{array}{l} (1) \ f(x) = -3 \left(x + \frac{5}{6} \right)^2 - \frac{23}{12} \\ \quad f(4) \end{array}$$

$$\begin{array}{l} (6) \ f(x) = 3 \left(x + \frac{2}{3} \right)^2 - \frac{1}{3} \\ \quad f(5) \end{array}$$

$$\begin{array}{l} (2) \ f(x) = -4 \left(x + \frac{1}{2} \right)^2 + 4 \\ \quad f(-1) \end{array}$$

$$\begin{array}{l} (7) \ f(x) = -3 \left(x + \frac{5}{6} \right)^2 + \frac{73}{12} \\ \quad f(0) \end{array}$$

$$\begin{array}{l} (3) \ f(x) = -2 \left(x - \frac{1}{2} \right)^2 - \frac{1}{2} \\ \quad f(-4) \end{array}$$

$$\begin{array}{l} (8) \ f(x) = 2(x-1)^2 + 1 \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (4) \ f(x) = \left(x - \frac{5}{2} \right)^2 - \frac{13}{4} \\ \quad f(3) \end{array}$$

$$\begin{array}{l} (9) \ f(x) = -5 \left(x + \frac{1}{2} \right)^2 + \frac{9}{4} \\ \quad f(-4) \end{array}$$

$$\begin{array}{l} (5) \ f(x) = 5 \left(x - \frac{2}{5} \right)^2 - \frac{14}{5} \\ \quad f(0) \end{array}$$

$$\begin{array}{l} (10) \ f(x) = -5 \left(x + \frac{2}{5} \right)^2 - \frac{11}{5} \\ \quad f(0) \end{array}$$

$$(1) \begin{array}{l} f(x) = -3 \left(x + \frac{1}{6} \right)^2 - \frac{23}{12} \\ f(-1) \end{array}$$

$$(6) \begin{array}{l} f(x) = 4 \left(x + \frac{1}{8} \right)^2 + \frac{47}{16} \\ f(4) \end{array}$$

$$(2) \begin{array}{l} f(x) = - \left(x - \frac{5}{2} \right)^2 + \frac{9}{4} \\ f(5) \end{array}$$

$$(7) \begin{array}{l} f(x) = -5 \left(x - \frac{3}{10} \right)^2 + \frac{69}{20} \\ f(-3) \end{array}$$

$$(3) \begin{array}{l} f(x) = 2(x-1)^2 - 7 \\ f(-3) \end{array}$$

$$(8) \begin{array}{l} f(x) = 5 \left(x - \frac{2}{5} \right)^2 - \frac{19}{5} \\ f(-3) \end{array}$$

$$(4) \begin{array}{l} f(x) = 4 \left(x - \frac{1}{2} \right)^2 + 4 \\ f(0) \end{array}$$

$$(9) \begin{array}{l} f(x) = -5 \left(x - \frac{1}{10} \right)^2 - \frac{39}{20} \\ f(-3) \end{array}$$

$$(5) \begin{array}{l} f(x) = -3 \left(x - \frac{1}{6} \right)^2 - \frac{35}{12} \\ f(2) \end{array}$$

$$(10) \begin{array}{l} f(x) = 5 \left(x + \frac{1}{2} \right)^2 - \frac{9}{4} \\ f(4) \end{array}$$

$$(1) \begin{array}{l} f(x) = -3 \left(x - \frac{1}{2} \right)^2 + \frac{3}{4} \\ f(-2) \end{array}$$

$$(6) \begin{array}{l} f(x) = 5 \left(x + \frac{1}{10} \right)^2 + \frac{59}{20} \\ f(5) \end{array}$$

$$(2) \begin{array}{l} f(x) = 4 \left(x + \frac{3}{8} \right)^2 - \frac{89}{16} \\ f(4) \end{array}$$

$$(7) \begin{array}{l} f(x) = 5 \left(x + \frac{1}{5} \right)^2 + \frac{14}{5} \\ f(2) \end{array}$$

$$(3) \begin{array}{l} f(x) = -4 \left(x - \frac{1}{8} \right)^2 + \frac{1}{16} \\ f(1) \end{array}$$

$$(8) \begin{array}{l} f(x) = 4 \left(x + \frac{1}{4} \right)^2 + \frac{3}{4} \\ f(2) \end{array}$$

$$(4) \begin{array}{l} f(x) = 5 \left(x + \frac{1}{2} \right)^2 + \frac{15}{4} \\ f(-3) \end{array}$$

$$(9) \begin{array}{l} f(x) = -2 \left(x + \frac{1}{2} \right)^2 + \frac{7}{2} \\ f(3) \end{array}$$

$$(5) \begin{array}{l} f(x) = 5 \left(x + \frac{1}{10} \right)^2 - \frac{61}{20} \\ f(-3) \end{array}$$

$$(10) \begin{array}{l} f(x) = -5 \left(x + \frac{2}{5} \right)^2 + \frac{4}{5} \\ f(-3) \end{array}$$

$$(1) \quad f(x) = (x-1)^2 + 2 \\ f(5)$$

$$(6) \quad f(x) = -(x+1)^2 + 1 \\ f(0)$$

$$(2) \quad f(x) = -3\left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \\ f(-1)$$

$$(7) \quad f(x) = -5\left(x - \frac{2}{5}\right)^2 - \frac{1}{5} \\ f(4)$$

$$(3) \quad f(x) = 3\left(x + \frac{2}{3}\right)^2 + \frac{8}{3} \\ f(0)$$

$$(8) \quad f(x) = 4\left(x + \frac{1}{2}\right)^2 \\ f(-4)$$

$$(4) \quad f(x) = 3\left(x - \frac{2}{3}\right)^2 - \frac{10}{3} \\ f(-3)$$

$$(9) \quad f(x) = -3\left(x - \frac{1}{6}\right)^2 + \frac{49}{12} \\ f(-2)$$

$$(5) \quad f(x) = -4\left(x - \frac{1}{2}\right)^2 + 4 \\ f(-4)$$

$$(10) \quad f(x) = 2\left(x + \frac{1}{2}\right)^2 - \frac{5}{2} \\ f(3)$$

$$\begin{array}{l} (1) \ f(x) = 2 \left(x + \frac{1}{4} \right)^2 - \frac{41}{8} \\ f(4) \end{array}$$

$$\begin{array}{l} (6) \ f(x) = -3 \left(x - \frac{5}{6} \right)^2 - \frac{11}{12} \\ f(-1) \end{array}$$

$$\begin{array}{l} (2) \ f(x) = 3 \left(x + \frac{2}{3} \right)^2 + \frac{8}{3} \\ f(4) \end{array}$$

$$\begin{array}{l} (7) \ f(x) = -2 \left(x + \frac{5}{4} \right)^2 + \frac{25}{8} \\ f(-2) \end{array}$$

$$\begin{array}{l} (3) \ f(x) = -5 \left(x + \frac{2}{5} \right)^2 + \frac{14}{5} \\ f(4) \end{array}$$

$$\begin{array}{l} (8) \ f(x) = -4 \left(x + \frac{1}{8} \right)^2 - \frac{63}{16} \\ f(2) \end{array}$$

$$\begin{array}{l} (4) \ f(x) = (x - 1)^2 + 1 \\ f(-5) \end{array}$$

$$\begin{array}{l} (9) \ f(x) = -4 \left(x + \frac{1}{8} \right)^2 + \frac{17}{16} \\ f(-4) \end{array}$$

$$\begin{array}{l} (5) \ f(x) = 4 \left(x + \frac{1}{4} \right)^2 + \frac{3}{4} \\ f(1) \end{array}$$

$$\begin{array}{l} (10) \ f(x) = -4 \left(x + \frac{1}{2} \right)^2 - 3 \\ f(0) \end{array}$$

$$(1) \begin{array}{l} f(x) = -3 \left(x + \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(-1) \end{array}$$

$$(6) \begin{array}{l} f(x) = -4 \left(x - \frac{3}{8} \right)^2 - \frac{55}{16} \\ f(-2) \end{array}$$

$$(2) \begin{array}{l} f(x) = -3 \left(x + \frac{2}{3} \right)^2 + \frac{10}{3} \\ f(0) \end{array}$$

$$(7) \begin{array}{l} f(x) = (x - 2)^2 - 1 \\ f(-3) \end{array}$$

$$(3) \begin{array}{l} f(x) = 3 \left(x - \frac{1}{2} \right)^2 + \frac{1}{4} \\ f(-4) \end{array}$$

$$(8) \begin{array}{l} f(x) = \left(x - \frac{3}{2} \right)^2 - \frac{13}{4} \\ f(-1) \end{array}$$

$$(4) \begin{array}{l} f(x) = -5 \left(x - \frac{3}{10} \right)^2 + \frac{109}{20} \\ f(2) \end{array}$$

$$(9) \begin{array}{l} f(x) = 5 \left(x + \frac{1}{10} \right)^2 + \frac{59}{20} \\ f(3) \end{array}$$

$$(5) \begin{array}{l} f(x) = -3 \left(x + \frac{5}{6} \right)^2 + \frac{61}{12} \\ f(-1) \end{array}$$

$$(10) \begin{array}{l} f(x) = 5 \left(x + \frac{1}{2} \right)^2 - \frac{9}{4} \\ f(-4) \end{array}$$

$$(1) \begin{array}{l} f(x) = 4 \left(x + \frac{5}{8} \right)^2 - \frac{89}{16} \\ f(4) \end{array}$$

$$(6) \begin{array}{l} f(x) = 5 \left(x - \frac{1}{10} \right)^2 - \frac{61}{20} \\ f(-5) \end{array}$$

$$(2) \begin{array}{l} f(x) = -2 \left(x - \frac{5}{4} \right)^2 - \frac{7}{8} \\ f(-3) \end{array}$$

$$(7) \begin{array}{l} f(x) = -2 (x - 1)^2 - 2 \\ f(0) \end{array}$$

$$(3) \begin{array}{l} f(x) = -5 \left(x - \frac{1}{2} \right)^2 - \frac{3}{4} \\ f(5) \end{array}$$

$$(8) \begin{array}{l} f(x) = 3 \left(x + \frac{1}{6} \right)^2 + \frac{23}{12} \\ f(5) \end{array}$$

$$(4) \begin{array}{l} f(x) = 3 \left(x + \frac{1}{6} \right)^2 - \frac{25}{12} \\ f(-3) \end{array}$$

$$(9) \begin{array}{l} f(x) = -2 \left(x + \frac{5}{4} \right)^2 + \frac{65}{8} \\ f(-1) \end{array}$$

$$(5) \begin{array}{l} f(x) = 3 \left(x + \frac{2}{3} \right)^2 - \frac{4}{3} \\ f(5) \end{array}$$

$$(10) \begin{array}{l} f(x) = -5 \left(x + \frac{3}{10} \right)^2 + \frac{69}{20} \\ f(-2) \end{array}$$

$$\begin{array}{l} (1) \ f(x) = \left(x + \frac{3}{2}\right)^2 - \frac{9}{4} \\ \quad f(5) \end{array}$$

$$\begin{array}{l} (6) \ f(x) = -3\left(x - \frac{1}{3}\right)^2 - \frac{5}{3} \\ \quad f(1) \end{array}$$

$$\begin{array}{l} (2) \ f(x) = -2\left(x + \frac{3}{4}\right)^2 + \frac{33}{8} \\ \quad f(-4) \end{array}$$

$$\begin{array}{l} (7) \ f(x) = 5\left(x + \frac{1}{5}\right)^2 + \frac{4}{5} \\ \quad f(4) \end{array}$$

$$\begin{array}{l} (3) \ f(x) = -\left(x + \frac{3}{2}\right)^2 + \frac{1}{4} \\ \quad f(-4) \end{array}$$

$$\begin{array}{l} (8) \ f(x) = 2\left(x + \frac{1}{4}\right)^2 + \frac{31}{8} \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (4) \ f(x) = -5\left(x + \frac{2}{5}\right)^2 - \frac{16}{5} \\ \quad f(5) \end{array}$$

$$\begin{array}{l} (9) \ f(x) = -2(x + 1)^2 + 5 \\ \quad f(5) \end{array}$$

$$\begin{array}{l} (5) \ f(x) = -2\left(x - \frac{1}{2}\right)^2 + \frac{9}{2} \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (10) \ f(x) = -4\left(x + \frac{1}{4}\right)^2 - \frac{19}{4} \\ \quad f(2) \end{array}$$

$$(1) \begin{array}{l} f(x) = (x+2)^2 - 3 \\ f(2) \end{array}$$

$$(6) \begin{array}{l} f(x) = 3\left(x + \frac{1}{2}\right)^2 + \frac{9}{4} \\ f(1) \end{array}$$

$$(2) \begin{array}{l} f(x) = 3\left(x + \frac{5}{6}\right)^2 - \frac{37}{12} \\ f(5) \end{array}$$

$$(7) \begin{array}{l} f(x) = 2(x+1)^2 - 6 \\ f(3) \end{array}$$

$$(3) \begin{array}{l} f(x) = -2\left(x + \frac{5}{4}\right)^2 + \frac{65}{8} \\ f(-1) \end{array}$$

$$(8) \begin{array}{l} f(x) = -4\left(x - \frac{3}{8}\right)^2 + \frac{73}{16} \\ f(5) \end{array}$$

$$(4) \begin{array}{l} f(x) = 2\left(x + \frac{1}{2}\right)^2 - \frac{1}{2} \\ f(3) \end{array}$$

$$(9) \begin{array}{l} f(x) = -2\left(x + \frac{1}{4}\right)^2 - \frac{7}{8} \\ f(-4) \end{array}$$

$$(5) \begin{array}{l} f(x) = 3\left(x + \frac{5}{6}\right)^2 - \frac{85}{12} \\ f(-5) \end{array}$$

$$(10) \begin{array}{l} f(x) = 5\left(x - \frac{1}{2}\right)^2 - \frac{17}{4} \\ f(5) \end{array}$$

$$\begin{array}{l} (1) \ f(x) = 4\left(x + \frac{3}{8}\right)^2 - \frac{89}{16} \\ \quad f(1) \end{array}$$

$$\begin{array}{l} (6) \ f(x) = -3\left(x - \frac{5}{6}\right)^2 + \frac{73}{12} \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (2) \ f(x) = -(x + 1)^2 + 1 \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (7) \ f(x) = -3\left(x - \frac{5}{6}\right)^2 + \frac{25}{12} \\ \quad f(4) \end{array}$$

$$\begin{array}{l} (3) \ f(x) = -5\left(x - \frac{1}{5}\right)^2 + \frac{21}{5} \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (8) \ f(x) = 3\left(x + \frac{1}{3}\right)^2 + \frac{14}{3} \\ \quad f(4) \end{array}$$

$$\begin{array}{l} (4) \ f(x) = 2(x - 1)^2 - 4 \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (9) \ f(x) = -5\left(x - \frac{1}{5}\right)^2 - \frac{19}{5} \\ \quad f(5) \end{array}$$

$$\begin{array}{l} (5) \ f(x) = -2\left(x + \frac{1}{4}\right)^2 - \frac{7}{8} \\ \quad f(-2) \end{array}$$

$$\begin{array}{l} (10) \ f(x) = 4\left(x + \frac{1}{4}\right)^2 - \frac{9}{4} \\ \quad f(5) \end{array}$$

$$(1) \begin{array}{l} f(x) = \left(x - \frac{1}{2}\right)^2 + \frac{19}{4} \\ f(0) \end{array}$$

$$(6) \begin{array}{l} f(x) = -5\left(x - \frac{2}{5}\right)^2 + \frac{29}{5} \\ f(1) \end{array}$$

$$(2) \begin{array}{l} f(x) = \left(x + \frac{1}{2}\right)^2 - \frac{1}{4} \\ f(-5) \end{array}$$

$$(7) \begin{array}{l} f(x) = 3\left(x + \frac{5}{6}\right)^2 - \frac{25}{12} \\ f(-1) \end{array}$$

$$(3) \begin{array}{l} f(x) = 3\left(x + \frac{2}{3}\right)^2 + \frac{5}{3} \\ f(1) \end{array}$$

$$(8) \begin{array}{l} f(x) = 4\left(x + \frac{1}{4}\right)^2 - \frac{17}{4} \\ f(-4) \end{array}$$

$$(4) \begin{array}{l} f(x) = 4\left(x + \frac{1}{2}\right)^2 - 2 \\ f(-4) \end{array}$$

$$(9) \begin{array}{l} f(x) = -4\left(x - \frac{5}{8}\right)^2 + \frac{41}{16} \\ f(5) \end{array}$$

$$(5) \begin{array}{l} f(x) = 4\left(x - \frac{5}{8}\right)^2 - \frac{41}{16} \\ f(-2) \end{array}$$

$$(10) \begin{array}{l} f(x) = (x + 2)^2 \\ f(1) \end{array}$$

$$(1) \begin{array}{l} f(x) = -4 \left(x - \frac{1}{2} \right)^2 - 4 \\ f(0) \end{array}$$

$$(6) \begin{array}{l} f(x) = -2(x+1)^2 - 2 \\ f(1) \end{array}$$

$$(2) \begin{array}{l} f(x) = -4 \left(x - \frac{3}{8} \right)^2 - \frac{71}{16} \\ f(-2) \end{array}$$

$$(7) \begin{array}{l} f(x) = (x+1)^2 - 4 \\ f(3) \end{array}$$

$$(3) \begin{array}{l} f(x) = -2 \left(x - \frac{5}{4} \right)^2 - \frac{7}{8} \\ f(1) \end{array}$$

$$(8) \begin{array}{l} f(x) = -(x-2)^2 + 5 \\ f(1) \end{array}$$

$$(4) \begin{array}{l} f(x) = \left(x - \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(-5) \end{array}$$

$$(9) \begin{array}{l} f(x) = - \left(x - \frac{3}{2} \right)^2 + \frac{25}{4} \\ f(-2) \end{array}$$

$$(5) \begin{array}{l} f(x) = - \left(x - \frac{3}{2} \right)^2 - \frac{11}{4} \\ f(-2) \end{array}$$

$$(10) \begin{array}{l} f(x) = - \left(x - \frac{1}{2} \right)^2 - \frac{3}{4} \\ f(-1) \end{array}$$

$$(1) \begin{array}{l} f(x) = -5 \left(x + \frac{2}{5} \right)^2 + \frac{9}{5} \\ f(-4) \end{array}$$

$$(6) \begin{array}{l} f(x) = 3 \left(x + \frac{1}{2} \right)^2 - \frac{7}{4} \\ f(-4) \end{array}$$

$$(2) \begin{array}{l} f(x) = -3 \left(x + \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(5) \end{array}$$

$$(7) \begin{array}{l} f(x) = \left(x - \frac{3}{2} \right)^2 - \frac{25}{4} \\ f(0) \end{array}$$

$$(3) \begin{array}{l} f(x) = 5 \left(x - \frac{1}{2} \right)^2 - \frac{9}{4} \\ f(-2) \end{array}$$

$$(8) \begin{array}{l} f(x) = 3 \left(x + \frac{1}{3} \right)^2 + \frac{5}{3} \\ f(5) \end{array}$$

$$(4) \begin{array}{l} f(x) = -5 \left(x - \frac{1}{10} \right)^2 + \frac{41}{20} \\ f(2) \end{array}$$

$$(9) \begin{array}{l} f(x) = 5 \left(x - \frac{2}{5} \right)^2 + \frac{1}{5} \\ f(-5) \end{array}$$

$$(5) \begin{array}{l} f(x) = -2 \left(x - \frac{1}{2} \right)^2 + \frac{3}{2} \\ f(2) \end{array}$$

$$(10) \begin{array}{l} f(x) = 2 \left(x + \frac{1}{2} \right)^2 + \frac{1}{2} \\ f(3) \end{array}$$

$$(1) \begin{array}{l} f(x) = -\left(x + \frac{1}{2}\right)^2 - \frac{15}{4} \\ f(-2) \end{array}$$

$$(6) \begin{array}{l} f(x) = 3\left(x - \frac{5}{6}\right)^2 + \frac{23}{12} \\ f(-1) \end{array}$$

$$(2) \begin{array}{l} f(x) = (x - 1)^2 + 2 \\ f(1) \end{array}$$

$$(7) \begin{array}{l} f(x) = -3\left(x + \frac{5}{6}\right)^2 + \frac{25}{12} \\ f(-4) \end{array}$$

$$(3) \begin{array}{l} f(x) = 4\left(x + \frac{1}{4}\right)^2 + \frac{11}{4} \\ f(-4) \end{array}$$

$$(8) \begin{array}{l} f(x) = -4\left(x + \frac{5}{8}\right)^2 + \frac{25}{16} \\ f(2) \end{array}$$

$$(4) \begin{array}{l} f(x) = 2\left(x + \frac{5}{4}\right)^2 - \frac{25}{8} \\ f(-3) \end{array}$$

$$(9) \begin{array}{l} f(x) = -3\left(x - \frac{2}{3}\right)^2 + \frac{19}{3} \\ f(3) \end{array}$$

$$(5) \begin{array}{l} f(x) = -5\left(x - \frac{1}{5}\right)^2 + \frac{16}{5} \\ f(4) \end{array}$$

$$(10) \begin{array}{l} f(x) = -2(x + 1)^2 + 2 \\ f(-3) \end{array}$$

$$(1) \begin{aligned} f(x) &= -(x-1)^2 + 4 \\ f(0) \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -2(x+1)^2 + 4 \\ f(-1) \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 5\left(x + \frac{1}{2}\right)^2 - \frac{13}{4} \\ f(-4) \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 5\left(x - \frac{2}{5}\right)^2 - \frac{24}{5} \\ f(0) \end{aligned}$$

$$(3) \begin{aligned} f(x) &= \left(x - \frac{1}{2}\right)^2 + \frac{15}{4} \\ f(3) \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 4\left(x + \frac{5}{8}\right)^2 + \frac{39}{16} \\ f(-3) \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -3\left(x - \frac{1}{3}\right)^2 - \frac{11}{3} \\ f(0) \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 5\left(x - \frac{1}{5}\right)^2 + \frac{19}{5} \\ f(-5) \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -2\left(x + \frac{5}{4}\right)^2 - \frac{7}{8} \\ f(-3) \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -2\left(x + \frac{3}{4}\right)^2 - \frac{23}{8} \\ f(-4) \end{aligned}$$

$$(1) \begin{array}{l} f(x) = 3\left(x + \frac{5}{6}\right)^2 - \frac{13}{12} \\ f(-4) \end{array}$$

$$(6) \begin{array}{l} f(x) = -2\left(x - \frac{1}{2}\right)^2 + \frac{1}{2} \\ f(5) \end{array}$$

$$(2) \begin{array}{l} f(x) = 3\left(x - \frac{1}{3}\right)^2 + \frac{14}{3} \\ f(0) \end{array}$$

$$(7) \begin{array}{l} f(x) = 2\left(x - \frac{1}{4}\right)^2 + \frac{7}{8} \\ f(0) \end{array}$$

$$(3) \begin{array}{l} f(x) = 4\left(x + \frac{1}{4}\right)^2 - \frac{13}{4} \\ f(3) \end{array}$$

$$(8) \begin{array}{l} f(x) = 3\left(x - \frac{1}{2}\right)^2 - \frac{23}{4} \\ f(-2) \end{array}$$

$$(4) \begin{array}{l} f(x) = 2\left(x - \frac{5}{4}\right)^2 - \frac{57}{8} \\ f(-2) \end{array}$$

$$(9) \begin{array}{l} f(x) = \left(x - \frac{3}{2}\right)^2 - \frac{29}{4} \\ f(4) \end{array}$$

$$(5) \begin{array}{l} f(x) = 5\left(x - \frac{1}{2}\right)^2 + \frac{15}{4} \\ f(3) \end{array}$$

$$(10) \begin{array}{l} f(x) = 4\left(x - \frac{3}{8}\right)^2 - \frac{89}{16} \\ f(2) \end{array}$$

$$(1) \quad f(x) = (x+2)^2 - 9 \\ f(4) = 27$$

$$(6) \quad f(x) = -5 \left(x + \frac{1}{2} \right)^2 + \frac{9}{4} \\ f(-3) = -29$$

$$(2) \quad f(x) = -2 \left(x - \frac{5}{4} \right)^2 - \frac{15}{8} \\ f(-5) = -80$$

$$(7) \quad f(x) = -3 \left(x - \frac{2}{3} \right)^2 + \frac{1}{3} \\ f(3) = -16$$

$$(3) \quad f(x) = -5 \left(x - \frac{1}{5} \right)^2 - \frac{9}{5} \\ f(-2) = -26$$

$$(8) \quad f(x) = (x+2)^2 - 1 \\ f(1) = 8$$

$$(4) \quad f(x) = 4 \left(x - \frac{3}{8} \right)^2 - \frac{9}{16} \\ f(-3) = 45$$

$$(9) \quad f(x) = 3 \left(x + \frac{1}{2} \right)^2 + \frac{13}{4} \\ f(4) = 64$$

$$(5) \quad f(x) = -5 \left(x + \frac{1}{2} \right)^2 - \frac{15}{4} \\ f(5) = -155$$

$$(10) \quad f(x) = -2 \left(x - \frac{3}{4} \right)^2 + \frac{9}{8} \\ f(3) = -9$$

$$(1) \begin{aligned} f(x) &= \left(x - \frac{3}{2}\right)^2 + \frac{3}{4} \\ f(2) &= 1 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= \left(x + \frac{1}{2}\right)^2 - \frac{21}{4} \\ f(-2) &= -3 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 4\left(x + \frac{1}{2}\right)^2 + 4 \\ f(4) &= 85 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= \left(x - \frac{1}{2}\right)^2 - \frac{17}{4} \\ f(-1) &= -2 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -3\left(x + \frac{2}{3}\right)^2 + \frac{4}{3} \\ f(-2) &= -4 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -(x + 2)^2 + 7 \\ f(4) &= -29 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -4\left(x - \frac{1}{2}\right)^2 + 6 \\ f(-5) &= -115 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 3\left(x - \frac{1}{3}\right)^2 + \frac{5}{3} \\ f(0) &= 2 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -3\left(x - \frac{1}{2}\right)^2 + \frac{11}{4} \\ f(0) &= 2 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 2(x - 1)^2 - 1 \\ f(-5) &= 71 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 3\left(x + \frac{1}{3}\right)^2 - \frac{10}{3} \\ f(-3) &= 18 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -5\left(x + \frac{1}{10}\right)^2 - \frac{99}{20} \\ f(-2) &= -23 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 5\left(x - \frac{1}{2}\right)^2 + \frac{7}{4} \\ f(-5) &= 153 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= (x - 1)^2 + 3 \\ f(-3) &= 19 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -\left(x - \frac{5}{2}\right)^2 + \frac{37}{4} \\ f(1) &= 7 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2\left(x + \frac{3}{4}\right)^2 + \frac{9}{8} \\ f(1) &= -5 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 2\left(x + \frac{1}{4}\right)^2 + \frac{23}{8} \\ f(3) &= 24 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4\left(x + \frac{1}{2}\right)^2 - 3 \\ f(2) &= -28 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -2\left(x - \frac{5}{4}\right)^2 - \frac{7}{8} \\ f(2) &= -2 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 3\left(x - \frac{2}{3}\right)^2 - \frac{10}{3} \\ f(4) &= 30 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -4 \left(x + \frac{3}{8} \right)^2 - \frac{71}{16} \\ f(-3) &= -32 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= \left(x - \frac{3}{2} \right)^2 + \frac{11}{4} \\ f(-4) &= 33 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 5 \left(x - \frac{2}{5} \right)^2 - \frac{19}{5} \\ f(-2) &= 25 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 3 \left(x - \frac{5}{6} \right)^2 + \frac{23}{12} \\ f(5) &= 54 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -3 \left(x - \frac{1}{3} \right)^2 + \frac{7}{3} \\ f(-1) &= -3 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= (x + 2)^2 + 1 \\ f(3) &= 26 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -(x - 1)^2 \\ f(0) &= -1 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4 \left(x + \frac{1}{4} \right)^2 - \frac{19}{4} \\ f(5) &= -115 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -4 \left(x - \frac{1}{2} \right)^2 + 6 \\ f(-4) &= -75 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 2 \left(x - \frac{5}{4} \right)^2 - \frac{49}{8} \\ f(5) &= 22 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3 \left(x + \frac{5}{6} \right)^2 - \frac{23}{12} \\ f(4) &= -72 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3 \left(x + \frac{2}{3} \right)^2 - \frac{1}{3} \\ f(5) &= 96 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -4 \left(x + \frac{1}{2} \right)^2 + 4 \\ f(-1) &= 3 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -3 \left(x + \frac{5}{6} \right)^2 + \frac{73}{12} \\ f(0) &= 4 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -2 \left(x - \frac{1}{2} \right)^2 - \frac{1}{2} \\ f(-4) &= -41 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 2(x-1)^2 + 1 \\ f(-3) &= 33 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= \left(x - \frac{5}{2} \right)^2 - \frac{13}{4} \\ f(3) &= -3 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -5 \left(x + \frac{1}{2} \right)^2 + \frac{9}{4} \\ f(-4) &= -59 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 5 \left(x - \frac{2}{5} \right)^2 - \frac{14}{5} \\ f(0) &= -2 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -5 \left(x + \frac{2}{5} \right)^2 - \frac{11}{5} \\ f(0) &= -3 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3\left(x + \frac{1}{6}\right)^2 - \frac{23}{12} \\ f(-1) &= -4 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 4\left(x + \frac{1}{8}\right)^2 + \frac{47}{16} \\ f(4) &= 71 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -\left(x - \frac{5}{2}\right)^2 + \frac{9}{4} \\ f(5) &= -4 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -5\left(x - \frac{3}{10}\right)^2 + \frac{69}{20} \\ f(-3) &= -51 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 2(x - 1)^2 - 7 \\ f(-3) &= 25 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 5\left(x - \frac{2}{5}\right)^2 - \frac{19}{5} \\ f(-3) &= 54 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 4\left(x - \frac{1}{2}\right)^2 + 4 \\ f(0) &= 5 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -5\left(x - \frac{1}{10}\right)^2 - \frac{39}{20} \\ f(-3) &= -50 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -3\left(x - \frac{1}{6}\right)^2 - \frac{35}{12} \\ f(2) &= -13 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 5\left(x + \frac{1}{2}\right)^2 - \frac{9}{4} \\ f(4) &= 99 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3\left(x - \frac{1}{2}\right)^2 + \frac{3}{4} \\ f(-2) &= -18 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 5\left(x + \frac{1}{10}\right)^2 + \frac{59}{20} \\ f(5) &= 133 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 4\left(x + \frac{3}{8}\right)^2 - \frac{89}{16} \\ f(4) &= 71 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 5\left(x + \frac{1}{5}\right)^2 + \frac{14}{5} \\ f(2) &= 27 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4\left(x - \frac{1}{8}\right)^2 + \frac{1}{16} \\ f(1) &= -3 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 4\left(x + \frac{1}{4}\right)^2 + \frac{3}{4} \\ f(2) &= 21 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 5\left(x + \frac{1}{2}\right)^2 + \frac{15}{4} \\ f(-3) &= 35 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -2\left(x + \frac{1}{2}\right)^2 + \frac{7}{2} \\ f(3) &= -21 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 5\left(x + \frac{1}{10}\right)^2 - \frac{61}{20} \\ f(-3) &= 39 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -5\left(x + \frac{2}{5}\right)^2 + \frac{4}{5} \\ f(-3) &= -33 \end{aligned}$$

$$(1) \quad f(x) = (x-1)^2 + 2 \\ f(5) = 18$$

$$(6) \quad f(x) = -(x+1)^2 + 1 \\ f(0) = 0$$

$$(2) \quad f(x) = -3\left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \\ f(-1) = 0$$

$$(7) \quad f(x) = -5\left(x - \frac{2}{5}\right)^2 - \frac{1}{5} \\ f(4) = -65$$

$$(3) \quad f(x) = 3\left(x + \frac{2}{3}\right)^2 + \frac{8}{3} \\ f(0) = 4$$

$$(8) \quad f(x) = 4\left(x + \frac{1}{2}\right)^2 \\ f(-4) = 49$$

$$(4) \quad f(x) = 3\left(x - \frac{2}{3}\right)^2 - \frac{10}{3} \\ f(-3) = 37$$

$$(9) \quad f(x) = -3\left(x - \frac{1}{6}\right)^2 + \frac{49}{12} \\ f(-2) = -10$$

$$(5) \quad f(x) = -4\left(x - \frac{1}{2}\right)^2 + 4 \\ f(-4) = -77$$

$$(10) \quad f(x) = 2\left(x + \frac{1}{2}\right)^2 - \frac{5}{2} \\ f(3) = 22$$

$$(1) \begin{aligned} f(x) &= 2\left(x + \frac{1}{4}\right)^2 - \frac{41}{8} \\ f(4) &= 31 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -3\left(x - \frac{5}{6}\right)^2 - \frac{11}{12} \\ f(-1) &= -11 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 3\left(x + \frac{2}{3}\right)^2 + \frac{8}{3} \\ f(4) &= 68 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -2\left(x + \frac{5}{4}\right)^2 + \frac{25}{8} \\ f(-2) &= 2 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -5\left(x + \frac{2}{5}\right)^2 + \frac{14}{5} \\ f(4) &= -94 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -4\left(x + \frac{1}{8}\right)^2 - \frac{63}{16} \\ f(2) &= -22 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= (x - 1)^2 + 1 \\ f(-5) &= 37 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4\left(x + \frac{1}{8}\right)^2 + \frac{17}{16} \\ f(-4) &= -59 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 4\left(x + \frac{1}{4}\right)^2 + \frac{3}{4} \\ f(1) &= 7 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -4\left(x + \frac{1}{2}\right)^2 - 3 \\ f(0) &= -4 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3 \left(x + \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(-1) &= 1 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -4 \left(x - \frac{3}{8} \right)^2 - \frac{55}{16} \\ f(-2) &= -26 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -3 \left(x + \frac{2}{3} \right)^2 + \frac{10}{3} \\ f(0) &= 2 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= (x - 2)^2 - 1 \\ f(-3) &= 24 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 3 \left(x - \frac{1}{2} \right)^2 + \frac{1}{4} \\ f(-4) &= 61 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= \left(x - \frac{3}{2} \right)^2 - \frac{13}{4} \\ f(-1) &= 3 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5 \left(x - \frac{3}{10} \right)^2 + \frac{109}{20} \\ f(2) &= -9 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 5 \left(x + \frac{1}{10} \right)^2 + \frac{59}{20} \\ f(3) &= 51 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -3 \left(x + \frac{5}{6} \right)^2 + \frac{61}{12} \\ f(-1) &= 5 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 5 \left(x + \frac{1}{2} \right)^2 - \frac{9}{4} \\ f(-4) &= 59 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 4 \left(x + \frac{5}{8} \right)^2 - \frac{89}{16} \\ f(4) &= 80 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 5 \left(x - \frac{1}{10} \right)^2 - \frac{61}{20} \\ f(-5) &= 127 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -2 \left(x - \frac{5}{4} \right)^2 - \frac{7}{8} \\ f(-3) &= -37 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -2(x-1)^2 - 2 \\ f(0) &= -4 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -5 \left(x - \frac{1}{2} \right)^2 - \frac{3}{4} \\ f(5) &= -102 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 3 \left(x + \frac{1}{6} \right)^2 + \frac{23}{12} \\ f(5) &= 82 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3 \left(x + \frac{1}{6} \right)^2 - \frac{25}{12} \\ f(-3) &= 22 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -2 \left(x + \frac{5}{4} \right)^2 + \frac{65}{8} \\ f(-1) &= 8 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 3 \left(x + \frac{2}{3} \right)^2 - \frac{4}{3} \\ f(5) &= 95 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -5 \left(x + \frac{3}{10} \right)^2 + \frac{69}{20} \\ f(-2) &= -11 \end{aligned}$$

$$(1) \quad f(x) = \left(x + \frac{3}{2}\right)^2 - \frac{9}{4} \\ f(5) = 40$$

$$(6) \quad f(x) = -3\left(x - \frac{1}{3}\right)^2 - \frac{5}{3} \\ f(1) = -3$$

$$(2) \quad f(x) = -2\left(x + \frac{3}{4}\right)^2 + \frac{33}{8} \\ f(-4) = -17$$

$$(7) \quad f(x) = 5\left(x + \frac{1}{5}\right)^2 + \frac{4}{5} \\ f(4) = 89$$

$$(3) \quad f(x) = -\left(x + \frac{3}{2}\right)^2 + \frac{1}{4} \\ f(-4) = -6$$

$$(8) \quad f(x) = 2\left(x + \frac{1}{4}\right)^2 + \frac{31}{8} \\ f(-3) = 19$$

$$(4) \quad f(x) = -5\left(x + \frac{2}{5}\right)^2 - \frac{16}{5} \\ f(5) = -149$$

$$(9) \quad f(x) = -2(x + 1)^2 + 5 \\ f(5) = -67$$

$$(5) \quad f(x) = -2\left(x - \frac{1}{2}\right)^2 + \frac{9}{2} \\ f(-3) = -20$$

$$(10) \quad f(x) = -4\left(x + \frac{1}{4}\right)^2 - \frac{19}{4} \\ f(2) = -25$$

$$(1) \quad f(x) = (x+2)^2 - 3 \\ f(2) = 13$$

$$(6) \quad f(x) = 3\left(x + \frac{1}{2}\right)^2 + \frac{9}{4} \\ f(1) = 9$$

$$(2) \quad f(x) = 3\left(x + \frac{5}{6}\right)^2 - \frac{37}{12} \\ f(5) = 99$$

$$(7) \quad f(x) = 2(x+1)^2 - 6 \\ f(3) = 26$$

$$(3) \quad f(x) = -2\left(x + \frac{5}{4}\right)^2 + \frac{65}{8} \\ f(-1) = 8$$

$$(8) \quad f(x) = -4\left(x - \frac{3}{8}\right)^2 + \frac{73}{16} \\ f(5) = -81$$

$$(4) \quad f(x) = 2\left(x + \frac{1}{2}\right)^2 - \frac{1}{2} \\ f(3) = 24$$

$$(9) \quad f(x) = -2\left(x + \frac{1}{4}\right)^2 - \frac{7}{8} \\ f(-4) = -29$$

$$(5) \quad f(x) = 3\left(x + \frac{5}{6}\right)^2 - \frac{85}{12} \\ f(-5) = 45$$

$$(10) \quad f(x) = 5\left(x - \frac{1}{2}\right)^2 - \frac{17}{4} \\ f(5) = 97$$

$$(1) \quad f(x) = 4 \left(x + \frac{3}{8} \right)^2 - \frac{89}{16}$$
$$f(1) = 2$$

$$(6) \quad f(x) = -3 \left(x - \frac{5}{6} \right)^2 + \frac{73}{12}$$
$$f(-3) = -38$$

$$(2) \quad f(x) = -(x+1)^2 + 1$$
$$f(-3) = -3$$

$$(7) \quad f(x) = -3 \left(x - \frac{5}{6} \right)^2 + \frac{25}{12}$$
$$f(4) = -28$$

$$(3) \quad f(x) = -5 \left(x - \frac{1}{5} \right)^2 + \frac{21}{5}$$
$$f(-3) = -47$$

$$(8) \quad f(x) = 3 \left(x + \frac{1}{3} \right)^2 + \frac{14}{3}$$
$$f(4) = 61$$

$$(4) \quad f(x) = 2(x-1)^2 - 4$$
$$f(-3) = 28$$

$$(9) \quad f(x) = -5 \left(x - \frac{1}{5} \right)^2 - \frac{19}{5}$$
$$f(5) = -119$$

$$(5) \quad f(x) = -2 \left(x + \frac{1}{4} \right)^2 - \frac{7}{8}$$
$$f(-2) = -7$$

$$(10) \quad f(x) = 4 \left(x + \frac{1}{4} \right)^2 - \frac{9}{4}$$
$$f(5) = 108$$

$$(1) \begin{aligned} f(x) &= \left(x - \frac{1}{2}\right)^2 + \frac{19}{4} \\ f(0) &= 5 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -5 \left(x - \frac{2}{5}\right)^2 + \frac{29}{5} \\ f(1) &= 4 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= \left(x + \frac{1}{2}\right)^2 - \frac{1}{4} \\ f(-5) &= 20 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 3 \left(x + \frac{5}{6}\right)^2 - \frac{25}{12} \\ f(-1) &= -2 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 3 \left(x + \frac{2}{3}\right)^2 + \frac{5}{3} \\ f(1) &= 10 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 4 \left(x + \frac{1}{4}\right)^2 - \frac{17}{4} \\ f(-4) &= 52 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 4 \left(x + \frac{1}{2}\right)^2 - 2 \\ f(-4) &= 47 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4 \left(x - \frac{5}{8}\right)^2 + \frac{41}{16} \\ f(5) &= -74 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 4 \left(x - \frac{5}{8}\right)^2 - \frac{41}{16} \\ f(-2) &= 25 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= (x + 2)^2 \\ f(1) &= 9 \end{aligned}$$

$$(1) \quad f(x) = -4 \left(x - \frac{1}{2} \right)^2 - 4 \\ f(0) = -5$$

$$(6) \quad f(x) = -2(x+1)^2 - 2 \\ f(1) = -10$$

$$(2) \quad f(x) = -4 \left(x - \frac{3}{8} \right)^2 - \frac{71}{16} \\ f(-2) = -27$$

$$(7) \quad f(x) = (x+1)^2 - 4 \\ f(3) = 12$$

$$(3) \quad f(x) = -2 \left(x - \frac{5}{4} \right)^2 - \frac{7}{8} \\ f(1) = -1$$

$$(8) \quad f(x) = -(x-2)^2 + 5 \\ f(1) = 4$$

$$(4) \quad f(x) = \left(x - \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(-5) = 32$$

$$(9) \quad f(x) = - \left(x - \frac{3}{2} \right)^2 + \frac{25}{4} \\ f(-2) = -6$$

$$(5) \quad f(x) = - \left(x - \frac{3}{2} \right)^2 - \frac{11}{4} \\ f(-2) = -15$$

$$(10) \quad f(x) = - \left(x - \frac{1}{2} \right)^2 - \frac{3}{4} \\ f(-1) = -3$$

$$(1) \begin{aligned} f(x) &= -5 \left(x + \frac{2}{5} \right)^2 + \frac{9}{5} \\ f(-4) &= -63 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3 \left(x + \frac{1}{2} \right)^2 - \frac{7}{4} \\ f(-4) &= 35 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -3 \left(x + \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(5) &= -89 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= \left(x - \frac{3}{2} \right)^2 - \frac{25}{4} \\ f(0) &= -4 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 5 \left(x - \frac{1}{2} \right)^2 - \frac{9}{4} \\ f(-2) &= 29 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 3 \left(x + \frac{1}{3} \right)^2 + \frac{5}{3} \\ f(5) &= 87 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5 \left(x - \frac{1}{10} \right)^2 + \frac{41}{20} \\ f(2) &= -16 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 5 \left(x - \frac{2}{5} \right)^2 + \frac{1}{5} \\ f(-5) &= 146 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -2 \left(x - \frac{1}{2} \right)^2 + \frac{3}{2} \\ f(2) &= -3 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 2 \left(x + \frac{1}{2} \right)^2 + \frac{1}{2} \\ f(3) &= 25 \end{aligned}$$

$$(1) \quad f(x) = -\left(x + \frac{1}{2}\right)^2 - \frac{15}{4}$$
$$f(-2) = -6$$

$$(6) \quad f(x) = 3\left(x - \frac{5}{6}\right)^2 + \frac{23}{12}$$
$$f(-1) = 12$$

$$(2) \quad f(x) = (x - 1)^2 + 2$$
$$f(1) = 2$$

$$(7) \quad f(x) = -3\left(x + \frac{5}{6}\right)^2 + \frac{25}{12}$$
$$f(-4) = -28$$

$$(3) \quad f(x) = 4\left(x + \frac{1}{4}\right)^2 + \frac{11}{4}$$
$$f(-4) = 59$$

$$(8) \quad f(x) = -4\left(x + \frac{5}{8}\right)^2 + \frac{25}{16}$$
$$f(2) = -26$$

$$(4) \quad f(x) = 2\left(x + \frac{5}{4}\right)^2 - \frac{25}{8}$$
$$f(-3) = 3$$

$$(9) \quad f(x) = -3\left(x - \frac{2}{3}\right)^2 + \frac{19}{3}$$
$$f(3) = -10$$

$$(5) \quad f(x) = -5\left(x - \frac{1}{5}\right)^2 + \frac{16}{5}$$
$$f(4) = -69$$

$$(10) \quad f(x) = -2(x + 1)^2 + 2$$
$$f(-3) = -6$$

$$(1) \quad f(x) = -(x-1)^2 + 4 \\ f(0) = 3$$

$$(6) \quad f(x) = -2(x+1)^2 + 4 \\ f(-1) = 4$$

$$(2) \quad f(x) = 5\left(x + \frac{1}{2}\right)^2 - \frac{13}{4} \\ f(-4) = 58$$

$$(7) \quad f(x) = 5\left(x - \frac{2}{5}\right)^2 - \frac{24}{5} \\ f(0) = -4$$

$$(3) \quad f(x) = \left(x - \frac{1}{2}\right)^2 + \frac{15}{4} \\ f(3) = 10$$

$$(8) \quad f(x) = 4\left(x + \frac{5}{8}\right)^2 + \frac{39}{16} \\ f(-3) = 25$$

$$(4) \quad f(x) = -3\left(x - \frac{1}{3}\right)^2 - \frac{11}{3} \\ f(0) = -4$$

$$(9) \quad f(x) = 5\left(x - \frac{1}{5}\right)^2 + \frac{19}{5} \\ f(-5) = 139$$

$$(5) \quad f(x) = -2\left(x + \frac{5}{4}\right)^2 - \frac{7}{8} \\ f(-3) = -7$$

$$(10) \quad f(x) = -2\left(x + \frac{3}{4}\right)^2 - \frac{23}{8} \\ f(-4) = -24$$

$$(1) \quad f(x) = 3 \left(x + \frac{5}{6} \right)^2 - \frac{13}{12}$$
$$f(-4) = 29$$

$$(6) \quad f(x) = -2 \left(x - \frac{1}{2} \right)^2 + \frac{1}{2}$$
$$f(5) = -40$$

$$(2) \quad f(x) = 3 \left(x - \frac{1}{3} \right)^2 + \frac{14}{3}$$
$$f(0) = 5$$

$$(7) \quad f(x) = 2 \left(x - \frac{1}{4} \right)^2 + \frac{7}{8}$$
$$f(0) = 1$$

$$(3) \quad f(x) = 4 \left(x + \frac{1}{4} \right)^2 - \frac{13}{4}$$
$$f(3) = 39$$

$$(8) \quad f(x) = 3 \left(x - \frac{1}{2} \right)^2 - \frac{23}{4}$$
$$f(-2) = 13$$

$$(4) \quad f(x) = 2 \left(x - \frac{5}{4} \right)^2 - \frac{57}{8}$$
$$f(-2) = 14$$

$$(9) \quad f(x) = \left(x - \frac{3}{2} \right)^2 - \frac{29}{4}$$
$$f(4) = -1$$

$$(5) \quad f(x) = 5 \left(x - \frac{1}{2} \right)^2 + \frac{15}{4}$$
$$f(3) = 35$$

$$(10) \quad f(x) = 4 \left(x - \frac{3}{8} \right)^2 - \frac{89}{16}$$
$$f(2) = 5$$

2.8 1 次方程式

1 次方程式の問題。

(1) $-6x - 9 = -5x - 3$

(16) $8x + 8 = -3x + 8$

(31) $5x - 4 = 8x + 4$

(2) $3x + 7 = -4x - 8$

(17) $-7x + 4 = 9x - 3$

(32) $2x - 3 = 9x - 1$

(3) $7x - 8 = -7x - 5$

(18) $4x + 5 = -7x - 1$

(33) $-9x - 6 = 8x + 2$

(4) $-2x + 9 = -4x - 3$

(19) $-6x - 3 = 6x + 3$

(34) $3x + 4 = 5x + 1$

(5) $-4x + 2 = 2x + 6$

(20) $-6x - 2 = 3x + 8$

(35) $6x + 4 = -2x - 5$

(6) $8x - 3 = 3x - 6$

(21) $-6x + 5 = 4x + 4$

(36) $7x - 4 = 6x + 2$

(7) $6x - 9 = -7x - 5$

(22) $9x - 8 = 5x + 8$

(37) $5x - 9 = -5x - 6$

(8) $5x - 9 = x + 1$

(23) $-5x + 5 = 6x + 9$

(38) $x - 2 = 2x - 8$

(9) $-9x + 7 = x - 1$

(24) $-8x + 5 = 8x - 4$

(39) $3x + 3 = x + 5$

(10) $-6x - 7 = 3x - 5$

(25) $3x - 2 = 7x + 2$

(40) $9x + 4 = -3x + 7$

(11) $-7x - 7 = 3x - 5$

(26) $-x + 8 = 6x + 7$

(41) $-8x + 3 = -7x - 1$

(12) $5x + 7 = -4x - 9$

(27) $-9x - 3 = 8x + 2$

(42) $-2x - 1 = -3x + 2$

(13) $6x - 5 = -9x - 3$

(28) $6x + 1 = 5x + 4$

(43) $-2x - 7 = 3x - 4$

(14) $-2x + 5 = -5x - 8$

(29) $-7x + 8 = -3x + 5$

(44) $2x + 3 = -4x - 1$

(15) $-9x - 1 = -5x - 4$

(30) $-7x + 5 = 7x + 3$

(45) $4x + 4 = x - 6$

(1) $-8x + 6 = -3x - 4$

(16) $-7x - 6 = 4x - 4$

(31) $7x - 4 = -6x + 6$

(2) $-3x - 1 = 9x + 3$

(17) $-5x + 6 = -6x - 7$

(32) $7x - 6 = 2x - 9$

(3) $7x + 8 = -8x - 4$

(18) $2x - 3 = 9x - 2$

(33) $6x - 4 = -7x - 8$

(4) $3x - 9 = -x + 9$

(19) $-x - 9 = -2x + 2$

(34) $x + 4 = 4x - 7$

(5) $2x - 5 = x - 4$

(20) $-5x - 8 = 3x - 7$

(35) $-6x - 4 = -x + 8$

(6) $-5x + 8 = -6x - 5$

(21) $-8x - 9 = 4x + 2$

(36) $7x + 7 = 5x - 4$

(7) $7x + 5 = 5x - 1$

(22) $5x + 3 = 2x - 3$

(37) $6x + 4 = 5x + 2$

(8) $-2x - 1 = -7x - 8$

(23) $3x + 7 = -x + 5$

(38) $8x - 3 = -4x + 7$

(9) $5x - 2 = 7x + 1$

(24) $-4x + 2 = 4x + 3$

(39) $6x + 7 = -7x + 9$

(10) $-x - 9 = x + 1$

(25) $9x - 7 = -2x + 2$

(40) $4x + 5 = -8x - 1$

(11) $4x + 7 = -6x - 3$

(26) $4x - 4 = -4x + 3$

(41) $-8x + 9 = 2x + 3$

(12) $7x + 8 = 5x + 6$

(27) $-3x + 4 = -8x + 4$

(42) $8x - 9 = 7x + 3$

(13) $7x - 5 = x - 9$

(28) $-x - 1 = 3x + 6$

(43) $7x - 9 = -2x - 2$

(14) $7x + 9 = -3x - 3$

(29) $7x - 7 = -9x - 6$

(44) $6x - 3 = -2x - 6$

(15) $4x + 5 = -8x - 3$

(30) $8x + 5 = x + 4$

(45) $4x + 9 = -x - 3$

(1) $-7x + 5 = -8x - 8$

(16) $x - 4 = -x - 2$

(31) $-5x - 1 = 5x + 1$

(2) $8x - 9 = -7x - 7$

(17) $-6x - 5 = 7x - 8$

(32) $-3x - 1 = -7x + 4$

(3) $9x - 7 = -6x + 6$

(18) $3x + 5 = x - 4$

(33) $x - 5 = 8x - 3$

(4) $-x - 9 = 2x - 8$

(19) $8x + 4 = -3x - 6$

(34) $4x - 5 = -4x - 1$

(5) $-3x + 2 = -8x - 4$

(20) $-4x - 7 = -6x - 4$

(35) $5x + 5 = -4x + 7$

(6) $3x + 8 = 7x - 3$

(21) $7x - 8 = 3x - 1$

(36) $7x - 2 = -9x + 3$

(7) $5x + 5 = 2x - 2$

(22) $-7x - 8 = 4x + 8$

(37) $3x + 3 = -8x + 4$

(8) $-3x + 3 = 4x - 5$

(23) $-6x - 5 = 8x - 7$

(38) $9x - 6 = -x + 5$

(9) $7x + 8 = -6x + 1$

(24) $4x - 1 = -2x - 1$

(39) $2x + 1 = -8x - 2$

(10) $6x - 3 = -9x - 8$

(25) $2x - 2 = -x + 1$

(40) $8x + 3 = 9x + 1$

(11) $9x - 5 = 4x + 2$

(26) $-7x + 5 = -6x + 5$

(41) $-8x - 9 = -2x - 7$

(12) $-3x - 2 = -6x + 3$

(27) $3x + 6 = -4x + 9$

(42) $6x + 9 = -6x - 9$

(13) $-2x - 5 = 6x + 7$

(28) $x + 5 = 7x + 6$

(43) $5x + 2 = 7x + 6$

(14) $5x + 9 = 3x + 4$

(29) $-4x - 1 = -x + 7$

(44) $-5x - 8 = 9x - 9$

(15) $2x + 6 = 8x - 1$

(30) $x - 4 = 5x - 3$

(45) $x + 5 = 8x + 3$

(1) $6x + 1 = -x - 5$

(16) $3x + 3 = -8x - 1$

(31) $-6x + 6 = -x + 9$

(2) $-9x + 7 = 7x + 2$

(17) $-x - 6 = 8x + 1$

(32) $9x - 2 = -8x + 4$

(3) $4x - 9 = 2x + 6$

(18) $5x + 1 = -7x + 9$

(33) $-2x - 5 = 7x + 9$

(4) $-8x - 4 = 8x + 3$

(19) $-4x - 1 = -9x - 3$

(34) $8x - 8 = 6x - 7$

(5) $x - 6 = 8x + 5$

(20) $-6x - 1 = x + 7$

(35) $-7x - 5 = 6x - 2$

(6) $-8x + 9 = -5x + 7$

(21) $3x - 1 = -8x - 4$

(36) $-6x + 8 = -4x + 1$

(7) $9x - 2 = 2x - 1$

(22) $-3x - 1 = 3x - 9$

(37) $-6x + 9 = -3x + 8$

(8) $-9x - 9 = 2x - 9$

(23) $-4x + 3 = -2x - 2$

(38) $5x - 4 = 8x - 4$

(9) $-x - 6 = 6x + 9$

(24) $3x - 2 = 5x + 7$

(39) $4x + 9 = 5x - 9$

(10) $x + 5 = 9x + 9$

(25) $-2x - 2 = x - 2$

(40) $5x + 7 = -5x + 3$

(11) $5x - 6 = -8x - 3$

(26) $-x - 2 = -2x - 7$

(41) $5x + 2 = -9x - 6$

(12) $7x - 3 = -6x - 4$

(27) $x - 3 = 9x - 7$

(42) $6x + 1 = 7x - 6$

(13) $-7x - 2 = 6x - 6$

(28) $6x - 1 = 3x + 5$

(43) $7x - 3 = 5x + 1$

(14) $-3x + 4 = -7x - 2$

(29) $-2x + 9 = 8x - 8$

(44) $-5x + 8 = x + 6$

(15) $-5x - 5 = -x - 8$

(30) $-2x + 6 = -5x - 8$

(45) $-8x + 6 = 4x - 9$

(1) $8x - 9 = -7x - 3$

(16) $-8x + 7 = x + 1$

(31) $5x + 3 = -7x + 4$

(2) $-8x - 5 = -4x - 8$

(17) $-4x + 6 = 8x - 4$

(32) $6x - 6 = -2x - 3$

(3) $-6x + 3 = 5x + 9$

(18) $-7x + 7 = 5x - 7$

(33) $-5x + 8 = 8x - 5$

(4) $7x + 6 = 5x - 2$

(19) $-3x - 3 = -x + 4$

(34) $-9x - 3 = 8x - 6$

(5) $7x + 6 = -3x + 1$

(20) $9x + 3 = x + 5$

(35) $5x - 8 = 4x + 7$

(6) $2x - 7 = -7x - 1$

(21) $8x - 4 = 9x + 1$

(36) $8x + 2 = 9x - 6$

(7) $5x - 9 = x + 4$

(22) $-4x + 5 = 2x - 9$

(37) $5x - 9 = 6x - 4$

(8) $-5x + 8 = 6x - 3$

(23) $x - 4 = -6x - 6$

(38) $5x + 8 = -7x - 4$

(9) $-8x - 6 = 8x - 1$

(24) $-5x - 6 = 3x + 5$

(39) $9x - 7 = -x - 7$

(10) $-2x + 7 = 6x + 6$

(25) $-5x - 4 = -4x + 1$

(40) $-2x + 3 = 4x - 5$

(11) $-6x + 6 = 4x + 8$

(26) $8x + 5 = -5x - 3$

(41) $3x - 9 = -x - 2$

(12) $4x - 7 = 2x + 9$

(27) $2x + 9 = -9x + 3$

(42) $-7x + 8 = -8x + 3$

(13) $x + 8 = -5x + 3$

(28) $4x + 5 = 7x - 1$

(43) $4x - 7 = x + 4$

(14) $2x - 5 = -x + 2$

(29) $6x + 1 = 3x - 2$

(44) $3x + 7 = 5x + 6$

(15) $5x + 4 = -8x + 7$

(30) $-7x - 5 = 8x - 3$

(45) $4x - 9 = -6x - 9$

(1) $8x - 2 = 4x + 3$

(16) $-4x + 2 = 3x - 7$

(31) $-3x + 5 = 7x - 2$

(2) $x - 4 = 7x - 4$

(17) $-2x - 7 = 9x + 3$

(32) $6x - 8 = 9x - 8$

(3) $-6x - 8 = 8x - 3$

(18) $-4x - 7 = x + 5$

(33) $9x + 2 = -8x + 7$

(4) $-5x - 1 = 2x + 8$

(19) $9x + 3 = 2x - 5$

(34) $-6x - 6 = -2x + 3$

(5) $4x + 5 = -3x + 4$

(20) $-4x - 9 = 2x - 7$

(35) $6x - 7 = 2x - 4$

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(8) $-4x - 9 = 7x - 3$

(23) $5x + 5 = 7x + 8$

(38) $-3x + 4 = -4x - 1$

(9) $-2x - 2 = 2x + 8$

(24) $-7x - 8 = 5x - 6$

(39) $7x + 4 = -8x + 5$

(10) $-x - 8 = -3x - 2$

(25) $4x + 5 = 3x - 6$

(40) $5x + 7 = -2x + 5$

(11) $x + 2 = 4x - 8$

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(27) $6x - 9 = -2x - 8$

(42) $-x - 6 = 2x + 7$

(13) $4x + 6 = 7x + 4$

(28) $4x + 1 = x - 4$

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(16) $6x - 1 = 7x + 5$

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(26) $4x + 4 = 7x - 3$

(41) $x + 6 = 4x - 4$

(12) $-7x - 4 = 8x - 8$

(27) $3x - 3 = -2x + 6$

(42) $6x + 4 = -9x + 9$

(13) $-9x - 9 = -6x - 9$

(28) $x - 1 = -4x - 4$

(43) $-5x - 9 = -8x + 1$

(14) $8x + 4 = -2x + 7$

(29) $-6x + 5 = 3x + 3$

(44) $4x + 4 = 7x + 3$

(15) $2x - 4 = x - 6$

(30) $x - 7 = -x - 1$

(45) $-2x + 6 = x - 6$

(1) $x + 5 = -4x + 2$

(16) $7x - 3 = 5x + 2$

(31) $8x + 6 = 5x + 1$

(2) $2x - 9 = -5x + 5$

(17) $4x - 7 = -9x - 8$

(32) $5x - 4 = -3x - 8$

(3) $-8x - 6 = 2x + 4$

(18) $-7x - 8 = 7x + 7$

(33) $-9x - 8 = -2x - 7$

(4) $7x + 1 = 4x - 1$

(19) $-x - 6 = 3x + 8$

(34) $-4x + 2 = 4x - 3$

(5) $8x + 6 = x - 6$

(20) $9x + 1 = 5x - 9$

(35) $-x + 9 = x + 8$

(6) $5x + 2 = -2x + 7$

(21) $4x + 9 = -3x + 8$

(36) $9x - 5 = -x + 7$

(7) $6x - 1 = 5x - 4$

(22) $6x - 2 = -4x - 8$

(37) $-4x - 2 = -7x - 6$

(8) $-2x + 7 = -7x + 5$

(23) $-4x - 2 = -8x - 3$

(38) $-6x + 8 = -9x - 6$

(9) $4x + 9 = -9x - 9$

(24) $4x - 1 = 6x + 6$

(39) $6x + 6 = 9x - 1$

(10) $2x - 3 = 3x + 1$

(25) $5x - 7 = 2x - 8$

(40) $-8x + 4 = 7x - 9$

(11) $-8x - 1 = -9x - 8$

(26) $-3x + 4 = 8x - 1$

(41) $-3x - 5 = 6x + 3$

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(15) $5x + 5 = 2x + 9$

(30) $8x + 4 = -8x - 1$

(45) $6x - 3 = -8x - 5$

(1) $6x + 5 = x - 8$

(16) $8x - 3 = -2x - 3$

(31) $2x + 5 = x + 1$

(2) $4x + 5 = -7x - 4$

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(32) $-2x - 9 = x + 2$

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(5) $6x - 5 = 8x - 4$

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(35) $-7x + 7 = -2x + 4$

(6) $-x + 5 = x - 3$

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(44) $-7x + 4 = -x - 5$

(15) $-6x + 5 = 5x - 2$

(30) $7x - 3 = 4x - 1$

(45) $-3x + 8 = x + 7$

(1) $-7x - 7 = x - 5$

(16) $-3x + 7 = 5x - 7$

(31) $7x + 9 = 8x - 4$

(2) $8x - 1 = 2x - 3$

(17) $6x + 7 = -2x + 3$

(32) $3x - 9 = -6x + 5$

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(7) $-3x + 5 = 4x + 5$

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(10) $2x - 8 = -6x + 1$

(25) $-7x - 1 = 5x - 6$

(40) $3x + 3 = 9x + 5$

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(44) $-4x - 3 = 5x + 6$

(15) $-x - 7 = -6x + 3$

(30) $-6x + 4 = -3x - 7$

(45) $8x - 6 = -x - 9$

$$(1) \quad -6x - 9 = -5x - 3$$

$$x = -6$$

$$(2) \quad 3x + 7 = -4x - 8$$

$$x = -\frac{15}{7}$$

$$(3) \quad 7x - 8 = -7x - 5$$

$$x = \frac{3}{14}$$

$$(4) \quad -2x + 9 = -4x - 3$$

$$x = -6$$

$$(5) \quad -4x + 2 = 2x + 6$$

$$x = -\frac{2}{3}$$

$$(6) \quad 8x - 3 = 3x - 6$$

$$x = -\frac{3}{5}$$

$$(7) \quad 6x - 9 = -7x - 5$$

$$x = \frac{4}{13}$$

$$(8) \quad 5x - 9 = x + 1$$

$$x = \frac{5}{2}$$

$$(9) \quad -9x + 7 = x - 1$$

$$x = \frac{4}{5}$$

$$(10) \quad -6x - 7 = 3x - 5$$

$$x = -\frac{2}{9}$$

$$(11) \quad -7x - 7 = 3x - 5$$

$$x = -\frac{1}{5}$$

$$(12) \quad 5x + 7 = -4x - 9$$

$$x = -\frac{16}{9}$$

$$(13) \quad 6x - 5 = -9x - 3$$

$$x = \frac{2}{15}$$

$$(14) \quad -2x + 5 = -5x - 8$$

$$x = -\frac{13}{3}$$

$$(15) \quad -9x - 1 = -5x - 4$$

$$x = \frac{3}{4}$$

$$(16) \quad 8x + 8 = -3x + 8$$

$$x = 0$$

$$(17) \quad -7x + 4 = 9x - 3$$

$$x = \frac{7}{16}$$

$$(18) \quad 4x + 5 = -7x - 1$$

$$x = -\frac{6}{11}$$

$$(19) \quad -6x - 3 = 6x + 3$$

$$x = -\frac{1}{2}$$

$$(20) \quad -6x - 2 = 3x + 8$$

$$x = -\frac{10}{9}$$

$$(21) \quad -6x + 5 = 4x + 4$$

$$x = \frac{1}{10}$$

$$(22) \quad 9x - 8 = 5x + 8$$

$$x = 4$$

$$(23) \quad -5x + 5 = 6x + 9$$

$$x = -\frac{4}{11}$$

$$(24) \quad -8x + 5 = 8x - 4$$

$$x = \frac{9}{16}$$

$$(25) \quad 3x - 2 = 7x + 2$$

$$x = -1$$

$$(26) \quad -x + 8 = 6x + 7$$

$$x = \frac{1}{7}$$

$$(27) \quad -9x - 3 = 8x + 2$$

$$x = -\frac{5}{17}$$

$$(28) \quad 6x + 1 = 5x + 4$$

$$x = 3$$

$$(29) \quad -7x + 8 = -3x + 5$$

$$x = \frac{3}{4}$$

$$(30) \quad -7x + 5 = 7x + 3$$

$$x = \frac{1}{7}$$

$$(31) \quad 5x - 4 = 8x + 4$$

$$x = -\frac{8}{3}$$

$$(32) \quad 2x - 3 = 9x - 1$$

$$x = -\frac{2}{7}$$

$$(33) \quad -9x - 6 = 8x + 2$$

$$x = -\frac{8}{17}$$

$$(34) \quad 3x + 4 = 5x + 1$$

$$x = \frac{3}{2}$$

$$(35) \quad 6x + 4 = -2x - 5$$

$$x = -\frac{9}{8}$$

$$(36) \quad 7x - 4 = 6x + 2$$

$$x = 6$$

$$(37) \quad 5x - 9 = -5x - 6$$

$$x = \frac{3}{10}$$

$$(38) \quad x - 2 = 2x - 8$$

$$x = 6$$

$$(39) \quad 3x + 3 = x + 5$$

$$x = 1$$

$$(40) \quad 9x + 4 = -3x + 7$$

$$x = \frac{1}{4}$$

$$(41) \quad -8x + 3 = -7x - 1$$

$$x = 4$$

$$(42) \quad -2x - 1 = -3x + 2$$

$$x = 3$$

$$(43) \quad -2x - 7 = 3x - 4$$

$$x = -\frac{3}{5}$$

$$(44) \quad 2x + 3 = -4x - 1$$

$$x = -\frac{2}{3}$$

$$(45) \quad 4x + 4 = x - 6$$

$$x = -\frac{10}{3}$$

$$(1) \quad -8x + 6 = -3x - 4$$

$$x = 2$$

$$(2) \quad -3x - 1 = 9x + 3$$

$$x = -\frac{1}{3}$$

$$(3) \quad 7x + 8 = -8x - 4$$

$$x = -\frac{4}{5}$$

$$(4) \quad 3x - 9 = -x + 9$$

$$x = \frac{9}{2}$$

$$(5) \quad 2x - 5 = x - 4$$

$$x = 1$$

$$(6) \quad -5x + 8 = -6x - 5$$

$$x = -13$$

$$(7) \quad 7x + 5 = 5x - 1$$

$$x = -3$$

$$(8) \quad -2x - 1 = -7x - 8$$

$$x = -\frac{7}{5}$$

$$(9) \quad 5x - 2 = 7x + 1$$

$$x = -\frac{3}{2}$$

$$(10) \quad -x - 9 = x + 1$$

$$x = -5$$

$$(11) \quad 4x + 7 = -6x - 3$$

$$x = -1$$

$$(12) \quad 7x + 8 = 5x + 6$$

$$x = -1$$

$$(13) \quad 7x - 5 = x - 9$$

$$x = -\frac{2}{3}$$

$$(14) \quad 7x + 9 = -3x - 3$$

$$x = -\frac{6}{5}$$

$$(15) \quad 4x + 5 = -8x - 3$$

$$x = -\frac{2}{3}$$

$$(16) \quad -7x - 6 = 4x - 4$$

$$x = -\frac{2}{11}$$

$$(17) \quad -5x + 6 = -6x - 7$$

$$x = -13$$

$$(18) \quad 2x - 3 = 9x - 2$$

$$x = -\frac{1}{7}$$

$$(19) \quad -x - 9 = -2x + 2$$

$$x = 11$$

$$(20) \quad -5x - 8 = 3x - 7$$

$$x = -\frac{1}{8}$$

$$(21) \quad -8x - 9 = 4x + 2$$

$$x = -\frac{11}{12}$$

$$(22) \quad 5x + 3 = 2x - 3$$

$$x = -2$$

$$(23) \quad 3x + 7 = -x + 5$$

$$x = -\frac{1}{2}$$

$$(24) \quad -4x + 2 = 4x + 3$$

$$x = -\frac{1}{8}$$

$$(25) \quad 9x - 7 = -2x + 2$$

$$x = \frac{9}{11}$$

$$(26) \quad 4x - 4 = -4x + 3$$

$$x = \frac{7}{8}$$

$$(27) \quad -3x + 4 = -8x + 4$$

$$x = 0$$

$$(28) \quad -x - 1 = 3x + 6$$

$$x = -\frac{7}{4}$$

$$(29) \quad 7x - 7 = -9x - 6$$

$$x = \frac{1}{16}$$

$$(30) \quad 8x + 5 = x + 4$$

$$x = -\frac{1}{7}$$

$$(31) \quad 7x - 4 = -6x + 6$$

$$x = \frac{10}{13}$$

$$(32) \quad 7x - 6 = 2x - 9$$

$$x = -\frac{3}{5}$$

$$(33) \quad 6x - 4 = -7x - 8$$

$$x = -\frac{4}{13}$$

$$(34) \quad x + 4 = 4x - 7$$

$$x = \frac{11}{3}$$

$$(35) \quad -6x - 4 = -x + 8$$

$$x = -\frac{12}{5}$$

$$(36) \quad 7x + 7 = 5x - 4$$

$$x = -\frac{11}{2}$$

$$(37) \quad 6x + 4 = 5x + 2$$

$$x = -2$$

$$(38) \quad 8x - 3 = -4x + 7$$

$$x = \frac{5}{6}$$

$$(39) \quad 6x + 7 = -7x + 9$$

$$x = \frac{2}{13}$$

$$(40) \quad 4x + 5 = -8x - 1$$

$$x = -\frac{1}{2}$$

$$(41) \quad -8x + 9 = 2x + 3$$

$$x = \frac{3}{5}$$

$$(42) \quad 8x - 9 = 7x + 3$$

$$x = 12$$

$$(43) \quad 7x - 9 = -2x - 2$$

$$x = \frac{7}{9}$$

$$(44) \quad 6x - 3 = -2x - 6$$

$$x = -\frac{3}{8}$$

$$(45) \quad 4x + 9 = -x - 3$$

$$x = -\frac{12}{5}$$

$$(1) \quad -7x + 5 = -8x - 8$$

$$x = -13$$

$$(2) \quad 8x - 9 = -7x - 7$$

$$x = \frac{2}{15}$$

$$(3) \quad 9x - 7 = -6x + 6$$

$$x = \frac{13}{15}$$

$$(4) \quad -x - 9 = 2x - 8$$

$$x = -\frac{1}{3}$$

$$(5) \quad -3x + 2 = -8x - 4$$

$$x = -\frac{6}{5}$$

$$(6) \quad 3x + 8 = 7x - 3$$

$$x = \frac{11}{4}$$

$$(7) \quad 5x + 5 = 2x - 2$$

$$x = -\frac{7}{3}$$

$$(8) \quad -3x + 3 = 4x - 5$$

$$x = \frac{8}{7}$$

$$(9) \quad 7x + 8 = -6x + 1$$

$$x = -\frac{7}{13}$$

$$(10) \quad 6x - 3 = -9x - 8$$

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$$(11) \quad 9x - 5 = 4x + 2$$

$$x = \frac{7}{5}$$

$$(12) \quad -3x - 2 = -6x + 3$$

$$x = \frac{5}{3}$$

$$(13) \quad -2x - 5 = 6x + 7$$

$$x = -\frac{3}{2}$$

$$(14) \quad 5x + 9 = 3x + 4$$

$$x = -\frac{5}{2}$$

$$(15) \quad 2x + 6 = 8x - 1$$

$$x = \frac{7}{6}$$

$$(16) \quad x - 4 = -x - 2$$

$$x = 1$$

$$(17) \quad -6x - 5 = 7x - 8$$

$$x = \frac{3}{13}$$

$$(18) \quad 3x + 5 = x - 4$$

$$x = -\frac{9}{2}$$

$$(19) \quad 8x + 4 = -3x - 6$$

$$x = -\frac{10}{11}$$

$$(20) \quad -4x - 7 = -6x - 4$$

$$x = \frac{3}{2}$$

$$(21) \quad 7x - 8 = 3x - 1$$

$$x = \frac{7}{4}$$

$$(22) \quad -7x - 8 = 4x + 8$$

$$x = -\frac{16}{11}$$

$$(23) \quad -6x - 5 = 8x - 7$$

$$x = \frac{1}{7}$$

$$(24) \quad 4x - 1 = -2x - 1$$

$$x = 0$$

$$(25) \quad 2x - 2 = -x + 1$$

$$x = 1$$

$$(26) \quad -7x + 5 = -6x + 5$$

$$x = 0$$

$$(27) \quad 3x + 6 = -4x + 9$$

$$x = \frac{3}{7}$$

$$(28) \quad x + 5 = 7x + 6$$

$$x = -\frac{1}{6}$$

$$(29) \quad -4x - 1 = -x + 7$$

$$x = -\frac{8}{3}$$

$$(30) \quad x - 4 = 5x - 3$$

$$x = -\frac{1}{4}$$

$$(31) \quad -5x - 1 = 5x + 1$$

$$x = -\frac{1}{5}$$

$$(32) \quad -3x - 1 = -7x + 4$$

$$x = \frac{5}{4}$$

$$(33) \quad x - 5 = 8x - 3$$

$$x = -\frac{2}{7}$$

$$(34) \quad 4x - 5 = -4x - 1$$

$$x = \frac{1}{2}$$

$$(35) \quad 5x + 5 = -4x + 7$$

$$x = \frac{2}{9}$$

$$(36) \quad 7x - 2 = -9x + 3$$

$$x = \frac{5}{16}$$

$$(37) \quad 3x + 3 = -8x + 4$$

$$x = \frac{1}{11}$$

$$(38) \quad 9x - 6 = -x + 5$$

$$x = \frac{11}{10}$$

$$(39) \quad 2x + 1 = -8x - 2$$

$$x = -\frac{3}{10}$$

$$(40) \quad 8x + 3 = 9x + 1$$

$$x = 2$$

$$(41) \quad -8x - 9 = -2x - 7$$

$$x = -\frac{1}{3}$$

$$(42) \quad 6x + 9 = -6x - 9$$

$$x = -\frac{3}{2}$$

$$(43) \quad 5x + 2 = 7x + 6$$

$$x = -2$$

$$(44) \quad -5x - 8 = 9x - 9$$

$$x = \frac{1}{14}$$

$$(45) \quad x + 5 = 8x + 3$$

$$x = \frac{2}{7}$$

$$(1) \quad 6x + 1 = -x - 5$$

$$x = -\frac{6}{7}$$

$$(2) \quad -9x + 7 = 7x + 2$$

$$x = \frac{5}{16}$$

$$(3) \quad 4x - 9 = 2x + 6$$

$$x = \frac{15}{2}$$

$$(4) \quad -8x - 4 = 8x + 3$$

$$x = -\frac{7}{16}$$

$$(5) \quad x - 6 = 8x + 5$$

$$x = -\frac{11}{7}$$

$$(6) \quad -8x + 9 = -5x + 7$$

$$x = \frac{2}{3}$$

$$(7) \quad 9x - 2 = 2x - 1$$

$$x = \frac{1}{7}$$

$$(8) \quad -9x - 9 = 2x - 9$$

$$x = 0$$

$$(9) \quad -x - 6 = 6x + 9$$

$$x = -\frac{15}{7}$$

$$(10) \quad x + 5 = 9x + 9$$

$$x = -\frac{1}{2}$$

$$(11) \quad 5x - 6 = -8x - 3$$

$$x = \frac{3}{13}$$

$$(12) \quad 7x - 3 = -6x - 4$$

$$x = -\frac{1}{13}$$

$$(13) \quad -7x - 2 = 6x - 6$$

$$x = \frac{4}{13}$$

$$(14) \quad -3x + 4 = -7x - 2$$

$$x = -\frac{3}{2}$$

$$(15) \quad -5x - 5 = -x - 8$$

$$x = \frac{3}{4}$$

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$$(17) \quad -x - 6 = 8x + 1$$

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$$(20) \quad -6x - 1 = x + 7$$

$$x = -\frac{8}{7}$$

$$(21) \quad 3x - 1 = -8x - 4$$

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$$(23) \quad -4x + 3 = -2x - 2$$

$$x = \frac{5}{2}$$

$$(24) \quad 3x - 2 = 5x + 7$$

$$x = -\frac{9}{2}$$

$$(25) \quad -2x - 2 = x - 2$$

$$x = 0$$

$$(26) \quad -x - 2 = -2x - 7$$

$$x = -5$$

$$(27) \quad x - 3 = 9x - 7$$

$$x = \frac{1}{2}$$

$$(28) \quad 6x - 1 = 3x + 5$$

$$x = 2$$

$$(29) \quad -2x + 9 = 8x - 8$$

$$x = \frac{17}{10}$$

$$(30) \quad -2x + 6 = -5x - 8$$

$$x = -\frac{14}{3}$$

$$(31) \quad -6x + 6 = -x + 9$$

$$x = -\frac{3}{5}$$

$$(32) \quad 9x - 2 = -8x + 4$$

$$x = \frac{6}{17}$$

$$(33) \quad -2x - 5 = 7x + 9$$

$$x = -\frac{14}{9}$$

$$(34) \quad 8x - 8 = 6x - 7$$

$$x = \frac{1}{2}$$

$$(35) \quad -7x - 5 = 6x - 2$$

$$x = -\frac{3}{13}$$

$$(36) \quad -6x + 8 = -4x + 1$$

$$x = \frac{7}{2}$$

$$(37) \quad -6x + 9 = -3x + 8$$

$$x = \frac{1}{3}$$

$$(38) \quad 5x - 4 = 8x - 4$$

$$x = 0$$

$$(39) \quad 4x + 9 = 5x - 9$$

$$x = 18$$

$$(40) \quad 5x + 7 = -5x + 3$$

$$x = -\frac{2}{5}$$

$$(41) \quad 5x + 2 = -9x - 6$$

$$x = -\frac{4}{7}$$

$$(42) \quad 6x + 1 = 7x - 6$$

$$x = 7$$

$$(43) \quad 7x - 3 = 5x + 1$$

$$x = 2$$

$$(44) \quad -5x + 8 = x + 6$$

$$x = \frac{1}{3}$$

$$(45) \quad -8x + 6 = 4x - 9$$

$$x = \frac{5}{4}$$

$$(1) \quad 8x - 9 = -7x - 3$$

$$x = \frac{2}{5}$$

$$(2) \quad -8x - 5 = -4x - 8$$

$$x = \frac{3}{4}$$

$$(3) \quad -6x + 3 = 5x + 9$$

$$x = -\frac{6}{11}$$

$$(4) \quad 7x + 6 = 5x - 2$$

$$x = -4$$

$$(5) \quad 7x + 6 = -3x + 1$$

$$x = -\frac{1}{2}$$

$$(6) \quad 2x - 7 = -7x - 1$$

$$x = \frac{2}{3}$$

$$(7) \quad 5x - 9 = x + 4$$

$$x = \frac{13}{4}$$

$$(8) \quad -5x + 8 = 6x - 3$$

$$x = 1$$

$$(9) \quad -8x - 6 = 8x - 1$$

$$x = -\frac{5}{16}$$

$$(10) \quad -2x + 7 = 6x + 6$$

$$x = \frac{1}{8}$$

$$(11) \quad -6x + 6 = 4x + 8$$

$$x = -\frac{1}{5}$$

$$(12) \quad 4x - 7 = 2x + 9$$

$$x = 8$$

$$(13) \quad x + 8 = -5x + 3$$

$$x = -\frac{5}{6}$$

$$(14) \quad 2x - 5 = -x + 2$$

$$x = \frac{7}{3}$$

$$(15) \quad 5x + 4 = -8x + 7$$

$$x = \frac{3}{13}$$

$$(16) \quad -8x + 7 = x + 1$$

$$x = \frac{2}{3}$$

$$(17) \quad -4x + 6 = 8x - 4$$

$$x = \frac{5}{6}$$

$$(18) \quad -7x + 7 = 5x - 7$$

$$x = \frac{7}{6}$$

$$(19) \quad -3x - 3 = -x + 4$$

$$x = -\frac{7}{2}$$

$$(20) \quad 9x + 3 = x + 5$$

$$x = \frac{1}{4}$$

$$(21) \quad 8x - 4 = 9x + 1$$

$$x = -5$$

$$(22) \quad -4x + 5 = 2x - 9$$

$$x = \frac{7}{3}$$

$$(23) \quad x - 4 = -6x - 6$$

$$x = -\frac{2}{7}$$

$$(24) \quad -5x - 6 = 3x + 5$$

$$x = -\frac{11}{8}$$

$$(25) \quad -5x - 4 = -4x + 1$$

$$x = -5$$

$$(26) \quad 8x + 5 = -5x - 3$$

$$x = -\frac{8}{13}$$

$$(27) \quad 2x + 9 = -9x + 3$$

$$x = -\frac{6}{11}$$

$$(28) \quad 4x + 5 = 7x - 1$$

$$x = 2$$

$$(29) \quad 6x + 1 = 3x - 2$$

$$x = -1$$

$$(30) \quad -7x - 5 = 8x - 3$$

$$x = -\frac{2}{15}$$

$$(31) \quad 5x + 3 = -7x + 4$$

$$x = \frac{1}{12}$$

$$(32) \quad 6x - 6 = -2x - 3$$

$$x = \frac{3}{8}$$

$$(33) \quad -5x + 8 = 8x - 5$$

$$x = 1$$

$$(34) \quad -9x - 3 = 8x - 6$$

$$x = \frac{3}{17}$$

$$(35) \quad 5x - 8 = 4x + 7$$

$$x = 15$$

$$(36) \quad 8x + 2 = 9x - 6$$

$$x = 8$$

$$(37) \quad 5x - 9 = 6x - 4$$

$$x = -5$$

$$(38) \quad 5x + 8 = -7x - 4$$

$$x = -1$$

$$(39) \quad 9x - 7 = -x - 7$$

$$x = 0$$

$$(40) \quad -2x + 3 = 4x - 5$$

$$x = \frac{4}{3}$$

$$(41) \quad 3x - 9 = -x - 2$$

$$x = \frac{7}{4}$$

$$(42) \quad -7x + 8 = -8x + 3$$

$$x = -5$$

$$(43) \quad 4x - 7 = x + 4$$

$$x = \frac{11}{3}$$

$$(44) \quad 3x + 7 = 5x + 6$$

$$x = \frac{1}{2}$$

$$(45) \quad 4x - 9 = -6x - 9$$

$$x = 0$$

$$(1) \quad 8x - 2 = 4x + 3$$

$$x = \frac{5}{4}$$

$$(2) \quad x - 4 = 7x - 4$$

$$x = 0$$

$$(3) \quad -6x - 8 = 8x - 3$$

$$x = -\frac{5}{14}$$

$$(4) \quad -5x - 1 = 2x + 8$$

$$x = -\frac{9}{7}$$

$$(5) \quad 4x + 5 = -3x + 4$$

$$x = -\frac{1}{7}$$

$$(6) \quad 3x - 3 = -5x - 7$$

$$x = -\frac{1}{2}$$

$$(7) \quad -8x + 9 = 3x + 9$$

$$x = 0$$

$$(8) \quad -4x - 9 = 7x - 3$$

$$x = -\frac{6}{11}$$

$$(9) \quad -2x - 2 = 2x + 8$$

$$x = -\frac{5}{2}$$

$$(10) \quad -x - 8 = -3x - 2$$

$$x = 3$$

$$(11) \quad x + 2 = 4x - 8$$

$$x = \frac{10}{3}$$

$$(12) \quad 8x + 5 = -2x - 8$$

$$x = -\frac{13}{10}$$

$$(13) \quad 4x + 6 = 7x + 4$$

$$x = \frac{2}{3}$$

$$(14) \quad 3x - 2 = -6x - 5$$

$$x = -\frac{1}{3}$$

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$$x = -\frac{9}{14}$$

$$(45) \quad 4x - 6 = -7x + 5$$

$$x = 1$$

$$(1) -7x - 6 = 7x + 9$$

$$x = -\frac{15}{14}$$

$$(2) -2x + 8 = -3x - 1$$

$$x = -9$$

$$(3) 9x + 3 = 2x + 5$$

$$x = \frac{2}{7}$$

$$(4) 3x + 6 = 5x + 5$$

$$x = \frac{1}{2}$$

$$(5) -2x - 3 = -5x - 9$$

$$x = -2$$

$$(6) 2x - 2 = -9x + 2$$

$$x = \frac{4}{11}$$

$$(7) -2x - 3 = -x - 3$$

$$x = 0$$

$$(8) -5x - 8 = 5x - 8$$

$$x = 0$$

$$(9) 7x + 2 = x - 1$$

$$x = -\frac{1}{2}$$

$$(10) 3x - 8 = -3x - 7$$

$$x = \frac{1}{6}$$

$$(11) 2x - 9 = 3x - 8$$

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$$(12) -9x + 7 = -2x - 5$$

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$$(4) \quad 5x - 1 = 4x + 4$$

$$x = 5$$

$$(5) \quad 3x + 2 = -3x + 2$$

$$x = 0$$

$$(6) \quad 3x - 8 = 6x + 2$$

$$x = -\frac{10}{3}$$

$$(7) \quad 3x - 2 = -9x + 1$$

$$x = \frac{1}{4}$$

$$(8) \quad x + 9 = -4x + 1$$

$$x = -\frac{8}{5}$$

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$$(3) \quad 8x + 6 = -8x + 8$$

$$x = \frac{1}{8}$$

$$(4) \quad 9x + 6 = -7x - 5$$

$$x = -\frac{11}{16}$$

$$(5) \quad -x + 9 = 3x + 3$$

$$x = \frac{3}{2}$$

$$(6) \quad -x - 4 = -9x + 8$$

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$$(23) \quad 7x - 2 = 2x + 3$$

$$x = 1$$

$$(24) \quad 3x - 5 = -2x + 6$$

$$x = \frac{11}{5}$$

$$(25) \quad -5x + 5 = 8x - 8$$

$$x = 1$$

$$(26) \quad 2x + 6 = 5x + 2$$

$$x = \frac{4}{3}$$

$$(27) \quad -8x + 8 = -2x - 2$$

$$x = \frac{5}{3}$$

$$(28) \quad -8x - 5 = 3x - 5$$

$$x = 0$$

$$(29) \quad 5x + 7 = -4x + 9$$

$$x = \frac{2}{9}$$

$$(30) \quad -9x + 5 = x - 2$$

$$x = \frac{7}{10}$$

$$(31) \quad -6x - 8 = 3x - 5$$

$$x = -\frac{1}{3}$$

$$(32) \quad -8x - 2 = 9x - 2$$

$$x = 0$$

$$(33) \quad 7x + 5 = -9x - 3$$

$$x = -\frac{1}{2}$$

$$(34) \quad -5x - 8 = -4x + 2$$

$$x = -10$$

$$(35) \quad 6x + 6 = -5x - 5$$

$$x = -1$$

$$(36) \quad -2x + 3 = 5x - 3$$

$$x = \frac{6}{7}$$

$$(37) \quad 8x - 3 = 3x + 8$$

$$x = \frac{11}{5}$$

$$(38) \quad 5x + 3 = 2x - 5$$

$$x = -\frac{8}{3}$$

$$(39) \quad -2x + 2 = 7x - 5$$

$$x = \frac{7}{9}$$

$$(40) \quad 4x + 6 = 9x + 4$$

$$x = \frac{2}{5}$$

$$(41) \quad 4x + 1 = -5x - 6$$

$$x = -\frac{7}{9}$$

$$(42) \quad 5x - 2 = -9x - 6$$

$$x = -\frac{2}{7}$$

$$(43) \quad 3x - 1 = -3x - 8$$

$$x = -\frac{7}{6}$$

$$(44) \quad 8x - 6 = 4x - 7$$

$$x = -\frac{1}{4}$$

$$(45) \quad -5x + 1 = -3x + 9$$

$$x = -4$$

$$(1) \quad -2x - 4 = 5x - 3$$

$$x = -\frac{1}{7}$$

$$(2) \quad 7x + 9 = -5x + 1$$

$$x = -\frac{2}{3}$$

$$(3) \quad -2x + 6 = -6x - 4$$

$$x = -\frac{5}{2}$$

$$(4) \quad -7x + 8 = -4x + 3$$

$$x = \frac{5}{3}$$

$$(5) \quad x - 7 = 5x + 4$$

$$x = -\frac{11}{4}$$

$$(6) \quad 8x - 8 = -9x - 1$$

$$x = \frac{7}{17}$$

$$(7) \quad -5x + 2 = 7x - 3$$

$$x = \frac{5}{12}$$

$$(8) \quad -7x + 2 = -4x - 1$$

$$x = 1$$

$$(9) \quad -8x + 4 = 2x - 7$$

$$x = \frac{11}{10}$$

$$(10) \quad -2x + 5 = -9x + 9$$

$$x = \frac{4}{7}$$

$$(11) \quad -6x - 3 = -2x + 5$$

$$x = -2$$

$$(12) \quad -9x - 7 = 4x + 8$$

$$x = -\frac{15}{13}$$

$$(13) \quad -x - 9 = -5x + 2$$

$$x = \frac{11}{4}$$

$$(14) \quad 3x + 5 = 4x + 8$$

$$x = -3$$

$$(15) \quad -9x - 6 = 5x - 3$$

$$x = -\frac{3}{14}$$

$$(16) \quad -8x - 2 = -5x + 7$$

$$x = -3$$

$$(17) \quad -3x - 5 = -x - 9$$

$$x = 2$$

$$(18) \quad -6x - 4 = -5x + 2$$

$$x = -6$$

$$(19) \quad -5x - 6 = -9x - 7$$

$$x = -\frac{1}{4}$$

$$(20) \quad -7x + 1 = -4x + 4$$

$$x = -1$$

$$(21) \quad 8x + 3 = 9x + 9$$

$$x = -6$$

$$(22) \quad 8x - 5 = -6x + 5$$

$$x = \frac{5}{7}$$

$$(23) \quad 6x - 1 = 3x + 6$$

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$$(28) \quad 4x - 1 = 9x - 7$$

$$x = \frac{6}{5}$$

$$(29) \quad -5x + 4 = 6x + 7$$

$$x = -\frac{3}{11}$$

$$(30) \quad 3x + 5 = -6x + 6$$

$$x = \frac{1}{9}$$

$$(31) \quad 2x - 4 = -6x - 4$$

$$x = 0$$

$$(32) \quad 8x + 6 = 7x - 8$$

$$x = -14$$

$$(33) \quad -2x + 4 = -7x - 4$$

$$x = -\frac{8}{5}$$

$$(34) \quad -4x - 4 = -x - 3$$

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$$(35) \quad -6x - 3 = -8x - 1$$

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$$x = -\frac{2}{9}$$

$$(45) \quad 9x + 6 = -7x + 2$$

$$x = -\frac{1}{4}$$

$$(1) \quad 3x - 5 = -x - 1$$

$$x = 1$$

$$(2) \quad -5x - 1 = -6x + 9$$

$$x = 10$$

$$(3) \quad -9x + 4 = -6x + 1$$

$$x = 1$$

$$(4) \quad 9x + 8 = 8x + 7$$

$$x = -1$$

$$(5) \quad 9x + 8 = 6x - 4$$

$$x = -4$$

$$(6) \quad -x - 6 = x + 9$$

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$$(2) \quad -8x + 8 = 4x + 7$$

$$x = \frac{1}{12}$$

$$(3) \quad 3x + 3 = -2x + 8$$

$$x = 1$$

$$(4) \quad -4x - 4 = -5x - 3$$

$$x = 1$$

$$(5) \quad -5x + 4 = x - 5$$

$$x = \frac{3}{2}$$

$$(6) \quad -9x + 2 = 5x - 4$$

$$x = \frac{3}{7}$$

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$$x = \frac{1}{2}$$

$$(6) \quad 9x + 9 = 2x + 7$$

$$x = -\frac{2}{7}$$

$$(7) \quad 6x + 8 = -5x - 9$$

$$x = -\frac{17}{11}$$

$$(8) \quad 2x + 3 = -6x + 6$$

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$$x = \frac{5}{4}$$

$$(7) 9x - 2 = -7x + 6$$

$$x = \frac{1}{2}$$

$$(8) -2x - 3 = 7x + 4$$

$$x = -\frac{7}{9}$$

$$(9) 5x + 6 = 2x + 8$$

$$x = \frac{2}{3}$$

$$(10) x - 5 = -x - 1$$

$$x = 2$$

$$(11) -x - 3 = -2x - 3$$

$$x = 0$$

$$(12) -7x - 4 = 8x - 8$$

$$x = \frac{4}{15}$$

$$(13) -9x - 9 = -6x - 9$$

$$x = 0$$

$$(14) 8x + 4 = -2x + 7$$

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$$x = 1$$

$$(34) 2x + 5 = -4x - 2$$

$$x = -\frac{7}{6}$$

$$(35) -3x + 7 = 9x - 6$$

$$x = \frac{13}{12}$$

$$(36) -6x + 9 = -7x + 7$$

$$x = -2$$

$$(37) 2x + 2 = -5x - 5$$

$$x = -1$$

$$(38) -x - 9 = 2x + 1$$

$$x = -\frac{10}{3}$$

$$(39) -5x + 6 = -6x - 4$$

$$x = -10$$

$$(40) 6x - 3 = 9x + 7$$

$$x = -\frac{10}{3}$$

$$(41) x + 6 = 4x - 4$$

$$x = \frac{10}{3}$$

$$(42) 6x + 4 = -9x + 9$$

$$x = \frac{1}{3}$$

$$(43) -5x - 9 = -8x + 1$$

$$x = \frac{10}{3}$$

$$(44) 4x + 4 = 7x + 3$$

$$x = \frac{1}{3}$$

$$(45) -2x + 6 = x - 6$$

$$x = 4$$

$$(1) \quad x + 5 = -4x + 2$$

$$x = -\frac{3}{5}$$

$$(2) \quad 2x - 9 = -5x + 5$$

$$x = 2$$

$$(3) \quad -8x - 6 = 2x + 4$$

$$x = -1$$

$$(4) \quad 7x + 1 = 4x - 1$$

$$x = -\frac{2}{3}$$

$$(5) \quad 8x + 6 = x - 6$$

$$x = -\frac{12}{7}$$

$$(6) \quad 5x + 2 = -2x + 7$$

$$x = \frac{5}{7}$$

$$(7) \quad 6x - 1 = 5x - 4$$

$$x = -3$$

$$(8) \quad -2x + 7 = -7x + 5$$

$$x = -\frac{2}{5}$$

$$(9) \quad 4x + 9 = -9x - 9$$

$$x = -\frac{18}{13}$$

$$(10) \quad 2x - 3 = 3x + 1$$

$$x = -4$$

$$(11) \quad -8x - 1 = -9x - 8$$

$$x = -7$$

$$(12) \quad 6x + 8 = 4x - 6$$

$$x = -7$$

$$(13) \quad 9x - 6 = -4x + 4$$

$$x = \frac{10}{13}$$

$$(14) \quad -4x - 9 = 2x - 3$$

$$x = -1$$

$$(15) \quad 5x + 5 = 2x + 9$$

$$x = \frac{4}{3}$$

$$(16) \quad 7x - 3 = 5x + 2$$

$$x = \frac{5}{2}$$

$$(17) \quad 4x - 7 = -9x - 8$$

$$x = -\frac{1}{13}$$

$$(18) \quad -7x - 8 = 7x + 7$$

$$x = -\frac{15}{14}$$

$$(19) \quad -x - 6 = 3x + 8$$

$$x = -\frac{7}{2}$$

$$(20) \quad 9x + 1 = 5x - 9$$

$$x = -\frac{5}{2}$$

$$(21) \quad 4x + 9 = -3x + 8$$

$$x = -\frac{1}{7}$$

$$(22) \quad 6x - 2 = -4x - 8$$

$$x = -\frac{3}{5}$$

$$(23) \quad -4x - 2 = -8x - 3$$

$$x = -\frac{1}{4}$$

$$(24) \quad 4x - 1 = 6x + 6$$

$$x = -\frac{7}{2}$$

$$(25) \quad 5x - 7 = 2x - 8$$

$$x = -\frac{1}{3}$$

$$(26) \quad -3x + 4 = 8x - 1$$

$$x = \frac{5}{11}$$

$$(27) \quad 8x - 4 = 5x + 7$$

$$x = \frac{11}{3}$$

$$(28) \quad -6x + 9 = -5x - 1$$

$$x = 10$$

$$(29) \quad 2x - 6 = -2x + 2$$

$$x = 2$$

$$(30) \quad 8x + 4 = -8x - 1$$

$$x = -\frac{5}{16}$$

$$(31) \quad 8x + 6 = 5x + 1$$

$$x = -\frac{5}{3}$$

$$(32) \quad 5x - 4 = -3x - 8$$

$$x = -\frac{1}{2}$$

$$(33) \quad -9x - 8 = -2x - 7$$

$$x = -\frac{1}{7}$$

$$(34) \quad -4x + 2 = 4x - 3$$

$$x = \frac{5}{8}$$

$$(35) \quad -x + 9 = x + 8$$

$$x = \frac{1}{2}$$

$$(36) \quad 9x - 5 = -x + 7$$

$$x = \frac{6}{5}$$

$$(37) \quad -4x - 2 = -7x - 6$$

$$x = -\frac{4}{3}$$

$$(38) \quad -6x + 8 = -9x - 6$$

$$x = -\frac{14}{3}$$

$$(39) \quad 6x + 6 = 9x - 1$$

$$x = \frac{7}{3}$$

$$(40) \quad -8x + 4 = 7x - 9$$

$$x = \frac{13}{15}$$

$$(41) \quad -3x - 5 = 6x + 3$$

$$x = -\frac{8}{9}$$

$$(42) \quad -7x + 6 = 4x + 9$$

$$x = -\frac{3}{11}$$

$$(43) \quad 7x + 9 = 8x + 4$$

$$x = 5$$

$$(44) \quad 9x + 9 = 5x + 2$$

$$x = -\frac{7}{4}$$

$$(45) \quad 6x - 3 = -8x - 5$$

$$x = -\frac{1}{7}$$

$$(1) \quad 6x + 5 = x - 8$$

$$x = -\frac{13}{5}$$

$$(2) \quad 4x + 5 = -7x - 4$$

$$x = -\frac{9}{11}$$

$$(3) \quad 3x - 8 = 6x + 9$$

$$x = -\frac{17}{3}$$

$$(4) \quad 6x + 8 = -8x + 1$$

$$x = -\frac{1}{2}$$

$$(5) \quad 6x - 5 = 8x - 4$$

$$x = -\frac{1}{2}$$

$$(6) \quad -x + 5 = x - 3$$

$$x = 4$$

$$(7) \quad 2x + 2 = -7x + 2$$

$$x = 0$$

$$(8) \quad -9x + 3 = -2x - 9$$

$$x = \frac{12}{7}$$

$$(9) \quad 7x - 1 = -8x - 9$$

$$x = -\frac{8}{15}$$

$$(10) \quad -4x - 6 = -9x + 2$$

$$x = \frac{8}{5}$$

$$(11) \quad -2x + 8 = 6x - 2$$

$$x = \frac{5}{4}$$

$$(12) \quad -7x + 3 = x - 4$$

$$x = \frac{7}{8}$$

$$(13) \quad -3x - 1 = -2x + 6$$

$$x = -7$$

$$(14) \quad 9x - 4 = -x - 6$$

$$x = -\frac{1}{5}$$

$$(15) \quad -6x + 5 = 5x - 2$$

$$x = \frac{7}{11}$$

$$(16) \quad 8x - 3 = -2x - 3$$

$$x = 0$$

$$(17) \quad -8x + 5 = -5x + 3$$

$$x = \frac{2}{3}$$

$$(18) \quad -2x - 1 = -x + 9$$

$$x = -10$$

$$(19) \quad 5x - 4 = 8x - 3$$

$$x = -\frac{1}{3}$$

$$(20) \quad 3x - 5 = 8x - 6$$

$$x = \frac{1}{5}$$

$$(21) \quad -9x - 4 = 6x - 5$$

$$x = \frac{1}{15}$$

$$(22) \quad 6x + 1 = -7x - 5$$

$$x = -\frac{6}{13}$$

$$(23) \quad -3x - 4 = 3x - 1$$

$$x = -\frac{1}{2}$$

$$(24) \quad 3x + 2 = 2x + 5$$

$$x = 3$$

$$(25) \quad 6x - 9 = 3x - 9$$

$$x = 0$$

$$(26) \quad -9x - 3 = -x - 9$$

$$x = \frac{3}{4}$$

$$(27) \quad -4x - 3 = 9x - 9$$

$$x = \frac{6}{13}$$

$$(28) \quad 4x - 9 = -9x - 9$$

$$x = 0$$

$$(29) \quad 3x + 8 = -6x + 4$$

$$x = -\frac{4}{9}$$

$$(30) \quad 7x - 3 = 4x - 1$$

$$x = \frac{2}{3}$$

$$(31) \quad 2x + 5 = x + 1$$

$$x = -4$$

$$(32) \quad -2x - 9 = x + 2$$

$$x = -\frac{11}{3}$$

$$(33) \quad 3x + 7 = -4x + 1$$

$$x = -\frac{6}{7}$$

$$(34) \quad 2x + 9 = -8x + 5$$

$$x = -\frac{2}{5}$$

$$(35) \quad -7x + 7 = -2x + 4$$

$$x = \frac{3}{5}$$

$$(36) \quad -4x + 6 = x + 3$$

$$x = \frac{3}{5}$$

$$(37) \quad 5x + 6 = 9x + 4$$

$$x = \frac{1}{2}$$

$$(38) \quad -6x + 1 = 3x + 2$$

$$x = -\frac{1}{9}$$

$$(39) \quad 3x + 9 = x + 9$$

$$x = 0$$

$$(40) \quad x + 5 = 8x - 6$$

$$x = \frac{11}{7}$$

$$(41) \quad -2x + 3 = 2x + 2$$

$$x = \frac{1}{4}$$

$$(42) \quad 4x - 6 = 8x - 3$$

$$x = -\frac{3}{4}$$

$$(43) \quad -9x + 8 = -2x - 1$$

$$x = \frac{9}{7}$$

$$(44) \quad -7x + 4 = -x - 5$$

$$x = \frac{3}{2}$$

$$(45) \quad -3x + 8 = x + 7$$

$$x = \frac{1}{4}$$

$$(1) \quad -7x - 7 = x - 5$$

$$x = -\frac{1}{4}$$

$$(2) \quad 8x - 1 = 2x - 3$$

$$x = -\frac{1}{3}$$

$$(3) \quad 6x + 7 = 2x - 3$$

$$x = -\frac{5}{2}$$

$$(4) \quad -5x + 4 = -3x + 2$$

$$x = 1$$

$$(5) \quad 9x + 8 = x - 7$$

$$x = -\frac{15}{8}$$

$$(6) \quad x - 4 = 6x - 3$$

$$x = -\frac{1}{5}$$

$$(7) \quad -3x + 5 = 4x + 5$$

$$x = 0$$

$$(8) \quad 5x + 8 = -3x + 9$$

$$x = \frac{1}{8}$$

$$(9) \quad 9x - 2 = -3x - 9$$

$$x = -\frac{7}{12}$$

$$(10) \quad 2x - 8 = -6x + 1$$

$$x = \frac{9}{8}$$

$$(11) \quad -2x + 5 = 2x + 6$$

$$x = -\frac{1}{4}$$

$$(12) \quad -2x + 9 = 2x - 1$$

$$x = \frac{5}{2}$$

$$(13) \quad x + 3 = -9x + 2$$

$$x = -\frac{1}{10}$$

$$(14) \quad 7x + 5 = -2x - 9$$

$$x = -\frac{14}{9}$$

$$(15) \quad -x - 7 = -6x + 3$$

$$x = 2$$

$$(16) \quad -3x + 7 = 5x - 7$$

$$x = \frac{7}{4}$$

$$(17) \quad 6x + 7 = -2x + 3$$

$$x = -\frac{1}{2}$$

$$(18) \quad 7x + 8 = -3x - 1$$

$$x = -\frac{9}{10}$$

$$(19) \quad 8x + 6 = x - 3$$

$$x = -\frac{9}{7}$$

$$(20) \quad -8x - 5 = 2x - 3$$

$$x = -\frac{1}{5}$$

$$(21) \quad 3x - 8 = 6x + 7$$

$$x = -5$$

$$(22) \quad -5x + 6 = -6x + 6$$

$$x = 0$$

$$(23) \quad 7x - 3 = 6x + 7$$

$$x = 10$$

$$(24) \quad 5x - 3 = 9x - 4$$

$$x = \frac{1}{4}$$

$$(25) \quad -7x - 1 = 5x - 6$$

$$x = \frac{5}{12}$$

$$(26) \quad -8x + 8 = 2x - 5$$

$$x = \frac{13}{10}$$

$$(27) \quad 2x - 6 = 8x - 9$$

$$x = \frac{1}{2}$$

$$(28) \quad 2x + 3 = 3x + 8$$

$$x = -5$$

$$(29) \quad -3x + 1 = -4x - 3$$

$$x = -4$$

$$(30) \quad -6x + 4 = -3x - 7$$

$$x = \frac{11}{3}$$

$$(31) \quad 7x + 9 = 8x - 4$$

$$x = 13$$

$$(32) \quad 3x - 9 = -6x + 5$$

$$x = \frac{14}{9}$$

$$(33) \quad 9x - 1 = -4x + 1$$

$$x = \frac{2}{13}$$

$$(34) \quad 7x - 3 = -2x - 9$$

$$x = -\frac{2}{3}$$

$$(35) \quad -9x - 5 = -8x + 2$$

$$x = -7$$

$$(36) \quad -5x + 6 = 5x - 3$$

$$x = \frac{9}{10}$$

$$(37) \quad 5x + 9 = 6x - 1$$

$$x = 10$$

$$(38) \quad -2x + 8 = -9x + 9$$

$$x = \frac{1}{7}$$

$$(39) \quad -2x - 8 = -3x - 5$$

$$x = 3$$

$$(40) \quad 3x + 3 = 9x + 5$$

$$x = -\frac{1}{3}$$

$$(41) \quad -9x + 4 = 3x - 5$$

$$x = \frac{3}{4}$$

$$(42) \quad 9x + 4 = -7x - 5$$

$$x = -\frac{9}{16}$$

$$(43) \quad 5x - 6 = 3x + 9$$

$$x = \frac{15}{2}$$

$$(44) \quad -4x - 3 = 5x + 6$$

$$x = -1$$

$$(45) \quad 8x - 6 = -x - 9$$

$$x = -\frac{1}{3}$$

2.9 2 次方程式 因数分解利用

2 次式の因数分解を利用した方程式の問題。

(1) $6x^2 - 7x - 3 = 0$

(11) $6x^2 + 35x + 25 = 0$

(21) $x^2 + x - 56 = 0$

(2) $12x^2 + 29x + 14 = 0$

(12) $8x^2 + 22x + 5 = 0$

(22) $4x^2 + 5x - 9 = 0$

(3) $45x^2 + 98x + 49 = 0$

(13) $4x^2 + 3x - 1 = 0$

(23) $10x^2 - 17x + 6 = 0$

(4) $16x^2 + 66x + 35 = 0$

(14) $15x^2 - 4x - 4 = 0$

(24) $3x^2 - 11x + 6 = 0$

(5) $18x^2 - 21x + 5 = 0$

(15) $24x^2 - 14x - 3 = 0$

(25) $24x^2 - 11x + 1 = 0$

(6) $20x^2 - 11x - 45 = 0$

(16) $2x^2 - 9x + 10 = 0$

(26) $49x^2 + 21x - 54 = 0$

(7) $5x^2 - 17x + 6 = 0$

(17) $25x^2 - 30x - 27 = 0$

(27) $9x^2 - 37x + 4 = 0$

(8) $x^2 - 5x - 6 = 0$

(18) $15x^2 - 4x - 3 = 0$

(28) $16x^2 + 14x - 15 = 0$

(9) $7x^2 + 13x + 6 = 0$

(19) $16x^2 + 6x - 1 = 0$

(29) $5x^2 - 2x - 16 = 0$

(10) $3x^2 - 26x + 16 = 0$

(20) $2x^2 - 15x + 18 = 0$

(30) $4x^2 + 3x - 7 = 0$

(1) $40x^2 + 31x - 35 = 0$

(11) $7x^2 - 12x + 5 = 0$

(21) $9x^2 + 10x + 1 = 0$

(2) $30x^2 - 43x - 8 = 0$

(12) $6x^2 - 11x + 3 = 0$

(22) $10x^2 + 13x + 4 = 0$

(3) $9x^2 + 46x + 5 = 0$

(13) $5x^2 + 8x + 3 = 0$

(23) $54x^2 - 87x + 35 = 0$

(4) $9x^2 - 28x + 3 = 0$

(14) $4x^2 - 7x + 3 = 0$

(24) $18x^2 + 7x - 8 = 0$

(5) $28x^2 - 41x - 14 = 0$

(15) $18x^2 + 27x - 56 = 0$

(25) $32x^2 - 36x - 5 = 0$

(6) $27x^2 + 15x + 2 = 0$

(16) $16x^2 + 40x + 9 = 0$

(26) $56x^2 - 37x + 6 = 0$

(7) $5x^2 + 19x + 12 = 0$

(17) $2x^2 - 5x - 12 = 0$

(27) $8x^2 + 39x + 28 = 0$

(8) $14x^2 + 47x - 72 = 0$

(18) $4x^2 - 11x - 20 = 0$

(28) $8x^2 - 57x + 54 = 0$

(9) $6x^2 + 13x - 15 = 0$

(19) $21x^2 + 10x - 16 = 0$

(29) $36x^2 + 85x + 9 = 0$

(10) $14x^2 + 17x - 6 = 0$

(20) $7x^2 + 22x + 3 = 0$

(30) $63x^2 + 25x + 2 = 0$

(1) $6x^2 - 5x - 21 = 0$

(11) $3x^2 - 8x + 4 = 0$

(21) $25x^2 - 50x + 21 = 0$

(2) $12x^2 + 35x + 8 = 0$

(12) $15x^2 - 7x - 36 = 0$

(22) $45x^2 + 31x - 4 = 0$

(3) $4x^2 - 11x + 7 = 0$

(13) $7x^2 - 8x + 1 = 0$

(23) $6x^2 - 11x + 3 = 0$

(4) $21x^2 - 64x + 35 = 0$

(14) $56x^2 + 57x + 7 = 0$

(24) $x^2 - x - 2 = 0$

(5) $2x^2 - 13x + 21 = 0$

(15) $7x^2 + 27x + 18 = 0$

(25) $8x^2 - 6x + 1 = 0$

(6) $28x^2 + 29x - 9 = 0$

(16) $15x^2 - 7x - 2 = 0$

(26) $16x^2 + 26x + 9 = 0$

(7) $28x^2 + 57x + 27 = 0$

(17) $10x^2 - 7x - 12 = 0$

(27) $64x^2 + 96x + 35 = 0$

(8) $2x^2 + 17x + 8 = 0$

(18) $x^2 - 1 = 0$

(28) $8x^2 - 7x - 1 = 0$

(9) $3x^2 + 4x + 1 = 0$

(19) $64x^2 + 32x - 21 = 0$

(29) $5x^2 - 6x + 1 = 0$

(10) $63x^2 - 23x + 2 = 0$

(20) $9x^2 + 29x + 6 = 0$

(30) $3x^2 + 7x + 4 = 0$

(1) $7x^2 - 13x - 24 = 0$

(11) $16x^2 - 6x - 7 = 0$

(21) $24x^2 - 47x + 20 = 0$

(2) $x^2 + 7x + 12 = 0$

(12) $5x^2 + 14x - 3 = 0$

(22) $x^2 - 8x + 15 = 0$

(3) $12x^2 + 13x + 3 = 0$

(13) $72x^2 - 119x + 49 = 0$

(23) $x^2 - 1 = 0$

(4) $24x^2 - 23x + 5 = 0$

(14) $15x^2 - 29x - 14 = 0$

(24) $20x^2 - 43x + 21 = 0$

(5) $6x^2 - 25x + 14 = 0$

(15) $15x^2 - 34x + 15 = 0$

(25) $49x^2 - 25 = 0$

(6) $45x^2 + 17x - 14 = 0$

(16) $6x^2 + 13x - 5 = 0$

(26) $24x^2 + 25x + 6 = 0$

(7) $8x^2 - 18x + 9 = 0$

(17) $10x^2 - 33x - 54 = 0$

(27) $x^2 - 2x - 8 = 0$

(8) $18x^2 - 13x + 2 = 0$

(18) $8x^2 - 27x - 20 = 0$

(28) $7x^2 - 13x - 2 = 0$

(9) $5x^2 + 16x + 12 = 0$

(19) $42x^2 + 5x - 2 = 0$

(29) $9x^2 - 24x + 7 = 0$

(10) $2x^2 - 17x + 35 = 0$

(20) $56x^2 + 53x + 12 = 0$

(30) $9x^2 + 14x + 5 = 0$

(1) $8x^2 - 5x - 3 = 0$

(11) $3x^2 + 22x + 24 = 0$

(21) $21x^2 + x - 2 = 0$

(2) $24x^2 - 7x - 6 = 0$

(12) $8x^2 + 22x + 5 = 0$

(22) $63x^2 + 82x + 24 = 0$

(3) $36x^2 - 11x - 12 = 0$

(13) $2x^2 - 17x - 9 = 0$

(23) $14x^2 - 31x - 63 = 0$

(4) $10x^2 + 29x + 10 = 0$

(14) $63x^2 - 68x + 12 = 0$

(24) $5x^2 + 13x - 6 = 0$

(5) $x^2 + 11x + 18 = 0$

(15) $21x^2 - 2x - 8 = 0$

(25) $14x^2 - 27x + 9 = 0$

(6) $2x^2 + 7x - 4 = 0$

(16) $63x^2 - 73x + 20 = 0$

(26) $3x^2 - 14x + 8 = 0$

(7) $4x^2 + 5x + 1 = 0$

(17) $9x^2 + 10x + 1 = 0$

(27) $5x^2 + 41x + 42 = 0$

(8) $4x^2 - 8x + 3 = 0$

(18) $x^2 - x - 6 = 0$

(28) $24x^2 - 31x + 10 = 0$

(9) $6x^2 + 11x + 3 = 0$

(19) $5x^2 - 6x + 1 = 0$

(29) $35x^2 + 58x - 9 = 0$

(10) $12x^2 - 4x - 1 = 0$

(20) $12x^2 - 17x - 7 = 0$

(30) $x^2 - 8x + 12 = 0$

(1) $21x^2 - 46x + 24 = 0$

(11) $14x^2 - 3x - 27 = 0$

(21) $x^2 + 6x + 5 = 0$

(2) $63x^2 - 20x - 3 = 0$

(12) $56x^2 - 111x + 54 = 0$

(22) $72x^2 + x - 56 = 0$

(3) $63x^2 + 53x - 36 = 0$

(13) $2x^2 + x - 1 = 0$

(23) $18x^2 + 7x - 1 = 0$

(4) $35x^2 + 38x - 9 = 0$

(14) $4x^2 + 21x - 18 = 0$

(24) $16x^2 - 46x - 35 = 0$

(5) $40x^2 - 9x - 9 = 0$

(15) $7x^2 + 12x - 4 = 0$

(25) $2x^2 + 7x - 72 = 0$

(6) $x^2 - 7x - 18 = 0$

(16) $48x^2 + 50x - 7 = 0$

(26) $8x^2 + 2x - 3 = 0$

(7) $4x^2 - 3x - 1 = 0$

(17) $18x^2 + 17x - 15 = 0$

(27) $21x^2 - 4x - 32 = 0$

(8) $4x^2 + x - 3 = 0$

(18) $3x^2 - x - 2 = 0$

(28) $x^2 - 6x + 8 = 0$

(9) $64x^2 + 112x + 45 = 0$

(19) $27x^2 - 3x - 14 = 0$

(29) $6x^2 - 5x - 4 = 0$

(10) $x^2 - 2x + 1 = 0$

(20) $8x^2 - 17x + 9 = 0$

(30) $12x^2 + 23x - 24 = 0$

(1) $4x^2 - 15x - 4 = 0$

(11) $42x^2 + 17x - 4 = 0$

(21) $14x^2 - 17x - 6 = 0$

(2) $12x^2 - 13x - 35 = 0$

(12) $2x^2 + 3x - 5 = 0$

(22) $3x^2 + 2x - 8 = 0$

(3) $30x^2 + 13x - 3 = 0$

(13) $2x^2 - 13x - 24 = 0$

(23) $15x^2 + 32x + 9 = 0$

(4) $72x^2 + 23x - 4 = 0$

(14) $x^2 + 9x + 8 = 0$

(24) $7x^2 + 11x + 4 = 0$

(5) $x^2 - 1 = 0$

(15) $3x^2 - 2x - 1 = 0$

(25) $10x^2 - 21x + 8 = 0$

(6) $9x^2 + 30x + 16 = 0$

(16) $6x^2 + 53x + 40 = 0$

(26) $2x^2 - 3x + 1 = 0$

(7) $14x^2 + 37x + 24 = 0$

(17) $56x^2 + 65x + 14 = 0$

(27) $8x^2 - 63x - 81 = 0$

(8) $21x^2 - 76x + 63 = 0$

(18) $7x^2 + 20x - 32 = 0$

(28) $5x^2 + 14x - 3 = 0$

(9) $8x^2 + 13x + 5 = 0$

(19) $x^2 - 5x + 6 = 0$

(29) $72x^2 - 31x - 28 = 0$

(10) $4x^2 - x - 14 = 0$

(20) $35x^2 - 58x - 9 = 0$

(30) $4x^2 + x - 18 = 0$

(1) $12x^2 + x - 1 = 0$

(11) $2x^2 + 7x + 6 = 0$

(21) $21x^2 + 31x + 8 = 0$

(2) $7x^2 + 13x + 6 = 0$

(12) $6x^2 - 11x + 3 = 0$

(22) $16x^2 + 34x + 15 = 0$

(3) $x^2 + 2x - 15 = 0$

(13) $4x^2 + 13x + 9 = 0$

(23) $28x^2 + 43x - 45 = 0$

(4) $3x^2 + 16x + 5 = 0$

(14) $x^2 + 5x + 6 = 0$

(24) $3x^2 + 4x + 1 = 0$

(5) $3x^2 + 2x - 8 = 0$

(15) $27x^2 - 24x + 4 = 0$

(25) $x^2 + 6x - 7 = 0$

(6) $6x^2 + 31x + 28 = 0$

(16) $5x^2 + 14x + 8 = 0$

(26) $3x^2 + 7x - 6 = 0$

(7) $2x^2 + 3x - 2 = 0$

(17) $8x^2 + 6x - 5 = 0$

(27) $28x^2 + 11x + 1 = 0$

(8) $25x^2 + 35x + 6 = 0$

(18) $3x^2 + 13x + 4 = 0$

(28) $8x^2 - 34x + 21 = 0$

(9) $72x^2 + 79x + 14 = 0$

(19) $18x^2 + 11x - 24 = 0$

(29) $56x^2 + 57x - 8 = 0$

(10) $7x^2 + 66x + 27 = 0$

(20) $2x^2 + 7x + 5 = 0$

(30) $9x^2 - 4 = 0$

(1) $2x^2 + 5x + 3 = 0$

(11) $6x^2 - 17x + 12 = 0$

(21) $18x^2 - 37x + 15 = 0$

(2) $8x^2 + 22x - 21 = 0$

(12) $3x^2 + 20x + 12 = 0$

(22) $9x^2 + 18x + 5 = 0$

(3) $6x^2 + 13x + 7 = 0$

(13) $7x^2 - 11x - 6 = 0$

(23) $8x^2 + 59x + 21 = 0$

(4) $14x^2 - 55x + 21 = 0$

(14) $8x^2 - 10x - 7 = 0$

(24) $x^2 - 2x + 1 = 0$

(5) $21x^2 - 11x - 6 = 0$

(15) $8x^2 - 34x - 9 = 0$

(25) $15x^2 - 4x - 3 = 0$

(6) $14x^2 + 9x - 18 = 0$

(16) $49x^2 + 28x + 3 = 0$

(26) $2x^2 - 11x - 21 = 0$

(7) $24x^2 - 13x - 7 = 0$

(17) $63x^2 - 31x - 10 = 0$

(27) $14x^2 + 79x + 72 = 0$

(8) $5x^2 - 21x + 18 = 0$

(18) $10x^2 + 29x + 21 = 0$

(28) $3x^2 - 31x + 36 = 0$

(9) $36x^2 + 37x - 10 = 0$

(19) $72x^2 - 11x - 6 = 0$

(29) $8x^2 + x - 7 = 0$

(10) $16x^2 - 30x + 9 = 0$

(20) $10x^2 + 9x + 2 = 0$

(30) $9x^2 - 13x + 4 = 0$

(1) $2x^2 - 3x - 9 = 0$

(11) $x^2 - 2x - 8 = 0$

(21) $x^2 - 7x + 6 = 0$

(2) $x^2 + 2x + 1 = 0$

(12) $28x^2 + 3x - 18 = 0$

(22) $20x^2 - 7x - 6 = 0$

(3) $12x^2 - 25x + 7 = 0$

(13) $45x^2 - 79x + 30 = 0$

(23) $9x^2 - 15x - 14 = 0$

(4) $63x^2 + 10x - 25 = 0$

(14) $54x^2 + 15x - 4 = 0$

(24) $5x^2 + 8x - 21 = 0$

(5) $x^2 - 6x + 9 = 0$

(15) $3x^2 - 2x - 1 = 0$

(25) $7x^2 - 9x + 2 = 0$

(6) $36x^2 - 29x + 5 = 0$

(16) $3x^2 - 2x - 1 = 0$

(26) $7x^2 + 10x + 3 = 0$

(7) $9x^2 + 6x - 8 = 0$

(17) $2x^2 - 5x + 3 = 0$

(27) $32x^2 - 100x + 63 = 0$

(8) $7x^2 - 6x - 16 = 0$

(18) $2x^2 + 11x - 21 = 0$

(28) $12x^2 + x - 20 = 0$

(9) $2x^2 - 3x + 1 = 0$

(19) $7x^2 - 8x + 1 = 0$

(29) $10x^2 + 33x - 7 = 0$

(10) $8x^2 - 31x + 21 = 0$

(20) $32x^2 - 28x - 9 = 0$

(30) $14x^2 + 37x + 24 = 0$

(1) $5x^2 - 6x - 27 = 0$

(11) $x^2 - 9 = 0$

(21) $9x^2 + 25x - 6 = 0$

(2) $5x^2 - 9x + 4 = 0$

(12) $7x^2 + 39x + 20 = 0$

(22) $4x^2 + 15x + 14 = 0$

(3) $2x^2 - 17x + 36 = 0$

(13) $2x^2 - 5x + 3 = 0$

(23) $4x^2 - 4x - 3 = 0$

(4) $12x^2 + 37x + 21 = 0$

(14) $40x^2 + 107x + 63 = 0$

(24) $4x^2 - x - 3 = 0$

(5) $3x^2 - 8x + 4 = 0$

(15) $27x^2 + 30x + 7 = 0$

(25) $6x^2 - 7x - 3 = 0$

(6) $4x^2 + 15x + 14 = 0$

(16) $21x^2 + 17x - 30 = 0$

(26) $x^2 - 1 = 0$

(7) $10x^2 - 7x + 1 = 0$

(17) $32x^2 + 4x - 15 = 0$

(27) $45x^2 + 71x + 28 = 0$

(8) $5x^2 + 16x + 3 = 0$

(18) $3x^2 - 4x + 1 = 0$

(28) $2x^2 + 3x + 1 = 0$

(9) $12x^2 + 4x - 5 = 0$

(19) $24x^2 - 17x + 3 = 0$

(29) $4x^2 + 33x - 27 = 0$

(10) $10x^2 - 53x + 63 = 0$

(20) $6x^2 + 13x - 28 = 0$

(30) $x^2 - 4x + 3 = 0$

(1) $4x^2 - 3x - 1 = 0$

(11) $2x^2 - 7x + 6 = 0$

(21) $4x^2 + 11x + 7 = 0$

(2) $x^2 + 4x + 3 = 0$

(12) $16x^2 + 2x - 5 = 0$

(22) $48x^2 + 22x - 15 = 0$

(3) $4x^2 - 3x - 1 = 0$

(13) $40x^2 - 91x + 49 = 0$

(23) $x^2 - 7x + 12 = 0$

(4) $28x^2 - 43x + 9 = 0$

(14) $56x^2 + 15x - 54 = 0$

(24) $36x^2 + 73x - 18 = 0$

(5) $16x^2 + 2x - 3 = 0$

(15) $20x^2 + 37x + 8 = 0$

(25) $63x^2 - 46x + 8 = 0$

(6) $21x^2 + 44x - 32 = 0$

(16) $3x^2 - 8x + 5 = 0$

(26) $5x^2 - 14x + 8 = 0$

(7) $3x^2 + 4x + 1 = 0$

(17) $10x^2 + 3x - 4 = 0$

(27) $2x^2 + 3x - 2 = 0$

(8) $x^2 - 2x - 3 = 0$

(18) $5x^2 - 27x + 28 = 0$

(28) $28x^2 + 29x - 35 = 0$

(9) $2x^2 - x - 1 = 0$

(19) $x^2 + 3x + 2 = 0$

(29) $3x^2 + 22x - 16 = 0$

(10) $2x^2 + 5x - 12 = 0$

(20) $28x^2 + 45x + 18 = 0$

(30) $9x^2 - 34x + 21 = 0$

(1) $x^2 + 8x + 7 = 0$

(11) $3x^2 + 4x + 1 = 0$

(21) $18x^2 + 33x - 40 = 0$

(2) $36x^2 + 65x - 36 = 0$

(12) $7x^2 - 29x + 24 = 0$

(22) $5x^2 + 13x - 6 = 0$

(3) $32x^2 - 52x - 7 = 0$

(13) $9x^2 - 15x - 14 = 0$

(23) $x^2 - 3x + 2 = 0$

(4) $7x^2 - 2x - 5 = 0$

(14) $4x^2 + 20x + 9 = 0$

(24) $7x^2 + 8x - 12 = 0$

(5) $16x^2 - 18x + 5 = 0$

(15) $5x^2 - 29x + 36 = 0$

(25) $20x^2 - 3x - 35 = 0$

(6) $16x^2 - 1 = 0$

(16) $8x^2 + 59x + 21 = 0$

(26) $56x^2 + 73x + 21 = 0$

(7) $4x^2 + 11x + 7 = 0$

(17) $2x^2 + 17x - 9 = 0$

(27) $7x^2 - 25x + 12 = 0$

(8) $24x^2 + 22x - 7 = 0$

(18) $9x^2 - 65x - 56 = 0$

(28) $6x^2 - 23x - 18 = 0$

(9) $21x^2 + 17x - 8 = 0$

(19) $12x^2 + 25x + 12 = 0$

(29) $24x^2 + 58x + 9 = 0$

(10) $56x^2 - 57x + 7 = 0$

(20) $24x^2 + 26x + 5 = 0$

(30) $32x^2 - 44x - 21 = 0$

(1) $72x^2 + 29x - 10 = 0$

(11) $5x^2 - 14x + 9 = 0$

(21) $28x^2 + 61x + 21 = 0$

(2) $8x^2 - 17x + 2 = 0$

(12) $25x^2 - 80x + 64 = 0$

(22) $40x^2 - 27x - 4 = 0$

(3) $8x^2 - 9x + 1 = 0$

(13) $7x^2 - 8x + 1 = 0$

(23) $10x^2 - 21x + 9 = 0$

(4) $18x^2 + 21x + 5 = 0$

(14) $27x^2 + 69x + 14 = 0$

(24) $6x^2 + 5x - 4 = 0$

(5) $2x^2 + 5x + 3 = 0$

(15) $6x^2 - 7x + 1 = 0$

(25) $16x^2 - 50x + 25 = 0$

(6) $21x^2 + 31x + 8 = 0$

(16) $9x^2 - 25 = 0$

(26) $4x^2 - 12x - 27 = 0$

(7) $2x^2 + 5x + 2 = 0$

(17) $40x^2 + 79x + 24 = 0$

(27) $3x^2 - 4x - 4 = 0$

(8) $24x^2 - x - 10 = 0$

(18) $35x^2 - 6x - 8 = 0$

(28) $x^2 - 1 = 0$

(9) $35x^2 + 3x - 20 = 0$

(19) $x^2 - x - 2 = 0$

(29) $4x^2 + 19x + 12 = 0$

(10) $5x^2 + 12x + 7 = 0$

(20) $3x^2 - x - 2 = 0$

(30) $9x^2 - 43x + 28 = 0$

(1) $4x^2 + x - 3 = 0$

(11) $3x^2 + 7x - 6 = 0$

(21) $3x^2 + 11x + 6 = 0$

(2) $7x^2 + 13x - 24 = 0$

(12) $3x^2 + 4x - 4 = 0$

(22) $45x^2 + 83x + 28 = 0$

(3) $14x^2 + 13x + 3 = 0$

(13) $2x^2 + 3x - 9 = 0$

(23) $35x^2 + 9x - 2 = 0$

(4) $36x^2 - 84x + 49 = 0$

(14) $9x^2 + 23x + 10 = 0$

(24) $12x^2 - 11x - 5 = 0$

(5) $54x^2 - 39x - 5 = 0$

(15) $5x^2 - 4x - 1 = 0$

(25) $9x^2 + 16x + 7 = 0$

(6) $28x^2 - x - 15 = 0$

(16) $2x^2 + 5x + 3 = 0$

(26) $45x^2 - 92x + 32 = 0$

(7) $15x^2 - 7x - 2 = 0$

(17) $7x^2 - 25x - 12 = 0$

(27) $4x^2 + 17x - 15 = 0$

(8) $2x^2 - 13x + 18 = 0$

(18) $6x^2 + x - 7 = 0$

(28) $7x^2 - 22x + 3 = 0$

(9) $28x^2 + 9x - 9 = 0$

(19) $14x^2 - 33x - 56 = 0$

(29) $12x^2 + 41x + 24 = 0$

(10) $3x^2 - 16x - 12 = 0$

(20) $4x^2 + 17x + 4 = 0$

(30) $3x^2 - 4x + 1 = 0$

(1) $8x^2 - 39x - 54 = 0$

(11) $3x^2 - 13x + 4 = 0$

(21) $10x^2 - x - 24 = 0$

(2) $8x^2 - 15x - 2 = 0$

(12) $x^2 + 6x - 27 = 0$

(22) $2x^2 - 3x - 2 = 0$

(3) $x^2 - 2x + 1 = 0$

(13) $24x^2 - 31x + 10 = 0$

(23) $28x^2 - 3x - 18 = 0$

(4) $7x^2 + 5x - 2 = 0$

(14) $2x^2 - 9x - 18 = 0$

(24) $x^2 - 3x + 2 = 0$

(5) $2x^2 + 13x + 6 = 0$

(15) $6x^2 - 7x - 10 = 0$

(25) $27x^2 - 33x + 8 = 0$

(6) $6x^2 - 23x + 7 = 0$

(16) $3x^2 + 8x + 5 = 0$

(26) $28x^2 + 75x + 27 = 0$

(7) $35x^2 - 31x - 40 = 0$

(17) $9x^2 + 74x - 63 = 0$

(27) $4x^2 + 27x - 40 = 0$

(8) $9x^2 - 3x - 2 = 0$

(18) $5x^2 + 3x - 8 = 0$

(28) $5x^2 - 36x - 81 = 0$

(9) $7x^2 + 5x - 18 = 0$

(19) $x^2 - 3x - 4 = 0$

(29) $3x^2 - 8x + 4 = 0$

(10) $20x^2 - 43x + 14 = 0$

(20) $2x^2 + 3x - 20 = 0$

(30) $12x^2 + 11x - 5 = 0$

(1) $x^2 - 1 = 0$

(11) $7x^2 + 18x + 8 = 0$

(21) $x^2 + 7x + 12 = 0$

(2) $x^2 + x - 2 = 0$

(12) $12x^2 - 8x - 15 = 0$

(22) $3x^2 + x - 2 = 0$

(3) $7x^2 + 5x - 2 = 0$

(13) $36x^2 - 23x - 8 = 0$

(23) $4x^2 + 4x - 3 = 0$

(4) $8x^2 - 2x - 1 = 0$

(14) $12x^2 - 5x - 2 = 0$

(24) $8x^2 + 11x - 10 = 0$

(5) $4x^2 + 23x + 15 = 0$

(15) $21x^2 - 13x - 18 = 0$

(25) $8x^2 + 15x + 7 = 0$

(6) $2x^2 + 11x + 12 = 0$

(16) $5x^2 + 39x + 54 = 0$

(26) $30x^2 - 41x + 7 = 0$

(7) $4x^2 - 13x + 9 = 0$

(17) $21x^2 - 13x - 20 = 0$

(27) $12x^2 + 37x + 21 = 0$

(8) $3x^2 + 22x + 24 = 0$

(18) $x^2 - 1 = 0$

(28) $12x^2 - 5x - 25 = 0$

(9) $40x^2 + 73x + 30 = 0$

(19) $15x^2 + 7x - 36 = 0$

(29) $7x^2 + 12x + 5 = 0$

(10) $6x^2 - 19x + 14 = 0$

(20) $x^2 + 4x + 3 = 0$

(30) $49x^2 + 63x + 18 = 0$

(1) $8x^2 - 14x + 5 = 0$

(11) $28x^2 + 99x + 81 = 0$

(21) $2x^2 - 9x + 7 = 0$

(2) $27x^2 - 96x + 64 = 0$

(12) $15x^2 - 26x - 21 = 0$

(22) $5x^2 - 21x + 18 = 0$

(3) $16x^2 - 18x + 5 = 0$

(13) $40x^2 + 69x + 8 = 0$

(23) $6x^2 - 13x - 8 = 0$

(4) $2x^2 - x - 3 = 0$

(14) $49x^2 + 42x + 8 = 0$

(24) $9x^2 + 12x + 4 = 0$

(5) $9x^2 + 17x + 8 = 0$

(15) $5x^2 + 2x - 7 = 0$

(25) $81x^2 + 54x + 8 = 0$

(6) $7x^2 - 65x + 18 = 0$

(16) $27x^2 - 12x - 7 = 0$

(26) $49x^2 - 28x - 12 = 0$

(7) $16x^2 + 10x - 21 = 0$

(17) $18x^2 + 31x - 35 = 0$

(27) $12x^2 + 5x - 2 = 0$

(8) $10x^2 - x - 3 = 0$

(18) $3x^2 + 26x + 48 = 0$

(28) $9x^2 + 61x - 14 = 0$

(9) $5x^2 + 6x + 1 = 0$

(19) $9x^2 - 16 = 0$

(29) $24x^2 + 2x - 1 = 0$

(10) $2x^2 - 9x + 4 = 0$

(20) $21x^2 - 73x + 56 = 0$

(30) $6x^2 - 11x - 7 = 0$

(1) $x^2 - 4x + 3 = 0$

(11) $27x^2 - 12x - 32 = 0$

(21) $63x^2 - 40x - 7 = 0$

(2) $8x^2 + 47x - 6 = 0$

(12) $4x^2 - x - 3 = 0$

(22) $9x^2 + 17x + 8 = 0$

(3) $9x^2 - 56x - 49 = 0$

(13) $12x^2 - 7x - 10 = 0$

(23) $45x^2 + 46x + 8 = 0$

(4) $x^2 - 2x + 1 = 0$

(14) $6x^2 - 13x + 6 = 0$

(24) $2x^2 + 7x + 5 = 0$

(5) $72x^2 + 13x - 20 = 0$

(15) $x^2 + 5x + 4 = 0$

(25) $8x^2 + 2x - 1 = 0$

(6) $4x^2 + 7x - 15 = 0$

(16) $18x^2 - 51x + 35 = 0$

(26) $6x^2 - 11x - 2 = 0$

(7) $3x^2 - 5x + 2 = 0$

(17) $5x^2 - 46x + 9 = 0$

(27) $18x^2 - 29x + 3 = 0$

(8) $7x^2 - 30x - 25 = 0$

(18) $8x^2 - 10x - 7 = 0$

(28) $4x^2 + 5x + 1 = 0$

(9) $27x^2 + 3x - 10 = 0$

(19) $7x^2 - 23x + 18 = 0$

(29) $9x^2 + 44x - 5 = 0$

(10) $20x^2 - 9x - 18 = 0$

(20) $14x^2 + 25x + 6 = 0$

(30) $5x^2 + 29x - 6 = 0$

(1) $3x^2 + 5x - 12 = 0$

(11) $25x^2 - 10x - 63 = 0$

(21) $12x^2 + 5x - 3 = 0$

(2) $6x^2 + 13x - 5 = 0$

(12) $2x^2 + 15x - 8 = 0$

(22) $x^2 + 2x - 3 = 0$

(3) $35x^2 + 93x + 54 = 0$

(13) $56x^2 - 55x - 9 = 0$

(23) $42x^2 + 103x + 63 = 0$

(4) $2x^2 - 5x - 18 = 0$

(14) $x^2 - 10x + 16 = 0$

(24) $2x^2 + x - 21 = 0$

(5) $4x^2 - 23x + 28 = 0$

(15) $9x^2 + 9x - 10 = 0$

(25) $4x^2 - 12x + 5 = 0$

(6) $9x^2 + 9x + 2 = 0$

(16) $3x^2 - 29x + 18 = 0$

(26) $16x^2 + 6x - 1 = 0$

(7) $36x^2 - 73x + 35 = 0$

(17) $10x^2 - x - 3 = 0$

(27) $9x^2 + 25x + 14 = 0$

(8) $7x^2 - 65x + 18 = 0$

(18) $49x^2 + 7x - 20 = 0$

(28) $2x^2 + 5x - 3 = 0$

(9) $36x^2 + 5x - 24 = 0$

(19) $45x^2 + 8x - 21 = 0$

(29) $10x^2 - 11x - 8 = 0$

(10) $4x^2 - 9x - 28 = 0$

(20) $16x^2 - 38x + 21 = 0$

(30) $6x^2 - 13x - 15 = 0$

$$(1) \quad 6x^2 - 7x - 3 = 0$$

$$x = -\frac{1}{3}, \frac{3}{2}$$

$$(11) \quad 6x^2 + 35x + 25 = 0$$

$$x = -5, -\frac{5}{6}$$

$$(21) \quad x^2 + x - 56 = 0$$

$$x = -8, 7$$

$$(2) \quad 12x^2 + 29x + 14 = 0$$

$$x = -\frac{2}{3}, -\frac{7}{4}$$

$$(12) \quad 8x^2 + 22x + 5 = 0$$

$$x = -\frac{1}{4}, -\frac{5}{2}$$

$$(22) \quad 4x^2 + 5x - 9 = 0$$

$$x = -\frac{9}{4}, 1$$

$$(3) \quad 45x^2 + 98x + 49 = 0$$

$$x = -\frac{7}{9}, -\frac{7}{5}$$

$$(13) \quad 4x^2 + 3x - 1 = 0$$

$$x = \frac{1}{4}, -1$$

$$(23) \quad 10x^2 - 17x + 6 = 0$$

$$x = \frac{6}{5}, \frac{1}{2}$$

$$(4) \quad 16x^2 + 66x + 35 = 0$$

$$x = -\frac{5}{8}, -\frac{7}{2}$$

$$(14) \quad 15x^2 - 4x - 4 = 0$$

$$x = \frac{2}{3}, -\frac{2}{5}$$

$$(24) \quad 3x^2 - 11x + 6 = 0$$

$$x = \frac{2}{3}, 3$$

$$(5) \quad 18x^2 - 21x + 5 = 0$$

$$x = \frac{1}{3}, \frac{5}{6}$$

$$(15) \quad 24x^2 - 14x - 3 = 0$$

$$x = -\frac{1}{6}, \frac{3}{4}$$

$$(25) \quad 24x^2 - 11x + 1 = 0$$

$$x = \frac{1}{8}, \frac{1}{3}$$

$$(6) \quad 20x^2 - 11x - 45 = 0$$

$$x = \frac{9}{5}, -\frac{5}{4}$$

$$(16) \quad 2x^2 - 9x + 10 = 0$$

$$x = 2, \frac{5}{2}$$

$$(26) \quad 49x^2 + 21x - 54 = 0$$

$$x = \frac{6}{7}, -\frac{9}{7}$$

$$(7) \quad 5x^2 - 17x + 6 = 0$$

$$x = 3, \frac{2}{5}$$

$$(17) \quad 25x^2 - 30x - 27 = 0$$

$$x = -\frac{3}{5}, \frac{9}{5}$$

$$(27) \quad 9x^2 - 37x + 4 = 0$$

$$x = \frac{1}{9}, 4$$

$$(8) \quad x^2 - 5x - 6 = 0$$

$$x = -1, 6$$

$$(18) \quad 15x^2 - 4x - 3 = 0$$

$$x = \frac{3}{5}, -\frac{1}{3}$$

$$(28) \quad 16x^2 + 14x - 15 = 0$$

$$x = \frac{5}{8}, -\frac{3}{2}$$

$$(9) \quad 7x^2 + 13x + 6 = 0$$

$$x = -\frac{6}{7}, -1$$

$$(19) \quad 16x^2 + 6x - 1 = 0$$

$$x = \frac{1}{8}, -\frac{1}{2}$$

$$(29) \quad 5x^2 - 2x - 16 = 0$$

$$x = -\frac{8}{5}, 2$$

$$(10) \quad 3x^2 - 26x + 16 = 0$$

$$x = \frac{2}{3}, 8$$

$$(20) \quad 2x^2 - 15x + 18 = 0$$

$$x = 6, \frac{3}{2}$$

$$(30) \quad 4x^2 + 3x - 7 = 0$$

$$x = 1, -\frac{7}{4}$$

$$(1) 40x^2 + 31x - 35 = 0$$

$$x = -\frac{7}{5}, \frac{5}{8}$$

$$(11) 7x^2 - 12x + 5 = 0$$

$$x = 1, \frac{5}{7}$$

$$(21) 9x^2 + 10x + 1 = 0$$

$$x = -1, -\frac{1}{9}$$

$$(2) 30x^2 - 43x - 8 = 0$$

$$x = -\frac{1}{6}, \frac{8}{5}$$

$$(12) 6x^2 - 11x + 3 = 0$$

$$x = \frac{1}{3}, \frac{3}{2}$$

$$(22) 10x^2 + 13x + 4 = 0$$

$$x = -\frac{1}{2}, -\frac{4}{5}$$

$$(3) 9x^2 + 46x + 5 = 0$$

$$x = -\frac{1}{9}, -5$$

$$(13) 5x^2 + 8x + 3 = 0$$

$$x = -\frac{3}{5}, -1$$

$$(23) 54x^2 - 87x + 35 = 0$$

$$x = \frac{5}{6}, \frac{7}{9}$$

$$(4) 9x^2 - 28x + 3 = 0$$

$$x = 3, \frac{1}{9}$$

$$(14) 4x^2 - 7x + 3 = 0$$

$$x = 1, \frac{3}{4}$$

$$(24) 18x^2 + 7x - 8 = 0$$

$$x = -\frac{8}{9}, \frac{1}{2}$$

$$(5) 28x^2 - 41x - 14 = 0$$

$$x = \frac{7}{4}, -\frac{2}{7}$$

$$(15) 18x^2 + 27x - 56 = 0$$

$$x = -\frac{8}{3}, \frac{7}{6}$$

$$(25) 32x^2 - 36x - 5 = 0$$

$$x = -\frac{1}{8}, \frac{5}{4}$$

$$(6) 27x^2 + 15x + 2 = 0$$

$$x = -\frac{2}{9}, -\frac{1}{3}$$

$$(16) 16x^2 + 40x + 9 = 0$$

$$x = -\frac{9}{4}, -\frac{1}{4}$$

$$(26) 56x^2 - 37x + 6 = 0$$

$$x = \frac{3}{8}, \frac{2}{7}$$

$$(7) 5x^2 + 19x + 12 = 0$$

$$x = -\frac{4}{5}, -3$$

$$(17) 2x^2 - 5x - 12 = 0$$

$$x = -\frac{3}{2}, 4$$

$$(27) 8x^2 + 39x + 28 = 0$$

$$x = -4, -\frac{7}{8}$$

$$(8) 14x^2 + 47x - 72 = 0$$

$$x = -\frac{9}{2}, \frac{8}{7}$$

$$(18) 4x^2 - 11x - 20 = 0$$

$$x = 4, -\frac{5}{4}$$

$$(28) 8x^2 - 57x + 54 = 0$$

$$x = \frac{9}{8}, 6$$

$$(9) 6x^2 + 13x - 15 = 0$$

$$x = -3, \frac{5}{6}$$

$$(19) 21x^2 + 10x - 16 = 0$$

$$x = \frac{2}{3}, -\frac{8}{7}$$

$$(29) 36x^2 + 85x + 9 = 0$$

$$x = -\frac{9}{4}, -\frac{1}{9}$$

$$(10) 14x^2 + 17x - 6 = 0$$

$$x = \frac{2}{7}, -\frac{3}{2}$$

$$(20) 7x^2 + 22x + 3 = 0$$

$$x = -3, -\frac{1}{7}$$

$$(30) 63x^2 + 25x + 2 = 0$$

$$x = -\frac{2}{7}, -\frac{1}{9}$$

$$(1) \quad 6x^2 - 5x - 21 = 0$$

$$x = -\frac{3}{2}, \frac{7}{3}$$

$$(11) \quad 3x^2 - 8x + 4 = 0$$

$$x = \frac{2}{3}, 2$$

$$(21) \quad 25x^2 - 50x + 21 = 0$$

$$x = \frac{7}{5}, \frac{3}{5}$$

$$(2) \quad 12x^2 + 35x + 8 = 0$$

$$x = -\frac{8}{3}, -\frac{1}{4}$$

$$(12) \quad 15x^2 - 7x - 36 = 0$$

$$x = -\frac{4}{3}, \frac{9}{5}$$

$$(22) \quad 45x^2 + 31x - 4 = 0$$

$$x = \frac{1}{9}, -\frac{4}{5}$$

$$(3) \quad 4x^2 - 11x + 7 = 0$$

$$x = \frac{7}{4}, 1$$

$$(13) \quad 7x^2 - 8x + 1 = 0$$

$$x = 1, \frac{1}{7}$$

$$(23) \quad 6x^2 - 11x + 3 = 0$$

$$x = \frac{1}{3}, \frac{3}{2}$$

$$(4) \quad 21x^2 - 64x + 35 = 0$$

$$x = \frac{5}{7}, \frac{7}{3}$$

$$(14) \quad 56x^2 + 57x + 7 = 0$$

$$x = -\frac{7}{8}, -\frac{1}{7}$$

$$(24) \quad x^2 - x - 2 = 0$$

$$x = -1, 2$$

$$(5) \quad 2x^2 - 13x + 21 = 0$$

$$x = \frac{7}{2}, 3$$

$$(15) \quad 7x^2 + 27x + 18 = 0$$

$$x = -3, -\frac{6}{7}$$

$$(25) \quad 8x^2 - 6x + 1 = 0$$

$$x = \frac{1}{2}, \frac{1}{4}$$

$$(6) \quad 28x^2 + 29x - 9 = 0$$

$$x = \frac{1}{4}, -\frac{9}{7}$$

$$(16) \quad 15x^2 - 7x - 2 = 0$$

$$x = -\frac{1}{5}, \frac{2}{3}$$

$$(26) \quad 16x^2 + 26x + 9 = 0$$

$$x = -\frac{9}{8}, -\frac{1}{2}$$

$$(7) \quad 28x^2 + 57x + 27 = 0$$

$$x = -\frac{9}{7}, -\frac{3}{4}$$

$$(17) \quad 10x^2 - 7x - 12 = 0$$

$$x = -\frac{4}{5}, \frac{3}{2}$$

$$(27) \quad 64x^2 + 96x + 35 = 0$$

$$x = -\frac{5}{8}, -\frac{7}{8}$$

$$(8) \quad 2x^2 + 17x + 8 = 0$$

$$x = -\frac{1}{2}, -8$$

$$(18) \quad x^2 - 1 = 0$$

$$x = 1, -1$$

$$(28) \quad 8x^2 - 7x - 1 = 0$$

$$x = 1, -\frac{1}{8}$$

$$(9) \quad 3x^2 + 4x + 1 = 0$$

$$x = -1, -\frac{1}{3}$$

$$(19) \quad 64x^2 + 32x - 21 = 0$$

$$x = -\frac{7}{8}, \frac{3}{8}$$

$$(29) \quad 5x^2 - 6x + 1 = 0$$

$$x = \frac{1}{5}, 1$$

$$(10) \quad 63x^2 - 23x + 2 = 0$$

$$x = \frac{1}{7}, \frac{2}{9}$$

$$(20) \quad 9x^2 + 29x + 6 = 0$$

$$x = -\frac{2}{9}, -3$$

$$(30) \quad 3x^2 + 7x + 4 = 0$$

$$x = -1, -\frac{4}{3}$$

$$(1) \quad 7x^2 - 13x - 24 = 0$$

$$x = 3, -\frac{8}{7}$$

$$(11) \quad 16x^2 - 6x - 7 = 0$$

$$x = \frac{7}{8}, -\frac{1}{2}$$

$$(21) \quad 24x^2 - 47x + 20 = 0$$

$$x = \frac{4}{3}, \frac{5}{8}$$

$$(2) \quad x^2 + 7x + 12 = 0$$

$$x = -3, -4$$

$$(12) \quad 5x^2 + 14x - 3 = 0$$

$$x = \frac{1}{5}, -3$$

$$(22) \quad x^2 - 8x + 15 = 0$$

$$x = 5, 3$$

$$(3) \quad 12x^2 + 13x + 3 = 0$$

$$x = -\frac{1}{3}, -\frac{3}{4}$$

$$(13) \quad 72x^2 - 119x + 49 = 0$$

$$x = \frac{7}{9}, \frac{7}{8}$$

$$(23) \quad x^2 - 1 = 0$$

$$x = -1, 1$$

$$(4) \quad 24x^2 - 23x + 5 = 0$$

$$x = \frac{1}{3}, \frac{5}{8}$$

$$(14) \quad 15x^2 - 29x - 14 = 0$$

$$x = \frac{7}{3}, -\frac{2}{5}$$

$$(24) \quad 20x^2 - 43x + 21 = 0$$

$$x = \frac{7}{5}, \frac{3}{4}$$

$$(5) \quad 6x^2 - 25x + 14 = 0$$

$$x = \frac{2}{3}, \frac{7}{2}$$

$$(15) \quad 15x^2 - 34x + 15 = 0$$

$$x = \frac{5}{3}, \frac{3}{5}$$

$$(25) \quad 49x^2 - 25 = 0$$

$$x = -\frac{5}{7}, \frac{5}{7}$$

$$(6) \quad 45x^2 + 17x - 14 = 0$$

$$x = \frac{2}{5}, -\frac{7}{9}$$

$$(16) \quad 6x^2 + 13x - 5 = 0$$

$$x = \frac{1}{3}, -\frac{5}{2}$$

$$(26) \quad 24x^2 + 25x + 6 = 0$$

$$x = -\frac{3}{8}, -\frac{2}{3}$$

$$(7) \quad 8x^2 - 18x + 9 = 0$$

$$x = \frac{3}{4}, \frac{3}{2}$$

$$(17) \quad 10x^2 - 33x - 54 = 0$$

$$x = \frac{9}{2}, -\frac{6}{5}$$

$$(27) \quad x^2 - 2x - 8 = 0$$

$$x = 4, -2$$

$$(8) \quad 18x^2 - 13x + 2 = 0$$

$$x = \frac{1}{2}, \frac{2}{9}$$

$$(18) \quad 8x^2 - 27x - 20 = 0$$

$$x = -\frac{5}{8}, 4$$

$$(28) \quad 7x^2 - 13x - 2 = 0$$

$$x = -\frac{1}{7}, 2$$

$$(9) \quad 5x^2 + 16x + 12 = 0$$

$$x = -2, -\frac{6}{5}$$

$$(19) \quad 42x^2 + 5x - 2 = 0$$

$$x = -\frac{2}{7}, \frac{1}{6}$$

$$(29) \quad 9x^2 - 24x + 7 = 0$$

$$x = \frac{7}{3}, \frac{1}{3}$$

$$(10) \quad 2x^2 - 17x + 35 = 0$$

$$x = \frac{7}{2}, 5$$

$$(20) \quad 56x^2 + 53x + 12 = 0$$

$$x = -\frac{3}{8}, -\frac{4}{7}$$

$$(30) \quad 9x^2 + 14x + 5 = 0$$

$$x = -1, -\frac{5}{9}$$

$$(1) \quad 8x^2 - 5x - 3 = 0 \\ x = 1, -\frac{3}{8}$$

$$(11) \quad 3x^2 + 22x + 24 = 0 \\ x = -6, -\frac{4}{3}$$

$$(21) \quad 21x^2 + x - 2 = 0 \\ x = \frac{2}{7}, -\frac{1}{3}$$

$$(2) \quad 24x^2 - 7x - 6 = 0 \\ x = -\frac{3}{8}, \frac{2}{3}$$

$$(12) \quad 8x^2 + 22x + 5 = 0 \\ x = -\frac{1}{4}, -\frac{5}{2}$$

$$(22) \quad 63x^2 + 82x + 24 = 0 \\ x = -\frac{6}{7}, -\frac{4}{9}$$

$$(3) \quad 36x^2 - 11x - 12 = 0 \\ x = -\frac{4}{9}, \frac{3}{4}$$

$$(13) \quad 2x^2 - 17x - 9 = 0 \\ x = 9, -\frac{1}{2}$$

$$(23) \quad 14x^2 - 31x - 63 = 0 \\ x = -\frac{9}{7}, \frac{7}{2}$$

$$(4) \quad 10x^2 + 29x + 10 = 0 \\ x = -\frac{2}{5}, -\frac{5}{2}$$

$$(14) \quad 63x^2 - 68x + 12 = 0 \\ x = \frac{2}{9}, \frac{6}{7}$$

$$(24) \quad 5x^2 + 13x - 6 = 0 \\ x = -3, \frac{2}{5}$$

$$(5) \quad x^2 + 11x + 18 = 0 \\ x = -9, -2$$

$$(15) \quad 21x^2 - 2x - 8 = 0 \\ x = -\frac{4}{7}, \frac{2}{3}$$

$$(25) \quad 14x^2 - 27x + 9 = 0 \\ x = \frac{3}{2}, \frac{3}{7}$$

$$(6) \quad 2x^2 + 7x - 4 = 0 \\ x = -4, \frac{1}{2}$$

$$(16) \quad 63x^2 - 73x + 20 = 0 \\ x = \frac{5}{7}, \frac{4}{9}$$

$$(26) \quad 3x^2 - 14x + 8 = 0 \\ x = \frac{2}{3}, 4$$

$$(7) \quad 4x^2 + 5x + 1 = 0 \\ x = -\frac{1}{4}, -1$$

$$(17) \quad 9x^2 + 10x + 1 = 0 \\ x = -1, -\frac{1}{9}$$

$$(27) \quad 5x^2 + 41x + 42 = 0 \\ x = -\frac{6}{5}, -7$$

$$(8) \quad 4x^2 - 8x + 3 = 0 \\ x = \frac{1}{2}, \frac{3}{2}$$

$$(18) \quad x^2 - x - 6 = 0 \\ x = -2, 3$$

$$(28) \quad 24x^2 - 31x + 10 = 0 \\ x = \frac{5}{8}, \frac{2}{3}$$

$$(9) \quad 6x^2 + 11x + 3 = 0 \\ x = -\frac{3}{2}, -\frac{1}{3}$$

$$(19) \quad 5x^2 - 6x + 1 = 0 \\ x = 1, \frac{1}{5}$$

$$(29) \quad 35x^2 + 58x - 9 = 0 \\ x = -\frac{9}{5}, \frac{1}{7}$$

$$(10) \quad 12x^2 - 4x - 1 = 0 \\ x = \frac{1}{2}, -\frac{1}{6}$$

$$(20) \quad 12x^2 - 17x - 7 = 0 \\ x = -\frac{1}{3}, \frac{7}{4}$$

$$(30) \quad x^2 - 8x + 12 = 0 \\ x = 6, 2$$

$$(1) \quad 21x^2 - 46x + 24 = 0$$
$$x = \frac{6}{7}, \frac{4}{3}$$

$$(11) \quad 14x^2 - 3x - 27 = 0$$
$$x = -\frac{9}{7}, \frac{3}{2}$$

$$(21) \quad x^2 + 6x + 5 = 0$$
$$x = -1, -5$$

$$(2) \quad 63x^2 - 20x - 3 = 0$$
$$x = -\frac{1}{9}, \frac{3}{7}$$

$$(12) \quad 56x^2 - 111x + 54 = 0$$
$$x = \frac{9}{8}, \frac{6}{7}$$

$$(22) \quad 72x^2 + x - 56 = 0$$
$$x = -\frac{8}{9}, \frac{7}{8}$$

$$(3) \quad 63x^2 + 53x - 36 = 0$$
$$x = \frac{4}{9}, -\frac{9}{7}$$

$$(13) \quad 2x^2 + x - 1 = 0$$
$$x = \frac{1}{2}, -1$$

$$(23) \quad 18x^2 + 7x - 1 = 0$$
$$x = -\frac{1}{2}, \frac{1}{9}$$

$$(4) \quad 35x^2 + 38x - 9 = 0$$
$$x = \frac{1}{5}, -\frac{9}{7}$$

$$(14) \quad 4x^2 + 21x - 18 = 0$$
$$x = \frac{3}{4}, -6$$

$$(24) \quad 16x^2 - 46x - 35 = 0$$
$$x = -\frac{5}{8}, \frac{7}{2}$$

$$(5) \quad 40x^2 - 9x - 9 = 0$$
$$x = \frac{3}{5}, -\frac{3}{8}$$

$$(15) \quad 7x^2 + 12x - 4 = 0$$
$$x = -2, \frac{2}{7}$$

$$(25) \quad 2x^2 + 7x - 72 = 0$$
$$x = -8, \frac{9}{2}$$

$$(6) \quad x^2 - 7x - 18 = 0$$
$$x = 9, -2$$

$$(16) \quad 48x^2 + 50x - 7 = 0$$
$$x = -\frac{7}{6}, \frac{1}{8}$$

$$(26) \quad 8x^2 + 2x - 3 = 0$$
$$x = -\frac{3}{4}, \frac{1}{2}$$

$$(7) \quad 4x^2 - 3x - 1 = 0$$
$$x = -\frac{1}{4}, 1$$

$$(17) \quad 18x^2 + 17x - 15 = 0$$
$$x = \frac{5}{9}, -\frac{3}{2}$$

$$(27) \quad 21x^2 - 4x - 32 = 0$$
$$x = \frac{4}{3}, -\frac{8}{7}$$

$$(8) \quad 4x^2 + x - 3 = 0$$
$$x = -1, \frac{3}{4}$$

$$(18) \quad 3x^2 - x - 2 = 0$$
$$x = 1, -\frac{2}{3}$$

$$(28) \quad x^2 - 6x + 8 = 0$$
$$x = 2, 4$$

$$(9) \quad 64x^2 + 112x + 45 = 0$$
$$x = -\frac{9}{8}, -\frac{5}{8}$$

$$(19) \quad 27x^2 - 3x - 14 = 0$$
$$x = \frac{7}{9}, -\frac{2}{3}$$

$$(29) \quad 6x^2 - 5x - 4 = 0$$
$$x = \frac{4}{3}, -\frac{1}{2}$$

$$(10) \quad x^2 - 2x + 1 = 0$$
$$x = 1$$

$$(20) \quad 8x^2 - 17x + 9 = 0$$
$$x = \frac{9}{8}, 1$$

$$(30) \quad 12x^2 + 23x - 24 = 0$$
$$x = \frac{3}{4}, -\frac{8}{3}$$

$$(1) \quad 4x^2 - 15x - 4 = 0 \\ x = -\frac{1}{4}, 4$$

$$(11) \quad 42x^2 + 17x - 4 = 0 \\ x = -\frac{4}{7}, \frac{1}{6}$$

$$(21) \quad 14x^2 - 17x - 6 = 0 \\ x = \frac{3}{2}, -\frac{2}{7}$$

$$(2) \quad 12x^2 - 13x - 35 = 0 \\ x = -\frac{5}{4}, \frac{7}{3}$$

$$(12) \quad 2x^2 + 3x - 5 = 0 \\ x = -\frac{5}{2}, 1$$

$$(22) \quad 3x^2 + 2x - 8 = 0 \\ x = -2, \frac{4}{3}$$

$$(3) \quad 30x^2 + 13x - 3 = 0 \\ x = \frac{1}{6}, -\frac{3}{5}$$

$$(13) \quad 2x^2 - 13x - 24 = 0 \\ x = 8, -\frac{3}{2}$$

$$(23) \quad 15x^2 + 32x + 9 = 0 \\ x = -\frac{9}{5}, -\frac{1}{3}$$

$$(4) \quad 72x^2 + 23x - 4 = 0 \\ x = -\frac{4}{9}, \frac{1}{8}$$

$$(14) \quad x^2 + 9x + 8 = 0 \\ x = -8, -1$$

$$(24) \quad 7x^2 + 11x + 4 = 0 \\ x = -\frac{4}{7}, -1$$

$$(5) \quad x^2 - 1 = 0 \\ x = -1, 1$$

$$(15) \quad 3x^2 - 2x - 1 = 0 \\ x = -\frac{1}{3}, 1$$

$$(25) \quad 10x^2 - 21x + 8 = 0 \\ x = \frac{8}{5}, \frac{1}{2}$$

$$(6) \quad 9x^2 + 30x + 16 = 0 \\ x = -\frac{8}{3}, -\frac{2}{3}$$

$$(16) \quad 6x^2 + 53x + 40 = 0 \\ x = -8, -\frac{5}{6}$$

$$(26) \quad 2x^2 - 3x + 1 = 0 \\ x = 1, \frac{1}{2}$$

$$(7) \quad 14x^2 + 37x + 24 = 0 \\ x = -\frac{8}{7}, -\frac{3}{2}$$

$$(17) \quad 56x^2 + 65x + 14 = 0 \\ x = -\frac{7}{8}, -\frac{2}{7}$$

$$(27) \quad 8x^2 - 63x - 81 = 0 \\ x = 9, -\frac{9}{8}$$

$$(8) \quad 21x^2 - 76x + 63 = 0 \\ x = \frac{9}{7}, \frac{7}{3}$$

$$(18) \quad 7x^2 + 20x - 32 = 0 \\ x = \frac{8}{7}, -4$$

$$(28) \quad 5x^2 + 14x - 3 = 0 \\ x = \frac{1}{5}, -3$$

$$(9) \quad 8x^2 + 13x + 5 = 0 \\ x = -1, -\frac{5}{8}$$

$$(19) \quad x^2 - 5x + 6 = 0 \\ x = 2, 3$$

$$(29) \quad 72x^2 - 31x - 28 = 0 \\ x = \frac{7}{8}, -\frac{4}{9}$$

$$(10) \quad 4x^2 - x - 14 = 0 \\ x = 2, -\frac{7}{4}$$

$$(20) \quad 35x^2 - 58x - 9 = 0 \\ x = \frac{9}{5}, -\frac{1}{7}$$

$$(30) \quad 4x^2 + x - 18 = 0 \\ x = 2, -\frac{9}{4}$$

$$(1) \quad 12x^2 + x - 1 = 0$$

$$x = \frac{1}{4}, -\frac{1}{3}$$

$$(11) \quad 2x^2 + 7x + 6 = 0$$

$$x = -2, -\frac{3}{2}$$

$$(21) \quad 21x^2 + 31x + 8 = 0$$

$$x = -\frac{1}{3}, -\frac{8}{7}$$

$$(2) \quad 7x^2 + 13x + 6 = 0$$

$$x = -1, -\frac{6}{7}$$

$$(12) \quad 6x^2 - 11x + 3 = 0$$

$$x = \frac{3}{2}, \frac{1}{3}$$

$$(22) \quad 16x^2 + 34x + 15 = 0$$

$$x = -\frac{3}{2}, -\frac{5}{8}$$

$$(3) \quad x^2 + 2x - 15 = 0$$

$$x = -5, 3$$

$$(13) \quad 4x^2 + 13x + 9 = 0$$

$$x = -1, -\frac{9}{4}$$

$$(23) \quad 28x^2 + 43x - 45 = 0$$

$$x = \frac{5}{7}, -\frac{9}{4}$$

$$(4) \quad 3x^2 + 16x + 5 = 0$$

$$x = -\frac{1}{3}, -5$$

$$(14) \quad x^2 + 5x + 6 = 0$$

$$x = -3, -2$$

$$(24) \quad 3x^2 + 4x + 1 = 0$$

$$x = -\frac{1}{3}, -1$$

$$(5) \quad 3x^2 + 2x - 8 = 0$$

$$x = \frac{4}{3}, -2$$

$$(15) \quad 27x^2 - 24x + 4 = 0$$

$$x = \frac{2}{9}, \frac{2}{3}$$

$$(25) \quad x^2 + 6x - 7 = 0$$

$$x = 1, -7$$

$$(6) \quad 6x^2 + 31x + 28 = 0$$

$$x = -4, -\frac{7}{6}$$

$$(16) \quad 5x^2 + 14x + 8 = 0$$

$$x = -2, -\frac{4}{5}$$

$$(26) \quad 3x^2 + 7x - 6 = 0$$

$$x = -3, \frac{2}{3}$$

$$(7) \quad 2x^2 + 3x - 2 = 0$$

$$x = \frac{1}{2}, -2$$

$$(17) \quad 8x^2 + 6x - 5 = 0$$

$$x = \frac{1}{2}, -\frac{5}{4}$$

$$(27) \quad 28x^2 + 11x + 1 = 0$$

$$x = -\frac{1}{4}, -\frac{1}{7}$$

$$(8) \quad 25x^2 + 35x + 6 = 0$$

$$x = -\frac{6}{5}, -\frac{1}{5}$$

$$(18) \quad 3x^2 + 13x + 4 = 0$$

$$x = -4, -\frac{1}{3}$$

$$(28) \quad 8x^2 - 34x + 21 = 0$$

$$x = \frac{7}{2}, \frac{3}{4}$$

$$(9) \quad 72x^2 + 79x + 14 = 0$$

$$x = -\frac{7}{8}, -\frac{2}{9}$$

$$(19) \quad 18x^2 + 11x - 24 = 0$$

$$x = -\frac{3}{2}, \frac{8}{9}$$

$$(29) \quad 56x^2 + 57x - 8 = 0$$

$$x = -\frac{8}{7}, \frac{1}{8}$$

$$(10) \quad 7x^2 + 66x + 27 = 0$$

$$x = -\frac{3}{7}, -9$$

$$(20) \quad 2x^2 + 7x + 5 = 0$$

$$x = -1, -\frac{5}{2}$$

$$(30) \quad 9x^2 - 4 = 0$$

$$x = \frac{2}{3}, -\frac{2}{3}$$

$$(1) \quad 2x^2 + 5x + 3 = 0$$

$$x = -1, -\frac{3}{2}$$

$$(11) \quad 6x^2 - 17x + 12 = 0$$

$$x = \frac{4}{3}, \frac{3}{2}$$

$$(21) \quad 18x^2 - 37x + 15 = 0$$

$$x = \frac{3}{2}, \frac{5}{9}$$

$$(2) \quad 8x^2 + 22x - 21 = 0$$

$$x = \frac{3}{4}, -\frac{7}{2}$$

$$(12) \quad 3x^2 + 20x + 12 = 0$$

$$x = -\frac{2}{3}, -6$$

$$(22) \quad 9x^2 + 18x + 5 = 0$$

$$x = -\frac{5}{3}, -\frac{1}{3}$$

$$(3) \quad 6x^2 + 13x + 7 = 0$$

$$x = -1, -\frac{7}{6}$$

$$(13) \quad 7x^2 - 11x - 6 = 0$$

$$x = 2, -\frac{3}{7}$$

$$(23) \quad 8x^2 + 59x + 21 = 0$$

$$x = -\frac{3}{8}, -7$$

$$(4) \quad 14x^2 - 55x + 21 = 0$$

$$x = \frac{7}{2}, \frac{3}{7}$$

$$(14) \quad 8x^2 - 10x - 7 = 0$$

$$x = \frac{7}{4}, -\frac{1}{2}$$

$$(24) \quad x^2 - 2x + 1 = 0$$

$$x = 1$$

$$(5) \quad 21x^2 - 11x - 6 = 0$$

$$x = -\frac{1}{3}, \frac{6}{7}$$

$$(15) \quad 8x^2 - 34x - 9 = 0$$

$$x = \frac{9}{2}, -\frac{1}{4}$$

$$(25) \quad 15x^2 - 4x - 3 = 0$$

$$x = -\frac{1}{3}, \frac{3}{5}$$

$$(6) \quad 14x^2 + 9x - 18 = 0$$

$$x = \frac{6}{7}, -\frac{3}{2}$$

$$(16) \quad 49x^2 + 28x + 3 = 0$$

$$x = -\frac{1}{7}, -\frac{3}{7}$$

$$(26) \quad 2x^2 - 11x - 21 = 0$$

$$x = -\frac{3}{2}, 7$$

$$(7) \quad 24x^2 - 13x - 7 = 0$$

$$x = -\frac{1}{3}, \frac{7}{8}$$

$$(17) \quad 63x^2 - 31x - 10 = 0$$

$$x = \frac{5}{7}, -\frac{2}{9}$$

$$(27) \quad 14x^2 + 79x + 72 = 0$$

$$x = -\frac{9}{2}, -\frac{8}{7}$$

$$(8) \quad 5x^2 - 21x + 18 = 0$$

$$x = \frac{6}{5}, 3$$

$$(18) \quad 10x^2 + 29x + 21 = 0$$

$$x = -\frac{7}{5}, -\frac{3}{2}$$

$$(28) \quad 3x^2 - 31x + 36 = 0$$

$$x = 9, \frac{4}{3}$$

$$(9) \quad 36x^2 + 37x - 10 = 0$$

$$x = \frac{2}{9}, -\frac{5}{4}$$

$$(19) \quad 72x^2 - 11x - 6 = 0$$

$$x = -\frac{2}{9}, \frac{3}{8}$$

$$(29) \quad 8x^2 + x - 7 = 0$$

$$x = -1, \frac{7}{8}$$

$$(10) \quad 16x^2 - 30x + 9 = 0$$

$$x = \frac{3}{2}, \frac{3}{8}$$

$$(20) \quad 10x^2 + 9x + 2 = 0$$

$$x = -\frac{1}{2}, -\frac{2}{5}$$

$$(30) \quad 9x^2 - 13x + 4 = 0$$

$$x = 1, \frac{4}{9}$$

$$(1) \quad 2x^2 - 3x - 9 = 0$$
$$x = 3, -\frac{3}{2}$$

$$(11) \quad x^2 - 2x - 8 = 0$$
$$x = -2, 4$$

$$(21) \quad x^2 - 7x + 6 = 0$$
$$x = 6, 1$$

$$(2) \quad x^2 + 2x + 1 = 0$$
$$x = -1$$

$$(12) \quad 28x^2 + 3x - 18 = 0$$
$$x = -\frac{6}{7}, \frac{3}{4}$$

$$(22) \quad 20x^2 - 7x - 6 = 0$$
$$x = -\frac{2}{5}, \frac{3}{4}$$

$$(3) \quad 12x^2 - 25x + 7 = 0$$
$$x = \frac{7}{4}, \frac{1}{3}$$

$$(13) \quad 45x^2 - 79x + 30 = 0$$
$$x = \frac{5}{9}, \frac{6}{5}$$

$$(23) \quad 9x^2 - 15x - 14 = 0$$
$$x = -\frac{2}{3}, \frac{7}{3}$$

$$(4) \quad 63x^2 + 10x - 25 = 0$$
$$x = \frac{5}{9}, -\frac{5}{7}$$

$$(14) \quad 54x^2 + 15x - 4 = 0$$
$$x = \frac{1}{6}, -\frac{4}{9}$$

$$(24) \quad 5x^2 + 8x - 21 = 0$$
$$x = \frac{7}{5}, -3$$

$$(5) \quad x^2 - 6x + 9 = 0$$
$$x = 3$$

$$(15) \quad 3x^2 - 2x - 1 = 0$$
$$x = 1, -\frac{1}{3}$$

$$(25) \quad 7x^2 - 9x + 2 = 0$$
$$x = 1, \frac{2}{7}$$

$$(6) \quad 36x^2 - 29x + 5 = 0$$
$$x = \frac{5}{9}, \frac{1}{4}$$

$$(16) \quad 3x^2 - 2x - 1 = 0$$
$$x = 1, -\frac{1}{3}$$

$$(26) \quad 7x^2 + 10x + 3 = 0$$
$$x = -1, -\frac{3}{7}$$

$$(7) \quad 9x^2 + 6x - 8 = 0$$
$$x = \frac{2}{3}, -\frac{4}{3}$$

$$(17) \quad 2x^2 - 5x + 3 = 0$$
$$x = \frac{3}{2}, 1$$

$$(27) \quad 32x^2 - 100x + 63 = 0$$
$$x = \frac{9}{4}, \frac{7}{8}$$

$$(8) \quad 7x^2 - 6x - 16 = 0$$
$$x = -\frac{8}{7}, 2$$

$$(18) \quad 2x^2 + 11x - 21 = 0$$
$$x = -7, \frac{3}{2}$$

$$(28) \quad 12x^2 + x - 20 = 0$$
$$x = \frac{5}{4}, -\frac{4}{3}$$

$$(9) \quad 2x^2 - 3x + 1 = 0$$
$$x = 1, \frac{1}{2}$$

$$(19) \quad 7x^2 - 8x + 1 = 0$$
$$x = \frac{1}{7}, 1$$

$$(29) \quad 10x^2 + 33x - 7 = 0$$
$$x = \frac{1}{5}, -\frac{7}{2}$$

$$(10) \quad 8x^2 - 31x + 21 = 0$$
$$x = \frac{7}{8}, 3$$

$$(20) \quad 32x^2 - 28x - 9 = 0$$
$$x = \frac{9}{8}, -\frac{1}{4}$$

$$(30) \quad 14x^2 + 37x + 24 = 0$$
$$x = -\frac{3}{2}, -\frac{8}{7}$$

$$(1) \quad 5x^2 - 6x - 27 = 0$$

$$x = 3, -\frac{9}{5}$$

$$(11) \quad x^2 - 9 = 0$$

$$x = -3, 3$$

$$(21) \quad 9x^2 + 25x - 6 = 0$$

$$x = -3, \frac{2}{9}$$

$$(2) \quad 5x^2 - 9x + 4 = 0$$

$$x = \frac{4}{5}, 1$$

$$(12) \quad 7x^2 + 39x + 20 = 0$$

$$x = -\frac{4}{7}, -5$$

$$(22) \quad 4x^2 + 15x + 14 = 0$$

$$x = -2, -\frac{7}{4}$$

$$(3) \quad 2x^2 - 17x + 36 = 0$$

$$x = \frac{9}{2}, 4$$

$$(13) \quad 2x^2 - 5x + 3 = 0$$

$$x = 1, \frac{3}{2}$$

$$(23) \quad 4x^2 - 4x - 3 = 0$$

$$x = \frac{3}{2}, -\frac{1}{2}$$

$$(4) \quad 12x^2 + 37x + 21 = 0$$

$$x = -\frac{7}{3}, -\frac{3}{4}$$

$$(14) \quad 40x^2 + 107x + 63 = 0$$

$$x = -\frac{9}{5}, -\frac{7}{8}$$

$$(24) \quad 4x^2 - x - 3 = 0$$

$$x = 1, -\frac{3}{4}$$

$$(5) \quad 3x^2 - 8x + 4 = 0$$

$$x = 2, \frac{2}{3}$$

$$(15) \quad 27x^2 + 30x + 7 = 0$$

$$x = -\frac{1}{3}, -\frac{7}{9}$$

$$(25) \quad 6x^2 - 7x - 3 = 0$$

$$x = -\frac{1}{3}, \frac{3}{2}$$

$$(6) \quad 4x^2 + 15x + 14 = 0$$

$$x = -\frac{7}{4}, -2$$

$$(16) \quad 21x^2 + 17x - 30 = 0$$

$$x = \frac{6}{7}, -\frac{5}{3}$$

$$(26) \quad x^2 - 1 = 0$$

$$x = 1, -1$$

$$(7) \quad 10x^2 - 7x + 1 = 0$$

$$x = \frac{1}{2}, \frac{1}{5}$$

$$(17) \quad 32x^2 + 4x - 15 = 0$$

$$x = \frac{5}{8}, -\frac{3}{4}$$

$$(27) \quad 45x^2 + 71x + 28 = 0$$

$$x = -\frac{4}{5}, -\frac{7}{9}$$

$$(8) \quad 5x^2 + 16x + 3 = 0$$

$$x = -3, -\frac{1}{5}$$

$$(18) \quad 3x^2 - 4x + 1 = 0$$

$$x = 1, \frac{1}{3}$$

$$(28) \quad 2x^2 + 3x + 1 = 0$$

$$x = -\frac{1}{2}, -1$$

$$(9) \quad 12x^2 + 4x - 5 = 0$$

$$x = \frac{1}{2}, -\frac{5}{6}$$

$$(19) \quad 24x^2 - 17x + 3 = 0$$

$$x = \frac{3}{8}, \frac{1}{3}$$

$$(29) \quad 4x^2 + 33x - 27 = 0$$

$$x = -9, \frac{3}{4}$$

$$(10) \quad 10x^2 - 53x + 63 = 0$$

$$x = \frac{9}{5}, \frac{7}{2}$$

$$(20) \quad 6x^2 + 13x - 28 = 0$$

$$x = -\frac{7}{2}, \frac{4}{3}$$

$$(30) \quad x^2 - 4x + 3 = 0$$

$$x = 1, 3$$

$$(1) \quad 4x^2 - 3x - 1 = 0$$

$$x = -\frac{1}{4}, 1$$

$$(11) \quad 2x^2 - 7x + 6 = 0$$

$$x = \frac{3}{2}, 2$$

$$(21) \quad 4x^2 + 11x + 7 = 0$$

$$x = -1, -\frac{7}{4}$$

$$(2) \quad x^2 + 4x + 3 = 0$$

$$x = -1, -3$$

$$(12) \quad 16x^2 + 2x - 5 = 0$$

$$x = -\frac{5}{8}, \frac{1}{2}$$

$$(22) \quad 48x^2 + 22x - 15 = 0$$

$$x = -\frac{5}{6}, \frac{3}{8}$$

$$(3) \quad 4x^2 - 3x - 1 = 0$$

$$x = -\frac{1}{4}, 1$$

$$(13) \quad 40x^2 - 91x + 49 = 0$$

$$x = \frac{7}{8}, \frac{7}{5}$$

$$(23) \quad x^2 - 7x + 12 = 0$$

$$x = 4, 3$$

$$(4) \quad 28x^2 - 43x + 9 = 0$$

$$x = \frac{9}{7}, \frac{1}{4}$$

$$(14) \quad 56x^2 + 15x - 54 = 0$$

$$x = \frac{6}{7}, -\frac{9}{8}$$

$$(24) \quad 36x^2 + 73x - 18 = 0$$

$$x = \frac{2}{9}, -\frac{9}{4}$$

$$(5) \quad 16x^2 + 2x - 3 = 0$$

$$x = -\frac{1}{2}, \frac{3}{8}$$

$$(15) \quad 20x^2 + 37x + 8 = 0$$

$$x = -\frac{8}{5}, -\frac{1}{4}$$

$$(25) \quad 63x^2 - 46x + 8 = 0$$

$$x = \frac{2}{7}, \frac{4}{9}$$

$$(6) \quad 21x^2 + 44x - 32 = 0$$

$$x = \frac{4}{7}, -\frac{8}{3}$$

$$(16) \quad 3x^2 - 8x + 5 = 0$$

$$x = 1, \frac{5}{3}$$

$$(26) \quad 5x^2 - 14x + 8 = 0$$

$$x = \frac{4}{5}, 2$$

$$(7) \quad 3x^2 + 4x + 1 = 0$$

$$x = -1, -\frac{1}{3}$$

$$(17) \quad 10x^2 + 3x - 4 = 0$$

$$x = -\frac{4}{5}, \frac{1}{2}$$

$$(27) \quad 2x^2 + 3x - 2 = 0$$

$$x = \frac{1}{2}, -2$$

$$(8) \quad x^2 - 2x - 3 = 0$$

$$x = -1, 3$$

$$(18) \quad 5x^2 - 27x + 28 = 0$$

$$x = \frac{7}{5}, 4$$

$$(28) \quad 28x^2 + 29x - 35 = 0$$

$$x = \frac{5}{7}, -\frac{7}{4}$$

$$(9) \quad 2x^2 - x - 1 = 0$$

$$x = -\frac{1}{2}, 1$$

$$(19) \quad x^2 + 3x + 2 = 0$$

$$x = -2, -1$$

$$(29) \quad 3x^2 + 22x - 16 = 0$$

$$x = -8, \frac{2}{3}$$

$$(10) \quad 2x^2 + 5x - 12 = 0$$

$$x = \frac{3}{2}, -4$$

$$(20) \quad 28x^2 + 45x + 18 = 0$$

$$x = -\frac{3}{4}, -\frac{6}{7}$$

$$(30) \quad 9x^2 - 34x + 21 = 0$$

$$x = 3, \frac{7}{9}$$

$$(1) \quad x^2 + 8x + 7 = 0$$

$$x = -7, -1$$

$$(11) \quad 3x^2 + 4x + 1 = 0$$

$$x = -\frac{1}{3}, -1$$

$$(21) \quad 18x^2 + 33x - 40 = 0$$

$$x = -\frac{8}{3}, \frac{5}{6}$$

$$(2) \quad 36x^2 + 65x - 36 = 0$$

$$x = \frac{4}{9}, -\frac{9}{4}$$

$$(12) \quad 7x^2 - 29x + 24 = 0$$

$$x = 3, \frac{8}{7}$$

$$(22) \quad 5x^2 + 13x - 6 = 0$$

$$x = -3, \frac{2}{5}$$

$$(3) \quad 32x^2 - 52x - 7 = 0$$

$$x = -\frac{1}{8}, \frac{7}{4}$$

$$(13) \quad 9x^2 - 15x - 14 = 0$$

$$x = -\frac{2}{3}, \frac{7}{3}$$

$$(23) \quad x^2 - 3x + 2 = 0$$

$$x = 1, 2$$

$$(4) \quad 7x^2 - 2x - 5 = 0$$

$$x = -\frac{5}{7}, 1$$

$$(14) \quad 4x^2 + 20x + 9 = 0$$

$$x = -\frac{9}{2}, -\frac{1}{2}$$

$$(24) \quad 7x^2 + 8x - 12 = 0$$

$$x = -2, \frac{6}{7}$$

$$(5) \quad 16x^2 - 18x + 5 = 0$$

$$x = \frac{1}{2}, \frac{5}{8}$$

$$(15) \quad 5x^2 - 29x + 36 = 0$$

$$x = \frac{9}{5}, 4$$

$$(25) \quad 20x^2 - 3x - 35 = 0$$

$$x = \frac{7}{5}, -\frac{5}{4}$$

$$(6) \quad 16x^2 - 1 = 0$$

$$x = -\frac{1}{4}, \frac{1}{4}$$

$$(16) \quad 8x^2 + 59x + 21 = 0$$

$$x = -7, -\frac{3}{8}$$

$$(26) \quad 56x^2 + 73x + 21 = 0$$

$$x = -\frac{7}{8}, -\frac{3}{7}$$

$$(7) \quad 4x^2 + 11x + 7 = 0$$

$$x = -\frac{7}{4}, -1$$

$$(17) \quad 2x^2 + 17x - 9 = 0$$

$$x = \frac{1}{2}, -9$$

$$(27) \quad 7x^2 - 25x + 12 = 0$$

$$x = 3, \frac{4}{7}$$

$$(8) \quad 24x^2 + 22x - 7 = 0$$

$$x = -\frac{7}{6}, \frac{1}{4}$$

$$(18) \quad 9x^2 - 65x - 56 = 0$$

$$x = -\frac{7}{9}, 8$$

$$(28) \quad 6x^2 - 23x - 18 = 0$$

$$x = \frac{9}{2}, -\frac{2}{3}$$

$$(9) \quad 21x^2 + 17x - 8 = 0$$

$$x = \frac{1}{3}, -\frac{8}{7}$$

$$(19) \quad 12x^2 + 25x + 12 = 0$$

$$x = -\frac{3}{4}, -\frac{4}{3}$$

$$(29) \quad 24x^2 + 58x + 9 = 0$$

$$x = -\frac{1}{6}, -\frac{9}{4}$$

$$(10) \quad 56x^2 - 57x + 7 = 0$$

$$x = \frac{1}{7}, \frac{7}{8}$$

$$(20) \quad 24x^2 + 26x + 5 = 0$$

$$x = -\frac{5}{6}, -\frac{1}{4}$$

$$(30) \quad 32x^2 - 44x - 21 = 0$$

$$x = -\frac{3}{8}, \frac{7}{4}$$

$$(1) \quad 72x^2 + 29x - 10 = 0 \\ x = -\frac{5}{8}, \frac{2}{9}$$

$$(11) \quad 5x^2 - 14x + 9 = 0 \\ x = \frac{9}{5}, 1$$

$$(21) \quad 28x^2 + 61x + 21 = 0 \\ x = -\frac{7}{4}, -\frac{3}{7}$$

$$(2) \quad 8x^2 - 17x + 2 = 0 \\ x = \frac{1}{8}, 2$$

$$(12) \quad 25x^2 - 80x + 64 = 0 \\ x = \frac{8}{5}$$

$$(22) \quad 40x^2 - 27x - 4 = 0 \\ x = \frac{4}{5}, -\frac{1}{8}$$

$$(3) \quad 8x^2 - 9x + 1 = 0 \\ x = \frac{1}{8}, 1$$

$$(13) \quad 7x^2 - 8x + 1 = 0 \\ x = 1, \frac{1}{7}$$

$$(23) \quad 10x^2 - 21x + 9 = 0 \\ x = \frac{3}{2}, \frac{3}{5}$$

$$(4) \quad 18x^2 + 21x + 5 = 0 \\ x = -\frac{5}{6}, -\frac{1}{3}$$

$$(14) \quad 27x^2 + 69x + 14 = 0 \\ x = -\frac{7}{3}, -\frac{2}{9}$$

$$(24) \quad 6x^2 + 5x - 4 = 0 \\ x = -\frac{4}{3}, \frac{1}{2}$$

$$(5) \quad 2x^2 + 5x + 3 = 0 \\ x = -1, -\frac{3}{2}$$

$$(15) \quad 6x^2 - 7x + 1 = 0 \\ x = 1, \frac{1}{6}$$

$$(25) \quad 16x^2 - 50x + 25 = 0 \\ x = \frac{5}{8}, \frac{5}{2}$$

$$(6) \quad 21x^2 + 31x + 8 = 0 \\ x = -\frac{1}{3}, -\frac{8}{7}$$

$$(16) \quad 9x^2 - 25 = 0 \\ x = \frac{5}{3}, -\frac{5}{3}$$

$$(26) \quad 4x^2 - 12x - 27 = 0 \\ x = -\frac{3}{2}, \frac{9}{2}$$

$$(7) \quad 2x^2 + 5x + 2 = 0 \\ x = -2, -\frac{1}{2}$$

$$(17) \quad 40x^2 + 79x + 24 = 0 \\ x = -\frac{8}{5}, -\frac{3}{8}$$

$$(27) \quad 3x^2 - 4x - 4 = 0 \\ x = -\frac{2}{3}, 2$$

$$(8) \quad 24x^2 - x - 10 = 0 \\ x = \frac{2}{3}, -\frac{5}{8}$$

$$(18) \quad 35x^2 - 6x - 8 = 0 \\ x = \frac{4}{7}, -\frac{2}{5}$$

$$(28) \quad x^2 - 1 = 0 \\ x = 1, -1$$

$$(9) \quad 35x^2 + 3x - 20 = 0 \\ x = \frac{5}{7}, -\frac{4}{5}$$

$$(19) \quad x^2 - x - 2 = 0 \\ x = 2, -1$$

$$(29) \quad 4x^2 + 19x + 12 = 0 \\ x = -4, -\frac{3}{4}$$

$$(10) \quad 5x^2 + 12x + 7 = 0 \\ x = -\frac{7}{5}, -1$$

$$(20) \quad 3x^2 - x - 2 = 0 \\ x = 1, -\frac{2}{3}$$

$$(30) \quad 9x^2 - 43x + 28 = 0 \\ x = \frac{7}{9}, 4$$

$$(1) 4x^2 + x - 3 = 0$$
$$x = -1, \frac{3}{4}$$

$$(11) 3x^2 + 7x - 6 = 0$$
$$x = \frac{2}{3}, -3$$

$$(21) 3x^2 + 11x + 6 = 0$$
$$x = -\frac{2}{3}, -3$$

$$(2) 7x^2 + 13x - 24 = 0$$
$$x = -3, \frac{8}{7}$$

$$(12) 3x^2 + 4x - 4 = 0$$
$$x = -2, \frac{2}{3}$$

$$(22) 45x^2 + 83x + 28 = 0$$
$$x = -\frac{4}{9}, -\frac{7}{5}$$

$$(3) 14x^2 + 13x + 3 = 0$$
$$x = -\frac{3}{7}, -\frac{1}{2}$$

$$(13) 2x^2 + 3x - 9 = 0$$
$$x = -3, \frac{3}{2}$$

$$(23) 35x^2 + 9x - 2 = 0$$
$$x = \frac{1}{7}, -\frac{2}{5}$$

$$(4) 36x^2 - 84x + 49 = 0$$
$$x = \frac{7}{6}$$

$$(14) 9x^2 + 23x + 10 = 0$$
$$x = -2, -\frac{5}{9}$$

$$(24) 12x^2 - 11x - 5 = 0$$
$$x = \frac{5}{4}, -\frac{1}{3}$$

$$(5) 54x^2 - 39x - 5 = 0$$
$$x = \frac{5}{6}, -\frac{1}{9}$$

$$(15) 5x^2 - 4x - 1 = 0$$
$$x = -\frac{1}{5}, 1$$

$$(25) 9x^2 + 16x + 7 = 0$$
$$x = -1, -\frac{7}{9}$$

$$(6) 28x^2 - x - 15 = 0$$
$$x = \frac{3}{4}, -\frac{5}{7}$$

$$(16) 2x^2 + 5x + 3 = 0$$
$$x = -\frac{3}{2}, -1$$

$$(26) 45x^2 - 92x + 32 = 0$$
$$x = \frac{8}{5}, \frac{4}{9}$$

$$(7) 15x^2 - 7x - 2 = 0$$
$$x = \frac{2}{3}, -\frac{1}{5}$$

$$(17) 7x^2 - 25x - 12 = 0$$
$$x = 4, -\frac{3}{7}$$

$$(27) 4x^2 + 17x - 15 = 0$$
$$x = -5, \frac{3}{4}$$

$$(8) 2x^2 - 13x + 18 = 0$$
$$x = \frac{9}{2}, 2$$

$$(18) 6x^2 + x - 7 = 0$$
$$x = -\frac{7}{6}, 1$$

$$(28) 7x^2 - 22x + 3 = 0$$
$$x = 3, \frac{1}{7}$$

$$(9) 28x^2 + 9x - 9 = 0$$
$$x = -\frac{3}{4}, \frac{3}{7}$$

$$(19) 14x^2 - 33x - 56 = 0$$
$$x = \frac{7}{2}, -\frac{8}{7}$$

$$(29) 12x^2 + 41x + 24 = 0$$
$$x = -\frac{8}{3}, -\frac{3}{4}$$

$$(10) 3x^2 - 16x - 12 = 0$$
$$x = -\frac{2}{3}, 6$$

$$(20) 4x^2 + 17x + 4 = 0$$
$$x = -\frac{1}{4}, -4$$

$$(30) 3x^2 - 4x + 1 = 0$$
$$x = \frac{1}{3}, 1$$

$$(1) \quad 8x^2 - 39x - 54 = 0$$

$$x = -\frac{9}{8}, 6$$

$$(11) \quad 3x^2 - 13x + 4 = 0$$

$$x = \frac{1}{3}, 4$$

$$(21) \quad 10x^2 - x - 24 = 0$$

$$x = -\frac{3}{2}, \frac{8}{5}$$

$$(2) \quad 8x^2 - 15x - 2 = 0$$

$$x = -\frac{1}{8}, 2$$

$$(12) \quad x^2 + 6x - 27 = 0$$

$$x = -9, 3$$

$$(22) \quad 2x^2 - 3x - 2 = 0$$

$$x = 2, -\frac{1}{2}$$

$$(3) \quad x^2 - 2x + 1 = 0$$

$$x = 1$$

$$(13) \quad 24x^2 - 31x + 10 = 0$$

$$x = \frac{5}{8}, \frac{2}{3}$$

$$(23) \quad 28x^2 - 3x - 18 = 0$$

$$x = -\frac{3}{4}, \frac{6}{7}$$

$$(4) \quad 7x^2 + 5x - 2 = 0$$

$$x = \frac{2}{7}, -1$$

$$(14) \quad 2x^2 - 9x - 18 = 0$$

$$x = 6, -\frac{3}{2}$$

$$(24) \quad x^2 - 3x + 2 = 0$$

$$x = 2, 1$$

$$(5) \quad 2x^2 + 13x + 6 = 0$$

$$x = -6, -\frac{1}{2}$$

$$(15) \quad 6x^2 - 7x - 10 = 0$$

$$x = -\frac{5}{6}, 2$$

$$(25) \quad 27x^2 - 33x + 8 = 0$$

$$x = \frac{1}{3}, \frac{8}{9}$$

$$(6) \quad 6x^2 - 23x + 7 = 0$$

$$x = \frac{7}{2}, \frac{1}{3}$$

$$(16) \quad 3x^2 + 8x + 5 = 0$$

$$x = -\frac{5}{3}, -1$$

$$(26) \quad 28x^2 + 75x + 27 = 0$$

$$x = -\frac{9}{4}, -\frac{3}{7}$$

$$(7) \quad 35x^2 - 31x - 40 = 0$$

$$x = -\frac{5}{7}, \frac{8}{5}$$

$$(17) \quad 9x^2 + 74x - 63 = 0$$

$$x = \frac{7}{9}, -9$$

$$(27) \quad 4x^2 + 27x - 40 = 0$$

$$x = \frac{5}{4}, -8$$

$$(8) \quad 9x^2 - 3x - 2 = 0$$

$$x = -\frac{1}{3}, \frac{2}{3}$$

$$(18) \quad 5x^2 + 3x - 8 = 0$$

$$x = 1, -\frac{8}{5}$$

$$(28) \quad 5x^2 - 36x - 81 = 0$$

$$x = -\frac{9}{5}, 9$$

$$(9) \quad 7x^2 + 5x - 18 = 0$$

$$x = \frac{9}{7}, -2$$

$$(19) \quad x^2 - 3x - 4 = 0$$

$$x = -1, 4$$

$$(29) \quad 3x^2 - 8x + 4 = 0$$

$$x = \frac{2}{3}, 2$$

$$(10) \quad 20x^2 - 43x + 14 = 0$$

$$x = \frac{2}{5}, \frac{7}{4}$$

$$(20) \quad 2x^2 + 3x - 20 = 0$$

$$x = -4, \frac{5}{2}$$

$$(30) \quad 12x^2 + 11x - 5 = 0$$

$$x = \frac{1}{3}, -\frac{5}{4}$$

$$(1) \quad x^2 - 1 = 0$$

$$x = 1, -1$$

$$(11) \quad 7x^2 + 18x + 8 = 0$$

$$x = -\frac{4}{7}, -2$$

$$(21) \quad x^2 + 7x + 12 = 0$$

$$x = -3, -4$$

$$(2) \quad x^2 + x - 2 = 0$$

$$x = 1, -2$$

$$(12) \quad 12x^2 - 8x - 15 = 0$$

$$x = \frac{3}{2}, -\frac{5}{6}$$

$$(22) \quad 3x^2 + x - 2 = 0$$

$$x = -1, \frac{2}{3}$$

$$(3) \quad 7x^2 + 5x - 2 = 0$$

$$x = \frac{2}{7}, -1$$

$$(13) \quad 36x^2 - 23x - 8 = 0$$

$$x = \frac{8}{9}, -\frac{1}{4}$$

$$(23) \quad 4x^2 + 4x - 3 = 0$$

$$x = -\frac{3}{2}, \frac{1}{2}$$

$$(4) \quad 8x^2 - 2x - 1 = 0$$

$$x = -\frac{1}{4}, \frac{1}{2}$$

$$(14) \quad 12x^2 - 5x - 2 = 0$$

$$x = -\frac{1}{4}, \frac{2}{3}$$

$$(24) \quad 8x^2 + 11x - 10 = 0$$

$$x = \frac{5}{8}, -2$$

$$(5) \quad 4x^2 + 23x + 15 = 0$$

$$x = -5, -\frac{3}{4}$$

$$(15) \quad 21x^2 - 13x - 18 = 0$$

$$x = -\frac{2}{3}, \frac{9}{7}$$

$$(25) \quad 8x^2 + 15x + 7 = 0$$

$$x = -\frac{7}{8}, -1$$

$$(6) \quad 2x^2 + 11x + 12 = 0$$

$$x = -4, -\frac{3}{2}$$

$$(16) \quad 5x^2 + 39x + 54 = 0$$

$$x = -6, -\frac{9}{5}$$

$$(26) \quad 30x^2 - 41x + 7 = 0$$

$$x = \frac{7}{6}, \frac{1}{5}$$

$$(7) \quad 4x^2 - 13x + 9 = 0$$

$$x = 1, \frac{9}{4}$$

$$(17) \quad 21x^2 - 13x - 20 = 0$$

$$x = -\frac{5}{7}, \frac{4}{3}$$

$$(27) \quad 12x^2 + 37x + 21 = 0$$

$$x = -\frac{3}{4}, -\frac{7}{3}$$

$$(8) \quad 3x^2 + 22x + 24 = 0$$

$$x = -\frac{4}{3}, -6$$

$$(18) \quad x^2 - 1 = 0$$

$$x = 1, -1$$

$$(28) \quad 12x^2 - 5x - 25 = 0$$

$$x = -\frac{5}{4}, \frac{5}{3}$$

$$(9) \quad 40x^2 + 73x + 30 = 0$$

$$x = -\frac{6}{5}, -\frac{5}{8}$$

$$(19) \quad 15x^2 + 7x - 36 = 0$$

$$x = \frac{4}{3}, -\frac{9}{5}$$

$$(29) \quad 7x^2 + 12x + 5 = 0$$

$$x = -1, -\frac{5}{7}$$

$$(10) \quad 6x^2 - 19x + 14 = 0$$

$$x = \frac{7}{6}, 2$$

$$(20) \quad x^2 + 4x + 3 = 0$$

$$x = -1, -3$$

$$(30) \quad 49x^2 + 63x + 18 = 0$$

$$x = -\frac{6}{7}, -\frac{3}{7}$$

$$(1) \quad 8x^2 - 14x + 5 = 0$$

$$x = \frac{1}{2}, \frac{5}{4}$$

$$(11) \quad 28x^2 + 99x + 81 = 0$$

$$x = -\frac{9}{7}, -\frac{9}{4}$$

$$(21) \quad 2x^2 - 9x + 7 = 0$$

$$x = \frac{7}{2}, 1$$

$$(2) \quad 27x^2 - 96x + 64 = 0$$

$$x = \frac{8}{9}, \frac{8}{3}$$

$$(12) \quad 15x^2 - 26x - 21 = 0$$

$$x = \frac{7}{3}, -\frac{3}{5}$$

$$(22) \quad 5x^2 - 21x + 18 = 0$$

$$x = 3, \frac{6}{5}$$

$$(3) \quad 16x^2 - 18x + 5 = 0$$

$$x = \frac{1}{2}, \frac{5}{8}$$

$$(13) \quad 40x^2 + 69x + 8 = 0$$

$$x = -\frac{1}{8}, -\frac{8}{5}$$

$$(23) \quad 6x^2 - 13x - 8 = 0$$

$$x = \frac{8}{3}, -\frac{1}{2}$$

$$(4) \quad 2x^2 - x - 3 = 0$$

$$x = -1, \frac{3}{2}$$

$$(14) \quad 49x^2 + 42x + 8 = 0$$

$$x = -\frac{4}{7}, -\frac{2}{7}$$

$$(24) \quad 9x^2 + 12x + 4 = 0$$

$$x = -\frac{2}{3}$$

$$(5) \quad 9x^2 + 17x + 8 = 0$$

$$x = -1, -\frac{8}{9}$$

$$(15) \quad 5x^2 + 2x - 7 = 0$$

$$x = -\frac{7}{5}, 1$$

$$(25) \quad 81x^2 + 54x + 8 = 0$$

$$x = -\frac{2}{9}, -\frac{4}{9}$$

$$(6) \quad 7x^2 - 65x + 18 = 0$$

$$x = \frac{2}{7}, 9$$

$$(16) \quad 27x^2 - 12x - 7 = 0$$

$$x = \frac{7}{9}, -\frac{1}{3}$$

$$(26) \quad 49x^2 - 28x - 12 = 0$$

$$x = \frac{6}{7}, -\frac{2}{7}$$

$$(7) \quad 16x^2 + 10x - 21 = 0$$

$$x = \frac{7}{8}, -\frac{3}{2}$$

$$(17) \quad 18x^2 + 31x - 35 = 0$$

$$x = \frac{7}{9}, -\frac{5}{2}$$

$$(27) \quad 12x^2 + 5x - 2 = 0$$

$$x = \frac{1}{4}, -\frac{2}{3}$$

$$(8) \quad 10x^2 - x - 3 = 0$$

$$x = -\frac{1}{2}, \frac{3}{5}$$

$$(18) \quad 3x^2 + 26x + 48 = 0$$

$$x = -\frac{8}{3}, -6$$

$$(28) \quad 9x^2 + 61x - 14 = 0$$

$$x = -7, \frac{2}{9}$$

$$(9) \quad 5x^2 + 6x + 1 = 0$$

$$x = -\frac{1}{5}, -1$$

$$(19) \quad 9x^2 - 16 = 0$$

$$x = \frac{4}{3}, \frac{4}{3}$$

$$(29) \quad 24x^2 + 2x - 1 = 0$$

$$x = -\frac{1}{4}, \frac{1}{6}$$

$$(10) \quad 2x^2 - 9x + 4 = 0$$

$$x = \frac{1}{2}, 4$$

$$(20) \quad 21x^2 - 73x + 56 = 0$$

$$x = \frac{8}{7}, \frac{7}{3}$$

$$(30) \quad 6x^2 - 11x - 7 = 0$$

$$x = -\frac{1}{2}, \frac{7}{3}$$

$$(1) \quad x^2 - 4x + 3 = 0$$

$$x = 1, 3$$

$$(11) \quad 27x^2 - 12x - 32 = 0$$

$$x = \frac{4}{3}, -\frac{8}{9}$$

$$(21) \quad 63x^2 - 40x - 7 = 0$$

$$x = \frac{7}{9}, -\frac{1}{7}$$

$$(2) \quad 8x^2 + 47x - 6 = 0$$

$$x = \frac{1}{8}, -6$$

$$(12) \quad 4x^2 - x - 3 = 0$$

$$x = -\frac{3}{4}, 1$$

$$(22) \quad 9x^2 + 17x + 8 = 0$$

$$x = -1, -\frac{8}{9}$$

$$(3) \quad 9x^2 - 56x - 49 = 0$$

$$x = -\frac{7}{9}, 7$$

$$(13) \quad 12x^2 - 7x - 10 = 0$$

$$x = -\frac{2}{3}, \frac{5}{4}$$

$$(23) \quad 45x^2 + 46x + 8 = 0$$

$$x = -\frac{2}{9}, -\frac{4}{5}$$

$$(4) \quad x^2 - 2x + 1 = 0$$

$$x = 1$$

$$(14) \quad 6x^2 - 13x + 6 = 0$$

$$x = \frac{2}{3}, \frac{3}{2}$$

$$(24) \quad 2x^2 + 7x + 5 = 0$$

$$x = -1, -\frac{5}{2}$$

$$(5) \quad 72x^2 + 13x - 20 = 0$$

$$x = \frac{4}{9}, -\frac{5}{8}$$

$$(15) \quad x^2 + 5x + 4 = 0$$

$$x = -1, -4$$

$$(25) \quad 8x^2 + 2x - 1 = 0$$

$$x = \frac{1}{4}, -\frac{1}{2}$$

$$(6) \quad 4x^2 + 7x - 15 = 0$$

$$x = -3, \frac{5}{4}$$

$$(16) \quad 18x^2 - 51x + 35 = 0$$

$$x = \frac{5}{3}, \frac{7}{6}$$

$$(26) \quad 6x^2 - 11x - 2 = 0$$

$$x = -\frac{1}{6}, 2$$

$$(7) \quad 3x^2 - 5x + 2 = 0$$

$$x = 1, \frac{2}{3}$$

$$(17) \quad 5x^2 - 46x + 9 = 0$$

$$x = 9, \frac{1}{5}$$

$$(27) \quad 18x^2 - 29x + 3 = 0$$

$$x = \frac{3}{2}, \frac{1}{9}$$

$$(8) \quad 7x^2 - 30x - 25 = 0$$

$$x = -\frac{5}{7}, 5$$

$$(18) \quad 8x^2 - 10x - 7 = 0$$

$$x = \frac{7}{4}, -\frac{1}{2}$$

$$(28) \quad 4x^2 + 5x + 1 = 0$$

$$x = -1, -\frac{1}{4}$$

$$(9) \quad 27x^2 + 3x - 10 = 0$$

$$x = \frac{5}{9}, -\frac{2}{3}$$

$$(19) \quad 7x^2 - 23x + 18 = 0$$

$$x = \frac{9}{7}, 2$$

$$(29) \quad 9x^2 + 44x - 5 = 0$$

$$x = -5, \frac{1}{9}$$

$$(10) \quad 20x^2 - 9x - 18 = 0$$

$$x = -\frac{3}{4}, \frac{6}{5}$$

$$(20) \quad 14x^2 + 25x + 6 = 0$$

$$x = -\frac{3}{2}, -\frac{2}{7}$$

$$(30) \quad 5x^2 + 29x - 6 = 0$$

$$x = \frac{1}{5}, -6$$

$$(1) \quad 3x^2 + 5x - 12 = 0$$

$$x = \frac{4}{3}, -3$$

$$(11) \quad 25x^2 - 10x - 63 = 0$$

$$x = \frac{9}{5}, -\frac{7}{5}$$

$$(21) \quad 12x^2 + 5x - 3 = 0$$

$$x = -\frac{3}{4}, \frac{1}{3}$$

$$(2) \quad 6x^2 + 13x - 5 = 0$$

$$x = \frac{1}{3}, -\frac{5}{2}$$

$$(12) \quad 2x^2 + 15x - 8 = 0$$

$$x = -8, \frac{1}{2}$$

$$(22) \quad x^2 + 2x - 3 = 0$$

$$x = 1, -3$$

$$(3) \quad 35x^2 + 93x + 54 = 0$$

$$x = -\frac{6}{7}, -\frac{9}{5}$$

$$(13) \quad 56x^2 - 55x - 9 = 0$$

$$x = \frac{9}{8}, -\frac{1}{7}$$

$$(23) \quad 42x^2 + 103x + 63 = 0$$

$$x = -\frac{7}{6}, -\frac{9}{7}$$

$$(4) \quad 2x^2 - 5x - 18 = 0$$

$$x = -2, \frac{9}{2}$$

$$(14) \quad x^2 - 10x + 16 = 0$$

$$x = 8, 2$$

$$(24) \quad 2x^2 + x - 21 = 0$$

$$x = -\frac{7}{2}, 3$$

$$(5) \quad 4x^2 - 23x + 28 = 0$$

$$x = 4, \frac{7}{4}$$

$$(15) \quad 9x^2 + 9x - 10 = 0$$

$$x = \frac{2}{3}, -\frac{5}{3}$$

$$(25) \quad 4x^2 - 12x + 5 = 0$$

$$x = \frac{1}{2}, \frac{5}{2}$$

$$(6) \quad 9x^2 + 9x + 2 = 0$$

$$x = -\frac{1}{3}, -\frac{2}{3}$$

$$(16) \quad 3x^2 - 29x + 18 = 0$$

$$x = 9, \frac{2}{3}$$

$$(26) \quad 16x^2 + 6x - 1 = 0$$

$$x = \frac{1}{8}, -\frac{1}{2}$$

$$(7) \quad 36x^2 - 73x + 35 = 0$$

$$x = \frac{7}{9}, \frac{5}{4}$$

$$(17) \quad 10x^2 - x - 3 = 0$$

$$x = \frac{3}{5}, -\frac{1}{2}$$

$$(27) \quad 9x^2 + 25x + 14 = 0$$

$$x = -2, -\frac{7}{9}$$

$$(8) \quad 7x^2 - 65x + 18 = 0$$

$$x = 9, \frac{2}{7}$$

$$(18) \quad 49x^2 + 7x - 20 = 0$$

$$x = \frac{4}{7}, -\frac{5}{7}$$

$$(28) \quad 2x^2 + 5x - 3 = 0$$

$$x = -3, \frac{1}{2}$$

$$(9) \quad 36x^2 + 5x - 24 = 0$$

$$x = -\frac{8}{9}, \frac{3}{4}$$

$$(19) \quad 45x^2 + 8x - 21 = 0$$

$$x = -\frac{7}{9}, \frac{3}{5}$$

$$(29) \quad 10x^2 - 11x - 8 = 0$$

$$x = -\frac{1}{2}, \frac{8}{5}$$

$$(10) \quad 4x^2 - 9x - 28 = 0$$

$$x = -\frac{7}{4}, 4$$

$$(20) \quad 16x^2 - 38x + 21 = 0$$

$$x = \frac{3}{2}, \frac{7}{8}$$

$$(30) \quad 6x^2 - 13x - 15 = 0$$

$$x = 3, -\frac{5}{6}$$

2.10 1 次不等式

1 次不等式の問題。

(1) $2x - 7 \geq -8x + 8$

(16) $2x - 3 \geq 9x - 8$

(31) $4x + 6 < -x + 7$

(2) $-6x - 2 \geq 4x - 5$

(17) $7x - 8 \geq 2x + 3$

(32) $-4x + 1 \leq 7x + 9$

(3) $3x + 9 < -3x + 5$

(18) $x - 5 \leq 6x - 3$

(33) $5x + 6 \geq -5x + 4$

(4) $9x + 3 \geq -5x - 9$

(19) $-8x - 2 > -3x - 4$

(34) $-9x + 3 > -8x - 2$

(5) $-6x + 8 < -4x - 9$

(20) $x + 3 \geq -4x + 8$

(35) $5x + 9 \geq -2x - 5$

(6) $-6x - 7 \leq 3x + 7$

(21) $5x - 9 > x + 9$

(36) $-5x - 5 > -6x + 7$

(7) $x - 9 \leq 2x - 7$

(22) $3x - 9 < 8x + 2$

(37) $3x + 8 \leq -2x - 9$

(8) $-6x + 3 \leq 9x + 4$

(23) $2x + 7 \geq -4x - 4$

(38) $5x + 6 > 3x - 6$

(9) $-2x - 8 > 5x + 8$

(24) $-8x - 3 > 2x - 9$

(39) $3x + 7 < 9x - 5$

(10) $9x + 6 \leq -5x + 9$

(25) $-7x - 9 < -2x - 1$

(40) $-6x - 6 \geq -8x - 7$

(11) $-2x - 8 < 6x - 6$

(26) $9x - 4 < -x + 7$

(41) $-7x - 3 > 5x + 9$

(12) $x - 5 \geq 3x - 8$

(27) $-5x - 8 \geq -4x - 9$

(42) $6x + 5 \leq 2x - 4$

(13) $-6x + 7 < -4x - 7$

(28) $-8x - 3 \geq 2x + 4$

(43) $-x + 2 \geq -2x - 2$

(14) $-2x - 5 > -9x + 8$

(29) $9x - 8 > -5x - 5$

(44) $-2x - 3 \leq x + 7$

(15) $-6x - 9 \leq 9x + 9$

(30) $5x - 8 \leq -3x - 1$

(45) $-9x + 8 < 2x + 7$

(1) $6x - 8 > 9x + 8$

(16) $4x - 7 < -7x + 5$

(31) $6x + 5 < 8x + 8$

(2) $5x - 4 \leq -3x + 1$

(17) $-2x + 2 > -3x + 1$

(32) $5x + 3 < -7x + 7$

(3) $-x + 8 \geq -4x + 4$

(18) $-x - 8 < 2x + 9$

(33) $7x - 7 < 9x - 3$

(4) $2x + 2 > 9x + 8$

(19) $-3x + 2 \leq 4x - 6$

(34) $-4x + 9 > 7x - 6$

(5) $5x + 5 \leq 3x - 8$

(20) $-3x - 5 \leq -4x - 3$

(35) $-5x + 7 \geq -7x + 3$

(6) $-7x - 3 \geq -4x - 9$

(21) $4x + 8 > -7x + 9$

(36) $2x + 5 < -8x + 2$

(7) $-4x - 6 \leq 3x - 5$

(22) $-x - 1 > -9x - 2$

(37) $4x + 9 \geq 7x - 3$

(8) $-4x + 1 > -2x + 8$

(23) $8x - 9 < 4x + 2$

(38) $7x - 4 \geq -9x + 3$

(9) $-4x - 2 > 5x + 8$

(24) $-4x - 7 \geq -3x + 9$

(39) $x - 4 \geq 2x - 6$

(10) $-3x + 5 > -5x + 9$

(25) $9x + 2 \geq 4x - 6$

(40) $-4x + 8 > -3x - 8$

(11) $-4x + 3 \geq 2x + 2$

(26) $-5x + 1 \geq -x - 3$

(41) $-2x + 3 \geq 9x + 4$

(12) $8x + 4 < x - 8$

(27) $-2x - 6 \geq -5x - 1$

(42) $8x + 4 > -7x - 2$

(13) $7x + 3 < 3x + 8$

(28) $3x - 3 < x - 6$

(43) $-x + 9 > -2x + 7$

(14) $-8x + 2 < -5x - 3$

(29) $-6x + 5 \leq -2x + 6$

(44) $-4x - 6 \leq -8x - 3$

(15) $-5x + 5 > -7x - 1$

(30) $-3x - 7 \geq -7x - 5$

(45) $-3x + 1 > 8x + 4$

(1) $x + 9 \geq 6x - 4$

(16) $4x + 6 \leq 7x + 6$

(31) $6x - 8 \geq -9x + 6$

(2) $-x + 9 \geq 9x + 4$

(17) $2x + 2 > -4x + 2$

(32) $-9x + 7 < -2x - 7$

(3) $5x - 4 \geq -x - 4$

(18) $9x - 8 \leq -9x - 8$

(33) $x - 8 > 9x - 2$

(4) $9x + 3 > 5x + 4$

(19) $-x - 3 > -8x + 5$

(34) $-6x - 3 \geq x - 9$

(5) $-9x + 3 \leq -2x + 5$

(20) $-3x - 2 > -9x - 8$

(35) $6x + 6 < -2x - 3$

(6) $x + 7 > 8x + 3$

(21) $-9x - 1 > 8x - 7$

(36) $-9x + 1 < x - 5$

(7) $6x + 4 \geq 5x + 2$

(22) $-2x + 7 < 7x + 3$

(37) $-8x + 5 \geq -4x - 7$

(8) $8x - 3 \geq 2x + 6$

(23) $-8x + 8 < 4x - 6$

(38) $-8x + 4 > -3x + 6$

(9) $-7x + 9 > -2x - 7$

(24) $7x + 9 \leq -2x + 3$

(39) $5x + 8 \geq x + 6$

(10) $x - 8 \geq -7x - 5$

(25) $-9x + 2 \leq -6x - 4$

(40) $x + 8 < 3x - 1$

(11) $5x - 9 \geq x + 3$

(26) $8x - 3 > -x - 8$

(41) $-2x + 9 > -7x - 7$

(12) $7x + 4 > 5x + 1$

(27) $-x - 7 > -8x - 7$

(42) $2x + 9 \geq 9x - 6$

(13) $3x - 3 < -8x - 9$

(28) $7x + 2 \leq -8x - 1$

(43) $4x - 3 < 7x - 4$

(14) $-8x - 4 > -2x + 4$

(29) $9x + 3 \geq 8x + 5$

(44) $7x - 5 < -5x + 5$

(15) $-5x + 7 > 6x + 5$

(30) $9x + 2 \leq 2x - 2$

(45) $9x - 3 < -2x + 5$

(1) $-8x - 3 \geq 3x - 6$

(16) $3x + 8 > -7x + 9$

(31) $-7x + 4 < 3x + 4$

(2) $7x - 7 > -6x - 6$

(17) $2x - 3 < x + 7$

(32) $-8x - 4 \geq 7x + 9$

(3) $4x - 7 > -8x - 8$

(18) $-2x - 5 \geq -7x + 7$

(33) $-3x + 6 \geq -6x - 7$

(4) $-8x + 5 \geq -x + 4$

(19) $-2x + 9 \geq -8x - 3$

(34) $7x - 2 > -3x - 9$

(5) $-8x + 1 > 9x + 4$

(20) $-x + 4 > 6x + 6$

(35) $-x + 3 \leq -7x + 4$

(6) $4x + 2 > -8x + 3$

(21) $5x - 3 \geq x + 1$

(36) $-5x + 5 > 5x + 4$

(7) $x + 9 \geq 9x - 3$

(22) $-2x - 3 \leq 9x + 1$

(37) $2x + 6 \leq 7x + 5$

(8) $6x - 8 < -2x - 4$

(23) $2x + 5 \geq -9x - 1$

(38) $5x + 7 < -6x + 8$

(9) $-7x + 6 \leq 3x - 2$

(24) $-8x - 7 \geq 4x + 1$

(39) $-x - 7 > 5x + 2$

(10) $-2x - 8 \leq 5x - 1$

(25) $9x - 1 \geq -2x - 6$

(40) $-6x - 6 \leq 9x - 9$

(11) $9x - 5 \leq 8x - 8$

(26) $2x + 8 < -4x + 5$

(41) $-8x + 1 \leq -4x + 1$

(12) $7x + 2 < 6x - 2$

(27) $2x + 4 \leq -x - 1$

(42) $-5x - 8 \geq -7x + 8$

(13) $-3x + 1 \geq -x + 7$

(28) $-4x - 9 < -5x - 2$

(43) $-x - 1 > -2x - 9$

(14) $2x - 3 > 8x + 1$

(29) $-3x + 1 \geq 8x - 5$

(44) $-9x - 6 < 4x + 5$

(15) $-8x - 2 \geq -2x - 4$

(30) $-3x - 7 \leq -x - 7$

(45) $-9x - 3 \geq x - 8$

(1) $-5x + 7 < -x - 4$

(16) $5x - 9 < 3x + 8$

(31) $2x + 5 > 8x + 1$

(2) $8x + 1 > 2x - 3$

(17) $7x - 3 < 3x - 7$

(32) $9x - 1 \geq -2x + 3$

(3) $-7x - 3 > 8x + 9$

(18) $-2x - 8 > -6x + 7$

(33) $3x - 3 \leq -8x + 2$

(4) $7x - 5 \leq 8x + 3$

(19) $x - 9 > -9x - 5$

(34) $-8x + 8 \leq x + 7$

(5) $x - 4 \geq 2x - 4$

(20) $7x + 6 \leq 3x + 9$

(35) $-8x - 8 < -6x + 7$

(6) $6x - 5 \geq 4x + 3$

(21) $x - 6 \leq -7x + 6$

(36) $9x + 4 < -x + 3$

(7) $x - 1 \leq 8x - 2$

(22) $8x - 3 > 3x - 3$

(37) $-6x - 7 < -5x + 8$

(8) $2x + 6 \leq -3x + 3$

(23) $3x + 9 \geq 8x + 1$

(38) $-6x - 2 \leq 9x + 3$

(9) $-x + 5 > -8x - 7$

(24) $-x - 2 \leq -4x + 7$

(39) $-5x + 6 > 7x + 7$

(10) $-9x - 8 \geq x - 1$

(25) $7x + 7 \geq 8x + 2$

(40) $4x + 1 < 2x - 7$

(11) $8x - 3 < 4x + 9$

(26) $-8x + 9 \leq -6x - 5$

(41) $-5x - 3 \geq -9x + 8$

(12) $9x + 1 < 7x + 5$

(27) $-8x - 4 \leq -3x - 7$

(42) $-7x + 2 \geq 2x + 9$

(13) $-8x + 9 > -5x + 5$

(28) $-5x + 4 \leq 8x - 4$

(43) $5x + 9 < -5x + 2$

(14) $7x - 8 < -5x + 6$

(29) $-x + 8 < -2x + 7$

(44) $8x - 6 < -6x - 2$

(15) $3x - 6 \geq -3x - 9$

(30) $3x + 3 > x + 5$

(45) $-2x + 6 < 4x - 6$

(1) $9x + 7 > 4x + 8$

(16) $-8x - 9 \geq -6x + 4$

(31) $2x - 9 \leq -7x + 3$

(2) $-3x + 4 > -8x - 2$

(17) $-3x - 5 \geq x + 7$

(32) $6x + 7 > -5x + 8$

(3) $4x + 8 < 9x + 9$

(18) $4x - 1 < -5x + 8$

(33) $-6x - 3 \geq 6x - 1$

(4) $-5x + 2 < -6x + 4$

(19) $2x + 1 > 7x - 6$

(34) $-6x + 9 \leq 7x + 9$

(5) $2x + 4 > 4x + 2$

(20) $-4x + 4 < -7x - 5$

(35) $2x - 6 < -7x - 2$

(6) $8x + 4 \leq -4x + 5$

(21) $4x - 4 \leq 9x - 9$

(36) $-9x + 1 \geq -x + 2$

(7) $-4x - 6 < 8x + 6$

(22) $7x + 5 > x - 4$

(37) $-3x + 3 \leq 2x + 8$

(8) $-8x - 5 > -4x - 7$

(23) $8x - 6 \geq x - 2$

(38) $x - 9 > 8x + 4$

(9) $5x + 5 \geq 4x + 4$

(24) $x + 3 \geq -5x - 8$

(39) $-7x + 2 > -6x + 9$

(10) $-5x - 3 \leq -6x + 4$

(25) $-3x - 3 > 2x + 8$

(40) $-8x - 9 < 7x + 4$

(11) $5x - 8 \geq 4x + 4$

(26) $9x - 3 > 5x + 3$

(41) $2x - 2 < -2x + 3$

(12) $6x + 2 > 4x + 7$

(27) $6x - 2 < 2x + 1$

(42) $3x + 7 \leq -5x + 7$

(13) $8x + 2 \leq 4x - 7$

(28) $5x + 7 \geq -6x - 3$

(43) $-x - 2 \leq -3x - 7$

(14) $8x - 1 \geq -8x + 5$

(29) $6x + 2 > -4x - 6$

(44) $6x - 8 > -4x + 4$

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(30) $-5x + 1 \leq 4x - 3$

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(16) $-2x + 1 > 4x - 7$

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(21) $-7x + 6 < -9x + 6$

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(11) $8x + 6 < x - 9$

(26) $8x + 3 \leq -2x - 4$

(41) $-3x + 4 < -4x + 2$

(12) $5x - 6 \geq -2x - 9$

(27) $-8x - 7 \leq 2x + 6$

(42) $-3x - 5 < -x - 4$

(13) $7x + 1 \leq -4x - 3$

(28) $-7x + 9 > -5x + 1$

(43) $-7x + 3 \leq 2x + 5$

(14) $-2x + 8 < 7x - 1$

(29) $-6x + 9 \leq 9x - 9$

(44) $-9x + 6 \leq 6x - 7$

(15) $8x - 2 \leq -6x - 9$

(30) $-5x - 4 \geq 9x - 8$

(45) $5x - 2 \geq -8x - 5$

(1) $-9x - 9 \leq 9x + 2$

(16) $-9x + 5 < x + 4$

(31) $2x + 5 < -5x + 9$

(2) $-3x + 8 \geq -6x - 9$

(17) $-6x - 5 \leq 5x - 9$

(32) $-7x - 5 \geq -3x + 3$

(3) $-5x - 3 > x - 9$

(18) $-7x + 9 \geq 5x + 9$

(33) $-4x - 3 < -2x - 2$

(4) $x - 8 > 6x + 5$

(19) $-3x + 1 \leq -6x - 2$

(34) $-9x - 3 \leq 5x + 7$

(5) $4x - 9 > 8x + 2$

(20) $8x - 6 \geq -7x - 4$

(35) $-7x + 4 \leq 8x - 3$

(6) $9x - 4 < 6x + 9$

(21) $9x - 8 \geq 5x - 8$

(36) $-3x - 9 \leq -4x - 6$

(7) $2x - 5 \leq -2x + 4$

(22) $-x - 3 \leq -5x - 6$

(37) $7x + 3 < -2x - 6$

(8) $8x - 1 \leq 3x + 9$

(23) $-6x + 7 > x - 1$

(38) $-6x - 5 > -3x - 1$

(9) $-9x + 6 \geq 4x - 6$

(24) $x - 4 < 3x - 8$

(39) $3x + 7 \leq -9x - 1$

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(1) $-3x + 3 < -2x + 8$

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(23) $9x - 3 \geq 3x + 6$

(38) $-6x - 4 \leq 5x - 7$

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(24) $5x + 6 < 4x - 9$

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(29) $5x - 4 \geq x + 2$

(44) $-3x - 5 > -7x + 5$

(15) $6x - 3 < 4x - 9$

(30) $-6x - 5 < -9x + 3$

(45) $9x + 8 > -6x - 3$

$$(1) \quad 2x - 7 \geq -8x + 8$$

$$x \geq \frac{3}{2}$$

$$(2) \quad -6x - 2 \geq 4x - 5$$

$$x \leq \frac{3}{10}$$

$$(3) \quad 3x + 9 < -3x + 5$$

$$x < -\frac{2}{3}$$

$$(4) \quad 9x + 3 \geq -5x - 9$$

$$x \geq -\frac{6}{7}$$

$$(5) \quad -6x + 8 < -4x - 9$$

$$x > \frac{17}{2}$$

$$(6) \quad -6x - 7 \leq 3x + 7$$

$$x \geq -\frac{14}{9}$$

$$(7) \quad x - 9 \leq 2x - 7$$

$$x \geq -2$$

$$(8) \quad -6x + 3 \leq 9x + 4$$

$$x \geq -\frac{1}{15}$$

$$(9) \quad -2x - 8 > 5x + 8$$

$$x < -\frac{16}{7}$$

$$(10) \quad 9x + 6 \leq -5x + 9$$

$$x \leq \frac{3}{14}$$

$$(11) \quad -2x - 8 < 6x - 6$$

$$x > -\frac{1}{4}$$

$$(12) \quad x - 5 \geq 3x - 8$$

$$x \leq \frac{3}{2}$$

$$(13) \quad -6x + 7 < -4x - 7$$

$$x > 7$$

$$(14) \quad -2x - 5 > -9x + 8$$

$$x > \frac{13}{7}$$

$$(15) \quad -6x - 9 \leq 9x + 9$$

$$x \geq -\frac{6}{5}$$

$$(16) \quad 2x - 3 \geq 9x - 8$$

$$x \leq \frac{5}{7}$$

$$(17) \quad 7x - 8 \geq 2x + 3$$

$$x \geq \frac{11}{5}$$

$$(18) \quad x - 5 \leq 6x - 3$$

$$x \geq -\frac{2}{5}$$

$$(19) \quad -8x - 2 > -3x - 4$$

$$x < \frac{2}{5}$$

$$(20) \quad x + 3 \geq -4x + 8$$

$$x \geq 1$$

$$(21) \quad 5x - 9 > x + 9$$

$$x > \frac{9}{2}$$

$$(22) \quad 3x - 9 < 8x + 2$$

$$x > -\frac{11}{5}$$

$$(23) \quad 2x + 7 \geq -4x - 4$$

$$x \geq -\frac{11}{6}$$

$$(24) \quad -8x - 3 > 2x - 9$$

$$x < \frac{3}{5}$$

$$(25) \quad -7x - 9 < -2x - 1$$

$$x > -\frac{8}{5}$$

$$(26) \quad 9x - 4 < -x + 7$$

$$x < \frac{11}{10}$$

$$(27) \quad -5x - 8 \geq -4x - 9$$

$$x \leq 1$$

$$(28) \quad -8x - 3 \geq 2x + 4$$

$$x \leq -\frac{7}{10}$$

$$(29) \quad 9x - 8 > -5x - 5$$

$$x > \frac{3}{14}$$

$$(30) \quad 5x - 8 \leq -3x - 1$$

$$x \leq \frac{7}{8}$$

$$(31) \quad 4x + 6 < -x + 7$$

$$x < \frac{1}{5}$$

$$(32) \quad -4x + 1 \leq 7x + 9$$

$$x \geq -\frac{8}{11}$$

$$(33) \quad 5x + 6 \geq -5x + 4$$

$$x \geq -\frac{1}{5}$$

$$(34) \quad -9x + 3 > -8x - 2$$

$$x < 5$$

$$(35) \quad 5x + 9 \geq -2x - 5$$

$$x \geq -2$$

$$(36) \quad -5x - 5 > -6x + 7$$

$$x > 12$$

$$(37) \quad 3x + 8 \leq -2x - 9$$

$$x \leq -\frac{17}{5}$$

$$(38) \quad 5x + 6 > 3x - 6$$

$$x > -6$$

$$(39) \quad 3x + 7 < 9x - 5$$

$$x > 2$$

$$(40) \quad -6x - 6 \geq -8x - 7$$

$$x \geq -\frac{1}{2}$$

$$(41) \quad -7x - 3 > 5x + 9$$

$$x < -1$$

$$(42) \quad 6x + 5 \leq 2x - 4$$

$$x \leq -\frac{9}{4}$$

$$(43) \quad -x + 2 \geq -2x - 2$$

$$x \geq -4$$

$$(44) \quad -2x - 3 \leq x + 7$$

$$x \geq -\frac{10}{3}$$

$$(45) \quad -9x + 8 < 2x + 7$$

$$x > \frac{1}{11}$$

$$(1) \quad 6x - 8 > 9x + 8$$

$$x < -\frac{16}{3}$$

$$(2) \quad 5x - 4 \leq -3x + 1$$

$$x \leq \frac{5}{8}$$

$$(3) \quad -x + 8 \geq -4x + 4$$

$$x \geq -\frac{4}{3}$$

$$(4) \quad 2x + 2 > 9x + 8$$

$$x < -\frac{6}{7}$$

$$(5) \quad 5x + 5 \leq 3x - 8$$

$$x \leq -\frac{13}{2}$$

$$(6) \quad -7x - 3 \geq -4x - 9$$

$$x \leq 2$$

$$(7) \quad -4x - 6 \leq 3x - 5$$

$$x \geq -\frac{1}{7}$$

$$(8) \quad -4x + 1 > -2x + 8$$

$$x < -\frac{7}{2}$$

$$(9) \quad -4x - 2 > 5x + 8$$

$$x < -\frac{10}{9}$$

$$(10) \quad -3x + 5 > -5x + 9$$

$$x > 2$$

$$(11) \quad -4x + 3 \geq 2x + 2$$

$$x \leq \frac{1}{6}$$

$$(12) \quad 8x + 4 < x - 8$$

$$x < -\frac{12}{7}$$

$$(13) \quad 7x + 3 < 3x + 8$$

$$x < \frac{5}{4}$$

$$(14) \quad -8x + 2 < -5x - 3$$

$$x > \frac{5}{3}$$

$$(15) \quad -5x + 5 > -7x - 1$$

$$x > -3$$

$$(16) \quad 4x - 7 < -7x + 5$$

$$x < \frac{12}{11}$$

$$(17) \quad -2x + 2 > -3x + 1$$

$$x > -1$$

$$(18) \quad -x - 8 < 2x + 9$$

$$x > -\frac{17}{3}$$

$$(19) \quad -3x + 2 \leq 4x - 6$$

$$x \geq \frac{8}{7}$$

$$(20) \quad -3x - 5 \leq -4x - 3$$

$$x \leq 2$$

$$(21) \quad 4x + 8 > -7x + 9$$

$$x > \frac{1}{11}$$

$$(22) \quad -x - 1 > -9x - 2$$

$$x > -\frac{1}{8}$$

$$(23) \quad 8x - 9 < 4x + 2$$

$$x < \frac{11}{4}$$

$$(24) \quad -4x - 7 \geq -3x + 9$$

$$x \leq -16$$

$$(25) \quad 9x + 2 \geq 4x - 6$$

$$x \geq -\frac{8}{5}$$

$$(26) \quad -5x + 1 \geq -x - 3$$

$$x \leq 1$$

$$(27) \quad -2x - 6 \geq -5x - 1$$

$$x \geq \frac{5}{3}$$

$$(28) \quad 3x - 3 < x - 6$$

$$x < -\frac{3}{2}$$

$$(29) \quad -6x + 5 \leq -2x + 6$$

$$x \geq -\frac{1}{4}$$

$$(30) \quad -3x - 7 \geq -7x - 5$$

$$x \geq \frac{1}{2}$$

$$(31) \quad 6x + 5 < 8x + 8$$

$$x > -\frac{3}{2}$$

$$(32) \quad 5x + 3 < -7x + 7$$

$$x < \frac{1}{3}$$

$$(33) \quad 7x - 7 < 9x - 3$$

$$x > -2$$

$$(34) \quad -4x + 9 > 7x - 6$$

$$x < \frac{15}{11}$$

$$(35) \quad -5x + 7 \geq -7x + 3$$

$$x \geq -2$$

$$(36) \quad 2x + 5 < -8x + 2$$

$$x < -\frac{3}{10}$$

$$(37) \quad 4x + 9 \geq 7x - 3$$

$$x \leq 4$$

$$(38) \quad 7x - 4 \geq -9x + 3$$

$$x \geq \frac{7}{16}$$

$$(39) \quad x - 4 \geq 2x - 6$$

$$x \leq 2$$

$$(40) \quad -4x + 8 > -3x - 8$$

$$x < 16$$

$$(41) \quad -2x + 3 \geq 9x + 4$$

$$x \leq -\frac{1}{11}$$

$$(42) \quad 8x + 4 > -7x - 2$$

$$x > -\frac{2}{5}$$

$$(43) \quad -x + 9 > -2x + 7$$

$$x > -2$$

$$(44) \quad -4x - 6 \leq -8x - 3$$

$$x \leq \frac{3}{4}$$

$$(45) \quad -3x + 1 > 8x + 4$$

$$x < -\frac{3}{11}$$

$$(1) \quad x + 9 \geq 6x - 4$$

$$x \leq \frac{13}{5}$$

$$(2) \quad -x + 9 \geq 9x + 4$$

$$x \leq \frac{1}{2}$$

$$(3) \quad 5x - 4 \geq -x - 4$$

$$x \geq 0$$

$$(4) \quad 9x + 3 > 5x + 4$$

$$x > \frac{1}{4}$$

$$(5) \quad -9x + 3 \leq -2x + 5$$

$$x \geq -\frac{2}{7}$$

$$(6) \quad x + 7 > 8x + 3$$

$$x < \frac{4}{7}$$

$$(7) \quad 6x + 4 \geq 5x + 2$$

$$x \geq -2$$

$$(8) \quad 8x - 3 \geq 2x + 6$$

$$x \geq \frac{3}{2}$$

$$(9) \quad -7x + 9 > -2x - 7$$

$$x < \frac{16}{5}$$

$$(10) \quad x - 8 \geq -7x - 5$$

$$x \geq \frac{3}{8}$$

$$(11) \quad 5x - 9 \geq x + 3$$

$$x \geq 3$$

$$(12) \quad 7x + 4 > 5x + 1$$

$$x > -\frac{3}{2}$$

$$(13) \quad 3x - 3 < -8x - 9$$

$$x < -\frac{6}{11}$$

$$(14) \quad -8x - 4 > -2x + 4$$

$$x < -\frac{4}{3}$$

$$(15) \quad -5x + 7 > 6x + 5$$

$$x < \frac{2}{11}$$

$$(16) \quad 4x + 6 \leq 7x + 6$$

$$x \geq 0$$

$$(17) \quad 2x + 2 > -4x + 2$$

$$x > 0$$

$$(18) \quad 9x - 8 \leq -9x - 8$$

$$x \leq 0$$

$$(19) \quad -x - 3 > -8x + 5$$

$$x > \frac{8}{7}$$

$$(20) \quad -3x - 2 > -9x - 8$$

$$x > -1$$

$$(21) \quad -9x - 1 > 8x - 7$$

$$x < \frac{6}{17}$$

$$(22) \quad -2x + 7 < 7x + 3$$

$$x > \frac{4}{9}$$

$$(23) \quad -8x + 8 < 4x - 6$$

$$x > \frac{7}{6}$$

$$(24) \quad 7x + 9 \leq -2x + 3$$

$$x \leq -\frac{2}{3}$$

$$(25) \quad -9x + 2 \leq -6x - 4$$

$$x \geq 2$$

$$(26) \quad 8x - 3 > -x - 8$$

$$x > -\frac{5}{9}$$

$$(27) \quad -x - 7 > -8x - 7$$

$$x > 0$$

$$(28) \quad 7x + 2 \leq -8x - 1$$

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$$(29) \quad 9x + 3 \geq 8x + 5$$

$$x \geq 2$$

$$(30) \quad 9x + 2 \leq 2x - 2$$

$$x \leq -\frac{4}{7}$$

$$(31) \quad 6x - 8 \geq -9x + 6$$

$$x \geq \frac{14}{15}$$

$$(32) \quad -9x + 7 < -2x - 7$$

$$x > 2$$

$$(33) \quad x - 8 > 9x - 2$$

$$x < -\frac{3}{4}$$

$$(34) \quad -6x - 3 \geq x - 9$$

$$x \leq \frac{6}{7}$$

$$(35) \quad 6x + 6 < -2x - 3$$

$$x < -\frac{9}{8}$$

$$(36) \quad -9x + 1 < x - 5$$

$$x > \frac{3}{5}$$

$$(37) \quad -8x + 5 \geq -4x - 7$$

$$x \leq 3$$

$$(38) \quad -8x + 4 > -3x + 6$$

$$x < -\frac{2}{5}$$

$$(39) \quad 5x + 8 \geq x + 6$$

$$x \geq -\frac{1}{2}$$

$$(40) \quad x + 8 < 3x - 1$$

$$x > \frac{9}{2}$$

$$(41) \quad -2x + 9 > -7x - 7$$

$$x > -\frac{16}{5}$$

$$(42) \quad 2x + 9 \geq 9x - 6$$

$$x \leq \frac{15}{7}$$

$$(43) \quad 4x - 3 < 7x - 4$$

$$x > \frac{1}{3}$$

$$(44) \quad 7x - 5 < -5x + 5$$

$$x < \frac{5}{6}$$

$$(45) \quad 9x - 3 < -2x + 5$$

$$x < \frac{8}{11}$$

$$(1) -8x - 3 \geq 3x - 6$$

$$x \leq \frac{3}{11}$$

$$(2) 7x - 7 > -6x - 6$$

$$x > \frac{1}{13}$$

$$(3) 4x - 7 > -8x - 8$$

$$x > -\frac{1}{12}$$

$$(4) -8x + 5 \geq -x + 4$$

$$x \leq \frac{1}{7}$$

$$(5) -8x + 1 > 9x + 4$$

$$x < -\frac{3}{17}$$

$$(6) 4x + 2 > -8x + 3$$

$$x > \frac{1}{12}$$

$$(7) x + 9 \geq 9x - 3$$

$$x \leq \frac{3}{2}$$

$$(8) 6x - 8 < -2x - 4$$

$$x < \frac{1}{2}$$

$$(9) -7x + 6 \leq 3x - 2$$

$$x \geq \frac{4}{5}$$

$$(10) -2x - 8 \leq 5x - 1$$

$$x \geq -1$$

$$(11) 9x - 5 \leq 8x - 8$$

$$x \leq -3$$

$$(12) 7x + 2 < 6x - 2$$

$$x < -4$$

$$(13) -3x + 1 \geq -x + 7$$

$$x \leq -3$$

$$(14) 2x - 3 > 8x + 1$$

$$x < -\frac{2}{3}$$

$$(15) -8x - 2 \geq -2x - 4$$

$$x \leq \frac{1}{3}$$

$$(16) 3x + 8 > -7x + 9$$

$$x > \frac{1}{10}$$

$$(17) 2x - 3 < x + 7$$

$$x < 10$$

$$(18) -2x - 5 \geq -7x + 7$$

$$x \geq \frac{12}{5}$$

$$(19) -2x + 9 \geq -8x - 3$$

$$x \geq -2$$

$$(20) -x + 4 > 6x + 6$$

$$x < -\frac{2}{7}$$

$$(21) 5x - 3 \geq x + 1$$

$$x \geq 1$$

$$(22) -2x - 3 \leq 9x + 1$$

$$x \geq -\frac{4}{11}$$

$$(23) 2x + 5 \geq -9x - 1$$

$$x \geq -\frac{6}{11}$$

$$(24) -8x - 7 \geq 4x + 1$$

$$x \leq -\frac{2}{3}$$

$$(25) 9x - 1 \geq -2x - 6$$

$$x \geq -\frac{5}{11}$$

$$(26) 2x + 8 < -4x + 5$$

$$x < -\frac{1}{2}$$

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$$(28) -4x - 9 < -5x - 2$$

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$$(29) -3x + 1 \geq 8x - 5$$

$$x \leq \frac{6}{11}$$

$$(30) -3x - 7 \leq -x - 7$$

$$x \geq 0$$

$$(31) -7x + 4 < 3x + 4$$

$$x > 0$$

$$(32) -8x - 4 \geq 7x + 9$$

$$x \leq -\frac{13}{15}$$

$$(33) -3x + 6 \geq -6x - 7$$

$$x \geq -\frac{13}{3}$$

$$(34) 7x - 2 > -3x - 9$$

$$x > -\frac{7}{10}$$

$$(35) -x + 3 \leq -7x + 4$$

$$x \leq \frac{1}{6}$$

$$(36) -5x + 5 > 5x + 4$$

$$x < \frac{1}{10}$$

$$(37) 2x + 6 \leq 7x + 5$$

$$x \geq \frac{1}{5}$$

$$(38) 5x + 7 < -6x + 8$$

$$x < \frac{1}{11}$$

$$(39) -x - 7 > 5x + 2$$

$$x < -\frac{3}{2}$$

$$(40) -6x - 6 \leq 9x - 9$$

$$x \geq \frac{1}{5}$$

$$(41) -8x + 1 \leq -4x + 1$$

$$x \geq 0$$

$$(42) -5x - 8 \geq -7x + 8$$

$$x \geq 8$$

$$(43) -x - 1 > -2x - 9$$

$$x > -8$$

$$(44) -9x - 6 < 4x + 5$$

$$x > -\frac{11}{13}$$

$$(45) -9x - 3 \geq x - 8$$

$$x \leq \frac{1}{2}$$

$$(1) \quad -5x + 7 < -x - 4$$

$$x > \frac{11}{4}$$

$$(2) \quad 8x + 1 > 2x - 3$$

$$x > -\frac{2}{3}$$

$$(3) \quad -7x - 3 > 8x + 9$$

$$x < -\frac{4}{5}$$

$$(4) \quad 7x - 5 \leq 8x + 3$$

$$x \geq -8$$

$$(5) \quad x - 4 \geq 2x - 4$$

$$x \leq 0$$

$$(6) \quad 6x - 5 \geq 4x + 3$$

$$x \geq 4$$

$$(7) \quad x - 1 \leq 8x - 2$$

$$x \geq \frac{1}{7}$$

$$(8) \quad 2x + 6 \leq -3x + 3$$

$$x \leq -\frac{3}{5}$$

$$(9) \quad -x + 5 > -8x - 7$$

$$x > -\frac{12}{7}$$

$$(10) \quad -9x - 8 \geq x - 1$$

$$x \leq -\frac{7}{10}$$

$$(11) \quad 8x - 3 < 4x + 9$$

$$x < 3$$

$$(12) \quad 9x + 1 < 7x + 5$$

$$x < 2$$

$$(13) \quad -8x + 9 > -5x + 5$$

$$x < \frac{4}{3}$$

$$(14) \quad 7x - 8 < -5x + 6$$

$$x < \frac{7}{6}$$

$$(15) \quad 3x - 6 \geq -3x - 9$$

$$x \geq -\frac{1}{2}$$

$$(16) \quad 5x - 9 < 3x + 8$$

$$x < \frac{17}{2}$$

$$(17) \quad 7x - 3 < 3x - 7$$

$$x < -1$$

$$(18) \quad -2x - 8 > -6x + 7$$

$$x > \frac{15}{4}$$

$$(19) \quad x - 9 > -9x - 5$$

$$x > \frac{2}{5}$$

$$(20) \quad 7x + 6 \leq 3x + 9$$

$$x \leq \frac{3}{4}$$

$$(21) \quad x - 6 \leq -7x + 6$$

$$x \leq \frac{3}{2}$$

$$(22) \quad 8x - 3 > 3x - 3$$

$$x > 0$$

$$(23) \quad 3x + 9 \geq 8x + 1$$

$$x \leq \frac{8}{5}$$

$$(24) \quad -x - 2 \leq -4x + 7$$

$$x \leq 3$$

$$(25) \quad 7x + 7 \geq 8x + 2$$

$$x \leq 5$$

$$(26) \quad -8x + 9 \leq -6x - 5$$

$$x \geq 7$$

$$(27) \quad -8x - 4 \leq -3x - 7$$

$$x \geq \frac{3}{5}$$

$$(28) \quad -5x + 4 \leq 8x - 4$$

$$x \geq \frac{8}{13}$$

$$(29) \quad -x + 8 < -2x + 7$$

$$x < -1$$

$$(30) \quad 3x + 3 > x + 5$$

$$x > 1$$

$$(31) \quad 2x + 5 > 8x + 1$$

$$x < \frac{2}{3}$$

$$(32) \quad 9x - 1 \geq -2x + 3$$

$$x \geq \frac{4}{11}$$

$$(33) \quad 3x - 3 \leq -8x + 2$$

$$x \leq \frac{5}{11}$$

$$(34) \quad -8x + 8 \leq x + 7$$

$$x \geq \frac{1}{9}$$

$$(35) \quad -8x - 8 < -6x + 7$$

$$x > -\frac{15}{2}$$

$$(36) \quad 9x + 4 < -x + 3$$

$$x < -\frac{1}{10}$$

$$(37) \quad -6x - 7 < -5x + 8$$

$$x > -15$$

$$(38) \quad -6x - 2 \leq 9x + 3$$

$$x \geq -\frac{1}{3}$$

$$(39) \quad -5x + 6 > 7x + 7$$

$$x < -\frac{1}{12}$$

$$(40) \quad 4x + 1 < 2x - 7$$

$$x < -4$$

$$(41) \quad -5x - 3 \geq -9x + 8$$

$$x \geq \frac{11}{4}$$

$$(42) \quad -7x + 2 \geq 2x + 9$$

$$x \leq -\frac{7}{9}$$

$$(43) \quad 5x + 9 < -5x + 2$$

$$x < -\frac{7}{10}$$

$$(44) \quad 8x - 6 < -6x - 2$$

$$x < \frac{2}{7}$$

$$(45) \quad -2x + 6 < 4x - 6$$

$$x > 2$$

$$(1) \quad 9x + 7 > 4x + 8$$

$$x > \frac{1}{5}$$

$$(2) \quad -3x + 4 > -8x - 2$$

$$x > -\frac{6}{5}$$

$$(3) \quad 4x + 8 < 9x + 9$$

$$x > -\frac{1}{5}$$

$$(4) \quad -5x + 2 < -6x + 4$$

$$x < 2$$

$$(5) \quad 2x + 4 > 4x + 2$$

$$x < 1$$

$$(6) \quad 8x + 4 \leq -4x + 5$$

$$x \leq \frac{1}{12}$$

$$(7) \quad -4x - 6 < 8x + 6$$

$$x > -1$$

$$(8) \quad -8x - 5 > -4x - 7$$

$$x < \frac{1}{2}$$

$$(9) \quad 5x + 5 \geq 4x + 4$$

$$x \geq -1$$

$$(10) \quad -5x - 3 \leq -6x + 4$$

$$x \leq 7$$

$$(11) \quad 5x - 8 \geq 4x + 4$$

$$x \geq 12$$

$$(12) \quad 6x + 2 > 4x + 7$$

$$x > \frac{5}{2}$$

$$(13) \quad 8x + 2 \leq 4x - 7$$

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$$x > -\frac{1}{2}$$

$$(1) \quad -9x - 8 \leq -8x - 2$$

$$x \geq -6$$

$$(2) \quad 5x + 8 < -x + 9$$

$$x < \frac{1}{6}$$

$$(3) \quad x - 1 > 2x - 5$$

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$$(5) \quad -6x + 7 > x + 8$$

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$$(6) \quad 6x - 2 < x + 6$$

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$$(7) \quad x - 2 \leq -9x - 7$$

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$$x \leq \frac{5}{16}$$

$$(22) \quad -2x - 6 < 4x + 9$$

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$$(34) \quad 7x + 7 \geq 9x - 8$$

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$$x > -\frac{9}{10}$$

$$(4) \quad -5x + 5 \geq -9x - 6$$

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$$(5) \quad 4x - 2 > 2x + 7$$

$$x > \frac{9}{2}$$

$$(6) \quad -4x + 1 > 6x + 5$$

$$x < -\frac{2}{5}$$

$$(7) \quad x - 2 < -x - 8$$

$$x < -3$$

$$(8) \quad 3x - 8 \leq 5x + 3$$

$$x \geq -\frac{11}{2}$$

$$(9) \quad 8x - 7 \geq -8x - 8$$

$$x \geq -\frac{1}{16}$$

$$(10) \quad -8x + 2 \geq -7x - 3$$

$$x \leq 5$$

$$(11) \quad 8x - 9 < 9x + 7$$

$$x > -16$$

$$(12) \quad 7x - 2 < 2x - 2$$

$$x < 0$$

$$(13) \quad 4x - 9 \leq 2x + 5$$

$$x \leq 7$$

$$(14) \quad 2x + 9 < -6x + 8$$

$$x < -\frac{1}{8}$$

$$(15) \quad 2x + 8 > -6x + 8$$

$$x > 0$$

$$(16) \quad -7x + 2 > 8x + 5$$

$$x < -\frac{1}{5}$$

$$(17) \quad -4x + 6 \geq -5x - 5$$

$$x \geq -11$$

$$(18) \quad -8x - 1 \geq -x + 1$$

$$x \leq -\frac{2}{7}$$

$$(19) \quad -6x - 9 \leq -x - 5$$

$$x \geq -\frac{4}{5}$$

$$(20) \quad 4x - 5 \geq -5x - 2$$

$$x \geq \frac{1}{3}$$

$$(21) \quad 4x - 8 \geq 7x + 7$$

$$x \leq -5$$

$$(22) \quad 4x - 1 < 3x - 4$$

$$x < -3$$

$$(23) \quad -x + 7 \leq -8x - 1$$

$$x \leq -\frac{8}{7}$$

$$(24) \quad -8x - 2 \geq 9x + 6$$

$$x \leq -\frac{8}{17}$$

$$(25) \quad 3x + 3 < 2x + 8$$

$$x < 5$$

$$(26) \quad -3x - 6 \geq -6x - 3$$

$$x \geq 1$$

$$(27) \quad 9x + 1 > -2x + 3$$

$$x > \frac{2}{11}$$

$$(28) \quad 5x + 4 > -7x - 7$$

$$x > -\frac{11}{12}$$

$$(29) \quad -5x - 1 \leq 8x - 9$$

$$x \geq \frac{8}{13}$$

$$(30) \quad x + 7 \geq -x + 8$$

$$x \geq \frac{1}{2}$$

$$(31) \quad -9x - 7 < 9x - 3$$

$$x > -\frac{2}{9}$$

$$(32) \quad 2x + 3 \geq -8x + 3$$

$$x \geq 0$$

$$(33) \quad 3x + 9 < -x + 6$$

$$x < -\frac{3}{4}$$

$$(34) \quad 9x - 6 \leq 4x + 7$$

$$x \leq \frac{13}{5}$$

$$(35) \quad 2x + 2 \geq 5x - 5$$

$$x \leq \frac{7}{3}$$

$$(36) \quad -2x + 3 < -x - 7$$

$$x > 10$$

$$(37) \quad 6x - 7 > -9x - 8$$

$$x > -\frac{1}{15}$$

$$(38) \quad 2x + 4 > -7x - 8$$

$$x > -\frac{4}{3}$$

$$(39) \quad -x - 1 < 2x + 4$$

$$x > -\frac{5}{3}$$

$$(40) \quad -4x - 5 < -x - 1$$

$$x > -\frac{4}{3}$$

$$(41) \quad -7x + 9 < 8x - 3$$

$$x > \frac{4}{5}$$

$$(42) \quad -9x - 3 < -x + 8$$

$$x > -\frac{11}{8}$$

$$(43) \quad 3x + 7 \leq -8x + 9$$

$$x \leq \frac{2}{11}$$

$$(44) \quad -x + 9 \leq 8x - 2$$

$$x \geq \frac{11}{9}$$

$$(45) \quad 4x - 1 \leq -6x - 1$$

$$x \leq 0$$

$$(1) \quad 9x - 3 \leq 7x + 6$$

$$x \leq \frac{9}{2}$$

$$(2) \quad 2x + 4 \leq -x + 5$$

$$x \leq \frac{1}{3}$$

$$(3) \quad x - 5 > 7x + 7$$

$$x < -2$$

$$(4) \quad 4x - 5 > -9x + 3$$

$$x > \frac{8}{13}$$

$$(5) \quad -x - 3 \leq -3x + 4$$

$$x \leq \frac{7}{2}$$

$$(6) \quad 2x - 9 < -6x + 2$$

$$x < \frac{11}{8}$$

$$(7) \quad 3x - 2 \geq 9x + 6$$

$$x \leq -\frac{4}{3}$$

$$(8) \quad 4x - 9 \geq -9x - 1$$

$$x \geq \frac{8}{13}$$

$$(9) \quad -6x + 9 > 9x + 9$$

$$x < 0$$

$$(10) \quad -6x - 1 \leq 2x - 5$$

$$x \geq \frac{1}{2}$$

$$(11) \quad 8x - 4 \leq -2x + 9$$

$$x \leq \frac{13}{10}$$

$$(12) \quad 9x + 2 \geq -6x + 2$$

$$x \geq 0$$

$$(13) \quad -3x - 6 < x + 5$$

$$x > -\frac{11}{4}$$

$$(14) \quad 4x - 3 > -7x + 9$$

$$x > \frac{12}{11}$$

$$(15) \quad -9x + 9 < 9x + 5$$

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$$(24) \quad -3x - 5 < -8x + 5$$

$$x < 2$$

$$(25) \quad 9x + 2 > 7x + 6$$

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$$x < \frac{13}{7}$$

$$(42) \quad 5x + 2 \geq 6x + 1$$

$$x \leq 1$$

$$(43) \quad 9x + 2 \leq -x + 3$$

$$x \leq \frac{1}{10}$$

$$(44) \quad -8x - 6 \leq -5x - 5$$

$$x \geq -\frac{1}{3}$$

$$(45) \quad -5x - 5 > -7x - 5$$

$$x > 0$$

$$(1) \quad 5x - 3 > 2x - 6$$

$$x > -1$$

$$(2) \quad -2x - 1 \leq -7x + 4$$

$$x \leq 1$$

$$(3) \quad 3x + 9 < 5x - 4$$

$$x > \frac{13}{2}$$

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$$x > -1$$

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$$(7) \quad -3x - 5 \geq 2x - 4$$

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$$(8) \quad 5x - 6 < -4x + 5$$

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$$(10) \quad -9x - 8 > 5x + 1$$

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$$x \geq -3$$

$$(4) \quad -2x + 8 < -6x + 4$$

$$x < -1$$

$$(5) \quad 9x - 8 > -7x + 8$$

$$x > 1$$

$$(6) \quad -9x + 5 \geq 3x + 8$$

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$$(14) \quad -3x - 9 > -x + 9$$

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$$(15) \quad -3x + 6 \geq 2x - 2$$

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$$x \leq \frac{5}{8}$$

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$$(45) \quad -2x + 4 \leq -7x - 8$$

$$x \leq -\frac{12}{5}$$

$$(1) \quad 2x - 1 \geq -4x + 3$$

$$x \geq \frac{2}{3}$$

$$(2) \quad 7x + 1 > 4x - 8$$

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$$(3) \quad x - 7 > -8x + 9$$

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$$(4) \quad 3x + 3 > -2x + 8$$

$$x > 1$$

$$(5) \quad 3x - 2 < -x - 1$$

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$$(6) \quad 3x + 7 > -2x + 4$$

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$$(6) \quad -4x - 6 \geq -2x - 1$$

$$x \leq -\frac{5}{2}$$

$$(7) \quad -3x - 4 < 6x + 4$$

$$x > -\frac{8}{9}$$

$$(8) \quad -5x - 9 < 6x - 9$$

$$x > 0$$

$$(9) \quad 3x - 1 \leq -9x - 7$$

$$x \leq -\frac{1}{2}$$

$$(10) \quad 6x - 1 < -7x - 4$$

$$x < -\frac{3}{13}$$

$$(11) \quad 8x + 6 < x - 9$$

$$x < -\frac{15}{7}$$

$$(12) \quad 5x - 6 \geq -2x - 9$$

$$x \geq -\frac{3}{7}$$

$$(13) \quad 7x + 1 \leq -4x - 3$$

$$x \leq -\frac{4}{11}$$

$$(14) \quad -2x + 8 < 7x - 1$$

$$x > 1$$

$$(15) \quad 8x - 2 \leq -6x - 9$$

$$x \leq -\frac{1}{2}$$

$$(16) \quad -8x - 9 \geq 9x - 3$$

$$x \leq -\frac{6}{17}$$

$$(17) \quad -8x + 7 \leq -6x - 9$$

$$x \geq 8$$

$$(18) \quad 8x + 6 > -6x + 2$$

$$x > -\frac{2}{7}$$

$$(19) \quad -7x - 3 \geq -9x + 6$$

$$x \geq \frac{9}{2}$$

$$(20) \quad -8x + 2 \leq 7x + 2$$

$$x \geq 0$$

$$(21) \quad 9x - 5 < -5x + 6$$

$$x < \frac{11}{14}$$

$$(22) \quad x + 5 > 5x + 2$$

$$x < \frac{3}{4}$$

$$(23) \quad -x - 3 > 4x + 7$$

$$x < -2$$

$$(24) \quad 3x + 2 > 7x - 6$$

$$x < 2$$

$$(25) \quad 5x + 3 \geq -6x - 3$$

$$x \geq -\frac{6}{11}$$

$$(26) \quad 8x + 3 \leq -2x - 4$$

$$x \leq -\frac{7}{10}$$

$$(27) \quad -8x - 7 \leq 2x + 6$$

$$x \geq -\frac{13}{10}$$

$$(28) \quad -7x + 9 > -5x + 1$$

$$x < 4$$

$$(29) \quad -6x + 9 \leq 9x - 9$$

$$x \geq \frac{6}{5}$$

$$(30) \quad -5x - 4 \geq 9x - 8$$

$$x \leq \frac{2}{7}$$

$$(31) \quad 8x - 9 > -2x + 6$$

$$x > \frac{3}{2}$$

$$(32) \quad 9x - 9 > -6x - 6$$

$$x > \frac{1}{5}$$

$$(33) \quad -6x + 1 < -2x + 5$$

$$x > -1$$

$$(34) \quad -5x + 7 > 8x + 2$$

$$x < \frac{5}{13}$$

$$(35) \quad -9x + 7 \leq 7x - 3$$

$$x \geq \frac{5}{8}$$

$$(36) \quad -5x - 5 > 9x + 7$$

$$x < -\frac{6}{7}$$

$$(37) \quad 9x - 9 \geq 7x - 2$$

$$x \geq \frac{7}{2}$$

$$(38) \quad -2x - 7 > -6x - 6$$

$$x > \frac{1}{4}$$

$$(39) \quad -7x + 2 > 8x - 2$$

$$x < \frac{4}{15}$$

$$(40) \quad 4x - 8 \leq 8x - 9$$

$$x \geq \frac{1}{4}$$

$$(41) \quad -3x + 4 < -4x + 2$$

$$x < -2$$

$$(42) \quad -3x - 5 < -x - 4$$

$$x > -\frac{1}{2}$$

$$(43) \quad -7x + 3 \leq 2x + 5$$

$$x \geq -\frac{2}{9}$$

$$(44) \quad -9x + 6 \leq 6x - 7$$

$$x \geq \frac{13}{15}$$

$$(45) \quad 5x - 2 \geq -8x - 5$$

$$x \geq -\frac{3}{13}$$

$$(1) \quad -9x - 9 \leq 9x + 2$$

$$x \geq -\frac{11}{18}$$

$$(2) \quad -3x + 8 \geq -6x - 9$$

$$x \geq -\frac{17}{3}$$

$$(3) \quad -5x - 3 > x - 9$$

$$x < 1$$

$$(4) \quad x - 8 > 6x + 5$$

$$x < -\frac{13}{5}$$

$$(5) \quad 4x - 9 > 8x + 2$$

$$x < -\frac{11}{4}$$

$$(6) \quad 9x - 4 < 6x + 9$$

$$x < \frac{13}{3}$$

$$(7) \quad 2x - 5 \leq -2x + 4$$

$$x \leq \frac{9}{4}$$

$$(8) \quad 8x - 1 \leq 3x + 9$$

$$x \leq 2$$

$$(9) \quad -9x + 6 \geq 4x - 6$$

$$x \leq \frac{12}{13}$$

$$(10) \quad -2x + 4 > -4x + 1$$

$$x > -\frac{3}{2}$$

$$(11) \quad 9x - 4 \leq x - 2$$

$$x \leq \frac{1}{4}$$

$$(12) \quad 9x + 5 \geq 5x + 9$$

$$x \geq 1$$

$$(13) \quad 4x + 6 < x - 3$$

$$x < -3$$

$$(14) \quad 8x + 8 \leq -4x - 9$$

$$x \leq -\frac{17}{12}$$

$$(15) \quad -2x + 8 < 6x - 6$$

$$x > \frac{7}{4}$$

$$(16) \quad -9x + 5 < x + 4$$

$$x > \frac{1}{10}$$

$$(17) \quad -6x - 5 \leq 5x - 9$$

$$x \geq \frac{4}{11}$$

$$(18) \quad -7x + 9 \geq 5x + 9$$

$$x \leq 0$$

$$(19) \quad -3x + 1 \leq -6x - 2$$

$$x \leq -1$$

$$(20) \quad 8x - 6 \geq -7x - 4$$

$$x \geq \frac{2}{15}$$

$$(21) \quad 9x - 8 \geq 5x - 8$$

$$x \geq 0$$

$$(22) \quad -x - 3 \leq -5x - 6$$

$$x \leq -\frac{3}{4}$$

$$(23) \quad -6x + 7 > x - 1$$

$$x < \frac{8}{7}$$

$$(24) \quad x - 4 < 3x - 8$$

$$x > 2$$

$$(25) \quad -6x + 8 \leq -3x - 1$$

$$x \geq 3$$

$$(26) \quad -9x + 2 \geq 7x - 4$$

$$x \leq \frac{3}{8}$$

$$(27) \quad x + 5 > 4x + 7$$

$$x < -\frac{2}{3}$$

$$(28) \quad -4x - 7 \geq -2x - 8$$

$$x \leq \frac{1}{2}$$

$$(29) \quad -8x + 5 \leq -5x - 9$$

$$x \geq \frac{14}{3}$$

$$(30) \quad x + 1 > 8x + 1$$

$$x < 0$$

$$(31) \quad 2x + 5 < -5x + 9$$

$$x < \frac{4}{7}$$

$$(32) \quad -7x - 5 \geq -3x + 3$$

$$x \leq -2$$

$$(33) \quad -4x - 3 < -2x - 2$$

$$x > -\frac{1}{2}$$

$$(34) \quad -9x - 3 \leq 5x + 7$$

$$x \geq -\frac{5}{7}$$

$$(35) \quad -7x + 4 \leq 8x - 3$$

$$x \geq \frac{7}{15}$$

$$(36) \quad -3x - 9 \leq -4x - 6$$

$$x \leq 3$$

$$(37) \quad 7x + 3 < -2x - 6$$

$$x < -1$$

$$(38) \quad -6x - 5 > -3x - 1$$

$$x < -\frac{4}{3}$$

$$(39) \quad 3x + 7 \leq -9x - 1$$

$$x \leq -\frac{2}{3}$$

$$(40) \quad -6x + 4 > x + 9$$

$$x < -\frac{5}{7}$$

$$(41) \quad 3x - 4 \geq -4x + 4$$

$$x \geq \frac{8}{7}$$

$$(42) \quad 8x + 4 < -x + 5$$

$$x < \frac{1}{9}$$

$$(43) \quad 7x + 8 \geq 2x + 4$$

$$x \geq -\frac{4}{5}$$

$$(44) \quad 3x - 4 \geq 9x + 1$$

$$x \leq -\frac{5}{6}$$

$$(45) \quad -7x + 2 > -3x + 3$$

$$x < -\frac{1}{4}$$

$$(1) \quad -5x - 8 \geq -x + 2$$

$$x \leq -\frac{5}{2}$$

$$(2) \quad -5x + 7 \geq -3x - 1$$

$$x \leq 4$$

$$(3) \quad -4x + 7 \leq x + 3$$

$$x \geq \frac{4}{5}$$

$$(4) \quad -5x + 7 < 9x - 6$$

$$x > \frac{13}{14}$$

$$(5) \quad -x + 3 < -5x + 8$$

$$x < \frac{5}{4}$$

$$(6) \quad 3x - 9 \geq 6x - 8$$

$$x \leq -\frac{1}{3}$$

$$(7) \quad 8x + 1 \geq 9x + 6$$

$$x \leq -5$$

$$(8) \quad -x - 1 \geq x - 1$$

$$x \leq 0$$

$$(9) \quad -2x - 4 \leq 8x - 9$$

$$x \geq \frac{1}{2}$$

$$(10) \quad 7x - 5 < 3x + 4$$

$$x < \frac{9}{4}$$

$$(11) \quad x - 8 < -9x - 1$$

$$x < \frac{7}{10}$$

$$(12) \quad -8x - 5 > 9x - 4$$

$$x < -\frac{1}{17}$$

$$(13) \quad x + 2 < -2x - 5$$

$$x < -\frac{7}{3}$$

$$(14) \quad 3x + 5 \leq -8x - 5$$

$$x \leq -\frac{10}{11}$$

$$(15) \quad -2x + 1 > 5x + 9$$

$$x < -\frac{8}{7}$$

$$(16) \quad 6x - 4 \leq 7x - 3$$

$$x \geq -1$$

$$(17) \quad -3x + 8 \geq 6x - 7$$

$$x \leq \frac{5}{3}$$

$$(18) \quad -6x - 8 \leq -7x - 6$$

$$x \leq 2$$

$$(19) \quad 5x - 5 \geq 3x + 7$$

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$$(20) \quad 8x + 7 > -6x + 7$$

$$x > 0$$

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$$x \leq 0$$

$$(22) \quad 6x - 1 \geq -5x - 2$$

$$x \geq -\frac{1}{11}$$

$$(23) \quad 7x - 3 \leq -8x + 2$$

$$x \leq \frac{1}{3}$$

$$(24) \quad 3x - 8 < 7x - 3$$

$$x > -\frac{5}{4}$$

$$(25) \quad -5x - 8 \geq -7x + 5$$

$$x \geq \frac{13}{2}$$

$$(26) \quad -3x - 3 \geq -2x - 1$$

$$x \leq -2$$

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$$x > -\frac{3}{2}$$

$$(28) \quad -7x + 4 < -8x - 8$$

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$$(29) \quad 2x - 3 > -4x + 2$$

$$x > \frac{5}{6}$$

$$(30) \quad 5x + 4 \leq -9x - 2$$

$$x \leq -\frac{3}{7}$$

$$(31) \quad 9x - 5 \geq -3x - 3$$

$$x \geq \frac{1}{6}$$

$$(32) \quad -2x - 3 > -x + 4$$

$$x < -7$$

$$(33) \quad -8x + 9 \leq 6x + 2$$

$$x \geq \frac{1}{2}$$

$$(34) \quad 5x + 3 \geq -2x + 8$$

$$x \geq \frac{5}{7}$$

$$(35) \quad 5x - 1 \geq -9x + 9$$

$$x \geq \frac{5}{7}$$

$$(36) \quad 3x + 8 < 8x - 1$$

$$x > \frac{9}{5}$$

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$$x > 1$$

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$$x \leq -\frac{7}{10}$$

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$$x > \frac{7}{12}$$

$$(42) \quad -7x - 2 \leq -x - 3$$

$$x \geq \frac{1}{6}$$

$$(43) \quad -2x + 7 > -x - 2$$

$$x < 9$$

$$(44) \quad 4x - 9 \leq 6x + 9$$

$$x \geq -9$$

$$(45) \quad 5x + 4 \geq -2x + 5$$

$$x \geq \frac{1}{7}$$

$$(1) -3x + 3 < -2x + 8$$

$$x > -5$$

$$(2) -5x - 7 < -x + 5$$

$$x > -3$$

$$(3) 4x - 3 \leq -x + 9$$

$$x \leq \frac{12}{5}$$

$$(4) 7x + 4 \geq 2x + 6$$

$$x \geq \frac{2}{5}$$

$$(5) -3x - 8 \geq -x + 2$$

$$x \leq -5$$

$$(6) -8x + 8 < -6x + 3$$

$$x > \frac{5}{2}$$

$$(7) -4x + 9 \geq 5x - 7$$

$$x \leq \frac{16}{9}$$

$$(8) 6x - 4 < -4x - 8$$

$$x < -\frac{2}{5}$$

$$(9) -5x - 7 \geq 6x - 6$$

$$x \leq -\frac{1}{11}$$

$$(10) 2x + 8 < -x - 1$$

$$x < -3$$

$$(11) 9x + 1 < -2x + 2$$

$$x < \frac{1}{11}$$

$$(12) 9x + 6 > 5x - 3$$

$$x > -\frac{9}{4}$$

$$(13) x + 2 < 3x - 1$$

$$x > \frac{3}{2}$$

$$(14) 8x - 1 \leq 4x + 7$$

$$x \leq 2$$

$$(15) 6x - 3 < 4x - 9$$

$$x < -3$$

$$(16) 5x + 1 > -2x - 4$$

$$x > -\frac{5}{7}$$

$$(17) -6x - 3 \leq -x - 1$$

$$x \geq -\frac{2}{5}$$

$$(18) 7x - 6 \geq -8x - 9$$

$$x \geq -\frac{1}{5}$$

$$(19) -7x + 9 \geq 6x - 6$$

$$x \leq \frac{15}{13}$$

$$(20) 7x + 8 \geq 5x - 2$$

$$x \geq -5$$

$$(21) 5x + 1 \geq 4x - 5$$

$$x \geq -6$$

$$(22) -3x + 1 < 6x + 2$$

$$x > -\frac{1}{9}$$

$$(23) 9x - 3 \geq 3x + 6$$

$$x \geq \frac{3}{2}$$

$$(24) 5x + 6 < 4x - 9$$

$$x < -15$$

$$(25) -5x + 5 \geq 6x - 6$$

$$x \leq 1$$

$$(26) -2x + 4 \leq 9x + 8$$

$$x \geq -\frac{4}{11}$$

$$(27) -5x - 5 > -8x + 2$$

$$x > \frac{7}{3}$$

$$(28) 9x + 4 \geq -7x - 8$$

$$x \geq -\frac{3}{4}$$

$$(29) 5x - 4 \geq x + 2$$

$$x \geq \frac{3}{2}$$

$$(30) -6x - 5 < -9x + 3$$

$$x < \frac{8}{3}$$

$$(31) -8x - 5 < -5x - 8$$

$$x > 1$$

$$(32) 8x - 1 \leq 7x + 2$$

$$x \leq 3$$

$$(33) -6x + 4 \leq 5x + 6$$

$$x \geq -\frac{2}{11}$$

$$(34) -4x - 1 < -6x + 6$$

$$x < \frac{7}{2}$$

$$(35) -9x - 6 > 4x + 9$$

$$x < -\frac{15}{13}$$

$$(36) -4x + 2 > 6x + 6$$

$$x < -\frac{2}{5}$$

$$(37) -5x + 1 < 5x - 7$$

$$x > \frac{4}{5}$$

$$(38) -6x - 4 \leq 5x - 7$$

$$x \geq \frac{3}{11}$$

$$(39) 8x - 7 \leq -7x + 5$$

$$x \leq \frac{4}{5}$$

$$(40) 9x - 7 < 3x - 5$$

$$x < \frac{1}{3}$$

$$(41) -7x + 2 \geq -x - 6$$

$$x \leq \frac{4}{3}$$

$$(42) -3x + 7 < -4x - 7$$

$$x < -14$$

$$(43) -3x + 7 < -x - 3$$

$$x > 5$$

$$(44) -3x - 5 > -7x + 5$$

$$x > \frac{5}{2}$$

$$(45) 9x + 8 > -6x - 3$$

$$x > -\frac{11}{15}$$

2.11 円の方程式

円の方程式から中心点の座標と半径を求める問題。

(1) $x^2 + y^2 + 4x - 2y - 4 = 0$

(16) $x^2 + y^2 + 5x - 5y + 1 = 0$

(2) $x^2 + y^2 - x + 6y + 7 = 0$

(17) $x^2 + y^2 - x - 2 = 0$

(3) $x^2 + y^2 + 6x - 4y - 3 = 0$

(18) $x^2 + y^2 - 7x + 6y = 0$

(4) $x^2 + y^2 - 3x - 2y - 2 = 0$

(19) $x^2 + y^2 + 6x - 5 = 0$

(5) $x^2 + y^2 - 5x - 4y + 4 = 0$

(20) $x^2 + y^2 + 7x + 4y - 3 = 0$

(6) $x^2 + y^2 + 7x - 5 = 0$

(21) $x^2 + y^2 - x - 4y + 2 = 0$

(7) $x^2 + y^2 + 7x + 4y - 4 = 0$

(22) $x^2 + y^2 - x - 7y + 5 = 0$

(8) $x^2 + y^2 + 4x - 7y + 6 = 0$

(23) $x^2 + y^2 + 5x - 7y + 3 = 0$

(9) $x^2 + y^2 - 2x + y - 6 = 0$

(24) $x^2 + y^2 + 4x + 6y + 3 = 0$

(10) $x^2 + y^2 - 2x - 2y - 4 = 0$

(25) $x^2 + y^2 + 5x + y - 1 = 0$

(11) $x^2 + y^2 + 3x + y = 0$

(26) $x^2 + y^2 - 5x - 7y - 5 = 0$

(12) $x^2 + y^2 - 7y - 4 = 0$

(27) $x^2 + y^2 - 7x + y - 3 = 0$

(13) $x^2 + y^2 - 6x + 2y - 5 = 0$

(28) $x^2 + y^2 + 5y - 5 = 0$

(14) $x^2 + y^2 - 3x + y - 1 = 0$

(29) $x^2 + y^2 - x + 3y + 2 = 0$

(15) $x^2 + y^2 + x + 6y - 7 = 0$

(30) $x^2 + y^2 + 4x - y - 1 = 0$

(1) $x^2 + y^2 + 3x - 2y - 7 = 0$

(16) $x^2 + y^2 - 7x + 6y + 6 = 0$

(2) $x^2 + y^2 - 5y - 1 = 0$

(17) $x^2 + y^2 + 6x - 5y - 3 = 0$

(3) $x^2 + y^2 - x - 6 = 0$

(18) $x^2 + y^2 - 4y - 1 = 0$

(4) $x^2 + y^2 + 6x + 4y + 1 = 0$

(19) $x^2 + y^2 - 5x - y - 2 = 0$

(5) $x^2 + y^2 + 7y + 7 = 0$

(20) $x^2 + y^2 + 4x + 5y + 6 = 0$

(6) $x^2 + y^2 + 7x - 6y - 6 = 0$

(21) $x^2 + y^2 + 2x + y - 2 = 0$

(7) $x^2 + y^2 - 6x - 4y - 6 = 0$

(22) $x^2 + y^2 - 4x + y - 7 = 0$

(8) $x^2 + y^2 - x + y - 1 = 0$

(23) $x^2 + y^2 - 4x - y - 1 = 0$

(9) $x^2 + y^2 + x - 6 = 0$

(24) $x^2 + y^2 + 5x - 6y - 1 = 0$

(10) $x^2 + y^2 - 2x - 3y - 1 = 0$

(25) $x^2 + y^2 - 5x + 3y + 4 = 0$

(11) $x^2 + y^2 - x + 3y - 5 = 0$

(26) $x^2 + y^2 + x - 2y - 4 = 0$

(12) $x^2 + y^2 - 6x + y - 1 = 0$

(27) $x^2 + y^2 + x + 4y - 5 = 0$

(13) $x^2 + y^2 - 7x - 5y + 6 = 0$

(28) $x^2 + y^2 + 6x - 3y + 5 = 0$

(14) $x^2 + y^2 - 2x - 2y - 1 = 0$

(29) $x^2 + y^2 + 6x = 0$

(15) $x^2 + y^2 + 6x - 6y - 4 = 0$

(30) $x^2 + y^2 - 4x - 3y + 3 = 0$

(1) $x^2 + y^2 - 6x - 7y + 6 = 0$

(16) $x^2 + y^2 - 7x + 7y - 3 = 0$

(2) $x^2 + y^2 - 5x + y - 5 = 0$

(17) $x^2 + y^2 - x + 6y - 5 = 0$

(3) $x^2 + y^2 + 6x - 6y - 7 = 0$

(18) $x^2 + y^2 + 4x + 6y + 3 = 0$

(4) $x^2 + y^2 + x + 2y - 1 = 0$

(19) $x^2 + y^2 + 5x - 7y + 5 = 0$

(5) $x^2 + y^2 + 3x - 5y + 6 = 0$

(20) $x^2 + y^2 - 6x - 2y + 6 = 0$

(6) $x^2 + y^2 - 4x - 5y - 4 = 0$

(21) $x^2 + y^2 - 4x + y + 2 = 0$

(7) $x^2 + y^2 + 5x + y + 1 = 0$

(22) $x^2 + y^2 - 4y - 3 = 0$

(8) $x^2 + y^2 - 5x - y + 4 = 0$

(23) $x^2 + y^2 + 3x - 5y + 7 = 0$

(9) $x^2 + y^2 - x - y - 5 = 0$

(24) $x^2 + y^2 + 3x - 7y - 7 = 0$

(10) $x^2 + y^2 + 6x - 2y + 4 = 0$

(25) $x^2 + y^2 + 7x - 7y - 1 = 0$

(11) $x^2 + y^2 - 4x + 2y - 7 = 0$

(26) $x^2 + y^2 - 6x - 2y + 1 = 0$

(12) $x^2 + y^2 + 7x - 2y + 3 = 0$

(27) $x^2 + y^2 + 7x + 4y - 3 = 0$

(13) $x^2 + y^2 + 3x + 6y - 5 = 0$

(28) $x^2 + y^2 - 4x - 6y + 5 = 0$

(14) $x^2 + y^2 - 4x - 5y + 1 = 0$

(29) $x^2 + y^2 - 5x - 3y - 3 = 0$

(15) $x^2 + y^2 - 7x + 4y + 7 = 0$

(30) $x^2 + y^2 + 4x - 3y - 7 = 0$

(1) $x^2 + y^2 - 2x + 4y - 5 = 0$

(16) $x^2 + y^2 - 5x + 5y + 5 = 0$

(2) $x^2 + y^2 - 3x - 7y = 0$

(17) $x^2 + y^2 - y - 5 = 0$

(3) $x^2 + y^2 + 2x + 5y - 5 = 0$

(18) $x^2 + y^2 + 6x - 5y - 6 = 0$

(4) $x^2 + y^2 - x + 3y - 1 = 0$

(19) $x^2 + y^2 - 5x - 2y - 2 = 0$

(5) $x^2 + y^2 - 7x + 5y - 1 = 0$

(20) $x^2 + y^2 - 3x - 3y + 3 = 0$

(6) $x^2 + y^2 - 3x - 6y - 3 = 0$

(21) $x^2 + y^2 - 3x - 1 = 0$

(7) $x^2 + y^2 + 5y - 7 = 0$

(22) $x^2 + y^2 - 6x + 2y + 5 = 0$

(8) $x^2 + y^2 - 5x + y - 3 = 0$

(23) $x^2 + y^2 + 5y - 1 = 0$

(9) $x^2 + y^2 + 6x - 6y + 6 = 0$

(24) $x^2 + y^2 + x + 7y - 3 = 0$

(10) $x^2 + y^2 - 6x + y - 1 = 0$

(25) $x^2 + y^2 + 7x - 4y + 5 = 0$

(11) $x^2 + y^2 - 6x + 2y - 5 = 0$

(26) $x^2 + y^2 - 5x - y + 5 = 0$

(12) $x^2 + y^2 + x + 3y - 7 = 0$

(27) $x^2 + y^2 - 4x - 4y - 3 = 0$

(13) $x^2 + y^2 - x - 6 = 0$

(28) $x^2 + y^2 + x - 2y - 3 = 0$

(14) $x^2 + y^2 + 4x + 4y + 1 = 0$

(29) $x^2 + y^2 + 3x - 4y + 4 = 0$

(15) $x^2 + y^2 - 6x + y - 6 = 0$

(30) $x^2 + y^2 - 7x + 3y + 3 = 0$

(1) $x^2 + y^2 + x - 5y - 3 = 0$

(16) $x^2 + y^2 + 4x - 7y + 4 = 0$

(2) $x^2 + y^2 + 2x - 5y + 6 = 0$

(17) $x^2 + y^2 + 6x - 5y - 4 = 0$

(3) $x^2 + y^2 - 2x - 6y - 6 = 0$

(18) $x^2 + y^2 - 5x + 2y + 3 = 0$

(4) $x^2 + y^2 + 7y - 5 = 0$

(19) $x^2 + y^2 + x + 6y + 7 = 0$

(5) $x^2 + y^2 - x - 5y - 2 = 0$

(20) $x^2 + y^2 + 5x - 6y + 1 = 0$

(6) $x^2 + y^2 + 6x - 5y - 2 = 0$

(21) $x^2 + y^2 + 2x - 7y + 6 = 0$

(7) $x^2 + y^2 - 3x + 4y - 2 = 0$

(22) $x^2 + y^2 + 5x + y - 5 = 0$

(8) $x^2 + y^2 - 4x + 5y = 0$

(23) $x^2 + y^2 - 4x - 2y - 7 = 0$

(9) $x^2 + y^2 - 6x + 7y + 2 = 0$

(24) $x^2 + y^2 + 6x - y - 3 = 0$

(10) $x^2 + y^2 + x + 7y + 5 = 0$

(25) $x^2 + y^2 - x + y - 3 = 0$

(11) $x^2 + y^2 - 5x - 7y + 4 = 0$

(26) $x^2 + y^2 - 6x + y = 0$

(12) $x^2 + y^2 - 4x + 5y + 4 = 0$

(27) $x^2 + y^2 - 2x + 7y - 3 = 0$

(13) $x^2 + y^2 - x + y - 6 = 0$

(28) $x^2 + y^2 + 5x - 6y + 3 = 0$

(14) $x^2 + y^2 - 6x - 7y + 2 = 0$

(29) $x^2 + y^2 - 2x + 7y - 6 = 0$

(15) $x^2 + y^2 - 6y + 6 = 0$

(30) $x^2 + y^2 - 3x - 2y - 7 = 0$

(1) $x^2 + y^2 - 3x + 6y + 1 = 0$

(16) $x^2 + y^2 + 4x - 3y + 2 = 0$

(2) $x^2 + y^2 + x + 3y - 5 = 0$

(17) $x^2 + y^2 - 6x + 7y - 7 = 0$

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(20) $x^2 + y^2 + x - 7y - 3 = 0$

(6) $x^2 + y^2 + 6x - 1 = 0$

(21) $x^2 + y^2 + 5x = 0$

(7) $x^2 + y^2 + x + 2y = 0$

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(12) $x^2 + y^2 + 7x - 4y - 2 = 0$

(27) $x^2 + y^2 - 7y - 4 = 0$

(13) $x^2 + y^2 + x - 7y + 7 = 0$

(28) $x^2 + y^2 + 7x - 4y + 3 = 0$

(14) $x^2 + y^2 - 2x + 5y - 1 = 0$

(29) $x^2 + y^2 + x - 7y - 3 = 0$

(15) $x^2 + y^2 + 5x + 4y + 2 = 0$

(30) $x^2 + y^2 + 7x + 7y + 1 = 0$

(1) $x^2 + y^2 - 5x - 3y + 4 = 0$

(16) $x^2 + y^2 + 6x - 2y - 7 = 0$

(2) $x^2 + y^2 - 4x - 2y + 3 = 0$

(17) $x^2 + y^2 + 4x - 3y + 2 = 0$

(3) $x^2 + y^2 + 4x + y + 1 = 0$

(18) $x^2 + y^2 + 2x - 6y + 1 = 0$

(4) $x^2 + y^2 - 6x + 7y - 4 = 0$

(19) $x^2 + y^2 - 6x - y - 3 = 0$

(5) $x^2 + y^2 - 5x + 6y + 7 = 0$

(20) $x^2 + y^2 + x - 3y + 1 = 0$

(6) $x^2 + y^2 + 7x - 5y - 5 = 0$

(21) $x^2 + y^2 - 7x + 7y + 6 = 0$

(7) $x^2 + y^2 - 6x - 2y - 4 = 0$

(22) $x^2 + y^2 + 6x - y - 7 = 0$

(8) $x^2 + y^2 - 6x = 0$

(23) $x^2 + y^2 - x + 2y - 5 = 0$

(9) $x^2 + y^2 - 4x - 3y = 0$

(24) $x^2 + y^2 - 2x + 7y = 0$

(10) $x^2 + y^2 + 6x - 3y - 2 = 0$

(25) $x^2 + y^2 - x + 6y - 3 = 0$

(11) $x^2 + y^2 + 2x - 2y = 0$

(26) $x^2 + y^2 + 4y - 1 = 0$

(12) $x^2 + y^2 + 7x + 2y + 2 = 0$

(27) $x^2 + y^2 - 3x - y - 3 = 0$

(13) $x^2 + y^2 - 4y + 1 = 0$

(28) $x^2 + y^2 - 7x - 3y = 0$

(14) $x^2 + y^2 - 4x - y + 4 = 0$

(29) $x^2 + y^2 + 3x + 3y - 5 = 0$

(15) $x^2 + y^2 + 4x - 4y - 4 = 0$

(30) $x^2 + y^2 - x - 7y = 0$

(1) $x^2 + y^2 - 7x + y - 6 = 0$

(16) $x^2 + y^2 + 6x - 5y = 0$

(2) $x^2 + y^2 - 6x - 6 = 0$

(17) $x^2 + y^2 - 5x - 7y + 3 = 0$

(3) $x^2 + y^2 + 4x + 2y - 7 = 0$

(18) $x^2 + y^2 - 7x + 7y - 4 = 0$

(4) $x^2 + y^2 - 4x + 2y - 5 = 0$

(19) $x^2 + y^2 - 7x + 7y - 4 = 0$

(5) $x^2 + y^2 + x - 5 = 0$

(20) $x^2 + y^2 + 3x - 5y - 7 = 0$

(6) $x^2 + y^2 + 5x - 5y - 7 = 0$

(21) $x^2 + y^2 + 2x - 5y - 3 = 0$

(7) $x^2 + y^2 + 6y + 6 = 0$

(22) $x^2 + y^2 - 5y - 5 = 0$

(8) $x^2 + y^2 + 7x + 7y - 3 = 0$

(23) $x^2 + y^2 - 6x + 7y + 5 = 0$

(9) $x^2 + y^2 - 6x - 2y + 3 = 0$

(24) $x^2 + y^2 + 2x - 3y = 0$

(10) $x^2 + y^2 - 4x - 4y - 1 = 0$

(25) $x^2 + y^2 - 2x - 7y - 1 = 0$

(11) $x^2 + y^2 + x - 2 = 0$

(26) $x^2 + y^2 - 5x + 4y = 0$

(12) $x^2 + y^2 - 7x + 5y - 1 = 0$

(27) $x^2 + y^2 - x + y - 2 = 0$

(13) $x^2 + y^2 + 7x - y - 6 = 0$

(28) $x^2 + y^2 - 5x - 7y - 4 = 0$

(14) $x^2 + y^2 - 7x + 5y + 7 = 0$

(29) $x^2 + y^2 + 5x - 6y = 0$

(15) $x^2 + y^2 + 3x + 5y + 3 = 0$

(30) $x^2 + y^2 - 6x - 2y - 5 = 0$

(1) $x^2 + y^2 - x + 5y + 2 = 0$

(16) $x^2 + y^2 + 4x - 2y - 6 = 0$

(2) $x^2 + y^2 + 3x + 6y + 4 = 0$

(17) $x^2 + y^2 + 7x + 7y - 7 = 0$

(3) $x^2 + y^2 + 3x - 3y - 6 = 0$

(18) $x^2 + y^2 + 3x + 5y - 7 = 0$

(4) $x^2 + y^2 + 5x + y + 2 = 0$

(19) $x^2 + y^2 + 4x + 5y = 0$

(5) $x^2 + y^2 + 5x - y - 6 = 0$

(20) $x^2 + y^2 + 7x - 3y + 2 = 0$

(6) $x^2 + y^2 - x - 7y - 5 = 0$

(21) $x^2 + y^2 + 7x + 7y - 3 = 0$

(7) $x^2 + y^2 - 6x - y - 6 = 0$

(22) $x^2 + y^2 - 6x + 4 = 0$

(8) $x^2 + y^2 - x - 5y - 2 = 0$

(23) $x^2 + y^2 + 4x - 2y - 5 = 0$

(9) $x^2 + y^2 - 7x + 3y + 3 = 0$

(24) $x^2 + y^2 - 3x + 5y + 4 = 0$

(10) $x^2 + y^2 - 3x + 4y - 4 = 0$

(25) $x^2 + y^2 + 7x - 3 = 0$

(11) $x^2 + y^2 + 4x - 7y + 3 = 0$

(26) $x^2 + y^2 - 4x + y + 3 = 0$

(12) $x^2 + y^2 + 2x + 7y - 6 = 0$

(27) $x^2 + y^2 + 6x + 6y - 6 = 0$

(13) $x^2 + y^2 + 7x - y + 4 = 0$

(28) $x^2 + y^2 + x - 4y - 1 = 0$

(14) $x^2 + y^2 - 5x + 5y + 3 = 0$

(29) $x^2 + y^2 + 7x - 5y + 5 = 0$

(15) $x^2 + y^2 + 5x - 2y - 5 = 0$

(30) $x^2 + y^2 - 5x + 5y - 7 = 0$

(1) $x^2 + y^2 - 4x + 1 = 0$

(16) $x^2 + y^2 - 3x + 2y = 0$

(2) $x^2 + y^2 + 7x + y - 7 = 0$

(17) $x^2 + y^2 + 2x - 7y - 6 = 0$

(3) $x^2 + y^2 - 4x + 6y - 4 = 0$

(18) $x^2 + y^2 - x + 7y + 1 = 0$

(4) $x^2 + y^2 - 4x - 5y + 5 = 0$

(19) $x^2 + y^2 - 5x + 5y - 3 = 0$

(5) $x^2 + y^2 - 6x - y = 0$

(20) $x^2 + y^2 + 3x + 5y - 4 = 0$

(6) $x^2 + y^2 - 7x + 5y + 5 = 0$

(21) $x^2 + y^2 - 7x + 3y - 6 = 0$

(7) $x^2 + y^2 + 7x - 3y + 4 = 0$

(22) $x^2 + y^2 + 5x - 7y - 2 = 0$

(8) $x^2 + y^2 - 2x + 3y - 5 = 0$

(23) $x^2 + y^2 + 2x - 2y - 7 = 0$

(9) $x^2 + y^2 + 4x + 6y - 2 = 0$

(24) $x^2 + y^2 + 2x + 5y + 6 = 0$

(10) $x^2 + y^2 - 7x - 5y + 2 = 0$

(25) $x^2 + y^2 - 2x + 6y - 7 = 0$

(11) $x^2 + y^2 + 7x - 2y + 5 = 0$

(26) $x^2 + y^2 + 7x + 4y + 3 = 0$

(12) $x^2 + y^2 + x - 7y - 6 = 0$

(27) $x^2 + y^2 + 2x + 3y - 4 = 0$

(13) $x^2 + y^2 - 2x - 5y + 4 = 0$

(28) $x^2 + y^2 - 5y - 3 = 0$

(14) $x^2 + y^2 - 5x + 3y - 6 = 0$

(29) $x^2 + y^2 + 2x - 2y - 5 = 0$

(15) $x^2 + y^2 + 5x - 5y + 3 = 0$

(30) $x^2 + y^2 - 5x - 6y + 6 = 0$

- (1) $x^2 + y^2 + 4x - 2y - 4 = 0$
中心 $(-2, 1)$, 半径 3
- (2) $x^2 + y^2 - x + 6y + 7 = 0$
中心 $\left(\frac{1}{2}, -3\right)$, 半径 $\frac{3}{2}$
- (3) $x^2 + y^2 + 6x - 4y - 3 = 0$
中心 $(-3, 2)$, 半径 4
- (4) $x^2 + y^2 - 3x - 2y - 2 = 0$
中心 $\left(\frac{3}{2}, 1\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (5) $x^2 + y^2 - 5x - 4y + 4 = 0$
中心 $\left(\frac{5}{2}, 2\right)$, 半径 $\frac{5}{2}$
- (6) $x^2 + y^2 + 7x - 5 = 0$
中心 $\left(-\frac{7}{2}, 0\right)$, 半径 $\frac{\sqrt{69}}{2}$
- (7) $x^2 + y^2 + 7x + 4y - 4 = 0$
中心 $\left(-\frac{7}{2}, -2\right)$, 半径 $\frac{9}{2}$
- (8) $x^2 + y^2 + 4x - 7y + 6 = 0$
中心 $\left(-2, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (9) $x^2 + y^2 - 2x + y - 6 = 0$
中心 $\left(1, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (10) $x^2 + y^2 - 2x - 2y - 4 = 0$
中心 $(1, 1)$, 半径 $\sqrt{6}$
- (11) $x^2 + y^2 + 3x + y = 0$
中心 $\left(-\frac{3}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{10}}{2}$
- (12) $x^2 + y^2 - 7y - 4 = 0$
中心 $\left(0, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (13) $x^2 + y^2 - 6x + 2y - 5 = 0$
中心 $(3, -1)$, 半径 $\sqrt{15}$
- (14) $x^2 + y^2 - 3x + y - 1 = 0$
中心 $\left(\frac{3}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{14}}{2}$
- (15) $x^2 + y^2 + x + 6y - 7 = 0$
中心 $\left(-\frac{1}{2}, -3\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (16) $x^2 + y^2 + 5x - 5y + 1 = 0$
中心 $\left(-\frac{5}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (17) $x^2 + y^2 - x - 2 = 0$
中心 $\left(\frac{1}{2}, 0\right)$, 半径 $\frac{3}{2}$
- (18) $x^2 + y^2 - 7x + 6y = 0$
中心 $\left(\frac{7}{2}, -3\right)$, 半径 $\frac{\sqrt{85}}{2}$
- (19) $x^2 + y^2 + 6x - 5 = 0$
中心 $(-3, 0)$, 半径 $\sqrt{14}$
- (20) $x^2 + y^2 + 7x + 4y - 3 = 0$
中心 $\left(-\frac{7}{2}, -2\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (21) $x^2 + y^2 - x - 4y + 2 = 0$
中心 $\left(\frac{1}{2}, 2\right)$, 半径 $\frac{3}{2}$
- (22) $x^2 + y^2 - x - 7y + 5 = 0$
中心 $\left(\frac{1}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{30}}{2}$
- (23) $x^2 + y^2 + 5x - 7y + 3 = 0$
中心 $\left(-\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (24) $x^2 + y^2 + 4x + 6y + 3 = 0$
中心 $(-2, -3)$, 半径 $\sqrt{10}$
- (25) $x^2 + y^2 + 5x + y - 1 = 0$
中心 $\left(-\frac{5}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{30}}{2}$
- (26) $x^2 + y^2 - 5x - 7y - 5 = 0$
中心 $\left(\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{94}}{2}$
- (27) $x^2 + y^2 - 7x + y - 3 = 0$
中心 $\left(\frac{7}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (28) $x^2 + y^2 + 5y - 5 = 0$
中心 $\left(0, -\frac{5}{2}\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (29) $x^2 + y^2 - x + 3y + 2 = 0$
中心 $\left(\frac{1}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{2}}{2}$
- (30) $x^2 + y^2 + 4x - y - 1 = 0$
中心 $\left(-2, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{21}}{2}$

- (1) $x^2 + y^2 + 3x - 2y - 7 = 0$
中心 $\left(-\frac{3}{2}, 1\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (2) $x^2 + y^2 - 5y - 1 = 0$
中心 $\left(0, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (3) $x^2 + y^2 - x - 6 = 0$
中心 $\left(\frac{1}{2}, 0\right)$, 半径 $\frac{5}{2}$
- (4) $x^2 + y^2 + 6x + 4y + 1 = 0$
中心 $(-3, -2)$, 半径 $2\sqrt{3}$
- (5) $x^2 + y^2 + 7y + 7 = 0$
中心 $\left(0, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (6) $x^2 + y^2 + 7x - 6y - 6 = 0$
中心 $\left(-\frac{7}{2}, 3\right)$, 半径 $\frac{\sqrt{109}}{2}$
- (7) $x^2 + y^2 - 6x - 4y - 6 = 0$
中心 $(3, 2)$, 半径 $\sqrt{19}$
- (8) $x^2 + y^2 - x + y - 1 = 0$
中心 $\left(\frac{1}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{6}}{2}$
- (9) $x^2 + y^2 + x - 6 = 0$
中心 $\left(-\frac{1}{2}, 0\right)$, 半径 $\frac{5}{2}$
- (10) $x^2 + y^2 - 2x - 3y - 1 = 0$
中心 $\left(1, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (11) $x^2 + y^2 - x + 3y - 5 = 0$
中心 $\left(\frac{1}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{30}}{2}$
- (12) $x^2 + y^2 - 6x + y - 1 = 0$
中心 $\left(3, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (13) $x^2 + y^2 - 7x - 5y + 6 = 0$
中心 $\left(\frac{7}{2}, \frac{5}{2}\right)$, 半径 $\frac{5\sqrt{2}}{2}$
- (14) $x^2 + y^2 - 2x - 2y - 1 = 0$
中心 $(1, 1)$, 半径 $\sqrt{3}$
- (15) $x^2 + y^2 + 6x - 6y - 4 = 0$
中心 $(-3, 3)$, 半径 $\sqrt{22}$
- (16) $x^2 + y^2 - 7x + 6y + 6 = 0$
中心 $\left(\frac{7}{2}, -3\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (17) $x^2 + y^2 + 6x - 5y - 3 = 0$
中心 $\left(-3, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{73}}{2}$
- (18) $x^2 + y^2 - 4y - 1 = 0$
中心 $(0, 2)$, 半径 $\sqrt{5}$
- (19) $x^2 + y^2 - 5x - y - 2 = 0$
中心 $\left(\frac{5}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{34}}{2}$
- (20) $x^2 + y^2 + 4x + 5y + 6 = 0$
中心 $\left(-2, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (21) $x^2 + y^2 + 2x + y - 2 = 0$
中心 $\left(-1, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (22) $x^2 + y^2 - 4x + y - 7 = 0$
中心 $\left(2, -\frac{1}{2}\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (23) $x^2 + y^2 - 4x - y - 1 = 0$
中心 $\left(2, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (24) $x^2 + y^2 + 5x - 6y - 1 = 0$
中心 $\left(-\frac{5}{2}, 3\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (25) $x^2 + y^2 - 5x + 3y + 4 = 0$
中心 $\left(\frac{5}{2}, -\frac{3}{2}\right)$, 半径 $\frac{3\sqrt{2}}{2}$
- (26) $x^2 + y^2 + x - 2y - 4 = 0$
中心 $\left(-\frac{1}{2}, 1\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (27) $x^2 + y^2 + x + 4y - 5 = 0$
中心 $\left(-\frac{1}{2}, -2\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (28) $x^2 + y^2 + 6x - 3y + 5 = 0$
中心 $\left(-3, \frac{3}{2}\right)$, 半径 $\frac{5}{2}$
- (29) $x^2 + y^2 + 6x = 0$
中心 $(-3, 0)$, 半径 3
- (30) $x^2 + y^2 - 4x - 3y + 3 = 0$
中心 $\left(2, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{13}}{2}$

- (1) $x^2 + y^2 - 6x - 7y + 6 = 0$
中心 $\left(3, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (2) $x^2 + y^2 - 5x + y - 5 = 0$
中心 $\left(\frac{5}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (3) $x^2 + y^2 + 6x - 6y - 7 = 0$
中心 $(-3, 3)$, 半径 5
- (4) $x^2 + y^2 + x + 2y - 1 = 0$
中心 $\left(-\frac{1}{2}, -1\right)$, 半径 $\frac{3}{2}$
- (5) $x^2 + y^2 + 3x - 5y + 6 = 0$
中心 $\left(-\frac{3}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{10}}{2}$
- (6) $x^2 + y^2 - 4x - 5y - 4 = 0$
中心 $\left(2, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{57}}{2}$
- (7) $x^2 + y^2 + 5x + y + 1 = 0$
中心 $\left(-\frac{5}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{22}}{2}$
- (8) $x^2 + y^2 - 5x - y + 4 = 0$
中心 $\left(\frac{5}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{10}}{2}$
- (9) $x^2 + y^2 - x - y - 5 = 0$
中心 $\left(\frac{1}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{22}}{2}$
- (10) $x^2 + y^2 + 6x - 2y + 4 = 0$
中心 $(-3, 1)$, 半径 $\sqrt{6}$
- (11) $x^2 + y^2 - 4x + 2y - 7 = 0$
中心 $(2, -1)$, 半径 $2\sqrt{3}$
- (12) $x^2 + y^2 + 7x - 2y + 3 = 0$
中心 $\left(-\frac{7}{2}, 1\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (13) $x^2 + y^2 + 3x + 6y - 5 = 0$
中心 $\left(-\frac{3}{2}, -3\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (14) $x^2 + y^2 - 4x - 5y + 1 = 0$
中心 $\left(2, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (15) $x^2 + y^2 - 7x + 4y + 7 = 0$
中心 $\left(\frac{7}{2}, -2\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (16) $x^2 + y^2 - 7x + 7y - 3 = 0$
中心 $\left(\frac{7}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{110}}{2}$
- (17) $x^2 + y^2 - x + 6y - 5 = 0$
中心 $\left(\frac{1}{2}, -3\right)$, 半径 $\frac{\sqrt{57}}{2}$
- (18) $x^2 + y^2 + 4x + 6y + 3 = 0$
中心 $(-2, -3)$, 半径 $\sqrt{10}$
- (19) $x^2 + y^2 + 5x - 7y + 5 = 0$
中心 $\left(-\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{3\sqrt{6}}{2}$
- (20) $x^2 + y^2 - 6x - 2y + 6 = 0$
中心 $(3, 1)$, 半径 2
- (21) $x^2 + y^2 - 4x + y + 2 = 0$
中心 $\left(2, -\frac{1}{2}\right)$, 半径 $\frac{3}{2}$
- (22) $x^2 + y^2 - 4y - 3 = 0$
中心 $(0, 2)$, 半径 $\sqrt{7}$
- (23) $x^2 + y^2 + 3x - 5y + 7 = 0$
中心 $\left(-\frac{3}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{6}}{2}$
- (24) $x^2 + y^2 + 3x - 7y - 7 = 0$
中心 $\left(-\frac{3}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{86}}{2}$
- (25) $x^2 + y^2 + 7x - 7y - 1 = 0$
中心 $\left(-\frac{7}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{102}}{2}$
- (26) $x^2 + y^2 - 6x - 2y + 1 = 0$
中心 $(3, 1)$, 半径 3
- (27) $x^2 + y^2 + 7x + 4y - 3 = 0$
中心 $\left(-\frac{7}{2}, -2\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (28) $x^2 + y^2 - 4x - 6y + 5 = 0$
中心 $(2, 3)$, 半径 $2\sqrt{2}$
- (29) $x^2 + y^2 - 5x - 3y - 3 = 0$
中心 $\left(\frac{5}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (30) $x^2 + y^2 + 4x - 3y - 7 = 0$
中心 $\left(-2, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{53}}{2}$

- (1) $x^2 + y^2 - 2x + 4y - 5 = 0$
中心 $(1, -2)$, 半径 $\sqrt{10}$
- (2) $x^2 + y^2 - 3x - 7y = 0$
中心 $\left(\frac{3}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{58}}{2}$
- (3) $x^2 + y^2 + 2x + 5y - 5 = 0$
中心 $\left(-1, -\frac{5}{2}\right)$, 半径 $\frac{7}{2}$
- (4) $x^2 + y^2 - x + 3y - 1 = 0$
中心 $\left(\frac{1}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{14}}{2}$
- (5) $x^2 + y^2 - 7x + 5y - 1 = 0$
中心 $\left(\frac{7}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{78}}{2}$
- (6) $x^2 + y^2 - 3x - 6y - 3 = 0$
中心 $\left(\frac{3}{2}, 3\right)$, 半径 $\frac{\sqrt{57}}{2}$
- (7) $x^2 + y^2 + 5y - 7 = 0$
中心 $\left(0, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (8) $x^2 + y^2 - 5x + y - 3 = 0$
中心 $\left(\frac{5}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (9) $x^2 + y^2 + 6x - 6y + 6 = 0$
中心 $(-3, 3)$, 半径 $2\sqrt{3}$
- (10) $x^2 + y^2 - 6x + y - 1 = 0$
中心 $\left(3, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (11) $x^2 + y^2 - 6x + 2y - 5 = 0$
中心 $(3, -1)$, 半径 $\sqrt{15}$
- (12) $x^2 + y^2 + x + 3y - 7 = 0$
中心 $\left(-\frac{1}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (13) $x^2 + y^2 - x - 6 = 0$
中心 $\left(\frac{1}{2}, 0\right)$, 半径 $\frac{5}{2}$
- (14) $x^2 + y^2 + 4x + 4y + 1 = 0$
中心 $(-2, -2)$, 半径 $\sqrt{7}$
- (15) $x^2 + y^2 - 6x + y - 6 = 0$
中心 $\left(3, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (16) $x^2 + y^2 - 5x + 5y + 5 = 0$
中心 $\left(\frac{5}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{30}}{2}$
- (17) $x^2 + y^2 - y - 5 = 0$
中心 $\left(0, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (18) $x^2 + y^2 + 6x - 5y - 6 = 0$
中心 $\left(-3, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{85}}{2}$
- (19) $x^2 + y^2 - 5x - 2y - 2 = 0$
中心 $\left(\frac{5}{2}, 1\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (20) $x^2 + y^2 - 3x - 3y + 3 = 0$
中心 $\left(\frac{3}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{6}}{2}$
- (21) $x^2 + y^2 - 3x - 1 = 0$
中心 $\left(\frac{3}{2}, 0\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (22) $x^2 + y^2 - 6x + 2y + 5 = 0$
中心 $(3, -1)$, 半径 $\sqrt{5}$
- (23) $x^2 + y^2 + 5y - 1 = 0$
中心 $\left(0, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (24) $x^2 + y^2 + x + 7y - 3 = 0$
中心 $\left(-\frac{1}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (25) $x^2 + y^2 + 7x - 4y + 5 = 0$
中心 $\left(-\frac{7}{2}, 2\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (26) $x^2 + y^2 - 5x - y + 5 = 0$
中心 $\left(\frac{5}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{6}}{2}$
- (27) $x^2 + y^2 - 4x - 4y - 3 = 0$
中心 $(2, 2)$, 半径 $\sqrt{11}$
- (28) $x^2 + y^2 + x - 2y - 3 = 0$
中心 $\left(-\frac{1}{2}, 1\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (29) $x^2 + y^2 + 3x - 4y + 4 = 0$
中心 $\left(-\frac{3}{2}, 2\right)$, 半径 $\frac{3}{2}$
- (30) $x^2 + y^2 - 7x + 3y + 3 = 0$
中心 $\left(\frac{7}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$

- (1) $x^2 + y^2 + x - 5y - 3 = 0$
中心 $\left(-\frac{1}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (2) $x^2 + y^2 + 2x - 5y + 6 = 0$
中心 $\left(-1, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (3) $x^2 + y^2 - 2x - 6y - 6 = 0$
中心 $(1, 3)$, 半径 4
- (4) $x^2 + y^2 + 7y - 5 = 0$
中心 $\left(0, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{69}}{2}$
- (5) $x^2 + y^2 - x - 5y - 2 = 0$
中心 $\left(\frac{1}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{34}}{2}$
- (6) $x^2 + y^2 + 6x - 5y - 2 = 0$
中心 $\left(-3, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{69}}{2}$
- (7) $x^2 + y^2 - 3x + 4y - 2 = 0$
中心 $\left(\frac{3}{2}, -2\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (8) $x^2 + y^2 - 4x + 5y = 0$
中心 $\left(2, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (9) $x^2 + y^2 - 6x + 7y + 2 = 0$
中心 $\left(3, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (10) $x^2 + y^2 + x + 7y + 5 = 0$
中心 $\left(-\frac{1}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{30}}{2}$
- (11) $x^2 + y^2 - 5x - 7y + 4 = 0$
中心 $\left(\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{58}}{2}$
- (12) $x^2 + y^2 - 4x + 5y + 4 = 0$
中心 $\left(2, -\frac{5}{2}\right)$, 半径 $\frac{5}{2}$
- (13) $x^2 + y^2 - x + y - 6 = 0$
中心 $\left(\frac{1}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{26}}{2}$
- (14) $x^2 + y^2 - 6x - 7y + 2 = 0$
中心 $\left(3, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (15) $x^2 + y^2 - 6y + 6 = 0$
中心 $(0, 3)$, 半径 $\sqrt{3}$
- (16) $x^2 + y^2 + 4x - 7y + 4 = 0$
中心 $\left(-2, \frac{7}{2}\right)$, 半径 $\frac{7}{2}$
- (17) $x^2 + y^2 + 6x - 5y - 4 = 0$
中心 $\left(-3, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (18) $x^2 + y^2 - 5x + 2y + 3 = 0$
中心 $\left(\frac{5}{2}, -1\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (19) $x^2 + y^2 + x + 6y + 7 = 0$
中心 $\left(-\frac{1}{2}, -3\right)$, 半径 $\frac{3}{2}$
- (20) $x^2 + y^2 + 5x - 6y + 1 = 0$
中心 $\left(-\frac{5}{2}, 3\right)$, 半径 $\frac{\sqrt{57}}{2}$
- (21) $x^2 + y^2 + 2x - 7y + 6 = 0$
中心 $\left(-1, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (22) $x^2 + y^2 + 5x + y - 5 = 0$
中心 $\left(-\frac{5}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (23) $x^2 + y^2 - 4x - 2y - 7 = 0$
中心 $(2, 1)$, 半径 $2\sqrt{3}$
- (24) $x^2 + y^2 + 6x - y - 3 = 0$
中心 $\left(-3, \frac{1}{2}\right)$, 半径 $\frac{7}{2}$
- (25) $x^2 + y^2 - x + y - 3 = 0$
中心 $\left(\frac{1}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{14}}{2}$
- (26) $x^2 + y^2 - 6x + y = 0$
中心 $\left(3, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (27) $x^2 + y^2 - 2x + 7y - 3 = 0$
中心 $\left(1, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (28) $x^2 + y^2 + 5x - 6y + 3 = 0$
中心 $\left(-\frac{5}{2}, 3\right)$, 半径 $\frac{7}{2}$
- (29) $x^2 + y^2 - 2x + 7y - 6 = 0$
中心 $\left(1, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (30) $x^2 + y^2 - 3x - 2y - 7 = 0$
中心 $\left(\frac{3}{2}, 1\right)$, 半径 $\frac{\sqrt{41}}{2}$

- (1) $x^2 + y^2 - 3x + 6y + 1 = 0$
中心 $\left(\frac{3}{2}, -3\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (2) $x^2 + y^2 + x + 3y - 5 = 0$
中心 $\left(-\frac{1}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{30}}{2}$
- (3) $x^2 + y^2 + 3x + 4y + 6 = 0$
中心 $\left(-\frac{3}{2}, -2\right)$, 半径 $\frac{1}{2}$
- (4) $x^2 + y^2 + x - 5y + 5 = 0$
中心 $\left(-\frac{1}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{6}}{2}$
- (5) $x^2 + y^2 + 5x + 3y - 4 = 0$
中心 $\left(-\frac{5}{2}, -\frac{3}{2}\right)$, 半径 $\frac{5\sqrt{2}}{2}$
- (6) $x^2 + y^2 + 6x - 1 = 0$
中心 $(-3, 0)$, 半径 $\sqrt{10}$
- (7) $x^2 + y^2 + x + 2y = 0$
中心 $\left(-\frac{1}{2}, -1\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (8) $x^2 + y^2 - x - 6y + 1 = 0$
中心 $\left(\frac{1}{2}, 3\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (9) $x^2 + y^2 - 4x + 7y - 2 = 0$
中心 $\left(2, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{73}}{2}$
- (10) $x^2 + y^2 + 7x - 4y = 0$
中心 $\left(-\frac{7}{2}, 2\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (11) $x^2 + y^2 - 6x - 6y + 3 = 0$
中心 $(3, 3)$, 半径 $\sqrt{15}$
- (12) $x^2 + y^2 - 3x - 7y = 0$
中心 $\left(\frac{3}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{58}}{2}$
- (13) $x^2 + y^2 + 7x - 2y - 6 = 0$
中心 $\left(-\frac{7}{2}, 1\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (14) $x^2 + y^2 - 6x - 5y - 5 = 0$
中心 $\left(3, \frac{5}{2}\right)$, 半径 $\frac{9}{2}$
- (15) $x^2 + y^2 - 2x - 3y - 4 = 0$
中心 $\left(1, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (16) $x^2 + y^2 + 4x - 3y + 2 = 0$
中心 $\left(-2, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (17) $x^2 + y^2 - 6x + 7y - 7 = 0$
中心 $\left(3, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{113}}{2}$
- (18) $x^2 + y^2 + 3x + y + 2 = 0$
中心 $\left(-\frac{3}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{2}}{2}$
- (19) $x^2 + y^2 + x + 7y + 3 = 0$
中心 $\left(-\frac{1}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (20) $x^2 + y^2 + x - 7y - 3 = 0$
中心 $\left(-\frac{1}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (21) $x^2 + y^2 + 5x = 0$
中心 $\left(-\frac{5}{2}, 0\right)$, 半径 $\frac{5}{2}$
- (22) $x^2 + y^2 - 3x + 2y - 7 = 0$
中心 $\left(\frac{3}{2}, -1\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (23) $x^2 + y^2 - 2x - 6y - 4 = 0$
中心 $(1, 3)$, 半径 $\sqrt{14}$
- (24) $x^2 + y^2 - 2x + 4y - 1 = 0$
中心 $(1, -2)$, 半径 $\sqrt{6}$
- (25) $x^2 + y^2 - 6x + 4y - 4 = 0$
中心 $(3, -2)$, 半径 $\sqrt{17}$
- (26) $x^2 + y^2 - 3y = 0$
中心 $\left(0, \frac{3}{2}\right)$, 半径 $\frac{3}{2}$
- (27) $x^2 + y^2 + 5x + 7y - 5 = 0$
中心 $\left(-\frac{5}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{94}}{2}$
- (28) $x^2 + y^2 + x + 4y - 7 = 0$
中心 $\left(-\frac{1}{2}, -2\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (29) $x^2 + y^2 + x - 6y - 4 = 0$
中心 $\left(-\frac{1}{2}, 3\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (30) $x^2 + y^2 + 3x - 7y - 5 = 0$
中心 $\left(-\frac{3}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{78}}{2}$

- (1) $x^2 + y^2 - 4x - y + 4 = 0$
中心 $\left(2, \frac{1}{2}\right)$, 半径 $\frac{1}{2}$
- (2) $x^2 + y^2 + 5x - 7y - 2 = 0$
中心 $\left(-\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{82}}{2}$
- (3) $x^2 + y^2 - 4x - 7y + 1 = 0$
中心 $\left(2, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (4) $x^2 + y^2 + 4x + 4y + 1 = 0$
中心 $(-2, -2)$, 半径 $\sqrt{7}$
- (5) $x^2 + y^2 + 3x + 4y + 5 = 0$
中心 $\left(-\frac{3}{2}, -2\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (6) $x^2 + y^2 + 2x + 5y - 6 = 0$
中心 $\left(-1, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (7) $x^2 + y^2 - x - 2 = 0$
中心 $\left(\frac{1}{2}, 0\right)$, 半径 $\frac{3}{2}$
- (8) $x^2 + y^2 - 4x - 3y + 5 = 0$
中心 $\left(2, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (9) $x^2 + y^2 + 2x + 2y - 7 = 0$
中心 $(-1, -1)$, 半径 3
- (10) $x^2 + y^2 + x + y = 0$
中心 $\left(-\frac{1}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{2}}{2}$
- (11) $x^2 + y^2 + 7x + 3y + 5 = 0$
中心 $\left(-\frac{7}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (12) $x^2 + y^2 + 2x + 6y + 7 = 0$
中心 $(-1, -3)$, 半径 $\sqrt{3}$
- (13) $x^2 + y^2 - 7x - y + 6 = 0$
中心 $\left(\frac{7}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{26}}{2}$
- (14) $x^2 + y^2 - 3x + 7y - 7 = 0$
中心 $\left(\frac{3}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{86}}{2}$
- (15) $x^2 + y^2 - 4x - 7y - 5 = 0$
中心 $\left(2, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{85}}{2}$
- (16) $x^2 + y^2 + 3x + 7y - 7 = 0$
中心 $\left(-\frac{3}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{86}}{2}$
- (17) $x^2 + y^2 + 5x - y + 1 = 0$
中心 $\left(-\frac{5}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{22}}{2}$
- (18) $x^2 + y^2 - 5x + 5y + 1 = 0$
中心 $\left(\frac{5}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (19) $x^2 + y^2 - 6x + 3y - 7 = 0$
中心 $\left(3, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{73}}{2}$
- (20) $x^2 + y^2 - x - 7y + 3 = 0$
中心 $\left(\frac{1}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (21) $x^2 + y^2 - 7y - 6 = 0$
中心 $\left(0, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{73}}{2}$
- (22) $x^2 + y^2 - 7x + 4y - 4 = 0$
中心 $\left(\frac{7}{2}, -2\right)$, 半径 $\frac{9}{2}$
- (23) $x^2 + y^2 - 7x - 4y + 3 = 0$
中心 $\left(\frac{7}{2}, 2\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (24) $x^2 + y^2 - 6x + 3y - 3 = 0$
中心 $\left(3, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{57}}{2}$
- (25) $x^2 + y^2 - x + 5y + 4 = 0$
中心 $\left(\frac{1}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{10}}{2}$
- (26) $x^2 + y^2 + 2x - 6y = 0$
中心 $(-1, 3)$, 半径 $\sqrt{10}$
- (27) $x^2 + y^2 + 5x + 2y + 1 = 0$
中心 $\left(-\frac{5}{2}, -1\right)$, 半径 $\frac{5}{2}$
- (28) $x^2 + y^2 - 7x - 4y + 7 = 0$
中心 $\left(\frac{7}{2}, 2\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (29) $x^2 + y^2 - 2x - 6y - 2 = 0$
中心 $(1, 3)$, 半径 $2\sqrt{3}$
- (30) $x^2 + y^2 - 3x + 4y - 1 = 0$
中心 $\left(\frac{3}{2}, -2\right)$, 半径 $\frac{\sqrt{29}}{2}$

- (1) $x^2 + y^2 + 5x + 5 = 0$
中心 $\left(-\frac{5}{2}, 0\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (2) $x^2 + y^2 - x - 5 = 0$
中心 $\left(\frac{1}{2}, 0\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (3) $x^2 + y^2 + x - 5y + 2 = 0$
中心 $\left(-\frac{1}{2}, \frac{5}{2}\right)$, 半径 $\frac{3\sqrt{2}}{2}$
- (4) $x^2 + y^2 - 4x + 4y - 2 = 0$
中心 $(2, -2)$, 半径 $\sqrt{10}$
- (5) $x^2 + y^2 - 3x + 4y + 2 = 0$
中心 $\left(\frac{3}{2}, -2\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (6) $x^2 + y^2 - 3x + 3y - 1 = 0$
中心 $\left(\frac{3}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{22}}{2}$
- (7) $x^2 + y^2 + 3x - 7y + 7 = 0$
中心 $\left(-\frac{3}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{30}}{2}$
- (8) $x^2 + y^2 - x + 4y + 1 = 0$
中心 $\left(\frac{1}{2}, -2\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (9) $x^2 + y^2 + 6x + 6 = 0$
中心 $(-3, 0)$, 半径 $\sqrt{3}$
- (10) $x^2 + y^2 - 5x + 7y - 1 = 0$
中心 $\left(\frac{5}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{78}}{2}$
- (11) $x^2 + y^2 - 5x - 3y - 7 = 0$
中心 $\left(\frac{5}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (12) $x^2 + y^2 - 7x - 6y = 0$
中心 $\left(\frac{7}{2}, 3\right)$, 半径 $\frac{\sqrt{85}}{2}$
- (13) $x^2 + y^2 - 4x - y - 1 = 0$
中心 $\left(2, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (14) $x^2 + y^2 + 4x + 2y - 5 = 0$
中心 $(-2, -1)$, 半径 $\sqrt{10}$
- (15) $x^2 + y^2 + 6x + 2y - 1 = 0$
中心 $(-3, -1)$, 半径 $\sqrt{11}$
- (16) $x^2 + y^2 + 4x + 5y + 6 = 0$
中心 $\left(-2, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (17) $x^2 + y^2 + 5x - 7y - 5 = 0$
中心 $\left(-\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{94}}{2}$
- (18) $x^2 + y^2 + x - 5y - 5 = 0$
中心 $\left(-\frac{1}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (19) $x^2 + y^2 - 4x - 3y - 6 = 0$
中心 $\left(2, \frac{3}{2}\right)$, 半径 $\frac{7}{2}$
- (20) $x^2 + y^2 - x - 3y + 2 = 0$
中心 $\left(\frac{1}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{2}}{2}$
- (21) $x^2 + y^2 + 6x + 4y - 5 = 0$
中心 $(-3, -2)$, 半径 $3\sqrt{2}$
- (22) $x^2 + y^2 - 6x - 2y + 4 = 0$
中心 $(3, 1)$, 半径 $\sqrt{6}$
- (23) $x^2 + y^2 + 3x + 5y - 5 = 0$
中心 $\left(-\frac{3}{2}, -\frac{5}{2}\right)$, 半径 $\frac{3\sqrt{6}}{2}$
- (24) $x^2 + y^2 + 3x - 3y = 0$
中心 $\left(-\frac{3}{2}, \frac{3}{2}\right)$, 半径 $\frac{3\sqrt{2}}{2}$
- (25) $x^2 + y^2 - 5x - 7y - 4 = 0$
中心 $\left(\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{3\sqrt{10}}{2}$
- (26) $x^2 + y^2 - 2x + 5y + 5 = 0$
中心 $\left(1, -\frac{5}{2}\right)$, 半径 $\frac{3}{2}$
- (27) $x^2 + y^2 + x - 2y = 0$
中心 $\left(-\frac{1}{2}, 1\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (28) $x^2 + y^2 - 4x - 4y - 3 = 0$
中心 $(2, 2)$, 半径 $\sqrt{11}$
- (29) $x^2 + y^2 + 7x + y - 4 = 0$
中心 $\left(-\frac{7}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{66}}{2}$
- (30) $x^2 + y^2 - 3x + 4y + 6 = 0$
中心 $\left(\frac{3}{2}, -2\right)$, 半径 $\frac{1}{2}$

- (1) $x^2 + y^2 - 5x + 6y + 6 = 0$
中心 $\left(\frac{5}{2}, -3\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (2) $x^2 + y^2 - 2x + 2y = 0$
中心 $(1, -1)$, 半径 $\sqrt{2}$
- (3) $x^2 + y^2 - 4x - 7 = 0$
中心 $(2, 0)$, 半径 $\sqrt{11}$
- (4) $x^2 + y^2 - x - 6y - 6 = 0$
中心 $\left(\frac{1}{2}, 3\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (5) $x^2 + y^2 - 5x - 5y + 6 = 0$
中心 $\left(\frac{5}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{26}}{2}$
- (6) $x^2 + y^2 + 3x - 4y = 0$
中心 $\left(-\frac{3}{2}, 2\right)$, 半径 $\frac{5}{2}$
- (7) $x^2 + y^2 - 5x - 3y - 3 = 0$
中心 $\left(\frac{5}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (8) $x^2 + y^2 + 4x + 2 = 0$
中心 $(-2, 0)$, 半径 $\sqrt{2}$
- (9) $x^2 + y^2 + 4x - 5y + 1 = 0$
中心 $\left(-2, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (10) $x^2 + y^2 + 7x + 5 = 0$
中心 $\left(-\frac{7}{2}, 0\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (11) $x^2 + y^2 + 3x + 2y - 1 = 0$
中心 $\left(-\frac{3}{2}, -1\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (12) $x^2 + y^2 + 6x - 7 = 0$
中心 $(-3, 0)$, 半径 4
- (13) $x^2 + y^2 + 5x + y - 5 = 0$
中心 $\left(-\frac{5}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (14) $x^2 + y^2 - 6x + y - 6 = 0$
中心 $\left(3, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (15) $x^2 + y^2 - 3x + 2y + 1 = 0$
中心 $\left(\frac{3}{2}, -1\right)$, 半径 $\frac{3}{2}$
- (16) $x^2 + y^2 - 4x + 6y + 6 = 0$
中心 $(2, -3)$, 半径 $\sqrt{7}$
- (17) $x^2 + y^2 + 3x + 7y + 3 = 0$
中心 $\left(-\frac{3}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (18) $x^2 + y^2 + 5x + 2y - 7 = 0$
中心 $\left(-\frac{5}{2}, -1\right)$, 半径 $\frac{\sqrt{57}}{2}$
- (19) $x^2 + y^2 - 5x - 6y - 2 = 0$
中心 $\left(\frac{5}{2}, 3\right)$, 半径 $\frac{\sqrt{69}}{2}$
- (20) $x^2 + y^2 + x - 5y + 5 = 0$
中心 $\left(-\frac{1}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{6}}{2}$
- (21) $x^2 + y^2 + x - 6y + 5 = 0$
中心 $\left(-\frac{1}{2}, 3\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (22) $x^2 + y^2 - 6x - 7 = 0$
中心 $(3, 0)$, 半径 4
- (23) $x^2 + y^2 - 4x + 4y - 3 = 0$
中心 $(2, -2)$, 半径 $\sqrt{11}$
- (24) $x^2 + y^2 - 4x - 7y - 6 = 0$
中心 $\left(2, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{89}}{2}$
- (25) $x^2 + y^2 + 3x + 3y + 2 = 0$
中心 $\left(-\frac{3}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{10}}{2}$
- (26) $x^2 + y^2 + 5y + 5 = 0$
中心 $\left(0, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (27) $x^2 + y^2 + 3x - 7y + 6 = 0$
中心 $\left(-\frac{3}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{34}}{2}$
- (28) $x^2 + y^2 - 4x + y + 1 = 0$
中心 $\left(2, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (29) $x^2 + y^2 - 3x - 6y + 6 = 0$
中心 $\left(\frac{3}{2}, 3\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (30) $x^2 + y^2 + 5x - 6y - 5 = 0$
中心 $\left(-\frac{5}{2}, 3\right)$, 半径 $\frac{9}{2}$

- (1) $x^2 + y^2 - x + y - 6 = 0$
中心 $\left(\frac{1}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{26}}{2}$
- (2) $x^2 + y^2 - 4x + 2 = 0$
中心 $(2, 0)$, 半径 $\sqrt{2}$
- (3) $x^2 + y^2 + 7x + 7y = 0$
中心 $\left(-\frac{7}{2}, -\frac{7}{2}\right)$, 半径 $\frac{7\sqrt{2}}{2}$
- (4) $x^2 + y^2 - 4y + 2 = 0$
中心 $(0, 2)$, 半径 $\sqrt{2}$
- (5) $x^2 + y^2 - 6x + 6y - 2 = 0$
中心 $(3, -3)$, 半径 $2\sqrt{5}$
- (6) $x^2 + y^2 + 7x - y - 6 = 0$
中心 $\left(-\frac{7}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{74}}{2}$
- (7) $x^2 + y^2 + 5x + 4y - 1 = 0$
中心 $\left(-\frac{5}{2}, -2\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (8) $x^2 + y^2 + 3x - 5y - 1 = 0$
中心 $\left(-\frac{3}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (9) $x^2 + y^2 + 6x + y = 0$
中心 $\left(-3, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (10) $x^2 + y^2 + 3x - 6y + 4 = 0$
中心 $\left(-\frac{3}{2}, 3\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (11) $x^2 + y^2 - 3x - 5y - 3 = 0$
中心 $\left(\frac{3}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (12) $x^2 + y^2 + 2x + 3y - 6 = 0$
中心 $\left(-1, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (13) $x^2 + y^2 - 6x + y - 4 = 0$
中心 $\left(3, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (14) $x^2 + y^2 + 5x - 6y + 4 = 0$
中心 $\left(-\frac{5}{2}, 3\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (15) $x^2 + y^2 + 7x - 7 = 0$
中心 $\left(-\frac{7}{2}, 0\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (16) $x^2 + y^2 - 3x + 4y + 3 = 0$
中心 $\left(\frac{3}{2}, -2\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (17) $x^2 + y^2 - 2x - 4y + 4 = 0$
中心 $(1, 2)$, 半径 1
- (18) $x^2 + y^2 - 4x + 5y + 1 = 0$
中心 $\left(2, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (19) $x^2 + y^2 - 7x - 6 = 0$
中心 $\left(\frac{7}{2}, 0\right)$, 半径 $\frac{\sqrt{73}}{2}$
- (20) $x^2 + y^2 + 2x - 2y - 5 = 0$
中心 $(-1, 1)$, 半径 $\sqrt{7}$
- (21) $x^2 + y^2 + 6x - 2y + 4 = 0$
中心 $(-3, 1)$, 半径 $\sqrt{6}$
- (22) $x^2 + y^2 - 5x + 5y = 0$
中心 $\left(\frac{5}{2}, -\frac{5}{2}\right)$, 半径 $\frac{5\sqrt{2}}{2}$
- (23) $x^2 + y^2 + 6x + 3y - 5 = 0$
中心 $\left(-3, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (24) $x^2 + y^2 - 2x - 4y + 4 = 0$
中心 $(1, 2)$, 半径 1
- (25) $x^2 + y^2 + 6x - 5y = 0$
中心 $\left(-3, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (26) $x^2 + y^2 - 6x - 3y + 7 = 0$
中心 $\left(3, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (27) $x^2 + y^2 - 5x + 6y - 1 = 0$
中心 $\left(\frac{5}{2}, -3\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (28) $x^2 + y^2 + 2x + 7y - 3 = 0$
中心 $\left(-1, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (29) $x^2 + y^2 - 2x + y + 1 = 0$
中心 $\left(1, -\frac{1}{2}\right)$, 半径 $\frac{1}{2}$
- (30) $x^2 + y^2 + 5x - 7y - 5 = 0$
中心 $\left(-\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{94}}{2}$

- (1) $x^2 + y^2 - 5x - 6y - 4 = 0$
中心 $\left(\frac{5}{2}, 3\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (2) $x^2 + y^2 + 4x + 5y + 2 = 0$
中心 $\left(-2, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (3) $x^2 + y^2 + 2x - 2 = 0$
中心 $(-1, 0)$, 半径 $\sqrt{3}$
- (4) $x^2 + y^2 + 3x - y - 7 = 0$
中心 $\left(-\frac{3}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (5) $x^2 + y^2 - x + 3y - 2 = 0$
中心 $\left(\frac{1}{2}, -\frac{3}{2}\right)$, 半径 $\frac{3\sqrt{2}}{2}$
- (6) $x^2 + y^2 - x - 5 = 0$
中心 $\left(\frac{1}{2}, 0\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (7) $x^2 + y^2 - 5x - 7y + 3 = 0$
中心 $\left(\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (8) $x^2 + y^2 - 2x + 4y - 7 = 0$
中心 $(1, -2)$, 半径 $2\sqrt{3}$
- (9) $x^2 + y^2 - 5x + 4y + 4 = 0$
中心 $\left(\frac{5}{2}, -2\right)$, 半径 $\frac{5}{2}$
- (10) $x^2 + y^2 + 2x + 3y + 2 = 0$
中心 $\left(-1, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (11) $x^2 + y^2 + 7x + 4y - 3 = 0$
中心 $\left(-\frac{7}{2}, -2\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (12) $x^2 + y^2 - 5x + 2y - 2 = 0$
中心 $\left(\frac{5}{2}, -1\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (13) $x^2 + y^2 + 4x + 3y + 3 = 0$
中心 $\left(-2, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (14) $x^2 + y^2 - 5x - y + 3 = 0$
中心 $\left(\frac{5}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{14}}{2}$
- (15) $x^2 + y^2 - 6x + 2y - 3 = 0$
中心 $(3, -1)$, 半径 $\sqrt{13}$
- (16) $x^2 + y^2 - 6x - 6 = 0$
中心 $(3, 0)$, 半径 $\sqrt{15}$
- (17) $x^2 + y^2 + 3x + 3y - 4 = 0$
中心 $\left(-\frac{3}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{34}}{2}$
- (18) $x^2 + y^2 + 3y + 2 = 0$
中心 $\left(0, -\frac{3}{2}\right)$, 半径 $\frac{1}{2}$
- (19) $x^2 + y^2 - 5x - 4y + 5 = 0$
中心 $\left(\frac{5}{2}, 2\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (20) $x^2 + y^2 + 6x - 2y + 2 = 0$
中心 $(-3, 1)$, 半径 $2\sqrt{2}$
- (21) $x^2 + y^2 + x + y - 5 = 0$
中心 $\left(-\frac{1}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{22}}{2}$
- (22) $x^2 + y^2 + 5x + 7y - 2 = 0$
中心 $\left(-\frac{5}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{82}}{2}$
- (23) $x^2 + y^2 + x - 4y + 3 = 0$
中心 $\left(-\frac{1}{2}, 2\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (24) $x^2 + y^2 - 5x + 5y - 7 = 0$
中心 $\left(\frac{5}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{78}}{2}$
- (25) $x^2 + y^2 + 7x - 4 = 0$
中心 $\left(-\frac{7}{2}, 0\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (26) $x^2 + y^2 + 7x + 2y + 2 = 0$
中心 $\left(-\frac{7}{2}, -1\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (27) $x^2 + y^2 + 3x + 2y = 0$
中心 $\left(-\frac{3}{2}, -1\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (28) $x^2 + y^2 + 7x - 2y + 2 = 0$
中心 $\left(-\frac{7}{2}, 1\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (29) $x^2 + y^2 - 5x - 5y - 6 = 0$
中心 $\left(\frac{5}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{74}}{2}$
- (30) $x^2 + y^2 - 7x - 6y + 4 = 0$
中心 $\left(\frac{7}{2}, 3\right)$, 半径 $\frac{\sqrt{69}}{2}$

- (1) $x^2 + y^2 - 3x + 6y - 1 = 0$
中心 $\left(\frac{3}{2}, -3\right)$, 半径 $\frac{7}{2}$
- (2) $x^2 + y^2 - x - 7y + 7 = 0$
中心 $\left(\frac{1}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{22}}{2}$
- (3) $x^2 + y^2 + 6x + 6y + 5 = 0$
中心 $(-3, -3)$, 半径 $\sqrt{13}$
- (4) $x^2 + y^2 + 4x - 4y - 7 = 0$
中心 $(-2, 2)$, 半径 $\sqrt{15}$
- (5) $x^2 + y^2 - 2x - 6y + 5 = 0$
中心 $(1, 3)$, 半径 $\sqrt{5}$
- (6) $x^2 + y^2 + x - 4y = 0$
中心 $\left(-\frac{1}{2}, 2\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (7) $x^2 + y^2 + 4x + 3y + 3 = 0$
中心 $\left(-2, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (8) $x^2 + y^2 - 4x + 5y + 7 = 0$
中心 $\left(2, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (9) $x^2 + y^2 - 2x + 3y - 7 = 0$
中心 $\left(1, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (10) $x^2 + y^2 + 7x + y - 1 = 0$
中心 $\left(-\frac{7}{2}, -\frac{1}{2}\right)$, 半径 $\frac{3\sqrt{6}}{2}$
- (11) $x^2 + y^2 - 4x - 4y = 0$
中心 $(2, 2)$, 半径 $2\sqrt{2}$
- (12) $x^2 + y^2 - 7x + 1 = 0$
中心 $\left(\frac{7}{2}, 0\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (13) $x^2 + y^2 + 2x - 7y + 7 = 0$
中心 $\left(-1, \frac{7}{2}\right)$, 半径 $\frac{5}{2}$
- (14) $x^2 + y^2 - 5x - 2y - 6 = 0$
中心 $\left(\frac{5}{2}, 1\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (15) $x^2 + y^2 + x - 3y - 7 = 0$
中心 $\left(-\frac{1}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (16) $x^2 + y^2 + 7x - 7y + 2 = 0$
中心 $\left(-\frac{7}{2}, \frac{7}{2}\right)$, 半径 $\frac{3\sqrt{10}}{2}$
- (17) $x^2 + y^2 - 7x - 2y = 0$
中心 $\left(\frac{7}{2}, 1\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (18) $x^2 + y^2 + 7x + 6y - 7 = 0$
中心 $\left(-\frac{7}{2}, -3\right)$, 半径 $\frac{\sqrt{113}}{2}$
- (19) $x^2 + y^2 - 7x + 3y - 6 = 0$
中心 $\left(\frac{7}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{82}}{2}$
- (20) $x^2 + y^2 + 6x - 4y - 6 = 0$
中心 $(-3, 2)$, 半径 $\sqrt{19}$
- (21) $x^2 + y^2 - 4x - 4y - 2 = 0$
中心 $(2, 2)$, 半径 $\sqrt{10}$
- (22) $x^2 + y^2 - 6y - 2 = 0$
中心 $(0, 3)$, 半径 $\sqrt{11}$
- (23) $x^2 + y^2 + 7x + 6y + 4 = 0$
中心 $\left(-\frac{7}{2}, -3\right)$, 半径 $\frac{\sqrt{69}}{2}$
- (24) $x^2 + y^2 + 7x - 6y + 7 = 0$
中心 $\left(-\frac{7}{2}, 3\right)$, 半径 $\frac{\sqrt{57}}{2}$
- (25) $x^2 + y^2 + 7x - 2y + 1 = 0$
中心 $\left(-\frac{7}{2}, 1\right)$, 半径 $\frac{7}{2}$
- (26) $x^2 + y^2 - 7x = 0$
中心 $\left(\frac{7}{2}, 0\right)$, 半径 $\frac{7}{2}$
- (27) $x^2 + y^2 + 7x - 3y + 3 = 0$
中心 $\left(-\frac{7}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (28) $x^2 + y^2 - 4x + 4y + 1 = 0$
中心 $(2, -2)$, 半径 $\sqrt{7}$
- (29) $x^2 + y^2 + 2x + 7y - 2 = 0$
中心 $\left(-1, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (30) $x^2 + y^2 - 4x + y + 4 = 0$
中心 $\left(2, -\frac{1}{2}\right)$, 半径 $\frac{1}{2}$

- (1) $x^2 + y^2 + 5x + y - 5 = 0$
中心 $\left(-\frac{5}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (2) $x^2 + y^2 + 4y - 5 = 0$
中心 $(0, -2)$, 半径 3
- (3) $x^2 + y^2 + 7x - 5y - 7 = 0$
中心 $\left(-\frac{7}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{102}}{2}$
- (4) $x^2 + y^2 - 3x - 4y + 5 = 0$
中心 $\left(\frac{3}{2}, 2\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (5) $x^2 + y^2 - 7x - y - 6 = 0$
中心 $\left(\frac{7}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{74}}{2}$
- (6) $x^2 + y^2 - x - 6y - 4 = 0$
中心 $\left(\frac{1}{2}, 3\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (7) $x^2 + y^2 + 4x + 6y - 3 = 0$
中心 $(-2, -3)$, 半径 4
- (8) $x^2 + y^2 + 7x + 3y + 1 = 0$
中心 $\left(-\frac{7}{2}, -\frac{3}{2}\right)$, 半径 $\frac{3\sqrt{6}}{2}$
- (9) $x^2 + y^2 + 3x + 4y + 5 = 0$
中心 $\left(-\frac{3}{2}, -2\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (10) $x^2 + y^2 - 2x - 4y - 3 = 0$
中心 $(1, 2)$, 半径 $2\sqrt{2}$
- (11) $x^2 + y^2 + 6x + 4y + 3 = 0$
中心 $(-3, -2)$, 半径 $\sqrt{10}$
- (12) $x^2 + y^2 - 2x - 6y = 0$
中心 $(1, 3)$, 半径 $\sqrt{10}$
- (13) $x^2 + y^2 - 7x + 5y + 1 = 0$
中心 $\left(\frac{7}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{70}}{2}$
- (14) $x^2 + y^2 + x + y - 3 = 0$
中心 $\left(-\frac{1}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{14}}{2}$
- (15) $x^2 + y^2 + 4x + 5y + 1 = 0$
中心 $\left(-2, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (16) $x^2 + y^2 + 6x + 6y - 4 = 0$
中心 $(-3, -3)$, 半径 $\sqrt{22}$
- (17) $x^2 + y^2 - 6x + 2y + 1 = 0$
中心 $(3, -1)$, 半径 3
- (18) $x^2 + y^2 - x + 4y = 0$
中心 $\left(\frac{1}{2}, -2\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (19) $x^2 + y^2 + 6x - 6y - 6 = 0$
中心 $(-3, 3)$, 半径 $2\sqrt{6}$
- (20) $x^2 + y^2 - 3x + 2y + 1 = 0$
中心 $\left(\frac{3}{2}, -1\right)$, 半径 $\frac{3}{2}$
- (21) $x^2 + y^2 - 2x - 5y - 1 = 0$
中心 $\left(1, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (22) $x^2 + y^2 - 6x + 3y - 4 = 0$
中心 $\left(3, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (23) $x^2 + y^2 - 3x + 7y + 4 = 0$
中心 $\left(\frac{3}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{42}}{2}$
- (24) $x^2 + y^2 + 4x + 7y + 6 = 0$
中心 $\left(-2, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (25) $x^2 + y^2 + x + 2y - 2 = 0$
中心 $\left(-\frac{1}{2}, -1\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (26) $x^2 + y^2 - 4x + 5y - 4 = 0$
中心 $\left(2, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{57}}{2}$
- (27) $x^2 + y^2 + x - 7y + 5 = 0$
中心 $\left(-\frac{1}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{30}}{2}$
- (28) $x^2 + y^2 + 6x + 2y + 1 = 0$
中心 $(-3, -1)$, 半径 3
- (29) $x^2 + y^2 + 3x + 2y - 3 = 0$
中心 $\left(-\frac{3}{2}, -1\right)$, 半径 $\frac{5}{2}$
- (30) $x^2 + y^2 + 4x - 3y - 5 = 0$
中心 $\left(-2, \frac{3}{2}\right)$, 半径 $\frac{3\sqrt{5}}{2}$

- (1) $x^2 + y^2 - 5x = 0$
中心 $\left(\frac{5}{2}, 0\right)$, 半径 $\frac{5}{2}$
- (2) $x^2 + y^2 + 6x + y + 7 = 0$
中心 $\left(-3, -\frac{1}{2}\right)$, 半径 $\frac{3}{2}$
- (3) $x^2 + y^2 + 5x + 5 = 0$
中心 $\left(-\frac{5}{2}, 0\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (4) $x^2 + y^2 - 1 = 0$
中心 $(0, 0)$, 半径 1
- (5) $x^2 + y^2 - 5x + 3y = 0$
中心 $\left(\frac{5}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{34}}{2}$
- (6) $x^2 + y^2 + 7x + 6y + 1 = 0$
中心 $\left(-\frac{7}{2}, -3\right)$, 半径 $\frac{9}{2}$
- (7) $x^2 + y^2 - 4y - 5 = 0$
中心 $(0, 2)$, 半径 3
- (8) $x^2 + y^2 + 7x + 3 = 0$
中心 $\left(-\frac{7}{2}, 0\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (9) $x^2 + y^2 - 3x + 7y - 2 = 0$
中心 $\left(\frac{3}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{66}}{2}$
- (10) $x^2 + y^2 - 7x + 5 = 0$
中心 $\left(\frac{7}{2}, 0\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (11) $x^2 + y^2 + 3x - 3y + 2 = 0$
中心 $\left(-\frac{3}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{10}}{2}$
- (12) $x^2 + y^2 - 6x - y + 1 = 0$
中心 $\left(3, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (13) $x^2 + y^2 + 2x + 3y - 5 = 0$
中心 $\left(-1, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (14) $x^2 + y^2 - 7 = 0$
中心 $(0, 0)$, 半径 $\sqrt{7}$
- (15) $x^2 + y^2 - 4x - 6 = 0$
中心 $(2, 0)$, 半径 $\sqrt{10}$
- (16) $x^2 + y^2 - 2x + 3y + 1 = 0$
中心 $\left(1, -\frac{3}{2}\right)$, 半径 $\frac{3}{2}$
- (17) $x^2 + y^2 - 2x - 5y - 2 = 0$
中心 $\left(1, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (18) $x^2 + y^2 + 4x + 3y = 0$
中心 $\left(-2, -\frac{3}{2}\right)$, 半径 $\frac{5}{2}$
- (19) $x^2 + y^2 + x - 3y - 2 = 0$
中心 $\left(-\frac{1}{2}, \frac{3}{2}\right)$, 半径 $\frac{3\sqrt{2}}{2}$
- (20) $x^2 + y^2 - 7x + 7y + 5 = 0$
中心 $\left(\frac{7}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{78}}{2}$
- (21) $x^2 + y^2 + 7x + 3y + 1 = 0$
中心 $\left(-\frac{7}{2}, -\frac{3}{2}\right)$, 半径 $\frac{3\sqrt{6}}{2}$
- (22) $x^2 + y^2 - 7x + 2y + 1 = 0$
中心 $\left(\frac{7}{2}, -1\right)$, 半径 $\frac{7}{2}$
- (23) $x^2 + y^2 + 3x - 2y - 3 = 0$
中心 $\left(-\frac{3}{2}, 1\right)$, 半径 $\frac{5}{2}$
- (24) $x^2 + y^2 + 5x - 3y + 3 = 0$
中心 $\left(-\frac{5}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{22}}{2}$
- (25) $x^2 + y^2 - x + 2y - 1 = 0$
中心 $\left(\frac{1}{2}, -1\right)$, 半径 $\frac{3}{2}$
- (26) $x^2 + y^2 - 5x + 3y + 4 = 0$
中心 $\left(\frac{5}{2}, -\frac{3}{2}\right)$, 半径 $\frac{3\sqrt{2}}{2}$
- (27) $x^2 + y^2 + 7x + 6y + 2 = 0$
中心 $\left(-\frac{7}{2}, -3\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (28) $x^2 + y^2 - 7x - 4y = 0$
中心 $\left(\frac{7}{2}, 2\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (29) $x^2 + y^2 - 6x - 6y = 0$
中心 $(3, 3)$, 半径 $3\sqrt{2}$
- (30) $x^2 + y^2 + 7x + 3y - 3 = 0$
中心 $\left(-\frac{7}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{70}}{2}$

- (1) $x^2 + y^2 + 2x - 7y + 2 = 0$
中心 $\left(-1, \frac{7}{2}\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (2) $x^2 + y^2 - 2x - 6y + 4 = 0$
中心 $(1, 3)$, 半径 $\sqrt{6}$
- (3) $x^2 + y^2 - 3x - 7y - 2 = 0$
中心 $\left(\frac{3}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{66}}{2}$
- (4) $x^2 + y^2 - 5x - 4y + 3 = 0$
中心 $\left(\frac{5}{2}, 2\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (5) $x^2 + y^2 + 5x - 3y - 7 = 0$
中心 $\left(-\frac{5}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (6) $x^2 + y^2 + 6x - y - 6 = 0$
中心 $\left(-3, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (7) $x^2 + y^2 + 4x + y - 3 = 0$
中心 $\left(-2, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (8) $x^2 + y^2 - 2x - y - 1 = 0$
中心 $\left(1, \frac{1}{2}\right)$, 半径 $\frac{3}{2}$
- (9) $x^2 + y^2 - 3y - 3 = 0$
中心 $\left(0, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (10) $x^2 + y^2 - 7x - 2y - 4 = 0$
中心 $\left(\frac{7}{2}, 1\right)$, 半径 $\frac{\sqrt{69}}{2}$
- (11) $x^2 + y^2 + 6x + y + 1 = 0$
中心 $\left(-3, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (12) $x^2 + y^2 + x + 7y + 4 = 0$
中心 $\left(-\frac{1}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{34}}{2}$
- (13) $x^2 + y^2 - 4x + 3y + 5 = 0$
中心 $\left(2, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (14) $x^2 + y^2 - 3x - 7y - 5 = 0$
中心 $\left(\frac{3}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{78}}{2}$
- (15) $x^2 + y^2 + 5x - y - 3 = 0$
中心 $\left(-\frac{5}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (16) $x^2 + y^2 - 6x - 7y - 4 = 0$
中心 $\left(3, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{101}}{2}$
- (17) $x^2 + y^2 + 7x - 2y = 0$
中心 $\left(-\frac{7}{2}, 1\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (18) $x^2 + y^2 + 6x - 3y + 7 = 0$
中心 $\left(-3, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (19) $x^2 + y^2 + 3x - 7y - 4 = 0$
中心 $\left(-\frac{3}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{74}}{2}$
- (20) $x^2 + y^2 - 6x + y + 2 = 0$
中心 $\left(3, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (21) $x^2 + y^2 + 4x + 2y - 5 = 0$
中心 $(-2, -1)$, 半径 $\sqrt{10}$
- (22) $x^2 + y^2 - 5x + 6y + 2 = 0$
中心 $\left(\frac{5}{2}, -3\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (23) $x^2 + y^2 - 5x + 6y - 2 = 0$
中心 $\left(\frac{5}{2}, -3\right)$, 半径 $\frac{\sqrt{69}}{2}$
- (24) $x^2 + y^2 - 5y + 2 = 0$
中心 $\left(0, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (25) $x^2 + y^2 + 2x - 5y + 3 = 0$
中心 $\left(-1, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (26) $x^2 + y^2 + 3x - 5y - 4 = 0$
中心 $\left(-\frac{3}{2}, \frac{5}{2}\right)$, 半径 $\frac{5\sqrt{2}}{2}$
- (27) $x^2 + y^2 + 5x - 6y + 6 = 0$
中心 $\left(-\frac{5}{2}, 3\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (28) $x^2 + y^2 - 5x - 7y + 3 = 0$
中心 $\left(\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (29) $x^2 + y^2 - 6x + 6y + 7 = 0$
中心 $(3, -3)$, 半径 $\sqrt{11}$
- (30) $x^2 + y^2 + 6x - 4y + 1 = 0$
中心 $(-3, 2)$, 半径 $2\sqrt{3}$

- (1) $x^2 + y^2 - 4x + 7y - 4 = 0$
中心 $\left(2, -\frac{7}{2}\right)$, 半径 $\frac{9}{2}$
- (2) $x^2 + y^2 + x - 7y + 4 = 0$
中心 $\left(-\frac{1}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{34}}{2}$
- (3) $x^2 + y^2 - 3x - y - 5 = 0$
中心 $\left(\frac{3}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{30}}{2}$
- (4) $x^2 + y^2 + x - 4y - 4 = 0$
中心 $\left(-\frac{1}{2}, 2\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (5) $x^2 + y^2 - 2y - 3 = 0$
中心 $(0, 1)$, 半径 2
- (6) $x^2 + y^2 - 6x - 4y + 6 = 0$
中心 $(3, 2)$, 半径 $\sqrt{7}$
- (7) $x^2 + y^2 - x + 6y - 6 = 0$
中心 $\left(\frac{1}{2}, -3\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (8) $x^2 + y^2 - 7x + 6y - 2 = 0$
中心 $\left(\frac{7}{2}, -3\right)$, 半径 $\frac{\sqrt{93}}{2}$
- (9) $x^2 + y^2 + 5x - 2y + 4 = 0$
中心 $\left(-\frac{5}{2}, 1\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (10) $x^2 + y^2 - x + y - 7 = 0$
中心 $\left(\frac{1}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{30}}{2}$
- (11) $x^2 + y^2 - x + 4y - 4 = 0$
中心 $\left(\frac{1}{2}, -2\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (12) $x^2 + y^2 + 7x - 4y - 2 = 0$
中心 $\left(-\frac{7}{2}, 2\right)$, 半径 $\frac{\sqrt{73}}{2}$
- (13) $x^2 + y^2 + x - 7y + 7 = 0$
中心 $\left(-\frac{1}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{22}}{2}$
- (14) $x^2 + y^2 - 2x + 5y - 1 = 0$
中心 $\left(1, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (15) $x^2 + y^2 + 5x + 4y + 2 = 0$
中心 $\left(-\frac{5}{2}, -2\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (16) $x^2 + y^2 - 2x + 3y + 3 = 0$
中心 $\left(1, -\frac{3}{2}\right)$, 半径 $\frac{1}{2}$
- (17) $x^2 + y^2 + x - 2y - 7 = 0$
中心 $\left(-\frac{1}{2}, 1\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (18) $x^2 + y^2 - 4x + 4y - 1 = 0$
中心 $(2, -2)$, 半径 3
- (19) $x^2 + y^2 + 7x - 4y - 7 = 0$
中心 $\left(-\frac{7}{2}, 2\right)$, 半径 $\frac{\sqrt{93}}{2}$
- (20) $x^2 + y^2 - 2x + y - 2 = 0$
中心 $\left(1, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (21) $x^2 + y^2 - 5x - 5y + 6 = 0$
中心 $\left(\frac{5}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{26}}{2}$
- (22) $x^2 + y^2 + 6x + 4y + 3 = 0$
中心 $(-3, -2)$, 半径 $\sqrt{10}$
- (23) $x^2 + y^2 + 7x - 2y - 7 = 0$
中心 $\left(-\frac{7}{2}, 1\right)$, 半径 $\frac{9}{2}$
- (24) $x^2 + y^2 - x + 3y - 7 = 0$
中心 $\left(\frac{1}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (25) $x^2 + y^2 + 2x + 4y - 6 = 0$
中心 $(-1, -2)$, 半径 $\sqrt{11}$
- (26) $x^2 + y^2 - 6x + 2y - 6 = 0$
中心 $(3, -1)$, 半径 4
- (27) $x^2 + y^2 - 7y - 4 = 0$
中心 $\left(0, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (28) $x^2 + y^2 + 7x - 4y + 3 = 0$
中心 $\left(-\frac{7}{2}, 2\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (29) $x^2 + y^2 + x - 7y - 3 = 0$
中心 $\left(-\frac{1}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (30) $x^2 + y^2 + 7x + 7y + 1 = 0$
中心 $\left(-\frac{7}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{94}}{2}$

- (1) $x^2 + y^2 - 5x - 3y + 4 = 0$
中心 $\left(\frac{5}{2}, \frac{3}{2}\right)$, 半径 $\frac{3\sqrt{2}}{2}$
- (2) $x^2 + y^2 - 4x - 2y + 3 = 0$
中心 $(2, 1)$, 半径 $\sqrt{2}$
- (3) $x^2 + y^2 + 4x + y + 1 = 0$
中心 $\left(-2, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (4) $x^2 + y^2 - 6x + 7y - 4 = 0$
中心 $\left(3, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{101}}{2}$
- (5) $x^2 + y^2 - 5x + 6y + 7 = 0$
中心 $\left(\frac{5}{2}, -3\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (6) $x^2 + y^2 + 7x - 5y - 5 = 0$
中心 $\left(-\frac{7}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{94}}{2}$
- (7) $x^2 + y^2 - 6x - 2y - 4 = 0$
中心 $(3, 1)$, 半径 $\sqrt{14}$
- (8) $x^2 + y^2 - 6x = 0$
中心 $(3, 0)$, 半径 3
- (9) $x^2 + y^2 - 4x - 3y = 0$
中心 $\left(2, \frac{3}{2}\right)$, 半径 $\frac{5}{2}$
- (10) $x^2 + y^2 + 6x - 3y - 2 = 0$
中心 $\left(-3, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (11) $x^2 + y^2 + 2x - 2y = 0$
中心 $(-1, 1)$, 半径 $\sqrt{2}$
- (12) $x^2 + y^2 + 7x + 2y + 2 = 0$
中心 $\left(-\frac{7}{2}, -1\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (13) $x^2 + y^2 - 4y + 1 = 0$
中心 $(0, 2)$, 半径 $\sqrt{3}$
- (14) $x^2 + y^2 - 4x - y + 4 = 0$
中心 $\left(2, \frac{1}{2}\right)$, 半径 $\frac{1}{2}$
- (15) $x^2 + y^2 + 4x - 4y - 4 = 0$
中心 $(-2, 2)$, 半径 $2\sqrt{3}$
- (16) $x^2 + y^2 + 6x - 2y - 7 = 0$
中心 $(-3, 1)$, 半径 $\sqrt{17}$
- (17) $x^2 + y^2 + 4x - 3y + 2 = 0$
中心 $\left(-2, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{17}}{2}$
- (18) $x^2 + y^2 + 2x - 6y + 1 = 0$
中心 $(-1, 3)$, 半径 3
- (19) $x^2 + y^2 - 6x - y - 3 = 0$
中心 $\left(3, \frac{1}{2}\right)$, 半径 $\frac{7}{2}$
- (20) $x^2 + y^2 + x - 3y + 1 = 0$
中心 $\left(-\frac{1}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{6}}{2}$
- (21) $x^2 + y^2 - 7x + 7y + 6 = 0$
中心 $\left(\frac{7}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{74}}{2}$
- (22) $x^2 + y^2 + 6x - y - 7 = 0$
中心 $\left(-3, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (23) $x^2 + y^2 - x + 2y - 5 = 0$
中心 $\left(\frac{1}{2}, -1\right)$, 半径 $\frac{5}{2}$
- (24) $x^2 + y^2 - 2x + 7y = 0$
中心 $\left(1, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (25) $x^2 + y^2 - x + 6y - 3 = 0$
中心 $\left(\frac{1}{2}, -3\right)$, 半径 $\frac{7}{2}$
- (26) $x^2 + y^2 + 4y - 1 = 0$
中心 $(0, -2)$, 半径 $\sqrt{5}$
- (27) $x^2 + y^2 - 3x - y - 3 = 0$
中心 $\left(\frac{3}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{22}}{2}$
- (28) $x^2 + y^2 - 7x - 3y = 0$
中心 $\left(\frac{7}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{58}}{2}$
- (29) $x^2 + y^2 + 3x + 3y - 5 = 0$
中心 $\left(-\frac{3}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (30) $x^2 + y^2 - x - 7y = 0$
中心 $\left(\frac{1}{2}, \frac{7}{2}\right)$, 半径 $\frac{5\sqrt{2}}{2}$

- (1) $x^2 + y^2 - 7x + y - 6 = 0$
中心 $\left(\frac{7}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{74}}{2}$
- (2) $x^2 + y^2 - 6x - 6 = 0$
中心 $(3, 0)$, 半径 $\sqrt{15}$
- (3) $x^2 + y^2 + 4x + 2y - 7 = 0$
中心 $(-2, -1)$, 半径 $2\sqrt{3}$
- (4) $x^2 + y^2 - 4x + 2y - 5 = 0$
中心 $(2, -1)$, 半径 $\sqrt{10}$
- (5) $x^2 + y^2 + x - 5 = 0$
中心 $\left(-\frac{1}{2}, 0\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (6) $x^2 + y^2 + 5x - 5y - 7 = 0$
中心 $\left(-\frac{5}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{78}}{2}$
- (7) $x^2 + y^2 + 6y + 6 = 0$
中心 $(0, -3)$, 半径 $\sqrt{3}$
- (8) $x^2 + y^2 + 7x + 7y - 3 = 0$
中心 $\left(-\frac{7}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{110}}{2}$
- (9) $x^2 + y^2 - 6x - 2y + 3 = 0$
中心 $(3, 1)$, 半径 $\sqrt{7}$
- (10) $x^2 + y^2 - 4x - 4y - 1 = 0$
中心 $(2, 2)$, 半径 3
- (11) $x^2 + y^2 + x - 2 = 0$
中心 $\left(-\frac{1}{2}, 0\right)$, 半径 $\frac{3}{2}$
- (12) $x^2 + y^2 - 7x + 5y - 1 = 0$
中心 $\left(\frac{7}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{78}}{2}$
- (13) $x^2 + y^2 + 7x - y - 6 = 0$
中心 $\left(-\frac{7}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{74}}{2}$
- (14) $x^2 + y^2 - 7x + 5y + 7 = 0$
中心 $\left(\frac{7}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (15) $x^2 + y^2 + 3x + 5y + 3 = 0$
中心 $\left(-\frac{3}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{22}}{2}$
- (16) $x^2 + y^2 + 6x - 5y = 0$
中心 $\left(-3, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (17) $x^2 + y^2 - 5x - 7y + 3 = 0$
中心 $\left(\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (18) $x^2 + y^2 - 7x + 7y - 4 = 0$
中心 $\left(\frac{7}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{114}}{2}$
- (19) $x^2 + y^2 - 7x + 7y - 4 = 0$
中心 $\left(\frac{7}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{114}}{2}$
- (20) $x^2 + y^2 + 3x - 5y - 7 = 0$
中心 $\left(-\frac{3}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (21) $x^2 + y^2 + 2x - 5y - 3 = 0$
中心 $\left(-1, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (22) $x^2 + y^2 - 5y - 5 = 0$
中心 $\left(0, \frac{5}{2}\right)$, 半径 $\frac{3\sqrt{5}}{2}$
- (23) $x^2 + y^2 - 6x + 7y + 5 = 0$
中心 $\left(3, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{65}}{2}$
- (24) $x^2 + y^2 + 2x - 3y = 0$
中心 $\left(-1, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (25) $x^2 + y^2 - 2x - 7y - 1 = 0$
中心 $\left(1, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{57}}{2}$
- (26) $x^2 + y^2 - 5x + 4y = 0$
中心 $\left(\frac{5}{2}, -2\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (27) $x^2 + y^2 - x + y - 2 = 0$
中心 $\left(\frac{1}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{10}}{2}$
- (28) $x^2 + y^2 - 5x - 7y - 4 = 0$
中心 $\left(\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{3\sqrt{10}}{2}$
- (29) $x^2 + y^2 + 5x - 6y = 0$
中心 $\left(-\frac{5}{2}, 3\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (30) $x^2 + y^2 - 6x - 2y - 5 = 0$
中心 $(3, 1)$, 半径 $\sqrt{15}$

- (1) $x^2 + y^2 - x + 5y + 2 = 0$
中心 $\left(\frac{1}{2}, -\frac{5}{2}\right)$, 半径 $\frac{3\sqrt{2}}{2}$
- (2) $x^2 + y^2 + 3x + 6y + 4 = 0$
中心 $\left(-\frac{3}{2}, -3\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (3) $x^2 + y^2 + 3x - 3y - 6 = 0$
中心 $\left(-\frac{3}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{42}}{2}$
- (4) $x^2 + y^2 + 5x + y + 2 = 0$
中心 $\left(-\frac{5}{2}, -\frac{1}{2}\right)$, 半径 $\frac{3\sqrt{2}}{2}$
- (5) $x^2 + y^2 + 5x - y - 6 = 0$
中心 $\left(-\frac{5}{2}, \frac{1}{2}\right)$, 半径 $\frac{5\sqrt{2}}{2}$
- (6) $x^2 + y^2 - x - 7y - 5 = 0$
中心 $\left(\frac{1}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{70}}{2}$
- (7) $x^2 + y^2 - 6x - y - 6 = 0$
中心 $\left(3, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (8) $x^2 + y^2 - x - 5y - 2 = 0$
中心 $\left(\frac{1}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{34}}{2}$
- (9) $x^2 + y^2 - 7x + 3y + 3 = 0$
中心 $\left(\frac{7}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (10) $x^2 + y^2 - 3x + 4y - 4 = 0$
中心 $\left(\frac{3}{2}, -2\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (11) $x^2 + y^2 + 4x - 7y + 3 = 0$
中心 $\left(-2, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (12) $x^2 + y^2 + 2x + 7y - 6 = 0$
中心 $\left(-1, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (13) $x^2 + y^2 + 7x - y + 4 = 0$
中心 $\left(-\frac{7}{2}, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{34}}{2}$
- (14) $x^2 + y^2 - 5x + 5y + 3 = 0$
中心 $\left(\frac{5}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (15) $x^2 + y^2 + 5x - 2y - 5 = 0$
中心 $\left(-\frac{5}{2}, 1\right)$, 半径 $\frac{7}{2}$
- (16) $x^2 + y^2 + 4x - 2y - 6 = 0$
中心 $(-2, 1)$, 半径 $\sqrt{11}$
- (17) $x^2 + y^2 + 7x + 7y - 7 = 0$
中心 $\left(-\frac{7}{2}, -\frac{7}{2}\right)$, 半径 $\frac{3\sqrt{14}}{2}$
- (18) $x^2 + y^2 + 3x + 5y - 7 = 0$
中心 $\left(-\frac{3}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (19) $x^2 + y^2 + 4x + 5y = 0$
中心 $\left(-2, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{41}}{2}$
- (20) $x^2 + y^2 + 7x - 3y + 2 = 0$
中心 $\left(-\frac{7}{2}, \frac{3}{2}\right)$, 半径 $\frac{5\sqrt{2}}{2}$
- (21) $x^2 + y^2 + 7x + 7y - 3 = 0$
中心 $\left(-\frac{7}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{110}}{2}$
- (22) $x^2 + y^2 - 6x + 4 = 0$
中心 $(3, 0)$, 半径 $\sqrt{5}$
- (23) $x^2 + y^2 + 4x - 2y - 5 = 0$
中心 $(-2, 1)$, 半径 $\sqrt{10}$
- (24) $x^2 + y^2 - 3x + 5y + 4 = 0$
中心 $\left(\frac{3}{2}, -\frac{5}{2}\right)$, 半径 $\frac{3\sqrt{2}}{2}$
- (25) $x^2 + y^2 + 7x - 3 = 0$
中心 $\left(-\frac{7}{2}, 0\right)$, 半径 $\frac{\sqrt{61}}{2}$
- (26) $x^2 + y^2 - 4x + y + 3 = 0$
中心 $\left(2, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (27) $x^2 + y^2 + 6x + 6y - 6 = 0$
中心 $(-3, -3)$, 半径 $2\sqrt{6}$
- (28) $x^2 + y^2 + x - 4y - 1 = 0$
中心 $\left(-\frac{1}{2}, 2\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (29) $x^2 + y^2 + 7x - 5y + 5 = 0$
中心 $\left(-\frac{7}{2}, \frac{5}{2}\right)$, 半径 $\frac{3\sqrt{6}}{2}$
- (30) $x^2 + y^2 - 5x + 5y - 7 = 0$
中心 $\left(\frac{5}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{78}}{2}$

- (1) $x^2 + y^2 - 4x + 1 = 0$
中心 $(2, 0)$, 半径 $\sqrt{3}$
- (2) $x^2 + y^2 + 7x + y - 7 = 0$
中心 $\left(-\frac{7}{2}, -\frac{1}{2}\right)$, 半径 $\frac{\sqrt{78}}{2}$
- (3) $x^2 + y^2 - 4x + 6y - 4 = 0$
中心 $(2, -3)$, 半径 $\sqrt{17}$
- (4) $x^2 + y^2 - 4x - 5y + 5 = 0$
中心 $\left(2, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{21}}{2}$
- (5) $x^2 + y^2 - 6x - y = 0$
中心 $\left(3, \frac{1}{2}\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (6) $x^2 + y^2 - 7x + 5y + 5 = 0$
中心 $\left(\frac{7}{2}, -\frac{5}{2}\right)$, 半径 $\frac{3\sqrt{6}}{2}$
- (7) $x^2 + y^2 + 7x - 3y + 4 = 0$
中心 $\left(-\frac{7}{2}, \frac{3}{2}\right)$, 半径 $\frac{\sqrt{42}}{2}$
- (8) $x^2 + y^2 - 2x + 3y - 5 = 0$
中心 $\left(1, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (9) $x^2 + y^2 + 4x + 6y - 2 = 0$
中心 $(-2, -3)$, 半径 $\sqrt{15}$
- (10) $x^2 + y^2 - 7x - 5y + 2 = 0$
中心 $\left(\frac{7}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{66}}{2}$
- (11) $x^2 + y^2 + 7x - 2y + 5 = 0$
中心 $\left(-\frac{7}{2}, 1\right)$, 半径 $\frac{\sqrt{33}}{2}$
- (12) $x^2 + y^2 + x - 7y - 6 = 0$
中心 $\left(-\frac{1}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{74}}{2}$
- (13) $x^2 + y^2 - 2x - 5y + 4 = 0$
中心 $\left(1, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (14) $x^2 + y^2 - 5x + 3y - 6 = 0$
中心 $\left(\frac{5}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{58}}{2}$
- (15) $x^2 + y^2 + 5x - 5y + 3 = 0$
中心 $\left(-\frac{5}{2}, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{38}}{2}$
- (16) $x^2 + y^2 - 3x + 2y = 0$
中心 $\left(\frac{3}{2}, -1\right)$, 半径 $\frac{\sqrt{13}}{2}$
- (17) $x^2 + y^2 + 2x - 7y - 6 = 0$
中心 $\left(-1, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{77}}{2}$
- (18) $x^2 + y^2 - x + 7y + 1 = 0$
中心 $\left(\frac{1}{2}, -\frac{7}{2}\right)$, 半径 $\frac{\sqrt{46}}{2}$
- (19) $x^2 + y^2 - 5x + 5y - 3 = 0$
中心 $\left(\frac{5}{2}, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{62}}{2}$
- (20) $x^2 + y^2 + 3x + 5y - 4 = 0$
中心 $\left(-\frac{3}{2}, -\frac{5}{2}\right)$, 半径 $\frac{5\sqrt{2}}{2}$
- (21) $x^2 + y^2 - 7x + 3y - 6 = 0$
中心 $\left(\frac{7}{2}, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{82}}{2}$
- (22) $x^2 + y^2 + 5x - 7y - 2 = 0$
中心 $\left(-\frac{5}{2}, \frac{7}{2}\right)$, 半径 $\frac{\sqrt{82}}{2}$
- (23) $x^2 + y^2 + 2x - 2y - 7 = 0$
中心 $(-1, 1)$, 半径 3
- (24) $x^2 + y^2 + 2x + 5y + 6 = 0$
中心 $\left(-1, -\frac{5}{2}\right)$, 半径 $\frac{\sqrt{5}}{2}$
- (25) $x^2 + y^2 - 2x + 6y - 7 = 0$
中心 $(1, -3)$, 半径 $\sqrt{17}$
- (26) $x^2 + y^2 + 7x + 4y + 3 = 0$
中心 $\left(-\frac{7}{2}, -2\right)$, 半径 $\frac{\sqrt{53}}{2}$
- (27) $x^2 + y^2 + 2x + 3y - 4 = 0$
中心 $\left(-1, -\frac{3}{2}\right)$, 半径 $\frac{\sqrt{29}}{2}$
- (28) $x^2 + y^2 - 5y - 3 = 0$
中心 $\left(0, \frac{5}{2}\right)$, 半径 $\frac{\sqrt{37}}{2}$
- (29) $x^2 + y^2 + 2x - 2y - 5 = 0$
中心 $(-1, 1)$, 半径 $\sqrt{7}$
- (30) $x^2 + y^2 - 5x - 6y + 6 = 0$
中心 $\left(\frac{5}{2}, 3\right)$, 半径 $\frac{\sqrt{37}}{2}$

2.12 点と直線の距離

点と直線の距離を求める問題。

(1) 直線 : $3x + y - 1 = 0$, 点(3, 3)

(11) 直線 : $-x + 3y + 2 = 0$, 点(2, -1)

(2) 直線 : $-3x - y - 1 = 0$, 点(-2, 1)

(12) 直線 : $2x - y + 1 = 0$, 点(3, 2)

(3) 直線 : $2x + 2y - 1 = 0$, 点(2, -2)

(13) 直線 : $2x + 3y - 2 = 0$, 点(-3, -1)

(4) 直線 : $-2x + 2y + 1 = 0$, 点(2, -2)

(14) 直線 : $-3x - y - 2 = 0$, 点(-1, -2)

(5) 直線 : $-x - 3y - 2 = 0$, 点(-2, 2)

(15) 直線 : $-2x - 2y + 2 = 0$, 点(3, 2)

(6) 直線 : $-x - 3y + 3 = 0$, 点(1, 1)

(16) 直線 : $-3x + y - 2 = 0$, 点(2, -2)

(7) 直線 : $-3x - 3y - 1 = 0$, 点(1, -3)

(17) 直線 : $-x + 2y + 3 = 0$, 点(-1, 2)

(8) 直線 : $-x + y - 2 = 0$, 点(-1, 1)

(18) 直線 : $3x + 2y + 2 = 0$, 点(3, -1)

(9) 直線 : $3x - 2y - 1 = 0$, 点(-1, 2)

(19) 直線 : $-x + y - 2 = 0$, 点(1, 1)

(10) 直線 : $-3x - y - 3 = 0$, 点(3, 2)

(20) 直線 : $x + 3y + 2 = 0$, 点(3, -2)

(1) 直線 : $-x - 2y + 3 = 0$, 点 $(-3, -2)$

(11) 直線 : $x - 3y - 3 = 0$, 点 $(-3, -3)$

(2) 直線 : $-2x + 3y - 1 = 0$, 点 $(-3, -1)$

(12) 直線 : $3x - 2y + 3 = 0$, 点 $(-1, 3)$

(3) 直線 : $-x - 3y + 2 = 0$, 点 $(-2, -3)$

(13) 直線 : $2x + y - 2 = 0$, 点 $(-2, -3)$

(4) 直線 : $-3x + 2y + 2 = 0$, 点 $(-2, 2)$

(14) 直線 : $-x + 2y + 3 = 0$, 点 $(1, 1)$

(5) 直線 : $-2x + 2y - 2 = 0$, 点 $(-1, 1)$

(15) 直線 : $2x - y + 1 = 0$, 点 $(-2, 3)$

(6) 直線 : $3x - 2y - 3 = 0$, 点 $(1, 1)$

(16) 直線 : $2x + 2y + 1 = 0$, 点 $(-1, -1)$

(7) 直線 : $2x + y - 1 = 0$, 点 $(-1, -1)$

(17) 直線 : $-3x - y + 3 = 0$, 点 $(3, 1)$

(8) 直線 : $x - 2y + 2 = 0$, 点 $(2, -3)$

(18) 直線 : $-3x + 2y + 1 = 0$, 点 $(-1, -1)$

(9) 直線 : $x - y - 2 = 0$, 点 $(-2, -3)$

(19) 直線 : $x - 3y + 1 = 0$, 点 $(-2, 2)$

(10) 直線 : $-3x + 2y - 1 = 0$, 点 $(-3, -1)$

(20) 直線 : $-x - 2y + 1 = 0$, 点 $(-2, 3)$

(1) 直線 : $-x + 3y - 1 = 0$, 点 $(-3, 3)$

(11) 直線 : $2x + 3y - 2 = 0$, 点 $(-2, 3)$

(2) 直線 : $-3x + 3y - 1 = 0$, 点 $(-3, 2)$

(12) 直線 : $-x - y + 2 = 0$, 点 $(-2, -3)$

(3) 直線 : $-2x + 2y - 3 = 0$, 点 $(-2, 1)$

(13) 直線 : $-x + y + 2 = 0$, 点 $(3, -3)$

(4) 直線 : $3x + 3y - 2 = 0$, 点 $(-2, -3)$

(14) 直線 : $-2x - 3y - 1 = 0$, 点 $(-3, 3)$

(5) 直線 : $2x + y - 2 = 0$, 点 $(1, 2)$

(15) 直線 : $x - 3y - 2 = 0$, 点 $(1, 3)$

(6) 直線 : $3x + y + 2 = 0$, 点 $(-2, 2)$

(16) 直線 : $2x + 2y - 2 = 0$, 点 $(-1, 1)$

(7) 直線 : $-x - 3y + 2 = 0$, 点 $(2, -2)$

(17) 直線 : $3x + 3y - 2 = 0$, 点 $(-2, -3)$

(8) 直線 : $-2x - y - 3 = 0$, 点 $(-1, -2)$

(18) 直線 : $x + 3y + 1 = 0$, 点 $(3, -3)$

(9) 直線 : $2x - 3y + 1 = 0$, 点 $(-1, -2)$

(19) 直線 : $-3x + 3y - 2 = 0$, 点 $(2, -2)$

(10) 直線 : $-x + 3y + 1 = 0$, 点 $(1, -2)$

(20) 直線 : $2x - 2y - 3 = 0$, 点 $(-3, -3)$

(1) 直線 : $3x - 3y + 2 = 0$, 点 $(1, -2)$

(11) 直線 : $3x + 2y - 1 = 0$, 点 $(3, -3)$

(2) 直線 : $-3x - y + 2 = 0$, 点 $(-3, -3)$

(12) 直線 : $-3x - 3y - 2 = 0$, 点 $(1, -2)$

(3) 直線 : $3x + 2y + 1 = 0$, 点 $(-2, -2)$

(13) 直線 : $3x - y + 1 = 0$, 点 $(1, 3)$

(4) 直線 : $2x - y - 1 = 0$, 点 $(2, 3)$

(14) 直線 : $-3x - 3y + 3 = 0$, 点 $(-2, 2)$

(5) 直線 : $-2x + y - 1 = 0$, 点 $(1, 3)$

(15) 直線 : $3x - 2y + 3 = 0$, 点 $(-3, -1)$

(6) 直線 : $-3x - 3y + 1 = 0$, 点 $(-3, -2)$

(16) 直線 : $2x + y + 1 = 0$, 点 $(-2, 3)$

(7) 直線 : $3x + 3y - 2 = 0$, 点 $(3, -2)$

(17) 直線 : $x - y + 3 = 0$, 点 $(-3, -3)$

(8) 直線 : $x + 2y + 2 = 0$, 点 $(-2, 2)$

(18) 直線 : $3x - y + 3 = 0$, 点 $(1, -3)$

(9) 直線 : $-3x - 2y + 3 = 0$, 点 $(1, 3)$

(19) 直線 : $2x + 2y + 3 = 0$, 点 $(-3, -3)$

(10) 直線 : $3x + y - 1 = 0$, 点 $(-2, 3)$

(20) 直線 : $2x + y - 1 = 0$, 点 $(-2, 2)$

(1) 直線 : $-3x + 3y + 2 = 0$, 点(1, 1)

(11) 直線 : $x + 2y + 1 = 0$, 点(1, 2)

(2) 直線 : $2x + y + 1 = 0$, 点(-3, 2)

(12) 直線 : $2x - 2y - 2 = 0$, 点(1, -3)

(3) 直線 : $x - y + 3 = 0$, 点(2, 1)

(13) 直線 : $-3x - 2y + 3 = 0$, 点(2, 2)

(4) 直線 : $3x + 3y - 1 = 0$, 点(-3, 2)

(14) 直線 : $x - 2y + 2 = 0$, 点(-1, 3)

(5) 直線 : $-x - 2y - 1 = 0$, 点(2, -1)

(15) 直線 : $3x - 2y - 2 = 0$, 点(3, 3)

(6) 直線 : $-2x + 2y - 3 = 0$, 点(-3, -3)

(16) 直線 : $3x - 3y + 3 = 0$, 点(-2, -1)

(7) 直線 : $-x + 3y - 2 = 0$, 点(3, -2)

(17) 直線 : $2x - y - 2 = 0$, 点(3, 3)

(8) 直線 : $2x - 2y - 3 = 0$, 点(3, -1)

(18) 直線 : $2x - y - 2 = 0$, 点(1, 1)

(9) 直線 : $3x + 2y + 3 = 0$, 点(-3, -2)

(19) 直線 : $3x + y + 1 = 0$, 点(-2, 3)

(10) 直線 : $-3x - 2y - 3 = 0$, 点(2, 1)

(20) 直線 : $-2x + 3y + 2 = 0$, 点(1, 3)

(1) 直線 : $-3x + 3y - 3 = 0$, 点(2, 3)

(11) 直線 : $3x - 3y + 3 = 0$, 点(-2, 2)

(2) 直線 : $3x - 3y + 3 = 0$, 点(2, 2)

(12) 直線 : $3x - y + 2 = 0$, 点(2, 1)

(3) 直線 : $-3x - y + 3 = 0$, 点(2, -2)

(13) 直線 : $-3x + y - 2 = 0$, 点(-2, 1)

(4) 直線 : $2x + y + 1 = 0$, 点(2, 1)

(14) 直線 : $x - 2y - 1 = 0$, 点(-1, 1)

(5) 直線 : $2x - 3y + 2 = 0$, 点(-2, -1)

(15) 直線 : $-3x + 3y + 1 = 0$, 点(-2, 1)

(6) 直線 : $-2x - 2y + 2 = 0$, 点(-1, -2)

(16) 直線 : $-2x + 3y - 1 = 0$, 点(1, -3)

(7) 直線 : $2x + 3y + 2 = 0$, 点(1, -2)

(17) 直線 : $x + y + 3 = 0$, 点(2, -3)

(8) 直線 : $x - y - 3 = 0$, 点(-3, 1)

(18) 直線 : $-x - y - 1 = 0$, 点(1, -3)

(9) 直線 : $3x + 2y - 1 = 0$, 点(-3, -1)

(19) 直線 : $-x + 2y + 1 = 0$, 点(3, -3)

(10) 直線 : $x - 3y + 1 = 0$, 点(3, -3)

(20) 直線 : $2x + y - 1 = 0$, 点(-3, -1)

(1) 直線 : $2x - 2y - 3 = 0$, 点 $(-3, 3)$

(11) 直線 : $-x - 2y - 3 = 0$, 点 $(2, -3)$

(2) 直線 : $-3x - y - 1 = 0$, 点 $(-1, 3)$

(12) 直線 : $3x + 2y - 2 = 0$, 点 $(1, -3)$

(3) 直線 : $-x + 3y + 3 = 0$, 点 $(-3, -1)$

(13) 直線 : $-2x + 2y + 1 = 0$, 点 $(1, 1)$

(4) 直線 : $3x + 2y - 1 = 0$, 点 $(1, 3)$

(14) 直線 : $2x - y - 2 = 0$, 点 $(-1, -3)$

(5) 直線 : $-x - y + 2 = 0$, 点 $(3, -2)$

(15) 直線 : $-3x - 2y - 3 = 0$, 点 $(-1, -1)$

(6) 直線 : $2x + 3y + 3 = 0$, 点 $(-2, 3)$

(16) 直線 : $x - 3y - 2 = 0$, 点 $(2, -2)$

(7) 直線 : $2x - 3y + 3 = 0$, 点 $(-1, -2)$

(17) 直線 : $x + 3y - 2 = 0$, 点 $(-2, 3)$

(8) 直線 : $2x - 2y + 2 = 0$, 点 $(-1, 1)$

(18) 直線 : $-2x + 3y + 3 = 0$, 点 $(-2, -1)$

(9) 直線 : $3x + 2y + 2 = 0$, 点 $(-3, 1)$

(19) 直線 : $2x + y - 1 = 0$, 点 $(3, -3)$

(10) 直線 : $-3x - 3y - 2 = 0$, 点 $(3, 3)$

(20) 直線 : $-3x + y + 2 = 0$, 点 $(-2, -3)$

(1) 直線 : $3x + 2y + 3 = 0$, 点(1, 2)

(11) 直線 : $3x + y - 2 = 0$, 点(-2, -1)

(2) 直線 : $-3x - y - 1 = 0$, 点(2, 2)

(12) 直線 : $3x + 3y - 2 = 0$, 点(-2, -1)

(3) 直線 : $-2x - 3y + 3 = 0$, 点(-3, -3)

(13) 直線 : $-x + 2y - 1 = 0$, 点(-3, 1)

(4) 直線 : $x - y - 1 = 0$, 点(-3, 2)

(14) 直線 : $x - 3y + 1 = 0$, 点(-2, 2)

(5) 直線 : $-x + 3y + 2 = 0$, 点(-2, 2)

(15) 直線 : $-x - y + 2 = 0$, 点(-1, 2)

(6) 直線 : $-3x + 3y - 2 = 0$, 点(-1, -2)

(16) 直線 : $2x + 3y - 1 = 0$, 点(3, 1)

(7) 直線 : $x - 3y - 2 = 0$, 点(-1, 1)

(17) 直線 : $-x + 3y + 1 = 0$, 点(-3, 2)

(8) 直線 : $-x + 2y + 1 = 0$, 点(-3, 1)

(18) 直線 : $2x + 3y + 3 = 0$, 点(-3, -2)

(9) 直線 : $-3x + 2y + 2 = 0$, 点(1, -2)

(19) 直線 : $-2x + 2y - 3 = 0$, 点(1, 1)

(10) 直線 : $x - y - 1 = 0$, 点(-3, -1)

(20) 直線 : $3x - 2y - 1 = 0$, 点(-1, -2)

(1) 直線 : $-2x - y + 3 = 0$, 点 $(-2, 1)$

(11) 直線 : $2x - y + 1 = 0$, 点 $(2, 2)$

(2) 直線 : $-x + 3y - 3 = 0$, 点 $(3, -3)$

(12) 直線 : $2x - 3y + 3 = 0$, 点 $(3, 3)$

(3) 直線 : $-3x + 2y - 2 = 0$, 点 $(-2, 3)$

(13) 直線 : $-3x - 2y + 3 = 0$, 点 $(2, 3)$

(4) 直線 : $3x - y + 2 = 0$, 点 $(2, -1)$

(14) 直線 : $-3x - 2y + 2 = 0$, 点 $(3, 2)$

(5) 直線 : $-x - y - 1 = 0$, 点 $(1, -2)$

(15) 直線 : $2x - y + 1 = 0$, 点 $(2, -2)$

(6) 直線 : $-2x + 3y + 2 = 0$, 点 $(-2, 1)$

(16) 直線 : $3x - y + 2 = 0$, 点 $(2, -3)$

(7) 直線 : $-x + 2y + 2 = 0$, 点 $(-3, 2)$

(17) 直線 : $2x - 2y - 1 = 0$, 点 $(2, 2)$

(8) 直線 : $3x - y + 1 = 0$, 点 $(-3, 1)$

(18) 直線 : $x - y + 1 = 0$, 点 $(3, 2)$

(9) 直線 : $-x - 3y - 2 = 0$, 点 $(-3, -1)$

(19) 直線 : $-x + 3y - 2 = 0$, 点 $(2, -2)$

(10) 直線 : $-x - 2y + 1 = 0$, 点 $(3, -1)$

(20) 直線 : $x - 3y - 2 = 0$, 点 $(-2, 1)$

(1) 直線 : $-x + y + 1 = 0$, 点(3, -3)

(11) 直線 : $2x + y - 2 = 0$, 点(-2, -2)

(2) 直線 : $-3x + y + 1 = 0$, 点(-3, -3)

(12) 直線 : $2x + y + 3 = 0$, 点(1, 2)

(3) 直線 : $-x - y - 3 = 0$, 点(-3, 1)

(13) 直線 : $-3x - 3y + 3 = 0$, 点(-3, 3)

(4) 直線 : $2x - y + 1 = 0$, 点(1, 3)

(14) 直線 : $x + y - 3 = 0$, 点(1, -1)

(5) 直線 : $3x - 2y - 1 = 0$, 点(2, 2)

(15) 直線 : $-x + 2y + 1 = 0$, 点(2, 2)

(6) 直線 : $2x - 3y + 3 = 0$, 点(-3, -3)

(16) 直線 : $-3x - 2y + 2 = 0$, 点(-2, -2)

(7) 直線 : $x + 3y - 2 = 0$, 点(-2, -1)

(17) 直線 : $-3x + y + 2 = 0$, 点(3, 1)

(8) 直線 : $2x - y + 1 = 0$, 点(3, 3)

(18) 直線 : $-3x - y - 3 = 0$, 点(2, 1)

(9) 直線 : $-3x - 2y - 3 = 0$, 点(1, 3)

(19) 直線 : $3x + 2y - 1 = 0$, 点(-3, -2)

(10) 直線 : $x + y - 2 = 0$, 点(3, -2)

(20) 直線 : $-3x + y - 1 = 0$, 点(3, 1)

(1) 直線 : $-x - 3y - 2 = 0$, 点(1, 1)

(11) 直線 : $x - 2y + 3 = 0$, 点(-3, 3)

(2) 直線 : $-3x - 2y + 1 = 0$, 点(2, 1)

(12) 直線 : $-2x + 2y - 3 = 0$, 点(-3, 1)

(3) 直線 : $2x + 3y + 3 = 0$, 点(1, -1)

(13) 直線 : $-3x + y + 3 = 0$, 点(-3, 3)

(4) 直線 : $2x - 3y + 3 = 0$, 点(-2, 3)

(14) 直線 : $x + 2y - 3 = 0$, 点(-1, 2)

(5) 直線 : $2x - 2y - 3 = 0$, 点(3, -2)

(15) 直線 : $-2x + y - 3 = 0$, 点(1, -2)

(6) 直線 : $-2x + 3y - 2 = 0$, 点(-1, 3)

(16) 直線 : $2x + 2y + 2 = 0$, 点(-3, 3)

(7) 直線 : $2x - 3y + 3 = 0$, 点(3, -3)

(17) 直線 : $3x - 3y + 2 = 0$, 点(-3, 3)

(8) 直線 : $2x + 2y - 3 = 0$, 点(3, -3)

(18) 直線 : $2x - 3y - 3 = 0$, 点(3, -3)

(9) 直線 : $-3x - y + 3 = 0$, 点(-1, -3)

(19) 直線 : $-3x + y - 1 = 0$, 点(1, -3)

(10) 直線 : $2x + 3y + 1 = 0$, 点(3, -1)

(20) 直線 : $-3x - 2y - 3 = 0$, 点(-2, -2)

(1) 直線 : $-3x - 2y + 1 = 0$, 点 $(3, -2)$

(11) 直線 : $2x + 2y + 3 = 0$, 点 $(3, -3)$

(2) 直線 : $-x - 2y + 2 = 0$, 点 $(1, -2)$

(12) 直線 : $-3x + y - 3 = 0$, 点 $(2, -2)$

(3) 直線 : $-x - y + 3 = 0$, 点 $(1, 3)$

(13) 直線 : $3x + 3y + 1 = 0$, 点 $(-2, 2)$

(4) 直線 : $x + 2y - 3 = 0$, 点 $(-2, 2)$

(14) 直線 : $x + 3y - 2 = 0$, 点 $(-1, -2)$

(5) 直線 : $2x - 2y - 3 = 0$, 点 $(1, 1)$

(15) 直線 : $-3x + 2y + 2 = 0$, 点 $(1, 1)$

(6) 直線 : $-x - y - 3 = 0$, 点 $(-2, -3)$

(16) 直線 : $-x - y - 1 = 0$, 点 $(-1, -1)$

(7) 直線 : $-x - 3y + 1 = 0$, 点 $(2, -1)$

(17) 直線 : $3x + y + 2 = 0$, 点 $(-3, -3)$

(8) 直線 : $-x - 2y - 2 = 0$, 点 $(-3, 2)$

(18) 直線 : $3x + y - 2 = 0$, 点 $(2, -2)$

(9) 直線 : $-3x - 3y + 2 = 0$, 点 $(-1, 2)$

(19) 直線 : $-2x - y - 3 = 0$, 点 $(2, 1)$

(10) 直線 : $3x - 2y - 1 = 0$, 点 $(-2, -1)$

(20) 直線 : $-x + 3y - 2 = 0$, 点 $(-2, -1)$

(1) 直線 : $-3x + 3y - 3 = 0$, 点(3,1)

(11) 直線 : $-x - 2y - 3 = 0$, 点(-2,1)

(2) 直線 : $3x + 2y + 2 = 0$, 点(-3,-3)

(12) 直線 : $3x + 3y + 1 = 0$, 点(-3,-2)

(3) 直線 : $x - 3y - 1 = 0$, 点(1,-2)

(13) 直線 : $2x - 2y - 2 = 0$, 点(3,-3)

(4) 直線 : $-3x - y + 2 = 0$, 点(-2,-1)

(14) 直線 : $3x - y + 2 = 0$, 点(1,2)

(5) 直線 : $-x + 2y + 3 = 0$, 点(-2,-3)

(15) 直線 : $-x + y + 2 = 0$, 点(-2,-2)

(6) 直線 : $2x - 2y - 1 = 0$, 点(-1,-3)

(16) 直線 : $x - 3y - 1 = 0$, 点(-3,1)

(7) 直線 : $2x - 2y - 1 = 0$, 点(-2,3)

(17) 直線 : $-2x + 2y - 1 = 0$, 点(-1,-1)

(8) 直線 : $-x - 3y + 2 = 0$, 点(1,-1)

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(20) 直線 : $3x - 2y - 2 = 0$, 点(1,-2)

(1) 直線 : $3x + 3y - 1 = 0$, 点(1, 2)

(11) 直線 : $-3x + 3y - 1 = 0$, 点(3, -3)

(2) 直線 : $-3x + 3y + 2 = 0$, 点(3, 1)

(12) 直線 : $-3x - 2y - 3 = 0$, 点(-1, -1)

(3) 直線 : $-x - y - 1 = 0$, 点(3, -1)

(13) 直線 : $2x - y + 1 = 0$, 点(1, -2)

(4) 直線 : $2x + y - 3 = 0$, 点(-3, 1)

(14) 直線 : $-x + y + 1 = 0$, 点(-1, -2)

(5) 直線 : $3x - 3y - 3 = 0$, 点(1, 2)

(15) 直線 : $3x - y + 3 = 0$, 点(-3, 3)

(6) 直線 : $-x - 2y - 2 = 0$, 点(2, -1)

(16) 直線 : $-x - y - 1 = 0$, 点(-1, -3)

(7) 直線 : $-x + y + 1 = 0$, 点(1, 2)

(17) 直線 : $x - y - 3 = 0$, 点(2, 3)

(8) 直線 : $3x + y + 3 = 0$, 点(3, -3)

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(19) 直線 : $2x + 2y + 2 = 0$, 点(2, -2)

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(20) 直線 : $-3x - y + 3 = 0$, 点(-2, -1)

(1) 直線 : $-2x + 2y - 2 = 0$, 点(1, 2)

(11) 直線 : $-2x - 3y + 1 = 0$, 点(2, 1)

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(12) 直線 : $-3x - 3y - 2 = 0$, 点(3, -1)

(3) 直線 : $-x - y - 3 = 0$, 点(3, -1)

(13) 直線 : $3x - 2y - 1 = 0$, 点(-3, -1)

(4) 直線 : $-x - 3y + 3 = 0$, 点(1, -3)

(14) 直線 : $x - y - 1 = 0$, 点(3, 2)

(5) 直線 : $-3x - 2y - 2 = 0$, 点(-2, 2)

(15) 直線 : $-x + 3y + 1 = 0$, 点(-3, 2)

(6) 直線 : $-x - y - 3 = 0$, 点(1, -1)

(16) 直線 : $x + y + 1 = 0$, 点(3, 1)

(7) 直線 : $-2x - 2y + 1 = 0$, 点(-1, -1)

(17) 直線 : $-x - 2y + 1 = 0$, 点(-2, -3)

(8) 直線 : $2x + 3y - 1 = 0$, 点(2, 3)

(18) 直線 : $-2x - y + 2 = 0$, 点(-3, 3)

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(20) 直線 : $-3x - y - 3 = 0$, 点(1, 3)

(1) 直線 : $x + y - 1 = 0$, 点(3, 1)

(11) 直線 : $-2x + y - 3 = 0$, 点(-2, 1)

(2) 直線 : $2x - 3y - 3 = 0$, 点(2, -3)

(12) 直線 : $-x + 2y - 2 = 0$, 点(2, -2)

(3) 直線 : $-2x + 2y + 1 = 0$, 点(1, -3)

(13) 直線 : $-2x - y + 3 = 0$, 点(2, 3)

(4) 直線 : $2x - 3y - 3 = 0$, 点(-3, 2)

(14) 直線 : $-x - 3y - 1 = 0$, 点(-1, 2)

(5) 直線 : $x + 2y - 3 = 0$, 点(3, -1)

(15) 直線 : $3x + 3y + 1 = 0$, 点(1, 3)

(6) 直線 : $-x - 2y - 3 = 0$, 点(-2, -1)

(16) 直線 : $-x + 3y + 2 = 0$, 点(-3, 1)

(7) 直線 : $x + 2y + 3 = 0$, 点(2, 3)

(17) 直線 : $-2x + y + 3 = 0$, 点(1, -1)

(8) 直線 : $-2x - 3y - 1 = 0$, 点(-3, -3)

(18) 直線 : $x - 3y - 2 = 0$, 点(-2, 2)

(9) 直線 : $x - y - 3 = 0$, 点(3, 2)

(19) 直線 : $x + y + 1 = 0$, 点(-2, -2)

(10) 直線 : $2x - 2y + 1 = 0$, 点(-1, 1)

(20) 直線 : $-x - y + 2 = 0$, 点(2, 3)

(1) 直線 : $-x - y + 2 = 0$, 点 $(-2, -3)$

(11) 直線 : $-x - 3y + 1 = 0$, 点 $(2, 1)$

(2) 直線 : $3x - 2y + 1 = 0$, 点 $(-3, -1)$

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(14) 直線 : $x + 3y + 1 = 0$, 点 $(-1, -1)$

(5) 直線 : $-2x + y - 3 = 0$, 点 $(2, 2)$

(15) 直線 : $x + y - 1 = 0$, 点 $(-3, -2)$

(6) 直線 : $3x + 2y - 2 = 0$, 点 $(1, -1)$

(16) 直線 : $-3x - y - 3 = 0$, 点 $(2, -3)$

(7) 直線 : $3x - y + 1 = 0$, 点 $(1, 1)$

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(9) 直線 : $x + 3y + 1 = 0$, 点 $(-2, 2)$

(19) 直線 : $x + 3y + 2 = 0$, 点 $(1, 3)$

(10) 直線 : $x + 3y - 3 = 0$, 点 $(1, 2)$

(20) 直線 : $-2x + y - 3 = 0$, 点 $(-3, -1)$

(1) 直線 : $x + 3y + 1 = 0$, 点 $(-3, -1)$

(11) 直線 : $x + 2y + 1 = 0$, 点 $(2, -3)$

(2) 直線 : $-x - 3y - 1 = 0$, 点 $(1, 2)$

(12) 直線 : $-2x - 3y - 3 = 0$, 点 $(3, 2)$

(3) 直線 : $-3x - 2y - 3 = 0$, 点 $(2, 3)$

(13) 直線 : $-3x - 2y - 2 = 0$, 点 $(-2, -2)$

(4) 直線 : $3x - y + 1 = 0$, 点 $(-1, 2)$

(14) 直線 : $-x - y + 3 = 0$, 点 $(2, 2)$

(5) 直線 : $-3x + y - 3 = 0$, 点 $(2, -3)$

(15) 直線 : $2x + 3y + 2 = 0$, 点 $(2, -2)$

(6) 直線 : $3x - 3y + 1 = 0$, 点 $(3, -2)$

(16) 直線 : $-2x - y - 3 = 0$, 点 $(-1, 3)$

(7) 直線 : $-x + 3y - 1 = 0$, 点 $(2, -1)$

(17) 直線 : $-2x + y + 3 = 0$, 点 $(-3, -3)$

(8) 直線 : $-3x + y - 2 = 0$, 点 $(-1, -2)$

(18) 直線 : $-x + 3y - 3 = 0$, 点 $(2, 2)$

(9) 直線 : $2x - 3y - 3 = 0$, 点 $(-1, -1)$

(19) 直線 : $-x - 3y - 1 = 0$, 点 $(1, 1)$

(10) 直線 : $x - y - 2 = 0$, 点 $(2, -3)$

(20) 直線 : $3x - 2y + 1 = 0$, 点 $(3, -1)$

(1) 直線 : $x + 3y + 1 = 0$, 点 $(-3, -3)$

(11) 直線 : $-3x - 3y - 3 = 0$, 点 $(-2, 1)$

(2) 直線 : $x + 2y + 1 = 0$, 点 $(1, 1)$

(12) 直線 : $3x - y - 1 = 0$, 点 $(-3, -2)$

(3) 直線 : $2x - 3y + 2 = 0$, 点 $(3, 1)$

(13) 直線 : $3x + 3y + 2 = 0$, 点 $(-1, -3)$

(4) 直線 : $3x + y - 2 = 0$, 点 $(-3, -1)$

(14) 直線 : $3x + 3y - 1 = 0$, 点 $(1, -3)$

(5) 直線 : $-3x + y + 1 = 0$, 点 $(-2, 1)$

(15) 直線 : $-3x - 2y - 3 = 0$, 点 $(1, -2)$

(6) 直線 : $-x - 2y - 2 = 0$, 点 $(-1, -2)$

(16) 直線 : $-2x + 3y + 3 = 0$, 点 $(-3, -2)$

(7) 直線 : $-x - 2y + 2 = 0$, 点 $(2, -1)$

(17) 直線 : $-x + y + 3 = 0$, 点 $(-2, 2)$

(8) 直線 : $x - 3y - 3 = 0$, 点 $(1, -3)$

(18) 直線 : $2x - 2y - 3 = 0$, 点 $(1, 3)$

(9) 直線 : $-2x - y - 3 = 0$, 点 $(-2, 1)$

(19) 直線 : $-3x + y - 3 = 0$, 点 $(-1, 1)$

(10) 直線 : $x - 2y - 2 = 0$, 点 $(-2, -3)$

(20) 直線 : $3x + y - 2 = 0$, 点 $(-3, 2)$

(1) 直線 : $-x - y + 3 = 0$, 点(2, -3)

(11) 直線 : $-3x + 3y + 1 = 0$, 点(3, 2)

(2) 直線 : $-3x + y - 3 = 0$, 点(-1, -2)

(12) 直線 : $3x - 3y - 3 = 0$, 点(3, 3)

(3) 直線 : $-3x - y + 2 = 0$, 点(1, 1)

(13) 直線 : $x - y - 1 = 0$, 点(2, -2)

(4) 直線 : $3x + 2y - 1 = 0$, 点(2, 2)

(14) 直線 : $3x - y + 3 = 0$, 点(-3, 3)

(5) 直線 : $2x - 2y + 3 = 0$, 点(3, -3)

(15) 直線 : $3x - 2y - 1 = 0$, 点(-1, 3)

(6) 直線 : $-2x + 2y - 2 = 0$, 点(-2, -1)

(16) 直線 : $-x + 2y - 3 = 0$, 点(2, -1)

(7) 直線 : $x + y - 3 = 0$, 点(-2, -2)

(17) 直線 : $3x - y - 2 = 0$, 点(-3, -2)

(8) 直線 : $2x - 2y - 1 = 0$, 点(-3, -1)

(18) 直線 : $-3x - 3y - 3 = 0$, 点(1, 3)

(9) 直線 : $2x + y + 2 = 0$, 点(1, 1)

(19) 直線 : $3x - 3y + 2 = 0$, 点(-1, 2)

(10) 直線 : $-2x - 2y + 3 = 0$, 点(-2, 1)

(20) 直線 : $-3x - y + 3 = 0$, 点(-1, 2)

(1) 直線 : $3x + y - 1 = 0$, 点(3, 3)
距離 : $\frac{11}{10}\sqrt{10}$

(11) 直線 : $-x + 3y + 2 = 0$, 点(2, -1)
距離 : $\frac{3}{10}\sqrt{10}$

(2) 直線 : $-3x - y - 1 = 0$, 点(-2, 1)
距離 : $\frac{2}{5}\sqrt{10}$

(12) 直線 : $2x - y + 1 = 0$, 点(3, 2)
距離 : $\sqrt{5}$

(3) 直線 : $2x + 2y - 1 = 0$, 点(2, -2)
距離 : $\frac{1}{4}\sqrt{2}$

(13) 直線 : $2x + 3y - 2 = 0$, 点(-3, -1)
距離 : $\frac{11}{13}\sqrt{13}$

(4) 直線 : $-2x + 2y + 1 = 0$, 点(2, -2)
距離 : $\frac{7}{4}\sqrt{2}$

(14) 直線 : $-3x - y - 2 = 0$, 点(-1, -2)
距離 : $\frac{3}{10}\sqrt{10}$

(5) 直線 : $-x - 3y - 2 = 0$, 点(-2, 2)
距離 : $\frac{3}{5}\sqrt{10}$

(15) 直線 : $-2x - 2y + 2 = 0$, 点(3, 2)
距離 : $2\sqrt{2}$

(6) 直線 : $-x - 3y + 3 = 0$, 点(1, 1)
距離 : $\frac{1}{10}\sqrt{10}$

(16) 直線 : $-3x + y - 2 = 0$, 点(2, -2)
距離 : $\sqrt{10}$

(7) 直線 : $-3x - 3y - 1 = 0$, 点(1, -3)
距離 : $\frac{5}{6}\sqrt{2}$

(17) 直線 : $-x + 2y + 3 = 0$, 点(-1, 2)
距離 : $\frac{8}{5}\sqrt{5}$

(8) 直線 : $-x + y - 2 = 0$, 点(-1, 1)
距離 : 0

(18) 直線 : $3x + 2y + 2 = 0$, 点(3, -1)
距離 : $\frac{9}{13}\sqrt{13}$

(9) 直線 : $3x - 2y - 1 = 0$, 点(-1, 2)
距離 : $\frac{8}{13}\sqrt{13}$

(19) 直線 : $-x + y - 2 = 0$, 点(1, 1)
距離 : $\sqrt{2}$

(10) 直線 : $-3x - y - 3 = 0$, 点(3, 2)
距離 : $\frac{7}{5}\sqrt{10}$

(20) 直線 : $x + 3y + 2 = 0$, 点(3, -2)
距離 : $\frac{1}{10}\sqrt{10}$

(1) 直線 : $-x - 2y + 3 = 0$, 点 $(-3, -2)$
距離 : $2\sqrt{5}$

(11) 直線 : $x - 3y - 3 = 0$, 点 $(-3, -3)$
距離 : $\frac{3}{10}\sqrt{10}$

(2) 直線 : $-2x + 3y - 1 = 0$, 点 $(-3, -1)$
距離 : $\frac{2}{13}\sqrt{13}$

(12) 直線 : $3x - 2y + 3 = 0$, 点 $(-1, 3)$
距離 : $\frac{6}{13}\sqrt{13}$

(3) 直線 : $-x - 3y + 2 = 0$, 点 $(-2, -3)$
距離 : $\frac{13}{10}\sqrt{10}$

(13) 直線 : $2x + y - 2 = 0$, 点 $(-2, -3)$
距離 : $\frac{9}{5}\sqrt{5}$

(4) 直線 : $-3x + 2y + 2 = 0$, 点 $(-2, 2)$
距離 : $\frac{12}{13}\sqrt{13}$

(14) 直線 : $-x + 2y + 3 = 0$, 点 $(1, 1)$
距離 : $\frac{4}{5}\sqrt{5}$

(5) 直線 : $-2x + 2y - 2 = 0$, 点 $(-1, 1)$
距離 : $\frac{1}{2}\sqrt{2}$

(15) 直線 : $2x - y + 1 = 0$, 点 $(-2, 3)$
距離 : $\frac{6}{5}\sqrt{5}$

(6) 直線 : $3x - 2y - 3 = 0$, 点 $(1, 1)$
距離 : $\frac{2}{13}\sqrt{13}$

(16) 直線 : $2x + 2y + 1 = 0$, 点 $(-1, -1)$
距離 : $\frac{3}{4}\sqrt{2}$

(7) 直線 : $2x + y - 1 = 0$, 点 $(-1, -1)$
距離 : $\frac{4}{5}\sqrt{5}$

(17) 直線 : $-3x - y + 3 = 0$, 点 $(3, 1)$
距離 : $\frac{7}{10}\sqrt{10}$

(8) 直線 : $x - 2y + 2 = 0$, 点 $(2, -3)$
距離 : $2\sqrt{5}$

(18) 直線 : $-3x + 2y + 1 = 0$, 点 $(-1, -1)$
距離 : $\frac{2}{13}\sqrt{13}$

(9) 直線 : $x - y - 2 = 0$, 点 $(-2, -3)$
距離 : $\frac{1}{2}\sqrt{2}$

(19) 直線 : $x - 3y + 1 = 0$, 点 $(-2, 2)$
距離 : $\frac{7}{10}\sqrt{10}$

(10) 直線 : $-3x + 2y - 1 = 0$, 点 $(-3, -1)$
距離 : $\frac{6}{13}\sqrt{13}$

(20) 直線 : $-x - 2y + 1 = 0$, 点 $(-2, 3)$
距離 : $\frac{3}{5}\sqrt{5}$

(1) 直線 : $-x + 3y - 1 = 0$, 点 $(-3, 3)$
距離 : $\frac{11}{10}\sqrt{10}$

(11) 直線 : $2x + 3y - 2 = 0$, 点 $(-2, 3)$
距離 : $\frac{3}{13}\sqrt{13}$

(2) 直線 : $-3x + 3y - 1 = 0$, 点 $(-3, 2)$
距離 : $\frac{7}{3}\sqrt{2}$

(12) 直線 : $-x - y + 2 = 0$, 点 $(-2, -3)$
距離 : $\frac{7}{2}\sqrt{2}$

(3) 直線 : $-2x + 2y - 3 = 0$, 点 $(-2, 1)$
距離 : $\frac{3}{4}\sqrt{2}$

(13) 直線 : $-x + y + 2 = 0$, 点 $(3, -3)$
距離 : $2\sqrt{2}$

(4) 直線 : $3x + 3y - 2 = 0$, 点 $(-2, -3)$
距離 : $\frac{17}{6}\sqrt{2}$

(14) 直線 : $-2x - 3y - 1 = 0$, 点 $(-3, 3)$
距離 : $\frac{4}{13}\sqrt{13}$

(5) 直線 : $2x + y - 2 = 0$, 点 $(1, 2)$
距離 : $\frac{2}{5}\sqrt{5}$

(15) 直線 : $x - 3y - 2 = 0$, 点 $(1, 3)$
距離 : $\sqrt{10}$

(6) 直線 : $3x + y + 2 = 0$, 点 $(-2, 2)$
距離 : $\frac{1}{5}\sqrt{10}$

(16) 直線 : $2x + 2y - 2 = 0$, 点 $(-1, 1)$
距離 : $\frac{1}{2}\sqrt{2}$

(7) 直線 : $-x - 3y + 2 = 0$, 点 $(2, -2)$
距離 : $\frac{3}{5}\sqrt{10}$

(17) 直線 : $3x + 3y - 2 = 0$, 点 $(-2, -3)$
距離 : $\frac{17}{6}\sqrt{2}$

(8) 直線 : $-2x - y - 3 = 0$, 点 $(-1, -2)$
距離 : $\frac{1}{5}\sqrt{5}$

(18) 直線 : $x + 3y + 1 = 0$, 点 $(3, -3)$
距離 : $\frac{1}{2}\sqrt{10}$

(9) 直線 : $2x - 3y + 1 = 0$, 点 $(-1, -2)$
距離 : $\frac{5}{13}\sqrt{13}$

(19) 直線 : $-3x + 3y - 2 = 0$, 点 $(2, -2)$
距離 : $\frac{7}{3}\sqrt{2}$

(10) 直線 : $-x + 3y + 1 = 0$, 点 $(1, -2)$
距離 : $\frac{3}{5}\sqrt{10}$

(20) 直線 : $2x - 2y - 3 = 0$, 点 $(-3, -3)$
距離 : $\frac{3}{4}\sqrt{2}$

(1) 直線 : $3x - 3y + 2 = 0$, 点 $(1, -2)$
距離 : $\frac{11}{6}\sqrt{2}$

(11) 直線 : $3x + 2y - 1 = 0$, 点 $(3, -3)$
距離 : $\frac{2}{13}\sqrt{13}$

(2) 直線 : $-3x - y + 2 = 0$, 点 $(-3, -3)$
距離 : $\frac{7}{5}\sqrt{10}$

(12) 直線 : $-3x - 3y - 2 = 0$, 点 $(1, -2)$
距離 : $\frac{1}{6}\sqrt{2}$

(3) 直線 : $3x + 2y + 1 = 0$, 点 $(-2, -2)$
距離 : $\frac{9}{13}\sqrt{13}$

(13) 直線 : $3x - y + 1 = 0$, 点 $(1, 3)$
距離 : $\frac{1}{10}\sqrt{10}$

(4) 直線 : $2x - y - 1 = 0$, 点 $(2, 3)$
距離 : 0

(14) 直線 : $-3x - 3y + 3 = 0$, 点 $(-2, 2)$
距離 : $\frac{1}{2}\sqrt{2}$

(5) 直線 : $-2x + y - 1 = 0$, 点 $(1, 3)$
距離 : 0

(15) 直線 : $3x - 2y + 3 = 0$, 点 $(-3, -1)$
距離 : $\frac{4}{13}\sqrt{13}$

(6) 直線 : $-3x - 3y + 1 = 0$, 点 $(-3, -2)$
距離 : $\frac{8}{3}\sqrt{2}$

(16) 直線 : $2x + y + 1 = 0$, 点 $(-2, 3)$
距離 : 0

(7) 直線 : $3x + 3y - 2 = 0$, 点 $(3, -2)$
距離 : $\frac{1}{6}\sqrt{2}$

(17) 直線 : $x - y + 3 = 0$, 点 $(-3, -3)$
距離 : $\frac{3}{2}\sqrt{2}$

(8) 直線 : $x + 2y + 2 = 0$, 点 $(-2, 2)$
距離 : $\frac{4}{5}\sqrt{5}$

(18) 直線 : $3x - y + 3 = 0$, 点 $(1, -3)$
距離 : $\frac{9}{10}\sqrt{10}$

(9) 直線 : $-3x - 2y + 3 = 0$, 点 $(1, 3)$
距離 : $\frac{6}{13}\sqrt{13}$

(19) 直線 : $2x + 2y + 3 = 0$, 点 $(-3, -3)$
距離 : $\frac{9}{4}\sqrt{2}$

(10) 直線 : $3x + y - 1 = 0$, 点 $(-2, 3)$
距離 : $\frac{2}{5}\sqrt{10}$

(20) 直線 : $2x + y - 1 = 0$, 点 $(-2, 2)$
距離 : $\frac{3}{5}\sqrt{5}$

(1) 直線 : $-3x + 3y + 2 = 0$, 点(1, 1)
距離 : $\frac{1}{3}\sqrt{2}$

(11) 直線 : $x + 2y + 1 = 0$, 点(1, 2)
距離 : $\frac{6}{5}\sqrt{5}$

(2) 直線 : $2x + y + 1 = 0$, 点(-3, 2)
距離 : $\frac{3}{5}\sqrt{5}$

(12) 直線 : $2x - 2y - 2 = 0$, 点(1, -3)
距離 : $\frac{3}{2}\sqrt{2}$

(3) 直線 : $x - y + 3 = 0$, 点(2, 1)
距離 : $2\sqrt{2}$

(13) 直線 : $-3x - 2y + 3 = 0$, 点(2, 2)
距離 : $\frac{7}{13}\sqrt{13}$

(4) 直線 : $3x + 3y - 1 = 0$, 点(-3, 2)
距離 : $\frac{2}{3}\sqrt{2}$

(14) 直線 : $x - 2y + 2 = 0$, 点(-1, 3)
距離 : $\sqrt{5}$

(5) 直線 : $-x - 2y - 1 = 0$, 点(2, -1)
距離 : $\frac{1}{5}\sqrt{5}$

(15) 直線 : $3x - 2y - 2 = 0$, 点(3, 3)
距離 : $\frac{1}{13}\sqrt{13}$

(6) 直線 : $-2x + 2y - 3 = 0$, 点(-3, -3)
距離 : $\frac{3}{4}\sqrt{2}$

(16) 直線 : $3x - 3y + 3 = 0$, 点(-2, -1)
距離 : 0

(7) 直線 : $-x + 3y - 2 = 0$, 点(3, -2)
距離 : $\frac{11}{10}\sqrt{10}$

(17) 直線 : $2x - y - 2 = 0$, 点(3, 3)
距離 : $\frac{1}{5}\sqrt{5}$

(8) 直線 : $2x - 2y - 3 = 0$, 点(3, -1)
距離 : $\frac{5}{4}\sqrt{2}$

(18) 直線 : $2x - y - 2 = 0$, 点(1, 1)
距離 : $\frac{1}{5}\sqrt{5}$

(9) 直線 : $3x + 2y + 3 = 0$, 点(-3, -2)
距離 : $\frac{10}{13}\sqrt{13}$

(19) 直線 : $3x + y + 1 = 0$, 点(-2, 3)
距離 : $\frac{1}{5}\sqrt{10}$

(10) 直線 : $-3x - 2y - 3 = 0$, 点(2, 1)
距離 : $\frac{11}{13}\sqrt{13}$

(20) 直線 : $-2x + 3y + 2 = 0$, 点(1, 3)
距離 : $\frac{9}{13}\sqrt{13}$

(1) 直線: $-3x + 3y - 3 = 0$, 点(2, 3)
距離: 0

(11) 直線: $3x - 3y + 3 = 0$, 点(-2, 2)
距離: $\frac{3}{2}\sqrt{2}$

(2) 直線: $3x - 3y + 3 = 0$, 点(2, 2)
距離: $\frac{1}{2}\sqrt{2}$

(12) 直線: $3x - y + 2 = 0$, 点(2, 1)
距離: $\frac{7}{10}\sqrt{10}$

(3) 直線: $-3x - y + 3 = 0$, 点(2, -2)
距離: $\frac{1}{10}\sqrt{10}$

(13) 直線: $-3x + y - 2 = 0$, 点(-2, 1)
距離: $\frac{1}{2}\sqrt{10}$

(4) 直線: $2x + y + 1 = 0$, 点(2, 1)
距離: $\frac{6}{5}\sqrt{5}$

(14) 直線: $x - 2y - 1 = 0$, 点(-1, 1)
距離: $\frac{4}{5}\sqrt{5}$

(5) 直線: $2x - 3y + 2 = 0$, 点(-2, -1)
距離: $\frac{1}{13}\sqrt{13}$

(15) 直線: $-3x + 3y + 1 = 0$, 点(-2, 1)
距離: $\frac{5}{3}\sqrt{2}$

(6) 直線: $-2x - 2y + 2 = 0$, 点(-1, -2)
距離: $2\sqrt{2}$

(16) 直線: $-2x + 3y - 1 = 0$, 点(1, -3)
距離: $\frac{12}{13}\sqrt{13}$

(7) 直線: $2x + 3y + 2 = 0$, 点(1, -2)
距離: $\frac{2}{13}\sqrt{13}$

(17) 直線: $x + y + 3 = 0$, 点(2, -3)
距離: $\sqrt{2}$

(8) 直線: $x - y - 3 = 0$, 点(-3, 1)
距離: $\frac{7}{2}\sqrt{2}$

(18) 直線: $-x - y - 1 = 0$, 点(1, -3)
距離: $\frac{1}{2}\sqrt{2}$

(9) 直線: $3x + 2y - 1 = 0$, 点(-3, -1)
距離: $\frac{12}{13}\sqrt{13}$

(19) 直線: $-x + 2y + 1 = 0$, 点(3, -3)
距離: $\frac{8}{5}\sqrt{5}$

(10) 直線: $x - 3y + 1 = 0$, 点(3, -3)
距離: $\frac{13}{10}\sqrt{10}$

(20) 直線: $2x + y - 1 = 0$, 点(-3, -1)
距離: $\frac{8}{5}\sqrt{5}$

(1) 直線 : $2x - 2y - 3 = 0$, 点 $(-3, 3)$
距離 : $\frac{15}{4}\sqrt{2}$

(11) 直線 : $-x - 2y - 3 = 0$, 点 $(2, -3)$
距離 : $\frac{1}{5}\sqrt{5}$

(2) 直線 : $-3x - y - 1 = 0$, 点 $(-1, 3)$
距離 : $\frac{1}{10}\sqrt{10}$

(12) 直線 : $3x + 2y - 2 = 0$, 点 $(1, -3)$
距離 : $\frac{5}{13}\sqrt{13}$

(3) 直線 : $-x + 3y + 3 = 0$, 点 $(-3, -1)$
距離 : $\frac{3}{10}\sqrt{10}$

(13) 直線 : $-2x + 2y + 1 = 0$, 点 $(1, 1)$
距離 : $\frac{1}{4}\sqrt{2}$

(4) 直線 : $3x + 2y - 1 = 0$, 点 $(1, 3)$
距離 : $\frac{8}{13}\sqrt{13}$

(14) 直線 : $2x - y - 2 = 0$, 点 $(-1, -3)$
距離 : $\frac{1}{5}\sqrt{5}$

(5) 直線 : $-x - y + 2 = 0$, 点 $(3, -2)$
距離 : $\frac{1}{2}\sqrt{2}$

(15) 直線 : $-3x - 2y - 3 = 0$, 点 $(-1, -1)$
距離 : $\frac{2}{13}\sqrt{13}$

(6) 直線 : $2x + 3y + 3 = 0$, 点 $(-2, 3)$
距離 : $\frac{8}{13}\sqrt{13}$

(16) 直線 : $x - 3y - 2 = 0$, 点 $(2, -2)$
距離 : $\frac{3}{5}\sqrt{10}$

(7) 直線 : $2x - 3y + 3 = 0$, 点 $(-1, -2)$
距離 : $\frac{7}{13}\sqrt{13}$

(17) 直線 : $x + 3y - 2 = 0$, 点 $(-2, 3)$
距離 : $\frac{1}{2}\sqrt{10}$

(8) 直線 : $2x - 2y + 2 = 0$, 点 $(-1, 1)$
距離 : $\frac{1}{2}\sqrt{2}$

(18) 直線 : $-2x + 3y + 3 = 0$, 点 $(-2, -1)$
距離 : $\frac{4}{13}\sqrt{13}$

(9) 直線 : $3x + 2y + 2 = 0$, 点 $(-3, 1)$
距離 : $\frac{5}{13}\sqrt{13}$

(19) 直線 : $2x + y - 1 = 0$, 点 $(3, -3)$
距離 : $\frac{2}{5}\sqrt{5}$

(10) 直線 : $-3x - 3y - 2 = 0$, 点 $(3, 3)$
距離 : $\frac{10}{3}\sqrt{2}$

(20) 直線 : $-3x + y + 2 = 0$, 点 $(-2, -3)$
距離 : $\frac{1}{2}\sqrt{10}$

(1) 直線 : $3x + 2y + 3 = 0$, 点(1, 2)
距離 : $\frac{10}{13}\sqrt{13}$

(11) 直線 : $3x + y - 2 = 0$, 点(-2, -1)
距離 : $\frac{9}{10}\sqrt{10}$

(2) 直線 : $-3x - y - 1 = 0$, 点(2, 2)
距離 : $\frac{9}{10}\sqrt{10}$

(12) 直線 : $3x + 3y - 2 = 0$, 点(-2, -1)
距離 : $\frac{11}{6}\sqrt{2}$

(3) 直線 : $-2x - 3y + 3 = 0$, 点(-3, -3)
距離 : $\frac{18}{13}\sqrt{13}$

(13) 直線 : $-x + 2y - 1 = 0$, 点(-3, 1)
距離 : $\frac{4}{5}\sqrt{5}$

(4) 直線 : $x - y - 1 = 0$, 点(-3, 2)
距離 : $3\sqrt{2}$

(14) 直線 : $x - 3y + 1 = 0$, 点(-2, 2)
距離 : $\frac{7}{10}\sqrt{10}$

(5) 直線 : $-x + 3y + 2 = 0$, 点(-2, 2)
距離 : $\sqrt{10}$

(15) 直線 : $-x - y + 2 = 0$, 点(-1, 2)
距離 : $\frac{1}{2}\sqrt{2}$

(6) 直線 : $-3x + 3y - 2 = 0$, 点(-1, -2)
距離 : $\frac{5}{6}\sqrt{2}$

(16) 直線 : $2x + 3y - 1 = 0$, 点(3, 1)
距離 : $\frac{8}{13}\sqrt{13}$

(7) 直線 : $x - 3y - 2 = 0$, 点(-1, 1)
距離 : $\frac{3}{5}\sqrt{10}$

(17) 直線 : $-x + 3y + 1 = 0$, 点(-3, 2)
距離 : $\sqrt{10}$

(8) 直線 : $-x + 2y + 1 = 0$, 点(-3, 1)
距離 : $\frac{6}{5}\sqrt{5}$

(18) 直線 : $2x + 3y + 3 = 0$, 点(-3, -2)
距離 : $\frac{9}{13}\sqrt{13}$

(9) 直線 : $-3x + 2y + 2 = 0$, 点(1, -2)
距離 : $\frac{5}{13}\sqrt{13}$

(19) 直線 : $-2x + 2y - 3 = 0$, 点(1, 1)
距離 : $\frac{3}{4}\sqrt{2}$

(10) 直線 : $x - y - 1 = 0$, 点(-3, -1)
距離 : $\frac{3}{2}\sqrt{2}$

(20) 直線 : $3x - 2y - 1 = 0$, 点(-1, -2)
距離 : 0

(1) 直線 : $-2x - y + 3 = 0$, 点 $(-2, 1)$
距離 : $\frac{6}{5}\sqrt{5}$

(11) 直線 : $2x - y + 1 = 0$, 点 $(2, 2)$
距離 : $\frac{3}{5}\sqrt{5}$

(2) 直線 : $-x + 3y - 3 = 0$, 点 $(3, -3)$
距離 : $\frac{3}{2}\sqrt{10}$

(12) 直線 : $2x - 3y + 3 = 0$, 点 $(3, 3)$
距離 : 0

(3) 直線 : $-3x + 2y - 2 = 0$, 点 $(-2, 3)$
距離 : $\frac{10}{13}\sqrt{13}$

(13) 直線 : $-3x - 2y + 3 = 0$, 点 $(2, 3)$
距離 : $\frac{9}{13}\sqrt{13}$

(4) 直線 : $3x - y + 2 = 0$, 点 $(2, -1)$
距離 : $\frac{9}{10}\sqrt{10}$

(14) 直線 : $-3x - 2y + 2 = 0$, 点 $(3, 2)$
距離 : $\frac{11}{13}\sqrt{13}$

(5) 直線 : $-x - y - 1 = 0$, 点 $(1, -2)$
距離 : 0

(15) 直線 : $2x - y + 1 = 0$, 点 $(2, -2)$
距離 : $\frac{7}{5}\sqrt{5}$

(6) 直線 : $-2x + 3y + 2 = 0$, 点 $(-2, 1)$
距離 : $\frac{9}{13}\sqrt{13}$

(16) 直線 : $3x - y + 2 = 0$, 点 $(2, -3)$
距離 : $\frac{11}{10}\sqrt{10}$

(7) 直線 : $-x + 2y + 2 = 0$, 点 $(-3, 2)$
距離 : $\frac{9}{5}\sqrt{5}$

(17) 直線 : $2x - 2y - 1 = 0$, 点 $(2, 2)$
距離 : $\frac{1}{4}\sqrt{2}$

(8) 直線 : $3x - y + 1 = 0$, 点 $(-3, 1)$
距離 : $\frac{9}{10}\sqrt{10}$

(18) 直線 : $x - y + 1 = 0$, 点 $(3, 2)$
距離 : $\sqrt{2}$

(9) 直線 : $-x - 3y - 2 = 0$, 点 $(-3, -1)$
距離 : $\frac{2}{5}\sqrt{10}$

(19) 直線 : $-x + 3y - 2 = 0$, 点 $(2, -2)$
距離 : $\sqrt{10}$

(10) 直線 : $-x - 2y + 1 = 0$, 点 $(3, -1)$
距離 : 0

(20) 直線 : $x - 3y - 2 = 0$, 点 $(-2, 1)$
距離 : $\frac{7}{10}\sqrt{10}$

(1) 直線 : $-x + y + 1 = 0$, 点 $(3, -3)$
距離 : $\frac{5}{2}\sqrt{2}$

(11) 直線 : $2x + y - 2 = 0$, 点 $(-2, -2)$
距離 : $\frac{8}{5}\sqrt{5}$

(2) 直線 : $-3x + y + 1 = 0$, 点 $(-3, -3)$
距離 : $\frac{7}{10}\sqrt{10}$

(12) 直線 : $2x + y + 3 = 0$, 点 $(1, 2)$
距離 : $\frac{7}{5}\sqrt{5}$

(3) 直線 : $-x - y - 3 = 0$, 点 $(-3, 1)$
距離 : $\frac{1}{2}\sqrt{2}$

(13) 直線 : $-3x - 3y + 3 = 0$, 点 $(-3, 3)$
距離 : $\frac{1}{2}\sqrt{2}$

(4) 直線 : $2x - y + 1 = 0$, 点 $(1, 3)$
距離 : 0

(14) 直線 : $x + y - 3 = 0$, 点 $(1, -1)$
距離 : $\frac{3}{2}\sqrt{2}$

(5) 直線 : $3x - 2y - 1 = 0$, 点 $(2, 2)$
距離 : $\frac{1}{13}\sqrt{13}$

(15) 直線 : $-x + 2y + 1 = 0$, 点 $(2, 2)$
距離 : $\frac{3}{5}\sqrt{5}$

(6) 直線 : $2x - 3y + 3 = 0$, 点 $(-3, -3)$
距離 : $\frac{6}{13}\sqrt{13}$

(16) 直線 : $-3x - 2y + 2 = 0$, 点 $(-2, -2)$
距離 : $\frac{12}{13}\sqrt{13}$

(7) 直線 : $x + 3y - 2 = 0$, 点 $(-2, -1)$
距離 : $\frac{7}{10}\sqrt{10}$

(17) 直線 : $-3x + y + 2 = 0$, 点 $(3, 1)$
距離 : $\frac{3}{5}\sqrt{10}$

(8) 直線 : $2x - y + 1 = 0$, 点 $(3, 3)$
距離 : $\frac{4}{5}\sqrt{5}$

(18) 直線 : $-3x - y - 3 = 0$, 点 $(2, 1)$
距離 : $\sqrt{10}$

(9) 直線 : $-3x - 2y - 3 = 0$, 点 $(1, 3)$
距離 : $\frac{12}{13}\sqrt{13}$

(19) 直線 : $3x + 2y - 1 = 0$, 点 $(-3, -2)$
距離 : $\frac{14}{13}\sqrt{13}$

(10) 直線 : $x + y - 2 = 0$, 点 $(3, -2)$
距離 : $\frac{1}{2}\sqrt{2}$

(20) 直線 : $-3x + y - 1 = 0$, 点 $(3, 1)$
距離 : $\frac{9}{10}\sqrt{10}$

(1) 直線 : $-x - 3y - 2 = 0$, 点(1, 1)
距離 : $\frac{3}{5}\sqrt{10}$

(11) 直線 : $x - 2y + 3 = 0$, 点(-3, 3)
距離 : $\frac{6}{5}\sqrt{5}$

(2) 直線 : $-3x - 2y + 1 = 0$, 点(2, 1)
距離 : $\frac{7}{13}\sqrt{13}$

(12) 直線 : $-2x + 2y - 3 = 0$, 点(-3, 1)
距離 : $\frac{5}{4}\sqrt{2}$

(3) 直線 : $2x + 3y + 3 = 0$, 点(1, -1)
距離 : $\frac{2}{13}\sqrt{13}$

(13) 直線 : $-3x + y + 3 = 0$, 点(-3, 3)
距離 : $\frac{3}{2}\sqrt{10}$

(4) 直線 : $2x - 3y + 3 = 0$, 点(-2, 3)
距離 : $\frac{10}{13}\sqrt{13}$

(14) 直線 : $x + 2y - 3 = 0$, 点(-1, 2)
距離 : 0

(5) 直線 : $2x - 2y - 3 = 0$, 点(3, -2)
距離 : $\frac{7}{4}\sqrt{2}$

(15) 直線 : $-2x + y - 3 = 0$, 点(1, -2)
距離 : $\frac{7}{5}\sqrt{5}$

(6) 直線 : $-2x + 3y - 2 = 0$, 点(-1, 3)
距離 : $\frac{9}{13}\sqrt{13}$

(16) 直線 : $2x + 2y + 2 = 0$, 点(-3, 3)
距離 : $\frac{1}{2}\sqrt{2}$

(7) 直線 : $2x - 3y + 3 = 0$, 点(3, -3)
距離 : $\frac{18}{13}\sqrt{13}$

(17) 直線 : $3x - 3y + 2 = 0$, 点(-3, 3)
距離 : $\frac{8}{3}\sqrt{2}$

(8) 直線 : $2x + 2y - 3 = 0$, 点(3, -3)
距離 : $\frac{3}{4}\sqrt{2}$

(18) 直線 : $2x - 3y - 3 = 0$, 点(3, -3)
距離 : $\frac{12}{13}\sqrt{13}$

(9) 直線 : $-3x - y + 3 = 0$, 点(-1, -3)
距離 : $\frac{9}{10}\sqrt{10}$

(19) 直線 : $-3x + y - 1 = 0$, 点(1, -3)
距離 : $\frac{7}{10}\sqrt{10}$

(10) 直線 : $2x + 3y + 1 = 0$, 点(3, -1)
距離 : $\frac{4}{13}\sqrt{13}$

(20) 直線 : $-3x - 2y - 3 = 0$, 点(-2, -2)
距離 : $\frac{7}{13}\sqrt{13}$

(1) 直線 : $-3x - 2y + 1 = 0$, 点(3, -2)
距離 : $\frac{4}{13}\sqrt{13}$

(11) 直線 : $2x + 2y + 3 = 0$, 点(3, -3)
距離 : $\frac{3}{4}\sqrt{2}$

(2) 直線 : $-x - 2y + 2 = 0$, 点(1, -2)
距離 : $\sqrt{5}$

(12) 直線 : $-3x + y - 3 = 0$, 点(2, -2)
距離 : $\frac{11}{10}\sqrt{10}$

(3) 直線 : $-x - y + 3 = 0$, 点(1, 3)
距離 : $\frac{1}{2}\sqrt{2}$

(13) 直線 : $3x + 3y + 1 = 0$, 点(-2, 2)
距離 : $\frac{1}{6}\sqrt{2}$

(4) 直線 : $x + 2y - 3 = 0$, 点(-2, 2)
距離 : $\frac{1}{5}\sqrt{5}$

(14) 直線 : $x + 3y - 2 = 0$, 点(-1, -2)
距離 : $\frac{9}{10}\sqrt{10}$

(5) 直線 : $2x - 2y - 3 = 0$, 点(1, 1)
距離 : $\frac{3}{4}\sqrt{2}$

(15) 直線 : $-3x + 2y + 2 = 0$, 点(1, 1)
距離 : $\frac{1}{13}\sqrt{13}$

(6) 直線 : $-x - y - 3 = 0$, 点(-2, -3)
距離 : $\sqrt{2}$

(16) 直線 : $-x - y - 1 = 0$, 点(-1, -1)
距離 : $\frac{1}{2}\sqrt{2}$

(7) 直線 : $-x - 3y + 1 = 0$, 点(2, -1)
距離 : $\frac{1}{5}\sqrt{10}$

(17) 直線 : $3x + y + 2 = 0$, 点(-3, -3)
距離 : $\sqrt{10}$

(8) 直線 : $-x - 2y - 2 = 0$, 点(-3, 2)
距離 : $\frac{3}{5}\sqrt{5}$

(18) 直線 : $3x + y - 2 = 0$, 点(2, -2)
距離 : $\frac{1}{5}\sqrt{10}$

(9) 直線 : $-3x - 3y + 2 = 0$, 点(-1, 2)
距離 : $\frac{1}{6}\sqrt{2}$

(19) 直線 : $-2x - y - 3 = 0$, 点(2, 1)
距離 : $\frac{8}{5}\sqrt{5}$

(10) 直線 : $3x - 2y - 1 = 0$, 点(-2, -1)
距離 : $\frac{5}{13}\sqrt{13}$

(20) 直線 : $-x + 3y - 2 = 0$, 点(-2, -1)
距離 : $\frac{3}{10}\sqrt{10}$

(1) 直線 : $-3x + 3y - 3 = 0$, 点(3, 1)
距離 : $\frac{3}{2}\sqrt{2}$

(11) 直線 : $-x - 2y - 3 = 0$, 点(-2, 1)
距離 : $\frac{3}{5}\sqrt{5}$

(2) 直線 : $3x + 2y + 2 = 0$, 点(-3, -3)
距離 : $\sqrt{13}$

(12) 直線 : $3x + 3y + 1 = 0$, 点(-3, -2)
距離 : $\frac{7}{3}\sqrt{2}$

(3) 直線 : $x - 3y - 1 = 0$, 点(1, -2)
距離 : $\frac{3}{5}\sqrt{10}$

(13) 直線 : $2x - 2y - 2 = 0$, 点(3, -3)
距離 : $\frac{5}{2}\sqrt{2}$

(4) 直線 : $-3x - y + 2 = 0$, 点(-2, -1)
距離 : $\frac{9}{10}\sqrt{10}$

(14) 直線 : $3x - y + 2 = 0$, 点(1, 2)
距離 : $\frac{3}{10}\sqrt{10}$

(5) 直線 : $-x + 2y + 3 = 0$, 点(-2, -3)
距離 : $\frac{1}{5}\sqrt{5}$

(15) 直線 : $-x + y + 2 = 0$, 点(-2, -2)
距離 : $\sqrt{2}$

(6) 直線 : $2x - 2y - 1 = 0$, 点(-1, -3)
距離 : $\frac{3}{4}\sqrt{2}$

(16) 直線 : $x - 3y - 1 = 0$, 点(-3, 1)
距離 : $\frac{7}{10}\sqrt{10}$

(7) 直線 : $2x - 2y - 1 = 0$, 点(-2, 3)
距離 : $\frac{11}{4}\sqrt{2}$

(17) 直線 : $-2x + 2y - 1 = 0$, 点(-1, -1)
距離 : $\frac{1}{4}\sqrt{2}$

(8) 直線 : $-x - 3y + 2 = 0$, 点(1, -1)
距離 : $\frac{2}{5}\sqrt{10}$

(18) 直線 : $-x - 2y + 1 = 0$, 点(3, 1)
距離 : $\frac{4}{5}\sqrt{5}$

(9) 直線 : $-x - 2y + 1 = 0$, 点(2, 2)
距離 : $\sqrt{5}$

(19) 直線 : $-3x + 3y + 3 = 0$, 点(-3, -3)
距離 : $\frac{1}{2}\sqrt{2}$

(10) 直線 : $-x + y + 1 = 0$, 点(2, 1)
距離 : 0

(20) 直線 : $3x - 2y - 2 = 0$, 点(1, -2)
距離 : $\frac{5}{13}\sqrt{13}$

(1) 直線 : $3x + 3y - 1 = 0$, 点(1, 2)
距離 : $\frac{4}{3}\sqrt{2}$

(11) 直線 : $-3x + 3y - 1 = 0$, 点(3, -3)
距離 : $\frac{19}{6}\sqrt{2}$

(2) 直線 : $-3x + 3y + 2 = 0$, 点(3, 1)
距離 : $\frac{2}{3}\sqrt{2}$

(12) 直線 : $-3x - 2y - 3 = 0$, 点(-1, -1)
距離 : $\frac{2}{13}\sqrt{13}$

(3) 直線 : $-x - y - 1 = 0$, 点(3, -1)
距離 : $\frac{3}{2}\sqrt{2}$

(13) 直線 : $2x - y + 1 = 0$, 点(1, -2)
距離 : $\sqrt{5}$

(4) 直線 : $2x + y - 3 = 0$, 点(-3, 1)
距離 : $\frac{8}{5}\sqrt{5}$

(14) 直線 : $-x + y + 1 = 0$, 点(-1, -2)
距離 : 0

(5) 直線 : $3x - 3y - 3 = 0$, 点(1, 2)
距離 : $\sqrt{2}$

(15) 直線 : $3x - y + 3 = 0$, 点(-3, 3)
距離 : $\frac{9}{10}\sqrt{10}$

(6) 直線 : $-x - 2y - 2 = 0$, 点(2, -1)
距離 : $\frac{2}{5}\sqrt{5}$

(16) 直線 : $-x - y - 1 = 0$, 点(-1, -3)
距離 : $\frac{3}{2}\sqrt{2}$

(7) 直線 : $-x + y + 1 = 0$, 点(1, 2)
距離 : $\sqrt{2}$

(17) 直線 : $x - y - 3 = 0$, 点(2, 3)
距離 : $2\sqrt{2}$

(8) 直線 : $3x + y + 3 = 0$, 点(3, -3)
距離 : $\frac{9}{10}\sqrt{10}$

(18) 直線 : $x + 2y - 2 = 0$, 点(2, 2)
距離 : $\frac{4}{5}\sqrt{5}$

(9) 直線 : $x - y + 3 = 0$, 点(2, -3)
距離 : $4\sqrt{2}$

(19) 直線 : $2x + 2y + 2 = 0$, 点(2, -2)
距離 : $\frac{1}{2}\sqrt{2}$

(10) 直線 : $-3x - 2y + 2 = 0$, 点(-3, -2)
距離 : $\frac{15}{13}\sqrt{13}$

(20) 直線 : $-3x - y + 3 = 0$, 点(-2, -1)
距離 : $\sqrt{10}$

(1) 直線 : $-2x + 2y - 2 = 0$, 点(1, 2)
距離 : 0

(11) 直線 : $-2x - 3y + 1 = 0$, 点(2, 1)
距離 : $\frac{6}{13}\sqrt{13}$

(2) 直線 : $-2x + 3y - 1 = 0$, 点(2, -3)
距離 : $\frac{14}{13}\sqrt{13}$

(12) 直線 : $-3x - 3y - 2 = 0$, 点(3, -1)
距離 : $\frac{4}{3}\sqrt{2}$

(3) 直線 : $-x - y - 3 = 0$, 点(3, -1)
距離 : $\frac{5}{2}\sqrt{2}$

(13) 直線 : $3x - 2y - 1 = 0$, 点(-3, -1)
距離 : $\frac{8}{13}\sqrt{13}$

(4) 直線 : $-x - 3y + 3 = 0$, 点(1, -3)
距離 : $\frac{11}{10}\sqrt{10}$

(14) 直線 : $x - y - 1 = 0$, 点(3, 2)
距離 : 0

(5) 直線 : $-3x - 2y - 2 = 0$, 点(-2, 2)
距離 : 0

(15) 直線 : $-x + 3y + 1 = 0$, 点(-3, 2)
距離 : $\sqrt{10}$

(6) 直線 : $-x - y - 3 = 0$, 点(1, -1)
距離 : $\frac{3}{2}\sqrt{2}$

(16) 直線 : $x + y + 1 = 0$, 点(3, 1)
距離 : $\frac{5}{2}\sqrt{2}$

(7) 直線 : $-2x - 2y + 1 = 0$, 点(-1, -1)
距離 : $\frac{5}{4}\sqrt{2}$

(17) 直線 : $-x - 2y + 1 = 0$, 点(-2, -3)
距離 : $\frac{9}{5}\sqrt{5}$

(8) 直線 : $2x + 3y - 1 = 0$, 点(2, 3)
距離 : $\frac{12}{13}\sqrt{13}$

(18) 直線 : $-2x - y + 2 = 0$, 点(-3, 3)
距離 : $\sqrt{5}$

(9) 直線 : $-x - 2y + 3 = 0$, 点(2, 3)
距離 : $\sqrt{5}$

(19) 直線 : $3x - y - 3 = 0$, 点(-2, -1)
距離 : $\frac{4}{5}\sqrt{10}$

(10) 直線 : $2x - y + 1 = 0$, 点(-3, 1)
距離 : $\frac{6}{5}\sqrt{5}$

(20) 直線 : $-3x - y - 3 = 0$, 点(1, 3)
距離 : $\frac{9}{10}\sqrt{10}$

(1) 直線 : $x + y - 1 = 0$, 点(3, 1)
距離 : $\frac{3}{2}\sqrt{2}$

(11) 直線 : $-2x + y - 3 = 0$, 点(-2, 1)
距離 : $\frac{2}{5}\sqrt{5}$

(2) 直線 : $2x - 3y - 3 = 0$, 点(2, -3)
距離 : $\frac{10}{13}\sqrt{13}$

(12) 直線 : $-x + 2y - 2 = 0$, 点(2, -2)
距離 : $\frac{8}{5}\sqrt{5}$

(3) 直線 : $-2x + 2y + 1 = 0$, 点(1, -3)
距離 : $\frac{7}{4}\sqrt{2}$

(13) 直線 : $-2x - y + 3 = 0$, 点(2, 3)
距離 : $\frac{4}{5}\sqrt{5}$

(4) 直線 : $2x - 3y - 3 = 0$, 点(-3, 2)
距離 : $\frac{15}{13}\sqrt{13}$

(14) 直線 : $-x - 3y - 1 = 0$, 点(-1, 2)
距離 : $\frac{3}{5}\sqrt{10}$

(5) 直線 : $x + 2y - 3 = 0$, 点(3, -1)
距離 : $\frac{2}{5}\sqrt{5}$

(15) 直線 : $3x + 3y + 1 = 0$, 点(1, 3)
距離 : $\frac{13}{6}\sqrt{2}$

(6) 直線 : $-x - 2y - 3 = 0$, 点(-2, -1)
距離 : $\frac{1}{5}\sqrt{5}$

(16) 直線 : $-x + 3y + 2 = 0$, 点(-3, 1)
距離 : $\frac{4}{5}\sqrt{10}$

(7) 直線 : $x + 2y + 3 = 0$, 点(2, 3)
距離 : $\frac{11}{5}\sqrt{5}$

(17) 直線 : $-2x + y + 3 = 0$, 点(1, -1)
距離 : 0

(8) 直線 : $-2x - 3y - 1 = 0$, 点(-3, -3)
距離 : $\frac{14}{13}\sqrt{13}$

(18) 直線 : $x - 3y - 2 = 0$, 点(-2, 2)
距離 : $\sqrt{10}$

(9) 直線 : $x - y - 3 = 0$, 点(3, 2)
距離 : $\sqrt{2}$

(19) 直線 : $x + y + 1 = 0$, 点(-2, -2)
距離 : $\frac{3}{2}\sqrt{2}$

(10) 直線 : $2x - 2y + 1 = 0$, 点(-1, 1)
距離 : $\frac{3}{4}\sqrt{2}$

(20) 直線 : $-x - y + 2 = 0$, 点(2, 3)
距離 : $\frac{3}{2}\sqrt{2}$

(1) 直線 : $-x - y + 2 = 0$, 点 $(-2, -3)$
距離 : $\frac{7}{2}\sqrt{2}$

(11) 直線 : $-x - 3y + 1 = 0$, 点 $(2, 1)$
距離 : $\frac{2}{5}\sqrt{10}$

(2) 直線 : $3x - 2y + 1 = 0$, 点 $(-3, -1)$
距離 : $\frac{6}{13}\sqrt{13}$

(12) 直線 : $2x + y + 3 = 0$, 点 $(2, 3)$
距離 : $2\sqrt{5}$

(3) 直線 : $-3x - 2y + 1 = 0$, 点 $(-3, -2)$
距離 : $\frac{14}{13}\sqrt{13}$

(13) 直線 : $x + 3y - 2 = 0$, 点 $(-2, 1)$
距離 : $\frac{1}{10}\sqrt{10}$

(4) 直線 : $x + 3y - 1 = 0$, 点 $(-2, -3)$
距離 : $\frac{6}{5}\sqrt{10}$

(14) 直線 : $x + 3y + 1 = 0$, 点 $(-1, -1)$
距離 : $\frac{3}{10}\sqrt{10}$

(5) 直線 : $-2x + y - 3 = 0$, 点 $(2, 2)$
距離 : $\sqrt{5}$

(15) 直線 : $x + y - 1 = 0$, 点 $(-3, -2)$
距離 : $3\sqrt{2}$

(6) 直線 : $3x + 2y - 2 = 0$, 点 $(1, -1)$
距離 : $\frac{1}{13}\sqrt{13}$

(16) 直線 : $-3x - y - 3 = 0$, 点 $(2, -3)$
距離 : $\frac{3}{5}\sqrt{10}$

(7) 直線 : $3x - y + 1 = 0$, 点 $(1, 1)$
距離 : $\frac{3}{10}\sqrt{10}$

(17) 直線 : $2x + 2y - 2 = 0$, 点 $(3, -3)$
距離 : $\frac{1}{2}\sqrt{2}$

(8) 直線 : $x - 3y + 2 = 0$, 点 $(-2, 3)$
距離 : $\frac{9}{10}\sqrt{10}$

(18) 直線 : $x + y + 1 = 0$, 点 $(2, -2)$
距離 : $\frac{1}{2}\sqrt{2}$

(9) 直線 : $x + 3y + 1 = 0$, 点 $(-2, 2)$
距離 : $\frac{1}{2}\sqrt{10}$

(19) 直線 : $x + 3y + 2 = 0$, 点 $(1, 3)$
距離 : $\frac{6}{5}\sqrt{10}$

(10) 直線 : $x + 3y - 3 = 0$, 点 $(1, 2)$
距離 : $\frac{2}{5}\sqrt{10}$

(20) 直線 : $-2x + y - 3 = 0$, 点 $(-3, -1)$
距離 : $\frac{2}{5}\sqrt{5}$

(1) 直線 : $x + 3y + 1 = 0$, 点 $(-3, -1)$
距離 : $\frac{1}{2}\sqrt{10}$

(11) 直線 : $x + 2y + 1 = 0$, 点 $(2, -3)$
距離 : $\frac{3}{5}\sqrt{5}$

(2) 直線 : $-x - 3y - 1 = 0$, 点 $(1, 2)$
距離 : $\frac{4}{5}\sqrt{10}$

(12) 直線 : $-2x - 3y - 3 = 0$, 点 $(3, 2)$
距離 : $\frac{15}{13}\sqrt{13}$

(3) 直線 : $-3x - 2y - 3 = 0$, 点 $(2, 3)$
距離 : $\frac{15}{13}\sqrt{13}$

(13) 直線 : $-3x - 2y - 2 = 0$, 点 $(-2, -2)$
距離 : $\frac{8}{13}\sqrt{13}$

(4) 直線 : $3x - y + 1 = 0$, 点 $(-1, 2)$
距離 : $\frac{2}{5}\sqrt{10}$

(14) 直線 : $-x - y + 3 = 0$, 点 $(2, 2)$
距離 : $\frac{1}{2}\sqrt{2}$

(5) 直線 : $-3x + y - 3 = 0$, 点 $(2, -3)$
距離 : $\frac{6}{5}\sqrt{10}$

(15) 直線 : $2x + 3y + 2 = 0$, 点 $(2, -2)$
距離 : 0

(6) 直線 : $3x - 3y + 1 = 0$, 点 $(3, -2)$
距離 : $\frac{8}{3}\sqrt{2}$

(16) 直線 : $-2x - y - 3 = 0$, 点 $(-1, 3)$
距離 : $\frac{4}{5}\sqrt{5}$

(7) 直線 : $-x + 3y - 1 = 0$, 点 $(2, -1)$
距離 : $\frac{3}{5}\sqrt{10}$

(17) 直線 : $-2x + y + 3 = 0$, 点 $(-3, -3)$
距離 : $\frac{6}{5}\sqrt{5}$

(8) 直線 : $-3x + y - 2 = 0$, 点 $(-1, -2)$
距離 : $\frac{1}{10}\sqrt{10}$

(18) 直線 : $-x + 3y - 3 = 0$, 点 $(2, 2)$
距離 : $\frac{1}{10}\sqrt{10}$

(9) 直線 : $2x - 3y - 3 = 0$, 点 $(-1, -1)$
距離 : $\frac{2}{13}\sqrt{13}$

(19) 直線 : $-x - 3y - 1 = 0$, 点 $(1, 1)$
距離 : $\frac{1}{2}\sqrt{10}$

(10) 直線 : $x - y - 2 = 0$, 点 $(2, -3)$
距離 : $\frac{3}{2}\sqrt{2}$

(20) 直線 : $3x - 2y + 1 = 0$, 点 $(3, -1)$
距離 : $\frac{12}{13}\sqrt{13}$

(1) 直線 : $x + 3y + 1 = 0$, 点 $(-3, -3)$
距離 : $\frac{11}{10}\sqrt{10}$

(11) 直線 : $-3x - 3y - 3 = 0$, 点 $(-2, 1)$
距離 : 0

(2) 直線 : $x + 2y + 1 = 0$, 点 $(1, 1)$
距離 : $\frac{4}{5}\sqrt{5}$

(12) 直線 : $3x - y - 1 = 0$, 点 $(-3, -2)$
距離 : $\frac{4}{5}\sqrt{10}$

(3) 直線 : $2x - 3y + 2 = 0$, 点 $(3, 1)$
距離 : $\frac{5}{13}\sqrt{13}$

(13) 直線 : $3x + 3y + 2 = 0$, 点 $(-1, -3)$
距離 : $\frac{5}{3}\sqrt{2}$

(4) 直線 : $3x + y - 2 = 0$, 点 $(-3, -1)$
距離 : $\frac{6}{5}\sqrt{10}$

(14) 直線 : $3x + 3y - 1 = 0$, 点 $(1, -3)$
距離 : $\frac{7}{6}\sqrt{2}$

(5) 直線 : $-3x + y + 1 = 0$, 点 $(-2, 1)$
距離 : $\frac{4}{5}\sqrt{10}$

(15) 直線 : $-3x - 2y - 3 = 0$, 点 $(1, -2)$
距離 : $\frac{2}{13}\sqrt{13}$

(6) 直線 : $-x - 2y - 2 = 0$, 点 $(-1, -2)$
距離 : $\frac{3}{5}\sqrt{5}$

(16) 直線 : $-2x + 3y + 3 = 0$, 点 $(-3, -2)$
距離 : $\frac{3}{13}\sqrt{13}$

(7) 直線 : $-x - 2y + 2 = 0$, 点 $(2, -1)$
距離 : $\frac{2}{5}\sqrt{5}$

(17) 直線 : $-x + y + 3 = 0$, 点 $(-2, 2)$
距離 : $\frac{7}{2}\sqrt{2}$

(8) 直線 : $x - 3y - 3 = 0$, 点 $(1, -3)$
距離 : $\frac{7}{10}\sqrt{10}$

(18) 直線 : $2x - 2y - 3 = 0$, 点 $(1, 3)$
距離 : $\frac{7}{4}\sqrt{2}$

(9) 直線 : $-2x - y - 3 = 0$, 点 $(-2, 1)$
距離 : 0

(19) 直線 : $-3x + y - 3 = 0$, 点 $(-1, 1)$
距離 : $\frac{1}{10}\sqrt{10}$

(10) 直線 : $x - 2y - 2 = 0$, 点 $(-2, -3)$
距離 : $\frac{2}{5}\sqrt{5}$

(20) 直線 : $3x + y - 2 = 0$, 点 $(-3, 2)$
距離 : $\frac{9}{10}\sqrt{10}$

(1) 直線: $-x - y + 3 = 0$, 点(2, -3)
距離: $2\sqrt{2}$

(11) 直線: $-3x + 3y + 1 = 0$, 点(3, 2)
距離: $\frac{1}{3}\sqrt{2}$

(2) 直線: $-3x + y - 3 = 0$, 点(-1, -2)
距離: $\frac{1}{5}\sqrt{10}$

(12) 直線: $3x - 3y - 3 = 0$, 点(3, 3)
距離: $\frac{1}{2}\sqrt{2}$

(3) 直線: $-3x - y + 2 = 0$, 点(1, 1)
距離: $\frac{1}{5}\sqrt{10}$

(13) 直線: $x - y - 1 = 0$, 点(2, -2)
距離: $\frac{3}{2}\sqrt{2}$

(4) 直線: $3x + 2y - 1 = 0$, 点(2, 2)
距離: $\frac{9}{13}\sqrt{13}$

(14) 直線: $3x - y + 3 = 0$, 点(-3, 3)
距離: $\frac{9}{10}\sqrt{10}$

(5) 直線: $2x - 2y + 3 = 0$, 点(3, -3)
距離: $\frac{15}{4}\sqrt{2}$

(15) 直線: $3x - 2y - 1 = 0$, 点(-1, 3)
距離: $\frac{10}{13}\sqrt{13}$

(6) 直線: $-2x + 2y - 2 = 0$, 点(-2, -1)
距離: 0

(16) 直線: $-x + 2y - 3 = 0$, 点(2, -1)
距離: $\frac{7}{5}\sqrt{5}$

(7) 直線: $x + y - 3 = 0$, 点(-2, -2)
距離: $\frac{7}{2}\sqrt{2}$

(17) 直線: $3x - y - 2 = 0$, 点(-3, -2)
距離: $\frac{9}{10}\sqrt{10}$

(8) 直線: $2x - 2y - 1 = 0$, 点(-3, -1)
距離: $\frac{5}{4}\sqrt{2}$

(18) 直線: $-3x - 3y - 3 = 0$, 点(1, 3)
距離: $\frac{5}{2}\sqrt{2}$

(9) 直線: $2x + y + 2 = 0$, 点(1, 1)
距離: $\sqrt{5}$

(19) 直線: $3x - 3y + 2 = 0$, 点(-1, 2)
距離: $\frac{7}{6}\sqrt{2}$

(10) 直線: $-2x - 2y + 3 = 0$, 点(-2, 1)
距離: $\frac{5}{4}\sqrt{2}$

(20) 直線: $-3x - y + 3 = 0$, 点(-1, 2)
距離: $\frac{2}{5}\sqrt{10}$

第 3 章

関数

3.1 2 次関数

3.1.1 平方完成 (易しい)

2 次多項式を平方完成する問題。
式に現れる数は整数となる。

(1) $2x^2 - 8x + 4$

(11) $-4x^2 - 40x - 98$

(21) $-4x^2 - 8x - 6$

(2) $x^2 - 8x + 19$

(12) $-5x^2 + 50x - 129$

(22) $-2x^2 + 20x - 52$

(3) $x^2 + 6x + 11$

(13) $-5x^2 - 40x - 85$

(23) $4x^2 - 40x + 101$

(4) $-2x^2 - 20x - 45$

(14) $5x^2 - 30x + 45$

(24) $-x^2 - 6x - 8$

(5) $3x^2 + 18x + 22$

(15) $5x^2 + 50x + 122$

(25) $-4x^2 + 32x - 65$

(6) $2x^2 + 16x + 29$

(16) $-5x^2 - 30x - 43$

(26) $-5x^2 - 30x - 45$

(7) $5x^2 + 50x + 130$

(17) $5x^2 + 30x + 43$

(27) $2x^2 + 4x + 6$

(8) $5x^2 + 30x + 48$

(18) $x^2 - 8x + 14$

(28) $x^2 - 4x + 4$

(9) $-2x^2 + 16x - 35$

(19) $-4x^2 - 16x - 17$

(29) $2x^2 - 8x + 10$

(10) $x^2 + 4x + 8$

(20) $-5x^2 + 10x - 8$

(30) $-5x^2 - 30x - 46$

(1) $x^2 - 6x + 8$

(11) $3x^2 - 6x + 3$

(21) $2x^2 - 12x + 22$

(2) $4x^2 + 24x + 36$

(12) $3x^2 + 24x + 48$

(22) $-4x^2 - 16x - 20$

(3) $5x^2 + 30x + 49$

(13) $-x^2 - 10x - 28$

(23) $-3x^2 + 12x - 15$

(4) $4x^2 + 40x + 104$

(14) $5x^2 + 20x + 21$

(24) $2x^2 - 20x + 54$

(5) $5x^2 + 30x + 47$

(15) $-5x^2 - 50x - 130$

(25) $-5x^2 + 50x - 123$

(6) $2x^2 + 4x - 3$

(16) $-2x^2 + 4x + 3$

(26) $4x^2 - 24x + 34$

(7) $-5x^2 - 30x - 45$

(17) $-3x^2 + 30x - 80$

(27) $3x^2 - 30x + 71$

(8) $4x^2 - 16x + 18$

(18) $2x^2 - 4x + 1$

(28) $2x^2 + 8x + 8$

(9) $-4x^2 + 8x + 1$

(19) $4x^2 + 8x + 6$

(29) $-2x^2 + 8x - 3$

(10) $2x^2 + 20x + 53$

(20) $x^2 - 4x + 4$

(30) $2x^2 + 4x - 3$

(1) $x^2 - 10x + 21$

(11) $5x^2 + 10x + 1$

(21) $-4x^2 + 40x - 95$

(2) $2x^2 - 4x + 1$

(12) $4x^2 - 8x + 1$

(22) $-x^2 + 6x - 6$

(3) $-4x^2 + 24x - 41$

(13) $3x^2 + 18x + 27$

(23) $2x^2 - 8x + 5$

(4) $4x^2 + 16x + 14$

(14) $3x^2 + 24x + 43$

(24) $3x^2 + 6x - 1$

(5) $x^2 + 10x + 30$

(15) $2x^2 + 16x + 27$

(25) $-5x^2 + 30x - 47$

(6) $x^2 - 10x + 24$

(16) $-3x^2 - 6x - 3$

(26) $-x^2 - 6x - 10$

(7) $-x^2 + 6x - 7$

(17) $-5x^2 - 30x - 46$

(27) $-2x^2 + 4x - 5$

(8) $x^2 + 2x + 1$

(18) $-4x^2 - 8x - 9$

(28) $4x^2 + 8x + 2$

(9) $4x^2 + 24x + 33$

(19) $4x^2 + 8x + 4$

(29) $-x^2 - 10x - 28$

(10) $-3x^2 + 18x - 23$

(20) $-3x^2 + 12x - 10$

(30) $-2x^2 - 12x - 18$

(1) $x^2 + 6x + 5$

(11) $-x^2 + 6x - 5$

(21) $5x^2 - 30x + 49$

(2) $5x^2 - 40x + 83$

(12) $-2x^2 - 4x + 2$

(22) $x^2 + 2x - 2$

(3) $-3x^2 + 6x - 1$

(13) $x^2 - 4x + 7$

(23) $3x^2 - 6x + 7$

(4) $-3x^2 - 6x + 2$

(14) $-2x^2 + 16x - 35$

(24) $-5x^2 - 10x - 4$

(5) $-4x^2 + 8x - 3$

(15) $5x^2 + 50x + 125$

(25) $x^2 + 6x + 13$

(6) $2x^2 - 8x + 6$

(16) $4x^2 - 32x + 61$

(26) $-4x^2 + 32x - 65$

(7) $-3x^2 + 18x - 31$

(17) $x^2 - 10x + 20$

(27) $-2x^2 + 4x - 6$

(8) $-5x^2 + 30x - 44$

(18) $-2x^2 + 4x - 3$

(28) $-4x^2 - 8x$

(9) $-5x^2 - 40x - 80$

(19) $5x^2 + 10x + 9$

(29) $-x^2 - 10x - 20$

(10) $x^2 - 4x + 2$

(20) $x^2 - 2x + 3$

(30) $3x^2 - 18x + 26$

(1) $-2x^2 + 4x - 3$

(11) $5x^2 + 30x + 42$

(21) $-5x^2 - 40x - 75$

(2) $-3x^2 - 6x - 8$

(12) $-4x^2 - 16x - 17$

(22) $-2x^2 + 4x - 5$

(3) $-3x^2 - 6x$

(13) $5x^2 - 40x + 82$

(23) $3x^2 - 24x + 51$

(4) $-4x^2 + 16x - 18$

(14) $5x^2 + 20x + 16$

(24) $3x^2 - 30x + 70$

(5) $-4x^2 - 32x - 62$

(15) $3x^2 - 12x + 13$

(25) $3x^2 + 6x + 5$

(6) $x^2 + 2x - 3$

(16) $x^2 + 8x + 17$

(26) $-5x^2 - 50x - 125$

(7) $-x^2 + 2x + 3$

(17) $5x^2 + 40x + 81$

(27) $-5x^2 - 20x - 21$

(8) $-3x^2 - 24x - 43$

(18) $-5x^2 - 30x - 40$

(28) $2x^2 + 16x + 36$

(9) $-2x^2 - 4x + 3$

(19) $3x^2 - 24x + 53$

(29) $-2x^2 - 12x - 18$

(10) $-2x^2 + 16x - 34$

(20) $3x^2 + 24x + 45$

(30) $-5x^2 - 40x - 81$

(1) $-4x^2 + 24x - 32$

(11) $4x^2 + 40x + 105$

(21) $-3x^2 - 18x - 24$

(2) $-2x^2 - 4x - 6$

(12) $5x^2 + 30x + 49$

(22) $-5x^2 - 20x - 18$

(3) $2x^2 + 12x + 13$

(13) $-4x^2 + 40x - 104$

(23) $-4x^2 + 40x - 97$

(4) $4x^2 + 40x + 104$

(14) $-4x^2 - 40x - 103$

(24) $-5x^2 + 50x - 129$

(5) $3x^2 - 24x + 45$

(15) $5x^2 - 40x + 80$

(25) $-5x^2 + 50x - 123$

(6) $-x^2 + 6x - 8$

(16) $5x^2 + 20x + 24$

(26) $-5x^2 + 30x - 41$

(7) $x^2 - 4x + 6$

(17) $-2x^2 - 16x - 28$

(27) $2x^2 + 8x + 7$

(8) $-5x^2 + 50x - 124$

(18) $3x^2 + 18x + 32$

(28) $-5x^2 + 30x - 46$

(9) $-x^2 + 2x - 1$

(19) $5x^2 + 20x + 16$

(29) $-x^2 + 6x - 6$

(10) $-3x^2 - 12x - 9$

(20) $-5x^2 - 10x$

(30) $x^2 + 4x + 1$

(1) $3x^2 + 12x + 10$

(11) $-2x^2 + 12x - 21$

(21) $-2x^2 + 16x - 31$

(2) $4x^2 + 40x + 104$

(12) $3x^2 - 6x + 4$

(22) $x^2 - 2x - 1$

(3) $-4x^2 + 40x - 105$

(13) $4x^2 + 8x + 4$

(23) $-x^2 + 4x - 5$

(4) $-2x^2 - 12x - 18$

(14) $3x^2 + 12x + 16$

(24) $-3x^2 - 18x - 29$

(5) $-x^2 - 10x - 30$

(15) $2x^2 + 4x + 6$

(25) $4x^2 - 24x + 38$

(6) $3x^2 + 30x + 75$

(16) $3x^2 - 6x + 2$

(26) $-x^2 + 10x - 23$

(7) $-x^2 + 10x - 25$

(17) $-4x^2 - 16x - 11$

(27) $x^2 + 6x + 12$

(8) $4x^2 + 24x + 38$

(18) $x^2 - 6x + 13$

(28) $x^2 - 6x + 11$

(9) $-2x^2 - 4x - 5$

(19) $2x^2 - 8x + 6$

(29) $3x^2 - 18x + 28$

(10) $-x^2 - 2x + 3$

(20) $4x^2 - 32x + 67$

(30) $3x^2 - 30x + 80$

(1) $x^2 - 10x + 26$

(11) $-4x^2 - 8x - 1$

(21) $x^2 - 6x + 12$

(2) $x^2 + 4x + 2$

(12) $5x^2 + 10x + 7$

(22) $x^2 + 6x + 9$

(3) $-4x^2 + 32x - 61$

(13) $-2x^2 + 20x - 46$

(23) $-x^2 - 4x - 1$

(4) $4x^2 + 8x + 9$

(14) $-x^2 + 4x - 2$

(24) $4x^2 + 8x + 3$

(5) $-2x^2 + 8x - 3$

(15) $-3x^2 + 30x - 72$

(25) $5x^2 + 40x + 76$

(6) $3x^2 + 6x - 2$

(16) $5x^2 + 20x + 21$

(26) $x^2 - 4x + 8$

(7) $x^2 + 10x + 23$

(17) $2x^2 - 20x + 47$

(27) $4x^2 + 32x + 62$

(8) $-4x^2 + 40x - 100$

(18) $-2x^2 - 16x - 32$

(28) $-5x^2 - 10x - 2$

(9) $x^2 - 6x + 6$

(19) $x^2 + 6x + 14$

(29) $5x^2 + 50x + 129$

(10) $x^2 + 8x + 16$

(20) $-x^2 - 8x - 15$

(30) $x^2 - 2x + 3$

(1) $4x^2 + 40x + 103$

(11) $-3x^2 - 24x - 45$

(21) $-x^2 + 4x + 1$

(2) $3x^2 - 24x + 47$

(12) $-x^2 - 6x - 9$

(22) $-3x^2 + 12x - 10$

(3) $3x^2 + 30x + 76$

(13) $-2x^2 + 16x - 33$

(23) $x^2 - 4x + 7$

(4) $2x^2 - 4x + 1$

(14) $-x^2 - 10x - 28$

(24) $4x^2 + 32x + 62$

(5) $-x^2 + 8x - 14$

(15) $3x^2 + 12x + 7$

(25) $-4x^2 - 40x - 101$

(6) $4x^2 - 24x + 40$

(16) $3x^2 + 24x + 43$

(26) $5x^2 + 40x + 82$

(7) $x^2 - 10x + 26$

(17) $3x^2 + 18x + 32$

(27) $-5x^2 - 40x - 82$

(8) $x^2 - 2x + 5$

(18) $5x^2 + 40x + 80$

(28) $-5x^2 + 10x - 6$

(9) $-x^2 - 8x - 18$

(19) $4x^2 + 8x + 3$

(29) $x^2 - 2x + 5$

(10) $x^2 + 10x + 30$

(20) $4x^2 + 8x + 9$

(30) $-4x^2 + 32x - 62$

(1) $-x^2 - 8x - 18$

(11) $-5x^2 - 10x - 5$

(21) $2x^2 + 20x + 46$

(2) $-5x^2 + 40x - 76$

(12) $-3x^2 - 6x - 4$

(22) $4x^2 + 40x + 96$

(3) $-3x^2 + 24x - 46$

(13) $-5x^2 + 30x - 43$

(23) $x^2 - 6x + 7$

(4) $-4x^2 + 16x - 11$

(14) $5x^2 - 50x + 124$

(24) $3x^2 - 18x + 29$

(5) $4x^2 + 8x$

(15) $-4x^2 - 24x - 39$

(25) $-4x^2 - 8x - 5$

(6) $x^2 - 6x + 4$

(16) $x^2 + 10x + 22$

(26) $-3x^2 - 30x - 77$

(7) $-x^2 - 4x + 1$

(17) $-2x^2 - 8x - 11$

(27) $5x^2 + 40x + 84$

(8) $-3x^2 + 30x - 73$

(18) $-2x^2 - 8x - 5$

(28) $-x^2 + 10x - 24$

(9) $-x^2 + 4x - 3$

(19) $-5x^2 + 30x - 45$

(29) $5x^2 + 30x + 45$

(10) $-x^2 + 6x - 14$

(20) $4x^2 - 32x + 60$

(30) $4x^2 + 24x + 33$

(1) $4x^2 + 16x + 17$

(11) $3x^2 - 24x + 43$

(21) $-3x^2 - 12x - 7$

(2) $-5x^2 + 30x - 49$

(12) $-x^2 + 4x - 3$

(22) $3x^2 - 30x + 70$

(3) $5x^2 + 40x + 78$

(13) $-3x^2 - 12x - 17$

(23) $-x^2 + 6x - 14$

(4) $-4x^2 + 24x - 39$

(14) $-x^2 - 10x - 22$

(24) $-5x^2 - 20x - 22$

(5) $5x^2 + 20x + 22$

(15) $-5x^2 + 40x - 82$

(25) $-x^2 + 6x - 9$

(6) $5x^2 + 20x + 24$

(16) $-x^2 + 8x - 17$

(26) $-3x^2 - 30x - 71$

(7) $2x^2 - 4x - 3$

(17) $-3x^2 + 6x - 8$

(27) $-2x^2 - 12x - 15$

(8) $2x^2 - 12x + 16$

(18) $5x^2 - 10x + 4$

(28) $-4x^2 - 24x - 35$

(9) $5x^2 + 30x + 45$

(19) $-2x^2 - 4x + 1$

(29) $4x^2 - 40x + 104$

(10) $2x^2 - 16x + 36$

(20) $-3x^2 - 24x - 45$

(30) $-5x^2 - 40x - 79$

(1) $-3x^2 - 24x - 52$

(11) $x^2 + 4x - 1$

(21) $5x^2 + 10x + 3$

(2) $3x^2 + 6x - 2$

(12) $-2x^2 + 16x - 33$

(22) $-2x^2 + 12x - 21$

(3) $-4x^2 + 40x - 104$

(13) $4x^2 - 40x + 100$

(23) $2x^2 + 8x + 12$

(4) $3x^2 + 6x + 7$

(14) $x^2 - 4x + 2$

(24) $x^2 + 2x + 2$

(5) $-4x^2 + 16x - 16$

(15) $-4x^2 + 32x - 59$

(25) $x^2 - 8x + 18$

(6) $-x^2 - 2x + 1$

(16) $-x^2 - 2x + 2$

(26) $x^2 + 10x + 23$

(7) $-4x^2 + 8x - 1$

(17) $5x^2 + 40x + 85$

(27) $-5x^2 + 10x - 8$

(8) $-5x^2 + 50x - 127$

(18) $4x^2 - 32x + 69$

(28) $4x^2 - 32x + 64$

(9) $-2x^2 - 16x - 27$

(19) $-5x^2 - 30x - 49$

(29) $-2x^2 + 20x - 55$

(10) $-2x^2 + 20x - 46$

(20) $-5x^2 - 50x - 122$

(30) $-4x^2 + 40x - 95$

(1) $x^2 + 2x + 5$

(11) $5x^2 - 20x + 17$

(21) $2x^2 + 16x + 31$

(2) $-3x^2 - 24x - 49$

(12) $2x^2 - 12x + 15$

(22) $x^2 - 8x + 18$

(3) $4x^2 - 16x + 12$

(13) $3x^2 - 24x + 45$

(23) $-2x^2 - 4x + 3$

(4) $-3x^2 + 24x - 47$

(14) $-2x^2 + 12x - 20$

(24) $3x^2 - 30x + 74$

(5) $5x^2 + 20x + 24$

(15) $-3x^2 + 24x - 47$

(25) $x^2 + 2x + 5$

(6) $x^2 + 6x + 11$

(16) $-3x^2 - 30x - 71$

(26) $-3x^2 - 30x - 77$

(7) $-4x^2 - 32x - 68$

(17) $-x^2 - 2x$

(27) $x^2 + 6x + 4$

(8) $4x^2 + 16x + 16$

(18) $x^2 + 4x - 1$

(28) $-x^2 + 10x - 22$

(9) $-4x^2 - 32x - 65$

(19) $2x^2 + 20x + 48$

(29) $4x^2 - 24x + 34$

(10) $2x^2 - 8x + 9$

(20) $-5x^2 + 10x - 6$

(30) $-5x^2 - 30x - 49$

(1) $x^2 + 10x + 24$

(11) $4x^2 - 24x + 36$

(21) $-4x^2 + 32x - 60$

(2) $-5x^2 - 10x$

(12) $-5x^2 + 50x - 122$

(22) $-x^2 + 2x + 1$

(3) $4x^2 - 16x + 15$

(13) $-2x^2 + 8x - 3$

(23) $2x^2 - 16x + 31$

(4) $4x^2 + 32x + 68$

(14) $x^2 + 2x - 4$

(24) $-2x^2 - 12x - 23$

(5) $5x^2 + 10x + 2$

(15) $-x^2 - 2x + 3$

(25) $2x^2 - 12x + 16$

(6) $3x^2 + 12x + 10$

(16) $4x^2 + 32x + 69$

(26) $5x^2 + 30x + 50$

(7) $-5x^2 - 20x - 16$

(17) $2x^2 + 4x$

(27) $-5x^2 - 40x - 75$

(8) $-x^2 + 2x - 1$

(18) $-3x^2 + 12x - 9$

(28) $3x^2 - 24x + 50$

(9) $3x^2 + 18x + 25$

(19) $-3x^2 + 12x - 8$

(29) $3x^2 - 18x + 23$

(10) $-3x^2 - 12x - 17$

(20) $2x^2 + 4x - 3$

(30) $x^2 - 6x + 12$

(1) $5x^2 + 20x + 25$

(11) $-x^2 - 6x - 11$

(21) $-2x^2 + 20x - 55$

(2) $-x^2 + 6x - 11$

(12) $x^2 + 6x + 8$

(22) $5x^2 + 20x + 25$

(3) $5x^2 + 30x + 46$

(13) $-4x^2 - 8x - 2$

(23) $2x^2 + 20x + 49$

(4) $-x^2 + 4x - 8$

(14) $3x^2 - 12x + 13$

(24) $4x^2 - 32x + 61$

(5) $-x^2 + 2x + 3$

(15) $-4x^2 - 8x - 3$

(25) $-3x^2 + 12x - 16$

(6) $-4x^2 + 24x - 35$

(16) $-4x^2 + 40x - 98$

(26) $-5x^2 + 10x - 4$

(7) $5x^2 + 30x + 41$

(17) $-4x^2 - 16x - 11$

(27) $-5x^2 - 50x - 129$

(8) $-4x^2 - 32x - 60$

(18) $-4x^2 + 24x - 38$

(28) $-2x^2 + 16x - 37$

(9) $-2x^2 - 16x - 34$

(19) $-2x^2 - 20x - 54$

(29) $2x^2 + 16x + 31$

(10) $2x^2 - 16x + 32$

(20) $4x^2 - 32x + 66$

(30) $-5x^2 + 40x - 83$

(1) $5x^2 - 20x + 15$

(11) $-4x^2 - 16x - 12$

(21) $5x^2 + 50x + 124$

(2) $5x^2 + 10x + 2$

(12) $-3x^2 - 30x - 74$

(22) $3x^2 + 6x + 6$

(3) $3x^2 - 18x + 26$

(13) $-x^2 - 4x - 2$

(23) $4x^2 - 24x + 36$

(4) $2x^2 + 12x + 23$

(14) $2x^2 - 12x + 14$

(24) $-3x^2 - 6x + 1$

(5) $-x^2 + 10x - 30$

(15) $4x^2 - 16x + 17$

(25) $4x^2 + 24x + 36$

(6) $-5x^2 + 50x - 126$

(16) $-5x^2 - 10x - 8$

(26) $-2x^2 + 4x - 3$

(7) $-4x^2 - 32x - 60$

(17) $2x^2 - 12x + 19$

(27) $-2x^2 + 4x - 5$

(8) $-x^2 + 10x - 23$

(18) $2x^2 - 8x + 12$

(28) $4x^2 + 32x + 64$

(9) $3x^2 - 6x - 2$

(19) $3x^2 + 30x + 73$

(29) $3x^2 + 24x + 50$

(10) $4x^2 + 32x + 63$

(20) $-x^2 + 10x - 25$

(30) $4x^2 + 24x + 39$

(1) $5x^2 + 40x + 77$

(11) $-2x^2 + 20x - 48$

(21) $-4x^2 + 40x - 104$

(2) $-5x^2 + 40x - 78$

(12) $5x^2 - 50x + 128$

(22) $2x^2 - 16x + 37$

(3) $-3x^2 + 6x - 3$

(13) $-4x^2 + 16x - 20$

(23) $4x^2 - 24x + 32$

(4) $5x^2 - 50x + 130$

(14) $-4x^2 - 16x - 12$

(24) $-5x^2 - 20x - 18$

(5) $2x^2 + 20x + 49$

(15) $2x^2 + 8x + 4$

(25) $-x^2 + 6x - 9$

(6) $-3x^2 - 12x - 8$

(16) $-4x^2 + 40x - 98$

(26) $x^2 - 4x + 1$

(7) $5x^2 + 40x + 75$

(17) $3x^2 + 6x$

(27) $-x^2 - 10x - 30$

(8) $2x^2 + 4x - 2$

(18) $-3x^2 - 12x - 9$

(28) $-x^2 + 10x - 26$

(9) $3x^2 + 12x + 9$

(19) $-2x^2 - 12x - 14$

(29) $-4x^2 + 32x - 60$

(10) $-5x^2 + 50x - 129$

(20) $3x^2 + 24x + 47$

(30) $x^2 + 2x - 1$

(1) $-4x^2 - 40x - 101$

(11) $-3x^2 + 18x - 30$

(21) $3x^2 - 30x + 73$

(2) $-4x^2 + 32x - 64$

(12) $-2x^2 - 16x - 31$

(22) $-x^2 - 8x - 11$

(3) $-5x^2 + 30x - 47$

(13) $-x^2 + 6x - 9$

(23) $-2x^2 + 16x - 32$

(4) $3x^2 + 12x + 14$

(14) $x^2 - 10x + 23$

(24) $4x^2 - 40x + 98$

(5) $x^2 + 8x + 16$

(15) $-4x^2 + 8x$

(25) $-x^2 - 6x - 8$

(6) $-5x^2 + 40x - 75$

(16) $-x^2 + 6x - 5$

(26) $4x^2 + 40x + 98$

(7) $-x^2 - 6x - 5$

(17) $x^2 + 2x + 4$

(27) $-x^2 - 10x - 27$

(8) $-5x^2 + 10x - 1$

(18) $5x^2 - 30x + 46$

(28) $x^2 - 2x - 1$

(9) $-5x^2 + 30x - 47$

(19) $5x^2 - 40x + 79$

(29) $-5x^2 + 30x - 50$

(10) $-5x^2 - 30x - 48$

(20) $x^2 - 6x + 10$

(30) $5x^2 + 20x + 22$

(1) $-2x^2 - 16x - 27$

(11) $-5x^2 + 20x - 15$

(21) $3x^2 - 12x + 17$

(2) $2x^2 - 8x + 6$

(12) $2x^2 - 8x + 3$

(22) $x^2 + 2x + 2$

(3) $-x^2 - 8x - 11$

(13) $-2x^2 - 16x - 35$

(23) $x^2 + 8x + 15$

(4) $2x^2 + 4x + 4$

(14) $-4x^2 - 40x - 105$

(24) $-x^2 - 6x - 6$

(5) $2x^2 - 8x + 6$

(15) $-3x^2 + 30x - 76$

(25) $5x^2 + 40x + 84$

(6) $-x^2 - 2x - 5$

(16) $-4x^2 + 16x - 17$

(26) $-5x^2 - 50x - 130$

(7) $-3x^2 - 18x - 24$

(17) $4x^2 - 24x + 31$

(27) $-4x^2 - 24x - 32$

(8) $x^2 + 10x + 23$

(18) $-x^2 + 8x - 11$

(28) $5x^2 - 30x + 46$

(9) $-5x^2 - 20x - 17$

(19) $2x^2 + 4x + 2$

(29) $x^2 + 4x + 6$

(10) $2x^2 + 4x + 2$

(20) $3x^2 - 30x + 78$

(30) $2x^2 - 8x + 5$

(1) $-x^2 + 6x - 8$

(11) $-5x^2 + 20x - 19$

(21) $-4x^2 - 32x - 69$

(2) $x^2 + 10x + 30$

(12) $3x^2 - 24x + 52$

(22) $3x^2 + 24x + 47$

(3) $-3x^2 - 12x - 12$

(13) $4x^2 + 24x + 35$

(23) $x^2 + 10x + 27$

(4) $4x^2 + 8x - 1$

(14) $-4x^2 - 16x - 15$

(24) $x^2 - 6x + 10$

(5) $-2x^2 + 12x - 20$

(15) $-2x^2 - 8x - 5$

(25) $-3x^2 + 18x - 32$

(6) $4x^2 + 24x + 41$

(16) $-3x^2 - 18x - 25$

(26) $5x^2 - 50x + 123$

(7) $-3x^2 + 24x - 46$

(17) $2x^2 + 12x + 20$

(27) $-3x^2 + 12x - 12$

(8) $2x^2 - 20x + 52$

(18) $4x^2 + 40x + 95$

(28) $4x^2 - 8x + 8$

(9) $-3x^2 - 30x - 72$

(19) $-5x^2 - 40x - 84$

(29) $-3x^2 - 30x - 70$

(10) $3x^2 - 18x + 30$

(20) $x^2 - 6x + 14$

(30) $3x^2 - 6x + 1$

$$\begin{aligned}(1) \quad & 2x^2 - 8x + 4 \\ &= 2(x-2)^2 - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & -4x^2 - 40x - 98 \\ &= -4(x+5)^2 + 2\end{aligned}$$

$$\begin{aligned}(21) \quad & -4x^2 - 8x - 6 \\ &= -4(x+1)^2 - 2\end{aligned}$$

$$\begin{aligned}(2) \quad & x^2 - 8x + 19 \\ &= (x-4)^2 + 3\end{aligned}$$

$$\begin{aligned}(12) \quad & -5x^2 + 50x - 129 \\ &= -5(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(22) \quad & -2x^2 + 20x - 52 \\ &= -2(x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(3) \quad & x^2 + 6x + 11 \\ &= (x+3)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & -5x^2 - 40x - 85 \\ &= -5(x+4)^2 - 5\end{aligned}$$

$$\begin{aligned}(23) \quad & 4x^2 - 40x + 101 \\ &= 4(x-5)^2 + 1\end{aligned}$$

$$\begin{aligned}(4) \quad & -2x^2 - 20x - 45 \\ &= -2(x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & 5x^2 - 30x + 45 \\ &= 5(x-3)^2\end{aligned}$$

$$\begin{aligned}(24) \quad & -x^2 - 6x - 8 \\ &= -(x+3)^2 + 1\end{aligned}$$

$$\begin{aligned}(5) \quad & 3x^2 + 18x + 22 \\ &= 3(x+3)^2 - 5\end{aligned}$$

$$\begin{aligned}(15) \quad & 5x^2 + 50x + 122 \\ &= 5(x+5)^2 - 3\end{aligned}$$

$$\begin{aligned}(25) \quad & -4x^2 + 32x - 65 \\ &= -4(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(6) \quad & 2x^2 + 16x + 29 \\ &= 2(x+4)^2 - 3\end{aligned}$$

$$\begin{aligned}(16) \quad & -5x^2 - 30x - 43 \\ &= -5(x+3)^2 + 2\end{aligned}$$

$$\begin{aligned}(26) \quad & -5x^2 - 30x - 45 \\ &= -5(x+3)^2\end{aligned}$$

$$\begin{aligned}(7) \quad & 5x^2 + 50x + 130 \\ &= 5(x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(17) \quad & 5x^2 + 30x + 43 \\ &= 5(x+3)^2 - 2\end{aligned}$$

$$\begin{aligned}(27) \quad & 2x^2 + 4x + 6 \\ &= 2(x+1)^2 + 4\end{aligned}$$

$$\begin{aligned}(8) \quad & 5x^2 + 30x + 48 \\ &= 5(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(18) \quad & x^2 - 8x + 14 \\ &= (x-4)^2 - 2\end{aligned}$$

$$\begin{aligned}(28) \quad & x^2 - 4x + 4 \\ &= (x-2)^2\end{aligned}$$

$$\begin{aligned}(9) \quad & -2x^2 + 16x - 35 \\ &= -2(x-4)^2 - 3\end{aligned}$$

$$\begin{aligned}(19) \quad & -4x^2 - 16x - 17 \\ &= -4(x+2)^2 - 1\end{aligned}$$

$$\begin{aligned}(29) \quad & 2x^2 - 8x + 10 \\ &= 2(x-2)^2 + 2\end{aligned}$$

$$\begin{aligned}(10) \quad & x^2 + 4x + 8 \\ &= (x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(20) \quad & -5x^2 + 10x - 8 \\ &= -5(x-1)^2 - 3\end{aligned}$$

$$\begin{aligned}(30) \quad & -5x^2 - 30x - 46 \\ &= -5(x+3)^2 - 1\end{aligned}$$

$$\begin{aligned}(1) \quad & x^2 - 6x + 8 \\ &= (x - 3)^2 - 1\end{aligned}$$

$$\begin{aligned}(11) \quad & 3x^2 - 6x + 3 \\ &= 3(x - 1)^2\end{aligned}$$

$$\begin{aligned}(21) \quad & 2x^2 - 12x + 22 \\ &= 2(x - 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(2) \quad & 4x^2 + 24x + 36 \\ &= 4(x + 3)^2\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 + 24x + 48 \\ &= 3(x + 4)^2\end{aligned}$$

$$\begin{aligned}(22) \quad & -4x^2 - 16x - 20 \\ &= -4(x + 2)^2 - 4\end{aligned}$$

$$\begin{aligned}(3) \quad & 5x^2 + 30x + 49 \\ &= 5(x + 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 - 10x - 28 \\ &= -(x + 5)^2 - 3\end{aligned}$$

$$\begin{aligned}(23) \quad & -3x^2 + 12x - 15 \\ &= -3(x - 2)^2 - 3\end{aligned}$$

$$\begin{aligned}(4) \quad & 4x^2 + 40x + 104 \\ &= 4(x + 5)^2 + 4\end{aligned}$$

$$\begin{aligned}(14) \quad & 5x^2 + 20x + 21 \\ &= 5(x + 2)^2 + 1\end{aligned}$$

$$\begin{aligned}(24) \quad & 2x^2 - 20x + 54 \\ &= 2(x - 5)^2 + 4\end{aligned}$$

$$\begin{aligned}(5) \quad & 5x^2 + 30x + 47 \\ &= 5(x + 3)^2 + 2\end{aligned}$$

$$\begin{aligned}(15) \quad & -5x^2 - 50x - 130 \\ &= -5(x + 5)^2 - 5\end{aligned}$$

$$\begin{aligned}(25) \quad & -5x^2 + 50x - 123 \\ &= -5(x - 5)^2 + 2\end{aligned}$$

$$\begin{aligned}(6) \quad & 2x^2 + 4x - 3 \\ &= 2(x + 1)^2 - 5\end{aligned}$$

$$\begin{aligned}(16) \quad & -2x^2 + 4x + 3 \\ &= -2(x - 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(26) \quad & 4x^2 - 24x + 34 \\ &= 4(x - 3)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -5x^2 - 30x - 45 \\ &= -5(x + 3)^2\end{aligned}$$

$$\begin{aligned}(17) \quad & -3x^2 + 30x - 80 \\ &= -3(x - 5)^2 - 5\end{aligned}$$

$$\begin{aligned}(27) \quad & 3x^2 - 30x + 71 \\ &= 3(x - 5)^2 - 4\end{aligned}$$

$$\begin{aligned}(8) \quad & 4x^2 - 16x + 18 \\ &= 4(x - 2)^2 + 2\end{aligned}$$

$$\begin{aligned}(18) \quad & 2x^2 - 4x + 1 \\ &= 2(x - 1)^2 - 1\end{aligned}$$

$$\begin{aligned}(28) \quad & 2x^2 + 8x + 8 \\ &= 2(x + 2)^2\end{aligned}$$

$$\begin{aligned}(9) \quad & -4x^2 + 8x + 1 \\ &= -4(x - 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(19) \quad & 4x^2 + 8x + 6 \\ &= 4(x + 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(29) \quad & -2x^2 + 8x - 3 \\ &= -2(x - 2)^2 + 5\end{aligned}$$

$$\begin{aligned}(10) \quad & 2x^2 + 20x + 53 \\ &= 2(x + 5)^2 + 3\end{aligned}$$

$$\begin{aligned}(20) \quad & x^2 - 4x + 4 \\ &= (x - 2)^2\end{aligned}$$

$$\begin{aligned}(30) \quad & 2x^2 + 4x - 3 \\ &= 2(x + 1)^2 - 5\end{aligned}$$

$$\begin{aligned}(1) \quad & x^2 - 10x + 21 \\ &= (x - 5)^2 - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & 5x^2 + 10x + 1 \\ &= 5(x + 1)^2 - 4\end{aligned}$$

$$\begin{aligned}(21) \quad & -4x^2 + 40x - 95 \\ &= -4(x - 5)^2 + 5\end{aligned}$$

$$\begin{aligned}(2) \quad & 2x^2 - 4x + 1 \\ &= 2(x - 1)^2 - 1\end{aligned}$$

$$\begin{aligned}(12) \quad & 4x^2 - 8x + 1 \\ &= 4(x - 1)^2 - 3\end{aligned}$$

$$\begin{aligned}(22) \quad & -x^2 + 6x - 6 \\ &= -(x - 3)^2 + 3\end{aligned}$$

$$\begin{aligned}(3) \quad & -4x^2 + 24x - 41 \\ &= -4(x - 3)^2 - 5\end{aligned}$$

$$\begin{aligned}(13) \quad & 3x^2 + 18x + 27 \\ &= 3(x + 3)^2\end{aligned}$$

$$\begin{aligned}(23) \quad & 2x^2 - 8x + 5 \\ &= 2(x - 2)^2 - 3\end{aligned}$$

$$\begin{aligned}(4) \quad & 4x^2 + 16x + 14 \\ &= 4(x + 2)^2 - 2\end{aligned}$$

$$\begin{aligned}(14) \quad & 3x^2 + 24x + 43 \\ &= 3(x + 4)^2 - 5\end{aligned}$$

$$\begin{aligned}(24) \quad & 3x^2 + 6x - 1 \\ &= 3(x + 1)^2 - 4\end{aligned}$$

$$\begin{aligned}(5) \quad & x^2 + 10x + 30 \\ &= (x + 5)^2 + 5\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 + 16x + 27 \\ &= 2(x + 4)^2 - 5\end{aligned}$$

$$\begin{aligned}(25) \quad & -5x^2 + 30x - 47 \\ &= -5(x - 3)^2 - 2\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 - 10x + 24 \\ &= (x - 5)^2 - 1\end{aligned}$$

$$\begin{aligned}(16) \quad & -3x^2 - 6x - 3 \\ &= -3(x + 1)^2\end{aligned}$$

$$\begin{aligned}(26) \quad & -x^2 - 6x - 10 \\ &= -(x + 3)^2 - 1\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 + 6x - 7 \\ &= -(x - 3)^2 + 2\end{aligned}$$

$$\begin{aligned}(17) \quad & -5x^2 - 30x - 46 \\ &= -5(x + 3)^2 - 1\end{aligned}$$

$$\begin{aligned}(27) \quad & -2x^2 + 4x - 5 \\ &= -2(x - 1)^2 - 3\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 + 2x + 1 \\ &= (x + 1)^2\end{aligned}$$

$$\begin{aligned}(18) \quad & -4x^2 - 8x - 9 \\ &= -4(x + 1)^2 - 5\end{aligned}$$

$$\begin{aligned}(28) \quad & 4x^2 + 8x + 2 \\ &= 4(x + 1)^2 - 2\end{aligned}$$

$$\begin{aligned}(9) \quad & 4x^2 + 24x + 33 \\ &= 4(x + 3)^2 - 3\end{aligned}$$

$$\begin{aligned}(19) \quad & 4x^2 + 8x + 4 \\ &= 4(x + 1)^2\end{aligned}$$

$$\begin{aligned}(29) \quad & -x^2 - 10x - 28 \\ &= -(x + 5)^2 - 3\end{aligned}$$

$$\begin{aligned}(10) \quad & -3x^2 + 18x - 23 \\ &= -3(x - 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(20) \quad & -3x^2 + 12x - 10 \\ &= -3(x - 2)^2 + 2\end{aligned}$$

$$\begin{aligned}(30) \quad & -2x^2 - 12x - 18 \\ &= -2(x + 3)^2\end{aligned}$$

$$\begin{aligned}(1) \quad & x^2 + 6x + 5 \\ &= (x + 3)^2 - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & -x^2 + 6x - 5 \\ &= -(x - 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(21) \quad & 5x^2 - 30x + 49 \\ &= 5(x - 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(2) \quad & 5x^2 - 40x + 83 \\ &= 5(x - 4)^2 + 3\end{aligned}$$

$$\begin{aligned}(12) \quad & -2x^2 - 4x + 2 \\ &= -2(x + 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(22) \quad & x^2 + 2x - 2 \\ &= (x + 1)^2 - 3\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 + 6x - 1 \\ &= -3(x - 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 - 4x + 7 \\ &= (x - 2)^2 + 3\end{aligned}$$

$$\begin{aligned}(23) \quad & 3x^2 - 6x + 7 \\ &= 3(x - 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(4) \quad & -3x^2 - 6x + 2 \\ &= -3(x + 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & -2x^2 + 16x - 35 \\ &= -2(x - 4)^2 - 3\end{aligned}$$

$$\begin{aligned}(24) \quad & -5x^2 - 10x - 4 \\ &= -5(x + 1)^2 + 1\end{aligned}$$

$$\begin{aligned}(5) \quad & -4x^2 + 8x - 3 \\ &= -4(x - 1)^2 + 1\end{aligned}$$

$$\begin{aligned}(15) \quad & 5x^2 + 50x + 125 \\ &= 5(x + 5)^2\end{aligned}$$

$$\begin{aligned}(25) \quad & x^2 + 6x + 13 \\ &= (x + 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(6) \quad & 2x^2 - 8x + 6 \\ &= 2(x - 2)^2 - 2\end{aligned}$$

$$\begin{aligned}(16) \quad & 4x^2 - 32x + 61 \\ &= 4(x - 4)^2 - 3\end{aligned}$$

$$\begin{aligned}(26) \quad & -4x^2 + 32x - 65 \\ &= -4(x - 4)^2 - 1\end{aligned}$$

$$\begin{aligned}(7) \quad & -3x^2 + 18x - 31 \\ &= -3(x - 3)^2 - 4\end{aligned}$$

$$\begin{aligned}(17) \quad & x^2 - 10x + 20 \\ &= (x - 5)^2 - 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -2x^2 + 4x - 6 \\ &= -2(x - 1)^2 - 4\end{aligned}$$

$$\begin{aligned}(8) \quad & -5x^2 + 30x - 44 \\ &= -5(x - 3)^2 + 1\end{aligned}$$

$$\begin{aligned}(18) \quad & -2x^2 + 4x - 3 \\ &= -2(x - 1)^2 - 1\end{aligned}$$

$$\begin{aligned}(28) \quad & -4x^2 - 8x \\ &= -4(x + 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(9) \quad & -5x^2 - 40x - 80 \\ &= -5(x + 4)^2\end{aligned}$$

$$\begin{aligned}(19) \quad & 5x^2 + 10x + 9 \\ &= 5(x + 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(29) \quad & -x^2 - 10x - 20 \\ &= -(x + 5)^2 + 5\end{aligned}$$

$$\begin{aligned}(10) \quad & x^2 - 4x + 2 \\ &= (x - 2)^2 - 2\end{aligned}$$

$$\begin{aligned}(20) \quad & x^2 - 2x + 3 \\ &= (x - 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(30) \quad & 3x^2 - 18x + 26 \\ &= 3(x - 3)^2 - 1\end{aligned}$$

$$\begin{aligned}(1) \quad & -2x^2 + 4x - 3 \\ & = -2(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(11) \quad & 5x^2 + 30x + 42 \\ & = 5(x+3)^2 - 3\end{aligned}$$

$$\begin{aligned}(21) \quad & -5x^2 - 40x - 75 \\ & = -5(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(2) \quad & -3x^2 - 6x - 8 \\ & = -3(x+1)^2 - 5\end{aligned}$$

$$\begin{aligned}(12) \quad & -4x^2 - 16x - 17 \\ & = -4(x+2)^2 - 1\end{aligned}$$

$$\begin{aligned}(22) \quad & -2x^2 + 4x - 5 \\ & = -2(x-1)^2 - 3\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 - 6x \\ & = -3(x+1)^2 + 3\end{aligned}$$

$$\begin{aligned}(13) \quad & 5x^2 - 40x + 82 \\ & = 5(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(23) \quad & 3x^2 - 24x + 51 \\ & = 3(x-4)^2 + 3\end{aligned}$$

$$\begin{aligned}(4) \quad & -4x^2 + 16x - 18 \\ & = -4(x-2)^2 - 2\end{aligned}$$

$$\begin{aligned}(14) \quad & 5x^2 + 20x + 16 \\ & = 5(x+2)^2 - 4\end{aligned}$$

$$\begin{aligned}(24) \quad & 3x^2 - 30x + 70 \\ & = 3(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(5) \quad & -4x^2 - 32x - 62 \\ & = -4(x+4)^2 + 2\end{aligned}$$

$$\begin{aligned}(15) \quad & 3x^2 - 12x + 13 \\ & = 3(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(25) \quad & 3x^2 + 6x + 5 \\ & = 3(x+1)^2 + 2\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 + 2x - 3 \\ & = (x+1)^2 - 4\end{aligned}$$

$$\begin{aligned}(16) \quad & x^2 + 8x + 17 \\ & = (x+4)^2 + 1\end{aligned}$$

$$\begin{aligned}(26) \quad & -5x^2 - 50x - 125 \\ & = -5(x+5)^2\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 + 2x + 3 \\ & = -(x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(17) \quad & 5x^2 + 40x + 81 \\ & = 5(x+4)^2 + 1\end{aligned}$$

$$\begin{aligned}(27) \quad & -5x^2 - 20x - 21 \\ & = -5(x+2)^2 - 1\end{aligned}$$

$$\begin{aligned}(8) \quad & -3x^2 - 24x - 43 \\ & = -3(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(18) \quad & -5x^2 - 30x - 40 \\ & = -5(x+3)^2 + 5\end{aligned}$$

$$\begin{aligned}(28) \quad & 2x^2 + 16x + 36 \\ & = 2(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(9) \quad & -2x^2 - 4x + 3 \\ & = -2(x+1)^2 + 5\end{aligned}$$

$$\begin{aligned}(19) \quad & 3x^2 - 24x + 53 \\ & = 3(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(29) \quad & -2x^2 - 12x - 18 \\ & = -2(x+3)^2\end{aligned}$$

$$\begin{aligned}(10) \quad & -2x^2 + 16x - 34 \\ & = -2(x-4)^2 - 2\end{aligned}$$

$$\begin{aligned}(20) \quad & 3x^2 + 24x + 45 \\ & = 3(x+4)^2 - 3\end{aligned}$$

$$\begin{aligned}(30) \quad & -5x^2 - 40x - 81 \\ & = -5(x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(1) \quad & -4x^2 + 24x - 32 \\ & = -4(x-3)^2 + 4\end{aligned}$$

$$\begin{aligned}(11) \quad & 4x^2 + 40x + 105 \\ & = 4(x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(21) \quad & -3x^2 - 18x - 24 \\ & = -3(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(2) \quad & -2x^2 - 4x - 6 \\ & = -2(x+1)^2 - 4\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 + 30x + 49 \\ & = 5(x+3)^2 + 4\end{aligned}$$

$$\begin{aligned}(22) \quad & -5x^2 - 20x - 18 \\ & = -5(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(3) \quad & 2x^2 + 12x + 13 \\ & = 2(x+3)^2 - 5\end{aligned}$$

$$\begin{aligned}(13) \quad & -4x^2 + 40x - 104 \\ & = -4(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(23) \quad & -4x^2 + 40x - 97 \\ & = -4(x-5)^2 + 3\end{aligned}$$

$$\begin{aligned}(4) \quad & 4x^2 + 40x + 104 \\ & = 4(x+5)^2 + 4\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 - 40x - 103 \\ & = -4(x+5)^2 - 3\end{aligned}$$

$$\begin{aligned}(24) \quad & -5x^2 + 50x - 129 \\ & = -5(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(5) \quad & 3x^2 - 24x + 45 \\ & = 3(x-4)^2 - 3\end{aligned}$$

$$\begin{aligned}(15) \quad & 5x^2 - 40x + 80 \\ & = 5(x-4)^2\end{aligned}$$

$$\begin{aligned}(25) \quad & -5x^2 + 50x - 123 \\ & = -5(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(6) \quad & -x^2 + 6x - 8 \\ & = -(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(16) \quad & 5x^2 + 20x + 24 \\ & = 5(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(26) \quad & -5x^2 + 30x - 41 \\ & = -5(x-3)^2 + 4\end{aligned}$$

$$\begin{aligned}(7) \quad & x^2 - 4x + 6 \\ & = (x-2)^2 + 2\end{aligned}$$

$$\begin{aligned}(17) \quad & -2x^2 - 16x - 28 \\ & = -2(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(27) \quad & 2x^2 + 8x + 7 \\ & = 2(x+2)^2 - 1\end{aligned}$$

$$\begin{aligned}(8) \quad & -5x^2 + 50x - 124 \\ & = -5(x-5)^2 + 1\end{aligned}$$

$$\begin{aligned}(18) \quad & 3x^2 + 18x + 32 \\ & = 3(x+3)^2 + 5\end{aligned}$$

$$\begin{aligned}(28) \quad & -5x^2 + 30x - 46 \\ & = -5(x-3)^2 - 1\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 + 2x - 1 \\ & = -(x-1)^2\end{aligned}$$

$$\begin{aligned}(19) \quad & 5x^2 + 20x + 16 \\ & = 5(x+2)^2 - 4\end{aligned}$$

$$\begin{aligned}(29) \quad & -x^2 + 6x - 6 \\ & = -(x-3)^2 + 3\end{aligned}$$

$$\begin{aligned}(10) \quad & -3x^2 - 12x - 9 \\ & = -3(x+2)^2 + 3\end{aligned}$$

$$\begin{aligned}(20) \quad & -5x^2 - 10x \\ & = -5(x+1)^2 + 5\end{aligned}$$

$$\begin{aligned}(30) \quad & x^2 + 4x + 1 \\ & = (x+2)^2 - 3\end{aligned}$$

$$\begin{aligned}(1) \quad & 3x^2 + 12x + 10 \\ &= 3(x+2)^2 - 2\end{aligned}$$

$$\begin{aligned}(11) \quad & -2x^2 + 12x - 21 \\ &= -2(x-3)^2 - 3\end{aligned}$$

$$\begin{aligned}(21) \quad & -2x^2 + 16x - 31 \\ &= -2(x-4)^2 + 1\end{aligned}$$

$$\begin{aligned}(2) \quad & 4x^2 + 40x + 104 \\ &= 4(x+5)^2 + 4\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 - 6x + 4 \\ &= 3(x-1)^2 + 1\end{aligned}$$

$$\begin{aligned}(22) \quad & x^2 - 2x - 1 \\ &= (x-1)^2 - 2\end{aligned}$$

$$\begin{aligned}(3) \quad & -4x^2 + 40x - 105 \\ &= -4(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(13) \quad & 4x^2 + 8x + 4 \\ &= 4(x+1)^2\end{aligned}$$

$$\begin{aligned}(23) \quad & -x^2 + 4x - 5 \\ &= -(x-2)^2 - 1\end{aligned}$$

$$\begin{aligned}(4) \quad & -2x^2 - 12x - 18 \\ &= -2(x+3)^2\end{aligned}$$

$$\begin{aligned}(14) \quad & 3x^2 + 12x + 16 \\ &= 3(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(24) \quad & -3x^2 - 18x - 29 \\ &= -3(x+3)^2 - 2\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 - 10x - 30 \\ &= -(x+5)^2 - 5\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 + 4x + 6 \\ &= 2(x+1)^2 + 4\end{aligned}$$

$$\begin{aligned}(25) \quad & 4x^2 - 24x + 38 \\ &= 4(x-3)^2 + 2\end{aligned}$$

$$\begin{aligned}(6) \quad & 3x^2 + 30x + 75 \\ &= 3(x+5)^2\end{aligned}$$

$$\begin{aligned}(16) \quad & 3x^2 - 6x + 2 \\ &= 3(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(26) \quad & -x^2 + 10x - 23 \\ &= -(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 + 10x - 25 \\ &= -(x-5)^2\end{aligned}$$

$$\begin{aligned}(17) \quad & -4x^2 - 16x - 11 \\ &= -4(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(27) \quad & x^2 + 6x + 12 \\ &= (x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(8) \quad & 4x^2 + 24x + 38 \\ &= 4(x+3)^2 + 2\end{aligned}$$

$$\begin{aligned}(18) \quad & x^2 - 6x + 13 \\ &= (x-3)^2 + 4\end{aligned}$$

$$\begin{aligned}(28) \quad & x^2 - 6x + 11 \\ &= (x-3)^2 + 2\end{aligned}$$

$$\begin{aligned}(9) \quad & -2x^2 - 4x - 5 \\ &= -2(x+1)^2 - 3\end{aligned}$$

$$\begin{aligned}(19) \quad & 2x^2 - 8x + 6 \\ &= 2(x-2)^2 - 2\end{aligned}$$

$$\begin{aligned}(29) \quad & 3x^2 - 18x + 28 \\ &= 3(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(10) \quad & -x^2 - 2x + 3 \\ &= -(x+1)^2 + 4\end{aligned}$$

$$\begin{aligned}(20) \quad & 4x^2 - 32x + 67 \\ &= 4(x-4)^2 + 3\end{aligned}$$

$$\begin{aligned}(30) \quad & 3x^2 - 30x + 80 \\ &= 3(x-5)^2 + 5\end{aligned}$$

$$(1) \ x^2 - 10x + 26 \\ = (x - 5)^2 + 1$$

$$(11) \ -4x^2 - 8x - 1 \\ = -4(x + 1)^2 + 3$$

$$(21) \ x^2 - 6x + 12 \\ = (x - 3)^2 + 3$$

$$(2) \ x^2 + 4x + 2 \\ = (x + 2)^2 - 2$$

$$(12) \ 5x^2 + 10x + 7 \\ = 5(x + 1)^2 + 2$$

$$(22) \ x^2 + 6x + 9 \\ = (x + 3)^2$$

$$(3) \ -4x^2 + 32x - 61 \\ = -4(x - 4)^2 + 3$$

$$(13) \ -2x^2 + 20x - 46 \\ = -2(x - 5)^2 + 4$$

$$(23) \ -x^2 - 4x - 1 \\ = -(x + 2)^2 + 3$$

$$(4) \ 4x^2 + 8x + 9 \\ = 4(x + 1)^2 + 5$$

$$(14) \ -x^2 + 4x - 2 \\ = -(x - 2)^2 + 2$$

$$(24) \ 4x^2 + 8x + 3 \\ = 4(x + 1)^2 - 1$$

$$(5) \ -2x^2 + 8x - 3 \\ = -2(x - 2)^2 + 5$$

$$(15) \ -3x^2 + 30x - 72 \\ = -3(x - 5)^2 + 3$$

$$(25) \ 5x^2 + 40x + 76 \\ = 5(x + 4)^2 - 4$$

$$(6) \ 3x^2 + 6x - 2 \\ = 3(x + 1)^2 - 5$$

$$(16) \ 5x^2 + 20x + 21 \\ = 5(x + 2)^2 + 1$$

$$(26) \ x^2 - 4x + 8 \\ = (x - 2)^2 + 4$$

$$(7) \ x^2 + 10x + 23 \\ = (x + 5)^2 - 2$$

$$(17) \ 2x^2 - 20x + 47 \\ = 2(x - 5)^2 - 3$$

$$(27) \ 4x^2 + 32x + 62 \\ = 4(x + 4)^2 - 2$$

$$(8) \ -4x^2 + 40x - 100 \\ = -4(x - 5)^2$$

$$(18) \ -2x^2 - 16x - 32 \\ = -2(x + 4)^2$$

$$(28) \ -5x^2 - 10x - 2 \\ = -5(x + 1)^2 + 3$$

$$(9) \ x^2 - 6x + 6 \\ = (x - 3)^2 - 3$$

$$(19) \ x^2 + 6x + 14 \\ = (x + 3)^2 + 5$$

$$(29) \ 5x^2 + 50x + 129 \\ = 5(x + 5)^2 + 4$$

$$(10) \ x^2 + 8x + 16 \\ = (x + 4)^2$$

$$(20) \ -x^2 - 8x - 15 \\ = -(x + 4)^2 + 1$$

$$(30) \ x^2 - 2x + 3 \\ = (x - 1)^2 + 2$$

$$\begin{aligned}(1) \quad & 4x^2 + 40x + 103 \\ &= 4(x+5)^2 + 3\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 - 24x - 45 \\ &= -3(x+4)^2 + 3\end{aligned}$$

$$\begin{aligned}(21) \quad & -x^2 + 4x + 1 \\ &= -(x-2)^2 + 5\end{aligned}$$

$$\begin{aligned}(2) \quad & 3x^2 - 24x + 47 \\ &= 3(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(12) \quad & -x^2 - 6x - 9 \\ &= -(x+3)^2\end{aligned}$$

$$\begin{aligned}(22) \quad & -3x^2 + 12x - 10 \\ &= -3(x-2)^2 + 2\end{aligned}$$

$$\begin{aligned}(3) \quad & 3x^2 + 30x + 76 \\ &= 3(x+5)^2 + 1\end{aligned}$$

$$\begin{aligned}(13) \quad & -2x^2 + 16x - 33 \\ &= -2(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(23) \quad & x^2 - 4x + 7 \\ &= (x-2)^2 + 3\end{aligned}$$

$$\begin{aligned}(4) \quad & 2x^2 - 4x + 1 \\ &= 2(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(14) \quad & -x^2 - 10x - 28 \\ &= -(x+5)^2 - 3\end{aligned}$$

$$\begin{aligned}(24) \quad & 4x^2 + 32x + 62 \\ &= 4(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 + 8x - 14 \\ &= -(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(15) \quad & 3x^2 + 12x + 7 \\ &= 3(x+2)^2 - 5\end{aligned}$$

$$\begin{aligned}(25) \quad & -4x^2 - 40x - 101 \\ &= -4(x+5)^2 - 1\end{aligned}$$

$$\begin{aligned}(6) \quad & 4x^2 - 24x + 40 \\ &= 4(x-3)^2 + 4\end{aligned}$$

$$\begin{aligned}(16) \quad & 3x^2 + 24x + 43 \\ &= 3(x+4)^2 - 5\end{aligned}$$

$$\begin{aligned}(26) \quad & 5x^2 + 40x + 82 \\ &= 5(x+4)^2 + 2\end{aligned}$$

$$\begin{aligned}(7) \quad & x^2 - 10x + 26 \\ &= (x-5)^2 + 1\end{aligned}$$

$$\begin{aligned}(17) \quad & 3x^2 + 18x + 32 \\ &= 3(x+3)^2 + 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -5x^2 - 40x - 82 \\ &= -5(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 - 2x + 5 \\ &= (x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(18) \quad & 5x^2 + 40x + 80 \\ &= 5(x+4)^2\end{aligned}$$

$$\begin{aligned}(28) \quad & -5x^2 + 10x - 6 \\ &= -5(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 - 8x - 18 \\ &= -(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(19) \quad & 4x^2 + 8x + 3 \\ &= 4(x+1)^2 - 1\end{aligned}$$

$$\begin{aligned}(29) \quad & x^2 - 2x + 5 \\ &= (x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(10) \quad & x^2 + 10x + 30 \\ &= (x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(20) \quad & 4x^2 + 8x + 9 \\ &= 4(x+1)^2 + 5\end{aligned}$$

$$\begin{aligned}(30) \quad & -4x^2 + 32x - 62 \\ &= -4(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(1) \quad & -x^2 - 8x - 18 \\ & = -(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(11) \quad & -5x^2 - 10x - 5 \\ & = -5(x+1)^2\end{aligned}$$

$$\begin{aligned}(21) \quad & 2x^2 + 20x + 46 \\ & = 2(x+5)^2 - 4\end{aligned}$$

$$\begin{aligned}(2) \quad & -5x^2 + 40x - 76 \\ & = -5(x-4)^2 + 4\end{aligned}$$

$$\begin{aligned}(12) \quad & -3x^2 - 6x - 4 \\ & = -3(x+1)^2 - 1\end{aligned}$$

$$\begin{aligned}(22) \quad & 4x^2 + 40x + 96 \\ & = 4(x+5)^2 - 4\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 + 24x - 46 \\ & = -3(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & -5x^2 + 30x - 43 \\ & = -5(x-3)^2 + 2\end{aligned}$$

$$\begin{aligned}(23) \quad & x^2 - 6x + 7 \\ & = (x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(4) \quad & -4x^2 + 16x - 11 \\ & = -4(x-2)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & 5x^2 - 50x + 124 \\ & = 5(x-5)^2 - 1\end{aligned}$$

$$\begin{aligned}(24) \quad & 3x^2 - 18x + 29 \\ & = 3(x-3)^2 + 2\end{aligned}$$

$$\begin{aligned}(5) \quad & 4x^2 + 8x \\ & = 4(x+1)^2 - 4\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 - 24x - 39 \\ & = -4(x+3)^2 - 3\end{aligned}$$

$$\begin{aligned}(25) \quad & -4x^2 - 8x - 5 \\ & = -4(x+1)^2 - 1\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 - 6x + 4 \\ & = (x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(16) \quad & x^2 + 10x + 22 \\ & = (x+5)^2 - 3\end{aligned}$$

$$\begin{aligned}(26) \quad & -3x^2 - 30x - 77 \\ & = -3(x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 - 4x + 1 \\ & = -(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(17) \quad & -2x^2 - 8x - 11 \\ & = -2(x+2)^2 - 3\end{aligned}$$

$$\begin{aligned}(27) \quad & 5x^2 + 40x + 84 \\ & = 5(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(8) \quad & -3x^2 + 30x - 73 \\ & = -3(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(18) \quad & -2x^2 - 8x - 5 \\ & = -2(x+2)^2 + 3\end{aligned}$$

$$\begin{aligned}(28) \quad & -x^2 + 10x - 24 \\ & = -(x-5)^2 + 1\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 + 4x - 3 \\ & = -(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(19) \quad & -5x^2 + 30x - 45 \\ & = -5(x-3)^2\end{aligned}$$

$$\begin{aligned}(29) \quad & 5x^2 + 30x + 45 \\ & = 5(x+3)^2\end{aligned}$$

$$\begin{aligned}(10) \quad & -x^2 + 6x - 14 \\ & = -(x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(20) \quad & 4x^2 - 32x + 60 \\ & = 4(x-4)^2 - 4\end{aligned}$$

$$\begin{aligned}(30) \quad & 4x^2 + 24x + 33 \\ & = 4(x+3)^2 - 3\end{aligned}$$

$$\begin{aligned}(1) \quad & 4x^2 + 16x + 17 \\ &= 4(x+2)^2 + 1\end{aligned}$$

$$\begin{aligned}(11) \quad & 3x^2 - 24x + 43 \\ &= 3(x-4)^2 - 5\end{aligned}$$

$$\begin{aligned}(21) \quad & -3x^2 - 12x - 7 \\ &= -3(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(2) \quad & -5x^2 + 30x - 49 \\ &= -5(x-3)^2 - 4\end{aligned}$$

$$\begin{aligned}(12) \quad & -x^2 + 4x - 3 \\ &= -(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(22) \quad & 3x^2 - 30x + 70 \\ &= 3(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(3) \quad & 5x^2 + 40x + 78 \\ &= 5(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(13) \quad & -3x^2 - 12x - 17 \\ &= -3(x+2)^2 - 5\end{aligned}$$

$$\begin{aligned}(23) \quad & -x^2 + 6x - 14 \\ &= -(x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(4) \quad & -4x^2 + 24x - 39 \\ &= -4(x-3)^2 - 3\end{aligned}$$

$$\begin{aligned}(14) \quad & -x^2 - 10x - 22 \\ &= -(x+5)^2 + 3\end{aligned}$$

$$\begin{aligned}(24) \quad & -5x^2 - 20x - 22 \\ &= -5(x+2)^2 - 2\end{aligned}$$

$$\begin{aligned}(5) \quad & 5x^2 + 20x + 22 \\ &= 5(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(15) \quad & -5x^2 + 40x - 82 \\ &= -5(x-4)^2 - 2\end{aligned}$$

$$\begin{aligned}(25) \quad & -x^2 + 6x - 9 \\ &= -(x-3)^2\end{aligned}$$

$$\begin{aligned}(6) \quad & 5x^2 + 20x + 24 \\ &= 5(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(16) \quad & -x^2 + 8x - 17 \\ &= -(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(26) \quad & -3x^2 - 30x - 71 \\ &= -3(x+5)^2 + 4\end{aligned}$$

$$\begin{aligned}(7) \quad & 2x^2 - 4x - 3 \\ &= 2(x-1)^2 - 5\end{aligned}$$

$$\begin{aligned}(17) \quad & -3x^2 + 6x - 8 \\ &= -3(x-1)^2 - 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -2x^2 - 12x - 15 \\ &= -2(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(8) \quad & 2x^2 - 12x + 16 \\ &= 2(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(18) \quad & 5x^2 - 10x + 4 \\ &= 5(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(28) \quad & -4x^2 - 24x - 35 \\ &= -4(x+3)^2 + 1\end{aligned}$$

$$\begin{aligned}(9) \quad & 5x^2 + 30x + 45 \\ &= 5(x+3)^2\end{aligned}$$

$$\begin{aligned}(19) \quad & -2x^2 - 4x + 1 \\ &= -2(x+1)^2 + 3\end{aligned}$$

$$\begin{aligned}(29) \quad & 4x^2 - 40x + 104 \\ &= 4(x-5)^2 + 4\end{aligned}$$

$$\begin{aligned}(10) \quad & 2x^2 - 16x + 36 \\ &= 2(x-4)^2 + 4\end{aligned}$$

$$\begin{aligned}(20) \quad & -3x^2 - 24x - 45 \\ &= -3(x+4)^2 + 3\end{aligned}$$

$$\begin{aligned}(30) \quad & -5x^2 - 40x - 79 \\ &= -5(x+4)^2 + 1\end{aligned}$$

$$\begin{aligned}(1) \quad & -3x^2 - 24x - 52 \\ & = -3(x+4)^2 - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & x^2 + 4x - 1 \\ & = (x+2)^2 - 5\end{aligned}$$

$$\begin{aligned}(21) \quad & 5x^2 + 10x + 3 \\ & = 5(x+1)^2 - 2\end{aligned}$$

$$\begin{aligned}(2) \quad & 3x^2 + 6x - 2 \\ & = 3(x+1)^2 - 5\end{aligned}$$

$$\begin{aligned}(12) \quad & -2x^2 + 16x - 33 \\ & = -2(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(22) \quad & -2x^2 + 12x - 21 \\ & = -2(x-3)^2 - 3\end{aligned}$$

$$\begin{aligned}(3) \quad & -4x^2 + 40x - 104 \\ & = -4(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(13) \quad & 4x^2 - 40x + 100 \\ & = 4(x-5)^2\end{aligned}$$

$$\begin{aligned}(23) \quad & 2x^2 + 8x + 12 \\ & = 2(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(4) \quad & 3x^2 + 6x + 7 \\ & = 3(x+1)^2 + 4\end{aligned}$$

$$\begin{aligned}(14) \quad & x^2 - 4x + 2 \\ & = (x-2)^2 - 2\end{aligned}$$

$$\begin{aligned}(24) \quad & x^2 + 2x + 2 \\ & = (x+1)^2 + 1\end{aligned}$$

$$\begin{aligned}(5) \quad & -4x^2 + 16x - 16 \\ & = -4(x-2)^2\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 + 32x - 59 \\ & = -4(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(25) \quad & x^2 - 8x + 18 \\ & = (x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(6) \quad & -x^2 - 2x + 1 \\ & = -(x+1)^2 + 2\end{aligned}$$

$$\begin{aligned}(16) \quad & -x^2 - 2x + 2 \\ & = -(x+1)^2 + 3\end{aligned}$$

$$\begin{aligned}(26) \quad & x^2 + 10x + 23 \\ & = (x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -4x^2 + 8x - 1 \\ & = -4(x-1)^2 + 3\end{aligned}$$

$$\begin{aligned}(17) \quad & 5x^2 + 40x + 85 \\ & = 5(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -5x^2 + 10x - 8 \\ & = -5(x-1)^2 - 3\end{aligned}$$

$$\begin{aligned}(8) \quad & -5x^2 + 50x - 127 \\ & = -5(x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(18) \quad & 4x^2 - 32x + 69 \\ & = 4(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(28) \quad & 4x^2 - 32x + 64 \\ & = 4(x-4)^2\end{aligned}$$

$$\begin{aligned}(9) \quad & -2x^2 - 16x - 27 \\ & = -2(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(19) \quad & -5x^2 - 30x - 49 \\ & = -5(x+3)^2 - 4\end{aligned}$$

$$\begin{aligned}(29) \quad & -2x^2 + 20x - 55 \\ & = -2(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(10) \quad & -2x^2 + 20x - 46 \\ & = -2(x-5)^2 + 4\end{aligned}$$

$$\begin{aligned}(20) \quad & -5x^2 - 50x - 122 \\ & = -5(x+5)^2 + 3\end{aligned}$$

$$\begin{aligned}(30) \quad & -4x^2 + 40x - 95 \\ & = -4(x-5)^2 + 5\end{aligned}$$

$$\begin{aligned}(1) \quad & x^2 + 2x + 5 \\ &= (x + 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(11) \quad & 5x^2 - 20x + 17 \\ &= 5(x - 2)^2 - 3\end{aligned}$$

$$\begin{aligned}(21) \quad & 2x^2 + 16x + 31 \\ &= 2(x + 4)^2 - 1\end{aligned}$$

$$\begin{aligned}(2) \quad & -3x^2 - 24x - 49 \\ &= -3(x + 4)^2 - 1\end{aligned}$$

$$\begin{aligned}(12) \quad & 2x^2 - 12x + 15 \\ &= 2(x - 3)^2 - 3\end{aligned}$$

$$\begin{aligned}(22) \quad & x^2 - 8x + 18 \\ &= (x - 4)^2 + 2\end{aligned}$$

$$\begin{aligned}(3) \quad & 4x^2 - 16x + 12 \\ &= 4(x - 2)^2 - 4\end{aligned}$$

$$\begin{aligned}(13) \quad & 3x^2 - 24x + 45 \\ &= 3(x - 4)^2 - 3\end{aligned}$$

$$\begin{aligned}(23) \quad & -2x^2 - 4x + 3 \\ &= -2(x + 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(4) \quad & -3x^2 + 24x - 47 \\ &= -3(x - 4)^2 + 1\end{aligned}$$

$$\begin{aligned}(14) \quad & -2x^2 + 12x - 20 \\ &= -2(x - 3)^2 - 2\end{aligned}$$

$$\begin{aligned}(24) \quad & 3x^2 - 30x + 74 \\ &= 3(x - 5)^2 - 1\end{aligned}$$

$$\begin{aligned}(5) \quad & 5x^2 + 20x + 24 \\ &= 5(x + 2)^2 + 4\end{aligned}$$

$$\begin{aligned}(15) \quad & -3x^2 + 24x - 47 \\ &= -3(x - 4)^2 + 1\end{aligned}$$

$$\begin{aligned}(25) \quad & x^2 + 2x + 5 \\ &= (x + 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 + 6x + 11 \\ &= (x + 3)^2 + 2\end{aligned}$$

$$\begin{aligned}(16) \quad & -3x^2 - 30x - 71 \\ &= -3(x + 5)^2 + 4\end{aligned}$$

$$\begin{aligned}(26) \quad & -3x^2 - 30x - 77 \\ &= -3(x + 5)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -4x^2 - 32x - 68 \\ &= -4(x + 4)^2 - 4\end{aligned}$$

$$\begin{aligned}(17) \quad & -x^2 - 2x \\ &= -(x + 1)^2 + 1\end{aligned}$$

$$\begin{aligned}(27) \quad & x^2 + 6x + 4 \\ &= (x + 3)^2 - 5\end{aligned}$$

$$\begin{aligned}(8) \quad & 4x^2 + 16x + 16 \\ &= 4(x + 2)^2\end{aligned}$$

$$\begin{aligned}(18) \quad & x^2 + 4x - 1 \\ &= (x + 2)^2 - 5\end{aligned}$$

$$\begin{aligned}(28) \quad & -x^2 + 10x - 22 \\ &= -(x - 5)^2 + 3\end{aligned}$$

$$\begin{aligned}(9) \quad & -4x^2 - 32x - 65 \\ &= -4(x + 4)^2 - 1\end{aligned}$$

$$\begin{aligned}(19) \quad & 2x^2 + 20x + 48 \\ &= 2(x + 5)^2 - 2\end{aligned}$$

$$\begin{aligned}(29) \quad & 4x^2 - 24x + 34 \\ &= 4(x - 3)^2 - 2\end{aligned}$$

$$\begin{aligned}(10) \quad & 2x^2 - 8x + 9 \\ &= 2(x - 2)^2 + 1\end{aligned}$$

$$\begin{aligned}(20) \quad & -5x^2 + 10x - 6 \\ &= -5(x - 1)^2 - 1\end{aligned}$$

$$\begin{aligned}(30) \quad & -5x^2 - 30x - 49 \\ &= -5(x + 3)^2 - 4\end{aligned}$$

$$(1) \quad x^2 + 10x + 24 \\ = (x + 5)^2 - 1$$

$$(11) \quad 4x^2 - 24x + 36 \\ = 4(x - 3)^2$$

$$(21) \quad -4x^2 + 32x - 60 \\ = -4(x - 4)^2 + 4$$

$$(2) \quad -5x^2 - 10x \\ = -5(x + 1)^2 + 5$$

$$(12) \quad -5x^2 + 50x - 122 \\ = -5(x - 5)^2 + 3$$

$$(22) \quad -x^2 + 2x + 1 \\ = -(x - 1)^2 + 2$$

$$(3) \quad 4x^2 - 16x + 15 \\ = 4(x - 2)^2 - 1$$

$$(13) \quad -2x^2 + 8x - 3 \\ = -2(x - 2)^2 + 5$$

$$(23) \quad 2x^2 - 16x + 31 \\ = 2(x - 4)^2 - 1$$

$$(4) \quad 4x^2 + 32x + 68 \\ = 4(x + 4)^2 + 4$$

$$(14) \quad x^2 + 2x - 4 \\ = (x + 1)^2 - 5$$

$$(24) \quad -2x^2 - 12x - 23 \\ = -2(x + 3)^2 - 5$$

$$(5) \quad 5x^2 + 10x + 2 \\ = 5(x + 1)^2 - 3$$

$$(15) \quad -x^2 - 2x + 3 \\ = -(x + 1)^2 + 4$$

$$(25) \quad 2x^2 - 12x + 16 \\ = 2(x - 3)^2 - 2$$

$$(6) \quad 3x^2 + 12x + 10 \\ = 3(x + 2)^2 - 2$$

$$(16) \quad 4x^2 + 32x + 69 \\ = 4(x + 4)^2 + 5$$

$$(26) \quad 5x^2 + 30x + 50 \\ = 5(x + 3)^2 + 5$$

$$(7) \quad -5x^2 - 20x - 16 \\ = -5(x + 2)^2 + 4$$

$$(17) \quad 2x^2 + 4x \\ = 2(x + 1)^2 - 2$$

$$(27) \quad -5x^2 - 40x - 75 \\ = -5(x + 4)^2 + 5$$

$$(8) \quad -x^2 + 2x - 1 \\ = -(x - 1)^2$$

$$(18) \quad -3x^2 + 12x - 9 \\ = -3(x - 2)^2 + 3$$

$$(28) \quad 3x^2 - 24x + 50 \\ = 3(x - 4)^2 + 2$$

$$(9) \quad 3x^2 + 18x + 25 \\ = 3(x + 3)^2 - 2$$

$$(19) \quad -3x^2 + 12x - 8 \\ = -3(x - 2)^2 + 4$$

$$(29) \quad 3x^2 - 18x + 23 \\ = 3(x - 3)^2 - 4$$

$$(10) \quad -3x^2 - 12x - 17 \\ = -3(x + 2)^2 - 5$$

$$(20) \quad 2x^2 + 4x - 3 \\ = 2(x + 1)^2 - 5$$

$$(30) \quad x^2 - 6x + 12 \\ = (x - 3)^2 + 3$$

$$\begin{aligned}(1) \quad & 5x^2 + 20x + 25 \\ &= 5(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(11) \quad & -x^2 - 6x - 11 \\ &= -(x+3)^2 - 2\end{aligned}$$

$$\begin{aligned}(21) \quad & -2x^2 + 20x - 55 \\ &= -2(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(2) \quad & -x^2 + 6x - 11 \\ &= -(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(12) \quad & x^2 + 6x + 8 \\ &= (x+3)^2 - 1\end{aligned}$$

$$\begin{aligned}(22) \quad & 5x^2 + 20x + 25 \\ &= 5(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(3) \quad & 5x^2 + 30x + 46 \\ &= 5(x+3)^2 + 1\end{aligned}$$

$$\begin{aligned}(13) \quad & -4x^2 - 8x - 2 \\ &= -4(x+1)^2 + 2\end{aligned}$$

$$\begin{aligned}(23) \quad & 2x^2 + 20x + 49 \\ &= 2(x+5)^2 - 1\end{aligned}$$

$$\begin{aligned}(4) \quad & -x^2 + 4x - 8 \\ &= -(x-2)^2 - 4\end{aligned}$$

$$\begin{aligned}(14) \quad & 3x^2 - 12x + 13 \\ &= 3(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(24) \quad & 4x^2 - 32x + 61 \\ &= 4(x-4)^2 - 3\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 + 2x + 3 \\ &= -(x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 - 8x - 3 \\ &= -4(x+1)^2 + 1\end{aligned}$$

$$\begin{aligned}(25) \quad & -3x^2 + 12x - 16 \\ &= -3(x-2)^2 - 4\end{aligned}$$

$$\begin{aligned}(6) \quad & -4x^2 + 24x - 35 \\ &= -4(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(16) \quad & -4x^2 + 40x - 98 \\ &= -4(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(26) \quad & -5x^2 + 10x - 4 \\ &= -5(x-1)^2 + 1\end{aligned}$$

$$\begin{aligned}(7) \quad & 5x^2 + 30x + 41 \\ &= 5(x+3)^2 - 4\end{aligned}$$

$$\begin{aligned}(17) \quad & -4x^2 - 16x - 11 \\ &= -4(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -5x^2 - 50x - 129 \\ &= -5(x+5)^2 - 4\end{aligned}$$

$$\begin{aligned}(8) \quad & -4x^2 - 32x - 60 \\ &= -4(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(18) \quad & -4x^2 + 24x - 38 \\ &= -4(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(28) \quad & -2x^2 + 16x - 37 \\ &= -2(x-4)^2 - 5\end{aligned}$$

$$\begin{aligned}(9) \quad & -2x^2 - 16x - 34 \\ &= -2(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(19) \quad & -2x^2 - 20x - 54 \\ &= -2(x+5)^2 - 4\end{aligned}$$

$$\begin{aligned}(29) \quad & 2x^2 + 16x + 31 \\ &= 2(x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(10) \quad & 2x^2 - 16x + 32 \\ &= 2(x-4)^2\end{aligned}$$

$$\begin{aligned}(20) \quad & 4x^2 - 32x + 66 \\ &= 4(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(30) \quad & -5x^2 + 40x - 83 \\ &= -5(x-4)^2 - 3\end{aligned}$$

$$\begin{aligned}(1) \quad & 5x^2 - 20x + 15 \\ &= 5(x-2)^2 - 5\end{aligned}$$

$$\begin{aligned}(11) \quad & -4x^2 - 16x - 12 \\ &= -4(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(21) \quad & 5x^2 + 50x + 124 \\ &= 5(x+5)^2 - 1\end{aligned}$$

$$\begin{aligned}(2) \quad & 5x^2 + 10x + 2 \\ &= 5(x+1)^2 - 3\end{aligned}$$

$$\begin{aligned}(12) \quad & -3x^2 - 30x - 74 \\ &= -3(x+5)^2 + 1\end{aligned}$$

$$\begin{aligned}(22) \quad & 3x^2 + 6x + 6 \\ &= 3(x+1)^2 + 3\end{aligned}$$

$$\begin{aligned}(3) \quad & 3x^2 - 18x + 26 \\ &= 3(x-3)^2 - 1\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 - 4x - 2 \\ &= -(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(23) \quad & 4x^2 - 24x + 36 \\ &= 4(x-3)^2\end{aligned}$$

$$\begin{aligned}(4) \quad & 2x^2 + 12x + 23 \\ &= 2(x+3)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & 2x^2 - 12x + 14 \\ &= 2(x-3)^2 - 4\end{aligned}$$

$$\begin{aligned}(24) \quad & -3x^2 - 6x + 1 \\ &= -3(x+1)^2 + 4\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 + 10x - 30 \\ &= -(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(15) \quad & 4x^2 - 16x + 17 \\ &= 4(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(25) \quad & 4x^2 + 24x + 36 \\ &= 4(x+3)^2\end{aligned}$$

$$\begin{aligned}(6) \quad & -5x^2 + 50x - 126 \\ &= -5(x-5)^2 - 1\end{aligned}$$

$$\begin{aligned}(16) \quad & -5x^2 - 10x - 8 \\ &= -5(x+1)^2 - 3\end{aligned}$$

$$\begin{aligned}(26) \quad & -2x^2 + 4x - 3 \\ &= -2(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(7) \quad & -4x^2 - 32x - 60 \\ &= -4(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(17) \quad & 2x^2 - 12x + 19 \\ &= 2(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(27) \quad & -2x^2 + 4x - 5 \\ &= -2(x-1)^2 - 3\end{aligned}$$

$$\begin{aligned}(8) \quad & -x^2 + 10x - 23 \\ &= -(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(18) \quad & 2x^2 - 8x + 12 \\ &= 2(x-2)^2 + 4\end{aligned}$$

$$\begin{aligned}(28) \quad & 4x^2 + 32x + 64 \\ &= 4(x+4)^2\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 - 6x - 2 \\ &= 3(x-1)^2 - 5\end{aligned}$$

$$\begin{aligned}(19) \quad & 3x^2 + 30x + 73 \\ &= 3(x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(29) \quad & 3x^2 + 24x + 50 \\ &= 3(x+4)^2 + 2\end{aligned}$$

$$\begin{aligned}(10) \quad & 4x^2 + 32x + 63 \\ &= 4(x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(20) \quad & -x^2 + 10x - 25 \\ &= -(x-5)^2\end{aligned}$$

$$\begin{aligned}(30) \quad & 4x^2 + 24x + 39 \\ &= 4(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(1) \quad & 5x^2 + 40x + 77 \\ &= 5(x+4)^2 - 3\end{aligned}$$

$$\begin{aligned}(11) \quad & -2x^2 + 20x - 48 \\ &= -2(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(21) \quad & -4x^2 + 40x - 104 \\ &= -4(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(2) \quad & -5x^2 + 40x - 78 \\ &= -5(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 - 50x + 128 \\ &= 5(x-5)^2 + 3\end{aligned}$$

$$\begin{aligned}(22) \quad & 2x^2 - 16x + 37 \\ &= 2(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 + 6x - 3 \\ &= -3(x-1)^2\end{aligned}$$

$$\begin{aligned}(13) \quad & -4x^2 + 16x - 20 \\ &= -4(x-2)^2 - 4\end{aligned}$$

$$\begin{aligned}(23) \quad & 4x^2 - 24x + 32 \\ &= 4(x-3)^2 - 4\end{aligned}$$

$$\begin{aligned}(4) \quad & 5x^2 - 50x + 130 \\ &= 5(x-5)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 - 16x - 12 \\ &= -4(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(24) \quad & -5x^2 - 20x - 18 \\ &= -5(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(5) \quad & 2x^2 + 20x + 49 \\ &= 2(x+5)^2 - 1\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 + 8x + 4 \\ &= 2(x+2)^2 - 4\end{aligned}$$

$$\begin{aligned}(25) \quad & -x^2 + 6x - 9 \\ &= -(x-3)^2\end{aligned}$$

$$\begin{aligned}(6) \quad & -3x^2 - 12x - 8 \\ &= -3(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(16) \quad & -4x^2 + 40x - 98 \\ &= -4(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(26) \quad & x^2 - 4x + 1 \\ &= (x-2)^2 - 3\end{aligned}$$

$$\begin{aligned}(7) \quad & 5x^2 + 40x + 75 \\ &= 5(x+4)^2 - 5\end{aligned}$$

$$\begin{aligned}(17) \quad & 3x^2 + 6x \\ &= 3(x+1)^2 - 3\end{aligned}$$

$$\begin{aligned}(27) \quad & -x^2 - 10x - 30 \\ &= -(x+5)^2 - 5\end{aligned}$$

$$\begin{aligned}(8) \quad & 2x^2 + 4x - 2 \\ &= 2(x+1)^2 - 4\end{aligned}$$

$$\begin{aligned}(18) \quad & -3x^2 - 12x - 9 \\ &= -3(x+2)^2 + 3\end{aligned}$$

$$\begin{aligned}(28) \quad & -x^2 + 10x - 26 \\ &= -(x-5)^2 - 1\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 + 12x + 9 \\ &= 3(x+2)^2 - 3\end{aligned}$$

$$\begin{aligned}(19) \quad & -2x^2 - 12x - 14 \\ &= -2(x+3)^2 + 4\end{aligned}$$

$$\begin{aligned}(29) \quad & -4x^2 + 32x - 60 \\ &= -4(x-4)^2 + 4\end{aligned}$$

$$\begin{aligned}(10) \quad & -5x^2 + 50x - 129 \\ &= -5(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(20) \quad & 3x^2 + 24x + 47 \\ &= 3(x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(30) \quad & x^2 + 2x - 1 \\ &= (x+1)^2 - 2\end{aligned}$$

$$\begin{aligned}(1) \quad & -4x^2 - 40x - 101 \\ & = -4(x+5)^2 - 1\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 + 18x - 30 \\ & = -3(x-3)^2 - 3\end{aligned}$$

$$\begin{aligned}(21) \quad & 3x^2 - 30x + 73 \\ & = 3(x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(2) \quad & -4x^2 + 32x - 64 \\ & = -4(x-4)^2\end{aligned}$$

$$\begin{aligned}(12) \quad & -2x^2 - 16x - 31 \\ & = -2(x+4)^2 + 1\end{aligned}$$

$$\begin{aligned}(22) \quad & -x^2 - 8x - 11 \\ & = -(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(3) \quad & -5x^2 + 30x - 47 \\ & = -5(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 + 6x - 9 \\ & = -(x-3)^2\end{aligned}$$

$$\begin{aligned}(23) \quad & -2x^2 + 16x - 32 \\ & = -2(x-4)^2\end{aligned}$$

$$\begin{aligned}(4) \quad & 3x^2 + 12x + 14 \\ & = 3(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(14) \quad & x^2 - 10x + 23 \\ & = (x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(24) \quad & 4x^2 - 40x + 98 \\ & = 4(x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(5) \quad & x^2 + 8x + 16 \\ & = (x+4)^2\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 + 8x \\ & = -4(x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(25) \quad & -x^2 - 6x - 8 \\ & = -(x+3)^2 + 1\end{aligned}$$

$$\begin{aligned}(6) \quad & -5x^2 + 40x - 75 \\ & = -5(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(16) \quad & -x^2 + 6x - 5 \\ & = -(x-3)^2 + 4\end{aligned}$$

$$\begin{aligned}(26) \quad & 4x^2 + 40x + 98 \\ & = 4(x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 - 6x - 5 \\ & = -(x+3)^2 + 4\end{aligned}$$

$$\begin{aligned}(17) \quad & x^2 + 2x + 4 \\ & = (x+1)^2 + 3\end{aligned}$$

$$\begin{aligned}(27) \quad & -x^2 - 10x - 27 \\ & = -(x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(8) \quad & -5x^2 + 10x - 1 \\ & = -5(x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(18) \quad & 5x^2 - 30x + 46 \\ & = 5(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(28) \quad & x^2 - 2x - 1 \\ & = (x-1)^2 - 2\end{aligned}$$

$$\begin{aligned}(9) \quad & -5x^2 + 30x - 47 \\ & = -5(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(19) \quad & 5x^2 - 40x + 79 \\ & = 5(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(29) \quad & -5x^2 + 30x - 50 \\ & = -5(x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(10) \quad & -5x^2 - 30x - 48 \\ & = -5(x+3)^2 - 3\end{aligned}$$

$$\begin{aligned}(20) \quad & x^2 - 6x + 10 \\ & = (x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(30) \quad & 5x^2 + 20x + 22 \\ & = 5(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(1) \quad & -2x^2 - 16x - 27 \\ & = -2(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(11) \quad & -5x^2 + 20x - 15 \\ & = -5(x-2)^2 + 5\end{aligned}$$

$$\begin{aligned}(21) \quad & 3x^2 - 12x + 17 \\ & = 3(x-2)^2 + 5\end{aligned}$$

$$\begin{aligned}(2) \quad & 2x^2 - 8x + 6 \\ & = 2(x-2)^2 - 2\end{aligned}$$

$$\begin{aligned}(12) \quad & 2x^2 - 8x + 3 \\ & = 2(x-2)^2 - 5\end{aligned}$$

$$\begin{aligned}(22) \quad & x^2 + 2x + 2 \\ & = (x+1)^2 + 1\end{aligned}$$

$$\begin{aligned}(3) \quad & -x^2 - 8x - 11 \\ & = -(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(13) \quad & -2x^2 - 16x - 35 \\ & = -2(x+4)^2 - 3\end{aligned}$$

$$\begin{aligned}(23) \quad & x^2 + 8x + 15 \\ & = (x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(4) \quad & 2x^2 + 4x + 4 \\ & = 2(x+1)^2 + 2\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 - 40x - 105 \\ & = -4(x+5)^2 - 5\end{aligned}$$

$$\begin{aligned}(24) \quad & -x^2 - 6x - 6 \\ & = -(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(5) \quad & 2x^2 - 8x + 6 \\ & = 2(x-2)^2 - 2\end{aligned}$$

$$\begin{aligned}(15) \quad & -3x^2 + 30x - 76 \\ & = -3(x-5)^2 - 1\end{aligned}$$

$$\begin{aligned}(25) \quad & 5x^2 + 40x + 84 \\ & = 5(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(6) \quad & -x^2 - 2x - 5 \\ & = -(x+1)^2 - 4\end{aligned}$$

$$\begin{aligned}(16) \quad & -4x^2 + 16x - 17 \\ & = -4(x-2)^2 - 1\end{aligned}$$

$$\begin{aligned}(26) \quad & -5x^2 - 50x - 130 \\ & = -5(x+5)^2 - 5\end{aligned}$$

$$\begin{aligned}(7) \quad & -3x^2 - 18x - 24 \\ & = -3(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(17) \quad & 4x^2 - 24x + 31 \\ & = 4(x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -4x^2 - 24x - 32 \\ & = -4(x+3)^2 + 4\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 + 10x + 23 \\ & = (x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(18) \quad & -x^2 + 8x - 11 \\ & = -(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(28) \quad & 5x^2 - 30x + 46 \\ & = 5(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(9) \quad & -5x^2 - 20x - 17 \\ & = -5(x+2)^2 + 3\end{aligned}$$

$$\begin{aligned}(19) \quad & 2x^2 + 4x + 2 \\ & = 2(x+1)^2\end{aligned}$$

$$\begin{aligned}(29) \quad & x^2 + 4x + 6 \\ & = (x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(10) \quad & 2x^2 + 4x + 2 \\ & = 2(x+1)^2\end{aligned}$$

$$\begin{aligned}(20) \quad & 3x^2 - 30x + 78 \\ & = 3(x-5)^2 + 3\end{aligned}$$

$$\begin{aligned}(30) \quad & 2x^2 - 8x + 5 \\ & = 2(x-2)^2 - 3\end{aligned}$$

$$\begin{aligned}(1) \quad & -x^2 + 6x - 8 \\ & = -(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(11) \quad & -5x^2 + 20x - 19 \\ & = -5(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(21) \quad & -4x^2 - 32x - 69 \\ & = -4(x+4)^2 - 5\end{aligned}$$

$$\begin{aligned}(2) \quad & x^2 + 10x + 30 \\ & = (x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 - 24x + 52 \\ & = 3(x-4)^2 + 4\end{aligned}$$

$$\begin{aligned}(22) \quad & 3x^2 + 24x + 47 \\ & = 3(x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 - 12x - 12 \\ & = -3(x+2)^2\end{aligned}$$

$$\begin{aligned}(13) \quad & 4x^2 + 24x + 35 \\ & = 4(x+3)^2 - 1\end{aligned}$$

$$\begin{aligned}(23) \quad & x^2 + 10x + 27 \\ & = (x+5)^2 + 2\end{aligned}$$

$$\begin{aligned}(4) \quad & 4x^2 + 8x - 1 \\ & = 4(x+1)^2 - 5\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 - 16x - 15 \\ & = -4(x+2)^2 + 1\end{aligned}$$

$$\begin{aligned}(24) \quad & x^2 - 6x + 10 \\ & = (x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(5) \quad & -2x^2 + 12x - 20 \\ & = -2(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(15) \quad & -2x^2 - 8x - 5 \\ & = -2(x+2)^2 + 3\end{aligned}$$

$$\begin{aligned}(25) \quad & -3x^2 + 18x - 32 \\ & = -3(x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(6) \quad & 4x^2 + 24x + 41 \\ & = 4(x+3)^2 + 5\end{aligned}$$

$$\begin{aligned}(16) \quad & -3x^2 - 18x - 25 \\ & = -3(x+3)^2 + 2\end{aligned}$$

$$\begin{aligned}(26) \quad & 5x^2 - 50x + 123 \\ & = 5(x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -3x^2 + 24x - 46 \\ & = -3(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(17) \quad & 2x^2 + 12x + 20 \\ & = 2(x+3)^2 + 2\end{aligned}$$

$$\begin{aligned}(27) \quad & -3x^2 + 12x - 12 \\ & = -3(x-2)^2\end{aligned}$$

$$\begin{aligned}(8) \quad & 2x^2 - 20x + 52 \\ & = 2(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(18) \quad & 4x^2 + 40x + 95 \\ & = 4(x+5)^2 - 5\end{aligned}$$

$$\begin{aligned}(28) \quad & 4x^2 - 8x + 8 \\ & = 4(x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(9) \quad & -3x^2 - 30x - 72 \\ & = -3(x+5)^2 + 3\end{aligned}$$

$$\begin{aligned}(19) \quad & -5x^2 - 40x - 84 \\ & = -5(x+4)^2 - 4\end{aligned}$$

$$\begin{aligned}(29) \quad & -3x^2 - 30x - 70 \\ & = -3(x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(10) \quad & 3x^2 - 18x + 30 \\ & = 3(x-3)^2 + 3\end{aligned}$$

$$\begin{aligned}(20) \quad & x^2 - 6x + 14 \\ & = (x-3)^2 + 5\end{aligned}$$

$$\begin{aligned}(30) \quad & 3x^2 - 6x + 1 \\ & = 3(x-1)^2 - 2\end{aligned}$$

3.1.2 平方完成 (難しい)

2 次多項式を平方完成する問題。
式に現れる数は有理数となる。

(1) $-5x^2 - 4x - 4$

(6) $2x^2 + x + 3$

(11) $-3x^2 - 3x + 2$

(2) $-2x^2 + 3x - 3$

(7) $-5x^2 + 4x - 1$

(12) $-4x^2 - 4x + 4$

(3) $-x^2 - 2x - 5$

(8) $-4x^2 + 5x - 1$

(13) $x^2 - x - 3$

(4) $-5x^2 + 2x$

(9) $3x^2 - 5x + 5$

(14) $-x^2 - 4x - 2$

(5) $x^2 - 2x + 5$

(10) $-3x^2 - 4x + 2$

(15) $3x^2 + 3x - 5$

(1) $2x^2 + 2x + 5$

(6) $-x^2 + 4x - 1$

(11) $-3x^2 - 2x + 1$

(2) $-x^2 - 2x - 3$

(7) $4x^2 - 3x - 2$

(12) $3x^2 - 2x$

(3) $5x^2 + 2x + 4$

(8) $4x^2 - 4x + 3$

(13) $x^2 - x + 3$

(4) $-5x^2 - 2x - 1$

(9) $-x^2 + 2x + 4$

(14) $3x^2 + 3x + 1$

(5) $-4x^2 - 2x$

(10) $4x^2 - 3x - 2$

(15) $-5x^2 + 4x + 2$

(1) $-3x^2 + 2x + 5$

(6) $-3x^2 + 2x + 4$

(11) $-3x^2 - 5x - 4$

(2) $x^2 - 3x - 1$

(7) $-4x^2 - 3x - 5$

(12) $-5x^2 - 3x$

(3) $-2x^2 + 4x - 1$

(8) $x^2 - x - 5$

(13) $x^2 - 3x + 4$

(4) $-2x^2 + 4x - 5$

(9) $3x^2 + 5x + 1$

(14) $-2x^2 - 5x + 3$

(5) $-5x^2 + x - 2$

(10) $-5x^2 + 2x - 1$

(15) $-2x^2 + x + 1$

(1) $-4x^2 - 4x - 3$

(6) $x^2 - 2x + 5$

(11) $-2x^2 + 3x - 2$

(2) $-5x^2 - 4x$

(7) $x^2 + 3x + 2$

(12) $5x^2 - 3x - 2$

(3) $-3x^2 + 3x - 2$

(8) $2x^2 - 4x - 4$

(13) $2x^2 + x + 4$

(4) $-x^2 - x$

(9) $x^2 + 2x - 5$

(14) $2x^2 - x$

(10) $3x^2 + 4x - 5$

(5) $-5x^2 + 3x$

(15) $3x^2 - x + 5$

(1) $5x^2 + x + 3$

(6) $-4x^2 - 5x - 1$

(11) $-x^2 + x - 4$

(2) $5x^2 + x + 1$

(7) $-3x^2 + 2x + 4$

(12) $2x^2 - 5x + 3$

(3) $x^2 - 3x + 1$

(8) $-4x^2 + x + 1$

(13) $-2x^2 + x + 2$

(4) $-2x^2 - 2x + 2$

(9) $-5x^2 + x$

(14) $-3x^2 + 4x$

(5) $-2x^2 - 5x + 4$

(10) $3x^2 - 2x + 5$

(15) $3x^2 - 5x + 5$

(1) $-4x^2 + 4x - 5$

(6) $3x^2 + 4x + 3$

(11) $4x^2 - 5x + 5$

(2) $-3x^2 + x + 4$

(7) $3x^2 - 2x + 5$

(12) $5x^2 + 2x - 2$

(3) $-2x^2 + 2x + 5$

(8) $5x^2 + 5x$

(13) $5x^2 - 2x - 5$

(4) $2x^2 - 3x + 4$

(9) $2x^2 + x - 5$

(14) $-2x^2 + 5x + 3$

(5) $3x^2 + 3x - 3$

(10) $-4x^2 + 5x - 5$

(15) $2x^2 + 5x + 3$

(1) $-5x^2 + 4x + 4$

(6) $-5x^2 + x - 4$

(11) $2x^2 + 5x - 3$

(2) $-4x^2 + 5x + 1$

(7) $5x^2 - 2x - 4$

(12) $-2x^2 + x + 5$

(3) $2x^2 - 4x + 2$

(8) $-3x^2 + 5x - 4$

(13) $-x^2 + 4x$

(4) $2x^2 + x - 1$

(9) $-3x^2 + 5x + 4$

(14) $-5x^2 + 3x + 4$

(5) $4x^2 - 3x + 1$

(10) $-x^2 + 5x - 1$

(15) $-5x^2 + 4x$

(1) $-2x^2 + x + 4$

(6) $3x^2 - 2x - 5$

(11) $-4x^2 - 2x - 4$

(2) $3x^2 + 5x - 5$

(7) $-x^2 + 5x - 3$

(12) $2x^2 + 5x - 2$

(3) $-5x^2 + x - 3$

(8) $-4x^2 - x - 1$

(13) $-x^2 - x - 1$

(4) $3x^2 + 2x + 4$

(9) $3x^2 + x - 4$

(14) $4x^2 + 5x - 1$

(5) $-2x^2 - 4x$

(10) $-x^2 - 5x - 4$

(15) $-x^2 - x + 1$

(1) $-2x^2 + 2x - 2$

(6) $2x^2 + x + 2$

(11) $-3x^2 + x + 5$

(2) $-4x^2 - 5x + 3$

(7) $-4x^2 - 3x + 1$

(12) $3x^2 - x + 4$

(3) $-x^2 - 2x + 4$

(8) $-3x^2 - 2x - 4$

(13) $2x^2 + 5x - 4$

(4) $-3x^2 + 4x - 4$

(9) $-x^2 - 3x - 2$

(14) $-2x^2 - x - 1$

(5) $-x^2 + 2x - 3$

(10) $-x^2 - 3x + 2$

(15) $4x^2 + 4x + 3$

(1) $x^2 + x - 5$

(6) $-3x^2 - 4x$

(11) $-3x^2 - 3x + 4$

(2) $5x^2 + x - 5$

(7) $2x^2 + 4x - 1$

(12) $-4x^2 - 2x - 3$

(3) $-2x^2 - 2x + 1$

(8) $x^2 + 5x - 2$

(13) $-2x^2 + 4x - 4$

(4) $2x^2 - x - 5$

(9) $-3x^2 - 4x + 1$

(14) $3x^2 + x + 3$

(5) $x^2 - 5x - 4$

(10) $-3x^2 + 3x - 1$

(15) $5x^2 - 2x - 3$

(1) $3x^2 + 5x + 2$

(6) $5x^2 + 3x + 1$

(11) $-4x^2 - x + 4$

(2) $x^2 - 3x + 4$

(7) $-x^2 - 2x - 5$

(12) $-x^2 - 2x + 1$

(3) $3x^2 + 4x - 3$

(8) $-2x^2 + 4x$

(13) $-x^2 + 4x - 5$

(4) $-2x^2 + 3x + 3$

(9) $4x^2 + 2x - 3$

(14) $5x^2 - x - 5$

(5) $2x^2 + 2x$

(10) $x^2 + 5x - 4$

(15) $-4x^2 + 5x - 4$

(1) $-3x^2 + 5x - 2$

(6) $-4x^2 + 4x - 5$

(11) $5x^2 - 5x + 2$

(2) $-2x^2 + 3x$

(7) $5x^2 + x + 5$

(12) $-4x^2 - x - 3$

(3) $-2x^2 - 4x - 1$

(8) $3x^2 - 2x - 2$

(13) $x^2 + 4x + 4$

(4) $4x^2 + 4x - 1$

(9) $2x^2 - 5x - 2$

(14) $-3x^2 + 5x + 1$

(5) $-4x^2 - 4x - 5$

(10) $-3x^2 - 2x + 1$

(15) $5x^2 - 2x - 4$

(1) $-4x^2 + 4x + 2$

(6) $4x^2 - 4x + 2$

(11) $-x^2 - x$

(2) $5x^2 + 3x$

(7) $4x^2 - 2x - 5$

(12) $-4x^2 - 5x - 2$

(3) $-5x^2 + 3x - 5$

(8) $-x^2 + 3x - 5$

(13) $-3x^2 - x + 5$

(4) $x^2 + 5x + 5$

(9) $4x^2 + 3x - 1$

(14) $-3x^2 - x - 4$

(5) $2x^2 + 3x - 3$

(10) $-3x^2 - 3x - 5$

(15) $x^2 - x + 5$

(1) $-5x^2 + 4x + 1$

(6) $5x^2 - x + 1$

(11) $x^2 + 2x$

(2) $-2x^2 - 2x - 1$

(7) $-3x^2 - 3x - 2$

(12) $5x^2 + 4x + 4$

(3) $3x^2 + 2x$

(8) $x^2 - 4x$

(13) $3x^2 - 2x - 5$

(4) $-2x^2 + 3x - 3$

(9) $4x^2 - 4x + 3$

(14) $2x^2 - 3x + 2$

(5) $-x^2 - 3x - 4$

(10) $3x^2 - 4x$

(15) $-3x^2 + 4x - 3$

(1) $-5x^2 - 3x + 4$

(6) $5x^2 - x - 3$

(11) $3x^2 - x + 5$

(2) $-x^2 + 3x + 4$

(7) $-5x^2 - 3x + 5$

(12) $-3x^2 + 5x + 3$

(3) $-3x^2 - 3x - 2$

(8) $x^2 - 3x + 5$

(13) $5x^2 - 2x + 2$

(4) $-5x^2 + 3x - 4$

(9) $-x^2 - 3x - 2$

(14) $-4x^2 + 5x$

(5) $5x^2 - x + 1$

(10) $-2x^2 - x - 2$

(15) $2x^2 + 3x + 5$

(1) $3x^2 + 4x + 4$

(6) $x^2 - 3x + 3$

(11) $-5x^2 + x + 1$

(2) $3x^2 - x - 1$

(7) $-2x^2 + 5x + 2$

(12) $5x^2 - x - 3$

(3) $-3x^2 + 5x - 5$

(8) $2x^2 + 4x + 4$

(13) $x^2 + 3x + 1$

(4) $5x^2 - 4x - 5$

(9) $3x^2 + 5x - 3$

(14) $-4x^2 + 2x - 2$

(5) $2x^2 - 2x - 4$

(10) $-4x^2 + 3x - 3$

(15) $2x^2 - 2x + 3$

(1) $3x^2 + 2x + 1$

(6) $4x^2 - 5x - 3$

(11) $-5x^2 + x - 4$

(2) $3x^2 - x - 4$

(7) $4x^2 - 4x - 4$

(12) $3x^2 + 3x - 1$

(3) $-4x^2 + 4x + 5$

(8) $-4x^2 + 2x$

(13) $-x^2 + 3x - 4$

(4) $-4x^2 + 5x$

(9) $5x^2 + 3x + 1$

(14) $4x^2 - 3x$

(5) $3x^2 + x + 3$

(10) $x^2 - x + 1$

(15) $-4x^2 - x - 1$

(1) $2x^2 - 3x$

(6) $2x^2 + 5x - 1$

(11) $-x^2 + x - 3$

(2) $x^2 + 5x - 3$

(7) $-2x^2 + 2x$

(12) $2x^2 - x - 4$

(3) $3x^2 + 2x + 5$

(8) $-3x^2 + 5x + 3$

(13) $-4x^2 - 5x - 3$

(4) $-5x^2 + 4x - 5$

(9) $-4x^2 - x - 2$

(14) $-2x^2 + 3x - 3$

(5) $4x^2 - 3x + 3$

(10) $3x^2 - 2x - 2$

(15) $-5x^2 + 4x + 5$

(1) $2x^2 + 3x + 2$

(6) $x^2 + 3x + 5$

(11) $3x^2 + 4x - 5$

(2) $3x^2 + 5x + 3$

(7) $5x^2 - 2x + 5$

(12) $-x^2 - 5x - 4$

(3) $-x^2 - 3x + 4$

(8) $-3x^2 + 5x - 3$

(13) $3x^2 + x$

(4) $-x^2 - 2x + 1$

(9) $-3x^2 + 3x + 2$

(14) $4x^2 + 4x - 4$

(5) $x^2 + 5x - 2$

(10) $-2x^2 + 3x + 3$

(15) $-x^2 + 5x + 5$

(1) $x^2 + 3x + 3$

(6) $-x^2 + 5x + 1$

(11) $-x^2 - 3x - 1$

(2) $x^2 + x - 5$

(7) $-3x^2 - 3x$

(12) $5x^2 - 4x - 1$

(3) $5x^2 + 4x + 1$

(8) $3x^2 + 4x - 5$

(13) $-5x^2 + 5x + 2$

(4) $-2x^2 - 2x - 1$

(9) $4x^2 + 4x$

(14) $-4x^2 - x + 5$

(5) $2x^2 + 2x + 4$

(10) $3x^2 - x - 4$

(15) $-5x^2 + 2x + 2$

$$\begin{aligned}(1) \quad & -5x^2 - 4x - 4 \\ & = -5 \left(x + \frac{2}{5} \right)^2 - \frac{16}{5}\end{aligned}$$

$$\begin{aligned}(6) \quad & 2x^2 + x + 3 \\ & = 2 \left(x + \frac{1}{4} \right)^2 + \frac{23}{8}\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 - 3x + 2 \\ & = -3 \left(x + \frac{1}{2} \right)^2 + \frac{11}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & -2x^2 + 3x - 3 \\ & = -2 \left(x - \frac{3}{4} \right)^2 - \frac{15}{8}\end{aligned}$$

$$\begin{aligned}(7) \quad & -5x^2 + 4x - 1 \\ & = -5 \left(x - \frac{2}{5} \right)^2 - \frac{1}{5}\end{aligned}$$

$$\begin{aligned}(12) \quad & -4x^2 - 4x + 4 \\ & = -4 \left(x + \frac{1}{2} \right)^2 + 5\end{aligned}$$

$$\begin{aligned}(3) \quad & -x^2 - 2x - 5 \\ & = -(x + 1)^2 - 4\end{aligned}$$

$$\begin{aligned}(8) \quad & -4x^2 + 5x - 1 \\ & = -4 \left(x - \frac{5}{8} \right)^2 + \frac{9}{16}\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 - x - 3 \\ & = \left(x - \frac{1}{2} \right)^2 - \frac{13}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & -5x^2 + 2x \\ & = -5 \left(x - \frac{1}{5} \right)^2 + \frac{1}{5}\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 - 5x + 5 \\ & = 3 \left(x - \frac{5}{6} \right)^2 + \frac{35}{12}\end{aligned}$$

$$\begin{aligned}(14) \quad & -x^2 - 4x - 2 \\ & = -(x + 2)^2 + 2\end{aligned}$$

$$\begin{aligned}(5) \quad & x^2 - 2x + 5 \\ & = (x - 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(10) \quad & -3x^2 - 4x + 2 \\ & = -3 \left(x + \frac{2}{3} \right)^2 + \frac{10}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & 3x^2 + 3x - 5 \\ & = 3 \left(x + \frac{1}{2} \right)^2 - \frac{23}{4}\end{aligned}$$

$$\begin{aligned}(1) \quad & 2x^2 + 2x + 5 \\ &= 2 \left(x + \frac{1}{2} \right)^2 + \frac{9}{2}\end{aligned}$$

$$\begin{aligned}(6) \quad & -x^2 + 4x - 1 \\ &= -(x-2)^2 + 3\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 - 2x + 1 \\ &= -3 \left(x + \frac{1}{3} \right)^2 + \frac{4}{3}\end{aligned}$$

$$\begin{aligned}(2) \quad & -x^2 - 2x - 3 \\ &= -(x+1)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & 4x^2 - 3x - 2 \\ &= 4 \left(x - \frac{3}{8} \right)^2 - \frac{41}{16}\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 - 2x \\ &= 3 \left(x - \frac{1}{3} \right)^2 - \frac{1}{3}\end{aligned}$$

$$\begin{aligned}(3) \quad & 5x^2 + 2x + 4 \\ &= 5 \left(x + \frac{1}{5} \right)^2 + \frac{19}{5}\end{aligned}$$

$$\begin{aligned}(8) \quad & 4x^2 - 4x + 3 \\ &= 4 \left(x - \frac{1}{2} \right)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 - x + 3 \\ &= \left(x - \frac{1}{2} \right)^2 + \frac{11}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & -5x^2 - 2x - 1 \\ &= -5 \left(x + \frac{1}{5} \right)^2 - \frac{4}{5}\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 + 2x + 4 \\ &= -(x-1)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & 3x^2 + 3x + 1 \\ &= 3 \left(x + \frac{1}{2} \right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(5) \quad & -4x^2 - 2x \\ &= -4 \left(x + \frac{1}{4} \right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(10) \quad & 4x^2 - 3x - 2 \\ &= 4 \left(x - \frac{3}{8} \right)^2 - \frac{41}{16}\end{aligned}$$

$$\begin{aligned}(15) \quad & -5x^2 + 4x + 2 \\ &= -5 \left(x - \frac{2}{5} \right)^2 + \frac{14}{5}\end{aligned}$$

$$\begin{aligned}(1) \quad & -3x^2 + 2x + 5 \\ & = -3\left(x - \frac{1}{3}\right)^2 + \frac{16}{3}\end{aligned}$$

$$\begin{aligned}(6) \quad & -3x^2 + 2x + 4 \\ & = -3\left(x - \frac{1}{3}\right)^2 + \frac{13}{3}\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 - 5x - 4 \\ & = -3\left(x + \frac{5}{6}\right)^2 - \frac{23}{12}\end{aligned}$$

$$\begin{aligned}(2) \quad & x^2 - 3x - 1 \\ & = \left(x - \frac{3}{2}\right)^2 - \frac{13}{4}\end{aligned}$$

$$\begin{aligned}(7) \quad & -4x^2 - 3x - 5 \\ & = -4\left(x + \frac{3}{8}\right)^2 - \frac{71}{16}\end{aligned}$$

$$\begin{aligned}(12) \quad & -5x^2 - 3x \\ & = -5\left(x + \frac{3}{10}\right)^2 + \frac{9}{20}\end{aligned}$$

$$\begin{aligned}(3) \quad & -2x^2 + 4x - 1 \\ & = -2(x - 1)^2 + 1\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 - x - 5 \\ & = \left(x - \frac{1}{2}\right)^2 - \frac{21}{4}\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 - 3x + 4 \\ & = \left(x - \frac{3}{2}\right)^2 + \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & -2x^2 + 4x - 5 \\ & = -2(x - 1)^2 - 3\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 + 5x + 1 \\ & = 3\left(x + \frac{5}{6}\right)^2 - \frac{13}{12}\end{aligned}$$

$$\begin{aligned}(14) \quad & -2x^2 - 5x + 3 \\ & = -2\left(x + \frac{5}{4}\right)^2 + \frac{49}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & -5x^2 + x - 2 \\ & = -5\left(x - \frac{1}{10}\right)^2 - \frac{39}{20}\end{aligned}$$

$$\begin{aligned}(10) \quad & -5x^2 + 2x - 1 \\ & = -5\left(x - \frac{1}{5}\right)^2 - \frac{4}{5}\end{aligned}$$

$$\begin{aligned}(15) \quad & -2x^2 + x + 1 \\ & = -2\left(x - \frac{1}{4}\right)^2 + \frac{9}{8}\end{aligned}$$

$$\begin{aligned}(1) \quad & -4x^2 - 4x - 3 \\ & = -4\left(x + \frac{1}{2}\right)^2 - 2\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 - 2x + 5 \\ & = (x - 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(11) \quad & -2x^2 + 3x - 2 \\ & = -2\left(x - \frac{3}{4}\right)^2 - \frac{7}{8}\end{aligned}$$

$$\begin{aligned}(2) \quad & -5x^2 - 4x \\ & = -5\left(x + \frac{2}{5}\right)^2 + \frac{4}{5}\end{aligned}$$

$$\begin{aligned}(7) \quad & x^2 + 3x + 2 \\ & = \left(x + \frac{3}{2}\right)^2 - \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 - 3x - 2 \\ & = 5\left(x - \frac{3}{10}\right)^2 - \frac{49}{20}\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 + 3x - 2 \\ & = -3\left(x - \frac{1}{2}\right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(8) \quad & 2x^2 - 4x - 4 \\ & = 2(x - 1)^2 - 6\end{aligned}$$

$$\begin{aligned}(13) \quad & 2x^2 + x + 4 \\ & = 2\left(x + \frac{1}{4}\right)^2 + \frac{31}{8}\end{aligned}$$

$$\begin{aligned}(4) \quad & -x^2 - x \\ & = -\left(x + \frac{1}{2}\right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(9) \quad & x^2 + 2x - 5 \\ & = (x + 1)^2 - 6\end{aligned}$$

$$\begin{aligned}(14) \quad & 2x^2 - x \\ & = 2\left(x - \frac{1}{4}\right)^2 - \frac{1}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & -5x^2 + 3x \\ & = -5\left(x - \frac{3}{10}\right)^2 + \frac{9}{20}\end{aligned}$$

$$\begin{aligned}(10) \quad & 3x^2 + 4x - 5 \\ & = 3\left(x + \frac{2}{3}\right)^2 - \frac{19}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & 3x^2 - x + 5 \\ & = 3\left(x - \frac{1}{6}\right)^2 + \frac{59}{12}\end{aligned}$$

$$(1) \quad 5x^2 + x + 3 \\ = 5 \left(x + \frac{1}{10} \right)^2 + \frac{59}{20}$$

$$(6) \quad -4x^2 - 5x - 1 \\ = -4 \left(x + \frac{5}{8} \right)^2 + \frac{9}{16}$$

$$(11) \quad -x^2 + x - 4 \\ = - \left(x - \frac{1}{2} \right)^2 - \frac{15}{4}$$

$$(2) \quad 5x^2 + x + 1 \\ = 5 \left(x + \frac{1}{10} \right)^2 + \frac{19}{20}$$

$$(7) \quad -3x^2 + 2x + 4 \\ = -3 \left(x - \frac{1}{3} \right)^2 + \frac{13}{3}$$

$$(12) \quad 2x^2 - 5x + 3 \\ = 2 \left(x - \frac{5}{4} \right)^2 - \frac{1}{8}$$

$$(3) \quad x^2 - 3x + 1 \\ = \left(x - \frac{3}{2} \right)^2 - \frac{5}{4}$$

$$(8) \quad -4x^2 + x + 1 \\ = -4 \left(x - \frac{1}{8} \right)^2 + \frac{17}{16}$$

$$(13) \quad -2x^2 + x + 2 \\ = -2 \left(x - \frac{1}{4} \right)^2 + \frac{17}{8}$$

$$(4) \quad -2x^2 - 2x + 2 \\ = -2 \left(x + \frac{1}{2} \right)^2 + \frac{5}{2}$$

$$(9) \quad -5x^2 + x \\ = -5 \left(x - \frac{1}{10} \right)^2 + \frac{1}{20}$$

$$(14) \quad -3x^2 + 4x \\ = -3 \left(x - \frac{2}{3} \right)^2 + \frac{4}{3}$$

$$(5) \quad -2x^2 - 5x + 4 \\ = -2 \left(x + \frac{5}{4} \right)^2 + \frac{57}{8}$$

$$(10) \quad 3x^2 - 2x + 5 \\ = 3 \left(x - \frac{1}{3} \right)^2 + \frac{14}{3}$$

$$(15) \quad 3x^2 - 5x + 5 \\ = 3 \left(x - \frac{5}{6} \right)^2 + \frac{35}{12}$$

$$\begin{aligned}(1) \quad & -4x^2 + 4x - 5 \\ & = -4 \left(x - \frac{1}{2} \right)^2 - 4\end{aligned}$$

$$\begin{aligned}(6) \quad & 3x^2 + 4x + 3 \\ & = 3 \left(x + \frac{2}{3} \right)^2 + \frac{5}{3}\end{aligned}$$

$$\begin{aligned}(11) \quad & 4x^2 - 5x + 5 \\ & = 4 \left(x - \frac{5}{8} \right)^2 + \frac{55}{16}\end{aligned}$$

$$\begin{aligned}(2) \quad & -3x^2 + x + 4 \\ & = -3 \left(x - \frac{1}{6} \right)^2 + \frac{49}{12}\end{aligned}$$

$$\begin{aligned}(7) \quad & 3x^2 - 2x + 5 \\ & = 3 \left(x - \frac{1}{3} \right)^2 + \frac{14}{3}\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 + 2x - 2 \\ & = 5 \left(x + \frac{1}{5} \right)^2 - \frac{11}{5}\end{aligned}$$

$$\begin{aligned}(3) \quad & -2x^2 + 2x + 5 \\ & = -2 \left(x - \frac{1}{2} \right)^2 + \frac{11}{2}\end{aligned}$$

$$\begin{aligned}(8) \quad & 5x^2 + 5x \\ & = 5 \left(x + \frac{1}{2} \right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(13) \quad & 5x^2 - 2x - 5 \\ & = 5 \left(x - \frac{1}{5} \right)^2 - \frac{26}{5}\end{aligned}$$

$$\begin{aligned}(4) \quad & 2x^2 - 3x + 4 \\ & = 2 \left(x - \frac{3}{4} \right)^2 + \frac{23}{8}\end{aligned}$$

$$\begin{aligned}(9) \quad & 2x^2 + x - 5 \\ & = 2 \left(x + \frac{1}{4} \right)^2 - \frac{41}{8}\end{aligned}$$

$$\begin{aligned}(14) \quad & -2x^2 + 5x + 3 \\ & = -2 \left(x - \frac{5}{4} \right)^2 + \frac{49}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & 3x^2 + 3x - 3 \\ & = 3 \left(x + \frac{1}{2} \right)^2 - \frac{15}{4}\end{aligned}$$

$$\begin{aligned}(10) \quad & -4x^2 + 5x - 5 \\ & = -4 \left(x - \frac{5}{8} \right)^2 - \frac{55}{16}\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 + 5x + 3 \\ & = 2 \left(x + \frac{5}{4} \right)^2 - \frac{1}{8}\end{aligned}$$

$$\begin{aligned}(1) \quad & -5x^2 + 4x + 4 \\ & = -5\left(x - \frac{2}{5}\right)^2 + \frac{24}{5}\end{aligned}$$

$$\begin{aligned}(6) \quad & -5x^2 + x - 4 \\ & = -5\left(x - \frac{1}{10}\right)^2 - \frac{79}{20}\end{aligned}$$

$$\begin{aligned}(11) \quad & 2x^2 + 5x - 3 \\ & = 2\left(x + \frac{5}{4}\right)^2 - \frac{49}{8}\end{aligned}$$

$$\begin{aligned}(2) \quad & -4x^2 + 5x + 1 \\ & = -4\left(x - \frac{5}{8}\right)^2 + \frac{41}{16}\end{aligned}$$

$$\begin{aligned}(7) \quad & 5x^2 - 2x - 4 \\ & = 5\left(x - \frac{1}{5}\right)^2 - \frac{21}{5}\end{aligned}$$

$$\begin{aligned}(12) \quad & -2x^2 + x + 5 \\ & = -2\left(x - \frac{1}{4}\right)^2 + \frac{41}{8}\end{aligned}$$

$$\begin{aligned}(3) \quad & 2x^2 - 4x + 2 \\ & = 2(x - 1)^2\end{aligned}$$

$$\begin{aligned}(8) \quad & -3x^2 + 5x - 4 \\ & = -3\left(x - \frac{5}{6}\right)^2 - \frac{23}{12}\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 + 4x \\ & = -(x - 2)^2 + 4\end{aligned}$$

$$\begin{aligned}(4) \quad & 2x^2 + x - 1 \\ & = 2\left(x + \frac{1}{4}\right)^2 - \frac{9}{8}\end{aligned}$$

$$\begin{aligned}(9) \quad & -3x^2 + 5x + 4 \\ & = -3\left(x - \frac{5}{6}\right)^2 + \frac{73}{12}\end{aligned}$$

$$\begin{aligned}(14) \quad & -5x^2 + 3x + 4 \\ & = -5\left(x - \frac{3}{10}\right)^2 + \frac{89}{20}\end{aligned}$$

$$\begin{aligned}(5) \quad & 4x^2 - 3x + 1 \\ & = 4\left(x - \frac{3}{8}\right)^2 + \frac{7}{16}\end{aligned}$$

$$\begin{aligned}(10) \quad & -x^2 + 5x - 1 \\ & = -\left(x - \frac{5}{2}\right)^2 + \frac{21}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & -5x^2 + 4x \\ & = -5\left(x - \frac{2}{5}\right)^2 + \frac{4}{5}\end{aligned}$$

$$\begin{aligned}(1) \quad & -2x^2 + x + 4 \\ & = -2\left(x - \frac{1}{4}\right)^2 + \frac{33}{8}\end{aligned}$$

$$\begin{aligned}(6) \quad & 3x^2 - 2x - 5 \\ & = 3\left(x - \frac{1}{3}\right)^2 - \frac{16}{3}\end{aligned}$$

$$\begin{aligned}(11) \quad & -4x^2 - 2x - 4 \\ & = -4\left(x + \frac{1}{4}\right)^2 - \frac{15}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & 3x^2 + 5x - 5 \\ & = 3\left(x + \frac{5}{6}\right)^2 - \frac{85}{12}\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 + 5x - 3 \\ & = -\left(x - \frac{5}{2}\right)^2 + \frac{13}{4}\end{aligned}$$

$$\begin{aligned}(12) \quad & 2x^2 + 5x - 2 \\ & = 2\left(x + \frac{5}{4}\right)^2 - \frac{41}{8}\end{aligned}$$

$$\begin{aligned}(3) \quad & -5x^2 + x - 3 \\ & = -5\left(x - \frac{1}{10}\right)^2 - \frac{59}{20}\end{aligned}$$

$$\begin{aligned}(8) \quad & -4x^2 - x - 1 \\ & = -4\left(x + \frac{1}{8}\right)^2 - \frac{15}{16}\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 - x - 1 \\ & = -\left(x + \frac{1}{2}\right)^2 - \frac{3}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & 3x^2 + 2x + 4 \\ & = 3\left(x + \frac{1}{3}\right)^2 + \frac{11}{3}\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 + x - 4 \\ & = 3\left(x + \frac{1}{6}\right)^2 - \frac{49}{12}\end{aligned}$$

$$\begin{aligned}(14) \quad & 4x^2 + 5x - 1 \\ & = 4\left(x + \frac{5}{8}\right)^2 - \frac{41}{16}\end{aligned}$$

$$\begin{aligned}(5) \quad & -2x^2 - 4x \\ & = -2(x + 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(10) \quad & -x^2 - 5x - 4 \\ & = -\left(x + \frac{5}{2}\right)^2 + \frac{9}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & -x^2 - x + 1 \\ & = -\left(x + \frac{1}{2}\right)^2 + \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(1) \quad & -2x^2 + 2x - 2 \\ & = -2 \left(x - \frac{1}{2} \right)^2 - \frac{3}{2}\end{aligned}$$

$$\begin{aligned}(6) \quad & 2x^2 + x + 2 \\ & = 2 \left(x + \frac{1}{4} \right)^2 + \frac{15}{8}\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 + x + 5 \\ & = -3 \left(x - \frac{1}{6} \right)^2 + \frac{61}{12}\end{aligned}$$

$$\begin{aligned}(2) \quad & -4x^2 - 5x + 3 \\ & = -4 \left(x + \frac{5}{8} \right)^2 + \frac{73}{16}\end{aligned}$$

$$\begin{aligned}(7) \quad & -4x^2 - 3x + 1 \\ & = -4 \left(x + \frac{3}{8} \right)^2 + \frac{25}{16}\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 - x + 4 \\ & = 3 \left(x - \frac{1}{6} \right)^2 + \frac{47}{12}\end{aligned}$$

$$\begin{aligned}(3) \quad & -x^2 - 2x + 4 \\ & = -(x + 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(8) \quad & -3x^2 - 2x - 4 \\ & = -3 \left(x + \frac{1}{3} \right)^2 - \frac{11}{3}\end{aligned}$$

$$\begin{aligned}(13) \quad & 2x^2 + 5x - 4 \\ & = 2 \left(x + \frac{5}{4} \right)^2 - \frac{57}{8}\end{aligned}$$

$$\begin{aligned}(4) \quad & -3x^2 + 4x - 4 \\ & = -3 \left(x - \frac{2}{3} \right)^2 - \frac{8}{3}\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 - 3x - 2 \\ & = - \left(x + \frac{3}{2} \right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(14) \quad & -2x^2 - x - 1 \\ & = -2 \left(x + \frac{1}{4} \right)^2 - \frac{7}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 + 2x - 3 \\ & = -(x - 1)^2 - 2\end{aligned}$$

$$\begin{aligned}(10) \quad & -x^2 - 3x + 2 \\ & = - \left(x + \frac{3}{2} \right)^2 + \frac{17}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & 4x^2 + 4x + 3 \\ & = 4 \left(x + \frac{1}{2} \right)^2 + 2\end{aligned}$$

$$(1) \quad x^2 + x - 5 \\ = \left(x + \frac{1}{2}\right)^2 - \frac{21}{4}$$

$$(6) \quad -3x^2 - 4x \\ = -3\left(x + \frac{2}{3}\right)^2 + \frac{4}{3}$$

$$(11) \quad -3x^2 - 3x + 4 \\ = -3\left(x + \frac{1}{2}\right)^2 + \frac{19}{4}$$

$$(2) \quad 5x^2 + x - 5 \\ = 5\left(x + \frac{1}{10}\right)^2 - \frac{101}{20}$$

$$(7) \quad 2x^2 + 4x - 1 \\ = 2(x + 1)^2 - 3$$

$$(12) \quad -4x^2 - 2x - 3 \\ = -4\left(x + \frac{1}{4}\right)^2 - \frac{11}{4}$$

$$(3) \quad -2x^2 - 2x + 1 \\ = -2\left(x + \frac{1}{2}\right)^2 + \frac{3}{2}$$

$$(8) \quad x^2 + 5x - 2 \\ = \left(x + \frac{5}{2}\right)^2 - \frac{33}{4}$$

$$(13) \quad -2x^2 + 4x - 4 \\ = -2(x - 1)^2 - 2$$

$$(4) \quad 2x^2 - x - 5 \\ = 2\left(x - \frac{1}{4}\right)^2 - \frac{41}{8}$$

$$(9) \quad -3x^2 - 4x + 1 \\ = -3\left(x + \frac{2}{3}\right)^2 + \frac{7}{3}$$

$$(14) \quad 3x^2 + x + 3 \\ = 3\left(x + \frac{1}{6}\right)^2 + \frac{35}{12}$$

$$(5) \quad x^2 - 5x - 4 \\ = \left(x - \frac{5}{2}\right)^2 - \frac{41}{4}$$

$$(10) \quad -3x^2 + 3x - 1 \\ = -3\left(x - \frac{1}{2}\right)^2 - \frac{1}{4}$$

$$(15) \quad 5x^2 - 2x - 3 \\ = 5\left(x - \frac{1}{5}\right)^2 - \frac{16}{5}$$

$$\begin{aligned}(1) \quad & 3x^2 + 5x + 2 \\ &= 3 \left(x + \frac{5}{6} \right)^2 - \frac{1}{12}\end{aligned}$$

$$\begin{aligned}(6) \quad & 5x^2 + 3x + 1 \\ &= 5 \left(x + \frac{3}{10} \right)^2 + \frac{11}{20}\end{aligned}$$

$$\begin{aligned}(11) \quad & -4x^2 - x + 4 \\ &= -4 \left(x + \frac{1}{8} \right)^2 + \frac{65}{16}\end{aligned}$$

$$\begin{aligned}(2) \quad & x^2 - 3x + 4 \\ &= \left(x - \frac{3}{2} \right)^2 + \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 - 2x - 5 \\ &= -(x + 1)^2 - 4\end{aligned}$$

$$\begin{aligned}(12) \quad & -x^2 - 2x + 1 \\ &= -(x + 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(3) \quad & 3x^2 + 4x - 3 \\ &= 3 \left(x + \frac{2}{3} \right)^2 - \frac{13}{3}\end{aligned}$$

$$\begin{aligned}(8) \quad & -2x^2 + 4x \\ &= -2(x - 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 + 4x - 5 \\ &= -(x - 2)^2 - 1\end{aligned}$$

$$\begin{aligned}(4) \quad & -2x^2 + 3x + 3 \\ &= -2 \left(x - \frac{3}{4} \right)^2 + \frac{33}{8}\end{aligned}$$

$$\begin{aligned}(9) \quad & 4x^2 + 2x - 3 \\ &= 4 \left(x + \frac{1}{4} \right)^2 - \frac{13}{4}\end{aligned}$$

$$\begin{aligned}(14) \quad & 5x^2 - x - 5 \\ &= 5 \left(x - \frac{1}{10} \right)^2 - \frac{101}{20}\end{aligned}$$

$$\begin{aligned}(5) \quad & 2x^2 + 2x \\ &= 2 \left(x + \frac{1}{2} \right)^2 - \frac{1}{2}\end{aligned}$$

$$\begin{aligned}(10) \quad & x^2 + 5x - 4 \\ &= \left(x + \frac{5}{2} \right)^2 - \frac{41}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 + 5x - 4 \\ &= -4 \left(x - \frac{5}{8} \right)^2 - \frac{39}{16}\end{aligned}$$

$$\begin{aligned}(1) \quad & -3x^2 + 5x - 2 \\ & = -3\left(x - \frac{5}{6}\right)^2 + \frac{1}{12}\end{aligned}$$

$$\begin{aligned}(6) \quad & -4x^2 + 4x - 5 \\ & = -4\left(x - \frac{1}{2}\right)^2 - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & 5x^2 - 5x + 2 \\ & = 5\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & -2x^2 + 3x \\ & = -2\left(x - \frac{3}{4}\right)^2 + \frac{9}{8}\end{aligned}$$

$$\begin{aligned}(7) \quad & 5x^2 + x + 5 \\ & = 5\left(x + \frac{1}{10}\right)^2 + \frac{99}{20}\end{aligned}$$

$$\begin{aligned}(12) \quad & -4x^2 - x - 3 \\ & = -4\left(x + \frac{1}{8}\right)^2 - \frac{47}{16}\end{aligned}$$

$$\begin{aligned}(3) \quad & -2x^2 - 4x - 1 \\ & = -2(x + 1)^2 + 1\end{aligned}$$

$$\begin{aligned}(8) \quad & 3x^2 - 2x - 2 \\ & = 3\left(x - \frac{1}{3}\right)^2 - \frac{7}{3}\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 + 4x + 4 \\ & = (x + 2)^2\end{aligned}$$

$$\begin{aligned}(4) \quad & 4x^2 + 4x - 1 \\ & = 4\left(x + \frac{1}{2}\right)^2 - 2\end{aligned}$$

$$\begin{aligned}(9) \quad & 2x^2 - 5x - 2 \\ & = 2\left(x - \frac{5}{4}\right)^2 - \frac{41}{8}\end{aligned}$$

$$\begin{aligned}(14) \quad & -3x^2 + 5x + 1 \\ & = -3\left(x - \frac{5}{6}\right)^2 + \frac{37}{12}\end{aligned}$$

$$\begin{aligned}(5) \quad & -4x^2 - 4x - 5 \\ & = -4\left(x + \frac{1}{2}\right)^2 - 4\end{aligned}$$

$$\begin{aligned}(10) \quad & -3x^2 - 2x + 1 \\ & = -3\left(x + \frac{1}{3}\right)^2 + \frac{4}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & 5x^2 - 2x - 4 \\ & = 5\left(x - \frac{1}{5}\right)^2 - \frac{21}{5}\end{aligned}$$

$$\begin{aligned}(1) \quad & -4x^2 + 4x + 2 \\ & = -4\left(x - \frac{1}{2}\right)^2 + 3\end{aligned}$$

$$\begin{aligned}(6) \quad & 4x^2 - 4x + 2 \\ & = 4\left(x - \frac{1}{2}\right)^2 + 1\end{aligned}$$

$$\begin{aligned}(11) \quad & -x^2 - x \\ & = -\left(x + \frac{1}{2}\right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & 5x^2 + 3x \\ & = 5\left(x + \frac{3}{10}\right)^2 - \frac{9}{20}\end{aligned}$$

$$\begin{aligned}(7) \quad & 4x^2 - 2x - 5 \\ & = 4\left(x - \frac{1}{4}\right)^2 - \frac{21}{4}\end{aligned}$$

$$\begin{aligned}(12) \quad & -4x^2 - 5x - 2 \\ & = -4\left(x + \frac{5}{8}\right)^2 - \frac{7}{16}\end{aligned}$$

$$\begin{aligned}(3) \quad & -5x^2 + 3x - 5 \\ & = -5\left(x - \frac{3}{10}\right)^2 - \frac{91}{20}\end{aligned}$$

$$\begin{aligned}(8) \quad & -x^2 + 3x - 5 \\ & = -\left(x - \frac{3}{2}\right)^2 - \frac{11}{4}\end{aligned}$$

$$\begin{aligned}(13) \quad & -3x^2 - x + 5 \\ & = -3\left(x + \frac{1}{6}\right)^2 + \frac{61}{12}\end{aligned}$$

$$\begin{aligned}(4) \quad & x^2 + 5x + 5 \\ & = \left(x + \frac{5}{2}\right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(9) \quad & 4x^2 + 3x - 1 \\ & = 4\left(x + \frac{3}{8}\right)^2 - \frac{25}{16}\end{aligned}$$

$$\begin{aligned}(14) \quad & -3x^2 - x - 4 \\ & = -3\left(x + \frac{1}{6}\right)^2 - \frac{47}{12}\end{aligned}$$

$$\begin{aligned}(5) \quad & 2x^2 + 3x - 3 \\ & = 2\left(x + \frac{3}{4}\right)^2 - \frac{33}{8}\end{aligned}$$

$$\begin{aligned}(10) \quad & -3x^2 - 3x - 5 \\ & = -3\left(x + \frac{1}{2}\right)^2 - \frac{17}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & x^2 - x + 5 \\ & = \left(x - \frac{1}{2}\right)^2 + \frac{19}{4}\end{aligned}$$

$$\begin{aligned}(1) \quad & -5x^2 + 4x + 1 \\ & = -5 \left(x - \frac{2}{5} \right)^2 + \frac{9}{5}\end{aligned}$$

$$\begin{aligned}(6) \quad & 5x^2 - x + 1 \\ & = 5 \left(x - \frac{1}{10} \right)^2 + \frac{19}{20}\end{aligned}$$

$$\begin{aligned}(11) \quad & x^2 + 2x \\ & = (x + 1)^2 - 1\end{aligned}$$

$$\begin{aligned}(2) \quad & -2x^2 - 2x - 1 \\ & = -2 \left(x + \frac{1}{2} \right)^2 - \frac{1}{2}\end{aligned}$$

$$\begin{aligned}(7) \quad & -3x^2 - 3x - 2 \\ & = -3 \left(x + \frac{1}{2} \right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 + 4x + 4 \\ & = 5 \left(x + \frac{2}{5} \right)^2 + \frac{16}{5}\end{aligned}$$

$$\begin{aligned}(3) \quad & 3x^2 + 2x \\ & = 3 \left(x + \frac{1}{3} \right)^2 - \frac{1}{3}\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 - 4x \\ & = (x - 2)^2 - 4\end{aligned}$$

$$\begin{aligned}(13) \quad & 3x^2 - 2x - 5 \\ & = 3 \left(x - \frac{1}{3} \right)^2 - \frac{16}{3}\end{aligned}$$

$$\begin{aligned}(4) \quad & -2x^2 + 3x - 3 \\ & = -2 \left(x - \frac{3}{4} \right)^2 - \frac{15}{8}\end{aligned}$$

$$\begin{aligned}(9) \quad & 4x^2 - 4x + 3 \\ & = 4 \left(x - \frac{1}{2} \right)^2 + 2\end{aligned}$$

$$\begin{aligned}(14) \quad & 2x^2 - 3x + 2 \\ & = 2 \left(x - \frac{3}{4} \right)^2 + \frac{7}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 - 3x - 4 \\ & = - \left(x + \frac{3}{2} \right)^2 - \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(10) \quad & 3x^2 - 4x \\ & = 3 \left(x - \frac{2}{3} \right)^2 - \frac{4}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & -3x^2 + 4x - 3 \\ & = -3 \left(x - \frac{2}{3} \right)^2 - \frac{5}{3}\end{aligned}$$

$$\begin{aligned}(1) \quad & -5x^2 - 3x + 4 \\ & = -5 \left(x + \frac{3}{10} \right)^2 + \frac{89}{20}\end{aligned}$$

$$\begin{aligned}(6) \quad & 5x^2 - x - 3 \\ & = 5 \left(x - \frac{1}{10} \right)^2 - \frac{61}{20}\end{aligned}$$

$$\begin{aligned}(11) \quad & 3x^2 - x + 5 \\ & = 3 \left(x - \frac{1}{6} \right)^2 + \frac{59}{12}\end{aligned}$$

$$\begin{aligned}(2) \quad & -x^2 + 3x + 4 \\ & = - \left(x - \frac{3}{2} \right)^2 + \frac{25}{4}\end{aligned}$$

$$\begin{aligned}(7) \quad & -5x^2 - 3x + 5 \\ & = -5 \left(x + \frac{3}{10} \right)^2 + \frac{109}{20}\end{aligned}$$

$$\begin{aligned}(12) \quad & -3x^2 + 5x + 3 \\ & = -3 \left(x - \frac{5}{6} \right)^2 + \frac{61}{12}\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 - 3x - 2 \\ & = -3 \left(x + \frac{1}{2} \right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 - 3x + 5 \\ & = \left(x - \frac{3}{2} \right)^2 + \frac{11}{4}\end{aligned}$$

$$\begin{aligned}(13) \quad & 5x^2 - 2x + 2 \\ & = 5 \left(x - \frac{1}{5} \right)^2 + \frac{9}{5}\end{aligned}$$

$$\begin{aligned}(4) \quad & -5x^2 + 3x - 4 \\ & = -5 \left(x - \frac{3}{10} \right)^2 - \frac{71}{20}\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 - 3x - 2 \\ & = - \left(x + \frac{3}{2} \right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 + 5x \\ & = -4 \left(x - \frac{5}{8} \right)^2 + \frac{25}{16}\end{aligned}$$

$$\begin{aligned}(5) \quad & 5x^2 - x + 1 \\ & = 5 \left(x - \frac{1}{10} \right)^2 + \frac{19}{20}\end{aligned}$$

$$\begin{aligned}(10) \quad & -2x^2 - x - 2 \\ & = -2 \left(x + \frac{1}{4} \right)^2 - \frac{15}{8}\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 + 3x + 5 \\ & = 2 \left(x + \frac{3}{4} \right)^2 + \frac{31}{8}\end{aligned}$$

$$\begin{aligned}(1) \quad & 3x^2 + 4x + 4 \\ &= 3\left(x + \frac{2}{3}\right)^2 + \frac{8}{3}\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 - 3x + 3 \\ &= \left(x - \frac{3}{2}\right)^2 + \frac{3}{4}\end{aligned}$$

$$\begin{aligned}(11) \quad & -5x^2 + x + 1 \\ &= -5\left(x - \frac{1}{10}\right)^2 + \frac{21}{20}\end{aligned}$$

$$\begin{aligned}(2) \quad & 3x^2 - x - 1 \\ &= 3\left(x - \frac{1}{6}\right)^2 - \frac{13}{12}\end{aligned}$$

$$\begin{aligned}(7) \quad & -2x^2 + 5x + 2 \\ &= -2\left(x - \frac{5}{4}\right)^2 + \frac{41}{8}\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 - x - 3 \\ &= 5\left(x - \frac{1}{10}\right)^2 - \frac{61}{20}\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 + 5x - 5 \\ &= -3\left(x - \frac{5}{6}\right)^2 - \frac{35}{12}\end{aligned}$$

$$\begin{aligned}(8) \quad & 2x^2 + 4x + 4 \\ &= 2(x + 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 + 3x + 1 \\ &= \left(x + \frac{3}{2}\right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & 5x^2 - 4x - 5 \\ &= 5\left(x - \frac{2}{5}\right)^2 - \frac{29}{5}\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 + 5x - 3 \\ &= 3\left(x + \frac{5}{6}\right)^2 - \frac{61}{12}\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 + 2x - 2 \\ &= -4\left(x - \frac{1}{4}\right)^2 - \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(5) \quad & 2x^2 - 2x - 4 \\ &= 2\left(x - \frac{1}{2}\right)^2 - \frac{9}{2}\end{aligned}$$

$$\begin{aligned}(10) \quad & -4x^2 + 3x - 3 \\ &= -4\left(x - \frac{3}{8}\right)^2 - \frac{39}{16}\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 - 2x + 3 \\ &= 2\left(x - \frac{1}{2}\right)^2 + \frac{5}{2}\end{aligned}$$

$$\begin{aligned}(1) \quad & 3x^2 + 2x + 1 \\ &= 3\left(x + \frac{1}{3}\right)^2 + \frac{2}{3}\end{aligned}$$

$$\begin{aligned}(6) \quad & 4x^2 - 5x - 3 \\ &= 4\left(x - \frac{5}{8}\right)^2 - \frac{73}{16}\end{aligned}$$

$$\begin{aligned}(11) \quad & -5x^2 + x - 4 \\ &= -5\left(x - \frac{1}{10}\right)^2 - \frac{79}{20}\end{aligned}$$

$$\begin{aligned}(2) \quad & 3x^2 - x - 4 \\ &= 3\left(x - \frac{1}{6}\right)^2 - \frac{49}{12}\end{aligned}$$

$$\begin{aligned}(7) \quad & 4x^2 - 4x - 4 \\ &= 4\left(x - \frac{1}{2}\right)^2 - 5\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 + 3x - 1 \\ &= 3\left(x + \frac{1}{2}\right)^2 - \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(3) \quad & -4x^2 + 4x + 5 \\ &= -4\left(x - \frac{1}{2}\right)^2 + 6\end{aligned}$$

$$\begin{aligned}(8) \quad & -4x^2 + 2x \\ &= -4\left(x - \frac{1}{4}\right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 + 3x - 4 \\ &= -\left(x - \frac{3}{2}\right)^2 - \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & -4x^2 + 5x \\ &= -4\left(x - \frac{5}{8}\right)^2 + \frac{25}{16}\end{aligned}$$

$$\begin{aligned}(9) \quad & 5x^2 + 3x + 1 \\ &= 5\left(x + \frac{3}{10}\right)^2 + \frac{11}{20}\end{aligned}$$

$$\begin{aligned}(14) \quad & 4x^2 - 3x \\ &= 4\left(x - \frac{3}{8}\right)^2 - \frac{9}{16}\end{aligned}$$

$$\begin{aligned}(5) \quad & 3x^2 + x + 3 \\ &= 3\left(x + \frac{1}{6}\right)^2 + \frac{35}{12}\end{aligned}$$

$$\begin{aligned}(10) \quad & x^2 - x + 1 \\ &= \left(x - \frac{1}{2}\right)^2 + \frac{3}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 - x - 1 \\ &= -4\left(x + \frac{1}{8}\right)^2 - \frac{15}{16}\end{aligned}$$

$$(1) \quad 2x^2 - 3x \\ = 2 \left(x - \frac{3}{4} \right)^2 - \frac{9}{8}$$

$$(6) \quad 2x^2 + 5x - 1 \\ = 2 \left(x + \frac{5}{4} \right)^2 - \frac{33}{8}$$

$$(11) \quad -x^2 + x - 3 \\ = - \left(x - \frac{1}{2} \right)^2 - \frac{11}{4}$$

$$(2) \quad x^2 + 5x - 3 \\ = \left(x + \frac{5}{2} \right)^2 - \frac{37}{4}$$

$$(7) \quad -2x^2 + 2x \\ = -2 \left(x - \frac{1}{2} \right)^2 + \frac{1}{2}$$

$$(12) \quad 2x^2 - x - 4 \\ = 2 \left(x - \frac{1}{4} \right)^2 - \frac{33}{8}$$

$$(3) \quad 3x^2 + 2x + 5 \\ = 3 \left(x + \frac{1}{3} \right)^2 + \frac{14}{3}$$

$$(8) \quad -3x^2 + 5x + 3 \\ = -3 \left(x - \frac{5}{6} \right)^2 + \frac{61}{12}$$

$$(13) \quad -4x^2 - 5x - 3 \\ = -4 \left(x + \frac{5}{8} \right)^2 - \frac{23}{16}$$

$$(4) \quad -5x^2 + 4x - 5 \\ = -5 \left(x - \frac{2}{5} \right)^2 - \frac{21}{5}$$

$$(9) \quad -4x^2 - x - 2 \\ = -4 \left(x + \frac{1}{8} \right)^2 - \frac{31}{16}$$

$$(14) \quad -2x^2 + 3x - 3 \\ = -2 \left(x - \frac{3}{4} \right)^2 - \frac{15}{8}$$

$$(5) \quad 4x^2 - 3x + 3 \\ = 4 \left(x - \frac{3}{8} \right)^2 + \frac{39}{16}$$

$$(10) \quad 3x^2 - 2x - 2 \\ = 3 \left(x - \frac{1}{3} \right)^2 - \frac{7}{3}$$

$$(15) \quad -5x^2 + 4x + 5 \\ = -5 \left(x - \frac{2}{5} \right)^2 + \frac{29}{5}$$

$$(1) \quad 2x^2 + 3x + 2 \\ = 2 \left(x + \frac{3}{4} \right)^2 + \frac{7}{8}$$

$$(6) \quad x^2 + 3x + 5 \\ = \left(x + \frac{3}{2} \right)^2 + \frac{11}{4}$$

$$(11) \quad 3x^2 + 4x - 5 \\ = 3 \left(x + \frac{2}{3} \right)^2 - \frac{19}{3}$$

$$(2) \quad 3x^2 + 5x + 3 \\ = 3 \left(x + \frac{5}{6} \right)^2 + \frac{11}{12}$$

$$(7) \quad 5x^2 - 2x + 5 \\ = 5 \left(x - \frac{1}{5} \right)^2 + \frac{24}{5}$$

$$(12) \quad -x^2 - 5x - 4 \\ = - \left(x + \frac{5}{2} \right)^2 + \frac{9}{4}$$

$$(3) \quad -x^2 - 3x + 4 \\ = - \left(x + \frac{3}{2} \right)^2 + \frac{25}{4}$$

$$(8) \quad -3x^2 + 5x - 3 \\ = -3 \left(x - \frac{5}{6} \right)^2 - \frac{11}{12}$$

$$(13) \quad 3x^2 + x \\ = 3 \left(x + \frac{1}{6} \right)^2 - \frac{1}{12}$$

$$(4) \quad -x^2 - 2x + 1 \\ = - (x + 1)^2 + 2$$

$$(9) \quad -3x^2 + 3x + 2 \\ = -3 \left(x - \frac{1}{2} \right)^2 + \frac{11}{4}$$

$$(14) \quad 4x^2 + 4x - 4 \\ = 4 \left(x + \frac{1}{2} \right)^2 - 5$$

$$(5) \quad x^2 + 5x - 2 \\ = \left(x + \frac{5}{2} \right)^2 - \frac{33}{4}$$

$$(10) \quad -2x^2 + 3x + 3 \\ = -2 \left(x - \frac{3}{4} \right)^2 + \frac{33}{8}$$

$$(15) \quad -x^2 + 5x + 5 \\ = - \left(x - \frac{5}{2} \right)^2 + \frac{45}{4}$$

$$(1) \quad x^2 + 3x + 3 \\ = \left(x + \frac{3}{2}\right)^2 + \frac{3}{4}$$

$$(6) \quad -x^2 + 5x + 1 \\ = -\left(x - \frac{5}{2}\right)^2 + \frac{29}{4}$$

$$(11) \quad -x^2 - 3x - 1 \\ = -\left(x + \frac{3}{2}\right)^2 + \frac{5}{4}$$

$$(2) \quad x^2 + x - 5 \\ = \left(x + \frac{1}{2}\right)^2 - \frac{21}{4}$$

$$(7) \quad -3x^2 - 3x \\ = -3\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$(12) \quad 5x^2 - 4x - 1 \\ = 5\left(x - \frac{2}{5}\right)^2 - \frac{9}{5}$$

$$(3) \quad 5x^2 + 4x + 1 \\ = 5\left(x + \frac{2}{5}\right)^2 + \frac{1}{5}$$

$$(8) \quad 3x^2 + 4x - 5 \\ = 3\left(x + \frac{2}{3}\right)^2 - \frac{19}{3}$$

$$(13) \quad -5x^2 + 5x + 2 \\ = -5\left(x - \frac{1}{2}\right)^2 + \frac{13}{4}$$

$$(4) \quad -2x^2 - 2x - 1 \\ = -2\left(x + \frac{1}{2}\right)^2 - \frac{1}{2}$$

$$(9) \quad 4x^2 + 4x \\ = 4\left(x + \frac{1}{2}\right)^2 - 1$$

$$(14) \quad -4x^2 - x + 5 \\ = -4\left(x + \frac{1}{8}\right)^2 + \frac{81}{16}$$

$$(5) \quad 2x^2 + 2x + 4 \\ = 2\left(x + \frac{1}{2}\right)^2 + \frac{7}{2}$$

$$(10) \quad 3x^2 - x - 4 \\ = 3\left(x - \frac{1}{6}\right)^2 - \frac{49}{12}$$

$$(15) \quad -5x^2 + 2x + 2 \\ = -5\left(x - \frac{1}{5}\right)^2 + \frac{11}{5}$$

3.1.3 頂点の座標 (易しい 1)

2 次関数の頂点を平方完成している式から求める問題。

頂点の座標は 0 を含めた整数。

(1) $y = -6(x + 7)^2 - 5$

(20) $y = -3(x - 9)^2 + 10$

(39) $y = -4(x - 8)^2 + 2$

(2) $y = -3(x + 2)^2 + 9$

(21) $y = 4(x - 8)^2 + 2$

(40) $y = -4(x + 8)^2 - 5$

(3) $y = 3(x - 4)^2 - 10$

(22) $y = -7(x - 6)^2 + 4$

(41) $y = 2(x - 7)^2 + 7$

(4) $y = -4(x + 8)^2 + 5$

(23) $y = 7(x + 9)^2 + 7$

(42) $y = -6(x - 4)^2 - 8$

(5) $y = 7(x - 5)^2 + 4$

(24) $y = -2(x - 6)^2 - 1$

(43) $y = -10(x + 8)^2 - 5$

(6) $y = -(x + 5)^2 - 4$

(25) $y = (x - 9)^2 + 4$

(44) $y = -10(x - 9)^2 + 9$

(7) $y = 9(x - 9)^2 + 10$

(26) $y = -10(x - 5)^2 - 7$

(45) $y = -10(x - 4)^2 + 6$

(8) $y = -8(x - 5)^2 - 8$

(27) $y = -8(x + 5)^2 + 6$

(46) $y = 7(x - 10)^2 + 8$

(9) $y = 5(x - 9)^2 + 6$

(28) $y = 8(x + 7)^2 - 8$

(47) $y = -2(x - 8)^2 + 9$

(10) $y = -9x^2 - 8$

(29) $y = -3(x + 7)^2 + 1$

(48) $y = 6(x + 4)^2 - 1$

(11) $y = 2(x - 6)^2 + 4$

(30) $y = 4(x - 4)^2 + 7$

(49) $y = -3(x - 3)^2 - 3$

(12) $y = -10(x - 7)^2 - 1$

(31) $y = 5(x - 10)^2 - 3$

(50) $y = 5(x + 10)^2 + 4$

(13) $y = 3(x + 8)^2 + 10$

(32) $y = -5(x + 8)^2 + 10$

(51) $y = -10(x - 4)^2 - 7$

(14) $y = -(x + 6)^2 + 10$

(33) $y = -4(x - 4)^2 + 6$

(52) $y = 9(x - 10)^2 + 3$

(15) $y = 3(x - 10)^2 - 9$

(34) $y = 8(x + 8)^2 + 10$

(53) $y = -9(x + 1)^2 - 2$

(16) $y = 6(x - 3)^2 - 1$

(35) $y = -6(x + 3)^2 + 3$

(54) $y = -4(x + 10)^2 + 4$

(17) $y = 8(x - 1)^2 + 8$

(36) $y = -3x^2 - 9$

(55) $y = 8(x - 5)^2 - 9$

(18) $y = -6(x + 2)^2 - 9$

(37) $y = -8(x - 2)^2 + 7$

(56) $y = -5(x - 8)^2 + 6$

(19) $y = 7(x - 6)^2 + 6$

(38) $y = -5(x - 8)^2 + 4$

(57) $y = 9(x - 4)^2 - 7$

(1) $y = -6(x - 3)^2 + 7$

(20) $y = 7(x - 3)^2 - 2$

(39) $y = -5(x - 8)^2 + 7$

(2) $y = 5(x + 1)^2 - 7$

(21) $y = 2(x + 10)^2 - 6$

(40) $y = -6(x - 7)^2 + 10$

(3) $y = -5(x - 10)^2 + 4$

(22) $y = 4(x - 8)^2 + 1$

(41) $y = -2(x + 1)^2 + 1$

(4) $y = 4(x + 5)^2 - 1$

(23) $y = 6(x - 7)^2 + 6$

(42) $y = 3(x - 7)^2 - 6$

(5) $y = -7(x - 2)^2 + 3$

(24) $y = 10(x + 3)^2 - 10$

(43) $y = 2(x + 4)^2 + 8$

(6) $y = -3(x + 9)^2 - 5$

(25) $y = 7(x - 6)^2 + 8$

(44) $y = 9(x + 9)^2 - 10$

(7) $y = 9(x - 10)^2 - 8$

(26) $y = -8(x - 10)^2 - 5$

(45) $y = 3(x - 9)^2 - 4$

(8) $y = 5(x - 10)^2 + 6$

(27) $y = -3(x - 2)^2 + 1$

(46) $y = 3(x + 3)^2 - 10$

(9) $y = 7(x + 8)^2 - 4$

(28) $y = -7(x - 5)^2 - 4$

(47) $y = 2(x - 5)^2 + 1$

(10) $y = -8(x - 5)^2 + 6$

(29) $y = -6(x + 3)^2 + 7$

(48) $y = 5(x - 9)^2 + 7$

(11) $y = -8(x - 1)^2$

(30) $y = 8(x + 9)^2 - 8$

(49) $y = -4(x + 3)^2 + 2$

(12) $y = -7(x - 9)^2 - 2$

(31) $y = 6(x + 10)^2 + 5$

(50) $y = -10(x + 3)^2 + 2$

(13) $y = -2(x + 10)^2 + 4$

(32) $y = -10(x - 6)^2 + 5$

(51) $y = -9(x - 3)^2 - 7$

(14) $y = 10(x + 4)^2 - 3$

(33) $y = -(x + 10)^2 + 7$

(52) $y = -8(x - 4)^2 - 1$

(15) $y = -7(x - 2)^2 - 10$

(34) $y = -7(x + 4)^2 - 4$

(53) $y = 5(x - 3)^2 + 6$

(16) $y = -4(x - 5)^2$

(35) $y = -8(x - 6)^2 - 7$

(54) $y = -9x^2 + 6$

(17) $y = 7(x + 5)^2 + 10$

(36) $y = -(x + 1)^2 - 2$

(55) $y = -5(x - 8)^2 - 10$

(18) $y = 4(x - 2)^2 + 10$

(37) $y = -7(x - 3)^2 + 2$

(56) $y = 6(x - 7)^2 + 9$

(19) $y = 8(x + 3)^2 + 3$

(38) $y = -3x^2$

(57) $y = 2(x + 10)^2 + 4$

(1) $y = -2(x+5)^2 + 3$

(20) $y = -6(x-4)^2 - 10$

(39) $y = -2(x-9)^2 - 1$

(2) $y = -2(x+10)^2 + 5$

(21) $y = 4(x-2)^2 + 3$

(40) $y = 3(x-2)^2 + 10$

(3) $y = -(x-5)^2 + 10$

(22) $y = -5(x+7)^2 - 8$

(41) $y = 2(x+5)^2 - 6$

(4) $y = 4(x-4)^2 + 10$

(23) $y = -5(x+2)^2 + 8$

(42) $y = 10x^2 + 4$

(5) $y = 7(x+7)^2 + 3$

(24) $y = -9(x-1)^2 - 7$

(43) $y = 9(x+2)^2 - 9$

(6) $y = -10(x+7)^2 + 10$

(25) $y = (x+8)^2 + 4$

(44) $y = 4(x+10)^2 + 9$

(7) $y = -10(x-8)^2 - 7$

(26) $y = -8(x-4)^2 + 1$

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(28) $y = 5(x+3)^2 - 4$

(47) $y = 2(x-8)^2 + 6$

(10) $y = 5(x+8)^2 - 9$

(29) $y = 3(x-6)^2 - 1$

(48) $y = -10(x-10)^2 - 10$

(11) $y = -3(x+2)^2 + 1$

(30) $y = 9(x+7)^2 - 2$

(49) $y = -4(x+2)^2 - 2$

(12) $y = 4(x-9)^2 - 4$

(31) $y = -9(x+9)^2 + 6$

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(13) $y = 4(x+4)^2 + 3$

(32) $y = 3(x-8)^2 + 2$

(51) $y = -3(x-2)^2$

(14) $y = 5(x+10)^2 + 7$

(33) $y = -5(x+4)^2 + 5$

(52) $y = 7x^2 - 7$

(15) $y = -9x^2 + 7$

(34) $y = (x+7)^2 - 3$

(53) $y = -7(x-1)^2 - 2$

(16) $y = 3(x-2)^2 + 10$

(35) $y = 10(x-8)^2 + 8$

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(1) $y = 8(x + 5)^2 - 1$

(20) $y = (x + 4)^2 + 7$

(39) $y = 6(x - 9)^2 + 6$

(2) $y = 9(x + 8)^2 + 5$

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(34) $y = -3(x - 7)^2 + 5$

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(1) $y = 9(x + 8)^2 + 5$

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(2) $y = -10(x + 1)^2 - 6$

(21) $y = -10(x - 5)^2 + 5$

(40) $y = -10(x - 4)^2 + 7$

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(25) $y = 4(x + 5)^2 - 4$

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(7) $y = 7(x - 1)^2 - 6$

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(47) $y = -3(x - 9)^2 + 1$

(10) $y = -3(x + 3)^2 + 10$

(29) $y = 8(x - 1)^2 + 3$

(48) $y = -10(x + 5)^2 + 5$

(11) $y = 2(x + 3)^2 + 1$

(30) $y = -10(x - 2)^2 - 7$

(49) $y = 4(x + 5)^2 - 4$

(12) $y = 3(x + 3)^2 + 7$

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(50) $y = (x + 1)^2 + 2$

(13) $y = -(x + 7)^2 + 2$

(32) $y = -9(x + 9)^2 - 1$

(51) $y = -(x + 3)^2 - 1$

(14) $y = -10(x + 8)^2 + 3$

(33) $y = -2(x - 4)^2 + 8$

(52) $y = 6(x + 7)^2 + 9$

(15) $y = 9(x + 5)^2 + 7$

(34) $y = -(x - 10)^2 + 1$

(53) $y = -(x - 10)^2 - 10$

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(54) $y = -3(x - 8)^2 + 1$

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(36) $y = -4(x + 10)^2 - 6$

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(37) $y = 7(x - 4)^2 + 9$

(56) $y = -3(x + 9)^2 + 2$

(19) $y = 10(x + 5)^2 - 1$

(38) $y = 7(x + 7)^2 + 3$

(57) $y = -2(x - 5)^2 + 9$

(1) $y = 9x^2 - 7$

(20) $y = 7(x - 9)^2 + 7$

(39) $y = (x - 4)^2 - 8$

(2) $y = -(x + 9)^2 - 5$

(21) $y = 8x^2 + 6$

(40) $y = -10(x - 8)^2 - 3$

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(26) $y = -5(x - 2)^2 - 5$

(45) $y = 3(x - 6)^2 + 1$

(8) $y = 9(x + 10)^2 + 8$

(27) $y = 6x^2$

(46) $y = -8(x + 9)^2 - 9$

(9) $y = -7(x - 6)^2 + 2$

(28) $y = 2(x - 6)^2 + 9$

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(10) $y = -(x + 8)^2 + 10$

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(12) $y = 9(x + 10)^2 + 7$

(31) $y = 10(x + 7)^2$

(50) $y = -9(x - 1)^2 - 2$

(13) $y = -9(x + 6)^2 - 1$

(32) $y = x^2 + 5$

(51) $y = -5(x + 4)^2 - 8$

(14) $y = -6(x - 1)^2$

(33) $y = -7(x - 10)^2 - 6$

(52) $y = -10(x + 1)^2 - 2$

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(36) $y = -10(x + 9)^2 - 8$

(55) $y = -10x^2 + 7$

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(1) $y = -(x+4)^2 + 9$

(20) $y = -5(x-4)^2 + 3$

(39) $y = -5(x+1)^2 - 1$

(2) $y = 10(x-8)^2 - 3$

(21) $y = -8(x-7)^2 + 10$

(40) $y = 5(x+2)^2 - 7$

(3) $y = 9(x+9)^2 - 8$

(22) $y = -2(x-6)^2 - 4$

(41) $y = 3(x+3)^2 + 9$

(4) $y = -4(x-7)^2 - 9$

(23) $y = 8(x+5)^2 + 2$

(42) $y = 7(x+9)^2 - 8$

(5) $y = -3(x+7)^2 - 6$

(24) $y = -(x+4)^2 + 9$

(43) $y = -2(x+6)^2 + 8$

(6) $y = 9(x-4)^2 - 6$

(25) $y = 4(x-9)^2 - 3$

(44) $y = -4(x-1)^2 + 4$

(7) $y = 10(x+5)^2 - 7$

(26) $y = -(x-9)^2 + 4$

(45) $y = -5(x-5)^2 - 6$

(8) $y = -(x+10)^2 - 10$

(27) $y = 4(x-5)^2 - 9$

(46) $y = -4(x+1)^2 + 2$

(9) $y = 5(x-4)^2$

(28) $y = (x-2)^2 - 1$

(47) $y = -3(x+9)^2 + 10$

(10) $y = 5x^2 + 5$

(29) $y = 9(x+4)^2 - 10$

(48) $y = -4(x+9)^2 + 7$

(11) $y = -5(x+9)^2 + 10$

(30) $y = -4(x+7)^2 - 9$

(49) $y = -x^2 - 8$

(12) $y = -(x+4)^2 + 3$

(31) $y = (x-4)^2 + 3$

(50) $y = x^2 + 10$

(13) $y = -4(x+5)^2 - 8$

(32) $y = -4(x+9)^2 + 1$

(51) $y = -4(x-4)^2 - 4$

(14) $y = -2(x-10)^2$

(33) $y = -9(x+2)^2 + 6$

(52) $y = -4(x-2)^2 - 2$

(15) $y = -9(x-7)^2 + 2$

(34) $y = 5(x+2)^2 + 2$

(53) $y = 3(x-7)^2 - 8$

(16) $y = -8(x+4)^2 + 2$

(35) $y = (x+7)^2 - 3$

(54) $y = -8(x+10)^2 + 9$

(17) $y = 8(x+1)^2 + 1$

(36) $y = 9(x+1)^2 + 2$

(55) $y = (x-1)^2 - 8$

(18) $y = 3x^2 - 7$

(37) $y = -4(x-1)^2 + 4$

(56) $y = -3(x-9)^2 - 10$

(19) $y = (x-2)^2 - 1$

(38) $y = 2(x-7)^2 - 9$

(57) $y = -10(x+3)^2 + 9$

(1) $y = 8(x + 3)^2 + 4$

(20) $y = (x - 3)^2 + 8$

(39) $y = -6(x + 9)^2 - 4$

(2) $y = 7(x + 6)^2 - 1$

(21) $y = 3(x + 5)^2 - 5$

(40) $y = -7(x - 1)^2 - 3$

(3) $y = 4(x + 10)^2 - 9$

(22) $y = 3(x + 8)^2 + 1$

(41) $y = -8(x + 5)^2 - 2$

(4) $y = -3(x - 8)^2 - 7$

(23) $y = -2(x - 2)^2 + 5$

(42) $y = -9(x + 4)^2 - 2$

(5) $y = -(x + 9)^2 + 10$

(24) $y = x^2 - 2$

(43) $y = -3(x - 10)^2 + 5$

(6) $y = -3(x - 1)^2 + 9$

(25) $y = 7(x - 2)^2 + 10$

(44) $y = 8(x + 10)^2 + 2$

(7) $y = -6(x - 8)^2 + 10$

(26) $y = -6x^2 + 10$

(45) $y = (x + 7)^2 + 1$

(8) $y = 10(x - 3)^2 + 9$

(27) $y = 5(x + 7)^2 - 5$

(46) $y = 8x^2 - 5$

(9) $y = 8(x - 5)^2 - 5$

(28) $y = -3(x - 7)^2 - 10$

(47) $y = 5(x + 7)^2 + 2$

(10) $y = -2(x + 2)^2 - 9$

(29) $y = -7(x - 8)^2 - 8$

(48) $y = 3(x + 7)^2 + 4$

(11) $y = -(x + 1)^2 + 2$

(30) $y = (x - 5)^2 + 7$

(49) $y = 8(x - 2)^2 - 5$

(12) $y = -8(x + 3)^2 + 1$

(31) $y = 2(x - 7)^2 - 1$

(50) $y = -2(x - 9)^2 - 1$

(13) $y = (x + 9)^2 - 7$

(32) $y = -3(x + 5)^2 + 3$

(51) $y = -10(x + 2)^2 + 5$

(14) $y = 2(x + 8)^2 + 5$

(33) $y = (x + 4)^2 - 7$

(52) $y = 5(x + 9)^2 + 4$

(15) $y = -8(x - 9)^2 - 2$

(34) $y = -6(x - 1)^2 - 2$

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(19) $y = -6(x + 1)^2 - 6$

(38) $y = -9(x - 2)^2 - 10$

(57) $y = -8(x - 9)^2 + 1$

(1) $y = -7x^2 + 9$

(20) $y = 6(x - 2)^2 - 7$

(39) $y = 7(x + 3)^2 + 1$

(2) $y = -10(x - 1)^2 + 5$

(21) $y = -2(x + 5)^2 - 7$

(40) $y = 7(x + 5)^2$

(3) $y = 5(x - 2)^2 + 8$

(22) $y = 6x^2 + 5$

(41) $y = -10(x - 8)^2 - 8$

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(48) $y = -7(x - 8)^2 + 2$

(11) $y = -5(x + 1)^2 - 5$

(30) $y = -9(x + 9)^2 - 8$

(49) $y = 2(x + 7)^2 + 5$

(12) $y = -9(x - 8)^2 + 2$

(31) $y = -(x + 10)^2 + 7$

(50) $y = 9x^2 - 10$

(13) $y = -(x - 9)^2 + 6$

(32) $y = 9x^2 - 1$

(51) $y = -(x - 6)^2 - 3$

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(33) $y = 4(x - 3)^2 - 7$

(52) $y = 6(x - 6)^2 + 7$

(15) $y = 10(x - 4)^2 - 2$

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(16) $y = 5(x - 8)^2 + 5$

(35) $y = 7(x + 1)^2 + 1$

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(5) $y = 4(x - 6)^2 - 1$

(24) $y = 10(x + 9)^2 + 3$

(43) $y = 10(x + 7)^2 - 4$

(6) $y = 7(x - 6)^2 - 3$

(25) $y = 6(x - 3)^2 + 9$

(44) $y = 2(x + 3)^2 + 5$

(7) $y = 10(x + 6)^2 + 9$

(26) $y = 4(x + 7)^2 - 3$

(45) $y = 6(x - 6)^2 + 10$

(8) $y = 2(x - 2)^2 + 9$

(27) $y = 7(x - 9)^2 - 3$

(46) $y = -5x^2 + 8$

(9) $y = 7(x + 5)^2 + 4$

(28) $y = 2(x + 4)^2 + 9$

(47) $y = 7(x + 8)^2 - 7$

(10) $y = -2(x - 6)^2 + 9$

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(56) $y = -5(x-9)^2 + 7$

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(1) $y = 8(x - 6)^2 - 4$

(20) $y = 3(x - 2)^2 + 9$

(39) $y = -3(x - 3)^2 + 5$

(2) $y = -2(x + 8)^2 - 2$

(21) $y = -7(x + 2)^2 + 5$

(40) $y = -7(x + 3)^2 + 4$

(3) $y = -10x^2 - 4$

(22) $y = x^2$

(41) $y = -(x + 6)^2 + 5$

(4) $y = -9(x - 6)^2 + 5$

(23) $y = -7(x + 6)^2 + 9$

(42) $y = 8(x - 10)^2 - 1$

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(8) $y = -4(x + 7)^2 - 9$

(27) $y = -(x - 1)^2 + 7$

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(45) $y = (x + 9)^2 + 3$

(8) $y = 4(x - 9)^2 - 4$

(27) $y = -(x + 5)^2 + 4$

(46) $y = -6(x + 1)^2 - 7$

(9) $y = -(x - 5)^2 + 7$

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(15) $y = 9x^2 - 4$

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(19) $y = 2(x - 5)^2 + 6$

(38) $y = 8(x + 4)^2 + 6$

(57) $y = 6(x - 7)^2 + 1$

(1) $y = -4(x + 4)^2 + 5$

(20) $y = -6x^2 + 6$

(39) $y = 8(x - 5)^2 - 2$

(2) $y = -9(x + 10)^2 + 7$

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(30) $y = 2(x + 2)^2 + 6$

(49) $y = -3x^2 + 5$

(12) $y = -4(x + 3)^2 - 2$

(31) $y = 4(x + 4)^2 - 9$

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(55) $y = -3(x - 4)^2 - 8$

(18) $y = 7(x + 10)^2 + 2$

(37) $y = -5(x - 1)^2 - 10$

(56) $y = -(x - 5)^2 - 3$

(19) $y = -(x - 10)^2 + 7$

(38) $y = -(x + 9)^2 - 2$

(57) $y = 9(x - 2)^2 - 8$

(1) $y = -4(x + 4)^2 + 6$

(20) $y = -7(x - 1)^2 - 1$

(39) $y = -7(x - 10)^2 + 7$

(2) $y = -6(x - 10)^2 - 2$

(21) $y = -10(x + 7)^2 + 3$

(40) $y = 4(x - 5)^2 - 6$

(3) $y = -8(x + 4)^2 + 5$

(22) $y = -3x^2 - 1$

(41) $y = 10(x - 1)^2 + 8$

(4) $y = 2(x + 7)^2 + 10$

(23) $y = 6(x + 7)^2 + 6$

(42) $y = -4(x - 1)^2 + 6$

(5) $y = -6(x + 6)^2 + 6$

(24) $y = 10x^2 + 3$

(43) $y = 8(x - 6)^2 + 7$

(6) $y = -4(x + 3)^2 - 7$

(25) $y = -3(x + 10)^2 - 2$

(44) $y = 5(x + 5)^2 - 10$

(7) $y = 2(x + 5)^2 + 6$

(26) $y = -7(x + 4)^2 - 5$

(45) $y = -10(x + 7)^2 - 4$

(8) $y = 6(x - 7)^2 - 10$

(27) $y = -6(x - 3)^2 + 10$

(46) $y = 10(x - 8)^2 - 9$

(9) $y = -8(x + 4)^2$

(28) $y = -(x + 7)^2 + 3$

(47) $y = -5(x + 1)^2 + 2$

(10) $y = -6(x + 2)^2 - 8$

(29) $y = -(x + 4)^2 - 8$

(48) $y = -(x + 4)^2 - 8$

(11) $y = 9(x + 9)^2$

(30) $y = 3(x - 1)^2 + 2$

(49) $y = -7(x + 3)^2 - 7$

(12) $y = 3(x + 1)^2$

(31) $y = 5(x - 8)^2 - 2$

(50) $y = 6(x - 4)^2 + 4$

(13) $y = -9(x + 10)^2 + 5$

(32) $y = 9(x + 3)^2 - 7$

(51) $y = -5(x - 3)^2 - 8$

(14) $y = -3(x - 8)^2 + 7$

(33) $y = -8(x - 6)^2 - 7$

(52) $y = -8(x + 1)^2 - 5$

(15) $y = -5(x - 7)^2 - 7$

(34) $y = -7(x - 3)^2 + 5$

(53) $y = 2(x + 7)^2 - 6$

(16) $y = 8(x + 2)^2 - 10$

(35) $y = -4(x + 9)^2 - 8$

(54) $y = 5(x - 10)^2 + 1$

(17) $y = 2(x - 8)^2 - 6$

(36) $y = 9(x - 3)^2 + 2$

(55) $y = 4(x - 4)^2 - 8$

(18) $y = -8(x - 9)^2 - 6$

(37) $y = 5(x - 10)^2 - 2$

(56) $y = 2(x - 6)^2 + 8$

(19) $y = 10(x + 6)^2 + 7$

(38) $y = -10(x - 4)^2 - 8$

(57) $y = -8(x + 5)^2 - 4$

(1) $y = -8(x - 1)^2 + 10$

(20) $y = -6(x - 6)^2 - 4$

(39) $y = -2(x - 6)^2 + 7$

(2) $y = 4(x - 5)^2 - 3$

(21) $y = -3(x - 9)^2 + 1$

(40) $y = 2(x + 3)^2 - 10$

(3) $y = 6(x - 6)^2 - 7$

(22) $y = 5(x + 4)^2 + 7$

(41) $y = -4(x + 5)^2 + 9$

(4) $y = 10(x - 1)^2$

(23) $y = (x - 6)^2 - 1$

(42) $y = 4(x - 10)^2 + 9$

(5) $y = -7(x + 8)^2 + 7$

(24) $y = -10(x - 6)^2 - 5$

(43) $y = -5(x + 8)^2 - 2$

(6) $y = 7(x + 10)^2 + 4$

(25) $y = 3(x + 4)^2 + 2$

(44) $y = -10(x + 9)^2 - 10$

(7) $y = 4(x - 4)^2 - 4$

(26) $y = 4(x - 2)^2 - 6$

(45) $y = 2(x - 1)^2 - 3$

(8) $y = -5(x + 6)^2 - 9$

(27) $y = -4(x - 6)^2 + 3$

(46) $y = -7(x - 2)^2 + 3$

(9) $y = 7(x + 1)^2 + 8$

(28) $y = -(x + 6)^2 + 10$

(47) $y = 4(x - 1)^2 + 1$

(10) $y = -4(x - 5)^2 + 5$

(29) $y = 6(x - 8)^2 - 2$

(48) $y = 9(x + 3)^2 - 3$

(11) $y = 3(x + 8)^2 - 6$

(30) $y = -10(x + 5)^2 + 10$

(49) $y = 7(x - 9)^2 + 4$

(12) $y = 4(x - 7)^2 - 5$

(31) $y = 10(x + 4)^2 - 10$

(50) $y = -4(x + 6)^2 + 9$

(13) $y = 9(x + 4)^2 - 5$

(32) $y = -2(x + 2)^2$

(51) $y = 4(x - 10)^2 + 1$

(14) $y = 6(x + 8)^2 - 9$

(33) $y = 3(x - 1)^2 - 1$

(52) $y = 10(x + 10)^2 - 5$

(15) $y = (x + 7)^2 + 10$

(34) $y = -5(x - 2)^2 - 8$

(53) $y = 2(x - 10)^2 + 4$

(16) $y = 3(x - 4)^2 - 9$

(35) $y = -5(x + 2)^2 + 9$

(54) $y = -9(x + 8)^2 + 6$

(17) $y = -4(x - 9)^2 + 10$

(36) $y = -2(x + 3)^2 - 5$

(55) $y = -6(x + 9)^2 + 7$

(18) $y = -2(x + 2)^2 - 2$

(37) $y = 6(x - 7)^2 - 9$

(56) $y = (x - 4)^2 - 5$

(19) $y = 5(x + 9)^2 - 2$

(38) $y = -10(x - 6)^2 + 10$

(57) $y = -10(x + 1)^2 + 2$

(1) $y = -2(x - 7)^2 + 10$

(20) $y = -2(x - 3)^2 + 10$

(39) $y = 2(x - 1)^2 + 6$

(2) $y = 5(x - 2)^2 + 10$

(21) $y = -8(x + 6)^2 + 9$

(40) $y = 7(x - 6)^2 - 5$

(3) $y = 9(x - 3)^2 + 6$

(22) $y = -6(x + 2)^2 - 5$

(41) $y = 10(x + 4)^2 + 3$

(4) $y = -8(x + 7)^2 + 9$

(23) $y = -10(x - 6)^2 - 1$

(42) $y = 7(x - 4)^2 + 2$

(5) $y = -2(x - 10)^2 + 7$

(24) $y = 4(x - 1)^2 + 10$

(43) $y = 2(x + 7)^2 - 6$

(6) $y = -5(x - 3)^2 + 6$

(25) $y = 9(x - 4)^2 - 4$

(44) $y = -8(x - 9)^2 - 8$

(7) $y = 4(x - 2)^2 - 6$

(26) $y = -5(x - 10)^2 - 9$

(45) $y = 3(x + 9)^2 - 2$

(8) $y = -8(x + 8)^2 - 1$

(27) $y = 2(x - 2)^2 - 1$

(46) $y = -4(x - 6)^2 - 10$

(9) $y = 7(x + 8)^2 - 4$

(28) $y = 7(x - 4)^2 - 3$

(47) $y = -3(x - 4)^2 - 2$

(10) $y = -7x^2 - 2$

(29) $y = -3x^2 + 5$

(48) $y = -(x - 5)^2 - 7$

(11) $y = -2(x + 10)^2 - 1$

(30) $y = 5(x - 7)^2 + 3$

(49) $y = 5(x + 7)^2 + 3$

(12) $y = 10(x + 5)^2 - 6$

(31) $y = 6(x - 10)^2 - 6$

(50) $y = -10(x + 10)^2 - 2$

(13) $y = 7(x - 5)^2 + 9$

(32) $y = 8(x + 3)^2 + 7$

(51) $y = 7(x + 10)^2 - 10$

(14) $y = -9(x - 6)^2 + 10$

(33) $y = 4(x - 2)^2 + 8$

(52) $y = -7(x - 4)^2 + 9$

(15) $y = -7(x - 8)^2 + 10$

(34) $y = -6(x + 5)^2 - 1$

(53) $y = -10(x + 3)^2 - 8$

(16) $y = 8(x + 7)^2$

(35) $y = 2(x - 5)^2 + 3$

(54) $y = 8(x + 6)^2 + 2$

(17) $y = 8(x - 6)^2 - 7$

(36) $y = 8(x - 6)^2 - 2$

(55) $y = -9(x + 7)^2 - 6$

(18) $y = 6(x - 1)^2 + 4$

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(38) $y = -5(x - 9)^2 + 7$

(57) $y = 10(x - 6)^2 + 2$

(1) $y = 5(x - 2)^2 - 3$

(20) $y = (x + 5)^2 + 1$

(39) $y = 4x^2 - 4$

(2) $y = -4(x - 6)^2 - 7$

(21) $y = -2(x + 6)^2 + 1$

(40) $y = -4(x + 2)^2 - 1$

(3) $y = -5(x - 5)^2 - 3$

(22) $y = 10(x + 6)^2 + 6$

(41) $y = -9x^2 + 3$

(4) $y = 5(x + 10)^2 - 10$

(23) $y = -3(x + 2)^2 - 3$

(42) $y = -2(x + 2)^2 + 2$

(5) $y = 2x^2$

(24) $y = -2(x - 1)^2 - 3$

(43) $y = 8(x + 2)^2 - 2$

(6) $y = -5(x + 6)^2 + 1$

(25) $y = -2(x - 10)^2 + 5$

(44) $y = 3(x - 4)^2 - 4$

(7) $y = 5(x + 4)^2 - 3$

(26) $y = -2(x - 7)^2 - 7$

(45) $y = 10(x + 9)^2$

(8) $y = 3(x + 4)^2 - 9$

(27) $y = -8(x + 4)^2 + 10$

(46) $y = 2(x - 2)^2$

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(25) $y = -(x-9)^2 + 3$

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(10) $y = -(x-6)^2 - 8$

(29) $y = (x-9)^2 - 7$

(48) $y = -9(x+6)^2 + 10$

(11) $y = -5(x+7)^2 + 9$

(30) $y = 2(x-8)^2 - 9$

(49) $y = (x-4)^2 + 9$

(12) $y = (x+8)^2 + 4$

(31) $y = -6(x-5)^2 - 10$

(50) $y = -10(x+5)^2$

(13) $y = -2(x+6)^2 + 1$

(32) $y = -8x^2 + 10$

(51) $y = 9(x-4)^2 + 1$

(14) $y = 7(x-9)^2 + 3$

(33) $y = -5(x+10)^2 + 5$

(52) $y = 10(x-9)^2 + 7$

(15) $y = 2(x-4)^2 - 10$

(34) $y = 4(x-3)^2 - 2$

(53) $y = 5(x-10)^2 + 6$

(16) $y = 10(x+4)^2 + 2$

(35) $y = 10(x+3)^2 + 10$

(54) $y = 4(x+10)^2 - 2$

(17) $y = -10(x+7)^2 - 1$

(36) $y = -6(x+9)^2 + 10$

(55) $y = -8(x-2)^2 - 3$

(18) $y = 10(x-2)^2 - 2$

(37) $y = -10(x+9)^2 + 4$

(56) $y = -2(x-6)^2 + 3$

(19) $y = -2(x+10)^2$

(38) $y = 3(x+9)^2 + 4$

(57) $y = -7(x-2)^2 + 4$

(1) $y = 10(x + 3)^2 + 10$

(20) $y = -3(x - 10)^2 - 8$

(39) $y = 2(x - 5)^2 - 7$

(2) $y = 5(x - 1)^2 - 1$

(21) $y = -9(x - 9)^2 - 8$

(40) $y = 8(x + 6)^2 - 1$

(3) $y = (x + 7)^2 + 9$

(22) $y = -7(x - 9)^2 - 3$

(41) $y = 2(x - 1)^2$

(4) $y = 4(x - 7)^2 - 8$

(23) $y = -6(x + 5)^2 - 6$

(42) $y = 2(x + 7)^2 + 7$

(5) $y = -6(x + 5)^2 - 1$

(24) $y = -3(x - 4)^2 + 4$

(43) $y = 10(x + 3)^2 - 9$

(6) $y = -9(x - 8)^2 + 7$

(25) $y = 8(x + 9)^2 - 5$

(44) $y = 10(x - 2)^2 + 5$

(7) $y = -4(x - 4)^2 - 2$

(26) $y = -7(x + 6)^2 - 9$

(45) $y = -2(x - 5)^2 + 9$

(8) $y = 2(x + 2)^2 - 4$

(27) $y = -7(x + 3)^2 + 8$

(46) $y = -(x - 2)^2 + 3$

(9) $y = -8x^2 + 5$

(28) $y = 9(x - 6)^2 + 6$

(47) $y = -6(x + 3)^2 + 7$

(10) $y = 10(x - 9)^2 + 8$

(29) $y = -(x - 9)^2 + 9$

(48) $y = 10(x - 10)^2 - 1$

(11) $y = 5(x - 2)^2 + 5$

(30) $y = 6(x - 4)^2$

(49) $y = -4(x - 1)^2 - 7$

(12) $y = 2(x - 4)^2 + 3$

(31) $y = -(x - 6)^2$

(50) $y = -5(x + 1)^2$

(13) $y = -5(x + 5)^2 + 2$

(32) $y = 3(x + 2)^2 - 4$

(51) $y = -8(x + 4)^2 - 10$

(14) $y = (x + 3)^2 + 4$

(33) $y = -9(x - 9)^2 - 2$

(52) $y = -2(x + 10)^2 - 3$

(15) $y = -(x + 10)^2 - 3$

(34) $y = 5(x + 9)^2 - 1$

(53) $y = 10(x + 6)^2 + 9$

(16) $y = -3(x + 9)^2 + 7$

(35) $y = -10(x - 8)^2 - 2$

(54) $y = -10(x - 2)^2 - 7$

(17) $y = 10(x + 8)^2 + 7$

(36) $y = -6(x - 7)^2 - 10$

(55) $y = -(x + 6)^2 + 5$

(18) $y = 7(x + 8)^2 - 10$

(37) $y = -4(x + 2)^2 + 6$

(56) $y = -3x^2 + 7$

(19) $y = -9(x + 10)^2 + 1$

(38) $y = 10(x - 3)^2 - 5$

(57) $y = 9(x + 6)^2 - 8$

- | | | |
|---|--|--|
| (1) $y = -6(x+7)^2 - 5$
頂点 $(-7, -5)$ | (20) $y = -3(x-9)^2 + 10$
頂点 $(9, 10)$ | (39) $y = -4(x-8)^2 + 2$
頂点 $(8, 2)$ |
| (2) $y = -3(x+2)^2 + 9$
頂点 $(-2, 9)$ | (21) $y = 4(x-8)^2 + 2$
頂点 $(8, 2)$ | (40) $y = -4(x+8)^2 - 5$
頂点 $(-8, -5)$ |
| (3) $y = 3(x-4)^2 - 10$
頂点 $(4, -10)$ | (22) $y = -7(x-6)^2 + 4$
頂点 $(6, 4)$ | (41) $y = 2(x-7)^2 + 7$
頂点 $(7, 7)$ |
| (4) $y = -4(x+8)^2 + 5$
頂点 $(-8, 5)$ | (23) $y = 7(x+9)^2 + 7$
頂点 $(-9, 7)$ | (42) $y = -6(x-4)^2 - 8$
頂点 $(4, -8)$ |
| (5) $y = 7(x-5)^2 + 4$
頂点 $(5, 4)$ | (24) $y = -2(x-6)^2 - 1$
頂点 $(6, -1)$ | (43) $y = -10(x+8)^2 - 5$
頂点 $(-8, -5)$ |
| (6) $y = -(x+5)^2 - 4$
頂点 $(-5, -4)$ | (25) $y = (x-9)^2 + 4$
頂点 $(9, 4)$ | (44) $y = -10(x-9)^2 + 9$
頂点 $(9, 9)$ |
| (7) $y = 9(x-9)^2 + 10$
頂点 $(9, 10)$ | (26) $y = -10(x-5)^2 - 7$
頂点 $(5, -7)$ | (45) $y = -10(x-4)^2 + 6$
頂点 $(4, 6)$ |
| (8) $y = -8(x-5)^2 - 8$
頂点 $(5, -8)$ | (27) $y = -8(x+5)^2 + 6$
頂点 $(-5, 6)$ | (46) $y = 7(x-10)^2 + 8$
頂点 $(10, 8)$ |
| (9) $y = 5(x-9)^2 + 6$
頂点 $(9, 6)$ | (28) $y = 8(x+7)^2 - 8$
頂点 $(-7, -8)$ | (47) $y = -2(x-8)^2 + 9$
頂点 $(8, 9)$ |
| (10) $y = -9x^2 - 8$
頂点 $(0, -8)$ | (29) $y = -3(x+7)^2 + 1$
頂点 $(-7, 1)$ | (48) $y = 6(x+4)^2 - 1$
頂点 $(-4, -1)$ |
| (11) $y = 2(x-6)^2 + 4$
頂点 $(6, 4)$ | (30) $y = 4(x-4)^2 + 7$
頂点 $(4, 7)$ | (49) $y = -3(x-3)^2 - 3$
頂点 $(3, -3)$ |
| (12) $y = -10(x-7)^2 - 1$
頂点 $(7, -1)$ | (31) $y = 5(x-10)^2 - 3$
頂点 $(10, -3)$ | (50) $y = 5(x+10)^2 + 4$
頂点 $(-10, 4)$ |
| (13) $y = 3(x+8)^2 + 10$
頂点 $(-8, 10)$ | (32) $y = -5(x+8)^2 + 10$
頂点 $(-8, 10)$ | (51) $y = -10(x-4)^2 - 7$
頂点 $(4, -7)$ |
| (14) $y = -(x+6)^2 + 10$
頂点 $(-6, 10)$ | (33) $y = -4(x-4)^2 + 6$
頂点 $(4, 6)$ | (52) $y = 9(x-10)^2 + 3$
頂点 $(10, 3)$ |
| (15) $y = 3(x-10)^2 - 9$
頂点 $(10, -9)$ | (34) $y = 8(x+8)^2 + 10$
頂点 $(-8, 10)$ | (53) $y = -9(x+1)^2 - 2$
頂点 $(-1, -2)$ |
| (16) $y = 6(x-3)^2 - 1$
頂点 $(3, -1)$ | (35) $y = -6(x+3)^2 + 3$
頂点 $(-3, 3)$ | (54) $y = -4(x+10)^2 + 4$
頂点 $(-10, 4)$ |
| (17) $y = 8(x-1)^2 + 8$
頂点 $(1, 8)$ | (36) $y = -3x^2 - 9$
頂点 $(0, -9)$ | (55) $y = 8(x-5)^2 - 9$
頂点 $(5, -9)$ |
| (18) $y = -6(x+2)^2 - 9$
頂点 $(-2, -9)$ | (37) $y = -8(x-2)^2 + 7$
頂点 $(2, 7)$ | (56) $y = -5(x-8)^2 + 6$
頂点 $(8, 6)$ |
| (19) $y = 7(x-6)^2 + 6$
頂点 $(6, 6)$ | (38) $y = -5(x-8)^2 + 4$
頂点 $(8, 4)$ | (57) $y = 9(x-4)^2 - 7$
頂点 $(4, -7)$ |

- | | | |
|--|---|--|
| (1) $y = -6(x - 3)^2 + 7$
頂点 (3, 7) | (20) $y = 7(x - 3)^2 - 2$
頂点 (3, -2) | (39) $y = -5(x - 8)^2 + 7$
頂点 (8, 7) |
| (2) $y = 5(x + 1)^2 - 7$
頂点 (-1, -7) | (21) $y = 2(x + 10)^2 - 6$
頂点 (-10, -6) | (40) $y = -6(x - 7)^2 + 10$
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頂点 (0, 9) | (20) $y = 6(x - 2)^2 - 7$
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| (19) $y = 5(x+9)^2 - 2$
頂点 (-9, -2) | (38) $y = -10(x-6)^2 + 10$
頂点 (6, 10) | (57) $y = -10(x+1)^2 + 2$
頂点 (-1, 2) |

- | | | |
|---|--|--|
| (1) $y = -2(x - 7)^2 + 10$
頂点 (7, 10) | (20) $y = -2(x - 3)^2 + 10$
頂点 (3, 10) | (39) $y = 2(x - 1)^2 + 6$
頂点 (1, 6) |
| (2) $y = 5(x - 2)^2 + 10$
頂点 (2, 10) | (21) $y = -8(x + 6)^2 + 9$
頂点 (-6, 9) | (40) $y = 7(x - 6)^2 - 5$
頂点 (6, -5) |
| (3) $y = 9(x - 3)^2 + 6$
頂点 (3, 6) | (22) $y = -6(x + 2)^2 - 5$
頂点 (-2, -5) | (41) $y = 10(x + 4)^2 + 3$
頂点 (-4, 3) |
| (4) $y = -8(x + 7)^2 + 9$
頂点 (-7, 9) | (23) $y = -10(x - 6)^2 - 1$
頂点 (6, -1) | (42) $y = 7(x - 4)^2 + 2$
頂点 (4, 2) |
| (5) $y = -2(x - 10)^2 + 7$
頂点 (10, 7) | (24) $y = 4(x - 1)^2 + 10$
頂点 (1, 10) | (43) $y = 2(x + 7)^2 - 6$
頂点 (-7, -6) |
| (6) $y = -5(x - 3)^2 + 6$
頂点 (3, 6) | (25) $y = 9(x - 4)^2 - 4$
頂点 (4, -4) | (44) $y = -8(x - 9)^2 - 8$
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| (7) $y = 4(x - 2)^2 - 6$
頂点 (2, -6) | (26) $y = -5(x - 10)^2 - 9$
頂点 (10, -9) | (45) $y = 3(x + 9)^2 - 2$
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頂点 (-8, -1) | (27) $y = 2(x - 2)^2 - 1$
頂点 (2, -1) | (46) $y = -4(x - 6)^2 - 10$
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頂点 (-8, -4) | (28) $y = 7(x - 4)^2 - 3$
頂点 (4, -3) | (47) $y = -3(x - 4)^2 - 2$
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頂点 (0, -2) | (29) $y = -3x^2 + 5$
頂点 (0, 5) | (48) $y = -(x - 5)^2 - 7$
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頂点 (10, -6) | (50) $y = -10(x + 10)^2 - 2$
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頂点 (8, 10) | (34) $y = -6(x + 5)^2 - 1$
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| (16) $y = 8(x + 7)^2$
頂点 (-7, 0) | (35) $y = 2(x - 5)^2 + 3$
頂点 (5, 3) | (54) $y = 8(x + 6)^2 + 2$
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| (17) $y = 8(x - 6)^2 - 7$
頂点 (6, -7) | (36) $y = 8(x - 6)^2 - 2$
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|---|---|--|
| (1) $y = 5(x - 2)^2 - 3$
頂点 (2, -3) | (20) $y = (x + 5)^2 + 1$
頂点 (-5, 1) | (39) $y = 4x^2 - 4$
頂点 (0, -4) |
| (2) $y = -4(x - 6)^2 - 7$
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頂点 (-6, 1) | (40) $y = -4(x + 2)^2 - 1$
頂点 (-2, -1) |
| (3) $y = -5(x - 5)^2 - 3$
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頂点 (-2, -3) | (42) $y = -2(x + 2)^2 + 2$
頂点 (-2, 2) |
| (5) $y = 2x^2$
頂点 (0, 0) | (24) $y = -2(x - 1)^2 - 3$
頂点 (1, -3) | (43) $y = 8(x + 2)^2 - 2$
頂点 (-2, -2) |
| (6) $y = -5(x + 6)^2 + 1$
頂点 (-6, 1) | (25) $y = -2(x - 10)^2 + 5$
頂点 (10, 5) | (44) $y = 3(x - 4)^2 - 4$
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|--|--|---|
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頂点 $(10, -1)$ | (39) $y = -7(x-1)^2 + 8$
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頂点 $(2, -8)$ | (23) $y = -10(x+4)^2 - 10$
頂点 $(-4, -10)$ | (42) $y = -7(x+7)^2 - 7$
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頂点 $(-6, -9)$ | (43) $y = 4(x-9)^2 + 6$
頂点 $(9, 6)$ |
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頂点 $(-8, 7)$ | (25) $y = -(x-9)^2 + 3$
頂点 $(9, 3)$ | (44) $y = 8(x-6)^2 - 6$
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頂点 $(2, 5)$ | (46) $y = 6(x-8)^2 - 8$
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| (9) $y = -6(x-6)^2 + 9$
頂点 $(6, 9)$ | (28) $y = (x-5)^2 + 2$
頂点 $(5, 2)$ | (47) $y = 9(x-10)^2 + 2$
頂点 $(10, 2)$ |
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頂点 $(6, -8)$ | (29) $y = (x-9)^2 - 7$
頂点 $(9, -7)$ | (48) $y = -9(x+6)^2 + 10$
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| (11) $y = -5(x+7)^2 + 9$
頂点 $(-7, 9)$ | (30) $y = 2(x-8)^2 - 9$
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頂点 $(5, -10)$ | (50) $y = -10(x+5)^2$
頂点 $(-5, 0)$ |
| (13) $y = -2(x+6)^2 + 1$
頂点 $(-6, 1)$ | (32) $y = -8x^2 + 10$
頂点 $(0, 10)$ | (51) $y = 9(x-4)^2 + 1$
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| (19) $y = -2(x+10)^2$
頂点 $(-10, 0)$ | (38) $y = 3(x+9)^2 + 4$
頂点 $(-9, 4)$ | (57) $y = -7(x-2)^2 + 4$
頂点 $(2, 4)$ |

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|--|--|---|
| (1) $y = 10(x + 3)^2 + 10$
頂点 $(-3, 10)$ | (20) $y = -3(x - 10)^2 - 8$
頂点 $(10, -8)$ | (39) $y = 2(x - 5)^2 - 7$
頂点 $(5, -7)$ |
| (2) $y = 5(x - 1)^2 - 1$
頂点 $(1, -1)$ | (21) $y = -9(x - 9)^2 - 8$
頂点 $(9, -8)$ | (40) $y = 8(x + 6)^2 - 1$
頂点 $(-6, -1)$ |
| (3) $y = (x + 7)^2 + 9$
頂点 $(-7, 9)$ | (22) $y = -7(x - 9)^2 - 3$
頂点 $(9, -3)$ | (41) $y = 2(x - 1)^2$
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頂点 $(-5, -6)$ | (42) $y = 2(x + 7)^2 + 7$
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頂点 $(-5, -1)$ | (24) $y = -3(x - 4)^2 + 4$
頂点 $(4, 4)$ | (43) $y = 10(x + 3)^2 - 9$
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頂点 $(8, 7)$ | (25) $y = 8(x + 9)^2 - 5$
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頂点 $(-6, -9)$ | (45) $y = -2(x - 5)^2 + 9$
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頂点 $(-3, 8)$ | (46) $y = -(x - 2)^2 + 3$
頂点 $(2, 3)$ |
| (9) $y = -8x^2 + 5$
頂点 $(0, 5)$ | (28) $y = 9(x - 6)^2 + 6$
頂点 $(6, 6)$ | (47) $y = -6(x + 3)^2 + 7$
頂点 $(-3, 7)$ |
| (10) $y = 10(x - 9)^2 + 8$
頂点 $(9, 8)$ | (29) $y = -(x - 9)^2 + 9$
頂点 $(9, 9)$ | (48) $y = 10(x - 10)^2 - 1$
頂点 $(10, -1)$ |
| (11) $y = 5(x - 2)^2 + 5$
頂点 $(2, 5)$ | (30) $y = 6(x - 4)^2$
頂点 $(4, 0)$ | (49) $y = -4(x - 1)^2 - 7$
頂点 $(1, -7)$ |
| (12) $y = 2(x - 4)^2 + 3$
頂点 $(4, 3)$ | (31) $y = -(x - 6)^2$
頂点 $(6, 0)$ | (50) $y = -5(x + 1)^2$
頂点 $(-1, 0)$ |
| (13) $y = -5(x + 5)^2 + 2$
頂点 $(-5, 2)$ | (32) $y = 3(x + 2)^2 - 4$
頂点 $(-2, -4)$ | (51) $y = -8(x + 4)^2 - 10$
頂点 $(-4, -10)$ |
| (14) $y = (x + 3)^2 + 4$
頂点 $(-3, 4)$ | (33) $y = -9(x - 9)^2 - 2$
頂点 $(9, -2)$ | (52) $y = -2(x + 10)^2 - 3$
頂点 $(-10, -3)$ |
| (15) $y = -(x + 10)^2 - 3$
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頂点 $(-9, -1)$ | (53) $y = 10(x + 6)^2 + 9$
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頂点 $(-9, 7)$ | (35) $y = -10(x - 8)^2 - 2$
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頂点 $(-8, 7)$ | (36) $y = -6(x - 7)^2 - 10$
頂点 $(7, -10)$ | (55) $y = -(x + 6)^2 + 5$
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| (18) $y = 7(x + 8)^2 - 10$
頂点 $(-8, -10)$ | (37) $y = -4(x + 2)^2 + 6$
頂点 $(-2, 6)$ | (56) $y = -3x^2 + 7$
頂点 $(0, 7)$ |
| (19) $y = -9(x + 10)^2 + 1$
頂点 $(-10, 1)$ | (38) $y = 10(x - 3)^2 - 5$
頂点 $(3, -5)$ | (57) $y = 9(x + 6)^2 - 8$
頂点 $(-6, -8)$ |

3.1.4 頂点の座標 (易しい 2)

2 次関数の頂点を平方完成している式から求める問題。

頂点の座標は 0 を含めた有理数。

$$(1) y = 9 \left(x - \frac{1}{2} \right)^2 + \frac{27}{4}$$

$$(12) y = -9 \left(x + \frac{1}{6} \right)^2 - \frac{23}{4}$$

$$(23) y = 9 \left(x - \frac{1}{6} \right)^2 + \frac{31}{4}$$

$$(2) y = 8 \left(x - \frac{7}{16} \right)^2 - \frac{241}{32}$$

$$(13) y = -7 \left(x - \frac{3}{14} \right)^2 + \frac{121}{28}$$

$$(24) y = -3 \left(x - \frac{3}{2} \right)^2 + \frac{23}{4}$$

$$(3) y = 9 \left(x - \frac{1}{18} \right)^2 - \frac{181}{36}$$

$$(14) y = 6 \left(x + \frac{7}{12} \right)^2 - \frac{265}{24}$$

$$(25) y = -2 \left(x - \frac{5}{4} \right)^2 + \frac{9}{8}$$

$$(4) y = 6 \left(x + \frac{7}{12} \right)^2 + \frac{47}{24}$$

$$(15) y = - \left(x - \frac{7}{2} \right)^2 + \frac{65}{4}$$

$$(26) y = - (x + 3)^2 + 6$$

$$(5) y = 8 \left(x - \frac{1}{8} \right)^2 - \frac{33}{8}$$

$$(16) y = -6 \left(x + \frac{1}{3} \right)^2 - \frac{22}{3}$$

$$(27) y = -6 \left(x + \frac{1}{4} \right)^2 + \frac{35}{8}$$

$$(6) y = 5 \left(x + \frac{3}{10} \right)^2 + \frac{151}{20}$$

$$(17) y = 3 \left(x - \frac{4}{3} \right)^2 - \frac{19}{3}$$

$$(28) y = -4 \left(x - \frac{1}{8} \right)^2 - \frac{15}{16}$$

$$(7) y = -6x^2 + 8$$

$$(18) y = -5 \left(x - \frac{1}{10} \right)^2 + \frac{21}{20}$$

$$(29) y = -2 \left(x + \frac{1}{2} \right)^2 - \frac{9}{2}$$

$$(8) y = 4 \left(x - \frac{1}{8} \right)^2 + \frac{15}{16}$$

$$(19) y = -3 \left(x - \frac{7}{6} \right)^2 + \frac{25}{12}$$

$$(30) y = 9 \left(x + \frac{1}{2} \right)^2 - \frac{5}{4}$$

$$(9) y = -6 \left(x - \frac{2}{3} \right)^2 + \frac{20}{3}$$

$$(20) y = 9 \left(x - \frac{5}{18} \right)^2 + \frac{191}{36}$$

$$(31) y = -7 \left(x - \frac{4}{7} \right)^2 - \frac{40}{7}$$

$$(10) y = 7 \left(x - \frac{5}{14} \right)^2 + \frac{171}{28}$$

$$(21) y = -7 \left(x + \frac{3}{14} \right)^2 + \frac{149}{28}$$

$$(32) y = 9 \left(x - \frac{1}{2} \right)^2 - \frac{5}{4}$$

$$(11) y = -3(x + 1)^2 + 5$$

$$(33) y = 4x^2 - 6$$

$$(22) y = -4(x + 1)^2 + 7$$

$$(1) y = 7 \left(x + \frac{1}{7} \right)^2 + \frac{13}{7}$$

$$(12) y = 5 \left(x + \frac{7}{10} \right)^2 - \frac{169}{20}$$

$$(23) y = -8 \left(x - \frac{3}{8} \right)^2 + \frac{73}{8}$$

$$(2) y = 9 \left(x + \frac{1}{9} \right)^2 - \frac{28}{9}$$

$$(13) y = 3 \left(x + \frac{5}{6} \right)^2 + \frac{23}{12}$$

$$(24) y = 5x^2 - 8$$

$$(3) y = 3 \left(x + \frac{2}{3} \right)^2 + \frac{14}{3}$$

$$(14) y = 9 \left(x + \frac{1}{18} \right)^2 - \frac{289}{36}$$

$$(25) y = 7 \left(x - \frac{9}{14} \right)^2 - \frac{25}{28}$$

$$(4) y = 3 \left(x - \frac{7}{6} \right)^2 + \frac{11}{12}$$

$$(15) y = -4 \left(x - \frac{3}{8} \right)^2 - \frac{119}{16}$$

$$(26) y = -7 \left(x - \frac{1}{14} \right)^2 - \frac{223}{28}$$

$$(5) y = 6 \left(x + \frac{1}{4} \right)^2 + \frac{21}{8}$$

$$(16) y = -8 \left(x - \frac{9}{16} \right)^2 + \frac{241}{32}$$

$$(27) y = -4 \left(x + \frac{1}{8} \right)^2 + \frac{1}{16}$$

$$(6) y = -6 \left(x - \frac{1}{4} \right)^2 - \frac{37}{8}$$

$$(17) y = -5 \left(x + \frac{4}{5} \right)^2 + \frac{1}{5}$$

$$(28) y = 2 \left(x - \frac{3}{4} \right)^2 + \frac{55}{8}$$

$$(7) y = -(x-1)^2 - 8$$

$$(18) y = -6 \left(x + \frac{1}{4} \right)^2 - \frac{69}{8}$$

$$(29) y = -(x+4)^2 + 8$$

$$(8) y = 6 \left(x - \frac{7}{12} \right)^2 - \frac{73}{24}$$

$$(19) y = -2 \left(x + \frac{3}{2} \right)^2 + \frac{13}{2}$$

$$(30) y = 3 \left(x + \frac{1}{6} \right)^2 + \frac{47}{12}$$

$$(9) y = 8 \left(x + \frac{1}{16} \right)^2 - \frac{193}{32}$$

$$(20) y = 3 \left(x + \frac{7}{6} \right)^2 - \frac{85}{12}$$

$$(31) y = -2 \left(x + \frac{3}{2} \right)^2 + \frac{9}{2}$$

$$(10) y = -2 \left(x - \frac{3}{2} \right)^2 + \frac{1}{2}$$

$$(21) y = -8 \left(x + \frac{3}{16} \right)^2 + \frac{265}{32}$$

$$(32) y = -9 \left(x + \frac{7}{18} \right)^2 - \frac{131}{36}$$

$$(11) y = 6 \left(x - \frac{1}{12} \right)^2 - \frac{73}{24}$$

$$(22) y = 4 \left(x + \frac{1}{2} \right)^2 - 6$$

$$(33) y = -(x+1)^2 - 5$$

(1) $y = 5 \left(x - \frac{3}{5} \right)^2 + \frac{11}{5}$

(12) $y = 6 \left(x + \frac{2}{3} \right)^2 - \frac{29}{3}$

(23) $y = -3 \left(x + \frac{7}{6} \right)^2 + \frac{133}{12}$

(2) $y = 8 \left(x + \frac{3}{16} \right)^2 - \frac{265}{32}$

(13) $y = 9 \left(x - \frac{1}{18} \right)^2 - \frac{1}{36}$

(24) $y = 4 \left(x - \frac{3}{4} \right)^2 + \frac{15}{4}$

(3) $y = 3 \left(x - \frac{1}{3} \right)^2 - \frac{19}{3}$

(14) $y = -9 \left(x + \frac{2}{9} \right)^2 - \frac{59}{9}$

(25) $y = -4(x+1)^2 - 5$

(4) $y = (x+4)^2 - 9$

(15) $y = -2(x+2)^2 + 10$

(26) $y = -5 \left(x + \frac{4}{5} \right)^2 + \frac{11}{5}$

(5) $y = -5 \left(x + \frac{1}{2} \right)^2 + \frac{29}{4}$

(16) $y = 7 \left(x + \frac{1}{2} \right)^2 - \frac{7}{4}$

(27) $y = -3 \left(x - \frac{3}{2} \right)^2 + \frac{19}{4}$

(6) $y = 3 \left(x - \frac{1}{2} \right)^2 + \frac{1}{4}$

(17) $y = 8 \left(x + \frac{7}{16} \right)^2 - \frac{177}{32}$

(28) $y = -5 \left(x - \frac{7}{10} \right)^2 + \frac{49}{20}$

(7) $y = 3 \left(x - \frac{7}{6} \right)^2 + \frac{23}{12}$

(18) $y = 8 \left(x + \frac{1}{4} \right)^2 - \frac{5}{2}$

(29) $y = 6 \left(x + \frac{1}{6} \right)^2 - \frac{25}{6}$

(8) $y = 3 \left(x + \frac{2}{3} \right)^2 - \frac{13}{3}$

(19) $y = -8 \left(x + \frac{1}{16} \right)^2 - \frac{127}{32}$

(30) $y = \left(x - \frac{9}{2} \right)^2 - \frac{57}{4}$

(9) $y = -9 \left(x - \frac{5}{18} \right)^2 + \frac{205}{36}$

(20) $y = 9x^2 + 7$

(31) $y = 3 \left(x - \frac{7}{6} \right)^2 - \frac{1}{12}$

(10) $y = 6 \left(x - \frac{3}{4} \right)^2 - \frac{3}{8}$

(21) $y = 9 \left(x - \frac{1}{9} \right)^2 + \frac{17}{9}$

(32) $y = 3 \left(x - \frac{3}{2} \right)^2 + \frac{9}{4}$

(11) $y = 4 \left(x - \frac{3}{4} \right)^2 - \frac{1}{4}$

(22) $y = -8 \left(x - \frac{5}{16} \right)^2 - \frac{167}{32}$

(33) $y = 8 \left(x - \frac{3}{16} \right)^2 - \frac{233}{32}$

$$(1) y = -9 \left(x - \frac{1}{2} \right)^2 + \frac{45}{4}$$

$$(12) y = -8 \left(x + \frac{1}{2} \right)^2 - 5$$

$$(23) y = 5 \left(x + \frac{2}{5} \right)^2 - \frac{4}{5}$$

$$(2) y = -5 \left(x + \frac{4}{5} \right)^2 + \frac{36}{5}$$

$$(13) y = 8 \left(x + \frac{9}{16} \right)^2 - \frac{145}{32}$$

$$(24) y = 2(x+1)^2 - 2$$

$$(3) y = 8 \left(x - \frac{3}{16} \right)^2 + \frac{151}{32}$$

$$(14) y = 4 \left(x + \frac{3}{8} \right)^2 + \frac{7}{16}$$

$$(25) y = -9 \left(x - \frac{1}{2} \right)^2 - \frac{27}{4}$$

$$(4) y = -6 \left(x + \frac{1}{3} \right)^2 - \frac{7}{3}$$

$$(15) y = -8 \left(x + \frac{9}{16} \right)^2 - \frac{79}{32}$$

$$(26) y = 2 \left(x + \frac{3}{4} \right)^2 - \frac{57}{8}$$

$$(5) y = 2x^2 - 7$$

$$(16) y = -9 \left(x + \frac{4}{9} \right)^2 - \frac{47}{9}$$

$$(27) y = -5 \left(x - \frac{3}{5} \right)^2 + \frac{34}{5}$$

$$(6) y = 7 \left(x + \frac{5}{14} \right)^2 + \frac{143}{28}$$

$$(17) y = -3 \left(x - \frac{1}{6} \right)^2 - \frac{83}{12}$$

$$(28) y = -4 \left(x + \frac{1}{4} \right)^2 + \frac{17}{4}$$

$$(7) y = -2(x-2)^2 + 9$$

$$(18) y = 9 \left(x - \frac{2}{9} \right)^2 + \frac{41}{9}$$

$$(29) y = -7 \left(x - \frac{3}{14} \right)^2 - \frac{47}{28}$$

$$(8) y = -3 \left(x - \frac{3}{2} \right)^2 + \frac{11}{4}$$

$$(19) y = 7 \left(x + \frac{9}{14} \right)^2 + \frac{171}{28}$$

$$(30) y = 3 \left(x - \frac{1}{6} \right)^2 - \frac{1}{12}$$

$$(9) y = 8 \left(x + \frac{1}{2} \right)^2 + 7$$

$$(20) y = -2 \left(x + \frac{3}{4} \right)^2 - \frac{31}{8}$$

$$(31) y = -3 \left(x + \frac{5}{6} \right)^2 - \frac{83}{12}$$

$$(10) y = -6 \left(x + \frac{1}{3} \right)^2 + \frac{20}{3}$$

$$(21) y = 8 \left(x + \frac{9}{16} \right)^2 - \frac{305}{32}$$

$$(32) y = -2 \left(x + \frac{5}{4} \right)^2 + \frac{65}{8}$$

$$(11) y = 8 \left(x - \frac{3}{16} \right)^2 + \frac{215}{32}$$

$$(22) y = -8 \left(x + \frac{1}{16} \right)^2 + \frac{33}{32}$$

$$(33) y = -8 \left(x - \frac{3}{16} \right)^2 + \frac{265}{32}$$

(1) $y = x^2 - 5$

(12) $y = -4 \left(x + \frac{1}{8} \right)^2 + \frac{1}{16}$

(23) $y = -4 \left(x - \frac{5}{8} \right)^2 - \frac{87}{16}$

(2) $y = -(x - 2)^2 + 5$

(13) $y = -7x^2 + 4$

(24) $y = 7 \left(x + \frac{1}{2} \right)^2 - \frac{43}{4}$

(3) $y = -6 \left(x + \frac{2}{3} \right)^2 + \frac{29}{3}$

(14) $y = - \left(x - \frac{1}{2} \right)^2 - \frac{23}{4}$

(25) $y = -3 \left(x + \frac{1}{3} \right)^2 + \frac{13}{3}$

(4) $y = 3 \left(x + \frac{4}{3} \right)^2 - \frac{37}{3}$

(15) $y = 6 \left(x - \frac{3}{4} \right)^2 + \frac{37}{8}$

(26) $y = -4(x + 1)^2 - 1$

(5) $y = 3 \left(x - \frac{2}{3} \right)^2 - \frac{4}{3}$

(16) $y = 5 \left(x - \frac{1}{10} \right)^2 + \frac{19}{20}$

(27) $y = -7 \left(x - \frac{3}{7} \right)^2 - \frac{33}{7}$

(6) $y = -5 \left(x + \frac{7}{10} \right)^2 - \frac{111}{20}$

(17) $y = 5 \left(x + \frac{3}{5} \right)^2 + \frac{31}{5}$

(28) $y = 6 \left(x + \frac{2}{3} \right)^2 - \frac{2}{3}$

(7) $y = -5 \left(x - \frac{1}{2} \right)^2 + \frac{33}{4}$

(18) $y = (x - 2)^2 - 3$

(29) $y = 2 \left(x + \frac{7}{4} \right)^2 - \frac{17}{8}$

(8) $y = 3 \left(x - \frac{3}{2} \right)^2 - \frac{3}{4}$

(19) $y = 5 \left(x + \frac{3}{10} \right)^2 - \frac{9}{20}$

(30) $y = -8 \left(x + \frac{1}{4} \right)^2 + \frac{5}{2}$

(9) $y = 5 \left(x - \frac{1}{2} \right)^2 + \frac{7}{4}$

(20) $y = - \left(x - \frac{1}{2} \right)^2 + \frac{25}{4}$

(31) $y = 8 \left(x - \frac{7}{16} \right)^2 - \frac{113}{32}$

(10) $y = 4 \left(x - \frac{1}{4} \right)^2 - \frac{13}{4}$

(21) $y = -2(x - 1)^2 + 4$

(32) $y = -6 \left(x + \frac{5}{12} \right)^2 - \frac{191}{24}$

(11) $y = -6 \left(x + \frac{2}{3} \right)^2 - \frac{1}{3}$

(22) $y = -5 \left(x + \frac{4}{5} \right)^2 + \frac{36}{5}$

(33) $y = 2(x + 1)^2 - 3$

$$(1) y = 3 \left(x + \frac{3}{2} \right)^2 - \frac{63}{4}$$

$$(12) y = -(x+1)^2 - 8$$

$$(23) y = 7 \left(x - \frac{9}{14} \right)^2 + \frac{31}{28}$$

$$(2) y = 6 \left(x + \frac{1}{3} \right)^2 - \frac{26}{3}$$

$$(13) y = 3 \left(x - \frac{7}{6} \right)^2 + \frac{47}{12}$$

$$(24) y = 9 \left(x + \frac{2}{9} \right)^2 - \frac{22}{9}$$

$$(3) y = 5 \left(x - \frac{1}{2} \right)^2 + \frac{11}{4}$$

$$(14) y = -6 \left(x + \frac{1}{12} \right)^2 - \frac{47}{24}$$

$$(25) y = -4 \left(x - \frac{7}{8} \right)^2 - \frac{79}{16}$$

$$(15) y = -2(x+1)^2 + 4$$

$$(4) y = -6 \left(x - \frac{3}{4} \right)^2 - \frac{13}{8}$$

$$(16) y = 7 \left(x - \frac{3}{7} \right)^2 - \frac{44}{7}$$

$$(26) y = 5 \left(x - \frac{9}{10} \right)^2 - \frac{81}{20}$$

$$(5) y = 5 \left(x + \frac{3}{10} \right)^2 - \frac{69}{20}$$

$$(17) y = -6 \left(x + \frac{7}{12} \right)^2 - \frac{143}{24}$$

$$(27) y = 8 \left(x - \frac{1}{2} \right)^2 + 5$$

$$(6) y = 6 \left(x - \frac{1}{4} \right)^2 + \frac{61}{8}$$

$$(18) y = 6 \left(x - \frac{2}{3} \right)^2 + \frac{7}{3}$$

$$(28) y = -2(x+1)^2 + 8$$

$$(7) y = 7 \left(x + \frac{1}{2} \right)^2 + \frac{25}{4}$$

$$(19) y = 4 \left(x - \frac{1}{8} \right)^2 - \frac{49}{16}$$

$$(29) y = 8 \left(x - \frac{3}{8} \right)^2 - \frac{81}{8}$$

$$(8) y = -8 \left(x + \frac{5}{16} \right)^2 + \frac{313}{32}$$

$$(20) y = -8 \left(x + \frac{1}{4} \right)^2 - \frac{7}{2}$$

$$(30) y = 9 \left(x - \frac{5}{18} \right)^2 - \frac{205}{36}$$

$$(9) y = 6 \left(x + \frac{1}{2} \right)^2 + \frac{9}{2}$$

$$(21) y = -6x^2 + 9$$

$$(31) y = -2 \left(x - \frac{7}{4} \right)^2 + \frac{113}{8}$$

$$(10) y = -4 \left(x + \frac{3}{8} \right)^2 - \frac{7}{16}$$

$$(22) y = -2(x+2)^2 + 17$$

$$(32) y = -8 \left(x - \frac{3}{16} \right)^2 + \frac{9}{32}$$

$$(11) y = 7 \left(x - \frac{2}{7} \right)^2 + \frac{38}{7}$$

$$(33) y = -4 \left(x + \frac{1}{2} \right)^2 + 7$$

$$(1) y = -6 \left(x - \frac{2}{3} \right)^2 - \frac{10}{3}$$

$$(12) y = 7 \left(x - \frac{2}{7} \right)^2 + \frac{3}{7}$$

$$(23) y = 7 \left(x - \frac{9}{14} \right)^2 - \frac{249}{28}$$

$$(2) y = -7 \left(x - \frac{1}{7} \right)^2 + \frac{8}{7}$$

$$(13) y = -5x^2 + 6$$

$$(24) y = -6 \left(x - \frac{1}{6} \right)^2 - \frac{17}{6}$$

$$(3) y = -(x+1)^2 + 3$$

$$(14) y = -7 \left(x - \frac{9}{14} \right)^2 - \frac{115}{28}$$

$$(25) y = 8 \left(x + \frac{7}{16} \right)^2 - \frac{177}{32}$$

$$(4) y = \left(x + \frac{7}{2} \right)^2 - \frac{13}{4}$$

$$(15) y = -8 \left(x - \frac{1}{8} \right)^2 + \frac{9}{8}$$

$$(26) y = 6 \left(x + \frac{2}{3} \right)^2 - \frac{8}{3}$$

$$(5) y = -7 \left(x - \frac{9}{14} \right)^2 - \frac{59}{28}$$

$$(16) y = -8 \left(x - \frac{3}{8} \right)^2 + \frac{1}{8}$$

$$(27) y = -3 \left(x + \frac{4}{3} \right)^2 + \frac{31}{3}$$

$$(6) y = -5x^2 - 9$$

$$(17) y = -3(x-1)^2 + 10$$

$$(28) y = -2(x-2)^2 + 7$$

$$(7) y = -6 \left(x - \frac{1}{6} \right)^2 - \frac{11}{6}$$

$$(18) y = -6 \left(x - \frac{1}{3} \right)^2 + \frac{2}{3}$$

$$(29) y = -4 \left(x - \frac{1}{8} \right)^2 + \frac{145}{16}$$

$$(8) y = 7 \left(x + \frac{2}{7} \right)^2 + \frac{3}{7}$$

$$(19) y = -8 \left(x + \frac{1}{16} \right)^2 - \frac{191}{32}$$

$$(30) y = 3 \left(x + \frac{2}{3} \right)^2 + \frac{20}{3}$$

$$(9) y = -3 \left(x + \frac{1}{3} \right)^2 + \frac{13}{3}$$

$$(20) y = 6 \left(x - \frac{1}{12} \right)^2 + \frac{47}{24}$$

$$(31) y = -6 \left(x - \frac{1}{3} \right)^2 + \frac{23}{3}$$

$$(10) y = -6 \left(x - \frac{1}{3} \right)^2 + \frac{23}{3}$$

$$(21) y = - \left(x + \frac{7}{2} \right)^2 + \frac{65}{4}$$

$$(32) y = 2 \left(x - \frac{5}{4} \right)^2 + \frac{31}{8}$$

$$(11) y = -7 \left(x - \frac{1}{2} \right)^2 + \frac{31}{4}$$

$$(22) y = 9 \left(x + \frac{1}{18} \right)^2 + \frac{215}{36}$$

$$(33) y = - \left(x + \frac{3}{2} \right)^2 + \frac{1}{4}$$

$$(1) y = 8 \left(x - \frac{1}{4} \right)^2 + \frac{13}{2}$$

$$(12) y = 3 \left(x + \frac{3}{2} \right)^2 - \frac{15}{4}$$

$$(23) y = 9 \left(x - \frac{1}{9} \right)^2 + \frac{71}{9}$$

$$(2) y = 6 \left(x - \frac{2}{3} \right)^2 + \frac{19}{3}$$

$$(13) y = -9 \left(x - \frac{4}{9} \right)^2 + \frac{43}{9}$$

$$(24) y = -8 \left(x + \frac{1}{16} \right)^2 + \frac{161}{32}$$

$$(3) y = 4 \left(x - \frac{7}{8} \right)^2 - \frac{81}{16}$$

$$(14) y = \left(x + \frac{3}{2} \right)^2 - \frac{33}{4}$$

$$(25) y = -8 \left(x - \frac{5}{16} \right)^2 - \frac{167}{32}$$

$$(4) y = 4x^2 - 9$$

$$(15) y = 2 \left(x + \frac{3}{4} \right)^2 - \frac{41}{8}$$

$$(26) y = 5 \left(x + \frac{2}{5} \right)^2 + \frac{6}{5}$$

$$(5) y = - \left(x + \frac{3}{2} \right)^2 - \frac{23}{4}$$

$$(16) y = 4 \left(x - \frac{1}{8} \right)^2 + \frac{79}{16}$$

$$(27) y = 3 \left(x + \frac{1}{6} \right)^2 - \frac{1}{12}$$

$$(6) y = -3 \left(x - \frac{7}{6} \right)^2 + \frac{133}{12}$$

$$(17) y = -7 \left(x - \frac{2}{7} \right)^2 - \frac{59}{7}$$

$$(28) y = -8 \left(x + \frac{7}{16} \right)^2 + \frac{49}{32}$$

$$(7) y = (x - 3)^2 - 6$$

$$(18) y = -6 \left(x + \frac{5}{12} \right)^2 - \frac{191}{24}$$

$$(29) y = -7x^2 - 8$$

$$(8) y = -6 \left(x + \frac{1}{2} \right)^2 + \frac{1}{2}$$

$$(19) y = 8 \left(x - \frac{1}{16} \right)^2 - \frac{97}{32}$$

$$(30) y = 7 \left(x - \frac{1}{2} \right)^2 - \frac{23}{4}$$

$$(9) y = -5 \left(x + \frac{1}{5} \right)^2 - \frac{4}{5}$$

$$(19) y = 8 \left(x - \frac{1}{16} \right)^2 - \frac{97}{32}$$

$$(31) y = -(x - 4)^2 + 23$$

$$(10) y = 2 \left(x - \frac{7}{4} \right)^2 - \frac{81}{8}$$

$$(20) y = 5 \left(x + \frac{4}{5} \right)^2 - \frac{36}{5}$$

$$(32) y = -2 \left(x - \frac{5}{4} \right)^2 + \frac{17}{8}$$

$$(11) y = 2x^2 - 6$$

$$(21) y = -8 \left(x - \frac{1}{2} \right)^2 - 6$$

$$(33) y = 5 \left(x + \frac{1}{5} \right)^2 + \frac{9}{5}$$

$$(22) y = 4 \left(x + \frac{3}{4} \right)^2 + \frac{27}{4}$$

$$(1) y = -7 \left(x - \frac{1}{2} \right)^2 - \frac{29}{4}$$

$$(12) y = 3 \left(x + \frac{1}{2} \right)^2 - \frac{19}{4}$$

$$(23) y = -3 \left(x - \frac{1}{6} \right)^2 - \frac{83}{12}$$

$$(2) y = 3x^2 - 7$$

$$(13) y = 7 \left(x - \frac{2}{7} \right)^2 + \frac{31}{7}$$

$$(24) y = - \left(x + \frac{3}{2} \right)^2 + \frac{1}{4}$$

$$(3) y = 5 \left(x - \frac{9}{10} \right)^2 + \frac{59}{20}$$

$$(14) y = -3 \left(x + \frac{7}{6} \right)^2 - \frac{23}{12}$$

$$(25) y = 9x^2 - 3$$

$$(4) y = - (x + 4)^2 + 14$$

$$(26) y = 9 \left(x - \frac{4}{9} \right)^2 - \frac{43}{9}$$

$$(5) y = -2 (x + 2)^2 + 16$$

$$(15) y = 2 \left(x - \frac{1}{4} \right)^2 + \frac{63}{8}$$

$$(27) y = 2x^2 + 9$$

$$(6) y = 4 \left(x + \frac{7}{8} \right)^2 + \frac{63}{16}$$

$$(16) y = -5 \left(x + \frac{1}{2} \right)^2 - \frac{23}{4}$$

$$(28) y = -6 \left(x - \frac{7}{12} \right)^2 + \frac{1}{24}$$

$$(7) y = 2 \left(x + \frac{1}{4} \right)^2 - \frac{33}{8}$$

$$(17) y = -9 \left(x - \frac{2}{9} \right)^2 - \frac{59}{9}$$

$$(29) y = -9 \left(x - \frac{1}{6} \right)^2 + \frac{17}{4}$$

$$(8) y = 5 \left(x + \frac{3}{5} \right)^2 - \frac{34}{5}$$

$$(18) y = -9 \left(x - \frac{1}{2} \right)^2 - \frac{27}{4}$$

$$(30) y = 5 \left(x + \frac{3}{10} \right)^2 - \frac{189}{20}$$

$$(9) y = -8 \left(x - \frac{1}{8} \right)^2 - \frac{7}{8}$$

$$(19) y = 3 \left(x + \frac{5}{6} \right)^2 - \frac{121}{12}$$

$$(31) y = 6x^2$$

$$(10) y = -7 \left(x - \frac{3}{7} \right)^2 + \frac{30}{7}$$

$$(20) y = 3 \left(x - \frac{7}{6} \right)^2 - \frac{1}{12}$$

$$(32) y = -6 \left(x - \frac{1}{3} \right)^2 - \frac{16}{3}$$

$$(11) y = 7x^2$$

$$(21) y = -9 \left(x + \frac{1}{9} \right)^2 + \frac{64}{9}$$

$$(33) y = 5x^2$$

$$(22) y = -2 \left(x - \frac{3}{4} \right)^2 + \frac{57}{8}$$

(1) $y = 5 \left(x - \frac{7}{10} \right)^2 + \frac{131}{20}$

(12) $y = -9 \left(x + \frac{4}{9} \right)^2 + \frac{61}{9}$

(23) $y = 3 \left(x + \frac{1}{3} \right)^2 - \frac{4}{3}$

(2) $y = 6 \left(x + \frac{2}{3} \right)^2 - \frac{29}{3}$

(13) $y = \left(x + \frac{1}{2} \right)^2 - \frac{9}{4}$

(24) $y = 4 \left(x - \frac{1}{8} \right)^2 + \frac{111}{16}$

(3) $y = -3 \left(x - \frac{7}{6} \right)^2 - \frac{59}{12}$

(14) $y = -6 \left(x + \frac{1}{4} \right)^2 + \frac{43}{8}$

(25) $y = -9 \left(x + \frac{1}{3} \right)^2 - 3$

(4) $y = 6 \left(x + \frac{1}{2} \right)^2 - \frac{21}{2}$

(15) $y = -8 \left(x - \frac{3}{16} \right)^2 + \frac{265}{32}$

(26) $y = - \left(x + \frac{1}{2} \right)^2 + \frac{25}{4}$

(5) $y = 4 \left(x - \frac{1}{2} \right)^2 - 7$

(16) $y = (x - 2)^2 + 2$

(27) $y = \left(x - \frac{7}{2} \right)^2 - \frac{29}{4}$

(6) $y = -7x^2 + 3$

(17) $y = -4 \left(x - \frac{3}{8} \right)^2 - \frac{135}{16}$

(28) $y = 2(x + 1)^2 - 8$

(7) $y = 4 \left(x + \frac{9}{8} \right)^2 - \frac{65}{16}$

(18) $y = - \left(x - \frac{1}{2} \right)^2 + \frac{33}{4}$

(29) $y = -2 \left(x - \frac{3}{2} \right)^2 + \frac{13}{2}$

(8) $y = 2 \left(x - \frac{3}{4} \right)^2 + \frac{15}{8}$

(19) $y = 5 \left(x + \frac{1}{5} \right)^2 - \frac{21}{5}$

(30) $y = - \left(x - \frac{5}{2} \right)^2 + \frac{37}{4}$

(9) $y = (x + 3)^2 - 15$

(20) $y = 5x^2 - 3$

(31) $y = - \left(x + \frac{1}{2} \right)^2 - \frac{7}{4}$

(10) $y = -4 \left(x + \frac{1}{8} \right)^2 + \frac{49}{16}$

(21) $y = \left(x - \frac{5}{2} \right)^2 - \frac{49}{4}$

(32) $y = -3 \left(x + \frac{3}{2} \right)^2 - \frac{5}{4}$

(11) $y = -2(x - 1)^2 + 7$

(22) $y = 9 \left(x - \frac{1}{9} \right)^2 - \frac{46}{9}$

(33) $y = \left(x - \frac{1}{2} \right)^2 - \frac{5}{4}$

(1) $y = -6x^2 + 6$

(12) $y = \left(x + \frac{7}{2}\right)^2 - \frac{81}{4}$

(23) $y = -8\left(x + \frac{3}{16}\right)^2 + \frac{9}{32}$

(2) $y = 3\left(x + \frac{5}{6}\right)^2 + \frac{47}{12}$

(13) $y = \left(x - \frac{9}{2}\right)^2 - \frac{61}{4}$

(24) $y = 2\left(x + \frac{7}{4}\right)^2 - \frac{41}{8}$

(3) $y = -2(x - 1)^2 - 4$

(14) $y = -7\left(x + \frac{1}{7}\right)^2 + \frac{50}{7}$

(25) $y = 4\left(x + \frac{3}{8}\right)^2 - \frac{41}{16}$

(4) $y = 6\left(x - \frac{1}{3}\right)^2 - \frac{8}{3}$

(15) $y = 9\left(x - \frac{1}{9}\right)^2 - \frac{46}{9}$

(26) $y = 7\left(x - \frac{4}{7}\right)^2 - \frac{2}{7}$

(5) $y = -8\left(x + \frac{3}{16}\right)^2 - \frac{215}{32}$

(16) $y = 4\left(x + \frac{7}{8}\right)^2 - \frac{177}{16}$

(27) $y = 7\left(x + \frac{1}{7}\right)^2 - \frac{57}{7}$

(6) $y = 6\left(x + \frac{1}{4}\right)^2 - \frac{43}{8}$

(17) $y = \left(x - \frac{3}{2}\right)^2 + \frac{7}{4}$

(28) $y = -8\left(x - \frac{3}{16}\right)^2 - \frac{23}{32}$

(7) $y = 3\left(x - \frac{2}{3}\right)^2 + \frac{5}{3}$

(8) $y = -7\left(x - \frac{3}{14}\right)^2 - \frac{243}{28}$

(18) $y = 4\left(x - \frac{1}{8}\right)^2 - \frac{65}{16}$

(29) $y = -8\left(x - \frac{7}{16}\right)^2 + \frac{81}{32}$

(9) $y = -8\left(x + \frac{3}{16}\right)^2 - \frac{55}{32}$

(19) $y = -2\left(x - \frac{3}{4}\right)^2 - \frac{31}{8}$

(30) $y = \left(x - \frac{3}{2}\right)^2 + \frac{27}{4}$

(10) $y = -5\left(x + \frac{3}{10}\right)^2 + \frac{29}{20}$

(20) $y = 4\left(x - \frac{1}{4}\right)^2 + \frac{7}{4}$

(31) $y = -8\left(x - \frac{7}{16}\right)^2 + \frac{81}{32}$

(11) $y = 2(x - 2)^2 - 2$

(21) $y = 9\left(x + \frac{1}{3}\right)^2 - 1$

(32) $y = -4\left(x - \frac{3}{8}\right)^2 + \frac{41}{16}$

(22) $y = -6\left(x + \frac{1}{6}\right)^2 + \frac{19}{6}$

(33) $y = 6\left(x + \frac{1}{6}\right)^2 - \frac{43}{6}$

$$(1) y = 7 \left(x - \frac{4}{7} \right)^2 - \frac{2}{7}$$

$$(12) y = 8 \left(x - \frac{1}{4} \right)^2 - \frac{9}{2}$$

$$(23) y = - \left(x + \frac{3}{2} \right)^2 + \frac{41}{4}$$

$$(2) y = -7x^2 - 5$$

$$(13) y = -8 \left(x - \frac{1}{16} \right)^2 + \frac{289}{32}$$

$$(24) y = - (x + 3)^2 + 8$$

$$(3) y = 6 \left(x + \frac{1}{4} \right)^2 + \frac{21}{8}$$

$$(14) y = 9 \left(x - \frac{2}{9} \right)^2 + \frac{5}{9}$$

$$(25) y = 9 \left(x + \frac{2}{9} \right)^2 - \frac{22}{9}$$

$$(4) y = 2 \left(x + \frac{1}{4} \right)^2 + \frac{47}{8}$$

$$(15) y = -2 (x + 1)^2 + 5$$

$$(26) y = -5 \left(x + \frac{9}{10} \right)^2 + \frac{241}{20}$$

$$(5) y = -2 (x - 1)^2 + 5$$

$$(16) y = -9x^2 - 2$$

$$(27) y = -4 \left(x - \frac{1}{8} \right)^2 + \frac{33}{16}$$

$$(6) y = 4 (x - 1)^2 - 11$$

$$(17) y = -6 \left(x + \frac{1}{4} \right)^2 - \frac{29}{8}$$

$$(28) y = (x + 3)^2 - 12$$

$$(7) y = 6 \left(x - \frac{1}{12} \right)^2 - \frac{49}{24}$$

$$(18) y = -6 \left(x - \frac{7}{12} \right)^2 - \frac{95}{24}$$

$$(29) y = -5 \left(x + \frac{1}{5} \right)^2 + \frac{6}{5}$$

$$(8) y = 3 (x - 1)^2 - 10$$

$$(9) y = 7 \left(x + \frac{3}{14} \right)^2 + \frac{103}{28}$$

$$(19) y = -8 \left(x + \frac{7}{16} \right)^2 + \frac{337}{32}$$

$$(30) y = 5 \left(x + \frac{4}{5} \right)^2 - \frac{16}{5}$$

$$(10) y = -7 \left(x + \frac{1}{14} \right)^2 - \frac{195}{28}$$

$$(20) y = -4 (x + 1)^2 + 11$$

$$(31) y = -3 \left(x + \frac{1}{3} \right)^2 - \frac{20}{3}$$

$$(21) y = 3 \left(x - \frac{2}{3} \right)^2 - \frac{10}{3}$$

$$(32) y = 2 \left(x - \frac{5}{4} \right)^2 + \frac{15}{8}$$

$$(11) y = -3 \left(x - \frac{1}{6} \right)^2 - \frac{47}{12}$$

$$(22) y = -8 \left(x - \frac{9}{16} \right)^2 + \frac{145}{32}$$

$$(33) y = 3 \left(x + \frac{2}{3} \right)^2 - \frac{19}{3}$$

$$(1) y = 7 \left(x - \frac{1}{2} \right)^2 - \frac{43}{4}$$

$$(12) y = 2(x-1)^2 + 2$$

$$(23) y = 3 \left(x - \frac{7}{6} \right)^2 + \frac{11}{12}$$

$$(2) y = 2 \left(x - \frac{9}{4} \right)^2 - \frac{33}{8}$$

$$(13) y = 2 \left(x + \frac{3}{4} \right)^2 - \frac{41}{8}$$

$$(24) y = -2x^2 + 4$$

$$(3) y = 6 \left(x - \frac{7}{12} \right)^2 + \frac{143}{24}$$

$$(14) y = 6 \left(x + \frac{1}{4} \right)^2 + \frac{21}{8}$$

$$(25) y = 4x^2 + 3$$

$$(4) y = 9 \left(x + \frac{2}{9} \right)^2 + \frac{32}{9}$$

$$(15) y = -9 \left(x + \frac{1}{6} \right)^2 + \frac{9}{4}$$

$$(26) y = -7 \left(x + \frac{1}{7} \right)^2 - \frac{6}{7}$$

$$(5) y = 8 \left(x - \frac{3}{8} \right)^2 - \frac{33}{8}$$

$$(16) y = -2 \left(x - \frac{1}{4} \right)^2 + \frac{57}{8}$$

$$(27) y = -7 \left(x + \frac{1}{14} \right)^2 + \frac{85}{28}$$

$$(6) y = \left(x + \frac{9}{2} \right)^2 - \frac{73}{4}$$

$$(17) y = 8 \left(x + \frac{3}{16} \right)^2 + \frac{247}{32}$$

$$(28) y = -6 \left(x - \frac{1}{12} \right)^2 - \frac{191}{24}$$

$$(7) y = 3 \left(x + \frac{7}{6} \right)^2 - \frac{13}{12}$$

$$(18) y = 2(x+2)^2 - 16$$

$$(29) y = -7 \left(x - \frac{1}{7} \right)^2 + \frac{8}{7}$$

$$(8) y = 6 \left(x - \frac{3}{4} \right)^2 - \frac{99}{8}$$

$$(19) y = 5 \left(x + \frac{2}{5} \right)^2 + \frac{41}{5}$$

$$(30) y = 3 \left(x + \frac{3}{2} \right)^2 - \frac{23}{4}$$

$$(9) y = -3 \left(x - \frac{1}{6} \right)^2 - \frac{83}{12}$$

$$(20) y = -9 \left(x + \frac{4}{9} \right)^2 + \frac{52}{9}$$

$$(31) y = -6 \left(x - \frac{1}{2} \right)^2 + \frac{15}{2}$$

$$(10) y = -3 \left(x - \frac{4}{3} \right)^2 + \frac{16}{3}$$

$$(21) y = 8 \left(x + \frac{1}{2} \right)^2 - 5$$

$$(32) y = 3 \left(x - \frac{4}{3} \right)^2 - \frac{40}{3}$$

$$(11) y = -5 \left(x - \frac{1}{5} \right)^2 + \frac{26}{5}$$

$$(22) y = -2 \left(x - \frac{1}{2} \right)^2 + \frac{17}{2}$$

$$(33) y = 2(x-2)^2 - 12$$

$$(1) y = 3 \left(x - \frac{1}{6} \right)^2 - \frac{25}{12}$$

$$(12) y = 6 \left(x - \frac{2}{3} \right)^2 - \frac{35}{3}$$

$$(23) y = 8 \left(x + \frac{9}{16} \right)^2 - \frac{81}{32}$$

$$(2) y = 8 \left(x + \frac{7}{16} \right)^2 - \frac{209}{32}$$

$$(13) y = -8 \left(x + \frac{1}{2} \right)^2 + 9$$

$$(24) y = 4x^2 + 5$$

$$(3) y = 7 \left(x + \frac{4}{7} \right)^2 - \frac{44}{7}$$

$$(14) y = -(x+1)^2 - 2$$

$$(25) y = -5 \left(x + \frac{9}{10} \right)^2 - \frac{39}{20}$$

$$(4) y = 5 \left(x - \frac{3}{5} \right)^2 + \frac{1}{5}$$

$$(15) y = 4(x-1)^2$$

$$(26) y = -7 \left(x + \frac{1}{7} \right)^2 + \frac{36}{7}$$

$$(16) y = 5 \left(x + \frac{1}{2} \right)^2 + \frac{19}{4}$$

$$(27) y = -2 \left(x - \frac{9}{4} \right)^2 + \frac{33}{8}$$

$$(5) y = 5 \left(x - \frac{4}{5} \right)^2 + \frac{19}{5}$$

$$(17) y = -8 \left(x - \frac{3}{8} \right)^2 + \frac{57}{8}$$

$$(28) y = 2 \left(x - \frac{1}{4} \right)^2 + \frac{31}{8}$$

$$(6) y = -5 \left(x - \frac{3}{10} \right)^2 + \frac{169}{20}$$

$$(18) y = 3 \left(x - \frac{2}{3} \right)^2 - \frac{16}{3}$$

$$(29) y = 5 \left(x + \frac{1}{5} \right)^2 - \frac{1}{5}$$

$$(7) y = 8 \left(x - \frac{1}{16} \right)^2 - \frac{129}{32}$$

$$(19) y = - \left(x + \frac{7}{2} \right)^2 + \frac{41}{4}$$

$$(30) y = -6 \left(x + \frac{1}{2} \right)^2 + \frac{19}{2}$$

$$(8) y = -4 \left(x + \frac{1}{2} \right)^2 + 1$$

$$(20) y = -8 \left(x + \frac{1}{4} \right)^2 - \frac{13}{2}$$

$$(31) y = 3 \left(x + \frac{2}{3} \right)^2 + \frac{23}{3}$$

$$(9) y = -4 \left(x - \frac{1}{4} \right)^2 - \frac{23}{4}$$

$$(21) y = 5 \left(x - \frac{1}{5} \right)^2 - \frac{36}{5}$$

$$(32) y = 6 \left(x + \frac{1}{12} \right)^2 + \frac{71}{24}$$

$$(10) y = 3 \left(x + \frac{4}{3} \right)^2 - \frac{7}{3}$$

$$(22) y = - \left(x - \frac{7}{2} \right)^2 + \frac{41}{4}$$

$$(33) y = 2(x-1)^2 - 10$$

$$(11) y = 5 \left(x - \frac{3}{10} \right)^2 + \frac{131}{20}$$

(1) $y = 9x^2 - 3$

(12) $y = 6\left(x - \frac{3}{4}\right)^2 - \frac{75}{8}$

(23) $y = -2\left(x + \frac{9}{4}\right)^2 + \frac{113}{8}$

(2) $y = (x + 1)^2$

(13) $y = 5\left(x + \frac{1}{10}\right)^2 + \frac{79}{20}$

(24) $y = (x - 2)^2 - 1$

(3) $y = -2\left(x + \frac{1}{4}\right)^2 - \frac{7}{8}$

(14) $y = -5\left(x - \frac{7}{10}\right)^2 - \frac{131}{20}$

(25) $y = -8\left(x - \frac{3}{16}\right)^2 - \frac{119}{32}$

(4) $y = 7\left(x - \frac{4}{7}\right)^2 + \frac{33}{7}$

(15) $y = 9\left(x - \frac{1}{3}\right)^2 - 7$

(26) $y = -7\left(x + \frac{9}{14}\right)^2 - \frac{31}{28}$

(5) $y = -4\left(x + \frac{9}{8}\right)^2 + \frac{177}{16}$

(16) $y = 3\left(x + \frac{2}{3}\right)^2 - \frac{7}{3}$

(27) $y = -4\left(x + \frac{1}{4}\right)^2 - \frac{27}{4}$

(6) $y = -7\left(x + \frac{9}{14}\right)^2 + \frac{193}{28}$

(17) $y = -8\left(x - \frac{9}{16}\right)^2 - \frac{111}{32}$

(28) $y = 7\left(x + \frac{1}{7}\right)^2 - \frac{29}{7}$

(7) $y = -3(x - 1)^2 + 12$

(18) $y = 9\left(x + \frac{5}{18}\right)^2 - \frac{313}{36}$

(29) $y = -\left(x + \frac{3}{2}\right)^2 - \frac{19}{4}$

(8) $y = 7\left(x + \frac{4}{7}\right)^2 + \frac{19}{7}$

(19) $y = 3\left(x + \frac{1}{2}\right)^2 + \frac{25}{4}$

(30) $y = (x - 4)^2 - 15$

(9) $y = -4\left(x + \frac{7}{8}\right)^2 + \frac{33}{16}$

(20) $y = -2\left(x - \frac{1}{4}\right)^2 + \frac{33}{8}$

(31) $y = -4x^2 - 4$

(10) $y = -x^2 + 4$

(21) $y = -6\left(x + \frac{5}{12}\right)^2 + \frac{1}{24}$

(32) $y = 9\left(x + \frac{2}{9}\right)^2 + \frac{23}{9}$

(11) $y = -(x - 3)^2 + 7$

(22) $y = -4\left(x + \frac{7}{8}\right)^2 + \frac{81}{16}$

(33) $y = -4\left(x + \frac{7}{8}\right)^2 + \frac{81}{16}$

$$(1) y = 7 \left(x + \frac{2}{7} \right)^2 + \frac{38}{7}$$

$$(12) y = 2 \left(x + \frac{1}{2} \right)^2 - \frac{1}{2}$$

$$(23) y = -5 \left(x - \frac{3}{10} \right)^2 - \frac{171}{20}$$

$$(2) y = 2 \left(x + \frac{9}{4} \right)^2 - \frac{81}{8}$$

$$(13) y = -4 \left(x - \frac{7}{8} \right)^2 - \frac{15}{16}$$

$$(24) y = - \left(x - \frac{1}{2} \right)^2 + \frac{13}{4}$$

$$(3) y = -2 \left(x - \frac{3}{4} \right)^2 - \frac{7}{8}$$

$$(14) y = 4 \left(x - \frac{9}{8} \right)^2 - \frac{161}{16}$$

$$(25) y = 8 \left(x + \frac{9}{16} \right)^2 - \frac{337}{32}$$

$$(4) y = -8 \left(x - \frac{1}{8} \right)^2 - \frac{63}{8}$$

$$(15) y = -6 \left(x - \frac{2}{3} \right)^2 + \frac{35}{3}$$

$$(26) y = 4 \left(x - \frac{7}{8} \right)^2 + \frac{63}{16}$$

$$(5) y = -5 \left(x - \frac{3}{5} \right)^2 - \frac{21}{5}$$

$$(16) y = - \left(x - \frac{1}{2} \right)^2 - \frac{11}{4}$$

$$(27) y = 9 \left(x - \frac{4}{9} \right)^2 - \frac{61}{9}$$

$$(6) y = 5 \left(x - \frac{1}{2} \right)^2 - \frac{33}{4}$$

$$(17) y = 3 \left(x + \frac{1}{3} \right)^2 - \frac{16}{3}$$

$$(28) y = -8 \left(x + \frac{5}{16} \right)^2 + \frac{153}{32}$$

$$(7) y = -4 \left(x + \frac{1}{2} \right)^2 + 8$$

$$(18) y = \left(x - \frac{3}{2} \right)^2 - \frac{37}{4}$$

$$(29) y = -9 \left(x + \frac{5}{18} \right)^2 - \frac{155}{36}$$

$$(8) y = 6 \left(x + \frac{1}{6} \right)^2 - \frac{37}{6}$$

$$(19) y = 9 \left(x - \frac{1}{18} \right)^2 + \frac{179}{36}$$

$$(30) y = 2 \left(x - \frac{1}{4} \right)^2 - \frac{41}{8}$$

$$(9) y = 6 \left(x + \frac{1}{6} \right)^2 - \frac{49}{6}$$

$$(20) y = 5x^2 - 8$$

$$(31) y = 8 \left(x + \frac{1}{8} \right)^2 - \frac{73}{8}$$

$$(21) y = 2x^2 + 1$$

$$(10) y = -3 \left(x + \frac{2}{3} \right)^2 + \frac{28}{3}$$

$$(22) y = -9 \left(x - \frac{1}{18} \right)^2 + \frac{253}{36}$$

$$(32) y = -7 \left(x - \frac{4}{7} \right)^2 + \frac{2}{7}$$

$$(11) y = -3 \left(x + \frac{5}{6} \right)^2 + \frac{121}{12}$$

$$(33) y = 4 \left(x + \frac{3}{8} \right)^2 - \frac{137}{16}$$

$$(1) y = -\left(x - \frac{3}{2}\right)^2 + \frac{13}{4}$$

$$(12) y = -\left(x + \frac{3}{2}\right)^2 + \frac{33}{4}$$

$$(23) y = -5\left(x - \frac{4}{5}\right)^2 + \frac{6}{5}$$

$$(2) y = \left(x + \frac{9}{2}\right)^2 - \frac{97}{4}$$

$$(13) y = -2x^2 - 2$$

$$(24) y = 9\left(x - \frac{1}{3}\right)^2 + 5$$

$$(3) y = -6\left(x - \frac{1}{6}\right)^2 + \frac{31}{6}$$

$$(14) y = -9\left(x + \frac{1}{18}\right)^2 + \frac{217}{36}$$

$$(25) y = 3\left(x + \frac{1}{3}\right)^2 + \frac{17}{3}$$

$$(4) y = -8x^2 + 3$$

$$(15) y = -3\left(x + \frac{4}{3}\right)^2 + \frac{19}{3}$$

$$(26) y = 5\left(x + \frac{4}{5}\right)^2 + \frac{24}{5}$$

$$(5) y = -6\left(x + \frac{1}{2}\right)^2 - \frac{15}{2}$$

$$(16) y = -3\left(x - \frac{1}{3}\right)^2 - \frac{11}{3}$$

$$(27) y = 2(x - 1)^2 + 3$$

$$(6) y = 6\left(x - \frac{1}{6}\right)^2 - \frac{25}{6}$$

$$(17) y = 7\left(x - \frac{5}{14}\right)^2 - \frac{53}{28}$$

$$(28) y = -5\left(x - \frac{4}{5}\right)^2 + \frac{51}{5}$$

$$(7) y = 8x^2$$

$$(18) y = -8\left(x - \frac{9}{16}\right)^2 + \frac{17}{32}$$

$$(29) y = -7\left(x + \frac{3}{7}\right)^2 - \frac{5}{7}$$

$$(8) y = 6\left(x - \frac{1}{12}\right)^2 - \frac{1}{24}$$

$$(19) y = -9\left(x + \frac{2}{9}\right)^2 + \frac{13}{9}$$

$$(30) y = -8\left(x - \frac{1}{2}\right)^2 - 5$$

$$(9) y = 5\left(x - \frac{1}{5}\right)^2 + \frac{24}{5}$$

$$(20) y = 9\left(x + \frac{2}{9}\right)^2 + \frac{77}{9}$$

$$(31) y = \left(x + \frac{3}{2}\right)^2 + \frac{19}{4}$$

$$(10) y = -8\left(x - \frac{1}{4}\right)^2 + \frac{19}{2}$$

$$(21) y = -6\left(x - \frac{1}{2}\right)^2 + \frac{3}{2}$$

$$(32) y = 8\left(x + \frac{9}{16}\right)^2 - \frac{273}{32}$$

$$(11) y = 7\left(x + \frac{1}{2}\right)^2 + \frac{5}{4}$$

$$(22) y = 6\left(x - \frac{1}{2}\right)^2 + \frac{15}{2}$$

$$(33) y = 6\left(x + \frac{1}{3}\right)^2 + \frac{1}{3}$$

$$(1) y = -4 \left(x - \frac{5}{8} \right)^2 + \frac{153}{16}$$

$$(12) y = -5 \left(x - \frac{2}{5} \right)^2 - \frac{36}{5}$$

$$(23) y = 8 \left(x - \frac{7}{16} \right)^2 - \frac{81}{32}$$

$$(2) y = 3 \left(x + \frac{2}{3} \right)^2 - \frac{1}{3}$$

$$(13) y = 7 \left(x + \frac{9}{14} \right)^2 - \frac{109}{28}$$

$$(24) y = 8 \left(x + \frac{9}{16} \right)^2 + \frac{143}{32}$$

$$(3) y = 4 \left(x - \frac{1}{2} \right)^2 - 1$$

$$(14) y = -8 \left(x + \frac{3}{8} \right)^2 - \frac{31}{8}$$

$$(25) y = 5 \left(x - \frac{3}{5} \right)^2 + \frac{16}{5}$$

$$(4) y = 5 \left(x + \frac{2}{5} \right)^2 + \frac{26}{5}$$

$$(15) y = \left(x - \frac{1}{2} \right)^2 + \frac{19}{4}$$

$$(26) y = 2 \left(x + \frac{9}{4} \right)^2 - \frac{89}{8}$$

$$(5) y = 8 \left(x + \frac{3}{16} \right)^2 - \frac{297}{32}$$

$$(16) y = 7 \left(x + \frac{3}{14} \right)^2 + \frac{103}{28}$$

$$(27) y = -6 \left(x - \frac{1}{4} \right)^2 - \frac{29}{8}$$

$$(6) y = 8 \left(x - \frac{5}{16} \right)^2 - \frac{153}{32}$$

$$(17) y = -7 \left(x + \frac{5}{14} \right)^2 - \frac{199}{28}$$

$$(28) y = 3 \left(x + \frac{1}{2} \right)^2 + \frac{33}{4}$$

$$(7) y = \left(x - \frac{9}{2} \right)^2 - \frac{81}{4}$$

$$(18) y = 4(x+1)^2 - 6$$

$$(29) y = 4 \left(x - \frac{1}{8} \right)^2 - \frac{17}{16}$$

$$(8) y = 8 \left(x + \frac{1}{8} \right)^2 + \frac{31}{8}$$

$$(19) y = -4 \left(x - \frac{1}{2} \right)^2 - 5$$

$$(30) y = -5 \left(x + \frac{4}{5} \right)^2 + \frac{16}{5}$$

$$(9) y = -(x-4)^2 + 9$$

$$(20) y = 7 \left(x - \frac{5}{14} \right)^2 + \frac{143}{28}$$

$$(31) y = 7 \left(x + \frac{3}{7} \right)^2 + \frac{19}{7}$$

$$(10) y = 7 \left(x - \frac{1}{14} \right)^2 - \frac{253}{28}$$

$$(21) y = -4x^2 - 9$$

$$(32) y = 9 \left(x + \frac{4}{9} \right)^2 - \frac{70}{9}$$

$$(11) y = 9 \left(x + \frac{5}{18} \right)^2 - \frac{241}{36}$$

$$(22) y = 8 \left(x - \frac{1}{4} \right)^2 - \frac{5}{2}$$

$$(33) y = -8 \left(x + \frac{1}{8} \right)^2 - \frac{7}{8}$$

(1) $y = 3(x+1)^2 - 11$

(12) $y = 2\left(x - \frac{9}{4}\right)^2 - \frac{33}{8}$

(23) $y = -9\left(x + \frac{1}{9}\right)^2 - \frac{35}{9}$

(2) $y = 4(x+1)^2 - 3$

(13) $y = -7\left(x - \frac{1}{7}\right)^2 - \frac{20}{7}$

(24) $y = -6\left(x - \frac{3}{4}\right)^2 - \frac{13}{8}$

(3) $y = -4\left(x + \frac{5}{8}\right)^2 - \frac{23}{16}$

(14) $y = -8\left(x - \frac{3}{8}\right)^2 + \frac{49}{8}$

(25) $y = 9\left(x + \frac{5}{18}\right)^2 - \frac{205}{36}$

(4) $y = 3\left(x - \frac{1}{3}\right)^2 - \frac{4}{3}$

(15) $y = \left(x - \frac{5}{2}\right)^2 - \frac{33}{4}$

(26) $y = \left(x - \frac{1}{2}\right)^2 - \frac{33}{4}$

(5) $y = 5\left(x + \frac{1}{5}\right)^2 + \frac{34}{5}$

(16) $y = -6\left(x - \frac{1}{4}\right)^2 - \frac{37}{8}$

(27) $y = 8\left(x - \frac{1}{4}\right)^2 + \frac{11}{2}$

(6) $y = 2\left(x + \frac{3}{4}\right)^2 - \frac{17}{8}$

(17) $y = -(x+2)^2 + 4$

(28) $y = 4\left(x + \frac{5}{8}\right)^2 - \frac{57}{16}$

(7) $y = 6\left(x - \frac{1}{2}\right)^2 + \frac{11}{2}$

(18) $y = 8\left(x + \frac{1}{2}\right)^2 - 2$

(29) $y = -7\left(x + \frac{3}{7}\right)^2 + \frac{44}{7}$

(8) $y = 8x^2 + 2$

(19) $y = 8\left(x + \frac{1}{4}\right)^2 - \frac{9}{2}$

(30) $y = 9x^2 + 6$

(9) $y = -3\left(x - \frac{1}{2}\right)^2 + \frac{31}{4}$

(20) $y = 8\left(x + \frac{3}{16}\right)^2 - \frac{265}{32}$

(31) $y = 2\left(x + \frac{9}{4}\right)^2 - \frac{153}{8}$

(10) $y = -9\left(x - \frac{1}{2}\right)^2 + \frac{21}{4}$

(21) $y = -\left(x + \frac{9}{2}\right)^2 + \frac{109}{4}$

(32) $y = -9\left(x - \frac{7}{18}\right)^2 + \frac{157}{36}$

(11) $y = 8\left(x + \frac{1}{2}\right)^2 + 6$

(22) $y = -7\left(x - \frac{1}{7}\right)^2 - \frac{41}{7}$

(33) $y = -6x^2$

$$(1) y = 3 \left(x + \frac{1}{3} \right)^2 - \frac{25}{3}$$

$$(12) y = 7 \left(x - \frac{3}{7} \right)^2 + \frac{54}{7}$$

$$(23) y = 7x^2 - 9$$

$$(2) y = -(x+2)^2 - 5$$

$$(13) y = 9 \left(x + \frac{7}{18} \right)^2 - \frac{49}{36}$$

$$(24) y = 7 \left(x - \frac{1}{2} \right)^2 - \frac{31}{4}$$

$$(3) y = -9 \left(x + \frac{1}{9} \right)^2 + \frac{28}{9}$$

$$(14) y = -5 \left(x - \frac{1}{5} \right)^2 - \frac{4}{5}$$

$$(25) y = -9 \left(x + \frac{4}{9} \right)^2 - \frac{2}{9}$$

$$(4) y = -7 \left(x + \frac{1}{14} \right)^2 - \frac{167}{28}$$

$$(15) y = 7x^2 + 3$$

$$(26) y = 5 \left(x - \frac{7}{10} \right)^2 - \frac{209}{20}$$

$$(5) y = 3 \left(x + \frac{3}{2} \right)^2 - \frac{19}{4}$$

$$(16) y = -3 \left(x - \frac{7}{6} \right)^2 + \frac{97}{12}$$

$$(27) y = -7 \left(x + \frac{1}{2} \right)^2 - \frac{1}{4}$$

$$(6) y = -9 \left(x - \frac{1}{18} \right)^2 - \frac{215}{36}$$

$$(17) y = -6x^2 + 4$$

$$(28) y = -8 \left(x - \frac{1}{8} \right)^2 - \frac{71}{8}$$

$$(18) y = -5 \left(x + \frac{1}{2} \right)^2 + \frac{5}{4}$$

$$(7) y = -5 \left(x + \frac{9}{10} \right)^2 + \frac{61}{20}$$

$$(19) y = -(x+1)^2 + 3$$

$$(29) y = - \left(x - \frac{1}{2} \right)^2 + \frac{21}{4}$$

$$(8) y = 9 \left(x - \frac{7}{18} \right)^2 + \frac{95}{36}$$

$$(20) y = -9 \left(x + \frac{5}{18} \right)^2 - \frac{83}{36}$$

$$(30) y = -(x+3)^2 + 9$$

$$(9) y = 7x^2 - 9$$

$$(21) y = 8 \left(x - \frac{3}{16} \right)^2 + \frac{247}{32}$$

$$(31) y = 3x^2 - 1$$

$$(10) y = -7 \left(x - \frac{9}{14} \right)^2 + \frac{305}{28}$$

$$(22) y = 6 \left(x + \frac{3}{4} \right)^2 + \frac{45}{8}$$

$$(32) y = -7 \left(x + \frac{3}{14} \right)^2 + \frac{37}{28}$$

$$(11) y = -9x^2 - 8$$

$$(33) y = -9 \left(x - \frac{4}{9} \right)^2 + \frac{16}{9}$$

- (1) $y = 9\left(x - \frac{1}{2}\right)^2 + \frac{27}{4}$
頂点 $\left(\frac{1}{2}, \frac{27}{4}\right)$
- (2) $y = 8\left(x - \frac{7}{16}\right)^2 - \frac{241}{32}$
頂点 $\left(\frac{7}{16}, -\frac{241}{32}\right)$
- (3) $y = 9\left(x - \frac{1}{18}\right)^2 - \frac{181}{36}$
頂点 $\left(\frac{1}{18}, -\frac{181}{36}\right)$
- (4) $y = 6\left(x + \frac{7}{12}\right)^2 + \frac{47}{24}$
頂点 $\left(-\frac{7}{12}, \frac{47}{24}\right)$
- (5) $y = 8\left(x - \frac{1}{8}\right)^2 - \frac{33}{8}$
頂点 $\left(\frac{1}{8}, -\frac{33}{8}\right)$
- (6) $y = 5\left(x + \frac{3}{10}\right)^2 + \frac{151}{20}$
頂点 $\left(-\frac{3}{10}, \frac{151}{20}\right)$
- (7) $y = -6x^2 + 8$
頂点 $(0, 8)$
- (8) $y = 4\left(x - \frac{1}{8}\right)^2 + \frac{15}{16}$
頂点 $\left(\frac{1}{8}, \frac{15}{16}\right)$
- (9) $y = -6\left(x - \frac{2}{3}\right)^2 + \frac{20}{3}$
頂点 $\left(\frac{2}{3}, \frac{20}{3}\right)$
- (10) $y = 7\left(x - \frac{5}{14}\right)^2 + \frac{171}{28}$
頂点 $\left(\frac{5}{14}, \frac{171}{28}\right)$
- (11) $y = -3(x+1)^2 + 5$
頂点 $(-1, 5)$
- (12) $y = -9\left(x + \frac{1}{6}\right)^2 - \frac{23}{4}$
頂点 $\left(-\frac{1}{6}, -\frac{23}{4}\right)$
- (13) $y = -7\left(x - \frac{3}{14}\right)^2 + \frac{121}{28}$
頂点 $\left(\frac{3}{14}, \frac{121}{28}\right)$
- (14) $y = 6\left(x + \frac{7}{12}\right)^2 - \frac{265}{24}$
頂点 $\left(-\frac{7}{12}, -\frac{265}{24}\right)$
- (15) $y = -\left(x - \frac{7}{2}\right)^2 + \frac{65}{4}$
頂点 $\left(\frac{7}{2}, \frac{65}{4}\right)$
- (16) $y = -6\left(x + \frac{1}{3}\right)^2 - \frac{22}{3}$
頂点 $\left(-\frac{1}{3}, -\frac{22}{3}\right)$
- (17) $y = 3\left(x - \frac{4}{3}\right)^2 - \frac{19}{3}$
頂点 $\left(\frac{4}{3}, -\frac{19}{3}\right)$
- (18) $y = -5\left(x - \frac{1}{10}\right)^2 + \frac{21}{20}$
頂点 $\left(\frac{1}{10}, \frac{21}{20}\right)$
- (19) $y = -3\left(x - \frac{7}{6}\right)^2 + \frac{25}{12}$
頂点 $\left(\frac{7}{6}, \frac{25}{12}\right)$
- (20) $y = 9\left(x - \frac{5}{18}\right)^2 + \frac{191}{36}$
頂点 $\left(\frac{5}{18}, \frac{191}{36}\right)$
- (21) $y = -7\left(x + \frac{3}{14}\right)^2 + \frac{149}{28}$
頂点 $\left(-\frac{3}{14}, \frac{149}{28}\right)$
- (22) $y = -4(x+1)^2 + 7$
頂点 $(-1, 7)$
- (23) $y = 9\left(x - \frac{1}{6}\right)^2 + \frac{31}{4}$
頂点 $\left(\frac{1}{6}, \frac{31}{4}\right)$
- (24) $y = -3\left(x - \frac{3}{2}\right)^2 + \frac{23}{4}$
頂点 $\left(\frac{3}{2}, \frac{23}{4}\right)$
- (25) $y = -2\left(x - \frac{5}{4}\right)^2 + \frac{9}{8}$
頂点 $\left(\frac{5}{4}, \frac{9}{8}\right)$
- (26) $y = -(x+3)^2 + 6$
頂点 $(-3, 6)$
- (27) $y = -6\left(x + \frac{1}{4}\right)^2 + \frac{35}{8}$
頂点 $\left(-\frac{1}{4}, \frac{35}{8}\right)$
- (28) $y = -4\left(x - \frac{1}{8}\right)^2 - \frac{15}{16}$
頂点 $\left(\frac{1}{8}, -\frac{15}{16}\right)$
- (29) $y = -2\left(x + \frac{1}{2}\right)^2 - \frac{9}{2}$
頂点 $\left(-\frac{1}{2}, -\frac{9}{2}\right)$
- (30) $y = 9\left(x + \frac{1}{2}\right)^2 - \frac{5}{4}$
頂点 $\left(-\frac{1}{2}, -\frac{5}{4}\right)$
- (31) $y = -7\left(x - \frac{4}{7}\right)^2 - \frac{40}{7}$
頂点 $\left(\frac{4}{7}, -\frac{40}{7}\right)$
- (32) $y = 9\left(x - \frac{1}{2}\right)^2 - \frac{5}{4}$
頂点 $\left(\frac{1}{2}, -\frac{5}{4}\right)$
- (33) $y = 4x^2 - 6$
頂点 $(0, -6)$

$$(1) y = 7 \left(x + \frac{1}{7} \right)^2 + \frac{13}{7}$$

頂点 $\left(-\frac{1}{7}, \frac{13}{7} \right)$

$$(2) y = 9 \left(x + \frac{1}{9} \right)^2 - \frac{28}{9}$$

頂点 $\left(-\frac{1}{9}, -\frac{28}{9} \right)$

$$(3) y = 3 \left(x + \frac{2}{3} \right)^2 + \frac{14}{3}$$

頂点 $\left(-\frac{2}{3}, \frac{14}{3} \right)$

$$(4) y = 3 \left(x - \frac{7}{6} \right)^2 + \frac{11}{12}$$

頂点 $\left(\frac{7}{6}, \frac{11}{12} \right)$

$$(5) y = 6 \left(x + \frac{1}{4} \right)^2 + \frac{21}{8}$$

頂点 $\left(-\frac{1}{4}, \frac{21}{8} \right)$

$$(6) y = -6 \left(x - \frac{1}{4} \right)^2 - \frac{37}{8}$$

頂点 $\left(\frac{1}{4}, -\frac{37}{8} \right)$

$$(7) y = -(x-1)^2 - 8$$

頂点 $(1, -8)$

$$(8) y = 6 \left(x - \frac{7}{12} \right)^2 - \frac{73}{24}$$

頂点 $\left(\frac{7}{12}, -\frac{73}{24} \right)$

$$(9) y = 8 \left(x + \frac{1}{16} \right)^2 - \frac{193}{32}$$

頂点 $\left(-\frac{1}{16}, -\frac{193}{32} \right)$

$$(10) y = -2 \left(x - \frac{3}{2} \right)^2 + \frac{1}{2}$$

頂点 $\left(\frac{3}{2}, \frac{1}{2} \right)$

$$(11) y = 6 \left(x - \frac{1}{12} \right)^2 - \frac{73}{24}$$

頂点 $\left(\frac{1}{12}, -\frac{73}{24} \right)$

$$(12) y = 5 \left(x + \frac{7}{10} \right)^2 - \frac{169}{20}$$

頂点 $\left(-\frac{7}{10}, -\frac{169}{20} \right)$

$$(13) y = 3 \left(x + \frac{5}{6} \right)^2 + \frac{23}{12}$$

頂点 $\left(-\frac{5}{6}, \frac{23}{12} \right)$

$$(14) y = 9 \left(x + \frac{1}{18} \right)^2 - \frac{289}{36}$$

頂点 $\left(-\frac{1}{18}, -\frac{289}{36} \right)$

$$(15) y = -4 \left(x - \frac{3}{8} \right)^2 - \frac{119}{16}$$

頂点 $\left(\frac{3}{8}, -\frac{119}{16} \right)$

$$(16) y = -8 \left(x - \frac{9}{16} \right)^2 + \frac{241}{32}$$

頂点 $\left(\frac{9}{16}, \frac{241}{32} \right)$

$$(17) y = -5 \left(x + \frac{4}{5} \right)^2 + \frac{1}{5}$$

頂点 $\left(-\frac{4}{5}, \frac{1}{5} \right)$

$$(18) y = -6 \left(x + \frac{1}{4} \right)^2 - \frac{69}{8}$$

頂点 $\left(-\frac{1}{4}, -\frac{69}{8} \right)$

$$(19) y = -2 \left(x + \frac{3}{2} \right)^2 + \frac{13}{2}$$

頂点 $\left(-\frac{3}{2}, \frac{13}{2} \right)$

$$(20) y = 3 \left(x + \frac{7}{6} \right)^2 - \frac{85}{12}$$

頂点 $\left(-\frac{7}{6}, -\frac{85}{12} \right)$

$$(21) y = -8 \left(x + \frac{3}{16} \right)^2 + \frac{265}{32}$$

頂点 $\left(-\frac{3}{16}, \frac{265}{32} \right)$

$$(22) y = 4 \left(x + \frac{1}{2} \right)^2 - 6$$

頂点 $\left(-\frac{1}{2}, -6 \right)$

$$(23) y = -8 \left(x - \frac{3}{8} \right)^2 + \frac{73}{8}$$

頂点 $\left(\frac{3}{8}, \frac{73}{8} \right)$

$$(24) y = 5x^2 - 8$$

頂点 $(0, -8)$

$$(25) y = 7 \left(x - \frac{9}{14} \right)^2 - \frac{25}{28}$$

頂点 $\left(\frac{9}{14}, -\frac{25}{28} \right)$

$$(26) y = -7 \left(x - \frac{1}{14} \right)^2 - \frac{223}{28}$$

頂点 $\left(\frac{1}{14}, -\frac{223}{28} \right)$

$$(27) y = -4 \left(x + \frac{1}{8} \right)^2 + \frac{1}{16}$$

頂点 $\left(-\frac{1}{8}, \frac{1}{16} \right)$

$$(28) y = 2 \left(x - \frac{3}{4} \right)^2 + \frac{55}{8}$$

頂点 $\left(\frac{3}{4}, \frac{55}{8} \right)$

$$(29) y = -(x+4)^2 + 8$$

頂点 $(-4, 8)$

$$(30) y = 3 \left(x + \frac{1}{6} \right)^2 + \frac{47}{12}$$

頂点 $\left(-\frac{1}{6}, \frac{47}{12} \right)$

$$(31) y = -2 \left(x + \frac{3}{2} \right)^2 + \frac{9}{2}$$

頂点 $\left(-\frac{3}{2}, \frac{9}{2} \right)$

$$(32) y = -9 \left(x + \frac{7}{18} \right)^2 - \frac{131}{36}$$

頂点 $\left(-\frac{7}{18}, -\frac{131}{36} \right)$

$$(33) y = -(x+1)^2 - 5$$

頂点 $(-1, -5)$

$$(1) y = 5 \left(x - \frac{3}{5} \right)^2 + \frac{11}{5}$$

頂点 $\left(\frac{3}{5}, \frac{11}{5} \right)$

$$(2) y = 8 \left(x + \frac{3}{16} \right)^2 - \frac{265}{32}$$

頂点 $\left(-\frac{3}{16}, -\frac{265}{32} \right)$

$$(3) y = 3 \left(x - \frac{1}{3} \right)^2 - \frac{19}{3}$$

頂点 $\left(\frac{1}{3}, -\frac{19}{3} \right)$

$$(4) y = (x + 4)^2 - 9$$

頂点 $(-4, -9)$

$$(5) y = -5 \left(x + \frac{1}{2} \right)^2 + \frac{29}{4}$$

頂点 $\left(-\frac{1}{2}, \frac{29}{4} \right)$

$$(6) y = 3 \left(x - \frac{1}{2} \right)^2 + \frac{1}{4}$$

頂点 $\left(\frac{1}{2}, \frac{1}{4} \right)$

$$(7) y = 3 \left(x - \frac{7}{6} \right)^2 + \frac{23}{12}$$

頂点 $\left(\frac{7}{6}, \frac{23}{12} \right)$

$$(8) y = 3 \left(x + \frac{2}{3} \right)^2 - \frac{13}{3}$$

頂点 $\left(-\frac{2}{3}, -\frac{13}{3} \right)$

$$(9) y = -9 \left(x - \frac{5}{18} \right)^2 + \frac{205}{36}$$

頂点 $\left(\frac{5}{18}, \frac{205}{36} \right)$

$$(10) y = 6 \left(x - \frac{3}{4} \right)^2 - \frac{3}{8}$$

頂点 $\left(\frac{3}{4}, -\frac{3}{8} \right)$

$$(11) y = 4 \left(x - \frac{3}{4} \right)^2 - \frac{1}{4}$$

頂点 $\left(\frac{3}{4}, -\frac{1}{4} \right)$

$$(12) y = 6 \left(x + \frac{2}{3} \right)^2 - \frac{29}{3}$$

頂点 $\left(-\frac{2}{3}, -\frac{29}{3} \right)$

$$(13) y = 9 \left(x - \frac{1}{18} \right)^2 - \frac{1}{36}$$

頂点 $\left(\frac{1}{18}, -\frac{1}{36} \right)$

$$(14) y = -9 \left(x + \frac{2}{9} \right)^2 - \frac{59}{9}$$

頂点 $\left(-\frac{2}{9}, -\frac{59}{9} \right)$

$$(15) y = -2(x + 2)^2 + 10$$

頂点 $(-2, 10)$

$$(16) y = 7 \left(x + \frac{1}{2} \right)^2 - \frac{7}{4}$$

頂点 $\left(-\frac{1}{2}, -\frac{7}{4} \right)$

$$(17) y = 8 \left(x + \frac{7}{16} \right)^2 - \frac{177}{32}$$

頂点 $\left(-\frac{7}{16}, -\frac{177}{32} \right)$

$$(18) y = 8 \left(x + \frac{1}{4} \right)^2 - \frac{5}{2}$$

頂点 $\left(-\frac{1}{4}, -\frac{5}{2} \right)$

$$(19) y = -8 \left(x + \frac{1}{16} \right)^2 - \frac{127}{32}$$

頂点 $\left(-\frac{1}{16}, -\frac{127}{32} \right)$

$$(20) y = 9x^2 + 7$$

頂点 $(0, 7)$

$$(21) y = 9 \left(x - \frac{1}{9} \right)^2 + \frac{17}{9}$$

頂点 $\left(\frac{1}{9}, \frac{17}{9} \right)$

$$(22) y = -8 \left(x - \frac{5}{16} \right)^2 - \frac{167}{32}$$

頂点 $\left(\frac{5}{16}, -\frac{167}{32} \right)$

$$(23) y = -3 \left(x + \frac{7}{6} \right)^2 + \frac{133}{12}$$

頂点 $\left(-\frac{7}{6}, \frac{133}{12} \right)$

$$(24) y = 4 \left(x - \frac{3}{4} \right)^2 + \frac{15}{4}$$

頂点 $\left(\frac{3}{4}, \frac{15}{4} \right)$

$$(25) y = -4(x + 1)^2 - 5$$

頂点 $(-1, -5)$

$$(26) y = -5 \left(x + \frac{4}{5} \right)^2 + \frac{11}{5}$$

頂点 $\left(-\frac{4}{5}, \frac{11}{5} \right)$

$$(27) y = -3 \left(x - \frac{3}{2} \right)^2 + \frac{19}{4}$$

頂点 $\left(\frac{3}{2}, \frac{19}{4} \right)$

$$(28) y = -5 \left(x - \frac{7}{10} \right)^2 + \frac{49}{20}$$

頂点 $\left(\frac{7}{10}, \frac{49}{20} \right)$

$$(29) y = 6 \left(x + \frac{1}{6} \right)^2 - \frac{25}{6}$$

頂点 $\left(-\frac{1}{6}, -\frac{25}{6} \right)$

$$(30) y = \left(x - \frac{9}{2} \right)^2 - \frac{57}{4}$$

頂点 $\left(\frac{9}{2}, -\frac{57}{4} \right)$

$$(31) y = 3 \left(x - \frac{7}{6} \right)^2 - \frac{1}{12}$$

頂点 $\left(\frac{7}{6}, -\frac{1}{12} \right)$

$$(32) y = 3 \left(x - \frac{3}{2} \right)^2 + \frac{9}{4}$$

頂点 $\left(\frac{3}{2}, \frac{9}{4} \right)$

$$(33) y = 8 \left(x - \frac{3}{16} \right)^2 - \frac{233}{32}$$

頂点 $\left(\frac{3}{16}, -\frac{233}{32} \right)$

- (1) $y = -9\left(x - \frac{1}{2}\right)^2 + \frac{45}{4}$
頂点 $\left(\frac{1}{2}, \frac{45}{4}\right)$
- (2) $y = -5\left(x + \frac{4}{5}\right)^2 + \frac{36}{5}$
頂点 $\left(-\frac{4}{5}, \frac{36}{5}\right)$
- (3) $y = 8\left(x - \frac{3}{16}\right)^2 + \frac{151}{32}$
頂点 $\left(\frac{3}{16}, \frac{151}{32}\right)$
- (4) $y = -6\left(x + \frac{1}{3}\right)^2 - \frac{7}{3}$
頂点 $\left(-\frac{1}{3}, -\frac{7}{3}\right)$
- (5) $y = 2x^2 - 7$
頂点 $(0, -7)$
- (6) $y = 7\left(x + \frac{5}{14}\right)^2 + \frac{143}{28}$
頂点 $\left(-\frac{5}{14}, \frac{143}{28}\right)$
- (7) $y = -2(x - 2)^2 + 9$
頂点 $(2, 9)$
- (8) $y = -3\left(x - \frac{3}{2}\right)^2 + \frac{11}{4}$
頂点 $\left(\frac{3}{2}, \frac{11}{4}\right)$
- (9) $y = 8\left(x + \frac{1}{2}\right)^2 + 7$
頂点 $\left(-\frac{1}{2}, 7\right)$
- (10) $y = -6\left(x + \frac{1}{3}\right)^2 + \frac{20}{3}$
頂点 $\left(-\frac{1}{3}, \frac{20}{3}\right)$
- (11) $y = 8\left(x - \frac{3}{16}\right)^2 + \frac{215}{32}$
頂点 $\left(\frac{3}{16}, \frac{215}{32}\right)$
- (12) $y = -8\left(x + \frac{1}{2}\right)^2 - 5$
頂点 $\left(-\frac{1}{2}, -5\right)$
- (13) $y = 8\left(x + \frac{9}{16}\right)^2 - \frac{145}{32}$
頂点 $\left(-\frac{9}{16}, -\frac{145}{32}\right)$
- (14) $y = 4\left(x + \frac{3}{8}\right)^2 + \frac{7}{16}$
頂点 $\left(-\frac{3}{8}, \frac{7}{16}\right)$
- (15) $y = -8\left(x + \frac{9}{16}\right)^2 - \frac{79}{32}$
頂点 $\left(-\frac{9}{16}, -\frac{79}{32}\right)$
- (16) $y = -9\left(x + \frac{4}{9}\right)^2 - \frac{47}{9}$
頂点 $\left(-\frac{4}{9}, -\frac{47}{9}\right)$
- (17) $y = -3\left(x - \frac{1}{6}\right)^2 - \frac{83}{12}$
頂点 $\left(\frac{1}{6}, -\frac{83}{12}\right)$
- (18) $y = 9\left(x - \frac{2}{9}\right)^2 + \frac{41}{9}$
頂点 $\left(\frac{2}{9}, \frac{41}{9}\right)$
- (19) $y = 7\left(x + \frac{9}{14}\right)^2 + \frac{171}{28}$
頂点 $\left(-\frac{9}{14}, \frac{171}{28}\right)$
- (20) $y = -2\left(x + \frac{3}{4}\right)^2 - \frac{31}{8}$
頂点 $\left(-\frac{3}{4}, -\frac{31}{8}\right)$
- (21) $y = 8\left(x + \frac{9}{16}\right)^2 - \frac{305}{32}$
頂点 $\left(-\frac{9}{16}, -\frac{305}{32}\right)$
- (22) $y = -8\left(x + \frac{1}{16}\right)^2 + \frac{33}{32}$
頂点 $\left(-\frac{1}{16}, \frac{33}{32}\right)$
- (23) $y = 5\left(x + \frac{2}{5}\right)^2 - \frac{4}{5}$
頂点 $\left(-\frac{2}{5}, -\frac{4}{5}\right)$
- (24) $y = 2(x + 1)^2 - 2$
頂点 $(-1, -2)$
- (25) $y = -9\left(x - \frac{1}{2}\right)^2 - \frac{27}{4}$
頂点 $\left(\frac{1}{2}, -\frac{27}{4}\right)$
- (26) $y = 2\left(x + \frac{3}{4}\right)^2 - \frac{57}{8}$
頂点 $\left(-\frac{3}{4}, -\frac{57}{8}\right)$
- (27) $y = -5\left(x - \frac{3}{5}\right)^2 + \frac{34}{5}$
頂点 $\left(\frac{3}{5}, \frac{34}{5}\right)$
- (28) $y = -4\left(x + \frac{1}{4}\right)^2 + \frac{17}{4}$
頂点 $\left(-\frac{1}{4}, \frac{17}{4}\right)$
- (29) $y = -7\left(x - \frac{3}{14}\right)^2 - \frac{47}{28}$
頂点 $\left(\frac{3}{14}, -\frac{47}{28}\right)$
- (30) $y = 3\left(x - \frac{1}{6}\right)^2 - \frac{1}{12}$
頂点 $\left(\frac{1}{6}, -\frac{1}{12}\right)$
- (31) $y = -3\left(x + \frac{5}{6}\right)^2 - \frac{83}{12}$
頂点 $\left(-\frac{5}{6}, -\frac{83}{12}\right)$
- (32) $y = -2\left(x + \frac{5}{4}\right)^2 + \frac{65}{8}$
頂点 $\left(-\frac{5}{4}, \frac{65}{8}\right)$
- (33) $y = -8\left(x - \frac{3}{16}\right)^2 + \frac{265}{32}$
頂点 $\left(\frac{3}{16}, \frac{265}{32}\right)$

- (1) $y = x^2 - 5$
頂点 $(0, -5)$
- (2) $y = -(x - 2)^2 + 5$
頂点 $(2, 5)$
- (3) $y = -6\left(x + \frac{2}{3}\right)^2 + \frac{29}{3}$
頂点 $\left(-\frac{2}{3}, \frac{29}{3}\right)$
- (4) $y = 3\left(x + \frac{4}{3}\right)^2 - \frac{37}{3}$
頂点 $\left(-\frac{4}{3}, -\frac{37}{3}\right)$
- (5) $y = 3\left(x - \frac{2}{3}\right)^2 - \frac{4}{3}$
頂点 $\left(\frac{2}{3}, -\frac{4}{3}\right)$
- (6) $y = -5\left(x + \frac{7}{10}\right)^2 - \frac{111}{20}$
頂点 $\left(-\frac{7}{10}, -\frac{111}{20}\right)$
- (7) $y = -5\left(x - \frac{1}{2}\right)^2 + \frac{33}{4}$
頂点 $\left(\frac{1}{2}, \frac{33}{4}\right)$
- (8) $y = 3\left(x - \frac{3}{2}\right)^2 - \frac{3}{4}$
頂点 $\left(\frac{3}{2}, -\frac{3}{4}\right)$
- (9) $y = 5\left(x - \frac{1}{2}\right)^2 + \frac{7}{4}$
頂点 $\left(\frac{1}{2}, \frac{7}{4}\right)$
- (10) $y = 4\left(x - \frac{1}{4}\right)^2 - \frac{13}{4}$
頂点 $\left(\frac{1}{4}, -\frac{13}{4}\right)$
- (11) $y = -6\left(x + \frac{2}{3}\right)^2 - \frac{1}{3}$
頂点 $\left(-\frac{2}{3}, -\frac{1}{3}\right)$
- (12) $y = -4\left(x + \frac{1}{8}\right)^2 + \frac{1}{16}$
頂点 $\left(-\frac{1}{8}, \frac{1}{16}\right)$
- (13) $y = -7x^2 + 4$
頂点 $(0, 4)$
- (14) $y = -\left(x - \frac{1}{2}\right)^2 - \frac{23}{4}$
頂点 $\left(\frac{1}{2}, -\frac{23}{4}\right)$
- (15) $y = 6\left(x - \frac{3}{4}\right)^2 + \frac{37}{8}$
頂点 $\left(\frac{3}{4}, \frac{37}{8}\right)$
- (16) $y = 5\left(x - \frac{1}{10}\right)^2 + \frac{19}{20}$
頂点 $\left(\frac{1}{10}, \frac{19}{20}\right)$
- (17) $y = 5\left(x + \frac{3}{5}\right)^2 + \frac{31}{5}$
頂点 $\left(-\frac{3}{5}, \frac{31}{5}\right)$
- (18) $y = (x - 2)^2 - 3$
頂点 $(2, -3)$
- (19) $y = 5\left(x + \frac{3}{10}\right)^2 - \frac{9}{20}$
頂点 $\left(-\frac{3}{10}, -\frac{9}{20}\right)$
- (20) $y = -\left(x - \frac{1}{2}\right)^2 + \frac{25}{4}$
頂点 $\left(\frac{1}{2}, \frac{25}{4}\right)$
- (21) $y = -2(x - 1)^2 + 4$
頂点 $(1, 4)$
- (22) $y = -5\left(x + \frac{4}{5}\right)^2 + \frac{36}{5}$
頂点 $\left(-\frac{4}{5}, \frac{36}{5}\right)$
- (23) $y = -4\left(x - \frac{5}{8}\right)^2 - \frac{87}{16}$
頂点 $\left(\frac{5}{8}, -\frac{87}{16}\right)$
- (24) $y = 7\left(x + \frac{1}{2}\right)^2 - \frac{43}{4}$
頂点 $\left(-\frac{1}{2}, -\frac{43}{4}\right)$
- (25) $y = -3\left(x + \frac{1}{3}\right)^2 + \frac{13}{3}$
頂点 $\left(-\frac{1}{3}, \frac{13}{3}\right)$
- (26) $y = -4(x + 1)^2 - 1$
頂点 $(-1, -1)$
- (27) $y = -7\left(x - \frac{3}{7}\right)^2 - \frac{33}{7}$
頂点 $\left(\frac{3}{7}, -\frac{33}{7}\right)$
- (28) $y = 6\left(x + \frac{2}{3}\right)^2 - \frac{2}{3}$
頂点 $\left(-\frac{2}{3}, -\frac{2}{3}\right)$
- (29) $y = 2\left(x + \frac{7}{4}\right)^2 - \frac{17}{8}$
頂点 $\left(-\frac{7}{4}, -\frac{17}{8}\right)$
- (30) $y = -8\left(x + \frac{1}{4}\right)^2 + \frac{5}{2}$
頂点 $\left(-\frac{1}{4}, \frac{5}{2}\right)$
- (31) $y = 8\left(x - \frac{7}{16}\right)^2 - \frac{113}{32}$
頂点 $\left(\frac{7}{16}, -\frac{113}{32}\right)$
- (32) $y = -6\left(x + \frac{5}{12}\right)^2 - \frac{191}{24}$
頂点 $\left(-\frac{5}{12}, -\frac{191}{24}\right)$
- (33) $y = 2(x + 1)^2 - 3$
頂点 $(-1, -3)$

- (1) $y = 3\left(x + \frac{3}{2}\right)^2 - \frac{63}{4}$
頂点 $\left(-\frac{3}{2}, -\frac{63}{4}\right)$
- (2) $y = 6\left(x + \frac{1}{3}\right)^2 - \frac{26}{3}$
頂点 $\left(-\frac{1}{3}, -\frac{26}{3}\right)$
- (3) $y = 5\left(x - \frac{1}{2}\right)^2 + \frac{11}{4}$
頂点 $\left(\frac{1}{2}, \frac{11}{4}\right)$
- (4) $y = -6\left(x - \frac{3}{4}\right)^2 - \frac{13}{8}$
頂点 $\left(\frac{3}{4}, -\frac{13}{8}\right)$
- (5) $y = 5\left(x + \frac{3}{10}\right)^2 - \frac{69}{20}$
頂点 $\left(-\frac{3}{10}, -\frac{69}{20}\right)$
- (6) $y = 6\left(x - \frac{1}{4}\right)^2 + \frac{61}{8}$
頂点 $\left(\frac{1}{4}, \frac{61}{8}\right)$
- (7) $y = 7\left(x + \frac{1}{2}\right)^2 + \frac{25}{4}$
頂点 $\left(-\frac{1}{2}, \frac{25}{4}\right)$
- (8) $y = -8\left(x + \frac{5}{16}\right)^2 + \frac{313}{32}$
頂点 $\left(-\frac{5}{16}, \frac{313}{32}\right)$
- (9) $y = 6\left(x + \frac{1}{2}\right)^2 + \frac{9}{2}$
頂点 $\left(-\frac{1}{2}, \frac{9}{2}\right)$
- (10) $y = -4\left(x + \frac{3}{8}\right)^2 - \frac{7}{16}$
頂点 $\left(-\frac{3}{8}, -\frac{7}{16}\right)$
- (11) $y = 7\left(x - \frac{2}{7}\right)^2 + \frac{38}{7}$
頂点 $\left(\frac{2}{7}, \frac{38}{7}\right)$
- (12) $y = -(x+1)^2 - 8$
頂点 $(-1, -8)$
- (13) $y = 3\left(x - \frac{7}{6}\right)^2 + \frac{47}{12}$
頂点 $\left(\frac{7}{6}, \frac{47}{12}\right)$
- (14) $y = -6\left(x + \frac{1}{12}\right)^2 - \frac{47}{24}$
頂点 $\left(-\frac{1}{12}, -\frac{47}{24}\right)$
- (15) $y = -2(x+1)^2 + 4$
頂点 $(-1, 4)$
- (16) $y = 7\left(x - \frac{3}{7}\right)^2 - \frac{44}{7}$
頂点 $\left(\frac{3}{7}, -\frac{44}{7}\right)$
- (17) $y = -6\left(x + \frac{7}{12}\right)^2 - \frac{143}{24}$
頂点 $\left(-\frac{7}{12}, -\frac{143}{24}\right)$
- (18) $y = 6\left(x - \frac{2}{3}\right)^2 + \frac{7}{3}$
頂点 $\left(\frac{2}{3}, \frac{7}{3}\right)$
- (19) $y = 4\left(x - \frac{1}{8}\right)^2 - \frac{49}{16}$
頂点 $\left(\frac{1}{8}, -\frac{49}{16}\right)$
- (20) $y = -8\left(x + \frac{1}{4}\right)^2 - \frac{7}{2}$
頂点 $\left(-\frac{1}{4}, -\frac{7}{2}\right)$
- (21) $y = -6x^2 + 9$
頂点 $(0, 9)$
- (22) $y = -2(x+2)^2 + 17$
頂点 $(-2, 17)$
- (23) $y = 7\left(x - \frac{9}{14}\right)^2 + \frac{31}{28}$
頂点 $\left(\frac{9}{14}, \frac{31}{28}\right)$
- (24) $y = 9\left(x + \frac{2}{9}\right)^2 - \frac{22}{9}$
頂点 $\left(-\frac{2}{9}, -\frac{22}{9}\right)$
- (25) $y = -4\left(x - \frac{7}{8}\right)^2 - \frac{79}{16}$
頂点 $\left(\frac{7}{8}, -\frac{79}{16}\right)$
- (26) $y = 5\left(x - \frac{9}{10}\right)^2 - \frac{81}{20}$
頂点 $\left(\frac{9}{10}, -\frac{81}{20}\right)$
- (27) $y = 8\left(x - \frac{1}{2}\right)^2 + 5$
頂点 $\left(\frac{1}{2}, 5\right)$
- (28) $y = -2(x+1)^2 + 8$
頂点 $(-1, 8)$
- (29) $y = 8\left(x - \frac{3}{8}\right)^2 - \frac{81}{8}$
頂点 $\left(\frac{3}{8}, -\frac{81}{8}\right)$
- (30) $y = 9\left(x - \frac{5}{18}\right)^2 - \frac{205}{36}$
頂点 $\left(\frac{5}{18}, -\frac{205}{36}\right)$
- (31) $y = -2\left(x - \frac{7}{4}\right)^2 + \frac{113}{8}$
頂点 $\left(\frac{7}{4}, \frac{113}{8}\right)$
- (32) $y = -8\left(x - \frac{3}{16}\right)^2 + \frac{9}{32}$
頂点 $\left(\frac{3}{16}, \frac{9}{32}\right)$
- (33) $y = -4\left(x + \frac{1}{2}\right)^2 + 7$
頂点 $\left(-\frac{1}{2}, 7\right)$

- (1) $y = -6 \left(x - \frac{2}{3} \right)^2 - \frac{10}{3}$
頂点 $\left(\frac{2}{3}, -\frac{10}{3} \right)$
- (2) $y = -7 \left(x - \frac{1}{7} \right)^2 + \frac{8}{7}$
頂点 $\left(\frac{1}{7}, \frac{8}{7} \right)$
- (3) $y = -(x+1)^2 + 3$
頂点 $(-1, 3)$
- (4) $y = \left(x + \frac{7}{2} \right)^2 - \frac{13}{4}$
頂点 $\left(-\frac{7}{2}, -\frac{13}{4} \right)$
- (5) $y = -7 \left(x - \frac{9}{14} \right)^2 - \frac{59}{28}$
頂点 $\left(\frac{9}{14}, -\frac{59}{28} \right)$
- (6) $y = -5x^2 - 9$
頂点 $(0, -9)$
- (7) $y = -6 \left(x - \frac{1}{6} \right)^2 - \frac{11}{6}$
頂点 $\left(\frac{1}{6}, -\frac{11}{6} \right)$
- (8) $y = 7 \left(x + \frac{2}{7} \right)^2 + \frac{3}{7}$
頂点 $\left(-\frac{2}{7}, \frac{3}{7} \right)$
- (9) $y = -3 \left(x + \frac{1}{3} \right)^2 + \frac{13}{3}$
頂点 $\left(-\frac{1}{3}, \frac{13}{3} \right)$
- (10) $y = -6 \left(x - \frac{1}{3} \right)^2 + \frac{23}{3}$
頂点 $\left(\frac{1}{3}, \frac{23}{3} \right)$
- (11) $y = -7 \left(x - \frac{1}{2} \right)^2 + \frac{31}{4}$
頂点 $\left(\frac{1}{2}, \frac{31}{4} \right)$
- (12) $y = 7 \left(x - \frac{2}{7} \right)^2 + \frac{3}{7}$
頂点 $\left(\frac{2}{7}, \frac{3}{7} \right)$
- (13) $y = -5x^2 + 6$
頂点 $(0, 6)$
- (14) $y = -7 \left(x - \frac{9}{14} \right)^2 - \frac{115}{28}$
頂点 $\left(\frac{9}{14}, -\frac{115}{28} \right)$
- (15) $y = -8 \left(x - \frac{1}{8} \right)^2 + \frac{9}{8}$
頂点 $\left(\frac{1}{8}, \frac{9}{8} \right)$
- (16) $y = -8 \left(x - \frac{3}{8} \right)^2 + \frac{1}{8}$
頂点 $\left(\frac{3}{8}, \frac{1}{8} \right)$
- (17) $y = -3(x-1)^2 + 10$
頂点 $(1, 10)$
- (18) $y = -6 \left(x - \frac{1}{3} \right)^2 + \frac{2}{3}$
頂点 $\left(\frac{1}{3}, \frac{2}{3} \right)$
- (19) $y = -8 \left(x + \frac{1}{16} \right)^2 - \frac{191}{32}$
頂点 $\left(-\frac{1}{16}, -\frac{191}{32} \right)$
- (20) $y = 6 \left(x - \frac{1}{12} \right)^2 + \frac{47}{24}$
頂点 $\left(\frac{1}{12}, \frac{47}{24} \right)$
- (21) $y = - \left(x + \frac{7}{2} \right)^2 + \frac{65}{4}$
頂点 $\left(-\frac{7}{2}, \frac{65}{4} \right)$
- (22) $y = 9 \left(x + \frac{1}{18} \right)^2 + \frac{215}{36}$
頂点 $\left(-\frac{1}{18}, \frac{215}{36} \right)$
- (23) $y = 7 \left(x - \frac{9}{14} \right)^2 - \frac{249}{28}$
頂点 $\left(\frac{9}{14}, -\frac{249}{28} \right)$
- (24) $y = -6 \left(x - \frac{1}{6} \right)^2 - \frac{17}{6}$
頂点 $\left(\frac{1}{6}, -\frac{17}{6} \right)$
- (25) $y = 8 \left(x + \frac{7}{16} \right)^2 - \frac{177}{32}$
頂点 $\left(-\frac{7}{16}, -\frac{177}{32} \right)$
- (26) $y = 6 \left(x + \frac{2}{3} \right)^2 - \frac{8}{3}$
頂点 $\left(-\frac{2}{3}, -\frac{8}{3} \right)$
- (27) $y = -3 \left(x + \frac{4}{3} \right)^2 + \frac{31}{3}$
頂点 $\left(-\frac{4}{3}, \frac{31}{3} \right)$
- (28) $y = -2(x-2)^2 + 7$
頂点 $(2, 7)$
- (29) $y = -4 \left(x - \frac{1}{8} \right)^2 + \frac{145}{16}$
頂点 $\left(\frac{1}{8}, \frac{145}{16} \right)$
- (30) $y = 3 \left(x + \frac{2}{3} \right)^2 + \frac{20}{3}$
頂点 $\left(-\frac{2}{3}, \frac{20}{3} \right)$
- (31) $y = -6 \left(x - \frac{1}{3} \right)^2 + \frac{23}{3}$
頂点 $\left(\frac{1}{3}, \frac{23}{3} \right)$
- (32) $y = 2 \left(x - \frac{5}{4} \right)^2 + \frac{31}{8}$
頂点 $\left(\frac{5}{4}, \frac{31}{8} \right)$
- (33) $y = - \left(x + \frac{3}{2} \right)^2 + \frac{1}{4}$
頂点 $\left(-\frac{3}{2}, \frac{1}{4} \right)$

$$(1) y = 8 \left(x - \frac{1}{4} \right)^2 + \frac{13}{2}$$

頂点 $\left(\frac{1}{4}, \frac{13}{2} \right)$

$$(2) y = 6 \left(x - \frac{2}{3} \right)^2 + \frac{19}{3}$$

頂点 $\left(\frac{2}{3}, \frac{19}{3} \right)$

$$(3) y = 4 \left(x - \frac{7}{8} \right)^2 - \frac{81}{16}$$

頂点 $\left(\frac{7}{8}, -\frac{81}{16} \right)$

$$(4) y = 4x^2 - 9$$

頂点 $(0, -9)$

$$(5) y = - \left(x + \frac{3}{2} \right)^2 - \frac{23}{4}$$

頂点 $\left(-\frac{3}{2}, -\frac{23}{4} \right)$

$$(6) y = -3 \left(x - \frac{7}{6} \right)^2 + \frac{133}{12}$$

頂点 $\left(\frac{7}{6}, \frac{133}{12} \right)$

$$(7) y = (x - 3)^2 - 6$$

頂点 $(3, -6)$

$$(8) y = -6 \left(x + \frac{1}{2} \right)^2 + \frac{1}{2}$$

頂点 $\left(-\frac{1}{2}, \frac{1}{2} \right)$

$$(9) y = -5 \left(x + \frac{1}{5} \right)^2 - \frac{4}{5}$$

頂点 $\left(-\frac{1}{5}, -\frac{4}{5} \right)$

$$(10) y = 2 \left(x - \frac{7}{4} \right)^2 - \frac{81}{8}$$

頂点 $\left(\frac{7}{4}, -\frac{81}{8} \right)$

$$(11) y = 2x^2 - 6$$

頂点 $(0, -6)$

$$(12) y = 3 \left(x + \frac{3}{2} \right)^2 - \frac{15}{4}$$

頂点 $\left(-\frac{3}{2}, -\frac{15}{4} \right)$

$$(13) y = -9 \left(x - \frac{4}{9} \right)^2 + \frac{43}{9}$$

頂点 $\left(\frac{4}{9}, \frac{43}{9} \right)$

$$(14) y = \left(x + \frac{3}{2} \right)^2 - \frac{33}{4}$$

頂点 $\left(-\frac{3}{2}, -\frac{33}{4} \right)$

$$(15) y = 2 \left(x + \frac{3}{4} \right)^2 - \frac{41}{8}$$

頂点 $\left(-\frac{3}{4}, -\frac{41}{8} \right)$

$$(16) y = 4 \left(x - \frac{1}{8} \right)^2 + \frac{79}{16}$$

頂点 $\left(\frac{1}{8}, \frac{79}{16} \right)$

$$(17) y = -7 \left(x - \frac{2}{7} \right)^2 - \frac{59}{7}$$

頂点 $\left(\frac{2}{7}, -\frac{59}{7} \right)$

$$(18) y = -6 \left(x + \frac{5}{12} \right)^2 - \frac{191}{24}$$

頂点 $\left(-\frac{5}{12}, -\frac{191}{24} \right)$

$$(19) y = 8 \left(x - \frac{1}{16} \right)^2 - \frac{97}{32}$$

頂点 $\left(\frac{1}{16}, -\frac{97}{32} \right)$

$$(20) y = 5 \left(x + \frac{4}{5} \right)^2 - \frac{36}{5}$$

頂点 $\left(-\frac{4}{5}, -\frac{36}{5} \right)$

$$(21) y = -8 \left(x - \frac{1}{2} \right)^2 - 6$$

頂点 $\left(\frac{1}{2}, -6 \right)$

$$(22) y = 4 \left(x + \frac{3}{4} \right)^2 + \frac{27}{4}$$

頂点 $\left(-\frac{3}{4}, \frac{27}{4} \right)$

$$(23) y = 9 \left(x - \frac{1}{9} \right)^2 + \frac{71}{9}$$

頂点 $\left(\frac{1}{9}, \frac{71}{9} \right)$

$$(24) y = -8 \left(x + \frac{1}{16} \right)^2 + \frac{161}{32}$$

頂点 $\left(-\frac{1}{16}, \frac{161}{32} \right)$

$$(25) y = -8 \left(x - \frac{5}{16} \right)^2 - \frac{167}{32}$$

頂点 $\left(\frac{5}{16}, -\frac{167}{32} \right)$

$$(26) y = 5 \left(x + \frac{2}{5} \right)^2 + \frac{6}{5}$$

頂点 $\left(-\frac{2}{5}, \frac{6}{5} \right)$

$$(27) y = 3 \left(x + \frac{1}{6} \right)^2 - \frac{1}{12}$$

頂点 $\left(-\frac{1}{6}, -\frac{1}{12} \right)$

$$(28) y = -8 \left(x + \frac{7}{16} \right)^2 + \frac{49}{32}$$

頂点 $\left(-\frac{7}{16}, \frac{49}{32} \right)$

$$(29) y = -7x^2 - 8$$

頂点 $(0, -8)$

$$(30) y = 7 \left(x - \frac{1}{2} \right)^2 - \frac{23}{4}$$

頂点 $\left(\frac{1}{2}, -\frac{23}{4} \right)$

$$(31) y = -(x - 4)^2 + 23$$

頂点 $(4, 23)$

$$(32) y = -2 \left(x - \frac{5}{4} \right)^2 + \frac{17}{8}$$

頂点 $\left(\frac{5}{4}, \frac{17}{8} \right)$

$$(33) y = 5 \left(x + \frac{1}{5} \right)^2 + \frac{9}{5}$$

頂点 $\left(-\frac{1}{5}, \frac{9}{5} \right)$

- (1) $y = -7\left(x - \frac{1}{2}\right)^2 - \frac{29}{4}$
頂点 $\left(\frac{1}{2}, -\frac{29}{4}\right)$
- (2) $y = 3x^2 - 7$
頂点 $(0, -7)$
- (3) $y = 5\left(x - \frac{9}{10}\right)^2 + \frac{59}{20}$
頂点 $\left(\frac{9}{10}, \frac{59}{20}\right)$
- (4) $y = -(x + 4)^2 + 14$
頂点 $(-4, 14)$
- (5) $y = -2(x + 2)^2 + 16$
頂点 $(-2, 16)$
- (6) $y = 4\left(x + \frac{7}{8}\right)^2 + \frac{63}{16}$
頂点 $\left(-\frac{7}{8}, \frac{63}{16}\right)$
- (7) $y = 2\left(x + \frac{1}{4}\right)^2 - \frac{33}{8}$
頂点 $\left(-\frac{1}{4}, -\frac{33}{8}\right)$
- (8) $y = 5\left(x + \frac{3}{5}\right)^2 - \frac{34}{5}$
頂点 $\left(-\frac{3}{5}, -\frac{34}{5}\right)$
- (9) $y = -8\left(x - \frac{1}{8}\right)^2 - \frac{7}{8}$
頂点 $\left(\frac{1}{8}, -\frac{7}{8}\right)$
- (10) $y = -7\left(x - \frac{3}{7}\right)^2 + \frac{30}{7}$
頂点 $\left(\frac{3}{7}, \frac{30}{7}\right)$
- (11) $y = 7x^2$
頂点 $(0, 0)$
- (12) $y = 3\left(x + \frac{1}{2}\right)^2 - \frac{19}{4}$
頂点 $\left(-\frac{1}{2}, -\frac{19}{4}\right)$
- (13) $y = 7\left(x - \frac{2}{7}\right)^2 + \frac{31}{7}$
頂点 $\left(\frac{2}{7}, \frac{31}{7}\right)$
- (14) $y = -3\left(x + \frac{7}{6}\right)^2 - \frac{23}{12}$
頂点 $\left(-\frac{7}{6}, -\frac{23}{12}\right)$
- (15) $y = 2\left(x - \frac{1}{4}\right)^2 + \frac{63}{8}$
頂点 $\left(\frac{1}{4}, \frac{63}{8}\right)$
- (16) $y = -5\left(x + \frac{1}{2}\right)^2 - \frac{23}{4}$
頂点 $\left(-\frac{1}{2}, -\frac{23}{4}\right)$
- (17) $y = -9\left(x - \frac{2}{9}\right)^2 - \frac{59}{9}$
頂点 $\left(\frac{2}{9}, -\frac{59}{9}\right)$
- (18) $y = -9\left(x - \frac{1}{2}\right)^2 - \frac{27}{4}$
頂点 $\left(\frac{1}{2}, -\frac{27}{4}\right)$
- (19) $y = 3\left(x + \frac{5}{6}\right)^2 - \frac{121}{12}$
頂点 $\left(-\frac{5}{6}, -\frac{121}{12}\right)$
- (20) $y = 3\left(x - \frac{7}{6}\right)^2 - \frac{1}{12}$
頂点 $\left(\frac{7}{6}, -\frac{1}{12}\right)$
- (21) $y = -9\left(x + \frac{1}{9}\right)^2 + \frac{64}{9}$
頂点 $\left(-\frac{1}{9}, \frac{64}{9}\right)$
- (22) $y = -2\left(x - \frac{3}{4}\right)^2 + \frac{57}{8}$
頂点 $\left(\frac{3}{4}, \frac{57}{8}\right)$
- (23) $y = -3\left(x - \frac{1}{6}\right)^2 - \frac{83}{12}$
頂点 $\left(\frac{1}{6}, -\frac{83}{12}\right)$
- (24) $y = -\left(x + \frac{3}{2}\right)^2 + \frac{1}{4}$
頂点 $\left(-\frac{3}{2}, \frac{1}{4}\right)$
- (25) $y = 9x^2 - 3$
頂点 $(0, -3)$
- (26) $y = 9\left(x - \frac{4}{9}\right)^2 - \frac{43}{9}$
頂点 $\left(\frac{4}{9}, -\frac{43}{9}\right)$
- (27) $y = 2x^2 + 9$
頂点 $(0, 9)$
- (28) $y = -6\left(x - \frac{7}{12}\right)^2 + \frac{1}{24}$
頂点 $\left(\frac{7}{12}, \frac{1}{24}\right)$
- (29) $y = -9\left(x - \frac{1}{6}\right)^2 + \frac{17}{4}$
頂点 $\left(\frac{1}{6}, \frac{17}{4}\right)$
- (30) $y = 5\left(x + \frac{3}{10}\right)^2 - \frac{189}{20}$
頂点 $\left(-\frac{3}{10}, -\frac{189}{20}\right)$
- (31) $y = 6x^2$
頂点 $(0, 0)$
- (32) $y = -6\left(x - \frac{1}{3}\right)^2 - \frac{16}{3}$
頂点 $\left(\frac{1}{3}, -\frac{16}{3}\right)$
- (33) $y = 5x^2$
頂点 $(0, 0)$

$$(1) y = 5 \left(x - \frac{7}{10} \right)^2 + \frac{131}{20}$$

頂点 $\left(\frac{7}{10}, \frac{131}{20} \right)$

$$(2) y = 6 \left(x + \frac{2}{3} \right)^2 - \frac{29}{3}$$

頂点 $\left(-\frac{2}{3}, -\frac{29}{3} \right)$

$$(3) y = -3 \left(x - \frac{7}{6} \right)^2 - \frac{59}{12}$$

頂点 $\left(\frac{7}{6}, -\frac{59}{12} \right)$

$$(4) y = 6 \left(x + \frac{1}{2} \right)^2 - \frac{21}{2}$$

頂点 $\left(-\frac{1}{2}, -\frac{21}{2} \right)$

$$(5) y = 4 \left(x - \frac{1}{2} \right)^2 - 7$$

頂点 $\left(\frac{1}{2}, -7 \right)$

$$(6) y = -7x^2 + 3$$

頂点 $(0, 3)$

$$(7) y = 4 \left(x + \frac{9}{8} \right)^2 - \frac{65}{16}$$

頂点 $\left(-\frac{9}{8}, -\frac{65}{16} \right)$

$$(8) y = 2 \left(x - \frac{3}{4} \right)^2 + \frac{15}{8}$$

頂点 $\left(\frac{3}{4}, \frac{15}{8} \right)$

$$(9) y = (x + 3)^2 - 15$$

頂点 $(-3, -15)$

$$(10) y = -4 \left(x + \frac{1}{8} \right)^2 + \frac{49}{16}$$

頂点 $\left(-\frac{1}{8}, \frac{49}{16} \right)$

$$(11) y = -2(x - 1)^2 + 7$$

頂点 $(1, 7)$

$$(12) y = -9 \left(x + \frac{4}{9} \right)^2 + \frac{61}{9}$$

頂点 $\left(-\frac{4}{9}, \frac{61}{9} \right)$

$$(13) y = \left(x + \frac{1}{2} \right)^2 - \frac{9}{4}$$

頂点 $\left(-\frac{1}{2}, -\frac{9}{4} \right)$

$$(14) y = -6 \left(x + \frac{1}{4} \right)^2 + \frac{43}{8}$$

頂点 $\left(-\frac{1}{4}, \frac{43}{8} \right)$

$$(15) y = -8 \left(x - \frac{3}{16} \right)^2 + \frac{265}{32}$$

頂点 $\left(\frac{3}{16}, \frac{265}{32} \right)$

$$(16) y = (x - 2)^2 + 2$$

頂点 $(2, 2)$

$$(17) y = -4 \left(x - \frac{3}{8} \right)^2 - \frac{135}{16}$$

頂点 $\left(\frac{3}{8}, -\frac{135}{16} \right)$

$$(18) y = - \left(x - \frac{1}{2} \right)^2 + \frac{33}{4}$$

頂点 $\left(\frac{1}{2}, \frac{33}{4} \right)$

$$(19) y = 5 \left(x + \frac{1}{5} \right)^2 - \frac{21}{5}$$

頂点 $\left(-\frac{1}{5}, -\frac{21}{5} \right)$

$$(20) y = 5x^2 - 3$$

頂点 $(0, -3)$

$$(21) y = \left(x - \frac{5}{2} \right)^2 - \frac{49}{4}$$

頂点 $\left(\frac{5}{2}, -\frac{49}{4} \right)$

$$(22) y = 9 \left(x - \frac{1}{9} \right)^2 - \frac{46}{9}$$

頂点 $\left(\frac{1}{9}, -\frac{46}{9} \right)$

$$(23) y = 3 \left(x + \frac{1}{3} \right)^2 - \frac{4}{3}$$

頂点 $\left(-\frac{1}{3}, -\frac{4}{3} \right)$

$$(24) y = 4 \left(x - \frac{1}{8} \right)^2 + \frac{111}{16}$$

頂点 $\left(\frac{1}{8}, \frac{111}{16} \right)$

$$(25) y = -9 \left(x + \frac{1}{3} \right)^2 - 3$$

頂点 $\left(-\frac{1}{3}, -3 \right)$

$$(26) y = - \left(x + \frac{1}{2} \right)^2 + \frac{25}{4}$$

頂点 $\left(-\frac{1}{2}, \frac{25}{4} \right)$

$$(27) y = \left(x - \frac{7}{2} \right)^2 - \frac{29}{4}$$

頂点 $\left(\frac{7}{2}, -\frac{29}{4} \right)$

$$(28) y = 2(x + 1)^2 - 8$$

頂点 $(-1, -8)$

$$(29) y = -2 \left(x - \frac{3}{2} \right)^2 + \frac{13}{2}$$

頂点 $\left(\frac{3}{2}, \frac{13}{2} \right)$

$$(30) y = - \left(x - \frac{5}{2} \right)^2 + \frac{37}{4}$$

頂点 $\left(\frac{5}{2}, \frac{37}{4} \right)$

$$(31) y = - \left(x + \frac{1}{2} \right)^2 - \frac{7}{4}$$

頂点 $\left(-\frac{1}{2}, -\frac{7}{4} \right)$

$$(32) y = -3 \left(x + \frac{3}{2} \right)^2 - \frac{5}{4}$$

頂点 $\left(-\frac{3}{2}, -\frac{5}{4} \right)$

$$(33) y = \left(x - \frac{1}{2} \right)^2 - \frac{5}{4}$$

頂点 $\left(\frac{1}{2}, -\frac{5}{4} \right)$

- (1) $y = -6x^2 + 6$
頂点 $(0, 6)$
- (2) $y = 3\left(x + \frac{5}{6}\right)^2 + \frac{47}{12}$
頂点 $\left(-\frac{5}{6}, \frac{47}{12}\right)$
- (3) $y = -2(x - 1)^2 - 4$
頂点 $(1, -4)$
- (4) $y = 6\left(x - \frac{1}{3}\right)^2 - \frac{8}{3}$
頂点 $\left(\frac{1}{3}, -\frac{8}{3}\right)$
- (5) $y = -8\left(x + \frac{3}{16}\right)^2 - \frac{215}{32}$
頂点 $\left(-\frac{3}{16}, -\frac{215}{32}\right)$
- (6) $y = 6\left(x + \frac{1}{4}\right)^2 - \frac{43}{8}$
頂点 $\left(-\frac{1}{4}, -\frac{43}{8}\right)$
- (7) $y = 3\left(x - \frac{2}{3}\right)^2 + \frac{5}{3}$
頂点 $\left(\frac{2}{3}, \frac{5}{3}\right)$
- (8) $y = -7\left(x - \frac{3}{14}\right)^2 - \frac{243}{28}$
頂点 $\left(\frac{3}{14}, -\frac{243}{28}\right)$
- (9) $y = -8\left(x + \frac{3}{16}\right)^2 - \frac{55}{32}$
頂点 $\left(-\frac{3}{16}, -\frac{55}{32}\right)$
- (10) $y = -5\left(x + \frac{3}{10}\right)^2 + \frac{29}{20}$
頂点 $\left(-\frac{3}{10}, \frac{29}{20}\right)$
- (11) $y = 2(x - 2)^2 - 2$
頂点 $(2, -2)$
- (12) $y = \left(x + \frac{7}{2}\right)^2 - \frac{81}{4}$
頂点 $\left(-\frac{7}{2}, -\frac{81}{4}\right)$
- (13) $y = \left(x - \frac{9}{2}\right)^2 - \frac{61}{4}$
頂点 $\left(\frac{9}{2}, -\frac{61}{4}\right)$
- (14) $y = -7\left(x + \frac{1}{7}\right)^2 + \frac{50}{7}$
頂点 $\left(-\frac{1}{7}, \frac{50}{7}\right)$
- (15) $y = 9\left(x - \frac{1}{9}\right)^2 - \frac{46}{9}$
頂点 $\left(\frac{1}{9}, -\frac{46}{9}\right)$
- (16) $y = 4\left(x + \frac{7}{8}\right)^2 - \frac{177}{16}$
頂点 $\left(-\frac{7}{8}, -\frac{177}{16}\right)$
- (17) $y = \left(x - \frac{3}{2}\right)^2 + \frac{7}{4}$
頂点 $\left(\frac{3}{2}, \frac{7}{4}\right)$
- (18) $y = 4\left(x - \frac{1}{8}\right)^2 - \frac{65}{16}$
頂点 $\left(\frac{1}{8}, -\frac{65}{16}\right)$
- (19) $y = -2\left(x - \frac{3}{4}\right)^2 - \frac{31}{8}$
頂点 $\left(\frac{3}{4}, -\frac{31}{8}\right)$
- (20) $y = 4\left(x - \frac{1}{4}\right)^2 + \frac{7}{4}$
頂点 $\left(\frac{1}{4}, \frac{7}{4}\right)$
- (21) $y = 9\left(x + \frac{1}{3}\right)^2 - 1$
頂点 $\left(-\frac{1}{3}, -1\right)$
- (22) $y = -6\left(x + \frac{1}{6}\right)^2 + \frac{19}{6}$
頂点 $\left(-\frac{1}{6}, \frac{19}{6}\right)$
- (23) $y = -8\left(x + \frac{3}{16}\right)^2 + \frac{9}{32}$
頂点 $\left(-\frac{3}{16}, \frac{9}{32}\right)$
- (24) $y = 2\left(x + \frac{7}{4}\right)^2 - \frac{41}{8}$
頂点 $\left(-\frac{7}{4}, -\frac{41}{8}\right)$
- (25) $y = 4\left(x + \frac{3}{8}\right)^2 - \frac{41}{16}$
頂点 $\left(-\frac{3}{8}, -\frac{41}{16}\right)$
- (26) $y = 7\left(x - \frac{4}{7}\right)^2 - \frac{2}{7}$
頂点 $\left(\frac{4}{7}, -\frac{2}{7}\right)$
- (27) $y = 7\left(x + \frac{1}{7}\right)^2 - \frac{57}{7}$
頂点 $\left(-\frac{1}{7}, -\frac{57}{7}\right)$
- (28) $y = -8\left(x - \frac{3}{16}\right)^2 - \frac{23}{32}$
頂点 $\left(\frac{3}{16}, -\frac{23}{32}\right)$
- (29) $y = -8\left(x - \frac{7}{16}\right)^2 + \frac{81}{32}$
頂点 $\left(\frac{7}{16}, \frac{81}{32}\right)$
- (30) $y = \left(x - \frac{3}{2}\right)^2 + \frac{27}{4}$
頂点 $\left(\frac{3}{2}, \frac{27}{4}\right)$
- (31) $y = -8\left(x - \frac{7}{16}\right)^2 + \frac{81}{32}$
頂点 $\left(\frac{7}{16}, \frac{81}{32}\right)$
- (32) $y = -4\left(x - \frac{3}{8}\right)^2 + \frac{41}{16}$
頂点 $\left(\frac{3}{8}, \frac{41}{16}\right)$
- (33) $y = 6\left(x + \frac{1}{6}\right)^2 - \frac{43}{6}$
頂点 $\left(-\frac{1}{6}, -\frac{43}{6}\right)$

$$(1) y = 7 \left(x - \frac{4}{7} \right)^2 - \frac{2}{7}$$

頂点 $\left(\frac{4}{7}, -\frac{2}{7} \right)$

$$(2) y = -7x^2 - 5$$

頂点 $(0, -5)$

$$(3) y = 6 \left(x + \frac{1}{4} \right)^2 + \frac{21}{8}$$

頂点 $\left(-\frac{1}{4}, \frac{21}{8} \right)$

$$(4) y = 2 \left(x + \frac{1}{4} \right)^2 + \frac{47}{8}$$

頂点 $\left(-\frac{1}{4}, \frac{47}{8} \right)$

$$(5) y = -2(x-1)^2 + 5$$

頂点 $(1, 5)$

$$(6) y = 4(x-1)^2 - 11$$

頂点 $(1, -11)$

$$(7) y = 6 \left(x - \frac{1}{12} \right)^2 - \frac{49}{24}$$

頂点 $\left(\frac{1}{12}, -\frac{49}{24} \right)$

$$(8) y = 3(x-1)^2 - 10$$

頂点 $(1, -10)$

$$(9) y = 7 \left(x + \frac{3}{14} \right)^2 + \frac{103}{28}$$

頂点 $\left(-\frac{3}{14}, \frac{103}{28} \right)$

$$(10) y = -7 \left(x + \frac{1}{14} \right)^2 - \frac{195}{28}$$

頂点 $\left(-\frac{1}{14}, -\frac{195}{28} \right)$

$$(11) y = -3 \left(x - \frac{1}{6} \right)^2 - \frac{47}{12}$$

頂点 $\left(\frac{1}{6}, -\frac{47}{12} \right)$

$$(12) y = 8 \left(x - \frac{1}{4} \right)^2 - \frac{9}{2}$$

頂点 $\left(\frac{1}{4}, -\frac{9}{2} \right)$

$$(13) y = -8 \left(x - \frac{1}{16} \right)^2 + \frac{289}{32}$$

頂点 $\left(\frac{1}{16}, \frac{289}{32} \right)$

$$(14) y = 9 \left(x - \frac{2}{9} \right)^2 + \frac{5}{9}$$

頂点 $\left(\frac{2}{9}, \frac{5}{9} \right)$

$$(15) y = -2(x+1)^2 + 5$$

頂点 $(-1, 5)$

$$(16) y = -9x^2 - 2$$

頂点 $(0, -2)$

$$(17) y = -6 \left(x + \frac{1}{4} \right)^2 - \frac{29}{8}$$

頂点 $\left(-\frac{1}{4}, -\frac{29}{8} \right)$

$$(18) y = -6 \left(x - \frac{7}{12} \right)^2 - \frac{95}{24}$$

頂点 $\left(\frac{7}{12}, -\frac{95}{24} \right)$

$$(19) y = -8 \left(x + \frac{7}{16} \right)^2 + \frac{337}{32}$$

頂点 $\left(-\frac{7}{16}, \frac{337}{32} \right)$

$$(20) y = -4(x+1)^2 + 11$$

頂点 $(-1, 11)$

$$(21) y = 3 \left(x - \frac{2}{3} \right)^2 - \frac{10}{3}$$

頂点 $\left(\frac{2}{3}, -\frac{10}{3} \right)$

$$(22) y = -8 \left(x - \frac{9}{16} \right)^2 + \frac{145}{32}$$

頂点 $\left(\frac{9}{16}, \frac{145}{32} \right)$

$$(23) y = - \left(x + \frac{3}{2} \right)^2 + \frac{41}{4}$$

頂点 $\left(-\frac{3}{2}, \frac{41}{4} \right)$

$$(24) y = -(x+3)^2 + 8$$

頂点 $(-3, 8)$

$$(25) y = 9 \left(x + \frac{2}{9} \right)^2 - \frac{22}{9}$$

頂点 $\left(-\frac{2}{9}, -\frac{22}{9} \right)$

$$(26) y = -5 \left(x + \frac{9}{10} \right)^2 + \frac{241}{20}$$

頂点 $\left(-\frac{9}{10}, \frac{241}{20} \right)$

$$(27) y = -4 \left(x - \frac{1}{8} \right)^2 + \frac{33}{16}$$

頂点 $\left(\frac{1}{8}, \frac{33}{16} \right)$

$$(28) y = (x+3)^2 - 12$$

頂点 $(-3, -12)$

$$(29) y = -5 \left(x + \frac{1}{5} \right)^2 + \frac{6}{5}$$

頂点 $\left(-\frac{1}{5}, \frac{6}{5} \right)$

$$(30) y = 5 \left(x + \frac{4}{5} \right)^2 - \frac{16}{5}$$

頂点 $\left(-\frac{4}{5}, -\frac{16}{5} \right)$

$$(31) y = -3 \left(x + \frac{1}{3} \right)^2 - \frac{20}{3}$$

頂点 $\left(-\frac{1}{3}, -\frac{20}{3} \right)$

$$(32) y = 2 \left(x - \frac{5}{4} \right)^2 + \frac{15}{8}$$

頂点 $\left(\frac{5}{4}, \frac{15}{8} \right)$

$$(33) y = 3 \left(x + \frac{2}{3} \right)^2 - \frac{19}{3}$$

頂点 $\left(-\frac{2}{3}, -\frac{19}{3} \right)$

- (1) $y = 7\left(x - \frac{1}{2}\right)^2 - \frac{43}{4}$
頂点 $\left(\frac{1}{2}, -\frac{43}{4}\right)$
- (2) $y = 2\left(x - \frac{9}{4}\right)^2 - \frac{33}{8}$
頂点 $\left(\frac{9}{4}, -\frac{33}{8}\right)$
- (3) $y = 6\left(x - \frac{7}{12}\right)^2 + \frac{143}{24}$
頂点 $\left(\frac{7}{12}, \frac{143}{24}\right)$
- (4) $y = 9\left(x + \frac{2}{9}\right)^2 + \frac{32}{9}$
頂点 $\left(-\frac{2}{9}, \frac{32}{9}\right)$
- (5) $y = 8\left(x - \frac{3}{8}\right)^2 - \frac{33}{8}$
頂点 $\left(\frac{3}{8}, -\frac{33}{8}\right)$
- (6) $y = \left(x + \frac{9}{2}\right)^2 - \frac{73}{4}$
頂点 $\left(-\frac{9}{2}, -\frac{73}{4}\right)$
- (7) $y = 3\left(x + \frac{7}{6}\right)^2 - \frac{13}{12}$
頂点 $\left(-\frac{7}{6}, -\frac{13}{12}\right)$
- (8) $y = 6\left(x - \frac{3}{4}\right)^2 - \frac{99}{8}$
頂点 $\left(\frac{3}{4}, -\frac{99}{8}\right)$
- (9) $y = -3\left(x - \frac{1}{6}\right)^2 - \frac{83}{12}$
頂点 $\left(\frac{1}{6}, -\frac{83}{12}\right)$
- (10) $y = -3\left(x - \frac{4}{3}\right)^2 + \frac{16}{3}$
頂点 $\left(\frac{4}{3}, \frac{16}{3}\right)$
- (11) $y = -5\left(x - \frac{1}{5}\right)^2 + \frac{26}{5}$
頂点 $\left(\frac{1}{5}, \frac{26}{5}\right)$
- (12) $y = 2(x - 1)^2 + 2$
頂点 $(1, 2)$
- (13) $y = 2\left(x + \frac{3}{4}\right)^2 - \frac{41}{8}$
頂点 $\left(-\frac{3}{4}, -\frac{41}{8}\right)$
- (14) $y = 6\left(x + \frac{1}{4}\right)^2 + \frac{21}{8}$
頂点 $\left(-\frac{1}{4}, \frac{21}{8}\right)$
- (15) $y = -9\left(x + \frac{1}{6}\right)^2 + \frac{9}{4}$
頂点 $\left(-\frac{1}{6}, \frac{9}{4}\right)$
- (16) $y = -2\left(x - \frac{1}{4}\right)^2 + \frac{57}{8}$
頂点 $\left(\frac{1}{4}, \frac{57}{8}\right)$
- (17) $y = 8\left(x + \frac{3}{16}\right)^2 + \frac{247}{32}$
頂点 $\left(-\frac{3}{16}, \frac{247}{32}\right)$
- (18) $y = 2(x + 2)^2 - 16$
頂点 $(-2, -16)$
- (19) $y = 5\left(x + \frac{2}{5}\right)^2 + \frac{41}{5}$
頂点 $\left(-\frac{2}{5}, \frac{41}{5}\right)$
- (20) $y = -9\left(x + \frac{4}{9}\right)^2 + \frac{52}{9}$
頂点 $\left(-\frac{4}{9}, \frac{52}{9}\right)$
- (21) $y = 8\left(x + \frac{1}{2}\right)^2 - 5$
頂点 $\left(-\frac{1}{2}, -5\right)$
- (22) $y = -2\left(x - \frac{1}{2}\right)^2 + \frac{17}{2}$
頂点 $\left(\frac{1}{2}, \frac{17}{2}\right)$
- (23) $y = 3\left(x - \frac{7}{6}\right)^2 + \frac{11}{12}$
頂点 $\left(\frac{7}{6}, \frac{11}{12}\right)$
- (24) $y = -2x^2 + 4$
頂点 $(0, 4)$
- (25) $y = 4x^2 + 3$
頂点 $(0, 3)$
- (26) $y = -7\left(x + \frac{1}{7}\right)^2 - \frac{6}{7}$
頂点 $\left(-\frac{1}{7}, -\frac{6}{7}\right)$
- (27) $y = -7\left(x + \frac{1}{14}\right)^2 + \frac{85}{28}$
頂点 $\left(-\frac{1}{14}, \frac{85}{28}\right)$
- (28) $y = -6\left(x - \frac{1}{12}\right)^2 - \frac{191}{24}$
頂点 $\left(\frac{1}{12}, -\frac{191}{24}\right)$
- (29) $y = -7\left(x - \frac{1}{7}\right)^2 + \frac{8}{7}$
頂点 $\left(\frac{1}{7}, \frac{8}{7}\right)$
- (30) $y = 3\left(x + \frac{3}{2}\right)^2 - \frac{23}{4}$
頂点 $\left(-\frac{3}{2}, -\frac{23}{4}\right)$
- (31) $y = -6\left(x - \frac{1}{2}\right)^2 + \frac{15}{2}$
頂点 $\left(\frac{1}{2}, \frac{15}{2}\right)$
- (32) $y = 3\left(x - \frac{4}{3}\right)^2 - \frac{40}{3}$
頂点 $\left(\frac{4}{3}, -\frac{40}{3}\right)$
- (33) $y = 2(x - 2)^2 - 12$
頂点 $(2, -12)$

- (1) $y = 3\left(x - \frac{1}{6}\right)^2 - \frac{25}{12}$
頂点 $\left(\frac{1}{6}, -\frac{25}{12}\right)$
- (2) $y = 8\left(x + \frac{7}{16}\right)^2 - \frac{209}{32}$
頂点 $\left(-\frac{7}{16}, -\frac{209}{32}\right)$
- (3) $y = 7\left(x + \frac{4}{7}\right)^2 - \frac{44}{7}$
頂点 $\left(-\frac{4}{7}, -\frac{44}{7}\right)$
- (4) $y = 5\left(x - \frac{3}{5}\right)^2 + \frac{1}{5}$
頂点 $\left(\frac{3}{5}, \frac{1}{5}\right)$
- (5) $y = 5\left(x - \frac{4}{5}\right)^2 + \frac{19}{5}$
頂点 $\left(\frac{4}{5}, \frac{19}{5}\right)$
- (6) $y = -5\left(x - \frac{3}{10}\right)^2 + \frac{169}{20}$
頂点 $\left(\frac{3}{10}, \frac{169}{20}\right)$
- (7) $y = 8\left(x - \frac{1}{16}\right)^2 - \frac{129}{32}$
頂点 $\left(\frac{1}{16}, -\frac{129}{32}\right)$
- (8) $y = -4\left(x + \frac{1}{2}\right)^2 + 1$
頂点 $\left(-\frac{1}{2}, 1\right)$
- (9) $y = -4\left(x - \frac{1}{4}\right)^2 - \frac{23}{4}$
頂点 $\left(\frac{1}{4}, -\frac{23}{4}\right)$
- (10) $y = 3\left(x + \frac{4}{3}\right)^2 - \frac{7}{3}$
頂点 $\left(-\frac{4}{3}, -\frac{7}{3}\right)$
- (11) $y = 5\left(x - \frac{3}{10}\right)^2 + \frac{131}{20}$
頂点 $\left(\frac{3}{10}, \frac{131}{20}\right)$
- (12) $y = 6\left(x - \frac{2}{3}\right)^2 - \frac{35}{3}$
頂点 $\left(\frac{2}{3}, -\frac{35}{3}\right)$
- (13) $y = -8\left(x + \frac{1}{2}\right)^2 + 9$
頂点 $\left(-\frac{1}{2}, 9\right)$
- (14) $y = -(x+1)^2 - 2$
頂点 $(-1, -2)$
- (15) $y = 4(x-1)^2$
頂点 $(1, 0)$
- (16) $y = 5\left(x + \frac{1}{2}\right)^2 + \frac{19}{4}$
頂点 $\left(-\frac{1}{2}, \frac{19}{4}\right)$
- (17) $y = -8\left(x - \frac{3}{8}\right)^2 + \frac{57}{8}$
頂点 $\left(\frac{3}{8}, \frac{57}{8}\right)$
- (18) $y = 3\left(x - \frac{2}{3}\right)^2 - \frac{16}{3}$
頂点 $\left(\frac{2}{3}, -\frac{16}{3}\right)$
- (19) $y = -\left(x + \frac{7}{2}\right)^2 + \frac{41}{4}$
頂点 $\left(-\frac{7}{2}, \frac{41}{4}\right)$
- (20) $y = -8\left(x + \frac{1}{4}\right)^2 - \frac{13}{2}$
頂点 $\left(-\frac{1}{4}, -\frac{13}{2}\right)$
- (21) $y = 5\left(x - \frac{1}{5}\right)^2 - \frac{36}{5}$
頂点 $\left(\frac{1}{5}, -\frac{36}{5}\right)$
- (22) $y = -\left(x - \frac{7}{2}\right)^2 + \frac{41}{4}$
頂点 $\left(\frac{7}{2}, \frac{41}{4}\right)$
- (23) $y = 8\left(x + \frac{9}{16}\right)^2 - \frac{81}{32}$
頂点 $\left(-\frac{9}{16}, -\frac{81}{32}\right)$
- (24) $y = 4x^2 + 5$
頂点 $(0, 5)$
- (25) $y = -5\left(x + \frac{9}{10}\right)^2 - \frac{39}{20}$
頂点 $\left(-\frac{9}{10}, -\frac{39}{20}\right)$
- (26) $y = -7\left(x + \frac{1}{7}\right)^2 + \frac{36}{7}$
頂点 $\left(-\frac{1}{7}, \frac{36}{7}\right)$
- (27) $y = -2\left(x - \frac{9}{4}\right)^2 + \frac{33}{8}$
頂点 $\left(\frac{9}{4}, \frac{33}{8}\right)$
- (28) $y = 2\left(x - \frac{1}{4}\right)^2 + \frac{31}{8}$
頂点 $\left(\frac{1}{4}, \frac{31}{8}\right)$
- (29) $y = 5\left(x + \frac{1}{5}\right)^2 - \frac{1}{5}$
頂点 $\left(-\frac{1}{5}, -\frac{1}{5}\right)$
- (30) $y = -6\left(x + \frac{1}{2}\right)^2 + \frac{19}{2}$
頂点 $\left(-\frac{1}{2}, \frac{19}{2}\right)$
- (31) $y = 3\left(x + \frac{2}{3}\right)^2 + \frac{23}{3}$
頂点 $\left(-\frac{2}{3}, \frac{23}{3}\right)$
- (32) $y = 6\left(x + \frac{1}{12}\right)^2 + \frac{71}{24}$
頂点 $\left(-\frac{1}{12}, \frac{71}{24}\right)$
- (33) $y = 2(x-1)^2 - 10$
頂点 $(1, -10)$

- (1) $y = 9x^2 - 3$
頂点 $(0, -3)$
- (2) $y = (x + 1)^2$
頂点 $(-1, 0)$
- (3) $y = -2\left(x + \frac{1}{4}\right)^2 - \frac{7}{8}$
頂点 $\left(-\frac{1}{4}, -\frac{7}{8}\right)$
- (4) $y = 7\left(x - \frac{4}{7}\right)^2 + \frac{33}{7}$
頂点 $\left(\frac{4}{7}, \frac{33}{7}\right)$
- (5) $y = -4\left(x + \frac{9}{8}\right)^2 + \frac{177}{16}$
頂点 $\left(-\frac{9}{8}, \frac{177}{16}\right)$
- (6) $y = -7\left(x + \frac{9}{14}\right)^2 + \frac{193}{28}$
頂点 $\left(-\frac{9}{14}, \frac{193}{28}\right)$
- (7) $y = -3(x - 1)^2 + 12$
頂点 $(1, 12)$
- (8) $y = 7\left(x + \frac{4}{7}\right)^2 + \frac{19}{7}$
頂点 $\left(-\frac{4}{7}, \frac{19}{7}\right)$
- (9) $y = -4\left(x + \frac{7}{8}\right)^2 + \frac{33}{16}$
頂点 $\left(-\frac{7}{8}, \frac{33}{16}\right)$
- (10) $y = -x^2 + 4$
頂点 $(0, 4)$
- (11) $y = -(x - 3)^2 + 7$
頂点 $(3, 7)$
- (12) $y = 6\left(x - \frac{3}{4}\right)^2 - \frac{75}{8}$
頂点 $\left(\frac{3}{4}, -\frac{75}{8}\right)$
- (13) $y = 5\left(x + \frac{1}{10}\right)^2 + \frac{79}{20}$
頂点 $\left(-\frac{1}{10}, \frac{79}{20}\right)$
- (14) $y = -5\left(x - \frac{7}{10}\right)^2 - \frac{131}{20}$
頂点 $\left(\frac{7}{10}, -\frac{131}{20}\right)$
- (15) $y = 9\left(x - \frac{1}{3}\right)^2 - 7$
頂点 $\left(\frac{1}{3}, -7\right)$
- (16) $y = 3\left(x + \frac{2}{3}\right)^2 - \frac{7}{3}$
頂点 $\left(-\frac{2}{3}, -\frac{7}{3}\right)$
- (17) $y = -8\left(x - \frac{9}{16}\right)^2 - \frac{111}{32}$
頂点 $\left(\frac{9}{16}, -\frac{111}{32}\right)$
- (18) $y = 9\left(x + \frac{5}{18}\right)^2 - \frac{313}{36}$
頂点 $\left(-\frac{5}{18}, -\frac{313}{36}\right)$
- (19) $y = 3\left(x + \frac{1}{2}\right)^2 + \frac{25}{4}$
頂点 $\left(-\frac{1}{2}, \frac{25}{4}\right)$
- (20) $y = -2\left(x - \frac{1}{4}\right)^2 + \frac{33}{8}$
頂点 $\left(\frac{1}{4}, \frac{33}{8}\right)$
- (21) $y = 4\left(x - \frac{9}{8}\right)^2 - \frac{33}{16}$
頂点 $\left(\frac{9}{8}, -\frac{33}{16}\right)$
- (22) $y = -6\left(x + \frac{5}{12}\right)^2 + \frac{1}{24}$
頂点 $\left(-\frac{5}{12}, \frac{1}{24}\right)$
- (23) $y = -2\left(x + \frac{9}{4}\right)^2 + \frac{113}{8}$
頂点 $\left(-\frac{9}{4}, \frac{113}{8}\right)$
- (24) $y = (x - 2)^2 - 1$
頂点 $(2, -1)$
- (25) $y = -8\left(x - \frac{3}{16}\right)^2 - \frac{119}{32}$
頂点 $\left(\frac{3}{16}, -\frac{119}{32}\right)$
- (26) $y = -7\left(x + \frac{9}{14}\right)^2 - \frac{31}{28}$
頂点 $\left(-\frac{9}{14}, -\frac{31}{28}\right)$
- (27) $y = -4\left(x + \frac{1}{4}\right)^2 - \frac{27}{4}$
頂点 $\left(-\frac{1}{4}, -\frac{27}{4}\right)$
- (28) $y = 7\left(x + \frac{1}{7}\right)^2 - \frac{29}{7}$
頂点 $\left(-\frac{1}{7}, -\frac{29}{7}\right)$
- (29) $y = -\left(x + \frac{3}{2}\right)^2 - \frac{19}{4}$
頂点 $\left(-\frac{3}{2}, -\frac{19}{4}\right)$
- (30) $y = (x - 4)^2 - 15$
頂点 $(4, -15)$
- (31) $y = -4x^2 - 4$
頂点 $(0, -4)$
- (32) $y = 9\left(x + \frac{2}{9}\right)^2 + \frac{23}{9}$
頂点 $\left(-\frac{2}{9}, \frac{23}{9}\right)$
- (33) $y = -4\left(x + \frac{7}{8}\right)^2 + \frac{81}{16}$
頂点 $\left(-\frac{7}{8}, \frac{81}{16}\right)$

$$(1) y = 7 \left(x + \frac{2}{7} \right)^2 + \frac{38}{7}$$

頂点 $\left(-\frac{2}{7}, \frac{38}{7} \right)$

$$(2) y = 2 \left(x + \frac{9}{4} \right)^2 - \frac{81}{8}$$

頂点 $\left(-\frac{9}{4}, -\frac{81}{8} \right)$

$$(3) y = -2 \left(x - \frac{3}{4} \right)^2 - \frac{7}{8}$$

頂点 $\left(\frac{3}{4}, -\frac{7}{8} \right)$

$$(4) y = -8 \left(x - \frac{1}{8} \right)^2 - \frac{63}{8}$$

頂点 $\left(\frac{1}{8}, -\frac{63}{8} \right)$

$$(5) y = -5 \left(x - \frac{3}{5} \right)^2 - \frac{21}{5}$$

頂点 $\left(\frac{3}{5}, -\frac{21}{5} \right)$

$$(6) y = 5 \left(x - \frac{1}{2} \right)^2 - \frac{33}{4}$$

頂点 $\left(\frac{1}{2}, -\frac{33}{4} \right)$

$$(7) y = -4 \left(x + \frac{1}{2} \right)^2 + 8$$

頂点 $\left(-\frac{1}{2}, 8 \right)$

$$(8) y = 6 \left(x + \frac{1}{6} \right)^2 - \frac{37}{6}$$

頂点 $\left(-\frac{1}{6}, -\frac{37}{6} \right)$

$$(9) y = 6 \left(x + \frac{1}{6} \right)^2 - \frac{49}{6}$$

頂点 $\left(-\frac{1}{6}, -\frac{49}{6} \right)$

$$(10) y = -3 \left(x + \frac{2}{3} \right)^2 + \frac{28}{3}$$

頂点 $\left(-\frac{2}{3}, \frac{28}{3} \right)$

$$(11) y = -3 \left(x + \frac{5}{6} \right)^2 + \frac{121}{12}$$

頂点 $\left(-\frac{5}{6}, \frac{121}{12} \right)$

$$(12) y = 2 \left(x + \frac{1}{2} \right)^2 - \frac{1}{2}$$

頂点 $\left(-\frac{1}{2}, -\frac{1}{2} \right)$

$$(13) y = -4 \left(x - \frac{7}{8} \right)^2 - \frac{15}{16}$$

頂点 $\left(\frac{7}{8}, -\frac{15}{16} \right)$

$$(14) y = 4 \left(x - \frac{9}{8} \right)^2 - \frac{161}{16}$$

頂点 $\left(\frac{9}{8}, -\frac{161}{16} \right)$

$$(15) y = -6 \left(x - \frac{2}{3} \right)^2 + \frac{35}{3}$$

頂点 $\left(\frac{2}{3}, \frac{35}{3} \right)$

$$(16) y = - \left(x - \frac{1}{2} \right)^2 - \frac{11}{4}$$

頂点 $\left(\frac{1}{2}, -\frac{11}{4} \right)$

$$(17) y = 3 \left(x + \frac{1}{3} \right)^2 - \frac{16}{3}$$

頂点 $\left(-\frac{1}{3}, -\frac{16}{3} \right)$

$$(18) y = \left(x - \frac{3}{2} \right)^2 - \frac{37}{4}$$

頂点 $\left(\frac{3}{2}, -\frac{37}{4} \right)$

$$(19) y = 9 \left(x - \frac{1}{18} \right)^2 + \frac{179}{36}$$

頂点 $\left(\frac{1}{18}, \frac{179}{36} \right)$

$$(20) y = 5x^2 - 8$$

頂点 $(0, -8)$

$$(21) y = 2x^2 + 1$$

頂点 $(0, 1)$

$$(22) y = -9 \left(x - \frac{1}{18} \right)^2 + \frac{253}{36}$$

頂点 $\left(\frac{1}{18}, \frac{253}{36} \right)$

$$(23) y = -5 \left(x - \frac{3}{10} \right)^2 - \frac{171}{20}$$

頂点 $\left(\frac{3}{10}, -\frac{171}{20} \right)$

$$(24) y = - \left(x - \frac{1}{2} \right)^2 + \frac{13}{4}$$

頂点 $\left(\frac{1}{2}, \frac{13}{4} \right)$

$$(25) y = 8 \left(x + \frac{9}{16} \right)^2 - \frac{337}{32}$$

頂点 $\left(-\frac{9}{16}, -\frac{337}{32} \right)$

$$(26) y = 4 \left(x - \frac{7}{8} \right)^2 + \frac{63}{16}$$

頂点 $\left(\frac{7}{8}, \frac{63}{16} \right)$

$$(27) y = 9 \left(x - \frac{4}{9} \right)^2 - \frac{61}{9}$$

頂点 $\left(\frac{4}{9}, -\frac{61}{9} \right)$

$$(28) y = -8 \left(x + \frac{5}{16} \right)^2 + \frac{153}{32}$$

頂点 $\left(-\frac{5}{16}, \frac{153}{32} \right)$

$$(29) y = -9 \left(x + \frac{5}{18} \right)^2 - \frac{155}{36}$$

頂点 $\left(-\frac{5}{18}, -\frac{155}{36} \right)$

$$(30) y = 2 \left(x - \frac{1}{4} \right)^2 - \frac{41}{8}$$

頂点 $\left(\frac{1}{4}, -\frac{41}{8} \right)$

$$(31) y = 8 \left(x + \frac{1}{8} \right)^2 - \frac{73}{8}$$

頂点 $\left(-\frac{1}{8}, -\frac{73}{8} \right)$

$$(32) y = -7 \left(x - \frac{4}{7} \right)^2 + \frac{2}{7}$$

頂点 $\left(\frac{4}{7}, \frac{2}{7} \right)$

$$(33) y = 4 \left(x + \frac{3}{8} \right)^2 - \frac{137}{16}$$

頂点 $\left(-\frac{3}{8}, -\frac{137}{16} \right)$

$$(1) y = -\left(x - \frac{3}{2}\right)^2 + \frac{13}{4}$$

頂点 $\left(\frac{3}{2}, \frac{13}{4}\right)$

$$(2) y = \left(x + \frac{9}{2}\right)^2 - \frac{97}{4}$$

頂点 $\left(-\frac{9}{2}, -\frac{97}{4}\right)$

$$(3) y = -6\left(x - \frac{1}{6}\right)^2 + \frac{31}{6}$$

頂点 $\left(\frac{1}{6}, \frac{31}{6}\right)$

$$(4) y = -8x^2 + 3$$

頂点 $(0, 3)$

$$(5) y = -6\left(x + \frac{1}{2}\right)^2 - \frac{15}{2}$$

頂点 $\left(-\frac{1}{2}, -\frac{15}{2}\right)$

$$(6) y = 6\left(x - \frac{1}{6}\right)^2 - \frac{25}{6}$$

頂点 $\left(\frac{1}{6}, -\frac{25}{6}\right)$

$$(7) y = 8x^2$$

頂点 $(0, 0)$

$$(8) y = 6\left(x - \frac{1}{12}\right)^2 - \frac{1}{24}$$

頂点 $\left(\frac{1}{12}, -\frac{1}{24}\right)$

$$(9) y = 5\left(x - \frac{1}{5}\right)^2 + \frac{24}{5}$$

頂点 $\left(\frac{1}{5}, \frac{24}{5}\right)$

$$(10) y = -8\left(x - \frac{1}{4}\right)^2 + \frac{19}{2}$$

頂点 $\left(\frac{1}{4}, \frac{19}{2}\right)$

$$(11) y = 7\left(x + \frac{1}{2}\right)^2 + \frac{5}{4}$$

頂点 $\left(-\frac{1}{2}, \frac{5}{4}\right)$

$$(12) y = -\left(x + \frac{3}{2}\right)^2 + \frac{33}{4}$$

頂点 $\left(-\frac{3}{2}, \frac{33}{4}\right)$

$$(13) y = -2x^2 - 2$$

頂点 $(0, -2)$

$$(14) y = -9\left(x + \frac{1}{18}\right)^2 + \frac{217}{36}$$

頂点 $\left(-\frac{1}{18}, \frac{217}{36}\right)$

$$(15) y = -3\left(x + \frac{4}{3}\right)^2 + \frac{19}{3}$$

頂点 $\left(-\frac{4}{3}, \frac{19}{3}\right)$

$$(16) y = -3\left(x - \frac{1}{3}\right)^2 - \frac{11}{3}$$

頂点 $\left(\frac{1}{3}, -\frac{11}{3}\right)$

$$(17) y = 7\left(x - \frac{5}{14}\right)^2 - \frac{53}{28}$$

頂点 $\left(\frac{5}{14}, -\frac{53}{28}\right)$

$$(18) y = -8\left(x - \frac{9}{16}\right)^2 + \frac{17}{32}$$

頂点 $\left(\frac{9}{16}, \frac{17}{32}\right)$

$$(19) y = -9\left(x + \frac{2}{9}\right)^2 + \frac{13}{9}$$

頂点 $\left(-\frac{2}{9}, \frac{13}{9}\right)$

$$(20) y = 9\left(x + \frac{2}{9}\right)^2 + \frac{77}{9}$$

頂点 $\left(-\frac{2}{9}, \frac{77}{9}\right)$

$$(21) y = -6\left(x - \frac{1}{2}\right)^2 + \frac{3}{2}$$

頂点 $\left(\frac{1}{2}, \frac{3}{2}\right)$

$$(22) y = 6\left(x - \frac{1}{2}\right)^2 + \frac{15}{2}$$

頂点 $\left(\frac{1}{2}, \frac{15}{2}\right)$

$$(23) y = -5\left(x - \frac{4}{5}\right)^2 + \frac{6}{5}$$

頂点 $\left(\frac{4}{5}, \frac{6}{5}\right)$

$$(24) y = 9\left(x - \frac{1}{3}\right)^2 + 5$$

頂点 $\left(\frac{1}{3}, 5\right)$

$$(25) y = 3\left(x + \frac{1}{3}\right)^2 + \frac{17}{3}$$

頂点 $\left(-\frac{1}{3}, \frac{17}{3}\right)$

$$(26) y = 5\left(x + \frac{4}{5}\right)^2 + \frac{24}{5}$$

頂点 $\left(-\frac{4}{5}, \frac{24}{5}\right)$

$$(27) y = 2(x - 1)^2 + 3$$

頂点 $(1, 3)$

$$(28) y = -5\left(x - \frac{4}{5}\right)^2 + \frac{51}{5}$$

頂点 $\left(\frac{4}{5}, \frac{51}{5}\right)$

$$(29) y = -7\left(x + \frac{3}{7}\right)^2 - \frac{5}{7}$$

頂点 $\left(-\frac{3}{7}, -\frac{5}{7}\right)$

$$(30) y = -8\left(x - \frac{1}{2}\right)^2 - 5$$

頂点 $\left(\frac{1}{2}, -5\right)$

$$(31) y = \left(x + \frac{3}{2}\right)^2 + \frac{19}{4}$$

頂点 $\left(-\frac{3}{2}, \frac{19}{4}\right)$

$$(32) y = 8\left(x + \frac{9}{16}\right)^2 - \frac{273}{32}$$

頂点 $\left(-\frac{9}{16}, -\frac{273}{32}\right)$

$$(33) y = 6\left(x + \frac{1}{3}\right)^2 + \frac{1}{3}$$

頂点 $\left(-\frac{1}{3}, \frac{1}{3}\right)$

$$(1) y = -4 \left(x - \frac{5}{8} \right)^2 + \frac{153}{16}$$

頂点 $\left(\frac{5}{8}, \frac{153}{16} \right)$

$$(2) y = 3 \left(x + \frac{2}{3} \right)^2 - \frac{1}{3}$$

頂点 $\left(-\frac{2}{3}, -\frac{1}{3} \right)$

$$(3) y = 4 \left(x - \frac{1}{2} \right)^2 - 1$$

頂点 $\left(\frac{1}{2}, -1 \right)$

$$(4) y = 5 \left(x + \frac{2}{5} \right)^2 + \frac{26}{5}$$

頂点 $\left(-\frac{2}{5}, \frac{26}{5} \right)$

$$(5) y = 8 \left(x + \frac{3}{16} \right)^2 - \frac{297}{32}$$

頂点 $\left(-\frac{3}{16}, -\frac{297}{32} \right)$

$$(6) y = 8 \left(x - \frac{5}{16} \right)^2 - \frac{153}{32}$$

頂点 $\left(\frac{5}{16}, -\frac{153}{32} \right)$

$$(7) y = \left(x - \frac{9}{2} \right)^2 - \frac{81}{4}$$

頂点 $\left(\frac{9}{2}, -\frac{81}{4} \right)$

$$(8) y = 8 \left(x + \frac{1}{8} \right)^2 + \frac{31}{8}$$

頂点 $\left(-\frac{1}{8}, \frac{31}{8} \right)$

$$(9) y = -(x-4)^2 + 9$$

頂点 $(4, 9)$

$$(10) y = 7 \left(x - \frac{1}{14} \right)^2 - \frac{253}{28}$$

頂点 $\left(\frac{1}{14}, -\frac{253}{28} \right)$

$$(11) y = 9 \left(x + \frac{5}{18} \right)^2 - \frac{241}{36}$$

頂点 $\left(-\frac{5}{18}, -\frac{241}{36} \right)$

$$(12) y = -5 \left(x - \frac{2}{5} \right)^2 - \frac{36}{5}$$

頂点 $\left(\frac{2}{5}, -\frac{36}{5} \right)$

$$(13) y = 7 \left(x + \frac{9}{14} \right)^2 - \frac{109}{28}$$

頂点 $\left(-\frac{9}{14}, -\frac{109}{28} \right)$

$$(14) y = -8 \left(x + \frac{3}{8} \right)^2 - \frac{31}{8}$$

頂点 $\left(-\frac{3}{8}, -\frac{31}{8} \right)$

$$(15) y = \left(x - \frac{1}{2} \right)^2 + \frac{19}{4}$$

頂点 $\left(\frac{1}{2}, \frac{19}{4} \right)$

$$(16) y = 7 \left(x + \frac{3}{14} \right)^2 + \frac{103}{28}$$

頂点 $\left(-\frac{3}{14}, \frac{103}{28} \right)$

$$(17) y = -7 \left(x + \frac{5}{14} \right)^2 - \frac{199}{28}$$

頂点 $\left(-\frac{5}{14}, -\frac{199}{28} \right)$

$$(18) y = 4(x+1)^2 - 6$$

頂点 $(-1, -6)$

$$(19) y = -4 \left(x - \frac{1}{2} \right)^2 - 5$$

頂点 $\left(\frac{1}{2}, -5 \right)$

$$(20) y = 7 \left(x - \frac{5}{14} \right)^2 + \frac{143}{28}$$

頂点 $\left(\frac{5}{14}, \frac{143}{28} \right)$

$$(21) y = -4x^2 - 9$$

頂点 $(0, -9)$

$$(22) y = 8 \left(x - \frac{1}{4} \right)^2 - \frac{5}{2}$$

頂点 $\left(\frac{1}{4}, -\frac{5}{2} \right)$

$$(23) y = 8 \left(x - \frac{7}{16} \right)^2 - \frac{81}{32}$$

頂点 $\left(\frac{7}{16}, -\frac{81}{32} \right)$

$$(24) y = 8 \left(x + \frac{9}{16} \right)^2 + \frac{143}{32}$$

頂点 $\left(-\frac{9}{16}, \frac{143}{32} \right)$

$$(25) y = 5 \left(x - \frac{3}{5} \right)^2 + \frac{16}{5}$$

頂点 $\left(\frac{3}{5}, \frac{16}{5} \right)$

$$(26) y = 2 \left(x + \frac{9}{4} \right)^2 - \frac{89}{8}$$

頂点 $\left(-\frac{9}{4}, -\frac{89}{8} \right)$

$$(27) y = -6 \left(x - \frac{1}{4} \right)^2 - \frac{29}{8}$$

頂点 $\left(\frac{1}{4}, -\frac{29}{8} \right)$

$$(28) y = 3 \left(x + \frac{1}{2} \right)^2 + \frac{33}{4}$$

頂点 $\left(-\frac{1}{2}, \frac{33}{4} \right)$

$$(29) y = 4 \left(x - \frac{1}{8} \right)^2 - \frac{17}{16}$$

頂点 $\left(\frac{1}{8}, -\frac{17}{16} \right)$

$$(30) y = -5 \left(x + \frac{4}{5} \right)^2 + \frac{16}{5}$$

頂点 $\left(-\frac{4}{5}, \frac{16}{5} \right)$

$$(31) y = 7 \left(x + \frac{3}{7} \right)^2 + \frac{19}{7}$$

頂点 $\left(-\frac{3}{7}, \frac{19}{7} \right)$

$$(32) y = 9 \left(x + \frac{4}{9} \right)^2 - \frac{70}{9}$$

頂点 $\left(-\frac{4}{9}, -\frac{70}{9} \right)$

$$(33) y = -8 \left(x + \frac{1}{8} \right)^2 - \frac{7}{8}$$

頂点 $\left(-\frac{1}{8}, -\frac{7}{8} \right)$

$$(1) y = 3(x+1)^2 - 11$$

頂点 $(-1, -11)$

$$(2) y = 4(x+1)^2 - 3$$

頂点 $(-1, -3)$

$$(3) y = -4\left(x + \frac{5}{8}\right)^2 - \frac{23}{16}$$

頂点 $\left(-\frac{5}{8}, -\frac{23}{16}\right)$

$$(4) y = 3\left(x - \frac{1}{3}\right)^2 - \frac{4}{3}$$

頂点 $\left(\frac{1}{3}, -\frac{4}{3}\right)$

$$(5) y = 5\left(x + \frac{1}{5}\right)^2 + \frac{34}{5}$$

頂点 $\left(-\frac{1}{5}, \frac{34}{5}\right)$

$$(6) y = 2\left(x + \frac{3}{4}\right)^2 - \frac{17}{8}$$

頂点 $\left(-\frac{3}{4}, -\frac{17}{8}\right)$

$$(7) y = 6\left(x - \frac{1}{2}\right)^2 + \frac{11}{2}$$

頂点 $\left(\frac{1}{2}, \frac{11}{2}\right)$

$$(8) y = 8x^2 + 2$$

頂点 $(0, 2)$

$$(9) y = -3\left(x - \frac{1}{2}\right)^2 + \frac{31}{4}$$

頂点 $\left(\frac{1}{2}, \frac{31}{4}\right)$

$$(10) y = -9\left(x - \frac{1}{2}\right)^2 + \frac{21}{4}$$

頂点 $\left(\frac{1}{2}, \frac{21}{4}\right)$

$$(11) y = 8\left(x + \frac{1}{2}\right)^2 + 6$$

頂点 $\left(-\frac{1}{2}, 6\right)$

$$(12) y = 2\left(x - \frac{9}{4}\right)^2 - \frac{33}{8}$$

頂点 $\left(\frac{9}{4}, -\frac{33}{8}\right)$

$$(13) y = -7\left(x - \frac{1}{7}\right)^2 - \frac{20}{7}$$

頂点 $\left(\frac{1}{7}, -\frac{20}{7}\right)$

$$(14) y = -8\left(x - \frac{3}{8}\right)^2 + \frac{49}{8}$$

頂点 $\left(\frac{3}{8}, \frac{49}{8}\right)$

$$(15) y = \left(x - \frac{5}{2}\right)^2 - \frac{33}{4}$$

頂点 $\left(\frac{5}{2}, -\frac{33}{4}\right)$

$$(16) y = -6\left(x - \frac{1}{4}\right)^2 - \frac{37}{8}$$

頂点 $\left(\frac{1}{4}, -\frac{37}{8}\right)$

$$(17) y = -(x+2)^2 + 4$$

頂点 $(-2, 4)$

$$(18) y = 8\left(x + \frac{1}{2}\right)^2 - 2$$

頂点 $\left(-\frac{1}{2}, -2\right)$

$$(19) y = 8\left(x + \frac{1}{4}\right)^2 - \frac{9}{2}$$

頂点 $\left(-\frac{1}{4}, -\frac{9}{2}\right)$

$$(20) y = 8\left(x + \frac{3}{16}\right)^2 - \frac{265}{32}$$

頂点 $\left(-\frac{3}{16}, -\frac{265}{32}\right)$

$$(21) y = -\left(x + \frac{9}{2}\right)^2 + \frac{109}{4}$$

頂点 $\left(-\frac{9}{2}, \frac{109}{4}\right)$

$$(22) y = -7\left(x - \frac{1}{7}\right)^2 - \frac{41}{7}$$

頂点 $\left(\frac{1}{7}, -\frac{41}{7}\right)$

$$(23) y = -9\left(x + \frac{1}{9}\right)^2 - \frac{35}{9}$$

頂点 $\left(-\frac{1}{9}, -\frac{35}{9}\right)$

$$(24) y = -6\left(x - \frac{3}{4}\right)^2 - \frac{13}{8}$$

頂点 $\left(\frac{3}{4}, -\frac{13}{8}\right)$

$$(25) y = 9\left(x + \frac{5}{18}\right)^2 - \frac{205}{36}$$

頂点 $\left(-\frac{5}{18}, -\frac{205}{36}\right)$

$$(26) y = \left(x - \frac{1}{2}\right)^2 - \frac{33}{4}$$

頂点 $\left(\frac{1}{2}, -\frac{33}{4}\right)$

$$(27) y = 8\left(x - \frac{1}{4}\right)^2 + \frac{11}{2}$$

頂点 $\left(\frac{1}{4}, \frac{11}{2}\right)$

$$(28) y = 4\left(x + \frac{5}{8}\right)^2 - \frac{57}{16}$$

頂点 $\left(-\frac{5}{8}, -\frac{57}{16}\right)$

$$(29) y = -7\left(x + \frac{3}{7}\right)^2 + \frac{44}{7}$$

頂点 $\left(-\frac{3}{7}, \frac{44}{7}\right)$

$$(30) y = 9x^2 + 6$$

頂点 $(0, 6)$

$$(31) y = 2\left(x + \frac{9}{4}\right)^2 - \frac{153}{8}$$

頂点 $\left(-\frac{9}{4}, -\frac{153}{8}\right)$

$$(32) y = -9\left(x - \frac{7}{18}\right)^2 + \frac{157}{36}$$

頂点 $\left(\frac{7}{18}, \frac{157}{36}\right)$

$$(33) y = -6x^2$$

頂点 $(0, 0)$

- (1) $y = 3\left(x + \frac{1}{3}\right)^2 - \frac{25}{3}$
頂点 $\left(-\frac{1}{3}, -\frac{25}{3}\right)$
- (2) $y = -(x + 2)^2 - 5$
頂点 $(-2, -5)$
- (3) $y = -9\left(x + \frac{1}{9}\right)^2 + \frac{28}{9}$
頂点 $\left(-\frac{1}{9}, \frac{28}{9}\right)$
- (4) $y = -7\left(x + \frac{1}{14}\right)^2 - \frac{167}{28}$
頂点 $\left(-\frac{1}{14}, -\frac{167}{28}\right)$
- (5) $y = 3\left(x + \frac{3}{2}\right)^2 - \frac{19}{4}$
頂点 $\left(-\frac{3}{2}, -\frac{19}{4}\right)$
- (6) $y = -9\left(x - \frac{1}{18}\right)^2 - \frac{215}{36}$
頂点 $\left(\frac{1}{18}, -\frac{215}{36}\right)$
- (7) $y = -5\left(x + \frac{9}{10}\right)^2 + \frac{61}{20}$
頂点 $\left(-\frac{9}{10}, \frac{61}{20}\right)$
- (8) $y = 9\left(x - \frac{7}{18}\right)^2 + \frac{95}{36}$
頂点 $\left(\frac{7}{18}, \frac{95}{36}\right)$
- (9) $y = 7x^2 - 9$
頂点 $(0, -9)$
- (10) $y = -7\left(x - \frac{9}{14}\right)^2 + \frac{305}{28}$
頂点 $\left(\frac{9}{14}, \frac{305}{28}\right)$
- (11) $y = -9x^2 - 8$
頂点 $(0, -8)$
- (12) $y = 7\left(x - \frac{3}{7}\right)^2 + \frac{54}{7}$
頂点 $\left(\frac{3}{7}, \frac{54}{7}\right)$
- (13) $y = 9\left(x + \frac{7}{18}\right)^2 - \frac{49}{36}$
頂点 $\left(-\frac{7}{18}, -\frac{49}{36}\right)$
- (14) $y = -5\left(x - \frac{1}{5}\right)^2 - \frac{4}{5}$
頂点 $\left(\frac{1}{5}, -\frac{4}{5}\right)$
- (15) $y = 7x^2 + 3$
頂点 $(0, 3)$
- (16) $y = -3\left(x - \frac{7}{6}\right)^2 + \frac{97}{12}$
頂点 $\left(\frac{7}{6}, \frac{97}{12}\right)$
- (17) $y = -6x^2 + 4$
頂点 $(0, 4)$
- (18) $y = -5\left(x + \frac{1}{2}\right)^2 + \frac{5}{4}$
頂点 $\left(-\frac{1}{2}, \frac{5}{4}\right)$
- (19) $y = -(x + 1)^2 + 3$
頂点 $(-1, 3)$
- (20) $y = -9\left(x + \frac{5}{18}\right)^2 - \frac{83}{36}$
頂点 $\left(-\frac{5}{18}, -\frac{83}{36}\right)$
- (21) $y = 8\left(x - \frac{3}{16}\right)^2 + \frac{247}{32}$
頂点 $\left(\frac{3}{16}, \frac{247}{32}\right)$
- (22) $y = 6\left(x + \frac{3}{4}\right)^2 + \frac{45}{8}$
頂点 $\left(-\frac{3}{4}, \frac{45}{8}\right)$
- (23) $y = 7x^2 - 9$
頂点 $(0, -9)$
- (24) $y = 7\left(x - \frac{1}{2}\right)^2 - \frac{31}{4}$
頂点 $\left(\frac{1}{2}, -\frac{31}{4}\right)$
- (25) $y = -9\left(x + \frac{4}{9}\right)^2 - \frac{2}{9}$
頂点 $\left(-\frac{4}{9}, -\frac{2}{9}\right)$
- (26) $y = 5\left(x - \frac{7}{10}\right)^2 - \frac{209}{20}$
頂点 $\left(\frac{7}{10}, -\frac{209}{20}\right)$
- (27) $y = -7\left(x + \frac{1}{2}\right)^2 - \frac{1}{4}$
頂点 $\left(-\frac{1}{2}, -\frac{1}{4}\right)$
- (28) $y = -8\left(x - \frac{1}{8}\right)^2 - \frac{71}{8}$
頂点 $\left(\frac{1}{8}, -\frac{71}{8}\right)$
- (29) $y = -\left(x - \frac{1}{2}\right)^2 + \frac{21}{4}$
頂点 $\left(\frac{1}{2}, \frac{21}{4}\right)$
- (30) $y = -(x + 3)^2 + 9$
頂点 $(-3, 9)$
- (31) $y = 3x^2 - 1$
頂点 $(0, -1)$
- (32) $y = -7\left(x + \frac{3}{14}\right)^2 + \frac{37}{28}$
頂点 $\left(-\frac{3}{14}, \frac{37}{28}\right)$
- (33) $y = -9\left(x - \frac{4}{9}\right)^2 + \frac{16}{9}$
頂点 $\left(\frac{4}{9}, \frac{16}{9}\right)$

3.1.5 頂点の座標 (難しい)

2 次関数の頂点を平方完成を行ってから求める問題。

(1) $y = 2x^2 + x + 5$

(6) $y = 3x^2 - 3x - 3$

(11) $y = -5x^2 + 3x$

(2) $y = -2x^2 - 5x - 5$

(7) $y = -x^2 - 2x + 2$

(12) $y = -2x^2 - 4x - 3$

(3) $y = 3x^2 + x - 3$

(8) $y = -4x^2 + 5x + 5$

(13) $y = -2x^2 - 3x + 5$

(4) $y = -3x^2 - 3x + 3$

(9) $y = -4x^2 + 2x$

(14) $y = -4x^2 - 3x + 1$

(5) $y = -x^2 - 4x + 2$

(10) $y = x^2 - x + 4$

(15) $y = 5x^2 - 4x + 4$

(1) $y = -x^2 + 5x$

(6) $y = -5x^2 + x + 4$

(11) $y = 2x^2 - 3x$

(2) $y = -3x^2 + 5x + 5$

(7) $y = -3x^2 - 4x - 3$

(12) $y = 4x^2 - 5x - 3$

(3) $y = 2x^2 - 4x + 3$

(8) $y = 4x^2 + 4x - 3$

(13) $y = 5x^2 + x - 1$

(4) $y = x^2 - 5x$

(9) $y = -5x^2 - 2x + 1$

(14) $y = -2x^2 + 2x - 4$

(5) $y = -4x^2 - 4x$

(10) $y = -5x^2 - 4x - 3$

(15) $y = -5x^2 - 3x - 5$

(1) $y = -3x^2 + 2x + 4$

(6) $y = -5x^2 + 4x - 3$

(11) $y = -4x^2 + 3x - 2$

(2) $y = 2x^2 - 2x$

(7) $y = -x^2 + 5x + 4$

(12) $y = 4x^2 - 3x - 2$

(3) $y = -x^2 - 4x - 1$

(8) $y = -4x^2 + x - 1$

(13) $y = 5x^2 + 2x$

(4) $y = 2x^2 + x - 1$

(9) $y = -3x^2 - x - 4$

(14) $y = -2x^2 + 3x$

(5) $y = -3x^2 - 4x - 3$

(10) $y = 5x^2 - 2x + 3$

(15) $y = -x^2 - 3x + 1$

(1) $y = 5x^2 - 5x + 1$

(6) $y = -x^2 + 3x + 2$

(11) $y = 3x^2 + 2x - 5$

(2) $y = -x^2 - 2x + 4$

(7) $y = -3x^2 - 4x - 1$

(12) $y = 4x^2 - x + 5$

(3) $y = 4x^2 - x$

(8) $y = 3x^2 + 5x - 4$

(13) $y = x^2 + 2x - 3$

(4) $y = -4x^2 - 2x - 3$

(9) $y = x^2 - 5x - 5$

(14) $y = 4x^2 + x + 2$

(5) $y = -3x^2 - x + 1$

(10) $y = 4x^2 - 4x + 4$

(15) $y = 4x^2 - 4x + 4$

(1) $y = 5x^2 + 4x + 1$

(6) $y = 3x^2 + 5x - 3$

(11) $y = 5x^2 - 4x + 3$

(2) $y = -3x^2 - 2x + 1$

(7) $y = 4x^2 + x + 4$

(12) $y = -x^2 - 2x - 2$

(3) $y = x^2 + 4x - 4$

(8) $y = -3x^2 - 5x$

(13) $y = 2x^2 - 3x - 4$

(4) $y = 2x^2 + x + 4$

(9) $y = -3x^2 - 4x - 5$

(14) $y = 3x^2 + 4x - 4$

(5) $y = 4x^2 + 3x - 4$

(10) $y = 2x^2 + 4x + 3$

(15) $y = 5x^2 + 3x - 4$

(1) $y = -2x^2 + 2x$

(6) $y = -4x^2 - x - 2$

(11) $y = 4x^2 + 2x - 3$

(2) $y = 3x^2 + 4x - 5$

(7) $y = -3x^2 - 4x - 1$

(12) $y = 5x^2 - 5x + 3$

(3) $y = 2x^2 + 4x + 2$

(8) $y = 4x^2 + 4x + 1$

(13) $y = 3x^2 + 3x + 4$

(4) $y = -4x^2 - x + 4$

(9) $y = 5x^2 - 2x + 5$

(14) $y = -4x^2 + 3x + 4$

(5) $y = 3x^2 - x + 5$

(10) $y = 4x^2 - 5x + 4$

(15) $y = 3x^2 + 4x + 1$

(1) $y = x^2 - 4x - 3$

(6) $y = 3x^2 - 5x + 5$

(11) $y = 4x^2 - 3x + 5$

(2) $y = -5x^2 + 4x + 4$

(7) $y = -4x^2 + x - 3$

(12) $y = -2x^2 + 5x + 1$

(3) $y = -4x^2 + 4x - 2$

(8) $y = -x^2 - 5x - 2$

(13) $y = 5x^2 - 5x - 4$

(4) $y = x^2 + 3x - 5$

(9) $y = 2x^2 - 2x + 4$

(14) $y = -4x^2 - 5x$

(5) $y = -2x^2 - x - 1$

(10) $y = 4x^2 - 5x - 5$

(15) $y = -5x^2 - x + 5$

(1) $y = -4x^2 - 5x + 3$

(6) $y = -2x^2 - x$

(11) $y = -4x^2 + 4x + 2$

(2) $y = 3x^2 + 4x + 1$

(7) $y = -2x^2 - x - 1$

(12) $y = -2x^2 + 4x + 1$

(3) $y = -x^2 - x + 4$

(8) $y = -5x^2 + 2x - 4$

(13) $y = -5x^2 + 3x + 2$

(4) $y = 2x^2 + x$

(9) $y = -5x^2 + 4x + 5$

(14) $y = -5x^2 - x + 1$

(5) $y = -3x^2 + 5x - 4$

(10) $y = 4x^2 + x - 3$

(15) $y = 2x^2 - 2x + 4$

(1) $y = -2x^2 - x + 2$

(6) $y = 5x^2 + 5x + 1$

(11) $y = -x^2 + 4x - 5$

(2) $y = -2x^2 - 5x - 1$

(7) $y = -2x^2 + 5x - 3$

(12) $y = -5x^2 - 2x + 2$

(3) $y = 2x^2 - 3x + 5$

(8) $y = -3x^2 - 4x - 5$

(13) $y = 2x^2 - 2x + 4$

(4) $y = -2x^2 + x + 1$

(9) $y = 2x^2 - 5x - 2$

(14) $y = -3x^2 - 3x - 3$

(5) $y = -2x^2 - 4x - 4$

(10) $y = -3x^2 - 5x - 1$

(15) $y = 3x^2 + 3x + 4$

(1) $y = -5x^2 - 4x - 3$

(6) $y = -5x^2 + 2x - 1$

(11) $y = -4x^2 - 3x - 3$

(2) $y = 5x^2 - 3x + 5$

(7) $y = 4x^2 - 2x + 2$

(12) $y = -4x^2 - 4x - 4$

(3) $y = -3x^2 + 3x$

(8) $y = -3x^2 + 3x + 2$

(13) $y = -x^2 - x + 5$

(4) $y = -5x^2 + 4x + 4$

(9) $y = -5x^2 + 3x - 2$

(14) $y = -5x^2 + x - 5$

(5) $y = -4x^2 - x - 1$

(10) $y = x^2 - 4x$

(15) $y = 4x^2 - 5x + 1$

(1) $y = -2x^2 - x - 1$

(6) $y = -5x^2 - 5x - 2$

(11) $y = 5x^2 - 3x + 4$

(2) $y = 2x^2 + 5x - 4$

(7) $y = 4x^2 - 5x$

(12) $y = -2x^2 - 5x + 1$

(3) $y = 4x^2 + 4x - 2$

(8) $y = 4x^2 + x$

(13) $y = -3x^2 - 3x + 4$

(4) $y = 3x^2 + 2x + 4$

(9) $y = -x^2 + 4x - 1$

(14) $y = -x^2 - x + 1$

(5) $y = -4x^2 + 3x + 3$

(10) $y = 3x^2 + x - 5$

(15) $y = 4x^2 - 2x$

(1) $y = -4x^2 - 5x + 4$

(6) $y = x^2 + x$

(11) $y = 4x^2 + 2x + 3$

(2) $y = -x^2 + x - 2$

(7) $y = -5x^2 + 2x - 2$

(12) $y = -3x^2 + 3x$

(3) $y = 2x^2 + 2x - 2$

(8) $y = 2x^2 + 4x - 4$

(13) $y = -2x^2 - x + 5$

(4) $y = -3x^2 + 3x + 1$

(9) $y = x^2 - 4x + 3$

(14) $y = 3x^2 - 5x - 1$

(5) $y = 5x^2 - 2x + 4$

(10) $y = -x^2 - x + 3$

(15) $y = 3x^2 - 3x + 5$

(1) $y = -4x^2 - 5x - 4$

(6) $y = 5x^2 - 4x - 5$

(11) $y = 4x^2 + 3x - 4$

(2) $y = -x^2 + 3x - 5$

(7) $y = -3x^2 - x + 5$

(12) $y = -5x^2 + x + 1$

(3) $y = -3x^2 + 3x - 3$

(8) $y = 5x^2 - 4x + 4$

(13) $y = -3x^2 + 5x - 2$

(4) $y = 3x^2 - 5x$

(9) $y = -5x^2 + 2x - 5$

(14) $y = 5x^2 + 2x - 2$

(5) $y = -2x^2 - x + 4$

(10) $y = x^2 - 4x + 5$

(15) $y = -4x^2 + 5x - 1$

(1) $y = x^2 - 2x + 1$

(6) $y = -x^2 - 2x + 4$

(11) $y = -2x^2 - 4x - 4$

(2) $y = 4x^2 - 3x + 3$

(7) $y = 2x^2 - 5x + 4$

(12) $y = 3x^2 - 2x + 1$

(3) $y = x^2 + 5x + 4$

(8) $y = -x^2 + x - 3$

(13) $y = 4x^2 + x - 3$

(4) $y = 2x^2 - 4x + 2$

(9) $y = 2x^2 + x$

(14) $y = -3x^2 + x - 2$

(5) $y = 3x^2 + 5x - 2$

(10) $y = 4x^2 - x$

(15) $y = -2x^2 - 5x - 4$

(1) $y = x^2 + x - 1$

(6) $y = 2x^2 - x - 3$

(11) $y = 4x^2 + 3x - 4$

(2) $y = -4x^2 - 3x - 1$

(7) $y = 2x^2 + 4x - 5$

(12) $y = 4x^2 - 2x + 2$

(3) $y = 2x^2 - 3x + 4$

(8) $y = x^2 + 2x + 5$

(13) $y = 4x^2 + 2x - 1$

(4) $y = -5x^2 - x - 5$

(9) $y = 5x^2 + x + 2$

(14) $y = -3x^2 + 4x + 5$

(5) $y = x^2 - 5x - 4$

(10) $y = -x^2 + 2x - 4$

(15) $y = -2x^2 + 2x + 5$

(1) $y = -x^2 - 4x + 1$

(6) $y = x^2 - x + 1$

(11) $y = -2x^2 - 5x$

(2) $y = -5x^2 + 3x + 5$

(7) $y = -x^2 + 5x$

(12) $y = 4x^2 + 2x$

(3) $y = -3x^2 + 3x - 4$

(8) $y = -x^2 - 2x - 4$

(13) $y = -x^2 - 4x + 4$

(4) $y = -2x^2 + 4x - 2$

(9) $y = 5x^2 - 2x + 4$

(14) $y = 3x^2 - 3x + 2$

(5) $y = 2x^2 - 4x + 4$

(10) $y = x^2 - 3x + 1$

(15) $y = x^2 + 4x + 4$

(1) $y = -4x^2 - 2x + 3$

(6) $y = -4x^2 + x - 1$

(11) $y = -3x^2 - 2x - 3$

(2) $y = -2x^2 + 2x + 3$

(7) $y = 4x^2 - 4x$

(12) $y = x^2 + 2x - 3$

(3) $y = -x^2 + x + 2$

(8) $y = x^2 - 3x + 5$

(13) $y = -x^2 + 3x$

(4) $y = 5x^2 + 2x + 4$

(9) $y = -4x^2 + 2x + 2$

(14) $y = -2x^2 + 3x + 1$

(5) $y = 3x^2 + 3x + 4$

(10) $y = -2x^2 - 2x - 4$

(15) $y = -2x^2 + x$

(1) $y = -5x^2 - 5x + 5$

(6) $y = 5x^2 + 3x - 5$

(11) $y = -4x^2 - 3x$

(2) $y = -3x^2 + 2x + 5$

(7) $y = -4x^2 + 5x + 1$

(12) $y = 2x^2 + 2x - 3$

(3) $y = 2x^2 - 3x - 3$

(8) $y = -5x^2 - 3x + 4$

(13) $y = 3x^2 + 3x - 5$

(4) $y = 2x^2 + 2x - 5$

(9) $y = -4x^2 - 4x - 5$

(14) $y = 5x^2 - 4x + 3$

(5) $y = -4x^2 - 5x - 3$

(10) $y = 2x^2 - 4x + 4$

(15) $y = -3x^2 + x + 4$

(1) $y = -3x^2 - 3x - 4$

(6) $y = 4x^2 - 5x + 3$

(11) $y = 5x^2 - 5x - 4$

(2) $y = 3x^2 + 5x$

(7) $y = -2x^2 - x + 1$

(12) $y = x^2 - 5x - 2$

(3) $y = 5x^2 - 3x - 3$

(8) $y = 4x^2 + 4x + 3$

(13) $y = -5x^2 + 4x$

(4) $y = 3x^2 - 2x - 5$

(9) $y = 2x^2 + 5x - 3$

(14) $y = 5x^2 + 4x - 2$

(5) $y = 2x^2 - 4x + 5$

(10) $y = 5x^2 + 2x - 2$

(15) $y = -5x^2 - 5x + 2$

(1) $y = -2x^2 - 3x + 5$

(6) $y = 2x^2 + 3x$

(11) $y = 5x^2 - 5x$

(2) $y = -4x^2 + 5x - 2$

(7) $y = 2x^2 + x + 4$

(12) $y = -x^2 - x + 4$

(3) $y = -3x^2 - 3x - 3$

(8) $y = -2x^2 + 2x + 4$

(13) $y = 5x^2 - 5x + 4$

(4) $y = 2x^2 + 2x + 5$

(9) $y = x^2 + 4x - 3$

(14) $y = x^2 + 5x - 3$

(5) $y = -4x^2 + 5x - 4$

(10) $y = -2x^2 - 5x + 4$

(15) $y = -4x^2 - x - 1$

$$(1) y = 2x^2 + x + 5$$

頂点 $\left(-\frac{1}{4}, \frac{39}{8}\right)$

$$(6) y = 3x^2 - 3x - 3$$

頂点 $\left(\frac{1}{2}, -\frac{15}{4}\right)$

$$(11) y = -5x^2 + 3x$$

頂点 $\left(\frac{3}{10}, \frac{9}{20}\right)$

$$(2) y = -2x^2 - 5x - 5$$

頂点 $\left(-\frac{5}{4}, -\frac{15}{8}\right)$

$$(7) y = -x^2 - 2x + 2$$

頂点 $(-1, 3)$

$$(12) y = -2x^2 - 4x - 3$$

頂点 $(-1, -1)$

$$(3) y = 3x^2 + x - 3$$

頂点 $\left(-\frac{1}{6}, -\frac{37}{12}\right)$

$$(8) y = -4x^2 + 5x + 5$$

頂点 $\left(\frac{5}{8}, \frac{105}{16}\right)$

$$(13) y = -2x^2 - 3x + 5$$

頂点 $\left(-\frac{3}{4}, \frac{49}{8}\right)$

$$(4) y = -3x^2 - 3x + 3$$

頂点 $\left(-\frac{1}{2}, \frac{15}{4}\right)$

$$(9) y = -4x^2 + 2x$$

頂点 $\left(\frac{1}{4}, \frac{1}{4}\right)$

$$(14) y = -4x^2 - 3x + 1$$

頂点 $\left(-\frac{3}{8}, \frac{25}{16}\right)$

$$(5) y = -x^2 - 4x + 2$$

頂点 $(-2, 6)$

$$(10) y = x^2 - x + 4$$

頂点 $\left(\frac{1}{2}, \frac{15}{4}\right)$

$$(15) y = 5x^2 - 4x + 4$$

頂点 $\left(\frac{2}{5}, \frac{16}{5}\right)$

$$(1) y = -x^2 + 5x$$

頂点 $\left(\frac{5}{2}, \frac{25}{4}\right)$

$$(6) y = -5x^2 + x + 4$$

頂点 $\left(\frac{1}{10}, \frac{81}{20}\right)$

$$(11) y = 2x^2 - 3x$$

頂点 $\left(\frac{3}{4}, -\frac{9}{8}\right)$

$$(2) y = -3x^2 + 5x + 5$$

頂点 $\left(\frac{5}{6}, \frac{85}{12}\right)$

$$(7) y = -3x^2 - 4x - 3$$

頂点 $\left(-\frac{2}{3}, -\frac{5}{3}\right)$

$$(12) y = 4x^2 - 5x - 3$$

頂点 $\left(\frac{5}{8}, -\frac{73}{16}\right)$

$$(3) y = 2x^2 - 4x + 3$$

頂点 $(1, 1)$

$$(8) y = 4x^2 + 4x - 3$$

頂点 $\left(-\frac{1}{2}, -4\right)$

$$(13) y = 5x^2 + x - 1$$

頂点 $\left(-\frac{1}{10}, -\frac{21}{20}\right)$

$$(4) y = x^2 - 5x$$

頂点 $\left(\frac{5}{2}, -\frac{25}{4}\right)$

$$(9) y = -5x^2 - 2x + 1$$

頂点 $\left(-\frac{1}{5}, \frac{6}{5}\right)$

$$(14) y = -2x^2 + 2x - 4$$

頂点 $\left(\frac{1}{2}, -\frac{7}{2}\right)$

$$(5) y = -4x^2 - 4x$$

頂点 $\left(-\frac{1}{2}, 1\right)$

$$(10) y = -5x^2 - 4x - 3$$

頂点 $\left(-\frac{2}{5}, -\frac{11}{5}\right)$

$$(15) y = -5x^2 - 3x - 5$$

頂点 $\left(-\frac{3}{10}, -\frac{91}{20}\right)$

$$(1) y = -3x^2 + 2x + 4$$

頂点 $\left(\frac{1}{3}, \frac{13}{3}\right)$

$$(6) y = -5x^2 + 4x - 3$$

頂点 $\left(\frac{2}{5}, -\frac{11}{5}\right)$

$$(11) y = -4x^2 + 3x - 2$$

頂点 $\left(\frac{3}{8}, -\frac{23}{16}\right)$

$$(2) y = 2x^2 - 2x$$

頂点 $\left(\frac{1}{2}, -\frac{1}{2}\right)$

$$(7) y = -x^2 + 5x + 4$$

頂点 $\left(\frac{5}{2}, \frac{41}{4}\right)$

$$(12) y = 4x^2 - 3x - 2$$

頂点 $\left(\frac{3}{8}, -\frac{41}{16}\right)$

$$(3) y = -x^2 - 4x - 1$$

頂点 $(-2, 3)$

$$(8) y = -4x^2 + x - 1$$

頂点 $\left(\frac{1}{8}, -\frac{15}{16}\right)$

$$(13) y = 5x^2 + 2x$$

頂点 $\left(-\frac{1}{5}, -\frac{1}{5}\right)$

$$(4) y = 2x^2 + x - 1$$

頂点 $\left(-\frac{1}{4}, -\frac{9}{8}\right)$

$$(9) y = -3x^2 - x - 4$$

頂点 $\left(-\frac{1}{6}, -\frac{47}{12}\right)$

$$(14) y = -2x^2 + 3x$$

頂点 $\left(\frac{3}{4}, \frac{9}{8}\right)$

$$(5) y = -3x^2 - 4x - 3$$

頂点 $\left(-\frac{2}{3}, -\frac{5}{3}\right)$

$$(10) y = 5x^2 - 2x + 3$$

頂点 $\left(\frac{1}{5}, \frac{14}{5}\right)$

$$(15) y = -x^2 - 3x + 1$$

頂点 $\left(-\frac{3}{2}, \frac{13}{4}\right)$

$$(1) y = 5x^2 - 5x + 1$$

頂点 $\left(\frac{1}{2}, -\frac{1}{4}\right)$

$$(6) y = -x^2 + 3x + 2$$

頂点 $\left(\frac{3}{2}, \frac{17}{4}\right)$

$$(11) y = 3x^2 + 2x - 5$$

頂点 $\left(-\frac{1}{3}, -\frac{16}{3}\right)$

$$(2) y = -x^2 - 2x + 4$$

頂点 $(-1, 5)$

$$(7) y = -3x^2 - 4x - 1$$

頂点 $\left(-\frac{2}{3}, \frac{1}{3}\right)$

$$(12) y = 4x^2 - x + 5$$

頂点 $\left(\frac{1}{8}, \frac{79}{16}\right)$

$$(3) y = 4x^2 - x$$

頂点 $\left(\frac{1}{8}, -\frac{1}{16}\right)$

$$(8) y = 3x^2 + 5x - 4$$

頂点 $\left(-\frac{5}{6}, -\frac{73}{12}\right)$

$$(13) y = x^2 + 2x - 3$$

頂点 $(-1, -4)$

$$(4) y = -4x^2 - 2x - 3$$

頂点 $\left(-\frac{1}{4}, -\frac{11}{4}\right)$

$$(9) y = x^2 - 5x - 5$$

頂点 $\left(\frac{5}{2}, -\frac{45}{4}\right)$

$$(14) y = 4x^2 + x + 2$$

頂点 $\left(-\frac{1}{8}, \frac{31}{16}\right)$

$$(5) y = -3x^2 - x + 1$$

頂点 $\left(-\frac{1}{6}, \frac{13}{12}\right)$

$$(10) y = 4x^2 - 4x + 4$$

頂点 $\left(\frac{1}{2}, 3\right)$

$$(15) y = 4x^2 - 4x + 4$$

頂点 $\left(\frac{1}{2}, 3\right)$

$$(1) y = 5x^2 + 4x + 1$$

頂点 $\left(-\frac{2}{5}, \frac{1}{5}\right)$

$$(6) y = 3x^2 + 5x - 3$$

頂点 $\left(-\frac{5}{6}, -\frac{61}{12}\right)$

$$(11) y = 5x^2 - 4x + 3$$

頂点 $\left(\frac{2}{5}, \frac{11}{5}\right)$

$$(2) y = -3x^2 - 2x + 1$$

頂点 $\left(-\frac{1}{3}, \frac{4}{3}\right)$

$$(7) y = 4x^2 + x + 4$$

頂点 $\left(-\frac{1}{8}, \frac{63}{16}\right)$

$$(12) y = -x^2 - 2x - 2$$

頂点 $(-1, -1)$

$$(3) y = x^2 + 4x - 4$$

頂点 $(-2, -8)$

$$(8) y = -3x^2 - 5x$$

頂点 $\left(-\frac{5}{6}, \frac{25}{12}\right)$

$$(13) y = 2x^2 - 3x - 4$$

頂点 $\left(\frac{3}{4}, -\frac{41}{8}\right)$

$$(4) y = 2x^2 + x + 4$$

頂点 $\left(-\frac{1}{4}, \frac{31}{8}\right)$

$$(9) y = -3x^2 - 4x - 5$$

頂点 $\left(-\frac{2}{3}, -\frac{11}{3}\right)$

$$(14) y = 3x^2 + 4x - 4$$

頂点 $\left(-\frac{2}{3}, -\frac{16}{3}\right)$

$$(5) y = 4x^2 + 3x - 4$$

頂点 $\left(-\frac{3}{8}, -\frac{73}{16}\right)$

$$(10) y = 2x^2 + 4x + 3$$

頂点 $(-1, 1)$

$$(15) y = 5x^2 + 3x - 4$$

頂点 $\left(-\frac{3}{10}, -\frac{89}{20}\right)$

$$(1) y = -2x^2 + 2x$$

頂点 $\left(\frac{1}{2}, \frac{1}{2}\right)$

$$(6) y = -4x^2 - x - 2$$

頂点 $\left(-\frac{1}{8}, -\frac{31}{16}\right)$

$$(11) y = 4x^2 + 2x - 3$$

頂点 $\left(-\frac{1}{4}, -\frac{13}{4}\right)$

$$(2) y = 3x^2 + 4x - 5$$

頂点 $\left(-\frac{2}{3}, -\frac{19}{3}\right)$

$$(7) y = -3x^2 - 4x - 1$$

頂点 $\left(-\frac{2}{3}, \frac{1}{3}\right)$

$$(12) y = 5x^2 - 5x + 3$$

頂点 $\left(\frac{1}{2}, \frac{7}{4}\right)$

$$(3) y = 2x^2 + 4x + 2$$

頂点 $(-1, 0)$

$$(8) y = 4x^2 + 4x + 1$$

頂点 $\left(-\frac{1}{2}, 0\right)$

$$(13) y = 3x^2 + 3x + 4$$

頂点 $\left(-\frac{1}{2}, \frac{13}{4}\right)$

$$(4) y = -4x^2 - x + 4$$

頂点 $\left(-\frac{1}{8}, \frac{65}{16}\right)$

$$(9) y = 5x^2 - 2x + 5$$

頂点 $\left(\frac{1}{5}, \frac{24}{5}\right)$

$$(14) y = -4x^2 + 3x + 4$$

頂点 $\left(\frac{3}{8}, \frac{73}{16}\right)$

$$(5) y = 3x^2 - x + 5$$

頂点 $\left(\frac{1}{6}, \frac{59}{12}\right)$

$$(10) y = 4x^2 - 5x + 4$$

頂点 $\left(\frac{5}{8}, \frac{39}{16}\right)$

$$(15) y = 3x^2 + 4x + 1$$

頂点 $\left(-\frac{2}{3}, -\frac{1}{3}\right)$

$$(1) y = x^2 - 4x - 3$$

頂点 $(2, -7)$

$$(6) y = 3x^2 - 5x + 5$$

頂点 $\left(\frac{5}{6}, \frac{35}{12}\right)$

$$(11) y = 4x^2 - 3x + 5$$

頂点 $\left(\frac{3}{8}, \frac{71}{16}\right)$

$$(2) y = -5x^2 + 4x + 4$$

頂点 $\left(\frac{2}{5}, \frac{24}{5}\right)$

$$(7) y = -4x^2 + x - 3$$

頂点 $\left(\frac{1}{8}, -\frac{47}{16}\right)$

$$(12) y = -2x^2 + 5x + 1$$

頂点 $\left(\frac{5}{4}, \frac{33}{8}\right)$

$$(3) y = -4x^2 + 4x - 2$$

頂点 $\left(\frac{1}{2}, -1\right)$

$$(8) y = -x^2 - 5x - 2$$

頂点 $\left(-\frac{5}{2}, \frac{17}{4}\right)$

$$(13) y = 5x^2 - 5x - 4$$

頂点 $\left(\frac{1}{2}, -\frac{21}{4}\right)$

$$(4) y = x^2 + 3x - 5$$

頂点 $\left(-\frac{3}{2}, -\frac{29}{4}\right)$

$$(9) y = 2x^2 - 2x + 4$$

頂点 $\left(\frac{1}{2}, \frac{7}{2}\right)$

$$(14) y = -4x^2 - 5x$$

頂点 $\left(-\frac{5}{8}, \frac{25}{16}\right)$

$$(5) y = -2x^2 - x - 1$$

頂点 $\left(-\frac{1}{4}, -\frac{7}{8}\right)$

$$(10) y = 4x^2 - 5x - 5$$

頂点 $\left(\frac{5}{8}, -\frac{105}{16}\right)$

$$(15) y = -5x^2 - x + 5$$

頂点 $\left(-\frac{1}{10}, \frac{101}{20}\right)$

$$(1) y = -4x^2 - 5x + 3$$

頂点 $\left(-\frac{5}{8}, \frac{73}{16}\right)$

$$(6) y = -2x^2 - x$$

頂点 $\left(-\frac{1}{4}, \frac{1}{8}\right)$

$$(11) y = -4x^2 + 4x + 2$$

頂点 $\left(\frac{1}{2}, 3\right)$

$$(2) y = 3x^2 + 4x + 1$$

頂点 $\left(-\frac{2}{3}, -\frac{1}{3}\right)$

$$(7) y = -2x^2 - x - 1$$

頂点 $\left(-\frac{1}{4}, -\frac{7}{8}\right)$

$$(12) y = -2x^2 + 4x + 1$$

頂点 $(1, 3)$

$$(3) y = -x^2 - x + 4$$

頂点 $\left(-\frac{1}{2}, \frac{17}{4}\right)$

$$(8) y = -5x^2 + 2x - 4$$

頂点 $\left(\frac{1}{5}, -\frac{19}{5}\right)$

$$(13) y = -5x^2 + 3x + 2$$

頂点 $\left(\frac{3}{10}, \frac{49}{20}\right)$

$$(4) y = 2x^2 + x$$

頂点 $\left(-\frac{1}{4}, -\frac{1}{8}\right)$

$$(9) y = -5x^2 + 4x + 5$$

頂点 $\left(\frac{2}{5}, \frac{29}{5}\right)$

$$(14) y = -5x^2 - x + 1$$

頂点 $\left(-\frac{1}{10}, \frac{21}{20}\right)$

$$(5) y = -3x^2 + 5x - 4$$

頂点 $\left(\frac{5}{6}, -\frac{23}{12}\right)$

$$(10) y = 4x^2 + x - 3$$

頂点 $\left(-\frac{1}{8}, -\frac{49}{16}\right)$

$$(15) y = 2x^2 - 2x + 4$$

頂点 $\left(\frac{1}{2}, \frac{7}{2}\right)$

$$(1) y = -2x^2 - x + 2$$

頂点 $\left(-\frac{1}{4}, \frac{17}{8}\right)$

$$(6) y = 5x^2 + 5x + 1$$

頂点 $\left(-\frac{1}{2}, -\frac{1}{4}\right)$

$$(11) y = -x^2 + 4x - 5$$

頂点 $(2, -1)$

$$(2) y = -2x^2 - 5x - 1$$

頂点 $\left(-\frac{5}{4}, \frac{17}{8}\right)$

$$(7) y = -2x^2 + 5x - 3$$

頂点 $\left(\frac{5}{4}, \frac{1}{8}\right)$

$$(12) y = -5x^2 - 2x + 2$$

頂点 $\left(-\frac{1}{5}, \frac{11}{5}\right)$

$$(3) y = 2x^2 - 3x + 5$$

頂点 $\left(\frac{3}{4}, \frac{31}{8}\right)$

$$(8) y = -3x^2 - 4x - 5$$

頂点 $\left(-\frac{2}{3}, -\frac{11}{3}\right)$

$$(13) y = 2x^2 - 2x + 4$$

頂点 $\left(\frac{1}{2}, \frac{7}{2}\right)$

$$(4) y = -2x^2 + x + 1$$

頂点 $\left(\frac{1}{4}, \frac{9}{8}\right)$

$$(9) y = 2x^2 - 5x - 2$$

頂点 $\left(\frac{5}{4}, -\frac{41}{8}\right)$

$$(14) y = -3x^2 - 3x - 3$$

頂点 $\left(-\frac{1}{2}, -\frac{9}{4}\right)$

$$(5) y = -2x^2 - 4x - 4$$

頂点 $(-1, -2)$

$$(10) y = -3x^2 - 5x - 1$$

頂点 $\left(-\frac{5}{6}, \frac{13}{12}\right)$

$$(15) y = 3x^2 + 3x + 4$$

頂点 $\left(-\frac{1}{2}, \frac{13}{4}\right)$

$$(1) y = -5x^2 - 4x - 3$$

頂点 $\left(-\frac{2}{5}, -\frac{11}{5}\right)$

$$(6) y = -5x^2 + 2x - 1$$

頂点 $\left(\frac{1}{5}, -\frac{4}{5}\right)$

$$(11) y = -4x^2 - 3x - 3$$

頂点 $\left(-\frac{3}{8}, -\frac{39}{16}\right)$

$$(2) y = 5x^2 - 3x + 5$$

頂点 $\left(\frac{3}{10}, \frac{91}{20}\right)$

$$(7) y = 4x^2 - 2x + 2$$

頂点 $\left(\frac{1}{4}, \frac{7}{4}\right)$

$$(12) y = -4x^2 - 4x - 4$$

頂点 $\left(-\frac{1}{2}, -3\right)$

$$(3) y = -3x^2 + 3x$$

頂点 $\left(\frac{1}{2}, \frac{3}{4}\right)$

$$(8) y = -3x^2 + 3x + 2$$

頂点 $\left(\frac{1}{2}, \frac{11}{4}\right)$

$$(13) y = -x^2 - x + 5$$

頂点 $\left(-\frac{1}{2}, \frac{21}{4}\right)$

$$(4) y = -5x^2 + 4x + 4$$

頂点 $\left(\frac{2}{5}, \frac{24}{5}\right)$

$$(9) y = -5x^2 + 3x - 2$$

頂点 $\left(\frac{3}{10}, -\frac{31}{20}\right)$

$$(14) y = -5x^2 + x - 5$$

頂点 $\left(\frac{1}{10}, -\frac{99}{20}\right)$

$$(5) y = -4x^2 - x - 1$$

頂点 $\left(-\frac{1}{8}, -\frac{15}{16}\right)$

$$(10) y = x^2 - 4x$$

頂点 $(2, -4)$

$$(15) y = 4x^2 - 5x + 1$$

頂点 $\left(\frac{5}{8}, -\frac{9}{16}\right)$

$$(1) y = -2x^2 - x - 1$$

頂点 $\left(-\frac{1}{4}, -\frac{7}{8}\right)$

$$(6) y = -5x^2 - 5x - 2$$

頂点 $\left(-\frac{1}{2}, -\frac{3}{4}\right)$

$$(11) y = 5x^2 - 3x + 4$$

頂点 $\left(\frac{3}{10}, \frac{71}{20}\right)$

$$(2) y = 2x^2 + 5x - 4$$

頂点 $\left(-\frac{5}{4}, -\frac{57}{8}\right)$

$$(7) y = 4x^2 - 5x$$

頂点 $\left(\frac{5}{8}, -\frac{25}{16}\right)$

$$(12) y = -2x^2 - 5x + 1$$

頂点 $\left(-\frac{5}{4}, \frac{33}{8}\right)$

$$(3) y = 4x^2 + 4x - 2$$

頂点 $\left(-\frac{1}{2}, -3\right)$

$$(8) y = 4x^2 + x$$

頂点 $\left(-\frac{1}{8}, -\frac{1}{16}\right)$

$$(13) y = -3x^2 - 3x + 4$$

頂点 $\left(-\frac{1}{2}, \frac{19}{4}\right)$

$$(4) y = 3x^2 + 2x + 4$$

頂点 $\left(-\frac{1}{3}, \frac{11}{3}\right)$

$$(9) y = -x^2 + 4x - 1$$

頂点 $(2, 3)$

$$(14) y = -x^2 - x + 1$$

頂点 $\left(-\frac{1}{2}, \frac{5}{4}\right)$

$$(5) y = -4x^2 + 3x + 3$$

頂点 $\left(\frac{3}{8}, \frac{57}{16}\right)$

$$(10) y = 3x^2 + x - 5$$

頂点 $\left(-\frac{1}{6}, -\frac{61}{12}\right)$

$$(15) y = 4x^2 - 2x$$

頂点 $\left(\frac{1}{4}, -\frac{1}{4}\right)$

$$(1) y = -4x^2 - 5x + 4$$

頂点 $\left(-\frac{5}{8}, \frac{89}{16}\right)$

$$(6) y = x^2 + x$$

頂点 $\left(-\frac{1}{2}, -\frac{1}{4}\right)$

$$(11) y = 4x^2 + 2x + 3$$

頂点 $\left(-\frac{1}{4}, \frac{11}{4}\right)$

$$(2) y = -x^2 + x - 2$$

頂点 $\left(\frac{1}{2}, -\frac{7}{4}\right)$

$$(7) y = -5x^2 + 2x - 2$$

頂点 $\left(\frac{1}{5}, -\frac{9}{5}\right)$

$$(12) y = -3x^2 + 3x$$

頂点 $\left(\frac{1}{2}, \frac{3}{4}\right)$

$$(3) y = 2x^2 + 2x - 2$$

頂点 $\left(-\frac{1}{2}, -\frac{5}{2}\right)$

$$(8) y = 2x^2 + 4x - 4$$

頂点 $(-1, -6)$

$$(13) y = -2x^2 - x + 5$$

頂点 $\left(-\frac{1}{4}, \frac{41}{8}\right)$

$$(4) y = -3x^2 + 3x + 1$$

頂点 $\left(\frac{1}{2}, \frac{7}{4}\right)$

$$(9) y = x^2 - 4x + 3$$

頂点 $(2, -1)$

$$(14) y = 3x^2 - 5x - 1$$

頂点 $\left(\frac{5}{6}, -\frac{37}{12}\right)$

$$(5) y = 5x^2 - 2x + 4$$

頂点 $\left(\frac{1}{5}, \frac{19}{5}\right)$

$$(10) y = -x^2 - x + 3$$

頂点 $\left(-\frac{1}{2}, \frac{13}{4}\right)$

$$(15) y = 3x^2 - 3x + 5$$

頂点 $\left(\frac{1}{2}, \frac{17}{4}\right)$

$$(1) y = -4x^2 - 5x - 4$$

頂点 $\left(-\frac{5}{8}, -\frac{39}{16}\right)$

$$(6) y = 5x^2 - 4x - 5$$

頂点 $\left(\frac{2}{5}, -\frac{29}{5}\right)$

$$(11) y = 4x^2 + 3x - 4$$

頂点 $\left(-\frac{3}{8}, -\frac{73}{16}\right)$

$$(2) y = -x^2 + 3x - 5$$

頂点 $\left(\frac{3}{2}, -\frac{11}{4}\right)$

$$(7) y = -3x^2 - x + 5$$

頂点 $\left(-\frac{1}{6}, \frac{61}{12}\right)$

$$(12) y = -5x^2 + x + 1$$

頂点 $\left(\frac{1}{10}, \frac{21}{20}\right)$

$$(3) y = -3x^2 + 3x - 3$$

頂点 $\left(\frac{1}{2}, -\frac{9}{4}\right)$

$$(8) y = 5x^2 - 4x + 4$$

頂点 $\left(\frac{2}{5}, \frac{16}{5}\right)$

$$(13) y = -3x^2 + 5x - 2$$

頂点 $\left(\frac{5}{6}, \frac{1}{12}\right)$

$$(4) y = 3x^2 - 5x$$

頂点 $\left(\frac{5}{6}, -\frac{25}{12}\right)$

$$(9) y = -5x^2 + 2x - 5$$

頂点 $\left(\frac{1}{5}, -\frac{24}{5}\right)$

$$(14) y = 5x^2 + 2x - 2$$

頂点 $\left(-\frac{1}{5}, -\frac{11}{5}\right)$

$$(5) y = -2x^2 - x + 4$$

頂点 $\left(-\frac{1}{4}, \frac{33}{8}\right)$

$$(10) y = x^2 - 4x + 5$$

頂点 $(2, 1)$

$$(15) y = -4x^2 + 5x - 1$$

頂点 $\left(\frac{5}{8}, \frac{9}{16}\right)$

(1) $y = x^2 - 2x + 1$
頂点 $(1, 0)$

(6) $y = -x^2 - 2x + 4$
頂点 $(-1, 5)$

(11) $y = -2x^2 - 4x - 4$
頂点 $(-1, -2)$

(2) $y = 4x^2 - 3x + 3$
頂点 $\left(\frac{3}{8}, \frac{39}{16}\right)$

(7) $y = 2x^2 - 5x + 4$
頂点 $\left(\frac{5}{4}, \frac{7}{8}\right)$

(12) $y = 3x^2 - 2x + 1$
頂点 $\left(\frac{1}{3}, \frac{2}{3}\right)$

(3) $y = x^2 + 5x + 4$
頂点 $\left(-\frac{5}{2}, -\frac{9}{4}\right)$

(8) $y = -x^2 + x - 3$
頂点 $\left(\frac{1}{2}, -\frac{11}{4}\right)$

(13) $y = 4x^2 + x - 3$
頂点 $\left(-\frac{1}{8}, -\frac{49}{16}\right)$

(4) $y = 2x^2 - 4x + 2$
頂点 $(1, 0)$

(9) $y = 2x^2 + x$
頂点 $\left(-\frac{1}{4}, -\frac{1}{8}\right)$

(14) $y = -3x^2 + x - 2$
頂点 $\left(\frac{1}{6}, -\frac{23}{12}\right)$

(5) $y = 3x^2 + 5x - 2$
頂点 $\left(-\frac{5}{6}, -\frac{49}{12}\right)$

(10) $y = 4x^2 - x$
頂点 $\left(\frac{1}{8}, -\frac{1}{16}\right)$

(15) $y = -2x^2 - 5x - 4$
頂点 $\left(-\frac{5}{4}, -\frac{7}{8}\right)$

$$(1) y = x^2 + x - 1$$

頂点 $\left(-\frac{1}{2}, -\frac{5}{4}\right)$

$$(6) y = 2x^2 - x - 3$$

頂点 $\left(\frac{1}{4}, -\frac{25}{8}\right)$

$$(11) y = 4x^2 + 3x - 4$$

頂点 $\left(-\frac{3}{8}, -\frac{73}{16}\right)$

$$(2) y = -4x^2 - 3x - 1$$

頂点 $\left(-\frac{3}{8}, -\frac{7}{16}\right)$

$$(7) y = 2x^2 + 4x - 5$$

頂点 $(-1, -7)$

$$(12) y = 4x^2 - 2x + 2$$

頂点 $\left(\frac{1}{4}, \frac{7}{4}\right)$

$$(3) y = 2x^2 - 3x + 4$$

頂点 $\left(\frac{3}{4}, \frac{23}{8}\right)$

$$(8) y = x^2 + 2x + 5$$

頂点 $(-1, 4)$

$$(13) y = 4x^2 + 2x - 1$$

頂点 $\left(-\frac{1}{4}, -\frac{5}{4}\right)$

$$(4) y = -5x^2 - x - 5$$

頂点 $\left(-\frac{1}{10}, -\frac{99}{20}\right)$

$$(9) y = 5x^2 + x + 2$$

頂点 $\left(-\frac{1}{10}, \frac{39}{20}\right)$

$$(14) y = -3x^2 + 4x + 5$$

頂点 $\left(\frac{2}{3}, \frac{19}{3}\right)$

$$(5) y = x^2 - 5x - 4$$

頂点 $\left(\frac{5}{2}, -\frac{41}{4}\right)$

$$(10) y = -x^2 + 2x - 4$$

頂点 $(1, -3)$

$$(15) y = -2x^2 + 2x + 5$$

頂点 $\left(\frac{1}{2}, \frac{11}{2}\right)$

(1) $y = -x^2 - 4x + 1$
頂点 $(-2, 5)$

(6) $y = x^2 - x + 1$
頂点 $\left(\frac{1}{2}, \frac{3}{4}\right)$

(11) $y = -2x^2 - 5x$
頂点 $\left(-\frac{5}{4}, \frac{25}{8}\right)$

(2) $y = -5x^2 + 3x + 5$
頂点 $\left(\frac{3}{10}, \frac{109}{20}\right)$

(7) $y = -x^2 + 5x$
頂点 $\left(\frac{5}{2}, \frac{25}{4}\right)$

(12) $y = 4x^2 + 2x$
頂点 $\left(-\frac{1}{4}, -\frac{1}{4}\right)$

(3) $y = -3x^2 + 3x - 4$
頂点 $\left(\frac{1}{2}, -\frac{13}{4}\right)$

(8) $y = -x^2 - 2x - 4$
頂点 $(-1, -3)$

(13) $y = -x^2 - 4x + 4$
頂点 $(-2, 8)$

(4) $y = -2x^2 + 4x - 2$
頂点 $(1, 0)$

(9) $y = 5x^2 - 2x + 4$
頂点 $\left(\frac{1}{5}, \frac{19}{5}\right)$

(14) $y = 3x^2 - 3x + 2$
頂点 $\left(\frac{1}{2}, \frac{5}{4}\right)$

(5) $y = 2x^2 - 4x + 4$
頂点 $(1, 2)$

(10) $y = x^2 - 3x + 1$
頂点 $\left(\frac{3}{2}, -\frac{5}{4}\right)$

(15) $y = x^2 + 4x + 4$
頂点 $(-2, 0)$

$$(1) y = -4x^2 - 2x + 3$$

頂点 $\left(-\frac{1}{4}, \frac{13}{4}\right)$

$$(6) y = -4x^2 + x - 1$$

頂点 $\left(\frac{1}{8}, -\frac{15}{16}\right)$

$$(11) y = -3x^2 - 2x - 3$$

頂点 $\left(-\frac{1}{3}, -\frac{8}{3}\right)$

$$(2) y = -2x^2 + 2x + 3$$

頂点 $\left(\frac{1}{2}, \frac{7}{2}\right)$

$$(7) y = 4x^2 - 4x$$

頂点 $\left(\frac{1}{2}, -1\right)$

$$(12) y = x^2 + 2x - 3$$

頂点 $(-1, -4)$

$$(3) y = -x^2 + x + 2$$

頂点 $\left(\frac{1}{2}, \frac{9}{4}\right)$

$$(8) y = x^2 - 3x + 5$$

頂点 $\left(\frac{3}{2}, \frac{11}{4}\right)$

$$(13) y = -x^2 + 3x$$

頂点 $\left(\frac{3}{2}, \frac{9}{4}\right)$

$$(4) y = 5x^2 + 2x + 4$$

頂点 $\left(-\frac{1}{5}, \frac{19}{5}\right)$

$$(9) y = -4x^2 + 2x + 2$$

頂点 $\left(\frac{1}{4}, \frac{9}{4}\right)$

$$(14) y = -2x^2 + 3x + 1$$

頂点 $\left(\frac{3}{4}, \frac{17}{8}\right)$

$$(5) y = 3x^2 + 3x + 4$$

頂点 $\left(-\frac{1}{2}, \frac{13}{4}\right)$

$$(10) y = -2x^2 - 2x - 4$$

頂点 $\left(-\frac{1}{2}, -\frac{7}{2}\right)$

$$(15) y = -2x^2 + x$$

頂点 $\left(\frac{1}{4}, \frac{1}{8}\right)$

$$(1) y = -5x^2 - 5x + 5$$

頂点 $\left(-\frac{1}{2}, \frac{25}{4}\right)$

$$(6) y = 5x^2 + 3x - 5$$

頂点 $\left(-\frac{3}{10}, -\frac{109}{20}\right)$

$$(11) y = -4x^2 - 3x$$

頂点 $\left(-\frac{3}{8}, \frac{9}{16}\right)$

$$(2) y = -3x^2 + 2x + 5$$

頂点 $\left(\frac{1}{3}, \frac{16}{3}\right)$

$$(7) y = -4x^2 + 5x + 1$$

頂点 $\left(\frac{5}{8}, \frac{41}{16}\right)$

$$(12) y = 2x^2 + 2x - 3$$

頂点 $\left(-\frac{1}{2}, -\frac{7}{2}\right)$

$$(3) y = 2x^2 - 3x - 3$$

頂点 $\left(\frac{3}{4}, -\frac{33}{8}\right)$

$$(8) y = -5x^2 - 3x + 4$$

頂点 $\left(-\frac{3}{10}, \frac{89}{20}\right)$

$$(13) y = 3x^2 + 3x - 5$$

頂点 $\left(-\frac{1}{2}, -\frac{23}{4}\right)$

$$(4) y = 2x^2 + 2x - 5$$

頂点 $\left(-\frac{1}{2}, -\frac{11}{2}\right)$

$$(9) y = -4x^2 - 4x - 5$$

頂点 $\left(-\frac{1}{2}, -4\right)$

$$(14) y = 5x^2 - 4x + 3$$

頂点 $\left(\frac{2}{5}, \frac{11}{5}\right)$

$$(5) y = -4x^2 - 5x - 3$$

頂点 $\left(-\frac{5}{8}, -\frac{23}{16}\right)$

$$(10) y = 2x^2 - 4x + 4$$

頂点 $(1, 2)$

$$(15) y = -3x^2 + x + 4$$

頂点 $\left(\frac{1}{6}, \frac{49}{12}\right)$

$$(1) y = -3x^2 - 3x - 4$$

頂点 $\left(-\frac{1}{2}, -\frac{13}{4}\right)$

$$(6) y = 4x^2 - 5x + 3$$

頂点 $\left(\frac{5}{8}, \frac{23}{16}\right)$

$$(11) y = 5x^2 - 5x - 4$$

頂点 $\left(\frac{1}{2}, -\frac{21}{4}\right)$

$$(2) y = 3x^2 + 5x$$

頂点 $\left(-\frac{5}{6}, -\frac{25}{12}\right)$

$$(7) y = -2x^2 - x + 1$$

頂点 $\left(-\frac{1}{4}, \frac{9}{8}\right)$

$$(12) y = x^2 - 5x - 2$$

頂点 $\left(\frac{5}{2}, -\frac{33}{4}\right)$

$$(3) y = 5x^2 - 3x - 3$$

頂点 $\left(\frac{3}{10}, -\frac{69}{20}\right)$

$$(8) y = 4x^2 + 4x + 3$$

頂点 $\left(-\frac{1}{2}, 2\right)$

$$(13) y = -5x^2 + 4x$$

頂点 $\left(\frac{2}{5}, \frac{4}{5}\right)$

$$(4) y = 3x^2 - 2x - 5$$

頂点 $\left(\frac{1}{3}, -\frac{16}{3}\right)$

$$(9) y = 2x^2 + 5x - 3$$

頂点 $\left(-\frac{5}{4}, -\frac{49}{8}\right)$

$$(14) y = 5x^2 + 4x - 2$$

頂点 $\left(-\frac{2}{5}, -\frac{14}{5}\right)$

$$(5) y = 2x^2 - 4x + 5$$

頂点 $(1, 3)$

$$(10) y = 5x^2 + 2x - 2$$

頂点 $\left(-\frac{1}{5}, -\frac{11}{5}\right)$

$$(15) y = -5x^2 - 5x + 2$$

頂点 $\left(-\frac{1}{2}, \frac{13}{4}\right)$

$$(1) y = -2x^2 - 3x + 5$$

頂点 $\left(-\frac{3}{4}, \frac{49}{8}\right)$

$$(6) y = 2x^2 + 3x$$

頂点 $\left(-\frac{3}{4}, -\frac{9}{8}\right)$

$$(11) y = 5x^2 - 5x$$

頂点 $\left(\frac{1}{2}, -\frac{5}{4}\right)$

$$(2) y = -4x^2 + 5x - 2$$

頂点 $\left(\frac{5}{8}, -\frac{7}{16}\right)$

$$(7) y = 2x^2 + x + 4$$

頂点 $\left(-\frac{1}{4}, \frac{31}{8}\right)$

$$(12) y = -x^2 - x + 4$$

頂点 $\left(-\frac{1}{2}, \frac{17}{4}\right)$

$$(3) y = -3x^2 - 3x - 3$$

頂点 $\left(-\frac{1}{2}, -\frac{9}{4}\right)$

$$(8) y = -2x^2 + 2x + 4$$

頂点 $\left(\frac{1}{2}, \frac{9}{2}\right)$

$$(13) y = 5x^2 - 5x + 4$$

頂点 $\left(\frac{1}{2}, \frac{11}{4}\right)$

$$(4) y = 2x^2 + 2x + 5$$

頂点 $\left(-\frac{1}{2}, \frac{9}{2}\right)$

$$(9) y = x^2 + 4x - 3$$

頂点 $(-2, -7)$

$$(14) y = x^2 + 5x - 3$$

頂点 $\left(-\frac{5}{2}, -\frac{37}{4}\right)$

$$(5) y = -4x^2 + 5x - 4$$

頂点 $\left(\frac{5}{8}, -\frac{39}{16}\right)$

$$(10) y = -2x^2 - 5x + 4$$

頂点 $\left(-\frac{5}{4}, \frac{57}{8}\right)$

$$(15) y = -4x^2 - x - 1$$

頂点 $\left(-\frac{1}{8}, -\frac{15}{16}\right)$

3.2 三角関数

3.2.1 三角比 $30^\circ, 45^\circ, 60^\circ$

三角比 $30^\circ, 45^\circ, 60^\circ$ の値。

(1) $\tan 45^\circ$	(16) $\cos 60^\circ$	(31) $\sin 30^\circ$	(46) $\tan 30^\circ$	(61) $\tan 60^\circ$	(76) $\tan 60^\circ$
(2) $\cos 45^\circ$	(17) $\sin 30^\circ$	(32) $\cos 60^\circ$	(47) $\sin 45^\circ$	(62) $\sin 60^\circ$	(77) $\sin 45^\circ$
(3) $\sin 45^\circ$	(18) $\cos 60^\circ$	(33) $\cos 45^\circ$	(48) $\cos 60^\circ$	(63) $\sin 45^\circ$	(78) $\tan 30^\circ$
(4) $\cos 45^\circ$	(19) $\tan 60^\circ$	(34) $\tan 60^\circ$	(49) $\tan 60^\circ$	(64) $\sin 60^\circ$	(79) $\tan 30^\circ$
(5) $\tan 60^\circ$	(20) $\sin 60^\circ$	(35) $\cos 30^\circ$	(50) $\sin 45^\circ$	(65) $\cos 30^\circ$	(80) $\tan 60^\circ$
(6) $\cos 30^\circ$	(21) $\cos 45^\circ$	(36) $\sin 60^\circ$	(51) $\cos 60^\circ$	(66) $\cos 60^\circ$	(81) $\sin 45^\circ$
(7) $\sin 30^\circ$	(22) $\tan 30^\circ$	(37) $\cos 60^\circ$	(52) $\sin 30^\circ$	(67) $\cos 45^\circ$	(82) $\cos 30^\circ$
(8) $\sin 45^\circ$	(23) $\cos 30^\circ$	(38) $\cos 60^\circ$	(53) $\tan 30^\circ$	(68) $\tan 45^\circ$	(83) $\tan 30^\circ$
(9) $\cos 45^\circ$	(24) $\sin 45^\circ$	(39) $\tan 45^\circ$	(54) $\cos 60^\circ$	(69) $\cos 45^\circ$	(84) $\sin 30^\circ$
(10) $\cos 30^\circ$	(25) $\tan 30^\circ$	(40) $\tan 30^\circ$	(55) $\cos 30^\circ$	(70) $\sin 60^\circ$	(85) $\cos 30^\circ$
(11) $\tan 30^\circ$	(26) $\sin 30^\circ$	(41) $\cos 30^\circ$	(56) $\tan 30^\circ$	(71) $\sin 30^\circ$	(86) $\tan 60^\circ$
(12) $\cos 45^\circ$	(27) $\sin 30^\circ$	(42) $\cos 60^\circ$	(57) $\cos 60^\circ$	(72) $\tan 60^\circ$	(87) $\tan 30^\circ$
(13) $\tan 30^\circ$	(28) $\cos 30^\circ$	(43) $\tan 60^\circ$	(58) $\tan 30^\circ$	(73) $\sin 60^\circ$	(88) $\tan 45^\circ$
(14) $\sin 45^\circ$	(29) $\tan 30^\circ$	(44) $\sin 30^\circ$	(59) $\cos 60^\circ$	(74) $\cos 45^\circ$	(89) $\cos 30^\circ$
(15) $\tan 45^\circ$	(30) $\cos 45^\circ$	(45) $\cos 30^\circ$	(60) $\cos 30^\circ$	(75) $\cos 60^\circ$	(90) $\tan 45^\circ$

(1) $\sin 60^\circ$	(16) $\cos 60^\circ$	(31) $\sin 30^\circ$	(46) $\tan 30^\circ$	(61) $\tan 60^\circ$	(76) $\tan 30^\circ$
(2) $\sin 30^\circ$	(17) $\tan 45^\circ$	(32) $\cos 45^\circ$	(47) $\sin 30^\circ$	(62) $\sin 30^\circ$	(77) $\tan 30^\circ$
(3) $\cos 45^\circ$	(18) $\cos 30^\circ$	(33) $\tan 60^\circ$	(48) $\tan 45^\circ$	(63) $\cos 30^\circ$	(78) $\cos 60^\circ$
(4) $\sin 60^\circ$	(19) $\sin 45^\circ$	(34) $\tan 45^\circ$	(49) $\tan 30^\circ$	(64) $\cos 60^\circ$	(79) $\cos 45^\circ$
(5) $\sin 45^\circ$	(20) $\tan 45^\circ$	(35) $\tan 60^\circ$	(50) $\cos 60^\circ$	(65) $\sin 30^\circ$	(80) $\sin 45^\circ$
(6) $\cos 45^\circ$	(21) $\tan 45^\circ$	(36) $\cos 30^\circ$	(51) $\cos 60^\circ$	(66) $\tan 30^\circ$	(81) $\sin 60^\circ$
(7) $\cos 60^\circ$	(22) $\tan 45^\circ$	(37) $\cos 30^\circ$	(52) $\cos 60^\circ$	(67) $\cos 45^\circ$	(82) $\sin 45^\circ$
(8) $\tan 60^\circ$	(23) $\sin 45^\circ$	(38) $\sin 60^\circ$	(53) $\cos 60^\circ$	(68) $\sin 60^\circ$	(83) $\sin 45^\circ$
(9) $\sin 45^\circ$	(24) $\tan 45^\circ$	(39) $\cos 60^\circ$	(54) $\cos 45^\circ$	(69) $\sin 45^\circ$	(84) $\cos 60^\circ$
(10) $\tan 30^\circ$	(25) $\sin 45^\circ$	(40) $\sin 30^\circ$	(55) $\cos 30^\circ$	(70) $\cos 45^\circ$	(85) $\sin 60^\circ$
(11) $\tan 45^\circ$	(26) $\tan 45^\circ$	(41) $\tan 60^\circ$	(56) $\cos 60^\circ$	(71) $\cos 45^\circ$	(86) $\tan 60^\circ$
(12) $\sin 60^\circ$	(27) $\sin 60^\circ$	(42) $\cos 60^\circ$	(57) $\sin 45^\circ$	(72) $\cos 30^\circ$	(87) $\tan 60^\circ$
(13) $\cos 60^\circ$	(28) $\sin 60^\circ$	(43) $\sin 60^\circ$	(58) $\cos 60^\circ$	(73) $\sin 45^\circ$	(88) $\sin 60^\circ$
(14) $\sin 60^\circ$	(29) $\tan 60^\circ$	(44) $\sin 60^\circ$	(59) $\tan 60^\circ$	(74) $\tan 30^\circ$	(89) $\cos 60^\circ$
(15) $\sin 45^\circ$	(30) $\tan 45^\circ$	(45) $\cos 45^\circ$	(60) $\tan 30^\circ$	(75) $\tan 60^\circ$	(90) $\cos 60^\circ$

(1) $\tan 45^\circ$	(16) $\cos 30^\circ$	(31) $\tan 60^\circ$	(46) $\tan 30^\circ$	(61) $\tan 60^\circ$	(76) $\sin 30^\circ$
(2) $\sin 45^\circ$	(17) $\tan 30^\circ$	(32) $\tan 30^\circ$	(47) $\tan 45^\circ$	(62) $\tan 30^\circ$	(77) $\tan 60^\circ$
(3) $\tan 45^\circ$	(18) $\tan 45^\circ$	(33) $\tan 45^\circ$	(48) $\sin 60^\circ$	(63) $\sin 45^\circ$	(78) $\sin 30^\circ$
(4) $\tan 60^\circ$	(19) $\tan 30^\circ$	(34) $\tan 30^\circ$	(49) $\tan 30^\circ$	(64) $\sin 30^\circ$	(79) $\cos 60^\circ$
(5) $\sin 30^\circ$	(20) $\cos 30^\circ$	(35) $\cos 60^\circ$	(50) $\cos 60^\circ$	(65) $\tan 30^\circ$	(80) $\cos 30^\circ$
(6) $\cos 45^\circ$	(21) $\cos 45^\circ$	(36) $\sin 45^\circ$	(51) $\sin 60^\circ$	(66) $\tan 45^\circ$	(81) $\cos 30^\circ$
(7) $\tan 45^\circ$	(22) $\cos 30^\circ$	(37) $\tan 45^\circ$	(52) $\cos 45^\circ$	(67) $\sin 60^\circ$	(82) $\tan 60^\circ$
(8) $\sin 60^\circ$	(23) $\cos 45^\circ$	(38) $\sin 60^\circ$	(53) $\tan 60^\circ$	(68) $\cos 60^\circ$	(83) $\sin 60^\circ$
(9) $\sin 45^\circ$	(24) $\tan 30^\circ$	(39) $\cos 60^\circ$	(54) $\tan 60^\circ$	(69) $\cos 60^\circ$	(84) $\sin 45^\circ$
(10) $\sin 45^\circ$	(25) $\tan 45^\circ$	(40) $\cos 60^\circ$	(55) $\sin 60^\circ$	(70) $\tan 30^\circ$	(85) $\cos 60^\circ$
(11) $\tan 30^\circ$	(26) $\sin 45^\circ$	(41) $\sin 30^\circ$	(56) $\sin 60^\circ$	(71) $\tan 60^\circ$	(86) $\sin 45^\circ$
(12) $\sin 45^\circ$	(27) $\cos 45^\circ$	(42) $\sin 30^\circ$	(57) $\cos 45^\circ$	(72) $\cos 30^\circ$	(87) $\cos 60^\circ$
(13) $\tan 30^\circ$	(28) $\tan 60^\circ$	(43) $\sin 60^\circ$	(58) $\cos 30^\circ$	(73) $\sin 45^\circ$	(88) $\cos 45^\circ$
(14) $\tan 60^\circ$	(29) $\tan 45^\circ$	(44) $\cos 30^\circ$	(59) $\cos 60^\circ$	(74) $\tan 45^\circ$	(89) $\sin 30^\circ$
(15) $\cos 60^\circ$	(30) $\sin 30^\circ$	(45) $\tan 30^\circ$	(60) $\tan 60^\circ$	(75) $\tan 30^\circ$	(90) $\cos 60^\circ$

(1) $\sin 60^\circ$	(16) $\sin 45^\circ$	(31) $\cos 60^\circ$	(46) $\sin 30^\circ$	(61) $\cos 30^\circ$	(76) $\sin 30^\circ$
(2) $\cos 30^\circ$	(17) $\cos 60^\circ$	(32) $\sin 45^\circ$	(47) $\tan 45^\circ$	(62) $\tan 60^\circ$	(77) $\cos 45^\circ$
(3) $\cos 60^\circ$	(18) $\tan 60^\circ$	(33) $\tan 30^\circ$	(48) $\cos 30^\circ$	(63) $\sin 30^\circ$	(78) $\tan 45^\circ$
(4) $\cos 30^\circ$	(19) $\sin 30^\circ$	(34) $\cos 60^\circ$	(49) $\cos 45^\circ$	(64) $\sin 30^\circ$	(79) $\sin 60^\circ$
(5) $\tan 45^\circ$	(20) $\tan 60^\circ$	(35) $\cos 30^\circ$	(50) $\tan 45^\circ$	(65) $\sin 30^\circ$	(80) $\tan 60^\circ$
(6) $\cos 30^\circ$	(21) $\sin 60^\circ$	(36) $\tan 45^\circ$	(51) $\sin 30^\circ$	(66) $\tan 30^\circ$	(81) $\tan 45^\circ$
(7) $\cos 30^\circ$	(22) $\cos 45^\circ$	(37) $\tan 60^\circ$	(52) $\sin 30^\circ$	(67) $\cos 60^\circ$	(82) $\cos 45^\circ$
(8) $\cos 60^\circ$	(23) $\cos 60^\circ$	(38) $\tan 60^\circ$	(53) $\cos 60^\circ$	(68) $\tan 60^\circ$	(83) $\tan 45^\circ$
(9) $\tan 60^\circ$	(24) $\sin 45^\circ$	(39) $\sin 45^\circ$	(54) $\cos 45^\circ$	(69) $\tan 30^\circ$	(84) $\sin 30^\circ$
(10) $\sin 30^\circ$	(25) $\cos 30^\circ$	(40) $\sin 30^\circ$	(55) $\sin 45^\circ$	(70) $\tan 30^\circ$	(85) $\sin 60^\circ$
(11) $\cos 45^\circ$	(26) $\tan 30^\circ$	(41) $\tan 60^\circ$	(56) $\tan 60^\circ$	(71) $\tan 60^\circ$	(86) $\tan 30^\circ$
(12) $\tan 30^\circ$	(27) $\cos 45^\circ$	(42) $\sin 60^\circ$	(57) $\sin 30^\circ$	(72) $\tan 30^\circ$	(87) $\sin 60^\circ$
(13) $\sin 45^\circ$	(28) $\cos 45^\circ$	(43) $\cos 60^\circ$	(58) $\sin 60^\circ$	(73) $\sin 45^\circ$	(88) $\tan 45^\circ$
(14) $\sin 60^\circ$	(29) $\cos 60^\circ$	(44) $\tan 45^\circ$	(59) $\cos 60^\circ$	(74) $\tan 45^\circ$	(89) $\tan 45^\circ$
(15) $\tan 30^\circ$	(30) $\sin 60^\circ$	(45) $\tan 45^\circ$	(60) $\cos 60^\circ$	(75) $\cos 60^\circ$	(90) $\cos 30^\circ$

(1) $\sin 60^\circ$	(16) $\sin 30^\circ$	(31) $\sin 60^\circ$	(46) $\sin 60^\circ$	(61) $\tan 45^\circ$	(76) $\tan 30^\circ$
(2) $\cos 60^\circ$	(17) $\tan 30^\circ$	(32) $\cos 30^\circ$	(47) $\sin 45^\circ$	(62) $\cos 45^\circ$	(77) $\cos 30^\circ$
(3) $\sin 60^\circ$	(18) $\cos 45^\circ$	(33) $\sin 60^\circ$	(48) $\tan 30^\circ$	(63) $\cos 60^\circ$	(78) $\tan 60^\circ$
(4) $\tan 30^\circ$	(19) $\cos 30^\circ$	(34) $\sin 30^\circ$	(49) $\tan 60^\circ$	(64) $\cos 30^\circ$	(79) $\sin 30^\circ$
(5) $\sin 30^\circ$	(20) $\tan 60^\circ$	(35) $\tan 30^\circ$	(50) $\cos 30^\circ$	(65) $\tan 45^\circ$	(80) $\tan 30^\circ$
(6) $\tan 45^\circ$	(21) $\tan 60^\circ$	(36) $\sin 45^\circ$	(51) $\tan 60^\circ$	(66) $\tan 45^\circ$	(81) $\tan 30^\circ$
(7) $\tan 45^\circ$	(22) $\tan 30^\circ$	(37) $\cos 30^\circ$	(52) $\sin 30^\circ$	(67) $\tan 45^\circ$	(82) $\tan 30^\circ$
(8) $\tan 60^\circ$	(23) $\cos 30^\circ$	(38) $\tan 30^\circ$	(53) $\tan 30^\circ$	(68) $\cos 45^\circ$	(83) $\sin 30^\circ$
(9) $\cos 45^\circ$	(24) $\tan 45^\circ$	(39) $\sin 45^\circ$	(54) $\cos 60^\circ$	(69) $\tan 60^\circ$	(84) $\tan 30^\circ$
(10) $\tan 45^\circ$	(25) $\cos 30^\circ$	(40) $\cos 60^\circ$	(55) $\tan 30^\circ$	(70) $\tan 45^\circ$	(85) $\sin 45^\circ$
(11) $\cos 30^\circ$	(26) $\cos 45^\circ$	(41) $\sin 60^\circ$	(56) $\sin 45^\circ$	(71) $\tan 60^\circ$	(86) $\tan 30^\circ$
(12) $\tan 60^\circ$	(27) $\tan 30^\circ$	(42) $\cos 45^\circ$	(57) $\cos 45^\circ$	(72) $\tan 60^\circ$	(87) $\tan 60^\circ$
(13) $\sin 30^\circ$	(28) $\sin 60^\circ$	(43) $\cos 45^\circ$	(58) $\cos 30^\circ$	(73) $\sin 30^\circ$	(88) $\cos 60^\circ$
(14) $\sin 30^\circ$	(29) $\cos 45^\circ$	(44) $\sin 60^\circ$	(59) $\cos 45^\circ$	(74) $\tan 45^\circ$	(89) $\sin 45^\circ$
(15) $\cos 30^\circ$	(30) $\cos 45^\circ$	(45) $\sin 30^\circ$	(60) $\cos 60^\circ$	(75) $\tan 60^\circ$	(90) $\tan 45^\circ$

(1) $\cos 45^\circ$	(16) $\tan 30^\circ$	(31) $\sin 45^\circ$	(46) $\tan 45^\circ$	(61) $\cos 30^\circ$	(76) $\tan 60^\circ$
(2) $\tan 30^\circ$	(17) $\cos 45^\circ$	(32) $\cos 30^\circ$	(47) $\sin 30^\circ$	(62) $\cos 45^\circ$	(77) $\sin 45^\circ$
(3) $\sin 30^\circ$	(18) $\sin 30^\circ$	(33) $\sin 60^\circ$	(48) $\tan 45^\circ$	(63) $\sin 45^\circ$	(78) $\tan 45^\circ$
(4) $\cos 60^\circ$	(19) $\sin 60^\circ$	(34) $\tan 45^\circ$	(49) $\tan 30^\circ$	(64) $\tan 60^\circ$	(79) $\sin 30^\circ$
(5) $\cos 45^\circ$	(20) $\tan 30^\circ$	(35) $\sin 45^\circ$	(50) $\tan 60^\circ$	(65) $\sin 30^\circ$	(80) $\cos 60^\circ$
(6) $\sin 45^\circ$	(21) $\cos 30^\circ$	(36) $\cos 30^\circ$	(51) $\cos 60^\circ$	(66) $\sin 45^\circ$	(81) $\cos 30^\circ$
(7) $\sin 45^\circ$	(22) $\cos 30^\circ$	(37) $\tan 60^\circ$	(52) $\tan 30^\circ$	(67) $\cos 30^\circ$	(82) $\cos 45^\circ$
(8) $\tan 30^\circ$	(23) $\tan 60^\circ$	(38) $\tan 60^\circ$	(53) $\tan 45^\circ$	(68) $\tan 60^\circ$	(83) $\cos 30^\circ$
(9) $\cos 60^\circ$	(24) $\sin 45^\circ$	(39) $\cos 45^\circ$	(54) $\sin 60^\circ$	(69) $\sin 30^\circ$	(84) $\tan 30^\circ$
(10) $\cos 45^\circ$	(25) $\tan 45^\circ$	(40) $\sin 30^\circ$	(55) $\tan 45^\circ$	(70) $\cos 60^\circ$	(85) $\sin 60^\circ$
(11) $\sin 30^\circ$	(26) $\cos 60^\circ$	(41) $\tan 45^\circ$	(56) $\tan 45^\circ$	(71) $\sin 45^\circ$	(86) $\sin 60^\circ$
(12) $\cos 45^\circ$	(27) $\tan 30^\circ$	(42) $\sin 30^\circ$	(57) $\cos 60^\circ$	(72) $\sin 60^\circ$	(87) $\tan 45^\circ$
(13) $\cos 60^\circ$	(28) $\cos 45^\circ$	(43) $\tan 30^\circ$	(58) $\cos 30^\circ$	(73) $\tan 60^\circ$	(88) $\tan 45^\circ$
(14) $\sin 45^\circ$	(29) $\tan 60^\circ$	(44) $\tan 30^\circ$	(59) $\sin 60^\circ$	(74) $\sin 30^\circ$	(89) $\cos 60^\circ$
(15) $\tan 60^\circ$	(30) $\cos 30^\circ$	(45) $\sin 45^\circ$	(60) $\cos 45^\circ$	(75) $\sin 30^\circ$	(90) $\sin 60^\circ$

(1) $\sin 30^\circ$	(16) $\cos 60^\circ$	(31) $\cos 45^\circ$	(46) $\sin 60^\circ$	(61) $\tan 30^\circ$	(76) $\tan 45^\circ$
(2) $\cos 30^\circ$	(17) $\sin 45^\circ$	(32) $\cos 60^\circ$	(47) $\cos 45^\circ$	(62) $\tan 30^\circ$	(77) $\sin 45^\circ$
(3) $\cos 45^\circ$	(18) $\tan 45^\circ$	(33) $\cos 45^\circ$	(48) $\sin 60^\circ$	(63) $\sin 60^\circ$	(78) $\tan 60^\circ$
(4) $\cos 60^\circ$	(19) $\cos 45^\circ$	(34) $\sin 45^\circ$	(49) $\sin 30^\circ$	(64) $\tan 60^\circ$	(79) $\cos 60^\circ$
(5) $\tan 60^\circ$	(20) $\cos 60^\circ$	(35) $\cos 60^\circ$	(50) $\sin 45^\circ$	(65) $\tan 60^\circ$	(80) $\cos 30^\circ$
(6) $\sin 45^\circ$	(21) $\sin 60^\circ$	(36) $\tan 60^\circ$	(51) $\cos 60^\circ$	(66) $\cos 60^\circ$	(81) $\sin 45^\circ$
(7) $\tan 60^\circ$	(22) $\sin 60^\circ$	(37) $\cos 30^\circ$	(52) $\cos 60^\circ$	(67) $\sin 60^\circ$	(82) $\sin 30^\circ$
(8) $\tan 30^\circ$	(23) $\cos 45^\circ$	(38) $\sin 60^\circ$	(53) $\sin 30^\circ$	(68) $\sin 45^\circ$	(83) $\cos 30^\circ$
(9) $\tan 45^\circ$	(24) $\tan 30^\circ$	(39) $\sin 60^\circ$	(54) $\cos 60^\circ$	(69) $\cos 45^\circ$	(84) $\cos 60^\circ$
(10) $\sin 60^\circ$	(25) $\tan 30^\circ$	(40) $\tan 45^\circ$	(55) $\cos 45^\circ$	(70) $\cos 60^\circ$	(85) $\tan 45^\circ$
(11) $\tan 60^\circ$	(26) $\cos 45^\circ$	(41) $\cos 30^\circ$	(56) $\cos 30^\circ$	(71) $\sin 30^\circ$	(86) $\tan 60^\circ$
(12) $\tan 60^\circ$	(27) $\tan 30^\circ$	(42) $\cos 45^\circ$	(57) $\tan 60^\circ$	(72) $\sin 30^\circ$	(87) $\cos 45^\circ$
(13) $\tan 30^\circ$	(28) $\sin 45^\circ$	(43) $\sin 45^\circ$	(58) $\cos 30^\circ$	(73) $\cos 30^\circ$	(88) $\cos 45^\circ$
(14) $\cos 30^\circ$	(29) $\sin 60^\circ$	(44) $\sin 60^\circ$	(59) $\cos 60^\circ$	(74) $\cos 45^\circ$	(89) $\tan 30^\circ$
(15) $\tan 60^\circ$	(30) $\cos 30^\circ$	(45) $\sin 60^\circ$	(60) $\tan 60^\circ$	(75) $\tan 30^\circ$	(90) $\cos 30^\circ$

(1) $\cos 30^\circ$	(16) $\tan 30^\circ$	(31) $\tan 45^\circ$	(46) $\cos 30^\circ$	(61) $\cos 30^\circ$	(76) $\tan 60^\circ$
(2) $\sin 30^\circ$	(17) $\tan 45^\circ$	(32) $\tan 30^\circ$	(47) $\cos 30^\circ$	(62) $\tan 45^\circ$	(77) $\tan 60^\circ$
(3) $\tan 45^\circ$	(18) $\cos 45^\circ$	(33) $\cos 45^\circ$	(48) $\sin 30^\circ$	(63) $\cos 30^\circ$	(78) $\cos 60^\circ$
(4) $\cos 45^\circ$	(19) $\cos 30^\circ$	(34) $\tan 45^\circ$	(49) $\cos 30^\circ$	(64) $\sin 45^\circ$	(79) $\sin 45^\circ$
(5) $\cos 45^\circ$	(20) $\tan 60^\circ$	(35) $\cos 30^\circ$	(50) $\sin 60^\circ$	(65) $\sin 60^\circ$	(80) $\tan 45^\circ$
(6) $\tan 45^\circ$	(21) $\sin 30^\circ$	(36) $\tan 60^\circ$	(51) $\cos 30^\circ$	(66) $\sin 45^\circ$	(81) $\sin 45^\circ$
(7) $\sin 30^\circ$	(22) $\tan 30^\circ$	(37) $\tan 45^\circ$	(52) $\sin 60^\circ$	(67) $\cos 45^\circ$	(82) $\sin 60^\circ$
(8) $\tan 45^\circ$	(23) $\tan 60^\circ$	(38) $\cos 30^\circ$	(53) $\tan 60^\circ$	(68) $\cos 30^\circ$	(83) $\sin 45^\circ$
(9) $\cos 30^\circ$	(24) $\sin 45^\circ$	(39) $\cos 45^\circ$	(54) $\tan 60^\circ$	(69) $\cos 60^\circ$	(84) $\tan 30^\circ$
(10) $\tan 60^\circ$	(25) $\cos 60^\circ$	(40) $\sin 45^\circ$	(55) $\tan 45^\circ$	(70) $\sin 45^\circ$	(85) $\tan 60^\circ$
(11) $\cos 30^\circ$	(26) $\cos 60^\circ$	(41) $\tan 60^\circ$	(56) $\tan 45^\circ$	(71) $\cos 30^\circ$	(86) $\sin 60^\circ$
(12) $\sin 30^\circ$	(27) $\cos 30^\circ$	(42) $\tan 45^\circ$	(57) $\cos 45^\circ$	(72) $\tan 30^\circ$	(87) $\cos 60^\circ$
(13) $\tan 60^\circ$	(28) $\tan 30^\circ$	(43) $\tan 60^\circ$	(58) $\cos 30^\circ$	(73) $\sin 60^\circ$	(88) $\sin 60^\circ$
(14) $\cos 30^\circ$	(29) $\tan 30^\circ$	(44) $\cos 45^\circ$	(59) $\cos 45^\circ$	(74) $\tan 45^\circ$	(89) $\sin 45^\circ$
(15) $\tan 45^\circ$	(30) $\tan 30^\circ$	(45) $\sin 45^\circ$	(60) $\sin 30^\circ$	(75) $\tan 60^\circ$	(90) $\tan 45^\circ$

(1) $\sin 45^\circ$	(16) $\cos 45^\circ$	(31) $\cos 45^\circ$	(46) $\cos 30^\circ$	(61) $\cos 60^\circ$	(76) $\sin 60^\circ$
(2) $\sin 45^\circ$	(17) $\cos 30^\circ$	(32) $\sin 45^\circ$	(47) $\cos 45^\circ$	(62) $\cos 45^\circ$	(77) $\sin 30^\circ$
(3) $\sin 30^\circ$	(18) $\cos 45^\circ$	(33) $\tan 45^\circ$	(48) $\tan 30^\circ$	(63) $\cos 60^\circ$	(78) $\tan 60^\circ$
(4) $\cos 30^\circ$	(19) $\sin 60^\circ$	(34) $\tan 30^\circ$	(49) $\tan 45^\circ$	(64) $\tan 30^\circ$	(79) $\tan 60^\circ$
(5) $\tan 60^\circ$	(20) $\tan 30^\circ$	(35) $\cos 60^\circ$	(50) $\cos 60^\circ$	(65) $\cos 45^\circ$	(80) $\tan 45^\circ$
(6) $\cos 60^\circ$	(21) $\cos 60^\circ$	(36) $\cos 45^\circ$	(51) $\tan 60^\circ$	(66) $\tan 60^\circ$	(81) $\tan 60^\circ$
(7) $\cos 30^\circ$	(22) $\sin 30^\circ$	(37) $\cos 30^\circ$	(52) $\tan 30^\circ$	(67) $\tan 30^\circ$	(82) $\tan 60^\circ$
(8) $\cos 30^\circ$	(23) $\tan 60^\circ$	(38) $\sin 30^\circ$	(53) $\tan 45^\circ$	(68) $\tan 45^\circ$	(83) $\cos 60^\circ$
(9) $\sin 60^\circ$	(24) $\tan 45^\circ$	(39) $\sin 45^\circ$	(54) $\cos 45^\circ$	(69) $\tan 45^\circ$	(84) $\tan 30^\circ$
(10) $\cos 60^\circ$	(25) $\tan 30^\circ$	(40) $\cos 45^\circ$	(55) $\cos 45^\circ$	(70) $\sin 45^\circ$	(85) $\cos 30^\circ$
(11) $\sin 30^\circ$	(26) $\sin 60^\circ$	(41) $\sin 30^\circ$	(56) $\sin 30^\circ$	(71) $\cos 30^\circ$	(86) $\cos 45^\circ$
(12) $\cos 30^\circ$	(27) $\tan 45^\circ$	(42) $\tan 60^\circ$	(57) $\tan 30^\circ$	(72) $\cos 45^\circ$	(87) $\cos 45^\circ$
(13) $\tan 30^\circ$	(28) $\cos 45^\circ$	(43) $\sin 45^\circ$	(58) $\tan 30^\circ$	(73) $\tan 60^\circ$	(88) $\sin 30^\circ$
(14) $\tan 30^\circ$	(29) $\tan 45^\circ$	(44) $\tan 30^\circ$	(59) $\tan 30^\circ$	(74) $\tan 60^\circ$	(89) $\sin 30^\circ$
(15) $\sin 45^\circ$	(30) $\tan 30^\circ$	(45) $\tan 60^\circ$	(60) $\tan 30^\circ$	(75) $\cos 45^\circ$	(90) $\sin 30^\circ$

(1) $\sin 45^\circ$	(16) $\sin 45^\circ$	(31) $\cos 60^\circ$	(46) $\sin 60^\circ$	(61) $\sin 30^\circ$	(76) $\sin 45^\circ$
(2) $\tan 60^\circ$	(17) $\tan 30^\circ$	(32) $\sin 45^\circ$	(47) $\cos 30^\circ$	(62) $\tan 30^\circ$	(77) $\cos 45^\circ$
(3) $\tan 45^\circ$	(18) $\sin 60^\circ$	(33) $\sin 30^\circ$	(48) $\cos 30^\circ$	(63) $\tan 30^\circ$	(78) $\cos 30^\circ$
(4) $\tan 45^\circ$	(19) $\sin 30^\circ$	(34) $\sin 30^\circ$	(49) $\tan 60^\circ$	(64) $\tan 45^\circ$	(79) $\tan 60^\circ$
(5) $\cos 60^\circ$	(20) $\tan 45^\circ$	(35) $\cos 60^\circ$	(50) $\tan 30^\circ$	(65) $\cos 45^\circ$	(80) $\sin 30^\circ$
(6) $\sin 45^\circ$	(21) $\sin 30^\circ$	(36) $\cos 60^\circ$	(51) $\cos 45^\circ$	(66) $\tan 60^\circ$	(81) $\cos 30^\circ$
(7) $\tan 45^\circ$	(22) $\cos 45^\circ$	(37) $\cos 45^\circ$	(52) $\sin 30^\circ$	(67) $\tan 45^\circ$	(82) $\tan 60^\circ$
(8) $\tan 30^\circ$	(23) $\cos 60^\circ$	(38) $\sin 30^\circ$	(53) $\tan 60^\circ$	(68) $\tan 30^\circ$	(83) $\sin 30^\circ$
(9) $\cos 30^\circ$	(24) $\cos 30^\circ$	(39) $\sin 45^\circ$	(54) $\cos 60^\circ$	(69) $\sin 45^\circ$	(84) $\tan 60^\circ$
(10) $\cos 45^\circ$	(25) $\tan 30^\circ$	(40) $\cos 45^\circ$	(55) $\sin 45^\circ$	(70) $\tan 30^\circ$	(85) $\tan 45^\circ$
(11) $\tan 30^\circ$	(26) $\cos 60^\circ$	(41) $\sin 45^\circ$	(56) $\tan 60^\circ$	(71) $\tan 30^\circ$	(86) $\sin 45^\circ$
(12) $\cos 30^\circ$	(27) $\tan 60^\circ$	(42) $\tan 45^\circ$	(57) $\cos 30^\circ$	(72) $\sin 30^\circ$	(87) $\tan 60^\circ$
(13) $\cos 45^\circ$	(28) $\cos 60^\circ$	(43) $\sin 60^\circ$	(58) $\sin 60^\circ$	(73) $\cos 30^\circ$	(88) $\tan 30^\circ$
(14) $\sin 45^\circ$	(29) $\tan 45^\circ$	(44) $\sin 30^\circ$	(59) $\tan 30^\circ$	(74) $\tan 30^\circ$	(89) $\tan 45^\circ$
(15) $\tan 45^\circ$	(30) $\cos 60^\circ$	(45) $\sin 60^\circ$	(60) $\cos 30^\circ$	(75) $\cos 60^\circ$	(90) $\cos 45^\circ$

(1) $\sin 30^\circ$	(16) $\cos 60^\circ$	(31) $\cos 45^\circ$	(46) $\tan 30^\circ$	(61) $\tan 30^\circ$	(76) $\tan 30^\circ$
(2) $\cos 60^\circ$	(17) $\tan 60^\circ$	(32) $\cos 45^\circ$	(47) $\cos 60^\circ$	(62) $\cos 45^\circ$	(77) $\sin 60^\circ$
(3) $\sin 30^\circ$	(18) $\tan 45^\circ$	(33) $\cos 60^\circ$	(48) $\tan 60^\circ$	(63) $\tan 60^\circ$	(78) $\sin 30^\circ$
(4) $\cos 60^\circ$	(19) $\sin 60^\circ$	(34) $\cos 60^\circ$	(49) $\sin 30^\circ$	(64) $\cos 60^\circ$	(79) $\cos 60^\circ$
(5) $\tan 45^\circ$	(20) $\tan 30^\circ$	(35) $\sin 45^\circ$	(50) $\sin 45^\circ$	(65) $\sin 60^\circ$	(80) $\cos 45^\circ$
(6) $\tan 60^\circ$	(21) $\tan 30^\circ$	(36) $\tan 45^\circ$	(51) $\tan 60^\circ$	(66) $\cos 30^\circ$	(81) $\sin 45^\circ$
(7) $\cos 45^\circ$	(22) $\sin 30^\circ$	(37) $\cos 60^\circ$	(52) $\cos 45^\circ$	(67) $\cos 45^\circ$	(82) $\tan 45^\circ$
(8) $\sin 45^\circ$	(23) $\cos 60^\circ$	(38) $\sin 60^\circ$	(53) $\sin 30^\circ$	(68) $\sin 60^\circ$	(83) $\cos 45^\circ$
(9) $\sin 60^\circ$	(24) $\sin 60^\circ$	(39) $\sin 30^\circ$	(54) $\sin 45^\circ$	(69) $\sin 45^\circ$	(84) $\sin 30^\circ$
(10) $\cos 45^\circ$	(25) $\tan 45^\circ$	(40) $\sin 45^\circ$	(55) $\tan 60^\circ$	(70) $\cos 45^\circ$	(85) $\cos 45^\circ$
(11) $\sin 60^\circ$	(26) $\sin 60^\circ$	(41) $\tan 30^\circ$	(56) $\sin 45^\circ$	(71) $\tan 45^\circ$	(86) $\sin 60^\circ$
(12) $\sin 30^\circ$	(27) $\tan 45^\circ$	(42) $\cos 60^\circ$	(57) $\sin 30^\circ$	(72) $\tan 60^\circ$	(87) $\sin 60^\circ$
(13) $\sin 60^\circ$	(28) $\sin 30^\circ$	(43) $\cos 45^\circ$	(58) $\sin 30^\circ$	(73) $\tan 60^\circ$	(88) $\sin 30^\circ$
(14) $\cos 60^\circ$	(29) $\tan 60^\circ$	(44) $\cos 30^\circ$	(59) $\tan 60^\circ$	(74) $\sin 45^\circ$	(89) $\cos 60^\circ$
(15) $\tan 45^\circ$	(30) $\sin 30^\circ$	(45) $\tan 60^\circ$	(60) $\sin 60^\circ$	(75) $\tan 60^\circ$	(90) $\sin 30^\circ$

(1) $\tan 60^\circ$	(16) $\cos 30^\circ$	(31) $\tan 30^\circ$	(46) $\sin 60^\circ$	(61) $\sin 45^\circ$	(76) $\sin 45^\circ$
(2) $\sin 60^\circ$	(17) $\tan 30^\circ$	(32) $\tan 45^\circ$	(47) $\cos 60^\circ$	(62) $\tan 30^\circ$	(77) $\tan 30^\circ$
(3) $\cos 30^\circ$	(18) $\sin 45^\circ$	(33) $\sin 45^\circ$	(48) $\cos 30^\circ$	(63) $\tan 60^\circ$	(78) $\sin 60^\circ$
(4) $\tan 30^\circ$	(19) $\tan 30^\circ$	(34) $\tan 60^\circ$	(49) $\cos 30^\circ$	(64) $\sin 30^\circ$	(79) $\tan 60^\circ$
(5) $\tan 30^\circ$	(20) $\sin 60^\circ$	(35) $\sin 30^\circ$	(50) $\tan 30^\circ$	(65) $\sin 30^\circ$	(80) $\tan 60^\circ$
(6) $\tan 45^\circ$	(21) $\sin 45^\circ$	(36) $\cos 45^\circ$	(51) $\sin 45^\circ$	(66) $\sin 60^\circ$	(81) $\cos 60^\circ$
(7) $\tan 60^\circ$	(22) $\tan 30^\circ$	(37) $\sin 30^\circ$	(52) $\tan 60^\circ$	(67) $\sin 45^\circ$	(82) $\tan 45^\circ$
(8) $\sin 30^\circ$	(23) $\sin 30^\circ$	(38) $\cos 45^\circ$	(53) $\cos 60^\circ$	(68) $\cos 30^\circ$	(83) $\sin 45^\circ$
(9) $\cos 60^\circ$	(24) $\sin 45^\circ$	(39) $\cos 45^\circ$	(54) $\sin 45^\circ$	(69) $\tan 30^\circ$	(84) $\sin 45^\circ$
(10) $\cos 30^\circ$	(25) $\tan 45^\circ$	(40) $\tan 45^\circ$	(55) $\sin 45^\circ$	(70) $\cos 30^\circ$	(85) $\cos 45^\circ$
(11) $\tan 45^\circ$	(26) $\sin 60^\circ$	(41) $\cos 30^\circ$	(56) $\sin 30^\circ$	(71) $\sin 45^\circ$	(86) $\sin 60^\circ$
(12) $\sin 45^\circ$	(27) $\sin 30^\circ$	(42) $\sin 45^\circ$	(57) $\sin 30^\circ$	(72) $\tan 30^\circ$	(87) $\tan 30^\circ$
(13) $\cos 60^\circ$	(28) $\cos 60^\circ$	(43) $\tan 60^\circ$	(58) $\cos 30^\circ$	(73) $\sin 60^\circ$	(88) $\sin 60^\circ$
(14) $\tan 60^\circ$	(29) $\sin 30^\circ$	(44) $\cos 30^\circ$	(59) $\tan 45^\circ$	(74) $\sin 30^\circ$	(89) $\cos 45^\circ$
(15) $\tan 60^\circ$	(30) $\cos 45^\circ$	(45) $\sin 60^\circ$	(60) $\cos 30^\circ$	(75) $\cos 30^\circ$	(90) $\cos 30^\circ$

(1) $\tan 60^\circ$	(16) $\sin 45^\circ$	(31) $\tan 60^\circ$	(46) $\tan 45^\circ$	(61) $\tan 60^\circ$	(76) $\sin 30^\circ$
(2) $\sin 30^\circ$	(17) $\tan 45^\circ$	(32) $\tan 30^\circ$	(47) $\sin 60^\circ$	(62) $\tan 45^\circ$	(77) $\cos 30^\circ$
(3) $\sin 45^\circ$	(18) $\cos 60^\circ$	(33) $\sin 60^\circ$	(48) $\tan 30^\circ$	(63) $\tan 60^\circ$	(78) $\tan 60^\circ$
(4) $\cos 60^\circ$	(19) $\tan 60^\circ$	(34) $\cos 45^\circ$	(49) $\sin 30^\circ$	(64) $\sin 45^\circ$	(79) $\tan 30^\circ$
(5) $\cos 45^\circ$	(20) $\tan 45^\circ$	(35) $\sin 60^\circ$	(50) $\sin 60^\circ$	(65) $\sin 60^\circ$	(80) $\tan 60^\circ$
(6) $\tan 30^\circ$	(21) $\tan 60^\circ$	(36) $\tan 45^\circ$	(51) $\tan 60^\circ$	(66) $\cos 60^\circ$	(81) $\tan 30^\circ$
(7) $\tan 30^\circ$	(22) $\sin 60^\circ$	(37) $\tan 60^\circ$	(52) $\cos 45^\circ$	(67) $\cos 60^\circ$	(82) $\tan 60^\circ$
(8) $\sin 45^\circ$	(23) $\cos 45^\circ$	(38) $\cos 60^\circ$	(53) $\cos 30^\circ$	(68) $\sin 45^\circ$	(83) $\cos 30^\circ$
(9) $\tan 45^\circ$	(24) $\tan 60^\circ$	(39) $\cos 60^\circ$	(54) $\tan 60^\circ$	(69) $\cos 30^\circ$	(84) $\cos 30^\circ$
(10) $\sin 45^\circ$	(25) $\cos 60^\circ$	(40) $\sin 60^\circ$	(55) $\tan 45^\circ$	(70) $\tan 60^\circ$	(85) $\sin 60^\circ$
(11) $\tan 60^\circ$	(26) $\cos 30^\circ$	(41) $\sin 45^\circ$	(56) $\tan 30^\circ$	(71) $\tan 30^\circ$	(86) $\cos 30^\circ$
(12) $\sin 45^\circ$	(27) $\sin 30^\circ$	(42) $\cos 60^\circ$	(57) $\tan 30^\circ$	(72) $\sin 45^\circ$	(87) $\cos 60^\circ$
(13) $\tan 60^\circ$	(28) $\tan 60^\circ$	(43) $\tan 30^\circ$	(58) $\tan 30^\circ$	(73) $\tan 45^\circ$	(88) $\sin 30^\circ$
(14) $\sin 60^\circ$	(29) $\cos 45^\circ$	(44) $\cos 60^\circ$	(59) $\sin 30^\circ$	(74) $\cos 60^\circ$	(89) $\sin 60^\circ$
(15) $\sin 30^\circ$	(30) $\cos 60^\circ$	(45) $\tan 30^\circ$	(60) $\tan 30^\circ$	(75) $\sin 60^\circ$	(90) $\tan 30^\circ$

(1) $\sin 60^\circ$	(16) $\cos 30^\circ$	(31) $\cos 30^\circ$	(46) $\cos 60^\circ$	(61) $\cos 30^\circ$	(76) $\tan 60^\circ$
(2) $\tan 60^\circ$	(17) $\cos 60^\circ$	(32) $\tan 45^\circ$	(47) $\sin 45^\circ$	(62) $\sin 60^\circ$	(77) $\tan 60^\circ$
(3) $\cos 45^\circ$	(18) $\sin 30^\circ$	(33) $\tan 60^\circ$	(48) $\cos 60^\circ$	(63) $\cos 60^\circ$	(78) $\cos 45^\circ$
(4) $\tan 60^\circ$	(19) $\cos 60^\circ$	(34) $\sin 60^\circ$	(49) $\sin 60^\circ$	(64) $\cos 60^\circ$	(79) $\tan 30^\circ$
(5) $\sin 30^\circ$	(20) $\cos 45^\circ$	(35) $\tan 45^\circ$	(50) $\sin 60^\circ$	(65) $\sin 30^\circ$	(80) $\sin 30^\circ$
(6) $\tan 30^\circ$	(21) $\cos 45^\circ$	(36) $\sin 60^\circ$	(51) $\cos 60^\circ$	(66) $\sin 60^\circ$	(81) $\tan 60^\circ$
(7) $\sin 30^\circ$	(22) $\tan 45^\circ$	(37) $\cos 60^\circ$	(52) $\cos 60^\circ$	(67) $\cos 60^\circ$	(82) $\tan 60^\circ$
(8) $\tan 45^\circ$	(23) $\tan 45^\circ$	(38) $\tan 60^\circ$	(53) $\sin 45^\circ$	(68) $\tan 60^\circ$	(83) $\cos 60^\circ$
(9) $\tan 30^\circ$	(24) $\sin 45^\circ$	(39) $\sin 45^\circ$	(54) $\tan 30^\circ$	(69) $\sin 60^\circ$	(84) $\tan 45^\circ$
(10) $\tan 45^\circ$	(25) $\tan 60^\circ$	(40) $\sin 30^\circ$	(55) $\cos 30^\circ$	(70) $\cos 30^\circ$	(85) $\cos 60^\circ$
(11) $\cos 45^\circ$	(26) $\tan 60^\circ$	(41) $\tan 60^\circ$	(56) $\cos 45^\circ$	(71) $\cos 45^\circ$	(86) $\tan 60^\circ$
(12) $\tan 45^\circ$	(27) $\cos 60^\circ$	(42) $\cos 30^\circ$	(57) $\sin 30^\circ$	(72) $\cos 45^\circ$	(87) $\cos 45^\circ$
(13) $\tan 60^\circ$	(28) $\tan 60^\circ$	(43) $\cos 45^\circ$	(58) $\tan 60^\circ$	(73) $\sin 30^\circ$	(88) $\sin 45^\circ$
(14) $\cos 45^\circ$	(29) $\cos 60^\circ$	(44) $\cos 60^\circ$	(59) $\tan 45^\circ$	(74) $\sin 45^\circ$	(89) $\tan 60^\circ$
(15) $\cos 45^\circ$	(30) $\cos 45^\circ$	(45) $\tan 60^\circ$	(60) $\tan 60^\circ$	(75) $\sin 45^\circ$	(90) $\cos 45^\circ$

(1) $\cos 60^\circ$	(16) $\tan 45^\circ$	(31) $\tan 30^\circ$	(46) $\cos 30^\circ$	(61) $\cos 60^\circ$	(76) $\sin 60^\circ$
(2) $\sin 60^\circ$	(17) $\sin 30^\circ$	(32) $\sin 60^\circ$	(47) $\cos 45^\circ$	(62) $\tan 45^\circ$	(77) $\cos 30^\circ$
(3) $\cos 60^\circ$	(18) $\sin 60^\circ$	(33) $\tan 30^\circ$	(48) $\sin 30^\circ$	(63) $\sin 45^\circ$	(78) $\cos 30^\circ$
(4) $\tan 30^\circ$	(19) $\tan 60^\circ$	(34) $\sin 45^\circ$	(49) $\tan 60^\circ$	(64) $\sin 60^\circ$	(79) $\tan 45^\circ$
(5) $\tan 45^\circ$	(20) $\sin 60^\circ$	(35) $\sin 45^\circ$	(50) $\tan 45^\circ$	(65) $\cos 30^\circ$	(80) $\sin 45^\circ$
(6) $\sin 45^\circ$	(21) $\sin 45^\circ$	(36) $\tan 30^\circ$	(51) $\sin 30^\circ$	(66) $\cos 45^\circ$	(81) $\cos 60^\circ$
(7) $\sin 60^\circ$	(22) $\cos 60^\circ$	(37) $\tan 60^\circ$	(52) $\tan 30^\circ$	(67) $\sin 60^\circ$	(82) $\sin 60^\circ$
(8) $\tan 60^\circ$	(23) $\cos 30^\circ$	(38) $\tan 60^\circ$	(53) $\tan 45^\circ$	(68) $\tan 30^\circ$	(83) $\cos 30^\circ$
(9) $\sin 30^\circ$	(24) $\sin 60^\circ$	(39) $\cos 60^\circ$	(54) $\tan 45^\circ$	(69) $\cos 45^\circ$	(84) $\tan 30^\circ$
(10) $\sin 30^\circ$	(25) $\sin 45^\circ$	(40) $\cos 60^\circ$	(55) $\cos 60^\circ$	(70) $\tan 45^\circ$	(85) $\tan 45^\circ$
(11) $\tan 45^\circ$	(26) $\cos 45^\circ$	(41) $\cos 45^\circ$	(56) $\tan 30^\circ$	(71) $\cos 45^\circ$	(86) $\tan 30^\circ$
(12) $\tan 60^\circ$	(27) $\tan 60^\circ$	(42) $\cos 60^\circ$	(57) $\tan 30^\circ$	(72) $\tan 30^\circ$	(87) $\cos 60^\circ$
(13) $\tan 60^\circ$	(28) $\tan 60^\circ$	(43) $\sin 30^\circ$	(58) $\cos 30^\circ$	(73) $\sin 30^\circ$	(88) $\tan 45^\circ$
(14) $\tan 30^\circ$	(29) $\sin 60^\circ$	(44) $\sin 45^\circ$	(59) $\tan 45^\circ$	(74) $\tan 60^\circ$	(89) $\tan 60^\circ$
(15) $\cos 45^\circ$	(30) $\cos 30^\circ$	(45) $\sin 30^\circ$	(60) $\sin 60^\circ$	(75) $\cos 45^\circ$	(90) $\cos 60^\circ$

(1) $\tan 60^\circ$	(16) $\cos 45^\circ$	(31) $\sin 30^\circ$	(46) $\cos 30^\circ$	(61) $\cos 45^\circ$	(76) $\sin 30^\circ$
(2) $\sin 60^\circ$	(17) $\sin 30^\circ$	(32) $\sin 60^\circ$	(47) $\cos 45^\circ$	(62) $\tan 30^\circ$	(77) $\sin 60^\circ$
(3) $\cos 60^\circ$	(18) $\cos 45^\circ$	(33) $\sin 30^\circ$	(48) $\tan 45^\circ$	(63) $\cos 60^\circ$	(78) $\tan 30^\circ$
(4) $\sin 30^\circ$	(19) $\sin 30^\circ$	(34) $\sin 45^\circ$	(49) $\cos 45^\circ$	(64) $\tan 60^\circ$	(79) $\sin 45^\circ$
(5) $\tan 45^\circ$	(20) $\cos 30^\circ$	(35) $\tan 30^\circ$	(50) $\sin 30^\circ$	(65) $\tan 60^\circ$	(80) $\cos 30^\circ$
(6) $\tan 30^\circ$	(21) $\tan 45^\circ$	(36) $\cos 30^\circ$	(51) $\cos 60^\circ$	(66) $\cos 30^\circ$	(81) $\tan 45^\circ$
(7) $\tan 30^\circ$	(22) $\cos 45^\circ$	(37) $\sin 60^\circ$	(52) $\sin 30^\circ$	(67) $\sin 45^\circ$	(82) $\tan 60^\circ$
(8) $\sin 45^\circ$	(23) $\sin 45^\circ$	(38) $\cos 60^\circ$	(53) $\tan 45^\circ$	(68) $\cos 45^\circ$	(83) $\tan 45^\circ$
(9) $\tan 30^\circ$	(24) $\tan 45^\circ$	(39) $\tan 30^\circ$	(54) $\sin 30^\circ$	(69) $\cos 60^\circ$	(84) $\cos 60^\circ$
(10) $\tan 45^\circ$	(25) $\sin 60^\circ$	(40) $\tan 30^\circ$	(55) $\sin 30^\circ$	(70) $\sin 60^\circ$	(85) $\sin 30^\circ$
(11) $\sin 45^\circ$	(26) $\sin 60^\circ$	(41) $\sin 30^\circ$	(56) $\sin 45^\circ$	(71) $\sin 45^\circ$	(86) $\sin 45^\circ$
(12) $\tan 60^\circ$	(27) $\tan 30^\circ$	(42) $\tan 30^\circ$	(57) $\cos 45^\circ$	(72) $\tan 30^\circ$	(87) $\sin 30^\circ$
(13) $\tan 60^\circ$	(28) $\cos 60^\circ$	(43) $\tan 30^\circ$	(58) $\tan 30^\circ$	(73) $\cos 30^\circ$	(88) $\cos 60^\circ$
(14) $\sin 60^\circ$	(29) $\cos 30^\circ$	(44) $\cos 45^\circ$	(59) $\sin 45^\circ$	(74) $\cos 60^\circ$	(89) $\sin 60^\circ$
(15) $\tan 60^\circ$	(30) $\tan 60^\circ$	(45) $\sin 60^\circ$	(60) $\sin 45^\circ$	(75) $\tan 30^\circ$	(90) $\tan 30^\circ$

(1) $\sin 45^\circ$	(16) $\tan 30^\circ$	(31) $\sin 30^\circ$	(46) $\tan 60^\circ$	(61) $\tan 30^\circ$	(76) $\cos 30^\circ$
(2) $\tan 30^\circ$	(17) $\sin 45^\circ$	(32) $\sin 45^\circ$	(47) $\cos 30^\circ$	(62) $\tan 30^\circ$	(77) $\tan 30^\circ$
(3) $\tan 30^\circ$	(18) $\cos 30^\circ$	(33) $\cos 30^\circ$	(48) $\sin 30^\circ$	(63) $\tan 30^\circ$	(78) $\sin 45^\circ$
(4) $\tan 30^\circ$	(19) $\tan 45^\circ$	(34) $\cos 45^\circ$	(49) $\sin 30^\circ$	(64) $\tan 60^\circ$	(79) $\cos 45^\circ$
(5) $\cos 60^\circ$	(20) $\sin 60^\circ$	(35) $\sin 60^\circ$	(50) $\tan 30^\circ$	(65) $\tan 45^\circ$	(80) $\sin 60^\circ$
(6) $\tan 45^\circ$	(21) $\tan 60^\circ$	(36) $\tan 30^\circ$	(51) $\cos 45^\circ$	(66) $\sin 45^\circ$	(81) $\sin 30^\circ$
(7) $\sin 30^\circ$	(22) $\sin 30^\circ$	(37) $\sin 30^\circ$	(52) $\sin 60^\circ$	(67) $\cos 45^\circ$	(82) $\cos 60^\circ$
(8) $\sin 60^\circ$	(23) $\cos 45^\circ$	(38) $\cos 45^\circ$	(53) $\cos 45^\circ$	(68) $\tan 60^\circ$	(83) $\sin 45^\circ$
(9) $\cos 45^\circ$	(24) $\cos 45^\circ$	(39) $\cos 45^\circ$	(54) $\cos 60^\circ$	(69) $\cos 30^\circ$	(84) $\sin 30^\circ$
(10) $\tan 60^\circ$	(25) $\sin 45^\circ$	(40) $\sin 45^\circ$	(55) $\tan 45^\circ$	(70) $\sin 60^\circ$	(85) $\tan 60^\circ$
(11) $\tan 60^\circ$	(26) $\cos 30^\circ$	(41) $\tan 30^\circ$	(56) $\tan 60^\circ$	(71) $\sin 45^\circ$	(86) $\tan 45^\circ$
(12) $\sin 60^\circ$	(27) $\sin 30^\circ$	(42) $\tan 30^\circ$	(57) $\tan 60^\circ$	(72) $\tan 60^\circ$	(87) $\cos 45^\circ$
(13) $\cos 45^\circ$	(28) $\tan 60^\circ$	(43) $\cos 45^\circ$	(58) $\tan 60^\circ$	(73) $\tan 30^\circ$	(88) $\sin 30^\circ$
(14) $\sin 45^\circ$	(29) $\sin 30^\circ$	(44) $\tan 60^\circ$	(59) $\cos 30^\circ$	(74) $\tan 45^\circ$	(89) $\cos 30^\circ$
(15) $\sin 45^\circ$	(30) $\sin 45^\circ$	(45) $\tan 45^\circ$	(60) $\cos 60^\circ$	(75) $\tan 60^\circ$	(90) $\tan 45^\circ$

(1) $\sin 45^\circ$	(16) $\tan 60^\circ$	(31) $\sin 30^\circ$	(46) $\sin 45^\circ$	(61) $\tan 45^\circ$	(76) $\tan 30^\circ$
(2) $\tan 45^\circ$	(17) $\cos 45^\circ$	(32) $\cos 60^\circ$	(47) $\sin 30^\circ$	(62) $\tan 30^\circ$	(77) $\sin 45^\circ$
(3) $\sin 30^\circ$	(18) $\tan 45^\circ$	(33) $\sin 45^\circ$	(48) $\cos 45^\circ$	(63) $\cos 45^\circ$	(78) $\tan 60^\circ$
(4) $\cos 30^\circ$	(19) $\sin 30^\circ$	(34) $\cos 60^\circ$	(49) $\sin 45^\circ$	(64) $\tan 30^\circ$	(79) $\sin 60^\circ$
(5) $\tan 30^\circ$	(20) $\sin 45^\circ$	(35) $\tan 30^\circ$	(50) $\sin 45^\circ$	(65) $\sin 45^\circ$	(80) $\cos 30^\circ$
(6) $\tan 60^\circ$	(21) $\sin 30^\circ$	(36) $\tan 45^\circ$	(51) $\cos 60^\circ$	(66) $\sin 60^\circ$	(81) $\sin 45^\circ$
(7) $\sin 45^\circ$	(22) $\tan 60^\circ$	(37) $\sin 45^\circ$	(52) $\tan 60^\circ$	(67) $\cos 45^\circ$	(82) $\tan 45^\circ$
(8) $\sin 30^\circ$	(23) $\cos 60^\circ$	(38) $\sin 60^\circ$	(53) $\tan 60^\circ$	(68) $\tan 30^\circ$	(83) $\tan 60^\circ$
(9) $\cos 60^\circ$	(24) $\cos 45^\circ$	(39) $\tan 60^\circ$	(54) $\cos 30^\circ$	(69) $\cos 30^\circ$	(84) $\sin 30^\circ$
(10) $\tan 30^\circ$	(25) $\tan 30^\circ$	(40) $\sin 60^\circ$	(55) $\tan 45^\circ$	(70) $\cos 45^\circ$	(85) $\sin 60^\circ$
(11) $\tan 60^\circ$	(26) $\sin 45^\circ$	(41) $\cos 45^\circ$	(56) $\tan 30^\circ$	(71) $\sin 45^\circ$	(86) $\cos 45^\circ$
(12) $\tan 30^\circ$	(27) $\tan 60^\circ$	(42) $\sin 45^\circ$	(57) $\sin 45^\circ$	(72) $\sin 45^\circ$	(87) $\cos 60^\circ$
(13) $\sin 30^\circ$	(28) $\cos 45^\circ$	(43) $\sin 60^\circ$	(58) $\cos 30^\circ$	(73) $\sin 30^\circ$	(88) $\sin 30^\circ$
(14) $\cos 45^\circ$	(29) $\tan 60^\circ$	(44) $\sin 60^\circ$	(59) $\cos 30^\circ$	(74) $\cos 45^\circ$	(89) $\tan 60^\circ$
(15) $\sin 30^\circ$	(30) $\tan 30^\circ$	(45) $\tan 45^\circ$	(60) $\cos 60^\circ$	(75) $\cos 45^\circ$	(90) $\cos 30^\circ$

(1) $\sin 60^\circ$	(16) $\cos 60^\circ$	(31) $\sin 45^\circ$	(46) $\sin 45^\circ$	(61) $\cos 45^\circ$	(76) $\cos 60^\circ$
(2) $\sin 30^\circ$	(17) $\sin 30^\circ$	(32) $\tan 45^\circ$	(47) $\sin 45^\circ$	(62) $\sin 30^\circ$	(77) $\sin 60^\circ$
(3) $\tan 30^\circ$	(18) $\cos 45^\circ$	(33) $\sin 30^\circ$	(48) $\sin 30^\circ$	(63) $\cos 60^\circ$	(78) $\cos 30^\circ$
(4) $\sin 60^\circ$	(19) $\cos 60^\circ$	(34) $\sin 60^\circ$	(49) $\tan 60^\circ$	(64) $\sin 30^\circ$	(79) $\sin 45^\circ$
(5) $\sin 45^\circ$	(20) $\tan 45^\circ$	(35) $\sin 45^\circ$	(50) $\sin 45^\circ$	(65) $\tan 30^\circ$	(80) $\sin 60^\circ$
(6) $\cos 30^\circ$	(21) $\sin 30^\circ$	(36) $\tan 30^\circ$	(51) $\tan 30^\circ$	(66) $\cos 45^\circ$	(81) $\sin 30^\circ$
(7) $\cos 45^\circ$	(22) $\sin 30^\circ$	(37) $\sin 60^\circ$	(52) $\cos 60^\circ$	(67) $\sin 45^\circ$	(82) $\sin 30^\circ$
(8) $\sin 30^\circ$	(23) $\sin 30^\circ$	(38) $\tan 30^\circ$	(53) $\tan 45^\circ$	(68) $\sin 45^\circ$	(83) $\tan 60^\circ$
(9) $\cos 30^\circ$	(24) $\cos 30^\circ$	(39) $\tan 60^\circ$	(54) $\cos 60^\circ$	(69) $\cos 60^\circ$	(84) $\cos 60^\circ$
(10) $\sin 30^\circ$	(25) $\tan 60^\circ$	(40) $\cos 60^\circ$	(55) $\tan 30^\circ$	(70) $\tan 30^\circ$	(85) $\tan 60^\circ$
(11) $\sin 45^\circ$	(26) $\sin 45^\circ$	(41) $\tan 30^\circ$	(56) $\tan 30^\circ$	(71) $\cos 45^\circ$	(86) $\cos 60^\circ$
(12) $\sin 60^\circ$	(27) $\cos 45^\circ$	(42) $\cos 60^\circ$	(57) $\cos 60^\circ$	(72) $\cos 45^\circ$	(87) $\cos 45^\circ$
(13) $\tan 30^\circ$	(28) $\cos 30^\circ$	(43) $\tan 60^\circ$	(58) $\cos 60^\circ$	(73) $\tan 30^\circ$	(88) $\cos 60^\circ$
(14) $\tan 30^\circ$	(29) $\tan 45^\circ$	(44) $\sin 30^\circ$	(59) $\tan 45^\circ$	(74) $\sin 30^\circ$	(89) $\cos 45^\circ$
(15) $\sin 60^\circ$	(30) $\cos 30^\circ$	(45) $\sin 30^\circ$	(60) $\cos 30^\circ$	(75) $\tan 30^\circ$	(90) $\tan 45^\circ$

(1) $\cos 30^\circ$	(16) $\sin 45^\circ$	(31) $\cos 30^\circ$	(46) $\tan 45^\circ$	(61) $\cos 30^\circ$	(76) $\sin 30^\circ$
(2) $\tan 60^\circ$	(17) $\cos 30^\circ$	(32) $\tan 60^\circ$	(47) $\cos 30^\circ$	(62) $\tan 45^\circ$	(77) $\sin 30^\circ$
(3) $\sin 60^\circ$	(18) $\cos 30^\circ$	(33) $\sin 60^\circ$	(48) $\cos 45^\circ$	(63) $\cos 45^\circ$	(78) $\cos 60^\circ$
(4) $\cos 45^\circ$	(19) $\tan 45^\circ$	(34) $\tan 30^\circ$	(49) $\cos 60^\circ$	(64) $\cos 60^\circ$	(79) $\sin 30^\circ$
(5) $\sin 45^\circ$	(20) $\cos 30^\circ$	(35) $\sin 45^\circ$	(50) $\sin 60^\circ$	(65) $\cos 60^\circ$	(80) $\cos 60^\circ$
(6) $\cos 60^\circ$	(21) $\tan 30^\circ$	(36) $\tan 60^\circ$	(51) $\sin 30^\circ$	(66) $\sin 30^\circ$	(81) $\sin 30^\circ$
(7) $\tan 45^\circ$	(22) $\sin 60^\circ$	(37) $\cos 60^\circ$	(52) $\tan 30^\circ$	(67) $\tan 60^\circ$	(82) $\cos 30^\circ$
(8) $\sin 30^\circ$	(23) $\cos 60^\circ$	(38) $\sin 60^\circ$	(53) $\tan 60^\circ$	(68) $\sin 30^\circ$	(83) $\sin 45^\circ$
(9) $\sin 45^\circ$	(24) $\cos 30^\circ$	(39) $\sin 30^\circ$	(54) $\sin 60^\circ$	(69) $\tan 30^\circ$	(84) $\tan 30^\circ$
(10) $\sin 60^\circ$	(25) $\sin 45^\circ$	(40) $\tan 45^\circ$	(55) $\tan 30^\circ$	(70) $\cos 60^\circ$	(85) $\cos 45^\circ$
(11) $\cos 30^\circ$	(26) $\tan 45^\circ$	(41) $\sin 30^\circ$	(56) $\sin 60^\circ$	(71) $\tan 30^\circ$	(86) $\tan 30^\circ$
(12) $\cos 60^\circ$	(27) $\tan 30^\circ$	(42) $\tan 60^\circ$	(57) $\tan 30^\circ$	(72) $\tan 45^\circ$	(87) $\tan 45^\circ$
(13) $\sin 30^\circ$	(28) $\sin 45^\circ$	(43) $\cos 45^\circ$	(58) $\tan 60^\circ$	(73) $\sin 30^\circ$	(88) $\tan 30^\circ$
(14) $\sin 30^\circ$	(29) $\sin 45^\circ$	(44) $\tan 60^\circ$	(59) $\cos 60^\circ$	(74) $\sin 30^\circ$	(89) $\sin 30^\circ$
(15) $\tan 60^\circ$	(30) $\sin 30^\circ$	(45) $\sin 30^\circ$	(60) $\cos 30^\circ$	(75) $\cos 60^\circ$	(90) $\sin 45^\circ$

(1) $\tan 45^\circ$ $= 1$	(16) $\cos 60^\circ$ $= \frac{1}{2}$	(31) $\sin 30^\circ$ $= \frac{1}{2}$	(46) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(61) $\tan 60^\circ$ $= \sqrt{3}$	(76) $\tan 60^\circ$ $= \sqrt{3}$
(2) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(17) $\sin 30^\circ$ $= \frac{1}{2}$	(32) $\cos 60^\circ$ $= \frac{1}{2}$	(47) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(62) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(77) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(3) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(18) $\cos 60^\circ$ $= \frac{1}{2}$	(33) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(48) $\cos 60^\circ$ $= \frac{1}{2}$	(63) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(78) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(4) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(19) $\tan 60^\circ$ $= \sqrt{3}$	(34) $\tan 60^\circ$ $= \sqrt{3}$	(49) $\tan 60^\circ$ $= \sqrt{3}$	(64) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(79) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(5) $\tan 60^\circ$ $= \sqrt{3}$	(20) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(35) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(50) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(65) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(80) $\tan 60^\circ$ $= \sqrt{3}$
(6) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(21) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(36) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(51) $\cos 60^\circ$ $= \frac{1}{2}$	(66) $\cos 60^\circ$ $= \frac{1}{2}$	(81) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(7) $\sin 30^\circ$ $= \frac{1}{2}$	(22) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(37) $\cos 60^\circ$ $= \frac{1}{2}$	(52) $\sin 30^\circ$ $= \frac{1}{2}$	(67) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(82) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(8) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(38) $\cos 60^\circ$ $= \frac{1}{2}$	(53) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(68) $\tan 45^\circ$ $= 1$	(83) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(9) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(24) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(39) $\tan 45^\circ$ $= 1$	(54) $\cos 60^\circ$ $= \frac{1}{2}$	(69) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(84) $\sin 30^\circ$ $= \frac{1}{2}$
(10) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(25) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(40) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(55) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(70) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(85) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(11) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(26) $\sin 30^\circ$ $= \frac{1}{2}$	(41) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(56) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(71) $\sin 30^\circ$ $= \frac{1}{2}$	(86) $\tan 60^\circ$ $= \sqrt{3}$
(12) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(27) $\sin 30^\circ$ $= \frac{1}{2}$	(42) $\cos 60^\circ$ $= \frac{1}{2}$	(57) $\cos 60^\circ$ $= \frac{1}{2}$	(72) $\tan 60^\circ$ $= \sqrt{3}$	(87) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(13) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(28) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(43) $\tan 60^\circ$ $= \sqrt{3}$	(58) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(73) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(88) $\tan 45^\circ$ $= 1$
(14) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(29) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(44) $\sin 30^\circ$ $= \frac{1}{2}$	(59) $\cos 60^\circ$ $= \frac{1}{2}$	(74) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(89) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(15) $\tan 45^\circ$ $= 1$	(30) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(45) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(60) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(75) $\cos 60^\circ$ $= \frac{1}{2}$	(90) $\tan 45^\circ$ $= 1$

(1) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(16) $\cos 60^\circ$ $= \frac{1}{2}$	(31) $\sin 30^\circ$ $= \frac{1}{2}$	(46) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(61) $\tan 60^\circ$ $= \sqrt{3}$	(76) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(2) $\sin 30^\circ$ $= \frac{1}{2}$	(17) $\tan 45^\circ$ $= 1$	(32) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(47) $\sin 30^\circ$ $= \frac{1}{2}$	(62) $\sin 30^\circ$ $= \frac{1}{2}$	(77) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(3) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(18) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(33) $\tan 60^\circ$ $= \sqrt{3}$	(48) $\tan 45^\circ$ $= 1$	(63) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(78) $\cos 60^\circ$ $= \frac{1}{2}$
(4) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(19) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(34) $\tan 45^\circ$ $= 1$	(49) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(64) $\cos 60^\circ$ $= \frac{1}{2}$	(79) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(5) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(20) $\tan 45^\circ$ $= 1$	(35) $\tan 60^\circ$ $= \sqrt{3}$	(50) $\cos 60^\circ$ $= \frac{1}{2}$	(65) $\sin 30^\circ$ $= \frac{1}{2}$	(80) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(6) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(21) $\tan 45^\circ$ $= 1$	(36) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(51) $\cos 60^\circ$ $= \frac{1}{2}$	(66) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(81) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(7) $\cos 60^\circ$ $= \frac{1}{2}$	(22) $\tan 45^\circ$ $= 1$	(37) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(52) $\cos 60^\circ$ $= \frac{1}{2}$	(67) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(82) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(8) $\tan 60^\circ$ $= \sqrt{3}$	(23) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(38) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(53) $\cos 60^\circ$ $= \frac{1}{2}$	(68) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(83) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(9) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(24) $\tan 45^\circ$ $= 1$	(39) $\cos 60^\circ$ $= \frac{1}{2}$	(54) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(84) $\cos 60^\circ$ $= \frac{1}{2}$
(10) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(25) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(40) $\sin 30^\circ$ $= \frac{1}{2}$	(55) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(70) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(85) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(11) $\tan 45^\circ$ $= 1$	(26) $\tan 45^\circ$ $= 1$	(41) $\tan 60^\circ$ $= \sqrt{3}$	(56) $\cos 60^\circ$ $= \frac{1}{2}$	(71) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(86) $\tan 60^\circ$ $= \sqrt{3}$
(12) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(42) $\cos 60^\circ$ $= \frac{1}{2}$	(57) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(72) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(87) $\tan 60^\circ$ $= \sqrt{3}$
(13) $\cos 60^\circ$ $= \frac{1}{2}$	(28) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(43) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(58) $\cos 60^\circ$ $= \frac{1}{2}$	(73) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(88) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(14) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\tan 60^\circ$ $= \sqrt{3}$	(44) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(59) $\tan 60^\circ$ $= \sqrt{3}$	(74) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(89) $\cos 60^\circ$ $= \frac{1}{2}$
(15) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(30) $\tan 45^\circ$ $= 1$	(45) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(60) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(75) $\tan 60^\circ$ $= \sqrt{3}$	(90) $\cos 60^\circ$ $= \frac{1}{2}$

(1) $\tan 45^\circ$ $= 1$	(16) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(31) $\tan 60^\circ$ $= \sqrt{3}$	(46) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(61) $\tan 60^\circ$ $= \sqrt{3}$	(76) $\sin 30^\circ$ $= \frac{1}{2}$
(2) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(17) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(32) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(47) $\tan 45^\circ$ $= 1$	(62) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(77) $\tan 60^\circ$ $= \sqrt{3}$
(3) $\tan 45^\circ$ $= 1$	(18) $\tan 45^\circ$ $= 1$	(33) $\tan 45^\circ$ $= 1$	(48) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(63) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(78) $\sin 30^\circ$ $= \frac{1}{2}$
(4) $\tan 60^\circ$ $= \sqrt{3}$	(19) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(34) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(49) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(64) $\sin 30^\circ$ $= \frac{1}{2}$	(79) $\cos 60^\circ$ $= \frac{1}{2}$
(5) $\sin 30^\circ$ $= \frac{1}{2}$	(20) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(35) $\cos 60^\circ$ $= \frac{1}{2}$	(50) $\cos 60^\circ$ $= \frac{1}{2}$	(65) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(80) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(6) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(21) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(36) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(51) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(66) $\tan 45^\circ$ $= 1$	(81) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(7) $\tan 45^\circ$ $= 1$	(22) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(37) $\tan 45^\circ$ $= 1$	(52) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(67) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(82) $\tan 60^\circ$ $= \sqrt{3}$
(8) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(23) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(38) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(53) $\tan 60^\circ$ $= \sqrt{3}$	(68) $\cos 60^\circ$ $= \frac{1}{2}$	(83) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(9) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(24) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(39) $\cos 60^\circ$ $= \frac{1}{2}$	(54) $\tan 60^\circ$ $= \sqrt{3}$	(69) $\cos 60^\circ$ $= \frac{1}{2}$	(84) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(10) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\tan 45^\circ$ $= 1$	(40) $\cos 60^\circ$ $= \frac{1}{2}$	(55) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(70) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(85) $\cos 60^\circ$ $= \frac{1}{2}$
(11) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(26) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(41) $\sin 30^\circ$ $= \frac{1}{2}$	(56) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(71) $\tan 60^\circ$ $= \sqrt{3}$	(86) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(12) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(27) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(42) $\sin 30^\circ$ $= \frac{1}{2}$	(57) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(72) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(87) $\cos 60^\circ$ $= \frac{1}{2}$
(13) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(28) $\tan 60^\circ$ $= \sqrt{3}$	(43) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(58) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(88) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(14) $\tan 60^\circ$ $= \sqrt{3}$	(29) $\tan 45^\circ$ $= 1$	(44) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(59) $\cos 60^\circ$ $= \frac{1}{2}$	(74) $\tan 45^\circ$ $= 1$	(89) $\sin 30^\circ$ $= \frac{1}{2}$
(15) $\cos 60^\circ$ $= \frac{1}{2}$	(30) $\sin 30^\circ$ $= \frac{1}{2}$	(45) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(60) $\tan 60^\circ$ $= \sqrt{3}$	(75) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(90) $\cos 60^\circ$ $= \frac{1}{2}$

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(2) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(17) $\cos 60^\circ$ $= \frac{1}{2}$	(32) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(47) $\tan 45^\circ$ $= 1$	(62) $\tan 60^\circ$ $= \sqrt{3}$	(77) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(3) $\cos 60^\circ$ $= \frac{1}{2}$	(18) $\tan 60^\circ$ $= \sqrt{3}$	(33) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(48) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(63) $\sin 30^\circ$ $= \frac{1}{2}$	(78) $\tan 45^\circ$ $= 1$
(4) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(19) $\sin 30^\circ$ $= \frac{1}{2}$	(34) $\cos 60^\circ$ $= \frac{1}{2}$	(49) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(64) $\sin 30^\circ$ $= \frac{1}{2}$	(79) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(5) $\tan 45^\circ$ $= 1$	(20) $\tan 60^\circ$ $= \sqrt{3}$	(35) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(50) $\tan 45^\circ$ $= 1$	(65) $\sin 30^\circ$ $= \frac{1}{2}$	(80) $\tan 60^\circ$ $= \sqrt{3}$
(6) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(21) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(36) $\tan 45^\circ$ $= 1$	(51) $\sin 30^\circ$ $= \frac{1}{2}$	(66) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(81) $\tan 45^\circ$ $= 1$
(7) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(22) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(37) $\tan 60^\circ$ $= \sqrt{3}$	(52) $\sin 30^\circ$ $= \frac{1}{2}$	(67) $\cos 60^\circ$ $= \frac{1}{2}$	(82) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(8) $\cos 60^\circ$ $= \frac{1}{2}$	(23) $\cos 60^\circ$ $= \frac{1}{2}$	(38) $\tan 60^\circ$ $= \sqrt{3}$	(53) $\cos 60^\circ$ $= \frac{1}{2}$	(68) $\tan 60^\circ$ $= \sqrt{3}$	(83) $\tan 45^\circ$ $= 1$
(9) $\tan 60^\circ$ $= \sqrt{3}$	(24) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(39) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(84) $\sin 30^\circ$ $= \frac{1}{2}$
(10) $\sin 30^\circ$ $= \frac{1}{2}$	(25) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(40) $\sin 30^\circ$ $= \frac{1}{2}$	(55) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(70) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(85) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(11) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(26) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(41) $\tan 60^\circ$ $= \sqrt{3}$	(56) $\tan 60^\circ$ $= \sqrt{3}$	(71) $\tan 60^\circ$ $= \sqrt{3}$	(86) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(12) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(27) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(42) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(57) $\sin 30^\circ$ $= \frac{1}{2}$	(72) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(87) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(13) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(28) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(43) $\cos 60^\circ$ $= \frac{1}{2}$	(58) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(88) $\tan 45^\circ$ $= 1$
(14) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\cos 60^\circ$ $= \frac{1}{2}$	(44) $\tan 45^\circ$ $= 1$	(59) $\cos 60^\circ$ $= \frac{1}{2}$	(74) $\tan 45^\circ$ $= 1$	(89) $\tan 45^\circ$ $= 1$
(15) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(30) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(45) $\tan 45^\circ$ $= 1$	(60) $\cos 60^\circ$ $= \frac{1}{2}$	(75) $\cos 60^\circ$ $= \frac{1}{2}$	(90) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$

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(2) $\cos 60^\circ$ $= \frac{1}{2}$	(17) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(32) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(47) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(62) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(77) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(18) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(33) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(48) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(63) $\cos 60^\circ$ $= \frac{1}{2}$	(78) $\tan 60^\circ$ $= \sqrt{3}$
(4) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(19) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(34) $\sin 30^\circ$ $= \frac{1}{2}$	(49) $\tan 60^\circ$ $= \sqrt{3}$	(64) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(79) $\sin 30^\circ$ $= \frac{1}{2}$
(5) $\sin 30^\circ$ $= \frac{1}{2}$	(20) $\tan 60^\circ$ $= \sqrt{3}$	(35) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(50) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(65) $\tan 45^\circ$ $= 1$	(80) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(6) $\tan 45^\circ$ $= 1$	(21) $\tan 60^\circ$ $= \sqrt{3}$	(36) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(51) $\tan 60^\circ$ $= \sqrt{3}$	(66) $\tan 45^\circ$ $= 1$	(81) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(7) $\tan 45^\circ$ $= 1$	(22) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(37) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(52) $\sin 30^\circ$ $= \frac{1}{2}$	(67) $\tan 45^\circ$ $= 1$	(82) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(8) $\tan 60^\circ$ $= \sqrt{3}$	(23) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(38) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(53) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(68) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(83) $\sin 30^\circ$ $= \frac{1}{2}$
(9) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(24) $\tan 45^\circ$ $= 1$	(39) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\cos 60^\circ$ $= \frac{1}{2}$	(69) $\tan 60^\circ$ $= \sqrt{3}$	(84) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(10) $\tan 45^\circ$ $= 1$	(25) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(40) $\cos 60^\circ$ $= \frac{1}{2}$	(55) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(70) $\tan 45^\circ$ $= 1$	(85) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(11) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(26) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(41) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(56) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(71) $\tan 60^\circ$ $= \sqrt{3}$	(86) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(12) $\tan 60^\circ$ $= \sqrt{3}$	(27) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(42) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(57) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(72) $\tan 60^\circ$ $= \sqrt{3}$	(87) $\tan 60^\circ$ $= \sqrt{3}$
(13) $\sin 30^\circ$ $= \frac{1}{2}$	(28) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(43) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(58) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\sin 30^\circ$ $= \frac{1}{2}$	(88) $\cos 60^\circ$ $= \frac{1}{2}$
(14) $\sin 30^\circ$ $= \frac{1}{2}$	(29) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(44) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(59) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(74) $\tan 45^\circ$ $= 1$	(89) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(15) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(30) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(45) $\sin 30^\circ$ $= \frac{1}{2}$	(60) $\cos 60^\circ$ $= \frac{1}{2}$	(75) $\tan 60^\circ$ $= \sqrt{3}$	(90) $\tan 45^\circ$ $= 1$

(1) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(16) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(31) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(46) $\tan 45^\circ$ $= 1$	(61) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(76) $\tan 60^\circ$ $= \sqrt{3}$
(2) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(17) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(32) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(47) $\sin 30^\circ$ $= \frac{1}{2}$	(62) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(77) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(3) $\sin 30^\circ$ $= \frac{1}{2}$	(18) $\sin 30^\circ$ $= \frac{1}{2}$	(33) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(48) $\tan 45^\circ$ $= 1$	(63) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(78) $\tan 45^\circ$ $= 1$
(4) $\cos 60^\circ$ $= \frac{1}{2}$	(19) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(34) $\tan 45^\circ$ $= 1$	(49) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(64) $\tan 60^\circ$ $= \sqrt{3}$	(79) $\sin 30^\circ$ $= \frac{1}{2}$
(5) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(20) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(35) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(50) $\tan 60^\circ$ $= \sqrt{3}$	(65) $\sin 30^\circ$ $= \frac{1}{2}$	(80) $\cos 60^\circ$ $= \frac{1}{2}$
(6) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(21) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(36) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(51) $\cos 60^\circ$ $= \frac{1}{2}$	(66) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(81) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(7) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(22) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(37) $\tan 60^\circ$ $= \sqrt{3}$	(52) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(67) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(82) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(8) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(23) $\tan 60^\circ$ $= \sqrt{3}$	(38) $\tan 60^\circ$ $= \sqrt{3}$	(53) $\tan 45^\circ$ $= 1$	(68) $\tan 60^\circ$ $= \sqrt{3}$	(83) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(9) $\cos 60^\circ$ $= \frac{1}{2}$	(24) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(39) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(69) $\sin 30^\circ$ $= \frac{1}{2}$	(84) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(10) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\tan 45^\circ$ $= 1$	(40) $\sin 30^\circ$ $= \frac{1}{2}$	(55) $\tan 45^\circ$ $= 1$	(70) $\cos 60^\circ$ $= \frac{1}{2}$	(85) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(11) $\sin 30^\circ$ $= \frac{1}{2}$	(26) $\cos 60^\circ$ $= \frac{1}{2}$	(41) $\tan 45^\circ$ $= 1$	(56) $\tan 45^\circ$ $= 1$	(71) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(86) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(12) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(27) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(42) $\sin 30^\circ$ $= \frac{1}{2}$	(57) $\cos 60^\circ$ $= \frac{1}{2}$	(72) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(87) $\tan 45^\circ$ $= 1$
(13) $\cos 60^\circ$ $= \frac{1}{2}$	(28) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(43) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(58) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\tan 60^\circ$ $= \sqrt{3}$	(88) $\tan 45^\circ$ $= 1$
(14) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(29) $\tan 60^\circ$ $= \sqrt{3}$	(44) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(59) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(74) $\sin 30^\circ$ $= \frac{1}{2}$	(89) $\cos 60^\circ$ $= \frac{1}{2}$
(15) $\tan 60^\circ$ $= \sqrt{3}$	(30) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(45) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(60) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(75) $\sin 30^\circ$ $= \frac{1}{2}$	(90) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$

(1) $\sin 30^\circ$ $= \frac{1}{2}$	(16) $\cos 60^\circ$ $= \frac{1}{2}$	(31) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(46) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(61) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(76) $\tan 45^\circ$ $= 1$
(2) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(17) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(32) $\cos 60^\circ$ $= \frac{1}{2}$	(47) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(62) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(77) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(3) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(18) $\tan 45^\circ$ $= 1$	(33) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(48) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(63) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(78) $\tan 60^\circ$ $= \sqrt{3}$
(4) $\cos 60^\circ$ $= \frac{1}{2}$	(19) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(34) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(49) $\sin 30^\circ$ $= \frac{1}{2}$	(64) $\tan 60^\circ$ $= \sqrt{3}$	(79) $\cos 60^\circ$ $= \frac{1}{2}$
(5) $\tan 60^\circ$ $= \sqrt{3}$	(20) $\cos 60^\circ$ $= \frac{1}{2}$	(35) $\cos 60^\circ$ $= \frac{1}{2}$	(50) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(65) $\tan 60^\circ$ $= \sqrt{3}$	(80) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(6) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(21) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(36) $\tan 60^\circ$ $= \sqrt{3}$	(51) $\cos 60^\circ$ $= \frac{1}{2}$	(66) $\cos 60^\circ$ $= \frac{1}{2}$	(81) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(7) $\tan 60^\circ$ $= \sqrt{3}$	(22) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(37) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(52) $\cos 60^\circ$ $= \frac{1}{2}$	(67) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(82) $\sin 30^\circ$ $= \frac{1}{2}$
(8) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(23) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(38) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(53) $\sin 30^\circ$ $= \frac{1}{2}$	(68) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(83) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(9) $\tan 45^\circ$ $= 1$	(24) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(39) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(54) $\cos 60^\circ$ $= \frac{1}{2}$	(69) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(84) $\cos 60^\circ$ $= \frac{1}{2}$
(10) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(25) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(40) $\tan 45^\circ$ $= 1$	(55) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(70) $\cos 60^\circ$ $= \frac{1}{2}$	(85) $\tan 45^\circ$ $= 1$
(11) $\tan 60^\circ$ $= \sqrt{3}$	(26) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(41) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(56) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(71) $\sin 30^\circ$ $= \frac{1}{2}$	(86) $\tan 60^\circ$ $= \sqrt{3}$
(12) $\tan 60^\circ$ $= \sqrt{3}$	(27) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(42) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(57) $\tan 60^\circ$ $= \sqrt{3}$	(72) $\sin 30^\circ$ $= \frac{1}{2}$	(87) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(13) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(28) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(43) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(58) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(88) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(14) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(44) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(59) $\cos 60^\circ$ $= \frac{1}{2}$	(74) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(89) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(15) $\tan 60^\circ$ $= \sqrt{3}$	(30) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(45) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(60) $\tan 60^\circ$ $= \sqrt{3}$	(75) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(90) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$

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(2) $\sin 30^\circ$ $= \frac{1}{2}$	(17) $\tan 45^\circ$ $= 1$	(32) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(47) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(62) $\tan 45^\circ$ $= 1$	(77) $\tan 60^\circ$ $= \sqrt{3}$
(3) $\tan 45^\circ$ $= 1$	(18) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(33) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(48) $\sin 30^\circ$ $= \frac{1}{2}$	(63) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(78) $\cos 60^\circ$ $= \frac{1}{2}$
(4) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(19) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(34) $\tan 45^\circ$ $= 1$	(49) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(64) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(79) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(5) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(20) $\tan 60^\circ$ $= \sqrt{3}$	(35) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(50) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(65) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(80) $\tan 45^\circ$ $= 1$
(6) $\tan 45^\circ$ $= 1$	(21) $\sin 30^\circ$ $= \frac{1}{2}$	(36) $\tan 60^\circ$ $= \sqrt{3}$	(51) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(66) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(81) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(7) $\sin 30^\circ$ $= \frac{1}{2}$	(22) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(37) $\tan 45^\circ$ $= 1$	(52) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(67) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(82) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(8) $\tan 45^\circ$ $= 1$	(23) $\tan 60^\circ$ $= \sqrt{3}$	(38) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(53) $\tan 60^\circ$ $= \sqrt{3}$	(68) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(83) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(9) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(24) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(39) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\tan 60^\circ$ $= \sqrt{3}$	(69) $\cos 60^\circ$ $= \frac{1}{2}$	(84) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(10) $\tan 60^\circ$ $= \sqrt{3}$	(25) $\cos 60^\circ$ $= \frac{1}{2}$	(40) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(55) $\tan 45^\circ$ $= 1$	(70) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(85) $\tan 60^\circ$ $= \sqrt{3}$
(11) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(26) $\cos 60^\circ$ $= \frac{1}{2}$	(41) $\tan 60^\circ$ $= \sqrt{3}$	(56) $\tan 45^\circ$ $= 1$	(71) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(86) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(12) $\sin 30^\circ$ $= \frac{1}{2}$	(27) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(42) $\tan 45^\circ$ $= 1$	(57) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(72) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(87) $\cos 60^\circ$ $= \frac{1}{2}$
(13) $\tan 60^\circ$ $= \sqrt{3}$	(28) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(43) $\tan 60^\circ$ $= \sqrt{3}$	(58) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(88) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(14) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(44) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(59) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(74) $\tan 45^\circ$ $= 1$	(89) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(15) $\tan 45^\circ$ $= 1$	(30) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(45) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(60) $\sin 30^\circ$ $= \frac{1}{2}$	(75) $\tan 60^\circ$ $= \sqrt{3}$	(90) $\tan 45^\circ$ $= 1$

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(2) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(17) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(32) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(47) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(62) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(77) $\sin 30^\circ$ $= \frac{1}{2}$
(3) $\sin 30^\circ$ $= \frac{1}{2}$	(18) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(33) $\tan 45^\circ$ $= 1$	(48) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(63) $\cos 60^\circ$ $= \frac{1}{2}$	(78) $\tan 60^\circ$ $= \sqrt{3}$
(4) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(19) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(34) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(49) $\tan 45^\circ$ $= 1$	(64) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(79) $\tan 60^\circ$ $= \sqrt{3}$
(5) $\tan 60^\circ$ $= \sqrt{3}$	(20) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(35) $\cos 60^\circ$ $= \frac{1}{2}$	(50) $\cos 60^\circ$ $= \frac{1}{2}$	(65) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(80) $\tan 45^\circ$ $= 1$
(6) $\cos 60^\circ$ $= \frac{1}{2}$	(21) $\cos 60^\circ$ $= \frac{1}{2}$	(36) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(51) $\tan 60^\circ$ $= \sqrt{3}$	(66) $\tan 60^\circ$ $= \sqrt{3}$	(81) $\tan 60^\circ$ $= \sqrt{3}$
(7) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(22) $\sin 30^\circ$ $= \frac{1}{2}$	(37) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(52) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(67) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(82) $\tan 60^\circ$ $= \sqrt{3}$
(8) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(23) $\tan 60^\circ$ $= \sqrt{3}$	(38) $\sin 30^\circ$ $= \frac{1}{2}$	(53) $\tan 45^\circ$ $= 1$	(68) $\tan 45^\circ$ $= 1$	(83) $\cos 60^\circ$ $= \frac{1}{2}$
(9) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(24) $\tan 45^\circ$ $= 1$	(39) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\tan 45^\circ$ $= 1$	(84) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(10) $\cos 60^\circ$ $= \frac{1}{2}$	(25) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(40) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(55) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(70) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(85) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(11) $\sin 30^\circ$ $= \frac{1}{2}$	(26) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(41) $\sin 30^\circ$ $= \frac{1}{2}$	(56) $\sin 30^\circ$ $= \frac{1}{2}$	(71) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(86) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(12) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\tan 45^\circ$ $= 1$	(42) $\tan 60^\circ$ $= \sqrt{3}$	(57) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(72) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(87) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(13) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(28) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(43) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(58) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(73) $\tan 60^\circ$ $= \sqrt{3}$	(88) $\sin 30^\circ$ $= \frac{1}{2}$
(14) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(29) $\tan 45^\circ$ $= 1$	(44) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(59) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(74) $\tan 60^\circ$ $= \sqrt{3}$	(89) $\sin 30^\circ$ $= \frac{1}{2}$
(15) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(30) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(45) $\tan 60^\circ$ $= \sqrt{3}$	(60) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(75) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(90) $\sin 30^\circ$ $= \frac{1}{2}$

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(2) $\tan 60^\circ$ $= \sqrt{3}$	(17) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(32) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(47) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(62) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(77) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(3) $\tan 45^\circ$ $= 1$	(18) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(33) $\sin 30^\circ$ $= \frac{1}{2}$	(48) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(63) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(78) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(4) $\tan 45^\circ$ $= 1$	(19) $\sin 30^\circ$ $= \frac{1}{2}$	(34) $\sin 30^\circ$ $= \frac{1}{2}$	(49) $\tan 60^\circ$ $= \sqrt{3}$	(64) $\tan 45^\circ$ $= 1$	(79) $\tan 60^\circ$ $= \sqrt{3}$
(5) $\cos 60^\circ$ $= \frac{1}{2}$	(20) $\tan 45^\circ$ $= 1$	(35) $\cos 60^\circ$ $= \frac{1}{2}$	(50) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(65) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(80) $\sin 30^\circ$ $= \frac{1}{2}$
(6) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(21) $\sin 30^\circ$ $= \frac{1}{2}$	(36) $\cos 60^\circ$ $= \frac{1}{2}$	(51) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(66) $\tan 60^\circ$ $= \sqrt{3}$	(81) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(7) $\tan 45^\circ$ $= 1$	(22) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(37) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(52) $\sin 30^\circ$ $= \frac{1}{2}$	(67) $\tan 45^\circ$ $= 1$	(82) $\tan 60^\circ$ $= \sqrt{3}$
(8) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(23) $\cos 60^\circ$ $= \frac{1}{2}$	(38) $\sin 30^\circ$ $= \frac{1}{2}$	(53) $\tan 60^\circ$ $= \sqrt{3}$	(68) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(83) $\sin 30^\circ$ $= \frac{1}{2}$
(9) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(24) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(39) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\cos 60^\circ$ $= \frac{1}{2}$	(69) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(84) $\tan 60^\circ$ $= \sqrt{3}$
(10) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(40) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(55) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(70) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(85) $\tan 45^\circ$ $= 1$
(11) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(26) $\cos 60^\circ$ $= \frac{1}{2}$	(41) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(56) $\tan 60^\circ$ $= \sqrt{3}$	(71) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(86) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(12) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\tan 60^\circ$ $= \sqrt{3}$	(42) $\tan 45^\circ$ $= 1$	(57) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(72) $\sin 30^\circ$ $= \frac{1}{2}$	(87) $\tan 60^\circ$ $= \sqrt{3}$
(13) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(28) $\cos 60^\circ$ $= \frac{1}{2}$	(43) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(58) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(88) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(14) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(29) $\tan 45^\circ$ $= 1$	(44) $\sin 30^\circ$ $= \frac{1}{2}$	(59) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(74) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(89) $\tan 45^\circ$ $= 1$
(15) $\tan 45^\circ$ $= 1$	(30) $\cos 60^\circ$ $= \frac{1}{2}$	(45) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(60) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(75) $\cos 60^\circ$ $= \frac{1}{2}$	(90) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$

(1) $\sin 30^\circ$ $= \frac{1}{2}$	(16) $\cos 60^\circ$ $= \frac{1}{2}$	(31) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(46) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(61) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(76) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(2) $\cos 60^\circ$ $= \frac{1}{2}$	(17) $\tan 60^\circ$ $= \sqrt{3}$	(32) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(47) $\cos 60^\circ$ $= \frac{1}{2}$	(62) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(77) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\sin 30^\circ$ $= \frac{1}{2}$	(18) $\tan 45^\circ$ $= 1$	(33) $\cos 60^\circ$ $= \frac{1}{2}$	(48) $\tan 60^\circ$ $= \sqrt{3}$	(63) $\tan 60^\circ$ $= \sqrt{3}$	(78) $\sin 30^\circ$ $= \frac{1}{2}$
(4) $\cos 60^\circ$ $= \frac{1}{2}$	(19) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(34) $\cos 60^\circ$ $= \frac{1}{2}$	(49) $\sin 30^\circ$ $= \frac{1}{2}$	(64) $\cos 60^\circ$ $= \frac{1}{2}$	(79) $\cos 60^\circ$ $= \frac{1}{2}$
(5) $\tan 45^\circ$ $= 1$	(20) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(35) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(50) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(65) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(80) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(6) $\tan 60^\circ$ $= \sqrt{3}$	(21) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(36) $\tan 45^\circ$ $= 1$	(51) $\tan 60^\circ$ $= \sqrt{3}$	(66) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(81) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(7) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(22) $\sin 30^\circ$ $= \frac{1}{2}$	(37) $\cos 60^\circ$ $= \frac{1}{2}$	(52) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(67) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(82) $\tan 45^\circ$ $= 1$
(8) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\cos 60^\circ$ $= \frac{1}{2}$	(38) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(53) $\sin 30^\circ$ $= \frac{1}{2}$	(68) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(83) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(9) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(24) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(39) $\sin 30^\circ$ $= \frac{1}{2}$	(54) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(84) $\sin 30^\circ$ $= \frac{1}{2}$
(10) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\tan 45^\circ$ $= 1$	(40) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(55) $\tan 60^\circ$ $= \sqrt{3}$	(70) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(85) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(11) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(26) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(41) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(56) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(71) $\tan 45^\circ$ $= 1$	(86) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(12) $\sin 30^\circ$ $= \frac{1}{2}$	(27) $\tan 45^\circ$ $= 1$	(42) $\cos 60^\circ$ $= \frac{1}{2}$	(57) $\sin 30^\circ$ $= \frac{1}{2}$	(72) $\tan 60^\circ$ $= \sqrt{3}$	(87) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(13) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(28) $\sin 30^\circ$ $= \frac{1}{2}$	(43) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(58) $\sin 30^\circ$ $= \frac{1}{2}$	(73) $\tan 60^\circ$ $= \sqrt{3}$	(88) $\sin 30^\circ$ $= \frac{1}{2}$
(14) $\cos 60^\circ$ $= \frac{1}{2}$	(29) $\tan 60^\circ$ $= \sqrt{3}$	(44) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(59) $\tan 60^\circ$ $= \sqrt{3}$	(74) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(89) $\cos 60^\circ$ $= \frac{1}{2}$
(15) $\tan 45^\circ$ $= 1$	(30) $\sin 30^\circ$ $= \frac{1}{2}$	(45) $\tan 60^\circ$ $= \sqrt{3}$	(60) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(75) $\tan 60^\circ$ $= \sqrt{3}$	(90) $\sin 30^\circ$ $= \frac{1}{2}$

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(2) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(17) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(32) $\tan 45^\circ$ $= 1$	(47) $\cos 60^\circ$ $= \frac{1}{2}$	(62) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(77) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(3) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(18) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(33) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(48) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(63) $\tan 60^\circ$ $= \sqrt{3}$	(78) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(4) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(19) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(34) $\tan 60^\circ$ $= \sqrt{3}$	(49) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(64) $\sin 30^\circ$ $= \frac{1}{2}$	(79) $\tan 60^\circ$ $= \sqrt{3}$
(5) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(20) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(35) $\sin 30^\circ$ $= \frac{1}{2}$	(50) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(65) $\sin 30^\circ$ $= \frac{1}{2}$	(80) $\tan 60^\circ$ $= \sqrt{3}$
(6) $\tan 45^\circ$ $= 1$	(21) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(36) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(51) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(66) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(81) $\cos 60^\circ$ $= \frac{1}{2}$
(7) $\tan 60^\circ$ $= \sqrt{3}$	(22) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(37) $\sin 30^\circ$ $= \frac{1}{2}$	(52) $\tan 60^\circ$ $= \sqrt{3}$	(67) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(82) $\tan 45^\circ$ $= 1$
(8) $\sin 30^\circ$ $= \frac{1}{2}$	(23) $\sin 30^\circ$ $= \frac{1}{2}$	(38) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(53) $\cos 60^\circ$ $= \frac{1}{2}$	(68) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(83) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(9) $\cos 60^\circ$ $= \frac{1}{2}$	(24) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(39) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(84) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(10) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(25) $\tan 45^\circ$ $= 1$	(40) $\tan 45^\circ$ $= 1$	(55) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(70) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(85) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(11) $\tan 45^\circ$ $= 1$	(26) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(41) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(56) $\sin 30^\circ$ $= \frac{1}{2}$	(71) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(86) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(12) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(27) $\sin 30^\circ$ $= \frac{1}{2}$	(42) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(57) $\sin 30^\circ$ $= \frac{1}{2}$	(72) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(87) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(13) $\cos 60^\circ$ $= \frac{1}{2}$	(28) $\cos 60^\circ$ $= \frac{1}{2}$	(43) $\tan 60^\circ$ $= \sqrt{3}$	(58) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(88) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(14) $\tan 60^\circ$ $= \sqrt{3}$	(29) $\sin 30^\circ$ $= \frac{1}{2}$	(44) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(59) $\tan 45^\circ$ $= 1$	(74) $\sin 30^\circ$ $= \frac{1}{2}$	(89) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(15) $\tan 60^\circ$ $= \sqrt{3}$	(30) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(45) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(60) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(75) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(90) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$

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(2) $\sin 30^\circ$ $= \frac{1}{2}$	(17) $\tan 45^\circ$ $= 1$	(32) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(47) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(62) $\tan 45^\circ$ $= 1$	(77) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(18) $\cos 60^\circ$ $= \frac{1}{2}$	(33) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(48) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(63) $\tan 60^\circ$ $= \sqrt{3}$	(78) $\tan 60^\circ$ $= \sqrt{3}$
(4) $\cos 60^\circ$ $= \frac{1}{2}$	(19) $\tan 60^\circ$ $= \sqrt{3}$	(34) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(49) $\sin 30^\circ$ $= \frac{1}{2}$	(64) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(79) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(5) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(20) $\tan 45^\circ$ $= 1$	(35) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(50) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(65) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(80) $\tan 60^\circ$ $= \sqrt{3}$
(6) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(21) $\tan 60^\circ$ $= \sqrt{3}$	(36) $\tan 45^\circ$ $= 1$	(51) $\tan 60^\circ$ $= \sqrt{3}$	(66) $\cos 60^\circ$ $= \frac{1}{2}$	(81) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(7) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(22) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(37) $\tan 60^\circ$ $= \sqrt{3}$	(52) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(67) $\cos 60^\circ$ $= \frac{1}{2}$	(82) $\tan 60^\circ$ $= \sqrt{3}$
(8) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(38) $\cos 60^\circ$ $= \frac{1}{2}$	(53) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(68) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(83) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(9) $\tan 45^\circ$ $= 1$	(24) $\tan 60^\circ$ $= \sqrt{3}$	(39) $\cos 60^\circ$ $= \frac{1}{2}$	(54) $\tan 60^\circ$ $= \sqrt{3}$	(69) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(84) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(10) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\cos 60^\circ$ $= \frac{1}{2}$	(40) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(55) $\tan 45^\circ$ $= 1$	(70) $\tan 60^\circ$ $= \sqrt{3}$	(85) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(11) $\tan 60^\circ$ $= \sqrt{3}$	(26) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(41) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(56) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(71) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(86) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(12) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(27) $\sin 30^\circ$ $= \frac{1}{2}$	(42) $\cos 60^\circ$ $= \frac{1}{2}$	(57) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(72) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(87) $\cos 60^\circ$ $= \frac{1}{2}$
(13) $\tan 60^\circ$ $= \sqrt{3}$	(28) $\tan 60^\circ$ $= \sqrt{3}$	(43) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(58) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(73) $\tan 45^\circ$ $= 1$	(88) $\sin 30^\circ$ $= \frac{1}{2}$
(14) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(44) $\cos 60^\circ$ $= \frac{1}{2}$	(59) $\sin 30^\circ$ $= \frac{1}{2}$	(74) $\cos 60^\circ$ $= \frac{1}{2}$	(89) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(15) $\sin 30^\circ$ $= \frac{1}{2}$	(30) $\cos 60^\circ$ $= \frac{1}{2}$	(45) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(60) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(75) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(90) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$

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(2) $\tan 60^\circ$ $= \sqrt{3}$	(17) $\cos 60^\circ$ $= \frac{1}{2}$	(32) $\tan 45^\circ$ $= 1$	(47) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(62) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(77) $\tan 60^\circ$ $= \sqrt{3}$
(3) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(18) $\sin 30^\circ$ $= \frac{1}{2}$	(33) $\tan 60^\circ$ $= \sqrt{3}$	(48) $\cos 60^\circ$ $= \frac{1}{2}$	(63) $\cos 60^\circ$ $= \frac{1}{2}$	(78) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(4) $\tan 60^\circ$ $= \sqrt{3}$	(19) $\cos 60^\circ$ $= \frac{1}{2}$	(34) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(49) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(64) $\cos 60^\circ$ $= \frac{1}{2}$	(79) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(5) $\sin 30^\circ$ $= \frac{1}{2}$	(20) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(35) $\tan 45^\circ$ $= 1$	(50) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(65) $\sin 30^\circ$ $= \frac{1}{2}$	(80) $\sin 30^\circ$ $= \frac{1}{2}$
(6) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(21) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(36) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(51) $\cos 60^\circ$ $= \frac{1}{2}$	(66) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(81) $\tan 60^\circ$ $= \sqrt{3}$
(7) $\sin 30^\circ$ $= \frac{1}{2}$	(22) $\tan 45^\circ$ $= 1$	(37) $\cos 60^\circ$ $= \frac{1}{2}$	(52) $\cos 60^\circ$ $= \frac{1}{2}$	(67) $\cos 60^\circ$ $= \frac{1}{2}$	(82) $\tan 60^\circ$ $= \sqrt{3}$
(8) $\tan 45^\circ$ $= 1$	(23) $\tan 45^\circ$ $= 1$	(38) $\tan 60^\circ$ $= \sqrt{3}$	(53) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(68) $\tan 60^\circ$ $= \sqrt{3}$	(83) $\cos 60^\circ$ $= \frac{1}{2}$
(9) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(24) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(39) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(69) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(84) $\tan 45^\circ$ $= 1$
(10) $\tan 45^\circ$ $= 1$	(25) $\tan 60^\circ$ $= \sqrt{3}$	(40) $\sin 30^\circ$ $= \frac{1}{2}$	(55) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(70) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(85) $\cos 60^\circ$ $= \frac{1}{2}$
(11) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(26) $\tan 60^\circ$ $= \sqrt{3}$	(41) $\tan 60^\circ$ $= \sqrt{3}$	(56) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(71) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(86) $\tan 60^\circ$ $= \sqrt{3}$
(12) $\tan 45^\circ$ $= 1$	(27) $\cos 60^\circ$ $= \frac{1}{2}$	(42) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(57) $\sin 30^\circ$ $= \frac{1}{2}$	(72) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(87) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(13) $\tan 60^\circ$ $= \sqrt{3}$	(28) $\tan 60^\circ$ $= \sqrt{3}$	(43) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(58) $\tan 60^\circ$ $= \sqrt{3}$	(73) $\sin 30^\circ$ $= \frac{1}{2}$	(88) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(14) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(29) $\cos 60^\circ$ $= \frac{1}{2}$	(44) $\cos 60^\circ$ $= \frac{1}{2}$	(59) $\tan 45^\circ$ $= 1$	(74) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(89) $\tan 60^\circ$ $= \sqrt{3}$
(15) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(30) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(45) $\tan 60^\circ$ $= \sqrt{3}$	(60) $\tan 60^\circ$ $= \sqrt{3}$	(75) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(90) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$

(1) $\cos 60^\circ$ $= \frac{1}{2}$	(16) $\tan 45^\circ$ $= 1$	(31) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(46) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(61) $\cos 60^\circ$ $= \frac{1}{2}$	(76) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(2) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(17) $\sin 30^\circ$ $= \frac{1}{2}$	(32) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(47) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(62) $\tan 45^\circ$ $= 1$	(77) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\cos 60^\circ$ $= \frac{1}{2}$	(18) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(33) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(48) $\sin 30^\circ$ $= \frac{1}{2}$	(63) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(78) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(4) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(19) $\tan 60^\circ$ $= \sqrt{3}$	(34) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(49) $\tan 60^\circ$ $= \sqrt{3}$	(64) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(79) $\tan 45^\circ$ $= 1$
(5) $\tan 45^\circ$ $= 1$	(20) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(35) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(50) $\tan 45^\circ$ $= 1$	(65) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(80) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(6) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(21) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(36) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(51) $\sin 30^\circ$ $= \frac{1}{2}$	(66) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(81) $\cos 60^\circ$ $= \frac{1}{2}$
(7) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(22) $\cos 60^\circ$ $= \frac{1}{2}$	(37) $\tan 60^\circ$ $= \sqrt{3}$	(52) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(67) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(82) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(8) $\tan 60^\circ$ $= \sqrt{3}$	(23) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(38) $\tan 60^\circ$ $= \sqrt{3}$	(53) $\tan 45^\circ$ $= 1$	(68) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(83) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(9) $\sin 30^\circ$ $= \frac{1}{2}$	(24) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(39) $\cos 60^\circ$ $= \frac{1}{2}$	(54) $\tan 45^\circ$ $= 1$	(69) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(84) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(10) $\sin 30^\circ$ $= \frac{1}{2}$	(25) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(40) $\cos 60^\circ$ $= \frac{1}{2}$	(55) $\cos 60^\circ$ $= \frac{1}{2}$	(70) $\tan 45^\circ$ $= 1$	(85) $\tan 45^\circ$ $= 1$
(11) $\tan 45^\circ$ $= 1$	(26) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(41) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(56) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(71) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(86) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(12) $\tan 60^\circ$ $= \sqrt{3}$	(27) $\tan 60^\circ$ $= \sqrt{3}$	(42) $\cos 60^\circ$ $= \frac{1}{2}$	(57) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(72) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(87) $\cos 60^\circ$ $= \frac{1}{2}$
(13) $\tan 60^\circ$ $= \sqrt{3}$	(28) $\tan 60^\circ$ $= \sqrt{3}$	(43) $\sin 30^\circ$ $= \frac{1}{2}$	(58) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\sin 30^\circ$ $= \frac{1}{2}$	(88) $\tan 45^\circ$ $= 1$
(14) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(29) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(44) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(59) $\tan 45^\circ$ $= 1$	(74) $\tan 60^\circ$ $= \sqrt{3}$	(89) $\tan 60^\circ$ $= \sqrt{3}$
(15) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(30) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(45) $\sin 30^\circ$ $= \frac{1}{2}$	(60) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(75) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(90) $\cos 60^\circ$ $= \frac{1}{2}$

(1) $\tan 60^\circ$ $= \sqrt{3}$	(16) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(31) $\sin 30^\circ$ $= \frac{1}{2}$	(46) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(61) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(76) $\sin 30^\circ$ $= \frac{1}{2}$
(2) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(17) $\sin 30^\circ$ $= \frac{1}{2}$	(32) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(47) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(62) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(77) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\cos 60^\circ$ $= \frac{1}{2}$	(18) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(33) $\sin 30^\circ$ $= \frac{1}{2}$	(48) $\tan 45^\circ$ $= 1$	(63) $\cos 60^\circ$ $= \frac{1}{2}$	(78) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(4) $\sin 30^\circ$ $= \frac{1}{2}$	(19) $\sin 30^\circ$ $= \frac{1}{2}$	(34) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(49) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(64) $\tan 60^\circ$ $= \sqrt{3}$	(79) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(5) $\tan 45^\circ$ $= 1$	(20) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(35) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(50) $\sin 30^\circ$ $= \frac{1}{2}$	(65) $\tan 60^\circ$ $= \sqrt{3}$	(80) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(6) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(21) $\tan 45^\circ$ $= 1$	(36) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(51) $\cos 60^\circ$ $= \frac{1}{2}$	(66) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(81) $\tan 45^\circ$ $= 1$
(7) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(22) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(37) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(52) $\sin 30^\circ$ $= \frac{1}{2}$	(67) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(82) $\tan 60^\circ$ $= \sqrt{3}$
(8) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(38) $\cos 60^\circ$ $= \frac{1}{2}$	(53) $\tan 45^\circ$ $= 1$	(68) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(83) $\tan 45^\circ$ $= 1$
(9) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(24) $\tan 45^\circ$ $= 1$	(39) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(54) $\sin 30^\circ$ $= \frac{1}{2}$	(69) $\cos 60^\circ$ $= \frac{1}{2}$	(84) $\cos 60^\circ$ $= \frac{1}{2}$
(10) $\tan 45^\circ$ $= 1$	(25) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(40) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(55) $\sin 30^\circ$ $= \frac{1}{2}$	(70) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(85) $\sin 30^\circ$ $= \frac{1}{2}$
(11) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(26) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(41) $\sin 30^\circ$ $= \frac{1}{2}$	(56) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(71) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(86) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(12) $\tan 60^\circ$ $= \sqrt{3}$	(27) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(42) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(57) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(72) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(87) $\sin 30^\circ$ $= \frac{1}{2}$
(13) $\tan 60^\circ$ $= \sqrt{3}$	(28) $\cos 60^\circ$ $= \frac{1}{2}$	(43) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(58) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(73) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(88) $\cos 60^\circ$ $= \frac{1}{2}$
(14) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(44) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(59) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(74) $\cos 60^\circ$ $= \frac{1}{2}$	(89) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(15) $\tan 60^\circ$ $= \sqrt{3}$	(30) $\tan 60^\circ$ $= \sqrt{3}$	(45) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(60) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(75) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(90) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$

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(2) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(17) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(32) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(47) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(62) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(77) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(3) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(18) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(33) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(48) $\sin 30^\circ$ $= \frac{1}{2}$	(63) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(78) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(4) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(19) $\tan 45^\circ$ $= 1$	(34) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(49) $\sin 30^\circ$ $= \frac{1}{2}$	(64) $\tan 60^\circ$ $= \sqrt{3}$	(79) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(5) $\cos 60^\circ$ $= \frac{1}{2}$	(20) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(35) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(50) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(65) $\tan 45^\circ$ $= 1$	(80) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(6) $\tan 45^\circ$ $= 1$	(21) $\tan 60^\circ$ $= \sqrt{3}$	(36) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(51) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(66) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(81) $\sin 30^\circ$ $= \frac{1}{2}$
(7) $\sin 30^\circ$ $= \frac{1}{2}$	(22) $\sin 30^\circ$ $= \frac{1}{2}$	(37) $\sin 30^\circ$ $= \frac{1}{2}$	(52) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(67) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(82) $\cos 60^\circ$ $= \frac{1}{2}$
(8) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(23) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(38) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(53) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(68) $\tan 60^\circ$ $= \sqrt{3}$	(83) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(9) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(24) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(39) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\cos 60^\circ$ $= \frac{1}{2}$	(69) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(84) $\sin 30^\circ$ $= \frac{1}{2}$
(10) $\tan 60^\circ$ $= \sqrt{3}$	(25) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(40) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(55) $\tan 45^\circ$ $= 1$	(70) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(85) $\tan 60^\circ$ $= \sqrt{3}$
(11) $\tan 60^\circ$ $= \sqrt{3}$	(26) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(41) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(56) $\tan 60^\circ$ $= \sqrt{3}$	(71) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(86) $\tan 45^\circ$ $= 1$
(12) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\sin 30^\circ$ $= \frac{1}{2}$	(42) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(57) $\tan 60^\circ$ $= \sqrt{3}$	(72) $\tan 60^\circ$ $= \sqrt{3}$	(87) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(13) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(28) $\tan 60^\circ$ $= \sqrt{3}$	(43) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(58) $\tan 60^\circ$ $= \sqrt{3}$	(73) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(88) $\sin 30^\circ$ $= \frac{1}{2}$
(14) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(29) $\sin 30^\circ$ $= \frac{1}{2}$	(44) $\tan 60^\circ$ $= \sqrt{3}$	(59) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(74) $\tan 45^\circ$ $= 1$	(89) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(15) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(30) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(45) $\tan 45^\circ$ $= 1$	(60) $\cos 60^\circ$ $= \frac{1}{2}$	(75) $\tan 60^\circ$ $= \sqrt{3}$	(90) $\tan 45^\circ$ $= 1$

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(2) $\tan 45^\circ$ $= 1$	(17) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(32) $\cos 60^\circ$ $= \frac{1}{2}$	(47) $\sin 30^\circ$ $= \frac{1}{2}$	(62) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(77) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(3) $\sin 30^\circ$ $= \frac{1}{2}$	(18) $\tan 45^\circ$ $= 1$	(33) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(48) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(63) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(78) $\tan 60^\circ$ $= \sqrt{3}$
(4) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(19) $\sin 30^\circ$ $= \frac{1}{2}$	(34) $\cos 60^\circ$ $= \frac{1}{2}$	(49) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(64) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(79) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(5) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(20) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(35) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(50) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(65) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(80) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(6) $\tan 60^\circ$ $= \sqrt{3}$	(21) $\sin 30^\circ$ $= \frac{1}{2}$	(36) $\tan 45^\circ$ $= 1$	(51) $\cos 60^\circ$ $= \frac{1}{2}$	(66) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(81) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(7) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(22) $\tan 60^\circ$ $= \sqrt{3}$	(37) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(52) $\tan 60^\circ$ $= \sqrt{3}$	(67) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(82) $\tan 45^\circ$ $= 1$
(8) $\sin 30^\circ$ $= \frac{1}{2}$	(23) $\cos 60^\circ$ $= \frac{1}{2}$	(38) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(53) $\tan 60^\circ$ $= \sqrt{3}$	(68) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(83) $\tan 60^\circ$ $= \sqrt{3}$
(9) $\cos 60^\circ$ $= \frac{1}{2}$	(24) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(39) $\tan 60^\circ$ $= \sqrt{3}$	(54) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(69) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(84) $\sin 30^\circ$ $= \frac{1}{2}$
(10) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(25) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(40) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(55) $\tan 45^\circ$ $= 1$	(70) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(85) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(11) $\tan 60^\circ$ $= \sqrt{3}$	(26) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(41) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(56) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(71) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(86) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(12) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(27) $\tan 60^\circ$ $= \sqrt{3}$	(42) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(57) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(72) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(87) $\cos 60^\circ$ $= \frac{1}{2}$
(13) $\sin 30^\circ$ $= \frac{1}{2}$	(28) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(43) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(58) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\sin 30^\circ$ $= \frac{1}{2}$	(88) $\sin 30^\circ$ $= \frac{1}{2}$
(14) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(29) $\tan 60^\circ$ $= \sqrt{3}$	(44) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(59) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(74) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(89) $\tan 60^\circ$ $= \sqrt{3}$
(15) $\sin 30^\circ$ $= \frac{1}{2}$	(30) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(45) $\tan 45^\circ$ $= 1$	(60) $\cos 60^\circ$ $= \frac{1}{2}$	(75) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(90) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$

(1) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(16) $\cos 60^\circ$ $= \frac{1}{2}$	(31) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(46) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(61) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(76) $\cos 60^\circ$ $= \frac{1}{2}$
(2) $\sin 30^\circ$ $= \frac{1}{2}$	(17) $\sin 30^\circ$ $= \frac{1}{2}$	(32) $\tan 45^\circ$ $= 1$	(47) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(62) $\sin 30^\circ$ $= \frac{1}{2}$	(77) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(18) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(33) $\sin 30^\circ$ $= \frac{1}{2}$	(48) $\sin 30^\circ$ $= \frac{1}{2}$	(63) $\cos 60^\circ$ $= \frac{1}{2}$	(78) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(4) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(19) $\cos 60^\circ$ $= \frac{1}{2}$	(34) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(49) $\tan 60^\circ$ $= \sqrt{3}$	(64) $\sin 30^\circ$ $= \frac{1}{2}$	(79) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(5) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(20) $\tan 45^\circ$ $= 1$	(35) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(50) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(65) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(80) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(6) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(21) $\sin 30^\circ$ $= \frac{1}{2}$	(36) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(51) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(66) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(81) $\sin 30^\circ$ $= \frac{1}{2}$
(7) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(22) $\sin 30^\circ$ $= \frac{1}{2}$	(37) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(52) $\cos 60^\circ$ $= \frac{1}{2}$	(67) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(82) $\sin 30^\circ$ $= \frac{1}{2}$
(8) $\sin 30^\circ$ $= \frac{1}{2}$	(23) $\sin 30^\circ$ $= \frac{1}{2}$	(38) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(53) $\tan 45^\circ$ $= 1$	(68) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(83) $\tan 60^\circ$ $= \sqrt{3}$
(9) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(24) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(39) $\tan 60^\circ$ $= \sqrt{3}$	(54) $\cos 60^\circ$ $= \frac{1}{2}$	(69) $\cos 60^\circ$ $= \frac{1}{2}$	(84) $\cos 60^\circ$ $= \frac{1}{2}$
(10) $\sin 30^\circ$ $= \frac{1}{2}$	(25) $\tan 60^\circ$ $= \sqrt{3}$	(40) $\cos 60^\circ$ $= \frac{1}{2}$	(55) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(70) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(85) $\tan 60^\circ$ $= \sqrt{3}$
(11) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(26) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(41) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(56) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(71) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(86) $\cos 60^\circ$ $= \frac{1}{2}$
(12) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(42) $\cos 60^\circ$ $= \frac{1}{2}$	(57) $\cos 60^\circ$ $= \frac{1}{2}$	(72) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(87) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(13) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(28) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(43) $\tan 60^\circ$ $= \sqrt{3}$	(58) $\cos 60^\circ$ $= \frac{1}{2}$	(73) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(88) $\cos 60^\circ$ $= \frac{1}{2}$
(14) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(29) $\tan 45^\circ$ $= 1$	(44) $\sin 30^\circ$ $= \frac{1}{2}$	(59) $\tan 45^\circ$ $= 1$	(74) $\sin 30^\circ$ $= \frac{1}{2}$	(89) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(15) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(30) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(45) $\sin 30^\circ$ $= \frac{1}{2}$	(60) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(75) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(90) $\tan 45^\circ$ $= 1$

(1) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(16) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(31) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(46) $\tan 45^\circ$ $= 1$	(61) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(76) $\sin 30^\circ$ $= \frac{1}{2}$
(2) $\tan 60^\circ$ $= \sqrt{3}$	(17) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(32) $\tan 60^\circ$ $= \sqrt{3}$	(47) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(62) $\tan 45^\circ$ $= 1$	(77) $\sin 30^\circ$ $= \frac{1}{2}$
(3) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(18) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(33) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(48) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(63) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(78) $\cos 60^\circ$ $= \frac{1}{2}$
(4) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(19) $\tan 45^\circ$ $= 1$	(34) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(49) $\cos 60^\circ$ $= \frac{1}{2}$	(64) $\cos 60^\circ$ $= \frac{1}{2}$	(79) $\sin 30^\circ$ $= \frac{1}{2}$
(5) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(20) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(35) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(50) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(65) $\cos 60^\circ$ $= \frac{1}{2}$	(80) $\cos 60^\circ$ $= \frac{1}{2}$
(6) $\cos 60^\circ$ $= \frac{1}{2}$	(21) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(36) $\tan 60^\circ$ $= \sqrt{3}$	(51) $\sin 30^\circ$ $= \frac{1}{2}$	(66) $\sin 30^\circ$ $= \frac{1}{2}$	(81) $\sin 30^\circ$ $= \frac{1}{2}$
(7) $\tan 45^\circ$ $= 1$	(22) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(37) $\cos 60^\circ$ $= \frac{1}{2}$	(52) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(67) $\tan 60^\circ$ $= \sqrt{3}$	(82) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(8) $\sin 30^\circ$ $= \frac{1}{2}$	(23) $\cos 60^\circ$ $= \frac{1}{2}$	(38) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(53) $\tan 60^\circ$ $= \sqrt{3}$	(68) $\sin 30^\circ$ $= \frac{1}{2}$	(83) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(9) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(24) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(39) $\sin 30^\circ$ $= \frac{1}{2}$	(54) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(69) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(84) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(10) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(25) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(40) $\tan 45^\circ$ $= 1$	(55) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(70) $\cos 60^\circ$ $= \frac{1}{2}$	(85) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(11) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(26) $\tan 45^\circ$ $= 1$	(41) $\sin 30^\circ$ $= \frac{1}{2}$	(56) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(71) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(86) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(12) $\cos 60^\circ$ $= \frac{1}{2}$	(27) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(42) $\tan 60^\circ$ $= \sqrt{3}$	(57) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(72) $\tan 45^\circ$ $= 1$	(87) $\tan 45^\circ$ $= 1$
(13) $\sin 30^\circ$ $= \frac{1}{2}$	(28) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(43) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(58) $\tan 60^\circ$ $= \sqrt{3}$	(73) $\sin 30^\circ$ $= \frac{1}{2}$	(88) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(14) $\sin 30^\circ$ $= \frac{1}{2}$	(29) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(44) $\tan 60^\circ$ $= \sqrt{3}$	(59) $\cos 60^\circ$ $= \frac{1}{2}$	(74) $\sin 30^\circ$ $= \frac{1}{2}$	(89) $\sin 30^\circ$ $= \frac{1}{2}$
(15) $\tan 60^\circ$ $= \sqrt{3}$	(30) $\sin 30^\circ$ $= \frac{1}{2}$	(45) $\sin 30^\circ$ $= \frac{1}{2}$	(60) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(75) $\cos 60^\circ$ $= \frac{1}{2}$	(90) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$

3.2.2 三角比 $0^\circ \leq \theta \leq 180^\circ$

三角比 $0^\circ \leq \theta \leq 180^\circ$ の値。

(1) $\sin 90^\circ$	(16) $\tan 150^\circ$	(31) $\sin 60^\circ$	(46) $\cos 90^\circ$	(61) $\cos 90^\circ$
(2) $\tan 60^\circ$	(17) $\tan 30^\circ$	(32) $\cos 90^\circ$	(47) $\tan 135^\circ$	(62) $\tan 30^\circ$
(3) $\sin 90^\circ$	(18) $\tan 180^\circ$	(33) $\tan 135^\circ$	(48) $\tan 45^\circ$	(63) $\sin 60^\circ$
(4) $\sin 45^\circ$	(19) $\tan 30^\circ$	(34) $\sin 180^\circ$	(49) $\cos 90^\circ$	(64) $\tan 90^\circ$
(5) $\tan 120^\circ$	(20) $\sin 0^\circ$	(35) $\sin 30^\circ$	(50) $\tan 90^\circ$	(65) $\cos 150^\circ$
(6) $\sin 30^\circ$	(21) $\cos 180^\circ$	(36) $\tan 90^\circ$	(51) $\cos 180^\circ$	(66) $\cos 180^\circ$
(7) $\cos 150^\circ$	(22) $\sin 0^\circ$	(37) $\cos 180^\circ$	(52) $\cos 90^\circ$	(67) $\tan 180^\circ$
(8) $\sin 45^\circ$	(23) $\cos 90^\circ$	(38) $\cos 60^\circ$	(53) $\sin 60^\circ$	(68) $\tan 45^\circ$
(9) $\cos 180^\circ$	(24) $\tan 135^\circ$	(39) $\sin 150^\circ$	(54) $\sin 60^\circ$	(69) $\cos 120^\circ$
(10) $\sin 30^\circ$	(25) $\sin 150^\circ$	(40) $\cos 150^\circ$	(55) $\tan 90^\circ$	(70) $\tan 120^\circ$
(11) $\tan 90^\circ$	(26) $\tan 135^\circ$	(41) $\sin 180^\circ$	(56) $\cos 150^\circ$	(71) $\cos 135^\circ$
(12) $\sin 45^\circ$	(27) $\cos 90^\circ$	(42) $\tan 45^\circ$	(57) $\cos 30^\circ$	(72) $\sin 90^\circ$
(13) $\sin 0^\circ$	(28) $\cos 30^\circ$	(43) $\sin 60^\circ$	(58) $\sin 135^\circ$	(73) $\sin 120^\circ$
(14) $\tan 90^\circ$	(29) $\sin 120^\circ$	(44) $\tan 180^\circ$	(59) $\cos 180^\circ$	(74) $\sin 90^\circ$
(15) $\cos 120^\circ$	(30) $\cos 180^\circ$	(45) $\tan 135^\circ$	(60) $\sin 90^\circ$	(75) $\tan 30^\circ$

(1) $\tan 90^\circ$	(16) $\cos 150^\circ$	(31) $\tan 135^\circ$	(46) $\cos 60^\circ$	(61) $\cos 60^\circ$
(2) $\tan 90^\circ$	(17) $\tan 90^\circ$	(32) $\tan 150^\circ$	(47) $\tan 45^\circ$	(62) $\cos 150^\circ$
(3) $\sin 90^\circ$	(18) $\cos 90^\circ$	(33) $\cos 90^\circ$	(48) $\cos 30^\circ$	(63) $\cos 135^\circ$
(4) $\sin 45^\circ$	(19) $\cos 90^\circ$	(34) $\sin 45^\circ$	(49) $\tan 150^\circ$	(64) $\sin 150^\circ$
(5) $\sin 90^\circ$	(20) $\tan 60^\circ$	(35) $\tan 90^\circ$	(50) $\sin 120^\circ$	(65) $\cos 150^\circ$
(6) $\sin 180^\circ$	(21) $\tan 0^\circ$	(36) $\cos 135^\circ$	(51) $\tan 60^\circ$	(66) $\cos 120^\circ$
(7) $\tan 120^\circ$	(22) $\sin 0^\circ$	(37) $\cos 60^\circ$	(52) $\tan 180^\circ$	(67) $\sin 135^\circ$
(8) $\sin 0^\circ$	(23) $\cos 180^\circ$	(38) $\tan 180^\circ$	(53) $\sin 45^\circ$	(68) $\cos 135^\circ$
(9) $\sin 135^\circ$	(24) $\cos 90^\circ$	(39) $\sin 90^\circ$	(54) $\cos 60^\circ$	(69) $\cos 90^\circ$
(10) $\tan 0^\circ$	(25) $\sin 180^\circ$	(40) $\tan 0^\circ$	(55) $\sin 150^\circ$	(70) $\tan 150^\circ$
(11) $\sin 120^\circ$	(26) $\tan 135^\circ$	(41) $\tan 120^\circ$	(56) $\sin 0^\circ$	(71) $\tan 90^\circ$
(12) $\tan 150^\circ$	(27) $\tan 135^\circ$	(42) $\tan 60^\circ$	(57) $\cos 120^\circ$	(72) $\sin 30^\circ$
(13) $\sin 60^\circ$	(28) $\sin 0^\circ$	(43) $\cos 90^\circ$	(58) $\cos 135^\circ$	(73) $\cos 0^\circ$
(14) $\sin 30^\circ$	(29) $\tan 120^\circ$	(44) $\cos 150^\circ$	(59) $\tan 45^\circ$	(74) $\sin 60^\circ$
(15) $\sin 30^\circ$	(30) $\cos 30^\circ$	(45) $\cos 120^\circ$	(60) $\cos 0^\circ$	(75) $\sin 45^\circ$

(1) $\cos 135^\circ$	(16) $\cos 135^\circ$	(31) $\cos 150^\circ$	(46) $\sin 0^\circ$	(61) $\sin 135^\circ$
(2) $\tan 0^\circ$	(17) $\tan 180^\circ$	(32) $\sin 120^\circ$	(47) $\cos 60^\circ$	(62) $\cos 135^\circ$
(3) $\cos 180^\circ$	(18) $\tan 30^\circ$	(33) $\sin 135^\circ$	(48) $\cos 135^\circ$	(63) $\cos 30^\circ$
(4) $\sin 60^\circ$	(19) $\cos 180^\circ$	(34) $\cos 30^\circ$	(49) $\cos 0^\circ$	(64) $\cos 180^\circ$
(5) $\tan 180^\circ$	(20) $\tan 150^\circ$	(35) $\cos 90^\circ$	(50) $\cos 45^\circ$	(65) $\sin 30^\circ$
(6) $\cos 0^\circ$	(21) $\tan 135^\circ$	(36) $\cos 150^\circ$	(51) $\sin 150^\circ$	(66) $\cos 90^\circ$
(7) $\tan 150^\circ$	(22) $\tan 135^\circ$	(37) $\sin 0^\circ$	(52) $\sin 30^\circ$	(67) $\sin 120^\circ$
(8) $\sin 135^\circ$	(23) $\sin 90^\circ$	(38) $\sin 180^\circ$	(53) $\cos 0^\circ$	(68) $\cos 90^\circ$
(9) $\tan 45^\circ$	(24) $\sin 0^\circ$	(39) $\sin 90^\circ$	(54) $\sin 135^\circ$	(69) $\sin 60^\circ$
(10) $\cos 120^\circ$	(25) $\cos 45^\circ$	(40) $\cos 135^\circ$	(55) $\tan 60^\circ$	(70) $\cos 45^\circ$
(11) $\cos 180^\circ$	(26) $\sin 60^\circ$	(41) $\sin 45^\circ$	(56) $\cos 0^\circ$	(71) $\sin 90^\circ$
(12) $\tan 90^\circ$	(27) $\cos 90^\circ$	(42) $\tan 120^\circ$	(57) $\sin 120^\circ$	(72) $\cos 90^\circ$
(13) $\sin 150^\circ$	(28) $\cos 150^\circ$	(43) $\cos 60^\circ$	(58) $\cos 150^\circ$	(73) $\sin 45^\circ$
(14) $\tan 135^\circ$	(29) $\tan 150^\circ$	(44) $\tan 180^\circ$	(59) $\tan 0^\circ$	(74) $\sin 135^\circ$
(15) $\tan 60^\circ$	(30) $\tan 90^\circ$	(45) $\tan 150^\circ$	(60) $\cos 30^\circ$	(75) $\tan 90^\circ$

(1) $\sin 135^\circ$	(16) $\tan 180^\circ$	(31) $\sin 30^\circ$	(46) $\sin 0^\circ$	(61) $\tan 135^\circ$
(2) $\tan 180^\circ$	(17) $\cos 180^\circ$	(32) $\cos 120^\circ$	(47) $\cos 120^\circ$	(62) $\sin 120^\circ$
(3) $\sin 150^\circ$	(18) $\sin 90^\circ$	(33) $\sin 0^\circ$	(48) $\cos 180^\circ$	(63) $\tan 135^\circ$
(4) $\sin 30^\circ$	(19) $\sin 90^\circ$	(34) $\cos 45^\circ$	(49) $\tan 45^\circ$	(64) $\sin 150^\circ$
(5) $\tan 150^\circ$	(20) $\sin 90^\circ$	(35) $\cos 90^\circ$	(50) $\cos 60^\circ$	(65) $\cos 120^\circ$
(6) $\tan 180^\circ$	(21) $\tan 120^\circ$	(36) $\sin 150^\circ$	(51) $\sin 180^\circ$	(66) $\sin 0^\circ$
(7) $\sin 0^\circ$	(22) $\sin 45^\circ$	(37) $\sin 150^\circ$	(52) $\sin 45^\circ$	(67) $\sin 0^\circ$
(8) $\tan 180^\circ$	(23) $\tan 120^\circ$	(38) $\cos 120^\circ$	(53) $\sin 180^\circ$	(68) $\cos 150^\circ$
(9) $\sin 30^\circ$	(24) $\sin 30^\circ$	(39) $\tan 30^\circ$	(54) $\sin 0^\circ$	(69) $\tan 135^\circ$
(10) $\sin 45^\circ$	(25) $\cos 30^\circ$	(40) $\sin 135^\circ$	(55) $\sin 135^\circ$	(70) $\tan 0^\circ$
(11) $\cos 90^\circ$	(26) $\cos 90^\circ$	(41) $\tan 90^\circ$	(56) $\cos 180^\circ$	(71) $\cos 60^\circ$
(12) $\cos 180^\circ$	(27) $\cos 120^\circ$	(42) $\cos 180^\circ$	(57) $\sin 120^\circ$	(72) $\cos 180^\circ$
(13) $\tan 0^\circ$	(28) $\cos 60^\circ$	(43) $\tan 120^\circ$	(58) $\sin 45^\circ$	(73) $\sin 135^\circ$
(14) $\sin 180^\circ$	(29) $\tan 135^\circ$	(44) $\tan 30^\circ$	(59) $\cos 90^\circ$	(74) $\cos 150^\circ$
(15) $\cos 120^\circ$	(30) $\sin 150^\circ$	(45) $\sin 150^\circ$	(60) $\tan 30^\circ$	(75) $\tan 135^\circ$

(1) $\cos 30^\circ$	(16) $\tan 150^\circ$	(31) $\sin 0^\circ$	(46) $\cos 45^\circ$	(61) $\sin 135^\circ$
(2) $\sin 150^\circ$	(17) $\tan 30^\circ$	(32) $\sin 180^\circ$	(47) $\tan 0^\circ$	(62) $\tan 135^\circ$
(3) $\tan 150^\circ$	(18) $\cos 0^\circ$	(33) $\tan 90^\circ$	(48) $\tan 45^\circ$	(63) $\tan 30^\circ$
(4) $\tan 120^\circ$	(19) $\cos 60^\circ$	(34) $\sin 0^\circ$	(49) $\cos 135^\circ$	(64) $\sin 120^\circ$
(5) $\sin 45^\circ$	(20) $\cos 0^\circ$	(35) $\tan 30^\circ$	(50) $\tan 90^\circ$	(65) $\cos 45^\circ$
(6) $\sin 0^\circ$	(21) $\cos 0^\circ$	(36) $\cos 90^\circ$	(51) $\cos 90^\circ$	(66) $\tan 90^\circ$
(7) $\cos 135^\circ$	(22) $\sin 150^\circ$	(37) $\cos 180^\circ$	(52) $\sin 0^\circ$	(67) $\sin 150^\circ$
(8) $\cos 45^\circ$	(23) $\tan 60^\circ$	(38) $\tan 180^\circ$	(53) $\cos 180^\circ$	(68) $\tan 150^\circ$
(9) $\cos 45^\circ$	(24) $\sin 90^\circ$	(39) $\sin 135^\circ$	(54) $\tan 120^\circ$	(69) $\cos 90^\circ$
(10) $\sin 180^\circ$	(25) $\tan 120^\circ$	(40) $\sin 30^\circ$	(55) $\tan 180^\circ$	(70) $\sin 0^\circ$
(11) $\tan 0^\circ$	(26) $\cos 180^\circ$	(41) $\cos 60^\circ$	(56) $\sin 180^\circ$	(71) $\sin 60^\circ$
(12) $\tan 45^\circ$	(27) $\tan 45^\circ$	(42) $\cos 90^\circ$	(57) $\sin 90^\circ$	(72) $\sin 60^\circ$
(13) $\cos 120^\circ$	(28) $\tan 180^\circ$	(43) $\sin 135^\circ$	(58) $\cos 120^\circ$	(73) $\cos 120^\circ$
(14) $\cos 150^\circ$	(29) $\sin 150^\circ$	(44) $\cos 150^\circ$	(59) $\cos 120^\circ$	(74) $\cos 60^\circ$
(15) $\sin 0^\circ$	(30) $\sin 150^\circ$	(45) $\sin 90^\circ$	(60) $\sin 90^\circ$	(75) $\cos 0^\circ$

(1) $\sin 135^\circ$	(16) $\cos 45^\circ$	(31) $\tan 135^\circ$	(46) $\sin 45^\circ$	(61) $\sin 0^\circ$
(2) $\sin 135^\circ$	(17) $\cos 30^\circ$	(32) $\sin 45^\circ$	(47) $\cos 0^\circ$	(62) $\sin 135^\circ$
(3) $\cos 120^\circ$	(18) $\sin 135^\circ$	(33) $\tan 0^\circ$	(48) $\sin 45^\circ$	(63) $\cos 60^\circ$
(4) $\tan 150^\circ$	(19) $\cos 180^\circ$	(34) $\cos 180^\circ$	(49) $\tan 120^\circ$	(64) $\cos 135^\circ$
(5) $\tan 60^\circ$	(20) $\sin 120^\circ$	(35) $\cos 150^\circ$	(50) $\cos 30^\circ$	(65) $\cos 60^\circ$
(6) $\sin 120^\circ$	(21) $\sin 135^\circ$	(36) $\sin 60^\circ$	(51) $\tan 135^\circ$	(66) $\cos 60^\circ$
(7) $\cos 0^\circ$	(22) $\tan 135^\circ$	(37) $\sin 0^\circ$	(52) $\sin 90^\circ$	(67) $\sin 45^\circ$
(8) $\sin 60^\circ$	(23) $\tan 45^\circ$	(38) $\tan 180^\circ$	(53) $\cos 0^\circ$	(68) $\sin 0^\circ$
(9) $\sin 0^\circ$	(24) $\cos 180^\circ$	(39) $\cos 90^\circ$	(54) $\sin 0^\circ$	(69) $\sin 120^\circ$
(10) $\tan 30^\circ$	(25) $\sin 0^\circ$	(40) $\sin 90^\circ$	(55) $\sin 90^\circ$	(70) $\cos 30^\circ$
(11) $\tan 135^\circ$	(26) $\sin 30^\circ$	(41) $\sin 120^\circ$	(56) $\sin 120^\circ$	(71) $\sin 135^\circ$
(12) $\tan 0^\circ$	(27) $\cos 30^\circ$	(42) $\cos 135^\circ$	(57) $\sin 90^\circ$	(72) $\tan 120^\circ$
(13) $\cos 180^\circ$	(28) $\cos 45^\circ$	(43) $\sin 60^\circ$	(58) $\cos 45^\circ$	(73) $\cos 120^\circ$
(14) $\sin 135^\circ$	(29) $\tan 30^\circ$	(44) $\tan 120^\circ$	(59) $\tan 90^\circ$	(74) $\tan 0^\circ$
(15) $\sin 90^\circ$	(30) $\tan 180^\circ$	(45) $\cos 0^\circ$	(60) $\tan 0^\circ$	(75) $\cos 150^\circ$

(1) $\sin 90^\circ$	(16) $\sin 90^\circ$	(31) $\sin 90^\circ$	(46) $\sin 135^\circ$	(61) $\tan 90^\circ$
(2) $\sin 45^\circ$	(17) $\sin 120^\circ$	(32) $\tan 120^\circ$	(47) $\cos 30^\circ$	(62) $\tan 30^\circ$
(3) $\sin 0^\circ$	(18) $\tan 30^\circ$	(33) $\tan 180^\circ$	(48) $\cos 135^\circ$	(63) $\cos 120^\circ$
(4) $\sin 135^\circ$	(19) $\sin 135^\circ$	(34) $\cos 135^\circ$	(49) $\cos 45^\circ$	(64) $\cos 150^\circ$
(5) $\cos 0^\circ$	(20) $\cos 135^\circ$	(35) $\tan 45^\circ$	(50) $\cos 90^\circ$	(65) $\sin 135^\circ$
(6) $\sin 60^\circ$	(21) $\sin 120^\circ$	(36) $\cos 0^\circ$	(51) $\cos 0^\circ$	(66) $\sin 0^\circ$
(7) $\cos 0^\circ$	(22) $\tan 0^\circ$	(37) $\cos 90^\circ$	(52) $\sin 90^\circ$	(67) $\sin 180^\circ$
(8) $\sin 120^\circ$	(23) $\cos 180^\circ$	(38) $\tan 135^\circ$	(53) $\sin 135^\circ$	(68) $\tan 60^\circ$
(9) $\tan 60^\circ$	(24) $\cos 135^\circ$	(39) $\sin 30^\circ$	(54) $\cos 60^\circ$	(69) $\tan 180^\circ$
(10) $\tan 180^\circ$	(25) $\tan 30^\circ$	(40) $\tan 60^\circ$	(55) $\tan 150^\circ$	(70) $\cos 90^\circ$
(11) $\cos 45^\circ$	(26) $\tan 120^\circ$	(41) $\tan 45^\circ$	(56) $\sin 90^\circ$	(71) $\sin 135^\circ$
(12) $\sin 0^\circ$	(27) $\cos 45^\circ$	(42) $\sin 120^\circ$	(57) $\cos 180^\circ$	(72) $\cos 120^\circ$
(13) $\sin 135^\circ$	(28) $\sin 120^\circ$	(43) $\sin 60^\circ$	(58) $\tan 0^\circ$	(73) $\cos 135^\circ$
(14) $\cos 30^\circ$	(29) $\cos 30^\circ$	(44) $\sin 60^\circ$	(59) $\cos 90^\circ$	(74) $\sin 135^\circ$
(15) $\sin 60^\circ$	(30) $\sin 60^\circ$	(45) $\cos 90^\circ$	(60) $\tan 30^\circ$	(75) $\cos 150^\circ$

(1) $\sin 90^\circ$	(16) $\sin 60^\circ$	(31) $\sin 90^\circ$	(46) $\tan 30^\circ$	(61) $\sin 135^\circ$
(2) $\cos 45^\circ$	(17) $\tan 180^\circ$	(32) $\tan 150^\circ$	(47) $\cos 60^\circ$	(62) $\tan 120^\circ$
(3) $\sin 135^\circ$	(18) $\tan 45^\circ$	(33) $\cos 0^\circ$	(48) $\cos 45^\circ$	(63) $\cos 30^\circ$
(4) $\sin 150^\circ$	(19) $\cos 180^\circ$	(34) $\tan 150^\circ$	(49) $\cos 90^\circ$	(64) $\tan 60^\circ$
(5) $\sin 90^\circ$	(20) $\tan 180^\circ$	(35) $\tan 180^\circ$	(50) $\tan 180^\circ$	(65) $\sin 135^\circ$
(6) $\sin 0^\circ$	(21) $\sin 30^\circ$	(36) $\cos 60^\circ$	(51) $\sin 90^\circ$	(66) $\cos 90^\circ$
(7) $\tan 135^\circ$	(22) $\tan 45^\circ$	(37) $\tan 150^\circ$	(52) $\sin 135^\circ$	(67) $\cos 45^\circ$
(8) $\sin 45^\circ$	(23) $\tan 30^\circ$	(38) $\tan 45^\circ$	(53) $\tan 30^\circ$	(68) $\tan 150^\circ$
(9) $\cos 60^\circ$	(24) $\tan 60^\circ$	(39) $\cos 120^\circ$	(54) $\sin 45^\circ$	(69) $\sin 45^\circ$
(10) $\cos 90^\circ$	(25) $\tan 120^\circ$	(40) $\cos 0^\circ$	(55) $\tan 45^\circ$	(70) $\sin 60^\circ$
(11) $\tan 90^\circ$	(26) $\sin 180^\circ$	(41) $\cos 150^\circ$	(56) $\tan 90^\circ$	(71) $\cos 30^\circ$
(12) $\cos 0^\circ$	(27) $\tan 135^\circ$	(42) $\sin 45^\circ$	(57) $\tan 0^\circ$	(72) $\sin 45^\circ$
(13) $\tan 30^\circ$	(28) $\tan 45^\circ$	(43) $\tan 135^\circ$	(58) $\tan 0^\circ$	(73) $\sin 135^\circ$
(14) $\sin 0^\circ$	(29) $\sin 60^\circ$	(44) $\tan 0^\circ$	(59) $\cos 135^\circ$	(74) $\sin 180^\circ$
(15) $\sin 150^\circ$	(30) $\sin 135^\circ$	(45) $\sin 120^\circ$	(60) $\tan 150^\circ$	(75) $\cos 60^\circ$

(1) $\tan 180^\circ$	(16) $\cos 60^\circ$	(31) $\tan 30^\circ$	(46) $\tan 120^\circ$	(61) $\sin 45^\circ$
(2) $\tan 150^\circ$	(17) $\cos 45^\circ$	(32) $\tan 45^\circ$	(47) $\cos 60^\circ$	(62) $\sin 120^\circ$
(3) $\tan 135^\circ$	(18) $\cos 0^\circ$	(33) $\tan 90^\circ$	(48) $\sin 60^\circ$	(63) $\sin 0^\circ$
(4) $\sin 0^\circ$	(19) $\cos 90^\circ$	(34) $\tan 180^\circ$	(49) $\tan 135^\circ$	(64) $\cos 180^\circ$
(5) $\sin 180^\circ$	(20) $\tan 30^\circ$	(35) $\cos 45^\circ$	(50) $\cos 135^\circ$	(65) $\cos 45^\circ$
(6) $\sin 135^\circ$	(21) $\cos 180^\circ$	(36) $\cos 120^\circ$	(51) $\sin 180^\circ$	(66) $\cos 0^\circ$
(7) $\cos 180^\circ$	(22) $\tan 180^\circ$	(37) $\sin 30^\circ$	(52) $\cos 135^\circ$	(67) $\cos 0^\circ$
(8) $\cos 45^\circ$	(23) $\tan 90^\circ$	(38) $\sin 90^\circ$	(53) $\tan 0^\circ$	(68) $\tan 150^\circ$
(9) $\tan 60^\circ$	(24) $\sin 120^\circ$	(39) $\sin 150^\circ$	(54) $\tan 60^\circ$	(69) $\sin 120^\circ$
(10) $\sin 150^\circ$	(25) $\cos 0^\circ$	(40) $\tan 30^\circ$	(55) $\sin 90^\circ$	(70) $\tan 90^\circ$
(11) $\tan 135^\circ$	(26) $\tan 30^\circ$	(41) $\tan 150^\circ$	(56) $\sin 150^\circ$	(71) $\cos 0^\circ$
(12) $\cos 30^\circ$	(27) $\sin 90^\circ$	(42) $\cos 90^\circ$	(57) $\sin 180^\circ$	(72) $\cos 45^\circ$
(13) $\sin 30^\circ$	(28) $\cos 180^\circ$	(43) $\sin 60^\circ$	(58) $\tan 135^\circ$	(73) $\tan 135^\circ$
(14) $\sin 60^\circ$	(29) $\tan 30^\circ$	(44) $\sin 30^\circ$	(59) $\sin 120^\circ$	(74) $\sin 180^\circ$
(15) $\cos 45^\circ$	(30) $\sin 30^\circ$	(45) $\sin 90^\circ$	(60) $\cos 180^\circ$	(75) $\sin 45^\circ$

(1) $\tan 30^\circ$	(16) $\cos 135^\circ$	(31) $\tan 180^\circ$	(46) $\tan 0^\circ$	(61) $\sin 30^\circ$
(2) $\sin 180^\circ$	(17) $\sin 90^\circ$	(32) $\sin 0^\circ$	(47) $\cos 0^\circ$	(62) $\cos 180^\circ$
(3) $\cos 150^\circ$	(18) $\tan 150^\circ$	(33) $\tan 135^\circ$	(48) $\sin 30^\circ$	(63) $\cos 150^\circ$
(4) $\cos 90^\circ$	(19) $\tan 135^\circ$	(34) $\cos 45^\circ$	(49) $\sin 0^\circ$	(64) $\sin 180^\circ$
(5) $\sin 30^\circ$	(20) $\cos 120^\circ$	(35) $\cos 120^\circ$	(50) $\sin 30^\circ$	(65) $\sin 180^\circ$
(6) $\sin 90^\circ$	(21) $\cos 60^\circ$	(36) $\tan 150^\circ$	(51) $\cos 180^\circ$	(66) $\cos 135^\circ$
(7) $\tan 45^\circ$	(22) $\tan 150^\circ$	(37) $\tan 45^\circ$	(52) $\cos 45^\circ$	(67) $\sin 30^\circ$
(8) $\tan 30^\circ$	(23) $\cos 60^\circ$	(38) $\tan 150^\circ$	(53) $\cos 180^\circ$	(68) $\sin 135^\circ$
(9) $\tan 120^\circ$	(24) $\cos 45^\circ$	(39) $\tan 60^\circ$	(54) $\sin 45^\circ$	(69) $\cos 30^\circ$
(10) $\tan 45^\circ$	(25) $\cos 60^\circ$	(40) $\tan 90^\circ$	(55) $\cos 45^\circ$	(70) $\cos 60^\circ$
(11) $\tan 90^\circ$	(26) $\cos 60^\circ$	(41) $\tan 30^\circ$	(56) $\sin 60^\circ$	(71) $\cos 90^\circ$
(12) $\cos 180^\circ$	(27) $\cos 120^\circ$	(42) $\sin 180^\circ$	(57) $\tan 60^\circ$	(72) $\sin 30^\circ$
(13) $\tan 180^\circ$	(28) $\sin 0^\circ$	(43) $\sin 135^\circ$	(58) $\tan 150^\circ$	(73) $\tan 60^\circ$
(14) $\sin 60^\circ$	(29) $\sin 150^\circ$	(44) $\tan 60^\circ$	(59) $\cos 45^\circ$	(74) $\sin 180^\circ$
(15) $\sin 120^\circ$	(30) $\tan 150^\circ$	(45) $\tan 120^\circ$	(60) $\sin 0^\circ$	(75) $\tan 0^\circ$

(1) $\tan 90^\circ$	(16) $\tan 90^\circ$	(31) $\tan 135^\circ$	(46) $\sin 0^\circ$	(61) $\cos 90^\circ$
(2) $\sin 150^\circ$	(17) $\tan 90^\circ$	(32) $\sin 90^\circ$	(47) $\sin 45^\circ$	(62) $\tan 120^\circ$
(3) $\tan 135^\circ$	(18) $\sin 0^\circ$	(33) $\sin 45^\circ$	(48) $\sin 45^\circ$	(63) $\cos 60^\circ$
(4) $\tan 45^\circ$	(19) $\tan 120^\circ$	(34) $\tan 30^\circ$	(49) $\sin 60^\circ$	(64) $\sin 0^\circ$
(5) $\tan 120^\circ$	(20) $\cos 0^\circ$	(35) $\tan 135^\circ$	(50) $\sin 180^\circ$	(65) $\sin 135^\circ$
(6) $\tan 45^\circ$	(21) $\tan 150^\circ$	(36) $\tan 90^\circ$	(51) $\sin 90^\circ$	(66) $\tan 0^\circ$
(7) $\sin 0^\circ$	(22) $\tan 0^\circ$	(37) $\tan 150^\circ$	(52) $\sin 180^\circ$	(67) $\sin 0^\circ$
(8) $\cos 45^\circ$	(23) $\tan 150^\circ$	(38) $\sin 120^\circ$	(53) $\sin 0^\circ$	(68) $\sin 60^\circ$
(9) $\sin 120^\circ$	(24) $\cos 90^\circ$	(39) $\cos 0^\circ$	(54) $\cos 150^\circ$	(69) $\cos 120^\circ$
(10) $\sin 45^\circ$	(25) $\sin 180^\circ$	(40) $\sin 135^\circ$	(55) $\cos 150^\circ$	(70) $\sin 180^\circ$
(11) $\cos 45^\circ$	(26) $\sin 60^\circ$	(41) $\sin 60^\circ$	(56) $\sin 180^\circ$	(71) $\cos 60^\circ$
(12) $\sin 150^\circ$	(27) $\cos 150^\circ$	(42) $\tan 120^\circ$	(57) $\cos 90^\circ$	(72) $\sin 0^\circ$
(13) $\sin 150^\circ$	(28) $\sin 135^\circ$	(43) $\tan 60^\circ$	(58) $\tan 90^\circ$	(73) $\sin 60^\circ$
(14) $\tan 135^\circ$	(29) $\tan 60^\circ$	(44) $\cos 120^\circ$	(59) $\cos 135^\circ$	(74) $\sin 60^\circ$
(15) $\tan 30^\circ$	(30) $\sin 45^\circ$	(45) $\tan 60^\circ$	(60) $\sin 135^\circ$	(75) $\sin 150^\circ$

(1) $\sin 120^\circ$	(16) $\cos 90^\circ$	(31) $\sin 150^\circ$	(46) $\tan 135^\circ$	(61) $\sin 180^\circ$
(2) $\sin 60^\circ$	(17) $\sin 0^\circ$	(32) $\cos 45^\circ$	(47) $\sin 45^\circ$	(62) $\cos 30^\circ$
(3) $\sin 90^\circ$	(18) $\tan 0^\circ$	(33) $\cos 60^\circ$	(48) $\sin 135^\circ$	(63) $\tan 90^\circ$
(4) $\tan 60^\circ$	(19) $\cos 120^\circ$	(34) $\tan 180^\circ$	(49) $\tan 120^\circ$	(64) $\sin 0^\circ$
(5) $\cos 45^\circ$	(20) $\sin 30^\circ$	(35) $\sin 90^\circ$	(50) $\sin 45^\circ$	(65) $\tan 90^\circ$
(6) $\cos 30^\circ$	(21) $\sin 150^\circ$	(36) $\tan 90^\circ$	(51) $\tan 180^\circ$	(66) $\cos 90^\circ$
(7) $\cos 120^\circ$	(22) $\tan 45^\circ$	(37) $\tan 60^\circ$	(52) $\cos 45^\circ$	(67) $\tan 90^\circ$
(8) $\tan 120^\circ$	(23) $\cos 180^\circ$	(38) $\tan 150^\circ$	(53) $\cos 180^\circ$	(68) $\tan 45^\circ$
(9) $\sin 30^\circ$	(24) $\tan 60^\circ$	(39) $\sin 60^\circ$	(54) $\sin 135^\circ$	(69) $\tan 90^\circ$
(10) $\sin 120^\circ$	(25) $\cos 150^\circ$	(40) $\cos 30^\circ$	(55) $\cos 0^\circ$	(70) $\tan 135^\circ$
(11) $\cos 45^\circ$	(26) $\tan 150^\circ$	(41) $\cos 135^\circ$	(56) $\sin 150^\circ$	(71) $\cos 180^\circ$
(12) $\cos 45^\circ$	(27) $\cos 30^\circ$	(42) $\cos 0^\circ$	(57) $\cos 60^\circ$	(72) $\cos 45^\circ$
(13) $\sin 30^\circ$	(28) $\sin 0^\circ$	(43) $\sin 90^\circ$	(58) $\tan 60^\circ$	(73) $\sin 90^\circ$
(14) $\tan 120^\circ$	(29) $\tan 135^\circ$	(44) $\cos 45^\circ$	(59) $\tan 45^\circ$	(74) $\sin 30^\circ$
(15) $\tan 0^\circ$	(30) $\cos 30^\circ$	(45) $\sin 0^\circ$	(60) $\sin 30^\circ$	(75) $\cos 45^\circ$

(1) $\sin 180^\circ$	(16) $\sin 135^\circ$	(31) $\cos 60^\circ$	(46) $\tan 120^\circ$	(61) $\sin 45^\circ$
(2) $\tan 0^\circ$	(17) $\cos 45^\circ$	(32) $\cos 60^\circ$	(47) $\tan 30^\circ$	(62) $\tan 30^\circ$
(3) $\cos 120^\circ$	(18) $\tan 90^\circ$	(33) $\cos 90^\circ$	(48) $\cos 135^\circ$	(63) $\tan 45^\circ$
(4) $\cos 45^\circ$	(19) $\tan 60^\circ$	(34) $\cos 150^\circ$	(49) $\sin 135^\circ$	(64) $\sin 60^\circ$
(5) $\cos 45^\circ$	(20) $\cos 45^\circ$	(35) $\cos 90^\circ$	(50) $\sin 150^\circ$	(65) $\sin 180^\circ$
(6) $\tan 150^\circ$	(21) $\sin 120^\circ$	(36) $\sin 150^\circ$	(51) $\tan 30^\circ$	(66) $\cos 45^\circ$
(7) $\cos 90^\circ$	(22) $\cos 60^\circ$	(37) $\cos 90^\circ$	(52) $\tan 150^\circ$	(67) $\tan 180^\circ$
(8) $\tan 135^\circ$	(23) $\tan 180^\circ$	(38) $\cos 0^\circ$	(53) $\tan 150^\circ$	(68) $\sin 90^\circ$
(9) $\sin 180^\circ$	(24) $\tan 135^\circ$	(39) $\sin 0^\circ$	(54) $\cos 180^\circ$	(69) $\tan 150^\circ$
(10) $\sin 120^\circ$	(25) $\sin 30^\circ$	(40) $\sin 45^\circ$	(55) $\tan 0^\circ$	(70) $\cos 90^\circ$
(11) $\cos 180^\circ$	(26) $\tan 90^\circ$	(41) $\cos 90^\circ$	(56) $\sin 0^\circ$	(71) $\sin 45^\circ$
(12) $\sin 0^\circ$	(27) $\cos 180^\circ$	(42) $\sin 30^\circ$	(57) $\sin 30^\circ$	(72) $\cos 120^\circ$
(13) $\cos 60^\circ$	(28) $\sin 135^\circ$	(43) $\tan 0^\circ$	(58) $\tan 90^\circ$	(73) $\tan 150^\circ$
(14) $\sin 60^\circ$	(29) $\sin 150^\circ$	(44) $\sin 150^\circ$	(59) $\cos 45^\circ$	(74) $\tan 60^\circ$
(15) $\cos 45^\circ$	(30) $\tan 0^\circ$	(45) $\cos 60^\circ$	(60) $\tan 150^\circ$	(75) $\sin 0^\circ$

(1) $\tan 135^\circ$	(16) $\sin 90^\circ$	(31) $\cos 150^\circ$	(46) $\sin 180^\circ$	(61) $\sin 180^\circ$
(2) $\sin 135^\circ$	(17) $\tan 180^\circ$	(32) $\cos 150^\circ$	(47) $\tan 120^\circ$	(62) $\tan 30^\circ$
(3) $\cos 135^\circ$	(18) $\sin 150^\circ$	(33) $\tan 150^\circ$	(48) $\cos 180^\circ$	(63) $\cos 135^\circ$
(4) $\sin 45^\circ$	(19) $\sin 0^\circ$	(34) $\sin 150^\circ$	(49) $\sin 45^\circ$	(64) $\sin 30^\circ$
(5) $\cos 30^\circ$	(20) $\sin 120^\circ$	(35) $\sin 60^\circ$	(50) $\cos 180^\circ$	(65) $\sin 0^\circ$
(6) $\sin 135^\circ$	(21) $\tan 135^\circ$	(36) $\tan 135^\circ$	(51) $\tan 120^\circ$	(66) $\sin 45^\circ$
(7) $\cos 135^\circ$	(22) $\tan 45^\circ$	(37) $\sin 135^\circ$	(52) $\tan 60^\circ$	(67) $\tan 90^\circ$
(8) $\tan 90^\circ$	(23) $\tan 120^\circ$	(38) $\cos 150^\circ$	(53) $\sin 120^\circ$	(68) $\sin 45^\circ$
(9) $\cos 60^\circ$	(24) $\tan 180^\circ$	(39) $\cos 120^\circ$	(54) $\sin 135^\circ$	(69) $\sin 45^\circ$
(10) $\tan 90^\circ$	(25) $\sin 60^\circ$	(40) $\tan 135^\circ$	(55) $\sin 150^\circ$	(70) $\cos 45^\circ$
(11) $\sin 90^\circ$	(26) $\tan 60^\circ$	(41) $\sin 135^\circ$	(56) $\cos 135^\circ$	(71) $\tan 30^\circ$
(12) $\cos 120^\circ$	(27) $\tan 150^\circ$	(42) $\cos 90^\circ$	(57) $\tan 0^\circ$	(72) $\tan 180^\circ$
(13) $\cos 0^\circ$	(28) $\cos 30^\circ$	(43) $\cos 45^\circ$	(58) $\sin 60^\circ$	(73) $\tan 150^\circ$
(14) $\cos 0^\circ$	(29) $\tan 60^\circ$	(44) $\tan 120^\circ$	(59) $\sin 60^\circ$	(74) $\sin 180^\circ$
(15) $\tan 90^\circ$	(30) $\cos 135^\circ$	(45) $\cos 135^\circ$	(60) $\sin 30^\circ$	(75) $\sin 90^\circ$

(1) $\tan 90^\circ$	(16) $\tan 60^\circ$	(31) $\sin 120^\circ$	(46) $\cos 120^\circ$	(61) $\tan 45^\circ$
(2) $\sin 180^\circ$	(17) $\sin 30^\circ$	(32) $\cos 120^\circ$	(47) $\cos 150^\circ$	(62) $\tan 135^\circ$
(3) $\sin 60^\circ$	(18) $\cos 30^\circ$	(33) $\sin 0^\circ$	(48) $\sin 30^\circ$	(63) $\cos 45^\circ$
(4) $\cos 60^\circ$	(19) $\cos 90^\circ$	(34) $\tan 45^\circ$	(49) $\cos 120^\circ$	(64) $\tan 0^\circ$
(5) $\tan 30^\circ$	(20) $\sin 120^\circ$	(35) $\sin 180^\circ$	(50) $\cos 180^\circ$	(65) $\tan 60^\circ$
(6) $\tan 30^\circ$	(21) $\cos 150^\circ$	(36) $\tan 30^\circ$	(51) $\cos 0^\circ$	(66) $\sin 45^\circ$
(7) $\sin 45^\circ$	(22) $\tan 180^\circ$	(37) $\tan 150^\circ$	(52) $\cos 90^\circ$	(67) $\cos 30^\circ$
(8) $\tan 120^\circ$	(23) $\sin 135^\circ$	(38) $\sin 90^\circ$	(53) $\tan 45^\circ$	(68) $\sin 30^\circ$
(9) $\sin 135^\circ$	(24) $\tan 30^\circ$	(39) $\cos 135^\circ$	(54) $\cos 45^\circ$	(69) $\cos 0^\circ$
(10) $\sin 45^\circ$	(25) $\sin 45^\circ$	(40) $\cos 135^\circ$	(55) $\cos 90^\circ$	(70) $\tan 180^\circ$
(11) $\sin 45^\circ$	(26) $\tan 150^\circ$	(41) $\tan 0^\circ$	(56) $\cos 150^\circ$	(71) $\tan 60^\circ$
(12) $\sin 120^\circ$	(27) $\cos 180^\circ$	(42) $\tan 120^\circ$	(57) $\sin 135^\circ$	(72) $\cos 135^\circ$
(13) $\tan 30^\circ$	(28) $\sin 30^\circ$	(43) $\sin 0^\circ$	(58) $\sin 0^\circ$	(73) $\sin 0^\circ$
(14) $\cos 60^\circ$	(29) $\sin 180^\circ$	(44) $\tan 180^\circ$	(59) $\tan 60^\circ$	(74) $\sin 0^\circ$
(15) $\cos 150^\circ$	(30) $\tan 30^\circ$	(45) $\tan 30^\circ$	(60) $\cos 180^\circ$	(75) $\sin 60^\circ$

(1) $\tan 45^\circ$	(16) $\tan 90^\circ$	(31) $\cos 135^\circ$	(46) $\tan 120^\circ$	(61) $\sin 45^\circ$
(2) $\cos 90^\circ$	(17) $\sin 45^\circ$	(32) $\sin 60^\circ$	(47) $\cos 135^\circ$	(62) $\cos 30^\circ$
(3) $\sin 0^\circ$	(18) $\tan 90^\circ$	(33) $\tan 0^\circ$	(48) $\cos 45^\circ$	(63) $\sin 90^\circ$
(4) $\tan 120^\circ$	(19) $\tan 120^\circ$	(34) $\tan 45^\circ$	(49) $\cos 30^\circ$	(64) $\sin 30^\circ$
(5) $\cos 120^\circ$	(20) $\sin 180^\circ$	(35) $\tan 45^\circ$	(50) $\cos 0^\circ$	(65) $\sin 135^\circ$
(6) $\tan 0^\circ$	(21) $\tan 135^\circ$	(36) $\tan 135^\circ$	(51) $\cos 60^\circ$	(66) $\sin 30^\circ$
(7) $\sin 180^\circ$	(22) $\tan 0^\circ$	(37) $\tan 120^\circ$	(52) $\cos 45^\circ$	(67) $\sin 135^\circ$
(8) $\sin 90^\circ$	(23) $\tan 135^\circ$	(38) $\cos 135^\circ$	(53) $\sin 180^\circ$	(68) $\tan 180^\circ$
(9) $\sin 60^\circ$	(24) $\tan 135^\circ$	(39) $\tan 180^\circ$	(54) $\cos 180^\circ$	(69) $\sin 150^\circ$
(10) $\sin 45^\circ$	(25) $\cos 0^\circ$	(40) $\tan 120^\circ$	(55) $\cos 30^\circ$	(70) $\sin 60^\circ$
(11) $\sin 45^\circ$	(26) $\cos 120^\circ$	(41) $\tan 150^\circ$	(56) $\cos 30^\circ$	(71) $\cos 0^\circ$
(12) $\sin 120^\circ$	(27) $\tan 180^\circ$	(42) $\tan 135^\circ$	(57) $\tan 90^\circ$	(72) $\cos 60^\circ$
(13) $\sin 45^\circ$	(28) $\cos 90^\circ$	(43) $\tan 0^\circ$	(58) $\tan 120^\circ$	(73) $\sin 30^\circ$
(14) $\sin 120^\circ$	(29) $\sin 135^\circ$	(44) $\sin 120^\circ$	(59) $\cos 60^\circ$	(74) $\sin 45^\circ$
(15) $\tan 30^\circ$	(30) $\cos 135^\circ$	(45) $\cos 180^\circ$	(60) $\tan 135^\circ$	(75) $\cos 135^\circ$

(1) $\tan 30^\circ$	(16) $\sin 135^\circ$	(31) $\cos 90^\circ$	(46) $\sin 30^\circ$	(61) $\tan 30^\circ$
(2) $\sin 60^\circ$	(17) $\sin 90^\circ$	(32) $\sin 180^\circ$	(47) $\tan 90^\circ$	(62) $\cos 150^\circ$
(3) $\cos 150^\circ$	(18) $\tan 150^\circ$	(33) $\cos 120^\circ$	(48) $\tan 90^\circ$	(63) $\sin 30^\circ$
(4) $\tan 0^\circ$	(19) $\sin 180^\circ$	(34) $\cos 135^\circ$	(49) $\tan 135^\circ$	(64) $\cos 150^\circ$
(5) $\tan 30^\circ$	(20) $\cos 180^\circ$	(35) $\cos 135^\circ$	(50) $\cos 90^\circ$	(65) $\tan 120^\circ$
(6) $\sin 90^\circ$	(21) $\sin 45^\circ$	(36) $\sin 120^\circ$	(51) $\sin 90^\circ$	(66) $\tan 60^\circ$
(7) $\tan 90^\circ$	(22) $\cos 120^\circ$	(37) $\tan 60^\circ$	(52) $\sin 90^\circ$	(67) $\cos 180^\circ$
(8) $\sin 180^\circ$	(23) $\cos 180^\circ$	(38) $\sin 60^\circ$	(53) $\tan 180^\circ$	(68) $\cos 120^\circ$
(9) $\cos 0^\circ$	(24) $\sin 60^\circ$	(39) $\sin 120^\circ$	(54) $\cos 0^\circ$	(69) $\tan 0^\circ$
(10) $\tan 135^\circ$	(25) $\tan 120^\circ$	(40) $\sin 180^\circ$	(55) $\cos 30^\circ$	(70) $\cos 90^\circ$
(11) $\sin 135^\circ$	(26) $\tan 0^\circ$	(41) $\cos 180^\circ$	(56) $\cos 180^\circ$	(71) $\cos 120^\circ$
(12) $\cos 180^\circ$	(27) $\cos 45^\circ$	(42) $\sin 180^\circ$	(57) $\tan 180^\circ$	(72) $\cos 45^\circ$
(13) $\tan 45^\circ$	(28) $\sin 135^\circ$	(43) $\tan 0^\circ$	(58) $\tan 60^\circ$	(73) $\sin 180^\circ$
(14) $\sin 135^\circ$	(29) $\cos 45^\circ$	(44) $\cos 120^\circ$	(59) $\sin 30^\circ$	(74) $\tan 60^\circ$
(15) $\sin 0^\circ$	(30) $\tan 90^\circ$	(45) $\tan 30^\circ$	(60) $\cos 150^\circ$	(75) $\sin 60^\circ$

(1) $\sin 0^\circ$	(16) $\cos 150^\circ$	(31) $\tan 150^\circ$	(46) $\tan 90^\circ$	(61) $\cos 45^\circ$
(2) $\tan 0^\circ$	(17) $\tan 120^\circ$	(32) $\sin 180^\circ$	(47) $\sin 30^\circ$	(62) $\cos 135^\circ$
(3) $\cos 180^\circ$	(18) $\sin 60^\circ$	(33) $\tan 60^\circ$	(48) $\cos 180^\circ$	(63) $\tan 180^\circ$
(4) $\sin 45^\circ$	(19) $\sin 150^\circ$	(34) $\cos 120^\circ$	(49) $\cos 0^\circ$	(64) $\tan 30^\circ$
(5) $\cos 45^\circ$	(20) $\sin 30^\circ$	(35) $\cos 180^\circ$	(50) $\sin 135^\circ$	(65) $\sin 150^\circ$
(6) $\cos 150^\circ$	(21) $\cos 30^\circ$	(36) $\sin 30^\circ$	(51) $\tan 45^\circ$	(66) $\tan 135^\circ$
(7) $\cos 135^\circ$	(22) $\tan 45^\circ$	(37) $\tan 150^\circ$	(52) $\sin 90^\circ$	(67) $\sin 180^\circ$
(8) $\tan 150^\circ$	(23) $\cos 150^\circ$	(38) $\sin 90^\circ$	(53) $\sin 60^\circ$	(68) $\cos 135^\circ$
(9) $\tan 0^\circ$	(24) $\cos 0^\circ$	(39) $\sin 30^\circ$	(54) $\cos 90^\circ$	(69) $\sin 45^\circ$
(10) $\cos 30^\circ$	(25) $\sin 60^\circ$	(40) $\sin 120^\circ$	(55) $\sin 180^\circ$	(70) $\cos 30^\circ$
(11) $\tan 60^\circ$	(26) $\tan 60^\circ$	(41) $\sin 120^\circ$	(56) $\cos 30^\circ$	(71) $\cos 135^\circ$
(12) $\sin 120^\circ$	(27) $\cos 30^\circ$	(42) $\sin 60^\circ$	(57) $\cos 45^\circ$	(72) $\sin 90^\circ$
(13) $\cos 0^\circ$	(28) $\sin 0^\circ$	(43) $\cos 60^\circ$	(58) $\cos 45^\circ$	(73) $\cos 120^\circ$
(14) $\cos 60^\circ$	(29) $\tan 60^\circ$	(44) $\tan 30^\circ$	(59) $\tan 90^\circ$	(74) $\sin 180^\circ$
(15) $\cos 150^\circ$	(30) $\tan 0^\circ$	(45) $\cos 30^\circ$	(60) $\tan 0^\circ$	(75) $\tan 0^\circ$

(1) $\cos 60^\circ$	(16) $\sin 90^\circ$	(31) $\tan 135^\circ$	(46) $\tan 135^\circ$	(61) $\sin 30^\circ$
(2) $\tan 60^\circ$	(17) $\tan 135^\circ$	(32) $\sin 90^\circ$	(47) $\sin 120^\circ$	(62) $\sin 180^\circ$
(3) $\tan 150^\circ$	(18) $\tan 150^\circ$	(33) $\sin 180^\circ$	(48) $\sin 0^\circ$	(63) $\tan 0^\circ$
(4) $\sin 60^\circ$	(19) $\tan 135^\circ$	(34) $\tan 60^\circ$	(49) $\sin 135^\circ$	(64) $\cos 30^\circ$
(5) $\cos 0^\circ$	(20) $\sin 90^\circ$	(35) $\sin 120^\circ$	(50) $\tan 120^\circ$	(65) $\sin 180^\circ$
(6) $\tan 135^\circ$	(21) $\sin 150^\circ$	(36) $\tan 120^\circ$	(51) $\tan 30^\circ$	(66) $\sin 180^\circ$
(7) $\cos 90^\circ$	(22) $\sin 135^\circ$	(37) $\tan 120^\circ$	(52) $\sin 45^\circ$	(67) $\tan 90^\circ$
(8) $\sin 60^\circ$	(23) $\sin 150^\circ$	(38) $\cos 60^\circ$	(53) $\sin 90^\circ$	(68) $\sin 60^\circ$
(9) $\tan 30^\circ$	(24) $\tan 150^\circ$	(39) $\sin 150^\circ$	(54) $\cos 90^\circ$	(69) $\cos 60^\circ$
(10) $\cos 45^\circ$	(25) $\sin 60^\circ$	(40) $\tan 60^\circ$	(55) $\tan 135^\circ$	(70) $\sin 150^\circ$
(11) $\tan 30^\circ$	(26) $\sin 180^\circ$	(41) $\tan 0^\circ$	(56) $\cos 45^\circ$	(71) $\tan 45^\circ$
(12) $\tan 30^\circ$	(27) $\tan 0^\circ$	(42) $\tan 60^\circ$	(57) $\cos 120^\circ$	(72) $\sin 60^\circ$
(13) $\sin 45^\circ$	(28) $\cos 30^\circ$	(43) $\sin 180^\circ$	(58) $\cos 180^\circ$	(73) $\sin 0^\circ$
(14) $\sin 90^\circ$	(29) $\cos 45^\circ$	(44) $\sin 150^\circ$	(59) $\cos 30^\circ$	(74) $\cos 135^\circ$
(15) $\sin 45^\circ$	(30) $\cos 45^\circ$	(45) $\cos 135^\circ$	(60) $\cos 60^\circ$	(75) $\tan 150^\circ$

(1) $\sin 0^\circ$	(16) $\cos 135^\circ$	(31) $\sin 150^\circ$	(46) $\sin 0^\circ$	(61) $\tan 150^\circ$
(2) $\tan 150^\circ$	(17) $\tan 60^\circ$	(32) $\cos 120^\circ$	(47) $\tan 150^\circ$	(62) $\cos 30^\circ$
(3) $\tan 135^\circ$	(18) $\cos 30^\circ$	(33) $\sin 120^\circ$	(48) $\cos 90^\circ$	(63) $\tan 150^\circ$
(4) $\tan 60^\circ$	(19) $\cos 180^\circ$	(34) $\tan 150^\circ$	(49) $\cos 30^\circ$	(64) $\cos 135^\circ$
(5) $\cos 120^\circ$	(20) $\cos 90^\circ$	(35) $\sin 120^\circ$	(50) $\sin 150^\circ$	(65) $\sin 150^\circ$
(6) $\tan 180^\circ$	(21) $\cos 180^\circ$	(36) $\sin 30^\circ$	(51) $\tan 60^\circ$	(66) $\cos 135^\circ$
(7) $\sin 45^\circ$	(22) $\tan 120^\circ$	(37) $\sin 180^\circ$	(52) $\tan 180^\circ$	(67) $\sin 60^\circ$
(8) $\tan 30^\circ$	(23) $\cos 120^\circ$	(38) $\cos 90^\circ$	(53) $\sin 45^\circ$	(68) $\cos 0^\circ$
(9) $\cos 30^\circ$	(24) $\sin 180^\circ$	(39) $\tan 135^\circ$	(54) $\cos 45^\circ$	(69) $\sin 180^\circ$
(10) $\cos 0^\circ$	(25) $\tan 135^\circ$	(40) $\tan 120^\circ$	(55) $\cos 45^\circ$	(70) $\tan 150^\circ$
(11) $\cos 120^\circ$	(26) $\tan 90^\circ$	(41) $\tan 60^\circ$	(56) $\tan 90^\circ$	(71) $\sin 150^\circ$
(12) $\sin 135^\circ$	(27) $\tan 30^\circ$	(42) $\cos 60^\circ$	(57) $\tan 60^\circ$	(72) $\tan 30^\circ$
(13) $\tan 30^\circ$	(28) $\tan 180^\circ$	(43) $\tan 90^\circ$	(58) $\sin 45^\circ$	(73) $\tan 120^\circ$
(14) $\cos 0^\circ$	(29) $\cos 0^\circ$	(44) $\sin 45^\circ$	(59) $\cos 150^\circ$	(74) $\tan 150^\circ$
(15) $\tan 60^\circ$	(30) $\cos 135^\circ$	(45) $\tan 30^\circ$	(60) $\cos 150^\circ$	(75) $\sin 150^\circ$

(1) $\sin 90^\circ$ $= 1$	(16) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(31) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(46) $\cos 90^\circ$ $= 0$	(61) $\cos 90^\circ$ $= 0$
(2) $\tan 60^\circ$ $= \sqrt{3}$	(17) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(32) $\cos 90^\circ$ $= 0$	(47) $\tan 135^\circ$ $= -1$	(62) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(3) $\sin 90^\circ$ $= 1$	(18) $\tan 180^\circ$ $= 0$	(33) $\tan 135^\circ$ $= -1$	(48) $\tan 45^\circ$ $= 1$	(63) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(4) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(19) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(34) $\sin 180^\circ$ $= 0$	(49) $\cos 90^\circ$ $= 0$	(64) $\tan 90^\circ$ なし
(5) $\tan 120^\circ$ $= -\sqrt{3}$	(20) $\sin 0^\circ$ $= 0$	(35) $\sin 30^\circ$ $= \frac{1}{2}$	(50) $\tan 90^\circ$ なし	(65) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(6) $\sin 30^\circ$ $= \frac{1}{2}$	(21) $\cos 180^\circ$ $= -1$	(36) $\tan 90^\circ$ なし	(51) $\cos 180^\circ$ $= -1$	(66) $\cos 180^\circ$ $= -1$
(7) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(22) $\sin 0^\circ$ $= 0$	(37) $\cos 180^\circ$ $= -1$	(52) $\cos 90^\circ$ $= 0$	(67) $\tan 180^\circ$ $= 0$
(8) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\cos 90^\circ$ $= 0$	(38) $\cos 60^\circ$ $= \frac{1}{2}$	(53) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(68) $\tan 45^\circ$ $= 1$
(9) $\cos 180^\circ$ $= -1$	(24) $\tan 135^\circ$ $= -1$	(39) $\sin 150^\circ$ $= \frac{1}{2}$	(54) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(69) $\cos 120^\circ$ $= -\frac{1}{2}$
(10) $\sin 30^\circ$ $= \frac{1}{2}$	(25) $\sin 150^\circ$ $= \frac{1}{2}$	(40) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(55) $\tan 90^\circ$ なし	(70) $\tan 120^\circ$ $= -\sqrt{3}$
(11) $\tan 90^\circ$ なし	(26) $\tan 135^\circ$ $= -1$	(41) $\sin 180^\circ$ $= 0$	(56) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(71) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(12) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(27) $\cos 90^\circ$ $= 0$	(42) $\tan 45^\circ$ $= 1$	(57) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(72) $\sin 90^\circ$ $= 1$
(13) $\sin 0^\circ$ $= 0$	(28) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(43) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(58) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(73) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$
(14) $\tan 90^\circ$ なし	(29) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(44) $\tan 180^\circ$ $= 0$	(59) $\cos 180^\circ$ $= -1$	(74) $\sin 90^\circ$ $= 1$
(15) $\cos 120^\circ$ $= -\frac{1}{2}$	(30) $\cos 180^\circ$ $= -1$	(45) $\tan 135^\circ$ $= -1$	(60) $\sin 90^\circ$ $= 1$	(75) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$

(1) $\tan 90^\circ$ なし	(16) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(31) $\tan 135^\circ$ $= -1$	(46) $\cos 60^\circ$ $= \frac{1}{2}$	(61) $\cos 60^\circ$ $= \frac{1}{2}$
(2) $\tan 90^\circ$ なし	(17) $\tan 90^\circ$ なし	(32) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(47) $\tan 45^\circ$ $= 1$	(62) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(3) $\sin 90^\circ$ $= 1$	(18) $\cos 90^\circ$ $= 0$	(33) $\cos 90^\circ$ $= 0$	(48) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(63) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(4) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(19) $\cos 90^\circ$ $= 0$	(34) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(49) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(64) $\sin 150^\circ$ $= \frac{1}{2}$
(5) $\sin 90^\circ$ $= 1$	(20) $\tan 60^\circ$ $= \sqrt{3}$	(35) $\tan 90^\circ$ なし	(50) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(65) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(6) $\sin 180^\circ$ $= 0$	(21) $\tan 0^\circ$ $= 0$	(36) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(51) $\tan 60^\circ$ $= \sqrt{3}$	(66) $\cos 120^\circ$ $= -\frac{1}{2}$
(7) $\tan 120^\circ$ $= -\sqrt{3}$	(22) $\sin 0^\circ$ $= 0$	(37) $\cos 60^\circ$ $= \frac{1}{2}$	(52) $\tan 180^\circ$ $= 0$	(67) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(8) $\sin 0^\circ$ $= 0$	(23) $\cos 180^\circ$ $= -1$	(38) $\tan 180^\circ$ $= 0$	(53) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(68) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(9) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(24) $\cos 90^\circ$ $= 0$	(39) $\sin 90^\circ$ $= 1$	(54) $\cos 60^\circ$ $= \frac{1}{2}$	(69) $\cos 90^\circ$ $= 0$
(10) $\tan 0^\circ$ $= 0$	(25) $\sin 180^\circ$ $= 0$	(40) $\tan 0^\circ$ $= 0$	(55) $\sin 150^\circ$ $= \frac{1}{2}$	(70) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(11) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(26) $\tan 135^\circ$ $= -1$	(41) $\tan 120^\circ$ $= -\sqrt{3}$	(56) $\sin 0^\circ$ $= 0$	(71) $\tan 90^\circ$ なし
(12) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(27) $\tan 135^\circ$ $= -1$	(42) $\tan 60^\circ$ $= \sqrt{3}$	(57) $\cos 120^\circ$ $= -\frac{1}{2}$	(72) $\sin 30^\circ$ $= \frac{1}{2}$
(13) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(28) $\sin 0^\circ$ $= 0$	(43) $\cos 90^\circ$ $= 0$	(58) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(73) $\cos 0^\circ$ $= 1$
(14) $\sin 30^\circ$ $= \frac{1}{2}$	(29) $\tan 120^\circ$ $= -\sqrt{3}$	(44) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(59) $\tan 45^\circ$ $= 1$	(74) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(15) $\sin 30^\circ$ $= \frac{1}{2}$	(30) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(45) $\cos 120^\circ$ $= -\frac{1}{2}$	(60) $\cos 0^\circ$ $= 1$	(75) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$

(1) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(16) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(31) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(46) $\sin 0^\circ$ $= 0$	(61) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(2) $\tan 0^\circ$ $= 0$	(17) $\tan 180^\circ$ $= 0$	(32) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(47) $\cos 60^\circ$ $= \frac{1}{2}$	(62) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(3) $\cos 180^\circ$ $= -1$	(18) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(33) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(48) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(63) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(4) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(19) $\cos 180^\circ$ $= -1$	(34) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(49) $\cos 0^\circ$ $= 1$	(64) $\cos 180^\circ$ $= -1$
(5) $\tan 180^\circ$ $= 0$	(20) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(35) $\cos 90^\circ$ $= 0$	(50) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(65) $\sin 30^\circ$ $= \frac{1}{2}$
(6) $\cos 0^\circ$ $= 1$	(21) $\tan 135^\circ$ $= -1$	(36) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(51) $\sin 150^\circ$ $= \frac{1}{2}$	(66) $\cos 90^\circ$ $= 0$
(7) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(22) $\tan 135^\circ$ $= -1$	(37) $\sin 0^\circ$ $= 0$	(52) $\sin 30^\circ$ $= \frac{1}{2}$	(67) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$
(8) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\sin 90^\circ$ $= 1$	(38) $\sin 180^\circ$ $= 0$	(53) $\cos 0^\circ$ $= 1$	(68) $\cos 90^\circ$ $= 0$
(9) $\tan 45^\circ$ $= 1$	(24) $\sin 0^\circ$ $= 0$	(39) $\sin 90^\circ$ $= 1$	(54) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(10) $\cos 120^\circ$ $= -\frac{1}{2}$	(25) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(40) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(55) $\tan 60^\circ$ $= \sqrt{3}$	(70) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(11) $\cos 180^\circ$ $= -1$	(26) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(41) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(56) $\cos 0^\circ$ $= 1$	(71) $\sin 90^\circ$ $= 1$
(12) $\tan 90^\circ$ なし	(27) $\cos 90^\circ$ $= 0$	(42) $\tan 120^\circ$ $= -\sqrt{3}$	(57) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(72) $\cos 90^\circ$ $= 0$
(13) $\sin 150^\circ$ $= \frac{1}{2}$	(28) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(43) $\cos 60^\circ$ $= \frac{1}{2}$	(58) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(73) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(14) $\tan 135^\circ$ $= -1$	(29) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(44) $\tan 180^\circ$ $= 0$	(59) $\tan 0^\circ$ $= 0$	(74) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(15) $\tan 60^\circ$ $= \sqrt{3}$	(30) $\tan 90^\circ$ なし	(45) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(60) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(75) $\tan 90^\circ$ なし

(1) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(16) $\tan 180^\circ$ $= 0$	(31) $\sin 30^\circ$ $= \frac{1}{2}$	(46) $\sin 0^\circ$ $= 0$	(61) $\tan 135^\circ$ $= -1$
(2) $\tan 180^\circ$ $= 0$	(17) $\cos 180^\circ$ $= -1$	(32) $\cos 120^\circ$ $= -\frac{1}{2}$	(47) $\cos 120^\circ$ $= -\frac{1}{2}$	(62) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\sin 150^\circ$ $= \frac{1}{2}$	(18) $\sin 90^\circ$ $= 1$	(33) $\sin 0^\circ$ $= 0$	(48) $\cos 180^\circ$ $= -1$	(63) $\tan 135^\circ$ $= -1$
(4) $\sin 30^\circ$ $= \frac{1}{2}$	(19) $\sin 90^\circ$ $= 1$	(34) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(49) $\tan 45^\circ$ $= 1$	(64) $\sin 150^\circ$ $= \frac{1}{2}$
(5) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(20) $\sin 90^\circ$ $= 1$	(35) $\cos 90^\circ$ $= 0$	(50) $\cos 60^\circ$ $= \frac{1}{2}$	(65) $\cos 120^\circ$ $= -\frac{1}{2}$
(6) $\tan 180^\circ$ $= 0$	(21) $\tan 120^\circ$ $= -\sqrt{3}$	(36) $\sin 150^\circ$ $= \frac{1}{2}$	(51) $\sin 180^\circ$ $= 0$	(66) $\sin 0^\circ$ $= 0$
(7) $\sin 0^\circ$ $= 0$	(22) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(37) $\sin 150^\circ$ $= \frac{1}{2}$	(52) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(67) $\sin 0^\circ$ $= 0$
(8) $\tan 180^\circ$ $= 0$	(23) $\tan 120^\circ$ $= -\sqrt{3}$	(38) $\cos 120^\circ$ $= -\frac{1}{2}$	(53) $\sin 180^\circ$ $= 0$	(68) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(9) $\sin 30^\circ$ $= \frac{1}{2}$	(24) $\sin 30^\circ$ $= \frac{1}{2}$	(39) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(54) $\sin 0^\circ$ $= 0$	(69) $\tan 135^\circ$ $= -1$
(10) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(40) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(55) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(70) $\tan 0^\circ$ $= 0$
(11) $\cos 90^\circ$ $= 0$	(26) $\cos 90^\circ$ $= 0$	(41) $\tan 90^\circ$ なし	(56) $\cos 180^\circ$ $= -1$	(71) $\cos 60^\circ$ $= \frac{1}{2}$
(12) $\cos 180^\circ$ $= -1$	(27) $\cos 120^\circ$ $= -\frac{1}{2}$	(42) $\cos 180^\circ$ $= -1$	(57) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(72) $\cos 180^\circ$ $= -1$
(13) $\tan 0^\circ$ $= 0$	(28) $\cos 60^\circ$ $= \frac{1}{2}$	(43) $\tan 120^\circ$ $= -\sqrt{3}$	(58) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(73) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(14) $\sin 180^\circ$ $= 0$	(29) $\tan 135^\circ$ $= -1$	(44) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(59) $\cos 90^\circ$ $= 0$	(74) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(15) $\cos 120^\circ$ $= -\frac{1}{2}$	(30) $\sin 150^\circ$ $= \frac{1}{2}$	(45) $\sin 150^\circ$ $= \frac{1}{2}$	(60) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(75) $\tan 135^\circ$ $= -1$

- | | | | | |
|--|--|--|--|--|
| (1) $\cos 30^\circ$
$= \frac{\sqrt{3}}{2}$ | (16) $\tan 150^\circ$
$= -\frac{1}{\sqrt{3}}$ | (31) $\sin 0^\circ$
$= 0$ | (46) $\cos 45^\circ$
$= \frac{1}{\sqrt{2}}$ | (61) $\sin 135^\circ$
$= \frac{1}{\sqrt{2}}$ |
| (2) $\sin 150^\circ$
$= \frac{1}{2}$ | (17) $\tan 30^\circ$
$= \frac{1}{\sqrt{3}}$ | (32) $\sin 180^\circ$
$= 0$ | (47) $\tan 0^\circ$
$= 0$ | (62) $\tan 135^\circ$
$= -1$ |
| (3) $\tan 150^\circ$
$= -\frac{1}{\sqrt{3}}$ | (18) $\cos 0^\circ$
$= 1$ | (33) $\tan 90^\circ$
なし | (48) $\tan 45^\circ$
$= 1$ | (63) $\tan 30^\circ$
$= \frac{1}{\sqrt{3}}$ |
| (4) $\tan 120^\circ$
$= -\sqrt{3}$ | (19) $\cos 60^\circ$
$= \frac{1}{2}$ | (34) $\sin 0^\circ$
$= 0$ | (49) $\cos 135^\circ$
$= -\frac{1}{\sqrt{2}}$ | (64) $\sin 120^\circ$
$= \frac{\sqrt{3}}{2}$ |
| (5) $\sin 45^\circ$
$= \frac{1}{\sqrt{2}}$ | (20) $\cos 0^\circ$
$= 1$ | (35) $\tan 30^\circ$
$= \frac{1}{\sqrt{3}}$ | (50) $\tan 90^\circ$
なし | (65) $\cos 45^\circ$
$= \frac{1}{\sqrt{2}}$ |
| (6) $\sin 0^\circ$
$= 0$ | (21) $\cos 0^\circ$
$= 1$ | (36) $\cos 90^\circ$
$= 0$ | (51) $\cos 90^\circ$
$= 0$ | (66) $\tan 90^\circ$
なし |
| (7) $\cos 135^\circ$
$= -\frac{1}{\sqrt{2}}$ | (22) $\sin 150^\circ$
$= \frac{1}{2}$ | (37) $\cos 180^\circ$
$= -1$ | (52) $\sin 0^\circ$
$= 0$ | (67) $\sin 150^\circ$
$= \frac{1}{2}$ |
| (8) $\cos 45^\circ$
$= \frac{1}{\sqrt{2}}$ | (23) $\tan 60^\circ$
$= \sqrt{3}$ | (38) $\tan 180^\circ$
$= 0$ | (53) $\cos 180^\circ$
$= -1$ | (68) $\tan 150^\circ$
$= -\frac{1}{\sqrt{3}}$ |
| (9) $\cos 45^\circ$
$= \frac{1}{\sqrt{2}}$ | (24) $\sin 90^\circ$
$= 1$ | (39) $\sin 135^\circ$
$= \frac{1}{\sqrt{2}}$ | (54) $\tan 120^\circ$
$= -\sqrt{3}$ | (69) $\cos 90^\circ$
$= 0$ |
| (10) $\sin 180^\circ$
$= 0$ | (25) $\tan 120^\circ$
$= -\sqrt{3}$ | (40) $\sin 30^\circ$
$= \frac{1}{2}$ | (55) $\tan 180^\circ$
$= 0$ | (70) $\sin 0^\circ$
$= 0$ |
| (11) $\tan 0^\circ$
$= 0$ | (26) $\cos 180^\circ$
$= -1$ | (41) $\cos 60^\circ$
$= \frac{1}{2}$ | (56) $\sin 180^\circ$
$= 0$ | (71) $\sin 60^\circ$
$= \frac{\sqrt{3}}{2}$ |
| (12) $\tan 45^\circ$
$= 1$ | (27) $\tan 45^\circ$
$= 1$ | (42) $\cos 90^\circ$
$= 0$ | (57) $\sin 90^\circ$
$= 1$ | (72) $\sin 60^\circ$
$= \frac{\sqrt{3}}{2}$ |
| (13) $\cos 120^\circ$
$= -\frac{1}{2}$ | (28) $\tan 180^\circ$
$= 0$ | (43) $\sin 135^\circ$
$= \frac{1}{\sqrt{2}}$ | (58) $\cos 120^\circ$
$= -\frac{1}{2}$ | (73) $\cos 120^\circ$
$= -\frac{1}{2}$ |
| (14) $\cos 150^\circ$
$= -\frac{\sqrt{3}}{2}$ | (29) $\sin 150^\circ$
$= \frac{1}{2}$ | (44) $\cos 150^\circ$
$= -\frac{\sqrt{3}}{2}$ | (59) $\cos 120^\circ$
$= -\frac{1}{2}$ | (74) $\cos 60^\circ$
$= \frac{1}{2}$ |
| (15) $\sin 0^\circ$
$= 0$ | (30) $\sin 150^\circ$
$= \frac{1}{2}$ | (45) $\sin 90^\circ$
$= 1$ | (60) $\sin 90^\circ$
$= 1$ | (75) $\cos 0^\circ$
$= 1$ |

(1) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(16) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(31) $\tan 135^\circ$ $= -1$	(46) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(61) $\sin 0^\circ$ $= 0$
(2) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(17) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(32) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(47) $\cos 0^\circ$ $= 1$	(62) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(3) $\cos 120^\circ$ $= -\frac{1}{2}$	(18) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(33) $\tan 0^\circ$ $= 0$	(48) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(63) $\cos 60^\circ$ $= \frac{1}{2}$
(4) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(19) $\cos 180^\circ$ $= -1$	(34) $\cos 180^\circ$ $= -1$	(49) $\tan 120^\circ$ $= -\sqrt{3}$	(64) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(5) $\tan 60^\circ$ $= \sqrt{3}$	(20) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(35) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(50) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(65) $\cos 60^\circ$ $= \frac{1}{2}$
(6) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(21) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(36) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(51) $\tan 135^\circ$ $= -1$	(66) $\cos 60^\circ$ $= \frac{1}{2}$
(7) $\cos 0^\circ$ $= 1$	(22) $\tan 135^\circ$ $= -1$	(37) $\sin 0^\circ$ $= 0$	(52) $\sin 90^\circ$ $= 1$	(67) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(8) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(23) $\tan 45^\circ$ $= 1$	(38) $\tan 180^\circ$ $= 0$	(53) $\cos 0^\circ$ $= 1$	(68) $\sin 0^\circ$ $= 0$
(9) $\sin 0^\circ$ $= 0$	(24) $\cos 180^\circ$ $= -1$	(39) $\cos 90^\circ$ $= 0$	(54) $\sin 0^\circ$ $= 0$	(69) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$
(10) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(25) $\sin 0^\circ$ $= 0$	(40) $\sin 90^\circ$ $= 1$	(55) $\sin 90^\circ$ $= 1$	(70) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(11) $\tan 135^\circ$ $= -1$	(26) $\sin 30^\circ$ $= \frac{1}{2}$	(41) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(56) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(71) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(12) $\tan 0^\circ$ $= 0$	(27) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(42) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(57) $\sin 90^\circ$ $= 1$	(72) $\tan 120^\circ$ $= -\sqrt{3}$
(13) $\cos 180^\circ$ $= -1$	(28) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(43) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(58) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(73) $\cos 120^\circ$ $= -\frac{1}{2}$
(14) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(29) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(44) $\tan 120^\circ$ $= -\sqrt{3}$	(59) $\tan 90^\circ$ なし	(74) $\tan 0^\circ$ $= 0$
(15) $\sin 90^\circ$ $= 1$	(30) $\tan 180^\circ$ $= 0$	(45) $\cos 0^\circ$ $= 1$	(60) $\tan 0^\circ$ $= 0$	(75) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$

(1) $\sin 90^\circ$ $= 1$	(16) $\sin 90^\circ$ $= 1$	(31) $\sin 90^\circ$ $= 1$	(46) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(61) $\tan 90^\circ$ なし
(2) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(17) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(32) $\tan 120^\circ$ $= -\sqrt{3}$	(47) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(62) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(3) $\sin 0^\circ$ $= 0$	(18) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(33) $\tan 180^\circ$ $= 0$	(48) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(63) $\cos 120^\circ$ $= -\frac{1}{2}$
(4) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(19) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(34) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(49) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(64) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(5) $\cos 0^\circ$ $= 1$	(20) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(35) $\tan 45^\circ$ $= 1$	(50) $\cos 90^\circ$ $= 0$	(65) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(6) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(21) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(36) $\cos 0^\circ$ $= 1$	(51) $\cos 0^\circ$ $= 1$	(66) $\sin 0^\circ$ $= 0$
(7) $\cos 0^\circ$ $= 1$	(22) $\tan 0^\circ$ $= 0$	(37) $\cos 90^\circ$ $= 0$	(52) $\sin 90^\circ$ $= 1$	(67) $\sin 180^\circ$ $= 0$
(8) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(23) $\cos 180^\circ$ $= -1$	(38) $\tan 135^\circ$ $= -1$	(53) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(68) $\tan 60^\circ$ $= \sqrt{3}$
(9) $\tan 60^\circ$ $= \sqrt{3}$	(24) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(39) $\sin 30^\circ$ $= \frac{1}{2}$	(54) $\cos 60^\circ$ $= \frac{1}{2}$	(69) $\tan 180^\circ$ $= 0$
(10) $\tan 180^\circ$ $= 0$	(25) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(40) $\tan 60^\circ$ $= \sqrt{3}$	(55) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(70) $\cos 90^\circ$ $= 0$
(11) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(26) $\tan 120^\circ$ $= -\sqrt{3}$	(41) $\tan 45^\circ$ $= 1$	(56) $\sin 90^\circ$ $= 1$	(71) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(12) $\sin 0^\circ$ $= 0$	(27) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(42) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(57) $\cos 180^\circ$ $= -1$	(72) $\cos 120^\circ$ $= -\frac{1}{2}$
(13) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(28) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(43) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(58) $\tan 0^\circ$ $= 0$	(73) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(14) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(44) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(59) $\cos 90^\circ$ $= 0$	(74) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(15) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(30) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(45) $\cos 90^\circ$ $= 0$	(60) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(75) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$

(1) $\sin 90^\circ$ $= 1$	(16) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(31) $\sin 90^\circ$ $= 1$	(46) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(61) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(2) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(17) $\tan 180^\circ$ $= 0$	(32) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(47) $\cos 60^\circ$ $= \frac{1}{2}$	(62) $\tan 120^\circ$ $= -\sqrt{3}$
(3) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(18) $\tan 45^\circ$ $= 1$	(33) $\cos 0^\circ$ $= 1$	(48) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(63) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(4) $\sin 150^\circ$ $= \frac{1}{2}$	(19) $\cos 180^\circ$ $= -1$	(34) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(49) $\cos 90^\circ$ $= 0$	(64) $\tan 60^\circ$ $= \sqrt{3}$
(5) $\sin 90^\circ$ $= 1$	(20) $\tan 180^\circ$ $= 0$	(35) $\tan 180^\circ$ $= 0$	(50) $\tan 180^\circ$ $= 0$	(65) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(6) $\sin 0^\circ$ $= 0$	(21) $\sin 30^\circ$ $= \frac{1}{2}$	(36) $\cos 60^\circ$ $= \frac{1}{2}$	(51) $\sin 90^\circ$ $= 1$	(66) $\cos 90^\circ$ $= 0$
(7) $\tan 135^\circ$ $= -1$	(22) $\tan 45^\circ$ $= 1$	(37) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(52) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(67) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(8) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(38) $\tan 45^\circ$ $= 1$	(53) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(68) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(9) $\cos 60^\circ$ $= \frac{1}{2}$	(24) $\tan 60^\circ$ $= \sqrt{3}$	(39) $\cos 120^\circ$ $= -\frac{1}{2}$	(54) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(10) $\cos 90^\circ$ $= 0$	(25) $\tan 120^\circ$ $= -\sqrt{3}$	(40) $\cos 0^\circ$ $= 1$	(55) $\tan 45^\circ$ $= 1$	(70) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(11) $\tan 90^\circ$ なし	(26) $\sin 180^\circ$ $= 0$	(41) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(56) $\tan 90^\circ$ なし	(71) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(12) $\cos 0^\circ$ $= 1$	(27) $\tan 135^\circ$ $= -1$	(42) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(57) $\tan 0^\circ$ $= 0$	(72) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(13) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(28) $\tan 45^\circ$ $= 1$	(43) $\tan 135^\circ$ $= -1$	(58) $\tan 0^\circ$ $= 0$	(73) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(14) $\sin 0^\circ$ $= 0$	(29) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(44) $\tan 0^\circ$ $= 0$	(59) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(74) $\sin 180^\circ$ $= 0$
(15) $\sin 150^\circ$ $= \frac{1}{2}$	(30) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(45) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(60) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(75) $\cos 60^\circ$ $= \frac{1}{2}$

(1) $\tan 180^\circ$ $= 0$	(16) $\cos 60^\circ$ $= \frac{1}{2}$	(31) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(46) $\tan 120^\circ$ $= -\sqrt{3}$	(61) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(2) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(17) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(32) $\tan 45^\circ$ $= 1$	(47) $\cos 60^\circ$ $= \frac{1}{2}$	(62) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\tan 135^\circ$ $= -1$	(18) $\cos 0^\circ$ $= 1$	(33) $\tan 90^\circ$ なし	(48) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(63) $\sin 0^\circ$ $= 0$
(4) $\sin 0^\circ$ $= 0$	(19) $\cos 90^\circ$ $= 0$	(34) $\tan 180^\circ$ $= 0$	(49) $\tan 135^\circ$ $= -1$	(64) $\cos 180^\circ$ $= -1$
(5) $\sin 180^\circ$ $= 0$	(20) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(35) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(50) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(65) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(6) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(21) $\cos 180^\circ$ $= -1$	(36) $\cos 120^\circ$ $= -\frac{1}{2}$	(51) $\sin 180^\circ$ $= 0$	(66) $\cos 0^\circ$ $= 1$
(7) $\cos 180^\circ$ $= -1$	(22) $\tan 180^\circ$ $= 0$	(37) $\sin 30^\circ$ $= \frac{1}{2}$	(52) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(67) $\cos 0^\circ$ $= 1$
(8) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\tan 90^\circ$ なし	(38) $\sin 90^\circ$ $= 1$	(53) $\tan 0^\circ$ $= 0$	(68) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(9) $\tan 60^\circ$ $= \sqrt{3}$	(24) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(39) $\sin 150^\circ$ $= \frac{1}{2}$	(54) $\tan 60^\circ$ $= \sqrt{3}$	(69) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$
(10) $\sin 150^\circ$ $= \frac{1}{2}$	(25) $\cos 0^\circ$ $= 1$	(40) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(55) $\sin 90^\circ$ $= 1$	(70) $\tan 90^\circ$ なし
(11) $\tan 135^\circ$ $= -1$	(26) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(41) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(56) $\sin 150^\circ$ $= \frac{1}{2}$	(71) $\cos 0^\circ$ $= 1$
(12) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\sin 90^\circ$ $= 1$	(42) $\cos 90^\circ$ $= 0$	(57) $\sin 180^\circ$ $= 0$	(72) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(13) $\sin 30^\circ$ $= \frac{1}{2}$	(28) $\cos 180^\circ$ $= -1$	(43) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(58) $\tan 135^\circ$ $= -1$	(73) $\tan 135^\circ$ $= -1$
(14) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(44) $\sin 30^\circ$ $= \frac{1}{2}$	(59) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(74) $\sin 180^\circ$ $= 0$
(15) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(30) $\sin 30^\circ$ $= \frac{1}{2}$	(45) $\sin 90^\circ$ $= 1$	(60) $\cos 180^\circ$ $= -1$	(75) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$

(1) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(16) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(31) $\tan 180^\circ$ $= 0$	(46) $\tan 0^\circ$ $= 0$	(61) $\sin 30^\circ$ $= \frac{1}{2}$
(2) $\sin 180^\circ$ $= 0$	(17) $\sin 90^\circ$ $= 1$	(32) $\sin 0^\circ$ $= 0$	(47) $\cos 0^\circ$ $= 1$	(62) $\cos 180^\circ$ $= -1$
(3) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(18) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(33) $\tan 135^\circ$ $= -1$	(48) $\sin 30^\circ$ $= \frac{1}{2}$	(63) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(4) $\cos 90^\circ$ $= 0$	(19) $\tan 135^\circ$ $= -1$	(34) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(49) $\sin 0^\circ$ $= 0$	(64) $\sin 180^\circ$ $= 0$
(5) $\sin 30^\circ$ $= \frac{1}{2}$	(20) $\cos 120^\circ$ $= -\frac{1}{2}$	(35) $\cos 120^\circ$ $= -\frac{1}{2}$	(50) $\sin 30^\circ$ $= \frac{1}{2}$	(65) $\sin 180^\circ$ $= 0$
(6) $\sin 90^\circ$ $= 1$	(21) $\cos 60^\circ$ $= \frac{1}{2}$	(36) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(51) $\cos 180^\circ$ $= -1$	(66) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(7) $\tan 45^\circ$ $= 1$	(22) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(37) $\tan 45^\circ$ $= 1$	(52) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(67) $\sin 30^\circ$ $= \frac{1}{2}$
(8) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(23) $\cos 60^\circ$ $= \frac{1}{2}$	(38) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(53) $\cos 180^\circ$ $= -1$	(68) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(9) $\tan 120^\circ$ $= -\sqrt{3}$	(24) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(39) $\tan 60^\circ$ $= \sqrt{3}$	(54) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(10) $\tan 45^\circ$ $= 1$	(25) $\cos 60^\circ$ $= \frac{1}{2}$	(40) $\tan 90^\circ$ なし	(55) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(70) $\cos 60^\circ$ $= \frac{1}{2}$
(11) $\tan 90^\circ$ なし	(26) $\cos 60^\circ$ $= \frac{1}{2}$	(41) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(56) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(71) $\cos 90^\circ$ $= 0$
(12) $\cos 180^\circ$ $= -1$	(27) $\cos 120^\circ$ $= -\frac{1}{2}$	(42) $\sin 180^\circ$ $= 0$	(57) $\tan 60^\circ$ $= \sqrt{3}$	(72) $\sin 30^\circ$ $= \frac{1}{2}$
(13) $\tan 180^\circ$ $= 0$	(28) $\sin 0^\circ$ $= 0$	(43) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(58) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(73) $\tan 60^\circ$ $= \sqrt{3}$
(14) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\sin 150^\circ$ $= \frac{1}{2}$	(44) $\tan 60^\circ$ $= \sqrt{3}$	(59) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(74) $\sin 180^\circ$ $= 0$
(15) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(30) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(45) $\tan 120^\circ$ $= -\sqrt{3}$	(60) $\sin 0^\circ$ $= 0$	(75) $\tan 0^\circ$ $= 0$

(1) $\tan 90^\circ$ なし	(16) $\tan 90^\circ$ なし	(31) $\tan 135^\circ$ $= -1$	(46) $\sin 0^\circ$ $= 0$	(61) $\cos 90^\circ$ $= 0$
(2) $\sin 150^\circ$ $= \frac{1}{2}$	(17) $\tan 90^\circ$ なし	(32) $\sin 90^\circ$ $= 1$	(47) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(62) $\tan 120^\circ$ $= -\sqrt{3}$
(3) $\tan 135^\circ$ $= -1$	(18) $\sin 0^\circ$ $= 0$	(33) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(48) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(63) $\cos 60^\circ$ $= \frac{1}{2}$
(4) $\tan 45^\circ$ $= 1$	(19) $\tan 120^\circ$ $= -\sqrt{3}$	(34) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(49) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(64) $\sin 0^\circ$ $= 0$
(5) $\tan 120^\circ$ $= -\sqrt{3}$	(20) $\cos 0^\circ$ $= 1$	(35) $\tan 135^\circ$ $= -1$	(50) $\sin 180^\circ$ $= 0$	(65) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(6) $\tan 45^\circ$ $= 1$	(21) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(36) $\tan 90^\circ$ なし	(51) $\sin 90^\circ$ $= 1$	(66) $\tan 0^\circ$ $= 0$
(7) $\sin 0^\circ$ $= 0$	(22) $\tan 0^\circ$ $= 0$	(37) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(52) $\sin 180^\circ$ $= 0$	(67) $\sin 0^\circ$ $= 0$
(8) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(38) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(53) $\sin 0^\circ$ $= 0$	(68) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(9) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(24) $\cos 90^\circ$ $= 0$	(39) $\cos 0^\circ$ $= 1$	(54) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(69) $\cos 120^\circ$ $= -\frac{1}{2}$
(10) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\sin 180^\circ$ $= 0$	(40) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(55) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(70) $\sin 180^\circ$ $= 0$
(11) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(26) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(41) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(56) $\sin 180^\circ$ $= 0$	(71) $\cos 60^\circ$ $= \frac{1}{2}$
(12) $\sin 150^\circ$ $= \frac{1}{2}$	(27) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(42) $\tan 120^\circ$ $= -\sqrt{3}$	(57) $\cos 90^\circ$ $= 0$	(72) $\sin 0^\circ$ $= 0$
(13) $\sin 150^\circ$ $= \frac{1}{2}$	(28) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(43) $\tan 60^\circ$ $= \sqrt{3}$	(58) $\tan 90^\circ$ なし	(73) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(14) $\tan 135^\circ$ $= -1$	(29) $\tan 60^\circ$ $= \sqrt{3}$	(44) $\cos 120^\circ$ $= -\frac{1}{2}$	(59) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(74) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(15) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(30) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(45) $\tan 60^\circ$ $= \sqrt{3}$	(60) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(75) $\sin 150^\circ$ $= \frac{1}{2}$

(1) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(16) $\cos 90^\circ$ $= 0$	(31) $\sin 150^\circ$ $= \frac{1}{2}$	(46) $\tan 135^\circ$ $= -1$	(61) $\sin 180^\circ$ $= 0$
(2) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(17) $\sin 0^\circ$ $= 0$	(32) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(47) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(62) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\sin 90^\circ$ $= 1$	(18) $\tan 0^\circ$ $= 0$	(33) $\cos 60^\circ$ $= \frac{1}{2}$	(48) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(63) $\tan 90^\circ$ なし
(4) $\tan 60^\circ$ $= \sqrt{3}$	(19) $\cos 120^\circ$ $= -\frac{1}{2}$	(34) $\tan 180^\circ$ $= 0$	(49) $\tan 120^\circ$ $= -\sqrt{3}$	(64) $\sin 0^\circ$ $= 0$
(5) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(20) $\sin 30^\circ$ $= \frac{1}{2}$	(35) $\sin 90^\circ$ $= 1$	(50) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(65) $\tan 90^\circ$ なし
(6) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(21) $\sin 150^\circ$ $= \frac{1}{2}$	(36) $\tan 90^\circ$ なし	(51) $\tan 180^\circ$ $= 0$	(66) $\cos 90^\circ$ $= 0$
(7) $\cos 120^\circ$ $= -\frac{1}{2}$	(22) $\tan 45^\circ$ $= 1$	(37) $\tan 60^\circ$ $= \sqrt{3}$	(52) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(67) $\tan 90^\circ$ なし
(8) $\tan 120^\circ$ $= -\sqrt{3}$	(23) $\cos 180^\circ$ $= -1$	(38) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(53) $\cos 180^\circ$ $= -1$	(68) $\tan 45^\circ$ $= 1$
(9) $\sin 30^\circ$ $= \frac{1}{2}$	(24) $\tan 60^\circ$ $= \sqrt{3}$	(39) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(54) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\tan 90^\circ$ なし
(10) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(25) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(40) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(55) $\cos 0^\circ$ $= 1$	(70) $\tan 135^\circ$ $= -1$
(11) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(26) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(41) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(56) $\sin 150^\circ$ $= \frac{1}{2}$	(71) $\cos 180^\circ$ $= -1$
(12) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(27) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(42) $\cos 0^\circ$ $= 1$	(57) $\cos 60^\circ$ $= \frac{1}{2}$	(72) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(13) $\sin 30^\circ$ $= \frac{1}{2}$	(28) $\sin 0^\circ$ $= 0$	(43) $\sin 90^\circ$ $= 1$	(58) $\tan 60^\circ$ $= \sqrt{3}$	(73) $\sin 90^\circ$ $= 1$
(14) $\tan 120^\circ$ $= -\sqrt{3}$	(29) $\tan 135^\circ$ $= -1$	(44) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(59) $\tan 45^\circ$ $= 1$	(74) $\sin 30^\circ$ $= \frac{1}{2}$
(15) $\tan 0^\circ$ $= 0$	(30) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(45) $\sin 0^\circ$ $= 0$	(60) $\sin 30^\circ$ $= \frac{1}{2}$	(75) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$

(1) $\sin 180^\circ$ $= 0$	(16) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(31) $\cos 60^\circ$ $= \frac{1}{2}$	(46) $\tan 120^\circ$ $= -\sqrt{3}$	(61) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(2) $\tan 0^\circ$ $= 0$	(17) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(32) $\cos 60^\circ$ $= \frac{1}{2}$	(47) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(62) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(3) $\cos 120^\circ$ $= -\frac{1}{2}$	(18) $\tan 90^\circ$ なし	(33) $\cos 90^\circ$ $= 0$	(48) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(63) $\tan 45^\circ$ $= 1$
(4) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(19) $\tan 60^\circ$ $= \sqrt{3}$	(34) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(49) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(64) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(5) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(20) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(35) $\cos 90^\circ$ $= 0$	(50) $\sin 150^\circ$ $= \frac{1}{2}$	(65) $\sin 180^\circ$ $= 0$
(6) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(21) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(36) $\sin 150^\circ$ $= \frac{1}{2}$	(51) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(66) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(7) $\cos 90^\circ$ $= 0$	(22) $\cos 60^\circ$ $= \frac{1}{2}$	(37) $\cos 90^\circ$ $= 0$	(52) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(67) $\tan 180^\circ$ $= 0$
(8) $\tan 135^\circ$ $= -1$	(23) $\tan 180^\circ$ $= 0$	(38) $\cos 0^\circ$ $= 1$	(53) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(68) $\sin 90^\circ$ $= 1$
(9) $\sin 180^\circ$ $= 0$	(24) $\tan 135^\circ$ $= -1$	(39) $\sin 0^\circ$ $= 0$	(54) $\cos 180^\circ$ $= -1$	(69) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(10) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(25) $\sin 30^\circ$ $= \frac{1}{2}$	(40) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(55) $\tan 0^\circ$ $= 0$	(70) $\cos 90^\circ$ $= 0$
(11) $\cos 180^\circ$ $= -1$	(26) $\tan 90^\circ$ なし	(41) $\cos 90^\circ$ $= 0$	(56) $\sin 0^\circ$ $= 0$	(71) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(12) $\sin 0^\circ$ $= 0$	(27) $\cos 180^\circ$ $= -1$	(42) $\sin 30^\circ$ $= \frac{1}{2}$	(57) $\sin 30^\circ$ $= \frac{1}{2}$	(72) $\cos 120^\circ$ $= -\frac{1}{2}$
(13) $\cos 60^\circ$ $= \frac{1}{2}$	(28) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(43) $\tan 0^\circ$ $= 0$	(58) $\tan 90^\circ$ なし	(73) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(14) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\sin 150^\circ$ $= \frac{1}{2}$	(44) $\sin 150^\circ$ $= \frac{1}{2}$	(59) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(74) $\tan 60^\circ$ $= \sqrt{3}$
(15) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(30) $\tan 0^\circ$ $= 0$	(45) $\cos 60^\circ$ $= \frac{1}{2}$	(60) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(75) $\sin 0^\circ$ $= 0$

(1) $\tan 135^\circ$ $= -1$	(16) $\sin 90^\circ$ $= 1$	(31) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(46) $\sin 180^\circ$ $= 0$	(61) $\sin 180^\circ$ $= 0$
(2) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(17) $\tan 180^\circ$ $= 0$	(32) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(47) $\tan 120^\circ$ $= -\sqrt{3}$	(62) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(3) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(18) $\sin 150^\circ$ $= \frac{1}{2}$	(33) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(48) $\cos 180^\circ$ $= -1$	(63) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(4) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(19) $\sin 0^\circ$ $= 0$	(34) $\sin 150^\circ$ $= \frac{1}{2}$	(49) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(64) $\sin 30^\circ$ $= \frac{1}{2}$
(5) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(20) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(35) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(50) $\cos 180^\circ$ $= -1$	(65) $\sin 0^\circ$ $= 0$
(6) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(21) $\tan 135^\circ$ $= -1$	(36) $\tan 135^\circ$ $= -1$	(51) $\tan 120^\circ$ $= -\sqrt{3}$	(66) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(7) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(22) $\tan 45^\circ$ $= 1$	(37) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(52) $\tan 60^\circ$ $= \sqrt{3}$	(67) $\tan 90^\circ$ なし
(8) $\tan 90^\circ$ なし	(23) $\tan 120^\circ$ $= -\sqrt{3}$	(38) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(53) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(68) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(9) $\cos 60^\circ$ $= \frac{1}{2}$	(24) $\tan 180^\circ$ $= 0$	(39) $\cos 120^\circ$ $= -\frac{1}{2}$	(54) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(10) $\tan 90^\circ$ なし	(25) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(40) $\tan 135^\circ$ $= -1$	(55) $\sin 150^\circ$ $= \frac{1}{2}$	(70) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(11) $\sin 90^\circ$ $= 1$	(26) $\tan 60^\circ$ $= \sqrt{3}$	(41) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(56) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(71) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(12) $\cos 120^\circ$ $= -\frac{1}{2}$	(27) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(42) $\cos 90^\circ$ $= 0$	(57) $\tan 0^\circ$ $= 0$	(72) $\tan 180^\circ$ $= 0$
(13) $\cos 0^\circ$ $= 1$	(28) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(43) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(58) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(14) $\cos 0^\circ$ $= 1$	(29) $\tan 60^\circ$ $= \sqrt{3}$	(44) $\tan 120^\circ$ $= -\sqrt{3}$	(59) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(74) $\sin 180^\circ$ $= 0$
(15) $\tan 90^\circ$ なし	(30) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(45) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(60) $\sin 30^\circ$ $= \frac{1}{2}$	(75) $\sin 90^\circ$ $= 1$

(1) $\tan 90^\circ$ なし	(16) $\tan 60^\circ$ $= \sqrt{3}$	(31) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(46) $\cos 120^\circ$ $= -\frac{1}{2}$	(61) $\tan 45^\circ$ $= 1$
(2) $\sin 180^\circ$ $= 0$	(17) $\sin 30^\circ$ $= \frac{1}{2}$	(32) $\cos 120^\circ$ $= -\frac{1}{2}$	(47) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(62) $\tan 135^\circ$ $= -1$
(3) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(18) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(33) $\sin 0^\circ$ $= 0$	(48) $\sin 30^\circ$ $= \frac{1}{2}$	(63) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(4) $\cos 60^\circ$ $= \frac{1}{2}$	(19) $\cos 90^\circ$ $= 0$	(34) $\tan 45^\circ$ $= 1$	(49) $\cos 120^\circ$ $= -\frac{1}{2}$	(64) $\tan 0^\circ$ $= 0$
(5) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(20) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(35) $\sin 180^\circ$ $= 0$	(50) $\cos 180^\circ$ $= -1$	(65) $\tan 60^\circ$ $= \sqrt{3}$
(6) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(21) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(36) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(51) $\cos 0^\circ$ $= 1$	(66) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(7) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(22) $\tan 180^\circ$ $= 0$	(37) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(52) $\cos 90^\circ$ $= 0$	(67) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(8) $\tan 120^\circ$ $= -\sqrt{3}$	(23) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(38) $\sin 90^\circ$ $= 1$	(53) $\tan 45^\circ$ $= 1$	(68) $\sin 30^\circ$ $= \frac{1}{2}$
(9) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(24) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(39) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(54) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\cos 0^\circ$ $= 1$
(10) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(40) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(55) $\cos 90^\circ$ $= 0$	(70) $\tan 180^\circ$ $= 0$
(11) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(26) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(41) $\tan 0^\circ$ $= 0$	(56) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(71) $\tan 60^\circ$ $= \sqrt{3}$
(12) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\cos 180^\circ$ $= -1$	(42) $\tan 120^\circ$ $= -\sqrt{3}$	(57) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(72) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(13) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(28) $\sin 30^\circ$ $= \frac{1}{2}$	(43) $\sin 0^\circ$ $= 0$	(58) $\sin 0^\circ$ $= 0$	(73) $\sin 0^\circ$ $= 0$
(14) $\cos 60^\circ$ $= \frac{1}{2}$	(29) $\sin 180^\circ$ $= 0$	(44) $\tan 180^\circ$ $= 0$	(59) $\tan 60^\circ$ $= \sqrt{3}$	(74) $\sin 0^\circ$ $= 0$
(15) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(30) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(45) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(60) $\cos 180^\circ$ $= -1$	(75) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$

(1) $\tan 45^\circ$ $= 1$	(16) $\tan 90^\circ$ なし	(31) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(46) $\tan 120^\circ$ $= -\sqrt{3}$	(61) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(2) $\cos 90^\circ$ $= 0$	(17) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(32) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(47) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(62) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\sin 0^\circ$ $= 0$	(18) $\tan 90^\circ$ なし	(33) $\tan 0^\circ$ $= 0$	(48) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(63) $\sin 90^\circ$ $= 1$
(4) $\tan 120^\circ$ $= -\sqrt{3}$	(19) $\tan 120^\circ$ $= -\sqrt{3}$	(34) $\tan 45^\circ$ $= 1$	(49) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(64) $\sin 30^\circ$ $= \frac{1}{2}$
(5) $\cos 120^\circ$ $= -\frac{1}{2}$	(20) $\sin 180^\circ$ $= 0$	(35) $\tan 45^\circ$ $= 1$	(50) $\cos 0^\circ$ $= 1$	(65) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(6) $\tan 0^\circ$ $= 0$	(21) $\tan 135^\circ$ $= -1$	(36) $\tan 135^\circ$ $= -1$	(51) $\cos 60^\circ$ $= \frac{1}{2}$	(66) $\sin 30^\circ$ $= \frac{1}{2}$
(7) $\sin 180^\circ$ $= 0$	(22) $\tan 0^\circ$ $= 0$	(37) $\tan 120^\circ$ $= -\sqrt{3}$	(52) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(67) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(8) $\sin 90^\circ$ $= 1$	(23) $\tan 135^\circ$ $= -1$	(38) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(53) $\sin 180^\circ$ $= 0$	(68) $\tan 180^\circ$ $= 0$
(9) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(24) $\tan 135^\circ$ $= -1$	(39) $\tan 180^\circ$ $= 0$	(54) $\cos 180^\circ$ $= -1$	(69) $\sin 150^\circ$ $= \frac{1}{2}$
(10) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\cos 0^\circ$ $= 1$	(40) $\tan 120^\circ$ $= -\sqrt{3}$	(55) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(70) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(11) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(26) $\cos 120^\circ$ $= -\frac{1}{2}$	(41) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(56) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(71) $\cos 0^\circ$ $= 1$
(12) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\tan 180^\circ$ $= 0$	(42) $\tan 135^\circ$ $= -1$	(57) $\tan 90^\circ$ なし	(72) $\cos 60^\circ$ $= \frac{1}{2}$
(13) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(28) $\cos 90^\circ$ $= 0$	(43) $\tan 0^\circ$ $= 0$	(58) $\tan 120^\circ$ $= -\sqrt{3}$	(73) $\sin 30^\circ$ $= \frac{1}{2}$
(14) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(44) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(59) $\cos 60^\circ$ $= \frac{1}{2}$	(74) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(15) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(30) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(45) $\cos 180^\circ$ $= -1$	(60) $\tan 135^\circ$ $= -1$	(75) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$

(1) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(16) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(31) $\cos 90^\circ$ $= 0$	(46) $\sin 30^\circ$ $= \frac{1}{2}$	(61) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(2) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(17) $\sin 90^\circ$ $= 1$	(32) $\sin 180^\circ$ $= 0$	(47) $\tan 90^\circ$ なし	(62) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(3) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(18) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(33) $\cos 120^\circ$ $= -\frac{1}{2}$	(48) $\tan 90^\circ$ なし	(63) $\sin 30^\circ$ $= \frac{1}{2}$
(4) $\tan 0^\circ$ $= 0$	(19) $\sin 180^\circ$ $= 0$	(34) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(49) $\tan 135^\circ$ $= -1$	(64) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(5) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(20) $\cos 180^\circ$ $= -1$	(35) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(50) $\cos 90^\circ$ $= 0$	(65) $\tan 120^\circ$ $= -\sqrt{3}$
(6) $\sin 90^\circ$ $= 1$	(21) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(36) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(51) $\sin 90^\circ$ $= 1$	(66) $\tan 60^\circ$ $= \sqrt{3}$
(7) $\tan 90^\circ$ なし	(22) $\cos 120^\circ$ $= -\frac{1}{2}$	(37) $\tan 60^\circ$ $= \sqrt{3}$	(52) $\sin 90^\circ$ $= 1$	(67) $\cos 180^\circ$ $= -1$
(8) $\sin 180^\circ$ $= 0$	(23) $\cos 180^\circ$ $= -1$	(38) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(53) $\tan 180^\circ$ $= 0$	(68) $\cos 120^\circ$ $= -\frac{1}{2}$
(9) $\cos 0^\circ$ $= 1$	(24) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(39) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(54) $\cos 0^\circ$ $= 1$	(69) $\tan 0^\circ$ $= 0$
(10) $\tan 135^\circ$ $= -1$	(25) $\tan 120^\circ$ $= -\sqrt{3}$	(40) $\sin 180^\circ$ $= 0$	(55) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(70) $\cos 90^\circ$ $= 0$
(11) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(26) $\tan 0^\circ$ $= 0$	(41) $\cos 180^\circ$ $= -1$	(56) $\cos 180^\circ$ $= -1$	(71) $\cos 120^\circ$ $= -\frac{1}{2}$
(12) $\cos 180^\circ$ $= -1$	(27) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(42) $\sin 180^\circ$ $= 0$	(57) $\tan 180^\circ$ $= 0$	(72) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(13) $\tan 45^\circ$ $= 1$	(28) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(43) $\tan 0^\circ$ $= 0$	(58) $\tan 60^\circ$ $= \sqrt{3}$	(73) $\sin 180^\circ$ $= 0$
(14) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(29) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(44) $\cos 120^\circ$ $= -\frac{1}{2}$	(59) $\sin 30^\circ$ $= \frac{1}{2}$	(74) $\tan 60^\circ$ $= \sqrt{3}$
(15) $\sin 0^\circ$ $= 0$	(30) $\tan 90^\circ$ なし	(45) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(60) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(75) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$

(1) $\sin 0^\circ$ $= 0$	(16) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(31) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(46) $\tan 90^\circ$ なし	(61) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(2) $\tan 0^\circ$ $= 0$	(17) $\tan 120^\circ$ $= -\sqrt{3}$	(32) $\sin 180^\circ$ $= 0$	(47) $\sin 30^\circ$ $= \frac{1}{2}$	(62) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(3) $\cos 180^\circ$ $= -1$	(18) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(33) $\tan 60^\circ$ $= \sqrt{3}$	(48) $\cos 180^\circ$ $= -1$	(63) $\tan 180^\circ$ $= 0$
(4) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(19) $\sin 150^\circ$ $= \frac{1}{2}$	(34) $\cos 120^\circ$ $= -\frac{1}{2}$	(49) $\cos 0^\circ$ $= 1$	(64) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(5) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(20) $\sin 30^\circ$ $= \frac{1}{2}$	(35) $\cos 180^\circ$ $= -1$	(50) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(65) $\sin 150^\circ$ $= \frac{1}{2}$
(6) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(21) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(36) $\sin 30^\circ$ $= \frac{1}{2}$	(51) $\tan 45^\circ$ $= 1$	(66) $\tan 135^\circ$ $= -1$
(7) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(22) $\tan 45^\circ$ $= 1$	(37) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(52) $\sin 90^\circ$ $= 1$	(67) $\sin 180^\circ$ $= 0$
(8) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(23) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(38) $\sin 90^\circ$ $= 1$	(53) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(68) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(9) $\tan 0^\circ$ $= 0$	(24) $\cos 0^\circ$ $= 1$	(39) $\sin 30^\circ$ $= \frac{1}{2}$	(54) $\cos 90^\circ$ $= 0$	(69) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(10) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(25) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(40) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(55) $\sin 180^\circ$ $= 0$	(70) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(11) $\tan 60^\circ$ $= \sqrt{3}$	(26) $\tan 60^\circ$ $= \sqrt{3}$	(41) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(56) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(71) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(12) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(42) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(57) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(72) $\sin 90^\circ$ $= 1$
(13) $\cos 0^\circ$ $= 1$	(28) $\sin 0^\circ$ $= 0$	(43) $\cos 60^\circ$ $= \frac{1}{2}$	(58) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(73) $\cos 120^\circ$ $= -\frac{1}{2}$
(14) $\cos 60^\circ$ $= \frac{1}{2}$	(29) $\tan 60^\circ$ $= \sqrt{3}$	(44) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(59) $\tan 90^\circ$ なし	(74) $\sin 180^\circ$ $= 0$
(15) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(30) $\tan 0^\circ$ $= 0$	(45) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(60) $\tan 0^\circ$ $= 0$	(75) $\tan 0^\circ$ $= 0$

(1) $\cos 60^\circ$ $= \frac{1}{2}$	(16) $\sin 90^\circ$ $= 1$	(31) $\tan 135^\circ$ $= -1$	(46) $\tan 135^\circ$ $= -1$	(61) $\sin 30^\circ$ $= \frac{1}{2}$
(2) $\tan 60^\circ$ $= \sqrt{3}$	(17) $\tan 135^\circ$ $= -1$	(32) $\sin 90^\circ$ $= 1$	(47) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(62) $\sin 180^\circ$ $= 0$
(3) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(18) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(33) $\sin 180^\circ$ $= 0$	(48) $\sin 0^\circ$ $= 0$	(63) $\tan 0^\circ$ $= 0$
(4) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(19) $\tan 135^\circ$ $= -1$	(34) $\tan 60^\circ$ $= \sqrt{3}$	(49) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(64) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(5) $\cos 0^\circ$ $= 1$	(20) $\sin 90^\circ$ $= 1$	(35) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(50) $\tan 120^\circ$ $= -\sqrt{3}$	(65) $\sin 180^\circ$ $= 0$
(6) $\tan 135^\circ$ $= -1$	(21) $\sin 150^\circ$ $= \frac{1}{2}$	(36) $\tan 120^\circ$ $= -\sqrt{3}$	(51) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(66) $\sin 180^\circ$ $= 0$
(7) $\cos 90^\circ$ $= 0$	(22) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(37) $\tan 120^\circ$ $= -\sqrt{3}$	(52) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(67) $\tan 90^\circ$ なし
(8) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(23) $\sin 150^\circ$ $= \frac{1}{2}$	(38) $\cos 60^\circ$ $= \frac{1}{2}$	(53) $\sin 90^\circ$ $= 1$	(68) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(9) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(24) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(39) $\sin 150^\circ$ $= \frac{1}{2}$	(54) $\cos 90^\circ$ $= 0$	(69) $\cos 60^\circ$ $= \frac{1}{2}$
(10) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(40) $\tan 60^\circ$ $= \sqrt{3}$	(55) $\tan 135^\circ$ $= -1$	(70) $\sin 150^\circ$ $= \frac{1}{2}$
(11) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(26) $\sin 180^\circ$ $= 0$	(41) $\tan 0^\circ$ $= 0$	(56) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(71) $\tan 45^\circ$ $= 1$
(12) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(27) $\tan 0^\circ$ $= 0$	(42) $\tan 60^\circ$ $= \sqrt{3}$	(57) $\cos 120^\circ$ $= -\frac{1}{2}$	(72) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(13) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(28) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(43) $\sin 180^\circ$ $= 0$	(58) $\cos 180^\circ$ $= -1$	(73) $\sin 0^\circ$ $= 0$
(14) $\sin 90^\circ$ $= 1$	(29) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(44) $\sin 150^\circ$ $= \frac{1}{2}$	(59) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(74) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(15) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(30) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(45) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(60) $\cos 60^\circ$ $= \frac{1}{2}$	(75) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$

(1) $\sin 0^\circ$ $= 0$	(16) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(31) $\sin 150^\circ$ $= \frac{1}{2}$	(46) $\sin 0^\circ$ $= 0$	(61) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(2) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(17) $\tan 60^\circ$ $= \sqrt{3}$	(32) $\cos 120^\circ$ $= -\frac{1}{2}$	(47) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(62) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\tan 135^\circ$ $= -1$	(18) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(33) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(48) $\cos 90^\circ$ $= 0$	(63) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(4) $\tan 60^\circ$ $= \sqrt{3}$	(19) $\cos 180^\circ$ $= -1$	(34) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(49) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(64) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(5) $\cos 120^\circ$ $= -\frac{1}{2}$	(20) $\cos 90^\circ$ $= 0$	(35) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(50) $\sin 150^\circ$ $= \frac{1}{2}$	(65) $\sin 150^\circ$ $= \frac{1}{2}$
(6) $\tan 180^\circ$ $= 0$	(21) $\cos 180^\circ$ $= -1$	(36) $\sin 30^\circ$ $= \frac{1}{2}$	(51) $\tan 60^\circ$ $= \sqrt{3}$	(66) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(7) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(22) $\tan 120^\circ$ $= -\sqrt{3}$	(37) $\sin 180^\circ$ $= 0$	(52) $\tan 180^\circ$ $= 0$	(67) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(8) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(23) $\cos 120^\circ$ $= -\frac{1}{2}$	(38) $\cos 90^\circ$ $= 0$	(53) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(68) $\cos 0^\circ$ $= 1$
(9) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(24) $\sin 180^\circ$ $= 0$	(39) $\tan 135^\circ$ $= -1$	(54) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\sin 180^\circ$ $= 0$
(10) $\cos 0^\circ$ $= 1$	(25) $\tan 135^\circ$ $= -1$	(40) $\tan 120^\circ$ $= -\sqrt{3}$	(55) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(70) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(11) $\cos 120^\circ$ $= -\frac{1}{2}$	(26) $\tan 90^\circ$ なし	(41) $\tan 60^\circ$ $= \sqrt{3}$	(56) $\tan 90^\circ$ なし	(71) $\sin 150^\circ$ $= \frac{1}{2}$
(12) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(27) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(42) $\cos 60^\circ$ $= \frac{1}{2}$	(57) $\tan 60^\circ$ $= \sqrt{3}$	(72) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(13) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(28) $\tan 180^\circ$ $= 0$	(43) $\tan 90^\circ$ なし	(58) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(73) $\tan 120^\circ$ $= -\sqrt{3}$
(14) $\cos 0^\circ$ $= 1$	(29) $\cos 0^\circ$ $= 1$	(44) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(59) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(74) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(15) $\tan 60^\circ$ $= \sqrt{3}$	(30) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(45) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(60) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(75) $\sin 150^\circ$ $= \frac{1}{2}$

3.2.3 三角関数 $0^\circ \leq \theta \leq 360^\circ$

三角関数 $0^\circ \leq \theta \leq 360^\circ$ の値。

(1) $\sin 135^\circ$	(16) $\tan 135^\circ$	(31) $\cos 45^\circ$	(46) $\cos 135^\circ$	(61) $\cos 300^\circ$
(2) $\cos 270^\circ$	(17) $\sin 315^\circ$	(32) $\cos 135^\circ$	(47) $\sin 60^\circ$	(62) $\sin 30^\circ$
(3) $\tan 120^\circ$	(18) $\cos 315^\circ$	(33) $\cos 330^\circ$	(48) $\sin 60^\circ$	(63) $\sin 330^\circ$
(4) $\sin 330^\circ$	(19) $\sin 315^\circ$	(34) $\tan 150^\circ$	(49) $\cos 240^\circ$	(64) $\sin 225^\circ$
(5) $\cos 150^\circ$	(20) $\cos 270^\circ$	(35) $\tan 30^\circ$	(50) $\cos 90^\circ$	(65) $\tan 240^\circ$
(6) $\tan 0^\circ$	(21) $\tan 240^\circ$	(36) $\tan 315^\circ$	(51) $\tan 225^\circ$	(66) $\cos 300^\circ$
(7) $\sin 315^\circ$	(22) $\sin 0^\circ$	(37) $\sin 45^\circ$	(52) $\sin 270^\circ$	(67) $\cos 120^\circ$
(8) $\tan 180^\circ$	(23) $\cos 180^\circ$	(38) $\cos 240^\circ$	(53) $\tan 360^\circ$	(68) $\tan 270^\circ$
(9) $\tan 90^\circ$	(24) $\cos 315^\circ$	(39) $\cos 150^\circ$	(54) $\cos 60^\circ$	(69) $\cos 270^\circ$
(10) $\sin 330^\circ$	(25) $\cos 330^\circ$	(40) $\cos 300^\circ$	(55) $\cos 90^\circ$	(70) $\sin 0^\circ$
(11) $\sin 240^\circ$	(26) $\sin 90^\circ$	(41) $\sin 0^\circ$	(56) $\tan 150^\circ$	(71) $\sin 270^\circ$
(12) $\sin 120^\circ$	(27) $\sin 315^\circ$	(42) $\sin 30^\circ$	(57) $\tan 330^\circ$	(72) $\sin 300^\circ$
(13) $\cos 225^\circ$	(28) $\tan 120^\circ$	(43) $\tan 180^\circ$	(58) $\cos 0^\circ$	(73) $\cos 60^\circ$
(14) $\sin 0^\circ$	(29) $\sin 300^\circ$	(44) $\tan 300^\circ$	(59) $\sin 225^\circ$	(74) $\cos 0^\circ$
(15) $\cos 135^\circ$	(30) $\sin 0^\circ$	(45) $\sin 0^\circ$	(60) $\sin 330^\circ$	(75) $\tan 90^\circ$

(1) $\cos 30^\circ$	(16) $\sin 150^\circ$	(31) $\cos 30^\circ$	(46) $\cos 300^\circ$	(61) $\tan 45^\circ$
(2) $\tan 60^\circ$	(17) $\tan 60^\circ$	(32) $\sin 360^\circ$	(47) $\tan 180^\circ$	(62) $\cos 210^\circ$
(3) $\cos 210^\circ$	(18) $\cos 210^\circ$	(33) $\sin 60^\circ$	(48) $\cos 30^\circ$	(63) $\tan 45^\circ$
(4) $\cos 270^\circ$	(19) $\tan 315^\circ$	(34) $\sin 30^\circ$	(49) $\cos 90^\circ$	(64) $\cos 60^\circ$
(5) $\tan 150^\circ$	(20) $\sin 135^\circ$	(35) $\tan 300^\circ$	(50) $\cos 90^\circ$	(65) $\cos 0^\circ$
(6) $\cos 300^\circ$	(21) $\cos 300^\circ$	(36) $\tan 0^\circ$	(51) $\cos 300^\circ$	(66) $\cos 210^\circ$
(7) $\cos 0^\circ$	(22) $\cos 150^\circ$	(37) $\sin 90^\circ$	(52) $\cos 270^\circ$	(67) $\tan 120^\circ$
(8) $\cos 210^\circ$	(23) $\sin 60^\circ$	(38) $\tan 300^\circ$	(53) $\tan 360^\circ$	(68) $\sin 0^\circ$
(9) $\cos 135^\circ$	(24) $\sin 300^\circ$	(39) $\cos 240^\circ$	(54) $\sin 315^\circ$	(69) $\cos 300^\circ$
(10) $\sin 210^\circ$	(25) $\cos 210^\circ$	(40) $\tan 240^\circ$	(55) $\sin 120^\circ$	(70) $\cos 90^\circ$
(11) $\tan 315^\circ$	(26) $\tan 360^\circ$	(41) $\tan 150^\circ$	(56) $\cos 315^\circ$	(71) $\tan 225^\circ$
(12) $\sin 150^\circ$	(27) $\tan 315^\circ$	(42) $\tan 300^\circ$	(57) $\tan 120^\circ$	(72) $\tan 0^\circ$
(13) $\tan 30^\circ$	(28) $\cos 315^\circ$	(43) $\cos 150^\circ$	(58) $\cos 120^\circ$	(73) $\cos 0^\circ$
(14) $\cos 315^\circ$	(29) $\tan 150^\circ$	(44) $\cos 360^\circ$	(59) $\cos 210^\circ$	(74) $\cos 315^\circ$
(15) $\sin 60^\circ$	(30) $\sin 60^\circ$	(45) $\tan 45^\circ$	(60) $\sin 240^\circ$	(75) $\sin 240^\circ$

(1) $\sin 300^\circ$	(16) $\sin 270^\circ$	(31) $\tan 180^\circ$	(46) $\tan 120^\circ$	(61) $\tan 360^\circ$
(2) $\sin 45^\circ$	(17) $\tan 45^\circ$	(32) $\sin 210^\circ$	(47) $\tan 90^\circ$	(62) $\tan 300^\circ$
(3) $\tan 0^\circ$	(18) $\tan 45^\circ$	(33) $\sin 270^\circ$	(48) $\cos 120^\circ$	(63) $\sin 120^\circ$
(4) $\sin 300^\circ$	(19) $\sin 135^\circ$	(34) $\tan 225^\circ$	(49) $\tan 330^\circ$	(64) $\cos 90^\circ$
(5) $\sin 360^\circ$	(20) $\tan 60^\circ$	(35) $\sin 240^\circ$	(50) $\cos 90^\circ$	(65) $\sin 150^\circ$
(6) $\tan 300^\circ$	(21) $\tan 225^\circ$	(36) $\sin 135^\circ$	(51) $\sin 60^\circ$	(66) $\cos 225^\circ$
(7) $\tan 225^\circ$	(22) $\tan 90^\circ$	(37) $\cos 210^\circ$	(52) $\cos 330^\circ$	(67) $\tan 120^\circ$
(8) $\sin 135^\circ$	(23) $\tan 270^\circ$	(38) $\cos 120^\circ$	(53) $\sin 225^\circ$	(68) $\tan 90^\circ$
(9) $\tan 180^\circ$	(24) $\cos 120^\circ$	(39) $\cos 45^\circ$	(54) $\sin 120^\circ$	(69) $\tan 30^\circ$
(10) $\sin 300^\circ$	(25) $\tan 0^\circ$	(40) $\cos 135^\circ$	(55) $\cos 135^\circ$	(70) $\cos 60^\circ$
(11) $\cos 90^\circ$	(26) $\sin 120^\circ$	(41) $\sin 270^\circ$	(56) $\cos 300^\circ$	(71) $\sin 135^\circ$
(12) $\cos 120^\circ$	(27) $\tan 90^\circ$	(42) $\cos 360^\circ$	(57) $\tan 210^\circ$	(72) $\tan 240^\circ$
(13) $\cos 315^\circ$	(28) $\sin 225^\circ$	(43) $\tan 315^\circ$	(58) $\tan 180^\circ$	(73) $\tan 30^\circ$
(14) $\sin 360^\circ$	(29) $\sin 150^\circ$	(44) $\tan 120^\circ$	(59) $\sin 0^\circ$	(74) $\tan 150^\circ$
(15) $\cos 45^\circ$	(30) $\sin 270^\circ$	(45) $\cos 180^\circ$	(60) $\sin 315^\circ$	(75) $\tan 225^\circ$

(1) $\sin 120^\circ$	(16) $\tan 0^\circ$	(31) $\tan 0^\circ$	(46) $\sin 120^\circ$	(61) $\cos 330^\circ$
(2) $\sin 315^\circ$	(17) $\tan 240^\circ$	(32) $\tan 180^\circ$	(47) $\cos 60^\circ$	(62) $\sin 360^\circ$
(3) $\tan 90^\circ$	(18) $\cos 210^\circ$	(33) $\sin 90^\circ$	(48) $\tan 45^\circ$	(63) $\cos 150^\circ$
(4) $\tan 150^\circ$	(19) $\sin 240^\circ$	(34) $\sin 270^\circ$	(49) $\sin 360^\circ$	(64) $\sin 90^\circ$
(5) $\tan 135^\circ$	(20) $\cos 30^\circ$	(35) $\cos 30^\circ$	(50) $\tan 360^\circ$	(65) $\cos 300^\circ$
(6) $\cos 60^\circ$	(21) $\sin 300^\circ$	(36) $\tan 270^\circ$	(51) $\sin 90^\circ$	(66) $\tan 45^\circ$
(7) $\cos 60^\circ$	(22) $\cos 300^\circ$	(37) $\sin 135^\circ$	(52) $\tan 135^\circ$	(67) $\sin 210^\circ$
(8) $\sin 45^\circ$	(23) $\cos 135^\circ$	(38) $\tan 90^\circ$	(53) $\cos 240^\circ$	(68) $\tan 330^\circ$
(9) $\sin 300^\circ$	(24) $\tan 90^\circ$	(39) $\tan 225^\circ$	(54) $\tan 300^\circ$	(69) $\cos 30^\circ$
(10) $\sin 210^\circ$	(25) $\sin 90^\circ$	(40) $\cos 360^\circ$	(55) $\tan 315^\circ$	(70) $\sin 150^\circ$
(11) $\sin 240^\circ$	(26) $\cos 180^\circ$	(41) $\tan 90^\circ$	(56) $\tan 60^\circ$	(71) $\tan 225^\circ$
(12) $\cos 315^\circ$	(27) $\tan 30^\circ$	(42) $\tan 300^\circ$	(57) $\sin 225^\circ$	(72) $\sin 315^\circ$
(13) $\cos 330^\circ$	(28) $\sin 240^\circ$	(43) $\sin 270^\circ$	(58) $\cos 330^\circ$	(73) $\tan 180^\circ$
(14) $\cos 0^\circ$	(29) $\cos 360^\circ$	(44) $\cos 150^\circ$	(59) $\sin 225^\circ$	(74) $\tan 315^\circ$
(15) $\sin 360^\circ$	(30) $\tan 45^\circ$	(45) $\sin 360^\circ$	(60) $\tan 270^\circ$	(75) $\tan 210^\circ$

(1) $\cos 45^\circ$	(16) $\cos 330^\circ$	(31) $\cos 210^\circ$	(46) $\cos 270^\circ$	(61) $\cos 90^\circ$
(2) $\tan 270^\circ$	(17) $\sin 210^\circ$	(32) $\tan 180^\circ$	(47) $\cos 0^\circ$	(62) $\tan 180^\circ$
(3) $\tan 150^\circ$	(18) $\sin 60^\circ$	(33) $\sin 150^\circ$	(48) $\sin 90^\circ$	(63) $\sin 45^\circ$
(4) $\cos 45^\circ$	(19) $\cos 360^\circ$	(34) $\cos 90^\circ$	(49) $\cos 30^\circ$	(64) $\cos 270^\circ$
(5) $\sin 120^\circ$	(20) $\cos 135^\circ$	(35) $\cos 45^\circ$	(50) $\cos 150^\circ$	(65) $\cos 330^\circ$
(6) $\cos 60^\circ$	(21) $\cos 120^\circ$	(36) $\cos 135^\circ$	(51) $\sin 270^\circ$	(66) $\sin 360^\circ$
(7) $\cos 225^\circ$	(22) $\cos 0^\circ$	(37) $\cos 180^\circ$	(52) $\tan 315^\circ$	(67) $\cos 135^\circ$
(8) $\sin 330^\circ$	(23) $\cos 315^\circ$	(38) $\cos 330^\circ$	(53) $\sin 135^\circ$	(68) $\cos 90^\circ$
(9) $\cos 135^\circ$	(24) $\tan 300^\circ$	(39) $\tan 330^\circ$	(54) $\tan 135^\circ$	(69) $\cos 30^\circ$
(10) $\tan 0^\circ$	(25) $\sin 30^\circ$	(40) $\tan 180^\circ$	(55) $\cos 120^\circ$	(70) $\sin 60^\circ$
(11) $\sin 225^\circ$	(26) $\tan 30^\circ$	(41) $\cos 60^\circ$	(56) $\cos 135^\circ$	(71) $\cos 45^\circ$
(12) $\tan 150^\circ$	(27) $\tan 30^\circ$	(42) $\tan 60^\circ$	(57) $\sin 315^\circ$	(72) $\sin 0^\circ$
(13) $\tan 330^\circ$	(28) $\cos 45^\circ$	(43) $\tan 360^\circ$	(58) $\tan 315^\circ$	(73) $\cos 30^\circ$
(14) $\cos 60^\circ$	(29) $\tan 30^\circ$	(44) $\tan 330^\circ$	(59) $\sin 225^\circ$	(74) $\cos 135^\circ$
(15) $\sin 360^\circ$	(30) $\tan 240^\circ$	(45) $\cos 210^\circ$	(60) $\cos 240^\circ$	(75) $\cos 300^\circ$

(1) $\tan 0^\circ$	(16) $\cos 180^\circ$	(31) $\sin 300^\circ$	(46) $\tan 300^\circ$	(61) $\tan 360^\circ$
(2) $\cos 135^\circ$	(17) $\sin 30^\circ$	(32) $\sin 210^\circ$	(47) $\tan 225^\circ$	(62) $\tan 120^\circ$
(3) $\cos 150^\circ$	(18) $\cos 210^\circ$	(33) $\tan 135^\circ$	(48) $\tan 270^\circ$	(63) $\tan 315^\circ$
(4) $\sin 270^\circ$	(19) $\tan 60^\circ$	(34) $\tan 225^\circ$	(49) $\cos 60^\circ$	(64) $\sin 240^\circ$
(5) $\cos 300^\circ$	(20) $\tan 210^\circ$	(35) $\sin 0^\circ$	(50) $\sin 240^\circ$	(65) $\tan 90^\circ$
(6) $\sin 225^\circ$	(21) $\tan 330^\circ$	(36) $\cos 30^\circ$	(51) $\sin 90^\circ$	(66) $\tan 210^\circ$
(7) $\cos 45^\circ$	(22) $\tan 360^\circ$	(37) $\cos 180^\circ$	(52) $\tan 180^\circ$	(67) $\cos 135^\circ$
(8) $\sin 240^\circ$	(23) $\cos 225^\circ$	(38) $\tan 225^\circ$	(53) $\sin 300^\circ$	(68) $\tan 180^\circ$
(9) $\tan 240^\circ$	(24) $\tan 360^\circ$	(39) $\cos 60^\circ$	(54) $\sin 45^\circ$	(69) $\tan 210^\circ$
(10) $\cos 360^\circ$	(25) $\cos 120^\circ$	(40) $\tan 270^\circ$	(55) $\tan 30^\circ$	(70) $\cos 60^\circ$
(11) $\tan 300^\circ$	(26) $\sin 0^\circ$	(41) $\tan 330^\circ$	(56) $\cos 300^\circ$	(71) $\sin 180^\circ$
(12) $\tan 135^\circ$	(27) $\sin 135^\circ$	(42) $\sin 30^\circ$	(57) $\sin 315^\circ$	(72) $\cos 300^\circ$
(13) $\cos 90^\circ$	(28) $\cos 240^\circ$	(43) $\cos 150^\circ$	(58) $\cos 45^\circ$	(73) $\sin 120^\circ$
(14) $\cos 0^\circ$	(29) $\cos 135^\circ$	(44) $\cos 180^\circ$	(59) $\sin 270^\circ$	(74) $\cos 135^\circ$
(15) $\sin 45^\circ$	(30) $\tan 330^\circ$	(45) $\cos 210^\circ$	(60) $\sin 135^\circ$	(75) $\tan 270^\circ$

(1) $\cos 0^\circ$	(16) $\cos 210^\circ$	(31) $\cos 180^\circ$	(46) $\sin 150^\circ$	(61) $\tan 45^\circ$
(2) $\sin 120^\circ$	(17) $\cos 315^\circ$	(32) $\tan 315^\circ$	(47) $\cos 120^\circ$	(62) $\cos 315^\circ$
(3) $\tan 120^\circ$	(18) $\tan 300^\circ$	(33) $\sin 150^\circ$	(48) $\sin 30^\circ$	(63) $\sin 60^\circ$
(4) $\sin 0^\circ$	(19) $\cos 240^\circ$	(34) $\cos 225^\circ$	(49) $\sin 225^\circ$	(64) $\tan 180^\circ$
(5) $\sin 330^\circ$	(20) $\tan 150^\circ$	(35) $\sin 0^\circ$	(50) $\tan 0^\circ$	(65) $\tan 0^\circ$
(6) $\tan 45^\circ$	(21) $\tan 90^\circ$	(36) $\sin 45^\circ$	(51) $\cos 180^\circ$	(66) $\tan 120^\circ$
(7) $\sin 45^\circ$	(22) $\cos 300^\circ$	(37) $\sin 270^\circ$	(52) $\sin 90^\circ$	(67) $\tan 60^\circ$
(8) $\tan 360^\circ$	(23) $\sin 180^\circ$	(38) $\sin 120^\circ$	(53) $\sin 0^\circ$	(68) $\tan 120^\circ$
(9) $\cos 360^\circ$	(24) $\tan 270^\circ$	(39) $\tan 0^\circ$	(54) $\cos 225^\circ$	(69) $\tan 180^\circ$
(10) $\sin 315^\circ$	(25) $\cos 60^\circ$	(40) $\sin 0^\circ$	(55) $\sin 225^\circ$	(70) $\cos 210^\circ$
(11) $\cos 90^\circ$	(26) $\cos 315^\circ$	(41) $\tan 90^\circ$	(56) $\cos 120^\circ$	(71) $\cos 120^\circ$
(12) $\tan 60^\circ$	(27) $\sin 150^\circ$	(42) $\cos 225^\circ$	(57) $\cos 60^\circ$	(72) $\tan 60^\circ$
(13) $\sin 240^\circ$	(28) $\cos 150^\circ$	(43) $\tan 300^\circ$	(58) $\tan 300^\circ$	(73) $\tan 135^\circ$
(14) $\cos 210^\circ$	(29) $\tan 0^\circ$	(44) $\tan 270^\circ$	(59) $\cos 315^\circ$	(74) $\tan 180^\circ$
(15) $\cos 330^\circ$	(30) $\cos 45^\circ$	(45) $\cos 120^\circ$	(60) $\cos 45^\circ$	(75) $\cos 300^\circ$

(1) $\tan 270^\circ$	(16) $\cos 90^\circ$	(31) $\tan 45^\circ$	(46) $\tan 0^\circ$	(61) $\tan 300^\circ$
(2) $\tan 360^\circ$	(17) $\cos 135^\circ$	(32) $\tan 240^\circ$	(47) $\sin 240^\circ$	(62) $\tan 45^\circ$
(3) $\sin 300^\circ$	(18) $\sin 225^\circ$	(33) $\cos 240^\circ$	(48) $\cos 120^\circ$	(63) $\sin 330^\circ$
(4) $\sin 90^\circ$	(19) $\sin 240^\circ$	(34) $\sin 225^\circ$	(49) $\tan 360^\circ$	(64) $\tan 45^\circ$
(5) $\sin 270^\circ$	(20) $\cos 135^\circ$	(35) $\sin 210^\circ$	(50) $\sin 150^\circ$	(65) $\cos 240^\circ$
(6) $\cos 45^\circ$	(21) $\sin 30^\circ$	(36) $\sin 300^\circ$	(51) $\cos 360^\circ$	(66) $\cos 150^\circ$
(7) $\cos 270^\circ$	(22) $\tan 150^\circ$	(37) $\sin 240^\circ$	(52) $\cos 45^\circ$	(67) $\cos 150^\circ$
(8) $\cos 315^\circ$	(23) $\tan 315^\circ$	(38) $\sin 270^\circ$	(53) $\tan 135^\circ$	(68) $\tan 270^\circ$
(9) $\tan 330^\circ$	(24) $\sin 0^\circ$	(39) $\sin 0^\circ$	(54) $\cos 210^\circ$	(69) $\cos 315^\circ$
(10) $\sin 120^\circ$	(25) $\tan 0^\circ$	(40) $\sin 150^\circ$	(55) $\tan 360^\circ$	(70) $\tan 180^\circ$
(11) $\cos 360^\circ$	(26) $\cos 225^\circ$	(41) $\sin 30^\circ$	(56) $\sin 180^\circ$	(71) $\tan 300^\circ$
(12) $\cos 90^\circ$	(27) $\cos 240^\circ$	(42) $\tan 135^\circ$	(57) $\cos 360^\circ$	(72) $\sin 180^\circ$
(13) $\cos 150^\circ$	(28) $\cos 135^\circ$	(43) $\sin 330^\circ$	(58) $\tan 120^\circ$	(73) $\sin 0^\circ$
(14) $\sin 360^\circ$	(29) $\sin 180^\circ$	(44) $\tan 240^\circ$	(59) $\cos 30^\circ$	(74) $\sin 240^\circ$
(15) $\tan 360^\circ$	(30) $\sin 180^\circ$	(45) $\sin 60^\circ$	(60) $\sin 315^\circ$	(75) $\cos 60^\circ$

(1) $\tan 45^\circ$	(16) $\cos 330^\circ$	(31) $\sin 180^\circ$	(46) $\tan 360^\circ$	(61) $\cos 225^\circ$
(2) $\sin 270^\circ$	(17) $\cos 210^\circ$	(32) $\cos 150^\circ$	(47) $\cos 135^\circ$	(62) $\tan 240^\circ$
(3) $\cos 120^\circ$	(18) $\tan 330^\circ$	(33) $\cos 30^\circ$	(48) $\tan 120^\circ$	(63) $\tan 240^\circ$
(4) $\sin 45^\circ$	(19) $\cos 225^\circ$	(34) $\sin 270^\circ$	(49) $\tan 30^\circ$	(64) $\tan 180^\circ$
(5) $\sin 180^\circ$	(20) $\sin 135^\circ$	(35) $\tan 60^\circ$	(50) $\cos 30^\circ$	(65) $\tan 120^\circ$
(6) $\cos 150^\circ$	(21) $\cos 240^\circ$	(36) $\tan 90^\circ$	(51) $\sin 180^\circ$	(66) $\tan 360^\circ$
(7) $\sin 150^\circ$	(22) $\tan 150^\circ$	(37) $\cos 0^\circ$	(52) $\sin 315^\circ$	(67) $\sin 90^\circ$
(8) $\sin 300^\circ$	(23) $\cos 315^\circ$	(38) $\tan 240^\circ$	(53) $\cos 60^\circ$	(68) $\sin 150^\circ$
(9) $\sin 135^\circ$	(24) $\cos 270^\circ$	(39) $\cos 225^\circ$	(54) $\tan 120^\circ$	(69) $\tan 45^\circ$
(10) $\sin 180^\circ$	(25) $\sin 360^\circ$	(40) $\tan 45^\circ$	(55) $\sin 360^\circ$	(70) $\sin 240^\circ$
(11) $\sin 270^\circ$	(26) $\cos 360^\circ$	(41) $\cos 240^\circ$	(56) $\sin 360^\circ$	(71) $\tan 315^\circ$
(12) $\sin 120^\circ$	(27) $\sin 0^\circ$	(42) $\tan 45^\circ$	(57) $\cos 270^\circ$	(72) $\tan 30^\circ$
(13) $\cos 120^\circ$	(28) $\cos 120^\circ$	(43) $\tan 135^\circ$	(58) $\sin 90^\circ$	(73) $\sin 330^\circ$
(14) $\tan 270^\circ$	(29) $\sin 120^\circ$	(44) $\tan 30^\circ$	(59) $\cos 0^\circ$	(74) $\tan 210^\circ$
(15) $\cos 180^\circ$	(30) $\sin 180^\circ$	(45) $\tan 300^\circ$	(60) $\tan 315^\circ$	(75) $\tan 360^\circ$

(1) $\tan 180^\circ$	(16) $\cos 360^\circ$	(31) $\tan 240^\circ$	(46) $\cos 300^\circ$	(61) $\sin 60^\circ$
(2) $\tan 300^\circ$	(17) $\sin 330^\circ$	(32) $\cos 330^\circ$	(47) $\tan 360^\circ$	(62) $\tan 120^\circ$
(3) $\sin 45^\circ$	(18) $\tan 240^\circ$	(33) $\sin 150^\circ$	(48) $\cos 180^\circ$	(63) $\cos 120^\circ$
(4) $\cos 225^\circ$	(19) $\sin 90^\circ$	(34) $\cos 60^\circ$	(49) $\cos 120^\circ$	(64) $\sin 120^\circ$
(5) $\sin 120^\circ$	(20) $\cos 315^\circ$	(35) $\sin 330^\circ$	(50) $\sin 330^\circ$	(65) $\cos 180^\circ$
(6) $\sin 60^\circ$	(21) $\sin 150^\circ$	(36) $\tan 60^\circ$	(51) $\cos 180^\circ$	(66) $\sin 240^\circ$
(7) $\cos 240^\circ$	(22) $\cos 225^\circ$	(37) $\sin 120^\circ$	(52) $\cos 330^\circ$	(67) $\tan 90^\circ$
(8) $\cos 360^\circ$	(23) $\sin 225^\circ$	(38) $\tan 315^\circ$	(53) $\sin 315^\circ$	(68) $\sin 270^\circ$
(9) $\cos 270^\circ$	(24) $\sin 60^\circ$	(39) $\cos 330^\circ$	(54) $\tan 120^\circ$	(69) $\tan 315^\circ$
(10) $\tan 120^\circ$	(25) $\sin 300^\circ$	(40) $\sin 180^\circ$	(55) $\tan 300^\circ$	(70) $\tan 30^\circ$
(11) $\cos 150^\circ$	(26) $\sin 60^\circ$	(41) $\tan 330^\circ$	(56) $\cos 0^\circ$	(71) $\sin 90^\circ$
(12) $\cos 210^\circ$	(27) $\sin 45^\circ$	(42) $\cos 240^\circ$	(57) $\tan 60^\circ$	(72) $\cos 0^\circ$
(13) $\tan 270^\circ$	(28) $\tan 270^\circ$	(43) $\cos 120^\circ$	(58) $\cos 45^\circ$	(73) $\sin 90^\circ$
(14) $\sin 360^\circ$	(29) $\cos 45^\circ$	(44) $\tan 210^\circ$	(59) $\sin 120^\circ$	(74) $\tan 330^\circ$
(15) $\tan 0^\circ$	(30) $\tan 210^\circ$	(45) $\tan 210^\circ$	(60) $\tan 270^\circ$	(75) $\cos 30^\circ$

(1) $\tan 240^\circ$	(16) $\tan 330^\circ$	(31) $\sin 315^\circ$	(46) $\tan 120^\circ$	(61) $\sin 225^\circ$
(2) $\cos 180^\circ$	(17) $\cos 150^\circ$	(32) $\tan 30^\circ$	(47) $\tan 135^\circ$	(62) $\tan 300^\circ$
(3) $\sin 150^\circ$	(18) $\cos 150^\circ$	(33) $\cos 360^\circ$	(48) $\tan 210^\circ$	(63) $\tan 60^\circ$
(4) $\tan 120^\circ$	(19) $\tan 135^\circ$	(34) $\cos 120^\circ$	(49) $\tan 0^\circ$	(64) $\tan 240^\circ$
(5) $\sin 300^\circ$	(20) $\tan 210^\circ$	(35) $\tan 270^\circ$	(50) $\cos 150^\circ$	(65) $\tan 240^\circ$
(6) $\sin 240^\circ$	(21) $\tan 30^\circ$	(36) $\cos 0^\circ$	(51) $\sin 135^\circ$	(66) $\sin 360^\circ$
(7) $\cos 300^\circ$	(22) $\tan 240^\circ$	(37) $\sin 330^\circ$	(52) $\tan 240^\circ$	(67) $\tan 150^\circ$
(8) $\sin 90^\circ$	(23) $\sin 210^\circ$	(38) $\cos 330^\circ$	(53) $\sin 60^\circ$	(68) $\tan 45^\circ$
(9) $\tan 225^\circ$	(24) $\tan 210^\circ$	(39) $\sin 330^\circ$	(54) $\sin 240^\circ$	(69) $\cos 315^\circ$
(10) $\tan 135^\circ$	(25) $\sin 150^\circ$	(40) $\sin 210^\circ$	(55) $\tan 180^\circ$	(70) $\sin 45^\circ$
(11) $\sin 135^\circ$	(26) $\tan 210^\circ$	(41) $\cos 135^\circ$	(56) $\tan 60^\circ$	(71) $\tan 45^\circ$
(12) $\tan 60^\circ$	(27) $\sin 315^\circ$	(42) $\sin 315^\circ$	(57) $\sin 210^\circ$	(72) $\tan 90^\circ$
(13) $\sin 135^\circ$	(28) $\cos 270^\circ$	(43) $\cos 120^\circ$	(58) $\tan 150^\circ$	(73) $\cos 60^\circ$
(14) $\cos 150^\circ$	(29) $\sin 270^\circ$	(44) $\sin 135^\circ$	(59) $\tan 60^\circ$	(74) $\cos 210^\circ$
(15) $\cos 0^\circ$	(30) $\tan 360^\circ$	(45) $\sin 150^\circ$	(60) $\sin 210^\circ$	(75) $\cos 135^\circ$

(1) $\tan 135^\circ$	(16) $\tan 180^\circ$	(31) $\tan 0^\circ$	(46) $\tan 240^\circ$	(61) $\tan 180^\circ$
(2) $\tan 240^\circ$	(17) $\tan 90^\circ$	(32) $\cos 60^\circ$	(47) $\tan 0^\circ$	(62) $\sin 315^\circ$
(3) $\tan 90^\circ$	(18) $\tan 300^\circ$	(33) $\sin 60^\circ$	(48) $\tan 315^\circ$	(63) $\sin 135^\circ$
(4) $\tan 120^\circ$	(19) $\cos 225^\circ$	(34) $\cos 150^\circ$	(49) $\cos 315^\circ$	(64) $\cos 60^\circ$
(5) $\cos 0^\circ$	(20) $\sin 135^\circ$	(35) $\cos 60^\circ$	(50) $\tan 150^\circ$	(65) $\sin 180^\circ$
(6) $\tan 0^\circ$	(21) $\tan 30^\circ$	(36) $\cos 30^\circ$	(51) $\tan 180^\circ$	(66) $\tan 135^\circ$
(7) $\sin 315^\circ$	(22) $\cos 60^\circ$	(37) $\sin 360^\circ$	(52) $\cos 360^\circ$	(67) $\tan 150^\circ$
(8) $\cos 225^\circ$	(23) $\tan 210^\circ$	(38) $\tan 120^\circ$	(53) $\tan 330^\circ$	(68) $\sin 45^\circ$
(9) $\cos 30^\circ$	(24) $\tan 45^\circ$	(39) $\sin 90^\circ$	(54) $\cos 45^\circ$	(69) $\tan 240^\circ$
(10) $\tan 120^\circ$	(25) $\tan 300^\circ$	(40) $\tan 90^\circ$	(55) $\tan 330^\circ$	(70) $\cos 0^\circ$
(11) $\sin 225^\circ$	(26) $\cos 225^\circ$	(41) $\sin 90^\circ$	(56) $\tan 150^\circ$	(71) $\sin 240^\circ$
(12) $\sin 315^\circ$	(27) $\sin 150^\circ$	(42) $\tan 270^\circ$	(57) $\tan 60^\circ$	(72) $\cos 210^\circ$
(13) $\sin 240^\circ$	(28) $\sin 120^\circ$	(43) $\sin 315^\circ$	(58) $\sin 120^\circ$	(73) $\tan 90^\circ$
(14) $\sin 300^\circ$	(29) $\cos 150^\circ$	(44) $\cos 315^\circ$	(59) $\tan 45^\circ$	(74) $\cos 315^\circ$
(15) $\sin 225^\circ$	(30) $\sin 300^\circ$	(45) $\cos 150^\circ$	(60) $\sin 0^\circ$	(75) $\cos 315^\circ$

(1) $\cos 60^\circ$	(16) $\sin 120^\circ$	(31) $\tan 180^\circ$	(46) $\tan 330^\circ$	(61) $\sin 225^\circ$
(2) $\cos 0^\circ$	(17) $\cos 315^\circ$	(32) $\tan 300^\circ$	(47) $\sin 135^\circ$	(62) $\cos 180^\circ$
(3) $\cos 45^\circ$	(18) $\sin 150^\circ$	(33) $\tan 60^\circ$	(48) $\tan 0^\circ$	(63) $\tan 315^\circ$
(4) $\tan 360^\circ$	(19) $\sin 150^\circ$	(34) $\cos 30^\circ$	(49) $\sin 0^\circ$	(64) $\tan 120^\circ$
(5) $\sin 0^\circ$	(20) $\sin 225^\circ$	(35) $\sin 120^\circ$	(50) $\cos 120^\circ$	(65) $\cos 150^\circ$
(6) $\cos 45^\circ$	(21) $\tan 180^\circ$	(36) $\tan 360^\circ$	(51) $\sin 360^\circ$	(66) $\cos 315^\circ$
(7) $\cos 90^\circ$	(22) $\cos 315^\circ$	(37) $\cos 180^\circ$	(52) $\tan 150^\circ$	(67) $\cos 45^\circ$
(8) $\tan 135^\circ$	(23) $\sin 60^\circ$	(38) $\sin 315^\circ$	(53) $\cos 270^\circ$	(68) $\tan 90^\circ$
(9) $\cos 180^\circ$	(24) $\tan 330^\circ$	(39) $\sin 45^\circ$	(54) $\cos 240^\circ$	(69) $\tan 270^\circ$
(10) $\cos 300^\circ$	(25) $\sin 120^\circ$	(40) $\cos 270^\circ$	(55) $\tan 300^\circ$	(70) $\cos 60^\circ$
(11) $\tan 180^\circ$	(26) $\cos 135^\circ$	(41) $\cos 150^\circ$	(56) $\tan 225^\circ$	(71) $\cos 315^\circ$
(12) $\sin 330^\circ$	(27) $\cos 135^\circ$	(42) $\sin 30^\circ$	(57) $\sin 330^\circ$	(72) $\cos 150^\circ$
(13) $\sin 225^\circ$	(28) $\tan 90^\circ$	(43) $\cos 330^\circ$	(58) $\tan 0^\circ$	(73) $\cos 225^\circ$
(14) $\tan 60^\circ$	(29) $\sin 45^\circ$	(44) $\tan 225^\circ$	(59) $\sin 240^\circ$	(74) $\tan 90^\circ$
(15) $\cos 0^\circ$	(30) $\sin 225^\circ$	(45) $\tan 30^\circ$	(60) $\cos 300^\circ$	(75) $\sin 135^\circ$

(1) $\tan 225^\circ$	(16) $\tan 225^\circ$	(31) $\cos 135^\circ$	(46) $\tan 150^\circ$	(61) $\tan 180^\circ$
(2) $\tan 270^\circ$	(17) $\sin 30^\circ$	(32) $\sin 135^\circ$	(47) $\cos 0^\circ$	(62) $\tan 315^\circ$
(3) $\cos 270^\circ$	(18) $\tan 300^\circ$	(33) $\tan 240^\circ$	(48) $\tan 0^\circ$	(63) $\sin 225^\circ$
(4) $\sin 150^\circ$	(19) $\cos 120^\circ$	(34) $\tan 135^\circ$	(49) $\tan 60^\circ$	(64) $\cos 90^\circ$
(5) $\cos 360^\circ$	(20) $\tan 0^\circ$	(35) $\sin 315^\circ$	(50) $\cos 90^\circ$	(65) $\sin 135^\circ$
(6) $\tan 330^\circ$	(21) $\sin 330^\circ$	(36) $\cos 120^\circ$	(51) $\tan 330^\circ$	(66) $\sin 90^\circ$
(7) $\cos 120^\circ$	(22) $\tan 270^\circ$	(37) $\cos 0^\circ$	(52) $\tan 135^\circ$	(67) $\tan 225^\circ$
(8) $\sin 180^\circ$	(23) $\tan 360^\circ$	(38) $\sin 315^\circ$	(53) $\cos 300^\circ$	(68) $\tan 150^\circ$
(9) $\tan 330^\circ$	(24) $\cos 90^\circ$	(39) $\sin 360^\circ$	(54) $\cos 315^\circ$	(69) $\cos 135^\circ$
(10) $\sin 150^\circ$	(25) $\tan 90^\circ$	(40) $\cos 30^\circ$	(55) $\cos 120^\circ$	(70) $\tan 30^\circ$
(11) $\sin 180^\circ$	(26) $\tan 315^\circ$	(41) $\tan 315^\circ$	(56) $\tan 300^\circ$	(71) $\cos 270^\circ$
(12) $\tan 300^\circ$	(27) $\cos 150^\circ$	(42) $\tan 120^\circ$	(57) $\sin 120^\circ$	(72) $\sin 360^\circ$
(13) $\sin 135^\circ$	(28) $\cos 300^\circ$	(43) $\tan 240^\circ$	(58) $\sin 240^\circ$	(73) $\cos 240^\circ$
(14) $\tan 30^\circ$	(29) $\cos 270^\circ$	(44) $\cos 45^\circ$	(59) $\tan 90^\circ$	(74) $\sin 270^\circ$
(15) $\tan 30^\circ$	(30) $\cos 60^\circ$	(45) $\tan 240^\circ$	(60) $\sin 300^\circ$	(75) $\sin 0^\circ$

(1) $\tan 0^\circ$	(16) $\sin 240^\circ$	(31) $\tan 135^\circ$	(46) $\cos 30^\circ$	(61) $\tan 60^\circ$
(2) $\sin 330^\circ$	(17) $\tan 300^\circ$	(32) $\tan 120^\circ$	(47) $\sin 180^\circ$	(62) $\sin 90^\circ$
(3) $\tan 120^\circ$	(18) $\sin 90^\circ$	(33) $\tan 360^\circ$	(48) $\sin 0^\circ$	(63) $\tan 90^\circ$
(4) $\cos 90^\circ$	(19) $\tan 60^\circ$	(34) $\sin 210^\circ$	(49) $\cos 240^\circ$	(64) $\sin 0^\circ$
(5) $\cos 90^\circ$	(20) $\sin 150^\circ$	(35) $\tan 300^\circ$	(50) $\cos 300^\circ$	(65) $\cos 240^\circ$
(6) $\cos 0^\circ$	(21) $\tan 210^\circ$	(36) $\tan 315^\circ$	(51) $\sin 330^\circ$	(66) $\cos 45^\circ$
(7) $\cos 315^\circ$	(22) $\sin 45^\circ$	(37) $\tan 120^\circ$	(52) $\cos 60^\circ$	(67) $\cos 120^\circ$
(8) $\tan 225^\circ$	(23) $\cos 90^\circ$	(38) $\sin 360^\circ$	(53) $\cos 360^\circ$	(68) $\tan 60^\circ$
(9) $\sin 60^\circ$	(24) $\sin 300^\circ$	(39) $\cos 180^\circ$	(54) $\sin 30^\circ$	(69) $\sin 330^\circ$
(10) $\cos 315^\circ$	(25) $\sin 30^\circ$	(40) $\cos 180^\circ$	(55) $\cos 45^\circ$	(70) $\cos 135^\circ$
(11) $\tan 180^\circ$	(26) $\tan 270^\circ$	(41) $\sin 135^\circ$	(56) $\tan 225^\circ$	(71) $\cos 240^\circ$
(12) $\tan 30^\circ$	(27) $\cos 45^\circ$	(42) $\cos 210^\circ$	(57) $\tan 300^\circ$	(72) $\tan 0^\circ$
(13) $\tan 240^\circ$	(28) $\cos 300^\circ$	(43) $\sin 0^\circ$	(58) $\tan 60^\circ$	(73) $\cos 150^\circ$
(14) $\sin 60^\circ$	(29) $\tan 210^\circ$	(44) $\tan 210^\circ$	(59) $\sin 60^\circ$	(74) $\tan 45^\circ$
(15) $\tan 135^\circ$	(30) $\sin 60^\circ$	(45) $\tan 150^\circ$	(60) $\cos 210^\circ$	(75) $\tan 210^\circ$

(1) $\cos 120^\circ$	(16) $\tan 60^\circ$	(31) $\sin 210^\circ$	(46) $\cos 270^\circ$	(61) $\cos 300^\circ$
(2) $\sin 240^\circ$	(17) $\cos 300^\circ$	(32) $\cos 90^\circ$	(47) $\sin 330^\circ$	(62) $\tan 45^\circ$
(3) $\tan 270^\circ$	(18) $\tan 0^\circ$	(33) $\cos 150^\circ$	(48) $\tan 120^\circ$	(63) $\tan 30^\circ$
(4) $\sin 120^\circ$	(19) $\sin 225^\circ$	(34) $\cos 330^\circ$	(49) $\tan 360^\circ$	(64) $\tan 330^\circ$
(5) $\sin 90^\circ$	(20) $\sin 330^\circ$	(35) $\cos 45^\circ$	(50) $\sin 90^\circ$	(65) $\tan 225^\circ$
(6) $\cos 30^\circ$	(21) $\sin 315^\circ$	(36) $\tan 210^\circ$	(51) $\cos 330^\circ$	(66) $\tan 210^\circ$
(7) $\sin 315^\circ$	(22) $\sin 180^\circ$	(37) $\sin 240^\circ$	(52) $\tan 45^\circ$	(67) $\sin 60^\circ$
(8) $\tan 45^\circ$	(23) $\sin 240^\circ$	(38) $\tan 90^\circ$	(53) $\cos 150^\circ$	(68) $\sin 360^\circ$
(9) $\cos 45^\circ$	(24) $\sin 270^\circ$	(39) $\cos 225^\circ$	(54) $\sin 210^\circ$	(69) $\sin 150^\circ$
(10) $\cos 360^\circ$	(25) $\tan 60^\circ$	(40) $\cos 270^\circ$	(55) $\tan 0^\circ$	(70) $\sin 360^\circ$
(11) $\cos 210^\circ$	(26) $\cos 45^\circ$	(41) $\sin 225^\circ$	(56) $\sin 90^\circ$	(71) $\tan 210^\circ$
(12) $\sin 60^\circ$	(27) $\tan 240^\circ$	(42) $\cos 225^\circ$	(57) $\cos 330^\circ$	(72) $\cos 330^\circ$
(13) $\cos 60^\circ$	(28) $\sin 60^\circ$	(43) $\sin 90^\circ$	(58) $\cos 180^\circ$	(73) $\tan 210^\circ$
(14) $\sin 90^\circ$	(29) $\tan 60^\circ$	(44) $\tan 90^\circ$	(59) $\sin 240^\circ$	(74) $\tan 330^\circ$
(15) $\cos 30^\circ$	(30) $\cos 225^\circ$	(45) $\tan 315^\circ$	(60) $\tan 60^\circ$	(75) $\tan 330^\circ$

(1) $\tan 360^\circ$	(16) $\tan 90^\circ$	(31) $\tan 300^\circ$	(46) $\sin 135^\circ$	(61) $\sin 270^\circ$
(2) $\cos 300^\circ$	(17) $\tan 225^\circ$	(32) $\tan 30^\circ$	(47) $\tan 135^\circ$	(62) $\sin 210^\circ$
(3) $\sin 240^\circ$	(18) $\sin 150^\circ$	(33) $\tan 45^\circ$	(48) $\cos 135^\circ$	(63) $\tan 45^\circ$
(4) $\tan 150^\circ$	(19) $\cos 150^\circ$	(34) $\sin 270^\circ$	(49) $\sin 150^\circ$	(64) $\tan 360^\circ$
(5) $\tan 330^\circ$	(20) $\sin 210^\circ$	(35) $\sin 45^\circ$	(50) $\cos 270^\circ$	(65) $\sin 30^\circ$
(6) $\tan 360^\circ$	(21) $\cos 360^\circ$	(36) $\sin 150^\circ$	(51) $\cos 300^\circ$	(66) $\tan 315^\circ$
(7) $\sin 300^\circ$	(22) $\tan 135^\circ$	(37) $\tan 120^\circ$	(52) $\tan 360^\circ$	(67) $\sin 0^\circ$
(8) $\cos 135^\circ$	(23) $\cos 360^\circ$	(38) $\tan 30^\circ$	(53) $\sin 0^\circ$	(68) $\tan 270^\circ$
(9) $\sin 90^\circ$	(24) $\cos 360^\circ$	(39) $\cos 45^\circ$	(54) $\cos 270^\circ$	(69) $\cos 120^\circ$
(10) $\sin 30^\circ$	(25) $\cos 360^\circ$	(40) $\sin 0^\circ$	(55) $\tan 0^\circ$	(70) $\sin 180^\circ$
(11) $\sin 90^\circ$	(26) $\sin 0^\circ$	(41) $\sin 330^\circ$	(56) $\cos 0^\circ$	(71) $\tan 300^\circ$
(12) $\tan 0^\circ$	(27) $\sin 150^\circ$	(42) $\cos 45^\circ$	(57) $\cos 270^\circ$	(72) $\tan 120^\circ$
(13) $\tan 330^\circ$	(28) $\tan 315^\circ$	(43) $\sin 240^\circ$	(58) $\sin 135^\circ$	(73) $\cos 210^\circ$
(14) $\cos 60^\circ$	(29) $\tan 90^\circ$	(44) $\sin 60^\circ$	(59) $\sin 45^\circ$	(74) $\cos 45^\circ$
(15) $\tan 135^\circ$	(30) $\sin 90^\circ$	(45) $\cos 135^\circ$	(60) $\tan 45^\circ$	(75) $\sin 150^\circ$

(1) $\tan 120^\circ$	(16) $\cos 30^\circ$	(31) $\cos 360^\circ$	(46) $\tan 270^\circ$	(61) $\tan 300^\circ$
(2) $\cos 360^\circ$	(17) $\cos 330^\circ$	(32) $\cos 45^\circ$	(47) $\tan 240^\circ$	(62) $\sin 315^\circ$
(3) $\cos 150^\circ$	(18) $\tan 270^\circ$	(33) $\cos 180^\circ$	(48) $\cos 120^\circ$	(63) $\sin 30^\circ$
(4) $\cos 45^\circ$	(19) $\cos 315^\circ$	(34) $\tan 240^\circ$	(49) $\sin 45^\circ$	(64) $\cos 225^\circ$
(5) $\sin 45^\circ$	(20) $\cos 0^\circ$	(35) $\tan 225^\circ$	(50) $\sin 270^\circ$	(65) $\tan 135^\circ$
(6) $\tan 210^\circ$	(21) $\tan 210^\circ$	(36) $\sin 120^\circ$	(51) $\cos 0^\circ$	(66) $\sin 90^\circ$
(7) $\tan 225^\circ$	(22) $\tan 90^\circ$	(37) $\sin 150^\circ$	(52) $\cos 135^\circ$	(67) $\tan 330^\circ$
(8) $\cos 45^\circ$	(23) $\cos 240^\circ$	(38) $\cos 240^\circ$	(53) $\tan 270^\circ$	(68) $\cos 300^\circ$
(9) $\sin 330^\circ$	(24) $\sin 120^\circ$	(39) $\cos 360^\circ$	(54) $\tan 30^\circ$	(69) $\sin 60^\circ$
(10) $\cos 45^\circ$	(25) $\tan 240^\circ$	(40) $\tan 315^\circ$	(55) $\tan 60^\circ$	(70) $\cos 240^\circ$
(11) $\cos 120^\circ$	(26) $\cos 135^\circ$	(41) $\cos 150^\circ$	(56) $\tan 90^\circ$	(71) $\tan 60^\circ$
(12) $\tan 330^\circ$	(27) $\tan 270^\circ$	(42) $\sin 45^\circ$	(57) $\tan 135^\circ$	(72) $\cos 135^\circ$
(13) $\tan 60^\circ$	(28) $\tan 240^\circ$	(43) $\tan 60^\circ$	(58) $\tan 240^\circ$	(73) $\cos 360^\circ$
(14) $\sin 225^\circ$	(29) $\cos 210^\circ$	(44) $\cos 300^\circ$	(59) $\tan 300^\circ$	(74) $\sin 315^\circ$
(15) $\tan 210^\circ$	(30) $\tan 360^\circ$	(45) $\sin 150^\circ$	(60) $\cos 90^\circ$	(75) $\sin 45^\circ$

(1) $\tan 270^\circ$	(16) $\sin 150^\circ$	(31) $\sin 180^\circ$	(46) $\tan 330^\circ$	(61) $\sin 315^\circ$
(2) $\tan 120^\circ$	(17) $\sin 330^\circ$	(32) $\tan 240^\circ$	(47) $\cos 330^\circ$	(62) $\cos 180^\circ$
(3) $\sin 270^\circ$	(18) $\sin 360^\circ$	(33) $\tan 135^\circ$	(48) $\sin 330^\circ$	(63) $\cos 0^\circ$
(4) $\tan 30^\circ$	(19) $\tan 240^\circ$	(34) $\sin 120^\circ$	(49) $\tan 90^\circ$	(64) $\sin 135^\circ$
(5) $\tan 120^\circ$	(20) $\tan 270^\circ$	(35) $\sin 120^\circ$	(50) $\tan 360^\circ$	(65) $\tan 30^\circ$
(6) $\tan 315^\circ$	(21) $\tan 225^\circ$	(36) $\sin 270^\circ$	(51) $\sin 330^\circ$	(66) $\cos 45^\circ$
(7) $\cos 60^\circ$	(22) $\cos 270^\circ$	(37) $\sin 30^\circ$	(52) $\tan 60^\circ$	(67) $\tan 360^\circ$
(8) $\sin 210^\circ$	(23) $\tan 270^\circ$	(38) $\cos 180^\circ$	(53) $\sin 300^\circ$	(68) $\tan 225^\circ$
(9) $\tan 120^\circ$	(24) $\tan 330^\circ$	(39) $\sin 270^\circ$	(54) $\tan 315^\circ$	(69) $\sin 315^\circ$
(10) $\tan 360^\circ$	(25) $\sin 315^\circ$	(40) $\cos 330^\circ$	(55) $\cos 150^\circ$	(70) $\sin 225^\circ$
(11) $\tan 270^\circ$	(26) $\tan 360^\circ$	(41) $\tan 240^\circ$	(56) $\sin 330^\circ$	(71) $\sin 360^\circ$
(12) $\cos 60^\circ$	(27) $\cos 150^\circ$	(42) $\tan 180^\circ$	(57) $\sin 315^\circ$	(72) $\tan 330^\circ$
(13) $\cos 0^\circ$	(28) $\sin 300^\circ$	(43) $\cos 90^\circ$	(58) $\tan 135^\circ$	(73) $\tan 90^\circ$
(14) $\tan 180^\circ$	(29) $\tan 0^\circ$	(44) $\cos 300^\circ$	(59) $\tan 120^\circ$	(74) $\sin 150^\circ$
(15) $\sin 0^\circ$	(30) $\sin 30^\circ$	(45) $\cos 60^\circ$	(60) $\cos 270^\circ$	(75) $\tan 270^\circ$

(1) $\cos 150^\circ$	(16) $\cos 90^\circ$	(31) $\tan 180^\circ$	(46) $\sin 30^\circ$	(61) $\sin 270^\circ$
(2) $\tan 45^\circ$	(17) $\cos 240^\circ$	(32) $\sin 150^\circ$	(47) $\tan 135^\circ$	(62) $\cos 330^\circ$
(3) $\sin 45^\circ$	(18) $\cos 45^\circ$	(33) $\sin 90^\circ$	(48) $\cos 45^\circ$	(63) $\cos 30^\circ$
(4) $\sin 60^\circ$	(19) $\sin 150^\circ$	(34) $\tan 330^\circ$	(49) $\tan 120^\circ$	(64) $\sin 210^\circ$
(5) $\sin 240^\circ$	(20) $\cos 45^\circ$	(35) $\tan 0^\circ$	(50) $\sin 30^\circ$	(65) $\sin 150^\circ$
(6) $\sin 240^\circ$	(21) $\sin 60^\circ$	(36) $\cos 330^\circ$	(51) $\tan 270^\circ$	(66) $\cos 240^\circ$
(7) $\tan 0^\circ$	(22) $\tan 60^\circ$	(37) $\tan 0^\circ$	(52) $\cos 45^\circ$	(67) $\tan 180^\circ$
(8) $\cos 0^\circ$	(23) $\sin 180^\circ$	(38) $\cos 0^\circ$	(53) $\sin 315^\circ$	(68) $\cos 330^\circ$
(9) $\cos 0^\circ$	(24) $\cos 120^\circ$	(39) $\cos 360^\circ$	(54) $\sin 330^\circ$	(69) $\cos 150^\circ$
(10) $\tan 60^\circ$	(25) $\tan 45^\circ$	(40) $\sin 0^\circ$	(55) $\tan 225^\circ$	(70) $\tan 315^\circ$
(11) $\cos 315^\circ$	(26) $\cos 180^\circ$	(41) $\cos 30^\circ$	(56) $\cos 330^\circ$	(71) $\tan 90^\circ$
(12) $\tan 360^\circ$	(27) $\tan 60^\circ$	(42) $\sin 210^\circ$	(57) $\sin 0^\circ$	(72) $\tan 60^\circ$
(13) $\tan 315^\circ$	(28) $\tan 180^\circ$	(43) $\cos 120^\circ$	(58) $\sin 225^\circ$	(73) $\tan 135^\circ$
(14) $\tan 45^\circ$	(29) $\tan 45^\circ$	(44) $\tan 150^\circ$	(59) $\cos 0^\circ$	(74) $\tan 135^\circ$
(15) $\sin 30^\circ$	(30) $\cos 135^\circ$	(45) $\cos 135^\circ$	(60) $\cos 0^\circ$	(75) $\tan 90^\circ$

(1) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(16) $\tan 135^\circ$ $= -1$	(31) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(46) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(61) $\cos 300^\circ$ $= \frac{1}{2}$
(2) $\cos 270^\circ$ $= 0$	(17) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(32) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(47) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(62) $\sin 30^\circ$ $= \frac{1}{2}$
(3) $\tan 120^\circ$ $= -\sqrt{3}$	(18) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(33) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(48) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(63) $\sin 330^\circ$ $= -\frac{1}{2}$
(4) $\sin 330^\circ$ $= -\frac{1}{2}$	(19) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(34) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(49) $\cos 240^\circ$ $= -\frac{1}{2}$	(64) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$
(5) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(20) $\cos 270^\circ$ $= 0$	(35) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(50) $\cos 90^\circ$ $= 0$	(65) $\tan 240^\circ$ $= \sqrt{3}$
(6) $\tan 0^\circ$ $= 0$	(21) $\tan 240^\circ$ $= \sqrt{3}$	(36) $\tan 315^\circ$ $= -1$	(51) $\tan 225^\circ$ $= 1$	(66) $\cos 300^\circ$ $= \frac{1}{2}$
(7) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(22) $\sin 0^\circ$ $= 0$	(37) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(52) $\sin 270^\circ$ $= -1$	(67) $\cos 120^\circ$ $= -\frac{1}{2}$
(8) $\tan 180^\circ$ $= 0$	(23) $\cos 180^\circ$ $= -1$	(38) $\cos 240^\circ$ $= -\frac{1}{2}$	(53) $\tan 360^\circ$ $= 0$	(68) $\tan 270^\circ$ なし
(9) $\tan 90^\circ$ なし	(24) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(39) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(54) $\cos 60^\circ$ $= \frac{1}{2}$	(69) $\cos 270^\circ$ $= 0$
(10) $\sin 330^\circ$ $= -\frac{1}{2}$	(25) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(40) $\cos 300^\circ$ $= \frac{1}{2}$	(55) $\cos 90^\circ$ $= 0$	(70) $\sin 0^\circ$ $= 0$
(11) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(26) $\sin 90^\circ$ $= 1$	(41) $\sin 0^\circ$ $= 0$	(56) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(71) $\sin 270^\circ$ $= -1$
(12) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(42) $\sin 30^\circ$ $= \frac{1}{2}$	(57) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(72) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$
(13) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(28) $\tan 120^\circ$ $= -\sqrt{3}$	(43) $\tan 180^\circ$ $= 0$	(58) $\cos 0^\circ$ $= 1$	(73) $\cos 60^\circ$ $= \frac{1}{2}$
(14) $\sin 0^\circ$ $= 0$	(29) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(44) $\tan 300^\circ$ $= -\sqrt{3}$	(59) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(74) $\cos 0^\circ$ $= 1$
(15) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(30) $\sin 0^\circ$ $= 0$	(45) $\sin 0^\circ$ $= 0$	(60) $\sin 330^\circ$ $= -\frac{1}{2}$	(75) $\tan 90^\circ$ なし

- | | | | | |
|---|--|--|--|--|
| (1) $\cos 30^\circ$
$= \frac{\sqrt{3}}{2}$ | (16) $\sin 150^\circ$
$= \frac{1}{2}$ | (31) $\cos 30^\circ$
$= \frac{\sqrt{3}}{2}$ | (46) $\cos 300^\circ$
$= \frac{1}{2}$ | (61) $\tan 45^\circ$
$= 1$ |
| (2) $\tan 60^\circ$
$= \sqrt{3}$ | (17) $\tan 60^\circ$
$= \sqrt{3}$ | (32) $\sin 360^\circ$
$= 0$ | (47) $\tan 180^\circ$
$= 0$ | (62) $\cos 210^\circ$
$= -\frac{\sqrt{3}}{2}$ |
| (3) $\cos 210^\circ$
$= -\frac{\sqrt{3}}{2}$ | (18) $\cos 210^\circ$
$= -\frac{\sqrt{3}}{2}$ | (33) $\sin 60^\circ$
$= \frac{\sqrt{3}}{2}$ | (48) $\cos 30^\circ$
$= \frac{\sqrt{3}}{2}$ | (63) $\tan 45^\circ$
$= 1$ |
| (4) $\cos 270^\circ$
$= 0$ | (19) $\tan 315^\circ$
$= -1$ | (34) $\sin 30^\circ$
$= \frac{1}{2}$ | (49) $\cos 90^\circ$
$= 0$ | (64) $\cos 60^\circ$
$= \frac{1}{2}$ |
| (5) $\tan 150^\circ$
$= -\frac{1}{\sqrt{3}}$ | (20) $\sin 135^\circ$
$= \frac{1}{\sqrt{2}}$ | (35) $\tan 300^\circ$
$= -\sqrt{3}$ | (50) $\cos 90^\circ$
$= 0$ | (65) $\cos 0^\circ$
$= 1$ |
| (6) $\cos 300^\circ$
$= \frac{1}{2}$ | (21) $\cos 300^\circ$
$= \frac{1}{2}$ | (36) $\tan 0^\circ$
$= 0$ | (51) $\cos 300^\circ$
$= \frac{1}{2}$ | (66) $\cos 210^\circ$
$= -\frac{\sqrt{3}}{2}$ |
| (7) $\cos 0^\circ$
$= 1$ | (22) $\cos 150^\circ$
$= -\frac{\sqrt{3}}{2}$ | (37) $\sin 90^\circ$
$= 1$ | (52) $\cos 270^\circ$
$= 0$ | (67) $\tan 120^\circ$
$= -\sqrt{3}$ |
| (8) $\cos 210^\circ$
$= -\frac{\sqrt{3}}{2}$ | (23) $\sin 60^\circ$
$= \frac{\sqrt{3}}{2}$ | (38) $\tan 300^\circ$
$= -\sqrt{3}$ | (53) $\tan 360^\circ$
$= 0$ | (68) $\sin 0^\circ$
$= 0$ |
| (9) $\cos 135^\circ$
$= -\frac{1}{\sqrt{2}}$ | (24) $\sin 300^\circ$
$= -\frac{\sqrt{3}}{2}$ | (39) $\cos 240^\circ$
$= -\frac{1}{2}$ | (54) $\sin 315^\circ$
$= -\frac{1}{\sqrt{2}}$ | (69) $\cos 300^\circ$
$= \frac{1}{2}$ |
| (10) $\sin 210^\circ$
$= -\frac{1}{2}$ | (25) $\cos 210^\circ$
$= -\frac{\sqrt{3}}{2}$ | (40) $\tan 240^\circ$
$= \sqrt{3}$ | (55) $\sin 120^\circ$
$= \frac{\sqrt{3}}{2}$ | (70) $\cos 90^\circ$
$= 0$ |
| (11) $\tan 315^\circ$
$= -1$ | (26) $\tan 360^\circ$
$= 0$ | (41) $\tan 150^\circ$
$= -\frac{1}{\sqrt{3}}$ | (56) $\cos 315^\circ$
$= \frac{1}{\sqrt{2}}$ | (71) $\tan 225^\circ$
$= 1$ |
| (12) $\sin 150^\circ$
$= \frac{1}{2}$ | (27) $\tan 315^\circ$
$= -1$ | (42) $\tan 300^\circ$
$= -\sqrt{3}$ | (57) $\tan 120^\circ$
$= -\sqrt{3}$ | (72) $\tan 0^\circ$
$= 0$ |
| (13) $\tan 30^\circ$
$= \frac{1}{\sqrt{3}}$ | (28) $\cos 315^\circ$
$= \frac{1}{\sqrt{2}}$ | (43) $\cos 150^\circ$
$= -\frac{\sqrt{3}}{2}$ | (58) $\cos 120^\circ$
$= -\frac{1}{2}$ | (73) $\cos 0^\circ$
$= 1$ |
| (14) $\cos 315^\circ$
$= \frac{1}{\sqrt{2}}$ | (29) $\tan 150^\circ$
$= -\frac{1}{\sqrt{3}}$ | (44) $\cos 360^\circ$
$= 1$ | (59) $\cos 210^\circ$
$= -\frac{\sqrt{3}}{2}$ | (74) $\cos 315^\circ$
$= \frac{1}{\sqrt{2}}$ |
| (15) $\sin 60^\circ$
$= \frac{\sqrt{3}}{2}$ | (30) $\sin 60^\circ$
$= \frac{\sqrt{3}}{2}$ | (45) $\tan 45^\circ$
$= 1$ | (60) $\sin 240^\circ$
$= -\frac{\sqrt{3}}{2}$ | (75) $\sin 240^\circ$
$= -\frac{\sqrt{3}}{2}$ |

(1) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(16) $\sin 270^\circ$ $= -1$	(31) $\tan 180^\circ$ $= 0$	(46) $\tan 120^\circ$ $= -\sqrt{3}$	(61) $\tan 360^\circ$ $= 0$
(2) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(17) $\tan 45^\circ$ $= 1$	(32) $\sin 210^\circ$ $= -\frac{1}{2}$	(47) $\tan 90^\circ$ なし	(62) $\tan 300^\circ$ $= -\sqrt{3}$
(3) $\tan 0^\circ$ $= 0$	(18) $\tan 45^\circ$ $= 1$	(33) $\sin 270^\circ$ $= -1$	(48) $\cos 120^\circ$ $= -\frac{1}{2}$	(63) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$
(4) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(19) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(34) $\tan 225^\circ$ $= 1$	(49) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(64) $\cos 90^\circ$ $= 0$
(5) $\sin 360^\circ$ $= 0$	(20) $\tan 60^\circ$ $= \sqrt{3}$	(35) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(50) $\cos 90^\circ$ $= 0$	(65) $\sin 150^\circ$ $= \frac{1}{2}$
(6) $\tan 300^\circ$ $= -\sqrt{3}$	(21) $\tan 225^\circ$ $= 1$	(36) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(51) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(66) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$
(7) $\tan 225^\circ$ $= 1$	(22) $\tan 90^\circ$ なし	(37) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(52) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(67) $\tan 120^\circ$ $= -\sqrt{3}$
(8) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\tan 270^\circ$ なし	(38) $\cos 120^\circ$ $= -\frac{1}{2}$	(53) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(68) $\tan 90^\circ$ なし
(9) $\tan 180^\circ$ $= 0$	(24) $\cos 120^\circ$ $= -\frac{1}{2}$	(39) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(69) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(10) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(25) $\tan 0^\circ$ $= 0$	(40) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(55) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(70) $\cos 60^\circ$ $= \frac{1}{2}$
(11) $\cos 90^\circ$ $= 0$	(26) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(41) $\sin 270^\circ$ $= -1$	(56) $\cos 300^\circ$ $= \frac{1}{2}$	(71) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(12) $\cos 120^\circ$ $= -\frac{1}{2}$	(27) $\tan 90^\circ$ なし	(42) $\cos 360^\circ$ $= 1$	(57) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(72) $\tan 240^\circ$ $= \sqrt{3}$
(13) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(28) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(43) $\tan 315^\circ$ $= -1$	(58) $\tan 180^\circ$ $= 0$	(73) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(14) $\sin 360^\circ$ $= 0$	(29) $\sin 150^\circ$ $= \frac{1}{2}$	(44) $\tan 120^\circ$ $= -\sqrt{3}$	(59) $\sin 0^\circ$ $= 0$	(74) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(15) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(30) $\sin 270^\circ$ $= -1$	(45) $\cos 180^\circ$ $= -1$	(60) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(75) $\tan 225^\circ$ $= 1$

(1) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(16) $\tan 0^\circ$ $= 0$	(31) $\tan 0^\circ$ $= 0$	(46) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(61) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$
(2) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(17) $\tan 240^\circ$ $= \sqrt{3}$	(32) $\tan 180^\circ$ $= 0$	(47) $\cos 60^\circ$ $= \frac{1}{2}$	(62) $\sin 360^\circ$ $= 0$
(3) $\tan 90^\circ$ なし	(18) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(33) $\sin 90^\circ$ $= 1$	(48) $\tan 45^\circ$ $= 1$	(63) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(4) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(19) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(34) $\sin 270^\circ$ $= -1$	(49) $\sin 360^\circ$ $= 0$	(64) $\sin 90^\circ$ $= 1$
(5) $\tan 135^\circ$ $= -1$	(20) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(35) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(50) $\tan 360^\circ$ $= 0$	(65) $\cos 300^\circ$ $= \frac{1}{2}$
(6) $\cos 60^\circ$ $= \frac{1}{2}$	(21) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(36) $\tan 270^\circ$ なし	(51) $\sin 90^\circ$ $= 1$	(66) $\tan 45^\circ$ $= 1$
(7) $\cos 60^\circ$ $= \frac{1}{2}$	(22) $\cos 300^\circ$ $= \frac{1}{2}$	(37) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(52) $\tan 135^\circ$ $= -1$	(67) $\sin 210^\circ$ $= -\frac{1}{2}$
(8) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(38) $\tan 90^\circ$ なし	(53) $\cos 240^\circ$ $= -\frac{1}{2}$	(68) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$
(9) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(24) $\tan 90^\circ$ なし	(39) $\tan 225^\circ$ $= 1$	(54) $\tan 300^\circ$ $= -\sqrt{3}$	(69) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(10) $\sin 210^\circ$ $= -\frac{1}{2}$	(25) $\sin 90^\circ$ $= 1$	(40) $\cos 360^\circ$ $= 1$	(55) $\tan 315^\circ$ $= -1$	(70) $\sin 150^\circ$ $= \frac{1}{2}$
(11) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(26) $\cos 180^\circ$ $= -1$	(41) $\tan 90^\circ$ なし	(56) $\tan 60^\circ$ $= \sqrt{3}$	(71) $\tan 225^\circ$ $= 1$
(12) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(27) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(42) $\tan 300^\circ$ $= -\sqrt{3}$	(57) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(72) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$
(13) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(28) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(43) $\sin 270^\circ$ $= -1$	(58) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\tan 180^\circ$ $= 0$
(14) $\cos 0^\circ$ $= 1$	(29) $\cos 360^\circ$ $= 1$	(44) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(59) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(74) $\tan 315^\circ$ $= -1$
(15) $\sin 360^\circ$ $= 0$	(30) $\tan 45^\circ$ $= 1$	(45) $\sin 360^\circ$ $= 0$	(60) $\tan 270^\circ$ なし	(75) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$

- | | | | | |
|--|--|--|--|--|
| (1) $\cos 45^\circ$
$= \frac{1}{\sqrt{2}}$ | (16) $\cos 330^\circ$
$= \frac{\sqrt{3}}{2}$ | (31) $\cos 210^\circ$
$= -\frac{\sqrt{3}}{2}$ | (46) $\cos 270^\circ$
$= 0$ | (61) $\cos 90^\circ$
$= 0$ |
| (2) $\tan 270^\circ$
なし | (17) $\sin 210^\circ$
$= -\frac{1}{2}$ | (32) $\tan 180^\circ$
$= 0$ | (47) $\cos 0^\circ$
$= 1$ | (62) $\tan 180^\circ$
$= 0$ |
| (3) $\tan 150^\circ$
$= -\frac{1}{\sqrt{3}}$ | (18) $\sin 60^\circ$
$= \frac{\sqrt{3}}{2}$ | (33) $\sin 150^\circ$
$= \frac{1}{2}$ | (48) $\sin 90^\circ$
$= 1$ | (63) $\sin 45^\circ$
$= \frac{1}{\sqrt{2}}$ |
| (4) $\cos 45^\circ$
$= \frac{1}{\sqrt{2}}$ | (19) $\cos 360^\circ$
$= 1$ | (34) $\cos 90^\circ$
$= 0$ | (49) $\cos 30^\circ$
$= \frac{\sqrt{3}}{2}$ | (64) $\cos 270^\circ$
$= 0$ |
| (5) $\sin 120^\circ$
$= \frac{\sqrt{3}}{2}$ | (20) $\cos 135^\circ$
$= -\frac{1}{\sqrt{2}}$ | (35) $\cos 45^\circ$
$= \frac{1}{\sqrt{2}}$ | (50) $\cos 150^\circ$
$= -\frac{\sqrt{3}}{2}$ | (65) $\cos 330^\circ$
$= \frac{\sqrt{3}}{2}$ |
| (6) $\cos 60^\circ$
$= \frac{1}{2}$ | (21) $\cos 120^\circ$
$= -\frac{1}{2}$ | (36) $\cos 135^\circ$
$= -\frac{1}{\sqrt{2}}$ | (51) $\sin 270^\circ$
$= -1$ | (66) $\sin 360^\circ$
$= 0$ |
| (7) $\cos 225^\circ$
$= -\frac{1}{\sqrt{2}}$ | (22) $\cos 0^\circ$
$= 1$ | (37) $\cos 180^\circ$
$= -1$ | (52) $\tan 315^\circ$
$= -1$ | (67) $\cos 135^\circ$
$= -\frac{1}{\sqrt{2}}$ |
| (8) $\sin 330^\circ$
$= -\frac{1}{2}$ | (23) $\cos 315^\circ$
$= \frac{1}{\sqrt{2}}$ | (38) $\cos 330^\circ$
$= \frac{\sqrt{3}}{2}$ | (53) $\sin 135^\circ$
$= \frac{1}{\sqrt{2}}$ | (68) $\cos 90^\circ$
$= 0$ |
| (9) $\cos 135^\circ$
$= -\frac{1}{\sqrt{2}}$ | (24) $\tan 300^\circ$
$= -\sqrt{3}$ | (39) $\tan 330^\circ$
$= -\frac{1}{\sqrt{3}}$ | (54) $\tan 135^\circ$
$= -1$ | (69) $\cos 30^\circ$
$= \frac{\sqrt{3}}{2}$ |
| (10) $\tan 0^\circ$
$= 0$ | (25) $\sin 30^\circ$
$= \frac{1}{2}$ | (40) $\tan 180^\circ$
$= 0$ | (55) $\cos 120^\circ$
$= -\frac{1}{2}$ | (70) $\sin 60^\circ$
$= \frac{\sqrt{3}}{2}$ |
| (11) $\sin 225^\circ$
$= -\frac{1}{\sqrt{2}}$ | (26) $\tan 30^\circ$
$= \frac{1}{\sqrt{3}}$ | (41) $\cos 60^\circ$
$= \frac{1}{2}$ | (56) $\cos 135^\circ$
$= -\frac{1}{\sqrt{2}}$ | (71) $\cos 45^\circ$
$= \frac{1}{\sqrt{2}}$ |
| (12) $\tan 150^\circ$
$= -\frac{1}{\sqrt{3}}$ | (27) $\tan 30^\circ$
$= \frac{1}{\sqrt{3}}$ | (42) $\tan 60^\circ$
$= \sqrt{3}$ | (57) $\sin 315^\circ$
$= -\frac{1}{\sqrt{2}}$ | (72) $\sin 0^\circ$
$= 0$ |
| (13) $\tan 330^\circ$
$= -\frac{1}{\sqrt{3}}$ | (28) $\cos 45^\circ$
$= \frac{1}{\sqrt{2}}$ | (43) $\tan 360^\circ$
$= 0$ | (58) $\tan 315^\circ$
$= -1$ | (73) $\cos 30^\circ$
$= \frac{\sqrt{3}}{2}$ |
| (14) $\cos 60^\circ$
$= \frac{1}{2}$ | (29) $\tan 30^\circ$
$= \frac{1}{\sqrt{3}}$ | (44) $\tan 330^\circ$
$= -\frac{1}{\sqrt{3}}$ | (59) $\sin 225^\circ$
$= -\frac{1}{\sqrt{2}}$ | (74) $\cos 135^\circ$
$= -\frac{1}{\sqrt{2}}$ |
| (15) $\sin 360^\circ$
$= 0$ | (30) $\tan 240^\circ$
$= \sqrt{3}$ | (45) $\cos 210^\circ$
$= -\frac{\sqrt{3}}{2}$ | (60) $\cos 240^\circ$
$= -\frac{1}{2}$ | (75) $\cos 300^\circ$
$= \frac{1}{2}$ |

(1) $\tan 0^\circ$ $= 0$	(16) $\cos 180^\circ$ $= -1$	(31) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(46) $\tan 300^\circ$ $= -\sqrt{3}$	(61) $\tan 360^\circ$ $= 0$
(2) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(17) $\sin 30^\circ$ $= \frac{1}{2}$	(32) $\sin 210^\circ$ $= -\frac{1}{2}$	(47) $\tan 225^\circ$ $= 1$	(62) $\tan 120^\circ$ $= -\sqrt{3}$
(3) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(18) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(33) $\tan 135^\circ$ $= -1$	(48) $\tan 270^\circ$ なし	(63) $\tan 315^\circ$ $= -1$
(4) $\sin 270^\circ$ $= -1$	(19) $\tan 60^\circ$ $= \sqrt{3}$	(34) $\tan 225^\circ$ $= 1$	(49) $\cos 60^\circ$ $= \frac{1}{2}$	(64) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$
(5) $\cos 300^\circ$ $= \frac{1}{2}$	(20) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(35) $\sin 0^\circ$ $= 0$	(50) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(65) $\tan 90^\circ$ なし
(6) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(21) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(36) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(51) $\sin 90^\circ$ $= 1$	(66) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$
(7) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(22) $\tan 360^\circ$ $= 0$	(37) $\cos 180^\circ$ $= -1$	(52) $\tan 180^\circ$ $= 0$	(67) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(8) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(23) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(38) $\tan 225^\circ$ $= 1$	(53) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(68) $\tan 180^\circ$ $= 0$
(9) $\tan 240^\circ$ $= \sqrt{3}$	(24) $\tan 360^\circ$ $= 0$	(39) $\cos 60^\circ$ $= \frac{1}{2}$	(54) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$
(10) $\cos 360^\circ$ $= 1$	(25) $\cos 120^\circ$ $= -\frac{1}{2}$	(40) $\tan 270^\circ$ なし	(55) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(70) $\cos 60^\circ$ $= \frac{1}{2}$
(11) $\tan 300^\circ$ $= -\sqrt{3}$	(26) $\sin 0^\circ$ $= 0$	(41) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(56) $\cos 300^\circ$ $= \frac{1}{2}$	(71) $\sin 180^\circ$ $= 0$
(12) $\tan 135^\circ$ $= -1$	(27) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(42) $\sin 30^\circ$ $= \frac{1}{2}$	(57) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(72) $\cos 300^\circ$ $= \frac{1}{2}$
(13) $\cos 90^\circ$ $= 0$	(28) $\cos 240^\circ$ $= -\frac{1}{2}$	(43) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(58) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(73) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$
(14) $\cos 0^\circ$ $= 1$	(29) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(44) $\cos 180^\circ$ $= -1$	(59) $\sin 270^\circ$ $= -1$	(74) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(15) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(30) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(45) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(60) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(75) $\tan 270^\circ$ なし

(1) $\cos 0^\circ$ $= 1$	(16) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(31) $\cos 180^\circ$ $= -1$	(46) $\sin 150^\circ$ $= \frac{1}{2}$	(61) $\tan 45^\circ$ $= 1$
(2) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(17) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(32) $\tan 315^\circ$ $= -1$	(47) $\cos 120^\circ$ $= -\frac{1}{2}$	(62) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$
(3) $\tan 120^\circ$ $= -\sqrt{3}$	(18) $\tan 300^\circ$ $= -\sqrt{3}$	(33) $\sin 150^\circ$ $= \frac{1}{2}$	(48) $\sin 30^\circ$ $= \frac{1}{2}$	(63) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(4) $\sin 0^\circ$ $= 0$	(19) $\cos 240^\circ$ $= -\frac{1}{2}$	(34) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(49) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(64) $\tan 180^\circ$ $= 0$
(5) $\sin 330^\circ$ $= -\frac{1}{2}$	(20) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(35) $\sin 0^\circ$ $= 0$	(50) $\tan 0^\circ$ $= 0$	(65) $\tan 0^\circ$ $= 0$
(6) $\tan 45^\circ$ $= 1$	(21) $\tan 90^\circ$ なし	(36) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(51) $\cos 180^\circ$ $= -1$	(66) $\tan 120^\circ$ $= -\sqrt{3}$
(7) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(22) $\cos 300^\circ$ $= \frac{1}{2}$	(37) $\sin 270^\circ$ $= -1$	(52) $\sin 90^\circ$ $= 1$	(67) $\tan 60^\circ$ $= \sqrt{3}$
(8) $\tan 360^\circ$ $= 0$	(23) $\sin 180^\circ$ $= 0$	(38) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(53) $\sin 0^\circ$ $= 0$	(68) $\tan 120^\circ$ $= -\sqrt{3}$
(9) $\cos 360^\circ$ $= 1$	(24) $\tan 270^\circ$ なし	(39) $\tan 0^\circ$ $= 0$	(54) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(69) $\tan 180^\circ$ $= 0$
(10) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(25) $\cos 60^\circ$ $= \frac{1}{2}$	(40) $\sin 0^\circ$ $= 0$	(55) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(70) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$
(11) $\cos 90^\circ$ $= 0$	(26) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(41) $\tan 90^\circ$ なし	(56) $\cos 120^\circ$ $= -\frac{1}{2}$	(71) $\cos 120^\circ$ $= -\frac{1}{2}$
(12) $\tan 60^\circ$ $= \sqrt{3}$	(27) $\sin 150^\circ$ $= \frac{1}{2}$	(42) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(57) $\cos 60^\circ$ $= \frac{1}{2}$	(72) $\tan 60^\circ$ $= \sqrt{3}$
(13) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(28) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(43) $\tan 300^\circ$ $= -\sqrt{3}$	(58) $\tan 300^\circ$ $= -\sqrt{3}$	(73) $\tan 135^\circ$ $= -1$
(14) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(29) $\tan 0^\circ$ $= 0$	(44) $\tan 270^\circ$ なし	(59) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(74) $\tan 180^\circ$ $= 0$
(15) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(30) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(45) $\cos 120^\circ$ $= -\frac{1}{2}$	(60) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(75) $\cos 300^\circ$ $= \frac{1}{2}$

(1) $\tan 270^\circ$ なし	(16) $\cos 90^\circ$ $= 0$	(31) $\tan 45^\circ$ $= 1$	(46) $\tan 0^\circ$ $= 0$	(61) $\tan 300^\circ$ $= -\sqrt{3}$
(2) $\tan 360^\circ$ $= 0$	(17) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(32) $\tan 240^\circ$ $= \sqrt{3}$	(47) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(62) $\tan 45^\circ$ $= 1$
(3) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(18) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(33) $\cos 240^\circ$ $= -\frac{1}{2}$	(48) $\cos 120^\circ$ $= -\frac{1}{2}$	(63) $\sin 330^\circ$ $= -\frac{1}{2}$
(4) $\sin 90^\circ$ $= 1$	(19) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(34) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(49) $\tan 360^\circ$ $= 0$	(64) $\tan 45^\circ$ $= 1$
(5) $\sin 270^\circ$ $= -1$	(20) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(35) $\sin 210^\circ$ $= -\frac{1}{2}$	(50) $\sin 150^\circ$ $= \frac{1}{2}$	(65) $\cos 240^\circ$ $= -\frac{1}{2}$
(6) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(21) $\sin 30^\circ$ $= \frac{1}{2}$	(36) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(51) $\cos 360^\circ$ $= 1$	(66) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(7) $\cos 270^\circ$ $= 0$	(22) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(37) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(52) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(67) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(8) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\tan 315^\circ$ $= -1$	(38) $\sin 270^\circ$ $= -1$	(53) $\tan 135^\circ$ $= -1$	(68) $\tan 270^\circ$ なし
(9) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(24) $\sin 0^\circ$ $= 0$	(39) $\sin 0^\circ$ $= 0$	(54) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(69) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$
(10) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(25) $\tan 0^\circ$ $= 0$	(40) $\sin 150^\circ$ $= \frac{1}{2}$	(55) $\tan 360^\circ$ $= 0$	(70) $\tan 180^\circ$ $= 0$
(11) $\cos 360^\circ$ $= 1$	(26) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(41) $\sin 30^\circ$ $= \frac{1}{2}$	(56) $\sin 180^\circ$ $= 0$	(71) $\tan 300^\circ$ $= -\sqrt{3}$
(12) $\cos 90^\circ$ $= 0$	(27) $\cos 240^\circ$ $= -\frac{1}{2}$	(42) $\tan 135^\circ$ $= -1$	(57) $\cos 360^\circ$ $= 1$	(72) $\sin 180^\circ$ $= 0$
(13) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(28) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(43) $\sin 330^\circ$ $= -\frac{1}{2}$	(58) $\tan 120^\circ$ $= -\sqrt{3}$	(73) $\sin 0^\circ$ $= 0$
(14) $\sin 360^\circ$ $= 0$	(29) $\sin 180^\circ$ $= 0$	(44) $\tan 240^\circ$ $= \sqrt{3}$	(59) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(74) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$
(15) $\tan 360^\circ$ $= 0$	(30) $\sin 180^\circ$ $= 0$	(45) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(60) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(75) $\cos 60^\circ$ $= \frac{1}{2}$

(1) $\tan 45^\circ$ $= 1$	(16) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(31) $\sin 180^\circ$ $= 0$	(46) $\tan 360^\circ$ $= 0$	(61) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$
(2) $\sin 270^\circ$ $= -1$	(17) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(32) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(47) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(62) $\tan 240^\circ$ $= \sqrt{3}$
(3) $\cos 120^\circ$ $= -\frac{1}{2}$	(18) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(33) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(48) $\tan 120^\circ$ $= -\sqrt{3}$	(63) $\tan 240^\circ$ $= \sqrt{3}$
(4) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(19) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(34) $\sin 270^\circ$ $= -1$	(49) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(64) $\tan 180^\circ$ $= 0$
(5) $\sin 180^\circ$ $= 0$	(20) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(35) $\tan 60^\circ$ $= \sqrt{3}$	(50) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(65) $\tan 120^\circ$ $= -\sqrt{3}$
(6) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(21) $\cos 240^\circ$ $= -\frac{1}{2}$	(36) $\tan 90^\circ$ なし	(51) $\sin 180^\circ$ $= 0$	(66) $\tan 360^\circ$ $= 0$
(7) $\sin 150^\circ$ $= \frac{1}{2}$	(22) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(37) $\cos 0^\circ$ $= 1$	(52) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(67) $\sin 90^\circ$ $= 1$
(8) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(23) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(38) $\tan 240^\circ$ $= \sqrt{3}$	(53) $\cos 60^\circ$ $= \frac{1}{2}$	(68) $\sin 150^\circ$ $= \frac{1}{2}$
(9) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(24) $\cos 270^\circ$ $= 0$	(39) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(54) $\tan 120^\circ$ $= -\sqrt{3}$	(69) $\tan 45^\circ$ $= 1$
(10) $\sin 180^\circ$ $= 0$	(25) $\sin 360^\circ$ $= 0$	(40) $\tan 45^\circ$ $= 1$	(55) $\sin 360^\circ$ $= 0$	(70) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$
(11) $\sin 270^\circ$ $= -1$	(26) $\cos 360^\circ$ $= 1$	(41) $\cos 240^\circ$ $= -\frac{1}{2}$	(56) $\sin 360^\circ$ $= 0$	(71) $\tan 315^\circ$ $= -1$
(12) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\sin 0^\circ$ $= 0$	(42) $\tan 45^\circ$ $= 1$	(57) $\cos 270^\circ$ $= 0$	(72) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(13) $\cos 120^\circ$ $= -\frac{1}{2}$	(28) $\cos 120^\circ$ $= -\frac{1}{2}$	(43) $\tan 135^\circ$ $= -1$	(58) $\sin 90^\circ$ $= 1$	(73) $\sin 330^\circ$ $= -\frac{1}{2}$
(14) $\tan 270^\circ$ なし	(29) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(44) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(59) $\cos 0^\circ$ $= 1$	(74) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$
(15) $\cos 180^\circ$ $= -1$	(30) $\sin 180^\circ$ $= 0$	(45) $\tan 300^\circ$ $= -\sqrt{3}$	(60) $\tan 315^\circ$ $= -1$	(75) $\tan 360^\circ$ $= 0$

(1) $\tan 180^\circ$ $= 0$	(16) $\cos 360^\circ$ $= 1$	(31) $\tan 240^\circ$ $= \sqrt{3}$	(46) $\cos 300^\circ$ $= \frac{1}{2}$	(61) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(2) $\tan 300^\circ$ $= -\sqrt{3}$	(17) $\sin 330^\circ$ $= -\frac{1}{2}$	(32) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(47) $\tan 360^\circ$ $= 0$	(62) $\tan 120^\circ$ $= -\sqrt{3}$
(3) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(18) $\tan 240^\circ$ $= \sqrt{3}$	(33) $\sin 150^\circ$ $= \frac{1}{2}$	(48) $\cos 180^\circ$ $= -1$	(63) $\cos 120^\circ$ $= -\frac{1}{2}$
(4) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(19) $\sin 90^\circ$ $= 1$	(34) $\cos 60^\circ$ $= \frac{1}{2}$	(49) $\cos 120^\circ$ $= -\frac{1}{2}$	(64) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$
(5) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(20) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(35) $\sin 330^\circ$ $= -\frac{1}{2}$	(50) $\sin 330^\circ$ $= -\frac{1}{2}$	(65) $\cos 180^\circ$ $= -1$
(6) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(21) $\sin 150^\circ$ $= \frac{1}{2}$	(36) $\tan 60^\circ$ $= \sqrt{3}$	(51) $\cos 180^\circ$ $= -1$	(66) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$
(7) $\cos 240^\circ$ $= -\frac{1}{2}$	(22) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(37) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(52) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(67) $\tan 90^\circ$ なし
(8) $\cos 360^\circ$ $= 1$	(23) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(38) $\tan 315^\circ$ $= -1$	(53) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(68) $\sin 270^\circ$ $= -1$
(9) $\cos 270^\circ$ $= 0$	(24) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(39) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(54) $\tan 120^\circ$ $= -\sqrt{3}$	(69) $\tan 315^\circ$ $= -1$
(10) $\tan 120^\circ$ $= -\sqrt{3}$	(25) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(40) $\sin 180^\circ$ $= 0$	(55) $\tan 300^\circ$ $= -\sqrt{3}$	(70) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(11) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(26) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(41) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(56) $\cos 0^\circ$ $= 1$	(71) $\sin 90^\circ$ $= 1$
(12) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(27) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(42) $\cos 240^\circ$ $= -\frac{1}{2}$	(57) $\tan 60^\circ$ $= \sqrt{3}$	(72) $\cos 0^\circ$ $= 1$
(13) $\tan 270^\circ$ なし	(28) $\tan 270^\circ$ なし	(43) $\cos 120^\circ$ $= -\frac{1}{2}$	(58) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(73) $\sin 90^\circ$ $= 1$
(14) $\sin 360^\circ$ $= 0$	(29) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(44) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(59) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(74) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$
(15) $\tan 0^\circ$ $= 0$	(30) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(45) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(60) $\tan 270^\circ$ なし	(75) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$

- | | | | | |
|--|--|--|--|--|
| (1) $\tan 240^\circ$
$= \sqrt{3}$ | (16) $\tan 330^\circ$
$= -\frac{1}{\sqrt{3}}$ | (31) $\sin 315^\circ$
$= -\frac{1}{\sqrt{2}}$ | (46) $\tan 120^\circ$
$= -\sqrt{3}$ | (61) $\sin 225^\circ$
$= -\frac{1}{\sqrt{2}}$ |
| (2) $\cos 180^\circ$
$= -1$ | (17) $\cos 150^\circ$
$= -\frac{\sqrt{3}}{2}$ | (32) $\tan 30^\circ$
$= \frac{1}{\sqrt{3}}$ | (47) $\tan 135^\circ$
$= -1$ | (62) $\tan 300^\circ$
$= -\sqrt{3}$ |
| (3) $\sin 150^\circ$
$= \frac{1}{2}$ | (18) $\cos 150^\circ$
$= -\frac{\sqrt{3}}{2}$ | (33) $\cos 360^\circ$
$= 1$ | (48) $\tan 210^\circ$
$= \frac{1}{\sqrt{3}}$ | (63) $\tan 60^\circ$
$= \sqrt{3}$ |
| (4) $\tan 120^\circ$
$= -\sqrt{3}$ | (19) $\tan 135^\circ$
$= -1$ | (34) $\cos 120^\circ$
$= -\frac{1}{2}$ | (49) $\tan 0^\circ$
$= 0$ | (64) $\tan 240^\circ$
$= \sqrt{3}$ |
| (5) $\sin 300^\circ$
$= -\frac{\sqrt{3}}{2}$ | (20) $\tan 210^\circ$
$= \frac{1}{\sqrt{3}}$ | (35) $\tan 270^\circ$
なし | (50) $\cos 150^\circ$
$= -\frac{\sqrt{3}}{2}$ | (65) $\tan 240^\circ$
$= \sqrt{3}$ |
| (6) $\sin 240^\circ$
$= -\frac{\sqrt{3}}{2}$ | (21) $\tan 30^\circ$
$= \frac{1}{\sqrt{3}}$ | (36) $\cos 0^\circ$
$= 1$ | (51) $\sin 135^\circ$
$= \frac{1}{\sqrt{2}}$ | (66) $\sin 360^\circ$
$= 0$ |
| (7) $\cos 300^\circ$
$= \frac{1}{2}$ | (22) $\tan 240^\circ$
$= \sqrt{3}$ | (37) $\sin 330^\circ$
$= -\frac{1}{2}$ | (52) $\tan 240^\circ$
$= \sqrt{3}$ | (67) $\tan 150^\circ$
$= -\frac{1}{\sqrt{3}}$ |
| (8) $\sin 90^\circ$
$= 1$ | (23) $\sin 210^\circ$
$= -\frac{1}{2}$ | (38) $\cos 330^\circ$
$= \frac{\sqrt{3}}{2}$ | (53) $\sin 60^\circ$
$= \frac{\sqrt{3}}{2}$ | (68) $\tan 45^\circ$
$= 1$ |
| (9) $\tan 225^\circ$
$= 1$ | (24) $\tan 210^\circ$
$= \frac{1}{\sqrt{3}}$ | (39) $\sin 330^\circ$
$= -\frac{1}{2}$ | (54) $\sin 240^\circ$
$= -\frac{\sqrt{3}}{2}$ | (69) $\cos 315^\circ$
$= \frac{1}{\sqrt{2}}$ |
| (10) $\tan 135^\circ$
$= -1$ | (25) $\sin 150^\circ$
$= \frac{1}{2}$ | (40) $\sin 210^\circ$
$= -\frac{1}{2}$ | (55) $\tan 180^\circ$
$= 0$ | (70) $\sin 45^\circ$
$= \frac{1}{\sqrt{2}}$ |
| (11) $\sin 135^\circ$
$= \frac{1}{\sqrt{2}}$ | (26) $\tan 210^\circ$
$= \frac{1}{\sqrt{3}}$ | (41) $\cos 135^\circ$
$= -\frac{1}{\sqrt{2}}$ | (56) $\tan 60^\circ$
$= \sqrt{3}$ | (71) $\tan 45^\circ$
$= 1$ |
| (12) $\tan 60^\circ$
$= \sqrt{3}$ | (27) $\sin 315^\circ$
$= -\frac{1}{\sqrt{2}}$ | (42) $\sin 315^\circ$
$= -\frac{1}{\sqrt{2}}$ | (57) $\sin 210^\circ$
$= -\frac{1}{2}$ | (72) $\tan 90^\circ$
なし |
| (13) $\sin 135^\circ$
$= \frac{1}{\sqrt{2}}$ | (28) $\cos 270^\circ$
$= 0$ | (43) $\cos 120^\circ$
$= -\frac{1}{2}$ | (58) $\tan 150^\circ$
$= -\frac{1}{\sqrt{3}}$ | (73) $\cos 60^\circ$
$= \frac{1}{2}$ |
| (14) $\cos 150^\circ$
$= -\frac{\sqrt{3}}{2}$ | (29) $\sin 270^\circ$
$= -1$ | (44) $\sin 135^\circ$
$= \frac{1}{\sqrt{2}}$ | (59) $\tan 60^\circ$
$= \sqrt{3}$ | (74) $\cos 210^\circ$
$= -\frac{\sqrt{3}}{2}$ |
| (15) $\cos 0^\circ$
$= 1$ | (30) $\tan 360^\circ$
$= 0$ | (45) $\sin 150^\circ$
$= \frac{1}{2}$ | (60) $\sin 210^\circ$
$= -\frac{1}{2}$ | (75) $\cos 135^\circ$
$= -\frac{1}{\sqrt{2}}$ |

(1) $\tan 135^\circ$ $= -1$	(16) $\tan 180^\circ$ $= 0$	(31) $\tan 0^\circ$ $= 0$	(46) $\tan 240^\circ$ $= \sqrt{3}$	(61) $\tan 180^\circ$ $= 0$
(2) $\tan 240^\circ$ $= \sqrt{3}$	(17) $\tan 90^\circ$ なし	(32) $\cos 60^\circ$ $= \frac{1}{2}$	(47) $\tan 0^\circ$ $= 0$	(62) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$
(3) $\tan 90^\circ$ なし	(18) $\tan 300^\circ$ $= -\sqrt{3}$	(33) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(48) $\tan 315^\circ$ $= -1$	(63) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(4) $\tan 120^\circ$ $= -\sqrt{3}$	(19) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(34) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(49) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(64) $\cos 60^\circ$ $= \frac{1}{2}$
(5) $\cos 0^\circ$ $= 1$	(20) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(35) $\cos 60^\circ$ $= \frac{1}{2}$	(50) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(65) $\sin 180^\circ$ $= 0$
(6) $\tan 0^\circ$ $= 0$	(21) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(36) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(51) $\tan 180^\circ$ $= 0$	(66) $\tan 135^\circ$ $= -1$
(7) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(22) $\cos 60^\circ$ $= \frac{1}{2}$	(37) $\sin 360^\circ$ $= 0$	(52) $\cos 360^\circ$ $= 1$	(67) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(8) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(23) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(38) $\tan 120^\circ$ $= -\sqrt{3}$	(53) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(68) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$
(9) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(24) $\tan 45^\circ$ $= 1$	(39) $\sin 90^\circ$ $= 1$	(54) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\tan 240^\circ$ $= \sqrt{3}$
(10) $\tan 120^\circ$ $= -\sqrt{3}$	(25) $\tan 300^\circ$ $= -\sqrt{3}$	(40) $\tan 90^\circ$ なし	(55) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(70) $\cos 0^\circ$ $= 1$
(11) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(26) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(41) $\sin 90^\circ$ $= 1$	(56) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(71) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$
(12) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(27) $\sin 150^\circ$ $= \frac{1}{2}$	(42) $\tan 270^\circ$ なし	(57) $\tan 60^\circ$ $= \sqrt{3}$	(72) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$
(13) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(28) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(43) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(58) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(73) $\tan 90^\circ$ なし
(14) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(29) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(44) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(59) $\tan 45^\circ$ $= 1$	(74) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$
(15) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(30) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(45) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(60) $\sin 0^\circ$ $= 0$	(75) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$

(1) $\cos 60^\circ$ $= \frac{1}{2}$	(16) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(31) $\tan 180^\circ$ $= 0$	(46) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(61) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$
(2) $\cos 0^\circ$ $= 1$	(17) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(32) $\tan 300^\circ$ $= -\sqrt{3}$	(47) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(62) $\cos 180^\circ$ $= -1$
(3) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(18) $\sin 150^\circ$ $= \frac{1}{2}$	(33) $\tan 60^\circ$ $= \sqrt{3}$	(48) $\tan 0^\circ$ $= 0$	(63) $\tan 315^\circ$ $= -1$
(4) $\tan 360^\circ$ $= 0$	(19) $\sin 150^\circ$ $= \frac{1}{2}$	(34) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(49) $\sin 0^\circ$ $= 0$	(64) $\tan 120^\circ$ $= -\sqrt{3}$
(5) $\sin 0^\circ$ $= 0$	(20) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(35) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(50) $\cos 120^\circ$ $= -\frac{1}{2}$	(65) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(6) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(21) $\tan 180^\circ$ $= 0$	(36) $\tan 360^\circ$ $= 0$	(51) $\sin 360^\circ$ $= 0$	(66) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$
(7) $\cos 90^\circ$ $= 0$	(22) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(37) $\cos 180^\circ$ $= -1$	(52) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(67) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(8) $\tan 135^\circ$ $= -1$	(23) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(38) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(53) $\cos 270^\circ$ $= 0$	(68) $\tan 90^\circ$ なし
(9) $\cos 180^\circ$ $= -1$	(24) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(39) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\cos 240^\circ$ $= -\frac{1}{2}$	(69) $\tan 270^\circ$ なし
(10) $\cos 300^\circ$ $= \frac{1}{2}$	(25) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(40) $\cos 270^\circ$ $= 0$	(55) $\tan 300^\circ$ $= -\sqrt{3}$	(70) $\cos 60^\circ$ $= \frac{1}{2}$
(11) $\tan 180^\circ$ $= 0$	(26) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(41) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(56) $\tan 225^\circ$ $= 1$	(71) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$
(12) $\sin 330^\circ$ $= -\frac{1}{2}$	(27) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(42) $\sin 30^\circ$ $= \frac{1}{2}$	(57) $\sin 330^\circ$ $= -\frac{1}{2}$	(72) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(13) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(28) $\tan 90^\circ$ なし	(43) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(58) $\tan 0^\circ$ $= 0$	(73) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$
(14) $\tan 60^\circ$ $= \sqrt{3}$	(29) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(44) $\tan 225^\circ$ $= 1$	(59) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(74) $\tan 90^\circ$ なし
(15) $\cos 0^\circ$ $= 1$	(30) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(45) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(60) $\cos 300^\circ$ $= \frac{1}{2}$	(75) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$

(1) $\tan 225^\circ$ $= 1$	(16) $\tan 225^\circ$ $= 1$	(31) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(46) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(61) $\tan 180^\circ$ $= 0$
(2) $\tan 270^\circ$ なし	(17) $\sin 30^\circ$ $= \frac{1}{2}$	(32) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(47) $\cos 0^\circ$ $= 1$	(62) $\tan 315^\circ$ $= -1$
(3) $\cos 270^\circ$ $= 0$	(18) $\tan 300^\circ$ $= -\sqrt{3}$	(33) $\tan 240^\circ$ $= \sqrt{3}$	(48) $\tan 0^\circ$ $= 0$	(63) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$
(4) $\sin 150^\circ$ $= \frac{1}{2}$	(19) $\cos 120^\circ$ $= -\frac{1}{2}$	(34) $\tan 135^\circ$ $= -1$	(49) $\tan 60^\circ$ $= \sqrt{3}$	(64) $\cos 90^\circ$ $= 0$
(5) $\cos 360^\circ$ $= 1$	(20) $\tan 0^\circ$ $= 0$	(35) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(50) $\cos 90^\circ$ $= 0$	(65) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(6) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(21) $\sin 330^\circ$ $= -\frac{1}{2}$	(36) $\cos 120^\circ$ $= -\frac{1}{2}$	(51) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(66) $\sin 90^\circ$ $= 1$
(7) $\cos 120^\circ$ $= -\frac{1}{2}$	(22) $\tan 270^\circ$ なし	(37) $\cos 0^\circ$ $= 1$	(52) $\tan 135^\circ$ $= -1$	(67) $\tan 225^\circ$ $= 1$
(8) $\sin 180^\circ$ $= 0$	(23) $\tan 360^\circ$ $= 0$	(38) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(53) $\cos 300^\circ$ $= \frac{1}{2}$	(68) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$
(9) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(24) $\cos 90^\circ$ $= 0$	(39) $\sin 360^\circ$ $= 0$	(54) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(69) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(10) $\sin 150^\circ$ $= \frac{1}{2}$	(25) $\tan 90^\circ$ なし	(40) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(55) $\cos 120^\circ$ $= -\frac{1}{2}$	(70) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(11) $\sin 180^\circ$ $= 0$	(26) $\tan 315^\circ$ $= -1$	(41) $\tan 315^\circ$ $= -1$	(56) $\tan 300^\circ$ $= -\sqrt{3}$	(71) $\cos 270^\circ$ $= 0$
(12) $\tan 300^\circ$ $= -\sqrt{3}$	(27) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(42) $\tan 120^\circ$ $= -\sqrt{3}$	(57) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(72) $\sin 360^\circ$ $= 0$
(13) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(28) $\cos 300^\circ$ $= \frac{1}{2}$	(43) $\tan 240^\circ$ $= \sqrt{3}$	(58) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(73) $\cos 240^\circ$ $= -\frac{1}{2}$
(14) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(29) $\cos 270^\circ$ $= 0$	(44) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(59) $\tan 90^\circ$ なし	(74) $\sin 270^\circ$ $= -1$
(15) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(30) $\cos 60^\circ$ $= \frac{1}{2}$	(45) $\tan 240^\circ$ $= \sqrt{3}$	(60) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(75) $\sin 0^\circ$ $= 0$

(1) $\tan 0^\circ$ $= 0$	(16) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(31) $\tan 135^\circ$ $= -1$	(46) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(61) $\tan 60^\circ$ $= \sqrt{3}$
(2) $\sin 330^\circ$ $= -\frac{1}{2}$	(17) $\tan 300^\circ$ $= -\sqrt{3}$	(32) $\tan 120^\circ$ $= -\sqrt{3}$	(47) $\sin 180^\circ$ $= 0$	(62) $\sin 90^\circ$ $= 1$
(3) $\tan 120^\circ$ $= -\sqrt{3}$	(18) $\sin 90^\circ$ $= 1$	(33) $\tan 360^\circ$ $= 0$	(48) $\sin 0^\circ$ $= 0$	(63) $\tan 90^\circ$ なし
(4) $\cos 90^\circ$ $= 0$	(19) $\tan 60^\circ$ $= \sqrt{3}$	(34) $\sin 210^\circ$ $= -\frac{1}{2}$	(49) $\cos 240^\circ$ $= -\frac{1}{2}$	(64) $\sin 0^\circ$ $= 0$
(5) $\cos 90^\circ$ $= 0$	(20) $\sin 150^\circ$ $= \frac{1}{2}$	(35) $\tan 300^\circ$ $= -\sqrt{3}$	(50) $\cos 300^\circ$ $= \frac{1}{2}$	(65) $\cos 240^\circ$ $= -\frac{1}{2}$
(6) $\cos 0^\circ$ $= 1$	(21) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(36) $\tan 315^\circ$ $= -1$	(51) $\sin 330^\circ$ $= -\frac{1}{2}$	(66) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(7) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(22) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(37) $\tan 120^\circ$ $= -\sqrt{3}$	(52) $\cos 60^\circ$ $= \frac{1}{2}$	(67) $\cos 120^\circ$ $= -\frac{1}{2}$
(8) $\tan 225^\circ$ $= 1$	(23) $\cos 90^\circ$ $= 0$	(38) $\sin 360^\circ$ $= 0$	(53) $\cos 360^\circ$ $= 1$	(68) $\tan 60^\circ$ $= \sqrt{3}$
(9) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(24) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(39) $\cos 180^\circ$ $= -1$	(54) $\sin 30^\circ$ $= \frac{1}{2}$	(69) $\sin 330^\circ$ $= -\frac{1}{2}$
(10) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\sin 30^\circ$ $= \frac{1}{2}$	(40) $\cos 180^\circ$ $= -1$	(55) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(70) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(11) $\tan 180^\circ$ $= 0$	(26) $\tan 270^\circ$ なし	(41) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(56) $\tan 225^\circ$ $= 1$	(71) $\cos 240^\circ$ $= -\frac{1}{2}$
(12) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(27) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(42) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(57) $\tan 300^\circ$ $= -\sqrt{3}$	(72) $\tan 0^\circ$ $= 0$
(13) $\tan 240^\circ$ $= \sqrt{3}$	(28) $\cos 300^\circ$ $= \frac{1}{2}$	(43) $\sin 0^\circ$ $= 0$	(58) $\tan 60^\circ$ $= \sqrt{3}$	(73) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(14) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(29) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(44) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(59) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(74) $\tan 45^\circ$ $= 1$
(15) $\tan 135^\circ$ $= -1$	(30) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(45) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(60) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(75) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$

(1) $\cos 120^\circ$ $= -\frac{1}{2}$	(16) $\tan 60^\circ$ $= \sqrt{3}$	(31) $\sin 210^\circ$ $= -\frac{1}{2}$	(46) $\cos 270^\circ$ $= 0$	(61) $\cos 300^\circ$ $= \frac{1}{2}$
(2) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(17) $\cos 300^\circ$ $= \frac{1}{2}$	(32) $\cos 90^\circ$ $= 0$	(47) $\sin 330^\circ$ $= -\frac{1}{2}$	(62) $\tan 45^\circ$ $= 1$
(3) $\tan 270^\circ$ なし	(18) $\tan 0^\circ$ $= 0$	(33) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(48) $\tan 120^\circ$ $= -\sqrt{3}$	(63) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(4) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(19) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(34) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(49) $\tan 360^\circ$ $= 0$	(64) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$
(5) $\sin 90^\circ$ $= 1$	(20) $\sin 330^\circ$ $= -\frac{1}{2}$	(35) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(50) $\sin 90^\circ$ $= 1$	(65) $\tan 225^\circ$ $= 1$
(6) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(21) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(36) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(51) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(66) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$
(7) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(22) $\sin 180^\circ$ $= 0$	(37) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(52) $\tan 45^\circ$ $= 1$	(67) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(8) $\tan 45^\circ$ $= 1$	(23) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(38) $\tan 90^\circ$ なし	(53) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(68) $\sin 360^\circ$ $= 0$
(9) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(24) $\sin 270^\circ$ $= -1$	(39) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(54) $\sin 210^\circ$ $= -\frac{1}{2}$	(69) $\sin 150^\circ$ $= \frac{1}{2}$
(10) $\cos 360^\circ$ $= 1$	(25) $\tan 60^\circ$ $= \sqrt{3}$	(40) $\cos 270^\circ$ $= 0$	(55) $\tan 0^\circ$ $= 0$	(70) $\sin 360^\circ$ $= 0$
(11) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(26) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(41) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(56) $\sin 90^\circ$ $= 1$	(71) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$
(12) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(27) $\tan 240^\circ$ $= \sqrt{3}$	(42) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(57) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(72) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$
(13) $\cos 60^\circ$ $= \frac{1}{2}$	(28) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(43) $\sin 90^\circ$ $= 1$	(58) $\cos 180^\circ$ $= -1$	(73) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$
(14) $\sin 90^\circ$ $= 1$	(29) $\tan 60^\circ$ $= \sqrt{3}$	(44) $\tan 90^\circ$ なし	(59) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(74) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$
(15) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(30) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(45) $\tan 315^\circ$ $= -1$	(60) $\tan 60^\circ$ $= \sqrt{3}$	(75) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$

(1) $\tan 360^\circ$ $= 0$	(16) $\tan 90^\circ$ なし	(31) $\tan 300^\circ$ $= -\sqrt{3}$	(46) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(61) $\sin 270^\circ$ $= -1$
(2) $\cos 300^\circ$ $= \frac{1}{2}$	(17) $\tan 225^\circ$ $= 1$	(32) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(47) $\tan 135^\circ$ $= -1$	(62) $\sin 210^\circ$ $= -\frac{1}{2}$
(3) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(18) $\sin 150^\circ$ $= \frac{1}{2}$	(33) $\tan 45^\circ$ $= 1$	(48) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(63) $\tan 45^\circ$ $= 1$
(4) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(19) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(34) $\sin 270^\circ$ $= -1$	(49) $\sin 150^\circ$ $= \frac{1}{2}$	(64) $\tan 360^\circ$ $= 0$
(5) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(20) $\sin 210^\circ$ $= -\frac{1}{2}$	(35) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(50) $\cos 270^\circ$ $= 0$	(65) $\sin 30^\circ$ $= \frac{1}{2}$
(6) $\tan 360^\circ$ $= 0$	(21) $\cos 360^\circ$ $= 1$	(36) $\sin 150^\circ$ $= \frac{1}{2}$	(51) $\cos 300^\circ$ $= \frac{1}{2}$	(66) $\tan 315^\circ$ $= -1$
(7) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(22) $\tan 135^\circ$ $= -1$	(37) $\tan 120^\circ$ $= -\sqrt{3}$	(52) $\tan 360^\circ$ $= 0$	(67) $\sin 0^\circ$ $= 0$
(8) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(23) $\cos 360^\circ$ $= 1$	(38) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(53) $\sin 0^\circ$ $= 0$	(68) $\tan 270^\circ$ なし
(9) $\sin 90^\circ$ $= 1$	(24) $\cos 360^\circ$ $= 1$	(39) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(54) $\cos 270^\circ$ $= 0$	(69) $\cos 120^\circ$ $= -\frac{1}{2}$
(10) $\sin 30^\circ$ $= \frac{1}{2}$	(25) $\cos 360^\circ$ $= 1$	(40) $\sin 0^\circ$ $= 0$	(55) $\tan 0^\circ$ $= 0$	(70) $\sin 180^\circ$ $= 0$
(11) $\sin 90^\circ$ $= 1$	(26) $\sin 0^\circ$ $= 0$	(41) $\sin 330^\circ$ $= -\frac{1}{2}$	(56) $\cos 0^\circ$ $= 1$	(71) $\tan 300^\circ$ $= -\sqrt{3}$
(12) $\tan 0^\circ$ $= 0$	(27) $\sin 150^\circ$ $= \frac{1}{2}$	(42) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(57) $\cos 270^\circ$ $= 0$	(72) $\tan 120^\circ$ $= -\sqrt{3}$
(13) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(28) $\tan 315^\circ$ $= -1$	(43) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(58) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$	(73) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$
(14) $\cos 60^\circ$ $= \frac{1}{2}$	(29) $\tan 90^\circ$ なし	(44) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(59) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(74) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(15) $\tan 135^\circ$ $= -1$	(30) $\sin 90^\circ$ $= 1$	(45) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(60) $\tan 45^\circ$ $= 1$	(75) $\sin 150^\circ$ $= \frac{1}{2}$

(1) $\tan 120^\circ$ $= -\sqrt{3}$	(16) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(31) $\cos 360^\circ$ $= 1$	(46) $\tan 270^\circ$ なし	(61) $\tan 300^\circ$ $= -\sqrt{3}$
(2) $\cos 360^\circ$ $= 1$	(17) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(32) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(47) $\tan 240^\circ$ $= \sqrt{3}$	(62) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$
(3) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(18) $\tan 270^\circ$ なし	(33) $\cos 180^\circ$ $= -1$	(48) $\cos 120^\circ$ $= -\frac{1}{2}$	(63) $\sin 30^\circ$ $= \frac{1}{2}$
(4) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(19) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(34) $\tan 240^\circ$ $= \sqrt{3}$	(49) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(64) $\cos 225^\circ$ $= -\frac{1}{\sqrt{2}}$
(5) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(20) $\cos 0^\circ$ $= 1$	(35) $\tan 225^\circ$ $= 1$	(50) $\sin 270^\circ$ $= -1$	(65) $\tan 135^\circ$ $= -1$
(6) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(21) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(36) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(51) $\cos 0^\circ$ $= 1$	(66) $\sin 90^\circ$ $= 1$
(7) $\tan 225^\circ$ $= 1$	(22) $\tan 90^\circ$ なし	(37) $\sin 150^\circ$ $= \frac{1}{2}$	(52) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(67) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$
(8) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(23) $\cos 240^\circ$ $= -\frac{1}{2}$	(38) $\cos 240^\circ$ $= -\frac{1}{2}$	(53) $\tan 270^\circ$ なし	(68) $\cos 300^\circ$ $= \frac{1}{2}$
(9) $\sin 330^\circ$ $= -\frac{1}{2}$	(24) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(39) $\cos 360^\circ$ $= 1$	(54) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(69) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$
(10) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(25) $\tan 240^\circ$ $= \sqrt{3}$	(40) $\tan 315^\circ$ $= -1$	(55) $\tan 60^\circ$ $= \sqrt{3}$	(70) $\cos 240^\circ$ $= -\frac{1}{2}$
(11) $\cos 120^\circ$ $= -\frac{1}{2}$	(26) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(41) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(56) $\tan 90^\circ$ なし	(71) $\tan 60^\circ$ $= \sqrt{3}$
(12) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(27) $\tan 270^\circ$ なし	(42) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(57) $\tan 135^\circ$ $= -1$	(72) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$
(13) $\tan 60^\circ$ $= \sqrt{3}$	(28) $\tan 240^\circ$ $= \sqrt{3}$	(43) $\tan 60^\circ$ $= \sqrt{3}$	(58) $\tan 240^\circ$ $= \sqrt{3}$	(73) $\cos 360^\circ$ $= 1$
(14) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(29) $\cos 210^\circ$ $= -\frac{\sqrt{3}}{2}$	(44) $\cos 300^\circ$ $= \frac{1}{2}$	(59) $\tan 300^\circ$ $= -\sqrt{3}$	(74) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$
(15) $\tan 210^\circ$ $= \frac{1}{\sqrt{3}}$	(30) $\tan 360^\circ$ $= 0$	(45) $\sin 150^\circ$ $= \frac{1}{2}$	(60) $\cos 90^\circ$ $= 0$	(75) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$

(1) $\tan 270^\circ$ なし	(16) $\sin 150^\circ$ $= \frac{1}{2}$	(31) $\sin 180^\circ$ $= 0$	(46) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(61) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$
(2) $\tan 120^\circ$ $= -\sqrt{3}$	(17) $\sin 330^\circ$ $= -\frac{1}{2}$	(32) $\tan 240^\circ$ $= \sqrt{3}$	(47) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(62) $\cos 180^\circ$ $= -1$
(3) $\sin 270^\circ$ $= -1$	(18) $\sin 360^\circ$ $= 0$	(33) $\tan 135^\circ$ $= -1$	(48) $\sin 330^\circ$ $= -\frac{1}{2}$	(63) $\cos 0^\circ$ $= 1$
(4) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$	(19) $\tan 240^\circ$ $= \sqrt{3}$	(34) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(49) $\tan 90^\circ$ なし	(64) $\sin 135^\circ$ $= \frac{1}{\sqrt{2}}$
(5) $\tan 120^\circ$ $= -\sqrt{3}$	(20) $\tan 270^\circ$ なし	(35) $\sin 120^\circ$ $= \frac{\sqrt{3}}{2}$	(50) $\tan 360^\circ$ $= 0$	(65) $\tan 30^\circ$ $= \frac{1}{\sqrt{3}}$
(6) $\tan 315^\circ$ $= -1$	(21) $\tan 225^\circ$ $= 1$	(36) $\sin 270^\circ$ $= -1$	(51) $\sin 330^\circ$ $= -\frac{1}{2}$	(66) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$
(7) $\cos 60^\circ$ $= \frac{1}{2}$	(22) $\cos 270^\circ$ $= 0$	(37) $\sin 30^\circ$ $= \frac{1}{2}$	(52) $\tan 60^\circ$ $= \sqrt{3}$	(67) $\tan 360^\circ$ $= 0$
(8) $\sin 210^\circ$ $= -\frac{1}{2}$	(23) $\tan 270^\circ$ なし	(38) $\cos 180^\circ$ $= -1$	(53) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(68) $\tan 225^\circ$ $= 1$
(9) $\tan 120^\circ$ $= -\sqrt{3}$	(24) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(39) $\sin 270^\circ$ $= -1$	(54) $\tan 315^\circ$ $= -1$	(69) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$
(10) $\tan 360^\circ$ $= 0$	(25) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(40) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(55) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(70) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$
(11) $\tan 270^\circ$ なし	(26) $\tan 360^\circ$ $= 0$	(41) $\tan 240^\circ$ $= \sqrt{3}$	(56) $\sin 330^\circ$ $= -\frac{1}{2}$	(71) $\sin 360^\circ$ $= 0$
(12) $\cos 60^\circ$ $= \frac{1}{2}$	(27) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(42) $\tan 180^\circ$ $= 0$	(57) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(72) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$
(13) $\cos 0^\circ$ $= 1$	(28) $\sin 300^\circ$ $= -\frac{\sqrt{3}}{2}$	(43) $\cos 90^\circ$ $= 0$	(58) $\tan 135^\circ$ $= -1$	(73) $\tan 90^\circ$ なし
(14) $\tan 180^\circ$ $= 0$	(29) $\tan 0^\circ$ $= 0$	(44) $\cos 300^\circ$ $= \frac{1}{2}$	(59) $\tan 120^\circ$ $= -\sqrt{3}$	(74) $\sin 150^\circ$ $= \frac{1}{2}$
(15) $\sin 0^\circ$ $= 0$	(30) $\sin 30^\circ$ $= \frac{1}{2}$	(45) $\cos 60^\circ$ $= \frac{1}{2}$	(60) $\cos 270^\circ$ $= 0$	(75) $\tan 270^\circ$ なし

(1) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$	(16) $\cos 90^\circ$ $= 0$	(31) $\tan 180^\circ$ $= 0$	(46) $\sin 30^\circ$ $= \frac{1}{2}$	(61) $\sin 270^\circ$ $= -1$
(2) $\tan 45^\circ$ $= 1$	(17) $\cos 240^\circ$ $= -\frac{1}{2}$	(32) $\sin 150^\circ$ $= \frac{1}{2}$	(47) $\tan 135^\circ$ $= -1$	(62) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$
(3) $\sin 45^\circ$ $= \frac{1}{\sqrt{2}}$	(18) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(33) $\sin 90^\circ$ $= 1$	(48) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(63) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$
(4) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(19) $\sin 150^\circ$ $= \frac{1}{2}$	(34) $\tan 330^\circ$ $= -\frac{1}{\sqrt{3}}$	(49) $\tan 120^\circ$ $= -\sqrt{3}$	(64) $\sin 210^\circ$ $= -\frac{1}{2}$
(5) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(20) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(35) $\tan 0^\circ$ $= 0$	(50) $\sin 30^\circ$ $= \frac{1}{2}$	(65) $\sin 150^\circ$ $= \frac{1}{2}$
(6) $\sin 240^\circ$ $= -\frac{\sqrt{3}}{2}$	(21) $\sin 60^\circ$ $= \frac{\sqrt{3}}{2}$	(36) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(51) $\tan 270^\circ$ なし	(66) $\cos 240^\circ$ $= -\frac{1}{2}$
(7) $\tan 0^\circ$ $= 0$	(22) $\tan 60^\circ$ $= \sqrt{3}$	(37) $\tan 0^\circ$ $= 0$	(52) $\cos 45^\circ$ $= \frac{1}{\sqrt{2}}$	(67) $\tan 180^\circ$ $= 0$
(8) $\cos 0^\circ$ $= 1$	(23) $\sin 180^\circ$ $= 0$	(38) $\cos 0^\circ$ $= 1$	(53) $\sin 315^\circ$ $= -\frac{1}{\sqrt{2}}$	(68) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$
(9) $\cos 0^\circ$ $= 1$	(24) $\cos 120^\circ$ $= -\frac{1}{2}$	(39) $\cos 360^\circ$ $= 1$	(54) $\sin 330^\circ$ $= -\frac{1}{2}$	(69) $\cos 150^\circ$ $= -\frac{\sqrt{3}}{2}$
(10) $\tan 60^\circ$ $= \sqrt{3}$	(25) $\tan 45^\circ$ $= 1$	(40) $\sin 0^\circ$ $= 0$	(55) $\tan 225^\circ$ $= 1$	(70) $\tan 315^\circ$ $= -1$
(11) $\cos 315^\circ$ $= \frac{1}{\sqrt{2}}$	(26) $\cos 180^\circ$ $= -1$	(41) $\cos 30^\circ$ $= \frac{\sqrt{3}}{2}$	(56) $\cos 330^\circ$ $= \frac{\sqrt{3}}{2}$	(71) $\tan 90^\circ$ なし
(12) $\tan 360^\circ$ $= 0$	(27) $\tan 60^\circ$ $= \sqrt{3}$	(42) $\sin 210^\circ$ $= -\frac{1}{2}$	(57) $\sin 0^\circ$ $= 0$	(72) $\tan 60^\circ$ $= \sqrt{3}$
(13) $\tan 315^\circ$ $= -1$	(28) $\tan 180^\circ$ $= 0$	(43) $\cos 120^\circ$ $= -\frac{1}{2}$	(58) $\sin 225^\circ$ $= -\frac{1}{\sqrt{2}}$	(73) $\tan 135^\circ$ $= -1$
(14) $\tan 45^\circ$ $= 1$	(29) $\tan 45^\circ$ $= 1$	(44) $\tan 150^\circ$ $= -\frac{1}{\sqrt{3}}$	(59) $\cos 0^\circ$ $= 1$	(74) $\tan 135^\circ$ $= -1$
(15) $\sin 30^\circ$ $= \frac{1}{2}$	(30) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(45) $\cos 135^\circ$ $= -\frac{1}{\sqrt{2}}$	(60) $\cos 0^\circ$ $= 1$	(75) $\tan 90^\circ$ なし

3.3 微積分

3.3.1 微分 多項式

多項式の微分。

(1) $y = -x^3 - 4x^2 - 8x + 6$

(16) $y = 2x^3 + 7x^2 - x - 7$

(31) $y = 7x^3 - 3x^2 + 8x - 4$

(2) $y = 9x^3 - 6x^2 - 8$

(17) $y = -6x^3 + 4x^2 - 8x + 1$

(32) $y = -5x^3 - 3x^2 - 2$

(3) $y = -9x^3 + 2x^2 + 3x - 6$

(18) $y = -7x^3 - 7x^2 - 2x$

(33) $y = -5x^3 - 2x^2 - 3x + 8$

(4) $y = -5x^3 - 5x^2$

(19) $y = 5x^3 - 4x^2 - 8x - 1$

(34) $y = x^3 - 2x^2 - 2x + 7$

(5) $y = -7x^3 + 3x^2 - 8x - 2$

(20) $y = 5x^3 - 7x^2 + 4x + 1$

(35) $y = 9x^3 + x^2 + 2x - 1$

(6) $y = -8x^3 + 6x^2 - 5x - 2$

(21) $y = 9x^3 - 3x^2 + 6x + 4$

(36) $y = 9x^3 + 8x^2 - 4x - 6$

(7) $y = 3x^3 + 5x^2 + 8x + 7$

(22) $y = x^3 - 7x^2 - 4x - 5$

(37) $y = 8x^3 + 7x^2 + 5x + 2$

(8) $y = -9x^3 + x^2 + 6x + 9$

(23) $y = -7x^3 + 6x^2 + 9x + 6$

(38) $y = -8x^3 + 5x - 6$

(9) $y = 8x^3 + 4x^2 + 6x - 2$

(24) $y = -x^3 - x^2 + 9x - 7$

(39) $y = -6x^3 + 4x^2 - 7x - 9$

(10) $y = -9x^3 + 4x^2 + 6x - 6$

(25) $y = -5x^3 - 9x^2 - 6x - 4$

(40) $y = -6x^3 + 9x^2 - 8x - 3$

(11) $y = -2x^3 - 4x^2 + 6x + 9$

(26) $y = 3x^3 - 6x^2 - 3$

(41) $y = -7x^3 - 2x^2 - 6x - 2$

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(11) $y = -9x^3 - 3x + 6$

(26) $y = 5x^3 + x^2 + 4x - 4$

(41) $y = -5x^3 - 6x^2 - x + 2$

(12) $y = -4x^3 + 4x^2 - 2x + 7$

(27) $y = -4x^3 - 2x^2 - 9x - 7$

(42) $y = x^3 - 7x^2 + 7x + 8$

(13) $y = 9x^3 + 3x^2 - 8x - 1$

(28) $y = 7x^3 + 7x^2 + 6x + 1$

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(15) $y = -5x^3 - 6x - 3$

(30) $y = 2x^3 + 2x - 6$

(45) $y = -7x^3 + 7x^2 - 4x - 1$

(1) $y = 6x^3 - 5x^2 + x$

(16) $y = -6x^3 + 2x^2 - 4x - 9$

(31) $y = 9x^3 - 5x^2 - 3x - 4$

(2) $y = 4x^3 - 7x^2 - x - 2$

(17) $y = 5x^3 - 7x^2 + 2x + 1$

(32) $y = 6x^3 - 6x^2 - 2x - 1$

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(9) $y = 7x^3 + x - 8$

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(15) $y = -9x^3 + 4x^2 + 9x + 1$

(30) $y = -2x^3 - 8x^2 + 8x - 2$

(45) $y = 5x^3 + x + 6$

$$(1) \quad y = -x^3 - 4x^2 - 8x + 6 \\ y' = -3x^2 - 8x - 8$$

$$(2) \quad y = 9x^3 - 6x^2 - 8 \\ y' = 27x^2 - 12x$$

$$(3) \quad y = -9x^3 + 2x^2 + 3x - 6 \\ y' = -27x^2 + 4x + 3$$

$$(4) \quad y = -5x^3 - 5x^2 \\ y' = -15x^2 - 10x$$

$$(5) \quad y = -7x^3 + 3x^2 - 8x - 2 \\ y' = -21x^2 + 6x - 8$$

$$(6) \quad y = -8x^3 + 6x^2 - 5x - 2 \\ y' = -24x^2 + 12x - 5$$

$$(7) \quad y = 3x^3 + 5x^2 + 8x + 7 \\ y' = 9x^2 + 10x + 8$$

$$(8) \quad y = -9x^3 + x^2 + 6x + 9 \\ y' = -27x^2 + 2x + 6$$

$$(9) \quad y = 8x^3 + 4x^2 + 6x - 2 \\ y' = 24x^2 + 8x + 6$$

$$(10) \quad y = -9x^3 + 4x^2 + 6x - 6 \\ y' = -27x^2 + 8x + 6$$

$$(11) \quad y = -2x^3 - 4x^2 + 6x + 9 \\ y' = -6x^2 - 8x + 6$$

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$$(13) \quad y = -8x^3 + 7x^2 + 6x + 3 \\ y' = -24x^2 + 14x + 6$$

$$(14) \quad y = -5x^3 + 9x^2 + 5x + 7 \\ y' = -15x^2 + 18x + 5$$

$$(15) \quad y = 9x^3 + 8x^2 - 2x - 1 \\ y' = 27x^2 + 16x - 2$$

$$(16) \quad y = 2x^3 + 7x^2 - x - 7 \\ y' = 6x^2 + 14x - 1$$

$$(17) \quad y = -6x^3 + 4x^2 - 8x + 1 \\ y' = -18x^2 + 8x - 8$$

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$$(27) \quad y = -7x^3 - 7x^2 - 9x - 2 \\ y' = -21x^2 - 14x - 9$$

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$$(41) \quad y = -7x^3 - 2x^2 - 6x - 2 \\ y' = -21x^2 - 4x - 6$$

$$(42) \quad y = 4x^3 + 4x^2 + 9x - 7 \\ y' = 12x^2 + 8x + 9$$

$$(43) \quad y = 4x^3 + 2x^2 + 9x - 7 \\ y' = 12x^2 + 4x + 9$$

$$(44) \quad y = -3x^3 + 5x^2 - 6x - 9 \\ y' = -9x^2 + 10x - 6$$

$$(45) \quad y = -x^3 + 3x^2 - 9x - 2 \\ y' = -3x^2 + 6x - 9$$

- | | | |
|---|---|---|
| (1) $y = -5x^3 - 8x^2 + 4x - 6$
$y' = -15x^2 - 16x + 4$ | (16) $y = 6x^3 + 2x^2 + 4x + 9$
$y' = 18x^2 + 4x + 4$ | (31) $y = -9x^3 + x^2 + 6x - 7$
$y' = -27x^2 + 2x + 6$ |
| (2) $y = 4x^3 + 7x^2 - x + 8$
$y' = 12x^2 + 14x - 1$ | (17) $y = -4x^3 - 6x^2 + x + 2$
$y' = -12x^2 - 12x + 1$ | (32) $y = 2x^3 - 9x^2 - 8x + 8$
$y' = 6x^2 - 18x - 8$ |
| (3) $y = 8x^3 - 9x^2 + 8x - 6$
$y' = 24x^2 - 18x + 8$ | (18) $y = -2x^3 - 2x^2 - 8x + 1$
$y' = -6x^2 - 4x - 8$ | (33) $y = 5x^3 - 9x^2 - x - 7$
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| (4) $y = 9x^3 - 5x^2 - 6x - 4$
$y' = 27x^2 - 10x - 6$ | (19) $y = 5x^3 + 2x^2 - 9x - 7$
$y' = 15x^2 + 4x - 9$ | (34) $y = -9x^3 + 6x^2 + 5x + 5$
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$y' = 12x^2 - 16x - 8$ |
| (15) $y = 8x^3 - 5x^2 + 9x - 6$
$y' = 24x^2 - 10x + 9$ | (30) $y = 3x^3 + 5x^2 - x - 7$
$y' = 9x^2 + 10x - 1$ | (45) $y = -2x^3 - 7x^2 - 6x - 6$
$y' = -6x^2 - 14x - 6$ |

- | | | |
|---|---|---|
| (1) $y = -4x^3 - 4x^2 + 3x + 3$
$y' = -12x^2 - 8x + 3$ | (16) $y = -8x^3 - 9x^2 + 3x - 4$
$y' = -24x^2 - 18x + 3$ | (31) $y = 5x^3 + 6x^2 + 8$
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| (8) $y = 7x^3 + 8x^2 + 6x + 2$
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$y' = -9x^2 - 18x - 8$ | (40) $y = 6x^3 + 2x^2 + x - 1$
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$$(13) \quad y = -2x^3 + 3x^2 + x \\ y' = -6x^2 + 6x + 1$$

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$$(14) \quad y = -6x^3 + x^2 + 7x - 6 \\ y' = -18x^2 + 2x + 7$$

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- | | | |
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$y' = 18x^2 - 10x + 1$ | (16) $y = -6x^3 + 2x^2 - 4x - 9$
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- | | | |
|--|--|---|
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$y' = -9x^2 + 4x$ | (18) $y = 9x^3 + x - 4$
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$$(45) \quad y = 5x^3 + x + 6 \\ y' = 15x^2 + 1$$

3.3.2 不定積分

多項式の不定積分。

積分定数は C としている。

(1) $\int (-2x^2 - 4x + 3) dx$

(11) $\int (-9x^2 + 4x + 5) dx$

(21) $\int (-4x^2 - x + 9) dx$

(2) $\int (-x^2 + 2x - 1) dx$

(12) $\int (-4x^2 - 6x + 1) dx$

(22) $\int (3x^2 - 9x + 8) dx$

(3) $\int (-2x^2 - 9x + 3) dx$

(13) $\int (x^2 + 3x + 2) dx$

(23) $\int (5x^2 - 2x + 6) dx$

(4) $\int (7x^2 + 1) dx$

(14) $\int (-7x^2 - 9x - 9) dx$

(24) $\int (-8x^2 - 6x - 5) dx$

(5) $\int (-9x^2 + 3x + 8) dx$

(15) $\int (-7x^2 - 6x - 8) dx$

(25) $\int (3x^2 - 3x + 2) dx$

(6) $\int (7x^2 - 8x - 8) dx$

(16) $\int (7x^2 + 9x + 9) dx$

(26) $\int (9x^2 + 4x + 6) dx$

(7) $\int (-x^2 + 8x - 3) dx$

(17) $\int (-9x^2 - 9x - 9) dx$

(27) $\int (-7x^2 - 7x + 3) dx$

(8) $\int (-4x^2 - 7x + 8) dx$

(18) $\int (-4x^2 + 6x + 6) dx$

(28) $\int (9x^2 + 4x + 1) dx$

(9) $\int (-7x^2 + 2) dx$

(19) $\int (6x^2 + x + 7) dx$

(29) $\int (5x^2 + 7x - 9) dx$

(10) $\int (-3x^2 - 9x - 5) dx$

(20) $\int (7x^2 + x + 5) dx$

(30) $\int (-8x^2 - 8x) dx$

(1) $\int (-8x^2 + 7x - 3) dx$

(11) $\int (-8x^2 + 8x - 7) dx$

(21) $\int (-7x^2 + 5x + 1) dx$

(2) $\int (-8x^2 + 5x - 5) dx$

(12) $\int (-2x^2 + 5x + 4) dx$

(22) $\int (x^2 + 8x + 1) dx$

(3) $\int (-5x^2 - 9x - 4) dx$

(13) $\int (3x^2 - 9x) dx$

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(5) $\int (-9x^2 + x - 5) dx$

(15) $\int (3x^2 - 3x - 4) dx$

(25) $\int (-2x^2 + 4x - 6) dx$

(6) $\int (-4x^2 + 7x - 2) dx$

(16) $\int (-8x^2 + 9x - 2) dx$

(26) $\int (-x^2 + 5x - 8) dx$

(7) $\int (5x^2 - 9x - 5) dx$

(17) $\int (-7x^2 + 2x) dx$

(27) $\int (-4x^2 + 3x - 6) dx$

(8) $\int (9x^2 + 3x - 2) dx$

(18) $\int (-8x^2 - 4x - 4) dx$

(28) $\int (x^2 - 9x + 6) dx$

(9) $\int (x^2 + 4x - 8) dx$

(19) $\int (x^2 + 5) dx$

(29) $\int (6x^2 + x - 6) dx$

(10) $\int (-3x^2 + 5x - 3) dx$

(20) $\int (3x^2 + 2x) dx$

(30) $\int (7x^2 - 5x - 2) dx$

(1) $\int (-2x^2 + 5x - 8) dx$

(11) $\int (-x^2 + x - 5) dx$

(21) $\int (6x^2 - x - 4) dx$

(2) $\int (5x^2 + 3x - 2) dx$

(12) $\int (-x^2 + 8x) dx$

(22) $\int (-5x^2 + 4x - 1) dx$

(3) $\int (4x^2 - 5x - 2) dx$

(13) $\int (-3x^2 + 5x - 7) dx$

(23) $\int (-5x^2 + 6x + 7) dx$

(4) $\int (-4x^2 - 4x + 1) dx$

(14) $\int (-2x^2 - 9x + 4) dx$

(24) $\int (-8x^2 - 2x - 4) dx$

(5) $\int (8x^2 - 3x + 2) dx$

(15) $\int (x^2 + x - 7) dx$

(25) $\int (-4x^2 - 3) dx$

(6) $\int (8x^2 - x - 5) dx$

(16) $\int (-7x^2 + 7) dx$

(26) $\int (x^2 + 9x - 9) dx$

(7) $\int (-5x^2 - 7x + 3) dx$

(17) $\int (6x^2 + 3x - 1) dx$

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(9) $\int (-x^2 - 9x + 1) dx$

(19) $\int (7x^2 - 7x - 4) dx$

(29) $\int (5x^2 - 8x + 7) dx$

(10) $\int (-x^2 - 2x - 7) dx$

(20) $\int (7x^2 + 7x - 3) dx$

(30) $\int (-8x^2 + 3x - 1) dx$

(1) $\int (-2x^2 - 8x - 1) dx$

(11) $\int (9x^2 - x + 6) dx$

(21) $\int (3x^2 + x - 7) dx$

(2) $\int (-4x^2 + 5x - 1) dx$

(12) $\int (7x^2 + 3x + 5) dx$

(22) $\int (7x^2 - 2x - 8) dx$

(3) $\int (9x^2 + x + 7) dx$

(13) $\int (6x^2 + 2x - 9) dx$

(23) $\int (4x^2 - 4x - 2) dx$

(4) $\int (-x^2 - 8x) dx$

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(24) $\int (x^2 - 4x - 1) dx$

(5) $\int (-7x^2 + 9x - 4) dx$

(15) $\int (2x^2 - 5x - 8) dx$

(25) $\int (-x^2 - x - 6) dx$

(6) $\int (8x^2 - 4x - 4) dx$

(16) $\int (7x^2 - 6x + 2) dx$

(26) $\int (-4x^2 + x + 7) dx$

(7) $\int (-x^2 - 6x - 8) dx$

(17) $\int (6x^2 + x - 9) dx$

(27) $\int (8x^2 - 3x + 8) dx$

(8) $\int (7x^2 + 8x) dx$

(18) $\int (2x^2 - 8x + 5) dx$

(28) $\int (-2x^2 - 2x + 8) dx$

(9) $\int (3x^2 - 7x + 2) dx$

(19) $\int (-5x^2 + 4x - 6) dx$

(29) $\int (9x^2 + x + 1) dx$

(10) $\int (x^2 - 2x - 9) dx$

(20) $\int (-6x^2 + 6x - 7) dx$

(30) $\int (-9x^2 - x + 9) dx$

(1) $\int (x^2 + 3x - 7) \, dx$

(11) $\int (-3x^2 + 3x - 6) \, dx$

(21) $\int (-6x^2 + 4x) \, dx$

(2) $\int (7x^2 - 8x + 4) \, dx$

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$$(23) \int (-6x^2 - 9x + 7) dx \\ = -2x^3 - \frac{9}{2}x^2 + 7x + C$$

$$(4) \int (5x^2 - x + 2) dx \\ = \frac{5}{3}x^3 - \frac{1}{2}x^2 + 2x + C$$

$$(14) \int (-8x^2 + 7x + 7) dx \\ = -\frac{8}{3}x^3 + \frac{7}{2}x^2 + 7x + C$$

$$(24) \int (-9x^2 + 4x - 8) dx \\ = -3x^3 + 2x^2 - 8x + C$$

$$(5) \int (5x^2 + 3x + 8) dx \\ = \frac{5}{3}x^3 + \frac{3}{2}x^2 + 8x + C$$

$$(15) \int (-4x^2 - 6x - 9) dx \\ = -\frac{4}{3}x^3 - 3x^2 - 9x + C$$

$$(25) \int (8x^2 + 8x) dx \\ = \frac{8}{3}x^3 + 4x^2 + C$$

$$(6) \int (7x^2 + 4x + 7) dx \\ = \frac{7}{3}x^3 + 2x^2 + 7x + C$$

$$(16) \int (-2x^2 + 7x - 9) dx \\ = -\frac{2}{3}x^3 + \frac{7}{2}x^2 - 9x + C$$

$$(26) \int (-9x^2 - 4x - 5) dx \\ = -3x^3 - 2x^2 - 5x + C$$

$$(7) \int (9x^2 - 9x + 1) dx \\ = 3x^3 - \frac{9}{2}x^2 + x + C$$

$$(17) \int (-8x^2 + 6x - 3) dx \\ = -\frac{8}{3}x^3 + 3x^2 - 3x + C$$

$$(27) \int (2x^2 + 4x + 7) dx \\ = \frac{2}{3}x^3 + 2x^2 + 7x + C$$

$$(8) \int (-3x^2 + 2x - 8) dx \\ = -x^3 + x^2 - 8x + C$$

$$(18) \int (5x^2 + 9x + 5) dx \\ = \frac{5}{3}x^3 + \frac{9}{2}x^2 + 5x + C$$

$$(28) \int (-5x^2 - 8x - 2) dx \\ = -\frac{5}{3}x^3 - 4x^2 - 2x + C$$

$$(9) \int (3x^2 + 5x - 1) dx \\ = x^3 + \frac{5}{2}x^2 - x + C$$

$$(19) \int (x^2 + 6x + 2) dx \\ = \frac{1}{3}x^3 + 3x^2 + 2x + C$$

$$(29) \int (5x^2 + 5x - 9) dx \\ = \frac{5}{3}x^3 + \frac{5}{2}x^2 - 9x + C$$

$$(10) \int (-8x^2 - 9x + 8) dx \\ = -\frac{8}{3}x^3 - \frac{9}{2}x^2 + 8x + C$$

$$(20) \int (5x^2 + 3) dx \\ = \frac{5}{3}x^3 + 3x + C$$

$$(30) \int (2x^2 - x - 5) dx \\ = \frac{2}{3}x^3 - \frac{1}{2}x^2 - 5x + C$$

$$(1) \int (9x^2 + 2x - 6) dx \\ = 3x^3 + x^2 - 6x + C$$

$$(11) \int (7x^2 + 8x - 1) dx \\ = \frac{7}{3}x^3 + 4x^2 - x + C$$

$$(21) \int (-2x^2 + 7x - 7) dx \\ = -\frac{2}{3}x^3 + \frac{7}{2}x^2 - 7x + C$$

$$(2) \int 8x^2 dx \\ = \frac{8}{3}x^3 + C$$

$$(12) \int (-6x^2 + 2x - 1) dx \\ = -2x^3 + x^2 - x + C$$

$$(22) \int (-4x^2 + 4x + 5) dx \\ = -\frac{4}{3}x^3 + 2x^2 + 5x + C$$

$$(3) \int (7x^2 - 6x + 5) dx \\ = \frac{7}{3}x^3 - 3x^2 + 5x + C$$

$$(13) \int (-3x^2 - x - 6) dx \\ = -x^3 - \frac{1}{2}x^2 - 6x + C$$

$$(23) \int (-7x^2 + 7x + 6) dx \\ = -\frac{7}{3}x^3 + \frac{7}{2}x^2 + 6x + C$$

$$(4) \int (8x^2 + 8x - 9) dx \\ = \frac{8}{3}x^3 + 4x^2 - 9x + C$$

$$(14) \int (6x^2 - 7x - 3) dx \\ = 2x^3 - \frac{7}{2}x^2 - 3x + C$$

$$(24) \int (-x^2 + 5) dx \\ = -\frac{1}{3}x^3 + 5x + C$$

$$(5) \int (6x^2 - 5x + 1) dx \\ = 2x^3 - \frac{5}{2}x^2 + x + C$$

$$(15) \int (-4x^2 + 3x - 2) dx \\ = -\frac{4}{3}x^3 + \frac{3}{2}x^2 - 2x + C$$

$$(25) \int (7x^2 - x + 5) dx \\ = \frac{7}{3}x^3 - \frac{1}{2}x^2 + 5x + C$$

$$(6) \int (-7x^2 + 6x + 3) dx \\ = -\frac{7}{3}x^3 + 3x^2 + 3x + C$$

$$(16) \int (2x^2 - x - 6) dx \\ = \frac{2}{3}x^3 - \frac{1}{2}x^2 - 6x + C$$

$$(26) \int (-7x^2 - 3x + 1) dx \\ = -\frac{7}{3}x^3 - \frac{3}{2}x^2 + x + C$$

$$(7) \int (-4x^2 + x) dx \\ = -\frac{4}{3}x^3 + \frac{1}{2}x^2 + C$$

$$(17) \int (7x^2 + 6x - 4) dx \\ = \frac{7}{3}x^3 + 3x^2 - 4x + C$$

$$(27) \int (-8x^2 + 6x - 1) dx \\ = -\frac{8}{3}x^3 + 3x^2 - x + C$$

$$(8) \int (-8x^2 + 5x - 5) dx \\ = -\frac{8}{3}x^3 + \frac{5}{2}x^2 - 5x + C$$

$$(18) \int (-x^2 + 5x + 6) dx \\ = -\frac{1}{3}x^3 + \frac{5}{2}x^2 + 6x + C$$

$$(28) \int (3x^2 + 4x + 3) dx \\ = x^3 + 2x^2 + 3x + C$$

$$(9) \int (6x^2 - 3x + 2) dx \\ = 2x^3 - \frac{3}{2}x^2 + 2x + C$$

$$(19) \int (2x^2 - 6) dx \\ = \frac{2}{3}x^3 - 6x + C$$

$$(29) \int (-7x^2 - 4x + 3) dx \\ = -\frac{7}{3}x^3 - 2x^2 + 3x + C$$

$$(10) \int (-2x^2 + 7x + 4) dx \\ = -\frac{2}{3}x^3 + \frac{7}{2}x^2 + 4x + C$$

$$(20) \int (8x^2 + 7x + 3) dx \\ = \frac{8}{3}x^3 + \frac{7}{2}x^2 + 3x + C$$

$$(30) \int (-6x^2 - 2x + 4) dx \\ = -2x^3 - x^2 + 4x + C$$

$$(1) \int (-4x^2 - 4x + 5) dx \\ = -\frac{4}{3}x^3 - 2x^2 + 5x + C$$

$$(2) \int (-5x^2 + 7x + 7) dx \\ = -\frac{5}{3}x^3 + \frac{7}{2}x^2 + 7x + C$$

$$(3) \int (-x^2 + x + 1) dx \\ = -\frac{1}{3}x^3 + \frac{1}{2}x^2 + x + C$$

$$(4) \int (-8x^2 + 3x - 9) dx \\ = -\frac{8}{3}x^3 + \frac{3}{2}x^2 - 9x + C$$

$$(5) \int (-7x^2 - 9x - 9) dx \\ = -\frac{7}{3}x^3 - \frac{9}{2}x^2 - 9x + C$$

$$(6) \int (-9x^2 + x + 5) dx \\ = -3x^3 + \frac{1}{2}x^2 + 5x + C$$

$$(7) \int (-x^2 - 8x - 4) dx \\ = -\frac{1}{3}x^3 - 4x^2 - 4x + C$$

$$(8) \int (9x^2 - 8x - 6) dx \\ = 3x^3 - 4x^2 - 6x + C$$

$$(9) \int (4x^2 + 9x - 6) dx \\ = \frac{4}{3}x^3 + \frac{9}{2}x^2 - 6x + C$$

$$(10) \int (7x^2 - 4x + 8) dx \\ = \frac{7}{3}x^3 - 2x^2 + 8x + C$$

$$(11) \int (-6x^2 - 4x + 2) dx \\ = -2x^3 - 2x^2 + 2x + C$$

$$(12) \int (-8x^2 + 7x + 5) dx \\ = -\frac{8}{3}x^3 + \frac{7}{2}x^2 + 5x + C$$

$$(13) \int (x^2 + 4x + 9) dx \\ = \frac{1}{3}x^3 + 2x^2 + 9x + C$$

$$(14) \int (-4x^2 + 6x - 8) dx \\ = -\frac{4}{3}x^3 + 3x^2 - 8x + C$$

$$(15) \int (-9x^2 + 5x) dx \\ = -3x^3 + \frac{5}{2}x^2 + C$$

$$(16) \int (8x^2 + 8x) dx \\ = \frac{8}{3}x^3 + 4x^2 + C$$

$$(17) \int (6x^2 - 2x + 1) dx \\ = 2x^3 - x^2 + x + C$$

$$(18) \int (x^2 - 9x + 9) dx \\ = \frac{1}{3}x^3 - \frac{9}{2}x^2 + 9x + C$$

$$(19) \int (-4x^2 + 8x - 1) dx \\ = -\frac{4}{3}x^3 + 4x^2 - x + C$$

$$(20) \int (3x^2 - 2x + 5) dx \\ = x^3 - x^2 + 5x + C$$

$$(21) \int (-8x^2 + 9x + 9) dx \\ = -\frac{8}{3}x^3 + \frac{9}{2}x^2 + 9x + C$$

$$(22) \int (-x^2 + 3x + 3) dx \\ = -\frac{1}{3}x^3 + \frac{3}{2}x^2 + 3x + C$$

$$(23) \int (-7x^2 + 3x + 9) dx \\ = -\frac{7}{3}x^3 + \frac{3}{2}x^2 + 9x + C$$

$$(24) \int (8x^2 - 4x - 5) dx \\ = \frac{8}{3}x^3 - 2x^2 - 5x + C$$

$$(25) \int (4x^2 - 9x + 6) dx \\ = \frac{4}{3}x^3 - \frac{9}{2}x^2 + 6x + C$$

$$(26) \int (-x^2 - 4x - 4) dx \\ = -\frac{1}{3}x^3 - 2x^2 - 4x + C$$

$$(27) \int (2x^2 - 3) dx \\ = \frac{2}{3}x^3 - 3x + C$$

$$(28) \int (-x^2 - 9x + 3) dx \\ = -\frac{1}{3}x^3 - \frac{9}{2}x^2 + 3x + C$$

$$(29) \int (2x^2 + 7x - 5) dx \\ = \frac{2}{3}x^3 + \frac{7}{2}x^2 - 5x + C$$

$$(30) \int (3x^2 - 7x - 9) dx \\ = x^3 - \frac{7}{2}x^2 - 9x + C$$

3.3.3 定積分

多項式の定積分。

$$(1) \int_{-5}^{-2} (4x^2 + 5x + 3) \, dx$$

$$(6) \int_{-1}^4 (5x^2 + 3x - 4) \, dx$$

$$(11) \int_{-4}^{-1} (2x^2 - 5x) \, dx$$

$$(2) \int_{-1}^3 (-3x^2 + 2) \, dx$$

$$(7) \int_0^5 (-2x^2 + 4x + 5) \, dx$$

$$(12) \int_{-2}^1 (-3x^2 + 2x - 4) \, dx$$

$$(3) \int_{-5}^0 (-3x^2 - 3) \, dx$$

$$(8) \int_{-4}^{-2} (2x^2 - 2x - 2) \, dx$$

$$(13) \int_{-5}^4 (2x^2 + 5x + 1) \, dx$$

$$(4) \int_{-1}^5 (2x^2 + 1) \, dx$$

$$(9) \int_{-2}^{-1} (-x^2 + x - 4) \, dx$$

$$(14) \int_{-4}^2 (x^2 - 5) \, dx$$

$$(5) \int_{-4}^2 (4x^2 - 3x) \, dx$$

$$(10) \int_1^3 (-3x^2 + 2x - 3) \, dx$$

$$(15) \int_{-2}^3 (-x^2 - x - 2) \, dx$$

$$(1) \int_{-3}^{-2} (x^2 + 5x - 1) \, dx$$

$$(6) \int_{-2}^1 (-4x^2 - 4) \, dx$$

$$(11) \int_0^4 (-3x^2 - x + 5) \, dx$$

$$(2) \int_{-4}^{-3} (x^2 + 3x - 2) \, dx$$

$$(7) \int_{-1}^3 (x^2 + 3x - 3) \, dx$$

$$(12) \int_{-4}^{-1} (-4x^2 + x - 2) \, dx$$

$$(3) \int_{-3}^{-2} (-5x^2 + 3x - 4) \, dx$$

$$(8) \int_1^5 (-5x^2 - x + 4) \, dx$$

$$(13) \int_{-1}^0 (2x^2 - 5x + 1) \, dx$$

$$(4) \int_{-3}^{-2} (-3x^2 - 5x - 2) \, dx$$

$$(9) \int_{-4}^{-2} (x^2 - 5x - 4) \, dx$$

$$(14) \int_{-1}^3 (-2x^2 + 1) \, dx$$

$$(5) \int_{-4}^{-2} (x^2 - x - 3) \, dx$$

$$(10) \int_{-3}^{-2} (3x^2 + x - 5) \, dx$$

$$(15) \int_{-4}^0 (3x^2 + 3x - 5) \, dx$$

$$(1) \int_0^1 (-2x^2 - 5x - 4) \, dx$$

$$(6) \int_{-1}^1 (-4x^2 - 5x - 1) \, dx$$

$$(11) \int_{-5}^3 (3x^2 + 3x + 4) \, dx$$

$$(2) \int_{-2}^1 (-x^2 + 2x + 1) \, dx$$

$$(7) \int_{-5}^4 (-x^2 - 5x + 3) \, dx$$

$$(12) \int_{-2}^2 (-5x^2 + 4x + 1) \, dx$$

$$(3) \int_{-4}^{-3} (-5x^2 + 5x + 5) \, dx$$

$$(8) \int_{-2}^4 (x^2 + x + 4) \, dx$$

$$(13) \int_{-5}^5 (-2x^2 + 4x) \, dx$$

$$(4) \int_{-3}^3 (-x^2 - 4x + 4) \, dx$$

$$(9) \int_{-4}^0 (-2x^2 + 3x - 4) \, dx$$

$$(14) \int_{-3}^1 (-4x^2 - 2x) \, dx$$

$$(5) \int_{-2}^1 (-3x^2 - 3x - 3) \, dx$$

$$(10) \int_{-5}^{-1} (5x^2 + x - 4) \, dx$$

$$(15) \int_0^2 (-x^2 + x + 1) \, dx$$

$$(1) \int_{-1}^1 (-2x^2 - 5x + 4) \, dx$$

$$(6) \int_3^4 (-x^2 + 2) \, dx$$

$$(11) \int_{-5}^3 (-2x^2 + 2x - 3) \, dx$$

$$(2) \int_{-5}^{-4} (5x^2 - x - 5) \, dx$$

$$(7) \int_3^4 (-3x^2 - 5x) \, dx$$

$$(12) \int_{-2}^3 (-4x^2 + 2x - 5) \, dx$$

$$(3) \int_0^3 (-2x^2 - x) \, dx$$

$$(8) \int_{-1}^0 (-3x^2 + 2x + 2) \, dx$$

$$(13) \int_{-5}^3 (2x^2 + 3x - 2) \, dx$$

$$(4) \int_4^5 (2x^2 + 5x + 4) \, dx$$

$$(9) \int_{-3}^0 (-4x^2 - 5) \, dx$$

$$(14) \int_0^1 (-5x^2 + 2x - 1) \, dx$$

$$(5) \int_{-5}^1 (3x^2 - x - 2) \, dx$$

$$(10) \int_{-3}^5 (3x^2 + 4x - 2) \, dx$$

$$(15) \int_{-4}^4 (-x^2 - 3x + 4) \, dx$$

$$(1) \int_0^3 (-x^2 + 3x + 2) \, dx$$

$$(6) \int_0^4 (3x^2 - 4x - 2) \, dx$$

$$(11) \int_{-3}^{-2} (-4x^2 - 2) \, dx$$

$$(2) \int_0^2 (3x^2 + 2x + 1) \, dx$$

$$(7) \int_{-4}^5 (4x^2 + 3x - 5) \, dx$$

$$(12) \int_{-2}^{-1} (x^2 - 5x) \, dx$$

$$(3) \int_{-1}^2 (-3x^2 - 4x - 1) \, dx$$

$$(8) \int_{-5}^3 (-5x^2 - 2x + 2) \, dx$$

$$(13) \int_{-3}^{-2} (2x^2 - 2x + 3) \, dx$$

$$(4) \int_{-5}^{-2} (-x^2 + 5x + 4) \, dx$$

$$(9) \int_0^3 (4x^2 - x - 4) \, dx$$

$$(14) \int_{-2}^{-1} (-4x^2 - 5x) \, dx$$

$$(5) \int_2^4 (2x^2 - 5x + 2) \, dx$$

$$(10) \int_{-4}^3 (3x^2 - 2x + 5) \, dx$$

$$(15) \int_{-2}^1 (-4x^2 - 5x - 1) \, dx$$

$$(1) \int_1^4 (-x^2 + 2x - 4) \, dx$$

$$(6) \int_2^3 (-5x^2 - x + 1) \, dx$$

$$(11) \int_{-4}^{-2} (5x^2 + 2x + 2) \, dx$$

$$(2) \int_{-5}^{-2} (2x^2 - 3x - 4) \, dx$$

$$(7) \int_{-5}^5 (-3x^2 - 2x) \, dx$$

$$(12) \int_{-3}^{-1} (x^2 - 3x) \, dx$$

$$(3) \int_2^4 (x^2 + x) \, dx$$

$$(8) \int_{-4}^{-3} (-2x^2 - 4x + 5) \, dx$$

$$(13) \int_{-2}^3 (2x^2 + 3x + 5) \, dx$$

$$(4) \int_{-2}^1 (-5x^2 + 3x + 2) \, dx$$

$$(9) \int_{-5}^{-2} (-5x^2 + 5x - 3) \, dx$$

$$(14) \int_{-4}^0 (3x^2 - 4x - 1) \, dx$$

$$(5) \int_0^3 (4x^2 + 2x - 1) \, dx$$

$$(10) \int_{-5}^{-4} (3x^2 - 2) \, dx$$

$$(15) \int_{-4}^{-2} (5x^2 - 1) \, dx$$

$$(1) \int_{-3}^4 (2x^2 - 4x - 2) \, dx$$

$$(6) \int_{-4}^{-2} (5x^2 - 3x - 3) \, dx$$

$$(11) \int_{-4}^5 (4x^2 - 5x - 1) \, dx$$

$$(2) \int_0^2 (-3x^2 - x) \, dx$$

$$(7) \int_{-5}^0 (-4x^2 + 4x) \, dx$$

$$(12) \int_2^5 (3x^2 - 3x) \, dx$$

$$(3) \int_{-2}^{-1} (2x^2 + x - 1) \, dx$$

$$(8) \int_{-3}^1 (4x^2 + 4x + 1) \, dx$$

$$(13) \int_{-2}^5 (x^2 + 4x) \, dx$$

$$(4) \int_{-2}^1 (-5x^2 + x - 2) \, dx$$

$$(9) \int_{-2}^3 (-2x^2 + x + 5) \, dx$$

$$(14) \int_2^4 (2x^2 - 3x + 3) \, dx$$

$$(5) \int_{-5}^2 (2x^2 + x + 4) \, dx$$

$$(10) \int_0^1 (-x^2 + x - 1) \, dx$$

$$(15) \int_4^5 (-4x^2 + 3x + 3) \, dx$$

$$(1) \int_1^4 (4x^2 + 3x - 1) \, dx$$

$$(6) \int_{-4}^4 (-x^2 - 2x + 5) \, dx$$

$$(11) \int_0^1 (4x^2 - 3x) \, dx$$

$$(2) \int_1^2 (2x^2 - x - 3) \, dx$$

$$(7) \int_{-4}^1 (-3x^2 - 5x + 4) \, dx$$

$$(12) \int_2^4 (4x^2 - 3x + 3) \, dx$$

$$(3) \int_{-5}^{-4} (2x^2 + 2x + 4) \, dx$$

$$(8) \int_{-4}^{-2} (-2x^2 + x) \, dx$$

$$(13) \int_{-5}^3 (4x^2 - 5x - 5) \, dx$$

$$(4) \int_3^4 (-2x^2 - 4x + 1) \, dx$$

$$(9) \int_3^5 (x^2 - 4x - 4) \, dx$$

$$(14) \int_{-4}^1 (-5x^2 - 4x) \, dx$$

$$(5) \int_{-2}^2 (4x^2 - 3x - 5) \, dx$$

$$(10) \int_{-1}^3 (2x^2 + 3x + 5) \, dx$$

$$(15) \int_{-5}^0 (-3x^2 + 4x - 5) \, dx$$

$$(1) \int_{-4}^5 (-5x^2 + 4) \, dx$$

$$(6) \int_{-5}^5 (-3x^2 + x + 5) \, dx$$

$$(11) \int_{-3}^2 (3x^2 - x + 4) \, dx$$

$$(2) \int_{-2}^5 (3x^2 + 4x - 2) \, dx$$

$$(7) \int_{-5}^2 (-3x^2 - x) \, dx$$

$$(12) \int_{-4}^2 (4x^2 - 3x + 3) \, dx$$

$$(3) \int_{-5}^5 (2x^2 + 4x - 1) \, dx$$

$$(8) \int_{-3}^0 (-5x^2 - 3x - 1) \, dx$$

$$(13) \int_{-4}^3 (2x^2 - 4x) \, dx$$

$$(4) \int_{-5}^{-4} (5x^2 + 2x - 3) \, dx$$

$$(9) \int_0^3 (-2x^2 + 2x + 3) \, dx$$

$$(14) \int_{-4}^0 (-x^2 + 5) \, dx$$

$$(5) \int_3^5 (-x^2 + 5x + 2) \, dx$$

$$(10) \int_2^3 (-4x^2 + 5x + 1) \, dx$$

$$(15) \int_{-2}^0 (3x^2 - x) \, dx$$

$$(1) \int_1^5 (2x^2 + 2) \, dx$$

$$(6) \int_{-2}^0 (2x^2 + 2x) \, dx$$

$$(11) \int_{-3}^1 (-3x^2 - 2x - 4) \, dx$$

$$(2) \int_{-5}^{-4} (-4x^2 + 3x - 3) \, dx$$

$$(7) \int_{-2}^2 (-3x^2 + 3x + 2) \, dx$$

$$(12) \int_{-1}^0 (x^2 + 3x - 1) \, dx$$

$$(3) \int_{-3}^{-2} (2x^2 - 4x + 5) \, dx$$

$$(8) \int_{-5}^{-4} (x^2 + 4x + 3) \, dx$$

$$(13) \int_{-4}^{-3} (5x^2 + x + 1) \, dx$$

$$(4) \int_{-5}^4 (-3x^2 + 2x - 3) \, dx$$

$$(9) \int_2^5 (5x^2 - 4x - 2) \, dx$$

$$(14) \int_4^5 (-4x^2 - 5x) \, dx$$

$$(5) \int_{-2}^5 (-2x^2 + 2x + 2) \, dx$$

$$(10) \int_{-2}^5 3x^2 \, dx$$

$$(15) \int_4^5 (5x^2 + x - 4) \, dx$$

$$(1) \int_{-5}^{-2} (-5x^2 + x + 1) \, dx$$

$$(6) \int_2^5 (-5x^2 - 4x + 5) \, dx$$

$$(11) \int_{-5}^2 (x^2 + 2x + 4) \, dx$$

$$(2) \int_{-2}^0 (3x^2 + 2x - 3) \, dx$$

$$(7) \int_0^4 (-5x^2 + 5x + 1) \, dx$$

$$(12) \int_{-3}^4 (-5x^2 + 4x + 1) \, dx$$

$$(3) \int_{-4}^5 (3x^2 + 2x - 1) \, dx$$

$$(8) \int_{-5}^0 (x^2 - 2x + 4) \, dx$$

$$(13) \int_{-2}^1 (x^2 - x - 3) \, dx$$

$$(4) \int_{-3}^3 (-5x^2 + 2x + 3) \, dx$$

$$(9) \int_0^1 (4x^2 + x - 3) \, dx$$

$$(14) \int_1^4 5x^2 \, dx$$

$$(5) \int_{-5}^4 (3x^2 - 2x + 5) \, dx$$

$$(10) \int_{-4}^{-3} (4x^2 - 4x - 2) \, dx$$

$$(15) \int_{-1}^5 (-5x^2 + 3x - 5) \, dx$$

$$(1) \int_{-5}^3 (-5x^2 - 5x - 2) \, dx$$

$$(6) \int_1^2 (x^2 - x - 5) \, dx$$

$$(11) \int_{-5}^5 (4x^2 - 2x + 1) \, dx$$

$$(2) \int_0^3 (4x^2 + x + 4) \, dx$$

$$(7) \int_{-2}^{-1} (-4x^2 + 4x - 4) \, dx$$

$$(12) \int_{-1}^5 (2x^2 - x + 2) \, dx$$

$$(3) \int_{-2}^1 (3x^2 + 2x - 2) \, dx$$

$$(8) \int_{-5}^{-4} (-2x^2 - x + 4) \, dx$$

$$(13) \int_0^3 (-5x^2 + 5x - 3) \, dx$$

$$(4) \int_{-2}^2 (2x^2 - 3x) \, dx$$

$$(9) \int_{-2}^3 (-4x^2 - x + 3) \, dx$$

$$(14) \int_3^4 (5x^2 + x + 5) \, dx$$

$$(5) \int_{-4}^{-1} (5x^2 + 4x - 4) \, dx$$

$$(10) \int_2^4 (-2x^2 + 2x - 5) \, dx$$

$$(15) \int_{-3}^5 (-x^2 - x - 5) \, dx$$

$$(1) \int_{-2}^5 (-3x^2 + 4x - 4) \, dx$$

$$(6) \int_{-2}^4 (5x^2 + 4x - 2) \, dx$$

$$(11) \int_{-4}^{-3} (-4x^2 - 2x + 2) \, dx$$

$$(2) \int_{-3}^{-1} (-x^2 - 2) \, dx$$

$$(7) \int_{-5}^{-2} (2x^2 + x + 1) \, dx$$

$$(12) \int_{-5}^{-2} (-5x^2 - 5x - 1) \, dx$$

$$(3) \int_{-5}^1 (4x^2 + 4x + 4) \, dx$$

$$(8) \int_{-3}^2 (-5x^2 - 2x) \, dx$$

$$(13) \int_1^2 (-3x^2 + 5x - 2) \, dx$$

$$(4) \int_0^1 (-2x^2 + 1) \, dx$$

$$(9) \int_{-3}^0 (-5x^2 - 4x - 1) \, dx$$

$$(14) \int_{-3}^5 (3x^2 - 5x + 4) \, dx$$

$$(5) \int_{-3}^4 (-2x^2) \, dx$$

$$(10) \int_{-1}^2 (-x^2 - 5x + 2) \, dx$$

$$(15) \int_{-3}^0 (x^2 - 4x + 3) \, dx$$

$$(1) \int_{-4}^3 (-2x^2 - 5x) \, dx$$

$$(6) \int_3^4 (-3x^2 - 2x - 3) \, dx$$

$$(11) \int_{-5}^3 (-5x^2 - 4x - 1) \, dx$$

$$(2) \int_3^5 (5x^2 - 4x + 2) \, dx$$

$$(7) \int_1^2 (-3x^2 + 3x + 4) \, dx$$

$$(12) \int_{-3}^0 (3x^2 + 3x) \, dx$$

$$(3) \int_{-1}^0 (x^2 - 2x - 3) \, dx$$

$$(8) \int_{-2}^5 (-4x^2 - x - 3) \, dx$$

$$(13) \int_{-1}^3 (-x^2 + x - 2) \, dx$$

$$(4) \int_{-3}^3 (x^2 - 4x + 1) \, dx$$

$$(9) \int_3^4 (-x^2 - 3x + 4) \, dx$$

$$(14) \int_3^5 (2x^2 - 3x - 5) \, dx$$

$$(5) \int_0^5 (3x^2 + x - 4) \, dx$$

$$(10) \int_2^4 (-4x^2) \, dx$$

$$(15) \int_{-3}^1 (4x^2 - x + 1) \, dx$$

$$(1) \int_{-5}^1 (-3x^2 - 5x + 1) \, dx$$

$$(6) \int_{-3}^{-2} (-5x^2 + 4x - 5) \, dx$$

$$(11) \int_0^3 (4x^2 + 4x - 4) \, dx$$

$$(2) \int_{-1}^3 (2x^2 - 2x + 4) \, dx$$

$$(7) \int_{-1}^5 (-5x^2 - 3x - 3) \, dx$$

$$(12) \int_0^2 (4x^2 + 4x + 3) \, dx$$

$$(3) \int_{-1}^1 (-3x^2 - 3x - 2) \, dx$$

$$(8) \int_{-5}^2 (-4x^2 + 5x + 3) \, dx$$

$$(13) \int_2^3 (5x^2 - 2x + 4) \, dx$$

$$(4) \int_{-2}^0 (-3x^2 - 2x + 1) \, dx$$

$$(9) \int_0^3 (2x^2 + x) \, dx$$

$$(14) \int_{-5}^3 (-x^2 - 5x - 3) \, dx$$

$$(5) \int_{-4}^1 (4x^2 + 5x - 4) \, dx$$

$$(10) \int_{-1}^2 (-4x^2 - 2x + 5) \, dx$$

$$(15) \int_2^4 (5x^2 + 4x - 4) \, dx$$

$$(1) \int_0^4 (-3x^2 - 4x) \, dx$$

$$(6) \int_0^1 (2x^2 + x + 4) \, dx$$

$$(11) \int_{-3}^1 (x^2 - 2x - 5) \, dx$$

$$(2) \int_{-4}^{-3} (-4x^2 - 3x + 2) \, dx$$

$$(7) \int_{-4}^3 (2x^2 + x - 2) \, dx$$

$$(12) \int_{-4}^0 (x^2 - x + 4) \, dx$$

$$(3) \int_{-5}^5 (-3x^2 + 2x - 3) \, dx$$

$$(8) \int_{-5}^1 (-3x^2 + 2x - 4) \, dx$$

$$(13) \int_{-4}^4 (3x^2 + 4x + 4) \, dx$$

$$(4) \int_{-2}^5 (-2x^2 - 2x - 5) \, dx$$

$$(9) \int_{-4}^3 (-4x^2 - 2x + 5) \, dx$$

$$(14) \int_0^4 (-4x^2 + 2x + 1) \, dx$$

$$(5) \int_{-1}^4 (4x^2 + 4x - 5) \, dx$$

$$(10) \int_{-2}^3 (x^2 + 3x - 2) \, dx$$

$$(15) \int_{-1}^1 (-2x^2 + 5x + 5) \, dx$$

$$(1) \int_{-3}^{-2} (-2x^2 + 5x + 4) \, dx$$

$$(6) \int_0^1 (5x^2 - x - 2) \, dx$$

$$(11) \int_1^2 (2x^2 - 3x - 5) \, dx$$

$$(2) \int_4^5 (2x^2 - 2x - 1) \, dx$$

$$(7) \int_{-4}^{-1} (-x^2 - 2x - 4) \, dx$$

$$(12) \int_{-3}^{-2} (-2x^2 + 5x + 3) \, dx$$

$$(3) \int_4^5 (-5x^2 - 5x + 5) \, dx$$

$$(8) \int_{-4}^4 (3x^2 + 3x) \, dx$$

$$(13) \int_3^4 (x^2 + 4x + 5) \, dx$$

$$(4) \int_0^3 (4x^2 + x - 2) \, dx$$

$$(9) \int_{-4}^{-3} (2x^2 + 2x + 5) \, dx$$

$$(14) \int_{-5}^1 (-5x^2 + 3x + 2) \, dx$$

$$(5) \int_2^3 (2x^2 - 5x - 5) \, dx$$

$$(10) \int_2^5 (x^2 - 2x - 5) \, dx$$

$$(15) \int_{-5}^2 (-3x^2 - 3x - 3) \, dx$$

$$(1) \int_{-3}^3 (5x^2 - 1) \, dx$$

$$(6) \int_{-5}^2 (x^2 - 4x + 4) \, dx$$

$$(11) \int_2^5 (2x^2 + x - 2) \, dx$$

$$(2) \int_1^2 (2x^2 - 3x + 1) \, dx$$

$$(7) \int_{-5}^3 (-x^2 - 4) \, dx$$

$$(12) \int_1^2 (2x^2 + 4x - 4) \, dx$$

$$(3) \int_{-1}^0 (2x^2 - 1) \, dx$$

$$(8) \int_1^4 (-4x^2 + 5x + 1) \, dx$$

$$(13) \int_{-4}^{-3} (-x^2 - 3x - 3) \, dx$$

$$(4) \int_0^4 (-5x^2 - 4x + 4) \, dx$$

$$(9) \int_{-5}^2 (2x^2 + 5) \, dx$$

$$(14) \int_{-4}^{-1} (-4x^2 + 4) \, dx$$

$$(5) \int_{-1}^1 (-4x^2 + 3x - 3) \, dx$$

$$(10) \int_{-2}^2 (-x^2 - 2x - 1) \, dx$$

$$(15) \int_1^2 (5x^2 - 4x - 3) \, dx$$

$$(1) \int_{-1}^5 (3x^2 + 5x + 4) \, dx$$

$$(6) \int_{-2}^{-1} (x^2 + 3x + 3) \, dx$$

$$(11) \int_2^3 (5x^2 + 3x - 3) \, dx$$

$$(2) \int_{-4}^2 (2x^2 - 4) \, dx$$

$$(7) \int_{-5}^1 (-x^2 - 4x) \, dx$$

$$(12) \int_{-5}^{-4} (-3x^2 + 3) \, dx$$

$$(3) \int_{-4}^3 (x^2 - 3x - 1) \, dx$$

$$(8) \int_{-4}^4 (-4x^2 + 5x + 5) \, dx$$

$$(13) \int_{-1}^5 (5x^2 - 4x + 3) \, dx$$

$$(4) \int_{-2}^{-1} (3x^2 - x + 1) \, dx$$

$$(9) \int_0^2 (-x^2 - 4x + 5) \, dx$$

$$(14) \int_0^3 (x^2 + 3x + 4) \, dx$$

$$(5) \int_{-4}^2 (4x^2 - 1) \, dx$$

$$(10) \int_2^3 (x^2 + 4x - 2) \, dx$$

$$(15) \int_{-1}^1 (5x^2 + x + 5) \, dx$$

$$(1) \int_{-2}^4 (2x^2 - 4x + 5) \, dx$$

$$(6) \int_{-4}^0 (-5x^2 - 3x + 4) \, dx$$

$$(11) \int_0^5 (-3x^2 - 3x + 2) \, dx$$

$$(2) \int_1^3 (-2x^2 - x - 2) \, dx$$

$$(7) \int_4^5 (-5x^2 - 2x - 1) \, dx$$

$$(12) \int_{-3}^0 (-x^2 + 5x - 1) \, dx$$

$$(3) \int_{-4}^{-3} (-3x^2 - 5) \, dx$$

$$(8) \int_{-3}^1 (-3x^2 - 4x - 1) \, dx$$

$$(13) \int_{-4}^{-3} (-5x^2 + 1) \, dx$$

$$(4) \int_0^3 (-2x^2 - 3x + 2) \, dx$$

$$(9) \int_{-5}^4 (4x^2 - 2x) \, dx$$

$$(14) \int_{-1}^4 (4x^2 + 4x + 5) \, dx$$

$$(5) \int_0^4 (x^2 + 5x - 1) \, dx$$

$$(10) \int_{-3}^3 (3x^2 + x + 3) \, dx$$

$$(15) \int_{-4}^3 (5x^2 - 4x - 2) \, dx$$

$$(1) \int_{-5}^{-2} (4x^2 + 5x + 3) \, dx \\ = 105$$

$$(6) \int_{-1}^4 (5x^2 + 3x - 4) \, dx \\ = \frac{255}{2}$$

$$(11) \int_{-4}^{-1} (2x^2 - 5x) \, dx \\ = \frac{159}{2}$$

$$(2) \int_{-1}^3 (-3x^2 + 2) \, dx \\ = -\frac{80}{3}$$

$$(7) \int_0^5 (-2x^2 + 4x + 5) \, dx \\ = -\frac{175}{6}$$

$$(12) \int_{-2}^1 (-3x^2 + 2x - 4) \, dx \\ = -14$$

$$(3) \int_{-5}^0 (-3x^2 - 3) \, dx \\ = -\frac{255}{2}$$

$$(8) \int_{-4}^{-2} (2x^2 - 2x - 2) \, dx \\ = \frac{146}{3}$$

$$(13) \int_{-5}^4 (2x^2 + 5x + 1) \, dx \\ = 105$$

$$(4) \int_{-1}^5 (2x^2 + 1) \, dx \\ = 85$$

$$(9) \int_{-2}^{-1} (-x^2 + x - 4) \, dx \\ = -\frac{9}{2}$$

$$(14) \int_{-4}^2 (x^2 - 5) \, dx \\ = 19$$

$$(5) \int_{-4}^2 (4x^2 - 3x) \, dx \\ = 114$$

$$(10) \int_1^3 (-3x^2 + 2x - 3) \, dx \\ = -19$$

$$(15) \int_{-2}^3 (-x^2 - x - 2) \, dx \\ = -\frac{95}{6}$$

$$\begin{aligned}(1) \quad & \int_{-3}^{-2} (x^2 + 5x - 1) \, dx \\ &= -\frac{19}{3}\end{aligned}$$

$$\begin{aligned}(6) \quad & \int_{-2}^1 (-4x^2 - 4) \, dx \\ &= -14\end{aligned}$$

$$\begin{aligned}(11) \quad & \int_0^4 (-3x^2 - x + 5) \, dx \\ &= -\frac{206}{3}\end{aligned}$$

$$\begin{aligned}(2) \quad & \int_{-\frac{4}{3}}^{-3} (x^2 + 3x - 2) \, dx \\ &= \frac{3}{2}\end{aligned}$$

$$\begin{aligned}(7) \quad & \int_{-\frac{1}{58}}^3 (x^2 + 3x - 3) \, dx \\ &= \frac{3}{3}\end{aligned}$$

$$\begin{aligned}(12) \quad & \int_{-4}^{-1} (-4x^2 + x - 2) \, dx \\ &= -\frac{185}{2}\end{aligned}$$

$$\begin{aligned}(3) \quad & \int_{-3}^{-2} (-5x^2 + 3x - 4) \, dx \\ &= -\frac{239}{6}\end{aligned}$$

$$\begin{aligned}(8) \quad & \int_1^5 (-5x^2 - x + 4) \, dx \\ &= -216\end{aligned}$$

$$\begin{aligned}(13) \quad & \int_{-1}^0 (2x^2 - 5x + 1) \, dx \\ &= \frac{10}{3}\end{aligned}$$

$$\begin{aligned}(4) \quad & \int_{-3}^{-2} (-3x^2 - 5x - 2) \, dx \\ &= -\frac{41}{6}\end{aligned}$$

$$\begin{aligned}(9) \quad & \int_{-4}^{-2} (x^2 - 5x - 4) \, dx \\ &= \frac{142}{3}\end{aligned}$$

$$\begin{aligned}(14) \quad & \int_{-1}^3 (-2x^2 + 1) \, dx \\ &= -18\end{aligned}$$

$$\begin{aligned}(5) \quad & \int_{-\frac{4}{71}}^{-2} (x^2 - x - 3) \, dx \\ &= \frac{71}{3}\end{aligned}$$

$$\begin{aligned}(10) \quad & \int_{-\frac{3}{47}}^{-2} (3x^2 + x - 5) \, dx \\ &= \frac{47}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & \int_{-\frac{4}{110}}^0 (3x^2 + 3x - 5) \, dx \\ &= \frac{110}{3}\end{aligned}$$

$$(1) \int_0^1 (-2x^2 - 5x - 4) \, dx \\ = -\frac{23}{6}$$

$$(6) \int_{-1}^1 (-4x^2 - 5x - 1) \, dx \\ = -3$$

$$(11) \int_{-5}^3 (3x^2 + 3x + 4) \, dx \\ = \frac{400}{3}$$

$$(2) \int_{-2}^1 (-x^2 + 2x + 1) \, dx \\ = -\frac{11}{2}$$

$$(7) \int_{-5}^4 (-x^2 - 5x + 3) \, dx \\ = -36$$

$$(12) \int_{-2}^2 (-5x^2 + 4x + 1) \, dx \\ = -26$$

$$(3) \int_{-4}^{-3} (-5x^2 + 5x + 5) \, dx \\ = -\frac{235}{3}$$

$$(8) \int_{-2}^4 (x^2 + x + 4) \, dx \\ = 34$$

$$(13) \int_{-5}^5 (-2x^2 + 4x) \, dx \\ = -\frac{500}{3}$$

$$(4) \int_{-3}^3 (-x^2 - 4x + 4) \, dx \\ = -14$$

$$(9) \int_{-4}^0 (-2x^2 + 3x - 4) \, dx \\ = -\frac{208}{3}$$

$$(14) \int_{-3}^1 (-4x^2 - 2x) \, dx \\ = -\frac{88}{3}$$

$$(5) \int_{-2}^1 (-3x^2 - 3x - 3) \, dx \\ = -6$$

$$(10) \int_{-5}^{-1} (5x^2 + x - 4) \, dx \\ = 192$$

$$(15) \int_0^2 (-x^2 + x + 1) \, dx \\ = -\frac{1}{3}$$

$$(1) \int_{-1}^1 (-2x^2 - 5x + 4) \, dx \\ = 0$$

$$(6) \int_3^4 (-x^2 + 2) \, dx \\ = -12$$

$$(11) \int_{-5}^3 (-2x^2 + 2x - 3) \, dx \\ = -\frac{364}{3}$$

$$(2) \int_{-5}^{-4} (5x^2 - x - 5) \, dx \\ = \frac{316}{3}$$

$$(7) \int_3^4 (-3x^2 - 5x) \, dx \\ = -\frac{109}{2}$$

$$(12) \int_{-2}^3 (-4x^2 + 2x - 5) \, dx \\ = -\frac{275}{6}$$

$$(3) \int_0^3 (-2x^2 - x) \, dx \\ = -\frac{45}{2}$$

$$(8) \int_{-1}^0 (-3x^2 + 2x + 2) \, dx \\ = -\frac{5}{3}$$

$$(13) \int_{-5}^3 (2x^2 + 3x - 2) \, dx \\ = \frac{224}{3}$$

$$(4) \int_4^5 (2x^2 + 5x + 4) \, dx \\ = \frac{383}{6}$$

$$(9) \int_{-3}^0 (-4x^2 - 5) \, dx \\ = -\frac{77}{2}$$

$$(14) \int_0^1 (-5x^2 + 2x - 1) \, dx \\ = -\frac{5}{6}$$

$$(5) \int_{-5}^1 (3x^2 - x - 2) \, dx \\ = 136$$

$$(10) \int_{-3}^5 (3x^2 + 4x - 2) \, dx \\ = \frac{544}{3}$$

$$(15) \int_{-4}^4 (-x^2 - 3x + 4) \, dx \\ = -\frac{112}{3}$$

$$\begin{aligned}(1) \quad & \int_0^3 (-x^2 + 3x + 2) \, dx \\ &= \frac{11}{2}\end{aligned}$$

$$\begin{aligned}(6) \quad & \int_0^4 (3x^2 - 4x - 2) \, dx \\ &= \frac{92}{3}\end{aligned}$$

$$\begin{aligned}(11) \quad & \int_{-3}^{-2} (-4x^2 - 2) \, dx \\ &= -\frac{77}{3}\end{aligned}$$

$$\begin{aligned}(2) \quad & \int_0^2 (3x^2 + 2x + 1) \, dx \\ &= \frac{37}{3}\end{aligned}$$

$$\begin{aligned}(7) \quad & \int_{-4}^5 (4x^2 + 3x - 5) \, dx \\ &= 258\end{aligned}$$

$$\begin{aligned}(12) \quad & \int_{-2}^{-1} (x^2 - 5x) \, dx \\ &= \frac{59}{6}\end{aligned}$$

$$\begin{aligned}(3) \quad & \int_{-1}^2 (-3x^2 - 4x - 1) \, dx \\ &= -\frac{31}{2}\end{aligned}$$

$$\begin{aligned}(8) \quad & \int_{-5}^3 (-5x^2 - 2x + 2) \, dx \\ &= -\frac{704}{3}\end{aligned}$$

$$\begin{aligned}(13) \quad & \int_{-3}^{-2} (2x^2 - 2x + 3) \, dx \\ &= \frac{109}{6}\end{aligned}$$

$$\begin{aligned}(4) \quad & \int_{-5}^{-2} (-x^2 + 5x + 4) \, dx \\ &= -\frac{179}{2}\end{aligned}$$

$$\begin{aligned}(9) \quad & \int_0^3 (4x^2 - x - 4) \, dx \\ &= \frac{59}{2}\end{aligned}$$

$$\begin{aligned}(14) \quad & \int_{-2}^{-1} (-4x^2 - 5x) \, dx \\ &= -\frac{11}{6}\end{aligned}$$

$$\begin{aligned}(5) \quad & \int_2^4 (2x^2 - 5x + 2) \, dx \\ &= 8\end{aligned}$$

$$\begin{aligned}(10) \quad & \int_{-4}^3 (3x^2 - 2x + 5) \, dx \\ &= \frac{623}{6}\end{aligned}$$

$$\begin{aligned}(15) \quad & \int_{-2}^1 (-4x^2 - 5x - 1) \, dx \\ &= -5\end{aligned}$$

$$(1) \int_1^4 (-x^2 + 2x - 4) \, dx \\ = -8$$

$$(6) \int_2^3 (-5x^2 - x + 1) \, dx \\ = -34$$

$$(11) \int_{-4}^{-2} (5x^2 + 2x + 2) \, dx \\ = 82$$

$$(2) \int_{-5}^{-2} (2x^2 - 3x - 4) \, dx \\ = \frac{215}{2}$$

$$(7) \int_{-5}^5 (-3x^2 - 2x) \, dx \\ = -250$$

$$(12) \int_{-3}^{-1} (x^2 - 3x) \, dx \\ = \frac{62}{3}$$

$$(3) \int_2^4 (x^2 + x) \, dx \\ = \frac{74}{3}$$

$$(8) \int_{-4}^{-3} (-2x^2 - 4x + 5) \, dx \\ = -\frac{59}{6}$$

$$(13) \int_{-2}^3 (2x^2 + 3x + 5) \, dx \\ = 35$$

$$(4) \int_{-2}^1 (-5x^2 + 3x + 2) \, dx \\ = -\frac{37}{2}$$

$$(9) \int_{-5}^{-2} (-5x^2 + 5x - 3) \, dx \\ = -249$$

$$(14) \int_{-4}^0 (3x^2 - 4x - 1) \, dx \\ = \frac{286}{3}$$

$$(5) \int_0^3 (4x^2 + 2x - 1) \, dx \\ = \frac{89}{2}$$

$$(10) \int_{-5}^{-4} (3x^2 - 2) \, dx \\ = \frac{182}{3}$$

$$(15) \int_{-4}^{-2} (5x^2 - 1) \, dx \\ = 93$$

$$\begin{aligned} (1) \int_{-3}^4 (2x^2 - 4x - 2) \, dx \\ = \frac{133}{3} \end{aligned}$$

$$\begin{aligned} (6) \int_{-4}^{-2} (5x^2 - 3x - 3) \, dx \\ = \frac{331}{3} \end{aligned}$$

$$\begin{aligned} (11) \int_{-4}^5 (4x^2 - 5x - 1) \, dx \\ = 228 \end{aligned}$$

$$\begin{aligned} (2) \int_0^2 (-3x^2 - x) \, dx \\ = -10 \end{aligned}$$

$$\begin{aligned} (7) \int_{-5}^0 (-4x^2 + 4x) \, dx \\ = -\frac{650}{3} \end{aligned}$$

$$\begin{aligned} (12) \int_2^5 (3x^2 - 3x) \, dx \\ = \frac{171}{2} \end{aligned}$$

$$\begin{aligned} (3) \int_{-2}^{-1} (2x^2 + x - 1) \, dx \\ = \frac{3}{3} \end{aligned}$$

$$\begin{aligned} (8) \int_{-3}^1 (4x^2 + 4x + 1) \, dx \\ = \frac{22}{22} \end{aligned}$$

$$\begin{aligned} (13) \int_{-2}^5 (x^2 + 4x) \, dx \\ = \frac{259}{3} \end{aligned}$$

$$\begin{aligned} (4) \int_{-2}^1 (-5x^2 + x - 2) \, dx \\ = -\frac{35}{2} \end{aligned}$$

$$\begin{aligned} (9) \int_{-2}^3 (-2x^2 + x + 5) \, dx \\ = -\frac{50}{3} \end{aligned}$$

$$\begin{aligned} (14) \int_2^4 (2x^2 - 3x + 3) \, dx \\ = \frac{61}{3} \end{aligned}$$

$$\begin{aligned} (5) \int_{-5}^2 (2x^2 + x + 4) \, dx \\ = \frac{497}{6} \end{aligned}$$

$$\begin{aligned} (10) \int_0^1 (-x^2 + x - 1) \, dx \\ = 0 \end{aligned}$$

$$\begin{aligned} (15) \int_4^5 (-4x^2 + 3x + 3) \, dx \\ = -\frac{202}{3} \end{aligned}$$

$$(1) \int_1^4 (4x^2 + 3x - 1) \, dx \\ = 106$$

$$(6) \int_{-4}^4 (-x^2 - 2x + 5) \, dx \\ = -36$$

$$(11) \int_0^1 (4x^2 - 3x) \, dx \\ = -\frac{1}{6}$$

$$(2) \int_1^2 (2x^2 - x - 3) \, dx \\ = \frac{8}{3}$$

$$(7) \int_{-4}^1 (-3x^2 - 5x + 4) \, dx \\ = -\frac{145}{6}$$

$$(12) \int_2^4 (4x^2 - 3x + 3) \, dx \\ = \frac{173}{3}$$

$$(3) \int_{-5}^{-4} (2x^2 + 2x + 4) \, dx \\ = \frac{97}{3}$$

$$(8) \int_{-4}^{-2} (-2x^2 + x) \, dx \\ = -\frac{130}{3}$$

$$(13) \int_{-5}^3 (4x^2 - 5x - 5) \, dx \\ = 236$$

$$(4) \int_3^4 (-2x^2 - 4x + 1) \, dx \\ = -\frac{77}{2}$$

$$(9) \int_3^5 (x^2 - 4x - 4) \, dx \\ = -\frac{2}{3}$$

$$(14) \int_{-4}^1 (-5x^2 - 4x) \, dx \\ = -\frac{235}{3}$$

$$(5) \int_{-2}^2 (4x^2 - 3x - 5) \, dx \\ = 18$$

$$(10) \int_{-1}^3 (2x^2 + 3x + 5) \, dx \\ = 34$$

$$(15) \int_{-5}^0 (-3x^2 + 4x - 5) \, dx \\ = -\frac{1075}{6}$$

$$(1) \int_{-4}^5 (-5x^2 + 4) \, dx \\ = -309$$

$$(6) \int_{-5}^5 (-3x^2 + x + 5) \, dx \\ = -\frac{725}{3}$$

$$(11) \int_{-3}^2 (3x^2 - x + 4) \, dx \\ = \frac{245}{6}$$

$$(2) \int_{-2}^5 (3x^2 + 4x - 2) \, dx \\ = \frac{518}{3}$$

$$(7) \int_{-5}^2 (-3x^2 - x) \, dx \\ = -\frac{245}{2}$$

$$(12) \int_{-4}^2 (4x^2 - 3x + 3) \, dx \\ = 117$$

$$(3) \int_{-5}^5 (2x^2 + 4x - 1) \, dx \\ = 165$$

$$(8) \int_{-3}^0 (-5x^2 - 3x - 1) \, dx \\ = -32$$

$$(13) \int_{-4}^3 (2x^2 - 4x) \, dx \\ = \frac{224}{3}$$

$$(4) \int_{-5}^{-4} (5x^2 + 2x - 3) \, dx \\ = \frac{553}{6}$$

$$(9) \int_0^3 (-2x^2 + 2x + 3) \, dx \\ = -\frac{15}{2}$$

$$(14) \int_{-4}^0 (-x^2 + 5) \, dx \\ = -18$$

$$(5) \int_3^5 (-x^2 + 5x + 2) \, dx \\ = 8$$

$$(10) \int_2^3 (-4x^2 + 5x + 1) \, dx \\ = -\frac{38}{3}$$

$$(15) \int_{-2}^0 (3x^2 - x) \, dx \\ = 10$$

$$(1) \int_1^5 (2x^2 + 2) \, dx \\ = 84$$

$$(6) \int_{-\frac{2}{4}}^0 (2x^2 + 2x) \, dx \\ = \frac{4}{3}$$

$$(11) \int_{-3}^1 (-3x^2 - 2x - 4) \, dx \\ = -\frac{68}{3}$$

$$(2) \int_{-5}^{-4} (-4x^2 + 3x - 3) \, dx \\ = -\frac{286}{3}$$

$$(7) \int_{-2}^2 (-3x^2 + 3x + 2) \, dx \\ = -\frac{44}{3}$$

$$(12) \int_{-1}^0 (x^2 + 3x - 1) \, dx \\ = -\frac{4}{3}$$

$$(3) \int_{-\frac{3}{47}}^{-2} (2x^2 - 4x + 5) \, dx \\ = \frac{47}{2}$$

$$(8) \int_{-\frac{5}{17}}^{-4} (x^2 + 4x + 3) \, dx \\ = \frac{6}{17}$$

$$(13) \int_{-\frac{4}{175}}^{-3} (5x^2 + x + 1) \, dx \\ = \frac{175}{3}$$

$$(4) \int_{-5}^4 (-3x^2 + 2x - 3) \, dx \\ = -\frac{405}{2}$$

$$(9) \int_2^5 (5x^2 - 4x - 2) \, dx \\ = 152$$

$$(14) \int_4^5 (-4x^2 - 5x) \, dx \\ = -\frac{623}{6}$$

$$(5) \int_{-2}^5 (-2x^2 + 2x + 2) \, dx \\ = -\frac{196}{3}$$

$$(10) \int_{-\frac{2}{133}}^5 3x^2 \, dx \\ = 133$$

$$(15) \int_4^5 (5x^2 + x - 4) \, dx \\ = \frac{211}{2}$$

$$(1) \int_{-5}^{-2} (-5x^2 + x + 1) \, dx \\ = -205$$

$$(6) \int_2^5 (-5x^2 - 4x + 5) \, dx \\ = -\frac{469}{2}$$

$$(11) \int_{-5}^2 (x^2 + 2x + 4) \, dx \\ = 28$$

$$(2) \int_{-2}^0 (3x^2 + 2x - 3) \, dx \\ = 3$$

$$(7) \int_0^4 (-5x^2 + 5x + 1) \, dx \\ = -66$$

$$(12) \int_{-3}^4 (-5x^2 + 4x + 1) \, dx \\ = -\frac{273}{2}$$

$$(3) \int_{-4}^5 (3x^2 + 2x - 1) \, dx \\ = \frac{393}{2}$$

$$(8) \int_{-5}^0 (x^2 - 2x + 4) \, dx \\ = 70$$

$$(13) \int_{-2}^1 (x^2 - x - 3) \, dx \\ = 3$$

$$(4) \int_{-3}^3 (-5x^2 + 2x + 3) \, dx \\ = -87$$

$$(9) \int_0^1 (4x^2 + x - 3) \, dx \\ = \frac{4}{3}$$

$$(14) \int_1^4 5x^2 \, dx \\ = 105$$

$$(5) \int_{-5}^4 (3x^2 - 2x + 5) \, dx \\ = \frac{411}{2}$$

$$(10) \int_{-4}^{-3} (4x^2 - 4x - 2) \, dx \\ = 63$$

$$(15) \int_{-1}^5 (-5x^2 + 3x - 5) \, dx \\ = -179$$

$$(1) \int_{-5}^3 (-5x^2 - 5x - 2) \, dx \\ = -216$$

$$(6) \int_1^2 (x^2 - x - 5) \, dx \\ = 0$$

$$(11) \int_{-5}^5 (4x^2 - 2x + 1) \, dx \\ = 335$$

$$(2) \int_0^3 (4x^2 + x + 4) \, dx \\ = \frac{85}{2}$$

$$(7) \int_{-2}^{-1} (-4x^2 + 4x - 4) \, dx \\ = -16$$

$$(12) \int_{-1}^5 (2x^2 - x + 2) \, dx \\ = 74$$

$$(3) \int_{-2}^1 (3x^2 + 2x - 2) \, dx \\ = \frac{5}{5}$$

$$(8) \int_{-5}^{-4} (-2x^2 - x + 4) \, dx \\ = -\frac{71}{2}$$

$$(13) \int_0^3 (-5x^2 + 5x - 3) \, dx \\ = -24$$

$$(4) \int_{-2}^2 (2x^2 - 3x) \, dx \\ = \frac{32}{3}$$

$$(9) \int_{-2}^3 (-4x^2 - x + 3) \, dx \\ = -\frac{140}{3}$$

$$(14) \int_3^4 (5x^2 + x + 5) \, dx \\ = 66$$

$$(5) \int_{-4}^{-1} (5x^2 + 4x - 4) \, dx \\ = 73$$

$$(10) \int_2^4 (-2x^2 + 2x - 5) \, dx \\ = -27$$

$$(15) \int_{-3}^5 (-x^2 - x - 5) \, dx \\ = -\frac{196}{3}$$

$$\begin{aligned} (1) \quad & \int_{-2}^5 (-3x^2 + 4x - 4) \, dx \\ &= -\frac{287}{3} \end{aligned}$$

$$(6) \quad \int_{-2}^4 (5x^2 + 4x - 2) \, dx = 142$$

$$(11) \quad \int_{-4}^{-3} (-4x^2 - 2x + 2) \, dx = -42$$

$$\begin{aligned} (2) \quad & \int_{-3}^{-1} (-x^2 - 2) \, dx \\ &= -\frac{28}{3} \end{aligned}$$

$$(7) \quad \int_{-5}^{-2} (2x^2 + x + 1) \, dx = 68$$

$$(12) \quad \int_{-5}^{-2} (-5x^2 - 5x - 1) \, dx = -143$$

$$\begin{aligned} (3) \quad & \int_{-5}^1 (4x^2 + 4x + 4) \, dx \\ &= 124 \end{aligned}$$

$$\begin{aligned} (8) \quad & \int_{-3}^2 (-5x^2 - 2x) \, dx \\ &= -\frac{160}{3} \end{aligned}$$

$$\begin{aligned} (13) \quad & \int_1^2 (-3x^2 + 5x - 2) \, dx \\ &= \frac{1}{6} \end{aligned}$$

$$\begin{aligned} (4) \quad & \int_0^1 (-2x^2 + 1) \, dx \\ &= -\frac{1}{2} \end{aligned}$$

$$\begin{aligned} (9) \quad & \int_{-3}^0 (-5x^2 - 4x - 1) \, dx \\ &= -\frac{55}{2} \end{aligned}$$

$$\begin{aligned} (14) \quad & \int_{-3}^5 (3x^2 - 5x + 4) \, dx \\ &= \frac{352}{3} \end{aligned}$$

$$\begin{aligned} (5) \quad & \int_{-3}^4 (-2x^2) \, dx \\ &= -\frac{182}{3} \end{aligned}$$

$$\begin{aligned} (10) \quad & \int_{-1}^2 (-x^2 - 5x + 2) \, dx \\ &= -\frac{19}{2} \end{aligned}$$

$$\begin{aligned} (15) \quad & \int_{-3}^0 (x^2 - 4x + 3) \, dx \\ &= \frac{57}{2} \end{aligned}$$

$$\begin{aligned}(1) \int_{-4}^3 (-2x^2 - 5x) \, dx \\ = -\frac{259}{6}\end{aligned}$$

$$\begin{aligned}(6) \int_3^4 (-3x^2 - 2x - 3) \, dx \\ = -\frac{89}{2}\end{aligned}$$

$$\begin{aligned}(11) \int_{-5}^3 (-5x^2 - 4x - 1) \, dx \\ = -\frac{668}{3}\end{aligned}$$

$$\begin{aligned}(2) \int_3^5 (5x^2 - 4x + 2) \, dx \\ = 132\end{aligned}$$

$$\begin{aligned}(7) \int_1^2 (-3x^2 + 3x + 4) \, dx \\ = -\frac{11}{6}\end{aligned}$$

$$\begin{aligned}(12) \int_{-3}^0 (3x^2 + 3x) \, dx \\ = \frac{27}{2}\end{aligned}$$

$$\begin{aligned}(3) \int_{-\frac{1}{5}}^0 (x^2 - 2x - 3) \, dx \\ = \frac{6}{5}\end{aligned}$$

$$\begin{aligned}(8) \int_{-2}^5 (-4x^2 - x - 3) \, dx \\ = -\frac{574}{3}\end{aligned}$$

$$\begin{aligned}(13) \int_{-1}^3 (-x^2 + x - 2) \, dx \\ = -\frac{20}{3}\end{aligned}$$

$$\begin{aligned}(4) \int_{-\frac{3}{19}}^3 (x^2 - 4x + 1) \, dx \\ = 19\end{aligned}$$

$$\begin{aligned}(9) \int_3^4 (-x^2 - 3x + 4) \, dx \\ = -\frac{133}{6}\end{aligned}$$

$$\begin{aligned}(14) \int_3^5 (2x^2 - 3x - 5) \, dx \\ = \frac{119}{3}\end{aligned}$$

$$\begin{aligned}(5) \int_0^5 (3x^2 + x - 4) \, dx \\ = \frac{805}{6}\end{aligned}$$

$$\begin{aligned}(10) \int_2^4 (-4x^2) \, dx \\ = -\frac{224}{3}\end{aligned}$$

$$\begin{aligned}(15) \int_{-\frac{3}{42}}^1 (4x^2 - x + 1) \, dx \\ = 42\end{aligned}$$

$$(1) \int_{-5}^1 (-3x^2 - 5x + 1) dx \\ = -65$$

$$(6) \int_{-3}^{-2} (-5x^2 + 4x - 5) dx \\ = -\frac{85}{2}$$

$$(11) \int_0^3 (4x^2 + 4x - 4) dx \\ = 52$$

$$(2) \int_{-1}^3 (2x^2 - 2x + 4) dx \\ = \frac{40}{3}$$

$$(7) \int_{-1}^5 (-5x^2 - 3x - 3) dx \\ = -249$$

$$(12) \int_0^2 (4x^2 + 4x + 3) dx \\ = \frac{59}{3}$$

$$(3) \int_{-1}^1 (-3x^2 - 3x - 2) dx \\ = -\frac{8}{3}$$

$$(8) \int_{-5}^2 (-4x^2 + 5x + 3) dx \\ = -\frac{679}{3}$$

$$(13) \int_2^3 (5x^2 - 2x + 4) dx \\ = \frac{82}{3}$$

$$(4) \int_{-2}^0 (-3x^2 - 2x + 1) dx \\ = -\frac{11}{3}$$

$$(9) \int_0^3 (2x^2 + x) dx \\ = \frac{45}{2}$$

$$(14) \int_{-5}^3 (-x^2 - 5x - 3) dx \\ = -\frac{44}{3}$$

$$(5) \int_{-4}^1 (4x^2 + 5x - 4) dx \\ = \frac{275}{6}$$

$$(10) \int_{-1}^2 (-4x^2 - 2x + 5) dx \\ = -\frac{25}{2}$$

$$(15) \int_2^4 (5x^2 + 4x - 4) dx \\ = 116$$

$$(1) \int_0^4 (-3x^2 - 4x) \, dx \\ = -96$$

$$(6) \int_0^1 (2x^2 + x + 4) \, dx \\ = \frac{11}{6}$$

$$(11) \int_{-3}^1 (x^2 - 2x - 5) \, dx \\ = 14$$

$$(2) \int_{-4}^{-3} (-4x^2 - 3x + 2) \, dx \\ = -\frac{77}{2}$$

$$(7) \int_{-4}^3 (2x^2 + x - 2) \, dx \\ = \frac{329}{6}$$

$$(12) \int_{-4}^0 (x^2 - x + 4) \, dx \\ = 32$$

$$(3) \int_{-5}^5 (-3x^2 + 2x - 3) \, dx \\ = -255$$

$$(8) \int_{-5}^1 (-3x^2 + 2x - 4) \, dx \\ = -154$$

$$(13) \int_{-4}^4 (3x^2 + 4x + 4) \, dx \\ = \frac{400}{3}$$

$$(4) \int_{-2}^5 (-2x^2 - 2x - 5) \, dx \\ = -\frac{231}{2}$$

$$(9) \int_{-4}^3 (-4x^2 - 2x + 5) \, dx \\ = -\frac{217}{2}$$

$$(14) \int_0^4 (-4x^2 + 2x + 1) \, dx \\ = -\frac{206}{3}$$

$$(5) \int_{-1}^4 (4x^2 + 4x - 5) \, dx \\ = \frac{225}{2}$$

$$(10) \int_{-2}^3 (x^2 + 3x - 2) \, dx \\ = \frac{35}{2}$$

$$(15) \int_{-1}^1 (-2x^2 + 5x + 5) \, dx \\ = \frac{1}{3}$$

$$\begin{aligned}(1) \int_{-3}^{-2} (-2x^2 + 5x + 4) \, dx \\ = -\frac{49}{2}\end{aligned}$$

$$\begin{aligned}(6) \int_0^1 (5x^2 - x - 2) \, dx \\ = \frac{5}{6}\end{aligned}$$

$$\begin{aligned}(11) \int_1^2 (2x^2 - 3x - 5) \, dx \\ = -\frac{2}{3}\end{aligned}$$

$$\begin{aligned}(2) \int_4^5 (2x^2 - 2x - 1) \, dx \\ = \frac{63}{2}\end{aligned}$$

$$\begin{aligned}(7) \int_{-4}^{-1} (-x^2 - 2x - 4) \, dx \\ = -8\end{aligned}$$

$$\begin{aligned}(12) \int_{-3}^{-2} (-2x^2 + 5x + 3) \, dx \\ = -\frac{74}{3}\end{aligned}$$

$$\begin{aligned}(3) \int_4^5 (-5x^2 - 5x + 5) \, dx \\ = -\frac{370}{3}\end{aligned}$$

$$\begin{aligned}(8) \int_{-4}^4 (3x^2 + 3x) \, dx \\ = 128\end{aligned}$$

$$\begin{aligned}(13) \int_3^4 (x^2 + 4x + 5) \, dx \\ = \frac{163}{6}\end{aligned}$$

$$\begin{aligned}(4) \int_0^3 (4x^2 + x - 2) \, dx \\ = \frac{79}{2}\end{aligned}$$

$$\begin{aligned}(9) \int_{-4}^{-3} (2x^2 + 2x + 5) \, dx \\ = \frac{37}{2}\end{aligned}$$

$$\begin{aligned}(14) \int_{-5}^1 (-5x^2 + 3x + 2) \, dx \\ = -244\end{aligned}$$

$$\begin{aligned}(5) \int_2^3 (2x^2 - 5x - 5) \, dx \\ = -\frac{2}{3}\end{aligned}$$

$$\begin{aligned}(10) \int_2^5 (x^2 - 2x - 5) \, dx \\ = \frac{31}{2}\end{aligned}$$

$$\begin{aligned}(15) \int_{-5}^2 (-3x^2 - 3x - 3) \, dx \\ = -105\end{aligned}$$

$$(1) \int_{-3}^3 (5x^2 - 1) \, dx \\ = 89$$

$$(6) \int_{-5}^2 (x^2 - 4x + 4) \, dx \\ = 91$$

$$(11) \int_2^5 (2x^2 + x - 2) \, dx \\ = \frac{175}{2}$$

$$(2) \int_1^2 (2x^2 - 3x + 1) \, dx \\ = \frac{1}{3}$$

$$(7) \int_{-5}^3 (-x^2 - 4) \, dx \\ = -56$$

$$(12) \int_1^2 (2x^2 + 4x - 4) \, dx \\ = 10$$

$$(3) \int_{-1}^0 (2x^2 - 1) \, dx \\ = \frac{1}{2}$$

$$(8) \int_1^4 (-4x^2 + 5x + 1) \, dx \\ = -46$$

$$(13) \int_{-4}^{-3} (-x^2 - 3x - 3) \, dx \\ = -\frac{7}{3}$$

$$(4) \int_0^4 (-5x^2 - 4x + 4) \, dx \\ = -136$$

$$(9) \int_{-5}^2 (2x^2 + 5) \, dx \\ = \frac{189}{2}$$

$$(14) \int_{-4}^{-1} (-4x^2 + 4) \, dx \\ = -82$$

$$(5) \int_{-1}^1 (-4x^2 + 3x - 3) \, dx \\ = -\frac{11}{3}$$

$$(10) \int_{-2}^2 (-x^2 - 2x - 1) \, dx \\ = -6$$

$$(15) \int_1^2 (5x^2 - 4x - 3) \, dx \\ = \frac{31}{6}$$

$$(1) \int_{-1}^5 (3x^2 + 5x + 4) \, dx \\ = 190$$

$$(6) \int_{-2}^{-1} (x^2 + 3x + 3) \, dx \\ = -\frac{5}{3}$$

$$(11) \int_2^3 (5x^2 + 3x - 3) \, dx \\ = \frac{116}{3}$$

$$(2) \int_{-4}^2 (2x^2 - 4) \, dx \\ = 44$$

$$(7) \int_{-5}^1 (-x^2 - 4x) \, dx \\ = 6$$

$$(12) \int_{-5}^{-4} (-3x^2 + 3) \, dx \\ = -\frac{121}{2}$$

$$(3) \int_{-4}^3 (x^2 - 3x - 1) \, dx \\ = \frac{119}{3}$$

$$(8) \int_{-4}^4 (-4x^2 + 5x + 5) \, dx \\ = -164$$

$$(13) \int_{-1}^5 (5x^2 - 4x + 3) \, dx \\ = 165$$

$$(4) \int_{-2}^{-1} (3x^2 - x + 1) \, dx \\ = \frac{26}{3}$$

$$(9) \int_0^2 (-x^2 - 4x + 5) \, dx \\ = -9$$

$$(14) \int_0^3 (x^2 + 3x + 4) \, dx \\ = \frac{49}{2}$$

$$(5) \int_{-4}^2 (4x^2 - 1) \, dx \\ = 95$$

$$(10) \int_2^3 (x^2 + 4x - 2) \, dx \\ = 16$$

$$(15) \int_{-1}^1 (5x^2 + x + 5) \, dx \\ = 5$$

$$(1) \int_{-2}^4 (2x^2 - 4x + 5) dx \\ = 29$$

$$(6) \int_{-4}^0 (-5x^2 - 3x + 4) dx \\ = -80$$

$$(11) \int_0^5 (-3x^2 - 3x + 2) dx \\ = -\frac{965}{6}$$

$$(2) \int_1^3 (-2x^2 - x - 2) dx \\ = -22$$

$$(7) \int_4^5 (-5x^2 - 2x - 1) dx \\ = -\frac{665}{6}$$

$$(12) \int_{-3}^0 (-x^2 + 5x - 1) dx \\ = -32$$

$$(3) \int_{-4}^{-3} (-3x^2 - 5) dx \\ = -\frac{227}{6}$$

$$(8) \int_{-3}^1 (-3x^2 - 4x - 1) dx \\ = -\frac{38}{3}$$

$$(13) \int_{-4}^{-3} (-5x^2 + 1) dx \\ = -\frac{123}{2}$$

$$(4) \int_{-4}^3 (-2x^2 - 3x + 2) dx \\ = -\frac{287}{6}$$

$$(9) \int_{-5}^4 (4x^2 - 2x) dx \\ = 261$$

$$(14) \int_{-1}^4 (4x^2 + 4x + 5) dx \\ = \frac{725}{6}$$

$$(5) \int_0^4 (x^2 + 5x - 1) dx \\ = \frac{182}{3}$$

$$(10) \int_{-3}^3 (3x^2 + x + 3) dx \\ = 57$$

$$(15) \int_{-4}^3 (5x^2 - 4x - 2) dx \\ = \frac{490}{3}$$